NISTIR XXXX Draft

Face Recognition Technology Evaluation (FRTE)

Part 1: Verification

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This publication is available free of charge from:
https://www.nist.gov/programs-projects/face-recognition-vendor-test-frvt-ongoing

2026/09/01

U.S. Department of Commerce
Howard Lutnick, Secretary

National Institute of Standards and Technology
Arvind Raman, Under Secretary of Commerce for Standards and Technology and NIST Director

ACKNOWLEDGMENTS

The authors are grateful for the long-standing support and collaboration of the the Department of Homeland Security's Science & Technology Directorate (S&T) and the Office of Biometric Identity Management (OBIM).

Additionally, the authors are grateful to staff in the NIST Biometrics Research Laboratory for infrastructure supporting rapid evaluation of algorithms.

DISCLAIMER

Specific hardware and software products identified in this report were used in order to perform the evaluations described in this document. In no case does identification of any commercial product, trade name, or vendor, imply recommendation or endorsement by the National Institute of Standards and Technology, nor does it imply that the products and equipment identified are necessarily the best available for the purpose.

INSTITUTIONAL REVIEW BOARD

The National Institute of Standards and Technology's Research Protections Office reviewed the protocol for this project and determined it is not human subjects research as defined in Department of Commerce Regulations, 15 CFR 27, also known as the Common Rule for the Protection of Human Subjects (45 CFR 46, Subpart A).

FRTE STATUS

This report is a draft NIST Interagency Report, and is open for comment. It is the thirty seventh edition of the report since the first was published in June 2017. Prior editions of this report are maintained on the FRTE website, and may contain useful information about older algorithms and datasets no longer used in FRTE.

FRTE remains open: All tracks of the FRTE are open to new algorithm submissions.

2026-09-01 changes since 2026-07-31:

- We have added results for first algorithms from eight developers: Intpixel Labs Private Ltd, Integrated Software and Technologies Ltd, Norwegian University of Science and Technology, OCR Studio L.L.C.-FZ, OORU Digital Private Ltd, Security Pitch Co Ltd, Softtech Yazilim Teknolojileri Arastirma Gelistirme ve, and UniverseAI.
- We have added results for new algorithms from thirteen returning developers: Advance.AI, InfoCert, Innovatrics, Iris ID Systems, Miaxis Biometrics, Regula Forensics, ROC, Sensetime Group, Sum and Substance Ltd, Verigram, Viante.AI, Vision-Box, and Vsoft.

2026-07-31 changes since 2026-06-15:

- We have added results for first algorithms from two developers: Kaizen Secure Voiz, and Veertec SAL (Offshore).
- We have added results for new algorithms from fourteen returning developers: CAKE by VPBank, Compal Electronics, Dermalog, Facehawk Ltd, FRP LLC, Intozi Tech Pvt Ltd, IQSEC, Nominder, RealSense, Sparktek International Pte Ltd, STCON LLC, TAHAKOM, Technology Control Company, Techainer, and Veridium.

2026-06-15 changes since 2026-05-08:

- We have added results for first algorithms from five developers: AIPRACTIX, Bynn Intelligence Inc, ScanalitiX, Stellar Technologies, and TrueSight Laboratories Tech LLC.
- We have added results for new algorithms from nine returning developers: Cyberlink Corp, Enqura Ltd, Hyundai Motor Company, ID3 Technology, IDENTY, Matrix Comsec Pvt Ltd, Sightropic, Viettel Artificial Intelligence and Data Services Center, and Visica AI Solutions.

2026-05-08 changes since 2026-03-30:

- We have added results for first algorithms from four developers: Ethiopian Labor Market Information System, Face Int, Lishui Shuyan Huirong Technology Co, and YuanLi Technology.
- We have added results for new algorithms from twenty-one returning developers: Advance.AI, BioID GmbH, Tripleize, Dralmey Solutions, FaceLocate, HyperVerge Inc, InfoCert, ioNetworks Inc, Innov8tif Solutions, Kneron Inc, Mobai, Onfido, PAPIL11 S.R.O., PT Technology Biometric Indonesia, Shanghai Jiao Tong University, TNG Digital Sdn Bhd, TOPPAN Security Co, Unissey, UZINFOCOM, Viante.AI, and Vsoft.

2026-03-30 changes since 2026-03-05:

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have added results for first algorithms from three developers: Ice House, Shufti, and Smarg Technologies Pvt Ltd
- ▷ We have added results for new algorithms from four returning developers: AI Implementation Group, Daon, PNPSecure, and Sensetime Group

2026-03-05 changes since 2026-03-02:

- ▷ We have updated the missing results for VisionBox.

2026-03-02 changes since 2026-01-22:

- ▷ We have added results for first algorithms from two developers: FPT Telecom JSC, and Iris ID Systems
- ▷ We have added results for new algorithms from six returning developers: Biometrik Systems Private Limited, GTEL OTS JSC, ID3 Technology, NextMeta Labs, Nominder, and ROC.

2025-01-22 changes since 2025-12-19:

- ▷ We have added results for first algorithms from five developers: CAKE by VPBank, RealSense, Sum and Substance Ltd, Transmit Security Ltd, and Vaidio.
- ▷ We have added results for new algorithms from eleven returning developers: Beijing DeepSense Technologies, Compal Electronics, Candour Biometrics, Clearview AI Inc, IE SARSENOV, Miaxis Biometrics, Onfido, Paravision, Shanghai Jiao Tong University, STCON LLC, and Toshiba.

2025-12-19 changes since 2025-09-30:

- ▷ We have added results for first algorithms from one developers: Sparktek International Pte Ltd.
- ▷ We have added results for new algorithms from 13 returning developers: BioID GmbH, Chunghwa Telecom, Dermalog, Hyundai Motor Company, IDENTY, Innovatrics, InterLink labs, Matrix Comsec Pvt Ltd, Sightropic, Trust Stamp, Viettel Artificial Intelligence and Data Services Center, Viettel High Technology, and Vsoft.

2025-09-30 changes since 2025-08-13:

- ▷ We have added results for first algorithms from three developers: Biometrik Systems Private Limited, OOTO AI, and Toss.
- ▷ We have added results for new algorithms from four returning developers: Jaak IT, Online Mobile Services JSC, Regula Forensics, and Unissey

2025-08-13 changes since 2025-07-15:

- ▷ We have added results for first algorithms from four developers: BTS Digital, IE SARSENOV, UZINFOCOM, and Vsoft.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have added results for new algorithms from seventeen returning developers: 2060.io, Beijing DeepSense Technologies, ID R&D Inc, Innovatrics, IntelliVIX, Metsakuur, Mobai, NextMeta Labs, Neurotechnology, Onfido, QazSmartVision.AI, ROC, STCON LLC, Toshiba, VinBigData, Vision-Box, and Visica AI Solutions.

2025-07-15 changes since 2025-05-28:

- ▷ We have added results for first algorithms from three developers: Matrix Comsec Pvt Ltd, PNPSecure, and Urielsoft Co. Ltd.
- ▷ We have added results for new algorithms from eleven returning developers: Daon, Dermalog, Hyundai Motor Company, Incode Technologies Inc, Intel Research Group, Kloudspot Inc, Omnigarde Ltd, Paravision, Universidade de Coimbra, Viante.AI, and Vietnam Posts and Telecommunications Group.

2025-05-28 changes since 2025-04-25:

- ▷ We have added results for new algorithms from seven returning developers: BioID GmbH, Cubox, DiDi ChuXing Technology, InterLink labs, IDENTY, PAPIL11 S.R.O., and Trust Stamp.

2025-04-25 changes since 2025-03-18:

- ▷ We have added results for first algorithms from seven developers: ArVİs Teknoloji San.Tic.A.Ş., Compal Electronics, FCI Vietnam, GTEL OTS JSC, Shenzhen Polytechnic University, Sightropic, and TNG Digital Sdn Bhd.
- ▷ We have added results for new algorithms from eleven returning developers: Awidit Systems, Cyberextruder, FRP LLC, Kasikorn Labs, NCS Pte Ltd, Nominder, Shanghai Jiao Tong University, STCON LLC, Tech5 SA, Veridas Digital Authentication Solutions S.L., and Viettel High Technology.

2025-03-18 changes since 2025-03-05:

- ▷ We have added results for first algorithms from one developers: FaiceTech Ltd.
- ▷ We have added results for new algorithms from ten returning developers: Beijing DeepSense Technologies, BioID Technologies SA, EI Networks Private Ltd, Hangzhuo AIlu Network Information Technology, Innovatrics, Private Identity LLC, Samsung S1 Corp, TigerIT Bangladesh Ltd, Via Technologies Inc, and Viettel Cyberspace Center.

2025-03-05 changes since 2025-01-30:

- ▷ We have added results for first algorithms from three developers: 2060.io, MB Bank, and Visica AI Solutions
- ▷ We have added results for new algorithms from twelve returning developers: AFR Engine, AI Implementation Group, Authme, Hyundai Motor Company, FacePhi, Inspur (Beijing) Electronic Information Industry Co, LightVision Inc, QazSmartVision.AI, ROC, Toshiba, Techainer, and VinBigData.

2025-01-30 changes since 2024-12-18:

- ▷ We have added results for first algorithms from four developers: Innov8tif Solutions, Maverick Research, PT Technology Biometric Indonesia, and TTI Group.
- ▷ We have added results for new algorithms from ten returning developers: Coretech Knowledge Inc, Intel Research Group, IQSEC, i2v Systems, PAPIL11 S.R.O., Paravision, Taiwan Mobile, Touchless ID, useB, and Viante.AI.

2024-12-18 changes since 2024-11-08:

- ▷ We have added results for first algorithms from six developers: BioID GmbH, InterLink labs, KYB-AI Ltd, Keyless Technologies, Pyramid Cyber Security + Forensic (P),and Syscake S.r.l..
- ▷ We have added results for new algorithms from sixteen returning developers: Alchera Inc, Awidit Systems, Cubox, DeCloak Intellegences, Beijing DeepSense Technologies, Innovatrics, Kasikorn Labs, Mitek Systems, Mukh Technologies, NCS Pte Ltd, NextMeta Labs, Nominder, PT Autentika Digital Indonesia, Sensetime Group, Shanghai Jiao Tong University, and STCON LLC.

2024-11-08 changes since 2024-10-07:

- ▷ We have added results for first algorithms from two developers: Hyderabad-Centre for Development of Advanced Computing, and TAHAKOM.
- ▷ We have added results for new algorithms from ten returning developers: Candour Biometrics, Cyberextruder, DICIO, First Credit Bureau Kazakhstan, Metsakuur, Mobbeel Solutions, Samsung S1 Corp, Vietnam Payment Solution, VinBigData, and Yuan High-Tech Development.

2024-10-07 changes since 2024-09-16:

- ▷ We have added results for first algorithms from three developers: Enqura, Kloudspot, and Penacomp Systemy Informatyczne.
- ▷ We have added results for new algorithms from fourteen returning developers: Chunghwa Telecom, Cyberlink, Dactionable Technologies, Daon, FPT Information System, Hangzhuo AIlu Network Information Technology, Kakao Bank, Panasonic R+D Center Singapore, Paravision, Private Identity, QazSmartVision.AI, Toshiba, Vietnam Posts and Telecommunications Group, and UNICC-Solution Architecture Section.

2024-09-16 changes since 2024-08-09:

- ▷ We have added results for first algorithms from one developer: Taiwan Mobile.
- ▷ We have added results for new algorithms from eleven returning developers: Cognitec Systems GmbH, Coretech Knowledge Inc, FaceOnLive Inc, Idemia, Intel Research Group, Luxand Inc, Ovision, PAPIL11 S.R.O., Unissey, useB, and Vivotek.

2024-08-09 changes since 2024-07-03:

- ▷ We have added results for first algorithms from seven developers: Affinsys AI Ltd, AI Implementation Group, Celloscope Limited, Hyundai Motor Company, ID R&D Inc, LightVision Inc and NextMeta Labs.

- We have added results for new algorithms from eleven returning developers: AFR Engine, Antheus Technologia, CTBC Bank, Innovatrics, Mitek Systems, Neurotechnology, Nominder, Seventh Sense Artificial Intelligence, Universidade de Coimbra, University of Surrey-CVSSP, Viettel Cyberspace Center, Vision-Box and Yoti.

2024-07-03 changes since 2024-05-30:

- We have include all algorithms evaluated since the ongoing FRTE evaluation started in 2017. Previously we listed only results for each developer's two most recent implementations. The main results page has been revised with a "by developer" tab which lists only the most accurate algorithm from each developer, where accuracy is defined by the lowest FNMR value for the visa-border dataset.
- We have added results for first algorithms from two developers: Taiwan Computer Vision and Telexper International.
- We have added results for new algorithms from fifteen returning developers: Alchera Inc, Beijing DeepSense Technologies, Cyberlink Corp, Dermalog, FacePhi, Facia.ai, FRP LLC, NCS Pte Ltd, Omnigarde Ltd, Online Mobile Services JSC, Qnap Security, ROC, STCON LLC, Videmo Intelligente Videoanalyse, and Viettel Group.

2024-05-30 changes since 2024-04-17:

- We have added results for first algorithms from three developers: ALTTEKGlobal, Fourthline, and IQSEC
- We have added results for new algorithms from thirteen returning developers: Candour Biometrics, Chunghwa Telecom, Clearview AI Inc, Cognitec Systems GmbH, Incode Technologies Inc, ioNetworks Inc, HyperVerge Inc, Mobbeel Solutions, Panasonic R+D Center Singapore, QazSmartVision.AI, Vietnam Posts and Telecommunications Group, VinBigData and Yuan High-Tech Development
- We have retired results for eleven algorithms per our policy to only list results for eleven algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-04-17 changes since 2024-03-26:

- We have added results for first algorithms from four developers: Ovision, Vietnam National Cyber Security Technology Corporation, Vivotek, and Vocalize.
- We have added results for new algorithms from four returning developers: Intel Research Group, Kakao Bank, Luxand, and Turkcell Technology.
- We have retired results for nine algorithms per our policy to only list results for four algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-03-26 changes since 2024-02-21:

- We have added results for first algorithms from six developers: FaceLocate, FPT Education, FPT Information System, Kogniza Technology, i2v Systems, and Seamfix.
- We have added results for new algorithms from eleven returning developers: AFR Engine, Aware, Beijing Hisign Technology, Cu-Face, Dactionable Technologies, First Credit Bureau Kazakhstan, Innovatrics, Nominder, Securif AI, Veridium, and Viettel Cyberspace Center.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-02-21 changes since 2024-01-22:

- We discontinued the visa-visa benchmark in FRTE since Feburary 14th 2024. For highly performing algorithms, the only remaining false negative error in that set were in babies and in a few greyscale poor quality half-tone scans of paper photos.
- We have added results for first algorithms from four developers: Online Mobile Services JSC, Openedge Technologies, TNI Technoology, and Yoti.
- We have added results for new algorithms from nine returning developers: CMC Institute of Science and Technology, City and County of Honolulu, Dermalog, Euronovate SA, Maxvision Technology, Megvii/Face++, Neurotechnology, Toshiba, useB, and Universidade de Coimbra.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-01-22 changes since 2023-12-15:

- We have added results for first algorithms from six developer: GPS Vietnam Trading, iCOMM Media & Tech, PAPIL11 S.R.O., QazSmartVision.AI, Sparsh CCTV, and VNIS Joint Stock.
- We have added results for new algorithms from eight returning developers: Cyberlink Corp, Intellivision, Hangzhuo AIlu Network Information Technology, Lebentech Biometrics, Momentum Digital, Regular Biometrics Solutions, ROC, and Via Technologies Inc.
- We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-12-15 changes since 2023-11-21:

- We have added results for first algorithms from five developer: CMC University, Element System Solutions Company, Fraud.com, Techainer, and Vietnam Payment Solutions.
- We have added results for new algorithms from thirteen returning developers: Accurascan, Adera Global, Alchera, Aware, Clearview AI, Cognitec Systems, Incode Technologies Inc, Inspur (Beijing) Electronic Information, IntelliVIX, ioNetworks, NHN Corp, PT Autentika Digital Indonesia, and Qnap Security.
- We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-11-21 changes since 2023-10-27:

- We have added results for new algorithms from ten returning developers: Coretech Knowledge Inc, Beijing DeepSense Technologies, General Interface Solutions Holding Ltd, Glory, Kakao Brain, Nominder, Smarvist Teknoloji, STCON LLC, Suprema AI Inc, and Yuan High-Tech Development
- We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-10-27 changes since 2023-09-29:

- ▷ We have added results for first algorithms from two developer: Authme and Dactionable Technologies.
- ▷ We have added results for new algorithms from twelve returning developers: Intel Research Group, Guangzhou Pixel Solutions, Kasikorn Labs, Kakao Bank, Luxand, Megvii/Face++, Panasonic R+D Center Singapore, Private Identity LLC, Recognito, Seventh Sense Artificial Intelligence, UNICC-Solution Architecture Section, and Universidade de Coimbra.
- ▷ We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-09-29 changes since 2023-09-09:

- ▷ We have added results for first algorithms from one developer: Intozi Tech.
- ▷ We have added results for new algorithms from seven returning developers: Euronovate, Maxvision Technology, Miaxis Biometrics, Mukh Technologies, Toshiba, University of Surrey-CVSSP, and useB.
- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-09-09 changes since 2023-08-16:

- ▷ We have renamed and split FRVT into two sets of technology evaluations: The Face Recognition Technology Evaluation (FRTE) is the umbrella for the ongoing 1:1, 1:N and twins disambiguation tracks; the Face Analysis Technology Evaluation (FATE) holds the morph and attacked detection tracks, the quality assessment tracks, and the new age estimation tracks.
- ▷ We have added results for first algorithms from two developers: Facehawk and Regular Biometrics Solutions.
- ▷ We have added results for new algorithms from eight returning developers: CMC Institute of Science and Technology, Canon, Cu-Face, EI Networks, ICM Airport Technics, Securif AI, Securifai, Vietnam Posts and Telecommunications Group, and Vision Intelligence Center of Meituan
- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-08-16 changes since 2023-07-21:

- ▷ We have added results for first algorithms from one developers: Facia.ai.
- ▷ We have added results for new algorithms from fifteen returning developers: Accurascan, Alchera, FOO, Hangzhuo AIlu Network Information Technology, HyperVerge, ioNetworks, Innovatrics, Nominder, Rank One Computing, Neurotechnology, Tech5, Tripleize, Unissey, UX Labs, and Verigram.
- ▷ We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-07-21 changes since 2023-06-16:

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have added results for first algorithms from five developers: DeCloak Intellegences, Innominds Software SEZ India, Trust Stamp, Vcortex Labs, and Viettel Cyberspace Center
- ▷ We have added results for new algorithms from fourteen returning developers: AFR Engine, Alice Biometrics, Euronovate SA, IMDS Software, Idemia, IntelliVIX, Intellivision, NHN, Pangiam, STCON, Samtech InfoNet, Veridium, VinBigData, and Vision-Box
- ▷ We have retired results for eleven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have stopped using wild images in FRVT since May 1st 2023. All results for that set have been removed from the website, and will be removed from future PDF reports.

2023-06-16 changes since 2023-04-20:

- ▷ We have added results for first algorithms from seven developers: EI Networks Private Ltd, General Interface Solutions Holding Ltd, Recognito, Serendipity Ltd, ST Engineering, and useB.
- ▷ We have added results for new algorithms from eighteen returning developers: Advance.AI, AYF Technology, Cyberlink Corp, Incode Technologies Inc, Intel Research Group, Intellibrain Technological Projects, Maxvision Technology, Megvii/Face++, Omnigarde Ltd, Paravision, Qnap Security, Seventh Sense Artificial Intelligence, Shanghai Jiao Tong University, Veridas Digital Authentication Solutions S.L., Universidade de Coimbra, Yuan High-Tech Development, UNICC-Solution Architecture Section, and Verihubs,verihubs-inteligensia.
- ▷ We have retired results for 15 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-04-20 changes since 2023-04-04:

- ▷ We have added results for first algorithms from one developers: IDENTY.
- ▷ We have added results for new algorithms from three returning developers: Metsakuur, Autentika Digital Indonesia, and Verigram.
- ▷ FRVT will re-open 2023-05-01.
- ▷ We have retired results for 3 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-04-04 changes since 2023-03-09:

- ▷ We have added results for first algorithms from six developers: Aratek Biometrics City and County of Honolulu, FOO, Intelligent Control Technology NCS, and Swsam Solutions.
- ▷ We have added results for new algorithms from nine returning developers: Cubox, Glory, InsightFace AI, Miaxis Biometrics, Mukh Technologies, Turkcell Technology, Samsung S1, Via Technologies, and Vision Intelligence Center of Meituan
- ▷ We have retired results for 7 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-03-09 changes since 2023-02-01:

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have added results for first algorithms from eight developers: Biometric LLC (biometric.vision) KZ, Candour Biometrics, Fast Enterprises, KakaoBank, Mitek Systems, Nominder, Private Identity, and UNICC-Solution Architecture Section.
- ▷ We have added results for new algorithms from 22 returning developers: AFR Engine, Biocube Matrics, CMC Institute of Science and Technology, Cloudwalk - Moontime Smart Technology, Cyberlink Corp, Beijing DeepSense Technologies, First Credit Bureau Kazakhstan, Enface, Hangzhuo AIlu Network Information Technology, Herta Security, IMDS Software, Inspur (Beijing) Electronic Information Industry Co, Intellivision, MicroFocus, Neurotechnology, Pangiam, NSENSE Corp, STCON LLC, Touchless ID, University of Surrey-CVSSP, Vision-Box, and YooniK.
- ▷ We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have introduced new set of non-frontal portrait to border comparisons. The new images are described in section 2.3 and their use in seciton 3.2.

2023-02-01 changes since 2022-12-15:

- ▷ We have added results for first algorithms from four developers: CU-Face, Korea ID, Onfido, and TrueID-VNG.
- ▷ We have added results for new algorithms from 21 returning developers: Alchera, Armatura, Cogent-Thales, Dermalog, Didi ChuXing Global Face, Gorilla, Hyperverge, Innovatrics, Intel Research, IntelliVIX, Intema-LGL, Kasikorn Labs, Paravision, Rank One Computing, Sensetime Group, Suprema AI, Tech5, Unissey, U. Coimbra Visteam, Vixvizion (Imagus), and Yuan High-Tech Development.
- ▷ We have retired results for 20 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have introduced new set of non-frontal portrait to border comparisons. The new images are described in section 2.3 and their use in seciton 3.2.

2022-12-15 changes since 2022-11-06:

- ▷ We have added results for first algorithms from four developers: Miaxis Biometrics, PT Autentika Digital Indonesia, PT Qlue Performa Indonesia, and STCON.
- ▷ We have added results for new algorithms from 14 returning developers: Adera Global, Aiseemu Technology, Chunghwa Telecom,chtface, FRP, Griaule, Line Corporation, Maxvision Technology, Mukh Technologies, Papilon Savunma, Qnap Security, Realnetworks, Securif AI, SQIsoft, and Veridium.
- ▷ We have retired results for 10 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-11-06 changes since 2022-09-26:

- ▷ We have added results for first algorithms from six developers: AFR Engine, CMC Institute of Science and Technology, Saga Densan Center, Turkcell Technology, UXLabs, and Wise AI SDN BHD.
- ▷ We have added results for new algorithms from 14 returning developers: Coretech Knowledge, Cloudwalk - Moontime, Cloudmatrix, Deepglint, Guangzhou Pixel Solutions, Hangzhuo AIlu Network Information Technology, NEO Systems, One More Security, Palit Microsystems, Panasonic R+D Center Singapore, Samsung S1, Seventh Sense Artificial Intelligence, Touchless ID, and Veridas Digital Authentication Solutions S.L.

- We have retired results for 10 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-09-26 changes since 2022-08-30:

- We have added results for first algorithms from three developers: Codeline, First Credit Bureau Kazakhstan, and InfoCert.
- We have added results for new algorithms from 14 returning developers: Advancegroup, Armatura LLC, Beijing Hisign Technology, Cybercore, Cyberlink Corp, Herta Security, ICM Airport Technics, InsightFace AI, Metsakuur, NSENSE Corp, Samsung-SDS, Videmo Intelligente Videoanalyse, Vietnam Posts and Telecommunications Group, and Vision Intelligence Center of Meituan.
- We have retired results for 11 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-08-30 changes since 2022-07-29:

- We have added results for first algorithms from two developers: Aximetria, Intellibrain Technological Projects
- We have added results for new algorithms from twelve returning developers: Alchera Inc, Dermalog, Idemia, Incode Technologies Inc, Intellivision, Kasikorn Labs, Megvii/Face++, Techsign, TuringTech.vip, Universidade de Coimbra, Verijelais, Vixvizon
- We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-07-29 changes since 2022-06-27:

- We have added results for first algorithms from seven developers: FRP LLC (Hawaii), IMDS Software, Inspur (Beijing) Electronic Information Industry, Intema - LGL Group, PAPAGO, Qaz Biometric Systems, and VIDA-Digital Identity
- We have added results for new algorithms from nine returning developers: Cyberextruder, Glory, Maxvision Technology, Rank One Computing, Securif AI, Suprema AI, Suprema ID, Toshiba, and Yuan High-Tech Development.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-07-29 changes since 2022-06-27:

- We have added results for first algorithms from seven developers: FRP LLC (Hawaii), IMDS Software, Inspur (Beijing) Electronic Information Industry, Intema - LGL Group, PAPAGO, Qaz Biometric Systems, and VIDA-Digital Identity
- We have added results for new algorithms from nine returning developers: Cyberextruder, Glory, Maxvision Technology, Rank One Computing, Securif AI, Suprema AI, Suprema ID, Toshiba, and Yuan High-Tech Development.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-06-27 changes since 2022-06-03:

- ▷ We have added results for first algorithms from two developers: Krungthai Bank, and Smartbiometrik.
- ▷ We have added results for new algorithms from thirteen returning developers: Aiseemu, Corsight, Digidata, Griaule, Guangzhou Pixel Solutions, Hangzhuo AI Network Information Technology, Neurotechnology, Real Networks, Samsung S1, Sensetime Group, Smart Engines, Verihubs Inteligensia, and VinBigData.
- ▷ We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-06-03 changes since 2022-05-05:

- ▷ We have added results for first algorithms from seven developers: Jaak IT, Metsakuur, Palit Microsystems, Smarvist Teknoloji, and Touchless ID.
- ▷ We have added results for new algorithms from sixteen returning developers: Cyberlink, FaceOnLive, Kakao Enterprise, Line Corporation (Line Clova), Multi-Modality Intelligence, NEO Systems, and Unissey
- ▷ We have retired results for four algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have moved the results for the twenty human-difficult pairs used in the May 2018 paper *Face recognition accuracy of forensic examiners, superrecognizers, and face recognition algorithms* by Phillips et al. [1]. to the algorithm-specific report cards (example: PDF).
- ▷ Likewise, we have added figures showing impostor distribution shifts across demographics to the report card.

2022-05-05 changes since 2022-03-18:

- ▷ We have added results for first algorithms from seven developers: Accurascan, DICIO, FacePhi, Pangiam, University of Surrey-CVSSP, and Veridium.
- ▷ We have added results for new algorithms from sixteen returning developers: ACI Software, Canon Inc, Cloudwalk - Moontime Smart Technology, Cybercore,

2022-05-05 changes since 2022-03-18:

- ▷ We have added results for first algorithms from seven developers: Accurascan, DICIO, FacePhi, Pangiam, University of Surrey-CVSSP, and Veridium.
- ▷ We have added results for new algorithms from sixteen returning developers: ACI Software, Canon Inc, Cloudwalk - Moontime Smart Technology, Cybercore, Cyberextruder, Gemalto Cogent, HyperVerge Inc, KuKe3D Technology, Megvii/Face++, Mobbeel Solutions, Panasonic R+D Center Singapore, Qnap Security, Samsung-SDS, Vietnam Posts and Telecommunications Group, Viettel Group, and Vision Intelligence Center of Meituan.
- ▷ We have retired results for 12 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2022-03-18 changes since 2022-02-23:

- ▷ We have added support for the detection of multiple people in a single image (see Section 1.2). Specifically the API allows an algorithm to extract features from one or more faces it detects in an image. NIST scores such cases as a correct match when any detected face matches the reference photo, and as a false positive when either face matches a non-mated reference photo. The expected effect of doing this will be to improve reported false non-match rates, and to minimally elevate false match rates. This technique was only applied to images of type "border" and "kiosk".
- ▷ We have added results for first algorithms from four developers: IntelliVIX, Kasikorn Labs, Lebentech Biometrics, and Wicket.
- ▷ We have added results for new algorithms from 10 returning developers: Chunghwa Telecom, Cloudmatrix, Beijing DeepSense Technologies, FarBar Inc, Imagus Technology Pty, Intellivision, Maxvision Technology, NHN Corp, Seventh Sense Artificial Intelligence, and Verigram.
- ▷ We have retired results for 4 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-02-23 changes since 2022-01-24:

- ▷ We have added results for first algorithms from four developers: AFIS and Biometrics Consulting, Digidata, Graymatics, Hangzhuo AIlu Network Information Technology, KnowUTech LLC, Sukshi Technology Innovation, T4iSB, and TuringTech.vip
- ▷ We have added results for new algorithms from 18 returning developers: Cognitec Systems GmbH, GeoVision Inc, Glory, Herta Security, Intel Research Group, InsightFace AI, Kakao Enterprise, N-Tech Lab, Omnigarde Ltd, Papilon Savunma, Paravision, Realnetworks Inc, Reveal Media Ltd, Shenzhen Inst Adv Integrated Tech CAS, Suprema AI Inc, Toshiba, Universidade de Coimbra, and Yuan High-Tech Development
- ▷ We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-01-24 changes since 2022-01-20:

- ▷ We have added results for new algorithms from one returning developer: Vocord.

2022-01-20 changes since 2021-12-18:

- ▷ We have added results for first algorithms from four developers: Armatura, Beyne.AI, One More Security, and VinBigData
- ▷ We have added results for new algorithms from 19 returning developers: AuthenMetric, BOE Technology Group, Cybercore, Cyberlink, Dahua Technology, FaceTag Co, Innovatrics, Megvii, Mobbeel Solutions, Neurotechnology, Oz Forensics, Rank One Computing, Regula Forensics, Samsung S1, Securif AI, Sensetime Group, TigerIT Americas, Videmo Intelligente Videoanalyse, and YooniK.
- ▷ We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

: **2021-12-16** changes since 2021-11-22:

- We have added results for first algorithms from five developers: Alfabeta, Cloudmatrix, Euronovate SA, FaceOnLive Inc, and Mobipin Technology.
- We have added results for new algorithms from ten returning developers: ACI Software, ITMO University, NEO Systems, Guangzhou Pixel Solutions, Panasonic R+D Center Singapore, Qnap Security, Scanovate, Tevian, Unissey, and Vietnam Posts and Telecommunications Group.
- We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- We have revamped the figure showing performance on 20 pairs of open-source images. It now color-codes false negatives and postives against a default threshold value.

2021-11-22 changes since 2021-10-28:

- We have added results to the website for kiosk-collected images where the design and geometry configuration mean that many images have considerable downward pitch angle. In some images, the face is partially cropped. Some images have other background faces.
- We have stopped using child exploitation images in FRVT, as we lost access to the imagery. All results for that set have been removed from the website, and will be removed from future PDF reports.
- We have added results for first algorithms from seven new developers: CUDO Communication, Daon, KuKe3D Technology, Mantra Softech India, Maxvision Technology, Multi-Modality Intelligence, and Samsung-SDS.
- We have added results for new algorithms from seven returning developers: Acer Incorporated, Cloudwalk-Moontime Smart Technology, Gorilla Technology, ID3 Technology, Incode Technologies, NSENSE Corp., and SQIsoft.
- We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-10-28 changes since 2021-09-08:

- We have substantially revised the algorithm-specific report cards that are linked from the FRVT results page. (Example: HTML).
- We have added results for first algorithms from eight new developers: Beijing Mendaxia Technology, Beijing Hisign Technology, Biocube Matrics, Clearview AI, Reveal Media, Toppan ID Gate, Verigram, and Viettel High Technology.
- We have added results for new algorithms from thirty returning developers: 20Face, 3divi, Canon Inc Chunghwa Telecom, Corsight, Decatur Industries, Deepglint, Dermalog, FaceTag, Fiberhome Telecommunication Technologies, GeoVision, ICM Airport Technics, Imagus Technology, InsightFace AI, Kakao Enterprise, Kookmin University, Line Corporation, N-Tech Lab, NotionTag Technologies, Realnetworks, Suprema ID, Taiwan-Certificate Authority, Toshiba, Tripleize, Trueface.ai, Veridas Digital Authentication, Visidon, VisionLabs, YooniK, and Yuan High-Tech Development.
- We have retired results for twenty algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2021-09-08 changes since 2021-08-02:

- ▷ We have added results for first algorithms from seven new developers: Griaule, SQIsoft, Qnap Security, Techsign, Smart Engines, Verihubs, and Wuhan Tianyu Information Industry.
- ▷ We have added results for new algorithms from sixteen returning developers: ADVANCE.AI, AuthenMetric, CloudSmart Consulting, Code Everest Pvt, Cognitec Systems, Thales Gemalto Cogent, Intel Research Group, Omnigarde, Oz Forensics, Rank One Computing, Samsung S1 Corp, Securif AI, Tevian, TigerIT Americas, Universidade de Coimbra, and Vigilant Solutions
- ▷ We have retired results for eleven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-08-02 changes since 2021-06-25:

- ▷ We have added results for first algorithms from eight new developers: Bee the Data, Closeli Inc, Coretech Knowledge Inc, Deepsense (France), ioNetworks Inc, Kakao Pay Corp, Seventh Sense Artificial Intelligence, and SK Telecom.
- ▷ We have added results for new algorithms from fifteen returning developers: Alchera Inc, Adera Global PTE, Aware, Bresee Technology, Cyberlink Corp, Expasoft LLC, Fujitsu Research and Development Center, Gorilla Technology, Idemia, Neurotechnology, NEO Systems, NHN Corp, Paravision, Panasonic R+D Center Singapore, and Shenzhen University-Macau University of Science and Technology.
- ▷ We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-06-25 changes since 2021-05-21:

- ▷ We have added results for first algorithms from six new developers: Alice Biometrics, BOE Technology Group, Fincore, Neosecu, Sodec App, and Yuntu Data and Technology.
- ▷ We have added results for new algorithms from seven returning developers: Incode Technologies, HyperVerge, Mobbeel Solutions, Guangzhou Pixel Solutions, Remark Holdings, Sensetime, and Vietnam Posts and Telecommunications Group.
- ▷ We have retired results for four algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-05-21 changes since 2021-04-26:

- ▷ We have added results for first algorithms from five new developers: Ekin Smart City Technologies, Suprema ID, Tripleize, Taiwan-Certificate Authority, and Vision Intelligence Center of Meituan.
- ▷ We have added results for new algorithms from eight returning developers: ID3 Technology, Imagus Technology, Momentum Digital, N-Tech Lab, NSENSE, Shanghai Jiao Tong University, Vision-Box, and Yuan High-Tech Development
- ▷ We have retired results for seven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-04-26 changes since 2021-04-16:

- We have added results for first algorithms from three new developers: Quantasoft, Rendip, and NEO Systems.
- We have added results for new algorithms from four returning developers: 3Divi, Realnetworks, Veridas Digital Authentication Solutions, and Universidade de Coimbra.
- We have retired results for three algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-04-16 changes since 2021-03-19:

- We have added results for first algorithms from six new developers: 20Face, Beijing DeepSense Technologies, BitCenter UK, Enface, FaceTag, InsightFace AI, Line Corporation, Lema Labs, Nanjing Kiwi Network Technology, Omnigarde, Regula Forensics, and Suprema.
- We have added results for new algorithms from ten returning developers: CloudSmart Consulting, Dermalog, GeoVision, Neurotechnology, Panasonic R+D Center Singapore, Samsung S1, Securif AI, Trueface.ai, Vigilant Solutions, and Visidon.
- We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-03-19 changes since 2021-03-05:

- We have added results for first algorithms from six new developers: Ajou University, AuthenMetric, Code Everest, Corsight, Papilon Savunma, and NHN Corp
- We have added results for new algorithms from seven returning developers: Alchera, Deepglint, Fiberhome Telecommunication Technologies, Kakao Enterprise, Kookmin University, Megvii/Face++, and NotionTag Technologies.
- We have updated many of the hyperlinked HTML report-cards to include seven figures on demographic dependence. Figures of this kind first appeared, and are documented in, the December 2019 document, NIST Interagency Report 8280 on demographic differentials in face recognition. The figures quantify false negative dependence on demographics using "visa-border" comparisons, and false positive dependence using comparisons of "application" photos that uniformly of quality and similar to visa photos.

2021-03-05 changes since 2021-01-19:

- We have added results for first algorithms from three new developers: IVA Cognitive, Mobbeel, and MoreDian Technology.
- We have added results for new algorithms from returning developers: Ability Enterprise - Andro Video, ACI Software, Adera Global, AnyVision, BioID Technologies, China Electronics Import-Export, Cognitec Systems, Fujitsu Research and Development Center, Glory, Guangzhou Pixel Solutions, Hengrui AI Technology, Incode Technologies, Intel Research, iQIYI, Mobai, Oz Forensics, Paravision, VisionLabs, and Xforward AI Technology.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- We have added a new "resources" tab to the main webpage. It includes sortable columns for data related to speed, model size, storage, and memory consumption.
- We have retired results for 13 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-01-19 changes since 2020-12-18:

- This report adds results for first algorithms from four developers: Herta Security, Irex AI, Shenzhen University-Macau University of Science and Technology, and Vietnam Posts and Telecommunications Group. See Table 21 for more information.
- The report also includes results for thirteen developers who have previously submitted algorithms: Bresee Technology, Canon (previously Canon Information Technology (Beijing)), Cyberlink, CSA IntelliCloud Technology, Dahua Technology, ID3 Technology, Imagus Technology (Vixvizion), Moontime Smart Technology, N-Tech Lab, Thales Cogent, Veridas Digital Authentication Solutions, Vocord, and Yuan High-Tech Development.
- We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2020-12-18 changes since 2020-10-09:

- This report adds results for first algorithms from ten developers: BitCenter UK, CloudSmart Consulting, Cubox, Institute of Computing Technology, Naver Corp, Minivision, NSENSE Corp, Viettel Group, Visage Technologies, and Xiamen University. See Table 21 for more information.
- The report also includes results for eighteen developers who have previously submitted algorithms: ADVANCE.AI, Awidit Systems, Chosun University, Dermalog, GeoVision, ICM Airport Technics, Idemia, Institute of Information Technologies, Kakao Enterprise, Neurotechnology, Panasonic R+D Center Singapore, Rank One Computing, Sensetime Group, Shanghai Jiao Tong University, TigerIT Americas LLC, Vigilant Solutions, Winsense, and YooniK
- We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since September 18, 2020:

- This report adds results for first algorithms from five developers: Aigen, Cortica, Kookmin University, Securif AI and Vinai.
- The report also includes results for three developers who have previously submitted algorithms: Fujitsu Laboratories, Hengrui AI, and X-Forward AI.
- In the per-algorithm report-cards linked from tables and the main webpage, we have added a chart to showing reduction in error rates over the course of FRVT i.e. from 2017 onwards for all algorithms supplied by that developer. Similarly we have added a chart showing error rate reductions for our test of protective face mask verification.
- We plan to continue evaluating algorithms on various mask datasets. We hold that algorithms should be capable of detecting masks and verifying identity of all combinations of masked and unmasked faces. We have accordingly increased the amount to time allowed to extract those features from 1.0 to 1.5 seconds.

Changes since August 25, 2020:

- This report adds results for first algorithms from eight new developers. Akurat Satu Indonesia, Cybercore, Decatur Industries, Innef Labs, Satellite Innovation/Eocortex, Expasoft, and Mobai.
- The report includes results for seven developers who have previously submitted algorithms: 3Divi, BioID Technologies, Incode Technologies, Innovatrics, iSAP Solution, Synology, and Tevian.
- We have retired results for five algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since July 27, 2020:

- We have introduced per-algorithm report sheets. These are HTML documents linked from the accuracy tables in this report (i.e. Table 89) and on the FRVT 1:1 homepage. The sheets contain interactive graphics allowing, for example, mouseover exploration of FNMR(T) and FMR(T). Some of their content had previously appeared in this document.
- This report adds results for algorithms from six new developers. ACI Software, Bresee Technology, Fiberhome Telecommunication Technologies, Imageware Systems, Oz Forensics, and Pensees.
- The report includes results for thirteen developers who have previously submitted algorithms: Canon Information Technology (Beijing), Cyberlink, Dahua Technology, Gorilla Technology, ID3 Technology, Intel Research Group, iQIYI Inc, Momentum Digital, Netbridge Technology, Tech5 SA, Shenzhen AiMall Tech, Vigilant Solutions, and VisionLabs.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since May 18, 2020:

- The report is the first FRVT update since the pandemic closed it from March to June 2020.
- This report includes results for algorithms from nine new developers: GeoVision Inc, Su Zhou NaZhiTianDi Intelligent Technology, YooniK, AYF Technology, PXL Vision AG, Yuan High-Tech Development, Beihang University-ERCACAT, ICM Airport Technics, and Staqu Technologies
- This report includes results for algorithms from 15 returning developers Acer Incorporated, Antheus Technologia, Chosun University, Chunghwa Telecom, Idemia, Moontime Smart Technology, Neurotechnology, Guangzhou Pixel Solutions, Panasonic R+D Center Singapore, Rank One Computing, Scanovate, Shanghai Universiy - Shanghai Film Academy, Synesis, Trueface.ai, and Veridas Digital Authentication Solutions
- We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- We separated timing and other resource consumption from the main participation table. The new Table 55 includes template generation durations for four kinds of images, not just mugshots.
- We have published a separate report, NIST Interagency Report 8311 on accuracy of pre-pandemic algorithms on subjects wearing face masks. We plan to track improvements in accuracy on masked images going forward. In particular, we invite submission of algorithms that can detect whether a person is wearing a mask, extract features from the full face or the exposed periocular region, and do appropriate comparison. We do not intend to evaluate algorithms that assume 100% of images will be of masked individuals.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Changes since March 25, 2020:

- The report is a maintenance release - it does not add any new algorithms, and FRVT has been closed to new algorithms since mid March 2020.
- We modified the primary accuracy summary, Table 89, as follows:
  - For visa images, the column for FNMR at FMR = 0.0001 has been removed. The visa images are so highly controlled that the error rates for the most accurate algorithms are dominated by false rejection of very young children and by the presence of a few noisy greyscale images. For now, two visa columns remain: FNMR at FMR= 10^{-6} and, for matched covariates, FNMR at FMR= 10^{-4}.
  - We have inserted a new column labelled "BORDER" giving accuracy for comparison of moderately poor webcam border-crossing photos that exhibit pose variations, poor compression, and low contrast due to strong background illumination. The accuracies are the worst from all cooperative image datasets used in FRVT.
- Accordingly, we updated the failure-to-template rates in Table 115.
- We withdrew a figure showing how false matches are concentrated in certain visa images used in cross-comparison, because it didn't attempt to include demographic information.

Changes since February 27, 2020:

- The report adds results algorithms from two new developers: Beijing Alleyes Technology, and the Chinese University of Hong Kong. Results for newly submitted algorithms from two other developers will appear in the next report.
- The report adds results for algorithms from thirteen returning developers: ASUSTek Computer, Aware, Cyberlink Corp, Gorilla Technology, Innovative Technology, Kakao Enterprise, Lomonosov Moscow State University, Panasonic R+D Center Singapore, Shenzhen AiMall Technology, Shenzhen Intellifusion Technologies, Synology, Tech5 SA, and Via Technologies.
- Per policy to only list results for two algorithms per developer, we have dropped results for algorithms from Aware, Cyberlink, Gorilla Technology, Kakao Enterprise, Lomonosov Moscow State University, Panasonic R+D Center Singapore, and Tech5 SA.

Changes since January 20, 2020:

- The report adds results for five new developers: Ability Enterprise (Andro Video), Chosun University, Fujitsu Research and Development Center, University of Coimbra, and Xforward AI Technology.
- The report adds results for algorithms from six returning developers: AlphaSSTG, Incode Technologies, Kneron. Shanghai Jiao Tong University, Vocord, and X-Laboratory.
- We have corrected template comparison timing numbers for algorithms submitted September 2019 to January 2020. The values reported previously were slower due to a software bug.
- We have dropped results for algorithms from Vocord and Incode per policy to only list results for two algorithms per developer.
- The FRVT 1:1 homepage has been updated with latest accuracy results.
- The FRVT 1:N homepage now includes an update to the September 2019 NIST Interagency Report 8271. The new report adds results for one-to-many search algorithms submitted to NIST from June 2019 to January 2020.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Changes since January 6, 2020:

- Section 2 has been updated to better describe the Visa and Border images. The caption for Table 89 has been updated to better relate the accuracy values to particular image comparisons.
- The report adds results for five new developers: Acer, Advance.AI, Expasoft, Netbridge Technology, and Videmo Intelligente Videoanalyse.
- The report adds results for algorithms from 7 returning developers: China Electronics Import-Export Corp, Intel Research Group, ITMO University, Neurotechnology, N-Tech Lab, Rokid, and VisionLabs.
- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: N-Tech Lab, Neurotechnology, ITMO, Visionlabs, and CEIEC.
- The FRVT homepage has been updated with latest accuracy results.

Changes since November 11, 2019:

- Table 55 has been updated to include runtime memory usage. This is the first time such a quantity has been reported. The value is the peak size of the resident set size logged during enrollment of single images.
- We have migrated summary results table to a new platform that supports sortable tables: https://pages.nist.gov/frvt/html/frvt11.html
- The report adds results for four new developers: Antheus Technologia, BioID Technologies SA, Canon Information Tech. (Beijing), Samsung S1 (listed in the tables as S1), and Taiwan AI Labs.
- The report adds results for algorithms from 13 returning developers: Anke Investments, Chunghwa Telecom, Deepglint, Institute of Information Technologies, iQIYI, Kneron, Ping An Technology, Paravision, KanKan Ai, Rokid Corporation, Shanghai Universiy - Shanghai Film Academy, Veridas Digital Authentication Solutions, and Videonetics Technology.
- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: remarkai-000, veridas-001, sensetime-001, iit-000, anke-003, and everai-002. Results for these are available in prior editions of this report linked from the FRVT page.
- We issued NIST Interagency Report 8280: FRVT Part 3: Demographics on 2019-12-19. It includes results for many of the algorithms covered by this report.

Changes since October 16, 2019:

- The report adds results for ten new developers: Ai-Union Technology, ASUSTek Computer, DiDi ChuXing Technology, Innovative Technology, Luxand, MVision, Pyramid Cyber Security + Forensic, Scanovate, Shenzhen AiMall Tech, and TUPU Technology.
- The report adds results for 12 returning developers: CTBC Bank Glory Gorilla Technology Guangzhou Pixel Solutions Imagus Technology Incode Technologies Lomonosov Moscow State University Rank One Computing Samtech InfoNet Shanghai Ulucu Electronics Technology Synesis, and Winsense.
- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: glory-000, gorilla-002, incode-003, rankone-006, and synesis-004.
- Results for five recently submitted algorithms will appear in the next report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Changes since September 11, 2019:

- The report adds results for five new participants: Awidit Systems (Awiros), Momemtum Digital (Sertis), Trueface AI, Shanghai Jiao Tong University, and X-Laboratory.
- The reports adds results for five new algorithms from returning developers: Cyberlink, Hengrui AI Technology, Idemia, Panasonic R+D Singapore, and Tevian. This causes three algorithm, to be de-listed from the report per policy to list results for two algorithms per developer.

Changes since July 31 2019:

- The HTML table on the FRVT 1:1 homepage has been updated to include a column for cross-domain Visa-Border verification. Results for this new dataset appeared in the July 29 report under the name "CrossEV" - these are now renamed "Visa-Border".
- The FRVT 1:1 homepage lists algorithms according to lowest mean rank accuracy:
  Rank($\text{FNMR}_{\text{VISA}}$ at FMR = 0.000001) +
  Rank($\text{FNMR}_{\text{VISA-BORDER}}$ at FMR = 0.000001) +
  Rank($\text{FNMR}_{\text{MUGSHOT}}$ at FMR = 0.00001 after 14 years) +
  Rank($\text{FNMR}_{\text{WILD}}$ at FMR = 0.00001)
  This ordering rewards high accuracy across all datasets.
- The main results in Table 89 is now in landscape format to accomodate extra columns for the Visa-Border set, and mugshot comparisons after at least 12 years.
- The report adds results for nine new participants: Alpha SSTG, Intel Research, ULSee, Chungwa Telecon, iSAP Solution, Rokid, Shenzhen EI Networks, CSA Intellicloud, Shenzhen Intellifusion Technologies.
- The reports adds results for six new algorithms from returning developers: Innovatrics, Dahua Technology, Tech5 SA, Intellivision, Nodeflux and Imperial College, London. One algorithm, from Imperial has been retired, per policy to list results for two algorithms per developer.
- The cross-country false match rate heatmaps have been replotted to reveal more structure by listing countries by region instead of alphabetically.
- The next version of this report will be posted around October 18, 2019.

Changes since July 3 2019:

- The HTML table on the FRVT 1:1 homepage has been updated to list the 20 most accurate developers rather than algorithms, choosing the most accurate algorithm from each developer based on visa and mugshot results. Also, the algorithms are ordered in terms of lowest mean rank across mugshot, visa and wild datasets, rewarding broad accuracy over a good result on one particular dataset.
- This report includes results for a new dataset - see the column labelled "visa-border" in Table 5. It compares a new set of high quality visa-like portraits with a set webcam border-crossing photos that exhibit moderately poor pose variations and background illumination. The two new sets are described in sections 2.2 and 2.4. The comparisons are "cross-domain" in that the algorithm must compare "visa" and "wild" images. Results for other algorithms will be added in future reports as they become available.
- This report adds results for algorithms from 9 developers submitted in early July 2019. These are from 3DiVi, Camvi, EverAI-Paravision, Facesoft, Farbar (F8), Institute of Information Technologies, Shanghai U. Film Academy, Via Technologies, and Ulucu Electronics Tech. Six of these are new participants.
- Several other algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for September 5. That report will include results for new datasets.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- Older algorithms from Everai, Camvi and 3DiVi, have been retired, per the policy to list only two algorithms per developer.

Changes since June 20 2019:

- This report adds results for algorithms from 18 developers submitted in early June 2019. These are from CTBC Bank, Deep Glint, Thales Cogent, Ever AI Paravision, Gorilla Technology, Imagus, Incode, Kneron, N-Tech Lab, Neurotechnology, Notiontag Technologies, Star Hybrid, Videonetics, Vigilant Solutions, Winsense, Anke Investments, CEIEC, and DSK. Nine of these are new participants.
- Several other algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for August 1.
- Older algorithms from Everai, Thales Cogent, Gorilla Technology, Incode, Neurotechnology, N-Tech Lab and Vigilant Solutions have been retired, per the policy to list only two algorithms per developer.

Changes since April 2019:

- This report adds results for nine algorithms from nine developers submitted in early June 2019. These are from Tencent Deepsea, Hengrui, Kedacom, Moontime, Guangzhou Pixel, Rank One Computing, Synesis, Sensetime and Vocord.
- Another 23 algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for July 3.
- Older algorithms for Rank One, Synesis, and Vocord have been retired, per the policy to list only two algorithms per developer.

Changes since February 2019:

- This report adds results for 49 algorithms from 42 developers submitted in early March 2019.
- This report omits results for algorithms that we retired. We retired for three reasons: 1. The developer submitted a new algorithm, and we only list two. 2. The algorithm needs a GPU, and we no longer allow GPU-based algorithms. 3. Inoperable algorithms.
- Previous results for retired algorithms are available in older editions of this report linked here.
- The mugshot database used from February 2017 to January 2019 has been replaced with an extract of the mugshot database documented in NIST Interagency Report 8238, November 2018. The new mugshot set is described in section 2.5 and is adopted because:
  - It has much better identity label integrity, so that false non-match rates are substantially lower than those reported in FRVT 1:1 reports to date - see Figure 291.
  - It includes images collected over a 17 year period such that ageing can be much better characterized - - see Figure 799.
- Using the new mugshot database, Figure 799 shows accuracy for four demographic groups identified in the biographic metadata that accompanies the data: black females, black males, white females and white males.
- The report added a figure (now moved to web) with results for the twenty human-difficult pairs used in the May 2018 paper *Face recognition accuracy of forensic examiners, superrecognizers, and face recognition algorithms* by Phillips et al. [1].
- The report uses an update to the wild image database that corrects some ground truth labels.
- Some results for the child exploitation database are not complete. They are typically updated less frequently than for other image sets.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

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FNMR(T) "False non-match rate"
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664 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 829
665 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 830
666 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 831
667 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 832
668 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 833
669 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 834
670 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 835
671 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 836
672 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 837
673 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 838
674 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 839
675 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 840
676 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 841
677 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 842
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679 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 844
680 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 845
681 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 846
682 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 847
683 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 848
684 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 849
685 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 850
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687 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 852
688 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 853
689 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 854
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FNMR(T) "False non-match rate"
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692 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 857
693 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 858
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695 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 860
696 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 861
697 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 862
698 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 863
699 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 864
700 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 865
701 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 866
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710 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 875
711 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 876
712 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 877
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727 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 893
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733 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 899
734 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 900
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814 EFFECT OF SUBJECT AGE ON FNMR . . . 981
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816 EFFECT OF SUBJECT AGE ON FNMR . . . 983
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819 EFFECT OF SUBJECT AGE ON FNMR . . . 986
820 EFFECT OF SUBJECT AGE ON FNMR . . . 987
821 EFFECT OF SUBJECT AGE ON FNMR . . . 988
822 EFFECT OF SUBJECT AGE ON FNMR . . . 989
823 EFFECT OF SUBJECT AGE ON FNMR . . . 990
824 EFFECT OF SUBJECT AGE ON FNMR . . . 991
825 EFFECT OF SUBJECT AGE ON FNMR . . . 992
826 EFFECT OF SUBJECT AGE ON FNMR . . . 993
827 EFFECT OF SUBJECT AGE ON FNMR . . . 994
828 EFFECT OF SUBJECT AGE ON FNMR . . . 995
829 EFFECT OF SUBJECT AGE ON FNMR . . . 996
830 EFFECT OF SUBJECT AGE ON FNMR . . . 997
831 EFFECT OF SUBJECT AGE ON FNMR . . . 998
832 EFFECT OF SUBJECT AGE ON FNMR . . . 999
833 EFFECT OF SUBJECT AGE ON FNMR . . . 1000
834 EFFECT OF SUBJECT AGE ON FNMR . . . 1001
835 EFFECT OF SUBJECT AGE ON FNMR . . . 1002
836 EFFECT OF SUBJECT AGE ON FNMR . . . 1003
837 EFFECT OF SUBJECT AGE ON FNMR . . . 1004
838 EFFECT OF SUBJECT AGE ON FNMR . . . 1005
839 EFFECT OF SUBJECT AGE ON FNMR . . . 1006
840 EFFECT OF SUBJECT AGE ON FNMR . . . 1007
841 EFFECT OF SUBJECT AGE ON FNMR . . . 1008
842 EFFECT OF SUBJECT AGE ON FNMR . . . 1009
843 EFFECT OF SUBJECT AGE ON FNMR . . . 1010
844 EFFECT OF SUBJECT AGE ON FNMR . . . 1011
845 EFFECT OF SUBJECT AGE ON FNMR . . . 1012
846 EFFECT OF SUBJECT AGE ON FNMR . . . 1013
847 EFFECT OF SUBJECT AGE ON FNMR . . . 1014
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849 EFFECT OF SUBJECT AGE ON FNMR . . . 1016
850 EFFECT OF SUBJECT AGE ON FNMR . . . 1017
851 EFFECT OF SUBJECT AGE ON FNMR . . . 1018
852 EFFECT OF SUBJECT AGE ON FNMR . . . 1019
853 EFFECT OF SUBJECT AGE ON FNMR . . . 1020
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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

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856 EFFECT OF SUBJECT AGE ON FNMR 1023
857 EFFECT OF SUBJECT AGE ON FNMR 1024
858 EFFECT OF SUBJECT AGE ON FNMR 1025
859 EFFECT OF SUBJECT AGE ON FNMR 1026
860 EFFECT OF SUBJECT AGE ON FNMR 1027
861 EFFECT OF SUBJECT AGE ON FNMR 1028
862 EFFECT OF SUBJECT AGE ON FNMR 1029
863 EFFECT OF SUBJECT AGE ON FNMR 1030
864 EFFECT OF SUBJECT AGE ON FNMR 1031
865 EFFECT OF SUBJECT AGE ON FNMR 1032
866 EFFECT OF SUBJECT AGE ON FNMR 1033
867 EFFECT OF SUBJECT AGE ON FNMR 1034
868 EFFECT OF SUBJECT AGE ON FNMR 1035
869 EFFECT OF SUBJECT AGE ON FNMR 1036
870 EFFECT OF SUBJECT AGE ON FNMR 1037
871 EFFECT OF SUBJECT AGE ON FNMR 1038
872 EFFECT OF SUBJECT AGE ON FNMR 1039
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874 EFFECT OF SUBJECT AGE ON FNMR 1041
875 EFFECT OF SUBJECT AGE ON FNMR 1042
876 EFFECT OF SUBJECT AGE ON FNMR 1043
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878 EFFECT OF SUBJECT AGE ON FNMR 1045
879 EFFECT OF SUBJECT AGE ON FNMR 1046
880 EFFECT OF SUBJECT AGE ON FNMR 1047
881 EFFECT OF SUBJECT AGE ON FNMR 1048
882 EFFECT OF SUBJECT AGE ON FNMR 1049
883 EFFECT OF SUBJECT AGE ON FNMR 1050
884 EFFECT OF SUBJECT AGE ON FNMR 1051
885 EFFECT OF SUBJECT AGE ON FNMR 1052
886 EFFECT OF SUBJECT AGE ON FNMR 1053
887 EFFECT OF SUBJECT AGE ON FNMR 1054
888 EFFECT OF SUBJECT AGE ON FNMR 1055
889 EFFECT OF SUBJECT AGE ON FNMR 1056
890 EFFECT OF SUBJECT AGE ON FNMR 1057
891 IMPOSTOR COUNTS FOR CROSS COUNTRY FMR CALCULATIONS 1061

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1	KR	Urielsoft Co. Ltd	urielsoft-000	000	2025-06-30
2	EE	2060.io	2060io-001	001	2025-01-22
3	EE	2060.io	2060io-002	002	2025-07-23
4	NL	20Face	20face-000	000	2021-04-12
5	NL	20Face	20face-001	001	2021-09-29
6	US	3Divi	3divi-000	000	2017-03-16
7	US	3Divi	3divi-001	001	2017-06-22
8	US	3Divi	3divi-002	002	2017-10-20
9	US	3Divi	3divi-003	003	2018-10-09
10	US	3Divi	3divi-004	004	2019-07-22
11	US	3Divi	3divi-005	005	2020-08-28
12	US	3Divi	3divi-006	006	2021-04-14
13	US	3Divi	3divi-007	007	2021-09-27
14	TH	ACI Software	acisw-003	003	2020-08-03
15	TH	ACI Software	acisw-006	006	2021-02-25
16	TH	ACI Software	acisw-007	007	2021-11-15
17	TH	ACI Software	acisw-008	008	2022-03-22
18	US	AFIS and Biometrics Consulting	afisbiometrics-000	000	2022-01-27
19	US	AFR Engine	afrengine-000	000	2022-09-29
20	US	AFR Engine	afrengine-001	001	2023-01-26
21	US	AFR Engine	afrengine-002	002	2023-06-27
22	US	AFR Engine	afrengine-003	003	2024-03-15
23	US	AFR Engine	afrengine-004	004	2024-07-22
24	US	AFR Engine	afrengine-005	005	2025-02-18
25	UZ	AI Implementation Group	aigroup-000	000	2024-07-17
26	UZ	AI Implementation Group	aigroup-001	001	2025-02-18
27	UZ	AI Implementation Group	aigroup-002	002	2026-03-12
28	NV	AIPRACTIX	aipractix-000	000	2026-05-12
29	VN	ALTTEKGlobal	alttekglobal-000	000	2024-04-12
30	TW	ASUSTek Computer Inc	asusaics-000	000	2019-10-24
31	TW	ASUSTek Computer Inc	asusaics-001	001	2020-02-25
32	CN	AYF Technology	ayftech-001	001	2020-07-06
33	CN	AYF Technology	ayftech-003	003	2023-05-15
34	TW	Ability Enterprise - Andro Video	androvideo-000	000	2021-01-25
35	IN	Accurascan	accurascan-001	001	2023-0-15
36	IN	Accurascan	accurascan-002	002	2023-08-01
37	IN	Accurascan	accurascan-003	003	2023-12-06
38	TW	Acer Incorporated	acer-000	000	2020-01-08
39	TW	Acer Incorporated	acer-001	001	2020-06-30
40	TW	Acer Incorporated	acer-002	002	2021-11-10
41	SG	Adera Global PTE	adera-001	001	2019-06-17
42	SG	Adera Global PTE	adera-002	002	2021-02-16
43	SG	Adera Global PTE	adera-003	003	2021-07-12
44	SG	Adera Global PTE	adera-004	004	2022-11-14
45	SG	Adera Global PTE	adera-005	005	2023-11-17
46	SG	Advance.AI	advance-002	002	2019-12-19
47	SG	Advance.AI	advance-003	003	2021-08-05
48	SG	Advance.AI	advance-004	004	2022-09-06
49	SG	Advance.AI	advance-005	005	2023-05-25
50	SG	Advance.AI	advance-006	006	2026-04-03
51	SG	Advance.AI	advance-007	007	2026-08-07
52	AE	Affinsys AI Ltd	affinsys-ai-000	000	2024-07-11
53	TH	Ai First	aifirst-001	001	2019-11-21
54	TW	AiUnion Technology	aiunionface-000	000	2019-10-22
55	TH	Aigen	aigen-001	001	2020-10-06
56	TH	Aigen	aigen-002	002	2021-03-15
57	CN	Aiseemu Technology	aiseemu-001	001	2022-06-16
58	CN	Aiseemu Technology	aiseemu-002	002	2022-11-18
59	KR	Ajou University	ajou-001	001	2021-03-08
60	ID	Akurat Satu Indonesia	ptakuratsatu-000	000	2020-09-11
61	KR	Alchera Inc	alchera-000	000	2019-03-01
62	KR	Alchera Inc	alchera-001	001	2019-03-01
63	KR	Alchera Inc	alchera-002	002	2021-03-05
64	KR	Alchera Inc	alchera-003	003	2021-07-13
65	KR	Alchera Inc	alchera-004	004	2022-08-12
66	KR	Alchera Inc	alchera-005	005	2023-01-04
67	KR	Alchera Inc	alchera-006	006	2023-07-18
68	KR	Alchera Inc	alchera-007	007	2023-11-22
69	KR	Alchera Inc	alchera-008	008	2024-06-17
70	KR	Alchera Inc	alchera-009	009	2024-11-14

Table 1: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
71	ID	Alfabeta	alfabeta-001	001	2021-12-02
72	ES	Alice Biometrics	alice-000	000	2021-06-15
73	ES	Alice Biometrics	alice-001	001	2023-07-05
74	RU	Alivia / Innovation Sys	isystems-000	000	2018-01-11
75	RU	Alivia / Innovation Sys	isystems-001	001	2018-06-12
76	RU	Alivia / Innovation Sys	isystems-002	002	2018-10-18
77	IN	AllGoVision	allgovision-000	000	2019-03-01
78	CN	AlphaSSTG	alphaface-001	001	2019-09-03
79	CN	AlphaSSTG	alphaface-002	002	2020-02-20
80	GB	Amplified Group	amplifiedgroup-001	001	2019-03-01
81	CN	Anke Investments	anke-000	000	2018-06-26
82	CN	Anke Investments	anke-001	001	2018-06-26
83	CN	Anke Investments	anke-002	002	2018-10-19
84	CN	Anke Investments	anke-003	003	2019-02-27
85	CN	Anke Investments	anke-004	004	2019-06-27
86	CN	Anke Investments	anke-005	005	2019-11-21
87	BR	Antheus Technologia	antheus-000	000	2019-12-05
88	BR	Antheus Technologia	antheus-001	001	2020-06-25
89	BR	Antheus Technologia	antheus-002	002	2024-07-15
90	GB	AnyVision	anyvision-001	001	2018-01-31
91	GB	AnyVision	anyvision-002	002	2018-01-31
92	GB	AnyVision	anyvision-004	004	2018-06-15
93	GB	AnyVision	anyvision-005	005	2021-02-03
94	TR	ArVIs Teknoloji San.Tic.A.Ş.	arvis-000	000	2025-04-08
95	CN	Aratek Biometrics Co Ltd	aratek-001	001	2023-03-27
96	US	Armatura LLC	armatura-001	001	2022-01-04
97	US	Armatura LLC	armatura-002	002	2022-09-16
98	US	Armatura LLC	armatura-003	003	2023-01-13
99	CN	AuthenMetric	authenmetric-002	002	2021-03-10
100	CN	AuthenMetric	authenmetric-003	003	2021-08-09
101	CN	AuthenMetric	authenmetric-004	004	2022-01-03
102	TW	Authme	authme-001	001	2023-10-17
103	TW	Authme	authme-002	002	2025-02-10
104	US	Aware	aware-000	000	2017-10-27
105	US	Aware	aware-001	001	2017-10-27
106	US	Aware	aware-002	002	2018-06-15
107	US	Aware	aware-003	003	2018-10-19
108	US	Aware	aware-004	004	2019-03-01
109	US	Aware	aware-005	005	2020-02-27
110	US	Aware	aware-006	006	2021-07-03
111	US	Aware	aware-007	007	2023-11-15
112	US	Aware	aware-008	008	2024-03-15
113	IN	Awidit Systems	awiros-001	001	2019-09-23
114	IN	Awidit Systems	awiros-002	002	2020-10-28
115	IN	Awidit Systems	awiros-003	003	2024-11-01
116	IN	Awidit Systems	awiros-004	004	2025-03-31
117	CH	Aximetria	aximetria-001	001	2022-08-10
118	JP	Ayonix	ayonix-000	000	2017-06-22
119	CN	BOE Technology Group	boetech-001	001	2021-06-22
120	CN	BOE Technology Group	boetech-002	002	2021-12-21
121	KZ	BTS Digital	btsdigital-001	001	2025-07-31
122	ES	Bee the Data	beethedata-000	000	2021-07-26
123	CN	Beihang University-ERCACAT	ercacat-001	001	2020-07-06
124	CN	Beijing Alleyes Technology	alleyes-000	000	2020-03-09
125	CN	Beijing DeepSense Technologies	deepsense-000	000	2021-03-19
126	CN	Beijing DeepSense Technologies	deepsense-001	001	2022-03-11
127	CN	Beijing DeepSense Technologies	deepsense-002	002	2023-02-22
128	CN	Beijing DeepSense Technologies	deepsense-003	003	2023-10-23
129	CN	Beijing DeepSense Technologies	deepsense-004	004	2024-06-21
130	CN	Beijing DeepSense Technologies	deepsense-005	005	2024-10-28
131	CN	Beijing DeepSense Technologies	deepsense-006	006	2025-03-07
132	CN	Beijing DeepSense Technologies	deepsense-007	007	2025-07-08
133	CN	Beijing DeepSense Technologies	deepsense-008	008	2026-01-13
134	CN	Beijing Hisign Technology	hisign-001	001	2021-09-24
135	CN	Beijing Hisign Technology	hisign-002	002	2022-09-09
136	CN	Beijing Hisign Technology	hisign-003	003	2024-03-15
137	CN	Beijing Mendaxia Technology	mendaxiatech-000	000	2021-09-15
138	CN	Beijing Vion Technology Inc	vion-000	000	2018-10-19
139	DE	Betaface	betaface-000	000	2018-05-29
140	DE	Betaface	betaface-002	002	2018-05-29

Table 2: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
141	KZ	Beyne.AI	beyneai-000	000	2022-01-03
142	DE	BioID GmbH	bioid-001	001	2024-11-13
143	DE	BioID GmbH	bioid-002	002	2025-04-25
144	DE	BioID GmbH	bioid-003	003	2025-11-26
145	DE	BioID GmbH	bioid-004	004	2026-04-13
146	CH	BioID Technologies SA	bioidtechswiss-000	000	2019-11-15
147	CH	BioID Technologies SA	bioidtechswiss-001	001	2020-08-28
148	CH	BioID Technologies SA	bioidtechswiss-002	002	2021-02-17
149	CH	BioID Technologies SA	bioidtechswiss-003	003	2025-03-12
150	IN	Biocube Matrics	biocube-001	001	2021-09-08
151	IN	Biocube Matrics	biocube-002	002	2023-02-09
152	KZ	Biometric LLC	biometric-vision-000	000	2023-01-25
153	IN	Biometrik Systems Private Limited	biometrik-001	001	2025-09-18
154	IN	Biometrik Systems Private Limited	biometrik-002	002	2026-01-20
155	GB	BitCenter UK	farfaces-001	001	2021-04-09
156	CN	Bitmain	bm-001	001	2018-10-17
157	CN	Bresee Technology	bresee-000	000	2020-08-07
158	CN	Bresee Technology	bresee-001	001	2020-12-30
159	CN	Bresee Technology	bresee-002	002	2021-06-30
160	US	Bynn Intelligence Inc	bynn-000	000	2026-05-29
161	VN	CAKE by VPBank	cakevn-001	001	2026-01-12
162	VN	CAKE by VPBank	cakevn-002	002	2026-06-21
163	VN	CMC Institute of Science and Technology	cist-001	001	2022-10-20
164	VN	CMC Institute of Science and Technology	cist-002	002	2023-02-09
165	VN	CMC Institute of Science and Technology	cist-003	003	2023-08-14
166	VN	CMC Institute of Science and Technology	cist-004	004	2024-02-15
167	VN	CMC Univerisity	cmcuni-001	001	2023-12-05
168	CN	CSA IntelliCloud Technology	intellicloudai-001	001	2019-08-13
169	CN	CSA IntelliCloud Technology	intellicloudai-002	002	2020-12-17
170	TW	CTBC Bank	ctbcbank-000	000	2019-06-28
171	TW	CTBC Bank	ctbcbank-001	001	2019-10-28
172	TW	CTBC Bank	ctbcbank-002	002	2024-07-10
173	KR	CUDO Communication	cudocommunication-001	001	2021-10-20
174	US	Camvi Technologies	camvi-001	001	2017-09-13
175	US	Camvi Technologies	camvi-002	002	2018-10-19
176	US	Camvi Technologies	camvi-003	003	2019-03-01
177	US	Camvi Technologies	camvi-004	004	2019-07-12
178	FI	Candour Biometrics	candour-001	001	2023-02-10
179	FI	Candour Biometrics	candour-002	002	2024-03-07
180	FI	Candour Biometrics	candour-003	003	2024-10-23
181	FI	Candour Biometrics	candour-004	004	2026-01-08
182	CN	Canon Inc	canon-002	002	2020-12-29
183	JP	Canon Inc	canon-003	003	2021-09-15
184	JP	Canon Inc	canon-004	004	2022-04-25
185	JP	Canon Inc	canon-005	005	2023-09-05
186	CN	Canon Inc	cib-000	000	2019-12-11
187	CN	Canon Inc	cib-001	001	2020-08-05
188	BD	Celloscope Limited	celloscope-001	001	2024-07-15
189	CN	China Electronics Import-Export Corp	ceiec-001	001	2019-03-01
190	CN	China Electronics Import-Export Corp	ceiec-002	002	2019-06-12
191	CN	China Electronics Import-Export Corp	ceiec-003	003	2020-01-06
192	CN	China Electronics Import-Export Corp	ceiec-004	004	2021-01-18
193	CN	China University of Petroleum	upc-001	001	2019-06-05
194	CN	Chinese University of Hong Kong	cuhkee-001	001	2020-03-18
195	KR	Chosun University	chosun-000	000	2020-02-12
196	KR	Chosun University	chosun-001	001	2020-07-01
197	KR	Chosun University	chosun-002	002	2020-11-25
198	TW	Chunghwa Telecom	chtface-001	001	2019-08-06
199	TW	Chunghwa Telecom	chtface-002	002	2019-12-07
200	TW	Chunghwa Telecom	chtface-003	003	2020-06-24
201	TW	Chunghwa Telecom	chtface-004	004	2021-10-08
202	TW	Chunghwa Telecom	chtface-005	005	2022-03-09
203	TW	Chunghwa Telecom	chtface-006	006	2022-11-03
204	TW	Chunghwa Telecom	chtface-007	007	2024-04-23
205	TW	Chunghwa Telecom	chtface-008	008	2024-09-27
206	TW	Chunghwa Telecom	chtface-009	009	2025-10-21
207	US	City and County of Honolulu	cchonolulu-000	000	2023-03-27
208	US	City and County of Honolulu	cchonolulu-001	001	2024-01-18
209	US	Clearview AI Inc	clearviewai-000	000	2021-09-22
210	US	Clearview AI Inc	clearviewai-001	001	2023-12-05

Table 3: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
211	US	Clearview AI Inc	clearviewai-002	002	2024-05-22
212	US	Clearview AI Inc	clearviewai-003	003	2026-01-09
213	CN	Closeli Inc	closeli-001	001	2021-07-15
214	US	CloudSmart Consulting LLC	csc-001	001	2020-11-20
215	US	CloudSmart Consulting LLC	csc-002	002	2021-03-24
216	US	CloudSmart Consulting LLC	csc-003	003	2021-08-26
217	TW	Cloudmatrix	cloudmatrix-000	000	2021-10-22
218	TW	Cloudmatrix	cloudmatrix-001	001	2022-02-16
219	TW	Cloudmatrix	cloudmatrix-002	002	2022-10-17
220	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-000	000	2019-03-01
221	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-001	001	2019-06-04
222	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-002	002	2019-10-08
223	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-003	003	2020-09-25
224	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-004	004	2021-02-10
225	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-000	000	2019-06-03
226	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-002	002	2020-07-02
227	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-003	003	2020-12-22
228	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-004	004	2021-11-09
229	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-005	005	2022-03-29
230	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-006	006	2022-10-20
231	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-007	007	2023-02-21
232	IN	Code Everest Pvt	facex-001	001	2021-03-08
233	IN	Code Everest Pvt	facex-002	002	2021-08-24
234	KR	Codeline	codeline-000	000	2022-09-13
235	DE	Cognitec Systems GmbH	cognitec-000	000	2018-10-19
236	DE	Cognitec Systems GmbH	cognitec-001	001	2019-03-01
237	DE	Cognitec Systems GmbH	cognitec-002	002	2021-02-24
238	DE	Cognitec Systems GmbH	cognitec-003	003	2021-07-30
239	DE	Cognitec Systems GmbH	cognitec-004	004	2022-02-10
240	DE	Cognitec Systems GmbH	cognitec-005	005	2023-12-01
241	DE	Cognitec Systems GmbH	cognitec-006	006	2024-04-29
242	DE	Cognitec Systems GmbH	cognitec-007	007	2024-08-29
243	TW	Compal Electronics	compal-000	000	2025-03-20
244	TW	Compal Electronics	compal-001	001	2025-12-08
245	TW	Compal Electronics	compal-002	002	2026-07-09
246	TW	Coretech Knowledge Inc	coretech-000	000	2021-07-12
247	TW	Coretech Knowledge Inc	coretech-001	001	2022-09-29
248	TW	Coretech Knowledge Inc	coretech-002	002	2023-11-02
249	TW	Coretech Knowledge Inc	coretech-003	003	2024-08-07
250	TW	Coretech Knowledge Inc	coretech-004	004	2024-12-23
251	IL	Corsight	corsight-001	001	2021-03-11
252	IL	Corsight	corsight-002	002	2021-09-01
253	IL	Corsight	corsight-003	003	2022-06-09
254	IL	Cortica	cor-001	001	2020-09-24
255	TW	Cu-Face	cu-face-002	002	2023-01-05
256	TW	Cu-Face	cu-face-003	003	2023-08-28
257	TW	Cu-Face	cu-face-004	004	2024-03-13
258	KR	Cubox	cubox-001	001	2020-12-07
259	KR	Cubox	cubox-002	002	2021-08-24
260	KR	Cubox	cubox-003	003	2023-03-07
261	KR	Cubox	cubox-004	004	2024-12-05
262	KR	Cubox	cubox-005	005	2025-05-14
263	JP	Cybercore	cybercore-000	000	2020-08-26
264	JP	Cybercore	cybercore-001	001	2021-12-15
265	JP	Cybercore	cybercore-002	002	2022-04-25
266	JP	Cybercore	cybercore-003	003	2022-08-31
267	US	Cyberextruder	cyberextruder-001	001	2017-08-02
268	US	Cyberextruder	cyberextruder-002	002	2018-01-30
269	US	Cyberextruder	cyberextruder-003	003	2022-03-16
270	US	Cyberextruder	cyberextruder-004	004	2022-07-20
271	US	Cyberextruder	cyberextruder-005	005	2024-10-01
272	US	Cyberextruder	cyberextruder-006	006	2025-03-19
273	TW	Cyberlink Corp	cyberlink-000	000	2019-03-01
274	TW	Cyberlink Corp	cyberlink-001	001	2019-03-01
275	TW	Cyberlink Corp	cyberlink-002	002	2019-06-12
276	TW	Cyberlink Corp	cyberlink-003	003	2019-10-07
277	TW	Cyberlink Corp	cyberlink-004	004	2020-02-27
278	TW	Cyberlink Corp	cyberlink-005	005	2020-07-31
279	TW	Cyberlink Corp	cyberlink-006	006	2021-01-08
280	TW	Cyberlink Corp	cyberlink-007	007	2021-07-16

Table 4: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
281	TW	Cyberlink Corp	cyberlink-008	008	2022-01-07
282	TW	Cyberlink Corp	cyberlink-009	009	2022-05-12
283	TW	Cyberlink Corp	cyberlink-010	010	2022-09-16
284	TW	Cyberlink Corp	cyberlink-011	011	2023-01-30
285	TW	Cyberlink Corp	cyberlink-012	012	2023-06-02
286	TW	Cyberlink Corp	cyberlink-013	013	2023-12-15
287	TW	Cyberlink Corp	cyberlink-014	014	2024-05-23
288	TW	Cyberlink Corp	cyberlink-015	015	2024-09-26
289	TW	Cyberlink Corp	cyberlink-016	016	2026-06-02
290	MX	DICIO	dicio-001	001	2022-03-22
291	MX	DICIO	dicio-002	002	2024-10-03
292	CN	DSK	dsk-000	000	2019-06-28
293	VN	Dactionable Technologies	datech-000	000	2023-10-16
294	VN	Dactionable Technologies	datech-001	001	2024-02-21
295	VN	Dactionable Technologies	datech-002	002	2024-09-11
296	CN	Dahua Technology	dahua-000	000	2018-10-19
297	CN	Dahua Technology	dahua-001	001	2018-10-19
298	CN	Dahua Technology	dahua-002	002	2019-03-01
299	CN	Dahua Technology	dahua-003	003	2019-08-14
300	CN	Dahua Technology	dahua-004	004	2019-12-18
301	CN	Dahua Technology	dahua-005	005	2020-08-13
302	CN	Dahua Technology	dahua-006	006	2020-12-30
303	CN	Dahua Technology	dahua-007	007	2021-12-20
304	IE	Daon	daon-000	000	2021-11-03
305	IE	Daon	daon-001	001	2024-09-26
306	IE	Daon	daon-002	002	2025-07-01
307	IE	Daon	daon-003	003	2026-03-12
308	TW	DeCloak Intellegences	decloakface-001	001	2023-06-13
309	TW	DeCloak Intellegences	decloakface-002	002	2024-10-28
310	US	Decatur Industries Inc	decatur-000	000	2020-08-18
311	US	Decatur Industries Inc	decatur-001	001	2021-09-27
312	CN	Deepglint	deepglint-001	001	2019-06-21
313	CN	Deepglint	deepglint-002	002	2019-11-15
314	CN	Deepglint	deepglint-003	003	2021-03-03
315	CN	Deepglint	deepglint-004	004	2021-09-17
316	CN	Deepglint	deepglint-005	005	2022-10-17
317	FR	Deepsense	dps-000	000	2021-07-16
318	DE	Dermalog	dermalog-001	001	2017-02-22
319	DE	Dermalog	dermalog-002	002	2017-02-22
320	DE	Dermalog	dermalog-003	003	2017-07-10
321	DE	Dermalog	dermalog-004	004	2017-10-26
322	DE	Dermalog	dermalog-005	005	2018-02-02
323	DE	Dermalog	dermalog-006	006	2018-10-18
324	DE	Dermalog	dermalog-007	007	2020-11-25
325	DE	Dermalog	dermalog-008	008	2021-03-25
326	DE	Dermalog	dermalog-009	009	2021-10-06
327	DE	Dermalog	dermalog-010	010	2022-07-25
328	DE	Dermalog	dermalog-011	011	2022-12-12
329	DE	Dermalog	dermalog-012	012	2024-01-19
330	DE	Dermalog	dermalog-013	013	2024-06-18
331	DE	Dermalog	dermalog-014	014	2025-06-17
332	DE	Dermalog	dermalog-015	015	2025-11-26
333	DE	Dermalog	dermalog-016	016	2026-06-25
334	CN	DiDi ChuXing Technology	didiglobalface-001	001	2019-10-23
335	CN	DiDi ChuXing Technology	didiglobalface-002	002	2023-01-09
336	CN	DiDi ChuXing Technology	didiglobalface-003	003	2025-05-01
337	IN	Digidata	digidata-000	000	2022-01-27
338	IN	Digidata	digidata-001	001	2022-06-10
339	GB	Digital Barriers	digitalbarriers-000	000	2017-05-31
340	GB	Digital Barriers	digitalbarriers-001	001	2017-07-20
341	GB	Digital Barriers	digitalbarriers-002	002	2019-03-01
342	UY	Dralmey Solutions	dralmey-001	001	2026-03-27
343	IN	EI Networks Private Ltd	einetworksindia-000	000	2023-05-03
344	IN	EI Networks Private Ltd	einetworksindia-001	001	2023-09-05
345	IN	EI Networks Private Ltd	einetworksindia-002	002	2025-02-21
346	TR	Ekin Smart City Technologies	ekin-002	002	2021-05-04
347	TH	Element System Solutions Company Ltd	element-000	000	2023-11-14
348	AE	Enface	enface-000	000	2021-04-09
349	AE	Enface	enface-001	001	2021-12-17
350	AE	Enface	enface-002	002	2023-02-27

Table 5: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
351	GB	Enqura Ltd	enqura-001	001	2024-09-25
352	GB	Enqura Ltd	enqura-002	002	2026-05-19
353	ET	Ethiopian Labor Market Information System	ethiopianlmis-000	000	2026-03-18
354	CH	Euronovate SA	euronovate-001	001	2021-11-15
355	CH	Euronovate SA	euronovate-002	002	2023-06-28
356	CH	Euronovate SA	euronovate-003	003	2023-09-19
357	CH	Euronovate SA	euronovate-004	004	2024-01-19
358	RU	Expasoft LLC	expasoft-000	000	2020-01-06
359	RU	Expasoft LLC	expasoft-001	001	2020-09-03
360	RU	Expasoft LLC	expasoft-002	002	2021-07-26
361	VN	FCI Vietnam	fci-000	000	2025-04-01
362	LB	FOO	foomobi-001	001	2023-03-13
363	LB	FOO	foomobi-002	002	2023-08-07
364	VN	FPT Education	fedu-001	001	2024-03-15
365	VN	FPT Information System	fpt-000	000	2024-02-26
366	VN	FPT Information System	fpt-001	001	2024-09-16
367	VN	FPT Telecom JSC	fpttelecom-000	000	2026-02-03
368	US	FRP LLC	frpkauai-001	001	2022-07-18
369	US	FRP LLC	frpkauai-002	002	2022-11-21
370	US	FRP LLC	frpkauai-003	003	2024-05-30
371	US	FRP LLC	frpkauai-004	004	2025-03-25
372	US	FRP LLC	frpkauai-005	005	2026-06-22
373	IL	Face Int	faceint-001	001	2026-03-27
374	CA	FaceLocate	facelocate-001	001	2024-02-22
375	CA	FaceLocate	facelocate-002	002	2026-03-25
376	DE	FaceOnLive Inc	faceonlive-001	001	2021-11-23
377	DE	FaceOnLive Inc	faceonlive-002	002	2022-04-11
378	DE	FaceOnLive Inc	faceonlive-003	003	2024-08-12
379	ES	FacePhi	facephi-000	000	2022-04-06
380	ES	FacePhi	facephi-001	001	2024-05-24
381	ES	FacePhi	facephi-002	002	2025-02-11
382	GB	FaceSoft	facesoft-000	000	2019-07-10
383	KR	FaceTag Co	facetag-000	000	2021-03-22
384	KR	FaceTag Co	facetag-001	001	2021-08-17
385	KR	FaceTag Co	facetag-002	002	2022-01-06
386	GB	Facehawk Ltd	facehawk-000	000	2023-08-22
387	GB	Facehawk Ltd	facehawk-001	001	2026-06-17
388	GB	Facia.ai	facia-001	001	2023-07-21
389	GB	Facia.ai	facia-002	002	2024-06-21
390	GB	FaiceTech Ltd	faicetech-000	000	2025-03-11
391	TW	FarBar Inc	f8-001	001	2019-07-11
392	TW	FarBar Inc	f8-002	002	2022-03-02
393	US	Fast Enterprises	fastenterprises-000	000	2023-03-01
394	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-002	002	2020-08-10
395	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-003	003	2021-03-12
396	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-004	004	2021-09-14
397	GB	Fincore Ltd	fincore-000	000	2021-06-07
398	KZ	First Credit Bureau Kazakhstan	firstcreditkz-001	001	2022-08-22
399	KZ	First Credit Bureau Kazakhstan	firstcreditkz-002	002	2023-02-21
400	KZ	First Credit Bureau Kazakhstan	firstcreditkz-003	003	2024-02-20
401	KZ	First Credit Bureau Kazakhstan	firstcreditkz-004	004	2024-10-02
402	NL	Fourthline	fourthlinecom-001	001	2024-04-15
403	GB	Fraud.com	fraudcom-000	000	2023-12-01
404	CN	Fudan University	fdu-000	000	2017-11-22
405	CN	Fudan University	fdu-001	001	2018-01-11
406	CN	Fujitsu Research and Development Center	fujitsulab-000	000	2020-02-04
407	CN	Fujitsu Research and Development Center	fujitsulab-001	001	2020-09-30
408	CN	Fujitsu Research and Development Center	fujitsulab-002	002	2021-02-24
409	CN	Fujitsu Research and Development Center	fujitsulab-003	003	2021-07-12
410	VN	GPS Vietnam Trading	gpstechvn-000	000	2024-01-02
411	VN	GTEL OTS JSC	gtelots-000	000	2025-04-14
412	VN	GTEL OTS JSC	gtelots-001	001	2026-02-02
413	US	Gemalto Cogent	cogent-000	000	2018-01-12
414	US	Gemalto Cogent	cogent-001	001	2018-05-29
415	US	Gemalto Cogent	cogent-002	002	2018-10-19
416	US	Gemalto Cogent	cogent-003	003	2019-03-01
417	US	Gemalto Cogent	cogent-004	004	2019-06-14
418	US	Gemalto Cogent	cogent-005	005	2020-12-29
419	US	Gemalto Cogent	cogent-006	006	2021-07-28
420	US	Gemalto Cogent	cogent-007	007	2022-04-11

Table 6: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
421	US	Gemalto Cogent	cogent-008	008	2023-01-03
422	TW	General Interface Solutions Holding Ltd	gistouch-000	000	2023-05-08
423	TW	General Interface Solutions Holding Ltd	gistouch-001	001	2023-11-09
424	TW	GeoVision Inc	geo-000	000	2020-06-29
425	TW	GeoVision Inc	geo-001	001	2020-10-30
426	TW	GeoVision Inc	geo-002	002	2021-04-01
427	TW	GeoVision Inc	geo-003	003	2021-09-15
428	TW	GeoVision Inc	geo-004	004	2022-02-10
429	JP	Glory	glory-000	000	2018-06-06
430	JP	Glory	glory-001	001	2018-06-08
431	JP	Glory	glory-002	002	2019-11-12
432	JP	Glory	glory-003	003	2021-01-15
433	JP	Glory	glory-004	004	2022-02-08
434	JP	Glory	glory-005	005	2022-07-08
435	JP	Glory	glory-006	006	2023-03-23
436	JP	Glory	glory-007	007	2023-11-10
437	TW	Gorilla Technology	gorilla-000	000	2018-02-13
438	TW	Gorilla Technology	gorilla-001	001	2018-05-25
439	TW	Gorilla Technology	gorilla-002	002	2018-10-17
440	TW	Gorilla Technology	gorilla-003	003	2019-06-19
441	TW	Gorilla Technology	gorilla-004	004	2019-11-04
442	TW	Gorilla Technology	gorilla-005	005	2020-03-11
443	TW	Gorilla Technology	gorilla-006	006	2020-07-31
444	TW	Gorilla Technology	gorilla-007	007	2021-06-28
445	TW	Gorilla Technology	gorilla-008	008	2021-11-08
446	TW	Gorilla Technology	gorilla-009	009	2022-12-14
447	US	Graymatics	graymatics-001	001	2022-01-13
448	US	Griaule	griaule-000	000	2021-08-20
449	US	Griaule	griaule-001	001	2022-05-31
450	US	Griaule	griaule-002	002	2022-12-02
451	CN	Guangzhou Pixel Solutions	pixelall-002	002	2019-06-06
452	CN	Guangzhou Pixel Solutions	pixelall-003	003	2019-10-15
453	CN	Guangzhou Pixel Solutions	pixelall-004	004	2020-07-02
454	CN	Guangzhou Pixel Solutions	pixelall-005	005	2021-02-05
455	CN	Guangzhou Pixel Solutions	pixelall-006	006	2021-06-17
456	CN	Guangzhou Pixel Solutions	pixelall-007	007	2021-12-01
457	CN	Guangzhou Pixel Solutions	pixelall-008	008	2022-06-16
458	CN	Guangzhou Pixel Solutions	pixelall-009	009	2022-10-26
459	CN	Guangzhou Pixel Solutions	pixelall-010	010	2023-10-18
460	CN	Hangzhuo AIlu Network Information Technology	hzailu-001	001	2022-01-27
461	CN	Hangzhuo AIlu Network Information Technology	hzailu-002	002	2022-06-02
462	CN	Hangzhuo AIlu Network Information Technology	hzailu-003	003	2022-10-11
463	CN	Hangzhuo AIlu Network Information Technology	hzailu-004	004	2023-02-27
464	CN	Hangzhuo AIlu Network Information Technology	hzailu-005	005	2023-08-10
465	CN	Hangzhuo AIlu Network Information Technology	hzailu-006	006	2023-12-28
466	CN	Hangzhuo AIlu Network Information Technology	hzailu-007	007	2024-09-06
467	CN	Hangzhuo AIlu Network Information Technology	hzailu-008	008	2025-02-27
468	ES	Herta Security	hertasecurity-000	000	2021-01-05
469	ES	Herta Security	hertasecurity-001	001	2022-01-18
470	ES	Herta Security	hertasecurity-002	002	2022-09-02
471	ES	Herta Security	hertasecurity-003	003	2023-01-27
472	CN	Hikvision Research Institute	hik-001	001	2019-03-01
473	IN	Hyderabad-Centre for Development of Advanced Computing	cdachyd-001	001	2024-10-01
474	IN	HyperVerge Inc	hyperverge-001	001	2020-12-13
475	IN	HyperVerge Inc	hyperverge-002	002	2021-05-27
476	IN	HyperVerge Inc	hyperverge-003	003	2022-04-11
477	IN	HyperVerge Inc	hyperverge-004	004	2022-12-14
478	IN	HyperVerge Inc	hyperverge-005	005	2023-08-08
479	IN	HyperVerge Inc	hyperverge-006	006	2024-05-02
480	IN	HyperVerge Inc	hyperverge-007	007	2026-04-10
481	KR	Hyundai Motor Company	hyundai-roboticslab-000	000	2024-07-12
482	KR	Hyundai Motor Company	hyundai-roboticslab-001	001	2025-02-03
483	KR	Hyundai Motor Company	hyundai-roboticslab-002	002	2025-06-03
484	KR	Hyundai Motor Company	hyundai-roboticslab-003	003	2025-11-14
485	KR	Hyundai Motor Company	hyundai-roboticslab-004	004	2026-06-04
486	AU	ICM Airport Technics	icm-002	002	2020-11-13
487	AU	ICM Airport Technics	icm-003	003	2021-09-06
488	AU	ICM Airport Technics	icm-004	004	2022-09-07
489	AU	ICM Airport Technics	icm-005	005	2023-09-05
490	US	ID R&D Inc	idrnd-000	000	2024-07-26

Table 7: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
491	US	ID R&D Inc	idrnd-001	001	2025-07-31
492	FR	ID3 Technology	id3-001	001	2017-08-04
493	FR	ID3 Technology	id3-002	002	2017-08-04
494	FR	ID3 Technology	id3-003	003	2018-10-05
495	FR	ID3 Technology	id3-004	004	2019-03-01
496	FR	ID3 Technology	id3-005	005	2020-08-04
497	FR	ID3 Technology	id3-006	006	2020-12-17
498	FR	ID3 Technology	id3-007	007	2021-05-17
499	FR	ID3 Technology	id3-008	008	2021-11-10
500	FR	ID3 Technology	id3-009	009	2026-02-05
501	FR	ID3 Technology	id3-010	010	2026-06-05
502	GB	IDENTY	identy-000	000	2023-04-04
503	GB	IDENTY	identy-001	001	2025-05-21
504	GB	IDENTY	identy-002	002	2025-11-14
505	GB	IDENTY	identy-003	003	2026-05-15
506	KZ	IE SARSENOV	asr-000	000	2025-07-30
507	KZ	IE SARSENOV	asr-001	001	2025-12-01
508	CA	IMDS Software	imds-software-001	001	2022-07-06
509	CA	IMDS Software	imds-software-002	002	2023-02-10
510	CA	IMDS Software	imds-software-003	003	2023-06-20
511	MX	IQSEC	iqsec-001	001	2024-04-12
512	MX	IQSEC	iqsec-002	002	2025-01-08
513	MX	IQSEC	iqsec-003	003	2026-07-20
514	RU	ITMO University	itmo-001	001	2017-06-12
515	RU	ITMO University	itmo-002	002	2017-08-07
516	RU	ITMO University	itmo-003	003	2018-01-23
517	RU	ITMO University	itmo-004	004	2018-06-13
518	RU	ITMO University	itmo-005	005	2018-10-19
519	RU	ITMO University	itmo-006	006	2019-03-01
520	RU	ITMO University	itmo-007	007	2020-01-06
521	RU	ITMO University	itmo-008	008	2021-11-19
522	RU	IVA Cognitive	ivacognitive-001	001	2021-01-29
523	FR	Idemia	idemia-003	003	2018-10-19
524	FR	Idemia	idemia-004	004	2019-03-01
525	FR	Idemia	idemia-005	005	2019-10-11
526	FR	Idemia	idemia-006	006	2020-07-06
527	FR	Idemia	idemia-007	007	2020-12-04
528	FR	Idemia	idemia-008	008	2021-07-07
529	FR	Idemia	idemia-009	009	2022-07-27
530	FR	Idemia	idemia-010	010	2023-06-30
531	FR	Idemia	idemia-011	011	2024-08-06
532	FR	Idemia	morpho-000	000	2017-07-11
533	FR	Idemia	morpho-001	001	2017-07-10
534	FR	Idemia	morpho-002	002	2017-12-28
535	US	Imageware Systems	iws-000	000	2020-08-12
536	GB	Imperial College London	imperial-000	000	2019-03-01
537	GB	Imperial College London	imperial-001	001	2019-03-01
538	GB	Imperial College London	imperial-002	002	2019-08-28
539	US	Incode Technologies Inc	incode-000	000	2018-06-01
540	US	Incode Technologies Inc	incode-001	001	2018-06-13
541	US	Incode Technologies Inc	incode-002	002	2018-10-18
542	US	Incode Technologies Inc	incode-003	003	2019-03-01
543	US	Incode Technologies Inc	incode-004	004	2019-06-12
544	US	Incode Technologies Inc	incode-005	005	2019-10-17
545	US	Incode Technologies Inc	incode-006	006	2020-02-20
546	US	Incode Technologies Inc	incode-007	007	2020-08-25
547	US	Incode Technologies Inc	incode-008	008	2021-01-19
548	US	Incode Technologies Inc	incode-009	009	2021-06-22
549	US	Incode Technologies Inc	incode-010	010	2021-10-22
550	US	Incode Technologies Inc	incode-011	011	2022-08-10
551	US	Incode Technologies Inc	incode-012	012	2023-05-05
552	US	Incode Technologies Inc	incode-013	013	2023-12-08
553	US	Incode Technologies Inc	incode-014	014	2024-04-08
554	US	Incode Technologies Inc	incode-015	015	2025-06-12
555	IT	InfoCert	infocert-001	001	2022-09-08
556	IT	InfoCert	infocert-002	002	2026-03-18
557	IT	InfoCert	infocert-003	003	2026-07-20
558	IN	Innef Labs	innefulabs-000	000	2020-09-04
559	IN	Innominds Software SEZ India Pvt	innominds-001	001	2023-06-23
560	MY	Innov8tif Solutions	innov8tif-000	000	2024-12-10

Table 8: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
561	MY	Innov8tif Solutions	innov8tif-001	001	2026-04-22
562	GB	Innovative Technology	innovativetechnologyltd-001	001	2019-10-22
563	GB	Innovative Technology	innovativetechnologyltd-002	002	2020-02-26
564	SK	Innovatrics	innovatrics-000	000	2017-07-25
565	SK	Innovatrics	innovatrics-001	001	2017-07-25
566	SK	Innovatrics	innovatrics-002	002	2018-02-05
567	SK	Innovatrics	innovatrics-003	003	2018-06-14
568	SK	Innovatrics	innovatrics-004	004	2018-10-19
569	SK	Innovatrics	innovatrics-005	005	2019-03-01
570	SK	Innovatrics	innovatrics-006	006	2019-08-13
571	SK	Innovatrics	innovatrics-007	007	2020-08-19
572	SK	Innovatrics	innovatrics-008	008	2021-12-15
573	SK	Innovatrics	innovatrics-009	009	2022-01-19
574	SK	Innovatrics	innovatrics-010	010	2023-08-03
575	SK	Innovatrics	innovatrics-011	011	2024-03-08
576	SK	Innovatrics	innovatrics-012	012	2024-07-10
577	SK	Innovatrics	innovatrics-013	013	2024-11-13
578	SK	Innovatrics	innovatrics-014	014	2025-03-13
579	SK	Innovatrics	innovatrics-015	015	2025-07-24
580	SK	Innovatrics	innovatrics-016	016	2025-11-28
581	SK	Innovatrics	innovatrics-017	017	2026-07-30
582	CN	InsightFace AI	insightface-000	000	2021-03-17
583	CN	InsightFace AI	insightface-001	001	2021-09-27
584	CN	InsightFace AI	insightface-002	002	2022-01-31
585	CN	InsightFace AI	insightface-003	003	2022-08-23
586	CN	InsightFace AI	insightface-004	004	2023-03-31
587	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-000	000	2022-07-19
588	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-001	001	2023-02-24
589	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-002	002	2023-12-08
590	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-003	003	2025-02-11
591	CN	Institute of Computing Technology	icthtc-000	000	2020-11-29
592	RU	Institute of Information Technologies	iit-000	000	2019-03-01
593	RU	Institute of Information Technologies	iit-001	001	2019-07-05
594	RU	Institute of Information Technologies	iit-002	002	2019-12-04
595	RU	Institute of Information Technologies	iit-003	003	2020-12-01
596	BD	Integrated Software and Technologies Ltd	istl-001	001	2026-08-07
597	IS	Intel Research Group	intelresearch-000	000	2019-07-08
598	IS	Intel Research Group	intelresearch-001	001	2020-01-14
599	IS	Intel Research Group	intelresearch-002	002	2020-07-24
600	IS	Intel Research Group	intelresearch-003	003	2021-01-18
601	IS	Intel Research Group	intelresearch-004	004	2021-08-24
602	IS	Intel Research Group	intelresearch-005	005	2022-02-13
603	IS	Intel Research Group	intelresearch-006	006	2022-12-19
604	IS	Intel Research Group	intelresearch-007	007	2023-05-30
605	IS	Intel Research Group	intelresearch-008	008	2023-10-03
606	IS	Intel Research Group	intelresearch-009	009	2024-03-26
607	IS	Intel Research Group	intelresearch-010	010	2024-08-20
608	IS	Intel Research Group	intelresearch-011	011	2024-12-24
609	IS	Intel Research Group	intelresearch-012	012	2025-07-03
610	KR	IntelliVIX	intellivix-001	001	2022-02-25
611	KR	IntelliVIX	intellivix-002	002	2022-07-14
612	KR	IntelliVIX	intellivix-003	003	2022-12-12
613	KR	IntelliVIX	intellivix-004	004	2023-06-21
614	KR	IntelliVIX	intellivix-005	005	2023-11-14
615	KR	IntelliVIX	intellivix-006	006	2025-07-31
616	AE	Intellibrain Technological Projects	g42-intellibrain-001	001	2022-07-27
617	AE	Intellibrain Technological Projects	g42-intellibrain-002	002	2023-05-10
618	CN	Intelligent Control Technology Co Ltd - IGearx	igearx-face-000	000	2023-03-28
619	US	Intellivision	intellivision-001	001	2017-10-10
620	US	Intellivision	intellivision-002	002	2019-08-23
621	US	Intellivision	intellivision-003	003	2022-03-07
622	US	Intellivision	intellivision-004	004	2022-07-28
623	US	Intellivision	intellivision-005	005	2023-02-06
624	US	Intellivision	intellivision-006	006	2023-07-14
625	US	Intellivision	intellivision-007	007	2023-12-19
626	LU	Intema-LGL Group	intema-000	000	2022-07-15
627	LU	Intema-LGL Group	intema-001	001	2023-01-11
628	US	InterLink labs	interlinklabs-001	001	2024-11-08
629	US	InterLink labs	interlinklabs-002	002	2025-04-21
630	US	InterLink labs	interlinklabs-003	003	2025-11-18

Table 9: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
631	IN	Intozi Tech Pvt Ltd	intozi-001	001	2023-09-25
632	IN	Intozi Tech Pvt Ltd	intozi-002	002	2026-06-21
633	IN	Intpixel Labs Private Ltd	vogic-000	000	2026-07-30
634	US	IrexAI	irex-000	000	2020-12-17
635	US	Iris ID Systems	irisid-001	001	2026-01-30
636	US	Iris ID Systems	irisid-002	002	2026-08-05
637	IL	Is It You	isityou-000	000	2017-06-26
638	MX	Jaak IT	jaakit-001	001	2022-05-20
639	MX	Jaak IT	jaakit-002	002	2025-09-11
640	GB	KYB-AI Ltd	kbyai-000	000	2024-11-29
641	IN	Kaizen Secure Voiz	kaizenvoiz-001	001	2026-06-08
642	KR	Kakao Bank	kakaobank-000	000	2023-02-27
643	KR	Kakao Bank	kakaobank-001	001	2023-10-10
644	KR	Kakao Bank	kakaobank-002	002	2024-04-01
645	KR	Kakao Bank	kakaobank-003	003	2024-09-30
646	KR	Kakao Brain	kakao-001	001	2019-03-01
647	KR	Kakao Brain	kakao-002	002	2019-06-19
648	KR	Kakao Brain	kakao-003	003	2020-02-26
649	KR	Kakao Brain	kakao-004	004	2020-10-28
650	KR	Kakao Brain	kakao-005	005	2021-03-09
651	KR	Kakao Brain	kakao-007	007	2022-01-12
652	KR	Kakao Brain	kakao-008	008	2022-05-12
653	KR	Kakao Brain	kakao-009	009	2023-11-07
654	KR	Kakao Pay Corp	kakaopay-001	001	2021-07-06
655	TH	Kasikorn Labs	kasikornlabs-000	000	2022-03-02
656	TH	Kasikorn Labs	kasikornlabs-001	001	2022-07-26
657	TH	Kasikorn Labs	kasikornlabs-002	002	2022-12-13
658	TH	Kasikorn Labs	kasikornlabs-003	003	2023-10-05
659	TH	Kasikorn Labs	kasikornlabs-004	004	2024-11-04
660	TH	Kasikorn Labs	kasikornlabs-005	005	2025-04-04
661	SG	Kedacom International Pte	kedacom-000	000	2019-06-03
662	GB	Keyless Technologies	keyless-000	000	2024-11-21
663	US	Kloudspot Inc	kloudspot-001	001	2024-09-03
664	US	Kloudspot Inc	kloudspot-002	002	2025-06-16
665	US	Kneron Inc	kneron-003	003	2019-07-01
666	US	Kneron Inc	kneron-005	005	2020-02-21
667	US	Kneron Inc	kneron-006	006	2026-04-10
668	US	KnowUTech LLC	knowutech-000	000	2022-02-13
669	US	Kogniza Technology	kogniza-001	001	2024-03-01
670	KR	Kookmin University	kookmin-001	001	2020-09-28
671	KR	Kookmin University	kookmin-002	002	2021-03-05
672	KR	Korea Identification Inc	koreaid-001	001	2022-12-12
673	TH	Krungthai	krungthai-002	002	2022-06-21
674	CN	KuKe3D Technology	kuke3d-001	001	2021-10-28
675	CN	KuKe3D Technology	kuke3d-002	002	2022-04-14
676	MX	Lebentech Biometrics	lebentech-000	000	2022-02-16
677	MX	Lebentech Biometrics	lebentech-001	001	2023-12-12
678	IN	Lema Labs	lemalabs-001	001	2021-04-13
679	KR	LightVision Inc	lightvision-000	000	2024-06-27
680	KR	LightVision Inc	lightvision-001	001	2025-02-18
681	JP	Line Corporation	lineclova-000	000	2021-03-31
682	JP	Line Corporation	lineclova-001	001	2021-09-26
683	JP	Line Corporation	lineclova-002	002	2022-05-18
684	JP	Line Corporation	lineclova-003	003	2022-11-28
685	CN	Lishui Shuyan Huirong Technology Co	lssytech-000	000	2026-03-18
686	RU	Lomonosov Moscow State University	intsysmsu-000	000	2019-06-18
687	RU	Lomonosov Moscow State University	intsysmsu-001	001	2019-10-22
688	RU	Lomonosov Moscow State University	intsysmsu-002	002	2020-03-12
689	IN	Lookman Electroplast Industries	lookman-002	002	2018-06-13
690	IN	Lookman Electroplast Industries	lookman-004	004	2019-06-03
691	US	Luxand Inc	luxand-000	000	2019-11-07
692	US	Luxand Inc	luxand-001	001	2023-10-17
693	US	Luxand Inc	luxand-003	003	2024-03-25
694	US	Luxand Inc	luxand-004	004	2024-08-01
695	VN	MB Bank	mbbankvn-000	000	2025-02-12
696	RU	MVision	mvision-001	001	2019-11-12
697	IN	Mantra Softech India	mantra-000	000	2021-10-28
698	ID	Matrix Comsec Pvt Ltd	matrixcomsec-000	000	2025-06-11
699	ID	Matrix Comsec Pvt Ltd	matrixcomsec-001	001	2025-11-26
700	ID	Matrix Comsec Pvt Ltd	matrixcomsec-002	002	2026-05-21

Table 10: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
701	AE	Maverick Research	maverickresearch-000	000	2024-12-26
702	CN	Maxvision Technology	maxvision-000	000	2021-10-27
703	CN	Maxvision Technology	maxvision-001	001	2022-03-03
704	CN	Maxvision Technology	maxvision-002	002	2022-07-12
705	CN	Maxvision Technology	maxvision-003	003	2022-11-14
706	CN	Maxvision Technology	maxvision-004	004	2023-05-08
707	CN	Maxvision Technology	maxvision-005	005	2023-09-21
708	CN	Maxvision Technology	maxvision-006	006	2024-01-19
709	CN	Megvii/Face++	megvii-000	000	2018-02-13
710	CN	Megvii/Face++	megvii-001	001	2018-06-15
711	CN	Megvii/Face++	megvii-002	002	2018-10-19
712	CN	Megvii/Face++	megvii-003	003	2021-03-08
713	CN	Megvii/Face++	megvii-004	004	2021-11-19
714	CN	Megvii/Face++	megvii-005	005	2022-03-28
715	CN	Megvii/Face++	megvii-006	006	2022-08-08
716	CN	Megvii/Face++	megvii-007	007	2023-05-17
717	CN	Megvii/Face++	megvii-008	008	2023-10-12
718	CN	Megvii/Face++	megvii-009	009	2024-02-12
719	KR	Metsakuur	metsakuurcompany-001	001	2022-05-12
720	KR	Metsakuur	metsakuurcompany-002	002	2022-09-14
721	KR	Metsakuur	metsakuurcompany-003	003	2023-04-04
722	KR	Metsakuur	metsakuurcompany-004	004	2024-10-24
723	KR	Metsakuur	metsakuurcompany-005	005	2025-07-31
724	CN	Miaxis Biometrics	miaxis-001	001	2022-11-15
725	CN	Miaxis Biometrics	miaxis-002	002	2023-03-22
726	CN	Miaxis Biometrics	miaxis-003	003	2023-09-19
727	CN	Miaxis Biometrics	miaxis-004	004	2025-12-09
728	CN	Miaxis Biometrics	miaxis-005	005	2026-07-24
729	GB	MicroFocus	microfocus-001	001	2018-06-13
730	GB	MicroFocus	microfocus-002	002	2018-10-17
731	GB	MicroFocus	microfocus-003	003	2023-02-23
732	GB	Microfocus	microfocus-000	000	2018-06-13
733	CN	Minivision	minivision-000	000	2020-10-28
734	GB	Mitek Systems	mitek-000	000	2023-01-27
735	GB	Mitek Systems	mitek-001	001	2024-07-02
736	GB	Mitek Systems	mitek-002	002	2024-11-19
737	NO	Mobai	mobai-000	000	2020-08-26
738	NO	Mobai	mobai-001	001	2021-02-17
739	NO	Mobai	mobai-002	002	2025-07-07
740	NO	Mobai	mobai-003	003	2026-04-13
741	ES	Mobbeel Solutions	mobbl-000	000	2021-01-28
742	ES	Mobbeel Solutions	mobbl-001	001	2021-06-16
743	ES	Mobbeel Solutions	mobbl-002	002	2021-12-16
744	ES	Mobbeel Solutions	mobbl-003	003	2022-04-19
745	ES	Mobbeel Solutions	mobbl-004	004	2024-05-20
746	ES	Mobbeel Solutions	mobbl-005	005	2024-10-15
747	KR	Mobipin Technology	mobipintech-000	000	2021-11-23
748	TH	Momentum Digital	sertis-000	000	2019-10-07
749	TH	Momentum Digital	sertis-001	001	2020-07-30
750	TH	Momentum Digital	sertis-002	002	2021-05-13
751	TH	Momentum Digital	sertis-003	003	2023-12-27
752	CN	MoreDian Technology	moredian-000	000	2021-02-24
753	US	Mukh Technologies	mukh-001	001	2022-03-22
754	US	Mukh Technologies	mukh-002	002	2022-11-01
755	US	Mukh Technologies	mukh-003	003	2023-03-15
756	US	Mukh Technologies	mukh-004	004	2023-09-13
757	US	Mukh Technologies	mukh-005	005	2024-12-05
758	CN	Multi-Modality Intelligence	multimodality-000	000	2021-10-19
759	CN	Multi-Modality Intelligence	multimodality-001	001	2022-05-16
760	RU	N-Tech Lab	ntechlab-000	000	2017-03-13
761	RU	N-Tech Lab	ntechlab-001	001	2017-05-10
762	RU	N-Tech Lab	ntechlab-002	002	2017-08-23
763	RU	N-Tech Lab	ntechlab-003	003	2017-12-29
764	RU	N-Tech Lab	ntechlab-004	004	2018-06-14
765	RU	N-Tech Lab	ntechlab-005	005	2018-10-19
766	RU	N-Tech Lab	ntechlab-006	006	2019-03-01
767	RU	N-Tech Lab	ntechlab-007	007	2019-06-25
768	RU	N-Tech Lab	ntechlab-008	008	2020-01-06
769	RU	N-Tech Lab	ntechlab-009	009	2020-12-30
770	RU	N-Tech Lab	ntechlab-010	010	2021-04-30

Table 11: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
771	RU	N-Tech Lab	ntechlab-011	011	2021-09-13
772	RU	N-Tech Lab	ntechlab-012	012	2022-01-20
773	SG	NCS Pte Ltd	ncssg-001	001	2023-03-24
774	SG	NCS Pte Ltd	ncssg-002	002	2024-06-25
775	SG	NCS Pte Ltd	ncssg-003	003	2024-10-30
776	SG	NCS Pte Ltd	ncssg-004	004	2025-04-03
777	CA	NEO Systems	neosystems-001	001	2021-03-02
778	CA	NEO Systems	neosystems-002	002	2021-07-03
779	CA	NEO Systems	neosystems-003	003	2021-11-11
780	CA	NEO Systems	neosystems-004	004	2022-05-02
781	KR	NHN Corp	nhn-001	001	2021-03-15
782	KR	NHN Corp	nhn-002	002	2021-07-15
783	KR	NHN Corp	nhn-003	003	2022-02-22
784	KR	NHN Corp	nhn-004	004	2023-06-26
785	KR	NHN Corp	nhn-005	005	2023-12-06
786	KR	NSENSE Corp	nsensecorp-001	001	2020-10-20
787	KR	NSENSE Corp	nsensecorp-002	002	2021-05-06
788	KR	NSENSE Corp	nsensecorp-003	003	2021-10-29
789	KR	NSENSE Corp	nsensecorp-004	004	2022-09-08
790	KR	NSENSE Corp	nsensecorp-005	005	2023-02-08
791	CN	Nanjing Kiwi Network Technology	kiwitech-000	000	2021-03-19
792	KR	Naver Corp	clova-000	000	2020-10-21
793	KR	Neosecu Co	openface-001	001	2021-06-15
794	TW	Netbridge Technology Incoporation	netbridgetech-001	001	2020-01-08
795	TW	Netbridge Technology Incoporation	netbridgetech-002	002	2020-08-11
796	LT	Neurotechnology	neurotechnology-000	000	2017-03-22
797	LT	Neurotechnology	neurotechnology-001	001	2017-08-07
798	LT	Neurotechnology	neurotechnology-002	002	2017-11-02
799	LT	Neurotechnology	neurotechnology-003	003	2018-02-02
800	LT	Neurotechnology	neurotechnology-004	004	2018-10-19
801	LT	Neurotechnology	neurotechnology-005	005	2019-03-01
802	LT	Neurotechnology	neurotechnology-006	006	2019-06-26
803	LT	Neurotechnology	neurotechnology-008	008	2020-01-08
804	LT	Neurotechnology	neurotechnology-009	009	2020-07-07
805	LT	Neurotechnology	neurotechnology-010	010	2020-11-26
806	LT	Neurotechnology	neurotechnology-011	011	2021-03-26
807	LT	Neurotechnology	neurotechnology-012	012	2021-07-26
808	LT	Neurotechnology	neurotechnology-013	013	2022-01-07
809	LT	Neurotechnology	neurotechnology-015	015	2022-06-07
810	LT	Neurotechnology	neurotechnology-016	016	2023-02-01
811	LT	Neurotechnology	neurotechnology-017	017	2023-07-28
812	LT	Neurotechnology	neurotechnology-018	018	2024-02-09
813	LT	Neurotechnology	neurotechnology-019	019	2024-07-31
814	LT	Neurotechnology	neurotechnology-020	020	2025-08-01
815	IN	NextMeta Labs	nextmetalabs-001	001	2024-07-02
816	IN	NextMeta Labs	nextmetalabs-002	002	2024-12-04
817	IN	NextMeta Labs	nextmetalabs-003	003	2025-07-08
818	IN	NextMeta Labs	nextmetalabs-004	004	2026-01-27
819	US	Noblis	noblis-000	000	2017-07-05
820	ID	Nodeflux	nodeflux-000	000	2019-03-01
821	ID	Nodeflux	nodeflux-001	001	2019-03-01
822	ID	Nodeflux	nodeflux-002	002	2019-08-13
823	LT	Nominder	nominder-000	000	2023-02-06
824	LT	Nominder	nominder-001	001	2023-07-19
825	LT	Nominder	nominder-002	002	2023-11-04
826	LT	Nominder	nominder-003	003	2024-03-06
827	LT	Nominder	nominder-004	004	2024-07-11
828	LT	Nominder	nominder-005	005	2024-11-12
829	LT	Nominder	nominder-006	006	2025-04-07
830	LT	Nominder	nominder-007	007	2026-02-10
831	LT	Nominder	nominder-008	008	2026-07-20
832	NO	Norwegian University of Science and Technology	ntnu-000	000	2026-08-18
833	IN	NotionTag Technologies Private Limited	notiontag-000	000	2019-06-12
834	IN	NotionTag Technologies Private Limited	notiontag-001	001	2021-03-04
835	IN	NotionTag Technologies Private Limited	notiontag-002	002	2021-09-17
836	AE	OCR Studio L.L.C.-FZ	ocrstudio-001	001	2026-07-24
837	IN	OORU Digital Private Ltd	oorudigital-000	000	2026-07-22
838	US	OOTO AI	ooto-001	001	2025-09-11
839	US	Omnigarde Ltd	omnigarde-000	000	2021-04-05
840	US	Omnigarde Ltd	omnigarde-001	001	2021-08-23

Table 12: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
841	US	Omnigarde Ltd	omnigarde-002	002	2022-01-19
842	US	Omnigarde Ltd	omnigarde-003	003	2023-05-10
843	US	Omnigarde Ltd	omnigarde-004	004	2024-06-24
844	US	Omnigarde Ltd	omnigarde-005	005	2025-06-02
845	KR	One More Security	omface-000	000	2021-12-15
846	KR	One More Security	omface-001	001	2022-10-21
847	GB	Onfido	onfido-000	000	2022-12-13
848	GB	Onfido	onfido-001	001	2025-07-17
849	GB	Onfido	onfido-002	002	2025-12-16
850	GB	Onfido	onfido-003	003	2026-04-22
851	VN	Online Mobile Services JSC	momovn-001	001	2024-01-26
852	VN	Online Mobile Services JSC	momovn-002	002	2024-06-04
853	VN	Online Mobile Services JSC	momovn-003	003	2025-09-19
854	IN	Openedge Technologies Pvt	openedge-000	000	2024-01-29
855	AE	Ovision	ovisionllc-001	001	2024-04-03
856	AE	Ovision	ovisionllc-002	002	2024-08-05
857	RU	Oz Forensics LLC	oz-001	001	2020-07-29
858	RU	Oz Forensics LLC	oz-002	002	2021-01-18
859	RU	Oz Forensics LLC	oz-003	003	2021-08-09
860	RU	Oz Forensics LLC	oz-004	004	2021-12-13
861	TW	PAPAGO Inc	papago-001	001	2022-07-19
862	CZ	PAPIL11 S.R.O.	papil11-000	000	2024-01-04
863	CZ	PAPIL11 S.R.O.	papil11-001	001	2024-08-06
864	CZ	PAPIL11 S.R.O.	papil11-002	002	2025-01-02
865	CZ	PAPIL11 S.R.O.	papil11-003	003	2025-04-29
866	CZ	PAPIL11 S.R.O.	papil11-004	004	2026-03-31
867	KR	PNPSecure	pnpsecure-000	000	2025-06-05
868	KR	PNPSecure	pnpsecure-001	001	2026-03-03
869	ID	PT Autentika Digital Indonesia	autentika-000	000	2022-12-05
870	ID	PT Autentika Digital Indonesia	autentika-001	001	2023-04-05
871	ID	PT Autentika Digital Indonesia	autentika-002	002	2023-11-21
872	ID	PT Autentika Digital Indonesia	autentika-003	003	2024-11-12
873	ID	PT Ice House	icehouse-000	000	2026-02-27
874	ID	PT Qlue Performa Indonesia	qluevision-001	001	2022-11-15
875	ID	PT Technology Biometric Indonesia	techbiometricid-001	001	2025-01-02
876	ID	PT Technology Biometric Indonesia	techbiometricid-002	002	2026-04-09
877	CH	PXL Vision AG	pxl-001	001	2020-06-30
878	TW	Palit Microsystems	palit-000	000	2022-05-16
879	TW	Palit Microsystems	palit-001	001	2022-09-26
880	SG	Panasonic R+D Center Singapore	psl-000	000	2018-06-18
881	SG	Panasonic R+D Center Singapore	psl-001	001	2018-10-12
882	SG	Panasonic R+D Center Singapore	psl-002	002	2019-02-28
883	SG	Panasonic R+D Center Singapore	psl-003	003	2019-10-01
884	SG	Panasonic R+D Center Singapore	psl-004	004	2020-03-03
885	SG	Panasonic R+D Center Singapore	psl-005	005	2020-07-06
886	SG	Panasonic R+D Center Singapore	psl-006	006	2020-11-13
887	SG	Panasonic R+D Center Singapore	psl-007	007	2021-03-19
888	SG	Panasonic R+D Center Singapore	psl-008	008	2021-07-21
889	SG	Panasonic R+D Center Singapore	psl-009	009	2021-12-08
890	SG	Panasonic R+D Center Singapore	psl-010	010	2022-04-19
891	SG	Panasonic R+D Center Singapore	psl-011	011	2022-10-06
892	SG	Panasonic R+D Center Singapore	psl-012	012	2023-10-13
893	SG	Panasonic R+D Center Singapore	psl-013	013	2024-04-18
894	SG	Panasonic R+D Center Singapore	psl-014	014	2024-09-11
895	US	Pangiam	pangiam-000	000	2022-04-04
896	US	Pangiam	pangiam-001	001	2023-02-21
897	US	Pangiam	pangiam-002	002	2023-06-21
898	TR	Papilon Savunma	papsav1923-001	001	2021-03-10
899	TR	Papilon Savunma	papsav1923-002	002	2022-01-20
900	TR	Papilon Savunma	papsav1923-003	003	2022-11-25
901	US	Paravision	everai-000	000	2018-10-19
902	US	Paravision	everai-001	001	2018-10-19
903	US	Paravision	everai-002	002	2019-03-01
904	US	Paravision	paravision	003	2019-07-01
905	US	Paravision	paravision-004	004	2019-12-11
906	US	Paravision	paravision-006	006	2021-02-01
907	US	Paravision	paravision-008	008	2021-06-30
908	US	Paravision	paravision-010	010	2022-02-02
909	US	Paravision	paravision-011	011	2022-12-12
910	US	Paravision	paravision-013	013	2023-05-08

Table 13: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
911	US	Paravision	paravision-015	015	2024-09-09
912	US	Paravision	paravision-017	017	2025-01-13
913	US	Paravision	paravision-018	018	2025-06-12
914	US	Paravision	paravision-020	020	2025-12-11
915	SG	Pensees Pte	pensees-001	001	2020-08-17
916	PL	Pentacomp Systemy Informatyczne	pentacomp-000	000	2024-09-26
917	CN	Ping An Technology	pa-002	002	2017-09-25
918	CN	Ping An Technology	pa-003	003	2017-05-25
919	US	Private Identity LLC	privid-001	001	2023-02-06
920	US	Private Identity LLC	privid-002	002	2023-10-03
921	US	Private Identity LLC	privid-003	003	2024-09-24
922	US	Private Identity LLC	privid-004	004	2025-03-05
923	IN	Pyramid Cyber Security + Forensic (P)	pyramid-000	000	2019-11-04
924	IN	Pyramid Cyber Security + Forensic (P)	pyramid-001	001	2024-11-26
925	KZ	Qaz Biometric Systems	qazbs-000	000	2022-06-22
926	KZ	QazSmartVision.AI	qazsmartvisionai-000	000	2024-01-11
927	KZ	QazSmartVision.AI	qazsmartvisionai-001	001	2024-05-21
928	KZ	QazSmartVision.AI	qazsmartvisionai-002	002	2024-09-25
929	KZ	QazSmartVision.AI	qazsmartvisionai-003	003	2025-01-30
930	KZ	QazSmartVision.AI	qazsmartvisionai-004	004	2025-07-18
931	TW	Qnap Security	qnap-000	000	2021-08-09
932	TW	Qnap Security	qnap-001	001	2021-12-09
933	TW	Qnap Security	qnap-002	002	2022-04-15
934	TW	Qnap Security	qnap-003	003	2022-12-09
935	TW	Qnap Security	qnap-004	004	2023-05-05
936	TW	Qnap Security	qnap-005	005	2023-11-16
937	TW	Qnap Security	qnap-006	006	2024-06-14
938	CZ	Quantasoft	quantasoft-003	003	2021-04-19
939	US	ROC	rankone-000	000	2017-03-21
940	US	ROC	rankone-001	001	2017-04-12
941	US	ROC	rankone-002	002	2017-08-18
942	US	ROC	rankone-003	003	2017-11-28
943	US	ROC	rankone-004	004	2018-06-12
944	US	ROC	rankone-005	005	2018-10-09
945	US	ROC	rankone-006	006	2019-02-27
946	US	ROC	rankone-007	007	2019-06-03
947	US	ROC	rankone-008	008	2019-11-12
948	US	ROC	rankone-009	009	2020-06-26
949	US	ROC	rankone-010	010	2020-11-05
950	US	ROC	rankone-011	011	2021-08-27
951	US	ROC	rankone-012	012	2021-12-27
952	US	ROC	rankone-013	013	2022-07-09
953	US	ROC	rankone-014	014	2022-12-21
954	US	ROC	rankone-015	015	2023-07-20
955	US	ROC	roc-016	016	2023-12-19
956	US	ROC	roc-017	017	2024-06-24
957	US	ROC	roc-018	018	2025-01-30
958	US	ROC	roc-019	019	2025-07-21
959	US	ROC	roc-020	020	2026-02-06
960	US	ROC	roc-021	021	2026-08-05
961	US	RealSense	realsense-001	001	2025-12-01
962	US	RealSense	realsense-002	002	2026-06-09
963	US	Realnetworks Inc	realnetworks-000	000	2018-05-30
964	US	Realnetworks Inc	realnetworks-001	001	2018-10-19
965	US	Realnetworks Inc	realnetworks-002	002	2019-02-28
966	US	Realnetworks Inc	realnetworks-003	003	2019-06-12
967	US	Realnetworks Inc	realnetworks-004	004	2021-04-15
968	US	Realnetworks Inc	realnetworks-005	005	2021-09-27
969	US	Realnetworks Inc	realnetworks-006	006	2022-02-09
970	US	Realnetworks Inc	realnetworks-007	007	2022-06-14
971	US	Realnetworks Inc	realnetworks-008	008	2022-11-10
972	AE	Recognito	recognito-000	000	2023-05-24
973	AE	Recognito	recognito-001	001	2023-09-27
974	US	Regula Forensics	regula-000	000	2021-04-13
975	US	Regula Forensics	regula-001	001	2021-12-14
976	US	Regula Forensics	regula-002	002	2025-09-15
977	US	Regula Forensics	regula-003	003	2026-08-20
978	AM	Regular Biometrics Solutions	rebs-000	000	2023-08-22
979	AM	Regular Biometrics Solutions	rebs-001	001	2023-12-22
980	CN	Remark Holdings	remarkai-000	000	2019-03-01

Table 14: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
981	CN	Remark Holdings	remarkai-001	001	2019-03-01
982	CN	Remark Holdings	remarkai-002	002	2019-11-21
983	CN	Remark Holdings	remarkai-003	003	2021-06-22
984	SG	Rendip	rendip-000	000	2021-04-19
985	GB	Reveal Media Ltd	revealmedia-005	005	2021-09-24
986	GB	Reveal Media Ltd	revealmedia-006	006	2022-01-26
987	CN	Rokid Corporation	rokid-000	000	2019-08-01
988	CN	Rokid Corporation	rokid-001	001	2019-12-13
989	KR	SK Telecom	sktelecom-000	000	2021-07-09
990	KR	SQIsoft	sqisoft-001	001	2021-07-27
991	KR	SQIsoft	sqisoft-002	002	2021-11-03
992	KR	SQIsoft	sqisoft-003	003	2022-10-26
993	SG	ST Engineering	stengg-000	000	2023-06-06
994	SA	STCON LLC	stcon-000	000	2022-11-02
995	SA	STCON LLC	stcon-001	001	2023-03-03
996	SA	STCON LLC	stcon-002	002	2023-07-05
997	SA	STCON LLC	stcon-003	003	2023-11-07
998	SA	STCON LLC	stcon-004	004	2024-06-18
999	SA	STCON LLC	stcon-005	005	2024-11-13
1000	SA	STCON LLC	stcon-006	006	2025-03-17
1001	SA	STCON LLC	stcon-007	007	2025-07-18
1002	SA	STCON LLC	stcon-008	008	2025-12-30
1003	SA	STCON LLC	stcon-009	009	2026-05-01
1004	DE	Saffe	saffe-001	001	2018-10-19
1005	DE	Saffe	saffe-002	002	2019-03-01
1006	JP	Saga Densan Center Co Ltd	sdc-000	000	2022-10-18
1007	KR	Samsung S1 Corp	s1-001	001	2019-12-06
1008	KR	Samsung S1 Corp	s1-002	002	2021-03-24
1009	KR	Samsung S1 Corp	s1-003	003	2021-08-24
1010	KR	Samsung S1 Corp	s1-004	004	2022-01-04
1011	KR	Samsung S1 Corp	s1-005	005	2022-06-17
1012	KR	Samsung S1 Corp	s1-006	006	2022-10-17
1013	KR	Samsung S1 Corp	s1-007	007	2023-03-20
1014	KR	Samsung S1 Corp	s1-008	008	2024-10-16
1015	KR	Samsung S1 Corp	s1-009	009	2025-02-25
1016	KR	Samsung-SDS	samsungsds-000	000	2021-10-28
1017	KR	Samsung-SDS	samsungsds-001	001	2022-04-18
1018	KR	Samsung-SDS	samsungsds-002	002	2022-09-16
1019	IN	Samtech InfoNet Limited	samtech-000	000	2017-05-02
1020	IN	Samtech InfoNet Limited	samtech-001	001	2019-10-15
1021	IN	Samtech InfoNet Limited	samtech-002	002	2023-07-06
1022	RU	Satellite Innovation/Eocortex	eocortex-000	000	2020-08-26
1023	IN	ScanalitiX	scanalitix-001	001	2026-05-20
1024	IL	Scanovate	scanovate-001	001	2019-11-12
1025	IL	Scanovate	scanovate-002	002	2020-06-26
1026	IL	Scanovate	scanovate-003	003	2021-11-15
1027	NG	Seamfix	seamfix-001	001	2024-03-08
1028	RO	Securif AI	securifai-001	001	2020-10-06
1029	RO	Securif AI	securifai-002	002	2021-03-19
1030	RO	Securif AI	securifai-003	003	2021-08-03
1031	RO	Securif AI	securifai-004	004	2021-12-21
1032	RO	Securif AI	securifai-005	005	2022-05-16
1033	RO	Securif AI	securifai-006	006	2022-11-14
1034	RO	Securif AI	securifai-007	007	2023-08-17
1035	RO	Securif AI	securifai-008	008	2024-02-26
1036	TH	Security Pitch Co Ltd	securitypitch-000	000	2026-08-19
1037	CN	Sensetime Group	sensetime-001	001	2018-10-19
1038	CN	Sensetime Group	sensetime-002	002	2018-10-19
1039	CN	Sensetime Group	sensetime-003	003	2019-06-04
1040	CN	Sensetime Group	sensetime-004	004	2020-11-20
1041	CN	Sensetime Group	sensetime-005	005	2021-05-24
1042	CN	Sensetime Group	sensetime-006	006	2021-12-28
1043	CN	Sensetime Group	sensetime-007	007	2022-06-17
1044	CN	Sensetime Group	sensetime-008	008	2023-01-04
1045	CN	Sensetime Group	sensetime-010	010	2024-12-03
1046	CN	Sensetime Group	sensetime-011	011	2026-03-17
1047	CN	Sensetime Group	sensetime-012	012	2026-08-10
1048	UZ	Serendipity Ltd	serendipity-000	000	2023-05-31
1049	SG	Seventh Sense Artificial Intelligence	seventhsense-000	000	2021-06-29
1050	SG	Seventh Sense Artificial Intelligence	seventhsense-001	001	2022-03-04

Table 15: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1051	SG	Seventh Sense Artificial Intelligence	seventhsense-002	002	2022-10-17
1052	SG	Seventh Sense Artificial Intelligence	seventhsense-003	003	2023-05-31
1053	SG	Seventh Sense Artificial Intelligence	seventhsense-005	005	2023-10-05
1054	SG	Seventh Sense Artificial Intelligence	seventhsense-006	006	2024-07-09
1055	US	Shaman Software	shaman-000	000	2017-12-05
1056	US	Shaman Software	shaman-001	001	2018-01-13
1057	CN	Shanghai Jiao Tong University	sjtu-001	001	2019-09-27
1058	CN	Shanghai Jiao Tong University	sjtu-002	002	2020-02-12
1059	CN	Shanghai Jiao Tong University	sjtu-003	003	2020-11-02
1060	CN	Shanghai Jiao Tong University	sjtu-004	004	2021-05-13
1061	CN	Shanghai Jiao Tong University	sjtu-005	005	2023-05-05
1062	CN	Shanghai Jiao Tong University	sjtu-006	006	2024-12-06
1063	CN	Shanghai Jiao Tong University	sjtu-007	007	2025-04-16
1064	CN	Shanghai Jiao Tong University	sjtu-008	008	2025-12-11
1065	CN	Shanghai Jiao Tong University	sjtu-009	009	2026-04-13
1066	CN	Shanghai Ulucu Electronics Technology	uluface-002	002	2019-07-10
1067	CN	Shanghai Ulucu Electronics Technology	uluface-003	003	2019-11-12
1068	CN	Shanghai University - Shanghai Film Academy	shu-001	001	2019-06-17
1069	CN	Shanghai University - Shanghai Film Academy	shu-002	002	2019-12-10
1070	CN	Shanghai University - Shanghai Film Academy	shu-003	003	2020-06-24
1071	CN	Shanghai Yitu Technology	yitu-000	000	2017-05-23
1072	CN	Shanghai Yitu Technology	yitu-001	001	2018-06-12
1073	CN	Shanghai Yitu Technology	yitu-002	002	2018-10-19
1074	CN	Shanghai Yitu Technology	yitu-003	003	2019-03-01
1075	CN	Shenzhen AiMall Tech	aimall-001	001	2019-11-12
1076	CN	Shenzhen AiMall Tech	aimall-002	002	2020-03-12
1077	CN	Shenzhen AiMall Tech	aimall-003	003	2020-08-12
1078	CN	Shenzhen EI Networks	einetworks-000	000	2019-08-13
1079	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-002	002	2018-06-13
1080	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-004	004	2019-03-01
1081	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-005	005	2022-02-08
1082	CN	Shenzhen Intellifusion Technologies	intellifusion-001	001	2019-08-22
1083	CN	Shenzhen Intellifusion Technologies	intellifusion-002	002	2020-03-18
1084	CN	Shenzhen Polytechnic University	szpu-000	000	2025-03-28
1085	CN	Shenzhen University-Macau University of Science and Technology	sztu-000	000	2020-12-17
1086	CN	Shenzhen University-Macau University of Science and Technology	sztu-001	001	2021-07-13
1087	GB	Shufti	shufti-001	001	2026-03-17
1088	QA	Sightropic	sightropic-000	000	2025-03-26
1089	QA	Sightropic	sightropic-001	001	2025-11-25
1090	QA	Sightropic	sightropic-002	002	2026-06-04
1091	IN	Smarg Technologies Pvt Ltd	smargtech-001	001	2026-03-17
1092	RU	Smart Engines	smartengines-000	000	2021-08-25
1093	RU	Smart Engines	smartengines-001	001	2022-05-31
1094	ES	Smartbiometrik	smartbiometrik-001	001	2022-05-16
1095	TR	Smarvist Teknoloji	smartvist-000	000	2022-05-10
1096	TR	Smarvist Teknoloji	smartvist-001	001	2023-11-02
1097	DE	Smilart	smilart-001	001	2018-02-06
1098	DE	Smilart	smilart-002	002	2018-02-06
1099	DE	Smilart	smilart-003	003	2019-03-01
1100	TR	Sodec App Inc	sodec-000	000	2021-06-02
1101	TR	Softtech Yazilim Teknolojileri Arastirma Gelistirme ve	softtech-000	000	2026-07-30
1102	CN	Sparktek International Pte Ltd	sparktek-000	000	2025-11-24
1103	CN	Sparktek International Pte Ltd	sparktek-001	001	2026-06-23
1104	IN	Sparsh CCTV	sparsh-001	001	2024-01-02
1105	IN	Staqu Technologies	staqu-000	000	2020-07-15
1106	CN	Star Hybrid Limited	starhybrid-001	001	2019-06-19
1107	US	Stellar Technologies	stellartecs-001	001	2026-05-25
1108	CN	Su Zhou NaZhiTianDi intelligent technology	nazhiai-000	000	2020-06-25
1109	IN	Sukshi Technology Innovation	sukshi-000	000	2022-02-13
1110	GB	Sum and Substance Ltd	sumsub-000	000	2025-12-15
1111	GB	Sum and Substance Ltd	sumsub-001	001	2026-07-27
1112	KR	Suprema AI Inc	suprema-000	000	2021-03-31
1113	KR	Suprema AI Inc	suprema-001	001	2021-09-23
1114	KR	Suprema AI Inc	suprema-002	002	2022-02-11
1115	KR	Suprema AI Inc	suprema-003	003	2022-07-20
1116	KR	Suprema AI Inc	suprema-004	004	2023-01-09
1117	KR	Suprema AI Inc	suprema-005	005	2023-10-25
1118	KR	Suprema ID Inc	supremaid-001	001	2021-05-04
1119	KR	Suprema ID Inc	supremaid-002	002	2022-06-24
1120	GB	Swsam Solutions	swsam-001	001	2023-03-13

Table 16: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1121	RU	Synesis	synesis-000	000	2018-02-09
1122	RU	Synesis	synesis-001	001	2018-05-31
1123	RU	Synesis	synesis-003	003	2018-10-19
1124	RU	Synesis	synesis-004	004	2019-03-01
1125	RU	Synesis	synesis-005	005	2019-06-06
1126	RU	Synesis	synesis-006	006	2019-10-10
1127	RU	Synesis	synesis-007	007	2020-06-24
1128	TW	Synology Inc	synology-000	000	2019-10-23
1129	TW	Synology Inc	synology-001	001	2020-02-26
1130	TW	Synology Inc	synology-002	002	2020-08-20
1131	IT	Syscake S.r.l.	syscake-000	000	2024-11-26
1132	BR	T4iSB	t4isb-000	000	2022-01-28
1133	SA	TAHAKOM	tahakom-001	001	2024-10-17
1134	SA	TAHAKOM	tahakom-002	002	2026-06-15
1135	MY	TNG Digital Sdn Bhd	tngd-001	001	2025-03-17
1136	MY	TNG Digital Sdn Bhd	tngd-002	002	2026-03-18
1137	VN	TNI Tech Co	tnitech-000	000	2024-02-05
1138	AE	TOPPAN Security Co	toppansecurity-001	001	2026-03-30
1139	VN	TTI Group	ttigroup-000	000	2024-12-16
1140	CN	TUPU Technology	tuputech-000	000	2019-10-11
1141	TW	Taiwan AI Labs	ailabs-001	001	2019-12-18
1142	TW	Taiwan Computer Vision	tcv-001	001	2024-05-30
1143	TW	Taiwan Mobile	taiwanmobile-001	001	2024-08-06
1144	TW	Taiwan Mobile	taiwanmobile-002	002	2024-12-11
1145	TW	Taiwan-Certificate Authority Incorporation	twface-000	000	2021-05-14
1146	TW	Taiwan-Certificate Authority Incorporation	twface-001	001	2021-09-14
1147	CH	Tech5 SA	tech5-000	000	2018-10-09
1148	CH	Tech5 SA	tech5-001	001	2018-10-19
1149	CH	Tech5 SA	tech5-002	002	2019-03-01
1150	CH	Tech5 SA	tech5-003	003	2019-08-19
1151	CH	Tech5 SA	tech5-004	004	2020-03-09
1152	CH	Tech5 SA	tech5-005	005	2020-07-24
1153	CH	Tech5 SA	tech5-007	007	2022-12-30
1154	CH	Tech5 SA	tech5-008	008	2023-08-11
1155	CH	Tech5 SA	tech5-009	009	2025-04-17
1156	VN	Techainer	techainer-001	001	2023-11-30
1157	VN	Techainer	techainer-002	002	2025-02-14
1158	VN	Techainer	techainer-003	003	2026-06-08
1159	SA	Technology Control Company	tcc-000	000	2026-02-10
1160	SA	Technology Control Company	tcc-001	001	2026-07-13
1161	TR	Techsign	techsign-000	000	2021-08-25
1162	TR	Techsign	techsign-001	001	2022-07-01
1163	TW	Telexper International	telexper-000	000	2024-06-21
1164	CN	Tencent Deepsea Lab	deepsea-001	001	2019-06-03
1165	RU	Tevian	tevian-000	000	2017-12-10
1166	RU	Tevian	tevian-001	001	2018-01-19
1167	RU	Tevian	tevian-002	002	2018-06-14
1168	RU	Tevian	tevian-003	003	2018-10-19
1169	RU	Tevian	tevian-004	004	2019-03-01
1170	RU	Tevian	tevian-005	005	2019-09-21
1171	RU	Tevian	tevian-006	006	2020-09-11
1172	RU	Tevian	tevian-007	007	2021-08-06
1173	RU	Tevian	tevian-008	008	2021-12-06
1174	US	TigerIT Bangladesh Ltd	tiger-001	001	2018-01-23
1175	US	TigerIT Bangladesh Ltd	tiger-002	002	2018-06-13
1176	US	TigerIT Bangladesh Ltd	tiger-003	003	2018-10-16
1177	US	TigerIT Bangladesh Ltd	tiger-004	004	2020-12-01
1178	US	TigerIT Bangladesh Ltd	tiger-005	005	2021-07-29
1179	US	TigerIT Bangladesh Ltd	tiger-006	006	2021-12-13
1180	US	TigerIT Bangladesh Ltd	tiger-007	007	2025-03-05
1181	RU	Tinkoff Bank	tinkoff-001	001	2021-05-13
1182	CN	TongYi Transportation Technology	tongyi-004	004	2018-10-19
1183	CN	TongYi Transportation Technology	tongyi-005	005	2019-06-12
1184	CN	TongYi Transportation Technology	tongyitrans-001	001	2017-04-01
1185	CN	TongYi Transportation Technology	tongyitrans-002	002	2017-07-15
1186	CN	TongYi Transportation Technology	tongyitrans-003	003	2018-05-15
1187	TW	Toppan ID Gate	toppanidgate-000	000	2021-09-28
1188	JP	Toshiba	toshiba-000	000	2018-01-11
1189	JP	Toshiba	toshiba-001	001	2018-01-11
1190	JP	Toshiba	toshiba-002	002	2018-10-19

Table 17: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1191	JP	Toshiba	toshiba-003	003	2019-03-01
1192	JP	Toshiba	toshiba-004	004	2021-09-27
1193	JP	Toshiba	toshiba-005	005	2022-02-09
1194	JP	Toshiba	toshiba-006	006	2022-06-29
1195	JP	Toshiba	toshiba-007	007	2023-09-13
1196	JP	Toshiba	toshiba-008	008	2024-02-01
1197	JP	Toshiba	toshiba-009	009	2024-09-16
1198	JP	Toshiba	toshiba-010	010	2025-02-03
1199	JP	Toshiba	toshiba-011	011	2025-07-08
1200	JP	Toshiba	toshiba-012	012	2026-01-09
1201	KR	Toss	toss-000	000	2025-09-22
1202	ES	Touchless ID	touchlessid-000	000	2022-05-02
1203	ES	Touchless ID	touchlessid-001	001	2022-09-21
1204	ES	Touchless ID	touchlessid-002	002	2023-01-23
1205	ES	Touchless ID	touchlessid-003	003	2023-07-12
1206	ES	Touchless ID	touchlessid-004	004	2024-12-27
1207	Il	Transmit Security Ltd	transmitsecurity-001	001	2025-12-04
1208	JP	Tripleize	aize-001	001	2021-04-23
1209	JP	Tripleize	aize-002	002	2021-10-08
1210	JP	Tripleize	aize-003	003	2023-08-10
1211	JP	Tripleize	aize-006	006	2026-04-08
1212	VN	TrueID-VNG	trueidvng-001	001	2023-01-05
1213	AE	TrueSight Laboratories Tech LLC	tsight-000	000	2026-05-22
1214	US	Trueface.ai	trueface-000	000	2019-10-08
1215	US	Trueface.ai	trueface-001	001	2020-07-20
1216	US	Trueface.ai	trueface-002	002	2021-03-29
1217	US	Trueface.ai	trueface-003	003	2021-09-30
1218	GB	Trust Stamp	truststamp-001	001	2023-07-14
1219	GB	Trust Stamp	truststamp-002	002	2025-05-23
1220	GB	Trust Stamp	truststamp-003	003	2025-11-19
1221	CN	TuringTech.vip	turingtechvip-001	001	2022-02-03
1222	CN	TuringTech.vip	turingtechvip-002	002	2022-07-27
1223	TR	Turkcell Technology	turkcell-000	000	2022-10-11
1224	TR	Turkcell Technology	turkcell-001	001	2023-03-15
1225	TR	Turkcell Technology	turkcell-002	002	2024-03-18
1226	CN	ULSee Inc	ulsee-001	001	2019-07-31
1227	UN	UNICC-Solution Architecture Section	unicc-001	001	2023-02-17
1228	UN	UNICC-Solution Architecture Section	unicc-002	002	2023-06-07
1229	UN	UNICC-Solution Architecture Section	unicc-003	003	2023-10-16
1230	UN	UNICC-Solution Architecture Section	unicc-004	004	2024-09-23
1231	TW	UXLabs	uxlabs-001	001	2022-09-19
1232	TW	UXLabs	uxlabs-003	003	2023-07-28
1233	UZ	UZINFOCOM	uzinfocom-000	000	2025-07-28
1234	UZ	UZINFOCOM	uzinfocom-001	001	2026-04-08
1235	HU	Ultinous	ultinous-000	000	2017-12-18
1236	HU	Ultinous	ultinous-001	001	2018-02-02
1237	FR	Unissey	unissey-001	001	2021-11-29
1238	FR	Unissey	unissey-002	002	2022-04-29
1239	FR	Unissey	unissey-003	003	2022-12-19
1240	FR	Unissey	unissey-004	004	2023-07-19
1241	FR	Unissey	unissey-005	005	2024-08-06
1242	FR	Unissey	unissey-006	006	2025-09-11
1243	FR	Unissey	unissey-007	007	2026-03-25
1244	SG	UniverseAI	universeai-000	000	2026-08-11
1245	PT	Universidade de Coimbra	visteam-000	000	2020-01-14
1246	PT	Universidade de Coimbra	visteam-001	001	2021-03-16
1247	PT	Universidade de Coimbra	visteam-002	002	2021-08-20
1248	PT	Universidade de Coimbra	visteam-003	003	2022-01-31
1249	PT	Universidade de Coimbra	visteam-004	004	2022-08-01
1250	PT	Universidade de Coimbra	visteam-005	005	2023-01-04
1251	PT	Universidade de Coimbra	visteam-006	006	2023-05-25
1252	PT	Universidade de Coimbra	visteam-007	007	2023-10-18
1253	PT	Universidade de Coimbra	visteam-008	008	2024-02-06
1254	PT	Universidade de Coimbra	visteam-009	009	2024-07-26
1255	PT	Universidade de Coimbra	visteam-010	010	2025-05-28
1256	GB	University of Surrey-CVSSP	surrey-cvssp-000	000	2022-03-25
1257	GB	University of Surrey-CVSSP	surrey-cvssp-001	001	2022-09-22
1258	GB	University of Surrey-CVSSP	surrey-cvssp-002	002	2023-02-16
1259	GB	University of Surrey-CVSSP	surrey-cvssp-003	003	2023-09-13
1260	GB	University of Surrey-CVSSP	surrey-cvssp-004	004	2024-07-11

Table 18: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1261	US	VCognition	vcog-002	002	2017-06-12
1262	VN	VNIS Joint Stock	vnis-000	000	2024-01-08
1263	US	Vaidio	vaidio-001	001	2025-12-10
1264	US	Vcortex Labs	vcortex-001	001	2023-06-14
1265	LB	Veertec SAL (Offshore)	veertec-001	001	2026-07-07
1266	ES	Veridas Digital Authentication Solutions S.L.	veridas-001	001	2019-03-01
1267	ES	Veridas Digital Authentication Solutions S.L.	veridas-002	002	2019-03-01
1268	ES	Veridas Digital Authentication Solutions S.L.	veridas-003	003	2019-11-27
1269	ES	Veridas Digital Authentication Solutions S.L.	veridas-004	004	2020-07-21
1270	ES	Veridas Digital Authentication Solutions S.L.	veridas-005	005	2020-12-15
1271	ES	Veridas Digital Authentication Solutions S.L.	veridas-006	006	2021-04-15
1272	ES	Veridas Digital Authentication Solutions S.L.	veridas-007	007	2021-09-02
1273	ES	Veridas Digital Authentication Solutions S.L.	veridas-008	008	2022-10-17
1274	ES	Veridas Digital Authentication Solutions S.L.	veridas-009	009	2023-05-08
1275	ES	Veridas Digital Authentication Solutions S.L.	veridas-010	010	2025-03-18
1276	GB	Veridium	veridium-000	000	2022-03-28
1277	GB	Veridium	veridium-001	001	2022-11-03
1278	GB	Veridium	veridium-002	002	2023-06-30
1279	GB	Veridium	veridium-003	003	2024-02-21
1280	GB	Veridium	veridium-004	004	2026-07-07
1281	KZ	Verigram	verigram-000	000	2021-09-06
1282	KZ	Verigram	verigram-001	001	2022-03-09
1283	KZ	Verigram	verigram-002	002	2023-04-03
1284	KZ	Verigram	verigram-003	003	2023-08-09
1285	KZ	Verigram	verigram-004	004	2026-08-10
1286	ID	Verihubs	verihubs-inteligensia-000	000	2021-07-27
1287	ID	Verihubs	verihubs-inteligensia-001	001	2022-06-16
1288	ID	Verihubs	verihubs-inteligensia-002	002	2023-06-07
1289	ID	Verijelas	verijelas-000	000	2022-08-01
1290	TW	Via Technologies Inc	via-000	000	2019-07-08
1291	TW	Via Technologies Inc	via-001	001	2020-01-08
1292	TW	Via Technologies Inc	via-004	004	2023-02-23
1293	TW	Via Technologies Inc	via-005	005	2023-12-26
1294	TW	Via Technologies Inc	via-006	006	2025-02-24
1295	AE	Viante.AI	viante-000	000	2023-06-09
1296	AE	Viante.AI	viante-001	001	2024-12-30
1297	AE	Viante.AI	viante-002	002	2025-06-17
1298	AE	Viante.AI	viante-003	003	2026-03-27
1299	AE	Viante.AI	viante-004	004	2026-08-05
1300	DE	Videmo Intelligente Videoanalyse	videmo-000	000	2019-12-19
1301	DE	Videmo Intelligente Videoanalyse	videmo-001	001	2021-12-22
1302	DE	Videmo Intelligente Videoanalyse	videmo-002	002	2022-08-31
1303	DE	Videmo Intelligente Videoanalyse	videmo-003	003	2024-06-18
1304	IN	Videonetics Technology Pvt	videonetics-001	001	2019-06-19
1305	IN	Videonetics Technology Pvt	videonetics-002	002	2019-11-21
1306	VN	Vietnam National Cyber Security Technology Corporation	ncsgroup-000	000	2024-03-18
1307	VN	Vietnam Payment Solution	vnpay-000	000	2023-12-05
1308	VN	Vietnam Payment Solution	vnpay-001	001	2024-10-15
1309	VN	Vietnam Posts and Telecommunications Group	vnpt-001	001	2021-01-08
1310	VN	Vietnam Posts and Telecommunications Group	vnpt-002	002	2021-06-08
1311	VN	Vietnam Posts and Telecommunications Group	vnpt-003	003	2021-12-01
1312	VN	Vietnam Posts and Telecommunications Group	vnpt-004	004	2022-04-15
1313	VN	Vietnam Posts and Telecommunications Group	vnpt-005	005	2022-08-24
1314	VN	Vietnam Posts and Telecommunications Group	vnpt-006	006	2023-08-18
1315	VN	Vietnam Posts and Telecommunications Group	vnpt-007	007	2024-04-29
1316	VN	Vietnam Posts and Telecommunications Group	vnpt-008	008	2024-09-20
1317	VN	Vietnam Posts and Telecommunications Group	vnpt-009	009	2025-06-04
1318	VN	Viettel Artificial Intelligence	viettelai-000	000	2023-06-26
1319	VN	Viettel Artificial Intelligence	viettelai-001	001	2023-11-20
1320	VN	Viettel Artificial Intelligence	viettelai-002	002	2024-07-17
1321	VN	Viettel Artificial Intelligence	viettelai-003	003	2025-02-27
1322	VN	Viettel Artificial Intelligence and Data Services Center	viettelai-004	004	2025-11-26
1323	VN	Viettel Artificial Intelligence and Data Services Center	viettelai-005	005	2026-05-11
1324	VN	Viettel Group	vts-000	000	2020-11-04
1325	VN	Viettel Group	vts-001	001	2022-04-20
1326	VN	Viettel Group	vts-002	002	2024-06-04
1327	VN	Viettel High Technology	viettelhightech-000	000	2021-08-04
1328	VN	Viettel High Technology	viettelhightech-001	001	2025-03-18
1329	VN	Viettel High Technology	viettelhightech-002	002	2025-10-15
1330	US	Vigilant Solutions	vigilantsolutions-000	000	2017-03-30

Table 19: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1331	US	Vigilant Solutions	vigilantsolutions-001	001	2017-06-13
1332	US	Vigilant Solutions	vigilantsolutions-002	002	2017-09-28
1333	US	Vigilant Solutions	vigilantsolutions-003	003	2018-01-23
1334	US	Vigilant Solutions	vigilantsolutions-004	004	2018-06-14
1335	US	Vigilant Solutions	vigilantsolutions-005	005	2018-10-11
1336	US	Vigilant Solutions	vigilantsolutions-006	006	2019-03-01
1337	US	Vigilant Solutions	vigilantsolutions-007	007	2019-06-27
1338	US	Vigilant Solutions	vigilantsolutions-008	008	2020-08-03
1339	US	Vigilant Solutions	vigilantsolutions-009	009	2020-12-07
1340	US	Vigilant Solutions	vigilantsolutions-010	010	2021-04-07
1341	US	Vigilant Solutions	vigilantsolutions-011	011	2021-08-07
1342	VN	VinAI Research VietNam	vinai-000	000	2020-09-24
1343	VN	VinBigData	vinbigdata-001	001	2022-01-06
1344	VN	VinBigData	vinbigdata-002	002	2022-06-07
1345	VN	VinBigData	vinbigdata-003	003	2023-06-23
1346	VN	VinBigData	vinbigdata-004	004	2024-05-08
1347	VN	VinBigData	vinbigdata-005	005	2024-10-16
1348	VN	VinBigData	vinbigdata-006	006	2025-02-19
1349	VN	VinBigData	vinbigdata-007	007	2025-07-24
1350	SE	Visage Technologies	visage-000	000	2020-12-09
1351	BR	Visica AI Solutions	visica-001	001	2025-02-14
1352	BR	Visica AI Solutions	visica-002	002	2025-07-15
1353	BR	Visica AI Solutions	visica-003	003	2026-05-06
1354	FI	Visidon	vd-000	000	2018-06-13
1355	FI	Visidon	vd-001	001	2019-02-26
1356	FI	Visidon	vd-002	002	2021-04-12
1357	FI	Visidon	vd-003	003	2021-10-12
1358	CN	Vision Intelligence Center of Meituan	meituan-000	000	2021-05-14
1359	CN	Vision Intelligence Center of Meituan	meituan-001	001	2022-03-25
1360	CN	Vision Intelligence Center of Meituan	meituan-002	002	2022-09-14
1361	CN	Vision Intelligence Center of Meituan	meituan-003	003	2023-03-06
1362	CN	Vision Intelligence Center of Meituan	meituan-004	004	2023-08-16
1363	PT	Vision-Box	visionbox-000	000	2019-02-26
1364	PT	Vision-Box	visionbox-001	001	2019-03-01
1365	PT	Vision-Box	visionbox-002	002	2021-04-29
1366	PT	Vision-Box	visionbox-003	003	2023-02-01
1367	PT	Vision-Box	visionbox-004	004	2023-06-30
1368	PT	Vision-Box	visionbox-005	005	2024-07-29
1369	PT	Vision-Box	visionbox-006	006	2025-07-29
1370	PT	Vision-Box	visionbox-007	007	2026-02-03
1371	PT	Vision-Box	visionbox-008	008	2026-07-27
1372	RU	VisionLabs	visionlabs-001	001	2017-06-12
1373	RU	VisionLabs	visionlabs-002	002	2017-09-08
1374	RU	VisionLabs	visionlabs-003	003	2018-02-02
1375	RU	VisionLabs	visionlabs-004	004	2018-06-15
1376	RU	VisionLabs	visionlabs-005	005	2018-10-19
1377	RU	VisionLabs	visionlabs-006	006	2019-03-01
1378	RU	VisionLabs	visionlabs-007	007	2019-06-12
1379	RU	VisionLabs	visionlabs-008	008	2020-01-06
1380	RU	VisionLabs	visionlabs-009	009	2020-07-27
1381	RU	VisionLabs	visionlabs-010	010	2021-01-25
1382	RU	VisionLabs	visionlabs-011	011	2021-10-13
1383	TW	Vivotek	vivotek-001	001	2024-04-03
1384	TW	Vivotek	vivotek-002	002	2024-08-15
1385	AU	Vixvizon	vixvizion-000	000	2019-06-19
1386	AU	Vixvizon	vixvizion-001	001	2019-10-22
1387	AU	Vixvizon	vixvizion-002	002	2020-12-31
1388	AU	Vixvizon	vixvizion-003	003	2021-05-18
1389	AU	Vixvizon	vixvizion-004	004	2021-09-20
1390	AU	Vixvizon	vixvizion-005	005	2022-03-03
1391	AU	Vixvizon	vixvizion-006	006	2022-08-11
1392	AU	Vixvizon	vixvizion-007	007	2023-01-17
1393	US	Vocalize	vocalize-001	001	2024-03-27
1394	RU	Vocord	vocord-001	001	2017-04-21
1395	RU	Vocord	vocord-002	002	2017-06-07
1396	RU	Vocord	vocord-003	003	2018-01-23
1397	RU	Vocord	vocord-004	004	2018-06-15
1398	RU	Vocord	vocord-005	005	2018-10-19
1399	RU	Vocord	vocord-006	006	2019-03-01
1400	RU	Vocord	vocord-007	007	2019-06-06

Table 20: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1401	RU	Vocord	vocord-008	008	2020-01-31
1402	RU	Vocord	vocord-009	009	2020-12-28
1403	RU	Vocord	vocord-010	010	2021-12-20
1404	BR	Vsoft	vsoft-001	001	2025-08-01
1405	BR	Vsoft	vsoft-002	002	2025-11-28
1406	BR	Vsoft	vsoft-003	003	2026-03-30
1407	BR	Vsoft	vsoft-004	004	2026-07-31
1408	US	Wicket	wicket-000	000	2022-02-14
1409	CN	Winsense	winsense-000	000	2019-06-17
1410	CN	Winsense	winsense-001	001	2019-10-16
1411	CN	Winsense	winsense-002	002	2020-11-20
1412	MY	Wise AI SDN BHD	wiseai-001	001	2022-10-25
1413	CN	Wuhan Tianyu Information Industry	wuhantianyu-001	001	2021-08-05
1414	CN	X-Laboratory	x-laboratory-000	000	2019-09-03
1415	CN	X-Laboratory	x-laboratory-001	001	2020-01-21
1416	CN	Xforward AI Technology	xforwardai-000	000	2020-02-06
1417	CN	Xforward AI Technology	xforwardai-001	001	2020-09-25
1418	CN	Xforward AI Technology	xforwardai-002	002	2021-02-10
1419	CN	Xiamen Meiya Pico Information	meiya-001	001	2019-03-01
1420	CN	Xiamen University	xm-000	000	2020-10-19
1421	PT	YooniK	yoonik-000	000	2020-06-24
1422	PT	YooniK	yoonik-001	001	2020-10-26
1423	PT	YooniK	yoonik-002	002	2021-09-06
1424	PT	YooniK	yoonik-003	003	2022-01-06
1425	PT	YooniK	yoonik-004	004	2023-02-10
1426	GB	Yoti	yoti-001	001	2024-02-13
1427	GB	Yoti	yoti-002	002	2024-07-03
1428	TW	Yuan High-Tech Development	yuan-000	000	2020-06-30
1429	TW	Yuan High-Tech Development	yuan-001	001	2021-01-08
1430	TW	Yuan High-Tech Development	yuan-002	002	2021-05-17
1431	TW	Yuan High-Tech Development	yuan-003	003	2021-09-17
1432	TW	Yuan High-Tech Development	yuan-004	004	2022-01-14
1433	TW	Yuan High-Tech Development	yuan-005	005	2022-06-22
1434	TW	Yuan High-Tech Development	yuan-006	006	2022-12-14
1435	TW	Yuan High-Tech Development	yuan-007	007	2023-05-11
1436	TW	Yuan High-Tech Development	yuan-008	008	2023-11-06
1437	TW	Yuan High-Tech Development	yuan-009	009	2024-04-24
1438	TW	Yuan High-Tech Development	yuan-010	010	2024-10-22
1439	CN	YuanLi Technology	yuanli-001	001	2026-03-23
1440	CN	Yuntu Data and Technology	ytu-000	000	2021-06-16
1441	CN	Zhuhai Yisheng Electronics Technology	yisheng-000	000	2017-08-17
1442	CN	Zhuhai Yisheng Electronics Technology	yisheng-001	001	2017-08-17
1443	CN	Zhuhai Yisheng Electronics Technology	yisheng-002	002	2018-02-01
1444	CN	Zhuhai Yisheng Electronics Technology	yisheng-004	004	2018-06-12
1445	IN	i2v Systems	i2v-001	001	2024-03-11
1446	IN	i2v Systems	i2v-002	002	2025-01-15
1447	VN	iCOMM Media and Tech	icomm-000	000	2023-12-26
1448	CN	iQIYI Inc	iqface-000	000	2019-06-04
1449	CN	iQIYI Inc	iqface-001	001	2019-12-11
1450	CN	iQIYI Inc	iqface-003	003	2021-02-23
1451	TW	iSAP Solution Corporation	isap-001	001	2019-08-07
1452	TW	iSAP Solution Corporation	isap-002	002	2020-09-01
1453	TW	ioNetworks Inc	ionetworks-000	000	2021-07-20
1454	TW	ioNetworks Inc	ionetworks-001	001	2023-08-01
1455	TW	ioNetworks Inc	ionetworks-002	002	2023-12-08
1456	TW	ioNetworks Inc	ionetworks-003	003	2024-05-09
1457	TW	ioNetworks Inc	ionetworks-004	004	2026-04-22
1458	KR	useB	useb-000	000	2023-05-08
1459	KR	useB	useb-001	001	2023-09-12
1460	KR	useB	useb-002	002	2024-01-19
1461	KR	useB	useb-003	003	2024-08-09
1462	KR	useB	useb-004	004	2024-12-23

Table 21: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1	2060io-001	567279	43224	246 629	1447 10488 ± 0	1033 958 ± 22	757 983 ± 38	763 1045 ± 205	1075 1724 ± 310	1098 3474 ± 1170	1123 7105 ± 42	1119 7071 ± 47
2	2060io-002	579607	42980	649 1559	1448 10488 ± 0	352 398 ± 0	239 418 ± 1	303 525 ± 1	494 786 ± 1	-	1119 7000 ± 91	1113 6965 ± 85
3	20face-000	117155	324083	601 1417	759 2048 ± 0	164 232 ± 1	96 223 ± 1	77 226 ± 4	62 222 ± 1	44 224 ± 1	1372 44880 ± 134	1371 44462 ± 163
4	20face-001	226824	324119	629 1512	1242 4096 ± 0	206 279 ± 2	116 266 ± 1	90 266 ± 1	79 267 ± 1	60 267 ± 0	1067 5553 ± 54	1065 5541 ± 65
5	3divi-000	169360	139460	-	126 512 ± 0	216 285 ± 52	-	-	-	-	109 378 ± 20	113 375 ± 19
6	3divi-001	190867	139473	-	1300 4096 ± 0	199 274 ± 47	-	-	-	-	240 636 ± 19	251 634 ± 16
7	3divi-002	190867	175117	-	1211 4096 ± 0	207 279 ± 48	-	-	-	-	274 692 ± 22	299 707 ± 26
8	3divi-003	191636	55903	-	1292 4096 ± 0	667 650 ± 90	-	-	-	-	229 627 ± 11	238 623 ± 32
9	3divi-004	263670	50794	-	778 2048 ± 0	1061 984 ± 131	-	-	-	-	331 794 ± 35	344 801 ± 40
10	3divi-005	264098	53870	-	835 2048 ± 0	1070 993 ± 83	873 1136 ± 88	979 1309 ± 137	929 1348 ± 131	964 1671 ± 166	328 790 ± 20	335 791 ± 23
11	3divi-006	273866	52656	541 1252	637 2048 ± 0	677 654 ± 1	458 651 ± 0	427 660 ± 1	394 678 ± 2	381 759 ± 13	322 775 ± 19	327 770 ± 22
12	3divi-007	483115	24723	554 1288	341 2048 ± 0	626 615 ± 1	416 616 ± 1	384 623 ± 1	359 644 ± 1	356 727 ± 5	286 707 ± 31	304 712 ± 25
13	accurascan-001	14057	24437	403 944	141 512 ± 0	141 204 ± 27	91 215 ± 17	74 220 ± 16	70 237 ± 18	84 339 ± 19	1246 15634 ± 54	1248 15647 ± 77
14	accurascan-002	14057	24437	405 945	104 512 ± 0	100 173 ± 23	69 188 ± 14	61 191 ± 14	54 208 ± 16	71 317 ± 14	1207 12255 ± 69	1209 12231 ± 72
15	accurascan-003	998397	27018	758 1996	435 2048 ± 0	710 676 ± 1	486 677 ± 7	471 700 ± 8	461 751 ± 16	520 950 ± 47	165 540 ± 10	169 536 ± 10
16	acer-000	109735	88323	-	599 2048 ± 0	157 222 ± 0	102 233 ± 4	82 238 ± 4	77 262 ± 18	93 356 ± 46	427 1065 ± 40	455 1109 ± 35
17	acer-001	36650	66086	137 417	120 512 ± 0	137 199 ± 0	103 237 ± 28	78 229 ± 26	72 242 ± 37	56 259 ± 21	794 2453 ± 44	799 2461 ± 62
18	acer-002	43922	624858	71 192	559 2048 ± 0	115 184 ± 0	67 184 ± 0	52 185 ± 0	43 185 ± 0	30 186 ± 0	922 3370 ± 47	927 3350 ± 54
19	acisw-003	282029	35664	-	1455 18467 ± 8	163 232 ± 1	118 267 ± 22	268 488 ± 28	640 990 ± 24	1086 2977 ± 129	1458 847908 ± 16757	1458 851850 ± 17018
20	acisw-006	282029	36107	-	1454 18465 ± 8	155 219 ± 0	100 227 ± 0	192 410 ± 1	532 838 ± 1	1063 2532 ± 10	1454 548137 ± 16513	1454 549586 ± 9238
21	acisw-007	267619	36111	89 287	353 2048 ± 0	214 283 ± 0	137 293 ± 3	196 414 ± 0	167 404 ± 0	172 484 ± 1	513 1316 ± 22	518 1297 ± 23
22	acisw-008	171703	39359	481 1102	893 2048 ± 0	356 400 ± 1	189 362 ± 28	159 369 ± 9	97 300 ± 2	80 336 ± 5	515 1327 ± 19	526 1337 ± 32
23	adera-001	0	79272	-	1164 2560 ± 0	43 97 ± 0	-	-	-	-	602 1604 ± 71	614 1649 ± 56
24	adera-002	0	749797	-	1397 5120 ± 0	1354 1394 ± 11	1044 1381 ± 1	1037 1393 ± 1	976 1403 ± 1	884 1464 ± 2	731 2163 ± 32	731 2158 ± 28
25	adera-003	0	749778	393 917	1396 5120 ± 0	1340 1381 ± 12	1046 1385 ± 1	1038 1394 ± 1	972 1401 ± 1	885 1469 ± 1	729 2148 ± 34	726 2130 ± 32
26	adera-004	0	959123	703 1748	1404 6144 ± 0	1263 1246 ± 1	931 1204 ± 1	914 1230 ± 2	819 1207 ± 2	737 1254 ± 1	663 1840 ± 34	664 1828 ± 31
27	adera-005	0	959113	697 1734	1405 6144 ± 0	984 914 ± 1	662 879 ± 3	614 880 ± 2	567 886 ± 2	494 910 ± 2	679 1896 ± 14	675 1874 ± 20
28	advance-002	257173	20434	-	719 2048 ± 0	871 811 ± 2	601 803 ± 2	465 696 ± 2	416 699 ± 4	346 718 ± 1	394 987 ± 10	400 988 ± 45
29	advance-003	258867	78699	189 518	828 2048 ± 0	570 586 ± 0	-	346 583 ± 0	309 588 ± 0	249 591 ± 1	662 1813 ± 17	655 1788 ± 26
30	advance-004	803133	954494	272 681	703 2048 ± 0	1143 1099 ± 20	847 1107 ± 15	801 1093 ± 21	725 1103 ± 21	658 1138 ± 21	689 1935 ± 35	693 1936 ± 32
31	advance-005	277607	126218	242 624	544 2048 ± 0	551 573 ± 1	376 574 ± 2	336 573 ± 2	295 574 ± 2	259 601 ± 2	585 1517 ± 16	587 1518 ± 26
32	advance-006	472567	136481	227 605	700 2048 ± 0	703 673 ± 0	473 670 ± 0	442 673 ± 0	400 684 ± 0	373 745 ± 0	341 808 ± 4	350 808 ± 1
33	advance-007	1059702	136627	707 1768	588 2048 ± 0	1246 1224 ± 0	950 1228 ± 0	917 1234 ± 0	849 1236 ± 0	751 1277 ± 0	334 796 ± 2	342 800 ± 1
34	affinsys-ai-000	38016	46159	43 142	108 512 ± 0	15 48 ± 0	11 52 ± 0	25 106 ± 0	21 110 ± 0	35 194 ± 0	701 2007 ± 25	703 2006 ± 28
35	afisbiometrics-000	545886	32882	470 1085	130 512 ± 0	1238 1219 ± 1	871 1135 ± 1	840 1137 ± 2	761 1137 ± 1	665 1147 ± 1	535 1400 ± 29	532 1357 ± 32
36	afrengine-000	151875	382842	61 178	1204 4096 ± 0	52 107 ± 0	33 112 ± 0	100 284 ± 2	415 697 ± 2	1096 3299 ± 17	1393 54329 ± 140	1394 56195 ± 256
37	afrengine-001	410712	391289	429 1015	1207 4096 ± 0	1048 970 ± 0	755 982 ± 1	699 979 ± 2	645 996 ± 0	545 990 ± 4	1392 54240 ± 169	1393 56125 ± 179
38	afrengine-002	579618	239125	432 1018	770 2048 ± 0	919 851 ± 3	815 1065 ± 16	715 993 ± 7	544 855 ± 5	461 857 ± 9	496 1259 ± 13	509 1270 ± 23
39	afrengine-003	579563	239117	430 1017	907 2048 ± 0	1138 1088 ± 1	686 911 ± 15	787 1067 ± 5	526 833 ± 3	544 989 ± 2	509 1294 ± 48	515 1284 ± 29
40	afrengine-004	579587	239117	431 1018	426 2048 ± 0	946 875 ± 2	723 951 ± 119	737 1018 ± 5	573 895 ± 1	581 1043 ± 6	493 1254 ± 17	505 1258 ± 23
41	afrengine-005	694908	239155	433 1018	589 2048 ± 0	898 831 ± 10	744 972 ± 52	580 834 ± 2	570 891 ± 9	606 1065 ± 1	502 1276 ± 19	511 1273 ± 20
42	aifirst-001	224157	808777	166 485	652 2048 ± 0	577 587 ± 2	372 568 ± 2	348 584 ± 3	317 601 ± 6	377 755 ± 5	444 1099 ± 14	448 1087 ± 45
43	aigen-001	256958	595227	501 1154	427 2048 ± 0	1387 1448 ± 9	1096 1451 ± 8	1128 1759 ± 6	1150 2594 ± 4	1127 5691 ± 44	972 3772 ± 57	970 3736 ± 56
44	aigen-002	205300	1316138	387 896	886 2048 ± 0	573 586 ± 24	385 582 ± 4	645 920 ± 4	1081 1758 ± 5	1124 5427 ± 17	958 3678 ± 44	960 3646 ± 48

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 22: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
45	aigroup-000	357973	390046	130 398	533 2048 ± 0	292 351 ± 0	167 337 ± 1	144 353 ± 1	127 356 ± 1	110 385 ± 1	1168 9603 ± 216	1170 9689 ± 373
46	aigroup-001	298974	390068	217 587	414 2048 ± 0	507 535 ± 0	337 531 ± 0	307 533 ± 0	273 544 ± 5	253 595 ± 0	1162 9526 ± 170	1165 9562 ± 213
47	aigroup-002	271124	396396	384 888	1129 2078 ± 0	201 275 ± 0	127 276 ± 0	99 283 ± 2	104 318 ± 5	134 419 ± 0	1412 66379 ± 111	1412 66387 ± 124
48	ailabs-001	1054663	338989	540 1252	343 2048 ± 0	692 664 ± 4	580 774 ± 50	848 1145 ± 12	1117 1972 ± 74	1121 5205 ± 272	1441 104034 ± 661	1441 103415 ± 7722
49	aimall-001	128820	57257	-	236 1024 ± 0	266 326 ± 1	-	-	-	-	998 4127 ± 72	984 3909 ± 166
50	aimall-002	370156	25210	655 1576	577 2048 ± 0	831 776 ± 4	705 927 ± 27	661 940 ± 21	614 955 ± 34	558 1003 ± 75	1424 72811 ± 7399	1420 71216 ± 6286
51	aimall-003	504324	171935	741 1913	263 1024 ± 0	685 662 ± 1	550 740 ± 51	517 752 ± 62	455 741 ± 46	422 807 ± 47	1351 34565 ± 93	1353 34598 ± 118
52	aipractix-000	419503	113764	627 1504	1349 4112 ± 0	529 554 ± 0	358 553 ± 0	327 556 ± 0	279 548 ± 0	243 581 ± 0	748 2243 ± 37	754 2257 ± 40
53	aiseemu-001	0	1005354	873 2701	1299 4096 ± 0	1074 1001 ± 1	783 1017 ± 0	734 1014 ± 5	673 1022 ± 2	599 1059 ± 4	1036 4864 ± 25	1040 4855 ± 32
54	aiseemu-002	0	1216980	940 3453	1215 4096 ± 0	1297 1298 ± 5	1004 1303 ± 4	983 1313 ± 2	921 1329 ± 0	806 1348 ± 2	1041 4917 ± 37	1042 4916 ± 37
55	aiunionface-000	241642	840295	132 402	908 2048 ± 0	657 637 ± 13	565 754 ± 41	741 1025 ± 28	797 1179 ± 29	956 1639 ± 47	431 1072 ± 19	443 1080 ± 47
56	aize-001	268456	168970	640 1546	741 2048 ± 0	402 437 ± 10	262 440 ± 8	316 542 ± 17	465 756 ± 27	940 1583 ± 53	691 1937 ± 22	684 1919 ± 23
57	aize-002	257106	182517	216 586	702 2048 ± 0	434 467 ± 1	293 479 ± 1	522 756 ± 1	1028 1477 ± 1	1114 4617 ± 41	192 597 ± 16	210 598 ± 14
58	aize-003	257130	33045	201 546	557 2048 ± 0	120 187 ± 0	70 188 ± 0	49 182 ± 0	42 183 ± 0	36 195 ± 0	222 621 ± 37	202 592 ± 1
59	aize-006	276515	27909	174 497	1115 2060 ± 0	472 501 ± 0	318 504 ± 0	285 505 ± 0	245 508 ± 0	266 612 ± 0	943 3515 ± 35	946 3496 ± 40
60	ajou-001	363257	31734	148 442	944 2048 ± 0	496 530 ± 0	342 536 ± 0	309 535 ± 0	280 549 ± 0	242 577 ± 0	193 597 ± 19	206 596 ± 13
61	alchera-000	258450	18848	-	418 2048 ± 0	578 587 ± 13	518 706 ± 48	1030 1389 ± 58	1153 2722 ± 123	1144 9011 ± 455	903 3189 ± 32	891 3031 ± 142
62	alchera-001	174013	18848	-	541 2048 ± 0	641 627 ± 11	-	-	-	-	919 3342 ± 81	918 3243 ± 47
63	alchera-002	405409	22275	-	670 2048 ± 0	1045 968 ± 1	747 976 ± 2	698 979 ± 1	638 988 ± 1	572 1025 ± 2	938 3488 ± 63	937 3430 ± 63
64	alchera-003	487718	24613	591 1376	851 2048 ± 0	922 854 ± 3	644 862 ± 2	606 870 ± 1	562 882 ± 2	500 918 ± 1	929 3426 ± 57	930 3383 ± 53
65	alchera-004	1001019	388616	547 1271	690 2048 ± 0	1052 975 ± 0	727 955 ± 0	686 960 ± 0	639 989 ± 0	667 1152 ± 1	946 3529 ± 54	953 3530 ± 63
66	alchera-005	1001019	388616	548 1272	520 2048 ± 0	1046 969 ± 1	762 987 ± 3	705 985 ± 3	649 998 ± 0	675 1162 ± 2	935 3481 ± 59	936 3422 ± 57
67	alchera-006	1000967	493456	605 1426	692 2048 ± 0	492 522 ± 2	345 540 ± 3	319 544 ± 2	292 569 ± 3	368 739 ± 3	746 2233 ± 38	752 2253 ± 77
68	alchera-007	1000948	493456	611 1430	641 2048 ± 0	499 530 ± 2	361 556 ± 2	331 566 ± 2	298 581 ± 2	383 762 ± 4	736 2178 ± 35	734 2169 ± 44
69	alchera-008	1002024	493460	612 1430	643 2048 ± 0	580 589 ± 3	368 563 ± 4	377 614 ± 3	329 614 ± 1	424 810 ± 3	1049 5129 ± 35	1193 11226 ± 120
70	alchera-009	2086479	454616	887 2763	870 2048 ± 0	1408 1486 ± 4	1116 1507 ± 4	1066 1438 ± 9	1041 1508 ± 11	1009 2008 ± 4	758 2284 ± 44	758 2278 ± 69
71	alfabeta-001	128232	21780	18 74	128 512 ± 0	196 271 ± 0	126 276 ± 0	241 459 ± 2	566 886 ± 2	1064 2547 ± 9	136 470 ± 25	140 458 ± 20
72	alice-000	1741293	19355	930 3329	1266 4096 ± 0	1026 950 ± 2	710 933 ± 1	672 949 ± 1	668 1011 ± 3	743 1264 ± 8	1236 14975 ± 201	1238 14890 ± 229
73	alice-001	1428001	17074	278 690	468 2048 ± 0	1133 1080 ± 1	819 1075 ± 0	806 1095 ± 1	766 1143 ± 1	867 1435 ± 1	1288 21885 ± 278	1293 21921 ± 265
74	alleyes-000	507636	997090	367 857	604 2048 ± 0	839 784 ± 1	742 970 ± 61	694 974 ± 62	605 943 ± 69	593 1057 ± 23	510 1298 ± 34	520 1303 ± 51
75	allgovision-000	172509	155862	208 561	630 2048 ± 0	338 384 ± 8	220 395 ± 17	194 413 ± 14	217 471 ± 14	338 710 ± 21	1332 29903 ± 406	1333 29735 ± 194
76	alphaface-001	259849	81636	192 527	804 2048 ± 0	617 612 ± 1	411 613 ± 3	372 612 ± 1	335 619 ± 1	290 640 ± 2	402 1008 ± 10	409 1002 ± 19
77	alphaface-002	768995	70692	613 1434	412 2048 ± 0	644 628 ± 2	553 746 ± 19	515 751 ± 18	484 779 ± 22	435 828 ± 40	382 945 ± 25	387 935 ± 17
78	alttekglobal-000	649718	3060239	267 673	655 2048 ± 0	1125 1066 ± 2	892 1158 ± 2	879 1184 ± 1	787 1168 ± 1	699 1204 ± 2	524 1370 ± 23	536 1377 ± 26
79	amplifiedgroup-001	0	47053	20 81	209 866 ± 2	39 93 ± 0	-	-	-	-	1401 57803 ± 4210	1395 56365 ± 1196
80	androvideo-000	174847	585063	134 405	398 2048 ± 0	203 277 ± 0	133 285 ± 0	117 314 ± 0	138 372 ± 1	274 620 ± 0	860 2860 ± 28	860 2847 ± 22
81	anke-000	323402	39031	-	1128 2072 ± 0	541 566 ± 12	-	-	-	-	176 567 ± 20	173 545 ± 29
82	anke-001	642061	37364	-	1127 2072 ± 0	515 541 ± 4	-	-	-	-	166 540 ± 16	167 530 ± 20
83	anke-002	798686	28364	-	1126 2072 ± 0	380 419 ± 30	-	-	-	-	153 501 ± 17	153 492 ± 17
84	anke-003	340160	198638	-	1076 2056 ± 0	872 811 ± 23	-	-	-	-	119 425 ± 28	130 437 ± 32
85	anke-004	349388	410776	281 706	1099 2056 ± 0	638 625 ± 1	434 627 ± 2	408 635 ± 3	367 653 ± 2	539 982 ± 8	236 633 ± 22	246 632 ± 34
86	anke-005	328553	429160	493 1134	1060 2056 ± 0	581 590 ± 2	396 594 ± 5	362 601 ± 3	355 638 ± 4	430 821 ± 24	269 685 ± 19	285 687 ± 26
87	antheus-000	119453	41994	31 116	176 520 ± 0	53 109 ± 1	68 187 ± 1	57 189 ± 1	46 195 ± 1	48 236 ± 2	1114 6901 ± 268	1111 6936 ± 103
88	antheus-001	119453	41962	32 118	174 520 ± 0	60 120 ± 1	114 265 ± 13	252 468 ± 22	836 1223 ± 27	1066 2660 ± 87	1097 6218 ± 47	1092 6216 ± 45

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 23: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
89	antheus-002	271192	189635	373 875	644 2048 ± 0	1148 1104 ± 1	875 1138 ± 1	844 1140 ± 1	765 1142 ± 0	719 1240 ± 1	345 817 ± 5	354 824 ± 16
90	anyvision-001	662659	518032	-	222 1024 ± 0	174 244 ± 0	-	-	-	-	1429 74285 ± 212	1428 74145 ± 200
91	anyvision-002	662659	520039	-	251 1024 ± 0	179 248 ± 0	-	-	-	-	1428 74069 ± 188	1427 74019 ± 198
92	anyvision-004	401001	630797	482 1102	233 1024 ± 0	297 355 ± 1	-	-	-	-	677 1891 ± 51	666 1829 ± 85
93	anyvision-005	190979	116595	447 1048	239 1024 ± 0	1062 985 ± 1	771 997 ± 1	729 1004 ± 1	644 995 ± 1	550 995 ± 1	304 733 ± 14	317 733 ± 16
94	aratek-001	254521	15993	782 2135	774 2048 ± 0	182 251 ± 1	125 275 ± 0	88 252 ± 0	84 278 ± 0	64 286 ± 0	645 1776 ± 20	644 1744 ± 38
95	armatura-001	0	374608	945 3518	602 2048 ± 0	733 688 ± 1	499 689 ± 1	461 693 ± 1	426 708 ± 3	378 756 ± 13	65 270 ± 17	74 268 ± 11
96	armatura-002	0	1258644	533 1222	1410 6144 ± 0	1401 1476 ± 3	1101 1458 ± 5	1106 1535 ± 5	1039 1505 ± 12	947 1605 ± 26	603 1605 ± 26	603 1589 ± 25
97	armatura-003	0	836082	654 1572	1409 6144 ± 0	1099 1028 ± 1	797 1032 ± 1	744 1027 ± 0	678 1036 ± 1	579 1041 ± 3	1390 51850 ± 56	1390 51835 ± 48
98	arvis-000	529004	705484	375 878	247 1024 ± 0	145 209 ± 5	92 215 ± 1	135 342 ± 4	369 655 ± 13	1032 2158 ± 55	936 3484 ± 41	940 3461 ± 48
99	asr-000	461347	236001	408 951	1097 2056 ± 0	412 450 ± 0	284 462 ± 1	235 454 ± 0	197 453 ± 1	175 486 ± 0	490 1226 ± 14	503 1251 ± 23
100	asr-001	461347	236016	407 949	1055 2056 ± 0	755 701 ± 0	519 706 ± 0	474 704 ± 0	427 709 ± 0	418 800 ± 0	694 1951 ± 28	701 1982 ± 23
101	asusaics-000	257418	245320	229 605	779 2048 ± 0	456 484 ± 13	321 506 ± 21	591 850 ± 26	1088 1789 ± 61	1130 6305 ± 188	1058 5455 ± 78	1059 5422 ± 112
102	asusaics-001	257418	245330	223 595	1224 4096 ± 0	912 842 ± 17	778 1008 ± 20	1025 1377 ± 28	1147 2423 ± 90	1138 7284 ± 277	1155 8618 ± 42	1155 8638 ± 136
103	autentika-000	266093	3200425	742 1922	350 2048 ± 0	528 553 ± 1	402 605 ± 1	369 609 ± 2	321 608 ± 1	273 618 ± 2	1425 72833 ± 577	1421 71829 ± 541
104	autentika-001	0	3198534	674 1650	415 2048 ± 0	726 683 ± 2	687 912 ± 4	634 908 ± 3	587 919 ± 3	485 893 ± 4	1421 71804 ± 600	1419 70747 ± 582
105	autentika-002	0	3198136	680 1676	803 2048 ± 0	323 374 ± 1	323 511 ± 1	305 528 ± 1	265 534 ± 1	224 546 ± 1	1380 46941 ± 1172	1376 46351 ± 822
106	autentika-003	0	3198136	673 1648	896 2048 ± 0	401 437 ± 7	419 617 ± 14	347 584 ± 3	324 611 ± 8	267 614 ± 2	1382 47827 ± 1020	1378 46728 ± 916
107	authenmetric-002	449944	91489	-	917 2048 ± 0	1009 942 ± 1	722 950 ± 1	685 960 ± 1	618 960 ± 1	547 991 ± 2	626 1712 ± 20	636 1719 ± 19
108	authenmetric-003	293599	39492	370 866	697 2048 ± 0	1069 992 ± 1	776 1006 ± 1	727 1003 ± 2	655 1002 ± 1	577 1036 ± 1	640 1757 ± 19	645 1755 ± 19
109	authenmetric-004	381165	39492	531 1214	615 2048 ± 0	977 910 ± 1	684 909 ± 1	637 915 ± 1	588 921 ± 2	519 950 ± 1	631 1724 ± 14	625 1691 ± 29
110	authme-001	1015124	37791	777 2097	1232 4096 ± 0	1022 949 ± 1	739 969 ± 7	684 960 ± 6	620 961 ± 1	528 966 ± 4	854 2836 ± 34	856 2827 ± 54
111	authme-002	1016682	37819	772 2079	1236 4096 ± 0	1035 959 ± 1	734 959 ± 7	720 995 ± 5	626 966 ± 5	542 985 ± 6	848 2796 ± 14	847 2775 ± 34
112	aware-000	240240	27639	-	324 1572 ± 0	678 655 ± 26	-	-	-	-	994 4030 ± 83	884 2984 ± 152
113	aware-001	240240	27639	-	322 1572 ± 0	682 656 ± 26	-	-	-	-	869 2902 ± 51	855 2810 ± 111
114	aware-002	271312	36134	-	323 1572 ± 0	562 582 ± 14	-	-	-	-	342 810 ± 32	351 812 ± 28
115	aware-003	377729	27975	-	1188 3108 ± 0	838 783 ± 10	-	-	-	-	530 1392 ± 42	524 1334 ± 80
116	aware-004	427829	28219	-	1138 2084 ± 0	966 900 ± 10	844 1105 ± 10	911 1226 ± 25	1061 1585 ± 72	1084 2937 ± 166	504 1279 ± 50	516 1287 ± 100
117	aware-005	300017	26320	545 1265	321 1572 ± 0	961 886 ± 23	803 1038 ± 21	827 1121 ± 22	923 1337 ± 58	1033 2195 ± 144	564 1475 ± 63	550 1427 ± 115
118	aware-006	298543	14124	402 944	79 352 ± 0	1181 1148 ± 3	884 1146 ± 2	882 1190 ± 2	899 1306 ± 20	974 1754 ± 84	819 2598 ± 42	818 2559 ± 60
119	aware-007	135312	18711	127 388	91 480 ± 0	768 713 ± 1	529 716 ± 1	497 731 ± 3	493 785 ± 8	553 999 ± 30	421 1058 ± 23	449 1091 ± 37
120	aware-008	135312	18715	126 387	90 480 ± 0	806 751 ± 1	545 737 ± 0	507 739 ± 3	521 825 ± 9	561 1007 ± 27	454 1120 ± 25	475 1152 ± 35
121	awiros-001	15499	87480	23 88	125 512 ± 0	42 97 ± 6	27 98 ± 4	32 138 ± 6	65 225 ± 7	232 556 ± 8	434 1079 ± 44	431 1050 ± 45
122	awiros-002	289016	203723	204 556	503 2048 ± 0	448 479 ± 0	314 500 ± 0	308 534 ± 0	334 618 ± 0	516 946 ± 1	696 1966 ± 31	697 1957 ± 25
123	awiros-003	292302	598824	414 976	842 2048 ± 0	391 428 ± 2	264 442 ± 1	234 454 ± 2	190 442 ± 1	148 444 ± 1	814 2558 ± 20	819 2580 ± 42
124	awiros-004	489913	598874	559 1295	517 2048 ± 0	473 502 ± 0	328 517 ± 0	295 519 ± 3	254 518 ± 1	196 512 ± 0	808 2527 ± 17	814 2537 ± 20
125	aximetria-001	408902	487912	845 2539	441 2048 ± 0	1085 1013 ± 1	787 1023 ± 21	748 1029 ± 5	650 999 ± 2	623 1091 ± 5	1011 4401 ± 94	1015 4490 ± 80
126	ayftech-001	195423	43580	300 731	119 512 ± 0	368 408 ± 23	290 476 ± 52	555 814 ± 108	1091 1827 ± 384	1123 5412 ± 1029	218 615 ± 16	370 885 ± 44
127	ayftech-003	526164	127257	669 1640	883 2048 ± 0	1076 1003 ± 5	784 1017 ± 6	731 1006 ± 6	666 1009 ± 7	602 1063 ± 5	512 1301 ± 20	519 1300 ± 25
128	ayonix-000	58505	5252	15 69	277 1036 ± 0	2 18 ± 2	-	-	-	-	224 621 ± 23	234 620 ± 26
129	beethedata-000	227849	1087592	205 557	672 2048 ± 0	430 465 ± 0	287 467 ± 0	250 468 ± 0	215 467 ± 0	163 467 ± 0	723 2121 ± 34	724 2110 ± 38
130	betaface-000	0	1060599	-	1144 2116 ± 0	907 837 ± 9	-	-	-	-	583 1514 ± 28	589 1520 ± 66
131	betaface-002	0	1060599	-	1145 2116 ± 0	933 870 ± 13	-	-	-	-	563 1474 ± 50	577 1488 ± 41
132	beyneai-000	256958	591433	760 1998	428 2048 ± 0	414 451 ± 8	271 449 ± 1	530 767 ± 7	1064 1603 ± 25	1116 4669 ± 124	965 3730 ± 57	962 3668 ± 54

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
133	biocube-001	25030	6192987	153 457	1298 4096 ± 0	211 282 ± 22	136 292 ± 24	300 521 ± 57	401 684 ± 59	755 1282 ± 68	1287 21787 ± 96	1291 21812 ± 109
134	biocube-002	69898	10651580	159 468	1214 4096 ± 0	893 826 ± 4	628 838 ± 4	870 1175 ± 4	1156 2933 ± 1074	1145 9144 ± 1090	1299 23074 ± 94	1301 23036 ± 99
135	bioid-001	361549	144	245 629	1431 7680 ± 0	122 190 ± 1	78 192 ± 1	102 290 ± 2	284 556 ± 7	922 1528 ± 25	1045 5051 ± 35	1049 5042 ± 19
136	bioid-002	364015	149	599 1415	1432 7680 ± 0	119 187 ± 2	73 190 ± 3	105 295 ± 4	287 560 ± 13	955 1630 ± 43	1047 5052 ± 34	1048 5031 ± 23
137	bioid-003	530171	156	648 1558	1450 15368 ± 0	563 582 ± 2	391 587 ± 2	478 706 ± 3	643 994 ± 7	1026 2123 ± 33	1102 6396 ± 25	1098 6377 ± 29
138	bioid-004	530301	156	652 1568	1451 15368 ± 0	600 603 ± 1	408 611 ± 1	414 640 ± 1	440 726 ± 2	596 1058 ± 3	1116 6948 ± 25	1112 6948 ± 19
139	bioidtechswiss-000	758466	119955	-	31 256 ± 0	648 630 ± 2	-	-	-	-	1350 34416 ± 137	1352 34403 ± 126
140	bioidtechswiss-001	1178769	120811	617 1455	134 512 ± 0	1043 966 ± 4	981 1270 ± 270	971 1294 ± 96	981 1409 ± 157	982 1793 ± 79	820 2610 ± 25	822 2624 ± 32
141	bioidtechswiss-002	744786	114842	420 990	112 512 ± 0	988 917 ± 2	708 930 ± 2	673 952 ± 2	607 947 ± 3	595 1058 ± 11	735 2177 ± 29	735 2170 ± 31
142	bioidtechswiss-003	518317	10378	1001 4730	339 2048 ± 0	1131 1075 ± 1	877 1139 ± 7	1164 2207 ± 9	1106 1891 ± 11	1119 4787 ± 21	1385 50218 ± 204	1386 50184 ± 163
143	biometric-vision-000	445487	72132	392 915	793 2048 ± 0	904 836 ± 3	622 831 ± 3	604 869 ± 5	613 955 ± 8	818 1370 ± 19	1228 14437 ± 161	1229 14325 ± 167
144	biometrik-001	23995	68818	28 101	135 512 ± 0	1384 1446 ± 3	1094 1448 ± 3	1077 1449 ± 3	1012 1456 ± 3	899 1485 ± 3	403 1016 ± 5	404 999 ± 17
145	biometrik-002	171539	84430	114 353	347 2048 ± 0	1413 1499 ± 3	1113 1499 ± 3	1100 1501 ± 3	1037 1503 ± 3	909 1507 ± 2	516 1328 ± 19	523 1321 ± 19
146	bm-001	287734	38076	48 148	1 64 ± 0	409 444 ± 88	-	-	-	-	676 1887 ± 31	677 1877 ± 26
147	boetech-001	261376	88710	954 3658	706 2048 ± 0	195 271 ± 1	119 268 ± 1	91 273 ± 0	87 286 ± 1	73 318 ± 1	1415 68519 ± 1921	1414 67648 ± 822
148	boetech-002	294347	88710	767 2057	499 2048 ± 0	241 305 ± 4	142 296 ± 1	108 302 ± 1	102 313 ± 1	90 348 ± 2	1417 68921 ± 2137	1418 69473 ± 2104
149	bresee-000	287880	22872	-	906 2048 ± 0	1304 1309 ± 1	1173 2126 ± 1	993 1330 ± 0	909 1312 ± 0	800 1333 ± 0	1373 45317 ± 228	1383 48256 ± 425
150	bresee-001	287880	23227	534 1223	461 2048 ± 0	1243 1223 ± 3	937 1216 ± 1	996 1331 ± 1	842 1227 ± 1	815 1360 ± 1	1359 37240 ± 655	1362 37167 ± 584
151	bresee-002	313627	30902	749 1951	360 2048 ± 0	798 743 ± 4	880 1143 ± 2	850 1146 ± 2	770 1148 ± 2	686 1176 ± 2	646 1778 ± 22	650 1775 ± 23
152	btsdigital-001	362015	234409	409 955	386 2048 ± 0	784 728 ± 2	527 713 ± 1	493 726 ± 4	435 716 ± 2	363 738 ± 0	896 3123 ± 22	901 3134 ± 37
153	bynn-000	271083	37637	131 400	975 2049 ± 0	1441 1834 ± 0	1148 1835 ± 0	1139 1840 ± 0	1093 1838 ± 0	992 1875 ± 0	367 899 ± 7	382 915 ± 1
154	cakevn-001	271412	587471	303 739	838 2048 ± 0	575 587 ± 0	387 584 ± 0	349 585 ± 0	310 589 ± 0	304 664 ± 0	1432 78258 ± 72	1430 78329 ± 90
155	cakevn-002	1015434	73945	535 1228	935 2048 ± 0	969 905 ± 0	681 905 ± 0	631 907 ± 0	580 910 ± 0	496 916 ± 0	895 3115 ± 28	897 3086 ± 11
156	camvi-001	118759	21678	-	243 1024 ± 0	112 181 ± 6	-	-	-	-	141 481 ± 16	151 487 ± 23
157	camvi-002	236278	225285	302 737	234 1024 ± 0	712 677 ± 7	537 726 ± 36	603 869 ± 28	752 1129 ± 43	1075 2785 ± 113	213 612 ± 26	216 603 ± 20
158	camvi-003	285657	225320	-	257 1024 ± 0	803 750 ± 3	-	-	-	-	178 571 ± 23	182 565 ± 26
159	camvi-004	280733	615819	394 919	855 2048 ± 0	815 759 ± 10	643 861 ± 17	707 986 ± 34	887 1279 ± 51	1082 2891 ± 158	383 948 ± 40	391 963 ± 31
160	candour-001	150086	97632	164 477	876 2048 ± 0	1357 1400 ± 1	1058 1396 ± 1	1050 1408 ± 1	980 1407 ± 1	883 1464 ± 1	1185 10725 ± 133	1185 10712 ± 127
161	candour-002	149560	65228	129 392	460 2048 ± 0	79 144 ± 0	42 143 ± 0	36 145 ± 0	33 152 ± 1	39 206 ± 0	1140 7798 ± 56	1140 7781 ± 51
162	candour-003	339247	65570	348 833	1179 3072 ± 0	331 380 ± 0	218 392 ± 0	167 378 ± 0	156 393 ± 1	152 445 ± 0	1349 34039 ± 45	1351 34020 ± 44
163	candour-004	1192219	65226	840 2490	1216 4096 ± 0	1461 2243 ± 0	1175 2247 ± 0	1165 2223 ± 0	1140 2270 ± 1	1046 2330 ± 0	834 2714 ± 20	844 2755 ± 23
164	canon-002	446491	130232	-	1288 4096 ± 0	1303 1308 ± 2	1014 1315 ± 1	990 1326 ± 2	927 1345 ± 1	879 1452 ± 1	1096 6211 ± 25	1090 6194 ± 25
165	canon-003	2550850	101378	1018 5487	1412 6180 ± 0	1274 1263 ± 3	977 1263 ± 1	960 1283 ± 1	913 1320 ± 1	896 1482 ± 2	1031 4783 ± 17	1032 4780 ± 19
166	canon-004	2399160	114188	1041 7037	1414 6200 ± 0	1021 948 ± 4	728 955 ± 3	683 959 ± 3	629 977 ± 3	604 1064 ± 2	1127 7172 ± 63	1124 7169 ± 51
167	canon-005	2521818	80309	1036 6700	1365 4148 ± 0	1244 1223 ± 21	943 1221 ± 2	913 1230 ± 4	846 1234 ± 5	798 1329 ± 3	1092 6142 ± 99	1093 6226 ± 146
168	cchonolulu-000	727461	37284	136 416	103 512 ± 0	65 123 ± 13	39 126 ± 10	31 129 ± 10	28 134 ± 10	43 223 ± 10	632 1726 ± 35	626 1697 ± 35
169	cchonolulu-001	733744	37284	140 426	611 2048 ± 0	92 162 ± 21	81 197 ± 23	290 513 ± 96	737 1118 ± 267	1030 2142 ± 584	1012 4418 ± 37	1006 4361 ± 35
170	cdachyd-001	238593	1941345	38 135	152 512 ± 0	1424 1642 ± 2	1139 1749 ± 1	1177 6467 ± 3	1175 17252 ± 10	1166 58681 ± 45	1196 11594 ± 85	1201 11639 ± 130
171	ceiec-001	159618	46136	-	230 1024 ± 0	250 314 ± 3	-	-	-	-	1297 22831 ± 108	1300 22813 ± 120
172	ceiec-002	269063	90975	-	932 2048 ± 0	615 612 ± 17	-	-	-	-	737 2188 ± 57	763 2301 ± 56
173	ceiec-003	260371	88707	144 430	352 2048 ± 0	881 817 ± 4	667 883 ± 57	624 897 ± 60	575 899 ± 72	514 944 ± 72	751 2256 ± 38	747 2241 ± 54
174	ceiec-004	263476	67011	135 408	586 2048 ± 0	1093 1024 ± 1	790 1027 ± 1	745 1027 ± 1	676 1030 ± 1	590 1055 ± 1	665 1844 ± 26	667 1836 ± 20
175	celloscope-001	38016	59158	40 138	142 512 ± 0	16 49 ± 0	12 52 ± 0	56 188 ± 0	247 509 ± 1	-	840 2733 ± 35	840 2730 ± 35
176	chosun-000	167093	694	-	569 2048 ± 0	407 441 ± 1	-	-	-	-	392 983 ± 20	398 983 ± 29

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5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 25: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
177	chosun-001	765615	707	171 491	675 2048 ± 0	837 783 ± 2	614 826 ± 4	1121 1662 ± 13	1164 3679 ± 67	1159 11694 ± 243	398 998 ± 25	421 1035 ± 11
178	chosun-002	234001	31875	151 451	807 2048 ± 0	178 248 ± 3	123 273 ± 3	1097 1495 ± 14	1170 7920 ± 90	1167 80302 ± 1349	225 623 ± 17	252 634 ± 13
179	chtface-001	94088	369529	-	833 2048 ± 0	150 212 ± 10	-	-	-	-	752 2256 ± 26	749 2252 ± 91
180	chtface-002	363153	369529	-	404 2048 ± 0	567 584 ± 14	526 712 ± 41	757 1038 ± 42	1097 1861 ± 75	1115 4661 ± 232	754 2264 ± 26	746 2234 ± 103
181	chtface-003	363153	369529	-	516 2048 ± 0	589 594 ± 16	534 720 ± 33	766 1050 ± 41	1104 1884 ± 90	1125 5606 ± 334	721 2110 ± 37	742 2219 ± 65
182	chtface-004	409656	311027	607 1428	385 2048 ± 0	272 332 ± 0	156 323 ± 1	128 329 ± 1	116 335 ± 1	105 377 ± 1	633 1727 ± 17	637 1720 ± 16
183	chtface-005	408364	311100	598 1415	849 2048 ± 0	260 322 ± 0	151 316 ± 1	122 325 ± 2	111 324 ± 1	128 411 ± 2	682 1907 ± 19	680 1898 ± 23
184	chtface-006	733645	610439	823 2424	382 2048 ± 0	493 522 ± 1	325 514 ± 1	310 536 ± 2	288 561 ± 1	329 693 ± 2	706 2034 ± 41	713 2049 ± 29
185	chtface-007	733528	610443	817 2393	900 2048 ± 0	221 292 ± 0	135 290 ± 0	112 307 ± 1	114 328 ± 1	184 501 ± 1	595 1570 ± 53	598 1561 ± 34
186	chtface-008	810179	583222	1052 8766	898 2048 ± 0	1111 1043 ± 8	786 1019 ± 9	740 1022 ± 14	682 1039 ± 16	608 1066 ± 23	582 1513 ± 14	594 1535 ± 44
187	chtface-009	810179	342213	1056 9371	500 2048 ± 0	1391 1455 ± 0	1103 1468 ± 16	1099 1501 ± 26	1035 1501 ± 0	929 1544 ± 15	741 2213 ± 13	745 2226 ± 29
188	cib-000	340288	69515	-	656 2048 ± 0	1071 993 ± 40	-	-	-	-	1307 24340 ± 60	1318 25972 ± 97
189	cib-001	436253	133766	-	824 2048 ± 0	670 651 ± 2	522 707 ± 13	484 716 ± 15	443 728 ± 3	428 820 ± 5	973 3783 ± 38	973 3765 ± 37
190	cist-001	0	300551	214 582	354 2048 ± 0	1050 972 ± 0	750 977 ± 1	702 981 ± 0	633 983 ± 0	601 1061 ± 0	876 2947 ± 20	878 2940 ± 22
191	cist-002	0	1186073	694 1730	1194 3200 ± 0	1055 976 ± 0	766 992 ± 0	795 1085 ± 1	884 1279 ± 0	1021 2101 ± 1	1094 6165 ± 22	1091 6207 ± 38
192	cist-003	0	1267350	874 2706	1401 5248 ± 0	1292 1291 ± 1	1003 1300 ± 1	1010 1352 ± 3	892 1287 ± 2	782 1313 ± 1	1141 7838 ± 63	1141 7823 ± 40
193	cist-004	0	1603990	851 2563	1400 5248 ± 0	1260 1243 ± 1	982 1271 ± 1	943 1264 ± 3	860 1252 ± 1	758 1287 ± 1	908 3216 ± 25	915 3220 ± 32
194	clearviewai-000	342491	211852	881 2745	434 2048 ± 0	1360 1402 ± 1	1060 1403 ± 1	1056 1412 ± 1	988 1420 ± 1	853 1418 ± 1	600 1592 ± 37	599 1561 ± 37
195	clearviewai-001	613532	511807	580 1353	694 2048 ± 0	655 635 ± 2	452 642 ± 1	426 659 ± 1	379 664 ± 1	313 674 ± 1	348 827 ± 10	357 829 ± 20
196	clearviewai-002	713008	509504	645 1556	444 2048 ± 0	811 755 ± 2	578 773 ± 2	527 765 ± 3	489 783 ± 2	410 790 ± 1	357 864 ± 34	359 842 ± 29
197	clearviewai-003	712800	540093	572 1344	522 2048 ± 0	1151 1109 ± 0	832 1091 ± 0	797 1088 ± 0	718 1090 ± 0	647 1126 ± 0	849 2812 ± 45	857 2831 ± 50
198	closeli-001	420342	9851	318 773	1282 4096 ± 0	910 839 ± 1	632 843 ± 1	585 841 ± 1	536 845 ± 1	467 865 ± 1	1057 5404 ± 17	1058 5400 ± 25
199	cloudmatrix-000	309939	542141	-	581 2048 ± 0	808 754 ± 10	559 750 ± 2	519 754 ± 4	474 764 ± 1	413 793 ± 2	1383 49192 ± 206	1384 49275 ± 176
200	cloudmatrix-001	10390	542121	12 65	827 2048 ± 0	56 114 ± 1	34 117 ± 0	28 118 ± 0	26 123 ± 1	25 169 ± 1	1386 50263 ± 212	1387 50243 ± 237
201	cloudmatrix-002	256635	693318	444 1036	609 2048 ± 0	351 395 ± 1	226 398 ± 1	182 399 ± 1	165 402 ± 1	142 437 ± 20	1384 49578 ± 120	1385 49602 ± 180
202	cloudwalk-hr-000	284600	105489	-	1106 2057 ± 0	594 600 ± 2	-	-	-	-	1255 16747 ± 238	1257 16627 ± 220
203	cloudwalk-hr-001	346156	104483	-	1104 2057 ± 0	693 665 ± 3	-	-	-	-	1262 17816 ± 260	1267 17878 ± 464
204	cloudwalk-hr-002	390059	327494	-	1103 2057 ± 0	974 908 ± 3	-	-	-	-	1295 22530 ± 416	1289 21651 ± 533
205	cloudwalk-hr-003	383739	144263	417 983	1102 2057 ± 0	606 606 ± 0	392 588 ± 0	356 594 ± 0	326 612 ± 1	225 547 ± 2	1118 6982 ± 80	1115 6972 ± 84
206	cloudwalk-hr-004	502916	520169	595 1399	977 2049 ± 0	940 873 ± 1	660 877 ± 1	611 876 ± 1	559 879 ± 1	491 902 ± 3	1199 11652 ± 127	1199 11608 ± 123
207	cloudwalk-mt-000	372169	282036	-	979 2049 ± 0	779 724 ± 12	-	-	-	-	618 1678 ± 47	606 1614 ± 85
208	cloudwalk-mt-002	290753	145340	-	965 2049 ± 0	552 573 ± 1	530 717 ± 78	470 700 ± 66	459 749 ± 96	398 770 ± 80	1128 7205 ± 204	1125 7211 ± 244
209	cloudwalk-mt-003	490365	494959	-	969 2049 ± 0	996 923 ± 1	692 918 ± 1	654 926 ± 1	590 925 ± 1	511 936 ± 1	1198 11620 ± 179	1202 11661 ± 128
210	cloudwalk-mt-004	1384602	512628	-	699 2048 ± 0	995 923 ± 2	693 919 ± 1	641 918 ± 0	586 919 ± 0	505 927 ± 1	1200 11744 ± 170	1200 11631 ± 126
211	cloudwalk-mt-005	846026	573253	950 3615	714 2048 ± 0	1207 1179 ± 3	927 1200 ± 3	899 1209 ± 3	840 1226 ± 5	713 1229 ± 3	1215 12525 ± 225	1214 12394 ± 152
212	cloudwalk-mt-006	563322	480071	943 3499	950 2048 ± 0	1346 1385 ± 0	1052 1392 ± 1	1040 1398 ± 1	968 1397 ± 4	872 1444 ± 2	921 3364 ± 96	925 3324 ± 83
213	cloudwalk-mt-007	563322	480071	944 3499	573 2048 ± 0	1338 1379 ± 2	1079 1425 ± 7	1062 1427 ± 6	984 1417 ± 2	871 1443 ± 7	881 3000 ± 68	880 2971 ± 74
214	clova-000	198420	6824	-	597 2048 ± 0	404 437 ± 0	250 431 ± 0	216 435 ± 0	195 452 ± 2	194 508 ± 7	651 1794 ± 16	657 1795 ± 19
215	cmcuni-001	0	604227	511 1177	864 2048 ± 0	1190 1161 ± 1	916 1187 ± 1	886 1194 ± 5	812 1200 ± 1	703 1207 ± 1	900 3153 ± 22	905 3158 ± 26
216	codeline-000	361659	138388	526 1194	927 2048 ± 0	1389 1453 ± 0	1100 1456 ± 2	1081 1456 ± 0	1015 1457 ± 0	897 1483 ± 1	732 2171 ± 69	738 2194 ± 84
217	cogent-000	633812	18848	-	84 439 ± 0	481 511 ± 7	-	-	-	-	1163 9541 ± 54	1162 9500 ± 51
218	cogent-001	530508	18869	-	88 455 ± 0	532 557 ± 7	-	-	-	-	1181 10477 ± 278	1183 10392 ± 75
219	cogent-002	696959	18881	-	334 1979 ± 0	1006 941 ± 0	-	-	-	-	1227 14394 ± 134	1231 14385 ± 176
220	cogent-003	698290	40430	-	219 973 ± 0	1030 952 ± 0	-	-	-	-	1213 12496 ± 75	1204 11822 ± 163

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 26: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
221	cogent-004	722919	389164	-	[335]1983 ± 0	[1064]987 ± 50	-	-	-	-	[1245]15536 ± 75	[1250]15964 ± 708
222	cogent-005	1876796	75276	-	[1160]2523 ± 0	[1240]1221 ± 2	[955]1236 ± 1	[962]1289 ± 2	[989]1420 ± 4	[946]1602 ± 5	[1314]24854 ± 69	[1313]24858 ± 71
223	cogent-006	1078167	58108	[646]1557	[285]1062 ± 0	[825]768 ± 0	[590]789 ± 1	[576]831 ± 2	[593]930 ± 1	[532]971 ± 1	[657]1802 ± 17	[659]1797 ± 23
224	cogent-007	621565	72316	[738]1888	[195]550 ± 0	[1315]1329 ± 2	[1027]1333 ± 5	[1000]1337 ± 4	[932]1353 ± 5	[836]1390 ± 4	[99]355 ± 8	[110]367 ± 14
225	cogent-008	856817	73587	[789]2176	[194]550 ± 0	[1367]1412 ± 1	[1077]1419 ± 2	[1061]1426 ± 3	[1001]1437 ± 3	[894]1476 ± 1	[122]436 ± 14	[131]441 ± 23
226	cognitec-000	474759	27371	-	[999]2052 ± 0	[158]224 ± 1	-	-	-	-	[976]3835 ± 108	[975]3782 ± 83
227	cognitec-001	476809	27487	-	[1016]2052 ± 0	[222]293 ± 17	-	-	-	-	[1002]4253 ± 59	[1000]4102 ± 167
228	cognitec-002	394088	62354	-	[986]2052 ± 0	[126]192 ± 6	[93]219 ± 6	[81]233 ± 8	[71]241 ± 6	[70]314 ± 10	[914]3250 ± 41	[917]3241 ± 48
229	cognitec-003	471458	62502	[337]817	[984]2052 ± 0	[315]366 ± 9	[229]403 ± 9	[191]408 ± 9	[178]424 ± 9	[195]509 ± 13	[928]3417 ± 51	[938]3433 ± 53
230	cognitec-004	705645	62678	[212]580	[1003]2052 ± 0	[428]463 ± 9	[309]497 ± 9	[282]504 ± 10	[256]521 ± 10	[278]631 ± 12	[887]3028 ± 197	[895]3059 ± 238
231	cognitec-005	326804	198231	[340]819	[1007]2052 ± 0	[247]311 ± 10	[192]364 ± 12	[160]370 ± 10	[151]390 ± 10	[182]498 ± 16	[769]2332 ± 48	[772]2330 ± 40
232	cognitec-006	368383	40008	[354]844	[847]2048 ± 0	[279]335 ± 7	[206]381 ± 8	[178]390 ± 7	[159]396 ± 8	[157]455 ± 11	[767]2316 ± 60	[750]2252 ± 50
233	cognitec-007	368401	40132	[355]844	[954]2048 ± 0	[424]458 ± 20	[399]601 ± 17	[428]660 ± 17	[511]815 ± 23	[874]1445 ± 45	[710]2051 ± 26	[711]2045 ± 34
234	compal-000	1003818	186200	[800]2242	[258]1024 ± 0	[1269]1255 ± 1	[996]1293 ± 2	[972]1294 ± 2	[896]1300 ± 2	[788]1322 ± 4	[1347]33116 ± 349	[1349]33182 ± 252
235	compal-001	619796	186054	[934]3395	[968]2049 ± 0	[927]861 ± 0	[653]871 ± 0	[593]851 ± 0	[545]857 ± 0	[481]887 ± 0	[1446]149573 ± 724	[1446]149664 ± 582
236	compal-002	620531	185671	[920]3134	[967]2049 ± 0	[948]877 ± 0	[656]873 ± 1	[621]892 ± 1	[558]879 ± 0	[473]874 ± 0	[498]1264 ± 13	[510]1272 ± 16
237	cor-001	1194948	11240	[539]1249	[1113]2060 ± 0	[750]699 ± 3	[646]863 ± 76	[601]865 ± 80	[551]872 ± 89	[521]952 ± 39	[1451]270145 ± 2259	[1451]282686 ± 11788
238	coretech-000	186423	43964	[99]318	[109]512 ± 0	[598]602 ± 15	[462]659 ± 12	[842]1139 ± 24	[772]1149 ± 25	[679]1165 ± 23	[88]333 ± 14	[90]321 ± 13
239	coretech-001	235361	305490	[630]1523	[816]2048 ± 0	[731]688 ± 7	[506]695 ± 7	[605]870 ± 17	[557]879 ± 15	[475]877 ± 15	[226]625 ± 25	[259]641 ± 25
240	coretech-002	339881	296194	[928]3300	[590]2048 ± 0	[425]459 ± 1	[274]451 ± 1	[592]851 ± 1	[554]875 ± 2	[538]980 ± 2	[792]2443 ± 17	[790]2436 ± 13
241	coretech-003	339885	296194	[732]1861	[579]2048 ± 0	[878]815 ± 2	[630]838 ± 3	[950]1273 ± 6	[885]1279 ± 8	[819]1373 ± 6	[786]2416 ± 13	[787]2415 ± 19
242	coretech-004	339885	296194	[725]1828	[379]2048 ± 0	[390]427 ± 1	[265]443 ± 2	[588]847 ± 4	[546]858 ± 2	[515]945 ± 4	[788]2436 ± 35	[789]2432 ± 20
243	corsight-001	1437763	31525	-	[1119]2064 ± 0	[1293]1291 ± 3	[990]1285 ± 1	[967]1293 ± 1	[897]1303 ± 2	[827]1379 ± 3	[1450]249340 ± 1713	[1450]248929 ± 1909
244	corsight-002	1474921	32093	[739]1892	[1132]2080 ± 0	[1291]1290 ± 1	[992]1287 ± 1	[963]1290 ± 1	[900]1307 ± 2	[835]1388 ± 4	[1315]24953 ± 637	[1309]24263 ± 578
245	corsight-003	1413063	32198	[670]1642	[1133]2080 ± 0	[1227]1202 ± 2	[918]1190 ± 5	[893]1199 ± 3	[848]1236 ± 3	[808]1349 ± 7	[1328]28754 ± 434	[1330]28279 ± 446
246	csc-001	0	240698	-	[191]544 ± 0	[235]302 ± 1	[146]303 ± 0	[111]304 ± 0	[99]309 ± 1	[86]341 ± 4	[98]354 ± 8	[105]344 ± 11
247	csc-002	0	519768	[4]30	[190]544 ± 0	[443]473 ± 0	[308]494 ± 0	[258]481 ± 1	[231]490 ± 1	[201]514 ± 5	[104]367 ± 11	[112]371 ± 10
248	csc-003	0	400435	[5]31	[185]544 ± 0	[468]499 ± 0	[313]500 ± 1	[280]502 ± 0	[246]508 ± 1	[215]535 ± 4	[112]393 ± 8	[122]397 ± 7
249	ctbcbank-000	257208	599238	[211]570	[625]2048 ± 0	[546]568 ± 43	[405]606 ± 38	[458]690 ± 53	[428]711 ± 50	[440]831 ± 51	[948]3551 ± 87	[1035]4805 ± 209
250	ctbcbank-001	275511	599238	[225]603	[662]2048 ± 0	[673]652 ± 35	[586]781 ± 30	[610]875 ± 43	[574]898 ± 51	[574]1030 ± 47	[981]3926 ± 45	[985]3924 ± 56
251	ctbcbank-002	550523	253310	[469]1084	[101]512 ± 0	[747]694 ± 0	[520]707 ± 0	[494]727 ± 0	[458]744 ± 0	[380]758 ± 0	[1026]4656 ± 146	[1062]5486 ± 179
252	cu-face-002	812008	38655	[289]723	[1287]4096 ± 0	[1115]1054 ± 1	[813]1059 ± 0	[777]1060 ± 0	[695]1063 ± 1	[610]1070 ± 0	[1400]57287 ± 1750	[1399]57027 ± 945
253	cu-face-003	1015124	37791	[779]2116	[1259]4096 ± 0	[1016]945 ± 1	[726]955 ± 3	[679]957 ± 2	[610]949 ± 3	[526]963 ± 2	[845]2769 ± 19	[850]2782 ± 41
254	cu-face-004	1016682	37791	[773]2083	[1290]4096 ± 0	[1089]1022 ± 1	[794]1031 ± 1	[723]998 ± 3	[704]1070 ± 1	[568]1021 ± 2	[862]2884 ± 38	[863]2864 ± 48
255	cubox-001	369627	75427	[257]652	[934]2048 ± 0	[970]907 ± 1	[679]902 ± 1	[628]903 ± 0	[585]917 ± 0	[508]931 ± 0	[527]1379 ± 37	[544]1417 ± 38
256	cubox-002	542254	90975	[753]1966	[475]2048 ± 0	[993]921 ± 1	[697]921 ± 1	[648]922 ± 1	[596]933 ± 1	[559]1003 ± 1	[702]2008 ± 72	[700]1969 ± 57
257	cubox-003	1694397	209684	[942]3491	[820]2048 ± 0	[1205]1179 ± 0	[917]1189 ± 0	[892]1198 ± 0	[821]1208 ± 0	[742]1264 ± 1	[761]2289 ± 112	[761]2297 ± 68
258	cubox-004	1304512	209634	[883]2747	[749]2048 ± 0	[1038]963 ± 1	[759]986 ± 1	[713]992 ± 0	[663]1008 ± 0	[616]1087 ± 0	[565]1476 ± 62	[563]1460 ± 42
259	cubox-005	225237	209600	[592]1387	[868]2048 ± 0	[308]362 ± 0	[200]371 ± 0	[163]373 ± 0	[152]390 ± 0	[155]453 ± 0	[638]1748 ± 25	[634]1718 ± 40
260	cudocommunication-001	385258	341277	[467]1080	[629]2048 ± 0	[998]925 ± 1	[700]923 ± 1	[657]928 ± 1	[594]932 ± 0	[527]964 ± 1	[809]2534 ± 20	[813]2537 ± 20
261	cuhkee-001	787853	74917	[843]2515	[994]2052 ± 0	[1056]977 ± 31	-	-	-	-	[836]2719 ± 60	[851]2783 ± 56
262	cybercore-000	86008	55441	-	[117]512 ± 0	[680]655 ± 3	[498]689 ± 71	[421]649 ± 6	[363]648 ± 8	[316]680 ± 6	[1234]14800 ± 75	[1249]15757 ± 782
263	cybercore-001	166096	7791	-	[506]2048 ± 0	[458]487 ± 0	[301]486 ± 0	[266]488 ± 0	[230]487 ± 0	[186]502 ± 0	[1391]52119 ± 111	[1391]52127 ± 111
264	cybercore-002	166096	7374	[852]2565	[880]2048 ± 0	[460]489 ± 1	[315]500 ± 4	[279]500 ± 1	[238]499 ± 2	[211]528 ± 1	[1211]12389 ± 123	[1213]12352 ± 112

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Table 27: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
265	cybercore-003	289176	7969	984 4311	1281 4096 ± 0	914 844 ± 2	638 855 ± 4	600 864 ± 4	549 862 ± 4	477 878 ± 2	1074 5744 ± 41	1076 5737 ± 31
266	cyberextruder-001	121211	13629	-	36 256 ± 0	964 893 ± 25	-	-	-	-	436 1083 ± 16	442 1079 ± 19
267	cyberextruder-002	168909	13924	-	535 2048 ± 0	502 532 ± 6	-	-	-	-	658 1803 ± 14	653 1779 ± 22
268	cyberextruder-003	253300	12354	149 444	93 512 ± 0	346 390 ± 1	213 388 ± 1	180 393 ± 1	162 399 ± 1	140 435 ± 1	40 198 ± 4	38 189 ± 8
269	cyberextruder-004	169301	12354	96 302	6 128 ± 0	142 206 ± 0	86 208 ± 0	68 209 ± 0	68 229 ± 0	54 249 ± 1	24 145 ± 14	26 155 ± 14
270	cyberextruder-005	269610	45367	230 606	106 512 ± 0	97 168 ± 2	54 169 ± 1	66 205 ± 3	92 295 ± 5	449 835 ± 9	23 142 ± 8	24 144 ± 10
271	cyberextruder-006	184816	45435	125 382	25 256 ± 0	37 91 ± 1	22 89 ± 0	20 97 ± 0	17 95 ± 0	19 130 ± 1	9 98 ± 7	9 101 ± 11
272	cyberlink-000	221801	91440	-	1034 2052 ± 0	282 338 ± 6	-	-	-	-	720 2102 ± 40	722 2093 ± 32
273	cyberlink-001	222009	91440	-	1028 2052 ± 0	385 425 ± 29	-	-	-	-	709 2051 ± 32	716 2060 ± 31
274	cyberlink-002	222311	95086	-	998 2052 ± 0	681 656 ± 22	-	-	-	-	755 2264 ± 71	827 2649 ± 195
275	cyberlink-003	470949	104369	-	989 2052 ± 0	384 423 ± 1	-	-	-	-	776 2366 ± 38	777 2373 ± 45
276	cyberlink-004	340894	104662	-	1359 4140 ± 0	765 712 ± 1	-	-	-	-	963 3693 ± 51	982 3898 ± 71
277	cyberlink-005	341680	111358	-	1358 4140 ± 0	775 721 ± 1	561 752 ± 2	520 754 ± 2	464 755 ± 1	411 791 ± 3	960 3680 ± 37	998 4021 ± 97
278	cyberlink-006	341666	102456	-	1419 6212 ± 0	738 690 ± 1	514 702 ± 0	473 703 ± 0	430 712 ± 0	369 741 ± 0	66 270 ± 13	75 271 ± 13
279	cyberlink-007	380046	102446	-	1416 6212 ± 0	781 725 ± 1	542 732 ± 1	502 734 ± 1	452 736 ± 1	393 767 ± 1	79 304 ± 19	84 304 ± 16
280	cyberlink-008	380047	102470	-	1418 6212 ± 0	785 729 ± 1	535 725 ± 0	495 727 ± 0	448 732 ± 0	382 760 ± 0	62 263 ± 17	67 255 ± 13
281	cyberlink-009	588443	102201	687 1707	1367 4164 ± 0	1344 1384 ± 2	1057 1395 ± 2	1039 1398 ± 2	970 1401 ± 2	880 1456 ± 2	77 299 ± 17	83 304 ± 16
282	cyberlink-010	1590818	102180	956 3678	1441 8260 ± 0	1276 1265 ± 2	1012 1314 ± 5	970 1294 ± 2	880 1273 ± 2	775 1305 ± 2	138 476 ± 23	146 472 ± 14
283	cyberlink-011	1397496	102171	911 3029	1415 6212 ± 0	1220 1196 ± 3	920 1193 ± 2	889 1195 ± 3	828 1215 ± 6	716 1231 ± 5	971 3768 ± 75	969 3735 ± 57
284	cyberlink-012	1200318	102281	890 2789	1417 6212 ± 0	596 601 ± 1	424 621 ± 5	370 610 ± 2	328 614 ± 2	297 649 ± 5	772 2343 ± 26	778 2377 ± 34
285	cyberlink-013	893519	102285	751 1964	1420 6212 ± 0	599 603 ± 1	400 602 ± 2	366 606 ± 2	327 613 ± 2	292 642 ± 1	82 307 ± 7	86 306 ± 5
286	cyberlink-014	1190911	102304	897 2876	1442 8260 ± 0	774 720 ± 1	544 736 ± 5	504 735 ± 10	446 731 ± 3	384 762 ± 3	108 378 ± 8	115 377 ± 13
287	cyberlink-015	1190911	102309	896 2875	1443 8260 ± 0	787 730 ± 1	551 741 ± 6	503 735 ± 12	449 732 ± 2	405 778 ± 1	804 2497 ± 26	808 2493 ± 51
288	cyberlink-016	1503336	105850	948 3591	1446 10308 ± 0	1442 1846 ± 1	1159 1908 ± 1	1143 1863 ± 1	1099 1864 ± 0	1024 2115 ± 3	437 1085 ± 4	436 1069 ± 19
289	dahua-000	283642	171917	-	708 2048 ± 0	310 363 ± 7	-	-	-	-	107 373 ± 16	108 364 ± 17
290	dahua-001	283642	171917	-	358 2048 ± 0	311 363 ± 6	-	-	-	-	103 367 ± 10	107 354 ± 16
291	dahua-002	526452	171518	-	831 2048 ± 0	643 628 ± 7	-	-	-	-	130 461 ± 23	135 454 ± 20
292	dahua-003	605337	141332	-	484 2048 ± 0	509 535 ± 2	-	-	-	-	248 648 ± 20	245 631 ± 34
293	dahua-004	832455	141070	-	551 2048 ± 0	791 735 ± 3	-	-	-	-	302 730 ± 25	298 707 ± 44
294	dahua-005	1586899	169478	-	1209 4096 ± 0	1370 1418 ± 34	-	-	-	-	385 957 ± 23	395 969 ± 19
295	dahua-006	831641	119261	1011 5068	721 2048 ± 0	1356 1398 ± 2	1059 1397 ± 1	1047 1404 ± 1	973 1402 ± 1	841 1402 ± 1	58 249 ± 13	63 250 ± 11
296	dahua-007	1578737	119418	1043 7237	1303 4096 ± 0	1352 1393 ± 2	1042 1373 ± 1	1027 1378 ± 1	953 1378 ± 1	826 1379 ± 2	105 367 ± 102	127 434 ± 108
297	daon-000	280726	2307	762 2010	1121 2065 ± 0	537 562 ± 3	384 581 ± 5	540 791 ± 9	533 838 ± 15	589 1055 ± 32	1248 16052 ± 88	1251 16041 ± 85
298	daon-001	296035	69405	219 587	572 2048 ± 0	723 682 ± 0	466 661 ± 0	456 688 ± 0	397 683 ± 0	299 657 ± 0	338 803 ± 23	352 815 ± 22
299	daon-002	295360	66546	296 731	606 2048 ± 0	820 765 ± 0	575 767 ± 1	531 768 ± 6	483 776 ± 1	397 770 ± 0	352 840 ± 17	363 858 ± 26
300	daon-003	931820	66767	710 1776	1265 4096 ± 0	1459 2048 ± 0	1169 2046 ± 0	1159 2050 ± 0	1125 2069 ± 0	1015 2047 ± 0	738 2194 ± 59	740 2203 ± 28
301	datech-000	617605	139575	1020 5599	1254 4096 ± 0	550 573 ± 32	386 583 ± 44	355 593 ± 37	1174 14123 ± 9564	1162 15514 ± 2343	1142 7857 ± 29	1142 7852 ± 25
302	datech-001	835404	139615	1042 7179	1226 4096 ± 0	1087 1017 ± 83	809 1057 ± 65	782 1062 ± 88	731 1109 ± 81	769 1299 ± 92	1145 7944 ± 31	1146 7981 ± 56
303	datech-002	854184	139615	1046 7446	1237 4096 ± 0	1135 1086 ± 115	869 1130 ± 119	819 1113 ± 123	818 1204 ± 107	849 1413 ± 145	1143 7860 ± 20	1143 7857 ± 26
304	decatur-000	350495	171271	389 907	1325 4100 ± 0	1092 1024 ± 2	-	-	-	-	1193 11439 ± 80	1195 11418 ± 112
305	decatur-001	342866	253734	628 1508	1021 2052 ± 0	1146 1103 ± 2	814 1064 ± 2	783 1063 ± 2	699 1067 ± 2	614 1084 ± 2	209 610 ± 19	214 602 ± 8
306	decloakface-001	29693	170719	120 368	961 2048 ± 0	132 196 ± 58	82 201 ± 52	79 229 ± 53	95 298 ± 60	213 532 ± 61	1025 4638 ± 19	1025 4646 ± 22
307	decloakface-002	1692	143986	80 260	548 2048 ± 0	68 127 ± 0	41 135 ± 0	42 157 ± 0	61 218 ± 0	159 458 ± 2	1009 4380 ± 31	1010 4383 ± 45
308	deepglint-001	569802	206788	-	1260 4096 ± 0	777 721 ± 4	-	-	-	-	959 3680 ± 35	951 3517 ± 182

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 28: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
309	deepglint-002	459642	272878	-	1220 4096 ± 0	713 677 ± 2	615 826 ± 74	589 848 ± 42	537 849 ± 55	480 886 ± 27	1220 13633 ± 87	1219 12905 ± 440
310	deepglint-003	838065	262081	-	1407 6144 ± 0	1189 1159 ± 1	883 1145 ± 1	851 1148 ± 1	771 1148 ± 1	677 1163 ± 1	1258 17227 ± 41	1260 17210 ± 51
311	deepglint-004	1073382	261571	918 3116	767 2048 ± 0	1396 1470 ± 1	1107 1474 ± 1	1094 1485 ± 1	1025 1474 ± 1	903 1492 ± 2	1084 5961 ± 34	1082 5955 ± 29
312	deepglint-005	960326	213877	906 2991	909 2048 ± 0	1364 1408 ± 1	1085 1431 ± 2	1060 1424 ± 3	992 1424 ± 3	875 1446 ± 2	1110 6765 ± 38	1107 6765 ± 40
313	deepsea-001	147497	336250	115 358	256 1024 ± 0	647 630 ± 7	563 752 ± 37	512 746 ± 30	441 727 ± 32	429 820 ± 32	536 1401 ± 37	566 1467 ± 50
314	deepsense-000	357113	936618	-	783 2048 ± 0	688 664 ± 3	454 645 ± 1	429 660 ± 2	403 687 ± 2	423 808 ± 3	139 480 ± 22	141 459 ± 34
315	deepsense-001	73173	1288355	1014 5314	137 512 ± 0	1175 1142 ± 2	897 1164 ± 3	878 1183 ± 3	813 1201 ± 3	789 1323 ± 2	774 2356 ± 35	774 2354 ± 42
316	deepsense-002	73173	1266131	1012 5280	210 896 ± 0	1323 1350 ± 1	1002 1299 ± 8	988 1324 ± 0	928 1348 ± 0	892 1474 ± 4	939 3505 ± 32	950 3515 ± 40
317	deepsense-003	73173	1359497	1032 6555	213 901 ± 0	900 832 ± 2	715 940 ± 2	587 845 ± 4	634 983 ± 1	536 976 ± 7	652 1797 ± 25	658 1795 ± 19
318	deepsense-004	560774	1052093	1045 7310	215 916 ± 0	1272 1259 ± 2	997 1294 ± 9	946 1268 ± 7	869 1261 ± 8	801 1334 ± 7	674 1884 ± 19	678 1884 ± 25
319	deepsense-005	560774	1052093	1044 7309	216 916 ± 0	1374 1430 ± 1	978 1264 ± 1	927 1246 ± 1	863 1254 ± 1	810 1353 ± 4	686 1920 ± 26	691 1932 ± 40
320	deepsense-006	643809	1052098	1047 7719	214 916 ± 0	1319 1348 ± 2	1031 1347 ± 2	880 1184 ± 4	948 1369 ± 3	890 1473 ± 4	669 1860 ± 13	671 1853 ± 22
321	deepsense-007	643809	1070446	1048 7769	217 920 ± 0	1193 1163 ± 1	902 1173 ± 3	881 1190 ± 2	824 1209 ± 3	778 1310 ± 4	666 1847 ± 11	673 1860 ± 28
322	deepsense-008	643809	1070340	1049 7770	218 920 ± 0	1419 1571 ± 1	1125 1572 ± 0	1119 1610 ± 1	1076 1729 ± 1	1036 2216 ± 1	1016 4469 ± 109	1016 4498 ± 68
323	dermalog-001	0	287736	-	279 1043 ± 0	51 106 ± 1	-	-	-	-	1292 22160 ± 166	1296 22131 ± 131
324	dermalog-002	0	287738	-	281 1043 ± 0	30 81 ± 0	-	-	-	-	1293 22169 ± 114	1295 22105 ± 146
325	dermalog-003	0	309430	-	280 1043 ± 0	61 121 ± 22	-	-	-	-	1298 22957 ± 93	1299 22808 ± 131
326	dermalog-004	0	245770	-	8 128 ± 0	84 149 ± 5	-	-	-	-	147 490 ± 26	145 471 ± 26
327	dermalog-005	0	317687	-	9 128 ± 0	70 130 ± 11	-	-	-	-	151 499 ± 22	155 500 ± 22
328	dermalog-006	0	452387	-	7 128 ± 0	503 532 ± 12	-	-	-	-	154 506 ± 23	142 459 ± 23
329	dermalog-007	0	503332	-	114 512 ± 0	167 239 ± 9	-	-	-	-	364 887 ± 23	289 690 ± 25
330	dermalog-008	0	937895	-	132 512 ± 0	361 404 ± 2	233 410 ± 3	208 424 ± 5	181 430 ± 5	170 477 ± 5	134 468 ± 31	96 328 ± 13
331	dermalog-009	0	319363	588 1369	127 512 ± 0	290 349 ± 0	179 351 ± 0	142 352 ± 0	128 357 ± 0	115 389 ± 0	144 487 ± 34	118 385 ± 29
332	dermalog-010	0	525908	961 3789	136 512 ± 0	656 635 ± 0	450 640 ± 1	412 639 ± 4	362 647 ± 3	326 691 ± 5	126 444 ± 13	103 341 ± 26
333	dermalog-011	0	278395	287 716	4 128 ± 0	286 343 ± 0	173 345 ± 0	138 347 ± 0	122 351 ± 0	99 363 ± 0	76 299 ± 19	66 253 ± 14
334	dermalog-012	0	293431	290 723	11 128 ± 0	111 181 ± 0	59 179 ± 0	50 182 ± 0	44 186 ± 0	53 242 ± 0	26 163 ± 10	42 196 ± 16
335	dermalog-013	0	293658	982 4263	10 128 ± 0	113 183 ± 2	62 180 ± 2	48 181 ± 1	41 183 ± 1	37 205 ± 1	34 186 ± 8	51 221 ± 7
336	dermalog-014	0	756422	529 1201	241 1024 ± 0	140 204 ± 0	79 197 ± 0	63 198 ± 1	51 199 ± 0	51 241 ± 0	117 424 ± 17	114 377 ± 17
337	dermalog-015	0	909348	833 2452	246 1024 ± 0	1451 1929 ± 1	1157 1891 ± 0	1154 1955 ± 1	1109 1899 ± 1	997 1904 ± 1	101 360 ± 25	72 267 ± 14
338	dermalog-016	0	910481	832 2452	232 1024 ± 0	1449 1913 ± 2	1158 1908 ± 1	1146 1918 ± 1	1110 1917 ± 1	999 1927 ± 1	160 523 ± 23	139 457 ± 25
339	dicio-001	61751	119517	22 84	172 520 ± 0	512 538 ± 0	367 563 ± 10	638 915 ± 3	1089 1800 ± 7	1122 5286 ± 30	851 2818 ± 20	853 2807 ± 31
340	dicio-002	571566	85868	557 1290	1117 2060 ± 0	915 846 ± 1	569 761 ± 0	534 775 ± 1	491 784 ± 1	439 831 ± 0	1098 6242 ± 74	1096 6256 ± 80
341	didiglobalface-001	259849	70680	191 527	808 2048 ± 0	616 612 ± 1	446 633 ± 3	406 634 ± 3	365 650 ± 15	309 666 ± 4	390 973 ± 20	399 988 ± 20
342	didiglobalface-002	260054	161508	333 812	513 2048 ± 0	635 622 ± 1	447 633 ± 1	416 642 ± 2	375 659 ± 4	355 726 ± 15	171 560 ± 10	183 567 ± 13
343	didiglobalface-003	1001649	83986	549 1274	405 2048 ± 0	795 740 ± 2	547 740 ± 2	529 767 ± 5	485 779 ± 4	458 853 ± 17	591 1544 ± 13	595 1539 ± 17
344	digidata-000	133370	30249	79 259	534 2048 ± 0	307 361 ± 0	184 360 ± 0	152 361 ± 0	131 363 ± 0	107 380 ± 0	718 2084 ± 37	708 2039 ± 42
345	digidata-001	254564	33036	118 367	743 2048 ± 0	534 559 ± 1	365 561 ± 1	329 562 ± 1	289 564 ± 1	261 602 ± 1	1179 10308 ± 102	1180 10301 ± 121
346	digitalbarriers-000	157794	427472	-	1089 2056 ± 0	50 104 ± 0	-	-	-	-	1217 13232 ± 166	1221 13226 ± 146
347	digitalbarriers-001	236915	427537	-	1080 2056 ± 0	225 294 ± 1	-	-	-	-	1210 12311 ± 164	1212 12347 ± 197
348	digitalbarriers-002	83002	598577	745 1930	1091 2056 ± 0	144 209 ± 11	109 250 ± 19	193 411 ± 37	505 808 ± 72	1037 2236 ± 123	1218 13409 ± 228	1222 13267 ± 206
349	dps-000	0	2211812	597 1405	1264 4096 ± 0	930 868 ± 2	674 893 ± 6	1073 1445 ± 9	1154 2910 ± 38	1147 9345 ± 17	562 1473 ± 37	571 1479 ± 37
350	dralmey-001	571556	82266	556 1290	1116 2060 ± 0	1213 1186 ± 0	896 1162 ± 0	865 1170 ± 0	798 1180 ± 0	722 1243 ± 0	1224 13996 ± 56	1227 13971 ± 60
351	dsk-000	11967	782905	77 252	154 512 ± 0	238 304 ± 47	154 317 ± 33	725 1001 ± 96	1151 2660 ± 170	1155 10451 ± 832	1126 7152 ± 115	1121 7134 ± 111
352	einetworks-000	372608	219883	377 880	1078 2056 ± 0	664 645 ± 3	-	-	-	-	1038 4876 ± 66	1050 5156 ± 77

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Table 29: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
353	einetworksindia-000	372375	479472	569 1322	1077 2056 ± 0	326 378 ± 1	209 382 ± 1	174 383 ± 1	160 397 ± 0	158 456 ± 3	1241 15171 ± 41	1242 15138 ± 62
354	einetworksindia-001	562398	481792	719 1808	1088 2056 ± 0	446 479 ± 0	296 481 ± 1	263 485 ± 1	236 494 ± 1	221 541 ± 1	1239 15147 ± 63	1241 15134 ± 74
355	einetworksindia-002	1214466	143768	837 2462	776 2048 ± 0	1137 1087 ± 1	823 1082 ± 1	785 1065 ± 0	722 1098 ± 3	666 1150 ± 2	1221 13785 ± 34	1224 13767 ± 56
356	ekin-002	51434	278	41 139	1178 3072 ± 0	1214 1186 ± 13	909 1180 ± 12	874 1181 ± 11	810 1191 ± 11	702 1207 ± 8	1005 4294 ± 80	1069 5569 ± 112
357	element-000	721679	24895	552 1287	671 2048 ± 0	354 399 ± 1	237 417 ± 1	210 428 ± 9	225 478 ± 14	322 688 ± 45	182 583 ± 10	192 585 ± 13
358	enface-000	369598	153781	266 671	237 1024 ± 0	530 555 ± 4	363 558 ± 4	440 669 ± 6	637 987 ± 15	1048 2349 ± 54	1122 7059 ± 62	1117 6980 ± 65
359	enface-001	370710	173609	265 669	252 1024 ± 0	526 550 ± 4	360 555 ± 3	439 668 ± 7	631 981 ± 15	1058 2416 ± 59	1109 6734 ± 68	1108 6766 ± 69
360	enface-002	858356	98741	1054 9004	245 1024 ± 0	1095 1026 ± 1	800 1038 ± 3	775 1057 ± 7	742 1123 ± 4	802 1338 ± 8	588 1529 ± 23	592 1530 ± 35
361	enqura-001	0	6136892	542 1263	483 2048 ± 0	156 221 ± 5	94 220 ± 3	83 239 ± 3	76 262 ± 3	65 288 ± 3	1 0 ± 0	1 0 ± 0
362	enqura-002	724175	1441517	371 870	1101 2057 ± 0	421 456 ± 1	278 457 ± 1	264 486 ± 1	336 619 ± 1	923 1532 ± 3	1330 29132 ± 232	1332 28946 ± 171
363	eocortex-000	255937	59432	73 224	474 2048 ± 0	242 305 ± 22	169 341 ± 25	222 440 ± 47	213 464 ± 45	198 513 ± 44	377 923 ± 11	384 918 ± 11
364	ercacat-001	811623	58012	892 2816	990 2052 ± 0	1114 1052 ± 3	-	-	-	-	811 2551 ± 62	809 2501 ± 81
365	ethiopianlmis-000	309182	549917	165 481	476 2048 ± 0	954 882 ± 2	670 887 ± 1	567 824 ± 2	530 838 ± 2	493 903 ± 1	784 2400 ± 34	785 2407 ± 28
366	euronovate-001	0	1774966	574 1346	293 1177 ± 0	1105 1034 ± 2	898 1165 ± 3	860 1160 ± 3	794 1177 ± 3	684 1172 ± 2	1434 81294 ± 591	1432 81631 ± 931
367	euronovate-002	0	5476674	339 818	1383 4249 ± 0	223 293 ± 0	141 296 ± 0	107 298 ± 0	98 309 ± 0	83 339 ± 0	1345 32673 ± 391	1346 32587 ± 391
368	euronovate-003	0	5637394	525 1193	1154 2201 ± 0	234 300 ± 0	145 300 ± 0	109 303 ± 0	100 310 ± 0	85 340 ± 0	1361 37908 ± 968	1365 38285 ± 938
369	euronovate-004	0	1335993	448 1049	1155 2201 ± 0	218 290 ± 0	138 294 ± 0	104 294 ± 0	93 295 ± 0	66 294 ± 0	1375 46290 ± 490	1375 46199 ± 484
370	everai-000	229700	144802	-	550 2048 ± 0	419 455 ± 1	-	-	-	-	111 385 ± 10	117 382 ± 13
371	everai-001	449149	144802	-	661 2048 ± 0	754 701 ± 1	-	-	-	-	113 395 ± 11	123 404 ± 23
372	everai-002	561727	281965	-	1280 4096 ± 0	813 758 ± 0	-	-	-	-	245 644 ± 14	239 624 ± 35
373	everai-paravision-003	539802	118876	-	1278 4096 ± 0	706 674 ± 4	-	-	-	-	279 699 ± 20	305 713 ± 47
374	expasoft-000	15341	240451	-	711 2048 ± 0	22 68 ± 0	15 71 ± 0	11 75 ± 0	10 71 ± 0	7 72 ± 1	648 1779 ± 26	646 1757 ± 97
375	expasoft-001	39057	983064	44 142	593 2048 ± 0	23 70 ± 0	16 74 ± 0	12 77 ± 0	11 73 ± 0	9 74 ± 0	616 1660 ± 35	620 1676 ± 48
376	expasoft-002	38760	59825	58 170	821 2048 ± 0	7 34 ± 0	5 34 ± 0	5 34 ± 0	4 34 ± 0	2 34 ± 0	1158 8870 ± 78	1159 8838 ± 77
377	f8-001	272977	19668	550 1276	911 2048 ± 0	888 822 ± 39	-	-	-	-	1244 15262 ± 139	1247 15277 ± 212
378	f8-002	28278	215616	21 84	624 2048 ± 0	11 39 ± 0	7 41 ± 0	10 75 ± 0	50 197 ± 1	333 702 ± 1	1233 14765 ± 131	1236 14790 ± 133
379	facehawk-000	120009	7737	37 129	206 776 ± 1	355 399 ± 0	228 402 ± 0	195 414 ± 1	155 391 ± 2	143 440 ± 3	1188 10931 ± 155	1188 10915 ± 149
380	facehawk-001	173511	35863	110 343	1125 2070 ± 0	309 362 ± 0	190 363 ± 0	161 371 ± 0	142 375 ± 0	144 437 ± 0	558 1465 ± 19	568 1475 ± 11
381	faceint-001	110744	649612	902 2903	682 2048 ± 0	130 195 ± 0	74 191 ± 0	62 195 ± 0	53 203 ± 3	46 232 ± 10	547 1434 ± 11	551 1431 ± 19
382	facelocate-001	4739	11106	829 2449	19 168 ± 0	8 34 ± 1	6 38 ± 0	14 84 ± 1	22 112 ± 4	38 206 ± 13	1165 9565 ± 91	1169 9629 ± 131
383	facelocate-002	15198	486568	84 271	131 512 ± 0	152 214 ± 1	104 239 ± 0	203 422 ± 2	209 463 ± 5	239 571 ± 15	211 611 ± 8	235 620 ± 17
384	faceonlive-001	0	71529	95 300	1069 2056 ± 0	109 179 ± 0	60 179 ± 0	59 190 ± 0	59 217 ± 0	87 343 ± 1	424 1064 ± 37	420 1033 ± 35
385	faceonlive-002	155220	141019	422 997	879 2048 ± 0	836 783 ± 1	596 797 ± 2	543 794 ± 2	506 809 ± 3	489 901 ± 2	1222 13798 ± 197	1223 13743 ± 127
386	faceonlive-003	518321	560984	998 4691	262 1024 ± 0	1104 1032 ± 4	833 1091 ± 3	1152 1942 ± 6	1074 1723 ± 5	1110 4168 ± 3	1249 16149 ± 41	1252 16135 ± 54
387	facephi-000	148904	5219	1063 20248	477 2048 ± 0	935 871 ± 2	663 881 ± 3	615 880 ± 4	568 888 ± 4	518 949 ± 12	996 4067 ± 53	999 4047 ± 53
388	facephi-001	138133	282874	575 1347	121 512 ± 0	215 285 ± 0	121 272 ± 2	103 291 ± 5	94 296 ± 3	177 489 ± 5	904 3194 ± 41	909 3184 ± 54
389	facephi-002	1031362	112529	1029 6074	742 2048 ± 0	1259 1240 ± 5	966 1251 ± 5	937 1253 ± 21	895 1298 ± 5	969 1713 ± 19	1204 11961 ± 81	1207 11970 ± 83
390	facesoft-000	370120	10612	326 796	596 2048 ± 0	708 675 ± 18	472 669 ± 3	453 686 ± 3	390 675 ± 5	320 687 ± 2	747 2239 ± 28	757 2277 ± 96
391	facetag-000	1232331	4022	410 965	202 684 ± 0	296 355 ± 17	199 369 ± 8	709 989 ± 33	1146 2408 ± 91	1140 7930 ± 316	1422 72003 ± 625	1422 71912 ± 612
392	facetag-001	1258247	4022	-	1316 4100 ± 0	1068 991 ± 3	769 995 ± 3	738 1018 ± 3	701 1069 ± 5	757 1284 ± 8	917 3323 ± 28	923 3287 ± 34
393	facetag-002	819806	4021	282 706	948 2048 ± 0	520 544 ± 1	351 544 ± 0	315 542 ± 0	275 545 ± 0	230 554 ± 0	634 1730 ± 25	641 1733 ± 25
394	facex-001	305074	930372	931 3336	797 2048 ± 0	383 422 ± 4	254 434 ± 4	299 520 ± 7	453 737 ± 13	963 1670 ± 27	671 1871 ± 23	669 1846 ± 29
395	facex-002	305074	928334	910 3011	647 2048 ± 0	387 426 ± 5	248 429 ± 4	293 516 ± 8	445 730 ± 12	972 1738 ± 36	231 631 ± 25	231 614 ± 19
396	facia-001	296496	745403	385 889	375 2048 ± 0	1144 1099 ± 1	850 1109 ± 0	820 1113 ± 0	747 1125 ± 3	729 1250 ± 1	1408 62245 ± 397	1405 62249 ± 496

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1	The configuration size does not capture static data included in libraries.
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4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 30: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
397	facia-002	585227	694738	740 1907	1024 2052 ± 0	1318 1347 ± 4	1030 1337 ± 3	1103 1509 ± 7	1111 1919 ± 11	1104 3982 ± 34	1289 21963 ± 400	1292 21869 ± 318
398	faicetech-000	247687	34022	577 1349	469 2048 ± 0	1209 1179 ± 1	925 1199 ± 1	884 1193 ± 1	816 1202 ± 0	721 1240 ± 4	188 594 ± 4	203 593 ± 1
399	farfaces-001	346494	44581	81 262	151 512 ± 0	1206 1179 ± 1	910 1180 ± 1	873 1180 ± 0	804 1185 ± 1	706 1209 ± 2	356 855 ± 25	364 860 ± 31
400	fastenterprises-000	273365	235899	437 1030	525 2048 ± 0	686 662 ± 1	476 671 ± 0	436 664 ± 0	389 673 ± 3	321 688 ± 5	974 3806 ± 96	1345 32064 ± 185
401	fci-000	1486985	454513	919 3123	645 2048 ± 0	1395 1464 ± 12	1112 1494 ± 9	1083 1457 ± 10	1020 1462 ± 11	920 1525 ± 5	766 2309 ± 8	768 2314 ± 14
402	fdu-000	202296	1307	-	181 534 ± 0	33 84 ± 0	-	-	-	-	628 1713 ± 28	632 1715 ± 20
403	fdu-001	345583	1329	-	327 1808 ± 0	71 131 ± 0	-	-	-	-	508 1288 ± 26	513 1281 ± 22
404	fedu-001	466605	496375	647 1557	717 2048 ± 0	585 592 ± 1	426 623 ± 1	386 624 ± 0	342 623 ± 0	295 647 ± 0	775 2362 ± 13	775 2361 ± 22
405	fiberhome-nanjing-002	207398	596827	-	225 1024 ± 0	1236 1217 ± 2	998 1294 ± 30	981 1312 ± 12	1027 1477 ± 14	792 1326 ± 11	597 1582 ± 57	596 1560 ± 51
406	fiberhome-nanjing-003	352895	1482309	368 859	356 2048 ± 0	1170 1136 ± 7	870 1134 ± 4	838 1132 ± 3	763 1139 ± 3	669 1154 ± 5	443 1097 ± 38	446 1083 ± 42
407	fiberhome-nanjing-004	443779	1482313	465 1077	1267 4096 ± 0	1310 1321 ± 5	1007 1304 ± 3	978 1307 ± 2	904 1308 ± 3	795 1326 ± 5	503 1276 ± 40	507 1265 ± 38
408	fincore-000	256615	19409	196 536	936 2048 ± 0	479 508 ± 3	319 505 ± 0	287 508 ± 1	250 513 ± 2	214 535 ± 1	642 1765 ± 31	648 1763 ± 22
409	firstcreditkz-001	553811	24803	486 1113	395 2048 ± 0	865 808 ± 0	770 997 ± 0	781 1061 ± 1	792 1174 ± 1	978 1774 ± 54	369 904 ± 20	375 903 ± 23
410	firstcreditkz-002	977538	24879	748 1951	910 2048 ± 0	1275 1265 ± 1	985 1275 ± 4	976 1299 ± 7	925 1341 ± 17	914 1515 ± 40	317 761 ± 28	328 774 ± 32
411	firstcreditkz-003	1145884	24907	785 2144	806 2048 ± 0	1216 1188 ± 79	1111 1492 ± 77	1170 2566 ± 293	1167 4939 ± 888	1161 14753 ± 3518	219 616 ± 10	228 613 ± 16
412	firstcreditkz-004	1205746	38973	858 2587	832 2048 ± 0	1121 1060 ± 5	954 1236 ± 4	1003 1343 ± 6	1079 1752 ± 15	1103 3765 ± 42	162 525 ± 16	162 521 ± 14
413	foomobi-001	0	219961	66 189	634 2048 ± 0	301 357 ± 0	214 388 ± 2	841 1138 ± 1	1158 2974 ± 10	1152 10101 ± 10	1381 47409 ± 453	1381 47413 ± 421
414	foomobi-002	0	219961	65 189	747 2048 ± 0	232 299 ± 0	157 323 ± 0	681 957 ± 1	1148 2496 ± 2	1142 8549 ± 7	1335 30257 ± 724	1334 30010 ± 320
415	fourthlinecom-001	0	437655	818 2410	848 2048 ± 0	1333 1372 ± 5	973 1257 ± 4	959 1283 ± 2	856 1243 ± 3	723 1244 ± 4	783 2391 ± 17	783 2403 ± 26
416	fpt-000	19013	170633	72 219	837 2048 ± 0	59 118 ± 0	36 122 ± 1	29 122 ± 0	24 120 ± 0	28 178 ± 2	978 3871 ± 31	980 3883 ± 29
417	fpt-001	256529	142347	224 600	638 2048 ± 0	597 602 ± 0	404 606 ± 0	383 623 ± 1	340 621 ± 0	286 637 ± 0	983 3934 ± 10	986 3931 ± 14
418	fpttelecom-000	266955	273536	167 485	527 2048 ± 0	1422 1587 ± 0	1130 1589 ± 0	1118 1593 ± 0	1062 1591 ± 0	945 1599 ± 0	1144 7881 ± 40	1144 7901 ± 48
419	fraudcom-000	354020	561861	411 966	1013 2052 ± 0	536 561 ± 2	344 539 ± 2	313 541 ± 1	285 557 ± 5	252 594 ± 6	753 2258 ± 10	753 2256 ± 29
420	frpkauai-001	507771	24807	978 4143	421 2048 ± 0	736 689 ± 1	502 691 ± 0	467 697 ± 2	433 714 ± 6	403 775 ± 31	311 752 ± 29	325 764 ± 23
421	frpkauai-002	519141	24803	487 1113	799 2048 ± 0	860 799 ± 0	760 987 ± 0	765 1046 ± 1	779 1163 ± 2	977 1769 ± 4	371 907 ± 20	371 886 ± 28
422	frpkauai-003	1135621	24939	791 2184	784 2048 ± 0	1020 948 ± 3	851 1110 ± 2	910 1225 ± 5	946 1369 ± 16	1008 2007 ± 34	161 524 ± 14	166 529 ± 19
423	frpkauai-004	513748	3521	538 1248	73 296 ± 0	257 317 ± 1	152 316 ± 3	119 317 ± 1	110 322 ± 1	102 370 ± 9	1138 7700 ± 81	1137 7659 ± 78
424	frpkauai-005	1250936	39141	848 2544	1026 2052 ± 0	1423 1592 ± 2	1147 1814 ± 2	1145 1910 ± 6	1129 2094 ± 12	1083 2906 ± 39	478 1181 ± 71	485 1182 ± 53
425	fujitsulab-000	0	205894	-	149 512 ± 0	379 419 ± 1	-	-	-	-	954 3613 ± 37	958 3621 ± 29
426	fujitsulab-001	0	548917	-	1342 4104 ± 0	1298 1300 ± 5	995 1291 ± 2	973 1295 ± 2	898 1304 ± 2	830 1382 ± 2	858 2853 ± 16	859 2847 ± 14
427	fujitsulab-002	0	1088887	682 1685	1337 4104 ± 0	1255 1237 ± 2	944 1222 ± 2	919 1236 ± 1	858 1251 ± 2	796 1327 ± 2	853 2836 ± 25	854 2809 ± 44
428	fujitsulab-003	662263	318209	1035 6691	1336 4104 ± 0	1029 951 ± 20	716 941 ± 19	674 952 ± 19	627 971 ± 20	583 1045 ± 21	859 2855 ± 16	861 2849 ± 19
429	g42-intellibrain-001	1031335	235521	1066 25154	66 269 ± 0	1054 976 ± 6	746 975 ± 1	721 997 ± 2	700 1068 ± 3	816 1362 ± 8	1080 5878 ± 96	1080 5865 ± 71
430	g42-intellibrain-002	1043920	132389	1062 19747	65 269 ± 0	986 915 ± 2	695 920 ± 2	669 946 ± 2	648 998 ± 4	711 1224 ± 8	1231 14703 ± 56	1235 14658 ± 97
431	geo-000	114483	304983	-	865 2048 ± 0	918 851 ± 1	-	-	-	-	1445 137544 ± 6225	1445 137925 ± 6548
432	geo-001	258516	70163	-	583 2048 ± 0	1132 1079 ± 0	820 1076 ± 0	790 1076 ± 0	712 1078 ± 0	627 1102 ± 0	902 3163 ± 47	906 3158 ± 38
433	geo-002	369903	98667	434 1021	938 2048 ± 0	847 791 ± 1	593 793 ± 0	544 794 ± 0	497 795 ± 1	419 803 ± 1	927 3407 ± 45	935 3422 ± 65
434	geo-003	371712	102175	-	727 2048 ± 0	1285 1283 ± 1	994 1290 ± 1	961 1285 ± 1	894 1292 ± 1	772 1302 ± 1	397 997 ± 13	408 1001 ± 20
435	geo-004	168980	107714	553 1287	685 2048 ± 0	1277 1268 ± 1	986 1279 ± 1	953 1274 ± 0	868 1259 ± 1	766 1296 ± 1	408 1023 ± 20	418 1028 ± 22
436	gistouch-000	155338	444582	200 541	942 2048 ± 0	265 325 ± 1	160 326 ± 1	132 337 ± 1	125 354 ± 1	180 495 ± 1	1161 9505 ± 604	1160 9443 ± 388
437	gistouch-001	426635	444545	675 1651	364 2048 ± 0	1000 930 ± 3	694 919 ± 3	660 937 ± 4	625 966 ± 2	635 1113 ± 1	1151 8475 ± 133	1153 8460 ± 173
438	glory-000	0	98176	-	83 418 ± 0	94 165 ± 2	-	-	-	-	1120 7003 ± 84	1116 6978 ± 71
439	glory-001	0	144786	-	326 1726 ± 0	348 393 ± 2	-	-	-	-	1169 9607 ± 128	1164 9539 ± 182
440	glory-002	0	385177	-	1142 2106 ± 0	590 594 ± 3	548 740 ± 3	671 948 ± 3	1134 2168 ± 6	32 191 ± 15	1111 6787 ± 85	1101 6551 ± 249

Notes	
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2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 31: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
441	glory-003	0	536910	-	1379 4234 ± 0	459 489 ± 0	369 565 ± 0	500 732 ± 0	1103 1876 ± 2	1143 8941 ± 20	1086 6020 ± 90	1086 6003 ± 72
442	glory-004	0	999639	796 2200	1374 4182 ± 0	732 688 ± 0	567 759 ± 1	663 941 ± 1	1131 2134 ± 4	1148 9360 ± 47	1043 4982 ± 66	1045 4990 ± 63
443	glory-005	0	999999	830 2450	1375 4182 ± 0	758 703 ± 1	589 789 ± 0	692 972 ± 1	1136 2200 ± 25	1150 9679 ± 22	1053 5224 ± 93	1052 5176 ± 81
444	glory-006	0	1004784	826 2438	1373 4182 ± 0	759 704 ± 1	581 774 ± 0	690 968 ± 2	1142 2286 ± 18	1154 10381 ± 151	1078 5853 ± 77	1079 5819 ± 74
445	glory-007	0	1004846	827 2442	1372 4182 ± 0	340 385 ± 0	249 431 ± 0	339 575 ± 14	1022 1464 ± 24	1134 6782 ± 62	940 3506 ± 51	941 3489 ± 66
446	gorilla-000	98290	1122916	-	1377 4204 ± 0	184 256 ± 14	-	-	-	-	980 3875 ± 29	979 3872 ± 51
447	gorilla-001	93768	1101838	-	1149 2156 ± 0	90 160 ± 14	-	-	-	-	930 3429 ± 145	924 3288 ± 51
448	gorilla-002	93869	1281839	-	291 1132 ± 0	262 322 ± 14	-	-	-	-	835 2715 ± 68	820 2585 ± 84
449	gorilla-003	94409	1281823	-	292 1132 ± 0	278 334 ± 25	-	-	-	-	855 2840 ± 42	864 2865 ± 87
450	gorilla-004	186952	1274131	-	1152 2192 ± 0	344 389 ± 3	-	-	-	-	844 2768 ± 44	843 2745 ± 44
451	gorilla-005	100684	1297614	-	1153 2192 ± 0	363 407 ± 3	260 437 ± 2	227 448 ± 4	221 473 ± 4	247 589 ± 15	828 2678 ± 42	846 2770 ± 112
452	gorilla-006	172743	1318812	-	1381 4240 ± 0	418 454 ± 3	300 484 ± 3	275 497 ± 5	272 543 ± 25	506 928 ± 60	970 3755 ± 38	971 3737 ± 44
453	gorilla-007	441058	708166	681 1683	1421 6288 ± 0	584 592 ± 1	394 592 ± 1	363 603 ± 1	344 625 ± 2	350 722 ± 9	961 3686 ± 37	967 3709 ± 36
454	gorilla-008	450175	707000	721 1812	1444 8338 ± 0	591 595 ± 1	393 590 ± 0	361 600 ± 1	339 621 ± 2	349 720 ± 9	1021 4530 ± 44	1019 4524 ± 38
455	gorilla-009	329584	297395	566 1304	1382 4242 ± 0	965 899 ± 2	699 922 ± 1	627 901 ± 3	589 924 ± 4	575 1032 ± 12	762 2294 ± 74	765 2301 ± 66
456	gpstechvn-000	108365	345975	862 2646	931 2048 ± 0	227 296 ± 1	144 297 ± 1	110 304 ± 0	103 316 ± 0	108 382 ± 0	791 2442 ± 11	793 2440 ± 16
457	graymatics-001	13095	70406	35 128	1244 4096 ± 0	125 191 ± 1	83 203 ± 1	354 592 ± 5	1070 1698 ± 9	1136 7150 ± 34	1366 39874 ± 309	1368 39762 ± 295
458	griaule-000	0	598214	452 1055	993 2052 ± 0	378 416 ± 6	243 425 ± 7	533 770 ± 14	1078 1749 ± 43	1132 6406 ± 189	989 3987 ± 42	989 3938 ± 38
459	griaule-001	0	412061	546 1271	1004 2052 ± 0	1194 1164 ± 1	842 1096 ± 5	808 1099 ± 4	760 1136 ± 2	911 1509 ± 2	986 3948 ± 23	992 3957 ± 32
460	griaule-002	0	1320474	734 1870	1015 2052 ± 0	889 822 ± 1	701 924 ± 4	632 907 ± 1	681 1038 ± 21	862 1430 ± 9	990 4005 ± 32	996 4012 ± 31
461	gtelots-000	252631	59244	203 552	95 512 ± 0	842 785 ± 1	594 794 ± 0	552 810 ± 0	518 823 ± 2	474 875 ± 0	1290 22029 ± 130	1294 21988 ± 93
462	gtelots-001	442881	58709	412 966	1151 2176 ± 0	1097 1027 ± 0	792 1028 ± 0	756 1037 ± 0	702 1069 ± 0	680 1167 ± 1	149 493 ± 31	154 496 ± 38
463	hertasecurity-000	0	780014	-	30 256 ± 0	45 99 ± 0	26 98 ± 0	23 100 ± 0	20 107 ± 0	20 139 ± 0	290 710 ± 31	273 667 ± 28
464	hertasecurity-001	0	944427	520 1188	111 512 ± 0	288 346 ± 0	172 345 ± 0	139 349 ± 0	124 354 ± 0	113 388 ± 0	644 1770 ± 45	638 1726 ± 48
465	hertasecurity-002	0	944582	514 1182	145 512 ± 0	455 484 ± 7	292 478 ± 3	257 480 ± 3	237 495 ± 3	207 520 ± 3	760 2289 ± 40	756 2267 ± 48
466	hertasecurity-003	0	944583	513 1181	129 512 ± 0	464 495 ± 7	294 480 ± 3	265 486 ± 3	233 491 ± 3	208 520 ± 4	1136 7619 ± 65	1136 7597 ± 94
467	hik-001	667866	9290	1033 6597	303 1408 ± 0	672 651 ± 0	471 667 ± 8	448 677 ± 16	402 686 ± 13	362 737 ± 12	145 488 ± 19	148 477 ± 22
468	hisign-001	732412	167488	650 1560	1135 2080 ± 0	1301 1306 ± 1	1017 1320 ± 1	984 1315 ± 1	908 1312 ± 1	791 1325 ± 1	44 201 ± 10	35 185 ± 13
469	hisign-002	1014906	102652	781 2131	1136 2080 ± 0	855 797 ± 0	598 800 ± 5	547 800 ± 0	500 801 ± 0	420 803 ± 1	54 232 ± 11	45 207 ± 11
470	hisign-003	2232829	156367	662 1599	1134 2080 ± 0	1109 1040 ± 5	824 1083 ± 4	794 1085 ± 4	717 1088 ± 4	624 1093 ± 4	27 167 ± 8	25 150 ± 10
471	hyperverge-001	260819	88624	-	795 2048 ± 0	721 682 ± 20	505 695 ± 17	890 1196 ± 37	1145 2400 ± 68	1137 7178 ± 204	1088 6026 ± 40	1084 5984 ± 38
472	hyperverge-002	2951900	198832	755 1975	228 1024 ± 0	1004 938 ± 1	714 939 ± 1	664 941 ± 1	606 945 ± 1	535 975 ± 1	1087 6023 ± 37	1083 5966 ± 40
473	hyperverge-003	1167779	282156	868 2687	229 1024 ± 0	1402 1477 ± 2	1115 1503 ± 3	1105 1520 ± 3	1044 1525 ± 4	937 1565 ± 3	175 566 ± 11	179 561 ± 8
474	hyperverge-004	4924393	282156	967 3908	867 2048 ± 0	1397 1471 ± 2	1087 1434 ± 1	1074 1446 ± 6	1006 1445 ± 3	902 1491 ± 3	396 996 ± 10	406 1000 ± 19
475	hyperverge-005	3413303	284050	867 2686	244 1024 ± 0	734 689 ± 0	500 690 ± 0	462 693 ± 0	419 702 ± 6	375 752 ± 5	131 465 ± 2	144 464 ± 10
476	hyperverge-006	3901500	284050	770 2070	265 1024 ± 0	824 768 ± 1	577 770 ± 2	528 766 ± 1	481 773 ± 1	434 828 ± 6	132 466 ± 2	143 463 ± 4
477	hyperverge-007	2223041	169168	787 2159	224 1024 ± 0	1313 1327 ± 1	1000 1296 ± 0	957 1280 ± 0	891 1286 ± 0	805 1348 ± 0	256 659 ± 19	283 684 ± 26
478	hyundai-roboticslab-000	181134	191546	500 1149	607 2048 ± 0	325 376 ± 3	196 368 ± 1	157 367 ± 2	140 374 ± 0	126 408 ± 1	850 2815 ± 47	852 2790 ± 50
479	hyundai-roboticslab-001	361330	106869	342 820	920 2048 ± 0	302 358 ± 1	187 361 ± 1	153 362 ± 2	141 374 ± 3	118 394 ± 0	894 3115 ± 34	898 3117 ± 45
480	hyundai-roboticslab-002	181140	106752	338 817	498 2048 ± 0	289 347 ± 0	180 355 ± 0	147 357 ± 0	130 362 ± 3	112 387 ± 0	910 3243 ± 111	919 3245 ± 114
481	hyundai-roboticslab-003	484277	106656	764 2027	713 2048 ± 0	1437 1800 ± 0	1145 1792 ± 0	1134 1802 ± 0	1105 1889 ± 0	984 1810 ± 0	1157 8841 ± 42	1158 8819 ± 54
482	hyundai-roboticslab-004	484277	106656	763 2026	514 2048 ± 0	1445 1882 ± 1	1152 1842 ± 2	1138 1839 ± 0	1092 1837 ± 0	990 1871 ± 0	1187 10743 ± 201	1187 10767 ± 219
483	hzailu-001	0	372018	-	1092 2056 ± 0	1165 1126 ± 1	867 1128 ± 1	836 1130 ± 1	756 1132 ± 1	672 1159 ± 1	366 894 ± 19	373 899 ± 22
484	hzailu-002	1515880	74047	1000 4722	1090 2056 ± 0	1183 1150 ± 5	865 1127 ± 6	833 1129 ± 7	762 1137 ± 7	682 1172 ± 3	435 1079 ± 53	437 1070 ± 31

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5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 32: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
485	hzailu-003	1923030	222185	1006 4858	1186 3080 ± 0	1349 1389 ± 5	1025 1331 ± 7	997 1334 ± 2	930 1349 ± 6	858 1424 ± 8	568 1483 ± 35	565 1464 ± 31
486	hzailu-004	1938475	275639	973 4023	1187 3080 ± 0	1411 1495 ± 8	1110 1490 ± 6	1096 1493 ± 8	1036 1502 ± 8	939 1583 ± 17	522 1361 ± 32	545 1419 ± 44
487	hzailu-005	1573726	275643	927 3292	1398 5128 ± 0	727 684 ± 10	482 673 ± 7	382 622 ± 1	396 682 ± 6	343 715 ± 2	668 1853 ± 51	665 1829 ± 34
488	hzailu-006	2996729	275643	1028 6070	1430 7176 ± 0	1412 1498 ± 11	1109 1483 ± 15	1067 1439 ± 8	1004 1441 ± 10	935 1564 ± 40	787 2434 ± 19	796 2452 ± 32
489	hzailu-007	2906424	275643	1022 5897	1429 7176 ± 0	1376 1430 ± 19	1095 1448 ± 10	992 1328 ± 3	1038 1504 ± 17	865 1434 ± 4	801 2481 ± 28	804 2488 ± 34
490	hzailu-008	2906977	275643	1023 5898	1428 7176 ± 0	1388 1452 ± 13	1119 1557 ± 10	1107 1545 ± 12	1051 1561 ± 20	924 1533 ± 16	803 2492 ± 62	806 2489 ± 66
491	i2v-001	276820	26600	336 815	716 2048 ± 0	776 721 ± 0	539 729 ± 0	498 731 ± 0	447 731 ± 0	388 764 ± 0	1195 11523 ± 176	1197 11533 ± 142
492	i2v-002	339732	122488	691 1721	348 2048 ± 0	1381 1438 ± 1	1108 1476 ± 1	1070 1442 ± 1	1030 1480 ± 9	925 1533 ± 1	255 656 ± 35	220 604 ± 28
493	icehouse-000	287028	178880	383 887	949 2048 ± 0	646 629 ± 1	423 620 ± 1	420 648 ± 1	406 690 ± 4	525 961 ± 11	375 910 ± 11	388 938 ± 16
494	icm-002	621586	903	-	473 2048 ± 0	1103 1031 ± 7	-	-	-	-	1306 24052 ± 118	1307 24049 ± 124
495	icm-003	1513988	940	187 516	556 2048 ± 0	719 681 ± 6	480 672 ± 4	483 714 ± 11	529 837 ± 41	828 1381 ± 131	1308 24351 ± 161	1308 24227 ± 146
496	icm-004	2012129	1089	451 1052	518 2048 ± 0	381 419 ± 6	231 407 ± 6	233 454 ± 15	320 603 ± 51	921 1527 ± 235	1232 14730 ± 154	1234 14521 ± 152
497	icm-005	877530	1097	488 1116	794 2048 ± 0	116 185 ± 3	64 181 ± 3	67 208 ± 9	108 321 ± 36	566 1014 ± 185	1177 10245 ± 65	1179 10245 ± 69
498	icomm-000	807818	139431	1037 6728	1262 4096 ± 0	603 605 ± 26	407 610 ± 30	365 605 ± 27	323 610 ± 23	291 641 ± 32	1152 8507 ± 25	1154 8516 ± 26
499	icthtc-000	172459	1471004	505 1165	482 2048 ± 0	283 338 ± 11	168 338 ± 9	218 437 ± 16	422 705 ± 24	970 1719 ± 44	1056 5284 ± 63	1057 5290 ± 54
500	id3-001	225574	34433	-	175 520 ± 0	166 238 ± 19	-	-	-	-	420 1058 ± 32	429 1049 ± 28
501	id3-002	225574	34433	-	171 520 ± 0	453 482 ± 34	-	-	-	-	445 1100 ± 59	427 1048 ± 32
502	id3-003	265951	49731	-	58 264 ± 0	255 316 ± 19	-	-	-	-	517 1330 ± 25	530 1354 ± 28
503	id3-004	171526	49725	-	59 264 ± 0	514 541 ± 11	-	-	-	-	457 1135 ± 23	479 1156 ± 32
504	id3-005	153439	22125	-	62 264 ± 0	447 479 ± 1	-	-	-	-	1090 6072 ± 35	1089 6077 ± 38
505	id3-006	210116	7706	418 984	173 520 ± 0	725 683 ± 0	829 1088 ± 1	883 1192 ± 1	823 1209 ± 1	746 1270 ± 1	1066 5547 ± 34	1068 5563 ± 34
506	id3-007	185030	7728	-	61 264 ± 0	1086 1016 ± 1	876 1139 ± 1	1174 3911 ± 2	1172 10752 ± 12	1168 135182 ± 1912	1062 5500 ± 29	1061 5486 ± 35
507	id3-008	242416	8151	455 1059	63 264 ± 0	883 819 ± 0	934 1209 ± 2	975 1297 ± 2	920 1329 ± 1	864 1433 ± 1	1072 5658 ± 44	1073 5624 ± 40
508	id3-009	128638	9510	450 1051	60 264 ± 0	488 520 ± 1	329 518 ± 1	292 515 ± 2	255 519 ± 1	202 514 ± 2	264 682 ± 34	275 677 ± 44
509	id3-010	128638	9510	454 1056	64 264 ± 0	504 535 ± 2	341 535 ± 2	446 677 ± 2	769 1147 ± 4	1080 2889 ± 5	251 651 ± 34	236 621 ± 16
510	idemia-003	427244	106432	-	78 352 ± 0	318 368 ± 6	-	-	-	-	1106 6654 ± 75	1038 4835 ± 90
511	idemia-004	406924	106436	-	77 352 ± 0	243 306 ± 5	-	-	-	-	1069 5592 ± 518	1064 5533 ± 426
512	idemia-005	509824	116761	-	198 588 ± 0	484 514 ± 15	-	-	-	-	1107 6657 ± 54	1104 6616 ± 53
513	idemia-006	570566	115633	-	201 668 ± 0	717 679 ± 4	704 926 ± 54	496 728 ± 7	469 760 ± 29	438 830 ± 31	1052 5223 ± 80	1053 5193 ± 72
514	idemia-007	353242	67485	-	89 468 ± 0	337 384 ± 0	215 389 ± 0	181 393 ± 1	168 405 ± 2	146 441 ± 8	911 3243 ± 63	910 3202 ± 63
515	idemia-008	374017	69922	524 1192	75 348 ± 0	422 457 ± 1	282 461 ± 0	247 466 ± 1	223 476 ± 2	199 513 ± 10	890 3080 ± 41	892 3046 ± 56
516	idemia-009	1066728	70572	872 2700	200 636 ± 0	1229 1207 ± 1	940 1218 ± 1	906 1222 ± 2	835 1222 ± 3	753 1280 ± 10	1073 5664 ± 84	1071 5597 ± 90
517	idemia-010	996690	66244	811 2306	82 412 ± 0	593 596 ± 1	443 632 ± 22	385 624 ± 18	348 633 ± 7	347 719 ± 5	624 1699 ± 37	628 1703 ± 38
518	idemia-011	472694	65098	935 3401	199 612 ± 0	879 816 ± 25	637 853 ± 9	425 658 ± 2	492 785 ± 3	358 732 ± 8	817 2583 ± 29	823 2627 ± 48
519	identy-000	271397	14217	161 474	765 2048 ± 0	735 689 ± 0	513 702 ± 0	482 712 ± 3	467 757 ± 0	531 970 ± 1	254 655 ± 25	255 637 ± 23
520	identy-001	385977	17850	519 1187	635 2048 ± 0	830 772 ± 1	631 841 ± 1	773 1056 ± 1	1053 1567 ± 5	1107 4081 ± 9	569 1484 ± 16	572 1480 ± 17
521	identy-002	1136161	40480	819 2414	1177 3072 ± 0	1415 1548 ± 0	1128 1584 ± 0	1109 1549 ± 0	1080 1757 ± 1	1039 2246 ± 1	884 3020 ± 41	890 3007 ± 19
522	identy-003	1245276	74623	616 1444	1182 3072 ± 0	1430 1730 ± 1	1138 1739 ± 0	1133 1794 ± 1	1119 1996 ± 3	1073 2775 ± 11	481 1204 ± 11	491 1198 ± 5
523	idrnd-000	1069136	69522	933 3357	1338 4104 ± 0	870 810 ± 1	617 828 ± 11	574 830 ± 10	556 877 ± 5	460 857 ± 9	1194 11504 ± 47	1196 11494 ± 62
524	idrnd-001	1080396	107151	939 3437	1341 4104 ± 0	588 594 ± 1	413 615 ± 6	378 615 ± 6	332 616 ± 6	289 640 ± 8	641 1757 ± 8	647 1759 ± 16
525	igearx-face-000	665623	298245	382 885	902 2048 ± 0	925 860 ± 1	641 855 ± 4	586 844 ± 0	561 881 ± 0	504 924 ± 3	609 1628 ± 45	616 1657 ± 41
526	iit-000	237317	33898	-	255 1024 ± 0	134 197 ± 8	-	-	-	-	590 1537 ± 81	514 1282 ± 20
527	iit-001	269176	52055	-	369 2048 ± 0	752 699 ± 4	-	-	-	-	422 1060 ± 48	439 1074 ± 54
528	iit-002	259579	52070	298 731	724 2048 ± 0	483 514 ± 1	336 531 ± 2	321 547 ± 1	300 583 ± 1	359 733 ± 2	409 1023 ± 7	410 1011 ± 66

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
529	iit-003	261288	53791	343 821	930 2048 ± 0	451 482 ± 0	306 493 ± 0	288 509 ± 0	270 541 ± 0	303 661 ± 0	87 324 ± 17	93 326 ± 8
530	imds-software-001	373399	352623	321 785	424 2048 ± 0	431 465 ± 1	732 958 ± 6	837 1131 ± 5	758 1134 ± 2	642 1119 ± 10	1202 11885 ± 120	1203 11779 ± 174
531	imds-software-002	373632	352627	346 833	877 2048 ± 0	438 469 ± 1	725 954 ± 2	804 1094 ± 5	696 1064 ± 4	668 1153 ± 4	1190 11100 ± 155	1191 11090 ± 145
532	imds-software-003	373648	352628	315 769	775 2048 ± 0	258 317 ± 5	459 654 ± 4	542 793 ± 6	479 770 ± 4	456 848 ± 2	1105 6624 ± 66	1102 6603 ± 77
533	imperial-000	370120	10623	327 796	823 2048 ± 0	695 669 ± 1	485 675 ± 3	451 683 ± 17	391 676 ± 2	323 689 ± 2	725 2130 ± 32	714 2052 ± 100
534	imperial-001	370260	10623	-	470 2048 ± 0	699 671 ± 0	-	-	-	-	719 2090 ± 28	717 2062 ± 74
535	imperial-002	472327	16134	724 1826	712 2048 ± 0	547 569 ± 1	383 581 ± 15	340 575 ± 5	296 576 ± 2	246 588 ± 3	757 2278 ± 90	727 2131 ± 44
536	incode-000	23697	18014	-	264 1024 ± 0	131 196 ± 9	-	-	-	-	1050 5175 ± 81	1041 4907 ± 77
537	incode-001	155416	18020	-	449 2048 ± 0	469 499 ± 9	-	-	-	-	1093 6154 ± 96	1088 6045 ± 182
538	incode-002	73239	32435	-	384 2048 ± 0	210 281 ± 15	-	-	-	-	592 1544 ± 38	578 1488 ± 69
539	incode-003	170632	42857	-	1249 4096 ± 0	335 384 ± 11	-	-	-	-	688 1928 ± 44	676 1876 ± 81
540	incode-004	260224	51685	-	407 2048 ± 0	450 479 ± 23	-	-	-	-	683 1913 ± 60	794 2443 ± 114
541	incode-005	256242	51670	-	458 2048 ± 0	542 566 ± 1	-	-	-	-	623 1695 ± 38	617 1663 ± 85
542	incode-006	266095	63518	-	456 2048 ± 0	442 472 ± 1	-	-	-	-	649 1788 ± 41	660 1798 ± 59
543	incode-007	266103	63524	-	761 2048 ± 0	439 470 ± 1	326 516 ± 0	294 518 ± 1	262 530 ± 2	262 603 ± 4	654 1799 ± 35	656 1789 ± 59
544	incode-008	266103	21014	-	768 2048 ± 0	471 500 ± 0	305 493 ± 0	274 496 ± 0	244 506 ± 1	219 537 ± 0	430 1070 ± 28	445 1081 ± 34
545	incode-009	266103	21014	-	904 2048 ± 0	474 503 ± 0	304 490 ± 1	276 498 ± 0	243 505 ± 0	217 537 ± 0	447 1102 ± 28	457 1113 ± 29
546	incode-010	627808	21014	847 2543	595 2048 ± 0	1210 1180 ± 2	906 1178 ± 1	876 1182 ± 1	801 1184 ± 1	710 1221 ± 1	473 1164 ± 32	468 1144 ± 32
547	incode-011	477280	21781	623 1483	771 2048 ± 0	936 872 ± 0	658 875 ± 0	616 881 ± 1	571 892 ± 1	512 939 ± 0	452 1117 ± 31	454 1109 ± 37
548	incode-012	477371	21786	621 1479	678 2048 ± 0	489 521 ± 0	320 506 ± 0	281 503 ± 0	251 514 ± 0	240 571 ± 0	790 2441 ± 25	801 2466 ± 37
549	incode-013	483720	41434	672 1647	400 2048 ± 0	508 535 ± 1	355 548 ± 1	326 555 ± 4	274 545 ± 1	272 618 ± 1	468 1152 ± 37	477 1154 ± 41
550	incode-014	2365157	41516	1039 6927	1437 8196 ± 0	1123 1064 ± 4	831 1089 ± 2	810 1100 ± 5	728 1107 ± 7	690 1190 ± 3	608 1627 ± 42	610 1640 ± 59
551	incode-015	1808779	70449	1031 6430	1411 6148 ± 0	1118 1056 ± 2	913 1182 ± 12	793 1081 ± 2	715 1084 ± 1	689 1187 ± 2	526 1373 ± 22	528 1352 ± 35
552	infocert-001	1204340	38972	625 1490	748 2048 ± 0	943 874 ± 1	671 891 ± 1	767 1050 ± 5	1024 1473 ± 2	1092 3174 ± 8	1048 5055 ± 108	1047 5008 ± 100
553	infocert-002	186823	19501	251 641	349 2048 ± 0	797 742 ± 0	546 738 ± 0	506 738 ± 0	454 738 ± 0	367 739 ± 0	872 2917 ± 114	876 2930 ± 50
554	infocert-003	186823	17115	188 517	337 2048 ± 0	818 763 ± 0	570 764 ± 0	524 764 ± 0	473 763 ± 0	391 766 ± 0	407 1022 ± 7	415 1023 ± 19
555	innefulabs-000	370588	162172	146 439	576 2048 ± 0	1080 1006 ± 3	788 1025 ± 3	750 1030 ± 4	684 1041 ± 2	656 1135 ± 3	1076 5782 ± 41	1077 5741 ± 45
556	innominds-001	102207	777689	490 1124	377 2048 ± 0	275 333 ± 21	178 350 ± 19	130 335 ± 16	120 339 ± 17	89 346 ± 18	802 2482 ± 23	807 2491 ± 29
557	innov8tif-000	170383	2163199	626 1494	1208 4096 ± 0	280 335 ± 23	176 348 ± 21	635 912 ± 64	1152 2661 ± 142	1157 10665 ± 528	186 591 ± 11	191 582 ± 11
558	innov8tif-001	118502	1241515	181 508	1205 4096 ± 0	945 875 ± 83	688 915 ± 73	1169 2360 ± 185	1168 6410 ± 390	1163 24097 ± 1477	321 775 ± 31	329 778 ± 14
559	innovativetechnologyltd-001	177232	335757	108 341	650 2048 ± 0	396 433 ± 7	270 446 ± 8	220 439 ± 4	196 452 ± 4	173 485 ± 7	673 1877 ± 42	687 1924 ± 97
560	innovativetechnologyltd-002	173939	372324	390 912	686 2048 ± 0	684 661 ± 2	536 726 ± 4	701 981 ± 27	647 997 ± 40	392 766 ± 3	664 1841 ± 50	672 1857 ± 59
561	innovatrics-000	0	143500	-	15 146 ± 0	555 578 ± 5	-	-	-	-	1042 4964 ± 63	1026 4665 ± 262
562	innovatrics-001	0	143500	-	67 284 ± 0	665 645 ± 5	-	-	-	-	1063 5506 ± 131	1044 4975 ± 308
563	innovatrics-002	0	249195	-	179 530 ± 0	86 153 ± 0	-	-	-	-	1226 14384 ± 260	1233 14497 ± 207
564	innovatrics-003	0	538170	-	180 530 ± 0	181 250 ± 0	-	-	-	-	1247 15709 ± 134	1245 15193 ± 292
565	innovatrics-004	0	496988	-	288 1076 ± 0	347 391 ± 0	-	-	-	-	1153 8573 ± 274	1145 7929 ± 244
566	innovatrics-005	0	664175	-	289 1076 ± 0	554 577 ± 1	-	-	-	-	1256 16880 ± 194	1261 17232 ± 385
567	innovatrics-006	0	466269	-	183 538 ± 0	884 820 ± 5	597 799 ± 4	549 805 ± 3	498 796 ± 9	482 890 ± 15	1079 5855 ± 204	1055 5266 ± 118
568	innovatrics-007	0	493269	-	286 1064 ± 0	1407 1485 ± 7	1143 1785 ± 184	1161 2078 ± 24	1130 2123 ± 15	1034 2210 ± 42	1085 5978 ± 88	1075 5690 ± 102
569	innovatrics-008	307323	59842	609 1429	184 538 ± 0	832 778 ± 6	573 767 ± 3	532 770 ± 3	503 803 ± 3	457 853 ± 10	886 3021 ± 66	831 2673 ± 88
570	innovatrics-009	624485	105187	747 1947	1357 4136 ± 0	1160 1116 ± 1	848 1107 ± 5	815 1104 ± 5	732 1110 ± 5	664 1146 ± 6	1046 5051 ± 54	1029 4733 ± 102
571	innovatrics-010	604731	111234	689 1715	1356 4136 ± 0	608 611 ± 3	444 632 ± 2	489 723 ± 1	444 730 ± 1	395 768 ± 6	997 4077 ± 42	978 3857 ± 65
572	innovatrics-011	643997	50017	701 1745	1376 4184 ± 0	807 754 ± 2	574 767 ± 3	526 765 ± 3	490 783 ± 4	468 867 ± 3	999 4135 ± 57	988 3937 ± 91

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
573	innovatrics-012	0	680831	713 1789	1391 4721 ± 0	854 797 ± 3	627 837 ± 3	545 795 ± 1	577 904 ± 4	556 1001 ± 3	1433 80879 ± 981	1431 80521 ± 655
574	innovatrics-013	0	1665828	971 3931	1378 4212 ± 0	1219 1195 ± 2	952 1230 ± 11	908 1223 ± 4	844 1233 ± 5	797 1329 ± 3	1437 85968 ± 990	1435 85796 ± 836
575	innovatrics-014	0	2870004	1030 6138	1422 6303 ± 0	1090 1022 ± 6	795 1031 ± 2	768 1051 ± 1	688 1052 ± 5	678 1163 ± 11	1404 60311 ± 644	1401 60186 ± 499
576	innovatrics-015	0	5116879	1060 15251	1424 6327 ± 0	1322 1350 ± 5	1064 1405 ± 4	1020 1371 ± 3	978 1405 ± 21	906 1502 ± 17	1411 65175 ± 1027	1407 64294 ± 1390
577	innovatrics-016	0	3130540	1040 6982	1423 6303 ± 0	1460 2084 ± 0	1172 2097 ± 0	1162 2103 ± 1	1133 2139 ± 0	1035 2215 ± 0	1346 32998 ± 309	1348 32967 ± 351
578	innovatrics-017	2456649	28156	1013 5307	1380 4237 ± 0	1418 1569 ± 0	1127 1579 ± 0	1115 1570 ± 0	1056 1574 ± 0	941 1584 ± 0	1348 33161 ± 158	1350 33197 ± 143
579	insightface-000	806953	16606	-	1261 4096 ± 0	1082 1009 ± 1	785 1019 ± 2	735 1017 ± 2	671 1020 ± 2	576 1032 ± 2	647 1778 ± 31	649 1773 ± 35
580	insightface-001	776777	16606	963 3858	459 2048 ± 0	1329 1366 ± 2	1039 1368 ± 3	1021 1372 ± 3	952 1375 ± 5	833 1386 ± 4	453 1119 ± 29	453 1108 ± 34
581	insightface-002	800572	16606	-	715 2048 ± 0	1355 1396 ± 2	1048 1389 ± 4	1046 1403 ± 3	974 1402 ± 2	850 1413 ± 3	474 1169 ± 40	460 1118 ± 38
582	insightface-003	1016917	26668	644 1550	928 2048 ± 0	1130 1073 ± 0	835 1092 ± 2	788 1070 ± 1	714 1082 ± 1	626 1101 ± 1	194 597 ± 16	204 595 ± 17
583	insightface-004	928816	26668	717 1800	376 2048 ± 0	1296 1298 ± 1	1035 1356 ± 2	982 1313 ± 1	907 1311 ± 0	807 1349 ± 1	174 565 ± 7	178 559 ± 5
584	inspur-000	364844	91926	332 808	1241 4096 ± 0	1330 1367 ± 1	1026 1331 ± 2	1018 1368 ± 2	1023 1465 ± 1	989 1861 ± 3	1173 9831 ± 37	1173 9860 ± 40
585	inspur-001	364862	91934	347 833	754 2048 ± 0	1347 1385 ± 1	1037 1361 ± 4	1044 1401 ± 2	1034 1495 ± 1	996 1891 ± 1	1071 5641 ± 42	1072 5622 ± 40
586	inspur-002	447948	91934	427 1009	709 2048 ± 0	1117 1056 ± 5	808 1056 ± 7	807 1096 ± 7	1085 1776 ± 23	873 1444 ± 12	1017 4504 ± 32	1017 4515 ± 35
587	inspur-003	444594	91934	440 1033	346 2048 ± 0	1247 1225 ± 2	924 1199 ± 3	923 1241 ± 1	910 1316 ± 1	958 1645 ± 5	1020 4525 ± 16	1018 4523 ± 26
588	intellicloudai-001	220831	868246	259 655	830 2048 ± 0	436 468 ± 2	277 456 ± 1	246 466 ± 3	235 492 ± 1	279 632 ± 2	419 1056 ± 4	432 1051 ± 72
589	intellicloudai-002	259047	58559	947 3585	1319 4100 ± 0	916 847 ± 1	634 847 ± 2	590 849 ± 1	539 853 ± 1	476 878 ± 4	346 822 ± 28	353 818 ± 23
590	intellifusion-001	271872	289387	312 762	413 2048 ± 0	819 764 ± 38	579 774 ± 39	546 797 ± 42	501 803 ± 34	421 805 ± 33	448 1112 ± 28	462 1128 ± 41
591	intellifusion-002	762731	385841	401 941	1296 4096 ± 0	1025 950 ± 2	841 1096 ± 42	798 1088 ± 33	786 1168 ± 31	681 1171 ± 10	627 1713 ± 57	618 1665 ± 87
592	intellivision-001	43692	11649	-	1081 2056 ± 0	21 62 ± 2	-	-	-	-	815 2573 ± 91	815 2544 ± 38
593	intellivision-002	43692	14505	-	1068 2056 ± 0	261 322 ± 1	181 355 ± 2	162 372 ± 1	176 422 ± 2	257 600 ± 1	1219 13525 ± 134	1218 12782 ± 278
594	intellivision-003	64023	133748	328 800	1046 2056 ± 0	367 407 ± 3	224 398 ± 2	201 418 ± 2	194 450 ± 1	250 591 ± 4	1189 11069 ± 56	1190 11066 ± 75
595	intellivision-004	117727	131310	186 515	1058 2056 ± 0	268 330 ± 0	162 330 ± 0	137 347 ± 0	146 382 ± 0	200 514 ± 0	1191 11197 ± 63	1192 11165 ± 72
596	intellivision-005	372608	257259	276 686	1039 2056 ± 0	629 618 ± 0	427 623 ± 0	405 634 ± 0	373 659 ± 0	399 771 ± 0	1276 19244 ± 68	1278 19270 ± 77
597	intellivision-006	287736	262742	288 723	1051 2056 ± 0	153 214 ± 0	90 214 ± 0	75 221 ± 0	82 272 ± 0	95 357 ± 1	1240 15153 ± 32	1243 15142 ± 38
598	intellivision-007	192454	262742	138 419	1048 2056 ± 0	83 148 ± 0	49 157 ± 0	44 166 ± 0	40 180 ± 0	59 265 ± 0	1237 15100 ± 50	1239 15095 ± 45
599	intellivix-001	256654	111858	358 846	478 2048 ± 0	328 378 ± 1	204 379 ± 1	171 381 ± 1	147 384 ± 1	136 421 ± 3	446 1100 ± 16	456 1109 ± 22
600	intellivix-002	361566	116162	509 1175	753 2048 ± 0	1032 956 ± 0	720 947 ± 6	696 976 ± 0	635 984 ± 4	620 1089 ± 1	1334 30096 ± 128	1339 31287 ± 140
601	intellivix-003	234409	116167	564 1301	755 2048 ± 0	971 908 ± 0	689 916 ± 1	659 930 ± 0	619 961 ± 1	653 1129 ± 3	1333 30025 ± 137	1338 31190 ± 131
602	intellivix-004	361668	128097	523 1189	567 2048 ± 0	389 427 ± 0	261 437 ± 0	224 444 ± 0	206 461 ± 0	238 565 ± 2	578 1498 ± 37	570 1478 ± 16
603	intellivix-005	774405	128093	957 3692	357 2048 ± 0	704 673 ± 1	453 643 ± 3	423 654 ± 3	383 666 ± 3	396 769 ± 5	566 1481 ± 65	557 1451 ± 26
604	intellivix-006	792822	128093	959 3753	649 2048 ± 0	694 665 ± 1	461 657 ± 2	432 662 ± 2	393 677 ± 2	409 784 ± 1	573 1486 ± 13	574 1482 ± 16
605	intelresearch-000	388229	307985	-	736 2048 ± 0	968 902 ± 6	-	-	-	-	1034 4800 ± 152	1022 4561 ± 97
606	intelresearch-001	353997	272602	-	850 2048 ± 0	720 682 ± 4	-	-	-	-	949 3553 ± 57	941 3462 ± 161
607	intelresearch-002	452850	86454	-	446 2048 ± 0	762 707 ± 2	591 790 ± 33	538 788 ± 26	525 831 ± 29	464 862 ± 22	1000 4204 ± 91	1001 4153 ± 93
608	intelresearch-003	401343	85085	510 1177	654 2048 ± 0	1251 1232 ± 3	957 1237 ± 2	925 1242 ± 2	872 1263 ± 2	790 1324 ± 3	1014 4443 ± 75	1008 4374 ± 77
609	intelresearch-004	646918	85290	733 1866	361 2048 ± 0	1309 1319 ± 2	1019 1322 ± 3	994 1330 ± 3	926 1345 ± 3	847 1411 ± 5	1028 4696 ± 63	1028 4692 ± 66
610	intelresearch-005	398137	85290	504 1164	537 2048 ± 0	1314 1328 ± 1	1028 1334 ± 2	1004 1344 ± 2	933 1356 ± 2	856 1423 ± 4	1019 4524 ± 87	1012 4461 ± 74
611	intelresearch-006	445223	101126	462 1069	592 2048 ± 0	990 918 ± 1	664 881 ± 0	622 892 ± 0	582 913 ± 1	562 1008 ± 3	1091 6137 ± 410	1087 6024 ± 109
612	intelresearch-007	523218	101150	528 1197	600 2048 ± 0	582 590 ± 1	388 585 ± 1	350 588 ± 1	319 603 ± 4	310 668 ± 6	966 3738 ± 42	972 3752 ± 53
613	intelresearch-008	523218	101214	527 1196	387 2048 ± 0	654 634 ± 1	436 630 ± 1	400 632 ± 1	349 635 ± 2	341 711 ± 4	957 3661 ± 34	959 3641 ± 44
614	intelresearch-009	290599	120716	636 1535	570 2048 ± 0	535 560 ± 1	467 664 ± 3	410 638 ± 3	395 678 ± 13	327 692 ± 4	975 3829 ± 207	963 3679 ± 71
615	intelresearch-010	307886	120767	676 1655	575 2048 ± 0	1186 1153 ± 1	893 1158 ± 2	859 1160 ± 2	783 1166 ± 3	718 1237 ± 5	977 3850 ± 26	977 3822 ± 75
616	intelresearch-011	317134	120757	705 1758	396 2048 ± 0	1254 1236 ± 4	941 1219 ± 3	916 1231 ± 1	851 1237 ± 3	780 1312 ± 6	979 3873 ± 47	981 3890 ± 57

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	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
617	intelresearch-012	317135	120715	704 1758	905 2048 ± 0	1283 1280 ± 2	971 1256 ± 4	948 1270 ± 3	890 1284 ± 3	799 1331 ± 5	969 3744 ± 32	968 3722 ± 44
618	intema-000	1532392	19488	473 1088	155 513 ± 0	1084 1010 ± 0	773 1001 ± 4	718 994 ± 0	642 993 ± 5	592 1056 ± 1	374 910 ± 29	379 906 ± 32
619	intema-001	1122562	19536	618 1458	159 513 ± 0	1326 1354 ± 1	1016 1318 ± 5	999 1336 ± 4	919 1328 ± 2	823 1375 ± 0	391 977 ± 31	397 980 ± 31
620	interlinklabs-001	155897	1013105	305 746	763 2048 ± 0	267 328 ± 3	174 346 ± 8	136 343 ± 7	123 352 ± 3	133 418 ± 11	550 1441 ± 10	556 1450 ± 14
621	interlinklabs-002	286341	1013147	586 1364	1230 4096 ± 0	405 438 ± 1	256 435 ± 4	215 435 ± 5	189 439 ± 0	189 505 ± 2	830 2699 ± 35	833 2678 ± 25
622	interlinklabs-003	528962	951057	642 1549	1203 4096 ± 0	789 734 ± 0	558 750 ± 0	511 746 ± 0	477 769 ± 0	509 932 ± 0	942 3513 ± 50	947 3504 ± 26
623	intozi-001	21863	74210	42 140	667 2048 ± 0	34 85 ± 0	21 86 ± 0	16 86 ± 0	13 87 ± 0	11 87 ± 0	1166 9579 ± 252	1163 9530 ± 167
624	intozi-002	172046	95119	356 844	796 2048 ± 0	1420 1571 ± 0	1129 1585 ± 1	1117 1580 ± 0	1060 1585 ± 0	961 1665 ± 1	340 805 ± 14	345 803 ± 4
625	intsysmsu-000	650193	171481	-	738 2048 ± 0	506 535 ± 20	-	-	-	-	208 610 ± 22	227 613 ± 31
626	intsysmsu-001	384409	172480	324 789	800 2048 ± 0	620 614 ± 2	414 615 ± 2	417 642 ± 2	460 750 ± 3	673 1159 ± 4	223 621 ± 8	224 611 ± 31
627	intsysmsu-002	765921	172298	323 786	226 1024 ± 0	587 593 ± 1	592 793 ± 2	570 827 ± 1	553 875 ± 104	764 1293 ± 3	167 549 ± 25	174 548 ± 29
628	ionetworks-000	287609	51236	113 352	811 2048 ± 0	393 430 ± 0	257 435 ± 0	214 433 ± 0	182 432 ± 0	149 444 ± 0	1115 6913 ± 102	1122 7150 ± 160
629	ionetworks-001	39860	333205	93 298	691 2048 ± 0	406 439 ± 2	251 433 ± 2	244 461 ± 2	198 454 ± 2	176 488 ± 2	1100 6335 ± 103	1097 6358 ± 91
630	ionetworks-002	337801	73038	702 1746	423 2048 ± 0	737 689 ± 1	477 672 ± 5	437 665 ± 3	386 669 ± 3	305 664 ± 6	1113 6884 ± 154	1109 6808 ± 102
631	ionetworks-003	309918	68642	666 1631	982 2050 ± 0	769 713 ± 0	490 681 ± 1	438 665 ± 0	385 668 ± 0	311 671 ± 0	1182 10649 ± 225	1184 10658 ± 106
632	ionetworks-004	302266	135028	723 1813	983 2050 ± 0	1051 975 ± 0	685 910 ± 0	646 921 ± 0	583 916 ± 0	499 917 ± 1	1325 27437 ± 130	1327 27472 ± 114
633	iqface-000	268819	596337	280 704	1392 4750 ± 32	511 538 ± 26	307 494 ± 2	318 543 ± 3	450 734 ± 4	838 1393 ± 4	1457 636433 ± 38446	1457 632654 ± 85615
634	iqface-001	1145590	760941	-	1393 4752 ± 35	601 603 ± 1	-	-	-	-	1455 568761 ± 3659	1455 570335 ± 5122
635	iqface-003	370803	963398	344 821	1394 4763 ± 37	495 529 ± 1	338 532 ± 2	359 599 ± 8	538 850 ± 2	965 1694 ± 2	1456 575924 ± 2601	1456 576653 ± 2051
636	iqsec-001	172239	41022	363 850	773 2048 ± 0	185 258 ± 6	111 254 ± 4	256 474 ± 9	744 1124 ± 24	1109 4155 ± 119	1310 24586 ± 143	1316 25631 ± 157
637	iqsec-002	172239	41022	365 852	618 2048 ± 0	172 243 ± 6	113 259 ± 4	283 504 ± 10	802 1184 ± 25	1108 4081 ± 124	1311 24688 ± 149	1317 25737 ± 142
638	iqsec-003	293643	15245	241 622	1353 4116 ± 0	942 873 ± 0	654 872 ± 0	607 872 ± 0	560 880 ± 0	507 929 ± 0	1013 4441 ± 20	1013 4462 ± 41
639	irex-000	741899	47419	775 2090	1184 3080 ± 0	921 852 ± 2	636 850 ± 1	609 874 ± 2	603 939 ± 1	727 1249 ± 5	41 201 ± 11	47 208 ± 8
640	irisid-001	480885	100974	423 1000	972 2049 ± 0	1042 966 ± 0	743 971 ± 0	747 1028 ± 0	780 1163 ± 0	960 1662 ± 1	678 1892 ± 13	709 2040 ± 17
641	irisid-002	735508	101002	571 1334	974 2049 ± 0	1431 1732 ± 3	1137 1730 ± 2	1127 1752 ± 2	1112 1923 ± 2	1057 2410 ± 2	1130 7367 ± 16	1130 7410 ± 48
642	isap-001	99049	204201	1 18	1274 4096 ± 0	1 0 ± 0	-	-	-	-	129 459 ± 17	137 456 ± 11
643	isap-002	256765	49931	90 288	658 2048 ± 0	826 769 ± 3	789 1027 ± 2	613 877 ± 2	471 761 ± 1	495 912 ± 2	888 3045 ± 94	882 2973 ± 66
644	isityou-000	48010	36621	29 110	1456 19200 ± 0	55 113 ± 5	-	-	-	-	1449 237517 ± 1318	1449 237374 ± 1279
645	istl-001	271118	27325	209 561	1111 2060 ± 0	592 595 ± 0	398 598 ± 0	358 598 ± 0	318 603 ± 0	263 605 ± 0	715 2073 ± 19	719 2072 ± 34
646	isystems-000	209270	713870	-	223 1024 ± 0	87 154 ± 4	-	-	-	-	114 398 ± 14	121 396 ± 16
647	isystems-001	274621	639268	478 1091	875 2048 ± 0	219 291 ± 9	-	-	-	-	169 557 ± 16	180 564 ± 22
648	isystems-002	358984	803389	659 1595	501 2048 ± 0	890 822 ± 8	-	-	-	-	309 749 ± 31	248 632 ± 28
649	itmo-001	1923215	62191	-	1461 37997 ± 0	1034 959 ± 18	-	-	-	-	1329 28901 ± 1492	1328 27517 ± 186
650	itmo-002	1923215	62191	-	1366 4162 ± 0	609 611 ± 17	-	-	-	-	1133 7423 ± 96	1133 7451 ± 94
651	itmo-003	579961	82630	-	1143 2114 ± 0	29 79 ± 2	-	-	-	-	1175 10038 ± 180	1176 10030 ± 134
652	itmo-004	487023	79012	-	1141 2100 ± 0	399 436 ± 0	-	-	-	-	1146 8219 ± 100	1150 8214 ± 130
653	itmo-005	482155	81168	-	1370 4173 ± 0	816 759 ± 1	-	-	-	-	1216 13214 ± 164	1217 12576 ± 257
654	itmo-006	599187	96762	-	1148 2121 ± 0	875 814 ± 1	623 831 ± 26	575 830 ± 17	524 830 ± 3	522 952 ± 38	1318 26154 ± 148	1319 26217 ± 260
655	itmo-007	415979	245376	794 2199	873 2048 ± 0	796 741 ± 2	-	-	-	-	812 2551 ± 50	812 2529 ± 80
656	itmo-008	726866	318238	590 1376	1202 4096 ± 0	1122 1060 ± 1	810 1058 ± 1	776 1059 ± 1	706 1072 ± 4	630 1104 ± 1	951 3578 ± 25	957 3580 ± 28
657	ivacognitive-001	256958	62791	413 975	780 2048 ± 0	1294 1292 ± 3	993 1289 ± 4	965 1292 ± 4	893 1292 ± 3	786 1321 ± 4	1001 4228 ± 41	1002 4226 ± 41
658	iws-000	30875	3063	19 77	124 512 ± 0	202 277 ± 5	132 283 ± 1	271 494 ± 3	636 984 ± 3	1087 2987 ± 39	399 999 ± 40	402 992 ± 22
659	jaakit-001	99024	24754	76 251	122 512 ± 0	26 76 ± 0	17 77 ± 0	13 79 ± 0	12 81 ± 0	13 93 ± 0	798 2466 ± 57	800 2465 ± 66
660	jaakit-002	179256	75037	85 277	495 2048 ± 0	814 758 ± 0	568 760 ± 0	653 925 ± 0	911 1318 ± 1	1076 2799 ± 2	381 943 ± 7	393 963 ± 5

Notes	
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2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 36: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
661	kaizenvoiz-001	254811	74894	128 388	598 2048 ± 0	295 354 ± 0	185 360 ± 0	286 508 ± 0	-	-	931 3439 ± 19	948 3506 ± 31
662	kakao-001	107616	285676	-	253 1024 ± 0	330 379 ± 1	-	-	-	-	378 930 ± 22	389 948 ± 38
663	kakao-002	479406	412822	-	610 2048 ± 0	800 747 ± 6	-	-	-	-	630 1720 ± 62	633 1715 ± 83
664	kakao-003	414379	113944	-	729 2048 ± 0	950 878 ± 3	-	-	-	-	724 2128 ± 34	729 2134 ± 60
665	kakao-004	414411	135270	-	958 2048 ± 0	924 856 ± 1	640 855 ± 0	596 857 ± 0	547 858 ± 0	463 860 ± 1	695 1957 ± 28	696 1953 ± 37
666	kakao-005	414316	152216	-	1009 2052 ± 0	1126 1068 ± 1	818 1073 ± 1	792 1079 ± 0	708 1077 ± 1	619 1089 ± 1	713 2067 ± 26	710 2043 ± 34
667	kakao-007	526993	129545	922 3150	512 2048 ± 0	1031 952 ± 1	735 961 ± 1	682 958 ± 1	622 962 ± 1	530 968 ± 1	418 1056 ± 16	426 1047 ± 28
668	kakao-008	734583	104820	966 3879	745 2048 ± 0	1168 1135 ± 3	886 1148 ± 3	852 1150 ± 3	774 1156 ± 1	685 1175 ± 1	306 736 ± 23	314 727 ± 22
669	kakao-009	766989	104820	975 4042	809 2048 ± 0	586 593 ± 1	401 603 ± 4	375 613 ± 2	333 617 ± 2	284 636 ± 4	128 457 ± 20	133 448 ± 17
670	kakaobank-000	570796	98479	869 2690	1218 4096 ± 0	1252 1234 ± 1	968 1254 ± 7	930 1248 ± 3	961 1389 ± 3	986 1843 ± 11	1033 4790 ± 35	1034 4803 ± 35
671	kakaobank-001	780621	98829	793 2189	439 2048 ± 0	568 584 ± 0	406 608 ± 1	367 607 ± 5	398 684 ± 1	502 922 ± 2	1324 27092 ± 174	1326 27065 ± 158
672	kakaobank-002	780621	98833	836 2459	668 2048 ± 0	1078 1004 ± 1	780 1010 ± 0	742 1025 ± 5	710 1078 ± 1	783 1313 ± 1	1322 26972 ± 118	1324 27022 ± 139
673	kakaobank-003	1409862	98833	980 4186	859 2048 ± 0	1218 1194 ± 2	947 1224 ± 6	915 1231 ± 2	874 1269 ± 5	926 1535 ± 5	1323 26986 ± 183	1325 27026 ± 151
674	kakaopay-001	397864	179869	275 685	1269 4096 ± 0	411 448 ± 0	348 542 ± 0	317 542 ± 0	271 542 ± 0	229 553 ± 0	235 633 ± 22	242 630 ± 22
675	kasikornlabs-000	256471	61000	258 655	766 2048 ± 0	972 908 ± 36	661 878 ± 22	691 969 ± 39	800 1184 ± 54	1053 2382 ± 145	1339 31669 ± 188	1340 31714 ± 182
676	kasikornlabs-001	256471	61037	270 676	605 2048 ± 0	979 912 ± 38	649 868 ± 10	730 1005 ± 50	793 1176 ± 44	1054 2387 ± 147	1336 30759 ± 198	1336 30867 ± 174
677	kasikornlabs-002	256431	61063	235 613	554 2048 ± 0	989 917 ± 35	683 907 ± 13	687 963 ± 13	914 1320 ± 45	1065 2629 ± 178	1338 31025 ± 180	1337 31054 ± 186
678	kasikornlabs-003	256442	61063	308 751	481 2048 ± 0	674 652 ± 5	460 656 ± 5	521 755 ± 11	662 1007 ± 27	1038 2240 ± 168	1305 24022 ± 121	1306 23998 ± 97
679	kasikornlabs-004	466064	61063	446 1045	818 2048 ± 0	1023 950 ± 7	741 970 ± 4	751 1031 ± 11	950 1372 ± 30	1050 2368 ± 156	1302 23867 ± 56	1304 23919 ± 91
680	kasikornlabs-005	466202	61063	436 1028	869 2048 ± 0	1002 936 ± 8	731 956 ± 14	774 1057 ± 14	855 1243 ± 28	1042 2285 ± 145	1304 23900 ± 44	1305 23923 ± 56
681	kbyai-000	502373	34357	822 2417	72 296 ± 0	1226 1202 ± 3	1174 2167 ± 3	1168 2293 ± 9	1149 2586 ± 13	1102 3689 ± 20	1261 17785 ± 51	1265 17801 ± 125
682	kedacom-000	245292	37401	1065 23574	70 292 ± 0	476 506 ± 3	354 547 ± 10	376 614 ± 9	308 588 ± 10	307 665 ± 24	268 684 ± 14	279 682 ± 16
683	keyless-000	551302	27472	503 1157	781 2048 ± 0	623 615 ± 0	543 736 ± 5	1163 2122 ± 2	1128 2086 ± 1	1022 2110 ± 4	373 909 ± 8	377 905 ± 16
684	kiwitech-000	369711	21375	351 841	391 2048 ± 0	583 591 ± 0	395 594 ± 0	357 595 ± 1	315 596 ± 0	265 609 ± 0	639 1755 ± 20	643 1734 ± 16
685	kloudspot-001	261662	887420	379 882	553 2048 ± 0	269 330 ± 1	155 319 ± 6	123 325 ± 1	105 318 ± 0	72 318 ± 0	1065 5539 ± 48	1067 5555 ± 66
686	kloudspot-002	635738	460698	638 1540	1308 4096 ± 0	1166 1128 ± 1	866 1128 ± 2	829 1123 ± 0	776 1158 ± 2	654 1132 ± 0	1059 5456 ± 154	1060 5460 ± 108
687	kneron-003	58366	1747	64 188	733 2048 ± 0	208 281 ± 3	130 280 ± 1	118 315 ± 13	133 365 ± 7	712 1224 ± 30	1055 5237 ± 63	1056 5274 ± 99
688	kneron-005	375374	13633	154 457	388 2048 ± 0	487 518 ± 2	331 522 ± 4	328 556 ± 5	468 757 ± 19	975 1760 ± 25	687 1922 ± 11	688 1926 ± 20
689	kneron-006	590352	63741	596 1402	389 2048 ± 0	559 579 ± 0	382 580 ± 0	360 599 ± 0	439 726 ± 1	752 1278 ± 4	749 2247 ± 22	751 2252 ± 28
690	knowutech-000	808045	32886	567 1312	307 1536 ± 0	1371 1419 ± 2	1041 1372 ± 1	1026 1377 ± 1	954 1382 ± 2	832 1386 ± 2	968 3743 ± 31	965 3693 ± 38
691	kogniza-001	0	394421	555 1288	306 1536 ± 0	1288 1285 ± 2	1006 1304 ± 4	958 1280 ± 2	883 1278 ± 2	794 1326 ± 2	878 2968 ± 37	881 2971 ± 20
692	kookmin-001	234001	31875	-	471 2048 ± 0	57 114 ± 1	31 110 ± 1	27 116 ± 1	27 128 ± 1	27 172 ± 1	230 629 ± 35	233 616 ± 11
693	kookmin-002	371771	30734	345 827	895 2048 ± 0	1108 1038 ± 2	804 1047 ± 1	762 1045 ± 1	694 1061 ± 1	638 1116 ± 1	242 638 ± 19	253 636 ± 20
694	koreaid-001	256261	20152	720 1810	480 2048 ± 0	339 384 ± 2	216 390 ± 1	226 444 ± 2	283 556 ± 6	415 795 ± 5	239 636 ± 11	254 636 ± 10
695	krungthai-002	2360957	15033	518 1185	549 2048 ± 0	244 308 ± 0	149 314 ± 5	114 309 ± 0	107 319 ± 0	98 362 ± 0	882 3014 ± 20	883 2980 ± 22
696	kuke3d-001	403462	68786	194 530	1213 4096 ± 0	876 814 ± 2	604 811 ± 2	556 814 ± 2	508 814 ± 1	447 834 ± 1	1103 6412 ± 57	1099 6413 ± 51
697	kuke3d-002	270544	1227855	341 820	368 2048 ± 0	475 504 ± 3	317 504 ± 1	289 511 ± 1	258 523 ± 2	245 585 ± 1	875 2943 ± 22	879 2966 ± 38
698	lebentech-000	0	10360	2 26	110 512 ± 0	3 22 ± 0	1 22 ± 0	1 22 ± 0	1 23 ± 0	1 23 ± 0	337 801 ± 42	355 825 ± 51
699	lebentech-001	0	144456	651 1567	442 2048 ± 0	102 174 ± 0	58 175 ± 0	64 200 ± 1	66 227 ± 9	123 397 ± 20	143 485 ± 17	147 473 ± 28
700	lemalabs-001	748400	198794	882 2745	660 2048 ± 0	869 810 ± 0	606 812 ± 0	554 813 ± 0	514 819 ± 0	453 844 ± 1	1203 11930 ± 35	1206 11913 ± 37
701	lightvision-000	182569	62181	397 930	524 2048 ± 0	371 411 ± 1	207 381 ± 0	168 379 ± 1	149 386 ± 3	121 396 ± 2	570 1484 ± 13	569 1477 ± 13
702	lightvision-001	266840	62181	663 1604	874 2048 ± 0	497 530 ± 2	333 523 ± 2	314 542 ± 1	278 548 ± 10	216 537 ± 1	554 1456 ± 10	561 1457 ± 14
703	lineclova-000	264443	407003	-	960 2048 ± 0	572 586 ± 0	409 612 ± 0	368 609 ± 1	325 611 ± 0	271 618 ± 1	841 2753 ± 19	842 2745 ± 23
704	lineclova-001	944355	407058	799 2233	885 2048 ± 0	902 833 ± 10	621 830 ± 3	571 828 ± 4	531 838 ± 8	446 833 ± 4	829 2696 ± 23	832 2677 ± 35

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Table 37: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
705	lineclova-002	475779	406756	578 1352	509 2048 ± 0	1287 1284 ± 1	984 1275 ± 2	954 1275 ± 1	881 1273 ± 2	754 1281 ± 2	843 2765 ± 10	845 2767 ± 31
706	lineclova-003	585149	410482	693 1725	419 2048 ± 0	1383 1444 ± 1	1090 1438 ± 1	1068 1439 ± 2	1002 1440 ± 1	876 1446 ± 2	864 2890 ± 23	869 2899 ± 29
707	lookman-002	138200	25410	1061 16518	192 548 ± 0	99 173 ± 1	-	-	-	-	210 610 ± 19	226 612 ± 22
708	lookman-004	244775	37401	1064 23548	193 548 ± 0	478 507 ± 5	352 545 ± 12	373 613 ± 12	312 590 ± 11	298 656 ± 16	358 871 ± 29	367 878 ± 29
709	lssytech-000	1316125	197944	996 4622	1183 3076 ± 0	1187 1154 ± 0	861 1124 ± 0	835 1129 ± 0	751 1128 ± 0	651 1128 ± 0	305 733 ± 17	310 722 ± 13
710	luxand-000	0	57908	587 1366	278 1040 ± 0	365 407 ± 23	252 433 ± 11	225 444 ± 14	214 464 ± 14	235 562 ± 25	350 828 ± 28	356 828 ± 32
711	luxand-001	85194	72189	248 636	844 2048 ± 0	394 431 ± 0	266 443 ± 0	223 443 ± 2	192 448 ± 0	205 519 ± 0	546 1433 ± 16	553 1433 ± 16
712	luxand-003	127425	72189	361 847	640 2048 ± 0	760 706 ± 4	507 697 ± 1	468 698 ± 3	411 694 ± 0	408 782 ± 3	552 1443 ± 17	552 1431 ± 17
713	luxand-004	127425	72189	362 848	889 2048 ± 0	711 677 ± 0	492 682 ± 0	459 691 ± 0	423 705 ± 0	390 766 ± 0	551 1441 ± 10	554 1440 ± 11
714	mantra-000	471458	62566	306 749	1008 2052 ± 0	373 413 ± 18	303 487 ± 19	270 494 ± 18	249 511 ± 18	256 598 ± 19	898 3151 ± 51	900 3127 ± 63
715	matrixcomsec-000	114103	177320	83 264	887 2048 ± 0	40 94 ± 0	24 94 ± 0	19 95 ± 0	38 173 ± 1	21 140 ± 1	220 616 ± 22	241 630 ± 28
716	matrixcomsec-001	179657	168204	496 1141	431 2048 ± 0	95 166 ± 0	53 168 ± 0	46 169 ± 0	37 171 ± 0	33 191 ± 0	64 268 ± 53	71 261 ± 62
717	matrixcomsec-002	93740	168203	238 618	390 2048 ± 0	81 147 ± 0	45 146 ± 0	37 147 ± 0	30 148 ± 0	29 178 ± 0	329 791 ± 13	334 790 ± 16
718	maverickresearch-000	256262	20268	104 331	689 2048 ± 0	1072 995 ± 1	772 999 ± 1	724 999 ± 2	646 997 ± 0	570 1023 ± 2	1283 20268 ± 108	1285 20254 ± 121
719	maxvision-000	133114	56426	-	144 512 ± 0	304 359 ± 0	183 356 ± 0	150 359 ± 0	126 356 ± 0	101 370 ± 1	796 2461 ± 20	797 2452 ± 17
720	maxvision-001	256146	61793	912 3037	872 2048 ± 0	200 275 ± 3	124 274 ± 2	92 277 ± 4	85 280 ± 4	74 325 ± 3	292 714 ± 13	307 717 ± 13
721	maxvision-002	171894	60623	706 1765	642 2048 ± 0	98 172 ± 0	56 171 ± 0	47 172 ± 0	39 174 ± 0	42 221 ± 0	299 725 ± 5	312 725 ± 5
722	maxvision-003	234062	61252	831 2452	430 2048 ± 0	444 474 ± 0	288 468 ± 0	253 471 ± 0	222 475 ± 0	206 519 ± 0	800 2467 ± 28	805 2488 ± 23
723	maxvision-004	394121	61583	924 3217	901 2048 ± 0	662 644 ± 1	457 648 ± 0	419 647 ± 1	354 637 ± 1	331 695 ± 1	680 1897 ± 10	682 1903 ± 14
724	maxvision-005	309326	61739	784 2138	760 2048 ± 0	1081 1007 ± 1	962 1245 ± 5	736 1017 ± 1	683 1040 ± 4	770 1299 ± 3	693 1950 ± 38	695 1948 ± 34
725	maxvision-006	315987	61739	923 3172	915 2048 ± 0	1264 1247 ± 3	933 1208 ± 3	901 1211 ± 5	829 1215 ± 1	745 1268 ± 4	692 1938 ± 17	689 1929 ± 35
726	mbbankvn-000	472410	139431	981 4227	399 2048 ± 0	432 466 ± 17	295 480 ± 26	260 481 ± 27	226 479 ± 21	185 501 ± 29	1010 4388 ± 14	1009 4380 ± 17
727	megvii-000	1358915	16014	-	813 2048 ± 0	840 785 ± 0	-	-	-	-	1279 19816 ± 158	1281 19805 ± 174
728	megvii-001	1361523	16486	-	401 2048 ± 0	518 543 ± 0	-	-	-	-	1054 5228 ± 32	1054 5252 ± 60
729	megvii-002	1809564	16491	-	1314 4100 ± 0	663 644 ± 0	-	-	-	-	1388 50630 ± 183	1382 47591 ± 716
730	megvii-003	4430290	42790	-	1273 4096 ± 0	1230 1210 ± 1	946 1223 ± 0	1014 1356 ± 4	1059 1582 ± 7	1069 2727 ± 23	1447 225342 ± 3574	1448 225413 ± 6344
731	megvii-004	3962505	44019	-	1309 4097 ± 0	1290 1287 ± 1	1040 1369 ± 2	980 1310 ± 2	956 1384 ± 3	868 1436 ± 5	1378 46801 ± 204	1379 46832 ± 207
732	megvii-005	1378009	44038	974 4025	964 2049 ± 0	1308 1319 ± 5	964 1247 ± 6	921 1240 ± 2	857 1245 ± 2	768 1298 ± 3	1343 32025 ± 121	1344 32008 ± 114
733	megvii-006	1554938	44038	985 4346	963 2049 ± 0	1289 1287 ± 3	991 1286 ± 0	1036 1393 ± 5	912 1319 ± 1	814 1360 ± 1	1341 31845 ± 100	1342 31872 ± 118
734	megvii-007	1366028	44035	692 1723	973 2049 ± 0	1182 1148 ± 1	887 1153 ± 0	868 1172 ± 4	791 1172 ± 2	761 1288 ± 2	1282 20191 ± 1753	1283 20191 ± 1620
735	megvii-008	1644071	44035	774 2086	971 2049 ± 0	1306 1313 ± 1	1061 1404 ± 4	1017 1363 ± 2	940 1362 ± 3	854 1420 ± 2	1280 20059 ± 1598	1287 20448 ± 1633
736	megvii-009	1644071	44039	806 2294	978 2049 ± 0	1311 1322 ± 1	1024 1328 ± 1	1002 1339 ± 3	1127 2075 ± 3	842 1404 ± 4	1281 20081 ± 1621	1286 20260 ± 1577
737	meituan-000	259514	333178	-	437 2048 ± 0	400 436 ± 4	263 441 ± 1	391 626 ± 5	721 1098 ± 15	1090 3126 ± 53	241 638 ± 17	250 633 ± 16
738	meituan-001	615387	333249	485 1113	612 2048 ± 0	1088 1017 ± 4	779 1008 ± 3	733 1010 ± 2	667 1010 ± 3	564 1011 ± 4	252 654 ± 10	269 658 ± 14
739	meituan-002	686111	244091	795 2200	1283 4096 ± 0	1113 1052 ± 0	828 1086 ± 1	784 1064 ± 2	693 1060 ± 5	603 1063 ± 1	425 1064 ± 10	438 1070 ± 16
740	meituan-003	860010	244091	889 2772	1304 4096 ± 0	1409 1487 ± 1	1118 1518 ± 0	1088 1469 ± 6	1029 1478 ± 0	887 1471 ± 7	449 1112 ± 16	450 1097 ± 20
741	meituan-004	850507	244091	885 2759	1277 4096 ± 0	853 796 ± 1	612 822 ± 0	564 820 ± 11	509 815 ± 11	431 824 ± 0	462 1149 ± 10	471 1148 ± 5
742	meiya-001	280055	264913	179 507	981 2049 ± 0	634 622 ± 12	-	-	-	-	1148 8356 ± 615	1148 8134 ± 97
743	mendaxiatech-000	1941475	45484	989 4397	1310 4097 ± 0	1261 1243 ± 2	969 1255 ± 1	1023 1373 ± 2	1063 1598 ± 3	1067 2689 ± 8	1379 46906 ± 275	1380 46872 ± 217
744	metsakuurcompany-001	445177	1091558	656 1588	1052 2056 ± 0	556 578 ± 1	390 587 ± 3	351 590 ± 1	372 659 ± 1	459 854 ± 1	1154 8600 ± 192	1149 8155 ± 298
745	metsakuurcompany-002	0	957558	416 980	1063 2056 ± 0	1058 980 ± 1	752 978 ± 1	695 976 ± 2	658 1005 ± 1	628 1103 ± 2	1156 8766 ± 326	1157 8786 ± 324
746	metsakuurcompany-003	0	957562	426 1004	1064 2056 ± 0	1041 965 ± 2	740 970 ± 5	778 1060 ± 12	659 1005 ± 6	644 1121 ± 1	1147 8234 ± 145	1147 8111 ± 280
747	metsakuurcompany-004	0	1035754	453 1056	1061 2056 ± 0	1018 946 ± 10	642 861 ± 6	612 877 ± 3	591 926 ± 4	546 991 ± 5	1099 6281 ± 160	1095 6232 ± 191
748	metsakuurcompany-005	0	305952	264 661	657 2048 ± 0	441 472 ± 0	291 477 ± 0	272 494 ± 1	260 527 ± 1	404 778 ± 2	924 3376 ± 22	931 3383 ± 40

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Table 38: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
749	miaxis-001	0	215019	102 325	147 512 ± 0	205 279 ± 0	129 278 ± 0	93 278 ± 1	86 285 ± 0	67 297 ± 0	359 872 ± 14	369 881 ± 22
750	miaxis-002	0	216484	103 328	153 512 ± 0	220 292 ± 0	134 290 ± 1	106 297 ± 0	96 298 ± 0	96 359 ± 1	428 1065 ± 48	423 1040 ± 54
751	miaxis-003	0	439726	226 604	584 2048 ± 0	653 634 ± 0	430 624 ± 2	374 613 ± 0	259 524 ± 0	306 665 ± 0	712 2053 ± 20	712 2048 ± 28
752	miaxis-004	729303	89655	516 1183	980 2049 ± 0	1454 1941 ± 0	1164 1947 ± 0	1151 1940 ± 0	1162 3153 ± 0	1005 1963 ± 0	1440 103338 ± 44	1440 103361 ± 65
753	miaxis-005	738575	172996	990 4416	976 2049 ± 0	1433 1741 ± 1	1141 1776 ± 13	1130 1771 ± 4	1084 1774 ± 1	998 1925 ± 5	967 3738 ± 17	974 3774 ± 45
754	microfocus-000	103483	27242	-	38 256 ± 0	190 263 ± 18	-	-	-	-	45 202 ± 5	43 203 ± 7
755	microfocus-001	104524	27242	-	35 256 ± 0	191 264 ± 18	-	-	-	-	48 215 ± 8	49 217 ± 10
756	microfocus-002	96288	27362	60 176	28 256 ± 0	186 259 ± 18	-	-	-	-	90 337 ± 34	55 230 ± 25
757	microfocus-003	169603	27689	180 507	27 256 ± 0	461 490 ± 19	327 517 ± 30	518 752 ± 55	748 1126 ± 121	1068 2713 ± 345	20 121 ± 5	20 117 ± 5
758	minivision-000	836697	16597	983 4263	1239 4096 ± 0	1106 1035 ± 1	799 1033 ± 2	753 1035 ± 1	680 1037 ± 1	598 1059 ± 2	799 2466 ± 26	798 2460 ± 25
759	mitek-000	105584	44643	86 277	261 1024 ± 0	183 256 ± 21	115 265 ± 10	164 375 ± 24	276 545 ± 42	625 1098 ± 124	856 2846 ± 88	848 2777 ± 74
760	mitek-001	519928	69522	711 1777	1054 2056 ± 0	321 371 ± 0	222 397 ± 0	190 407 ± 4	158 394 ± 5	131 416 ± 2	1108 6682 ± 28	1106 6694 ± 48
761	mitek-002	534528	68069	709 1773	1075 2056 ± 0	312 365 ± 0	201 372 ± 0	156 366 ± 2	143 377 ± 2	119 394 ± 1	401 1005 ± 19	392 963 ± 23
762	mobai-000	365451	80573	322 786	1403 6144 ± 0	822 766 ± 8	650 869 ± 6	897 1205 ± 31	1101 1867 ± 45	1100 3549 ± 190	1253 16458 ± 333	1256 16423 ± 1473
763	mobai-001	265297	60164	195 534	740 2048 ± 0	618 612 ± 3	412 614 ± 3	455 687 ± 9	564 886 ± 31	968 1707 ± 103	529 1386 ± 25	535 1377 ± 26
764	mobai-002	254811	33242	622 1482	585 2048 ± 0	299 356 ± 0	195 366 ± 0	151 359 ± 4	134 367 ± 2	138 428 ± 0	1149 8387 ± 126	1151 8381 ± 105
765	mobai-003	1046772	427612	946 3555	1256 4096 ± 0	1427 1691 ± 7	1136 1698 ± 3	1123 1693 ± 1	1071 1705 ± 2	979 1786 ± 1	1442 105417 ± 231	1442 105475 ± 169
766	mobbl-000	182052	58727	-	786 2048 ± 0	187 261 ± 16	117 267 ± 22	165 375 ± 92	370 655 ± 273	1016 2059 ± 1129	1206 12061 ± 142	1208 12050 ± 133
767	mobbl-001	231160	58706	74 226	432 2048 ± 0	114 183 ± 32	66 184 ± 25	146 354 ± 76	517 823 ± 396	1074 2781 ± 1166	1201 11832 ± 109	1205 11851 ± 88
768	mobbl-002	242920	60119	-	888 2048 ± 0	687 663 ± 6	464 660 ± 5	431 662 ± 5	378 663 ± 5	315 676 ± 5	1197 11616 ± 78	1198 11588 ± 97
769	mobbl-003	172248	60960	87 282	378 2048 ± 0	691 664 ± 6	465 661 ± 5	435 663 ± 5	380 665 ± 6	325 691 ± 5	1214 12506 ± 111	1215 12509 ± 100
770	mobbl-004	432204	970955	372 872	619 2048 ± 0	217 288 ± 0	131 281 ± 0	95 281 ± 1	89 288 ± 2	68 312 ± 2	918 3327 ± 22	929 3369 ± 48
771	mobbl-005	352570	983522	415 978	359 2048 ± 0	1142 1098 ± 1	862 1125 ± 2	830 1123 ± 2	784 1167 ± 2	811 1354 ± 1	1061 5498 ± 62	1063 5492 ± 53
772	mobipintech-000	370514	303291	507 1173	680 2048 ± 0	1262 1245 ± 1	953 1234 ± 1	944 1264 ± 1	938 1360 ± 1	967 1707 ± 1	1230 14506 ± 214	1232 14433 ± 197
773	momovn-001	246938	1013076	316 769	594 2048 ± 0	165 237 ± 1	120 268 ± 7	113 308 ± 10	91 293 ± 5	116 390 ± 3	549 1438 ± 13	555 1443 ± 13
774	momovn-002	320036	1013104	307 750	532 2048 ± 0	171 243 ± 3	107 248 ± 0	85 245 ± 1	75 253 ± 0	77 328 ± 2	553 1454 ± 8	560 1456 ± 16
775	momovn-003	528962	951056	641 1548	1284 4096 ± 0	821 766 ± 1	583 777 ± 0	536 783 ± 0	512 815 ± 1	582 1045 ± 1	988 3979 ± 23	994 3984 ± 29
776	moredian-000	525259	21374	443 1035	764 2048 ± 0	745 694 ± 0	508 698 ± 0	469 699 ± 0	417 700 ± 0	342 713 ± 1	659 1803 ± 11	652 1779 ± 23
777	morpho-000	100806	107129	-	3 116 ± 0	54 109 ± 1	-	-	-	-	395 993 ± 31	405 1000 ± 34
778	morpho-001	100806	368573	-	2 116 ± 0	12 41 ± 1	-	-	-	-	410 1030 ± 38	424 1042 ± 37
779	morpho-002	383021	110995	-	76 352 ± 0	372 411 ± 7	-	-	-	-	955 3620 ± 94	803 2481 ± 65
780	mukh-001	866223	451194	671 1647	260 1024 ± 0	1334 1375 ± 17	1049 1390 ± 12	1048 1406 ± 8	965 1394 ± 10	813 1360 ± 11	121 433 ± 14	129 435 ± 14
781	mukh-002	693809	454936	468 1081	881 2048 ± 0	1044 968 ± 1	698 921 ± 12	678 957 ± 2	612 954 ± 6	523 953 ± 5	214 612 ± 13	223 611 ± 17
782	mukh-003	556405	180996	730 1853	452 2048 ± 0	1446 1896 ± 3	1165 1948 ± 1	1156 2003 ± 0	1118 1981 ± 0	1011 2016 ± 0	190 597 ± 13	219 604 ± 16
783	mukh-004	405118	181085	579 1353	536 2048 ± 0	533 558 ± 1	371 567 ± 1	334 569 ± 1	293 569 ± 3	251 593 ± 1	283 704 ± 1	296 703 ± 2
784	mukh-005	555239	264759	573 1346	962 2048 ± 0	398 436 ± 3	302 486 ± 6	273 495 ± 7	261 527 ± 16	255 597 ± 25	284 705 ± 1	297 706 ± 2
785	multimodality-000	0	503924	600 1417	627 2048 ± 0	377 416 ± 0	242 420 ± 0	204 423 ± 0	180 427 ± 0	162 463 ± 0	354 848 ± 25	343 800 ± 28
786	multimodality-001	185719	545045	593 1390	1200 4096 ± 0	1217 1190 ± 2	900 1169 ± 2	863 1165 ± 2	785 1167 ± 2	687 1177 ± 2	543 1424 ± 35	540 1384 ± 42
787	mvision-001	227502	149531	291 723	97 512 ± 0	743 691 ± 21	512 702 ± 19	466 697 ± 24	425 708 ± 29	340 710 ± 27	455 1123 ± 40	478 1154 ± 38
788	nazhiai-000	547484	16141	876 2716	756 2048 ± 0	724 683 ± 3	497 687 ± 2	581 835 ± 27	535 840 ± 31	448 834 ± 34	744 2230 ± 34	728 2133 ± 81
789	ncsgroup-000	95110	345861	445 1041	409 2048 ± 0	89 157 ± 0	47 153 ± 0	40 151 ± 0	34 157 ± 0	45 231 ± 0	785 2407 ± 16	786 2411 ± 22
790	ncssg-001	148743	490151	359 847	574 2048 ± 0	1003 936 ± 2	719 946 ± 3	667 944 ± 2	602 938 ± 5	533 972 ± 2	952 3579 ± 50	939 3459 ± 34
791	ncssg-002	148859	415290	406 945	734 2048 ± 0	343 389 ± 1	221 397 ± 0	184 401 ± 1	169 407 ± 0	164 469 ± 0	1413 66390 ± 246	1411 66363 ± 179
792	ncssg-003	148876	414848	311 761	538 2048 ± 0	375 416 ± 1	259 437 ± 2	217 437 ± 1	191 443 ± 2	181 497 ± 1	1312 24709 ± 68	1311 24768 ± 134

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
793	ncssg-004	521184	483011	805 2272	465 2048 ± 0	730 687 ± 1	496 687 ± 0	477 706 ± 5	409 693 ± 1	357 731 ± 21	1313 24815 ± 117	1312 24799 ± 117
794	neosystems-001	575295	349959	-	892 2048 ± 0	1171 1137 ± 4	843 1098 ± 1	1129 1767 ± 4	1082 1769 ± 3	976 1765 ± 4	1268 18557 ± 189	1273 18640 ± 192
795	neosystems-002	599441	349942	-	639 2048 ± 0	1169 1135 ± 2	1156 1855 ± 3	1166 2258 ± 5	1137 2238 ± 3	1040 2247 ± 3	1269 18752 ± 167	1272 18610 ± 213
796	neosystems-003	599442	349942	-	636 2048 ± 0	1176 1143 ± 2	1150 1836 ± 7	1167 2260 ± 3	1141 2273 ± 6	1041 2273 ± 3	1274 19130 ± 223	1277 19167 ± 186
797	neosystems-004	243546	352623	193 529	666 2048 ± 0	263 324 ± 0	525 711 ± 3	569 827 ± 7	543 854 ± 2	497 916 ± 2	1229 14437 ± 176	1230 14355 ± 173
798	netbridgetech-001	133108	205875	182 508	1279 4096 ± 0	35 85 ± 1	20 83 ± 0	15 84 ± 0	15 92 ± 0	15 113 ± 4	1159 9280 ± 74	1161 9446 ± 512
799	netbridgetech-002	257687	49931	94 299	510 2048 ± 0	908 838 ± 6	629 838 ± 2	583 839 ± 1	534 839 ± 3	462 859 ± 3	865 2893 ± 65	893 3050 ± 123
800	neurotechnology-000	62129	33905	-	1427 7148 ± 0	610 611 ± 48	-	-	-	-	1430 74288 ± 2194	1425 72879 ± 2640
801	neurotechnology-001	280771	35716	-	1173 2718 ± 0	953 881 ± 46	-	-	-	-	1420 69356 ± 684	1416 69140 ± 579
802	neurotechnology-002	280771	36036	-	1174 2718 ± 0	561 581 ± 6	-	-	-	-	1416 68738 ± 748	1415 68905 ± 993
803	neurotechnology-003	339962	36125	-	1399 5214 ± 0	560 580 ± 5	-	-	-	-	1426 73589 ± 2548	1424 72837 ± 1546
804	neurotechnology-004	293384	54399	-	26 256 ± 0	357 401 ± 0	-	-	-	-	50 219 ± 8	57 231 ± 13
805	neurotechnology-005	270450	58088	-	39 256 ± 0	353 399 ± 0	-	-	-	-	55 238 ± 10	61 237 ± 7
806	neurotechnology-006	525541	54522	-	143 512 ± 0	715 678 ± 56	-	-	-	-	158 513 ± 14	168 535 ± 26
807	neurotechnology-008	119718	53462	-	33 256 ± 0	284 339 ± 0	-	-	-	-	133 467 ± 19	150 486 ± 26
808	neurotechnology-009	137515	53213	-	37 256 ± 0	342 388 ± 1	-	-	-	-	159 513 ± 10	159 512 ± 16
809	neurotechnology-010	362144	50226	-	163 514 ± 0	852 796 ± 1	595 796 ± 1	558 815 ± 3	541 853 ± 1	560 1006 ± 11	19 120 ± 8	17 115 ± 7
810	neurotechnology-011	364137	51141	-	164 514 ± 0	858 798 ± 1	600 802 ± 1	568 827 ± 3	552 873 ± 2	597 1059 ± 15	16 114 ± 11	16 114 ± 8
811	neurotechnology-012	147830	51395	-	34 256 ± 0	336 384 ± 0	211 387 ± 0	189 404 ± 1	186 435 ± 1	244 583 ± 7	18 119 ± 7	18 116 ± 7
812	neurotechnology-013	474749	85552	1005 4825	161 514 ± 0	1073 1000 ± 1	777 1006 ± 2	739 1022 ± 2	691 1053 ± 2	691 1195 ± 8	14 109 ± 4	13 110 ± 4
813	neurotechnology-015	474782	86045	1002 4819	165 515 ± 0	1101 1028 ± 3	798 1033 ± 3	772 1055 ± 4	720 1097 ± 4	774 1304 ± 18	21 130 ± 2	21 130 ± 4
814	neurotechnology-016	661450	89972	977 4074	167 515 ± 0	1316 1339 ± 3	1033 1351 ± 2	1019 1369 ± 4	962 1390 ± 4	910 1508 ± 21	30 179 ± 20	29 171 ± 20
815	neurotechnology-017	371650	98242	816 2385	32 256 ± 0	480 508 ± 2	324 513 ± 0	296 519 ± 1	264 533 ± 1	275 622 ± 15	4 80 ± 1	4 79 ± 1
816	neurotechnology-018	698669	98271	986 4367	166 515 ± 0	782 725 ± 1	555 747 ± 1	513 747 ± 4	478 769 ± 2	444 833 ± 22	6 85 ± 2	7 87 ± 7
817	neurotechnology-019	365649	103237	813 2321	29 256 ± 0	639 625 ± 1	431 625 ± 0	403 633 ± 3	399 684 ± 2	371 742 ± 17	2 71 ± 2	2 72 ± 4
818	neurotechnology-020	807529	103241	1010 5010	157 513 ± 0	1224 1201 ± 4	922 1196 ± 5	888 1195 ± 8	839 1224 ± 7	820 1373 ± 22	11 102 ± 1	8 101 ± 1
819	nextmetalabs-001	361607	4729276	244 627	866 2048 ± 0	981 914 ± 0	706 928 ± 0	656 928 ± 0	608 948 ± 0	501 919 ± 0	1039 4876 ± 185	1031 4774 ± 72
820	nextmetalabs-002	571398	5046464	489 1123	912 2048 ± 0	1271 1259 ± 2	951 1229 ± 1	935 1253 ± 0	865 1257 ± 14	765 1296 ± 1	1032 4786 ± 103	1030 4753 ± 97
821	nextmetalabs-003	1105548	5046426	810 2301	687 2048 ± 0	1237 1218 ± 1	938 1217 ± 2	909 1225 ± 6	820 1207 ± 2	735 1253 ± 1	1037 4874 ± 57	1037 4833 ± 85
822	nextmetalabs-004	1105548	4174065	570 1332	626 2048 ± 0	1453 1940 ± 1	1166 1949 ± 0	1147 1919 ± 0	1113 1925 ± 1	1010 2011 ± 2	1212 12441 ± 88	1216 12539 ± 154
823	nhn-001	336391	817674	-	1217 4096 ± 0	1098 1027 ± 3	793 1029 ± 1	749 1029 ± 1	685 1044 ± 1	622 1090 ± 1	1397 56650 ± 260	1398 56639 ± 210
824	nhn-002	363471	817674	271 679	1272 4096 ± 0	1174 1141 ± 3	874 1138 ± 2	846 1141 ± 2	773 1151 ± 6	698 1203 ± 2	1396 56608 ± 579	1397 56549 ± 606
825	nhn-003	933665	432730	606 1427	1285 4096 ± 0	1248 1229 ± 2	976 1261 ± 1	942 1263 ± 3	886 1279 ± 2	822 1375 ± 3	1387 50560 ± 105	1388 50592 ± 142
826	nhn-004	931648	432753	895 2864	1291 4096 ± 0	613 612 ± 0	442 632 ± 0	409 636 ± 0	392 676 ± 6	427 820 ± 0	1259 17323 ± 99	1262 17334 ± 133
827	nhn-005	1278729	432776	898 2890	1210 4096 ± 0	873 811 ± 0	475 671 ± 13	422 652 ± 0	438 726 ± 12	483 890 ± 3	1263 18123 ± 400	1268 18214 ± 250
828	noblis-000	713573	253204	-	284 1061 ± 0	101 174 ± 20	-	-	-	-	519 1343 ± 45	534 1373 ± 56
829	nodeflux-000	347079	797842	-	519 2048 ± 0	517 542 ± 1	-	-	-	-	916 3283 ± 51	922 3285 ± 56
830	nodeflux-001	262553	797842	-	788 2048 ± 0	177 247 ± 1	-	-	-	-	909 3242 ± 81	921 3255 ± 93
831	nodeflux-002	774668	690213	157 466	402 2048 ± 0	763 708 ± 4	524 709 ± 4	485 716 ± 5	434 716 ± 7	361 736 ± 3	934 3475 ± 62	934 3408 ± 143
832	nominder-000	547519	102693	699 1741	41 258 ± 0	757 701 ± 0	510 701 ± 2	488 721 ± 2	437 720 ± 3	416 796 ± 1	83 314 ± 5	53 226 ± 45
833	nominder-001	332018	102774	506 1172	48 258 ± 0	118 186 ± 0	71 189 ± 1	55 188 ± 0	52 202 ± 1	61 271 ± 1	53 226 ± 1	30 172 ± 41
834	nominder-002	249049	116931	729 1852	44 258 ± 0	362 405 ± 1	244 425 ± 2	207 424 ± 1	185 434 ± 0	190 505 ± 0	10 101 ± 11	10 102 ± 14
835	nominder-003	249964	116931	821 2417	45 258 ± 0	707 675 ± 2	493 683 ± 5	445 677 ± 2	418 701 ± 3	402 775 ± 3	7 87 ± 4	5 84 ± 4
836	nominder-004	249964	114221	820 2416	43 258 ± 0	810 755 ± 2	560 751 ± 4	509 740 ± 3	476 768 ± 3	454 846 ± 1	5 81 ± 1	6 86 ± 8

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
837	nominder-005	289014	114221	871 2699	42 258 ± 0	1199 1170 ± 4	921 1196 ± 5	861 1161 ± 6	833 1221 ± 8	803 1343 ± 7	8 94 ± 14	19 117 ± 7
838	nominder-006	304867	114225	865 2660	46 258 ± 0	792 736 ± 2	554 746 ± 3	523 759 ± 3	472 762 ± 2	469 869 ± 5	13 105 ± 5	12 108 ± 10
839	nominder-007	249970	101106	398 932	47 258 ± 0	1134 1085 ± 0	817 1071 ± 0	809 1099 ± 1	735 1117 ± 2	717 1235 ± 5	28 171 ± 5	39 190 ± 7
840	nominder-008	484231	102193	683 1688	162 514 ± 0	1456 1982 ± 1	1167 1953 ± 0	1149 1937 ± 1	1116 1954 ± 3	1018 2074 ± 6	29 172 ± 2	41 191 ± 2
841	notiontag-000	92753	406791	-	196 584 ± 0	523 548 ± 64	356 548 ± 35	1078 1450 ± 99	1165 3771 ± 251	1160 13146 ± 792	1371 44672 ± 269	1372 44593 ± 358
842	notiontag-001	92753	427967	254 642	197 584 ± 0	999 929 ± 35	834 1092 ± 39	1173 3709 ± 81	1171 10233 ± 180	1165 31181 ± 521	1368 43636 ± 286	1369 43724 ± 330
843	notiontag-002	271987	967207	904 2965	1146 2120 ± 0	417 453 ± 2	275 453 ± 3	232 453 ± 3	200 458 ± 2	166 471 ± 3	1284 20278 ± 194	1284 20195 ± 186
844	nsensecorp-001	187421	258593	-	614 2048 ± 0	281 336 ± 0	166 335 ± 0	131 336 ± 0	117 335 ± 0	81 337 ± 0	1376 46605 ± 93	1377 46613 ± 112
845	nsensecorp-002	187421	122407	-	952 2048 ± 0	274 333 ± 0	165 333 ± 0	133 337 ± 0	119 338 ± 0	92 351 ± 0	1374 45965 ± 213	1374 45988 ± 158
846	nsensecorp-003	199895	117041	283 710	955 2048 ± 0	683 661 ± 0	469 664 ± 0	433 662 ± 1	374 659 ± 1	300 659 ± 0	1370 44658 ± 51	1373 44654 ± 72
847	nsensecorp-004	513276	139178	679 1675	355 2048 ± 0	1378 1433 ± 0	1092 1445 ± 7	1079 1450 ± 3	1033 1487 ± 5	-	781 2388 ± 42	781 2385 ± 63
848	nsensecorp-005	411845	138437	536 1240	566 2048 ± 0	1390 1455 ± 0	1098 1453 ± 0	1076 1449 ± 0	1016 1457 ± 2	901 1490 ± 4	726 2131 ± 51	733 2165 ± 66
849	ntechlab-000	191530	22085	-	1176 2906 ± 1	204 278 ± 13	-	-	-	-	1337 30787 ± 142	1335 30846 ± 77
850	ntechlab-001	691296	22848	-	1426 6744 ± 1	576 587 ± 11	-	-	-	-	1414 67692 ± 833	1413 67486 ± 244
851	ntechlab-002	894169	25023	-	1425 6744 ± 1	271 330 ± 11	-	-	-	-	1435 82508 ± 213	1433 82524 ± 220
852	ntechlab-003	1422738	24649	-	1175 2906 ± 1	69 128 ± 3	-	-	-	-	1353 35327 ± 117	1356 35371 ± 123
853	ntechlab-004	3752378	115782	-	1196 3482 ± 1	882 818 ± 11	-	-	-	-	1364 38929 ± 81	1367 38916 ± 111
854	ntechlab-005	1726214	116068	-	333 1940 ± 0	766 712 ± 1	-	-	-	-	363 882 ± 42	368 880 ± 29
855	ntechlab-006	7901590	43009	-	1171 2600 ± 0	801 749 ± 1	-	-	-	-	417 1055 ± 93	360 844 ± 48
856	ntechlab-007	2509686	52474	-	1195 3348 ± 0	849 792 ± 3	-	-	-	-	483 1209 ± 59	508 1267 ± 65
857	ntechlab-008	1138002	52483	-	300 1300 ± 0	531 556 ± 1	-	-	-	-	442 1095 ± 45	430 1049 ± 51
858	ntechlab-009	1748043	43730	-	332 1940 ± 0	1159 1115 ± 2	854 1114 ± 1	825 1119 ± 1	753 1130 ± 2	695 1202 ± 4	216 614 ± 20	205 595 ± 28
859	ntechlab-010	698591	217167	-	294 1280 ± 0	1204 1177 ± 2	908 1180 ± 2	891 1197 ± 2	838 1224 ± 1	793 1326 ± 3	116 405 ± 13	124 416 ± 31
860	ntechlab-011	786933	209458	1038 6867	298 1280 ± 0	1180 1148 ± 2	879 1142 ± 1	858 1159 ± 1	805 1185 ± 1	762 1290 ± 3	31 179 ± 11	32 173 ± 11
861	ntechlab-012	570796	212350	1016 5450	1165 2560 ± 0	1305 1309 ± 1	1022 1323 ± 1	995 1331 ± 1	939 1360 ± 1	882 1460 ± 3	47 211 ± 8	48 211 ± 7
862	ntnu-000	293085	433109	677 1665	1452 16896 ± 0	1382 1440 ± 0	1091 1441 ± 0	1064 1430 ± 0	1010 1452 ± 0	888 1473 ± 1	655 1799 ± 53	624 1690 ± 42
863	ocrstudio-001	9010	57372	13 66	24 240 ± 0	32 83 ± 0	19 83 ± 0	17 88 ± 0	19 104 ± 0	31 190 ± 2	1184 10725 ± 62	1189 11030 ± 130
864	omface-000	45945	844976	50 154	240 1024 ± 0	117 185 ± 1	85 206 ± 2	65 203 ± 1	47 195 ± 1	34 193 ± 1	140 481 ± 42	138 456 ± 20
865	omface-001	146370	1799745	47 145	248 1024 ± 0	129 194 ± 2	95 222 ± 2	69 209 ± 0	58 216 ± 1	47 233 ± 1	1267 18369 ± 19	1270 18366 ± 32
866	omnigarde-000	264057	32882	-	220 1024 ± 0	1015 944 ± 0	669 887 ± 0	620 888 ± 1	572 892 ± 0	492 902 ± 0	826 2671 ± 35	821 2620 ± 29
867	omnigarde-001	200523	32882	158 466	99 512 ± 0	1007 941 ± 0	666 883 ± 1	619 886 ± 1	569 891 ± 1	488 898 ± 0	538 1405 ± 31	538 1379 ± 26
868	omnigarde-002	368860	32882	310 760	254 1024 ± 0	1299 1303 ± 1	963 1246 ± 1	931 1249 ± 1	861 1253 ± 1	741 1261 ± 1	839 2727 ± 34	836 2686 ± 32
869	omnigarde-003	427751	35153	309 756	170 517 ± 0	702 673 ± 0	440 631 ± 1	402 633 ± 1	353 637 ± 1	294 647 ± 1	497 1259 ± 22	501 1244 ± 16
870	omnigarde-004	995019	35158	735 1873	169 517 ± 0	1225 1201 ± 1	975 1261 ± 1	898 1207 ± 1	827 1215 ± 0	714 1229 ± 0	499 1274 ± 22	499 1241 ± 26
871	omnigarde-005	1316067	35158	849 2545	178 525 ± 0	1198 1170 ± 1	894 1159 ± 0	864 1169 ± 7	807 1186 ± 1	692 1199 ± 1	574 1490 ± 40	573 1481 ± 38
872	onfido-000	273478	959781	391 914	488 2048 ± 0	426 459 ± 17	273 451 ± 15	231 451 ± 14	207 462 ± 15	192 505 ± 18	607 1617 ± 50	609 1637 ± 53
873	onfido-001	344394	1021960	884 2756	894 2048 ± 0	143 208 ± 4	87 208 ± 4	72 213 ± 4	63 223 ± 3	55 250 ± 4	362 874 ± 28	362 852 ± 20
874	onfido-002	852775	252949	425 1003	1294 4096 ± 0	1222 1199 ± 5	928 1202 ± 5	894 1203 ± 5	825 1213 ± 5	728 1250 ± 5	823 2625 ± 60	824 2628 ± 84
875	onfido-003	852775	252949	428 1010	1229 4096 ± 0	1228 1203 ± 5	929 1202 ± 5	896 1205 ± 5	826 1214 ± 5	732 1252 ± 5	797 2463 ± 14	829 2669 ± 37
876	oorudigital-000	1027828	75456	232 609	1122 2066 ± 0	690 664 ± 0	468 664 ± 0	460 693 ± 0	466 756 ± 0	607 1066 ± 8	604 1606 ± 13	615 1652 ± 23
877	ooto-001	382081	95138	744 1923	1181 3072 ± 0	1012 944 ± 0	724 953 ± 1	722 997 ± 0	616 957 ± 0	552 999 ± 0	1044 4993 ± 17	1046 5004 ± 50
878	openedge-000	447048	272235	589 1376	1040 2056 ± 0	317 367 ± 0	230 404 ± 1	143 352 ± 1	121 350 ± 0	111 385 ± 0	1238 15122 ± 41	1240 15110 ± 48
879	openface-001	0	40111	27 100	582 2048 ± 0	82 148 ± 1	48 154 ± 0	155 365 ± 3	172 409 ± 9	269 616 ± 31	206 608 ± 14	218 604 ± 13
880	ovisionllc-001	192059	252531	460 1066	231 1024 ± 0	104 175 ± 0	57 175 ± 0	51 183 ± 0	48 196 ± 0	52 241 ± 0	863 2889 ± 38	871 2901 ± 47

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 41: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
881	ovisionllc-002	247778	96584	620 1478	249 1024 ± 0	193 266 ± 0	112 256 ± 0	86 250 ± 0	78 263 ± 1	69 313 ± 0	842 2763 ± 72	838 2727 ± 29
882	oz-001	303723	238311	-	1354 4125 ± 0	1179 1147 ± 3	912 1182 ± 3	952 1273 ± 4	1065 1617 ± 7	1081 2890 ± 19	1448 228011 ± 5455	1447 220746 ± 5572
883	oz-002	716022	170261	-	1120 2065 ± 0	1124 1064 ± 3	901 1171 ± 3	1172 2953 ± 6	1169 7352 ± 13	1164 26658 ± 29	1444 131108 ± 1408	1444 126758 ± 913
884	oz-003	484147	519652	1059 11575	1037 2053 ± 0	1335 1375 ± 12	1047 1388 ± 3	1131 1773 ± 16	1123 2039 ± 6	1093 3209 ± 5	1427 73905 ± 456	1426 73892 ± 444
885	oz-004	373982	1075452	1050 7845	1036 2053 ± 0	901 832 ± 7	651 871 ± 6	625 899 ± 10	711 1078 ± 12	950 1608 ± 10	1406 61654 ± 418	1404 61749 ± 450
886	pa-002	442564	308954	-	1453 18436 ± 0	72 132 ± 7	-	-	-	-	891 3088 ± 32	894 3051 ± 28
887	pa-003	714102	1464254	-	491 2048 ± 0	899 831 ± 31	-	-	-	-	387 962 ± 23	380 914 ± 38
888	palit-000	428754	144958	583 1360	1234 4096 ± 0	549 570 ± 1	380 578 ± 1	341 576 ± 3	301 583 ± 1	268 614 ± 1	743 2227 ± 16	744 2226 ± 16
889	palit-001	173886	145564	218 587	959 2048 ± 0	159 227 ± 0	97 224 ± 1	76 224 ± 1	67 229 ± 3	58 262 ± 2	464 1150 ± 16	464 1135 ± 23
890	pangiam-000	464252	24512	972 3963	392 2048 ± 0	642 627 ± 5	420 618 ± 4	379 615 ± 3	337 620 ± 3	288 639 ± 3	17 118 ± 7	15 113 ± 7
891	pangiam-001	1015455	37259	653 1569	565 2048 ± 0	1049 972 ± 1	756 982 ± 3	700 981 ± 2	641 990 ± 7	587 1052 ± 2	22 136 ± 11	22 139 ± 13
892	pangiam-002	486035	39834	530 1205	696 2048 ± 0	254 316 ± 0	150 316 ± 0	120 317 ± 0	106 319 ± 1	76 326 ± 2	15 111 ± 13	11 103 ± 2
893	papago-001	669274	52817	814 2341	720 2048 ± 0	1279 1272 ± 6	1001 1296 ± 7	974 1295 ± 6	888 1281 ± 3	804 1345 ± 3	1243 15236 ± 169	1244 15184 ± 142
894	papil11-000	1146378	24895	792 2184	408 2048 ± 0	1010 942 ± 3	849 1108 ± 2	895 1204 ± 6	947 1369 ± 10	1012 2033 ± 35	212 612 ± 7	230 613 ± 11
895	papil11-001	1221825	38957	860 2618	531 2048 ± 0	1028 951 ± 2	872 1135 ± 2	905 1220 ± 4	958 1386 ± 10	1043 2292 ± 69	155 506 ± 22	161 519 ± 20
896	papil11-002	1208976	38985	846 2543	659 2048 ± 0	987 916 ± 2	837 1094 ± 2	862 1163 ± 1	901 1307 ± 4	1029 2140 ± 10	172 560 ± 22	188 577 ± 22
897	papil11-003	1263123	39005	864 2657	839 2048 ± 0	1107 1038 ± 1	945 1222 ± 1	966 1293 ± 2	998 1431 ± 3	1044 2298 ± 8	156 508 ± 10	157 501 ± 10
898	papil11-004	560851	18192	483 1104	71 296 ± 0	491 522 ± 0	332 522 ± 0	301 524 ± 0	263 531 ± 0	234 556 ± 2	1300 23204 ± 63	1302 23224 ± 68
899	papsav1923-001	279210	52652	162 474	840 2048 ± 0	640 626 ± 1	435 628 ± 1	396 630 ± 1	364 648 ± 2	372 744 ± 3	301 725 ± 25	315 731 ± 28
900	papsav1923-002	491185	24727	495 1139	997 2052 ± 0	850 792 ± 1	753 978 ± 1	759 1042 ± 1	777 1158 ± 1	957 1641 ± 19	482 1209 ± 29	493 1206 ± 38
901	papsav1923-003	515576	24803	484 1113	493 2048 ± 0	856 797 ± 0	761 987 ± 1	760 1043 ± 1	795 1178 ± 1	983 1809 ± 7	368 903 ± 26	378 905 ± 34
902	paravision-004	556670	145440	-	1221 4096 ± 0	896 829 ± 2	626 834 ± 6	578 832 ± 2	527 833 ± 4	445 833 ± 2	307 737 ± 31	308 718 ± 38
903	paravision-006	542423	202234	-	1235 4096 ± 0	753 700 ± 1	516 704 ± 0	475 705 ± 0	421 704 ± 0	336 705 ± 0	85 321 ± 16	87 315 ± 14
904	paravision-008	542190	204400	594 1399	1255 4096 ± 0	749 699 ± 0	509 700 ± 0	472 701 ± 0	420 702 ± 1	334 702 ± 0	91 337 ± 17	98 330 ± 13
905	paravision-010	688291	205854	786 2150	1323 4100 ± 0	652 634 ± 0	449 635 ± 0	407 635 ± 0	350 635 ± 0	283 635 ± 1	596 1577 ± 35	601 1571 ± 32
906	paravision-011	781138	95589	824 2424	1318 4100 ± 0	920 852 ± 0	652 871 ± 1	598 858 ± 1	542 854 ± 0	472 873 ± 1	605 1608 ± 35	607 1625 ± 32
907	paravision-013	1537125	95602	997 4648	1329 4100 ± 0	1066 991 ± 9	767 993 ± 5	714 992 ± 4	654 1002 ± 5	557 1002 ± 7	567 1481 ± 26	579 1490 ± 25
908	paravision-015	1584003	164658	1003 4820	1311 4100 ± 0	1332 1371 ± 12	1126 1577 ± 1	1098 1500 ± 4	1007 1449 ± 1	917 1521 ± 7	545 1432 ± 11	558 1455 ± 32
909	paravision-017	1584003	164639	1004 4820	1327 4100 ± 0	1295 1295 ± 11	885 1147 ± 4	856 1158 ± 8	782 1166 ± 7	661 1140 ± 12	556 1461 ± 10	562 1459 ± 16
910	paravision-018	1596372	164639	1007 4909	1326 4100 ± 0	1400 1474 ± 11	1106 1472 ± 12	1092 1475 ± 11	1017 1459 ± 11	919 1523 ± 16	589 1532 ± 68	588 1520 ± 40
911	paravision-020	1646513	155644	941 3487	1332 4100 ± 0	1458 2043 ± 0	1168 2043 ± 0	1157 2048 ± 0	1122 2038 ± 0	1020 2100 ± 0	722 2110 ± 10	723 2110 ± 17
912	pensees-001	1619431	408932	743 1922	1438 8200 ± 0	1150 1108 ± 3	1093 1448 ± 17	1069 1439 ± 10	1021 1464 ± 5	930 1546 ± 9	899 3151 ± 34	902 3143 ± 25
913	pentacomp-000	261914	192048	261 658	913 2048 ± 0	416 452 ± 1	268 444 ± 0	238 458 ± 4	199 456 ± 5	160 461 ± 3	438 1085 ± 25	444 1081 ± 38
914	pixelall-002	0	169030	-	1162 2560 ± 0	124 191 ± 1	-	-	-	-	488 1223 ± 56	496 1230 ± 47
915	pixelall-003	0	707030	-	1163 2560 ± 0	751 699 ± 8	-	-	-	-	475 1174 ± 28	465 1139 ± 68
916	pixelall-004	0	550919	-	1161 2560 ± 0	415 451 ± 11	272 449 ± 3	230 451 ± 1	202 459 ± 2	187 502 ± 4	458 1139 ± 65	488 1186 ± 31
917	pixelall-005	0	1001355	-	1395 5120 ± 0	1158 1112 ± 3	855 1115 ± 1	826 1120 ± 1	743 1124 ± 2	662 1143 ± 2	495 1259 ± 29	500 1243 ± 23
918	pixelall-006	0	746305	-	1166 2560 ± 0	1091 1024 ± 3	791 1028 ± 2	752 1033 ± 1	677 1032 ± 1	588 1054 ± 2	312 754 ± 14	311 722 ± 10
919	pixelall-007	0	444912	-	769 2048 ± 0	1096 1026 ± 4	801 1038 ± 2	799 1089 ± 2	716 1087 ± 2	646 1124 ± 2	288 708 ± 14	294 701 ± 19
920	pixelall-008	0	992249	700 1742	1435 8192 ± 0	1398 1471 ± 3	1063 1405 ± 4	1052 1409 ± 4	983 1413 ± 3	859 1426 ± 4	653 1799 ± 50	663 1807 ± 48
921	pixelall-009	0	1009114	695 1732	1436 8192 ± 0	1406 1484 ± 3	1056 1395 ± 3	1042 1400 ± 4	963 1391 ± 3	863 1433 ± 3	667 1848 ± 13	668 1842 ± 19
922	pixelall-010	0	1469903	870 2693	1434 8192 ± 0	1185 1153 ± 3	826 1084 ± 4	812 1103 ± 4	719 1093 ± 3	640 1118 ± 4	681 1906 ± 13	681 1898 ± 29
923	pnpsecure-000	9726	146266	24 93	722 2048 ± 0	9 35 ± 0	8 42 ± 0	34 145 ± 0	171 409 ± 1	936 1565 ± 5	685 1918 ± 11	683 1916 ± 16
924	pnpsecure-001	1015928	59243	768 2060	674 2048 ± 0	705 674 ± 0	511 701 ± 0	617 885 ± 0	1009 1452 ± 1	1095 3286 ± 3	822 2621 ± 19	826 2649 ± 41

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
925	privid-001	0	76008	46 143	1458 30720 ± 0	413 450 ± 0	276 454 ± 0	248 467 ± 1	220 472 ± 1	188 504 ± 1	1257 17041 ± 74	1259 17053 ± 100
926	privid-002	0	219733	439 1031	485 2048 ± 0	1273 1261 ± 3	889 1156 ± 4	854 1155 ± 2	778 1162 ± 3	773 1302 ± 4	270 686 ± 22	278 680 ± 20
927	privid-003	0	188341	456 1062	752 2048 ± 0	1375 1430 ± 2	1089 1437 ± 6	1085 1459 ± 7	1008 1451 ± 7	905 1499 ± 18	300 725 ± 7	309 720 ± 13
928	privid-004	0	992936	565 1304	466 2048 ± 0	1300 1303 ± 1	987 1282 ± 2	989 1324 ± 3	941 1363 ± 0	834 1387 ± 0	294 717 ± 29	292 699 ± 19
929	psl-000	260619	1537815	-	856 2048 ± 0	645 629 ± 11	-	-	-	-	69 280 ± 17	77 280 ± 17
930	psl-001	382035	1461947	-	1065 2056 ± 0	841 785 ± 16	-	-	-	-	75 298 ± 13	81 292 ± 14
931	psl-002	804934	1462048	-	1032 2052 ± 0	962 888 ± 9	-	-	-	-	599 1590 ± 48	463 1133 ± 78
932	psl-003	1159643	1434859	-	1147 2120 ± 0	928 865 ± 3	-	-	-	-	415 1052 ± 14	416 1025 ± 51
933	psl-004	771727	1560145	-	1193 3144 ± 0	909 839 ± 3	-	-	-	-	262 680 ± 34	274 670 ± 22
934	psl-005	774031	1560153	-	1191 3144 ± 0	994 922 ± 37	-	-	-	-	280 703 ± 32	300 708 ± 26
935	psl-006	1211577	524525	-	1192 3144 ± 0	1380 1438 ± 6	1086 1434 ± 4	1065 1434 ± 4	999 1434 ± 5	869 1437 ± 5	72 291 ± 34	73 268 ± 25
936	psl-007	954351	524521	-	1189 3144 ± 0	1365 1408 ± 5	1075 1417 ± 3	1058 1418 ± 3	986 1419 ± 2	855 1422 ± 3	63 265 ± 22	69 258 ± 17
937	psl-008	954351	524525	-	1190 3144 ± 0	1368 1412 ± 4	1073 1415 ± 3	1057 1416 ± 2	985 1418 ± 2	852 1418 ± 2	61 259 ± 22	65 252 ± 22
938	psl-009	411027	411504	-	1369 4168 ± 0	1341 1382 ± 2	1043 1381 ± 1	1029 1383 ± 1	955 1383 ± 2	831 1385 ± 1	84 316 ± 14	80 289 ± 14
939	psl-010	411027	591157	1015 5362	1368 4168 ± 0	1362 1403 ± 9	1054 1393 ± 3	1034 1392 ± 3	966 1395 ± 3	839 1396 ± 3	97 354 ± 53	97 329 ± 29
940	psl-011	814579	606050	1009 4959	1440 8248 ± 0	1312 1324 ± 2	1020 1323 ± 8	991 1326 ± 8	918 1324 ± 8	787 1322 ± 4	619 1680 ± 37	622 1688 ± 40
941	psl-012	1318147	622004	995 4535	1360 4144 ± 0	1156 1111 ± 2	859 1122 ± 2	816 1104 ± 2	739 1119 ± 1	639 1117 ± 1	181 583 ± 17	196 589 ± 17
942	psl-013	1318151	622036	992 4525	1361 4144 ± 0	1164 1119 ± 1	845 1105 ± 1	818 1110 ± 1	727 1104 ± 1	648 1126 ± 1	423 1062 ± 62	451 1098 ± 51
943	psl-014	915965	622044	994 4525	1362 4144 ± 0	1152 1109 ± 1	878 1139 ± 1	814 1104 ± 1	738 1118 ± 1	636 1114 ± 1	456 1132 ± 105	459 1114 ± 37
944	ptakuratsatu-000	0	585434	576 1347	182 538 ± 0	944 875 ± 3	645 863 ± 48	658 928 ± 9	617 958 ± 17	609 1066 ± 26	1082 5900 ± 103	1074 5687 ± 167
945	pxl-001	110116	78231	53 168	98 512 ± 0	47 101 ± 5	30 104 ± 5	58 189 ± 12	170 408 ± 27	886 1470 ± 144	1070 5598 ± 45	1070 5590 ± 68
946	pyramid-000	372608	219883	329 804	1045 2056 ± 0	566 583 ± 2	-	-	-	-	1125 7147 ± 59	1135 7586 ± 425
947	pyramid-001	1040862	143739	778 2098	608 2048 ± 0	867 809 ± 0	603 806 ± 0	557 815 ± 4	528 837 ± 0	478 881 ± 2	1208 12265 ± 48	1210 12276 ± 51
948	qazbs-000	362015	805258	441 1034	613 2048 ± 0	1302 1307 ± 1	961 1243 ± 0	929 1248 ± 9	862 1253 ± 1	747 1270 ± 0	1051 5181 ± 62	1051 5167 ± 93
949	qazsmartvisionai-000	560570	99515	815 2371	113 512 ± 0	746 694 ± 1	523 708 ± 7	486 717 ± 5	429 711 ± 5	352 723 ± 7	332 795 ± 29	330 781 ± 34
950	qazsmartvisionai-001	1417181	102251	905 2987	186 544 ± 0	1147 1103 ± 20	868 1129 ± 3	832 1124 ± 5	767 1145 ± 7	655 1134 ± 16	330 793 ± 29	347 804 ± 32
951	qazsmartvisionai-002	1365926	102252	900 2895	189 544 ± 0	1127 1068 ± 6	863 1125 ± 4	821 1114 ± 6	697 1066 ± 1	633 1113 ± 15	336 800 ± 31	338 798 ± 34
952	qazsmartvisionai-003	1365926	102265	899 2894	188 544 ± 0	1119 1056 ± 5	821 1078 ± 12	791 1079 ± 10	713 1078 ± 10	617 1088 ± 18	347 822 ± 28	346 803 ± 32
953	qazsmartvisionai-004	1564226	102265	926 3281	187 544 ± 0	1284 1280 ± 13	1009 1310 ± 14	1001 1338 ± 10	916 1323 ± 13	637 1115 ± 2	314 758 ± 38	322 758 ± 31
954	qluevision-001	173605	205230	123 376	1433 8192 ± 0	160 229 ± 1	101 230 ± 1	80 231 ± 1	69 233 ± 1	50 239 ± 1	923 3374 ± 38	928 3365 ± 41
955	qnap-000	186731	15598	-	451 2048 ± 0	783 726 ± 9	279 457 ± 1	239 458 ± 0	212 464 ± 1	171 482 ± 2	257 660 ± 25	268 654 ± 29
956	qnap-001	196210	13399	88 287	455 2048 ± 0	621 614 ± 1	415 615 ± 1	392 627 ± 1	341 623 ± 1	282 634 ± 2	249 649 ± 11	263 648 ± 14
957	qnap-002	346963	33284	378 882	416 2048 ± 0	887 821 ± 1	613 824 ± 1	566 824 ± 1	523 826 ± 1	442 832 ± 1	73 293 ± 13	78 287 ± 17
958	qnap-003	245476	61427	331 807	852 2048 ± 0	341 387 ± 0	219 393 ± 0	179 393 ± 0	157 393 ± 1	124 400 ± 2	265 683 ± 20	265 651 ± 17
959	qnap-004	371274	61422	424 1002	758 2048 ± 0	313 365 ± 0	193 365 ± 1	158 367 ± 0	136 368 ± 0	104 375 ± 3	263 681 ± 7	280 682 ± 13
960	qnap-005	977818	61429	893 2828	344 2048 ± 0	1047 969 ± 5	763 988 ± 8	712 992 ± 6	652 1000 ± 7	563 1008 ± 6	234 633 ± 1	249 632 ± 13
961	qnap-006	1244083	61533	932 3351	995 2052 ± 0	1094 1025 ± 5	806 1051 ± 6	758 1041 ± 6	692 1055 ± 8	594 1057 ± 10	272 688 ± 29	272 664 ± 10
962	quantasoft-003	370518	211354	457 1063	790 2048 ± 0	649 632 ± 2	448 634 ± 0	401 632 ± 0	346 631 ± 1	281 634 ± 0	42 201 ± 7	44 203 ± 8
963	rankone-000	0	96896	-	14 144 ± 0	31 82 ± 9	-	-	-	-	1367 39932 ± 468	1156 8722 ± 171
964	rankone-001	0	97092	-	21 208 ± 0	18 56 ± 4	-	-	-	-	1423 72754 ± 658	106 345 ± 29
965	rankone-002	0	98585	-	22 224 ± 0	25 75 ± 1	-	-	-	-	1418 69113 ± 3802	111 369 ± 26
966	rankone-003	0	93554	-	23 228 ± 0	10 37 ± 1	-	-	-	-	86 322 ± 20	91 324 ± 19
967	rankone-004	0	135727	-	13 133 ± 0	27 78 ± 1	-	-	-	-	93 340 ± 19	82 298 ± 17
968	rankone-005	0	104048	-	12 133 ± 0	24 71 ± 2	-	-	-	-	115 403 ± 10	101 337 ± 19

Notes	
1	The configuration size does not capture static data included in libraries.
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4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 43: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480X720	960X1440	1600X2400	3000X4500	GENUINE	IMPOSTOR
969	rankone-006	0	133045	-	[18]165 ± 0	[146]210 ± 1	-	-	-	-	[125]443 ± 26	[120]395 ± 22
970	rankone-007	0	136942	-	[17]165 ± 0	[175]245 ± 5	-	-	-	-	[273]688 ± 20	[213]601 ± 16
971	rankone-008	2	141234	-	[16]165 ± 0	[197]272 ± 3	-	-	-	-	[310]750 ± 20	[229]613 ± 23
972	rankone-009	0	107688	-	[50]260 ± 0	[108]179 ± 4	-	-	-	-	[291]710 ± 32	[175]552 ± 25
973	rankone-010	0	138435	-	[56]261 ± 0	[127]193 ± 1	-	-	-	-	[70]282 ± 13	[59]234 ± 16
974	rankone-011	0	179209	-	[55]261 ± 0	[543]567 ± 1	[362]557 ± 1	[333]567 ± 1	[305]586 ± 1	[317]682 ± 3	[71]283 ± 14	[50]220 ± 19
975	rankone-012	0	264182	-	[53]261 ± 0	[540]564 ± 3	[359]554 ± 1	[330]564 ± 1	[306]586 ± 1	[330]695 ± 1	[67]273 ± 17	[56]231 ± 14
976	rankone-013	0	228729	[49]150	[51]261 ± 0	[739]690 ± 5	[478]672 ± 1	[481]712 ± 1	[487]780 ± 1	[641]1118 ± 3	[100]356 ± 23	[85]304 ± 23
977	rankone-014	0	243130	[52]164	[52]261 ± 0	[756]701 ± 1	[517]705 ± 0	[499]732 ± 1	[499]800 ± 1	[634]1113 ± 1	[81]306 ± 16	[64]251 ± 13
978	rankone-015	0	289078	[68]190	[54]261 ± 0	[522]547 ± 5	[350]543 ± 6	[343]578 ± 2	[345]629 ± 7	[486]894 ± 9	[43]201 ± 19	[31]172 ± 16
979	realnetworks-000	99044	107893	-	[1315]4100 ± 0	[80]144 ± 2	-	-	-	-	[795]2458 ± 26	[802]2467 ± 35
980	realnetworks-001	99044	107893	-	[1334]4100 ± 0	[78]144 ± 2	-	-	-	-	[805]2500 ± 47	[788]2420 ± 34
981	realnetworks-002	95328	107088	-	[330]1848 ± 0	[180]250 ± 2	[105]242 ± 4	[98]282 ± 5	[145]381 ± 10	[400]774 ± 15	[507]1285 ± 17	[502]1247 ± 42
982	realnetworks-003	95334	104498	-	[329]1848 ± 0	[106]177 ± 10	-	-	-	-	[584]1516 ± 29	[590]1522 ± 60
983	realnetworks-004	172335	913988	-	[1057]2056 ± 0	[270]330 ± 4	[164]333 ± 3	[186]402 ± 7	[303]585 ± 15	[840]1402 ± 51	[484]1210 ± 29	[492]1202 ± 17
984	realnetworks-005	172253	56755	-	[1059]2056 ± 0	[148]211 ± 4	[84]205 ± 3	[101]290 ± 6	[252]515 ± 17	[779]1312 ± 78	[485]1213 ± 17	[494]1207 ± 16
985	realnetworks-006	466225	56771	[661]1596	[1049]2056 ± 0	[659]638 ± 4	[437]630 ± 3	[441]672 ± 5	[424]706 ± 5	[401]774 ± 5	[135]469 ± 19	[149]478 ± 25
986	realnetworks-007	570797	101527	[921]3145	[1082]2056 ± 0	[1320]1348 ± 2	[1036]1358 ± 11	[1016]1363 ± 10	[957]1386 ± 9	[915]1517 ± 6	[170]559 ± 31	[170]539 ± 35
987	realnetworks-008	73346	75421	[133]402	[1072]2056 ± 0	[229]296 ± 3	[139]294 ± 3	[145]353 ± 4	[129]361 ± 5	[174]485 ± 5	[164]539 ± 31	[171]543 ± 29
988	realsense-001	363245	116558	[757]1995	[1033]2052 ± 0	[1444]1867 ± 0	[1154]1845 ± 0	[1142]1854 ± 0	[1102]1875 ± 1	[1002]1952 ± 2	[1250]16252 ± 121	[1253]16305 ± 100
989	realsense-002	406105	116558	[728]1852	[560]2048 ± 0	[1452]1939 ± 0	[1160]1923 ± 0	[1150]1939 ± 0	[1114]1940 ± 1	[1014]2042 ± 2	[1251]16369 ± 206	[1254]16347 ± 167
990	rebs-000	1375350	62664	[903]2934	[1206]4096 ± 0	[636]623 ± 0	[445]633 ± 10	[390]626 ± 10	[331]616 ± 1	[270]617 ± 0	[463]1150 ± 1	[474]1150 ± 2
991	rebs-001	2038561	62673	[857]2582	[1313]4100 ± 0	[862]802 ± 1	[605]812 ± 8	[573]829 ± 1	[504]807 ± 2	[425]811 ± 4	[465]1151 ± 8	[469]1145 ± 4
992	recognito-000	549308	247223	[1024]5903	[1168]2564 ± 0	[1136]1086 ± 2	[840]1096 ± 1	[811]1102 ± 2	[764]1142 ± 3	[708]1219 ± 7	[36]193 ± 8	[40]190 ± 16
993	recognito-001	517547	62334	[962]3833	[1169]2564 ± 0	[1129]1072 ± 1	[830]1089 ± 1	[805]1095 ± 4	[733]1112 ± 2	[701]1205 ± 4	[32]183 ± 10	[34]183 ± 7
994	regula-000	262444	29384	[233]610	[447]2048 ± 0	[1215]1187 ± 1	[864]1126 ± 1	[834]1129 ± 0	[755]1132 ± 1	[674]1159 ± 1	[148]491 ± 16	[156]500 ± 22
995	regula-001	256075	25980	[419]987	[925]2048 ± 0	[1286]1284 ± 1	[942]1220 ± 1	[907]1222 ± 1	[841]1226 ± 1	[738]1255 ± 1	[102]361 ± 10	[104]342 ± 25
996	regula-002	355812	42119	[459]1066	[1020]2052 ± 0	[1036]962 ± 0	[691]918 ± 0	[640]917 ± 0	[595]932 ± 0	[555]1000 ± 1	[480]1194 ± 14	[497]1233 ± 53
997	regula-003	495453	49195	[608]1428	[1062]2056 ± 0	[1153]1110 ± 0	[805]1048 ± 0	[770]1055 ± 0	[709]1077 ± 1	[659]1140 ± 1	[1431]77846 ± 428	[1429]77865 ± 437
998	remarkai-000	240152	868276	-	[751]2048 ± 0	[895]829 ± 7	-	-	-	-	[360]873 ± 4	[358]835 ± 35
999	remarkai-001	241857	868314	[294]730	[1002]2052 ± 0	[897]831 ± 6	[635]849 ± 18	[769]1055 ± 25	[811]1198 ± 34	[916]1519 ± 38	[491]1229 ± 20	[349]805 ± 56
1000	remarkai-002	224157	808777	-	[943]2048 ± 0	[1024]950 ± 6	[712]935 ± 4	[676]953 ± 3	[623]964 ± 4	[612]1073 ± 4	[450]1115 ± 25	[435]1068 ± 54
1001	remarkai-003	280516	58559	[968]3909	[1335]4100 ± 0	[1063]986 ± 1	[768]993 ± 1	[711]992 ± 1	[651]999 ± 3	[567]1019 ± 2	[326]787 ± 20	[336]793 ± 22
1002	rendip-000	0	437653	[301]735	[338]2048 ± 0	[429]464 ± 2	[280]458 ± 0	[254]473 ± 0	[228]483 ± 1	[233]556 ± 4	[179]576 ± 13	[186]573 ± 11
1003	revealmedia-005	293933	202465	[314]767	[1328]4100 ± 0	[392]428 ± 0	[246]428 ± 0	[212]430 ± 0	[183]433 ± 0	[147]442 ± 0	[704]2023 ± 38	[704]2009 ± 26
1004	revealmedia-006	293933	200912	[304]745	[1010]2052 ± 0	[333]381 ± 0	[205]381 ± 0	[172]382 ± 0	[148]384 ± 0	[120]394 ± 0	[228]626 ± 35	[212]600 ± 2
1005	roc-016	0	674869	[69]191	[80]392 ± 0	[538]562 ± 1	[349]543 ± 2	[325]553 ± 2	[290]564 ± 2	[276]626 ± 2	[37]194 ± 8	[27]161 ± 26
1006	roc-017	0	565239	[91]291	[81]396 ± 0	[669]651 ± 2	[463]660 ± 1	[443]675 ± 1	[387]669 ± 2	[360]736 ± 2	[39]197 ± 4	[33]178 ± 7
1007	roc-018	0	932665	[155]459	[207]792 ± 0	[1249]1231 ± 3	[949]1227 ± 4	[936]1253 ± 2	[854]1243 ± 4	[777]1308 ± 3	[49]217 ± 10	[88]316 ± 14
1008	roc-019	0	944816	[116]360	[85]444 ± 0	[1221]1196 ± 4	[919]1191 ± 5	[904]1218 ± 6	[843]1232 ± 4	[748]1272 ± 5	[80]306 ± 7	[70]259 ± 8
1009	roc-020	0	1095173	[121]375	[86]444 ± 0	[1447]1906 ± 0	[1162]1936 ± 0	[1155]1985 ± 0	[1121]2030 ± 1	[1023]2112 ± 1	[204]602 ± 53	[181]564 ± 22
1010	roc-021	0	1004172	[122]375	[87]448 ± 0	[1432]1738 ± 0	[1142]1781 ± 0	[1141]1853 ± 0	[1098]1863 ± 0	[1004]1960 ± 23	[173]561 ± 11	[172]545 ± 11
1011	rokid-000	258612	396624	[532]1218	[1084]2056 ± 0	[521]546 ± 3	[347]542 ± 2	[320]545 ± 1	[257]522 ± 3	[236]563 ± 4	[933]3457 ± 62	[942]3463 ± 77
1012	rokid-001	641223	413733	[463]1071	[1112]2060 ± 0	[978]911 ± 2	[678]901 ± 5	[626]899 ± 2	[576]900 ± 3	[490]901 ± 3	[920]3345 ± 50	[926]3346 ± 149

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1013	s1-001	435491	844340	-	1140 2092 ± 0	604 605 ± 24	428 623 ± 20	643 920 ± 39	1054 1567 ± 92	-	544 1428 ± 34	543 1415 ± 85
1014	s1-002	520163	95479	-	1408 6144 ± 0	1270 1257 ± 1	974 1260 ± 1	939 1261 ± 1	871 1262 ± 1	750 1273 ± 1	1018 4513 ± 25	1014 4479 ± 25
1015	s1-003	145509	95446	-	1219 4096 ± 0	1019 947 ± 0	733 959 ± 0	675 952 ± 0	611 952 ± 1	524 955 ± 1	956 3657 ± 19	961 3652 ± 16
1016	s1-004	246514	202623	-	853 2048 ± 0	877 815 ± 0	608 818 ± 1	562 818 ± 1	516 820 ± 1	433 828 ± 1	912 3245 ± 100	907 3161 ± 88
1017	s1-005	482369	95685	499 1146	564 2048 ± 0	1075 1001 ± 0	775 1002 ± 0	728 1004 ± 0	664 1008 ± 0	573 1029 ± 2	227 626 ± 74	194 589 ± 14
1018	s1-006	482372	95681	498 1146	710 2048 ± 0	1027 951 ± 0	729 956 ± 0	680 957 ± 0	621 962 ± 0	540 983 ± 0	276 696 ± 23	290 696 ± 29
1019	s1-007	482385	59657	497 1144	443 2048 ± 0	1065 988 ± 1	736 965 ± 0	689 966 ± 0	628 972 ± 0	548 991 ± 0	247 648 ± 26	270 658 ± 25
1020	s1-008	786695	102436	853 2573	521 2048 ± 0	861 800 ± 5	607 814 ± 3	561 818 ± 10	507 809 ± 8	451 841 ± 3	250 651 ± 32	262 647 ± 34
1021	s1-009	1457134	102436	991 4435	860 2048 ± 0	1233 1213 ± 9	948 1224 ± 13	912 1230 ± 4	859 1251 ± 3	740 1258 ± 11	233 632 ± 13	237 622 ± 2
1022	saffe-001	85973	62488	55 168	297 1280 ± 0	209 281 ± 1	-	-	-	-	500 1274 ± 19	512 1277 ± 26
1023	saffe-002	260622	28285	366 855	665 2048 ± 0	880 817 ± 11	602 805 ± 15	551 809 ± 19	510 815 ± 29	426 813 ± 23	295 717 ± 7	306 714 ± 29
1024	samsungsds-000	0	307431	-	878 2048 ± 0	252 316 ± 0	161 326 ± 5	125 328 ± 4	113 327 ± 1	88 343 ± 0	1301 23722 ± 295	1303 23874 ± 305
1025	samsungsds-001	1189592	147444	970 3922	1270 4096 ± 0	1172 1140 ± 3	882 1145 ± 4	1005 1344 ± 5	943 1366 ± 5	913 1514 ± 7	1389 51559 ± 773	1389 51721 ± 1003
1026	samsungsds-002	1040732	147475	929 3321	1053 2056 ± 0	1162 1118 ± 1	904 1175 ± 12	1022 1372 ± 6	917 1324 ± 2	900 1489 ± 4	1356 35803 ± 266	1359 36181 ± 674
1027	samtech-000	109774	12807	-	1079 2056 ± 0	188 262 ± 2	-	-	-	-	1022 4550 ± 26	1020 4541 ± 28
1028	samtech-001	288082	219883	228 605	1085 2056 ± 0	224 294 ± 3	-	-	-	-	1137 7694 ± 59	1138 7678 ± 91
1029	samtech-002	192220	474779	286 714	1041 2056 ± 0	151 212 ± 0	89 212 ± 0	73 215 ± 0	64 225 ± 0	63 285 ± 1	1242 15231 ± 50	1246 15215 ± 59
1030	scanalitix-001	270730	152135	221 591	1006 2052 ± 0	1369 1418 ± 1	1078 1421 ± 1	1043 1401 ± 1	1005 1442 ± 1	908 1507 ± 3	1326 27691 ± 59	1329 27723 ± 65
1031	scanovate-001	257083	328532	-	882 2048 ± 0	553 577 ± 24	379 577 ± 21	397 632 ± 27	480 770 ± 28	843 1404 ± 32	1205 12054 ± 699	1225 13795 ± 705
1032	scanovate-002	256986	457227	364 850	505 2048 ± 0	748 696 ± 32	528 713 ± 33	505 738 ± 28	486 779 ± 32	683 1172 ± 53	885 3021 ± 38	899 3120 ± 163
1033	scanovate-003	135585	89469	330 805	425 2048 ± 0	569 585 ± 1	410 613 ± 12	352 591 ± 1	322 610 ± 2	318 684 ± 1	873 2926 ± 22	874 2925 ± 20
1034	sdc-000	256814	481583	320 784	792 2048 ± 0	980 913 ± 14	682 906 ± 9	847 1142 ± 19	1083 1774 ± 45	1117 4719 ± 222	1344 32645 ± 93	1347 32653 ± 112
1035	seamfix-001	429267	86391	101 320	453 2048 ± 0	773 718 ± 8	521 707 ± 25	670 947 ± 18	1042 1521 ± 55	1111 4179 ± 226	1081 5897 ± 118	1081 5867 ± 123
1036	securifai-001	120291	12456	-	1340 4104 ± 0	147 211 ± 1	88 211 ± 1	71 211 ± 1	57 211 ± 1	41 211 ± 1	620 1681 ± 29	627 1701 ± 25
1037	securifai-002	198824	13496	-	274 1032 ± 0	66 123 ± 0	37 123 ± 0	30 123 ± 0	25 123 ± 0	18 123 ± 0	217 615 ± 22	208 597 ± 19
1038	securifai-003	303794	13512	-	1339 4104 ± 0	525 549 ± 7	357 550 ± 7	322 549 ± 7	277 546 ± 6	223 546 ± 6	629 1714 ± 26	630 1713 ± 37
1039	securifai-004	282177	12027	249 636	777 2048 ± 0	931 869 ± 1	648 867 ± 1	602 867 ± 1	550 867 ± 1	466 865 ± 1	625 1711 ± 19	629 1705 ± 29
1040	securifai-005	252532	81777	190 526	923 2048 ± 0	1337 1377 ± 2	1034 1355 ± 1	1011 1353 ± 0	935 1357 ± 0	812 1356 ± 0	672 1873 ± 25	670 1847 ± 35
1041	securifai-006	452474	81856	317 771	1275 4096 ± 0	1139 1090 ± 2	827 1086 ± 3	802 1093 ± 1	726 1104 ± 10	621 1090 ± 2	925 3376 ± 42	933 3399 ± 40
1042	securifai-007	94589	88169	215 583	940 2048 ± 0	470 500 ± 0	322 506 ± 0	278 499 ± 1	240 501 ± 1	191 505 ± 1	571 1484 ± 11	575 1483 ± 13
1043	securifai-008	111759	84658	284 711	841 2048 ± 0	660 641 ± 4	455 647 ± 3	434 663 ± 3	361 645 ± 3	296 649 ± 1	579 1498 ± 11	576 1487 ± 13
1044	securitypitch-000	271112	45153	206 559	1123 2066 ± 0	802 750 ± 0	556 749 ± 0	514 750 ± 0	462 751 ± 0	379 758 ± 0	613 1644 ± 14	619 1669 ± 20
1045	sensetime-001	531783	7203	-	1019 2052 ± 0	780 725 ± 3	-	-	-	-	810 2546 ± 102	776 2371 ± 45
1046	sensetime-002	531783	7203	-	1005 2052 ± 0	857 797 ± 3	-	-	-	-	833 2713 ± 90	764 2301 ± 25
1047	sensetime-003	787853	74950	-	996 2052 ± 0	975 908 ± 4	-	-	-	-	807 2527 ± 65	888 3004 ± 174
1048	sensetime-004	954663	30733	-	273 1028 ± 0	1379 1437 ± 15	-	-	-	-	492 1239 ± 31	484 1171 ± 22
1049	sensetime-005	765353	37673	-	270 1028 ± 0	1328 1361 ± 27	1005 1304 ± 1	986 1319 ± 1	937 1360 ± 1	912 1514 ± 1	489 1223 ± 28	486 1184 ± 29
1050	sensetime-006	765353	37673	8 33	272 1028 ± 0	1325 1352 ± 17	1010 1311 ± 1	987 1323 ± 1	934 1357 ± 1	918 1523 ± 2	476 1179 ± 28	480 1157 ± 29
1051	sensetime-007	765353	37533	1026 5950	271 1028 ± 0	1348 1386 ± 41	1021 1323 ± 2	1007 1347 ± 2	945 1366 ± 2	944 1593 ± 8	555 1460 ± 29	549 1425 ± 26
1052	sensetime-008	1176483	60067	1021 5882	269 1028 ± 0	1404 1479 ± 31	1088 1436 ± 4	1093 1482 ± 4	1045 1525 ± 5	962 1669 ± 2	506 1283 ± 51	498 1240 ± 47
1053	sensetime-010	1175952	62384	1034 6671	267 1028 ± 0	1203 1176 ± 36	881 1144 ± 2	853 1153 ± 9	790 1172 ± 6	715 1229 ± 7	96 351 ± 1	92 324 ± 5
1054	sensetime-011	1176053	63554	1025 5909	283 1060 ± 0	1414 1531 ± 1	1114 1502 ± 0	1108 1547 ± 0	1040 1508 ± 0	943 1586 ± 0	152 501 ± 1	158 503 ± 2
1055	sensetime-012	1252842	63563	1027 6047	287 1064 ± 0	1429 1711 ± 0	1132 1651 ± 0	1120 1656 ± 0	1066 1651 ± 0	973 1753 ± 0	205 603 ± 77	266 652 ± 4
1056	serendipity-000	0	346443	156 462	486 2048 ± 0	967 900 ± 8	677 901 ± 3	649 922 ± 2	599 936 ± 1	543 987 ± 2	203 602 ± 1	221 607 ± 10

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1057	sertis-000	265572	68770	142 427	371 2048 ± 0	809 754 ± 0	566 759 ± 0	525 764 ± 0	470 760 ± 0	387 763 ± 0	576 1497 ± 29	602 1582 ± 38
1058	sertis-001	292591	68831	-	846 2048 ± 0	651 634 ± 1	-	-	-	-	793 2449 ± 23	835 2685 ± 41
1059	sertis-002	460790	68929	262 660	916 2048 ± 0	1212 1181 ± 1	905 1178 ± 0	877 1183 ± 0	808 1187 ± 0	709 1221 ± 0	439 1086 ± 32	440 1076 ± 31
1060	sertis-003	0	451507	752 1965	653 2048 ± 0	408 441 ± 0	258 436 ± 2	243 461 ± 2	210 463 ± 1	168 473 ± 4	376 915 ± 10	383 917 ± 8
1061	seventhsense-000	369850	1561668	-	992 2052 ± 0	1266 1250 ± 3	972 1257 ± 1	941 1261 ± 1	867 1259 ± 1	749 1272 ± 2	656 1800 ± 35	654 1787 ± 32
1062	seventhsense-001	369850	3183365	335 812	1029 2052 ± 0	1268 1255 ± 2	999 1294 ± 15	955 1277 ± 3	882 1275 ± 2	760 1288 ± 3	690 1936 ± 26	694 1943 ± 34
1063	seventhsense-002	452197	1567903	404 944	1027 2052 ± 0	1267 1252 ± 1	983 1271 ± 1	947 1269 ± 1	879 1272 ± 1	763 1290 ± 1	727 2131 ± 45	725 2123 ± 45
1064	seventhsense-003	603091	453847	476 1090	1017 2052 ± 0	1077 1003 ± 2	751 977 ± 0	572 828 ± 1	513 817 ± 1	432 826 ± 2	586 1523 ± 25	591 1530 ± 22
1065	seventhsense-005	4000865	453775	856 2581	1321 4100 ± 0	1163 1119 ± 1	858 1120 ± 1	839 1134 ± 4	754 1131 ± 2	663 1144 ± 1	847 2787 ± 28	849 2777 ± 22
1066	seventhsense-006	1444389	451757	854 2573	1317 4100 ± 0	1192 1162 ± 1	895 1161 ± 2	857 1158 ± 2	775 1156 ± 0	688 1182 ± 2	837 2724 ± 16	839 2728 ± 16
1067	shaman-000	0	120033	178 507	1238 4096 ± 0	675 653 ± 16	-	-	-	-	110 380 ± 25	116 379 ± 31
1068	shaman-001	0	174446	184 511	1268 4096 ± 0	226 294 ± 2	-	-	-	-	238 635 ± 19	132 441 ± 25
1069	shu-001	329513	62157	-	854 2048 ± 0	614 612 ± 5	-	-	-	-	821 2619 ± 19	885 2987 ± 143
1070	shu-002	731250	148309	386 890	1223 4096 ± 0	805 751 ± 2	576 769 ± 4	647 922 ± 4	997 1431 ± 9	1099 3489 ± 47	1462 2930763 ± 47355	1462 2929759 ± 39149
1071	shu-003	428774	146940	183 511	454 2048 ± 0	885 820 ± 6	619 828 ± 3	665 941 ± 9	903 1308 ± 15	1088 3045 ± 44	806 2506 ± 26	810 2512 ± 38
1072	shufti-001	442529	2108442	746 1941	1000 2052 ± 0	729 687 ± 0	503 694 ± 0	565 824 ± 1	830 1216 ± 4	1079 2875 ± 1	897 3133 ± 19	904 3153 ± 54
1073	siat-002	486842	7738	825 2434	1018 2052 ± 0	558 579 ± 0	-	-	-	-	320 769 ± 13	321 750 ± 13
1074	siat-004	940063	6984	-	1312 4100 ± 0	697 670 ± 0	474 671 ± 7	463 693 ± 10	456 742 ± 10	510 935 ± 17	991 4013 ± 45	976 3782 ± 173
1075	siat-005	380936	16935	562 1299	735 2048 ± 0	359 403 ± 0	227 400 ± 0	183 401 ± 0	160 403 ± 1	137 422 ± 7	180 577 ± 13	189 580 ± 17
1076	sightropic-000	355144	205017	360 847	438 2048 ± 0	486 515 ± 0	334 524 ± 0	298 519 ± 0	253 516 ± 0	222 544 ± 0	1176 10207 ± 35	1177 10214 ± 59
1077	sightropic-001	765429	490992	1008 4914	1066 2056 ± 0	1167 1134 ± 0	1018 1321 ± 6	843 1140 ± 1	759 1136 ± 2	676 1162 ± 6	1285 21280 ± 192	1288 21306 ± 176
1078	sightropic-002	611800	467645	750 1954	1035 2052 ± 0	1017 946 ± 0	721 947 ± 0	668 946 ± 0	615 955 ± 0	537 978 ± 0	1438 89034 ± 130	1437 89056 ± 133
1079	sjtu-001	347115	148322	-	897 2048 ± 0	668 650 ± 2	-	-	-	-	1461 2243804 ± 12751	1461 2249915 ± 19380
1080	sjtu-002	446215	148306	-	845 2048 ± 0	903 835 ± 2	-	-	-	-	1460 2211198 ± 26052	1460 2210428 ± 21877
1081	sjtu-003	480795	148243	198 539	422 2048 ± 0	886 821 ± 2	609 820 ± 2	650 923 ± 3	814 1201 ± 3	1051 2373 ± 9	594 1560 ± 20	597 1560 ± 14
1082	sjtu-004	1953267	241108	880 2741	1387 4608 ± 0	1253 1236 ± 2	935 1209 ± 2	969 1294 ± 4	1048 1554 ± 5	1071 2738 ± 8	889 3057 ± 14	896 3070 ± 20
1083	sjtu-005	1953267	241151	878 2734	1389 4608 ± 0	595 601 ± 1	418 617 ± 1	490 724 ± 2	750 1127 ± 3	1078 2855 ± 11	867 2898 ± 4	870 2900 ± 4
1084	sjtu-006	2685387	241394	913 3071	1390 4608 ± 0	671 651 ± 1	481 672 ± 4	537 788 ± 9	630 979 ± 2	613 1074 ± 7	866 2897 ± 4	868 2896 ± 4
1085	sjtu-007	1126310	244296	886 2759	1388 4608 ± 0	983 914 ± 12	730 956 ± 12	875 1181 ± 37	1072 1707 ± 169	1025 2118 ± 191	871 2903 ± 14	872 2902 ± 11
1086	sjtu-008	1451928	245324	909 2997	1386 4608 ± 0	1439 1832 ± 1	1149 1836 ± 2	1137 1835 ± 0	1094 1839 ± 1	993 1877 ± 1	1023 4621 ± 10	1023 4619 ± 25
1087	sjtu-009	894841	245301	908 2997	1385 4608 ± 0	1440 1832 ± 0	1153 1845 ± 1	1140 1846 ± 1	1095 1846 ± 1	991 1874 ± 2	962 3688 ± 17	966 3694 ± 11
1088	sktelecom-000	527132	298496	568 1321	304 1536 ± 0	1155 1110 ± 1	852 1113 ± 1	822 1114 ± 1	740 1120 ± 1	670 1155 ± 1	1321 26583 ± 128	1320 26508 ± 126
1089	smargtech-001	164944	16729	273 683	725 2048 ± 0	1426 1664 ± 6	1133 1659 ± 3	1124 1717 ± 11	1096 1847 ± 31	1052 2375 ± 105	932 3448 ± 10	945 3492 ± 17
1090	smartbiometrik-001	30875	92620	16 72	133 512 ± 0	633 620 ± 7	432 625 ± 7	415 640 ± 4	442 728 ± 6	585 1047 ± 8	281 703 ± 31	301 710 ± 40
1091	smartengines-000	1711	3025	10 50	69 288 ± 0	96 168 ± 7	61 180 ± 1	54 188 ± 3	60 217 ± 3	62 275 ± 1	38 197 ± 5	28 167 ± 11
1092	smartengines-001	7095	4601	9 47	68 288 ± 0	277 333 ± 89	232 408 ± 1	206 423 ± 1	203 460 ± 2	228 553 ± 5	25 153 ± 11	23 143 ± 13
1093	smartvist-000	5959	134084	57 169	123 512 ± 0	20 59 ± 0	14 56 ± 0	8 56 ± 0	8 58 ± 0	12 90 ± 1	548 1435 ± 31	547 1422 ± 48
1094	smartvist-001	349219	25644	388 904	591 2048 ± 0	320 370 ± 0	202 374 ± 0	166 377 ± 1	175 422 ± 0	151 445 ± 0	852 2835 ± 99	858 2836 ± 80
1095	smilart-001	67339	45737	-	115 512 ± 0	161 230 ± 11	-	-	-	-	525 1371 ± 29	531 1356 ± 57
1096	smilart-002	111826	87805	82 263	242 1024 ± 0	105 176 ± 16	-	-	-	-	1270 18784 ± 136	1274 18795 ± 151
1097	smilart-003	67339	91670	70 192	148 512 ± 0	110 180 ± 12	65 181 ± 10	116 313 ± 22	382 665 ± 49	1045 2299 ± 196	532 1395 ± 74	417 1027 ± 66
1098	sodec-000	836592	13142	901 2902	1198 4096 ± 0	1110 1041 ± 2	796 1032 ± 1	754 1035 ± 1	679 1037 ± 2	600 1061 ± 2	650 1794 ± 37	651 1775 ± 23
1099	softtech-000	656559	88676	492 1127	701 2048 ± 0	1112 1051 ± 0	807 1055 ± 0	771 1055 ± 0	690 1053 ± 0	605 1064 ± 0	598 1585 ± 29	604 1600 ± 40
1100	sparktek-000	2050067	115114	1057 9564	1289 4096 ± 0	1425 1656 ± 0	1134 1661 ± 0	1122 1671 ± 0	1067 1657 ± 0	959 1662 ± 0	150 495 ± 11	187 577 ± 32

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 46: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1101	sparktek-001	2095581	116978	1058 10113	1371 4180 ± 0	1438 1806 ± 10	1146 1799 ± 0	1136 1818 ± 0	1090 1820 ± 0	985 1818 ± 0	344 816 ± 13	348 804 ± 10
1102	sparsh-001	0	366290	993 4525	663 2048 ± 0	718 680 ± 2	456 648 ± 1	411 638 ± 0	358 642 ± 0	344 717 ± 0	1167 9600 ± 275	1166 9601 ± 290
1103	sqisoft-001	278968	386291	277 689	1098 2056 ± 0	445 477 ± 5	1032 1348 ± 18	1012 1353 ± 26	924 1340 ± 14	837 1393 ± 28	335 797 ± 22	332 788 ± 22
1104	sqisoft-002	278039	386291	269 675	1047 2056 ± 0	433 466 ± 8	286 466 ± 2	251 468 ± 11	205 461 ± 6	167 472 ± 4	313 758 ± 11	323 760 ± 23
1105	sqisoft-003	362737	607964	334 812	1073 2056 ± 0	658 638 ± 2	483 674 ± 7	487 718 ± 17	381 665 ± 6	348 720 ± 6	353 844 ± 11	361 844 ± 23
1106	staqu-000	879661	624676	458 1064	1246 4096 ± 0	874 813 ± 25	-	-	-	-	879 2979 ± 31	889 3007 ± 75
1107	starhybrid-001	100509	289356	357 845	890 2048 ± 0	303 358 ± 82	182 355 ± 49	169 379 ± 58	163 401 ± 79	117 393 ± 67	433 1075 ± 51	441 1078 ± 53
1108	stcon-000	408095	49619	494 1137	259 1024 ± 0	628 617 ± 1	441 632 ± 4	404 634 ± 1	360 645 ± 2	314 676 ± 6	123 437 ± 10	128 434 ± 11
1109	stcon-001	322871	64463	369 860	235 1024 ± 0	1059 980 ± 1	765 991 ± 1	719 995 ± 5	657 1003 ± 6	580 1042 ± 4	89 336 ± 14	99 334 ± 14
1110	stcon-002	1255490	62667	776 2095	1252 4096 ± 0	564 583 ± 0	374 570 ± 2	338 574 ± 2	299 582 ± 1	277 630 ± 0	459 1143 ± 2	467 1142 ± 4
1111	stcon-003	2119710	62673	964 3862	1324 4100 ± 0	891 823 ± 0	611 821 ± 1	577 831 ± 3	519 823 ± 1	450 838 ± 2	466 1151 ± 5	481 1157 ± 5
1112	stcon-004	2831397	64930	960 3781	988 2052 ± 0	1336 1375 ± 1	1045 1384 ± 4	1009 1348 ± 1	942 1365 ± 10	825 1375 ± 8	185 591 ± 4	197 589 ± 1
1113	stcon-005	3126319	50632	937 3424	1322 4100 ± 0	1359 1401 ± 2	1072 1415 ± 8	1041 1400 ± 10	1000 1435 ± 1	829 1382 ± 1	477 1181 ± 31	483 1162 ± 5
1114	stcon-006	3126319	50636	938 3426	1333 4100 ± 0	1343 1384 ± 1	1080 1426 ± 3	1054 1410 ± 1	987 1419 ± 21	848 1412 ± 7	461 1148 ± 1	470 1148 ± 1
1115	stcon-007	3126319	50636	936 3413	1330 4100 ± 0	1377 1431 ± 1	1081 1427 ± 9	1101 1503 ± 8	979 1406 ± 7	891 1473 ± 1	472 1163 ± 10	487 1185 ± 32
1116	stcon-008	2124613	56551	850 2552	1320 4100 ± 0	1416 1558 ± 0	1122 1564 ± 0	1112 1560 ± 0	1052 1565 ± 0	934 1561 ± 0	577 1497 ± 14	583 1505 ± 11
1117	stcon-009	2186212	56547	859 2599	1348 4108 ± 0	1417 1560 ± 0	1121 1561 ± 0	1114 1568 ± 2	1057 1580 ± 0	933 1555 ± 0	572 1485 ± 2	586 1515 ± 5
1118	stellartecs-001	1431086	33554	759 1997	1199 4096 ± 0	1324 1351 ± 0	1071 1414 ± 0	1053 1409 ± 2	1003 1441 ± 0	948 1605 ± 1	698 1987 ± 8	702 1986 ± 31
1119	stengg-000	361104	34674	292 724	372 2048 ± 0	868 809 ± 1	696 921 ± 2	726 1002 ± 1	496 794 ± 1	1028 2130 ± 5	700 2005 ± 22	705 2011 ± 34
1120	sukshi-000	94035	688738	124 382	1460 32768 ± 0	364 407 ± 11	234 413 ± 8	284 504 ± 8	405 689 ± 11	938 1574 ± 28	1172 9817 ± 50	1171 9787 ± 62
1121	sumsub-000	1120840	252	464 1074	871 2048 ± 0	1455 1947 ± 1	1163 1947 ± 0	1153 1947 ± 1	1115 1950 ± 1	1003 1958 ± 0	1419 69114 ± 201	1417 69212 ± 246
1122	sumsub-001	848331	252	515 1183	351 2048 ± 0	1450 1916 ± 0	1161 1934 ± 0	1148 1928 ± 0	-	1007 1978 ± 0	379 939 ± 5	385 926 ± 13
1123	suprema-000	246761	38507	-	705 2048 ± 0	827 771 ± 2	584 778 ± 1	599 864 ± 2	730 1109 ± 2	1031 2150 ± 4	622 1690 ± 17	623 1688 ± 13
1124	suprema-001	373423	41460	-	861 2048 ± 0	845 788 ± 1	616 826 ± 2	636 914 ± 2	768 1146 ± 7	1060 2443 ± 4	906 3212 ± 16	914 3220 ± 22
1125	suprema-002	373808	41473	696 1733	363 2048 ± 0	844 787 ± 3	625 833 ± 3	651 924 ± 4	803 1185 ± 6	1062 2479 ± 3	915 3255 ± 17	920 3253 ± 14
1126	suprema-003	498231	116054	537 1244	552 2048 ± 0	1386 1448 ± 1	1074 1417 ± 4	1059 1418 ± 3	990 1421 ± 4	878 1451 ± 5	739 2201 ± 10	739 2198 ± 13
1127	suprema-004	1430475	116085	916 3096	1201 4096 ± 0	1403 1478 ± 2	1104 1472 ± 2	1089 1469 ± 1	1026 1476 ± 1	904 1496 ± 1	905 3201 ± 14	911 3202 ± 22
1128	suprema-005	1430177	116089	915 3093	1286 4096 ± 0	851 794 ± 1	647 863 ± 1	550 808 ± 5	563 883 ± 1	437 830 ± 5	987 3950 ± 7	991 3948 ± 4
1129	supremaid-001	258193	23479	199 541	631 2048 ± 0	449 479 ± 1	297 481 ± 0	259 481 ± 0	232 490 ± 0	210 522 ± 0	282 704 ± 19	267 652 ± 19
1130	supremaid-002	256273	23899	106 334	843 2048 ± 0	454 483 ± 0	316 501 ± 0	267 488 ± 0	242 503 ± 0	237 565 ± 0	699 1990 ± 19	686 1923 ± 29
1131	surrey-cvssp-000	158030	70795	376 879	891 2048 ± 0	1173 1141 ± 3	891 1157 ± 3	855 1158 ± 4	781 1163 ± 3	725 1245 ± 3	1164 9557 ± 143	1167 9602 ± 186
1132	surrey-cvssp-001	900280	76392	686 1705	429 2048 ± 0	1241 1221 ± 1	959 1238 ± 2	922 1240 ± 0	853 1243 ± 0	739 1257 ± 0	1273 18970 ± 161	1276 18999 ± 176
1133	surrey-cvssp-002	903493	87776	690 1720	397 2048 ± 0	1234 1213 ± 1	936 1211 ± 1	956 1279 ± 0	889 1282 ± 0	771 1300 ± 0	1174 9906 ± 120	1174 9911 ± 167
1134	surrey-cvssp-003	1013099	109557	798 2208	633 2048 ± 0	846 789 ± 1	633 846 ± 1	595 854 ± 1	847 1234 ± 2	1094 3283 ± 3	1129 7358 ± 130	1127 7336 ± 93
1135	surrey-cvssp-004	1142444	108961	797 2203	679 2048 ± 0	799 747 ± 1	1124 1568 ± 2	1176 4350 ± 12	1173 11245 ± 7	-	1139 7794 ± 109	1139 7738 ± 118
1136	swsam-001	265887	95072	253 642	640 2048 ± 0	1120 1058 ± 34	822 1079 ± 31	800 1091 ± 35	707 1076 ± 9	631 1109 ± 45	870 2902 ± 31	867 2892 ± 32
1137	synesis-000	340264	76158	-	118 512 ± 0	58 117 ± 0	-	-	-	-	323 775 ± 206	184 568 ± 48
1138	synesis-001	340264	78195	-	96 512 ± 0	63 122 ± 0	-	-	-	-	351 828 ± 182	291 698 ± 44
1139	synesis-003	256748	155901	-	1257 4096 ± 0	36 90 ± 0	-	-	-	-	1124 7118 ± 121	1123 7162 ± 100
1140	synesis-004	270628	16252	-	829 2048 ± 0	790 735 ± 15	-	-	-	-	118 424 ± 14	126 430 ± 22
1141	synesis-005	146509	17726	-	374 2048 ± 0	149 211 ± 9	-	-	-	-	196 599 ± 23	190 581 ± 32
1142	synesis-006	731941	21817	619 1472	1344 4104 ± 0	524 549 ± 1	353 546 ± 1	323 552 ± 1	286 558 ± 2	287 639 ± 28	277 697 ± 32	286 688 ± 31
1143	synesis-007	1442961	24145	828 2443	1185 3080 ± 0	1235 1215 ± 5	979 1268 ± 30	977 1306 ± 67	906 1311 ± 58	857 1423 ± 52	266 684 ± 32	284 686 ± 25
1144	synology-000	221021	25809	152 453	366 2048 ± 0	366 407 ± 14	235 415 ± 14	464 694 ± 31	967 1396 ± 58	1113 4568 ± 211	1278 19720 ± 203	1279 19767 ± 379

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	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1145	synology-001	256895	25804	-	945 2048 ± 0	1008 942 ± 21	-	-	-	-	537 1402 ± 29	640 1730 ± 220
1146	synology-002	256713	25943	170 488	555 2048 ± 0	960 886 ± 4	673 892 ± 3	644 920 ± 2	653 1000 ± 5	784 1317 ± 12	559 1466 ± 32	581 1496 ± 45
1147	syscake-000	50022	12609	63 184	5 128 ± 0	173 243 ± 1	106 245 ± 0	84 241 ± 1	80 269 ± 0	94 357 ± 1	33 185 ± 1	36 185 ± 1
1148	szpu-000	283522	15225	866 2660	730 2048 ± 0	121 190 ± 1	77 192 ± 1	89 259 ± 6	188 438 ± 11	700 1204 ± 32	992 4026 ± 57	997 4021 ± 56
1149	sztu-000	338637	15871	561 1298	946 2048 ± 0	500 531 ± 0	339 532 ± 0	306 533 ± 0	268 537 ± 0	226 548 ± 0	183 585 ± 11	199 592 ± 13
1150	sztu-001	338650	15871	563 1299	956 2048 ± 0	505 535 ± 0	343 537 ± 0	312 538 ± 0	269 540 ± 0	227 553 ± 0	198 599 ± 10	209 598 ± 10
1151	t4isb-000	234227	115237	111 345	695 2048 ± 0	1079 1006 ± 5	774 1001 ± 1	732 1006 ± 1	665 1009 ± 1	569 1022 ± 2	953 3586 ± 34	954 3534 ± 34
1152	tahakom-001	198869	724103	213 581	529 2048 ± 0	236 302 ± 2	128 277 ± 3	154 364 ± 7	137 368 ± 4	114 389 ± 4	780 2387 ± 28	779 2381 ± 22
1153	tahakom-002	609724	37452	517 1184	1263 4096 ± 0	1250 1231 ± 1	965 1250 ± 1	918 1234 ± 0	831 1218 ± 0	744 1265 ± 0	581 1513 ± 2	585 1515 ± 2
1154	taiwanmobile-001	170523	1947147	210 570	1167 2560 ± 0	557 579 ± 8	582 775 ± 16	450 683 ± 4	436 718 ± 8	406 781 ± 6	1362 38086 ± 62	1364 38109 ± 66
1155	taiwanmobile-002	170523	1969728	92 295	929 2048 ± 0	88 156 ± 0	50 162 ± 0	96 281 ± 0	281 549 ± 3	1006 1978 ± 21	1394 55746 ± 469	1392 55598 ± 410
1156	tcc-000	54057	1188952	98 318	250 1024 ± 0	324 375 ± 93	191 363 ± 76	652 924 ± 279	1143 2339 ± 806	1146 9265 ± 3411	416 1053 ± 38	434 1064 ± 23
1157	tcc-001	989049	1238700	255 644	380 2048 ± 0	369 408 ± 0	236 417 ± 0	198 415 ± 0	173 413 ± 0	153 450 ± 0	441 1090 ± 5	452 1105 ± 29
1158	tcv-001	171624	816225	117 363	561 2048 ± 0	44 98 ± 0	25 98 ± 0	22 99 ± 0	18 100 ± 0	17 122 ± 0	429 1070 ± 31	425 1045 ± 17
1159	tech5-000	1451185	470097	-	238 1024 ± 0	466 498 ± 4	-	-	-	-	318 763 ± 23	326 766 ± 28
1160	tech5-001	636346	470101	-	1302 4096 ± 0	490 522 ± 5	-	-	-	-	440 1087 ± 34	339 799 ± 44
1161	tech5-002	1150887	111744	-	296 1280 ± 0	834 780 ± 10	-	-	-	-	539 1406 ± 120	428 1048 ± 57
1162	tech5-003	1427464	118901	-	308 1536 ± 0	957 885 ± 20	-	-	-	-	467 1151 ± 37	461 1128 ± 32
1163	tech5-004	2410272	118858	-	74 321 ± 0	937 872 ± 2	856 1117 ± 164	823 1114 ± 182	757 1134 ± 179	554 999 ± 44	191 597 ± 13	200 592 ± 16
1164	tech5-005	1178769	120517	604 1426	146 512 ± 0	1278 1272 ± 109	802 1038 ± 63	764 1046 ± 39	745 1124 ± 38	809 1351 ± 44	816 2573 ± 37	816 2545 ± 32
1165	tech5-007	0	340324	863 2647	94 512 ± 0	1327 1360 ± 0	1038 1366 ± 0	1024 1376 ± 0	951 1373 ± 0	870 1438 ± 6	163 538 ± 19	160 516 ± 22
1166	tech5-008	0	365877	1019 5567	899 2048 ± 0	679 655 ± 3	429 624 ± 11	387 625 ± 11	347 631 ± 11	339 710 ± 12	1131 7371 ± 81	1128 7367 ± 99
1167	tech5-009	0	361983	834 2453	951 2048 ± 0	1102 1029 ± 25	764 988 ± 21	746 1028 ± 23	661 1006 ± 20	632 1111 ± 22	1135 7478 ± 131	1134 7464 ± 120
1168	techainer-001	198738	703687	756 1977	362 2048 ± 0	334 381 ± 9	217 391 ± 5	187 402 ± 5	154 391 ± 1	161 461 ± 5	1186 10726 ± 189	1186 10714 ± 243
1169	techainer-002	286340	1013105	558 1293	1271 4096 ± 0	388 427 ± 2	247 429 ± 4	209 424 ± 1	187 438 ± 11	193 506 ± 6	561 1472 ± 10	564 1462 ± 16
1170	techainer-003	557482	1098186	643 1550	1307 4096 ± 0	817 763 ± 0	571 765 ± 0	539 789 ± 0	502 803 ± 0	571 1025 ± 2	985 3940 ± 17	993 3978 ± 37
1171	techbiometricid-001	172553	886937	141 427	526 2048 ± 0	154 218 ± 8	99 226 ± 6	236 455 ± 19	674 1023 ± 36	1101 3669 ± 176	1403 59682 ± 512	1403 61123 ± 662
1172	techbiometricid-002	601291	465337	147 439	448 2048 ± 0	519 543 ± 5	346 541 ± 4	324 552 ± 5	304 586 ± 5	389 764 ± 20	1252 16395 ± 120	1255 16374 ± 124
1173	techsign-000	0	1101622	754 1969	632 2048 ± 0	314 366 ± 1	225 398 ± 1	867 1172 ± 3	1161 3065 ± 18	1156 10460 ± 65	1029 4758 ± 112	1033 4789 ± 93
1174	techsign-001	0	586983	737 1881	669 2048 ± 0	829 772 ± 35	588 788 ± 23	548 802 ± 42	609 949 ± 10	845 1409 ± 26	187 592 ± 11	201 592 ± 13
1175	telexper-000	266638	74277	243 625	496 2048 ± 0	574 586 ± 1	377 576 ± 0	353 592 ± 1	356 639 ± 0	479 886 ± 0	1068 5559 ± 54	1066 5553 ± 47
1176	tevian-000	455746	16296	-	922 2048 ± 0	170 242 ± 14	-	-	-	-	120 430 ± 38	125 427 ± 23
1177	tevian-001	682319	16344	-	445 2048 ± 0	176 245 ± 13	-	-	-	-	94 350 ± 22	95 327 ± 13
1178	tevian-002	724437	16352	-	966 2049 ± 0	231 298 ± 15	-	-	-	-	106 369 ± 16	119 386 ± 25
1179	tevian-003	791725	16348	-	970 2049 ± 0	360 404 ± 15	-	-	-	-	95 350 ± 11	102 338 ± 25
1180	tevian-004	863474	16544	-	410 2048 ± 0	477 506 ± 30	-	-	-	-	137 474 ± 31	94 326 ± 20
1181	tevian-005	921043	16556	-	464 2048 ± 0	650 633 ± 21	479 672 ± 25	563 818 ± 37	736 1117 ± 64	1049 2364 ± 121	177 568 ± 22	222 607 ± 35
1182	tevian-006	692492	19339	-	463 2048 ± 0	612 611 ± 1	470 666 ± 41	430 661 ± 32	388 672 ± 37	353 723 ± 31	184 591 ± 19	185 573 ± 28
1183	tevian-007	779934	19523	688 1712	275 1032 ± 0	565 583 ± 1	381 579 ± 0	345 580 ± 0	307 588 ± 1	285 636 ± 0	1040 4894 ± 65	1039 4841 ± 83
1184	tevian-008	847177	19519	949 3610	276 1032 ± 0	955 884 ± 2	680 903 ± 1	629 903 ± 1	581 911 ± 1	517 946 ± 1	1035 4828 ± 40	1036 4811 ± 41
1185	tiger-001	171041	102515	-	1445 8632 ± 0	248 312 ± 12	-	-	-	-	1439 98339 ± 34	1439 98327 ± 32
1186	tiger-002	341638	178194	-	1038 2056 ± 0	349 393 ± 20	-	-	-	-	728 2135 ± 29	730 2137 ± 38
1187	tiger-003	426164	560292	-	1095 2056 ± 0	423 458 ± 21	-	-	-	-	705 2031 ± 35	707 2029 ± 38
1188	tiger-004	342866	253683	-	1014 2052 ± 0	859 798 ± 2	-	-	-	-	189 595 ± 5	211 598 ± 7

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 48: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480X720	960X1440	1600X2400	3000X4500	GENUINE	IMPOSTOR
1189	tiger-005	342866	253734	633 1532	1030 2052 ± 0	1141 1097 ± 2	816 1065 ± 2	786 1066 ± 2	698 1067 ± 3	618 1088 ± 3	221 620 ± 19	232 615 ± 16
1190	tiger-006	421186	394688	285 714	1001 2052 ± 0	1351 1392 ± 16	1069 1411 ± 10	1071 1444 ± 10	1046 1531 ± 11	987 1848 ± 10	661 1810 ± 20	662 1801 ± 13
1191	tiger-007	516470	192057	263 661	991 2052 ± 0	420 455 ± 1	285 464 ± 2	245 464 ± 3	219 472 ± 4	218 537 ± 1	411 1033 ± 8	422 1037 ± 19
1192	tinkoff-001	274660	389272	237 616	857 2048 ± 0	1202 1176 ± 3	907 1179 ± 3	872 1178 ± 3	788 1169 ± 2	696 1203 ± 3	1008 4361 ± 74	1007 4364 ± 75
1193	tngd-001	73167	519258	207 560	1025 2052 ± 0	103 175 ± 0	72 190 ± 0	188 404 ± 0	204 460 ± 1	139 432 ± 0	756 2274 ± 29	755 2265 ± 20
1194	tngd-002	134027	35925	353 843	919 2048 ± 0	744 694 ± 0	538 728 ± 0	1008 1348 ± 0	1157 2974 ± 1	1158 10990 ± 7	404 1016 ± 4	412 1019 ± 4
1195	tnitech-000	234032	13976	1017 5471	342 2048 ± 0	374 415 ± 1	245 427 ± 1	202 419 ± 4	179 427 ± 3	141 436 ± 2	789 2438 ± 14	791 2436 ± 19
1196	tongyi-004	947322	138901	-	1150 2167 ± 0	85 153 ± 0	-	-	-	-	1271 18916 ± 120	1275 18909 ± 126
1197	tongyi-005	1140701	138919	780 2121	1139 2089 ± 0	93 165 ± 1	-	-	-	-	1272 18924 ± 65	1282 20158 ± 103
1198	tongyitrans-001	625339	75069	-	1108 2058 ± 0	246 310 ± 20	-	-	-	-	1260 17769 ± 74	1263 17750 ± 63
1199	tongyitrans-002	625336	75081	-	1109 2058 ± 0	298 356 ± 35	-	-	-	-	1331 29816 ± 281	1264 17799 ± 127
1200	tongyitrans-003	1769974	138883	-	1384 4386 ± 0	437 468 ± 0	-	-	-	-	1407 61793 ± 154	1361 37122 ± 155
1201	toppanidgate-000	671181	711850	716 1798	1247 4096 ± 0	985 915 ± 1	690 916 ± 1	639 916 ± 1	584 917 ± 1	498 917 ± 1	1316 25262 ± 84	1314 25264 ± 97
1202	toppansecurity-001	1480690	130296	658 1595	1350 4113 ± 0	1257 1238 ± 3	939 1217 ± 0	964 1292 ± 0	837 1223 ± 0	726 1247 ± 1	892 3098 ± 28	912 3207 ± 38
1203	toshiba-000	3893310	235400	-	328 1812 ± 0	494 528 ± 3	-	-	-	-	750 2255 ± 60	748 2251 ± 53
1204	toshiba-001	3893310	235400	-	1170 2580 ± 0	622 615 ± 3	-	-	-	-	868 2900 ± 80	866 2881 ± 60
1205	toshiba-002	813606	114260	-	319 1560 ± 0	516 541 ± 0	-	-	-	-	944 3521 ± 369	795 2449 ± 124
1206	toshiba-003	984125	114264	-	320 1560 ± 0	513 540 ± 0	-	-	-	-	782 2390 ± 41	784 2407 ± 81
1207	toshiba-004	599297	27880	660 1596	1042 2056 ± 0	1385 1447 ± 3	1097 1453 ± 2	1084 1457 ± 9	1014 1457 ± 3	895 1479 ± 4	405 1020 ± 25	403 998 ± 32
1208	toshiba-005	599298	61113	-	1093 2056 ± 0	1392 1456 ± 4	1099 1454 ± 2	1086 1461 ± 2	1011 1455 ± 2	881 1459 ± 2	606 1613 ± 34	605 1607 ± 28
1209	toshiba-006	599566	44078	657 1588	1044 2056 ± 0	1405 1481 ± 16	1117 1515 ± 7	1102 1506 ± 6	1043 1521 ± 2	931 1546 ± 30	406 1022 ± 17	414 1022 ± 23
1210	toshiba-007	547545	76675	560 1297	1086 2056 ± 0	778 722 ± 1	587 784 ± 0	535 777 ± 4	475 766 ± 2	441 832 ± 15	289 709 ± 22	303 712 ± 26
1211	toshiba-008	707290	76687	714 1791	1087 2056 ± 0	1057 979 ± 1	748 976 ± 0	706 986 ± 3	660 1005 ± 2	591 1056 ± 14	258 674 ± 68	281 683 ± 71
1212	toshiba-009	651747	76688	668 1639	1071 2056 ± 0	992 921 ± 1	711 933 ± 1	642 919 ± 2	598 935 ± 3	551 996 ± 21	296 719 ± 48	293 700 ± 31
1213	toshiba-010	1091350	76688	875 2710	1067 2056 ± 0	1353 1393 ± 2	1131 1638 ± 15	1082 1457 ± 1	994 1426 ± 14	889 1473 ± 19	202 601 ± 25	217 604 ± 23
1214	toshiba-011	1106463	76900	877 2733	1043 2056 ± 0	1339 1379 ± 1	1065 1406 ± 1	1049 1407 ± 1	949 1370 ± 1	877 1448 ± 18	287 708 ± 56	260 641 ± 37
1215	toshiba-012	1151623	85416	803 2253	1096 2056 ± 0	1457 2020 ± 0	1170 2048 ± 0	1158 2049 ± 0	1126 2073 ± 4	1027 2125 ± 5	505 1281 ± 17	522 1315 ± 40
1216	toss-000	993640	94527	979 4182	393 2048 ± 0	1100 1028 ± 0	926 1200 ± 3	928 1246 ± 7	734 1116 ± 2	611 1072 ± 0	372 909 ± 78	381 914 ± 60
1217	touchlessid-000	92561	64467	107 341	417 2048 ± 0	245 309 ± 5	147 305 ± 2	115 312 ± 5	83 277 ± 4	91 349 ± 17	1342 31935 ± 292	1343 31958 ± 243
1218	touchlessid-001	255274	14355	197 538	782 2048 ± 0	287 344 ± 1	175 347 ± 1	197 414 ± 3	314 595 ± 10	971 1732 ± 61	660 1806 ± 35	661 1800 ± 35
1219	touchlessid-002	255586	14284	119 367	381 2048 ± 0	322 371 ± 1	203 375 ± 1	219 438 ± 3	357 640 ± 10	932 1548 ± 57	684 1915 ± 41	674 1871 ± 38
1220	touchlessid-003	271397	14226	185 512	622 2048 ± 0	370 410 ± 0	241 419 ± 0	211 429 ± 0	227 479 ± 1	319 685 ± 6	207 608 ± 17	207 596 ± 1
1221	touchlessid-004	386383	14727	252 642	718 2048 ± 0	923 854 ± 0	709 932 ± 2	926 1244 ± 6	1120 2001 ± 2	1120 5013 ± 7	587 1527 ± 28	582 1501 ± 14
1222	transmitsecurity-001	1786905	2031398	585 1363	1406 6144 ± 0	285 342 ± 13	171 344 ± 15	129 332 ± 22	101 312 ± 15	100 370 ± 19	1171 9692 ± 23	1172 9847 ± 34
1223	trueface-000	255123	796861	-	757 2048 ± 0	316 367 ± 8	-	-	-	-	142 482 ± 13	165 528 ± 20
1224	trueface-001	255123	186754	-	92 500 ± 0	345 390 ± 1	-	-	-	-	260 676 ± 26	176 558 ± 50
1225	trueface-002	253947	123116	169 486	336 2000 ± 0	306 360 ± 0	188 361 ± 0	205 423 ± 0	311 590 ± 1	-	35 192 ± 14	37 186 ± 19
1226	trueface-003	346530	24308	969 3916	941 2048 ± 0	1149 1107 ± 22	488 677 ± 3	501 732 ± 7	578 905 ± 5	-	12 103 ± 11	14 112 ± 29
1227	trueidvng-001	766071	37721	712 1781	1402 6144 ± 0	1053 975 ± 1	758 985 ± 1	710 989 ± 1	670 1016 ± 1	652 1128 ± 2	1358 37129 ± 216	1423 72067 ± 305
1228	truststamp-001	1795742	275073	631 1524	205 768 ± 0	128 194 ± 0	76 191 ± 0	53 187 ± 0	56 210 ± 1	82 338 ± 8	1150 8415 ± 66	1152 8392 ± 71
1229	truststamp-002	1060439	10841	769 2066	623 2048 ± 0	233 300 ± 0	163 330 ± 1	124 325 ± 0	118 335 ± 2	154 452 ± 0	1192 11281 ± 179	1194 11273 ± 203
1230	truststamp-003	1206119	10474	736 1875	620 2048 ± 0	410 445 ± 0	269 445 ± 0	228 449 ± 0	208 462 ± 1	220 539 ± 2	1235 14901 ± 619	1237 14813 ± 472
1231	tsight-000	1218024	107998	855 2579	221 1024 ± 0	1443 1861 ± 0	1155 1855 ± 1	1144 1874 ± 1	1100 1864 ± 1	994 1877 ± 1	777 2373 ± 4	792 2438 ± 13
1232	ttigroup-000	466606	454674	812 2320	345 2048 ± 0	976 909 ± 2	703 925 ± 8	623 897 ± 1	579 909 ± 3	503 923 ± 2	778 2382 ± 17	782 2387 ± 23

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5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 49: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1233	tuputech-000	11476	17185	[7]33	[704]2048 ± 0	[62]122 ± 4	[35]120 ± 1	[33]142 ± 2	[49]196 ± 5	[130]411 ± 14	[1303]23893 ± 406	[1315]25279 ± 406
1234	turingtechvip-001	399874	54535	[236]615	[616]2048 ± 0	[1345]1384 ± 4	[1051]1391 ± 1	[1035]1393 ± 1	[982]1411 ± 1	[893]1476 ± 2	[635]1733 ± 19	[642]1734 ± 20
1235	turingtechvip-002	167556	140995	[374]877	[698]2048 ± 0	[1410]1493 ± 2	[1008]1306 ± 1	[1028]1382 ± 1	[922]1337 ± 1	[860]1426 ± 3	[1223]13819 ± 103	[1226]13807 ± 137
1236	turkcell-000	271083	133553	[250]636	[523]2048 ± 0	[1154]1110 ± 1	[838]1094 ± 0	[813]1103 ± 0	[749]1126 ± 1	[693]1201 ± 1	[861]2866 ± 23	[865]2873 ± 40
1237	turkcell-001	287616	133870	[260]657	[787]2048 ± 0	[722]682 ± 0	[504]695 ± 0	[457]688 ± 1	[408]692 ± 0	[335]702 ± 0	[1006]4342 ± 71	[1004]4310 ± 72
1238	turkcell-002	542178	139150	[247]633	[403]2048 ± 0	[848]791 ± 1	[364]559 ± 1	[541]793 ± 3	[495]793 ± 2	[241]574 ± 0	[913]3249 ± 65	[916]3231 ± 53
1239	twface-000	661735	11782	[891]2796	[814]2048 ± 0	[934]871 ± 1	[657]873 ± 1	[608]873 ± 2	[555]876 ± 2	[487]898 ± 1	[580]1504 ± 29	[584]1510 ± 34
1240	twface-001	671511	11782	[894]2851	[411]2048 ± 0	[997]923 ± 1	[702]925 ± 2	[655]926 ± 1	[592]929 ± 2	[513]940 ± 2	[534]1400 ± 32	[541]1402 ± 37
1241	ulsee-001	370519	57261	-	[677]2048 ± 0	[676]654 ± 2	-	-	-	-	[1089]6065 ± 94	[1094]6228 ± 77
1242	ultinous-000	90803	503529	-	[1331]4100 ± 0	[17]53 ± 0	-	-	-	-	[1004]4263 ± 50	[1003]4262 ± 42
1243	ultinous-001	403572	501651	-	[1449]12304 ± 0	[138]200 ± 0	-	-	-	-	[1357]36357 ± 223	[1360]36332 ± 210
1244	uluface-002	0	480761	[474]1088	[862]2048 ± 0	[941]873 ± 42	[639]855 ± 9	[697]978 ± 24	[876]1271 ± 40	[1047]2333 ± 68	[1275]19207 ± 1114	[1271]18501 ± 274
1245	uluface-003	97357	529422	[544]1264	[1180]3072 ± 0	[1040]965 ± 11	[738]968 ± 10	[796]1087 ± 20	[960]1387 ± 36	[1061]2469 ± 86	[1317]26057 ± 195	[1323]26865 ± 566
1246	unicc-001	24840	5123261	[727]1849	[367]2048 ± 0	[256]316 ± 4	[177]349 ± 4	[582]838 ± 14	[724]1102 ± 11	[781]1312 ± 15	[877]2950 ± 29	[875]2929 ± 26
1247	unicc-002	277015	1705310	[678]1673	[836]2048 ± 0	[607]609 ± 1	[422]620 ± 1	[371]612 ± 0	[376]660 ± 2	[386]763 ± 3	[759]2289 ± 13	[760]2295 ± 22
1248	unicc-003	361274	1737967	[718]1806	[490]2048 ± 0	[1317]1343 ± 3	[1023]1328 ± 1	[1006]1345 ± 1	[944]1366 ± 1	[898]1484 ± 6	[771]2339 ± 74	[771]2327 ± 29
1249	unicc-004	361274	1737981	[715]1793	[563]2048 ± 0	[1321]1348 ± 2	[1029]1334 ± 1	[1013]1355 ± 1	[969]1397 ± 1	[951]1613 ± 4	[765]2306 ± 13	[766]2304 ± 16
1250	unissey-001	0	1956593	[399]934	[1231]4096 ± 0	[952]880 ± 3	[672]892 ± 3	[1080]1452 ± 8	[1160]3048 ± 12	[1151]10017 ± 387	[557]1463 ± 35	[567]1471 ± 34
1251	unissey-002	0	1443765	[313]765	[1253]4096 ± 0	[793]736 ± 1	[562]752 ± 1	[717]994 ± 1	[993]1426 ± 1	[1097]3331 ± 2	[1209]12308 ± 91	[1211]12302 ± 137
1252	unissey-003	0	814526	[256]650	[1293]4096 ± 0	[772]718 ± 1	[552]744 ± 0	[677]956 ± 1	[977]1403 ± 1	[1089]3055 ± 2	[601]1594 ± 20	[600]1570 ± 44
1253	unissey-004	1	1065839	[438]1030	[1250]4096 ± 0	[294]354 ± 2	[194]366 ± 0	[199]416 ± 0	[174]417 ± 0	[127]409 ± 0	[1003]4261 ± 124	[1021]4557 ± 480
1254	unissey-005	0	1125043	[449]1050	[1225]4096 ± 0	[293]352 ± 0	[198]369 ± 0	[185]402 ± 0	[161]397 ± 0	[125]408 ± 0	[1024]4630 ± 97	[1043]4964 ± 161
1255	unissey-006	0	1315567	[521]1188	[1251]4096 ± 0	[982]914 ± 0	[707]930 ± 0	[761]1043 ± 0	[686]1044 ± 1	[586]1048 ± 0	[1095]6189 ± 111	[1103]6605 ± 186
1256	unissey-007	1	1763338	[508]1173	[1245]4096 ± 0	[1245]1224 ± 22	[988]1283 ± 17	[1031]1390 ± 23	[971]1401 ± 23	[846]1410 ± 23	[1309]24488 ± 173	[1310]24524 ± 185
1257	universeai-000	341437	1395338	[100]320	[177]521 ± 0	[264]324 ± 0	[159]325 ± 0	[148]357 ± 0	[184]434 ± 0	[436]829 ± 0	[1225]14062 ± 60	[1228]14142 ± 74
1258	upc-001	0	89914	[466]1077	[282]1052 ± 0	[527]551 ± 15	[515]703 ± 56	[491]724 ± 51	[463]751 ± 49	[465]863 ± 33	[893]3114 ± 44	[908]3165 ± 97
1259	urielsoft-000	256261	67932	[999]4700	[726]2048 ± 0	[133]197 ± 0	[98]226 ± 0	[97]281 ± 2	[90]289 ± 2	[212]528 ± 2	[232]631 ± 10	[243]630 ± 2
1260	useb-000	1002886	493465	[615]1437	[810]2048 ± 0	[619]613 ± 2	[403]605 ± 1	[380]621 ± 2	[368]653 ± 2	[455]848 ± 3	[730]2153 ± 45	[743]2222 ± 115
1261	useb-001	1000960	493816	[610]1429	[817]2048 ± 0	[701]673 ± 4	[378]577 ± 4	[344]579 ± 8	[316]596 ± 9	[394]768 ± 2	[745]2230 ± 65	[762]2299 ± 90
1262	useb-002	1000967	493456	[603]1423	[933]2048 ± 0	[501]532 ± 2	[585]778 ± 2	[424]657 ± 4	[371]657 ± 2	[452]843 ± 3	[770]2336 ± 47	[769]2323 ± 60
1263	useb-003	1001019	493456	[624]1484	[489]2048 ± 0	[482]512 ± 2	[330]521 ± 3	[304]528 ± 12	[282]550 ± 2	[354]723 ± 2	[733]2172 ± 38	[736]2175 ± 75
1264	useb-004	2086479	454237	[888]2765	[507]2048 ± 0	[1331]1369 ± 2	[1082]1428 ± 12	[1063]1428 ± 5	[1049]1557 ± 10	[1000]1945 ± 4	[764]2303 ± 54	[773]2350 ± 94
1265	uxlabs-001	291127	39378	[279]700	[1295]4096 ± 0	[350]395 ± 0	[212]387 ± 0	[176]388 ± 0	[150]390 ± 0	[122]396 ± 0	[670]1863 ± 31	[685]1921 ± 45
1266	uxlabs-003	1051837	37767	[667]1638	[1228]4096 ± 0	[1350]1390 ± 1	[1068]1410 ± 5	[1032]1391 ± 3	[975]1402 ± 6	[866]1435 ± 1	[614]1645 ± 10	[611]1641 ± 37
1267	uzinfocom-000	271124	508385	[381]885	[1131]2078 ± 0	[107]178 ± 0	[63]180 ± 0	[60]191 ± 0	[55]208 ± 12	[75]325 ± 0	[426]1064 ± 41	[433]1055 ± 26
1268	uzinfocom-001	377667	396457	[477]1091	[1130]2078 ± 0	[397]435 ± 0	[253]433 ± 0	[229]450 ± 0	[248]511 ± 0	[248]589 ± 1	[1402]59486 ± 145	[1400]59551 ± 123
1269	vaidio-001	461288	112836	[461]1068	[884]2048 ± 0	[627]616 ± 0	[425]621 ± 0	[381]622 ± 0	[343]623 ± 0	[324]690 ± 0	[1395]56447 ± 44	[1396]56451 ± 63
1270	vcog-002	3229434	118946	[955]3666	[1462]61504 ± 5	[300]357 ± 25	-	-	-	-	[1452]296154 ± 3077	[1452]296436 ± 4183
1271	vcortex-001	105305	55240	[176]502	[212]896 ± 0	[139]201 ± 2	[80]197 ± 0	[70]209 ± 3	[73]244 ± 6	[103]371 ± 19	[68]279 ± 4	[76]277 ± 4
1272	vd-000	213686	12962	-	[268]1028 ± 0	[74]134 ± 3	-	-	-	-	[560]1467 ± 19	[559]1456 ± 16
1273	vd-001	170262	44058	-	[1012]2052 ± 0	[253]316 ± 6	-	-	-	-	[494]1258 ± 38	[472]1148 ± 109
1274	vd-002	254498	34389	[6]31	[168]516 ± 0	[728]684 ± 5	[489]679 ± 4	[444]676 ± 5	[410]693 ± 5	[376]754 ± 5	[78]300 ± 14	[89]319 ± 32
1275	vd-003	254505	44051	[3]29	[1031]2052 ± 0	[741]691 ± 5	[501]690 ± 5	[452]683 ± 4	[407]691 ± 5	[351]722 ± 5	[400]1003 ± 11	[407]1001 ± 7
1276	veertec-001	106668	28071	[268]674	[1114]2060 ± 0	[1434]1751 ± 0	[1140]1751 ± 0	[1126]1749 ± 0	[1077]1747 ± 0	[981]1792 ± 15	[1015]4453 ± 20	[1011]4449 ± 25

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 50: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1277	veridas-001	196540	190018	-	[648] 2048 ± 0	[698] 671 ± 21	-	-	-	-	[1075] 5748 ± 20	[1120] 7111 ± 148
1278	veridas-002	193466	190018	-	[138] 512 ± 0	[696] 669 ± 20	-	-	-	-	[636] 1733 ± 81	[692] 1934 ± 44
1279	veridas-003	293109	141587	-	[947] 2048 ± 0	[963] 890 ± 33	-	-	-	-	[1060] 5484 ± 42	[1126] 7306 ± 410
1280	veridas-004	196585	160684	-	[744] 2048 ± 0	[716] 678 ± 22	[557] 749 ± 27	[1090] 1470 ± 52	[1163] 3228 ± 95	[1153] 10349 ± 273	[1064] 5516 ± 42	[1131] 7425 ± 130
1281	veridas-005	196585	269478	-	[587] 2048 ± 0	[605] 606 ± 12	-	-	-	-	[1399] 56976 ± 200	[1410] 66298 ± 216
1282	veridas-006	355669	896424	-	[739] 2048 ± 0	[951] 880 ± 8	[668] 885 ± 8	[949] 1271 ± 18	[1139] 2242 ± 38	[1133] 6414 ± 156	[1398] 56940 ± 149	[1409] 66077 ± 194
1283	veridas-007	355105	891492	[844] 2522	[546] 2048 ± 0	[938] 872 ± 9	[659] 875 ± 8	[940] 1261 ± 18	[1138] 2238 ± 38	[1131] 6374 ± 147	[253] 655 ± 16	[271] 660 ± 19
1284	veridas-008	1100495	1190915	[1051] 8633	[542] 2048 ± 0	[1014] 944 ± 12	[718] 945 ± 11	[998] 1334 ± 27	[1144] 2382 ± 48	[1135] 6959 ± 172	[297] 723 ± 14	[316] 731 ± 16
1285	veridas-009	1100503	625120	[1055] 9163	[921] 2048 ± 0	[579] 589 ± 6	[370] 565 ± 5	[342] 576 ± 5	[302] 583 ± 5	[308] 666 ± 6	[243] 638 ± 5	[258] 638 ± 16
1286	veridas-010	1103722	623326	[1053] 8991	[903] 2048 ± 0	[770] 714 ± 4	[495] 686 ± 4	[476] 705 ± 6	[404] 687 ± 4	[328] 693 ± 5	[246] 644 ± 2	[261] 644 ± 4
1287	veridium-000	0	47198	[26] 98	[1457] 29399 ± 2045	[28] 79 ± 0	[18] 80 ± 0	[18] 89 ± 0	[14] 90 ± 0	[14] 111 ± 0	[1410] 64880 ± 171	[1408] 64697 ± 247
1288	veridium-001	0	40561	[45] 142	[1158] 2489 ± 0	[13] 44 ± 0	[9] 45 ± 0	[6] 48 ± 0	[6] 50 ± 0	[8] 72 ± 0	[1409] 63417 ± 1061	[1406] 63225 ± 2133
1289	veridium-002	0	52482	[51] 161	[1156] 2486 ± 0	[4] 25 ± 0	[3] 25 ± 0	[3] 29 ± 0	[3] 30 ± 0	[3] 51 ± 0	[1369] 43891 ± 177	[1370] 43918 ± 167
1290	veridium-003	0	25975	[39] 136	[1159] 2490 ± 0	[5] 27 ± 0	[2] 25 ± 0	[2] 28 ± 0	[2] 30 ± 0	[4] 53 ± 0	[1352] 34755 ± 140	[1354] 34828 ± 177
1291	veridium-004	0	30232	[62] 181	[1157] 2486 ± 0	[14] 46 ± 0	[10] 46 ± 0	[7] 51 ± 0	[7] 53 ± 0	[10] 87 ± 0	[1327] 28700 ± 167	[1331] 28723 ± 149
1292	verigram-000	256209	7798	[783] 2137	[732] 2048 ± 0	[864] 807 ± 1	[610] 821 ± 1	[693] 972 ± 2	[936] 1358 ± 3	[1077] 2848 ± 13	[487] 1222 ± 17	[495] 1219 ± 17
1293	verigram-001	282155	11773	[861] 2638	[539] 2048 ± 0	[689] 664 ± 2	[484] 675 ± 2	[579] 833 ± 4	[817] 1202 ± 7	[1070] 2733 ± 32	[617] 1664 ± 60	[612] 1648 ± 56
1294	verigram-002	1588249	20691	[766] 2057	[1305] 4096 ± 0	[1394] 1464 ± 3	[1067] 1406 ± 0	[1095] 1486 ± 1	[996] 1431 ± 0	[942] 1586 ± 5	[1453] 453119 ± 13595	[1453] 452140 ± 9654
1295	verigram-003	1590775	20692	[765] 2052	[1306] 4096 ± 0	[740] 691 ± 4	[487] 677 ± 1	[447] 677 ± 1	[412] 694 ± 1	[374] 746 ± 1	[1363] 38843 ± 1140	[1366] 38665 ± 1071
1296	verigram-004	1521843	23835	[802] 2247	[1439] 8216 ± 0	[1421] 1575 ± 2	[1123] 1565 ± 1	[1111] 1558 ± 1	[1050] 1558 ± 1	[949] 1606 ± 1	[874] 2928 ± 17	[873] 2915 ± 11
1297	verihubs-inteligensia-000	209562	51877	[143] 430	[504] 2048 ± 0	[544] 567 ± 0	[1120] 1558 ± 8	[1113] 1560 ± 8	[1055] 1568 ± 8	[953] 1621 ± 8	[1294] 22351 ± 91	[1298] 22371 ± 81
1298	verihubs-inteligensia-001	216524	51916	[145] 437	[798] 2048 ± 0	[539] 564 ± 0	[366] 562 ± 0	[332] 566 ± 1	[291] 566 ± 0	[258] 600 ± 0	[1286] 21770 ± 84	[1290] 21735 ± 102
1299	verihubs-inteligensia-002	361105	17045	[293] 727	[628] 2048 ± 0	[440] 471 ± 0	[281] 458 ± 0	[240] 459 ± 0	[211] 463 ± 1	[165] 471 ± 0	[1264] 18131 ± 558	[1266] 17851 ± 197
1300	verijelas-000	254540	10322	[698] 1736	[664] 2048 ± 0	[259] 321 ± 0	[158] 325 ± 1	[127] 329 ± 0	[115] 335 ± 5	[97] 360 ± 0	[1178] 10267 ± 143	[1178] 10218 ± 109
1301	via-000	124422	11151	-	[467] 2048 ± 0	[761] 707 ± 8	[549] 740 ± 5	[630] 906 ± 41	[604] 941 ± 40	[578] 1040 ± 5	[389] 966 ± 28	[413] 1021 ± 44
1302	via-001	370255	11151	[685] 1697	[683] 2048 ± 0	[1039] 964 ± 3	[781] 1011 ± 3	[743] 1026 ± 4	[687] 1045 ± 3	[657] 1137 ± 28	[393] 983 ± 31	[401] 989 ± 40
1303	via-004	530941	93634	[584] 1361	[580] 2048 ± 0	[319] 368 ± 0	[223] 397 ± 1	[141] 352 ± 1	[132] 364 ± 2	[178] 490 ± 1	[60] 255 ± 7	[68] 255 ± 8
1304	via-005	510729	93629	[804] 2257	[707] 2048 ± 0	[510] 538 ± 0	[335] 528 ± 0	[302] 525 ± 1	[266] 535 ± 2	[260] 601 ± 1	[51] 223 ± 2	[52] 222 ± 7
1305	via-006	1087232	93594	[976] 4074	[601] 2048 ± 0	[939] 872 ± 1	[676] 898 ± 1	[662] 941 ± 1	[600] 936 ± 2	[549] 993 ± 1	[52] 224 ± 2	[54] 230 ± 11
1306	viante-000	1244225	19718	[634] 1532	[1083] 2056 ± 0	[1116] 1054 ± 0	[812] 1059 ± 1	[789] 1075 ± 1	[703] 1070 ± 4	[615] 1087 ± 1	[1104] 6461 ± 44	[1100] 6424 ± 32
1307	viante-001	1401078	19722	[722] 1812	[1056] 2056 ± 0	[1140] 1094 ± 1	[857] 1117 ± 5	[824] 1119 ± 6	[741] 1120 ± 3	[629] 1103 ± 1	[901] 3158 ± 45	[903] 3143 ± 53
1308	viante-002	1247402	19718	[637] 1538	[1074] 2056 ± 0	[1184] 1152 ± 1	[853] 1113 ± 3	[817] 1109 ± 1	[729] 1107 ± 2	[645] 1121 ± 2	[907] 3213 ± 90	[913] 3207 ± 90
1309	viante-003	1247402	19705	[635] 1534	[1094] 2056 ± 0	[1436] 1768 ± 2	[1144] 1787 ± 0	[1132] 1775 ± 0	[1086] 1786 ± 8	[980] 1791 ± 0	[1027] 4666 ± 111	[1027] 4690 ± 83
1310	viante-004	1247402	19705	[639] 1542	[1070] 2056 ± 0	[1435] 1760 ± 0	[1151] 1840 ± 0	[1135] 1809 ± 0	[1087] 1786 ± 0	-	[1007] 4346 ± 41	[1005] 4357 ± 44
1311	videmo-000	139643	39470	-	[440] 2048 ± 0	[76] 142 ± 5	[46] 150 ± 4	[38] 150 ± 6	[32] 151 ± 4	[23] 155 ± 8	[157] 513 ± 16	[163] 523 ± 38
1312	videmo-001	212051	95063	[97] 309	[957] 2048 ± 0	[136] 199 ± 0	[52] 164 ± 0	[43] 164 ± 0	[35] 164 ± 0	[24] 165 ± 0	[74] 296 ± 17	[79] 288 ± 16
1313	videmo-002	212053	32963	[105] 333	[823] 2048 ± 0	[135] 199 ± 0	[55] 169 ± 0	[45] 169 ± 0	[36] 170 ± 0	[26] 170 ± 0	[46] 209 ± 7	[46] 208 ± 8
1314	videmo-003	173512	43964	[150] 447	[116] 512 ± 0	[73] 133 ± 0	[32] 111 ± 0	[26] 115 ± 0	[23] 114 ± 0	[16] 115 ± 0	[3] 75 ± 10	[3] 75 ± 16
1315	videonetics-001	30875	5963	[11] 61	[140] 512 ± 0	[189] 262 ± 3	[122] 273 ± 1	[221] 439 ± 3	[515] 820 ± 3	[1055] 2393 ± 43	[469] 1153 ± 38	[466] 1142 ± 65
1316	videonetics-002	121981	6289	[30] 115	[987] 2052 ± 0	[212] 282 ± 5	[140] 295 ± 1	[291] 513 ± 4	[675] 1029 ± 3	[1091] 3151 ± 46	[486] 1219 ± 57	[506] 1262 ± 56
1317	viettelai-000	284612	132031	[239] 619	[737] 2048 ± 0	[251] 315 ± 0	[153] 317 ± 0	[121] 321 ± 0	[109] 322 ± 0	[79] 334 ± 1	[1291] 22074 ± 112	[1297] 22173 ± 163
1318	viettelai-001	528186	132058	[522] 1189	[801] 2048 ± 0	[771] 715 ± 1	-	-	-	-	[1296] 22740 ± 167	[1436] 86287 ± 1073
1319	viettelai-002	528186	132049	[480] 1094	[406] 2048 ± 0	[767] 712 ± 1	[540] 729 ± 0	[492] 724 ± 0	[451] 735 ± 0	[364] 738 ± 5	[1319] 26231 ± 154	[1438] 89838 ± 1111
1320	viettelai-003	528138	130904	[479] 1093	[502] 2048 ± 0	[764] 710 ± 1	[533] 720 ± 0	[510] 743 ± 3	[488] 782 ± 0	[704] 1207 ± 1	[637] 1743 ± 11	[639] 1728 ± 7

Notes	
1	The configuration size does not capture static data included in libraries.
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4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 51: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1321	viettelai-004	1506350	203627	917 3115	1011 2052 ± 0	1462 2910 ± 0	1176 2922 ± 1	1171 2915 ± 0	1155 2917 ± 0	1085 2944 ± 0	501 1274 ± 17	517 1290 ± 32
1322	viettelai-005	1865780	365644	914 3089	1022 2052 ± 0	1448 1908 ± 0	1171 2054 ± 2	1160 2053 ± 2	1124 2051 ± 2	1019 2095 ± 4	541 1413 ± 14	548 1425 ± 20
1323	viettelhightech-000	259471	215557	139 421	578 2048 ± 0	427 461 ± 1	283 461 ± 2	242 461 ± 1	216 467 ± 2	179 494 ± 0	197 599 ± 11	198 591 ± 13
1324	viettelhightech-001	461523	147432	475 1088	462 2048 ± 0	1128 1068 ± 1	825 1084 ± 1	849 1145 ± 4	875 1270 ± 4	1013 2041 ± 2	195 598 ± 14	195 589 ± 5
1325	viettelhightech-002	461606	115515	435 1022	545 2048 ± 0	1428 1700 ± 1	1135 1665 ± 0	1125 1723 ± 1	1107 1893 ± 1	1072 2742 ± 1	324 778 ± 4	331 783 ± 8
1326	vigilantsolutions-000	352218	389974	-	1459 31540 ± 0	956 884 ± 23	-	-	-	-	1265 18201 ± 94	1220 13030 ± 83
1327	vigilantsolutions-001	344685	89145	-	309 1544 ± 0	991 921 ± 2	-	-	-	-	244 644 ± 13	264 649 ± 16
1328	vigilantsolutions-002	344137	89162	-	1118 2060 ± 0	913 844 ± 2	-	-	-	-	261 679 ± 10	277 680 ± 7
1329	vigilantsolutions-003	343048	94301	-	318 1548 ± 0	892 824 ± 3	-	-	-	-	237 634 ± 19	257 638 ± 17
1330	vigilantsolutions-004	343048	125719	-	311 1548 ± 0	905 837 ± 6	-	-	-	-	271 686 ± 14	287 688 ± 17
1331	vigilantsolutions-005	343048	125719	-	310 1548 ± 0	906 837 ± 13	-	-	-	-	361 874 ± 22	256 637 ± 16
1332	vigilantsolutions-006	343048	125719	-	316 1548 ± 0	911 841 ± 8	-	-	-	-	380 939 ± 32	302 711 ± 37
1333	vigilantsolutions-007	255600	125715	-	314 1548 ± 0	462 493 ± 6	-	-	-	-	339 803 ± 35	341 800 ± 40
1334	vigilantsolutions-008	441835	47824	-	315 1548 ± 0	666 647 ± 2	-	-	-	-	365 889 ± 23	374 903 ± 44
1335	vigilantsolutions-009	348798	49973	-	313 1548 ± 0	624 615 ± 0	438 631 ± 0	399 632 ± 1	352 636 ± 0	301 659 ± 0	127 452 ± 11	134 451 ± 11
1336	vigilantsolutions-010	348798	49973	350 841	312 1548 ± 0	625 615 ± 0	439 631 ± 0	398 632 ± 0	351 636 ± 0	302 659 ± 0	146 490 ± 13	152 488 ± 11
1337	vigilantsolutions-011	255661	49973	220 591	317 1548 ± 0	358 402 ± 0	240 418 ± 0	200 418 ± 0	177 422 ± 0	150 445 ± 0	92 339 ± 20	109 366 ± 37
1338	vinai-000	402391	866522	442 1034	492 2048 ± 0	1145 1099 ± 1	839 1095 ± 1	803 1093 ± 1	723 1099 ± 1	649 1126 ± 1	880 2996 ± 20	887 2993 ± 26
1339	vinbigdata-001	271405	44746	222 591	394 2048 ± 0	1358 1400 ± 5	1053 1393 ± 2	1033 1391 ± 2	964 1393 ± 1	844 1404 ± 1	520 1351 ± 50	521 1310 ± 38
1340	vinbigdata-002	256322	138864	231 607	673 2048 ± 0	548 569 ± 2	375 572 ± 1	335 571 ± 1	294 572 ± 1	254 596 ± 1	734 2175 ± 44	732 2160 ± 53
1341	vinbigdata-003	259968	183557	109 342	676 2048 ± 0	917 849 ± 6	1011 1312 ± 1	594 853 ± 2	915 1321 ± 2	443 832 ± 2	432 1074 ± 37	447 1086 ± 40
1342	vinbigdata-004	1570839	333050	925 3247	1240 4096 ± 0	1191 1162 ± 3	888 1155 ± 14	845 1141 ± 2	789 1171 ± 5	660 1140 ± 2	511 1299 ± 17	504 1252 ± 28
1343	vinbigdata-005	1106052	333050	807 2294	436 2048 ± 0	1232 1210 ± 1	956 1237 ± 1	932 1251 ± 1	845 1234 ± 1	756 1282 ± 7	59 251 ± 17	60 234 ± 16
1344	vinbigdata-006	1106052	333624	808 2294	543 2048 ± 0	1231 1210 ± 2	932 1205 ± 5	869 1175 ± 2	796 1178 ± 3	694 1202 ± 16	56 246 ± 16	58 232 ± 16
1345	vinbigdata-007	1106071	335755	809 2295	791 2048 ± 0	1223 1201 ± 10	899 1166 ± 3	934 1252 ± 6	864 1256 ± 15	736 1254 ± 20	57 249 ± 10	62 238 ± 16
1346	vion-000	228219	7533	175 498	985 2052 ± 0	276 333 ± 1	-	-	-	-	1365 39839 ± 3561	1322 26830 ± 2241
1347	visage-000	49218	70150	17 72	102 512 ± 0	6 27 ± 0	4 27 ± 0	4 31 ± 0	5 38 ± 0	6 63 ± 0	742 2220 ± 14	741 2218 ± 14
1348	visica-001	1	59680	14 68	266 1024 ± 0	77 144 ± 1	44 143 ± 1	39 151 ± 1	45 188 ± 2	109 383 ± 11	412 1035 ± 41	411 1017 ± 56
1349	visica-002	1	63831	34 126	815 2048 ± 0	67 124 ± 1	38 125 ± 1	41 156 ± 2	81 271 ± 4	484 892 ± 29	1030 4759 ± 31	1024 4630 ± 26
1350	visica-003	1	252112	160 470	540 2048 ± 0	230 297 ± 2	143 297 ± 1	134 341 ± 2	218 471 ± 5	767 1297 ± 32	1083 5939 ± 47	1085 5987 ± 81
1351	visionbox-000	176501	190645	-	450 2048 ± 0	240 304 ± 7	-	-	-	-	615 1648 ± 57	489 1192 ± 42
1352	visionbox-001	256869	190645	-	472 2048 ± 0	1060 983 ± 7	836 1093 ± 46	1015 1360 ± 68	1135 2181 ± 105	1128 5955 ± 281	471 1161 ± 22	476 1154 ± 20
1353	visionbox-002	259063	135281	234 613	1110 2059 ± 0	452 482 ± 1	299 482 ± 0	262 484 ± 1	234 492 ± 1	204 517 ± 3	697 1969 ± 44	690 1931 ± 42
1354	visionbox-003	259542	156891	543 1263	1124 2068 ± 0	714 678 ± 2	491 682 ± 1	449 682 ± 2	414 695 ± 2	366 739 ± 5	612 1643 ± 56	613 1649 ± 66
1355	visionbox-004	309730	154797	602 1422	1351 4116 ± 0	327 378 ± 1	197 369 ± 0	177 389 ± 5	164 402 ± 2	143 437 ± 4	714 2071 ± 31	720 2076 ± 41
1356	visionbox-005	309402	197528	708 1771	1352 4116 ± 0	291 350 ± 1	170 343 ± 1	140 351 ± 2	135 367 ± 5	156 454 ± 19	716 2074 ± 17	718 2067 ± 29
1357	visionbox-006	479723	170096	907 2995	1345 4107 ± 0	1001 932 ± 2	745 973 ± 4	716 994 ± 6	656 1003 ± 11	643 1120 ± 20	707 2038 ± 74	706 2022 ± 31
1358	visionbox-007	479723	364416	835 2456	1346 4107 ± 0	630 619 ± 1	417 616 ± 0	395 629 ± 2	384 666 ± 7	414 794 ± 32	945 3524 ± 29	952 3527 ± 40
1359	visionbox-008	479723	449470	838 2476	1347 4107 ± 0	632 619 ± 1	421 619 ± 1	394 628 ± 2	377 663 ± 7	417 799 ± 36	964 3704 ± 35	964 3692 ± 40
1360	visionlabs-001	343661	127922	-	20 204 ± 0	1011 943 ± 8	-	-	-	-	531 1395 ± 45	473 1148 ± 53
1361	visionlabs-002	591936	46855	-	57 264 ± 0	192 265 ± 8	-	-	-	-	333 796 ± 35	340 799 ± 31
1362	visionlabs-003	639849	15246	-	49 260 ± 0	162 231 ± 5	-	-	-	-	124 439 ± 23	136 454 ± 20
1363	visionlabs-004	313016	90858	-	40 256 ± 0	239 304 ± 0	-	-	-	-	349 827 ± 26	365 865 ± 32
1364	visionlabs-005	369602	16523	-	139 512 ± 0	249 313 ± 0	-	-	-	-	355 848 ± 26	372 889 ± 37

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
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Table 52: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1365	visionlabs-006	353044	16655	-	[150]512 ± 0	[194]270 ± 0	-	-	-	-	[278]698 ± 19	[318]734 ± 28
1366	visionlabs-007	357204	16643	-	[107]512 ± 0	[198]272 ± 0	-	-	-	-	[388]965 ± 41	[396]972 ± 31
1367	visionlabs-008	706099	19705	-	[105]512 ± 0	[435]467 ± 1	-	-	-	-	[384]955 ± 23	[390]962 ± 25
1368	visionlabs-009	706099	19862	-	[156]513 ± 0	[485]515 ± 41	[289]472 ± 1	[255]474 ± 1	[224]476 ± 1	[209]521 ± 1	[386]957 ± 28	[394]965 ± 32
1369	visionlabs-010	1067280	19357	[380]883	[160]513 ± 0	[786]730 ± 0	[532]717 ± 1	[479]709 ± 0	[431]713 ± 1	[365]739 ± 0	[200]600 ± 41	[240]626 ± 35
1370	visionlabs-011	1067280	19353	[396]923	[158]513 ± 0	[788]731 ± 1	[531]717 ± 1	[480]710 ± 1	[432]714 ± 1	[370]741 ± 1	[168]556 ± 26	[177]559 ± 25
1371	visteam-000	32729	17740	-	[305]1536 ± 0	[41]96 ± 7	[29]102 ± 8	[149]358 ± 19	[672]1022 ± 50	[1105]3987 ± 211	[1101]6361 ± 87	[1105]6668 ± 277
1372	visteam-001	186440	30878	-	[1233]4096 ± 0	[932]869 ± 7	[655]872 ± 6	[828]1121 ± 15	[1073]1719 ± 38	[1112]4375 ± 157	[1121]7054 ± 108	[1118]7025 ± 109
1373	visteam-002	186440	30888	-	[1276]4096 ± 0	[894]829 ± 5	[624]832 ± 6	[584]839 ± 7	[540]853 ± 6	[565]1013 ± 14	[1117]6952 ± 118	[1114]6970 ± 120
1374	visteam-003	215359	33730	[172]493	[1297]4096 ± 0	[1265]1249 ± 4	[967]1251 ± 4	[945]1266 ± 5	[878]1272 ± 5	[817]1370 ± 9	[1112]6816 ± 111	[1110]6816 ± 105
1375	visteam-004	61594	35369	[54]168	[805]2048 ± 0	[237]303 ± 5	[148]313 ± 6	[94]278 ± 4	[88]288 ± 4	[106]377 ± 7	[984]3936 ± 72	[990]3938 ± 79
1376	visteam-005	288140	35427	[112]347	[651]2048 ± 0	[1161]1117 ± 6	[846]1106 ± 6	[779]1060 ± 4	[705]1071 ± 4	[671]1156 ± 8	[982]3932 ± 97	[987]3932 ± 71
1377	visteam-006	339353	47921	[173]493	[802]2048 ± 0	[1083]1009 ± 1	[782]1015 ± 1	[755]1035 ± 2	[669]1014 ± 1	[584]1047 ± 2	[832]2711 ± 32	[837]2704 ± 41
1378	visteam-007	331673	264832	[168]485	[515]2048 ± 0	[611]611 ± 1	[433]625 ± 1	[388]626 ± 1	[330]615 ± 1	[293]647 ± 0	[831]2704 ± 38	[841]2732 ± 54
1379	visteam-008	582800	264832	[295]731	[511]2048 ± 0	[1208]1179 ± 1	[903]1173 ± 1	[900]1210 ± 1	[809]1190 ± 3	[705]1208 ± 1	[824]2632 ± 28	[825]2647 ± 42
1380	visteam-009	582793	48090	[297]731	[914]2048 ± 0	[1201]1174 ± 1	[914]1184 ± 1	[903]1218 ± 2	[832]1219 ± 2	[734]1253 ± 2	[825]2668 ± 34	[830]2671 ± 38
1381	visteam-010	582793	48090	[299]731	[939]2048 ± 0	[1200]1174 ± 1	[930]1203 ± 2	[902]1215 ± 1	[834]1221 ± 1	[733]1252 ± 1	[827]2675 ± 51	[834]2681 ± 57
1382	vivotek-001	15240	247926	[36]128	[547]2048 ± 0	[46]100 ± 0	[28]101 ± 0	[24]106 ± 0	[29]134 ± 0	[49]238 ± 1	[1320]26504 ± 57	[1321]26520 ± 69
1383	vivotek-002	12769	247949	[33]124	[681]2048 ± 0	[38]93 ± 0	[23]93 ± 0	[21]98 ± 0	[16]95 ± 0	[22]150 ± 0	[779]2384 ± 19	[780]2385 ± 20
1384	vixvizion-000	183453	599571	-	[746]2048 ± 0	[386]425 ± 24	-	-	-	-	[460]1145 ± 25	[635]1718 ± 63
1385	vixvizion-001	282680	256875	-	[785]2048 ± 0	[863]807 ± 29	-	-	-	-	[414]1045 ± 22	[386]934 ± 45
1386	vixvizion-002	227766	318409	-	[420]2048 ± 0	[843]786 ± 1	[572]766 ± 2	[618]885 ± 3	[995]1430 ± 3	[1106]4080 ± 10	[259]676 ± 16	[244]630 ± 20
1387	vixvizion-003	254860	378019	-	[508]2048 ± 0	[929]865 ± 2	[713]938 ± 1	[1116]1577 ± 1	[1159]3002 ± 3	[1149]9521 ± 16	[308]738 ± 26	[282]683 ± 25
1388	vixvizion-004	254405	380049	-	[693]2048 ± 0	[637]624 ± 1	[389]587 ± 10	[389]626 ± 3	[313]592 ± 3	[345]717 ± 6	[315]760 ± 22	[295]703 ± 28
1389	vixvizion-005	38886	534579	[319]776	[834]2048 ± 0	[395]433 ± 4	[208]381 ± 3	[173]383 ± 3	[139]373 ± 1	[129]411 ± 1	[303]731 ± 63	[247]632 ± 32
1390	vixvizion-006	594053	396294	[582]1359	[688]2048 ± 0	[947]876 ± 9	[618]828 ± 3	[560]817 ± 1	[520]825 ± 2	[471]871 ± 1	[199]600 ± 23	[225]611 ± 25
1391	vixvizion-007	594053	470119	[614]1435	[812]2048 ± 0	[958]885 ± 35	[620]828 ± 1	[559]816 ± 1	[522]825 ± 1	[470]870 ± 1	[201]600 ± 28	[215]602 ± 34
1392	vnis-000	285395	1404886	[202]546	[528]2048 ± 0	[403]437 ± 1	[267]443 ± 1	[237]456 ± 2	[201]458 ± 2	[169]477 ± 2	[528]1386 ± 10	[539]1382 ± 16
1393	vnpay-000	258919	1404844	[177]504	[562]2048 ± 0	[168]239 ± 1	[108]248 ± 0	[87]252 ± 0	[74]249 ± 0	[57]260 ± 0	[521]1353 ± 11	[529]1353 ± 23
1394	vnpay-001	1044099	3060240	[400]937	[822]2048 ± 0	[1188]1156 ± 1	[970]1255 ± 1	[933]1252 ± 2	[870]1261 ± 2	[759]1288 ± 10	[542]1421 ± 32	[546]1421 ± 26
1395	vnpt-001	266499	535529	-	[603]2048 ± 0	[467]499 ± 2	[312]499 ± 2	[269]494 ± 3	[241]502 ± 3	[197]512 ± 2	[708]2049 ± 29	[100]337 ± 121
1396	vnpt-002	271649	3203296	-	[571]2048 ± 0	[794]739 ± 2	[541]731 ± 2	[508]740 ± 1	[457]742 ± 2	[385]763 ± 2	[319]766 ± 13	[324]762 ± 13
1397	vnpt-003	369956	297799	-	[1258]4096 ± 0	[1307]1315 ± 4	[1013]1315 ± 4	[985]1318 ± 2	[931]1350 ± 3	[861]1428 ± 3	[1132]7397 ± 31	[1129]7384 ± 29
1398	vnpt-004	370110	240841	[421]992	[918]2048 ± 0	[1256]1238 ± 1	[960]1241 ± 1	[924]1242 ± 2	[902]1307 ± 2	[907]1505 ± 2	[995]4047 ± 48	[995]4008 ± 108
1399	vnpt-005	560630	240888	[502]1157	[494]2048 ± 0	[1361]1403 ± 0	[1062]1404 ± 6	[1045]1403 ± 6	[1013]1456 ± 0	[954]1630 ± 10	[950]3562 ± 23	[956]3554 ± 29
1400	vnpt-006	1114807	360198	[801]2245	[621]2048 ± 0	[1281]1277 ± 2	[1050]1390 ± 19	[1055]1410 ± 14	[1031]1481 ± 30	[966]1703 ± 69	[926]3400 ± 20	[932]3393 ± 22
1401	vnpt-007	2113252	360190	[951]3622	[340]2048 ± 0	[1177]1143 ± 1	[890]1157 ± 4	[866]1170 ± 2	[799]1183 ± 3	[707]1219 ± 3	[937]3486 ± 41	[943]3485 ± 35
1402	vnpt-008	1114270	360190	[953]3641	[863]2048 ± 0	[1195]1165 ± 1	[915]1186 ± 2	[871]1177 ± 9	[815]1201 ± 2	[724]1244 ± 2	[941]3506 ± 16	[949]3510 ± 23
1403	vnpt-009	1114270	360190	[952]3641	[772]2048 ± 0	[1196]1166 ± 1	[923]1197 ± 6	[885]1193 ± 6	[822]1209 ± 3	[730]1251 ± 4	[947]3539 ± 17	[955]3551 ± 31
1404	vocalize-001	0	38841	[25]96	[100]512 ± 0	[19]57 ± 0	[13]55 ± 0	[9]56 ± 0	[9]60 ± 0	[5]62 ± 0	[533]1400 ± 59	[533]1365 ± 26
1405	vocord-001	616989	109135	-	[1413]6194 ± 0	[973]908 ± 16	-	-	-	-	[1459]1094730 ± 64282	[1459]1107193 ± 66523
1406	vocord-002	918292	648259	-	[301]1330 ± 0	[835]782 ± 36	-	-	-	-	[1436]83063 ± 517	[1434]83072 ± 714
1407	vocord-003	900909	1140643	-	[208]817 ± 0	[228]296 ± 3	-	-	-	-	[1377]46683 ± 378	[1358]36007 ± 292
1408	vocord-004	1754886	129419	-	[211]896 ± 0	[631]619 ± 1	-	-	-	-	[675]1885 ± 66	[679]1889 ± 71

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 53: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1409	vocord-005	1060445	189974	-	203 768 ± 0	828 771 ± 1	-	-	-	-	643 1768 ± 48	631 1713 ± 77
1410	vocord-006	559457	189913	-	204 768 ± 0	959 886 ± 1	-	-	-	-	703 2020 ± 72	699 1969 ± 62
1411	vocord-007	587489	344995	-	325 1664 ± 0	833 780 ± 2	-	-	-	-	818 2593 ± 83	811 2526 ± 59
1412	vocord-008	603867	345047	-	1172 2688 ± 0	1037 962 ± 2	749 976 ± 2	780 1061 ± 3	850 1236 ± 23	988 1851 ± 9	883 3015 ± 50	886 2988 ± 62
1413	vocord-009	1380132	201560	988 4372	331 1920 ± 0	1399 1472 ± 2	1105 1472 ± 1	1110 1549 ± 1	1068 1667 ± 2	1017 2064 ± 2	711 2052 ± 50	715 2056 ± 39
1414	vocord-010	902552	206873	965 3866	290 1088 ± 0	1393 1459 ± 2	1102 1459 ± 1	1087 1463 ± 2	1032 1484 ± 1	927 1535 ± 3	838 2724 ± 31	828 2653 ± 45
1415	vogic-000	237025	23230	163 477	1343 4104 ± 0	465 497 ± 0	311 499 ± 0	277 499 ± 0	239 499 ± 1	183 499 ± 0	575 1490 ± 1	580 1495 ± 2
1416	vsoft-001	10202	61965	59 175	953 2048 ± 0	64 123 ± 8	40 131 ± 7	249 468 ± 19	991 1422 ± 48	1126 5622 ± 229	813 2553 ± 48	817 2559 ± 44
1417	vsoft-002	10202	62518	56 169	487 2048 ± 0	91 160 ± 10	51 164 ± 11	311 537 ± 21	1058 1581 ± 53	1129 6279 ± 228	993 4030 ± 260	983 3902 ± 137
1418	vsoft-003	10202	62894	78 258	1107 2057 ± 0	123 191 ± 14	75 191 ± 14	393 628 ± 27	1108 1895 ± 66	1139 7545 ± 376	1254 16609 ± 97	1258 16649 ± 154
1419	vsoft-004	488414	110735	512 1180	1105 2057 ± 0	382 420 ± 13	238 418 ± 15	597 857 ± 28	1132 2135 ± 73	1141 8068 ± 378	1277 19601 ± 136	1280 19776 ± 152
1420	vts-000	256589	169760	684 1697	617 2048 ± 0	457 486 ± 1	298 481 ± 0	261 484 ± 0	229 485 ± 1	203 517 ± 0	1443 124209 ± 352	1443 123652 ± 358
1421	vts-001	293000	475743	240 622	826 2048 ± 0	709 676 ± 1	494 683 ± 6	454 687 ± 3	413 695 ± 2	337 709 ± 2	1170 9620 ± 44	1168 9618 ± 54
1422	vts-002	361977	904603	349 840	558 2048 ± 0	376 416 ± 5	255 434 ± 3	213 432 ± 3	193 449 ± 3	231 556 ± 4	1134 7447 ± 34	1132 7450 ± 54
1423	wicket-000	826392	641802	771 2071	762 2048 ± 0	1372 1419 ± 2	1083 1429 ± 3	1072 1444 ± 4	1018 1460 ± 3	928 1537 ± 6	1405 60976 ± 232	1402 61096 ± 323
1424	winsense-000	270819	32034	-	299 1280 ± 0	213 283 ± 1	-	-	-	-	593 1551 ± 31	593 1532 ± 42
1425	winsense-001	264428	32035	395 922	295 1280 ± 0	823 766 ± 7	811 1058 ± 47	703 983 ± 97	689 1053 ± 119	785 1320 ± 84	610 1631 ± 28	698 1964 ± 171
1426	winsense-002	281379	25780	731 1855	684 2048 ± 0	463 494 ± 2	310 498 ± 1	297 519 ± 1	267 537 ± 1	280 634 ± 1	621 1683 ± 8	621 1685 ± 7
1427	wiseai-001	189467	60781	75 245	433 2048 ± 0	169 240 ± 0	110 251 ± 0	126 328 ± 1	112 327 ± 0	78 332 ± 0	857 2850 ± 29	862 2852 ± 31
1428	wuhantianyu-001	465118	66457	472 1086	370 2048 ± 0	661 642 ± 1	451 642 ± 1	418 644 ± 0	366 652 ± 0	332 697 ± 0	1160 9502 ± 151	1175 9920 ± 253
1429	x-laboratory-000	520020	197310	632 1524	1050 2056 ± 0	866 808 ± 7	675 897 ± 113	633 907 ± 103	565 886 ± 103	312 673 ± 39	298 725 ± 19	320 749 ± 34
1430	x-laboratory-001	625140	398792	726 1844	1100 2056 ± 0	571 586 ± 2	397 596 ± 5	364 603 ± 6	338 620 ± 7	412 793 ± 14	343 813 ± 28	366 872 ± 32
1431	xforwardai-000	242457	175556	-	568 2048 ± 0	812 757 ± 6	-	-	-	-	479 1185 ± 44	482 1157 ± 44
1432	xforwardai-001	340100	51163	790 2176	383 2048 ± 0	1211 1180 ± 2	911 1182 ± 1	887 1194 ± 1	806 1186 ± 2	697 1203 ± 1	325 779 ± 17	337 797 ± 13
1433	xforwardai-002	707715	51163	761 2003	1248 4096 ± 0	1013 944 ± 1	717 942 ± 1	666 943 ± 4	597 935 ± 1	529 967 ± 1	540 1406 ± 8	542 1405 ± 13
1434	xm-000	578041	148920	274 684	1023 2052 ± 0	949 878 ± 2	665 882 ± 1	708 988 ± 2	866 1258 ± 3	1059 2434 ± 7	611 1634 ± 17	608 1632 ± 20
1435	yisheng-000	122704	41760	-	789 2048 ± 0	49 103 ± 1	-	-	-	-	327 790 ± 23	333 789 ± 20
1436	yisheng-001	120112	41760	-	924 2048 ± 0	48 103 ± 1	-	-	-	-	370 906 ± 31	376 905 ± 25
1437	yisheng-002	604097	42014	-	302 1348 ± 0	273 332 ± 1	-	-	-	-	215 613 ± 40	193 588 ± 26
1438	yisheng-004	486351	38653	551 1279	1197 3704 ± 0	329 378 ± 12	-	-	-	-	275 693 ± 137	164 526 ± 34
1439	yitu-000	2211068	76146	-	1355 4130 ± 0	700 672 ± 2	-	-	-	-	1354 35352 ± 114	1363 37848 ± 1773
1440	yitu-001	2069440	138901	-	1363 4146 ± 0	804 751 ± 0	-	-	-	-	1360 37810 ± 127	1357 35486 ± 203
1441	yitu-002	1726673	138901	-	1364 4147 ± 0	742 691 ± 0	-	-	-	-	1355 35435 ± 111	1355 35325 ± 131
1442	yitu-003	1525719	138919	958 3737	1137 2082 ± 0	926 860 ± 0	-	-	-	-	1266 18305 ± 71	1269 18286 ± 62
1443	yoonik-000	290414	206059	-	497 2048 ± 0	1005 941 ± 3	737 965 ± 13	688 964 ± 10	624 964 ± 9	534 974 ± 23	451 1116 ± 34	458 1113 ± 54
1444	yoonik-001	346629	265353	-	723 2048 ± 0	1242 1223 ± 3	958 1238 ± 1	920 1238 ± 1	852 1240 ± 1	720 1240 ± 1	285 706 ± 29	288 690 ± 26
1445	yoonik-002	453720	265415	879 2737	731 2048 ± 0	1178 1145 ± 4	860 1123 ± 2	831 1124 ± 2	746 1125 ± 2	650 1126 ± 3	316 761 ± 32	319 736 ± 32
1446	yoonik-003	346691	265415	788 2161	457 2048 ± 0	1067 991 ± 3	754 980 ± 1	704 984 ± 4	632 982 ± 1	541 983 ± 1	267 684 ± 45	276 678 ± 41
1447	yoonik-004	469597	88673	325 792	750 2048 ± 0	1239 1219 ± 1	989 1284 ± 1	938 1255 ± 1	873 1266 ± 0	731 1251 ± 1	413 1039 ± 42	419 1030 ± 40
1448	yoti-001	267246	470114	491 1125	479 2048 ± 0	1258 1238 ± 4	1015 1316 ± 17	1175 4099 ± 13	1166 4160 ± 45	1118 4774 ± 95	1180 10352 ± 247	1182 10364 ± 246
1449	yoti-002	164043	470114	471 1086	365 2048 ± 0	602 604 ± 1	599 802 ± 3	553 812 ± 4	601 938 ± 15	821 1373 ± 4	1183 10662 ± 805	1181 10320 ± 265
1450	ytu-000	1477360	44032	839 2484	530 2048 ± 0	498 530 ± 0	340 533 ± 0	413 640 ± 0	548 861 ± 2	1001 1949 ± 8	1340 31797 ± 131	1341 31794 ± 133
1451	yuan-000	370472	331437	-	926 2048 ± 0	1157 1112 ± 6	-	-	-	-	846 2777 ± 44	877 2937 ± 87
1452	yuan-001	370473	189558	-	937 2048 ± 0	1342 1383 ± 3	1055 1394 ± 3	1051 1408 ± 3	1019 1461 ± 4	952 1615 ± 4	773 2344 ± 25	770 2325 ± 42

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Table 54: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1453	yuan-002	370472	165662	-	819 2048 ± 0	1373 1420 ± 3	1084 1429 ± 4	1104 1511 ± 4	1069 1695 ± 4	1056 2408 ± 5	763 2297 ± 23	767 2310 ± 31
1454	yuan-003	370419	147783	-	373 2048 ± 0	1363 1405 ± 2	1070 1413 ± 3	1075 1446 ± 3	1047 1547 ± 5	995 1878 ± 5	768 2320 ± 32	759 2287 ± 34
1455	yuan-004	428665	50011	581 1356	1243 4096 ± 0	545 567 ± 0	373 569 ± 0	337 573 ± 0	297 579 ± 0	264 607 ± 0	1077 5816 ± 35	1078 5800 ± 31
1456	yuan-005	258312	145564	352 843	858 2048 ± 0	332 381 ± 0	210 386 ± 0	175 387 ± 2	153 390 ± 4	135 421 ± 3	470 1156 ± 8	490 1196 ± 26
1457	yuan-006	1622733	145572	987 4371	1301 4096 ± 0	1282 1280 ± 2	980 1268 ± 1	951 1273 ± 0	877 1272 ± 2	776 1306 ± 3	740 2202 ± 19	737 2190 ± 20
1458	yuan-007	258312	145568	664 1614	728 2048 ± 0	305 359 ± 0	186 361 ± 1	170 379 ± 4	144 380 ± 0	132 417 ± 0	293 715 ± 4	313 726 ± 16
1459	yuan-008	620369	167833	665 1626	1222 4096 ± 0	1197 1169 ± 0	564 752 ± 0	516 752 ± 3	482 775 ± 4	407 782 ± 0	523 1362 ± 16	537 1377 ± 19
1460	yuan-009	1161457	167833	842 2507	1212 4096 ± 0	1366 1408 ± 2	1076 1418 ± 2	1091 1471 ± 1	959 1387 ± 8	851 1414 ± 12	514 1326 ± 11	525 1336 ± 29
1461	yuan-010	1054332	167833	841 2499	1227 4096 ± 0	1280 1276 ± 1	1066 1406 ± 8	968 1294 ± 1	905 1308 ± 2	824 1375 ± 2	518 1331 ± 22	527 1349 ± 47
1462	yuanli-001	136632	47927	67 190	227 1024 ± 0	75 142 ± 0	43 143 ± 0	35 145 ± 0	31 150 ± 0	40 210 ± 1	717 2080 ± 16	721 2083 ± 22

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Table 55: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1	2060io-001	-		-		0.9149	1331	0.9720	1327	0.9773	1286	0.9768	1133	0.8672	1164	0.9993	778		
2	2060io-002	-		-		0.8904	1329	0.9587	1325	0.9020	1272	1.0000	1282	1.0000	1303	1.0000	881		
3	20face-000	0.1268	936	0.1828	927	0.1748	1227	0.2768	1227	0.1765	1149	0.1864	837	0.0927	917	0.5943	572		
4	20face-001	0.0521	859	0.0732	852	0.1414	1220	0.2549	1223	0.0769	1095	0.1354	811	0.0419	825	0.8953	672		
5	3divi-000	-		0.1335	901	-		-		-		-		-		-			
6	3divi-001	0.0777	898	0.1540	914	-		-		-		-		-		-			
7	3divi-002	0.0788	900	0.1539	912	-		-		-		-		-		-			
8	3divi-003	0.0318	821	0.0588	831	0.0389	1129	0.0639	1123	0.0619	1078	0.0928	765	0.0555	865	-			
9	3divi-004	0.0095	551	0.0153	562	0.0097	882	0.0145	870	0.0175	863	0.0330	603	0.0242	726	-			
10	3divi-005	0.0094	548	0.0151	558	0.0078	803	0.0121	802	0.0135	785	0.0231	527	0.0156	602	-			
11	3divi-006	0.0064	430	0.0094	429	0.0047	617	0.0066	623	0.0091	657	0.0191	479	0.0113	495	0.2854	489		
12	3divi-007	0.0024	138	0.0038	152	0.0028	311	0.0034	311	0.0046	425	0.0101	310	0.0082	381	0.2515	469		
13	accurascan-001	0.3670	1002	0.4777	1008	0.5313	1281	0.6155	1273	0.5583	1220	0.6120	999	0.2569	1020	0.8833	663		
14	accurascan-002	0.4209	1012	0.5213	1018	0.5826	1289	0.6642	1278	0.6295	1224	0.6937	1017	0.3116	1042	0.9162	679		
15	accurascan-003	0.0017	82	0.0028	91	0.0024	147	0.0024	115	0.0029	213	0.0057	122	0.0048	159	0.0629	125		
16	acer-000	0.1393	941	0.9075	1068	0.9981	1346	1.0000	1353	1.0000	1310	1.0000	1255	0.9998	1274	-			
17	acer-001	0.0294	806	0.0504	811	0.0240	1079	0.0463	1086	0.0436	1037	0.0622	722	0.0360	805	-			
18	acer-002	0.0169	708	0.0262	708	0.0103	899	0.0167	919	0.0182	873	0.0281	565	0.0159	605	0.1555	393		
19	acisw-003	0.9682	1067	0.9971	1078	0.7892	1322	0.8738	1316	0.8752	1266	0.8275	1065	0.6698	1122	-			
20	acisw-006	0.2945	990	0.9788	1074	0.6044	1291	0.7588	1289	0.9900	1290	1.0000	1259	0.9999	1281	-			
21	acisw-007	0.4276	1014	0.5493	1025	0.8425	1325	0.9185	1323	0.8424	1257	0.9976	1193	0.9930	1235	0.9999	811		
22	acisw-008	0.0100	565	0.0147	551	0.0094	868	0.0126	812	0.1740	1148	0.6651	1012	0.4545	1081	0.9302	685		
23	adera-001	0.1021	918	0.1757	921	0.1823	1228	0.2967	1230	0.1714	1147	0.6357	1003	0.1127	934	-			
24	adera-002	0.0052	351	0.0071	340	0.0047	610	0.0064	614	0.0087	638	0.0159	426	0.0136	557	-			
25	adera-003	0.0043	293	0.0059	287	0.0036	484	0.0043	440	0.0076	594	0.0151	410	0.0128	536	0.1526	392		
26	adera-004	0.0014	52	0.0022	54	0.0035	474	0.0050	501	0.0023	94	0.0212	500	0.0058	232	0.1469	383		
27	adera-005	0.0017	78	0.0026	74	0.0027	282	0.0030	240	0.0024	107	0.0134	378	0.0049	164	0.1246	343		
28	advance-002	0.0089	527	0.0137	533	0.0073	782	0.0115	790	0.0400	1023	0.0722	745	0.0593	874	-			
29	advance-003	0.0060	410	0.0087	411	0.0052	654	0.0067	626	0.0389	1019	0.4914	963	0.1291	947	0.7998	629		
30	advance-004	0.0083	513	0.0101	462	0.0037	510	0.0054	538	0.0051	462	0.3555	923	0.1088	930	0.8285	636		
31	advance-005	0.0059	401	0.0084	399	0.0043	577	0.0060	582	0.0065	549	0.9998	1241	0.9993	1261	1.0000	898		
32	advance-006	-		-		0.0032	422	0.0042	428	0.0034	298	0.9982	1200	0.5024	1090	1.0000	883		
33	advance-007	-		-		0.0025	183	0.0028	199	0.0025	128	0.0801	754	0.0258	740	0.3283	502		
34	affinsys-ai-000	-		-		0.1154	1209	0.1788	1202	0.1886	1156	0.2159	857	0.1383	956	0.6126	577		
35	afisbiometrics-000	0.0051	349	0.0073	349	0.0030	370	0.0050	506	0.0044	399	0.0077	207	0.0057	229	0.0796	214		
36	afrengine-000	0.6244	1045	0.7336	1052	0.8318	1324	0.9083	1320	0.8122	1249	0.9980	1198	0.9895	1230	1.0000	816		
37	afrengine-001	0.0071	463	0.0129	521	0.0094	873	0.0201	959	0.0097	689	0.9737	1131	0.8179	1155	0.9999	795		
38	afrengine-002	0.0052	354	0.0071	338	0.0032	411	0.0043	445	0.0194	896	0.2125	855	0.0450	836	0.2330	461		
39	afrengine-003	-		-		0.0028	324	0.0031	273	0.0116	745	0.0224	517	0.0208	693	0.0860	236		
40	afrengine-004	-		-		0.0027	299	0.0028	210	0.1475	1137	0.2477	877	0.2470	1014	0.2062	445		
41	afrengine-005	-		-		0.0028	328	0.0031	258	0.0035	307	0.0078	212	0.0059	247	0.0768	201		
42	aifirst-001	0.0119	618	0.0170	595	0.0084	834	0.0127	822	0.0131	775	0.0212	499	0.0138	562	-			
43	aigen-001	0.0124	630	0.0219	661	0.0143	996	0.0217	971	0.0236	939	0.8960	1086	0.3255	1048	-			
44	aigen-002	0.0192	736	0.0343	752	0.0256	1083	0.0402	1063	0.0389	1018	0.9196	1095	0.3876	1065	0.9994	783		

Table 56: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
45	aigroup-000	-		-		0.0683	1179	0.0774	1143	0.0612	1074	0.9816	1140	0.9069	1181	1.0000	1320
46	aigroup-001	-		-		0.0114	941	0.0192	950	0.0179	866	0.9998	1243	0.9986	1257	1.0000	932
47	aigroup-002	-		-		0.0034	453	0.0049	499	0.0041	370	0.9104	1091	0.7156	1136	0.9949	749
48	ailabs-001	0.0158	695	0.0276	716	0.0192	1038	0.0317	1036	0.0352	1003	0.0608	718	0.0434	830	-	
49	aimall-001	0.0176	719	0.0324	741	0.0225	1068	0.0407	1067	-		-		-		-	
50	aimall-002	0.0119	619	0.0167	589	0.0224	1065	0.0411	1068	0.0233	931	0.0373	644	0.0235	719	-	
51	aimall-003	0.0033	214	0.0041	175	0.0033	445	0.0035	342	0.0056	499	0.0109	334	0.0087	401	-	
52	aipractix-000	-		-		0.0649	1173	0.0749	1139	0.0065	550	0.9940	1173	0.9055	1180	0.9999	807
53	aiseemu-001	0.0021	111	0.0029	98	0.0027	279	0.0033	297	0.0038	339	0.0339	616	0.0057	230	0.4819	543
54	aiseemu-002	0.0023	130	0.0032	112	0.0026	237	0.0027	177	0.0036	314	0.0439	662	0.0057	225	0.4259	530
55	aiunionface-000	0.0104	577	0.0154	564	0.0082	826	0.0122	805	0.0141	796	0.0243	534	0.0169	624	-	
56	aize-001	0.0223	762	0.0344	753	0.0199	1045	0.0313	1032	0.0367	1010	0.0522	695	0.0359	803	0.2506	468
57	aize-002	0.0210	752	0.0327	742	0.0280	1093	0.0489	1097	0.0504	1054	0.0692	740	0.0434	829	0.2657	481
58	aize-003	0.0086	520	0.0125	517	0.0072	773	0.0126	816	0.0516	1058	0.6137	1000	0.1038	926	0.9911	738
59	aize-006	-		-		0.0081	820	0.0160	902	0.0133	780	0.9925	1167	0.9473	1195	1.0000	822
60	ajou-001	0.0093	542	0.0147	552	0.0071	772	0.0126	813	0.0173	860	0.0274	559	0.0186	658	-	
61	alchera-000	0.0165	705	0.0243	687	0.0125	964	0.0186	942	0.0204	905	0.0349	623	0.0243	728	-	
62	alchera-001	0.0183	728	0.0299	727	0.0142	991	0.0234	988	0.0239	946	0.0388	651	0.0267	751	-	
63	alchera-002	0.0107	588	0.0157	574	0.0104	903	0.0229	983	0.0144	804	0.0246	538	0.0198	680	-	
64	alchera-003	0.0044	299	0.0055	269	0.0031	386	0.0039	390	0.0042	378	0.0077	209	0.0065	268	0.0777	205
65	alchera-004	0.0035	240	0.0052	257	0.0028	332	0.0039	392	0.0029	217	0.0075	197	0.0044	137	0.0742	181
66	alchera-005	0.0027	165	0.0040	166	0.0026	227	0.0030	234	0.0025	141	0.0055	118	0.0040	108	0.0642	138
67	alchera-006	0.0028	176	0.0039	161	0.0026	236	0.0031	264	0.0025	134	0.0090	267	0.0042	117	0.0867	238
68	alchera-007	0.0034	236	0.0049	235	0.0026	235	0.0031	266	0.0026	164	0.0096	293	0.0044	132	0.0834	232
69	alchera-008	-		-		0.0026	223	0.0030	242	0.0024	126	0.0099	301	0.0043	125	0.0951	280
70	alchera-009	-		-		0.0026	258	0.0029	224	0.0022	76	0.5197	973	0.2232	1010	0.8470	641
71	alfabeta-001	0.4867	1030	0.5831	1033	0.6855	1301	0.8156	1302	0.8253	1255	0.7765	1042	0.6416	1120	0.9627	699
72	alice-000	0.0119	621	0.0192	631	0.0106	911	0.0170	922	0.0167	846	0.0265	553	0.0150	592	0.1824	423
73	alice-001	0.0034	232	0.0046	218	0.0032	409	0.0037	369	0.0030	242	0.3164	908	0.1088	931	0.8336	637
74	alleyes-000	0.0058	392	0.0090	419	0.0055	677	0.0087	721	0.0068	558	0.0105	323	0.0076	350	-	
75	allgovision-000	0.0346	830	0.0527	818	0.0232	1073	0.0339	1039	0.0372	1014	0.0620	721	0.0443	835	-	
76	alphaface-001	0.0065	436	0.0097	445	0.0039	533	0.0063	610	0.0083	623	-		-		-	
77	alphaface-002	0.0052	357	0.0075	360	0.0030	357	0.0044	454	1.0000	1320	0.0115	352	0.0084	392	-	
78	alttekglobal-000	-		-		0.0031	376	0.0046	464	0.0051	461	0.0097	299	0.0075	338	0.0927	272
79	amplifiedgroup-001	0.5034	1032	0.5848	1034	0.6973	1305	0.8316	1303	0.7807	1244	0.7724	1039	0.6354	1115	-	
80	androvideo-000	0.0243	774	0.0438	789	0.0239	1077	0.0365	1054	0.0483	1048	0.1870	838	0.0635	879	-	
81	anke-000	0.0194	739	0.0260	705	-		-		-		-		-		-	
82	anke-001	0.0186	731	0.0238	684	-		-		-		-		-		-	
83	anke-002	0.0402	840	0.0759	858	0.0271	1091	0.0484	1095	-		-		-		-	
84	anke-003	0.0131	648	0.0213	655	0.0094	870	0.0175	925	0.0134	782	0.9951	1178	0.0150	591	-	
85	anke-004	0.0080	502	0.0154	565	0.0073	781	0.0112	784	0.0102	707	0.0178	460	0.0118	510	-	
86	anke-005	0.0070	455	0.0109	482	0.0059	708	0.0094	735	0.0105	714	0.0142	389	0.0102	458	-	
87	antheus-000	0.2564	982	0.3776	990	0.7240	1309	0.8699	1313	0.8899	1269	0.9872	1149	0.9483	1196	-	
88	antheus-001	0.1311	937	0.2306	948	0.5113	1279	0.6797	1280	0.8748	1265	0.9908	1159	0.9649	1209	-	

Table 57: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
89	antheus-002	-		-		0.0119	951	0.0157	892	0.0104	712	0.9974	1191	0.9791	1219	1.0000	849
90	anyvision-001	0.0716	891	0.0859	869	-		-		-		-		-		-	
91	anyvision-002	0.0660	886	0.0898	875	0.0928	1196	0.1512	1191	0.0899	1110	0.1191	797	0.0801	899	-	
92	anyvision-004	0.0267	792	0.0385	772	0.0258	1085	0.0487	1096	0.0234	936	0.0301	577	0.0191	665	-	
93	anyvision-005	0.0023	129	0.0037	147	0.0027	306	0.0035	332	0.0049	449	0.0084	246	0.0069	294	-	
94	aratek-001	0.0033	225	0.0045	210	0.0028	321	0.0031	274	0.0043	384	0.0092	276	0.0069	297	0.0897	252
95	armatura-001	0.0033	219	0.0042	182	0.0031	385	0.0037	362	0.0056	498	0.0110	335	0.0092	422	0.4625	539
96	armatura-002	0.0041	281	0.0052	251	0.0034	454	0.0044	447	0.0040	356	0.8502	1072	0.0275	756	0.9970	762
97	armatura-003	0.0020	106	0.0029	93	0.0026	251	0.0028	188	0.0025	146	0.0049	92	0.0043	126	0.0608	111
98	arvis-000	-		-		0.3652	1260	0.4777	1255	1.0000	1362	1.0000	1374	1.0000	1432	1.0000	1355
99	asr-000	-		-		0.5523	1283	0.7662	1290	0.8807	1267	1.0000	1289	1.0000	1288	1.0000	877
100	asr-001	-		-		0.0030	354	0.0039	388	0.0027	184	0.3390	914	0.1327	953	0.7167	604
101	asusaics-000	0.0125	637	0.0209	650	0.0085	836	0.0134	838	0.0143	802	0.7189	1023	0.0285	763	-	
102	asusaics-001	0.0125	638	0.0210	651	0.0085	838	0.0134	839	0.0143	803	0.7437	1029	0.0289	768	-	
103	autentika-000	0.1415	943	0.1916	932	0.4130	1268	0.5521	1264	0.4217	1208	0.9998	1239	0.9954	1242	1.0000	900
104	autentika-001	0.0444	850	0.0688	847	0.0987	1198	0.1531	1192	0.1547	1140	1.0000	1256	0.9997	1273	1.0000	868
105	autentika-002	0.0536	863	0.0866	871	0.0657	1176	0.1084	1170	0.1548	1141	0.9993	1217	0.9939	1238	1.0000	876
106	autentika-003	-		-		0.0203	1048	0.0314	1033	0.0404	1026	0.9791	1136	0.9294	1189	0.9994	782
107	authenmetric-002	0.0092	539	0.0134	527	0.0095	874	0.0177	927	0.0192	889	0.0463	675	0.0236	721	-	
108	authenmetric-003	0.0036	252	0.0053	263	0.0039	540	0.0051	512	0.0095	680	0.9930	1168	0.5932	1106	1.0000	841
109	authenmetric-004	0.0027	169	0.0042	184	0.0033	435	0.0036	358	0.0083	626	0.9879	1152	0.4058	1072	1.0000	828
110	authme-001	0.0036	243	0.0049	236	0.0037	512	0.0040	400	0.0030	239	0.3075	901	0.1310	950	0.7046	598
111	authme-002	-		-		0.0032	417	0.0035	336	0.0026	173	0.4284	949	0.2115	1004	0.7861	624
112	aware-000	0.1965	965	0.3090	972	-		-		-		-		-		-	
113	aware-001	0.1965	964	0.3092	973	-		-		-		-		-		-	
114	aware-002	0.1951	963	0.3202	976	-		-		-		-		-		-	
115	aware-003	0.0793	901	0.1161	895	0.1028	1200	0.1708	1198	0.1698	1146	0.2333	868	0.1174	940	-	
116	aware-004	0.0690	889	0.0949	881	0.0837	1191	0.1436	1189	0.1171	1124	0.8137	1055	0.1056	927	-	
117	aware-005	0.0457	853	0.0643	837	0.0603	1168	0.1094	1174	0.0613	1075	0.1075	785	0.0491	848	-	
118	aware-006	0.0487	856	0.0819	865	0.0529	1158	0.1090	1172	0.1011	1120	0.1058	780	0.0502	852	0.5309	552
119	aware-007	0.0101	570	0.0198	642	0.0060	719	0.0133	835	0.0224	922	0.5813	990	0.1673	985	0.8871	667
120	aware-008	-		-		0.0062	728	0.0143	867	0.0135	784	0.0332	606	0.0133	545	0.3622	508
121	awiros-001	0.4044	1008	0.4622	1005	0.5530	1284	0.6518	1276	0.2008	1159	0.1994	845	0.1386	958	-	
122	awiros-002	0.1990	968	0.2561	955	0.3319	1256	0.4411	1251	0.3821	1201	0.9938	1171	0.2634	1026	-	
123	awiros-003	-		-		0.0128	971	0.0158	897	0.9978	1294	1.0000	1280	1.0000	1295	1.0000	889
124	awiros-004	-		-		0.0158	1021	0.0255	999	0.9981	1297	1.0000	1283	1.0000	1300	1.0000	846
125	aximetria-001	0.0111	598	0.0186	621	0.0110	921	0.0148	878	0.0170	853	0.3928	939	0.2090	1002	0.7924	626
126	ayftech-001	0.0946	913	0.1941	934	0.2438	1242	0.3625	1240	0.1558	1142	0.1589	825	0.0936	919	-	
127	ayftech-003	0.0023	133	0.0037	143	0.0026	259	0.0030	255	0.0072	579	0.5411	980	0.2515	1016	0.8902	668
128	ayonix-000	0.4351	1017	0.4872	1011	0.6150	1293	0.7510	1288	0.6557	1230	0.6361	1004	0.4981	1088	-	
129	beethedata-000	0.0127	642	0.0195	634	0.0092	858	0.0157	893	0.0171	856	0.0306	581	0.0204	686	0.1617	405
130	betaface-000	0.8804	1060	0.8998	1064	-		-		-		-		-		-	
131	betaface-002	0.8813	1061	0.8998	1065	-		-		-		-		-		-	
132	beyneai-000	0.0071	467	0.0107	478	0.0104	904	0.0131	831	0.0170	854	0.9837	1143	0.6171	1111	1.0000	814

Table 58: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
133	biocube-001	0.5596	1040	0.6834	1046	0.7700	1320	0.8712	1314	0.8446	1258	0.9661	1126	0.7922	1151	0.9846	727
134	biocube-002	0.0612	875	0.0856	868	0.2330	1240	0.2972	1231	0.2365	1176	0.9327	1108	0.6947	1126	0.9989	773
135	bioid-001	-		-		0.0121	956	0.0197	957	0.0148	810	0.9883	1153	0.7807	1148	0.9997	789
136	bioid-002	-		-		0.0127	969	0.0205	962	0.0158	826	0.7684	1038	0.0224	708	0.9859	729
137	bioid-003	-		-		0.0083	830	0.0137	847	0.0112	736	0.1141	791	0.0135	550	0.3783	516
138	bioid-004	-		-		0.0083	833	0.0136	844	0.0109	728	0.1377	815	0.0139	566	0.5446	557
139	bioidtechswiss-000	0.0066	443	0.0082	393	0.0113	937	0.0225	978	0.0078	603	0.0139	385	0.0092	417	-	
140	bioidtechswiss-001	0.0054	367	0.0072	345	0.0069	762	0.0124	808	0.0060	520	0.0094	286	0.0065	276	-	
141	bioidtechswiss-002	0.0049	329	0.0067	322	0.0064	736	0.0116	792	0.0067	555	0.0117	358	0.0086	398	-	
142	bioidtechswiss-003	-		-		0.0045	595	0.0050	504	0.0193	893	0.0360	631	0.0356	799	0.1320	352
143	biometric-vision-000	0.0023	132	0.0036	134	0.0028	325	0.0034	325	0.0028	208	0.3109	904	0.1504	967	0.6849	594
144	biometrik-001	-		-		0.7415	1313	0.8509	1306	0.9980	1296	1.0000	1415	0.9877	1227	1.0000	939
145	biometrik-002	-		-		0.0641	1172	0.0984	1163	0.0419	1031	0.3401	915	0.0292	772	0.9219	684
146	bm-001	0.7431	1052	0.9494	1072	0.9586	1336	0.9843	1332	0.9049	1273	0.9021	1090	0.8395	1159	-	
147	boetech-001	0.0662	887	0.0802	863	0.0493	1151	0.0791	1145	0.0682	1087	0.1074	784	0.0758	893	0.9951	750
148	boetech-002	0.0535	862	0.0565	827	0.0114	939	0.0136	843	0.0403	1024	0.0650	725	0.0606	875	0.3470	506
149	bresee-000	0.8467	1058	0.9472	1070	0.9819	1337	0.9903	1333	0.9940	1292	0.9919	1162	0.9760	1216	-	
150	bresee-001	0.0085	517	0.0143	542	0.0086	844	0.0153	886	0.0108	723	0.0168	445	0.0115	502	-	
151	bresee-002	0.0079	500	0.0101	460	0.0065	745	0.0079	689	0.0129	767	0.0263	551	0.0224	709	0.2137	452
152	btsdigital-001	-		-		0.0046	604	0.0072	654	0.0077	597	0.0141	387	0.0092	420	0.1256	346
153	bynn-000	-		-		0.0193	1041	0.0234	987	0.1302	1128	0.9999	1247	0.9996	1269	1.0000	895
154	cakevn-001	-		-		0.0311	1110	0.0677	1128	0.0212	912	0.1427	817	0.0185	655	0.9170	681
155	cakevn-002	-		-		0.0027	287	0.0039	391	0.0026	176	0.0049	93	0.0037	70	0.0569	89
156	camvi-001	0.3822	1006	0.5375	1022	-		-		-		-		-		-	
157	camvi-002	0.0125	636	0.0221	665	0.0089	851	0.0145	872	0.0142	799	0.2650	885	0.0166	622	-	
158	camvi-003	0.0184	730	0.0320	739	0.0134	980	0.0255	1000	-		-		-		-	
159	camvi-004	0.0171	712	0.0316	738	0.0042	569	0.0049	495	0.0097	691	0.6636	1011	0.0141	570	-	
160	candour-001	0.1048	921	0.3189	975	0.0130	975	0.0182	934	0.3879	1203	0.9216	1097	0.7071	1131	0.9836	721
161	candour-002	-		-		0.0036	491	0.0061	597	0.1489	1139	0.4137	946	0.2751	1030	0.8795	659
162	candour-003	-		-		0.0040	551	0.0068	632	0.0041	371	0.1321	807	0.0225	711	0.6090	576
163	candour-004	-		-		0.0027	295	0.0039	397	0.0030	232	0.1049	779	0.0121	521	0.5691	564
164	canon-002	0.0034	231	0.0050	245	0.0026	260	0.0033	291	0.0043	393	0.0182	464	0.0065	274	-	
165	canon-003	0.0041	284	0.0059	282	0.0030	355	0.0040	402	0.0040	351	0.0073	191	0.0059	240	0.0735	178
166	canon-004	0.0052	355	0.0091	422	0.0033	443	0.0058	566	0.0037	323	0.0770	752	0.0494	849	0.3192	500
167	canon-005	0.0024	137	0.0035	132	0.0025	202	0.0033	308	0.0052	472	0.0096	296	0.0090	410	0.0542	73
168	cchonolulu-000	0.7910	1056	0.8611	1060	0.9824	1338	0.9926	1335	0.9940	1291	0.9861	1144	0.9534	1198	1.0000	817
169	cchonolulu-001	0.4202	1011	0.4853	1010	0.6298	1294	0.7694	1291	0.7402	1237	0.9720	1130	0.8870	1174	0.9951	751
170	cdachyd-001	-		-		0.7397	1312	0.8335	1304	0.7630	1239	0.7410	1027	0.5756	1104	0.9588	696
171	ceiec-001	0.0328	826	0.0475	802	0.0295	1107	0.0478	1093	0.0621	1080	0.1059	781	0.0932	918	-	
172	ceiec-002	0.0161	700	0.0193	632	0.0122	959	0.0164	915	0.0270	963	0.0555	702	0.0472	845	-	
173	ceiec-003	0.0071	464	0.0107	477	0.0061	724	0.0079	694	0.0160	831	0.0316	589	0.0260	744	-	
174	ceiec-004	0.0038	261	0.0051	250	0.0045	601	0.0053	526	0.0062	538	0.3939	941	0.0104	468	-	
175	celloscope-001	-		-		0.1245	1213	0.1840	1203	0.1886	1155	0.2636	884	0.1485	966	0.7055	601
176	chosun-000	0.8481	1059	1.0000	1149	1.0000	1451	0.9999	1351	1.0000	1385	1.0000	1336	1.0000	1396	-	

Table 59: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
177	chosun-001	0.0525	860	0.0936	879	0.0742	1183	0.1263	1183	0.0978	1117	1.0000	1284	0.9354	1190	-	
178	chosun-002	0.0390	839	0.0646	838	0.0339	1117	0.0576	1114	0.0455	1044	0.6904	1015	0.1746	986	-	
179	chtface-001	0.9993	1071	0.9994	1079	0.9999	1353	0.9996	1343	1.0000	1308	1.0000	1260	0.9999	1283	-	
180	chtface-002	0.0150	683	0.0268	710	0.0096	881	0.0140	858	0.0186	878	0.0320	595	0.0194	673	-	
181	chtface-003	0.0091	534	0.0146	548	0.0083	831	0.0128	824	0.0132	776	0.0220	512	0.0149	589	-	
182	chtface-004	0.0046	317	0.0062	305	0.0052	653	0.0080	696	0.0088	649	0.0152	411	0.0106	473	0.1152	323
183	chtface-005	0.0033	216	0.0049	230	0.0029	339	0.0041	416	0.0044	398	0.0317	592	0.0066	279	0.1952	438
184	chtface-006	0.0029	182	0.0043	193	0.0026	266	0.0034	326	0.0040	357	0.2701	889	0.0065	272	0.4464	534
185	chtface-007	-		-		0.0025	201	0.0032	281	0.0040	353	0.2018	847	0.0064	267	0.4751	542
186	chtface-008	-		-		0.0022	64	0.0026	146	0.0025	142	0.1246	801	0.0400	817	0.4384	533
187	chtface-009	-		-		0.0023	91	0.0027	167	0.0024	119	0.3659	929	0.1226	945	0.7714	616
188	canon-000	0.0084	515	0.0156	572	0.0049	632	0.0051	514	0.0275	971	0.1371	813	0.0138	564	-	
189	canon-001	0.0041	279	0.0061	296	0.0030	374	0.0041	414	0.0048	441	0.0578	707	0.0069	296	-	
190	cist-001	0.0046	311	0.0065	314	0.0042	573	0.0063	608	0.9675	1285	0.9997	1233	0.9994	1263	1.0000	832
191	cist-002	0.0073	475	0.0096	442	0.0085	839	0.0090	728	0.9990	1299	0.9997	1232	0.9996	1271	0.9998	791
192	cist-003	0.0065	437	0.0087	413	0.0046	608	0.0068	631	0.9993	1302	0.9994	1221	0.9993	1262	0.9995	785
193	cist-004	0.0043	296	0.0062	303	0.0031	383	0.0041	421	0.0031	248	0.0363	637	0.0069	299	0.2732	483
194	clearviewai-000	0.0010	27	0.0019	44	0.0024	129	0.0028	195	0.0030	227	0.0058	125	0.0050	168	0.0604	108
195	clearviewai-001	0.0015	58	0.0021	50	0.0024	134	0.0030	244	0.0022	86	0.0745	747	0.0194	672	0.2740	484
196	clearviewai-002	-		-		0.0024	131	0.0031	272	0.0022	84	0.0042	55	0.0037	73	0.0531	62
197	clearviewai-003	-		-		0.0023	73	0.0025	123	0.0022	79	0.0042	57	0.0038	76	0.0533	64
198	closeli-001	0.0136	655	0.0163	580	0.0039	537	0.0054	536	0.0072	577	1.0000	1261	0.0094	430	1.0000	890
199	cloudmatrix-000	0.0192	737	0.0340	749	0.0133	979	0.0220	974	0.9837	1288	1.0000	1267	0.0281	757	-	
200	cloudmatrix-001	0.0668	888	0.1141	894	0.0539	1159	0.0905	1159	0.3509	1197	0.9819	1141	0.9010	1178	0.9999	796
201	cloudmatrix-002	0.0075	483	0.0113	495	0.0084	835	0.0120	796	0.9248	1278	0.9997	1231	0.9985	1256	1.0000	896
202	cloudwalk-hr-000	0.0265	791	0.0434	787	0.0340	1118	0.0646	1125	-		-		-		-	
203	cloudwalk-hr-001	0.0044	298	0.0072	346	0.0073	784	0.0108	778	0.0125	760	0.0228	524	0.0145	581	-	
204	cloudwalk-hr-002	0.0043	292	0.0059	286	0.0054	669	0.0076	672	0.0076	595	0.5932	995	0.0093	425	-	
205	cloudwalk-hr-003	0.0026	159	0.0041	174	0.0040	547	0.0058	565	0.0060	530	0.9992	1214	0.0094	426	-	
206	cloudwalk-hr-004	0.0009	18	0.0018	38	0.0034	448	0.0028	206	0.0052	467	0.9992	1215	0.0093	424	-	
207	cloudwalk-mt-000	0.0100	564	0.0170	594	0.0074	789	0.0118	794	0.0127	763	0.0197	486	0.0129	538	-	
208	cloudwalk-mt-002	0.0064	434	0.0085	402	0.0054	672	0.0098	754	0.0070	569	0.0108	331	0.0076	349	-	
209	cloudwalk-mt-003	0.0013	49	0.0022	53	0.0026	225	0.0027	176	0.0039	343	0.0076	198	0.0067	282	-	
210	cloudwalk-mt-004	0.0009	23	0.0013	14	0.0024	144	0.0021	30	0.0028	197	0.0054	114	0.0050	170	0.0572	91
211	cloudwalk-mt-005	0.0006	9	0.0009	9	0.0025	197	0.0022	55	0.0017	31	0.9286	1103	0.5956	1107	0.9948	748
212	cloudwalk-mt-006	0.0006	11	0.0006	1	0.0023	87	0.0019	1	0.0016	18	0.0032	17	0.0030	32	0.0427	19
213	cloudwalk-mt-007	0.0006	8	0.0007	3	0.0023	94	0.0019	2	0.0016	16	0.0032	15	0.0030	30	0.0419	16
214	clova-000	0.0099	560	0.0150	554	0.0094	871	0.0147	876	0.0136	787	0.0213	503	0.0152	596	-	
215	cmcuni-001	0.0025	154	0.0038	154	0.0025	190	0.0025	140	0.0022	81	0.0052	106	0.0034	50	0.0660	152
216	codeline-000	0.0057	383	0.0079	379	0.0037	504	0.0053	534	0.2721	1182	1.0000	1264	0.9763	1217	1.0000	897
217	cogent-000	0.0435	848	0.0835	867	-		-		-		-		-		-	
218	cogent-001	0.0176	721	0.0385	773	-		-		-		-		-		-	
219	cogent-002	0.0148	678	0.0321	740	0.0095	876	0.0127	819	-		-		-		-	
220	cogent-003	0.0091	532	0.0188	624	0.0098	884	0.0132	832	0.0187	879	0.0319	593	0.0224	710	-	

Table 60: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
221	cogent-004	0.0064	433	0.0116	498	0.0096	879	0.0134	840	0.0157	823	0.0325	599	0.0204	685	-	
222	cogent-005	0.0060	407	0.0112	491	0.0064	740	0.0070	642	0.0095	678	0.0184	468	0.0135	551	-	
223	cogent-006	0.0046	320	0.0059	288	0.0036	476	0.0047	470	0.0058	516	0.0113	347	0.0091	413	0.1522	391
224	cogent-007	0.0022	124	0.0038	148	0.0028	329	0.0031	256	0.0040	359	0.0082	232	0.0067	281	0.1395	366
225	cogent-008	0.0015	61	0.0027	84	0.0023	96	0.0025	131	0.0033	282	0.0063	154	0.0055	212	0.0884	245
226	cognitec-000	0.0116	609	0.0177	608	0.0118	950	0.0167	918	0.0285	975	0.9924	1166	0.0435	831	-	
227	cognitec-001	0.0126	639	0.0185	620	0.0120	955	0.0168	920	0.0270	961	0.0554	700	0.0357	800	-	
228	cognitec-002	0.0066	441	0.0101	458	0.0079	806	0.0108	776	0.0181	870	0.0317	591	0.0237	722	-	
229	cognitec-003	0.0038	259	0.0052	253	0.0054	674	0.0057	560	0.0225	923	0.0416	660	0.0388	812	0.1397	367
230	cognitec-004	0.0036	245	0.0053	259	0.0053	660	0.0056	551	0.0098	693	0.0202	493	0.0154	597	0.3436	505
231	cognitec-005	0.0022	119	0.0030	105	0.0035	470	0.0041	408	0.0079	605	0.0480	681	0.0233	717	0.2134	450
232	cognitec-006	-		-		0.0034	458	0.0039	396	0.0069	561	0.0160	427	0.0108	479	0.1197	334
233	cognitec-007	-		-		0.0035	475	0.0042	432	0.0069	565	0.0114	350	0.0091	412	0.1112	316
234	compal-000	-		-		0.0028	336	0.0033	309	0.0039	347	0.0064	159	0.0051	182	0.0736	179
235	compal-001	-		-		0.0073	780	0.0053	529	0.0092	664	0.1965	844	0.0165	619	0.2610	479
236	compal-002	-		-		0.0073	779	0.0053	528	0.0092	663	0.1923	842	0.0165	618	0.2592	475
237	cor-001	0.0075	482	0.0113	494	0.0055	682	0.0084	709	0.0091	659	0.0148	402	0.0092	421	-	
238	coretech-000	0.7699	1055	1.0000	1346	1.0000	1404	0.9999	1352	1.0000	1453	1.0000	1409	1.0000	1309	1.0000	927
239	coretech-001	0.0052	353	0.0067	324	0.0083	832	0.0092	732	0.0346	1000	0.1363	812	0.0252	735	0.6210	582
240	coretech-002	0.0020	100	0.0036	135	0.0064	742	0.0064	613	0.0118	747	0.0157	424	0.0110	484	0.3091	495
241	coretech-003	-		-		0.0045	599	0.0051	511	0.1155	1123	0.1539	822	0.1515	968	0.1180	328
242	coretech-004	-		-		0.0044	587	0.0048	480	0.0051	465	0.0093	281	0.0070	304	0.0893	249
243	corsight-001	0.0040	275	0.0057	278	0.0033	444	0.0047	469	0.0045	415	0.0095	289	0.0063	259	-	
244	corsight-002	0.0053	360	0.0068	328	0.0030	364	0.0041	420	0.0039	348	0.0079	216	0.0054	205	0.0747	186
245	corsight-003	0.0026	163	0.0040	167	0.0028	313	0.0045	460	0.0035	311	0.0059	137	0.0046	144	0.0637	134
246	csc-001	0.9158	1063	0.9346	1069	0.9899	1339	0.9993	1341	0.9999	1305	0.9959	1181	0.9906	1232	-	
247	csc-002	0.0099	562	0.0132	524	0.0077	797	0.0142	866	0.0126	761	0.0195	485	0.0146	584	-	
248	csc-003	0.0053	362	0.0065	313	0.0037	506	0.0047	473	0.0074	588	0.0124	368	0.0112	493	0.0928	273
249	ctbcbank-000	0.0168	707	0.0250	697	0.0146	1001	0.0224	976	0.0211	911	0.8964	1088	0.3779	1062	-	
250	ctbcbank-001	0.0155	688	0.0235	682	0.0148	1008	0.0243	993	0.0207	906	0.9279	1102	0.3469	1053	-	
251	ctbcbank-002	-		-		0.0031	375	0.0049	491	0.0037	326	0.7785	1043	0.4878	1086	0.9727	706
252	cu-face-002	0.0105	582	0.0116	497	0.0650	1174	0.0568	1111	0.0271	965	0.0139	384	0.0076	345	0.1122	317
253	cu-face-003	0.0036	244	0.0049	233	0.0031	396	0.0037	368	0.0030	235	0.3075	900	0.1310	951	0.7046	599
254	cu-face-004	-		-		0.0032	416	0.0035	335	0.0026	172	0.4284	950	0.2115	1003	0.7861	625
255	cubox-001	0.0064	428	0.0080	381	0.0037	499	0.0055	544	0.0060	521	0.0111	340	0.0077	356	-	
256	cubox-002	0.0034	233	0.0041	171	0.0025	181	0.0025	142	0.0033	281	0.0064	157	0.0058	234	0.0519	57
257	cubox-003	0.0021	110	0.0028	92	0.0024	167	0.0026	157	0.0025	144	0.0046	72	0.0037	74	0.0495	47
258	cubox-004	-		-		0.0025	206	0.0027	181	0.0023	101	0.0044	68	0.0038	77	0.0490	44
259	cubox-005	-		-		0.0026	242	0.0030	251	0.0025	136	0.0047	82	0.0039	92	0.0515	56
260	cudocommunication-001	0.4777	1026	1.0000	1262	0.4373	1270	0.5360	1261	1.0000	1345	1.0000	1362	1.0000	1422	1.0000	1327
261	cuhkee-001	0.0036	253	0.0045	206	0.0031	398	0.0046	465	0.0051	466	0.0095	290	0.0079	362	-	
262	cybercore-000	0.0728	893	0.1110	890	0.1521	1223	0.2375	1217	0.1874	1152	0.1907	840	0.1178	942	-	
263	cybercore-001	0.3759	1004	0.5677	1029	0.6928	1304	0.7926	1295	0.8118	1248	0.9291	1106	0.7080	1132	0.9782	713
264	cybercore-002	0.0092	540	0.0119	504	0.0049	630	0.0072	651	0.9105	1276	1.0000	1281	1.0000	1290	1.0000	894

Table 61: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
265	cybercore-003	0.0155	687	0.0164	582	0.0032	421	0.0033	305	0.0024	110	0.9719	1129	0.8213	1157	0.9993	779
266	cyberextruder-001	0.1972	966	0.2547	954	0.4686	1277	0.6387	1275	0.3807	1200	0.3806	936	0.2582	1021	-	
267	cyberextruder-002	0.0811	906	0.1336	902	0.1465	1221	0.2266	1216	0.2086	1165	1.0000	1448	1.0000	1346	-	
268	cyberextruder-003	0.0109	593	0.0169	592	0.0071	770	0.0112	786	0.0165	842	0.0410	658	0.0272	755	0.2308	458
269	cyberextruder-004	0.0118	616	0.0181	613	0.0081	817	0.0133	834	0.0191	888	0.0329	602	0.0268	753	0.1904	430
270	cyberextruder-005	-		-		0.0049	629	0.0063	606	0.0055	485	0.0105	321	0.0079	367	0.0806	219
271	cyberextruder-006	-		-		0.0083	827	0.0138	850	0.0100	701	0.0180	462	0.0114	500	0.1276	348
272	cyberlink-000	0.0181	726	0.0274	714	0.0125	966	0.0180	932	-		-		-		-	
273	cyberlink-001	0.0131	647	0.0210	652	0.0439	1142	0.0517	1103	0.0173	861	0.0333	610	0.0209	694	-	
274	cyberlink-002	0.0114	604	0.0195	633	0.0101	894	0.0163	911	0.0160	832	0.0302	578	0.0192	668	-	
275	cyberlink-003	0.0118	614	0.0192	629	0.0098	887	0.0161	905	0.0153	816	0.0288	571	0.0159	607	-	
276	cyberlink-004	0.0074	477	0.0105	471	0.0068	758	0.0089	724	0.0094	674	0.0176	458	0.0117	507	-	
277	cyberlink-005	0.0060	409	0.0092	427	0.0058	703	0.0067	628	0.0074	584	0.0146	399	0.0105	472	-	
278	cyberlink-006	0.0042	289	0.0054	267	0.0043	576	0.0049	488	0.0052	471	0.0097	298	0.0077	357	-	
279	cyberlink-007	0.0032	212	0.0053	260	0.0041	557	0.0043	438	0.0052	470	0.0243	533	0.0084	393	0.3366	503
280	cyberlink-008	0.0042	288	0.0056	277	0.0038	529	0.0048	481	0.0053	476	0.0099	304	0.0074	332	0.1354	363
281	cyberlink-009	0.0018	87	0.0027	85	0.0047	609	0.0046	462	0.0040	363	0.0086	260	0.0062	256	0.1104	310
282	cyberlink-010	0.0011	37	0.0019	43	0.0041	554	0.0041	409	0.0039	342	0.1829	835	0.0054	206	0.8629	649
283	cyberlink-011	0.0010	29	0.0017	31	0.0035	471	0.0034	314	0.0045	407	0.0080	219	0.0070	307	0.0877	242
284	cyberlink-012	0.0010	28	0.0016	30	0.0036	480	0.0035	340	0.0047	437	0.0099	303	0.0074	333	0.1350	361
285	cyberlink-013	0.0011	36	0.0017	33	0.0037	501	0.0036	351	0.0045	409	0.0090	268	0.0080	371	0.0924	270
286	cyberlink-014	-		-		0.0026	263	0.0027	165	0.0030	225	0.0065	163	0.0052	189	0.0869	239
287	cyberlink-015	-		-		0.0025	192	0.0026	151	0.0028	199	0.0063	153	0.0050	173	0.0835	234
288	cyberlink-016	-		-		0.0025	179	0.0025	128	0.0028	193	0.0063	152	0.0050	169	0.0827	229
289	dahua-000	0.0249	778	0.0477	804	-		-		-		-		-		-	
290	dahua-001	0.0250	782	0.0466	799	0.0228	1071	0.0379	1057	-		-		-		-	
291	dahua-002	0.0129	643	0.0157	573	0.0116	945	0.0153	887	0.0134	781	0.0222	515	0.0170	627	-	
292	dahua-003	0.0052	356	0.0068	329	0.0056	686	0.0062	601	0.0113	739	0.0207	497	0.0185	657	-	
293	dahua-004	0.0045	305	0.0058	281	0.0036	486	0.0048	479	0.0051	463	0.0086	256	0.0070	308	-	
294	dahua-005	0.0031	198	0.0046	214	0.0035	473	0.0049	496	0.0046	419	0.0076	200	0.0062	255	-	
295	dahua-006	0.0027	166	0.0039	157	0.0031	391	0.0039	394	0.0039	346	0.0067	174	0.0058	231	-	
296	dahua-007	0.0017	74	0.0023	59	0.0026	248	0.0032	278	0.0033	274	0.0060	141	0.0054	204	0.0678	158
297	daon-000	0.0095	553	0.0117	500	0.0068	754	0.0077	681	0.0092	668	0.0174	454	0.0137	560	0.1430	376
298	daon-001	-		-		0.0036	481	0.0058	564	0.0050	455	0.0084	241	0.0057	226	0.0805	217
299	daon-002	-		-		0.0030	367	0.0042	427	0.0039	341	0.0146	400	0.0055	214	0.7444	610
300	daon-003	-		-		0.0024	165	0.0029	218	0.0030	233	0.0059	139	0.0048	160	0.0639	136
301	datech-000	0.0020	107	0.0031	111	0.0031	388	0.0041	417	0.0033	280	0.1646	827	0.0667	883	0.4898	546
302	datech-001	-		-		0.0028	316	0.0034	320	0.0026	179	0.2308	866	0.0818	904	0.5505	558
303	datech-002	-		-		0.0028	315	0.0034	322	0.0026	158	0.3779	934	0.1654	984	0.7940	627
304	decatur-000	0.0714	890	0.1115	892	0.0608	1169	0.1106	1175	0.0866	1106	1.0000	1275	0.0714	890	-	
305	decatur-001	0.0424	844	0.0711	849	0.0237	1074	0.0458	1082	0.0447	1042	1.0000	1252	0.9969	1248	1.0000	910
306	decloakface-001	0.0795	902	0.1131	893	0.3127	1252	0.4259	1247	0.2118	1168	0.7926	1047	0.6088	1110	0.9589	697
307	decloakface-002	-		-		0.0048	619	0.0077	680	1.0000	1318	0.9998	1240	0.9997	1272	1.0000	891
308	deepglint-001	0.0040	272	0.0062	302	0.0047	615	0.0067	627	0.0069	562	1.0000	1278	1.0000	1289	-	

Table 62: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
309	deepglint-002	0.0016	67	0.0027	82	0.0032	407	0.0033	307	0.0043	392	0.0084	247	0.0077	353	-	
310	deepglint-003	0.0027	168	0.0038	151	0.0030	363	0.0032	276	0.0043	382	0.0082	234	0.0076	346	-	
311	deepglint-004	0.0025	147	0.0034	123	0.0039	538	0.0061	596	0.0050	457	0.0091	272	0.0082	380	0.1003	287
312	deepglint-005	0.0052	350	0.0059	290	0.0030	356	0.0031	257	0.0033	286	0.7620	1036	0.1535	969	0.9912	739
313	deepsea-001	0.0136	657	0.0215	658	0.0142	993	0.0214	968	0.0163	836	0.0250	540	0.0192	669	-	
314	deepsense-000	0.0145	673	0.0265	709	0.0113	934	0.0196	955	0.0151	813	0.0215	506	0.0129	539	-	
315	deepsense-001	0.0013	48	0.0019	42	0.0024	155	0.0025	133	0.0027	188	0.0115	353	0.0053	195	0.1394	365
316	deepsense-002	0.0010	26	0.0016	29	0.0024	118	0.0023	84	0.0026	163	0.0052	108	0.0043	127	0.0781	206
317	deepsense-003	0.0009	25	0.0015	24	0.0025	203	0.0024	94	0.0023	99	0.0457	672	0.0097	440	0.2547	472
318	deepsense-004	-		-		0.0023	70	0.0021	36	0.0020	57	0.0111	339	0.0040	111	0.1195	332
319	deepsense-005	-		-		0.0023	68	0.0021	42	0.0019	46	0.0081	227	0.0035	60	0.0922	269
320	deepsense-006	-		-		0.0022	49	0.0022	61	0.0018	38	0.0336	613	0.0059	239	0.2813	488
321	deepsense-007	-		-		0.0022	51	0.0022	60	0.0018	37	0.0035	32	0.0030	33	0.0440	26
322	deepsense-008	-		-		0.0022	54	0.0022	63	0.0018	35	0.0034	31	0.0030	31	0.0434	21
323	dermalog-001	-		0.3049	971	-		-		-		-		-		-	
324	dermalog-002	-		0.3154	974	-		-		-		-		-		-	
325	dermalog-003	0.2355	976	0.2795	964	-		-		-		-		-		-	
326	dermalog-004	0.1810	959	0.2399	951	-		-		-		-		-		-	
327	dermalog-005	0.1526	947	0.1823	926	0.2580	1243	0.4018	1244	-		0.2651	886	0.1585	975	-	
328	dermalog-006	0.0253	784	0.0369	768	0.0171	1029	0.0283	1019	0.0217	918	0.0358	628	0.0230	713	-	
329	dermalog-007	0.0564	872	0.0798	862	0.1351	1217	0.2471	1219	0.2343	1175	0.2223	859	0.1397	959	-	
330	dermalog-008	0.0096	556	0.0166	588	0.0086	841	0.0133	833	0.0165	841	0.0586	712	0.0226	712	-	
331	dermalog-009	0.0067	445	0.0094	432	0.0051	648	0.0069	635	0.0116	746	0.0312	583	0.0177	643	0.3747	515
332	dermalog-010	0.0030	189	0.0041	173	0.0034	457	0.0037	372	0.0075	589	0.5181	971	0.2530	1017	0.8853	665
333	dermalog-011	0.0045	306	0.0062	307	0.0035	472	0.0059	575	0.0057	509	0.2242	861	0.0407	820	0.6329	583
334	dermalog-012	0.0080	503	0.0101	463	0.0037	509	0.0057	561	0.0055	491	0.0095	291	0.0068	290	0.0834	233
335	dermalog-013	-		-		0.0036	488	0.0057	558	0.0058	512	0.0103	313	0.0073	322	0.0894	250
336	dermalog-014	-		-		0.0029	346	0.0039	395	0.0037	328	0.0069	181	0.0057	224	0.0646	141
337	dermalog-015	-		-		0.0025	196	0.0025	119	0.0033	272	0.0064	160	0.0054	207	0.0587	101
338	dermalog-016	-		-		0.0023	97	0.0023	80	0.0025	154	0.0050	101	0.0040	102	0.0550	77
339	dicio-001	0.5486	1038	0.6442	1039	0.7516	1315	0.8607	1309	0.8678	1264	0.8268	1063	0.7034	1129	0.9752	711
340	dicio-002	-		-		0.0057	696	0.0095	742	0.0097	692	0.9998	1236	0.9989	1259	1.0000	850
341	didiglobalface-001	0.0055	372	0.0092	426	0.0030	361	0.0045	459	0.0088	642	0.0119	362	0.0085	396	-	
342	didiglobalface-002	0.0033	226	0.0051	248	0.0026	249	0.0034	328	0.0033	279	0.0085	249	0.0047	152	0.1135	320
343	didiglobalface-003	-		-		0.0023	78	0.0025	141	0.0027	186	0.1115	789	0.0052	192	0.5374	554
344	digidata-000	0.0967	914	0.1410	909	0.2596	1244	0.3462	1238	0.0293	981	0.0363	636	0.0212	696	0.1849	428
345	digidata-001	0.0224	763	0.0352	757	0.0330	1113	0.0570	1113	0.0109	729	0.0481	682	0.0123	526	0.3055	494
346	digitalbarriers-000	0.3683	1003	0.4630	1006	-		-		-		-		-		-	
347	digitalbarriers-001	0.4683	1024	0.5024	1015	-		-		-		-		-		-	
348	digitalbarriers-002	0.3360	998	0.3690	988	0.0877	1193	0.1557	1193	0.0971	1116	0.0951	771	0.0497	850	-	
349	dps-000	0.0115	607	0.0176	606	0.0149	1011	0.0185	941	0.0173	859	0.0275	561	0.0180	647	0.1757	418
350	dralmey-001	-		-		0.0036	485	0.0052	524	0.0057	504	0.9985	1206	0.9872	1226	1.0000	848
351	dsk-000	0.1526	946	0.2169	942	0.3787	1262	0.5426	1263	0.3115	1189	0.3089	902	0.1994	998	-	
352	einetworks-000	0.0099	563	0.0180	611	0.0088	849	0.0140	860	0.0130	770	0.0225	520	0.0147	586	-	

Table 63: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
353	einetworksindia-000	0.0058	393	0.0074	356	0.0051	647	0.0061	598	0.0074	585	0.0193	483	0.0102	459	0.1819	422
354	einetworksindia-001	0.0042	290	0.0062	298	0.0048	618	0.0064	611	0.0057	507	0.0473	679	0.0092	418	0.3984	526
355	einetworksindia-002	-		-		0.0024	143	0.0027	166	0.0023	91	0.1534	821	0.0513	854	0.4614	537
356	ekin-002	0.1168	929	0.2042	938	0.1530	1224	0.2524	1222	0.1777	1151	0.2773	891	0.1347	954	0.8954	673
357	element-000	0.0023	135	0.0035	133	0.0024	154	0.0026	148	0.0029	216	0.0060	144	0.0048	156	0.0631	129
358	enface-000	0.0028	178	0.0049	234	0.0043	582	0.0072	649	0.0058	517	0.0150	407	0.0090	411	0.7851	623
359	enface-001	0.0072	473	0.0107	475	0.0071	766	0.0138	851	0.0068	559	0.0515	692	0.0094	431	0.9942	746
360	enface-002	0.0033	220	0.0052	256	0.0038	524	0.0073	657	0.0026	178	0.9998	1242	0.9995	1267	1.0000	886
361	enqura-001	-		-		0.0155	1017	0.0144	869	0.0312	991	0.0597	714	0.0485	847	0.2970	492
362	enqura-002	-		-		0.0082	823	0.0150	881	0.0105	713	0.0193	480	0.0111	488	0.1302	350
363	eocortex-000	0.3485	1000	0.6943	1048	0.1122	1206	0.1574	1194	0.2155	1169	0.2257	865	0.1606	980	-	
364	ercacat-001	0.0036	248	0.0044	202	0.0033	434	0.0047	474	0.0106	717	0.0202	492	0.0184	653	-	
365	ethiopianlmis-000	-		-		0.2967	1248	0.4098	1245	0.9992	1301	1.0000	1257	0.9999	1279	1.0000	909
366	euronovate-001	0.2786	985	0.3608	987	0.4489	1273	0.6105	1272	0.5010	1214	0.5392	979	0.3769	1060	0.8802	661
367	euronovate-002	0.0256	787	0.0436	788	0.0226	1069	0.0425	1074	0.0301	986	0.0691	739	0.0319	781	0.3687	511
368	euronovate-003	0.0061	415	0.0088	414	0.0040	552	0.0060	584	0.0065	551	0.7815	1045	0.5034	1091	0.9823	717
369	euronovate-004	0.0082	508	0.0110	484	0.0050	642	0.0077	679	0.0074	586	0.6949	1019	0.3354	1051	0.9791	715
370	everai-000	0.0122	625	0.0225	670	-		-		-		-		-		-	
371	everai-001	0.0085	518	0.0156	570	0.0072	776	0.0114	789	-		-		-		-	
372	everai-002	0.0104	581	0.0159	575	0.0063	734	0.0112	785	0.0182	872	0.0353	625	0.0264	750	-	
373	paravision-003	0.0034	237	0.0050	244	0.0036	493	0.0052	523	0.0092	667	0.0193	482	0.0156	601	-	
374	expasoft-000	0.0427	845	0.0655	842	0.0239	1076	0.0393	1060	0.0673	1086	0.8963	1087	0.3832	1064	-	
375	expasoft-001	0.0328	825	0.0488	808	0.0211	1057	0.0342	1041	0.0629	1081	0.6483	1007	0.2816	1032	-	
376	expasoft-002	0.0170	710	0.0274	713	0.0787	1187	0.0768	1142	0.1629	1143	0.9996	1228	0.9631	1208	1.0000	853
377	f8-001	0.0249	781	0.0336	747	0.0178	1032	0.0232	985	0.0303	987	0.0615	720	0.0408	821	-	
378	f8-002	0.0340	828	0.0591	833	0.0213	1059	0.0374	1055	0.0452	1043	0.0760	748	0.0502	853	0.2708	482
379	facehawk-000	1.0000	1217	1.0000	1397	1.0000	1367	1.0000	1430	1.0000	1405	1.0000	1425	1.0000	1360	1.0000	1148
380	facehawk-001	-		-		0.0545	1162	0.0748	1138	0.0751	1093	0.9919	1163	0.9566	1202	1.0000	831
381	faceint-001	-		-		0.0105	908	0.0124	809	0.0313	992	0.3318	911	0.0557	867	0.9851	728
382	facelocate-001	-		-		0.0664	1177	0.1171	1178	0.0950	1113	0.9219	1098	0.7041	1130	0.9985	768
383	facelocate-002	-		-		0.0335	1114	0.0634	1120	0.0393	1020	0.9988	1212	0.8977	1177	1.0000	833
384	faceonlive-001	0.0269	794	0.0359	763	0.0387	1128	0.0721	1135	0.0246	951	0.0349	622	0.0220	703	0.2408	463
385	faceonlive-002	0.0121	622	0.0135	530	0.0033	439	0.0041	412	0.0037	330	0.9427	1111	0.7927	1152	0.9993	777
386	faceonlive-003	-		-		0.0045	594	0.0050	505	0.0193	894	0.0360	632	0.0356	798	0.1320	351
387	facephi-000	0.0044	300	0.0059	285	0.0047	611	0.0057	562	0.0088	648	1.0000	1293	1.0000	1296	1.0000	903
388	facephi-001	-		-		0.0031	392	0.0037	367	0.0058	515	0.0111	337	0.0098	445	0.1048	299
389	facephi-002	-		-		0.0024	119	0.0022	57	0.0036	317	0.0093	280	0.0070	301	0.1136	321
390	facesoft-000	0.0085	516	0.0112	489	0.0064	738	0.0107	773	0.0091	658	0.0171	450	0.0107	475	-	
391	facetag-000	0.2836	986	0.4081	997	0.2933	1247	0.4303	1248	0.3448	1195	0.6312	1002	0.3530	1054	0.9753	712
392	facetag-001	0.2920	988	0.4538	1003	0.3220	1254	0.4371	1249	1.0000	1454	1.0000	1411	0.3254	1047	1.0000	922
393	facetag-002	0.0098	559	0.0147	550	0.0064	743	0.0110	781	0.0116	744	0.0190	478	0.0119	516	0.1334	357
394	facex-001	1.0000	1376	1.0000	1201	1.0000	1354	1.0000	1355	1.0000	1332	1.0000	1359	1.0000	1307	-	
395	facex-002	0.0803	904	0.1404	907	0.1283	1214	0.1979	1210	0.1440	1136	0.1952	843	0.1299	949	0.7427	608
396	facia-001	0.0083	512	0.0124	509	0.0064	741	0.0097	747	0.0098	694	0.9983	1202	0.9849	1223	1.0000	875

Table 64: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
397	facia-002	-		-		0.0072	775	0.0109	779	0.0122	754	0.0460	673	0.0135	552	0.4464	535
398	faicetech-000	-		-		0.0026	264	0.0031	262	0.0023	102	0.0116	356	0.0037	72	0.0904	257
399	farfaces-001	0.4890	1031	0.5860	1035	0.5650	1285	0.7268	1285	0.8015	1247	0.7511	1032	0.5892	1105	0.9536	692
400	fastenterprises-000	0.0093	544	0.0151	560	0.0098	885	0.0140	861	0.0209	908	0.2991	896	0.1608	981	0.6727	591
401	fci-000	-		-		0.0027	278	0.0027	174	0.0028	200	0.1269	804	0.0563	868	0.3886	521
402	fdu-000	0.0386	838	0.0576	829	-		-		-		-		-		-	
403	fdu-001	0.0347	831	0.0439	791	-		-		-		-		-		-	
404	fedu-001	-		-		0.0033	428	0.0047	468	0.0128	764	0.9816	1139	0.9216	1186	0.9999	801
405	fiberhome-nanjing-002	0.0217	758	0.0381	771	0.0874	1192	0.1770	1201	0.0271	964	0.0351	624	0.0188	661	-	
406	fiberhome-nanjing-003	0.0090	528	0.0139	537	0.0082	824	0.0144	868	0.0110	731	0.0174	452	0.0107	477	-	
407	fiberhome-nanjing-004	0.0037	258	0.0056	276	0.0031	387	0.0043	439	0.0043	394	0.0083	240	0.0061	252	-	
408	fincore-000	0.0309	815	0.0502	810	0.0281	1094	0.0510	1101	0.0521	1060	0.0815	756	0.0522	856	0.3100	496
409	firstcreditkz-001	0.0024	143	0.0034	122	0.0024	162	0.0024	116	0.0034	287	1.0000	1419	1.0000	1315	1.0000	961
410	firstcreditkz-002	0.0018	86	0.0026	76	0.0024	145	0.0024	104	0.0029	223	0.0056	120	0.0049	163	0.0624	122
411	firstcreditkz-003	-		-		0.0023	104	0.0024	109	0.0026	168	0.0104	319	0.0047	149	0.0910	262
412	firstcreditkz-004	-		-		0.0023	86	0.0024	96	0.0024	112	0.0360	633	0.0061	251	0.2315	459
413	foomobi-001	0.4827	1028	0.5795	1031	0.6823	1299	0.8132	1299	0.8217	1253	1.0000	1455	1.0000	1353	1.0000	1098
414	foomobi-002	0.4827	1029	0.5795	1030	0.6823	1300	0.8132	1298	0.8217	1252	1.0000	1334	1.0000	1401	1.0000	1253
415	fourthlinecom-001	-		-		0.0442	1143	0.0696	1131	0.0609	1073	0.8783	1079	0.7523	1143	0.9917	740
416	fpt-000	-		-		0.0213	1060	0.0375	1056	0.0364	1009	0.9528	1115	0.8729	1167	0.9993	780
417	fpt-001	-		-		0.0099	888	0.0148	877	0.8158	1250	0.9986	1207	0.9934	1236	1.0000	907
418	fpttelecom-000	-		-		0.0586	1166	0.0761	1141	0.2308	1174	0.9988	1211	0.9937	1237	1.0000	865
419	fraudcom-000	0.0110	595	0.0173	597	0.0083	828	0.0139	856	0.0166	843	0.0258	550	0.0177	640	0.1488	387
420	frpkauai-001	0.0023	128	0.0035	131	0.0026	222	0.0030	250	0.0040	361	0.0080	223	0.0072	317	0.0706	167
421	frpkauai-002	0.0024	144	0.0035	128	0.0024	159	0.0024	112	0.0033	284	0.0065	166	0.0058	235	0.0599	104
422	frpkauai-003	-		-		0.0023	95	0.0023	89	0.0024	105	0.0075	196	0.0039	93	0.0762	196
423	frpkauai-004	-		-		0.0037	498	0.0036	359	0.0069	564	0.0226	521	0.0138	561	0.1354	362
424	frpkauai-005	-		-		0.0023	66	0.0023	73	0.0021	71	0.0082	230	0.0042	118	0.0760	195
425	fujitsulab-000	0.0123	627	0.0212	654	0.0091	856	0.0133	836	0.0251	953	0.4200	948	0.0360	804	-	
426	fujitsulab-001	0.0059	402	0.0082	391	0.0132	977	0.0258	1002	0.0140	795	0.0246	537	0.0142	573	-	
427	fujitsulab-002	0.0091	538	0.0124	512	0.0105	905	0.0156	890	0.0169	852	0.0345	620	0.0146	585	-	
428	fujitsulab-003	0.0045	308	0.0065	315	0.0057	698	0.0083	705	0.0080	613	0.0154	417	0.0101	451	0.5564	560
429	g42-intellibrain-001	0.0006	7	0.0009	10	0.0037	502	0.0044	448	0.0030	243	0.0059	134	0.0053	201	0.0750	192
430	g42-intellibrain-002	0.0006	10	0.0011	11	0.0029	337	0.0037	370	0.0028	196	0.0212	501	0.0091	414	0.1398	369
431	geo-000	0.0543	866	0.0814	864	0.1042	1202	0.1592	1195	0.1637	1144	0.8533	1073	0.1800	990	-	
432	geo-001	0.0180	724	0.0198	641	0.0037	503	0.0055	542	0.0070	567	0.0129	375	0.0094	427	-	
433	geo-002	0.0171	713	0.0187	623	0.0035	465	0.0051	516	0.0064	542	0.0117	359	0.0083	389	0.0938	276
434	geo-003	0.0180	725	0.0313	735	0.0239	1078	0.0552	1105	0.0319	995	0.0487	685	0.0222	706	0.1962	440
435	geo-004	0.0030	187	0.0041	172	0.0025	204	0.0030	231	0.0035	310	0.0065	164	0.0053	200	0.0648	143
436	gistouch-000	0.0296	808	0.0518	815	0.0342	1119	0.0609	1117	0.0493	1052	1.0000	1295	1.0000	1306	1.0000	913
437	gistouch-001	0.0176	720	0.0312	734	0.8931	1330	0.9735	1328	0.8625	1260	1.0000	1412	1.0000	1311	1.0000	923
438	glory-000	0.1094	925	0.1286	898	0.2179	1237	0.2656	1225	-		-		-		-	
439	glory-001	0.0902	910	0.1082	888	0.1642	1226	0.2065	1215	0.2186	1170	0.2669	887	0.2089	1001	-	
440	glory-002	0.0241	769	0.0311	733	0.0116	946	0.0151	885	0.0157	825	0.0264	552	0.0188	663	-	

Table 65: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
441	glory-003	0.0076	486	0.0125	516	0.0077	800	0.0103	767	0.0130	769	0.0205	494	0.0143	576	-	
442	glory-004	0.0077	488	0.0123	508	0.0074	790	0.0098	755	0.0122	757	0.0193	481	0.0134	549	0.1520	390
443	glory-005	0.0056	376	0.0076	364	0.0054	675	0.0072	652	0.0075	590	0.9237	1100	0.7853	1149	0.9974	764
444	glory-006	0.0050	335	0.0068	326	0.0051	650	0.0070	641	0.0069	566	0.7869	1046	0.5076	1092	0.9719	705
445	glory-007	0.0050	337	0.0068	330	0.0050	640	0.0069	637	0.0068	556	0.0126	373	0.0083	388	0.1066	301
446	gorilla-000	0.0802	903	0.1409	908	-		-		-		-		-		-	
447	gorilla-001	0.0488	857	0.0756	855	0.1218	1212	0.2034	1212	-		-		-		-	
448	gorilla-002	0.0256	788	0.0413	779	0.0478	1148	0.0912	1160	-		-		-		-	
449	gorilla-003	0.0165	704	0.0291	720	0.0205	1054	0.0437	1075	0.0297	984	0.9658	1125	0.0262	745	-	
450	gorilla-004	0.0138	660	0.0239	686	0.0243	1080	0.0444	1077	0.0275	970	0.0369	639	0.0204	688	-	
451	gorilla-005	-		-		0.0142	994	0.0267	1012	0.0228	927	0.0358	627	0.0195	674	-	
452	gorilla-006	0.0105	587	0.0152	561	0.0106	909	0.0203	961	0.0155	818	0.0218	510	0.0136	554	-	
453	gorilla-007	0.0074	480	0.0111	487	0.0065	746	0.0126	814	0.0100	704	0.0151	409	0.0102	455	0.1450	380
454	gorilla-008	0.0058	395	0.0091	421	0.0049	628	0.0079	693	0.0079	610	0.0126	372	0.0091	415	0.1217	339
455	gorilla-009	0.0049	326	0.0072	344	0.0038	519	0.0056	552	0.0065	546	0.0104	315	0.0070	302	0.1103	309
456	gpstechvn-000	0.0202	747	0.0355	760	0.0142	992	0.0248	995	0.0208	907	0.0495	687	0.0207	691	0.5098	550
457	graymatics-001	0.1039	919	0.1620	918	0.1344	1216	0.1917	1206	0.1648	1145	0.5160	969	0.2689	1027	0.9194	682
458	griaule-000	0.0071	466	0.0099	450	0.0050	635	0.0072	647	0.0160	829	0.0304	579	0.0267	752	0.1480	385
459	griaule-001	0.0057	382	0.0078	374	0.0045	600	0.0065	617	0.0070	570	0.7515	1033	0.5106	1093	0.9747	709
460	griaule-002	0.0021	117	0.0032	113	0.0025	216	0.0027	182	0.0034	299	0.0996	777	0.0201	682	0.3978	524
461	gtelots-000	-		-		0.0059	706	0.0098	751	0.0090	654	0.9995	1225	0.9975	1249	1.0000	862
462	gtelots-001	-		-		0.0053	662	0.0102	765	0.0071	571	0.0111	341	0.0070	300	0.1173	326
463	hertasecurity-000	0.0630	880	0.0780	861	0.0503	1154	0.0898	1155	0.0738	1090	0.0693	742	0.0420	826	-	
464	hertasecurity-001	0.0249	779	0.0309	732	0.0105	906	0.0161	904	0.0245	949	0.0447	667	0.0359	802	0.1850	429
465	hertasecurity-002	0.0206	750	0.0315	736	0.0060	712	0.0078	687	0.0253	956	0.0696	743	0.0457	837	0.4565	536
466	hertasecurity-003	0.0079	501	0.0104	470	0.0060	714	0.0078	683	0.0255	957	0.0732	746	0.0460	838	0.4731	540
467	hik-001	0.0096	555	0.0125	513	0.0093	865	0.0164	914	0.0108	724	0.0937	767	0.0127	533	-	
468	hisign-001	0.0036	251	0.0050	241	0.0034	446	0.0046	463	0.0079	609	0.0153	416	0.0133	546	0.1080	303
469	hisign-002	0.0029	185	0.0044	200	0.0027	302	0.0032	290	0.0028	206	0.0409	657	0.0132	544	0.2016	444
470	hisign-003	-		-		0.0030	359	0.0036	349	0.0036	320	0.0582	710	0.0201	683	0.2584	474
471	hyperverge-001	1.0000	1142	1.0000	1370	1.0000	1406	1.0000	1443	1.0000	1452	1.0000	1413	1.0000	1312	-	
472	hyperverge-002	0.0050	332	0.0066	320	0.0035	464	0.0051	510	0.0062	536	0.0107	327	0.0074	334	0.0921	268
473	hyperverge-003	0.0019	96	0.0030	101	0.0025	186	0.0029	220	0.0027	181	0.0049	99	0.0042	122	0.0576	93
474	hyperverge-004	0.0072	470	0.0116	499	0.0040	545	0.0071	646	0.0058	514	0.0080	221	0.0057	227	0.0621	119
475	hyperverge-005	0.0026	161	0.0039	164	0.0027	273	0.0031	271	0.0036	322	0.0066	168	0.0055	211	0.0658	150
476	hyperverge-006	-		-		0.0025	213	0.0030	236	0.0033	278	0.0060	143	0.0051	184	0.0601	106
477	hyperverge-007	-		-		0.0026	231	0.0030	246	0.0034	296	0.0061	146	0.0052	187	0.0605	109
478	hyundai-roboticslab-000	-		-		0.0037	500	0.0052	520	0.0063	540	1.0000	1329	0.0872	911	1.0000	1236
479	hyundai-roboticslab-001	-		-		0.0048	626	0.0066	624	0.0193	892	0.8620	1074	0.6861	1124	0.9861	730
480	hyundai-roboticslab-002	-		-		0.0039	543	0.0054	537	0.0086	637	0.9589	1118	0.8817	1172	0.9986	770
481	hyundai-roboticslab-003	-		-		0.0024	123	0.0028	191	0.0027	189	0.6384	1005	0.2934	1036	0.8692	652
482	hyundai-roboticslab-004	-		-		0.0023	112	0.0028	189	0.0028	209	0.2635	883	0.0802	900	0.5392	555
483	hzailu-001	0.0122	623	0.0164	584	0.0095	875	0.0196	953	0.0079	607	0.0118	360	0.0090	409	0.1173	327
484	hzailu-002	0.0051	345	0.0072	342	0.0038	531	0.0055	547	0.0040	358	0.0077	201	0.0062	257	0.0928	275

Table 66: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
485	hzailu-003	0.0178	722	0.0291	719	0.0031	403	0.0042	435	0.0035	301	0.0061	149	0.0052	186	0.0799	215
486	hzailu-004	0.0175	717	0.0197	639	0.0031	405	0.0042	430	0.0034	295	0.0060	145	0.0051	176	0.0775	204
487	hzailu-005	0.0172	715	0.0197	638	0.0028	326	0.0036	354	0.0032	269	0.0058	126	0.0048	157	0.0751	193
488	hzailu-006	0.0161	701	0.0174	599	0.0024	151	0.0027	161	0.0026	169	0.0049	98	0.0044	138	0.0654	147
489	hzailu-007	-		-		0.0025	178	0.0027	178	0.0026	159	0.0048	88	0.0044	135	0.0653	146
490	hzailu-008	-		-		0.0024	170	0.0026	160	0.0025	151	0.0048	86	0.0044	133	0.0652	145
491	i2v-001	-		-		0.0946	1197	0.1369	1185	0.3989	1205	0.9998	1238	0.9966	1245	1.0000	859
492	i2v-002	-		-		0.0035	468	0.0044	452	0.0037	325	0.9355	1109	0.4823	1084	0.9960	754
493	icehouse-000	-		-		0.0027	274	0.0036	357	0.0047	431	0.0088	265	0.0072	313	0.0726	173
494	icm-002	0.0143	670	0.0249	695	0.0144	999	0.0256	1001	0.0236	941	0.0386	650	0.0263	749	-	
495	icm-003	0.0138	662	0.0222	668	0.0149	1009	0.0282	1018	0.0227	926	0.0384	648	0.0257	739	0.1938	435
496	icm-004	0.0079	498	0.0120	505	0.0074	787	0.0107	772	0.0091	660	0.0281	566	0.0128	537	0.2778	486
497	icm-005	0.0100	566	0.0140	539	0.0091	852	0.0108	774	0.0169	851	0.0319	594	0.0290	769	0.1775	421
498	icomm-000	0.0019	97	0.0033	118	0.0033	438	0.0044	449	0.0031	259	0.1350	810	0.0552	864	0.4196	529
499	icthtc-000	0.0260	790	0.0396	777	0.0207	1055	0.0339	1040	0.0291	980	0.0474	680	0.0346	793	-	
500	id3-001	0.1698	953	0.2499	953	-		-		-		-		-		-	
501	id3-002	0.1615	952	0.2393	950	-		-		-		-		-		-	
502	id3-003	0.0361	835	0.0757	857	0.0292	1101	0.0476	1090	0.0447	1041	0.0674	734	0.0396	815	-	
503	id3-004	0.0198	744	0.0344	754	0.0238	1075	0.0423	1073	0.0289	979	0.0416	659	0.0257	738	-	
504	id3-005	0.0104	580	0.0169	593	0.0080	815	0.0133	837	0.0194	895	0.0334	611	0.0262	746	-	
505	id3-006	0.0072	472	0.0103	466	0.0049	631	0.0074	662	0.0095	676	0.0165	439	0.0119	515	-	
506	id3-007	0.0056	377	0.0078	371	0.0060	717	0.0072	656	0.0275	968	0.1374	814	0.0519	855	-	
507	id3-008	0.0039	265	0.0055	272	0.0032	419	0.0042	426	0.0081	618	0.0155	419	0.0134	547	0.1186	330
508	id3-009	-		-		0.0035	469	0.0049	497	0.0046	418	0.4689	958	0.0214	700	0.9377	687
509	id3-010	-		-		0.0032	425	0.0045	456	0.0042	381	0.8447	1071	0.2140	1005	0.9932	744
510	idemia-003	0.0222	760	0.0316	737	0.0188	1035	0.0325	1037	0.0548	1065	-		-		-	
511	idemia-004	0.0160	698	0.0244	690	0.0199	1046	0.0354	1046	0.0268	959	0.1489	820	0.0263	748	-	
512	idemia-005	0.0132	650	0.0216	660	0.0121	957	0.0218	973	0.0215	914	0.0323	598	0.0188	662	-	
513	idemia-006	0.0046	316	0.0062	304	0.0047	613	0.0066	619	0.0073	581	0.2882	893	0.0094	432	-	
514	idemia-007	0.0024	139	0.0039	163	0.0032	426	0.0038	382	0.0046	422	0.0092	277	0.0070	309	-	
515	idemia-008	0.0023	131	0.0032	114	0.0023	109	0.0028	185	0.0034	300	0.0067	173	0.0056	222	0.0765	198
516	idemia-009	0.0022	123	0.0030	102	0.0022	59	0.0023	91	0.0023	90	0.0046	73	0.0039	88	0.0533	65
517	idemia-010	0.0015	64	0.0026	71	0.0022	63	0.0020	7	0.0020	55	0.0040	47	0.0033	44	0.0465	34
518	idemia-011	-		-		0.0021	28	0.0021	37	0.0018	41	0.0077	206	0.0038	81	0.0782	209
519	identy-000	0.0073	474	0.0095	438	0.0050	636	0.0067	625	0.0071	574	0.8257	1062	0.4310	1076	0.9839	723
520	identy-001	-		-		0.0029	345	0.0034	310	0.0037	331	0.0066	169	0.0055	213	0.0748	191
521	identy-002	-		-		0.0027	283	0.0029	230	0.0030	226	0.0058	127	0.0051	179	0.0691	162
522	identy-003	-		-		0.0025	210	0.0030	238	0.0026	156	0.0050	104	0.0043	123	0.0547	74
523	idrnd-000	-		-		0.0041	565	0.0053	533	0.0050	452	0.0094	283	0.0075	336	0.1026	293
524	idrnd-001	-		-		0.0025	217	0.0033	301	0.0032	265	0.0528	697	0.0173	635	0.2243	454
525	igearx-face-000	0.0091	533	0.0146	549	0.0163	1024	0.0362	1051	0.0399	1022	0.6436	1006	0.3305	1049	0.7832	622
526	iit-000	0.1516	945	0.1981	935	0.0828	1190	0.1442	1190	0.2214	1171	0.2943	894	0.1877	993	-	
527	iit-001	0.0104	576	0.0179	610	0.0099	891	0.0142	865	0.1222	1125	0.2677	888	0.2171	1007	-	
528	iit-002	0.0111	600	0.0177	609	0.0085	837	0.0140	859	0.0193	891	0.0332	608	0.0260	742	-	

Table 67: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
529	iit-003	0.0082	511	0.0151	559	0.0053	663	0.0084	710	0.0122	756	0.0199	490	0.0137	559	-	
530	imds-software-001	0.0126	641	0.0228	672	0.0130	973	0.0221	975	0.0231	929	0.0469	678	0.0199	681	0.4260	531
531	imds-software-002	0.0048	324	0.0072	347	0.0036	483	0.0052	525	0.0047	435	0.9981	1199	0.0078	360	1.0000	864
532	imds-software-003	0.0032	205	0.0047	224	0.0028	309	0.0036	356	0.0036	321	0.9987	1208	0.0063	260	1.0000	852
533	imperial-000	0.0067	446	0.0108	480	0.0080	813	0.0134	841	0.0087	639	0.0581	709	0.0102	460	-	
534	imperial-001	0.0094	546	0.0154	566	0.0072	774	0.0120	798	0.0103	708	0.0190	476	0.0120	519	-	
535	imperial-002	0.0058	391	0.0081	387	0.0055	680	0.0085	713	0.0083	624	0.0157	421	0.0103	461	-	
536	incode-000	0.0766	896	0.1029	887	-		-		-		-		-		-	
537	incode-001	0.0550	869	0.0773	859	-		-		-		-		-		-	
538	incode-002	0.0293	804	0.0548	825	0.0436	1141	0.0785	1144	-		-		-		-	
539	incode-003	0.0142	669	0.0249	696	0.0448	1144	0.0869	1151	0.0354	1004	0.0661	729	0.0320	782	-	
540	incode-004	0.0077	490	0.0132	525	0.0096	880	0.0160	900	0.0171	855	0.0281	564	0.0179	645	-	
541	incode-005	0.0111	599	0.0189	625	0.0156	1019	0.0304	1028	0.0195	898	0.0307	582	0.0172	631	-	
542	incode-006	0.0156	689	0.0273	712	0.0117	947	0.0238	991	0.0202	903	0.0339	617	0.0164	616	-	
543	incode-007	0.0109	592	0.0155	568	0.0056	690	0.0099	756	0.0107	719	0.0168	444	0.0102	456	-	
544	incode-008	0.0063	427	0.0101	461	0.0046	605	0.0086	719	0.0057	505	0.0104	316	0.0074	327	-	
545	incode-009	0.0044	301	0.0067	325	0.0034	451	0.0051	509	0.0049	450	0.0091	271	0.0067	284	0.0892	248
546	incode-010	0.0041	280	0.0063	310	0.0028	330	0.0043	437	0.0047	439	0.0077	208	0.0061	250	0.0770	202
547	incode-011	0.0032	206	0.0044	201	0.0026	256	0.0034	323	0.0032	260	0.0359	629	0.0140	568	0.1840	427
548	incode-012	0.0027	167	0.0043	188	0.0025	189	0.0035	337	0.0034	292	0.0966	775	0.0385	810	0.3387	504
549	incode-013	0.0035	242	0.0049	231	0.0032	410	0.0039	393	0.0051	459	0.0094	284	0.0081	378	0.0768	200
550	incode-014	-		-		0.0022	60	0.0023	85	0.0034	291	0.0066	171	0.0062	258	0.0569	88
551	incode-015	-		-		0.0021	14	0.0022	62	0.0020	58	0.0040	49	0.0037	69	0.0537	69
552	infocert-001	0.0105	583	0.0172	596	0.0078	802	0.0125	811	0.0159	827	0.1573	824	0.0565	869	0.5716	565
553	infocert-002	-		-		1.0000	1395	0.9993	1340	1.0000	1442	1.0000	1393	1.0000	1324	1.0000	995
554	infocert-003	-		-		0.0202	1047	0.0284	1020	0.0620	1079	0.2624	882	0.1316	952	0.6450	585
555	innefulabs-000	0.0122	626	0.0199	643	0.0112	931	0.0197	956	0.0222	921	0.0372	643	0.0271	754	-	
556	innominds-001	0.5569	1039	0.6541	1042	0.7642	1319	0.8665	1312	0.7629	1238	0.8200	1057	0.6359	1116	0.9841	725
557	innov8tif-000	-		-		0.1842	1229	0.3028	1235	0.2056	1163	0.2348	870	0.1449	962	0.6142	578
558	innov8tif-001	-		-		0.1865	1231	0.2951	1229	0.1976	1158	0.2564	880	0.1593	976	0.6665	588
559	innovativetechnologyltd-001	0.0578	873	0.0938	880	0.0501	1153	0.0981	1162	0.0592	1072	0.0779	753	0.0422	827	-	
560	innovativetechnologyltd-002	0.0451	851	0.0716	851	0.0541	1160	0.1009	1168	0.0506	1055	0.0682	736	0.0371	807	-	
561	innovatrics-000	0.1105	926	0.1905	931	-		-		-		-		-		-	
562	innovatrics-001	0.1077	924	0.1828	928	-		-		-		-		-		-	
563	innovatrics-002	0.4049	1009	0.6142	1036	-		-		-		-		-		-	
564	innovatrics-003	0.0349	832	0.0483	806	-		-		-		-		-		-	
565	innovatrics-004	0.0194	740	0.0292	721	0.0344	1120	0.0617	1118	0.0562	1067	0.0762	749	0.0412	822	-	
566	innovatrics-005	0.0230	766	0.0353	758	0.0398	1132	0.0721	1136	-		-		-		-	
567	innovatrics-006	0.0058	389	0.0089	417	0.0061	725	0.0096	745	0.0096	686	0.0165	440	0.0103	462	-	
568	innovatrics-007	0.0040	271	0.0054	268	0.0057	693	0.0078	682	0.0079	608	0.0123	365	0.0088	404	-	
569	innovatrics-008	0.0047	322	0.0064	312	0.0038	527	0.0052	522	0.0053	477	0.0088	262	0.0069	298	0.0899	254
570	innovatrics-009	0.0022	120	0.0031	109	0.0028	308	0.0032	287	0.0034	290	0.1165	794	0.0326	783	0.3866	520
571	innovatrics-010	0.0017	84	0.0031	110	0.0027	298	0.0028	190	0.0041	366	0.0078	210	0.0066	280	0.0744	184
572	innovatrics-011	-		-		0.0026	257	0.0027	164	0.0032	267	0.0060	140	0.0044	134	0.0615	116

Table 68: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
573	innovatrics-012	-		-		0.0025	193	0.0026	145	0.0034	294	0.9309	1107	0.4977	1087	0.9960	753
574	innovatrics-013	-		-		0.0022	40	0.0021	39	0.0026	160	0.9984	1203	0.3780	1063	1.0000	856
575	innovatrics-014	-		-		0.0021	23	0.0020	5	0.0017	29	0.0033	23	0.0029	19	0.0417	15
576	innovatrics-015	-		-		0.0021	13	0.0020	20	0.0019	49	0.0036	35	0.0031	36	0.0468	35
577	innovatrics-016	-		-		0.0021	17	0.0020	21	0.0018	34	0.0034	29	0.0029	25	0.0451	29
578	innovatrics-017	-		-		0.0020	8	0.0021	34	0.0017	27	0.0033	19	0.0029	21	0.0437	23
579	insightface-000	0.0018	92	0.0027	86	0.0029	338	0.0030	253	0.0038	337	0.0077	205	0.0068	288	0.0765	197
580	insightface-001	0.0009	17	0.0014	19	0.0027	277	0.0024	114	0.0035	309	0.0070	183	0.0065	270	0.0692	163
581	insightface-002	0.0011	39	0.0019	40	0.0027	296	0.0026	150	0.0036	319	0.0069	179	0.0065	269	0.0694	164
582	insightface-003	0.0015	62	0.0021	48	0.0045	598	0.0034	327	0.1298	1127	1.0000	1299	0.9407	1192	1.0000	1276
583	insightface-004	0.0015	60	0.0023	58	0.0052	656	0.0036	355	0.1403	1134	0.9923	1165	0.9128	1182	0.9989	772
584	inspur-000	0.0060	411	0.0078	369	0.7415	1314	0.9093	1321	0.2838	1185	0.9996	1226	0.9976	1250	1.0000	899
585	inspur-001	0.0035	238	0.0049	232	0.0025	191	0.0030	241	0.0029	221	0.9951	1177	0.9592	1205	1.0000	837
586	inspur-002	0.0025	150	0.0039	159	0.0024	121	0.0027	163	0.0031	247	0.0062	151	0.0047	150	0.1112	313
587	inspur-003	-		-		0.0023	90	0.0026	154	0.0031	249	0.0060	142	0.0046	145	0.0812	224
588	intellicloudai-001	0.0142	668	0.0234	680	0.0092	860	0.0145	871	0.0162	834	0.0371	642	0.0171	630	-	
589	intellicloudai-002	0.0059	403	0.0085	400	0.0060	715	0.0069	638	0.0108	722	0.2477	878	0.0171	629	-	
590	intellifusion-001	0.0072	469	0.0094	434	0.0056	692	0.0085	714	0.0111	735	0.0212	502	0.0143	575	-	
591	intellifusion-002	0.0059	398	0.0077	365	0.0040	546	0.0074	661	0.0085	633	0.5352	978	0.0104	469	-	
592	intellivision-001	0.1335	938	0.2205	945	0.1090	1204	0.1670	1197	0.1385	1131	0.1676	829	0.1170	939	-	
593	intellivision-002	0.1000	915	0.1775	922	0.0610	1170	0.1009	1167	0.0805	1100	0.1074	783	0.0682	884	-	
594	intellivision-003	0.1177	931	0.2006	937	0.0760	1184	0.1244	1182	0.1069	1121	0.1431	818	0.0839	905	0.4940	548
595	intellivision-004	0.0271	796	0.0559	826	0.0294	1105	0.0503	1100	0.0327	996	0.0461	674	0.0293	774	0.2860	490
596	intellivision-005	0.0105	586	0.0182	616	0.0093	866	0.0141	862	0.0146	807	0.0252	542	0.0172	633	0.1763	420
597	intellivision-006	0.0111	601	0.0174	600	0.0110	922	0.0162	906	0.0181	868	0.0313	584	0.0191	666	0.2416	465
598	intellivision-007	0.0086	519	0.0128	520	0.0098	886	0.0159	898	0.0133	777	0.0218	511	0.0154	598	0.1751	417
599	intellivix-001	0.0064	435	0.0087	408	0.0046	603	0.0063	609	0.0072	575	0.9233	1099	0.7856	1150	0.9977	766
600	intellivix-002	0.0062	418	0.0085	401	0.0039	535	0.0056	550	0.0060	527	0.3464	918	0.0857	909	0.8759	655
601	intellivix-003	0.0075	481	0.0125	515	0.0052	655	0.0091	731	0.0066	553	0.0297	574	0.0096	439	0.1907	432
602	intellivix-004	0.0032	211	0.0045	212	0.0025	209	0.0031	270	0.0040	355	0.0074	193	0.0063	261	0.0662	153
603	intellivix-005	0.0034	235	0.0048	228	0.0024	130	0.0027	180	0.0035	308	0.0067	172	0.0059	248	0.0609	112
604	intellivix-006	-		-		0.0024	169	0.0030	248	0.0026	161	0.0085	248	0.0040	101	0.0918	265
605	intelresearch-000	0.0307	814	0.0578	830	0.0385	1126	0.0751	1140	0.0409	1028	0.9870	1148	0.5380	1098	-	
606	intelresearch-001	0.0242	772	0.0595	834	0.0129	972	0.0292	1024	0.0351	1002	0.9993	1216	0.1854	992	-	
607	intelresearch-002	0.0058	396	0.0082	388	0.0050	644	0.0086	716	0.0136	786	0.0434	661	0.0216	701	-	
608	intelresearch-003	0.0046	319	0.0062	299	0.0038	523	0.0060	585	0.0088	644	0.0168	446	0.0136	555	-	
609	intelresearch-004	0.0025	149	0.0035	129	0.0032	413	0.0038	377	0.0049	451	0.0094	282	0.0072	315	-	
610	intelresearch-005	0.0016	66	0.0023	56	0.0028	307	0.0034	317	0.0042	380	0.0084	243	0.0066	278	0.1433	377
611	intelresearch-006	0.0010	30	0.0015	25	0.0026	271	0.0028	201	0.0032	266	0.8123	1054	0.4742	1082	0.9907	736
612	intelresearch-007	0.0009	22	0.0013	17	0.0025	187	0.0025	120	0.0035	305	0.0069	180	0.0057	223	0.0746	185
613	intelresearch-008	0.0009	19	0.0013	16	0.0025	188	0.0025	121	0.0028	203	0.0304	580	0.0048	158	0.2375	462
614	intelresearch-009	-		-		0.0024	171	0.0025	126	0.0028	205	0.0058	130	0.0046	146	0.0665	154
615	intelresearch-010	-		-		0.0023	99	0.0024	98	0.0029	222	0.0058	131	0.0050	171	0.0683	161
616	intelresearch-011	-		-		0.0023	71	0.0023	65	0.0025	140	0.0049	96	0.0040	109	0.0549	76

Table 69: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
617	intelresearch-012	-		-		0.0023	85	0.0023	71	0.0025	132	0.0048	89	0.0040	107	0.0549	75		
618	intema-000	0.0012	43	0.0017	36	0.0023	65	0.0022	58	0.0022	83	0.0172	451	0.0061	249	0.0996	286		
619	intema-001	0.0010	32	0.0014	20	0.0021	21	0.0020	22	0.0019	51	0.0037	37	0.0030	35	0.0457	32		
620	interlinklabs-001	-		-		0.0102	897	0.0205	963	0.0195	897	0.0237	529	0.0115	504	0.1579	395		
621	interlinklabs-002	-		-		0.0033	440	0.0049	492	0.0040	352	0.9968	1188	0.9807	1220	1.0000	820		
622	interlinklabs-003	-		-		0.0029	349	0.0034	330	0.0046	423	0.0083	238	0.0074	326	0.0620	118		
623	intozi-001	0.9982	1070	0.9998	1081	0.9998	1351	1.0000	1356	1.0000	1317	1.0000	1286	1.0000	1299	1.0000	924		
624	intozi-002	-		-		0.0038	521	0.0050	507	0.0054	479	0.9998	1237	0.9989	1260	1.0000	880		
625	intsysmsu-000	0.0135	653	0.0204	644	0.0112	930	0.0161	903	0.0148	811	0.0244	535	0.0172	632	-			
626	intsysmsu-001	0.9543	1066	0.9888	1075	0.9923	1340	0.9948	1336	0.9977	1293	0.9955	1179	0.9892	1229	-			
627	intsysmsu-002	0.0130	644	0.0254	699	0.0137	985	0.0267	1013	0.0160	830	0.0267	555	0.0145	582	-			
628	ionetworks-000	0.0060	412	0.0087	407	0.0044	584	0.0058	570	0.0080	617	0.0144	395	0.0112	491	0.1149	322		
629	ionetworks-001	0.0254	785	0.0388	774	0.0435	1138	0.0662	1126	0.0426	1034	0.8936	1085	0.7391	1140	0.9964	757		
630	ionetworks-002	0.0045	304	0.0066	317	0.0033	433	0.0047	475	0.0544	1063	0.9648	1123	0.8765	1169	0.9994	781		
631	ionetworks-003	-		-		0.0034	459	0.0049	500	0.0043	389	0.0079	215	0.0056	219	0.0812	223		
632	ionetworks-004	-		-		0.0034	462	0.0050	503	0.0043	385	0.0079	214	0.0056	217	0.0808	221		
633	iqface-000	0.0091	536	0.0143	540	0.0075	791	0.0110	782	0.0171	857	0.2234	860	0.0359	801	-			
634	iqface-001	0.0088	524	0.0117	501	0.0177	1030	0.0230	984	0.0165	840	0.5514	982	0.0286	764	-			
635	iqface-003	0.0058	397	0.0079	378	0.0051	649	0.0058	571	0.0104	711	0.0200	491	0.0193	670	-			
636	iqsec-001	-		-		0.7050	1307	0.8569	1308	0.7728	1240	0.9962	1186	0.9840	1221	1.0000	838		
637	iqsec-002	-		-		0.7050	1306	0.8569	1307	0.7728	1241	0.9962	1185	0.9841	1222	1.0000	839		
638	iqsec-003	-		-		0.9924	1342	0.9984	1339	0.9998	1304	0.9996	1229	0.9981	1252	1.0000	871		
639	irex-000	0.0052	352	0.0099	451	0.0056	689	0.0083	707	0.0137	791	0.0163	436	0.0078	359	-			
640	irisid-001	-		-		0.0034	455	0.0048	478	0.0058	513	0.0108	332	0.0079	363	0.1024	291		
641	irisid-002	-		-		0.0031	401	0.0042	431	0.0055	489	0.0102	312	0.0077	355	0.1024	292		
642	isap-001	0.5092	1033	0.6588	1043	0.6899	1303	0.7978	1296	0.7200	1234	0.7253	1024	0.5373	1097	-			
643	isap-002	0.0114	606	0.0186	622	0.0087	846	0.0151	884	0.0156	822	0.5134	968	0.0333	787	-			
644	isityou-000	0.5682	1041	0.7033	1050	1.0000	1418	0.9999	1350	1.0000	1359	1.0000	1372	1.0000	1430	-			
645	istl-001	-		-		0.0041	564	0.0059	581	0.0053	475	0.5416	981	0.2534	1018	0.8765	656		
646	isystems-000	0.1267	935	0.1795	924	-		-		-		-		-		-			
647	isystems-001	0.0149	682	0.0245	692	0.0138	988	0.0210	966	0.0209	910	0.0332	607	0.0223	707	-			
648	isystems-002	0.0118	615	0.0182	615	0.0111	926	0.0162	909	0.0166	845	0.0284	568	0.0195	675	-			
649	itmo-001	-		0.4408	1001	-		-		-		-		-		-			
650	itmo-002	0.1607	950	0.2866	967	-		-		-		-		-		-			
651	itmo-003	0.0749	895	0.1398	906	-		-		-		-		-		-			
652	itmo-004	0.0255	786	0.0543	823	-		-		-		-		-		-			
653	itmo-005	0.0182	727	0.0345	755	0.0181	1033	0.0348	1045	0.0225	924	0.0378	646	0.0230	714	-			
654	itmo-006	0.0125	633	0.0220	664	0.0149	1010	0.0266	1009	0.0233	932	0.0383	647	0.0285	762	-			
655	itmo-007	0.0080	504	0.0125	514	0.0107	912	0.0185	939	0.0167	847	0.0222	516	0.0144	579	-			
656	itmo-008	0.0090	530	0.0150	555	0.0058	701	0.0059	578	0.0187	880	0.0355	626	0.0339	788	0.1758	419		
657	ivacognitive-001	0.0189	734	0.0351	756	0.0123	960	0.0235	989	0.0198	900	0.0274	560	0.0155	599	-			
658	iws-000	0.4824	1027	0.5801	1032	0.6859	1302	0.8155	1301	0.8251	1254	0.7756	1041	0.6400	1119	-			
659	jaakit-001	0.5830	1042	0.7146	1051	0.8173	1323	0.8893	1318	0.8950	1270	0.8387	1069	0.7091	1133	0.9733	708		
660	jaakit-002	-		-		0.5729	1287	0.6951	1282	0.6495	1228	0.9894	1157	0.9721	1213	0.9999	803		

Table 70: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
661	kaizenvoiz-001	-		-		0.4082	1266	0.5607	1265	0.4678	1211	0.4894	962	0.3770	1061	0.8558	645		
662	kakao-001	0.4553	1022	0.5532	1027	0.6580	1297	0.8150	1300	0.4831	1213	1.0000	1350	1.0000	1451	-			
663	kakao-002	0.0625	879	0.1779	923	0.0791	1188	0.1381	1187	0.0636	1082	1.0000	1406	1.0000	1337	-			
664	kakao-003	0.0130	646	0.0185	619	0.0261	1087	0.0464	1087	0.0252	955	0.2380	872	0.0284	761	-			
665	kakao-004	0.0078	493	0.0103	467	0.0059	710	0.0102	762	0.0155	819	0.1182	796	0.0230	715	-			
666	kakao-005	0.0040	270	0.0059	284	0.0036	492	0.0057	559	0.0085	631	0.0239	531	0.0125	529	-			
667	kakao-007	0.0019	98	0.0028	88	0.0024	128	0.0026	152	0.0033	277	0.0061	147	0.0053	196	0.0580	94		
668	kakao-008	0.0011	35	0.0018	39	0.0023	76	0.0023	77	0.0021	72	0.0041	53	0.0035	57	0.0475	40		
669	kakao-009	0.0009	24	0.0013	15	0.0023	83	0.0023	81	0.0020	59	0.0039	44	0.0034	49	0.0468	36		
670	kakaobank-000	0.0041	285	0.0058	280	0.0041	556	0.0058	573	0.0060	529	0.9988	1210	0.3243	1046	1.0000	851		
671	kakaobank-001	0.0030	192	0.0046	216	0.0032	423	0.0039	383	0.0049	448	0.9998	1244	0.9994	1264	1.0000	911		
672	kakaobank-002	-		-		0.0026	245	0.0036	347	0.0038	334	0.3184	909	0.1174	941	0.6931	596		
673	kakaobank-003	-		-		0.0023	103	0.0025	139	0.0029	218	0.0065	161	0.0052	191	0.0733	176		
674	kakaopay-001	0.0152	686	0.0252	698	0.0145	1000	0.0270	1014	0.0232	930	0.0344	619	0.0194	671	0.2005	443		
675	kasikornlabs-000	0.0112	602	0.0184	618	0.0086	842	0.0137	849	0.0130	772	0.0225	519	0.0148	588	0.1660	410		
676	kasikornlabs-001	0.0138	661	0.0206	646	0.0087	845	0.0139	857	0.0142	797	0.0236	528	0.0171	628	0.1684	413		
677	kasikornlabs-002	0.0069	453	0.0091	420	0.0048	621	0.0063	604	0.0076	593	0.0144	394	0.0110	486	0.1249	344		
678	kasikornlabs-003	0.0057	379	0.0076	362	0.0042	572	0.0056	553	0.0059	518	0.0123	364	0.0098	441	0.1212	338		
679	kasikornlabs-004	-		-		0.0041	562	0.0054	539	0.0057	501	0.0112	344	0.0095	433	0.1101	307		
680	kasikornlabs-005	-		-		0.0037	495	0.0043	442	0.0054	480	0.0107	328	0.0093	423	0.1069	302		
681	kbyai-000	-		-		0.0027	292	0.0032	285	0.0031	256	0.0079	217	0.0049	165	0.0655	148		
682	kedacom-000	0.0055	370	0.0081	385	0.0111	929	0.0120	797	0.0415	1029	0.0966	774	0.0686	885	-			
683	keyless-000	-		-		0.0025	205	0.0034	315	0.0031	255	0.1874	839	0.0568	870	0.5543	559		
684	kiwitech-000	0.0076	487	0.0105	472	0.0081	818	0.0128	826	0.0096	682	0.0163	435	0.0101	453	-			
685	kloudspot-001	-		-		0.0140	989	0.0259	1003	0.9999	1307	1.0000	1292	1.0000	1304	1.0000	861		
686	kloudspot-002	-		-		0.0255	1082	0.0423	1072	0.0192	890	0.8748	1077	0.7143	1135	0.9929	741		
687	kneron-003	0.0542	865	0.0902	876	0.0346	1121	0.0562	1109	0.0919	1111	0.1251	802	0.0973	921	-			
688	kneron-005	0.0157	691	0.0259	703	0.0126	968	0.0212	967	0.0406	1027	0.0693	741	0.0542	861	-			
689	kneron-006	-		-		0.0094	869	0.0125	810	0.1305	1129	0.9751	1132	0.9033	1179	0.9987	771		
690	knowutech-000	0.0039	267	0.0055	270	0.0028	335	0.0042	424	0.0042	373	0.0077	203	0.0059	242	0.0820	226		
691	kogniza-001	-		-		0.0028	334	0.0031	275	0.0025	139	0.0048	85	0.0042	115	0.0557	79		
692	kookmin-001	0.0462	855	0.0750	853	0.0489	1150	0.0842	1148	0.0659	1084	0.8380	1068	0.3212	1045	-			
693	kookmin-002	0.0054	366	0.0077	367	0.0043	579	0.0065	616	0.0123	758	0.7591	1035	0.0198	679	-			
694	koreaid-001	0.0031	199	0.0045	208	0.0026	246	0.0032	282	0.0043	388	0.0083	235	0.0068	292	0.0743	182		
695	krungthai-002	0.0105	584	0.0161	578	0.0091	857	0.0141	863	0.7350	1236	0.9889	1155	0.9605	1206	0.9999	800		
696	kuke3d-001	0.0058	388	0.0104	468	0.0083	829	0.0093	734	0.0270	962	0.9901	1158	0.8341	1158	1.0000	840		
697	kuke3d-002	0.0077	489	0.0135	529	0.0069	761	0.0098	752	0.0111	734	1.0000	1291	1.0000	1298	1.0000	872		
698	lebentech-000	0.5940	1043	0.7032	1049	0.8854	1328	0.9511	1324	0.9089	1275	0.9970	1189	0.9861	1225	0.9999	808		
699	lebentech-001	0.0304	812	0.9032	1066	0.9998	1352	1.0000	1354	1.0000	1314	1.0000	1272	0.9999	1287	1.0000	908		
700	lemalabs-001	0.0111	597	0.0175	603	0.0088	847	0.0142	864	0.0143	800	0.0228	523	0.0140	567	0.1646	407		
701	lightvision-000	-		-		0.0146	1004	0.0232	986	0.0237	942	0.9638	1122	0.7768	1147	0.9986	769		
702	lightvision-001	-		-		0.0068	756	0.0096	744	0.0084	629	0.9948	1176	0.9288	1188	1.0000	824		
703	lineclova-000	0.0172	714	0.0236	683	0.0109	918	0.0194	952	0.0183	874	0.0291	573	0.0204	687	0.1936	434		
704	lineclova-001	0.0025	148	0.0040	165	0.0026	270	0.0034	329	0.0045	417	0.4127	945	0.0080	369	0.9824	718		

Table 71: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
705	lineclova-002	0.0021	112	0.0035	125	0.0025	176	0.0027	171	0.0041	364	0.0086	254	0.0072	316	0.1595	399		
706	lineclova-003	0.0018	93	0.0030	100	0.0028	331	0.0031	259	0.0041	365	0.0083	239	0.0075	343	0.1014	290		
707	lookman-002	0.0297	809	0.0547	824	0.0339	1116	0.0562	1108	0.0614	1076	0.0960	773	0.0790	896	-			
708	lookman-004	0.0074	478	0.0099	454	0.0124	963	0.0149	879	0.0430	1036	0.0866	762	0.0694	887	-			
709	lssytech-000	-		-		0.0023	77	0.0023	86	0.0021	73	0.0040	51	0.0036	66	0.0574	92		
710	luxand-000	0.2056	970	0.2814	965	0.4053	1264	0.5365	1262	0.3497	1196	0.3743	932	0.2605	1024	-			
711	luxand-001	0.0116	608	0.0925	877	0.0144	998	0.0267	1010	1.0000	1396	1.0000	1328	1.0000	1395	1.0000	1235		
712	luxand-003	-		-		0.9938	1343	0.9999	1347	0.2905	1186	0.2256	864	0.0849	908	0.4876	544		
713	luxand-004	-		-		0.0031	399	0.0042	425	0.0110	730	0.9994	1222	0.9969	1247	1.0000	981		
714	mantra-000	0.0037	255	0.0052	255	0.0054	671	0.0056	555	0.0097	690	0.0181	463	0.0151	593	-			
715	matrixcomsec-000	-		-		0.1032	1201	0.0895	1154	1.0000	1358	1.0000	1373	1.0000	1431	1.0000	1363		
716	matrixcomsec-001	-		-		0.0070	763	0.0136	842	0.8826	1268	1.0000	1435	1.0000	1368	1.0000	1182		
717	matrixcomsec-002	-		-		0.0091	854	0.0172	923	0.0168	849	0.3770	933	0.1584	974	0.8585	646		
718	maverickresearch-000	-		-		0.0143	995	0.0192	948	0.0216	915	0.0362	635	0.0250	733	0.2293	455		
719	maxvision-000	0.0078	496	0.0106	474	0.0110	923	0.0147	874	0.0368	1012	1.0000	1367	0.1545	970	1.0000	1328		
720	maxvision-001	0.0305	813	0.0528	819	0.1028	1199	0.1921	1207	0.0650	1083	0.3001	897	0.1553	972	0.6188	581		
721	maxvision-002	0.0070	459	0.0107	476	0.0061	723	0.0093	733	0.0080	614	0.5726	987	0.2943	1037	0.9160	678		
722	maxvision-003	0.0056	375	0.0083	396	0.0038	528	0.0060	588	0.0061	532	0.2614	881	0.0650	880	0.5812	567		
723	maxvision-004	0.0024	145	0.0038	156	0.0025	221	0.0029	223	0.0029	214	0.4732	959	0.1965	995	0.8141	632		
724	maxvision-005	0.0014	55	0.0026	73	0.0024	137	0.0024	97	0.0025	129	0.1216	798	0.0332	785	0.3922	523		
725	maxvision-006	0.0016	69	0.0026	75	0.0024	142	0.0024	101	0.0024	124	0.4012	943	0.1799	989	0.7634	611		
726	mbbankvn-000	-		-		0.0030	358	0.0035	331	0.0026	157	0.3822	937	0.1130	935	0.7052	600		
727	megvii-000	0.0159	696	0.0246	694	-		-		-		-		-		-			
728	megvii-001	0.0157	692	0.0244	689	0.0392	1131	0.0671	1127	0.0168	850	0.0455	671	0.0164	615	-			
729	megvii-002	0.0104	579	0.0145	545	0.0225	1066	0.0345	1043	0.0099	700	0.0286	569	0.0240	725	-			
730	megvii-003	0.0064	431	0.0094	430	0.0136	984	0.0260	1005	0.0050	453	0.0080	222	0.0059	245	-			
731	megvii-004	0.0020	103	0.0033	119	0.0028	323	0.0035	343	0.0037	333	0.0074	194	0.0068	293	0.0736	180		
732	megvii-005	0.0010	31	0.0015	23	0.0026	238	0.0031	265	0.0019	47	0.0500	688	0.0057	228	0.1991	441		
733	megvii-006	0.0011	34	0.0016	28	0.0026	265	0.0033	302	0.0025	137	0.0050	100	0.0048	154	0.0588	103		
734	megvii-007	0.0006	12	0.0016	27	0.0025	207	0.0028	192	0.0024	120	0.0048	90	0.0047	151	0.0586	97		
735	megvii-008	0.0003	2	0.0008	7	0.0023	107	0.0021	45	0.0024	125	0.0049	94	0.0047	153	0.0541	71		
736	megvii-009	0.0003	3	0.0008	8	0.0023	106	0.0021	44	0.0018	32	0.0149	405	0.0036	62	0.1096	306		
737	meituan-000	0.0197	741	0.0424	784	0.0078	804	0.0074	660	0.0103	709	0.0193	484	0.0164	614	0.1133	319		
738	meituan-001	0.0164	703	0.1886	930	0.0025	195	0.0026	147	0.0030	231	0.0074	192	0.0051	180	0.0791	211		
739	meituan-002	0.0017	76	0.0025	65	0.0024	141	0.0023	72	0.0024	122	0.0067	175	0.0044	140	0.0773	203		
740	meituan-003	0.0017	73	0.0021	51	0.0024	133	0.0023	92	0.0024	117	0.0084	244	0.0046	143	0.0883	244		
741	meituan-004	0.0014	51	0.0021	47	0.0024	135	0.0024	110	0.0023	103	0.0070	182	0.0044	136	0.0813	225		
742	meiya-001	0.0171	711	0.0275	715	0.0159	1023	0.0261	1008	0.0311	989	0.2250	863	0.0245	730	-			
743	mendaxiatech-000	0.0027	170	0.0036	137	0.0029	340	0.0036	360	0.0031	252	0.0057	124	0.0051	181	-			
744	metsakuurcompany-001	0.0068	450	0.0087	412	0.0068	759	0.0078	685	0.0095	677	0.8972	1089	0.5635	1101	0.9970	761		
745	metsakuurcompany-002	0.0048	323	0.0071	339	0.0030	368	0.0043	444	0.0032	268	0.2059	852	0.0665	882	0.5374	553		
746	metsakuurcompany-003	0.0034	229	0.0051	249	0.0026	243	0.0036	345	0.0028	198	0.2418	873	0.1001	923	0.5866	571		
747	metsakuurcompany-004	-		-		0.0026	269	0.0032	277	0.0032	261	0.4488	955	0.2015	999	0.8129	631		
748	metsakuurcompany-005	-		-		0.0030	369	0.0041	413	0.0039	344	0.1725	831	0.0054	209	0.8274	635		

Table 72: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
749	miaxis-001	0.0068	449	0.0099	453	0.0059	709	0.0097	749	0.0096	685	0.0231	526	0.0109	483	0.1592	396
750	miaxis-002	0.0745	894	0.0774	860	0.3215	1253	0.4000	1243	0.1485	1138	0.2087	853	0.2058	1000	0.3846	519
751	miaxis-003	0.0045	303	0.0067	321	0.0037	511	0.0050	502	0.0061	534	0.0139	383	0.0088	402	0.1322	353
752	miaxis-004	-		-		0.0037	494	0.0046	466	0.0057	506	0.0107	329	0.0084	394	0.1261	347
753	miaxis-005	-		-		0.0033	436	0.0041	423	0.0053	474	0.0106	325	0.0079	365	0.1236	342
754	microfocus-000	0.4656	1023	0.5624	1028	-		-		-		-		-		-	
755	microfocus-001	0.4482	1020	0.5524	1026	0.7256	1310	0.8416	1305	0.7301	1235	0.6926	1016	0.5180	1094	-	
756	microfocus-002	0.3605	1001	0.5057	1016	0.5783	1288	0.7223	1284	0.5909	1221	0.5963	997	0.4160	1073	-	
757	microfocus-003	0.0416	842	0.0607	836	0.0652	1175	0.1025	1169	0.0793	1097	0.2248	862	0.0923	916	0.6153	579
758	minivision-000	0.0033	217	0.0048	227	0.0038	525	0.0049	487	0.0055	487	0.0094	288	0.0079	366	-	
759	mitek-000	0.6882	1050	0.8192	1058	0.9568	1335	0.9788	1331	0.9545	1283	0.9437	1112	0.8494	1161	0.9931	743
760	mitek-001	-		-		0.0050	639	0.0068	630	0.0053	473	0.0096	295	0.0077	354	0.1130	318
761	mitek-002	-		-		0.0039	541	0.0054	535	0.0044	401	0.0085	251	0.0065	273	0.1112	314
762	mobai-000	0.0360	834	0.0439	792	0.0372	1124	0.0700	1132	0.0367	1011	0.0939	768	0.0795	898	-	
763	mobai-001	0.0199	745	0.0219	662	0.0047	612	0.0061	593	0.0093	673	0.0174	453	0.0138	563	-	
764	mobai-002	-		-		0.0048	622	0.0070	640	0.0088	650	0.5904	993	0.3167	1044	0.8952	671
765	mobai-003	-		-		0.0053	666	0.0090	727	0.0689	1088	0.9994	1220	0.9982	1253	1.0000	879
766	mobbl-000	0.2938	989	0.3861	993	0.5391	1282	0.6888	1281	0.6545	1229	0.8027	1050	0.6207	1112	-	
767	mobbl-001	0.3208	993	0.4375	1000	0.5680	1286	0.7193	1283	0.6282	1223	0.5783	988	0.3984	1068	0.8864	666
768	mobbl-002	0.9914	1069	0.9970	1077	0.9355	1332	0.9923	1334	1.0000	1313	1.0000	1262	0.9999	1280	1.0000	878
769	mobbl-003	0.0087	522	0.0134	528	0.0062	727	0.0087	720	0.0099	696	0.0197	487	0.0122	525	0.1618	406
770	mobbl-004	-		-		0.0031	397	0.0043	441	0.0088	645	1.0000	1269	1.0000	1291	1.0000	874
771	mobbl-005	-		-		0.0031	389	0.0044	455	0.0045	405	0.4420	953	0.1465	963	0.7973	628
772	mobipintech-000	0.0090	531	0.0149	553	0.0039	544	0.0057	556	0.0115	743	0.0465	676	0.0182	650	0.6443	584
773	momovn-001	0.1044	920	0.1694	920	0.3253	1255	0.5244	1259	0.3861	1202	0.9995	1224	0.9985	1254	1.0000	888
774	momovn-002	-		-		0.0042	570	0.0060	590	0.0062	537	0.7672	1037	0.5456	1099	0.9731	707
775	momovn-003	-		-		0.0029	342	0.0033	295	0.0056	496	0.0093	279	0.0076	347	0.0612	114
776	moredian-000	0.3874	1007	0.4912	1012	0.9988	1349	0.9999	1349	0.9990	1300	0.9999	1249	0.9998	1278	-	
777	morpho-000	0.1020	917	0.1343	903	-		-		-		-		-		-	
778	morpho-001	-		0.2772	962	-		-		-		-		-		-	
779	morpho-002	0.0377	836	0.0680	846	0.0368	1123	0.0716	1134	-		-		-		-	
780	mukh-001	0.0170	709	0.0285	717	0.0225	1067	0.0405	1066	0.0272	966	0.0950	770	0.0291	771	0.5846	570
781	mukh-002	0.0269	795	0.0357	761	0.0435	1140	0.0799	1146	0.0143	801	0.0213	504	0.0122	524	0.1908	433
782	mukh-003	0.0077	491	0.0113	493	0.0111	927	0.0224	977	0.0061	533	0.7813	1044	0.4504	1079	0.9899	735
783	mukh-004	0.0050	336	0.0069	335	0.0051	646	0.0099	758	0.0045	408	0.0083	237	0.0059	244	0.1030	295
784	mukh-005	-		-		0.0023	74	0.0024	117	0.0056	495	0.0161	430	0.0067	283	0.1398	368
785	multimodality-000	0.0034	227	0.0047	221	0.0036	489	0.0044	453	0.0077	599	0.9976	1194	0.4456	1078	1.0000	860
786	multimodality-001	0.0029	181	0.0042	180	0.0031	381	0.0035	334	0.0038	335	0.0071	186	0.0059	241	0.0732	174
787	mvision-001	0.0191	735	0.0233	677	0.0204	1051	0.0356	1047	0.0198	901	0.0337	614	0.0242	727	-	
788	nazhiai-000	0.0040	276	0.0059	291	0.0036	477	0.0048	484	0.0057	508	0.0125	369	0.0083	387	-	
789	ncsgroup-000	-		-		0.0126	967	0.0226	979	0.0179	865	0.1715	830	0.0287	766	0.6160	580
790	ncssg-001	0.0207	751	0.1540	913	0.0045	592	0.0067	629	0.0056	500	0.1180	795	0.0301	775	0.3908	522
791	ncssg-002	-		-		0.0032	418	0.0038	379	0.0033	273	0.3144	906	0.1061	928	0.7441	609
792	ncssg-003	-		-		0.0033	437	0.0040	404	0.0033	285	0.2511	879	0.0959	920	0.6553	587

Table 73: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)													
		CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
793	ncssg-004	-		-		0.0033	431	0.0041	422	0.0029	224	0.2454	875	0.0860	910	0.6010	573
794	neosystems-001	1.0000	1084	1.0000	1288	0.2987	1250	0.4382	1250	0.5173	1216	0.6570	1009	0.4043	1071	-	
795	neosystems-002	0.2905	987	0.4077	996	0.2028	1235	0.3252	1236	0.4088	1207	0.5519	983	0.3331	1050	0.9561	694
796	neosystems-003	0.2429	978	0.3349	981	0.1844	1230	0.2999	1233	0.5942	1222	0.3936	940	0.2292	1011	0.8482	642
797	neosystems-004	0.0279	799	0.0495	809	0.0289	1099	0.0585	1116	0.0439	1039	0.9621	1120	0.1296	948	0.9996	787
798	netbridgetech-001	0.4749	1025	0.6599	1044	0.4438	1272	0.5676	1266	0.4491	1210	1.0000	1254	0.9541	1199	-	
799	netbridgetech-002	0.0101	569	0.0166	587	0.0077	799	0.0127	821	0.0133	779	0.8215	1060	0.0523	857	-	
800	neurotechnology-000	-		0.2371	949	-		-		-		-		-		-	
801	neurotechnology-001	0.1393	940	0.2218	946	-		-		-		-		-		-	
802	neurotechnology-002	0.1115	927	0.1663	919	-		-		-		-		-		-	
803	neurotechnology-003	0.0617	876	0.0880	872	-		-		-		-		-		-	
804	neurotechnology-004	0.0146	674	0.0297	725	0.0107	914	0.0164	913	-		-		-		-	
805	neurotechnology-005	0.0141	667	0.0300	728	0.0108	917	0.0163	912	0.0204	904	0.0875	763	0.0246	732	-	
806	neurotechnology-006	0.0098	558	0.0136	532	0.0105	907	0.0182	933	0.0164	838	0.0843	759	0.0214	697	-	
807	neurotechnology-008	0.0091	537	0.0184	617	0.0076	795	0.0116	791	0.0157	824	0.9973	1190	0.1803	991	-	
808	neurotechnology-009	0.0049	330	0.0087	410	0.0057	697	0.0094	736	0.0079	612	1.0000	1258	0.0101	452	-	
809	neurotechnology-010	0.0048	325	0.0085	403	0.0056	688	0.0090	726	0.0077	602	1.0000	1263	0.0113	494	-	
810	neurotechnology-011	0.0050	341	0.0087	405	0.0061	726	0.0097	748	0.0077	601	0.0404	656	0.0092	419	-	
811	neurotechnology-012	0.0051	347	0.0070	337	0.0038	515	0.0056	554	0.0066	552	0.0112	345	0.0075	340	0.1009	289
812	neurotechnology-013	0.0032	213	0.0045	207	0.0026	268	0.0036	346	0.0037	324	0.0068	176	0.0052	194	0.0747	187
813	neurotechnology-015	0.0022	121	0.0036	141	0.0024	126	0.0028	198	0.0030	228	0.0052	109	0.0041	114	0.0646	142
814	neurotechnology-016	0.0022	126	0.0032	115	0.0023	93	0.0027	172	0.0027	180	0.0049	95	0.0039	99	0.0613	115
815	neurotechnology-017	0.0017	80	0.0031	106	0.0023	72	0.0025	134	0.0025	145	0.0047	83	0.0038	85	0.0612	113
816	neurotechnology-018	0.0020	104	0.0029	94	0.0022	53	0.0024	105	0.0024	121	0.0046	74	0.0037	71	0.0586	98
817	neurotechnology-019	-		-		0.0022	50	0.0024	93	0.0022	78	0.0042	56	0.0034	46	0.0526	59
818	neurotechnology-020	-		-		0.0023	108	0.0028	216	0.0024	106	0.0043	63	0.0035	55	0.0562	83
819	nextmetalabs-001	-		-		0.1126	1207	0.1962	1209	0.1137	1122	1.0000	1268	0.9998	1275	1.0000	887
820	nextmetalabs-002	-		-		0.0066	750	0.0081	703	0.0097	688	0.9946	1175	0.9565	1201	1.0000	834
821	nextmetalabs-003	-		-		0.0023	116	0.0028	209	0.0028	191	0.9681	1127	0.8769	1170	0.9995	786
822	nextmetalabs-004	-		-		0.0023	117	0.0028	208	0.0029	212	0.0055	117	0.0048	155	0.0716	170
823	nhn-001	0.0066	442	0.0098	446	0.0053	664	0.0079	695	0.0093	669	0.0156	420	0.0109	481	-	
824	nhn-002	0.0068	452	0.0096	439	0.0057	699	0.0087	722	0.0136	790	0.0253	544	0.0186	660	0.1464	381
825	nhn-003	0.0033	215	0.0048	225	0.0027	291	0.0038	376	0.0036	316	0.0198	488	0.0071	311	0.1280	349
826	nhn-004	0.0030	188	0.0044	204	0.0025	208	0.0029	227	0.0030	244	0.0331	604	0.0113	497	0.1598	401
827	nhn-005	0.0028	174	0.0043	186	0.0023	98	0.0027	162	0.0025	130	0.0047	77	0.0038	84	0.0554	78
828	noblis-000	0.4424	1019	0.5415	1024	-		-		-		-		-		-	
829	nodeflux-000	1.0000	1318	1.0000	1153	1.0000	1455	-		0.9079	1274	-		-		-	
830	nodeflux-001	1.0000	1127	1.0000	1337	1.0000	1401	0.9998	1346	0.5169	1215	1.0000	1420	0.5654	1102	-	
831	nodeflux-002	0.0186	732	0.0340	748	0.0261	1086	0.0451	1079	0.0548	1064	1.0000	1279	1.0000	1297	-	
832	nominder-000	0.0025	152	0.0036	142	0.0029	348	0.0033	300	0.0026	171	0.9919	1164	0.8877	1175	1.0000	844
833	nominder-001	0.0029	184	0.0042	181	0.0031	402	0.0038	378	0.0031	251	0.9975	1192	0.9180	1184	0.9999	810
834	nominder-002	0.0025	153	0.0037	145	0.0033	430	0.0037	366	0.0026	177	0.5785	989	0.1435	961	0.8381	638
835	nominder-003	-		-		0.0031	379	0.0034	319	0.0026	162	0.7010	1020	0.2159	1006	0.8909	669
836	nominder-004	-		-		0.0028	327	0.0033	296	0.0024	108	0.0316	590	0.0045	141	0.1348	359

Table 74: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
837	nominder-005	-		-		0.0025	194	0.0029	222	0.0024	116	0.0046	75	0.0037	75	0.0537	70
838	nominder-006	-		-		0.0027	293	0.0032	280	0.0030	241	0.0059	133	0.0052	188	0.0644	140
839	nominder-007	-		-		0.0027	276	0.0031	261	0.0023	100	0.0072	189	0.0036	61	0.0946	278
840	nominder-008	-		-		0.0026	250	0.0028	214	0.0022	82	0.0069	178	0.0035	58	0.0928	274
841	notiontag-000	0.6669	1048	0.7885	1056	0.3715	1261	0.4978	1258	0.8571	1259	0.8102	1053	0.6460	1121	-	
842	notiontag-001	0.6846	1049	0.8006	1057	0.3955	1263	0.5247	1260	0.8669	1262	0.8313	1066	0.6362	1118	-	
843	notiontag-002	0.0066	439	0.0089	415	0.0045	591	0.0061	594	0.0077	600	0.0137	382	0.0104	466	0.1106	311
844	nsensecorp-001	0.9909	1068	0.9994	1080	0.9987	1348	0.9999	1348	1.0000	1316	1.0000	1290	1.0000	1301	-	
845	nsensecorp-002	0.4277	1015	0.5375	1021	0.6734	1298	0.7924	1294	0.7194	1233	0.6937	1018	0.5617	1100	0.9392	688
846	nsensecorp-003	0.0251	783	0.0295	724	0.0212	1058	0.0305	1030	0.0131	774	0.2139	856	0.0141	571	0.9968	760
847	nsensecorp-004	0.1370	939	0.1397	905	0.0066	749	0.0094	737	1.0000	1335	1.0000	1355	1.0000	1456	1.0000	1446
848	nsensecorp-005	0.0055	373	0.0080	383	0.0039	536	0.0058	572	0.2723	1183	0.9999	1248	0.9949	1239	1.0000	902
849	ntechlab-000	-		0.0862	870	-		-		-		-		-		-	
850	ntechlab-001	-		0.0825	866	-		-		-		-		-		-	
851	ntechlab-002	0.0558	871	0.0649	839	-		-		-		-		-		-	
852	ntechlab-003	0.0286	803	0.0390	775	-		-		-		-		-		-	
853	ntechlab-004	0.0090	529	0.0132	523	-		-		-		-		-		-	
854	ntechlab-005	0.0093	543	0.0145	544	0.0131	976	0.0275	1016	-		-		-		-	
855	ntechlab-006	0.0078	494	0.0111	485	0.0112	933	0.0227	982	0.0108	721	0.0149	406	0.0098	443	-	
856	ntechlab-007	0.0056	374	0.0076	361	0.0073	785	0.0128	825	0.0079	611	0.0144	393	0.0102	454	-	
857	ntechlab-008	0.0041	286	0.0061	295	0.0056	685	0.0108	775	0.0042	379	0.0080	220	0.0063	264	-	
858	ntechlab-009	0.0039	268	0.0054	266	0.0042	574	0.0063	605	0.0275	969	0.0674	733	0.0532	858	-	
859	ntechlab-010	0.0013	50	0.0017	32	0.0024	161	0.0029	225	0.0031	253	0.0058	132	0.0050	172	0.0656	149
860	ntechlab-011	0.0012	40	0.0019	41	0.0024	152	0.0028	212	0.0029	220	0.0055	115	0.0047	148	0.0630	127
861	ntechlab-012	0.0011	38	0.0016	26	0.0023	114	0.0030	239	0.0026	165	0.0050	103	0.0043	124	0.0587	100
862	ntnu-000	-		-		0.0040	550	0.0066	621	0.0069	563	0.9942	1174	0.9744	1215	0.9999	812
863	ocrstudio-001	-		-		0.4396	1271	0.6256	1274	0.3190	1190	0.3128	905	0.1754	987	0.7785	619
864	omface-000	0.2573	983	0.3835	992	0.3590	1259	0.4903	1256	0.3956	1204	0.5003	964	0.2595	1023	0.9137	677
865	omface-001	0.0137	658	0.0212	653	0.0114	943	0.0187	943	0.0174	862	1.0000	1403	0.0214	699	1.0000	1048
866	omnigarde-000	0.0633	881	0.1002	885	0.1109	1205	0.2042	1213	0.1288	1126	0.5113	967	0.1227	946	0.9604	698
867	omnigarde-001	0.0168	706	0.0260	704	0.0203	1049	0.0402	1062	0.0243	947	0.0327	600	0.0177	641	0.2164	453
868	omnigarde-002	0.0033	222	0.0046	217	0.0027	305	0.0039	385	0.0041	367	0.0076	199	0.0059	246	0.0806	220
869	omnigarde-003	0.0024	140	0.0038	153	0.0024	168	0.0029	217	0.0032	270	0.0059	136	0.0049	167	0.0706	168
870	omnigarde-004	-		-		0.0022	47	0.0024	95	0.0023	93	0.0044	67	0.0038	86	0.0559	81
871	omnigarde-005	-		-		0.0022	58	0.0024	103	0.0024	113	0.0045	71	0.0039	97	0.0582	95
872	onfido-000	0.1472	944	0.2881	968	0.0335	1115	0.0731	1137	0.0515	1057	0.9915	1161	0.9579	1203	1.0000	815
873	onfido-001	-		-		0.0146	1002	0.0294	1025	0.0190	885	0.0702	744	0.0352	796	0.3117	497
874	onfido-002	-		-		0.0067	751	0.0098	753	0.0062	539	0.0167	443	0.0091	416	0.1228	341
875	onfido-003	-		-		0.0045	596	0.0070	639	0.0043	386	0.0164	438	0.0076	348	0.1153	324
876	oorudigital-000	-		-		0.0106	910	0.0190	946	0.0146	808	0.9930	1169	0.9585	1204	1.0000	829
877	ooto-001	-		-		0.0193	1040	0.0290	1023	0.0529	1062	0.0272	557	0.0255	736	0.1956	439
878	openedge-000	0.0045	307	0.0062	308	0.0049	627	0.0069	636	0.0049	447	0.9889	1156	0.7991	1154	0.9999	805
879	openface-001	0.1804	958	0.2921	970	0.2878	1246	0.3906	1242	0.2054	1162	0.2338	869	0.1549	971	0.6531	586
880	ovisionllc-001	-		-		0.0058	704	0.0081	697	0.8171	1251	1.0000	1288	1.0000	1293	1.0000	863

Table 75: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
881	ovisionllc-002	-		-		0.1963	1232	0.2522	1221	1.0000	1319	1.0000	1294	1.0000	1305	1.0000	870
882	oz-001	0.0133	651	0.0215	659	0.0109	919	0.0160	901	0.0235	938	1.0000	1408	1.0000	1308	-	
883	oz-002	0.0071	465	0.0099	456	0.0099	892	0.0100	759	0.0139	793	0.0502	690	0.0202	684	-	
884	oz-003	0.0095	552	0.0143	541	0.0054	673	0.0077	678	0.0096	684	0.0175	457	0.0118	511	0.1404	370
885	oz-004	0.0033	221	0.0049	237	0.0038	530	0.0055	543	0.0081	620	0.0163	437	0.0142	574	-	
886	pa-002	0.1847	960	0.2859	966	-		-		-		-		-		-	
887	pa-003	0.0453	852	0.0679	845	-		-		-		-		-		-	
888	palit-000	0.0062	416	0.0084	397	0.0039	534	0.0055	540	0.0055	490	0.4610	956	0.2468	1013	0.8695	653
889	palit-001	0.0050	333	0.0068	331	0.0032	424	0.0047	476	0.0045	412	0.0110	336	0.0058	238	0.1446	378
890	pangiam-000	0.0031	201	0.0043	196	0.0026	228	0.0030	254	0.0038	336	0.0071	188	0.0061	254	0.0667	156
891	pangiam-001	0.0031	203	0.0044	199	0.0029	341	0.0040	407	0.0028	204	0.0362	634	0.0056	221	0.1711	415
892	pangiam-002	0.0030	190	0.0042	183	0.0026	262	0.0031	267	0.0030	238	0.1772	832	0.0385	809	0.6076	575
893	papago-001	0.0067	447	0.0096	444	0.0051	652	0.0077	676	0.0071	572	0.0126	370	0.0086	399	0.1033	297
894	papil11-000	0.0021	115	0.0030	103	0.0024	120	0.0024	113	0.0025	148	0.0047	80	0.0040	105	0.0536	67
895	papil11-001	-		-		0.0023	113	0.0024	111	0.0023	89	0.0185	470	0.0040	112	0.1449	379
896	papil11-002	-		-		0.0023	110	0.0024	106	0.0023	88	0.0130	376	0.0039	94	0.1112	315
897	papil11-003	-		-		0.0023	100	0.0024	107	0.0023	98	0.0042	58	0.0035	56	0.0529	61
898	papil11-004	-		-		0.0027	286	0.0029	228	0.0025	150	0.0153	414	0.0058	237	0.0972	282
899	papsav1923-001	0.0078	495	0.0130	522	0.0068	757	0.0105	769	0.0119	749	0.0221	514	0.0136	556	-	
900	papsav1923-002	0.0021	116	0.0034	120	0.0026	230	0.0030	252	0.0048	443	0.0093	278	0.0086	397	0.0630	126
901	papsav1923-003	0.0025	157	0.0035	124	0.0024	164	0.0025	122	0.0034	288	0.0066	167	0.0058	236	0.0603	107
902	paravision-004	0.0030	191	0.0046	215	0.0030	362	0.0036	350	0.0091	662	0.0188	474	0.0173	634	-	
903	paravision-006	0.0016	71	0.0025	67	0.0026	240	0.0026	159	0.0075	591	0.9933	1170	0.0142	572	-	
904	paravision-008	0.0018	85	0.0025	69	0.0024	140	0.0025	132	0.0036	315	0.0070	184	0.0063	263	0.0747	188
905	paravision-010	0.0012	46	0.0021	46	0.0022	46	0.0021	38	0.0027	187	0.0055	116	0.0050	174	0.0534	66
906	paravision-011	0.0008	15	0.0020	45	0.0021	31	0.0020	19	0.0026	166	0.0053	110	0.0049	166	0.0498	48
907	paravision-013	0.0010	33	0.0018	37	0.0021	22	0.0019	3	0.0019	53	0.0041	54	0.0036	65	0.0442	28
908	paravision-015	-		-		0.0021	29	0.0021	43	0.0021	62	0.0043	60	0.0039	90	0.0528	60
909	paravision-017	-		-		0.0021	19	0.0021	27	0.0019	43	0.0038	40	0.0035	52	0.0458	33
910	paravision-018	-		-		0.0020	6	0.0020	17	0.0016	17	0.0034	27	0.0029	26	0.0414	14
911	paravision-020	-		-		0.0020	2	0.0020	11	0.0015	13	0.0033	20	0.0028	17	0.0410	13
912	pensees-001	0.0087	523	0.0133	526	0.0071	768	0.0122	807	0.0145	805	0.0252	543	0.0195	676	-	
913	pentacomp-000	-		-		0.0041	563	0.0061	592	0.0047	433	0.3501	920	0.1211	944	0.7162	603
914	pixelall-002	0.0193	738	0.0340	750	0.0127	970	0.0209	964	0.0234	937	0.9182	1094	0.2728	1029	-	
915	pixelall-003	0.0074	479	0.0118	502	0.0057	694	0.0079	690	0.0121	753	0.0390	652	0.0170	626	-	
916	pixelall-004	0.0040	278	0.0061	294	0.0055	679	0.0069	634	0.0100	702	0.5663	986	0.0371	806	-	
917	pixelall-005	0.0038	260	0.0052	254	0.0043	580	0.0051	513	0.0077	598	0.0839	758	0.0136	558	-	
918	pixelall-006	0.0032	207	0.0042	179	0.0032	412	0.0039	387	0.0063	541	0.9960	1183	0.0723	891	1.0000	847
919	pixelall-007	0.0036	249	0.0049	238	0.0039	532	0.0044	446	0.0068	557	0.9873	1150	0.0217	702	1.0000	823
920	pixelall-008	0.0015	56	0.0023	61	0.0034	460	0.0049	486	0.0031	250	0.0057	123	0.0052	185	0.0682	160
921	pixelall-009	0.0018	88	0.0025	66	0.0024	157	0.0026	153	0.0031	258	0.3475	919	0.0053	199	0.9006	674
922	pixelall-010	0.0020	105	0.0030	104	0.0025	200	0.0028	215	0.0031	254	0.3512	921	0.0054	203	0.8813	662
923	pnpsecure-000	-		-		0.0190	1037	0.0274	1015	0.9980	1295	0.9997	1234	0.9996	1270	0.9997	790
924	pnpsecure-001	-		-		0.0050	641	0.0040	403	0.0037	332	0.0166	442	0.0075	339	0.1059	300

Table 76: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
925	privid-001	0.3350	997	0.5013	1014	0.4327	1269	0.5880	1267	0.9790	1287	1.0000	1270	0.9998	1276	1.0000	906
926	privid-002	0.0460	854	0.0652	841	0.0524	1157	0.0882	1153	0.0671	1085	0.9769	1134	0.8960	1176	0.9999	799
927	privid-003	-		-		0.0030	365	0.0039	384	0.0045	414	0.9996	1230	0.9985	1255	1.0000	858
928	privid-004	-		-		0.0026	261	0.0028	196	0.0034	289	0.9939	1172	0.9657	1210	1.0000	835
929	psl-000	0.0216	754	0.0333	745	-		-		-		-		-		-	
930	psl-001	0.0549	868	0.0927	878	0.0096	878	0.0149	880	-		-		-		-	
931	psl-002	0.0107	589	0.0180	612	0.0089	850	0.0120	800	0.0790	1096	0.0270	556	0.0192	667	-	
932	psl-003	0.0065	438	0.0099	449	0.0055	676	0.0075	668	1.0000	1315	0.0188	475	0.0129	540	-	
933	psl-004	0.0064	432	0.0094	435	0.0035	466	0.0050	508	0.0084	627	0.0145	398	0.0106	474	-	
934	psl-005	0.0060	408	0.0094	431	0.0034	461	0.0048	482	0.0081	619	0.0140	386	0.0103	463	-	
935	psl-006	0.0025	155	0.0039	162	0.0028	322	0.0030	247	0.0160	828	0.0482	683	0.0310	777	-	
936	psl-007	0.0026	158	0.0040	168	0.0027	280	0.0030	243	0.0054	478	0.0101	311	0.0081	376	-	
937	psl-008	0.0026	160	0.0040	169	0.0024	150	0.0028	207	0.0041	369	0.0077	202	0.0055	210	0.0719	171
938	psl-009	0.0161	699	0.0294	723	0.0023	88	0.0025	124	0.0036	318	0.0065	165	0.0048	161	0.0659	151
939	psl-010	0.0017	83	0.0035	126	0.0023	80	0.0025	118	0.0035	306	0.0104	317	0.0052	193	0.0994	285
940	psl-011	0.0013	47	0.0026	77	0.0021	9	0.0021	33	0.0024	111	0.0047	79	0.0035	59	0.0565	86
941	psl-012	0.0017	77	0.0027	81	0.0021	34	0.0023	66	0.0021	63	0.0039	43	0.0032	40	0.0494	46
942	psl-013	-		-		0.0022	39	0.0023	69	0.0021	70	0.0039	46	0.0033	42	0.0499	52
943	psl-014	-		-		0.0021	33	0.0023	67	0.0021	68	0.0039	45	0.0033	41	0.0498	49
944	ptakuratsatu-000	0.0060	405	0.0089	418	0.0070	765	0.0104	768	0.0096	687	0.0152	413	0.0100	449	-	
945	pxl-001	0.0488	858	0.0752	854	0.0586	1167	0.1087	1171	0.0946	1112	0.1065	782	0.0625	878	-	
946	pyramid-000	0.0136	654	0.0233	679	0.0117	948	0.0192	949	0.0185	877	0.0322	597	0.0206	690	-	
947	pyramid-001	-		-		0.0025	177	0.0030	249	0.0025	138	0.0099	302	0.0044	131	0.1031	296
948	qazbs-000	0.0058	387	0.0083	394	0.0046	607	0.0072	650	0.0130	771	0.0244	536	0.0196	677	0.7784	618
949	qazsmartvisionai-000	0.0008	16	0.0011	12	0.0021	12	0.0022	48	0.0018	39	0.0035	33	0.0030	29	0.0441	27
950	qazsmartvisionai-001	0.0003	1	0.0008	6	0.0020	7	0.0020	24	0.0015	12	0.0031	12	0.0027	12	0.0394	10
951	qazsmartvisionai-002	-		-		0.0020	3	0.0020	9	0.0014	3	0.0028	3	0.0025	3	0.0380	4
952	qazsmartvisionai-003	-		-		0.0020	4	0.0020	10	0.0015	5	0.0029	6	0.0025	5	0.0381	6
953	qazsmartvisionai-004	-		-		0.0020	5	0.0020	15	0.0014	2	0.0028	2	0.0025	2	0.0377	3
954	qluevision-001	0.0223	761	0.0419	781	0.0205	1053	0.0343	1042	0.0327	997	0.8762	1078	0.7413	1141	0.9908	737
955	qnap-000	0.0149	679	0.0228	673	0.0155	1016	0.0267	1011	0.0238	945	0.8329	1067	0.0396	814	0.9992	776
956	qnap-001	0.0148	677	0.0215	657	0.0103	900	0.0162	908	0.0183	876	0.0301	576	0.0186	659	0.1995	442
957	qnap-002	0.0122	624	0.0191	627	0.0075	794	0.0095	741	0.0146	806	0.0281	567	0.0184	652	0.1829	425
958	qnap-003	0.0637	882	0.0657	843	0.0058	702	0.0078	684	0.0082	622	0.9985	1205	0.9658	1211	1.0000	866
959	qnap-004	0.0046	313	0.0061	297	0.0038	518	0.0059	579	0.0042	377	0.9960	1182	0.9442	1193	1.0000	830
960	qnap-005	0.0033	224	0.0047	223	0.0068	755	0.0071	645	0.0096	683	0.0184	469	0.0163	612	0.0948	279
961	qnap-006	-		-		0.0030	371	0.0041	418	0.0044	400	0.0088	264	0.0065	275	0.0799	216
962	quantasoft-003	0.0081	507	0.0113	492	0.0056	687	0.0076	671	0.0091	661	0.0161	432	0.0107	478	0.1193	331
963	rankone-000	-		0.2760	960	-		-		-		-		-		-	
964	rankone-001	-		0.4198	998	-		-		-		-		-		-	
965	rankone-002	0.1547	948	0.2170	943	-		-		-		-		-		-	
966	rankone-003	0.1175	930	0.1838	929	-		-		-		-		-		-	
967	rankone-004	0.0548	867	0.1015	886	-		-		-		-		-		-	
968	rankone-005	0.0405	841	0.0757	856	0.0170	1028	0.0299	1027	-		-		-		-	

Table 77: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
969	rankone-006	0.0242	770	0.0460	797	0.0119	952	0.0188	945	0.0315	993	0.0539	699	0.0385	811	-	
970	rankone-007	0.0197	742	0.0366	766	0.0113	936	0.0177	928	0.0286	976	0.0454	670	0.0333	786	-	
971	rankone-008	0.0124	632	0.0232	676	0.0082	822	0.0107	771	0.0188	882	0.0320	596	0.0244	729	-	
972	rankone-009	0.0087	521	0.0119	503	0.0065	744	0.0086	718	0.0088	646	0.0161	429	0.0121	522	-	
973	rankone-010	0.0079	497	0.0112	490	0.0061	720	0.0081	701	0.0088	643	0.0149	403	0.0117	509	-	
974	rankone-011	0.0049	327	0.0075	358	0.0038	514	0.0048	483	0.0060	526	0.0143	392	0.0080	372	0.1839	426
975	rankone-012	0.0043	294	0.0058	279	0.0031	406	0.0038	374	0.0047	432	0.0081	226	0.0065	271	0.0842	235
976	rankone-013	0.0028	173	0.0041	170	0.0026	232	0.0033	298	0.0028	210	0.0055	119	0.0040	104	0.0725	172
977	rankone-014	0.0016	72	0.0021	49	0.0024	124	0.0027	170	0.0022	85	0.0047	81	0.0035	54	0.0666	155
978	rankone-015	0.0009	20	0.0014	21	0.0024	132	0.0026	149	0.0024	123	0.0044	69	0.0037	67	0.0635	133
979	realnetworks-000	0.0325	822	0.0463	798	-		-		-		-		-		-	
980	realnetworks-001	0.0315	820	0.0455	795	0.0920	1195	0.1938	1208	-		-		-		-	
981	realnetworks-002	0.0248	776	0.0358	762	0.0513	1155	0.1127	1177	0.0371	1013	0.0614	719	0.0316	780	-	
982	realnetworks-003	0.0259	789	0.0372	769	0.0541	1161	0.1208	1181	0.0378	1016	0.0578	708	0.0306	776	-	
983	realnetworks-004	0.0075	484	0.0101	459	0.0066	747	0.0097	750	0.0108	727	0.0187	472	0.0131	542	0.1430	375
984	realnetworks-005	0.0070	454	0.0093	428	0.0063	732	0.0089	725	0.0092	666	0.0161	433	0.0104	467	0.1249	345
985	realnetworks-006	0.0040	274	0.0056	275	0.8657	1327	0.9705	1326	0.0059	519	0.0112	342	0.0085	395	0.0911	263
986	realnetworks-007	0.0031	202	0.0051	246	0.0028	319	0.0035	338	0.0048	444	0.0091	270	0.0074	329	0.0945	277
987	realnetworks-008	0.0022	125	0.0039	158	0.0038	516	0.0045	457	0.0055	484	0.0100	309	0.0080	375	0.0889	246
988	realsense-001	-		-		0.0023	79	0.0025	136	0.0023	104	0.0046	76	0.0039	89	0.0537	68
989	realsense-002	-		-		0.0022	52	0.0023	82	0.0023	92	0.0045	70	0.0038	80	0.0531	63
990	rebs-000	0.0030	194	0.0043	197	0.0024	139	0.0032	279	0.0024	109	0.3069	899	0.1170	938	0.7667	613
991	rebs-001	0.0037	254	0.0049	239	0.0024	163	0.0033	294	0.0025	131	0.0257	549	0.0096	438	0.1487	386
992	recognito-000	0.0004	5	0.0006	2	0.0021	30	0.0022	59	0.0016	20	0.0555	701	0.0207	692	0.1943	436
993	recognito-001	0.0003	4	0.0007	4	0.0021	20	0.0022	52	0.0016	14	0.0663	730	0.0257	737	0.1825	424
994	regula-000	0.0184	729	0.0376	770	0.0103	901	0.0185	938	0.0120	751	0.9983	1201	0.0231	716	1.0000	869
995	regula-001	0.0072	471	0.0107	479	0.0102	898	0.0179	930	0.0123	759	0.0333	609	0.0174	636	0.3711	514
996	regula-002	-		-		0.0053	659	0.0058	569	0.0044	404	0.9999	1246	0.9995	1265	1.0000	867
997	regula-003	-		-		0.0056	683	0.0059	580	0.0060	528	0.5284	976	0.4298	1075	0.8213	634
998	remarkai-000	0.0147	676	0.0257	702	0.0102	895	0.0158	896	0.0163	837	0.0577	706	0.0185	654	-	
999	remarkai-001	0.0144	671	0.0256	700	0.0102	896	0.0159	899	0.0162	835	0.0582	711	0.0185	656	-	
1000	remarkai-002	0.0151	684	0.0197	640	0.0075	792	0.0108	777	0.0119	750	0.0187	473	0.0127	532	-	
1001	remarkai-003	0.0047	321	0.0063	309	0.0033	442	0.0049	490	0.0054	481	0.0100	308	0.0072	318	0.0908	260
1002	rendip-000	0.0055	371	0.0077	366	0.0048	624	0.0060	589	0.0080	615	0.0142	391	0.0110	487	0.1027	294
1003	revealmedia-005	0.0050	338	0.0074	354	0.0050	637	0.0068	633	0.0075	592	0.0124	366	0.0104	470	0.0895	251
1004	revealmedia-006	0.0040	273	0.0067	323	0.0041	559	0.0056	549	0.0056	493	0.0085	253	0.0068	287	0.0872	241
1005	roc-016	0.0009	21	0.0014	22	0.0023	84	0.0024	99	0.0024	114	0.0042	59	0.0035	53	0.0558	80
1006	roc-017	-		-		0.0023	75	0.0025	138	0.0021	69	0.0039	42	0.0032	37	0.0542	72
1007	roc-018	0.0007	13	0.0012	13	0.0022	48	0.0022	54	0.0018	40	0.0034	30	0.0029	22	0.0493	45
1008	roc-019	-		-		0.0021	36	0.0021	32	0.0016	21	0.0032	14	0.0027	14	0.0439	25
1009	roc-020	-		-		0.0021	15	0.0021	25	0.0016	19	0.0032	16	0.0027	13	0.0452	30
1010	roc-021	-		-		0.0021	10	0.0020	16	0.0015	9	0.0030	11	0.0026	8	0.0409	12
1011	rokid-000	0.0093	545	0.0145	546	0.0073	783	0.0102	764	0.0164	839	0.0280	563	0.0214	698	-	
1012	rokid-001	0.0105	585	0.0162	579	0.0094	872	0.0163	910	0.0181	869	0.0276	562	0.0165	620	-	

Table 78: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1013	s1-001	0.0314	819	0.0651	840	0.0252	1081	0.0357	1048	0.0444	1040	0.0653	726	0.0429	828	-	
1014	s1-002	0.0095	554	0.0144	543	0.0112	932	0.0196	954	0.0234	935	0.0371	641	0.0282	760	-	
1015	s1-003	0.0051	346	0.0073	350	0.0044	586	0.0063	607	0.0052	469	0.0096	294	0.0070	303	0.0870	240
1016	s1-004	0.0053	359	0.0080	384	0.0038	517	0.0059	577	0.0057	503	0.0103	314	0.0073	324	0.0899	255
1017	s1-005	0.0024	146	0.0036	138	0.0025	215	0.0029	226	0.0026	174	0.0048	91	0.0038	82	0.0638	135
1018	s1-006	0.0029	183	0.0044	198	0.0028	310	0.0033	292	0.0035	304	0.0073	190	0.0044	139	0.7160	602
1019	s1-007	0.0023	134	0.0056	274	0.0024	125	0.0028	186	0.0025	143	0.0048	87	0.0039	91	0.0608	110
1020	s1-008	-		-		0.0093	864	0.0137	846	0.0179	867	0.9118	1092	0.0286	765	0.9999	802
1021	s1-009	-		-		0.0023	115	0.0023	88	0.0022	77	0.0043	61	0.0037	68	0.0504	55
1022	saffe-001	0.4339	1016	0.5261	1019	0.7539	1317	0.8736	1315	0.7977	1246	0.9810	1138	0.7435	1142	-	
1023	saffe-002	0.0119	620	0.0206	645	0.0107	916	0.0177	926	0.0244	948	0.9998	1235	0.2785	1031	-	
1024	samsungsds-000	0.0046	315	0.0069	333	0.0132	978	0.0081	698	0.0099	697	0.0179	461	0.0162	611	0.1101	308
1025	samsungsds-001	0.0015	59	0.0026	78	0.0023	89	0.0023	87	0.0024	115	0.1660	828	0.0536	859	0.4995	549
1026	samsungsds-002	0.0017	81	0.0027	80	0.0023	82	0.0022	49	0.0021	67	0.0043	62	0.0036	64	0.0586	99
1027	samtech-000	0.3418	999	0.4425	1002	-		-		-		-		-		-	
1028	samtech-001	0.0197	743	0.0365	764	0.0146	1006	0.0241	992	0.0238	944	0.0394	653	0.0251	734	-	
1029	samtech-002	0.0084	514	0.0127	519	0.0093	862	0.0158	894	0.0113	740	0.0442	664	0.0132	543	0.4110	528
1030	scanalitix-001	-		-		0.9968	1345	0.9997	1344	1.0000	1309	1.0000	1276	1.0000	1292	1.0000	885
1031	scanovate-001	0.0175	716	0.0331	744	0.0163	1026	0.0248	994	0.2476	1177	0.3801	935	0.3740	1059	-	
1032	scanovate-002	0.0175	718	0.0355	759	0.0146	1003	0.0286	1021	0.0269	960	0.0301	575	0.0178	644	-	
1033	scanovate-003	0.0054	365	0.0080	382	0.0054	667	0.0072	655	0.0312	990	0.0599	715	0.0568	871	-	
1034	sdc-000	0.0303	811	0.0526	817	0.0572	1165	0.1094	1173	0.0867	1107	0.6230	1001	0.3682	1058	0.8797	660
1035	seamfix-001	-		-		0.1973	1233	0.2878	1228	0.2613	1180	0.3156	907	0.1976	997	0.8176	633
1036	securifai-001	0.4538	1021	0.6142	1037	0.5844	1290	0.7428	1286	0.7051	1232	0.9961	1184	0.9558	1200	-	
1037	securifai-002	0.7557	1054	0.8574	1059	0.4550	1274	0.5953	1268	0.9860	1289	0.9796	1137	0.9158	1183	-	
1038	securifai-003	0.4086	1010	0.7577	1054	0.7233	1308	0.8070	1297	0.7787	1243	1.0000	1285	0.9988	1258	1.0000	893
1039	securifai-004	0.0136	656	0.0192	630	0.0064	737	0.0099	757	0.0115	742	0.0272	558	0.0127	534	0.4881	545
1040	securifai-005	0.0125	635	0.0190	626	0.0080	814	0.0126	815	0.0134	783	0.9861	1145	0.9205	1185	1.0000	821
1041	securifai-006	0.0140	666	0.0196	637	0.0067	753	0.0102	761	0.0113	738	0.9888	1154	0.9239	1187	1.0000	826
1042	securifai-007	0.0124	629	0.0175	604	0.0064	739	0.0105	770	0.0108	720	0.9999	1245	0.9966	1246	1.0000	892
1043	securifai-008	-		-		0.0060	718	0.0102	763	0.2096	1166	0.9649	1124	0.8757	1168	0.9990	774
1044	securitypitch-000	-		-		0.0286	1097	0.0382	1058	0.0393	1021	0.9979	1197	0.9881	1228	1.0000	857
1045	sensetime-001	0.0063	426	0.0092	423	0.0130	974	-		0.0252	954	0.0886	764	0.0886	914	-	
1046	sensetime-002	0.0068	451	0.0098	448	0.0143	997	1.0000	1358	0.0278	973	0.0502	689	0.0502	851	-	
1047	sensetime-003	0.0021	114	0.0027	83	0.0027	272	0.0027	179	0.0051	464	0.0100	307	0.0089	406	-	
1048	sensetime-004	0.0026	162	0.0038	149	0.0022	55	0.0023	74	0.0042	374	0.0082	231	0.0078	358	-	
1049	sensetime-005	0.0019	94	0.0029	95	0.0022	43	0.0021	35	0.0023	96	0.0044	65	0.0039	87	0.0564	84
1050	sensetime-006	0.0014	54	0.0024	63	0.0021	32	0.0020	13	0.0021	60	0.0040	50	0.0036	63	0.0525	58
1051	sensetime-007	0.0012	45	0.0022	52	0.0021	35	0.0020	18	0.0018	36	0.0034	25	0.0029	28	0.0479	43
1052	sensetime-008	0.0008	14	0.0014	18	0.0021	16	0.0020	6	0.0018	42	0.0036	34	0.0033	43	0.0504	54
1053	sensetime-010	-		-		0.0022	38	0.0021	31	0.0018	33	0.0034	28	0.0029	27	0.0437	22
1054	sensetime-011	-		-		0.0022	37	0.0020	23	0.0017	26	0.0033	22	0.0029	20	0.0438	24
1055	sensetime-012	-		-		0.0022	45	0.0023	76	0.0015	8	0.0029	8	0.0026	7	0.0372	1
1056	serendipity-000	0.0046	310	0.0069	334	0.0042	575	0.0056	548	0.0073	582	0.9129	1093	0.7189	1138	0.9952	752

Table 79: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1057	sertis-000	0.0118	617	0.0208	648	0.0080	810	0.0127	818	0.0110	733	0.0176	459	0.0114	498	-			
1058	sertis-001	0.0113	603	0.0182	614	0.0204	1052	0.0403	1065	0.0216	916	0.0763	751	0.0282	759	-			
1059	sertis-002	0.0049	328	0.0061	292	0.0039	542	0.0061	599	0.0055	486	0.0099	305	0.0070	305	0.0921	267		
1060	sertis-003	0.0049	331	0.0059	289	0.0037	505	0.0058	568	0.0055	492	0.0100	306	0.0072	319	0.0881	243		
1061	seventhsense-000	0.0067	448	0.0099	455	0.0045	589	0.0065	618	0.0093	670	0.0169	449	0.0124	528	0.1325	355		
1062	seventhsense-001	0.0034	230	0.0047	222	0.0025	214	0.0031	263	0.0029	219	0.0338	615	0.0109	482	0.2135	451		
1063	seventhsense-002	0.0036	250	0.0050	242	0.0028	318	0.0036	348	0.0035	303	0.0811	755	0.0183	651	0.5444	556		
1064	seventhsense-003	0.0035	241	0.0046	213	0.0026	255	0.0033	293	0.0062	535	0.0448	668	0.0175	638	0.2594	476		
1065	seventhsense-005	0.0031	200	0.0045	209	0.0028	312	0.0030	232	0.0028	202	0.0466	677	0.0165	621	0.1949	437		
1066	seventhsense-006	-		-		0.0029	344	0.0034	324	0.0035	312	0.2052	851	0.0802	901	0.4742	541		
1067	shaman-000	0.9297	1065	0.9774	1073	0.9990	1350	0.9997	1345	0.9999	1306	1.0000	1265	0.9999	1285	-			
1068	shaman-001	0.3346	996	0.4616	1004	0.2368	1241	0.3723	1241	0.3574	1198	0.3527	922	0.2304	1012	-			
1069	shu-001	0.0103	574	0.0140	538	0.0293	1104	0.0688	1130	0.0172	858	0.0182	465	0.0120	518	-			
1070	shu-002	-		0.0079	377	0.0146	1005	0.0308	1031	1.0000	1312	0.0183	466	0.0115	501	-			
1071	shu-003	0.0028	172	0.0041	178	0.0050	634	0.0088	723	0.0081	621	0.0133	377	0.0094	428	-			
1072	shufti-001	-		-		0.0053	658	0.0083	706	0.0085	632	1.0000	1309	0.0098	444	1.0000	1309		
1073	siat-002	0.0091	535	0.0126	518	0.0109	920	0.0190	947	0.0276	972	0.0516	693	0.0464	843	-			
1074	siat-004	0.0067	444	0.0099	452	0.0152	1014	1.0000	1357	0.0275	967	0.4823	961	0.4823	1083	-			
1075	siat-005	0.0021	108	0.0038	150	0.0059	707	0.0049	493	0.0742	1091	0.9623	1121	0.6801	1123	0.9833	720		
1076	sightropic-000	-		-		0.0028	333	0.0038	381	0.0029	215	0.2207	858	0.0810	903	0.5137	551		
1077	sightropic-001	-		-		0.0052	657	0.0049	494	0.0050	456	0.0094	285	0.0081	379	0.0672	157		
1078	sightropic-002	-		-		0.0027	297	0.0037	365	0.0028	192	0.0051	105	0.0038	78	0.0599	105		
1079	sjtu-001	0.0051	344	0.0080	380	0.0211	1056	0.0446	1078	0.0131	773	0.0175	455	0.0110	485	-			
1080	sjtu-002	0.0053	364	0.0078	372	0.0138	986	0.0296	1026	0.0084	628	0.0136	380	0.0095	434	-			
1081	sjtu-003	0.0017	79	0.0033	116	0.0030	366	0.0037	363	0.0058	510	0.0104	318	0.0081	377	-			
1082	sjtu-004	0.0014	53	0.0025	64	0.0027	281	0.0028	213	0.0046	421	0.0086	257	0.0073	323	0.0906	258		
1083	sjtu-005	0.0029	179	0.0036	139	0.0025	219	0.0027	173	0.0044	396	0.0120	363	0.0072	314	0.1325	354		
1084	sjtu-006	-		-		0.0027	289	0.0028	193	0.0044	403	0.0081	229	0.0071	312	0.0715	169		
1085	sjtu-007	-		-		0.0023	111	0.0023	64	0.0040	354	0.0251	541	0.0126	531	0.1694	414		
1086	sjtu-008	-		-		0.0022	42	0.0023	68	0.0029	211	0.0078	213	0.0055	215	0.0748	189		
1087	sjtu-009	-		-		0.0022	44	0.0021	46	0.0028	195	0.5621	984	0.0397	816	0.6795	593		
1088	sktelecom-000	0.0038	263	0.0054	265	0.0031	377	0.0051	515	0.0042	372	0.3418	916	0.0061	253	0.9499	691		
1089	smargtech-001	-		-		0.4604	1276	0.6005	1270	0.5247	1217	0.5858	991	0.3098	1041	0.9826	719		
1090	smartbiometrik-001	0.5485	1037	0.6442	1040	0.7550	1318	0.8611	1311	0.8677	1263	0.8270	1064	0.7030	1128	0.9752	710		
1091	smartengines-000	0.6240	1044	0.7562	1053	0.9552	1334	0.9784	1330	0.9515	1282	0.9288	1105	0.8200	1156	0.9930	742		
1092	smartengines-001	0.6434	1046	0.7666	1055	0.9446	1333	0.9750	1329	0.9387	1280	0.9556	1116	0.8647	1163	0.9972	763		
1093	smartvist-000	0.0912	911	0.1587	915	0.1163	1210	0.1841	1204	0.1397	1133	0.9372	1110	0.7107	1134	0.9990	775		
1094	smartvist-001	0.0137	659	0.0244	688	0.0099	890	0.0158	895	0.0151	814	0.1349	809	0.0419	824	0.5589	562		
1095	smilart-001	0.6476	1047	0.8779	1061	-		-		-		-		-		-			
1096	smilart-002	0.2440	979	0.3532	986	-		-		0.3785	1199	0.4145	947	0.2611	1025	-			
1097	smilart-003	0.6944	1051	0.8836	1062	0.0695	1180	0.1193	1179	0.0894	1109	0.1221	799	0.0737	892	-			
1098	sodec-000	0.0033	223	0.0044	205	0.0040	548	0.0053	531	0.0054	482	0.0096	292	0.0080	368	0.0915	264		
1099	softtech-000	-		-		0.0024	158	0.0023	83	0.0022	80	0.0043	64	0.0038	79	0.0641	137		
1100	sparktek-000	-		-		0.0025	172	0.0025	130	0.0031	257	0.0058	129	0.0053	197	0.0643	139		

Table 80: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1101	sparktek-001	-		-		0.0021	26	0.0020	12	0.0017	28	0.0034	24	0.0029	23	0.0470	37		
1102	sparsh-001	0.0018	90	0.0029	99	0.0026	241	0.0028	204	0.0027	182	0.8806	1080	0.6313	1114	0.9963	755		
1103	sqisoft-001	0.1220	933	0.2088	939	0.1978	1234	0.3386	1237	0.2111	1167	0.2798	892	0.1474	965	0.7820	621		
1104	sqisoft-002	0.0082	509	0.0124	511	0.0051	645	0.0086	717	0.0102	706	0.0183	467	0.0122	523	0.1498	388		
1105	sqisoft-003	0.0041	287	0.0055	271	0.0026	226	0.0032	289	0.0039	349	1.0000	1365	1.0000	1425	1.0000	1319		
1106	staqu-000	0.0139	663	0.0208	649	0.0104	902	0.0145	873	0.0156	820	0.8063	1051	0.1408	960	-			
1107	starhybrid-001	0.0108	590	0.0138	534	0.0081	816	0.0113	787	0.0152	815	0.0265	554	0.0189	664	-			
1108	stcon-000	0.0040	277	0.0056	273	0.0031	393	0.0047	471	0.0048	445	0.9863	1146	0.3562	1056	0.9999	804		
1109	stcon-001	0.0040	269	0.0053	262	0.0030	372	0.0045	458	0.0047	430	0.8632	1076	0.1788	988	0.9892	733		
1110	stcon-002	0.0031	196	0.0043	187	0.0025	184	0.0034	318	0.0030	237	0.9988	1209	0.9855	1224	1.0000	842		
1111	stcon-003	0.0038	264	0.0050	243	0.0024	127	0.0032	286	0.0023	95	0.1082	787	0.0350	795	0.3982	525		
1112	stcon-004	-		-		0.0022	56	0.0025	137	0.0020	56	0.0519	694	0.0145	580	0.2564	473		
1113	stcon-005	-		-		0.0021	24	0.0021	28	0.0017	24	0.0336	612	0.0115	503	0.1725	416		
1114	stcon-006	-		-		0.0021	27	0.0021	29	0.0017	23	0.0034	26	0.0028	16	0.0425	18		
1115	stcon-007	-		-		0.0021	25	0.0021	26	0.0017	22	0.0033	21	0.0028	15	0.0424	17		
1116	stcon-008	-		-		0.0021	18	0.0020	4	0.0015	6	0.0029	10	0.0026	6	0.0390	9		
1117	stcon-009	-		-		0.0021	11	0.0020	8	0.0015	4	0.0029	4	0.0025	4	0.0380	5		
1118	stellartecs-001	-		-		0.0046	606	0.0060	587	0.0046	420	0.0658	728	0.0181	649	0.3818	518		
1119	stengg-000	0.0059	404	0.0082	389	0.0037	508	0.0060	586	0.0050	454	0.5065	966	0.0792	897	0.9348	686		
1120	sukshi-000	0.5409	1035	0.6612	1045	0.4556	1275	0.6567	1277	0.9296	1279	0.8898	1084	0.7384	1139	0.9899	734		
1121	sumsub-000	-		-		0.0041	560	0.0043	443	0.0038	338	0.5301	977	0.2536	1019	0.9068	675		
1122	sumsub-001	-		-		0.0023	81	0.0025	143	0.0028	207	0.0052	107	0.0044	128	0.0633	130		
1123	suprema-000	0.0064	429	0.0092	424	0.0081	819	0.0096	746	0.0139	794	0.0254	547	0.0220	704	-			
1124	suprema-001	0.0041	282	0.0053	258	0.0038	526	0.0047	472	0.0060	523	0.0111	338	0.0095	437	0.0907	259		
1125	suprema-002	0.0030	193	0.0041	177	0.0034	449	0.0040	399	0.0045	406	0.0085	250	0.0072	320	0.0781	208		
1126	suprema-003	0.0028	177	0.0041	176	0.0034	447	0.0039	386	0.0030	240	0.3095	903	0.0580	873	0.6672	589		
1127	suprema-004	0.0024	141	0.0035	130	0.0032	415	0.0036	352	0.0028	190	0.0053	111	0.0045	142	0.0625	123		
1128	suprema-005	0.0022	118	0.0036	136	0.0034	463	0.0031	269	0.0026	175	0.0050	102	0.0044	130	0.0588	102		
1129	supremaid-001	0.0053	363	0.0073	351	0.0045	590	0.0066	622	0.0099	699	0.0186	471	0.0148	587	0.1040	298		
1130	supremaid-002	0.0063	425	0.0094	436	0.0044	583	0.0062	602	0.0072	580	0.0229	525	0.0095	436	0.2986	493		
1131	surrey-cvssp-000	0.9084	1062	0.9909	1076	0.9923	1341	0.9950	1337	0.9981	1298	0.9994	1219	0.9979	1251	0.9998	794		
1132	surrey-cvssp-001	1.0000	1315	1.0000	1160	0.0077	801	0.0079	692	0.0266	958	0.3822	938	0.0551	863	0.7285	605		
1133	surrey-cvssp-002	0.0019	95	0.0028	89	0.0027	301	0.0028	194	0.0024	118	0.5978	998	0.1206	943	0.8488	643		
1134	surrey-cvssp-003	0.0016	70	0.0028	87	0.0022	57	0.0024	100	0.0022	87	0.3048	898	0.0539	860	0.5830	569		
1135	surrey-cvssp-004	-		-		0.0029	351	0.0025	144	0.0025	149	0.0690	738	0.0076	351	0.2741	485		
1136	swsam-001	0.0268	793	0.0476	803	0.0271	1090	0.0460	1083	0.0584	1071	0.7745	1040	0.5013	1089	0.9642	700		
1137	synesis-000	0.1752	955	0.3241	979	-		-		-		-		-		-			
1138	synesis-001	0.1777	956	1.0000	1413	1.0000	1369	-		-		-		-		-			
1139	synesis-003	0.8375	1057	0.8857	1063	0.8548	1326	0.9178	1322	-		-		-		-			
1140	synesis-004	0.0310	816	0.0480	805	0.0476	1147	0.0443	1076	0.0764	1094	-		-		-			
1141	synesis-005	0.0147	675	0.0226	671	0.0153	1015	0.0226	980	0.0213	913	0.0375	645	0.0262	747	-			
1142	synesis-006	0.0070	458	0.0096	440	0.0107	913	0.0166	917	-		0.0128	374	0.0089	405	-			
1143	synesis-007	0.0050	339	0.0073	352	0.0062	731	0.0076	670	-		0.0105	320	0.0080	373	-			
1144	synology-000	0.0149	681	0.0238	685	0.0148	1007	0.0261	1006	0.0221	920	0.0331	605	0.0209	695	-			

Table 81: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1145	synology-001	0.0214	753	0.0307	730	0.0266	1089	0.0513	1102	0.0283	974	0.8884	1082	0.2589	1022	-	
1146	synology-002	0.0104	578	0.0153	563	0.0107	915	0.0184	935	0.0189	884	0.2032	849	0.0180	646	-	
1147	syscake-000	-		-		0.0091	853	0.0155	889	0.0142	798	0.0287	570	0.0145	583	0.1682	412
1148	szpu-000	-		-		0.0034	450	0.0037	364	0.0039	345	0.8444	1070	0.3886	1066	0.8849	664
1149	sztu-000	0.0092	541	0.0139	536	0.0091	855	0.0201	960	0.0136	788	0.0685	737	0.0118	513	-	
1150	sztu-001	0.0031	197	0.0043	195	0.0025	199	0.0028	205	0.0051	458	0.0113	348	0.0089	407	0.9963	756
1151	t4isb-000	0.0058	385	0.0087	406	0.0041	568	0.0064	615	0.0083	625	0.0157	422	0.0103	464	0.1196	333
1152	tahakom-001	-		-		0.0223	1064	0.0364	1052	0.0361	1008	1.0000	1266	0.9998	1277	1.0000	873
1153	tahakom-002	-		-		0.0380	1125	0.0456	1080	0.0403	1025	0.4790	960	0.2494	1015	0.8529	644
1154	taiwanmobile-001	-		-		0.2136	1236	0.3014	1234	0.3408	1194	0.9439	1113	0.7559	1144	0.9966	759
1155	taiwanmobile-002	-		-		0.0088	848	0.0114	788	0.0108	725	0.9776	1135	0.8683	1165	0.9998	793
1156	tcc-000	-		-		0.1374	1218	0.2052	1214	0.6334	1225	0.9985	1204	0.9897	1231	0.9999	798
1157	tcc-001	-		-		0.0120	954	0.0217	972	0.2620	1181	0.9958	1180	0.9791	1218	1.0000	825
1158	tcv-001	-		-		0.0301	1108	0.0556	1106	0.0570	1069	0.1222	800	0.0786	894	0.3172	499
1159	tech5-000	0.0114	605	0.0160	577	-		-		-		-		-		-	
1160	tech5-001	0.0130	645	0.0176	607	0.0218	1062	0.0419	1070	0.0156	821	-		-		-	
1161	tech5-002	0.0046	314	0.0063	311	0.0113	938	0.0216	970	0.0233	933	0.9609	1119	0.7704	1146	-	
1162	tech5-003	0.0053	361	0.0070	336	0.0099	889	0.0185	937	0.0095	679	0.6670	1013	0.1024	925	-	
1163	tech5-004	0.0123	628	0.0234	681	0.0086	843	0.0162	907	0.0065	548	0.0112	343	0.0082	383	-	
1164	tech5-005	0.0054	368	0.0072	341	0.0069	760	0.0122	806	0.0060	522	0.0094	287	0.0066	277	-	
1165	tech5-007	0.0020	101	0.0029	96	0.0024	138	0.0028	187	0.0034	297	0.8622	1075	0.5335	1096	0.9867	731
1166	tech5-008	0.0020	102	0.0031	108	0.0026	239	0.0028	200	0.0026	167	0.9238	1101	0.7170	1137	0.9984	767
1167	tech5-009	-		-		0.0041	561	0.0040	398	0.0045	410	0.0087	261	0.0080	370	0.0821	227
1168	techainer-001	0.0061	414	0.0092	425	0.0046	602	0.0073	658	0.0060	524	0.9575	1117	0.8567	1162	0.9995	784
1169	techainer-002	-		-		0.0058	700	0.0121	803	0.0136	789	0.0853	761	0.0313	778	0.1610	403
1170	techainer-003	-		-		0.0029	343	0.0033	306	0.0046	429	0.0084	242	0.0074	330	0.0623	121
1171	techbiometricid-001	-		-		0.0877	1194	0.1663	1196	0.1000	1119	0.5946	996	0.3065	1039	0.9792	716
1172	techbiometricid-002	-		-		0.0060	713	0.0091	729	0.8984	1271	1.0000	1287	1.0000	1302	1.0000	904
1173	techsign-000	0.0325	823	0.0511	813	0.0435	1139	0.0710	1133	0.0746	1092	0.1104	788	0.0841	906	0.4288	532
1174	techsign-001	0.0110	594	0.0196	636	0.0067	752	0.0120	799	0.0087	641	0.2475	876	0.0883	913	0.6767	592
1175	telexper-000	-		-		0.0056	684	0.0077	675	0.0104	710	0.9976	1195	0.8691	1166	0.9999	806
1176	tevian-000	0.0871	908	0.1289	900	-		-		-		-		-		-	
1177	tevian-001	0.0813	907	0.1266	897	-		-		-		-		-		-	
1178	tevian-002	0.0279	800	0.0392	776	-		-		-		-		-		-	
1179	tevian-003	0.0217	756	0.0298	726	0.0230	1072	0.0484	1094	-		-		-		-	
1180	tevian-004	0.0228	765	0.0304	729	0.0226	1070	0.0478	1091	0.0128	766	0.0228	522	0.0138	565	-	
1181	tevian-005	0.0043	297	0.0062	300	0.0057	695	0.0085	715	0.0070	568	0.0135	379	0.0119	514	-	
1182	tevian-006	0.0045	309	0.0061	293	0.0045	597	0.0066	620	0.0046	427	0.0091	269	0.0075	342	-	
1183	tevian-007	0.0019	99	0.0027	79	0.0032	420	0.0041	411	0.0045	411	0.0086	255	0.0078	361	0.0925	271
1184	tevian-008	0.0012	41	0.0017	34	0.0033	427	0.0042	434	0.0042	375	0.0081	225	0.0068	289	0.0909	261
1185	tiger-001	0.3120	992	0.3974	994	-		-		-		-		-		-	
1186	tiger-002	0.0658	884	0.0889	874	0.1083	1203	0.1766	1200	0.0952	1114	0.1568	823	0.0661	881	-	
1187	tiger-003	0.0313	818	0.0602	835	0.0188	1036	0.0359	1050	0.0344	999	-		-		-	
1188	tiger-004	0.0779	899	0.1393	904	0.0488	1149	0.0905	1158	0.0800	1098	0.1640	826	0.0617	876	-	

Table 82: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1189	tiger-005	0.0624	878	0.2450	952	0.0292	1102	0.0556	1107	0.0430	1035	1.0000	1251	0.9964	1243	1.0000	843		
1190	tiger-006	0.0066	440	0.0101	464	0.0050	643	0.0075	669	0.0089	652	0.0158	425	0.0117	508	0.1466	382		
1191	tiger-007	-		-		0.0041	567	0.0063	603	0.0088	647	0.7340	1026	0.3935	1067	0.9464	690		
1192	tinkoff-001	0.0145	672	0.0244	691	0.0318	1111	0.0636	1121	0.0236	940	1.0000	1457	0.0339	789	1.0000	1093		
1193	tngd-001	-		-		0.0827	1189	0.1404	1188	0.0981	1118	0.1409	816	0.0808	902	0.7653	612		
1194	tngd-002	-		-		0.1471	1222	0.2504	1220	0.1879	1154	0.7463	1030	0.6359	1117	0.9555	693		
1195	tnitech-000	0.0025	151	0.0037	144	0.0033	432	0.0035	339	0.0025	135	0.8893	1083	0.6283	1113	0.9883	732		
1196	tongyi-004	0.0428	846	0.0540	820	-		-		-		-		-		-			
1197	tongyi-005	0.0073	476	0.0146	547	0.0187	1034	0.0421	1071	0.0161	833	0.0215	505	0.0149	590	-			
1198	tongyitrans-001	0.0617	877	0.0715	850	-		-		-		-		-		-			
1199	tongyitrans-002	0.0437	849	0.0657	844	-		-		-		-		-		-			
1200	tongyitrans-003	0.0249	780	0.0474	801	-		-		-		-		-		-			
1201	toppanidgate-000	0.0021	109	0.0033	117	0.0026	234	0.0028	197	0.0039	350	0.0075	195	0.0068	285	0.0631	128		
1202	toppansecurity-001	-		-		0.0025	180	0.0027	184	0.0032	262	0.0059	138	0.0051	178	0.0499	51		
1203	toshiba-000	0.1906	962	0.2896	969	-		-		-		-		-		-			
1204	toshiba-001	0.1063	923	0.1926	933	-		-		-		-		-		-			
1205	toshiba-002	0.0134	652	0.0222	669	0.0097	883	0.0154	888	-		0.0327	601	0.0158	604	-			
1206	toshiba-003	0.0125	634	0.0214	656	0.0085	840	0.0131	830	-		0.0241	532	0.0151	595	-			
1207	toshiba-004	0.0030	186	0.0042	185	0.0025	198	0.0027	183	0.0034	293	0.0063	156	0.0053	202	0.0700	166		
1208	toshiba-005	0.0023	136	0.0037	146	0.0024	146	0.0026	155	0.0072	578	0.0141	388	0.0130	541	0.0918	266		
1209	toshiba-006	0.0022	127	0.0035	127	0.0024	156	0.0025	135	0.0027	183	0.7425	1028	0.3070	1040	0.9691	703		
1210	toshiba-007	0.0016	68	0.0023	60	0.0024	166	0.0023	70	0.0021	64	0.0629	723	0.0056	218	0.2301	456		
1211	toshiba-008	0.0012	44	0.0017	35	0.0024	149	0.0022	56	0.0019	50	0.0108	330	0.0039	98	0.0899	253		
1212	toshiba-009	-		-		0.0024	136	0.0021	47	0.0019	48	0.0256	548	0.0055	216	0.1500	389		
1213	toshiba-010	-		-		0.0024	122	0.0021	40	0.0017	30	0.0316	588	0.0053	198	0.1331	356		
1214	toshiba-011	-		-		0.0024	148	0.0022	51	0.0019	45	0.0037	39	0.0033	45	0.0453	31		
1215	toshiba-012	-		-		0.0022	62	0.0022	50	0.0019	44	0.0040	48	0.0034	47	0.0503	53		
1216	toss-000	-		-		0.0023	67	0.0022	53	0.0021	66	0.0041	52	0.0035	51	0.0474	38		
1217	touchlessid-000	0.3296	995	0.4804	1009	0.4111	1267	0.6026	1271	0.5324	1219	0.9996	1227	0.9964	1244	1.0000	854		
1218	touchlessid-001	0.0076	485	0.0104	469	0.0680	1178	0.0842	1149	0.1358	1130	1.0000	1253	0.9995	1266	1.0000	912		
1219	touchlessid-002	0.0070	457	0.0096	443	0.0407	1134	0.0469	1088	0.4234	1209	0.9991	1213	0.9953	1241	1.0000	845		
1220	touchlessid-003	0.0061	413	0.0081	386	0.0040	553	0.0053	532	0.0056	497	0.0155	418	0.0082	382	0.1370	364		
1221	touchlessid-004	-		-		0.0035	467	0.0049	485	0.0043	390	0.0077	204	0.0058	233	0.0793	213		
1222	transmitsecurity-001	-		-		0.0051	651	0.0078	686	0.0089	653	0.0151	408	0.0111	490	0.1181	329		
1223	trueface-000	0.0249	777	0.4321	999	0.0119	953	0.0180	931	0.0297	985	0.7467	1031	0.1602	979	-			
1224	trueface-001	0.0204	749	0.0438	790	0.0095	877	0.0138	854	0.0154	817	0.0253	545	0.0169	625	-			
1225	trueface-002	0.0060	406	0.0096	441	0.0048	623	0.0061	595	0.0112	737	0.0198	489	0.0155	600	-			
1226	trueface-003	0.0070	461	0.0094	433	0.0053	661	0.0081	702	0.0122	755	0.0217	509	0.0159	606	0.1204	336		
1227	trueidvng-001	0.0063	421	0.0077	368	0.0033	441	0.0044	450	0.0046	424	0.0086	258	0.0069	295	0.0809	222		
1228	truststamp-001	0.0242	771	0.0484	807	0.0194	1042	0.0359	1049	0.0381	1017	0.8808	1081	0.6898	1125	0.9948	747		
1229	truststamp-002	-		-		0.0080	809	0.0138	855	0.0128	765	0.9718	1128	0.8779	1171	0.9998	792		
1230	truststamp-003	-		-		0.0047	616	0.0070	643	0.0073	583	0.7132	1022	0.4538	1080	0.9578	695		
1231	tsight-000	-		-		0.0020	1	0.0020	14	0.0014	1	0.0028	1	0.0025	1	0.0373	2		
1232	ttigroup-000	-		-		0.0028	320	0.0034	313	0.0043	395	0.9208	1096	0.7608	1145	0.9975	765		

Table 83: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1233	tuputech-000	0.3218	994	0.3696	989	-		-		0.3237	1191	0.4304	951	0.2973	1038	-	
1234	turingtechvip-001	0.0330	827	0.0540	821	0.0458	1146	0.1007	1165	0.4715	1212	0.9286	1104	0.8448	1160	0.9964	758
1235	turingtechvip-002	0.0126	640	0.0163	581	0.0092	861	0.0118	793	0.2264	1173	1.0000	1446	0.9925	1234	0.9999	809
1236	turkcell-000	0.1134	928	0.1288	899	0.0770	1186	0.1112	1176	0.2570	1178	1.0000	1250	0.9999	1284	1.0000	882
1237	turkcell-001	0.0043	295	0.0066	316	0.0037	513	0.0055	546	0.0043	383	0.0440	663	0.0082	384	0.3148	498
1238	turkcell-002	-		-		0.0036	482	0.0053	530	0.0044	397	0.0371	640	0.0075	337	0.2871	491
1239	twface-000	0.0051	343	0.0072	348	0.0041	558	0.0058	563	0.0071	573	0.0153	415	0.0100	446	0.1595	398
1240	twface-001	0.0036	246	0.0051	247	0.0031	394	0.0038	373	0.0049	446	0.0091	274	0.0075	341	0.0831	231
1241	ulsee-001	0.0151	685	0.0246	693	0.0113	935	0.0185	940	0.0187	881	0.6766	1014	0.0181	648	-	
1242	ultinous-000	0.2343	975	0.3484	984	-		-		-		-		-		-	
1243	ultinous-001	0.2485	980	0.4003	995	-		-		-		-		-		-	
1244	uluface-002	0.0081	506	0.0123	507	0.0071	767	0.0095	743	0.0107	718	1.0000	1316	0.0140	569	-	
1245	uluface-003	0.0100	567	0.0150	556	0.0079	805	0.0128	823	-		-		-		-	
1246	unicc-001	0.5101	1034	0.6167	1038	0.6577	1296	0.7894	1293	0.9208	1277	0.9866	1147	0.9386	1191	0.9999	813
1247	unicc-002	0.2153	972	0.2190	944	0.0417	1136	0.0526	1104	0.0891	1108	0.1321	808	0.1125	933	0.3813	517
1248	unicc-003	0.0217	757	0.0260	706	0.0093	867	0.0136	845	0.0129	768	0.0216	507	0.0143	577	0.2091	446
1249	unicc-004	-		-		0.0055	678	0.0077	677	0.0076	596	0.0147	401	0.0102	457	0.1421	374
1250	unissey-001	0.0095	550	0.0160	576	0.0134	981	0.0150	882	0.0147	809	0.0253	546	0.0163	613	0.1651	409
1251	unissey-002	0.0094	547	0.0151	557	0.0079	807	0.0110	780	0.0114	741	0.4424	954	0.1914	994	0.8051	630
1252	unissey-003	0.0057	380	0.0082	390	0.0047	614	0.0082	704	0.0067	554	0.5179	970	0.2863	1034	0.8684	651
1253	unissey-004	0.0043	291	0.0066	319	0.0037	497	0.0052	518	0.0048	442	0.5272	975	0.2903	1035	0.8777	658
1254	unissey-005	-		-		0.0037	496	0.0052	519	0.0086	636	0.0161	431	0.0134	548	0.1006	288
1255	unissey-006	-		-		0.0073	777	0.0121	804	0.0288	977	0.0529	698	0.0464	842	0.2808	487
1256	unissey-007	-		-		0.3124	1251	0.4571	1252	0.6912	1231	0.9823	1142	0.8827	1173	1.0000	827
1257	universeai-000	-		-		0.0023	105	0.0025	129	0.0043	391	0.0082	233	0.0075	335	0.0616	117
1258	upc-001	0.0234	767	0.0519	816	0.0291	1100	0.0490	1098	0.0294	982	0.2316	867	0.0389	813	-	
1259	urielsoft-000	-		-		0.0026	247	0.0032	283	0.0043	387	0.0083	236	0.0068	291	0.0744	183
1260	useb-000	0.0158	693	0.0222	667	0.0123	962	0.0168	921	0.0182	871	0.0289	572	0.0167	623	0.1614	404
1261	useb-001	0.0035	239	0.0053	261	0.0027	290	0.0033	304	0.0027	185	0.0085	252	0.0044	129	0.0806	218
1262	useb-002	0.0028	175	0.0039	160	0.0027	288	0.0031	268	0.0025	133	0.0090	266	0.0042	116	0.0867	237
1263	useb-003	-		-		0.0026	224	0.0030	233	0.0025	127	0.0091	273	0.0042	119	0.0901	256
1264	useb-004	-		-		0.0025	212	0.0028	211	0.0022	75	0.5197	974	0.2232	1009	0.8470	640
1265	uxlabs-001	0.0534	861	0.0570	828	0.0118	949	0.0131	829	0.0237	943	0.0399	655	0.0288	767	0.2097	448
1266	uxlabs-003	0.2133	971	0.2143	940	0.0386	1127	0.0457	1081	0.0374	1015	0.3343	912	0.1554	973	0.6941	597
1267	uzinfocom-000	-		-		0.0070	764	0.0120	801	0.0086	635	0.9874	1151	0.9469	1194	0.9999	797
1268	uzinfocom-001	-		-		0.0029	350	0.0038	375	0.0041	368	0.9963	1187	0.9725	1214	1.0000	836
1269	vaidio-001	-		-		0.0135	982	0.0166	916	0.0079	606	0.6587	1010	0.4279	1074	0.9123	676
1270	vcog-002	0.7522	1053	0.9033	1067	-		-		-		-		-		-	
1271	vcortex-001	0.0131	649	0.0207	647	0.0192	1039	0.0315	1034	0.2238	1172	0.3586	925	0.3536	1055	0.3569	507
1272	vd-000	0.3790	1005	0.4973	1013	-		-		-		-		-		-	
1273	vd-001	0.0243	773	0.0452	794	0.0271	1092	0.0402	1064	0.0424	1033	-		-		-	
1274	vd-002	0.0429	847	0.0704	848	0.0569	1164	0.0844	1150	0.0801	1099	0.0937	766	0.0577	872	-	
1275	vd-003	0.0199	746	0.0222	666	0.0115	944	0.0130	828	0.0138	792	0.0239	530	0.0177	642	0.1647	408
1276	veertec-001	-		-		0.0564	1163	0.0984	1164	0.0845	1104	0.1076	786	0.0691	886	0.4622	538

Table 84: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1277	veridas-001	0.1998	969	0.2724	959	0.2987	1249	0.4587	1253	0.2599	1179	0.2454	874	0.1357	955	-	
1278	veridas-002	0.1733	954	0.2257	947	0.2617	1245	0.4147	1246	0.2073	1164	0.2028	848	0.1112	932	-	
1279	veridas-003	0.0557	870	0.0983	883	0.0734	1182	0.1267	1184	0.0694	1089	0.0951	772	0.0480	846	-	
1280	veridas-004	0.0281	801	0.0467	800	0.0353	1122	0.0643	1124	0.0424	1032	0.0644	724	0.0342	790	-	
1281	veridas-005	0.0599	874	0.1110	891	0.0294	1106	0.0639	1122	0.0811	1102	0.1130	790	0.0548	862	-	
1282	veridas-006	0.0098	557	0.0167	590	0.0079	808	0.0127	817	0.0127	762	0.0217	508	0.0151	594	0.1573	394
1283	veridas-007	0.0063	424	0.0083	395	0.0044	585	0.0058	567	0.0080	616	0.0152	412	0.0120	520	0.1199	335
1284	veridas-008	0.0032	208	0.0045	211	0.0030	360	0.0033	299	0.0085	634	0.0206	495	0.0143	578	0.1228	340
1285	veridas-009	0.0031	195	0.0043	189	0.0030	373	0.0035	333	0.0055	483	0.0106	324	0.0100	447	0.0959	281
1286	veridas-010	-		-		0.0027	300	0.0033	303	0.0042	376	0.0080	224	0.0073	321	0.0649	144
1287	veridium-000	0.0726	892	0.1248	896	0.5226	1280	0.6652	1279	0.6425	1227	0.8150	1056	0.7989	1153	0.8772	657
1288	veridium-001	0.0274	797	0.0368	767	0.0292	1103	0.0475	1089	0.0488	1049	0.0673	732	0.0463	840	0.3691	513
1289	veridium-002	0.0343	829	0.0427	786	0.0283	1096	0.0460	1084	0.0489	1050	0.0673	731	0.0464	841	0.3658	509
1290	veridium-003	-		-		0.0286	1098	0.0462	1085	0.0492	1051	0.0679	735	0.0470	844	0.3672	510
1291	veridium-004	-		-		0.0141	990	0.0254	997	0.0217	917	0.0315	585	0.0197	678	0.2126	449
1292	verigram-000	0.0032	204	0.0043	192	0.0031	380	0.0034	316	0.0093	672	0.0175	456	0.0164	617	0.1158	325
1293	verigram-001	0.0032	209	0.0044	203	0.0027	284	0.0032	284	0.0030	229	0.9995	1223	0.9953	1240	1.0000	905
1294	verigram-002	0.0079	499	0.0089	416	0.0038	520	0.0040	401	0.0060	525	0.0113	349	0.0107	476	0.0634	131
1295	verigram-003	0.0101	568	0.0109	481	0.0034	452	0.0039	389	0.0037	327	0.0070	185	0.0063	262	0.0559	82
1296	verigram-004	-		-		0.0023	102	0.0023	75	0.0021	61	1.0000	1274	0.9999	1282	1.0000	884
1297	verihubs-inteligensia-000	0.0070	460	0.0098	447	0.0048	625	0.0076	674	0.0092	665	0.0160	428	0.0117	506	0.1348	360
1298	verihubs-inteligensia-001	0.0071	462	0.0114	496	0.0050	638	0.0076	673	0.0096	681	0.0165	441	0.0114	499	0.1409	372
1299	verihubs-inteligensia-002	0.0028	171	0.0043	190	0.0025	220	0.0030	235	0.0047	434	0.0092	275	0.0083	390	0.0983	284
1300	verijelas-000	0.2488	981	0.3431	982	0.4861	1278	0.6004	1269	0.0811	1101	0.1148	792	0.0440	832	0.4077	527
1301	via-000	0.0216	755	0.0365	765	0.0177	1031	0.0287	1022	0.0296	983	0.0572	704	0.0290	770	-	
1302	via-001	0.0149	680	0.0229	675	0.0114	942	0.0177	929	0.0183	875	0.4056	944	0.0176	639	-	
1303	via-004	0.0099	561	0.0135	531	0.0031	400	0.0044	451	0.0055	488	0.4009	942	0.0080	374	0.9840	724
1304	via-005	0.0063	423	0.0082	392	0.0031	382	0.0037	371	0.0032	264	0.3659	928	0.0068	286	0.8709	654
1305	via-006	-		-		0.0025	173	0.0027	168	0.0025	152	0.5904	994	0.0315	779	0.9455	689
1306	viante-000	0.0004	6	0.0007	5	0.0026	253	0.0026	158	0.0017	25	0.0033	18	0.0029	24	0.0475	41
1307	viante-001	-		-		0.0025	218	0.0025	125	0.0016	15	0.0031	13	0.0028	18	0.0432	20
1308	viante-002	-		-		0.0025	182	0.0024	102	0.0015	10	0.0029	9	0.0027	11	0.0399	11
1309	viante-003	-		-		0.0025	175	0.0023	78	0.0015	7	0.0029	5	0.0026	9	0.0388	7
1310	viante-004	-		-		0.0025	185	0.0023	79	0.0015	11	0.0029	7	0.0027	10	0.0390	8
1311	videmo-000	0.0298	810	0.0423	783	0.0155	1018	0.0260	1004	0.0246	950	0.0397	654	0.0239	724	-	
1312	videmo-001	0.0295	807	0.0417	780	0.0164	1027	0.0261	1007	0.0355	1005	0.0603	717	0.0442	834	0.2428	466
1313	videmo-002	0.0158	694	0.0288	718	0.0149	1012	0.0249	996	0.0230	928	0.3429	917	0.1468	964	0.8393	639
1314	videmo-003	-		-		0.0082	825	0.0138	853	0.0133	778	0.0220	513	0.0119	517	0.1596	400
1315	videonetics-001	0.5483	1036	0.6446	1041	0.7517	1316	0.8607	1310	0.8664	1261	0.8255	1061	0.6956	1127	-	
1316	videonetics-002	0.4274	1013	0.5329	1020	0.6081	1292	0.7438	1287	0.7775	1242	0.7297	1025	0.5756	1103	-	
1317	viettelai-000	0.0051	348	0.0068	327	0.0054	668	0.0064	612	0.9451	1281	1.0000	1313	1.0000	1415	1.0000	1310
1318	viettelai-001	0.0054	369	0.0076	363	0.0049	633	0.0079	691	0.0064	544	0.0225	518	0.0083	386	1.0000	1333
1319	viettelai-002	-		-		0.0029	347	0.0041	415	0.0069	560	0.0126	371	0.0074	331	0.1095	305
1320	viettelai-003	-		-		0.0034	456	0.0041	419	0.2038	1160	0.3193	910	0.1067	929	0.8681	650

Table 85: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1321	viettelai-004	-		-		0.0023	101	0.0028	203	0.0023	97	0.0044	66	0.0034	48	0.0566	87		
1322	viettelai-005	-		-		0.0023	92	0.0027	175	0.0021	65	0.0038	41	0.0030	34	0.0499	50		
1323	viettelhightech-000	0.0117	613	0.0166	586	0.0110	924	0.0198	958	0.0167	848	0.0249	539	0.0158	603	0.1604	402		
1324	viettelhightech-001	-		-		0.0031	395	0.0040	406	0.0032	271	0.0063	155	0.0051	177	0.0781	207		
1325	viettelhightech-002	-		-		0.0031	378	0.0042	436	0.0033	275	0.0061	150	0.0051	175	0.0758	194		
1326	vigilantsolutions-000	-		0.6881	1047	-		-		-		-		-		-			
1327	vigilantsolutions-001	-		0.3475	983	-		-		-		-		-		-			
1328	vigilantsolutions-002	0.2290	974	0.3214	977	-		-		-		-		-		-			
1329	vigilantsolutions-003	0.1878	961	0.2673	958	-		-		-		-		-		-			
1330	vigilantsolutions-004	0.1780	957	0.2598	956	-		-		-		-		-		-			
1331	vigilantsolutions-005	0.1613	951	0.4702	1007	0.0163	1025	0.0279	1017	-		-		-		-			
1332	vigilantsolutions-006	0.1264	934	0.3221	978	0.0150	1013	0.0254	998	0.0493	1053	0.2991	895	0.0403	819	-			
1333	vigilantsolutions-007	0.0202	748	0.0307	731	0.0136	983	0.0227	981	0.0356	1007	1.0000	1424	1.0000	1361	-			
1334	vigilantsolutions-008	0.0156	690	0.0233	678	0.0101	893	0.0147	875	0.0355	1006	0.0492	686	0.0293	773	-			
1335	vigilantsolutions-009	0.0117	612	0.0165	585	0.0075	793	0.0101	760	0.0219	919	0.0385	649	0.0238	723	-			
1336	vigilantsolutions-010	0.0109	591	0.0164	583	0.0074	788	0.0095	740	0.0209	909	0.0365	638	0.0233	718	0.2465	467		
1337	vigilantsolutions-011	0.0124	631	0.0176	605	0.0073	778	0.0095	739	0.0196	899	0.0360	630	0.0221	705	0.2633	480		
1338	vinai-000	0.0081	505	0.0124	510	0.0045	588	0.0072	653	0.0089	651	0.1814	834	0.0112	492	-			
1339	vinbigdata-001	0.2576	984	0.2763	961	0.1404	1219	0.1988	1211	0.1407	1135	0.1150	793	0.0703	889	0.3688	512		
1340	vinbigdata-002	0.0102	573	0.0175	602	0.0071	771	0.0084	711	0.0090	656	0.8017	1049	0.3134	1043	0.9936	745		
1341	vinbigdata-003	0.0771	897	0.0992	884	0.1202	1211	0.1756	1199	0.2762	1184	0.2369	871	0.0100	448	0.8914	670		
1342	vinbigdata-004	-		-		0.0031	384	0.0046	461	0.0037	329	0.1476	819	0.0063	265	0.4917	547		
1343	vinbigdata-005	-		-		0.0023	69	0.0024	108	0.0022	74	0.0068	177	0.0038	83	0.0699	165		
1344	vinbigdata-006	-		-		0.0022	61	0.0023	90	0.0019	54	0.0037	38	0.0032	38	0.0479	42		
1345	vinbigdata-007	-		-		0.0022	41	0.0021	41	0.0019	52	0.0036	36	0.0032	39	0.0475	39		
1346	vion-000	0.0419	843	0.0590	832	0.0422	1137	0.0478	1092	0.0581	1070	0.0968	776	0.0847	907	-			
1347	visage-000	0.0933	912	0.1441	911	0.1316	1215	0.2416	1218	0.1395	1132	0.1920	841	0.1001	922	-			
1348	visica-001	-		-		0.2293	1239	0.3588	1239	0.2936	1187	0.3564	924	0.1967	996	0.7795	620		
1349	visica-002	-		-		0.0520	1156	0.0898	1156	0.2960	1188	0.8202	1058	0.5956	1108	0.9838	722		
1350	visica-003	-		-		0.3326	1257	0.4772	1254	0.3237	1192	0.3591	926	0.2203	1008	0.7317	606		
1351	visionbox-000	0.0293	805	0.0541	822	0.0197	1044	0.0339	1038	0.0349	1001	0.0593	713	0.0329	784	-			
1352	visionbox-001	0.0159	697	0.0270	711	0.0111	928	0.0173	924	0.0190	886	0.0315	586	0.0205	689	-			
1353	visionbox-002	0.0058	386	0.0079	375	0.0060	711	0.0074	659	0.0084	630	0.0149	404	0.0113	496	0.0979	283		
1354	visionbox-003	0.0057	384	0.0075	359	0.0062	729	0.0083	708	0.0100	703	0.9915	1160	0.9625	1207	1.0000	819		
1355	visionbox-004	0.0046	318	0.0075	357	0.0060	716	0.0075	667	0.0056	494	0.0316	587	0.0070	306	0.2528	470		
1356	visionbox-005	-		-		0.0041	555	0.0057	557	0.0046	428	0.0109	333	0.0088	403	0.0767	199		
1357	visionbox-006	-		-		0.0063	733	0.0075	664	0.0048	440	0.0116	355	0.0094	429	0.0789	210		
1358	visionbox-007	-		-		0.0063	735	0.0075	663	0.0047	438	0.0116	357	0.0095	435	0.0793	212		
1359	visionbox-008	-		-		0.0056	691	0.0072	648	0.0044	402	0.0115	354	0.0089	408	0.0828	230		
1360	visionlabs-001	0.1049	922	0.1803	925	-		-		-		-		-		-			
1361	visionlabs-002	0.0326	824	0.0507	812	-		-		-		-		-		-			
1362	visionlabs-003	0.0178	723	0.0293	722	-		-		-		-		-		-			
1363	visionlabs-004	0.0219	759	0.0457	796	-		-		-		-		-		-			
1364	visionlabs-005	0.0088	525	0.0156	571	0.0048	620	0.0070	644	-		-		-		-			

Table 86: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1365	visionlabs-006	0.0037	256	0.0066	318	0.0041	566	0.0060	583	0.0061	531	0.0106	326	0.0076	352	-	
1366	visionlabs-007	0.0038	262	0.0048	226	0.0036	487	0.0048	477	0.0057	502	0.0097	297	0.0071	310	-	
1367	visionlabs-008	0.0026	164	0.0036	140	0.0031	404	0.0040	405	0.0045	413	0.0079	218	0.0064	266	-	
1368	visionlabs-009	0.0018	89	0.0025	68	0.0026	233	0.0029	229	0.0035	302	0.0064	158	0.0054	208	-	
1369	visionlabs-010	0.0017	75	0.0024	62	0.0026	252	0.0030	245	0.0033	283	0.0061	148	0.0052	190	-	
1370	visionlabs-011	0.0012	42	0.0022	55	0.0024	160	0.0026	156	0.0028	194	0.0053	112	0.0046	147	0.0635	132
1371	visteam-000	0.9200	1064	0.9489	1071	0.9959	1344	0.9984	1338	0.9994	1303	0.9978	1196	0.9914	1233	-	
1372	visteam-001	0.4417	1018	0.5385	1023	0.6410	1295	0.7788	1292	0.6386	1226	0.5904	992	0.4023	1070	-	
1373	visteam-002	0.1564	949	0.2789	963	0.1581	1225	0.2567	1224	0.1776	1150	0.2090	854	0.1021	924	0.5820	568
1374	visteam-003	0.0804	905	0.2166	941	0.0613	1171	0.1204	1180	0.0963	1115	0.1269	805	0.0441	833	0.6049	574
1375	visteam-004	0.0541	864	0.5202	1017	0.0406	1133	0.0827	1147	0.1879	1153	0.1795	833	0.0347	794	0.9673	702
1376	visteam-005	0.0235	768	0.0333	746	0.0265	1088	0.0583	1115	0.0341	998	0.0524	696	0.0259	741	0.2413	464
1377	visteam-006	0.0188	733	0.0256	701	0.0158	1022	0.0365	1053	0.0198	902	0.0340	618	0.0174	637	0.1904	431
1378	visteam-007	0.0312	817	0.0448	793	0.0416	1135	0.1008	1166	0.0466	1045	0.0602	716	0.0281	758	0.2601	477
1379	visteam-008	0.0378	837	0.0515	814	0.0391	1130	0.0937	1161	0.0437	1038	0.0567	703	0.0260	743	0.2533	471
1380	visteam-009	-		-		0.0138	987	0.0304	1029	0.0166	844	0.9993	1218	0.9719	1212	0.9997	788
1381	visteam-010	-		-		0.0156	1020	0.0347	1044	0.0189	883	1.0000	1271	0.9996	1268	1.0000	818
1382	vivotek-001	-		-		0.9982	1347	0.9994	1342	1.0000	1311	1.0000	1273	0.9999	1286	1.0000	855
1383	vivotek-002	-		-		0.0761	1185	0.1380	1186	0.9587	1284	1.0000	1277	1.0000	1294	1.0000	901
1384	vixvizion-000	0.0642	883	0.0882	873	0.0497	1152	0.0905	1157	0.0848	1105	0.1026	778	0.0618	877	-	
1385	vixvizion-001	0.0245	775	0.0407	778	0.0257	1084	0.0497	1099	0.0514	1056	0.0656	727	0.0400	818	-	
1386	vixvizion-002	0.0062	420	0.0086	404	0.0053	665	0.0075	665	0.0121	752	0.0207	496	0.0161	609	-	
1387	vixvizion-003	0.0059	400	0.0084	398	0.0059	705	0.0081	699	0.0119	748	0.0209	498	0.0162	610	-	
1388	vixvizion-004	0.0063	422	0.0094	437	0.0055	681	0.0081	700	0.0098	695	0.0157	423	0.0111	489	0.1417	373
1389	vixvizion-005	0.0276	798	0.0420	782	0.0302	1109	0.0629	1119	0.0288	978	0.0447	666	0.0235	720	0.2304	457
1390	vixvizion-006	0.0082	510	0.0122	506	0.0093	863	0.0194	951	0.0099	698	0.0169	448	0.0108	480	0.1338	358
1391	vixvizion-007	0.0110	596	0.0191	628	0.0080	812	0.0157	891	0.0101	705	0.0190	477	0.0118	512	0.1407	371
1392	vnis-000	0.1397	942	0.1608	917	0.2184	1238	0.2976	1232	0.5277	1218	0.3603	927	0.2714	1028	0.6929	595
1393	vnpay-000	0.2380	977	0.2614	957	0.4054	1265	0.4968	1257	0.3993	1206	0.4389	952	0.4006	1069	0.7693	615
1394	vnpay-001	-		-		0.0026	254	0.0030	237	0.0065	547	0.0124	367	0.0116	505	0.0826	228
1395	vnpt-001	0.3117	991	0.3523	985	0.3474	1258	0.2747	1226	0.3405	1193	0.5015	965	0.4827	1085	-	
1396	vnpt-002	0.0351	833	0.0424	785	0.0220	1063	0.0316	1035	0.0471	1046	0.0817	757	0.0698	888	0.2610	478
1397	vnpt-003	0.0117	611	0.0138	535	0.0040	549	0.0058	574	0.0087	640	0.0161	434	0.0126	530	0.1110	312
1398	vnpt-004	0.0058	390	0.0078	373	0.0037	507	0.0053	527	0.0051	460	0.4640	957	0.1384	957	0.7753	617
1399	vnpt-005	0.0036	247	0.0052	252	0.0027	285	0.0031	260	0.0036	313	0.0066	170	0.0056	220	0.0621	120
1400	vnpt-006	0.0034	234	0.0050	240	0.0027	275	0.0035	344	0.0026	170	0.0049	97	0.0040	110	0.0585	96
1401	vnpt-007	-		-		0.0025	211	0.0029	221	0.0025	147	0.0047	78	0.0039	96	0.0564	85
1402	vnpt-008	-		-		0.0026	229	0.0029	219	0.0025	153	0.0047	84	0.0040	100	0.0570	90
1403	vnpt-009	-		-		0.0030	353	0.0042	429	0.0030	236	0.0053	113	0.0042	120	0.0628	124
1404	vocalize-001	-		-		0.7356	1311	0.8750	1317	0.7830	1245	0.7067	1021	0.5268	1095	0.9164	680
1405	vocord-001	-		0.1415	910	-		-		-		-		-		-	
1406	vocord-002	0.0228	764	0.0341	751	-		-		-		-		-		-	
1407	vocord-003	0.1003	916	0.1092	889	-		-		-		-		-		-	
1408	vocord-004	0.0101	572	0.0174	601	0.0196	1043	0.0384	1059	-		-		-		-	

Table 87: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1409	vocord-005	0.0089	526	0.0155	569	0.0216	1061	0.0414	1069	-		-		-		-	
1410	vocord-006	0.0062	417	0.0102	465	0.0082	821	0.0151	883	0.0475	1047	0.0852	760	0.0788	895	-	
1411	vocord-007	0.0039	266	0.0053	264	0.0061	722	0.0094	738	0.0520	1059	0.0939	769	0.0910	915	-	
1412	vocord-008	0.0029	180	0.0038	155	0.0042	571	0.0055	541	0.0045	416	0.0086	259	0.0073	325	-	
1413	vocord-009	0.0022	122	0.0029	97	0.0036	478	0.0046	467	0.0052	468	0.0098	300	0.0086	400	-	
1414	vocord-010	0.0024	142	0.0031	107	0.0036	479	0.0049	498	0.0025	155	0.0065	162	0.0040	103	0.0679	159
1415	vogic-000	-		-		0.0043	578	0.0061	600	0.0078	604	0.8206	1059	0.6006	1109	0.9843	726
1416	vsoft-001	-		-		0.0707	1181	0.0877	1152	0.8390	1256	0.9519	1114	0.9506	1197	0.9655	701
1417	vsoft-002	-		-		0.0451	1145	0.0679	1129	0.0550	1066	0.3671	931	0.1633	982	0.8586	647
1418	vsoft-003	-		-		0.0326	1112	0.0563	1110	0.0526	1061	0.3671	930	0.1633	983	0.8586	648
1419	vsoft-004	-		-		0.0061	721	0.0103	766	0.0090	655	0.1830	836	0.0344	792	0.5669	563
1420	vts-000	0.0103	575	0.0174	598	0.0080	811	0.0129	827	0.0250	952	0.0450	669	0.0372	808	-	
1421	vts-001	0.0033	218	0.0048	229	0.0027	294	0.0036	353	0.0032	263	0.6519	1008	0.3563	1057	0.9218	683
1422	vts-002	-		-		0.0025	174	0.0032	288	0.0030	230	0.0113	346	0.0042	121	0.1593	397
1423	wicket-000	0.0018	91	0.0028	90	0.0024	153	0.0027	169	0.0031	246	0.7968	1048	0.4340	1077	0.9788	714
1424	winsense-000	0.0140	665	0.0228	674	0.0125	965	0.0215	969	0.0226	925	0.8091	1052	0.0461	839	-	
1425	winsense-001	0.0062	419	0.0099	457	0.0092	859	0.0210	965	0.0093	671	0.0144	397	0.0098	442	-	
1426	winsense-002	0.0050	340	0.0073	353	0.0038	522	0.0059	576	0.0064	543	0.0118	361	0.0084	391	-	
1427	wiseai-001	0.0658	885	0.0964	882	0.7743	1321	0.8956	1319	0.1967	1157	0.7526	1034	0.3419	1052	0.9695	704
1428	wuhantianyu-001	0.0163	702	0.0262	707	0.0281	1095	0.0569	1112	0.0316	994	0.0486	684	0.0344	791	0.2097	447
1429	x-laboratory-000	0.0071	468	0.0106	473	0.0123	961	0.0138	852	0.0419	1030	0.5629	985	0.2852	1033	-	
1430	x-laboratory-001	0.0059	399	0.0110	483	0.0054	670	0.0078	688	0.0094	675	0.0142	390	0.0100	450	-	
1431	xforwardai-000	0.0050	342	0.0072	343	0.0036	490	0.0051	517	0.0074	587	0.0136	381	0.0105	471	-	
1432	xforwardai-001	0.0021	113	0.0034	121	0.0027	303	0.0028	202	0.0046	426	0.0088	263	0.0079	364	-	
1433	xforwardai-002	0.0016	65	0.0023	57	0.0026	267	0.0025	127	0.0040	360	0.0081	228	0.0074	328	-	
1434	xm-000	0.0015	57	0.0026	72	0.0031	390	0.0038	380	0.0058	511	0.0105	322	0.0082	385	-	
1435	yisheng-000	0.1215	932	0.1990	936	-		-		-		-		-		-	
1436	yisheng-001	0.0892	909	0.1600	916	-		-		-		-		-		-	
1437	yisheng-002	0.2241	973	0.3795	991	-		-		-		-		-		-	
1438	yisheng-004	0.1988	967	0.3329	980	0.1147	1208	0.1849	1205	0.2044	1161	-		-		-	
1439	yitu-000	0.0284	802	0.0330	743	-		-		-		-		-		-	
1440	yitu-001	0.0050	334	0.0074	355	-		-		-		-		-		-	
1441	yitu-002	0.0025	156	0.0043	194	-		-		-		-		-		-	
1442	yitu-003	0.0015	63	0.0026	70	0.0066	748	0.0085	712	0.0064	545	0.0114	351	0.0103	465	-	
1443	yoonik-000	0.0070	456	0.0112	488	0.0074	786	0.0118	795	0.0564	1068	0.2013	846	0.1160	936	-	
1444	yoonik-001	0.0057	378	0.0079	376	0.0043	581	0.0061	591	0.0307	988	0.0762	750	0.0556	866	-	
1445	yoonik-002	0.0052	358	0.0062	306	0.0029	352	0.0034	321	0.0615	1077	0.1279	806	0.1166	937	0.2329	460
1446	yoonik-003	0.0034	228	0.0047	220	0.0032	414	0.0037	361	0.0816	1103	0.2033	850	0.1601	978	0.3210	501
1447	yoonik-004	0.0041	283	0.0059	283	0.0033	429	0.0052	521	0.0040	362	0.0071	187	0.0051	183	0.0732	175
1448	yoti-001	0.0139	664	0.0195	635	0.0110	925	0.0187	944	0.0191	887	0.1255	803	0.0415	823	0.5578	561
1449	yoti-002	-		-		0.0076	796	0.0137	848	0.0106	715	0.3375	913	0.1599	977	0.7672	614
1450	ytu-000	0.0057	381	0.0087	409	0.0121	958	0.0238	990	0.0047	436	0.0078	211	0.0059	243	-	
1451	yuan-000	0.0101	571	0.0168	591	0.0077	798	0.0127	820	0.0178	864	0.5189	972	0.0245	731	-	
1452	yuan-001	0.0116	610	0.0220	663	0.0114	940	0.0184	936	0.0149	812	0.0574	705	0.0160	608	-	

Table 88: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1453	yuan-002	0.0094	549	0.0154	567	0.0071	769	0.0110	783	0.0108	726	0.0348	621	0.0127	535	0.5738	566
1454	yuan-003	0.0078	492	0.0111	486	0.0062	730	0.0091	730	0.0106	716	0.0511	691	0.0123	527	0.7337	607
1455	yuan-004	0.0058	394	0.0078	370	0.0039	539	0.0055	545	0.0234	934	0.0442	665	0.0353	797	0.1673	411
1456	yuan-005	0.0037	257	0.0046	219	0.0027	304	0.0035	341	0.0033	276	0.2706	890	0.0876	912	0.6695	590
1457	yuan-006	0.0045	302	0.0062	301	0.0032	408	0.0049	489	0.0038	340	0.0084	245	0.0049	162	0.1211	337
1458	yuan-007	0.0046	312	0.0069	332	0.0045	593	0.0075	666	0.0072	576	0.0144	396	0.0076	344	0.1081	304
1459	yuan-008	0.0032	210	0.0043	191	0.0026	244	0.0034	312	0.0028	201	0.0059	135	0.0039	95	0.0891	247
1460	yuan-009	-		-		0.0028	317	0.0042	433	0.0030	234	0.0057	121	0.0040	106	0.0733	177
1461	yuan-010	-		-		0.0028	314	0.0041	410	0.0030	245	0.0058	128	0.0041	113	0.0748	190
1462	yuanli-001	-		-		0.0203	1050	0.0399	1061	0.0110	732	0.0168	447	0.0135	553	0.1476	384

Table 89: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

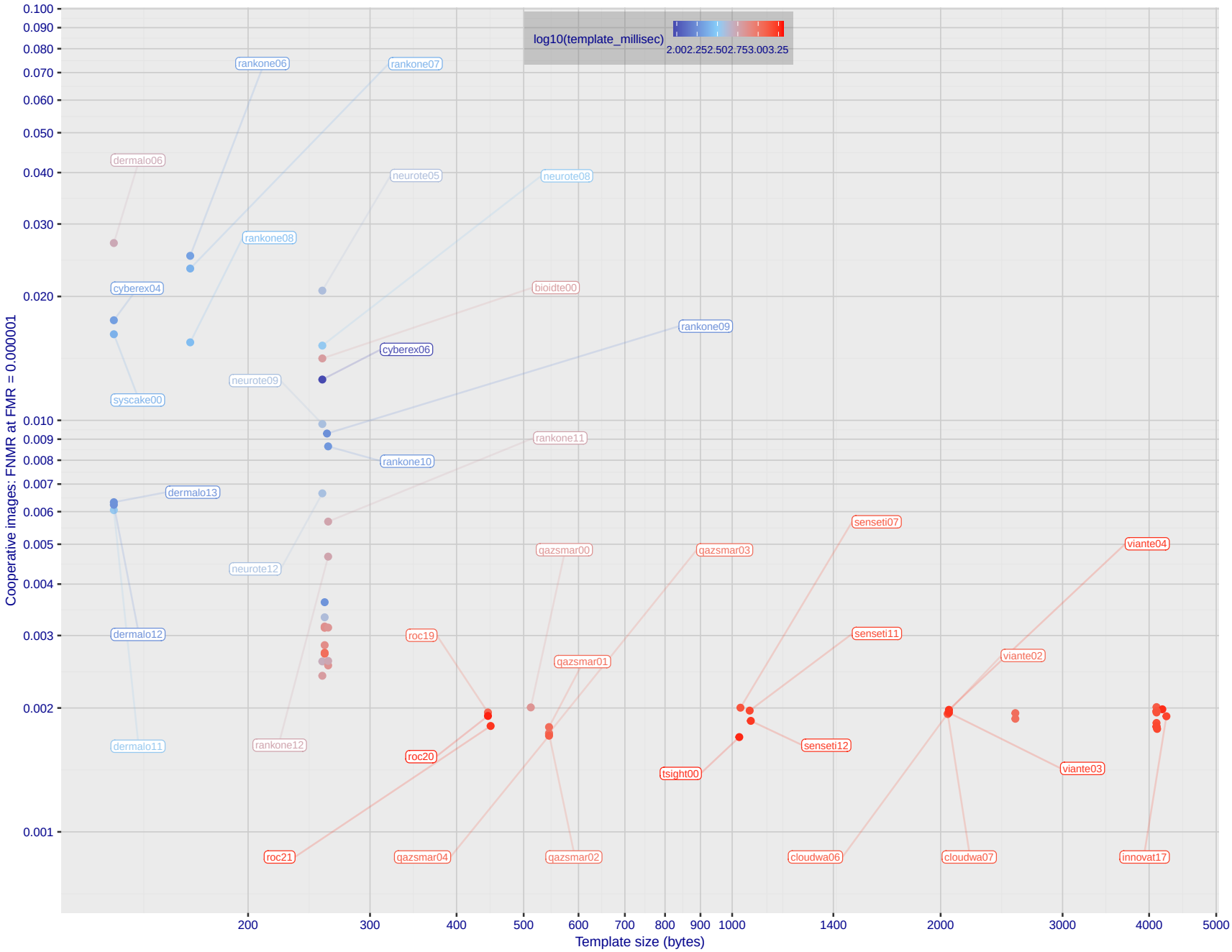


Figure 1: The points show false non-match rates (FNMR) versus the size of the encoded template. FNMR is the geometric mean of FNMR values for visa-border and mugshot images (from Figs. 222 and 291) at the false match rate (FMR) given in the y-axis label. The color of the points encodes template generation time - which spans at least one order of magnitude. Durations are measured on a single core of a c. 2016 Intel Xeon CPU E5-2630 v4 running at 2.20GHz. Algorithms with poor FNMR are omitted.

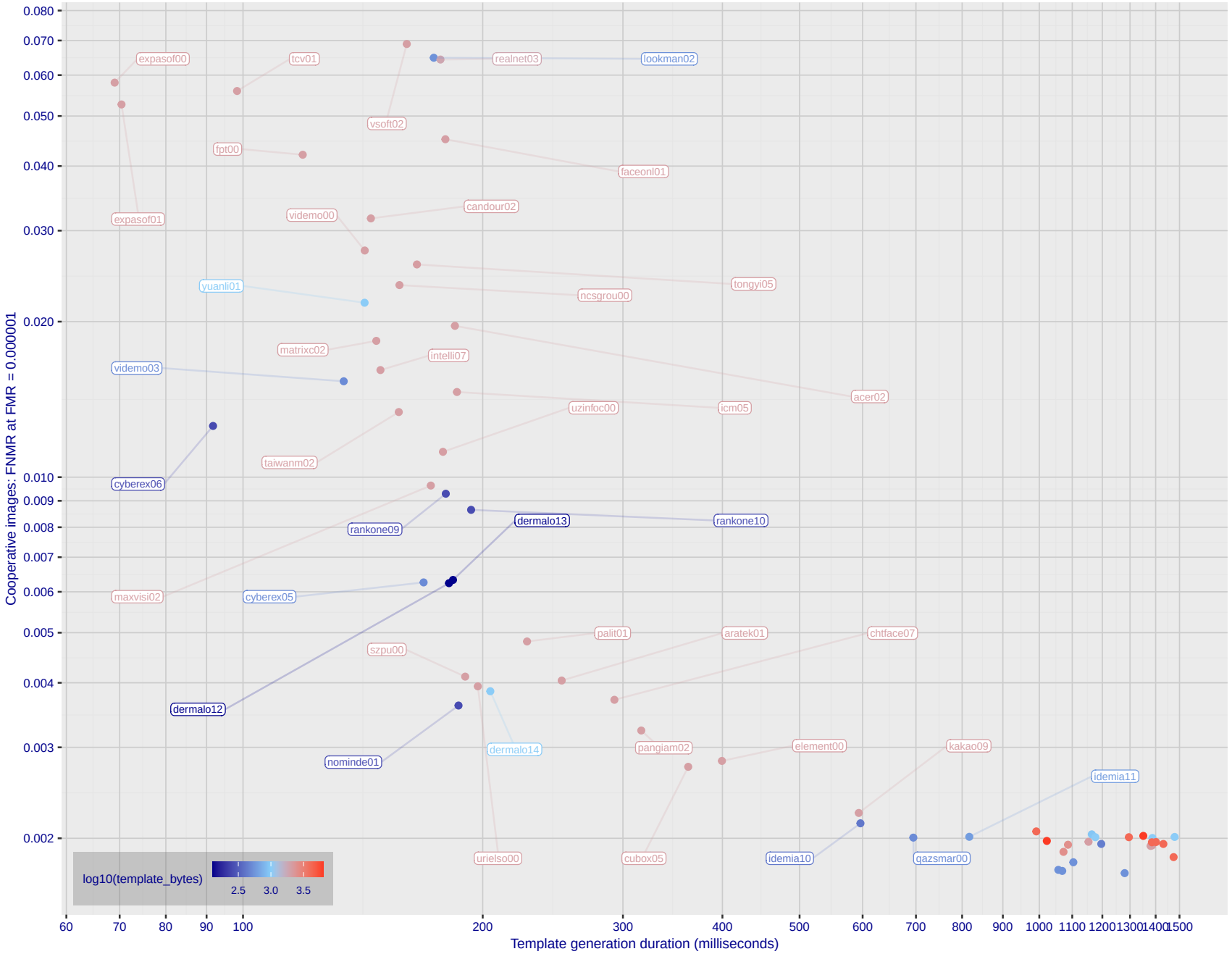


Figure 2: The points show false non-match rates (FNMR) versus the duration of the template generation operation. FNMR is the geometric mean of FNMR values for visa-border and mugshot images (from Figs. 222 and 291) at a false match rate (FMR) given in the y-axis label. Template generation time is a median estimated over 640 x 480 pixel portraits. It is measured on a single core of a c. 2016 Intel Xeon CPU E5-2630 v4 running at 2.20GHz. The color of the points encodes template size - which spans two orders of magnitude. Algorithms with poor FNMR are omitted.

1 Metrics

1.1 Core accuracy

Given a vector of N genuine scores, u, the false non-match rate (FNMR) is computed as the proportion below some threshold, T:

$$\text{FNMR}(T) = 1 - \frac{1}{N}\sum_{i=1}^{N} H(u_i - T) \tag{1}$$

where $H(x)$ is the unit step function, and $H(0)$ taken to be 1.

Similarly, given a vector of N impostor scores, v, the false match rate (FMR) is computed as the proportion above T:

$$\text{FMR}(T) = \frac{1}{N}\sum_{i=1}^{N} H(v_i - T) \tag{2}$$

The threshold, T, can take on any value. We typically generate a set of thresholds from quantiles of the observed impostor scores, v, as follows. Given some interesting false match rate range, $[\text{FMR}_L, \text{FMR}_U]$, we form a vector of K thresholds corresponding to FMR measurements evenly spaced on a logarithmic scale

$$T_k = Q_v(1 - \text{FMR}_k) \tag{3}$$

where Q is the quantile function, and FMR_k comes from

$$\log_{10}\text{FMR}_k = \log_{10}\text{FMR}_L + \frac{k}{K}\ [\log_{10}\text{FMR}_U - \log_{10}\text{FMR}_L] \tag{4}$$

Error tradeoff characteristics are plots of FNMR(T) vs. FMR(T). These are plotted with $\text{FMR}_U \to 1$ and FMR_L as low as is sustained by the number of impostor comparisons, N. This is somewhat higher than the "rule of three" limit $3/N$ because samples are not independent, due to re-use of images.

1.2 Multi-template scoring methodology

There are some scenarios when one or more people exist and are detected in an image, and some of the proposed test images include K $>$ 1 persons for some images and situations where the subject of interest may or may not be the foreground face (largest face in the image). The NIST FRTE 1:1 API supports this by allowing generation of multiple templates representing each person detected in an image. When this occurs, NIST will match all templates generated from the enrollment image with all templates generated from the verification image and use the **maximum** similarity score across all template comparisons. This scoring approach will be used in our calculation of FMR and FNMR (this applies to both genuine and imposter comparisons).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2 Datasets

2.1 Visa images

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^5.
- The number of subjects with two images is on the order of 10^4.
- The images have geometry in reasonable conformance with the ISO/IEC 19794-5 Full Frontal image type. Pose is generally excellent.
- The images are of size 252x300 pixels. The mean interocular distance (IOD) is 69 pixels.
- The images are of subjects from greater than 100 countries, with significant imbalance due to visa issuance patterns.
- The images are of subjects of all ages, including children, again with imbalance due to visa issuance demand.
- Many of the images are live capture. A substantial number of the images are photographs of paper photographs.
- When these images are input to the algorithm, they are labelled as being of type "ISO" - see Table 4 of the FRTE API.

2.2 Application images

- The number of images is on the order of 10^6.
- The number of subjects is on the order of 10^6.
- The number of subjects with two images is on the order of 10^6.
- The images have geometry in good conformance with the ISO/IEC 19794-5 Full Frontal image type. Pose is generally excellent.
- The images are of size 300x300 pixels. The mean interocular distance (IOD) is 61 pixels.
- The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.
- The images are of subjects of adults.
- All of the images are live capture.
- When these images are input to the algorithm, they are labelled as being of type "ISO" - see Table 4 of the FRTE API.

2.3 Application images with head yaw

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^5.
- The number of subjects with two images is on the order of 10^5.
- The images have geometry in good conformance with the ISO/IEC 19794-5 Full Frontal image type *except* the yaw angle is between 25 and 85 degrees. Our pose estimates are approximate, with an angular error that increases with yaw. The angular estimates will be improved over time.
- The images are of size 300x300 pixels. The mean interocular distance (IOD), if frontal, would be about pixels, but reduces with cosine of yaw.
- The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The images are of subjects of adults.
- ▷ All of the images are live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

2.4 Border crossing images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^6.
- ▷ The number of subjects with two images is on the order of 10^6.
- ▷ The images are taken with at camera oriented by an attendant toward a cooperating subject. This is done under time constraints so there are role, pitch and yaw angle variations. Also background illumination is sometimes strong, so the face is under-exposed. There is some perspective distortion due to close range images. Some faces are partially cropped.
- ▷ The images have mean IOD of 38 pixels.
- ▷ The images are of subjects of adults and childen aged 12 or above.
- ▷ The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.
- ▷ The images are all live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

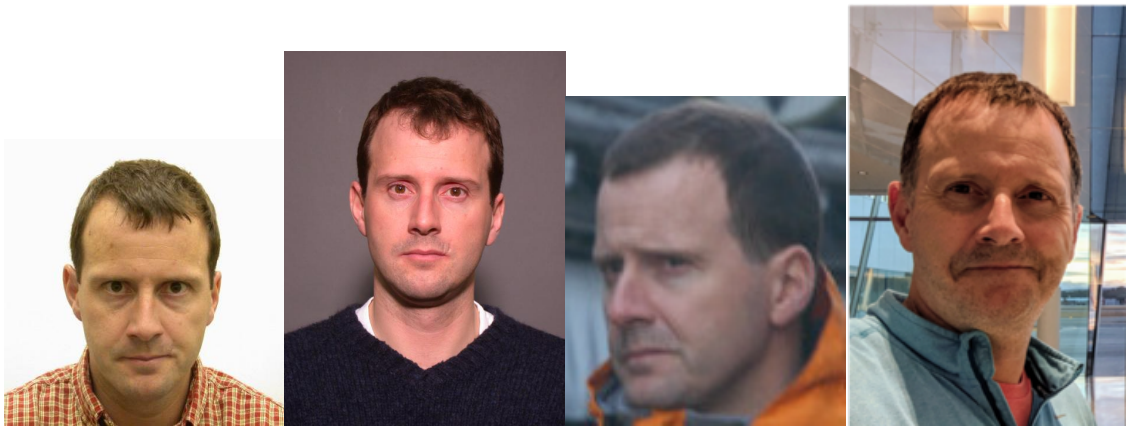
2.5 Mugshot images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^6.
- ▷ The number of subjects with two images is on the order of 10^6.
- ▷ The images have geometry in reasonable conformance with the ISO/IEC 19794-5 Full Frontal image type.
- ▷ The images are of variable sizes. The median IOD is 105 pixels. The mean IOD is 113 pixels. The 1-st, 5-th, 10-th, 25-th, 75-th, 90-th and 99-th percentiles are 34, 58, 70, 87, 121, 161 and 297 pixels.
- ▷ The images are of subjects from the United States.
- ▷ The images are of adults.
- ▷ The images are all live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”mugshot” - see Table 4 of the FRTE API.

2.6 Kiosk images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^5.
- ▷ The number of subjects with multiple images is the order of 10^5.

FNMR(T) “False non-match rate”
FMR(T) “False match rate”



(a) Visa (b) Mugshot (c) Wild (d) Border

Figure 3: The figure gives simulated samples of image types used in this report.

- The images are taken at kiosk equipped with a camera intended to capture a centered face. However the images have specific quality defects arising from the camera triggering before the subject looks at it. These are downward pitch of the face relative to the optical axis; cropping of the forehead; and cropping of left or right part of the face. Partial cropping affects perhaps 10% of the images. Resolution does not vary widely.
- The images are of adults.
- The images have mean IOD of 44 pixels, with maximum below 75, and minimum when both eyes are present above 25 pixels.
- All of the images are live capture, none are scanned.
- When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

2.7 Wild images

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^4.
- The number of subjects with two images on the order of 10^4.
- The images include many photojournalism-style images. Images are given to the algorithm using a variable but generally tight crop of the head. Resolution varies very widely. The images are very unconstrained, with wide yaw and pitch pose variation. Faces can be occluded, including hair and hands.
- The images are of adults.
- All of the images are live capture, none are scanned.
- When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

3 Results

3.1 Test goals

- To state absolute accuracy for different kinds of images, including those with and without subject cooperation.

FNMR(T) “False non-match rate”
FMR(T) “False match rate”

▷ To state comparative accuracy, across algorithms.

3.2 Test design

Method: For visa images:

▷ The comparisons are of visa photos against visa photos.
▷ The number of genuine comparisons is on the order of 10^4.
▷ The number of impostor comparisons is on the order of 10^{10}.
▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates. However, later analysis is conducted on subsets.
▷ The number of persons is on the order of 10^5.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For mugshot images:

▷ The comparisons are of mugshot photos against mugshot photos.
▷ The number of genuine comparisons is on the order of 10^6.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.
▷ The number of persons is on the order of 10^6.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For visa-border comparisons:

▷ The comparisons are of visa-like frontals against border crossing webcam photos.
▷ The number of genuine comparisons is on the order of 10^6.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.
▷ The number of persons is on the order of 10^6.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For visa-border non-frontal yaw comparisons:

▷ The comparisons are of visa-like images with yaw 25 to 85 degrees against border crossing webcam photos.
▷ The number of genuine comparisons is on the order of 10^5.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The number of persons is on the order of 10^5.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For kiosk-border comparisons:

- ▷ The comparisons are of visa-like frontals against kiosk-style photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The impostors are paired by sex, but not by age or other covariates.
- ▷ The number of persons is on the order of 10^5.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For border-border comparisons:

- ▷ The comparisons are of border crossing webcam photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The impostors are paired by sex, but not by age or other covariates.
- ▷ The number of persons is on the order of 10^6.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For wild images:

- ▷ The comparisons are of wild photos against wild photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates.
- ▷ The number of persons is on the order of 10^4.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For child exploitation images:

- ▷ The comparisons are of unconstrained child exploitation photos against others of the same type.
- ▷ The number of genuine comparisons is on the order of 10^4.
- ▷ The number of impostor comparisons is on the order of 10^7.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates.
- ▷ The number of persons is on the order of 10^3.
- ▷ The number of images used to make 1 template is 1.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.
- ▷ We produce two performance statements. First, is a DET as used for visa and mugshot images. The second is a cumulative match characteristic (CMC) summarizing a simulated one-to-many search process. This is done as follows.
  - We regard M enrollment templates as items in a gallery.
  - These M templates come from $M > N$ individuals, because multiple images of a subject are present in the gallery under separate identifiers.
  - We regard the verification templates as search templates.
  - For each search we compute the rank of the highest scoring mate.
  - This process should properly be conducted with a 1:N algorithm, such as those tested in NIST IR 8009. We use the 1:1 algorithms in a simulated 1:N mode here to a) better reflect what a child exploitation analyst does, and b) to show algorithm efficacy is better than that revealed in the verification DETs.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.3 Failure to enroll

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1	2060io-001	0.0160	1296	0.0998	1299	0.4698	942	0.0193	1356	-	1342
2	2060io-002	0.0000	1015	0.0015	929	0.0238	626	0.0028	1309	-	1382
3	20face-000	0.0000	848	0.0008	680	0.0217	576	0.0000	362	0.0004	618
4	20face-001	0.0000	604	0.0008	679	0.0000	95	0.0000	345	0.0004	608
5	3divi-000	-	1364	-	1364	-	1340	-	1414	0.0008	933
6	3divi-001	-	1403	-	1403	-	991	-	1445	0.0007	923
7	3divi-002	-	1404	-	1404	-	970	-	1442	0.0010	952
8	3divi-003	0.0001	1205	0.0059	1195	-	1253	0.0007	1190	0.0006	893
9	3divi-004	0.0001	1206	0.0059	1196	-	1391	0.0008	1205	0.0006	899
10	3divi-005	0.0000	915	0.0008	651	-	1430	0.0000	492	0.0002	272
11	3divi-006	0.0000	654	0.0007	597	0.0214	573	0.0001	697	0.0002	279
12	3divi-007	0.0000	645	0.0007	596	0.0214	572	0.0001	698	0.0002	277
13	accurascan-001	0.0001	1196	0.0060	1201	0.0838	902	0.0006	1179	0.0022	1031
14	accurascan-002	0.0001	1195	0.0060	1200	0.0838	903	0.0006	1178	0.0022	1030
15	accurascan-003	0.0000	621	0.0009	766	0.0226	594	0.0000	494	0.0003	397
16	acer-000	0.0000	1048	0.0024	1045	-	1355	0.0002	894	0.0004	732
17	acer-001	0.0000	520	0.0011	837	-	1254	0.0001	622	0.0004	638
18	acer-002	0.0000	1102	0.0008	663	0.0191	511	0.0003	985	0.0004	636
19	acisw-003	0.0000	264	0.0000	125	-	984	0.0000	209	0.0000	141
20	acisw-006	0.0000	475	0.0000	222	-	1168	0.0000	235	0.0000	224
21	acisw-007	0.0000	414	0.0000	205	0.0000	41	0.0000	247	0.0000	181
22	acisw-008	0.0000	693	0.0009	744	0.0173	453	0.0004	1085	0.0004	436
23	adera-001	0.0000	1047	0.0034	1121	-	1275	0.0003	1004	0.0005	858
24	adera-002	0.0000	1046	0.0034	1120	-	1301	0.0003	1005	0.0005	859
25	adera-003	0.0000	1050	0.0034	1122	0.0403	804	0.0003	1006	0.0005	861
26	adera-004	0.0000	863	0.0008	660	0.0202	531	0.0003	1037	0.0004	571
27	adera-005	0.0000	795	0.0008	659	0.0202	530	0.0003	1039	0.0004	568
28	advance-002	0.0000	674	0.0013	903	-	1417	0.0000	577	0.0004	646
29	advance-003	0.0000	1023	0.0012	863	0.0247	639	0.0001	770	0.0004	745
30	advance-004	0.0001	1202	0.0010	810	0.0157	419	0.0008	1216	0.0006	901
31	advance-005	0.0000	854	0.0003	316	0.0074	234	0.0002	885	0.0003	290
32	advance-006	0.0001	1198	0.0012	876	0.0202	529	0.0008	1212	-	1266
33	advance-007	0.0000	842	0.0007	600	0.0212	560	0.0002	835	-	1455
34	affinsys-ai-000	0.0000	1113	0.0068	1204	0.0641	862	0.0016	1273	-	1172
35	afisbiometrics-000	0.0000	835	0.0008	638	0.0213	570	0.0000	365	0.0004	657
36	afrengine-000	0.0000	810	0.0015	933	0.0254	662	0.0008	1213	0.0004	523
37	afrengine-001	0.0000	622	0.0007	616	0.0183	478	0.0005	1120	0.0004	741
38	afrengine-002	0.0000	706	0.0007	617	0.0183	477	0.0005	1123	0.0004	743
39	afrengine-003	0.0000	765	0.0099	1232	0.0384	788	0.0005	1121	-	1323
40	afrengine-004	0.0000	690	0.1602	1303	0.0528	833	0.0005	1122	-	1290
41	afrengine-005	0.0000	791	0.0012	873	0.0232	617	0.0006	1163	-	1392
42	aifirst-001	0.0000	377	0.0000	182	-	1128	0.0000	268	0.0000	195
43	aigen-001	0.0000	270	0.0000	127	-	982	0.0000	206	0.0000	142
44	aigen-002	0.0000	287	0.0000	135	0.0000	8	0.0000	179	0.0000	117
45	aigroup-000	0.0000	473	0.0005	420	0.0103	294	0.0000	236	-	1451
46	aigroup-001	0.0000	884	0.0005	435	0.0104	297	0.0001	823	-	1162
47	aigroup-002	0.0000	146	0.0003	346	0.0083	264	0.0001	708	-	1215
48	ailabs-001	0.0000	723	0.0090	1228	-	949	0.0007	1203	0.0005	798
49	aimall-001	0.0000	344	0.0000	166	-	1025	0.0000	149	0.0000	153
50	aimall-002	0.0000	1068	0.0043	1158	-	1081	0.0012	1249	0.0005	850
51	aimall-003	0.0000	993	0.0012	877	-	1085	0.0004	1061	0.0005	784
52	aipractix-000	0.0021	1268	0.0021	1020	0.0175	459	0.0390	1362	-	1314
53	aiseemu-001	0.0000	497	0.0000	233	0.0000	55	0.0000	219	0.0000	212
54	aiseemu-002	0.0000	295	0.0000	141	0.0000	1	0.0000	192	0.0000	129
55	aiunionface-000	0.0000	22	0.0000	9	-	1221	0.0000	69	0.0000	29
56	aize-001	0.0001	1204	0.0040	1140	0.0652	864	0.0026	1305	0.0022	1029
57	aize-002	0.0000	2	0.0014	921	0.0230	612	0.0005	1161	0.0004	605
58	aize-003	0.0000	244	0.0000	253	0.0001	155	0.0000	81	0.0000	241

Table 90: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
59	aize-006	0.0000	563	0.0004	365	0.0090	281	0.0001	822	-	1152
60	ajou-001	0.0000	703	0.0020	1008	-	988	0.0001	709	0.0004	753
61	alchera-000	0.0000	1162	0.0041	1153	-	980	0.0004	1083	0.0014	1007
62	alchera-001	0.0000	1161	0.0041	1152	-	1365	0.0004	1082	0.0014	1006
63	alchera-002	0.0000	584	0.0008	689	-	1300	0.0001	787	0.0004	486
64	alchera-003	0.0001	1225	0.0013	899	0.0317	720	0.0002	922	0.0004	669
65	alchera-004	0.0000	715	0.0009	739	0.0228	598	0.0001	795	0.0004	515
66	alchera-005	0.0000	740	0.0009	740	0.0228	601	0.0001	794	0.0004	519
67	alchera-006	0.0000	792	0.0009	743	0.0228	602	0.0001	797	0.0004	526
68	alchera-007	0.0000	756	0.0009	741	0.0228	599	0.0001	793	0.0004	518
69	alchera-008	0.0000	759	0.0009	742	0.0228	600	0.0001	792	-	1310
70	alchera-009	0.0000	534	0.0004	404	0.0112	326	0.0002	901	-	1123
71	alfabeta-001	0.0005	1249	0.0650	1288	0.2142	933	0.0024	1298	0.0018	1020
72	alice-000	0.0000	228	0.0006	489	0.0133	361	0.0000	426	0.0004	491
73	alice-001	0.0000	324	0.0006	490	0.0133	360	0.0000	434	0.0004	495
74	alleyes-000	0.0000	511	0.0010	783	-	1239	0.0002	828	0.0004	688
75	allgovision-000	0.0007	1258	0.0062	1202	-	1368	0.0026	1303	0.0052	1069
76	alphaface-001	0.0000	582	0.0012	850	-	1307	0.0000	583	0.0004	690
77	alphaface-002	0.0000	838	0.0012	853	-	1159	0.0000	589	0.0004	702
78	alttekglobal-000	0.0000	829	0.0020	1001	0.0471	820	0.0000	597	-	1437
79	amplifiedgroup-001	0.0114	1291	0.1023	1300	-	1298	0.0189	1355	0.0279	1084
80	androvideo-000	0.0000	173	0.0000	81	-	1350	0.0000	111	0.0000	65
81	anke-000	-	1387	-	1387	-	1420	-	1394	0.0021	1025
82	anke-001	-	1334	-	1331	-	1312	-	1373	0.0047	1068
83	anke-002	-	1343	-	1343	-	1261	0.0004	1067	0.0009	939
84	anke-003	-	1354	-	1352	-	1396	0.0001	729	0.0004	575
85	anke-004	0.0000	804	0.0011	832	-	1101	0.0001	736	0.0004	714
86	anke-005	0.0000	687	0.0012	859	-	1400	0.0001	773	0.0004	736
87	antheus-000	0.0000	120	0.0000	57	-	1271	0.0000	14	0.0000	40
88	antheus-001	0.0000	496	0.0000	232	-	1136	0.0000	220	0.0000	213
89	antheus-002	0.0000	1148	0.0015	931	0.0213	564	0.0044	1327	-	1284
90	anyvision-001	-	1423	-	1423	-	1029	-	1430	0.0091	1077
91	anyvision-002	0.0060	1280	0.0230	1260	-	1428	0.0070	1340	0.0090	1076
92	anyvision-004	0.0000	1009	0.0017	968	-	1404	0.0001	774	0.0004	625
93	anyvision-005	0.0000	585	0.0013	885	-	1278	0.0000	499	0.0004	481
94	aratek-001	0.0000	355	0.0000	172	0.0000	25	0.0000	140	0.0000	149
95	armatura-001	0.0000	1078	0.0021	1021	0.0257	668	0.0005	1144	0.0005	819
96	armatura-002	0.0000	1018	0.0018	975	0.0206	541	0.0003	1040	0.0004	744
97	armatura-003	0.0000	865	0.0012	864	0.0333	751	0.0004	1076	0.0004	603
98	arvis-000	0.0008	1262	0.0014	912	0.0195	519	0.0083	1342	-	1169
99	asr-000	0.0000	963	0.0007	601	0.0152	401	0.0002	948	-	1414
100	asr-001	0.0000	881	0.0006	533	0.0138	368	0.0002	966	-	1105
101	asusaics-000	0.0000	94	0.0000	41	-	1299	0.0000	28	0.0000	54
102	asusaics-001	0.0000	175	0.0000	82	-	1329	0.0000	125	0.0000	74
103	autentika-000	0.0000	216	0.0000	101	0.0000	140	0.0000	102	0.0000	114
104	autentika-001	0.0000	505	0.0000	237	0.0000	56	0.0000	215	0.0000	210
105	autentika-002	0.0000	112	0.0000	51	0.0000	91	0.0000	21	0.0000	46
106	autentika-003	0.0000	33	0.0000	18	0.0000	82	0.0000	61	-	1124
107	authenmetric-002	0.0000	126	0.0000	62	-	1384	0.0000	134	0.0000	80
108	authenmetric-003	0.0000	444	0.0000	212	0.0000	65	0.0000	227	0.0000	221
109	authenmetric-004	0.0000	240	0.0000	111	0.0000	124	0.0000	84	0.0000	101
110	authme-001	0.0000	1112	0.0009	755	0.0245	637	0.0003	1028	0.0005	777
111	authme-002	0.0000	903	0.0007	626	0.0179	467	0.0001	767	-	1185
112	aware-000	-	1428	-	1429	-	1005	-	1434	0.0005	803
113	aware-001	-	1382	-	1381	-	1457	-	1400	0.0002	261
114	aware-002	-	1391	-	1390	-	1406	-	1399	0.0006	872
115	aware-003	-	1454	-	1454	-	1169	0.0016	1271	0.0013	996
116	aware-004	0.0000	938	0.0023	1035	-	1047	0.0002	861	0.0005	783

Table 91: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
117	aware-005	0.0000	929	0.0020	1005	-	956	0.0001	825	0.0004	709
118	aware-006	0.0000	696	0.0009	724	0.0249	645	0.0000	517	0.0004	629
119	aware-007	0.0000	753	0.0012	865	0.0325	737	0.0000	529	0.0004	667
120	aware-008	0.0000	822	0.0012	871	0.0327	739	0.0000	532	-	1449
121	awiros-001	0.0039	1272	0.0369	1277	-	1071	0.0386	1361	0.0872	1089
122	awiros-002	0.0000	1110	0.0038	1136	-	1107	0.0007	1199	0.0012	995
123	awiros-003	0.0000	376	0.0000	184	0.0000	51	0.0000	269	-	1377
124	awiros-004	0.0000	222	0.0000	104	0.0000	133	0.0000	72	-	1231
125	aximetria-001	0.0000	951	0.0010	816	0.0217	577	0.0001	827	0.0004	617
126	ayftech-001	0.0002	1233	0.0046	1165	-	1328	0.0043	1326	0.0011	960
127	ayftech-003	0.0000	331	0.0006	546	0.0123	343	0.0000	171	0.0000	172
128	ayonix-000	0.0053	1277	0.0341	1273	-	1362	0.0113	1349	0.0137	1079
129	beethedata-000	0.0005	1246	0.0042	1155	0.0366	773	0.0002	869	0.0010	948
130	betaface-000	-	1461	-	1462	-	1140	-	1452	0.0011	965
131	betaface-002	-	1424	-	1425	-	1021	-	1429	0.0010	950
132	beyneai-000	0.0000	36	0.0000	16	0.0000	81	0.0000	60	0.0000	25
133	biocube-001	0.0006	1253	0.0391	1278	0.1207	914	0.0015	1265	0.0020	1024
134	biocube-002	0.0000	650	0.0046	1168	0.0721	890	0.0008	1211	0.0012	985
135	bioid-001	0.0005	1248	0.0019	997	0.0324	736	0.0005	1133	-	1424
136	bioid-002	0.0000	457	0.0000	217	0.0000	67	0.0000	221	-	1436
137	bioid-003	0.0000	226	0.0000	106	0.0000	135	0.0000	71	-	1227
138	bioid-004	0.0000	97	0.0000	44	0.0000	100	0.0000	9	-	1138
139	bioidtechswiss-000	0.0000	711	0.0008	643	-	969	0.0003	997	0.0004	752
140	bioidtechswiss-001	0.0000	587	0.0007	588	-	1283	0.0000	446	0.0004	676
141	bioidtechswiss-002	0.0000	512	0.0007	594	-	1242	0.0000	473	0.0004	660
142	bioidtechswiss-003	0.0000	989	0.0202	1256	0.0980	908	0.0005	1136	-	1324
143	biometric-vision-000	0.0000	744	0.0005	432	0.0107	313	0.0002	951	0.0004	551
144	biometrik-001	0.0000	98	0.0000	42	0.0000	99	0.0000	8	-	1140
145	biometrik-002	0.0000	493	0.0000	230	0.0000	59	0.0000	210	-	1406
146	bm-001	0.0000	332	0.0000	159	-	1057	0.0000	297	0.0000	171
147	boetech-001	0.0087	1287	0.0272	1265	0.2117	931	0.0032	1314	0.0160	1081
148	boetech-002	0.0087	1286	0.0272	1264	0.2117	930	0.0032	1313	0.0160	1080
149	bresee-000	0.0000	807	0.0010	803	-	1091	0.0002	870	0.0003	368
150	bresee-001	0.0000	551	0.0010	802	-	1182	0.0002	867	0.0003	354
151	bresee-002	0.0000	1035	0.0020	1013	0.0219	584	0.0008	1204	0.0004	578
152	btsdigital-001	0.0000	223	0.0009	721	0.0265	678	0.0000	406	-	1229
153	bynn-000	0.0130	1293	0.0048	1172	0.0089	276	0.0112	1348	-	1147
154	cakevn-001	0.0000	439	0.0007	605	0.0130	355	0.0000	230	-	1448
155	cakevn-002	0.0000	278	0.0008	637	0.0163	436	0.0000	202	-	1302
156	camvi-001	-	1400	-	1400	-	981	-	1447	0.0010	955
157	camvi-002	0.0000	286	0.0000	136	-	977	0.0000	180	0.0000	119
158	camvi-003	-	1311	-	1309	-	1243	0.0000	59	0.0000	19
159	camvi-004	0.0000	132	0.0000	249	-	1381	0.0000	132	0.0000	81
160	candour-001	0.0000	338	0.0000	161	0.0000	24	0.0000	152	0.0000	151
161	candour-002	0.0000	404	0.0000	197	0.0000	37	0.0000	254	-	1356
162	candour-003	0.0000	140	0.0000	68	0.0000	123	0.0000	128	-	1191
163	candour-004	0.0000	206	0.0000	98	0.0000	138	0.0000	108	-	1259
164	canon-002	0.0000	329	0.0000	157	-	1055	0.0000	172	0.0000	174
165	canon-003	0.0000	598	0.0008	635	0.0234	619	0.0000	566	0.0004	640
166	canon-004	0.0000	685	0.0008	636	0.0234	620	0.0000	567	0.0004	647
167	canon-005	0.0000	151	0.0044	1160	0.0269	683	0.0000	455	0.0004	541
168	cchonolulu-000	0.0054	1278	0.0395	1280	0.2802	938	0.0036	1317	0.0012	984
169	cchonolulu-001	0.0056	1279	0.0414	1281	0.2894	940	0.0038	1323	0.0012	987
170	cdachyd-001	0.0005	1247	0.0639	1285	0.2112	929	0.0023	1293	-	1219
171	ceiec-001	-	1443	-	1444	-	1089	0.0029	1311	0.0023	1035
172	ceiec-002	0.0002	1232	0.0056	1192	-	1109	0.0036	1318	0.0031	1051
173	ceiec-003	0.0000	180	0.0013	905	-	1326	0.0001	647	0.0004	679
174	ceiec-004	0.0000	55	0.0008	676	-	1191	0.0000	490	0.0004	501

Table 92: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
175	celloscope-001	0.0000	1099	0.0056	1191	0.0540	839	0.0022	1290	-	1415
176	chosun-000	0.0000	489	0.0000	231	-	1152	0.0000	211	0.0000	209
177	chosun-001	0.0000	114	0.0000	53	-	1256	0.0000	19	0.0000	48
178	chosun-002	0.0000	166	0.0000	79	-	1353	0.0000	115	0.0000	64
179	chtface-001	0.0000	59	0.0000	27	-	1197	0.0000	47	0.0000	6
180	chtface-002	0.0000	1104	0.0021	1023	-	1412	0.0002	976	0.0007	915
181	chtface-003	0.0000	1043	0.0018	979	-	1162	0.0001	664	0.0006	874
182	chtface-004	0.0000	380	0.0017	960	0.0320	724	0.0000	544	0.0004	696
183	chtface-005	0.0000	491	0.0017	962	0.0320	725	0.0000	540	0.0004	698
184	chtface-006	0.0000	297	0.0017	959	0.0320	723	0.0000	539	0.0004	692
185	chtface-007	0.0000	440	0.0017	961	0.0320	726	0.0000	543	-	1443
186	chtface-008	0.0000	827	0.0009	707	0.0221	586	0.0000	482	-	1439
187	chtface-009	0.0000	648	0.0009	706	0.0221	587	0.0000	477	-	1179
188	cib-000	0.0000	462	0.0000	219	-	1165	0.0001	799	0.0000	227
189	cib-001	0.0000	25	0.0000	11	-	1224	0.0000	68	0.0000	27
190	cist-001	0.0000	208	0.0005	472	0.0087	274	0.0000	107	0.0000	116
191	cist-002	0.0000	220	0.0048	1170	0.0128	349	0.0000	73	0.0000	89
192	cist-003	0.0000	28	0.0011	829	0.0343	762	0.0000	65	0.0000	23
193	cist-004	0.0000	302	0.0005	436	0.0108	316	0.0000	187	0.0000	124
194	clearviewai-000	0.0000	655	0.0003	323	0.0081	253	0.0000	546	0.0003	315
195	clearviewai-001	0.0000	543	0.0006	507	0.0174	455	0.0001	600	0.0004	447
196	clearviewai-002	0.0000	899	0.0009	763	0.0244	632	0.0001	629	-	1186
197	clearviewai-003	0.0000	964	0.0009	765	0.0244	631	0.0001	636	-	1420
198	closeli-001	0.0000	47	0.0000	22	0.0000	78	0.0000	37	0.0000	2
199	cloudmatrix-000	0.0000	913	0.0012	866	-	1423	0.0001	628	0.0004	459
200	cloudmatrix-001	0.0000	872	0.0028	1064	0.0225	590	0.0001	626	0.0004	450
201	cloudmatrix-002	0.0000	894	0.0028	1065	0.0225	591	0.0001	631	0.0004	458
202	cloudwalk-hr-000	-	1411	-	1412	-	946	0.0003	1021	0.0008	929
203	cloudwalk-hr-001	0.0000	539	0.0007	613	-	1203	0.0001	607	0.0004	553
204	cloudwalk-hr-002	0.0000	697	0.0007	610	-	987	0.0002	907	0.0004	711
205	cloudwalk-hr-003	0.0000	576	0.0008	681	-	1289	0.0001	660	0.0004	487
206	cloudwalk-hr-004	0.0000	757	0.0011	848	-	1023	0.0004	1070	0.0003	414
207	cloudwalk-mt-000	0.0000	671	0.0008	650	-	1424	0.0002	910	0.0004	708
208	cloudwalk-mt-002	0.0000	656	0.0003	315	-	1448	0.0001	608	0.0004	464
209	cloudwalk-mt-003	0.0000	779	0.0007	573	-	1124	0.0002	900	0.0004	717
210	cloudwalk-mt-004	0.0000	642	0.0009	696	0.0191	509	0.0002	925	0.0004	739
211	cloudwalk-mt-005	0.0000	859	0.0005	450	0.0130	356	0.0003	979	0.0004	718
212	cloudwalk-mt-006	0.0000	742	0.0006	496	0.0158	420	0.0002	897	0.0004	712
213	cloudwalk-mt-007	0.0000	817	0.0006	498	0.0158	421	0.0002	899	0.0004	715
214	clova-000	0.0000	1079	0.0022	1031	-	1215	0.0006	1176	0.0005	787
215	cmcuni-001	0.0000	327	0.0005	458	0.0119	339	0.0000	174	0.0000	177
216	codeline-000	0.0000	390	0.0000	193	0.0000	44	0.0000	281	0.0000	206
217	cogent-000	-	1457	-	1458	-	1146	-	1449	0.0005	836
218	cogent-001	-	1349	-	1347	-	1265	-	1365	0.0004	756
219	cogent-002	-	1362	-	1362	-	1345	0.0002	839	0.0004	757
220	cogent-003	0.0000	1033	0.0018	980	-	1335	0.0001	663	0.0004	609
221	cogent-004	0.0000	467	0.0000	220	-	1158	0.0000	240	0.0000	230
222	cogent-005	0.0000	378	0.0000	186	-	1131	0.0000	267	0.0000	193
223	cogent-006	0.0000	283	0.0000	132	0.0000	6	0.0000	183	0.0000	123
224	cogent-007	0.0000	1031	0.0000	246	0.0000	150	0.0000	505	0.0000	246
225	cogent-008	0.0000	446	0.0010	817	0.0304	711	0.0000	565	0.0004	445
226	cognitec-000	0.0005	1243	0.0112	1239	-	1277	0.0007	1195	0.0007	922
227	cognitec-001	0.0003	1239	0.0110	1238	-	1392	0.0008	1214	0.0010	944
228	cognitec-002	0.0001	1181	0.0069	1206	-	1414	0.0003	1035	0.0005	802
229	cognitec-003	0.0001	1182	0.0194	1254	0.0820	899	0.0003	1025	0.0005	807
230	cognitec-004	0.0001	1183	0.0037	1134	0.0580	846	0.0003	1024	0.0005	808
231	cognitec-005	0.0000	1094	0.0026	1050	0.0550	841	0.0002	852	0.0005	796
232	cognitec-006	0.0000	1084	0.0016	951	0.0511	826	0.0002	829	-	1170

Table 93: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
233	cognitec-007	0.0000	1174	0.0025	1049	0.0736	892	0.0002	878	-	1268
234	compal-000	0.0000	412	0.0000	203	0.0000	40	0.0000	249	-	1352
235	compal-001	0.0000	970	0.0060	1198	0.0320	727	0.0012	1246	-	1109
236	compal-002	0.0000	980	0.0060	1199	0.0320	728	0.0012	1247	-	1225
237	cor-001	0.0000	752	0.0006	512	-	1028	0.0002	953	0.0004	630
238	coretech-000	0.0000	333	0.0000	158	0.0000	31	0.0000	170	0.0000	173
239	coretech-001	0.0000	1157	0.0033	1116	0.0677	873	0.0005	1157	0.0011	974
240	coretech-002	0.0000	442	0.0013	909	0.0301	706	0.0000	363	0.0004	655
241	coretech-003	0.0000	418	0.0012	878	0.0282	688	0.0000	484	-	1350
242	coretech-004	0.0000	464	0.0013	910	0.0301	705	0.0000	366	-	1461
243	corsight-001	0.0000	678	0.0006	534	-	1434	0.0001	804	0.0004	577
244	corsight-002	0.0000	633	0.0005	478	0.0152	403	0.0001	739	0.0004	591
245	corsight-003	0.0000	682	0.0006	537	0.0175	458	0.0001	713	0.0004	613
246	csc-001	0.0000	1132	0.0030	1091	-	1418	0.0002	856	0.0005	789
247	csc-002	0.0015	1267	0.0033	1108	-	1456	0.0006	1184	0.0006	906
248	csc-003	0.0015	1266	0.0033	1107	0.0445	818	0.0006	1183	0.0006	904
249	ctbcbank-000	0.0001	1188	0.0051	1179	-	1166	0.0011	1236	0.0019	1021
250	ctbcbank-001	0.0000	1108	0.0036	1132	-	957	0.0005	1151	0.0010	945
251	ctbcbank-002	0.0000	315	0.0006	481	0.0150	396	0.0000	397	-	1328
252	cu-face-002	0.0000	269	0.0000	126	0.0000	12	0.0000	207	0.0000	143
253	cu-face-003	0.0000	1028	0.0009	754	0.0245	636	0.0003	988	0.0004	704
254	cu-face-004	0.0000	887	0.0007	624	0.0179	466	0.0001	765	-	1139
255	cubox-001	0.0000	314	0.0000	153	-	1072	0.0000	162	0.0000	165
256	cubox-002	0.0000	940	0.0006	524	0.0159	424	0.0002	954	0.0005	855
257	cubox-003	0.0000	93	0.0010	809	0.0210	556	0.0002	850	0.0004	677
258	cubox-004	0.0000	1007	0.0009	758	0.0193	515	0.0003	1045	-	1251
259	cubox-005	0.0000	1005	0.0009	756	0.0193	514	0.0003	1046	-	1178
260	cudocommunication-001	0.0000	7	0.0000	5	0.0000	85	0.0000	56	0.0000	20
261	cuhkee-001	0.0000	537	0.0011	843	-	1205	0.0000	348	0.0004	583
262	cybercore-000	0.0000	809	0.0073	1210	-	1094	0.0001	696	0.0005	805
263	cybercore-001	0.0000	1056	0.0001	269	0.0014	165	0.0002	844	0.0002	265
264	cybercore-002	0.0000	1060	0.0001	270	0.0014	164	0.0002	847	0.0002	267
265	cybercore-003	0.0000	762	0.0003	327	0.0060	202	0.0005	1159	0.0003	327
266	cyberextruder-001	0.0029	1271	0.0293	1266	-	975	0.0024	1296	0.0029	1048
267	cyberextruder-002	0.0013	1263	0.0840	1297	-	1056	0.0027	1306	0.0028	1044
268	cyberextruder-003	0.0000	1069	0.0077	1214	0.0887	905	0.0001	808	0.0006	894
269	cyberextruder-004	0.0000	1049	0.0097	1230	0.1025	911	0.0001	782	0.0007	907
270	cyberextruder-005	0.0000	254	0.0000	123	0.0000	17	0.0000	198	-	1293
271	cyberextruder-006	0.0000	391	0.0000	191	0.0000	43	0.0000	280	-	1399
272	cyberlink-000	-	1350	-	1348	-	1385	0.0006	1181	0.0008	927
273	cyberlink-001	-	1444	-	1443	-	1087	0.0073	1341	0.0005	791
274	cyberlink-002	-	1321	-	1319	-	1214	0.0004	1054	0.0004	766
275	cyberlink-003	0.0000	70	0.0007	604	-	1313	0.0001	620	0.0004	507
276	cyberlink-004	0.0000	545	0.0010	825	-	1186	0.0004	1059	0.0003	406
277	cyberlink-005	0.0000	635	0.0009	723	-	1347	0.0003	1041	0.0003	411
278	cyberlink-006	0.0000	1	0.0005	422	-	1245	0.0000	316	0.0003	355
279	cyberlink-007	0.0000	279	0.0003	294	0.0067	221	0.0000	332	0.0003	364
280	cyberlink-008	0.0000	155	0.0004	396	0.0106	306	0.0000	327	0.0003	362
281	cyberlink-009	0.0000	133	0.0004	395	0.0106	307	0.0000	326	0.0003	361
282	cyberlink-010	0.0000	108	0.0004	393	0.0106	303	0.0000	310	0.0003	356
283	cyberlink-011	0.0000	319	0.0004	402	0.0106	302	0.0000	329	0.0003	367
284	cyberlink-012	0.0000	189	0.0004	398	0.0106	310	0.0000	320	0.0003	363
285	cyberlink-013	0.0000	174	0.0004	397	0.0106	305	0.0000	323	0.0003	360
286	cyberlink-014	0.0000	217	0.0004	399	0.0106	309	0.0000	322	-	1255
287	cyberlink-015	0.0000	79	0.0004	392	0.0106	304	0.0000	312	-	1154
288	cyberlink-016	0.0000	231	0.0004	401	0.0106	308	0.0000	317	-	1224
289	dahua-000	-	1347	-	1346	-	1272	-	1367	0.0000	36
290	dahua-001	-	1450	-	1451	-	1173	0.0000	229	0.0000	220

Table 94: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
291	dahua-002	-	1462	-	1461	-	1138	0.0024	1299	0.0022	1032
292	dahua-003	0.0000	209	0.0019	986	-	1438	0.0002	912	0.0003	309
293	dahua-004	0.0000	81	0.0009	761	-	1292	0.0001	766	0.0003	287
294	dahua-005	0.0000	53	0.0000	240	-	1187	0.0000	501	0.0003	353
295	dahua-006	0.0000	34	0.0000	239	-	1232	0.0000	559	0.0003	408
296	dahua-007	0.0000	109	0.0000	242	0.0000	151	0.0000	555	0.0003	409
297	daon-000	0.0000	1120	0.0028	1073	0.0577	845	0.0014	1259	0.0015	1010
298	daon-001	0.0000	385	0.0009	705	0.0206	540	0.0000	487	-	1391
299	daon-002	0.0000	408	0.0004	413	0.0094	285	0.0000	252	-	1359
300	daon-003	0.0000	335	0.0008	656	0.0174	456	0.0000	382	-	1344
301	datech-000	0.0000	820	0.0004	352	0.0089	278	0.0000	508	0.0003	295
302	datech-001	0.0000	578	0.0004	350	0.0089	279	0.0000	500	-	1160
303	datech-002	0.0000	612	0.0004	351	0.0089	280	0.0000	504	-	1203
304	decatur-000	0.0000	946	0.0020	1004	-	1126	0.0004	1101	0.0005	786
305	decatur-001	0.0000	514	0.0009	749	0.0194	517	0.0001	668	0.0004	556
306	decloakface-001	0.0000	709	0.0016	950	0.0392	794	0.0001	724	0.0005	864
307	decloakface-002	0.0000	435	0.0011	833	0.0218	581	0.0001	612	-	1362
308	deepglint-001	0.0000	347	0.0000	165	-	1020	0.0000	146	0.0000	154
309	deepglint-002	0.0000	679	0.0004	400	-	1433	0.0002	924	0.0004	542
310	deepglint-003	0.0000	611	0.0004	394	-	1378	0.0002	926	0.0004	540
311	deepglint-004	0.0000	684	0.0005	425	0.0130	351	0.0002	945	0.0004	513
312	deepglint-005	0.0000	972	0.0019	989	0.0438	815	0.0006	1174	0.0006	902
313	deepsea-001	0.0000	358	0.0000	174	-	1010	0.0000	159	0.0000	160
314	deepsense-000	0.0000	453	0.0006	538	-	1177	0.0000	419	0.0004	430
315	deepsense-001	0.0000	230	0.0006	545	0.0191	504	0.0000	425	0.0004	435
316	deepsense-002	0.0000	339	0.0006	547	0.0191	503	0.0000	429	0.0004	438
317	deepsense-003	0.0000	262	0.0004	383	0.0110	318	0.0000	360	0.0003	324
318	deepsense-004	0.0000	777	0.0007	590	0.0179	468	0.0001	679	-	1317
319	deepsense-005	0.0000	490	0.0005	477	0.0144	383	0.0001	649	-	1404
320	deepsense-006	0.0000	320	0.0005	426	0.0113	327	0.0000	431	-	1331
321	deepsense-007	0.0000	111	0.0005	424	0.0113	329	0.0000	421	-	1133
322	deepsense-008	0.0000	465	0.0005	427	0.0113	328	0.0000	441	-	1460
323	dermalog-001	-	1338	-	1336	-	1290	-	1378	0.0013	998
324	dermalog-002	-	1317	-	1316	-	1208	-	1383	0.0013	997
325	dermalog-003	-	1335	-	1334	-	1321	-	1371	0.0025	1038
326	dermalog-004	-	1325	-	1323	-	1184	-	1386	0.0090	1075
327	dermalog-005	-	1433	0.0031	1096	-	1121	0.0013	1256	0.0041	1062
328	dermalog-006	0.0005	1245	0.0031	1100	-	1259	0.0013	1253	0.0041	1063
329	dermalog-007	0.0000	856	0.0006	493	-	1133	0.0000	542	0.0002	275
330	dermalog-008	0.0000	1086	0.0031	1097	-	1327	0.0006	1164	0.0003	296
331	dermalog-009	0.0000	1095	0.0031	1099	0.0148	391	0.0006	1167	0.0003	302
332	dermalog-010	0.0000	1092	0.0031	1098	0.0148	390	0.0006	1165	0.0003	299
333	dermalog-011	0.0000	1146	0.0005	429	0.0116	333	0.0001	624	0.0003	316
334	dermalog-012	0.0000	801	0.0003	336	0.0068	225	0.0003	993	0.0005	870
335	dermalog-013	0.0000	819	0.0002	291	0.0051	190	0.0002	840	-	1446
336	dermalog-014	0.0000	818	0.0003	337	0.0068	224	0.0003	994	-	1365
337	dermalog-015	0.0000	519	0.0003	335	0.0068	227	0.0003	992	-	1113
338	dermalog-016	0.0000	502	0.0003	338	0.0068	226	0.0001	604	-	1416
339	dicio-001	0.0005	1251	0.0649	1286	0.2136	932	0.0024	1295	0.0012	983
340	dicio-002	0.0000	145	0.0007	580	0.0185	486	0.0000	414	-	1216
341	didiglobalface-001	0.0000	780	0.0012	852	-	1123	0.0000	591	0.0004	697
342	didiglobalface-002	0.0000	850	0.0012	854	0.0247	642	0.0000	588	0.0004	700
343	didiglobalface-003	0.0000	628	0.0012	855	0.0247	643	0.0000	586	-	1218
344	digidata-000	0.0000	701	0.0023	1037	0.0375	782	0.0004	1112	0.0006	888
345	digidata-001	0.0000	857	0.0023	1038	0.0375	783	0.0004	1113	0.0006	889
346	digitalbarriers-000	-	1417	-	1418	-	1039	-	1437	0.0019	1022
347	digitalbarriers-001	-	1434	-	1434	-	1118	-	1459	0.0018	1019
348	digitalbarriers-002	0.0001	1214	0.0045	1162	-	1294	0.0028	1308	0.0027	1041

Table 95: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
349	dps-000	0.0000	188	0.0000	89	0.0000	144	0.0000	101	0.0000	107
350	dralmey-001	0.0000	741	0.0007	575	0.0185	485	0.0000	415	-	1327
351	dsk-000	0.0000	362	0.0000	176	-	1006	0.0000	156	0.0000	162
352	einetworks-000	0.0000	1109	0.0017	966	-	1045	0.0002	906	0.0005	843
353	einetworksindia-000	0.0000	1073	0.0018	977	0.0516	828	0.0002	973	0.0005	846
354	einetworksindia-001	0.0000	1077	0.0018	978	0.0516	829	0.0002	974	0.0005	847
355	einetworksindia-002	0.0000	548	0.0006	494	0.0159	426	0.0000	593	-	1111
356	ekin-002	0.0000	221	0.0000	250	0.0004	159	0.0000	319	0.0000	240
357	element-000	0.0000	513	0.0008	687	0.0192	512	0.0000	491	0.0003	370
358	enface-000	0.0000	6	0.0012	874	0.0305	713	0.0000	524	0.0004	606
359	enface-001	0.0000	398	0.0012	872	0.0304	712	0.0000	468	0.0004	602
360	enface-002	0.0000	730	0.0004	367	0.0084	267	0.0000	398	0.0004	440
361	enqura-001	0.0003	1234	0.0203	1257	0.1737	925	0.0023	1292	-	1150
362	enqura-002	0.0000	150	0.0000	72	0.0000	117	0.0000	138	-	1222
363	eocortex-000	0.0095	1289	0.0602	1284	-	1112	0.0094	1345	0.0059	1070
364	ercacat-001	0.0000	35	0.0005	460	-	1231	0.0000	525	0.0003	371
365	ethiopianlmis-000	0.0001	1185	0.0019	995	0.0159	428	0.0009	1221	-	1252
366	euronovate-001	0.0255	1304	0.0102	1235	0.0517	831	0.0021	1286	0.0004	758
367	euronovate-002	0.0000	837	0.0010	774	0.0265	676	0.0000	440	0.0004	703
368	euronovate-003	0.0000	721	0.0006	504	0.0159	423	0.0002	903	0.0004	693
369	euronovate-004	0.0000	1081	0.0016	943	0.0236	621	0.0015	1266	0.0006	890
370	everai-000	-	1319	-	1317	-	1216	-	1381	0.0004	446
371	everai-001	-	1448	-	1448	-	1075	0.0004	1087	0.0004	764
372	everai-002	-	1426	-	1428	-	1035	0.0002	915	0.0004	469
373	everai-paravision-003	0.0000	956	0.0008	690	-	1171	0.0002	848	0.0004	527
374	expasoft-000	0.0000	309	0.0000	151	-	1061	0.0000	167	0.0000	170
375	expasoft-001	0.0000	415	0.0000	204	-	1100	0.0000	246	0.0000	180
376	expasoft-002	0.0000	480	0.0000	225	0.0000	62	0.0000	232	0.0000	226
377	f8-001	0.0003	1236	0.0059	1194	-	1369	0.0035	1316	0.0030	1050
378	f8-002	0.0000	1165	0.0150	1248	0.0685	877	0.0005	1117	0.0013	1002
379	facehawk-000	0.0000	1042	0.0357	1274	0.0720	889	0.0021	1287	0.0009	938
380	facehawk-001	0.0299	1305	0.0098	1231	0.0236	622	0.0253	1359	-	1378
381	faceint-001	0.0001	1203	0.0190	1252	0.0898	906	0.0014	1264	-	1094
382	facelocate-001	0.0000	386	0.0019	992	0.0329	743	0.0000	554	-	1393
383	facelocate-002	0.0000	651	0.0025	1047	0.0387	792	0.0001	611	-	1181
384	faceonlive-001	0.0000	1138	0.0029	1080	0.0481	823	0.0013	1257	0.0011	961
385	faceonlive-002	0.0002	1230	0.0009	752	0.0075	240	0.0008	1208	0.0008	935
386	faceonlive-003	0.0000	978	0.0202	1255	0.0980	909	0.0005	1135	-	1248
387	facephi-000	0.0000	476	0.0004	354	0.0090	282	0.0001	742	0.0004	476
388	facephi-001	0.0000	883	0.0034	1119	0.0499	825	0.0003	1008	-	1164
389	facephi-002	0.0000	620	0.0016	948	0.0286	695	0.0002	921	-	1213
390	facesoft-000	0.0000	402	0.0000	199	-	1088	0.0000	257	0.0000	184
391	facetag-000	0.0000	367	0.0000	179	0.0000	50	0.0000	276	0.0000	198
392	facetag-001	0.0000	14	0.0000	6	0.0000	87	0.0000	52	0.0000	15
393	facetag-002	0.0000	192	0.0000	92	0.0000	148	0.0000	98	0.0000	104
394	facex-001	0.0001	1224	0.0360	1276	-	1007	0.0047	1329	0.0027	1042
395	facex-002	0.0001	1223	0.0360	1275	0.2663	936	0.0047	1330	0.0027	1043
396	facia-001	0.0000	458	0.0005	423	0.0122	342	0.0001	605	0.0004	499
397	facia-002	0.0000	1002	0.0017	964	0.0185	484	0.0002	962	-	1422
398	faicetech-000	0.0000	631	0.0007	592	0.0130	354	0.0002	957	-	1204
399	farfaces-001	0.0000	1101	0.0007	591	0.0061	204	0.0003	1012	0.0003	337
400	fastenterprises-000	0.0000	591	0.0082	1222	0.0169	449	0.0001	801	0.0003	357
401	fci-000	0.0000	305	0.0000	148	0.0000	3	0.0000	185	-	1281
402	fdu-000	-	1430	-	1431	-	1019	-	1433	0.0011	978
403	fdu-001	-	1374	-	1373	-	1334	-	1416	0.0009	943
404	fedu-001	0.0000	507	0.0000	238	0.0000	57	0.0000	213	-	1419
405	fiberhome-nanjing-002	0.0000	876	0.0006	541	-	1189	0.0001	687	0.0004	502
406	fiberhome-nanjing-003	0.0000	503	0.0004	375	-	1141	0.0000	216	0.0003	305

Table 96: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
407	fiberhome-nanjing-004	0.0000	198	0.0004	374	-	1458	0.0000	92	0.0003	303
408	fincore-000	0.0000	596	0.0008	684	0.0185	489	0.0001	606	0.0004	674
409	firstcreditkz-001	0.0000	984	0.0019	1000	0.0321	729	0.0000	580	0.0004	596
410	firstcreditkz-002	0.0000	731	0.0010	819	0.0232	618	0.0000	531	0.0004	520
411	firstcreditkz-003	0.0000	1114	0.0010	811	0.0219	582	0.0000	513	-	1440
412	firstcreditkz-004	0.0000	667	0.0004	388	0.0124	344	0.0000	321	-	1254
413	foomobi-001	0.0007	1256	0.0000	211	0.0000	66	0.0020	1285	0.0011	964
414	foomobi-002	0.0007	1255	0.0000	119	0.0000	18	0.0020	1284	0.0011	963
415	fourthlinecom-001	0.0000	880	0.0004	386	0.0078	247	0.0005	1130	-	1107
416	fpt-000	0.0001	1192	0.0007	599	0.0076	242	0.0006	1186	-	1118
417	fpt-001	0.0001	1194	0.0005	466	0.0067	222	0.0006	1170	-	1433
418	fpttelecom-000	0.0000	32	0.0000	14	0.0000	80	0.0000	62	-	1127
419	fraudcom-000	0.0000	906	0.0055	1188	0.0308	717	0.0002	916	0.0004	681
420	frpkauai-001	0.0000	967	0.0024	1046	0.0360	771	0.0001	625	0.0004	719
421	frpkauai-002	0.0000	971	0.0019	998	0.0321	730	0.0000	575	0.0004	589
422	frpkauai-003	0.0000	555	0.0006	513	0.0145	387	0.0000	471	-	1106
423	frpkauai-004	0.0001	1213	0.0051	1177	0.0730	891	0.0011	1237	-	1156
424	frpkauai-005	0.0000	858	0.0006	515	0.0145	386	0.0000	481	-	1428
425	fujitsulab-000	0.0000	1152	0.0033	1111	-	964	0.0005	1134	0.0002	263
426	fujitsulab-001	0.0000	19	0.0014	919	-	1226	0.0002	930	0.0003	392
427	fujitsulab-002	0.0000	303	0.0009	714	-	960	0.0001	769	0.0003	304
428	fujitsulab-003	0.0000	474	0.0008	658	0.0166	442	0.0001	731	0.0001	257
429	g42-intellibrain-001	0.0000	82	0.0000	35	0.0000	104	0.0000	34	0.0000	56
430	g42-intellibrain-002	0.0000	137	0.0000	65	0.0000	121	0.0000	129	0.0000	77
431	geo-000	0.0000	157	0.0000	73	-	1371	0.0000	137	0.0000	85
432	geo-001	0.0000	751	0.0011	831	-	1051	0.0000	579	0.0004	599
433	geo-002	0.0000	771	0.0015	930	0.0332	749	0.0001	603	0.0004	737
434	geo-003	0.0000	797	0.0010	790	0.0244	633	0.0000	390	0.0004	728
435	geo-004	0.0000	732	0.0005	474	0.0138	369	0.0001	700	0.0004	550
436	gistouch-000	0.0000	1163	0.0188	1251	0.1345	917	0.0018	1277	0.0007	908
437	gistouch-001	0.0000	39	0.0002	274	0.0027	176	0.0001	689	0.0003	313
438	glory-000	-	1441	-	1441	-	1110	0.0053	1335	0.0013	1000
439	glory-001	0.0160	1295	0.0314	1270	-	1200	0.0051	1334	0.0010	949
440	glory-002	0.0003	1235	0.0045	1161	-	1233	0.0015	1267	0.0011	976
441	glory-003	0.0000	983	0.0027	1063	-	952	0.0004	1069	0.0005	795
442	glory-004	0.0000	954	0.0020	1010	0.0345	763	0.0001	758	0.0004	726
443	glory-005	0.0000	959	0.0020	1011	0.0345	765	0.0001	756	0.0004	730
444	glory-006	0.0000	962	0.0020	1012	0.0345	764	0.0001	755	0.0004	729
445	glory-007	0.0000	992	0.0020	1018	0.0403	805	0.0002	927	0.0005	785
446	gorilla-000	-	1313	-	1311	-	1225	-	1392	0.0004	768
447	gorilla-001	-	1438	-	1439	-	1117	0.0003	1050	0.0004	775
448	gorilla-002	-	1418	-	1419	-	1054	0.0003	1047	0.0004	773
449	gorilla-003	-	1342	-	1340	-	1257	0.0003	1043	0.0004	769
450	gorilla-004	0.0000	46	0.0008	668	-	1211	0.0000	450	0.0003	334
451	gorilla-005	0.0000	80	0.0008	677	-	1318	0.0000	448	0.0003	339
452	gorilla-006	0.0000	149	0.0006	555	-	1360	0.0000	413	0.0003	341
453	gorilla-007	0.0000	586	0.0009	767	0.0252	657	0.0001	667	0.0004	662
454	gorilla-008	0.0000	595	0.0009	768	0.0259	669	0.0001	666	0.0004	675
455	gorilla-009	0.0000	699	0.0010	797	0.0276	686	0.0001	635	0.0004	649
456	gpstechvn-000	0.0000	163	0.0000	76	0.0000	116	0.0000	118	0.0000	66
457	graymatics-001	0.0000	63	0.0010	772	0.0210	552	0.0001	802	0.0004	554
458	griaule-000	0.0000	1127	0.0026	1061	0.0531	837	0.0004	1111	0.0010	946
459	griaule-001	0.0000	199	0.0012	875	0.0366	776	0.0000	407	0.0004	664
460	griaule-002	0.0000	15	0.0007	603	0.0209	544	0.0000	571	0.0004	573
461	gtelots-000	0.0000	987	0.0003	344	0.0081	260	0.0004	1066	-	1330
462	gtelots-001	0.0000	1006	0.0009	726	0.0177	462	0.0004	1080	-	1249
463	hertasecurity-000	0.0133	1294	0.0077	1215	-	1295	0.0025	1302	0.0243	1082
464	hertasecurity-001	0.0000	31	0.0000	252	0.0000	153	0.0000	374	0.0001	248

Table 97: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
465	hertasecurity-002	0.0000	105	0.0000	50	0.0000	101	0.0000	369	0.0000	233
466	hertasecurity-003	0.0000	227	0.0000	105	0.0000	134	0.0000	318	0.0000	235
467	hik-001	0.0000	284	0.0000	255	-	968	0.0000	182	0.0000	122
468	hisign-001	0.0000	119	0.0000	59	0.0000	96	0.0000	15	0.0000	37
469	hisign-002	0.0000	897	0.0006	525	0.0150	397	0.0001	759	0.0003	395
470	hisign-003	0.0000	902	0.0008	685	0.0237	624	0.0001	760	-	1184
471	hyperverge-001	0.0000	1160	0.0072	1207	-	1320	0.0015	1269	0.0014	1005
472	hyperverge-002	0.0000	342	0.0008	654	0.0210	557	0.0002	959	0.0004	565
473	hyperverge-003	0.0000	375	0.0008	648	0.0210	555	0.0002	964	0.0004	567
474	hyperverge-004	0.0000	605	0.0008	682	0.0218	578	0.0002	913	0.0004	588
475	hyperverge-005	0.0000	419	0.0008	649	0.0210	554	0.0002	963	0.0004	566
476	hyperverge-006	0.0000	268	0.0008	647	0.0210	553	0.0002	961	-	1308
477	hyperverge-007	0.0000	148	0.0008	653	0.0210	558	0.0002	958	-	1212
478	hyundai-roboticslab-000	0.0000	292	0.0011	845	0.0246	638	0.0001	730	-	1271
479	hyundai-roboticslab-001	0.0000	430	0.0105	1237	0.1972	926	0.0007	1198	-	1372
480	hyundai-roboticslab-002	0.0000	204	0.0020	1017	0.0299	704	0.0002	942	-	1263
481	hyundai-roboticslab-003	0.0000	210	0.0005	428	0.0136	364	0.0000	352	-	1261
482	hyundai-roboticslab-004	0.0000	147	0.0007	582	0.0200	521	0.0000	379	-	1211
483	hzailu-001	0.0000	1076	0.0016	949	0.0449	819	0.0003	1038	0.0005	851
484	hzailu-002	0.0000	1074	0.0015	937	0.0424	808	0.0003	1036	0.0005	849
485	hzailu-003	0.0000	768	0.0004	361	0.0081	254	0.0002	846	0.0003	389
486	hzailu-004	0.0000	535	0.0004	357	0.0081	256	0.0002	842	0.0003	379
487	hzailu-005	0.0000	556	0.0004	358	0.0081	258	0.0002	841	0.0003	382
488	hzailu-006	0.0000	680	0.0004	359	0.0081	259	0.0002	843	0.0003	385
489	hzailu-007	0.0000	935	0.0004	360	0.0081	255	0.0003	1001	-	1347
490	hzailu-008	0.0000	871	0.0004	356	0.0081	257	0.0003	999	-	1117
491	i2v-001	0.0000	1071	0.0018	983	0.0341	761	0.0001	688	-	1102
492	i2v-002	0.0000	966	0.0012	849	0.0244	634	0.0000	519	-	1115
493	icehouse-000	0.0000	9	0.0000	3	0.0000	84	0.0000	54	-	1121
494	icm-002	0.0000	43	0.0001	260	-	1202	0.0000	39	0.0000	231
495	icm-003	0.0000	196	0.0001	261	0.0023	171	0.0000	94	0.0000	232
496	icm-004	0.0000	1121	0.0033	1118	0.0698	881	0.0006	1180	0.0010	957
497	icm-005	0.0002	1229	0.0117	1242	0.1223	915	0.0018	1280	0.0014	1009
498	icomm-000	0.0000	834	0.0004	381	0.0098	292	0.0000	552	0.0003	344
499	icthtc-000	0.0001	1221	0.0047	1169	-	1012	0.0028	1310	0.0029	1045
500	id3-001	-	1452	-	1452	-	1176	-	1454	0.0043	1066
501	id3-002	-	1456	-	1456	-	1142	-	1450	0.0032	1052
502	id3-003	0.0001	1184	0.0052	1182	-	1000	0.0016	1272	0.0011	973
503	id3-004	0.0000	1176	0.0052	1180	-	1164	0.0015	1270	0.0011	975
504	id3-005	0.0000	593	0.0074	1212	-	1287	0.0002	866	0.0005	794
505	id3-006	0.0000	1000	0.0009	762	-	1145	0.0004	1086	0.0005	835
506	id3-007	0.0000	681	0.0041	1146	-	1408	0.0001	734	0.0004	624
507	id3-008	0.0000	75	0.0006	532	0.0184	481	0.0001	798	0.0004	432
508	id3-009	0.0000	870	0.0003	345	0.0088	275	0.0001	744	-	1119
509	id3-010	0.0000	307	0.0001	272	0.0030	177	0.0000	458	-	1337
510	idemia-003	-	1409	-	1409	-	965	0.0000	435	0.0004	466
511	idemia-004	0.0000	565	0.0006	509	-	1324	0.0000	470	0.0004	482
512	idemia-005	0.0000	23	0.0004	403	-	1218	0.0000	375	0.0003	381
513	idemia-006	0.0000	184	0.0004	406	-	1338	0.0000	377	0.0003	384
514	idemia-007	0.0000	250	0.0004	408	-	998	0.0000	384	0.0003	388
515	idemia-008	0.0000	90	0.0004	405	0.0078	250	0.0000	372	0.0003	383
516	idemia-009	0.0000	60	0.0004	389	0.0077	245	0.0000	373	0.0003	377
517	idemia-010	0.0000	443	0.0004	349	0.0074	235	0.0000	228	0.0003	343
518	idemia-011	0.0000	92	0.0003	296	0.0055	195	0.0000	29	-	1159
519	identy-000	0.0000	1164	0.0020	1014	0.0152	400	0.0012	1243	0.0004	722
520	identy-001	0.0000	1171	0.0013	908	0.0398	799	0.0002	952	-	1441
521	identy-002	0.0000	1172	0.0013	906	0.0401	801	0.0002	946	-	1300
522	identy-003	0.0000	1119	0.0011	839	0.0247	641	0.0001	701	-	1295

Table 98: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
523	idrnd-000	0.0000	922	0.0020	1003	0.0566	842	0.0003	1049	-	1297
524	idrnd-001	0.0000	161	0.0006	514	0.0182	472	0.0000	410	-	1173
525	igearx-face-000	0.0000	542	0.0006	488	0.0153	405	0.0004	1078	0.0004	658
526	iit-000	-	1421	-	1422	-	1052	0.0007	1189	0.0011	967
527	iit-001	0.0000	974	0.0806	1296	-	1372	0.0001	820	0.0004	644
528	iit-002	0.0000	1118	0.0021	1019	-	1331	0.0009	1223	0.0005	860
529	iit-003	0.0000	579	0.0008	683	-	1303	0.0000	488	0.0004	455
530	imds-software-001	0.0000	62	0.0000	28	0.0000	73	0.0000	45	0.0000	11
531	imds-software-002	0.0000	78	0.0000	33	0.0000	110	0.0000	22	0.0000	50
532	imds-software-003	0.0000	191	0.0000	90	0.0000	143	0.0000	99	0.0000	109
533	imperial-000	0.0000	190	0.0000	88	-	1454	0.0000	100	0.0000	108
534	imperial-001	-	1323	-	1320	-	1213	0.0000	35	0.0000	1
535	imperial-002	0.0000	89	0.0000	39	-	1304	0.0000	30	0.0000	53
536	incode-000	-	1393	-	1392	-	1398	-	1398	0.0035	1054
537	incode-001	-	1425	-	1426	-	1032	-	1428	0.0035	1055
538	incode-002	-	1447	-	1447	-	1074	0.0004	1108	0.0007	911
539	incode-003	-	1445	-	1445	-	1097	0.0004	1107	0.0007	910
540	incode-004	-	1379	-	1377	-	1452	0.0004	1105	0.0007	909
541	incode-005	0.0000	154	0.0008	646	-	1370	0.0001	677	0.0004	645
542	incode-006	0.0000	552	0.0005	441	-	1185	0.0001	781	0.0004	418
543	incode-007	0.0000	945	0.0009	734	-	1127	0.0002	881	0.0004	552
544	incode-008	0.0000	911	0.0009	731	-	1435	0.0002	874	0.0004	544
545	incode-009	0.0000	901	0.0009	730	0.0255	665	0.0002	875	0.0004	537
546	incode-010	0.0000	933	0.0009	733	0.0255	663	0.0002	877	0.0004	548
547	incode-011	0.0000	912	0.0009	732	0.0255	666	0.0002	873	0.0004	543
548	incode-012	0.0000	869	0.0009	729	0.0255	664	0.0002	871	0.0004	530
549	incode-013	0.0001	1226	0.0012	867	0.0254	661	0.0005	1143	0.0004	749
550	incode-014	0.0000	365	0.0006	543	0.0166	441	0.0001	812	-	1321
551	incode-015	0.0000	280	0.0007	621	0.0195	518	0.0000	436	-	1272
552	infocert-001	0.0000	1016	0.0059	1197	0.0424	809	0.0001	684	0.0006	879
553	infocert-002	0.0084	1285	0.0041	1154	0.0084	268	0.0085	1343	-	1411
554	infocert-003	0.0089	1288	0.0042	1156	0.0077	244	0.0087	1344	-	1379
555	innefulabs-000	0.0000	767	0.0024	1044	-	1011	0.0003	1014	0.0005	831
556	innominds-001	0.0000	1067	0.0030	1090	0.0637	861	0.0010	1233	0.0008	932
557	innov8tif-000	0.0000	1111	0.0029	1087	0.0657	865	0.0004	1099	-	1355
558	innov8tif-001	0.0000	1107	0.0029	1086	0.0657	866	0.0004	1098	-	1198
559	innovativetechnologyltd-001	0.0001	1218	0.0050	1176	-	1192	0.0024	1300	0.0025	1037
560	innovativetechnologyltd-002	0.0000	1001	0.0046	1164	-	1157	0.0057	1337	0.0005	833
561	innovatrics-000	-	1329	-	1327	-	1315	-	1377	0.0004	731
562	innovatrics-001	-	1371	-	1370	-	1349	-	1407	0.0004	733
563	innovatrics-002	-	1324	-	1322	-	1190	-	1387	0.0004	449
564	innovatrics-003	-	1431	-	1432	-	1013	-	1432	0.0005	781
565	innovatrics-004	0.0000	944	0.0017	965	-	1120	0.0000	582	0.0004	684
566	innovatrics-005	-	1340	-	1338	-	1279	0.0000	574	0.0004	663
567	innovatrics-006	0.0000	657	0.0009	757	-	1447	0.0000	515	0.0004	463
568	innovatrics-007	0.0000	541	0.0007	614	-	1201	0.0001	601	0.0003	335
569	innovatrics-008	0.0000	793	0.0009	747	0.0204	532	0.0000	521	0.0004	427
570	innovatrics-009	0.0000	182	0.0005	419	0.0142	378	0.0000	123	0.0000	239
571	innovatrics-010	0.0000	770	0.0019	991	0.0393	795	0.0001	670	0.0004	439
572	innovatrics-011	0.0000	796	0.0001	262	0.0017	169	0.0000	343	-	1384
573	innovatrics-012	0.0000	692	0.0004	417	0.0118	334	0.0000	506	-	1292
574	innovatrics-013	0.0000	564	0.0001	265	0.0027	175	0.0000	287	-	1148
575	innovatrics-014	0.0000	644	0.0007	581	0.0165	437	0.0000	428	-	1188
576	innovatrics-015	0.0000	746	0.0008	674	0.0236	623	0.0000	416	-	1348
577	innovatrics-016	0.0000	853	0.0006	491	0.0177	464	0.0000	306	-	1423
578	innovatrics-017	0.0000	698	0.0006	519	0.0189	495	0.0000	361	-	1304
579	insightface-000	0.0000	499	0.0000	234	0.0000	54	0.0000	218	0.0000	215
580	insightface-001	0.0000	273	0.0000	128	0.0000	14	0.0000	205	0.0000	138

Table 99: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
581	insightface-002	0.0000	373	0.0000	181	0.0000	47	0.0000	271	0.0000	200
582	insightface-003	0.0000	193	0.0000	93	0.0000	147	0.0000	97	0.0000	103
583	insightface-004	0.0000	52	0.0000	23	0.0000	71	0.0000	51	0.0000	13
584	inspur-000	0.0000	471	0.0000	223	0.0000	64	0.0000	237	0.0000	223
585	inspur-001	0.0000	69	0.0000	31	0.0000	109	0.0000	27	0.0000	51
586	inspur-002	0.0000	158	0.0000	74	0.0000	118	0.0000	136	0.0000	87
587	inspur-003	0.0000	134	0.0000	64	0.0000	120	0.0000	131	-	1201
588	intellicloudai-001	0.0000	387	0.0000	187	-	1105	0.0000	283	0.0000	205
589	intellicloudai-002	0.0000	501	0.0008	667	-	1134	0.0000	498	0.0004	443
590	intellifusion-001	0.0000	763	0.0005	468	-	1031	0.0001	655	0.0003	399
591	intellifusion-002	0.0000	275	0.0000	251	-	993	0.0000	300	0.0000	136
592	intellivision-001	0.0042	1275	0.0296	1267	-	1284	0.0048	1331	0.0042	1064
593	intellivision-002	0.0000	1168	0.0046	1163	-	1236	0.0012	1245	0.0005	868
594	intellivision-003	0.0000	627	0.0012	862	0.0308	715	0.0003	1000	0.0004	772
595	intellivision-004	0.0000	781	0.0011	834	0.0266	680	0.0002	970	0.0004	765
596	intellivision-005	0.0001	1209	0.0041	1147	0.0528	835	0.0005	1137	0.0007	918
597	intellivision-006	0.0001	1210	0.0041	1148	0.0528	836	0.0005	1138	0.0007	919
598	intellivision-007	0.0001	1212	0.0041	1150	0.0528	834	0.0005	1140	0.0007	920
599	intellivix-001	0.0000	83	0.0000	36	0.0000	105	0.0000	33	0.0000	55
600	intellivix-002	0.0000	364	0.0009	753	0.0184	482	0.0000	154	0.0000	158
601	intellivix-003	0.0000	397	0.0000	194	0.0000	46	0.0000	277	0.0000	201
602	intellivix-004	0.0000	100	0.0022	1026	0.0333	753	0.0000	7	0.0000	32
603	intellivix-005	0.0000	427	0.0022	1027	0.0333	752	0.0000	264	0.0000	191
604	intellivix-006	0.0000	257	0.0007	583	0.0180	469	0.0000	196	-	1286
605	intelresearch-000	0.0000	845	0.0006	482	-	1147	0.0000	550	0.0003	403
606	intelresearch-001	0.0000	1145	0.0082	1221	-	1090	0.0005	1152	0.0010	947
607	intelresearch-002	0.0000	840	0.0022	1024	-	1170	0.0000	551	0.0004	431
608	intelresearch-003	0.0000	747	0.0006	502	-	1043	0.0000	433	0.0004	521
609	intelresearch-004	0.0000	559	0.0006	505	-	1309	0.0000	423	0.0004	508
610	intelresearch-005	0.0000	783	0.0006	506	0.0144	385	0.0000	445	0.0004	525
611	intelresearch-006	0.0000	271	0.0000	248	0.0004	160	0.0000	336	0.0004	516
612	intelresearch-007	0.0000	45	0.0007	565	0.0168	445	0.0000	314	0.0004	500
613	intelresearch-008	0.0000	258	0.0006	556	0.0169	450	0.0000	333	0.0004	546
614	intelresearch-009	0.0000	110	0.0007	625	0.0232	616	0.0000	309	-	1132
615	intelresearch-010	0.0000	683	0.0010	818	0.0335	755	0.0000	503	-	1242
616	intelresearch-011	0.0000	649	0.0006	549	0.0175	457	0.0000	476	-	1180
617	intelresearch-012	0.0000	661	0.0007	593	0.0191	510	0.0000	475	-	1243
618	intema-000	0.0000	495	0.0005	438	0.0126	346	0.0000	564	0.0004	472
619	intema-001	0.0000	601	0.0004	387	0.0106	311	0.0000	311	0.0003	410
620	interlinklabs-001	0.0000	1124	0.0004	411	0.0085	271	0.0005	1126	-	1313
621	interlinklabs-002	0.0000	1123	0.0004	409	0.0085	269	0.0005	1128	-	1309
622	interlinklabs-003	0.0000	949	0.0031	1104	0.0249	646	0.0003	1018	-	1386
623	intozi-001	0.0000	861	0.0005	475	0.0126	345	0.0001	826	0.0004	429
624	intozi-002	0.0000	290	0.0000	138	0.0000	11	0.0000	177	-	1264
625	intsysmsu-000	-	1413	-	1414	-	1064	0.0004	1079	0.0012	980
626	intsysmsu-001	0.0000	99	0.0010	798	-	1276	0.0001	705	0.0004	619
627	intsysmsu-002	0.0000	492	0.0010	801	-	1151	0.0001	710	0.0004	634
628	ionetworks-000	0.0000	179	0.0016	956	0.0387	791	0.0004	1051	0.0005	797
629	ionetworks-001	0.0000	370	0.0081	1218	0.0666	867	0.0000	273	0.0000	199
630	ionetworks-002	0.0000	169	0.0003	308	0.0048	187	0.0000	113	0.0000	63
631	ionetworks-003	0.0000	76	0.0004	366	0.0079	252	0.0000	24	-	1151
632	ionetworks-004	0.0000	429	0.0004	355	0.0086	273	0.0000	263	-	1369
633	iqface-000	0.0000	185	0.0000	86	-	1337	0.0000	120	0.0000	69
634	iqface-001	0.0000	281	0.0000	131	-	972	0.0000	299	0.0000	121
635	iqface-003	0.0000	1115	0.0076	1213	-	1143	0.0006	1166	0.0005	867
636	iqsec-001	0.0000	359	0.0763	1295	0.1617	921	0.0000	158	-	1322
637	iqsec-002	0.0000	316	0.0763	1294	0.1617	922	0.0000	161	-	1332
638	iqsec-003	0.0000	101	0.0000	47	0.0000	97	0.0000	6	-	1141

Table 100: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
639	irex-000	0.0000	1008	0.0009	751	-	1431	0.0000	545	0.0005	780
640	irisid-001	0.0000	728	0.0016	946	0.0441	816	0.0000	595	-	1278
641	irisid-002	0.0000	847	0.0016	947	0.0441	817	0.0000	596	-	1408
642	isap-001	0.0000	77	0.0000	34	-	1325	0.0000	23	0.0000	49
643	isap-002	0.0000	125	0.0000	61	-	1266	0.0000	11	0.0000	43
644	isityou-000	0.0068	1283	0.0316	1271	-	1155	0.0023	1291	0.0010	954
645	istl-001	0.0000	1057	0.0005	479	0.0162	435	0.0010	1232	-	1175
646	isystems-000	-	1316	-	1314	-	1230	-	1389	0.0010	953
647	isystems-001	0.0000	1135	0.0035	1129	-	1403	0.0010	1229	0.0007	917
648	isystems-002	0.0000	1134	0.0035	1128	-	1383	0.0010	1230	0.0007	916
649	itmo-001	-	1355	-	1355	-	1390	-	1421	0.0047	1067
650	itmo-002	-	1372	-	1371	-	1336	-	1417	0.0029	1049
651	itmo-003	-	1330	-	1328	-	1316	-	1376	0.0008	925
652	itmo-004	-	1422	-	1424	-	1030	-	1431	0.0004	761
653	itmo-005	-	1327	-	1325	-	1193	0.0005	1124	0.0002	264
654	itmo-006	0.0000	238	0.0015	936	-	1405	0.0004	1104	0.0004	626
655	itmo-007	0.0000	357	0.0009	708	-	1009	0.0003	1042	0.0000	161
656	itmo-008	0.0000	170	0.0135	1244	0.1239	916	0.0024	1301	0.0000	62
657	ivacognitive-001	0.0000	914	0.0011	836	-	1427	0.0001	627	0.0004	735
658	iws-000	0.0005	1250	0.0650	1289	-	1104	0.0024	1297	0.0012	986
659	jaakit-001	0.0008	1260	0.0858	1298	0.2713	937	0.0042	1324	0.0021	1028
660	jaakit-002	0.0000	488	0.0000	229	0.0000	60	0.0000	212	-	1401
661	kaizenvoiz-001	0.0000	1100	0.0000	17	0.0000	83	0.0037	1319	-	1125
662	kakao-001	-	1398	-	1397	-	1001	0.0002	932	0.0005	814
663	kakao-002	0.0000	982	0.0057	1193	-	954	0.0002	931	0.0005	811
664	kakao-003	0.0000	27	0.0000	12	-	1234	0.0000	66	0.0000	24
665	kakao-004	0.0000	159	0.0000	75	-	1367	0.0000	135	0.0000	86
666	kakao-005	0.0000	249	0.0000	243	-	1411	0.0000	76	0.0000	243
667	kakao-007	0.0000	164	0.0007	561	0.0165	438	0.0001	694	0.0004	509
668	kakao-008	0.0000	11	0.0009	718	0.0209	550	0.0001	690	0.0004	504
669	kakao-009	0.0000	469	0.0009	720	0.0209	548	0.0001	695	0.0004	529
670	kakaobank-000	0.0000	348	0.0000	164	0.0000	23	0.0000	147	0.0000	155
671	kakaobank-001	0.0000	74	0.0006	503	0.0176	460	0.0000	370	0.0004	453
672	kakaobank-002	0.0000	826	0.0010	826	0.0303	710	0.0000	465	-	1429
673	kakaobank-003	0.0000	800	0.0013	907	0.0397	797	0.0000	443	-	1360
674	kakaopay-001	0.0000	905	0.0013	901	0.0322	733	0.0001	645	0.0004	742
675	kasikornlabs-000	0.0000	1150	0.0035	1126	0.0713	887	0.0004	1102	0.0012	993
676	kasikornlabs-001	0.0001	1220	0.0050	1175	0.0885	904	0.0006	1185	0.0035	1057
677	kasikornlabs-002	0.0000	1147	0.0033	1113	0.0698	883	0.0004	1094	0.0012	981
678	kasikornlabs-003	0.0000	1142	0.0033	1109	0.0698	884	0.0004	1092	0.0012	979
679	kasikornlabs-004	0.0000	1144	0.0033	1112	0.0698	882	0.0004	1096	-	1397
680	kasikornlabs-005	0.0000	1143	0.0033	1110	0.0698	885	0.0004	1091	-	1257
681	kbyai-000	0.0000	707	0.0013	886	0.0265	679	0.0001	623	-	1273
682	kedacom-000	0.0000	296	0.0000	143	-	955	0.0000	191	0.0000	128
683	keyless-000	0.0000	722	0.0010	795	0.0308	716	0.0000	461	-	1285
684	kiwitech-000	0.0000	787	0.0009	704	-	1130	0.0004	1095	0.0005	792
685	kloudspot-001	0.0000	975	0.0010	793	0.0211	559	0.0001	772	-	1209
686	kloudspot-002	0.0000	266	0.0006	550	0.0159	427	0.0002	919	-	1305
687	kneron-003	0.0239	1302	0.0306	1268	-	1161	0.0044	1328	0.0016	1017
688	kneron-005	0.0000	1140	0.0226	1259	-	1425	0.0006	1162	0.0005	822
689	kneron-006	0.0000	1103	0.1109	1301	0.0686	878	0.0012	1244	-	1155
690	knowutech-000	0.0000	607	0.0008	640	0.0215	574	0.0000	511	0.0004	643
691	kogniza-001	0.0000	625	0.0012	880	0.0337	760	0.0003	1030	-	1217
692	kookmin-001	0.0000	416	0.0000	207	-	1099	0.0000	245	0.0000	179
693	kookmin-002	0.0000	351	0.0000	168	-	1036	0.0000	143	0.0000	147
694	koreaid-001	0.0000	337	0.0023	1040	0.0371	779	0.0000	569	0.0005	782
695	krungthai-002	0.0000	550	0.0005	451	0.0111	321	0.0002	905	0.0003	407
696	kuke3d-001	0.0000	135	0.0000	67	0.0000	122	0.0000	130	0.0000	76

Table 101: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
697	kuke3d-002	0.0000	85	0.0000	37	0.0000	102	0.0000	32	0.0000	58
698	lebentech-000	0.0042	1273	0.0029	1088	0.0252	656	0.0051	1333	0.0066	1072
699	lebentech-001	0.0000	1175	0.0012	879	0.0317	721	0.0004	1103	0.0005	844
700	lemalabs-001	0.0000	211	0.0005	471	0.0141	376	0.0002	896	0.0004	465
701	lightvision-000	0.0000	425	0.0000	208	0.0000	34	0.0000	265	-	1370
702	lightvision-001	0.0000	334	0.0000	160	0.0000	30	0.0000	169	-	1343
703	lineclova-000	0.0000	115	0.0000	54	0.0000	89	0.0000	18	0.0000	47
704	lineclova-001	0.0000	328	0.0000	155	0.0000	28	0.0000	173	0.0000	176
705	lineclova-002	0.0000	215	0.0007	564	0.0181	471	0.0000	103	0.0000	111
706	lineclova-003	0.0000	1085	0.0023	1033	0.0700	886	0.0002	968	0.0005	779
707	lookman-002	0.0000	219	0.0000	103	-	1426	0.0000	74	0.0000	90
708	lookman-004	0.0000	393	0.0000	192	-	1103	0.0000	279	0.0000	207
709	lssytech-000	0.0000	653	0.0006	539	0.0177	461	0.0000	527	-	1183
710	luxand-000	0.0000	171	0.0000	80	-	1354	0.0000	112	0.0000	61
711	luxand-001	0.0000	1011	0.0016	953	0.0366	774	0.0006	1172	0.0005	842
712	luxand-003	0.0000	1025	0.0016	955	0.0366	775	0.0006	1173	-	1426
713	luxand-004	0.0000	629	0.0003	331	0.0061	206	0.0003	981	-	1205
714	mantra-000	0.0001	1186	0.0041	1151	0.0680	876	0.0003	1010	0.0004	760
715	matrixcomsec-000	0.0002	1227	1.0000	1360	1.0000	1348	0.0336	1360	-	1171
716	matrixcomsec-001	0.0000	434	0.0009	713	0.0252	655	0.0000	261	-	1371
717	matrixcomsec-002	0.0000	609	0.0009	715	0.0252	658	0.0001	699	-	1202
718	maverickresearch-000	0.0000	997	0.0096	1229	0.0574	844	0.0013	1255	-	1442
719	maxvision-000	0.0000	72	0.0000	241	0.0000	108	0.0000	26	0.0000	52
720	maxvision-001	0.0000	341	0.0000	163	0.0000	22	0.0000	151	0.0000	156
721	maxvision-002	0.0000	676	0.0009	699	0.0229	610	0.0002	832	0.0004	592
722	maxvision-003	0.0000	613	0.0009	698	0.0229	608	0.0002	833	0.0004	590
723	maxvision-004	0.0000	686	0.0009	700	0.0229	609	0.0002	838	0.0004	593
724	maxvision-005	0.0000	590	0.0009	697	0.0229	607	0.0002	836	0.0004	586
725	maxvision-006	0.0000	659	0.0007	630	0.0226	593	0.0002	858	0.0004	594
726	mbbankvn-000	0.0000	652	0.0004	379	0.0098	293	0.0000	547	-	1182
727	megvii-000	-	1396	-	1396	-	996	-	1444	0.0003	311
728	megvii-001	0.0000	160	0.0006	510	-	1366	0.0007	1196	0.0004	490
729	megvii-002	0.0000	71	0.0006	508	-	1314	0.0054	1336	0.0004	483
730	megvii-003	0.0000	580	0.0010	820	-	1302	0.0002	944	0.0004	706
731	megvii-004	0.0000	720	0.0010	794	-	945	0.0002	879	0.0004	648
732	megvii-005	0.0000	760	0.0010	780	0.0206	536	0.0002	937	0.0004	694
733	megvii-006	0.0000	823	0.0010	781	0.0206	538	0.0002	941	0.0004	701
734	megvii-007	0.0000	606	0.0010	775	0.0206	539	0.0002	936	0.0004	689
735	megvii-008	0.0000	712	0.0010	778	0.0206	535	0.0002	939	0.0004	691
736	megvii-009	0.0000	737	0.0010	779	0.0206	537	0.0002	938	0.0004	695
737	meituan-000	0.0000	485	0.0001	266	0.0013	163	0.0000	386	0.0002	274
738	meituan-001	0.0000	597	0.0014	922	0.0295	700	0.0001	712	0.0004	661
739	meituan-002	0.0000	638	0.0013	895	0.0251	653	0.0001	714	0.0004	641
740	meituan-003	0.0000	782	0.0013	896	0.0251	651	0.0001	716	0.0004	654
741	meituan-004	0.0000	839	0.0013	897	0.0251	652	0.0001	715	0.0004	656
742	meiya-001	0.0000	1128	0.0028	1075	-	1018	0.0004	1109	0.0010	956
743	mendaxiatech-000	0.0000	844	0.0010	782	0.0206	534	0.0002	947	0.0004	699
744	metsakuurcompany-001	0.0000	356	0.0011	828	0.0208	543	0.0002	935	0.0004	517
745	metsakuurcompany-002	0.0000	432	0.0000	209	0.0000	33	0.0000	262	0.0000	192
746	metsakuurcompany-003	0.0000	64	0.0000	30	0.0000	74	0.0000	44	0.0000	9
747	metsakuurcompany-004	0.0000	438	0.0000	210	0.0000	35	0.0000	260	-	1363
748	metsakuurcompany-005	0.0000	379	0.0000	185	0.0000	52	0.0000	266	-	1376
749	miaxis-001	0.0000	836	0.0013	892	0.0262	674	0.0001	807	0.0003	328
750	miaxis-002	0.0448	1309	0.1162	1302	0.2565	935	0.2128	1364	0.0347	1086
751	miaxis-003	0.0000	694	0.0013	891	0.0262	673	0.0001	806	0.0003	326
752	miaxis-004	0.0000	124	0.0000	60	0.0000	92	0.0000	12	-	1145
753	miaxis-005	0.0000	40	0.0000	19	0.0000	77	0.0000	42	-	1098
754	microfocus-000	-	1410	-	1410	-	979	-	1439	0.0016	1014

Table 102: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
755	microfocus-001	0.0001	1216	0.0053	1185	-	1228	0.0008	1209	0.0016	1013
756	microfocus-002	0.0001	1217	0.0053	1186	-	947	0.0008	1210	0.0016	1015
757	microfocus-003	0.0001	1208	0.0049	1174	0.0819	898	0.0007	1200	0.0015	1012
758	minivision-000	0.0000	304	0.0000	146	-	959	0.0000	186	0.0000	126
759	mitek-000	0.0000	1020	0.0029	1079	0.0521	832	0.0002	904	0.0003	351
760	mitek-001	0.0000	875	0.0020	1002	0.0566	843	0.0003	1044	-	1126
761	mitek-002	0.0000	37	0.0012	868	0.0373	781	0.0000	535	-	1100
762	mobai-000	0.0000	1039	0.0114	1240	-	995	0.0003	1022	0.0012	989
763	mobai-001	0.0000	904	0.0040	1139	-	1333	0.0001	738	0.0012	988
764	mobai-002	0.0000	892	0.0008	641	0.0149	392	0.0002	845	-	1200
765	mobai-003	0.0000	413	0.0000	245	0.0001	154	0.0000	248	-	1351
766	mobbl-000	0.0116	1292	0.0720	1293	-	1148	0.0119	1350	0.0063	1071
767	mobbl-001	0.0000	1116	0.0052	1181	0.0678	874	0.0002	855	0.0005	837
768	mobbl-002	0.0000	1139	0.0029	1081	0.0633	860	0.0002	920	0.0009	940
769	mobbl-003	0.0000	1141	0.0029	1082	0.0633	859	0.0002	923	0.0009	941
770	mobbl-004	0.0000	449	0.0012	870	0.0312	719	0.0000	466	-	1431
771	mobbl-005	0.0000	803	0.0009	710	0.0217	575	0.0000	442	-	1349
772	mobipintech-000	0.0000	13	0.0000	7	0.0000	88	0.0000	53	0.0000	14
773	momovn-001	0.0015	1265	0.0010	771	0.0103	296	0.0011	1239	0.0008	928
774	momovn-002	0.0015	1264	0.0010	770	0.0103	295	0.0011	1238	-	1137
775	momovn-003	0.0000	889	0.0031	1102	0.0249	647	0.0003	1016	-	1144
776	moredian-000	0.0000	738	0.0009	703	-	1063	0.0004	1093	0.0005	790
777	morpho-000	-	1402	-	1402	-	989	-	1446	0.0000	140
778	morpho-001	-	1352	-	1349	-	1376	-	1423	0.0000	84
779	morpho-002	-	1339	-	1337	-	1305	0.0000	534	0.0004	485
780	mukh-001	0.0000	424	0.0010	800	0.0154	414	0.0001	732	0.0003	293
781	mukh-002	0.0000	372	0.0022	1032	0.0513	827	0.0002	849	0.0004	581
782	mukh-003	0.0000	774	0.0003	303	0.0060	203	0.0001	824	0.0003	401
783	mukh-004	0.0000	583	0.0013	900	0.0327	738	0.0000	556	0.0004	420
784	mukh-005	0.0000	689	0.0009	764	0.0245	635	0.0000	496	-	1291
785	multimodality-000	0.0000	368	0.0000	178	0.0000	49	0.0000	275	0.0000	197
786	multimodality-001	0.0000	452	0.0009	695	0.0259	670	0.0000	223	0.0000	218
787	mvision-001	0.0000	407	0.0000	200	-	1082	0.0000	253	0.0000	190
788	nazhiai-000	0.0000	103	0.0000	46	-	1273	0.0000	4	0.0000	34
789	ncsgroup-000	0.0000	285	0.0000	137	0.0000	10	0.0000	181	-	1265
790	ncssg-001	0.0000	500	0.0000	236	0.0000	53	0.0000	217	0.0000	214
791	ncssg-002	0.0000	450	0.0000	215	0.0000	70	0.0000	224	-	1430
792	ncssg-003	0.0000	669	0.0006	529	0.0149	394	0.0003	990	-	1226
793	ncssg-004	0.0000	589	0.0006	527	0.0149	393	0.0003	989	-	1142
794	neosystems-001	0.0000	289	0.0000	244	-	976	0.0013	1254	0.9994	1093
795	neosystems-002	0.0000	4	0.0000	1	0.0000	86	0.0000	58	0.0000	18
796	neosystems-003	0.0000	218	0.0000	102	0.0000	136	0.0000	75	0.0000	91
797	neosystems-004	0.0000	276	0.0000	129	0.0000	15	0.0000	204	0.0000	137
798	netbridgetech-001	0.0000	318	0.0000	154	-	1065	0.0000	160	0.0000	166
799	netbridgetech-002	0.0000	301	0.0000	147	-	961	0.0000	188	0.0000	125
800	neurotechnology-000	-	1370	-	1369	-	1356	-	1408	0.0000	59
801	neurotechnology-001	-	1348	-	1345	-	1270	-	1366	0.0000	39
802	neurotechnology-002	-	1446	-	1446	-	1096	-	1456	0.0000	182
803	neurotechnology-003	-	1315	-	1313	-	1219	-	1390	0.0005	793
804	neurotechnology-004	-	1381	-	1380	-	1459	0.0000	408	0.0001	258
805	neurotechnology-005	-	1376	-	1376	-	1450	0.0004	1064	0.0004	560
806	neurotechnology-006	0.0000	1024	0.0006	526	-	1150	0.0004	1065	0.0004	570
807	neurotechnology-008	0.0000	973	0.0006	523	-	1380	0.0004	1062	0.0004	539
808	neurotechnology-009	0.0000	958	0.0008	645	-	1180	0.0002	863	0.0004	498
809	neurotechnology-010	0.0000	851	0.0001	263	-	1153	0.0000	549	0.0002	269
810	neurotechnology-011	0.0000	930	0.0013	883	-	958	0.0002	859	0.0003	398
811	neurotechnology-012	0.0000	1090	0.0010	827	0.0242	630	0.0001	771	0.0004	623
812	neurotechnology-013	0.0000	323	0.0008	688	0.0185	488	0.0000	358	0.0001	249

Table 103: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]							
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
813	neurotechnology-015	0.0000	371	0.0004	364	0.0082	261	0.0000	272	0.0000	238
814	neurotechnology-016	0.0000	168	0.0003	314	0.0069	229	0.0000	114	0.0000	234
815	neurotechnology-017	0.0000	252	0.0003	340	0.0065	218	0.0000	200	0.0000	237
816	neurotechnology-018	0.0000	821	0.0003	347	0.0074	236	0.0000	438	0.0004	444
817	neurotechnology-019	0.0000	617	0.0004	382	0.0115	331	0.0000	325	-	1193
818	neurotechnology-020	0.0000	786	0.0004	384	0.0115	332	0.0000	342	-	1375
819	nextmetalabs-001	0.0000	247	0.0000	116	0.0000	132	0.0000	79	-	1232
820	nextmetalabs-002	0.0000	448	0.0000	213	0.0000	68	0.0000	225	-	1435
821	nextmetalabs-003	0.0000	618	0.0006	559	0.0184	483	0.0000	378	-	1192
822	nextmetalabs-004	0.0000	816	0.0008	693	0.0255	667	0.0000	389	-	1367
823	nhn-001	0.0000	867	0.0019	985	-	1139	0.0001	672	0.0004	755
824	nhn-002	0.0000	468	0.0004	415	0.0091	283	0.0000	483	0.0003	317
825	nhn-003	0.0000	986	0.0000	149	0.0000	4	0.0001	817	0.0004	665
826	nhn-004	0.0000	999	0.0000	228	0.0000	58	0.0001	818	0.0004	672
827	nhn-005	0.0000	581	0.0000	40	0.0000	106	0.0000	514	0.0004	484
828	noblis-000	-	1375	-	1374	-	1455	-	1404	0.0000	106
829	nodeflux-000	-	1458	-	1457	-	1144	0.0001	741	0.0002	286
830	nodeflux-001	0.0000	256	0.0016	945	-	1003	0.0001	740	0.0002	284
831	nodeflux-002	0.0000	775	0.0261	1263	-	1015	0.0008	1207	0.0005	832
832	nominder-000	0.0000	828	0.0003	293	0.0053	193	0.0000	464	0.0003	391
833	nominder-001	0.0000	708	0.0003	292	0.0053	192	0.0000	460	0.0003	386
834	nominder-002	0.0000	745	0.0006	497	0.0119	335	0.0002	918	0.0003	402
835	nominder-003	0.0000	727	0.0005	457	0.0095	289	0.0002	862	-	1283
836	nominder-004	0.0000	735	0.0007	595	0.0130	353	0.0002	960	-	1333
837	nominder-005	0.0000	243	0.0009	711	0.0206	533	0.0000	474	-	1241
838	nominder-006	0.0000	919	0.0017	957	0.0379	787	0.0003	1033	-	1296
839	nominder-007	0.0000	893	0.0006	551	0.0119	337	0.0003	1029	-	1196
840	nominder-008	0.0000	955	0.0006	557	0.0119	336	0.0003	1031	-	1447
841	notiontag-000	0.0000	291	0.0000	139	-	974	0.0000	176	0.0000	120
842	notiontag-001	0.0000	16	0.0000	8	-	1249	0.0027	1307	0.0000	16
843	notiontag-002	0.0000	369	0.0000	180	0.0000	48	0.0000	274	0.0000	196
844	nsensecorp-001	0.0000	1158	0.0024	1042	-	1379	0.0014	1263	0.0101	1078
845	nsensecorp-002	0.0000	808	0.0009	709	0.0219	583	0.0003	986	0.0011	958
846	nsensecorp-003	0.0000	321	0.0000	257	0.0002	156	0.0000	430	0.0007	924
847	nsensecorp-004	0.0406	1308	0.0035	1125	0.0181	470	0.0016	1274	0.0760	1087
848	nsensecorp-005	0.0000	343	0.0000	162	0.0000	21	0.0000	150	0.0000	157
849	ntechlab-000	-	1346	-	1344	-	1255	-	1368	0.0016	1016
850	ntechlab-001	-	1361	-	1359	-	1373	-	1425	0.0005	800
851	ntechlab-002	-	1389	-	1389	-	1429	-	1393	0.0005	801
852	ntechlab-003	-	1351	-	1350	-	1377	-	1424	0.0005	799
853	ntechlab-004	-	1394	-	1394	-	1415	-	1397	0.0004	512
854	ntechlab-005	-	1415	-	1416	-	1070	0.0000	396	0.0004	493
855	ntechlab-006	-	1373	-	1372	-	1332	0.0000	324	0.0004	434
856	ntechlab-007	0.0000	927	0.0019	987	-	963	0.0001	634	0.0004	468
857	ntechlab-008	0.0000	890	0.0005	421	-	1269	0.0001	641	0.0004	451
858	ntechlab-009	0.0000	1012	0.0009	725	-	1068	0.0001	775	0.0004	425
859	ntechlab-010	0.0000	866	0.0005	467	0.0107	314	0.0001	727	0.0004	473
860	ntechlab-011	0.0000	263	0.0003	297	0.0057	198	0.0000	563	0.0004	423
861	ntechlab-012	0.0000	346	0.0003	298	0.0057	199	0.0000	562	0.0004	424
862	ntnu-000	0.0000	3	0.0004	353	0.0075	241	0.0001	809	-	1120
863	ocrstudio-001	0.0000	1129	0.0034	1123	0.0534	838	0.0001	777	-	1165
864	omface-000	0.0000	306	0.0000	150	0.0000	32	0.0000	168	0.0000	167
865	omface-001	0.0000	212	0.0000	247	0.0000	152	0.0000	106	0.0000	112
866	omnigarde-000	0.0000	769	0.0008	633	0.0213	567	0.0000	456	0.0004	631
867	omnigarde-001	0.0000	572	0.0008	632	0.0213	569	0.0000	449	0.0004	621
868	omnigarde-002	0.0000	843	0.0008	634	0.0213	568	0.0000	467	0.0004	635
869	omnigarde-003	0.0000	426	0.0007	628	0.0212	563	0.0000	341	0.0004	524
870	omnigarde-004	0.0000	441	0.0007	629	0.0213	566	0.0000	364	-	1444

Table 104: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
871	omnigarde-005	0.0000	325	0.0007	627	0.0213	565	0.0000	357	-	1345
872	onfido-000	0.0000	1131	0.0040	1138	0.0804	897	0.0004	1056	0.0012	982
873	onfido-001	0.0000	714	0.0006	552	0.0183	474	0.0000	298	-	1267
874	onfido-002	0.0000	855	0.0006	554	0.0183	475	0.0000	305	-	1427
875	onfido-003	0.0000	852	0.0006	553	0.0183	476	0.0000	302	-	1403
876	oorudigital-000	0.0000	991	0.0002	280	0.0016	168	0.0002	965	-	1383
877	ooto-001	0.0000	73	0.0000	32	0.0000	107	0.0000	25	-	1158
878	openedge-000	0.0000	569	0.0014	911	0.0212	561	0.0003	1007	0.0004	678
879	openface-001	0.0000	1055	0.0104	1236	0.0668	868	0.0004	1077	0.0006	903
880	ovisionllc-001	0.0000	242	0.0000	114	0.0000	126	0.0000	82	-	1239
881	ovisionllc-002	0.0000	479	0.0000	227	0.0000	63	0.0000	233	-	1454
882	oz-001	0.0000	1017	0.0011	846	-	1102	0.0006	1177	0.0004	569
883	oz-002	0.0000	455	0.0003	299	-	1178	0.0000	418	0.0003	352
884	oz-003	0.0000	433	0.0002	277	0.0042	182	0.0000	340	0.0003	289
885	oz-004	0.0000	1080	0.0003	311	0.0041	181	0.0000	344	0.0002	262
886	pa-002	-	1385	-	1384	-	1444	-	1405	0.0000	113
887	pa-003	-	1365	-	1363	-	1339	-	1413	0.0000	68
888	palit-000	0.0000	516	0.0005	459	0.0134	362	0.0002	884	0.0004	503
889	palit-001	0.0000	812	0.0007	631	0.0201	526	0.0002	886	0.0004	522
890	pangiam-000	0.0000	91	0.0021	1022	0.0364	772	0.0001	599	0.0005	788
891	pangiam-001	0.0000	459	0.0005	470	0.0128	348	0.0002	977	0.0003	416
892	pangiam-002	0.0000	267	0.0003	332	0.0072	231	0.0000	437	0.0003	350
893	papago-001	0.0000	924	0.0008	652	0.0159	425	0.0002	972	0.0004	580
894	papil11-000	0.0000	824	0.0008	686	0.0192	513	0.0000	507	0.0003	415
895	papil11-001	0.0000	568	0.0005	433	0.0133	359	0.0000	404	-	1153
896	papil11-002	0.0000	616	0.0004	363	0.0115	330	0.0000	356	-	1194
897	papil11-003	0.0000	406	0.0003	341	0.0077	246	0.0000	339	-	1361
898	papil11-004	0.0000	864	0.0008	644	0.0227	597	0.0003	1015	-	1417
899	papsav1923-001	0.0000	705	0.0007	598	-	990	0.0001	702	0.0002	280
900	papsav1923-002	0.0000	549	0.0018	981	0.0268	682	0.0000	523	0.0004	637
901	papsav1923-003	0.0000	979	0.0019	999	0.0321	731	0.0000	578	0.0004	595
902	paravision-004	0.0000	1013	0.0007	618	-	1067	0.0002	898	0.0004	597
903	paravision-006	0.0000	42	0.0000	20	-	1198	0.0000	40	0.0000	4
904	paravision-008	0.0000	388	0.0010	788	0.0201	523	0.0001	683	0.0004	442
905	paravision-010	0.0000	352	0.0010	786	0.0201	522	0.0001	678	0.0004	437
906	paravision-011	0.0000	95	0.0010	785	0.0201	525	0.0001	675	0.0004	433
907	paravision-013	0.0000	395	0.0010	789	0.0201	524	0.0001	682	0.0004	441
908	paravision-015	0.0000	436	0.0009	748	0.0266	681	0.0000	581	-	1366
909	paravision-017	0.0000	608	0.0009	701	0.0220	585	0.0001	661	-	1199
910	paravision-018	0.0000	181	0.0005	476	0.0142	382	0.0000	354	-	1190
911	paravision-020	0.0000	225	0.0004	370	0.0107	312	0.0000	350	-	1228
912	pensees-001	0.0000	637	0.0000	78	-	1342	0.0000	116	0.0000	67
913	pentacomp-000	0.0001	1197	0.0010	769	0.0206	542	0.0004	1068	-	1320
914	pixelall-002	0.0000	350	0.0000	169	-	1037	0.0000	144	0.0000	146
915	pixelall-003	0.0000	251	0.0000	120	-	997	0.0000	201	0.0000	134
916	pixelall-004	0.0000	366	0.0000	177	-	1014	0.0000	153	0.0000	159
917	pixelall-005	0.0000	195	0.0000	94	-	1462	0.0000	95	0.0000	102
918	pixelall-006	0.0000	299	0.0000	145	0.0000	5	0.0000	189	0.0000	127
919	pixelall-007	0.0000	187	0.0000	87	0.0000	112	0.0000	119	0.0000	72
920	pixelall-008	0.0000	506	0.0008	672	0.0247	640	0.0000	214	0.0000	211
921	pixelall-009	0.0000	194	0.0000	91	0.0000	146	0.0000	96	0.0000	105
922	pixelall-010	0.0000	121	0.0000	58	0.0000	94	0.0000	13	0.0000	38
923	pnpsecure-000	0.0000	237	0.0030	1089	0.0390	793	0.0003	980	-	1238
924	pnpsecure-001	0.0000	461	0.0023	1034	0.0328	740	0.0002	864	-	1457
925	privid-001	0.0001	1222	0.0176	1250	0.0598	853	0.0021	1288	0.0021	1027
926	privid-002	0.0198	1299	0.0040	1142	0.0386	790	0.0014	1260	0.0011	966
927	privid-003	0.0000	969	0.0002	282	0.0030	178	0.0001	788	-	1096
928	privid-004	0.0000	734	0.0001	273	0.0025	173	0.0001	656	-	1335

Table 105: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
929	psl-000	-	1369	-	1368	-	1357	-	1409	0.0000	60
930	psl-001	-	1427	-	1427	-	1034	0.0000	139	0.0000	148
931	psl-002	-	1414	-	1415	-	1062	0.0000	164	0.0000	168
932	psl-003	0.0000	805	0.0007	570	-	1092	0.0000	553	0.0004	616
933	psl-004	0.0000	764	0.0007	568	-	1008	0.0000	548	0.0004	615
934	psl-005	0.0000	660	0.0007	562	-	1461	0.0000	453	0.0004	559
935	psl-006	0.0000	736	0.0002	278	-	1059	0.0000	165	0.0003	288
936	psl-007	0.0000	571	0.0007	572	-	1293	0.0000	557	0.0003	358
937	psl-008	0.0000	672	0.0003	305	0.0068	223	0.0000	351	0.0003	348
938	psl-009	0.0000	640	0.0004	372	0.0095	288	0.0000	294	0.0004	421
939	psl-010	0.0000	846	0.0004	373	0.0095	287	0.0000	303	0.0004	428
940	psl-011	0.0000	540	0.0003	300	0.0063	208	0.0000	289	0.0003	376
941	psl-012	0.0000	523	0.0003	309	0.0067	220	0.0000	292	0.0003	378
942	psl-013	0.0000	675	0.0003	302	0.0063	209	0.0000	537	-	1230
943	psl-014	0.0000	553	0.0003	301	0.0063	207	0.0000	291	-	1104
944	ptakuratsatu-000	0.0000	529	0.0007	612	-	1220	0.0001	602	0.0003	338
945	pxl-001	0.0000	1180	0.0044	1159	-	999	0.0005	1142	0.0022	1033
946	pyramid-000	0.0001	1211	0.0041	1149	-	1174	0.0005	1141	0.0007	921
947	pyramid-001	0.0000	561	0.0008	655	0.0231	613	0.0001	776	-	1149
948	qazbs-000	0.0000	300	0.0009	722	0.0265	677	0.0000	417	0.0004	563
949	qazsmartvisionai-000	0.0000	702	0.0003	343	0.0083	265	0.0000	335	0.0003	349
950	qazsmartvisionai-001	0.0000	691	0.0003	320	0.0064	211	0.0000	334	0.0003	325
951	qazsmartvisionai-002	0.0000	841	0.0003	322	0.0064	213	0.0000	337	-	1450
952	qazsmartvisionai-003	0.0000	570	0.0003	319	0.0064	215	0.0000	313	-	1163
953	qazsmartvisionai-004	0.0000	831	0.0003	321	0.0064	212	0.0000	338	-	1459
954	qluevision-001	0.0000	1003	0.0008	661	0.0153	404	0.0008	1206	0.0004	767
955	qnap-000	0.0000	20	0.0007	615	0.0174	454	0.0002	872	0.0002	259
956	qnap-001	0.0000	643	0.0000	254	0.0002	157	0.0000	538	0.0001	252
957	qnap-002	0.0000	1117	0.0033	1106	0.0761	894	0.0004	1063	0.0002	260
958	qnap-003	0.0000	123	0.0016	941	0.0402	803	0.0000	570	0.0001	250
959	qnap-004	0.0000	336	0.0016	942	0.0402	802	0.0000	573	0.0001	251
960	qnap-005	0.0001	1193	0.0078	1216	0.0581	847	0.0010	1231	0.0005	863
961	qnap-006	0.0000	908	0.0018	973	0.0427	811	0.0002	911	-	1247
962	quantasoft-003	0.0000	1040	0.0015	939	0.0355	768	0.0005	1132	0.0006	891
963	rankone-000	-	1367	-	1366	-	1352	-	1411	0.0003	359
964	rankone-001	-	1405	-	1406	-	973	-	1441	0.0000	247
965	rankone-002	-	1455	-	1455	-	1149	-	1451	0.0000	245
966	rankone-003	-	1380	-	1379	-	1460	-	1401	0.0000	244
967	rankone-004	-	1360	-	1358	-	1359	-	1426	0.0000	242
968	rankone-005	-	1392	-	1393	-	1399	0.0000	86	0.0000	100
969	rankone-006	-	1320	-	1318	-	1210	0.0000	38	0.0000	3
970	rankone-007	0.0000	1179	0.0048	1171	-	1445	0.0003	1013	0.0004	724
971	rankone-008	0.0000	1159	0.0049	1173	-	1437	0.0003	978	0.0004	422
972	rankone-009	0.0000	400	0.0000	195	-	1086	0.0000	259	0.0000	188
973	rankone-010	0.0000	293	0.0000	140	-	953	0.0000	194	0.0000	131
974	rankone-011	0.0000	186	0.0000	85	0.0000	114	0.0000	121	0.0000	70
975	rankone-012	0.0000	117	0.0000	56	0.0000	93	0.0000	17	0.0000	41
976	rankone-013	0.0000	322	0.0005	439	0.0126	347	0.0000	399	0.0003	300
977	rankone-014	0.0000	259	0.0005	430	0.0129	350	0.0000	400	0.0002	266
978	rankone-015	0.0000	354	0.0000	173	0.0000	26	0.0000	141	0.0000	150
979	realnetworks-000	-	1312	-	1310	-	1240	-	1388	0.0003	319
980	realnetworks-001	-	1399	-	1399	-	986	0.0004	1058	0.0003	321
981	realnetworks-002	0.0000	968	0.0003	333	-	1204	0.0004	1055	0.0003	318
982	realnetworks-003	0.0000	988	0.0003	334	-	1026	0.0004	1057	0.0003	322
983	realnetworks-004	0.0000	574	0.0003	295	0.0052	191	0.0000	288	0.0002	281
984	realnetworks-005	0.0000	695	0.0002	286	0.0045	186	0.0000	301	0.0002	285
985	realnetworks-006	0.0000	802	0.0002	288	0.0045	184	0.0000	307	0.0002	282
986	realnetworks-007	0.0000	749	0.0013	902	0.0425	810	0.0000	331	0.0004	598

Table 106: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
987	realnetworks-008	0.0000	554	0.0002	287	0.0045	185	0.0000	290	0.0002	283
988	realsense-001	0.0000	526	0.0006	548	0.0177	463	0.0000	451	-	1128
989	realsense-002	0.0000	399	0.0006	518	0.0168	448	0.0000	420	-	1387
990	rebs-000	0.0000	44	0.0005	431	0.0130	352	0.0000	424	0.0004	477
991	rebs-001	0.0000	139	0.0005	447	0.0144	384	0.0000	493	0.0004	574
992	recognito-000	0.0000	718	0.0003	307	0.0056	196	0.0000	512	0.0003	387
993	recognito-001	0.0000	527	0.0003	306	0.0056	197	0.0000	509	0.0003	380
994	regula-000	0.0000	353	0.0000	171	0.0000	27	0.0000	142	0.0000	144
995	regula-001	0.0000	235	0.0000	110	0.0000	129	0.0000	88	0.0000	94
996	regula-002	0.0000	50	0.0009	759	0.0283	690	0.0000	36	-	1097
997	regula-003	0.0000	56	0.0009	760	0.0283	689	0.0000	49	-	1112
998	remarkai-000	-	1390	-	1391	-	1407	0.0000	89	0.0000	95
999	remarkai-001	0.0000	30	0.0000	15	-	1229	0.0000	63	0.0000	26
1000	remarkai-002	0.0000	106	0.0000	49	-	1288	0.0000	2	0.0000	30
1001	remarkai-003	0.0000	510	0.0007	589	0.0187	491	0.0000	536	0.0004	478
1002	rendip-000	0.0000	990	0.0016	952	0.0293	699	0.0002	876	0.0004	763
1003	revealmedia-005	0.0000	1004	0.0007	606	0.0189	497	0.0009	1222	0.0004	771
1004	revealmedia-006	0.0000	17	0.0009	746	0.0238	625	0.0001	718	0.0004	673
1005	roc-016	0.0000	236	0.0000	256	0.0006	161	0.0000	87	0.0000	236
1006	roc-017	0.0000	486	0.0003	348	0.0073	233	0.0000	402	-	1410
1007	roc-018	0.0000	743	0.0006	540	0.0160	430	0.0000	459	0.0004	470
1008	roc-019	0.0000	700	0.0007	587	0.0187	490	0.0000	497	-	1298
1009	roc-020	0.0000	739	0.0006	521	0.0157	418	0.0000	330	-	1338
1010	roc-021	0.0000	127	0.0000	259	0.0010	162	0.0000	296	-	1197
1011	rokid-000	0.0000	24	0.0072	1208	-	1217	0.0001	691	0.0005	820
1012	rokid-001	0.0000	423	0.0013	894	-	1093	0.0000	243	0.0000	183
1013	s1-001	0.0000	1177	0.0073	1211	-	1291	0.0013	1250	0.0007	912
1014	s1-002	0.0000	873	0.0089	1227	-	1248	0.0001	761	0.0005	828
1015	s1-003	0.0000	116	0.0002	281	0.0040	180	0.0007	1191	0.0003	308
1016	s1-004	0.0000	422	0.0000	258	0.0003	158	0.0000	590	0.0001	256
1017	s1-005	0.0000	308	0.0004	377	0.0120	340	0.0001	633	0.0002	268
1018	s1-006	0.0000	67	0.0003	304	0.0074	238	0.0001	621	0.0002	271
1019	s1-007	0.0000	451	0.0006	499	0.0110	319	0.0001	637	0.0002	270
1020	s1-008	0.0000	521	0.0018	984	0.0252	659	0.0005	1160	-	1116
1021	s1-009	0.0000	178	0.0006	495	0.0110	320	0.0001	630	-	1187
1022	saffe-001	0.0000	41	0.0000	21	-	1199	0.0000	41	0.0000	5
1023	saffe-002	0.0000	29	0.0000	13	-	1235	0.0000	64	0.0000	22
1024	samsungsds-000	0.0000	994	0.0055	1189	0.0646	863	0.0038	1321	0.0005	824
1025	samsungsds-001	0.0000	128	0.0005	465	0.0146	388	0.0001	676	0.0003	396
1026	samsungsds-002	0.0000	421	0.0004	385	0.0119	338	0.0001	681	0.0003	374
1027	samtech-000	-	1459	-	1459	-	1154	-	1448	0.0042	1065
1028	samtech-001	0.0001	1207	0.0032	1105	-	1282	0.0004	1090	0.0008	926
1029	samtech-002	0.0000	1072	0.0018	976	0.0516	830	0.0002	971	0.0005	845
1030	scanalitix-001	0.0000	733	0.0005	440	0.0095	290	0.0001	816	-	1336
1031	scanovate-001	0.0208	1300	0.2388	1305	-	1080	0.0024	1294	0.0014	1004
1032	scanovate-002	0.0000	878	0.0018	974	-	1183	0.0000	585	0.0004	746
1033	scanovate-003	0.0000	885	0.0233	1261	0.3371	941	0.0006	1168	0.0004	770
1034	sdc-000	0.0000	1155	0.0035	1124	0.0678	875	0.0005	1156	0.0011	971
1035	seamfix-001	0.0000	260	0.0000	124	0.0000	19	0.0000	195	-	1288
1036	securifai-001	0.0000	498	0.0000	235	-	1132	0.0000	304	0.0000	216
1037	securifai-002	0.0000	118	0.0000	55	-	1267	0.0000	16	0.0000	42
1038	securifai-003	0.0000	26	0.0000	10	0.0000	79	0.0000	67	0.0000	28
1039	securifai-004	0.0000	104	0.0000	48	0.0000	98	0.0000	3	0.0000	33
1040	securifai-005	0.0000	239	0.0000	112	0.0000	125	0.0000	85	0.0000	99
1041	securifai-006	0.0000	86	0.0000	38	0.0000	103	0.0000	31	0.0000	57
1042	securifai-007	0.0000	363	0.0000	175	0.0000	20	0.0000	155	0.0000	163
1043	securifai-008	0.0000	197	0.0000	95	0.0000	145	0.0000	93	-	1245
1044	securitypitch-000	0.0175	1298	0.0055	1190	0.0108	315	0.0137	1352	-	1329

Table 107: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1045	sensetime-001	-	1440	-	1438	-	1115	0.0000	486	0.0004	582
1046	sensetime-002	0.0004	1242	0.0082	1220	-	1223	0.0011	1242	0.0005	857
1047	sensetime-003	0.0000	603	0.0011	842	-	1268	0.0000	286	0.0004	587
1048	sensetime-004	0.0000	517	0.0011	840	-	1252	0.0000	293	0.0004	584
1049	sensetime-005	0.0000	156	0.0004	369	0.0106	301	0.0000	479	0.0003	340
1050	sensetime-006	0.0000	58	0.0004	368	0.0106	300	0.0000	472	0.0003	336
1051	sensetime-007	0.0000	274	0.0004	371	0.0106	299	0.0000	480	0.0003	342
1052	sensetime-008	0.0000	162	0.0007	611	0.0250	649	0.0000	295	0.0003	412
1053	sensetime-010	0.0000	84	0.0005	461	0.0111	323	0.0000	522	-	1167
1054	sensetime-011	0.0000	361	0.0005	464	0.0111	322	0.0000	530	-	1326
1055	sensetime-012	0.0000	246	0.0005	463	0.0111	324	0.0000	526	-	1234
1056	serendipity-000	0.0000	729	0.0008	657	0.0202	528	0.0001	813	0.0004	614
1057	sertis-000	0.0000	317	0.0007	602	-	1066	0.0000	594	0.0004	549
1058	sertis-001	0.0000	203	0.0004	416	-	1436	0.0002	917	0.0005	869
1059	sertis-002	0.0000	12	0.0007	579	0.0152	399	0.0000	576	0.0004	532
1060	sertis-003	0.0000	525	0.0008	670	0.0183	473	0.0000	502	0.0004	607
1061	seventhsense-000	0.0000	778	0.0006	544	0.0184	479	0.0001	615	0.0004	632
1062	seventhsense-001	0.0000	623	0.0006	542	0.0184	480	0.0001	614	0.0004	622
1063	seventhsense-002	0.0000	409	0.0003	342	0.0076	243	0.0000	598	0.0004	426
1064	seventhsense-003	0.0000	668	0.0003	328	0.0083	266	0.0000	394	0.0002	278
1065	seventhsense-005	0.0000	624	0.0002	290	0.0065	217	0.0000	412	0.0003	307
1066	seventhsense-006	0.0000	544	0.0002	289	0.0065	216	0.0000	405	-	1095
1067	shaman-000	0.0000	66	0.0000	29	-	1195	0.0000	43	0.0000	10
1068	shaman-001	0.0000	410	0.0000	202	-	1084	0.0000	251	0.0000	189
1069	shu-001	0.0000	1126	0.0022	1030	-	1258	0.0010	1228	0.0006	886
1070	shu-002	0.0000	886	0.0010	804	-	1306	0.0005	1116	0.0004	721
1071	shu-003	0.0000	176	0.0007	576	-	1330	0.0001	617	0.0003	330
1072	shufti-001	0.0000	811	0.0005	473	0.0131	357	0.0001	821	-	1368
1073	siat-002	0.0000	750	0.0012	869	-	1053	0.0000	520	0.0004	600
1074	siat-004	0.0000	614	0.0011	838	-	1382	0.0000	478	0.0004	576
1075	siat-005	0.0000	477	0.0000	224	0.0000	149	0.0000	234	0.0000	222
1076	sightropic-000	0.0000	961	0.0006	535	0.0138	367	0.0002	967	-	1418
1077	sightropic-001	0.0000	907	0.0037	1133	0.0261	672	0.0007	1202	-	1246
1078	sightropic-002	0.0000	849	0.0002	275	0.0022	170	0.0001	762	-	1409
1079	sjtu-001	0.0000	882	0.0010	822	-	1322	0.0005	1131	0.0004	750
1080	sjtu-002	0.0000	891	0.0010	805	-	1264	0.0005	1115	0.0004	720
1081	sjtu-003	0.0000	129	0.0005	480	-	1374	0.0000	560	0.0003	314
1082	sjtu-004	0.0000	411	0.0000	201	0.0000	36	0.0000	250	0.0003	312
1083	sjtu-005	0.0000	1166	0.0000	117	0.0000	131	0.0000	78	0.0006	885
1084	sjtu-006	0.0000	547	0.0026	1051	0.0298	701	0.0002	851	-	1110
1085	sjtu-007	0.0000	431	0.0002	285	0.0054	194	0.0000	308	-	1373
1086	sjtu-008	0.0000	107	0.0003	312	0.0074	239	0.0000	1	-	1134
1087	sjtu-009	0.0000	360	0.0003	313	0.0074	237	0.0000	157	-	1325
1088	sktelecom-000	0.0000	536	0.0008	671	0.0190	501	0.0000	558	0.0004	639
1089	smargtech-001	0.0000	466	0.0000	221	0.0000	61	0.0000	239	-	1462
1090	smartbiometrik-001	0.0005	1244	0.0649	1287	0.2147	934	0.0017	1276	0.0008	934
1091	smartengines-000	0.0066	1282	0.0150	1247	0.1656	923	0.0022	1289	0.0013	1001
1092	smartengines-001	0.0003	1237	0.0073	1209	0.0714	888	0.0007	1194	0.0005	834
1093	smartvist-000	0.0000	87	0.0026	1060	0.0357	770	0.0002	830	0.0011	970
1094	smartvist-001	0.0000	772	0.0005	437	0.0069	228	0.0001	784	0.0003	366
1095	smilart-001	-	1359	-	1357	-	1363	-	1427	0.0273	1083
1096	smilart-002	0.0000	1149	0.0036	1130	-	1297	-	1380	0.0011	968
1097	smilart-003	0.0003	1238	0.0100	1233	-	1017	0.0014	1258	0.0013	1003
1098	sodec-000	0.0000	214	0.0000	100	0.0000	142	0.0000	104	0.0000	110
1099	softtech-000	0.0000	122	0.0006	558	0.0166	440	0.0001	763	-	1146
1100	sparktek-000	0.0000	1063	0.0016	944	0.0356	769	0.0001	671	-	1299
1101	sparktek-001	0.0000	862	0.0004	376	0.0092	284	0.0000	541	-	1425
1102	sparsh-001	0.0000	381	0.0007	563	0.0168	443	0.0000	368	0.0004	496

Table 108: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1103	sqisoft-001	0.0000	478	0.0003	329	0.0078	248	0.0000	388	0.0003	405
1104	sqisoft-002	0.0000	272	0.0003	324	0.0078	249	0.0000	401	0.0003	413
1105	sqisoft-003	0.0000	51	0.0003	330	0.0078	251	0.0000	393	0.0003	345
1106	staqu-000	0.0000	345	0.0000	167	-	1024	0.0000	148	0.0000	152
1107	starhybrid-001	0.0001	1215	0.0033	1117	-	1222	0.0009	1220	0.0023	1036
1108	stcon-000	0.0000	860	0.0017	971	0.0301	707	0.0000	463	0.0003	375
1109	stcon-001	0.0000	567	0.0017	970	0.0301	709	0.0000	447	0.0003	372
1110	stcon-002	0.0000	599	0.0011	830	0.0226	595	0.0000	422	0.0003	320
1111	stcon-003	0.0000	330	0.0005	444	0.0142	379	0.0000	432	0.0004	494
1112	stcon-004	0.0000	482	0.0005	446	0.0142	381	0.0000	439	-	1453
1113	stcon-005	0.0000	437	0.0005	445	0.0142	380	0.0000	444	-	1364
1114	stcon-006	0.0000	382	0.0007	569	0.0191	506	0.0000	469	-	1396
1115	stcon-007	0.0000	483	0.0007	571	0.0191	507	0.0000	462	-	1407
1116	stcon-008	0.0000	312	0.0007	567	0.0191	505	0.0000	457	-	1339
1117	stcon-009	0.0000	152	0.0007	566	0.0191	508	0.0000	454	-	1220
1118	stellartecs-001	0.0000	925	0.0012	858	0.0324	735	0.0003	991	-	1269
1119	stengg-000	0.0000	49	0.0013	898	0.0350	766	0.0000	489	0.0004	604
1120	sukshi-000	0.0000	113	0.0000	52	0.0000	90	0.0000	20	0.0000	45
1121	sumsub-000	0.0000	1038	0.0018	982	0.0189	496	0.0011	1241	-	1289
1122	sumsub-001	0.0000	630	0.0007	585	0.0197	520	0.0000	328	-	1207
1123	suprema-000	0.0000	931	0.0017	963	-	1069	0.0002	914	0.0004	670
1124	suprema-001	0.0000	977	0.0027	1062	0.0471	821	0.0003	995	0.0004	751
1125	suprema-002	0.0000	868	0.0010	812	0.0271	685	0.0002	865	0.0004	533
1126	suprema-003	0.0000	716	0.0008	678	0.0231	614	0.0000	383	0.0004	545
1127	suprema-004	0.0000	766	0.0014	916	0.0299	703	0.0000	381	0.0004	547
1128	suprema-005	0.0000	921	0.0014	915	0.0299	702	0.0005	1129	0.0004	666
1129	supremaid-001	0.0000	533	0.0020	1007	0.0330	745	0.0001	706	0.0004	748
1130	supremaid-002	0.0000	794	0.0020	1009	0.0330	744	0.0001	711	0.0004	754
1131	surrey-cvssp-000	0.0000	241	0.0000	113	0.0000	127	0.0000	83	0.0000	98
1132	surrey-cvssp-001	0.0173	1297	0.0007	584	0.0179	465	0.0011	1240	0.0015	1011
1133	surrey-cvssp-002	0.0000	1088	0.0006	511	0.0156	415	0.0001	669	0.0004	579
1134	surrey-cvssp-003	0.0000	396	0.0009	712	0.0165	439	0.0000	278	0.0001	253
1135	surrey-cvssp-004	0.0000	636	0.0015	935	0.0311	718	0.0004	1060	-	1174
1136	swsam-001	0.0000	401	0.0012	881	0.0263	675	0.0000	258	0.0000	187
1137	synesis-000	-	1314	-	1312	-	1227	-	1391	0.0007	914
1138	synesis-001	-	1437	-	1437	-	1113	0.0035	1315	0.0021	1026
1139	synesis-003	-	1345	-	1341	-	1260	0.0102	1346	0.1184	1091
1140	synesis-004	-	1322	-	1321	-	1212	0.0164	1353	0.0035	1056
1141	synesis-005	0.0000	566	0.0019	988	-	1319	0.0001	685	0.0005	778
1142	synesis-006	0.0000	153	0.0003	339	-	1361	0.0000	561	0.0003	297
1143	synesis-007	0.0000	588	0.0013	893	-	1281	0.0002	934	0.0004	585
1144	synology-000	0.0000	310	0.0000	152	-	1060	0.0000	166	0.0000	169
1145	synology-001	0.0000	417	0.0000	206	-	1098	0.0000	244	0.0000	178
1146	synology-002	0.0000	481	0.0000	226	-	1167	0.0000	231	0.0000	225
1147	syscake-000	0.0008	1261	0.0017	969	0.0287	697	0.0007	1188	-	1262
1148	szpu-000	0.0000	392	0.0014	926	0.0330	746	0.0001	640	-	1400
1149	sztu-000	0.0000	349	0.0000	170	-	1038	0.0000	145	0.0000	145
1150	sztu-001	0.0000	447	0.0000	214	0.0000	69	0.0000	226	0.0000	217
1151	t4isb-000	0.0000	61	0.0000	26	0.0000	76	0.0000	46	0.0000	7
1152	tahakom-001	0.0025	1269	0.0020	1016	0.0139	370	0.0020	1282	-	1385
1153	tahakom-002	0.0329	1307	0.0083	1223	0.0185	487	0.0238	1358	-	1341
1154	taiwanmobile-001	0.0000	1022	0.0023	1041	0.0331	748	0.0006	1169	-	1432
1155	taiwanmobile-002	0.0000	1010	0.0015	928	0.0238	627	0.0004	1100	-	1233
1156	tcc-000	0.0000	663	0.0699	1292	0.0478	822	0.0000	353	-	1258
1157	tcc-001	0.0000	1169	0.0004	380	0.0089	277	0.0005	1139	-	1303
1158	tcv-001	0.0000	530	0.0013	882	0.0152	402	0.0001	803	-	1131
1159	tech5-000	-	1363	-	1361	-	1346	-	1415	0.0003	394
1160	tech5-001	-	1401	-	1401	-	992	0.0004	1081	0.0003	365

Table 109: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1161	tech5-002	-	1407	-	1407	-	967	0.0001	659	0.0003	292
1162	tech5-003	0.0000	874	0.0005	448	-	1250	0.0001	658	0.0003	306
1163	tech5-004	0.0000	615	0.0008	642	-	1394	0.0003	996	0.0004	740
1164	tech5-005	0.0000	677	0.0007	623	-	1432	0.0000	452	0.0004	680
1165	tech5-007	0.0000	646	0.0014	923	0.0305	714	0.0000	411	0.0004	510
1166	tech5-008	0.0000	18	0.0007	560	0.0168	444	0.0000	349	0.0004	479
1167	tech5-009	0.0000	996	0.0030	1094	0.0542	840	0.0002	908	-	1445
1168	techainer-001	0.0000	1125	0.0004	412	0.0085	270	0.0005	1127	0.0004	668
1169	techainer-002	0.0000	1122	0.0004	407	0.0085	272	0.0005	1125	-	1240
1170	techainer-003	0.0000	895	0.0031	1103	0.0249	648	0.0003	1017	-	1206
1171	techbiometricid-001	0.0000	201	0.0000	97	0.0000	139	0.0000	110	-	1256
1172	techbiometricid-002	0.0000	255	0.0000	122	0.0000	16	0.0000	197	-	1294
1173	techsign-000	0.0007	1254	0.0334	1272	0.2093	928	0.0020	1283	0.0011	962
1174	techsign-001	0.0000	610	0.0008	692	0.0253	660	0.0002	890	0.0004	610
1175	telexper-000	0.0000	879	0.0006	536	0.0147	389	0.0003	1032	-	1103
1176	tevian-000	-	1377	-	1375	-	1451	-	1403	0.0012	994
1177	tevian-001	-	1419	-	1420	-	1050	-	1436	0.0012	992
1178	tevian-002	-	1432	-	1433	-	1122	-	1460	0.0005	865
1179	tevian-003	-	1439	-	1440	-	1116	0.0003	998	0.0005	866
1180	tevian-004	0.0000	1037	0.0024	1043	-	1401	0.0002	860	0.0005	852
1181	tevian-005	0.0001	1219	0.0041	1144	-	1351	0.0006	1171	0.0006	905
1182	tevian-006	0.0000	10	0.0012	860	-	1241	0.0003	1009	0.0005	829
1183	tevian-007	0.0000	815	0.0015	940	0.0429	812	0.0002	909	0.0004	653
1184	tevian-008	0.0000	673	0.0006	492	0.0109	317	0.0000	427	0.0003	331
1185	tiger-001	-	1328	-	1326	-	1194	-	1384	0.0006	892
1186	tiger-002	0.0000	509	0.0010	792	-	1244	0.0001	745	0.0004	555
1187	tiger-003	0.0000	713	-	1411	-	978	0.0001	749	0.0004	561
1188	tiger-004	0.0000	1173	0.0022	1028	-	1095	0.0001	757	0.0004	738
1189	tiger-005	0.0000	830	0.0009	750	0.0194	516	0.0001	673	0.0004	572
1190	tiger-006	0.0000	960	0.0011	841	0.0396	796	0.0001	796	0.0004	776
1191	tiger-007	0.0000	947	0.0051	1178	0.0201	527	0.0002	882	-	1390
1192	tinkoff-001	0.0000	926	0.0008	669	0.0171	451	0.0001	768	0.0004	562
1193	tngd-001	0.0000	619	0.0009	694	0.0218	579	0.0000	528	-	1214
1194	tngd-002	0.0000	165	0.0000	77	0.0000	115	0.0000	117	-	1176
1195	tnitech-000	0.0000	528	0.0003	326	0.0060	201	0.0005	1158	0.0003	323
1196	tongyi-004	-	1356	-	1356	-	1389	-	1420	0.0006	896
1197	tongyi-005	0.0000	202	0.0000	96	-	1441	0.0000	109	0.0000	115
1198	tongyitrans-001	-	1420	-	1421	-	1048	-	1435	0.0040	1060
1199	tongyitrans-002	-	1451	-	1450	-	1172	-	1455	0.0040	1061
1200	tongyitrans-003	-	1358	-	1354	-	1386	-	1418	0.0006	897
1201	toppanidgate-000	0.0000	560	0.0008	662	0.0232	615	0.0004	1053	0.0004	620
1202	toppansecurity-001	0.0000	928	0.0020	1006	0.0161	434	0.0001	665	-	1282
1203	toshiba-000	-	1353	-	1351	-	1375	-	1422	0.0000	83
1204	toshiba-001	-	1397	-	1398	-	1002	-	1443	0.0000	133
1205	toshiba-002	0.0000	454	0.0000	216	-	1175	0.0000	222	0.0000	219
1206	toshiba-003	0.0000	224	0.0001	264	-	1416	0.0001	746	0.0001	255
1207	toshiba-004	0.0000	183	0.0000	84	0.0000	113	0.0000	122	0.0000	71
1208	toshiba-005	0.0000	664	0.0004	391	0.0050	189	0.0001	780	0.0003	332
1209	toshiba-006	0.0000	508	0.0004	390	0.0050	188	0.0001	778	0.0003	329
1210	toshiba-007	0.0000	917	0.0005	442	0.0082	262	0.0002	956	0.0004	612
1211	toshiba-008	0.0000	918	0.0005	443	0.0082	263	0.0002	955	0.0004	611
1212	toshiba-009	0.0000	788	0.0004	414	0.0070	230	0.0002	943	-	1380
1213	toshiba-010	0.0000	923	0.0004	378	0.0064	210	0.0002	933	-	1301
1214	toshiba-011	0.0000	626	0.0006	501	0.0137	366	0.0002	928	-	1221
1215	toshiba-012	0.0000	573	0.0004	418	0.0120	341	0.0000	584	-	1166
1216	toss-000	0.0000	21	0.0009	719	0.0209	549	0.0001	692	-	1130
1217	touchlessid-000	0.0042	1274	0.0133	1243	0.2009	927	0.0018	1279	0.0032	1053
1218	touchlessid-001	0.0000	313	0.0036	1131	0.0923	907	0.0000	163	0.0000	164

Table 110: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1219	touchlessid-002	0.0000	470	0.0043	1157	0.1087	912	0.0000	238	0.0000	229
1220	touchlessid-003	0.0000	632	0.0010	796	0.0287	696	0.0002	834	0.0004	734
1221	touchlessid-004	0.0000	1170	0.0013	904	0.0398	800	0.0002	950	-	1136
1222	transmitsecurity-001	0.0000	833	0.0009	717	0.0226	596	0.0000	518	-	1458
1223	trueface-000	0.0000	785	0.0000	183	-	1129	0.0000	270	0.0000	194
1224	trueface-001	0.0000	939	0.0038	1137	-	1027	0.0007	1201	0.0005	830
1225	trueface-002	0.0000	920	0.0046	1166	-	983	0.0003	983	0.0005	853
1226	trueface-003	0.0000	941	0.0046	1167	0.0397	798	0.0003	982	0.0005	854
1227	trueidvng-001	0.0000	932	0.0020	1015	0.0385	789	0.0002	895	0.0005	823
1228	truststamp-001	0.0000	577	0.0010	784	0.0189	500	0.0000	392	0.0003	373
1229	truststamp-002	0.0000	773	0.0010	787	0.0189	498	0.0000	395	-	1319
1230	truststamp-003	0.0000	798	0.0010	791	0.0189	499	0.0000	403	-	1388
1231	tsight-000	0.0000	546	0.0003	318	0.0064	214	0.0000	315	-	1108
1232	ttigroup-000	0.0000	282	0.0000	133	0.0000	7	0.0000	184	-	1275
1233	tuputech-000	0.0003	1240	0.0116	1241	-	1317	-	1370	0.0081	1074
1234	turingtechvip-001	0.0001	1200	0.0007	608	0.0061	205	0.0007	1192	0.0006	882
1235	turingtechvip-002	0.0001	1199	0.0017	967	0.0097	291	0.0007	1193	0.0006	881
1236	turkcell-000	0.0110	1290	0.0234	1262	0.0350	767	0.0103	1347	0.0306	1085
1237	turkcell-001	0.0000	88	0.0002	284	0.0034	179	0.0001	686	0.0003	291
1238	turkcell-002	0.0000	38	0.0002	276	0.0026	174	0.0001	613	-	1099
1239	twface-000	0.0000	130	0.0000	63	0.0000	119	0.0000	133	0.0000	82
1240	twface-001	0.0000	277	0.0000	130	0.0000	13	0.0000	203	0.0000	139
1241	ulsee-001	0.0000	5	0.0000	2	-	1246	0.0000	57	0.0000	17
1242	ultinous-000	-	1368	-	1367	-	1358	-	1410	0.0003	347
1243	ultinous-001	-	1331	-	1330	-	1310	-	1375	0.0003	346
1244	uluface-002	0.0000	253	0.0000	121	-	994	0.0000	199	0.0000	135
1245	uluface-003	0.0000	48	0.0001	271	-	1209	0.0002	837	0.0002	273
1246	unicc-001	0.0000	131	0.0012	857	0.0230	611	0.0004	1106	0.0004	558
1247	unicc-002	0.0316	1306	0.0082	1219	0.0188	492	0.0226	1357	0.0936	1090
1248	unicc-003	0.0007	1257	0.0022	1025	0.0094	286	0.0026	1304	0.0029	1046
1249	unicc-004	0.0000	1167	0.0012	856	0.0066	219	0.0012	1248	-	1307
1250	unissey-001	0.0000	54	0.0000	24	0.0000	72	0.0000	50	0.0000	12
1251	unissey-002	0.0000	233	0.0000	108	0.0000	128	0.0000	91	0.0000	97
1252	unissey-003	0.0000	472	0.0008	639	0.0191	502	0.0001	680	0.0004	528
1253	unissey-004	0.0000	374	0.0009	727	0.0240	628	0.0001	728	0.0004	601
1254	unissey-005	0.0000	484	0.0009	728	0.0240	629	0.0001	726	-	1412
1255	unissey-006	0.0000	726	0.0220	1258	0.0583	848	0.0003	1020	-	1279
1256	unissey-007	0.0000	825	0.0101	1234	0.0376	785	0.0004	1052	-	1434
1257	universeai-000	0.0000	1051	0.0030	1092	0.0321	732	0.0000	587	-	1405
1258	upc-001	0.0000	1075	0.0003	317	-	1083	0.0003	987	0.0003	390
1259	urielsoft-000	0.0000	167	0.0023	1039	0.0371	780	0.0000	568	-	1168
1260	useb-000	0.0000	602	0.0009	736	0.0228	604	0.0001	786	0.0004	505
1261	useb-001	0.0000	662	0.0009	737	0.0228	606	0.0001	791	0.0004	514
1262	useb-002	0.0000	670	0.0009	738	0.0228	605	0.0001	790	0.0004	511
1263	useb-003	0.0000	531	0.0009	735	0.0228	603	0.0001	789	-	1122
1264	useb-004	0.0000	724	0.0004	410	0.0112	325	0.0002	902	-	1276
1265	uxlabs-001	0.0000	248	0.0000	118	0.0000	130	0.0000	77	0.0000	93
1266	uxlabs-003	0.0000	57	0.0000	25	0.0000	75	0.0000	48	0.0000	8
1267	uzinfocom-000	0.0000	142	0.0005	434	0.0132	358	0.0001	735	-	1195
1268	uzinfocom-001	0.0000	420	0.0002	283	0.0044	183	0.0001	657	-	1354
1269	vaidio-001	0.0000	634	0.0006	528	0.0157	417	0.0043	1325	-	1210
1270	vcog-002	-	1406	-	1405	-	971	-	1440	0.0019	1023
1271	vcortex-001	0.0000	1106	0.2295	1304	0.0802	896	0.0017	1275	0.0006	880
1272	vd-000	-	1442	-	1442	-	1111	-	1461	0.0000	202
1273	vd-001	0.0000	1137	0.0030	1095	-	1440	0.0004	1084	0.0009	937
1274	vd-002	0.0000	141	0.0000	69	1.0000	1388	0.0000	127	0.0000	79
1275	vd-003	0.0001	1201	0.0041	1145	0.0676	872	0.0030	1312	0.0029	1047
1276	veertec-001	0.0006	1252	0.0140	1246	0.1098	913	0.0019	1281	-	1208

Table 111: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1277	veridas-001	-	1386	-	1385	-	1422	0.0001	747	0.0005	809
1278	veridas-002	0.0000	1053	0.0026	1059	-	1207	0.0001	743	0.0005	806
1279	veridas-003	0.0000	1061	0.0026	1052	-	1004	0.0001	751	0.0005	813
1280	veridas-004	0.0000	1066	0.0026	1058	-	1041	0.0001	748	0.0005	817
1281	veridas-005	0.0000	1065	0.0026	1055	-	948	0.0001	750	0.0005	812
1282	veridas-006	0.0000	1070	0.0026	1056	0.0595	851	0.0001	754	0.0005	818
1283	veridas-007	0.0000	1064	0.0026	1054	0.0595	850	0.0001	752	0.0005	815
1284	veridas-008	0.0000	1062	0.0026	1053	0.0595	849	0.0001	753	0.0005	816
1285	veridas-009	0.0000	1054	0.0026	1057	0.0597	852	0.0001	674	0.0005	821
1286	veridas-010	0.0000	916	0.0011	844	0.0212	562	0.0004	1097	-	1237
1287	veridium-000	0.0061	1281	0.5956	1307	0.2889	939	0.0050	1332	0.0009	936
1288	veridium-001	0.0001	1189	0.0087	1225	0.1615	920	0.0014	1261	0.0006	873
1289	veridium-002	0.0000	1178	0.0083	1224	0.1591	918	0.0010	1225	0.0005	848
1290	veridium-003	0.0001	1190	0.0087	1226	0.1615	919	0.0014	1262	-	1250
1291	veridium-004	0.0001	1191	0.0028	1074	0.0832	901	0.0005	1145	-	1177
1292	verigram-000	0.0000	981	0.0068	1203	0.0822	900	0.0003	1048	0.0005	840
1293	verigram-001	0.0000	957	0.0003	325	0.0060	200	0.0002	929	0.0003	404
1294	verigram-002	0.0000	1105	0.0052	1184	0.0378	786	0.0010	1227	0.0006	900
1295	verigram-003	0.0000	1153	0.0025	1048	0.0288	698	0.0009	1224	0.0008	930
1296	verigram-004	0.0000	953	0.0001	268	0.0016	167	0.0001	785	-	1358
1297	verihubs-inteligensia-000	0.0000	761	0.0029	1076	0.0669	869	0.0001	632	0.0004	682
1298	verihubs-inteligensia-001	0.0000	832	0.0029	1078	0.0669	871	0.0001	638	0.0004	685
1299	verihubs-inteligensia-002	0.0000	799	0.0029	1077	0.0669	870	0.0001	639	0.0004	683
1300	verijelas-000	0.0000	522	0.0023	1036	0.0375	784	0.0004	1110	0.0006	887
1301	via-000	0.0000	298	0.0000	144	-	950	0.0000	190	0.0000	132
1302	via-001	0.0000	234	0.0000	109	-	1409	0.0000	90	0.0000	96
1303	via-004	0.0000	229	0.0000	107	0.0000	137	0.0000	70	0.0000	88
1304	via-005	0.0000	326	0.0000	156	0.0000	29	0.0000	175	0.0000	175
1305	via-006	0.0000	213	0.0000	99	0.0000	141	0.0000	105	-	1253
1306	viante-000	0.0000	936	0.0008	675	0.0168	446	0.0001	723	0.0004	774
1307	viante-001	0.0000	910	0.0008	673	0.0168	447	0.0001	721	-	1244
1308	viante-002	0.0000	524	0.0007	620	0.0154	407	0.0001	720	-	1129
1309	viante-003	0.0000	518	0.0007	619	0.0154	408	0.0001	719	-	1114
1310	viante-004	0.0000	755	0.0007	622	0.0154	406	0.0001	722	-	1315
1311	videmo-000	0.0000	896	0.0019	993	-	1343	0.0003	1019	0.0012	991
1312	videmo-001	0.0000	1041	0.0170	1249	0.0332	750	0.0010	1235	0.0011	977
1313	videmo-002	0.0000	487	0.0006	530	0.0189	493	0.0001	703	0.0004	497
1314	videmo-003	0.0000	504	0.0006	531	0.0189	494	0.0001	704	-	1421
1315	videonetics-001	0.0004	1241	0.0309	1269	-	1040	0.0015	1268	0.0010	951
1316	videonetics-002	0.0000	998	0.0459	1282	-	1179	0.0006	1175	0.0005	862
1317	viettelai-000	0.0000	389	0.0000	190	0.0000	42	0.0000	282	0.0000	208
1318	viettelai-001	0.0000	900	0.0010	776	0.0161	433	0.0000	572	0.0004	489
1319	viettelai-002	0.0000	985	0.0019	990	0.0334	754	0.0002	940	-	1277
1320	viettelai-003	0.0000	937	0.0011	847	0.0218	580	0.0001	618	-	1346
1321	viettelai-004	0.0000	710	0.0005	456	0.0139	371	0.0000	495	-	1274
1322	viettelai-005	0.0000	814	0.0005	449	0.0137	365	0.0000	485	-	1374
1323	viettelhightech-000	0.0000	1097	0.0019	996	0.0368	777	0.0007	1197	0.0005	856
1324	viettelhightech-001	0.0000	754	0.0014	914	0.0434	813	0.0001	815	-	1311
1325	viettelhightech-002	0.0000	647	0.0014	913	0.0434	814	0.0001	814	-	1189
1326	vigilantsolutions-000	-	1395	-	1395	-	1410	-	1396	0.0007	913
1327	vigilantsolutions-001	-	1318	-	1315	-	1206	-	1382	0.0004	686
1328	vigilantsolutions-002	-	1326	-	1324	-	1196	-	1385	0.0004	687
1329	vigilantsolutions-003	-	1460	-	1460	-	1135	-	1453	0.0004	475
1330	vigilantsolutions-004	-	1378	-	1378	-	1453	-	1402	0.0004	462
1331	vigilantsolutions-005	-	1412	-	1413	-	962	0.0001	648	0.0004	467
1332	vigilantsolutions-006	0.0000	1045	0.0028	1071	-	1137	0.0001	651	0.0004	474
1333	vigilantsolutions-007	0.0000	1036	0.0028	1069	-	1397	0.0001	644	0.0004	461
1334	vigilantsolutions-008	0.0000	1044	0.0028	1070	-	1156	0.0001	650	0.0004	471

Table 112: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1335	vigilantsolutions-009	0.0000	1032	0.0028	1067	-	1344	0.0001	646	0.0004	456
1336	vigilantsolutions-010	0.0000	1034	0.0028	1068	0.0609	855	0.0001	643	0.0004	460
1337	vigilantsolutions-011	0.0000	1027	0.0028	1066	0.0609	854	0.0001	642	0.0004	448
1338	vinai-000	0.0000	384	0.0000	188	-	1108	0.0000	284	0.0000	203
1339	vinbigdata-001	0.0000	403	0.0000	198	0.0000	39	0.0000	256	0.0000	186
1340	vinbigdata-002	0.0000	138	0.0015	934	0.0250	650	0.0000	510	0.0004	707
1341	vinbigdata-003	0.0000	594	0.0007	578	0.0151	398	0.0000	592	0.0004	480
1342	vinbigdata-004	0.0000	789	0.0005	455	0.0141	373	0.0000	391	-	1395
1343	vinbigdata-005	0.0000	557	0.0005	452	0.0141	374	0.0000	371	-	1157
1344	vinbigdata-006	0.0000	688	0.0005	453	0.0141	375	0.0000	376	-	1235
1345	vinbigdata-007	0.0000	776	0.0005	454	0.0141	372	0.0000	380	-	1318
1346	vion-000	0.0050	1276	0.0392	1279	-	1033	0.0130	1351	0.0078	1073
1347	visage-000	0.0000	1083	0.0054	1187	-	1251	0.0009	1217	0.0006	883
1348	visica-001	0.0000	1136	0.0068	1205	0.0692	879	0.0010	1226	-	1334
1349	visica-002	0.0000	950	0.0080	1217	0.1006	910	0.0004	1089	-	1389
1350	visica-003	0.0000	943	0.0022	1029	0.0419	807	0.0004	1088	-	1381
1351	visionbox-000	0.0000	1154	0.0033	1114	-	1263	0.0005	1154	0.0011	969
1352	visionbox-001	0.0000	1156	0.0033	1115	-	1049	0.0005	1155	0.0011	972
1353	visionbox-002	0.0000	200	0.0017	958	0.0270	684	0.0000	516	0.0004	759
1354	visionbox-003	0.0000	976	0.0010	773	0.0209	551	0.0001	707	0.0004	642
1355	visionbox-004	0.0000	909	0.0013	884	0.0248	644	0.0001	779	0.0004	723
1356	visionbox-005	0.0000	1029	0.0037	1135	0.0301	708	0.0005	1114	-	1161
1357	visionbox-006	0.0000	1091	0.0040	1141	0.0331	747	0.0007	1187	-	1236
1358	visionbox-007	0.0000	1082	0.0041	1143	0.0324	734	0.0006	1182	-	1452
1359	visionbox-008	0.0000	1030	0.0035	1127	0.0280	687	0.0001	819	-	1143
1360	visionlabs-001	-	1435	-	1435	-	1125	-	1458	0.0014	1008
1361	visionlabs-002	-	1366	-	1365	-	1341	-	1412	0.0026	1040
1362	visionlabs-003	-	1332	-	1329	-	1308	-	1374	0.0025	1039
1363	visionlabs-004	-	1449	-	1449	-	1078	-	1457	0.0008	931
1364	visionlabs-005	-	1341	-	1339	-	1286	0.0002	975	0.0023	1034
1365	visionlabs-006	-	1383	-	1382	-	1439	0.0003	1027	0.0005	810
1366	visionlabs-007	0.0000	934	0.0014	920	-	1044	0.0003	1023	0.0005	804
1367	visionlabs-008	0.0000	758	0.0015	932	-	1022	0.0002	969	0.0004	762
1368	visionlabs-009	0.0000	665	0.0010	777	-	1443	0.0001	609	0.0004	627
1369	visionlabs-010	0.0000	1019	0.0009	716	-	1114	0.0001	783	0.0004	633
1370	visionlabs-011	0.0000	172	0.0006	522	0.0156	416	0.0001	654	0.0004	488
1371	visteam-000	0.0000	1151	0.0031	1101	-	1181	0.0005	1147	0.0011	959
1372	visteam-001	0.0000	1021	0.0014	917	-	1079	0.0002	880	0.0004	651
1373	visteam-002	0.0000	1014	0.0014	918	0.0285	694	0.0002	868	0.0004	650
1374	visteam-003	0.0000	575	0.0010	808	0.0225	592	0.0001	616	0.0004	536
1375	visteam-004	0.0000	515	0.0010	807	0.0225	589	0.0001	725	0.0004	531
1376	visteam-005	0.0000	592	0.0010	806	0.0224	588	0.0001	652	0.0004	534
1377	visteam-006	0.0000	641	0.0008	666	0.0209	547	0.0000	355	0.0004	457
1378	visteam-007	0.0000	558	0.0008	664	0.0209	545	0.0000	347	0.0004	454
1379	visteam-008	0.0000	562	0.0008	665	0.0209	546	0.0000	346	0.0004	452
1380	visteam-009	0.0000	725	0.0010	823	0.0328	741	0.0000	359	-	1280
1381	visteam-010	0.0000	806	0.0010	824	0.0328	742	0.0000	367	-	1353
1382	vivotek-001	0.0000	460	0.0006	517	0.0161	432	0.0000	242	-	1456
1383	vivotek-002	0.0000	265	0.0006	516	0.0161	431	0.0000	208	-	1306
1384	vixvizion-000	0.0002	1228	0.0028	1072	-	985	0.0010	1234	0.0012	990
1385	vixvizion-001	0.0000	877	0.0030	1093	-	1188	0.0001	733	0.0004	659
1386	vixvizion-002	0.0000	995	0.0018	972	-	1077	0.0000	533	0.0004	652
1387	vixvizion-003	0.0000	463	0.0000	218	-	1163	0.0000	241	0.0000	228
1388	vixvizion-004	0.0000	405	0.0000	196	0.0000	38	0.0000	255	0.0000	185
1389	vixvizion-005	0.0000	294	0.0000	142	0.0000	2	0.0000	193	0.0000	130
1390	vixvizion-006	0.0000	383	0.0000	189	0.0000	45	0.0000	285	0.0000	204
1391	vixvizion-007	0.0000	177	0.0000	83	0.0000	111	0.0000	124	0.0000	73
1392	vnis-000	0.0007	1259	0.0695	1291	0.0756	893	0.0038	1322	0.0009	942

Table 113: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1393	vnpay-000	0.0000	1026	0.0657	1290	0.0693	880	0.0037	1320	0.0006	875
1394	vnpay-001	0.0000	790	0.0052	1183	0.0497	824	0.0001	737	-	1394
1395	vnpt-001	0.0652	1310	0.2829	1306	-	1402	0.2116	1363	0.1598	1092
1396	vnpt-002	0.0000	658	0.0002	279	0.0024	172	0.0003	1026	0.0003	298
1397	vnpt-003	0.0000	65	0.0004	362	0.0105	298	0.0002	831	0.0004	417
1398	vnpt-004	0.0000	704	0.0006	500	0.0160	429	0.0002	857	0.0004	564
1399	vnpt-005	0.0000	68	0.0006	483	0.0154	411	0.0002	887	0.0004	535
1400	vnpt-006	0.0000	143	0.0006	484	0.0154	412	0.0002	889	0.0004	538
1401	vnpt-007	0.0000	261	0.0006	486	0.0154	409	0.0002	891	-	1287
1402	vnpt-008	0.0000	207	0.0006	485	0.0154	413	0.0002	888	-	1260
1403	vnpt-009	0.0000	456	0.0006	487	0.0154	410	0.0002	892	-	1438
1404	vocalize-001	0.0216	1301	0.0193	1253	0.1691	924	0.0068	1339	-	1270
1405	vocord-001	-	1357	-	1353	-	1387	-	1419	0.0038	1059
1406	vocord-002	-	1337	-	1335	-	1296	-	1379	0.0037	1058
1407	vocord-003	-	1336	-	1333	-	1323	-	1372	0.0776	1088
1408	vocord-004	-	1408	-	1408	-	966	0.0005	1146	0.0003	310
1409	vocord-005	-	1429	-	1430	-	1016	0.0013	1252	0.0003	400
1410	vocord-006	-	1453	-	1453	-	1160	0.0003	1011	0.0003	369
1411	vocord-007	0.0000	888	0.0535	1283	-	1285	0.0001	800	0.0004	419
1412	vocord-008	0.0000	719	0.0015	938	-	944	0.0003	1034	0.0001	254
1413	vocord-009	0.0000	748	0.0006	520	-	1042	0.0001	810	0.0003	301
1414	vocord-010	0.0000	898	0.0005	469	0.0141	377	0.0002	893	0.0003	393
1415	vogic-000	0.0000	538	0.0001	267	0.0015	166	0.0002	854	-	1101
1416	vsoft-001	0.0069	1284	0.8292	1308	0.8200	943	0.0173	1354	-	1316
1417	vsoft-002	0.0026	1270	0.0029	1085	0.0613	856	0.0059	1338	-	1340
1418	vsoft-003	0.0000	1130	0.0029	1083	0.0613	857	0.0005	1118	-	1135
1419	vsoft-004	0.0000	1133	0.0029	1084	0.0613	858	0.0005	1119	-	1223
1420	vts-000	0.0000	965	0.0011	835	-	1247	0.0001	811	0.0004	747
1421	vts-001	0.0000	445	0.0003	310	0.0073	232	0.0000	387	0.0003	294
1422	vts-002	0.0000	494	0.0008	691	0.0150	395	0.0000	385	-	1402
1423	wicket-000	0.0000	639	0.0009	702	0.0260	671	0.0000	409	0.0004	557
1424	winsense-000	0.0000	96	0.0000	43	-	1280	0.0000	10	0.0000	31
1425	winsense-001	0.0000	144	0.0000	70	-	1393	0.0000	126	0.0000	78
1426	winsense-002	0.0000	245	0.0000	115	-	1413	0.0000	80	0.0000	92
1427	wiseai-001	0.0001	1187	0.0137	1245	0.0768	895	0.0018	1278	0.0018	1018
1428	wuhantianyu-001	0.0000	205	0.0007	586	0.0159	422	0.0001	610	0.0004	628
1429	x-laboratory-000	0.0247	1303	0.0000	71	-	1364	0.0005	1153	0.0002	276
1430	x-laboratory-001	0.0000	784	0.0012	861	-	1119	0.0001	764	0.0004	727
1431	xforwardai-000	0.0000	8	0.0000	4	-	1238	0.0000	55	0.0000	21
1432	xforwardai-001	0.0000	717	0.0007	609	-	951	0.0003	1003	0.0004	710
1433	xforwardai-002	0.0000	532	0.0007	607	-	1237	0.0003	1002	0.0004	705
1434	xm-000	0.0000	428	0.0007	577	-	1076	0.0001	619	0.0003	333
1435	yisheng-000	-	1384	-	1383	-	1446	-	1406	0.0006	876
1436	yisheng-001	-	1416	-	1417	-	1046	-	1438	0.0006	877
1437	yisheng-002	-	1436	-	1436	-	1106	-	1462	0.0006	878
1438	yisheng-004	0.0002	1231	-	1388	-	1419	0.0013	1251	0.0006	898
1439	yitu-000	-	1388	-	1386	-	1421	-	1395	0.0013	999
1440	yitu-001	-	1333	-	1332	-	1311	0.0009	1219	0.0006	895
1441	yitu-002	-	1344	-	1342	-	1262	-	1369	0.0000	44
1442	yitu-003	0.0000	136	0.0000	66	-	1395	0.0009	1218	0.0000	75
1443	yoonik-000	0.0000	666	0.0019	994	-	1442	0.0001	693	0.0004	725
1444	yoonik-001	0.0000	311	0.0014	924	-	1058	0.0001	805	0.0004	713
1445	yoonik-002	0.0000	948	0.0010	799	0.0284	693	0.0003	984	0.0006	871
1446	yoonik-003	0.0000	942	0.0009	745	0.0214	571	0.0002	853	0.0004	671
1447	yoonik-004	0.0000	232	0.0012	851	0.0251	654	0.0001	653	0.0004	492
1448	yoti-001	0.0000	952	0.0014	925	0.0418	806	0.0008	1215	0.0006	884
1449	yoti-002	0.0000	394	0.0007	574	0.0171	452	0.0001	717	-	1398
1450	ytu-000	0.0000	813	0.0010	821	-	1073	0.0002	949	0.0004	716

Table 114: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1451	yuan-000	0.0000	102	0.0000	45	-	1274	0.0000	5	0.0000	35
1452	yuan-001	0.0000	1058	0.0010	814	-	1449	0.0005	1150	0.0005	827
1453	yuan-002	0.0000	1059	0.0010	815	0.0284	692	0.0005	1149	0.0005	826
1454	yuan-003	0.0000	1052	0.0010	813	0.0284	691	0.0005	1148	0.0005	825
1455	yuan-004	0.0000	288	0.0000	134	0.0000	9	0.0000	178	0.0000	118
1456	yuan-005	0.0000	600	0.0005	462	0.0134	363	0.0002	883	0.0004	506
1457	yuan-006	0.0000	1087	0.0014	927	0.0369	778	0.0004	1072	0.0005	838
1458	yuan-007	0.0000	1093	0.0013	888	0.0336	756	0.0004	1073	0.0005	841
1459	yuan-008	0.0000	1089	0.0013	887	0.0336	759	0.0004	1071	0.0005	839
1460	yuan-009	0.0000	1098	0.0013	890	0.0336	758	0.0004	1074	-	1413
1461	yuan-010	0.0000	1096	0.0013	889	0.0336	757	0.0004	1075	-	1357
1462	yuanli-001	0.0000	340	0.0016	954	0.0318	722	0.0001	662	-	1312

Table 115: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.4 Recognition accuracy

Core algorithm accuracy is stated via:

- ▷ **Cooperative subjects**
  - The summary table of Figure 89;
  - The visa image DETs of Figure 222;
  - The mugshot DETs of Figure 291;
  - The mugshot ageing profiles of Figure 799;
  - The human-difficult pairs of Figure 112
- ▷ **Non-cooperative subjects**
  - The photojournalism DET of Figure 327

Figure 624 shows dependence of false match rate on algorithm score threshold. This allows a deployer to set a threshold to target a particular false match rate appropriate to the security objectives of the application.

Figure 623 likewise shows FMR(T) but for mugshots, and specially four subsets of the population.

Note that in both the mugshot and visa sets false match rates vary with the ethnicity, age, and sex, of the enrollee and impostor. For example figure 395 summarizes FMR for impostors paired from four groups black females, black males, white females, white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

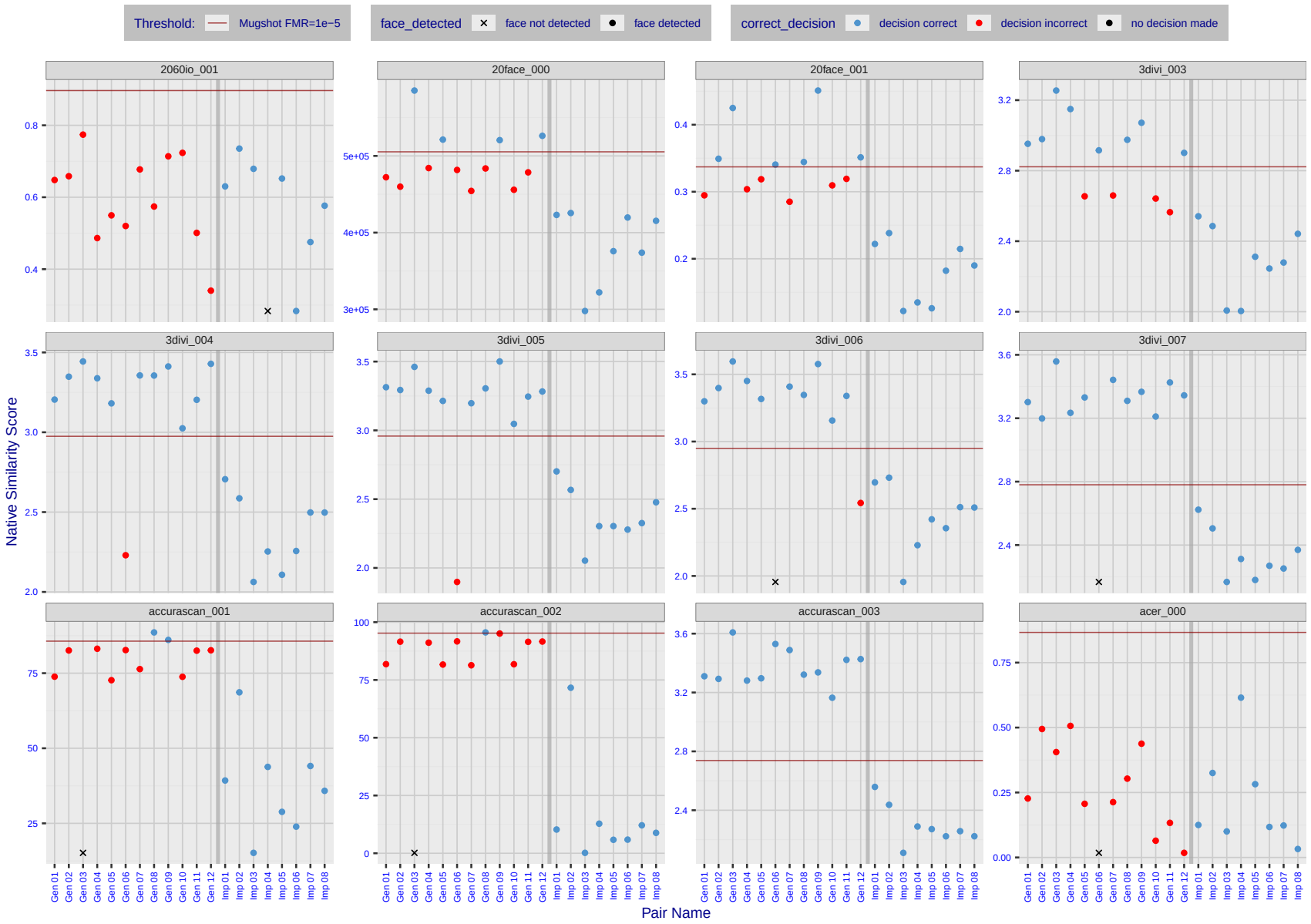


Figure 4: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

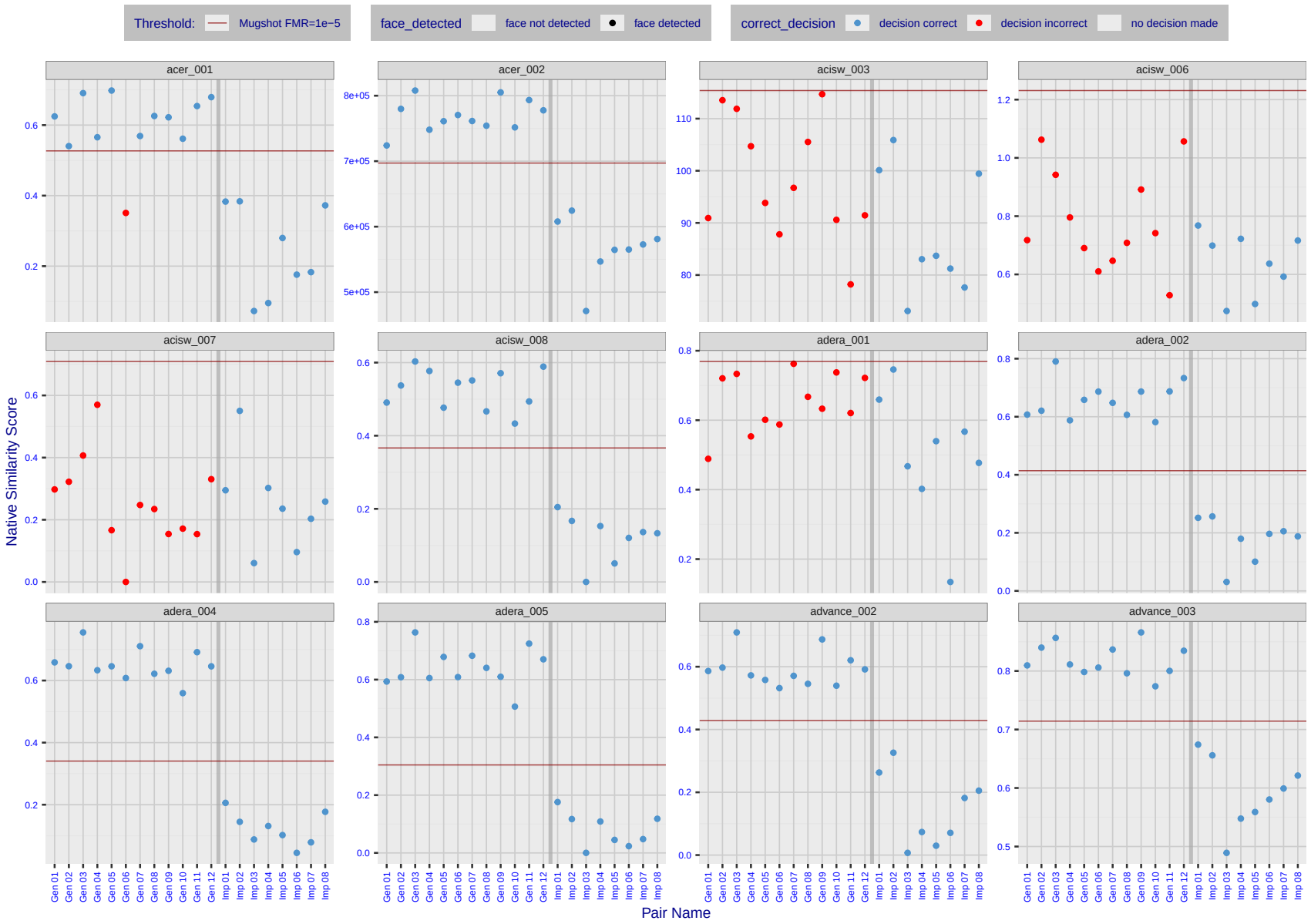


Figure 5: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

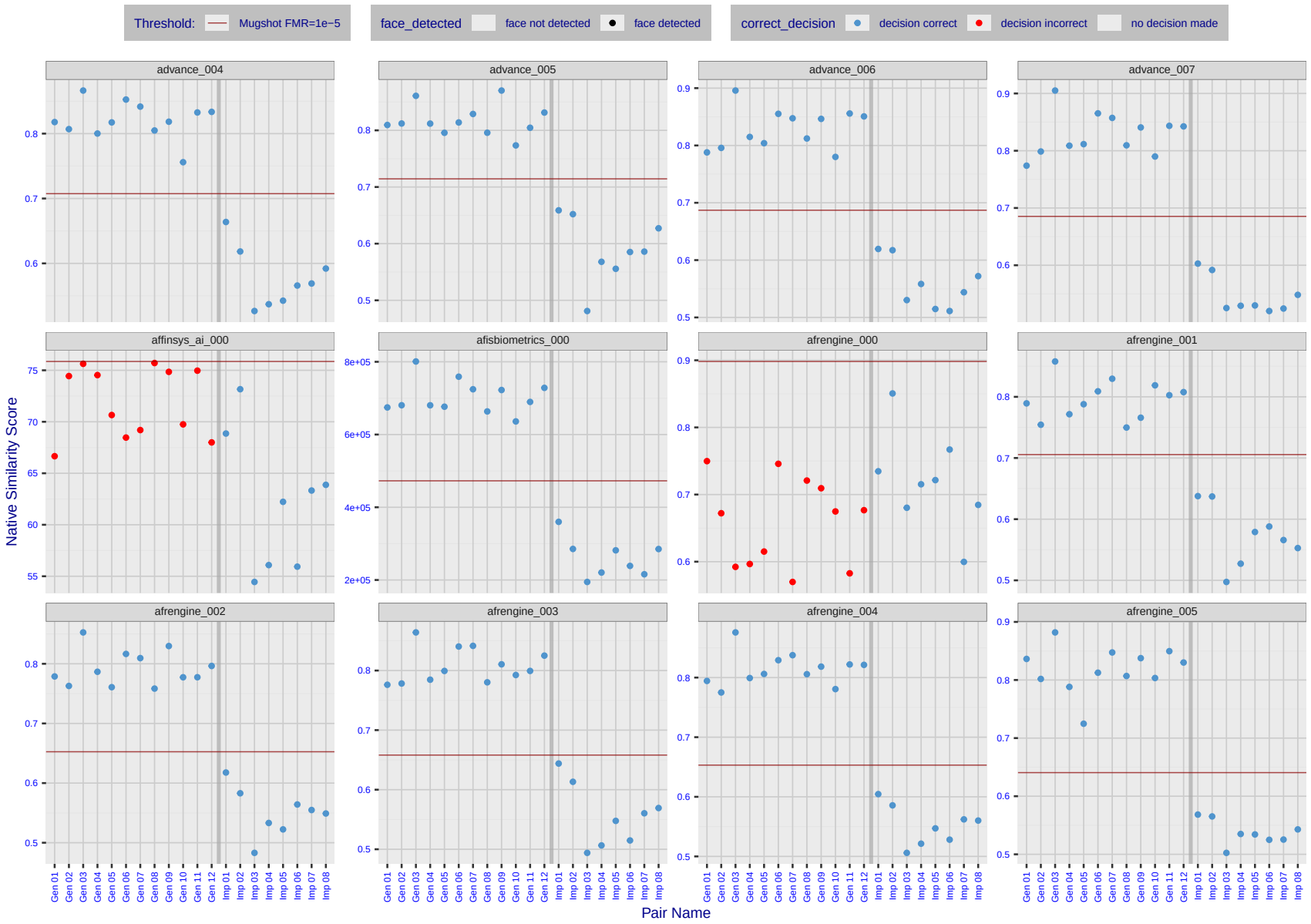


Figure 6: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

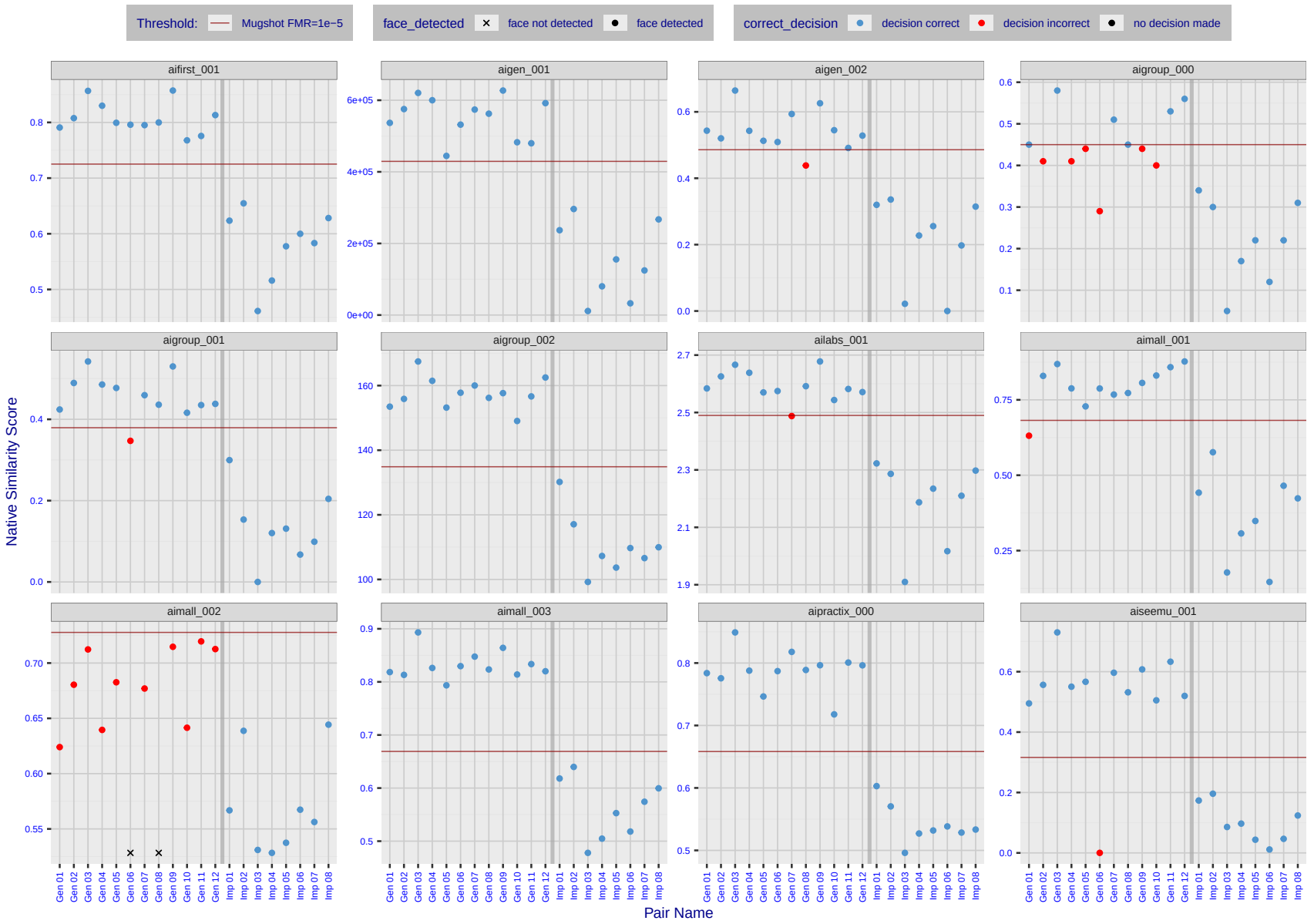


Figure 7: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

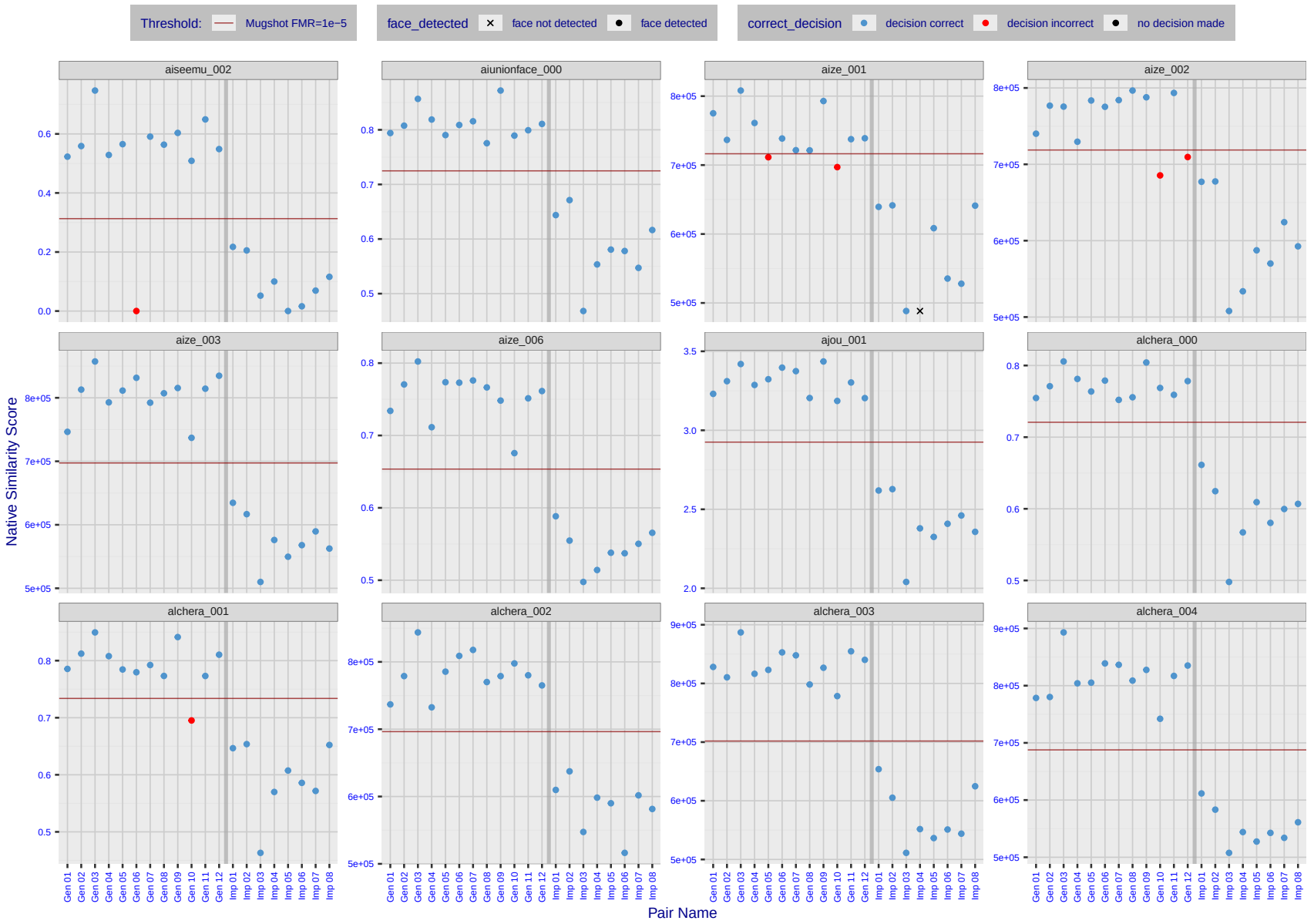


Figure 8: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

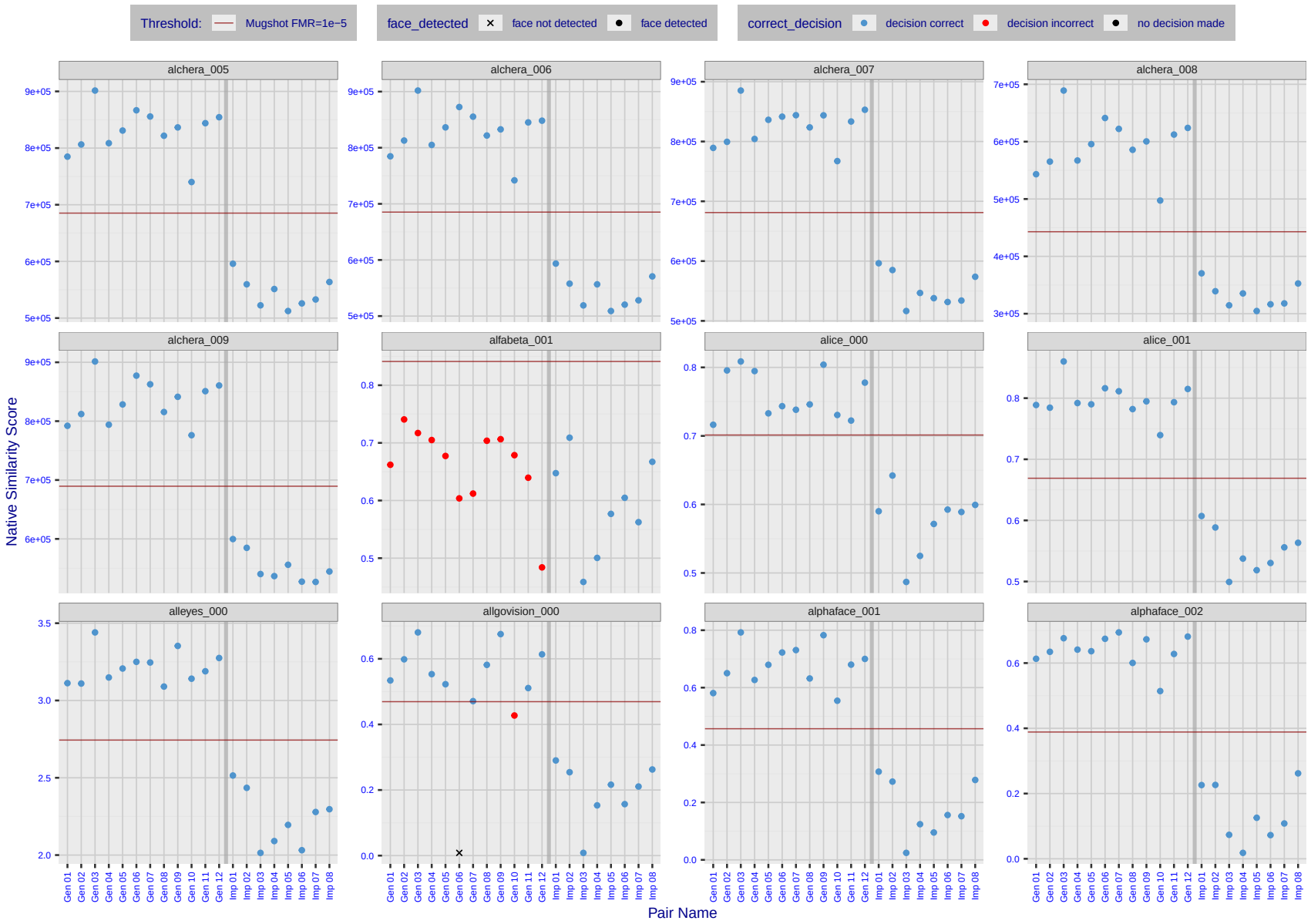


Figure 9: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

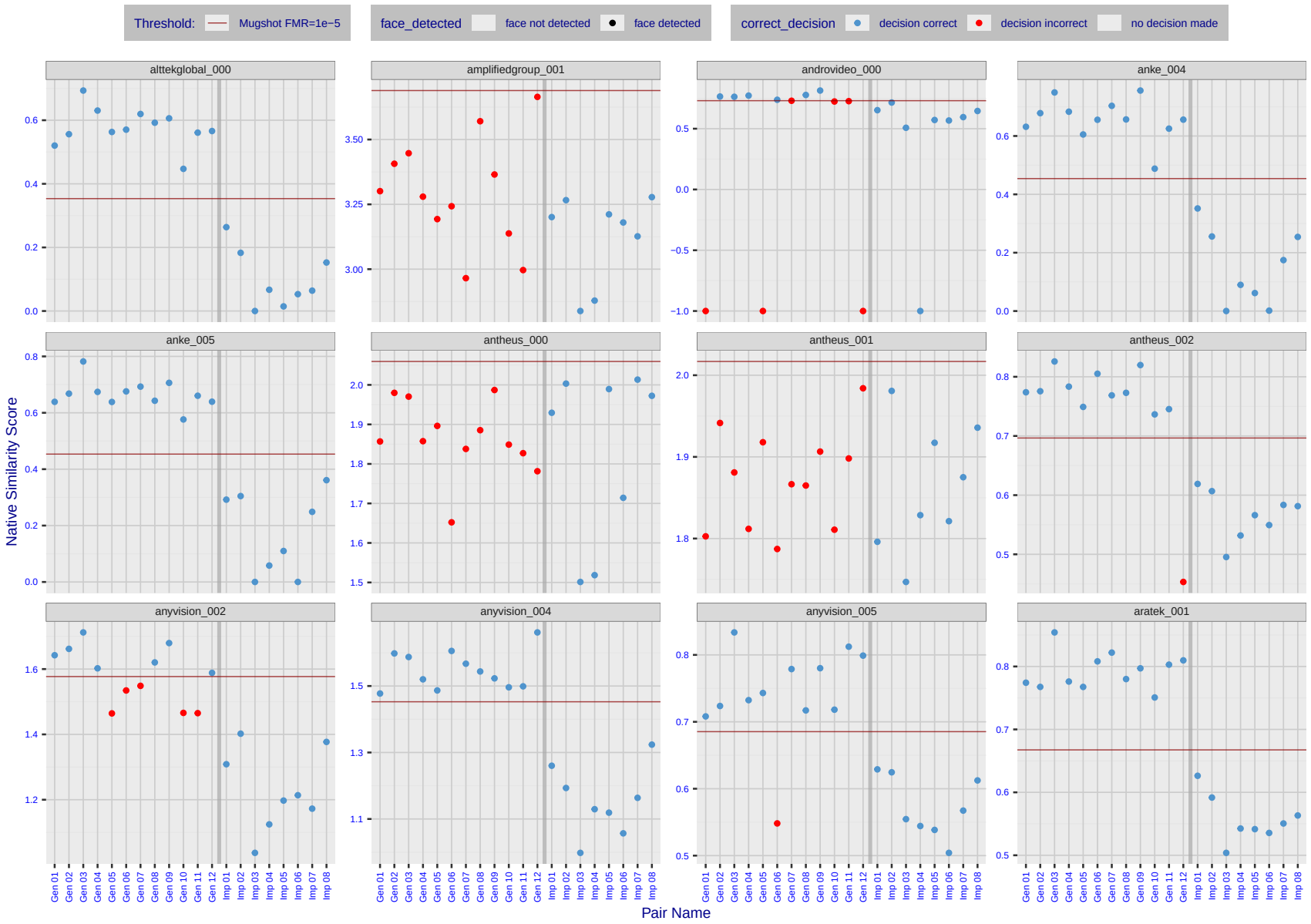


Figure 10: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

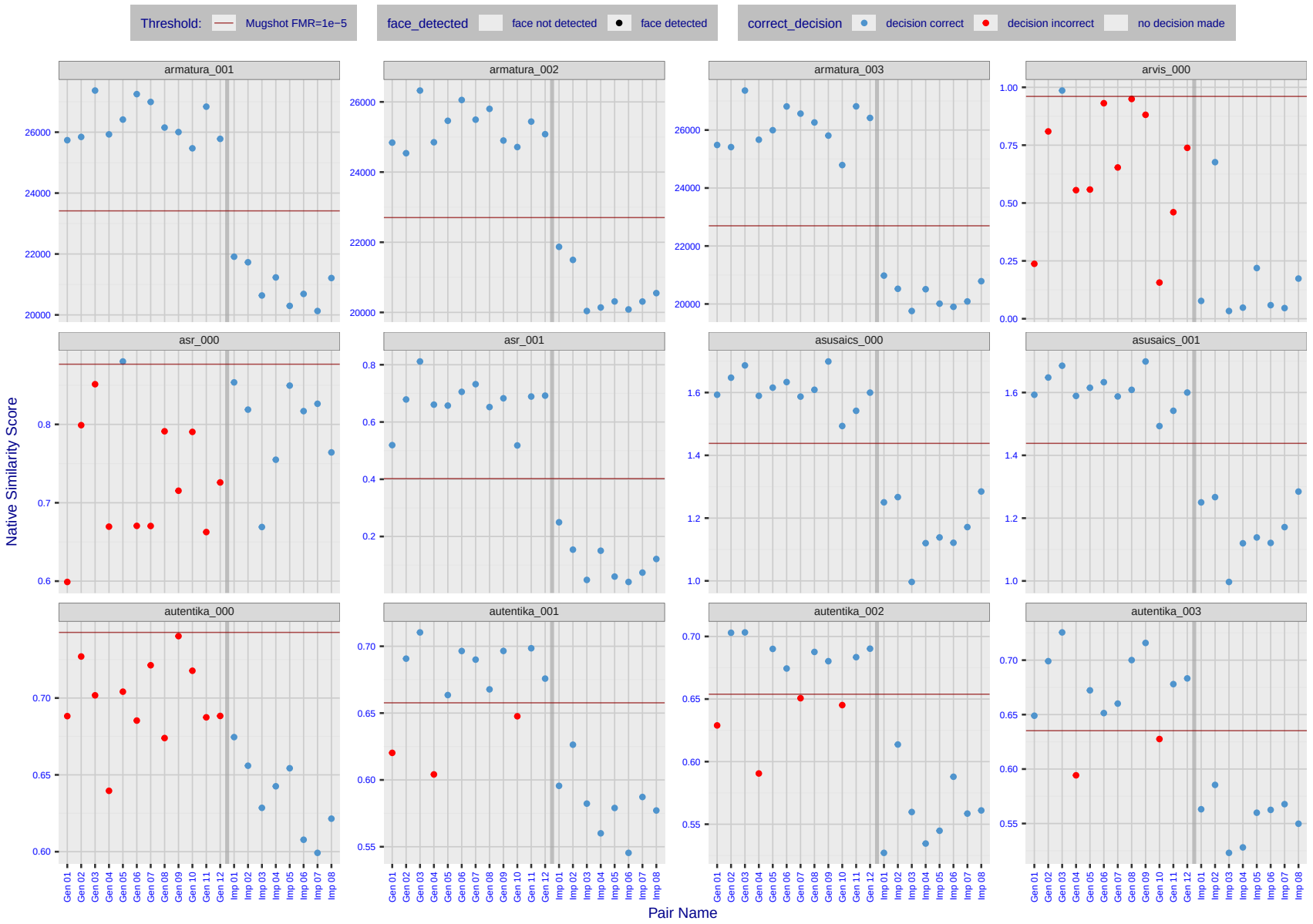


Figure 11: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

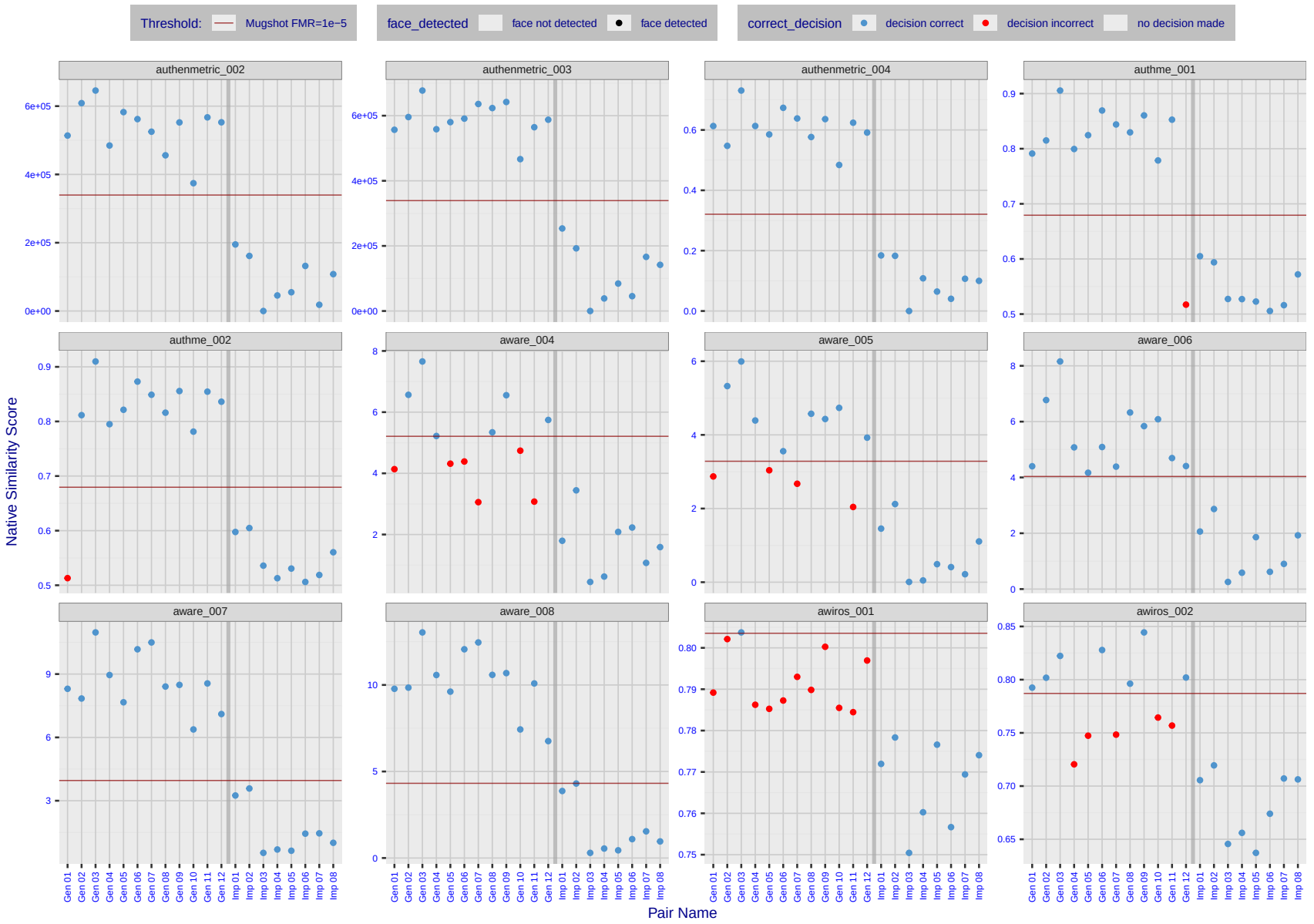


Figure 12: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

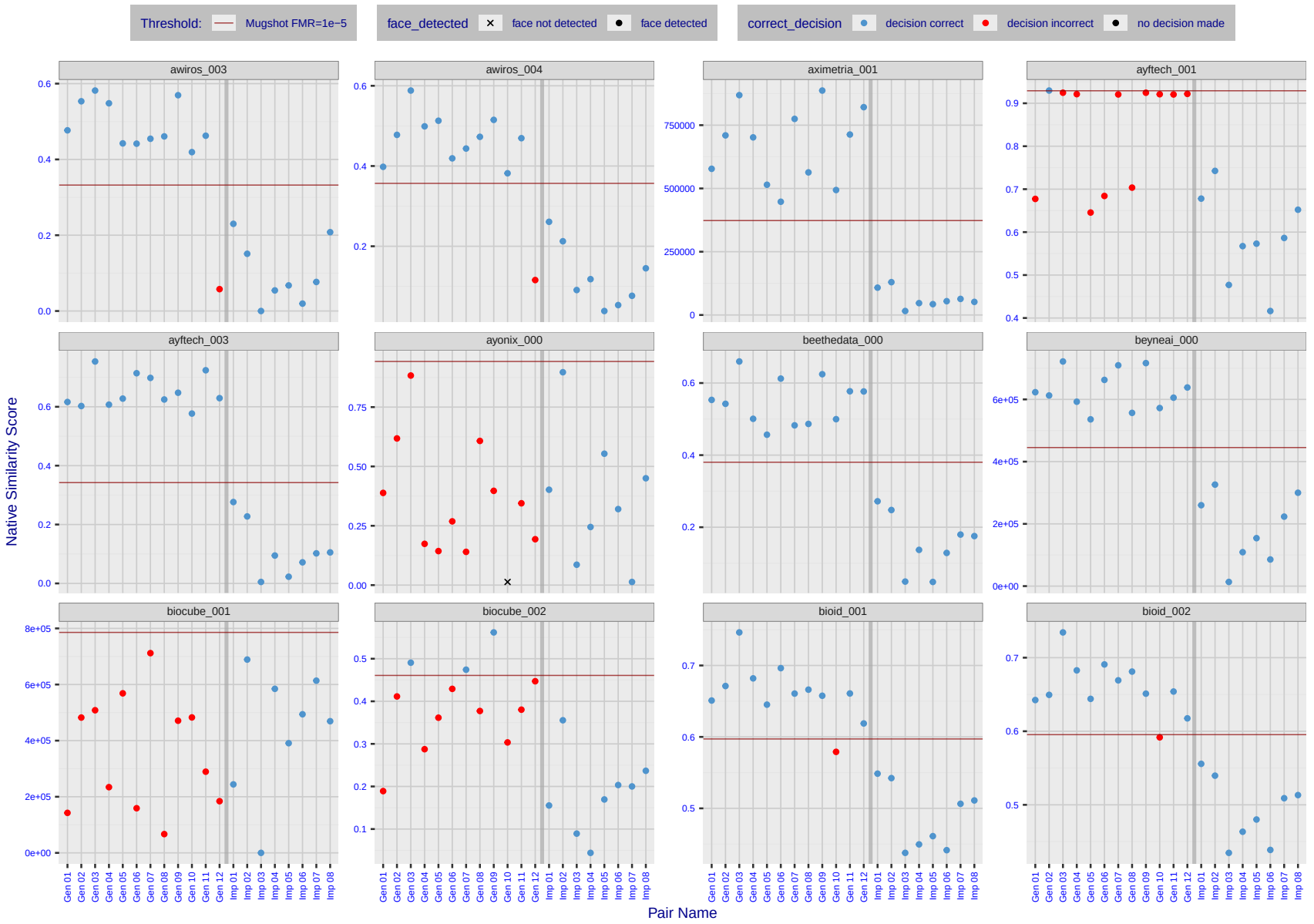


Figure 13: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

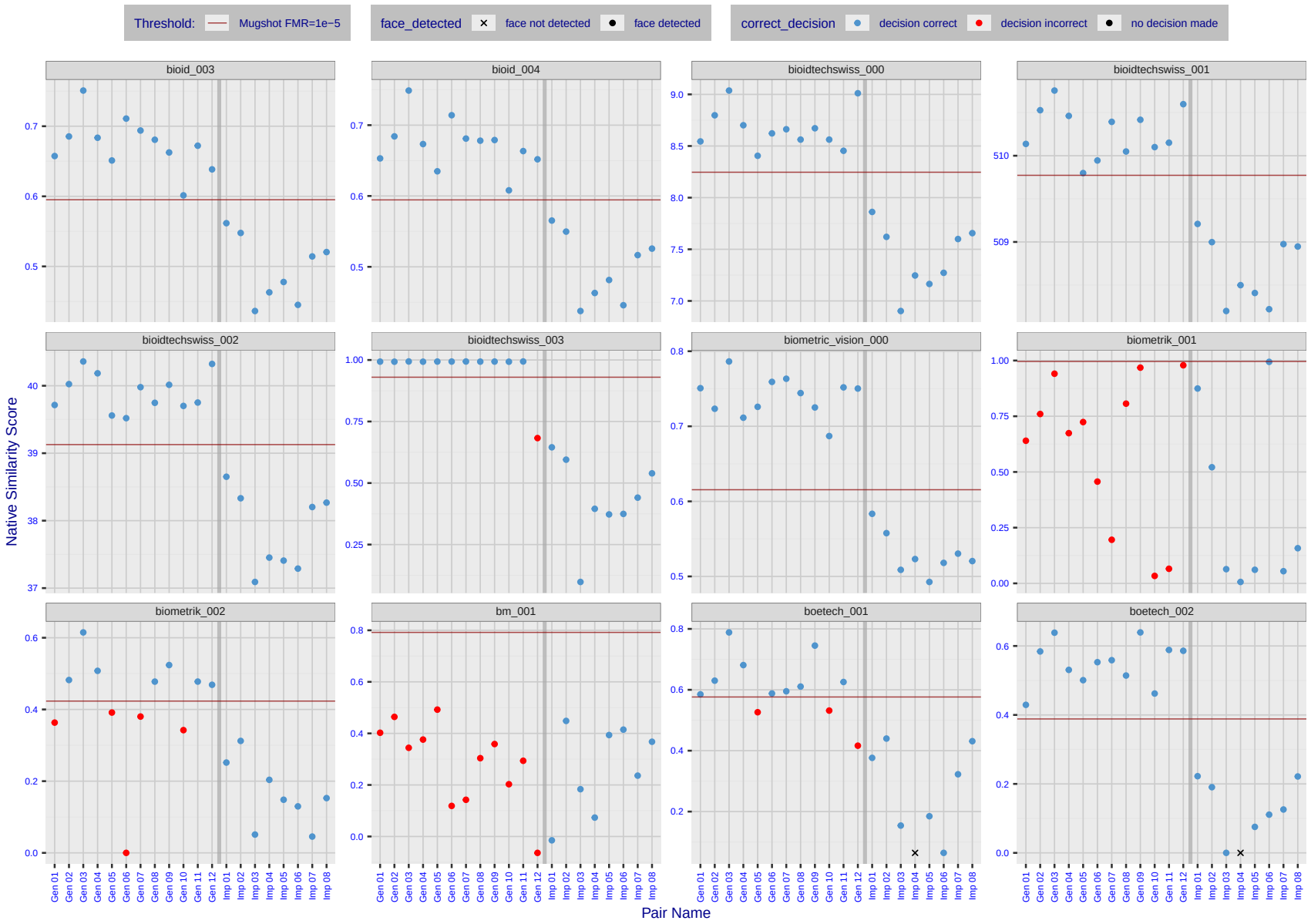


Figure 14: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

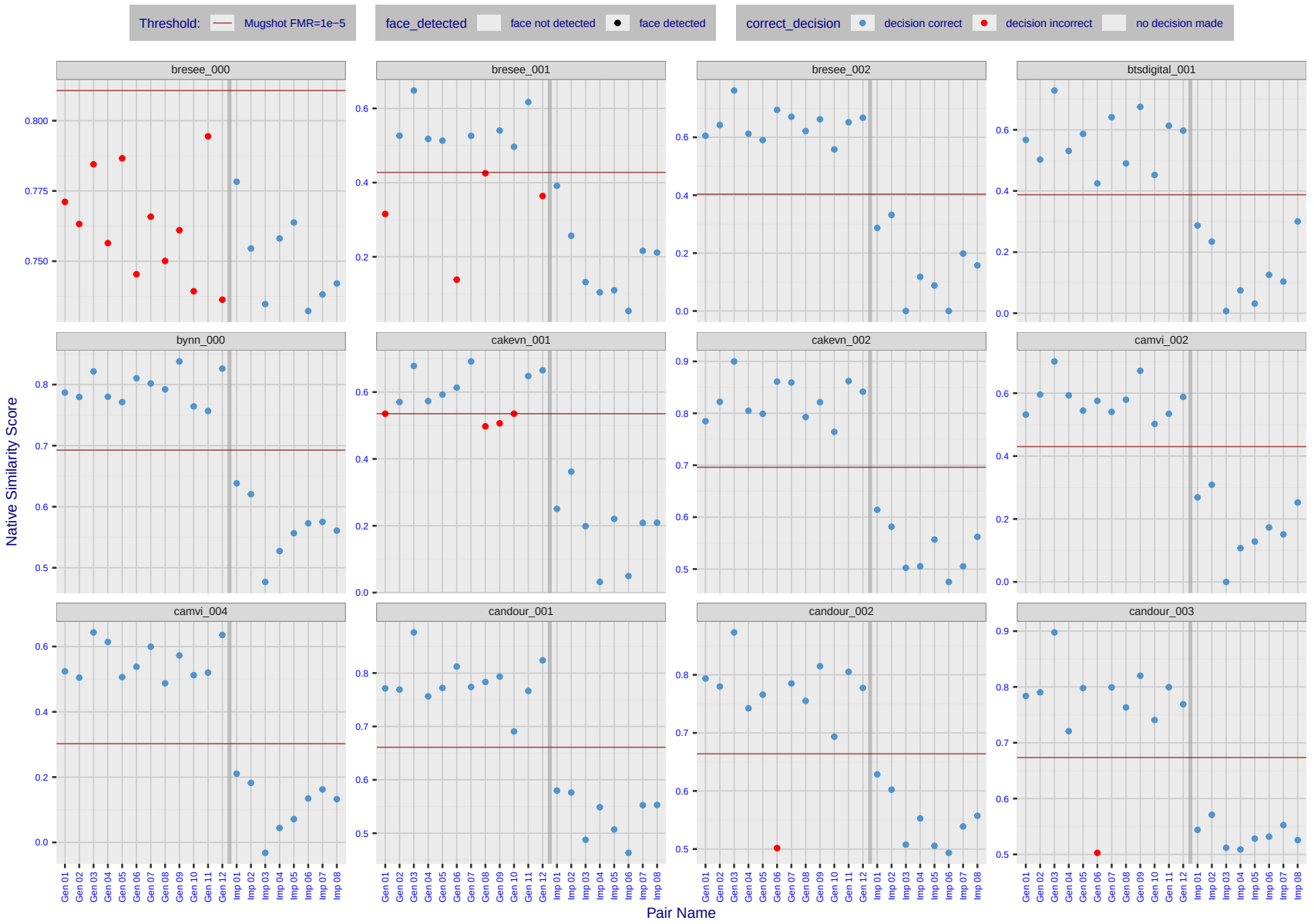


Figure 15: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

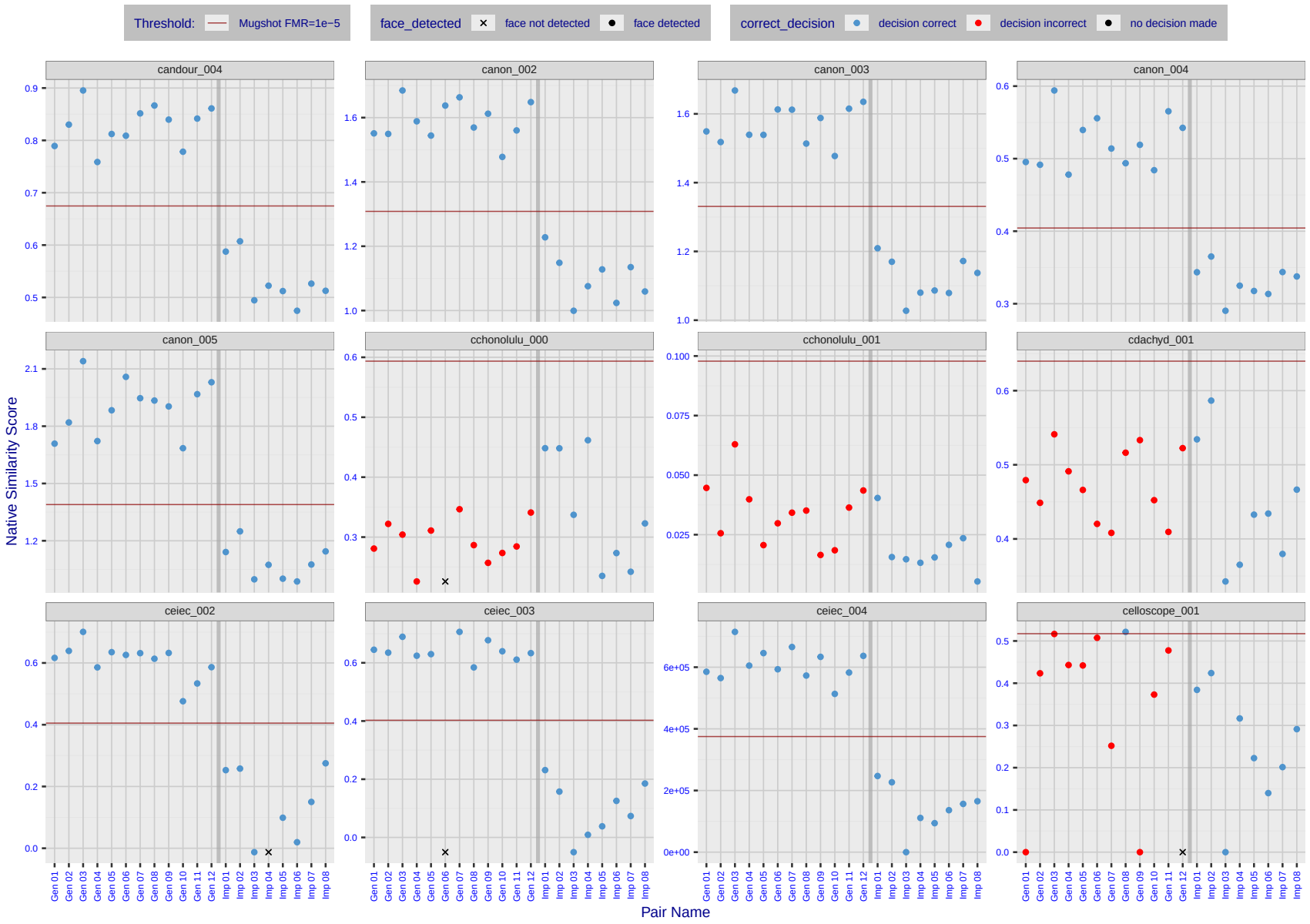


Figure 16: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

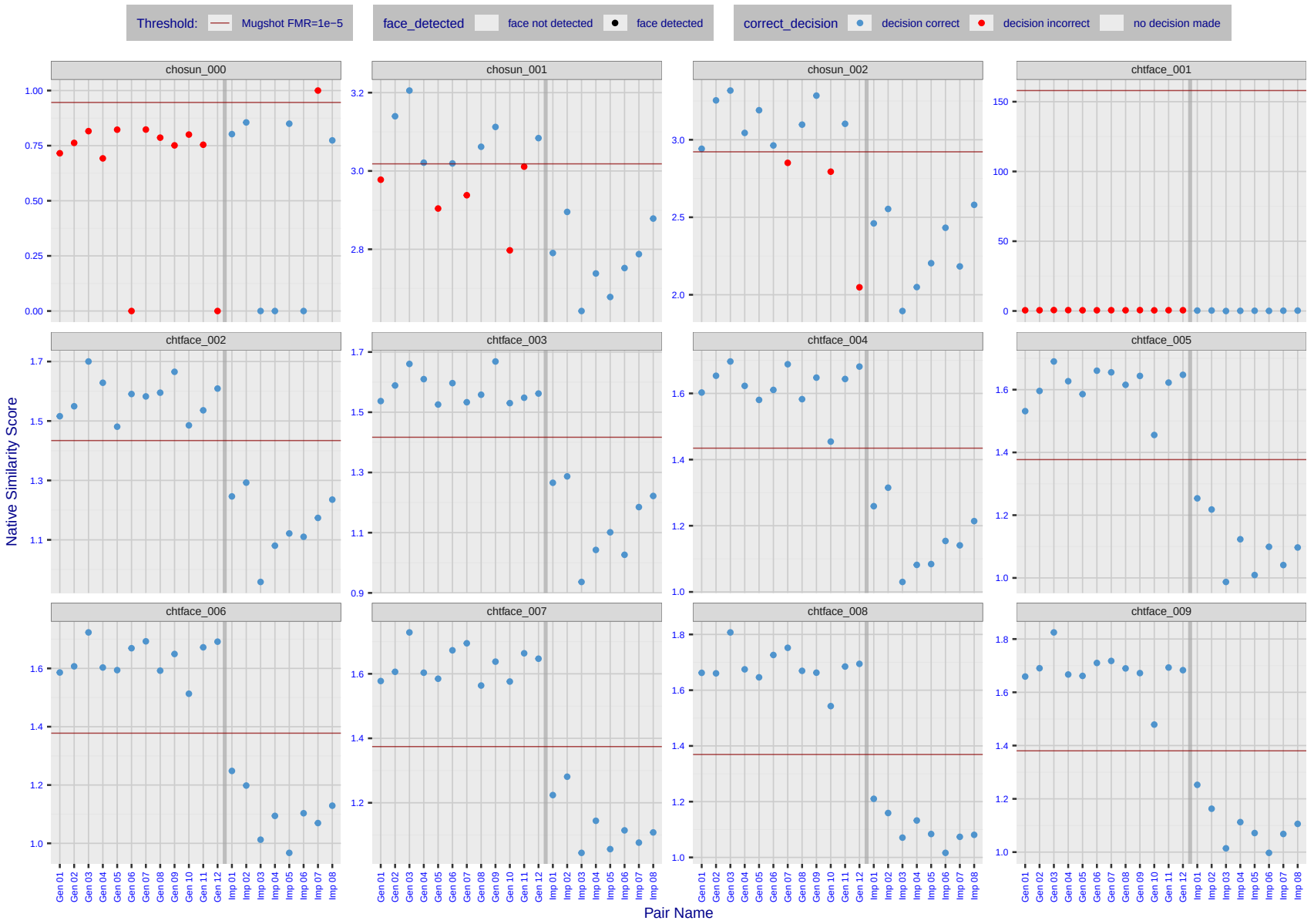


Figure 17: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

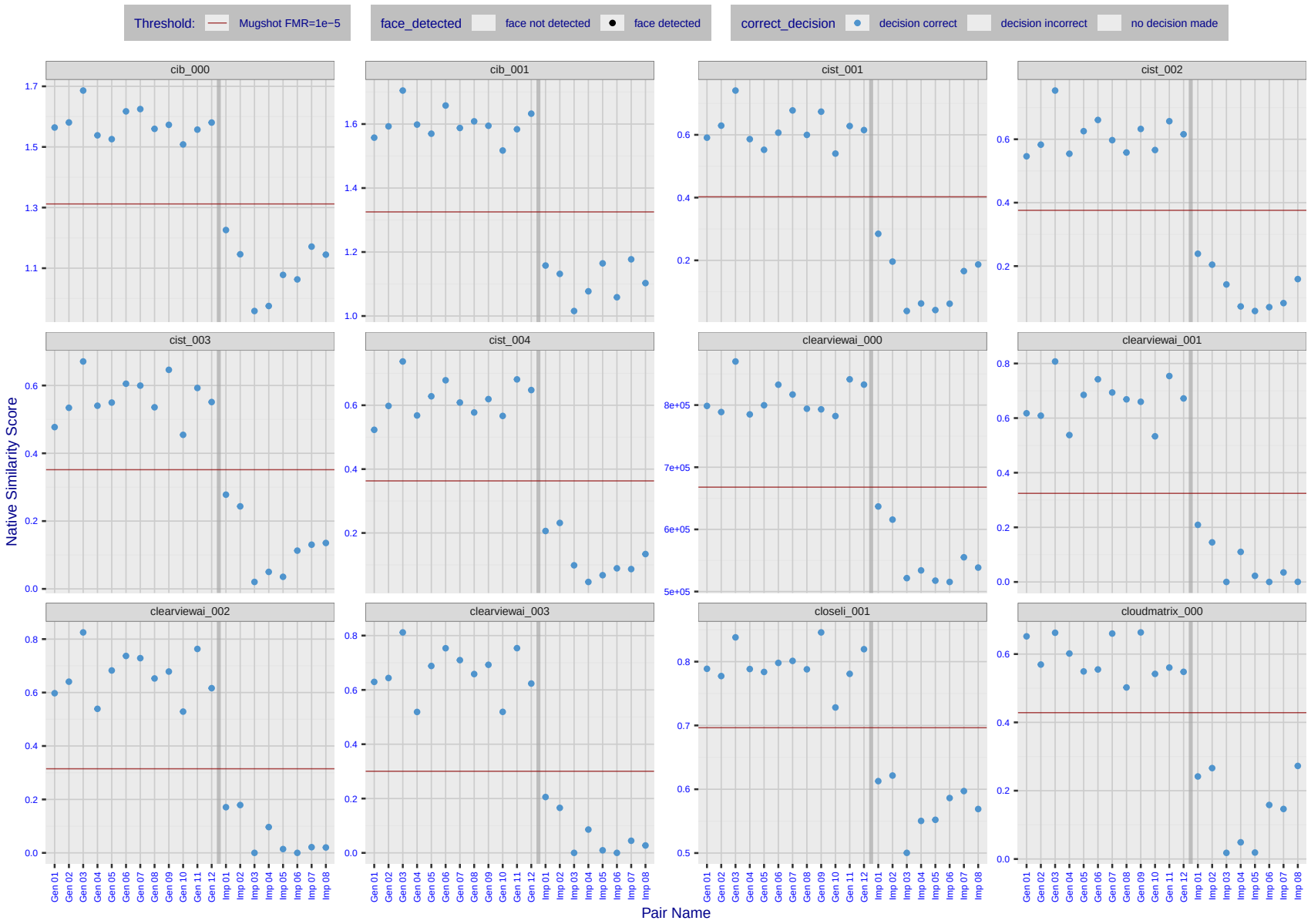


Figure 18: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

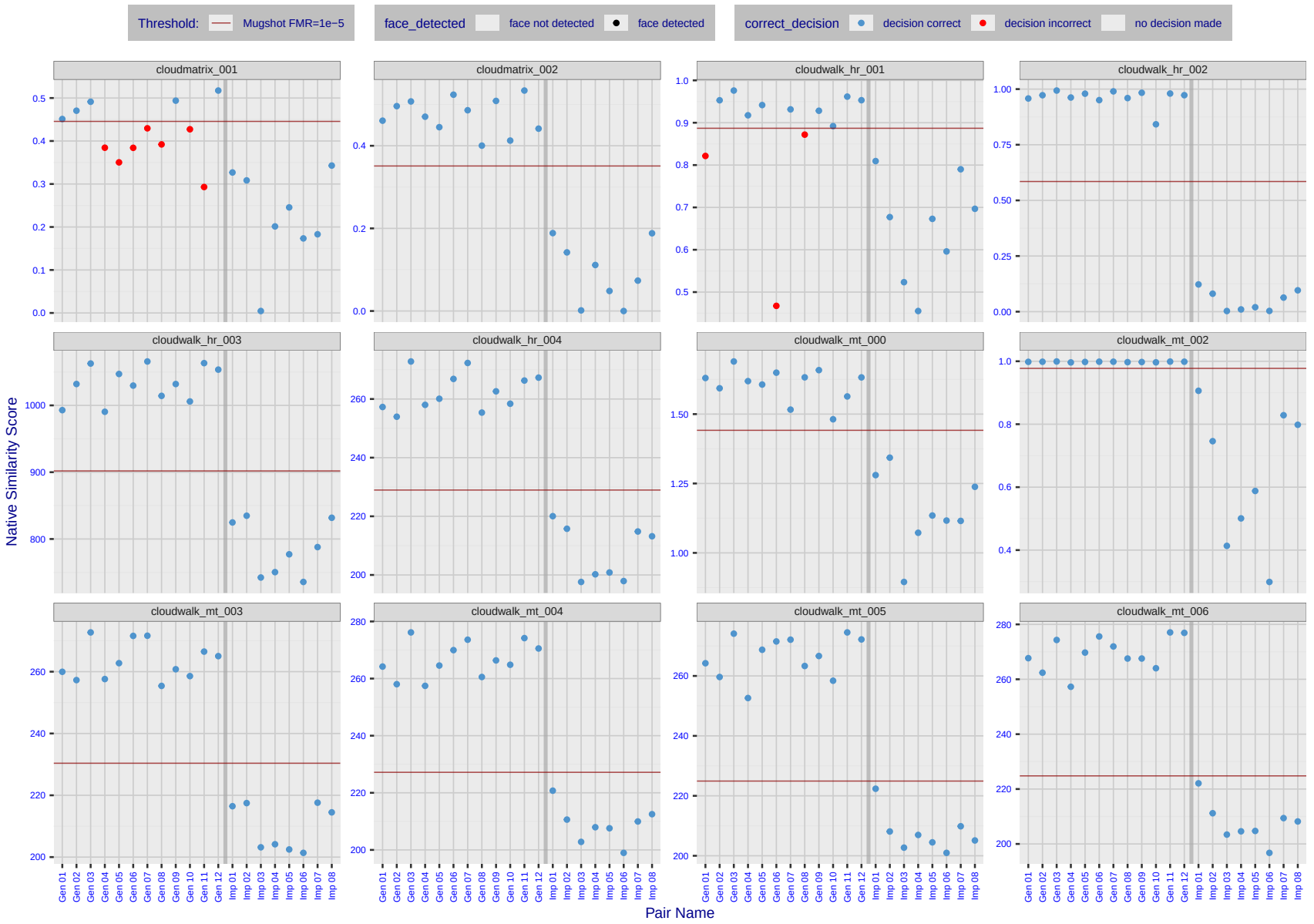


Figure 19: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

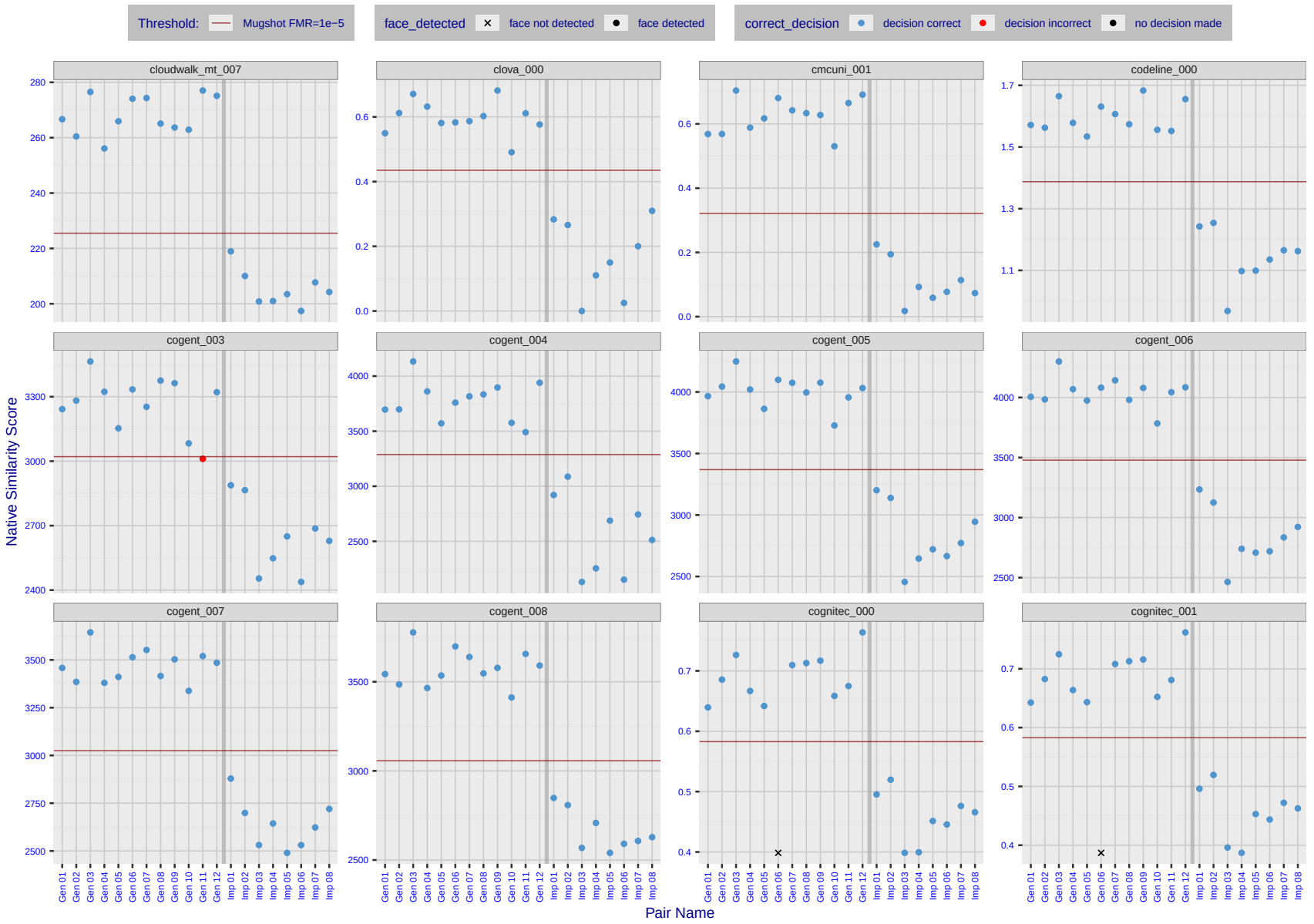


Figure 20: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

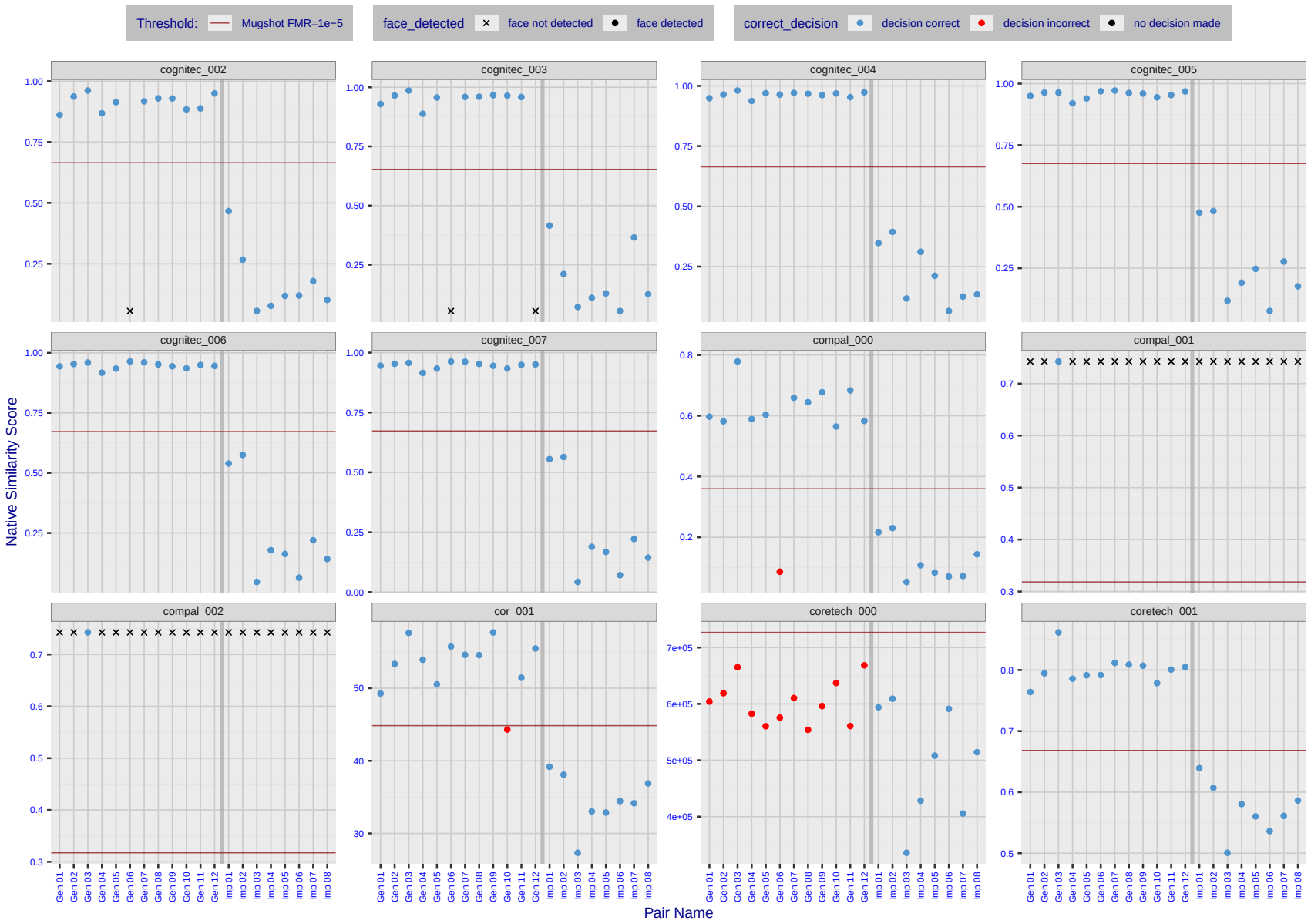


Figure 21: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

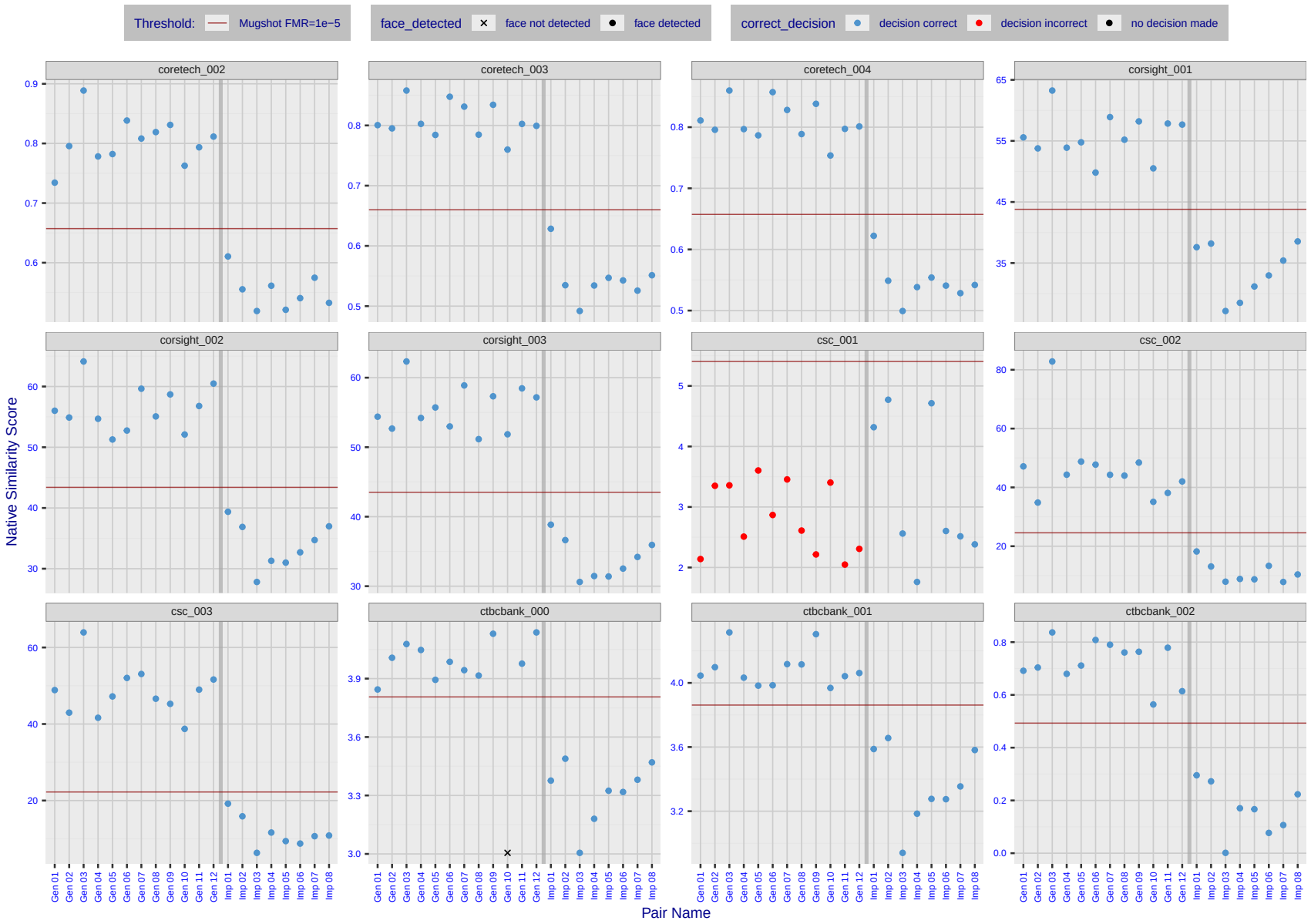


Figure 22: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

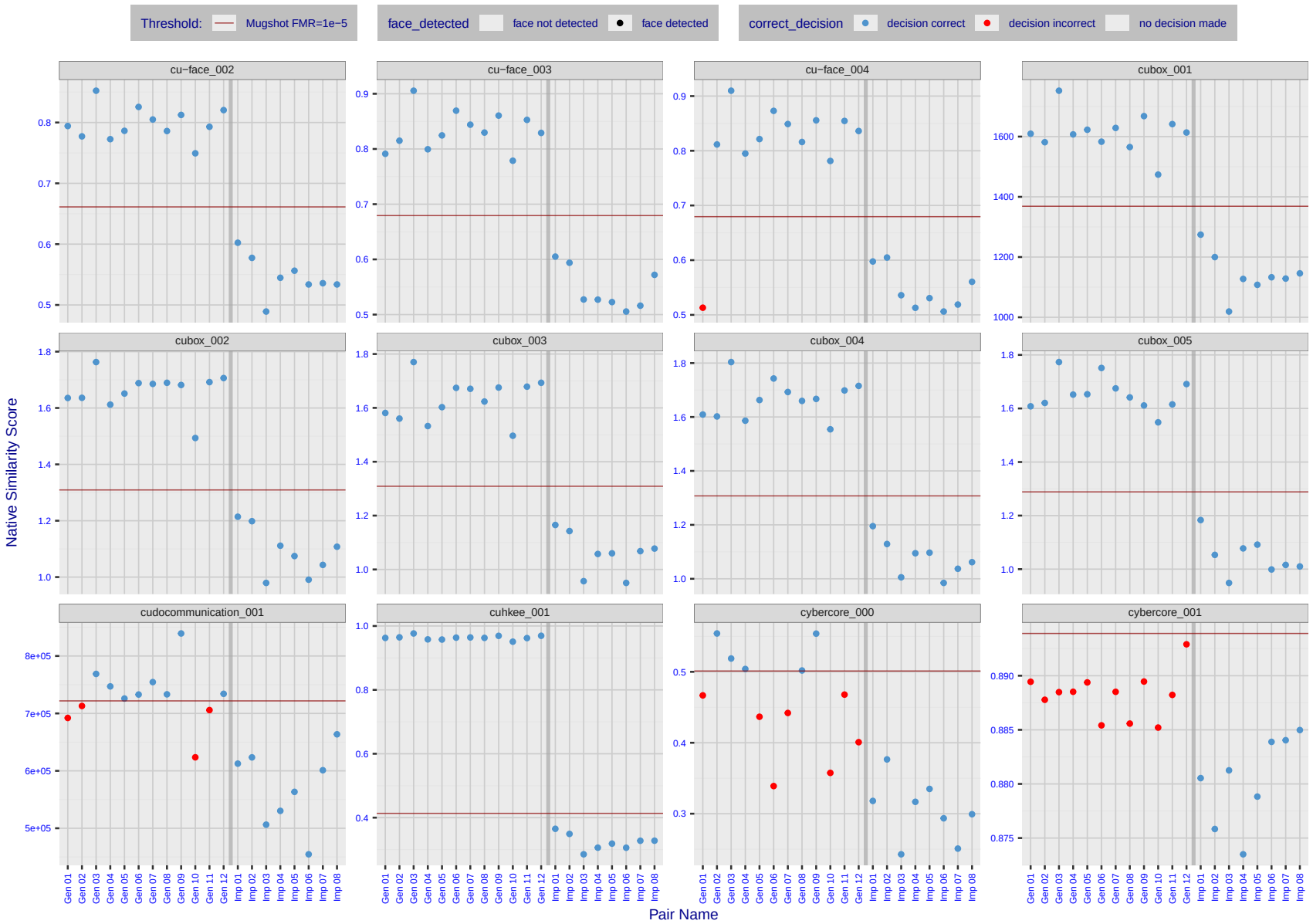


Figure 23: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

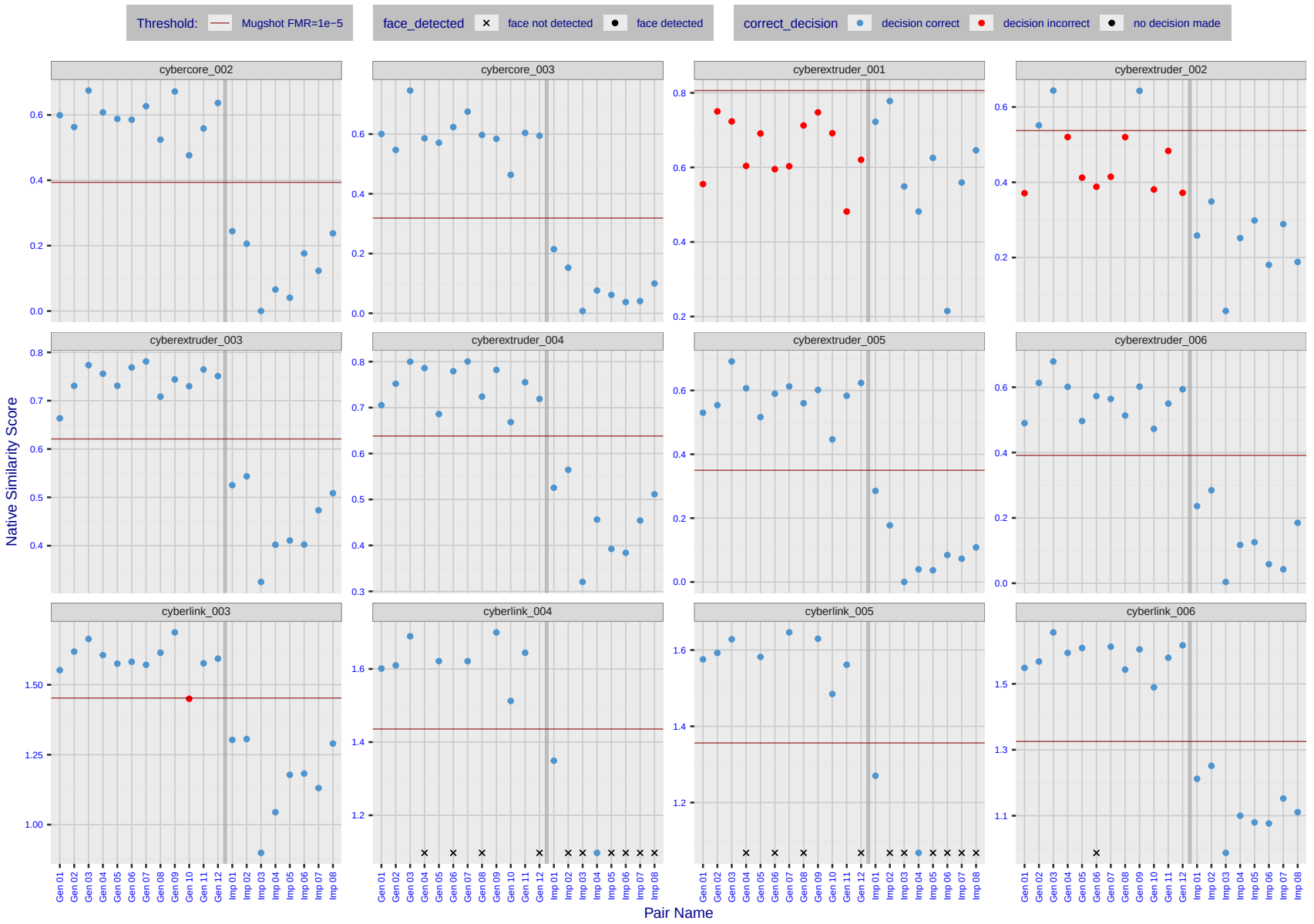


Figure 24: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

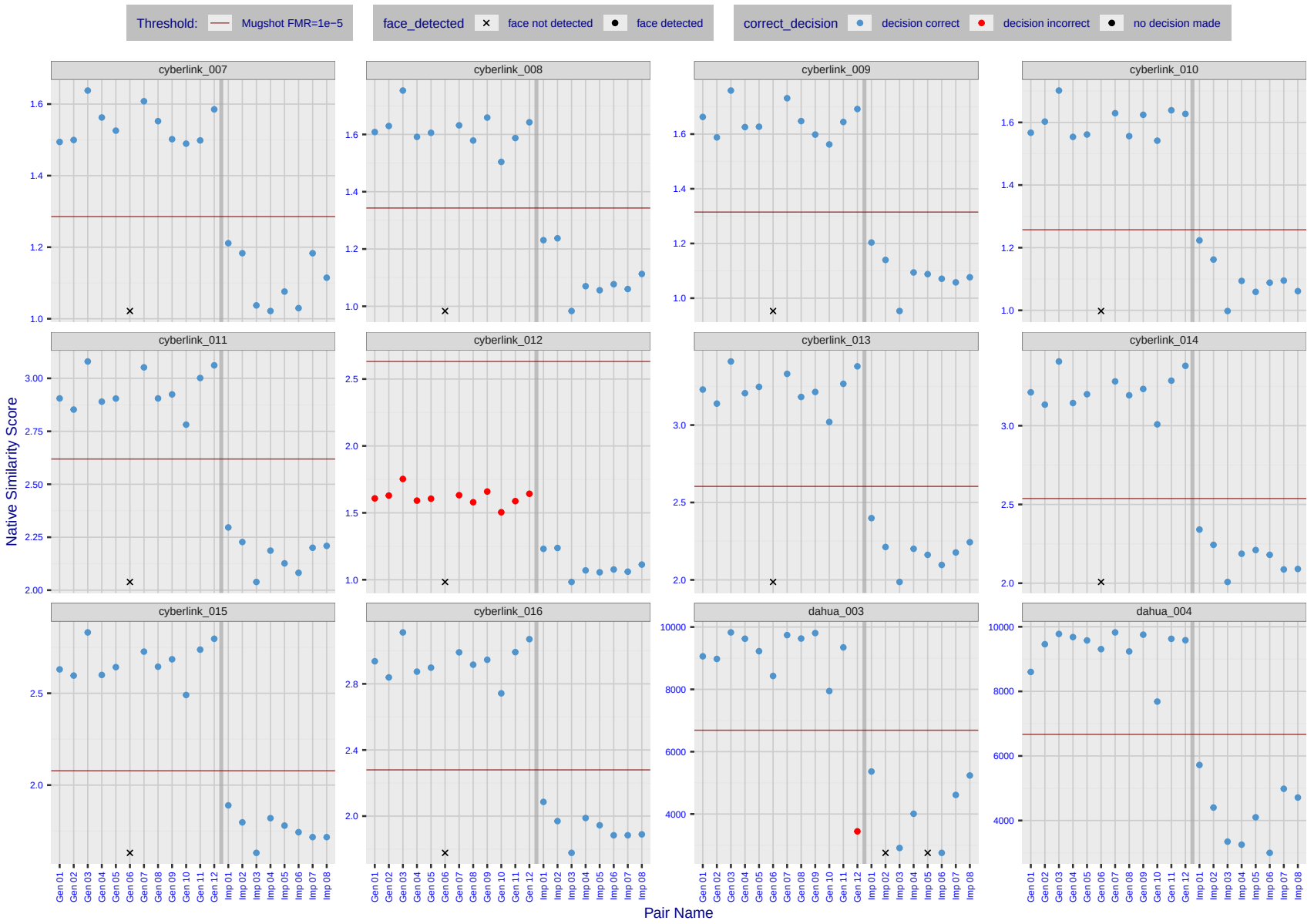


Figure 25: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

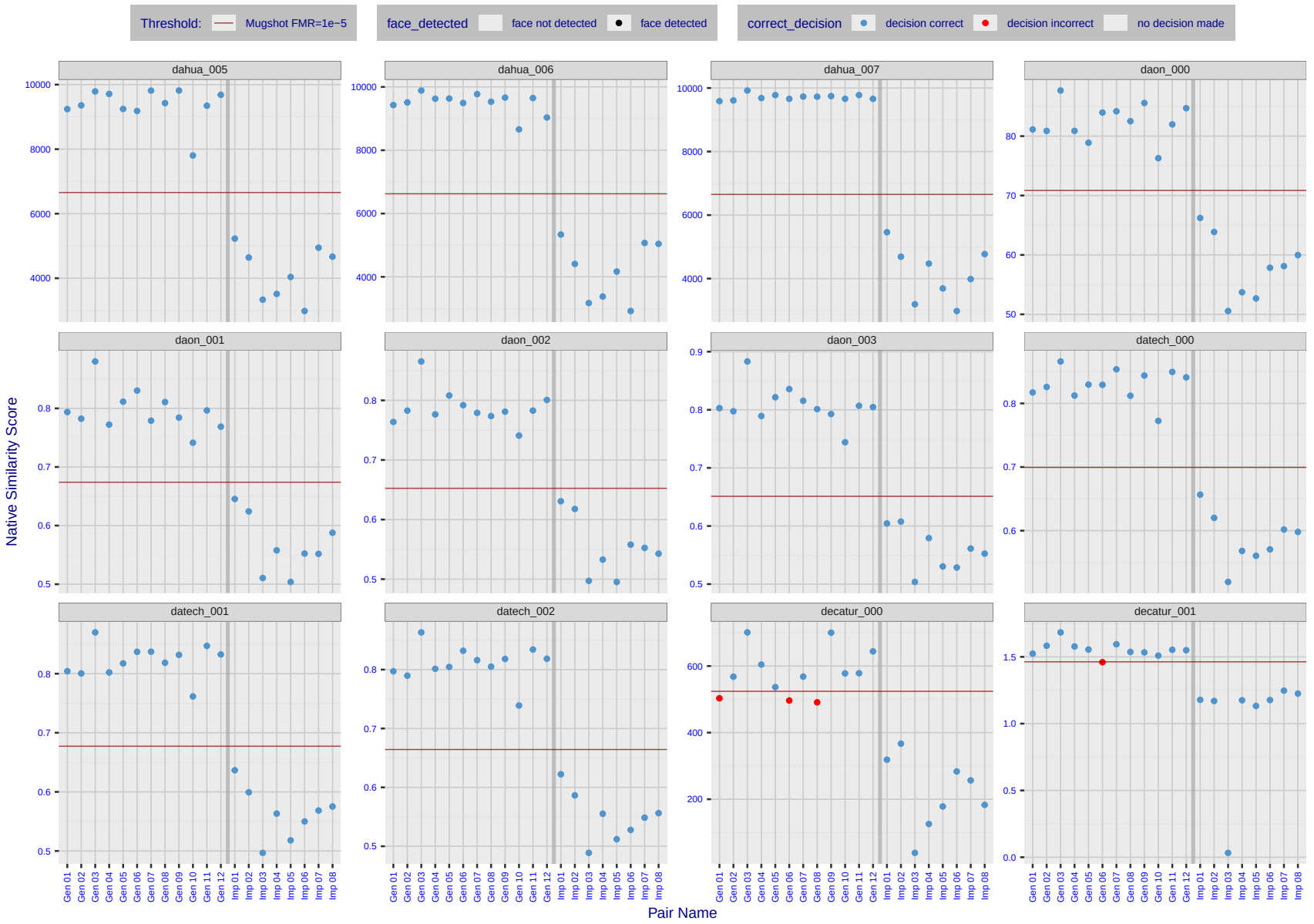


Figure 26: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

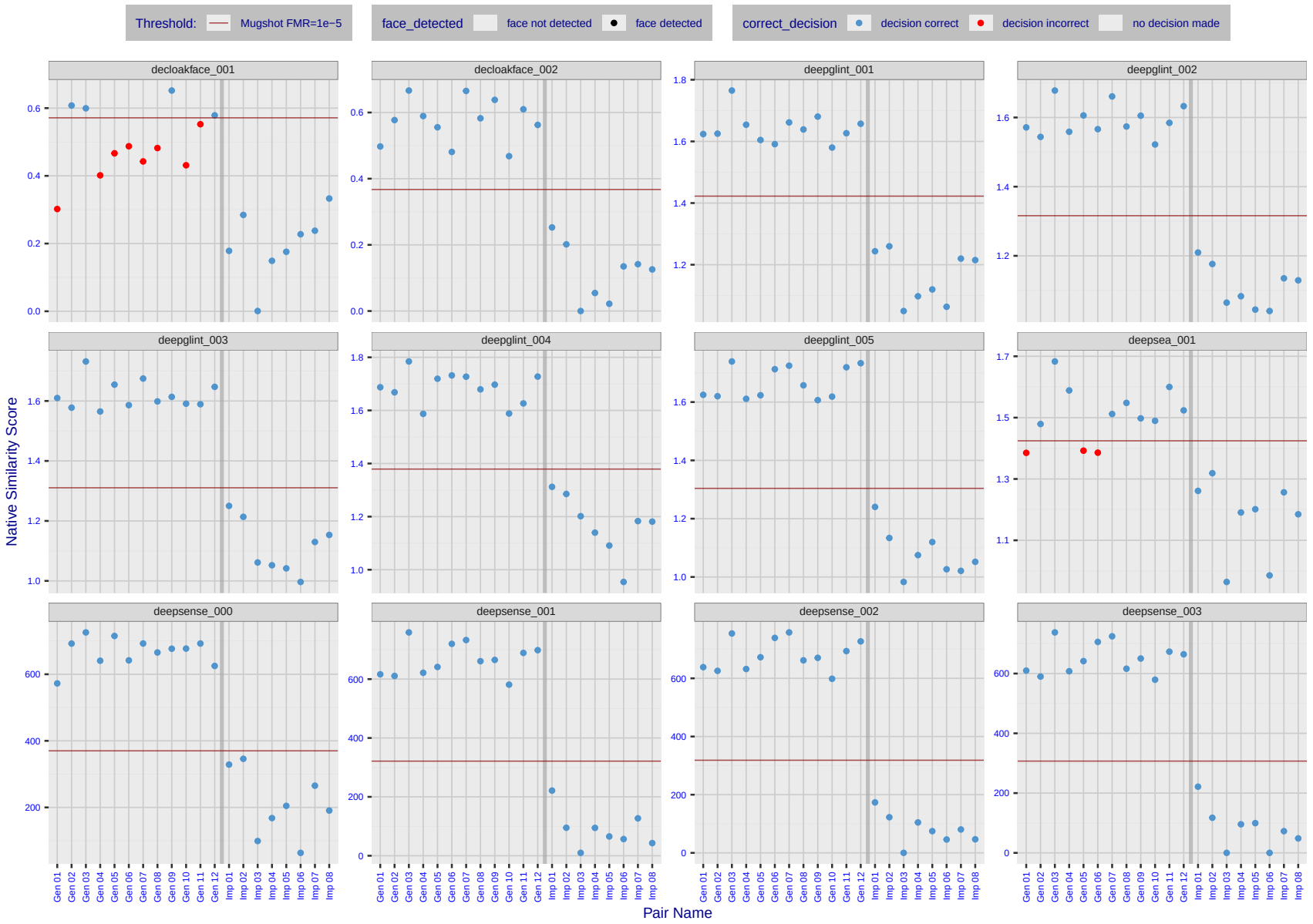


Figure 27: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

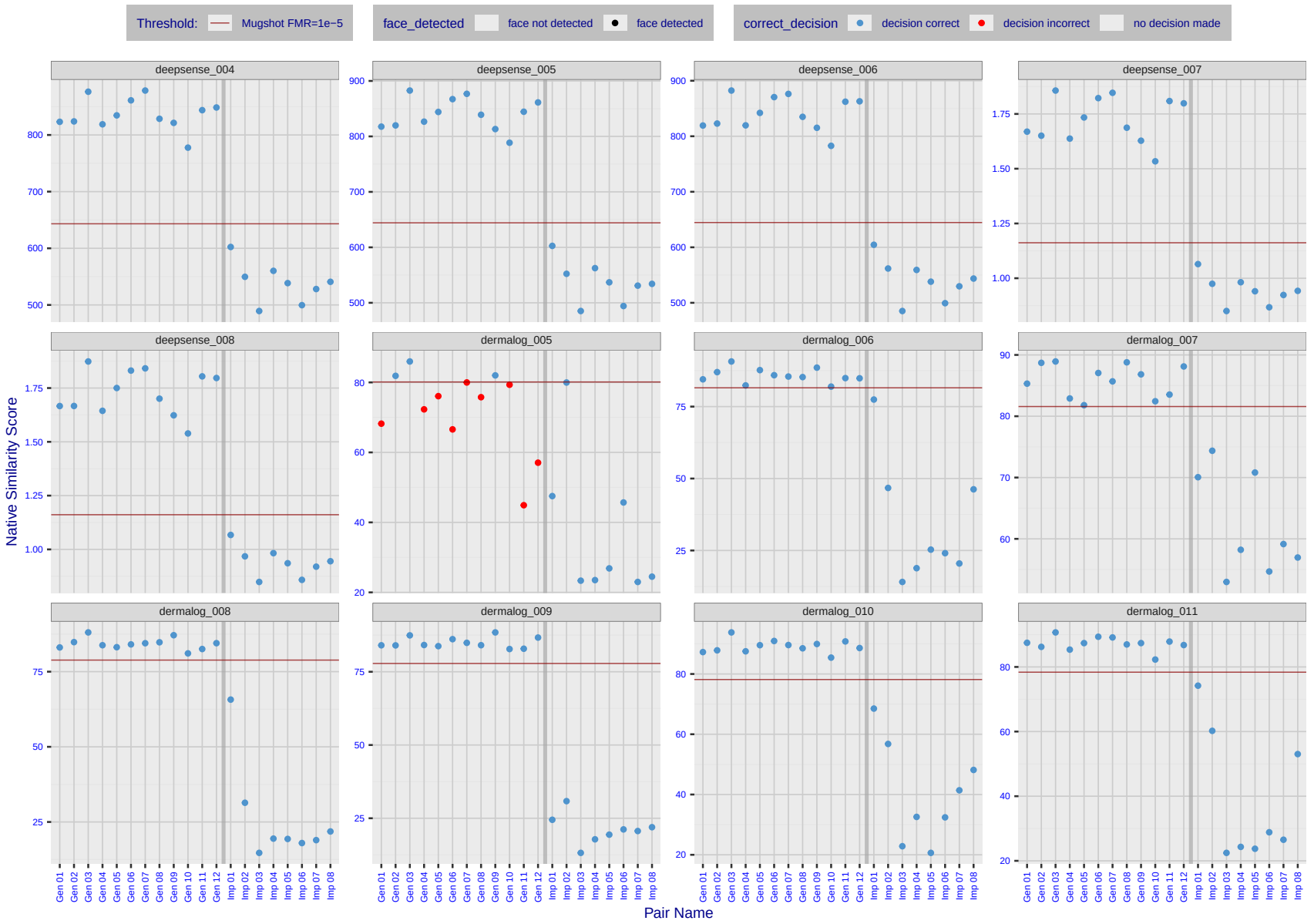


Figure 28: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

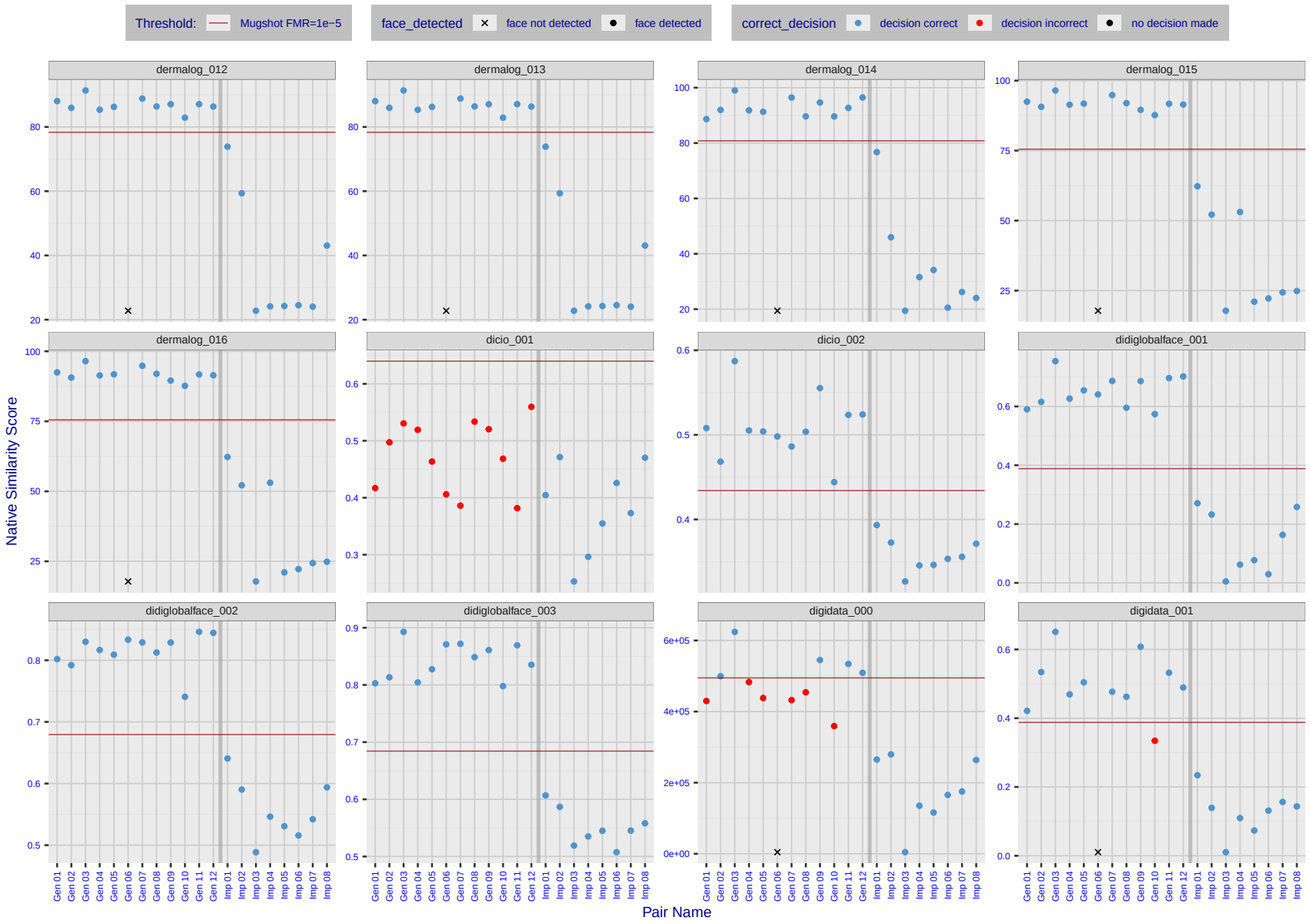


Figure 29: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

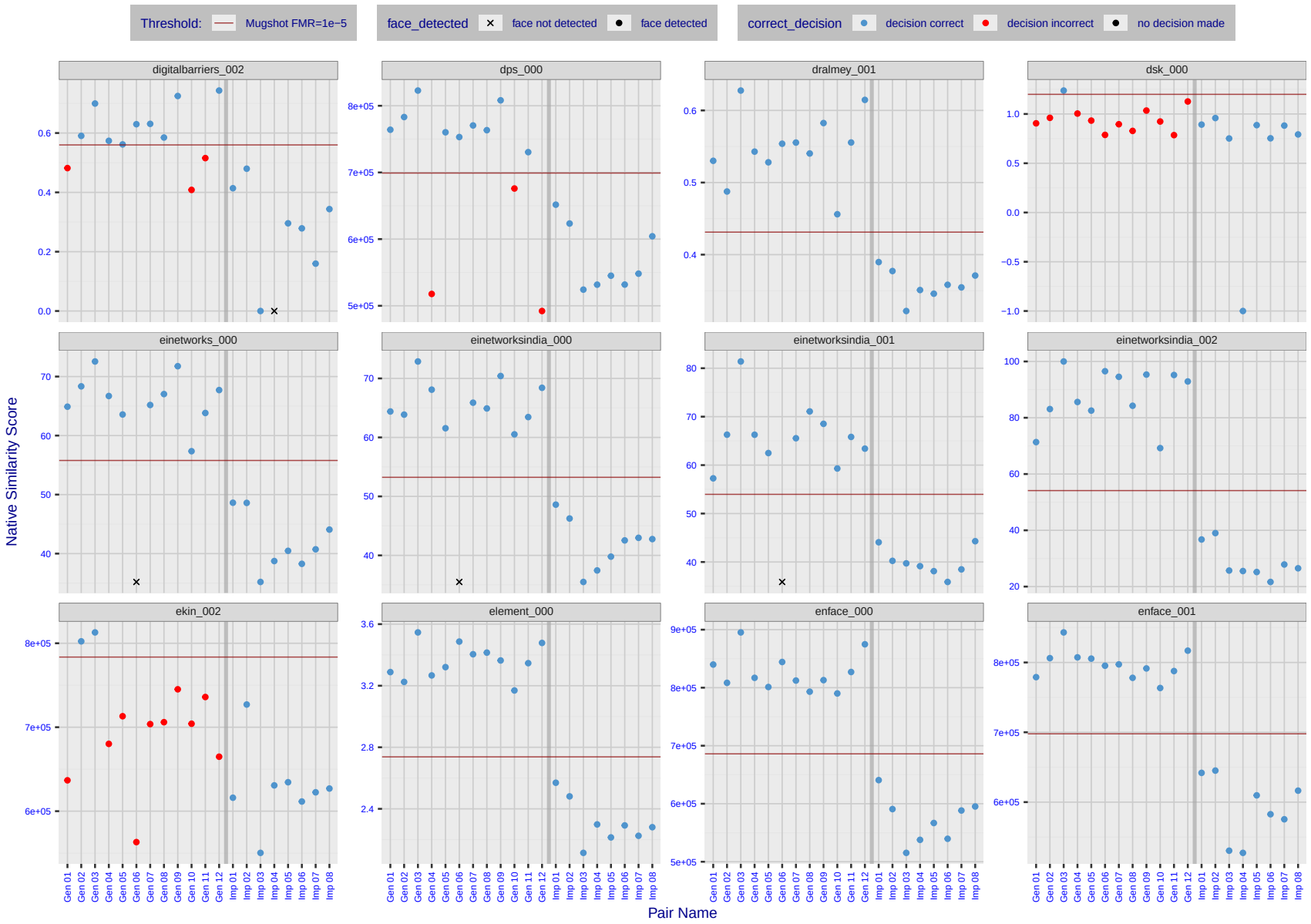


Figure 30: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

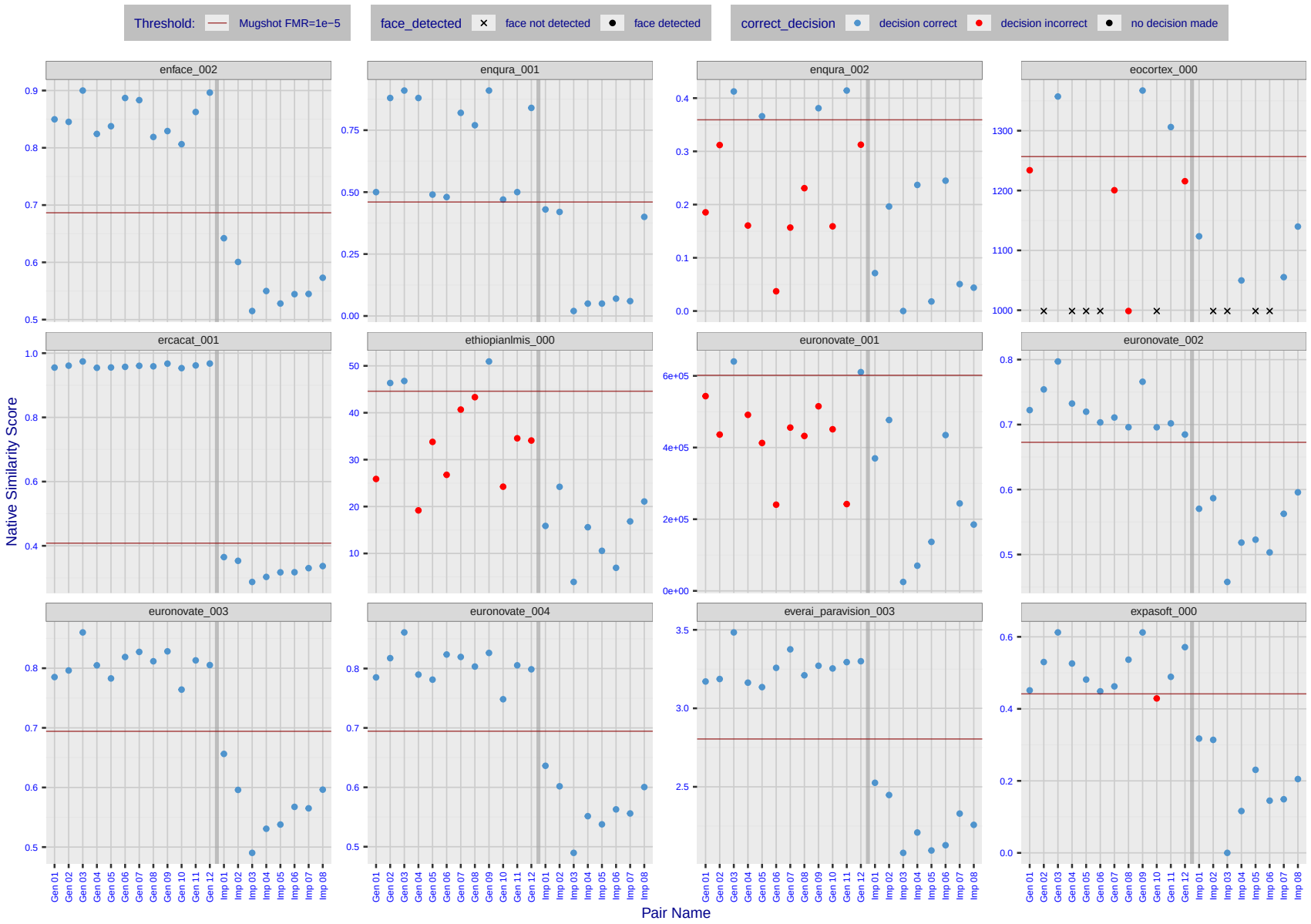


Figure 31: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

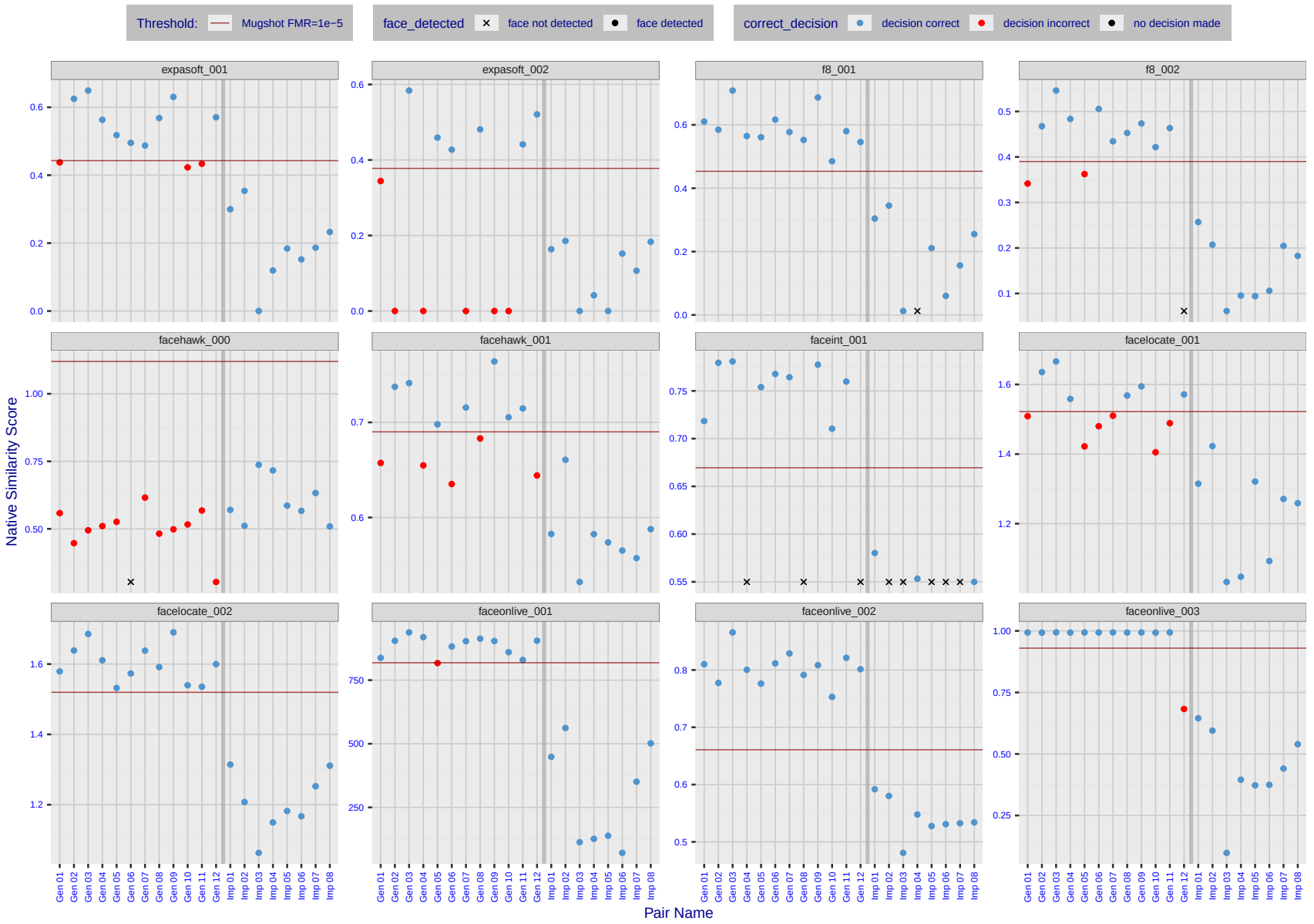


Figure 32: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

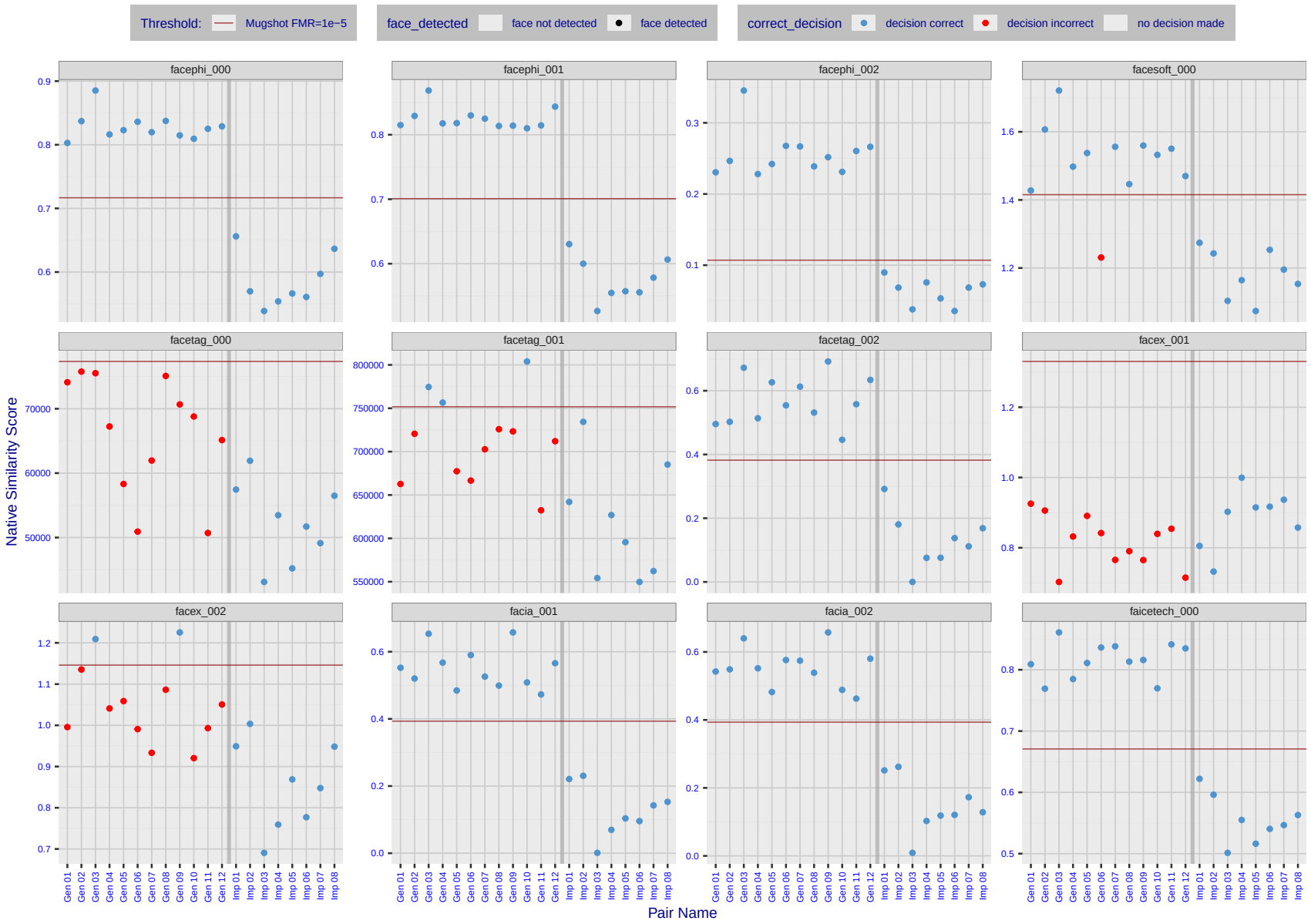


Figure 33: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

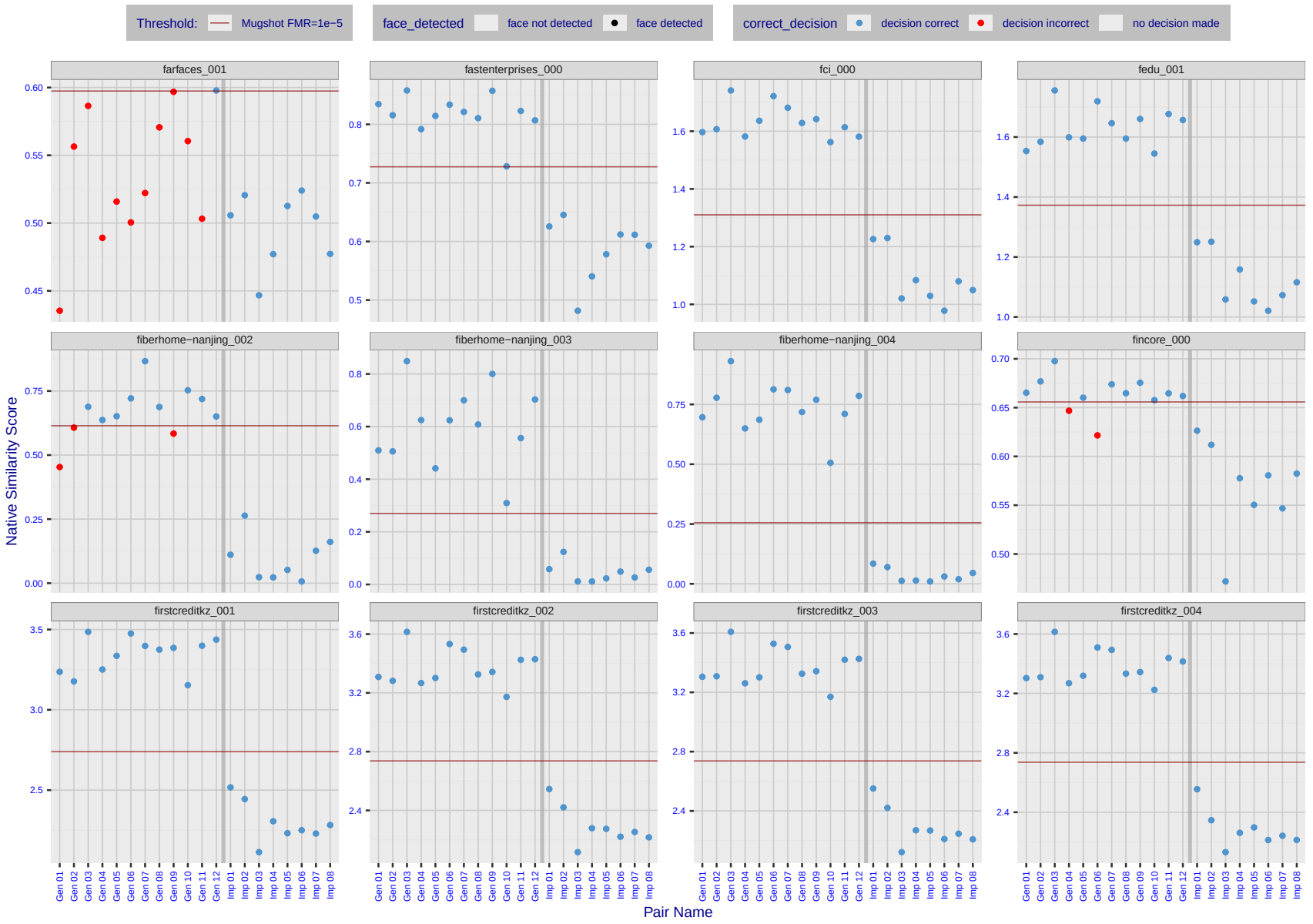


Figure 34: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

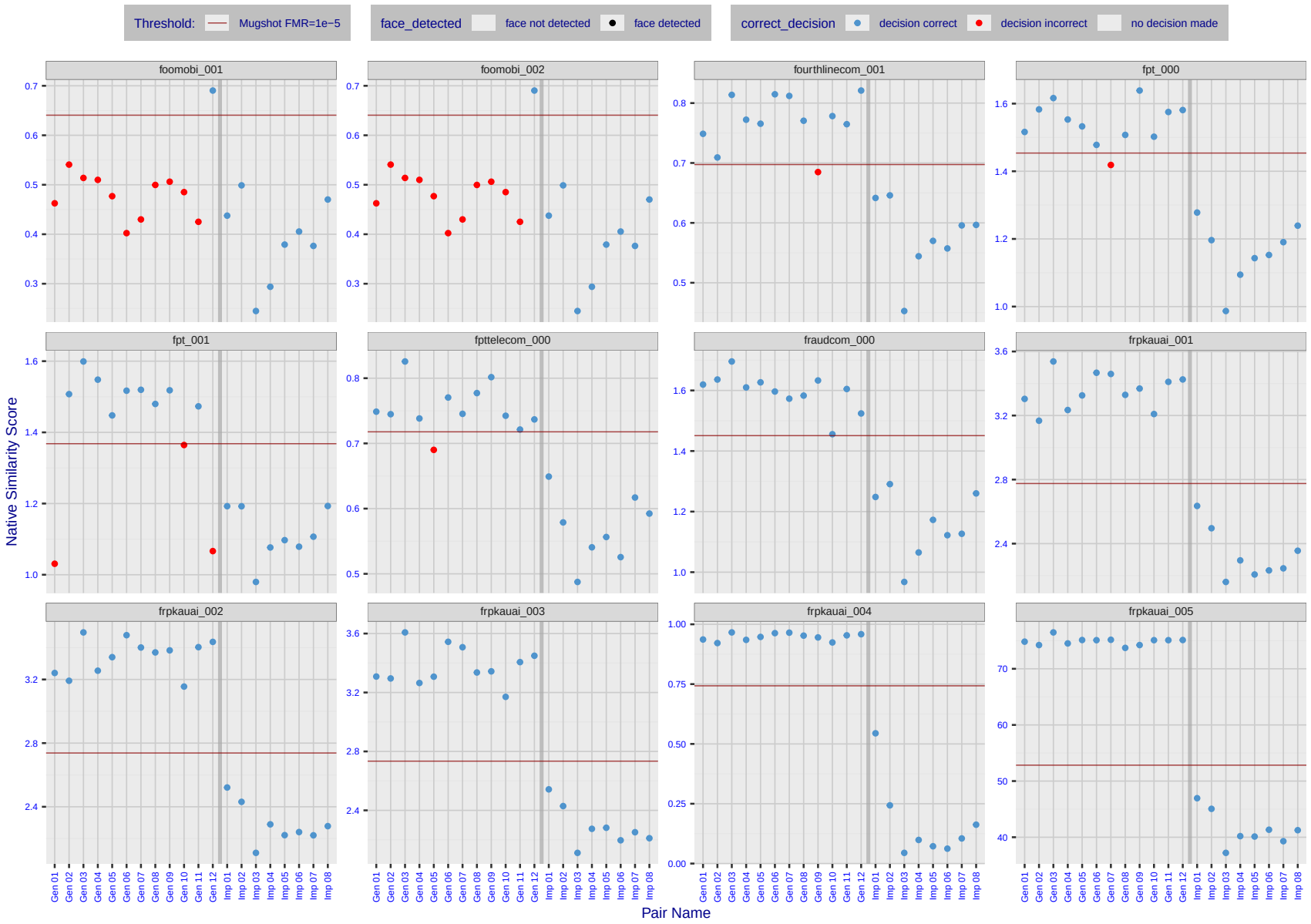


Figure 35: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

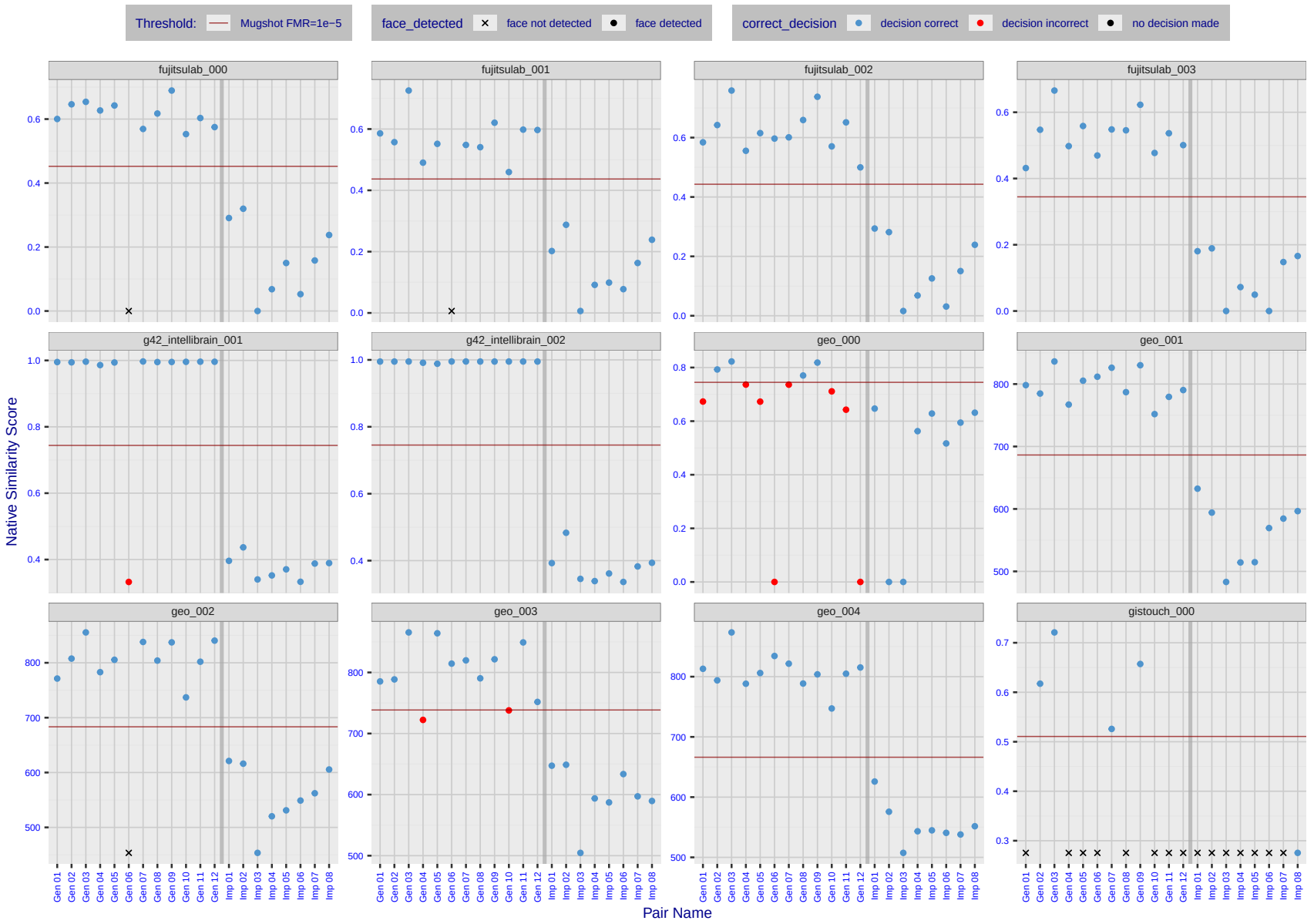


Figure 36: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

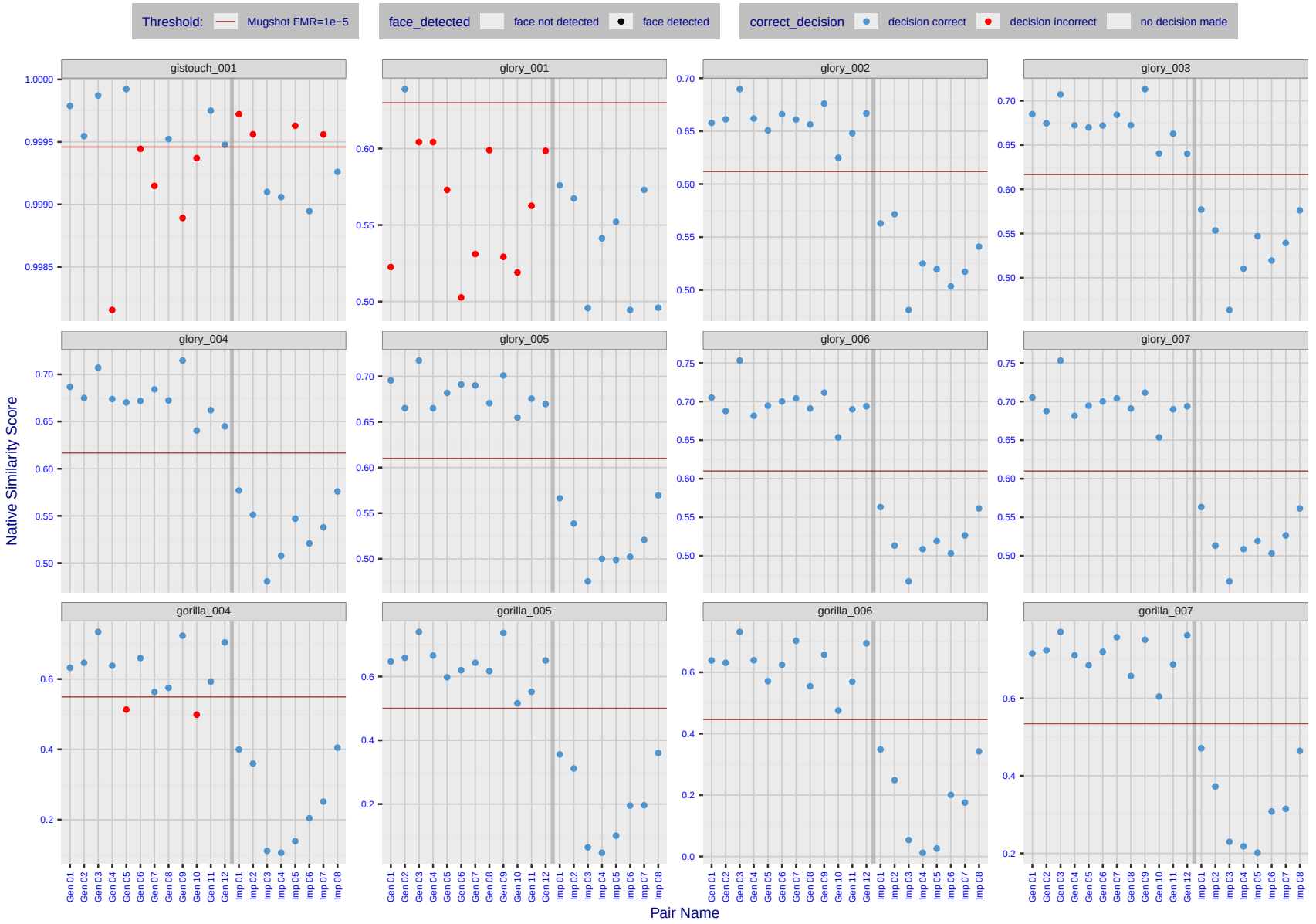


Figure 37: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

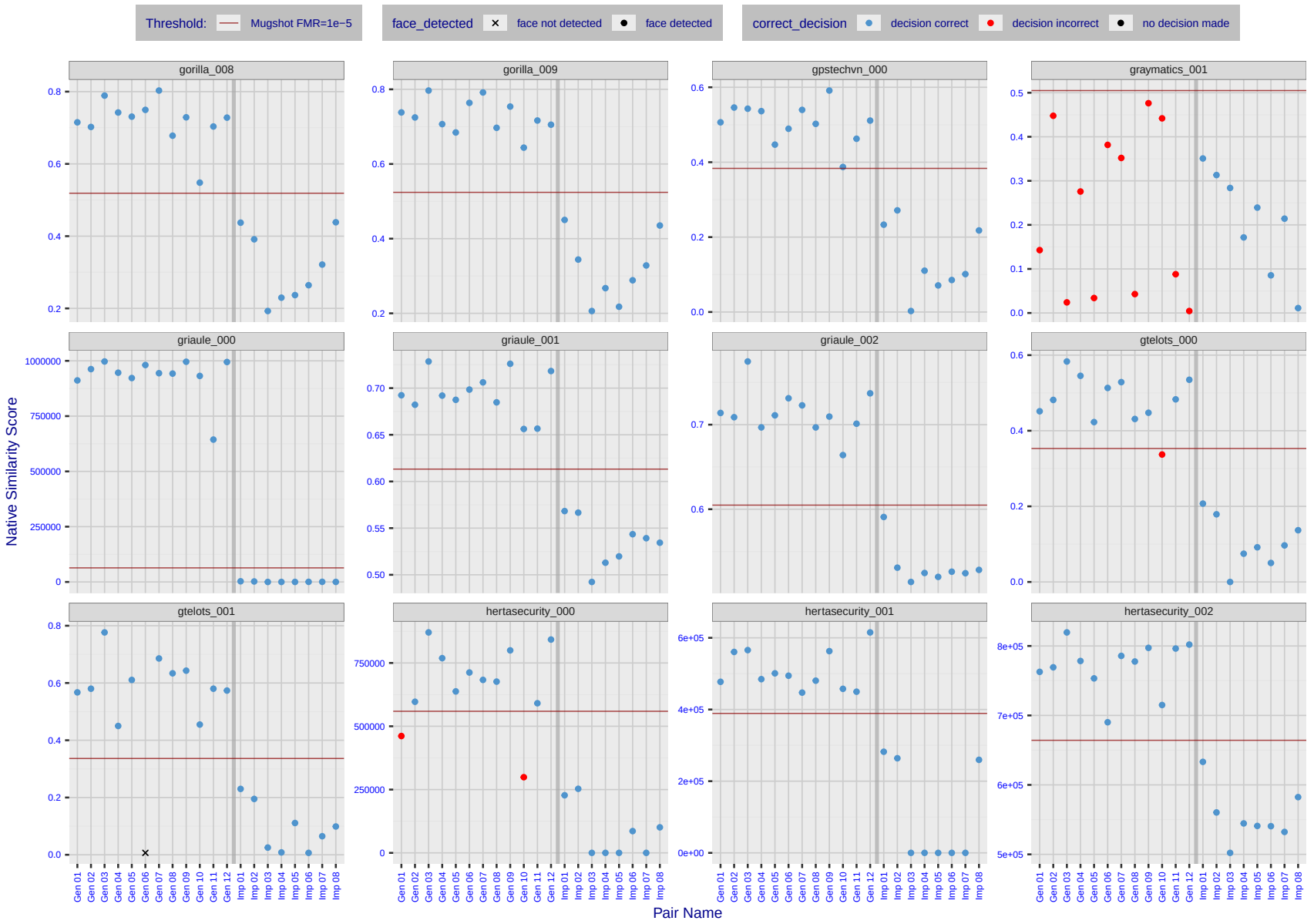


Figure 38: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

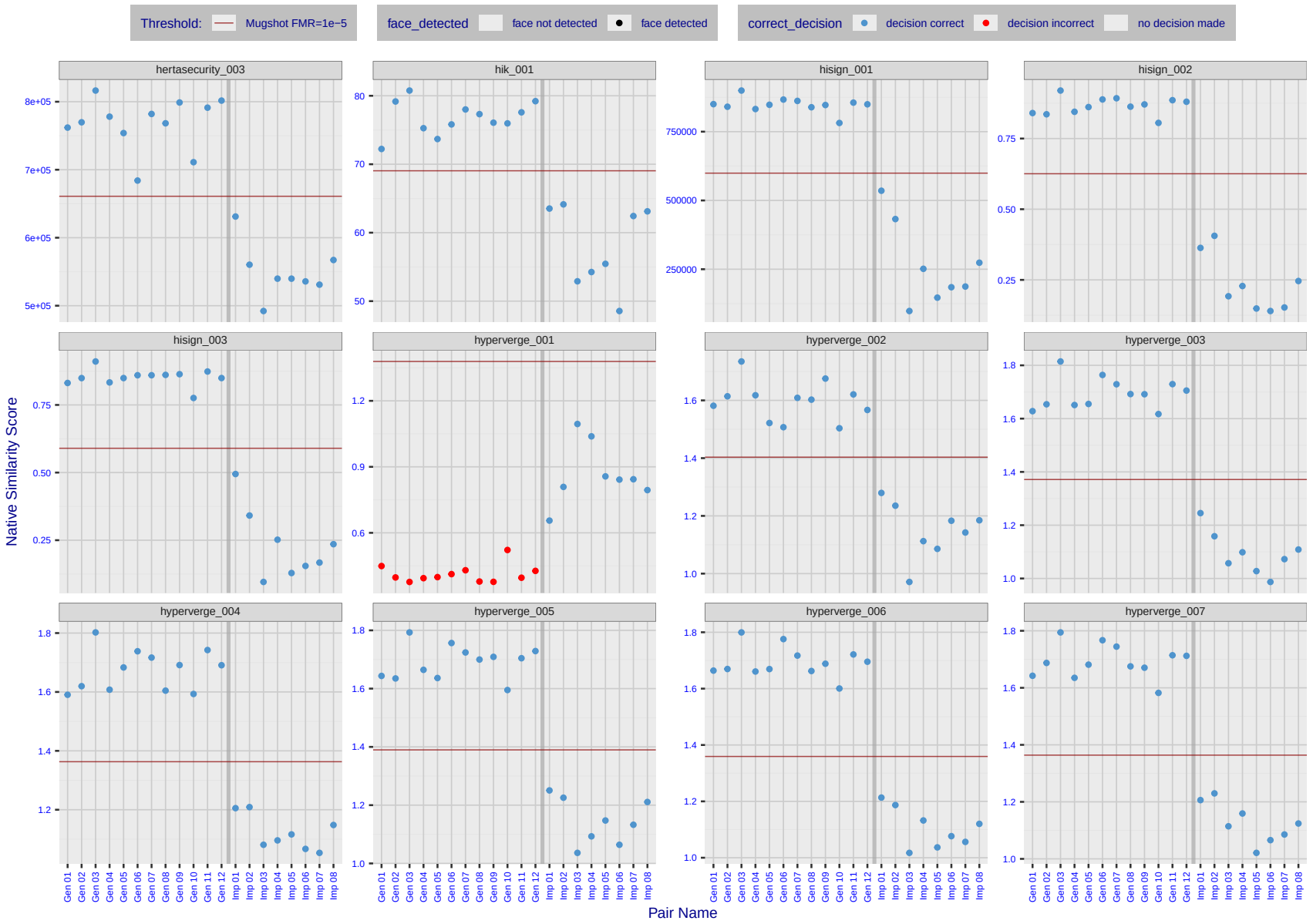


Figure 39: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

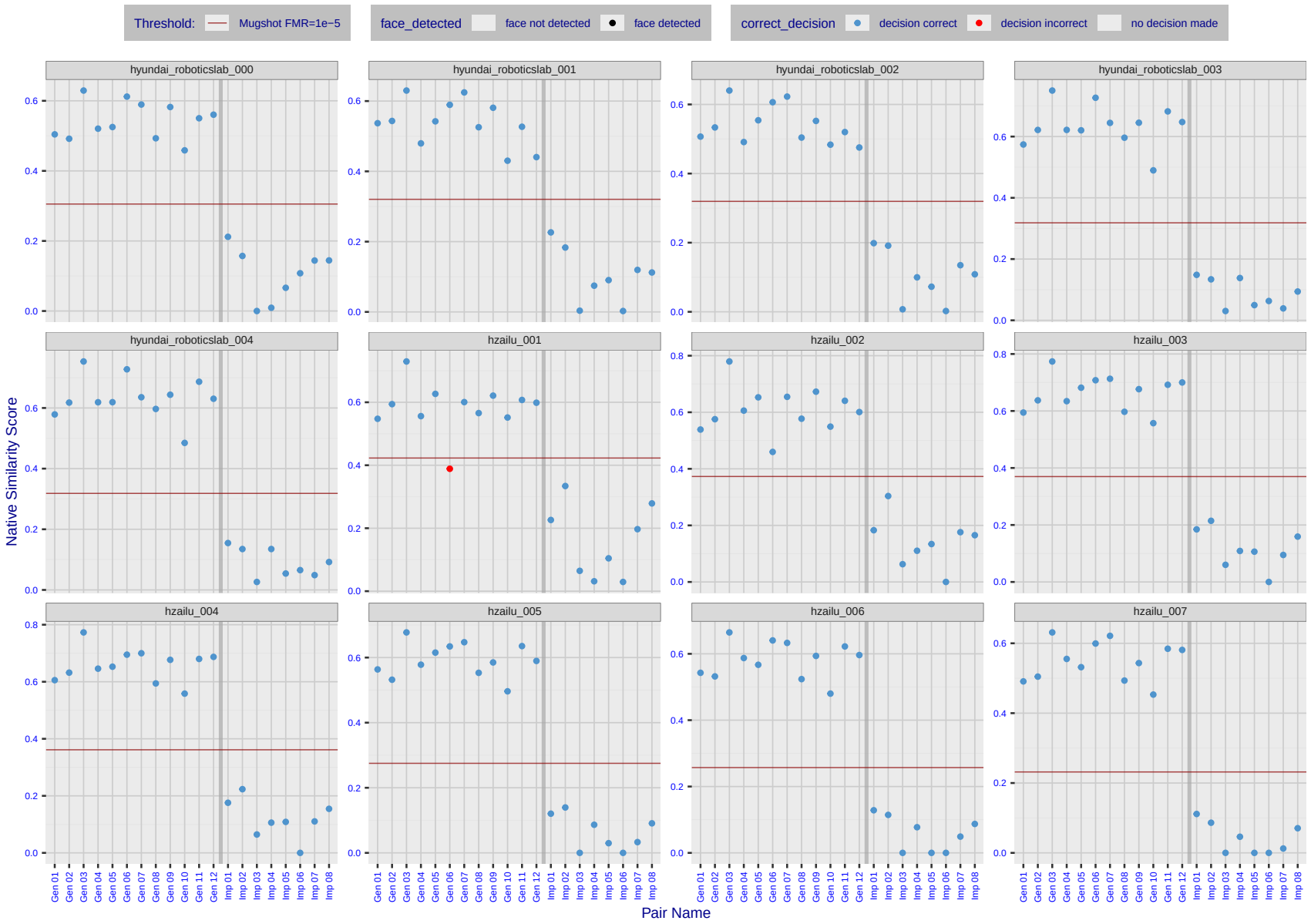


Figure 40: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

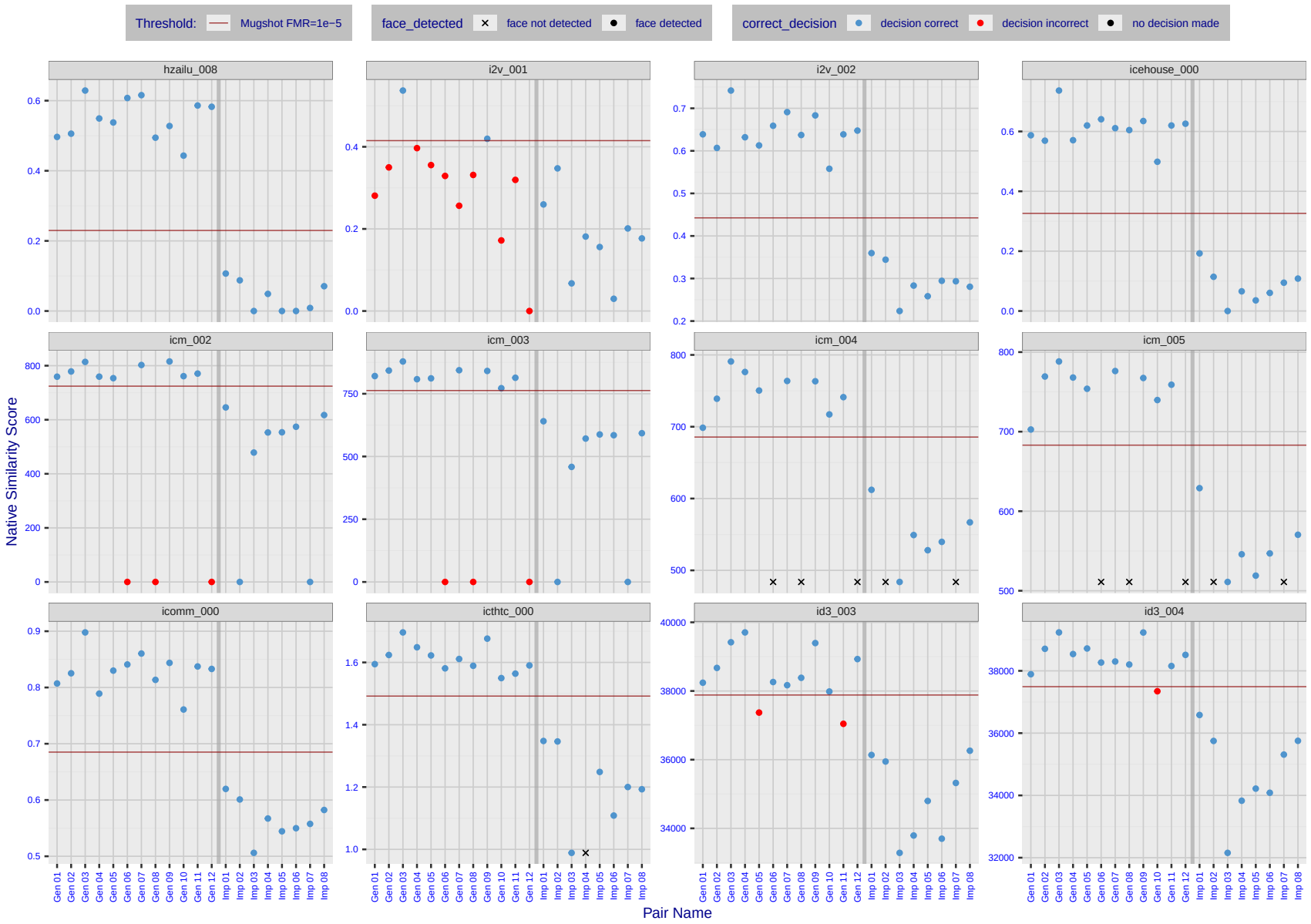


Figure 41: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

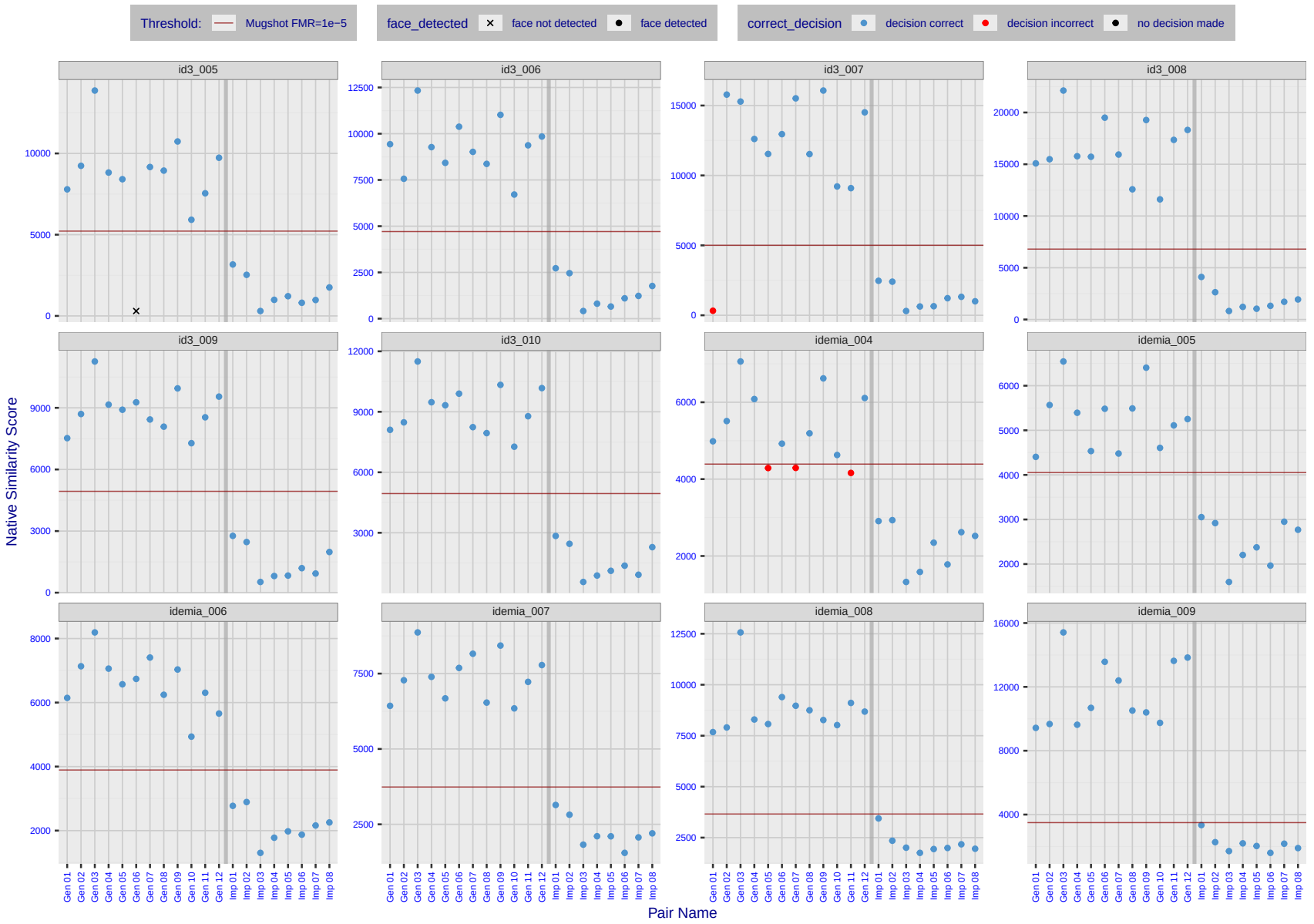


Figure 42: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

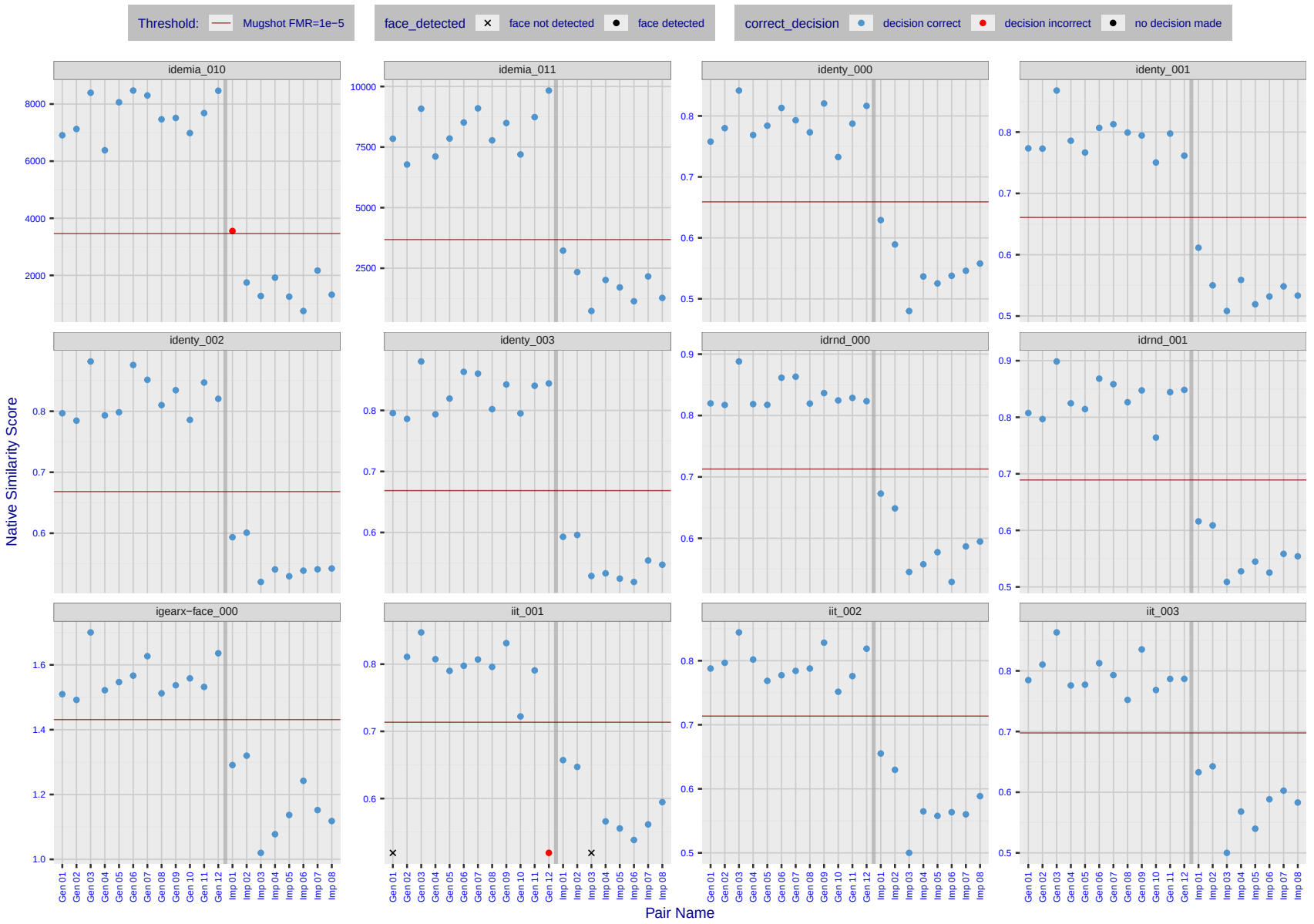


Figure 43: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

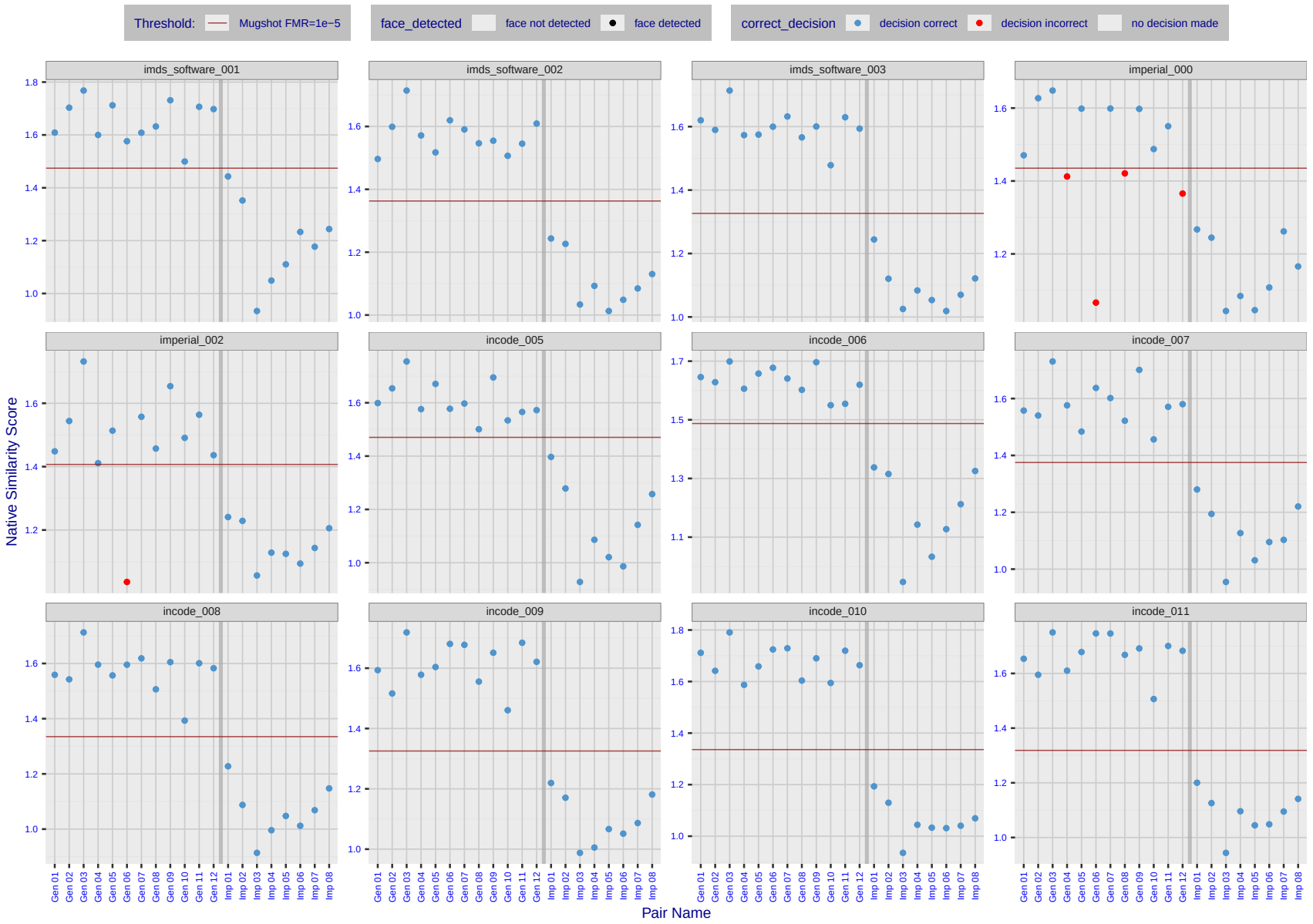


Figure 44: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

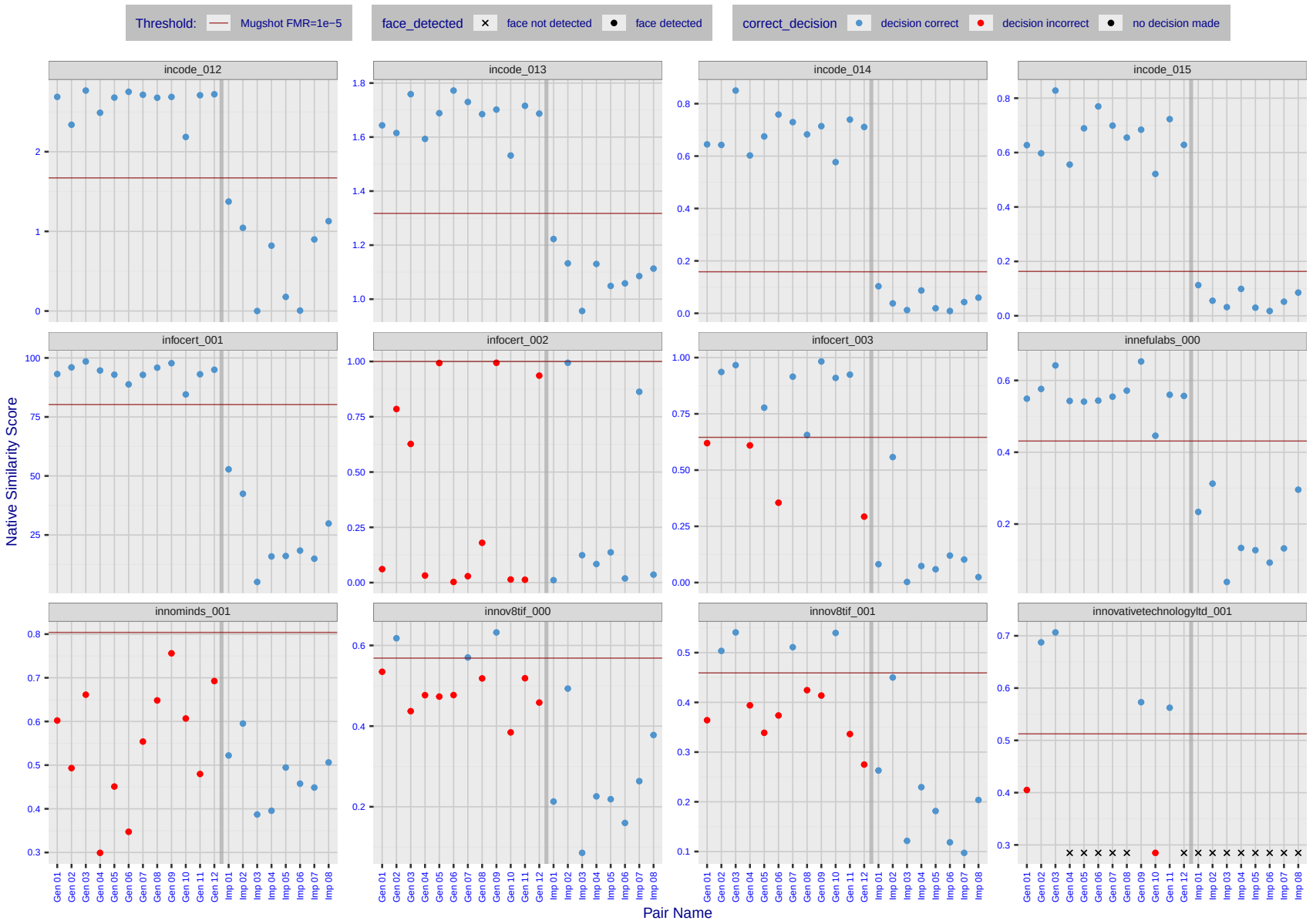


Figure 45: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

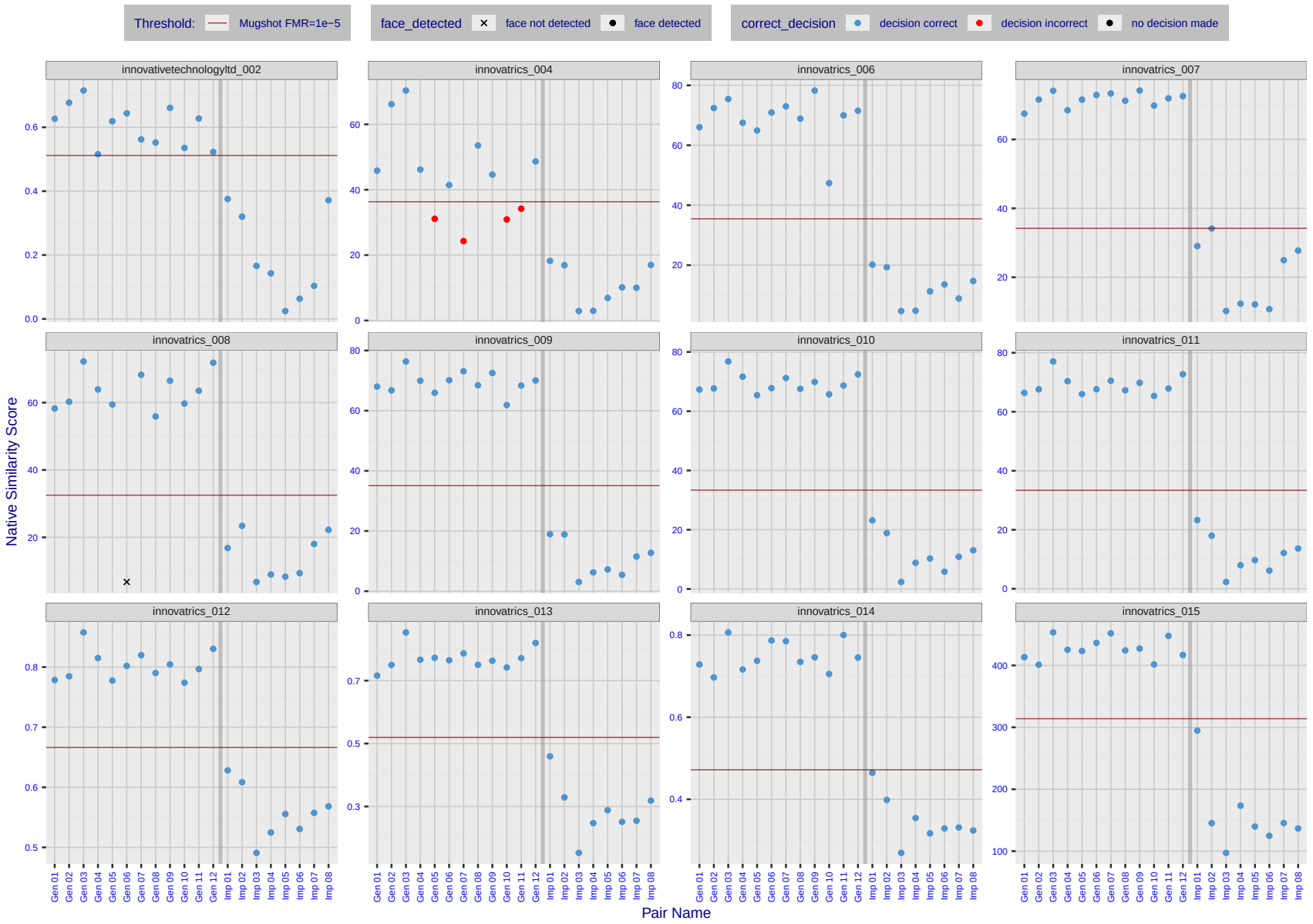


Figure 46: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

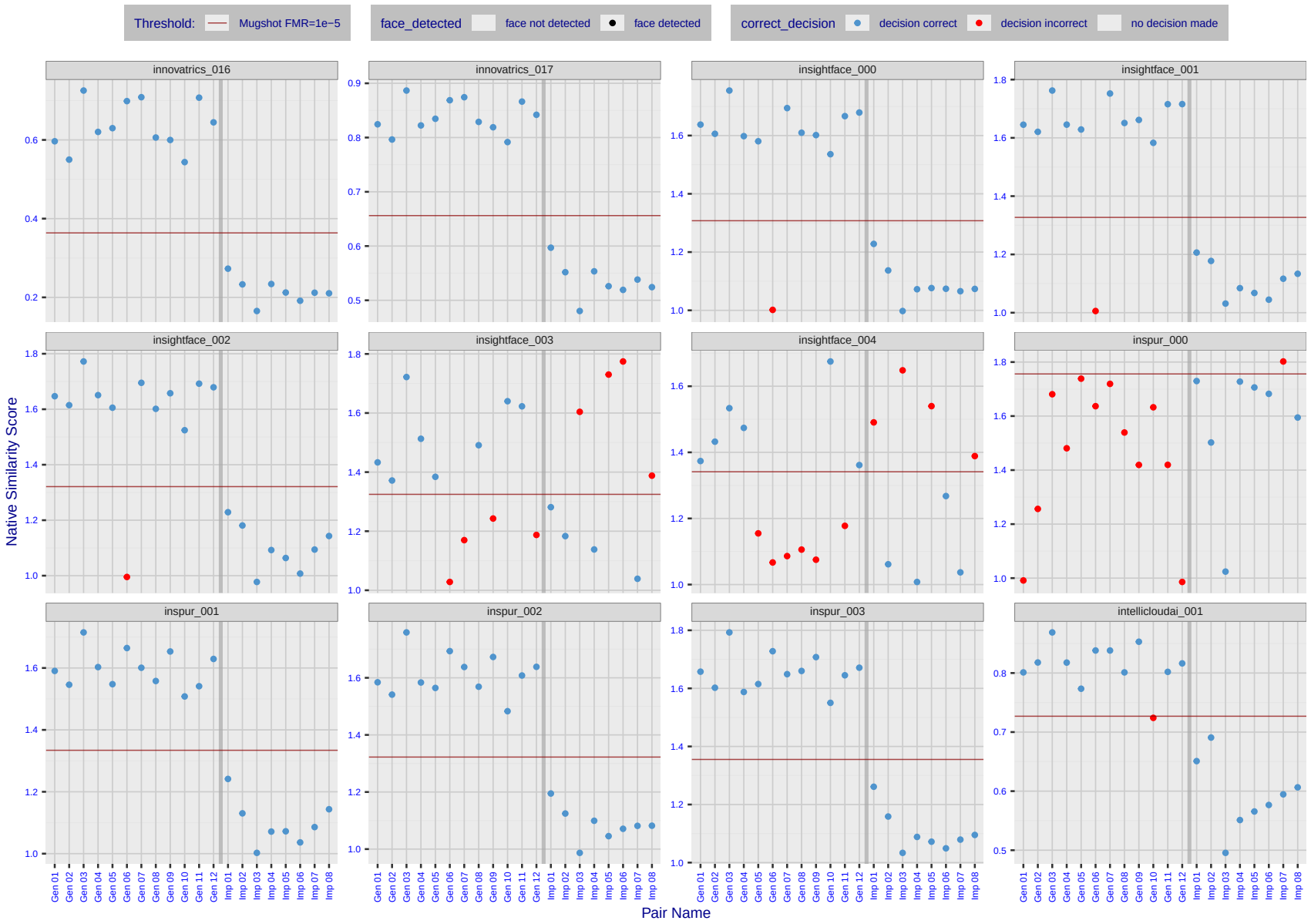


Figure 47: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

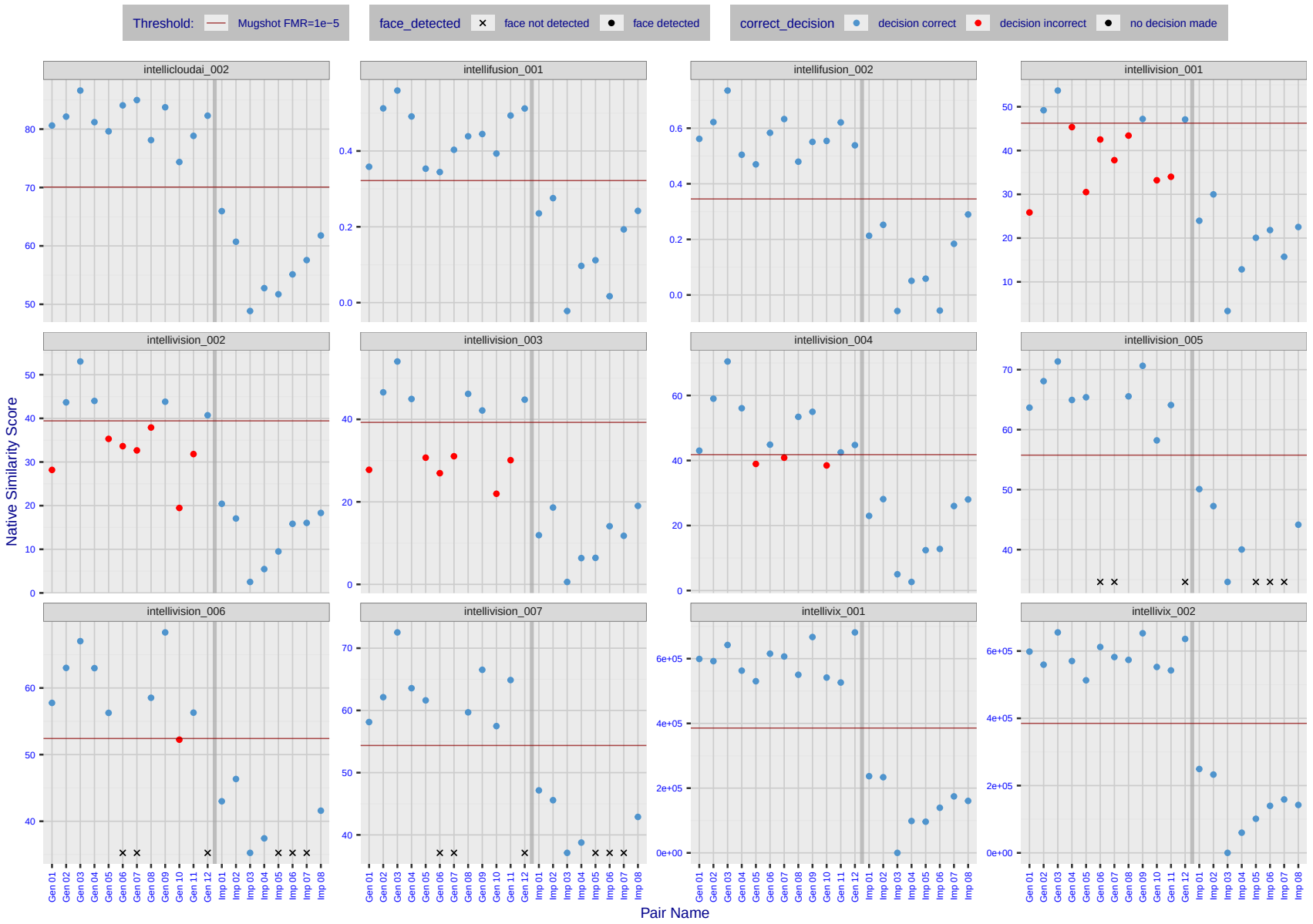


Figure 48: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

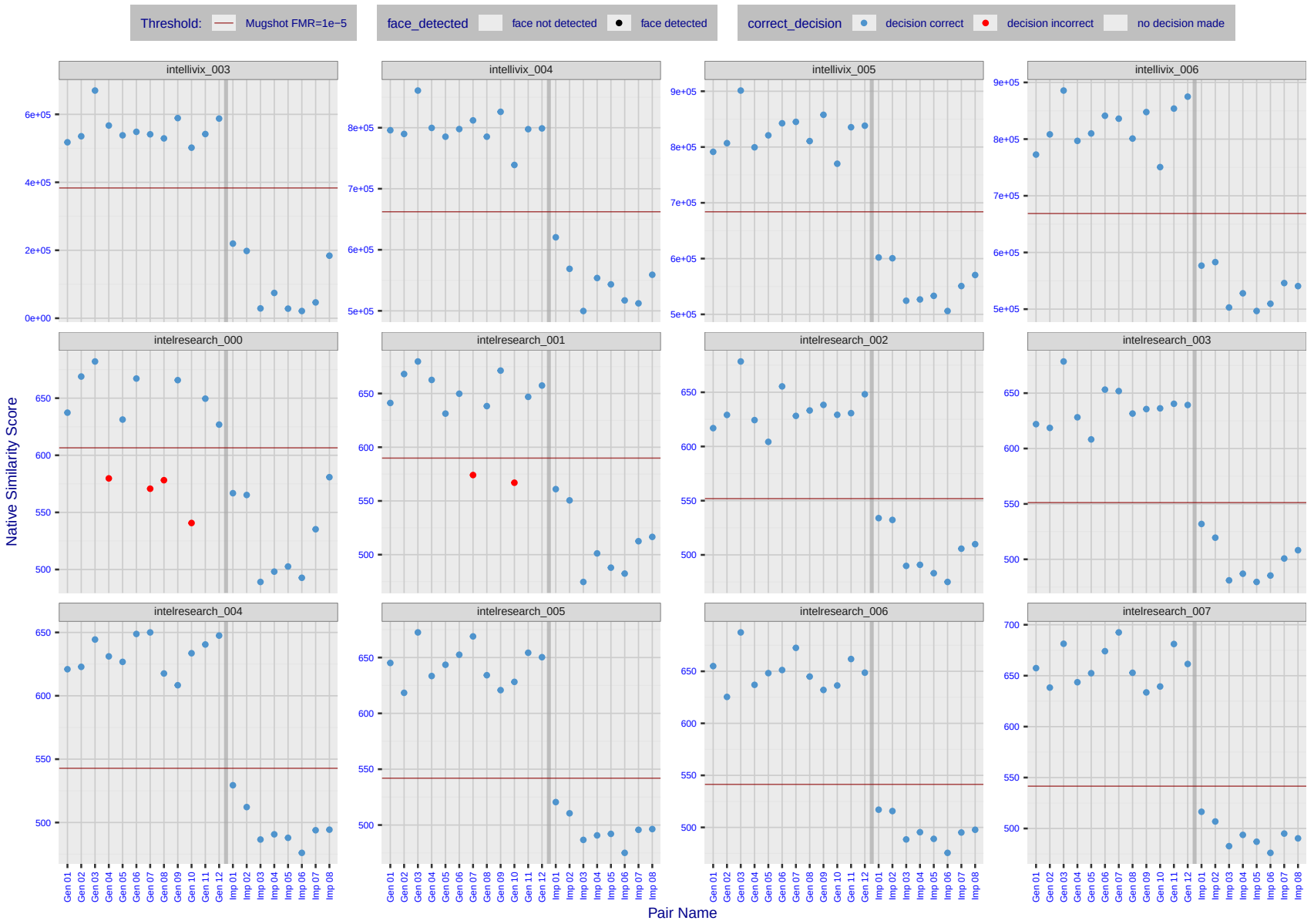


Figure 49: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

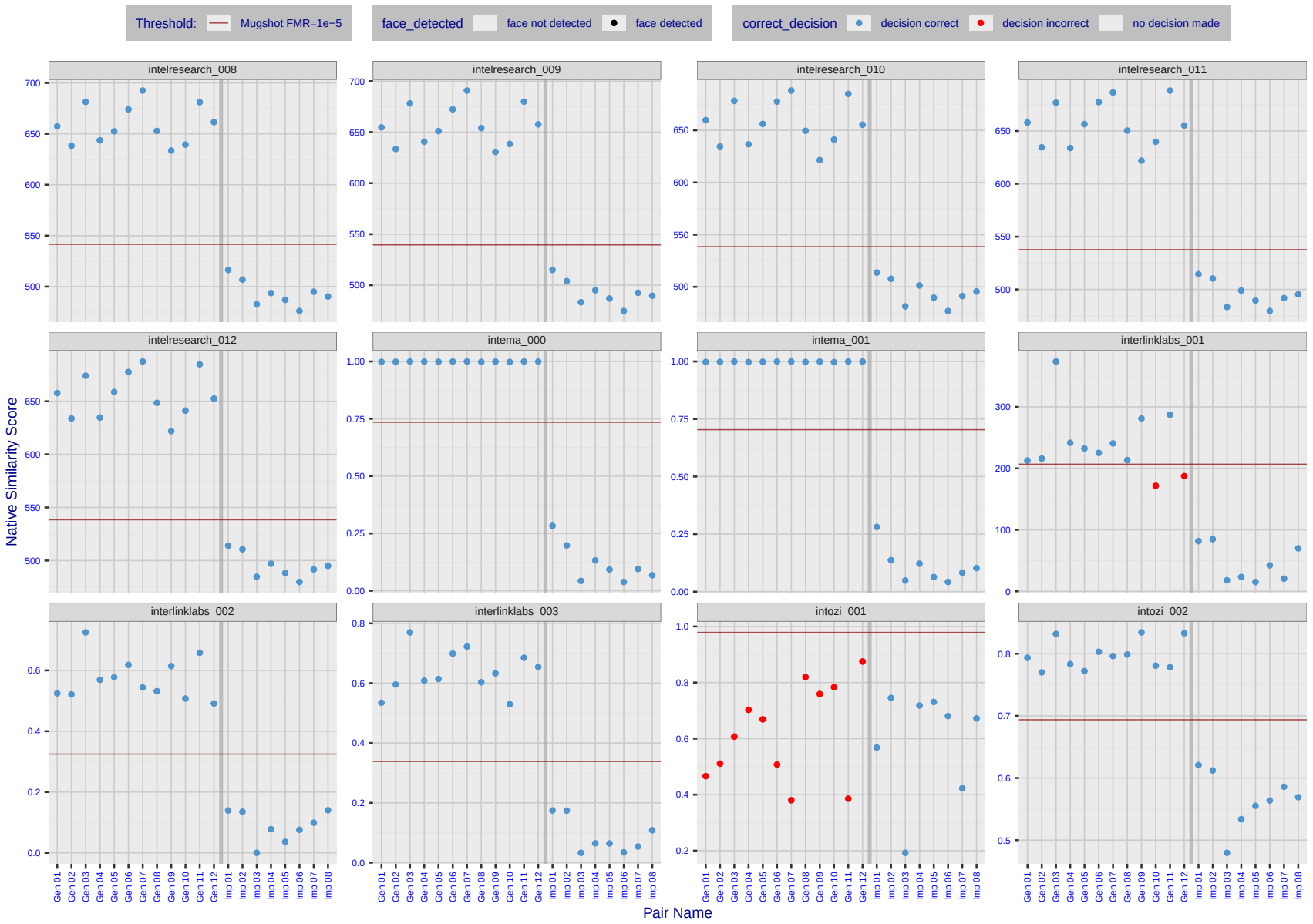


Figure 50: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

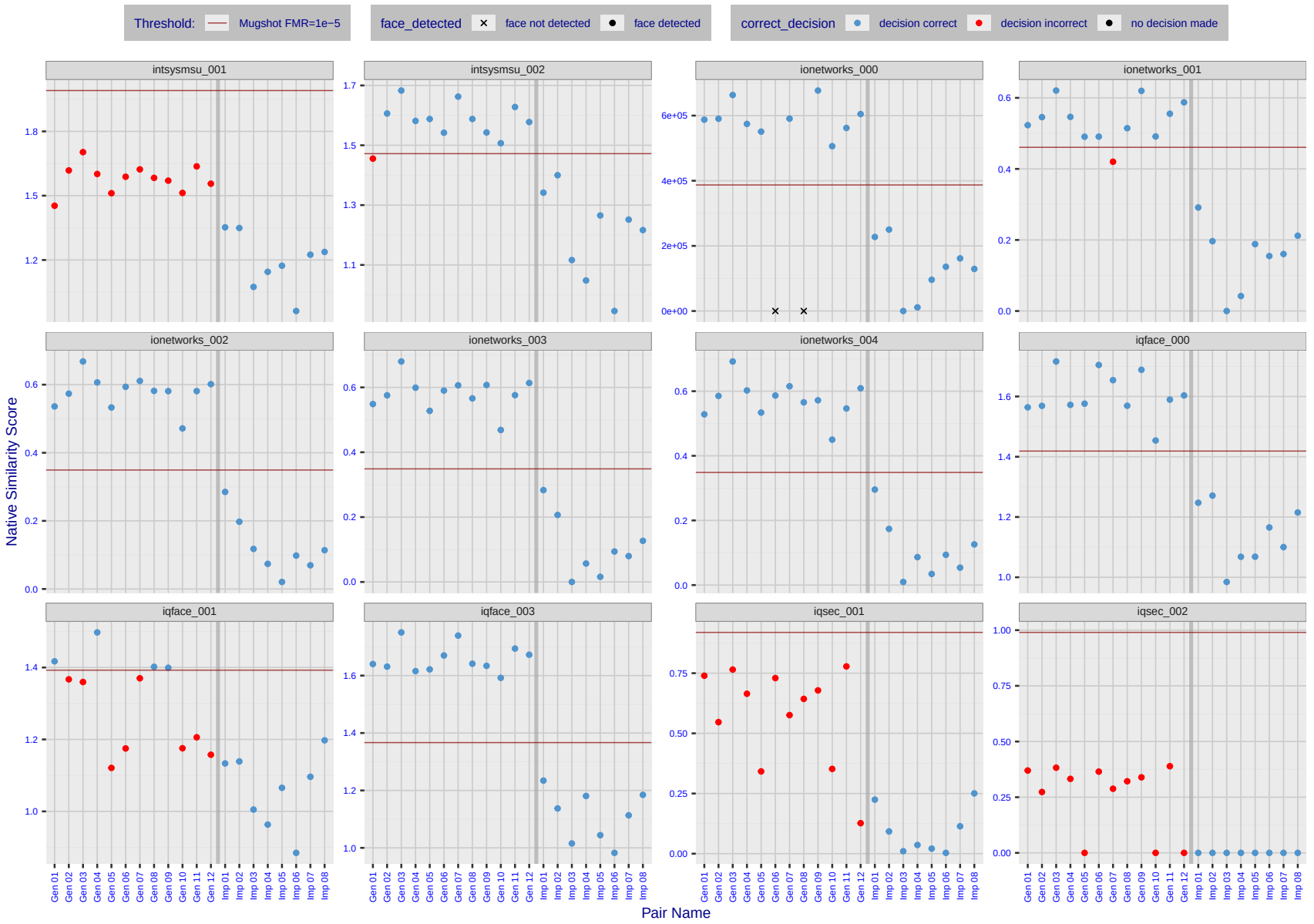


Figure 51: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

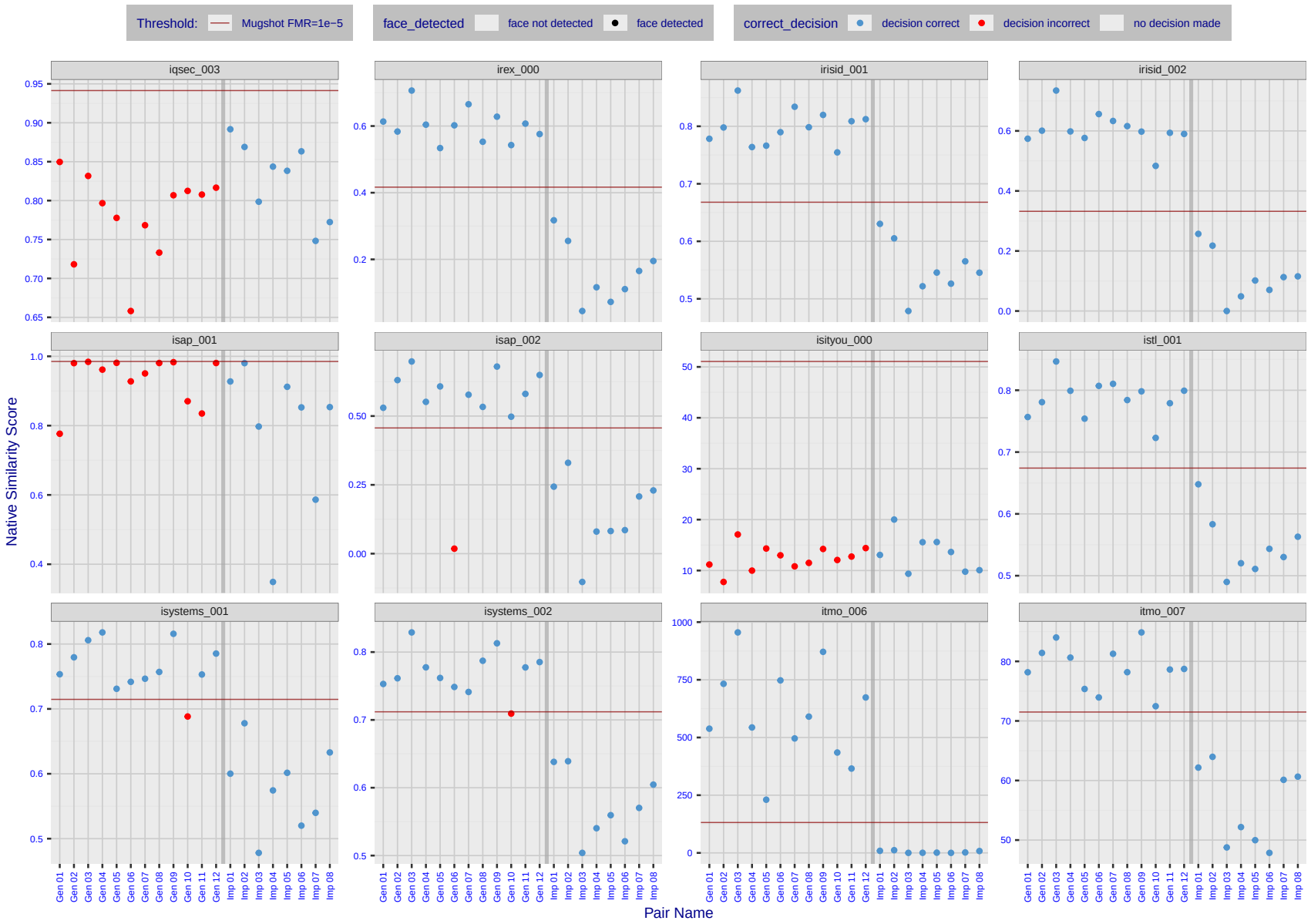


Figure 52: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

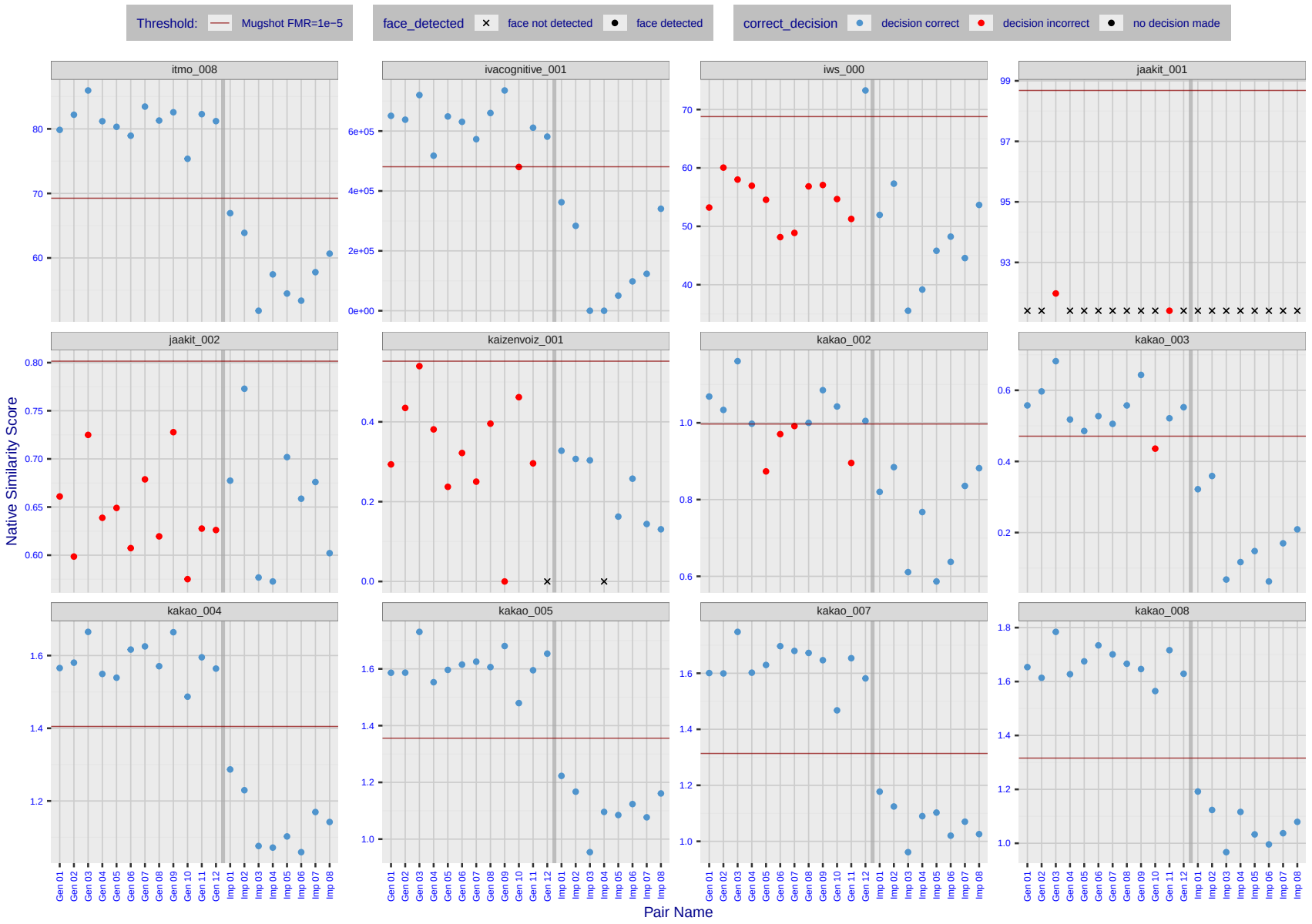


Figure 53: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

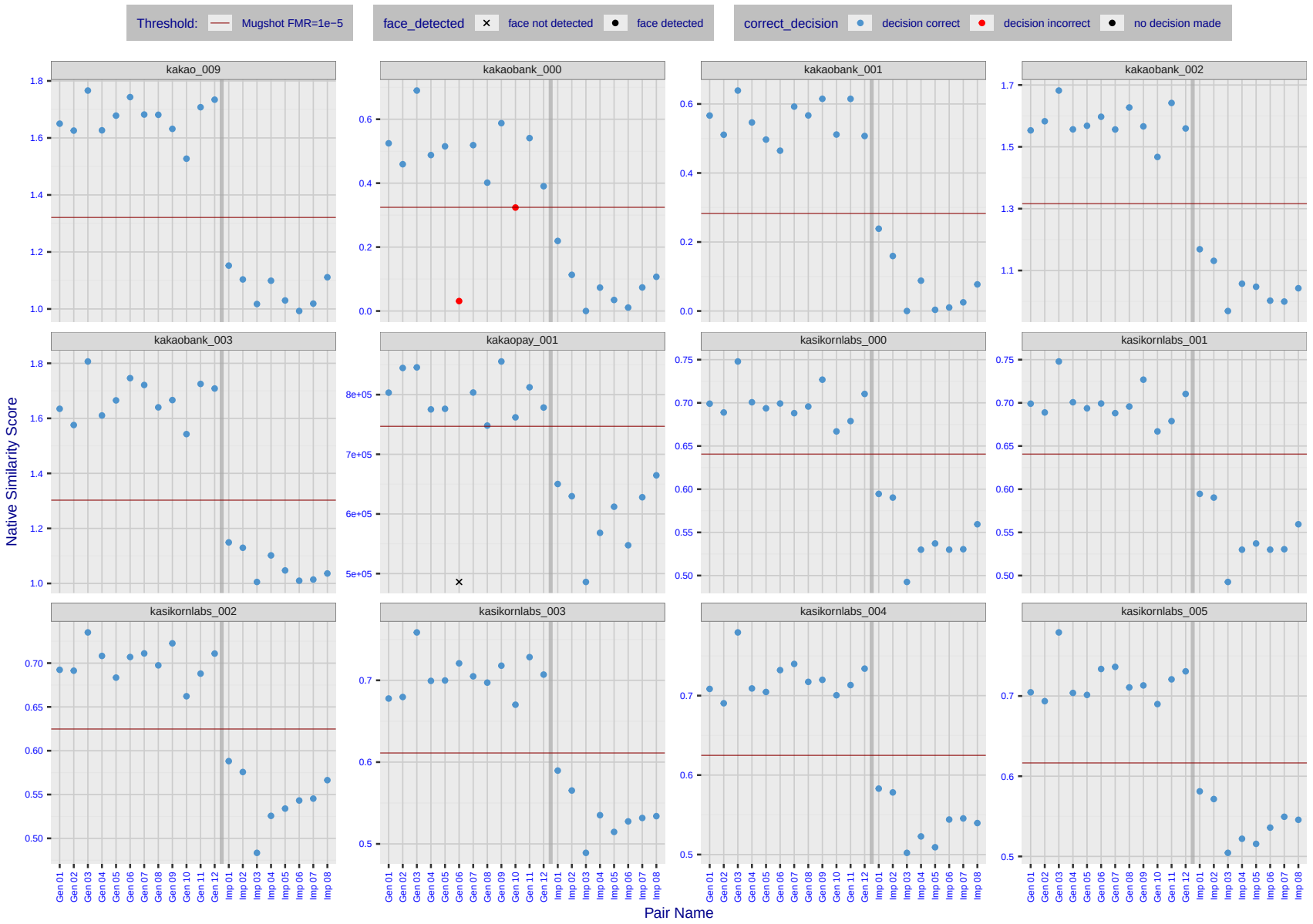


Figure 54: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

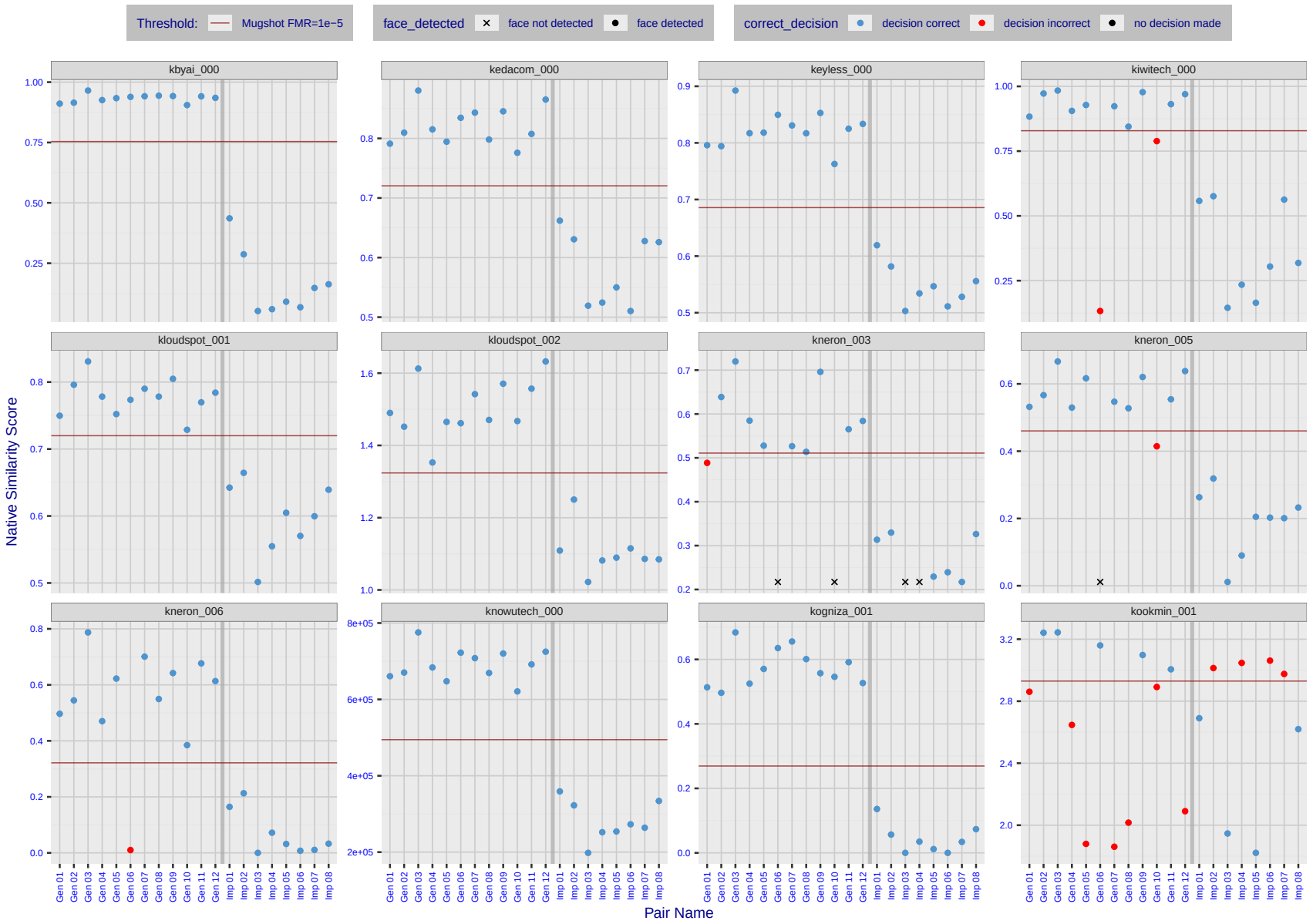


Figure 55: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

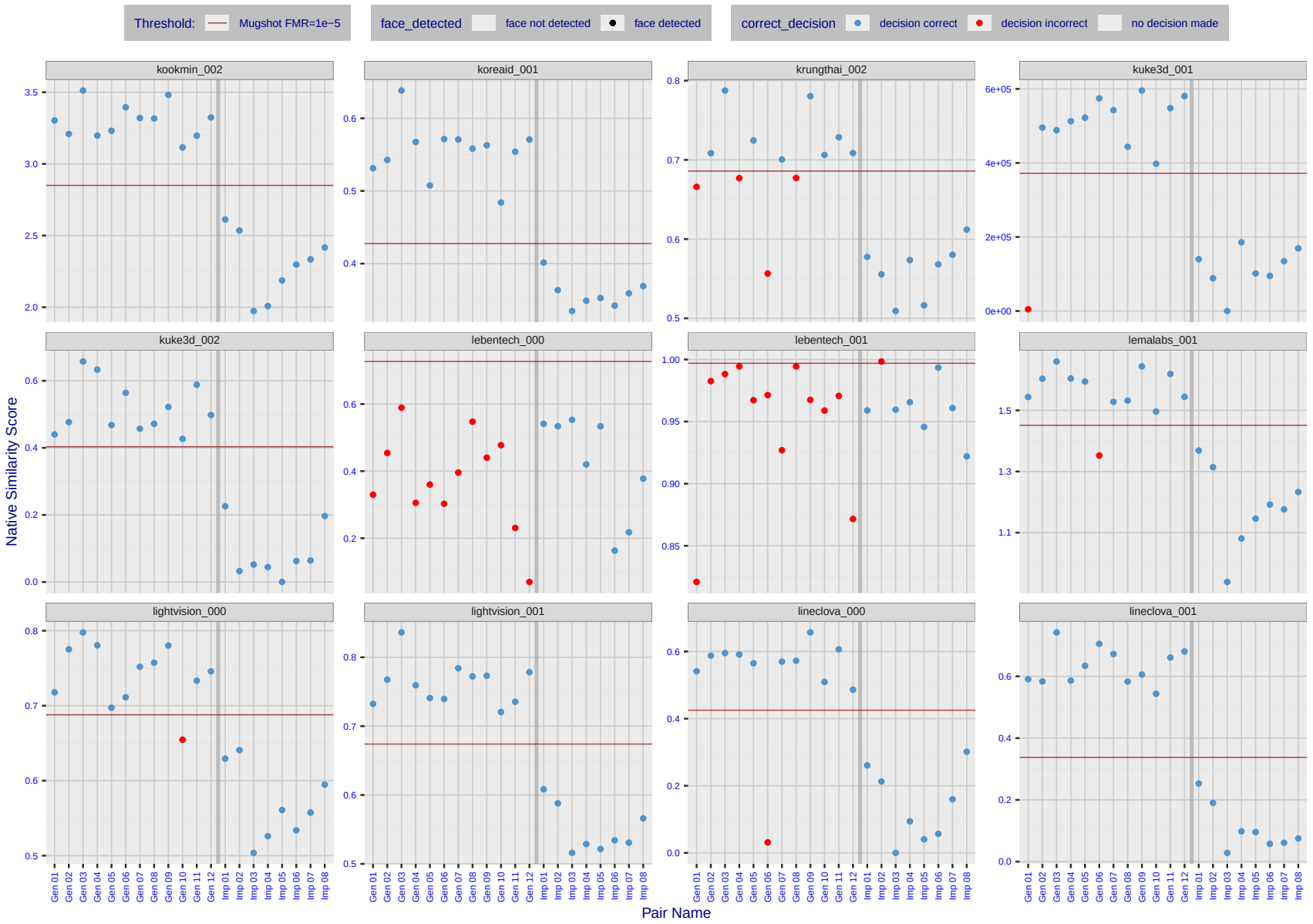


Figure 56: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

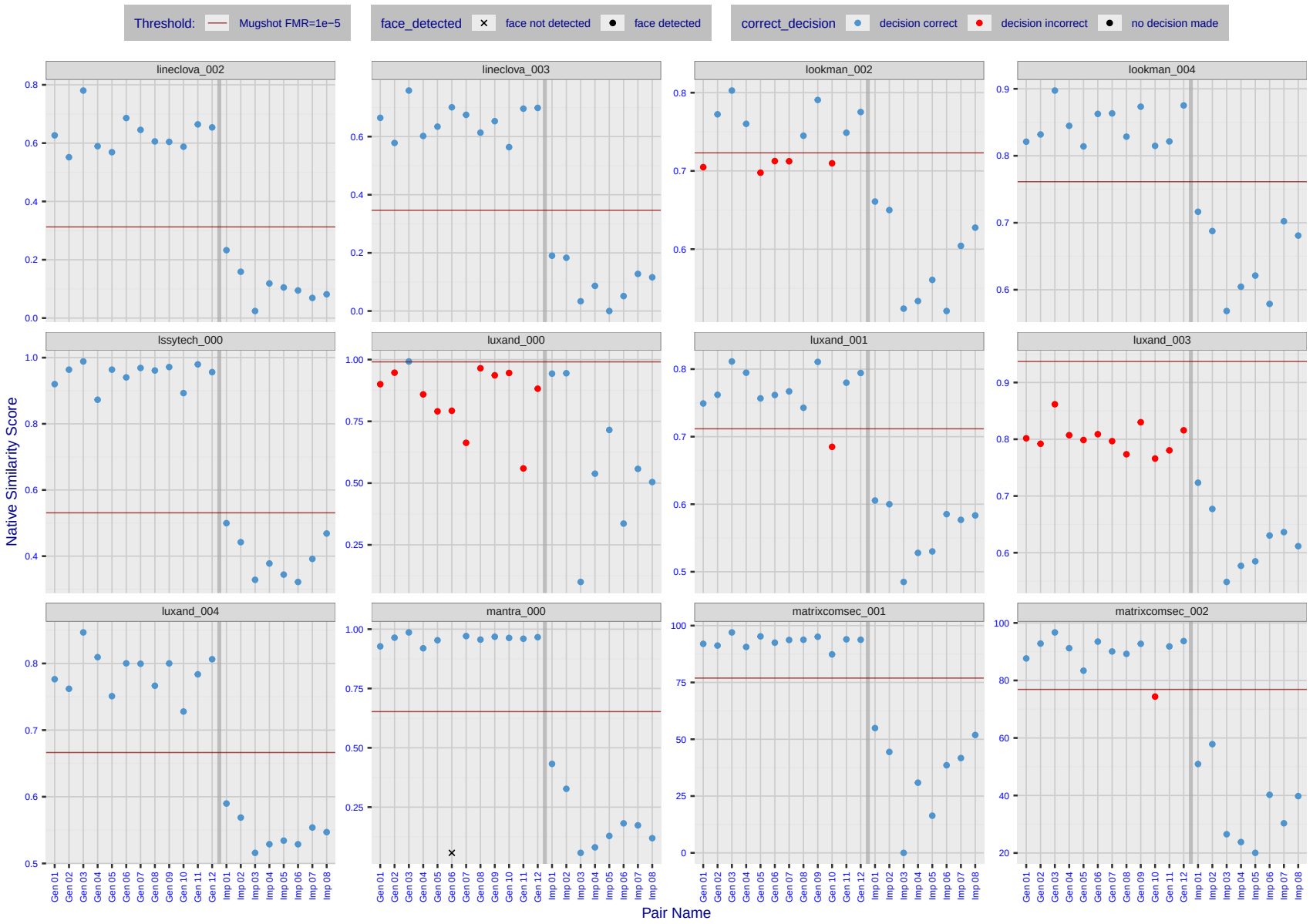


Figure 57: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

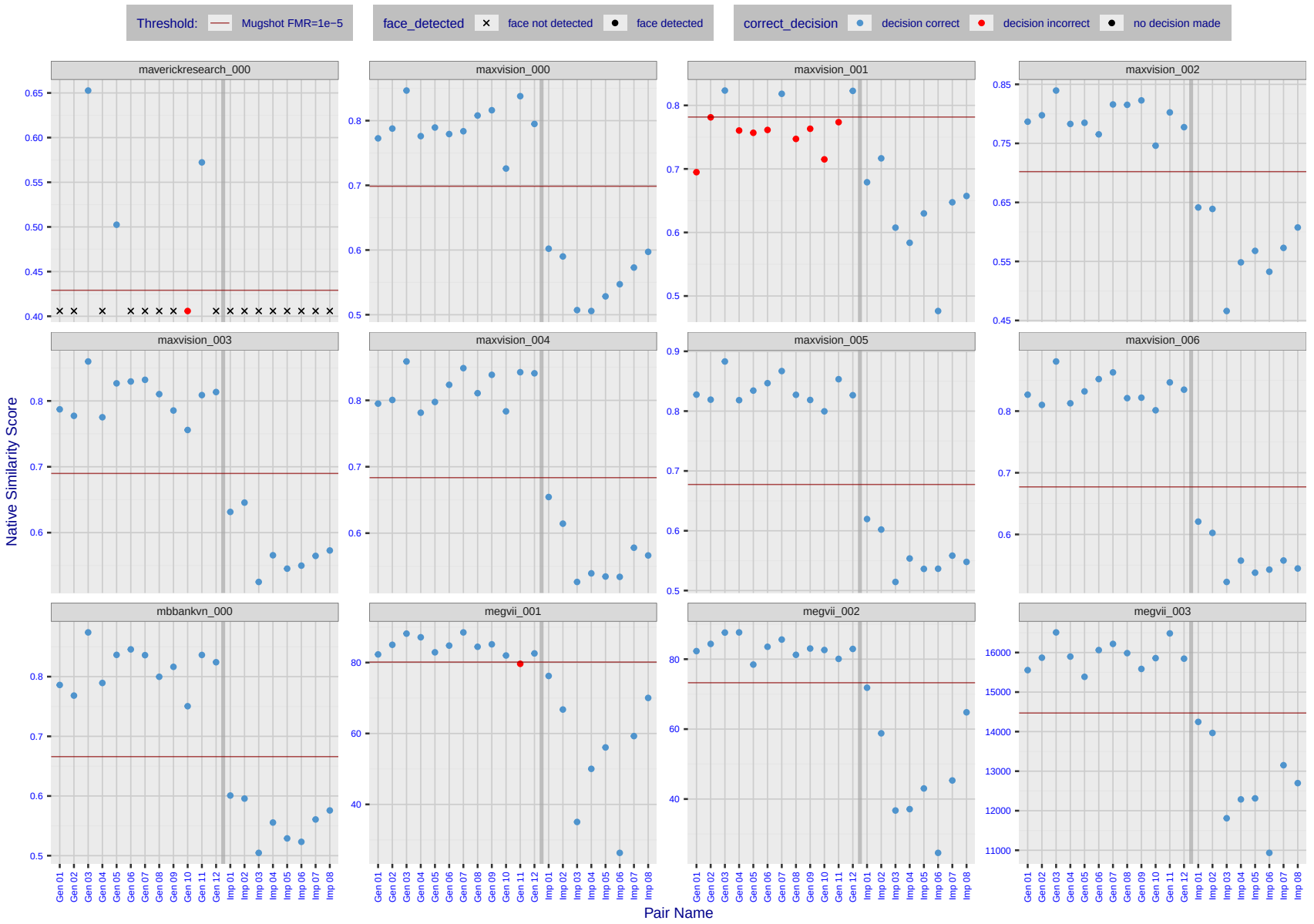


Figure 58: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

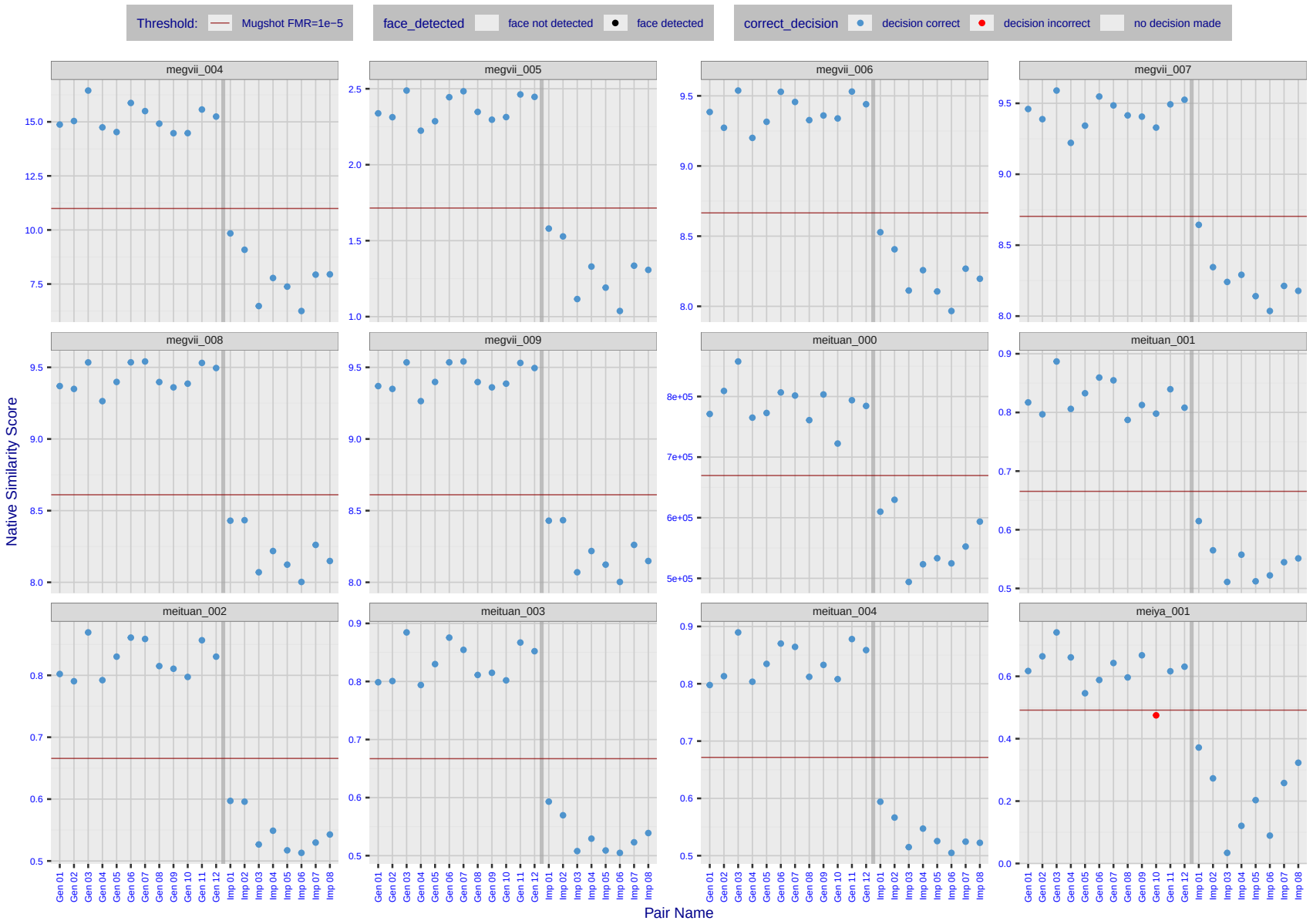


Figure 59: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

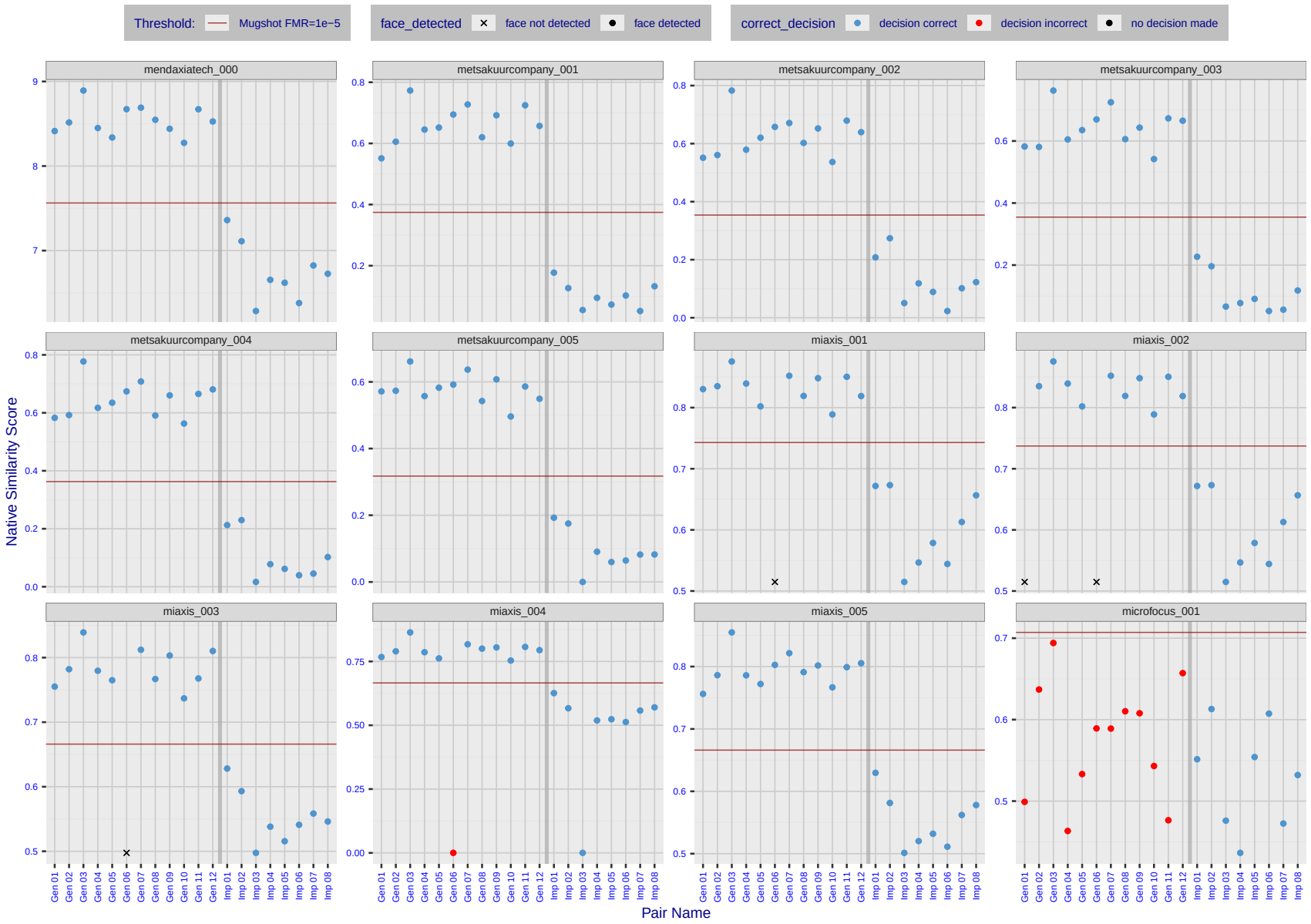


Figure 60: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

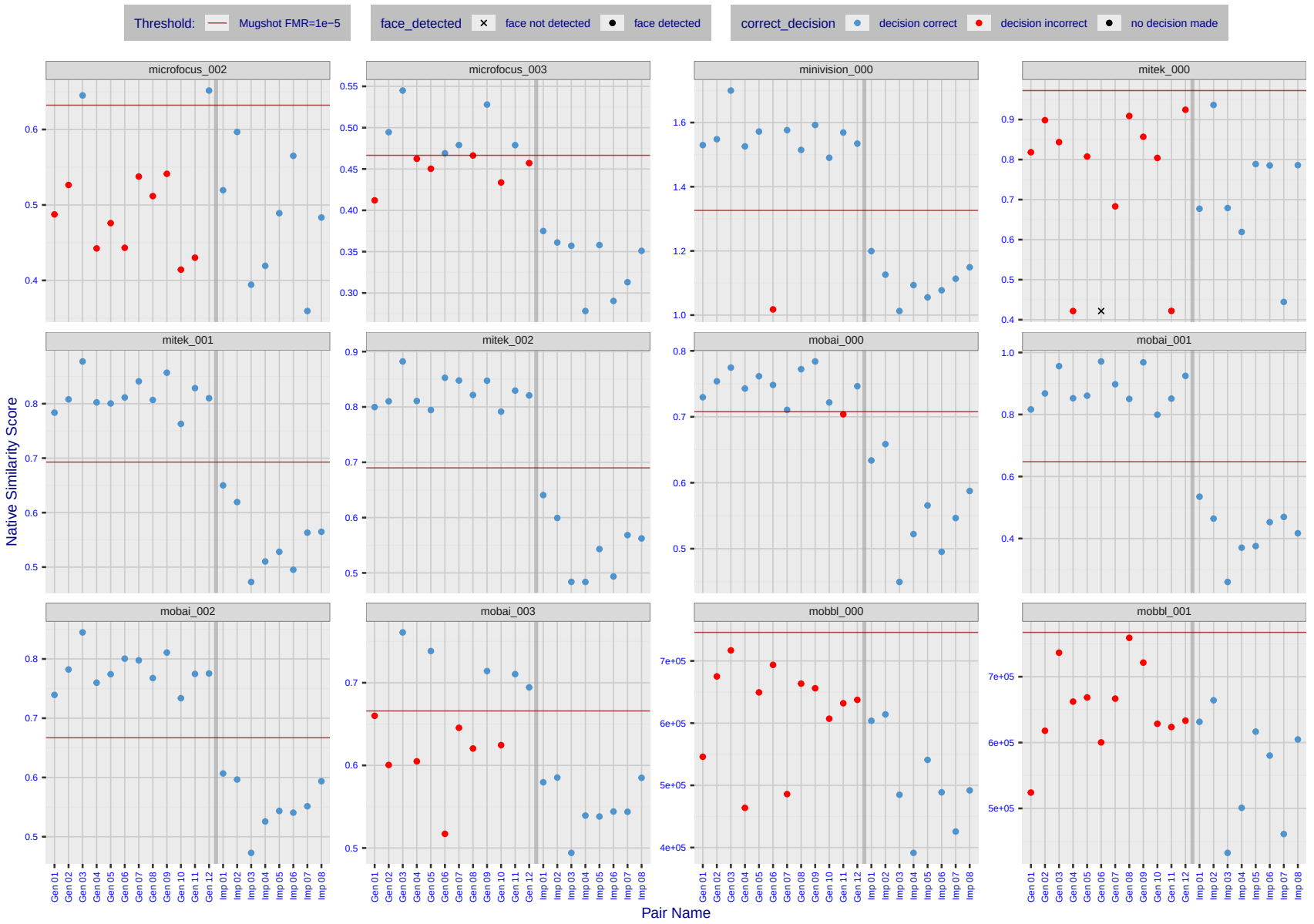


Figure 61: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

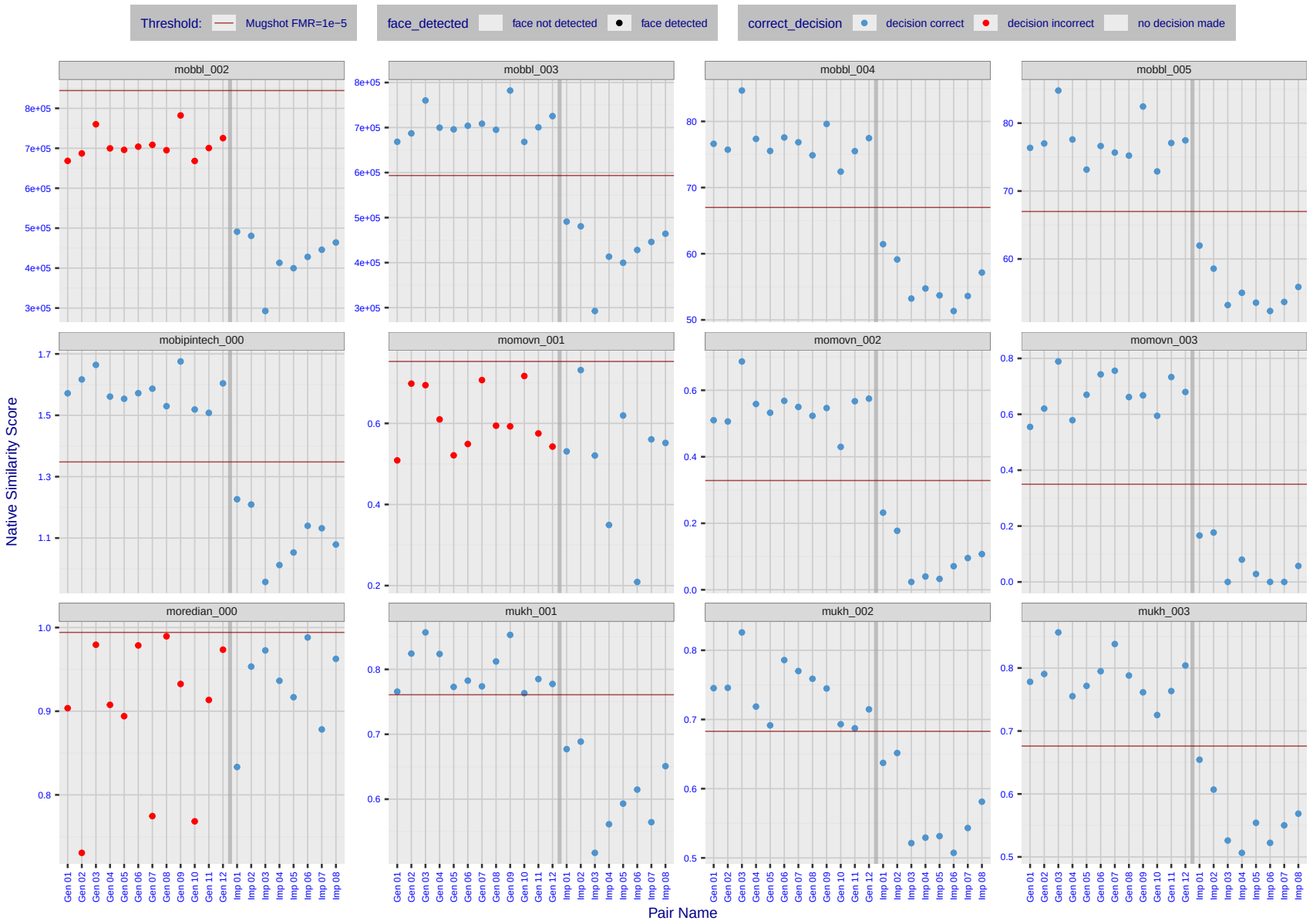


Figure 62: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

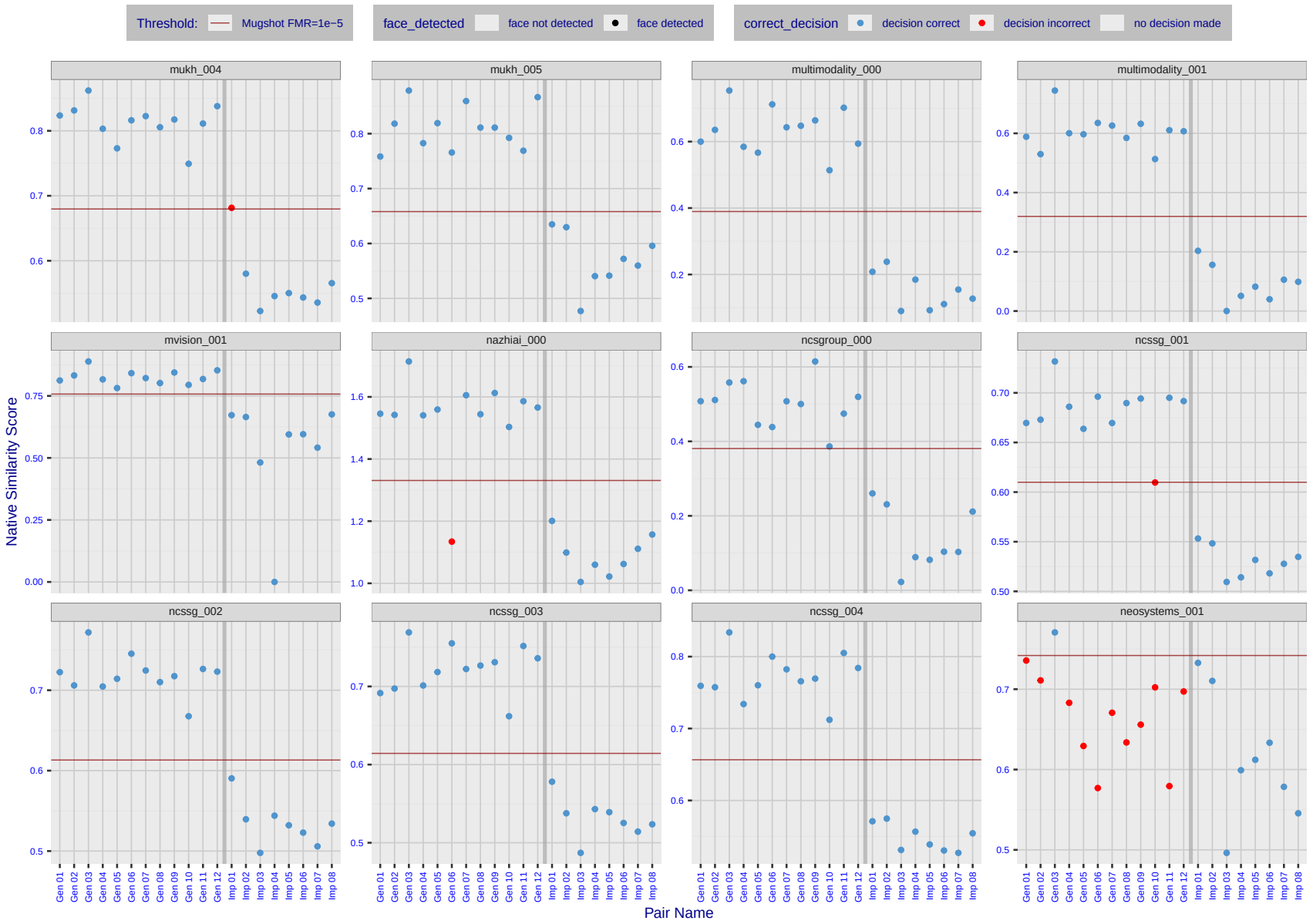


Figure 63: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

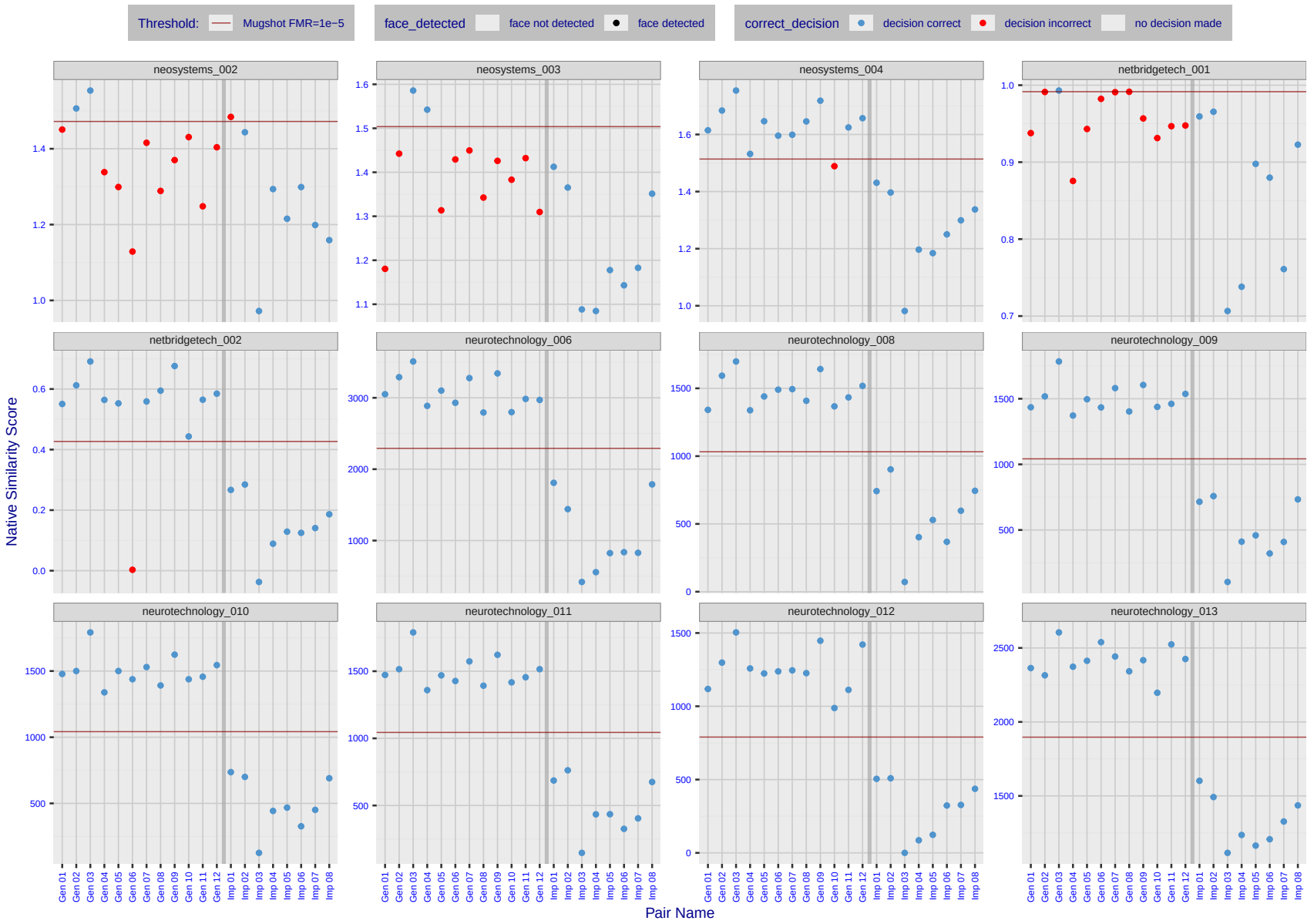


Figure 64: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

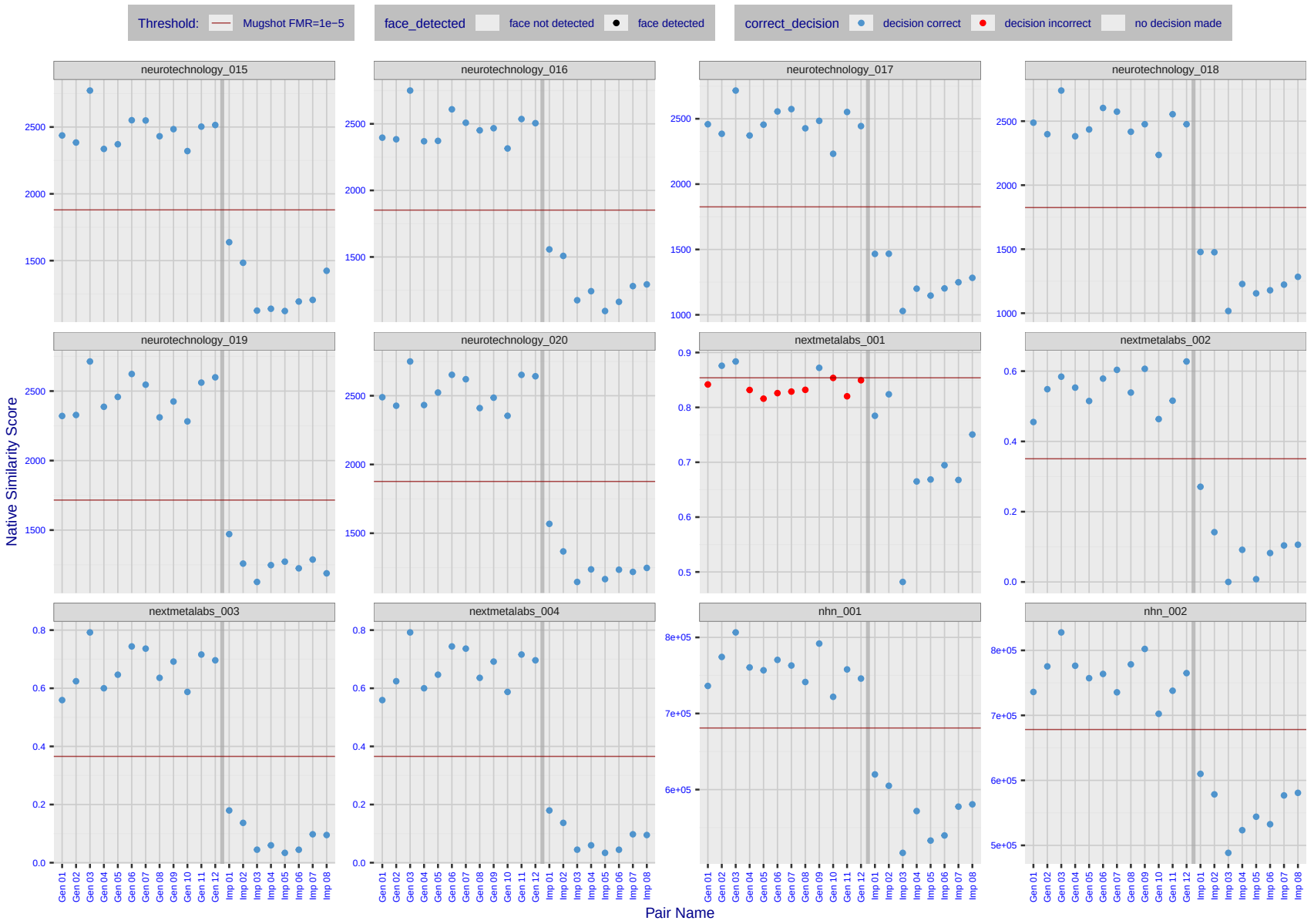


Figure 65: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

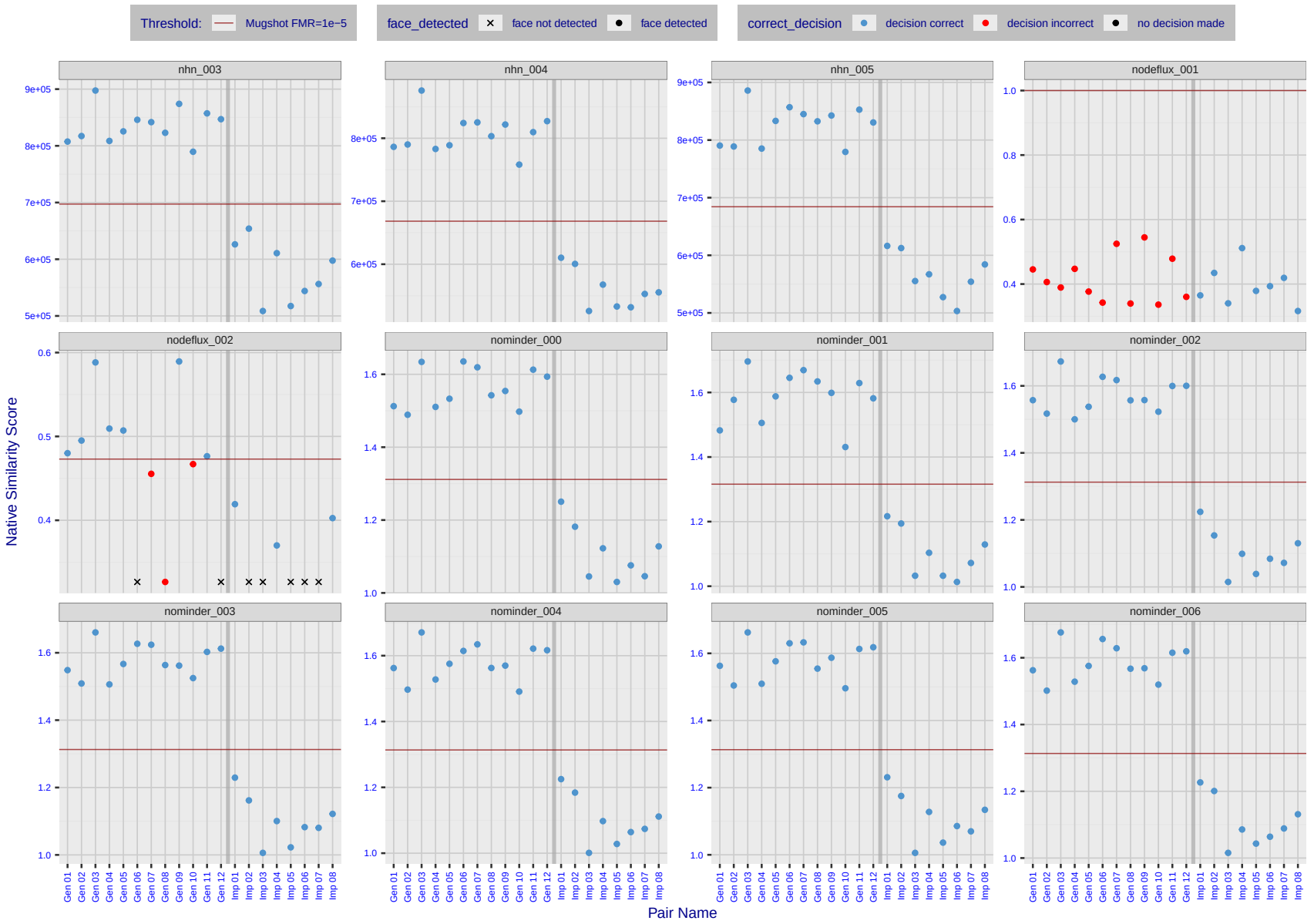


Figure 66: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

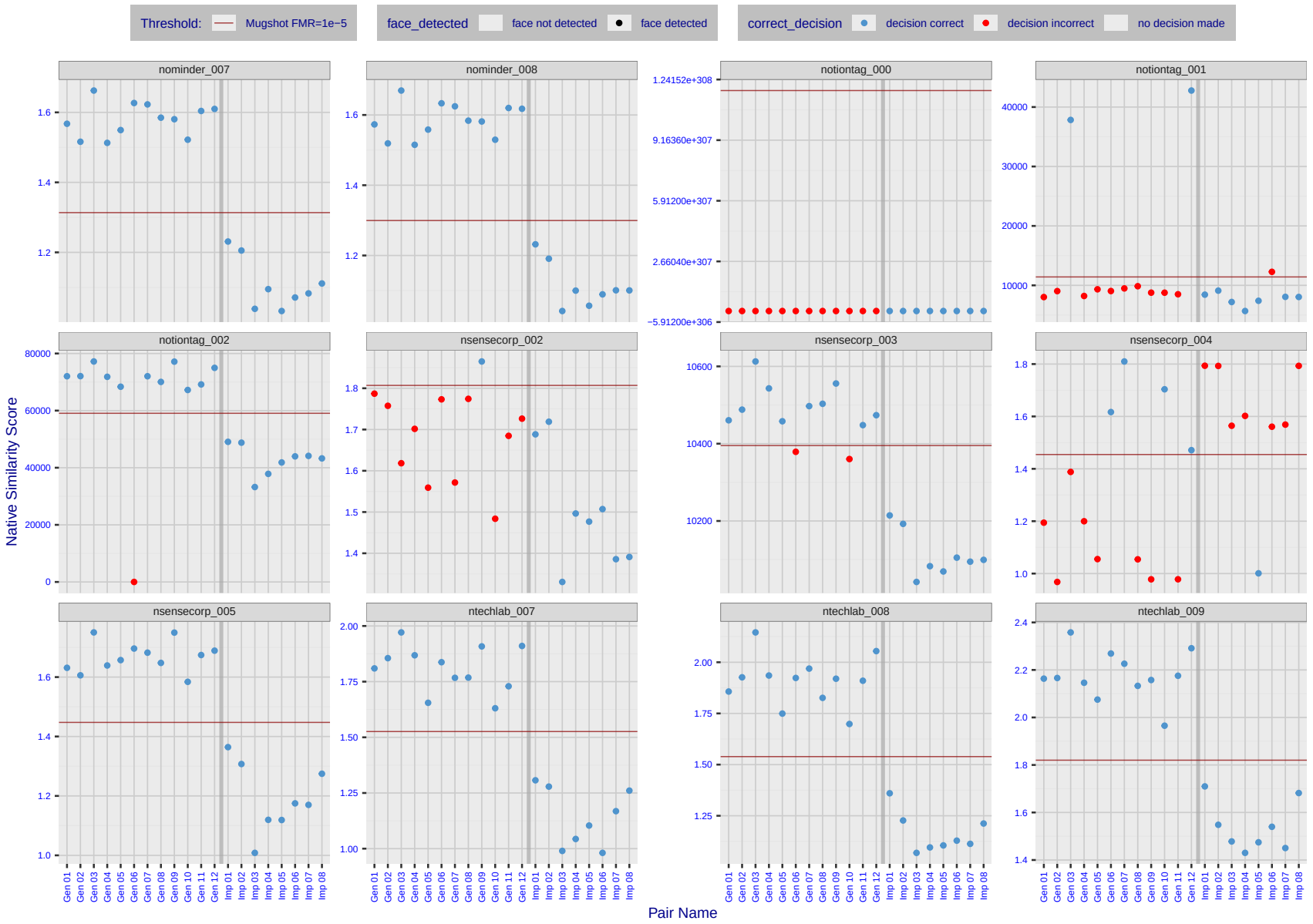


Figure 67: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

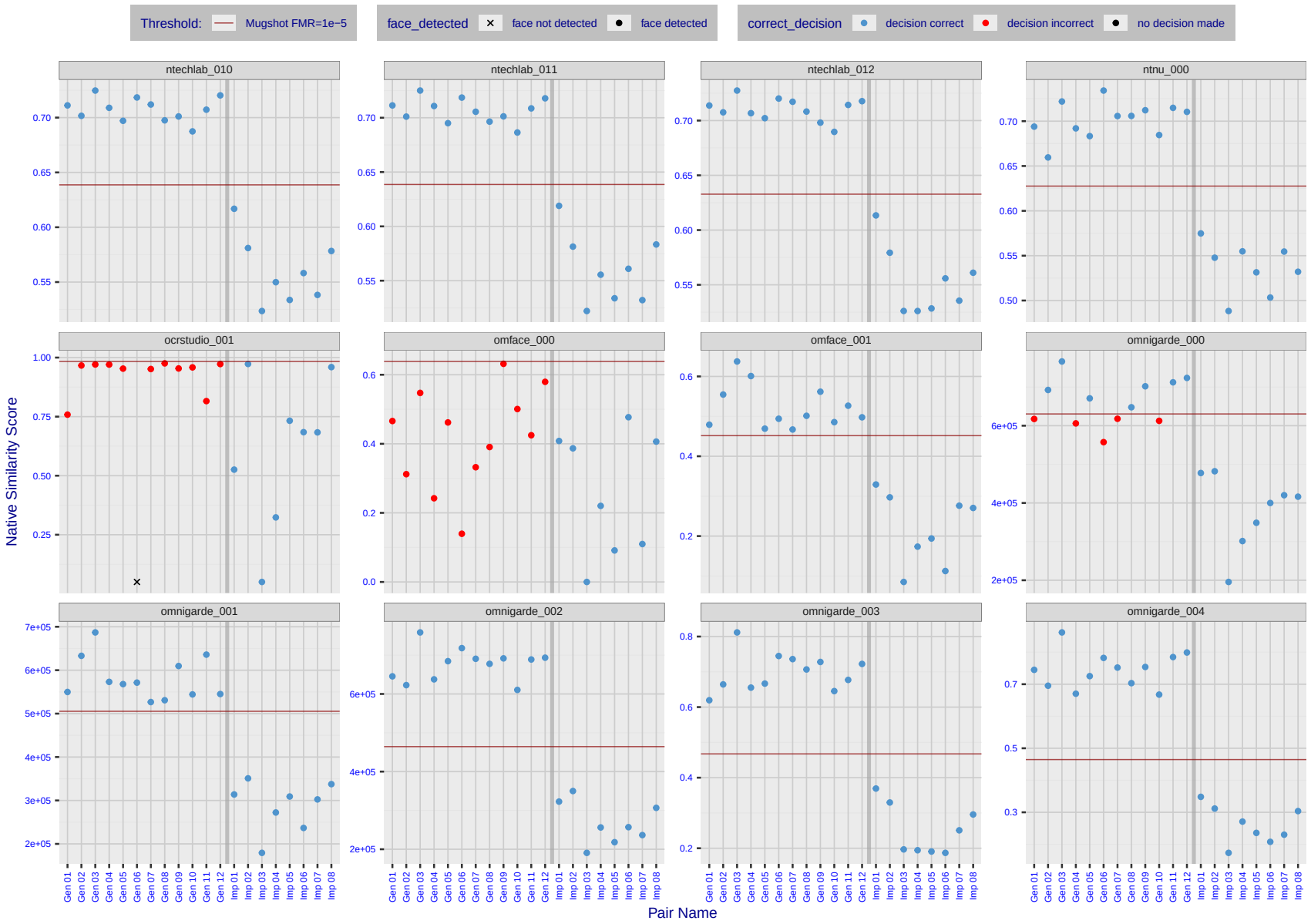


Figure 68: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

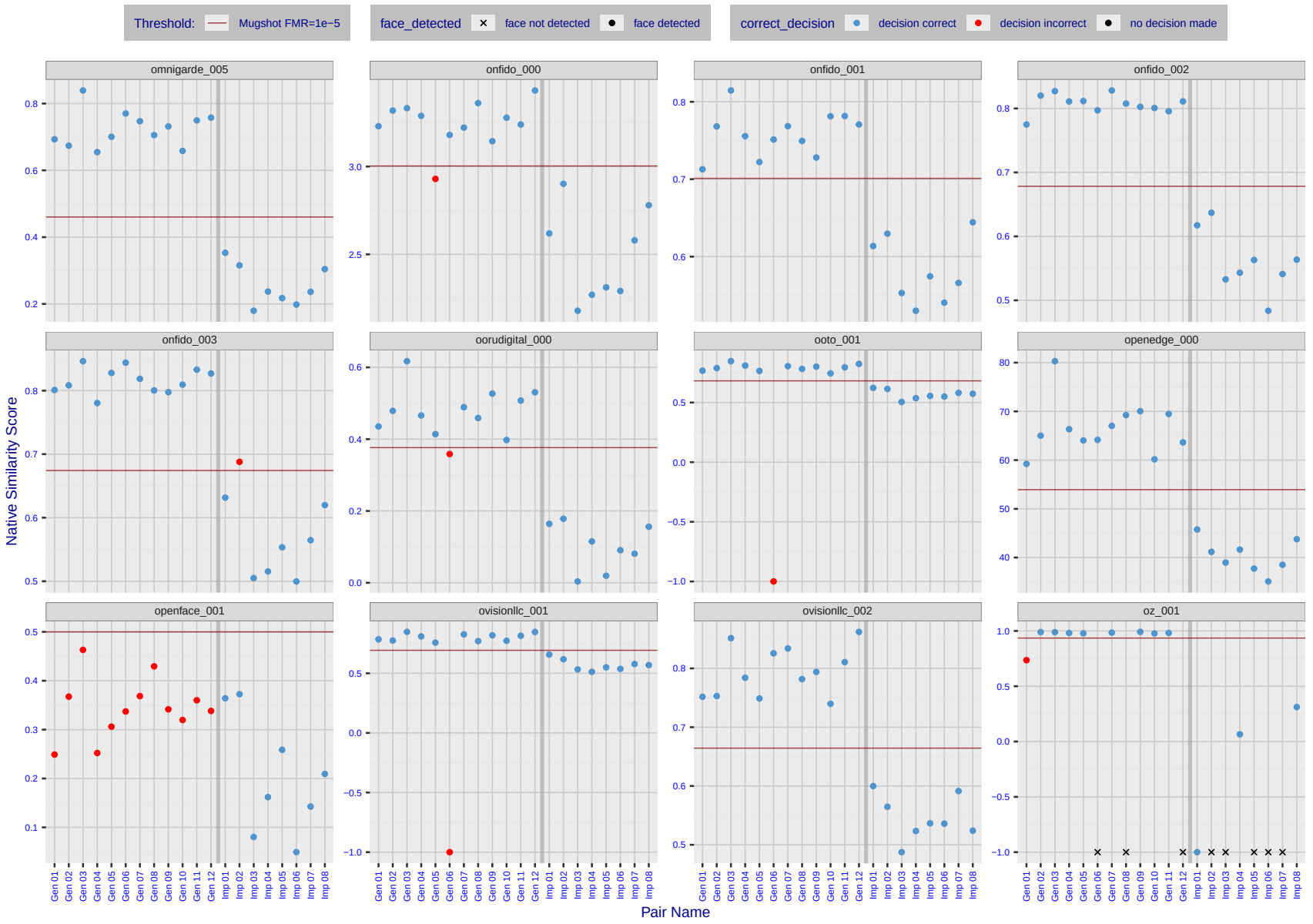


Figure 69: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

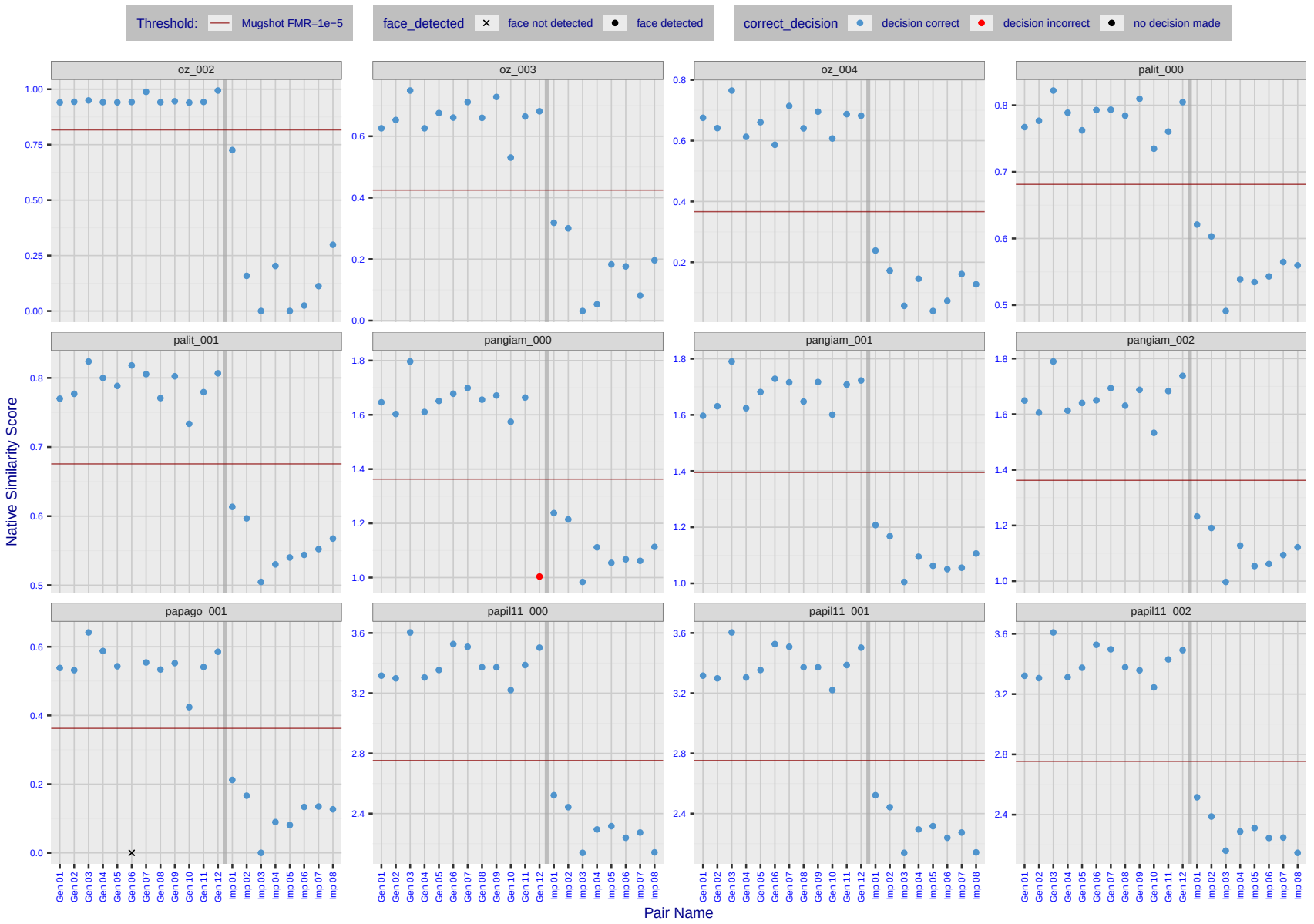


Figure 70: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

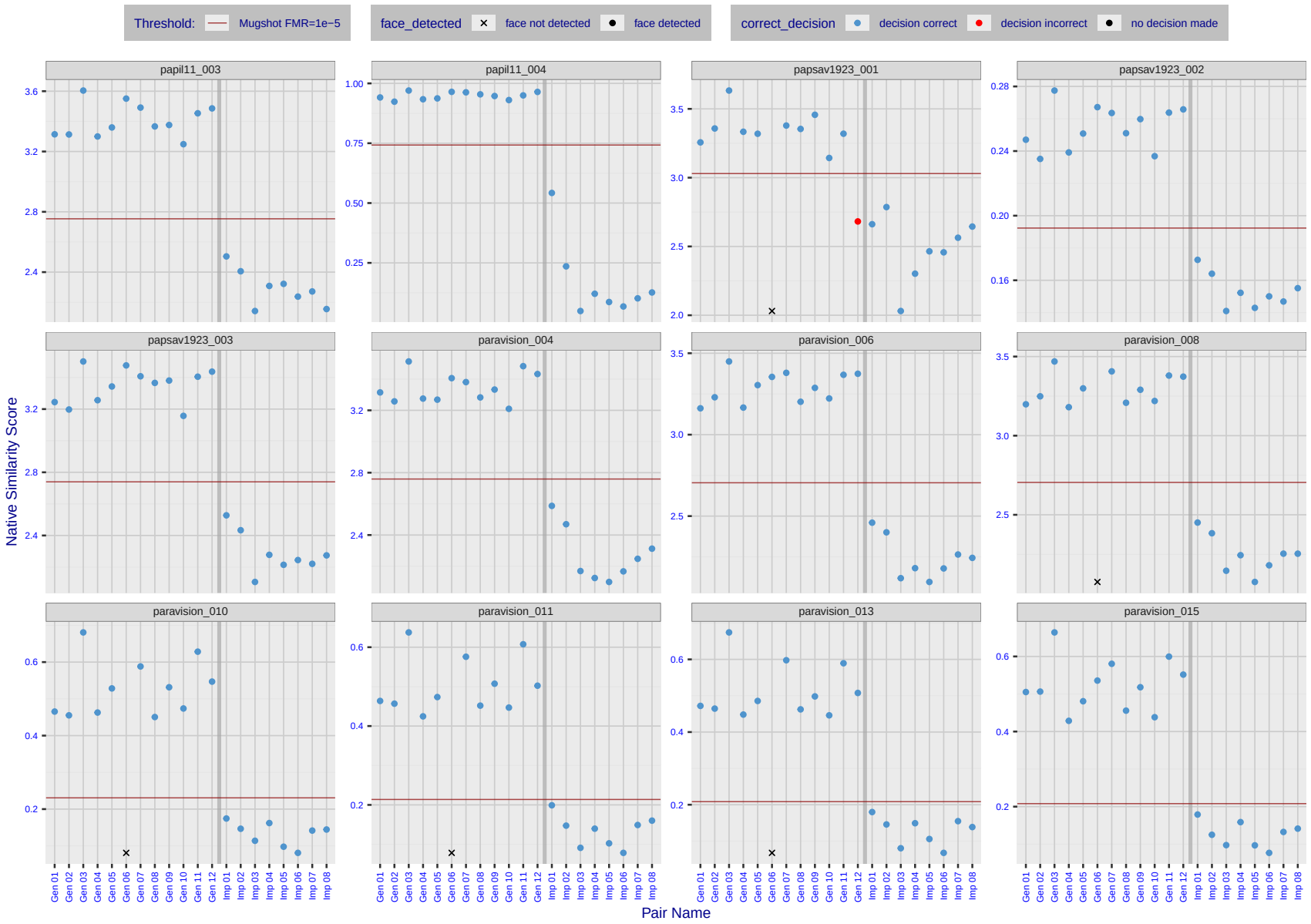


Figure 71: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

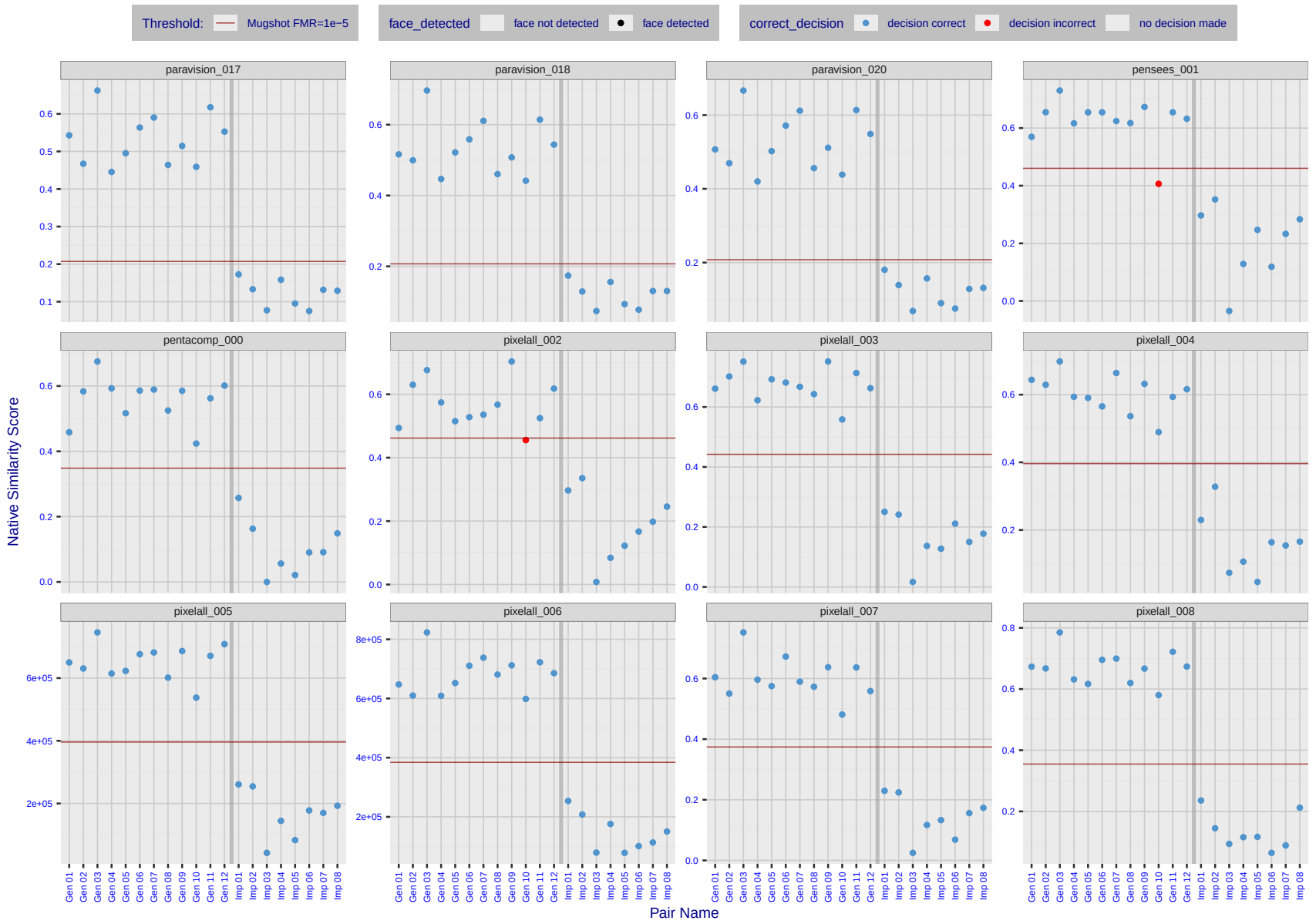


Figure 72: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

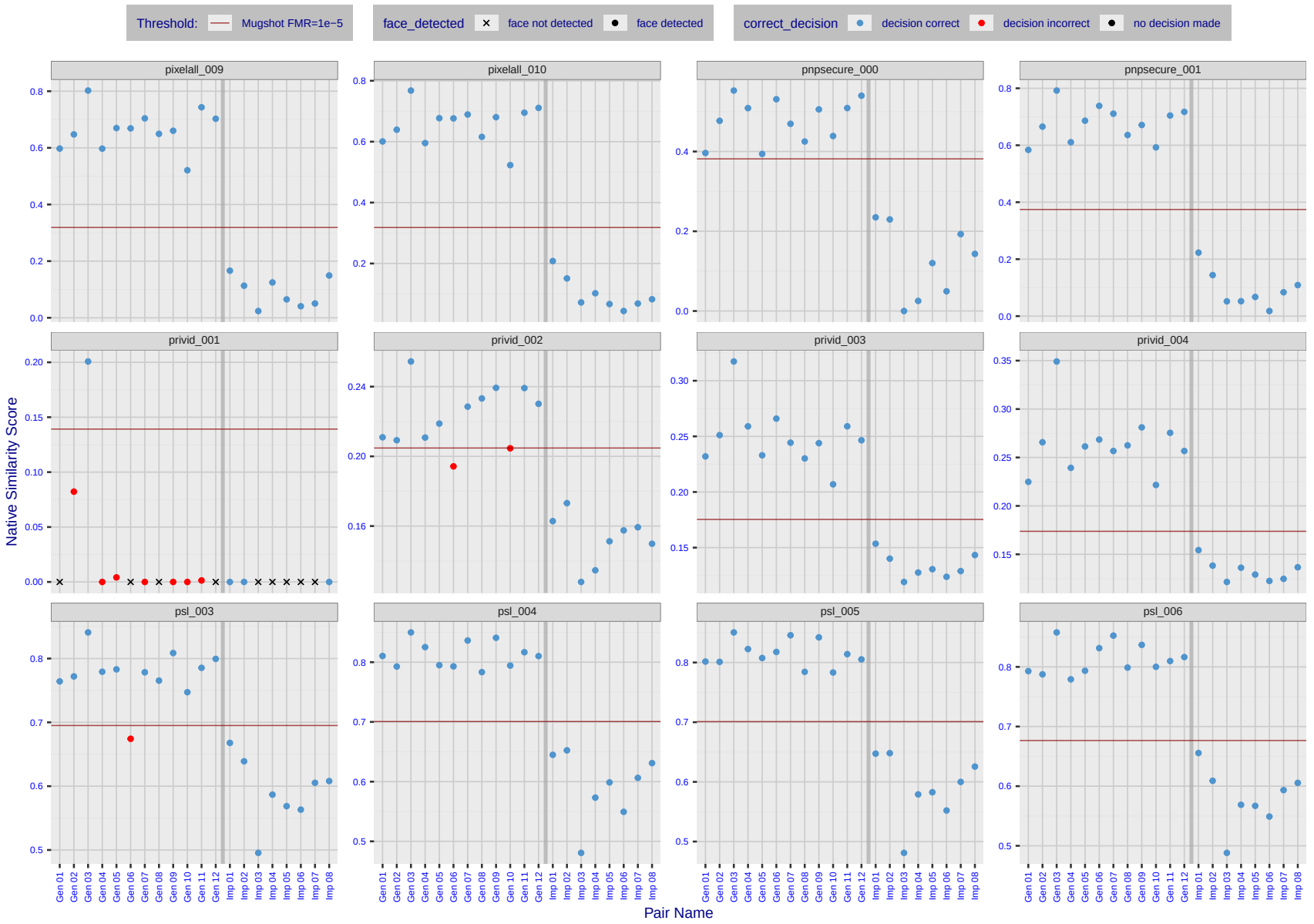


Figure 73: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

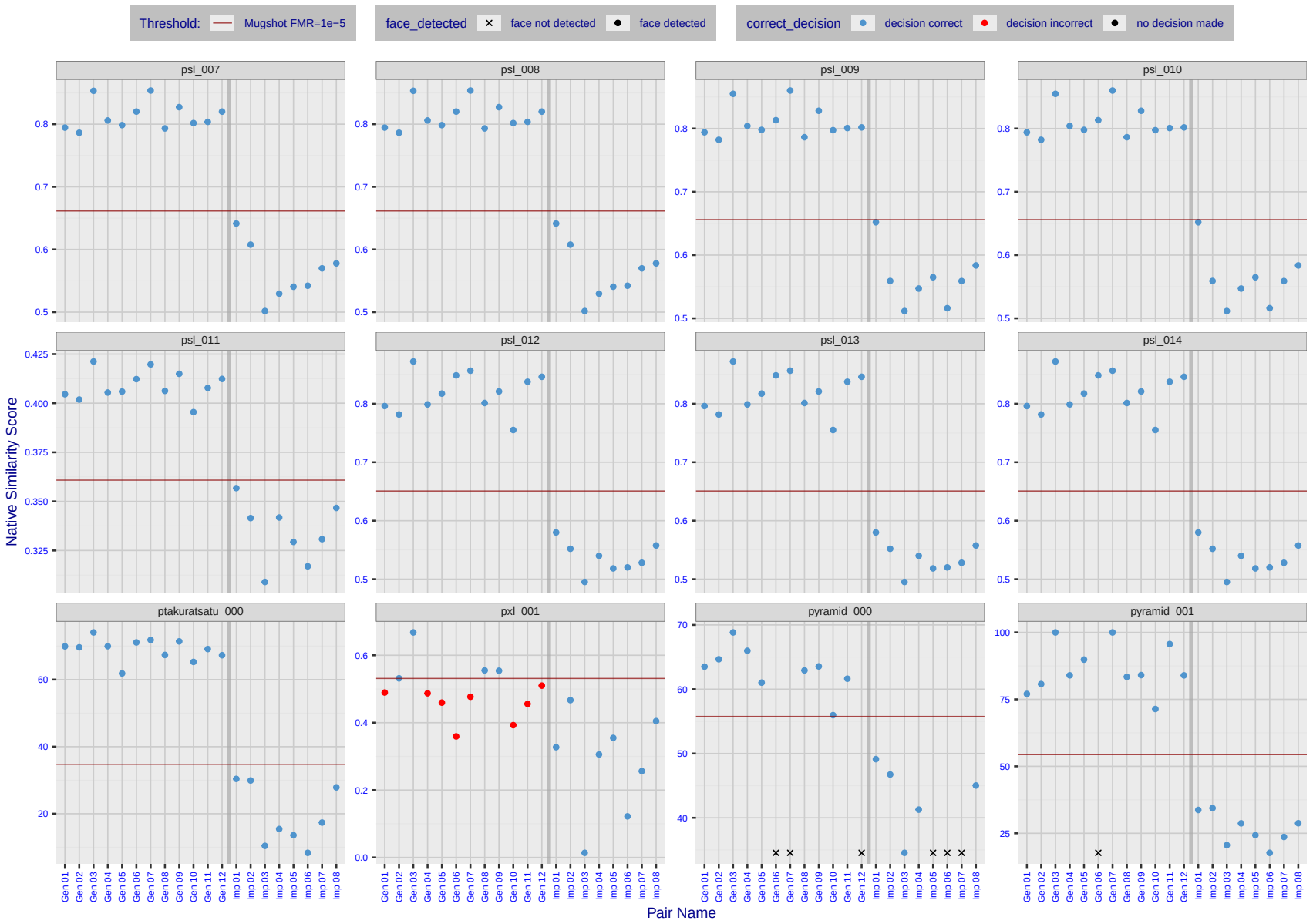


Figure 74: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

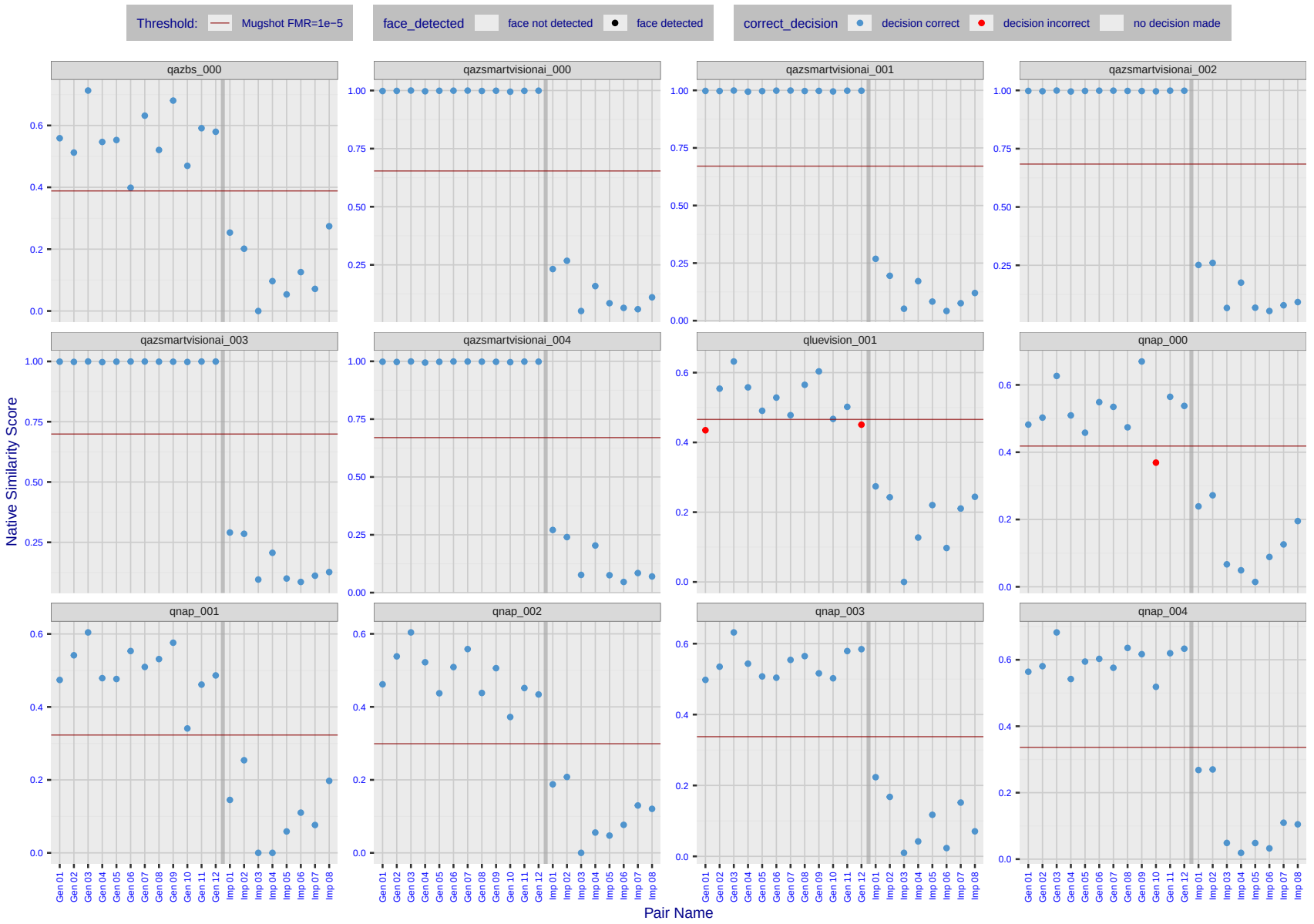


Figure 75: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

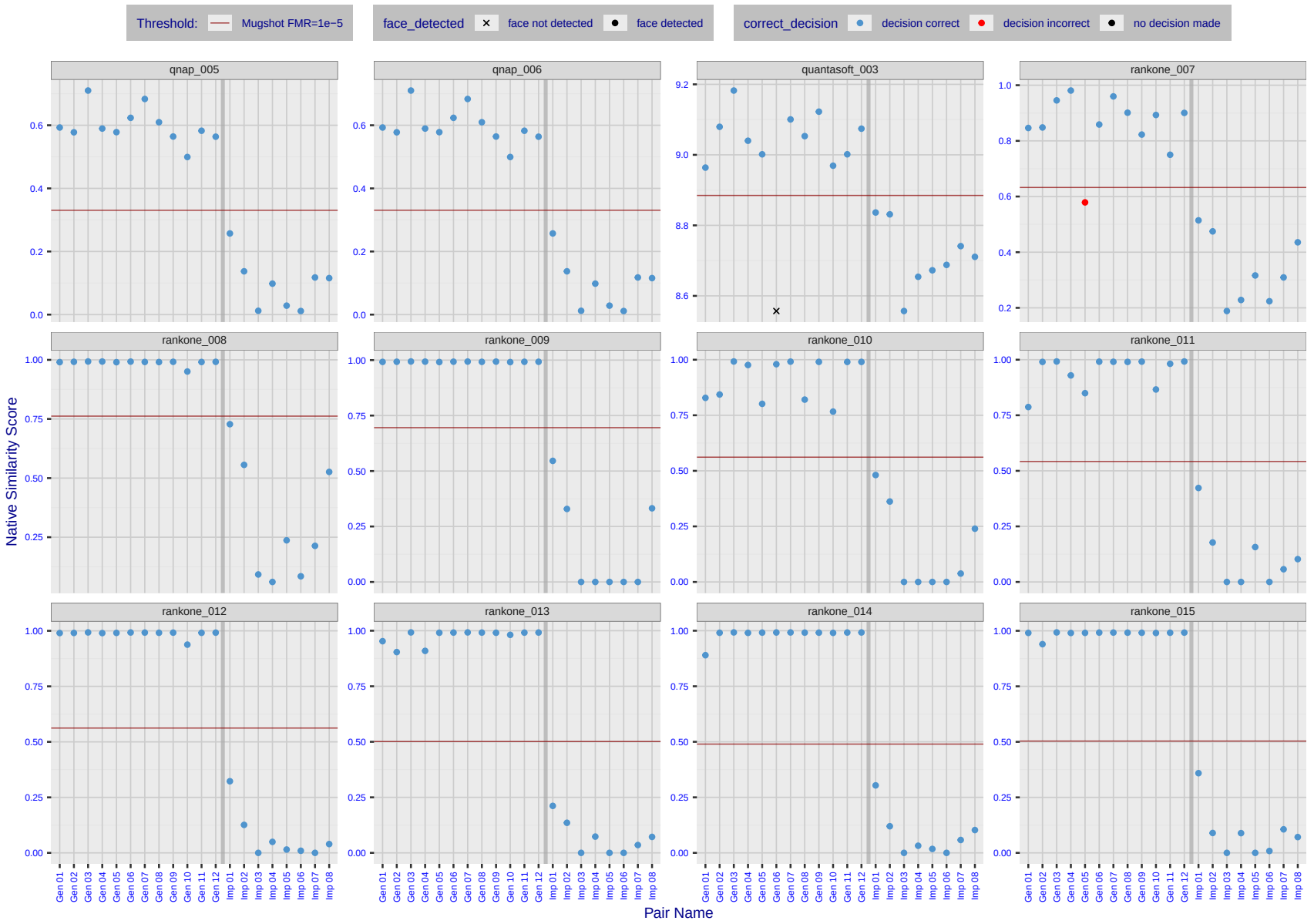


Figure 76: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

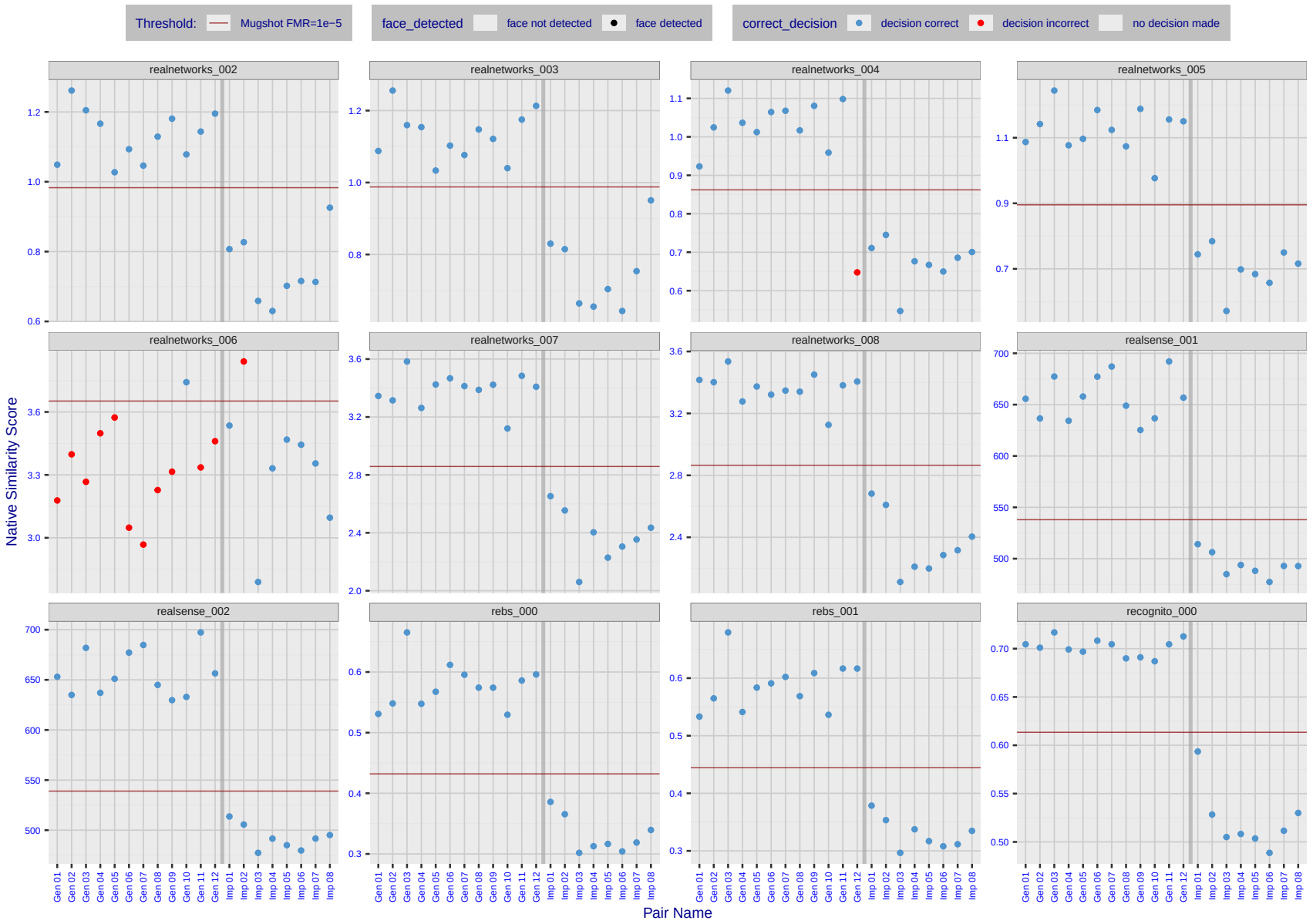


Figure 77: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

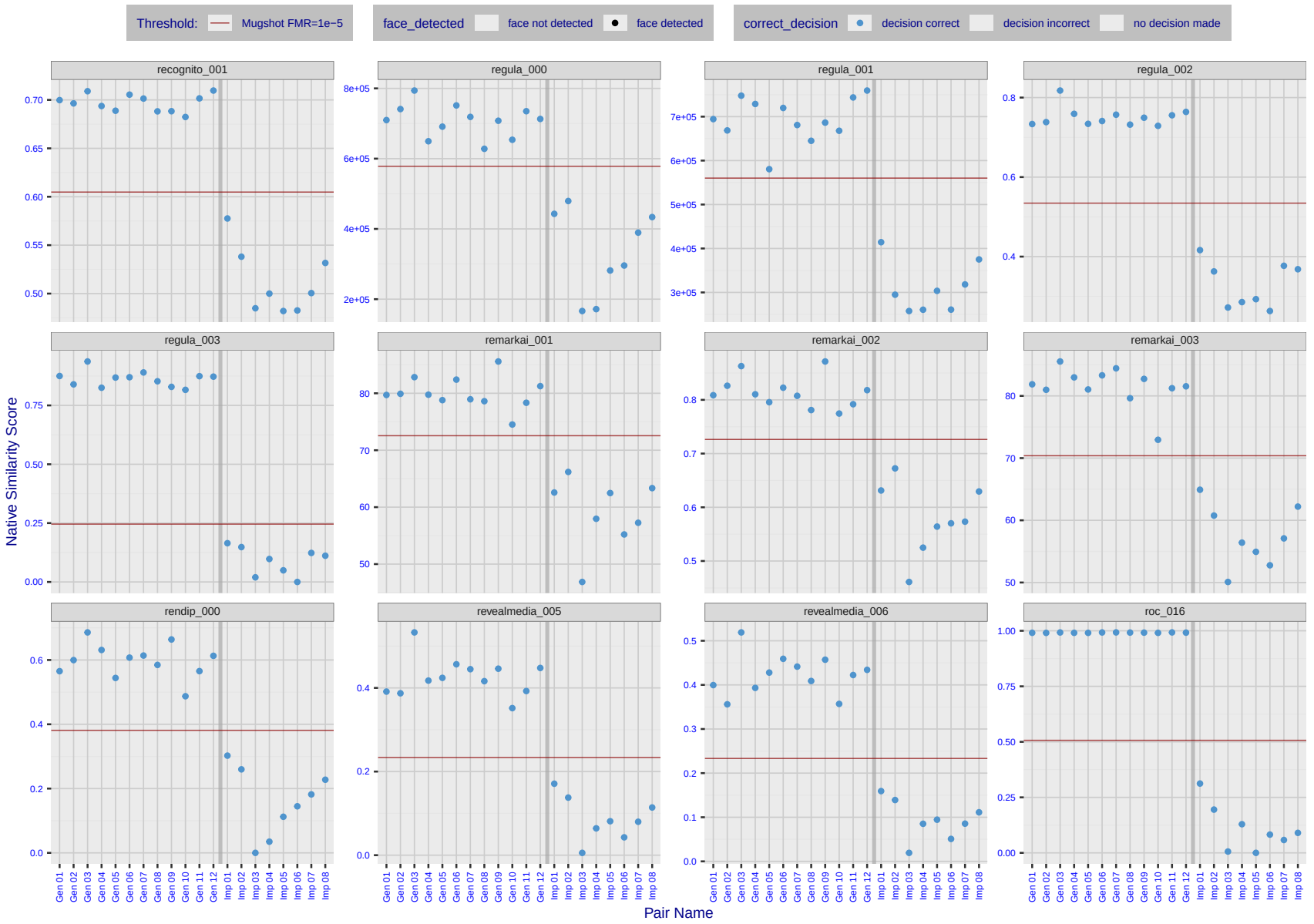


Figure 78: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

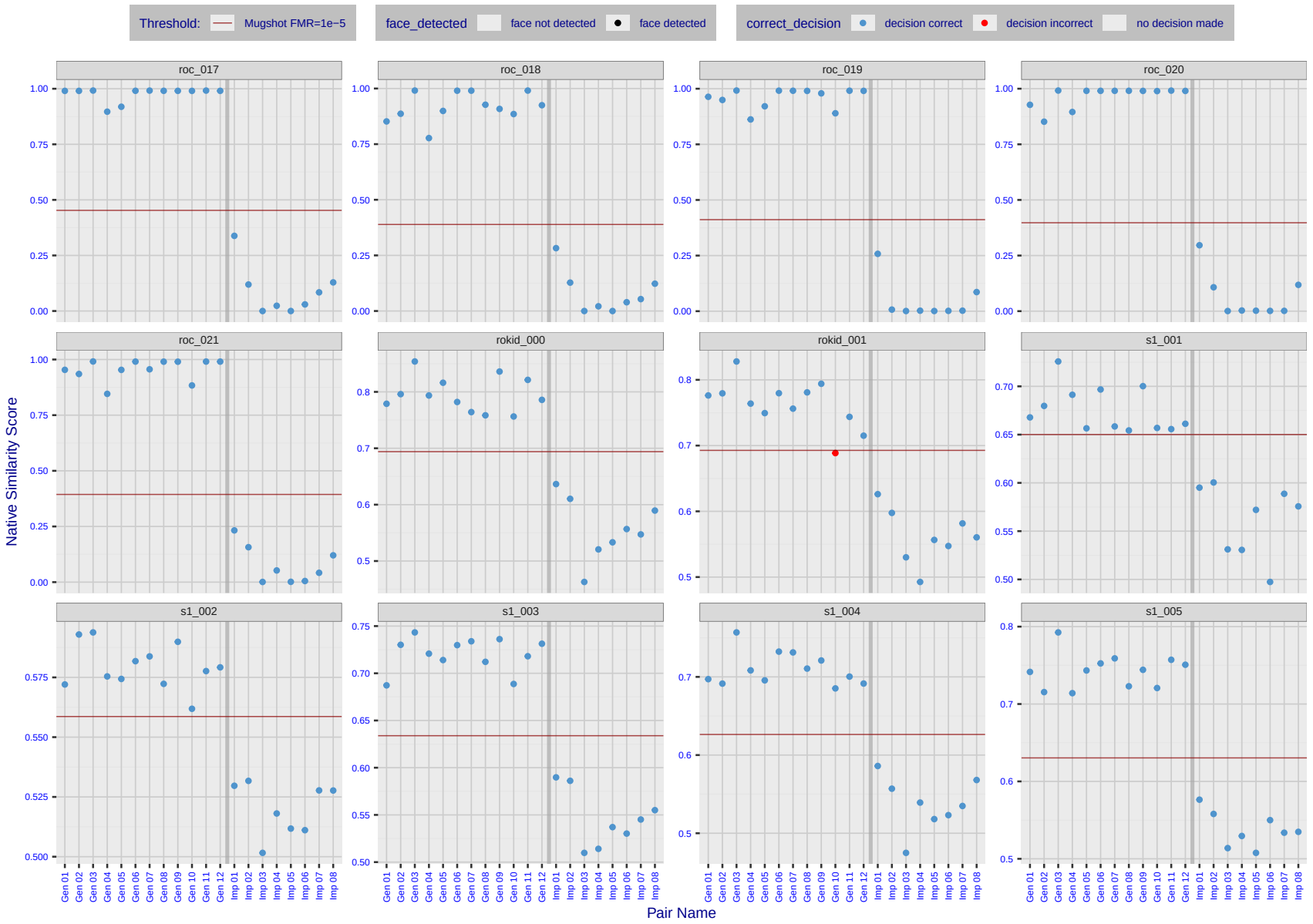


Figure 79: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

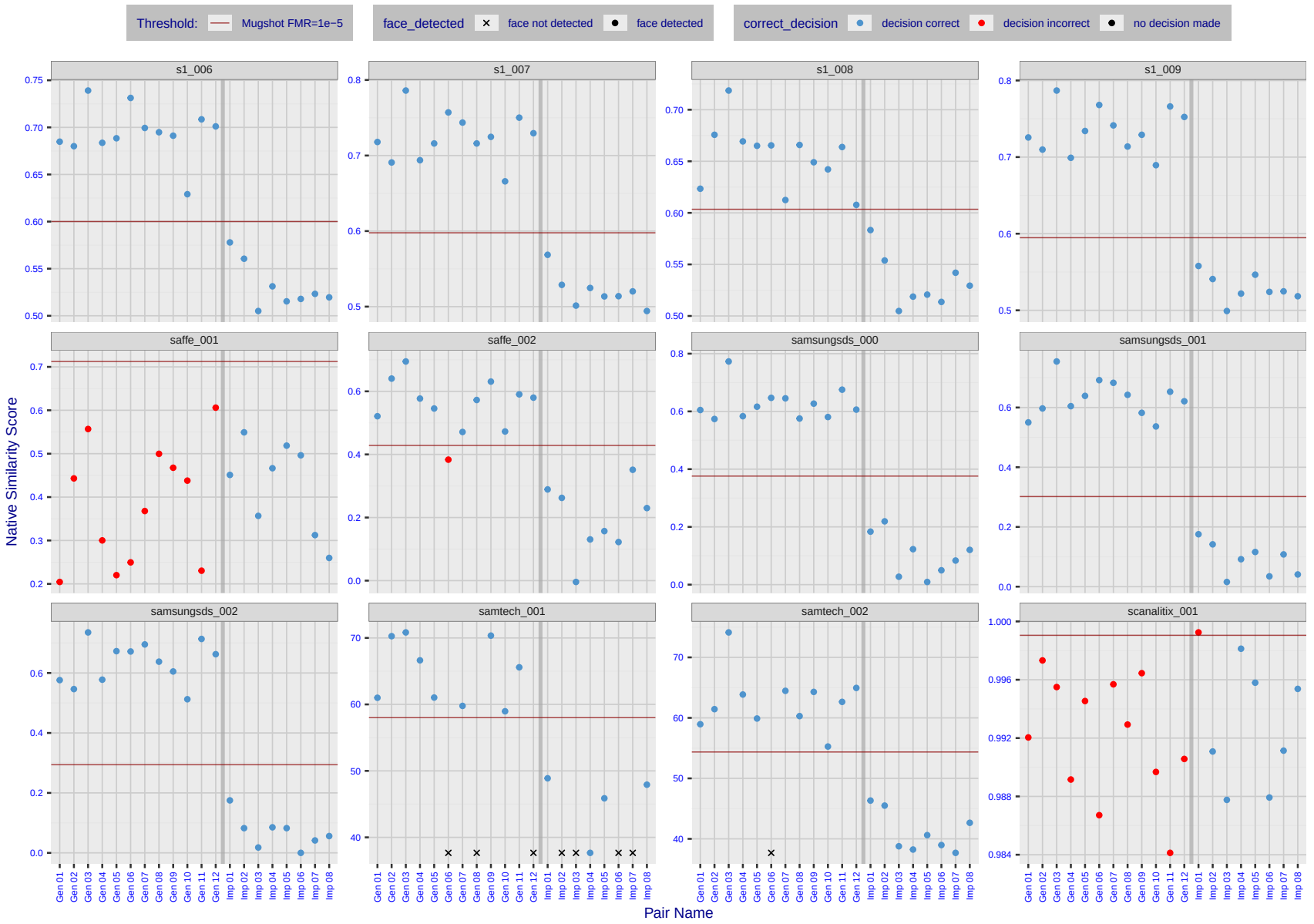


Figure 80: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

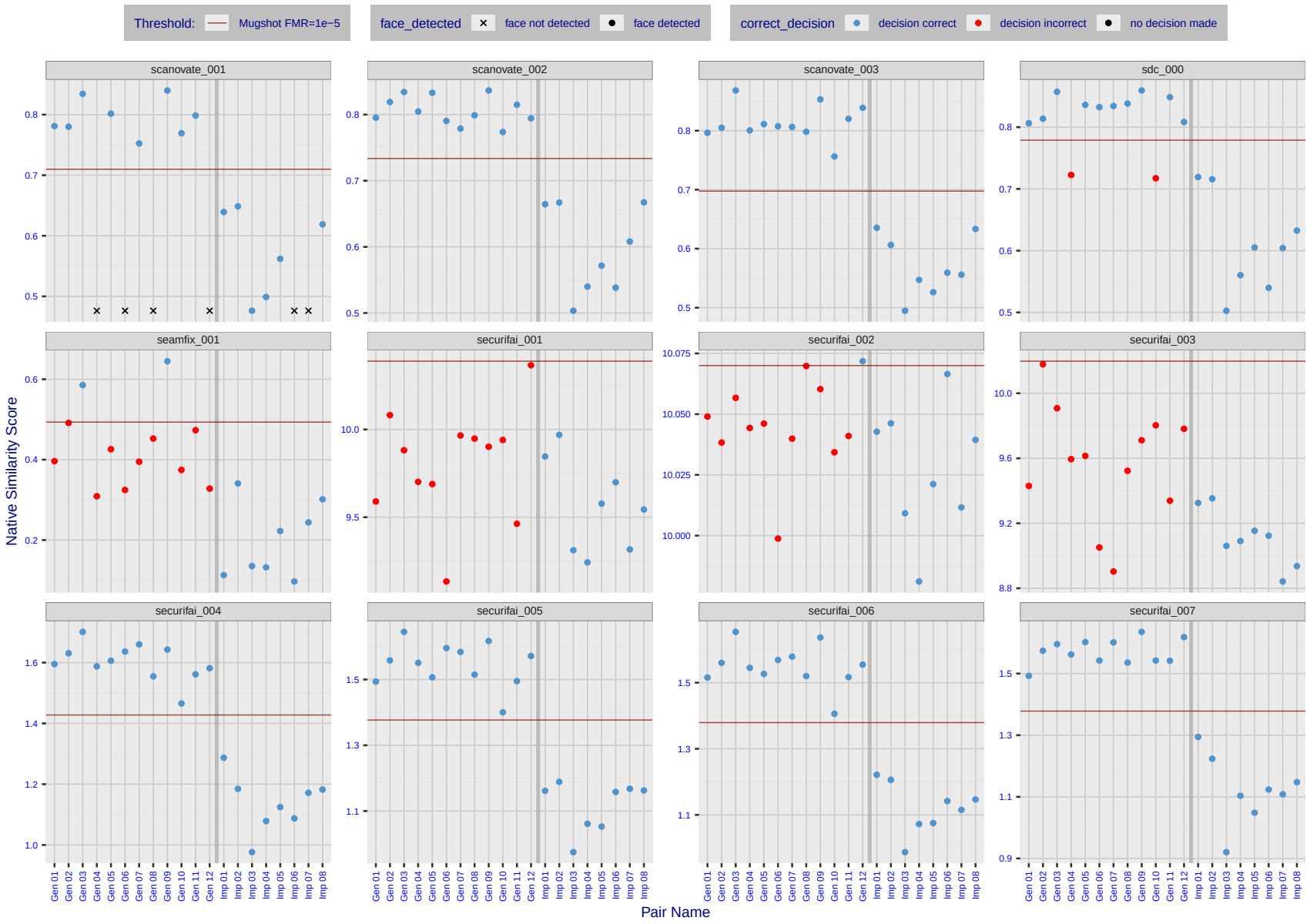


Figure 81: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

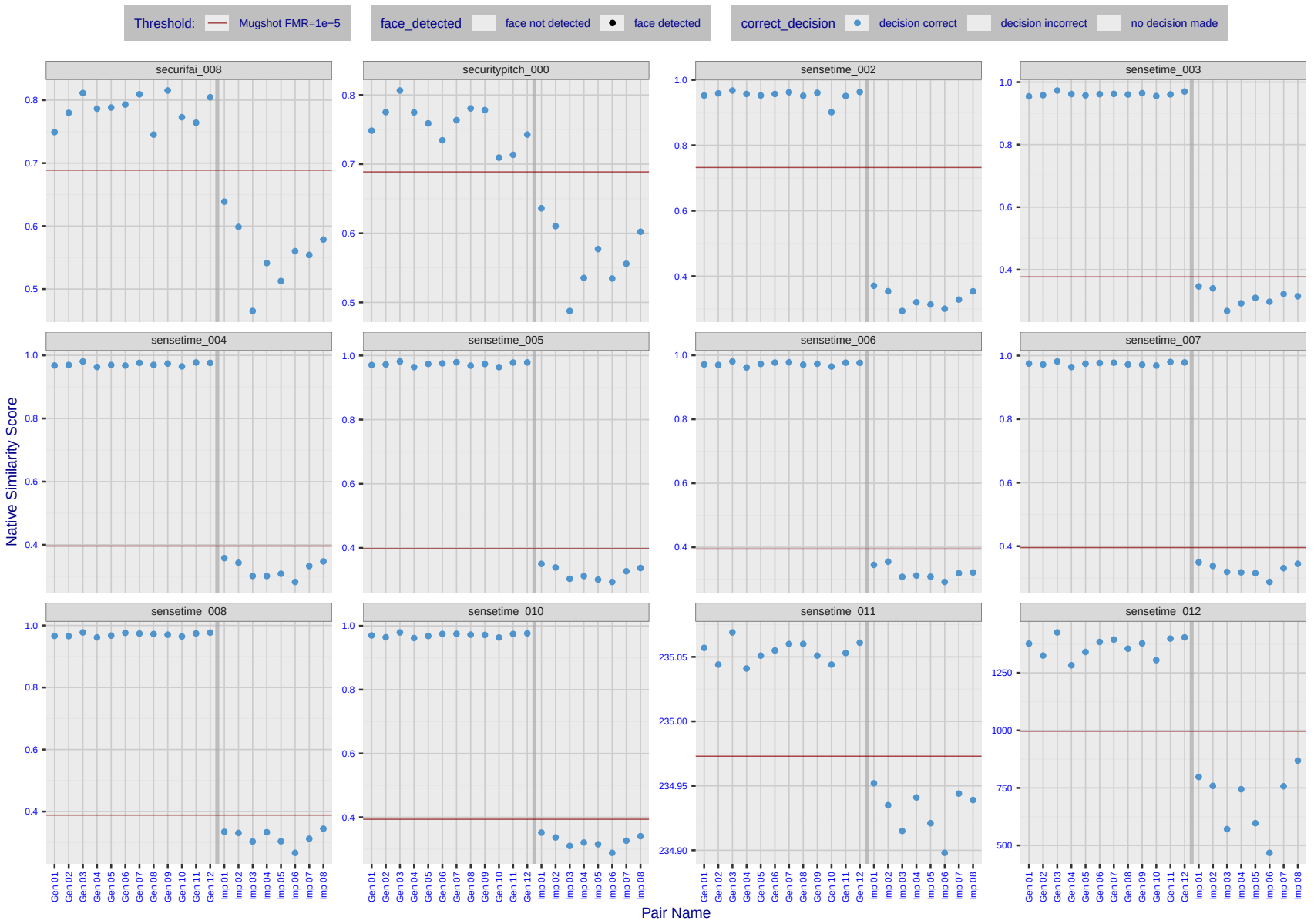


Figure 82: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

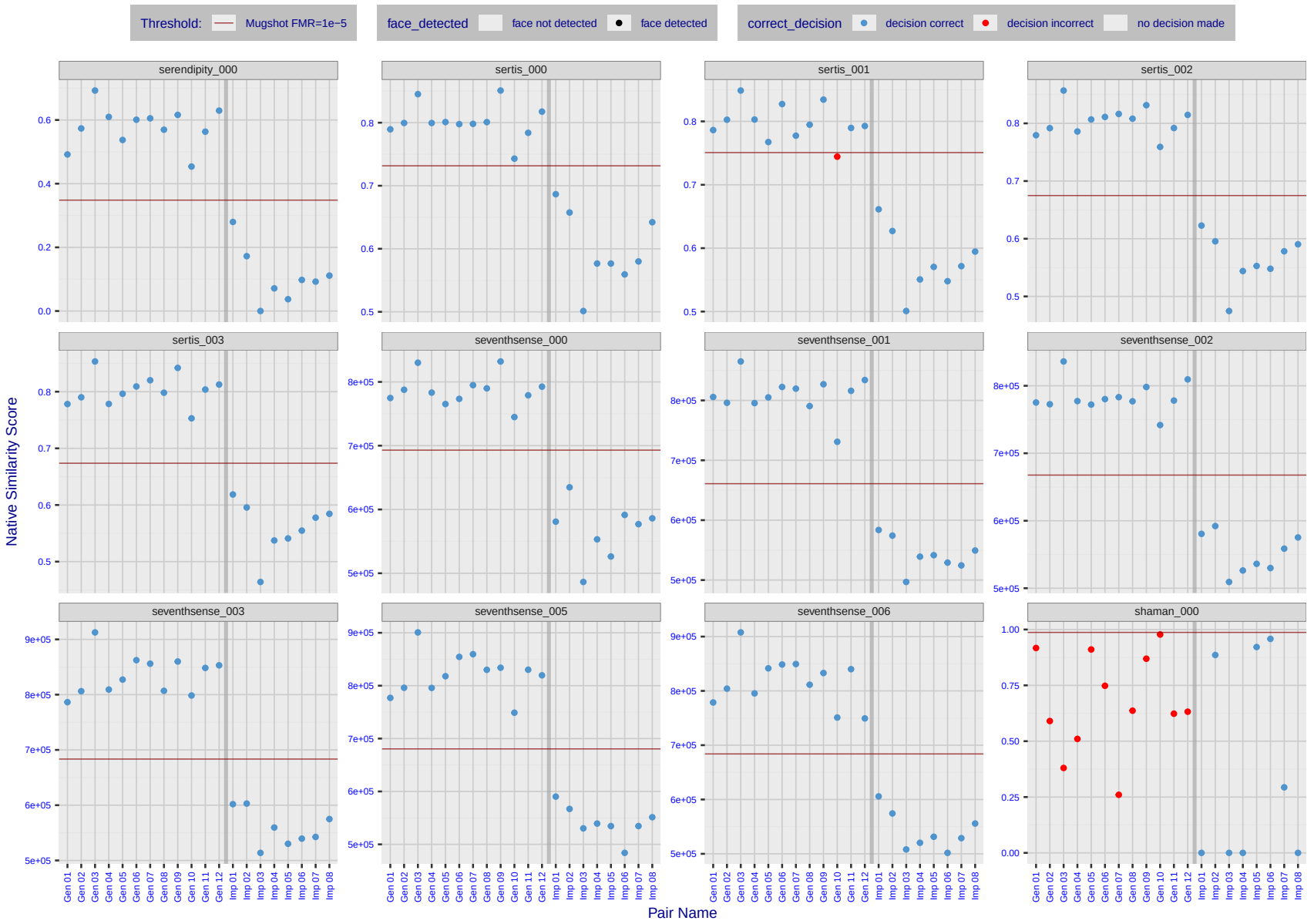


Figure 83: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

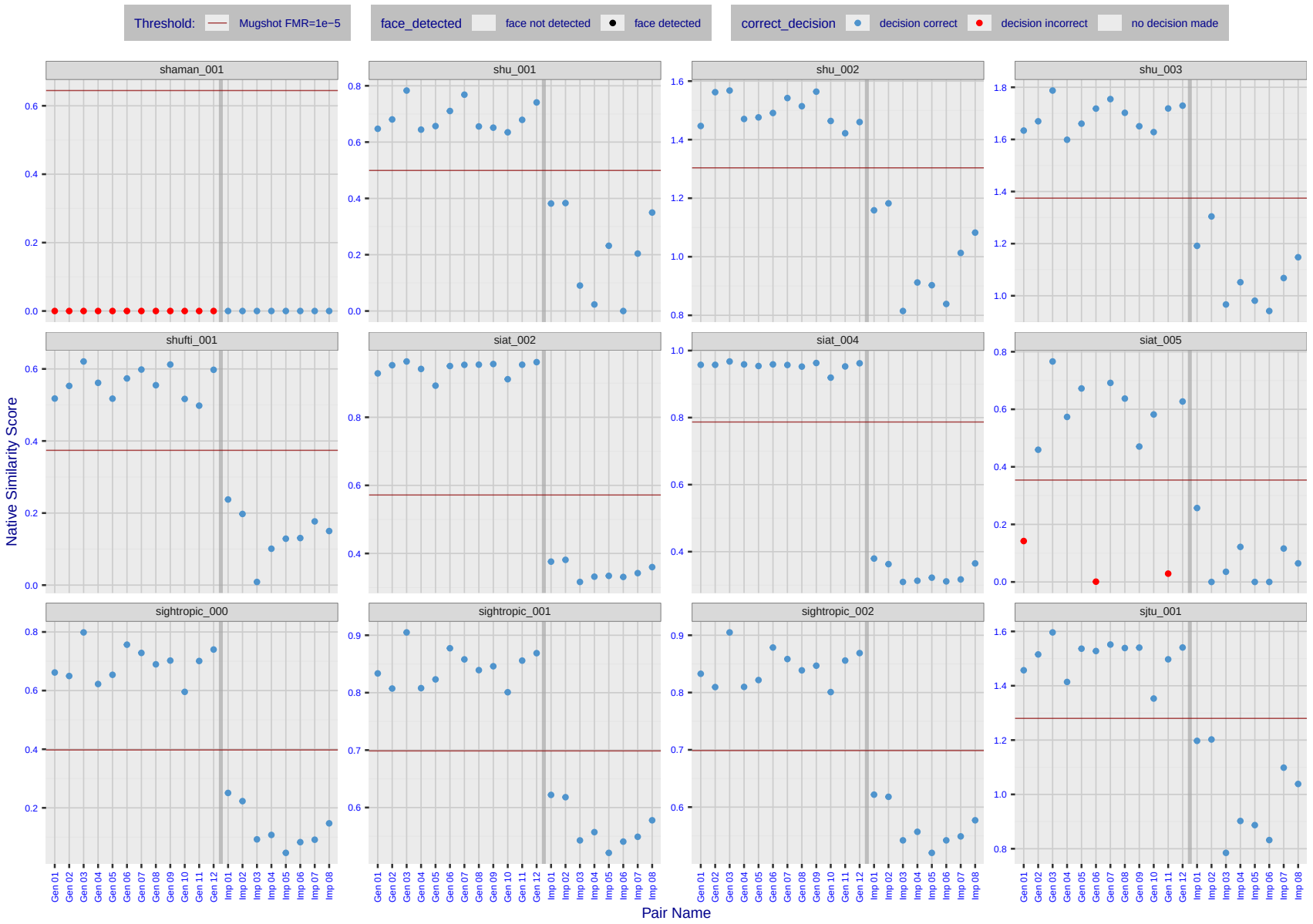


Figure 84: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

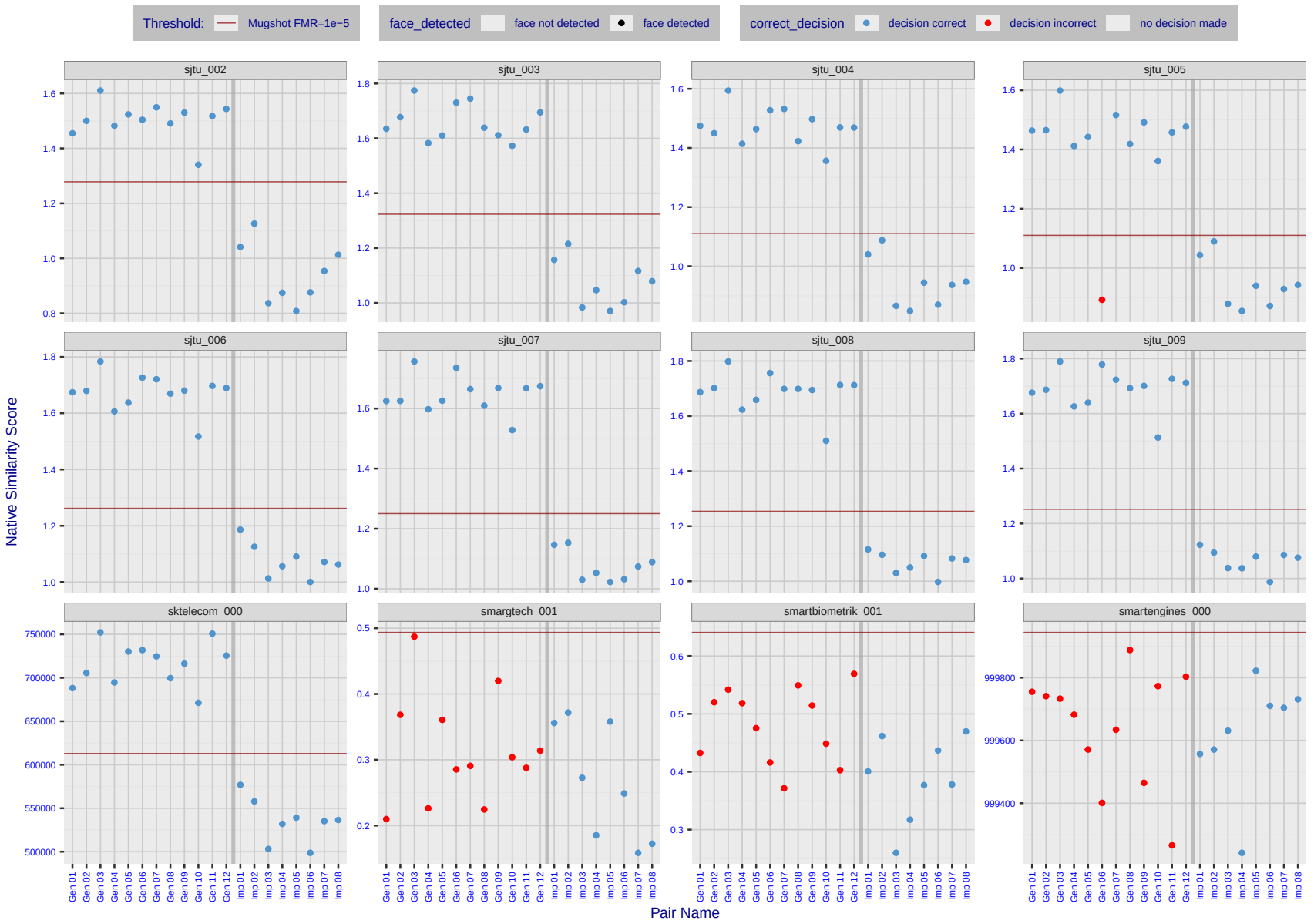


Figure 85: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

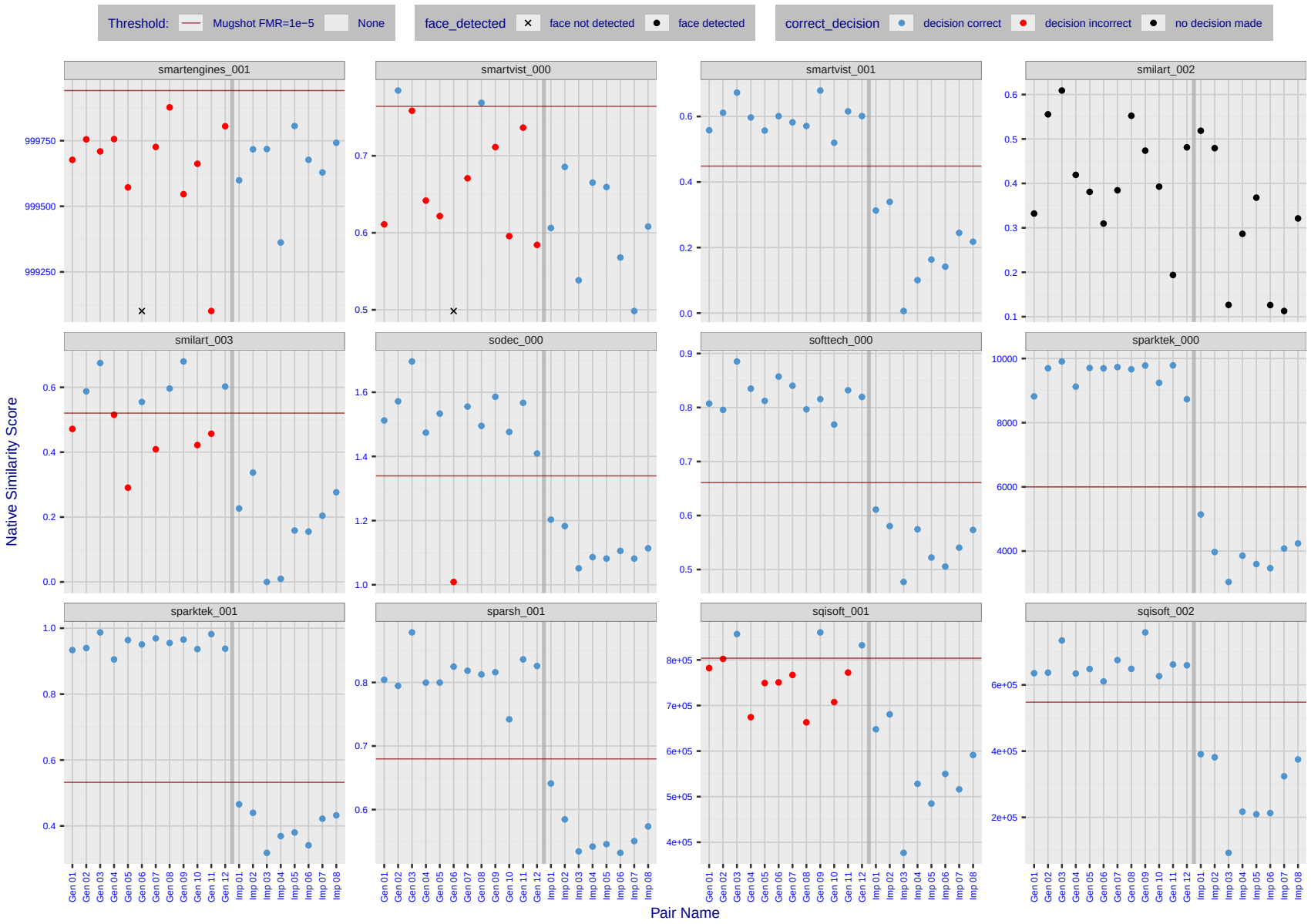


Figure 86: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

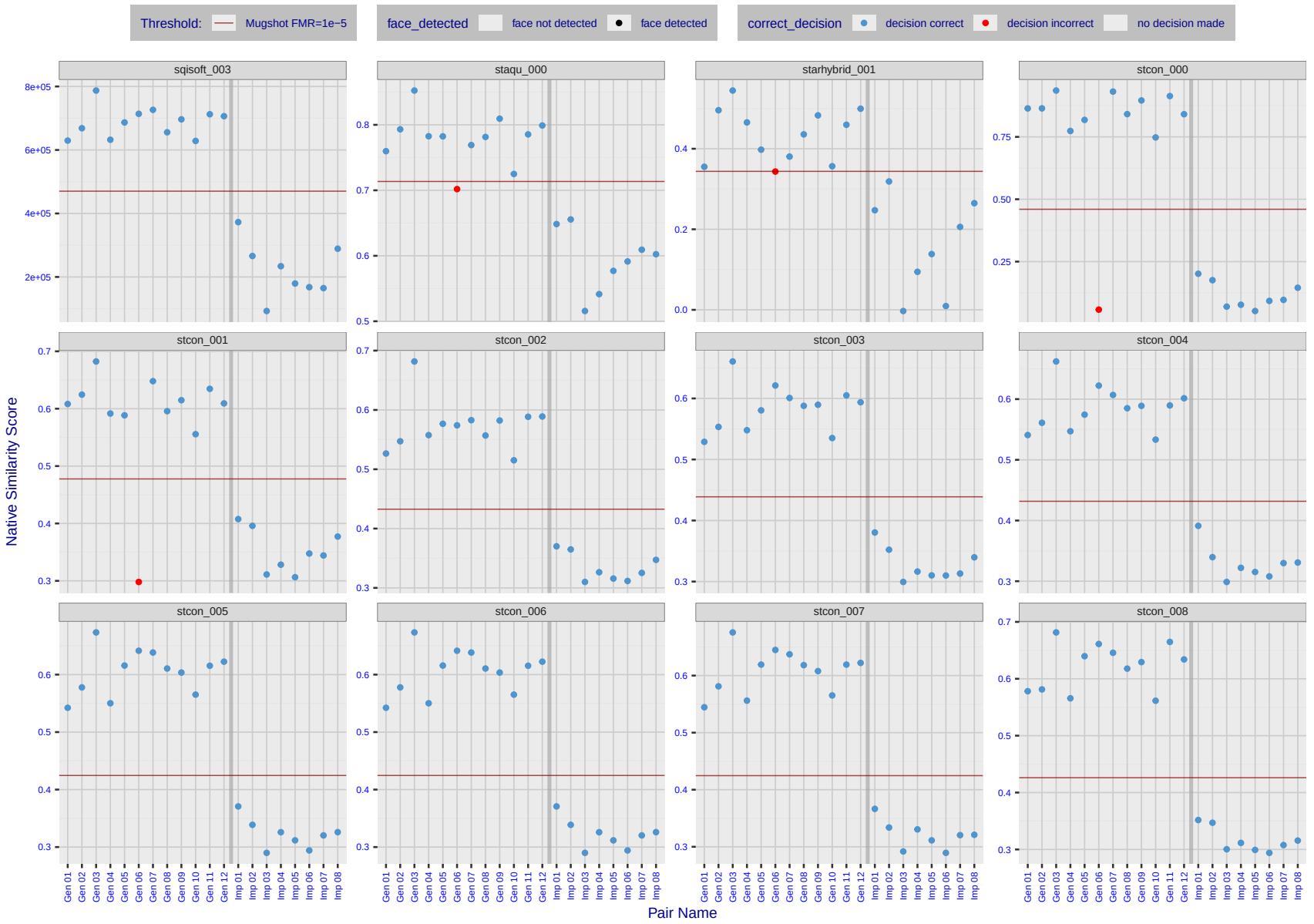


Figure 87: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

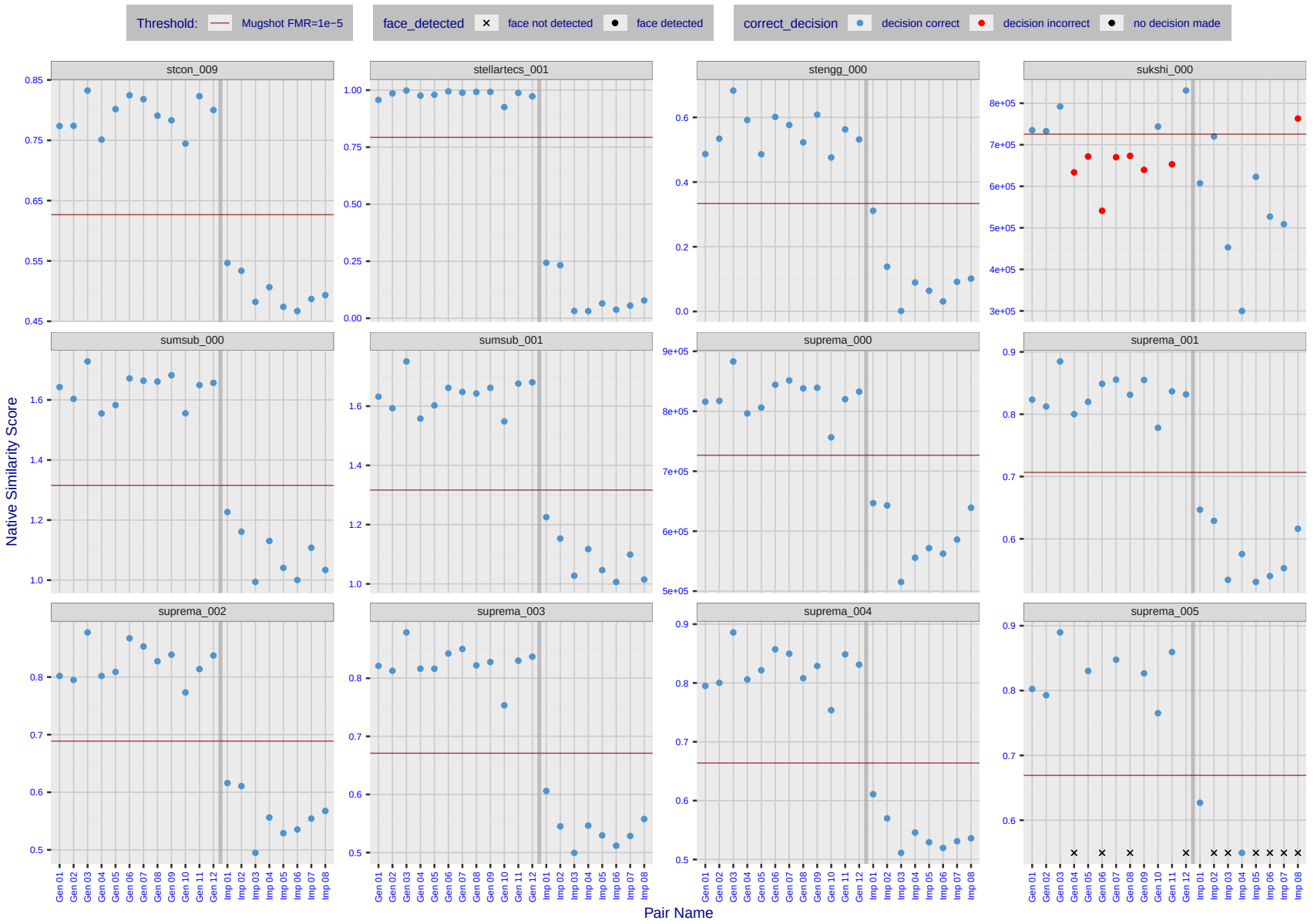


Figure 88: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

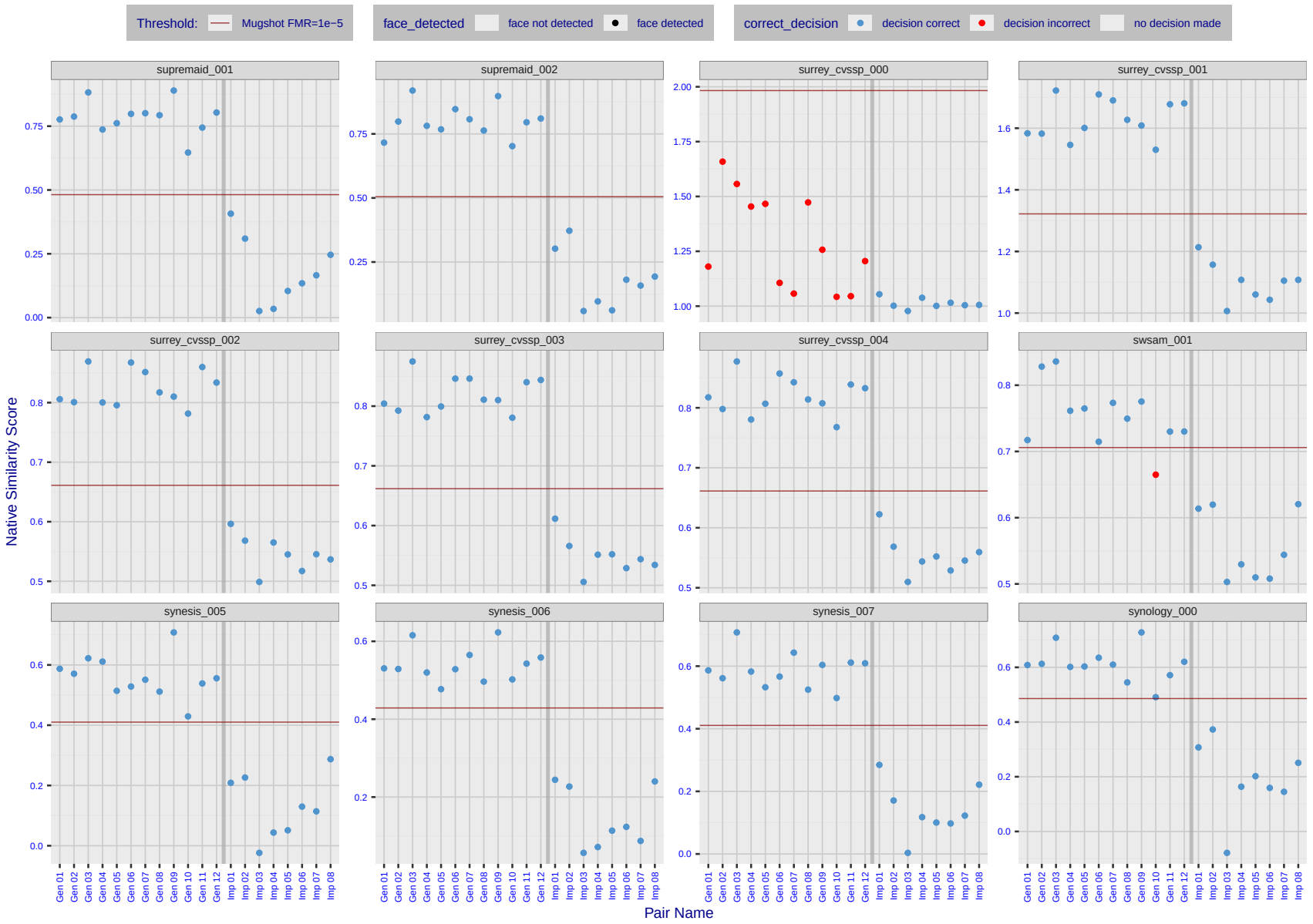


Figure 89: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

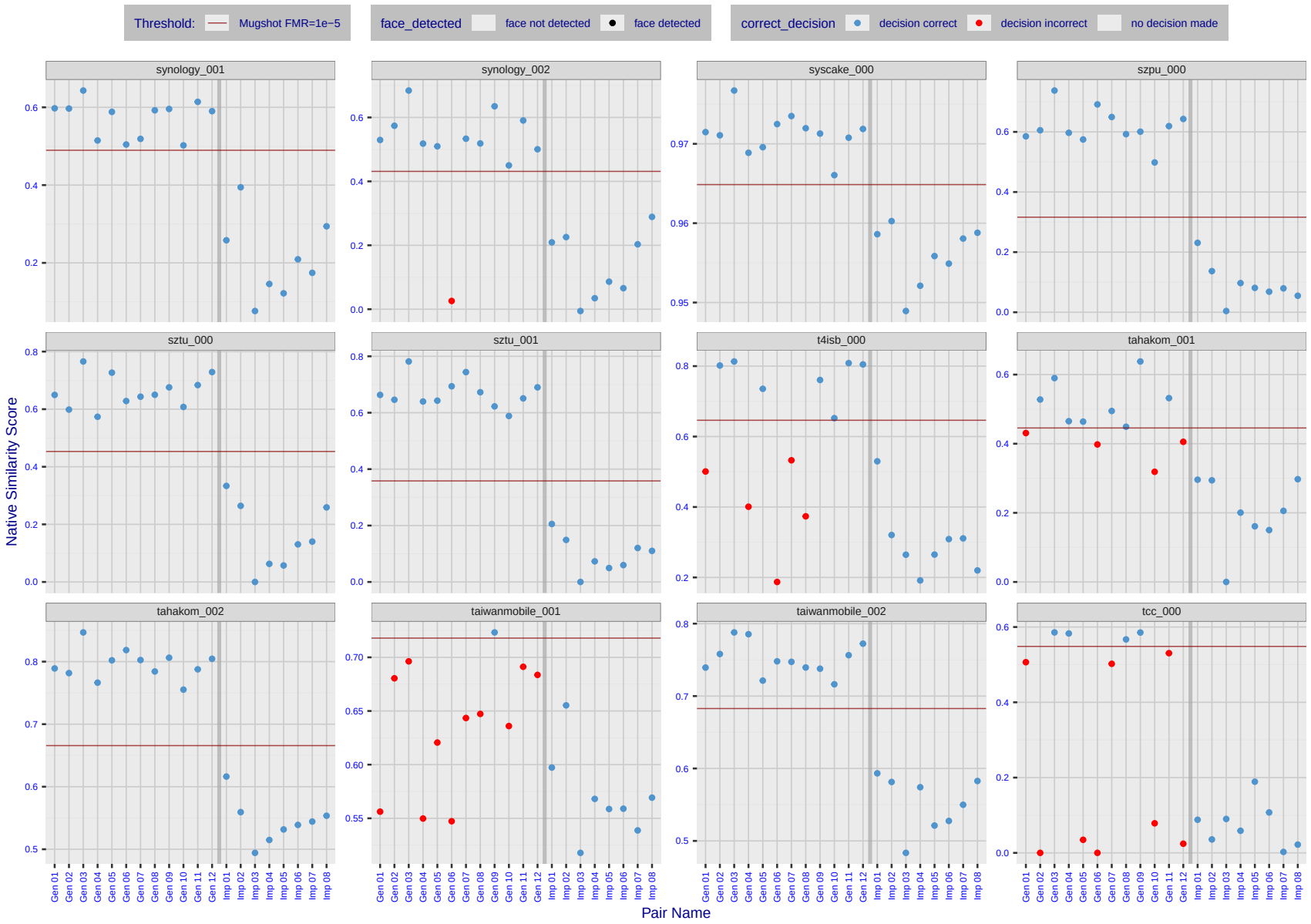


Figure 90: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

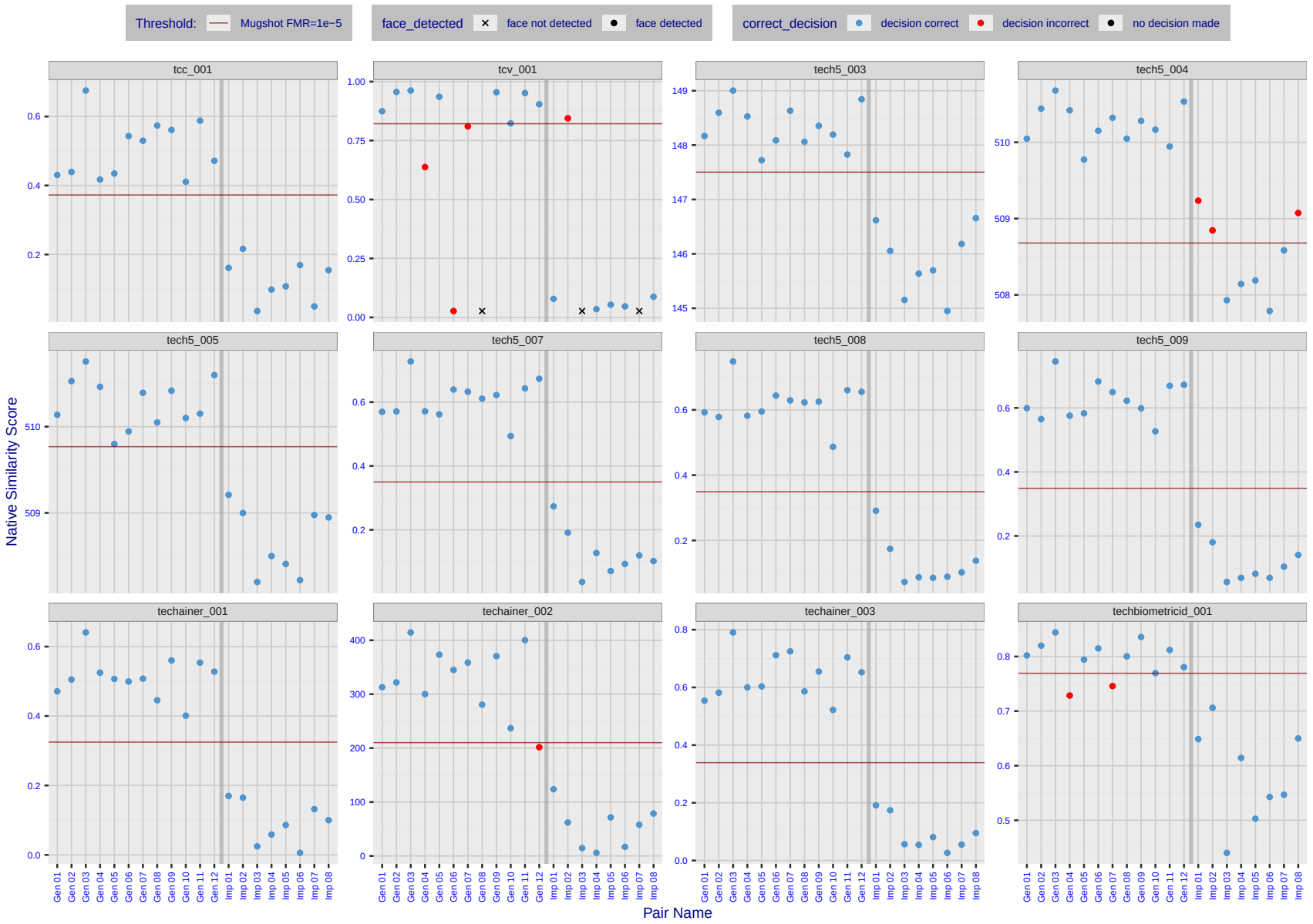


Figure 91: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

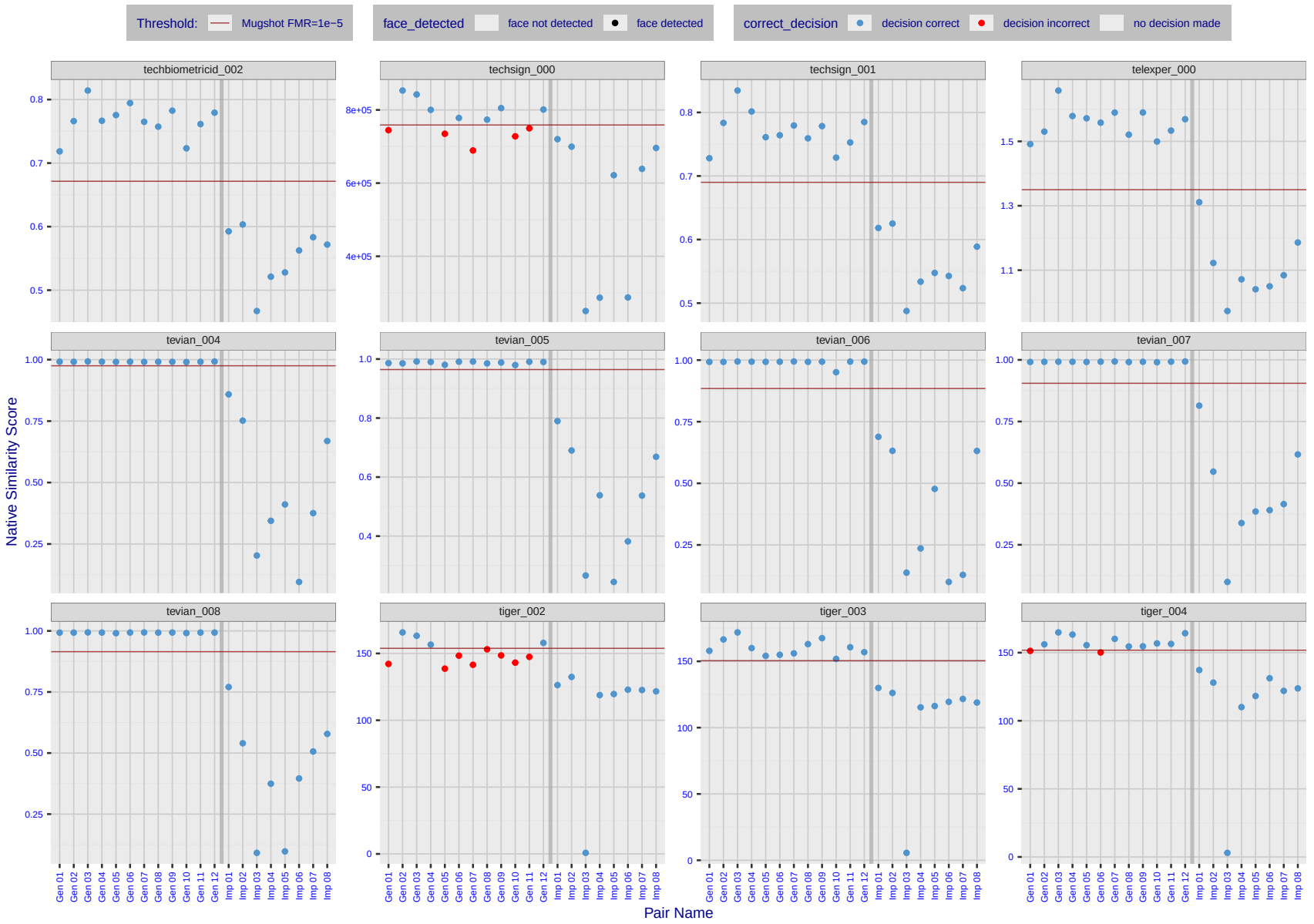


Figure 92: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

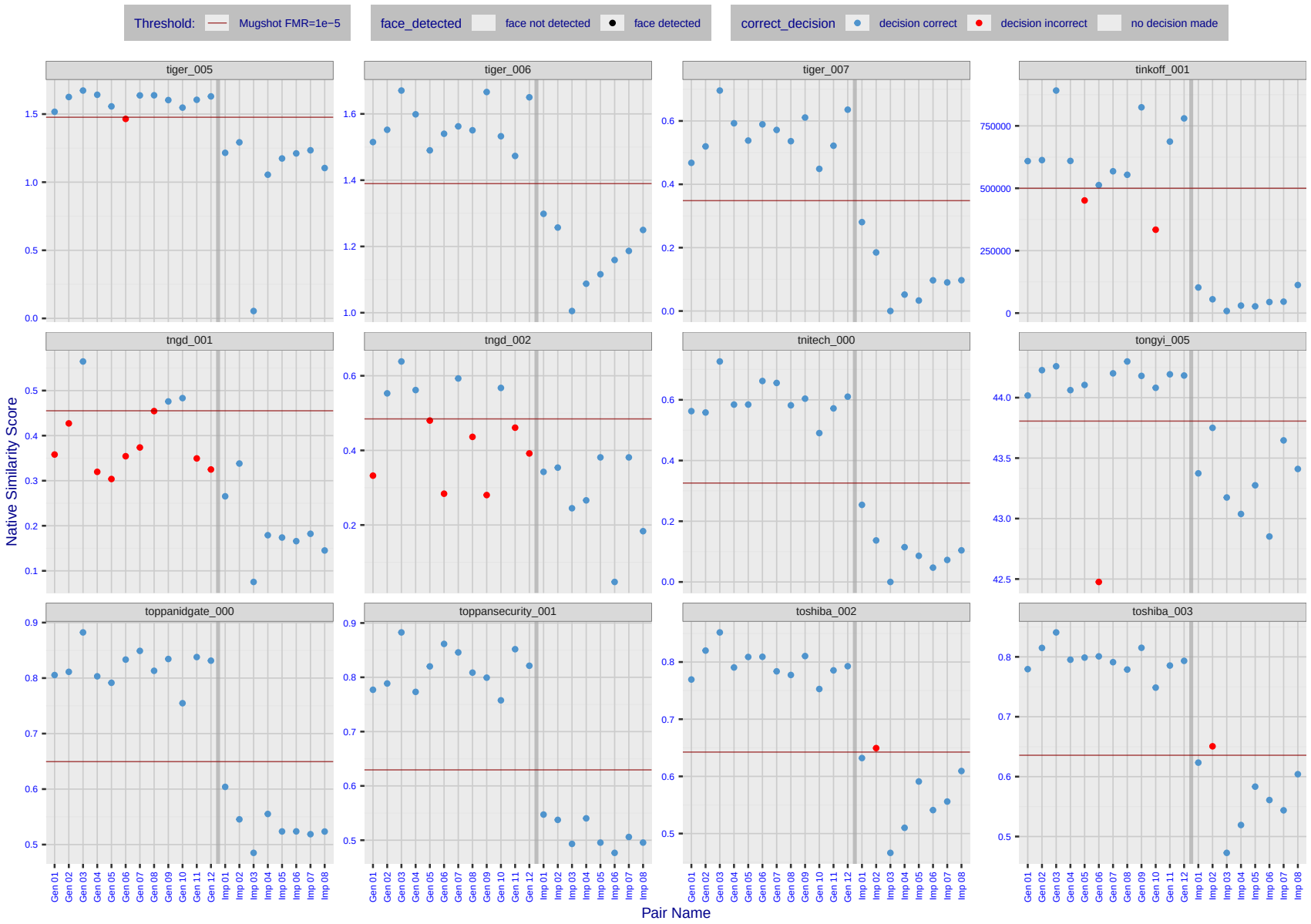


Figure 93: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

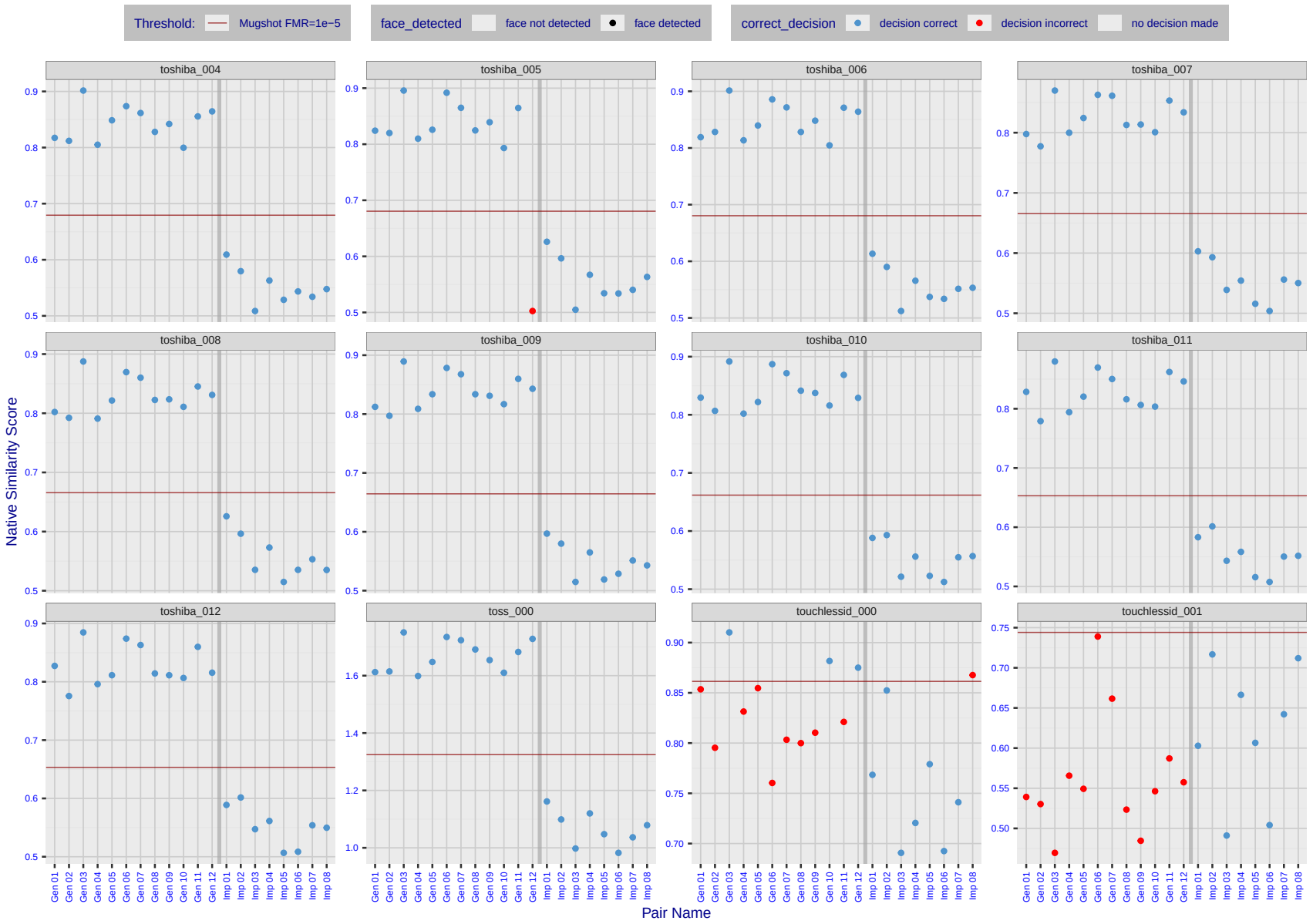


Figure 94: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

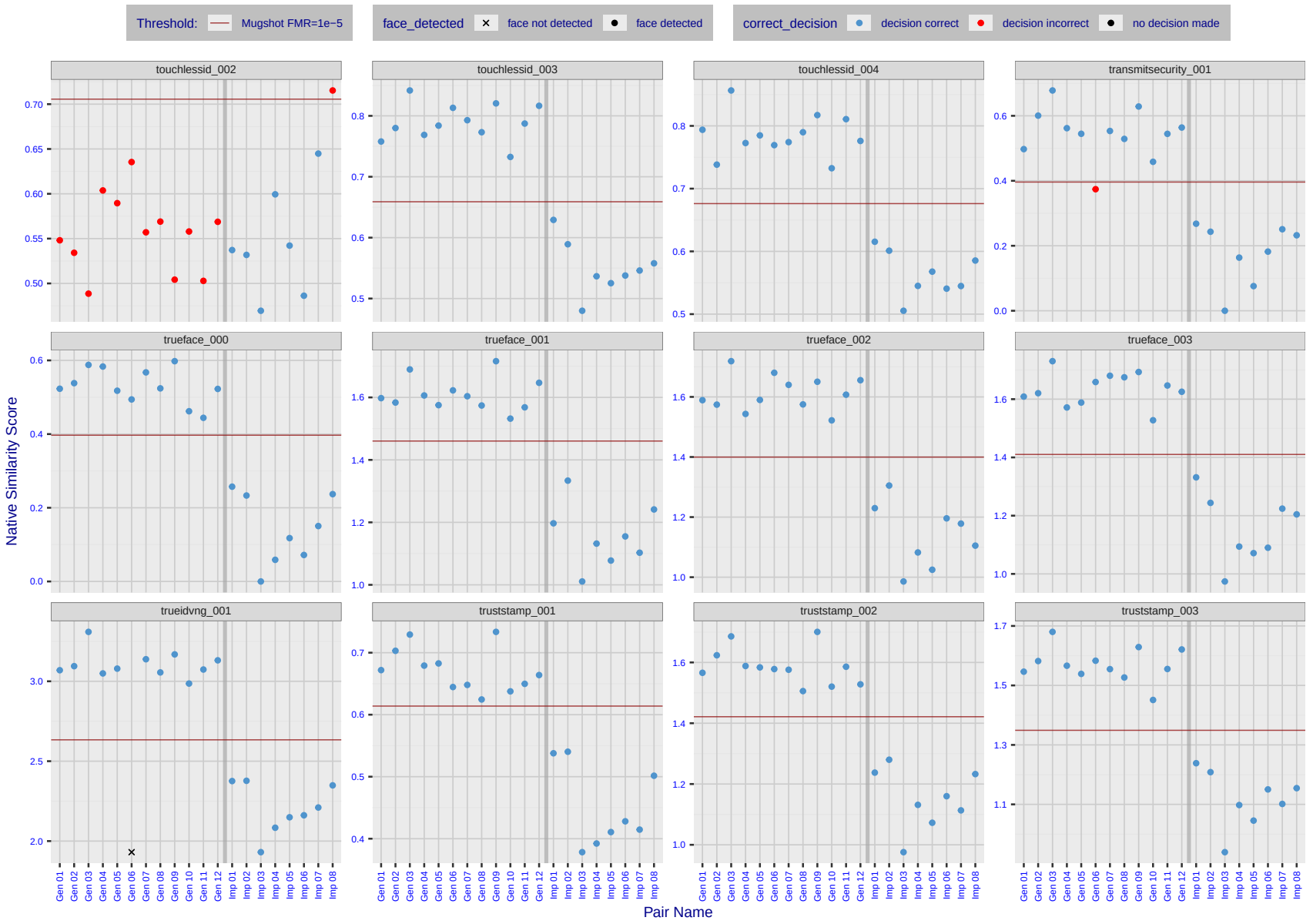


Figure 95: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

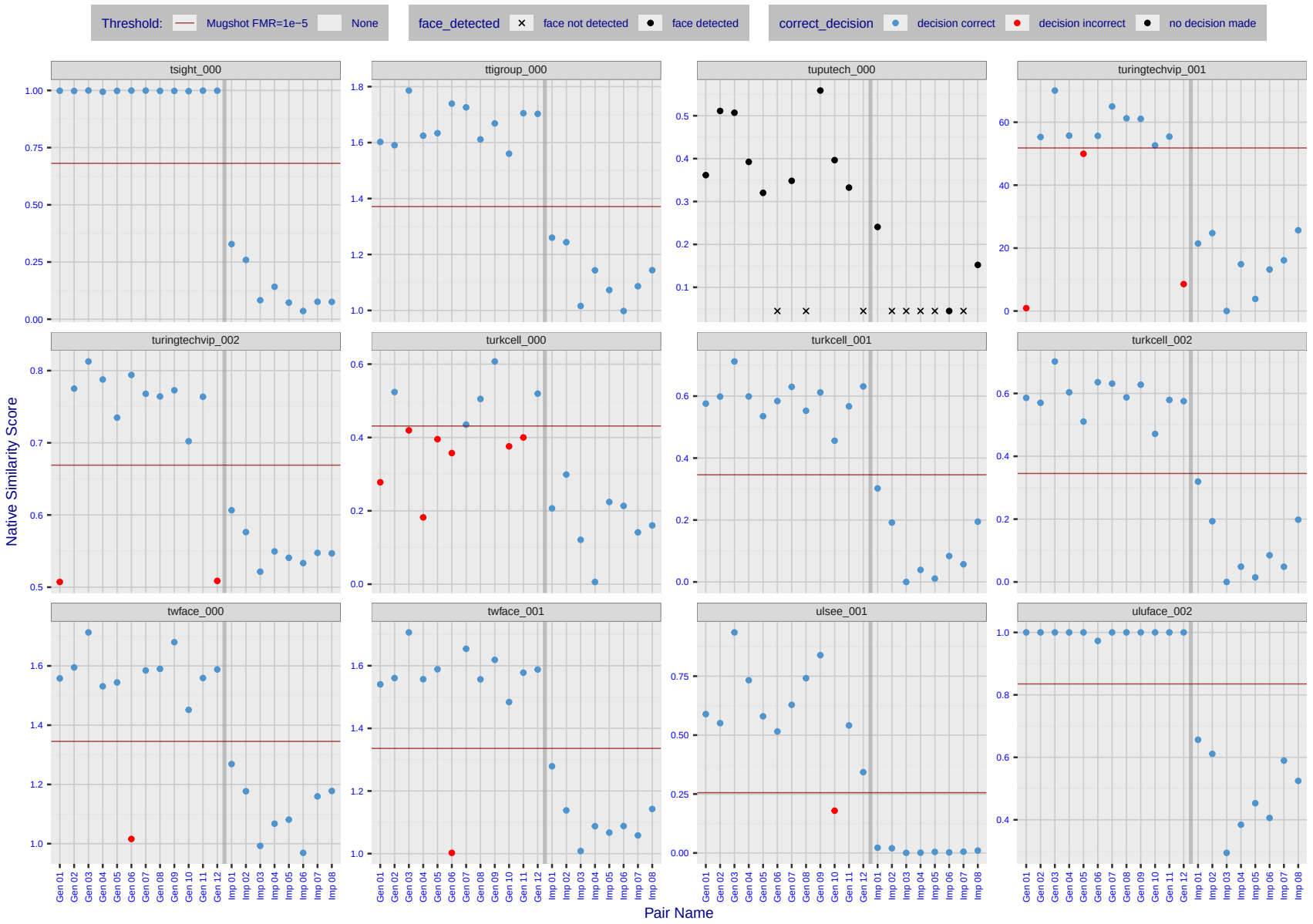


Figure 96: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

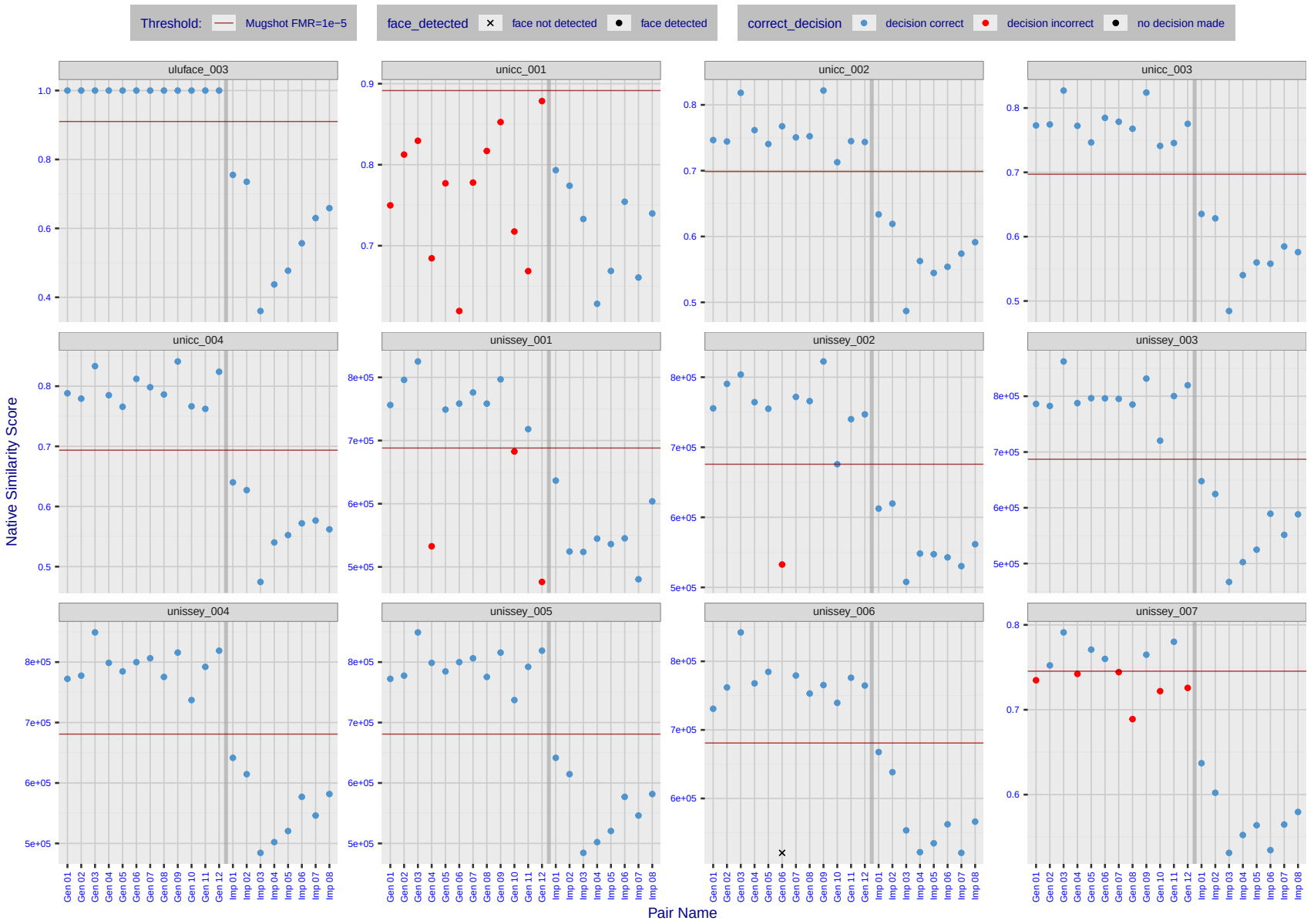


Figure 97: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

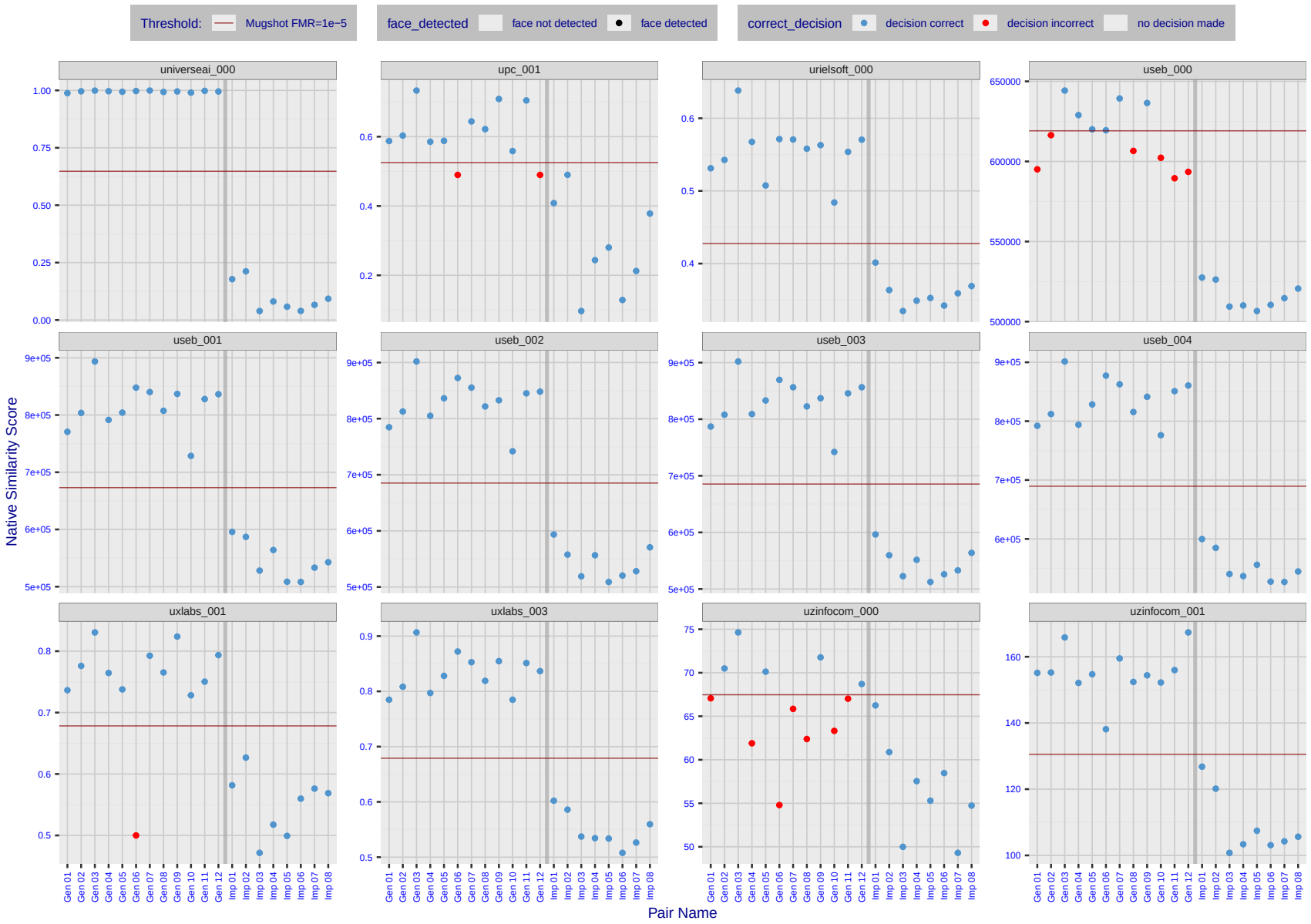


Figure 98: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

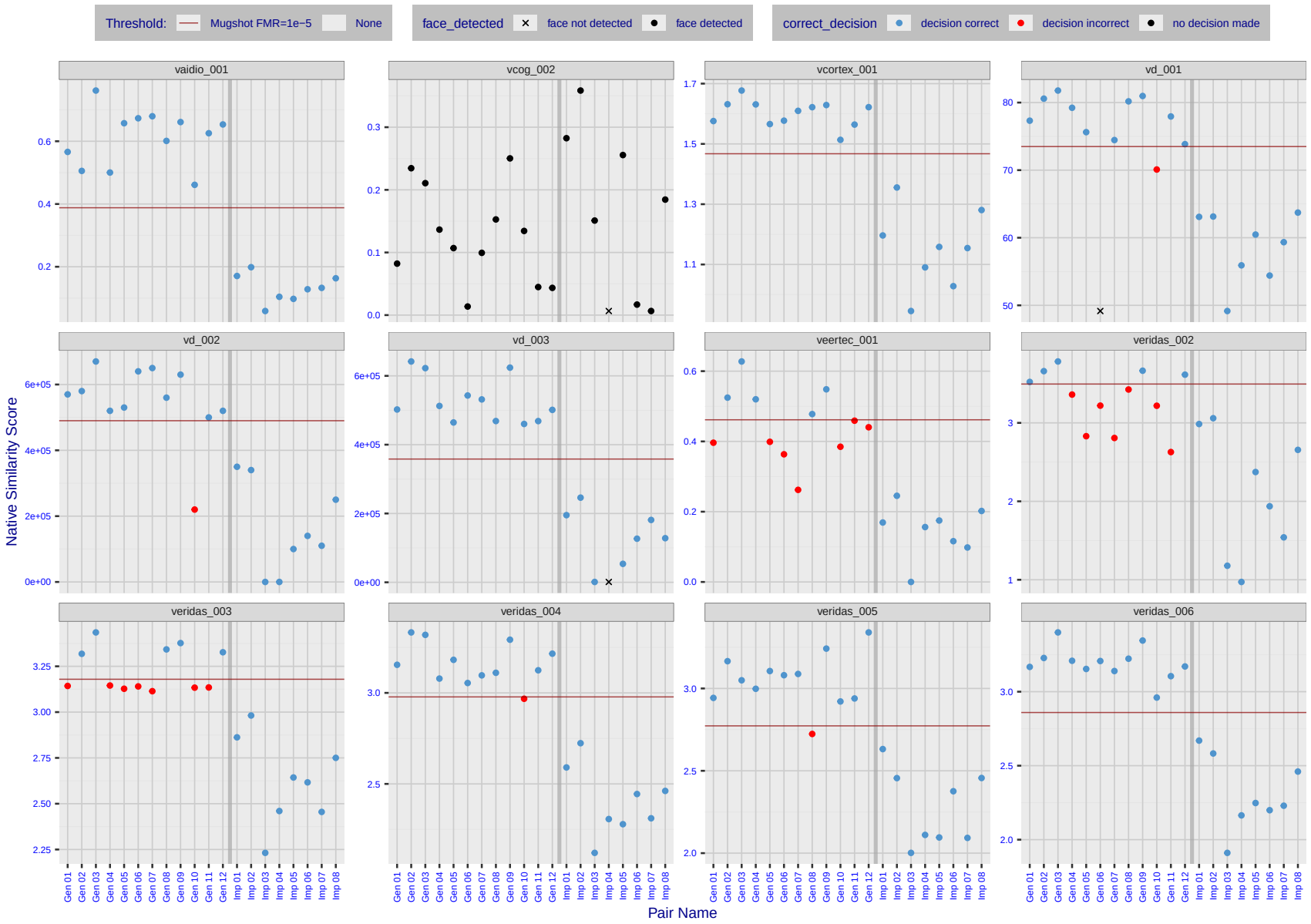


Figure 99: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

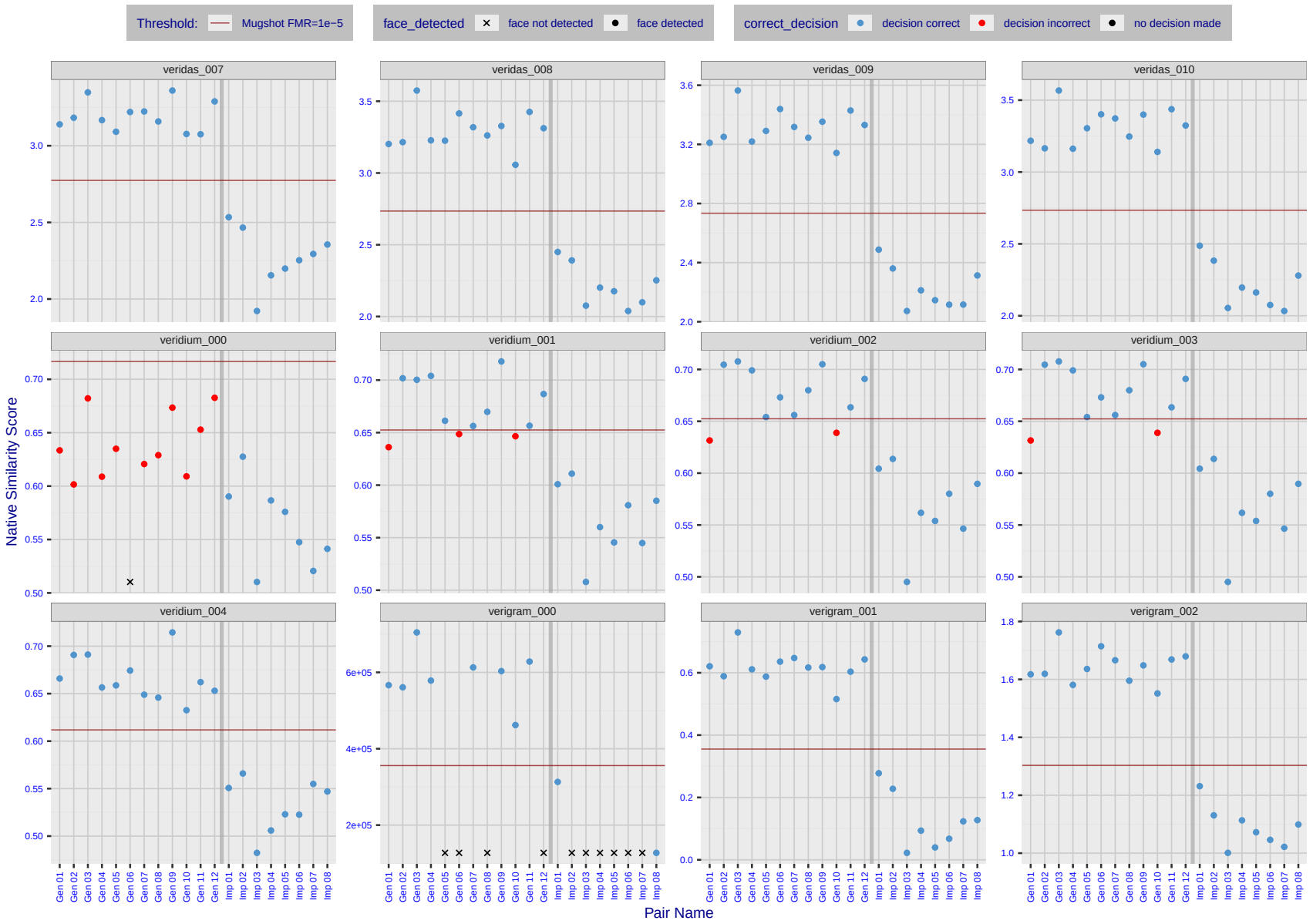


Figure 100: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

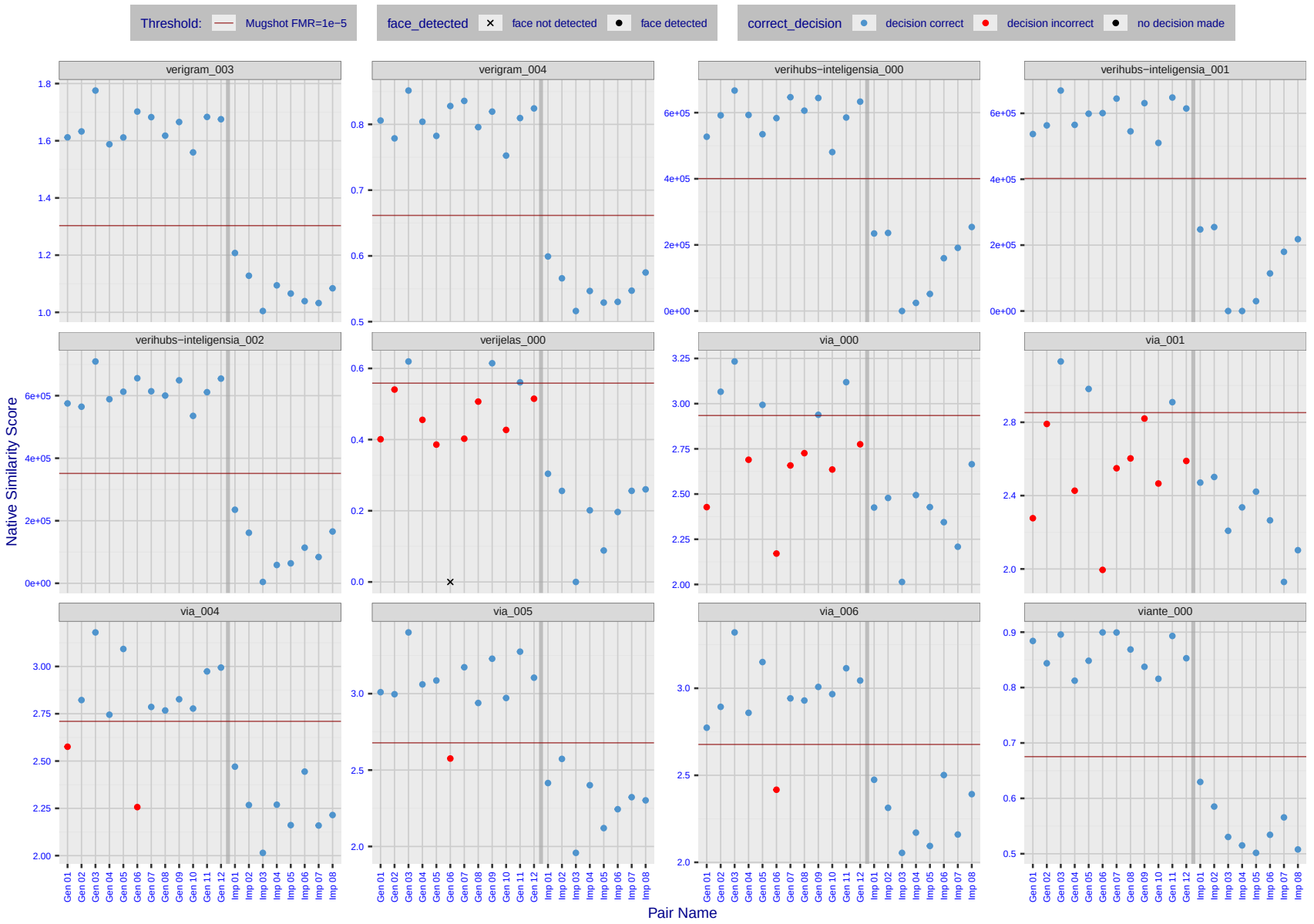


Figure 101: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

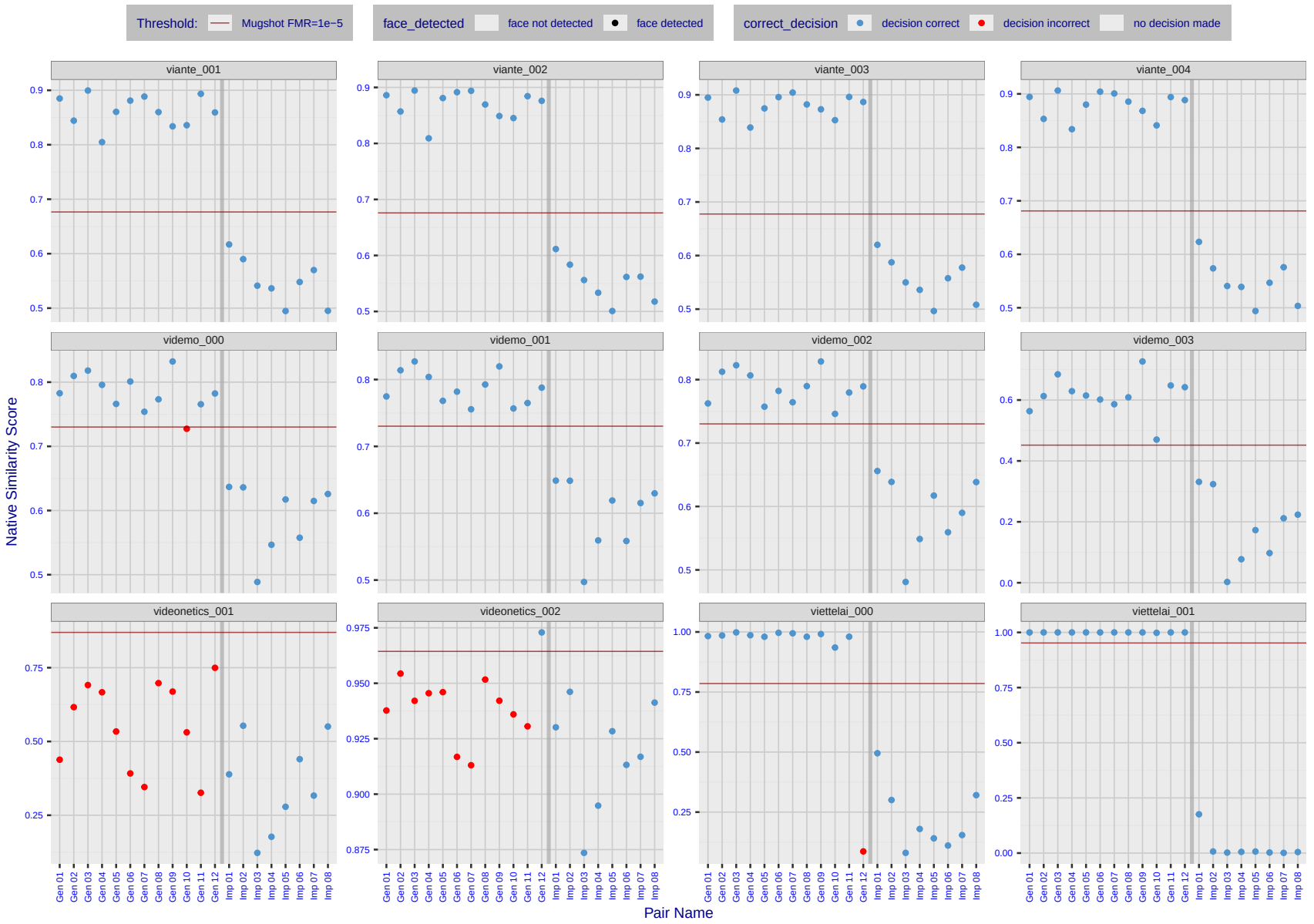


Figure 102: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

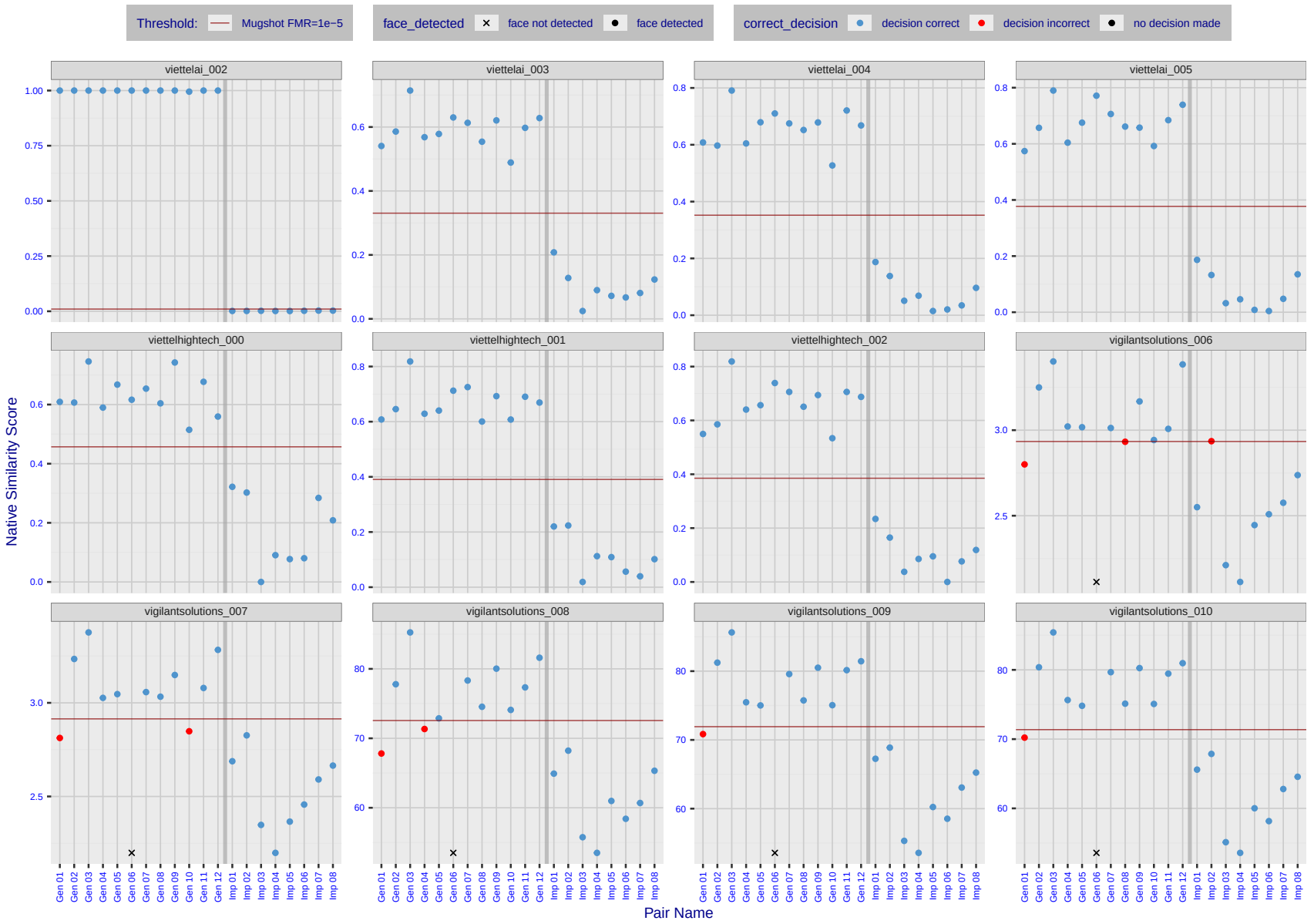


Figure 103: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

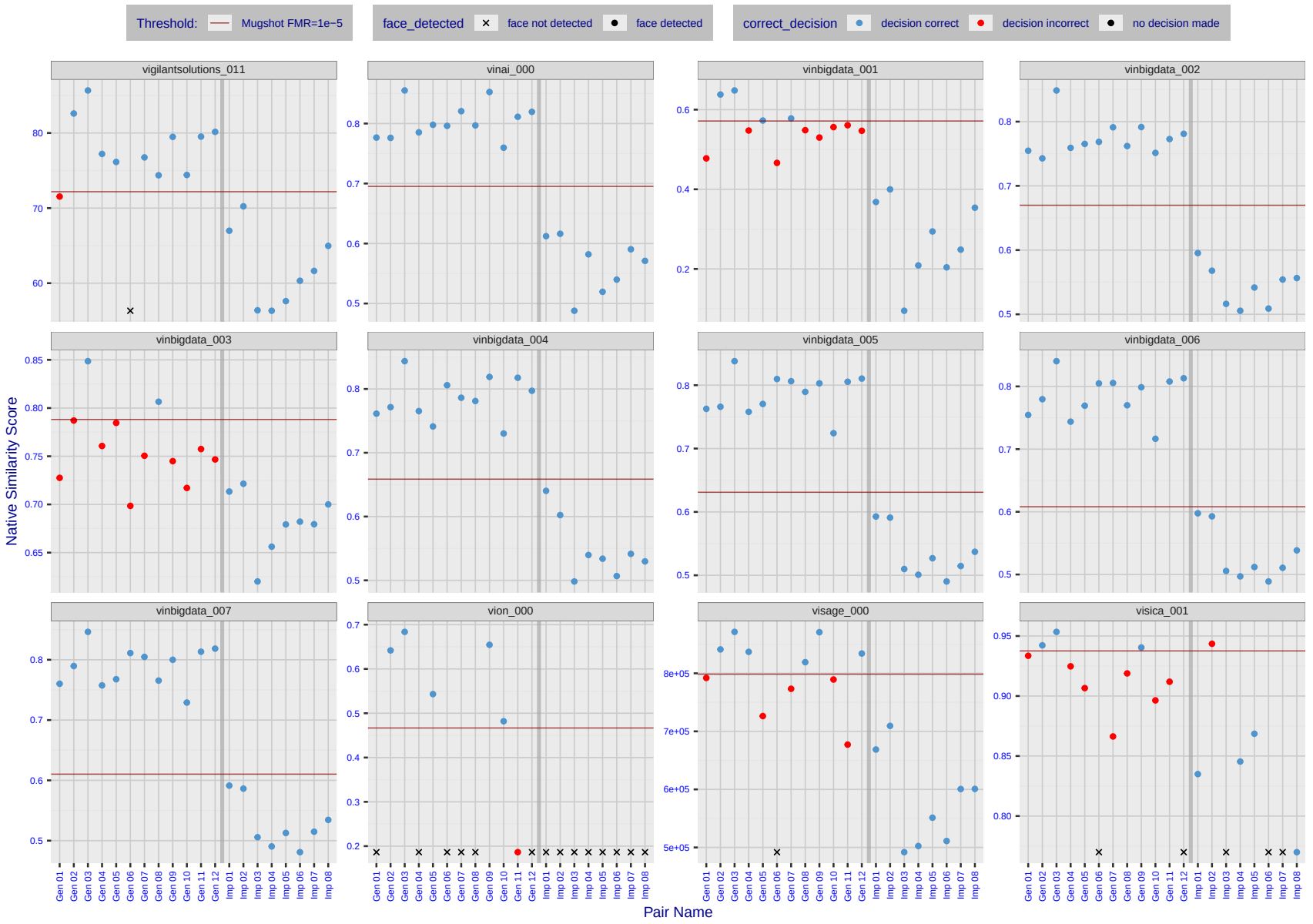


Figure 104: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

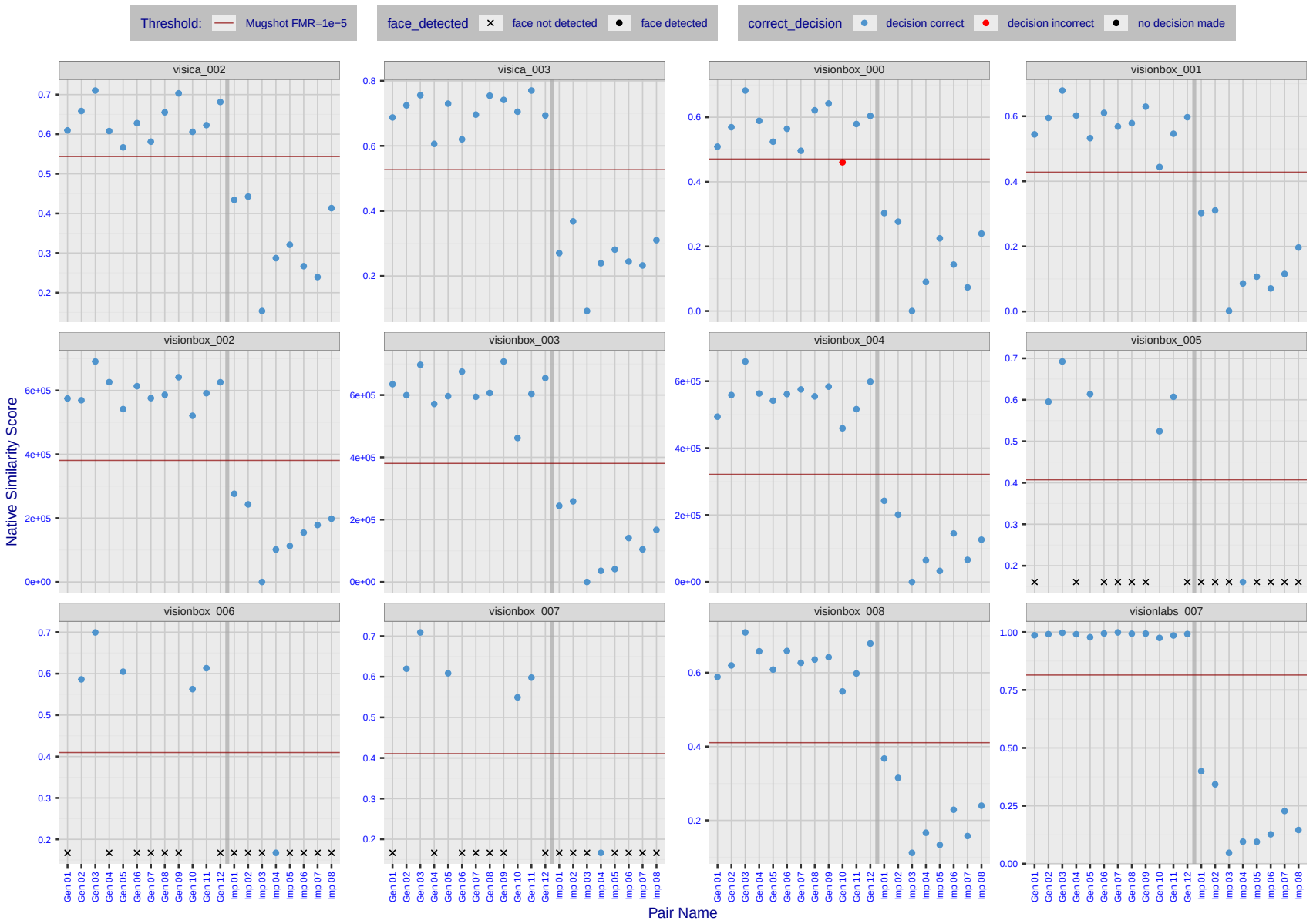


Figure 105: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

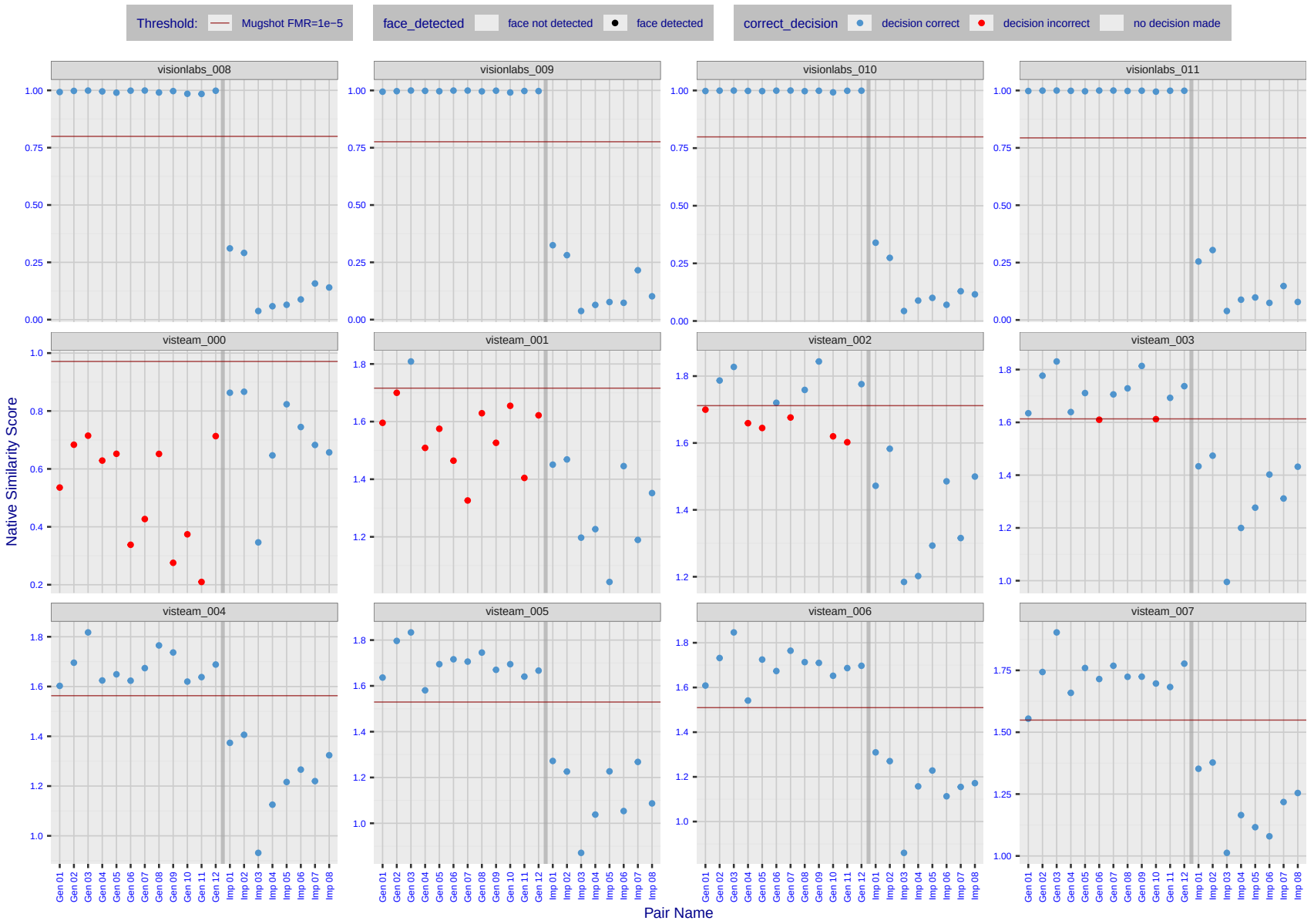


Figure 106: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

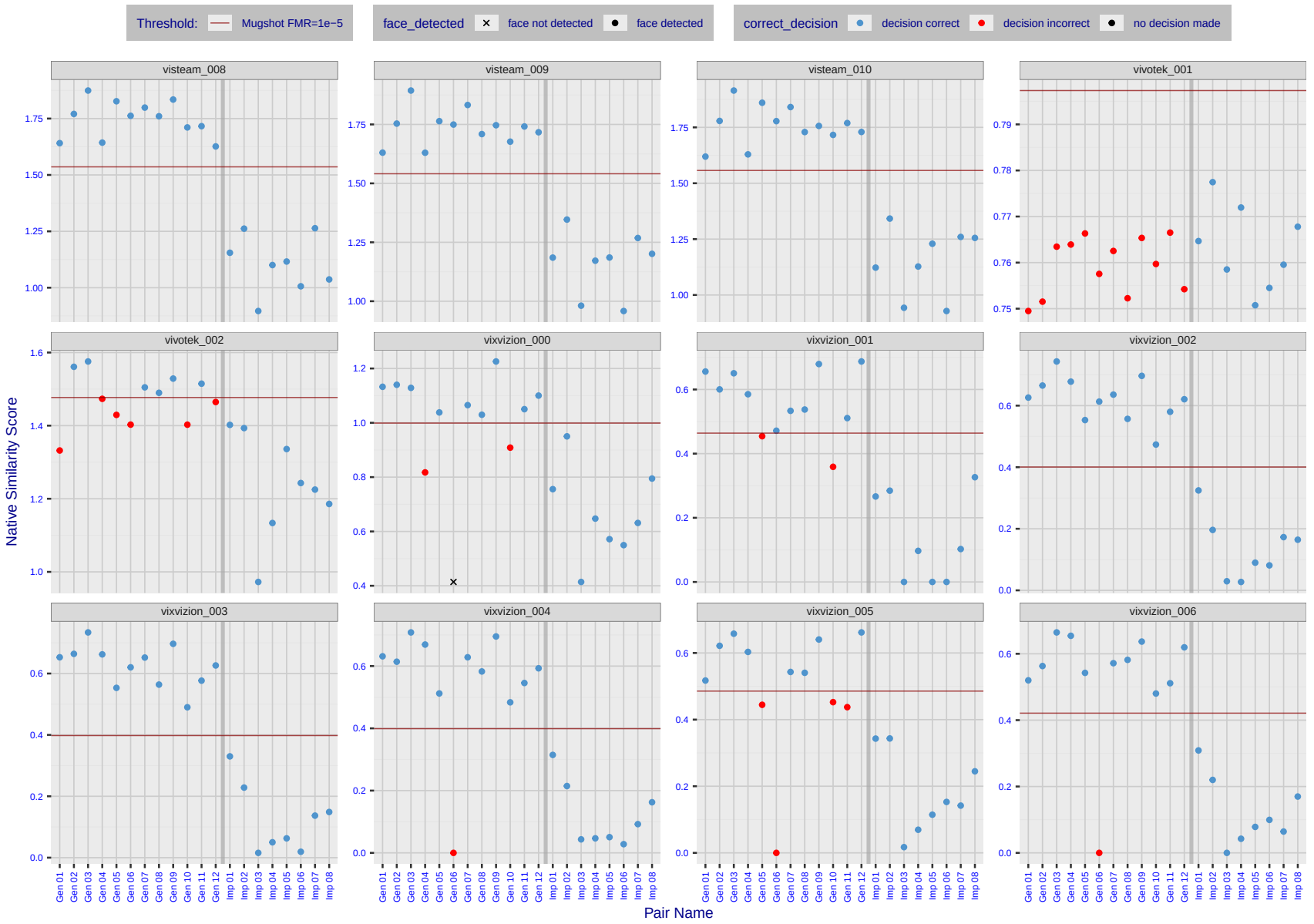


Figure 107: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

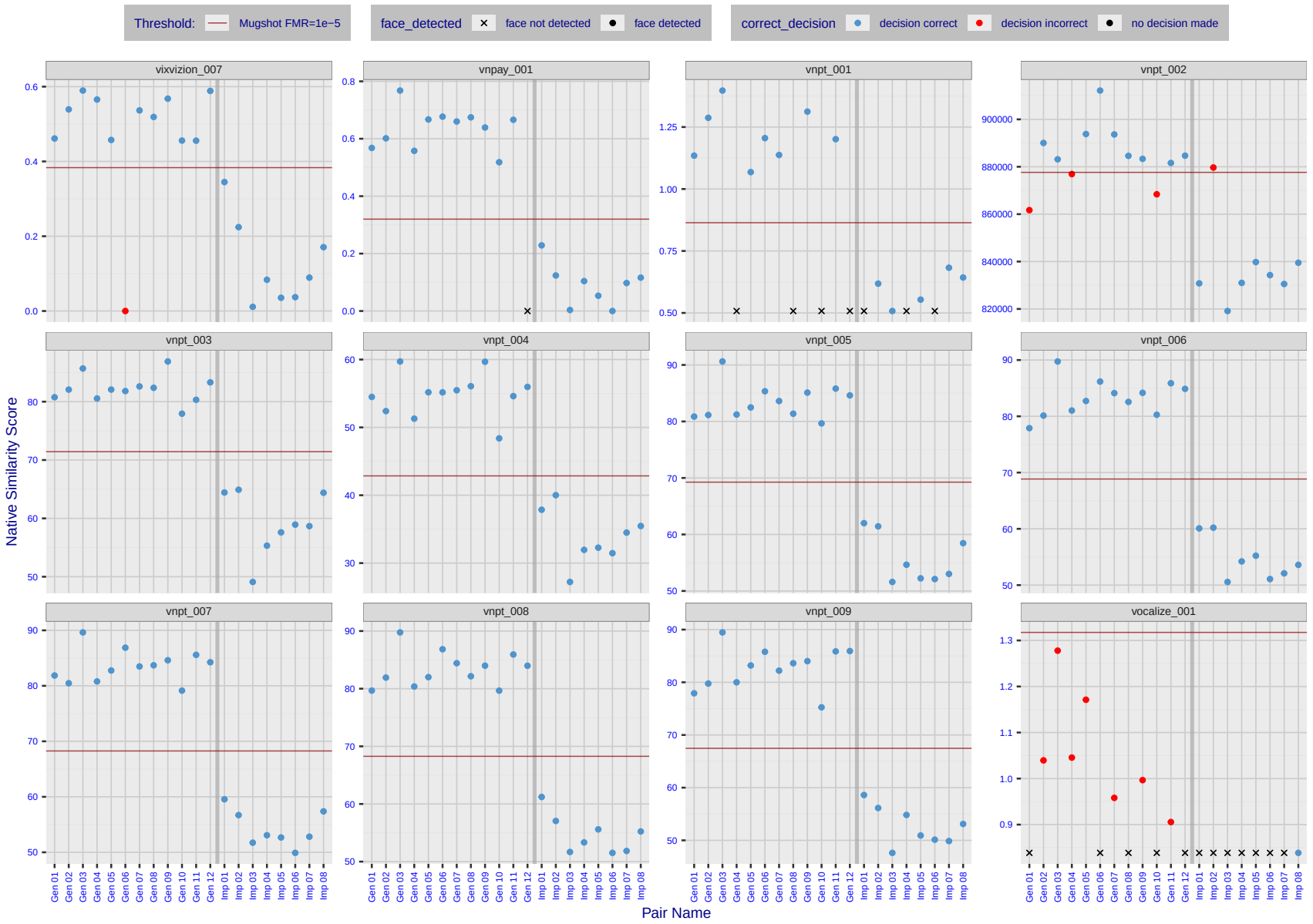


Figure 108: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

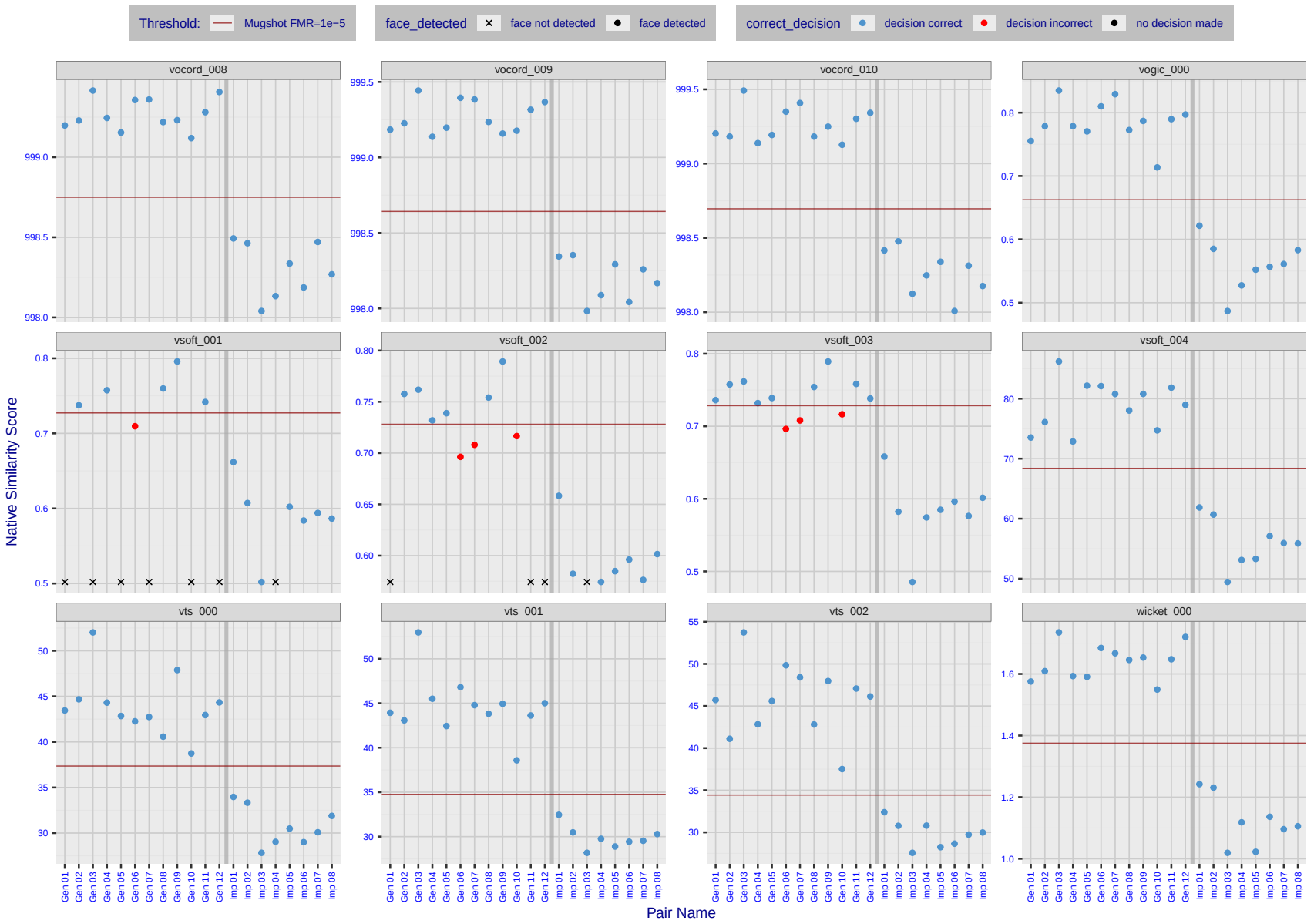


Figure 109: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

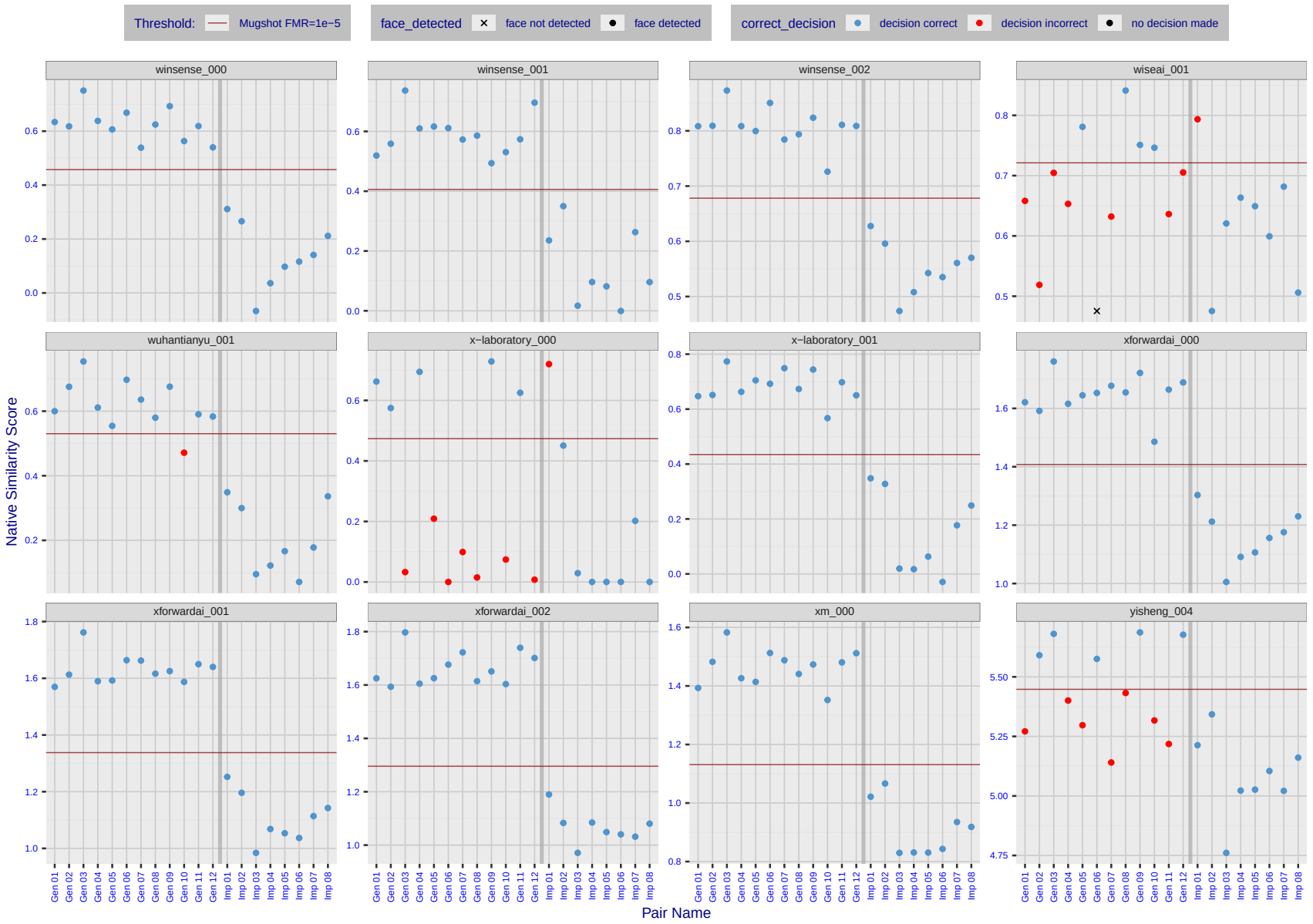


Figure 110: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

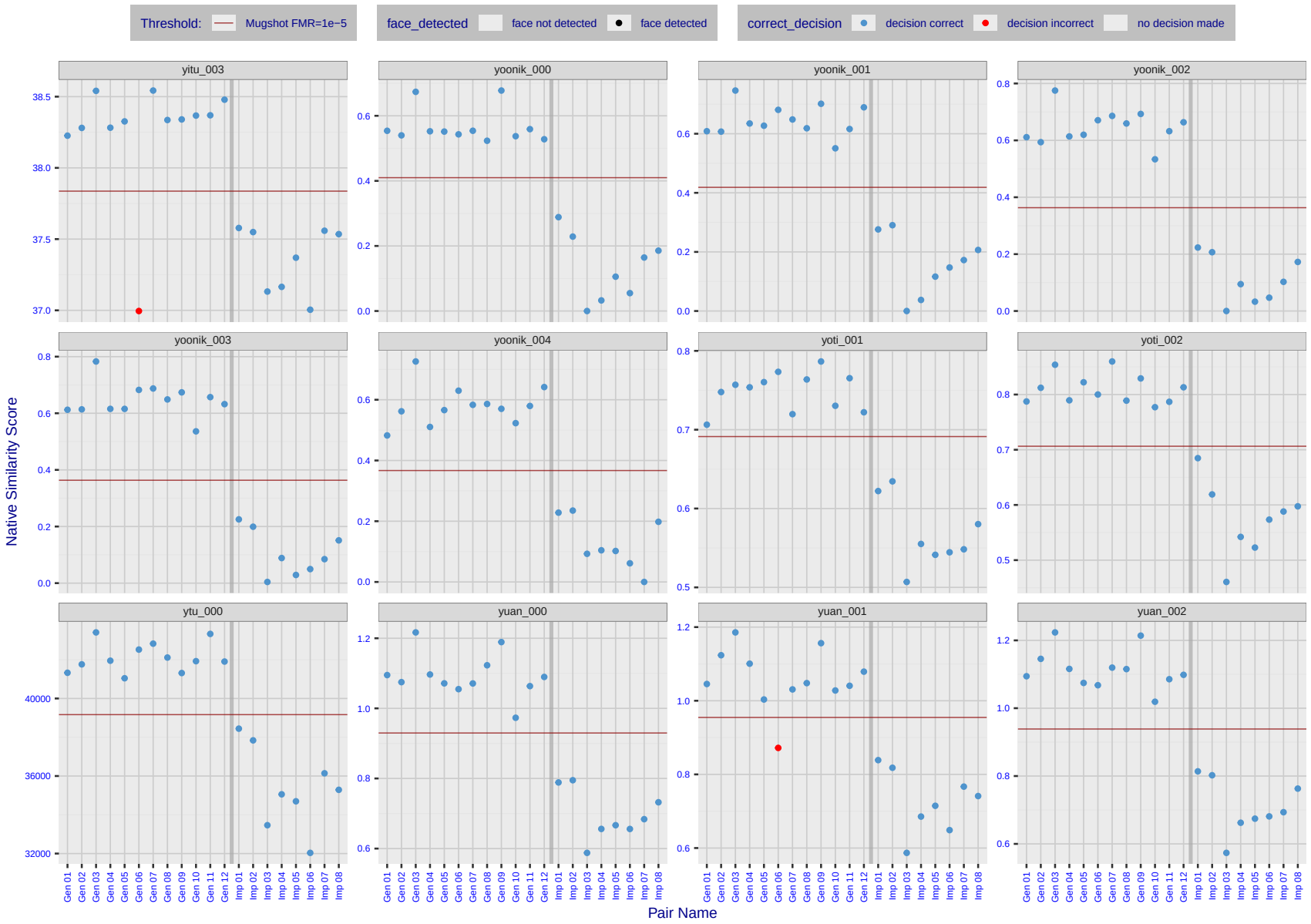


Figure 111: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

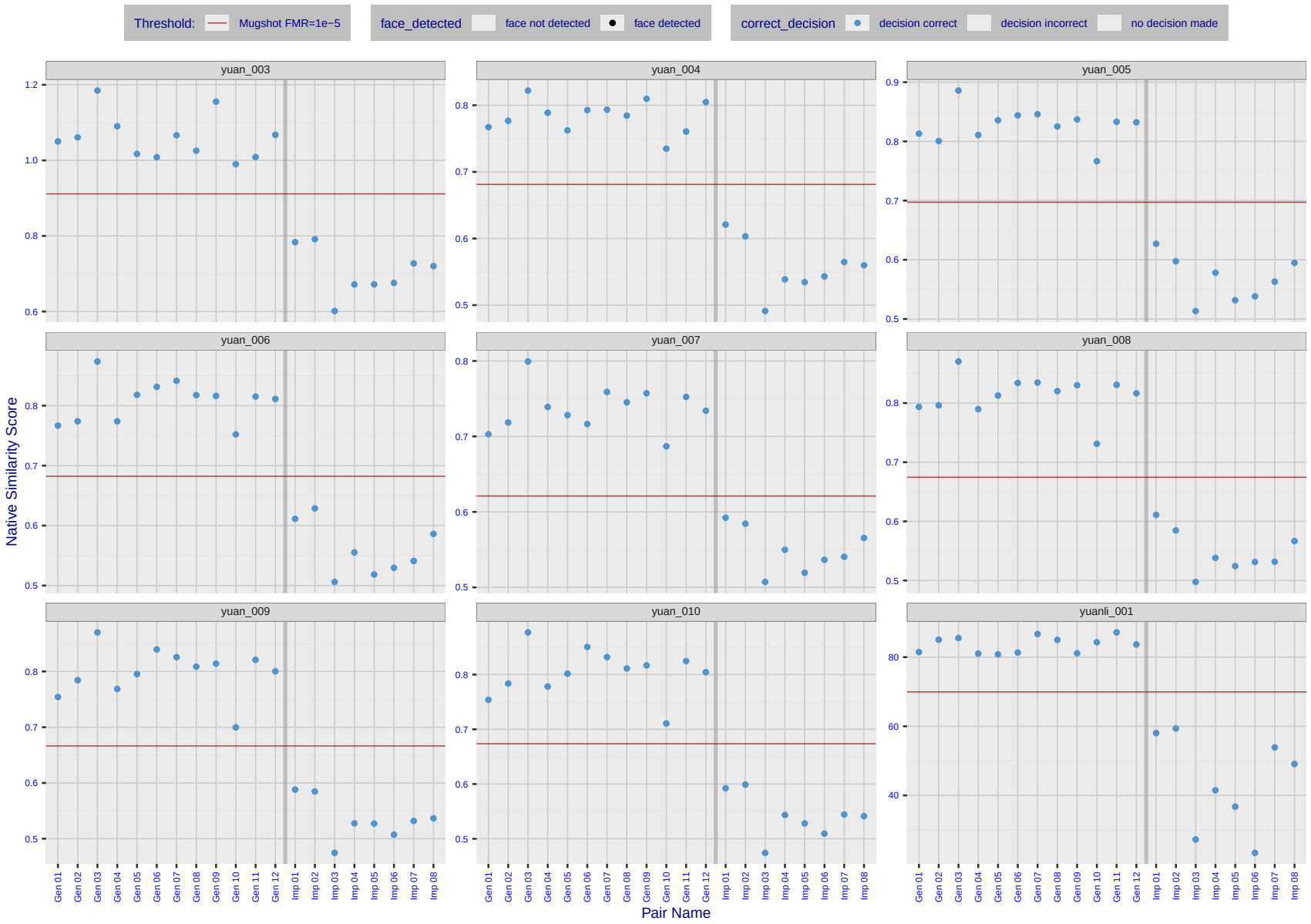


Figure 112: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

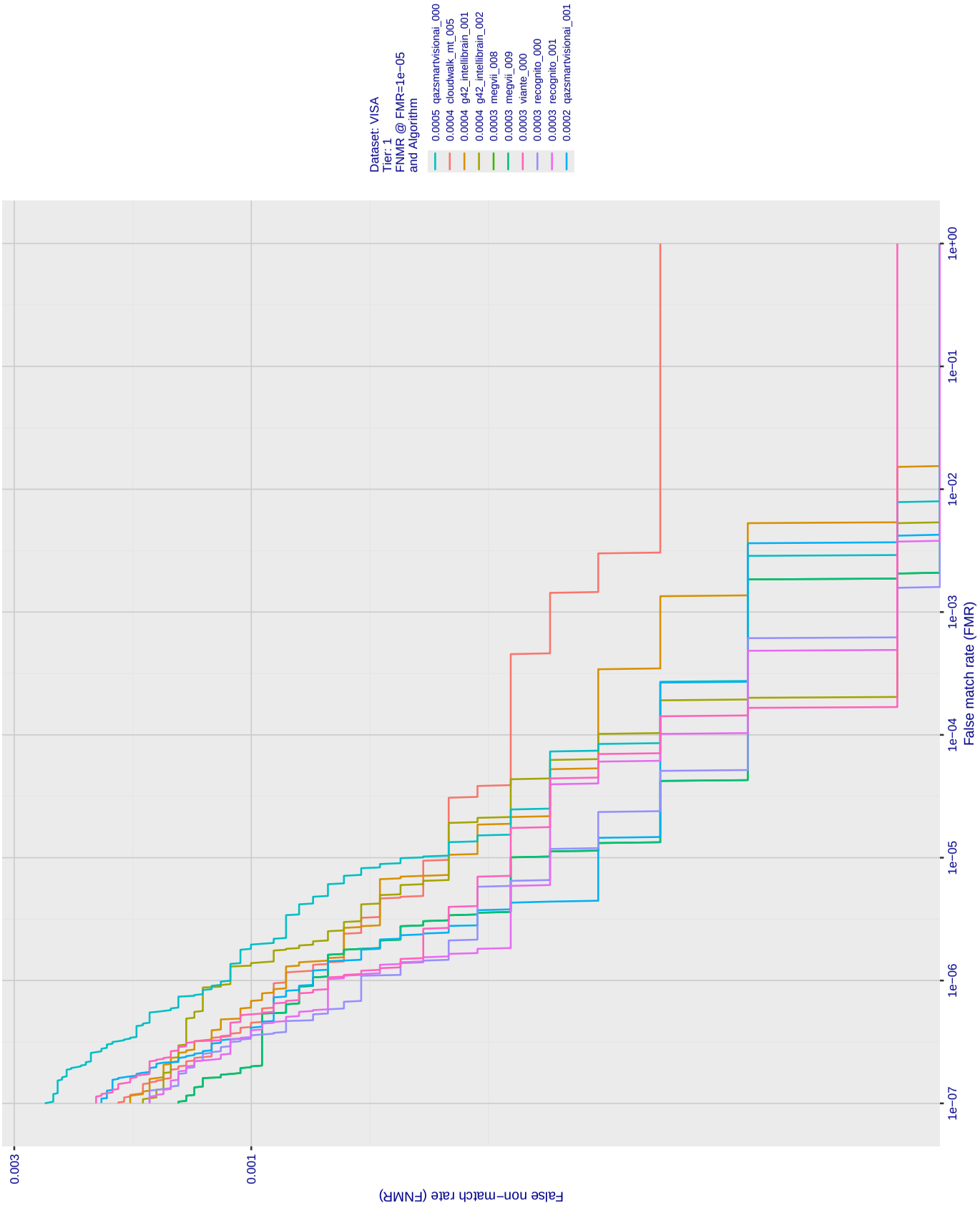


Figure 113: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

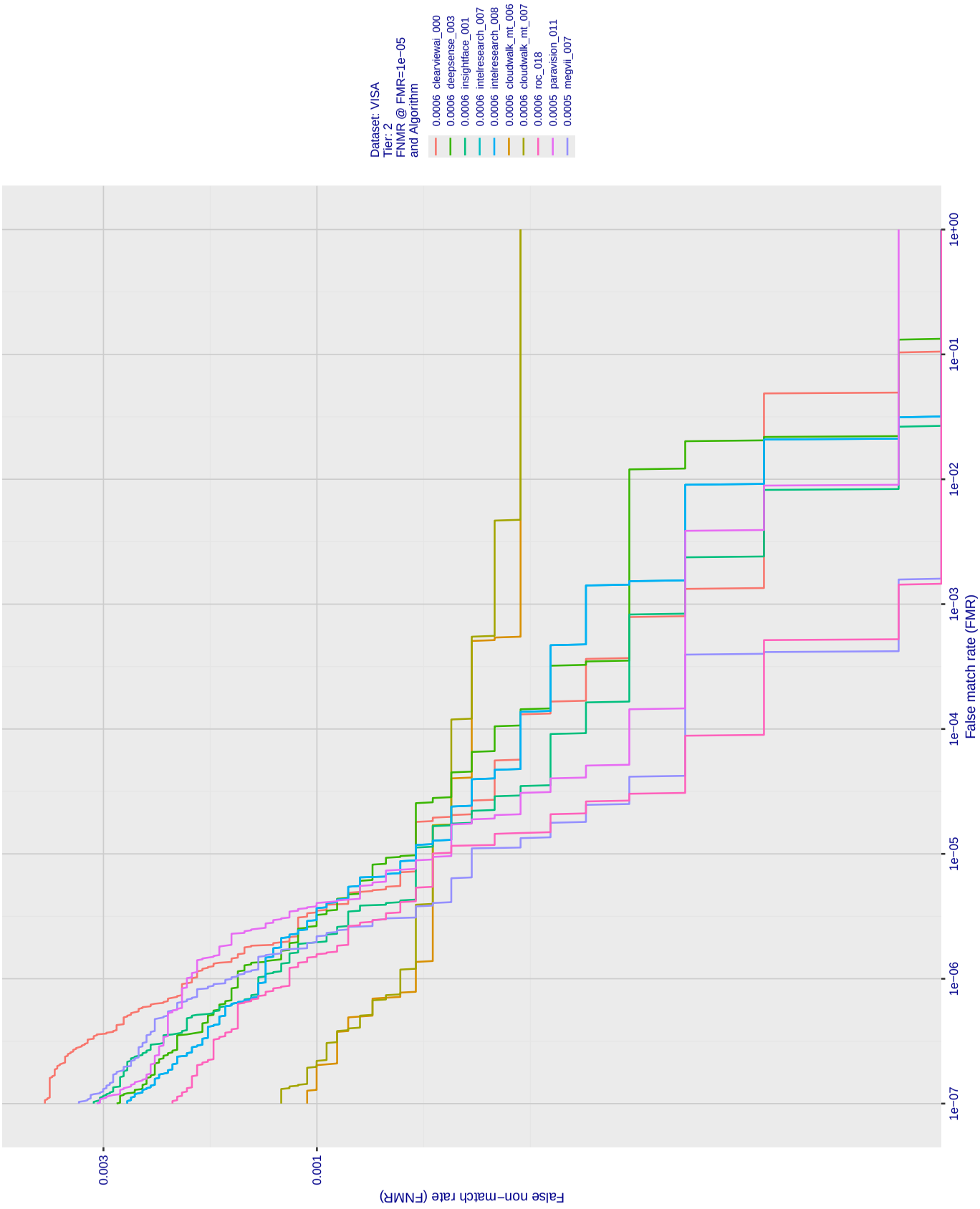


Figure 114: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

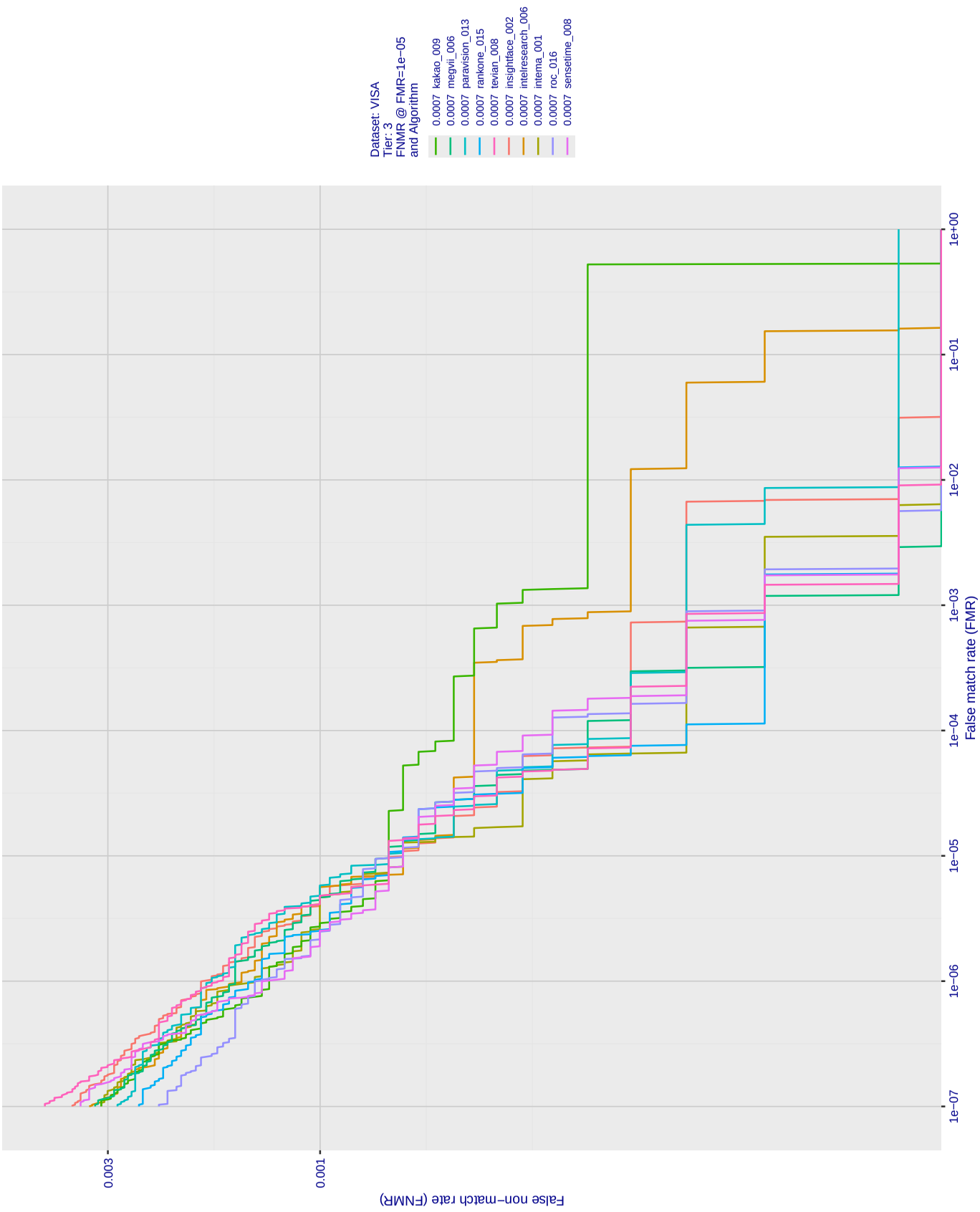


Figure 115: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

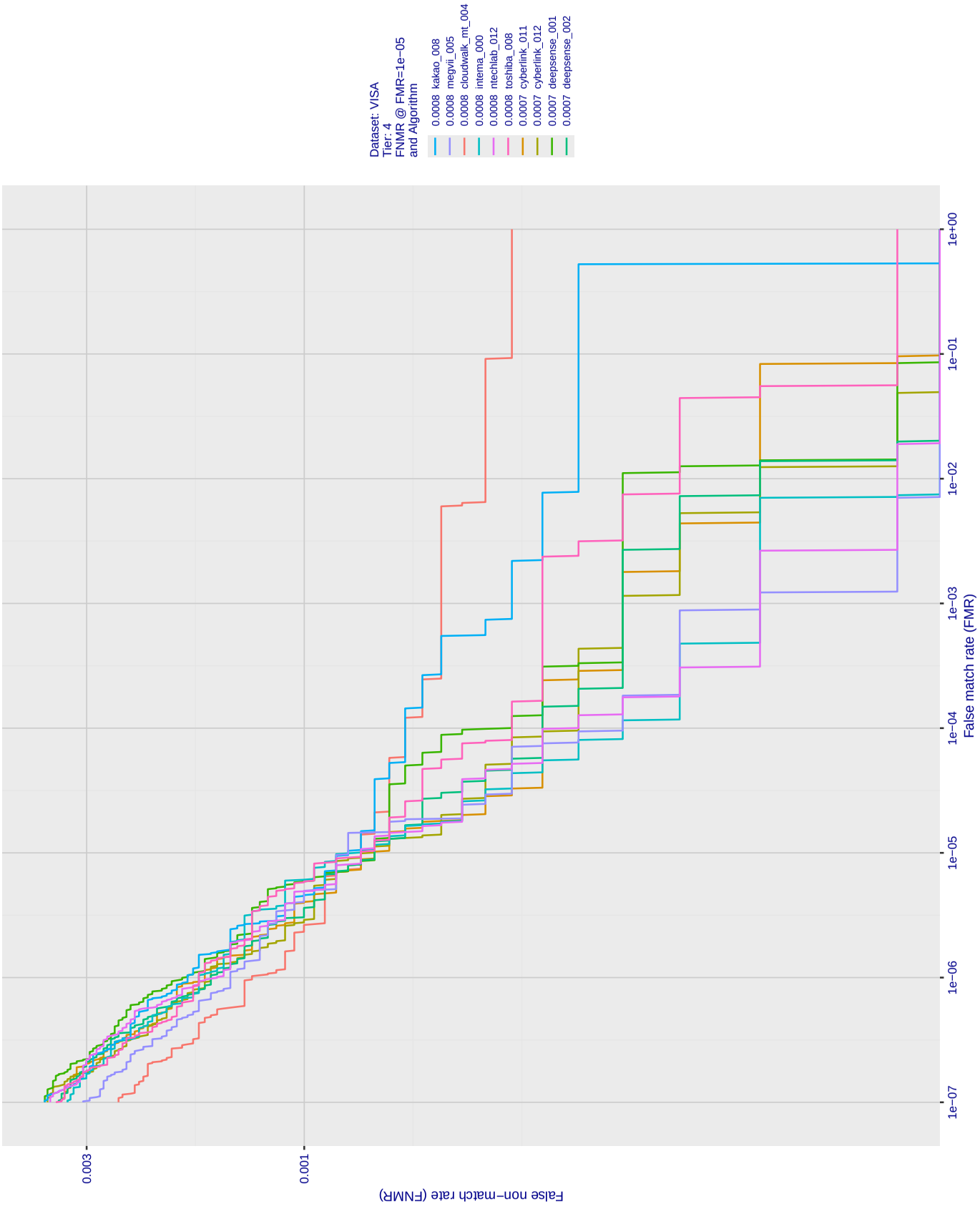


Figure 116: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

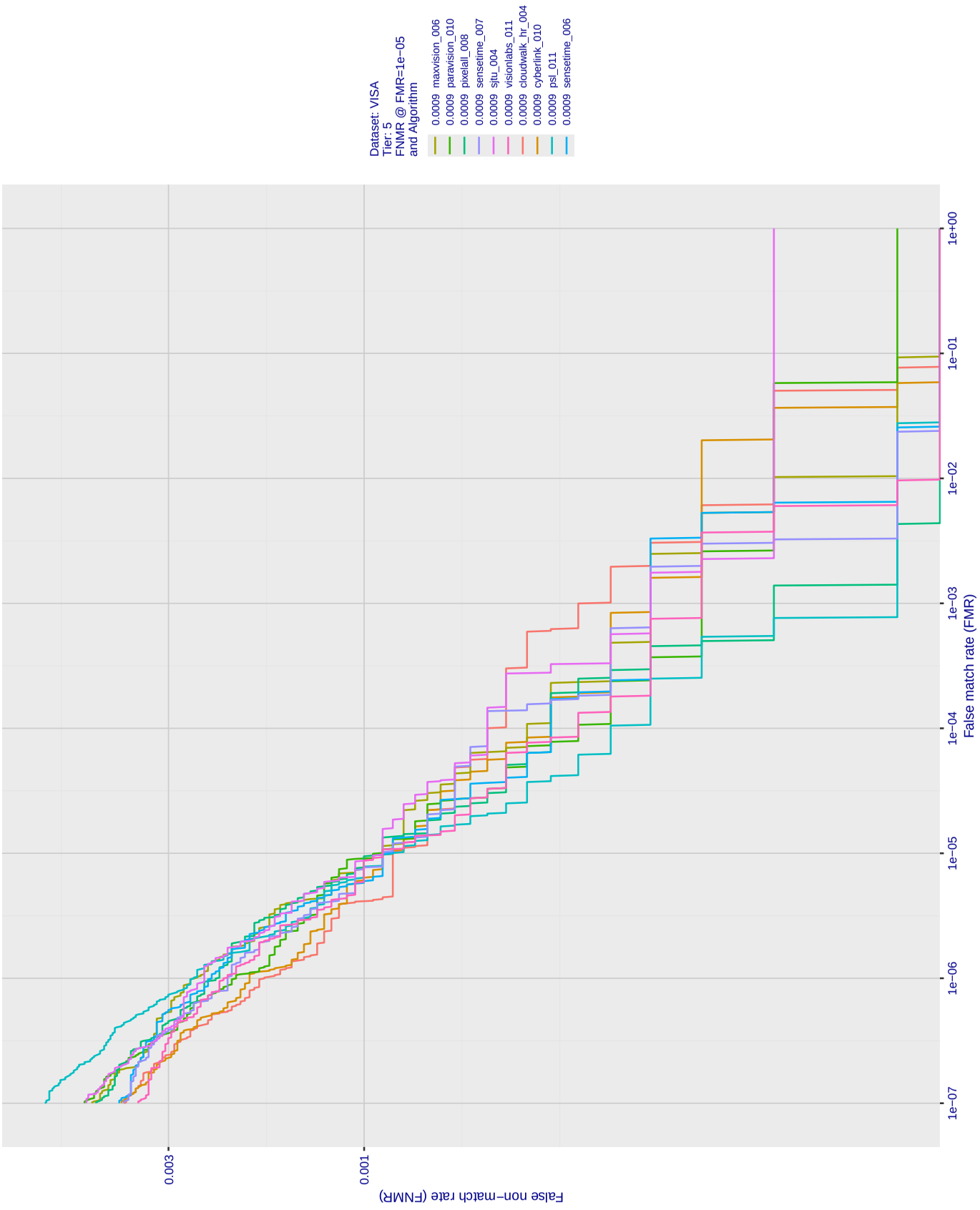


Figure 117: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

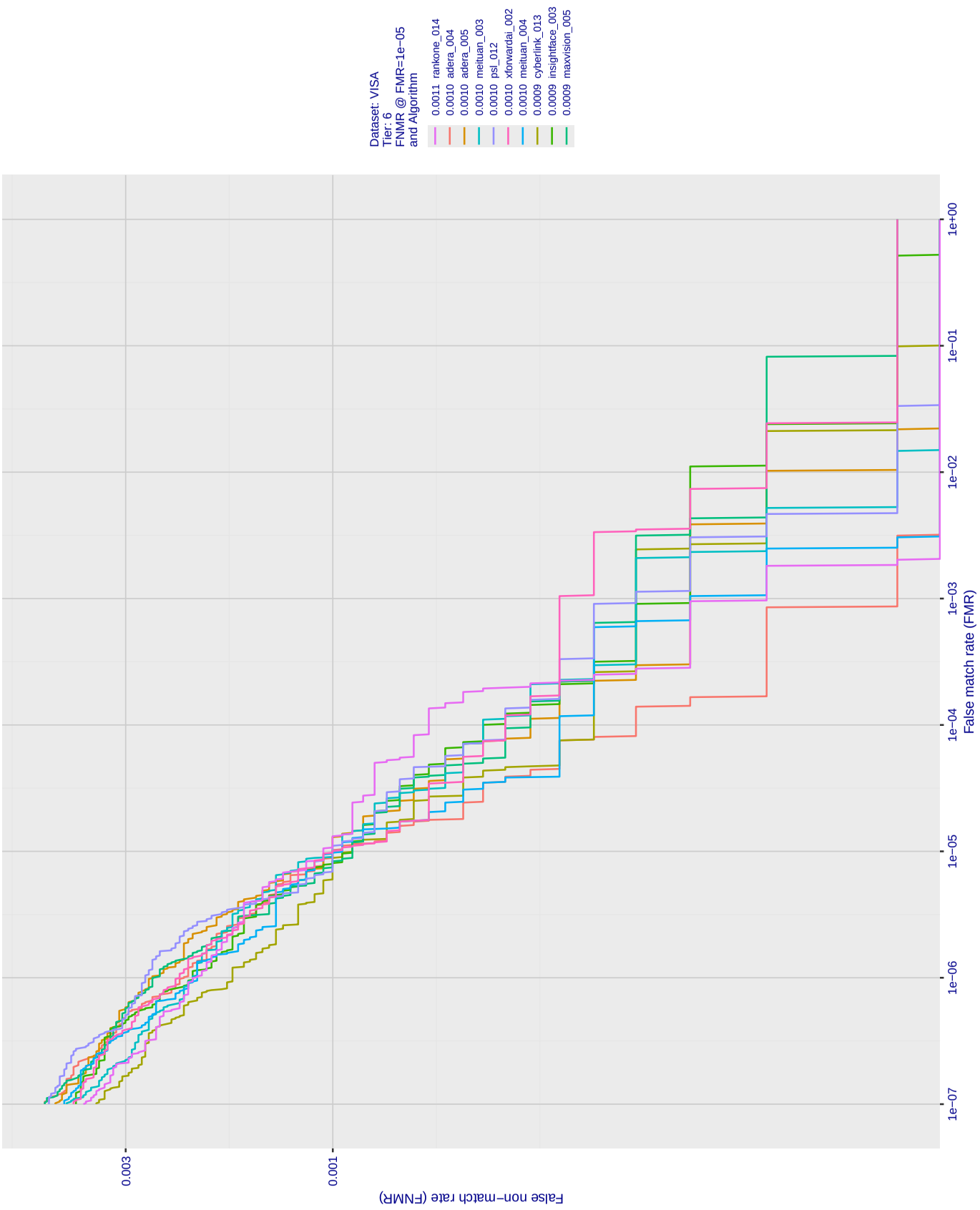


Figure 118: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

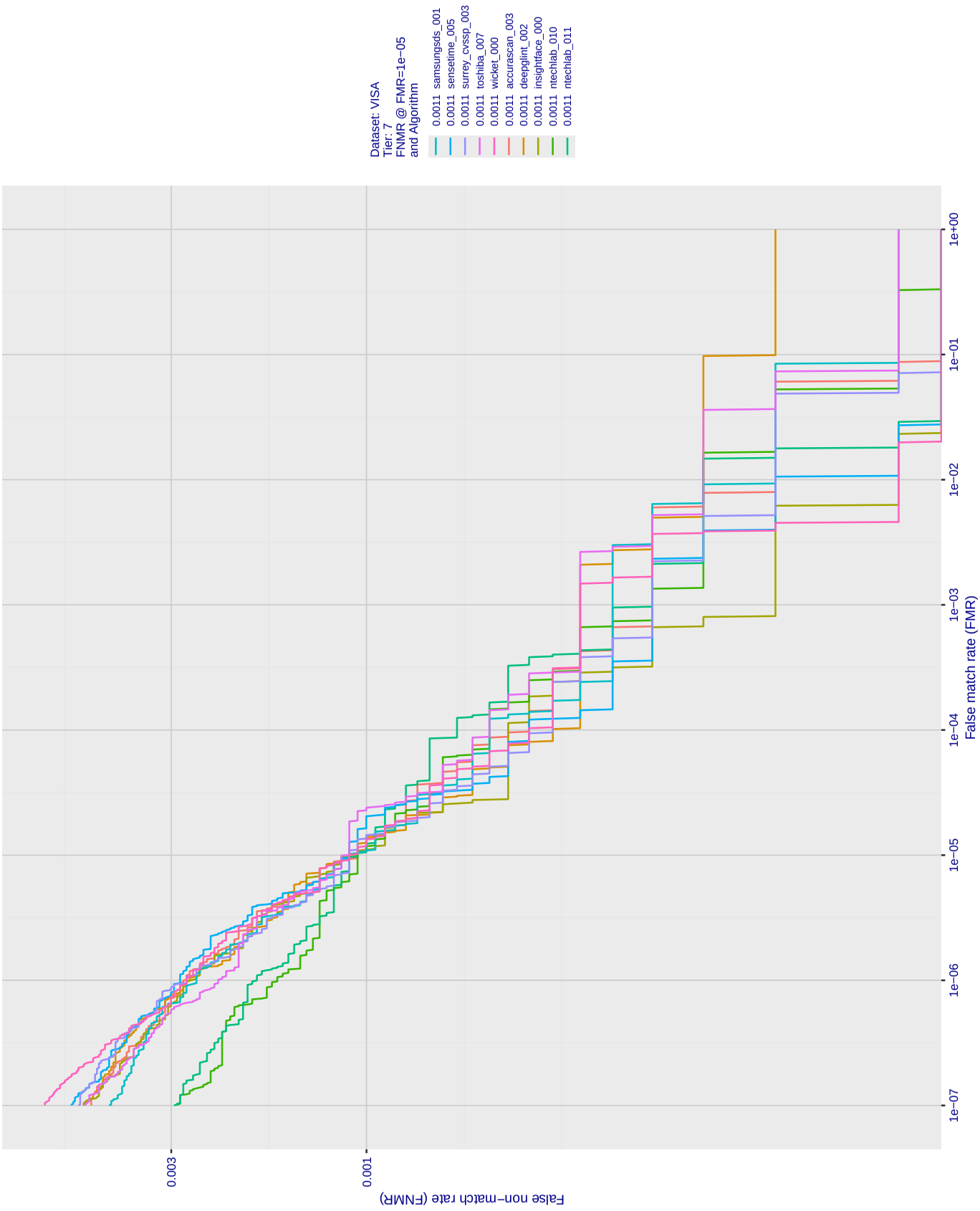


Figure 119: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

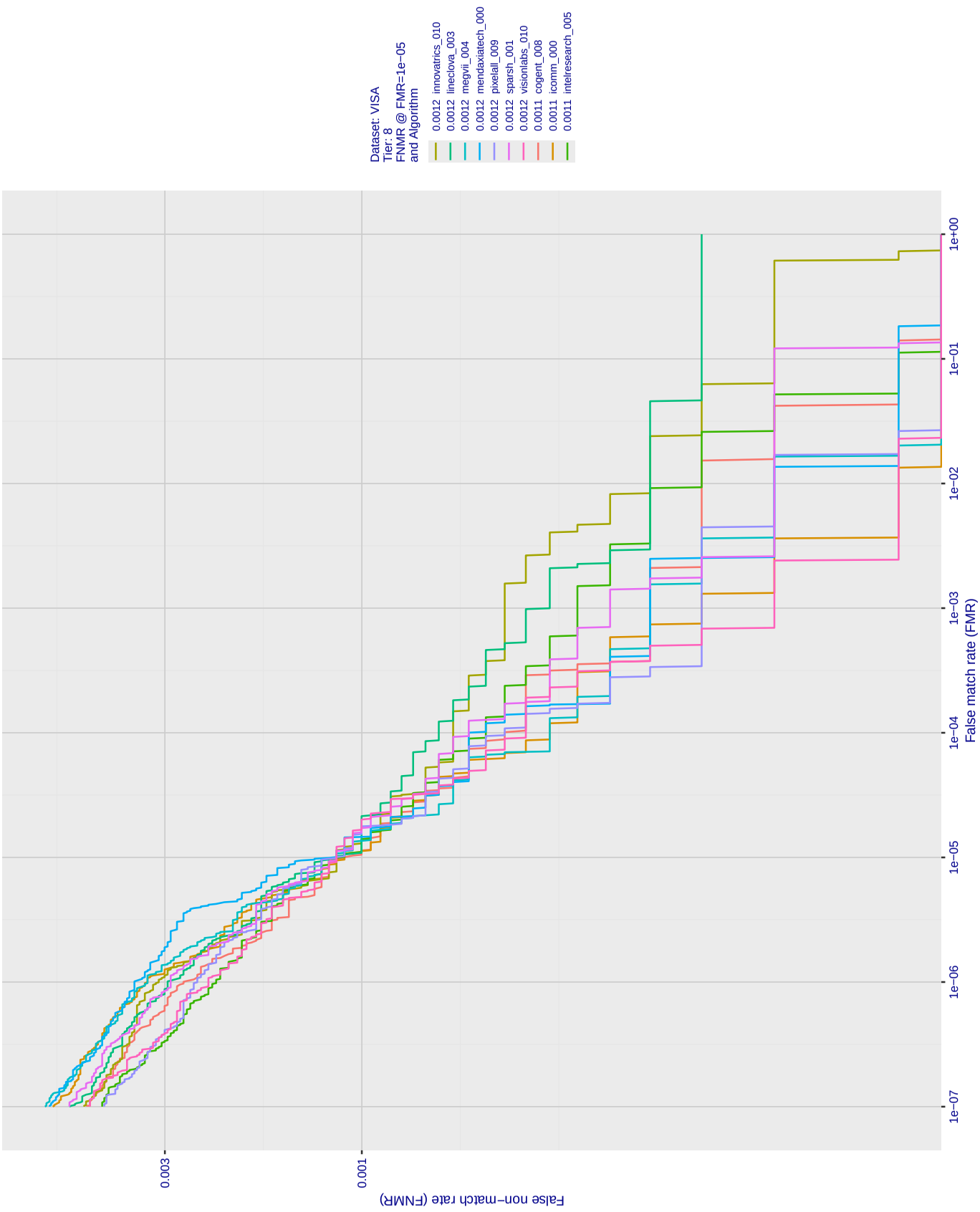


Figure 120: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

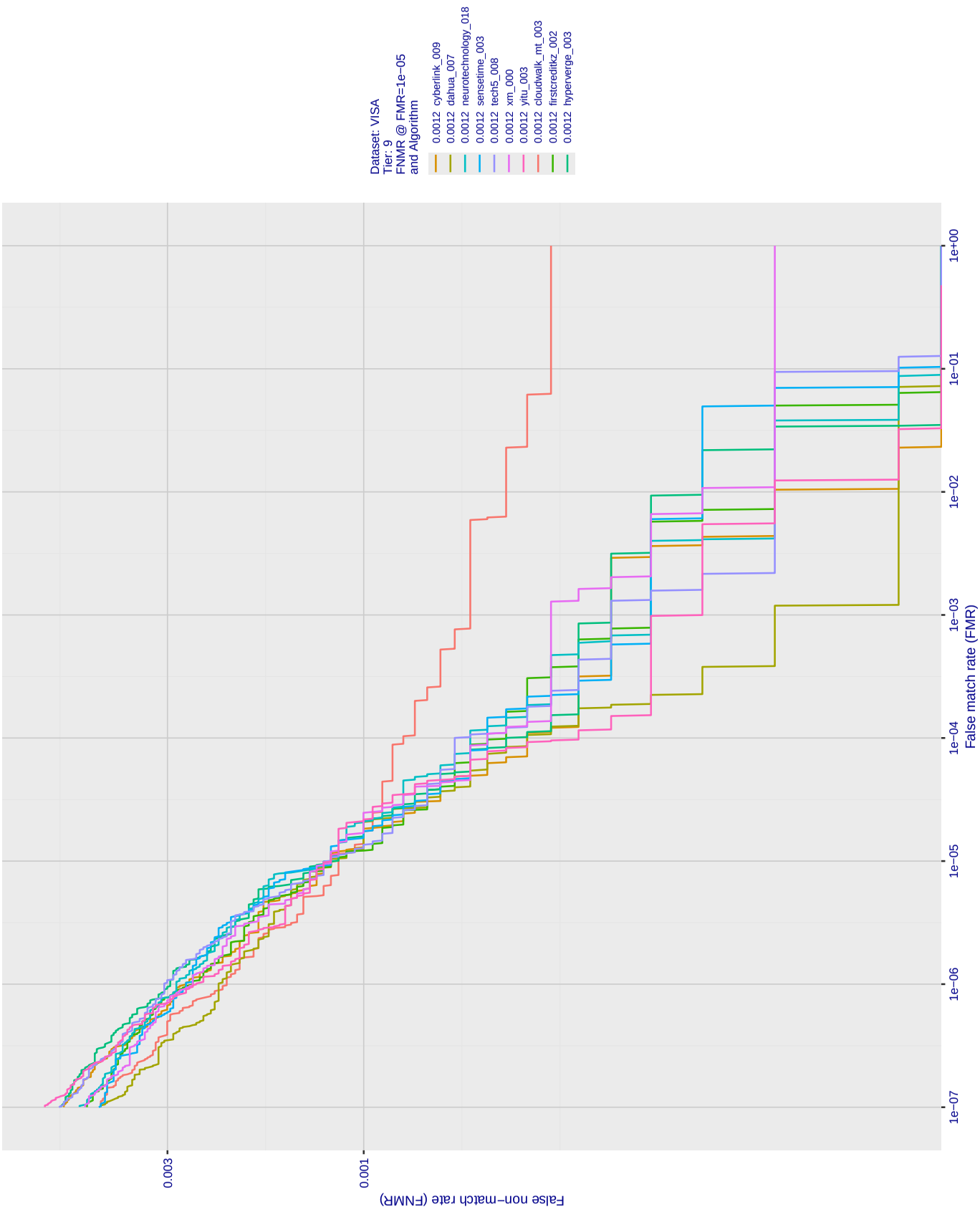


Figure 121: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

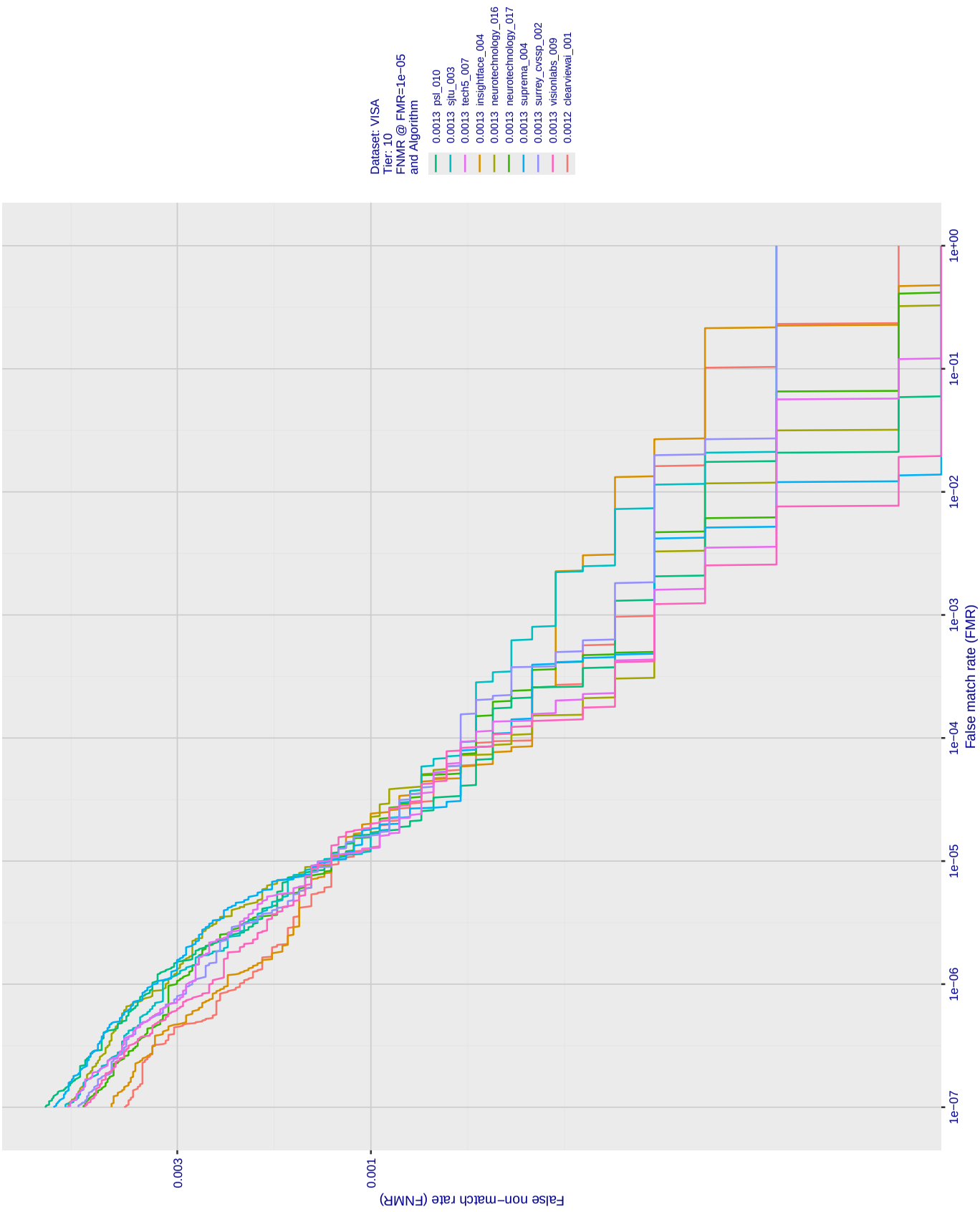


Figure 122: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

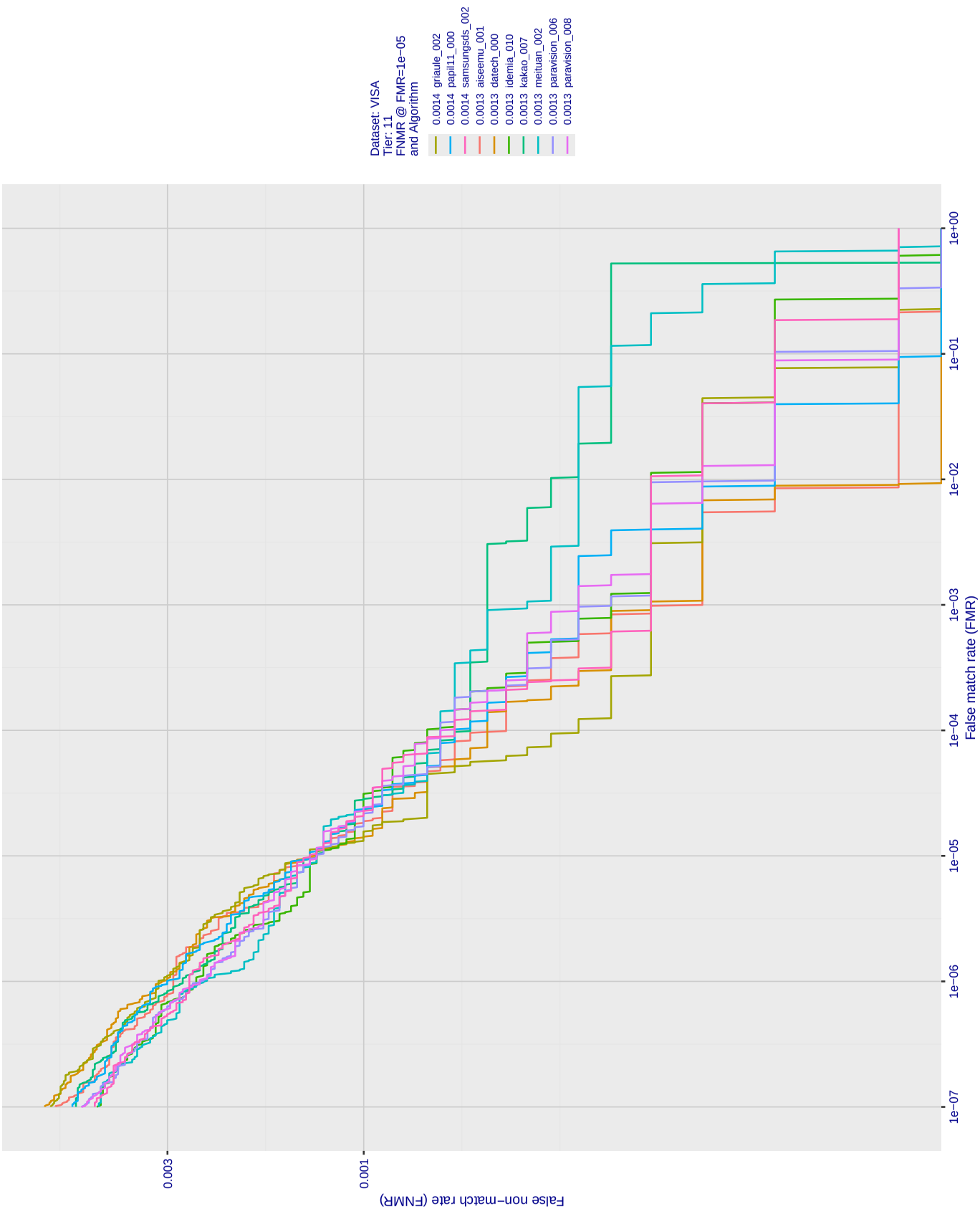


Figure 123: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

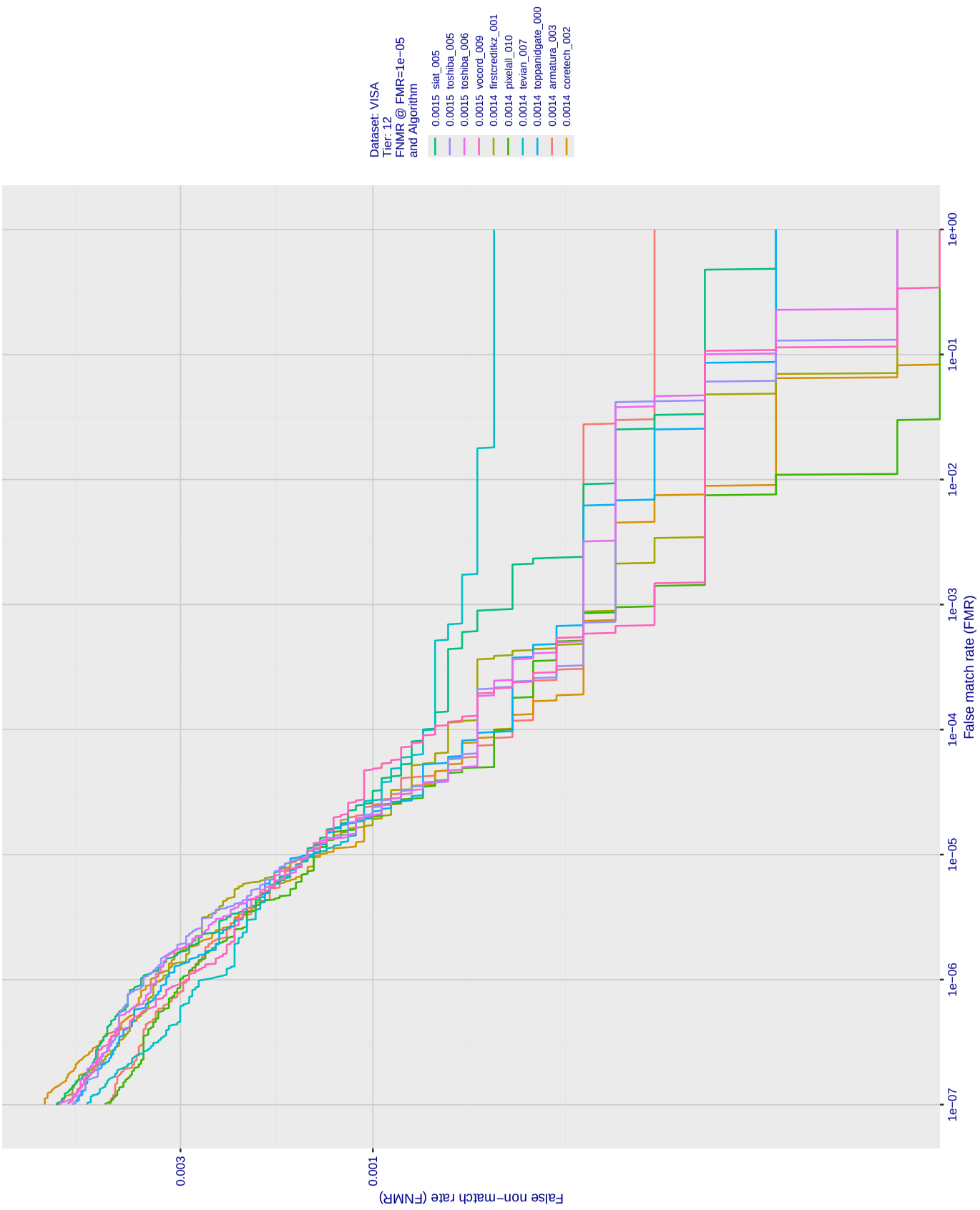


Figure 124: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

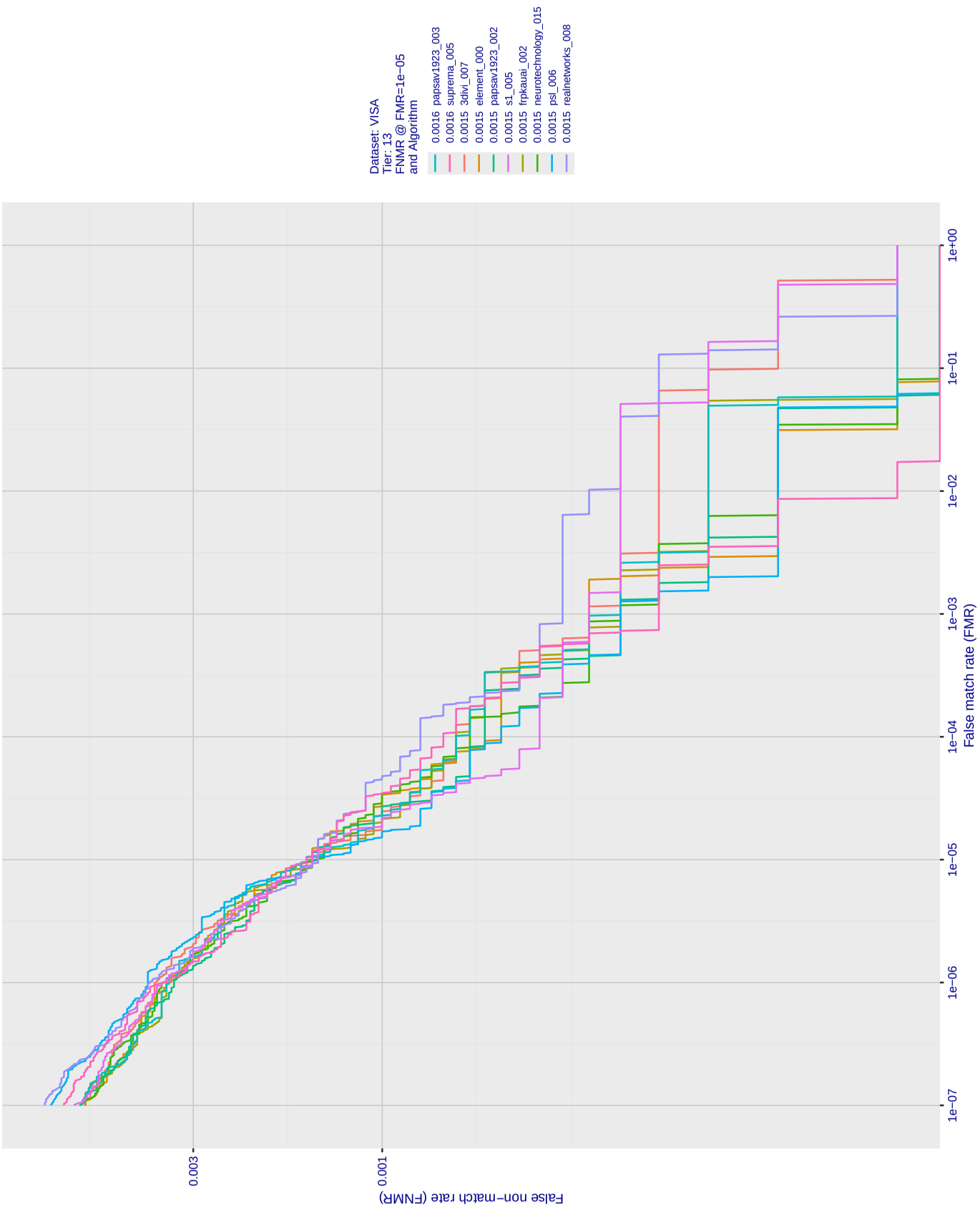


Figure 125: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

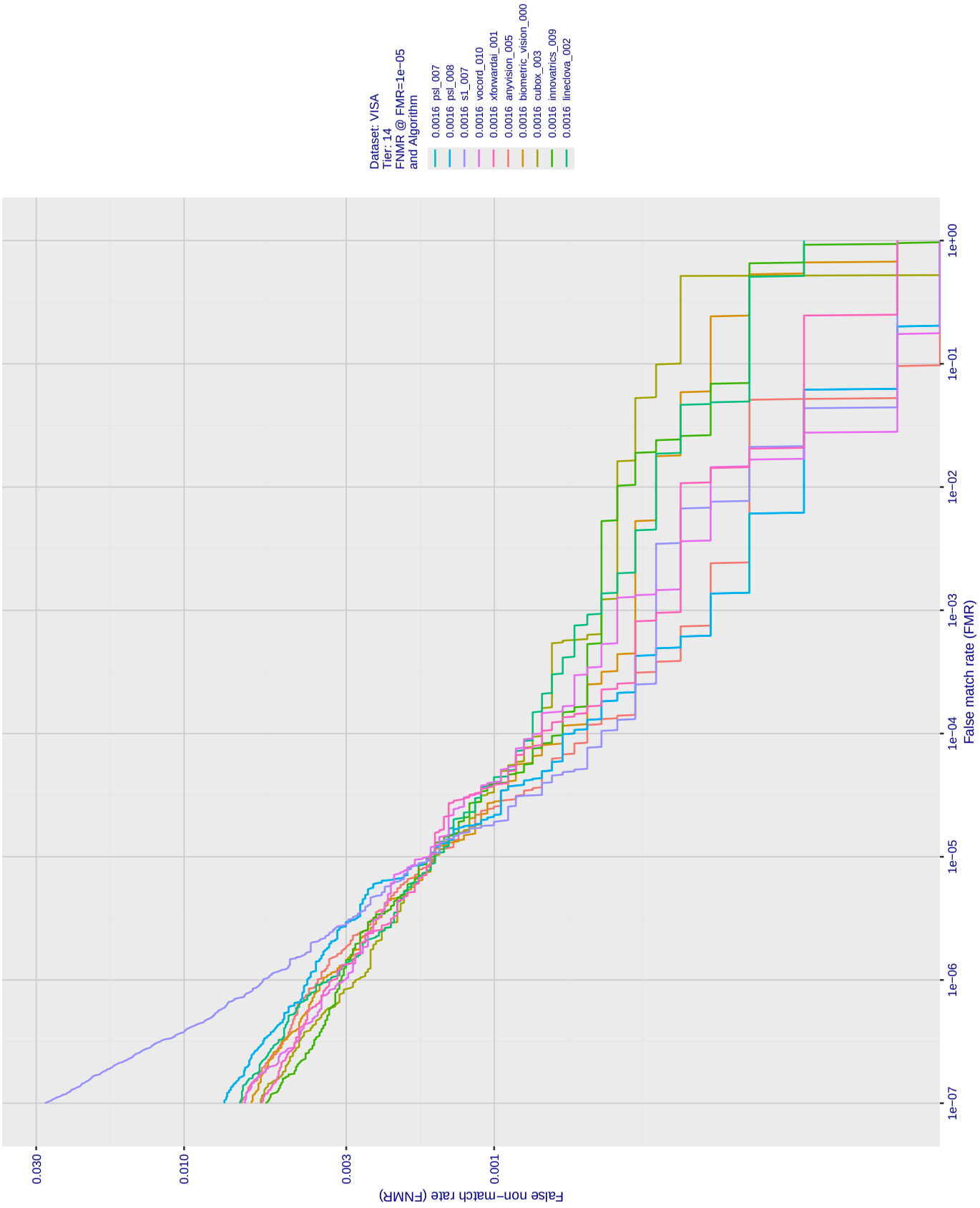


Figure 126: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

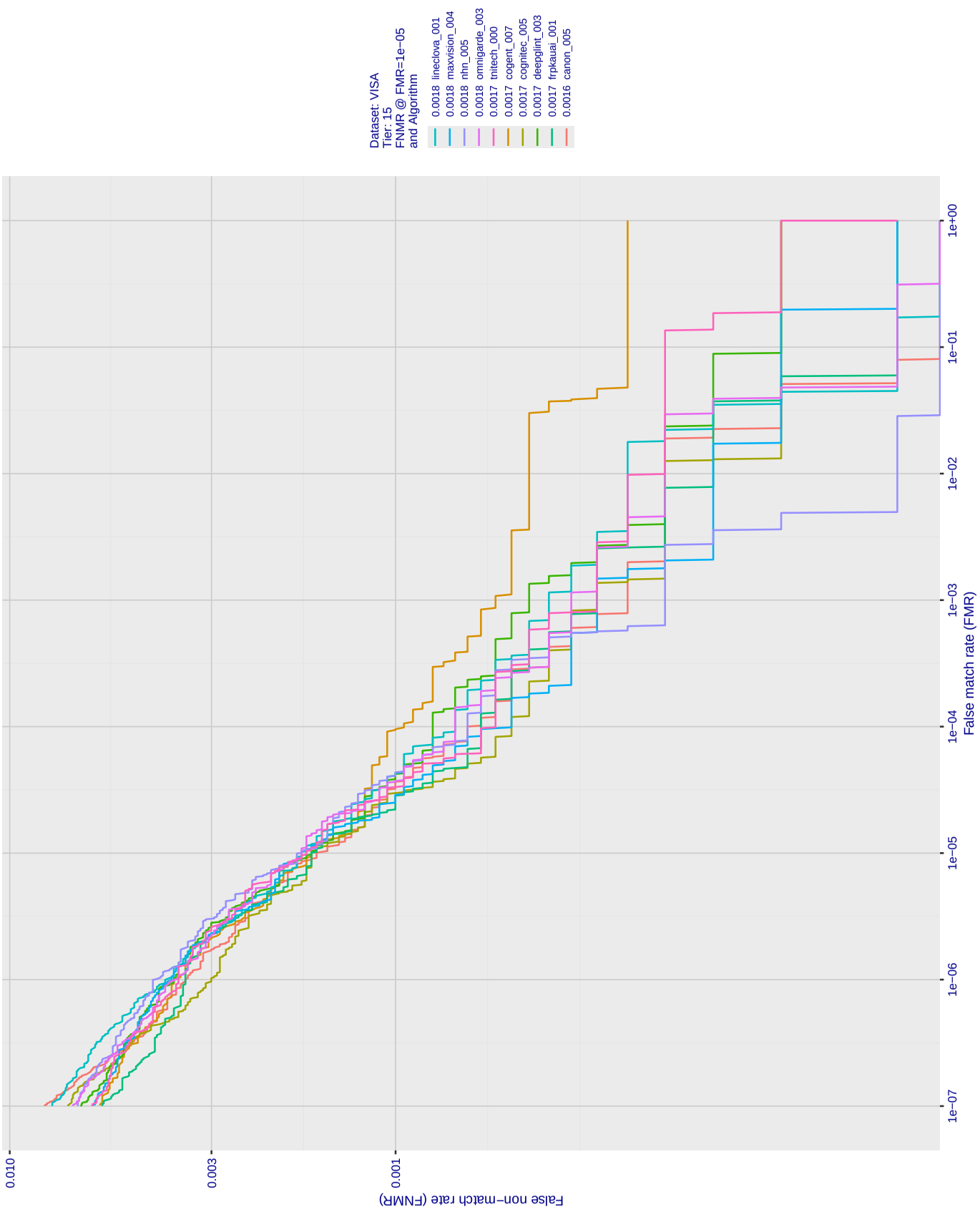


Figure 127: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

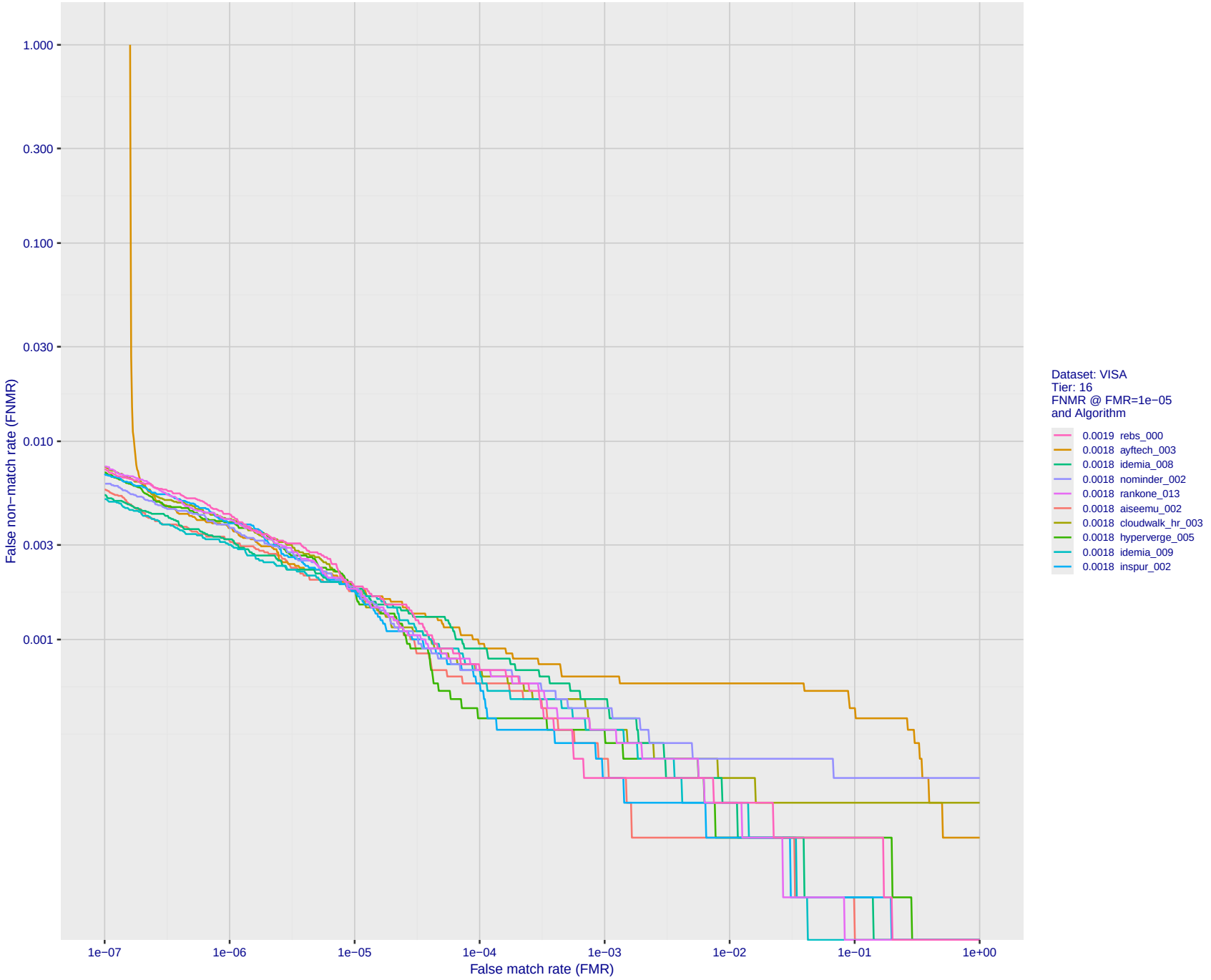


Figure 128: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

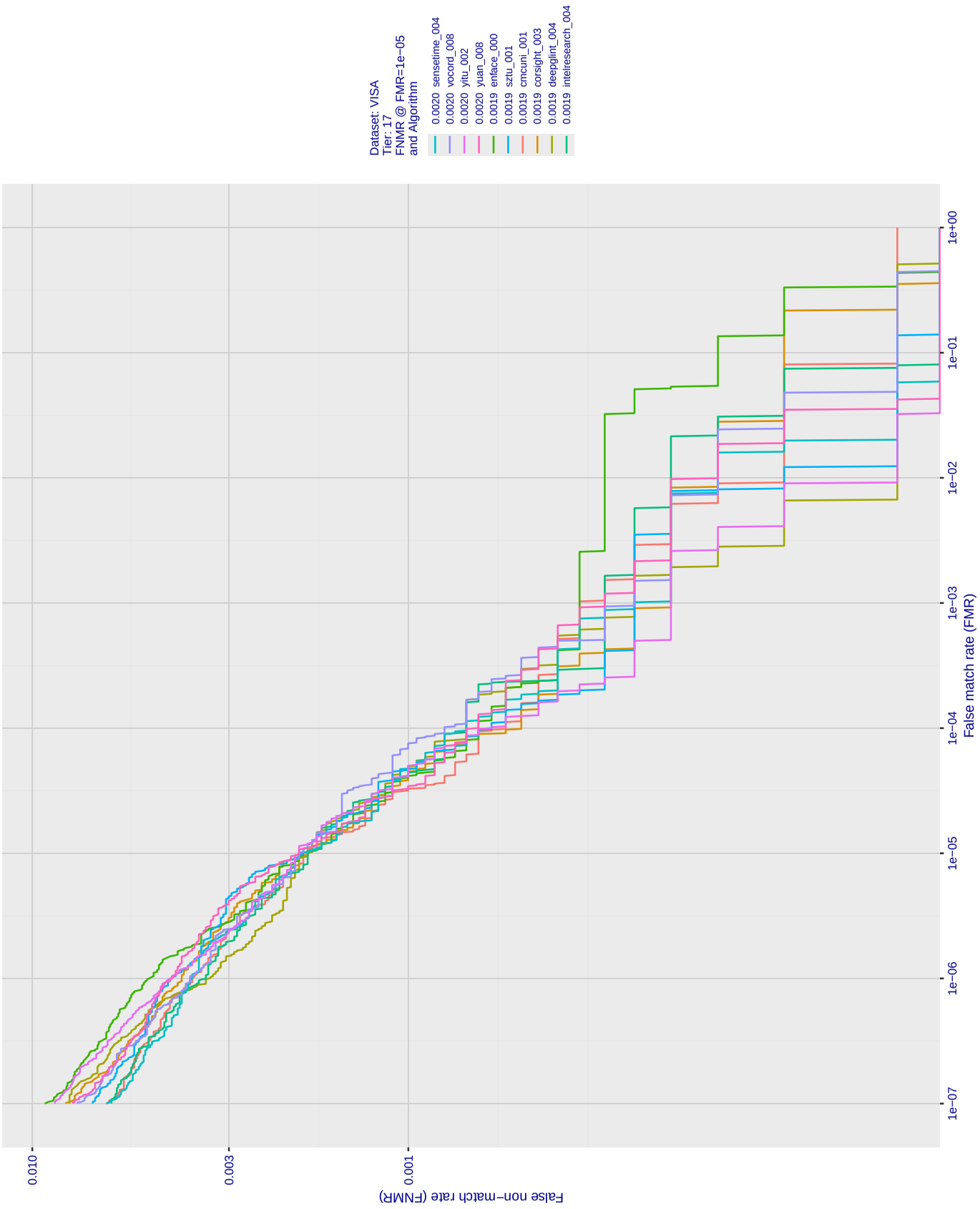


Figure 129: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

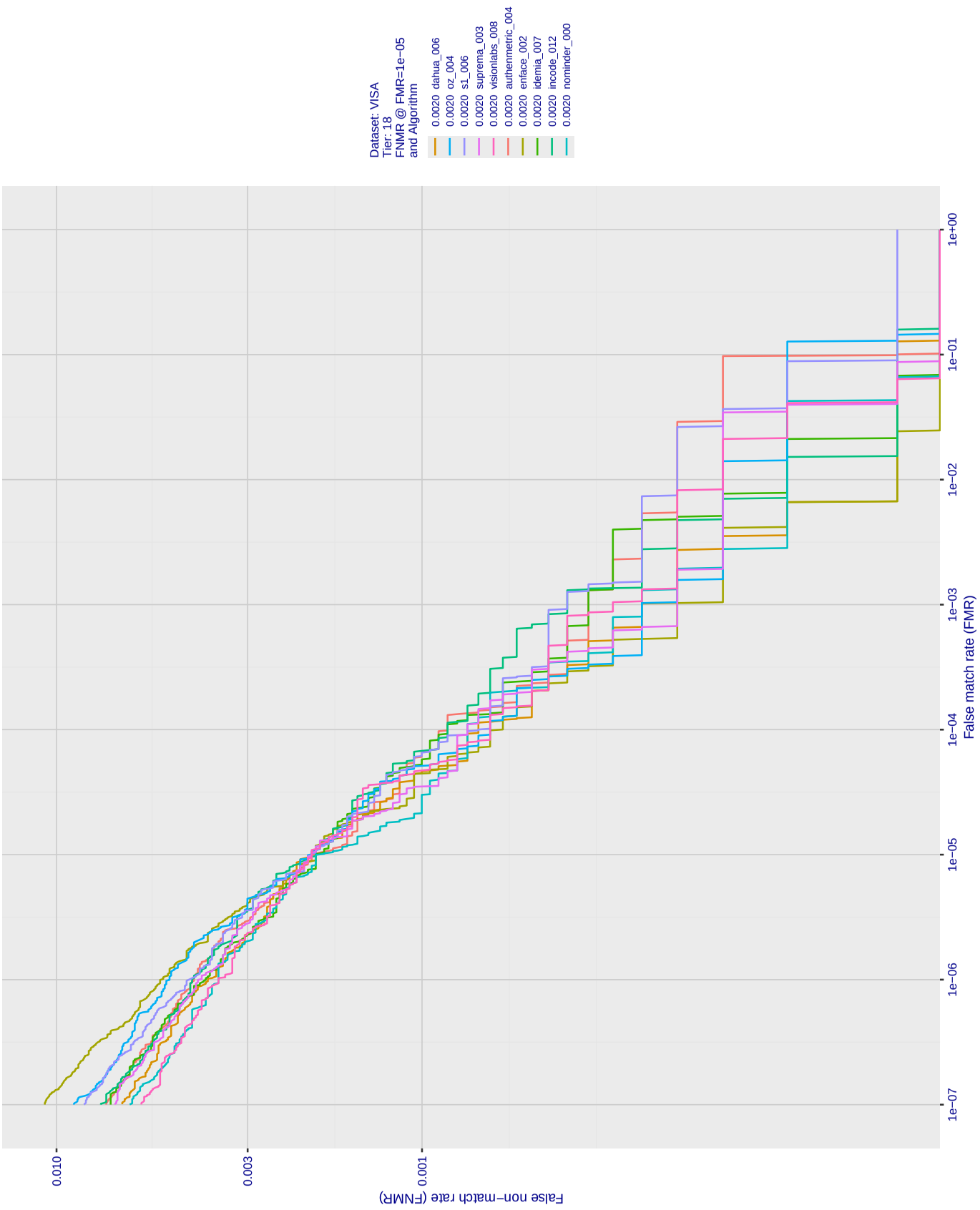


Figure 130: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

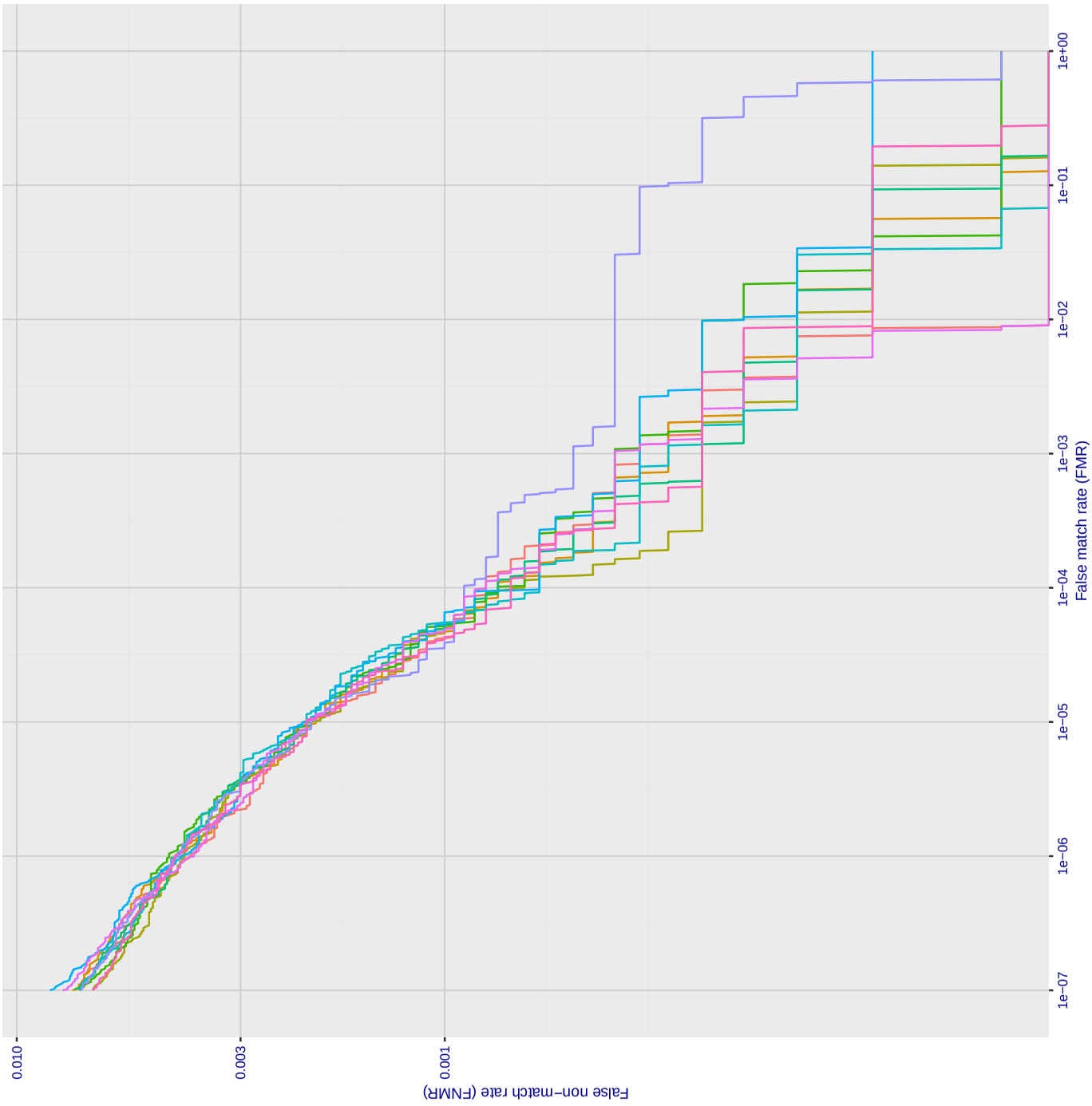


Figure 131: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

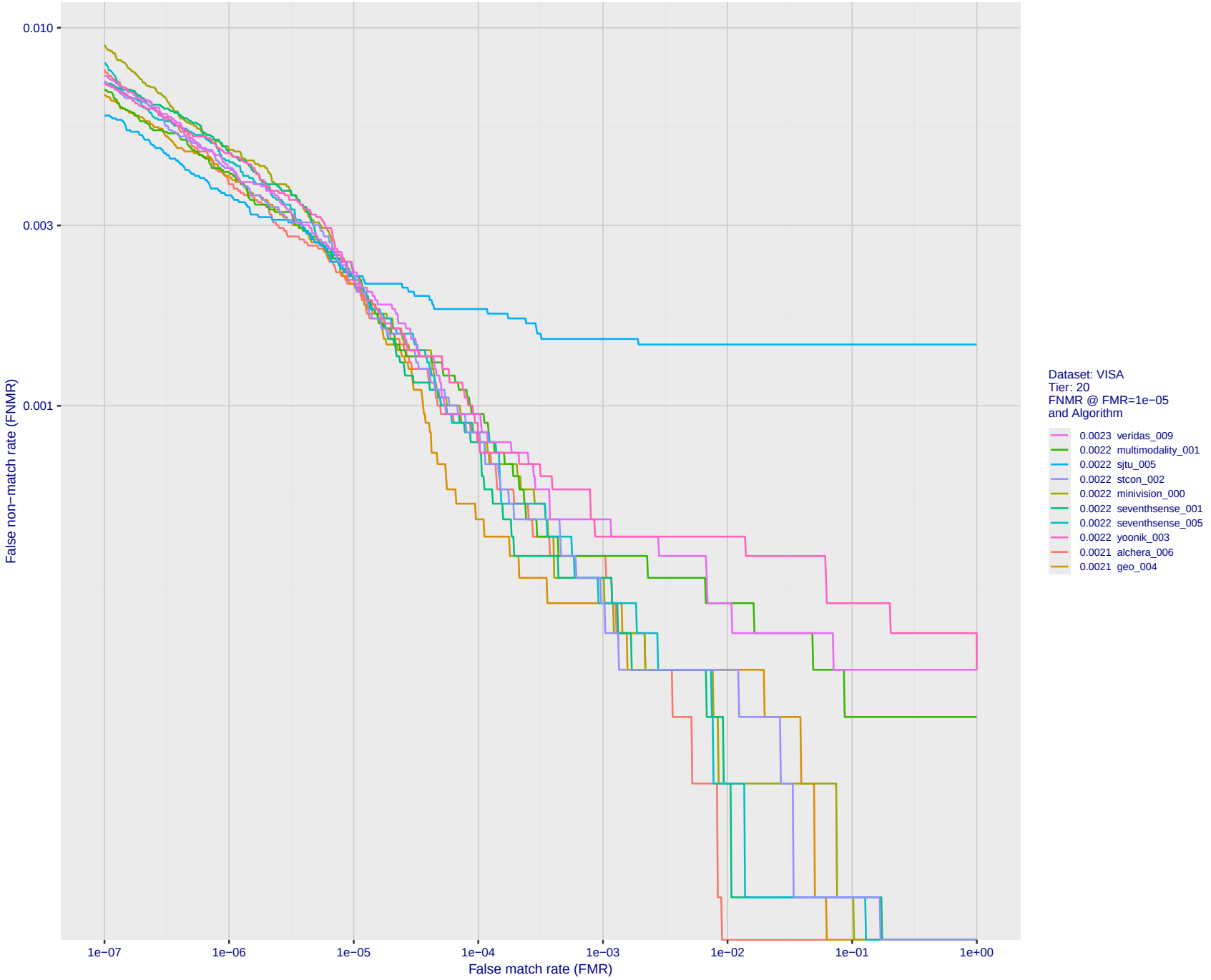


Figure 132: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

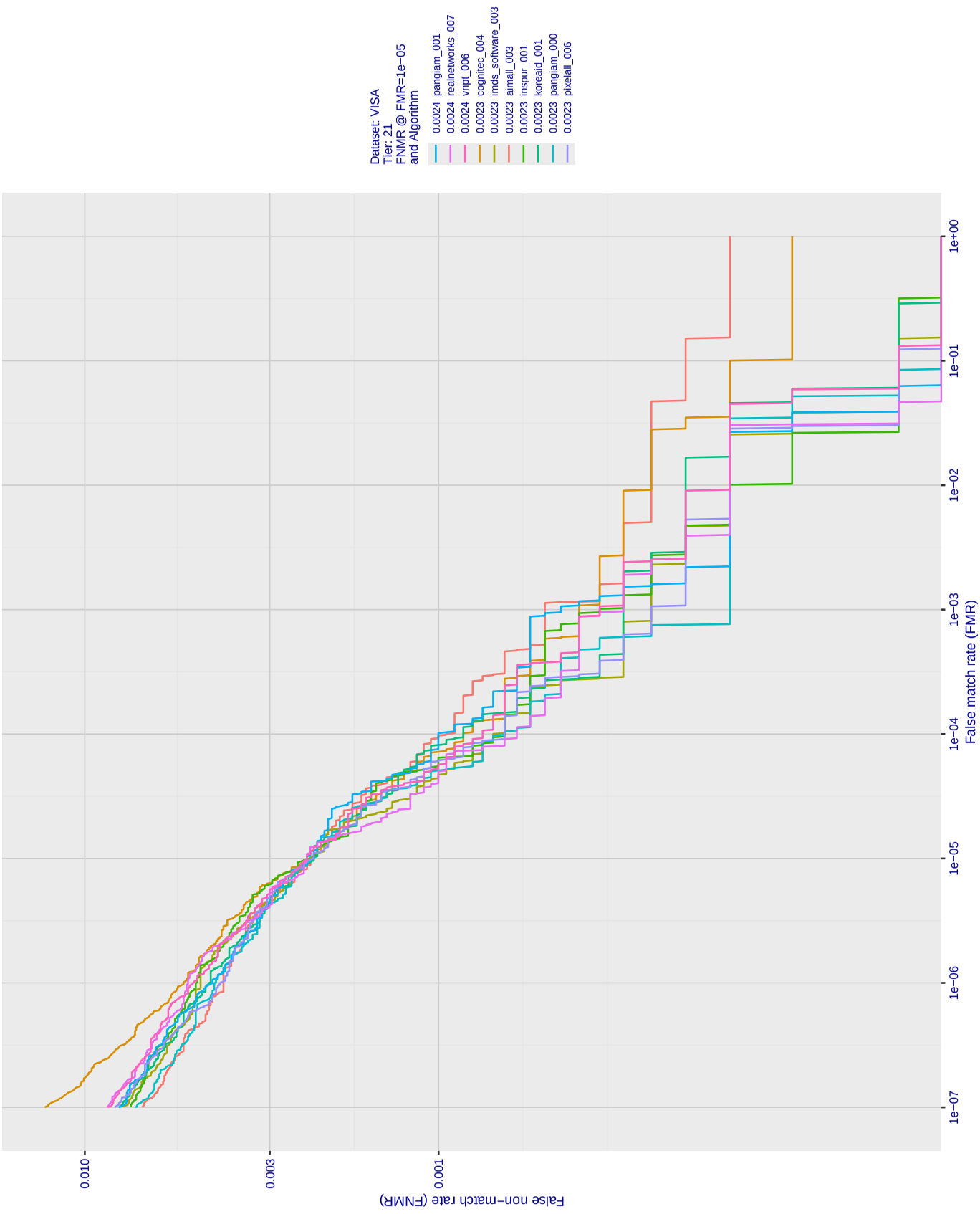


Figure 133: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

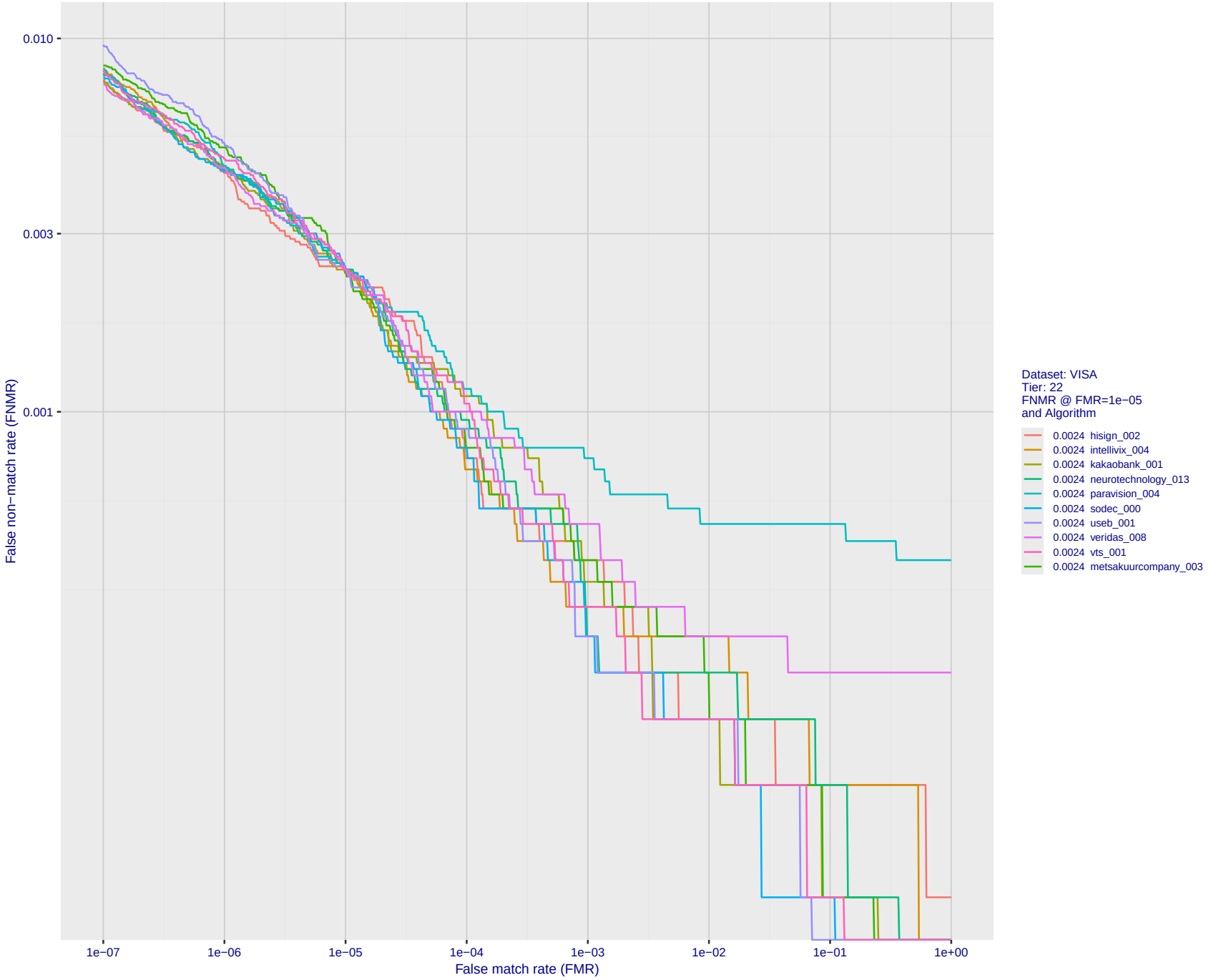


Figure 134: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

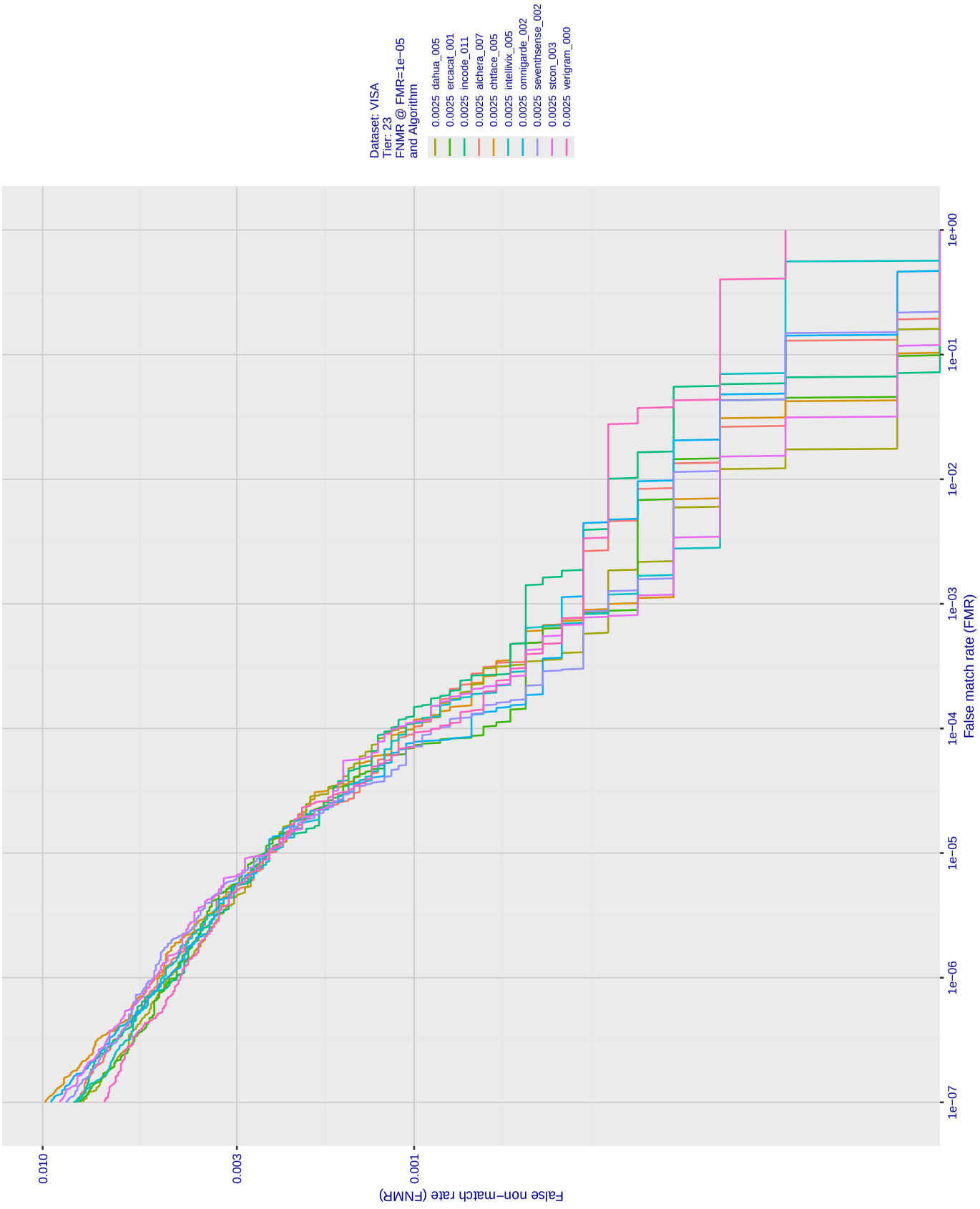


Figure 135: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

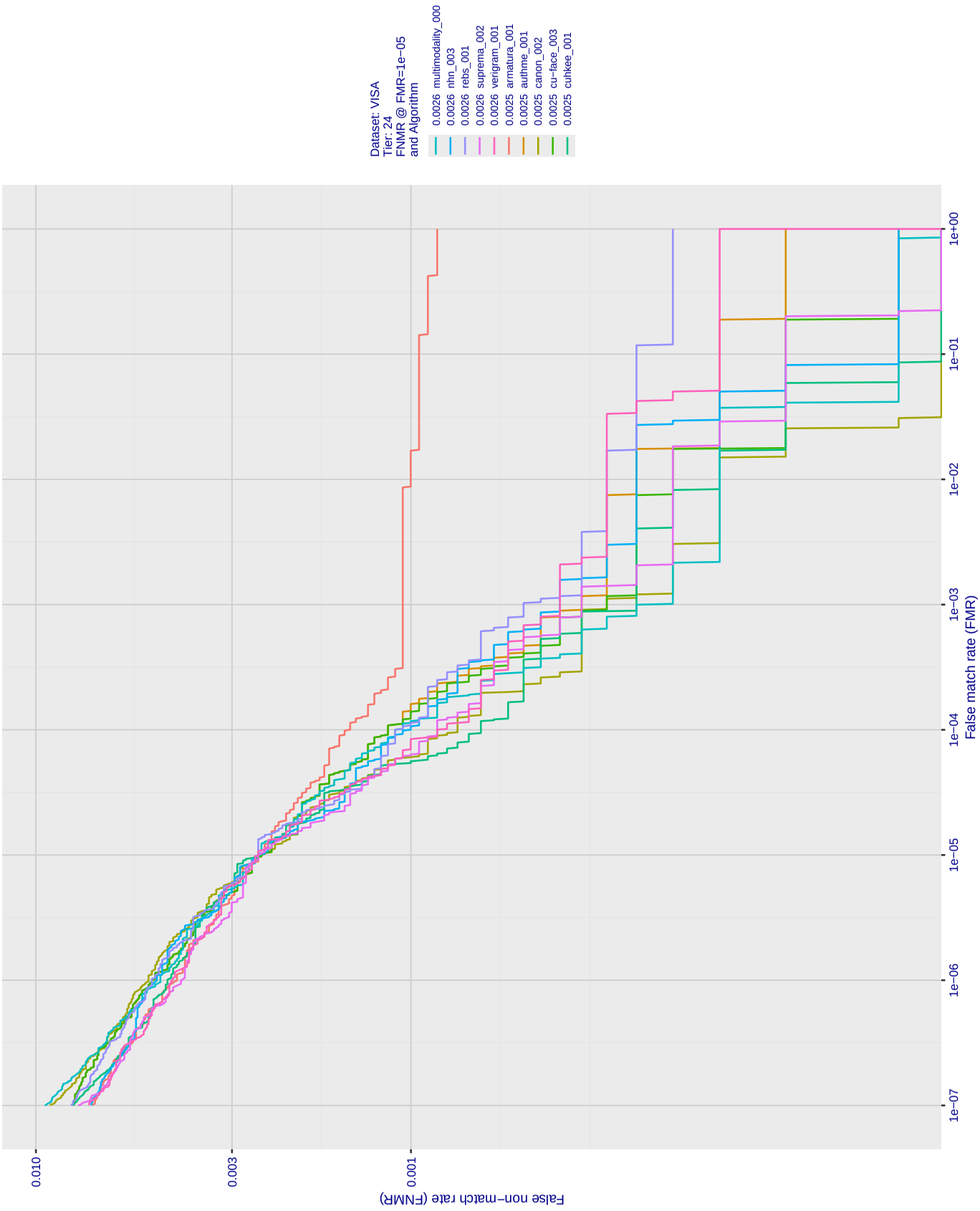


Figure 136: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

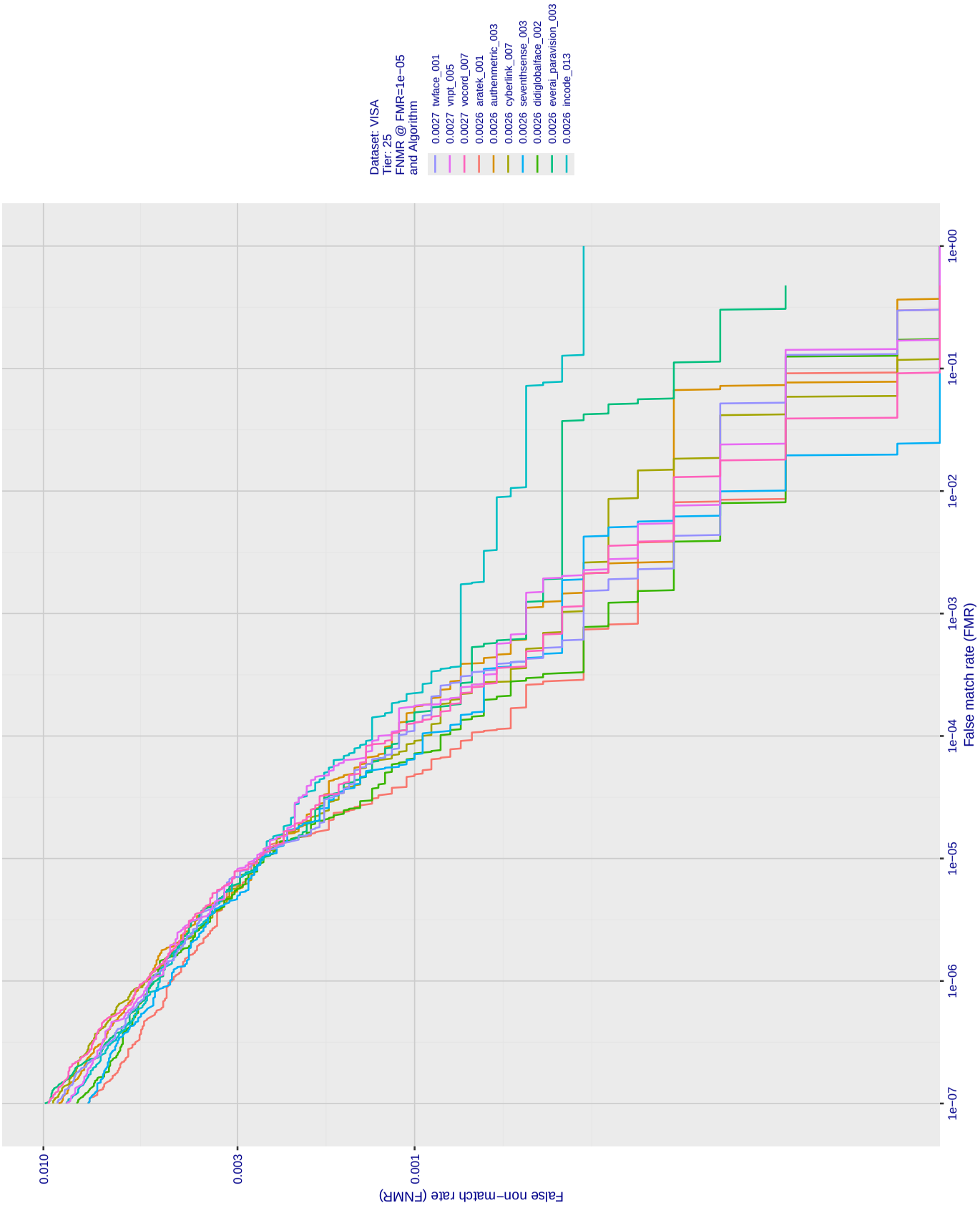


Figure 137: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

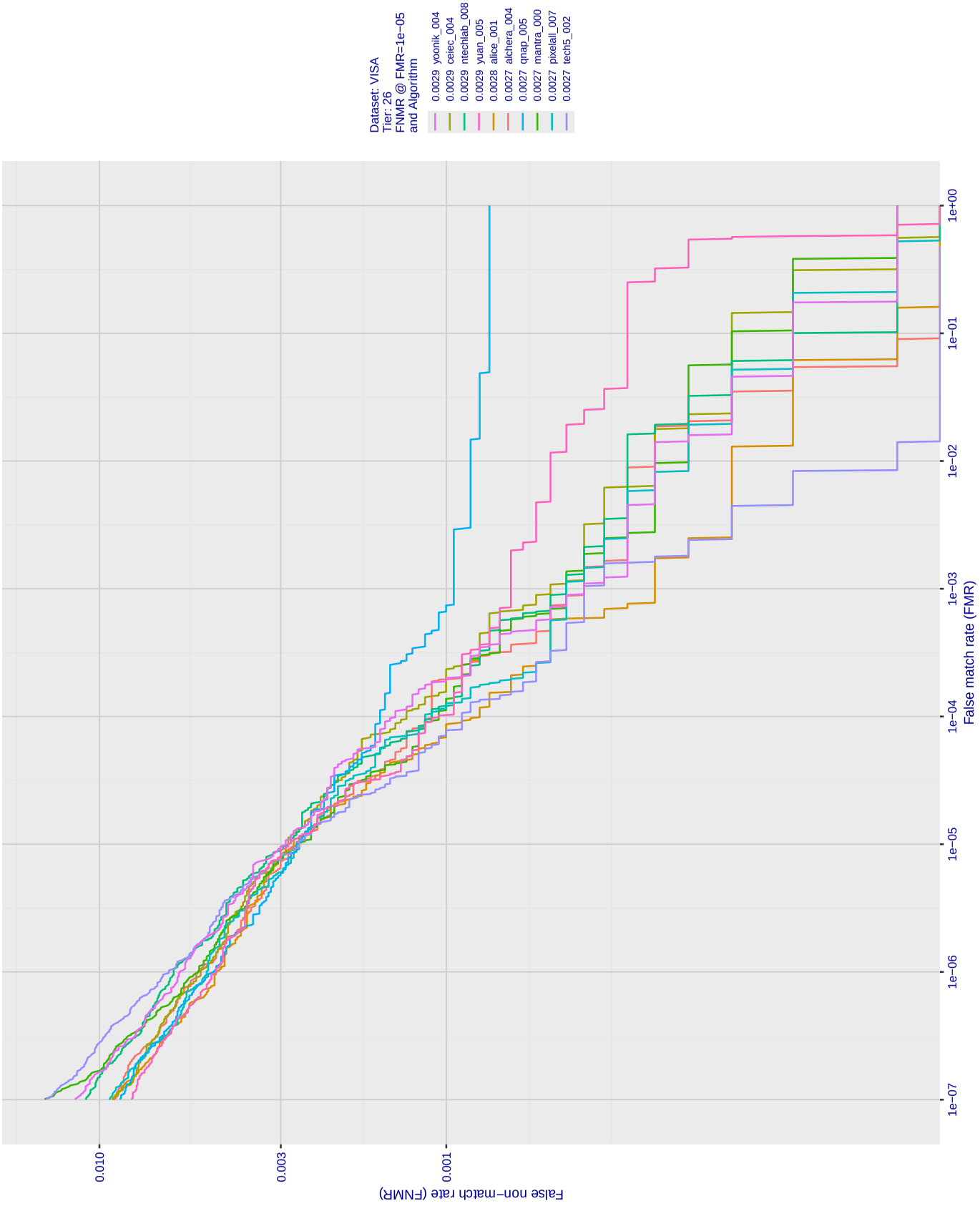


Figure 138: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

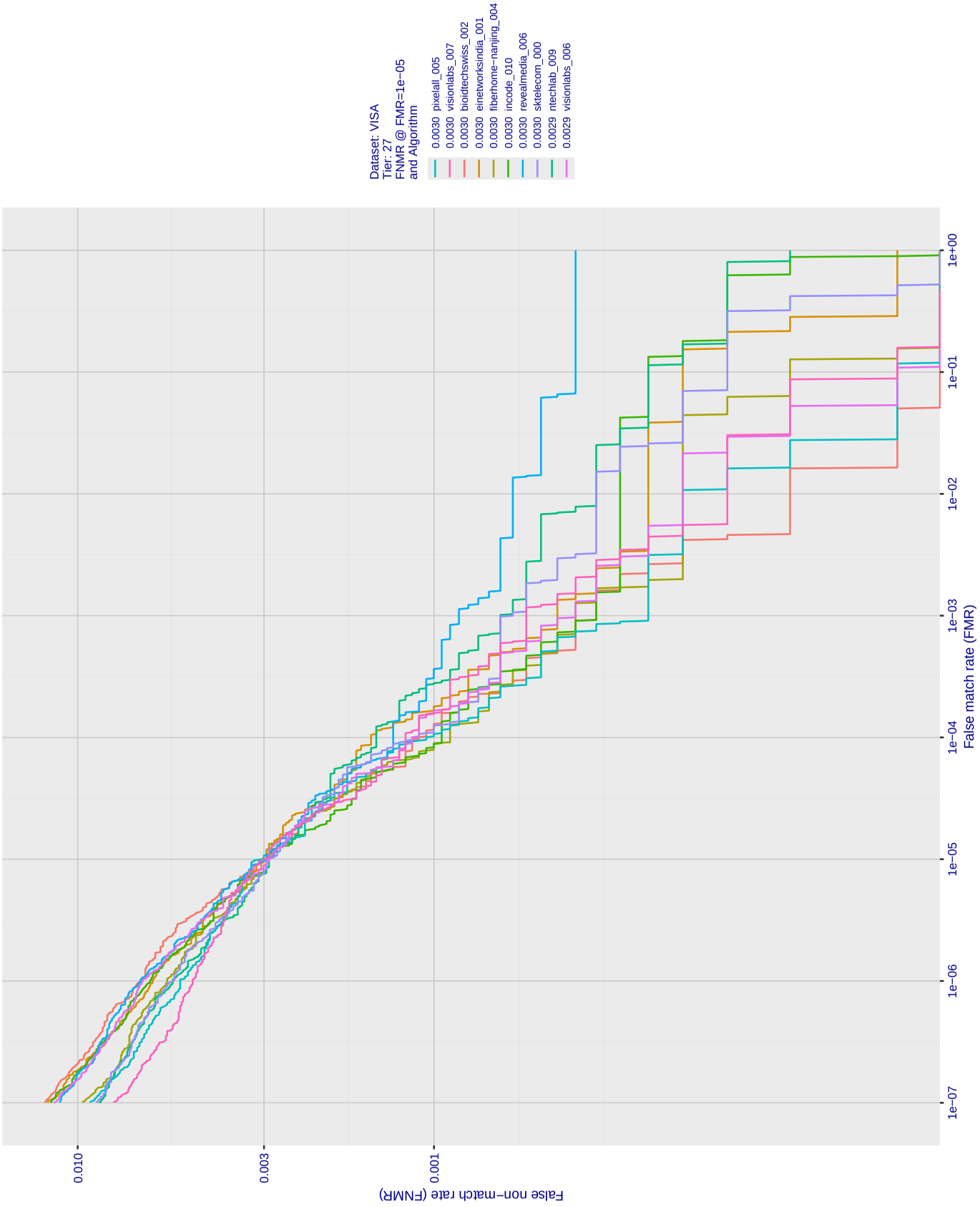


Figure 139: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

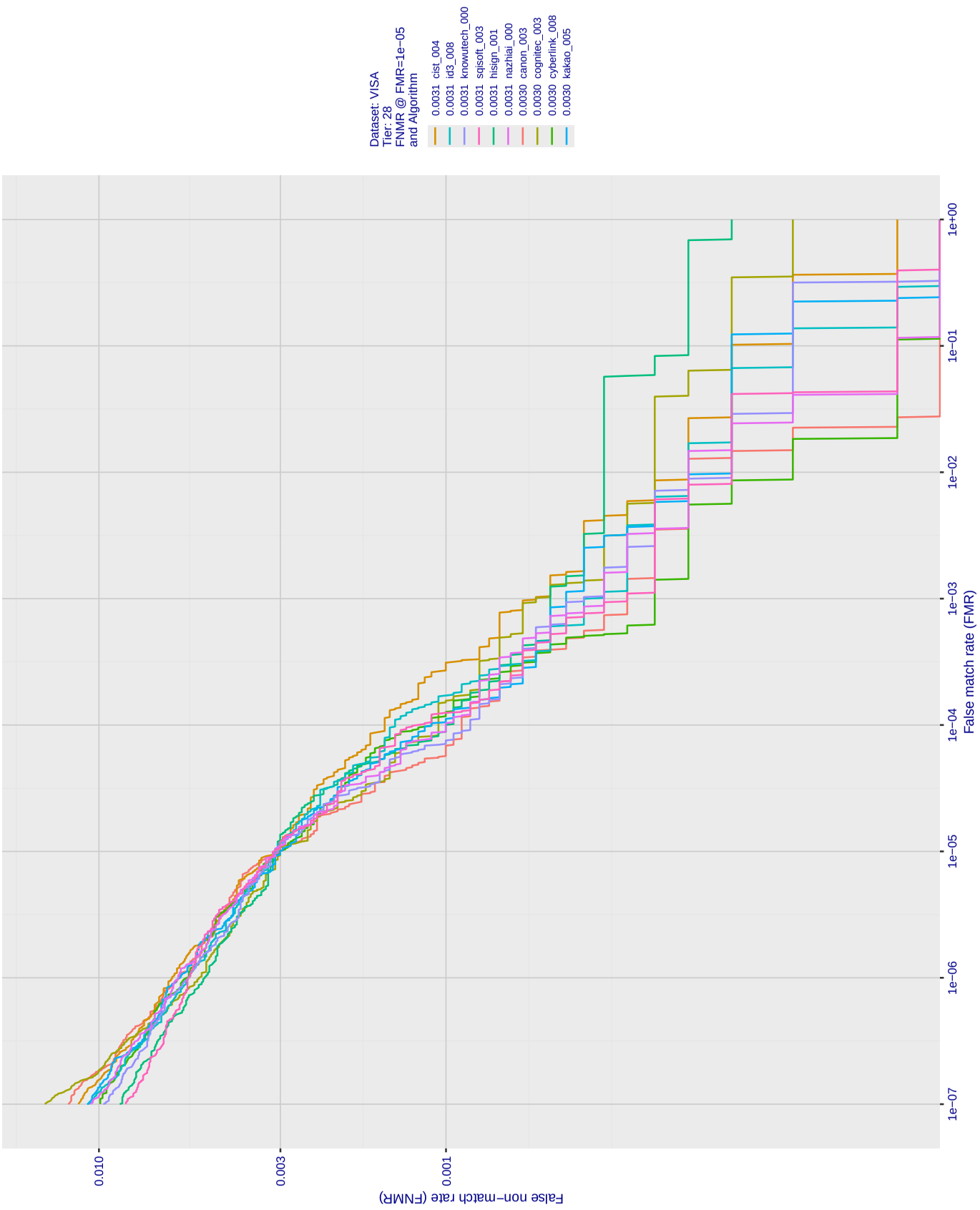


Figure 140: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

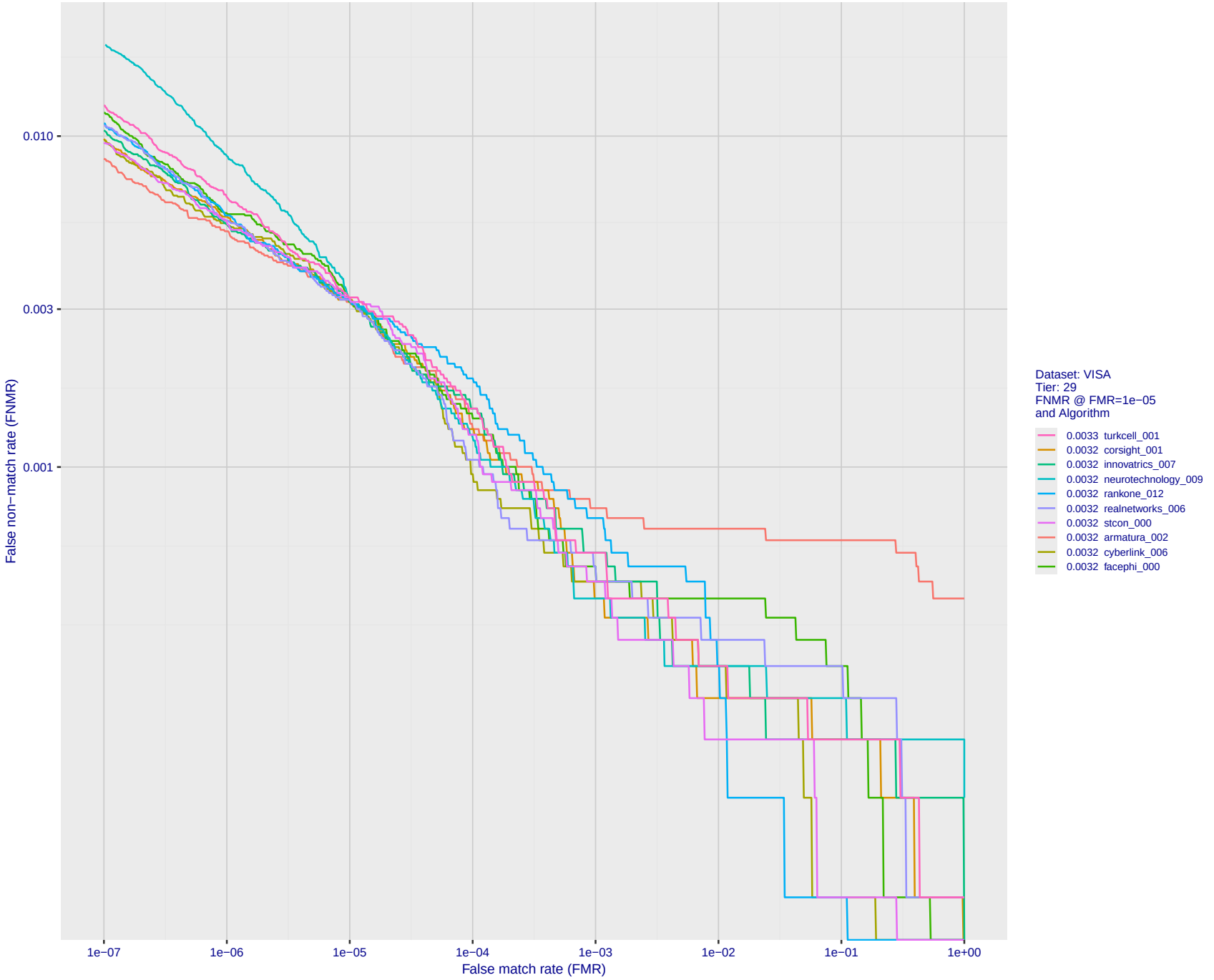


Figure 141: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

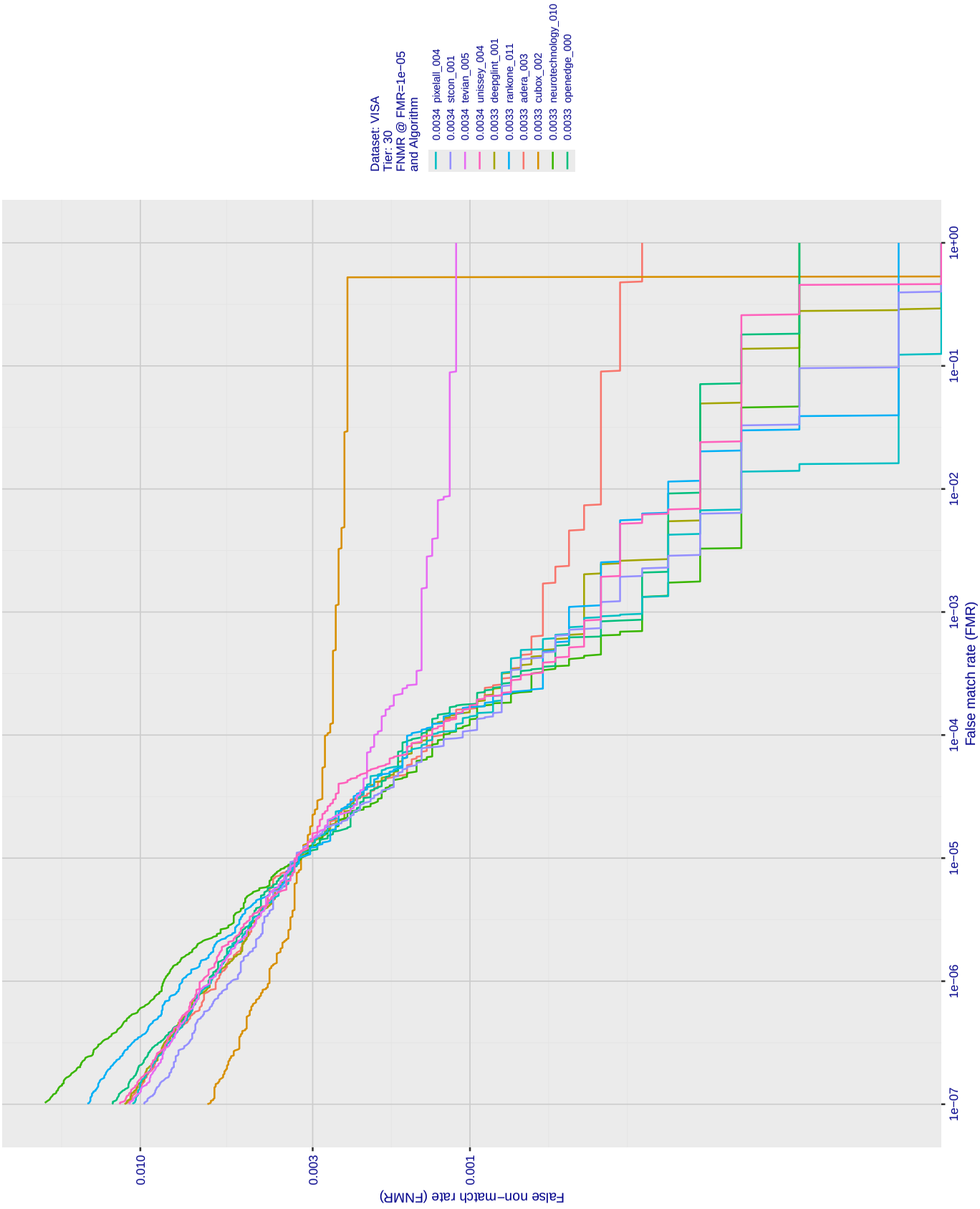


Figure 142: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

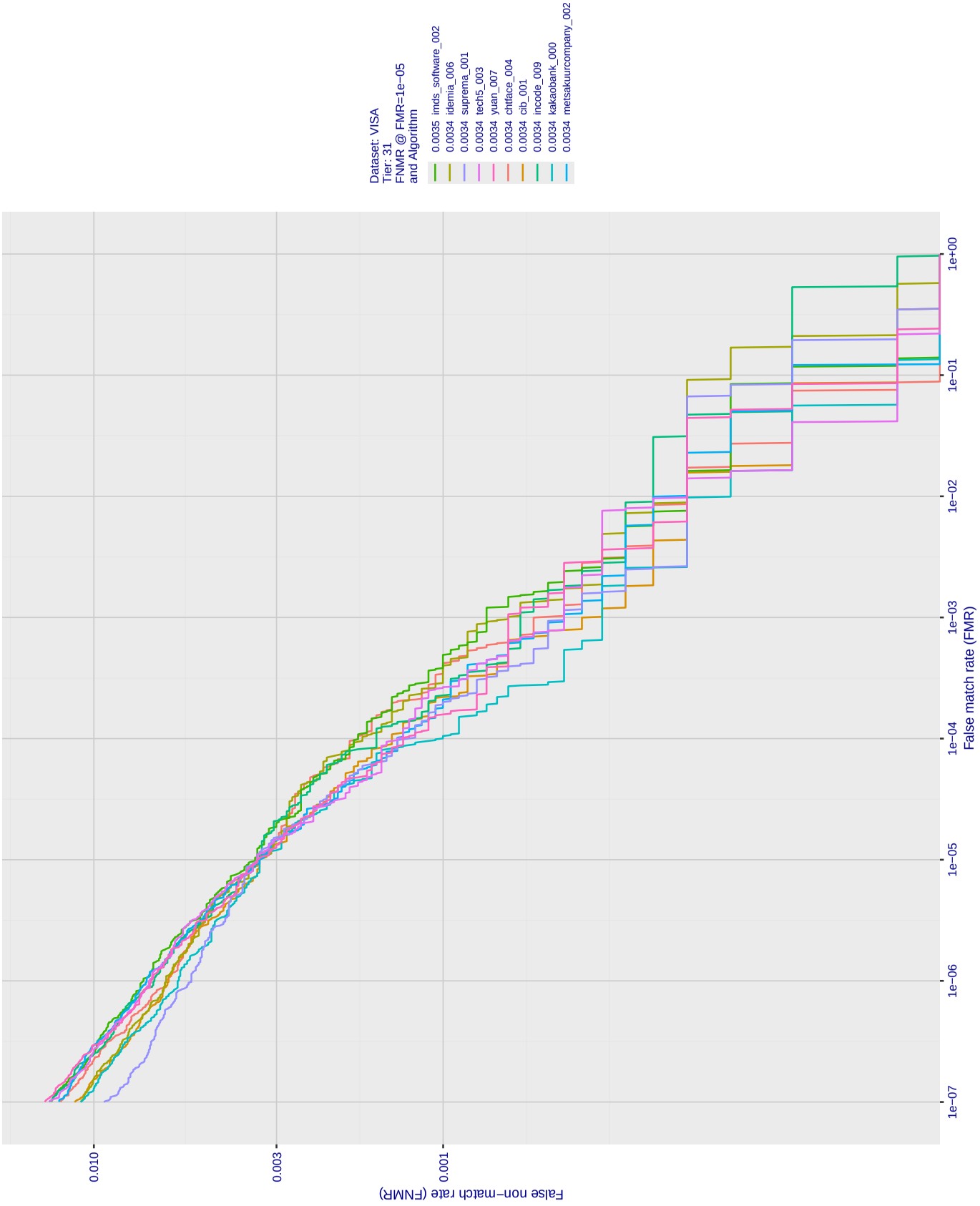


Figure 143: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

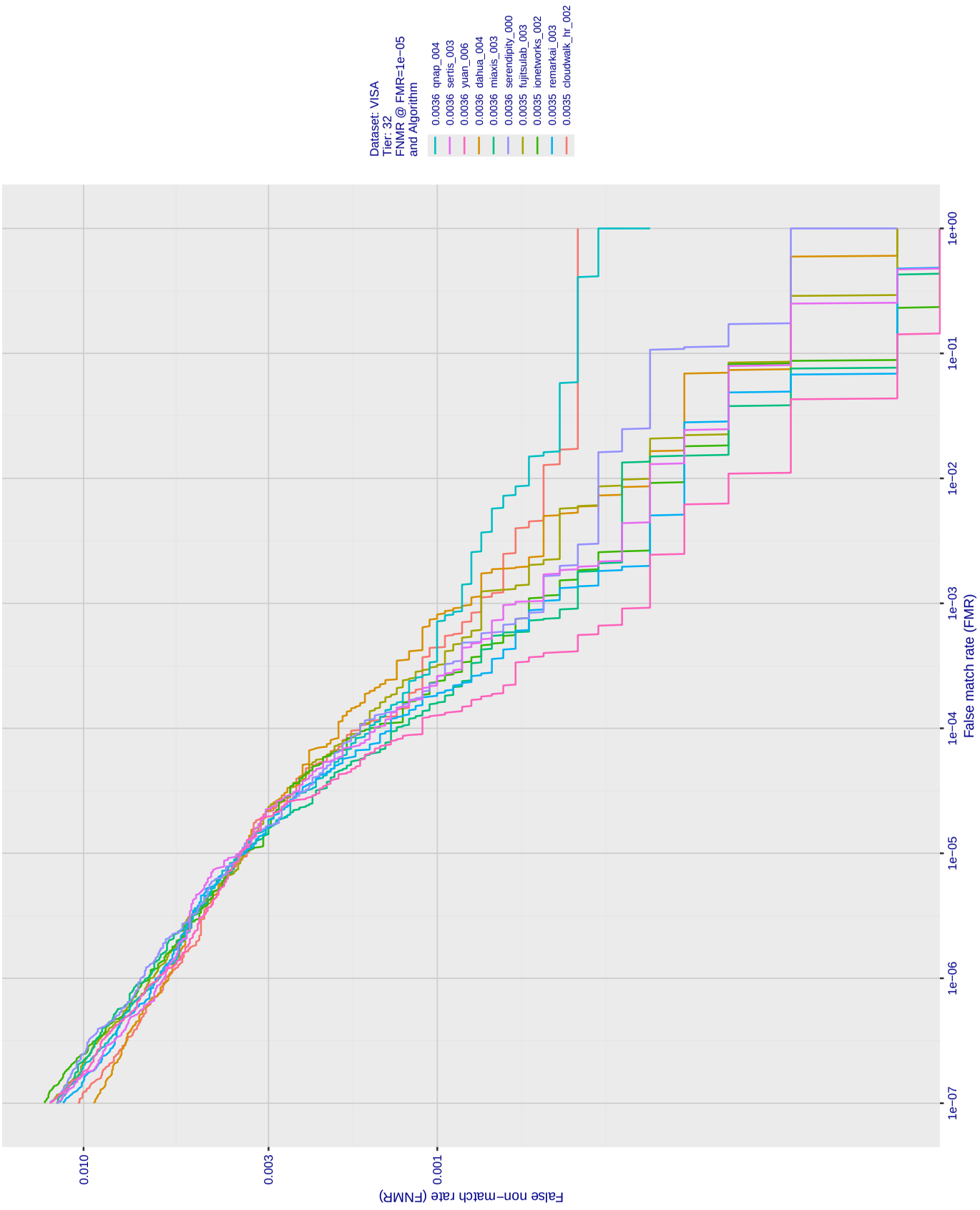


Figure 144: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

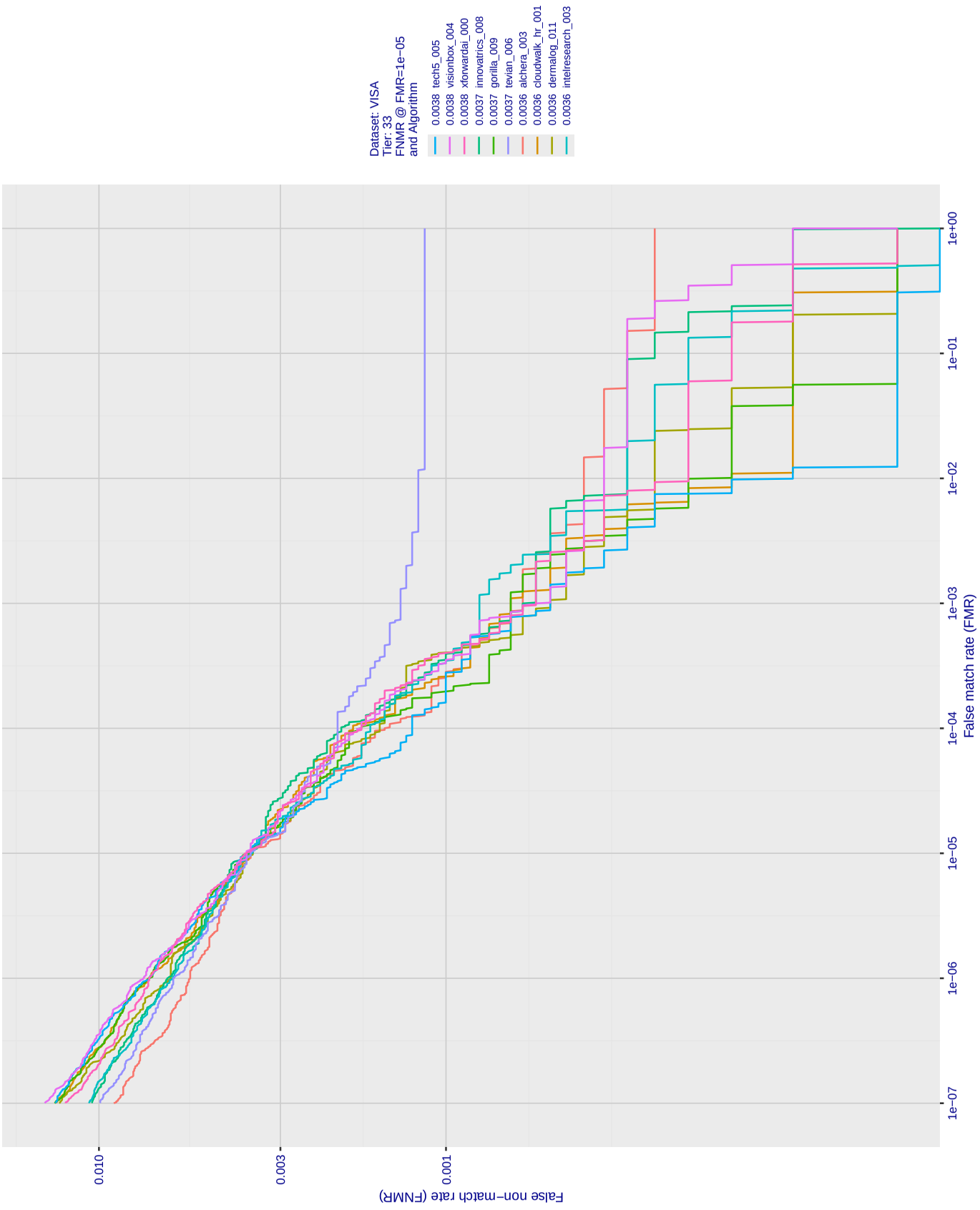


Figure 145: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

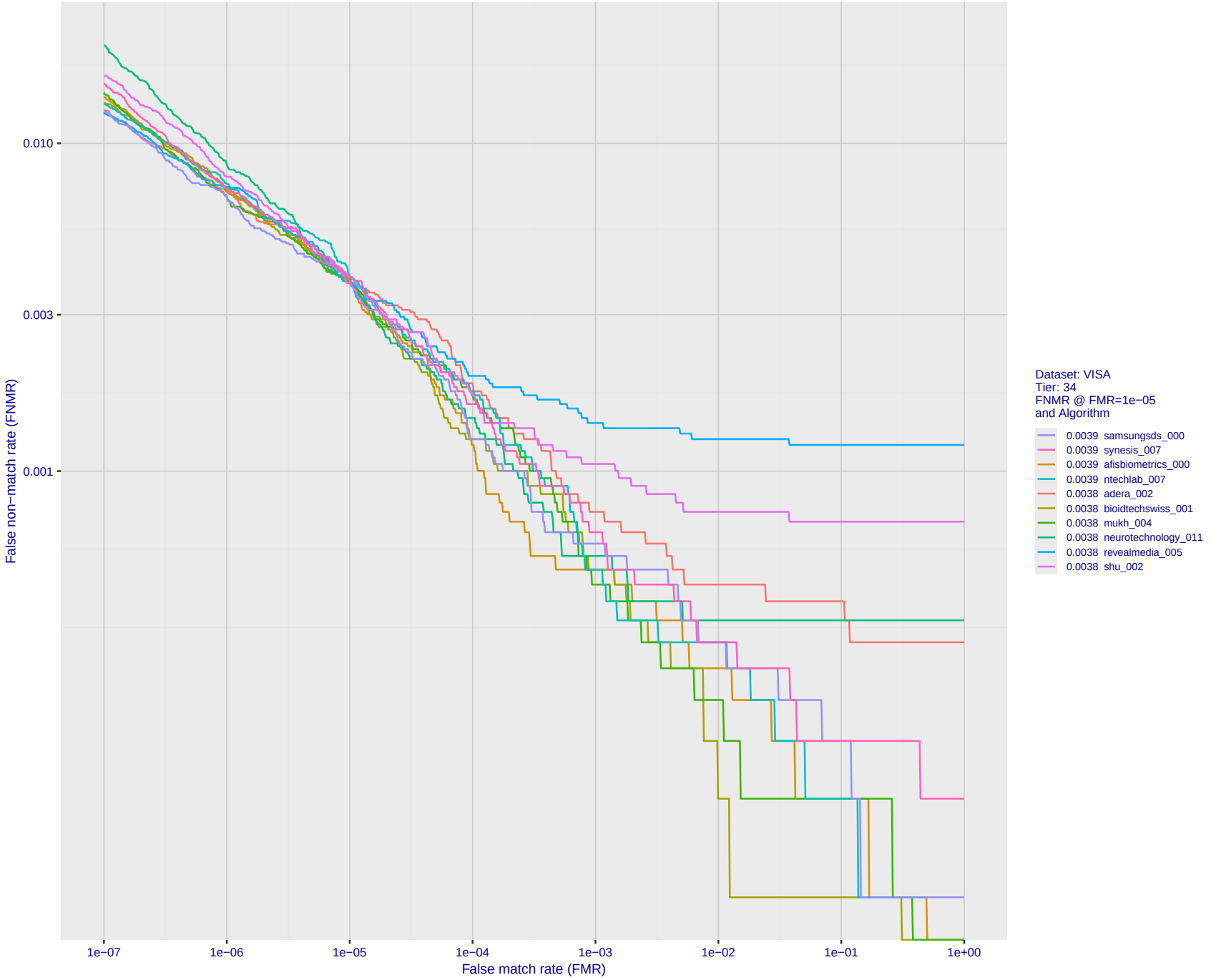


Figure 146: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

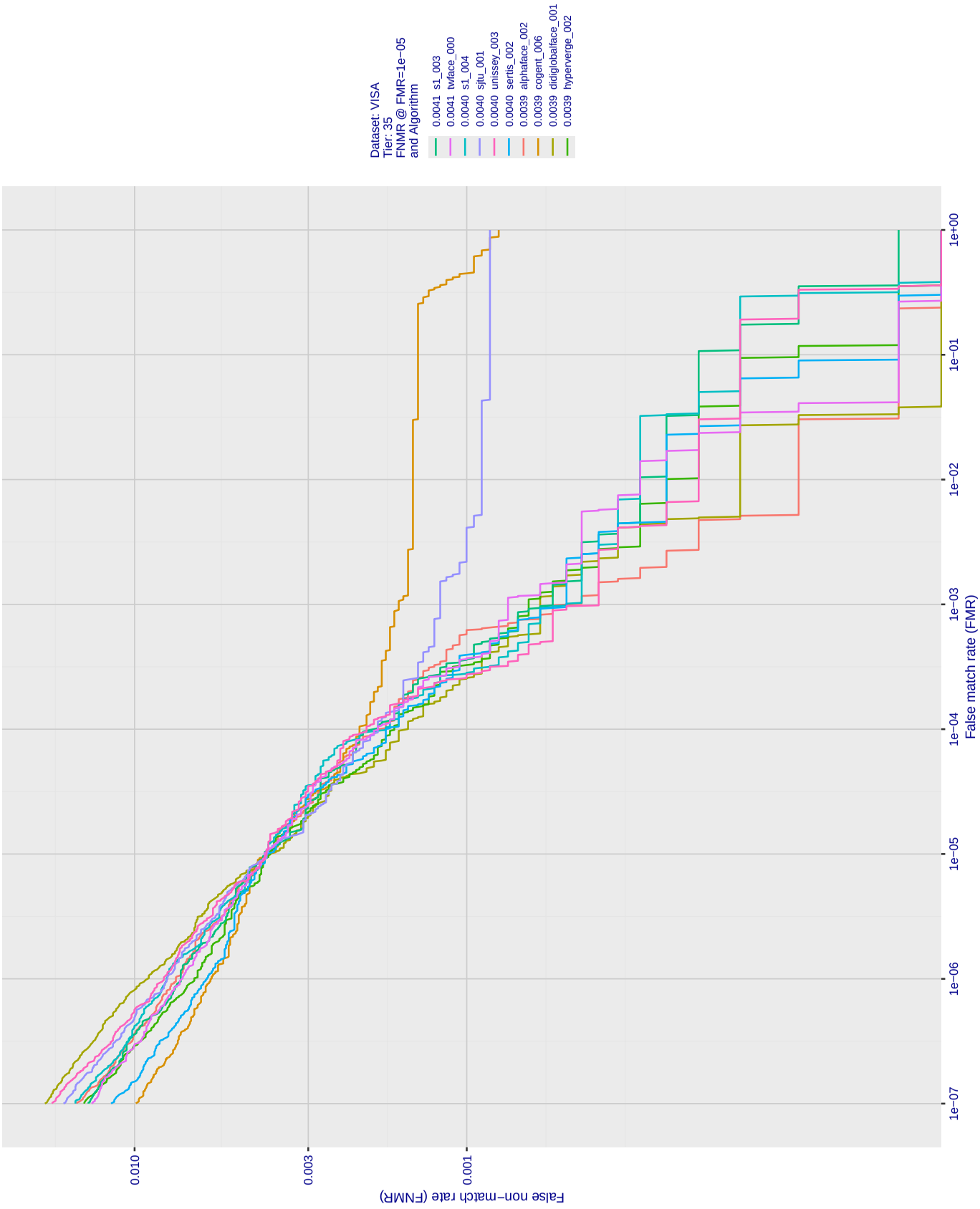


Figure 147: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

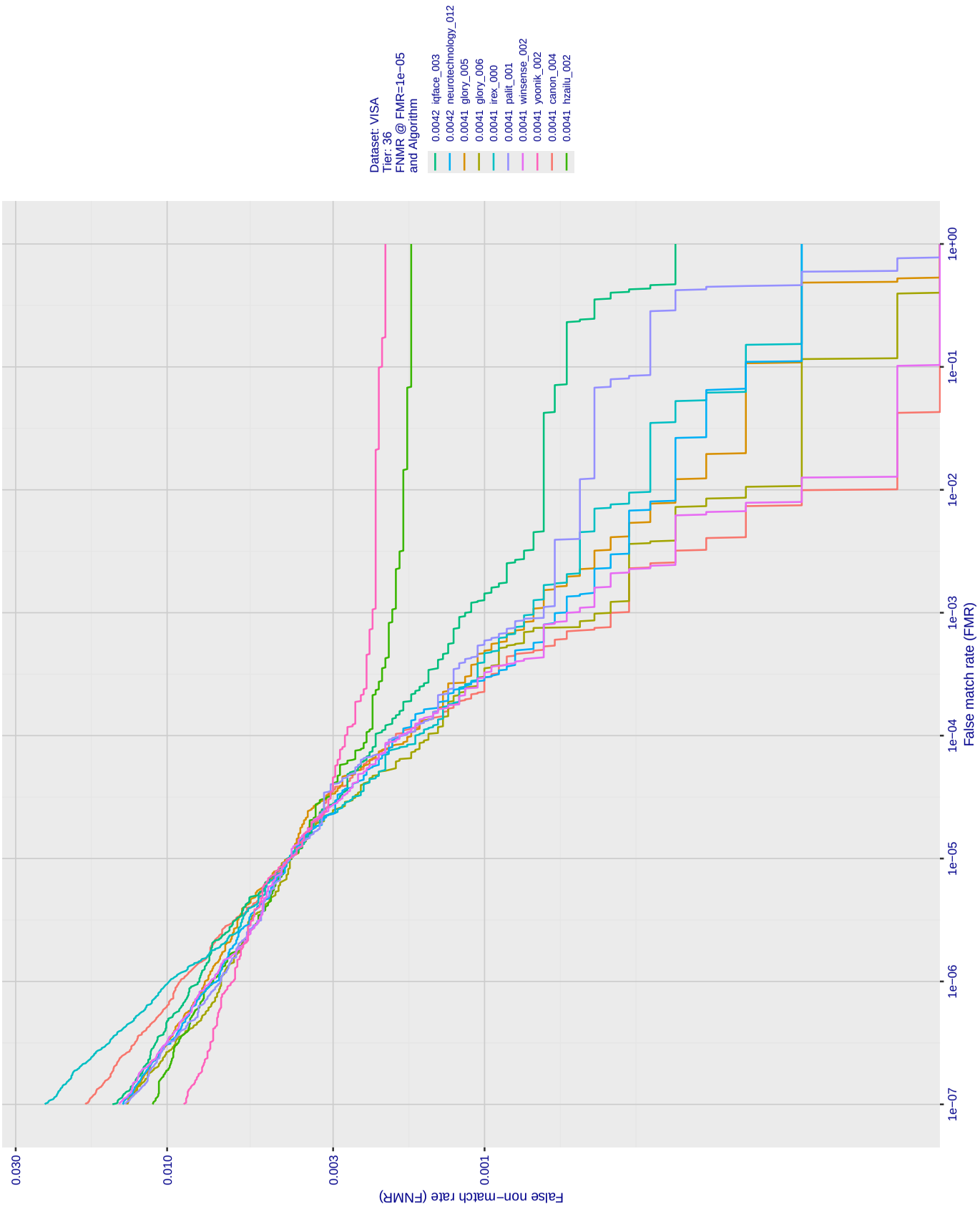


Figure 148: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

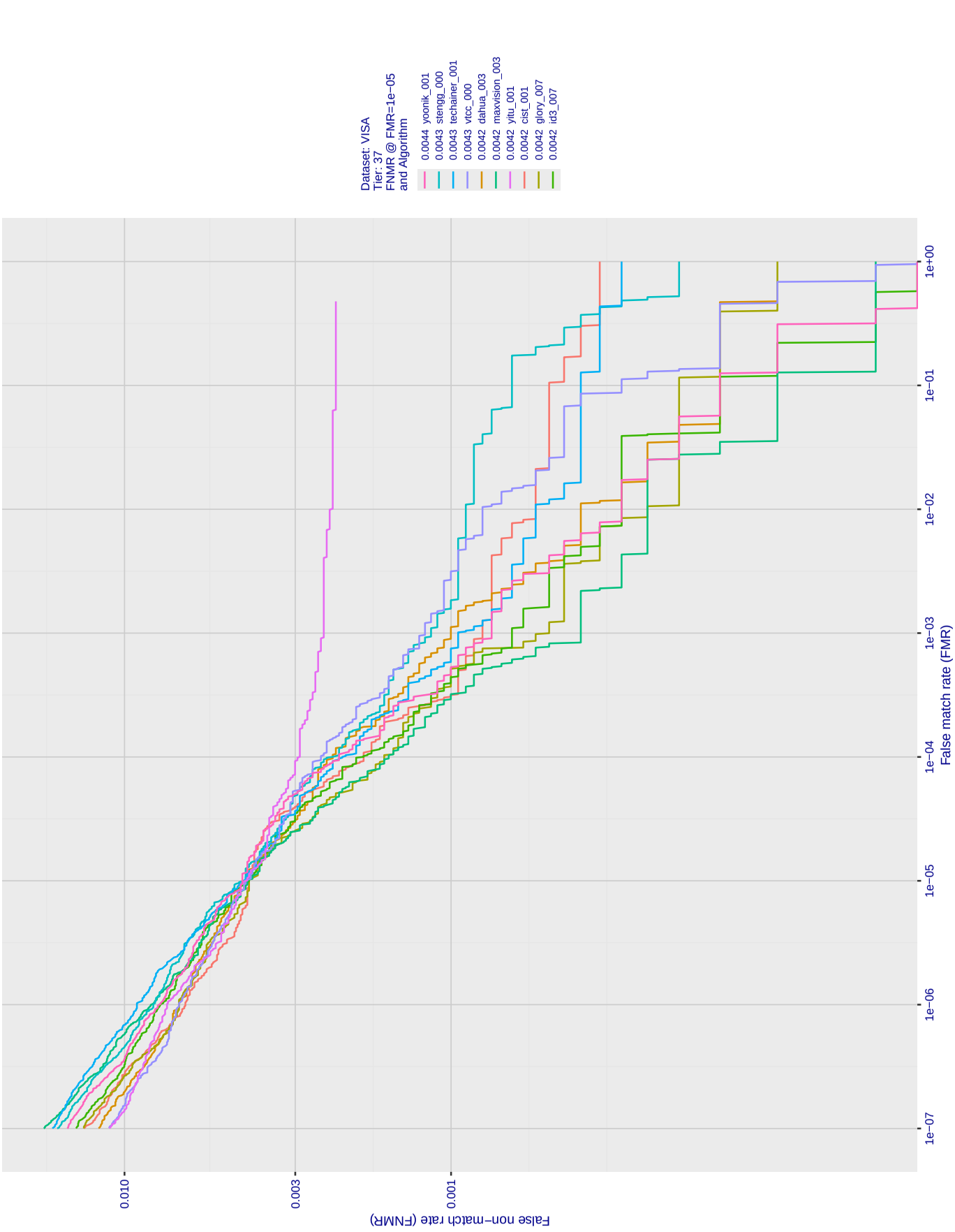


Figure 149: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

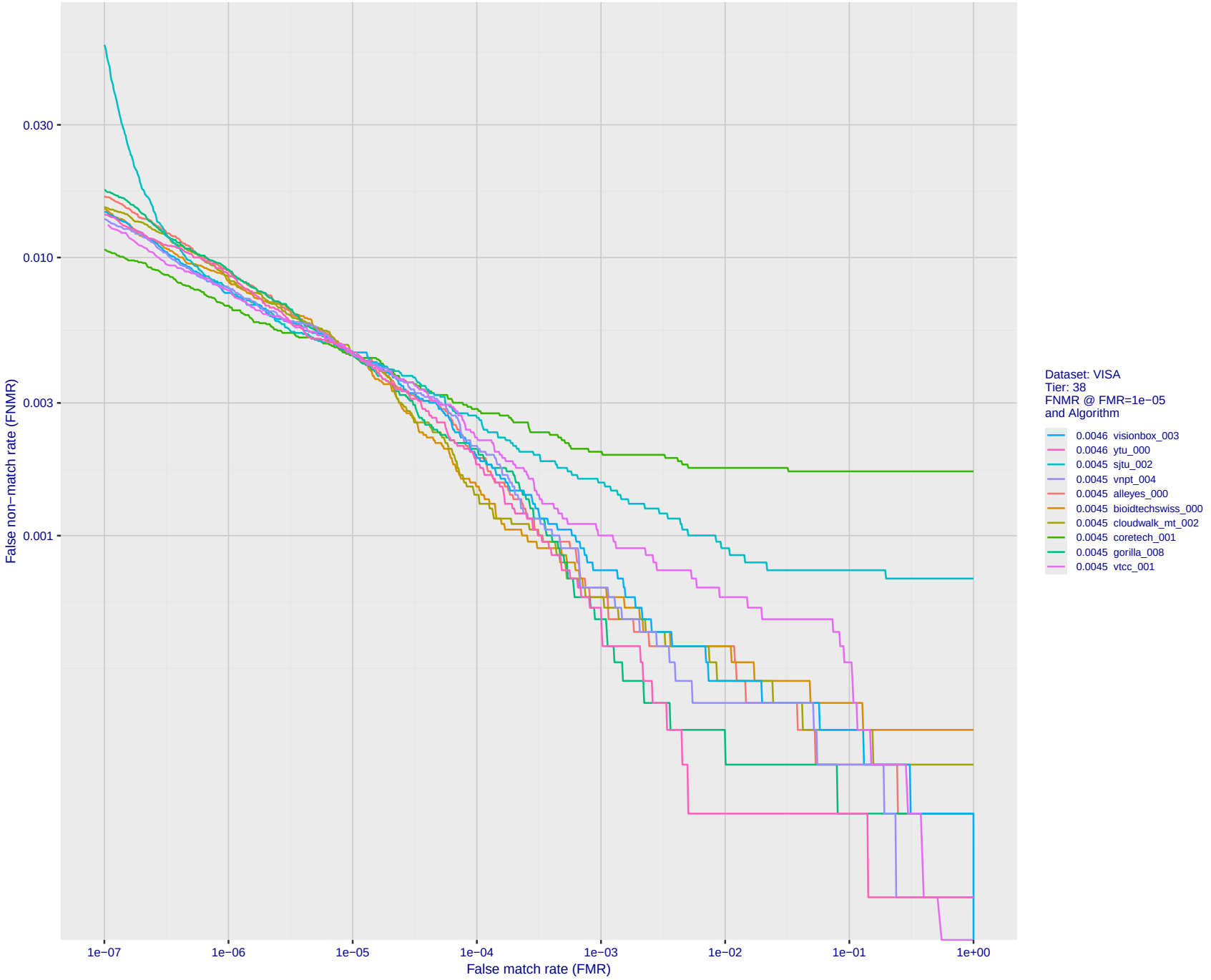


Figure 150: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

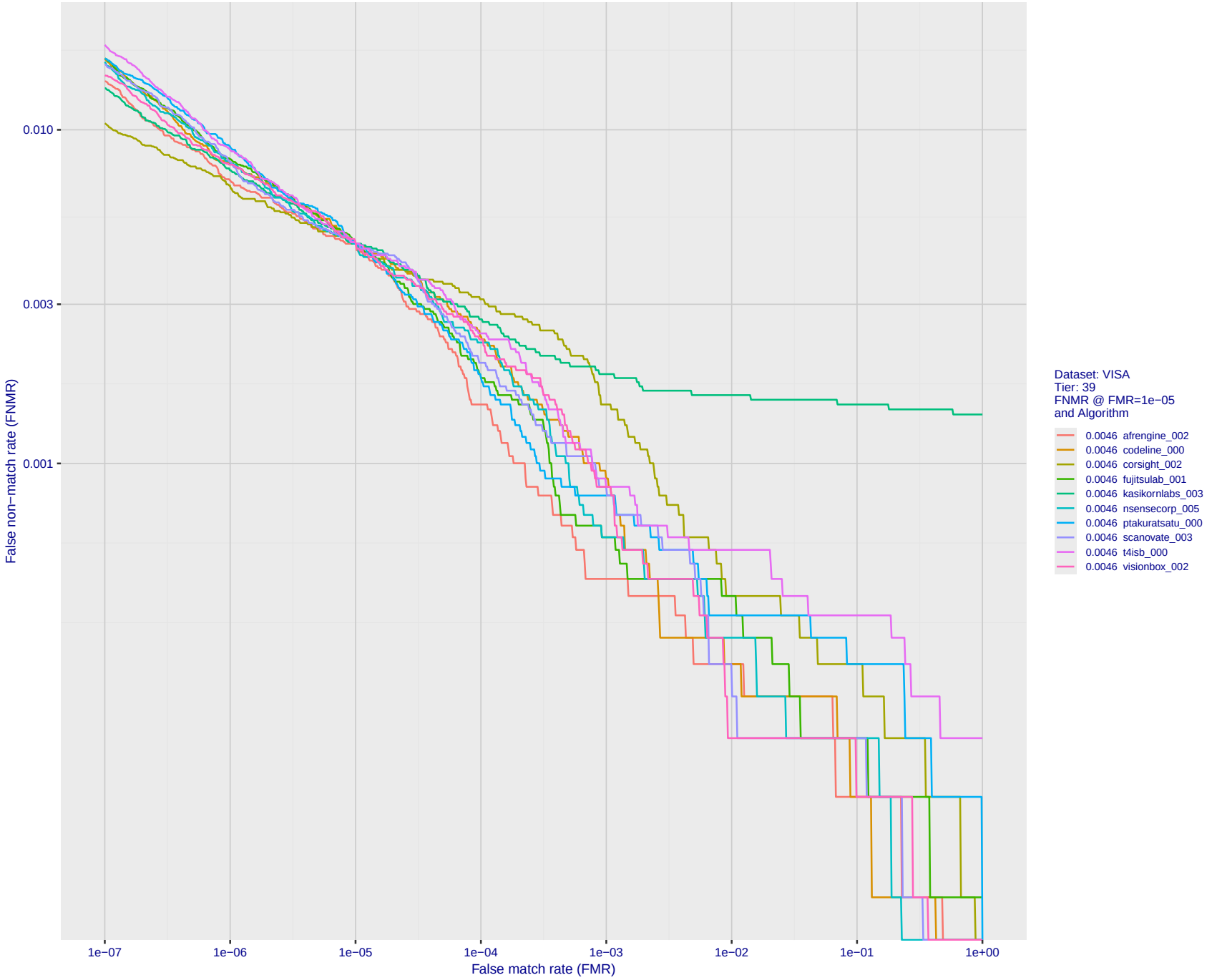


Figure 151: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

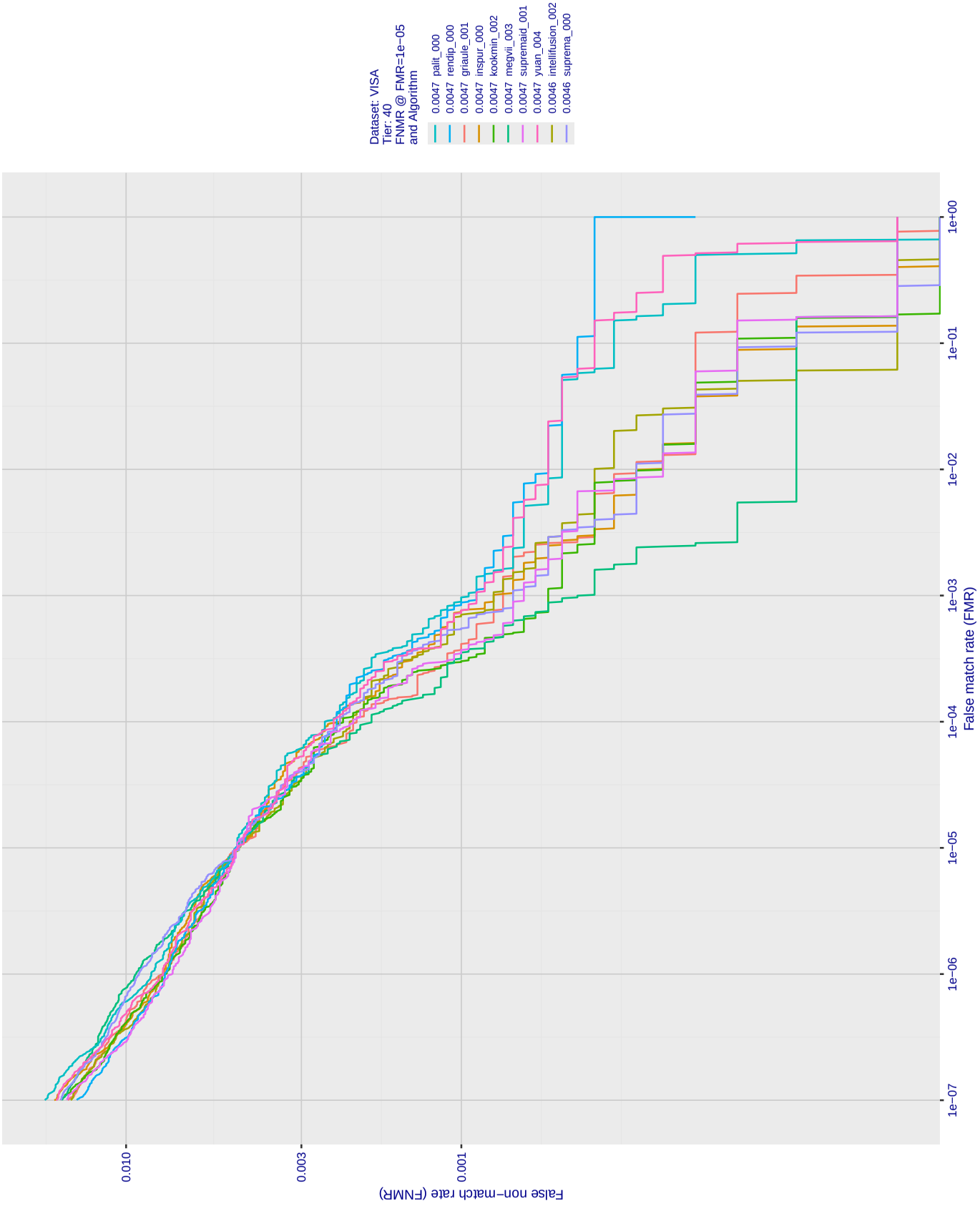


Figure 152: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

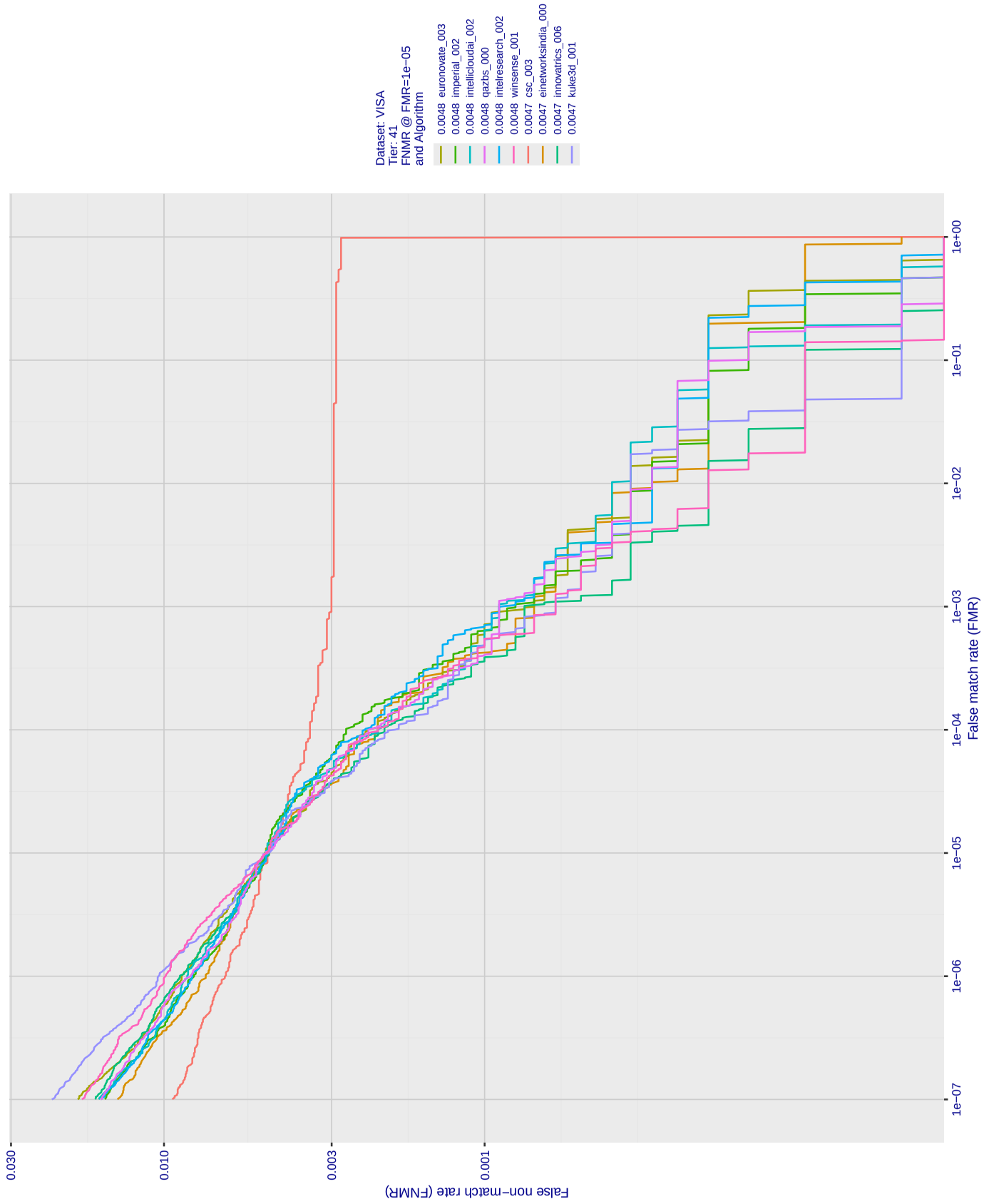


Figure 153: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

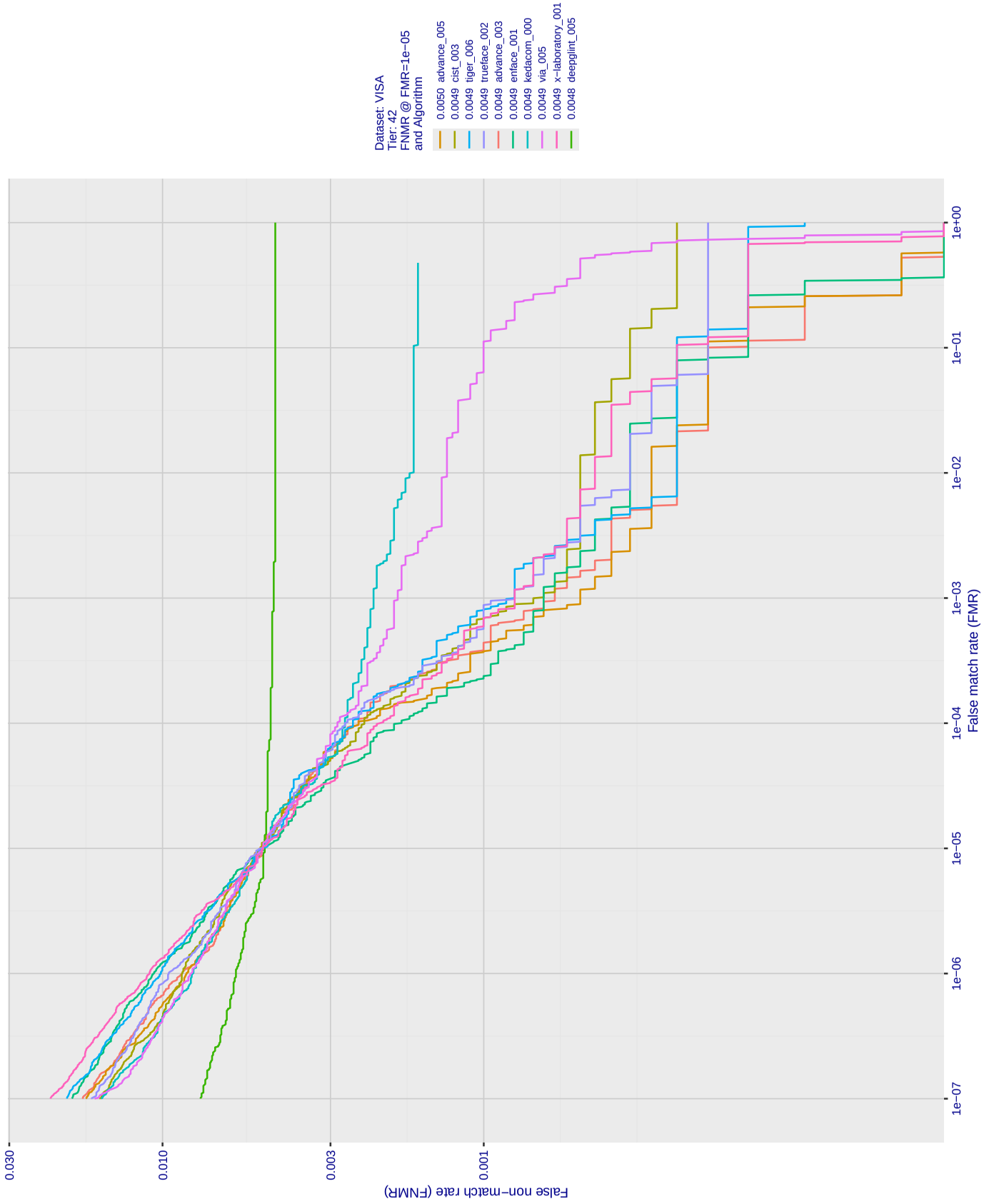


Figure 154: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

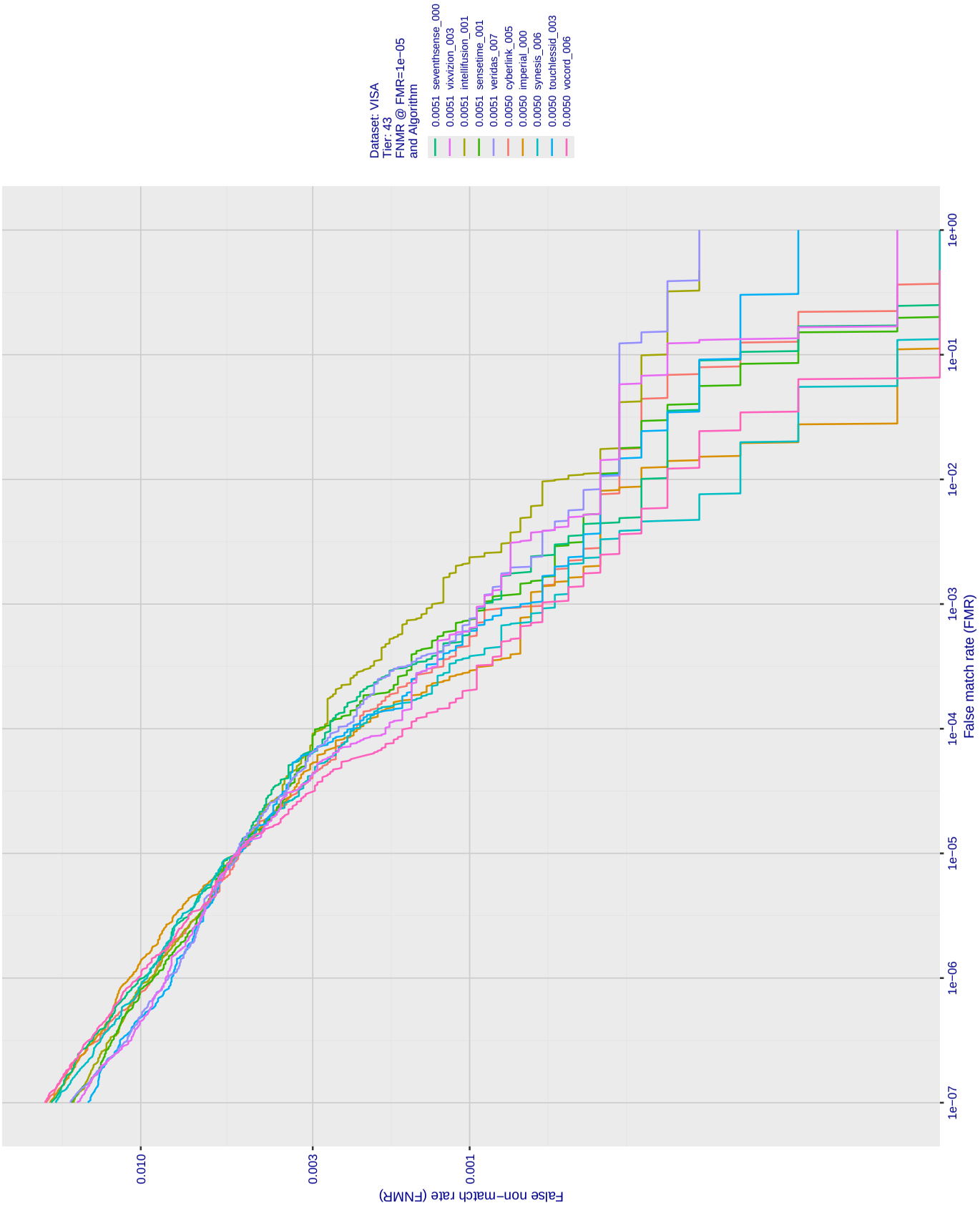


Figure 155: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

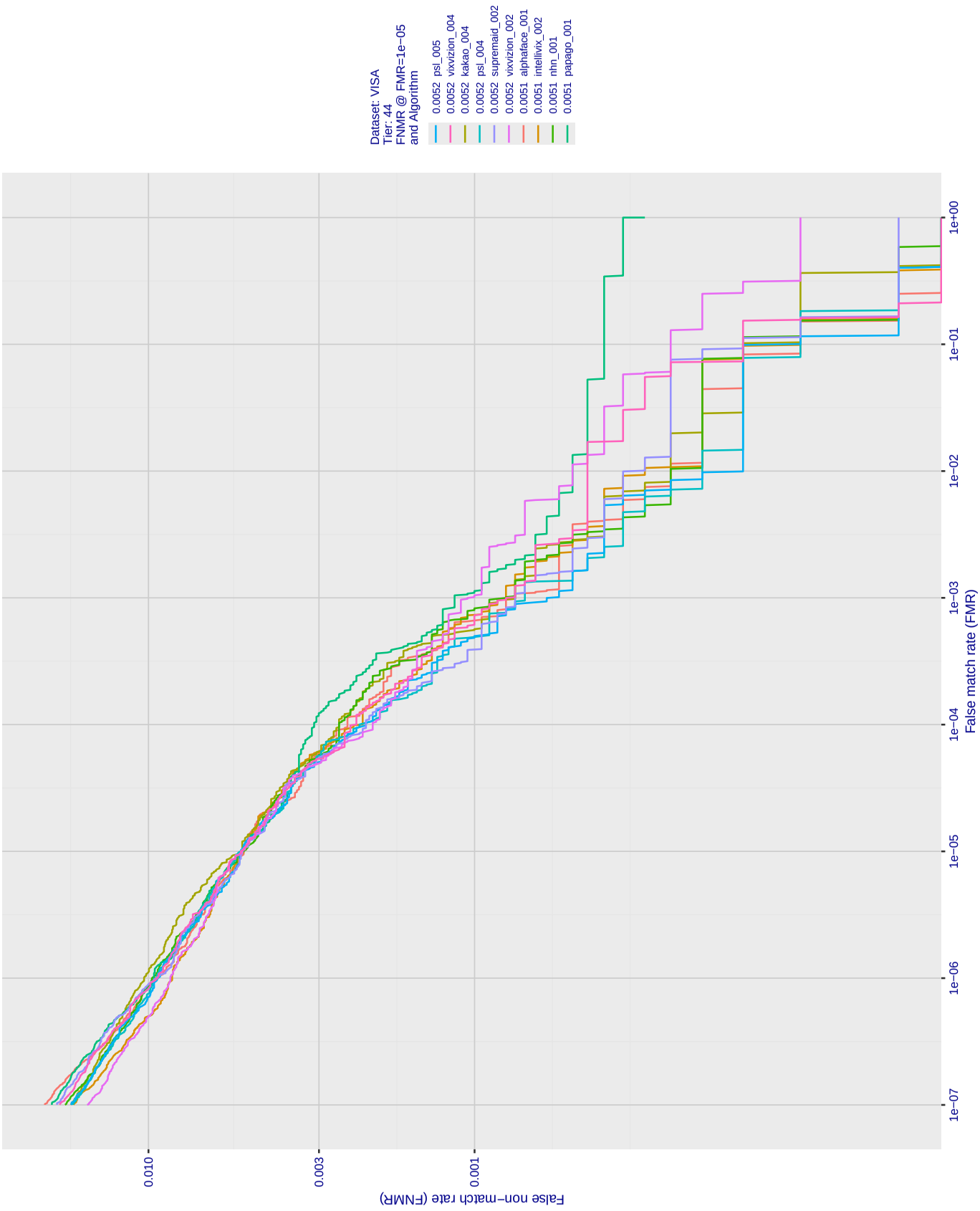


Figure 156: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

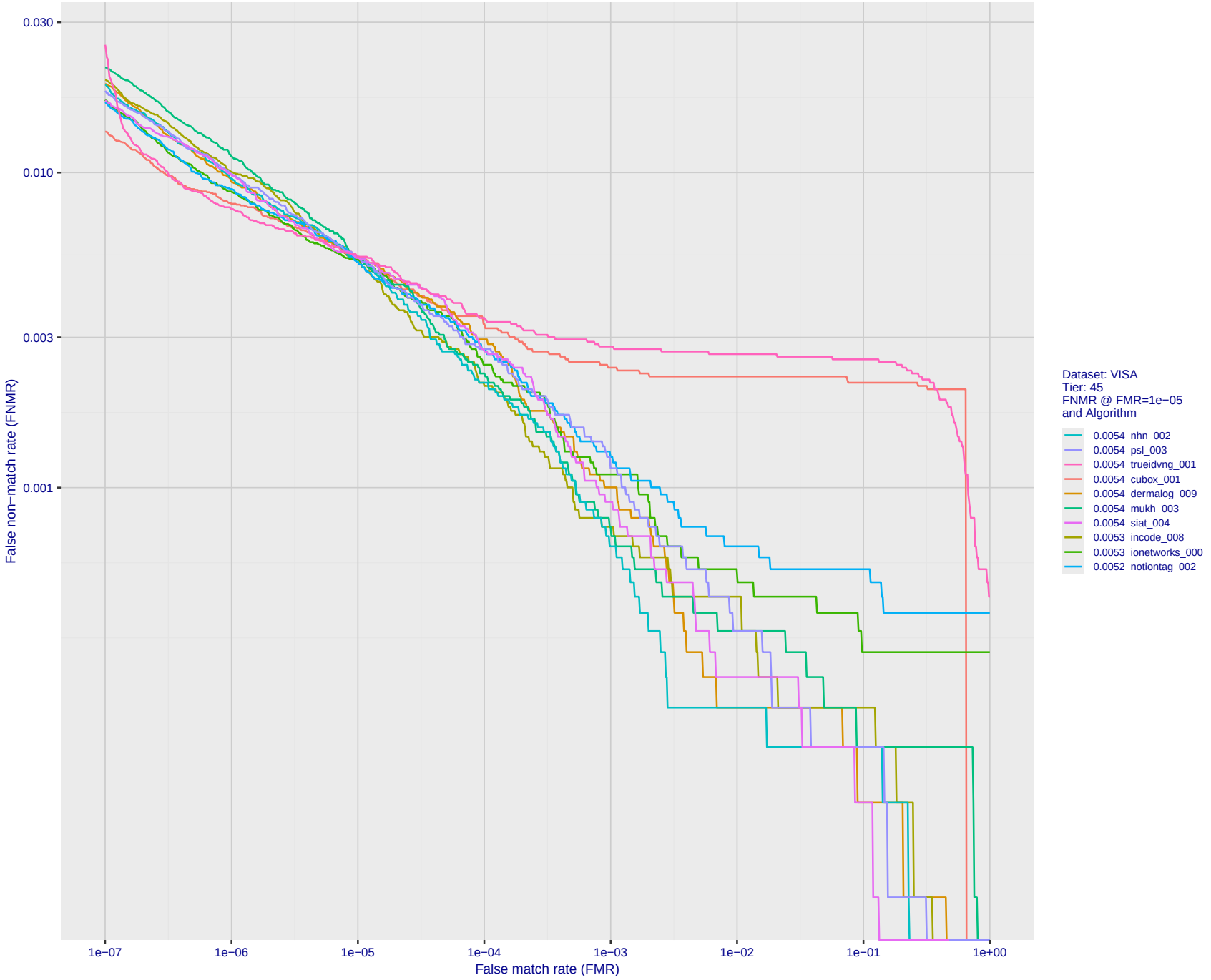


Figure 157: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

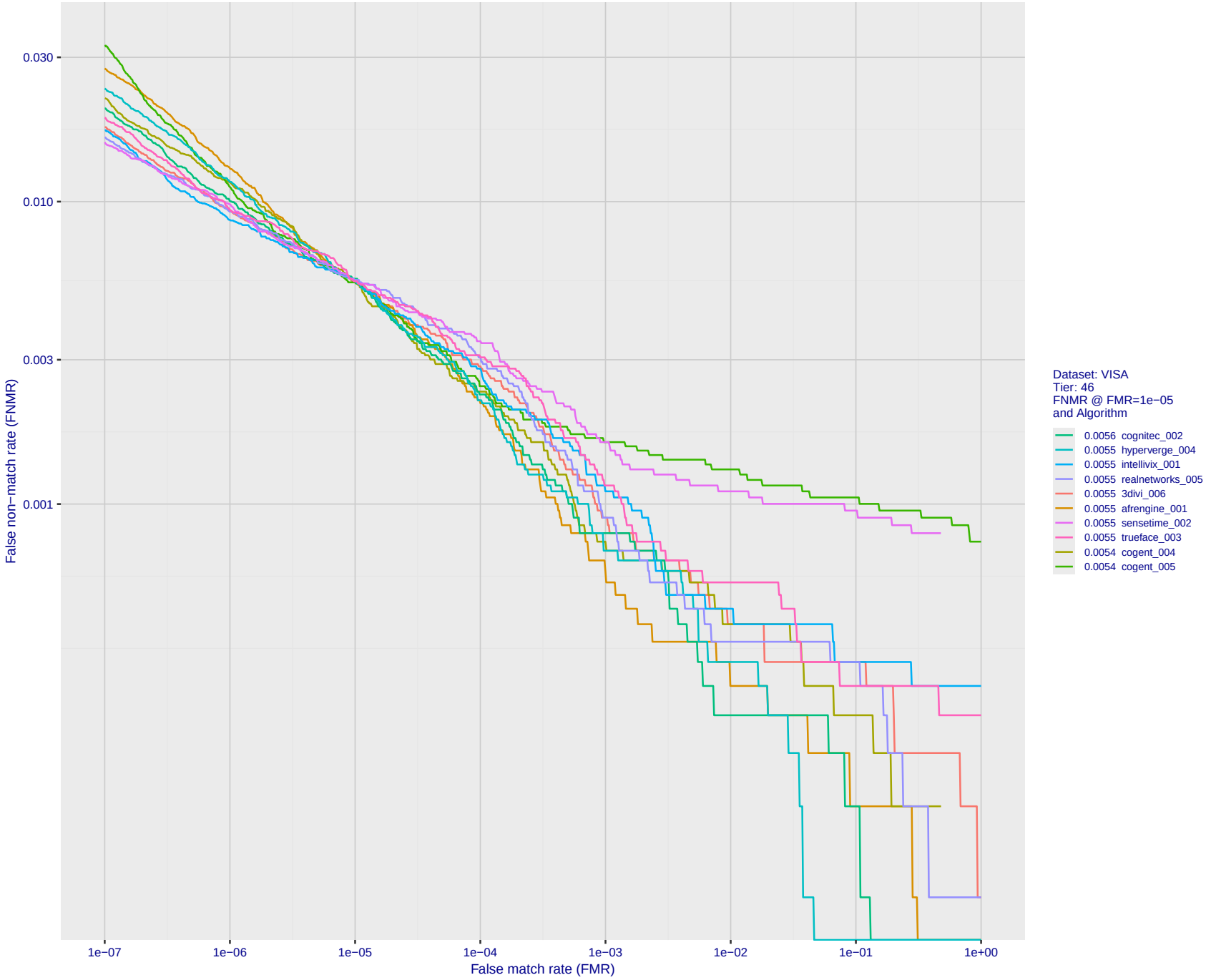


Figure 158: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

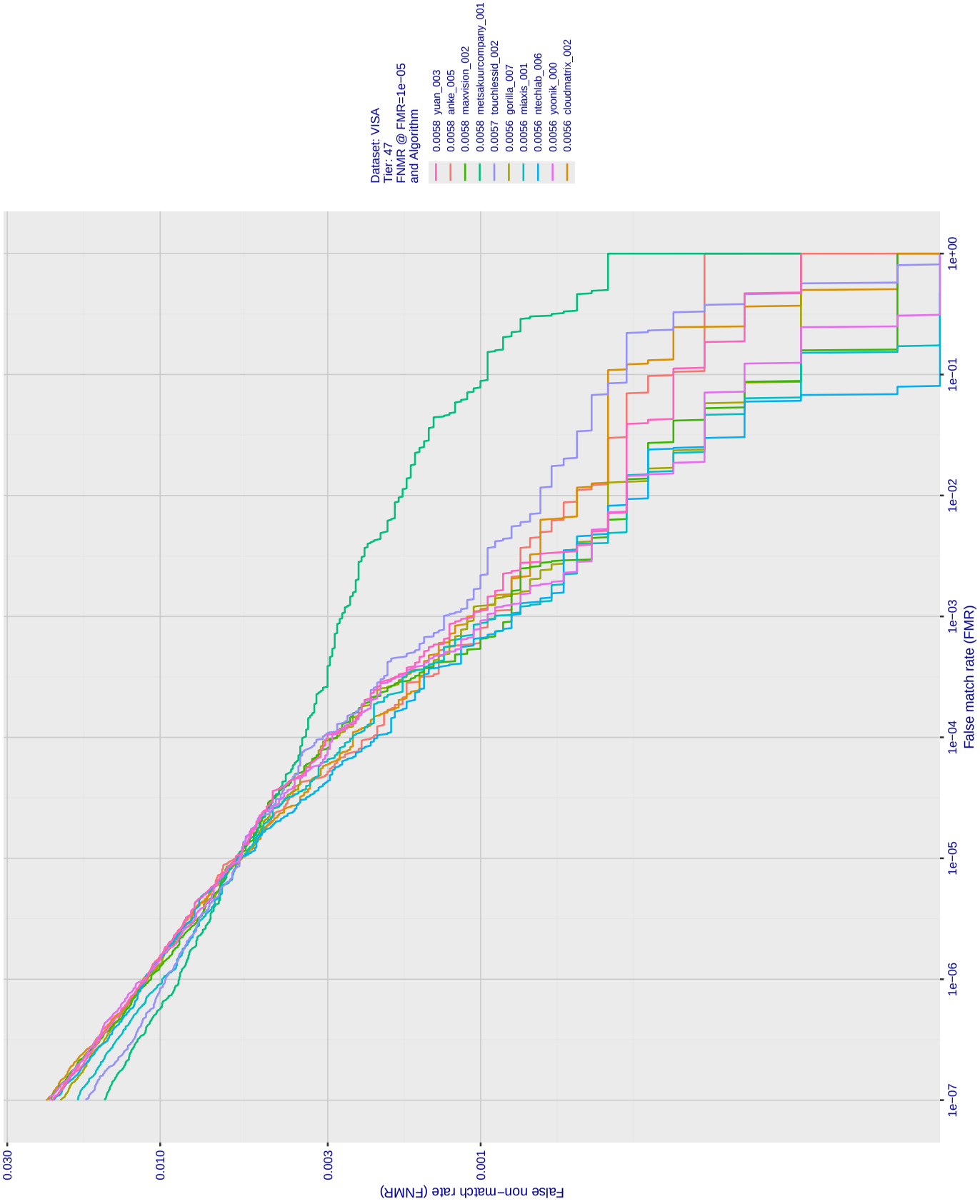


Figure 159: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

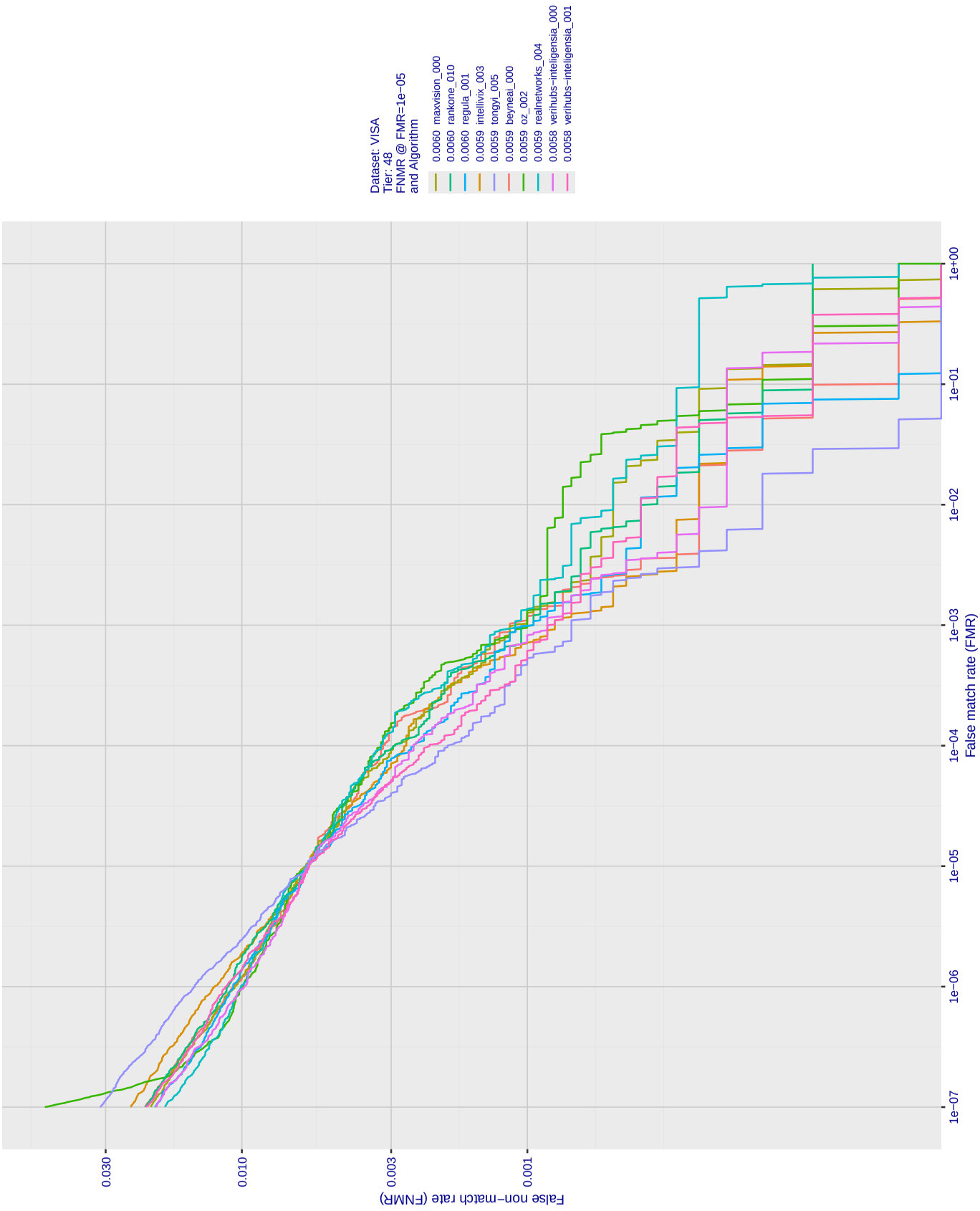


Figure 160: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

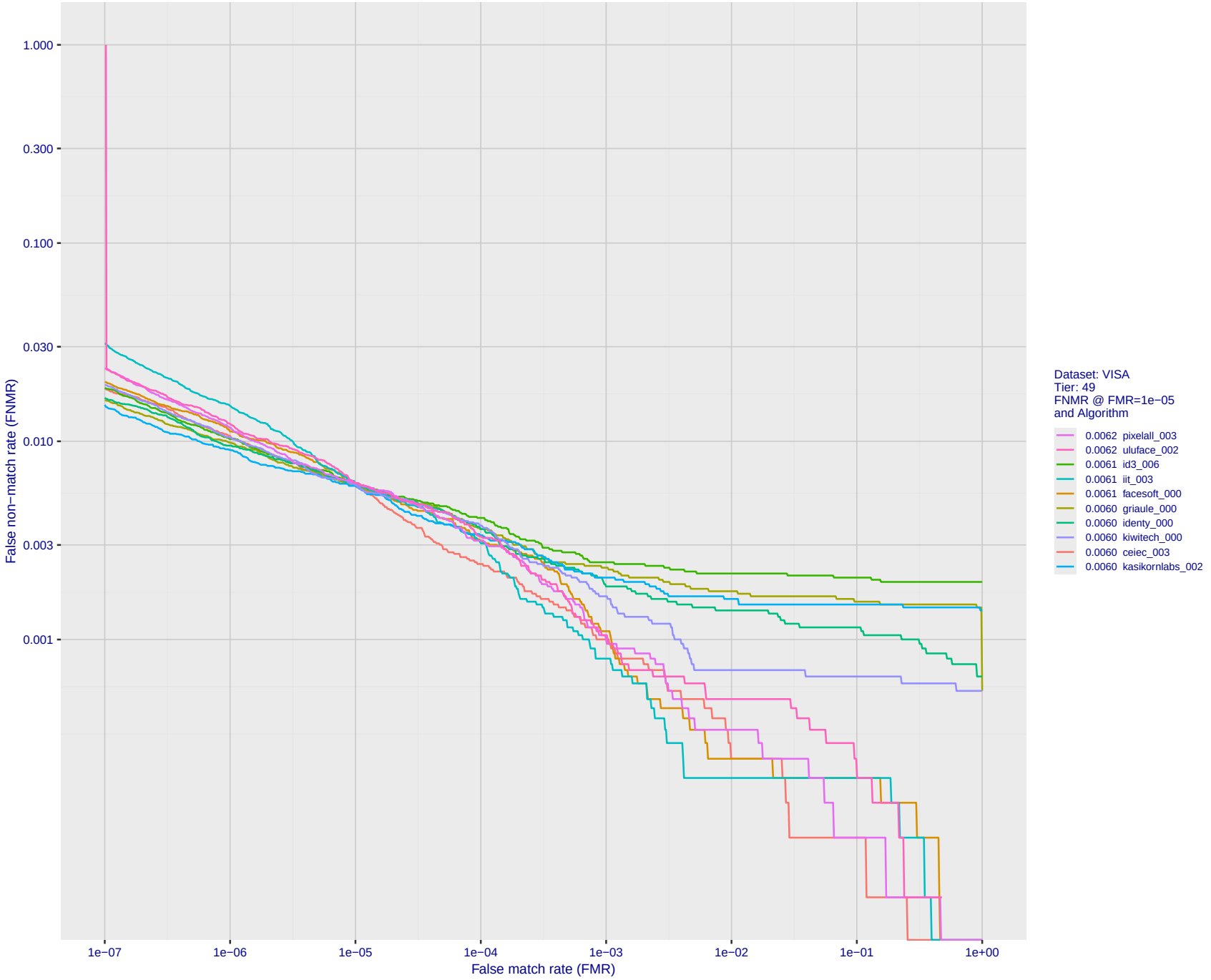


Figure 161: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

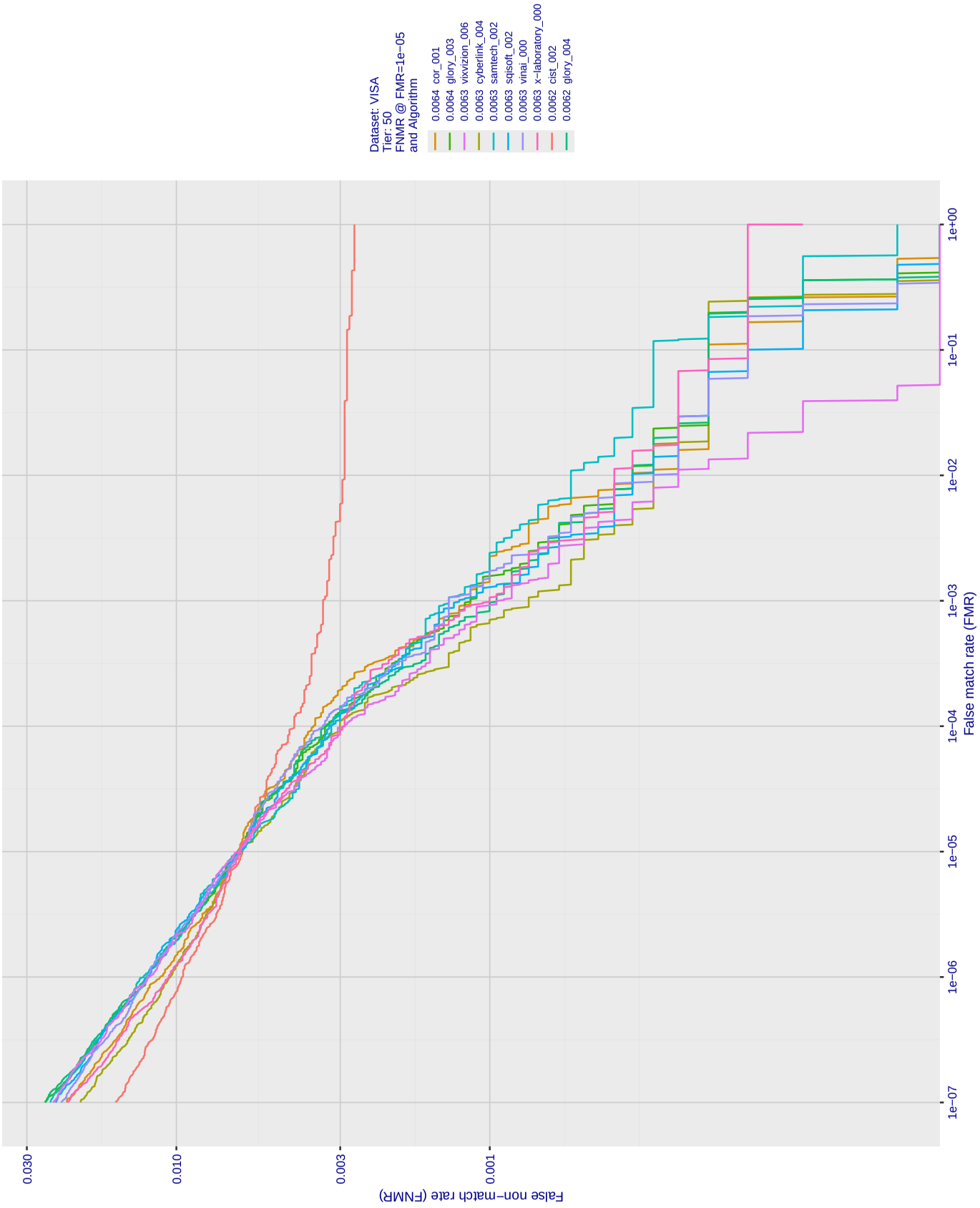


Figure 162: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

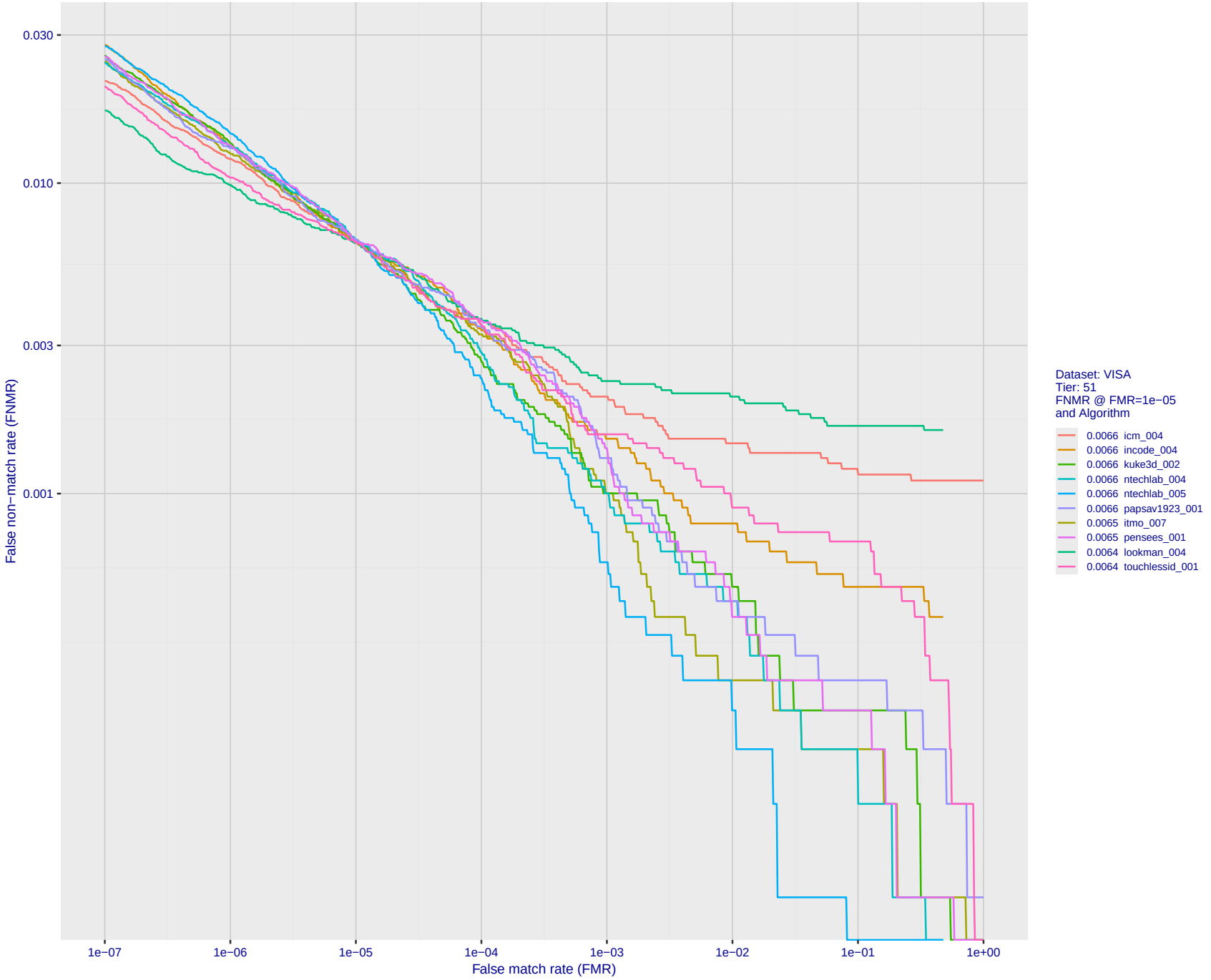


Figure 163: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

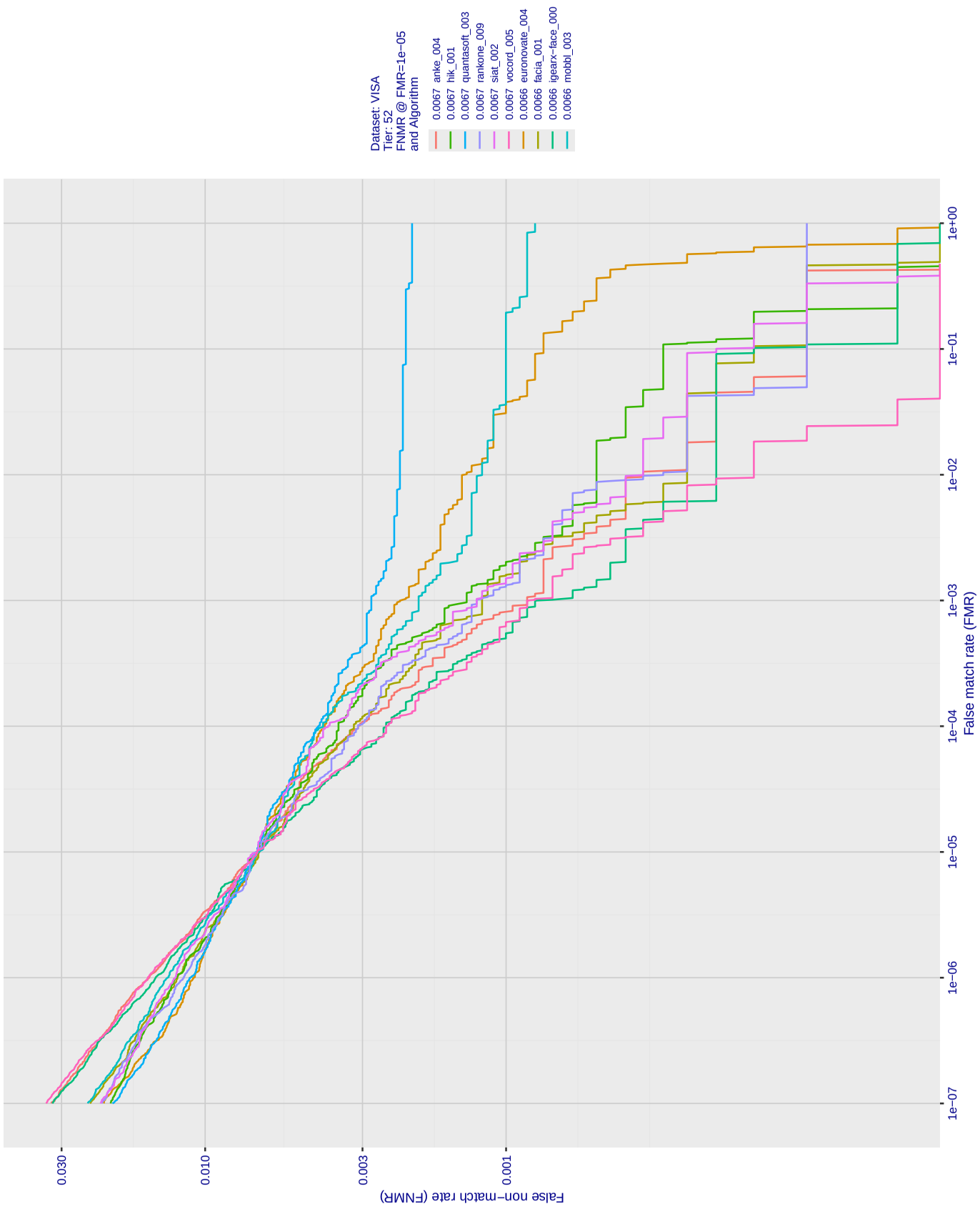


Figure 164: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

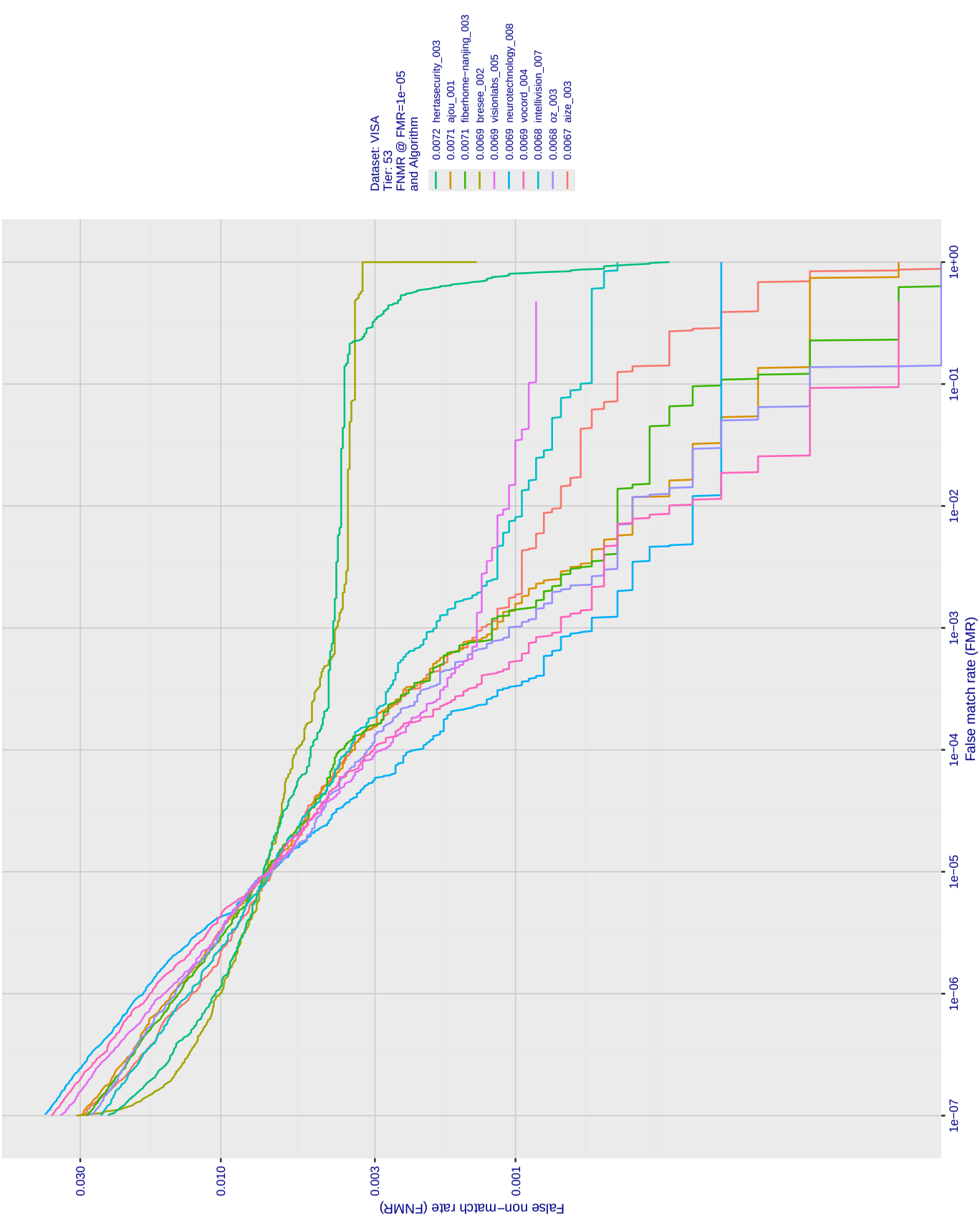


Figure 165: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

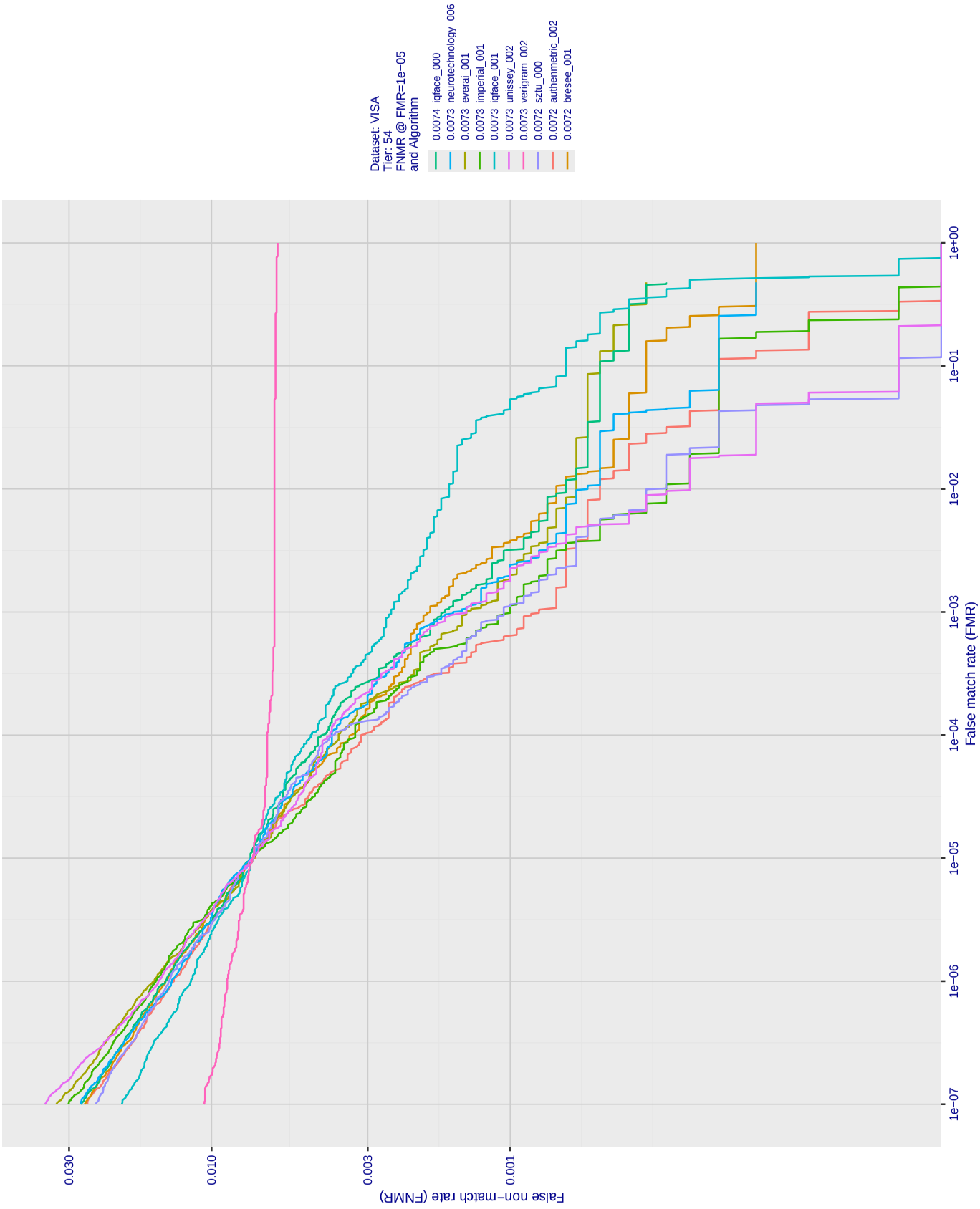


Figure 166: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

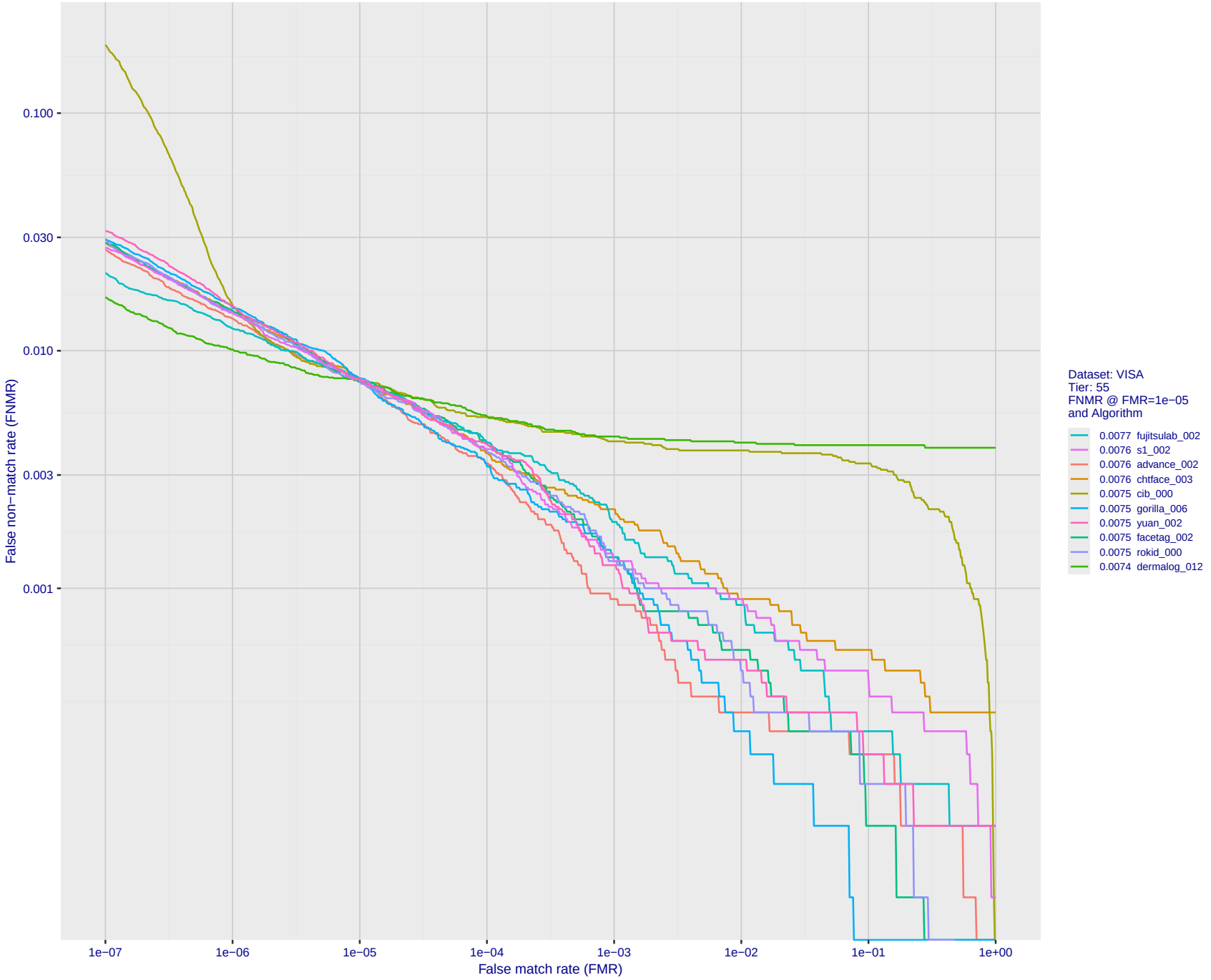


Figure 167: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

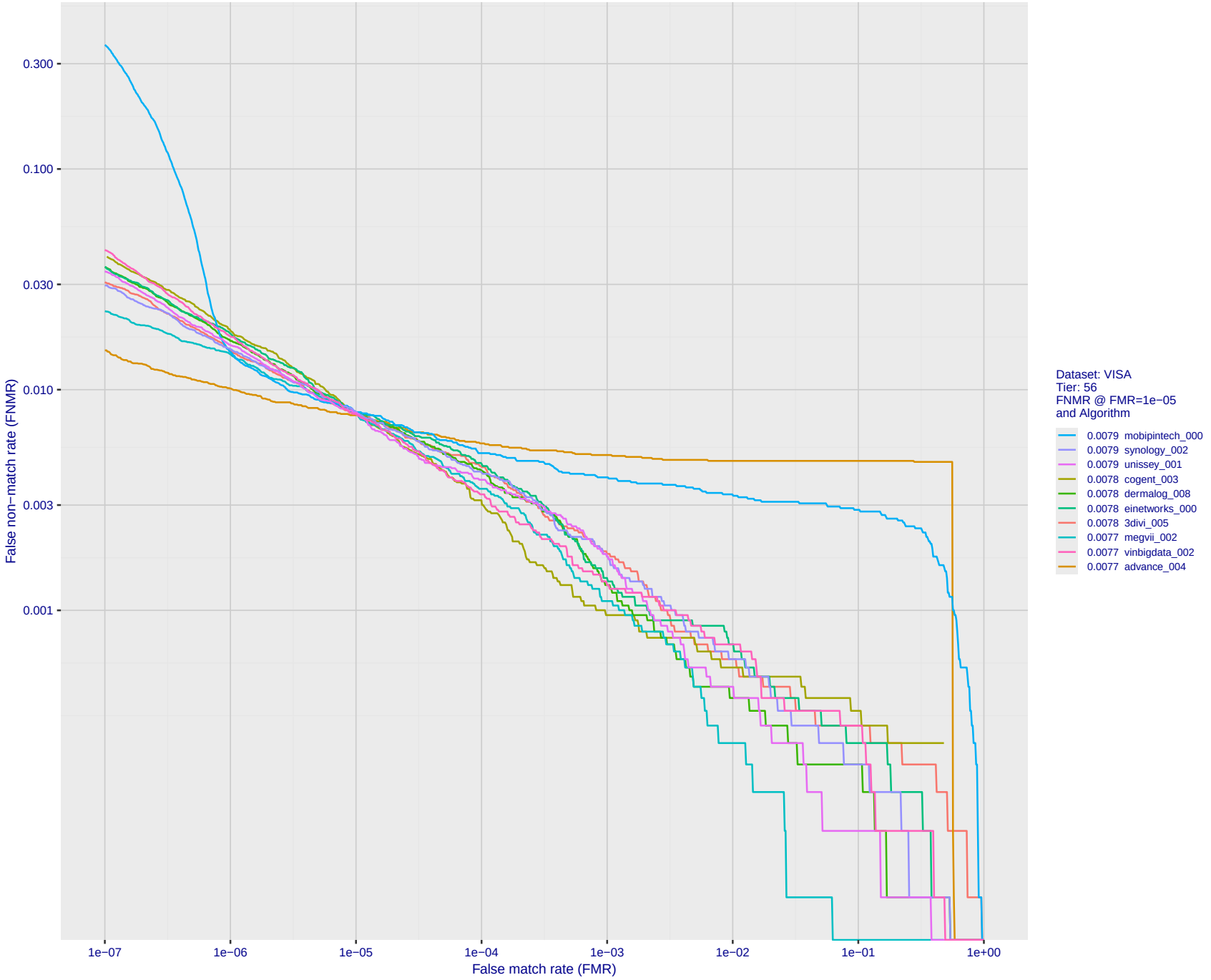


Figure 168: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

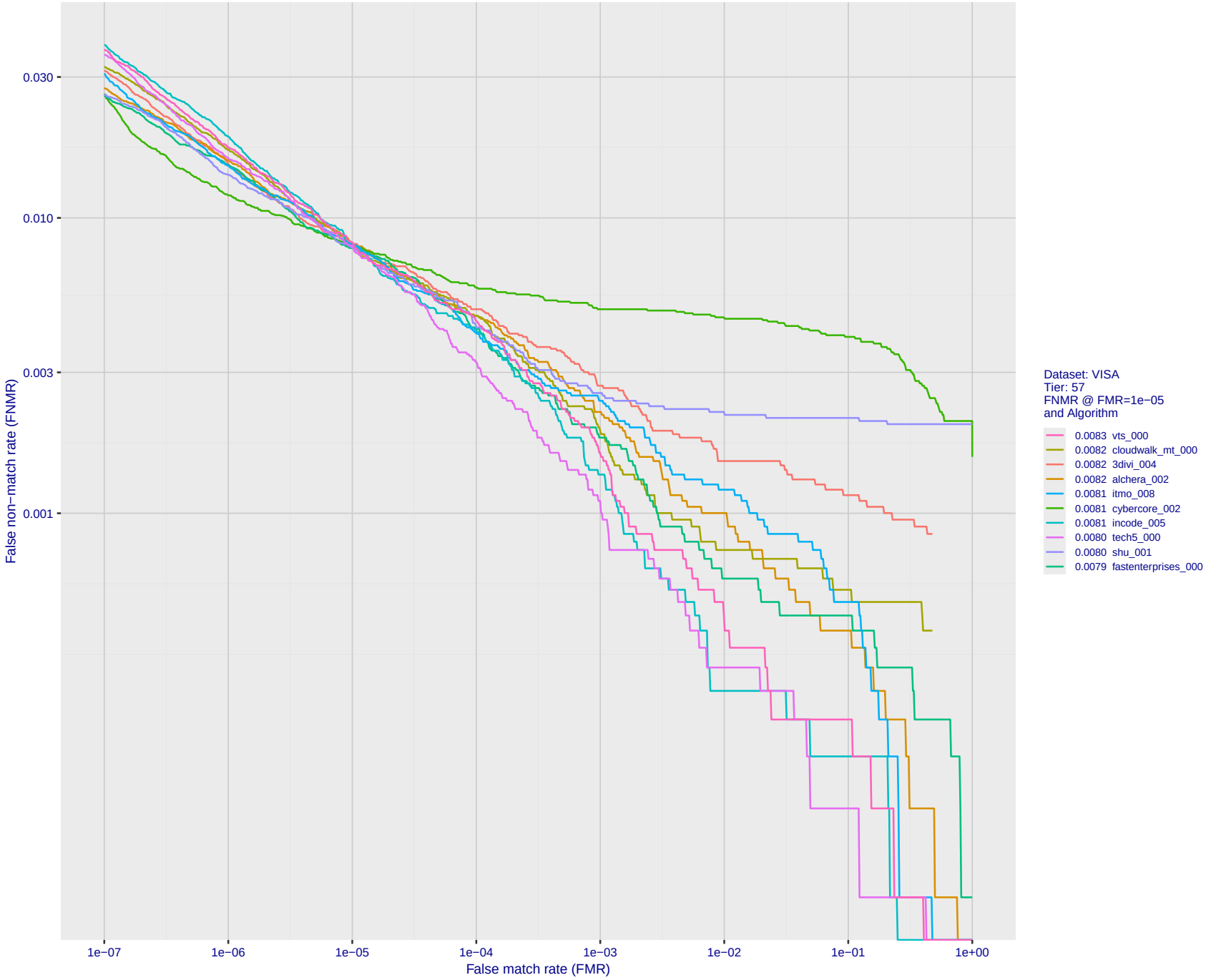


Figure 169: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

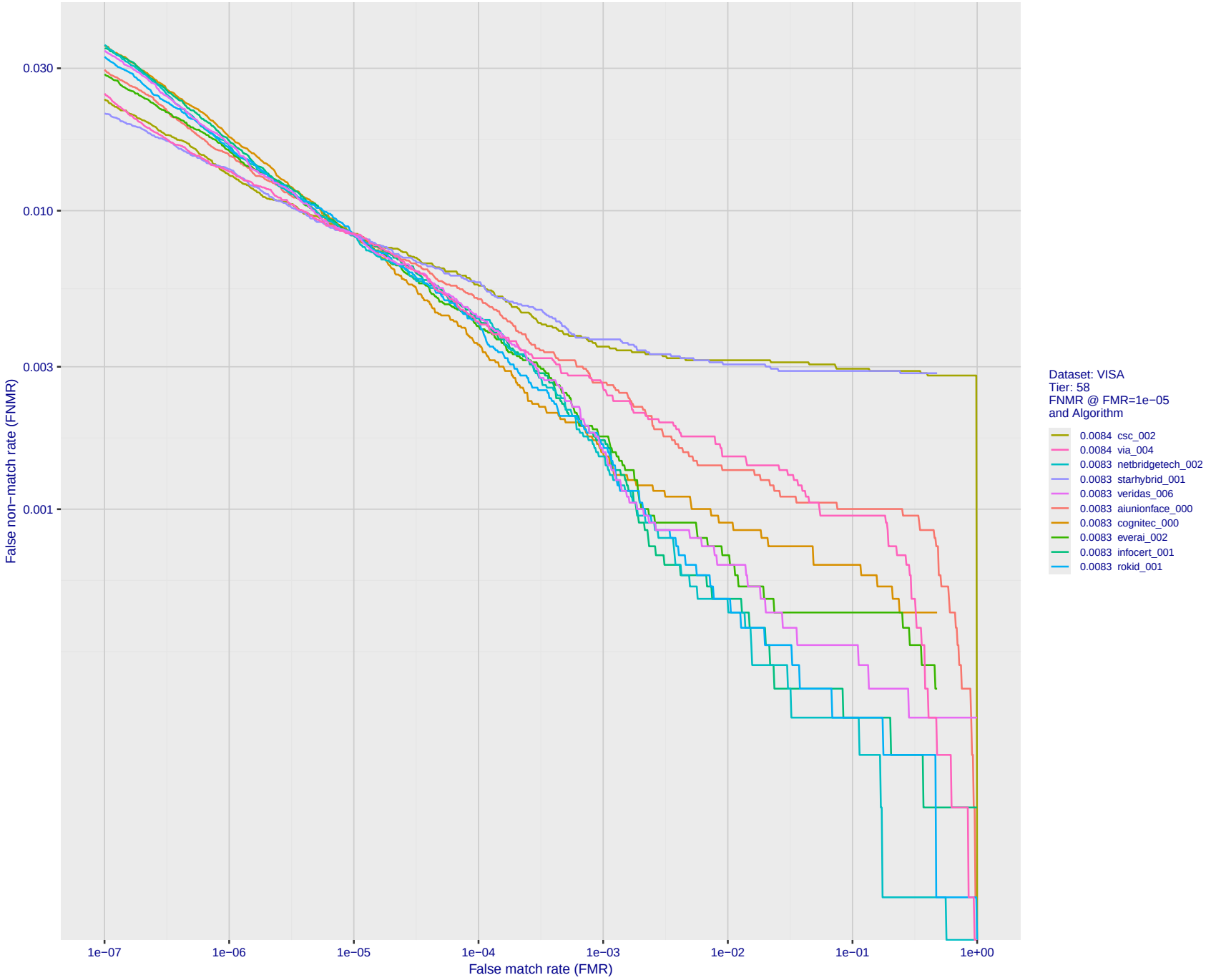


Figure 170: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

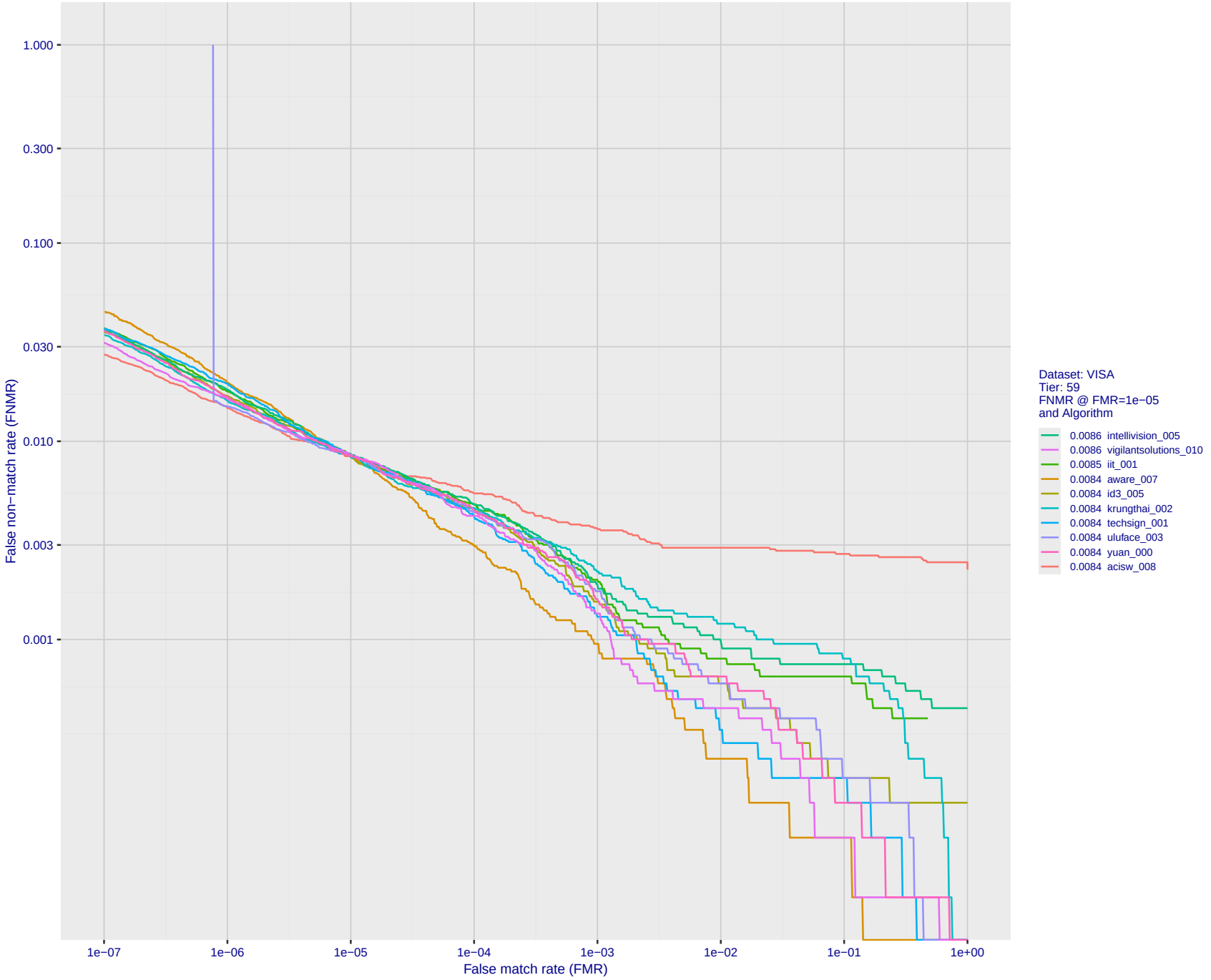


Figure 171: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

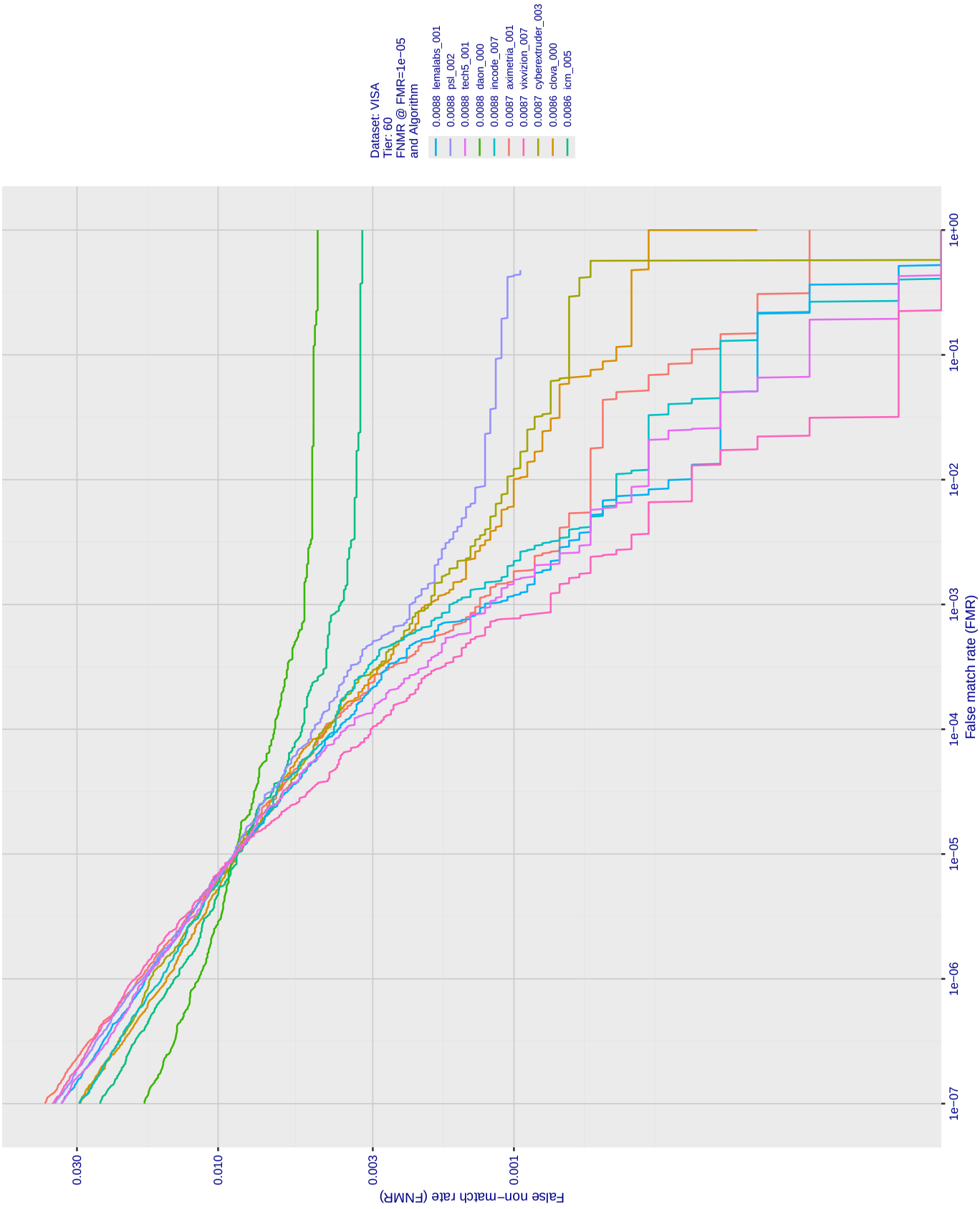


Figure 172: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

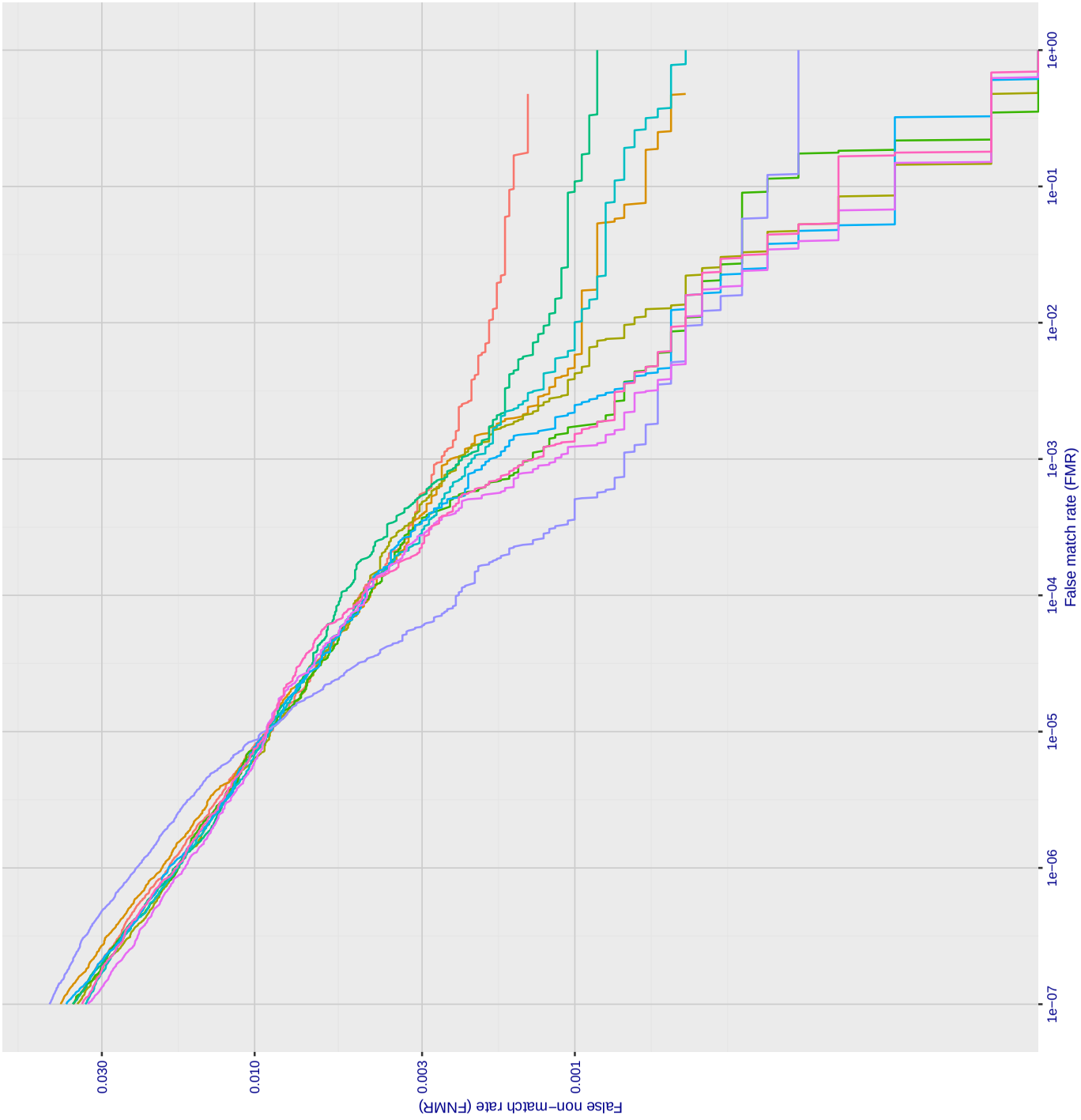


Figure 173: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

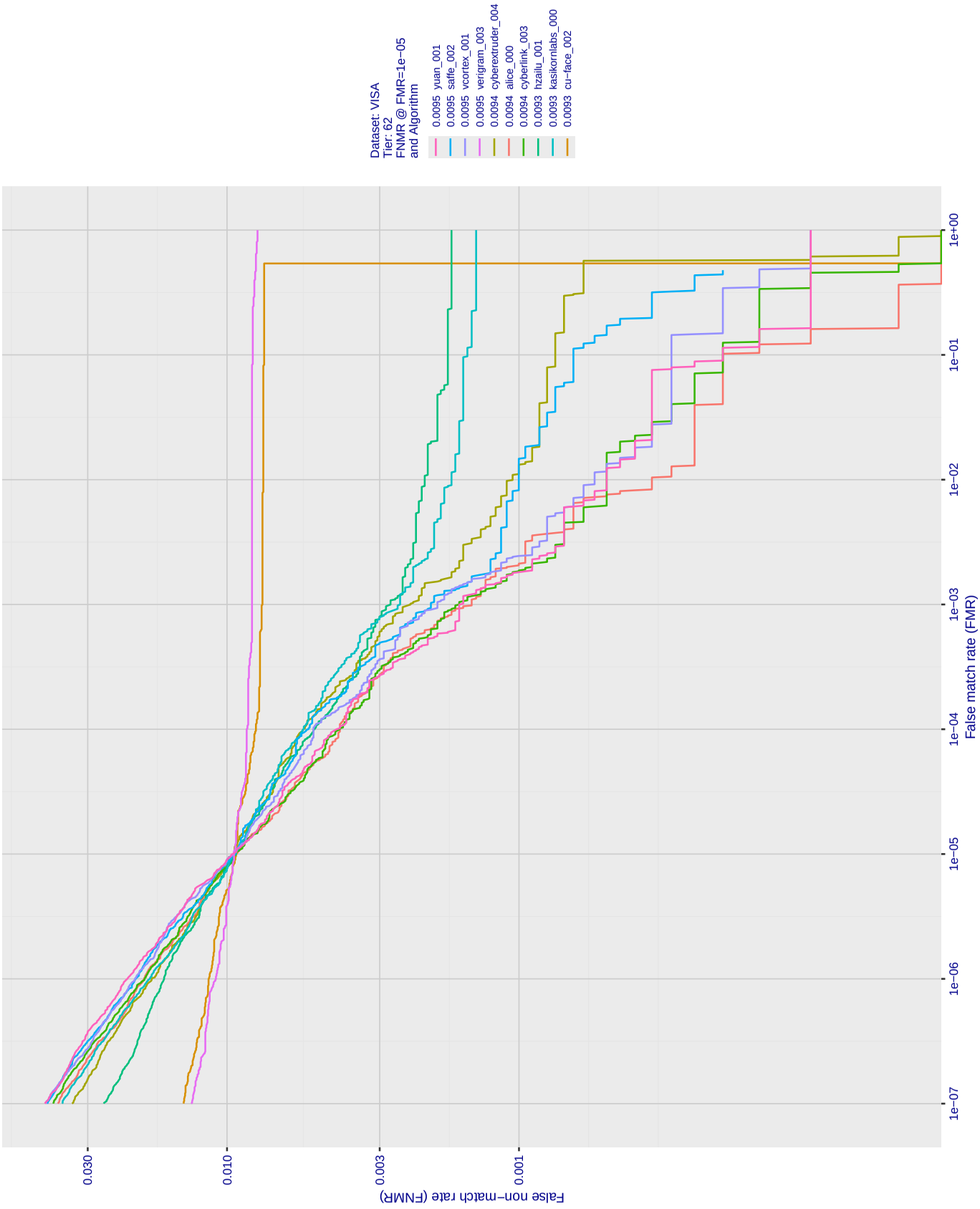


Figure 174: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

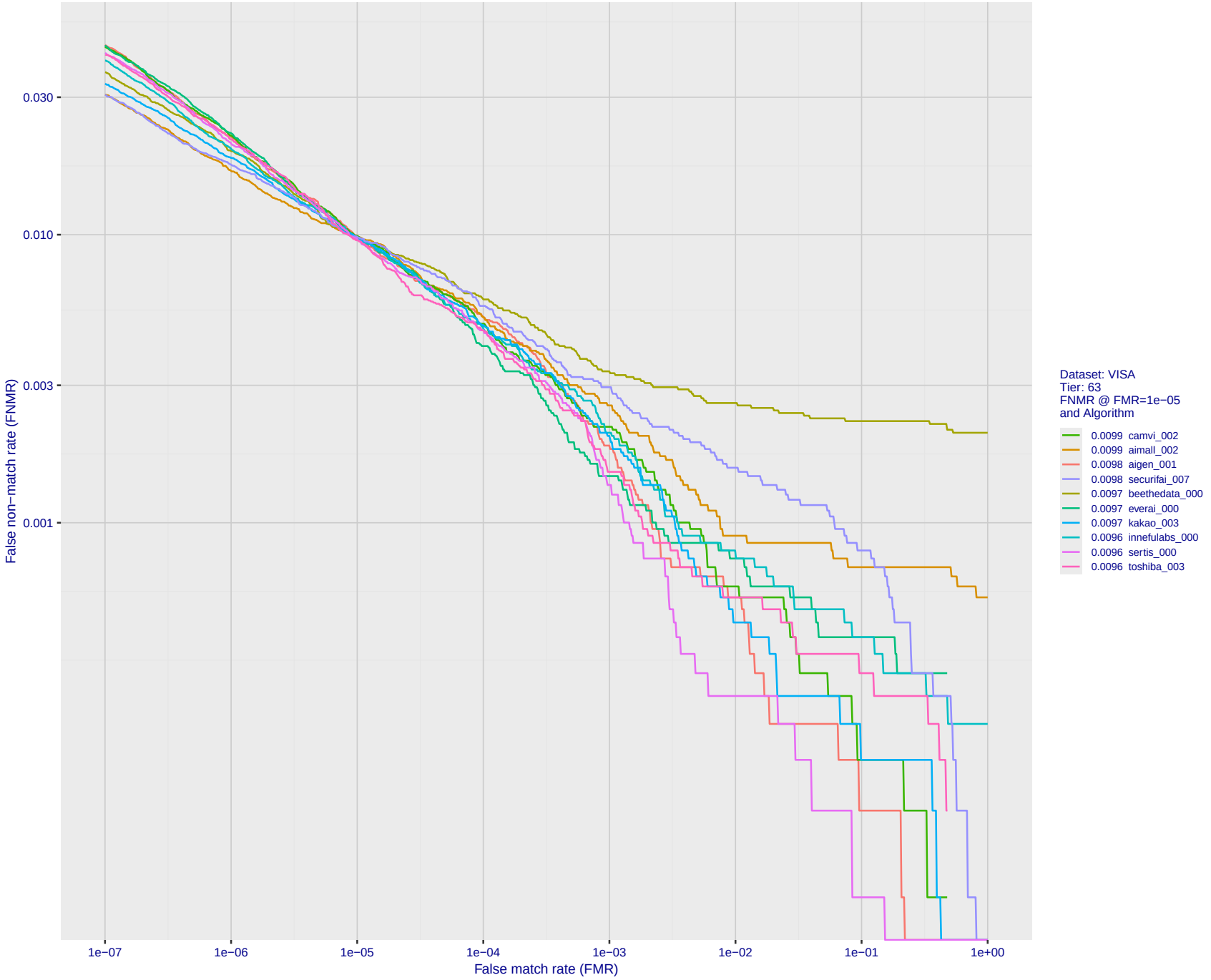


Figure 175: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

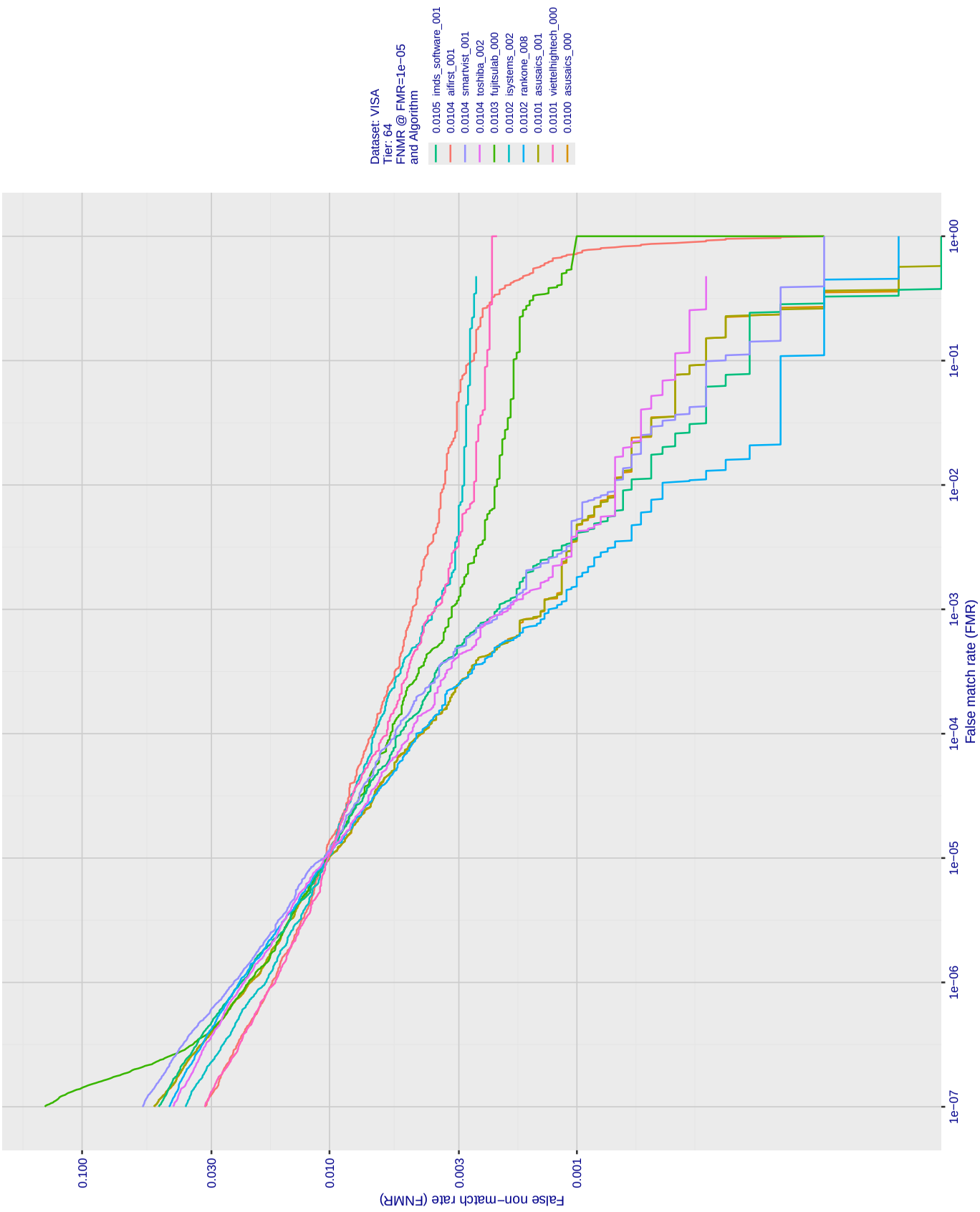


Figure 176: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

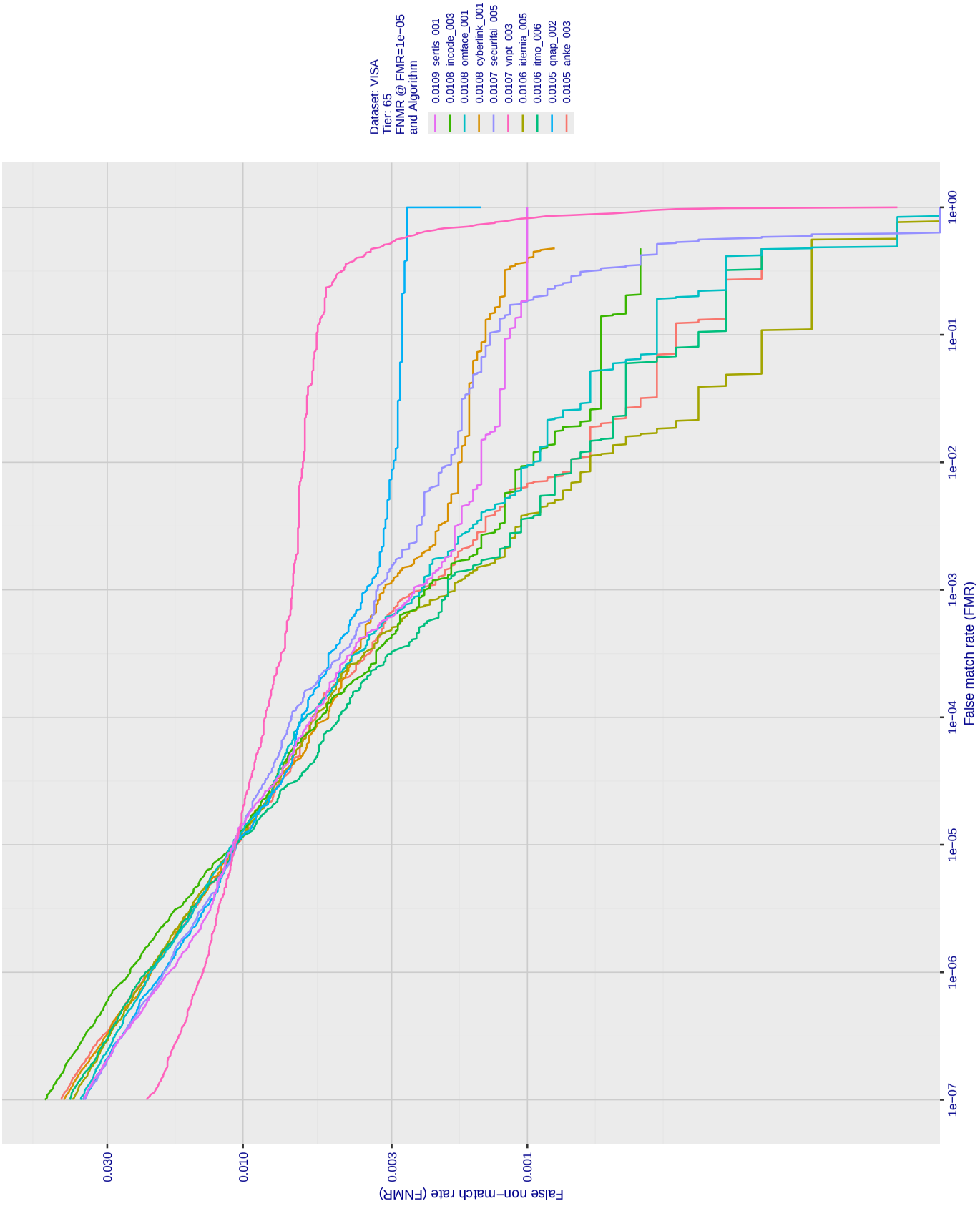


Figure 177: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

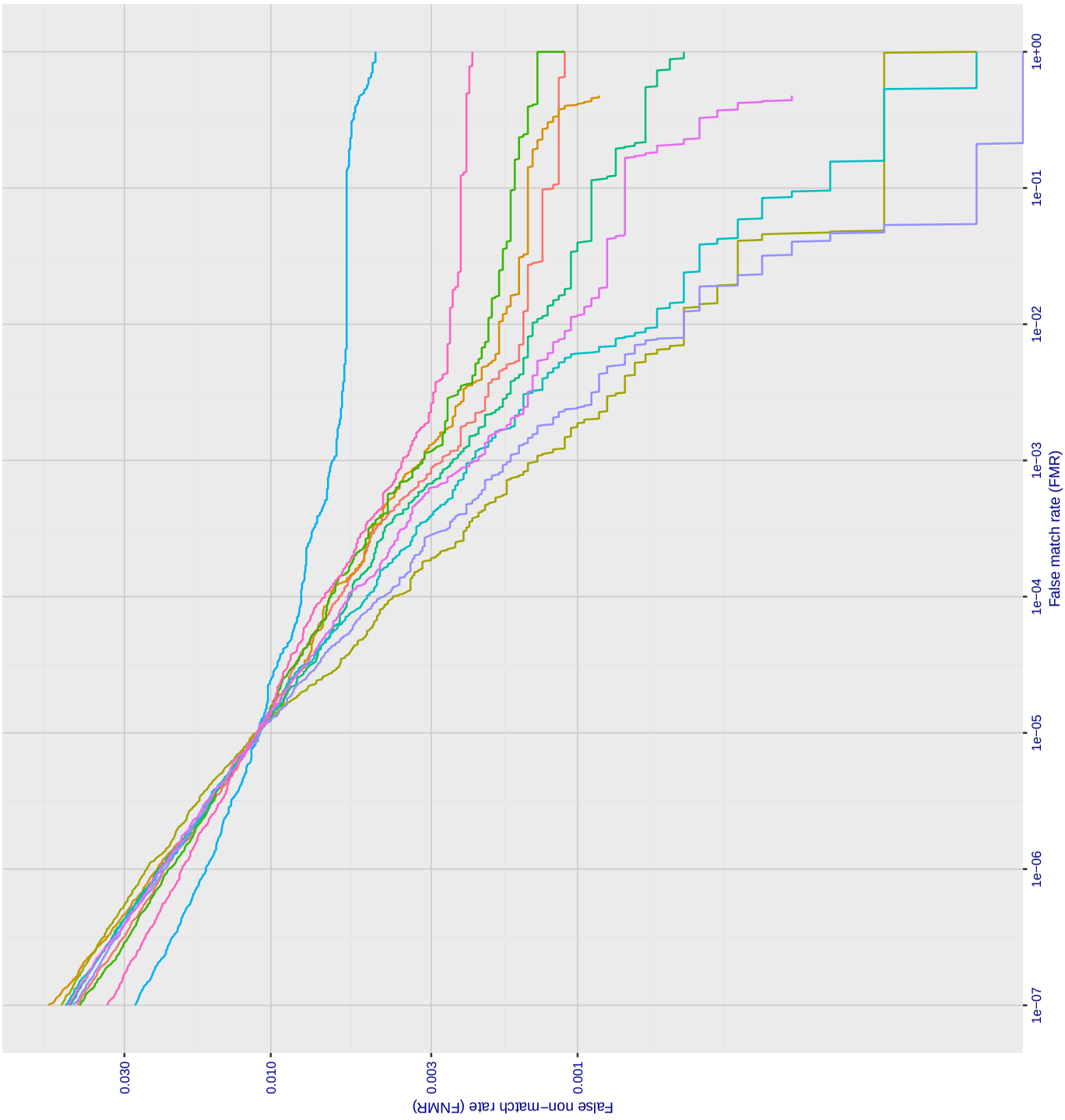


Figure 178: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

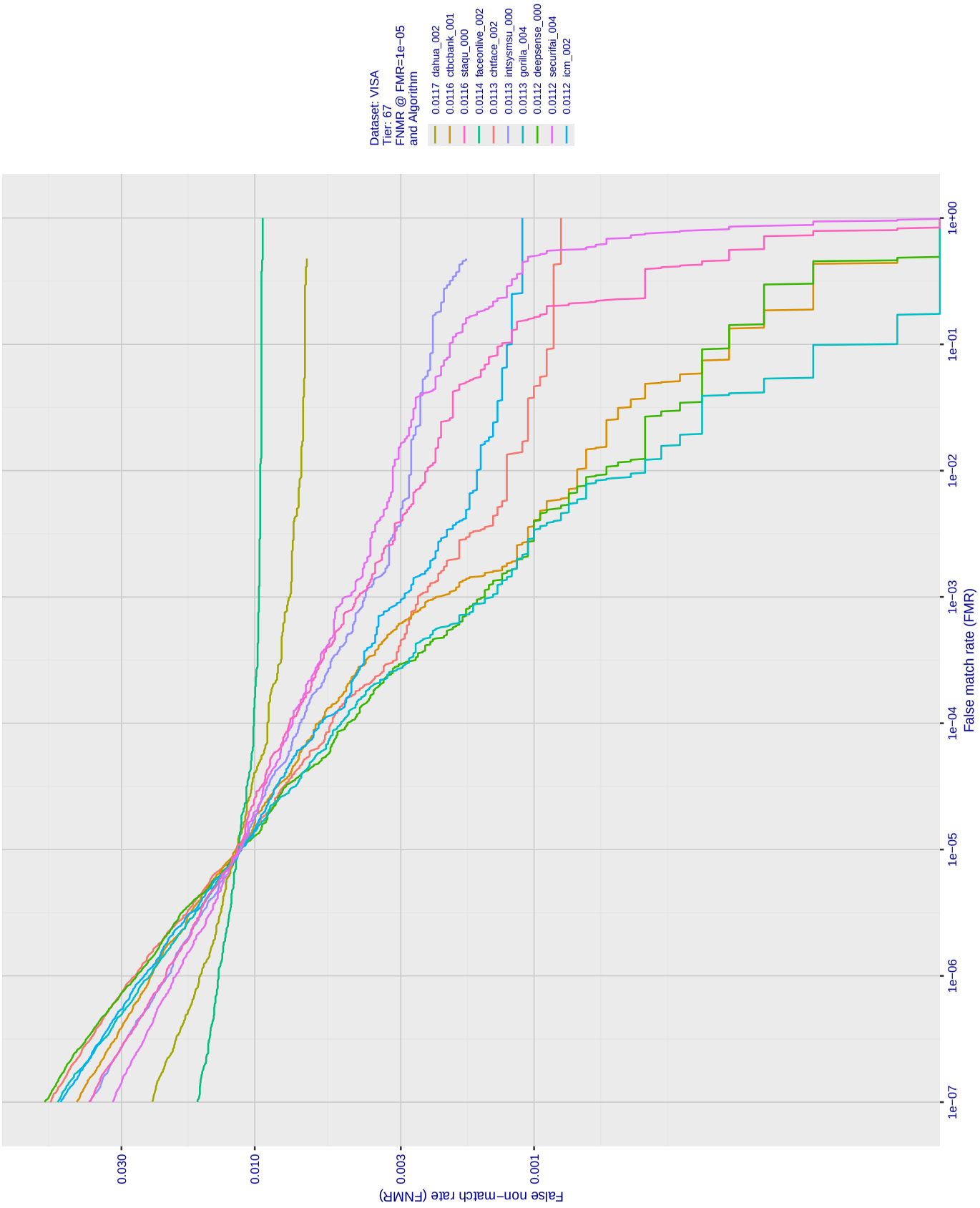


Figure 179: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

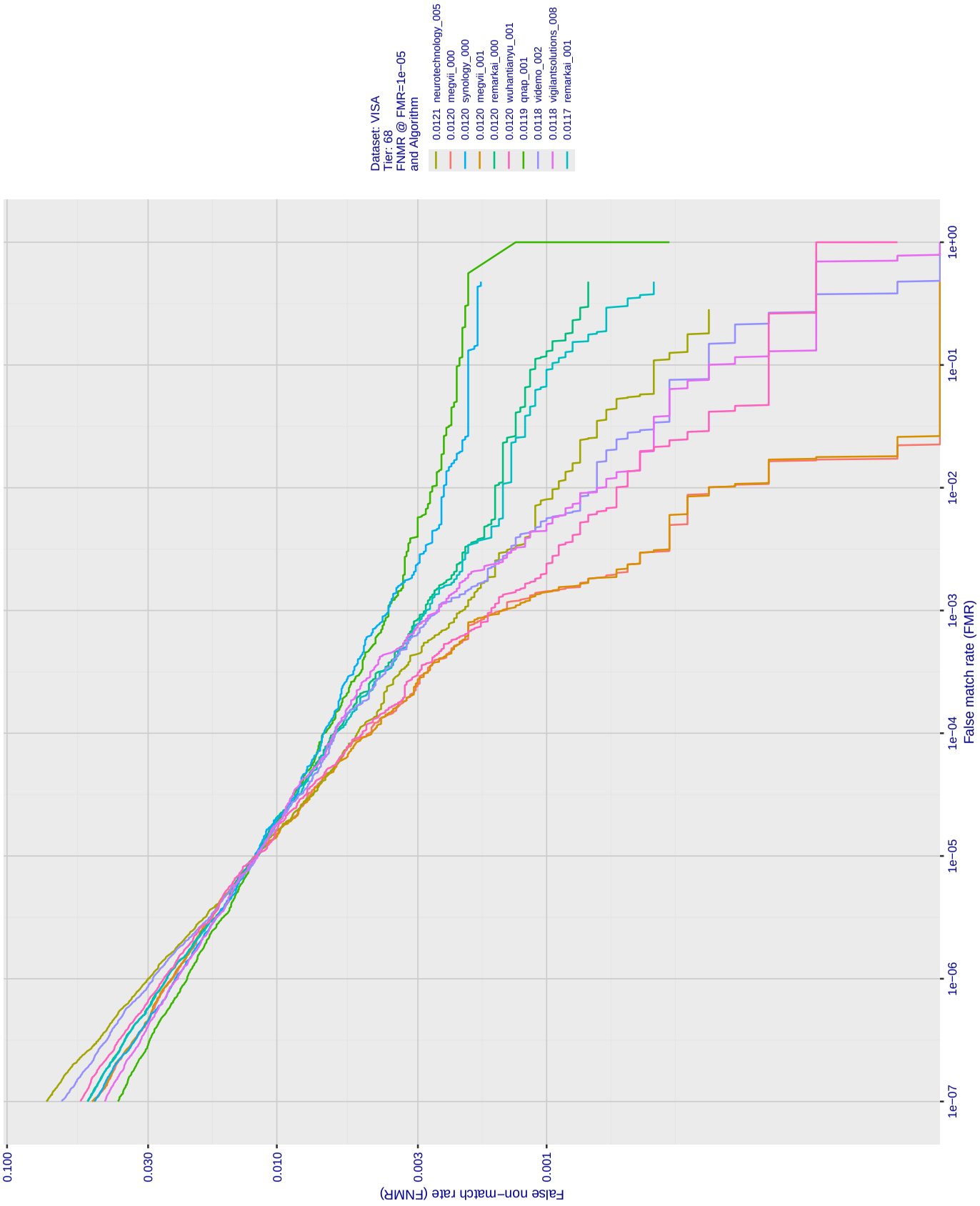


Figure 180: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

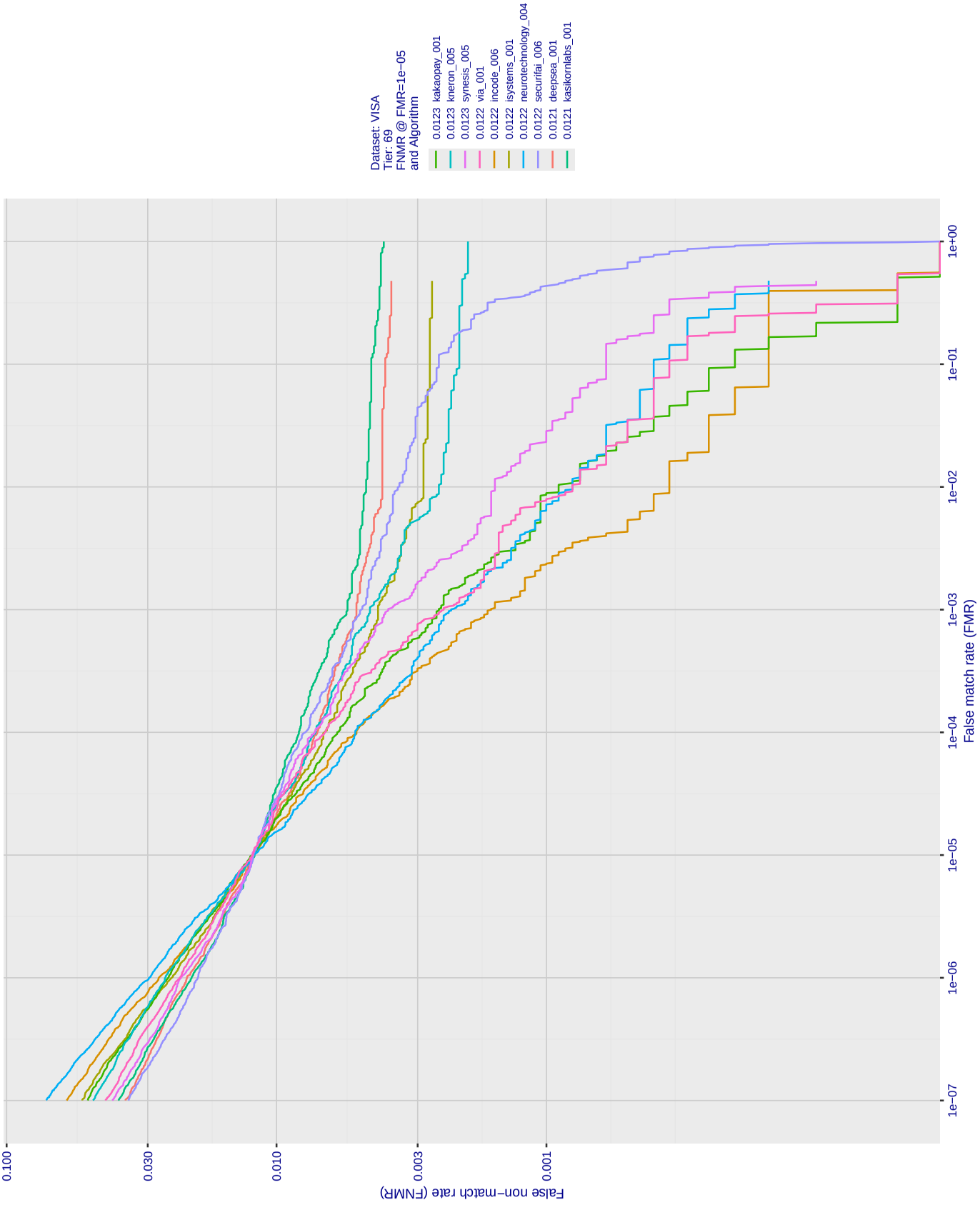


Figure 181: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

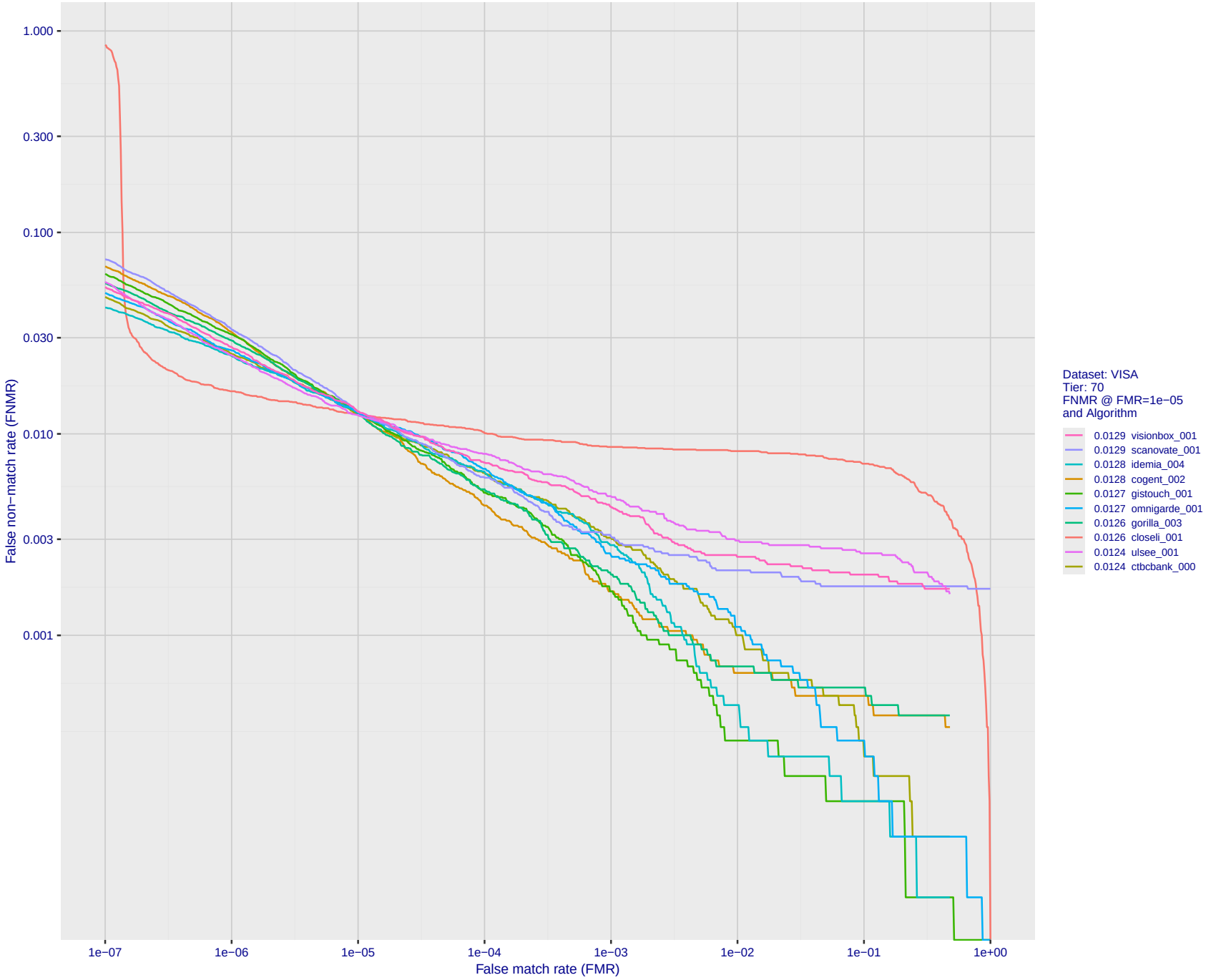


Figure 182: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

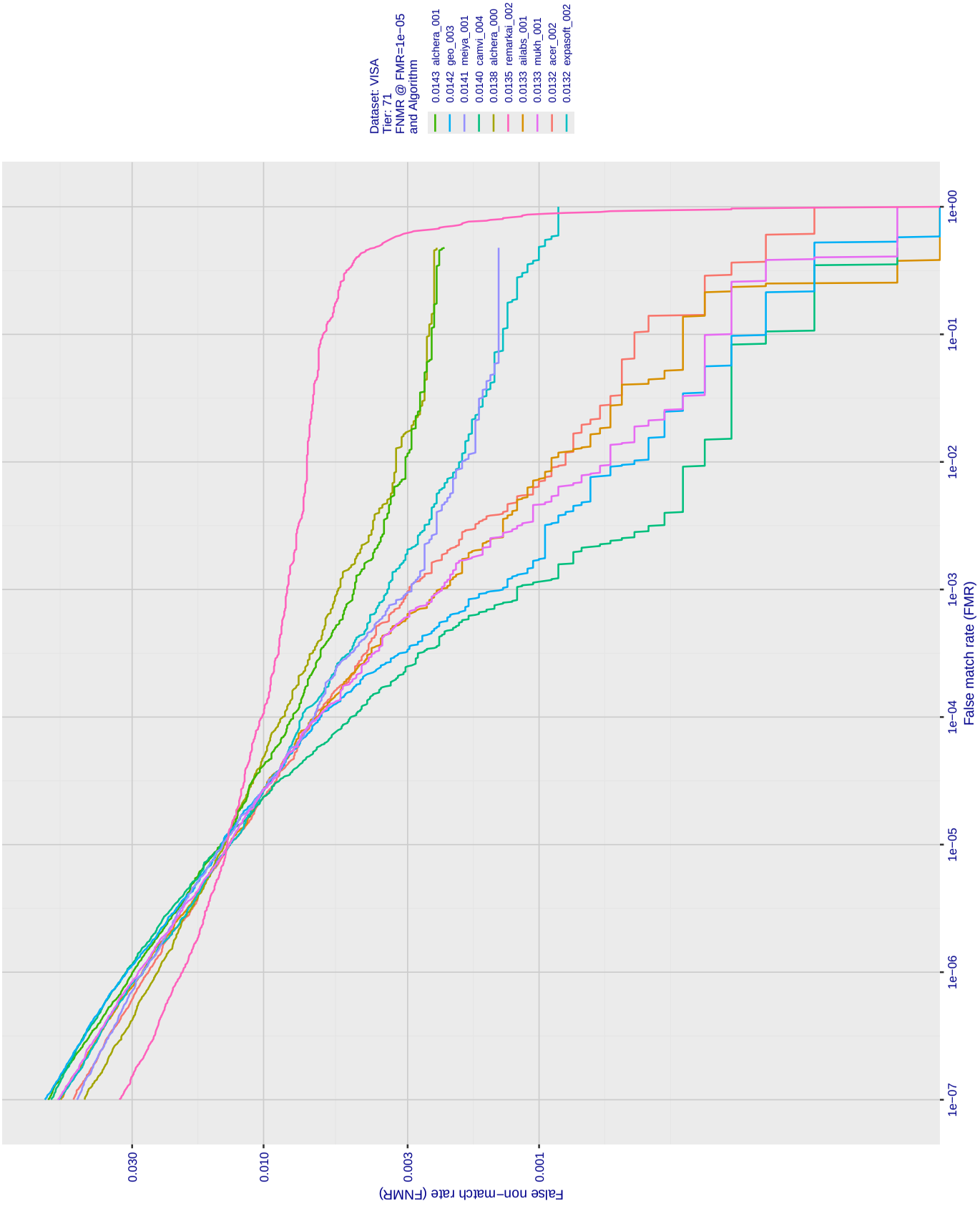


Figure 183: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

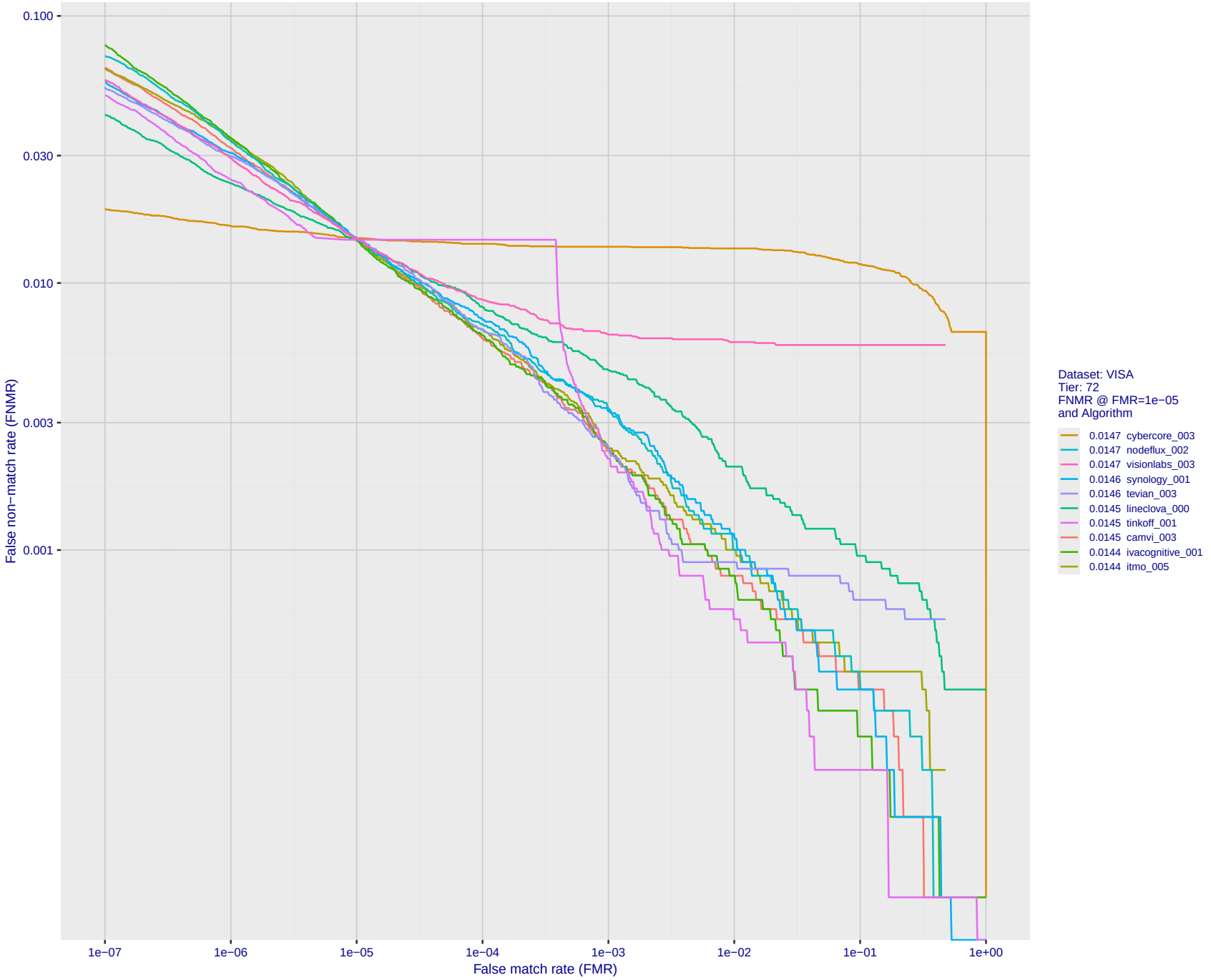


Figure 184: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

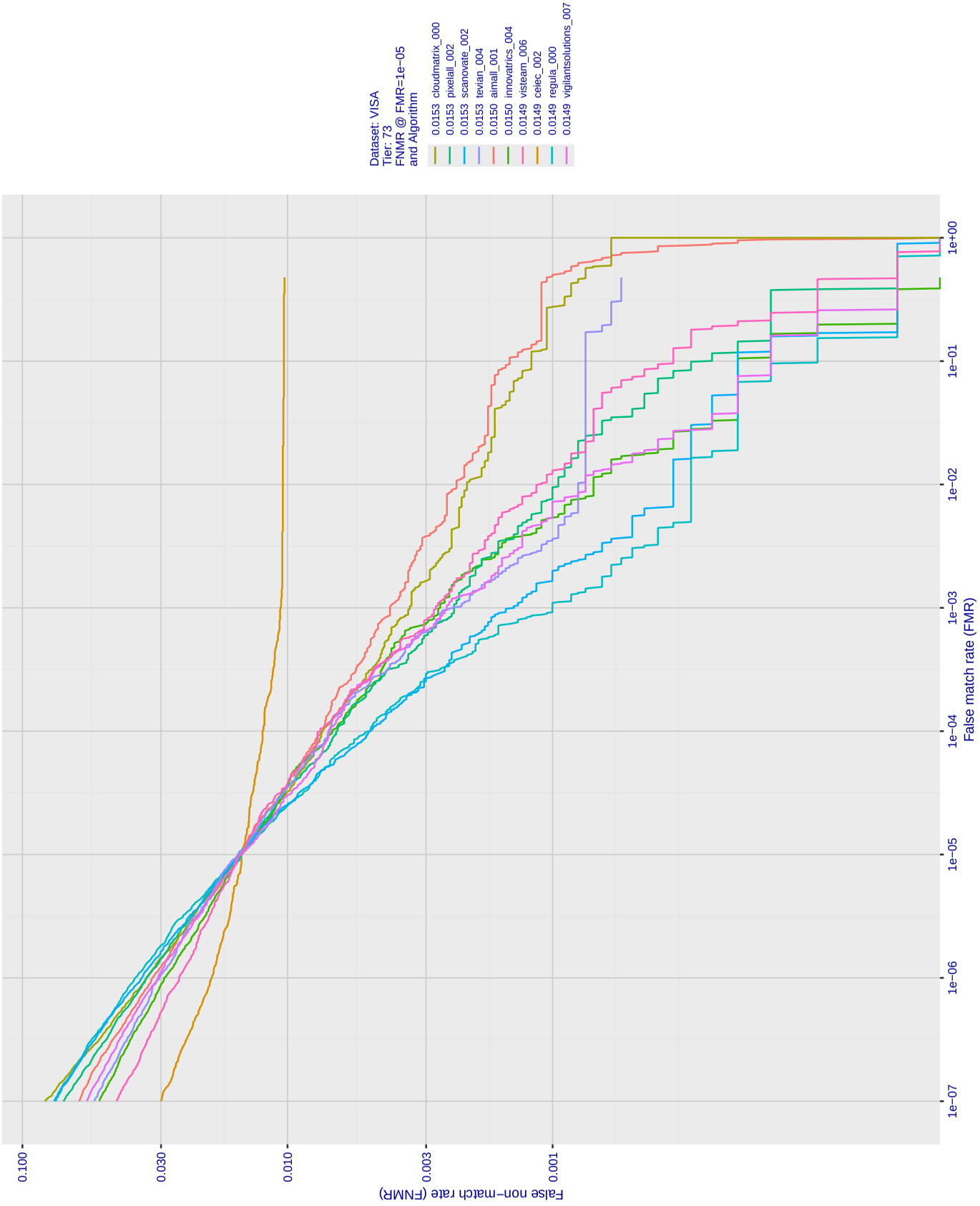


Figure 185: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

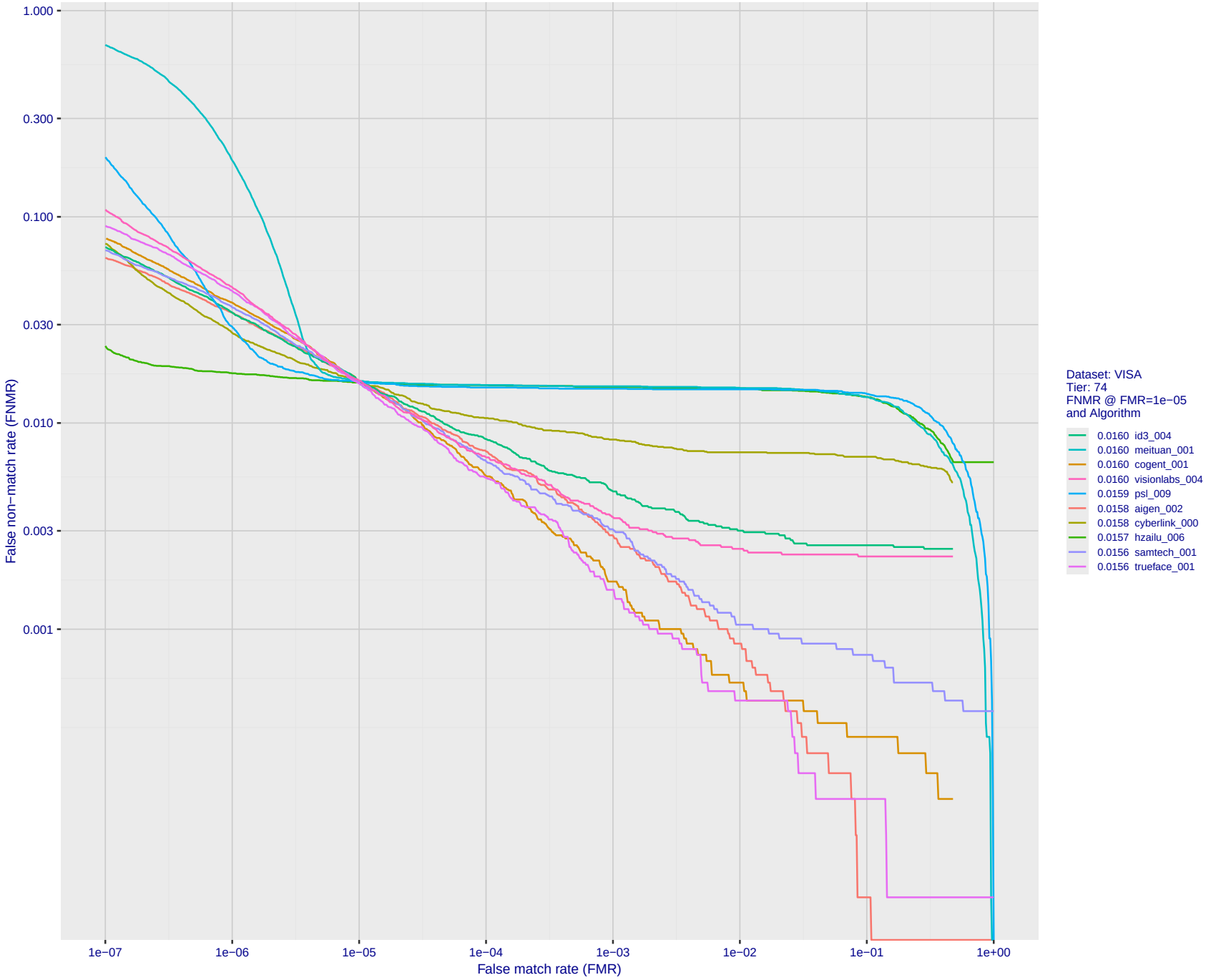


Figure 186: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

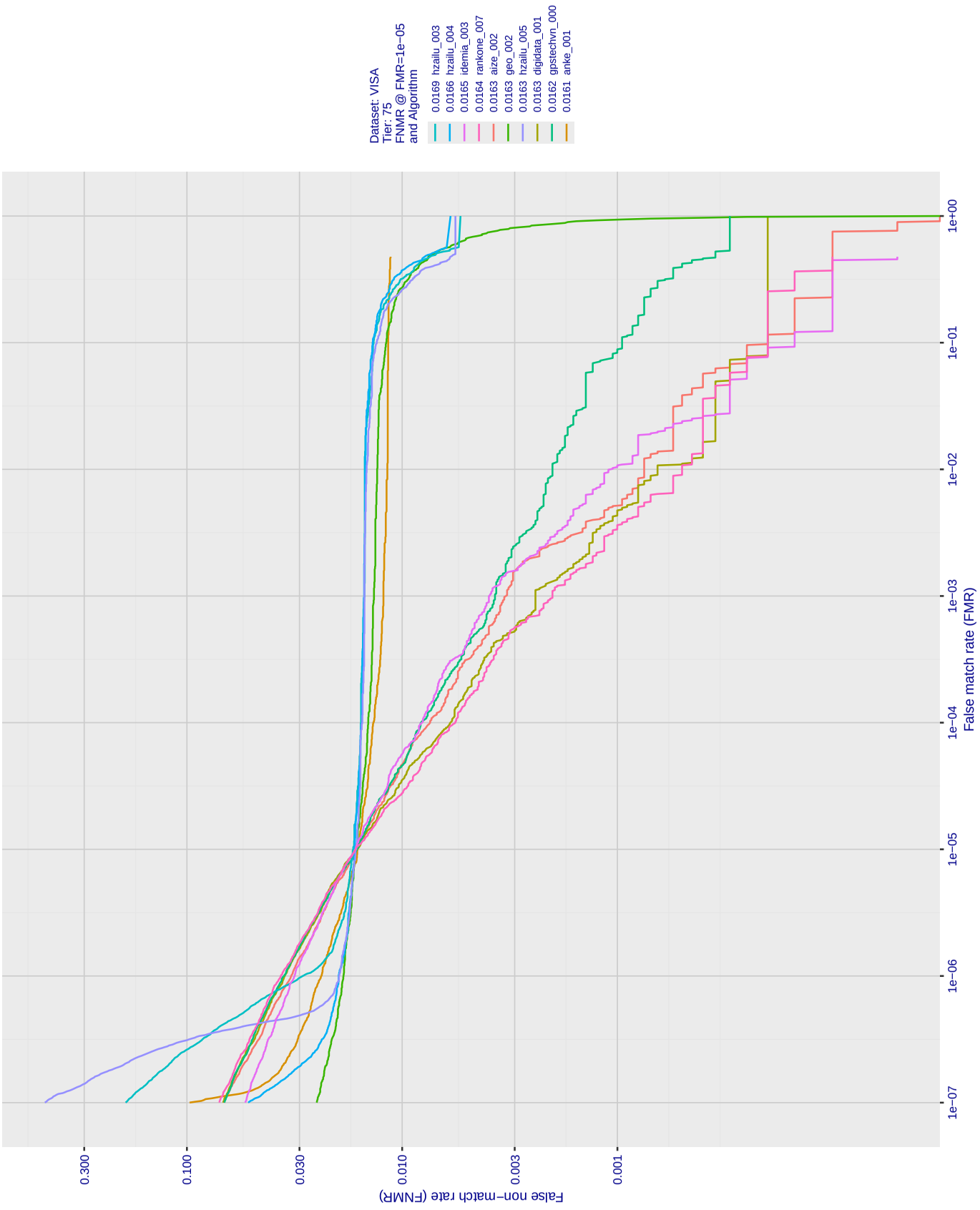


Figure 187: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

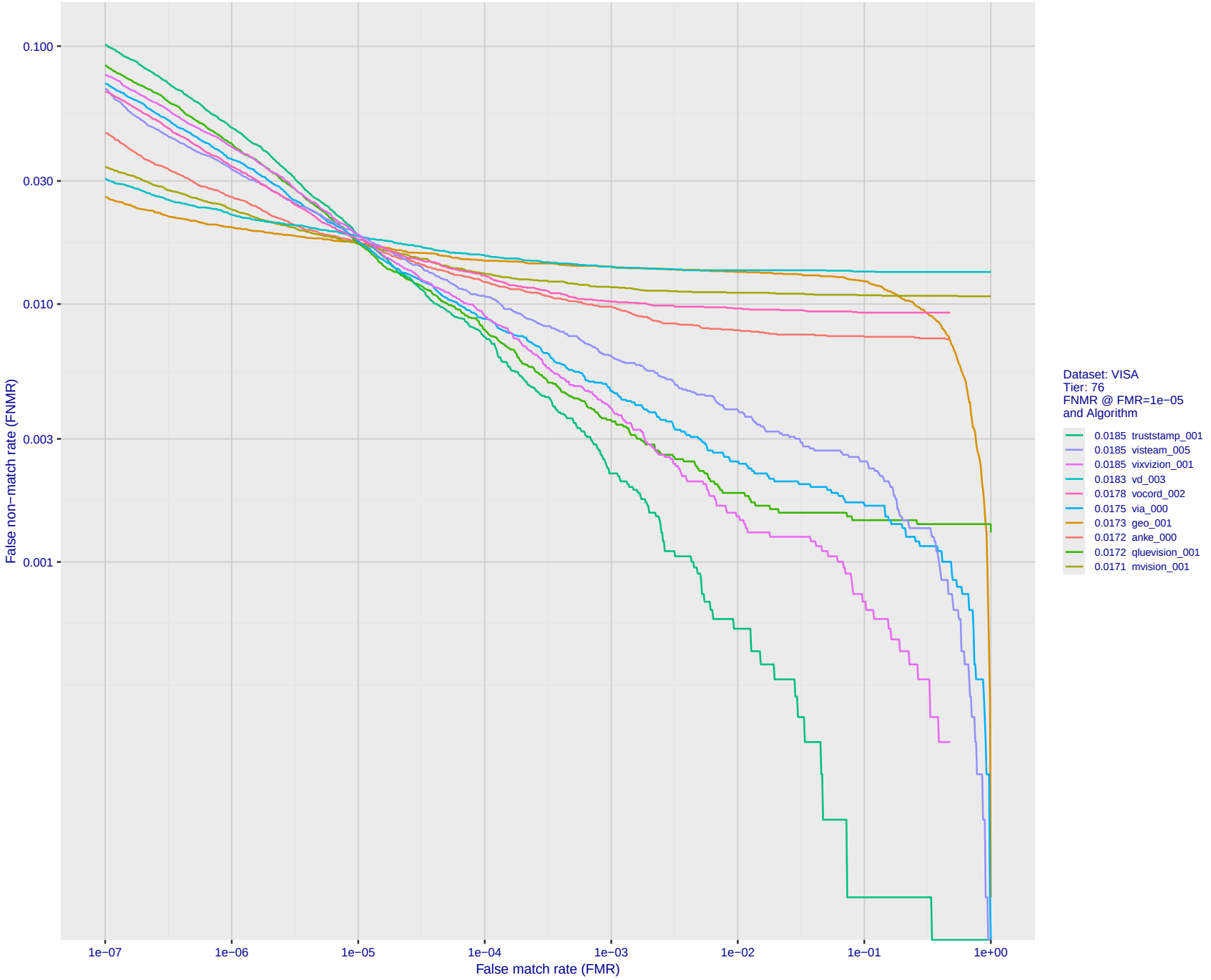


Figure 188: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

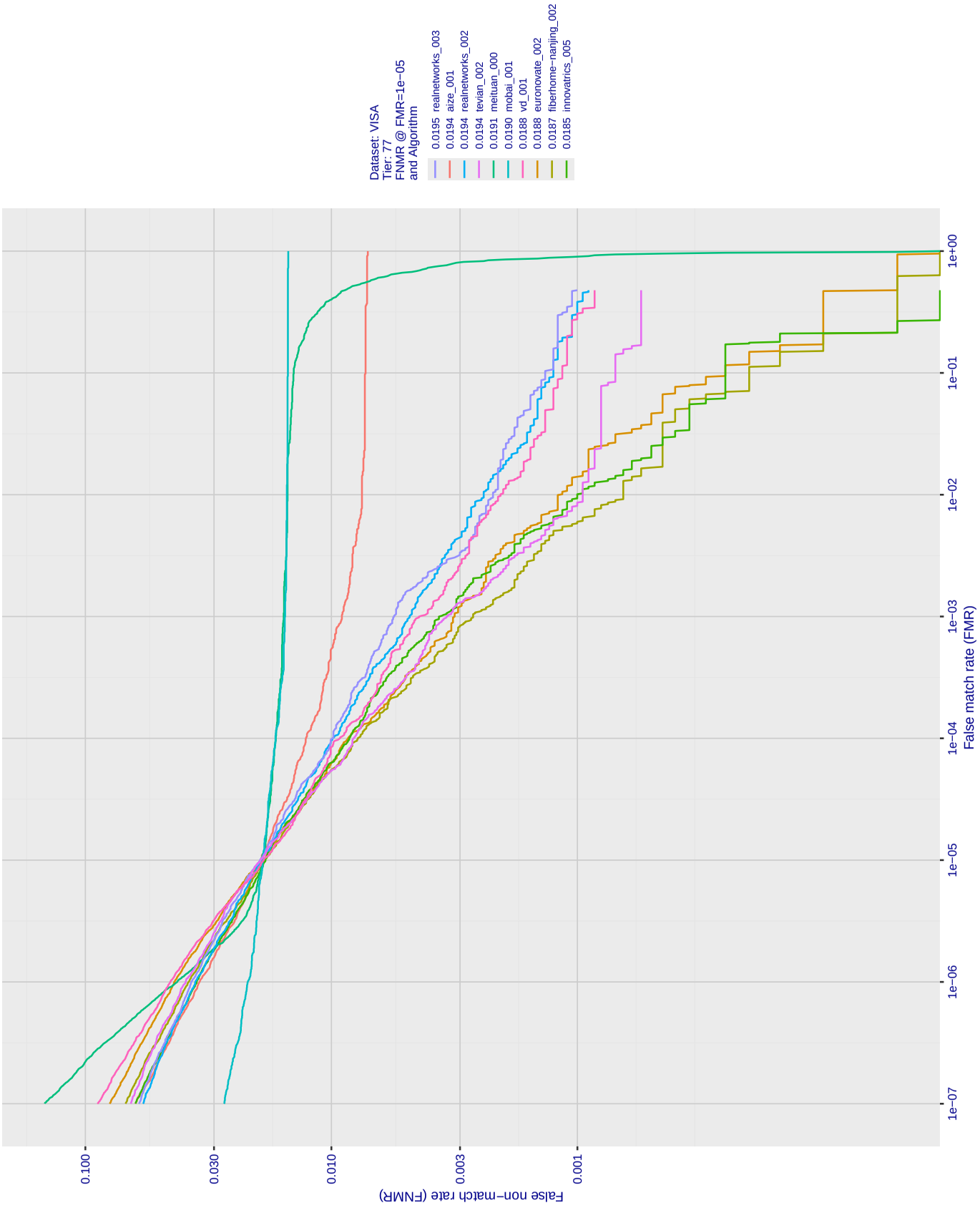

Figure 189: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

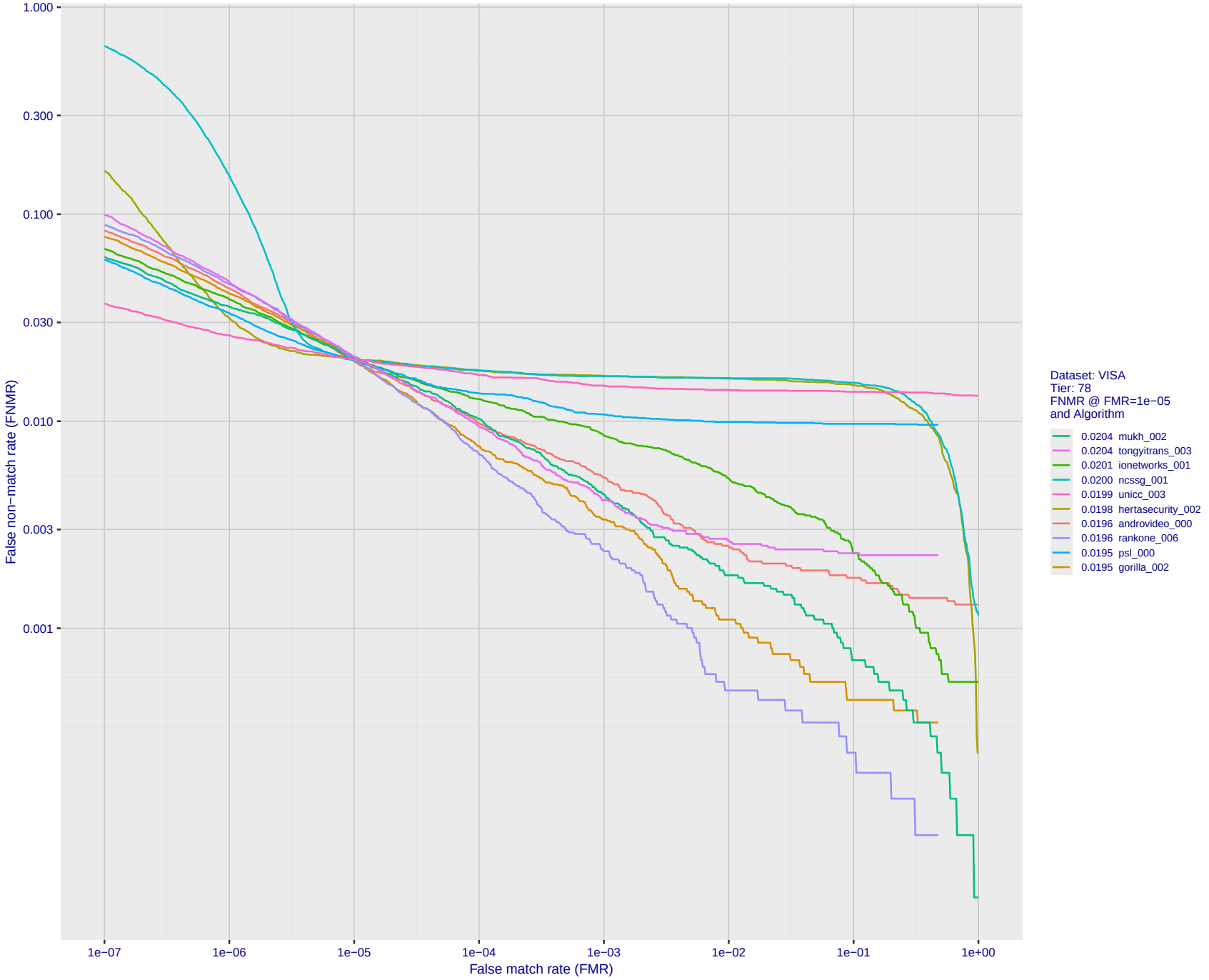

Figure 190: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

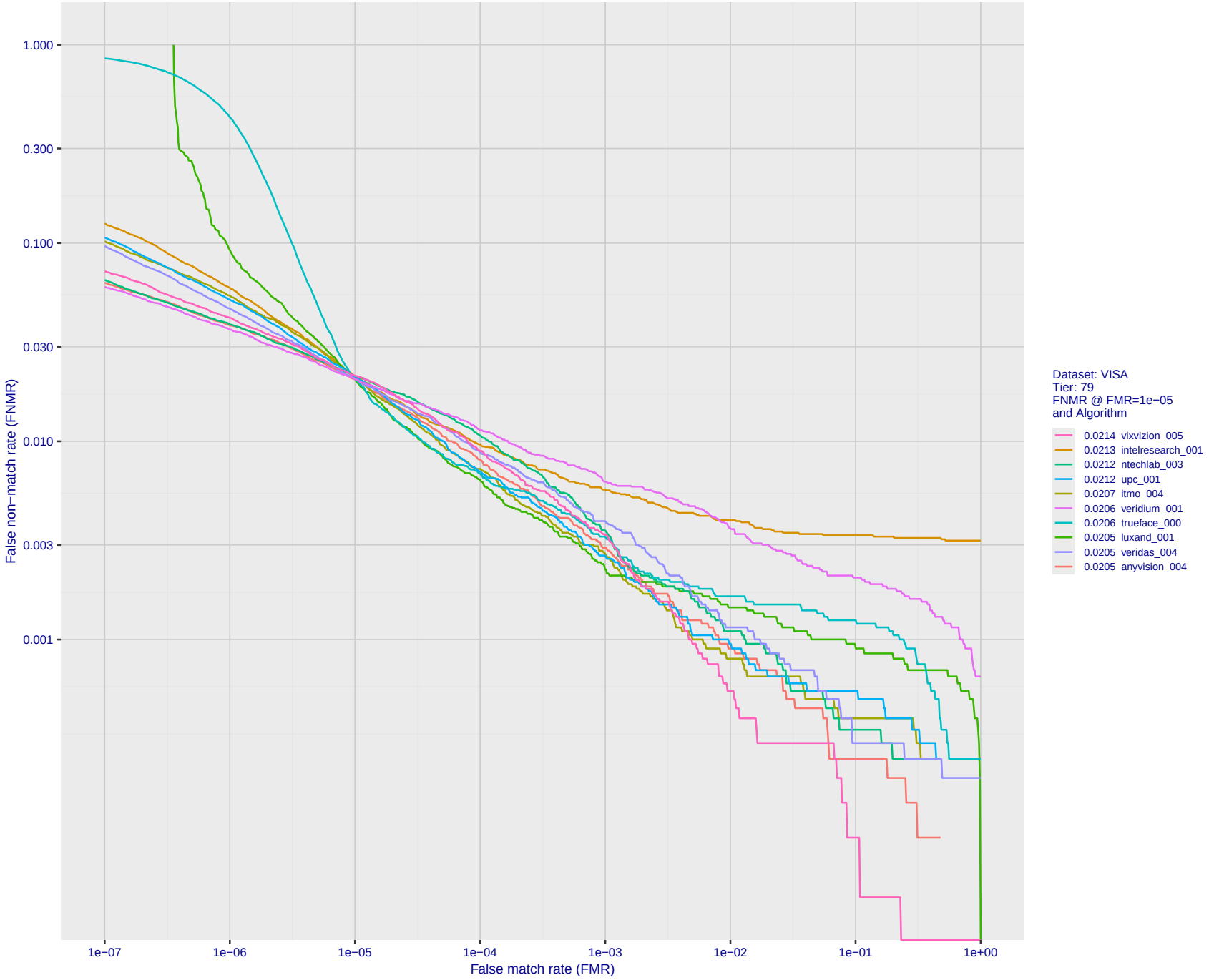

Figure 191: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

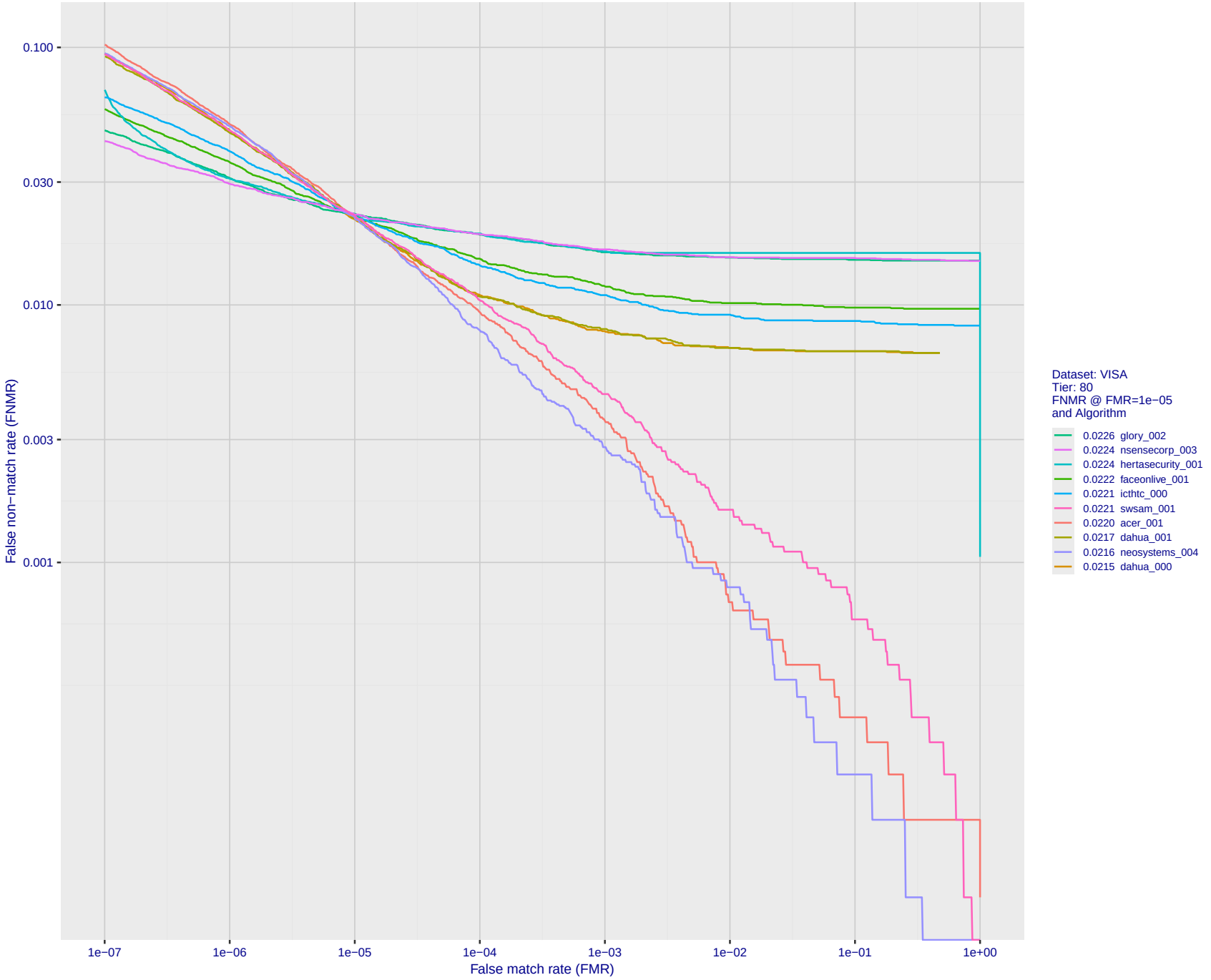

Figure 192: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

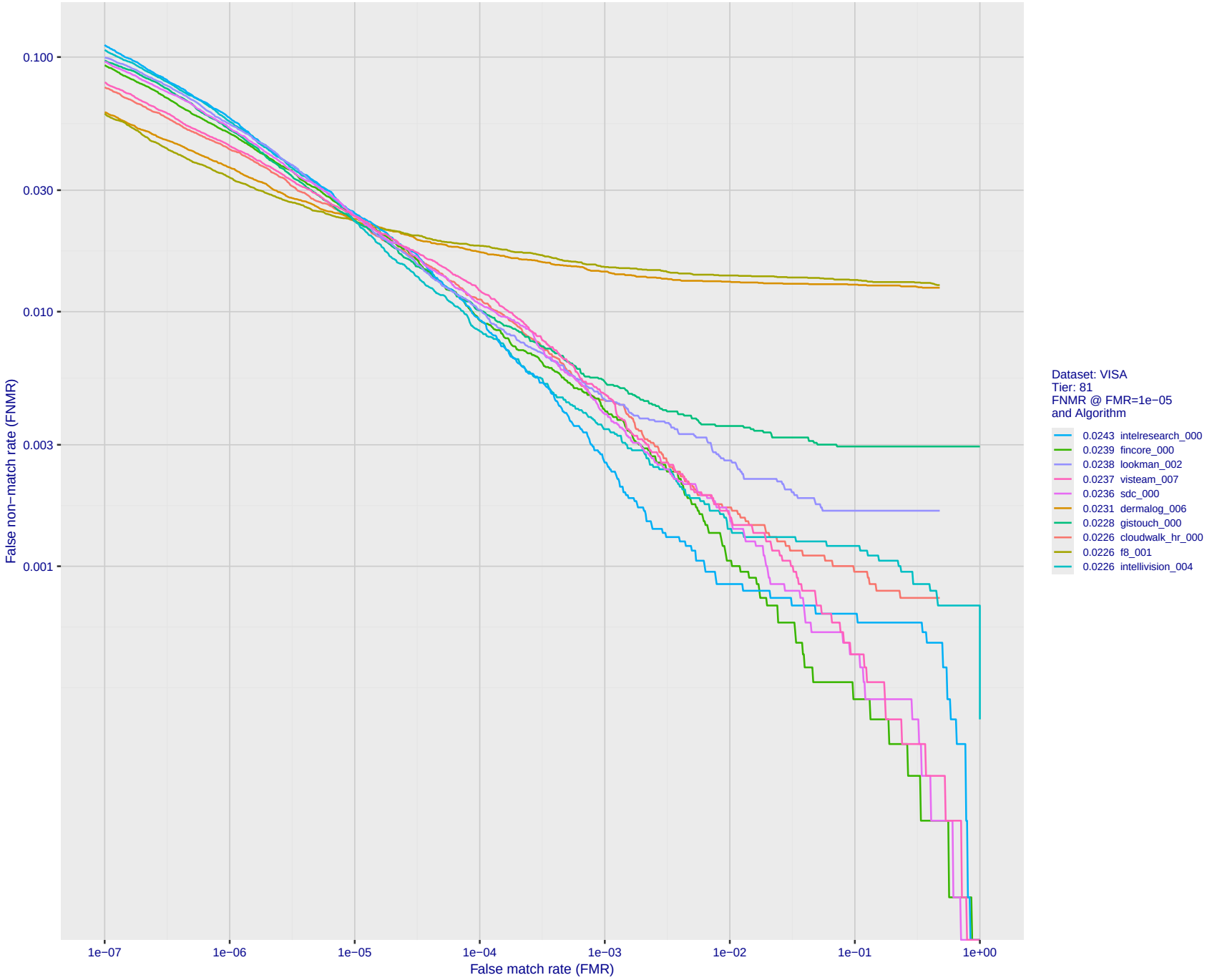

Figure 193: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

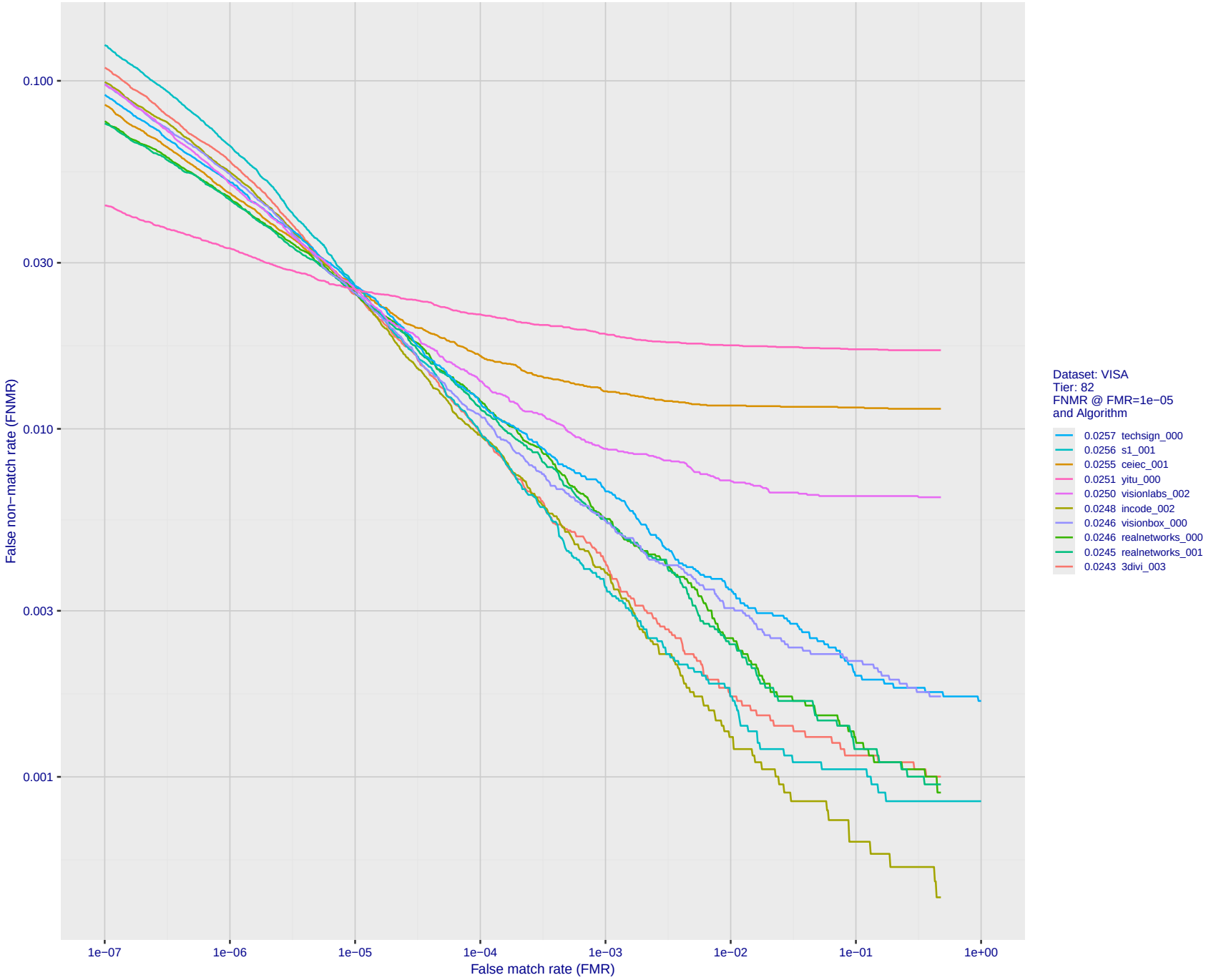

Figure 194: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

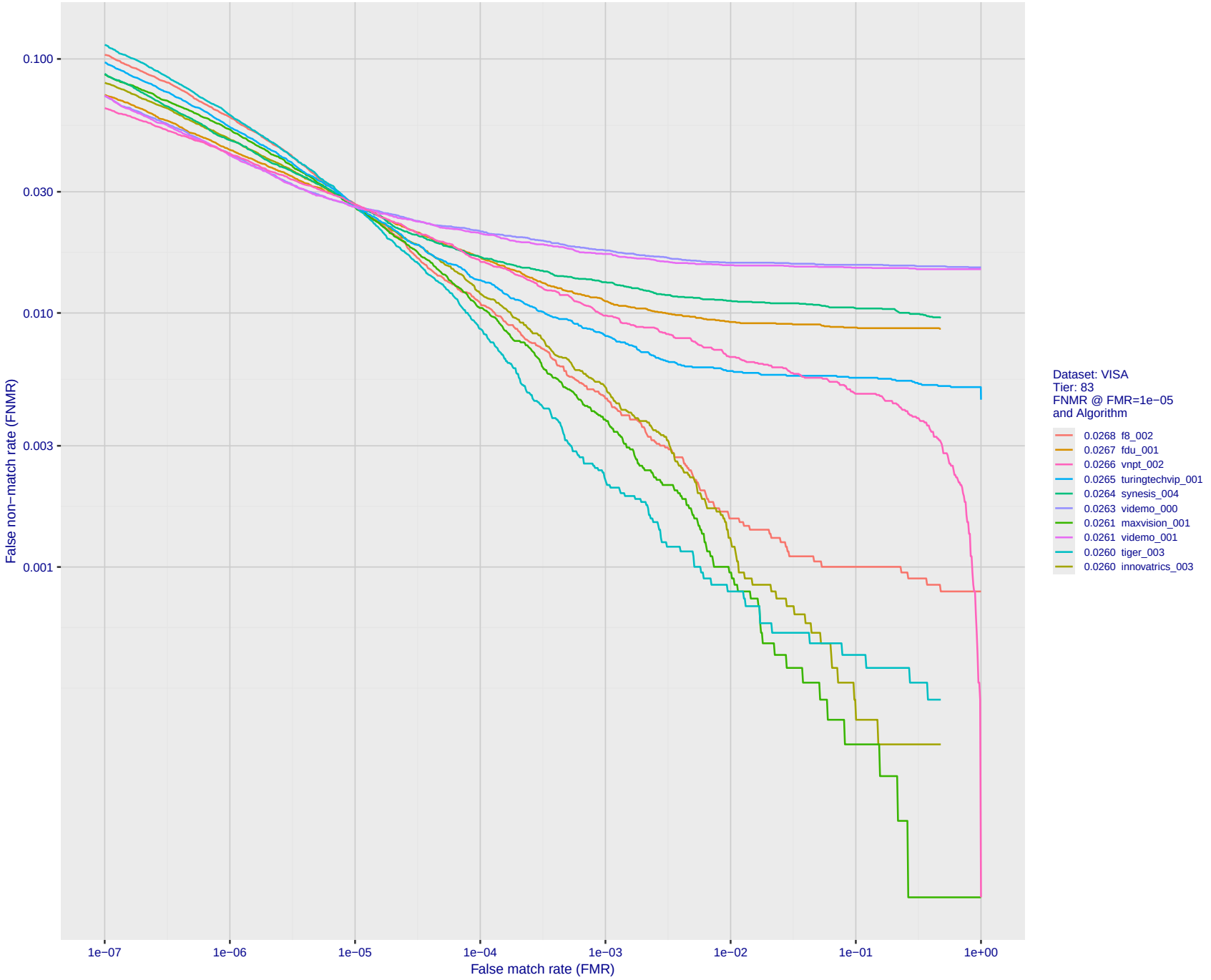

Figure 195: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

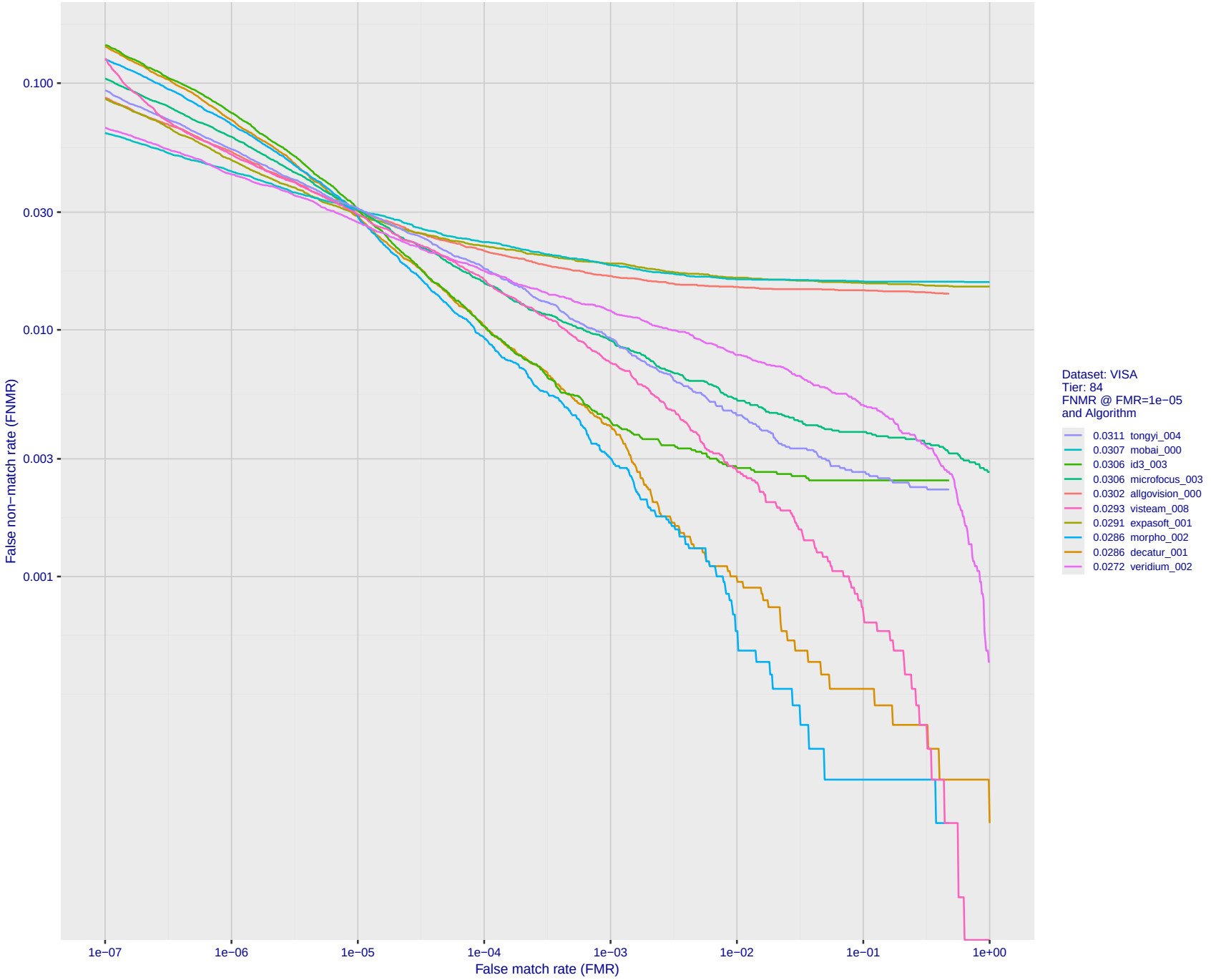

Figure 196: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

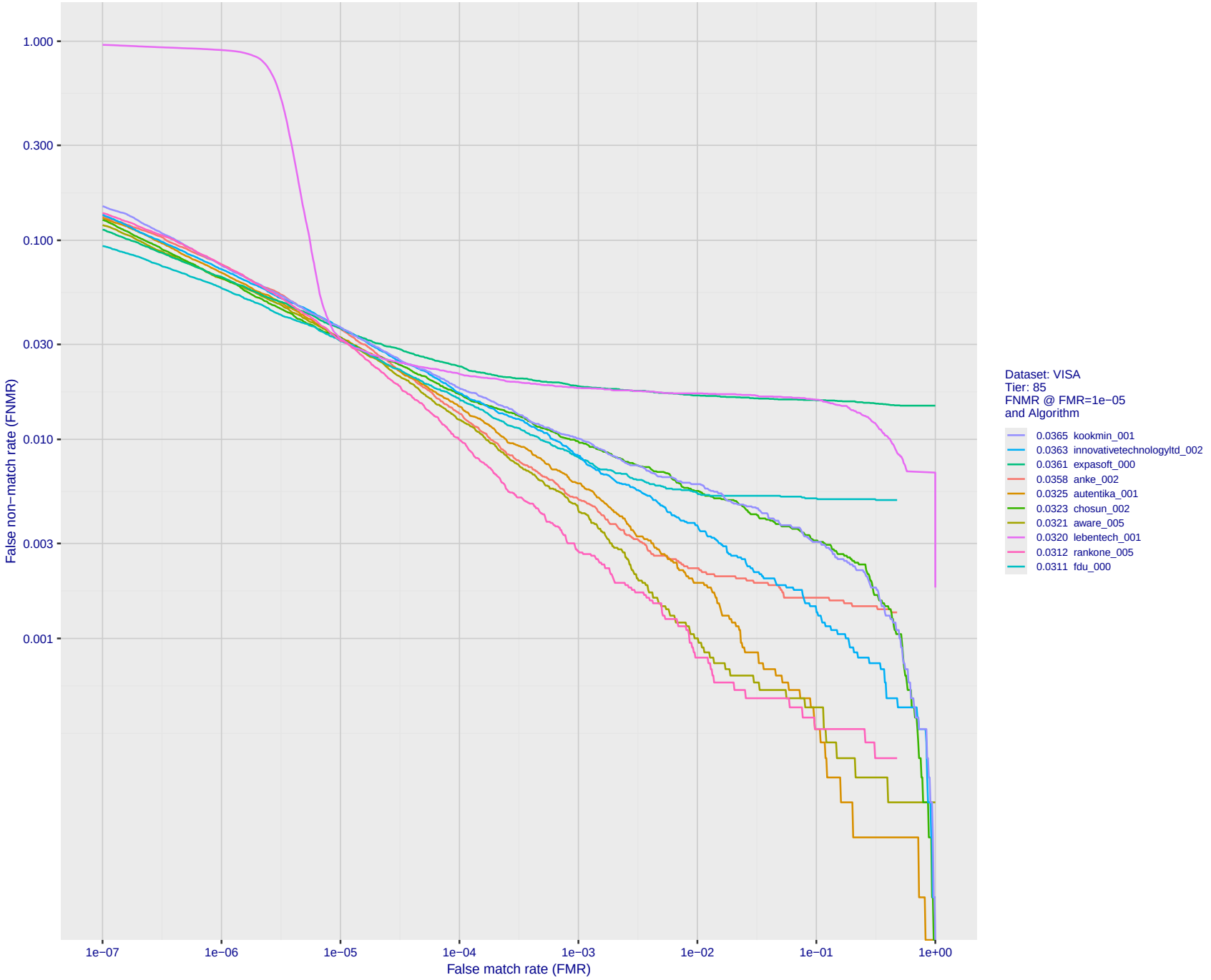

Figure 197: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

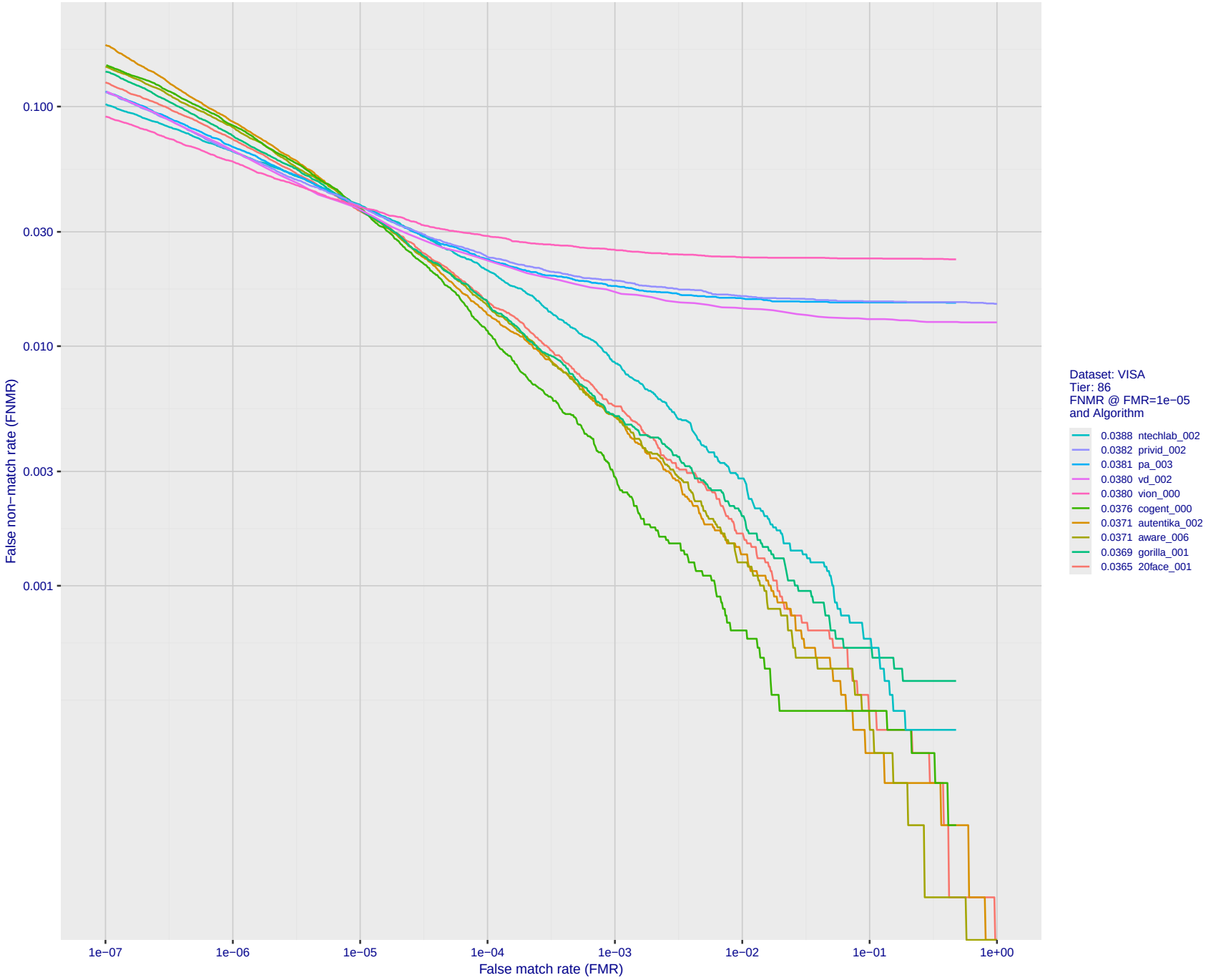

Figure 198: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

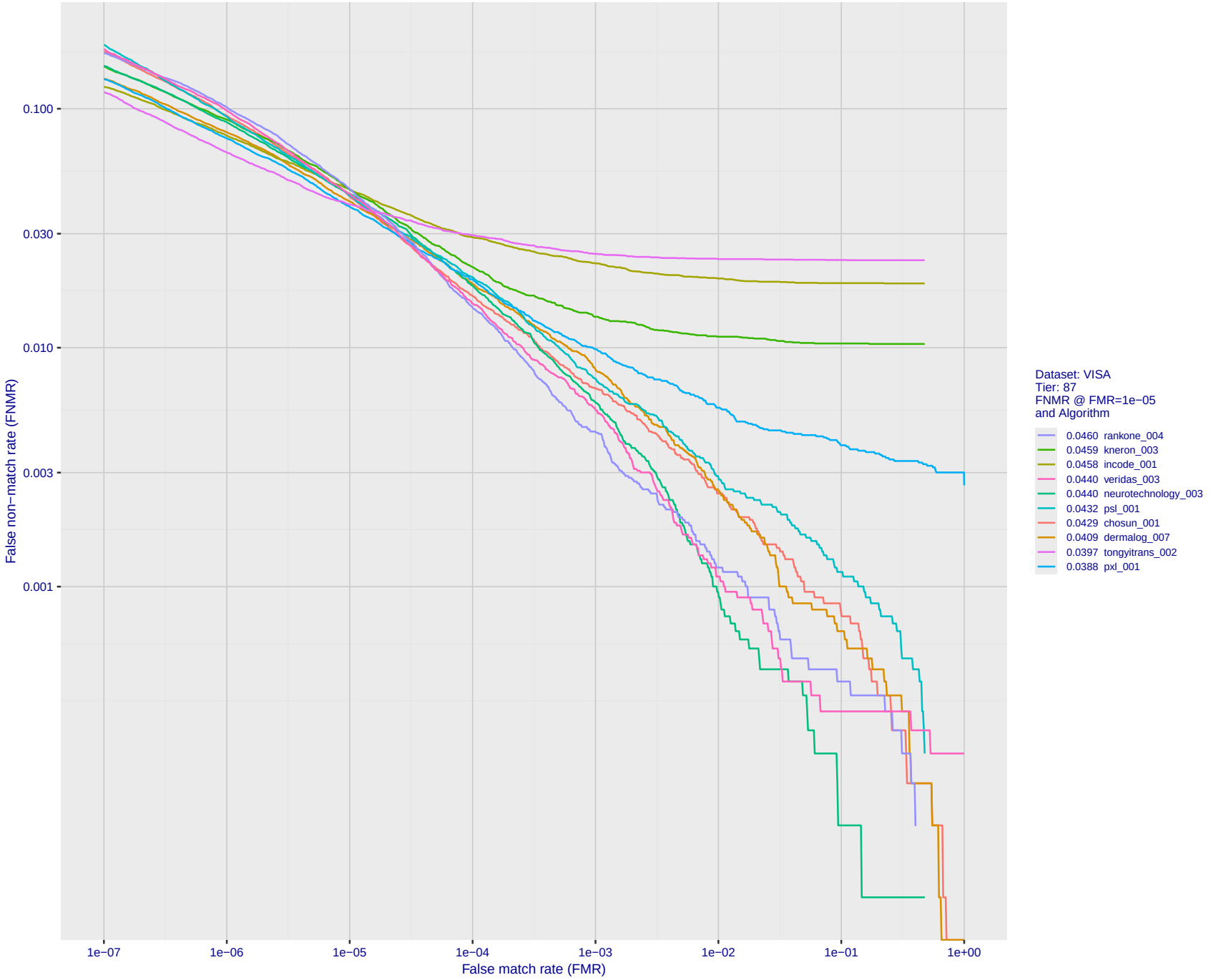

Figure 199: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

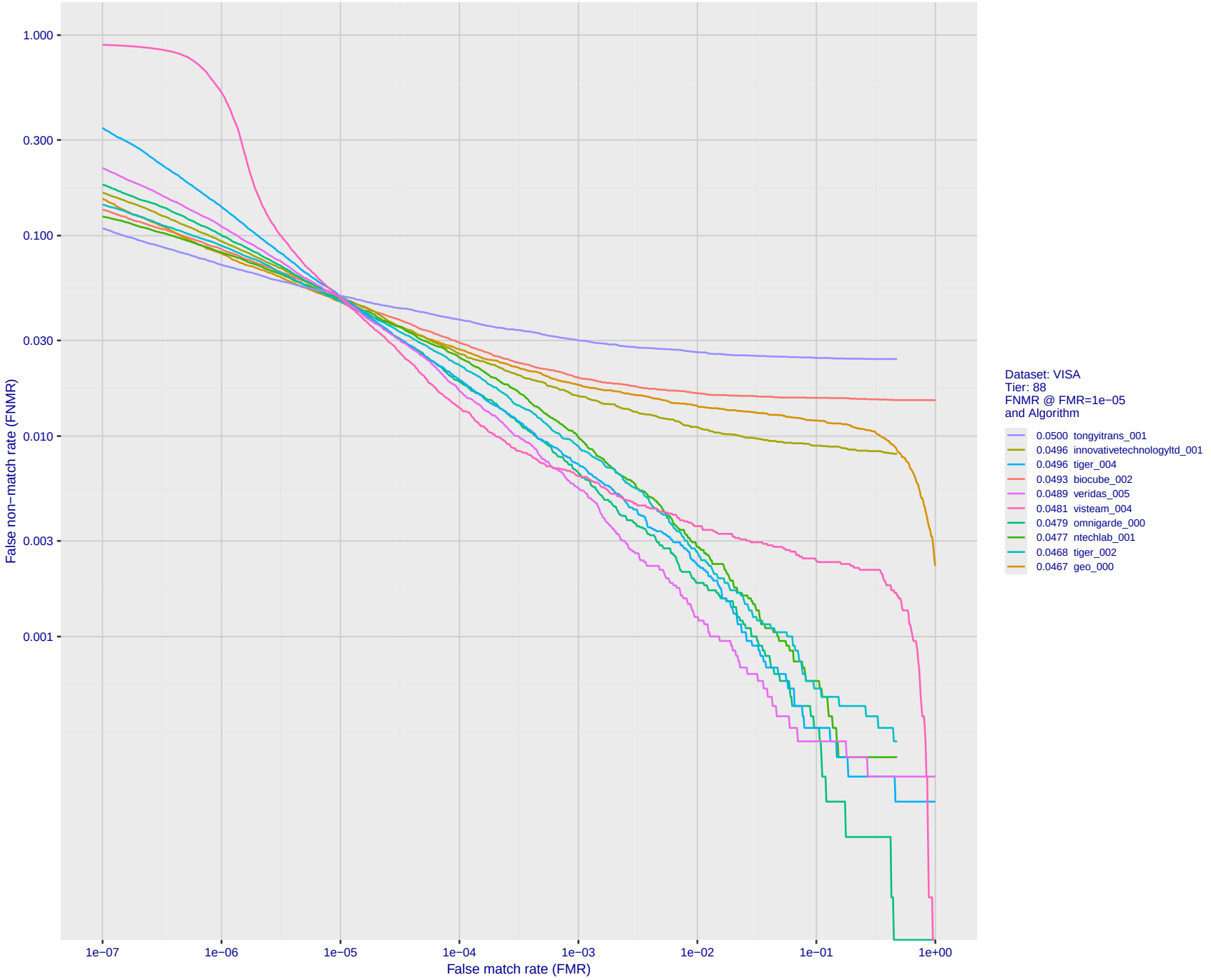

Figure 200: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

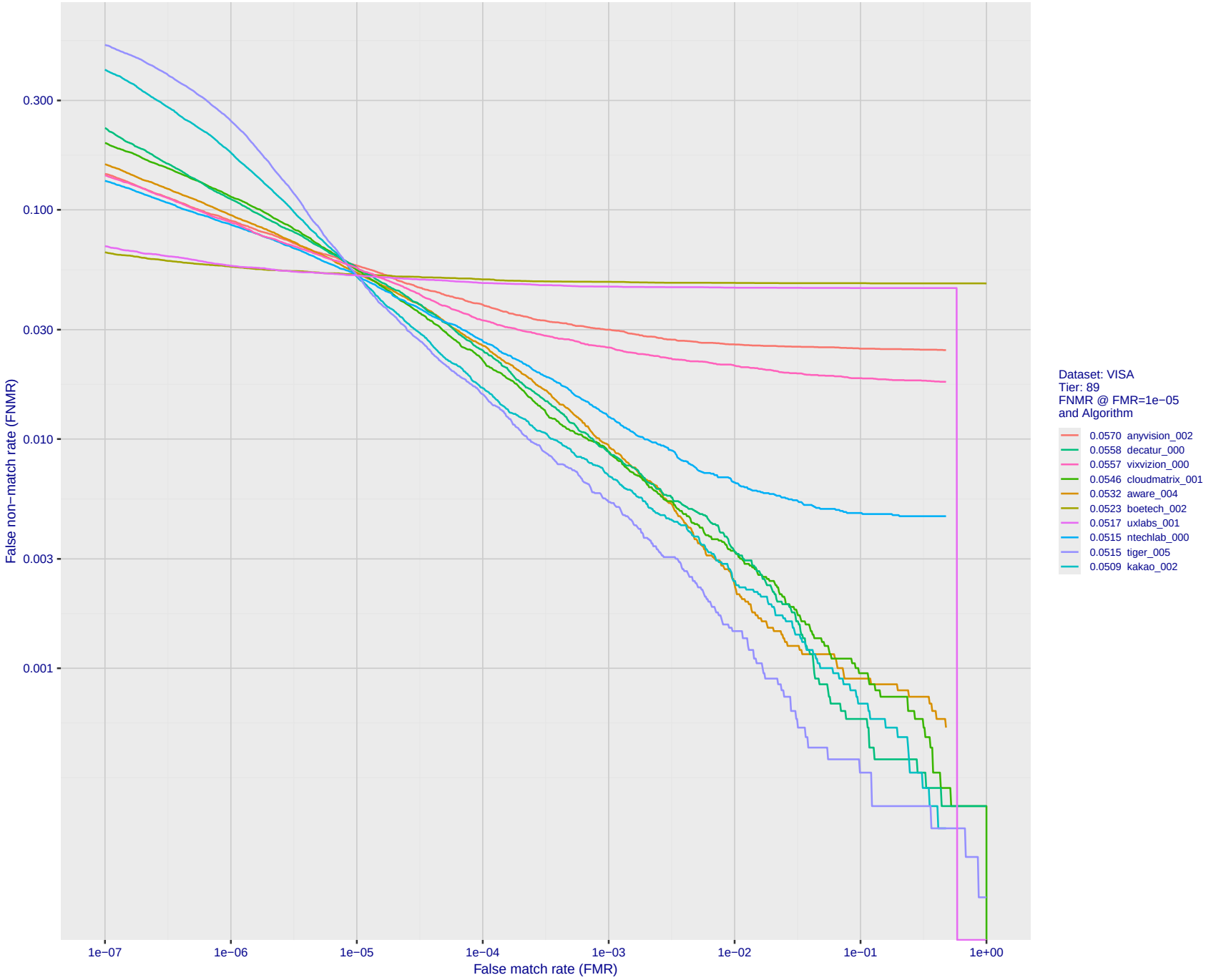

Figure 201: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

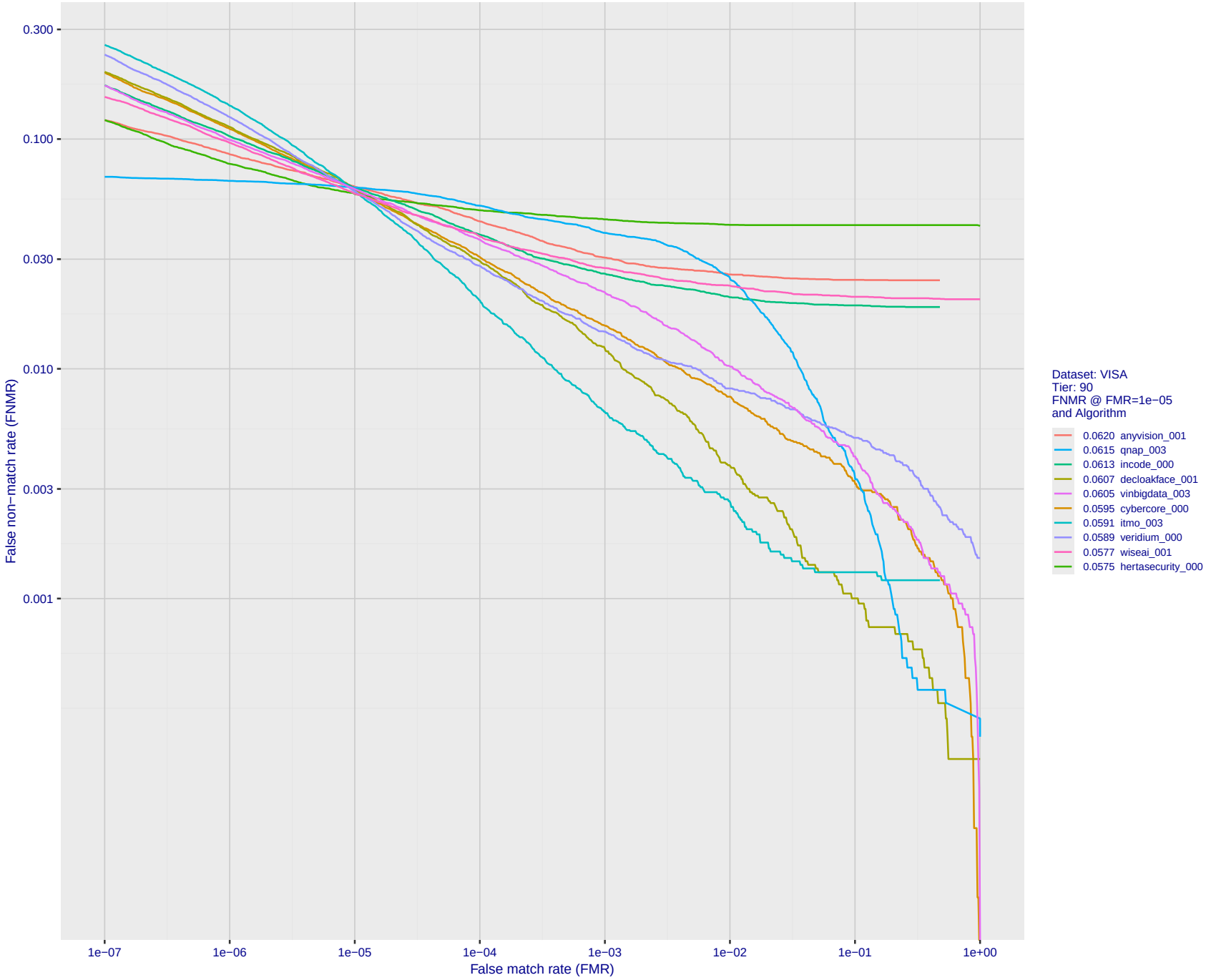

Figure 202: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

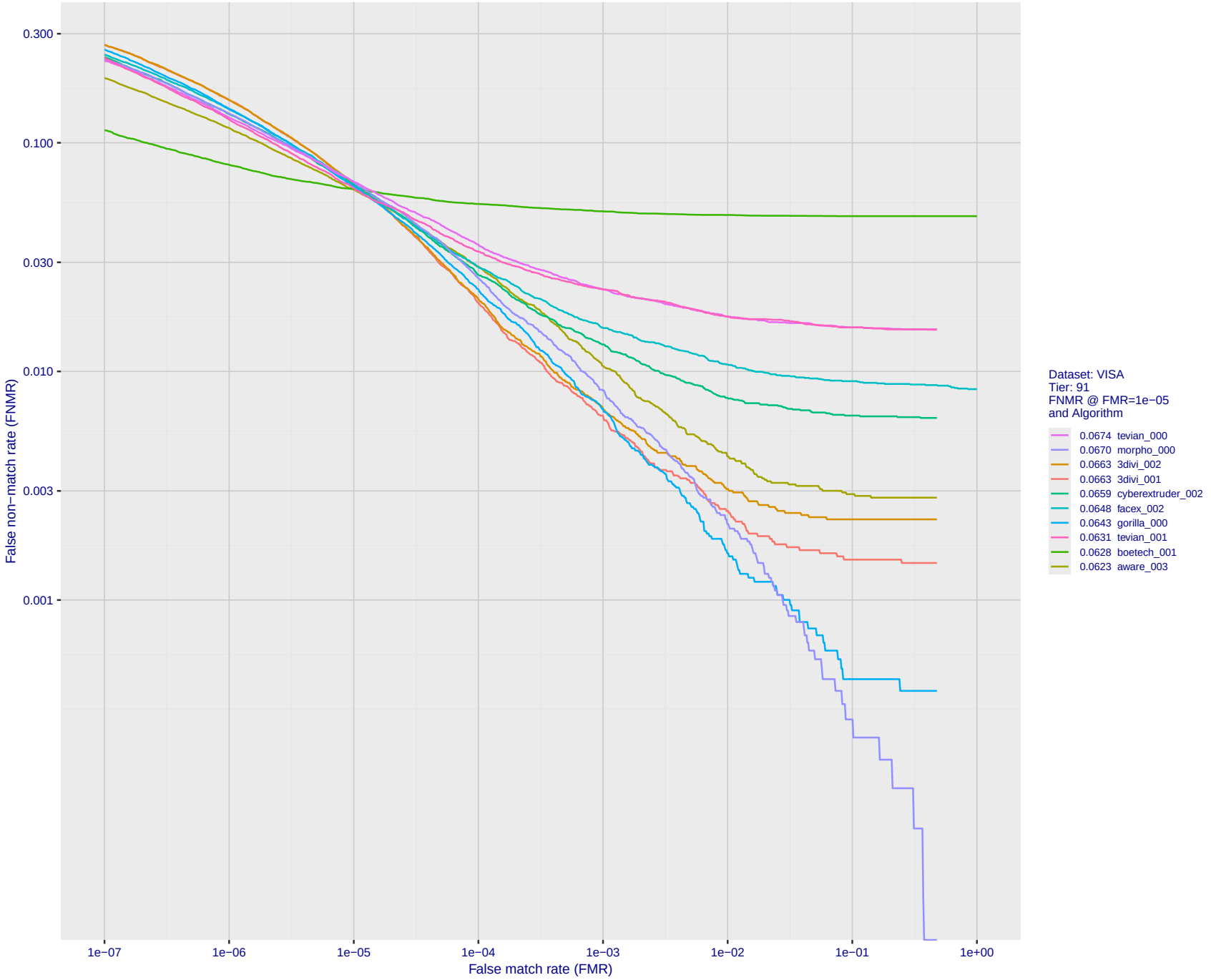

Figure 203: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

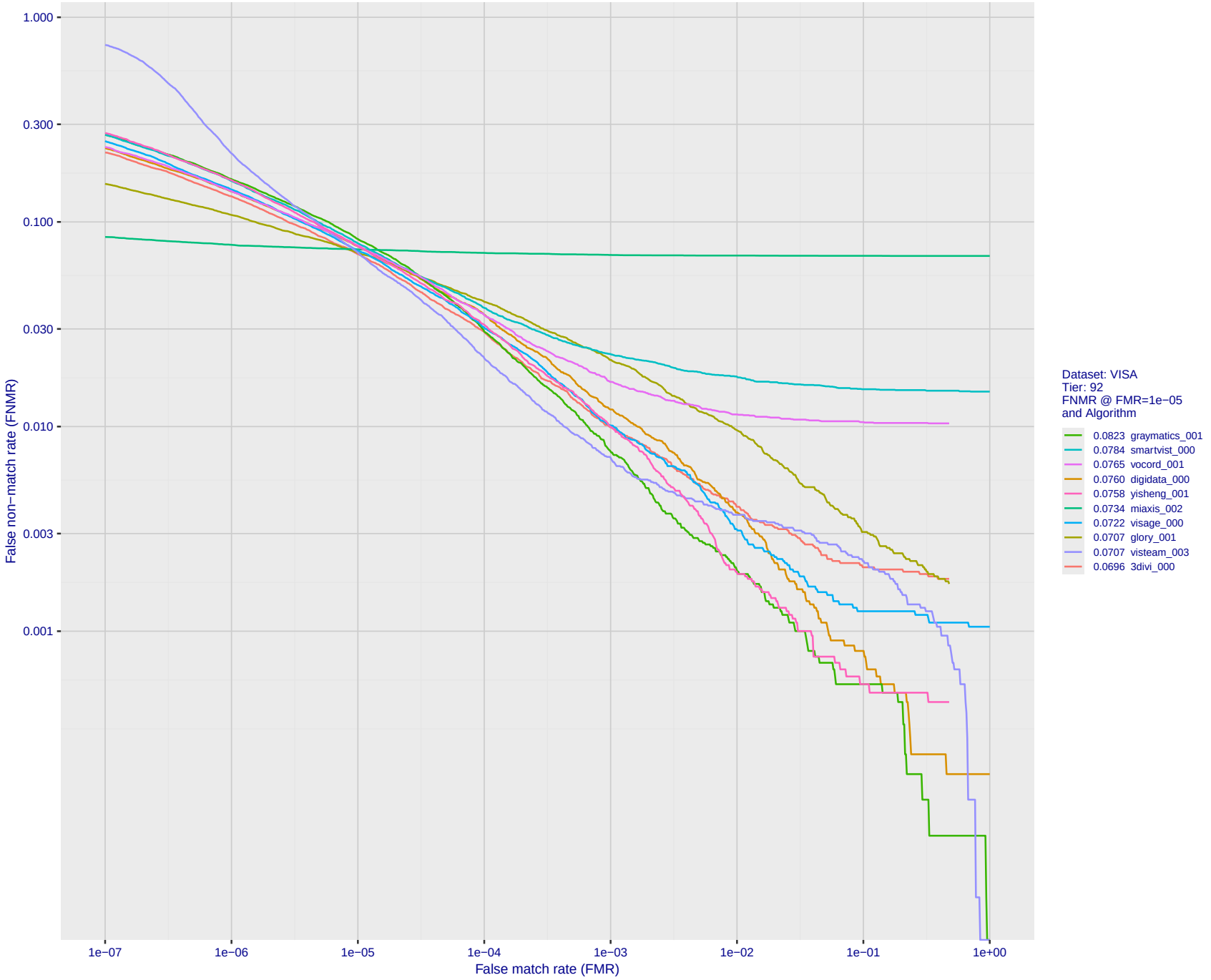

Figure 204: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

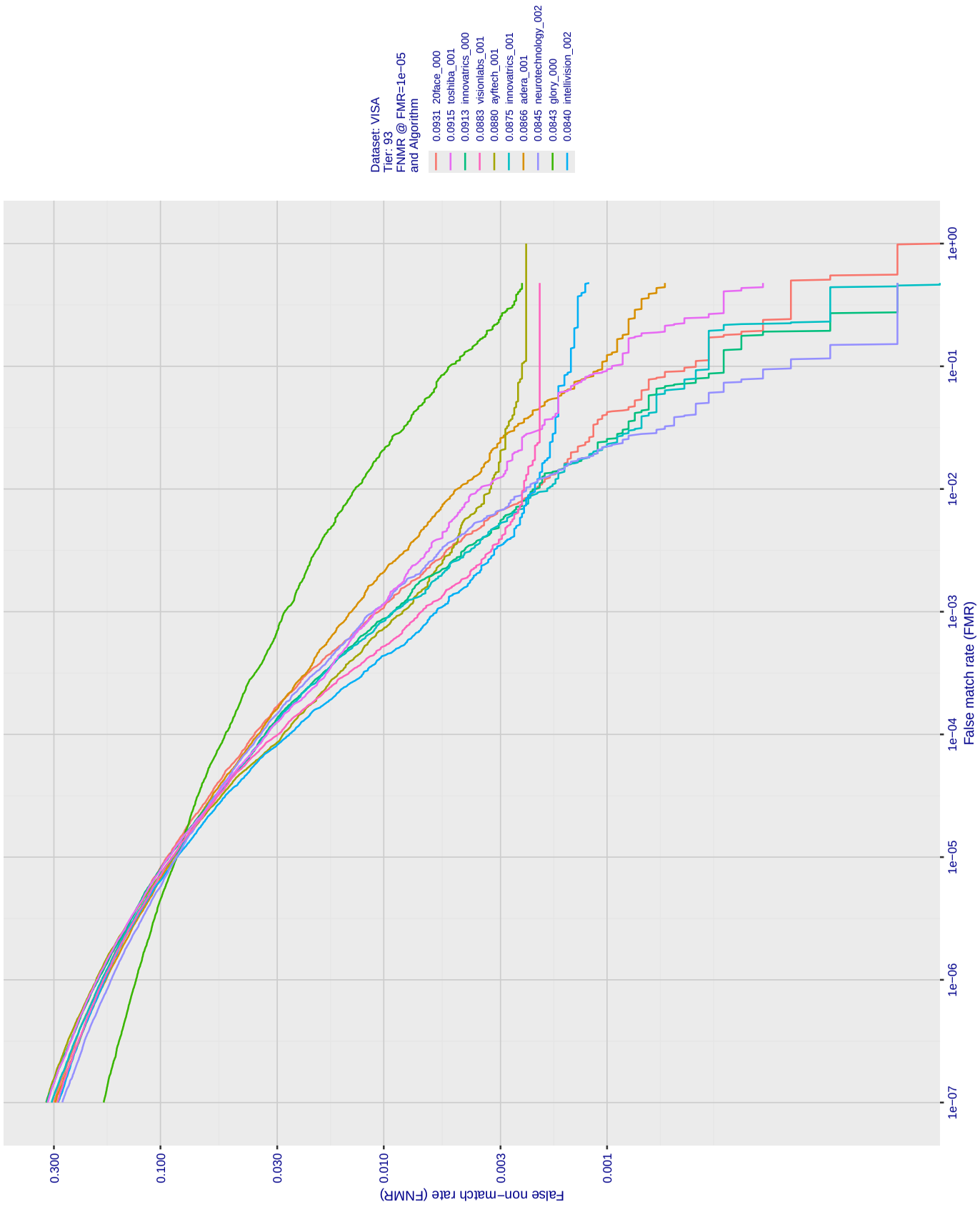

Figure 205: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

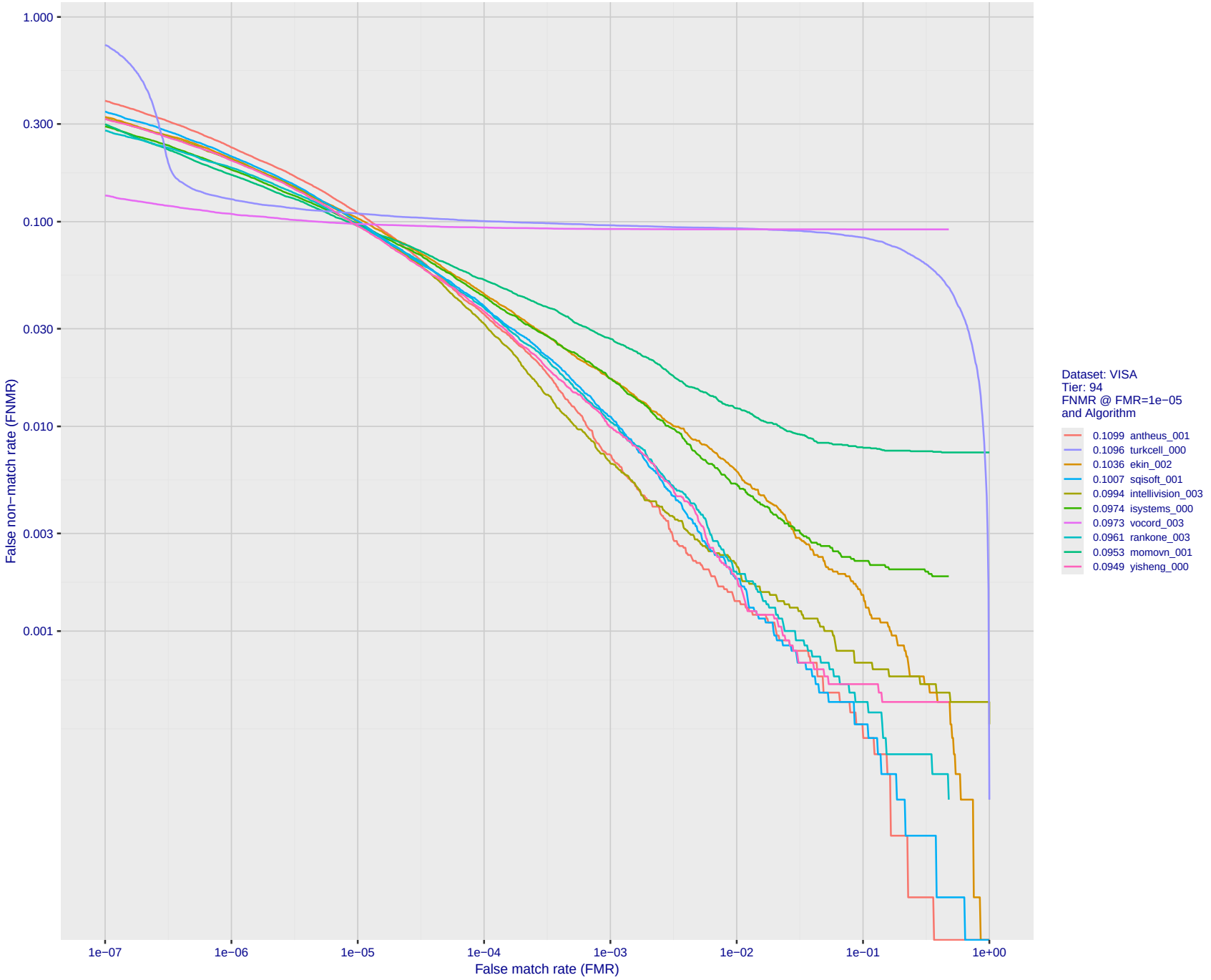

Figure 206: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

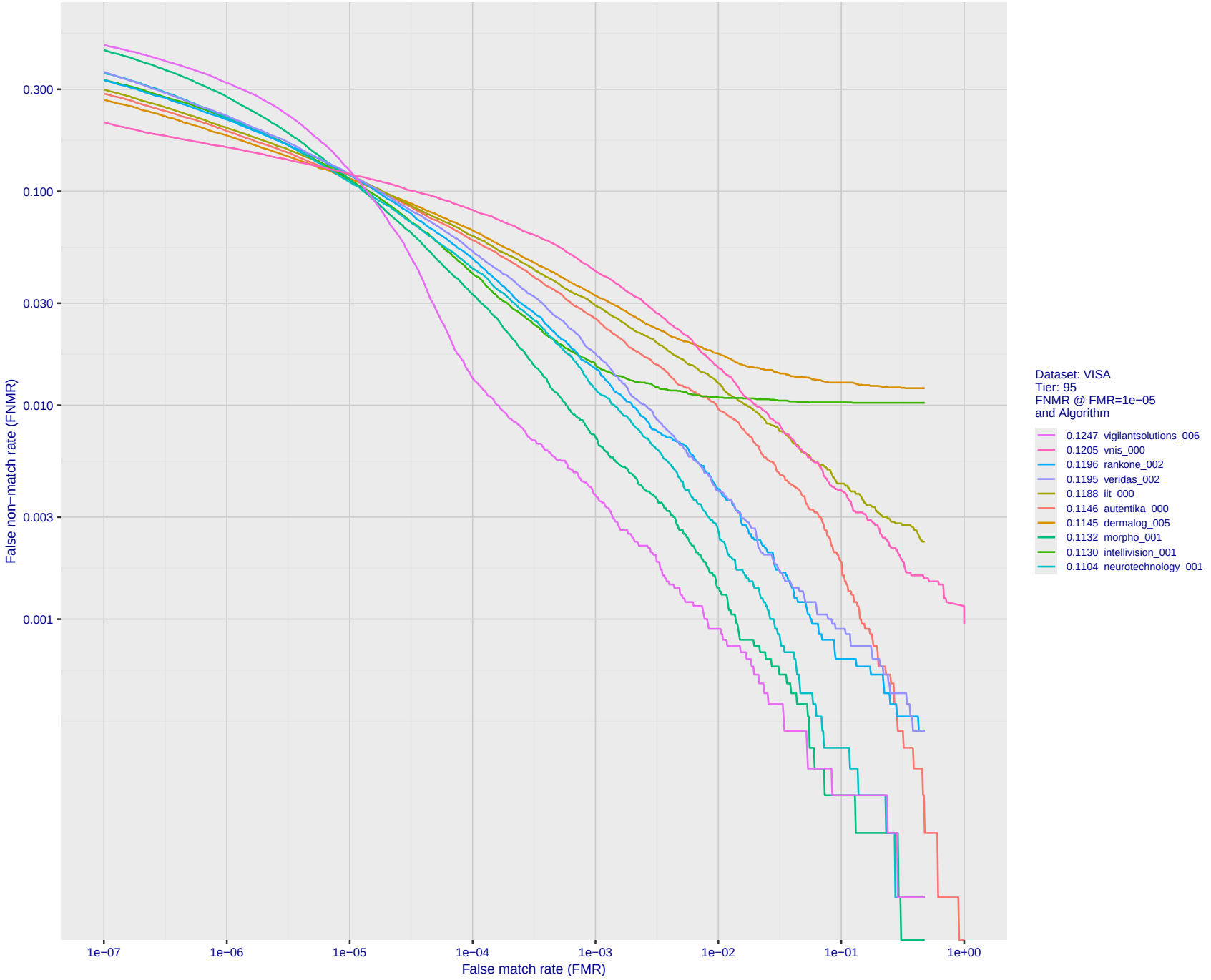

Figure 207: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

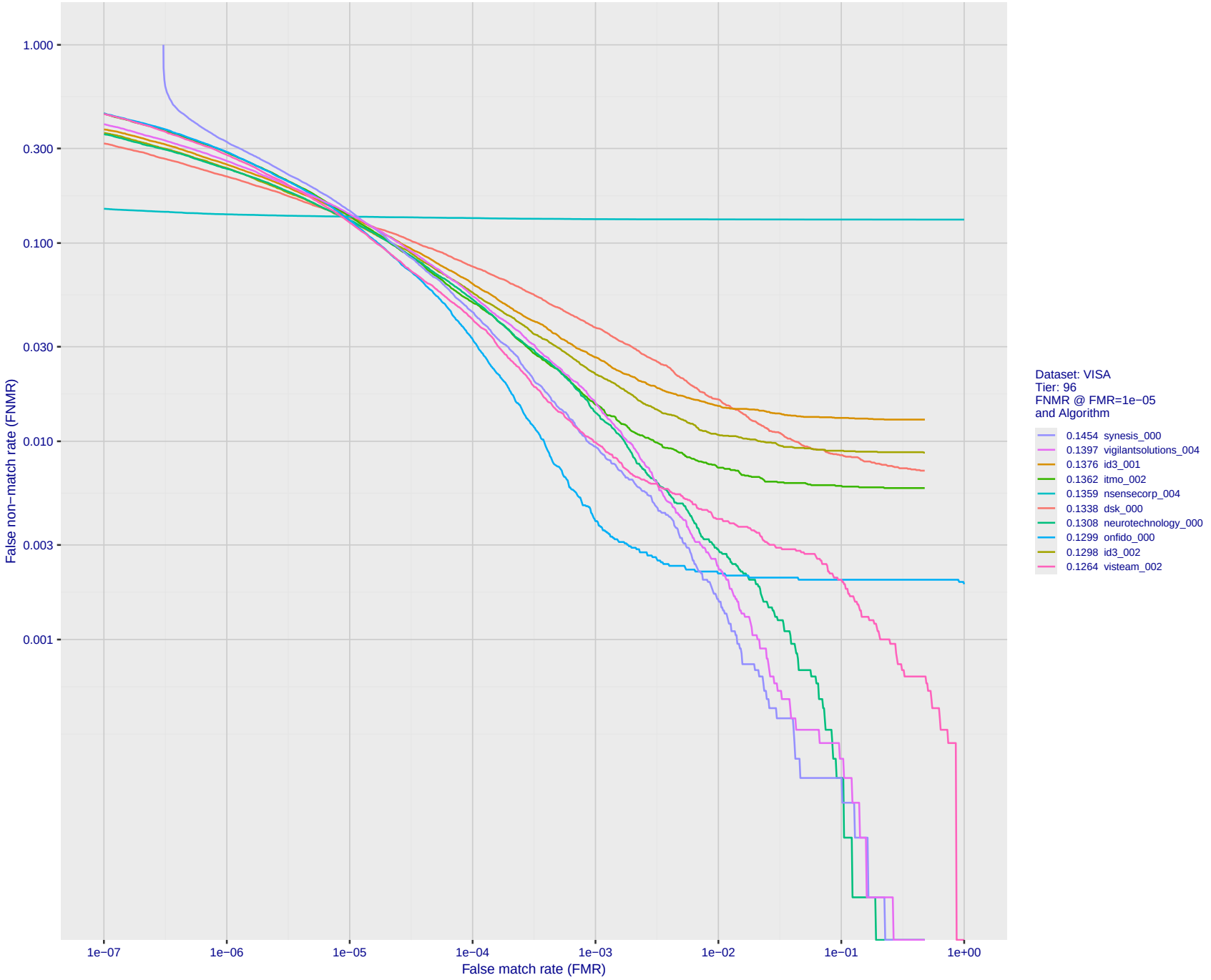

Figure 208: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

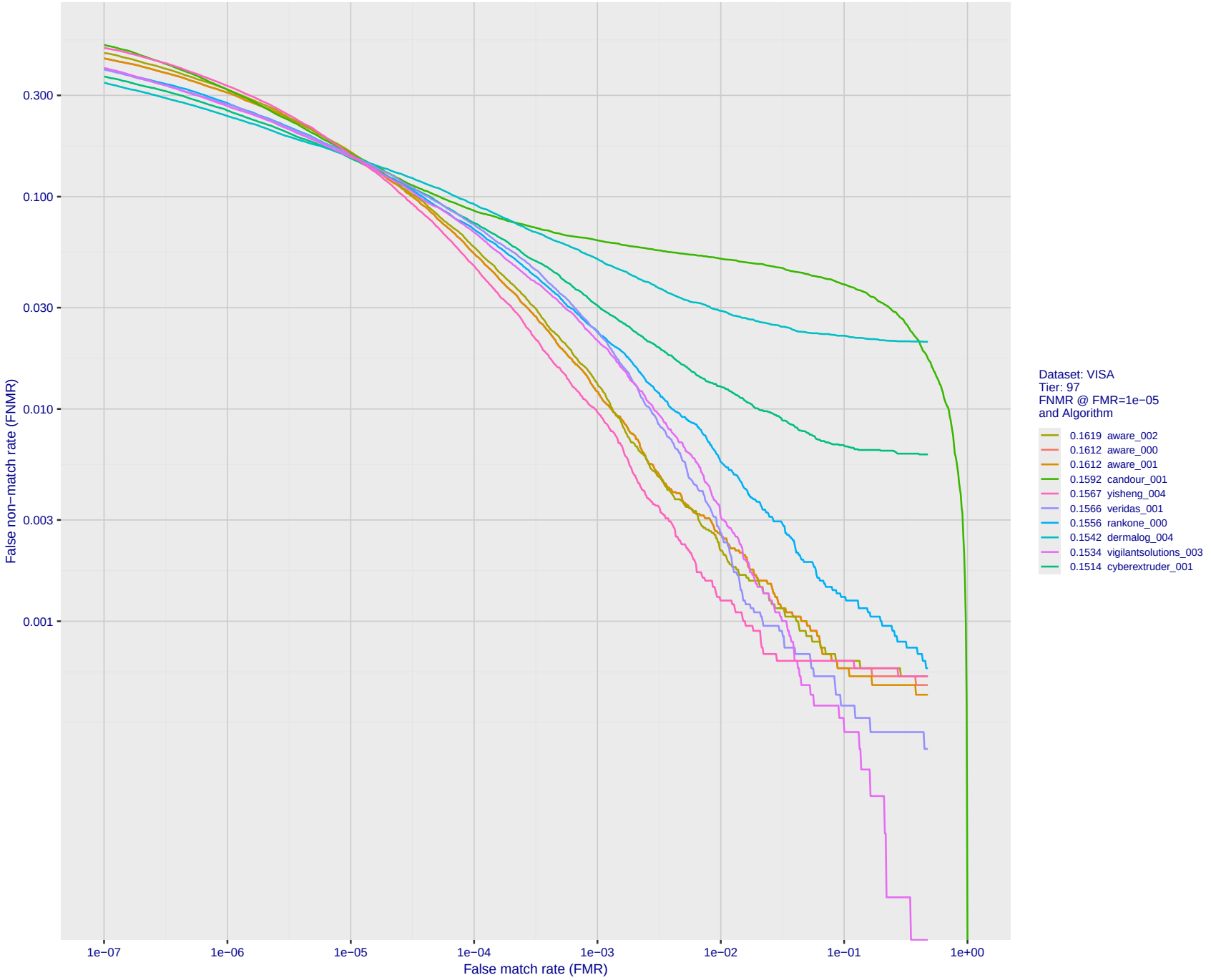

Figure 209: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

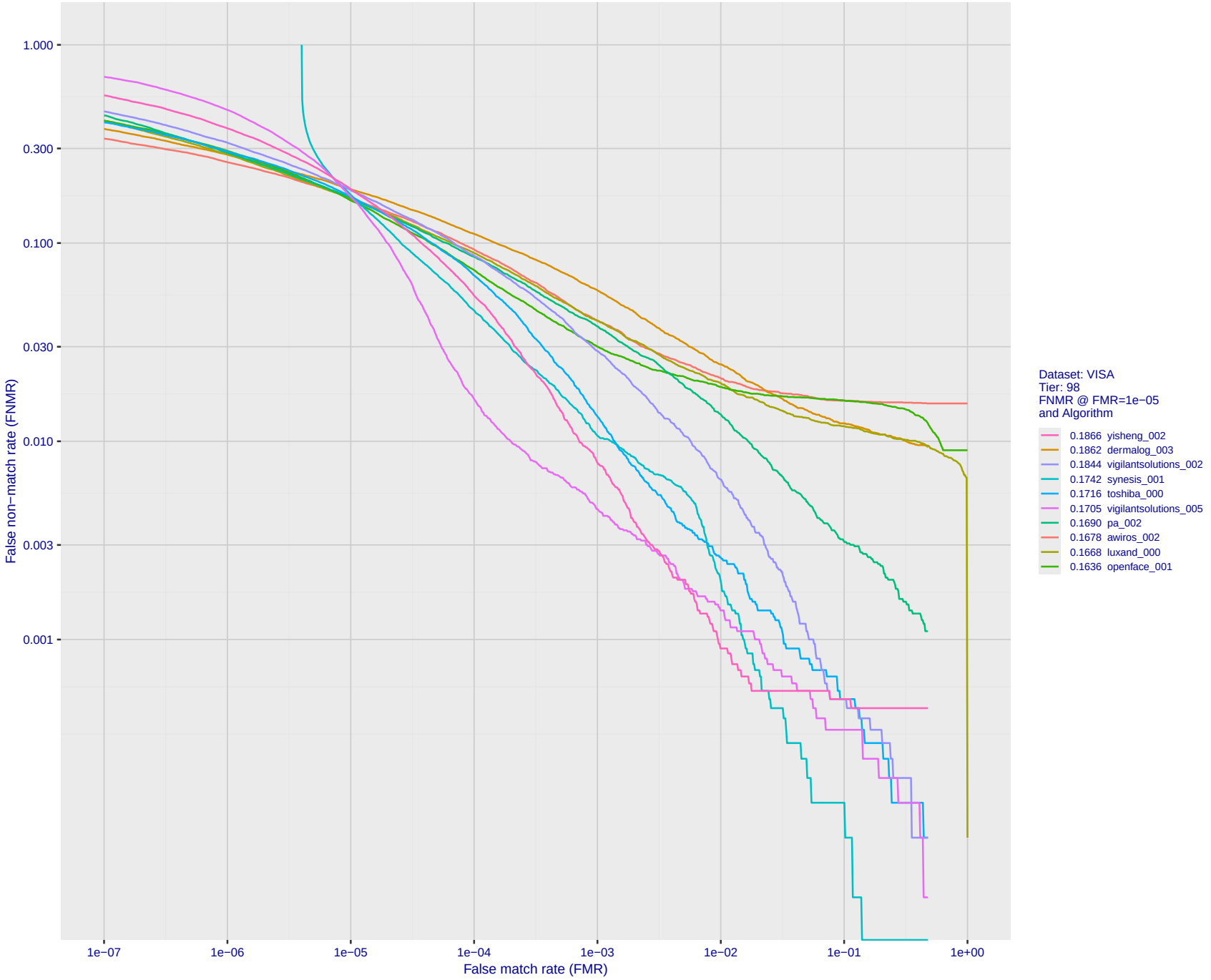

Figure 210: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

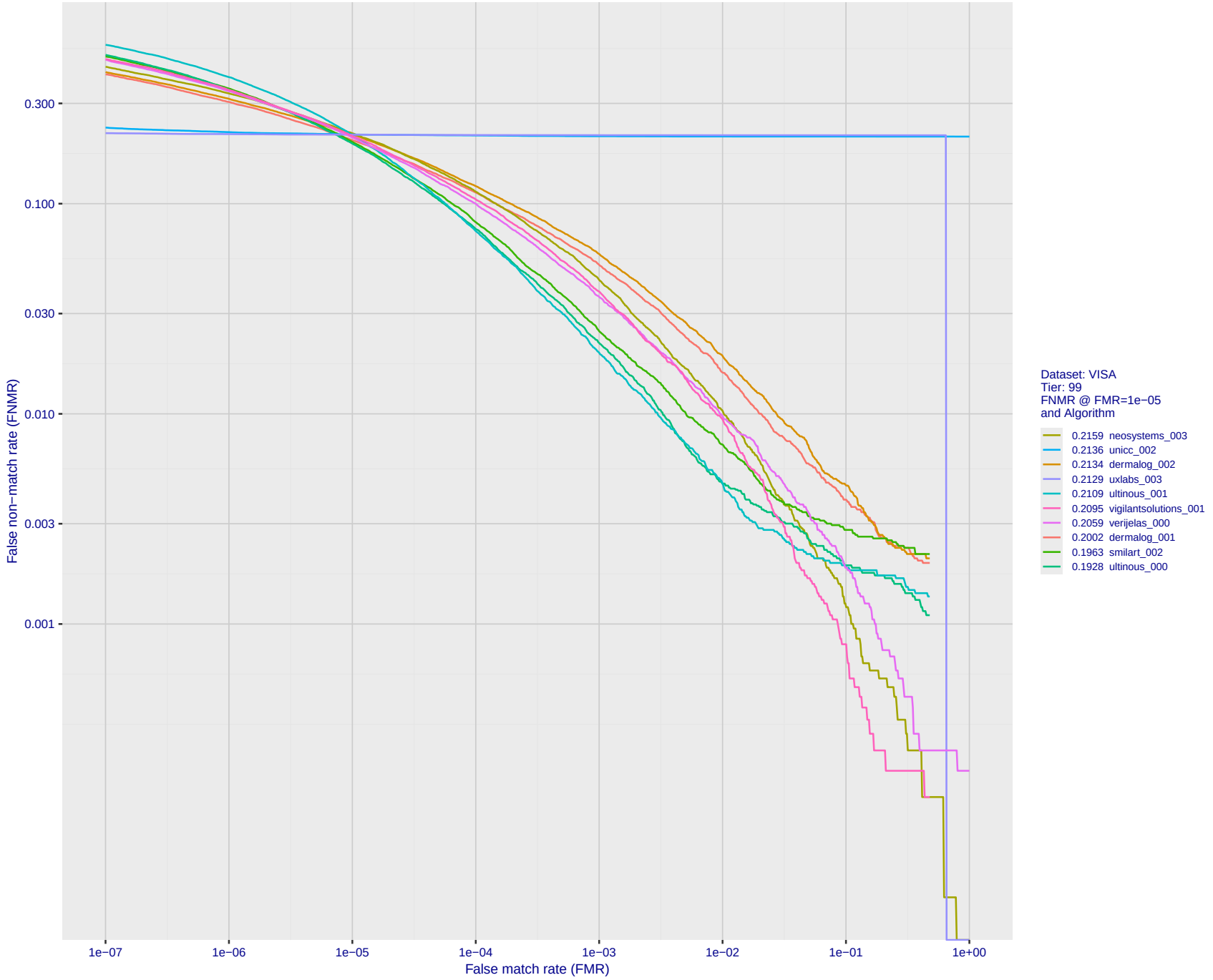

Figure 211: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

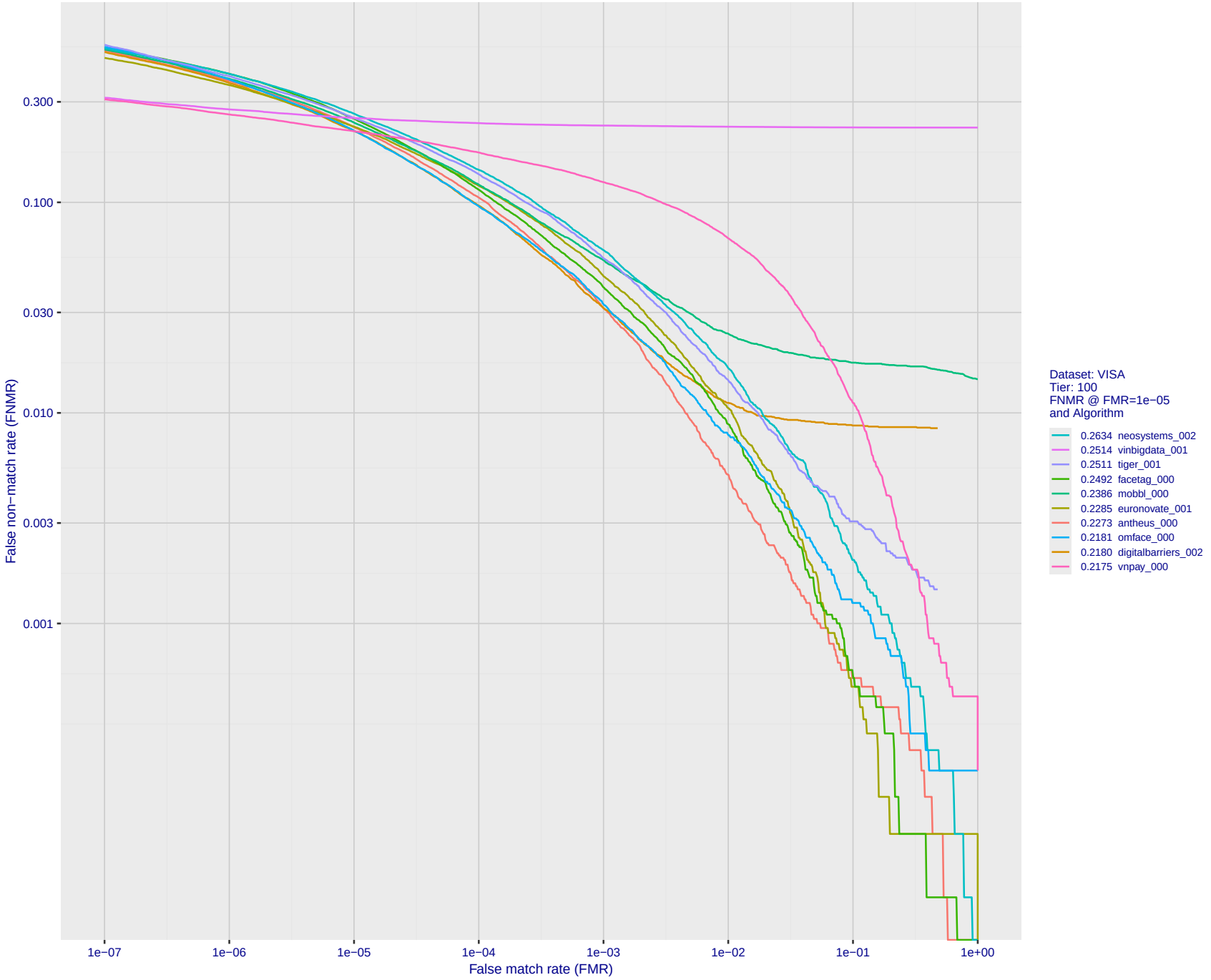

Figure 212: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

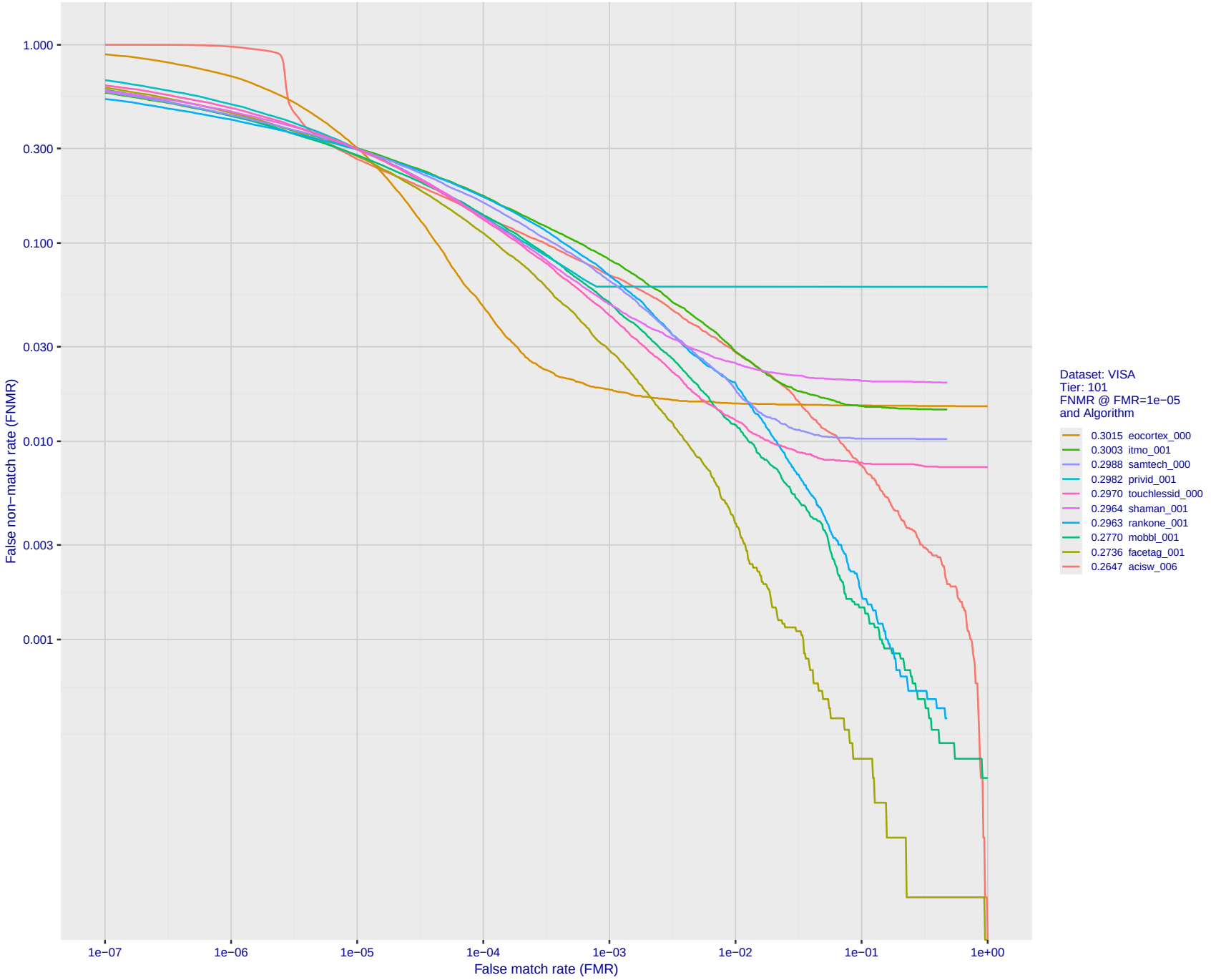

Figure 213: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

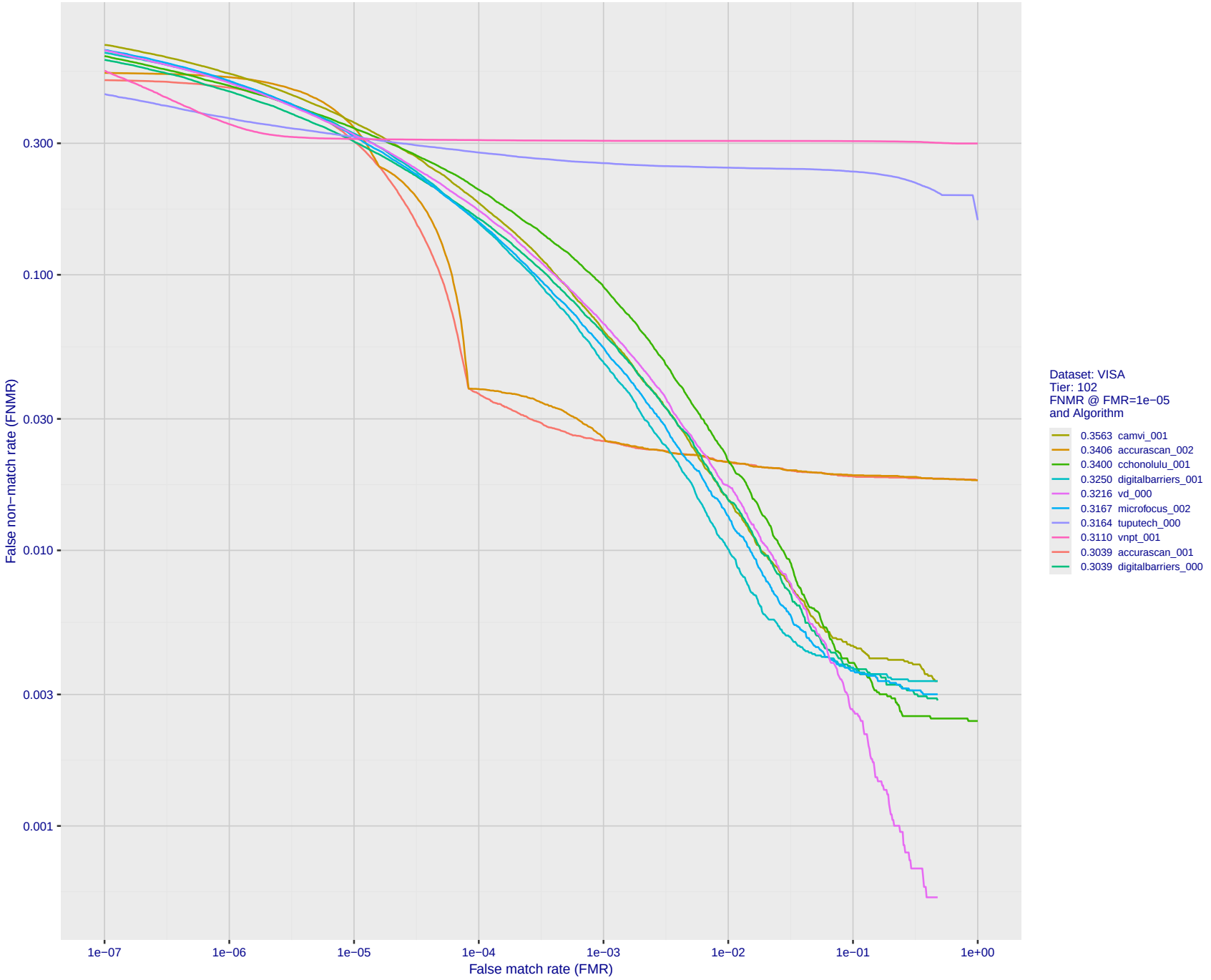

Figure 214: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

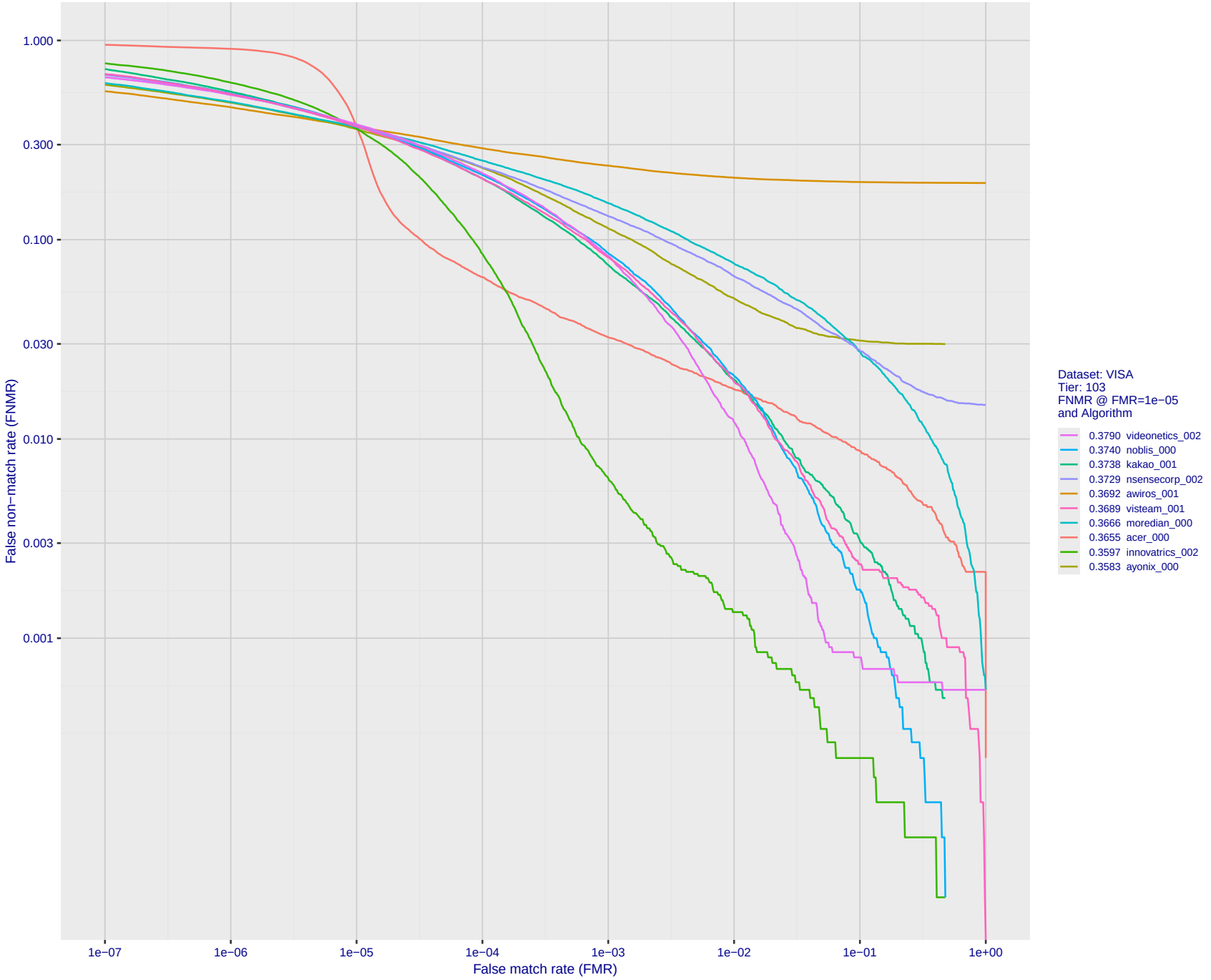

Figure 215: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

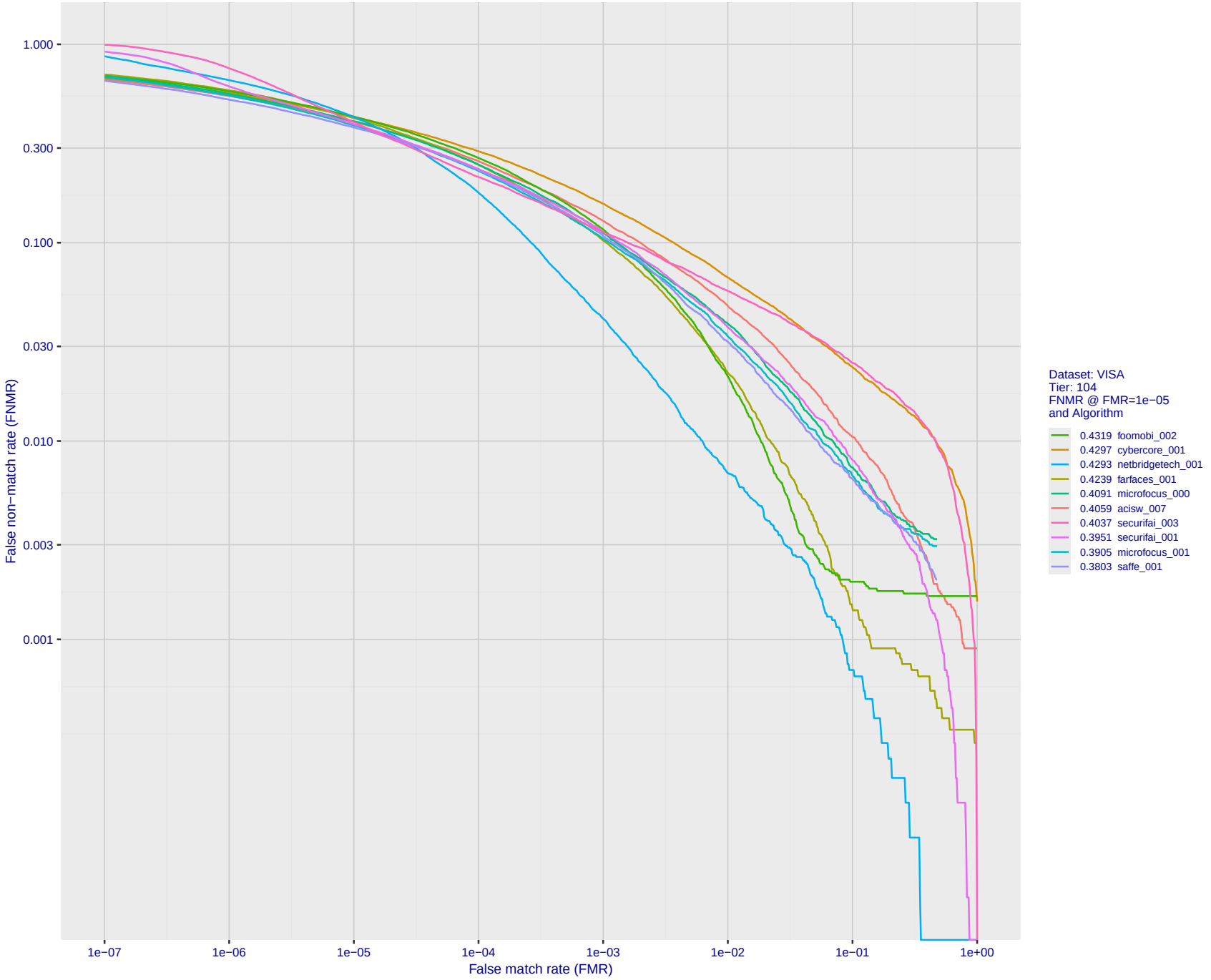

Figure 216: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

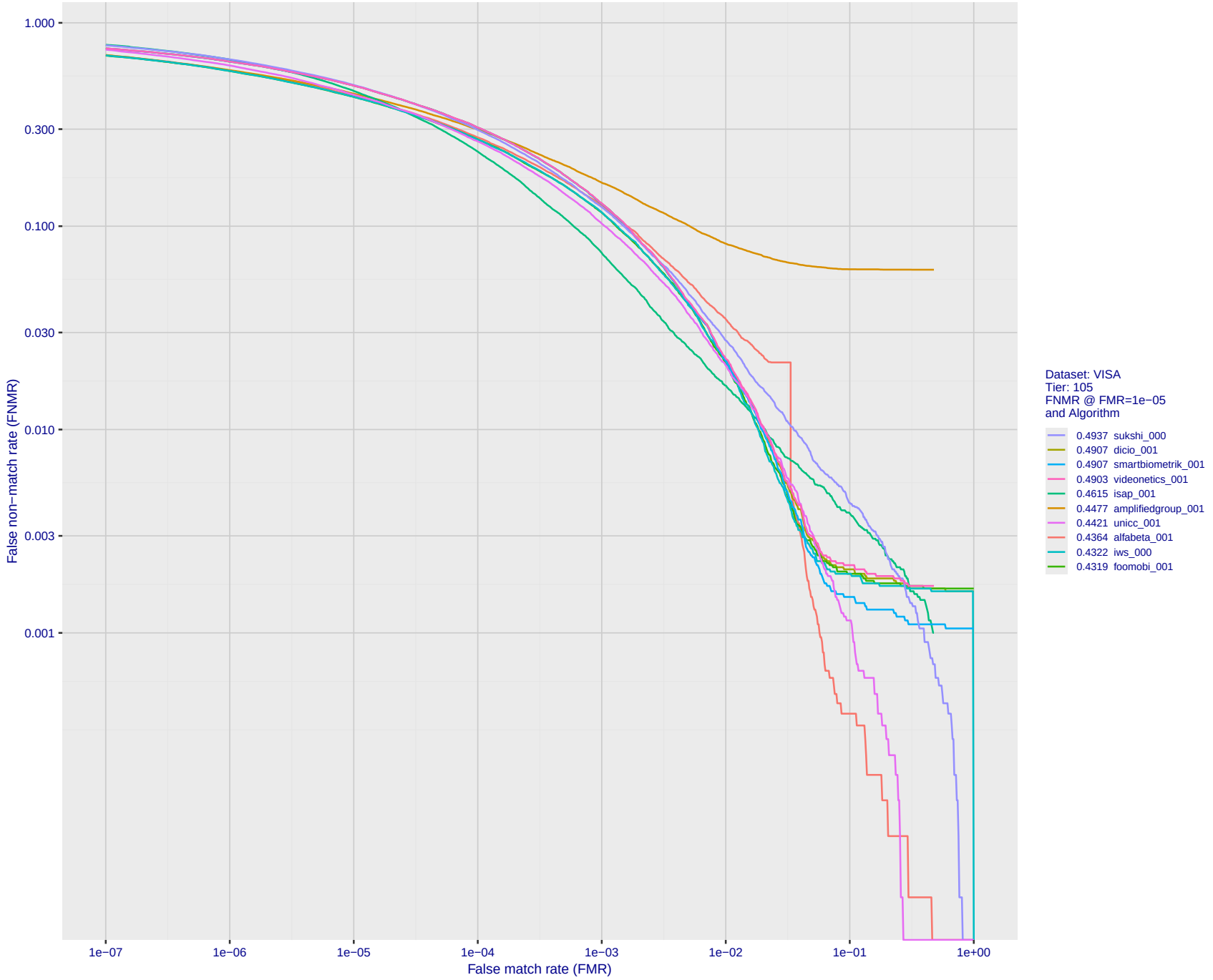

Figure 217: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

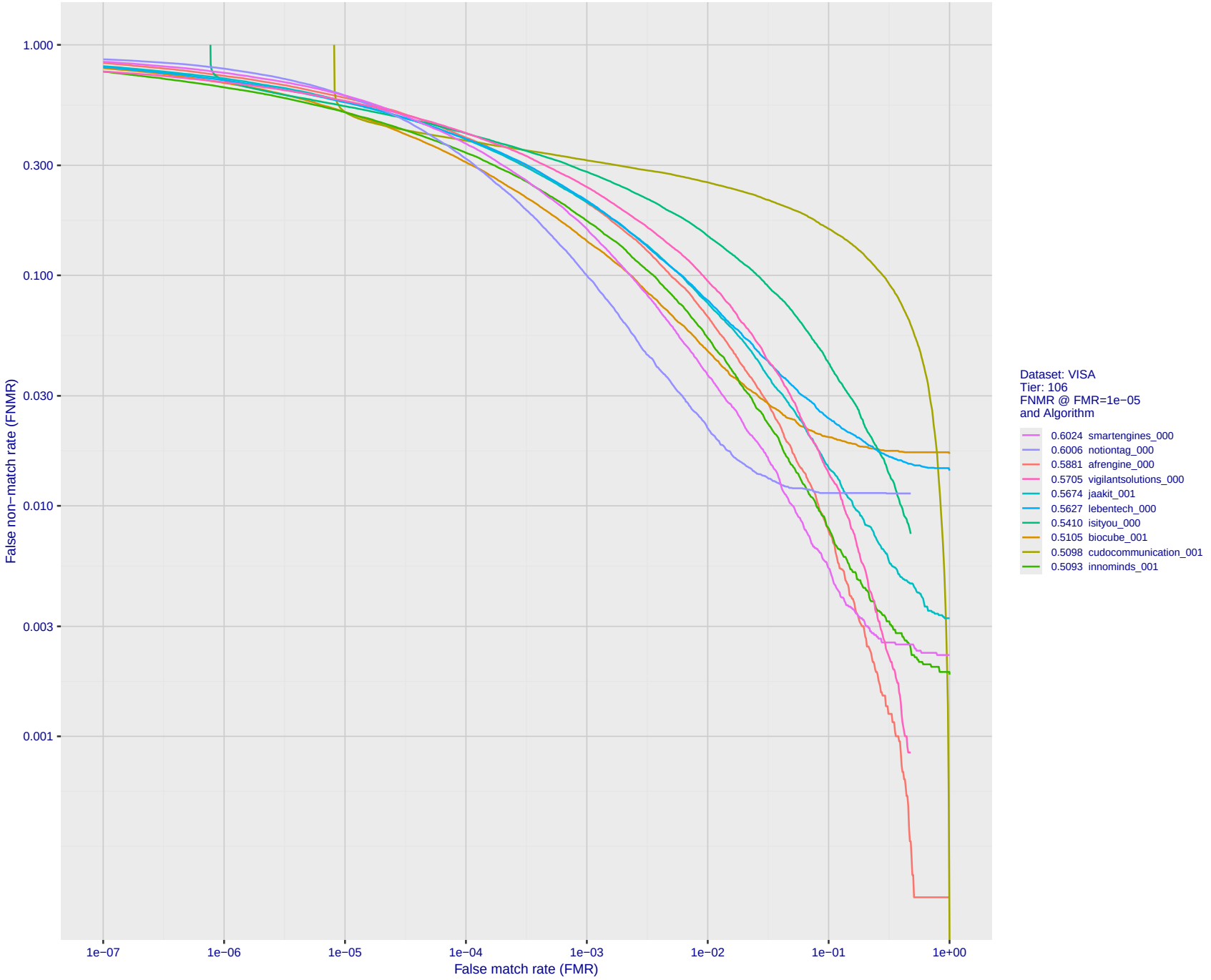

Figure 218: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

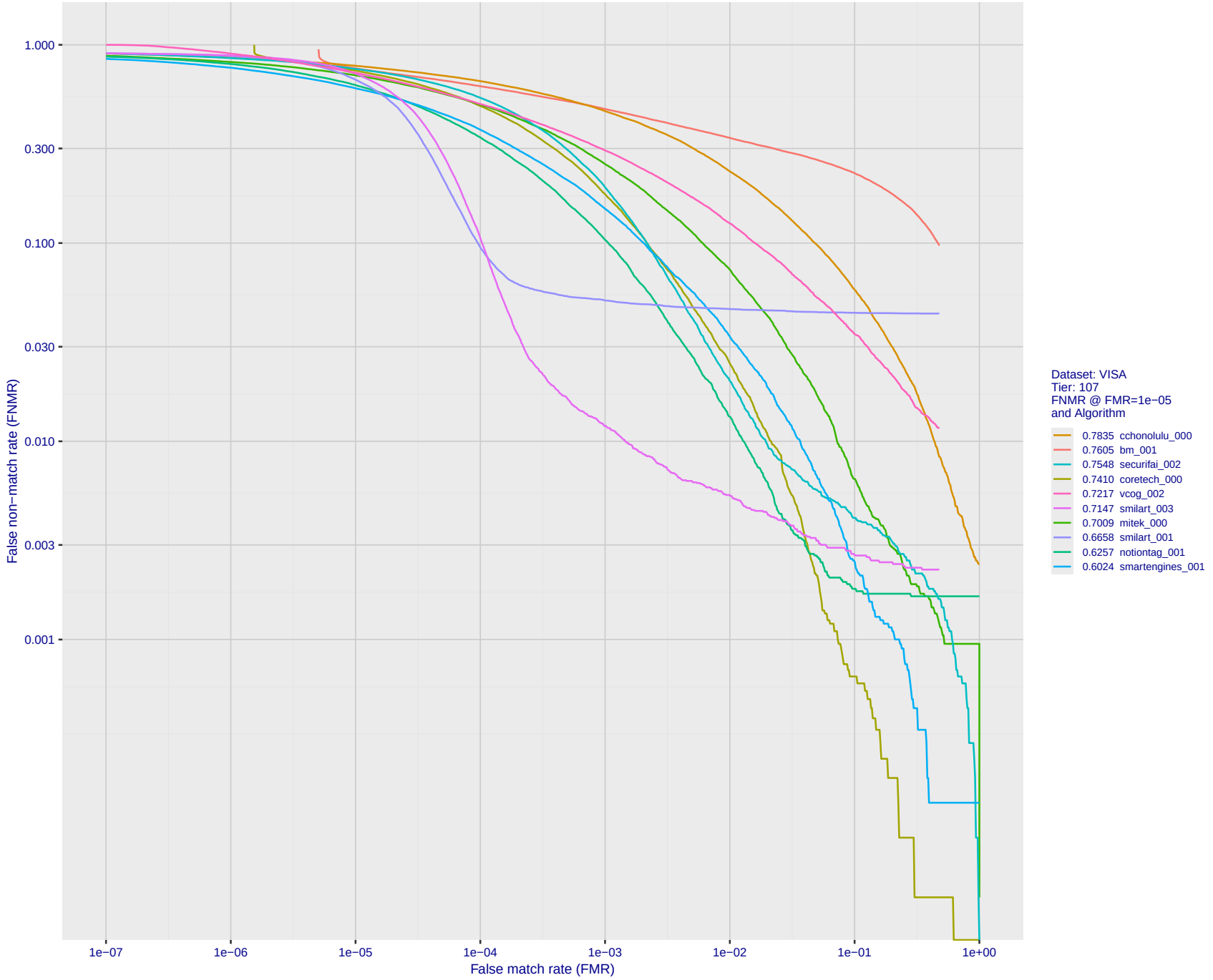

Figure 219: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

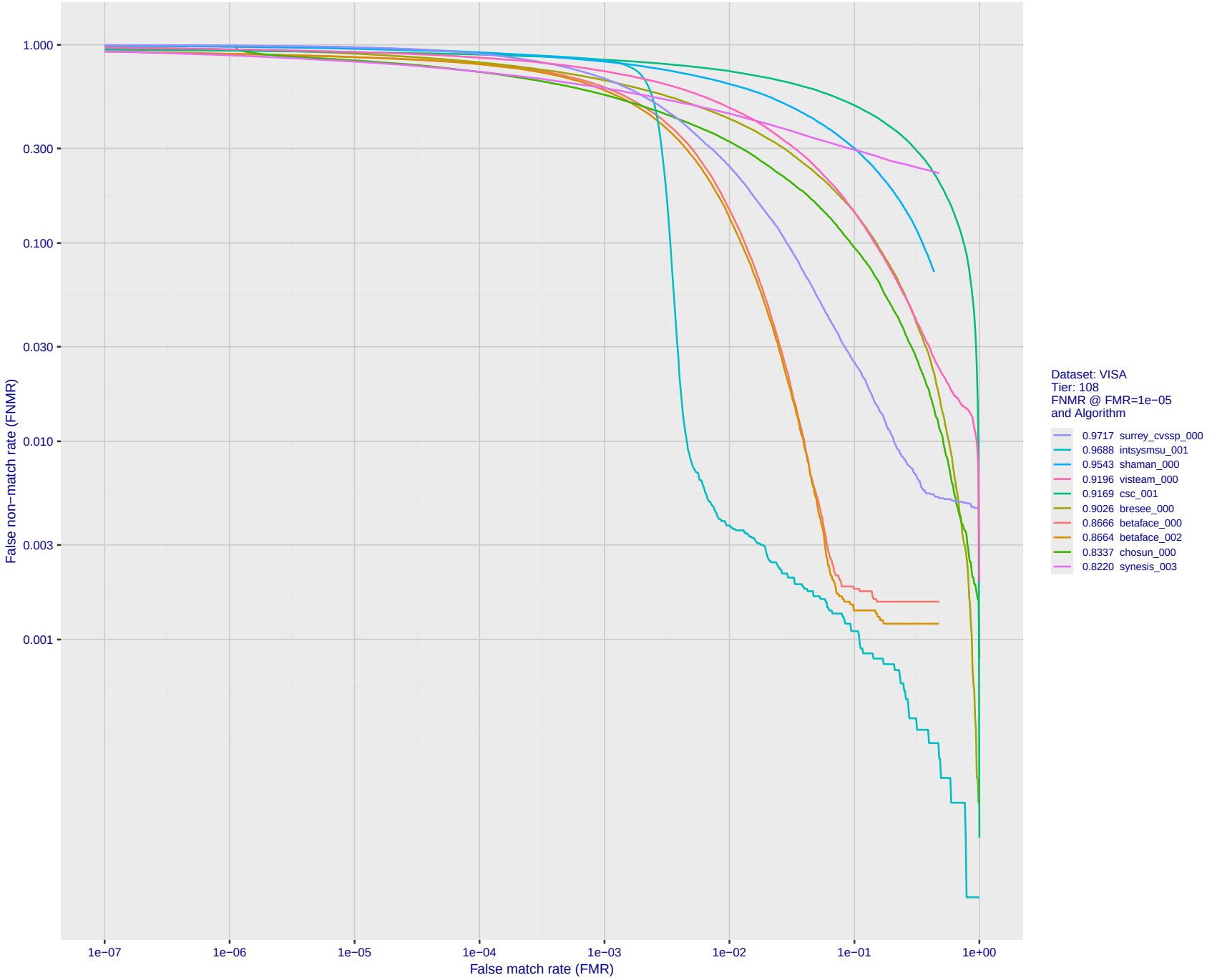

Figure 220: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

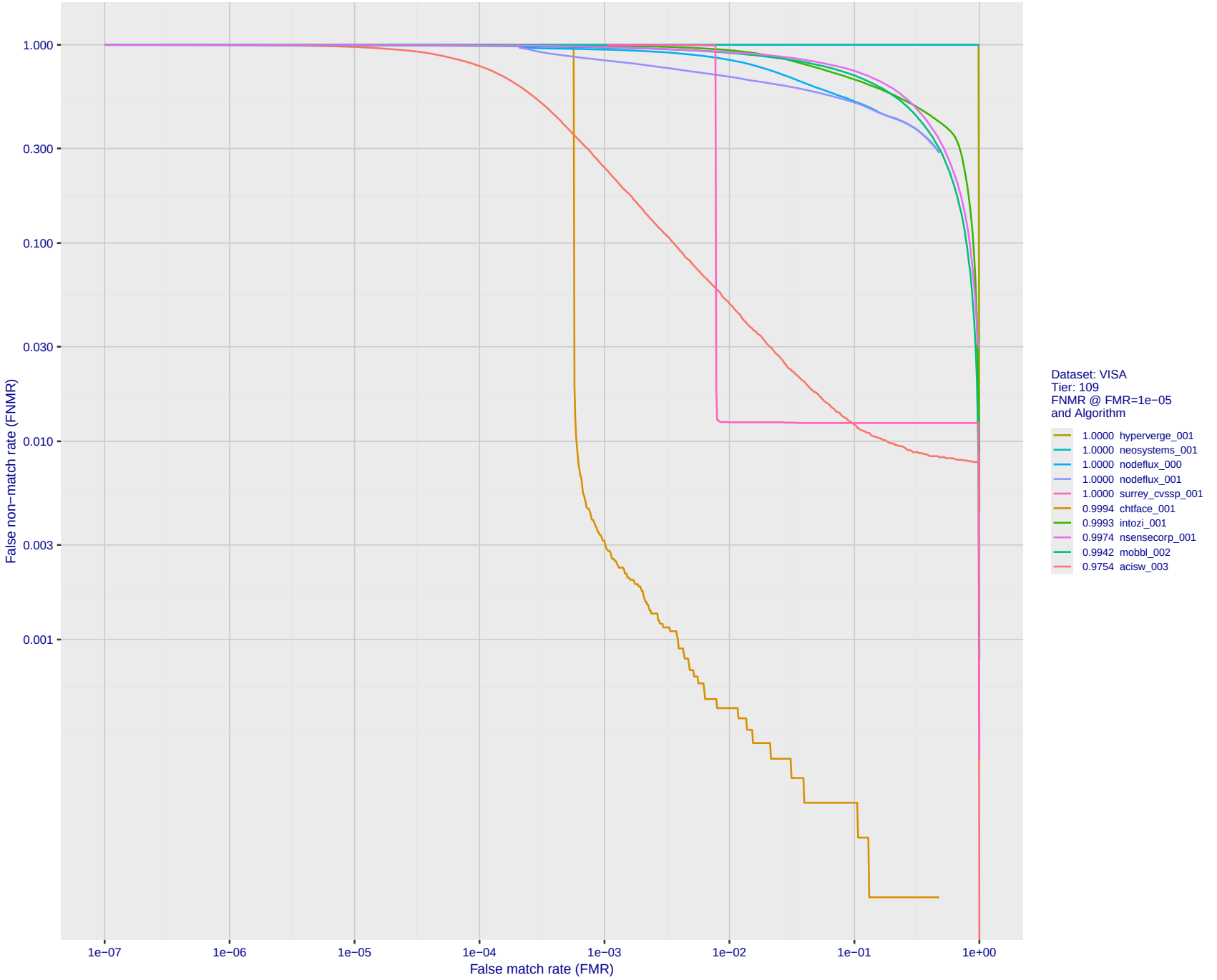

Figure 221: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

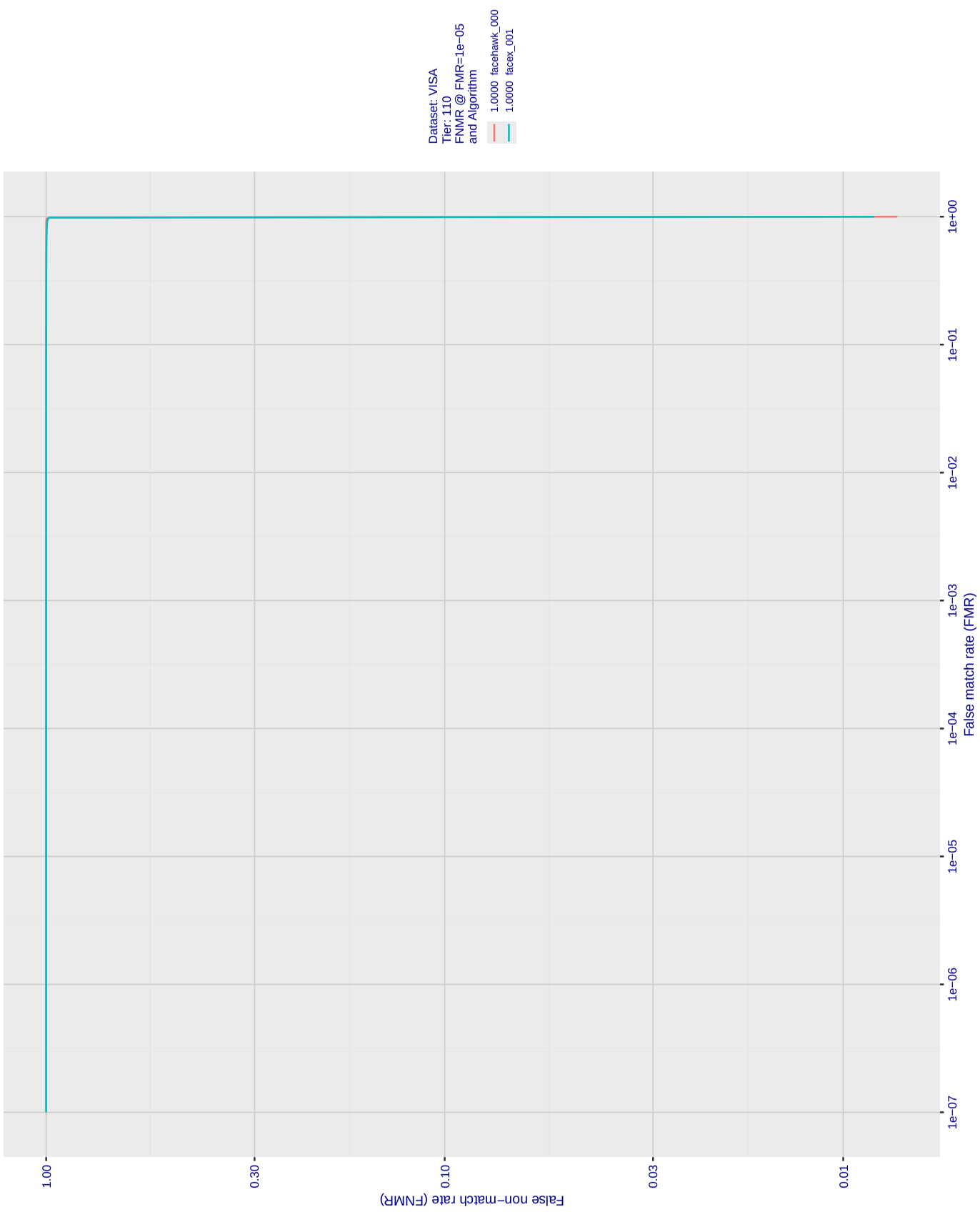

Figure 222: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

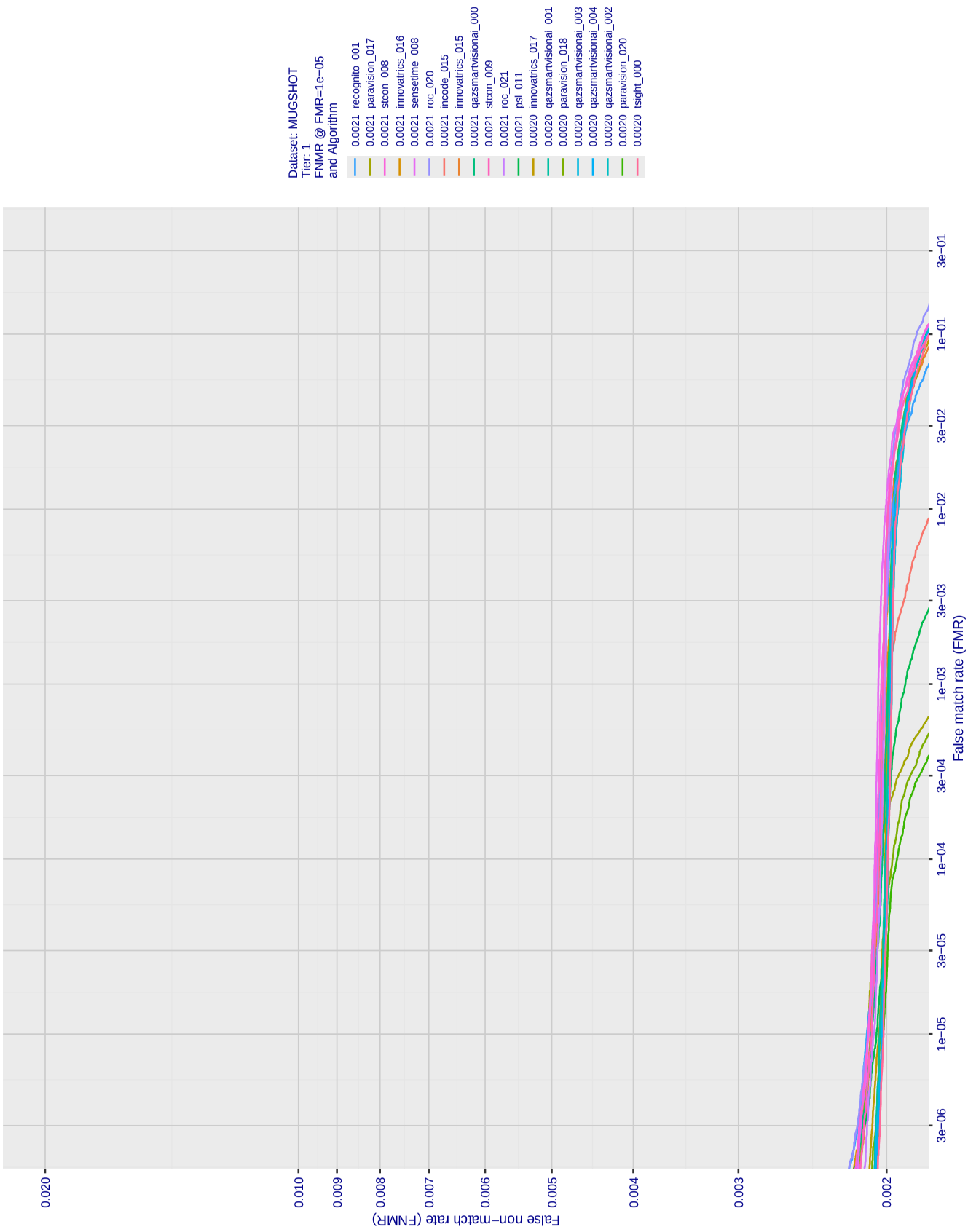

Figure 223: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

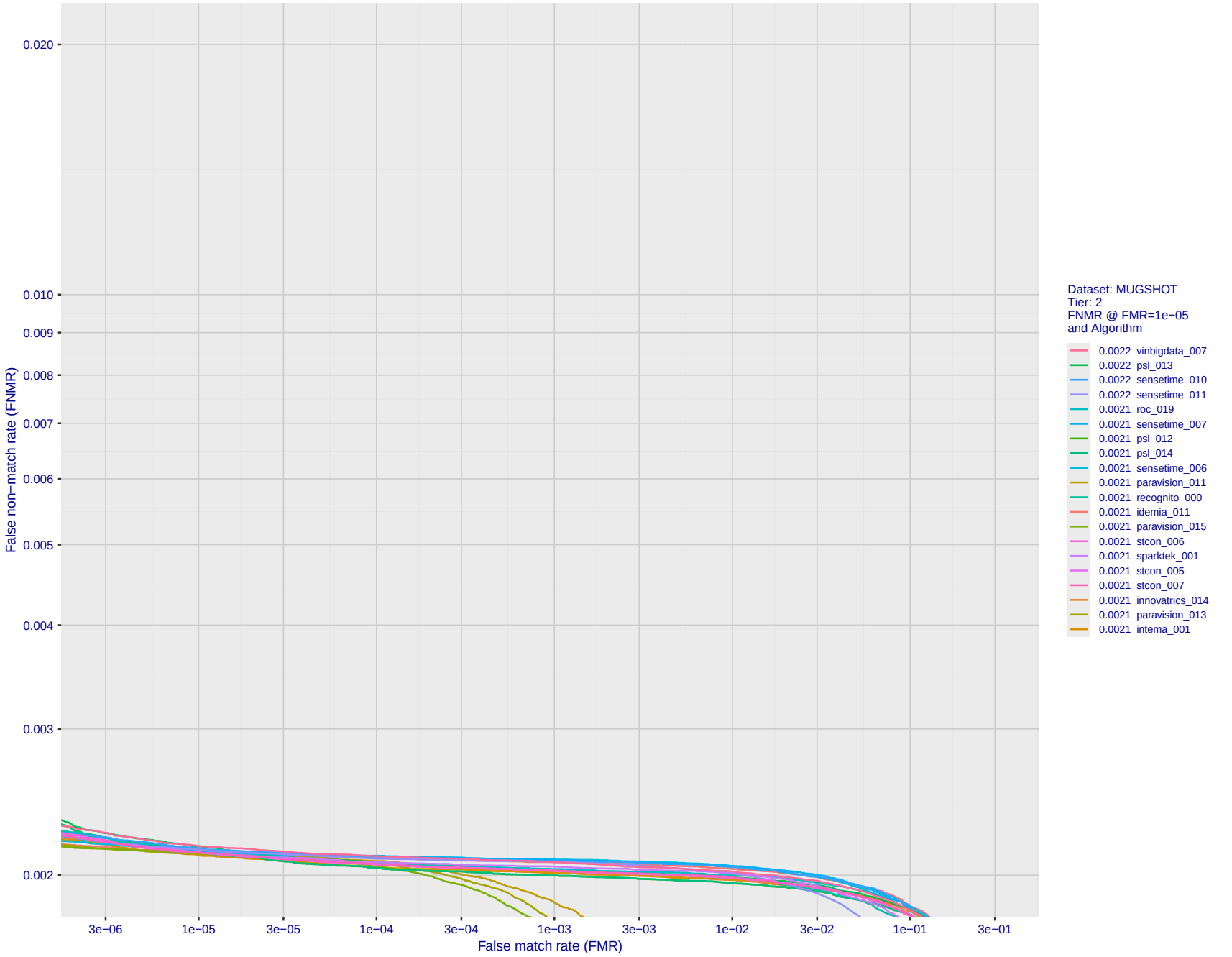

Figure 224: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

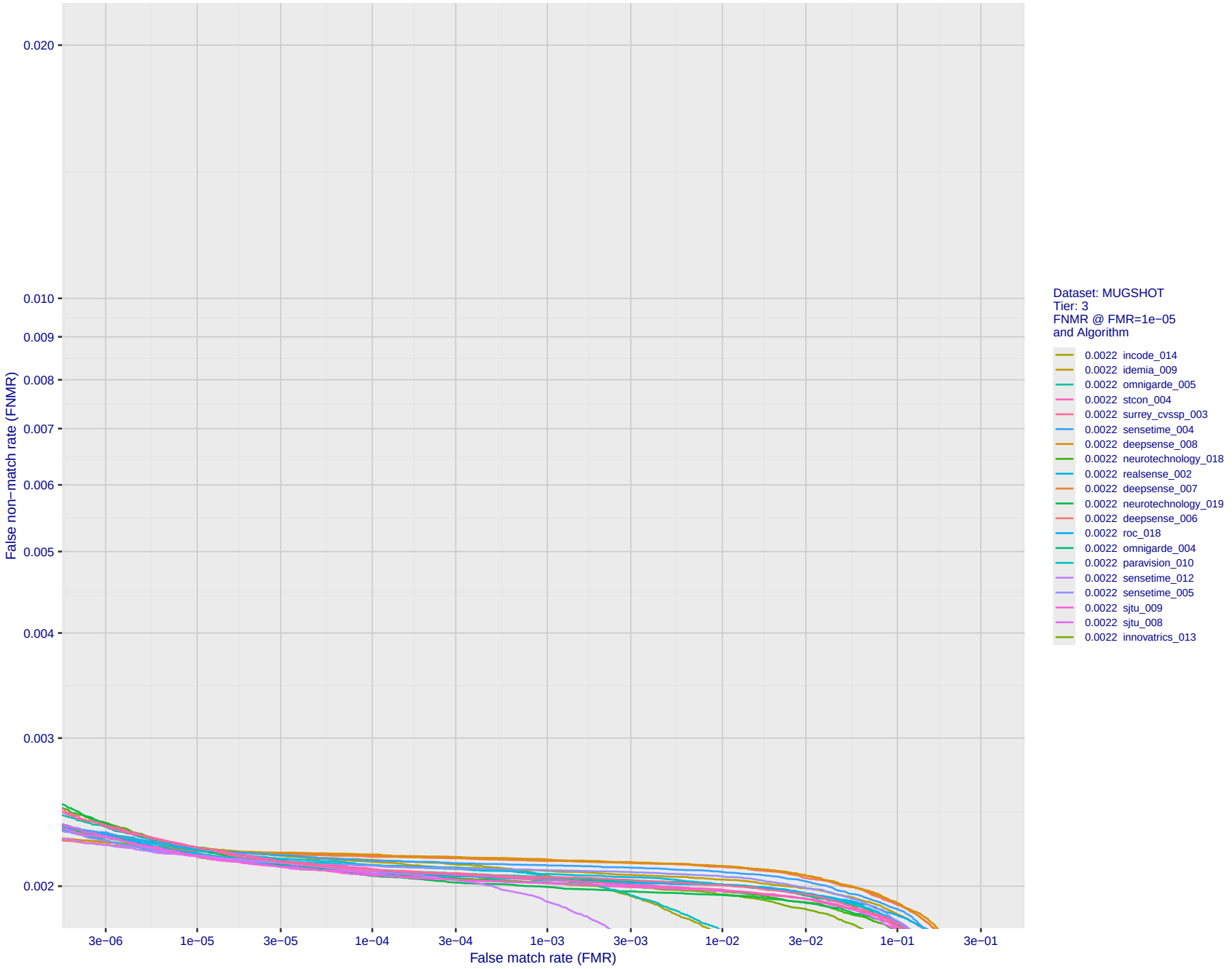

*Figure 225: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

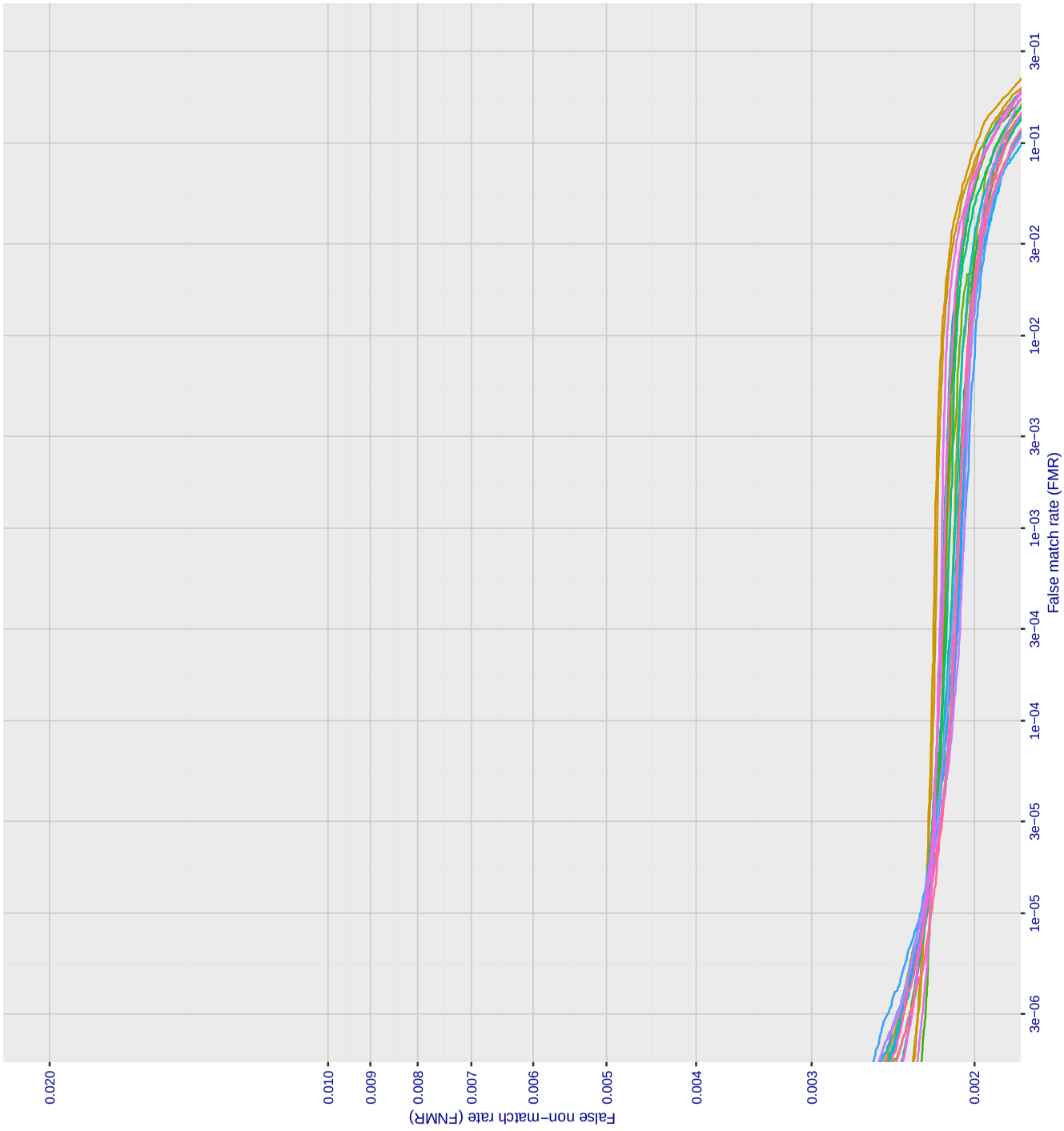

Figure 226: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

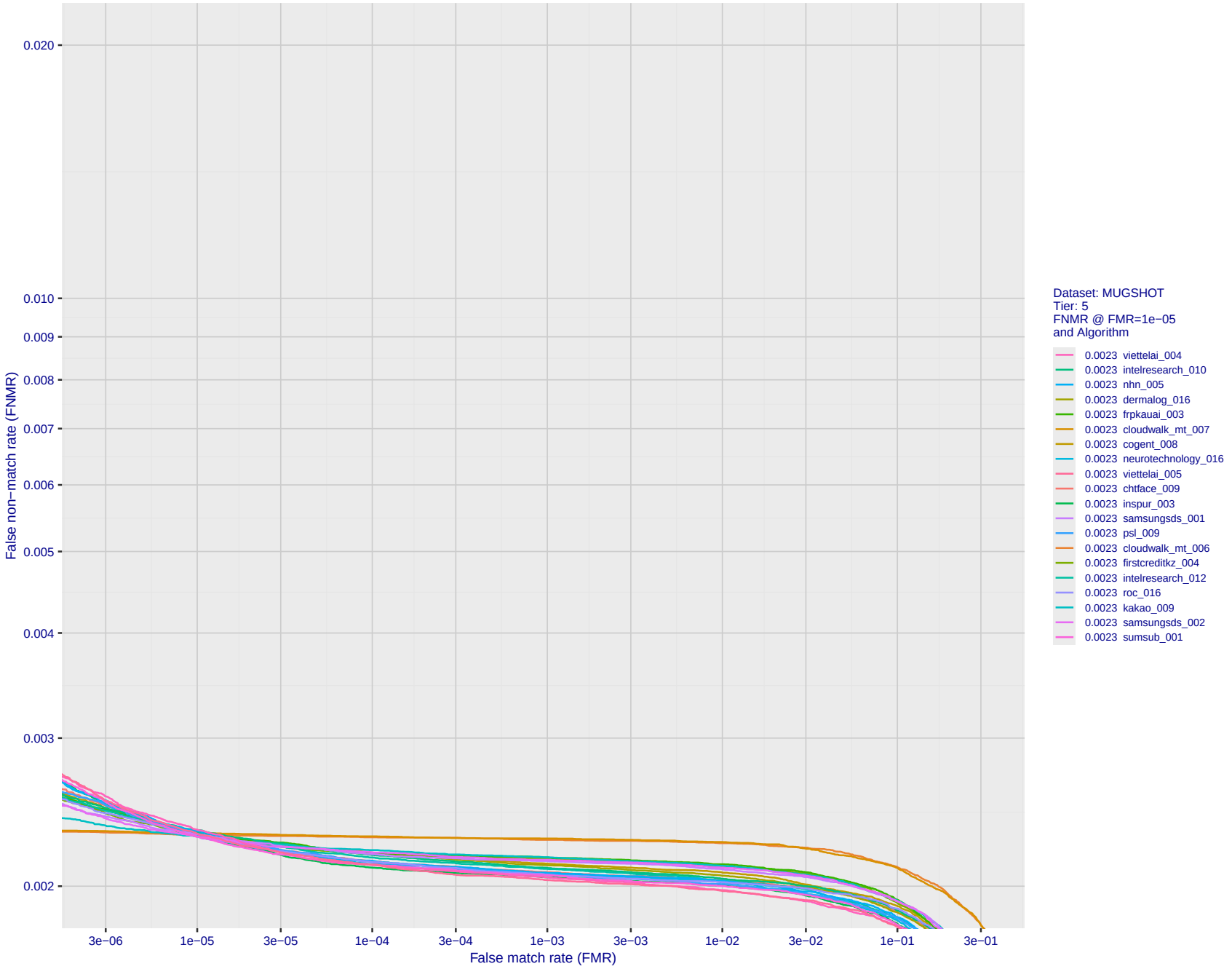

*Figure 227: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

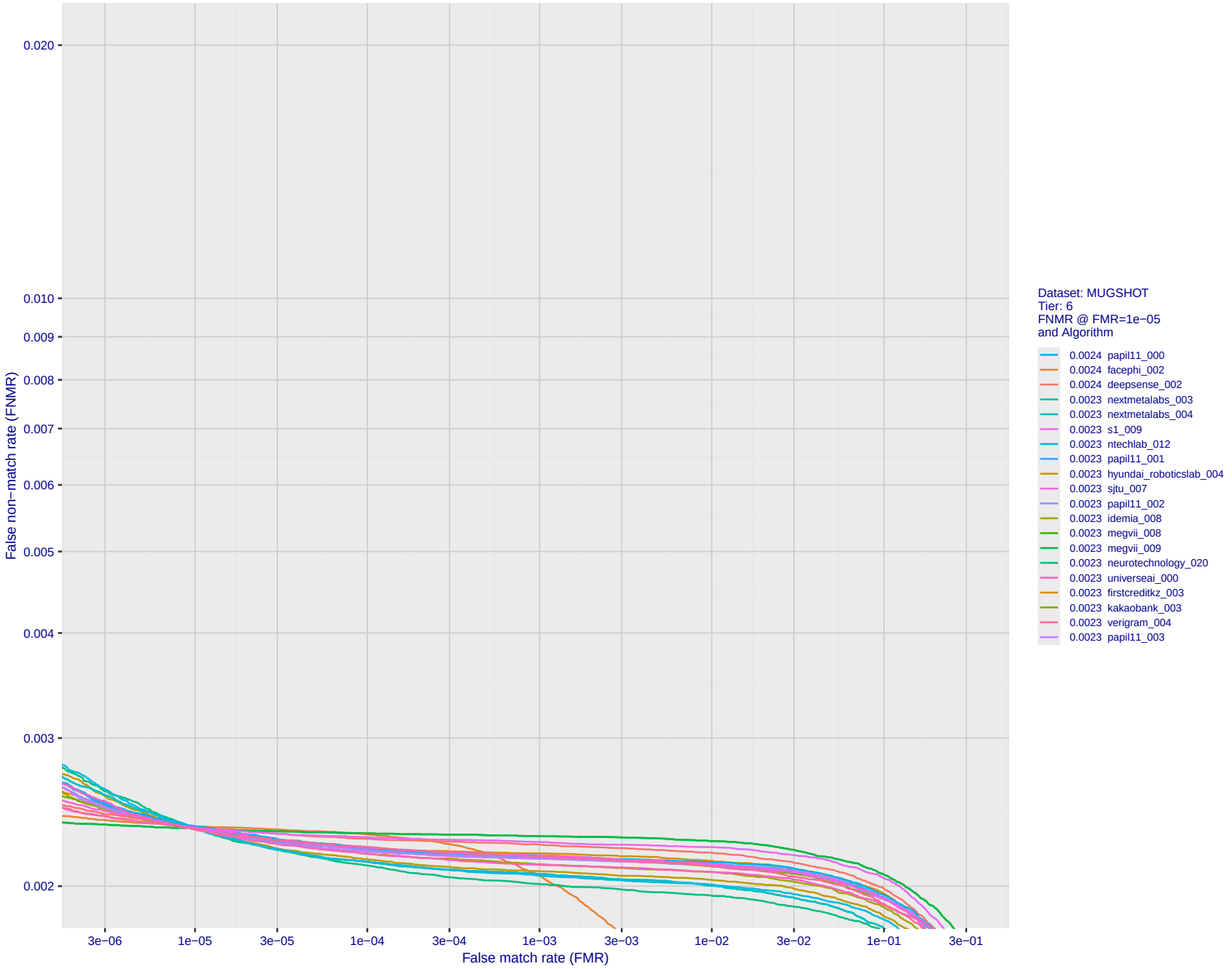

Figure 228: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

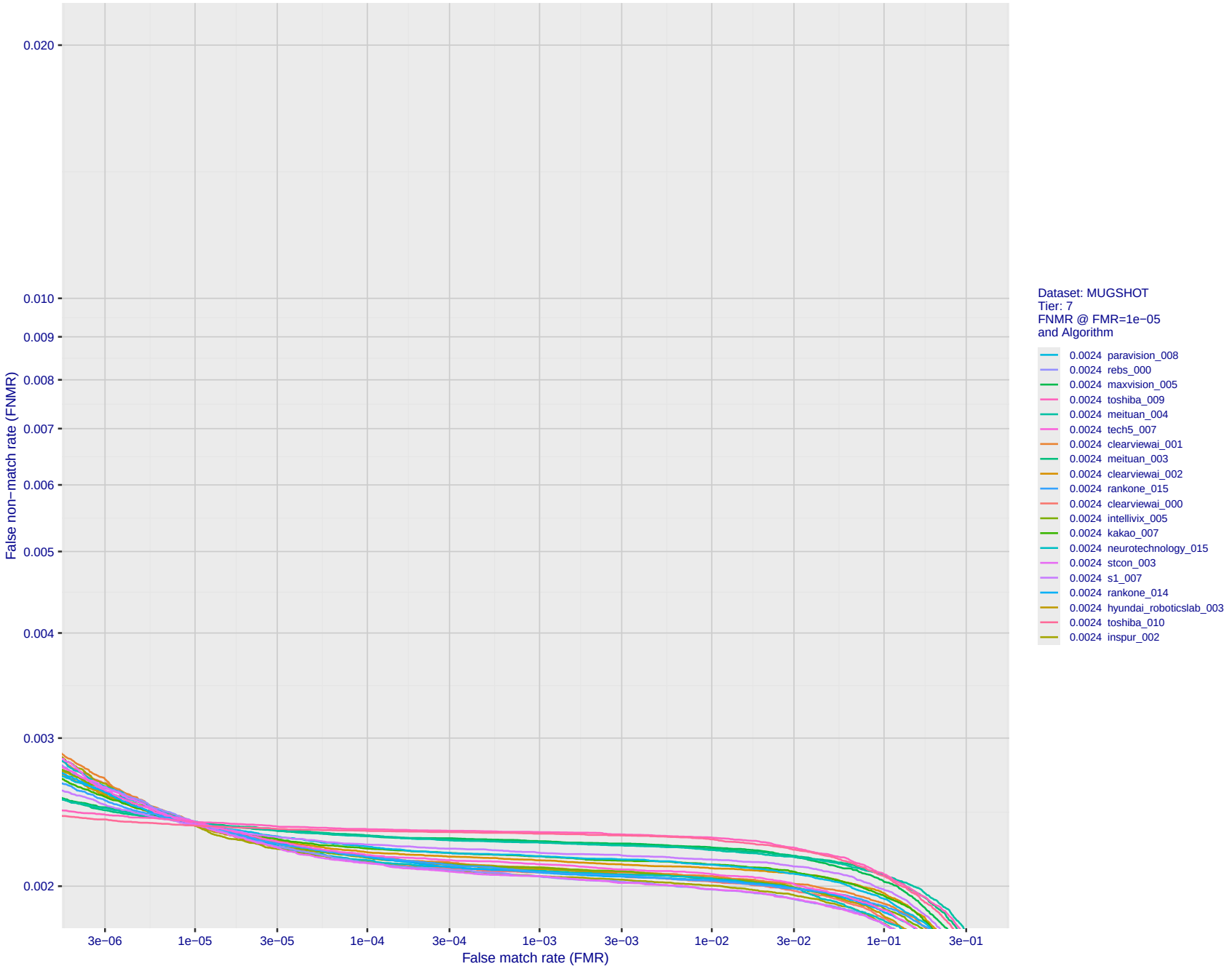

Figure 229: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

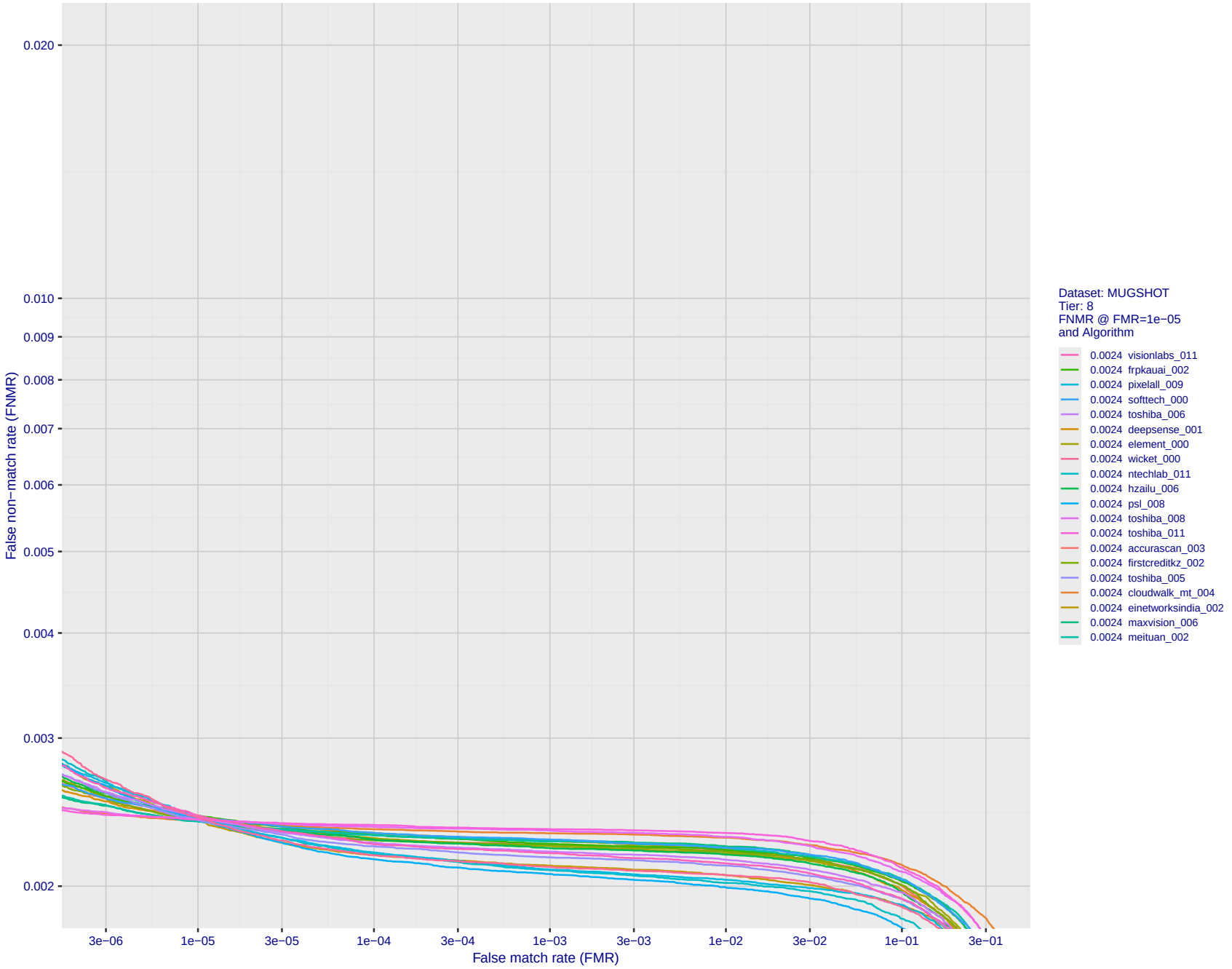

Figure 230: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

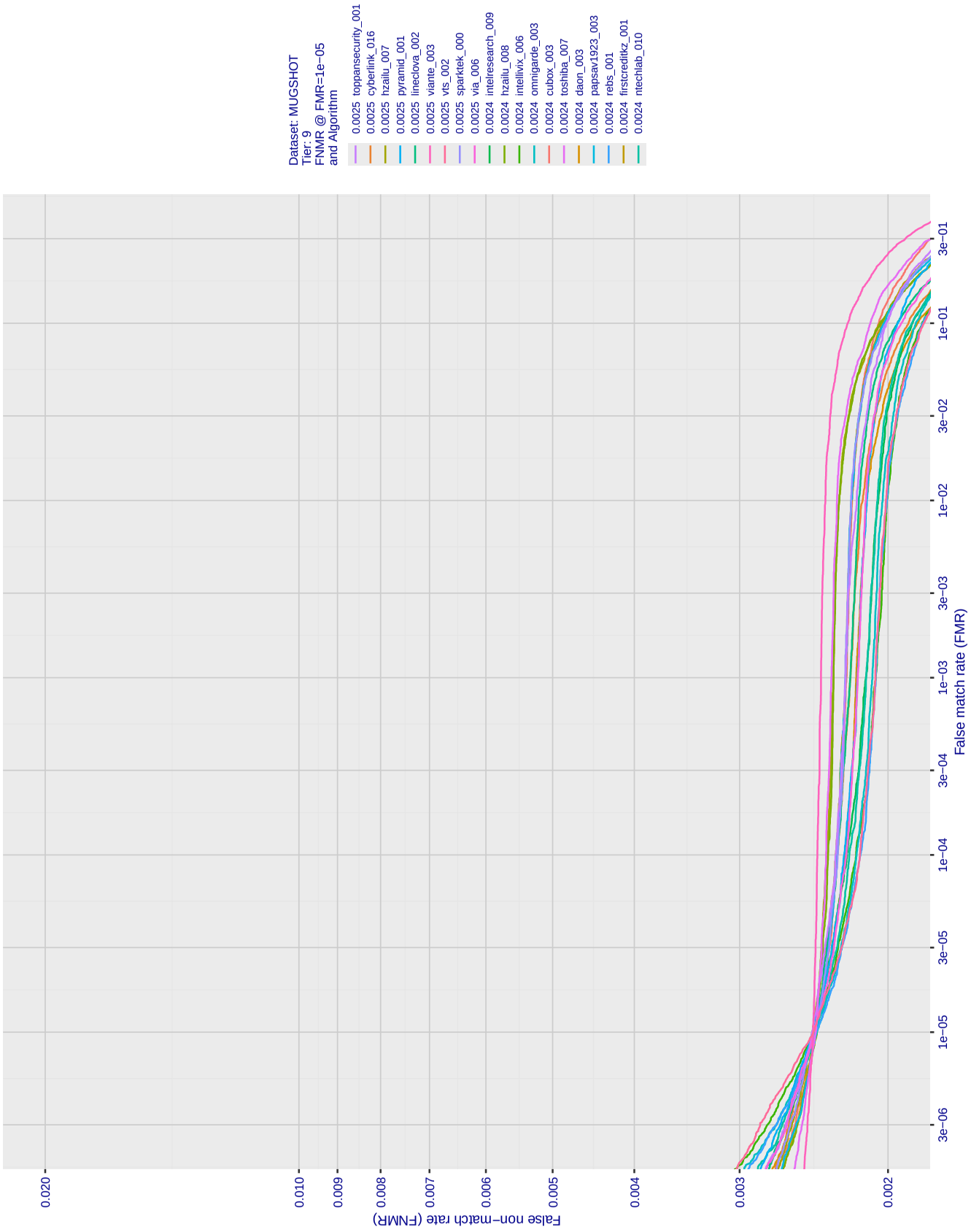

Figure 231: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

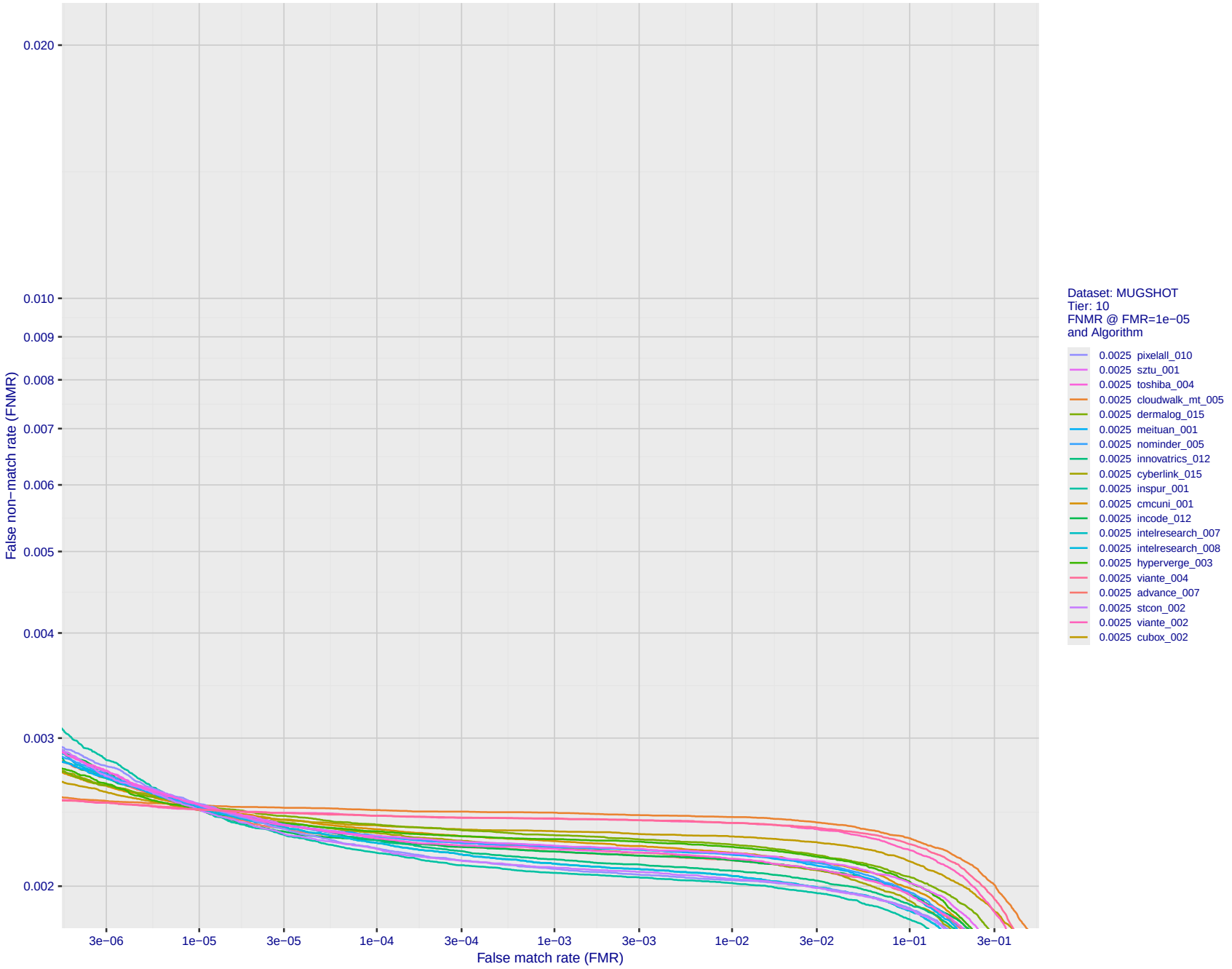

Figure 232: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

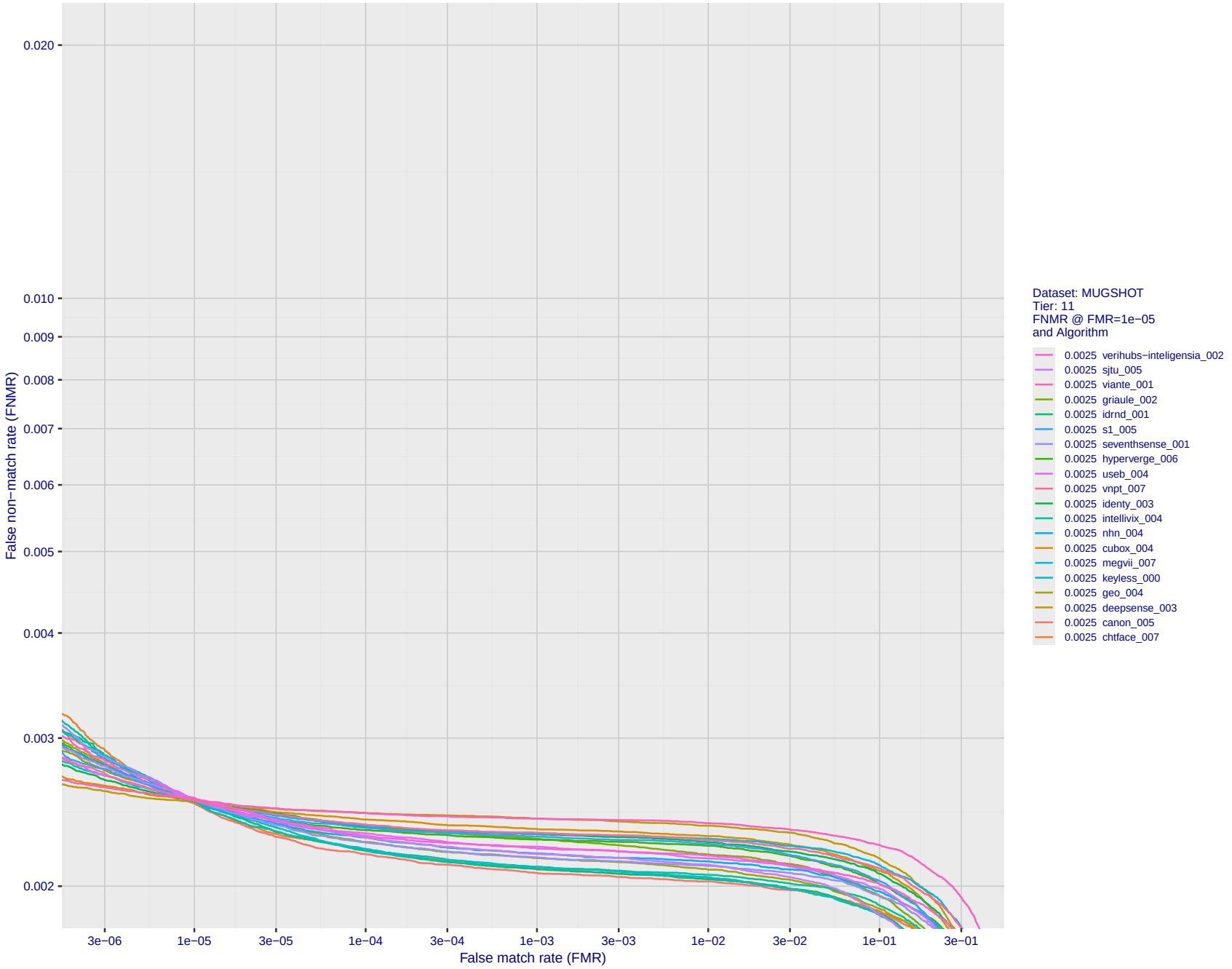

Figure 233: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

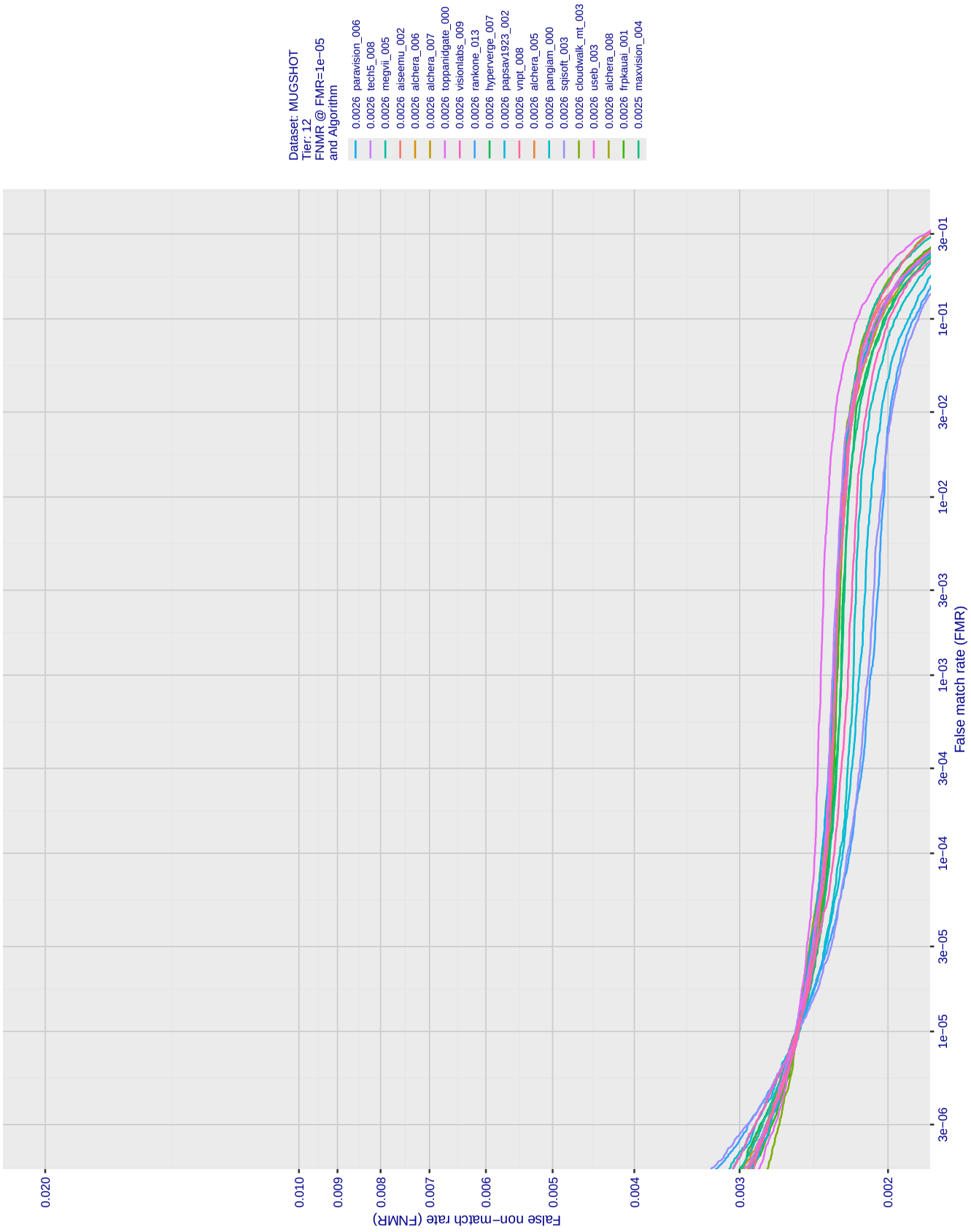

Figure 234: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

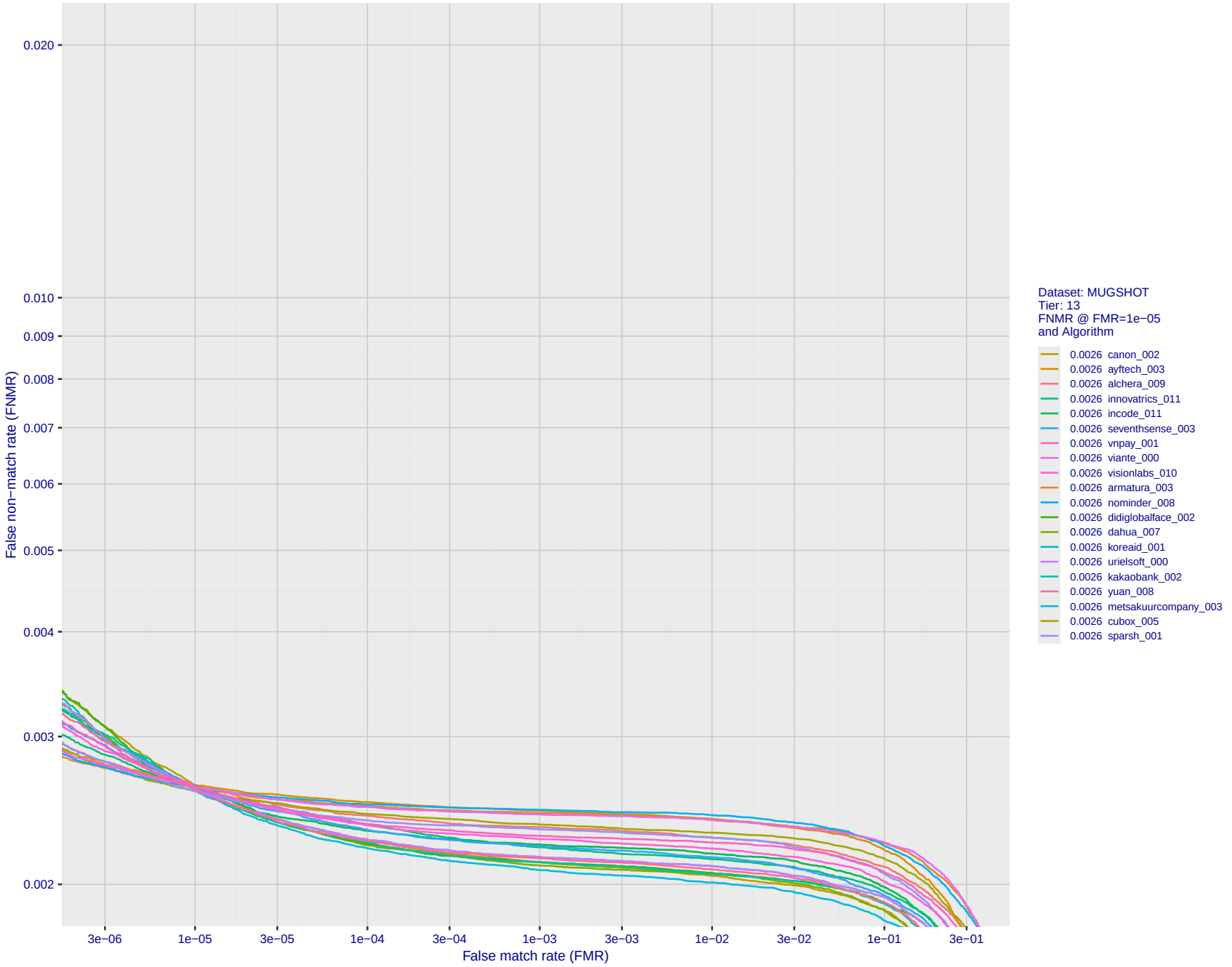

Figure 235: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

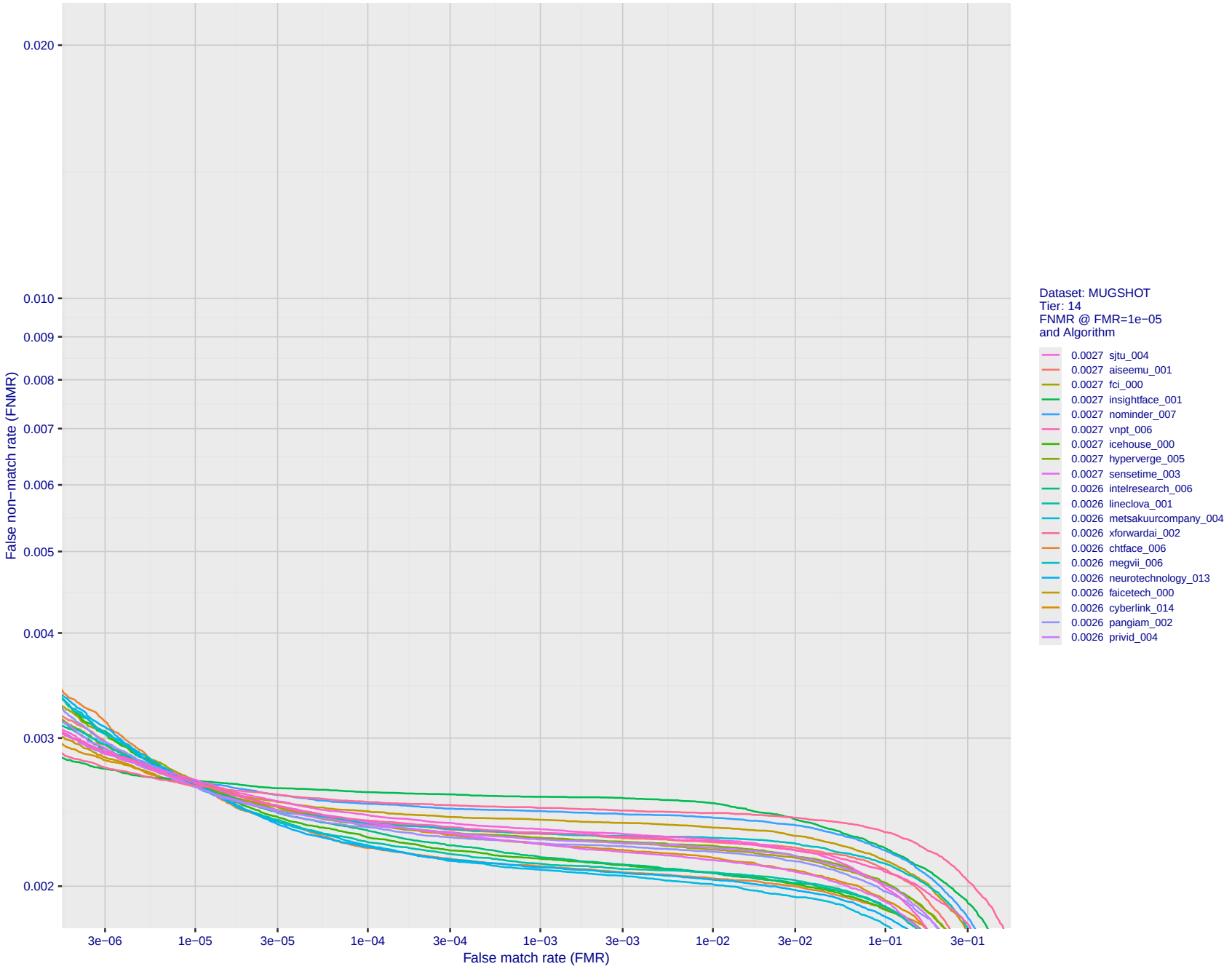

Figure 236: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

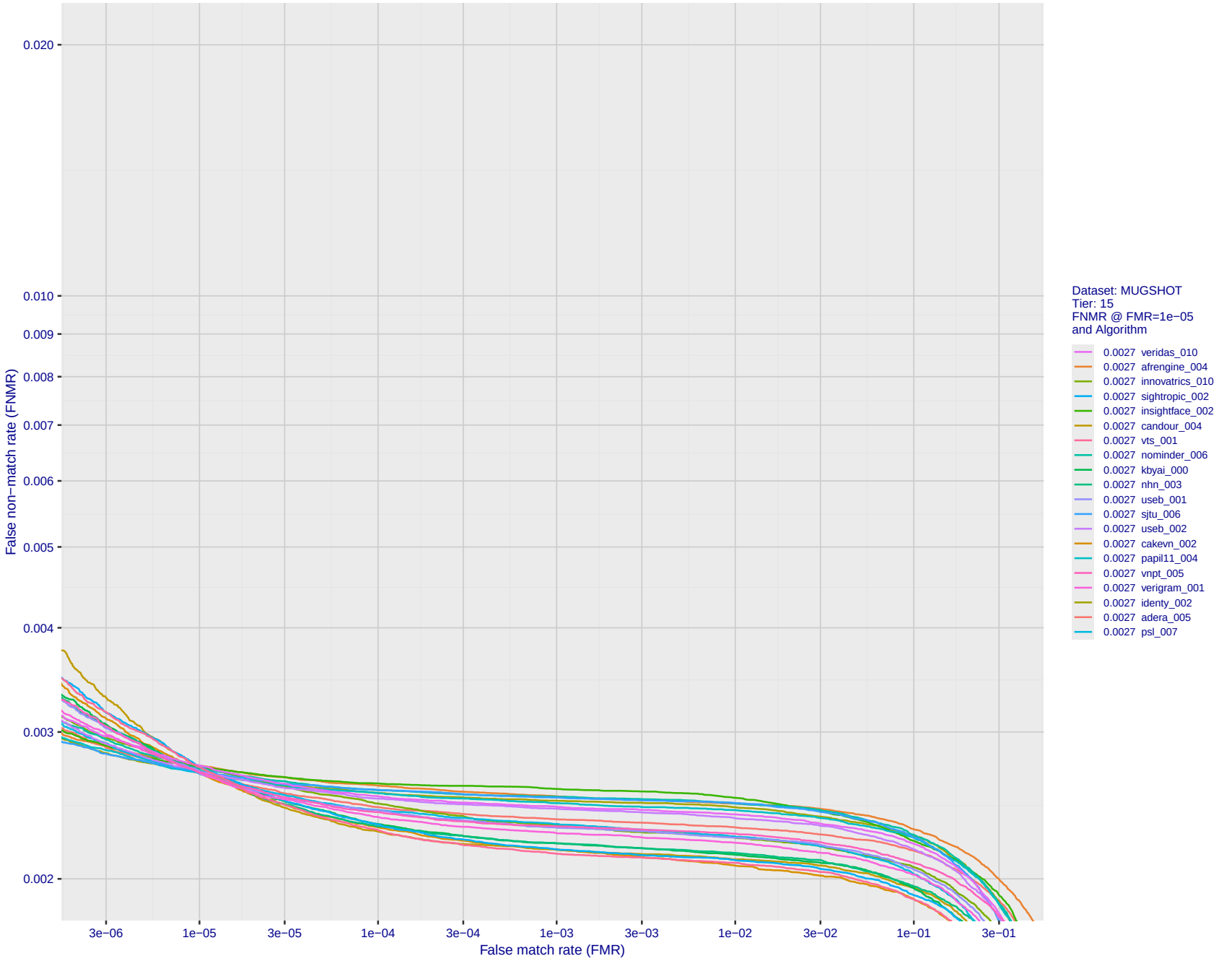

Figure 237: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

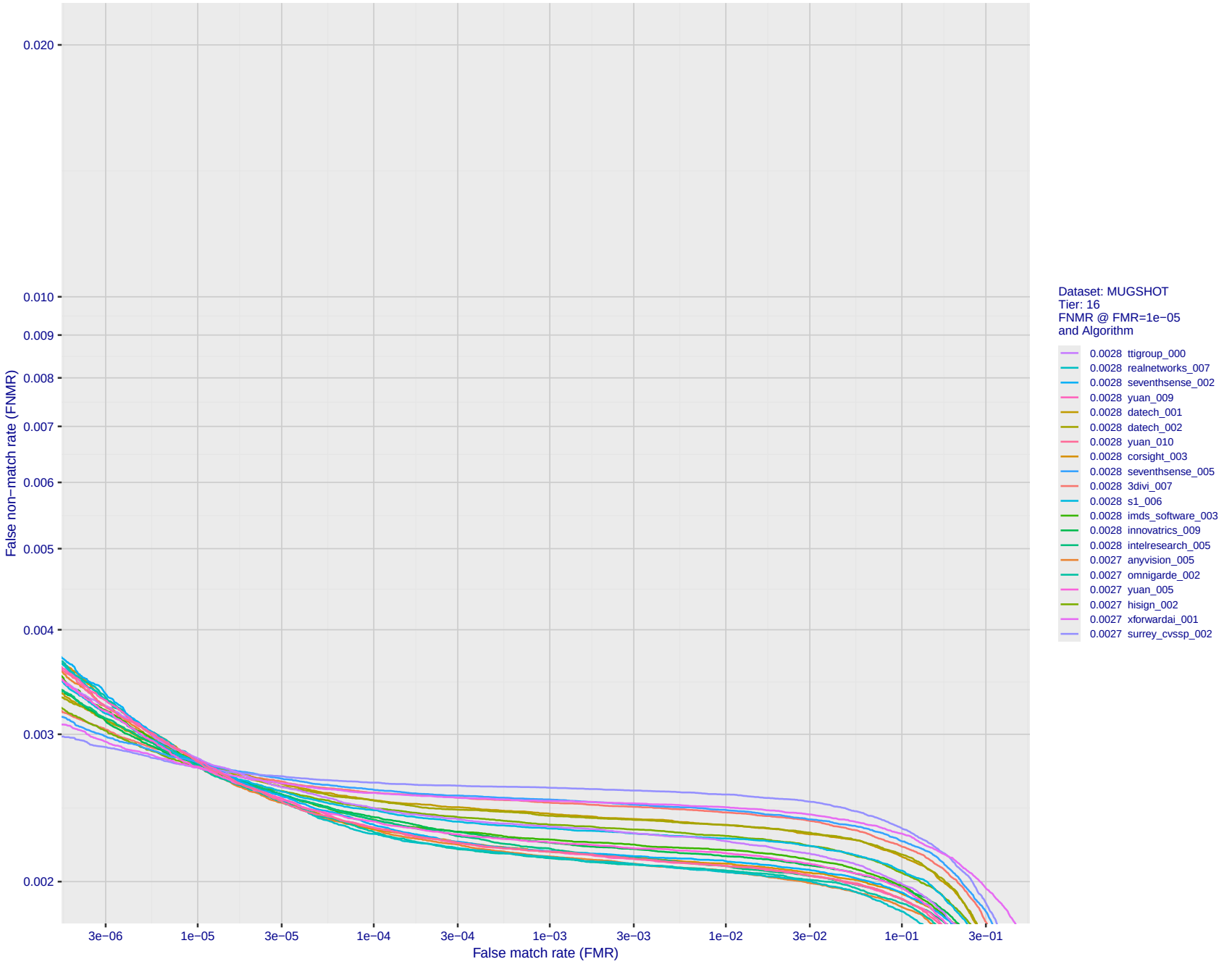

Figure 238: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

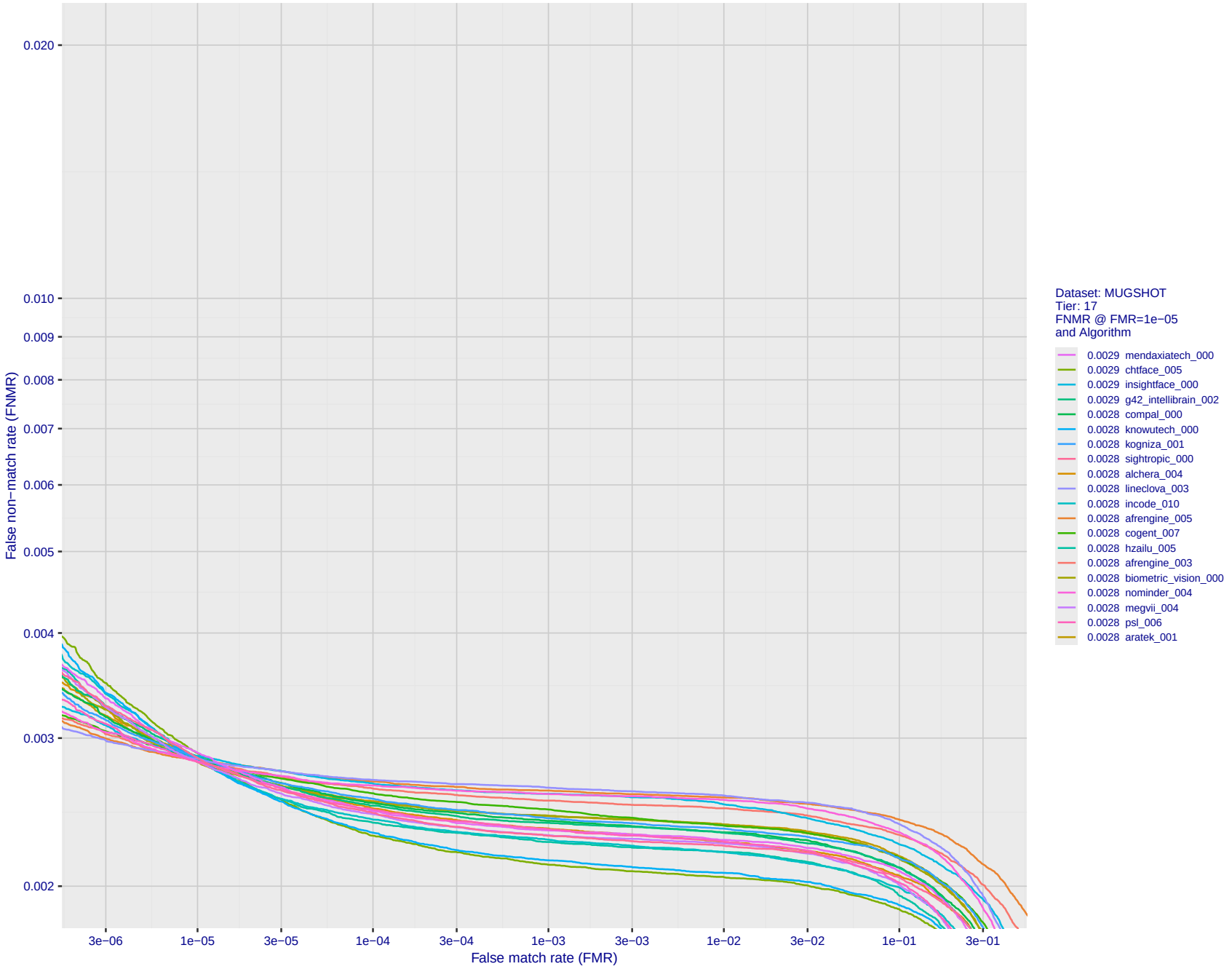

*Figure 239: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

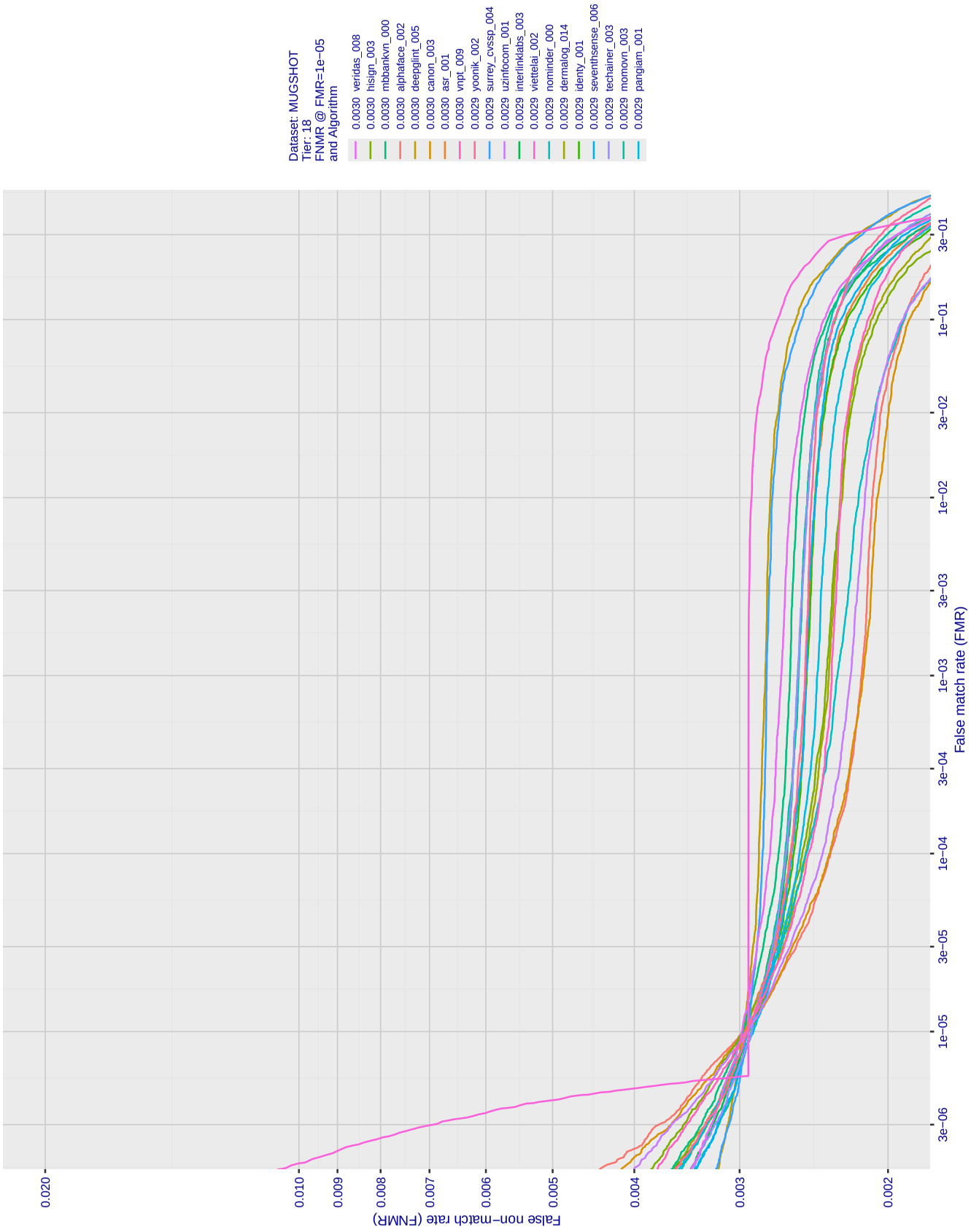

Figure 240: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

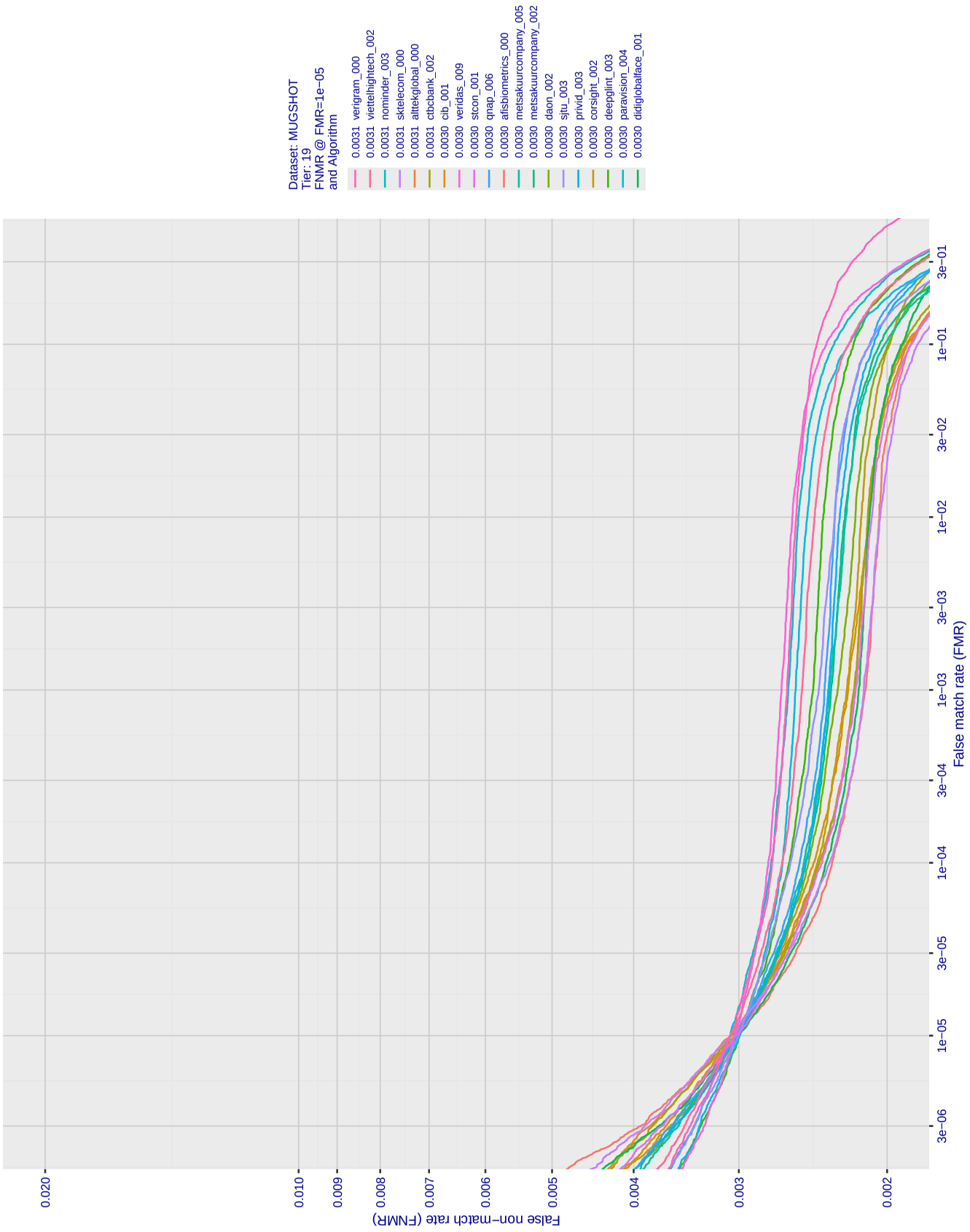

Figure 241: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

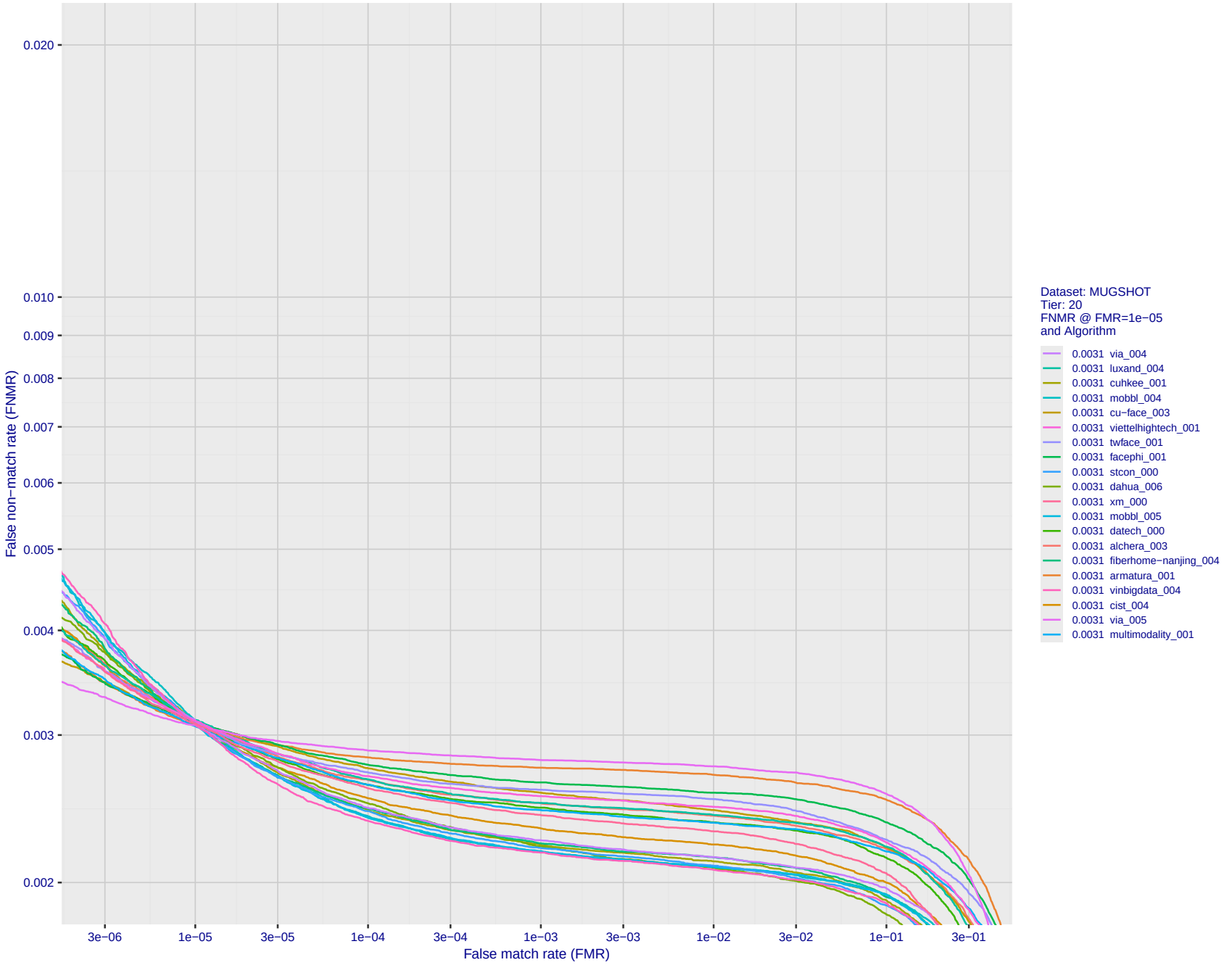

*Figure 242: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

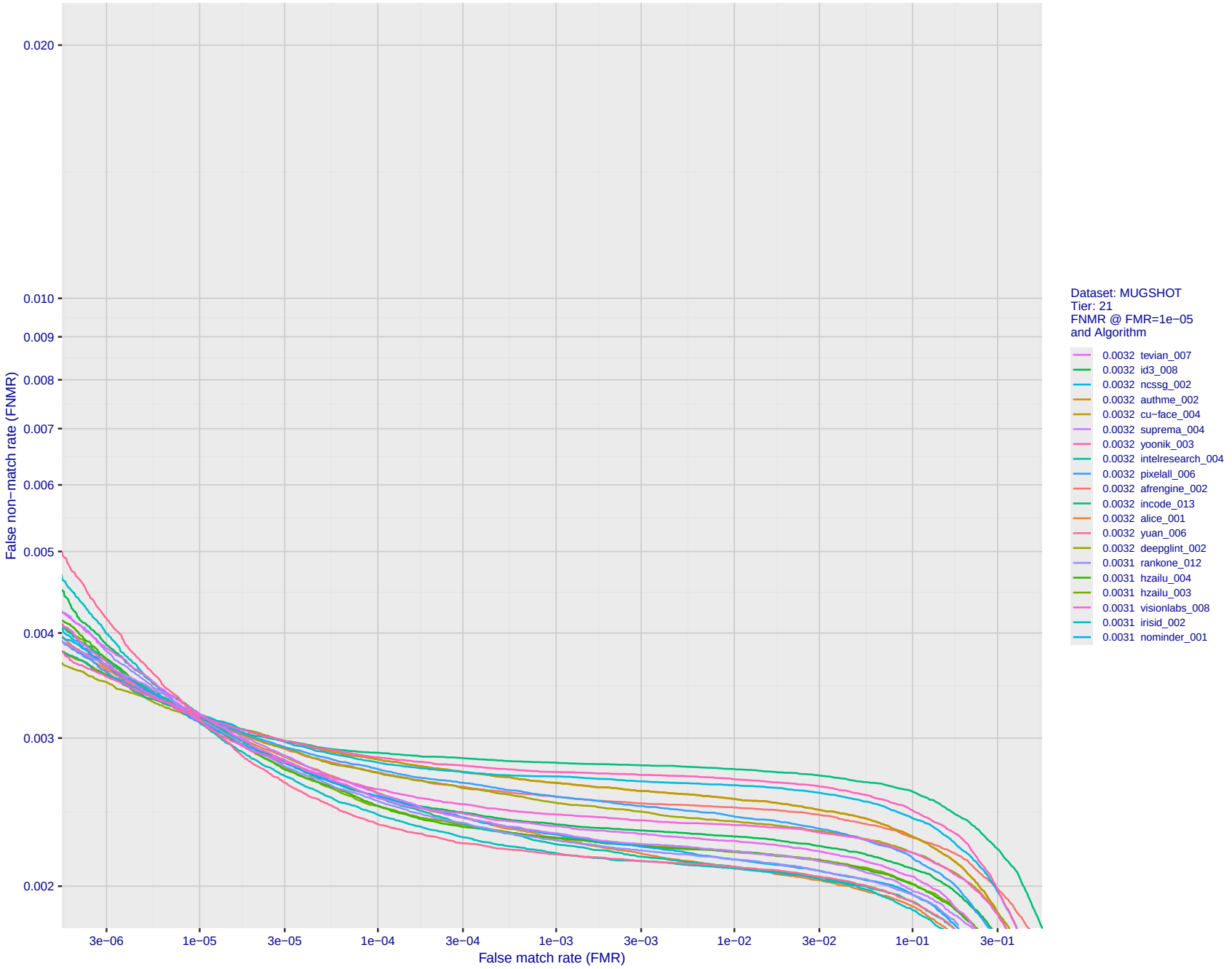

Figure 243: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

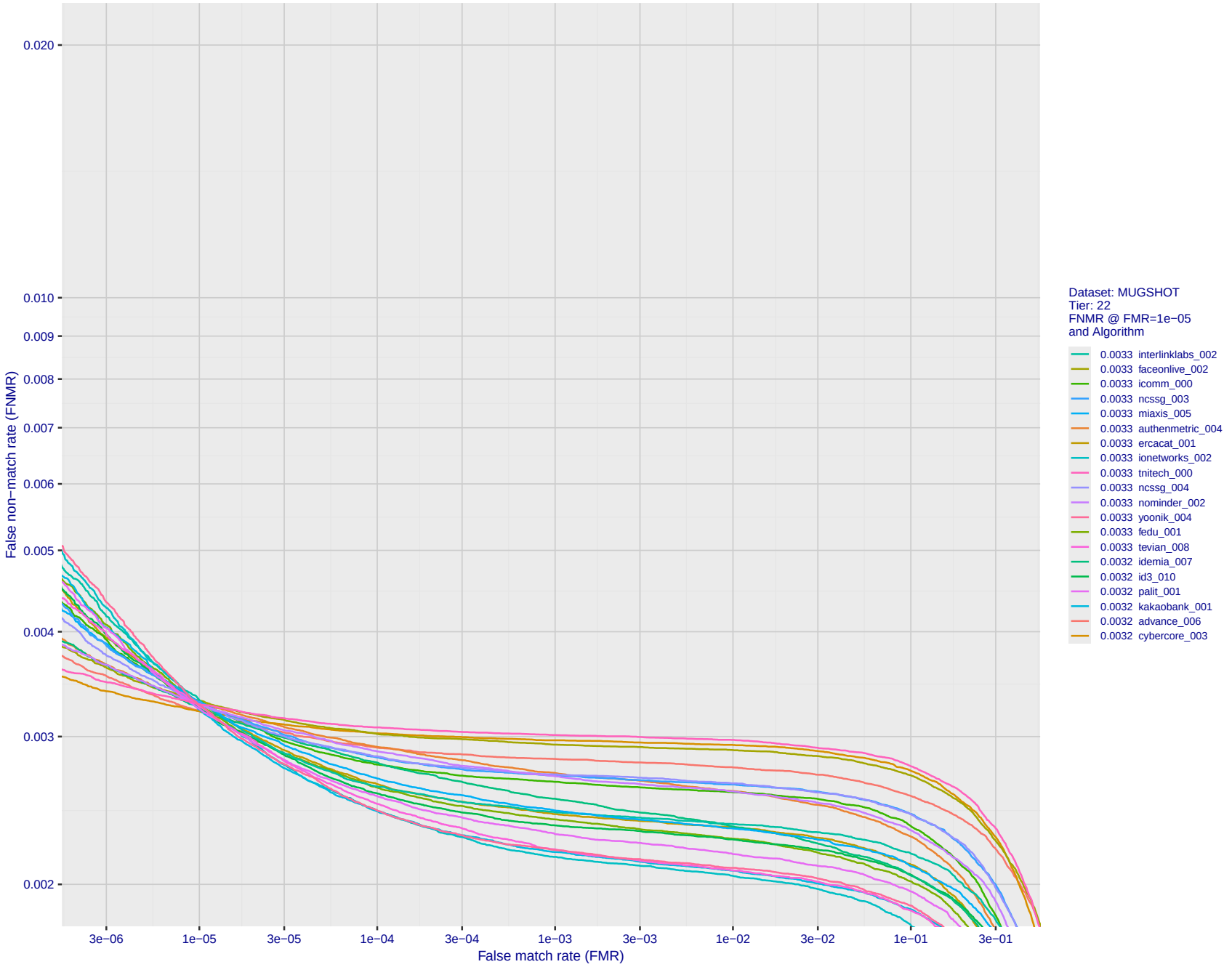

Figure 244: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

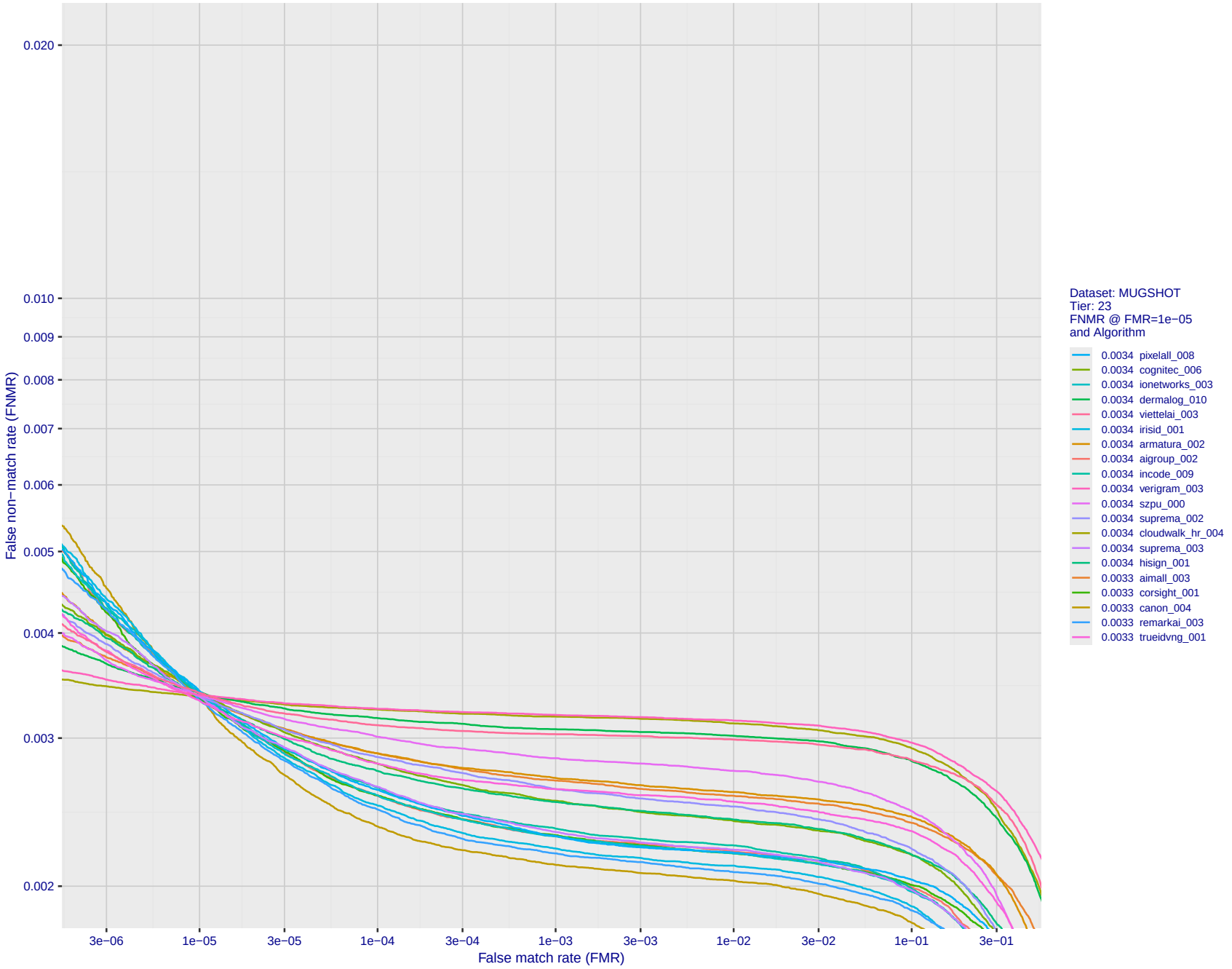

Figure 245: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

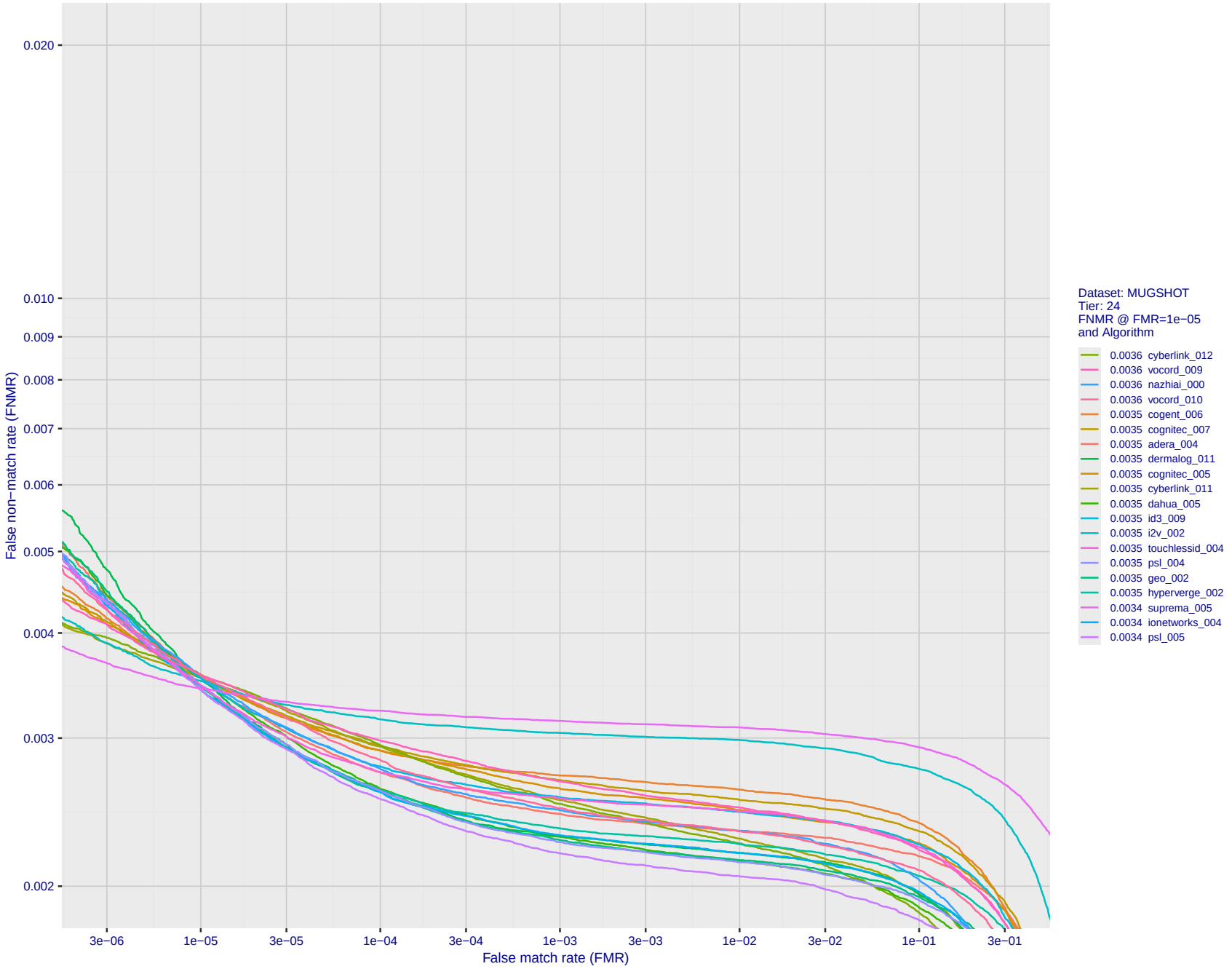

Figure 246: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

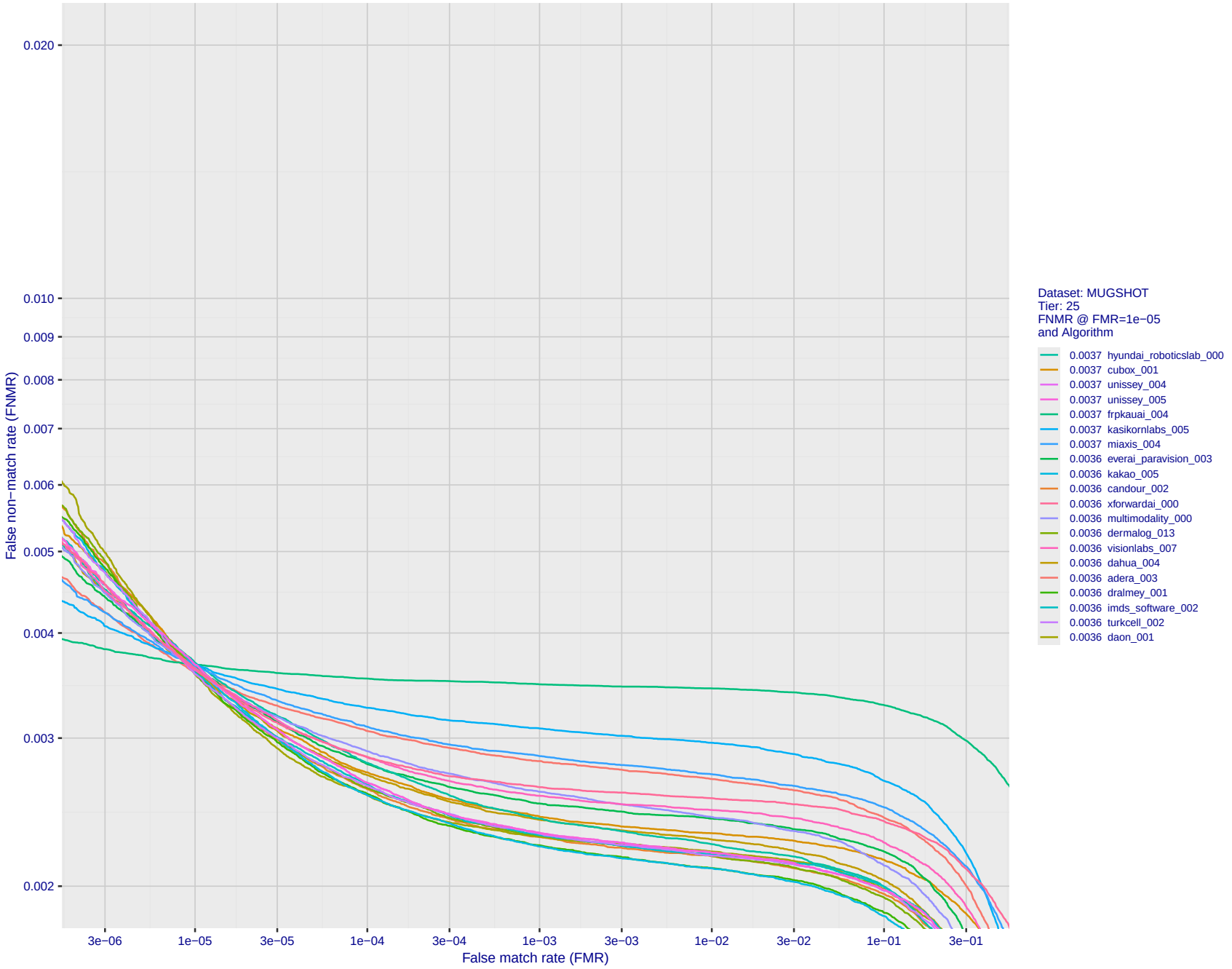

Figure 247: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

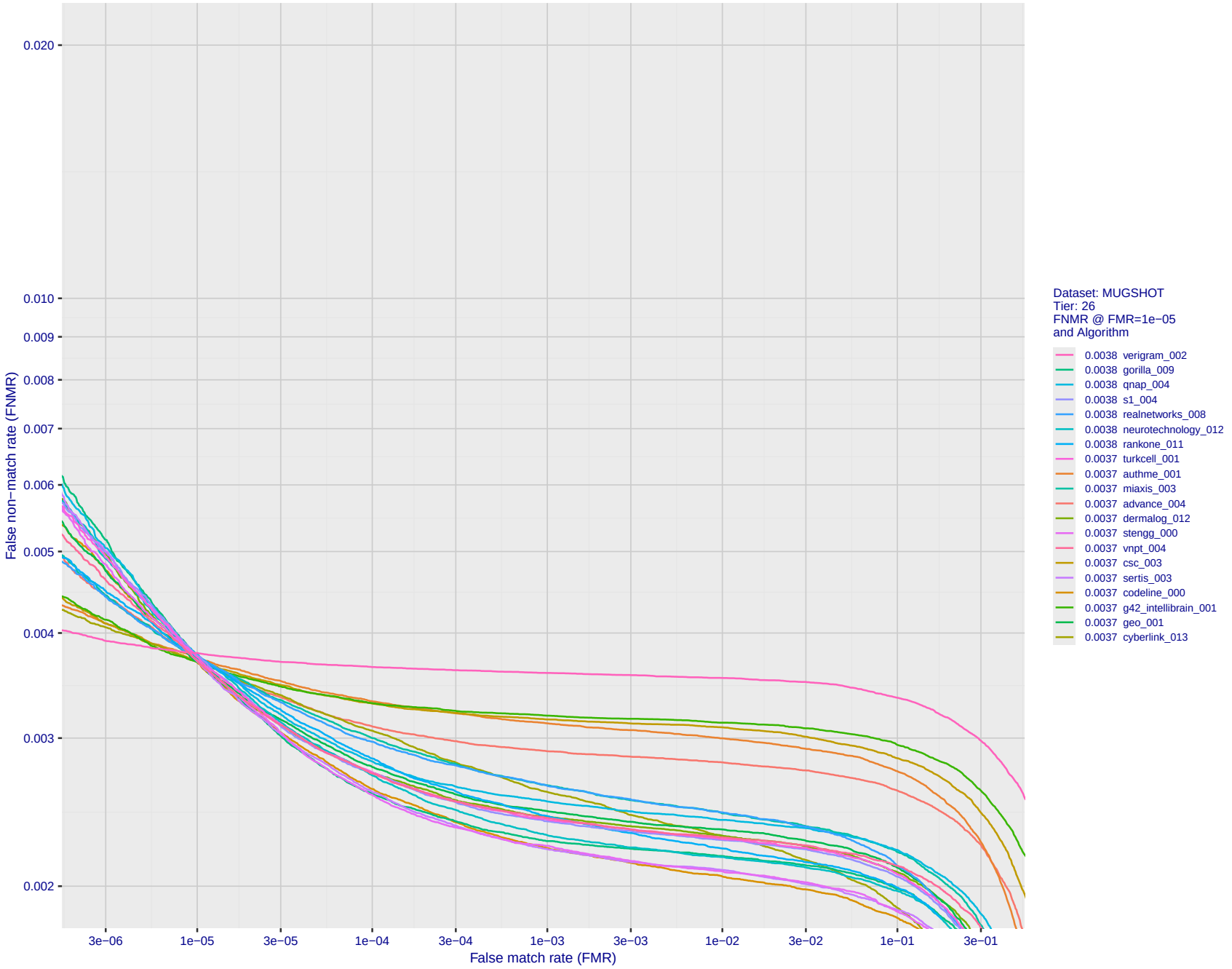

Figure 248: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

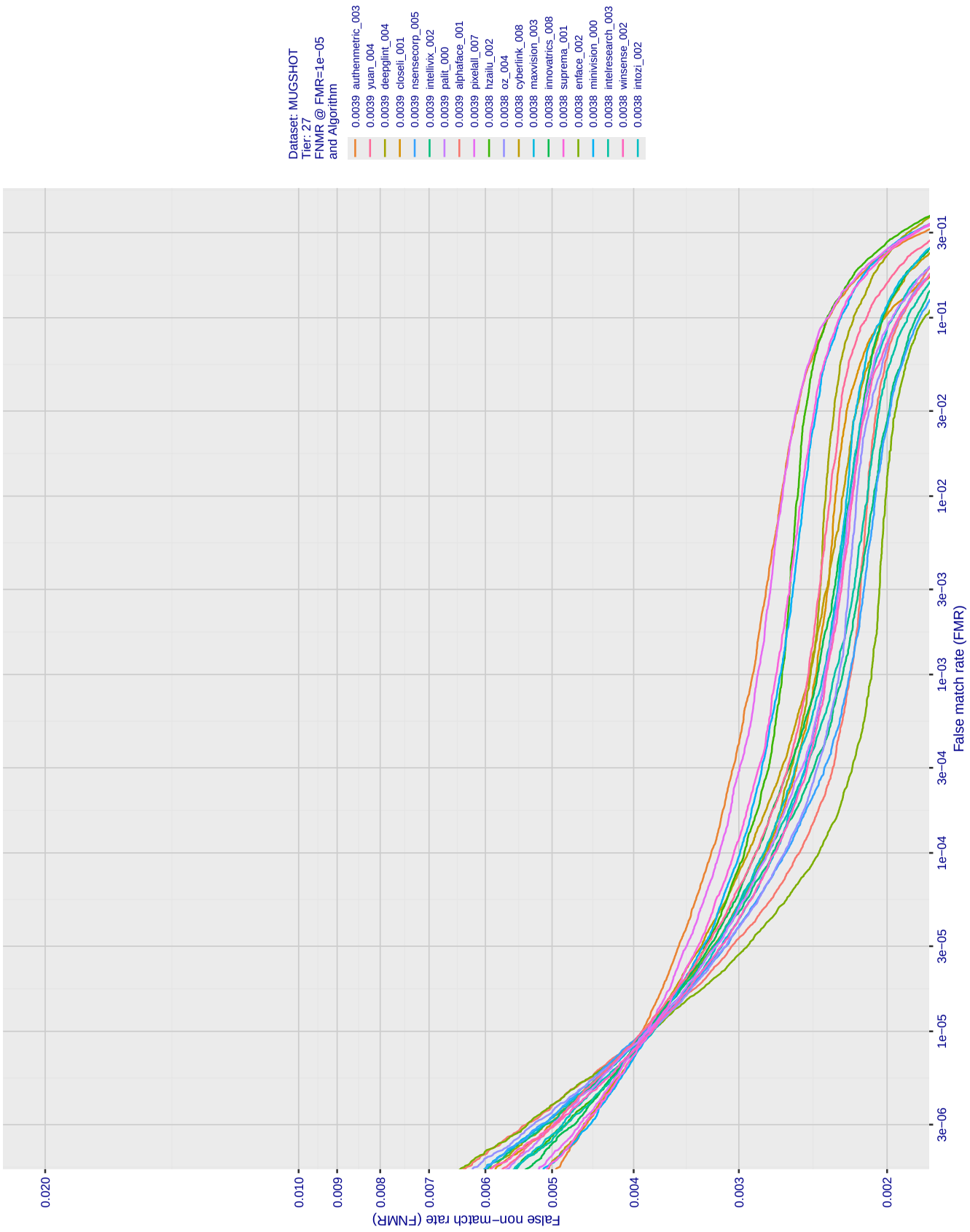

Figure 249: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

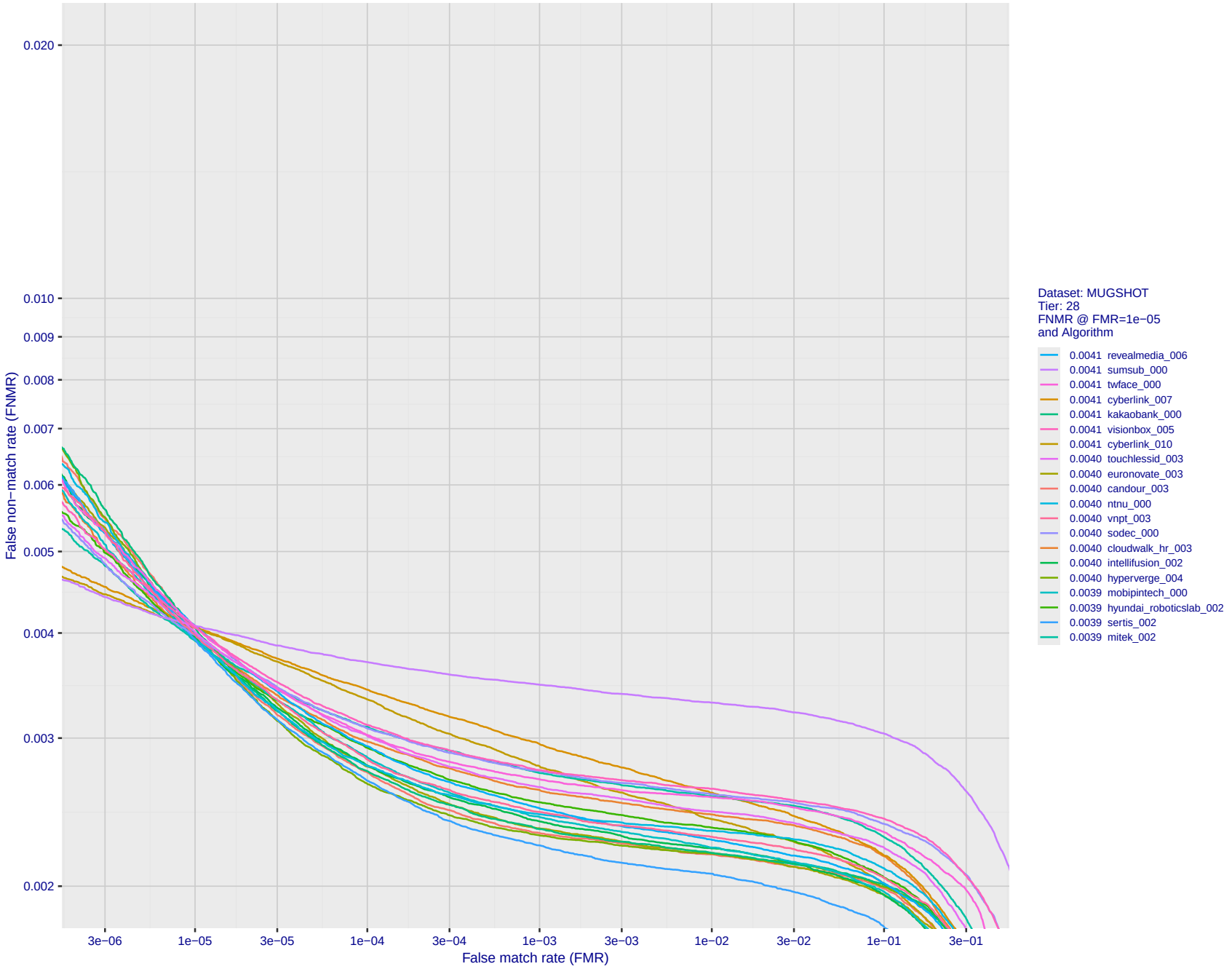

Figure 250: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

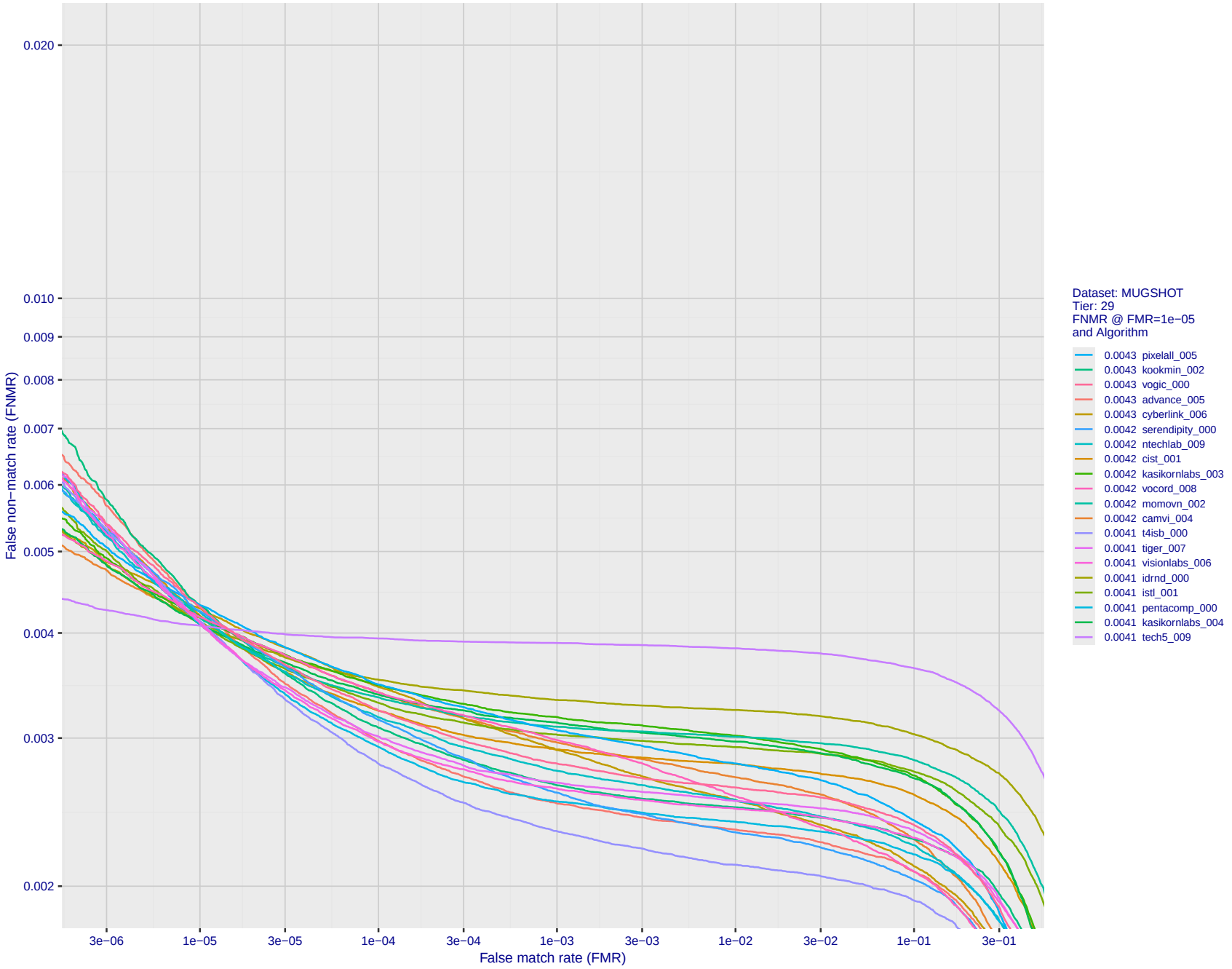

*Figure 251: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

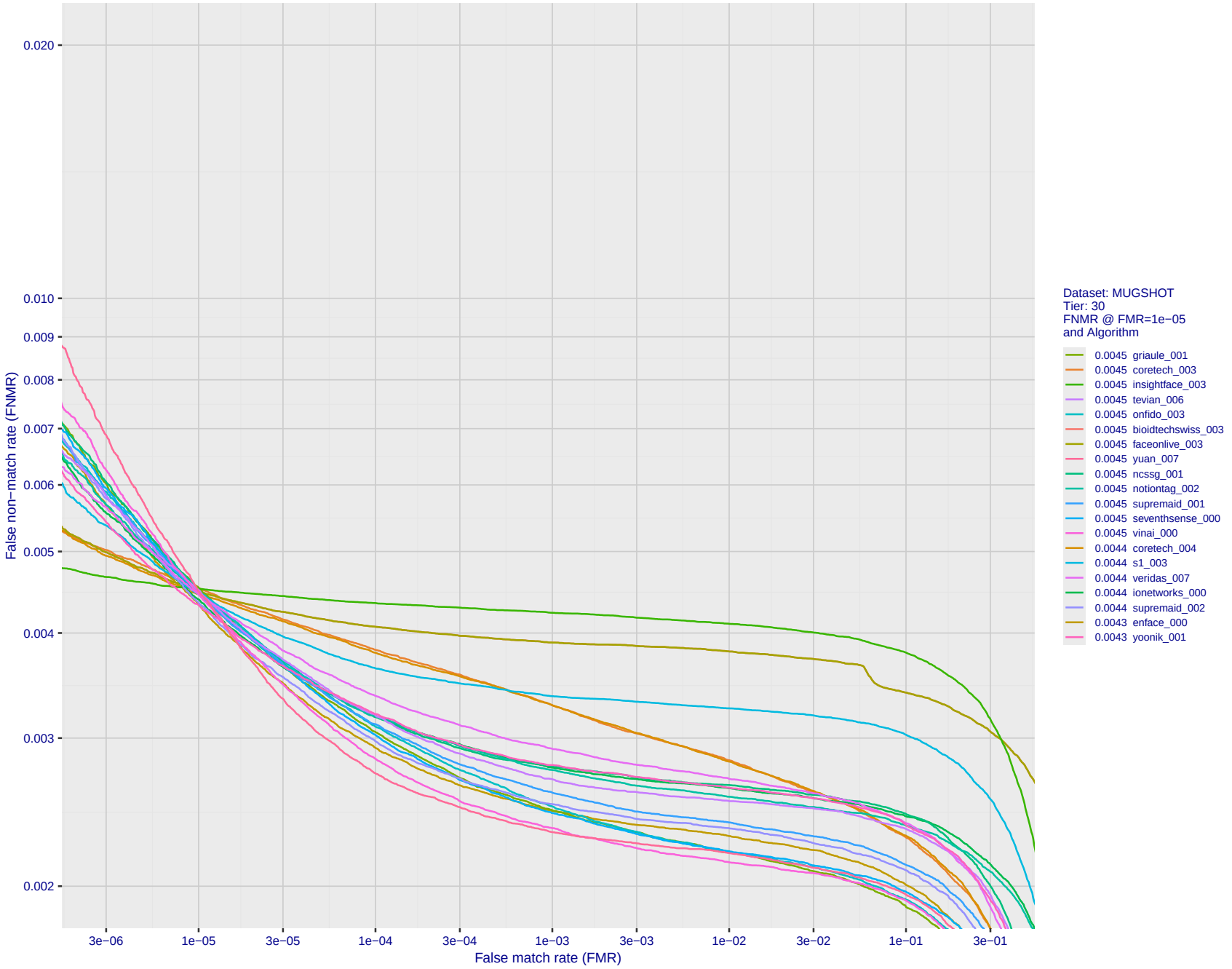

Figure 252: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

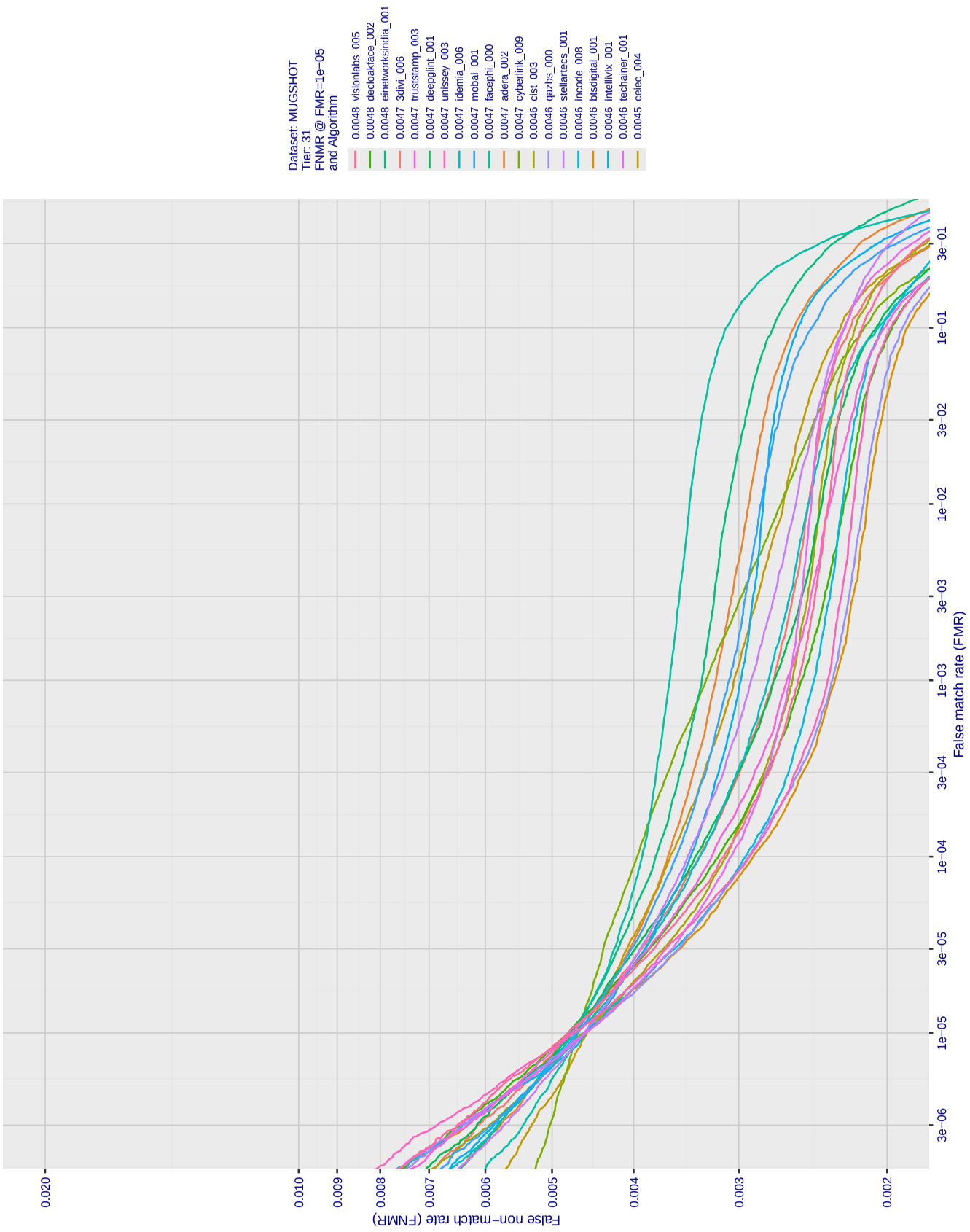

Figure 253: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

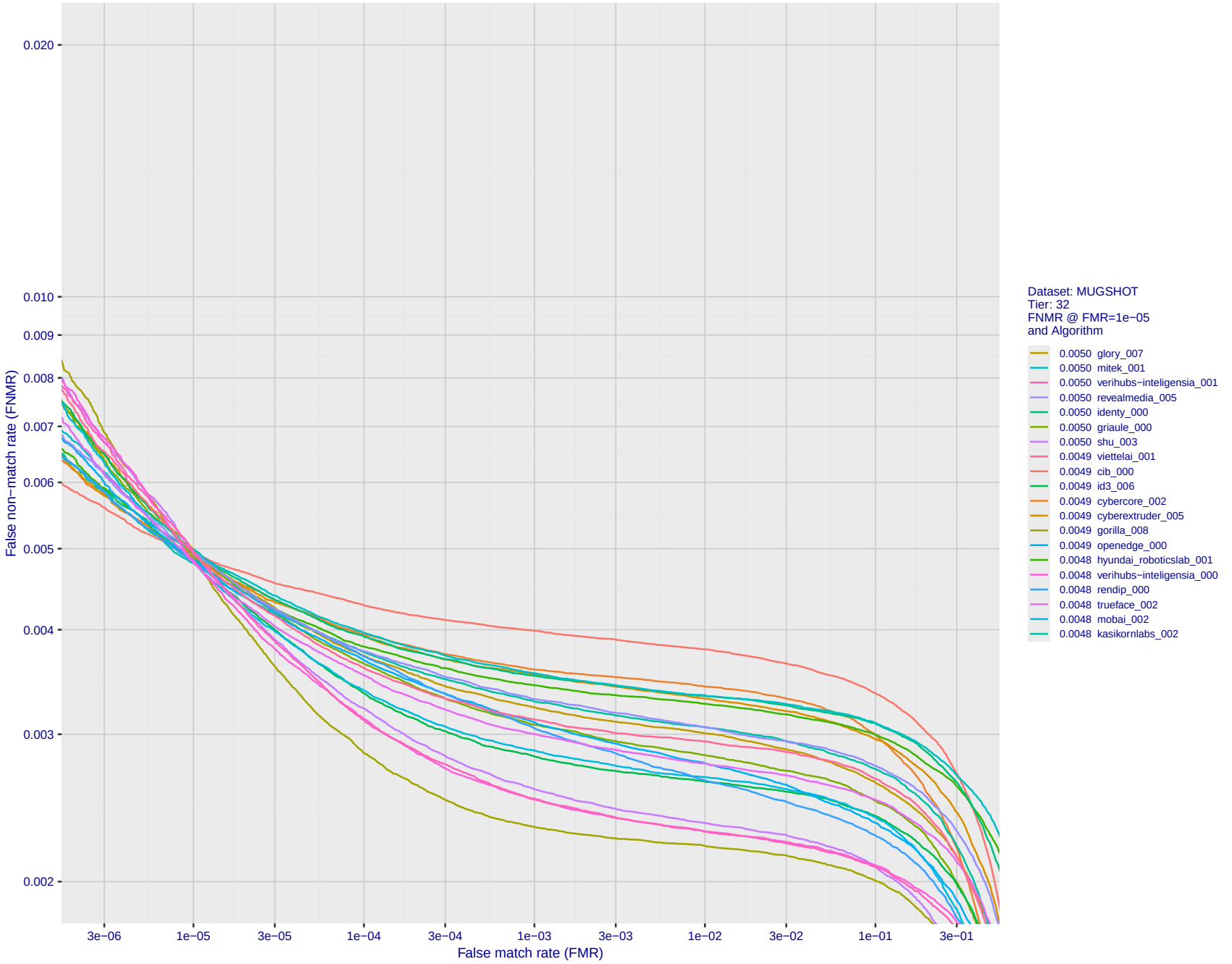

*Figure 254: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

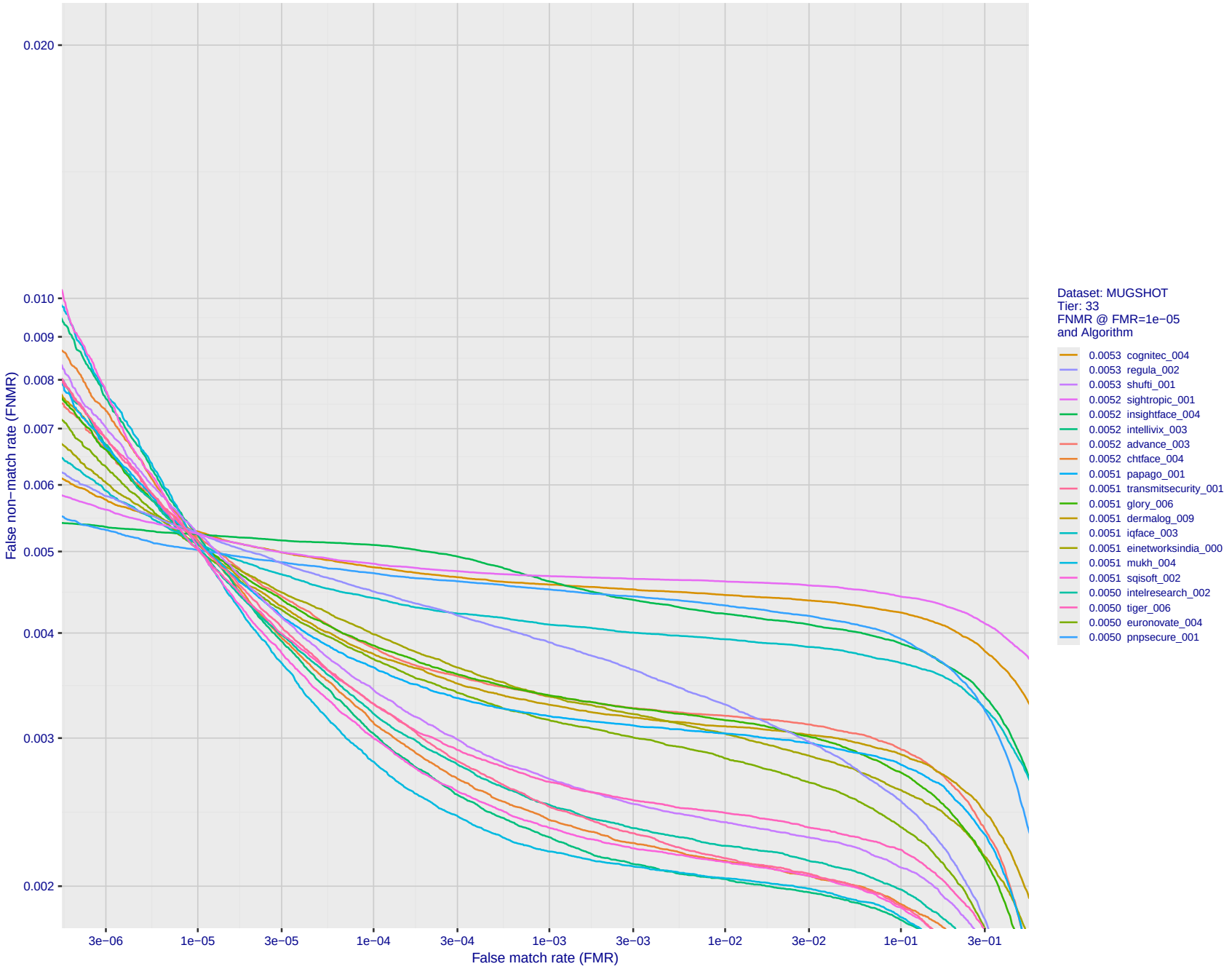

Figure 255: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

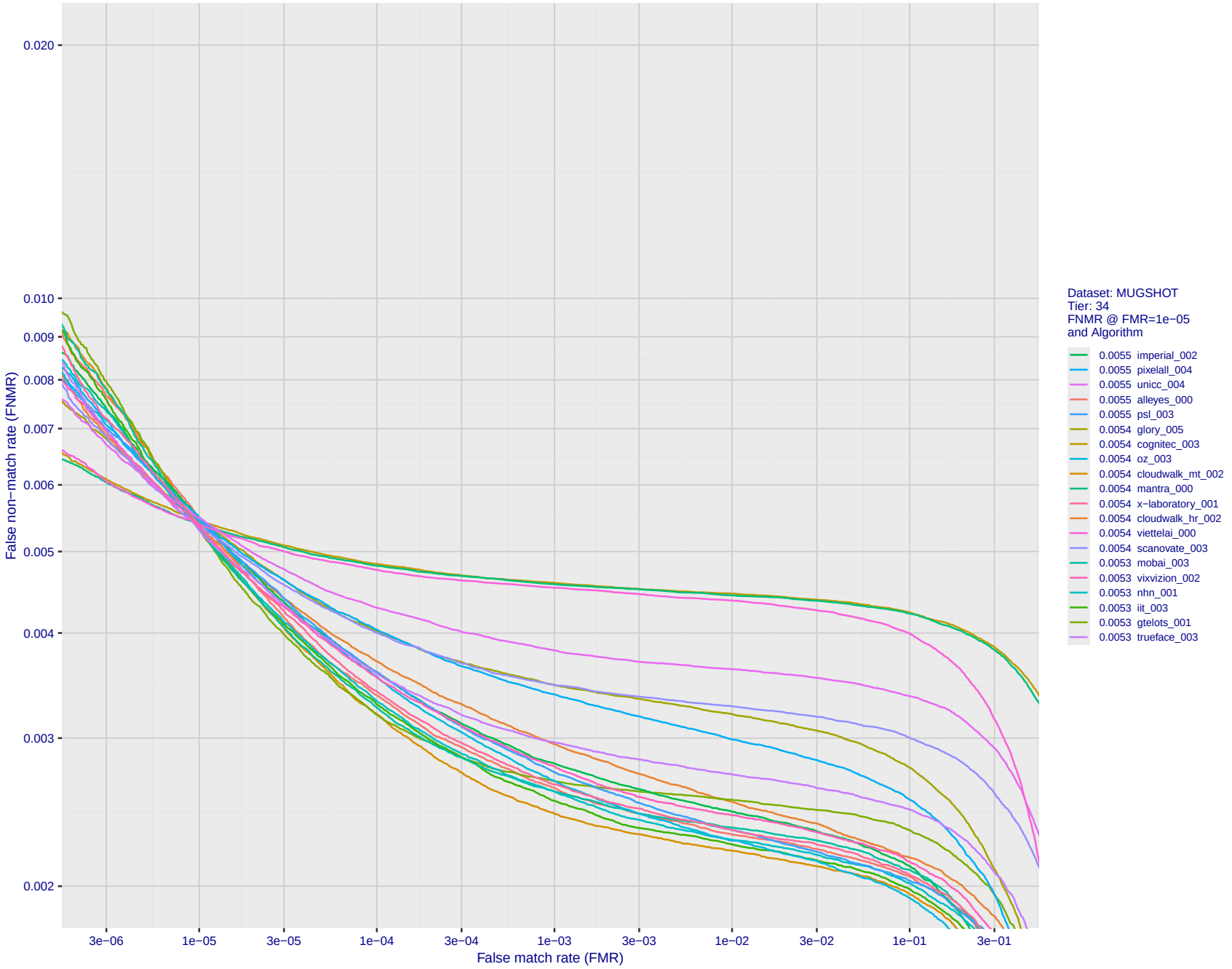

Figure 256: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

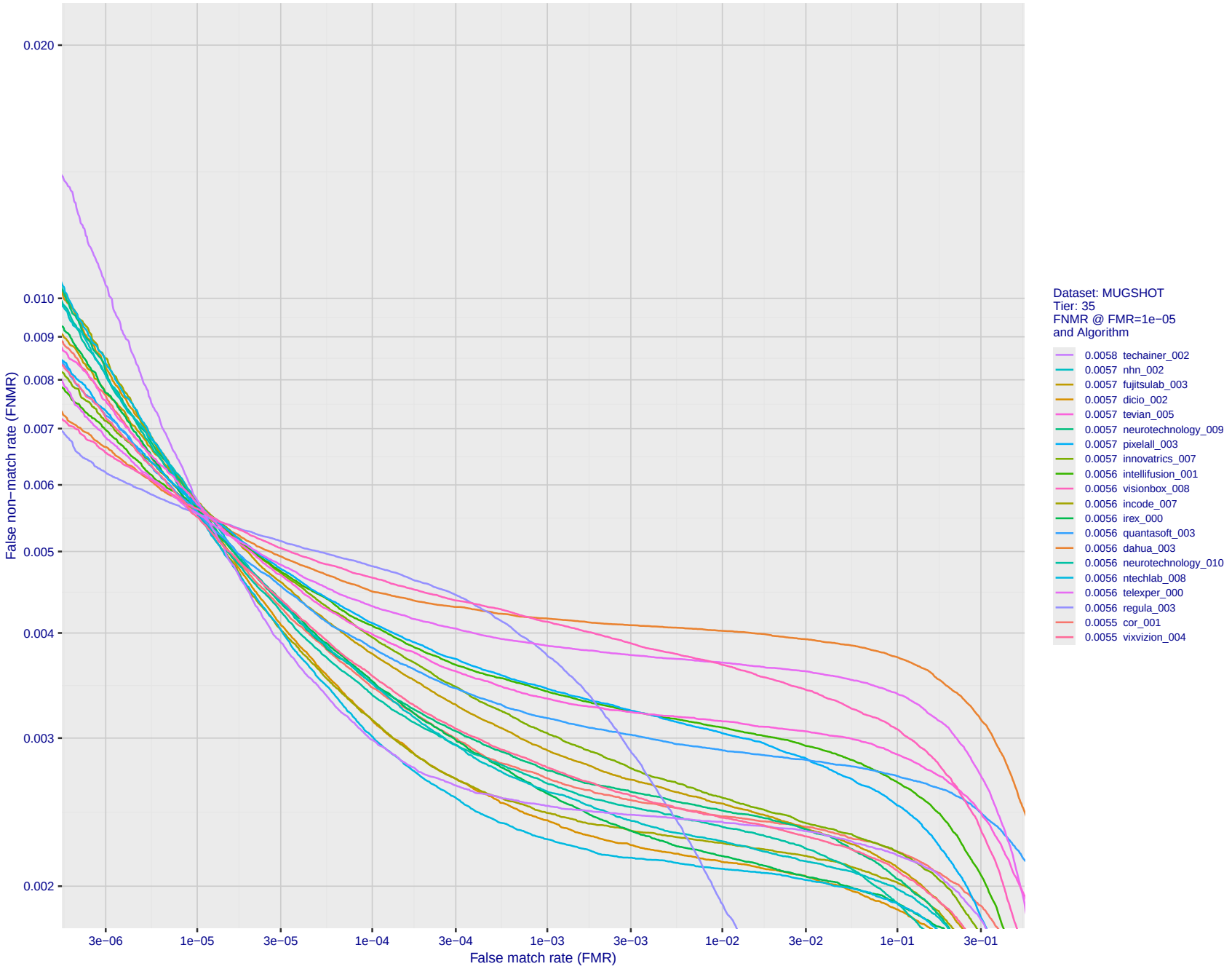

Figure 257: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

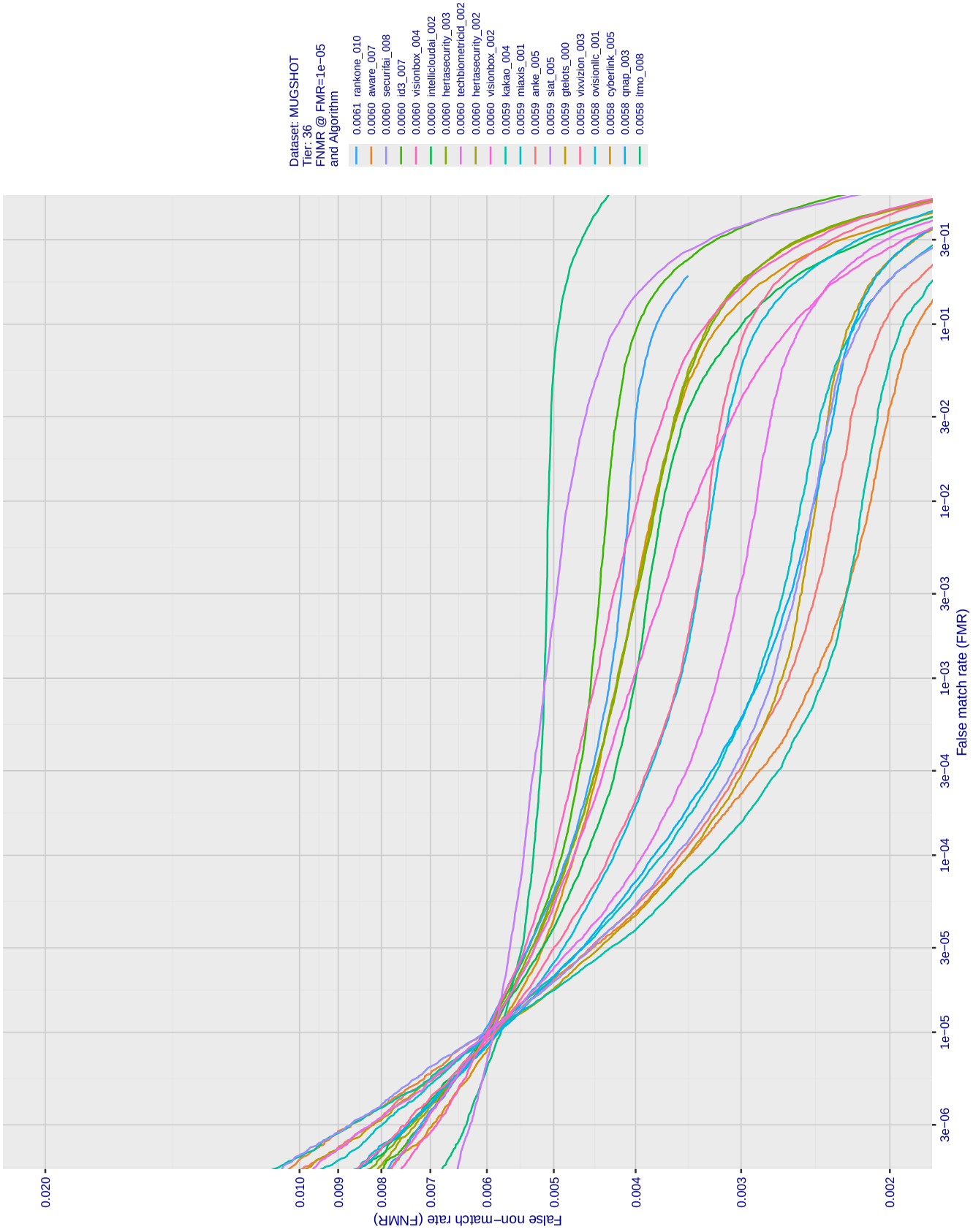

Figure 258: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

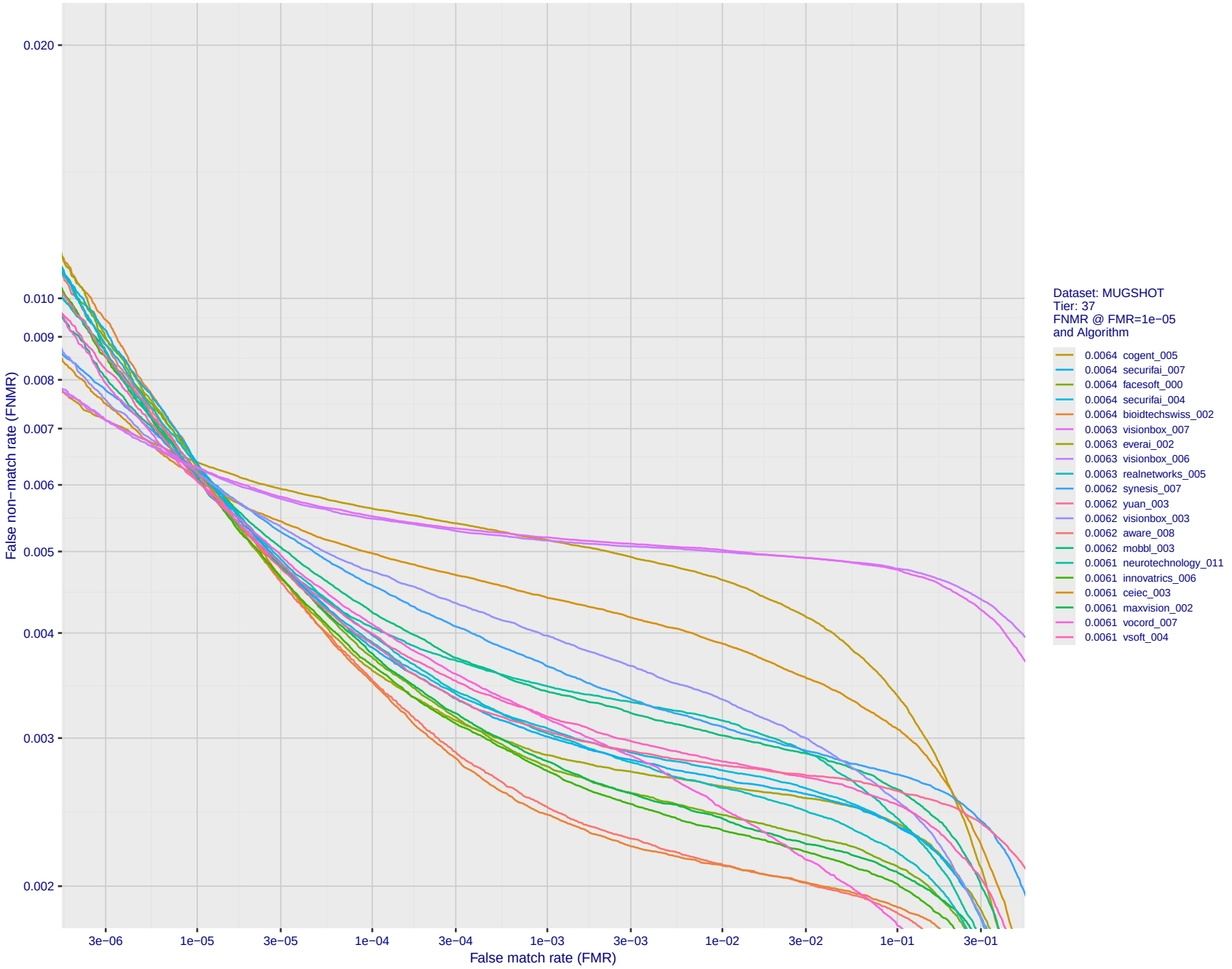

Figure 259: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

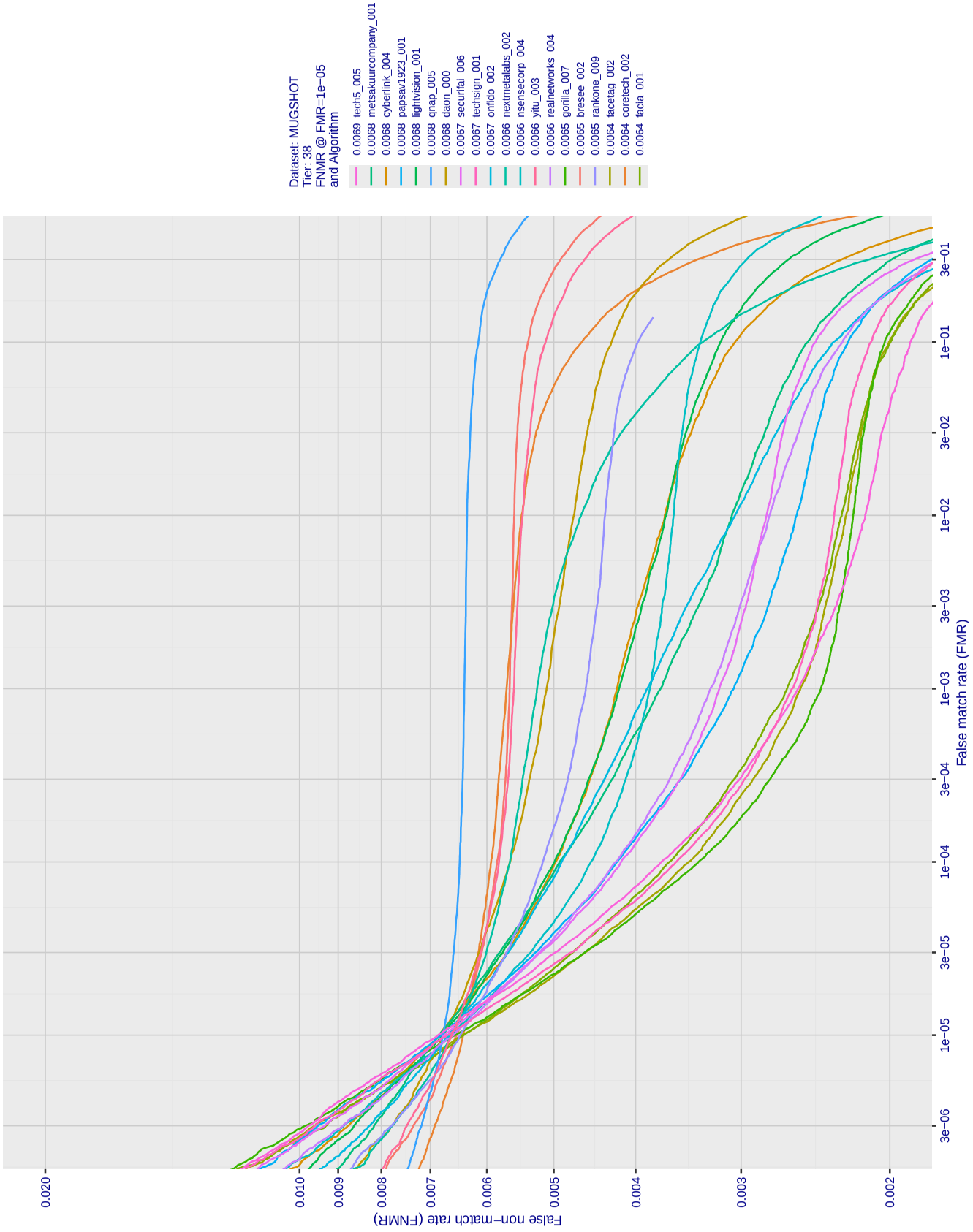

Figure 260: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

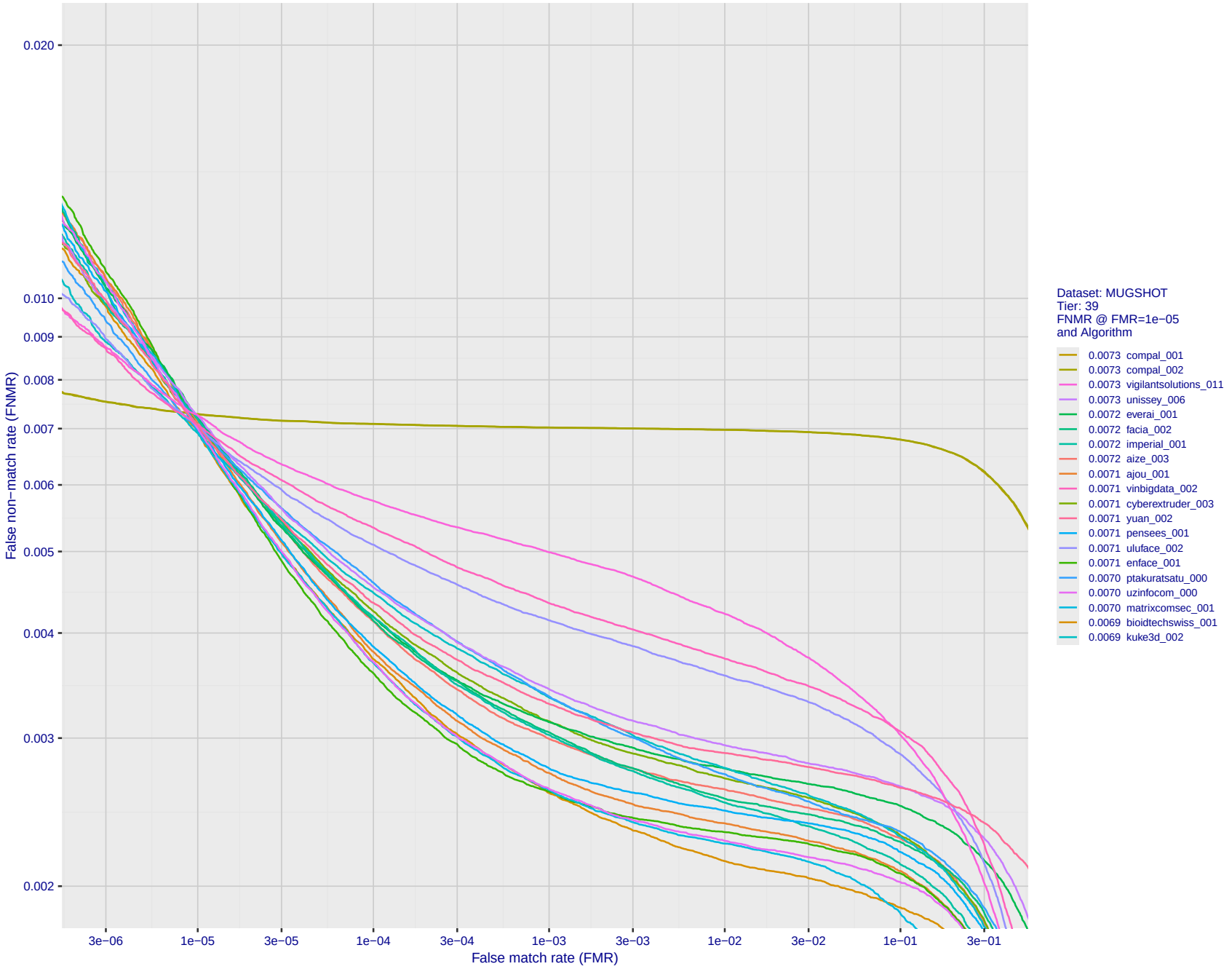

Figure 261: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

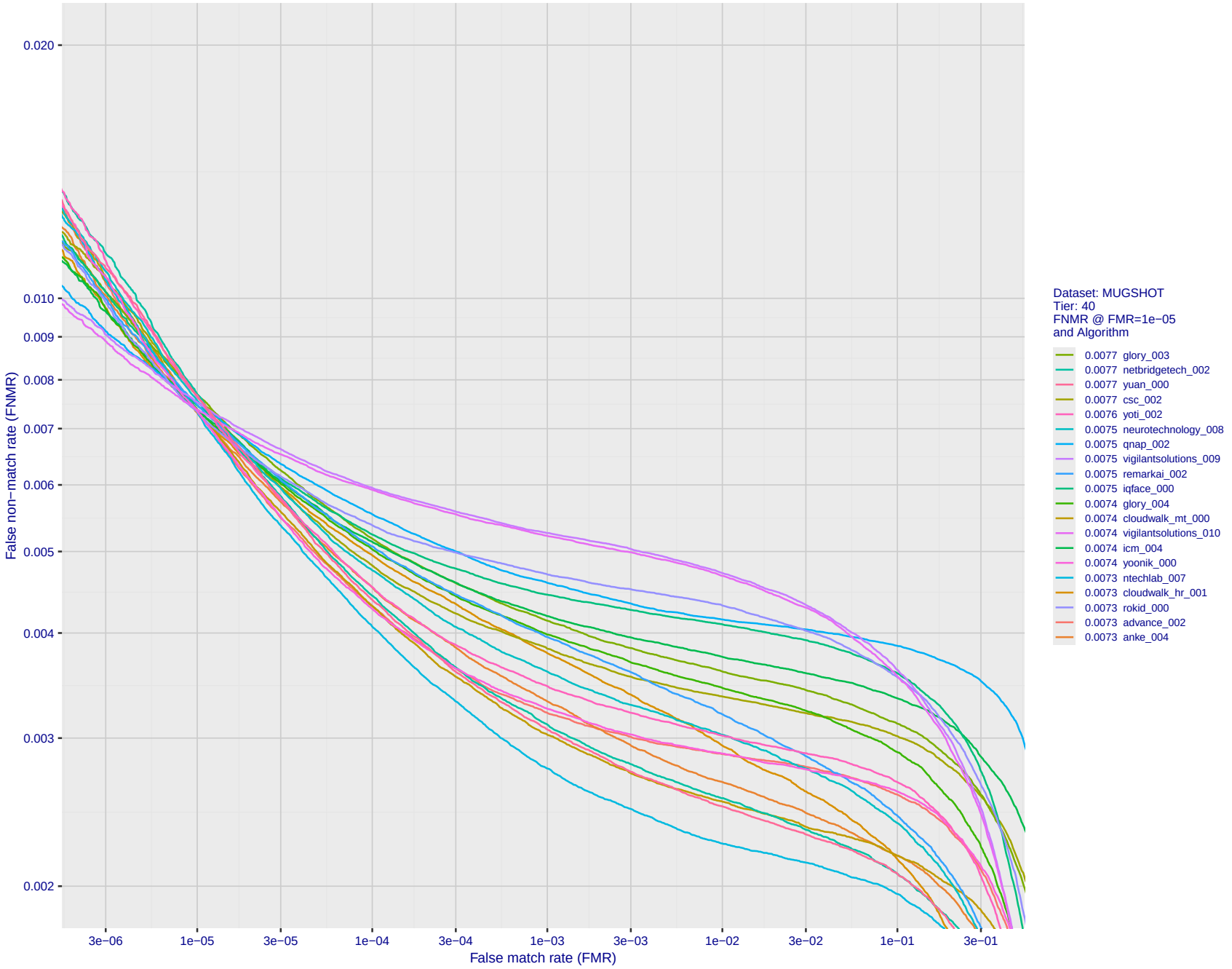

Figure 262: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

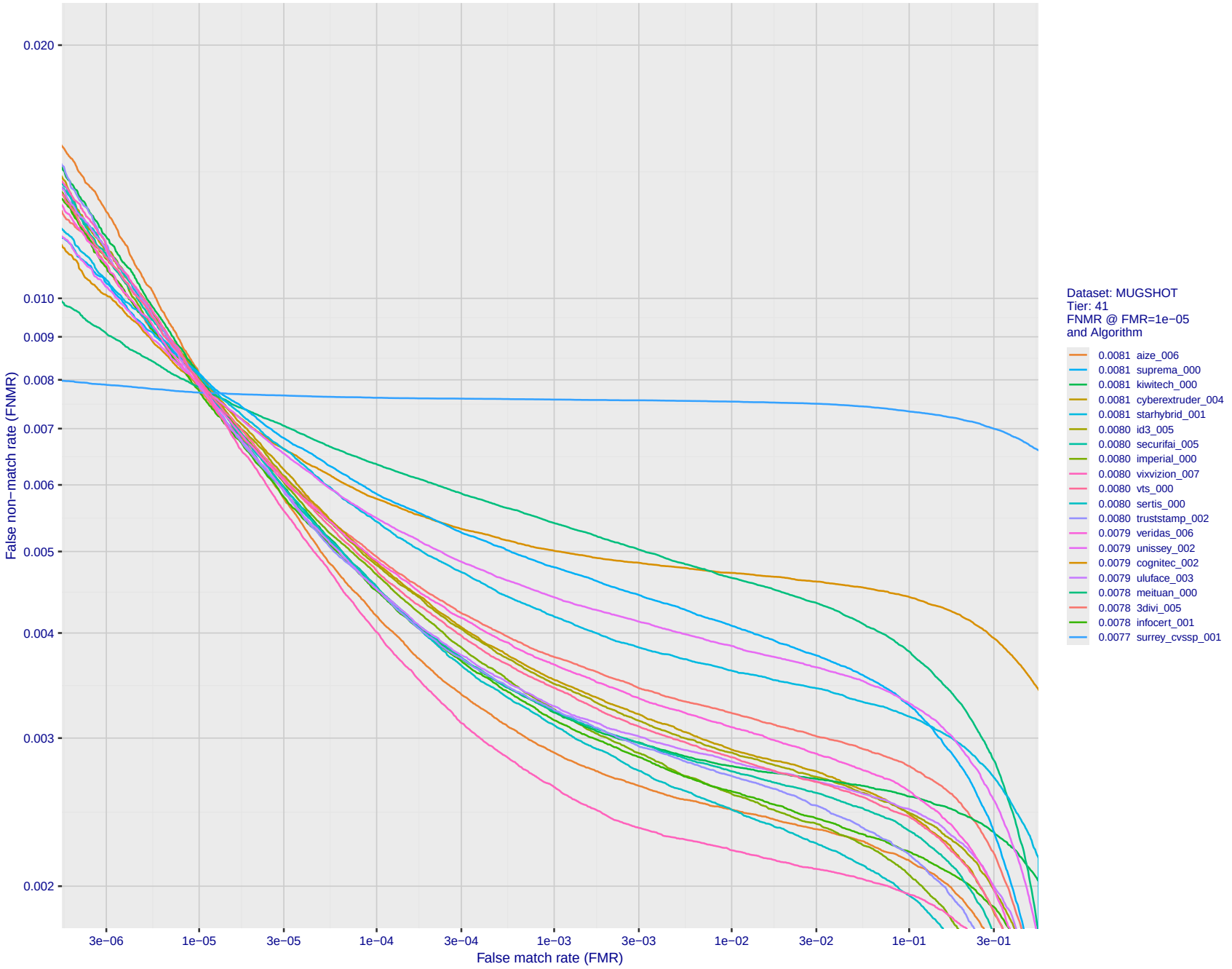

Figure 263: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

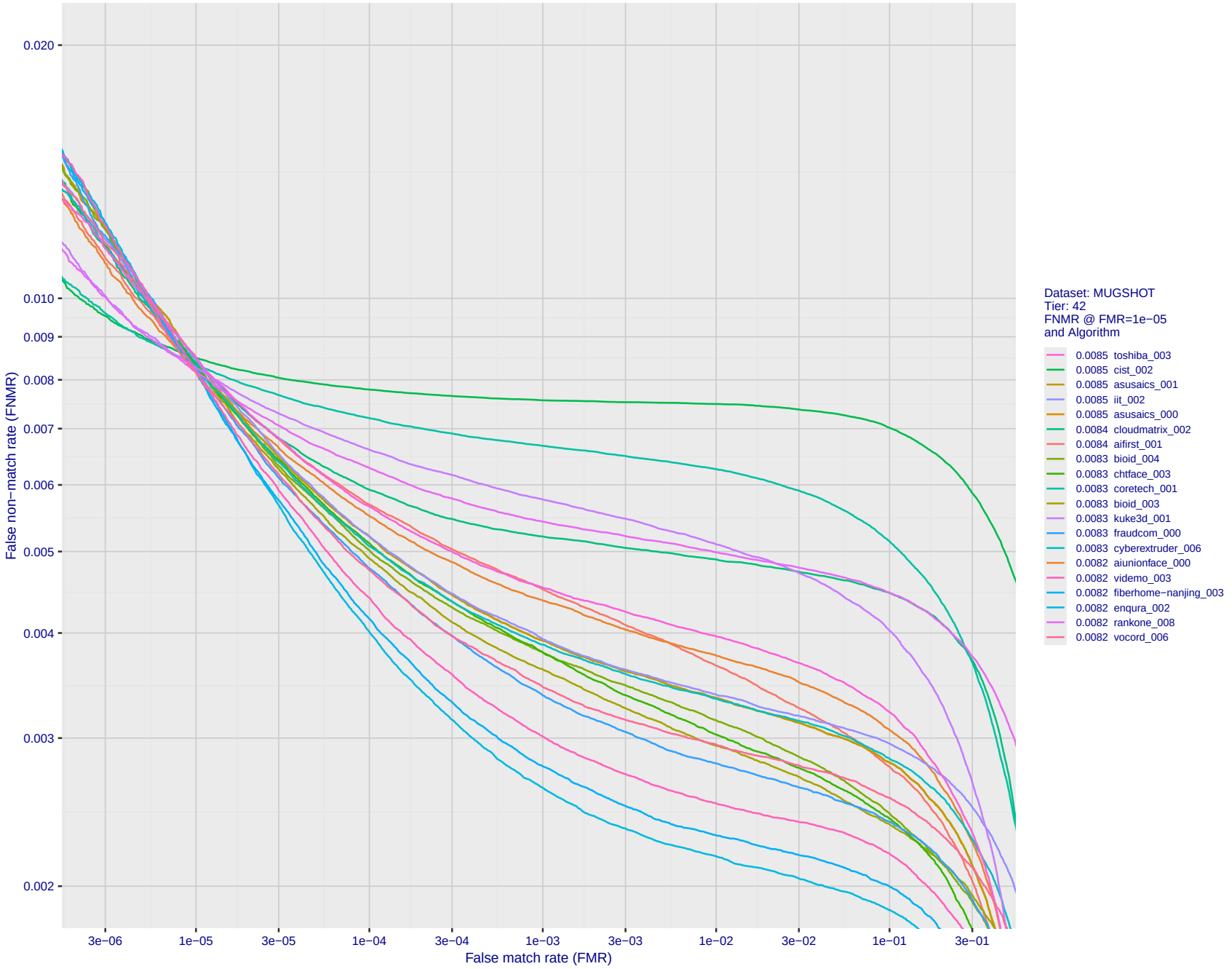

*Figure 264: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

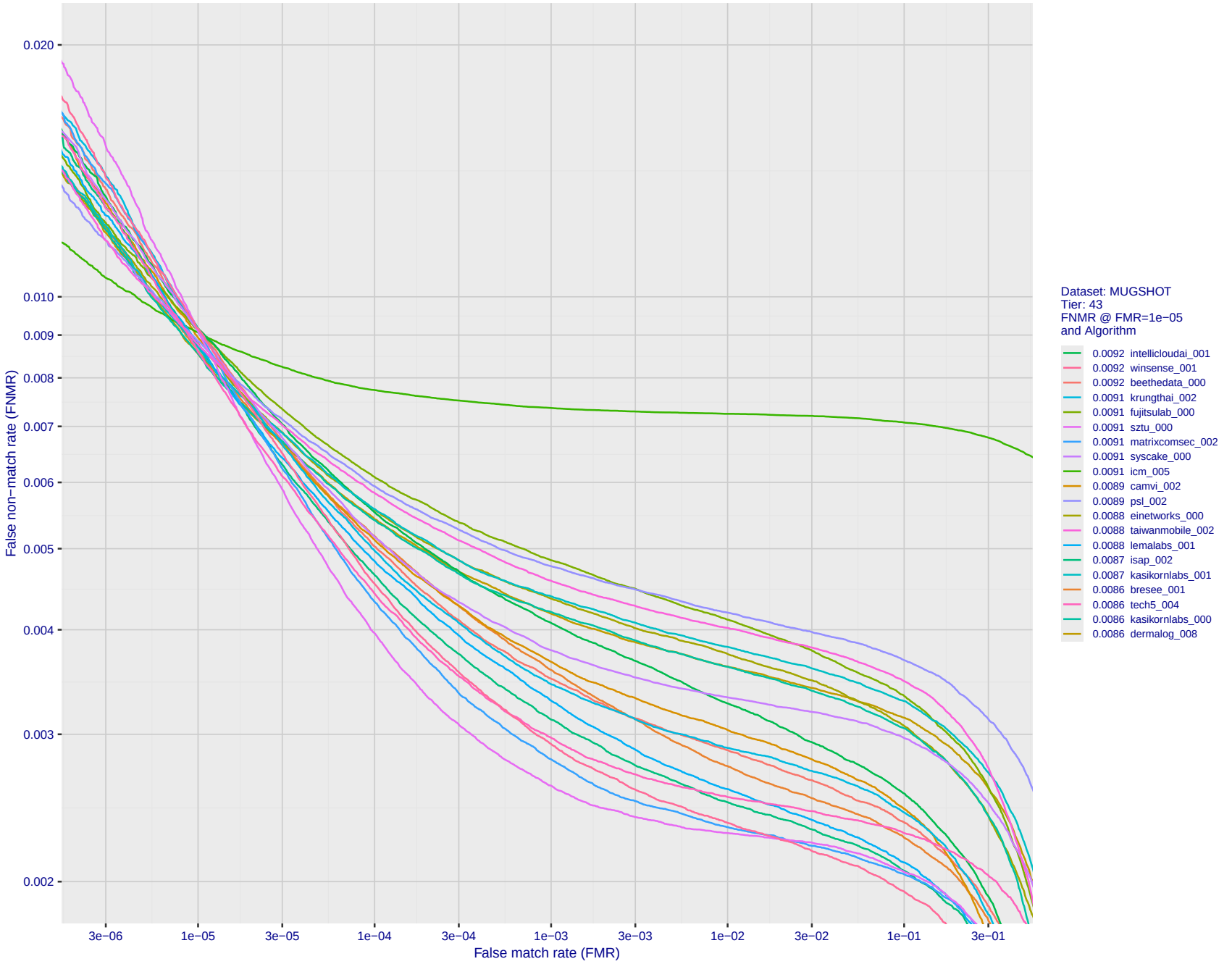

Figure 265: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

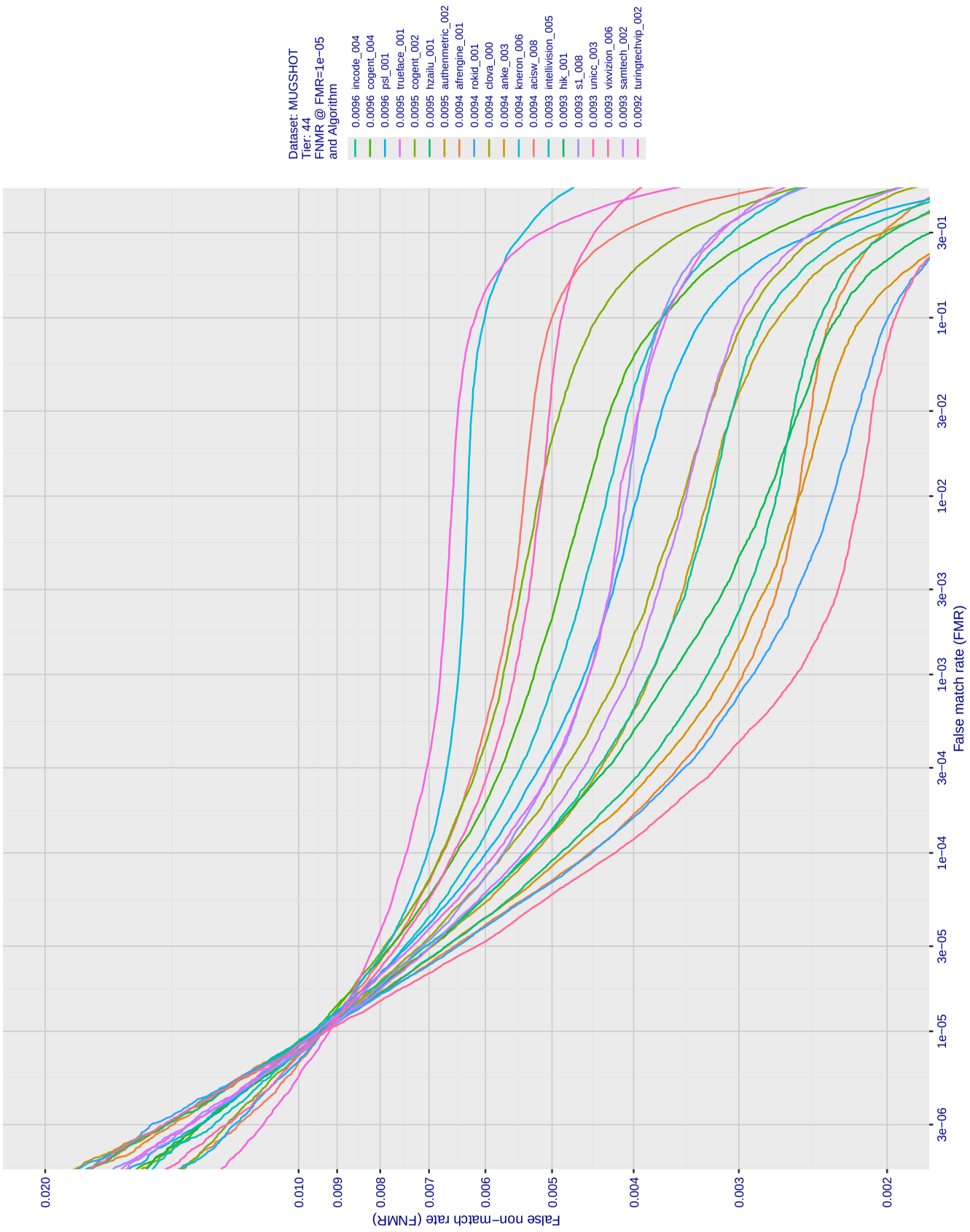

Figure 266: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

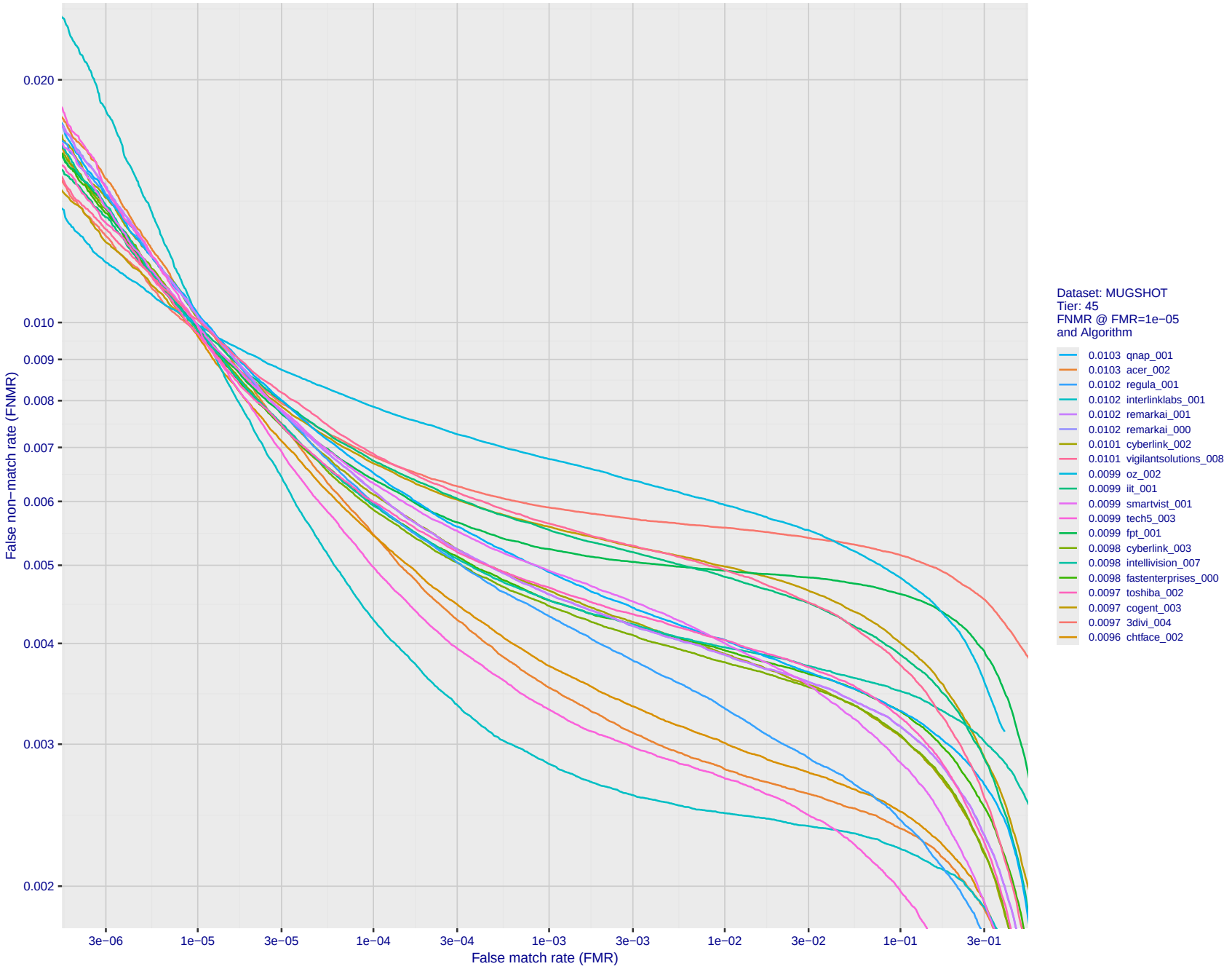

Figure 267: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

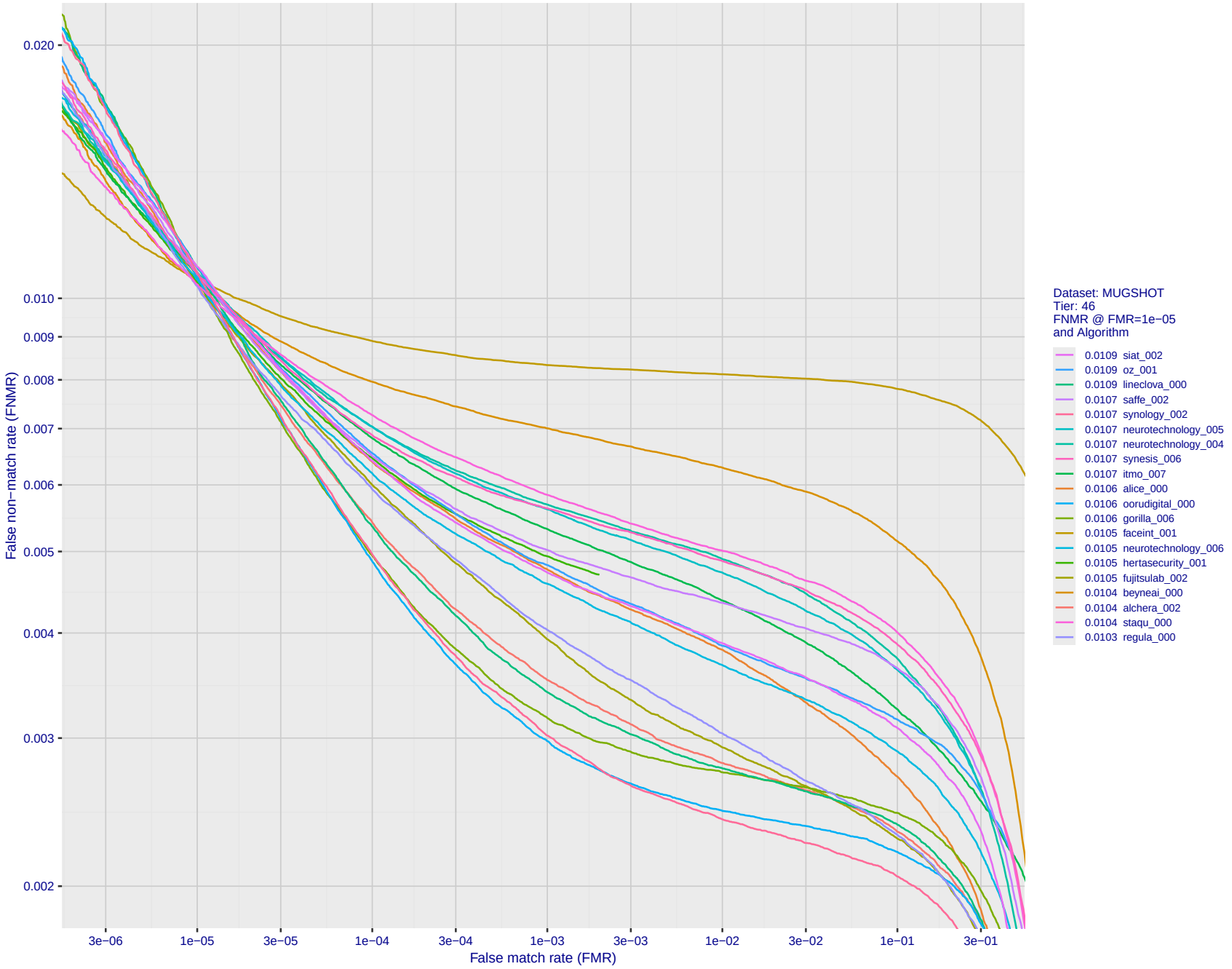

Figure 268: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

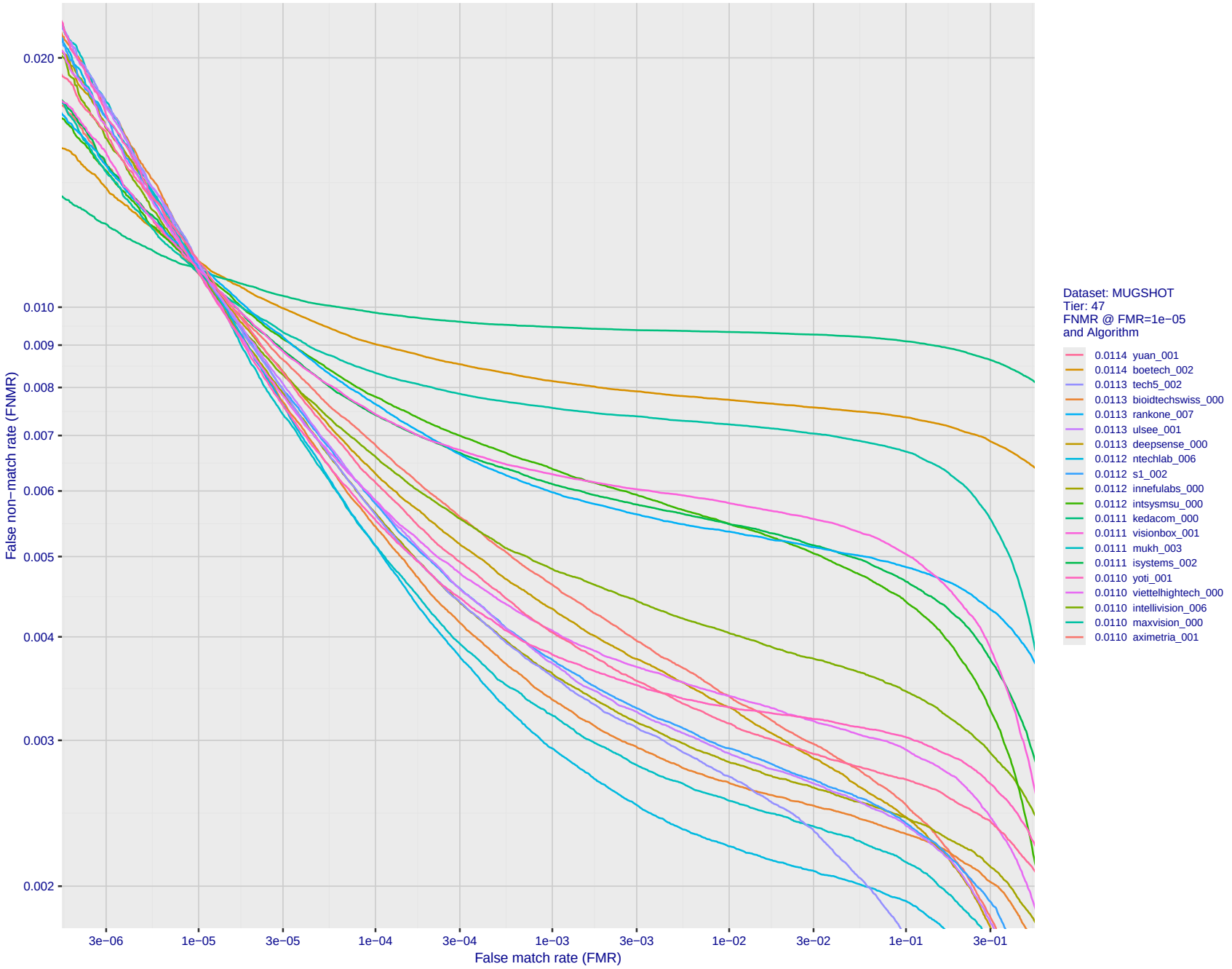

Figure 269: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

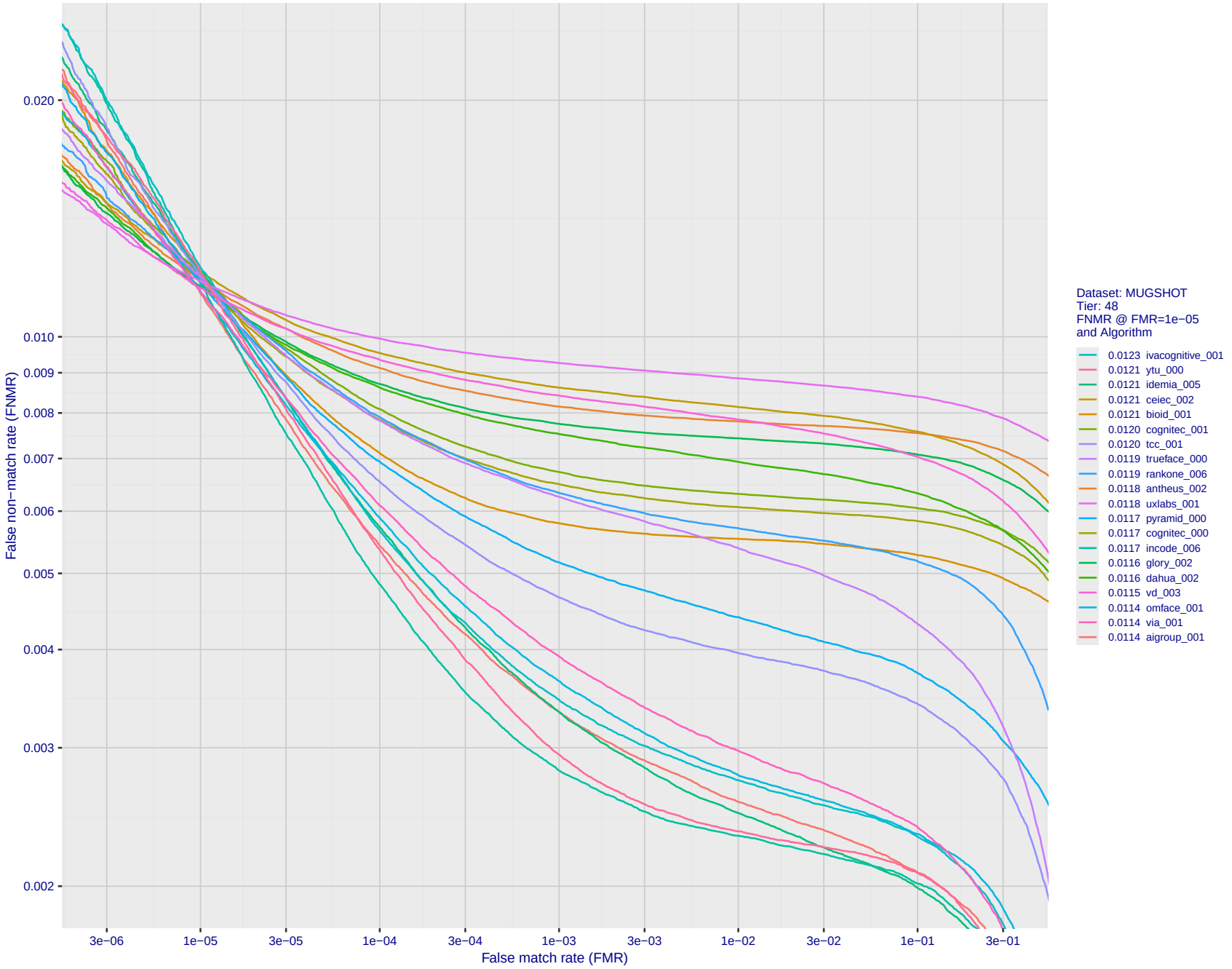

*Figure 270: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

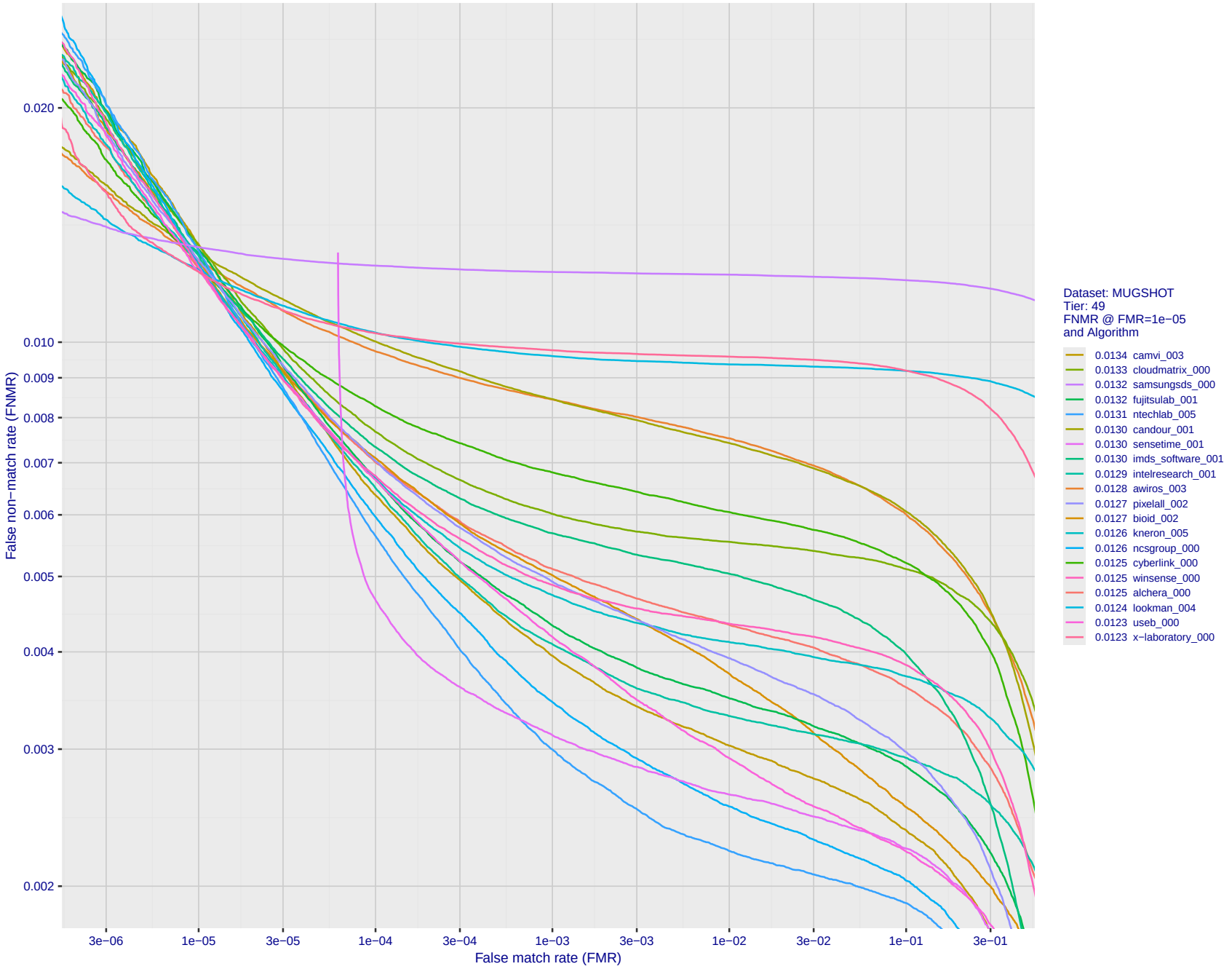

*Figure 271: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

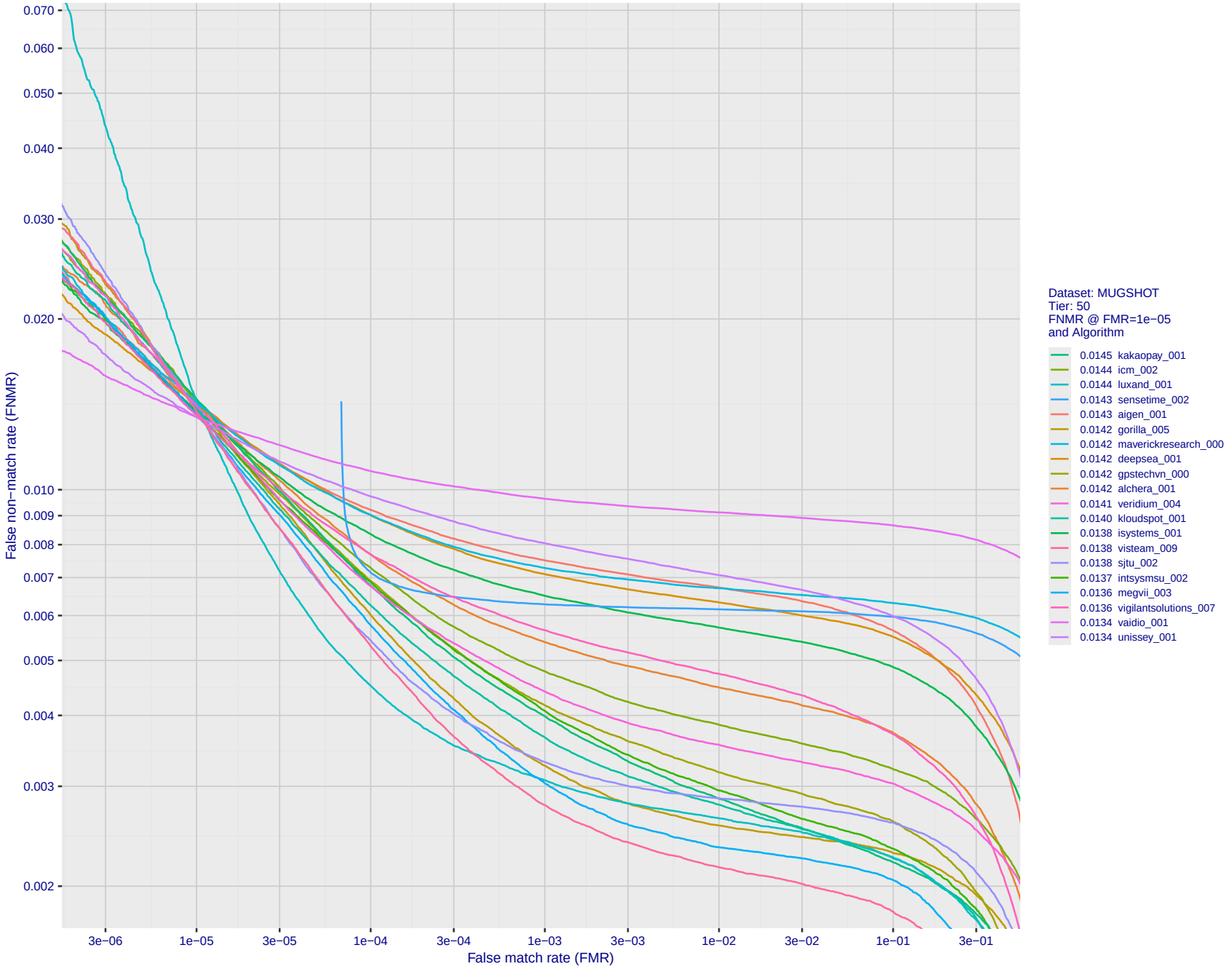

Figure 272: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

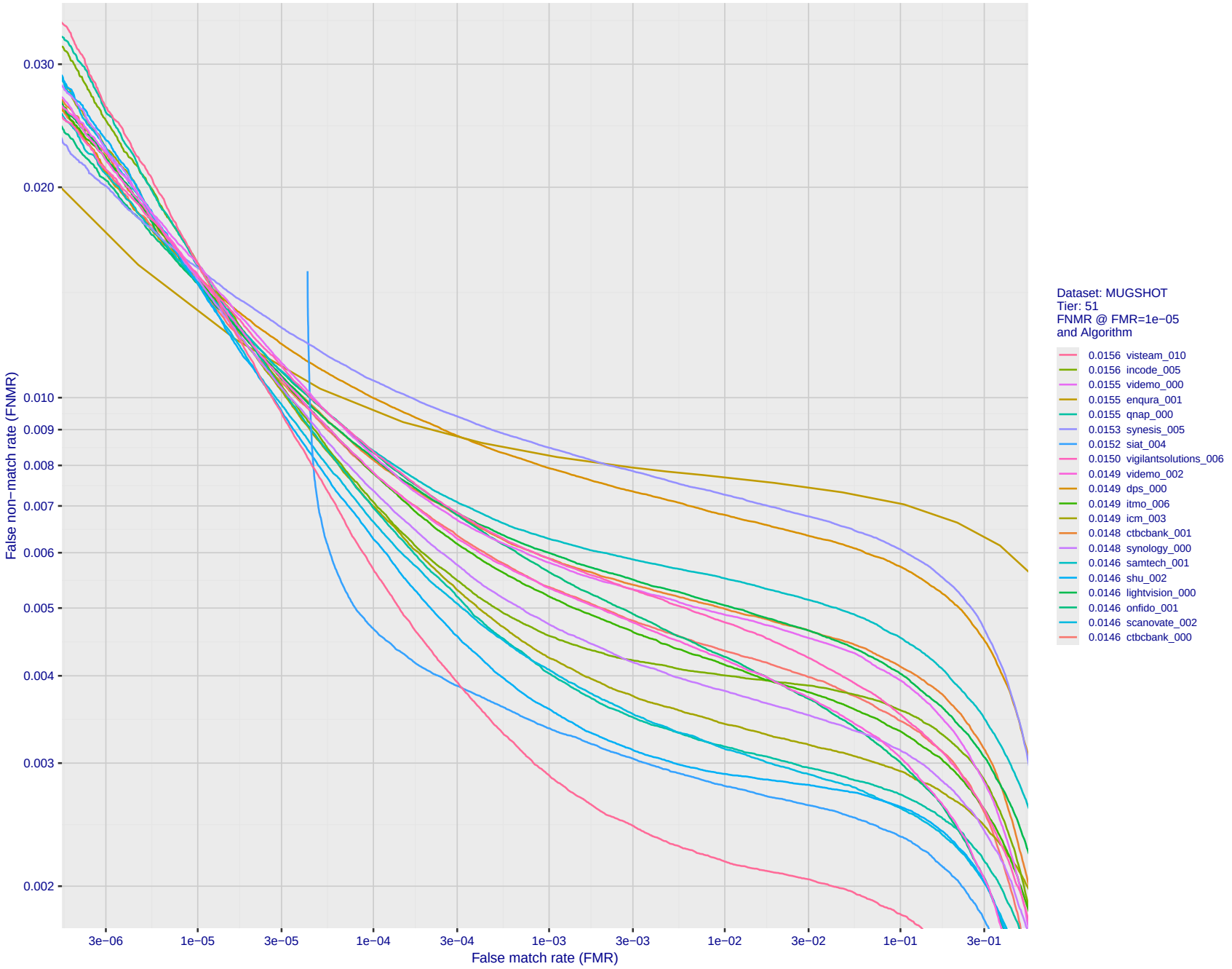

Figure 273: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

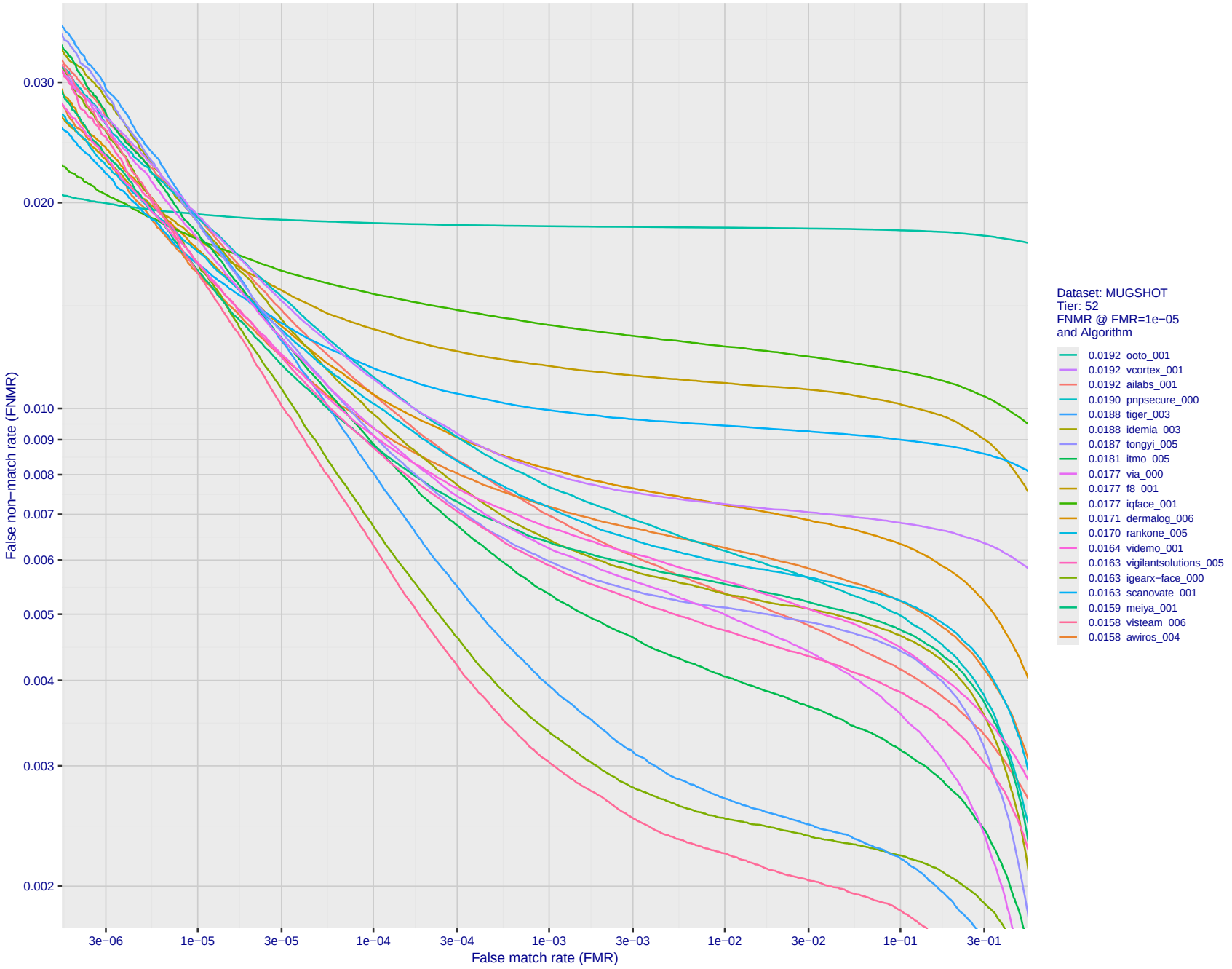

Figure 274: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

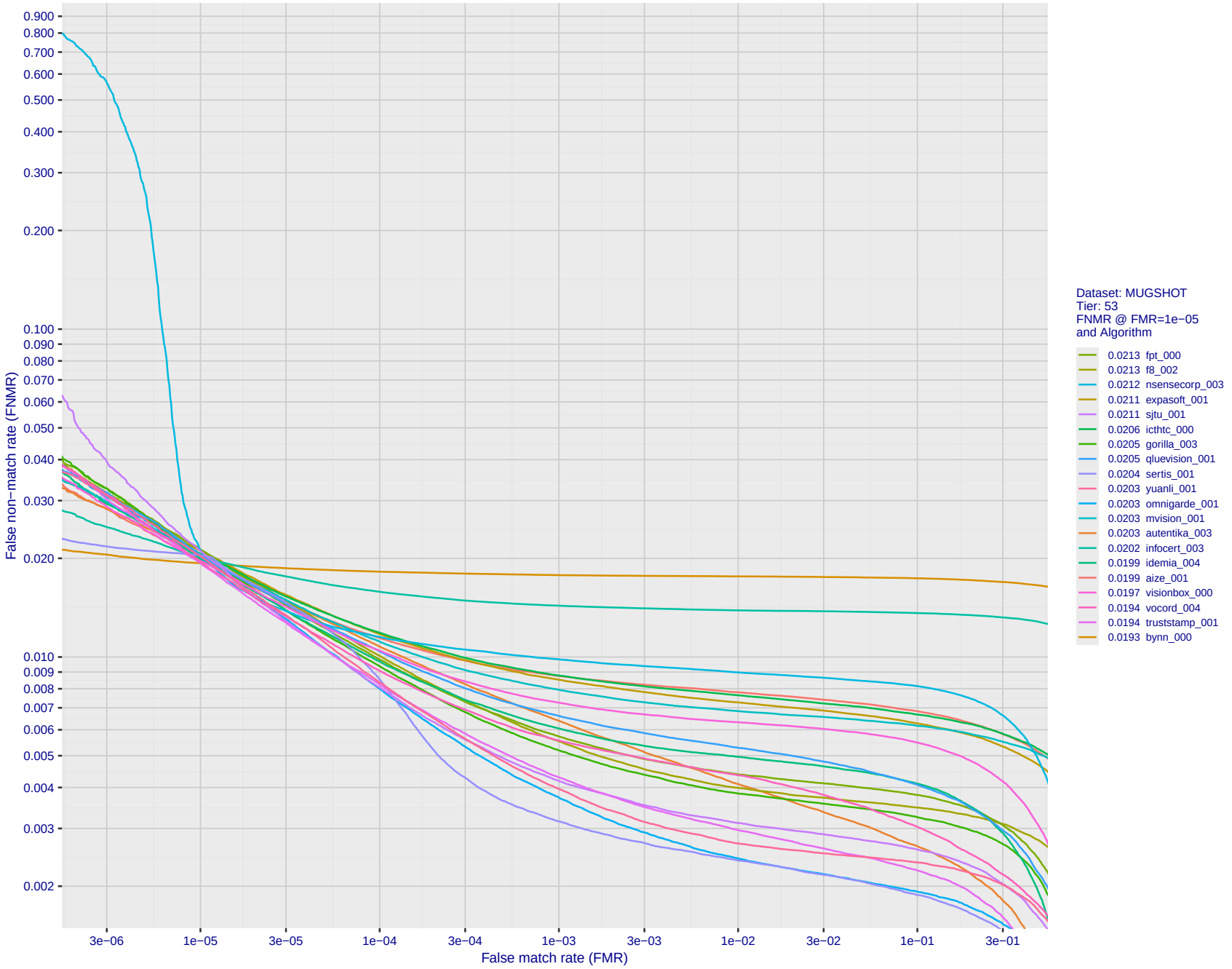

Figure 275: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

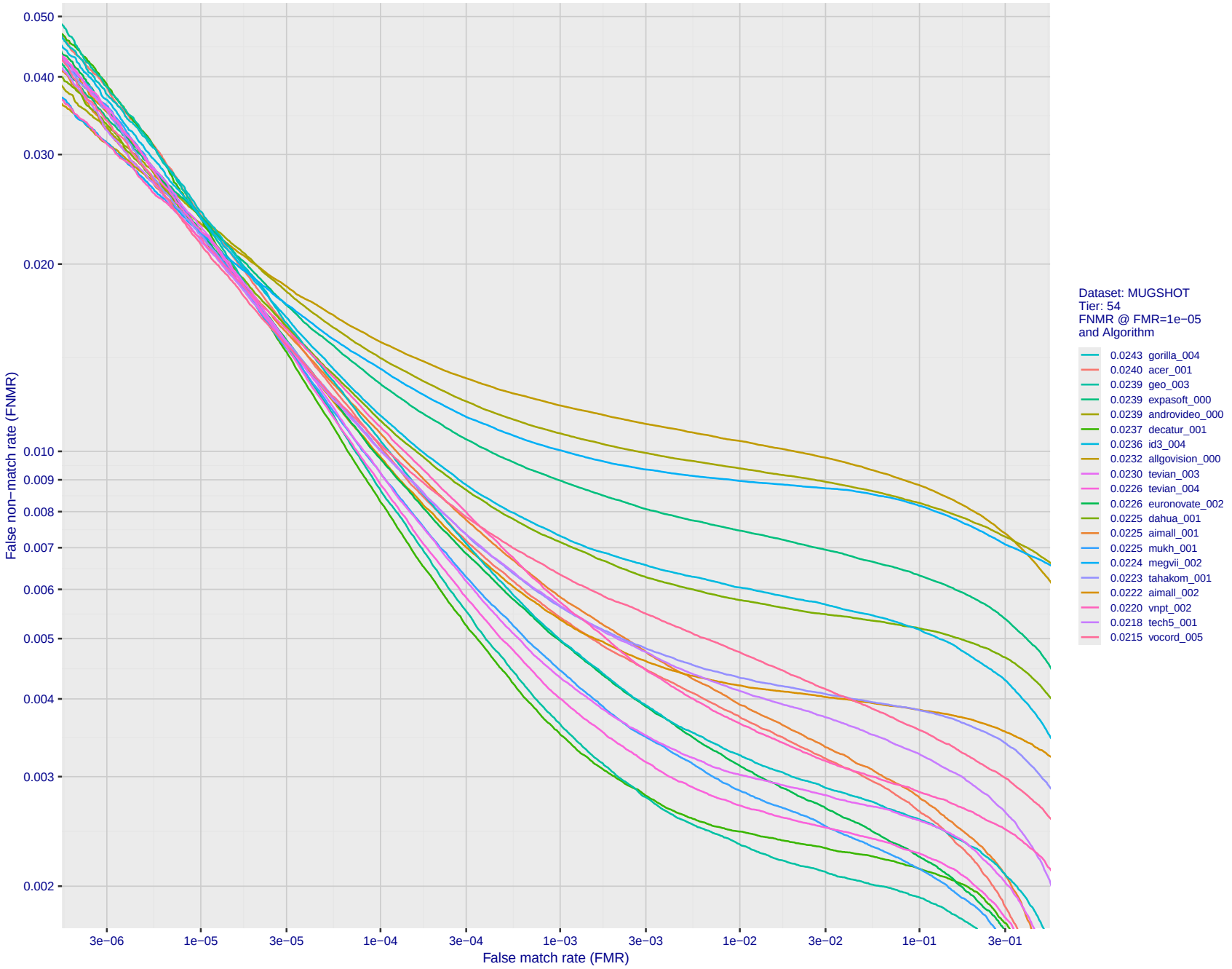

Figure 276: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

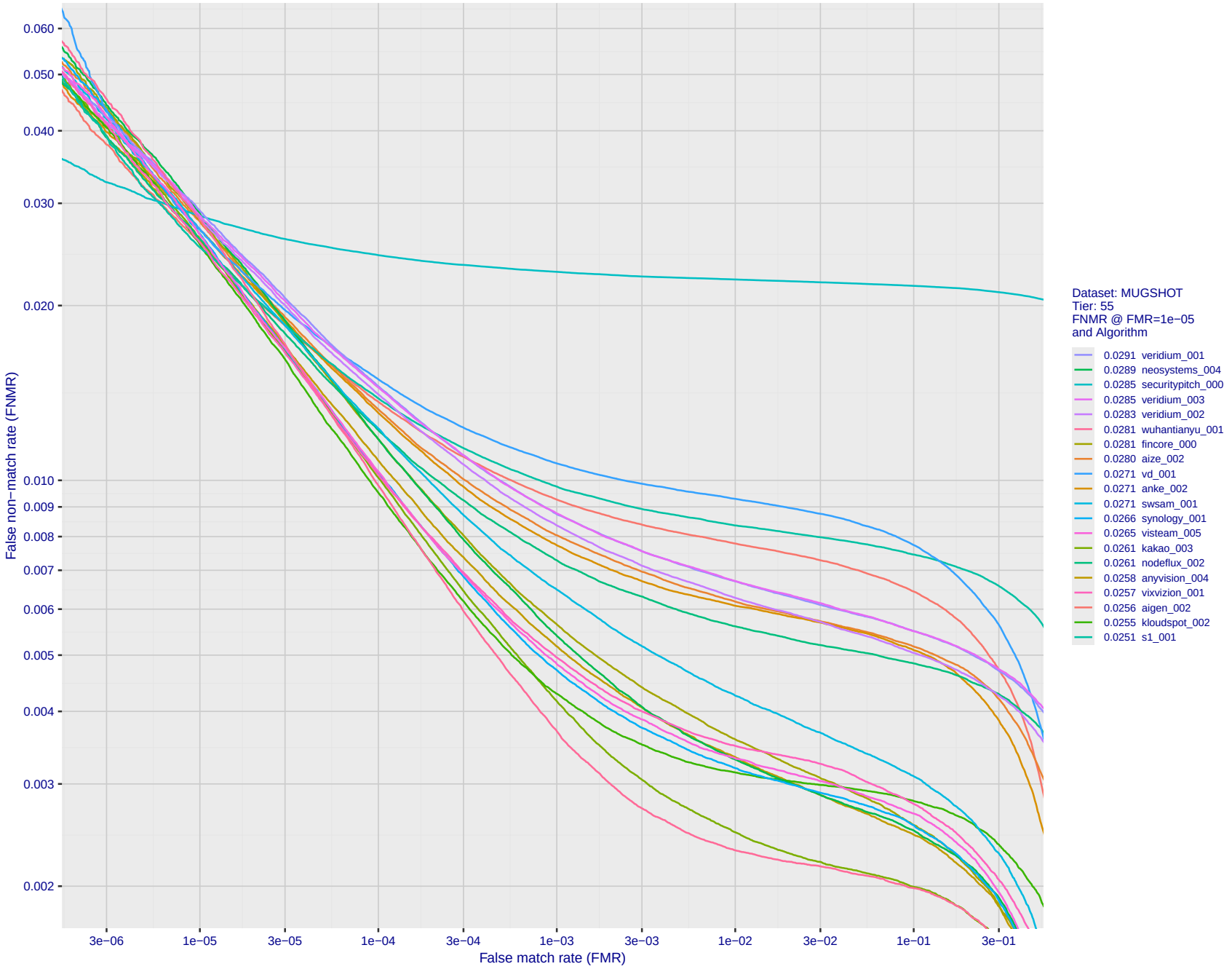

Figure 277: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

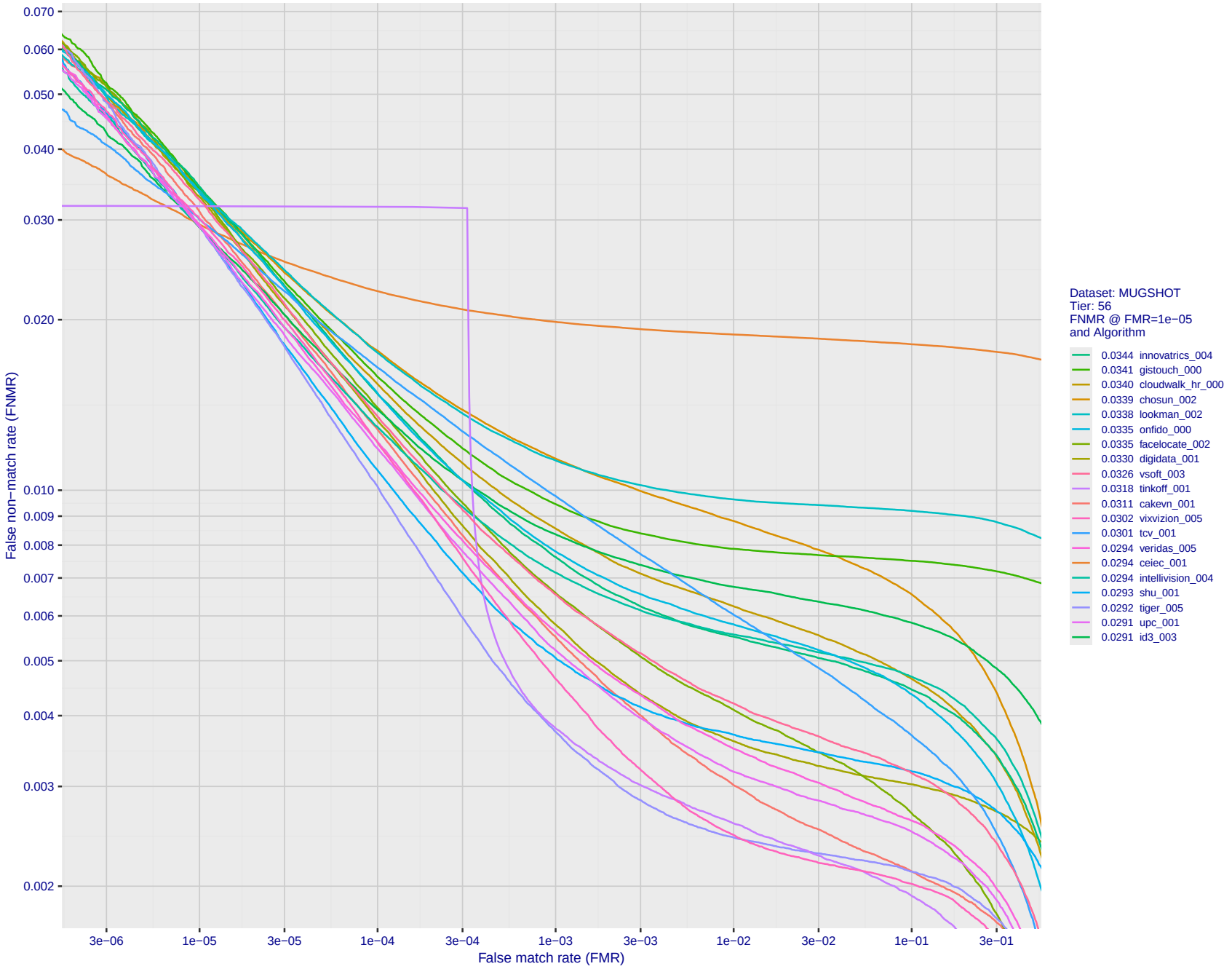

Figure 278: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

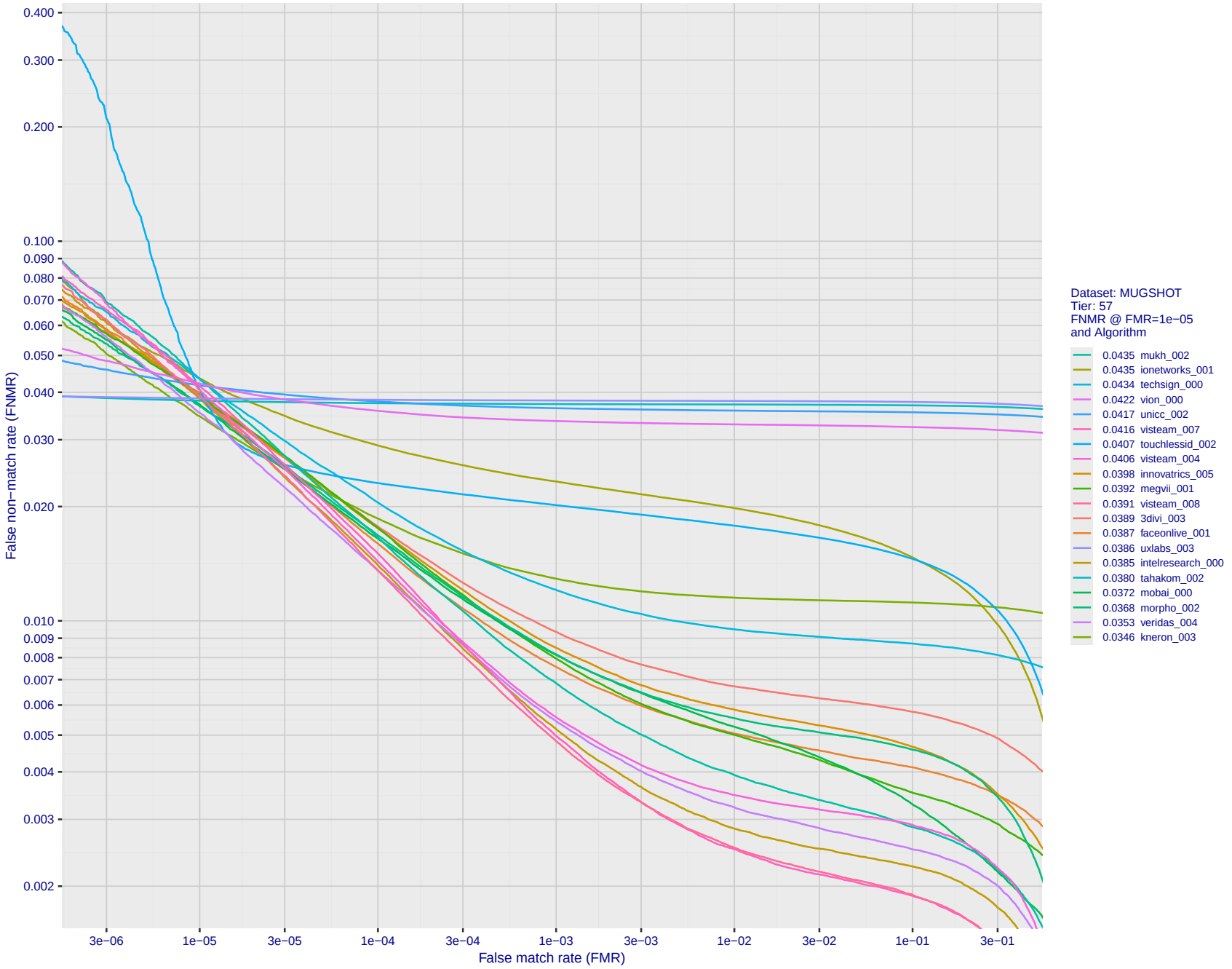

Figure 279: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

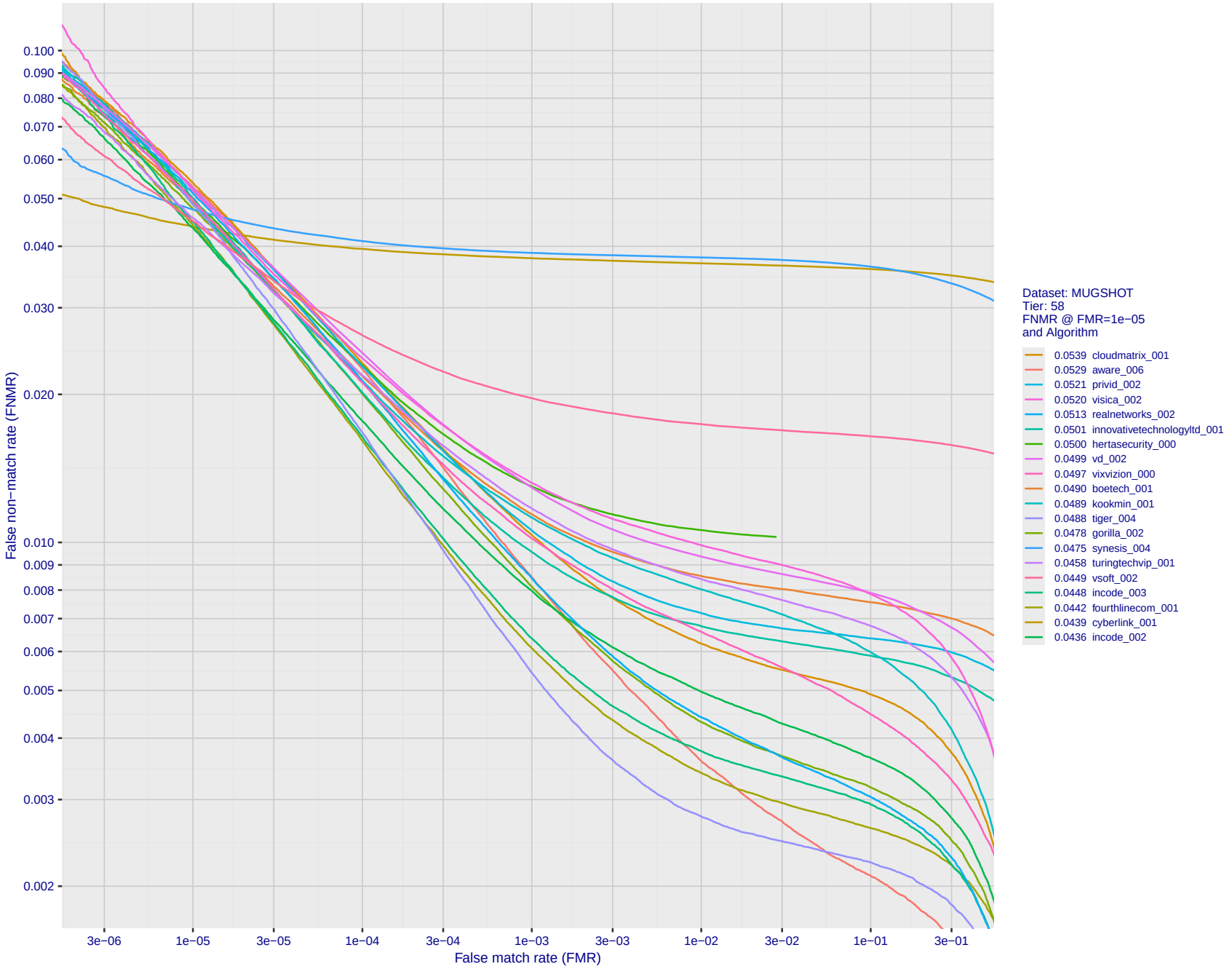

Figure 280: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

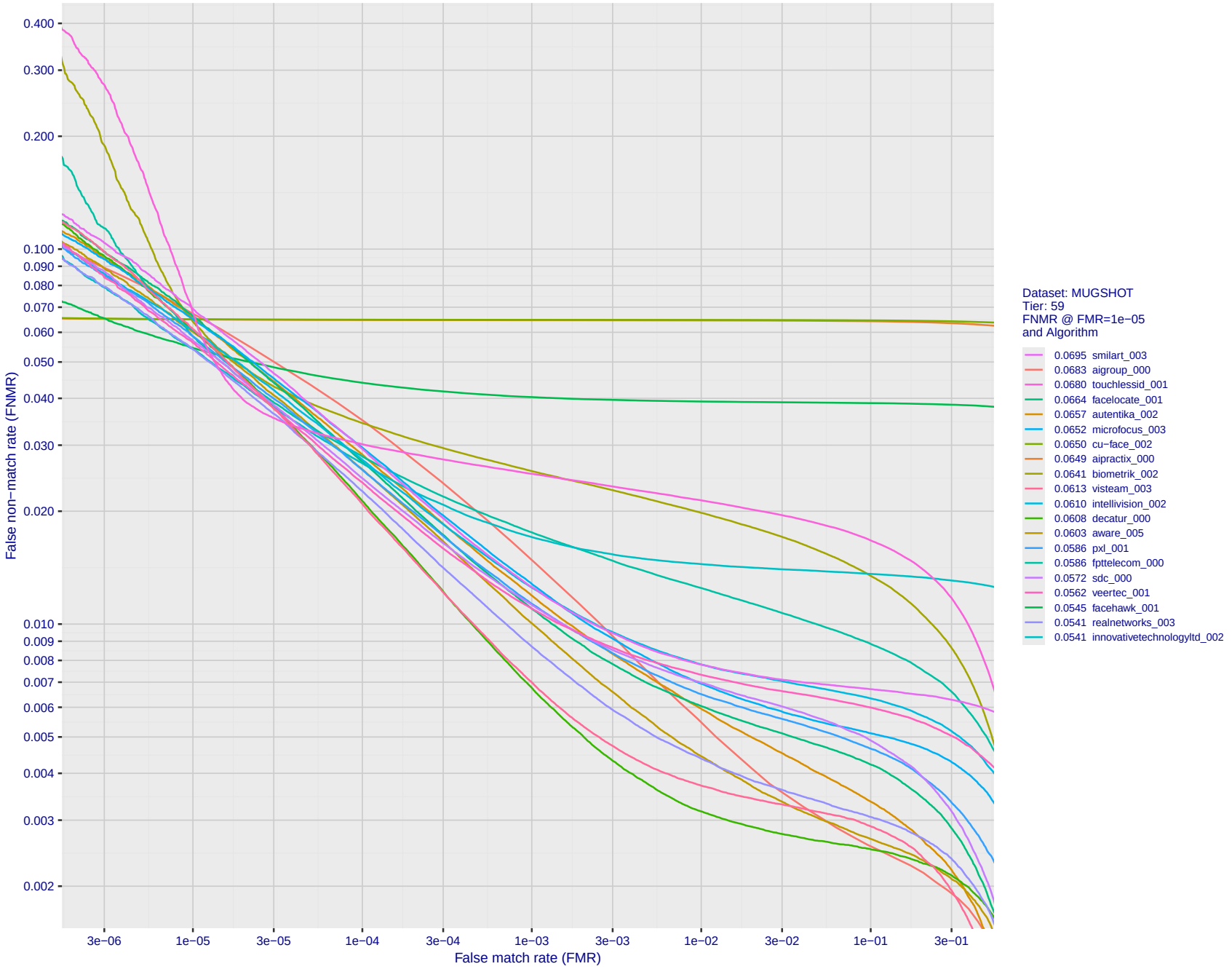

Figure 281: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

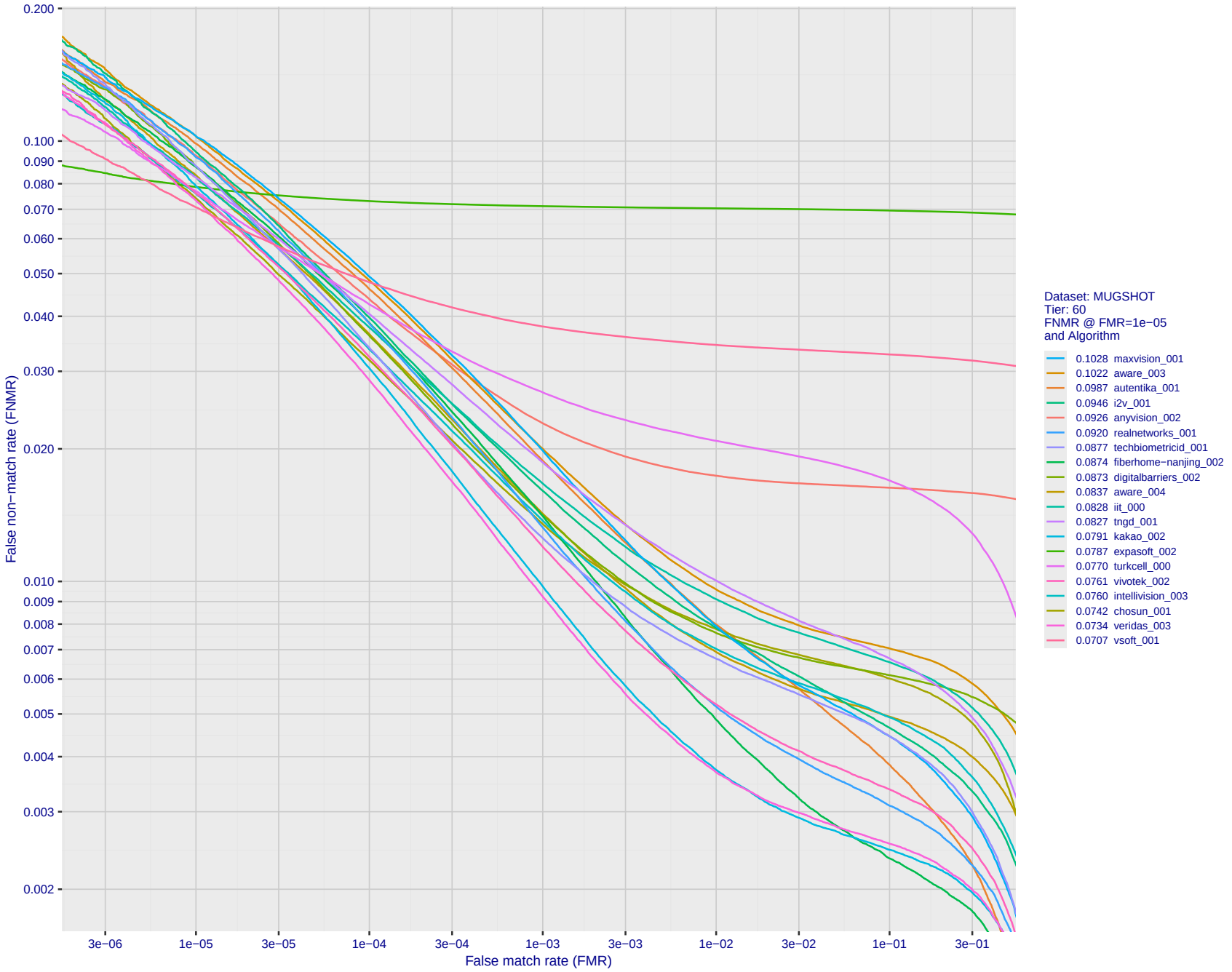

Figure 282: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

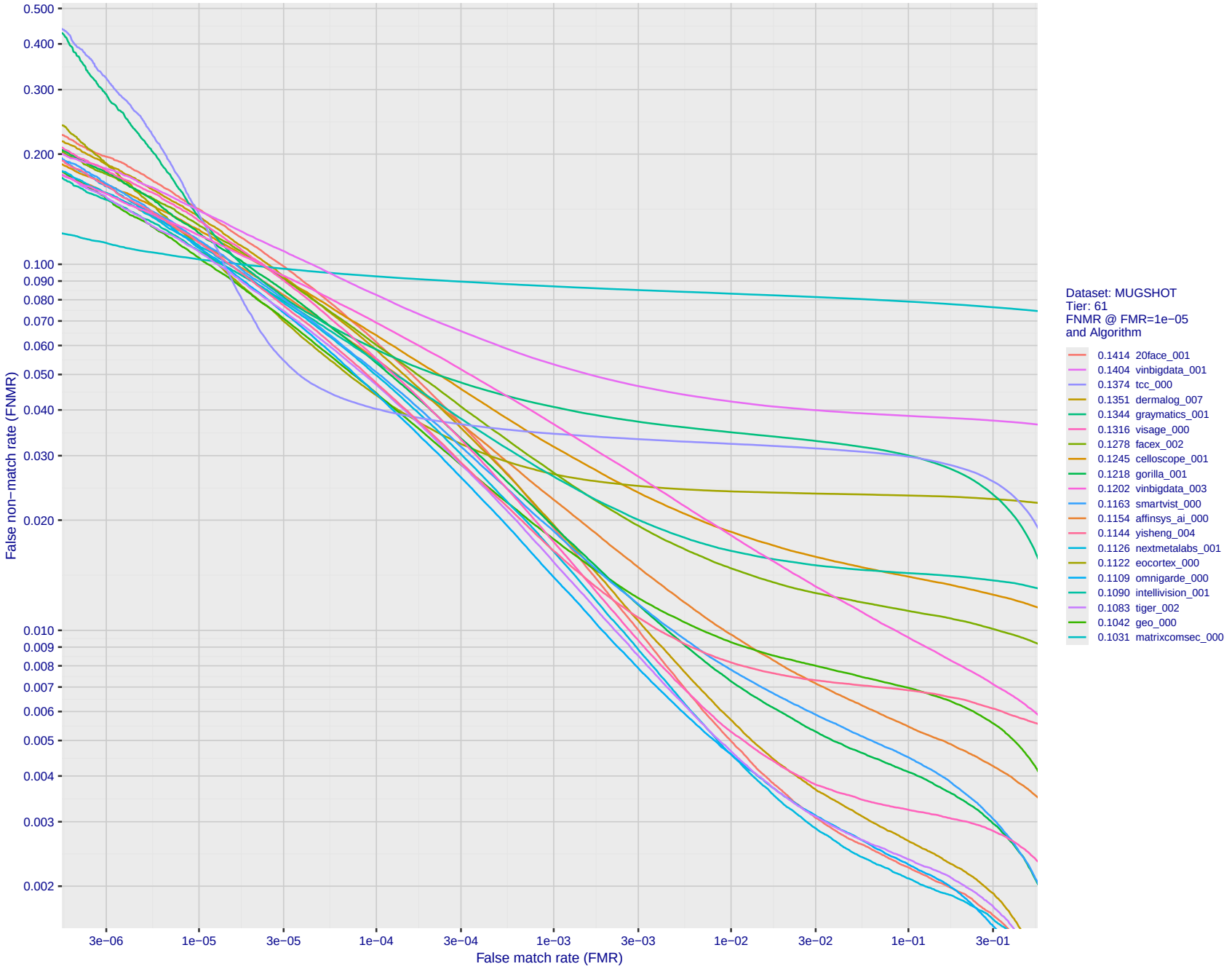

Figure 283: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

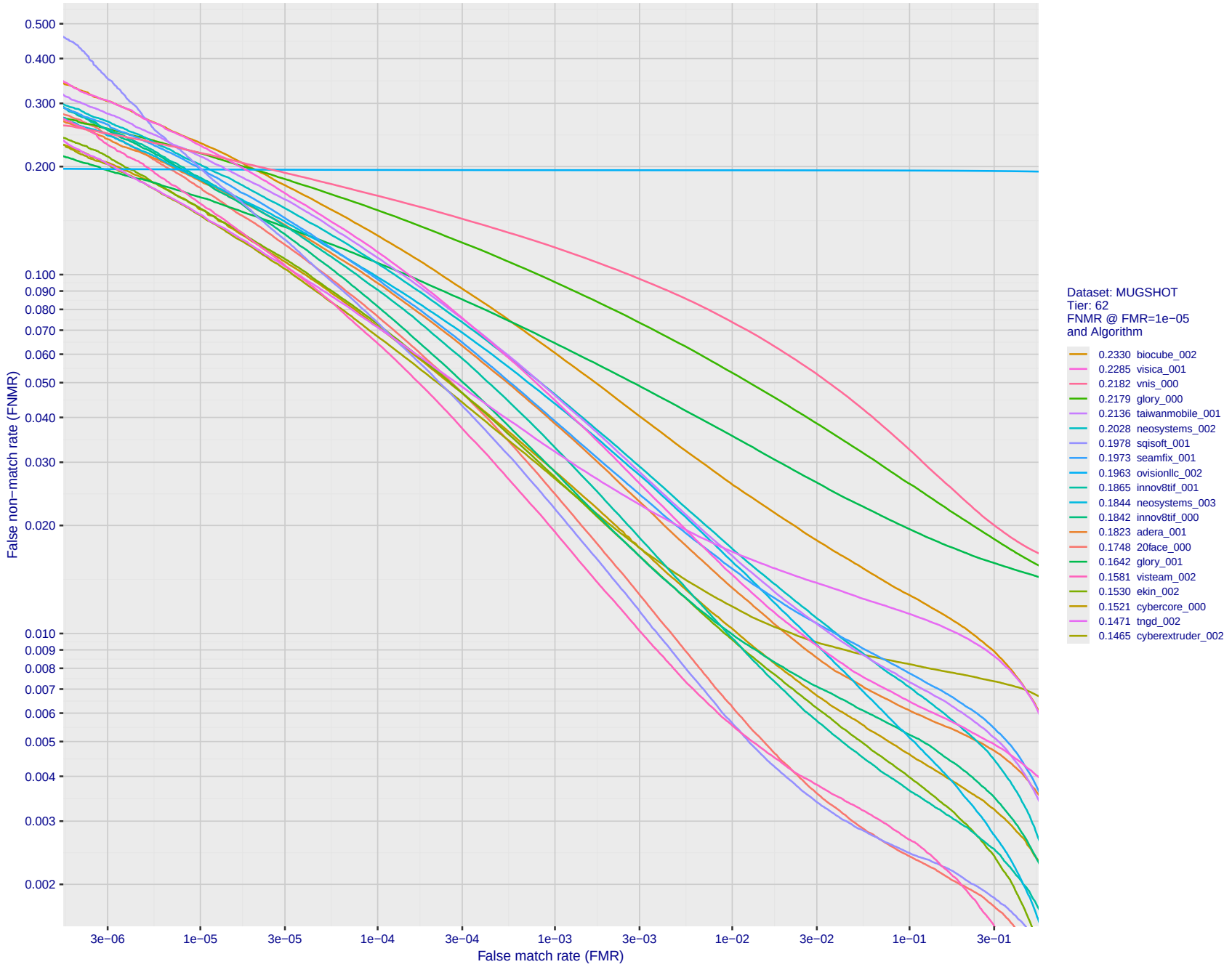

Figure 284: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

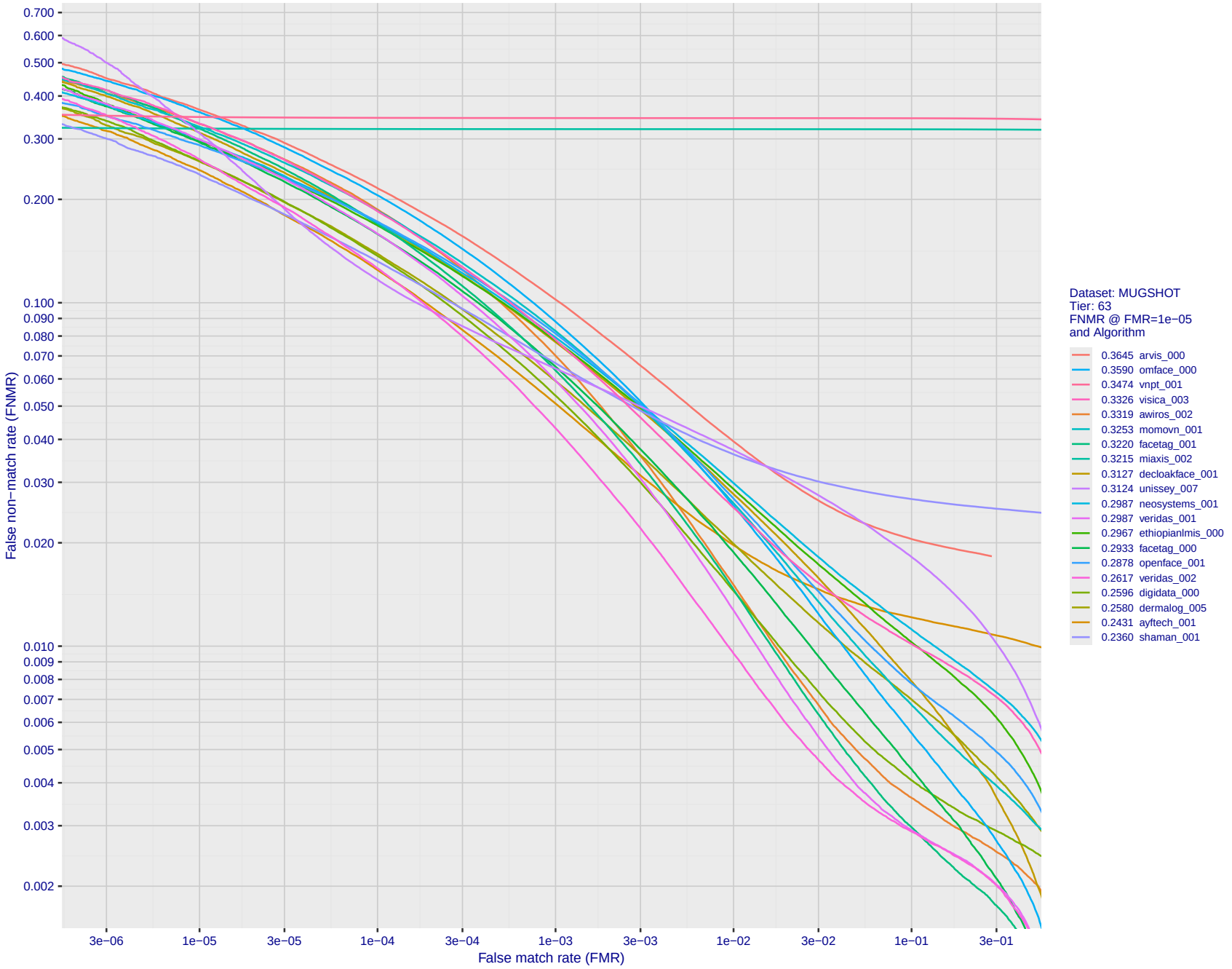

Figure 285: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

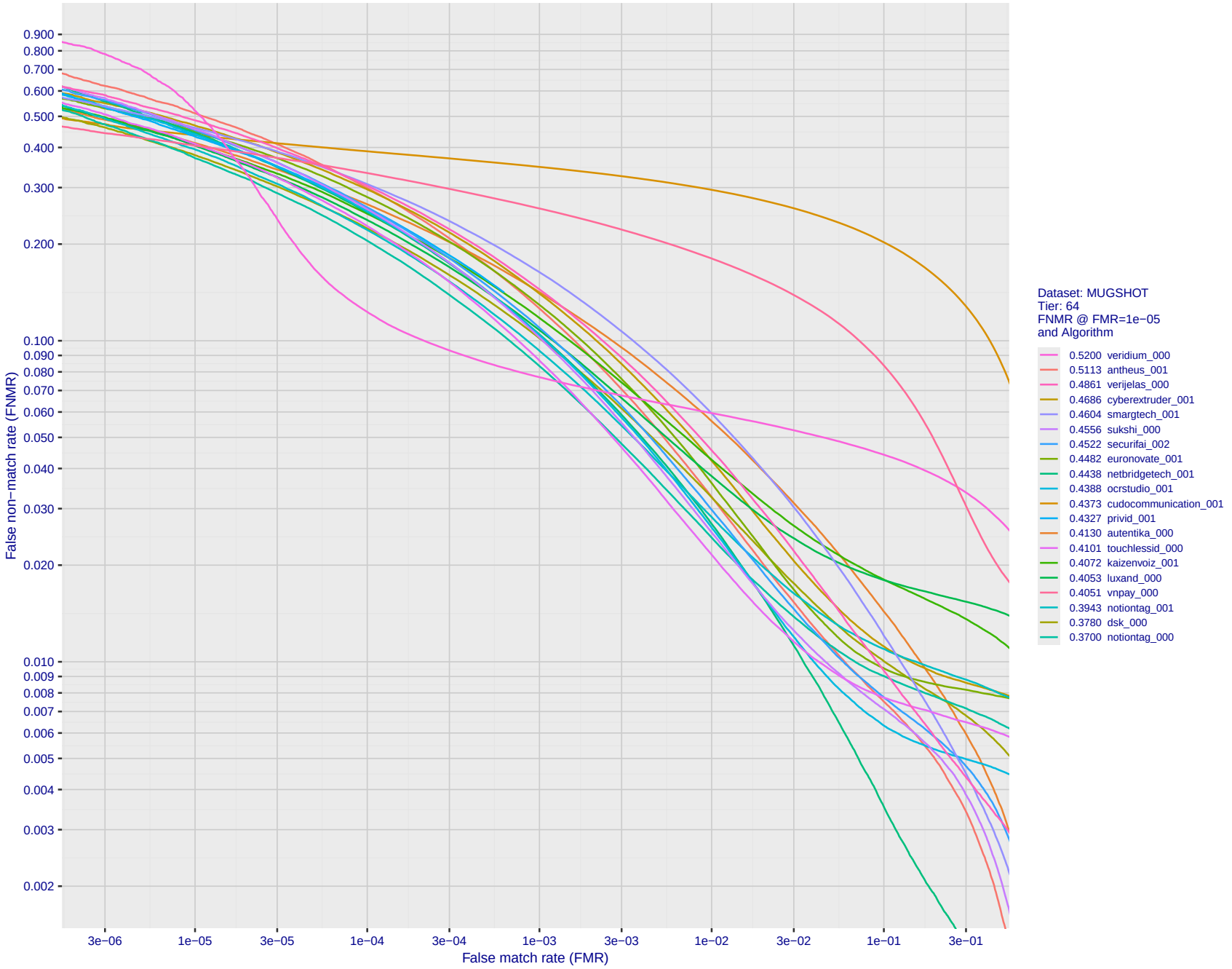

Figure 286: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

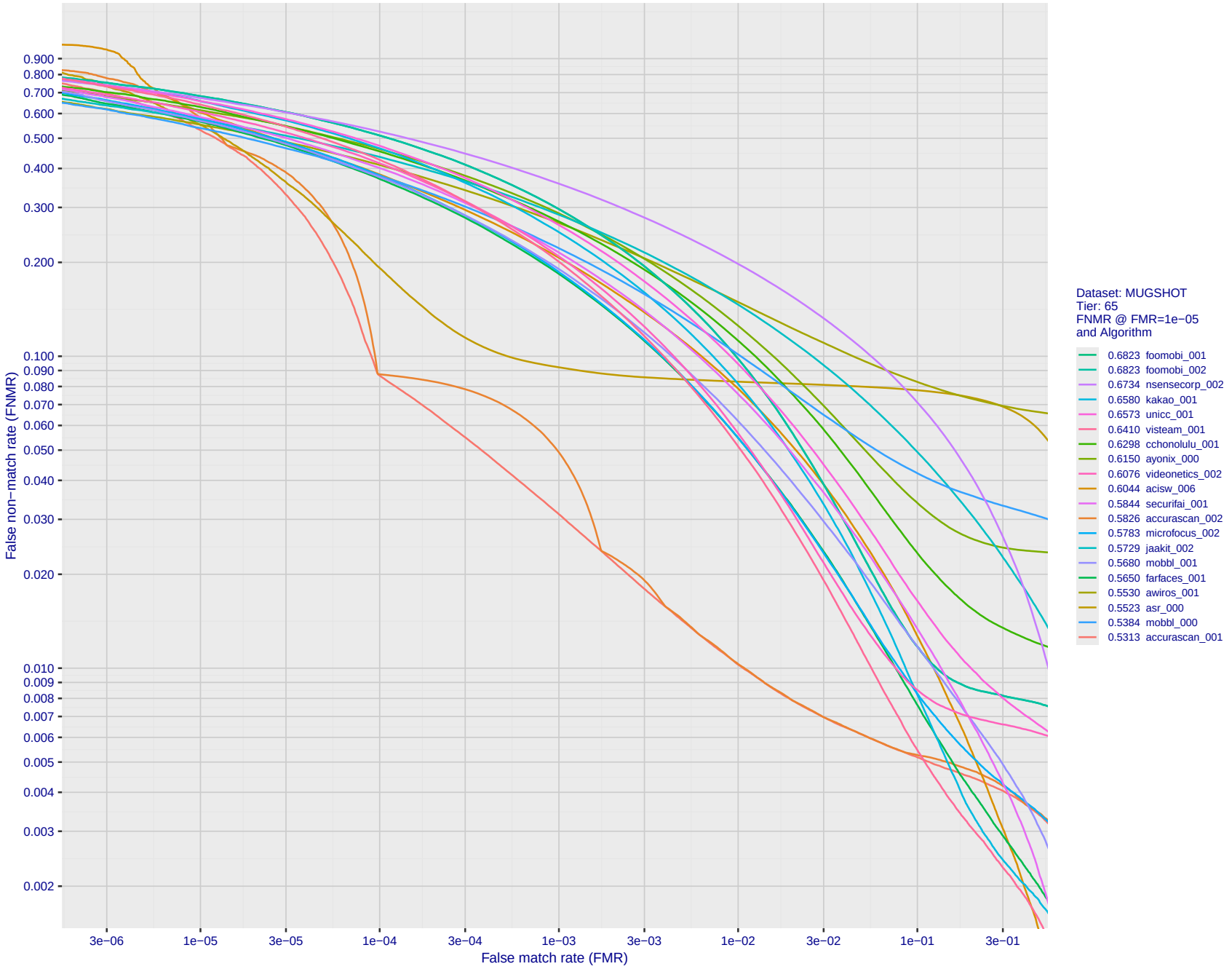

Figure 287: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

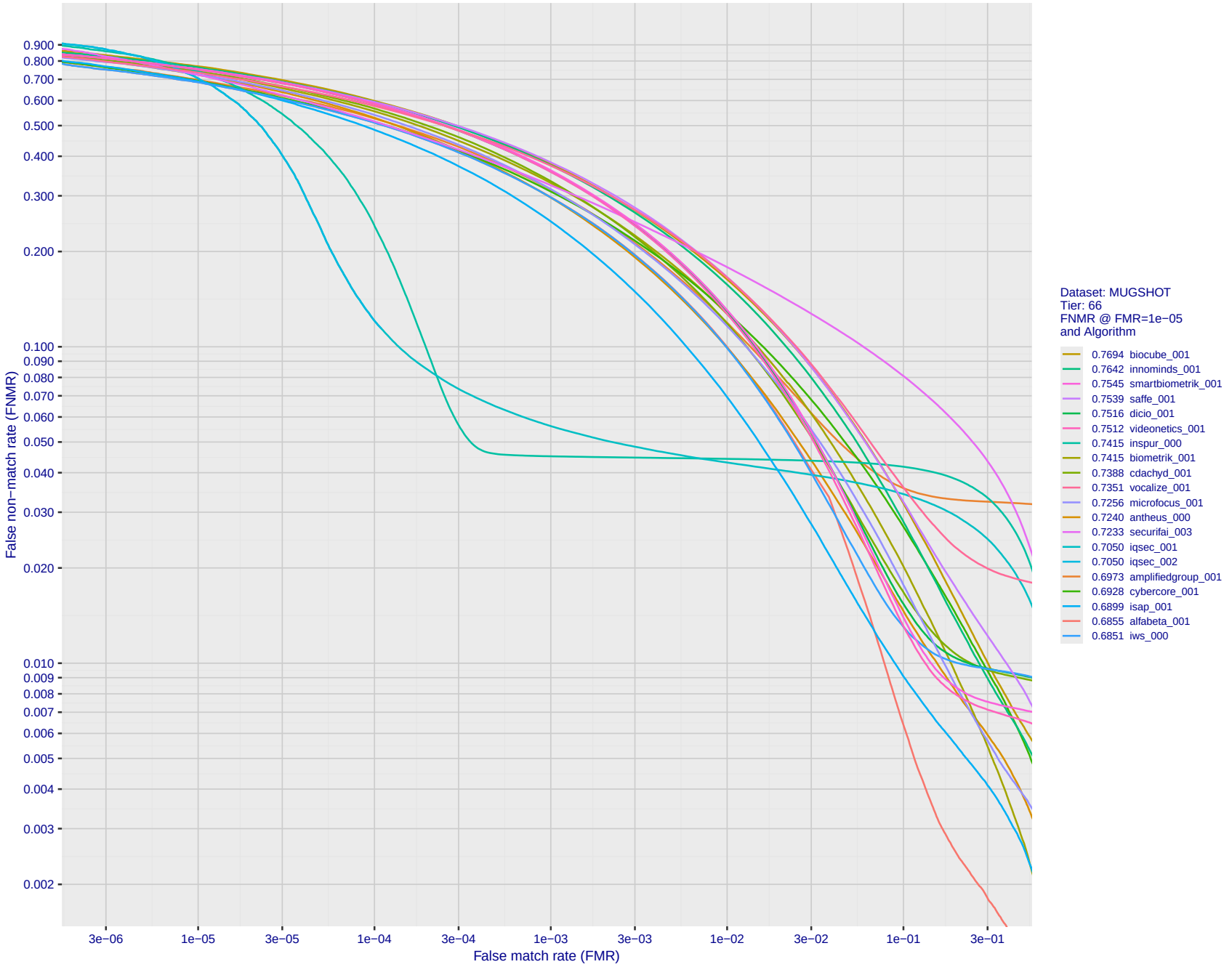

Figure 288: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

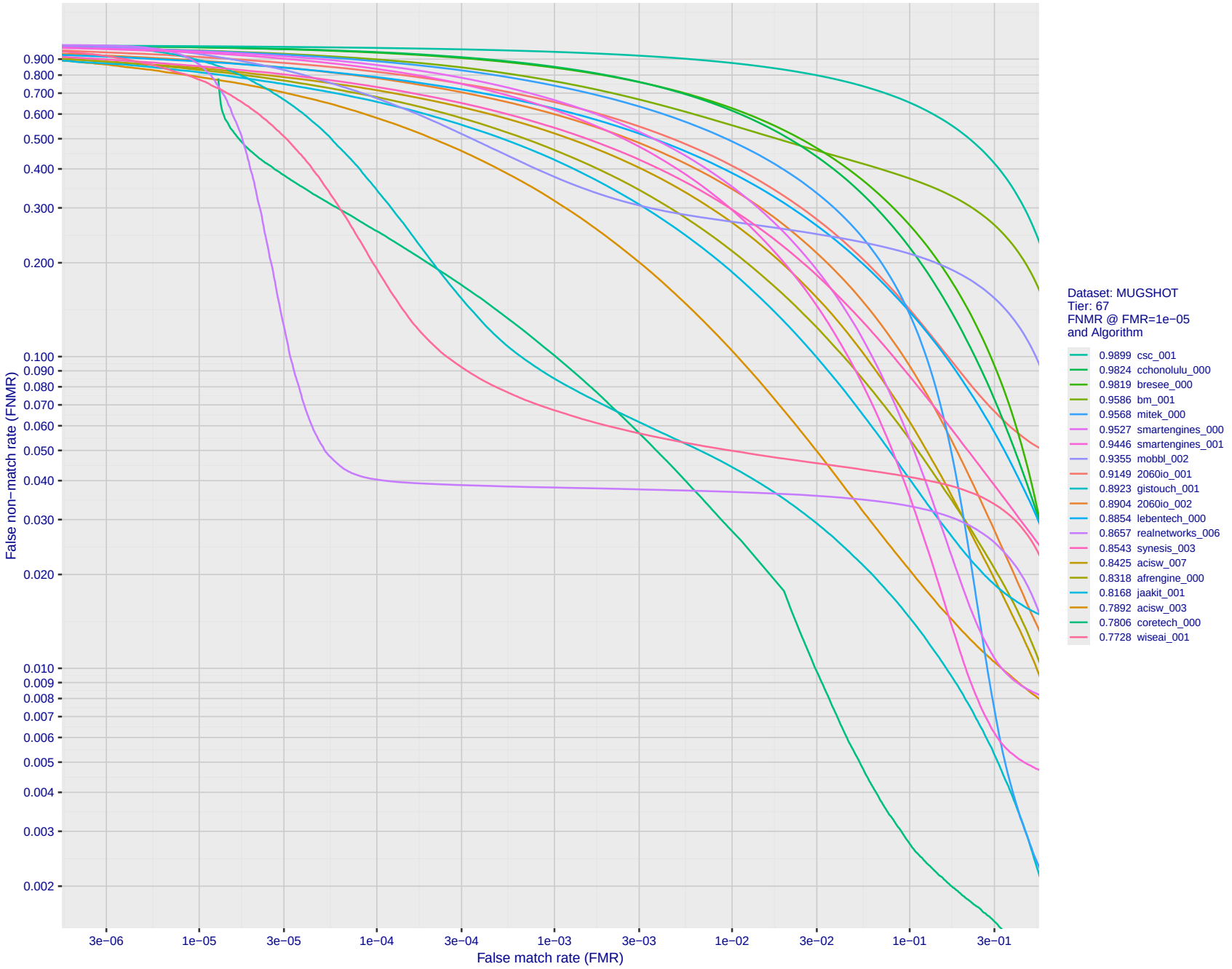

Figure 289: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

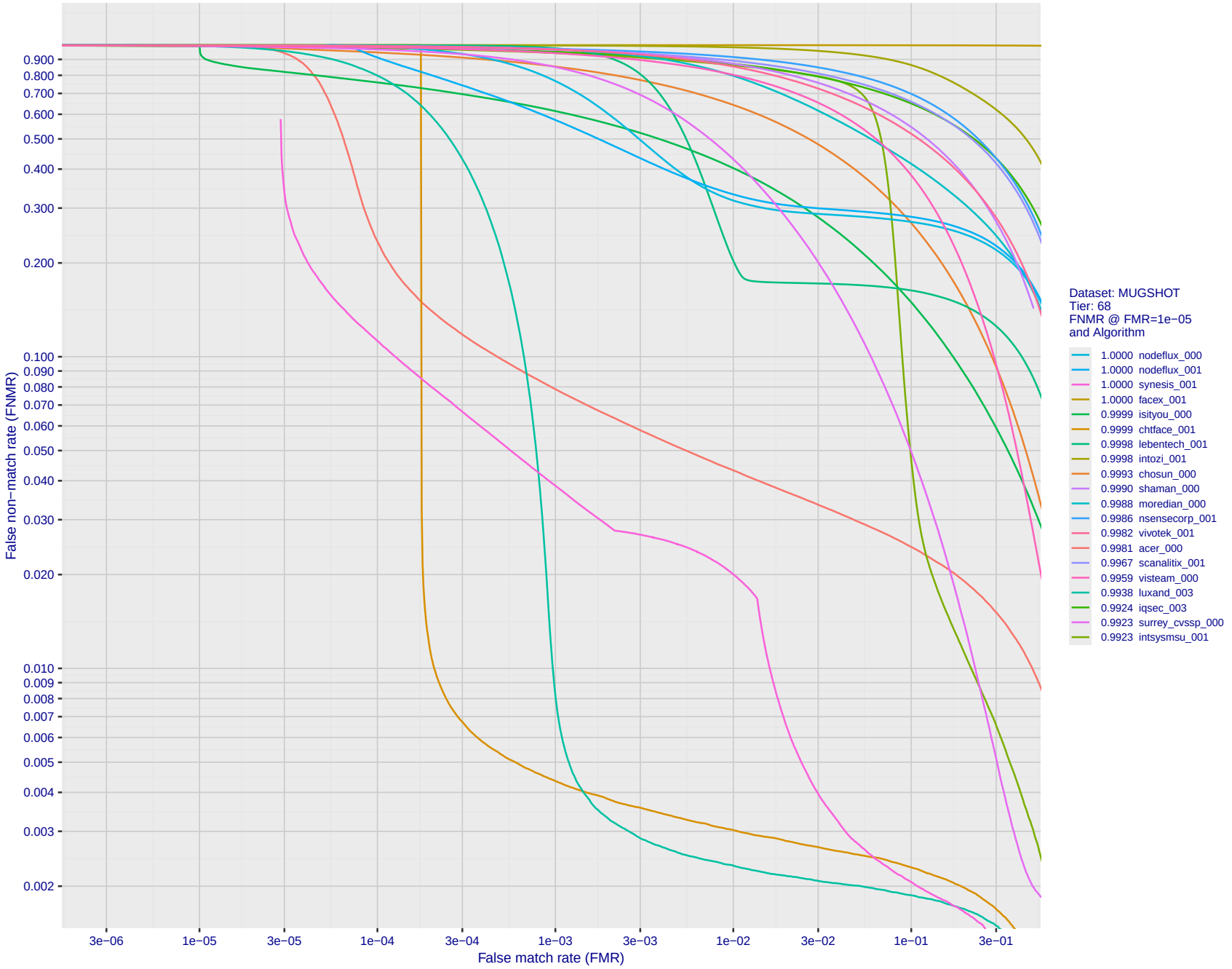

Figure 290: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

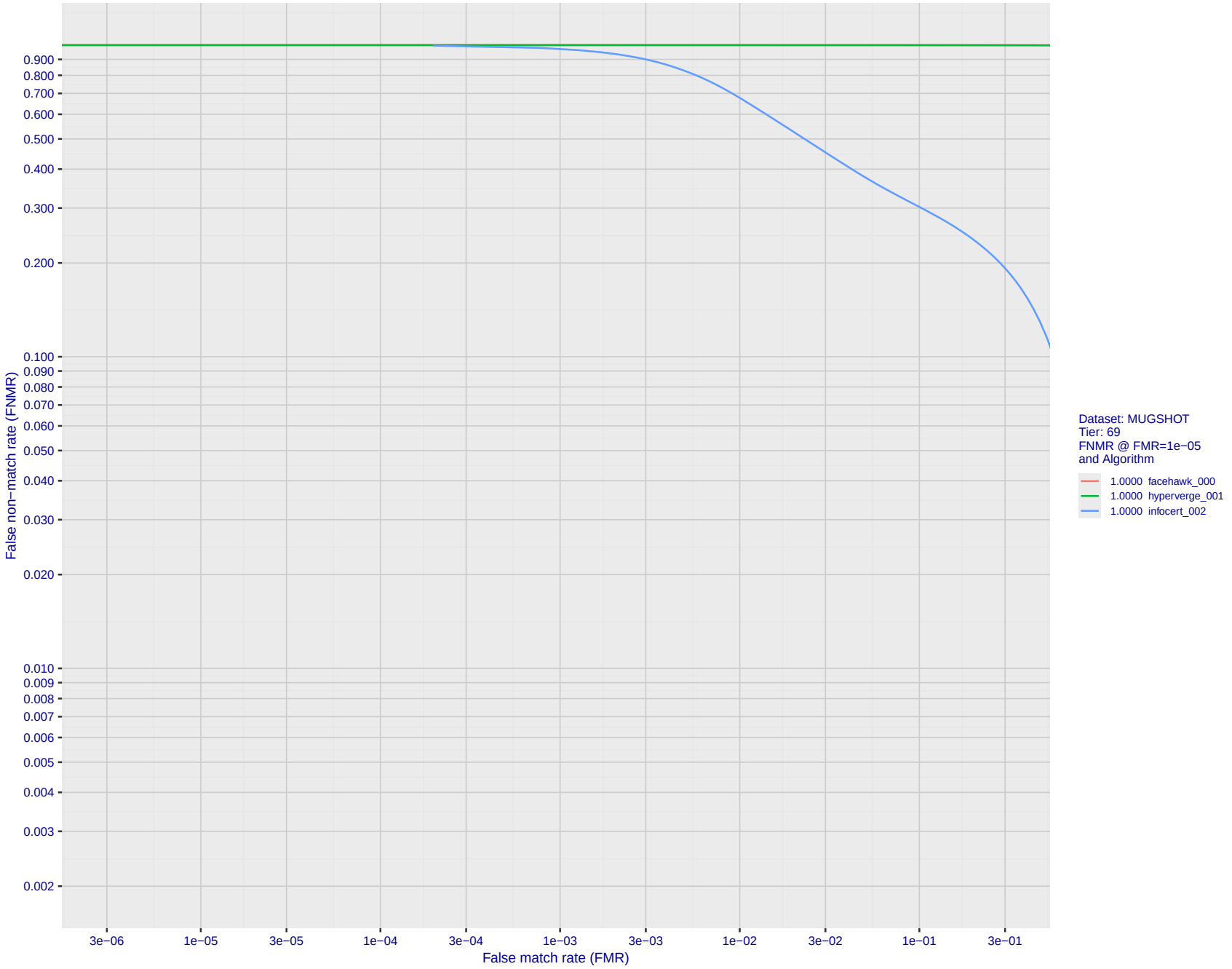

Figure 291: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

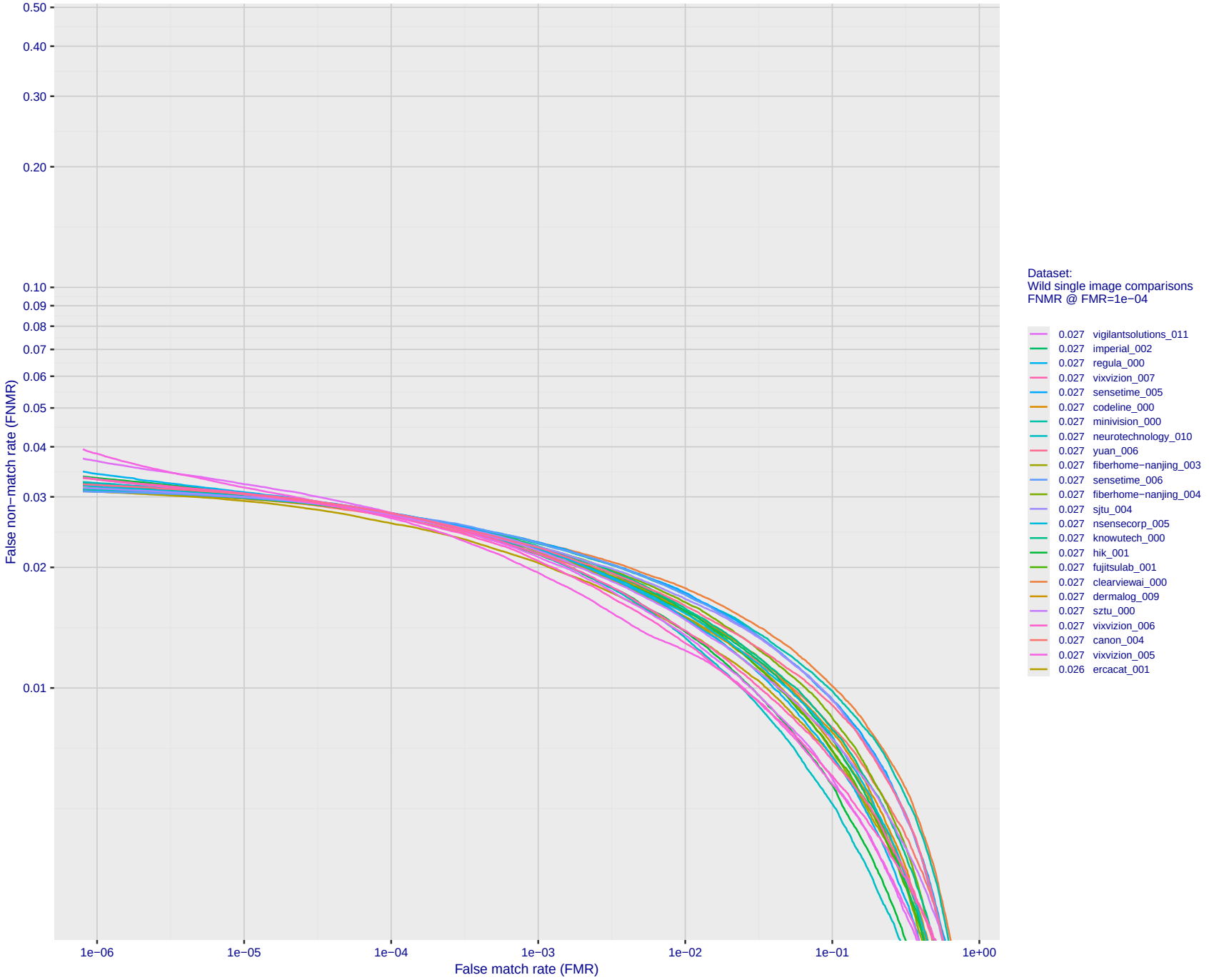

Figure 292: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

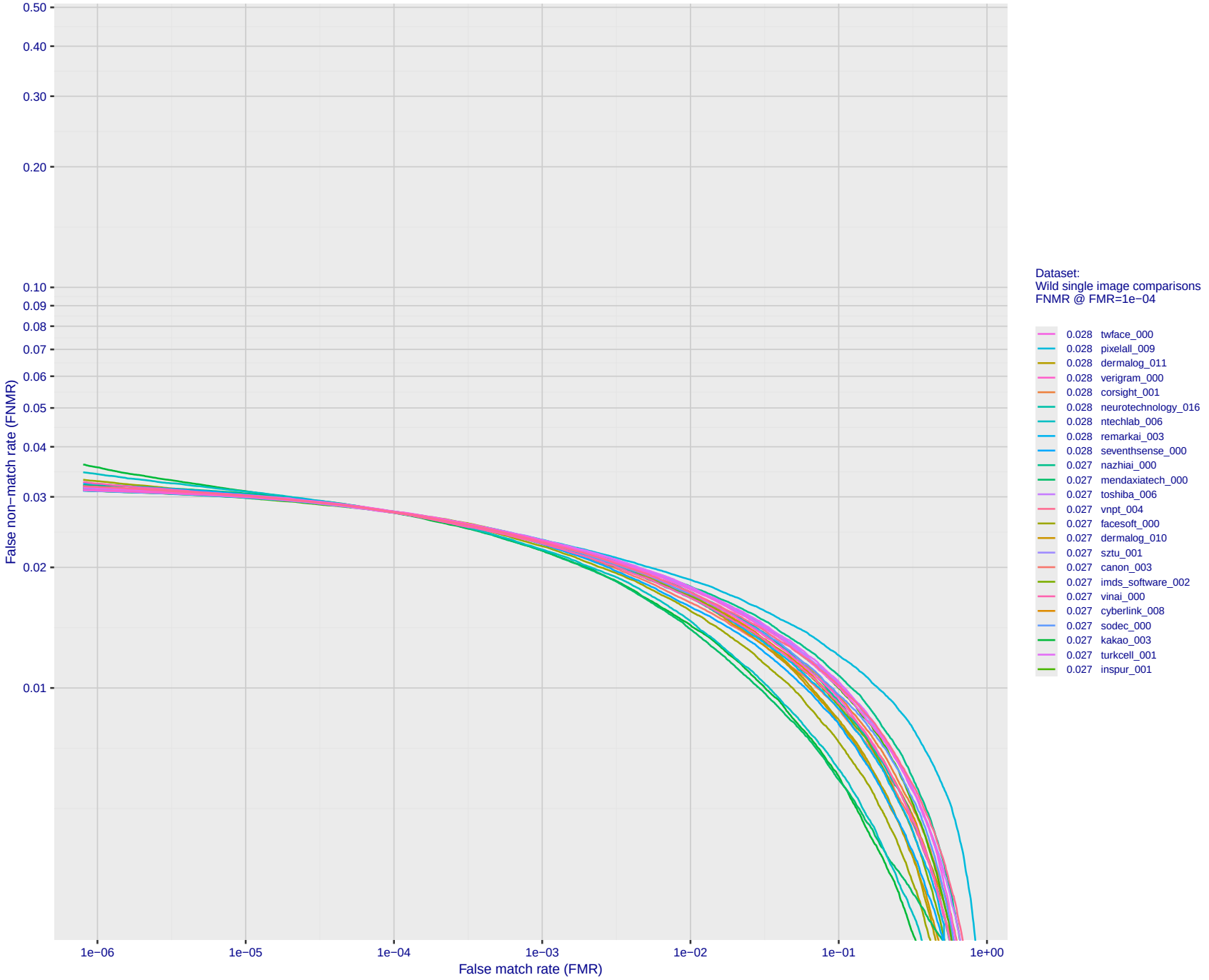

Figure 293: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

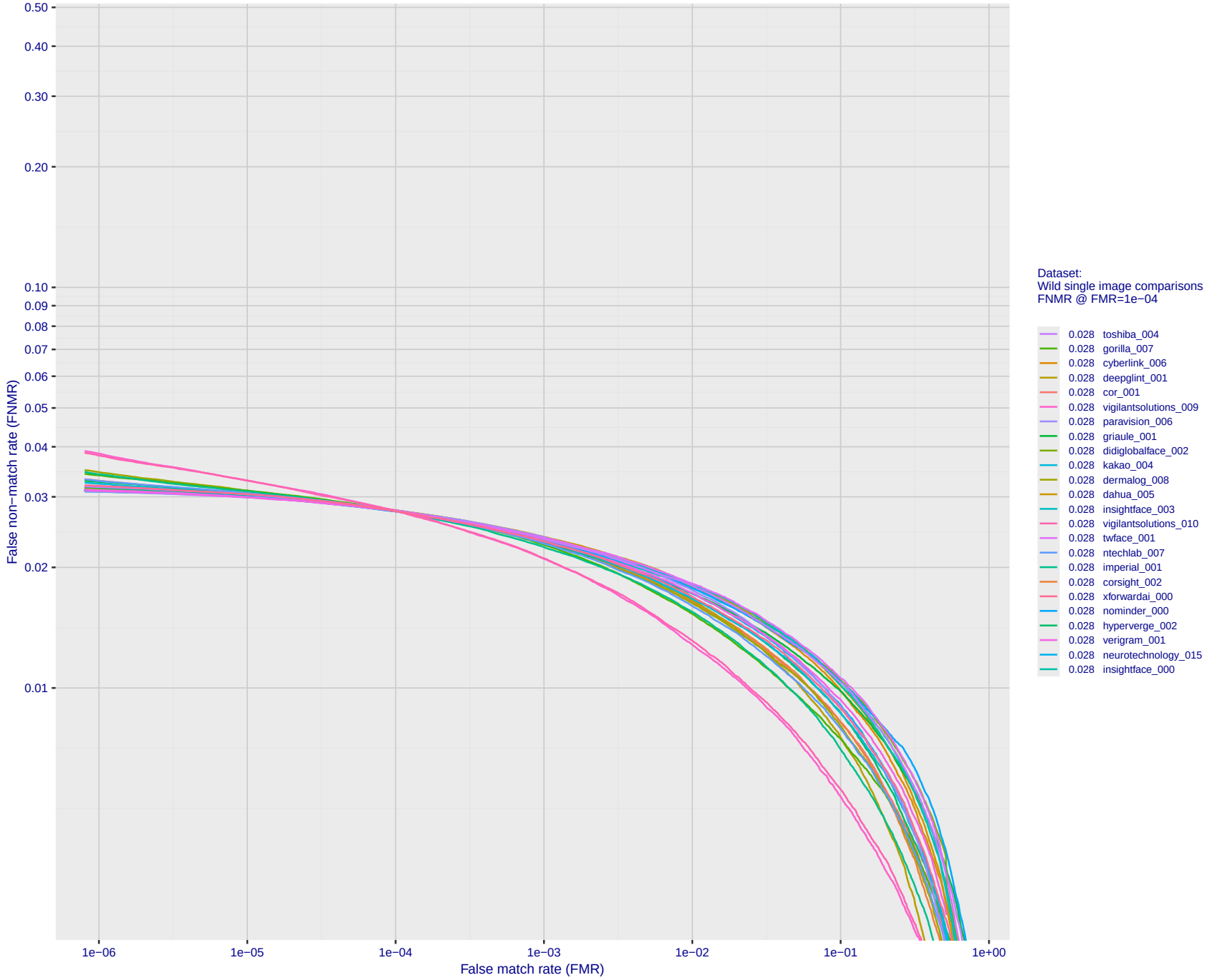

Figure 294: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

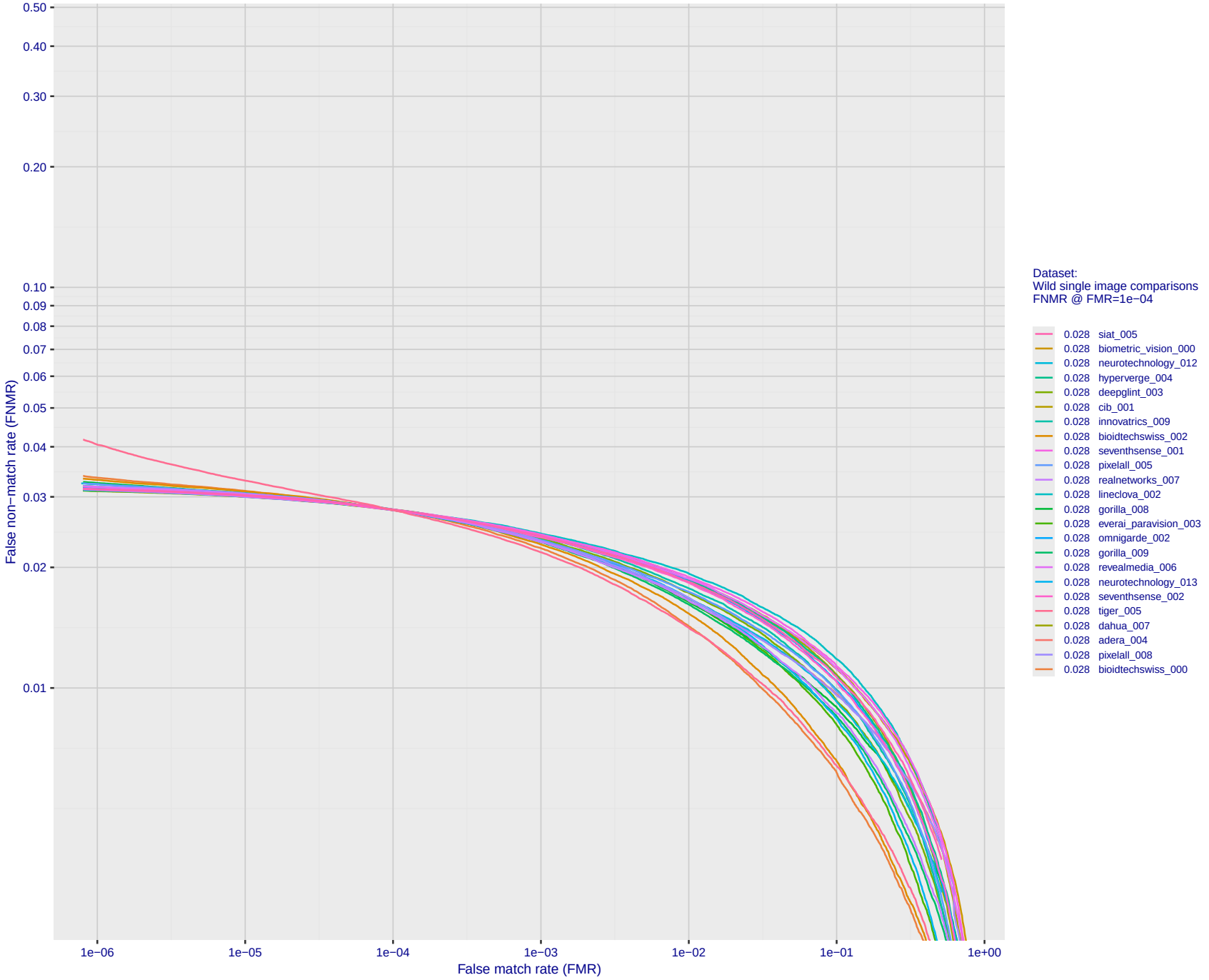

Figure 295: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

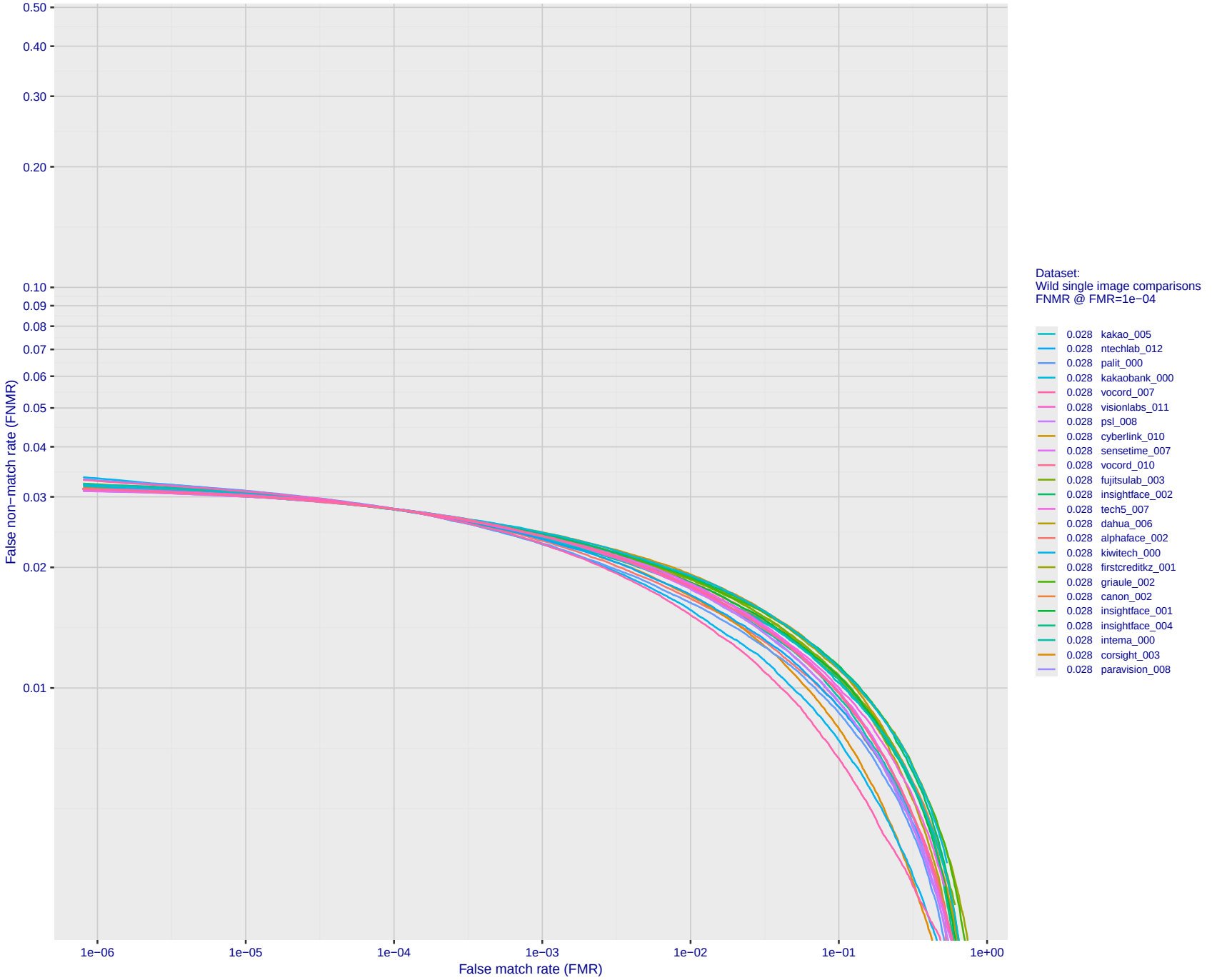

Figure 296: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

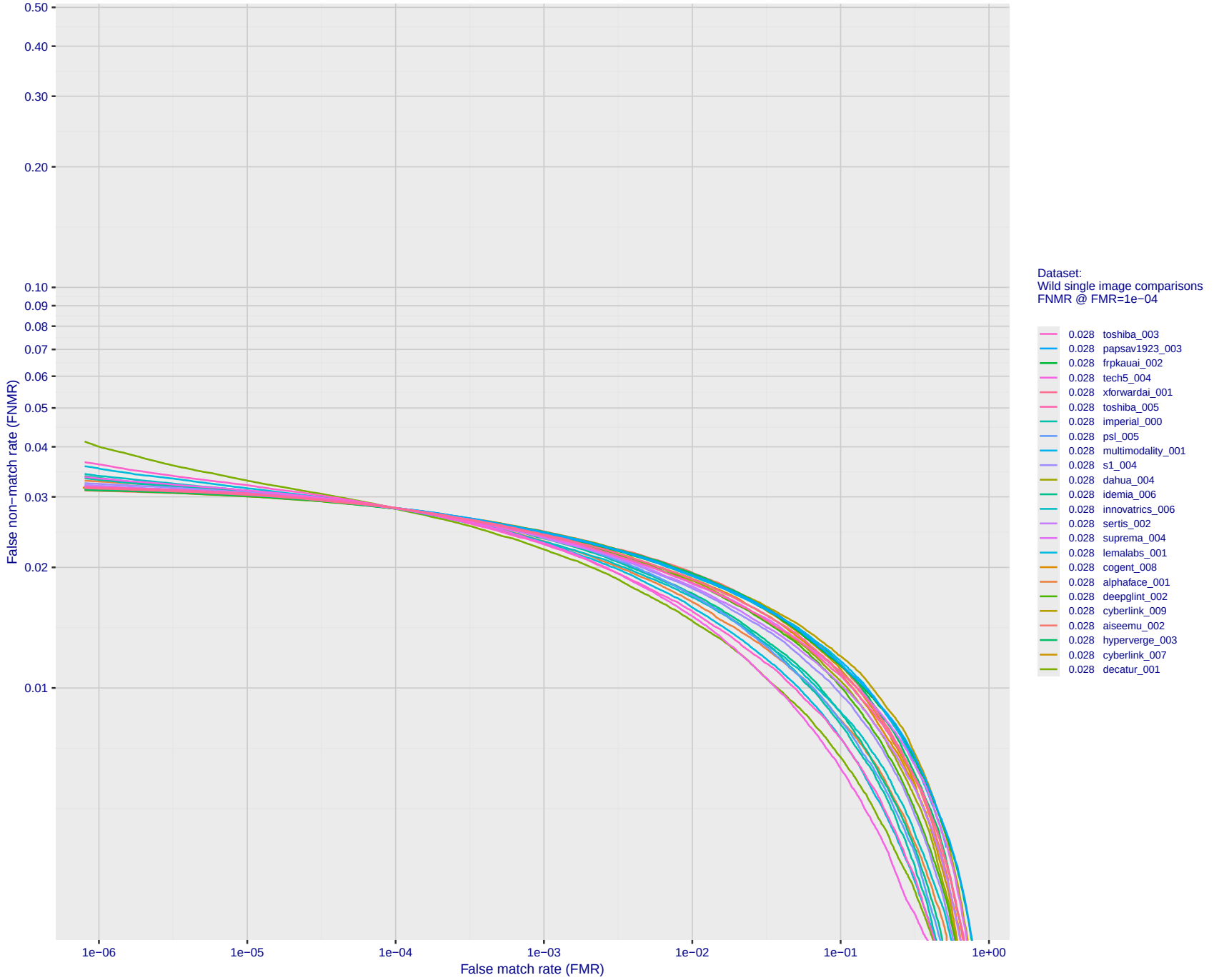

Figure 297: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

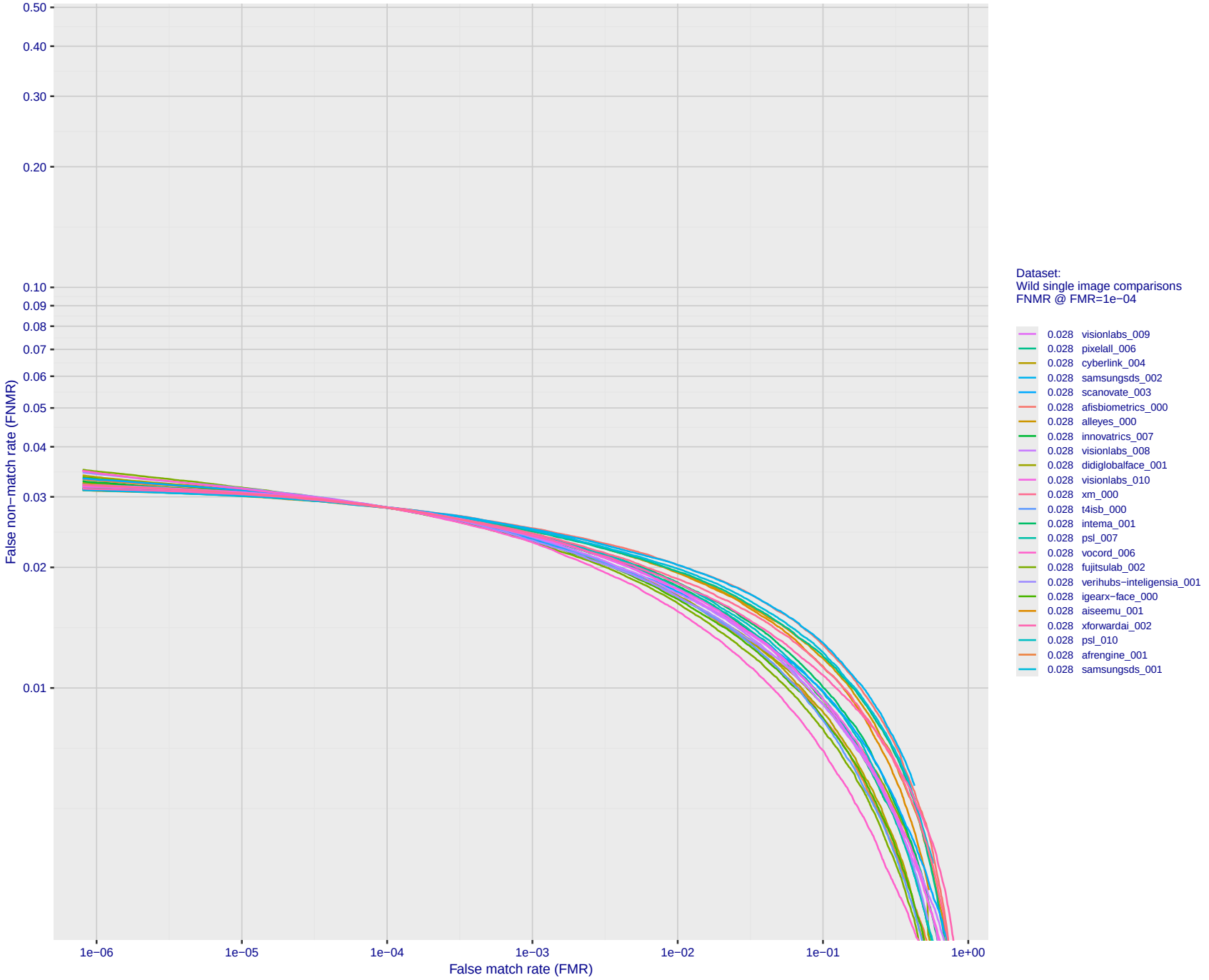

Figure 298: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

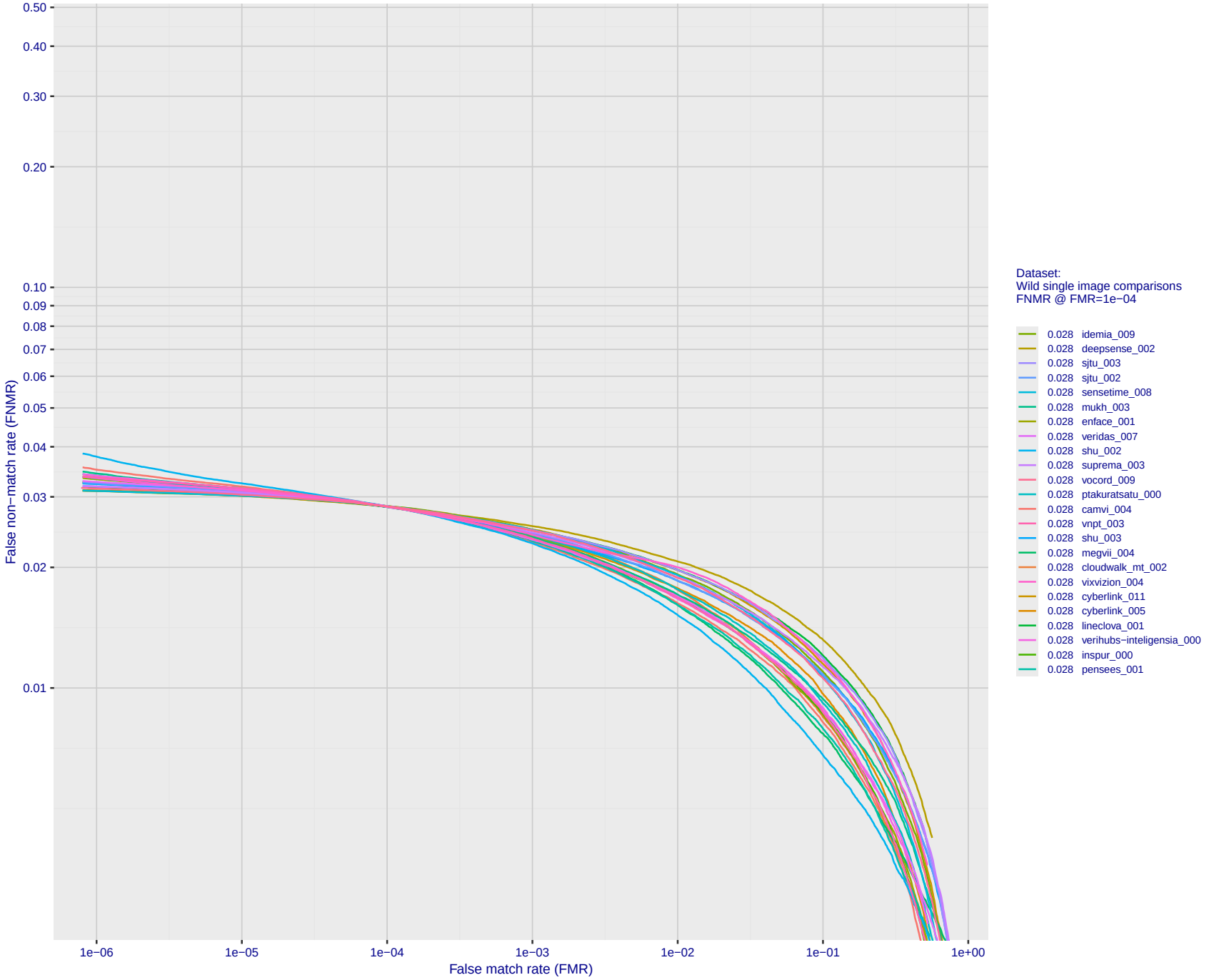

Figure 299: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

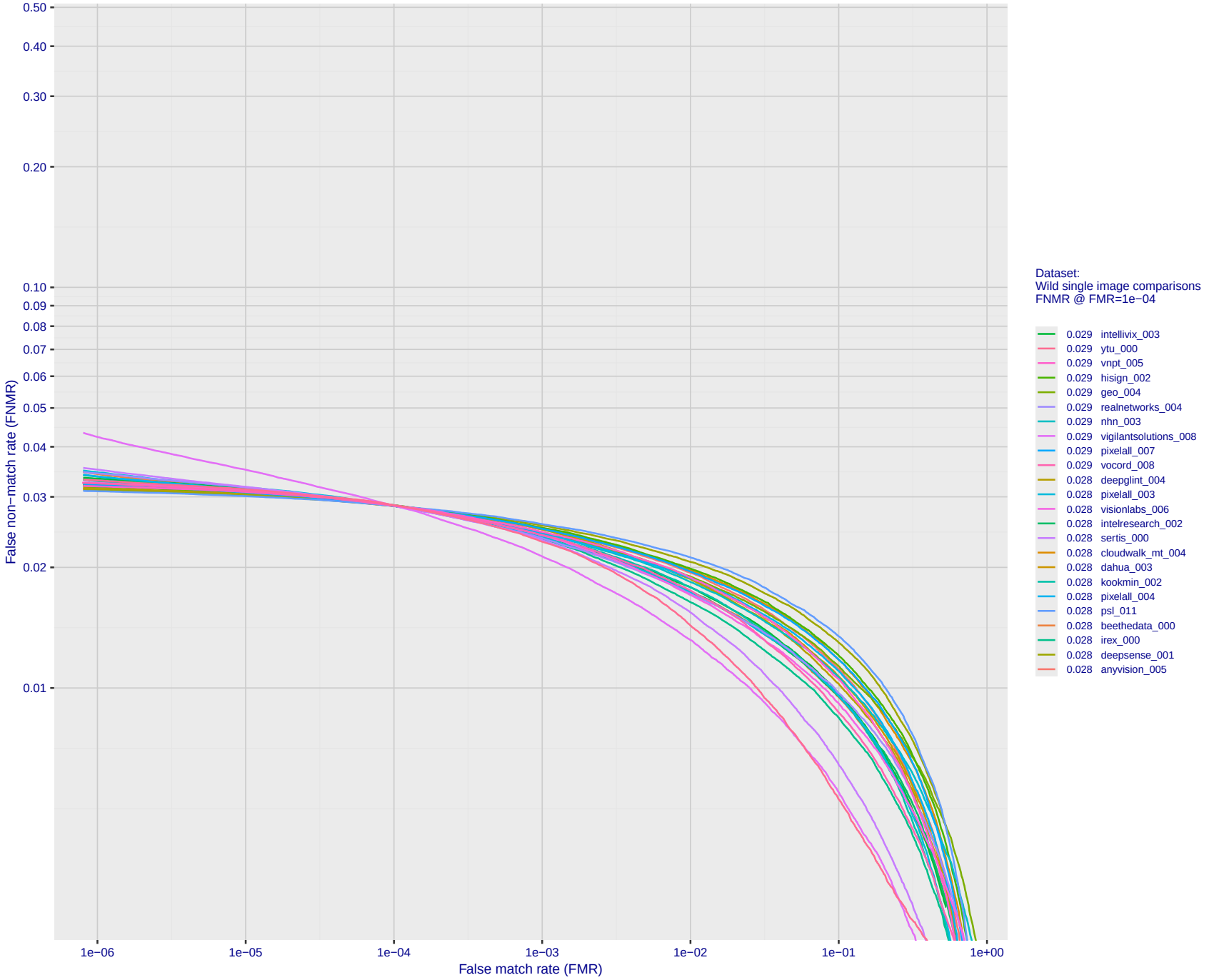

Figure 300: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

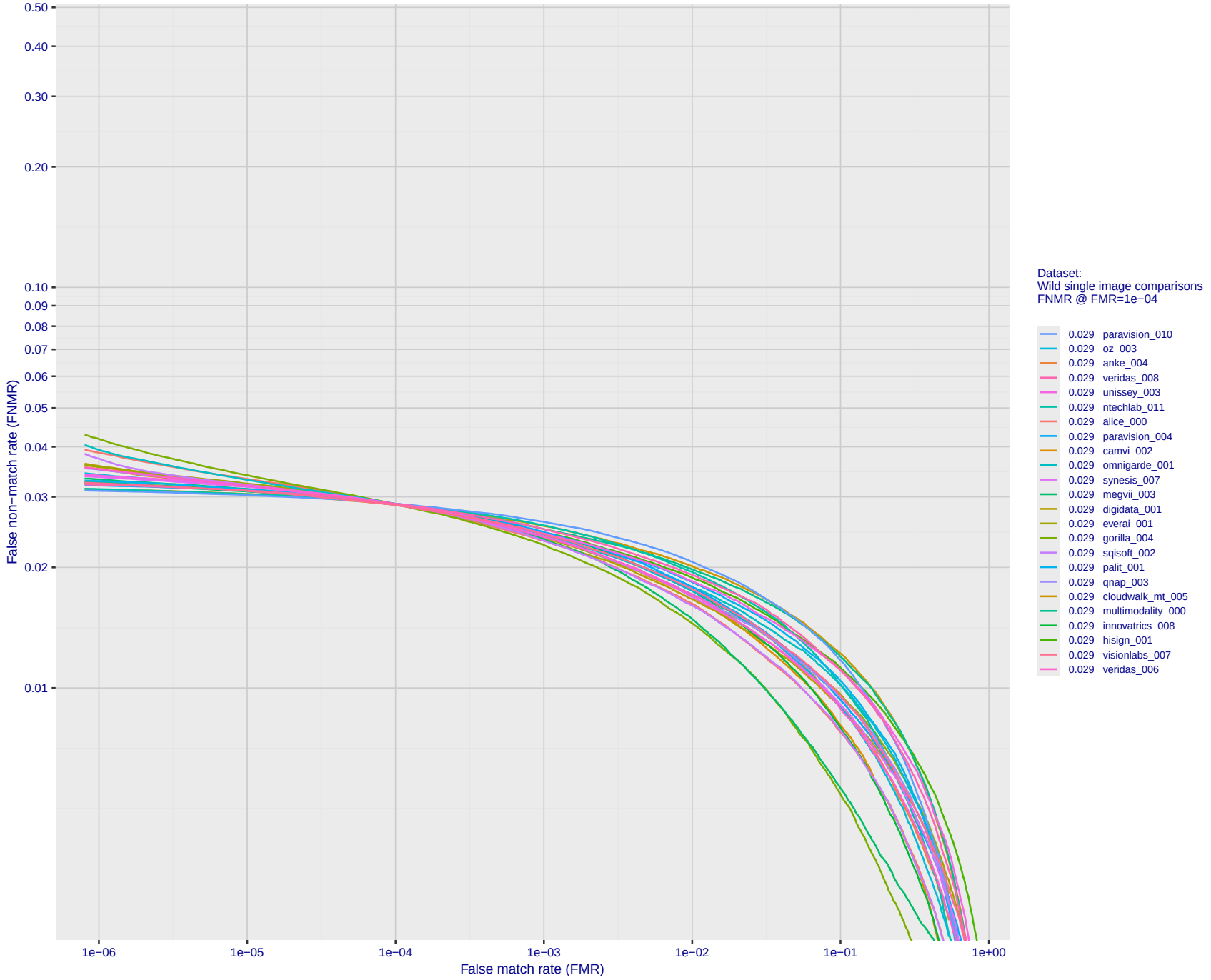

Figure 301: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

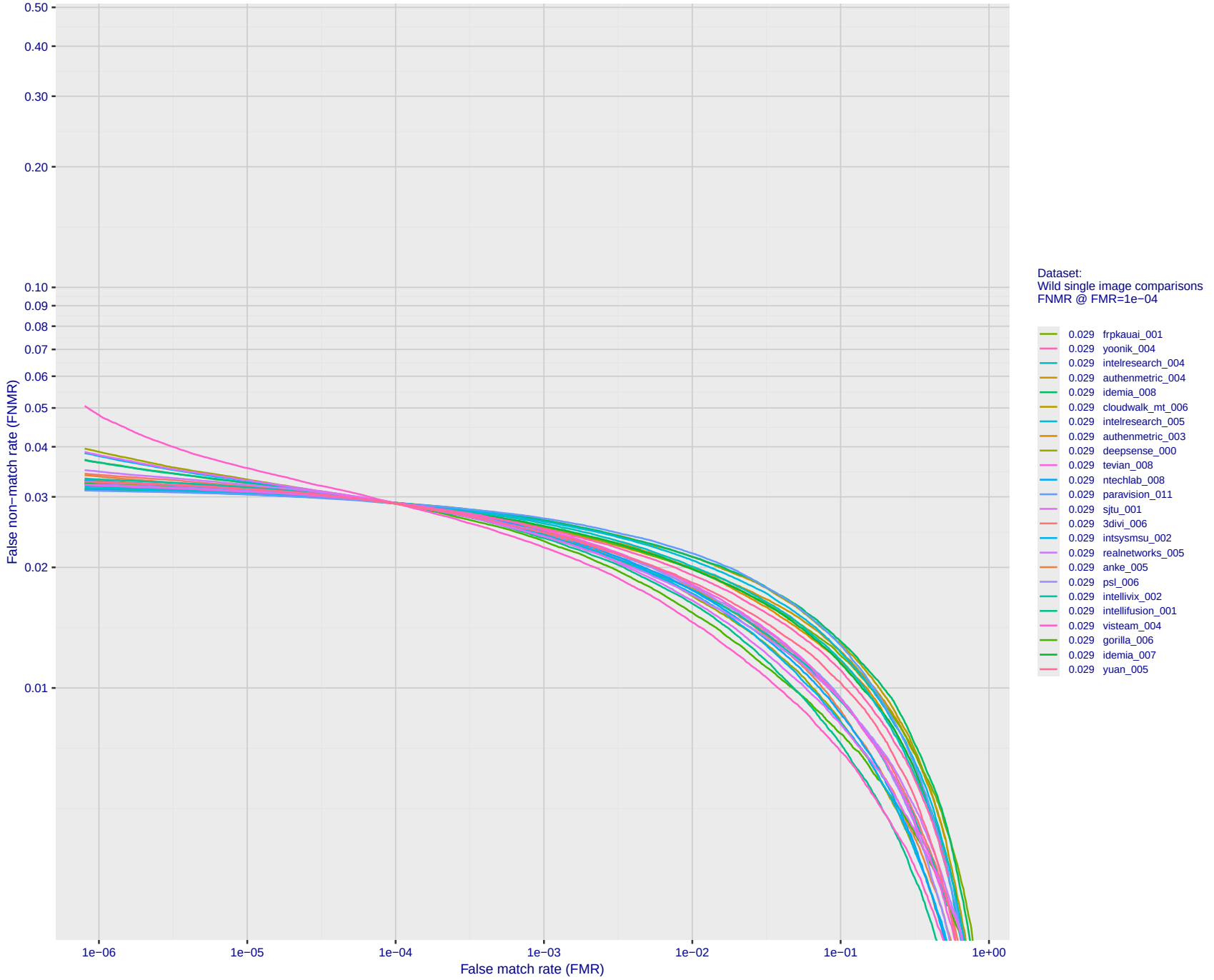

Figure 302: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

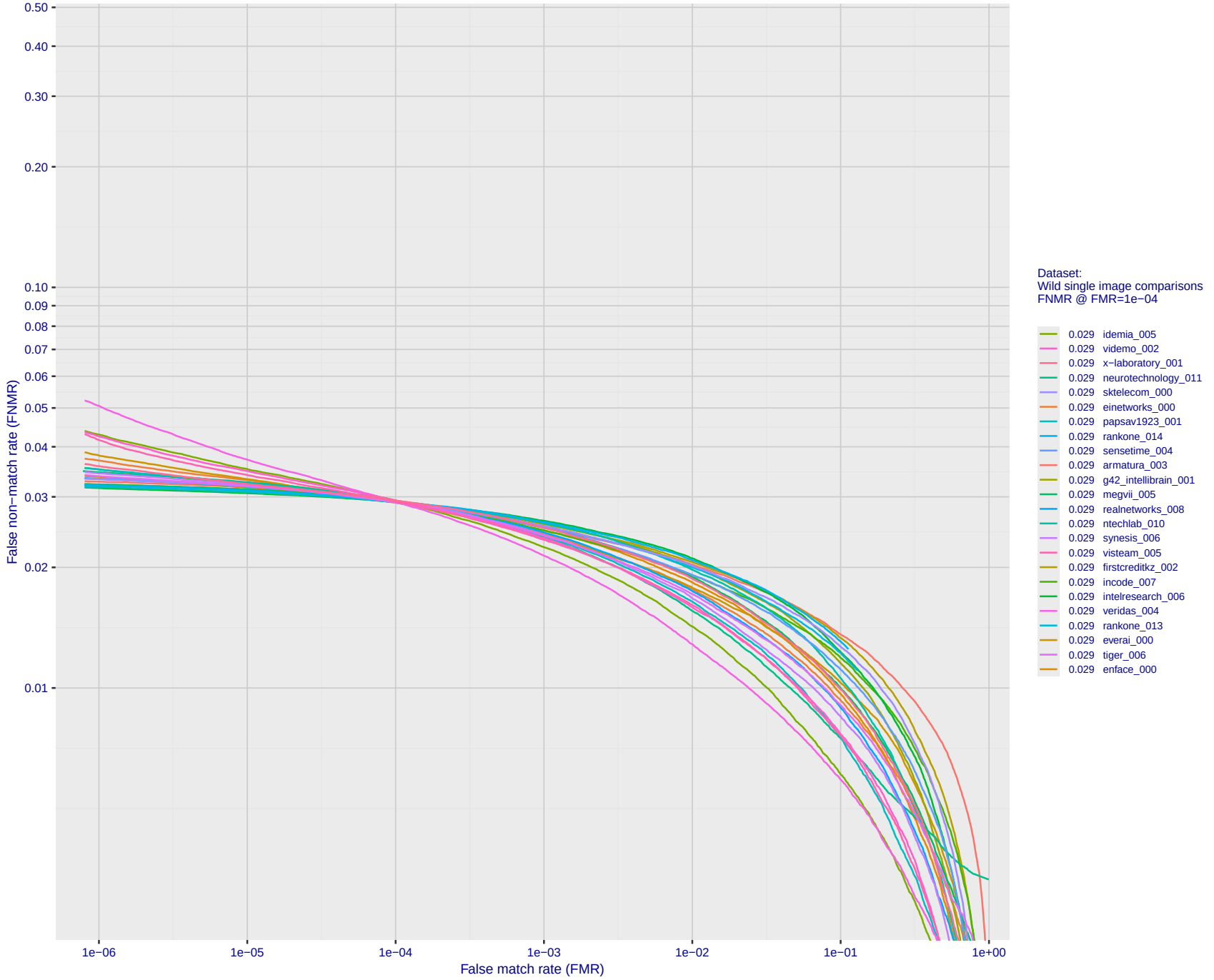

Figure 303: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

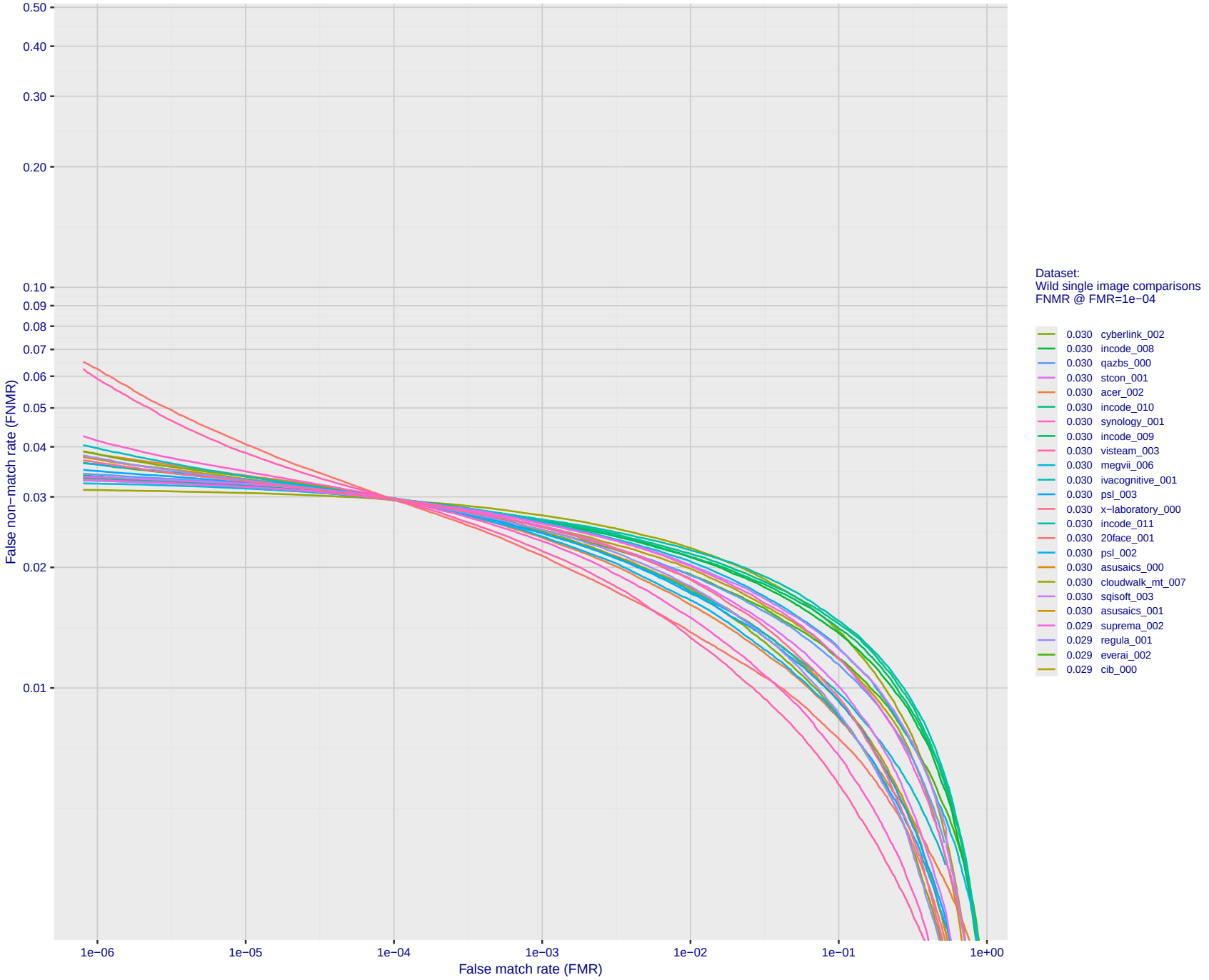

Figure 304: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

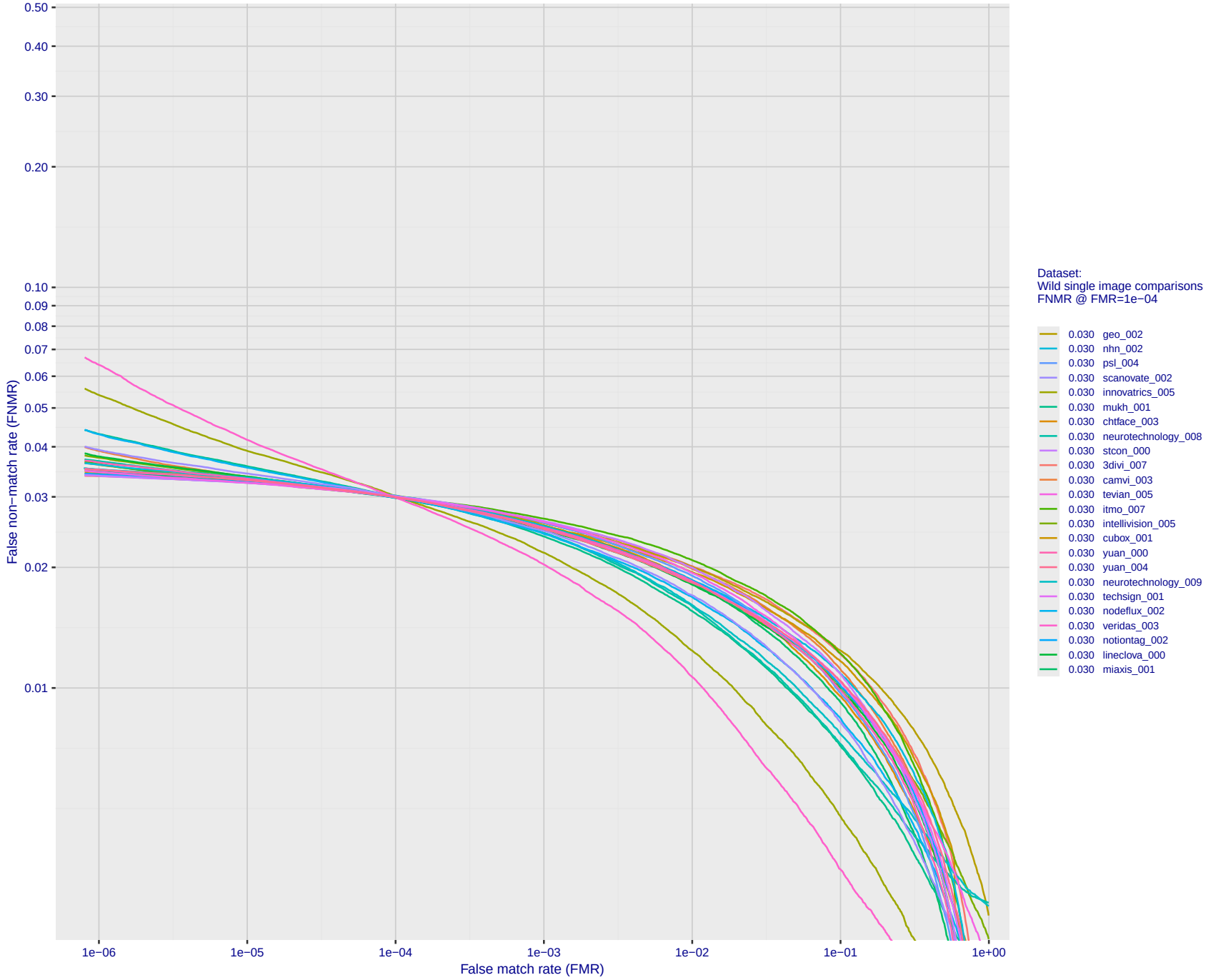

Figure 305: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

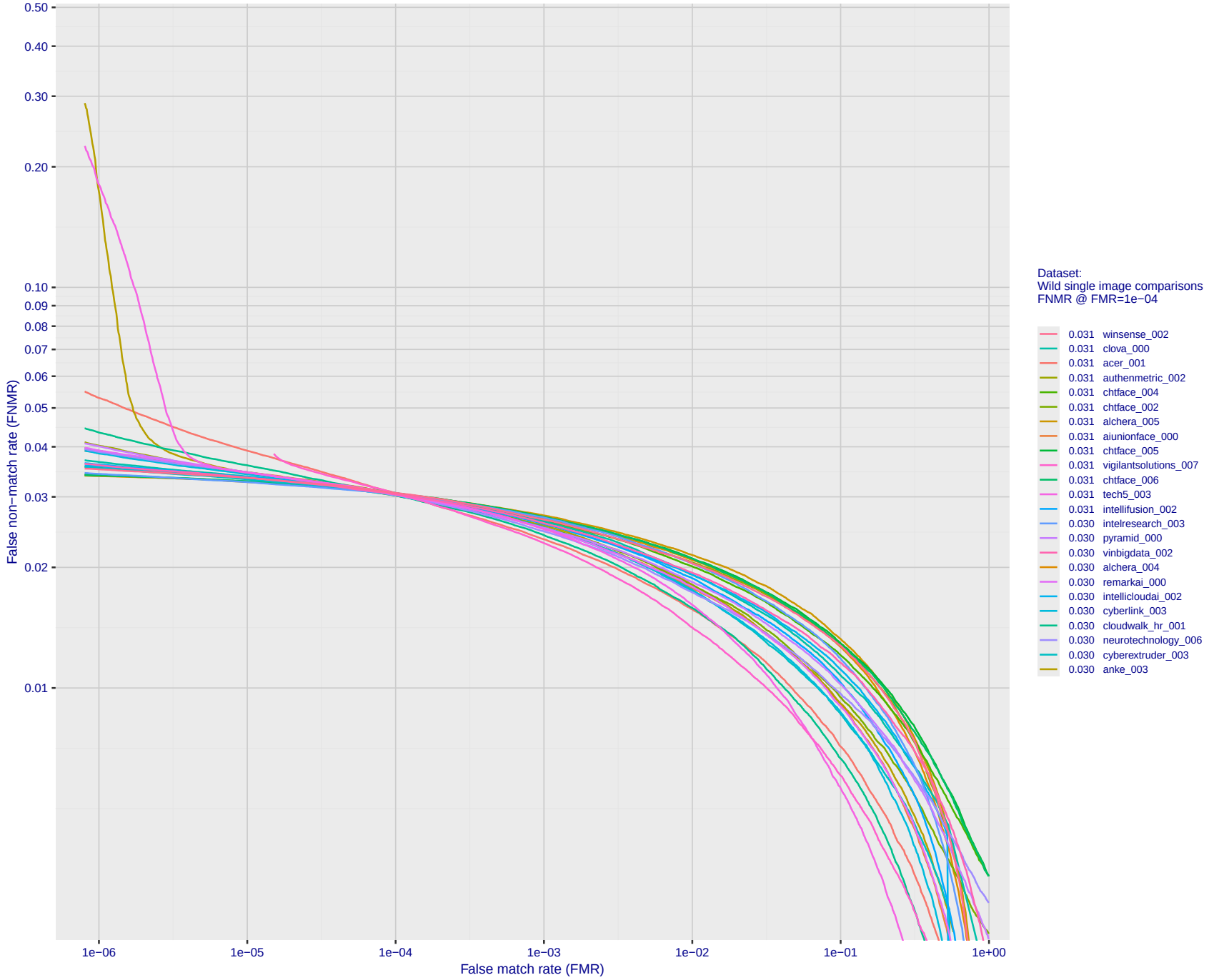

Figure 306: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

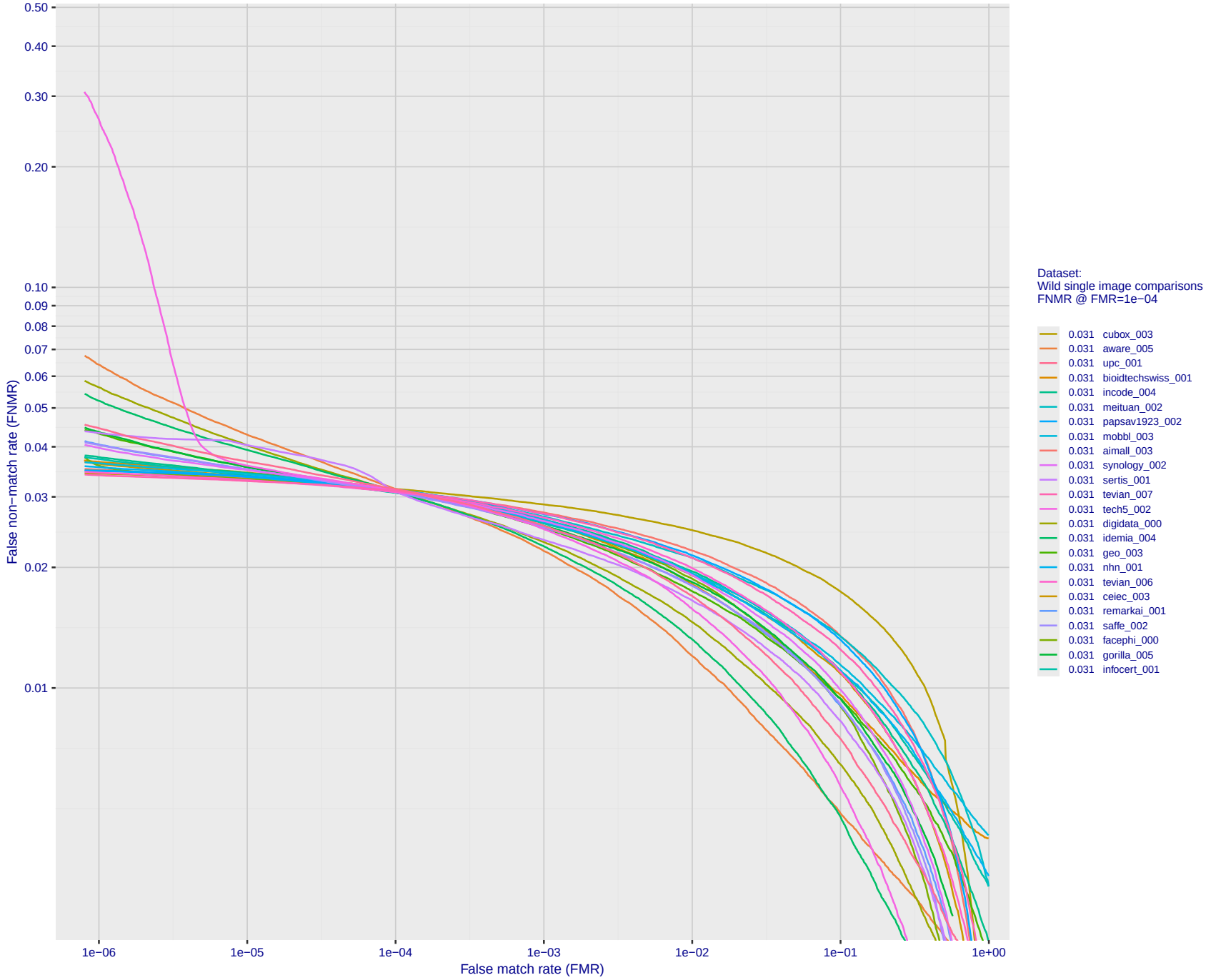

Figure 307: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

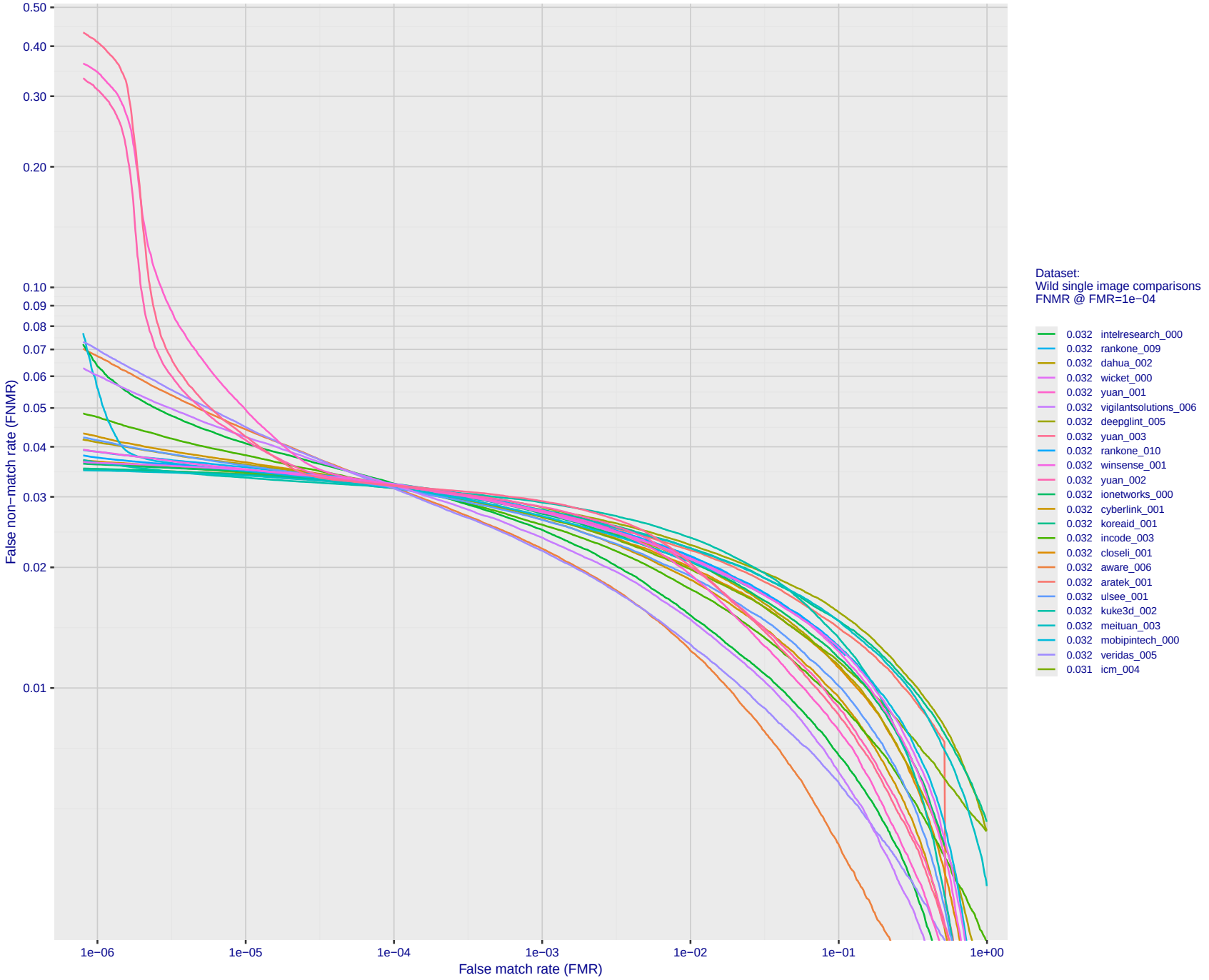

Figure 308: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

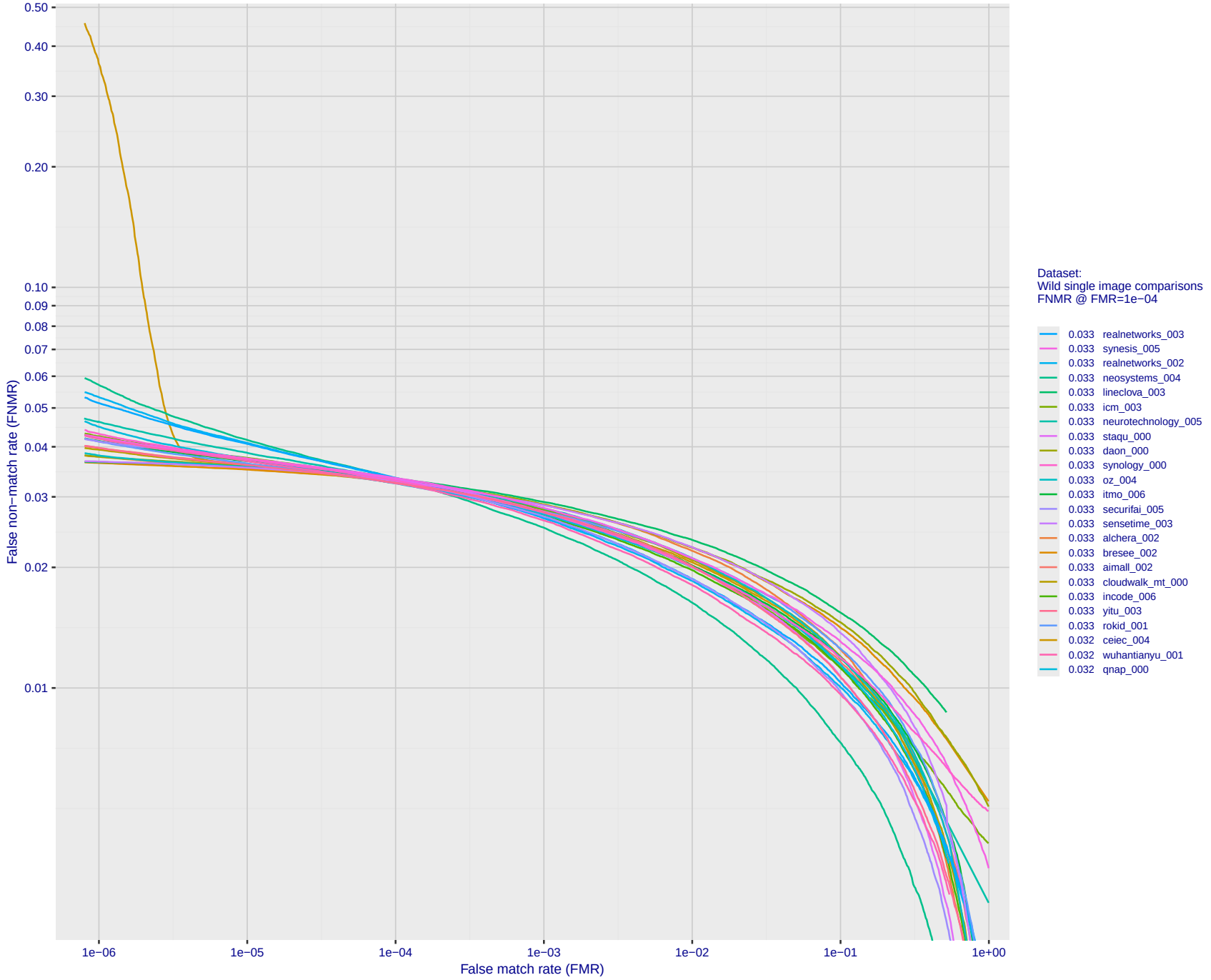

Figure 309: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

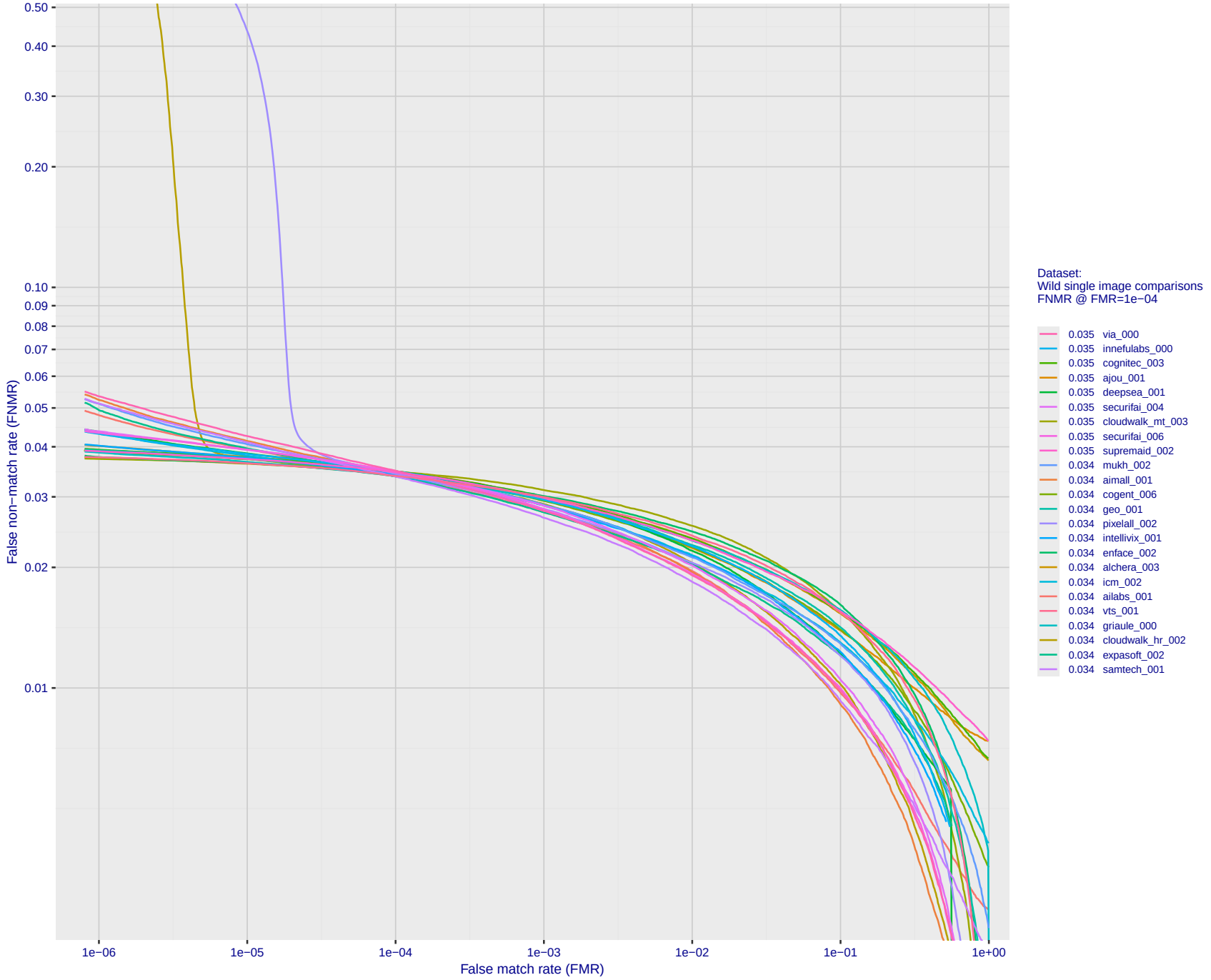

Figure 310: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

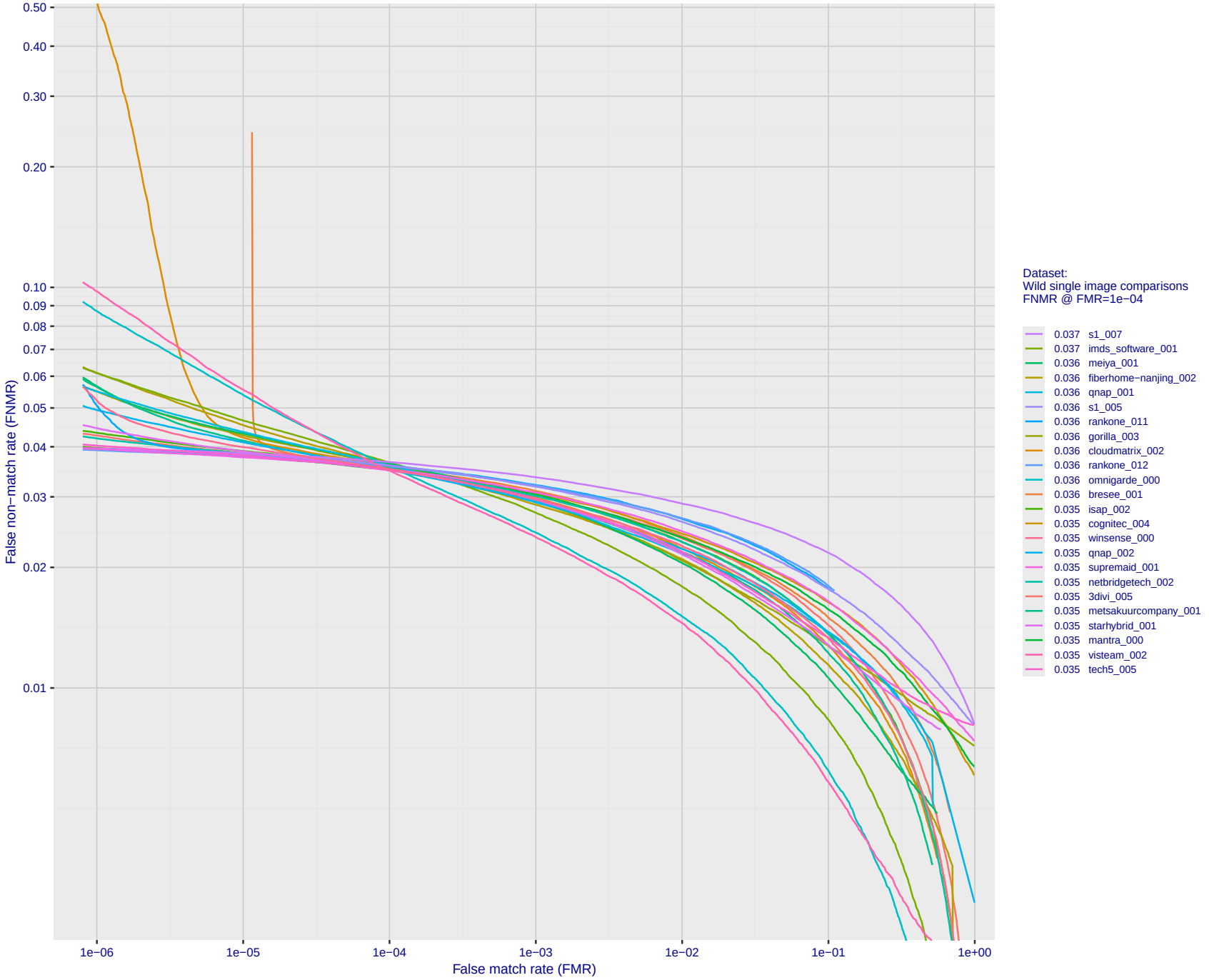

Figure 311: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

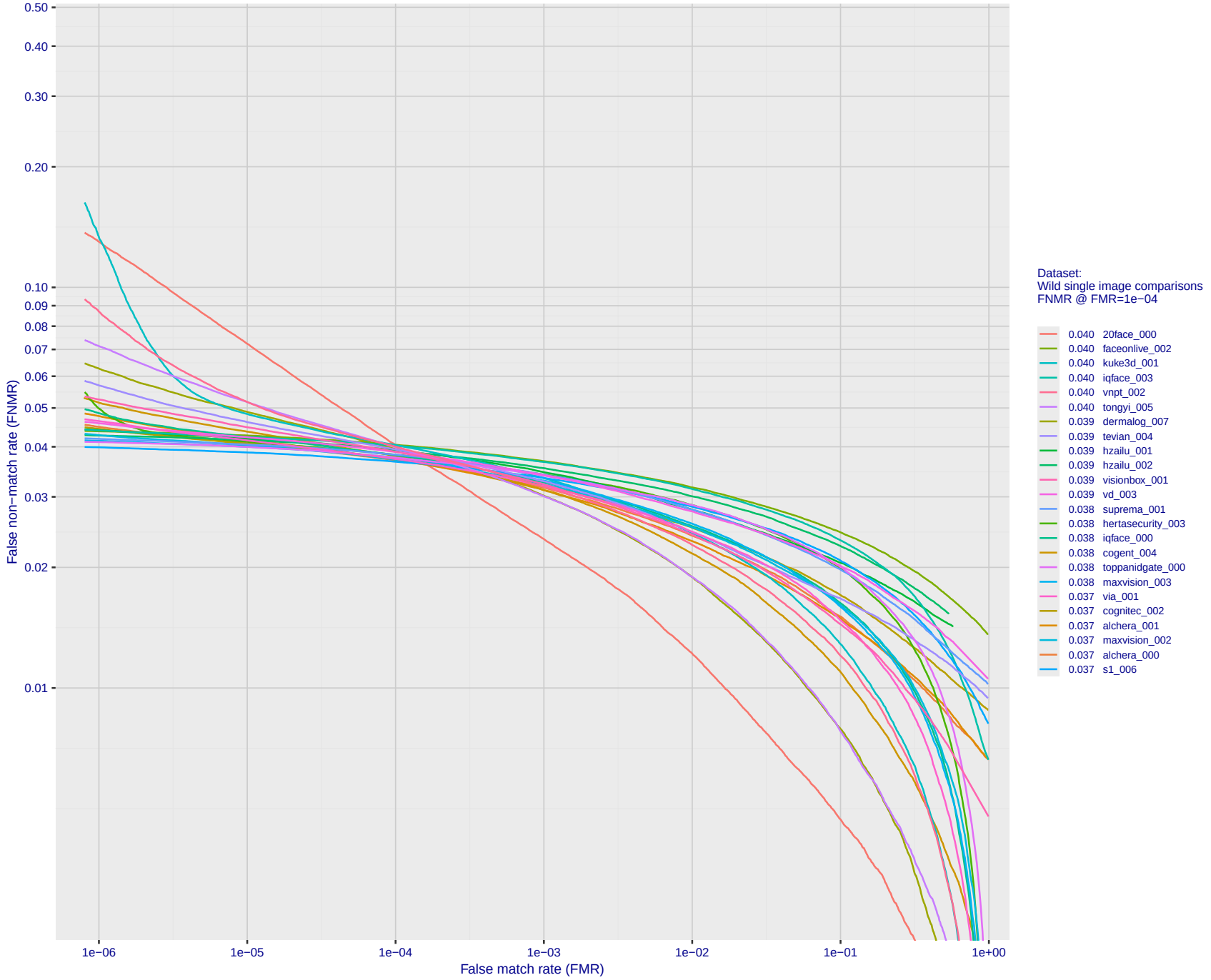

Figure 312: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

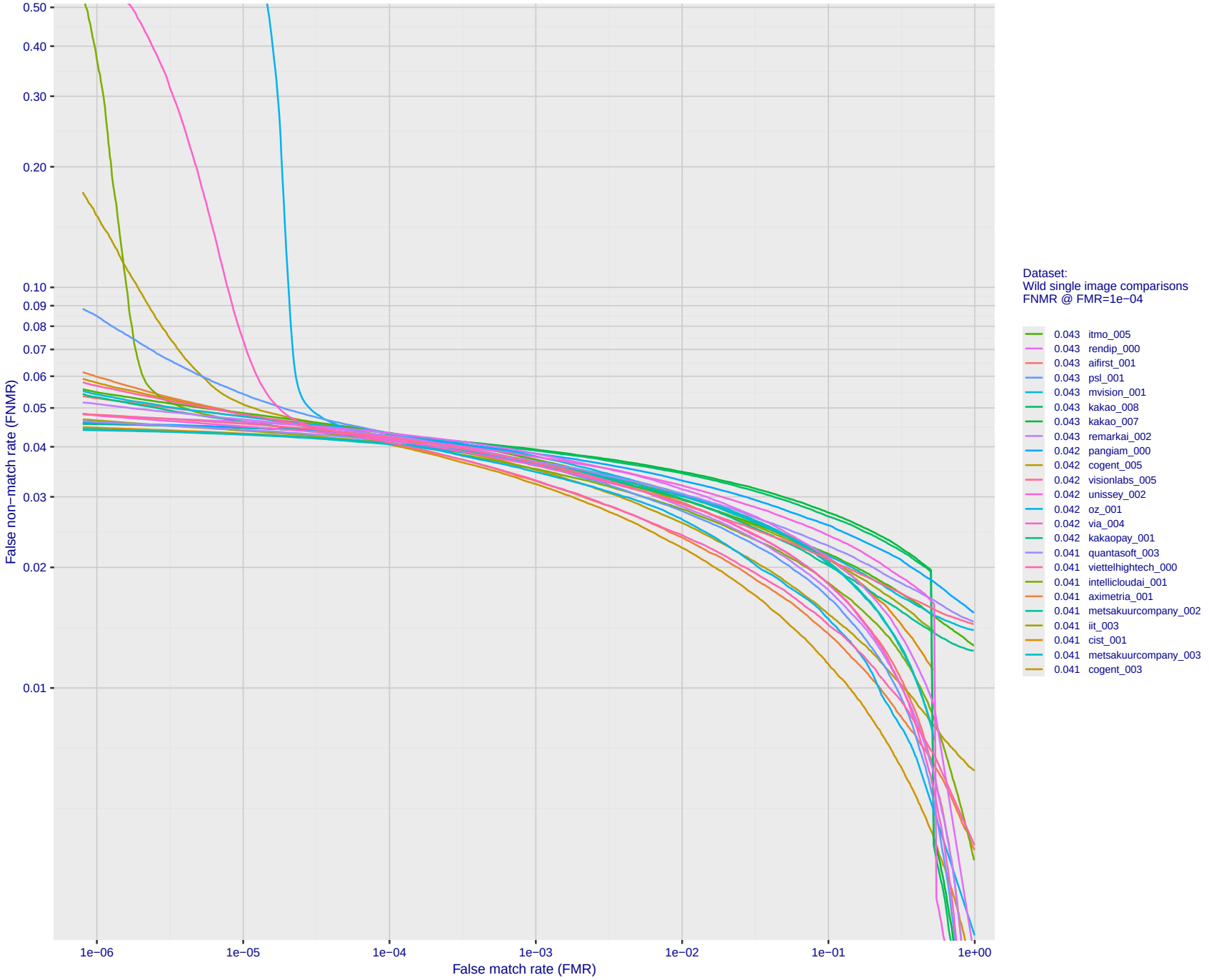

Figure 313: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

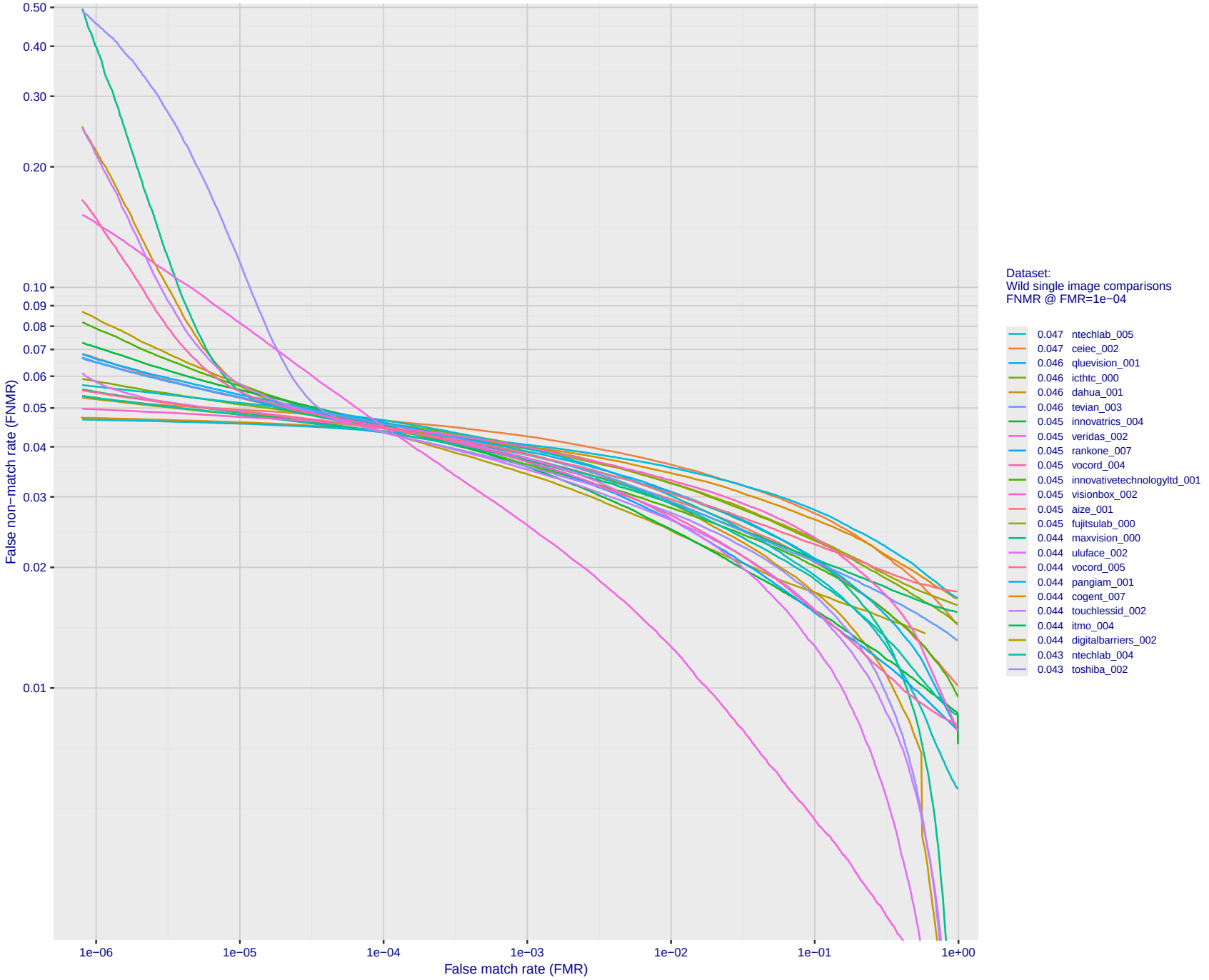

Figure 314: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

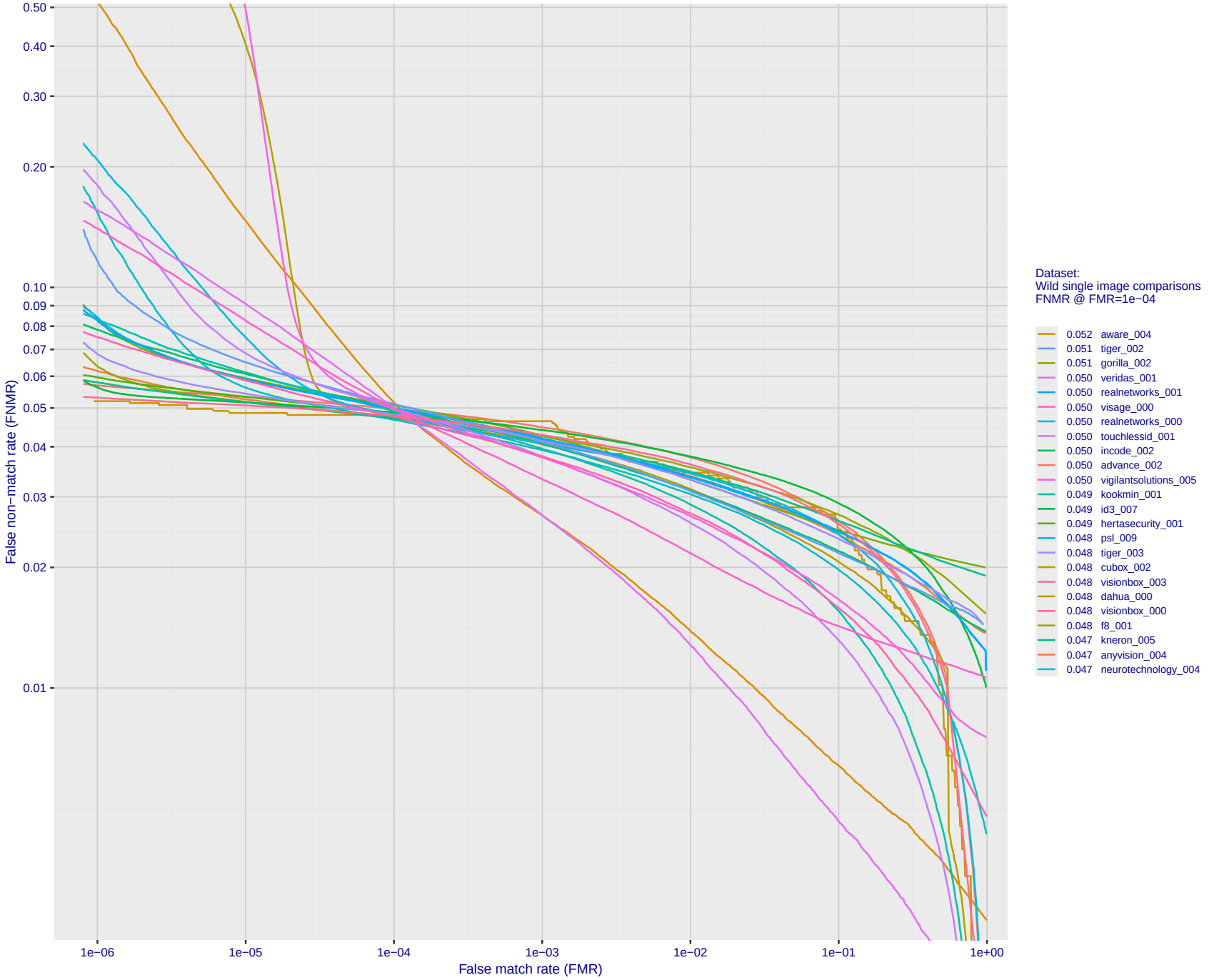

Figure 315: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

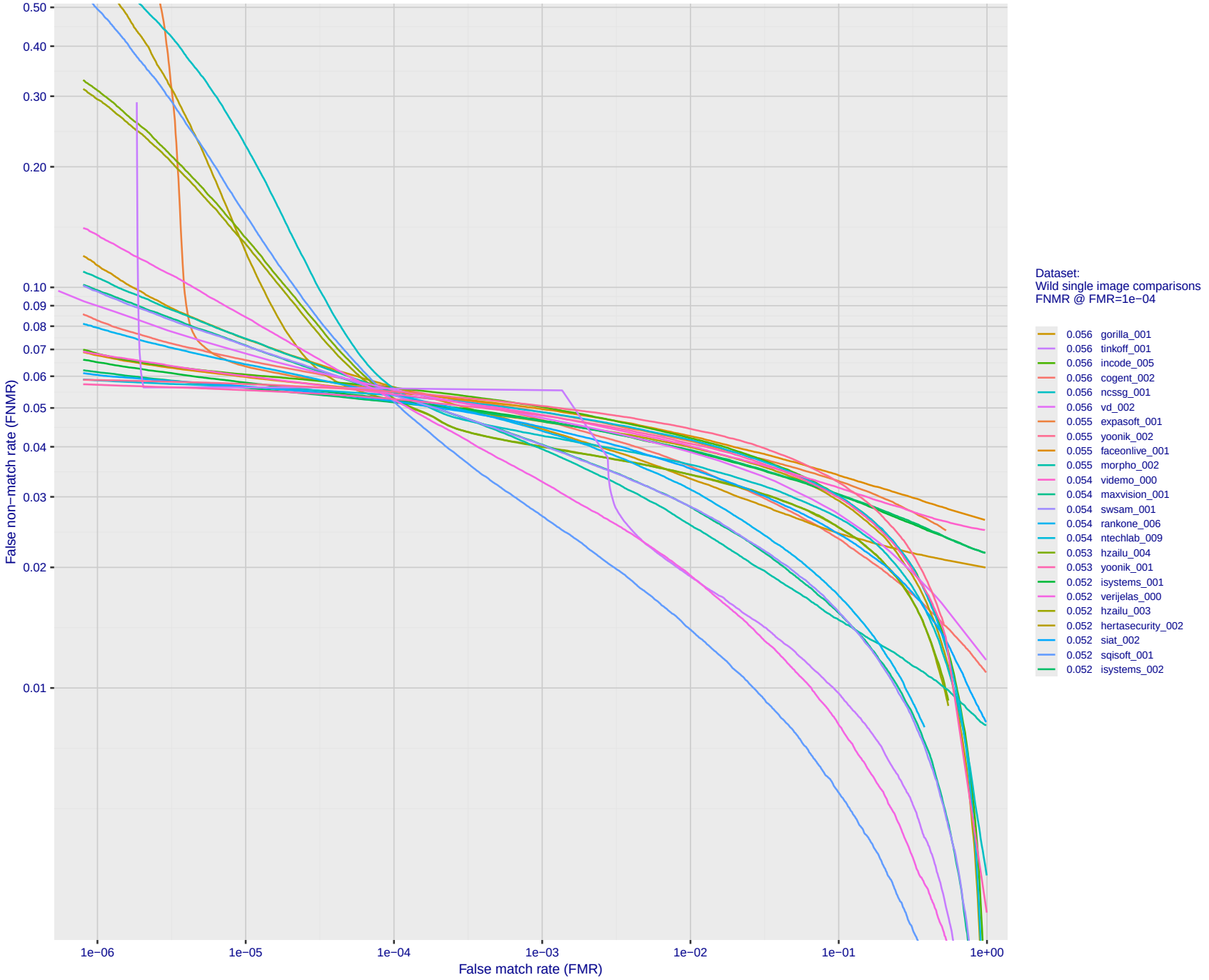

Figure 316: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

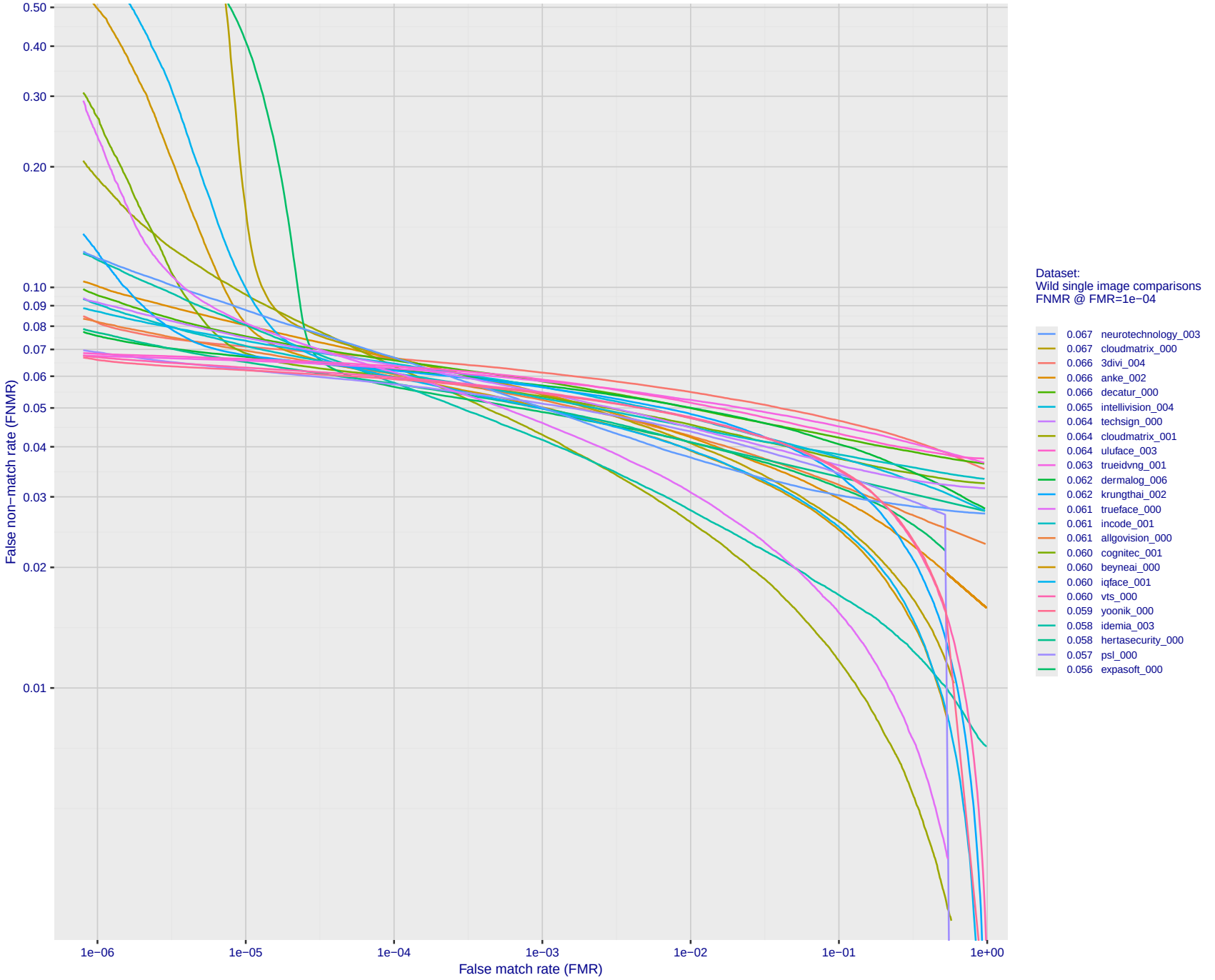

Figure 317: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

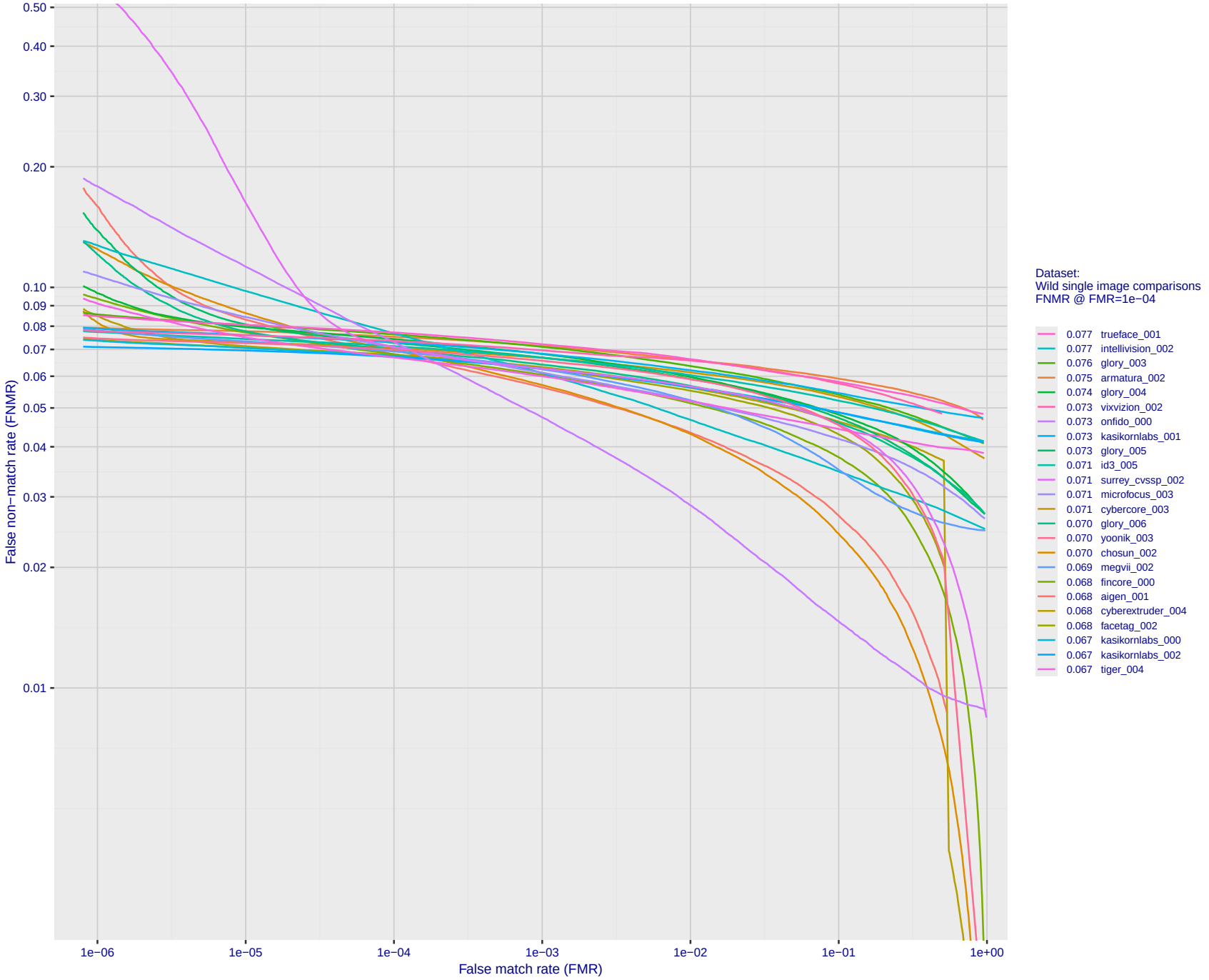

Figure 318: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

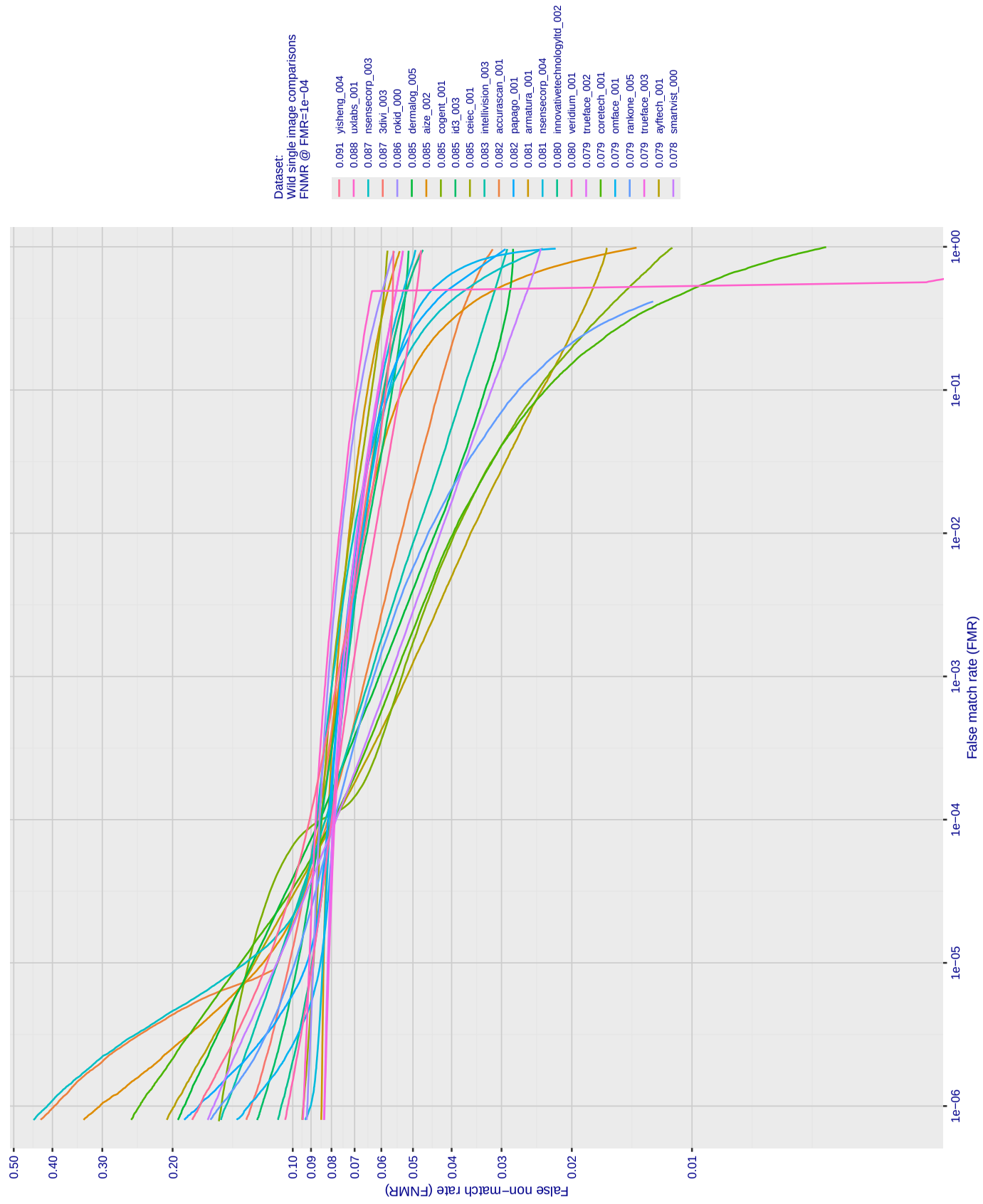

Figure 319: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

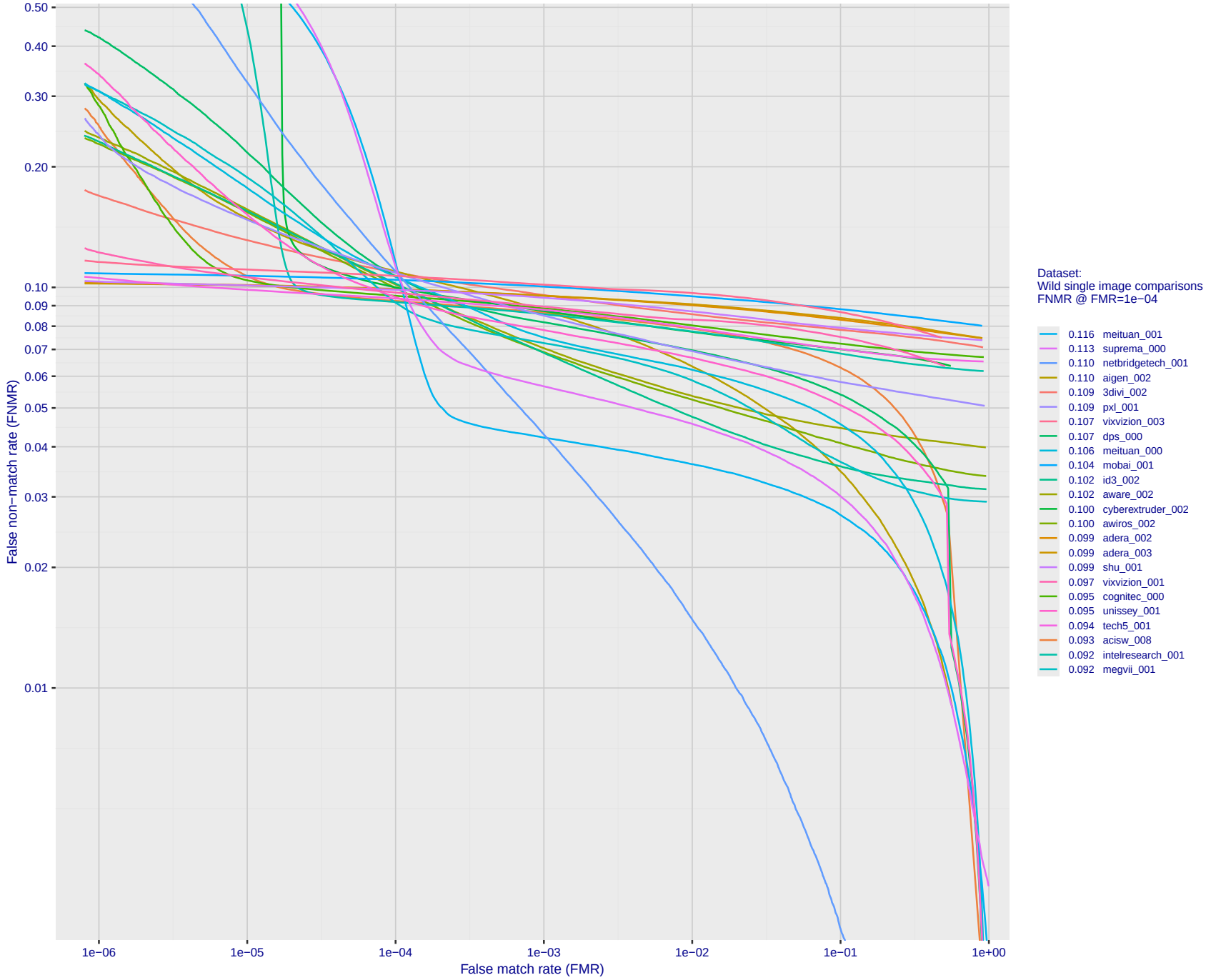

Figure 320: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

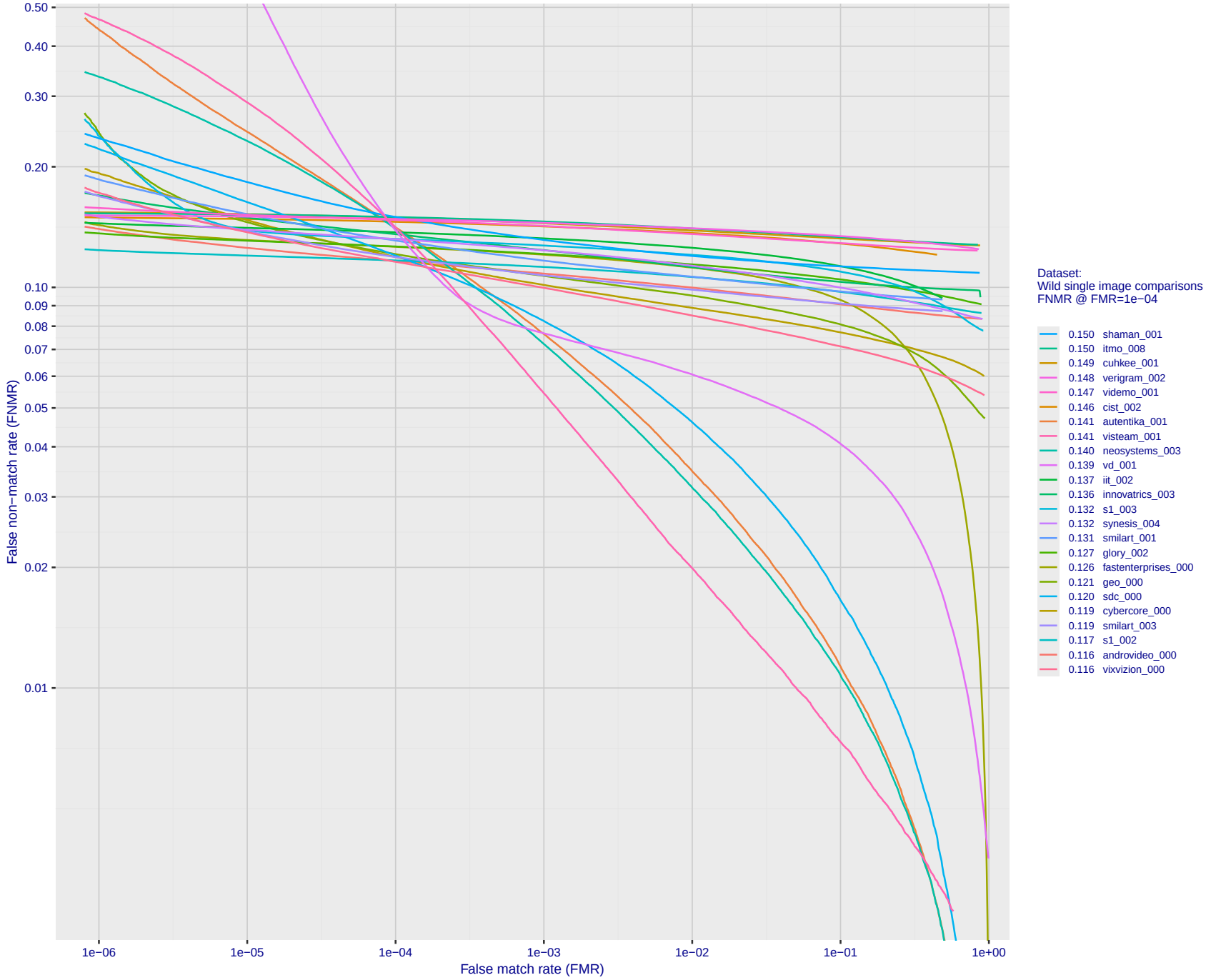

Figure 321: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

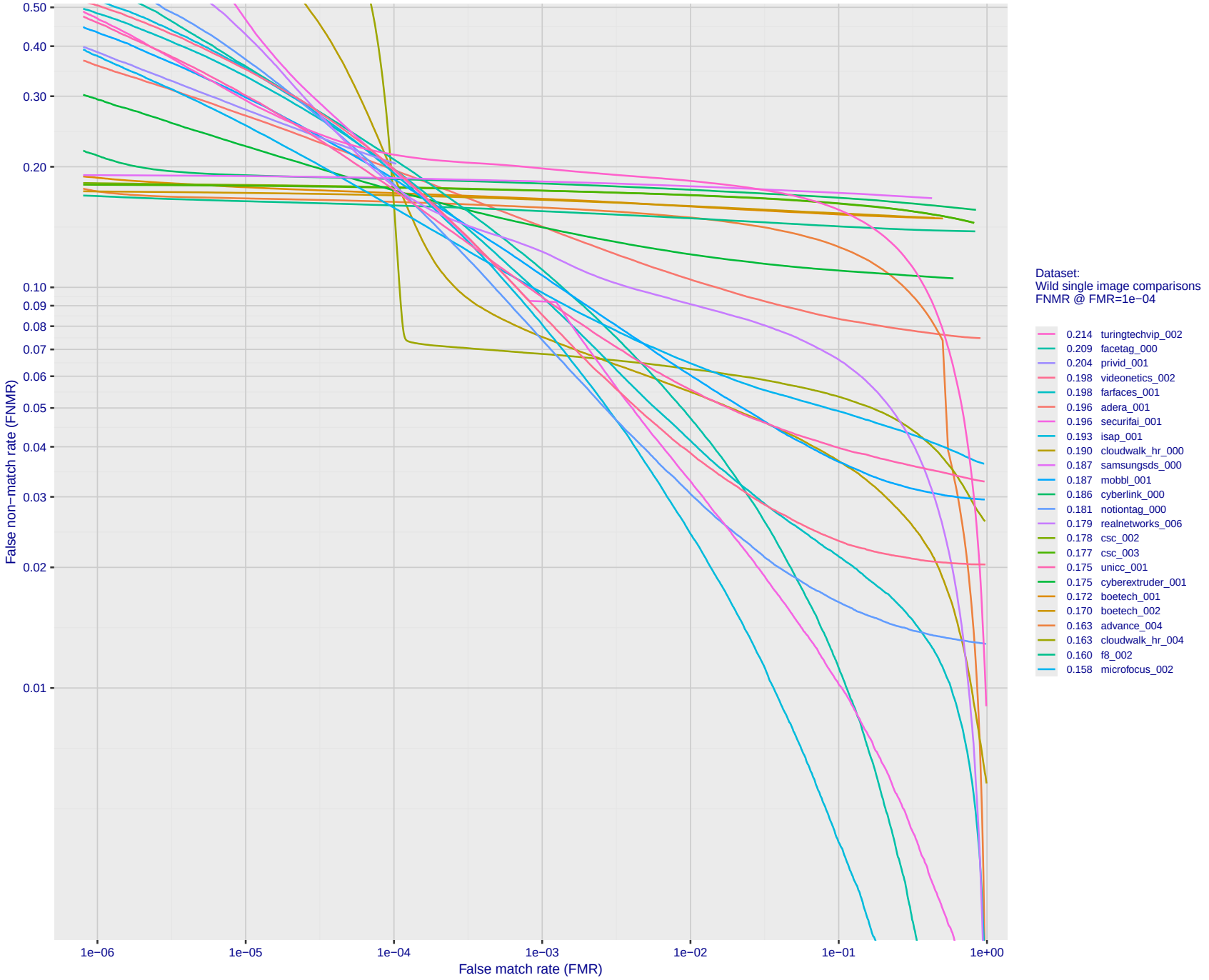

Figure 322: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

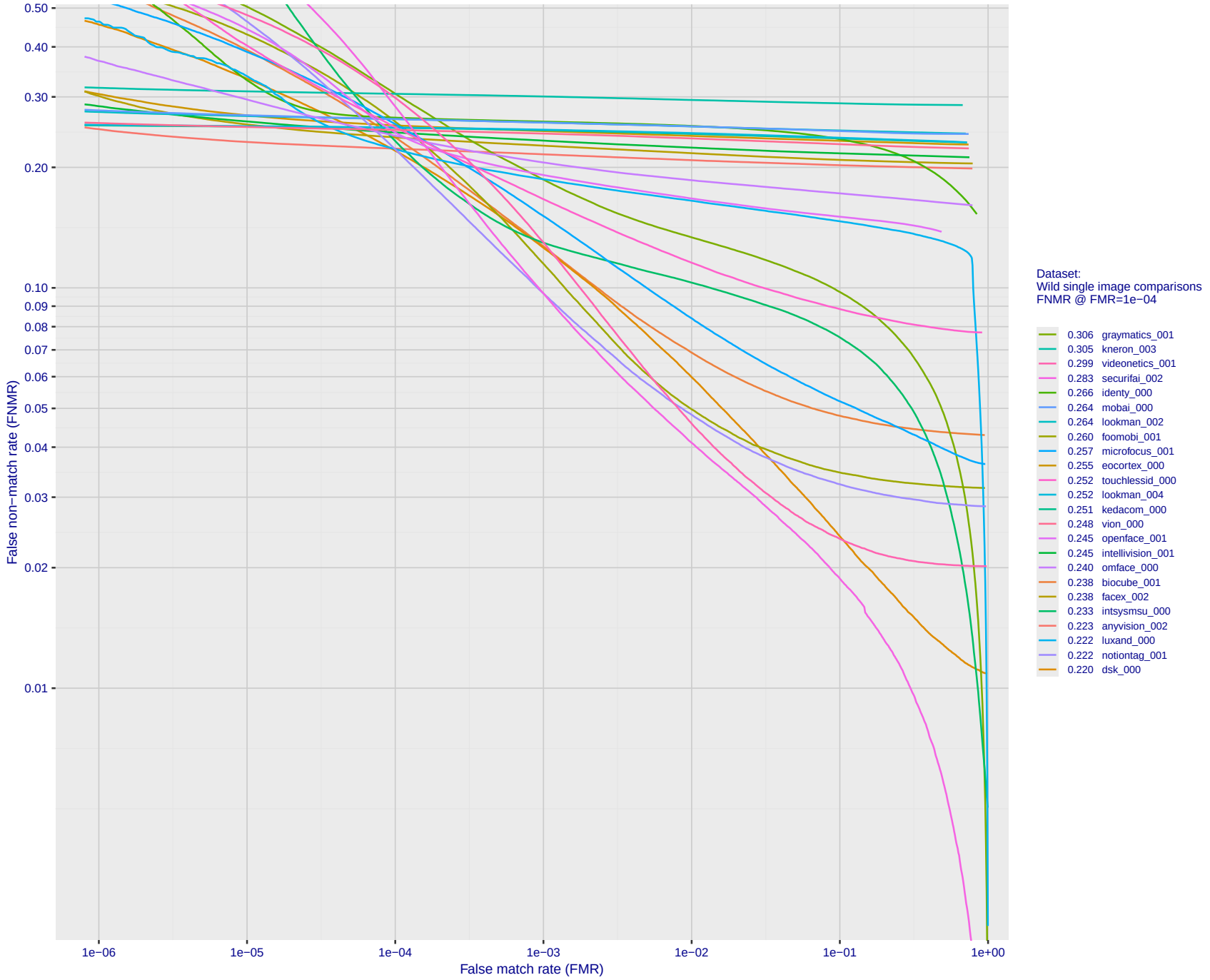

Figure 323: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43 FNMR(T) "False non-match rate" FMR(T) "False match rate"

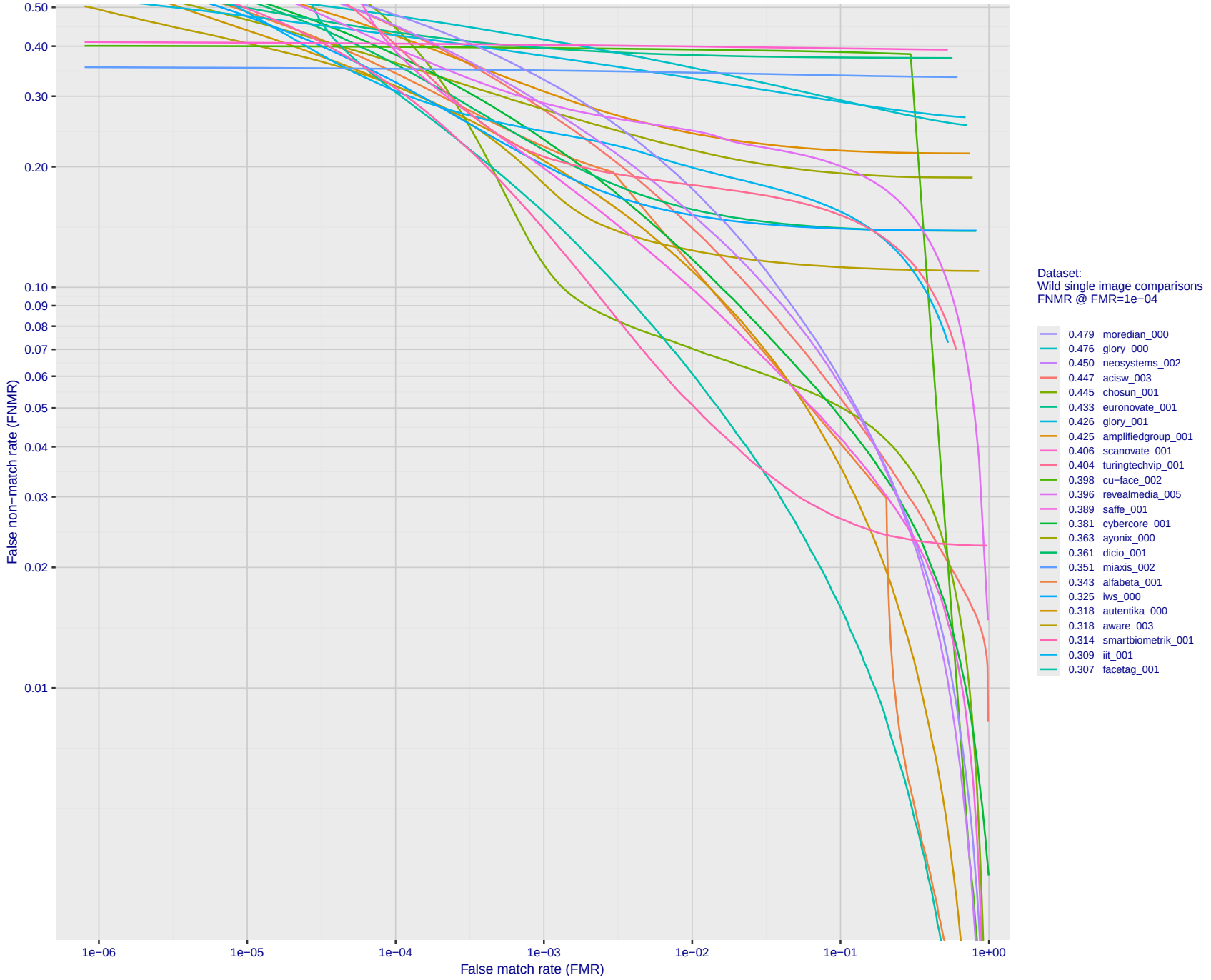

Figure 324: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

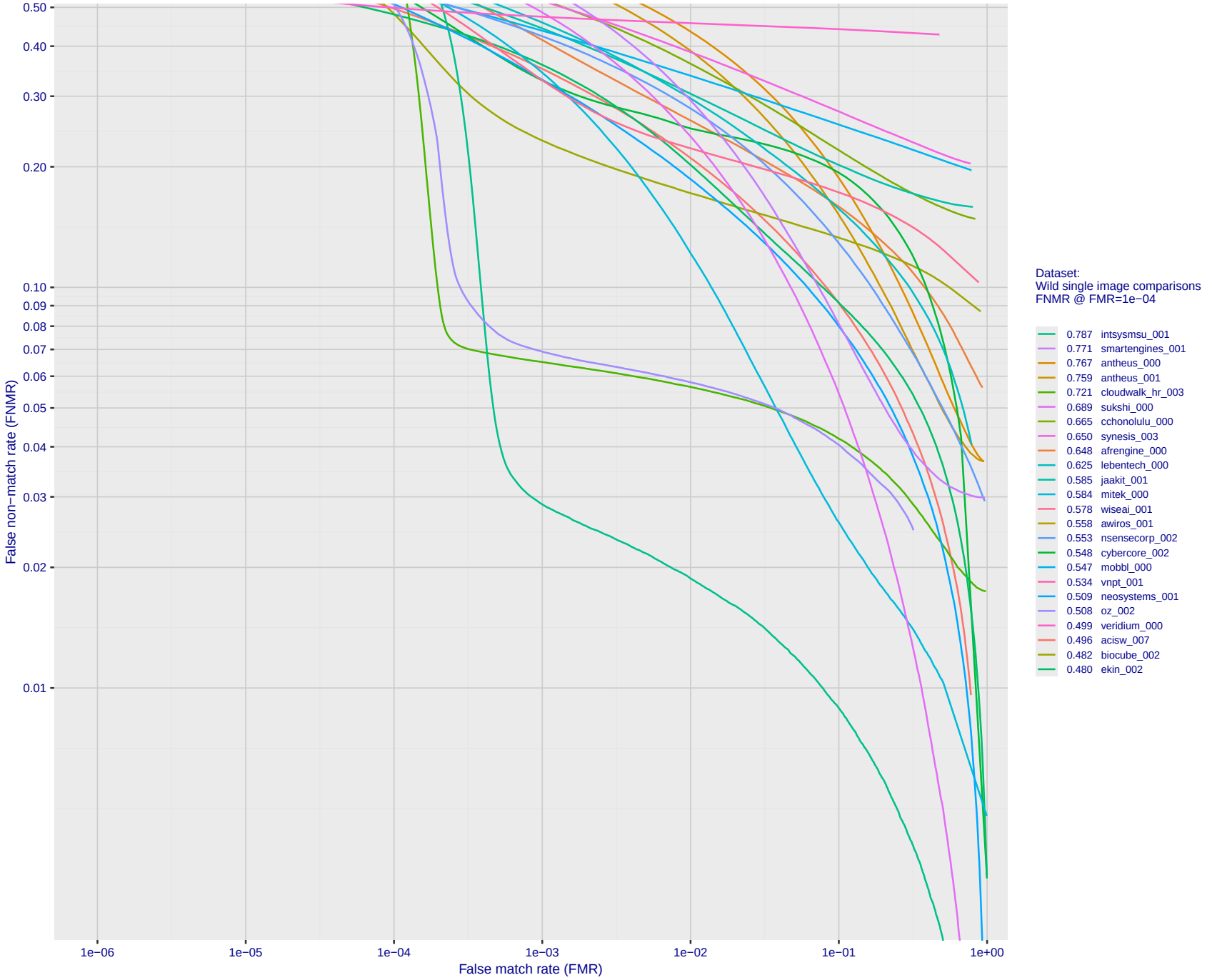

Figure 325: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

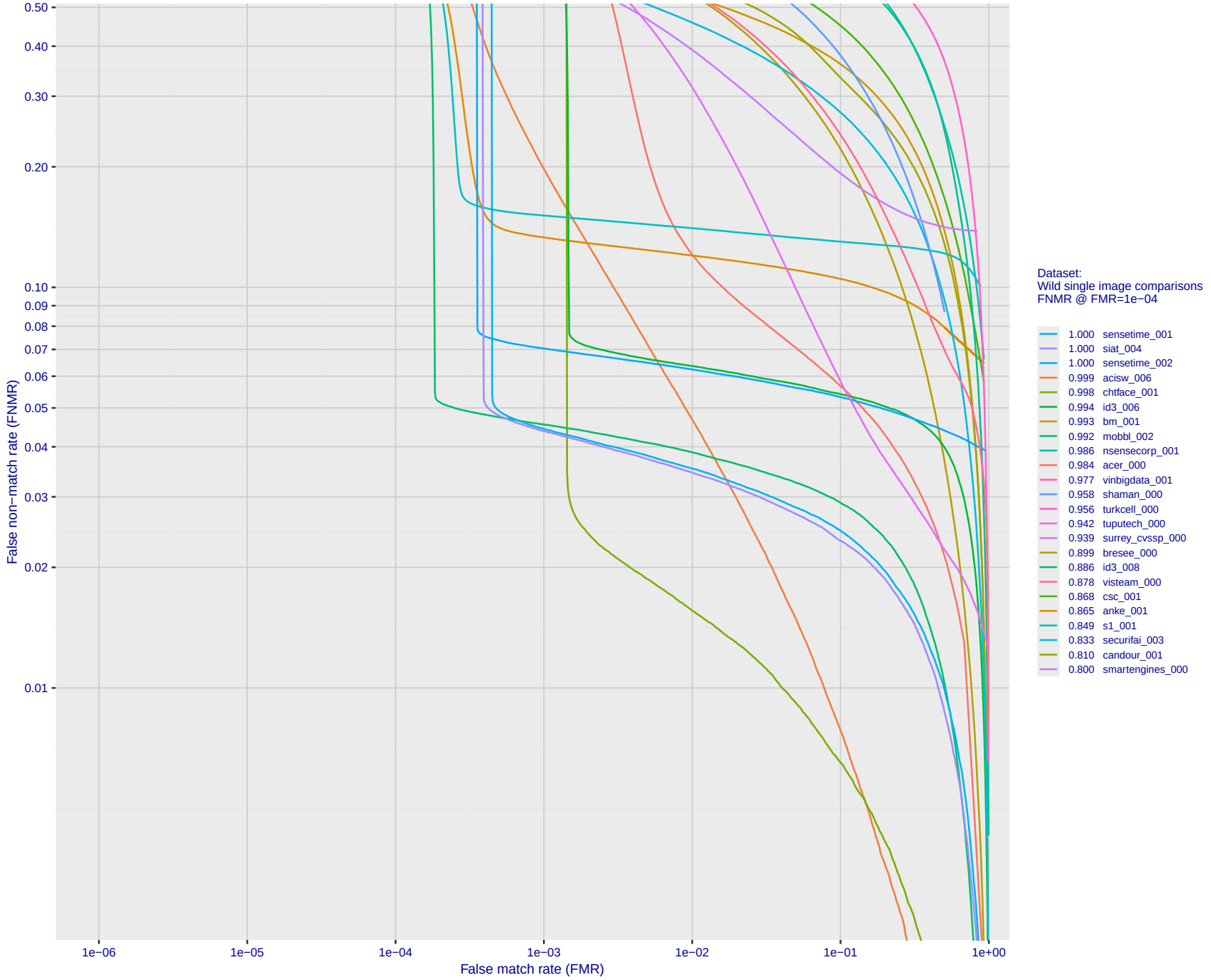

Figure 326: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

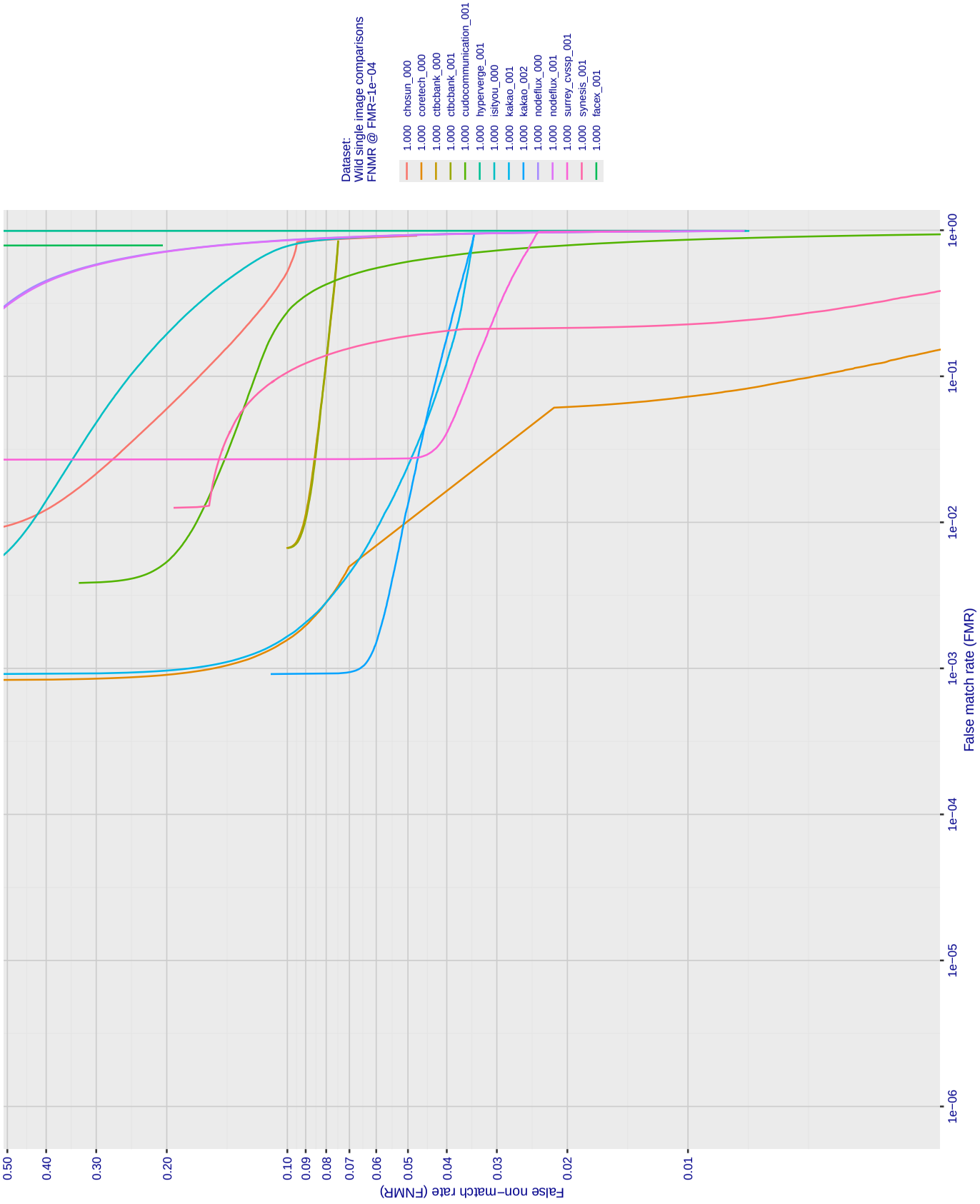

Figure 327: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

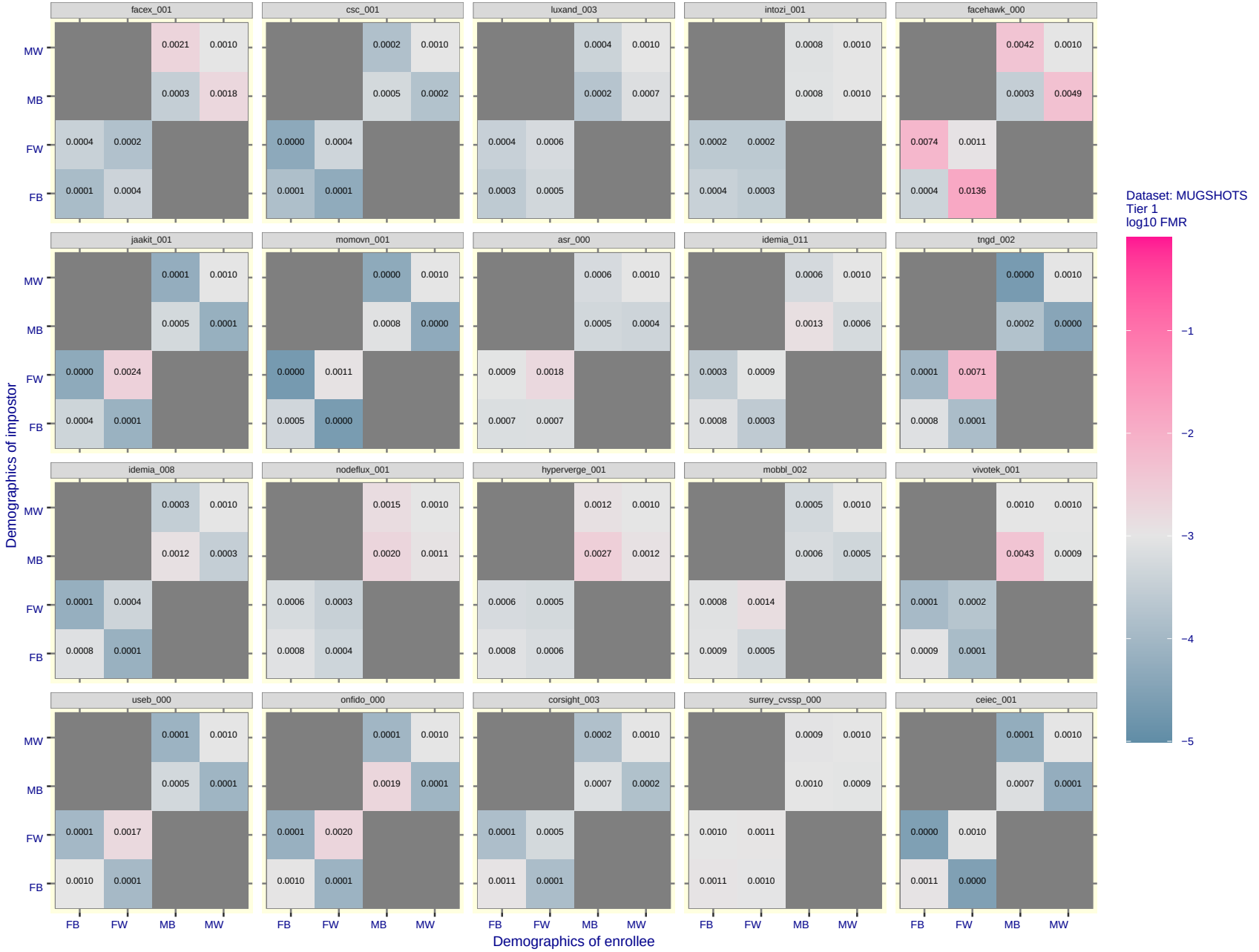

Figure 328: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

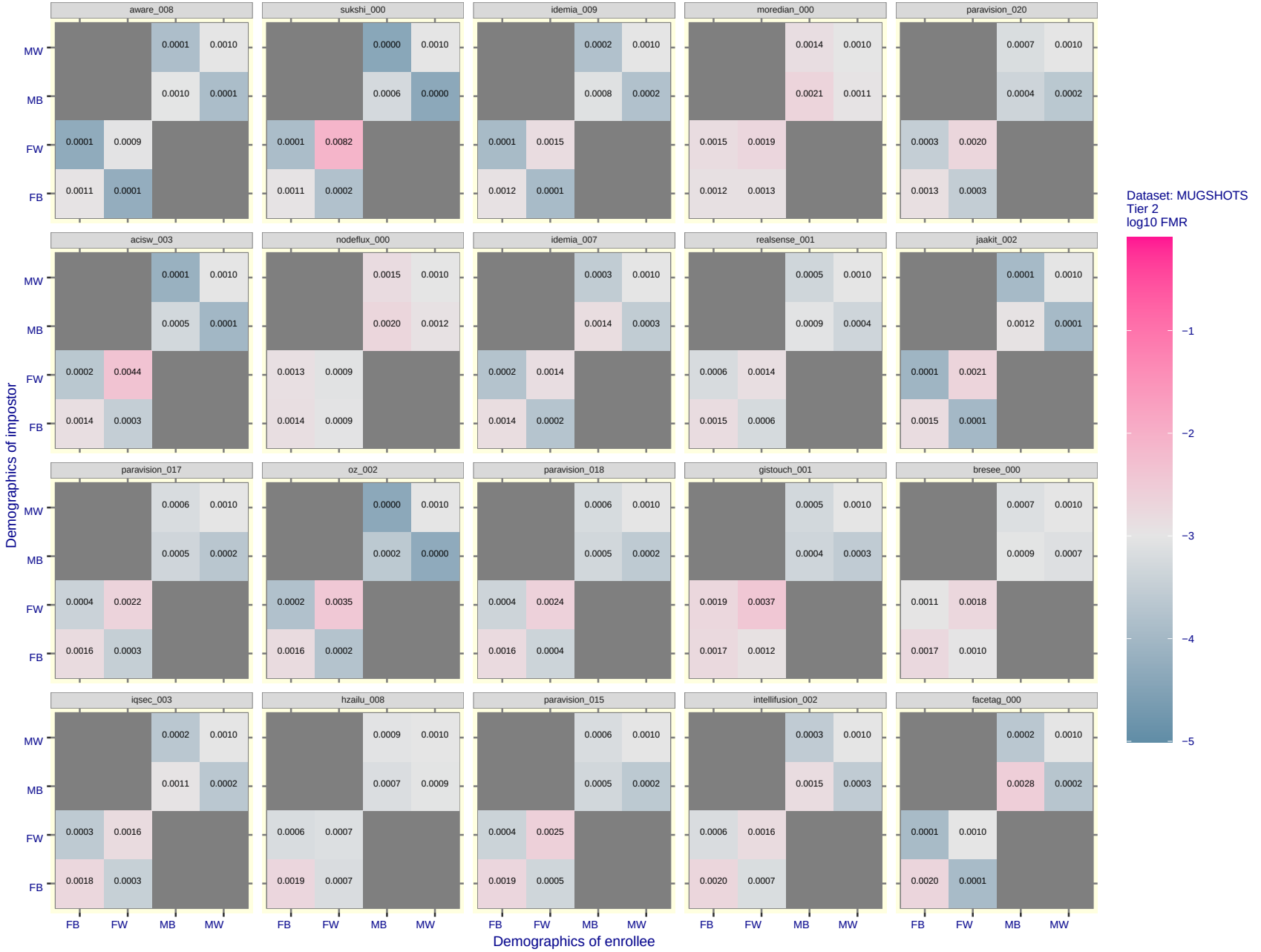

Figure 329: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

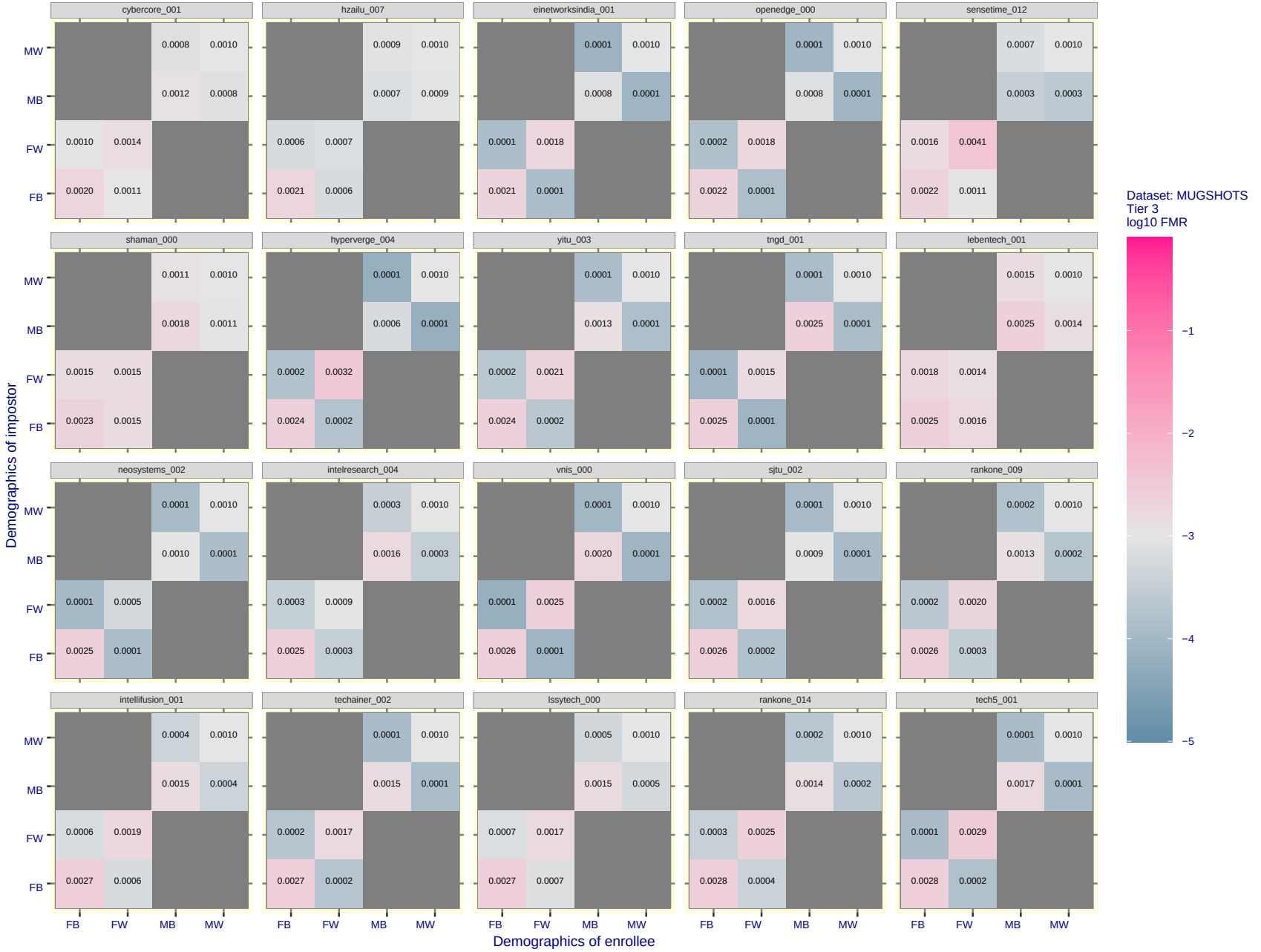

Figure 330: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

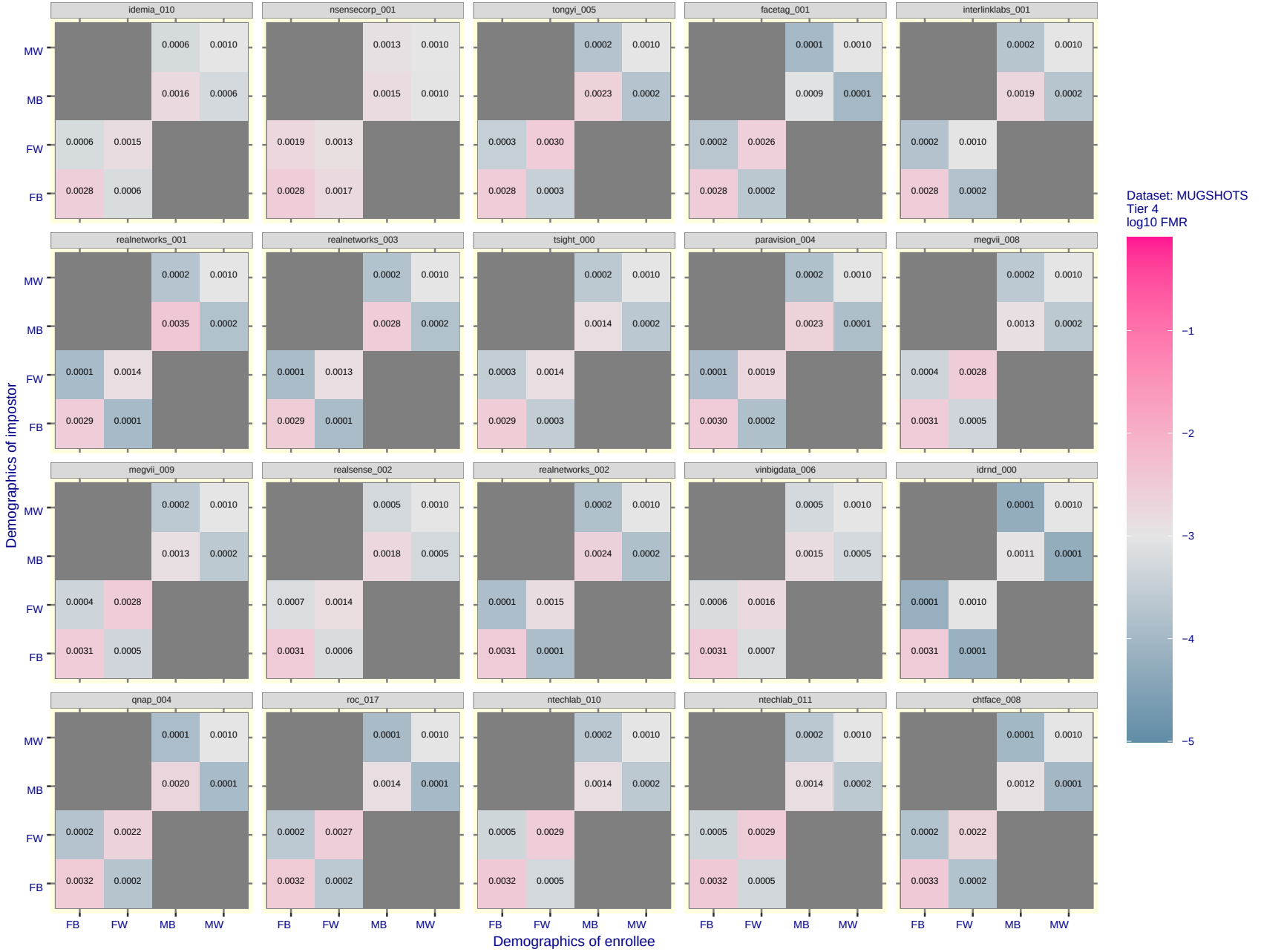

Figure 331: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

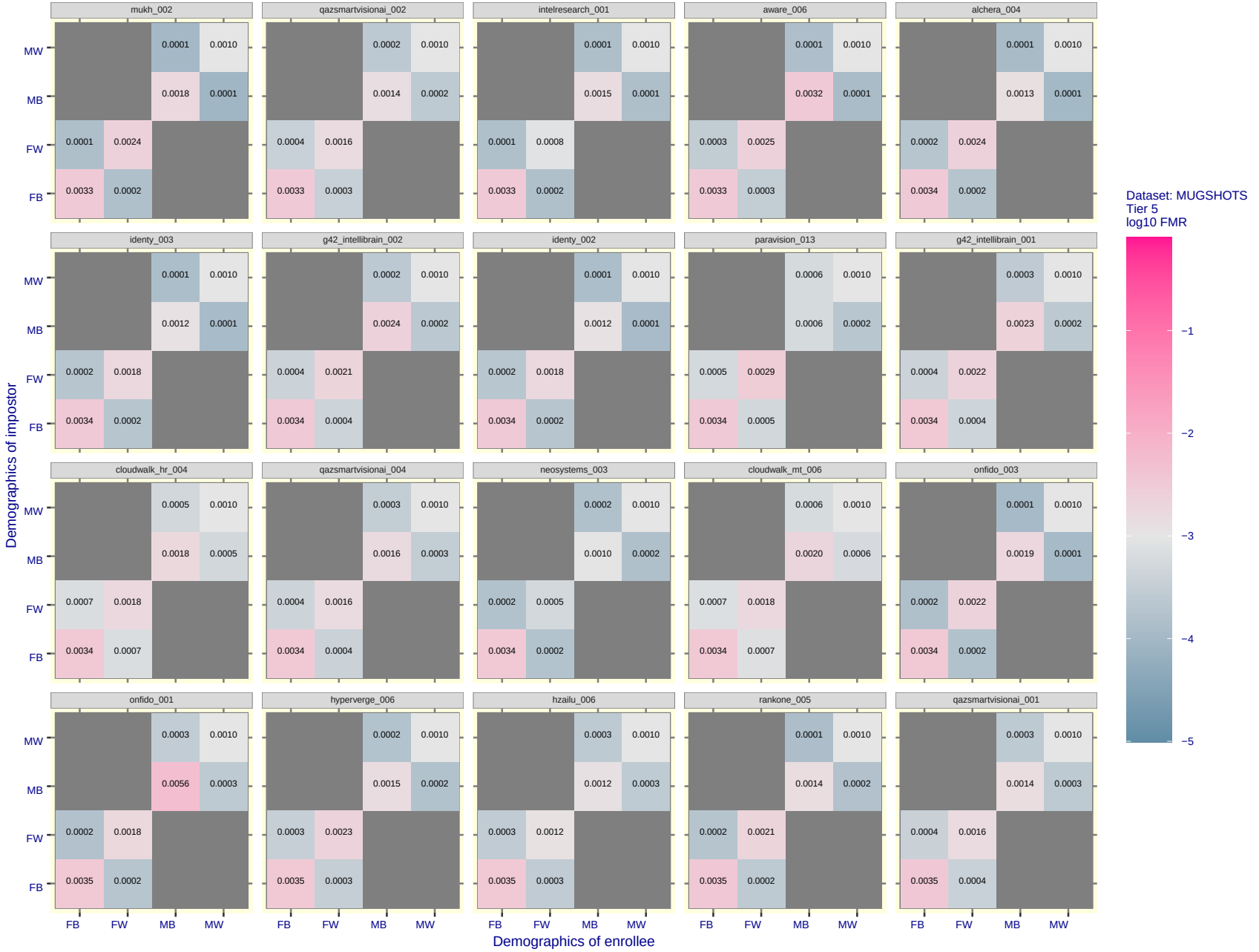

Figure 332: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

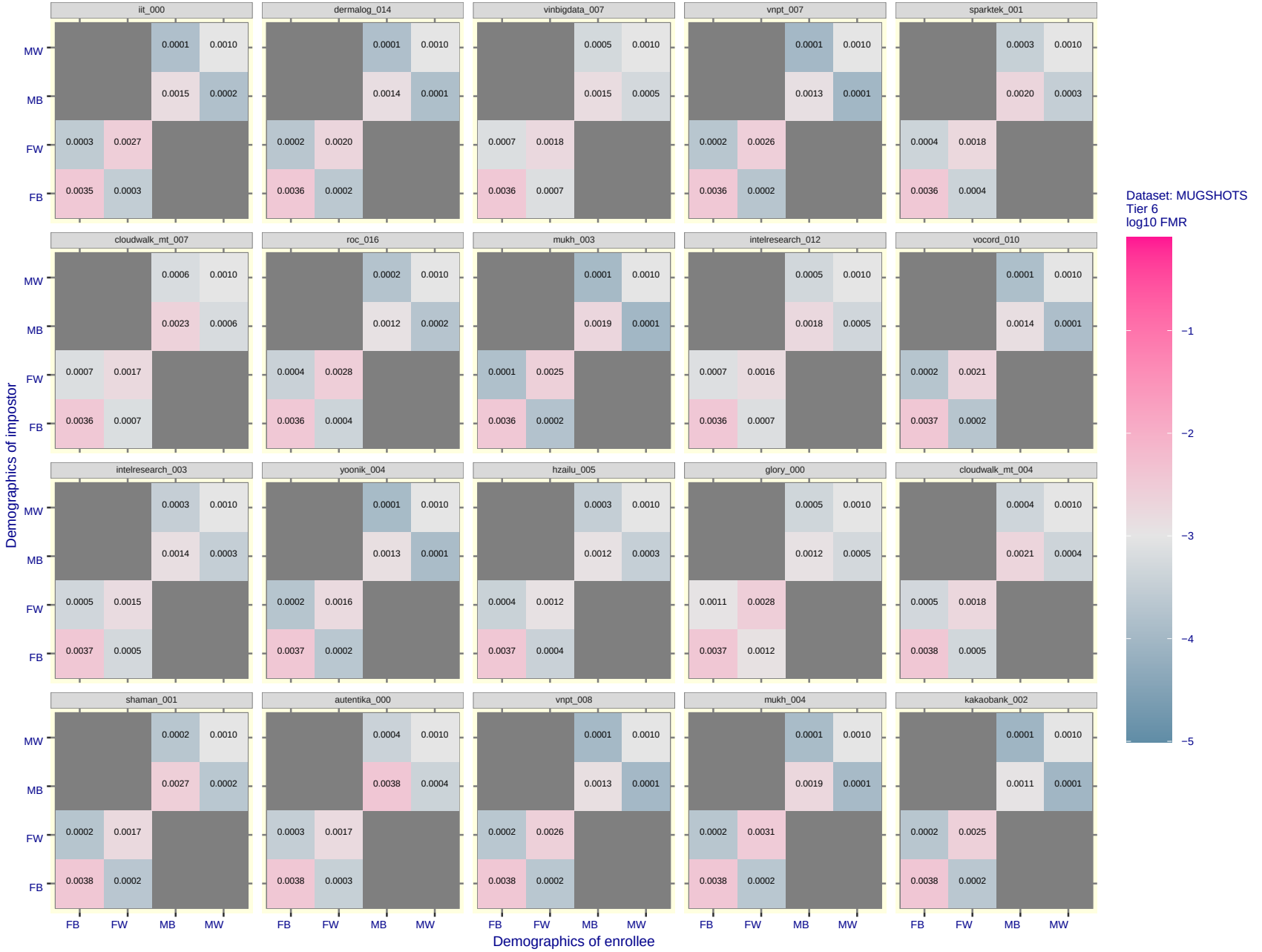

Figure 333: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

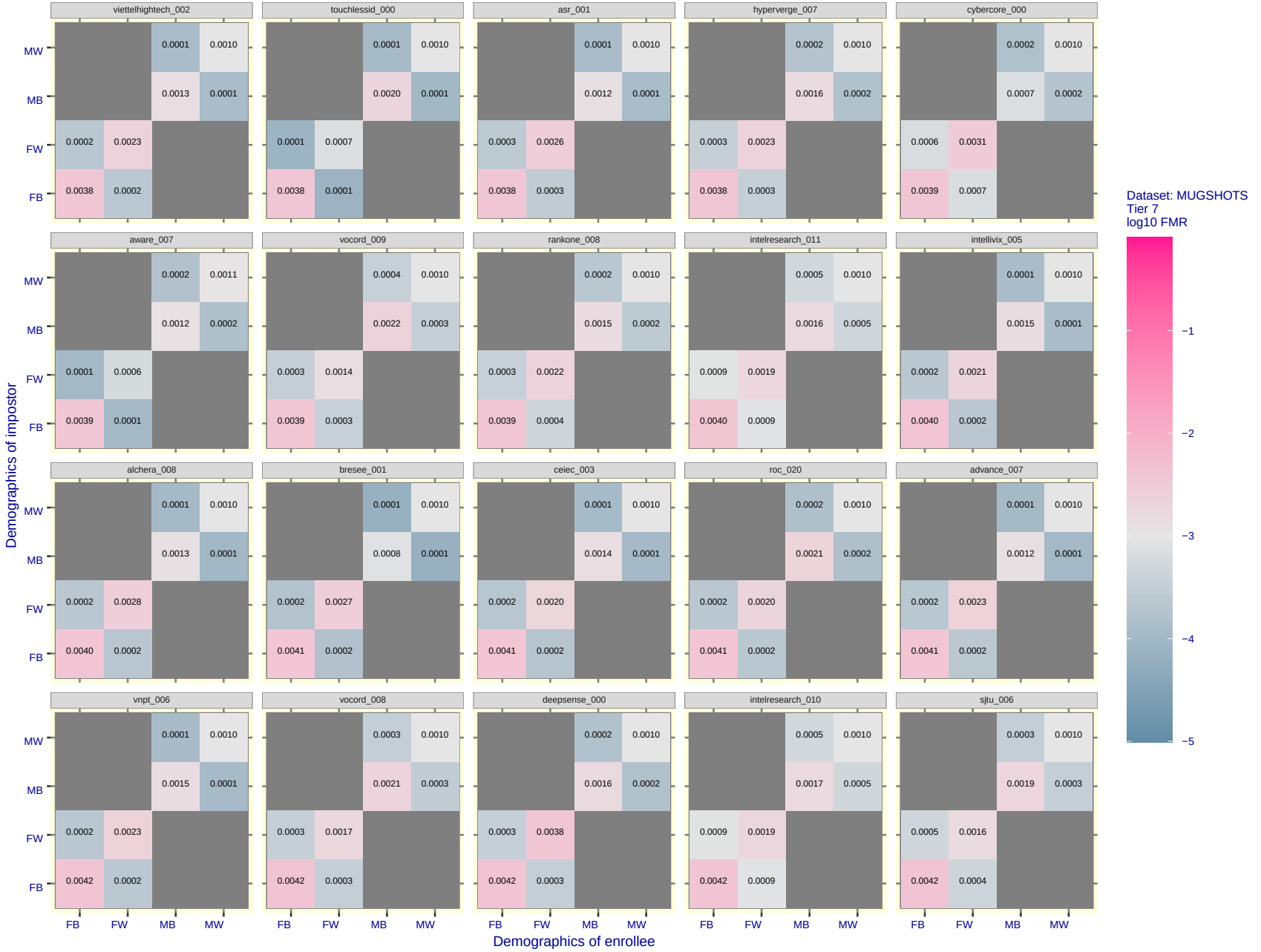

Figure 334: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

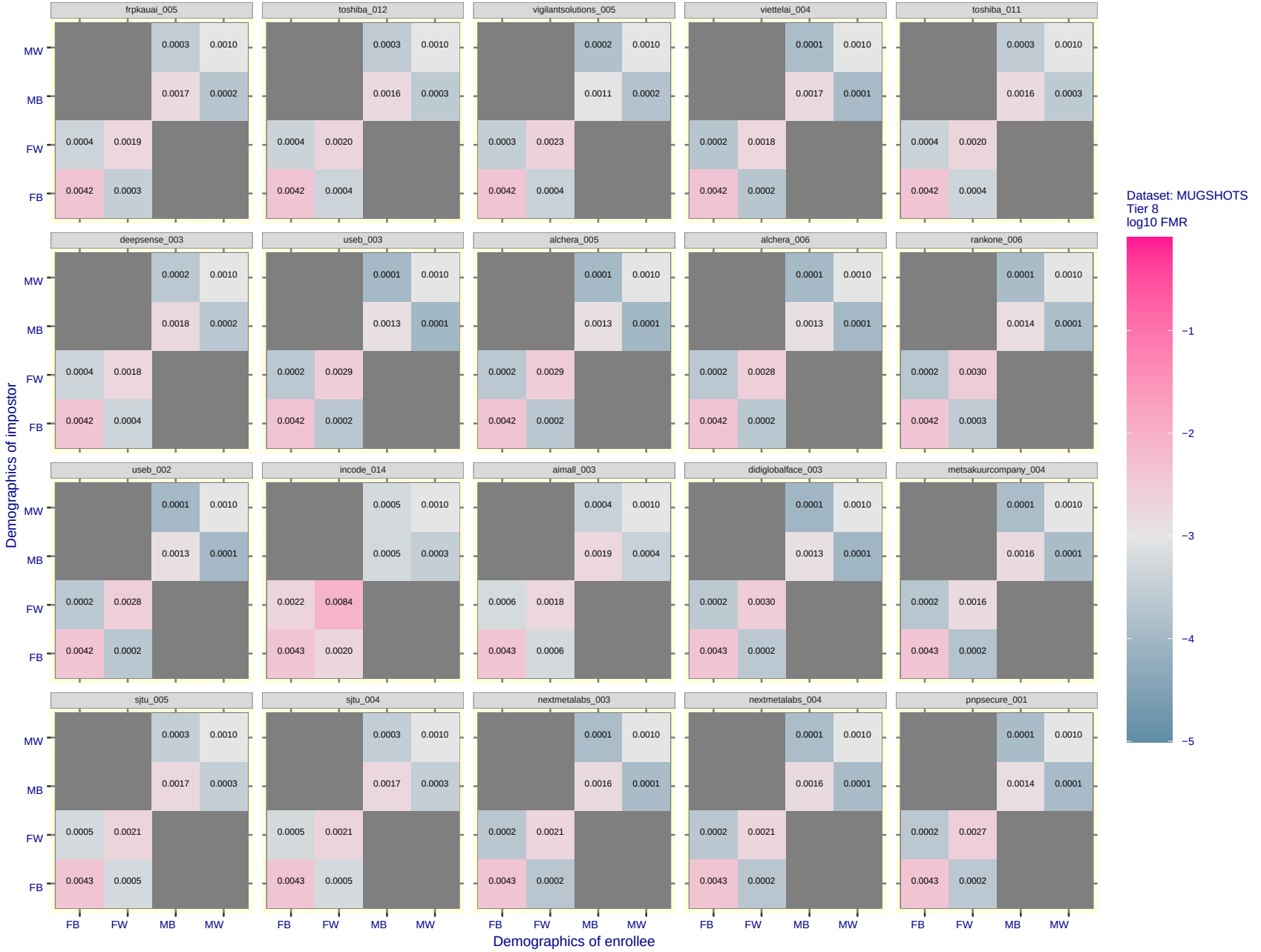

Figure 335: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

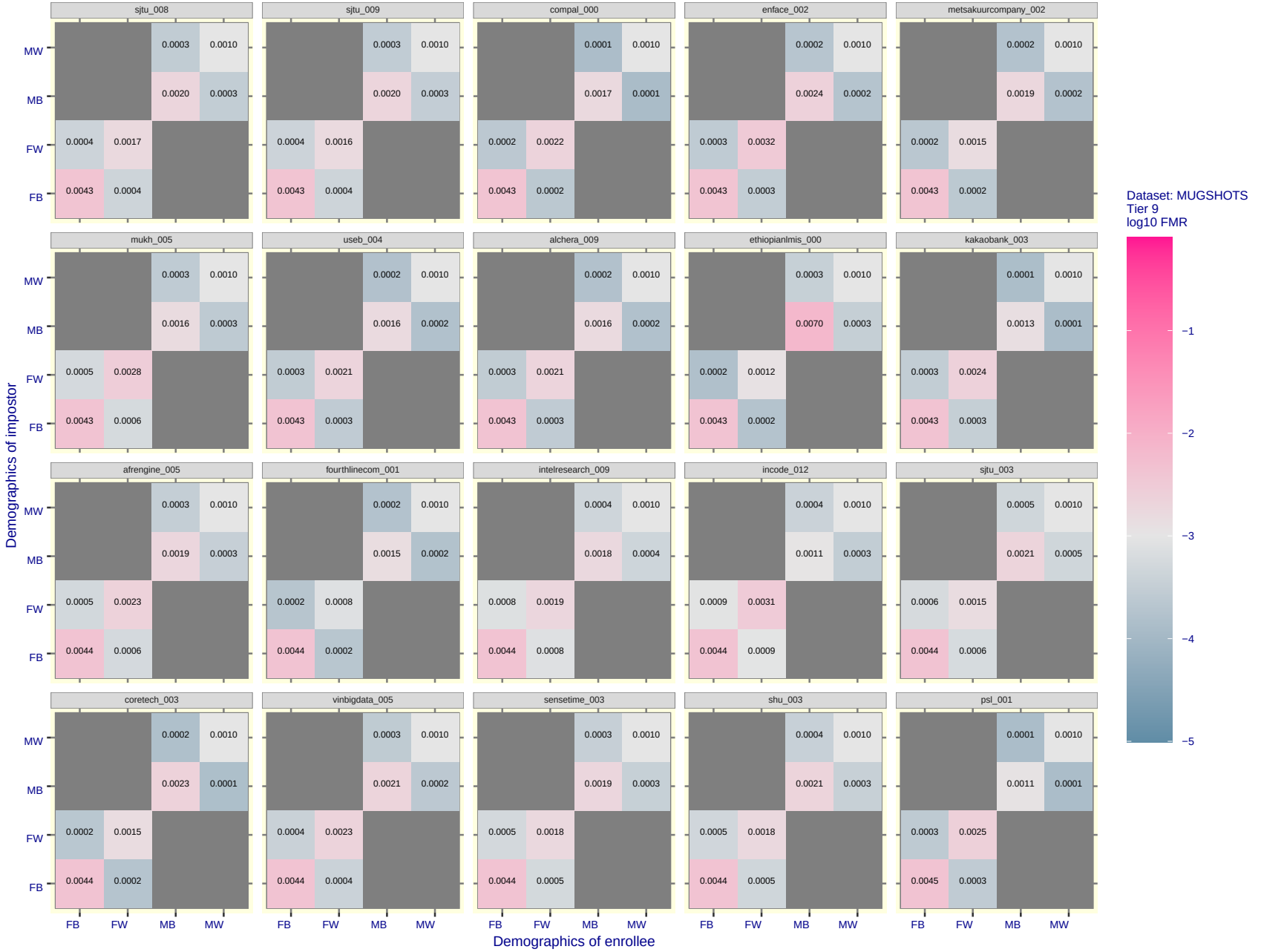

Figure 336: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

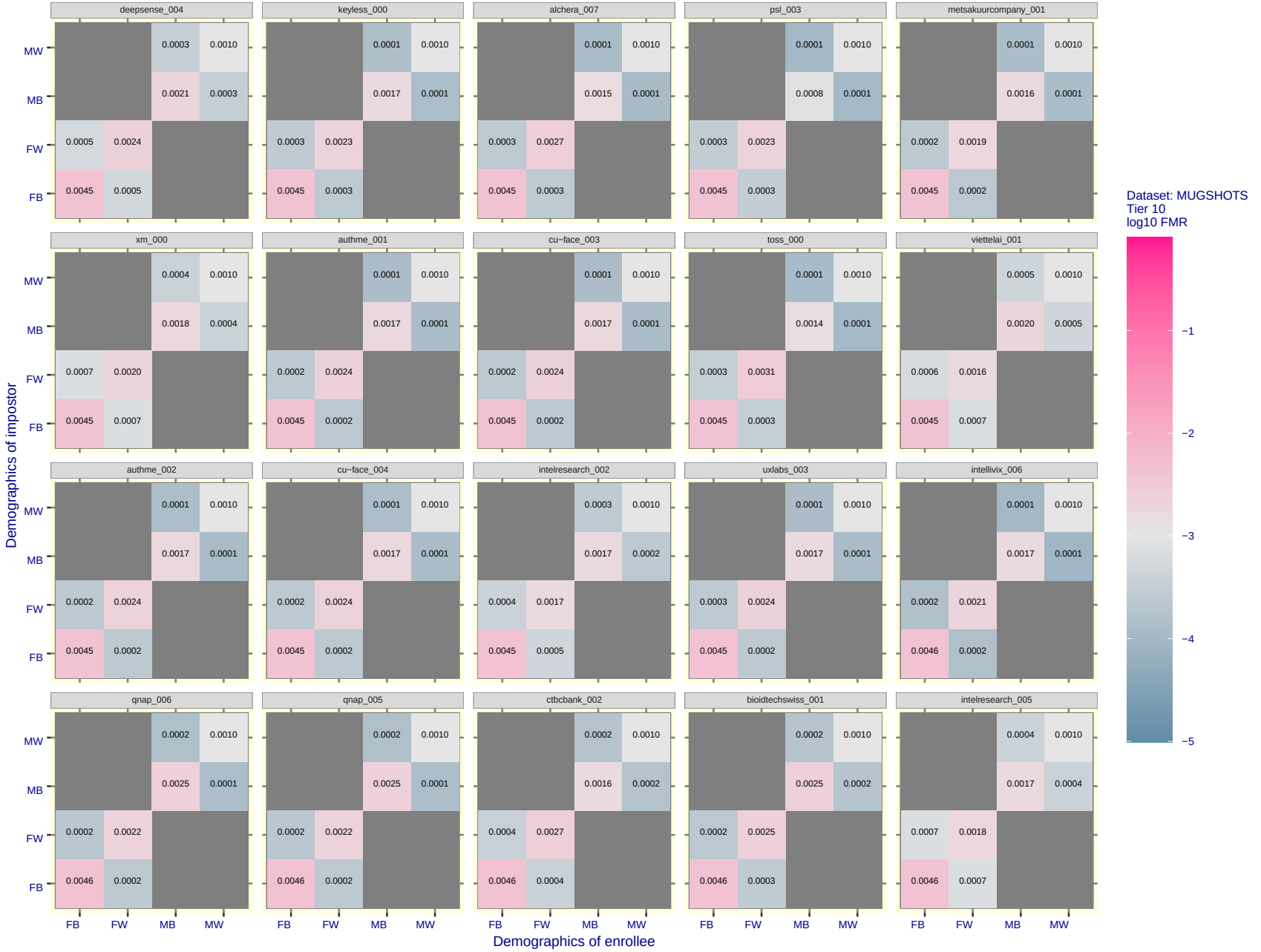

Figure 337: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

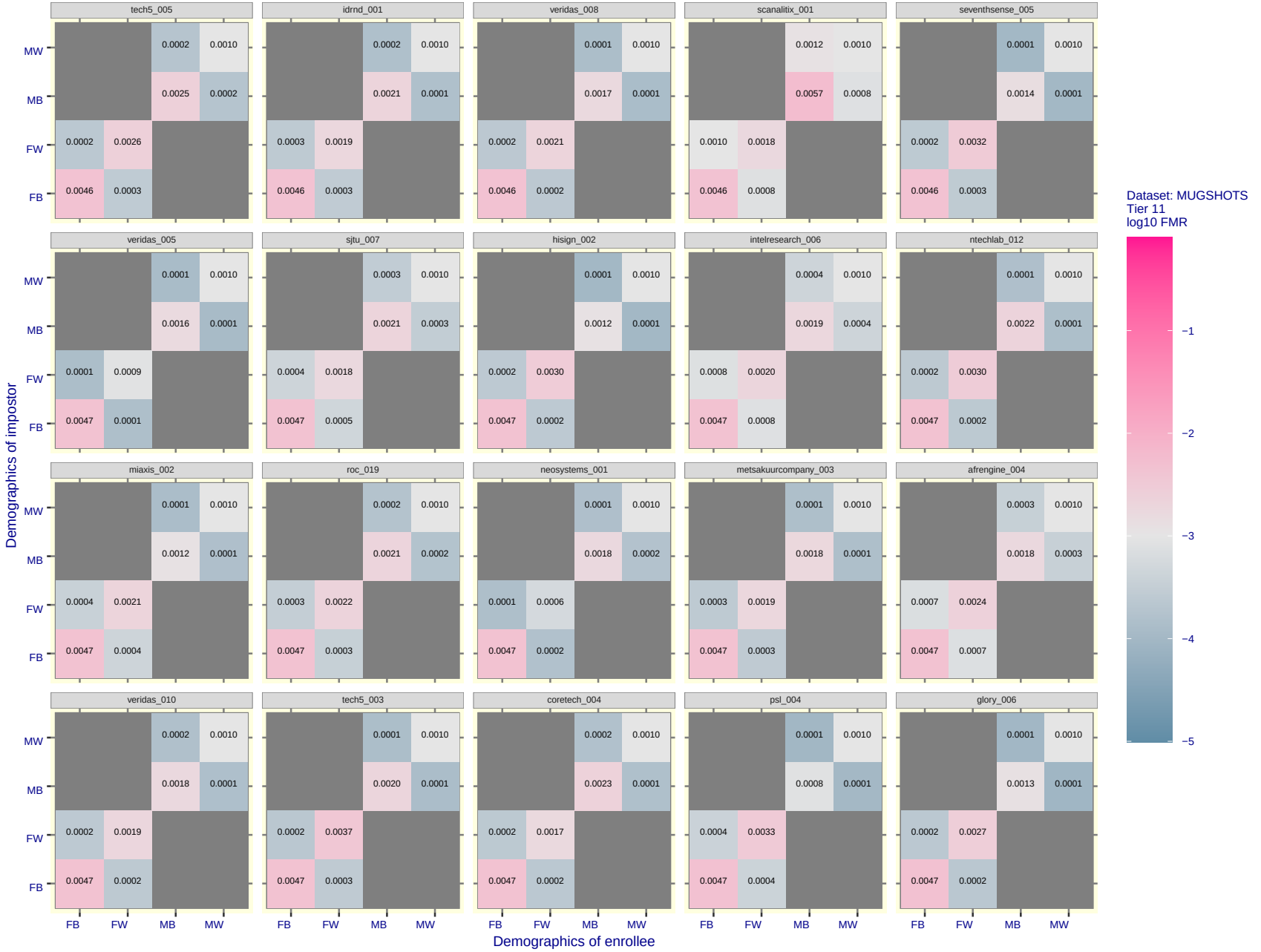

Figure 338: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

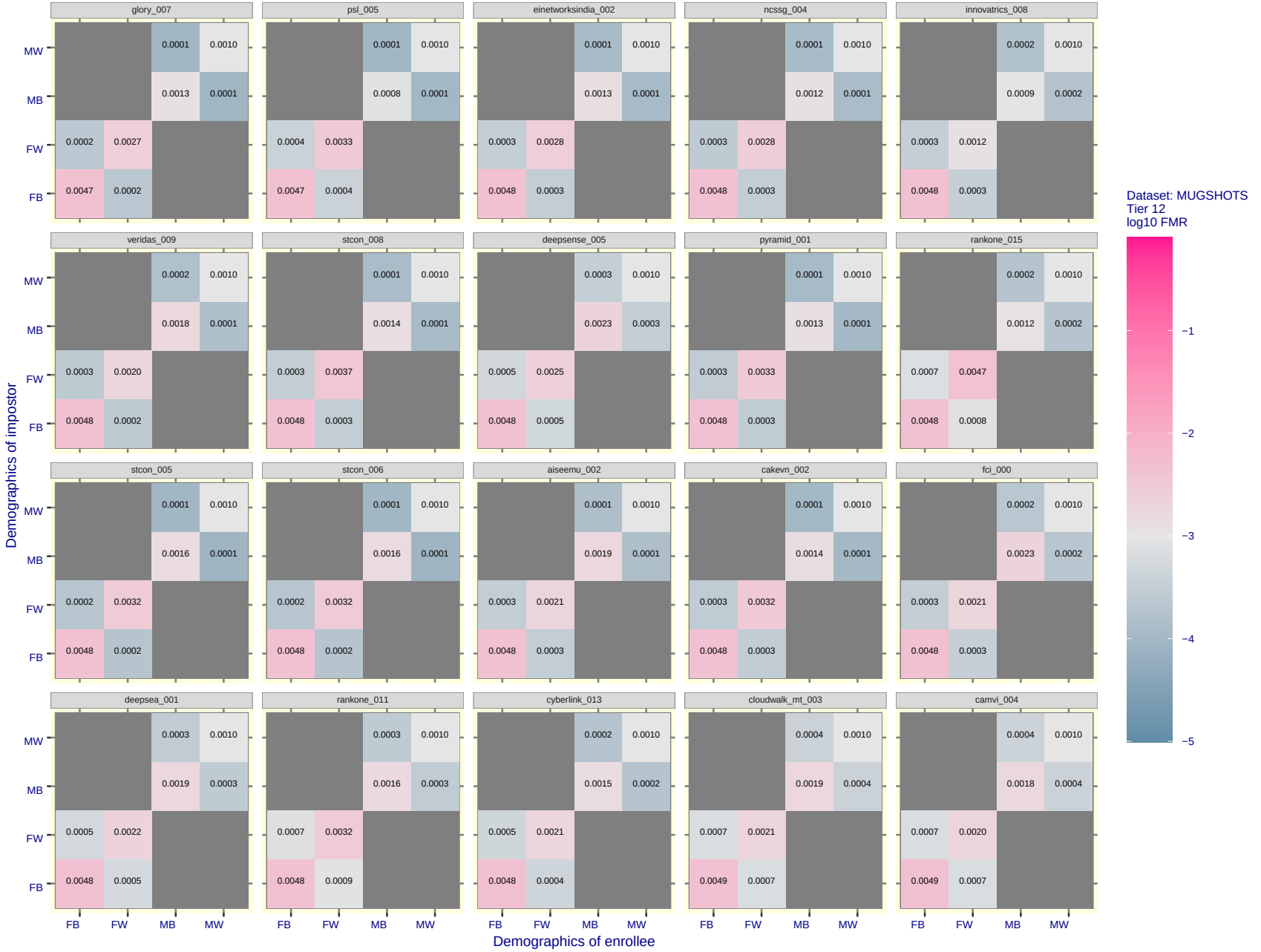

Figure 339: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

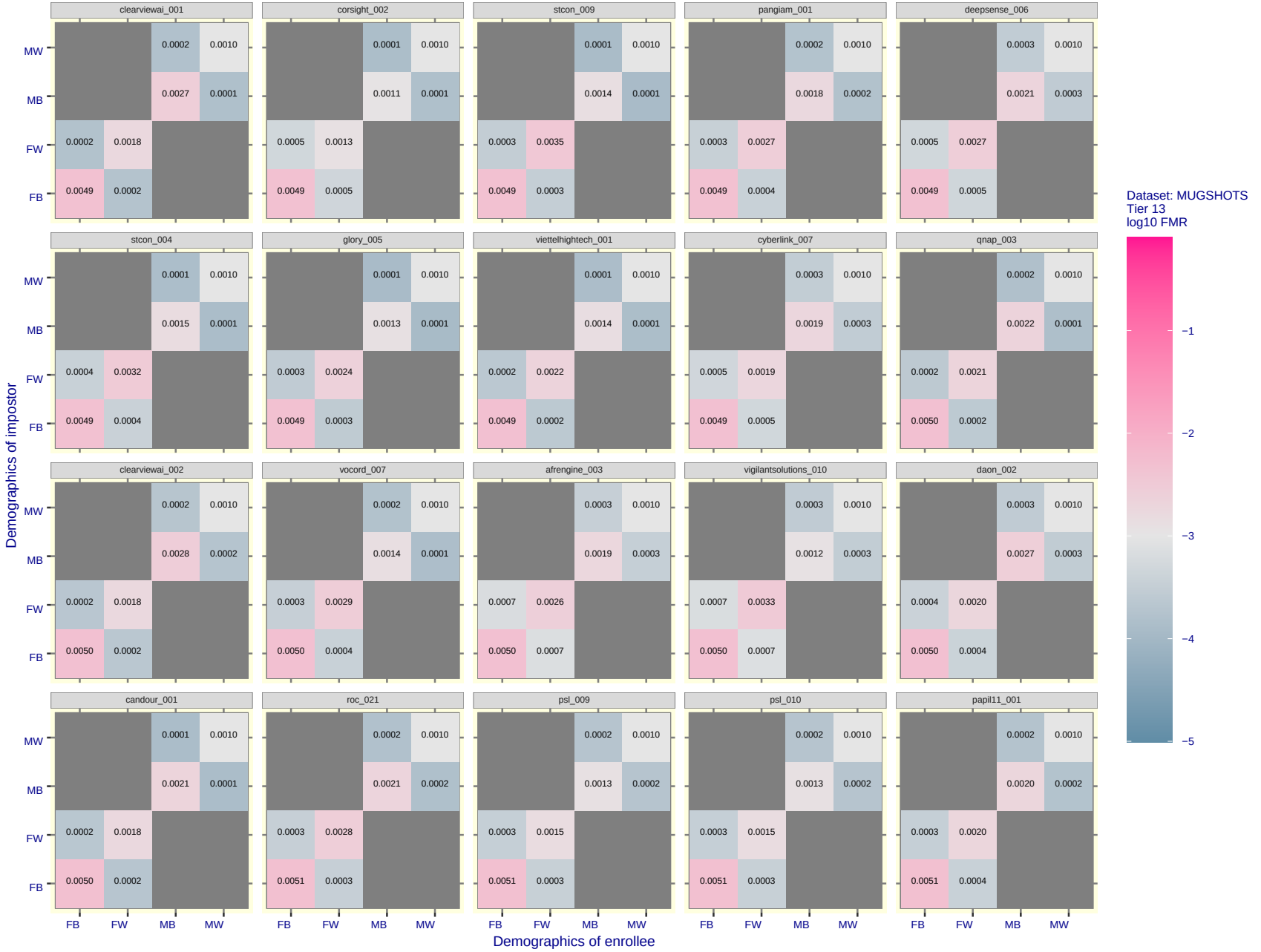

Figure 340: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

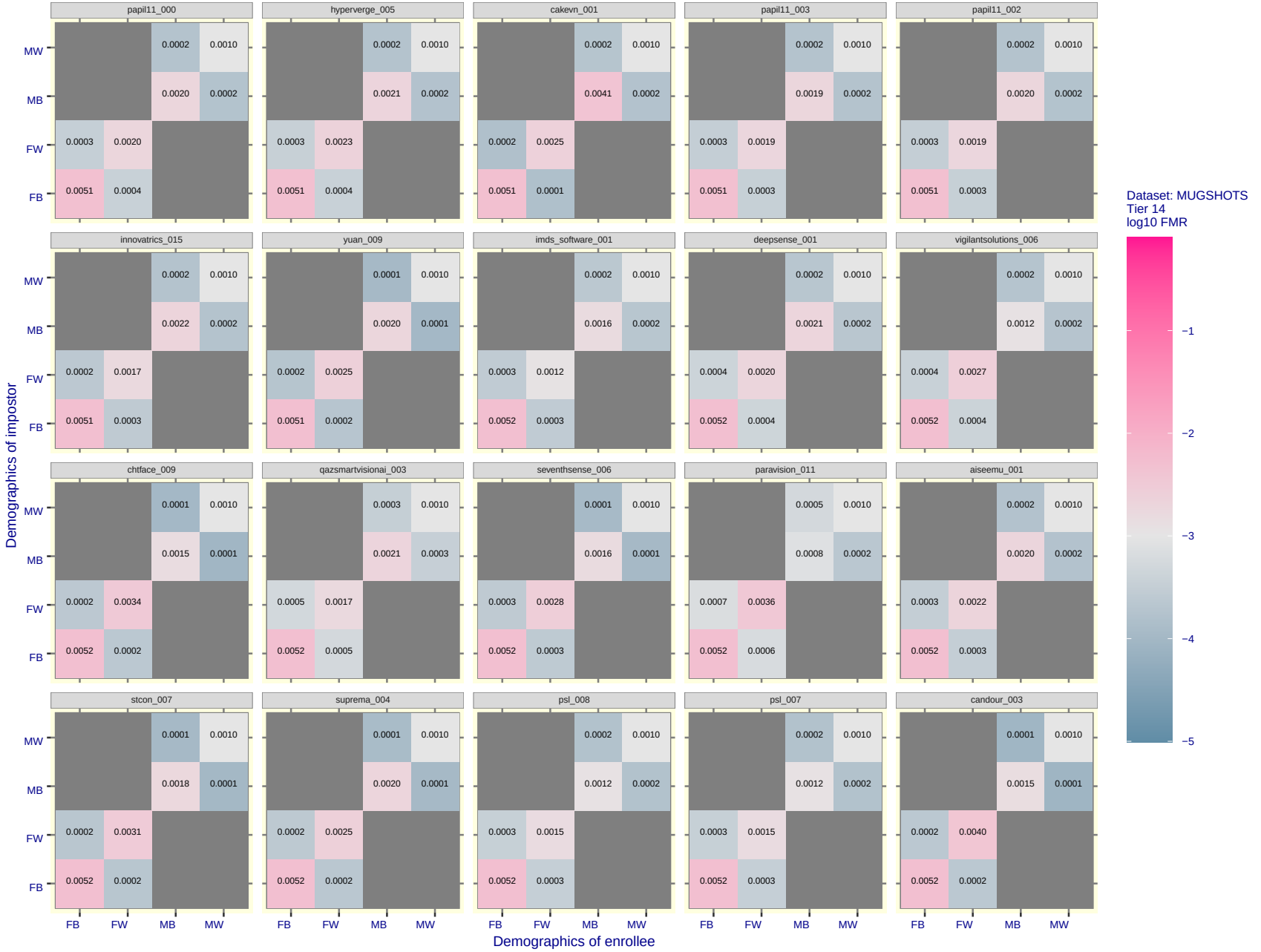

Figure 341: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

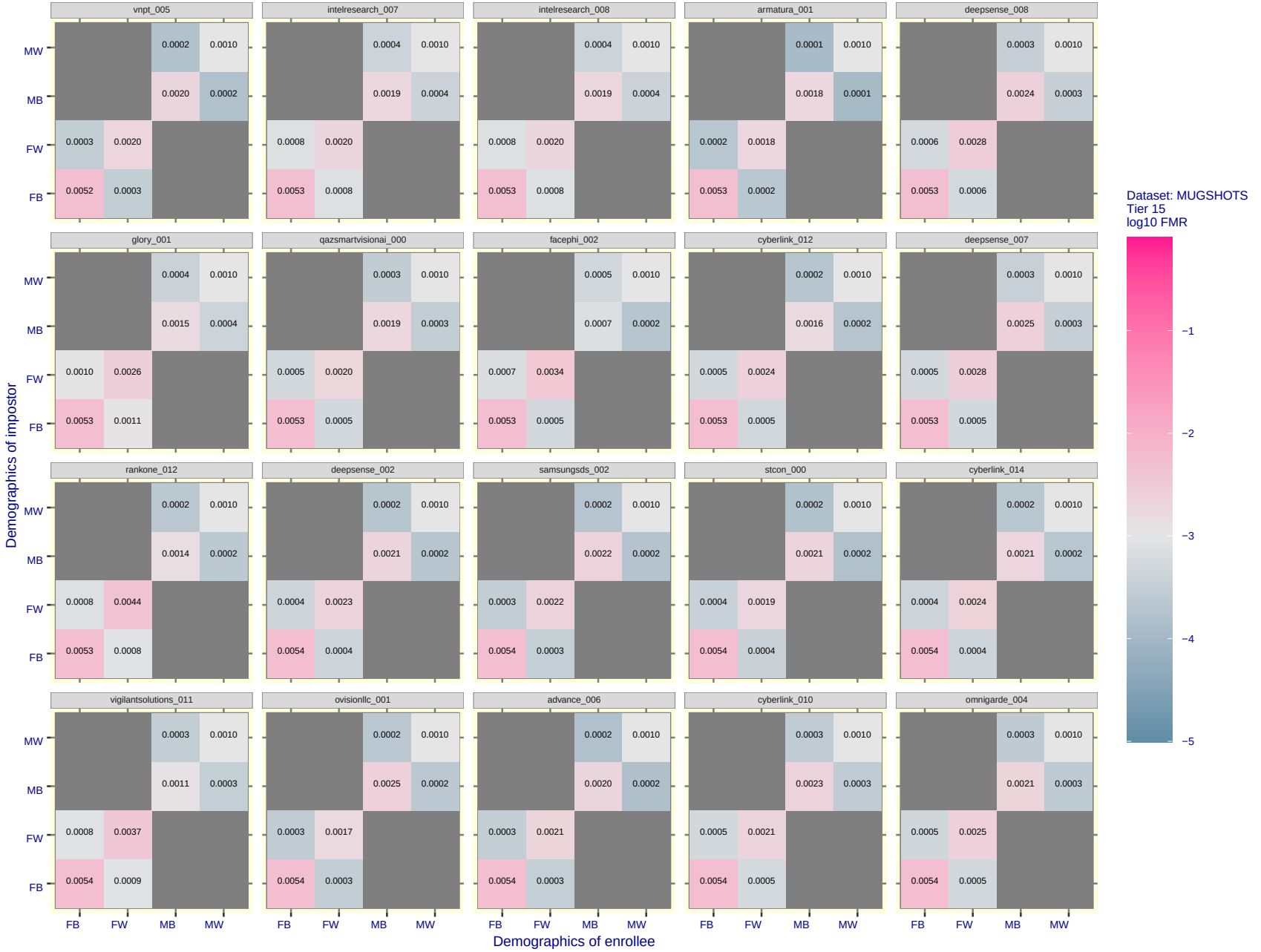

Figure 342: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

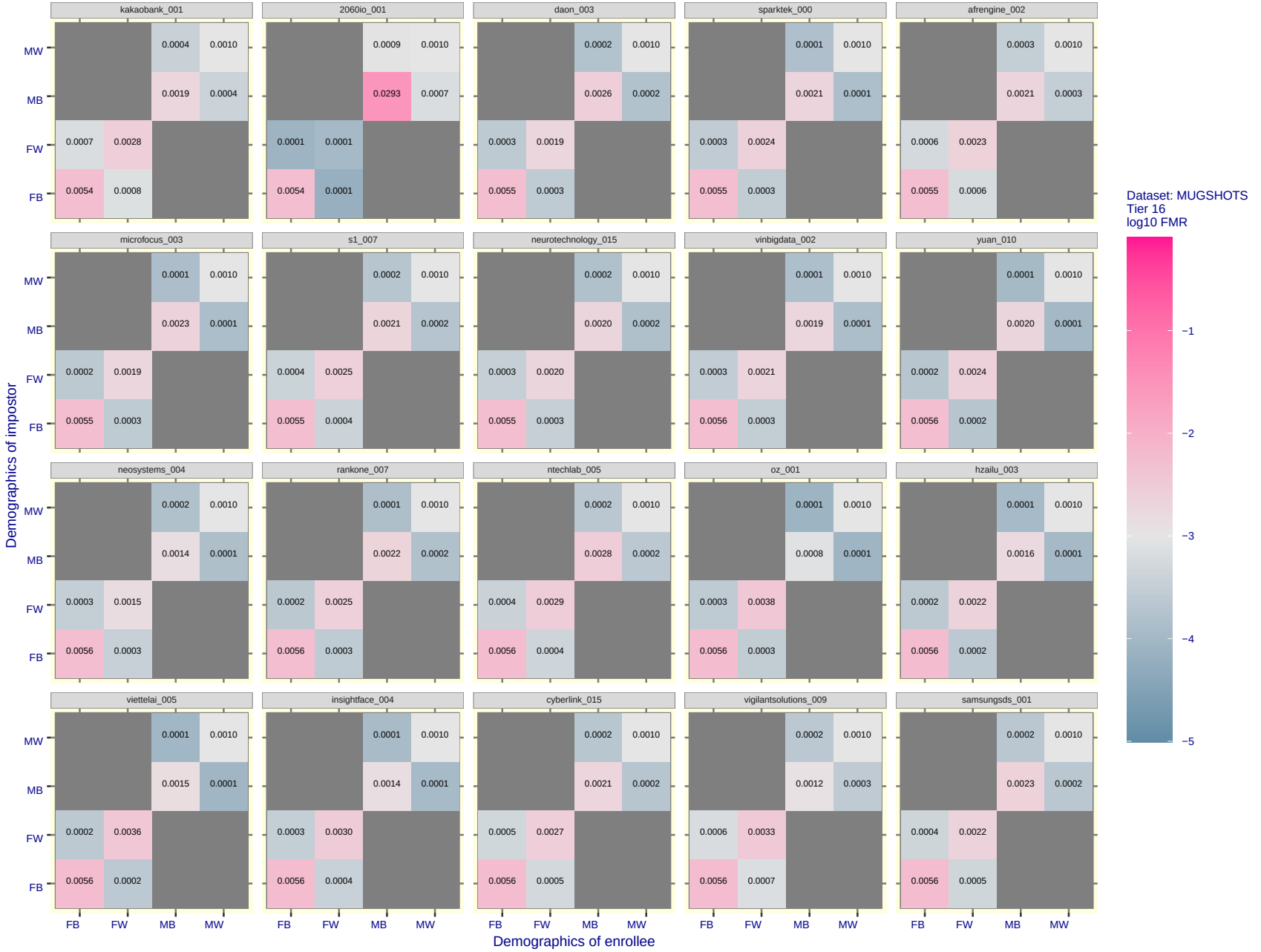

Figure 343: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

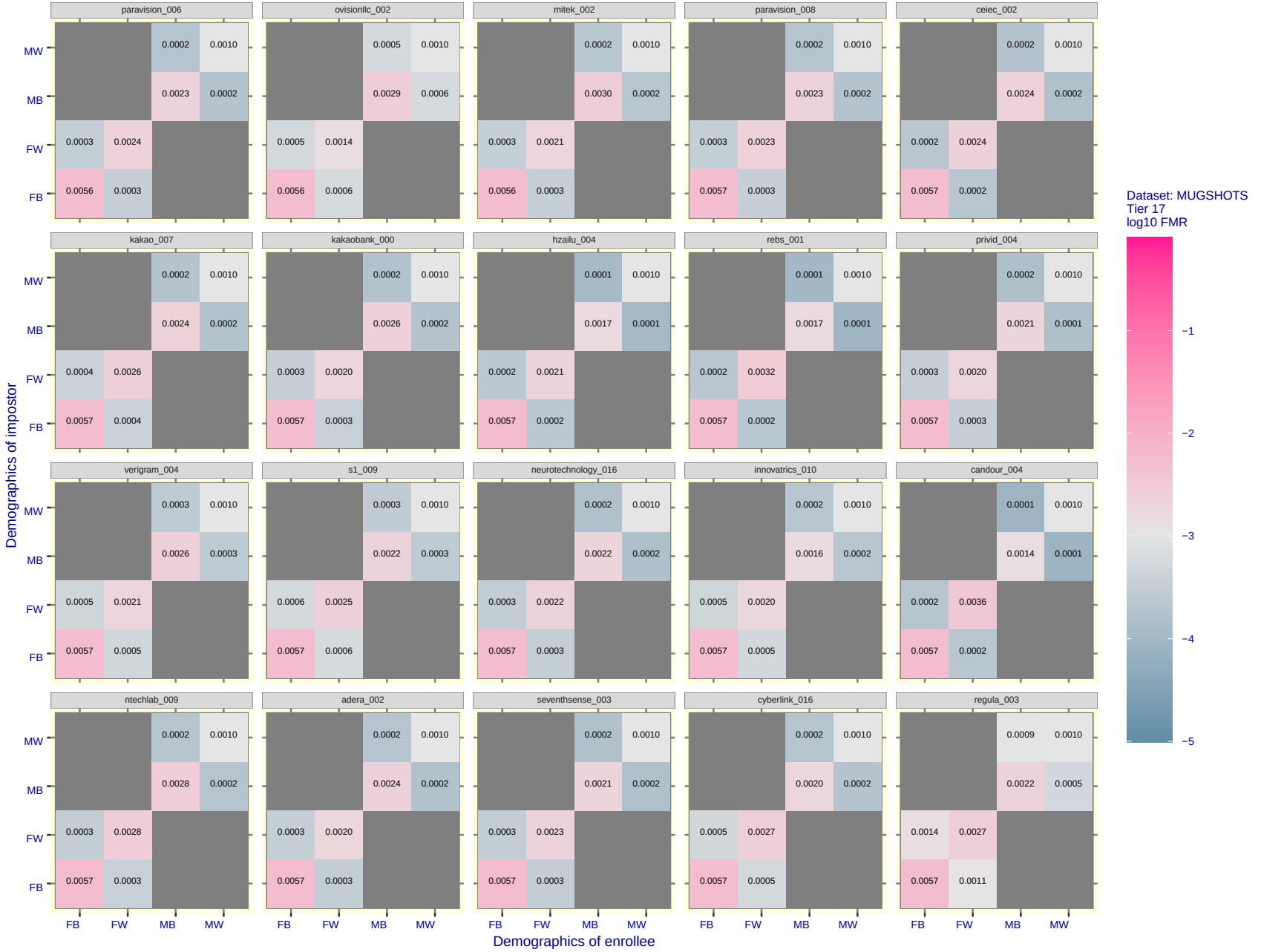

Figure 344: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

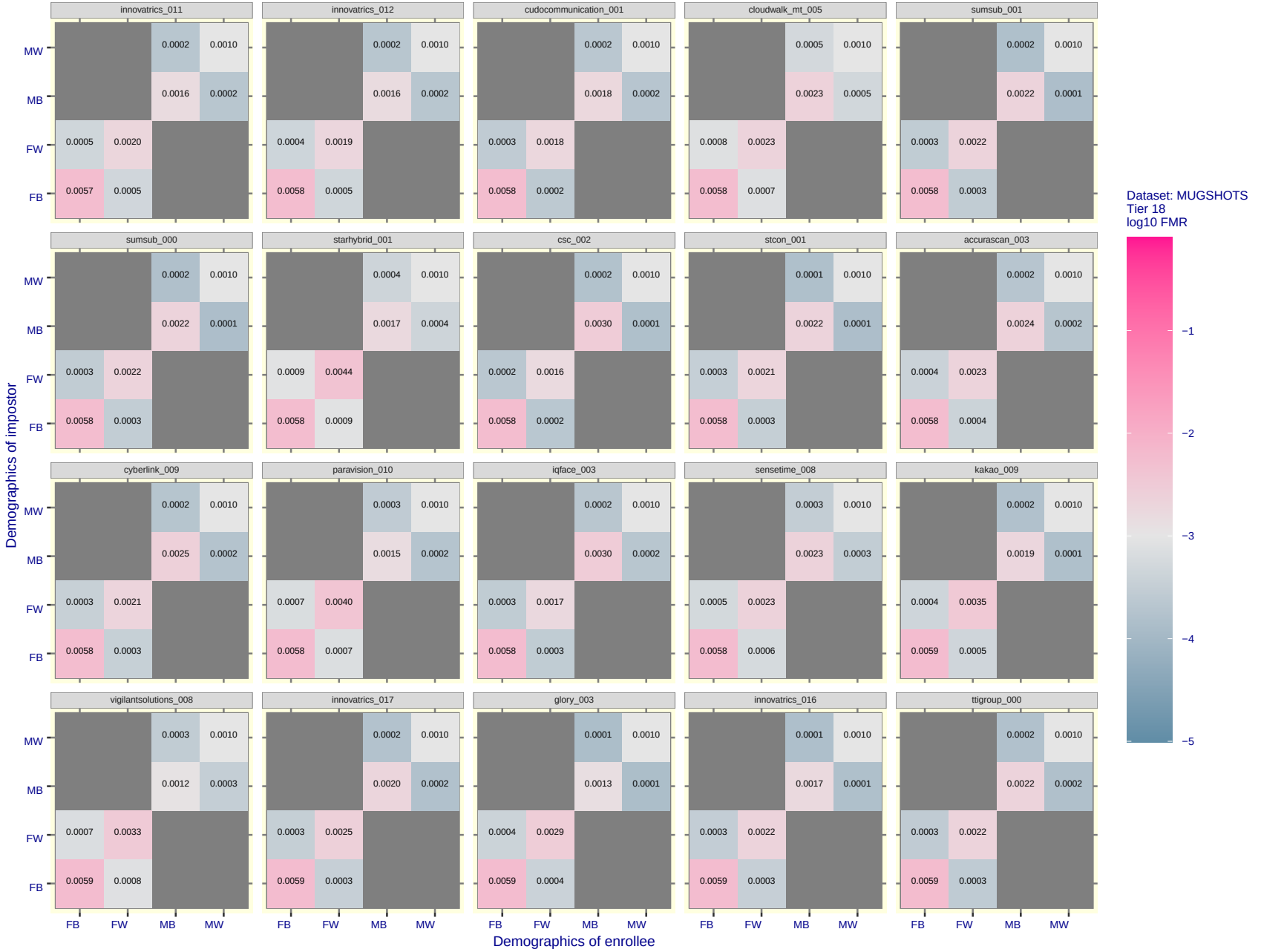

Figure 345: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

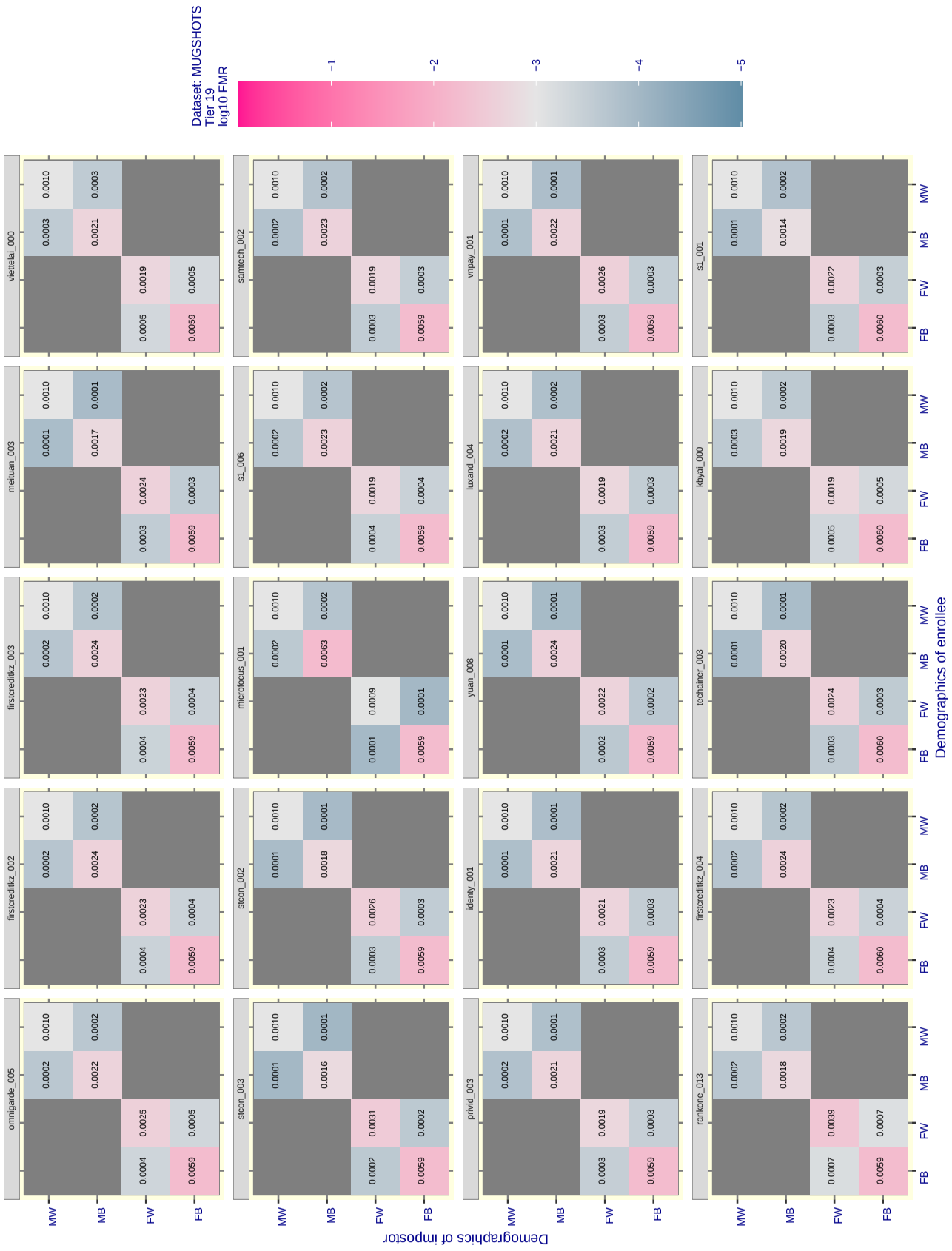

Figure 346: For the mugshot images, FMR for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give FMR = 0.001 for white males which is the demographic that usually gives the lowest FMR. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of FMR on black females, which is the demographc that usually gives the highest FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

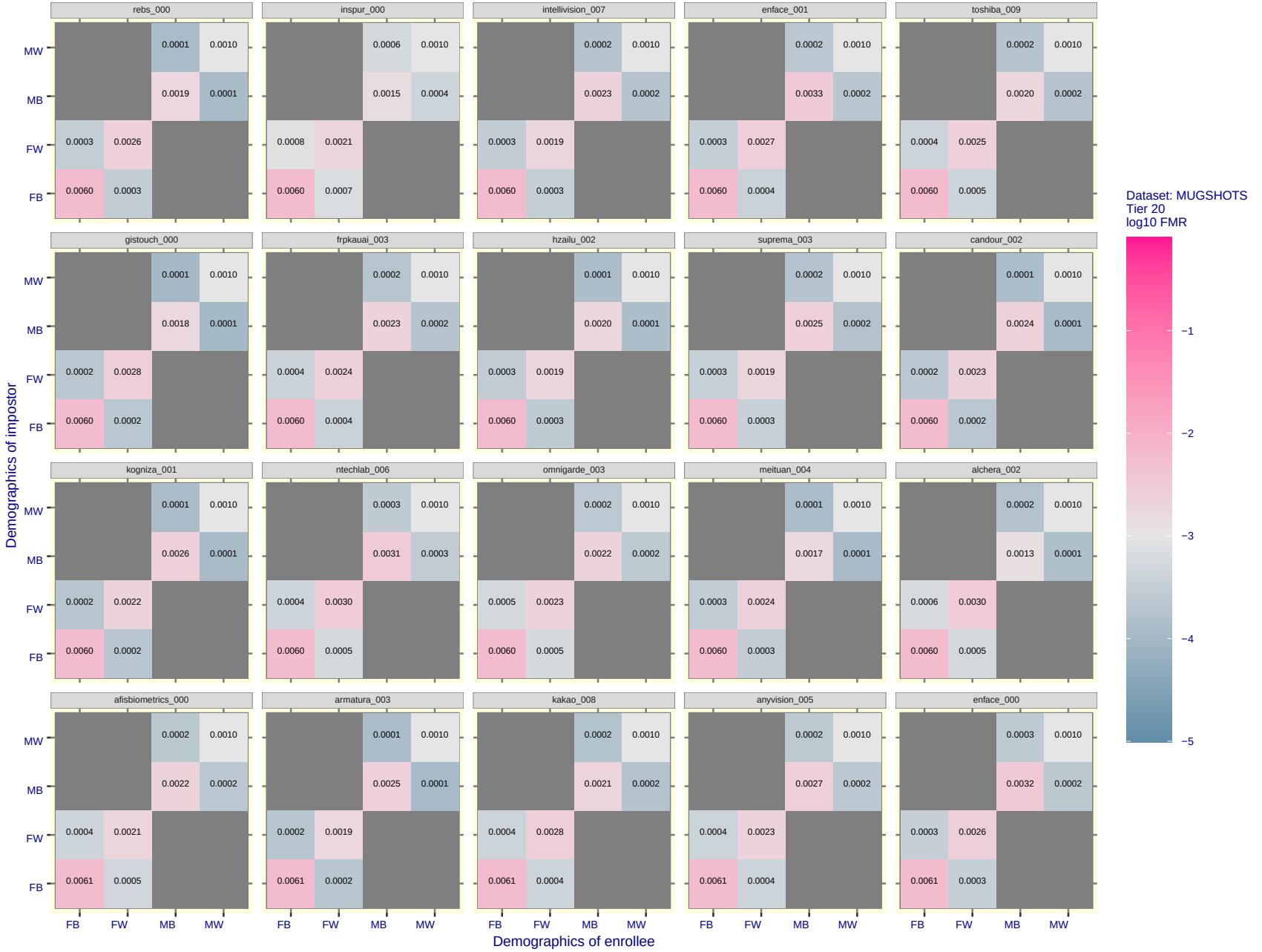

Figure 347: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

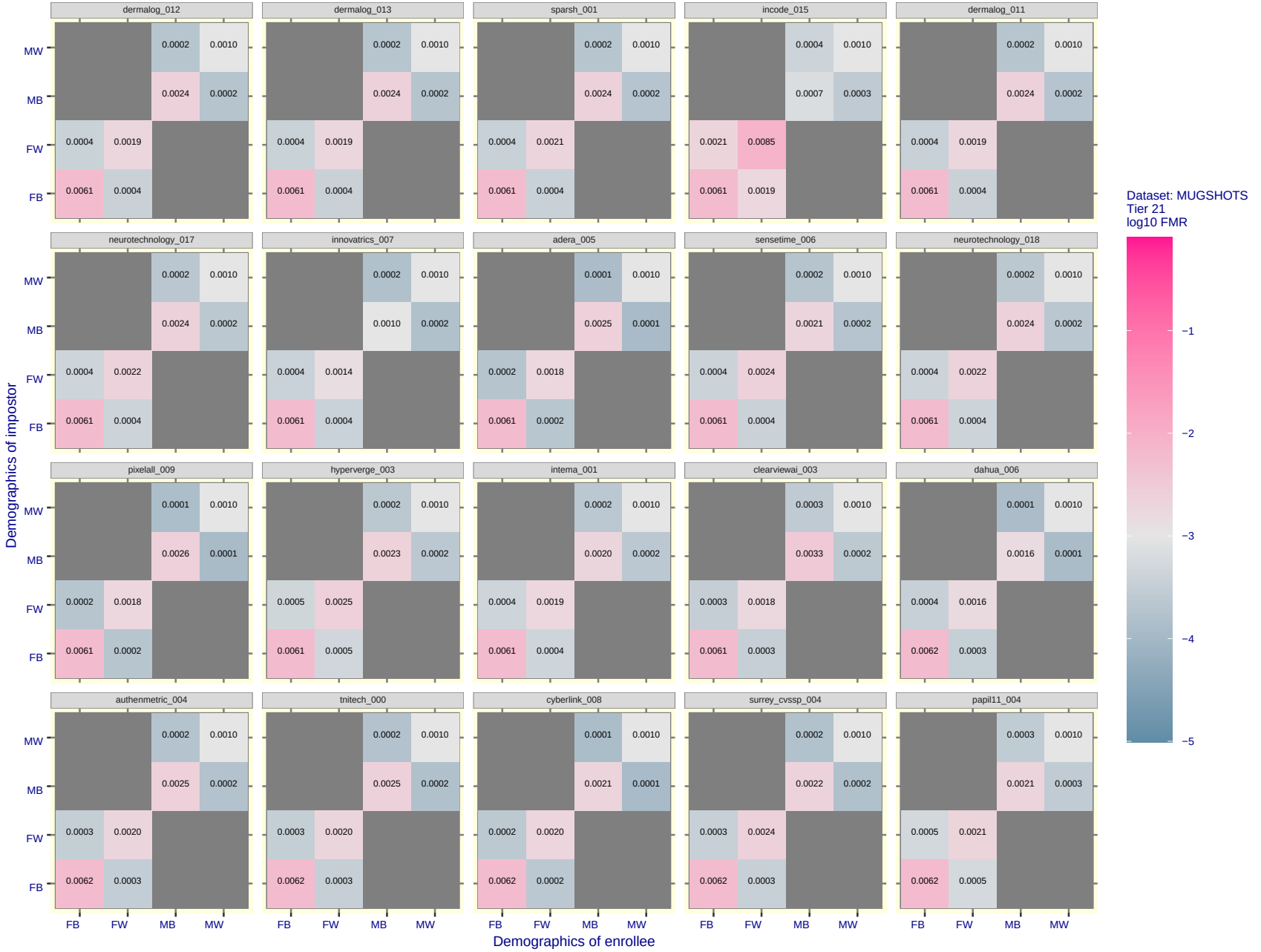

Figure 348: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

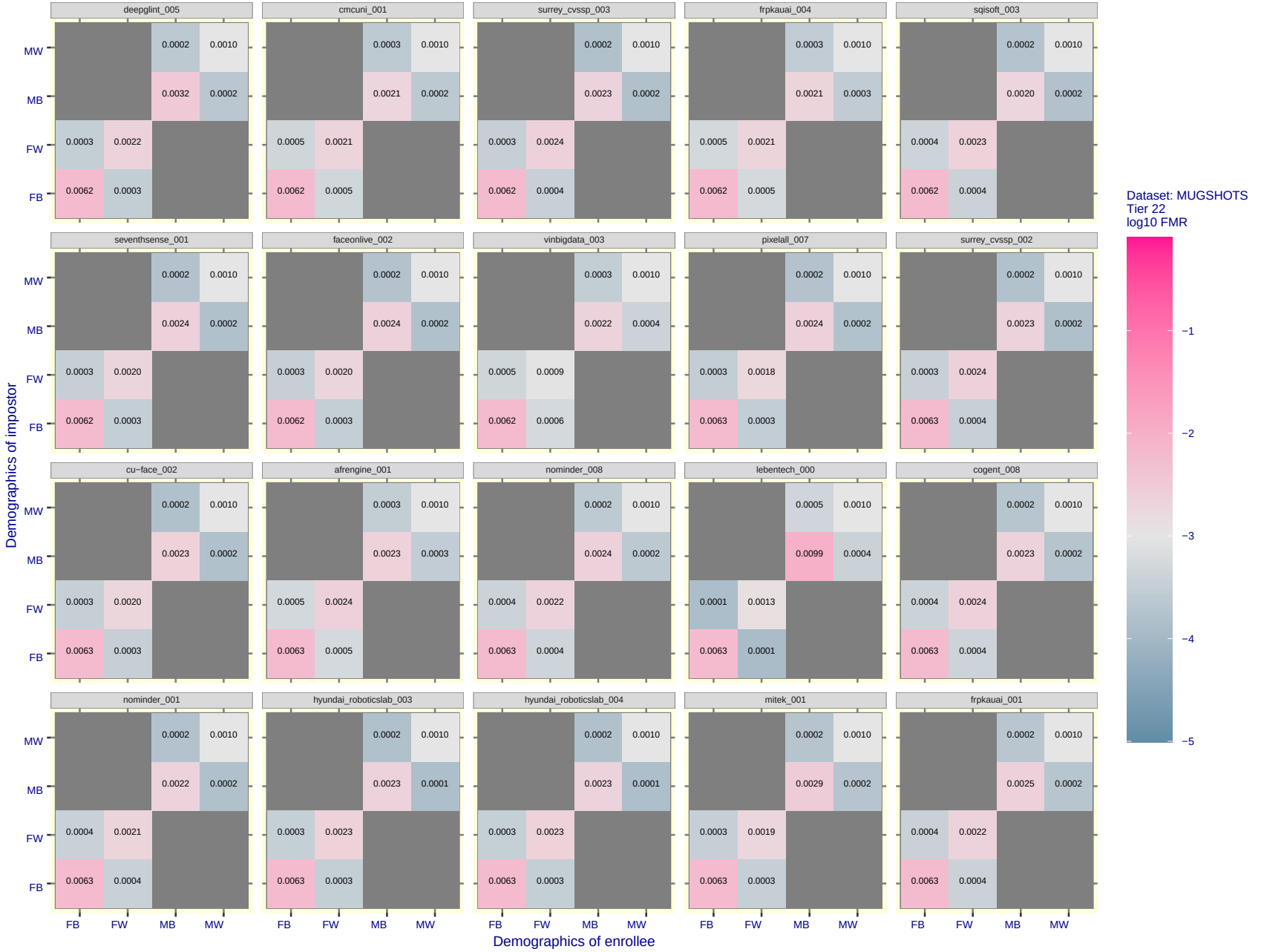

Figure 349: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

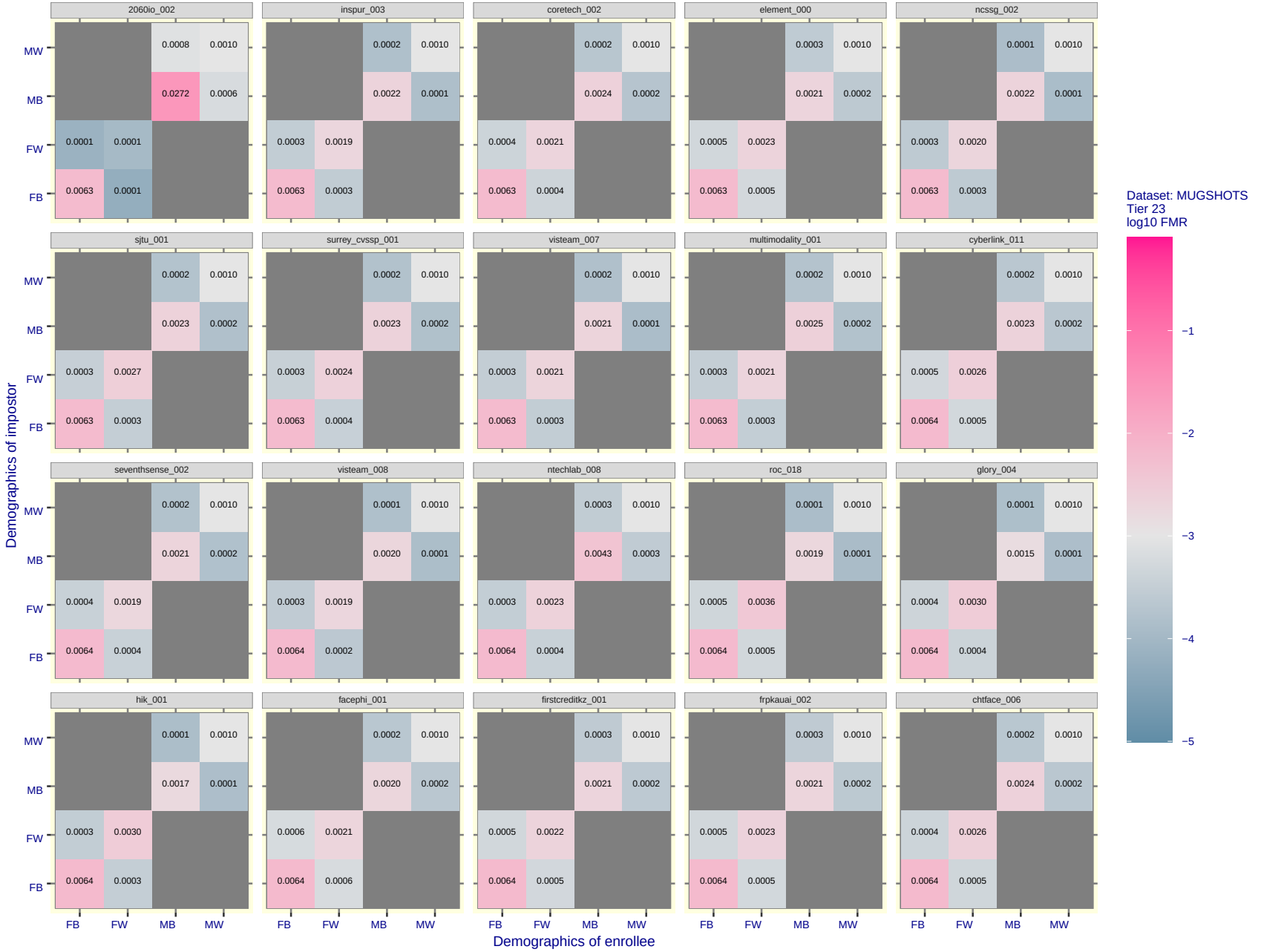

Figure 350: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

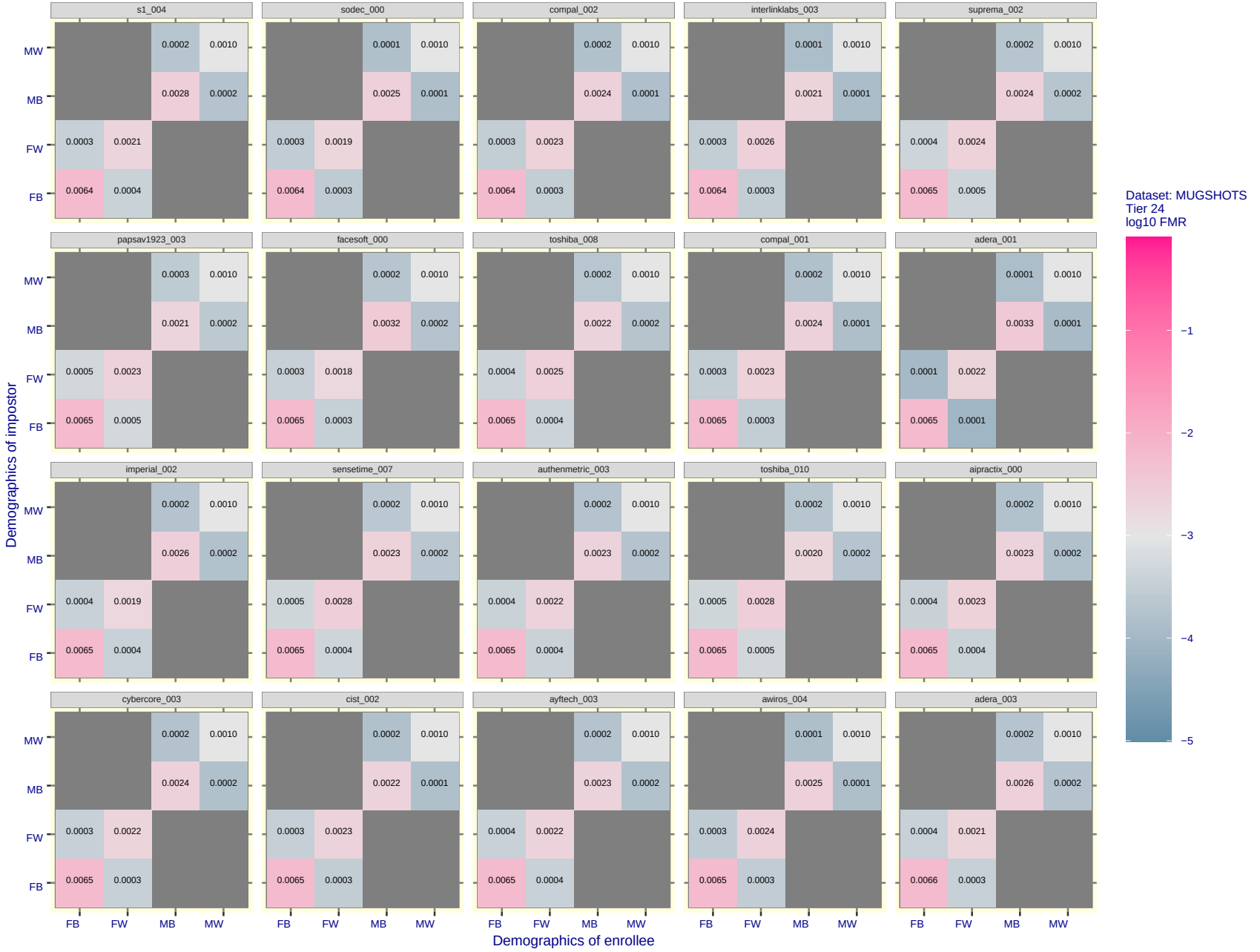

Figure 351: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

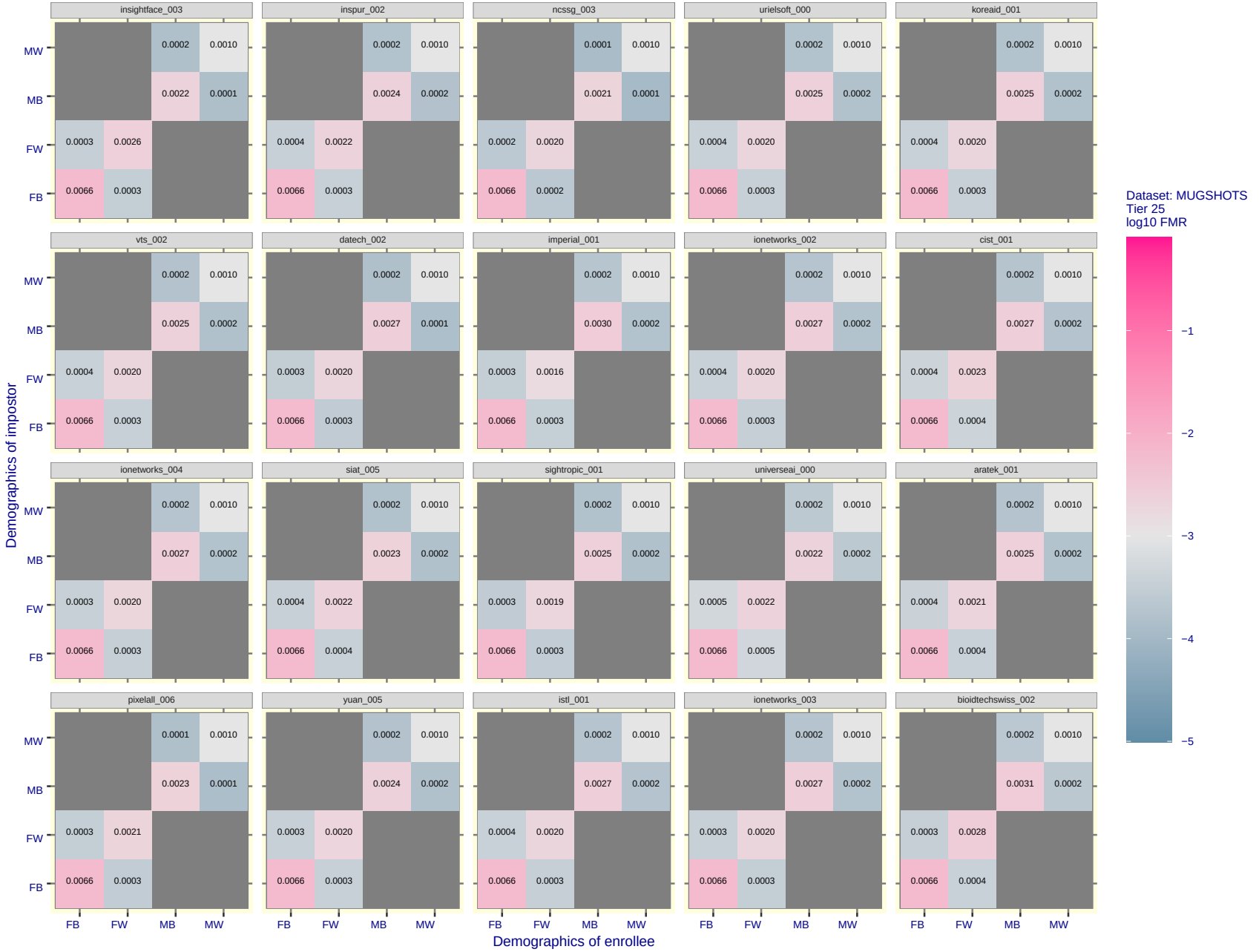

Figure 352: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

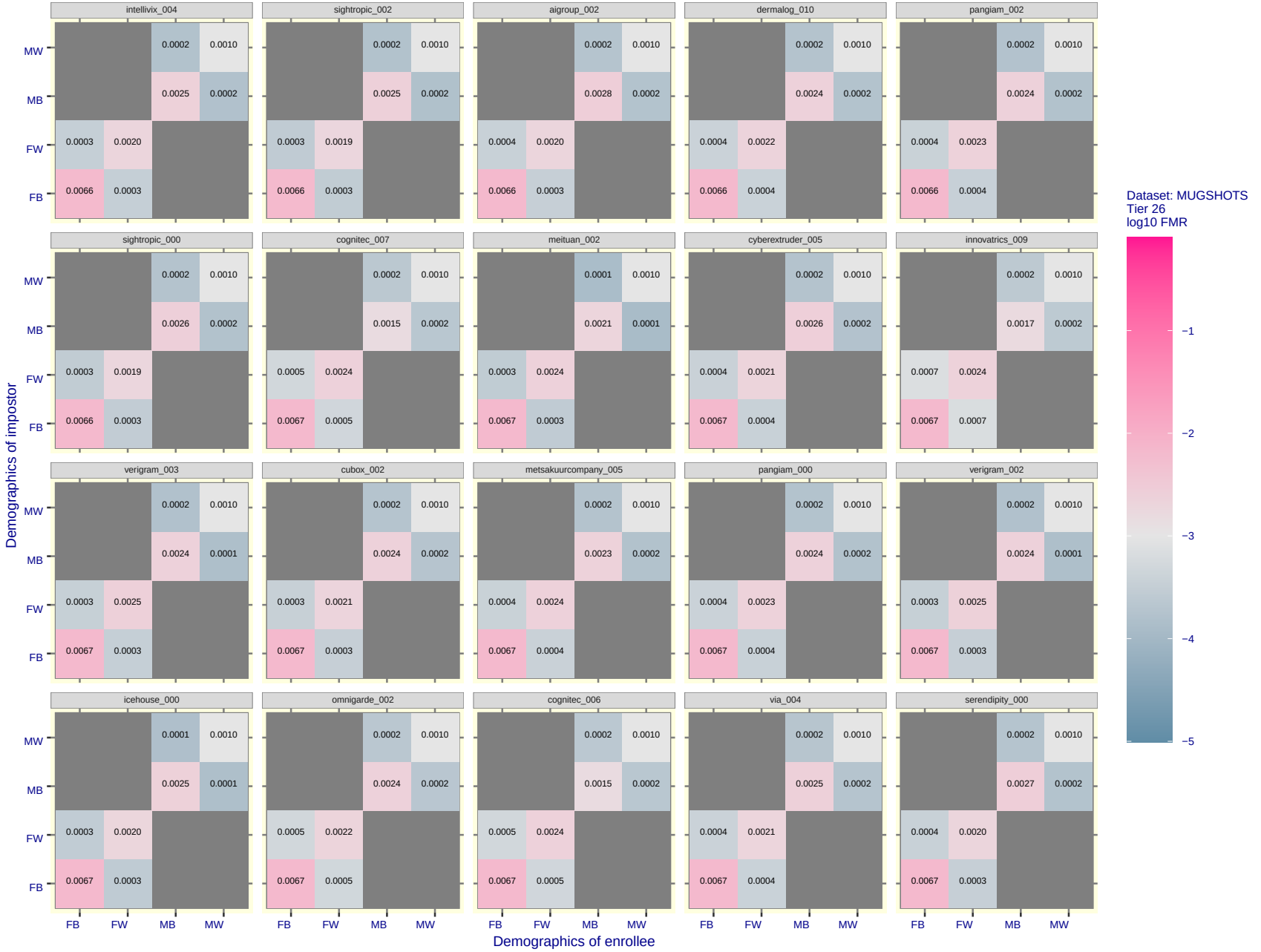

Figure 353: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

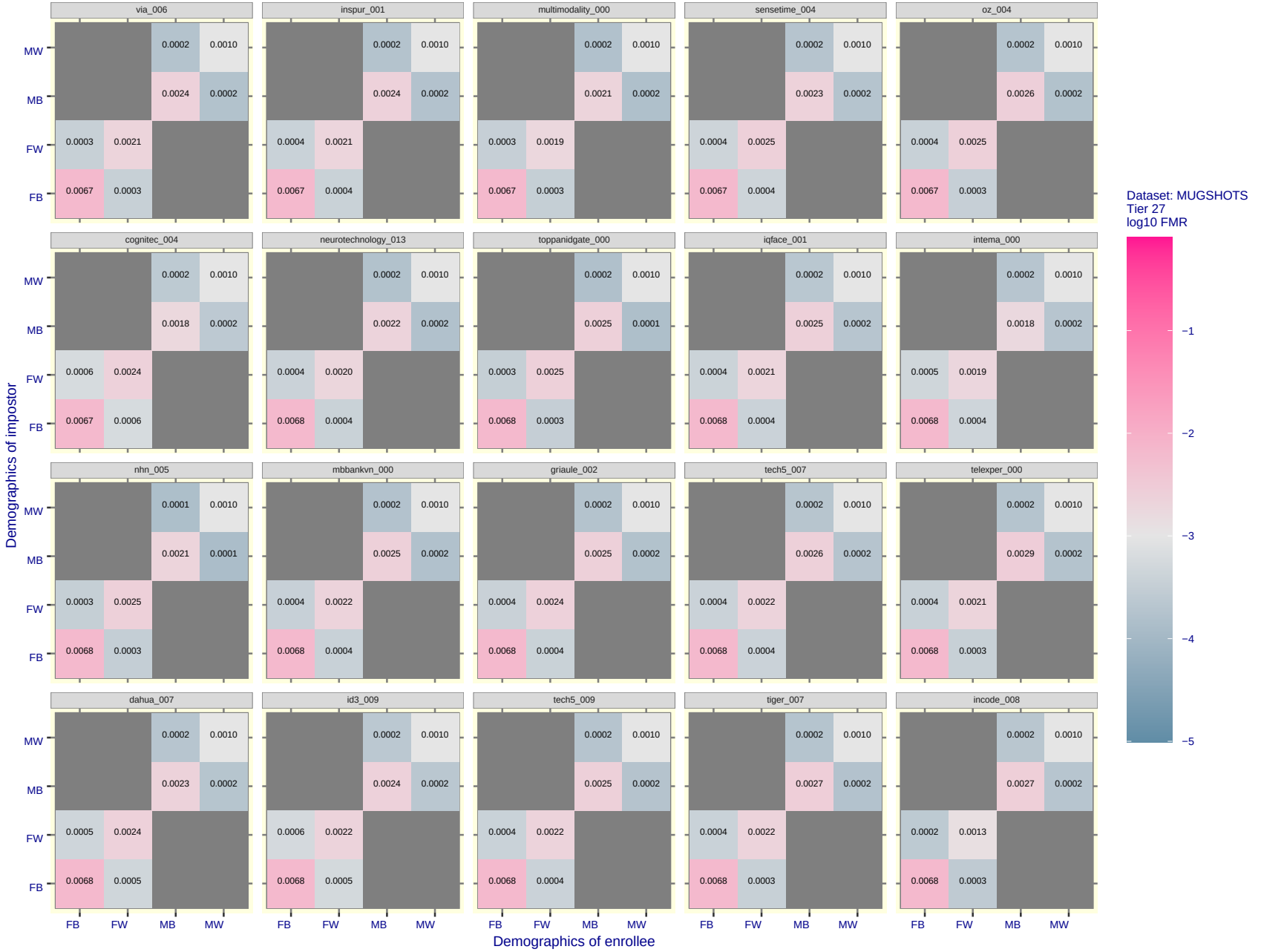

Figure 354: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

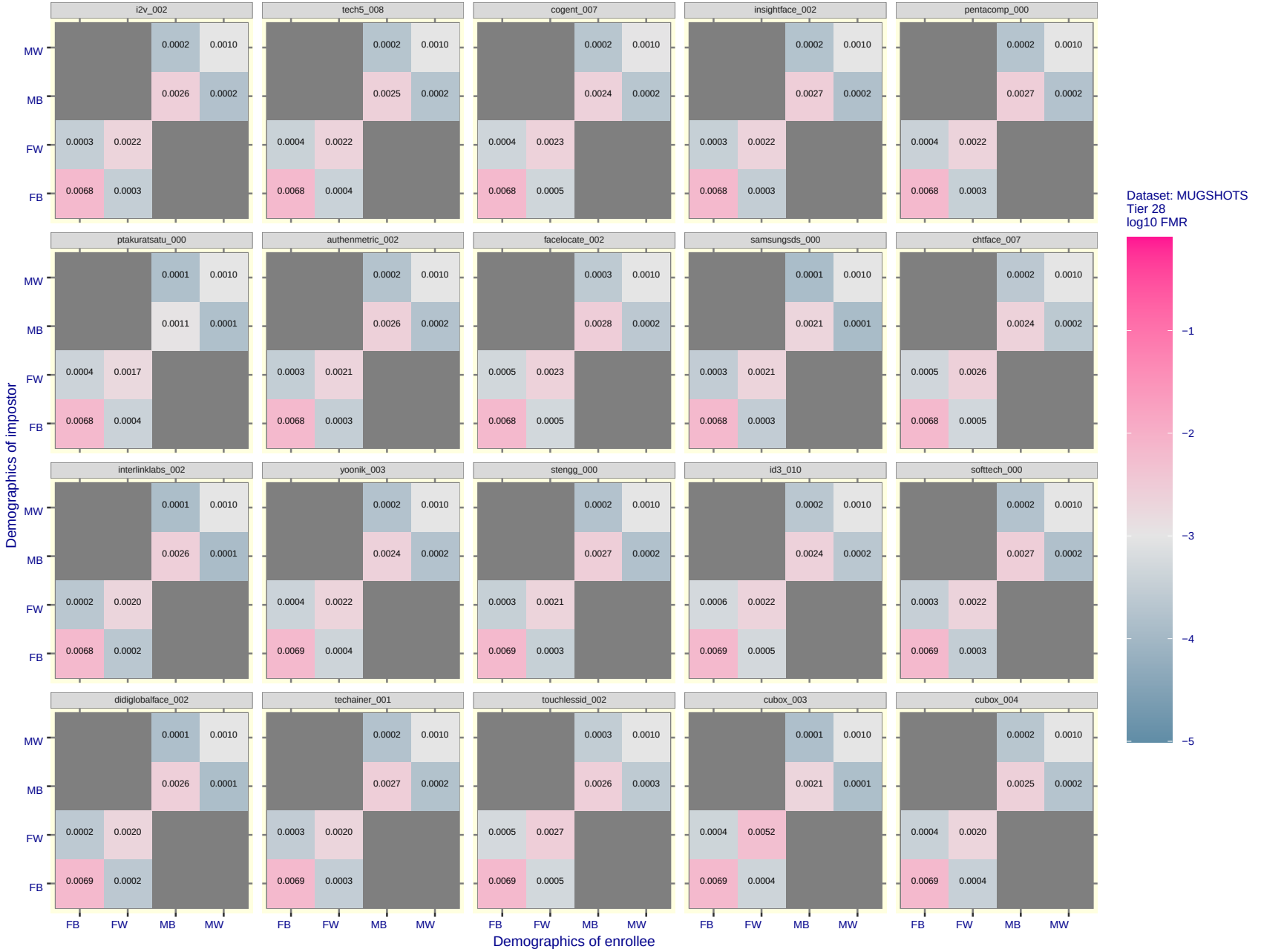

Figure 355: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

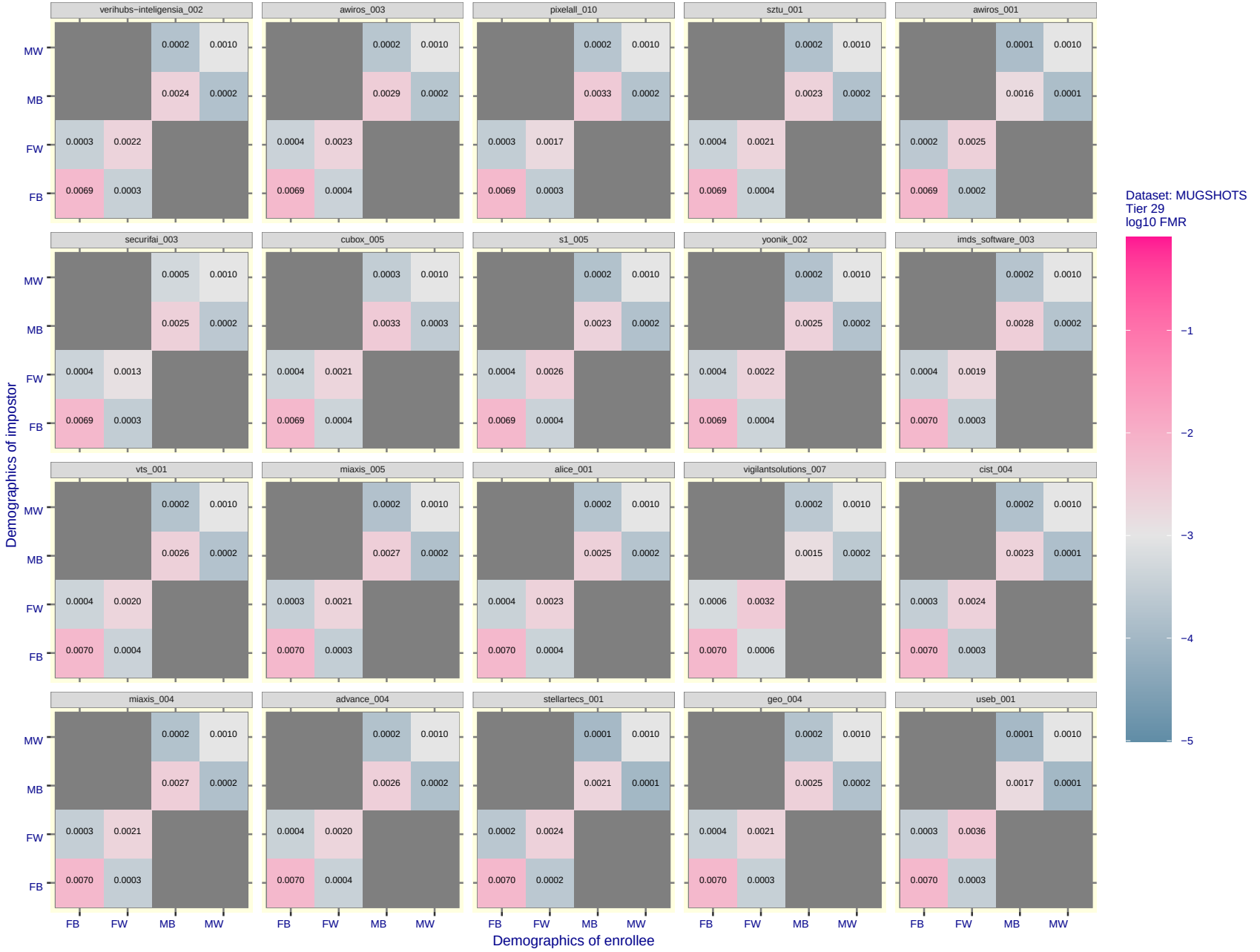

Figure 356: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

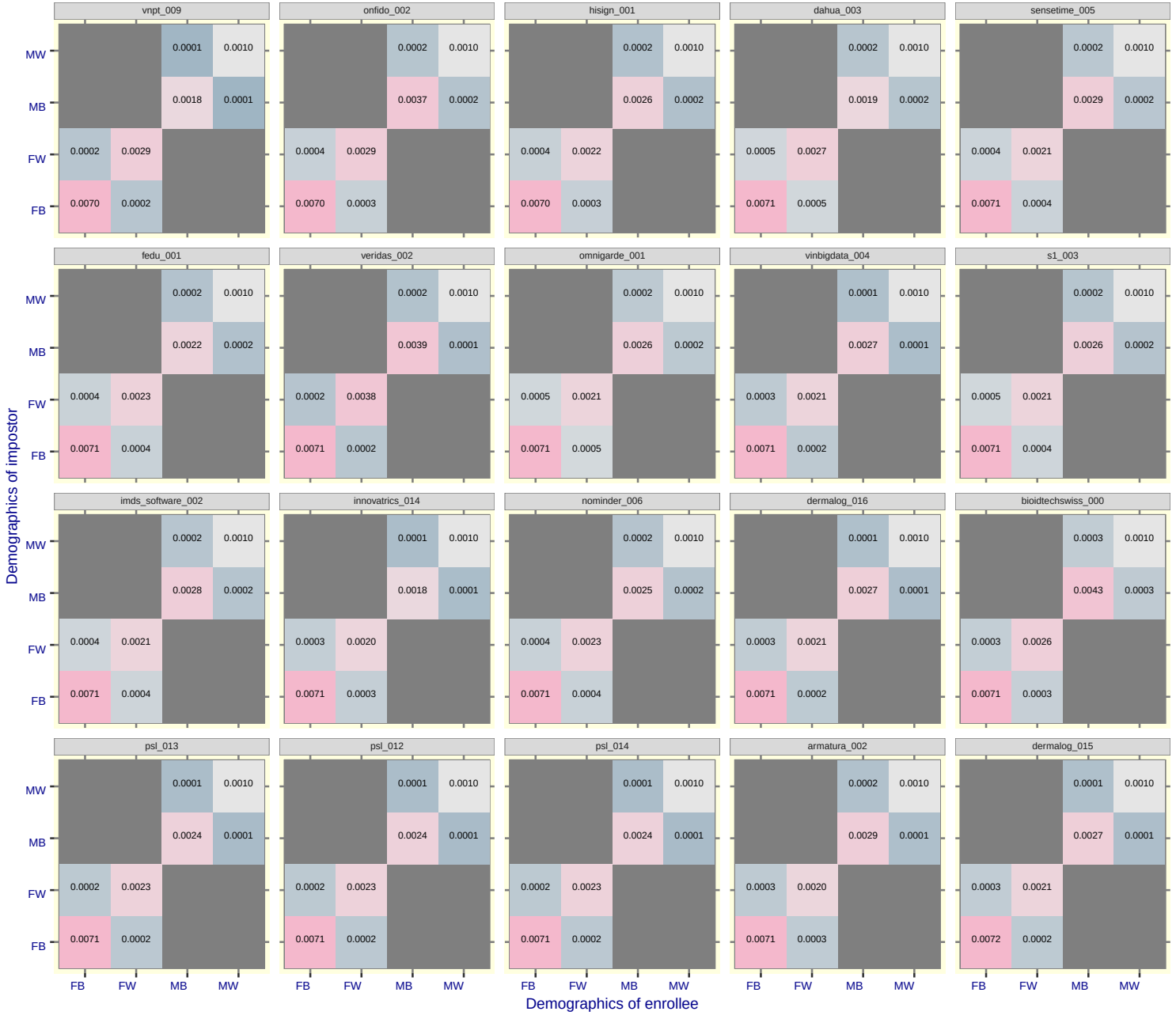

Figure 357: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

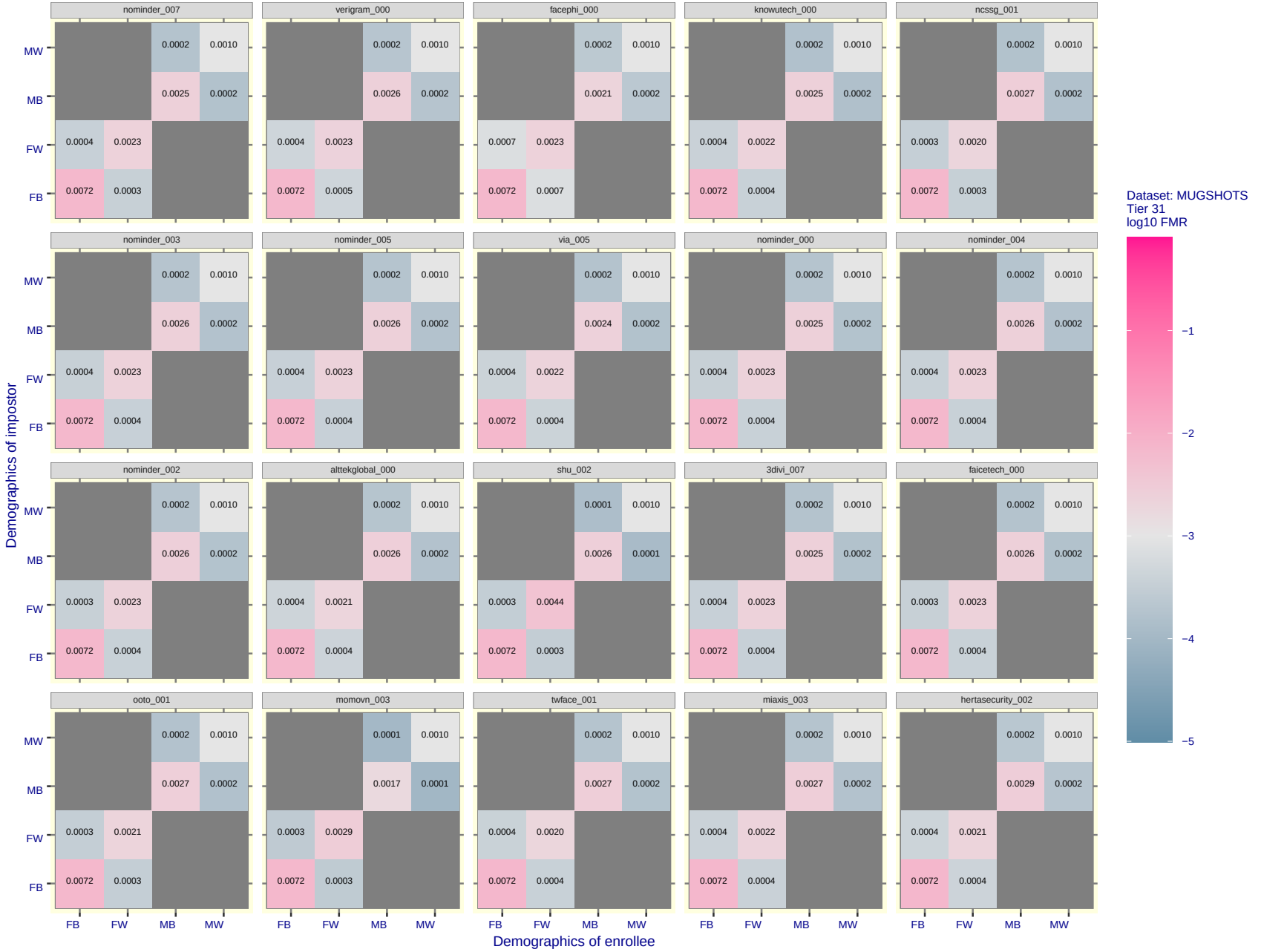

Figure 358: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

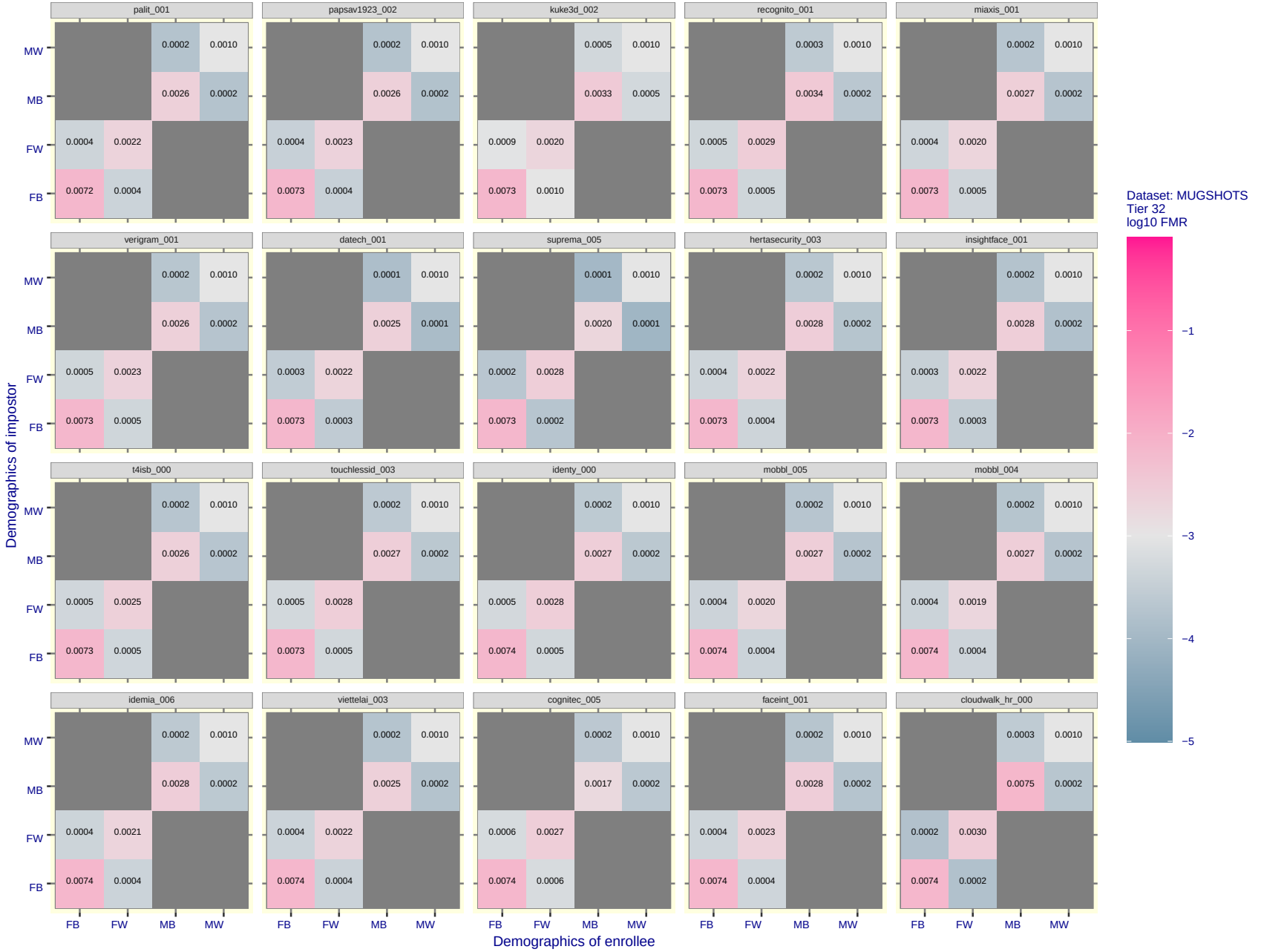

Figure 359: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

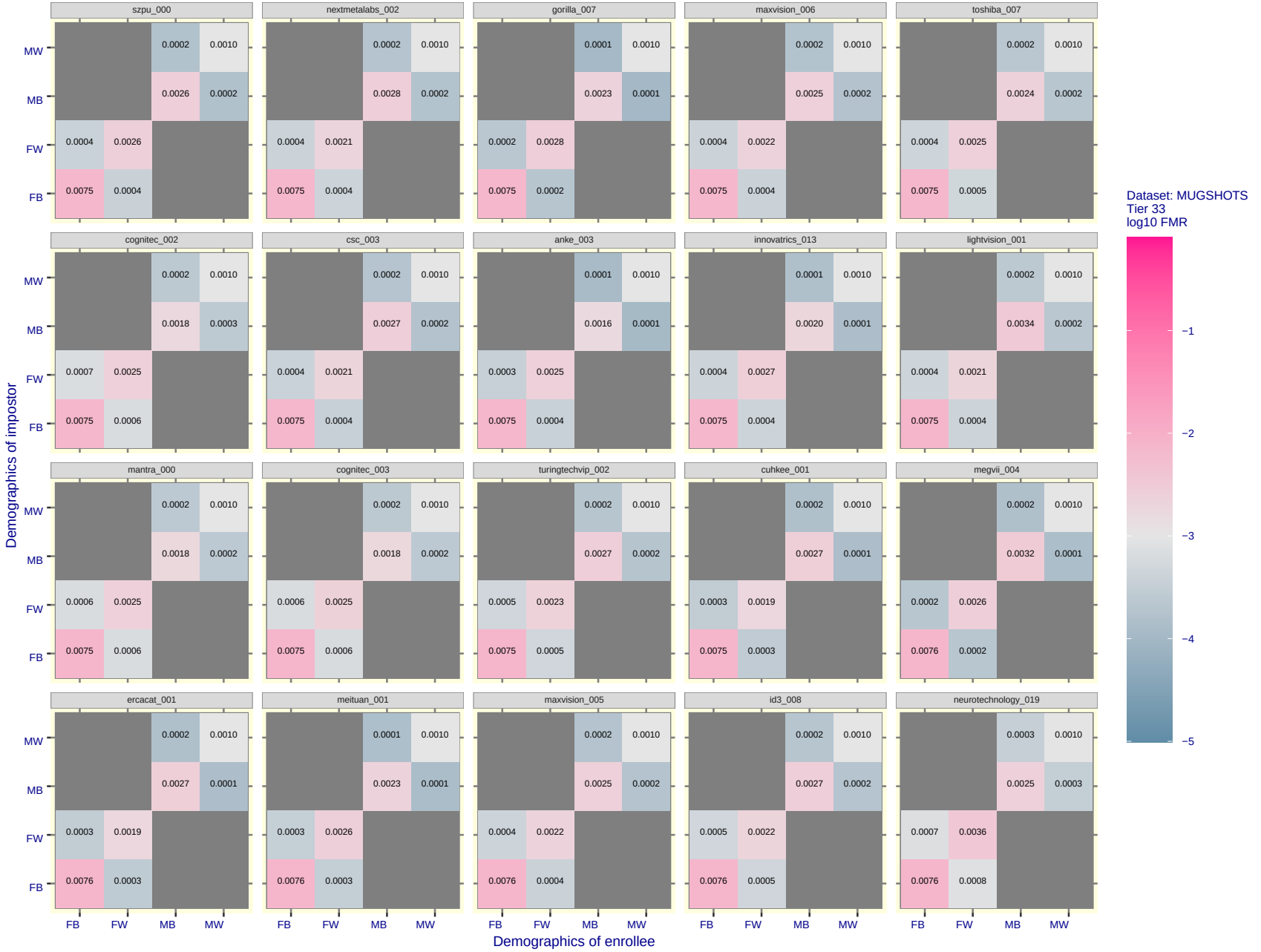

Figure 360: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

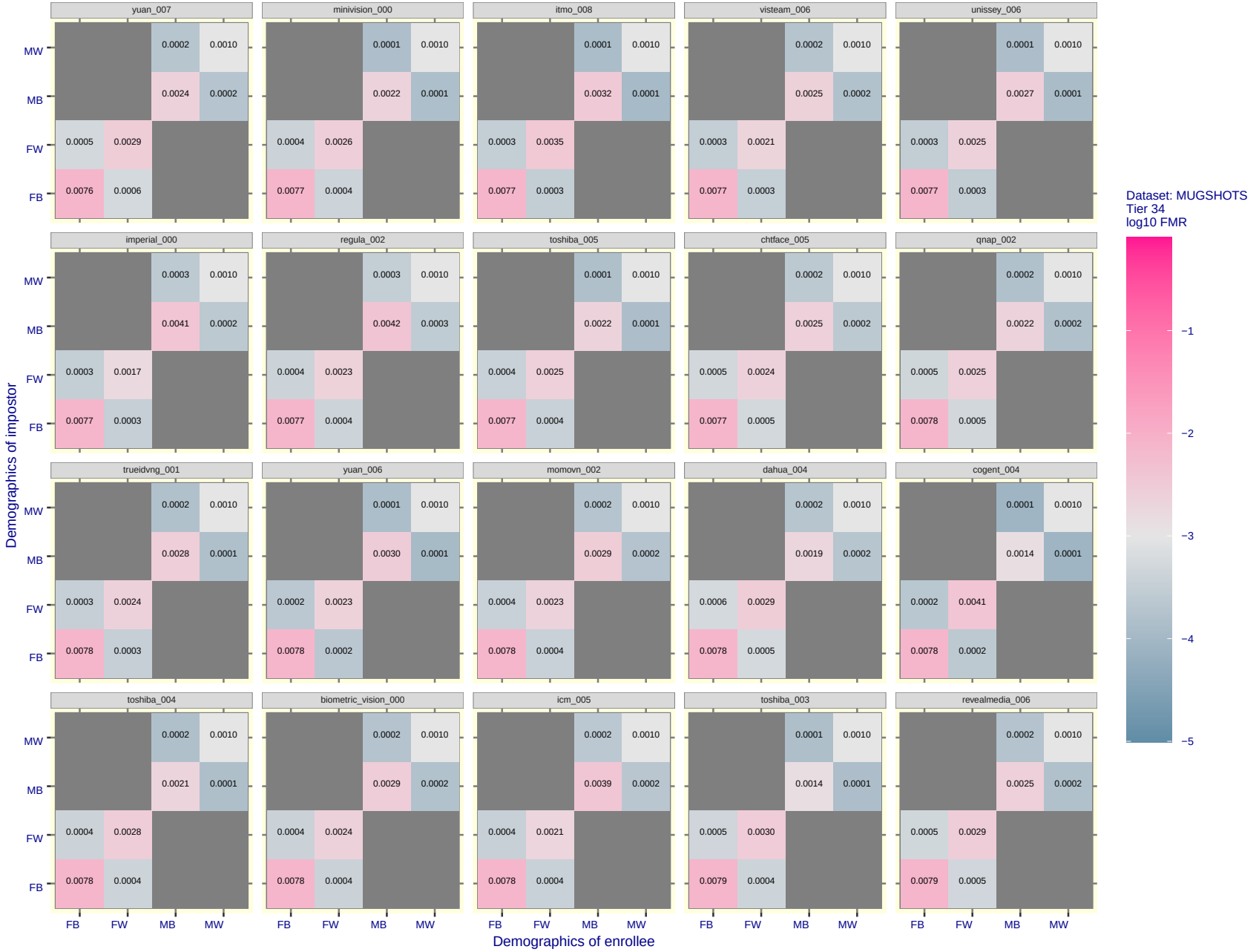

Figure 361: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

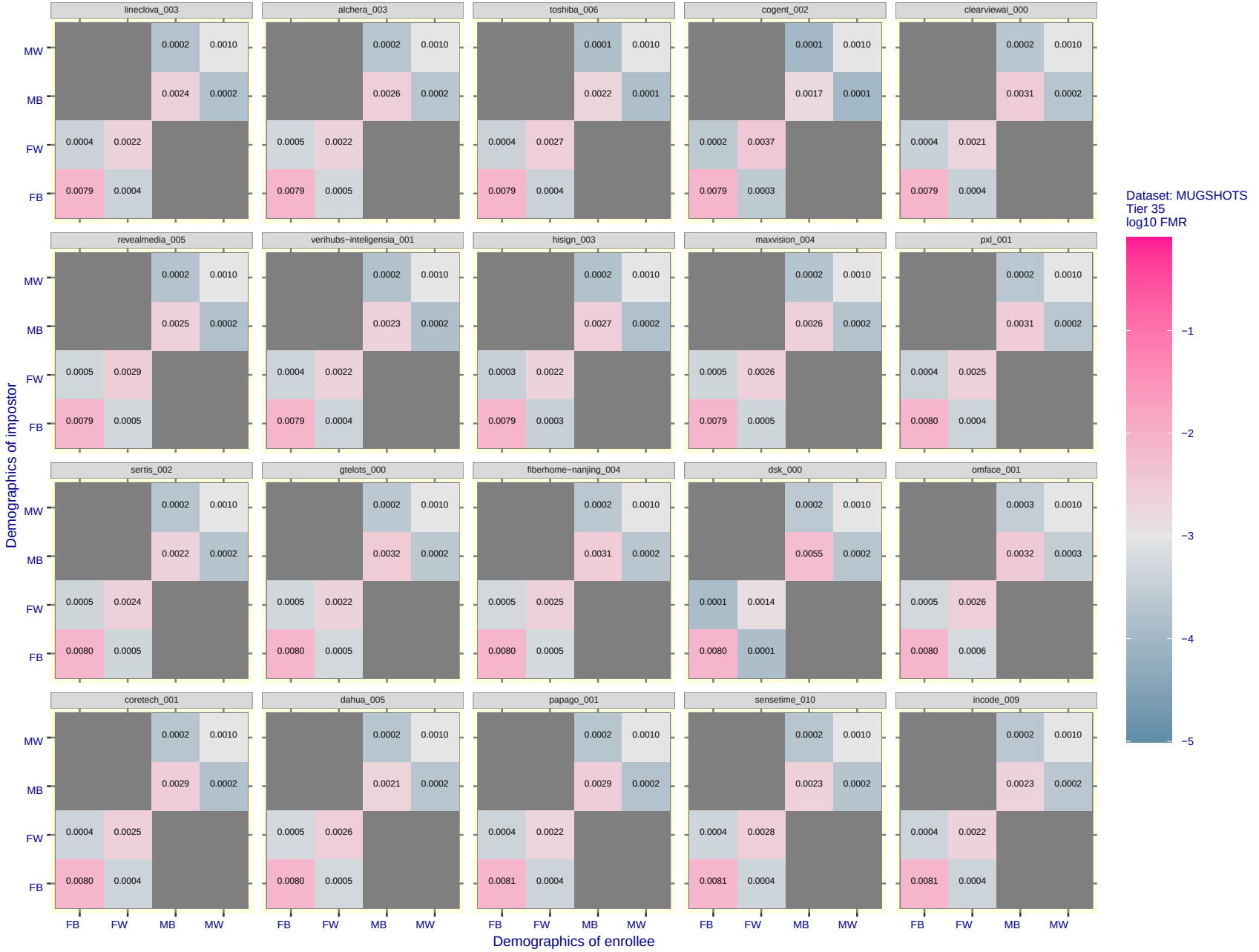

Figure 362: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

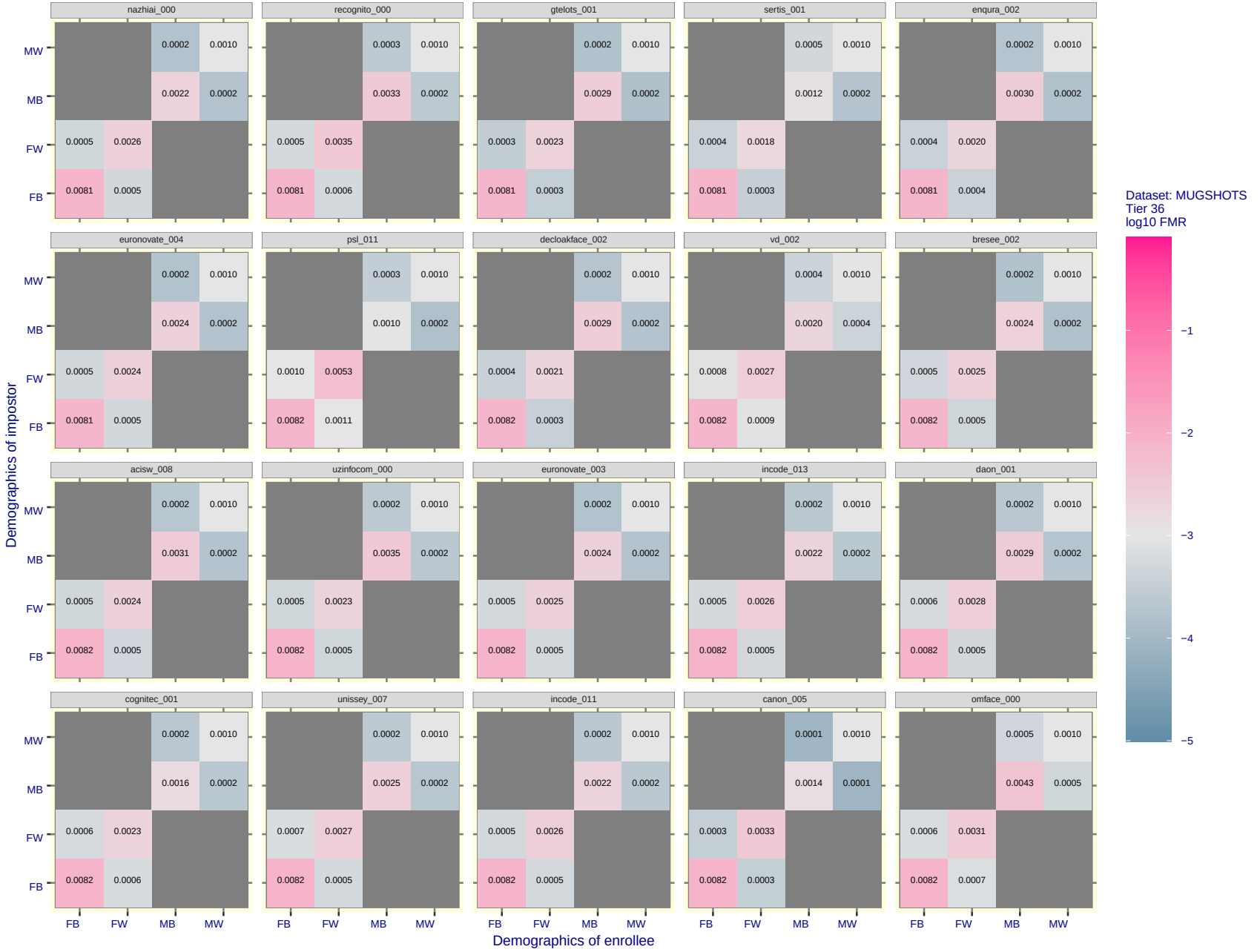

Figure 363: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

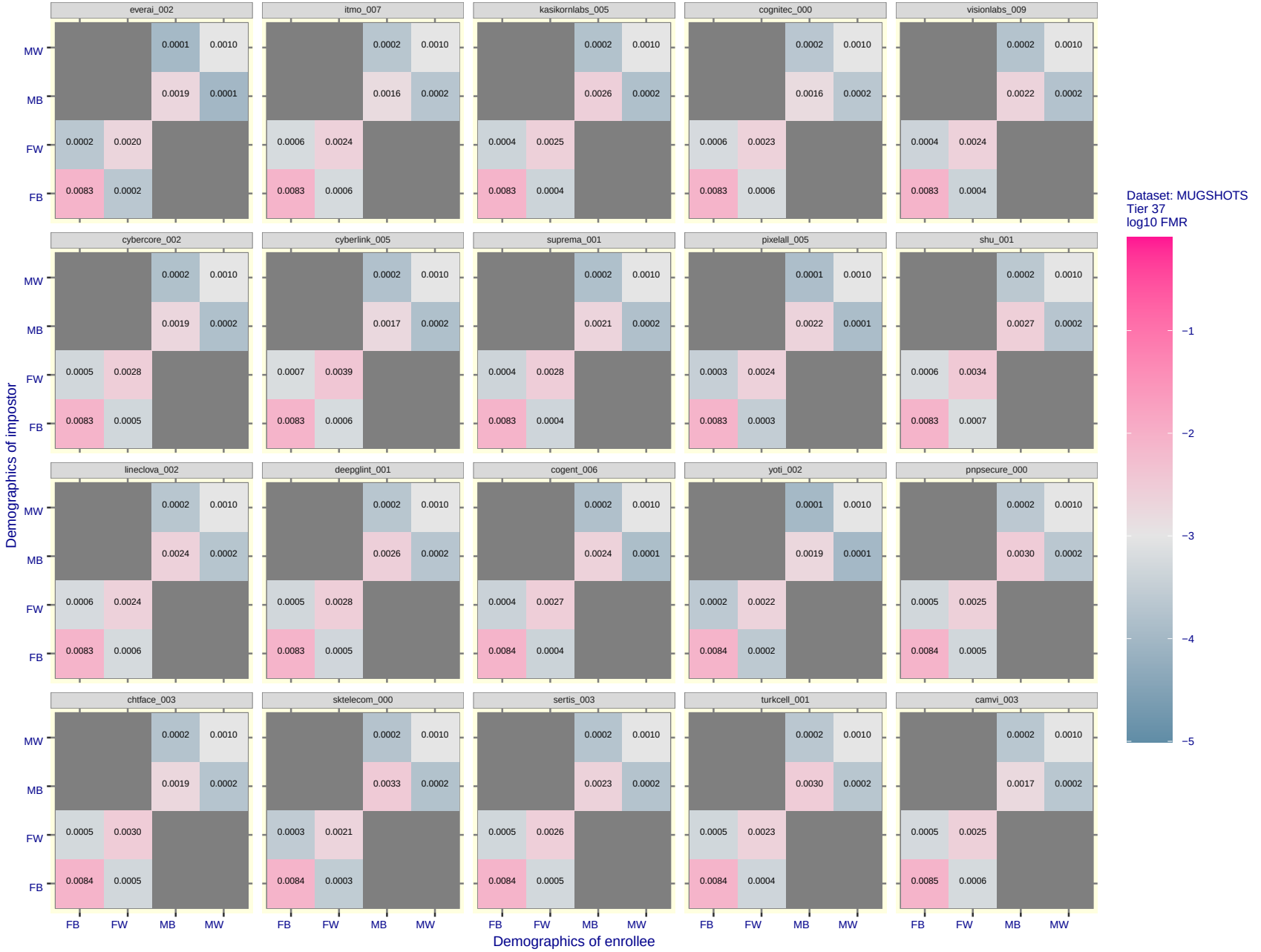

Figure 364: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

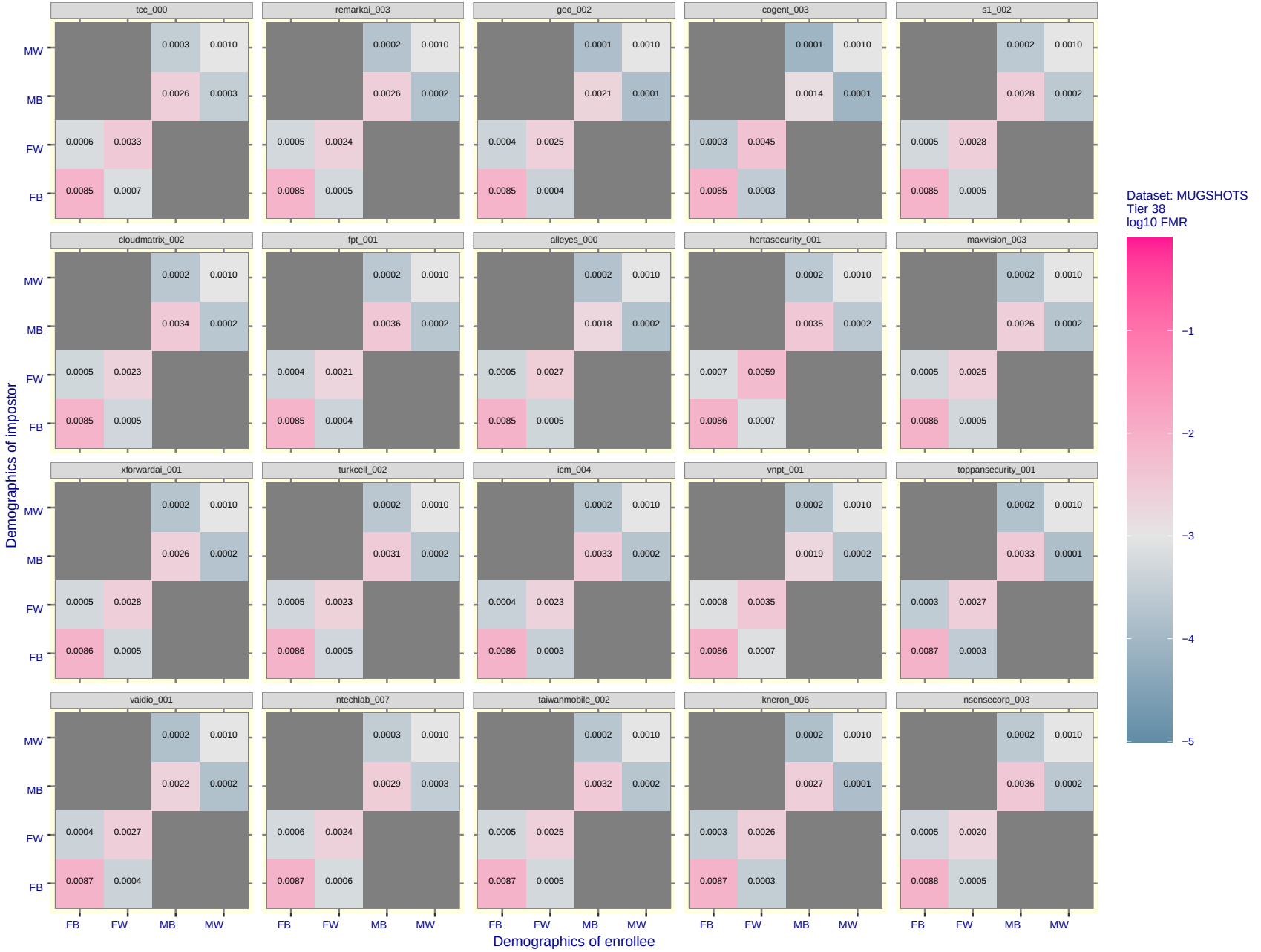

Figure 365: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

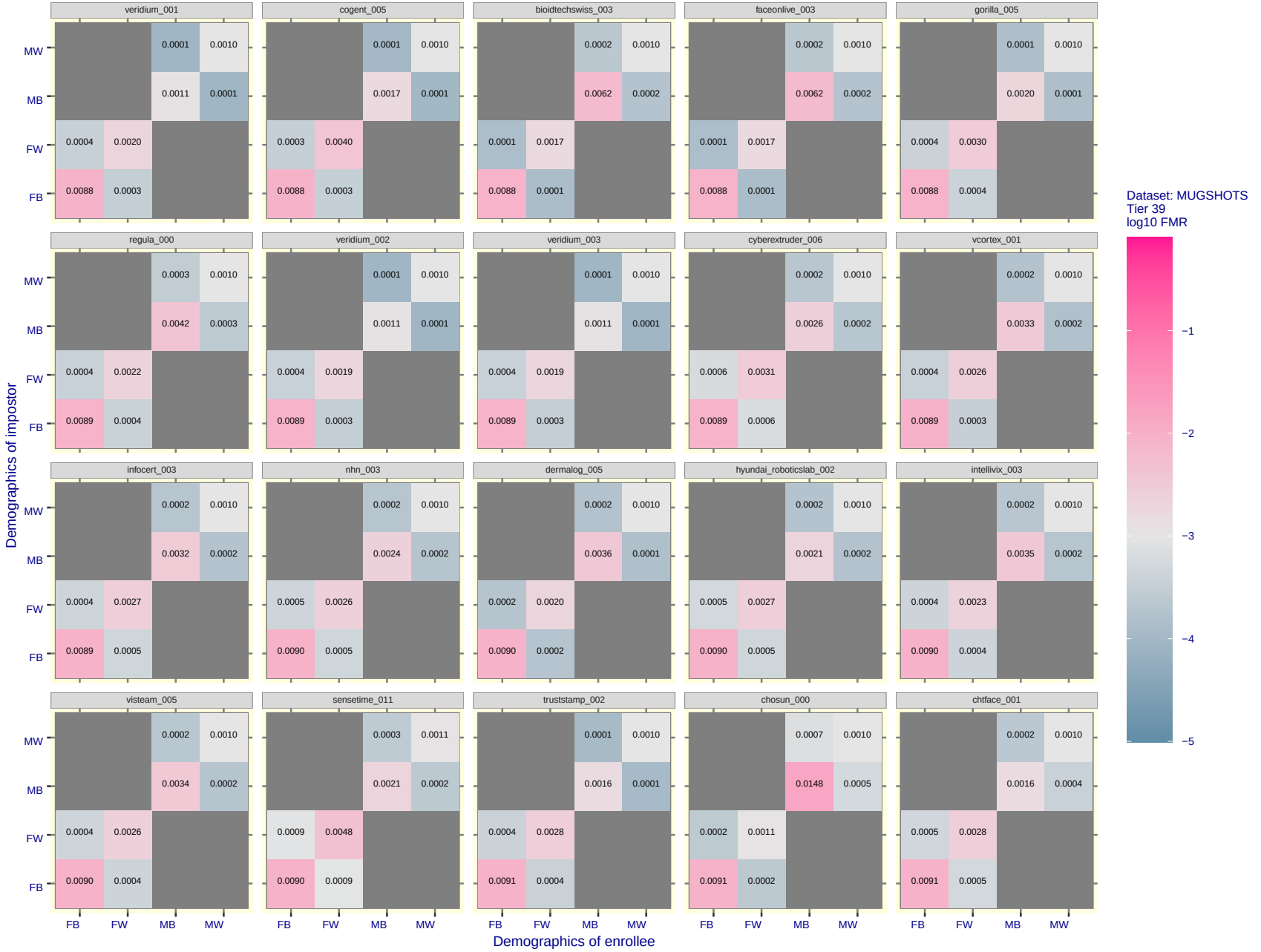

Figure 366: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

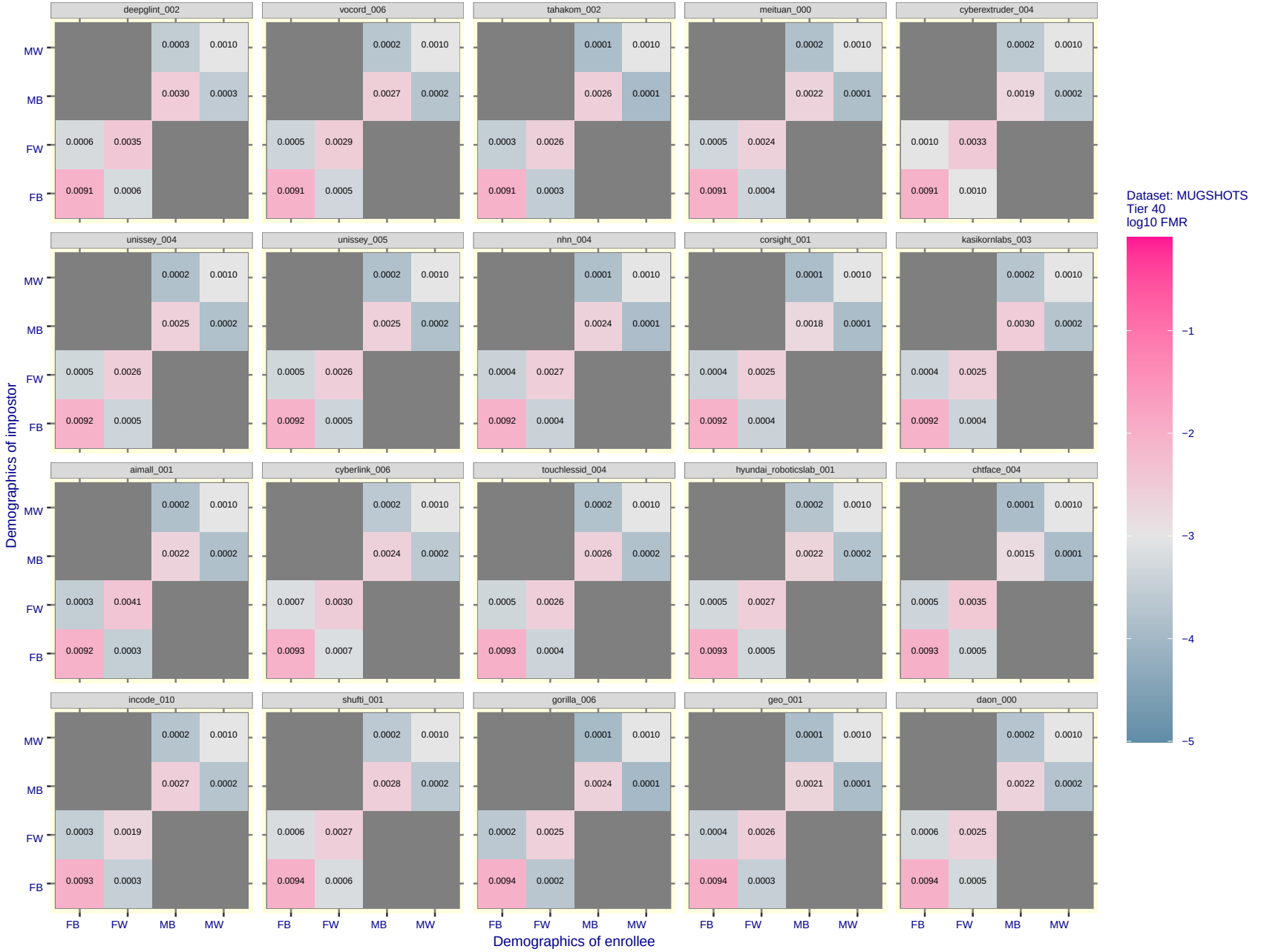

Figure 367: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

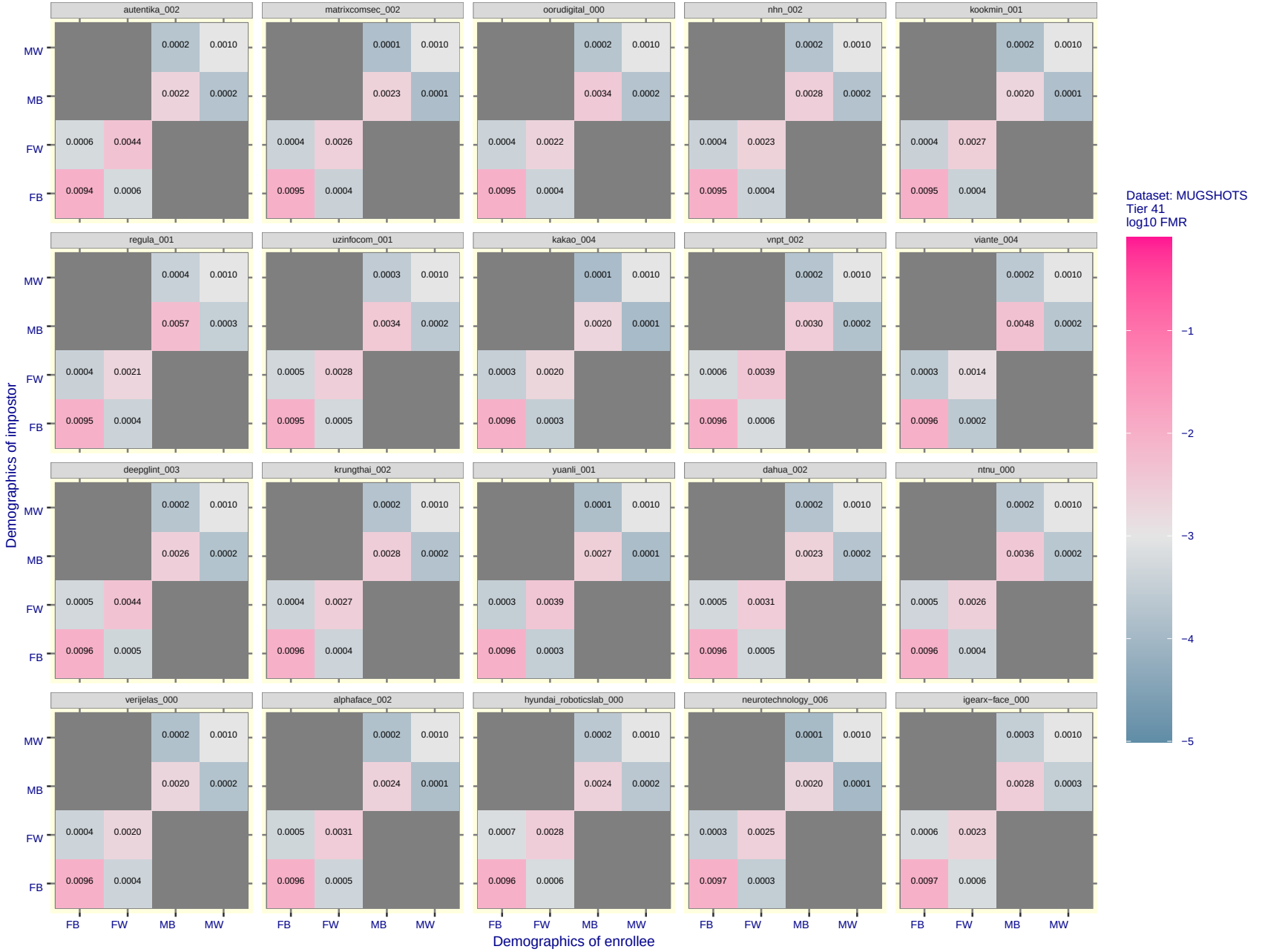

Figure 368: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

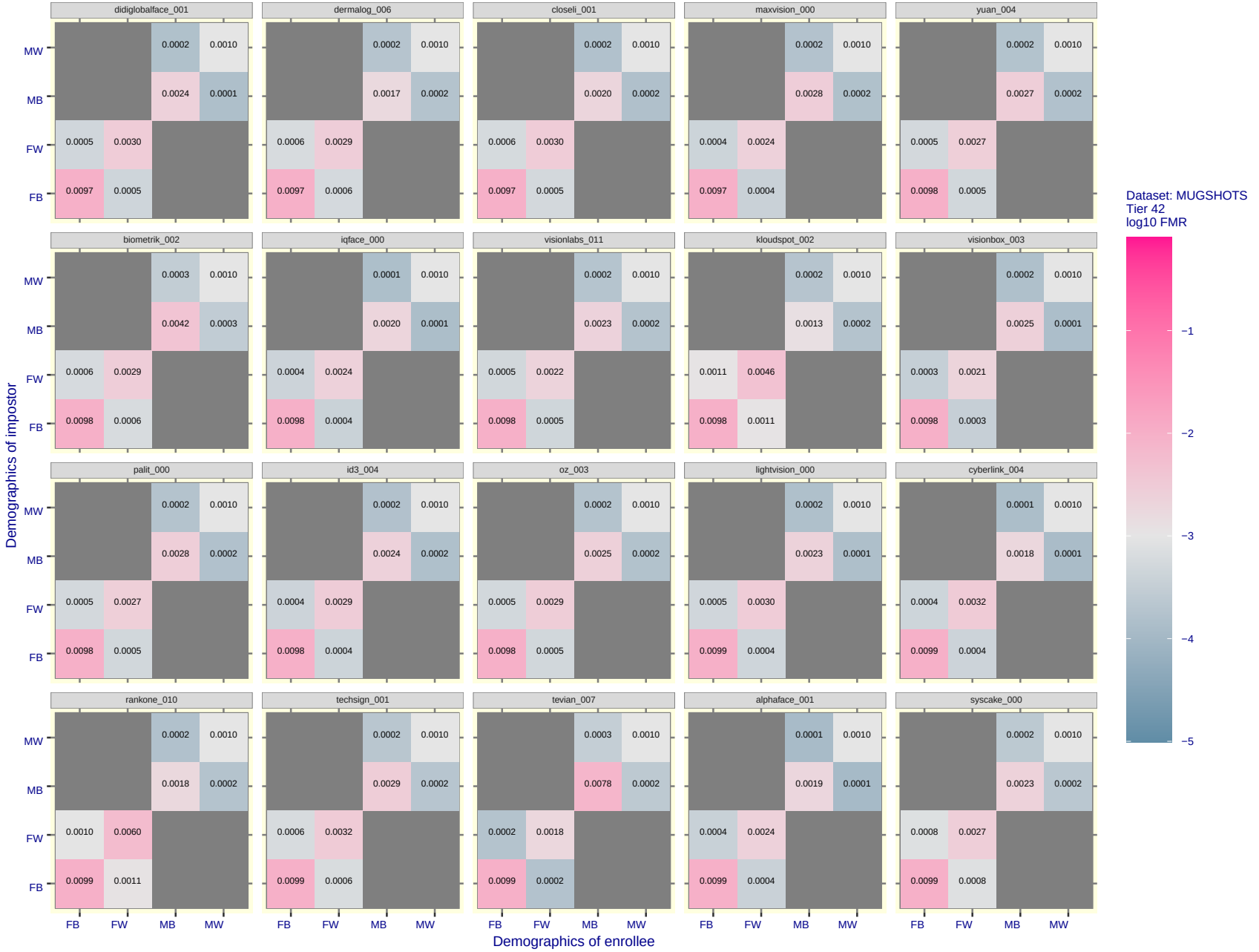

Figure 369: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/09/01 08:08:43
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

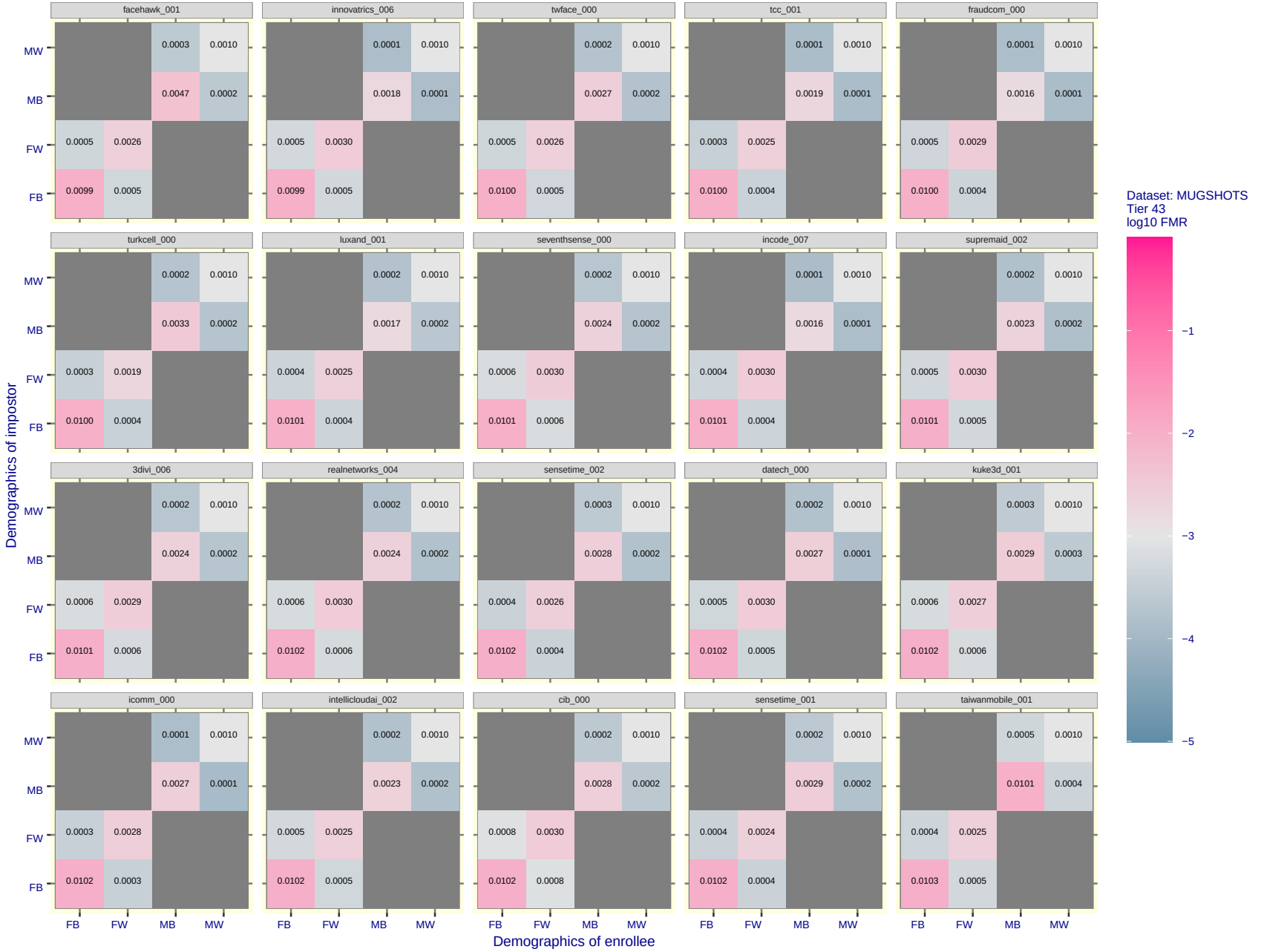

Figure 370: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

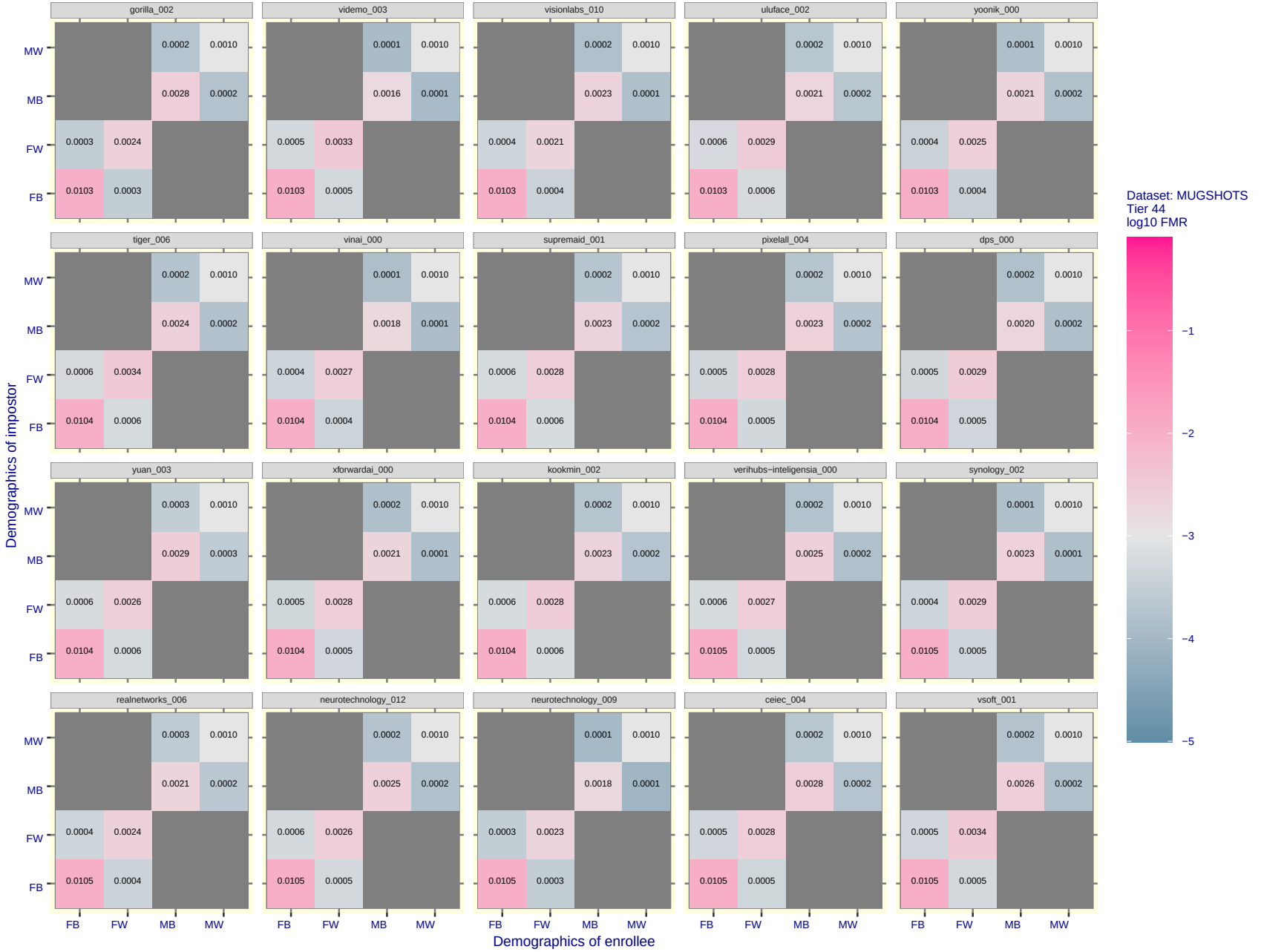

Figure 371: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

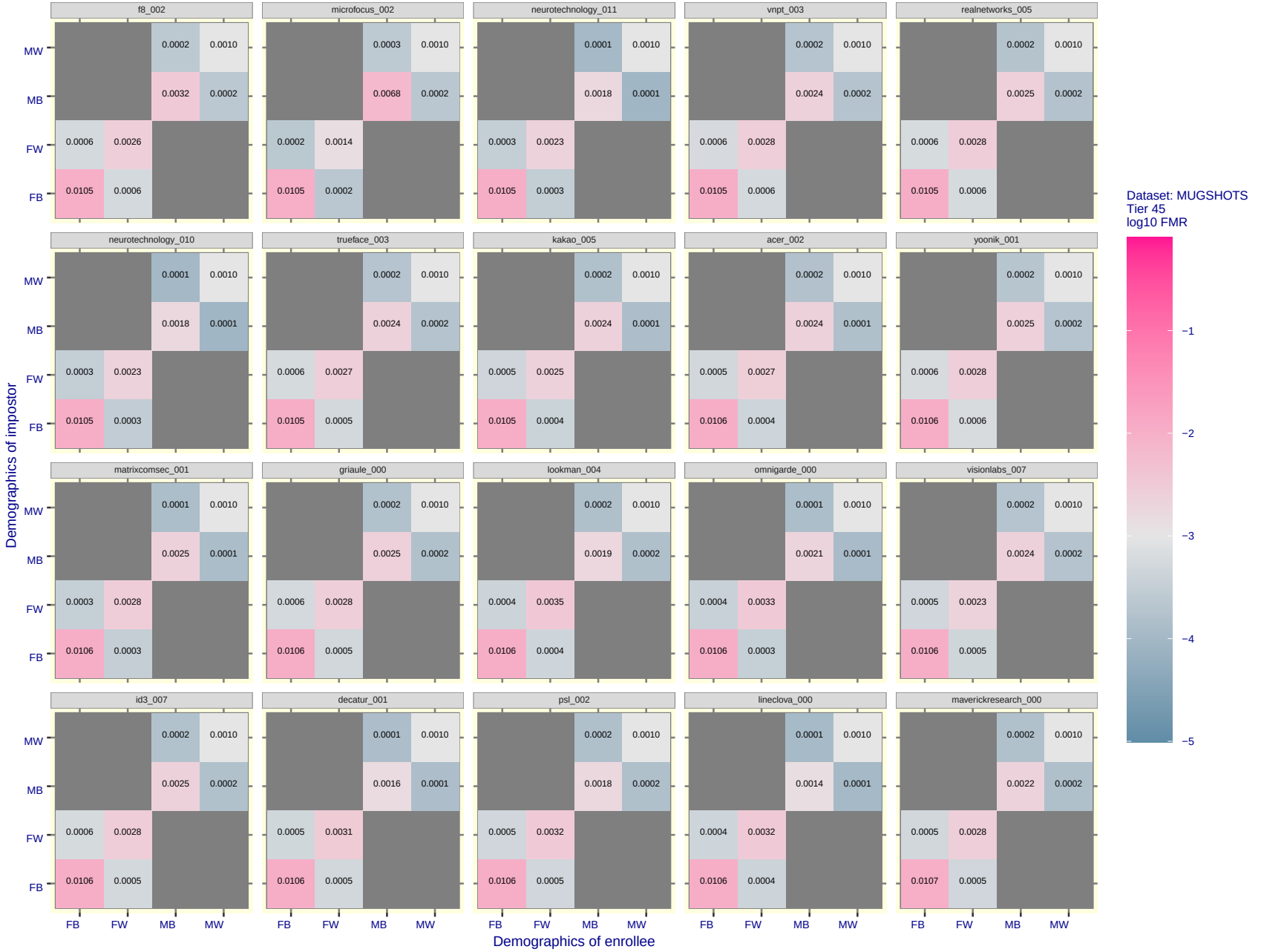

Figure 372: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

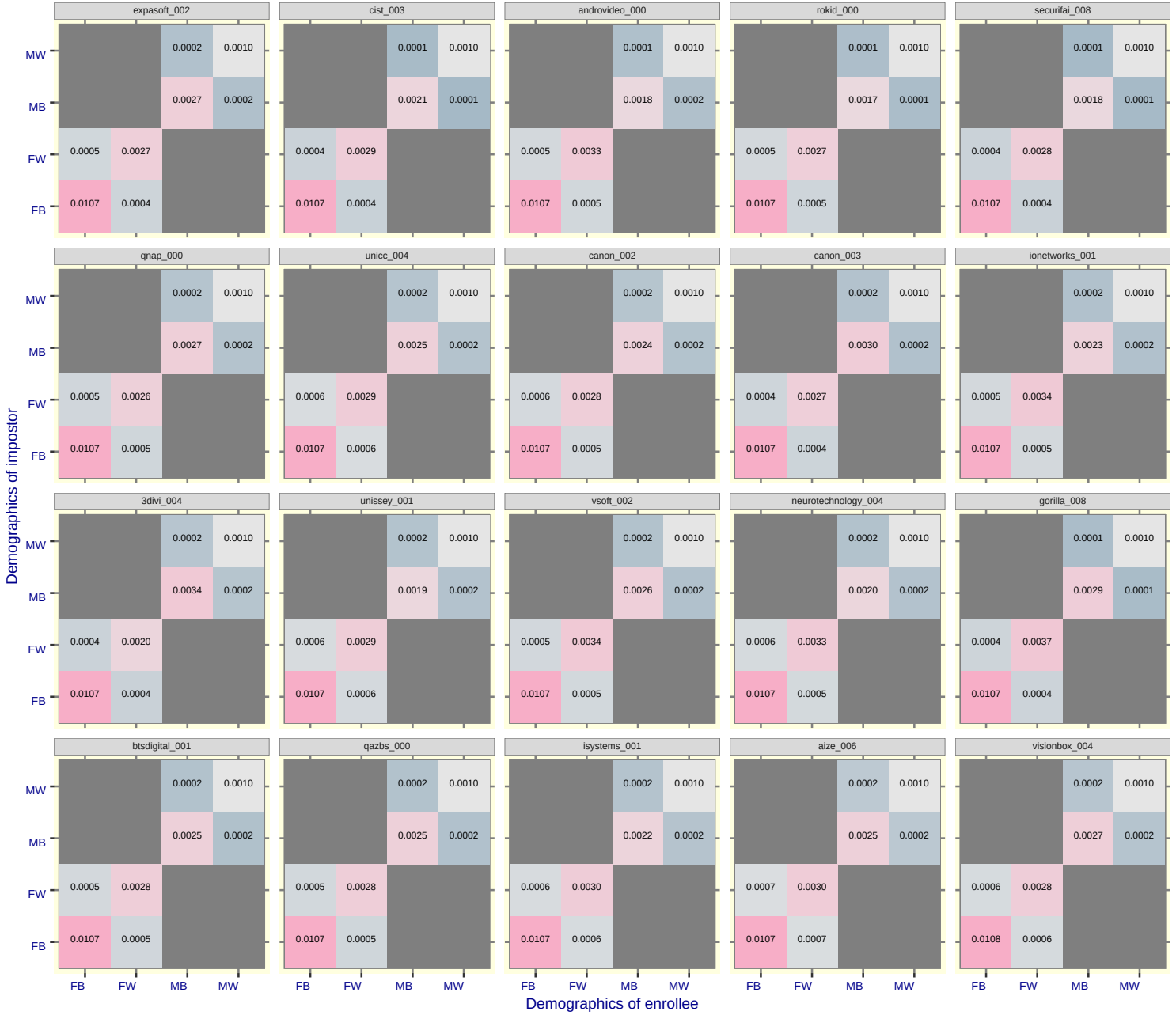

Figure 373: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

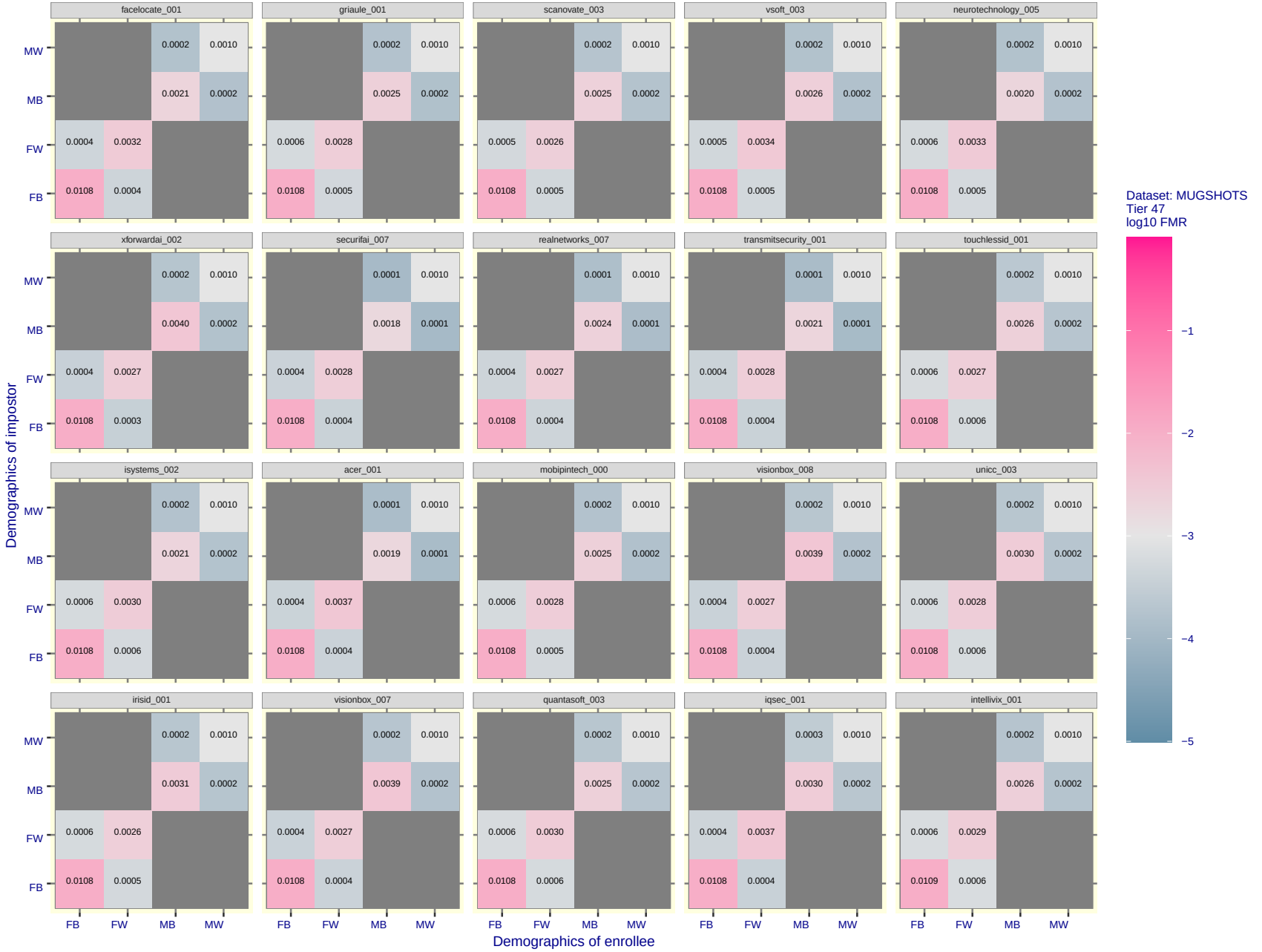

Figure 374: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

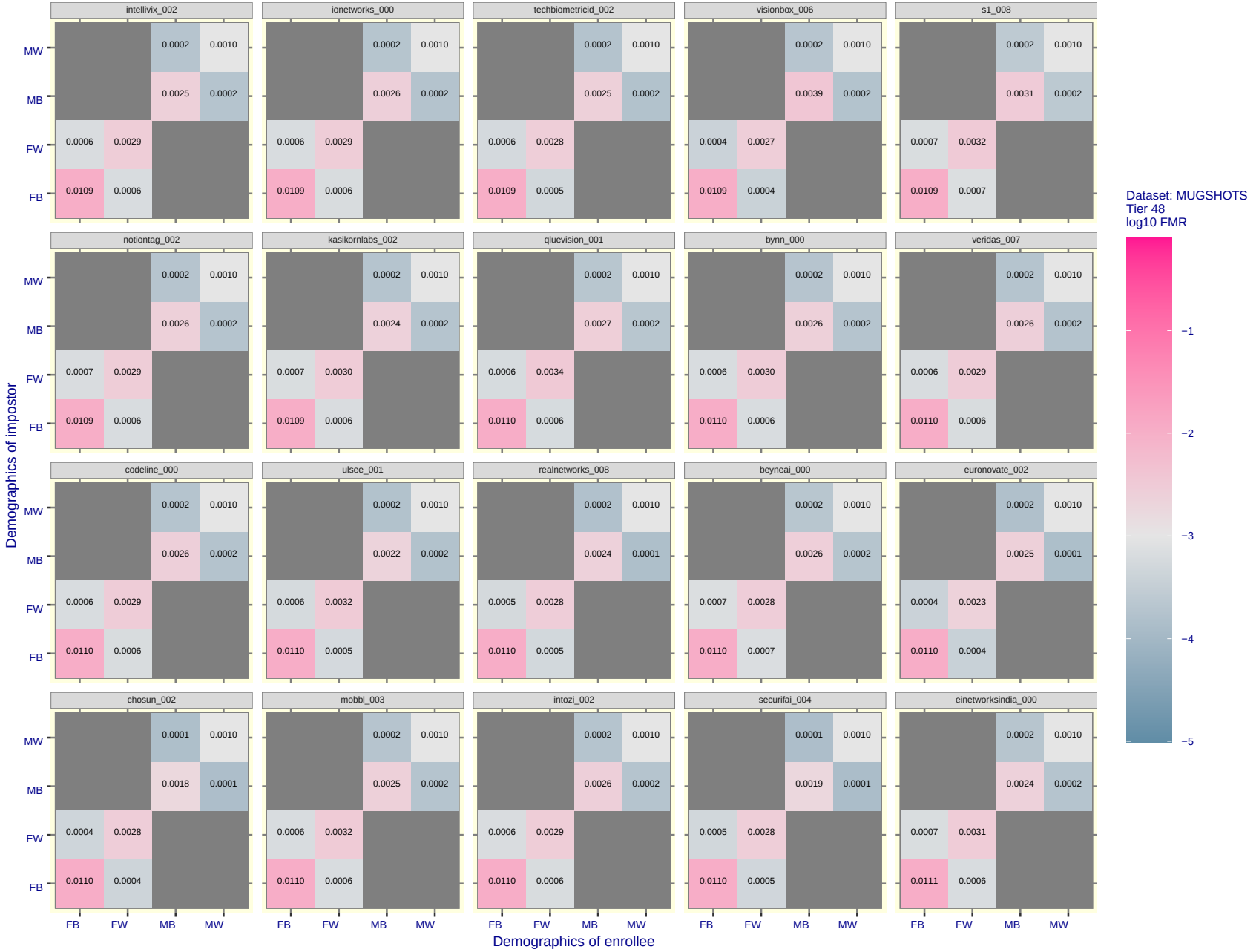

Figure 375: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

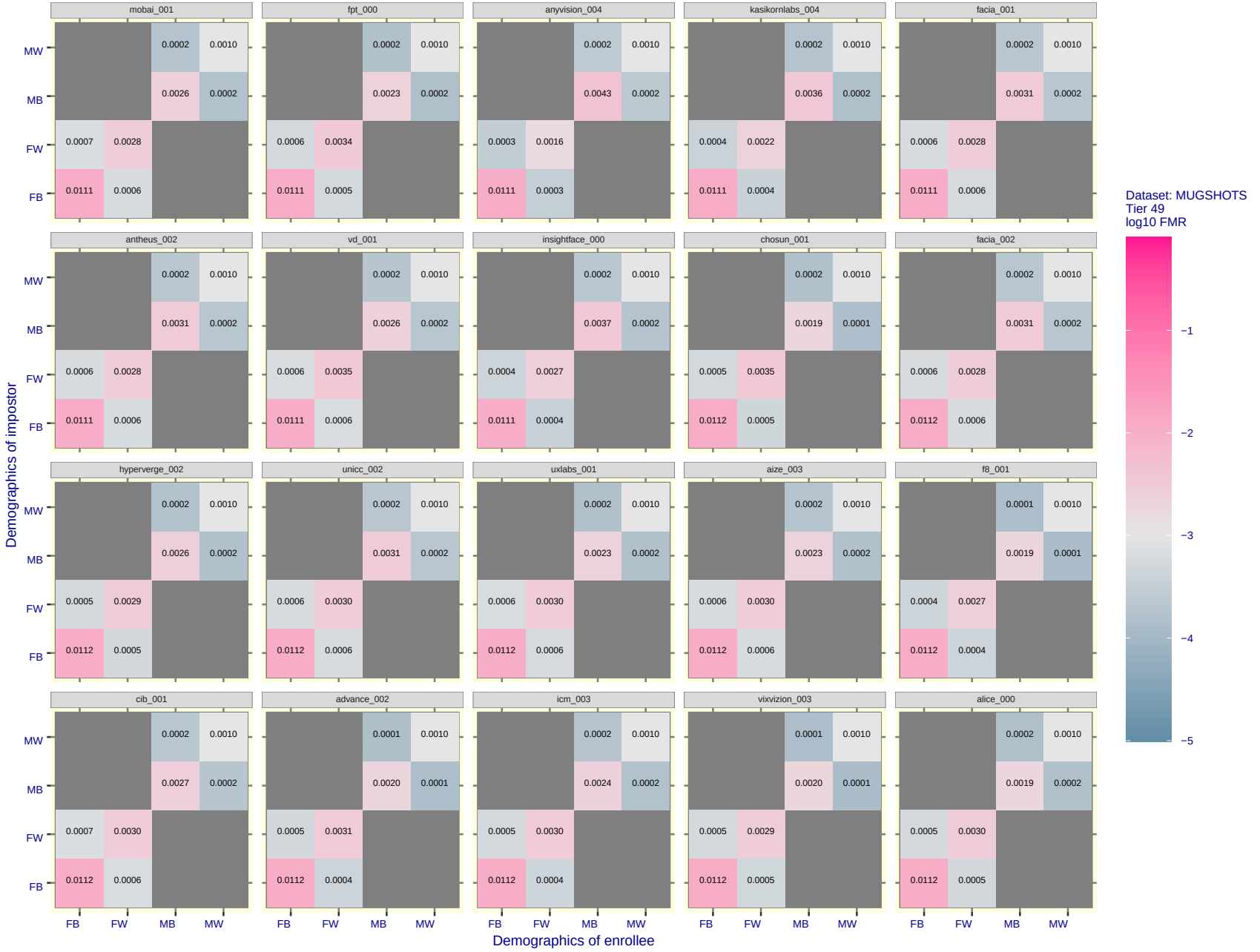

Figure 376: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

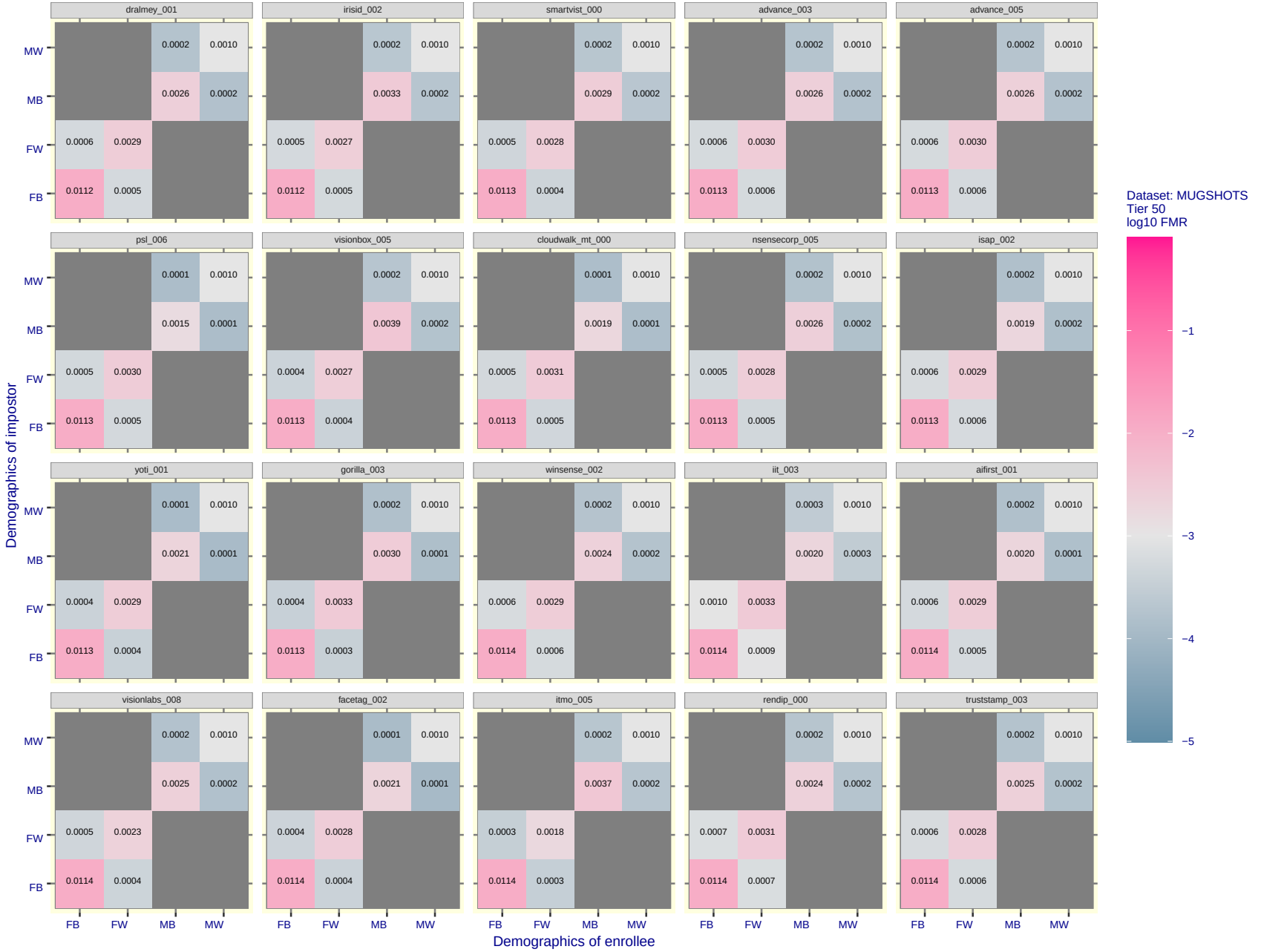

Figure 377: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

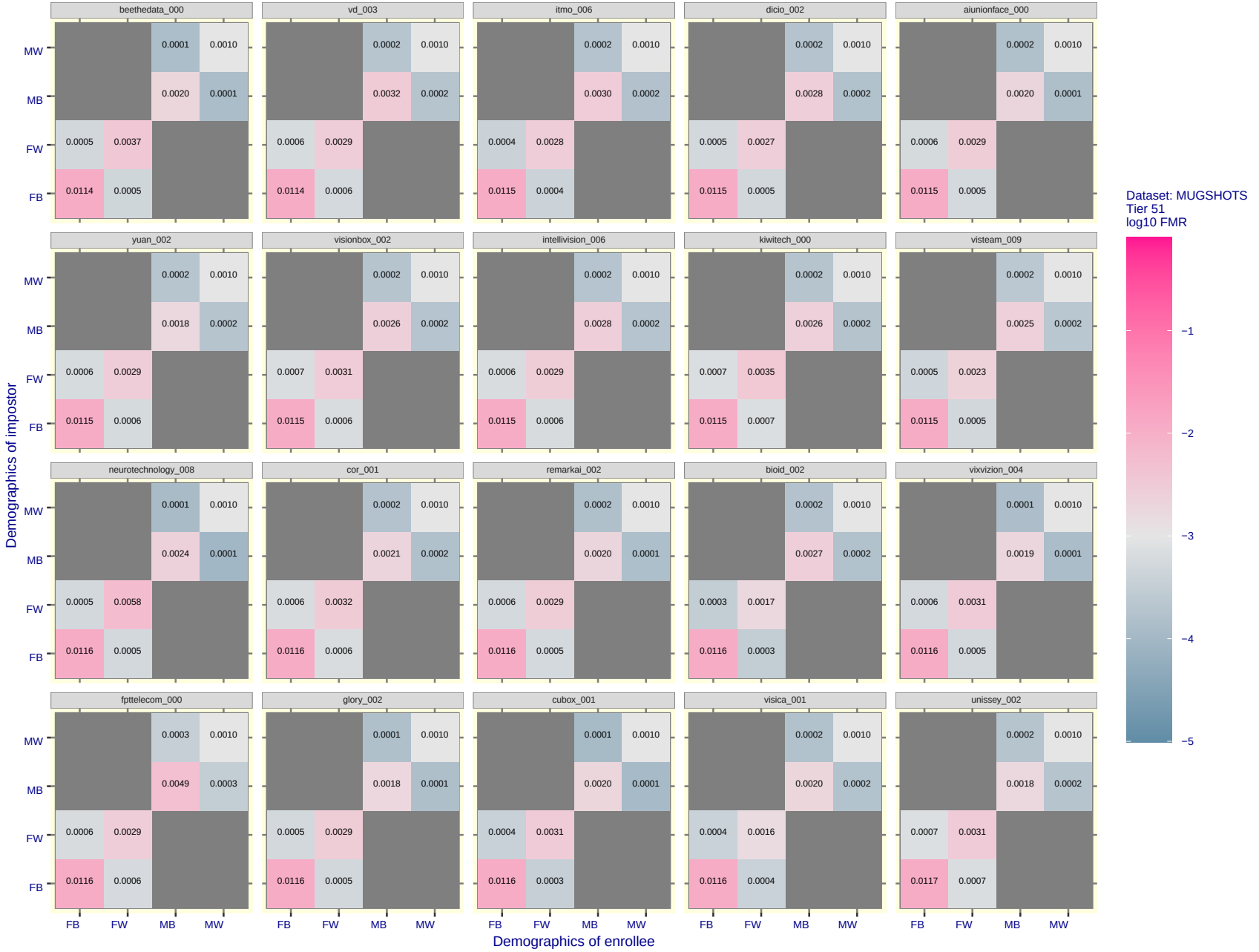

Figure 378: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

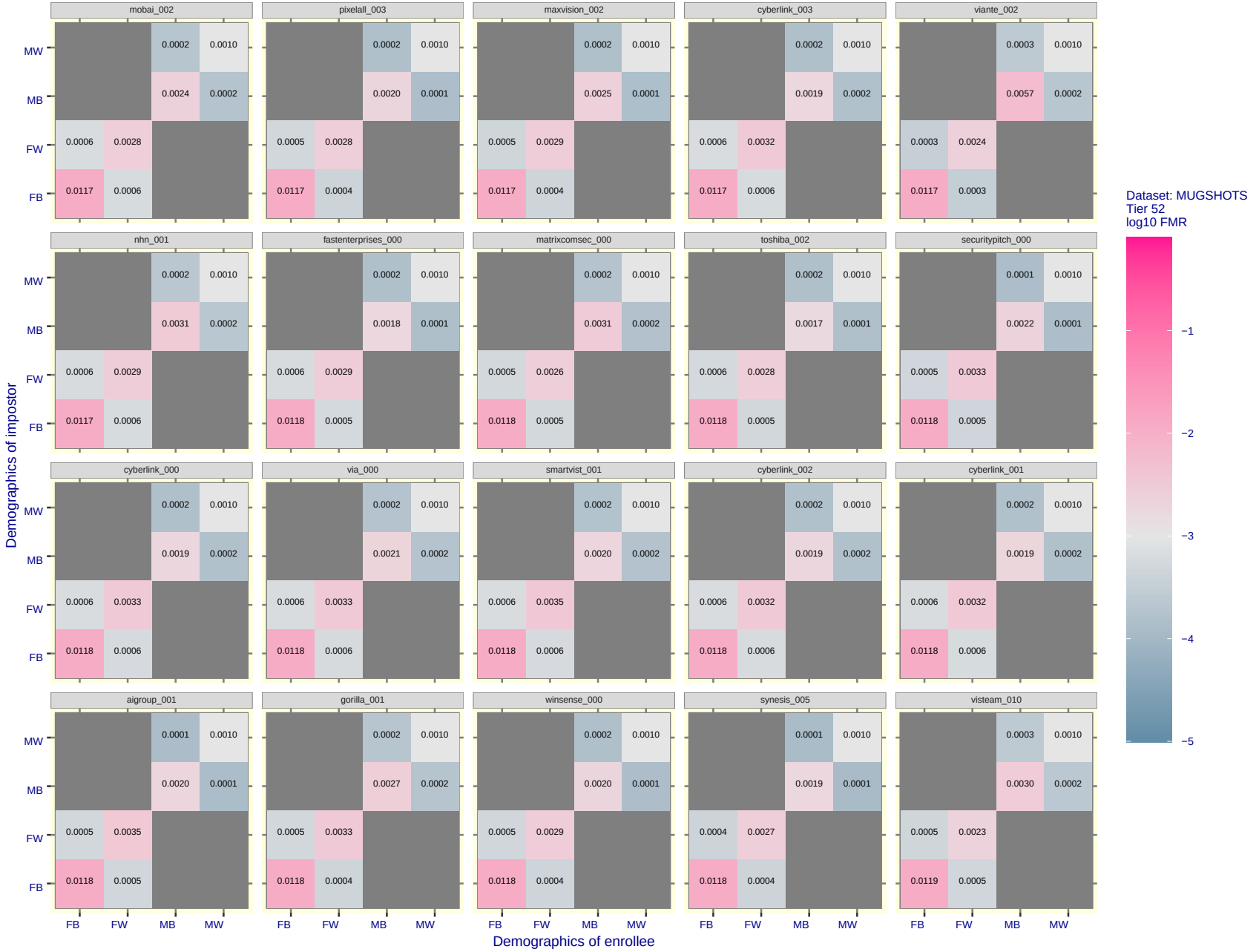

Figure 379: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

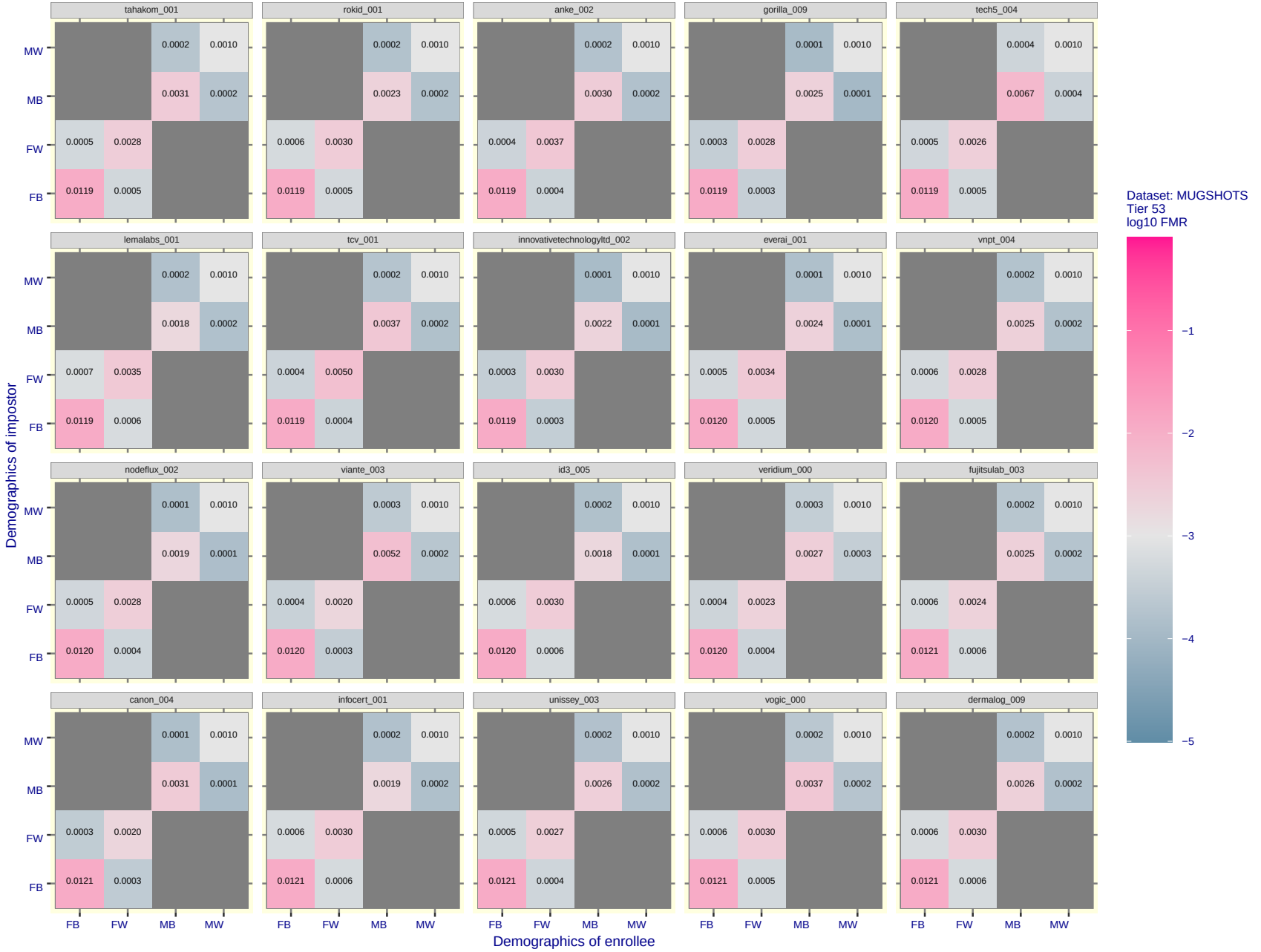

Figure 380: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

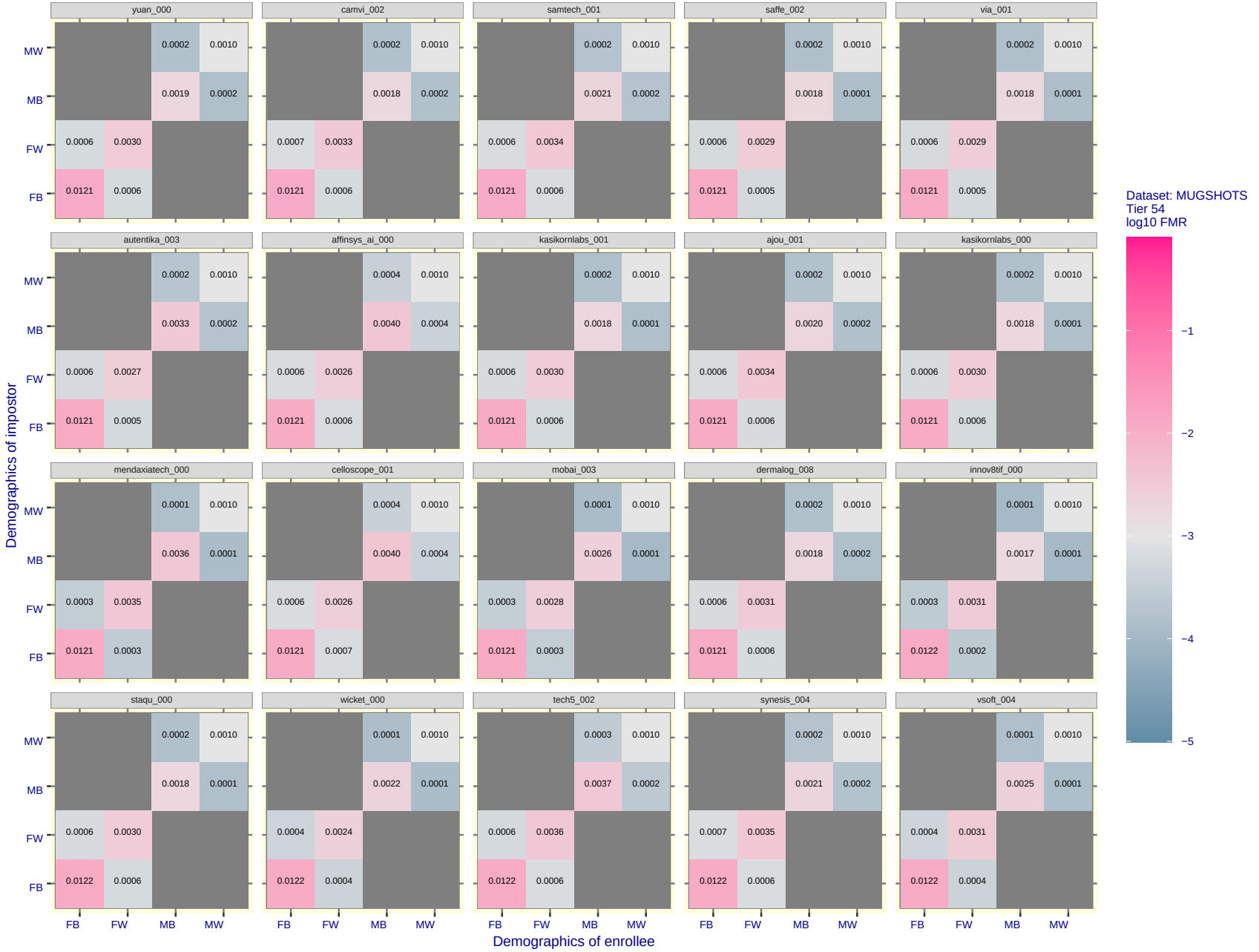

Figure 381: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

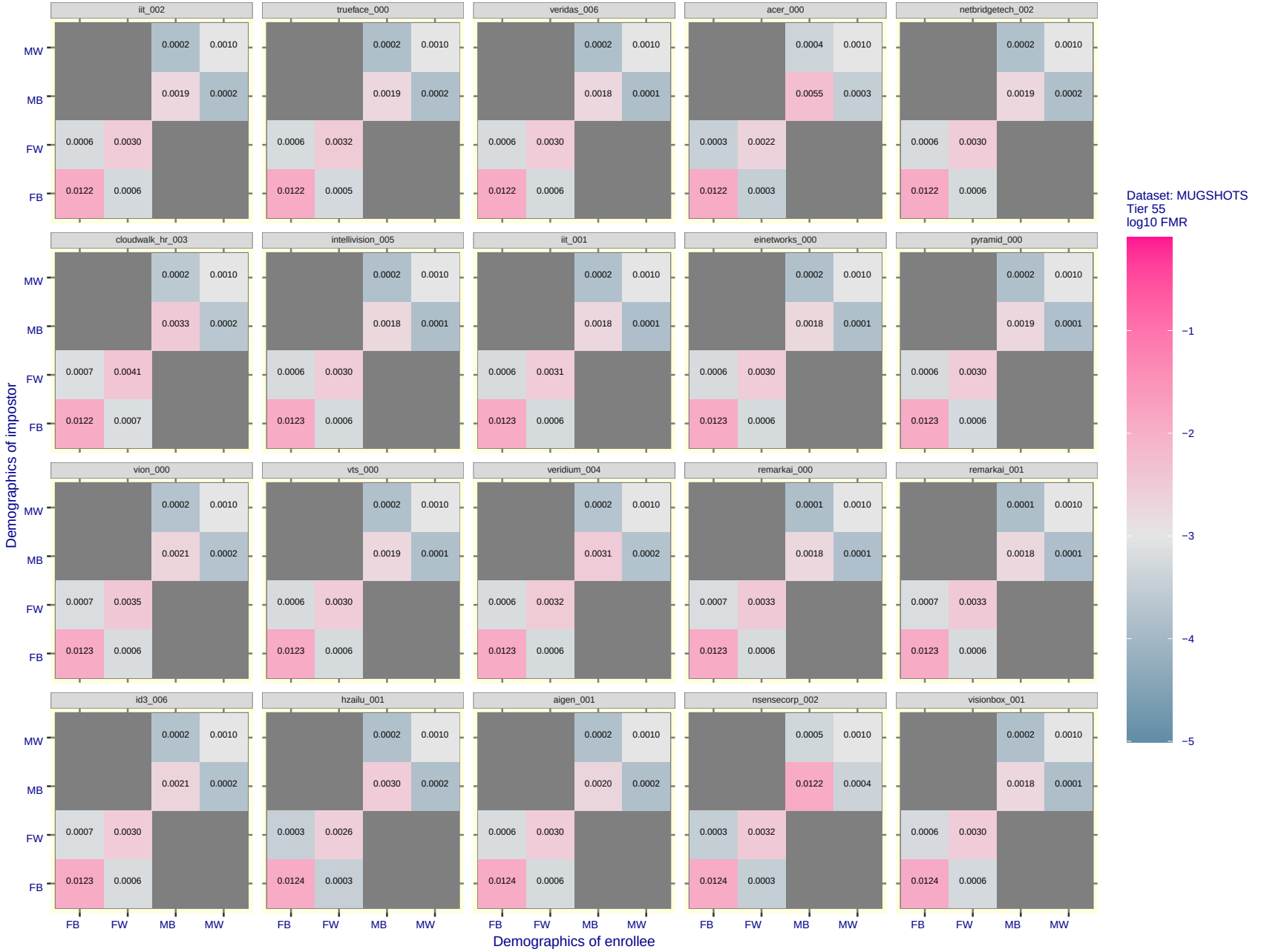

Figure 382: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

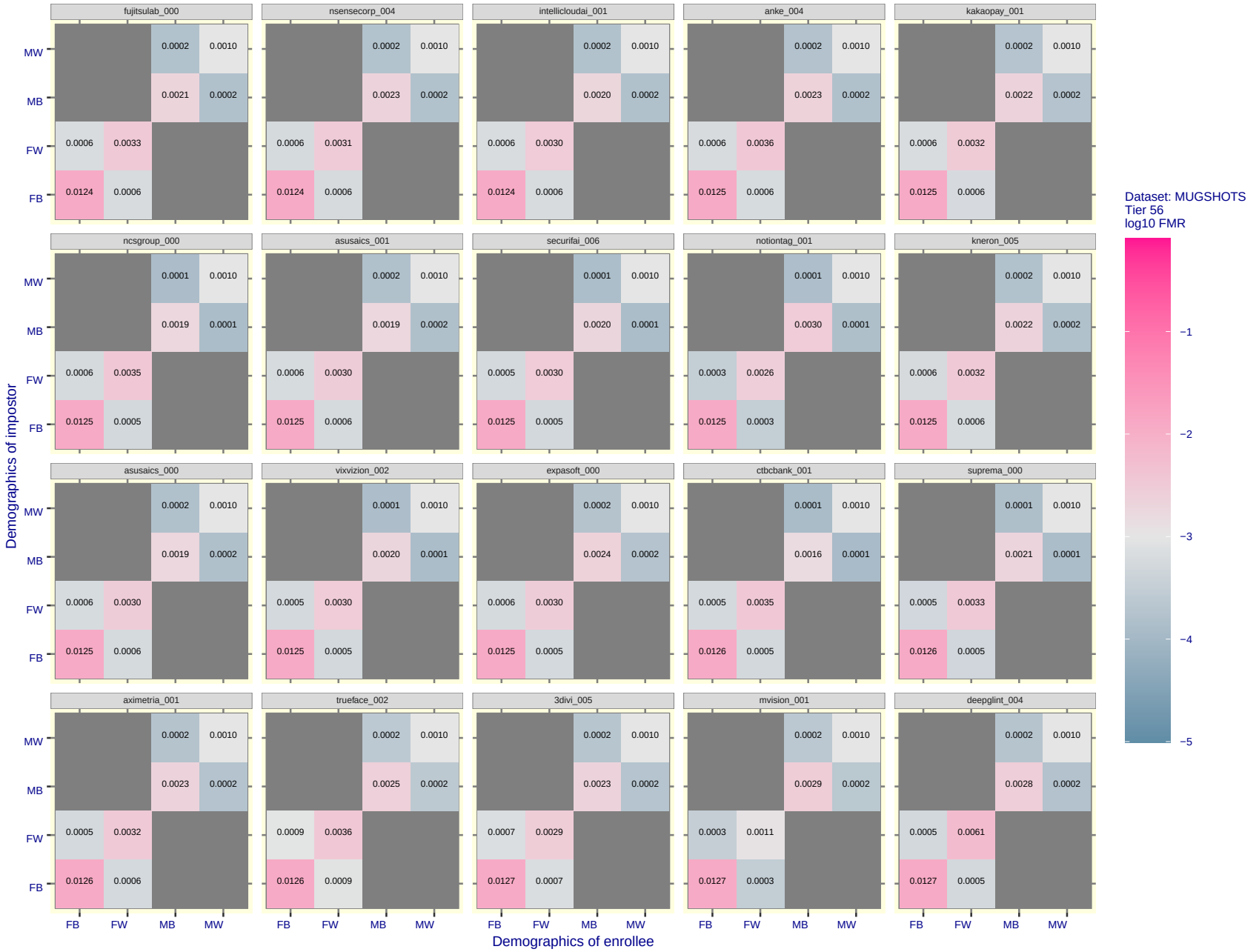

Figure 383: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

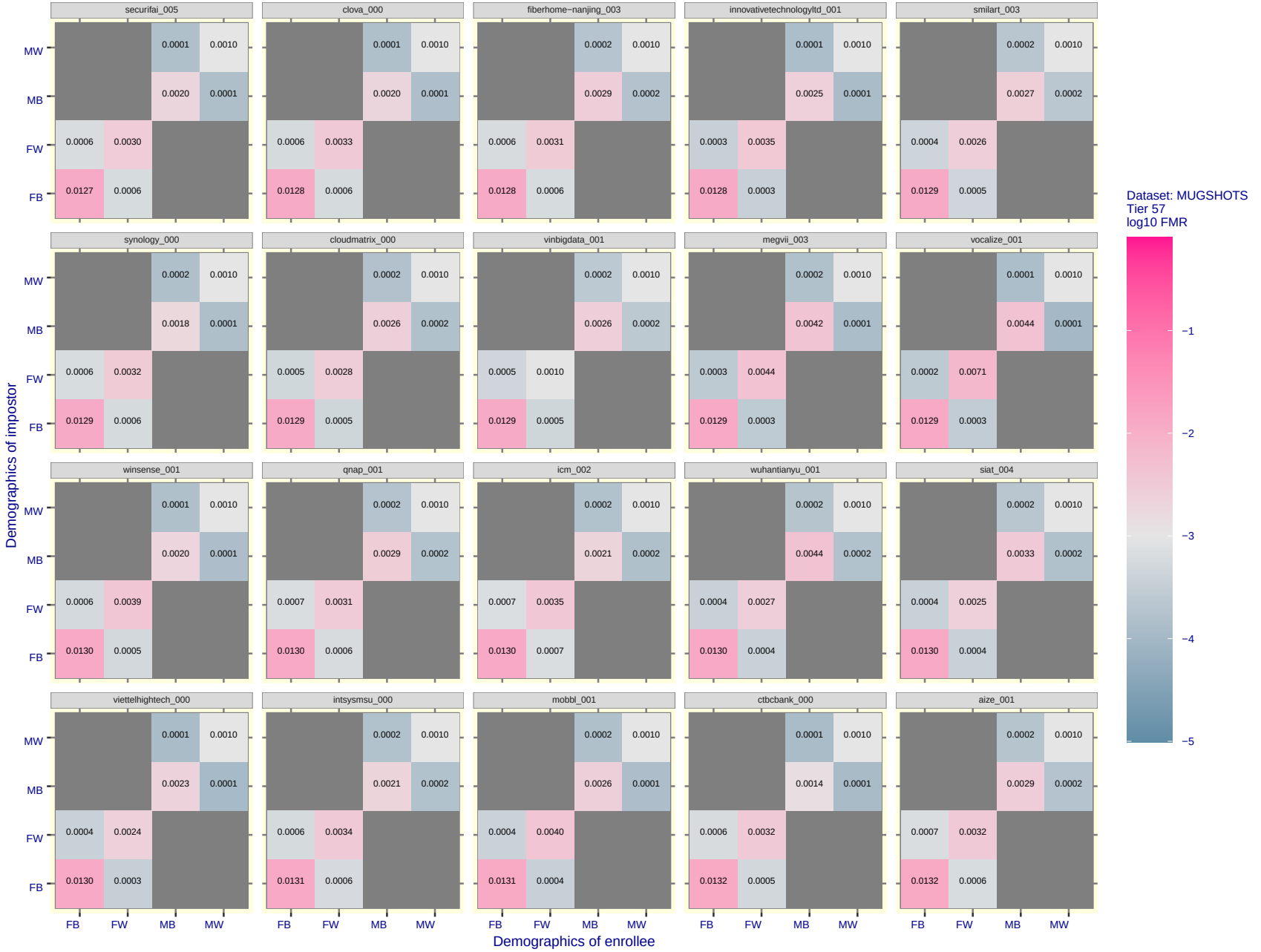

Figure 384: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

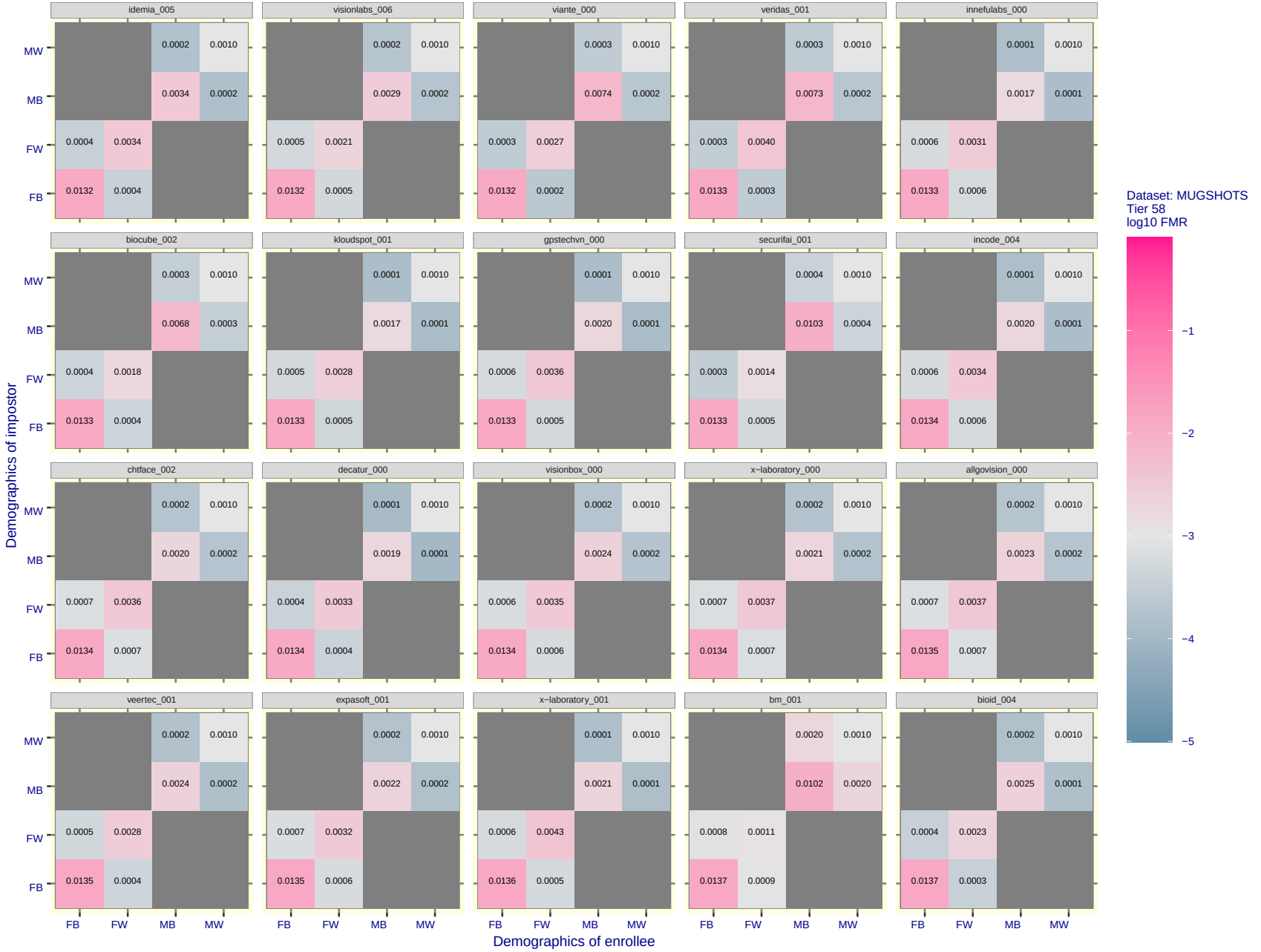

Figure 385: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

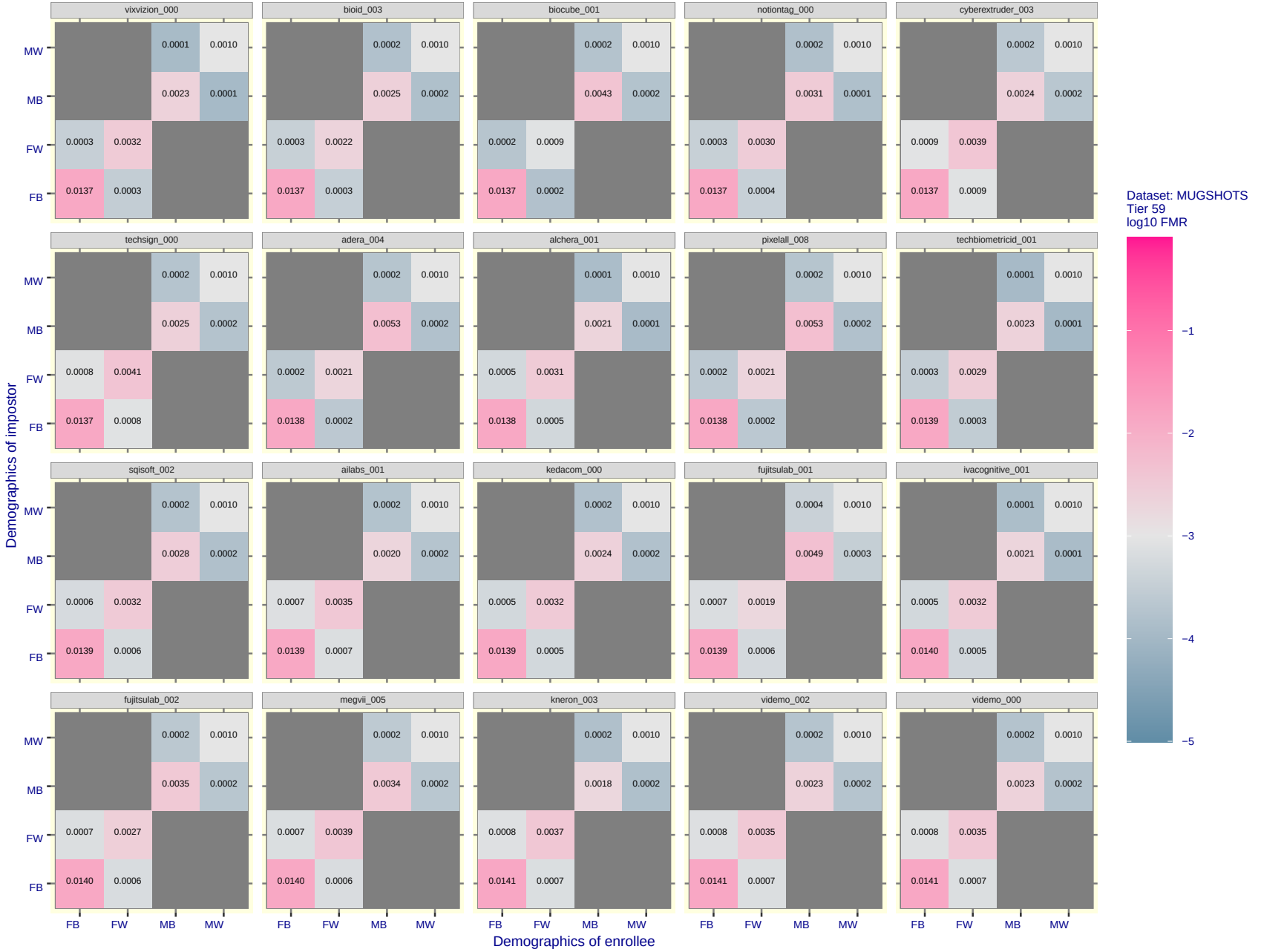

Figure 386: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

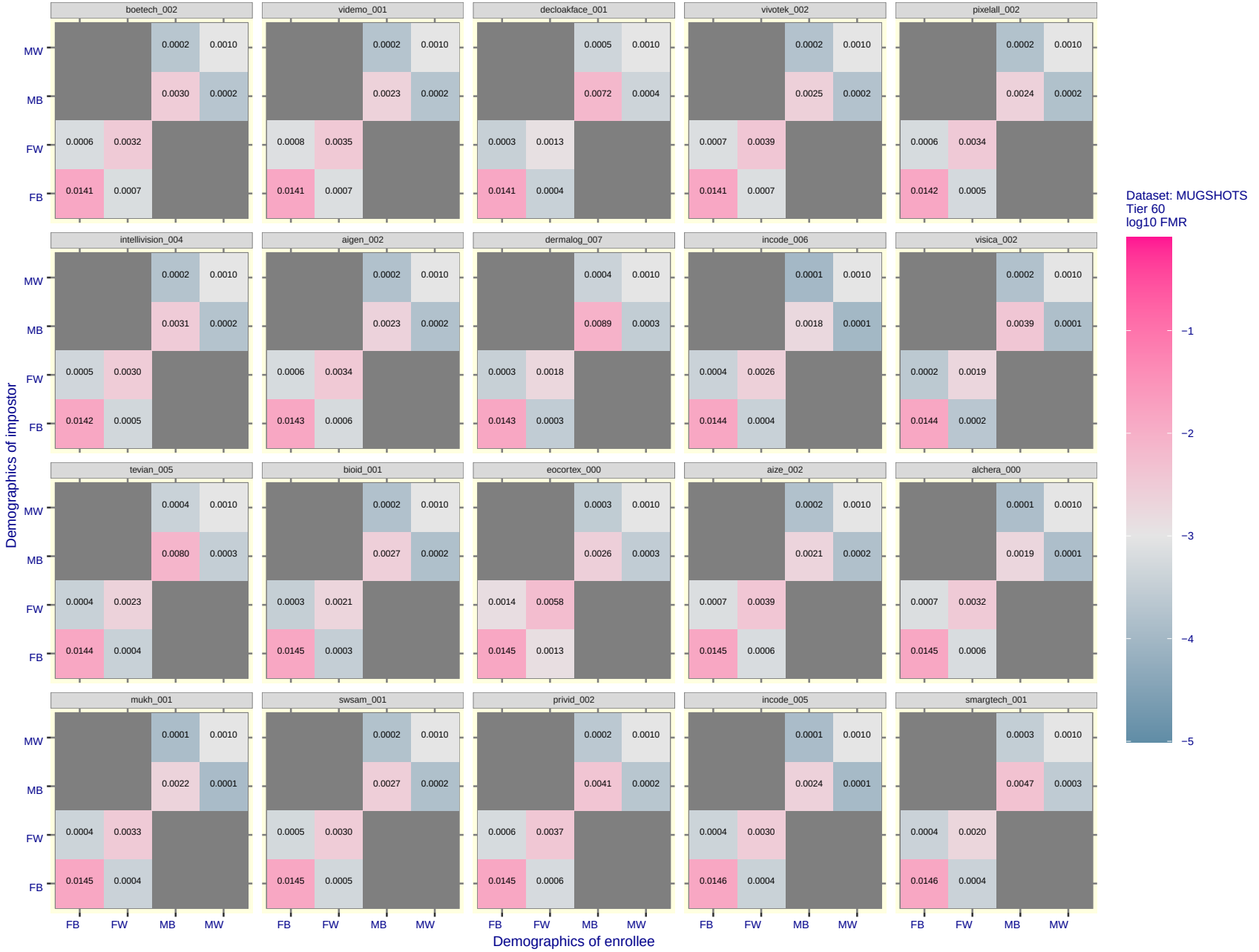

Figure 387: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

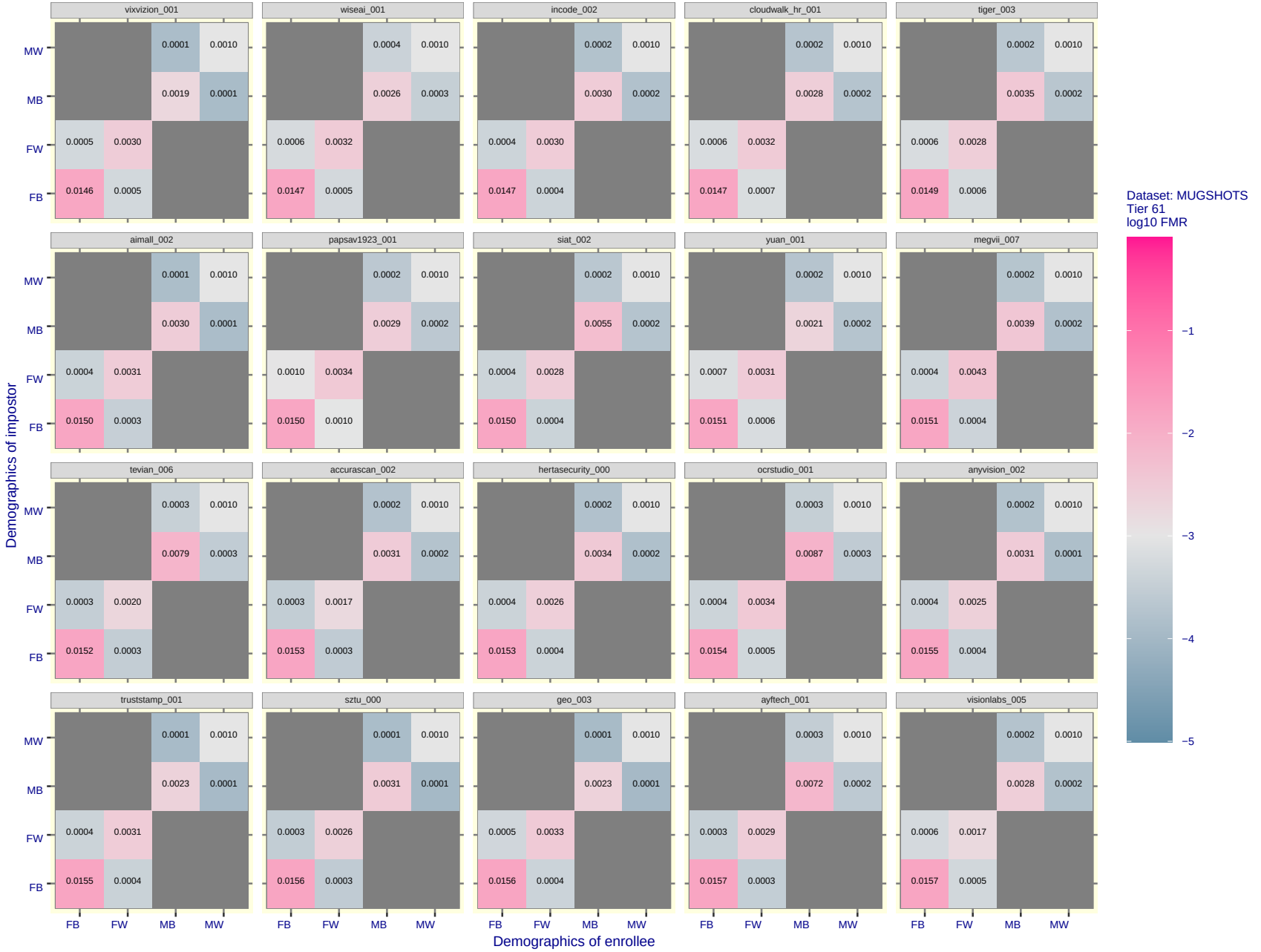

Figure 388: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

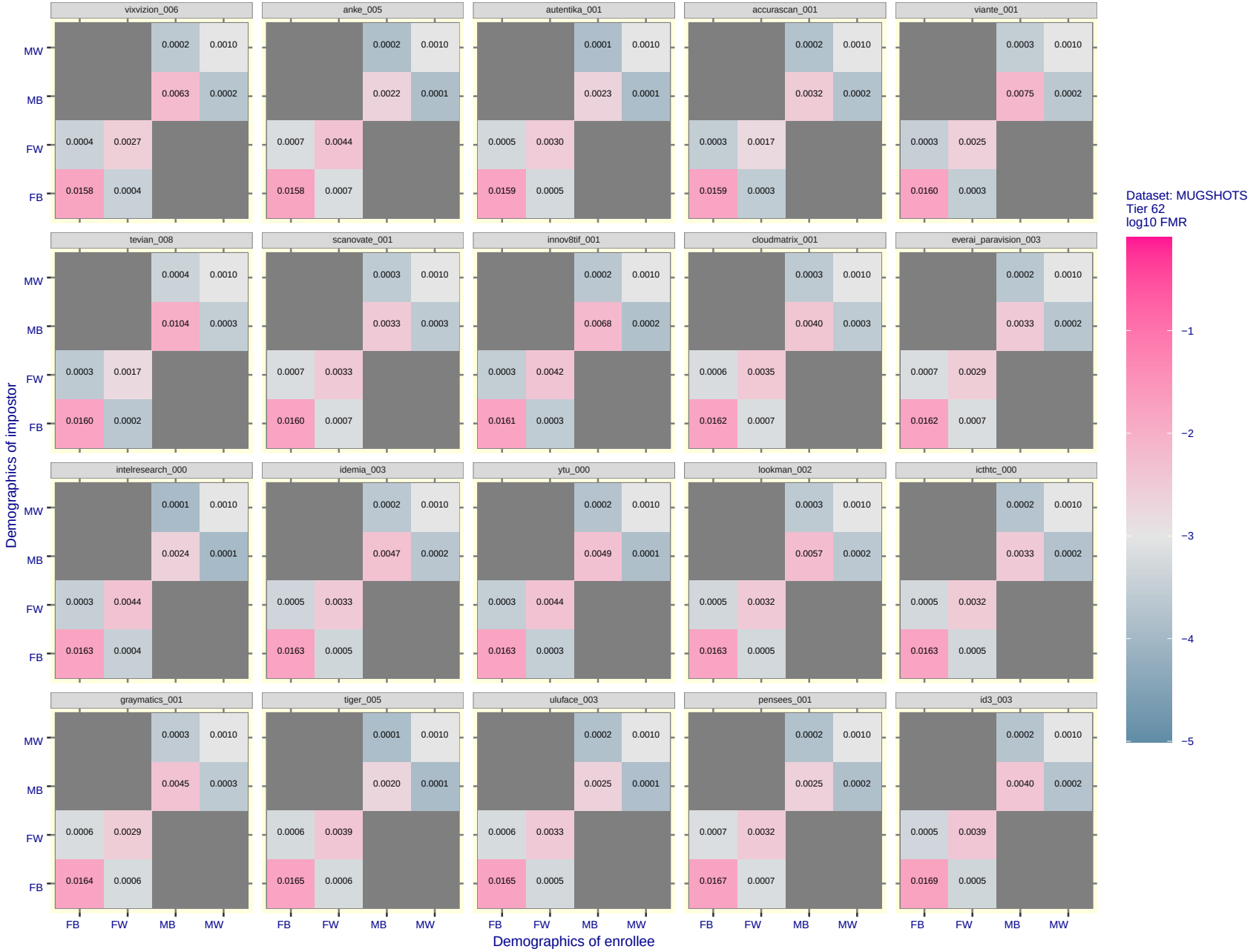

Figure 389: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

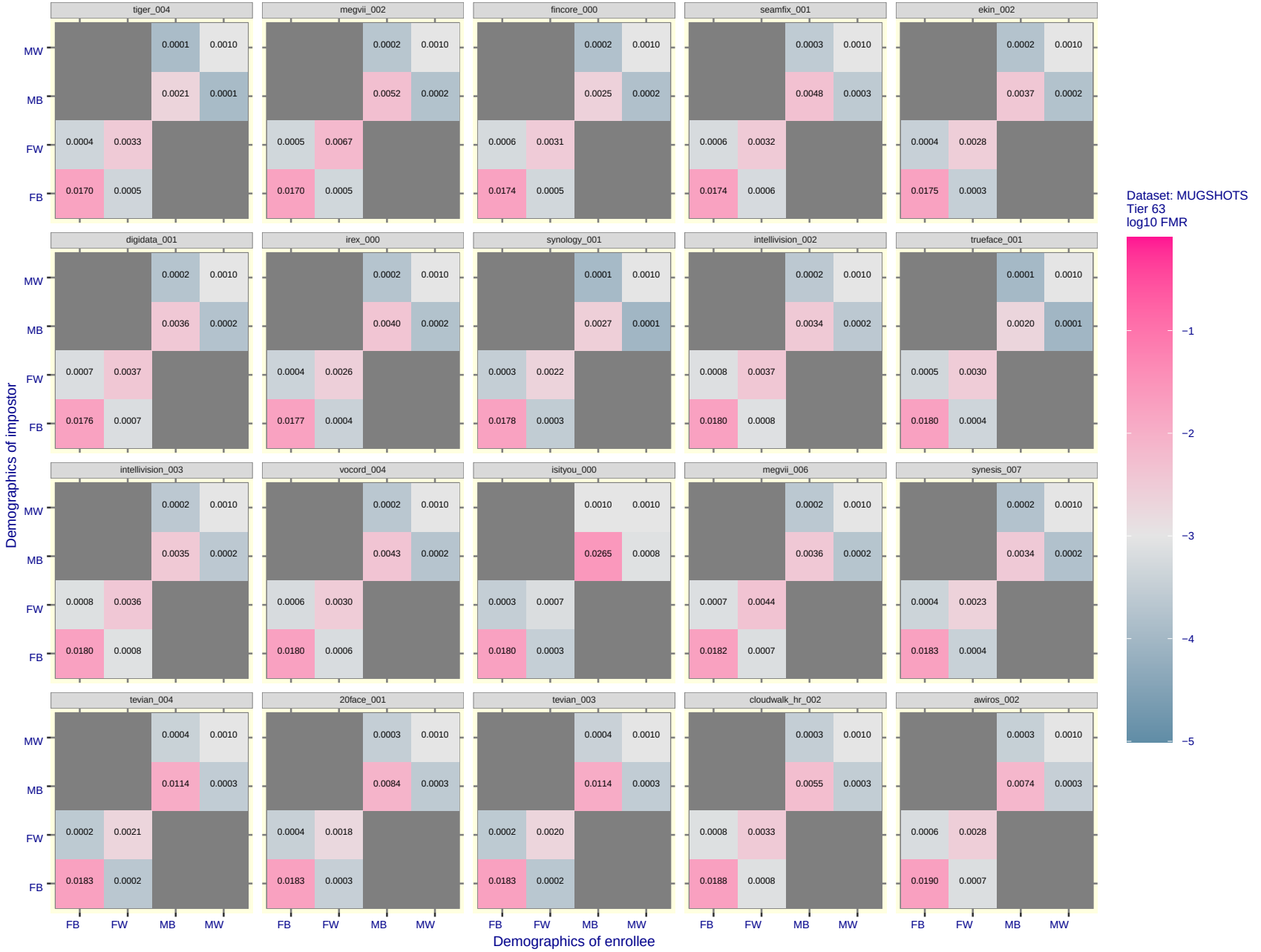

Figure 390: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

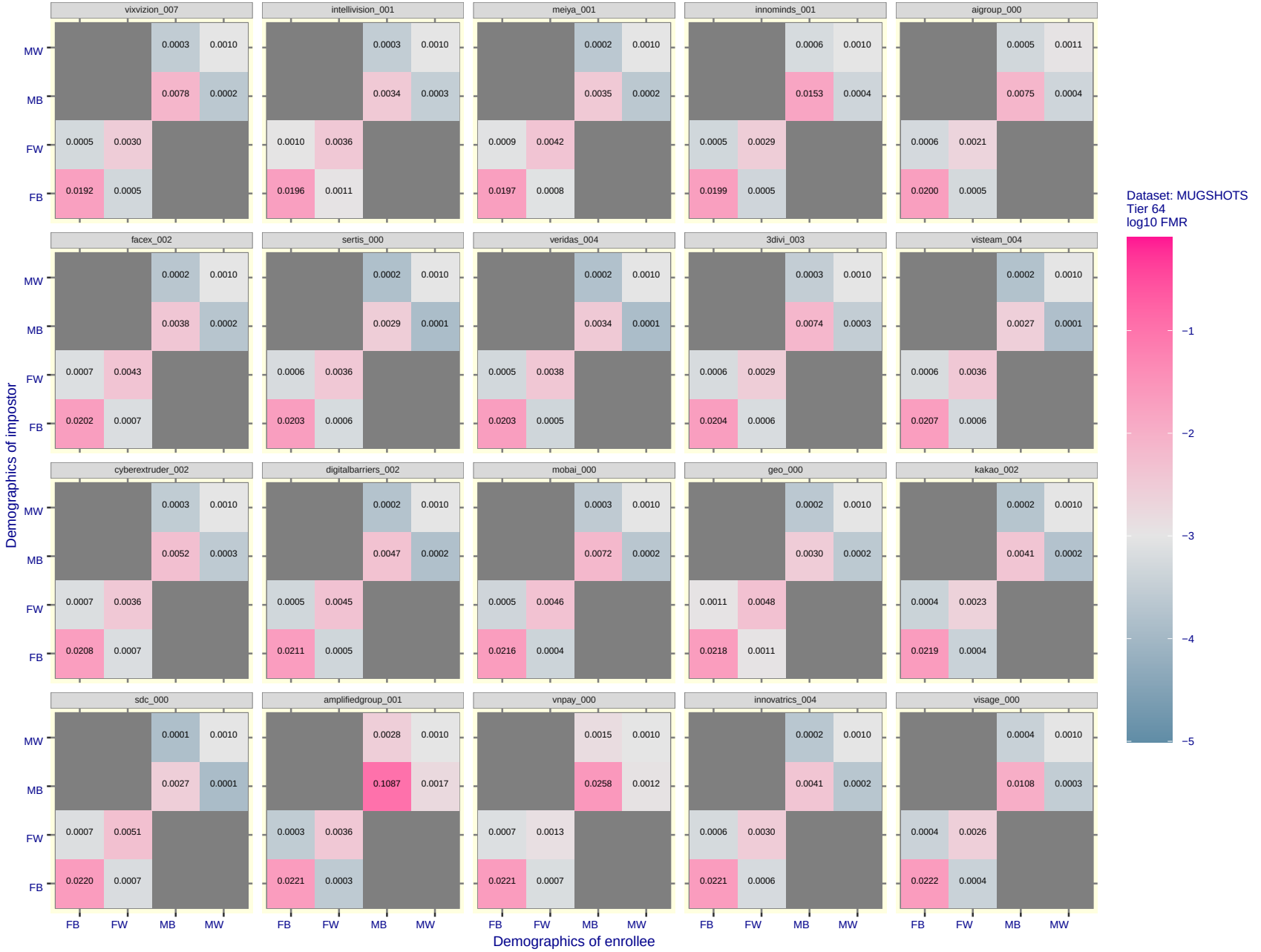

Figure 391: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

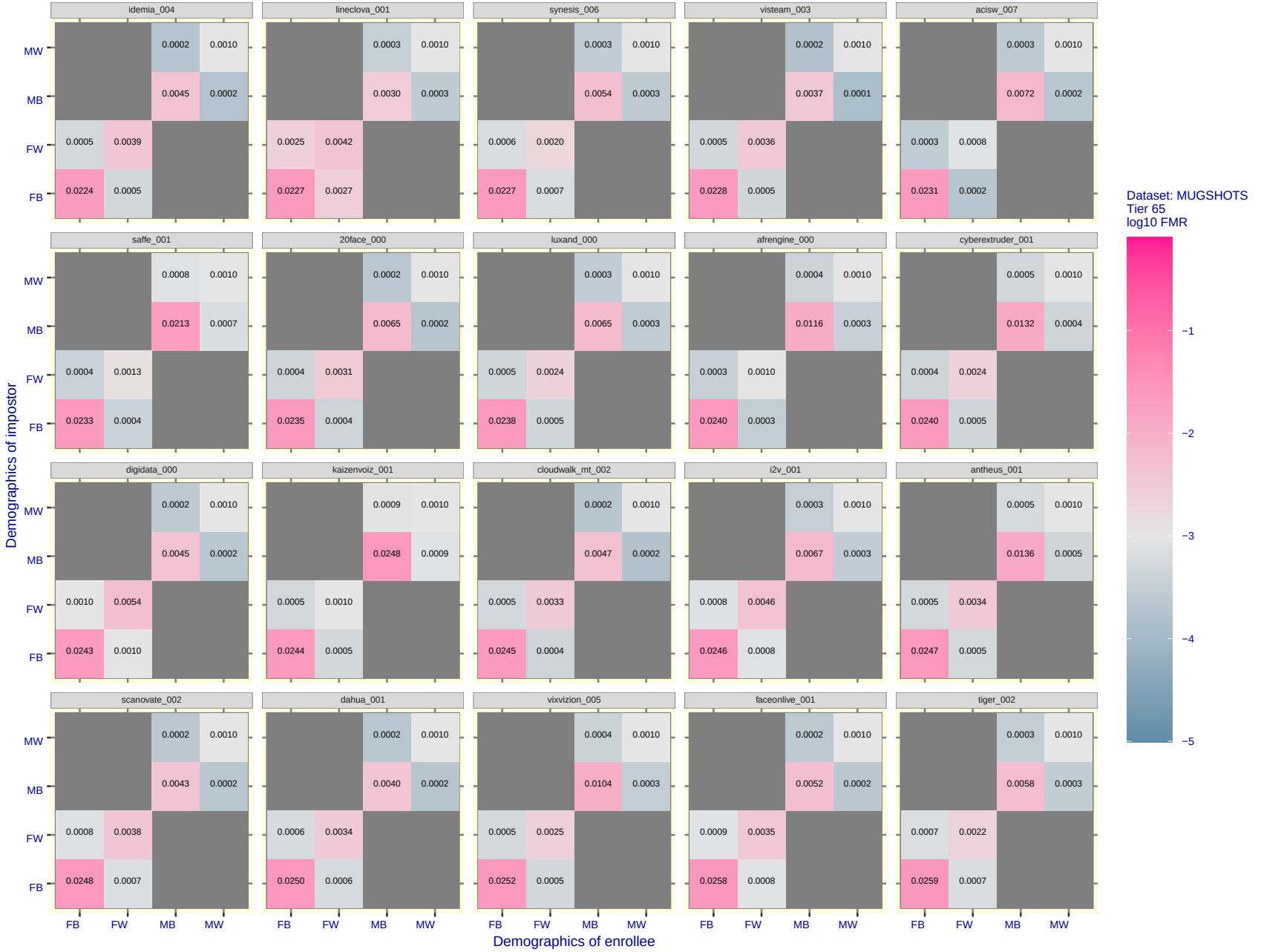

Figure 392: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

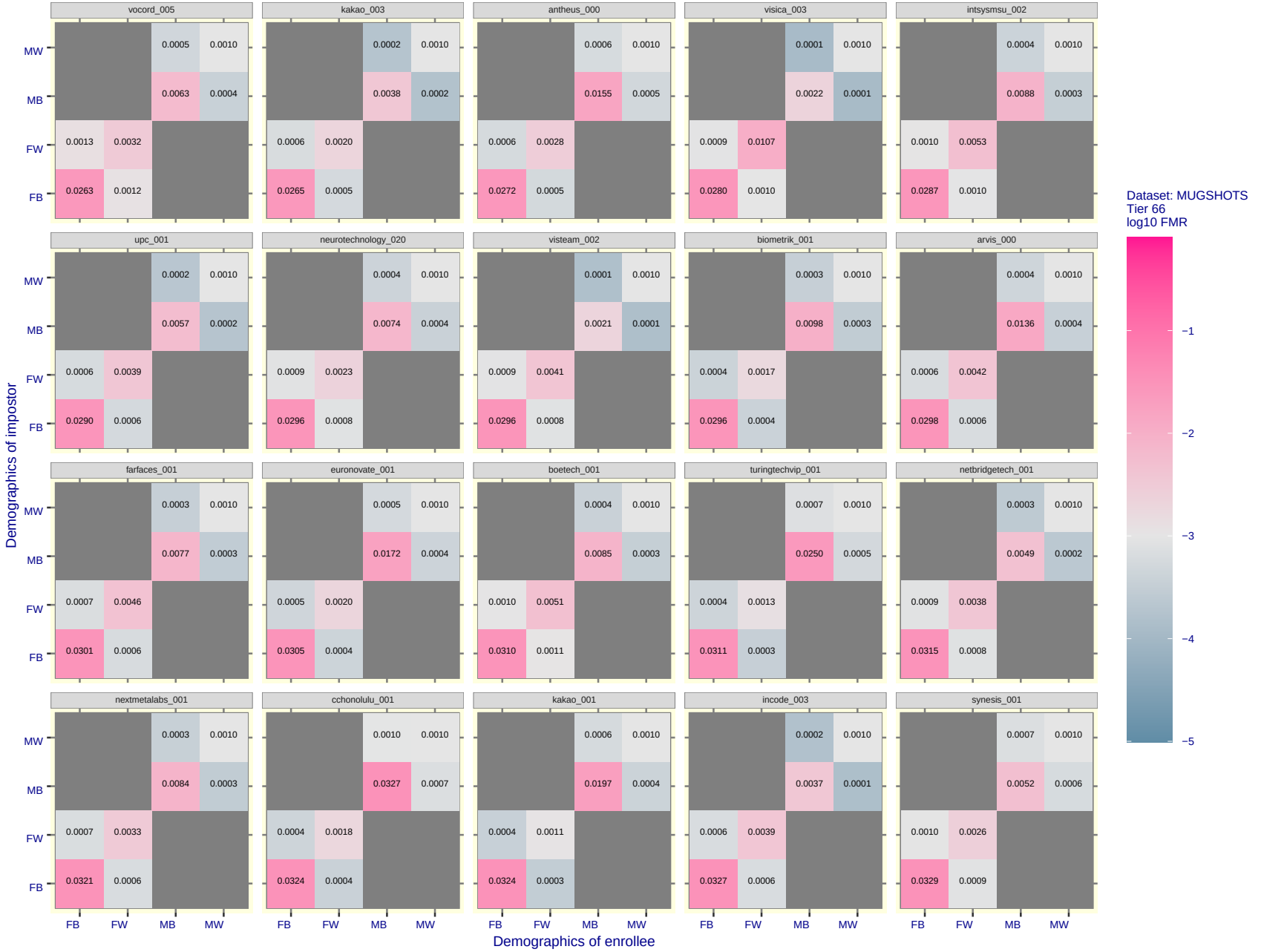

Figure 393: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

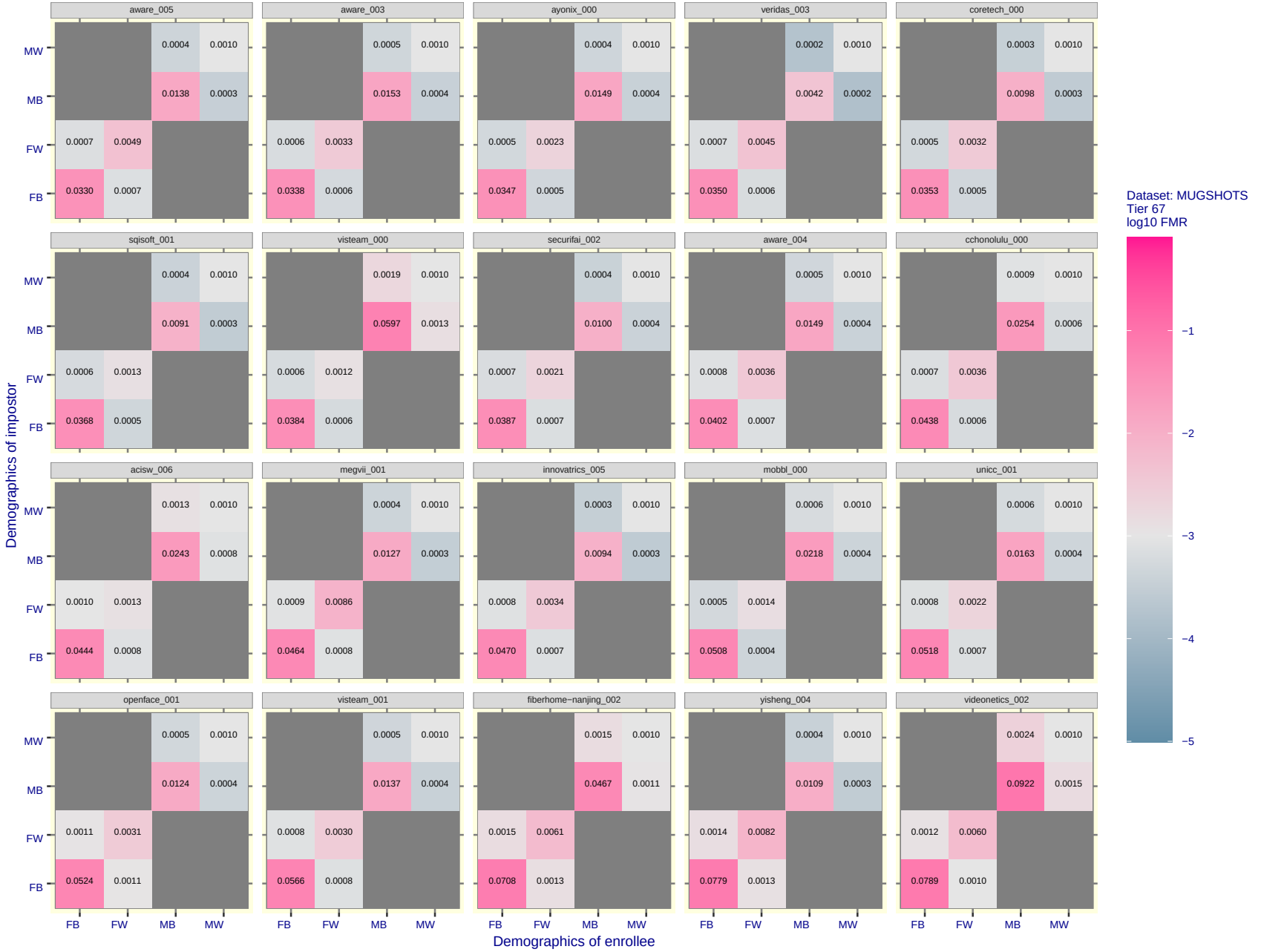

Figure 394: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 395: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

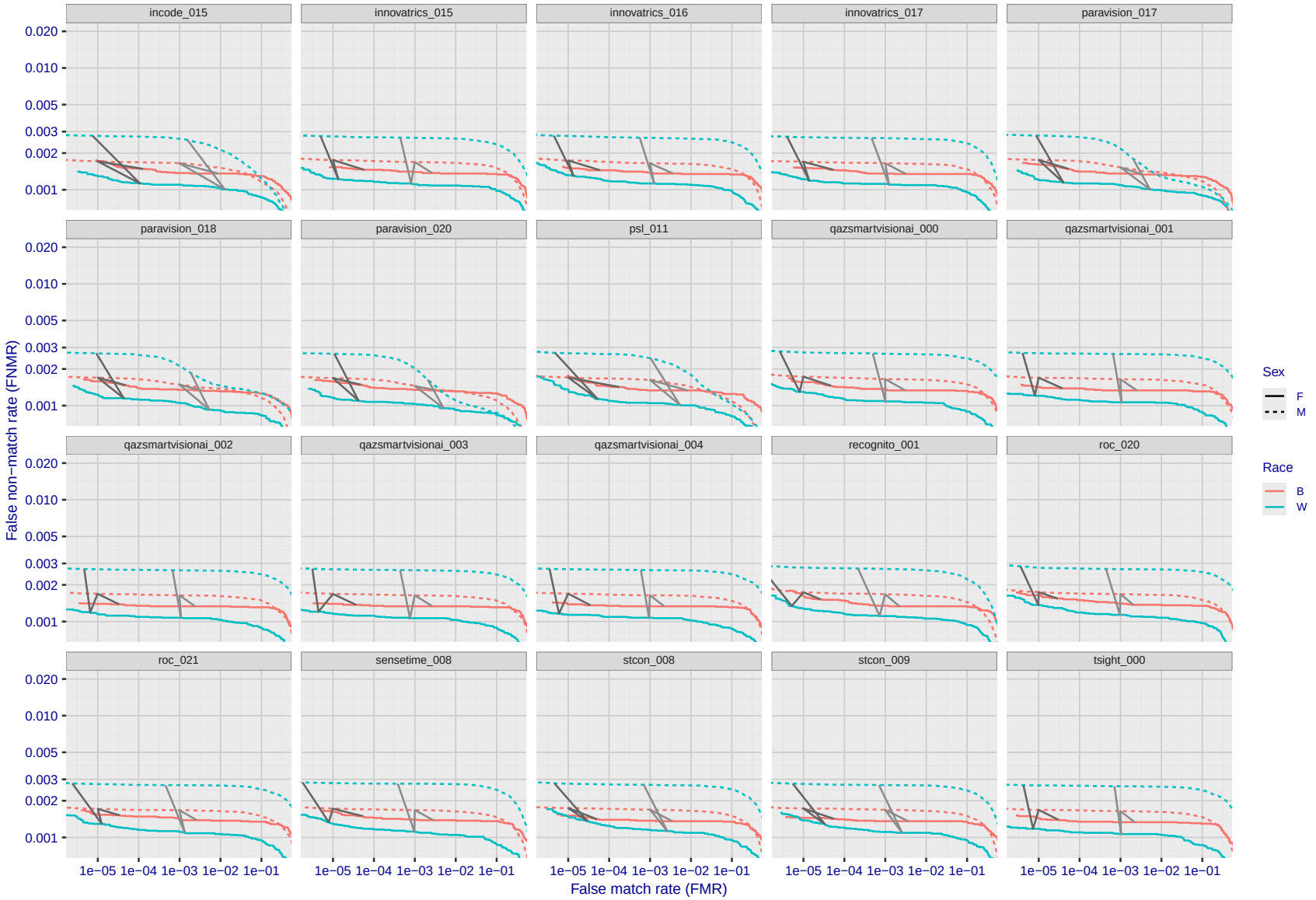


Figure 396: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

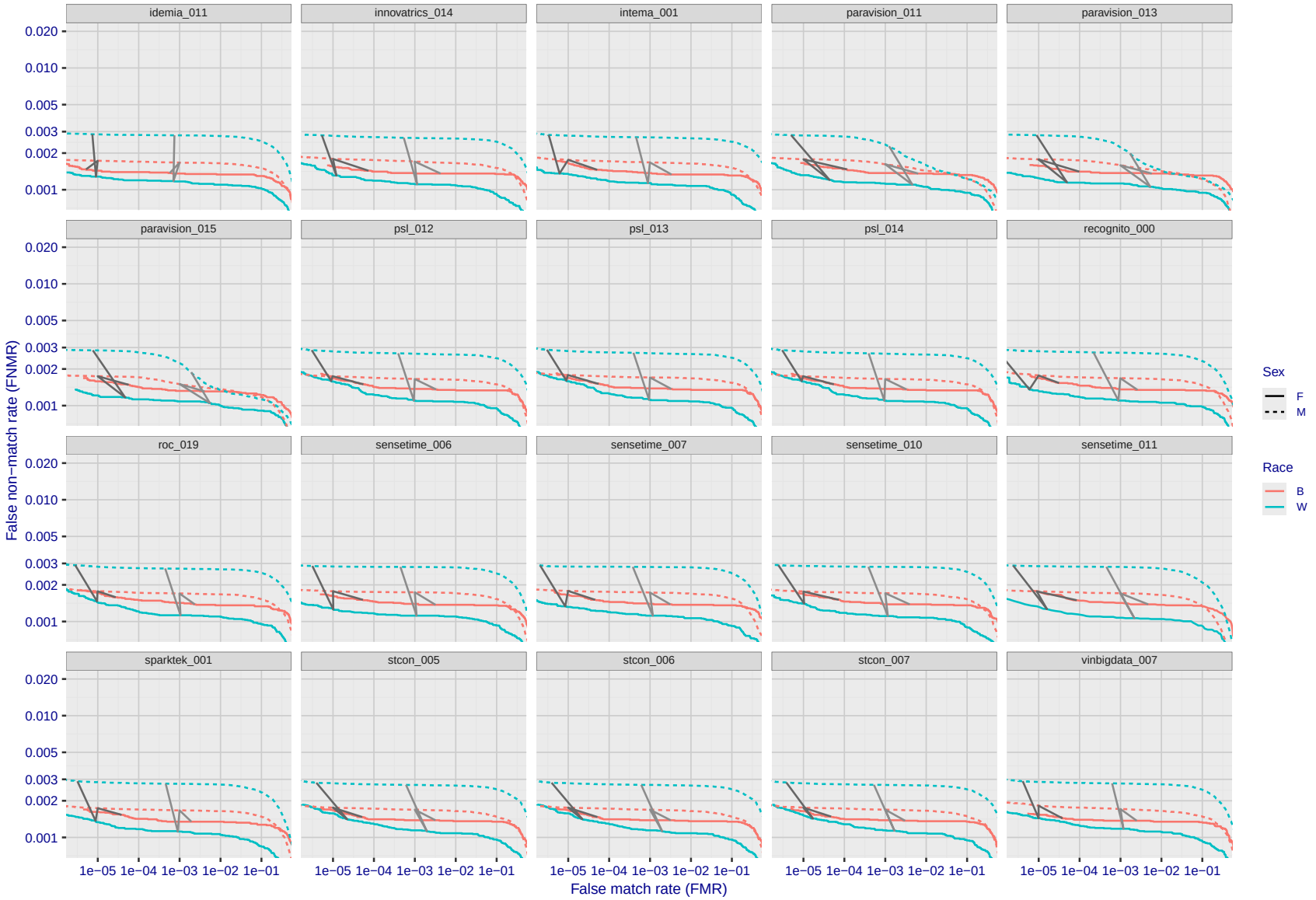


Figure 397: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

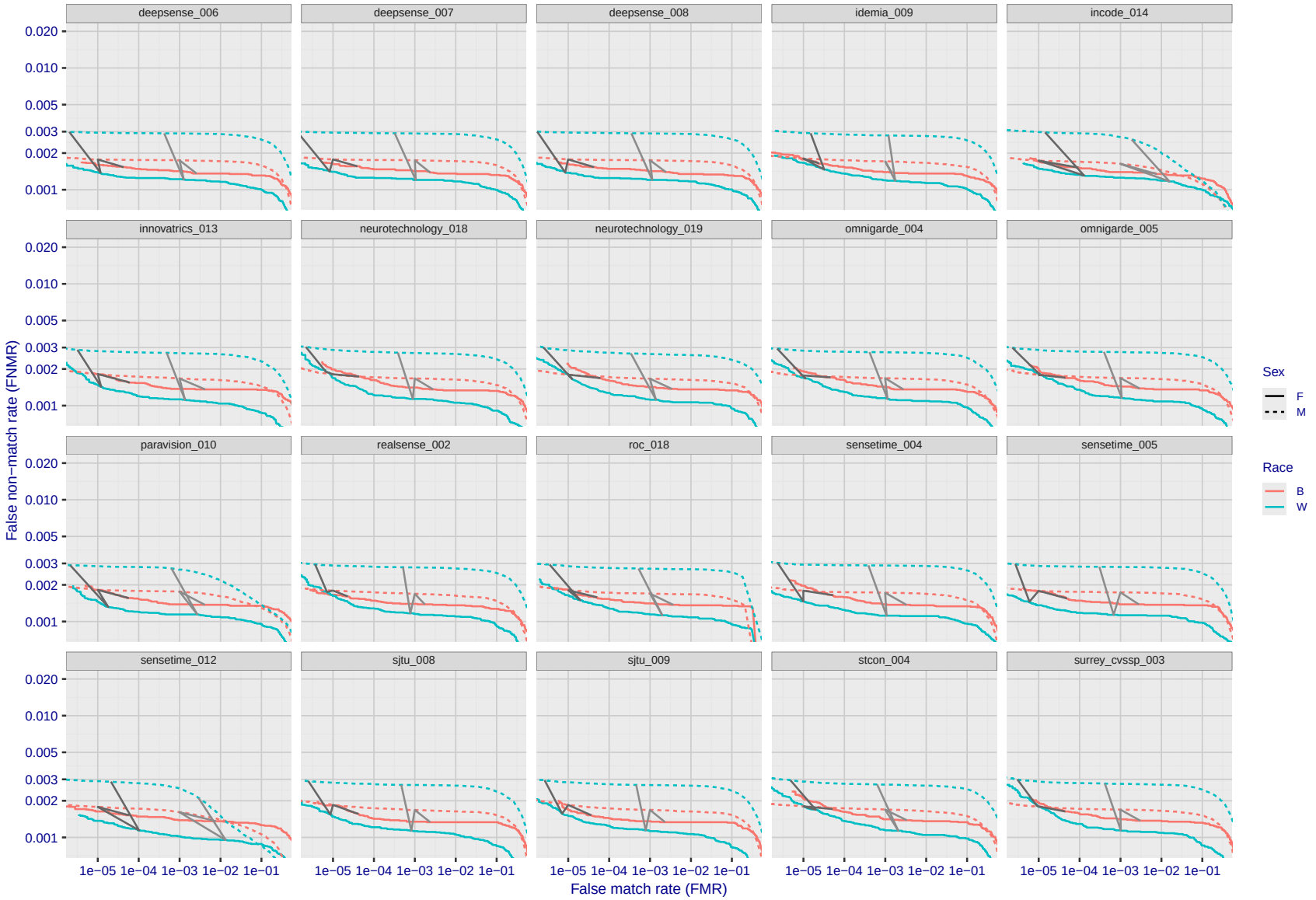


Figure 398: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

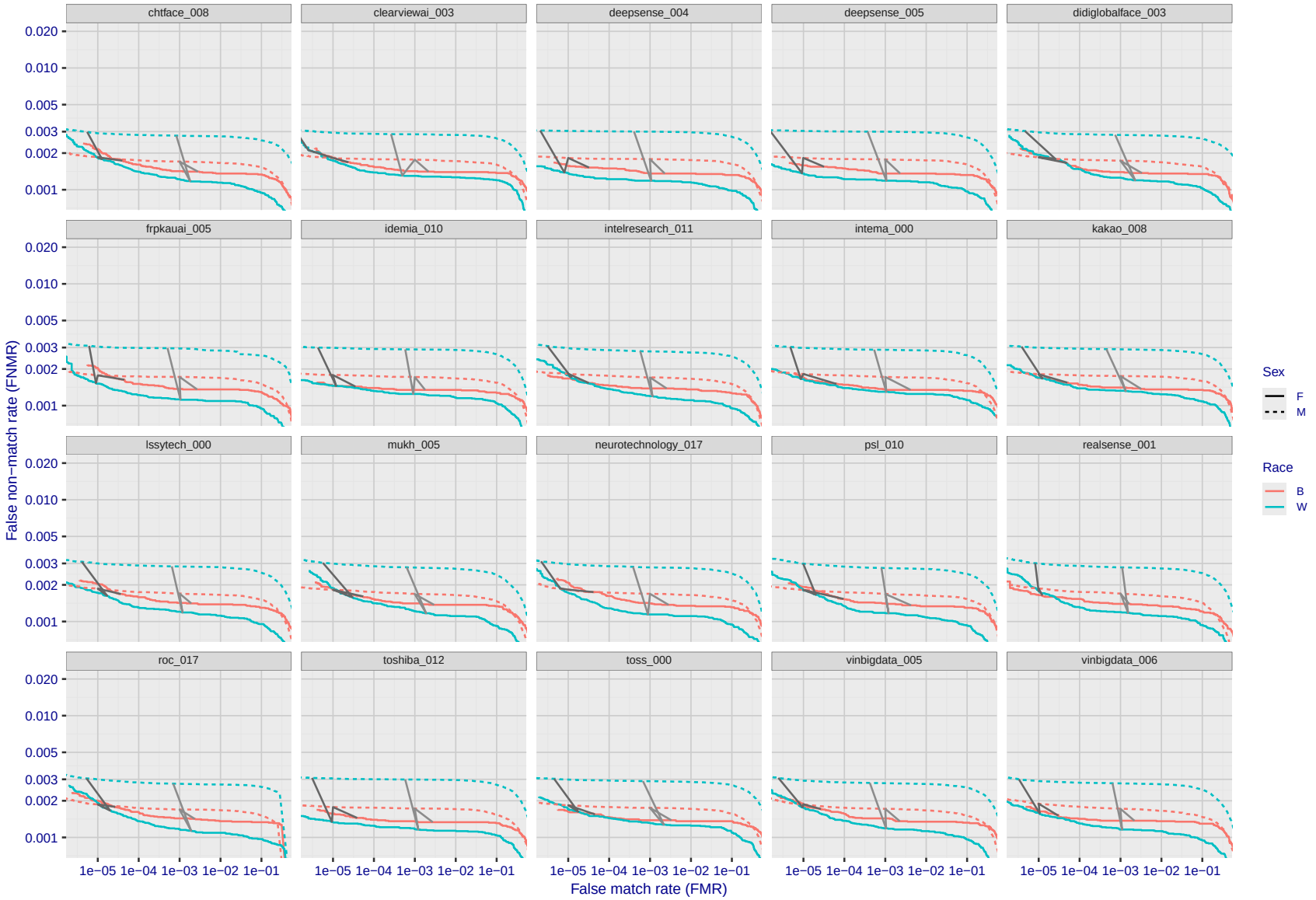


Figure 399: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

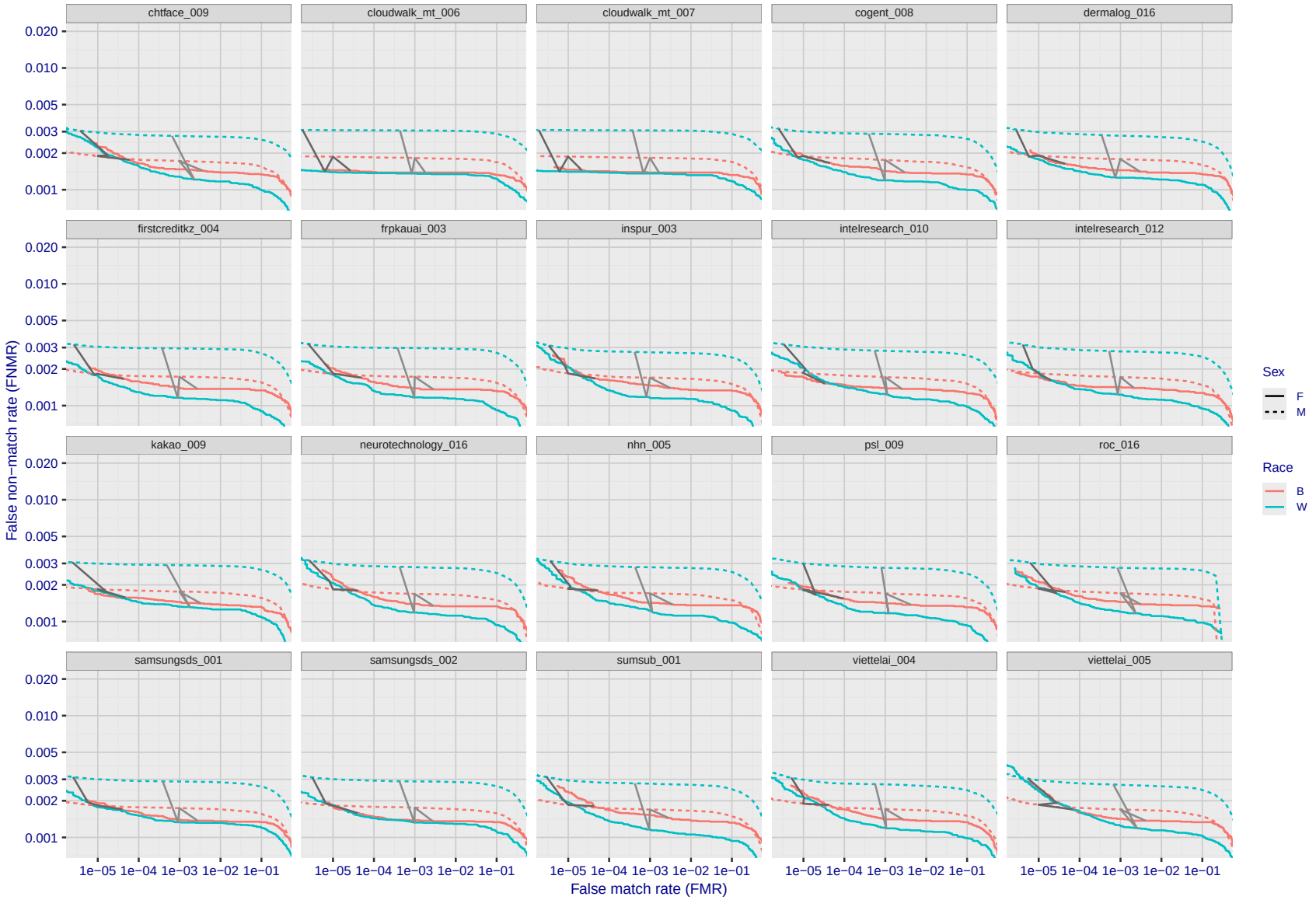


Figure 400: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

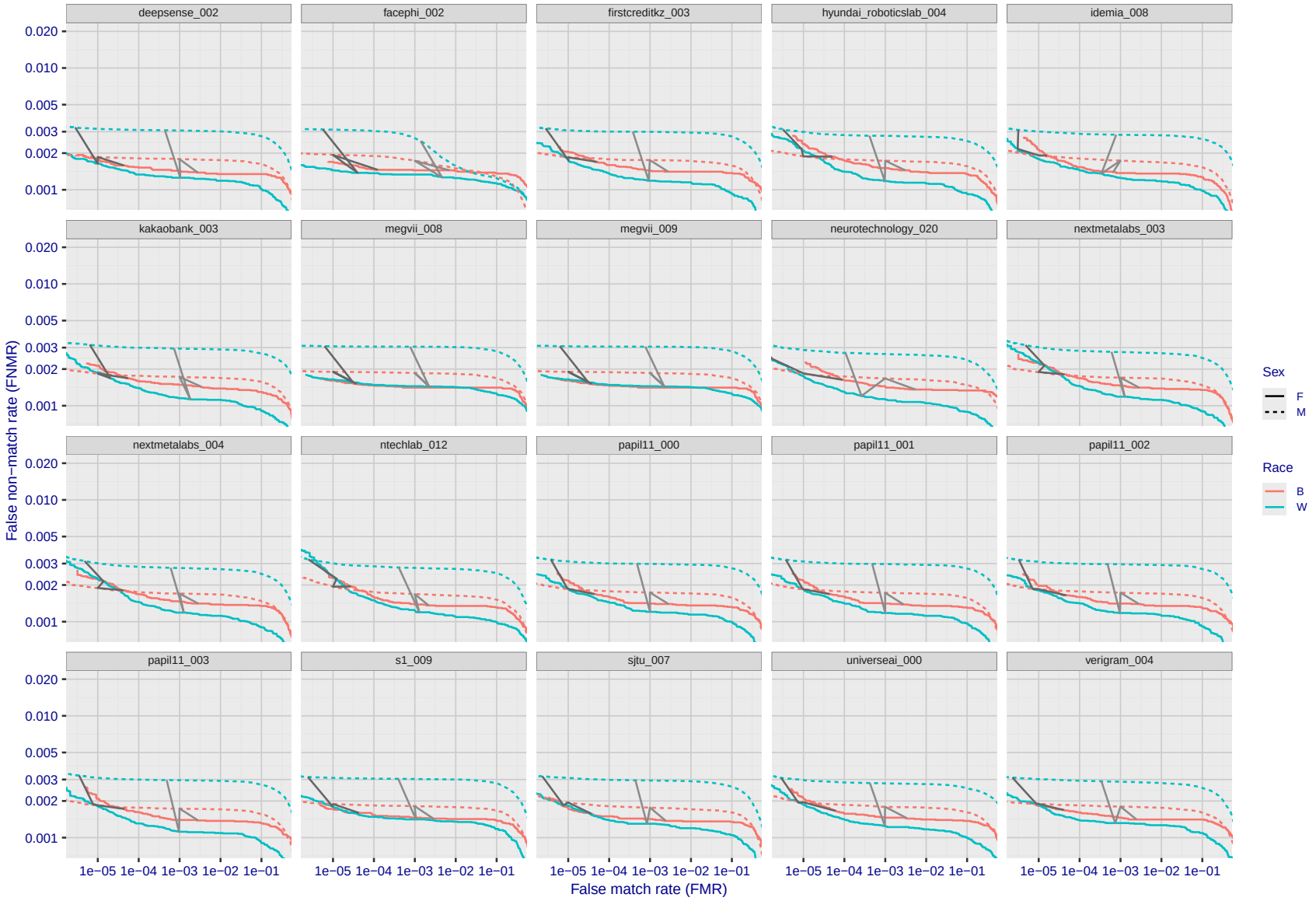


Figure 401: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

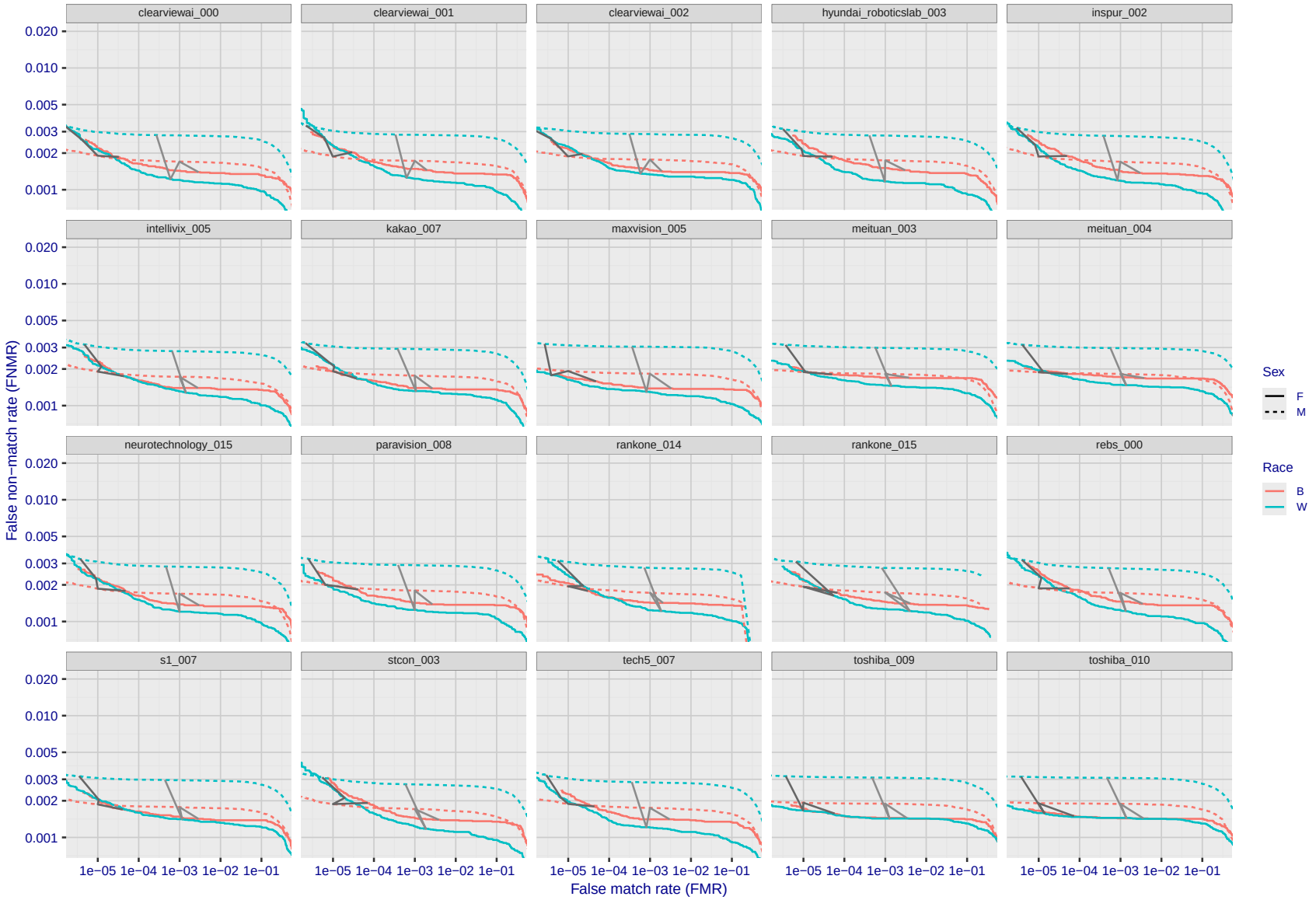


Figure 402: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

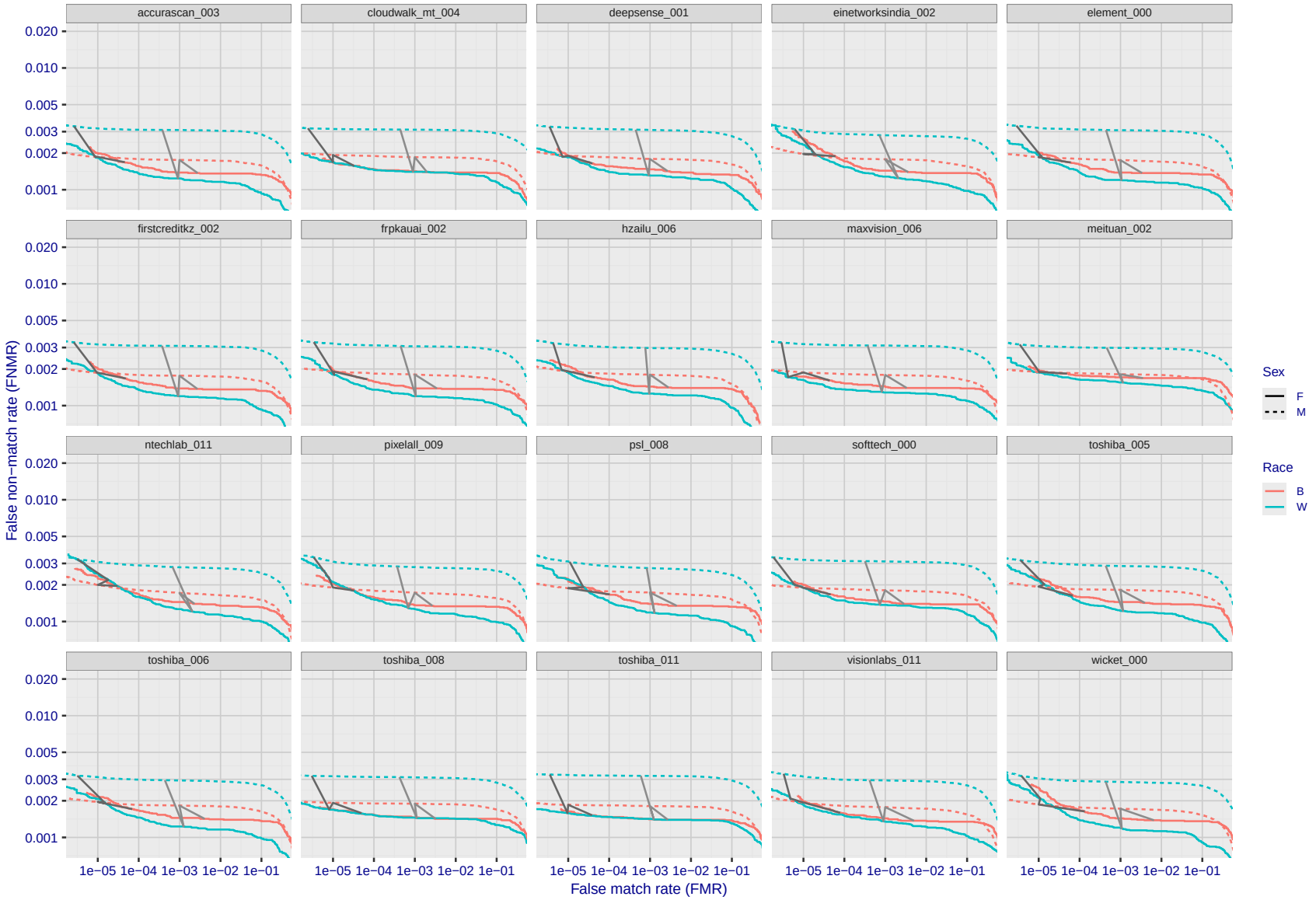


Figure 403: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

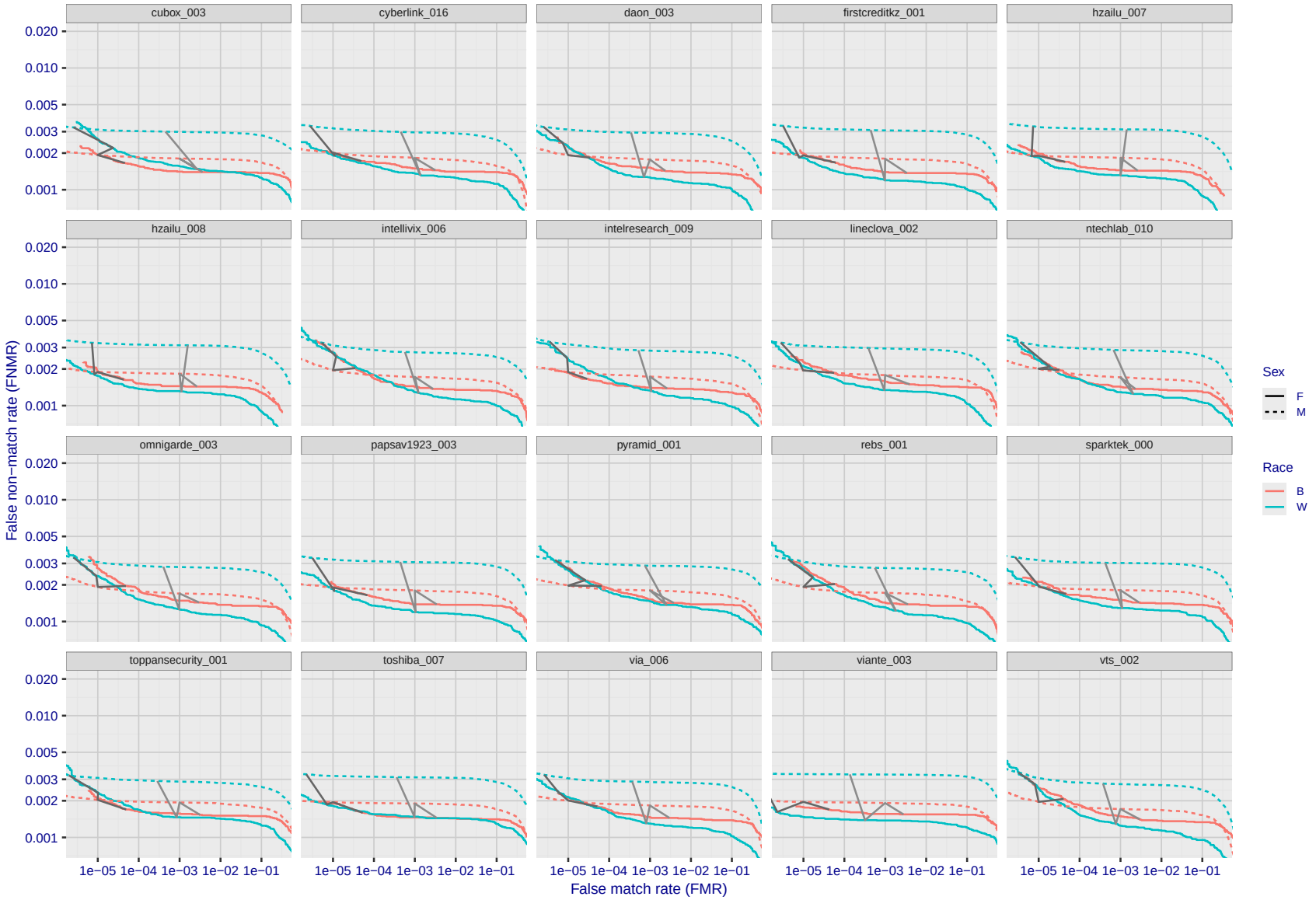


Figure 404: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

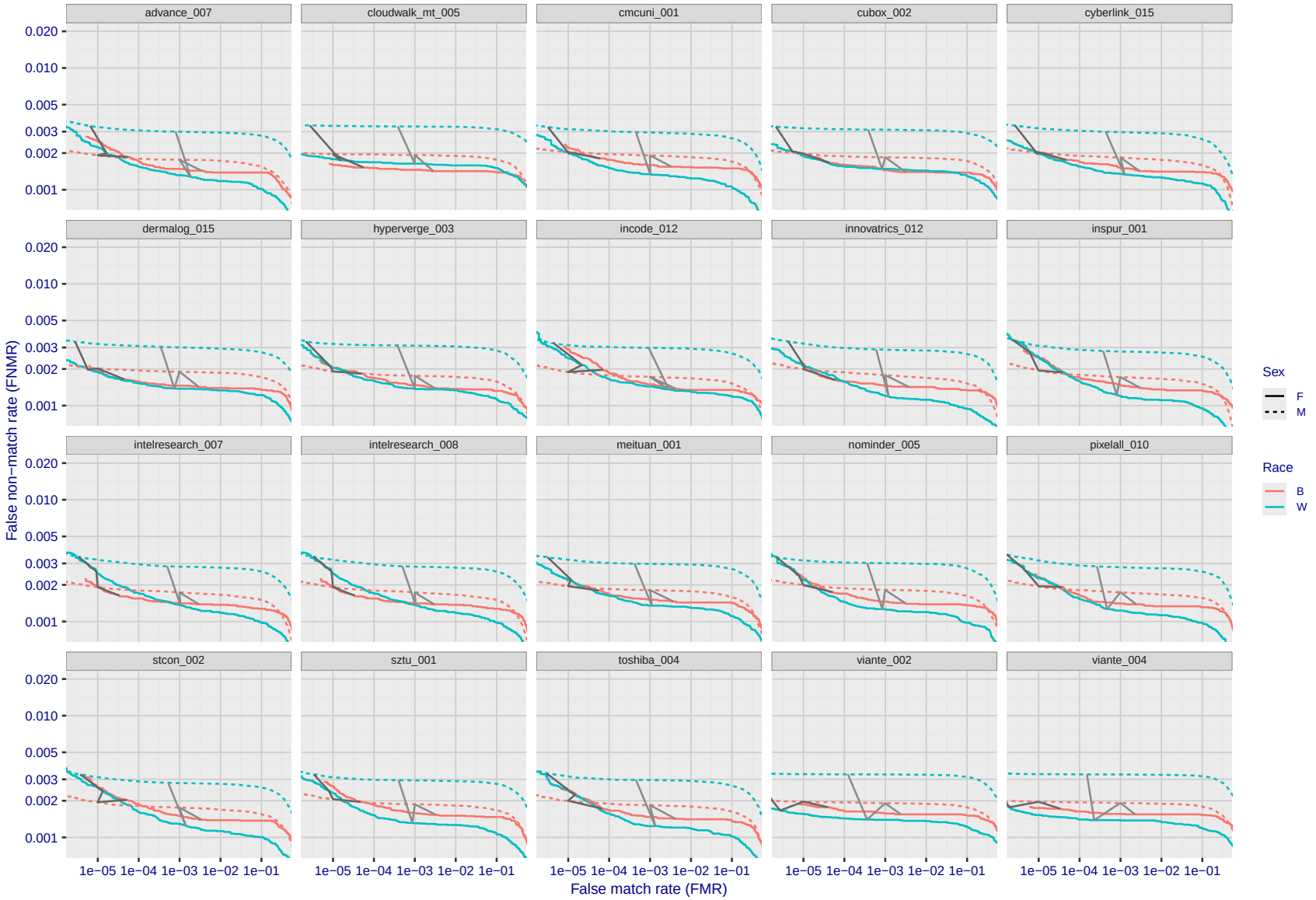


Figure 405: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

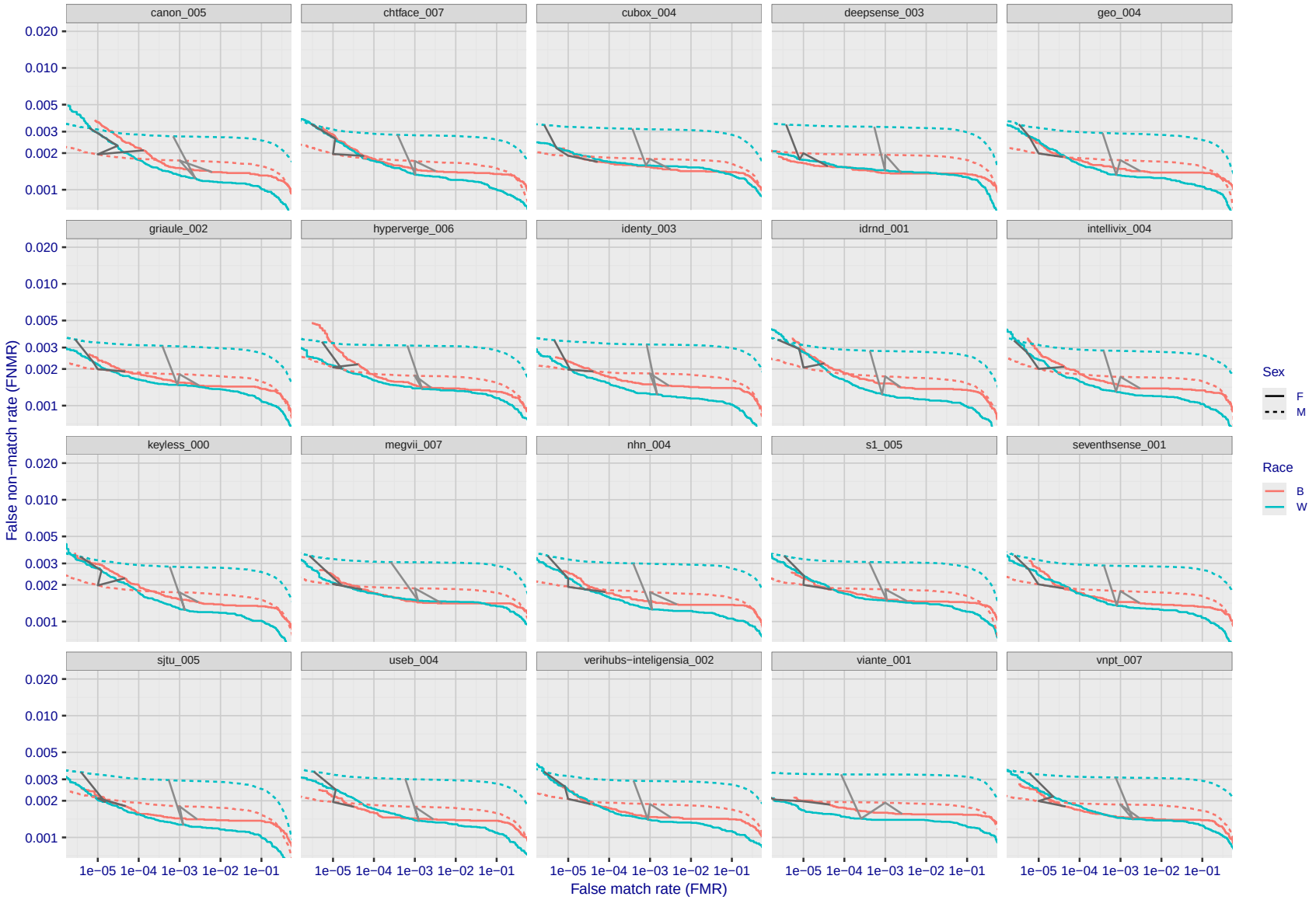


Figure 406: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

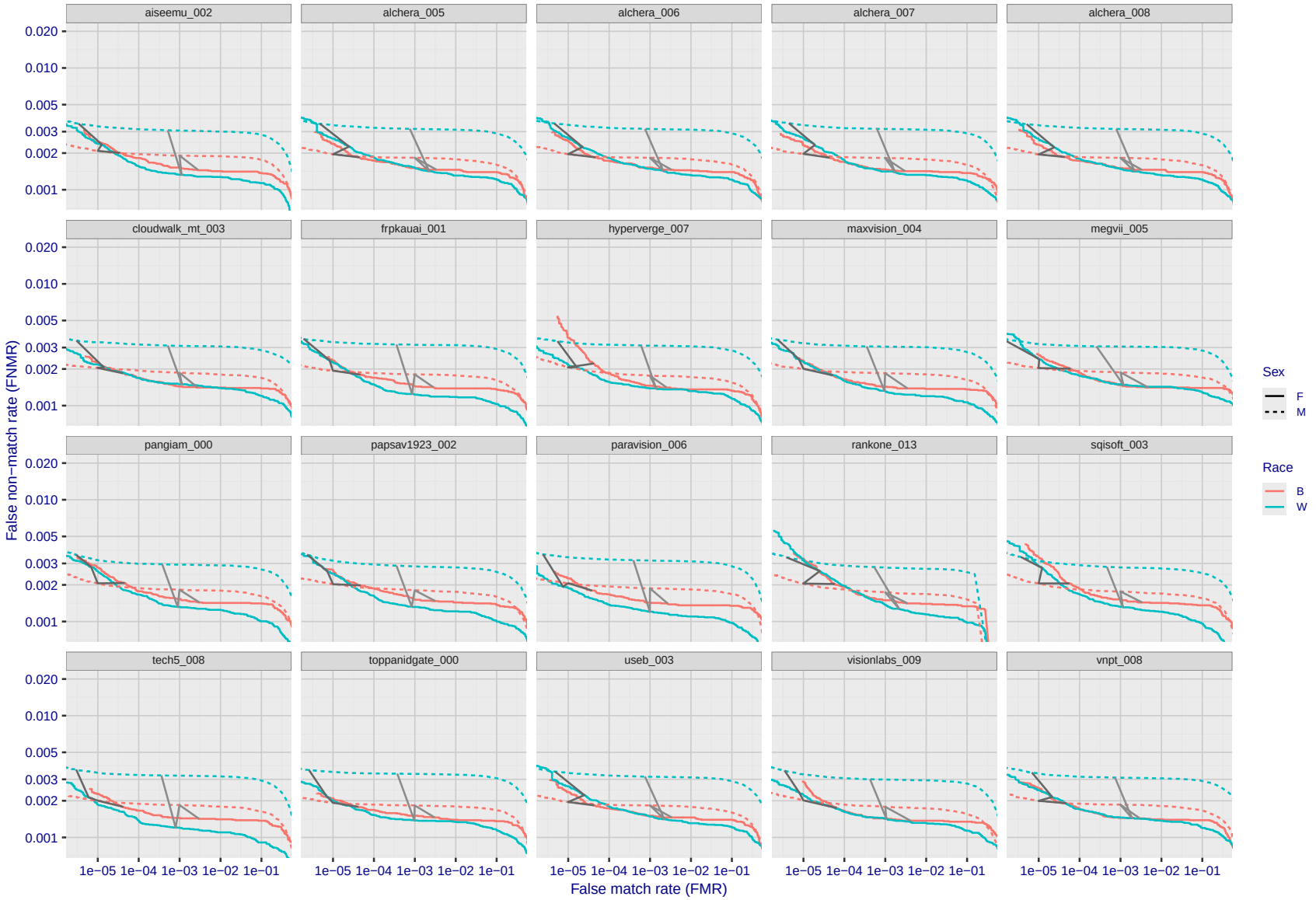


Figure 407: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

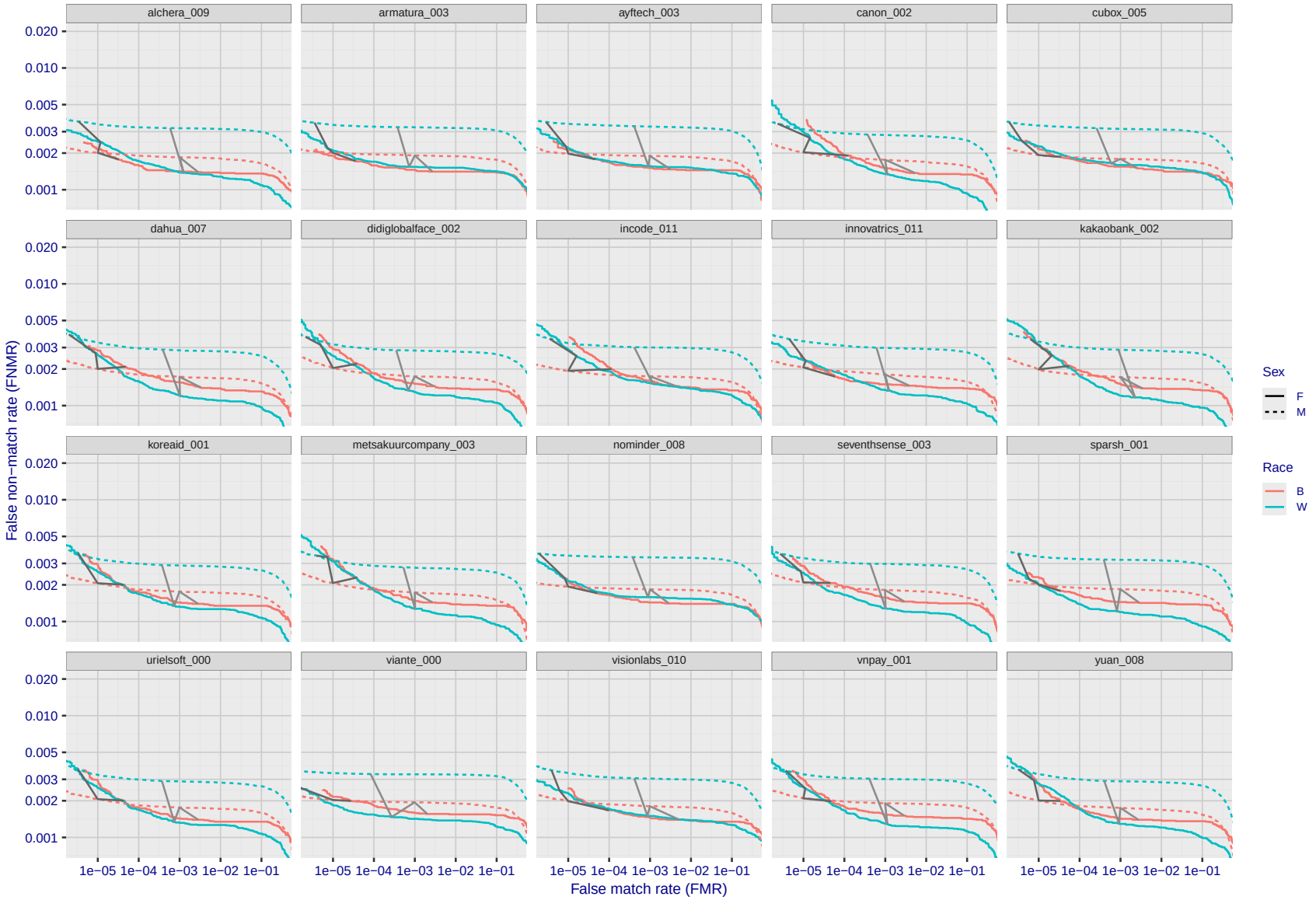


Figure 408: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

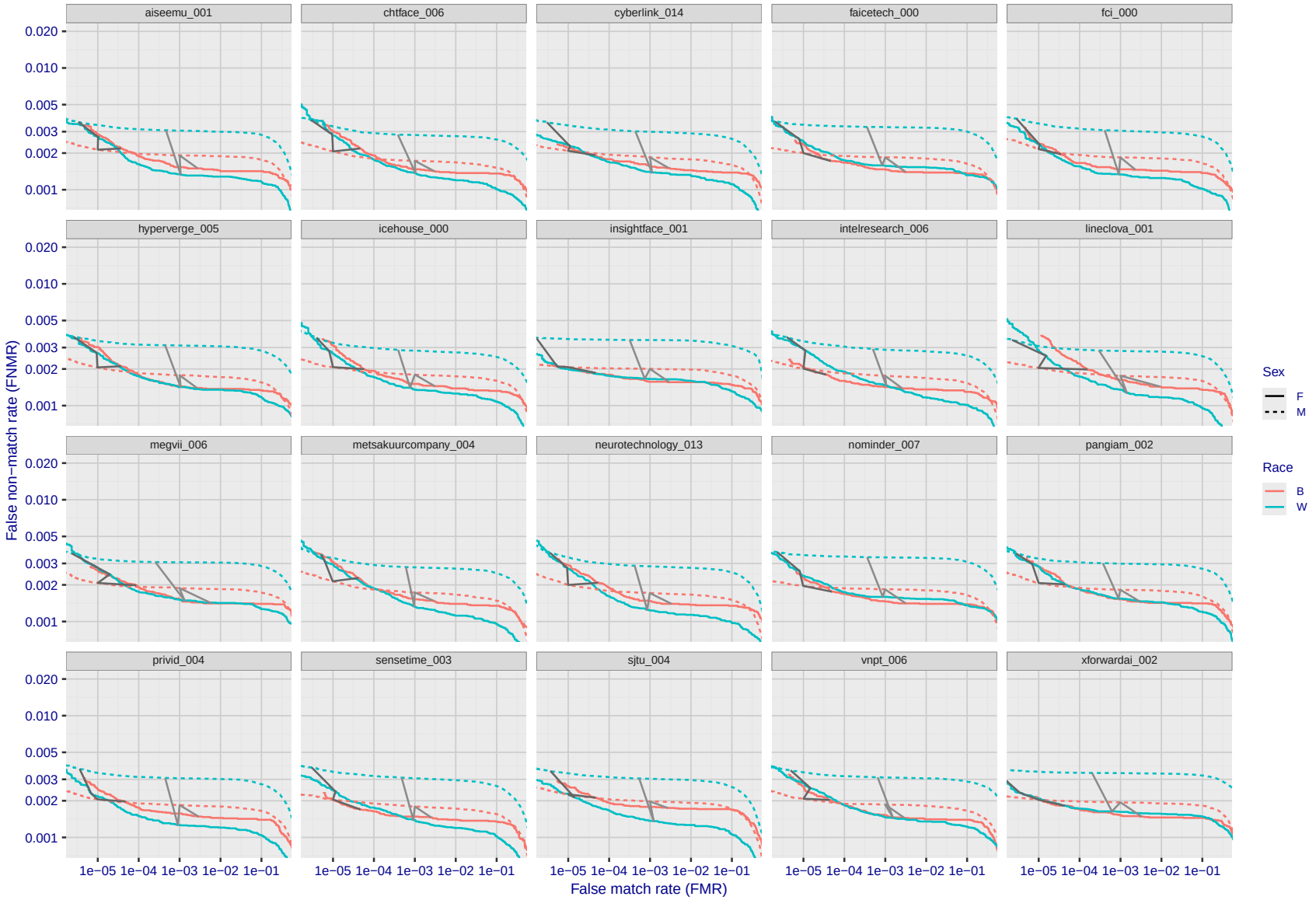


Figure 409: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

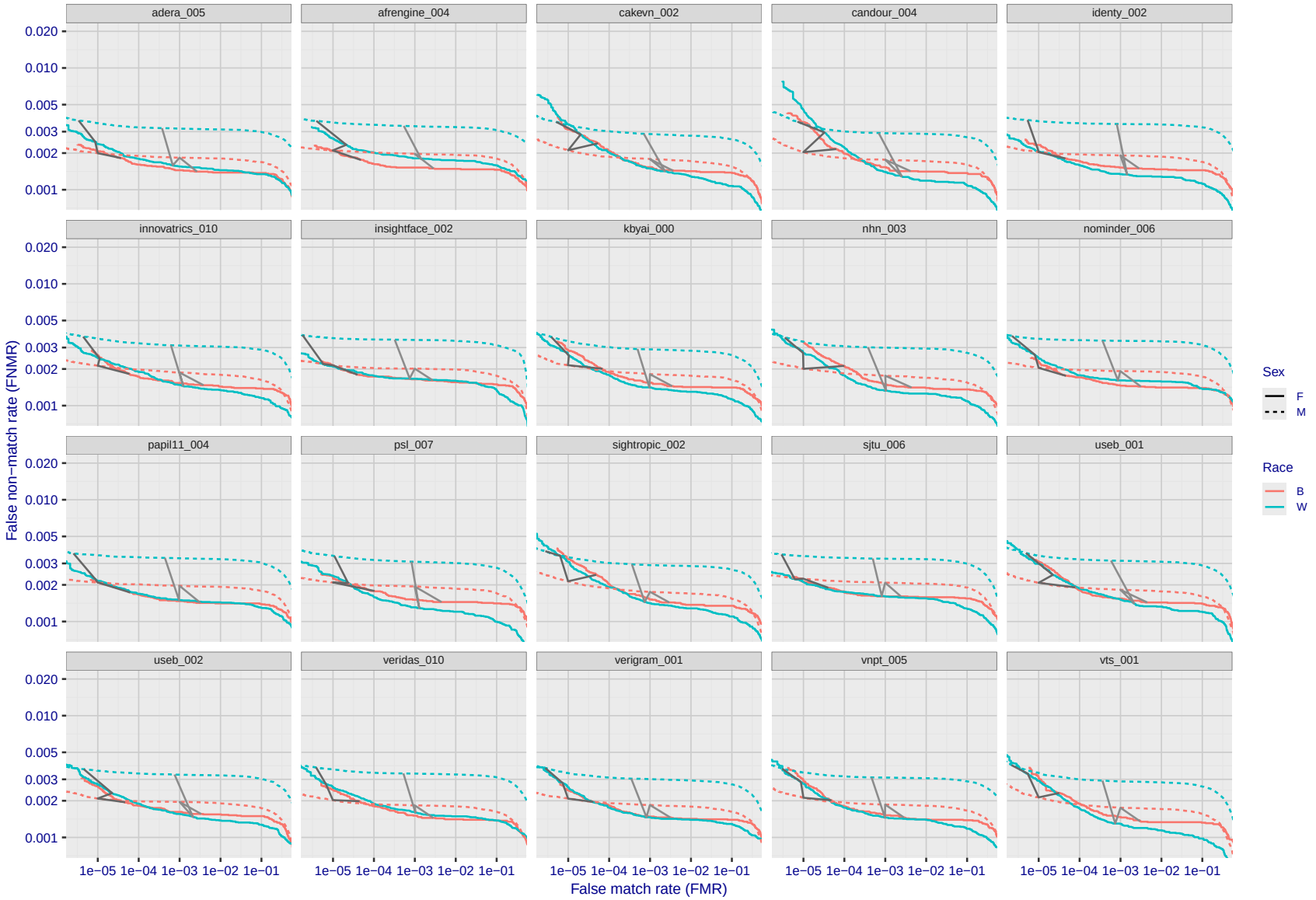


Figure 410: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

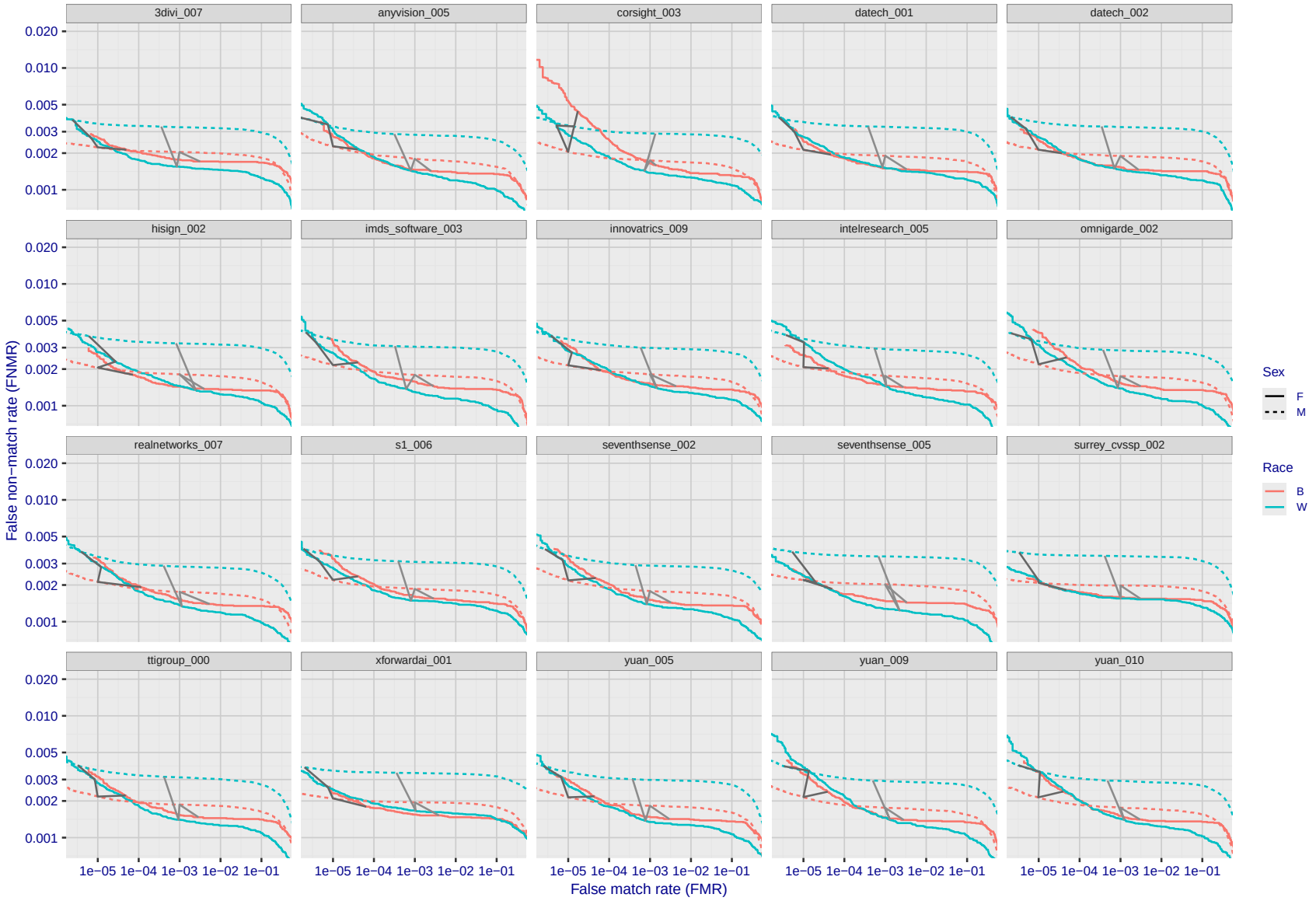


Figure 411: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

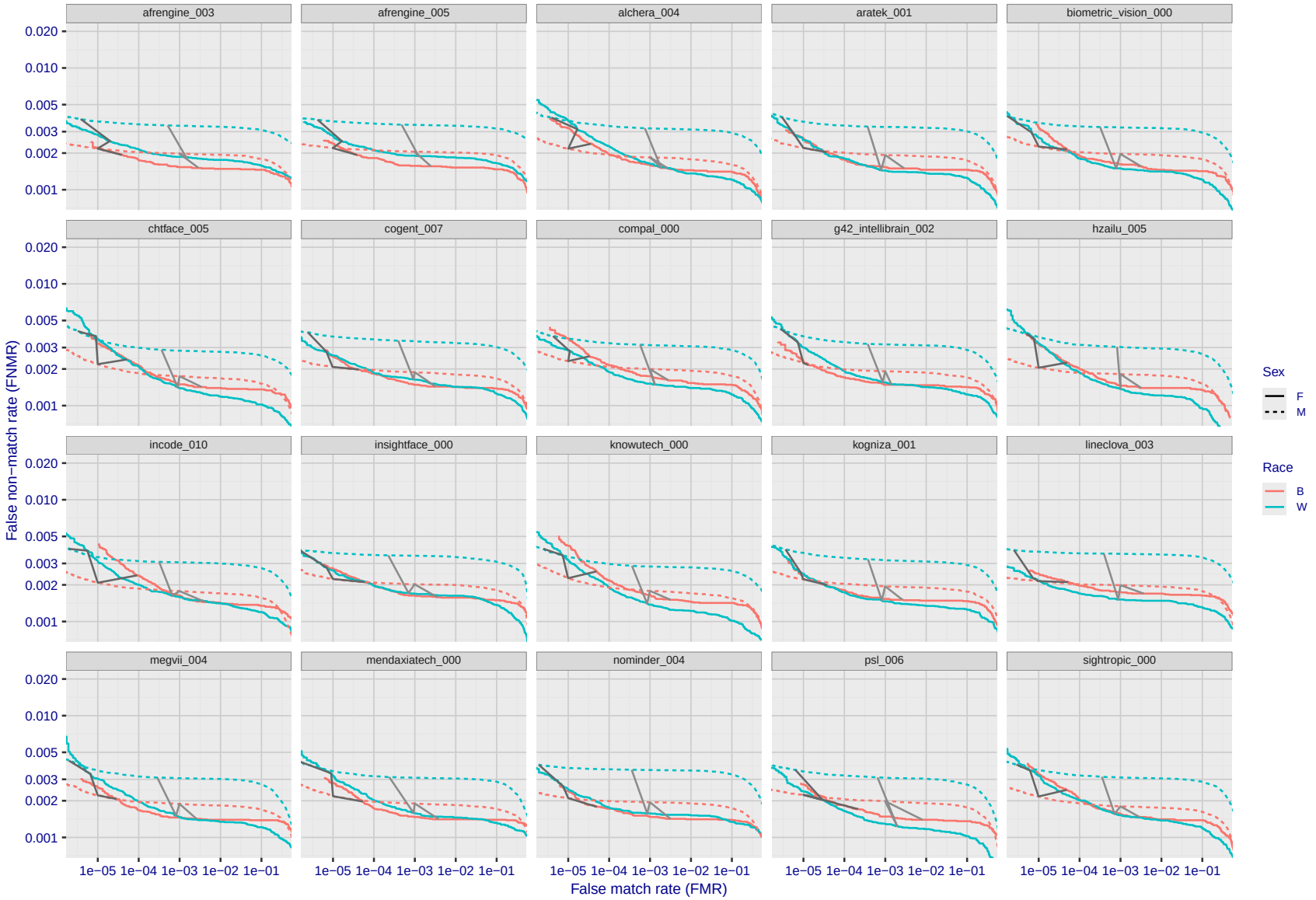


Figure 412: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

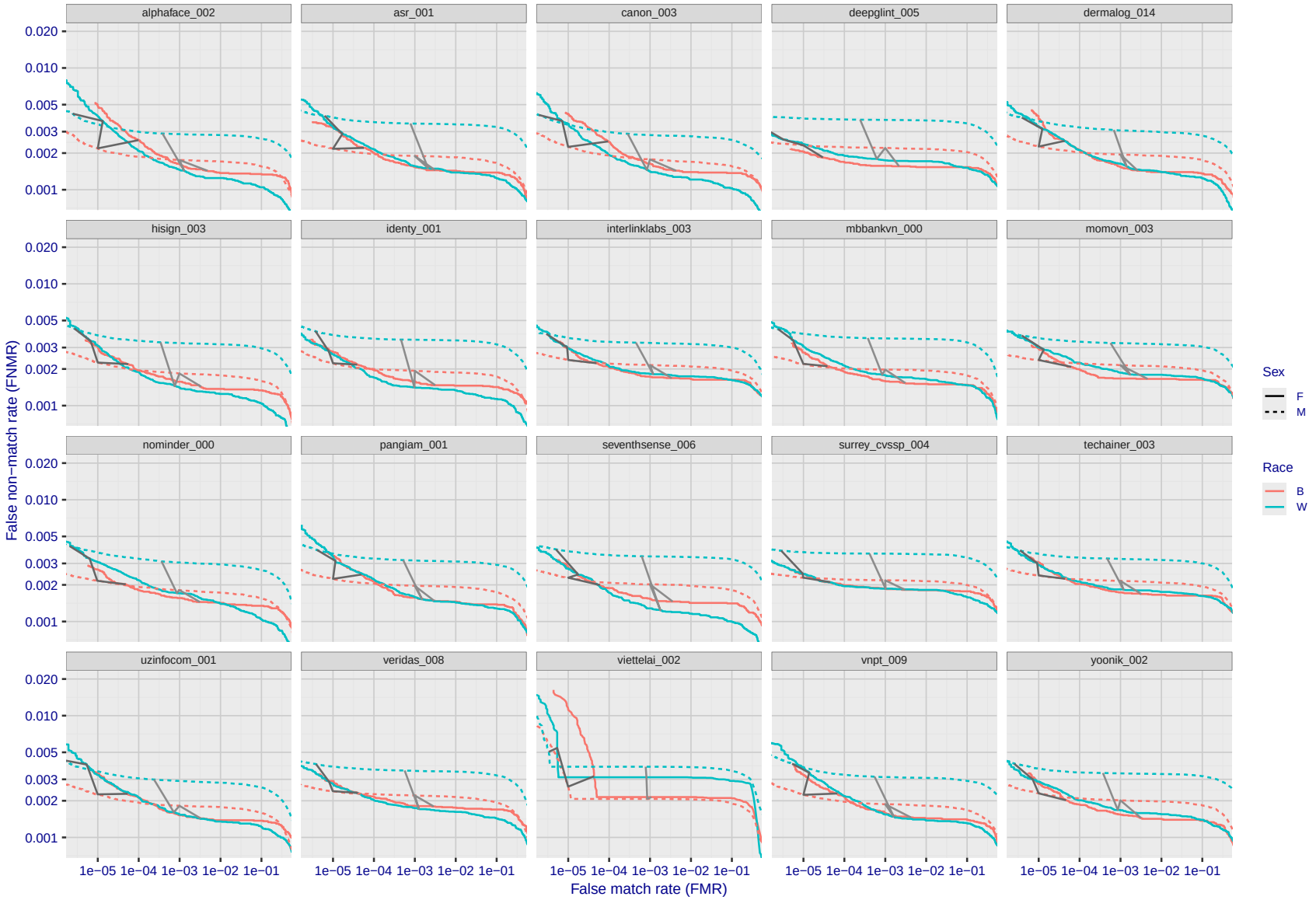


Figure 413: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

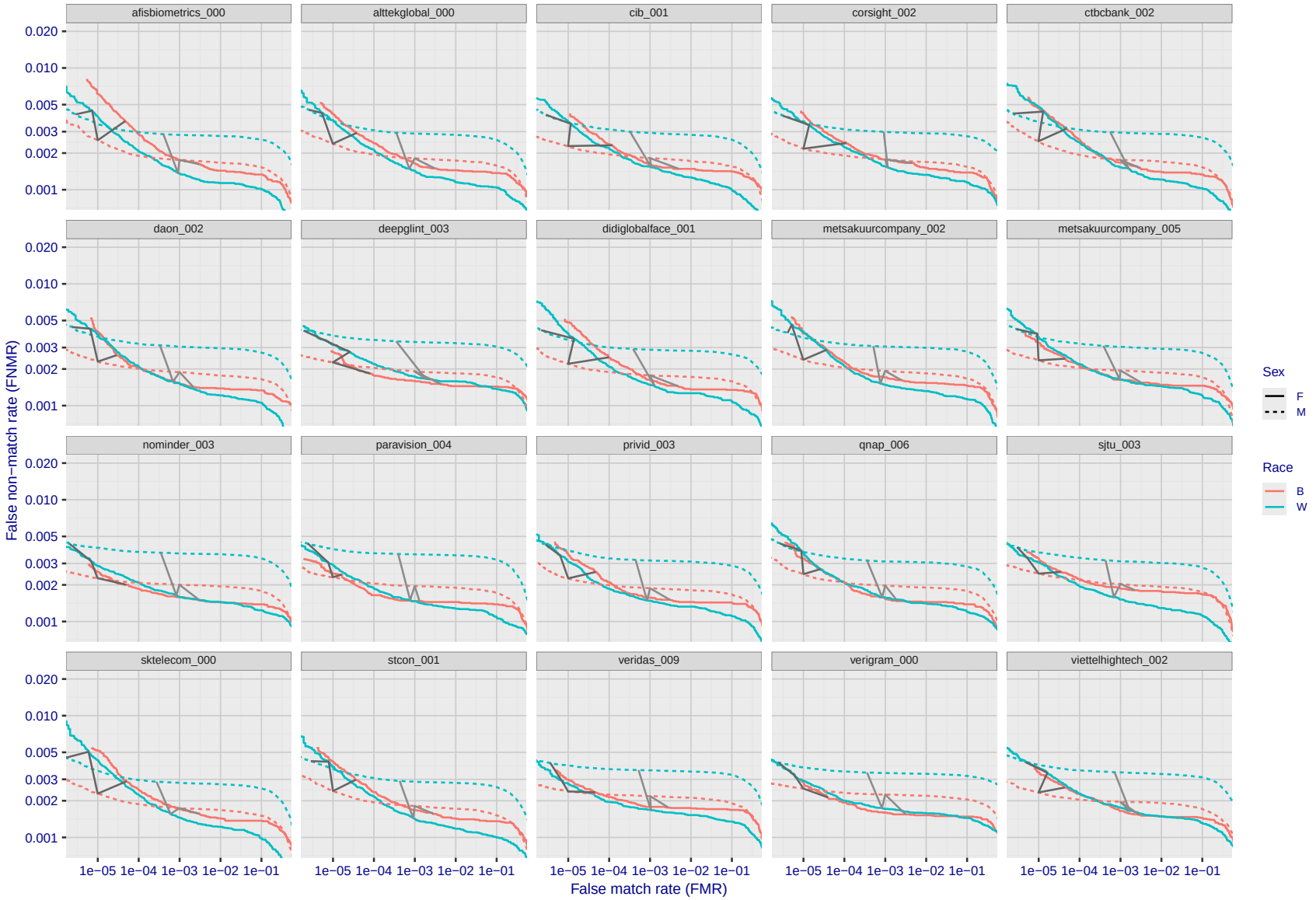


Figure 414: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

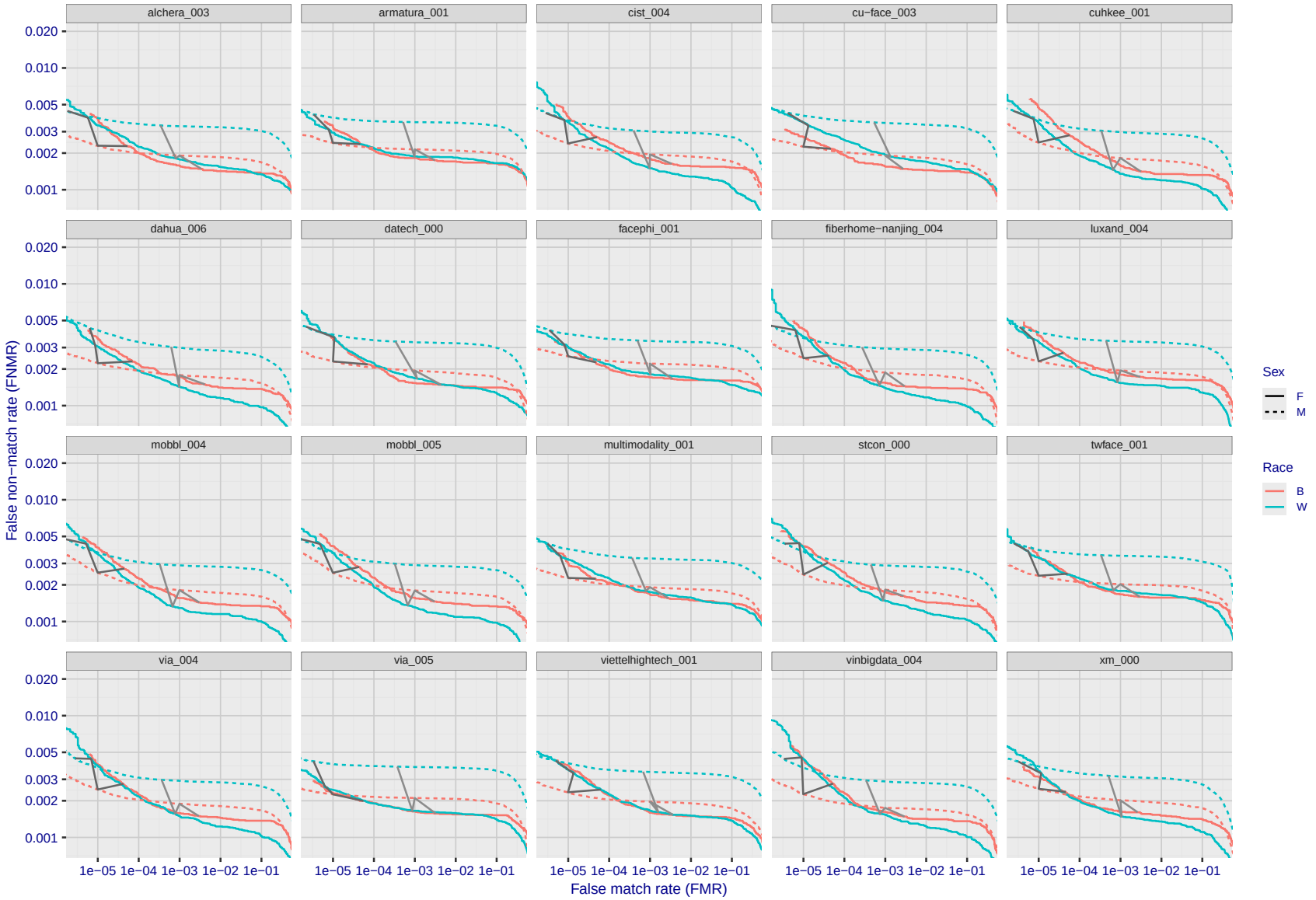


Figure 415: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

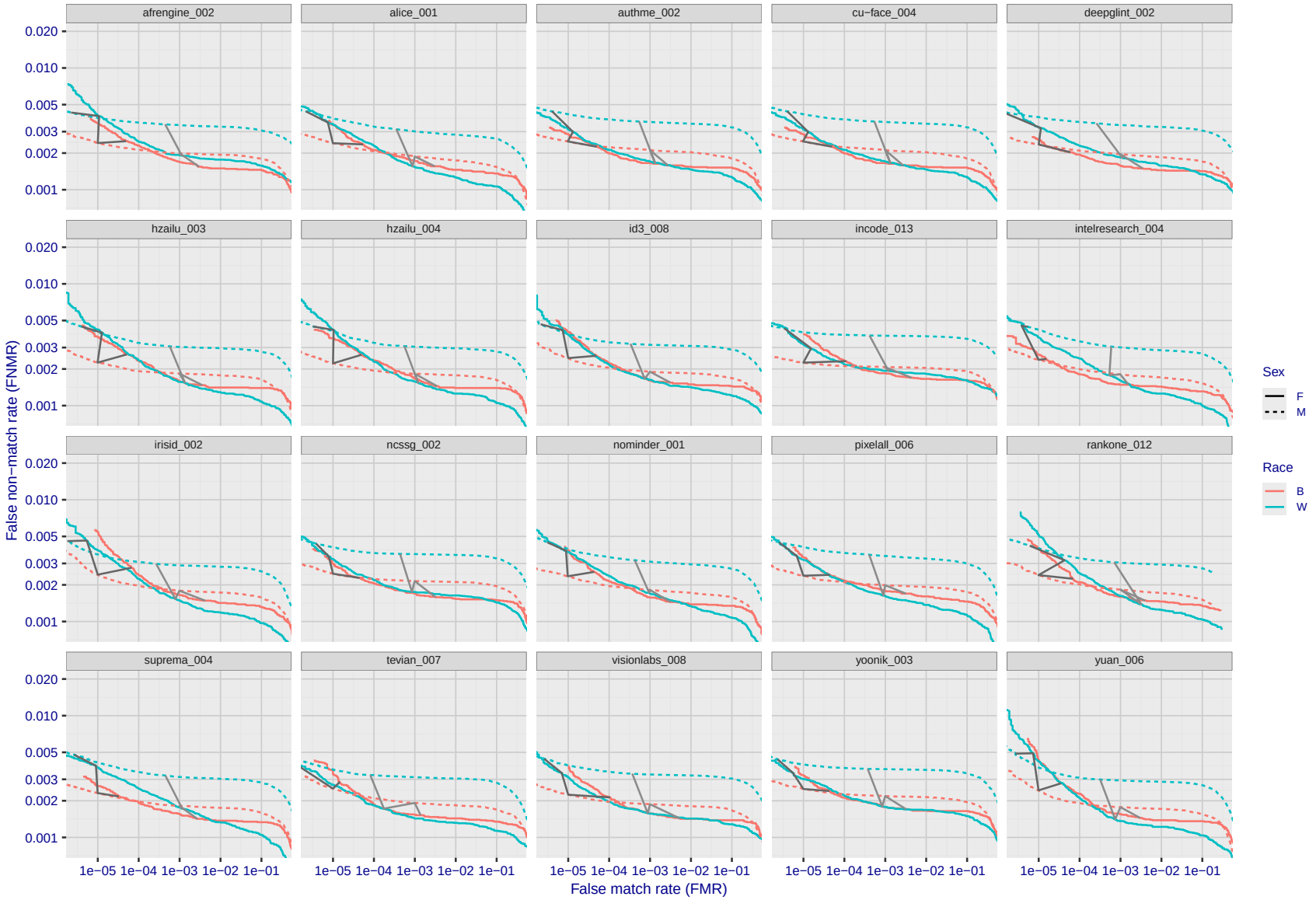


Figure 416: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

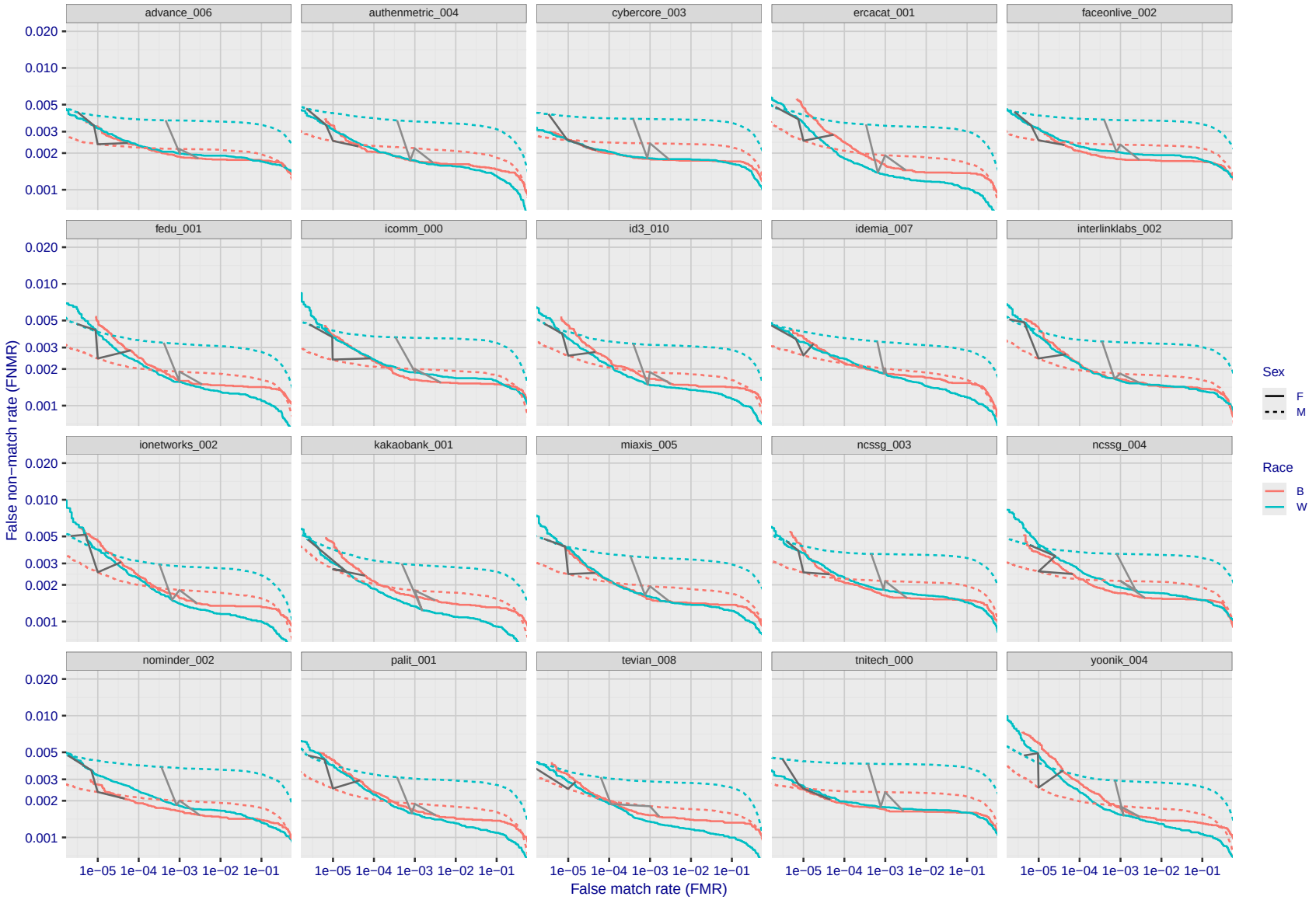


Figure 417: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

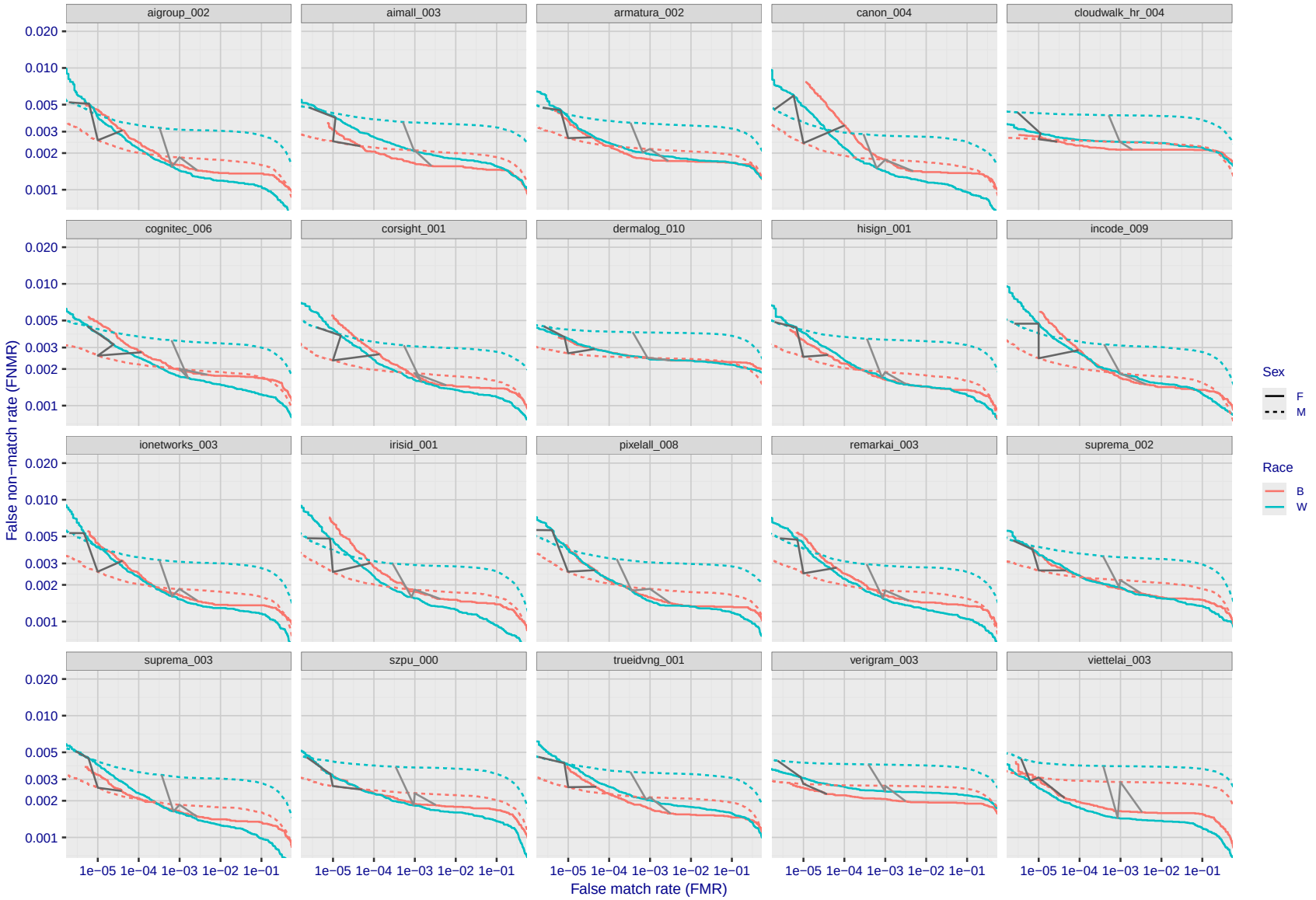


Figure 418: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

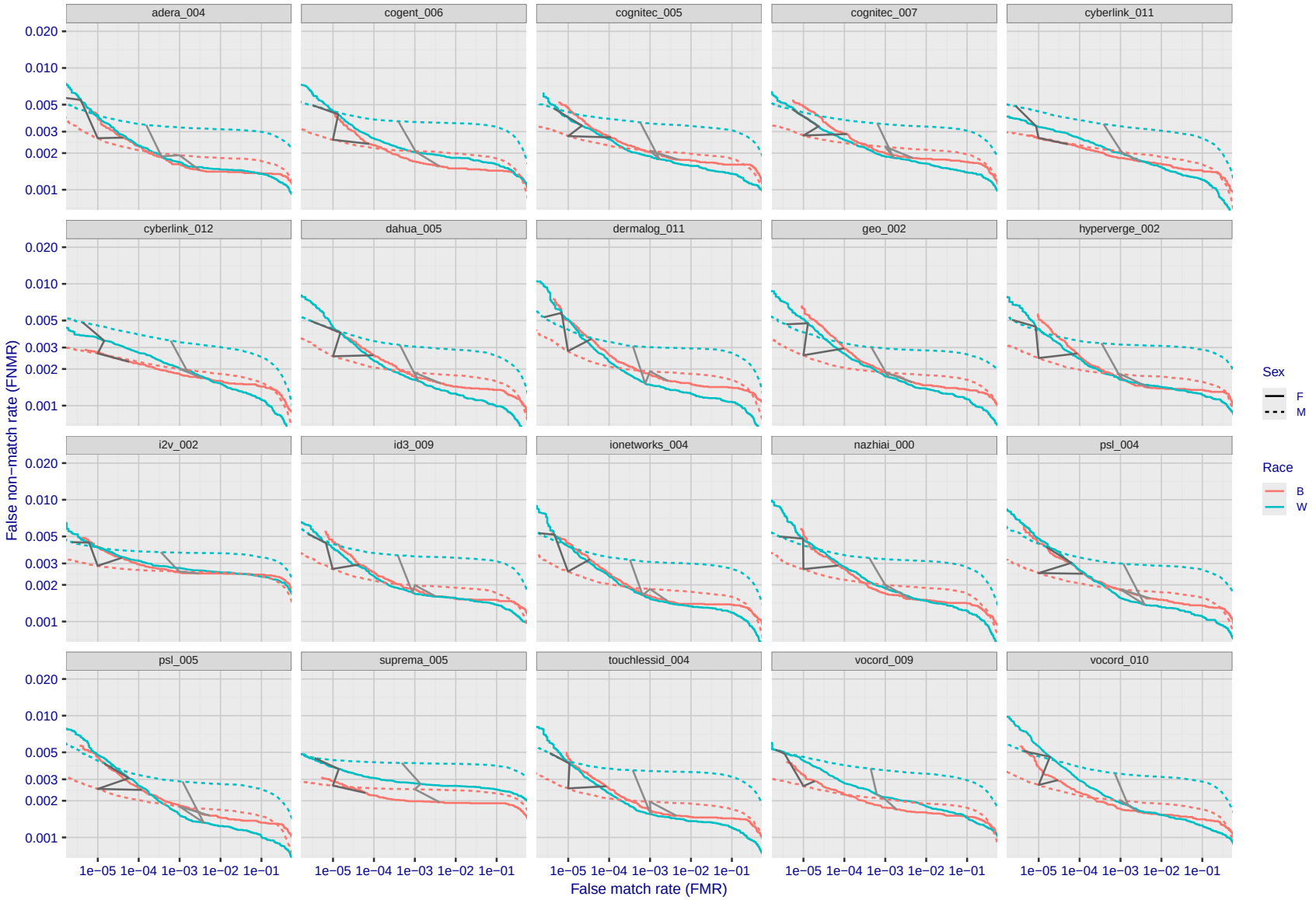


Figure 419: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

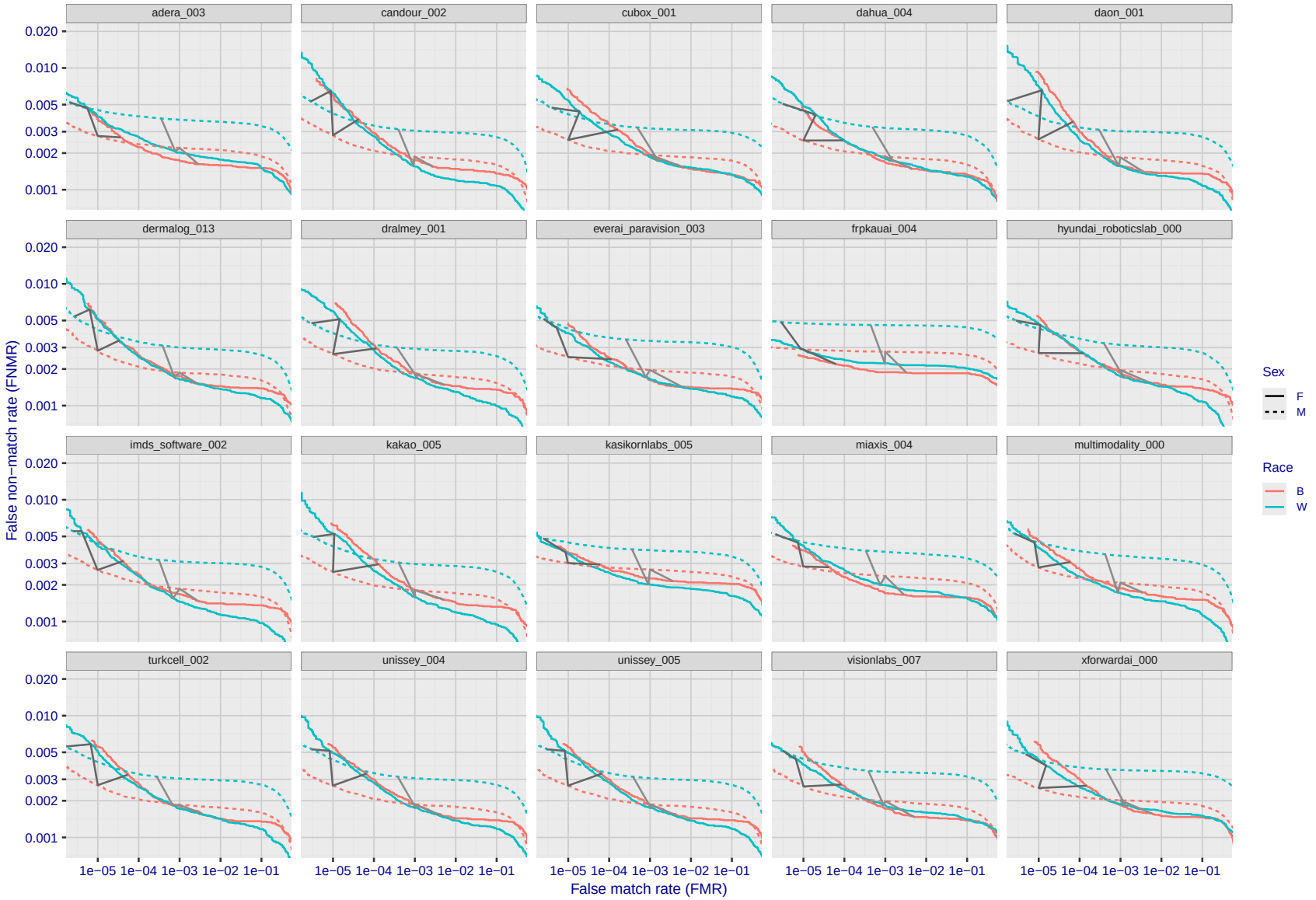


Figure 420: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

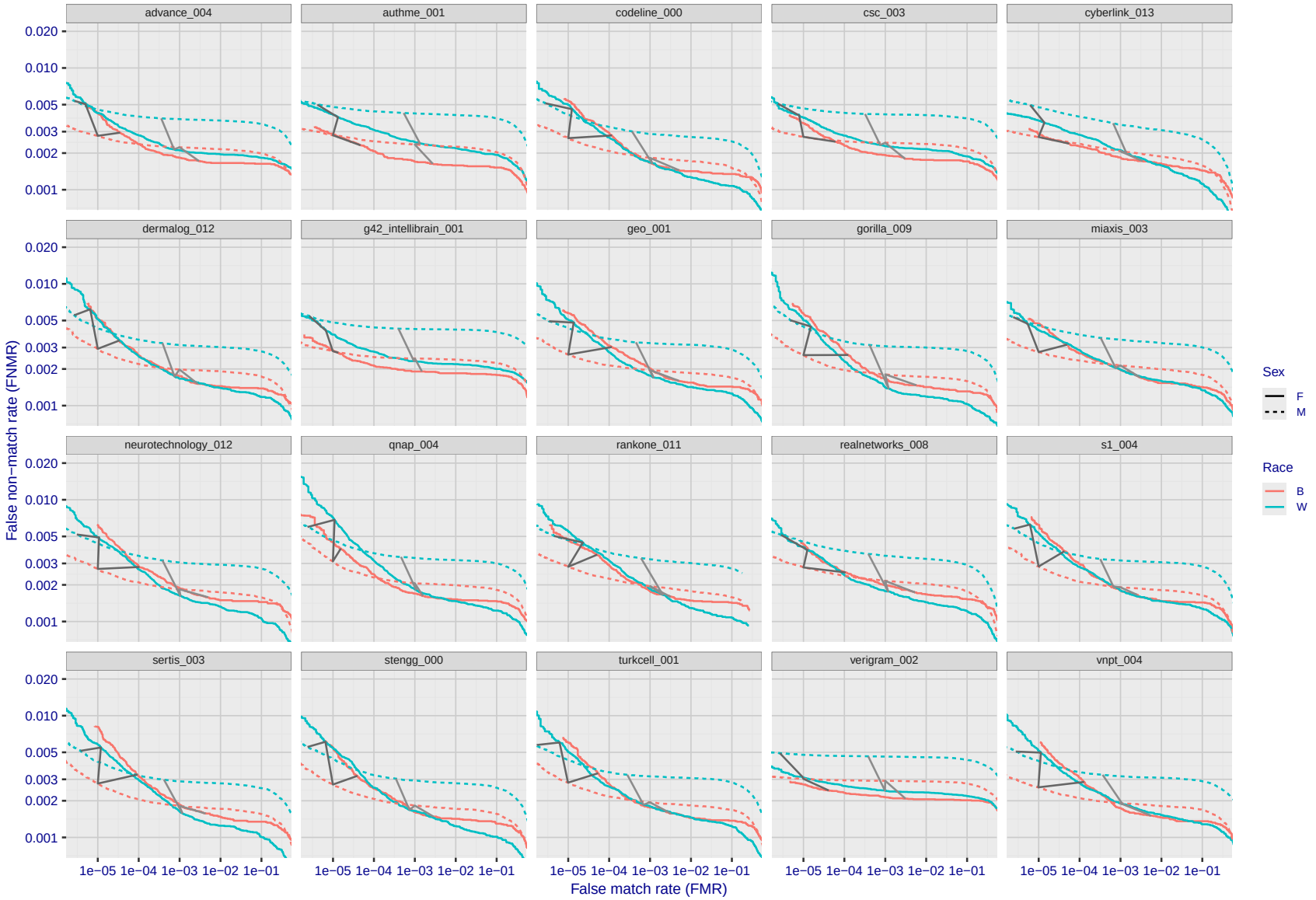


Figure 421: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

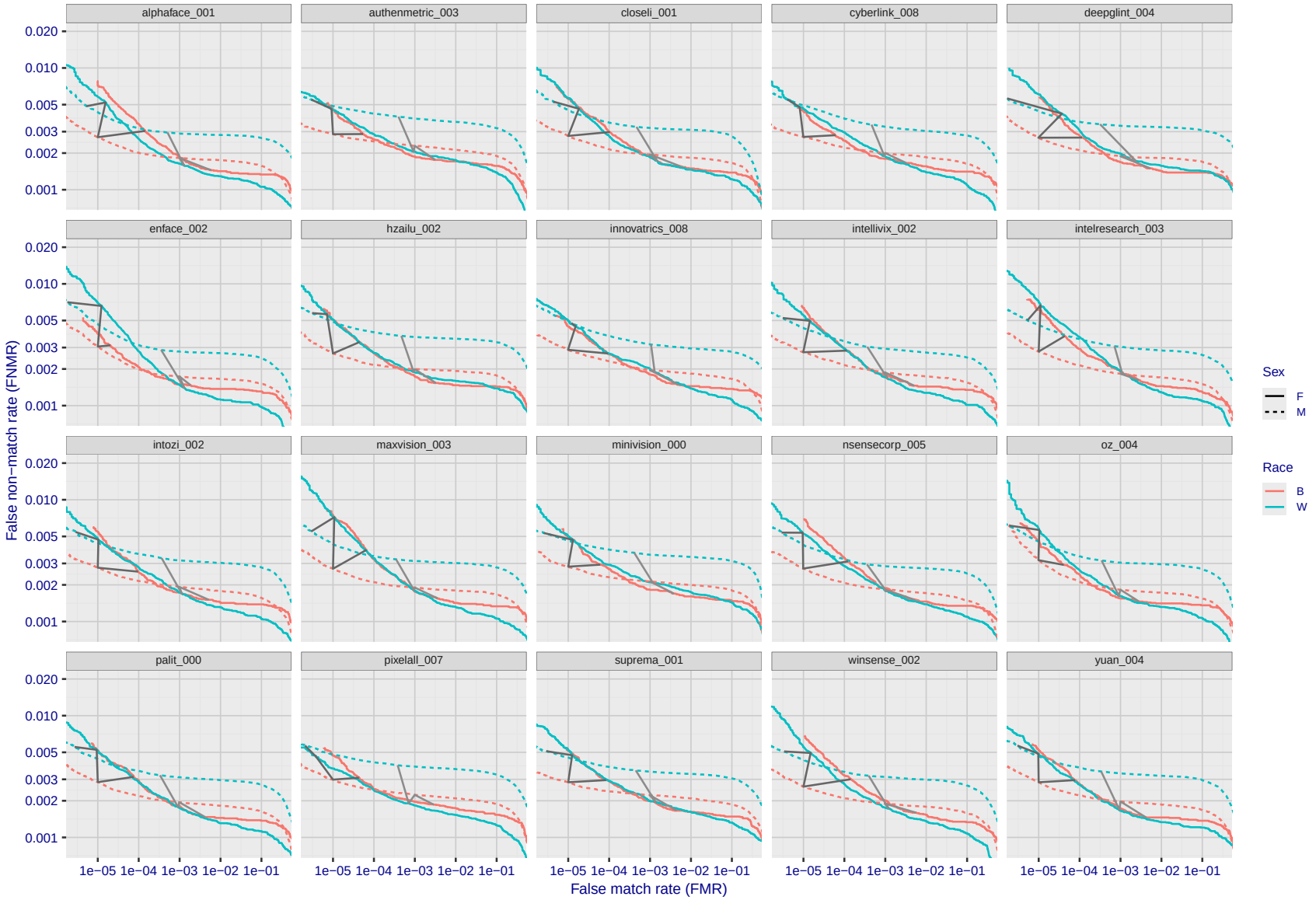


Figure 422: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

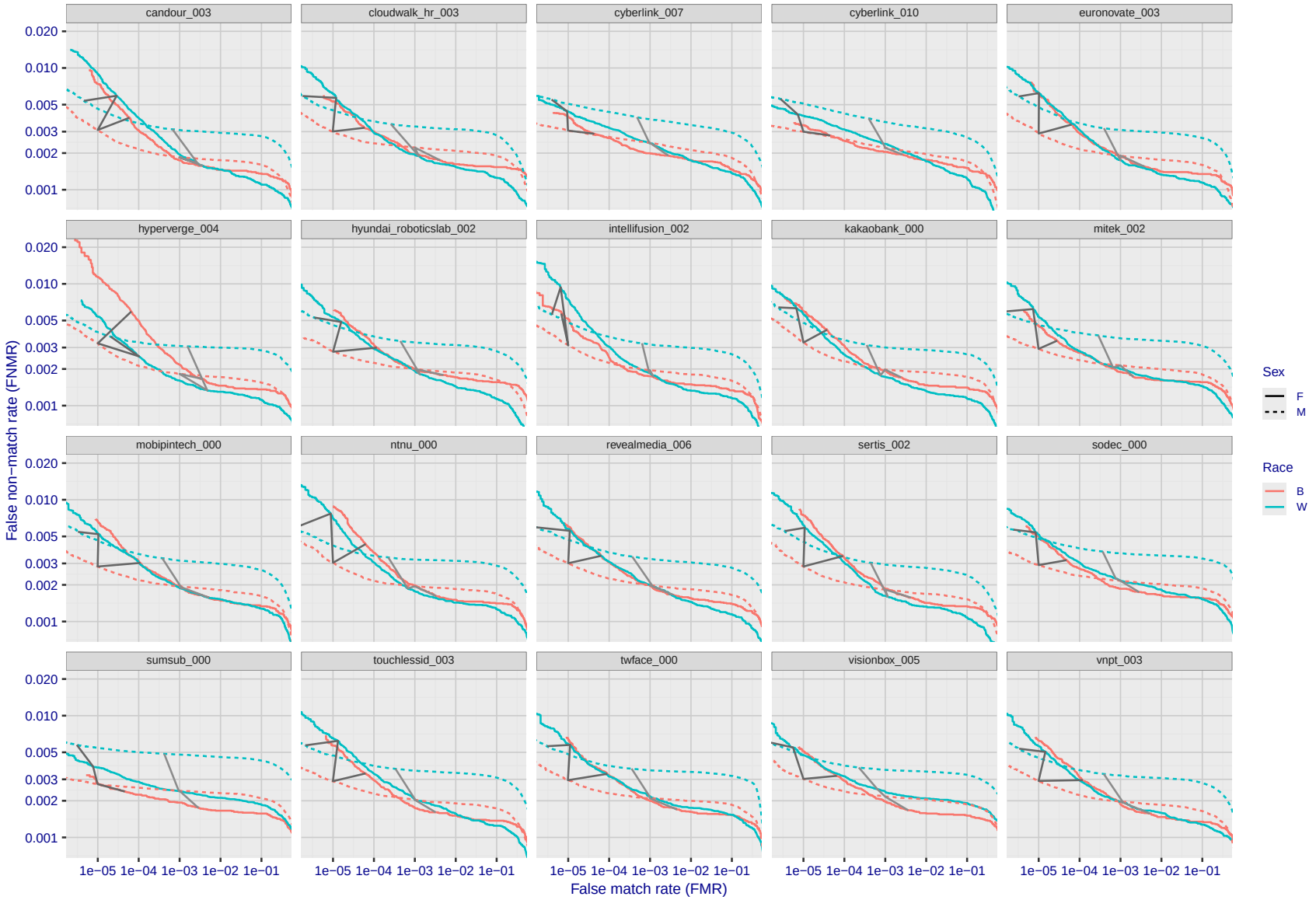


Figure 423: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

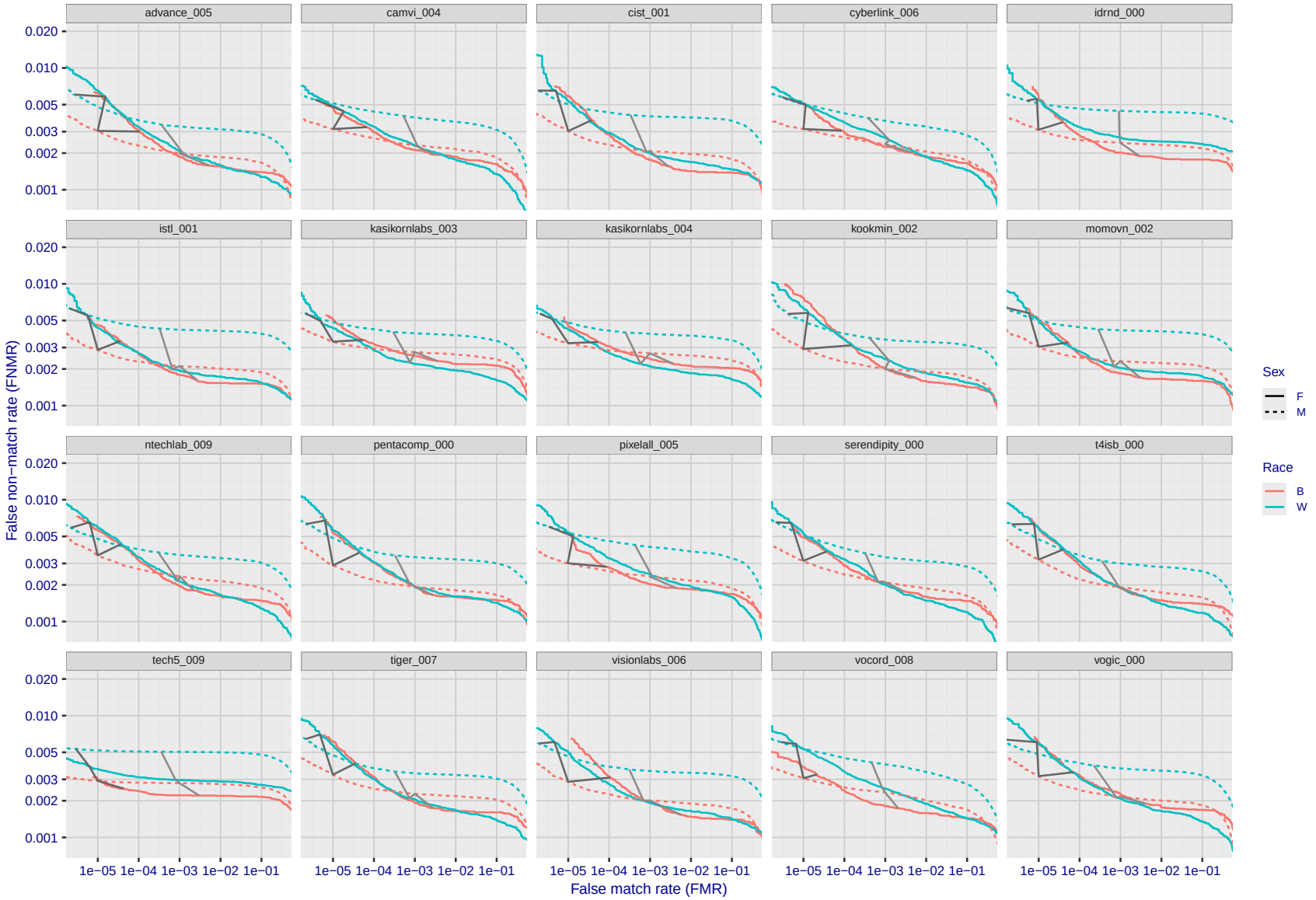


Figure 424: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

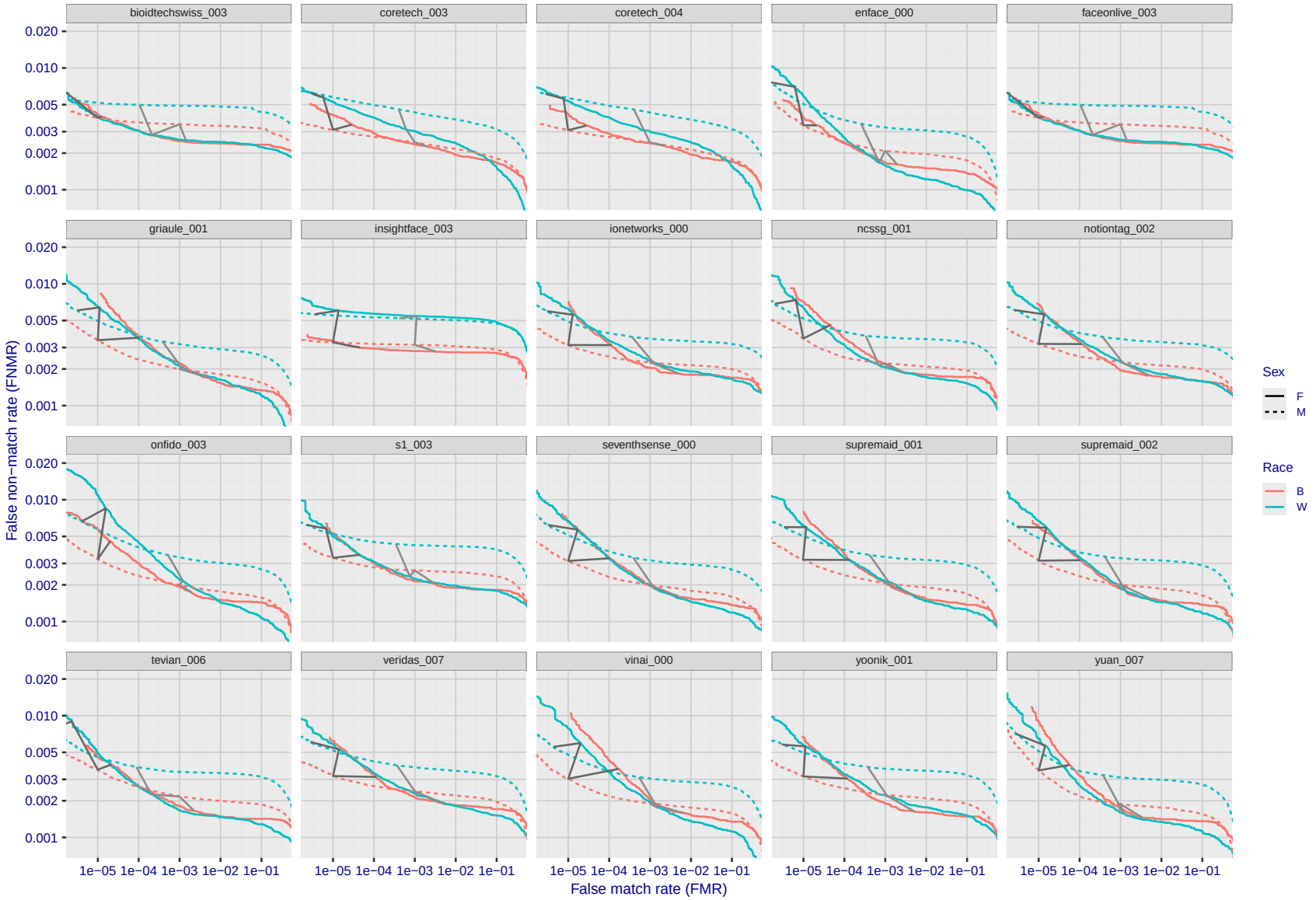


Figure 425: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

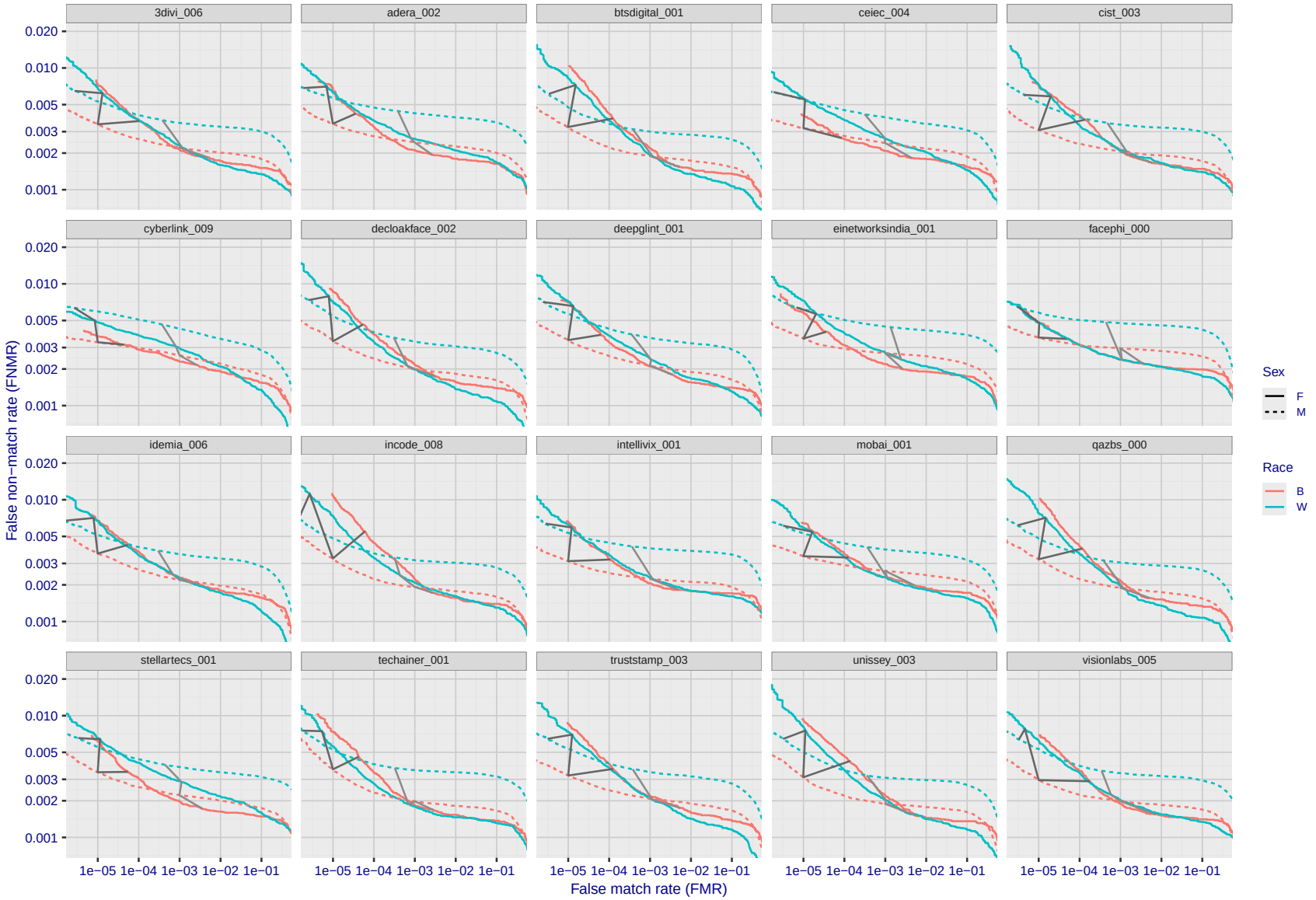


Figure 426: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

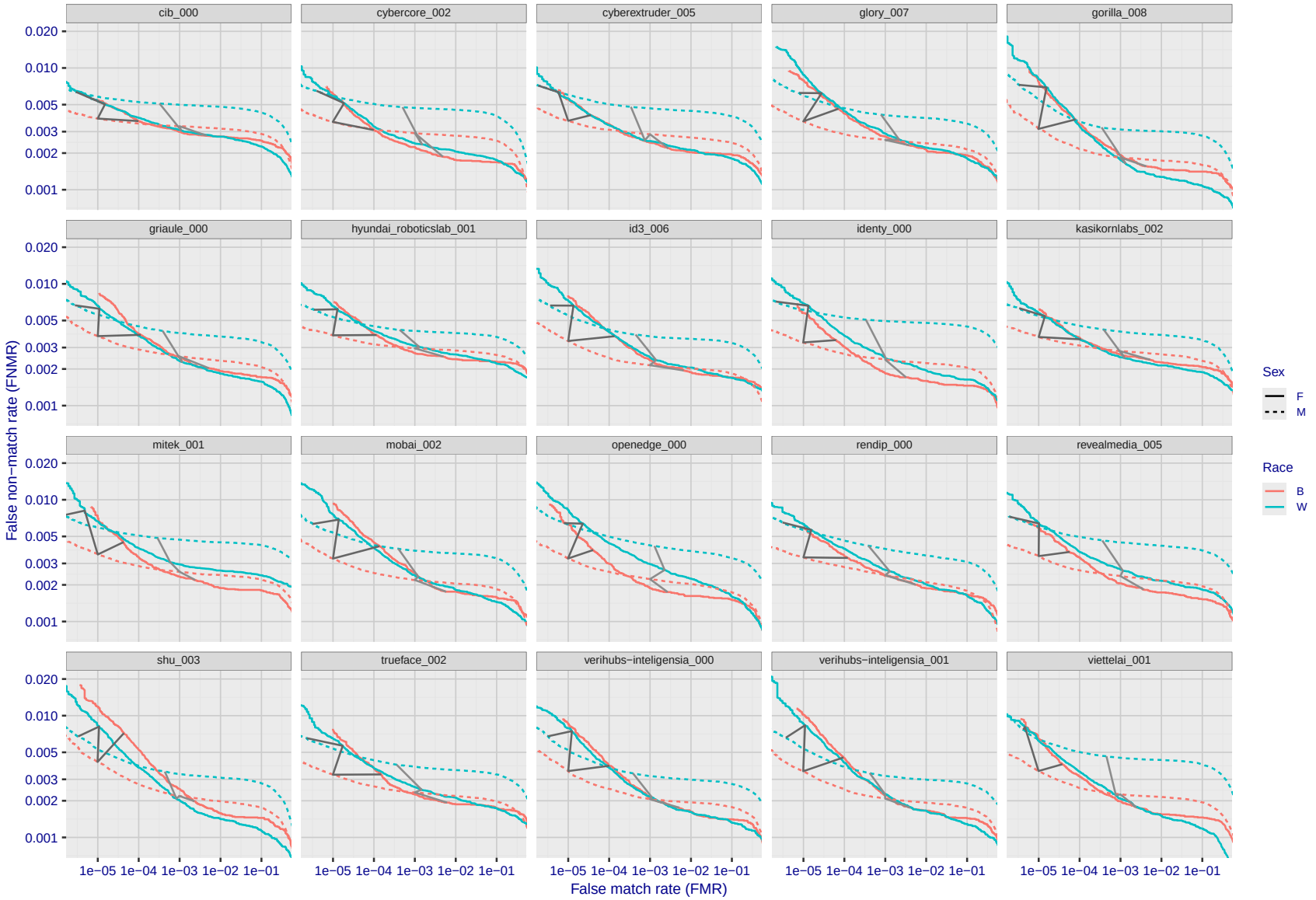


Figure 427: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

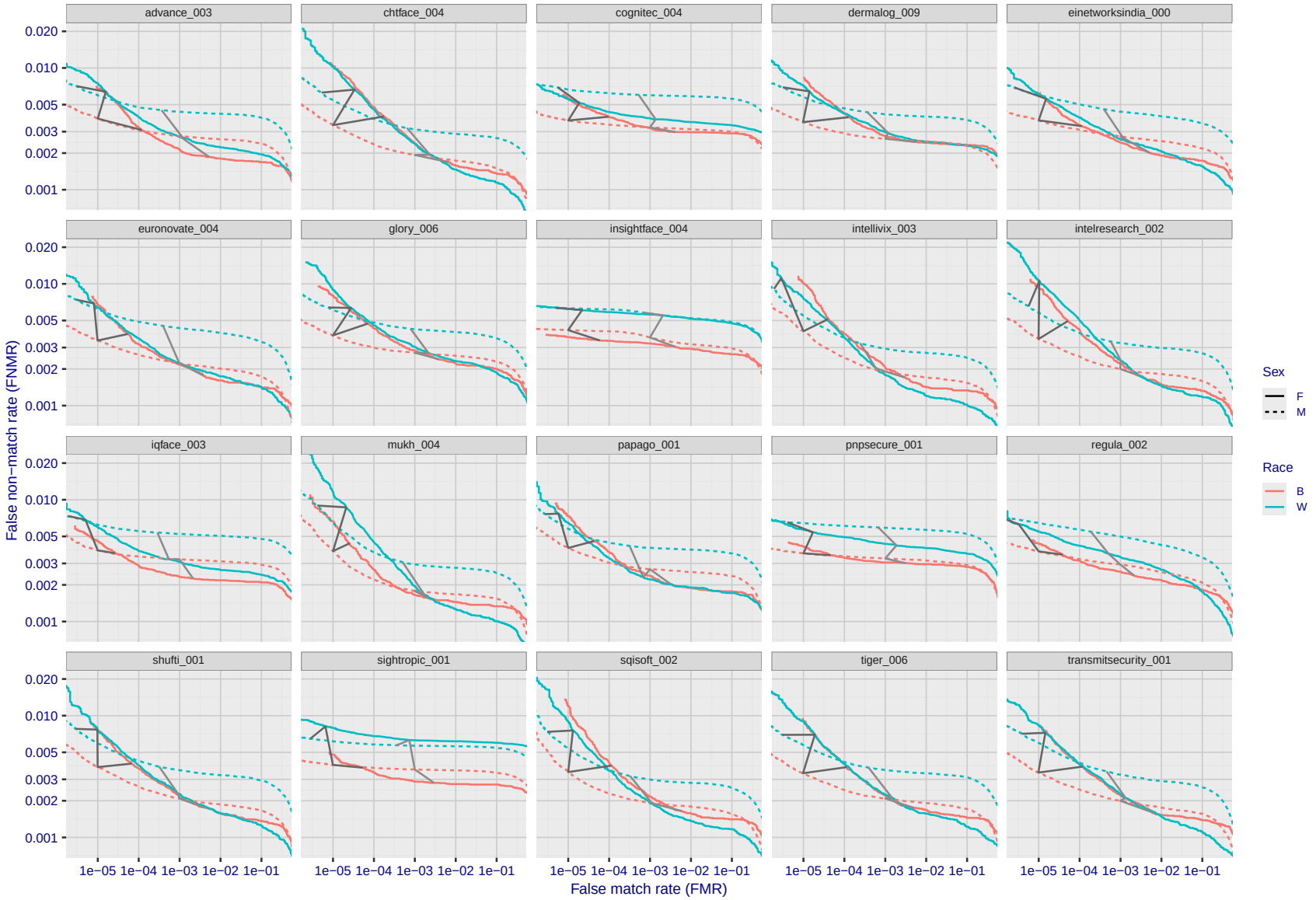


Figure 428: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

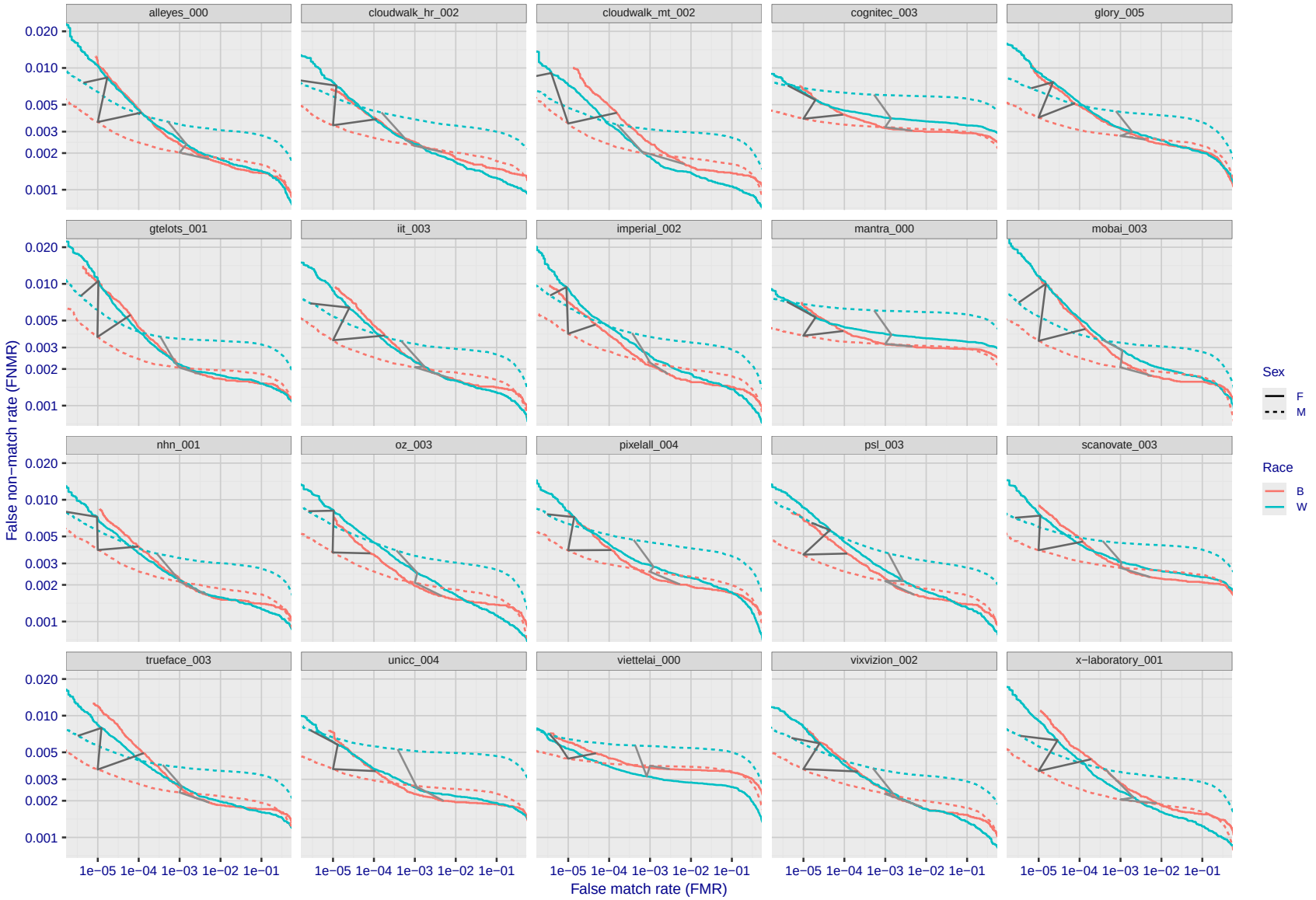


Figure 429: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

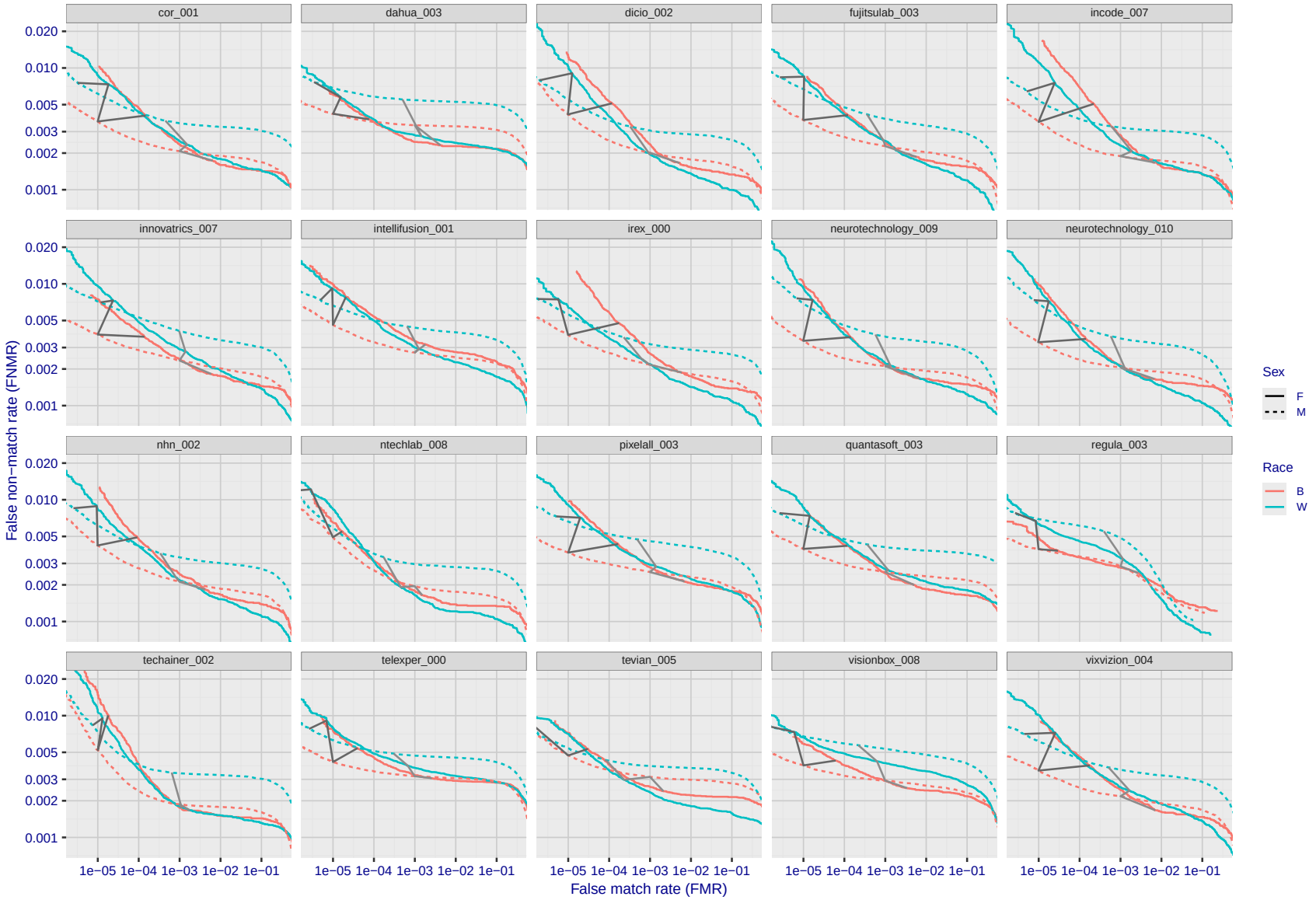


Figure 430: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

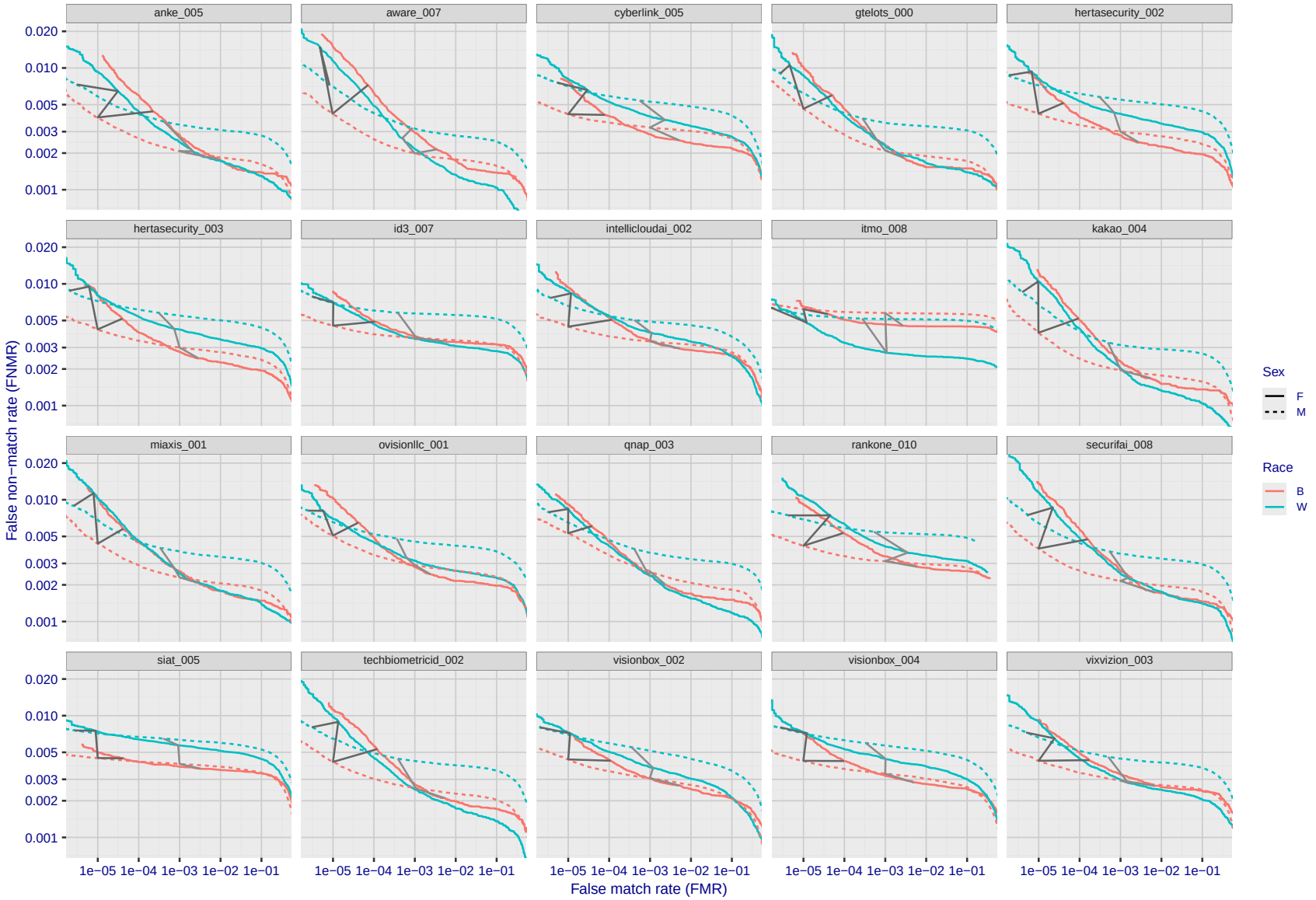


Figure 431: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

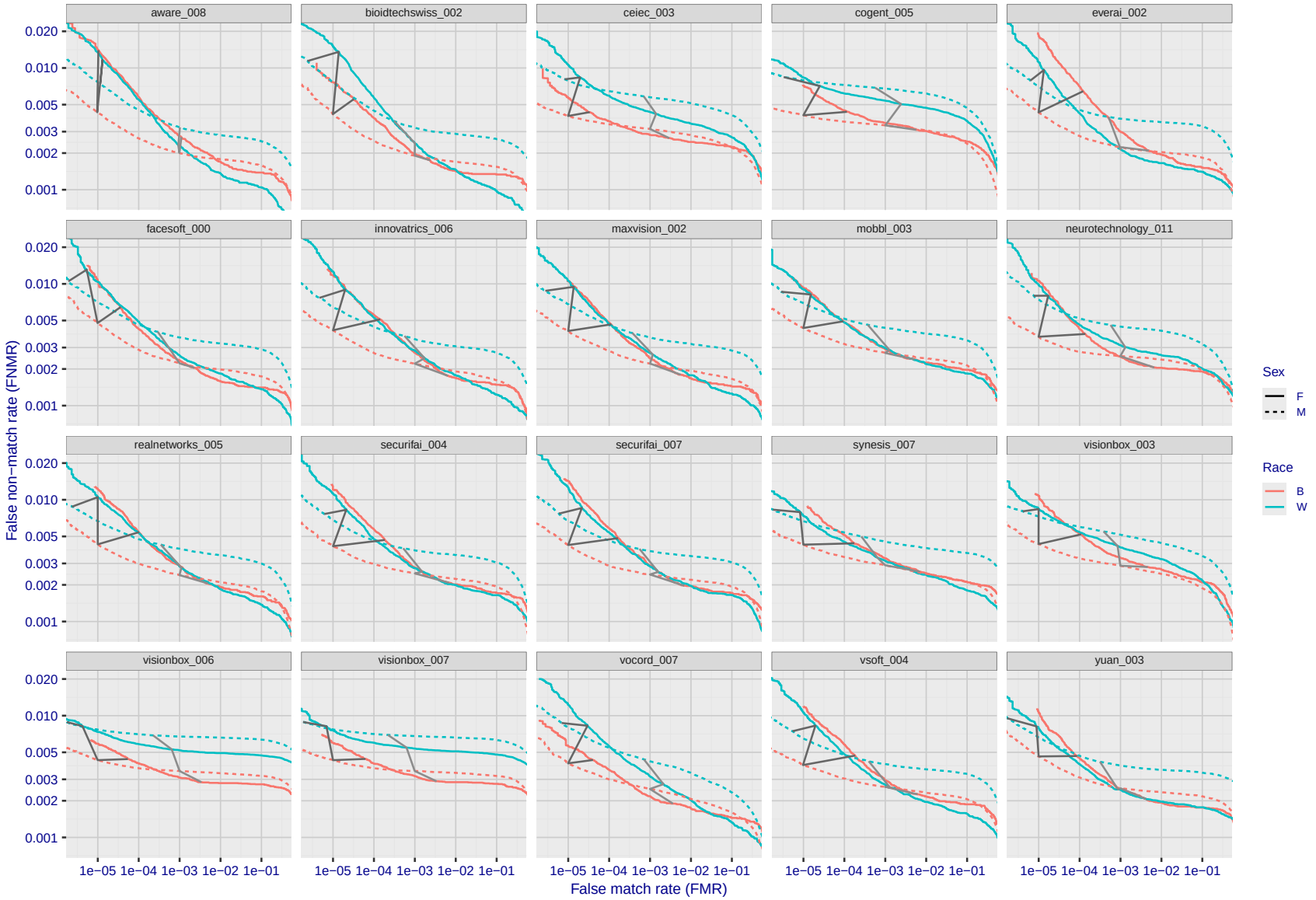


Figure 432: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

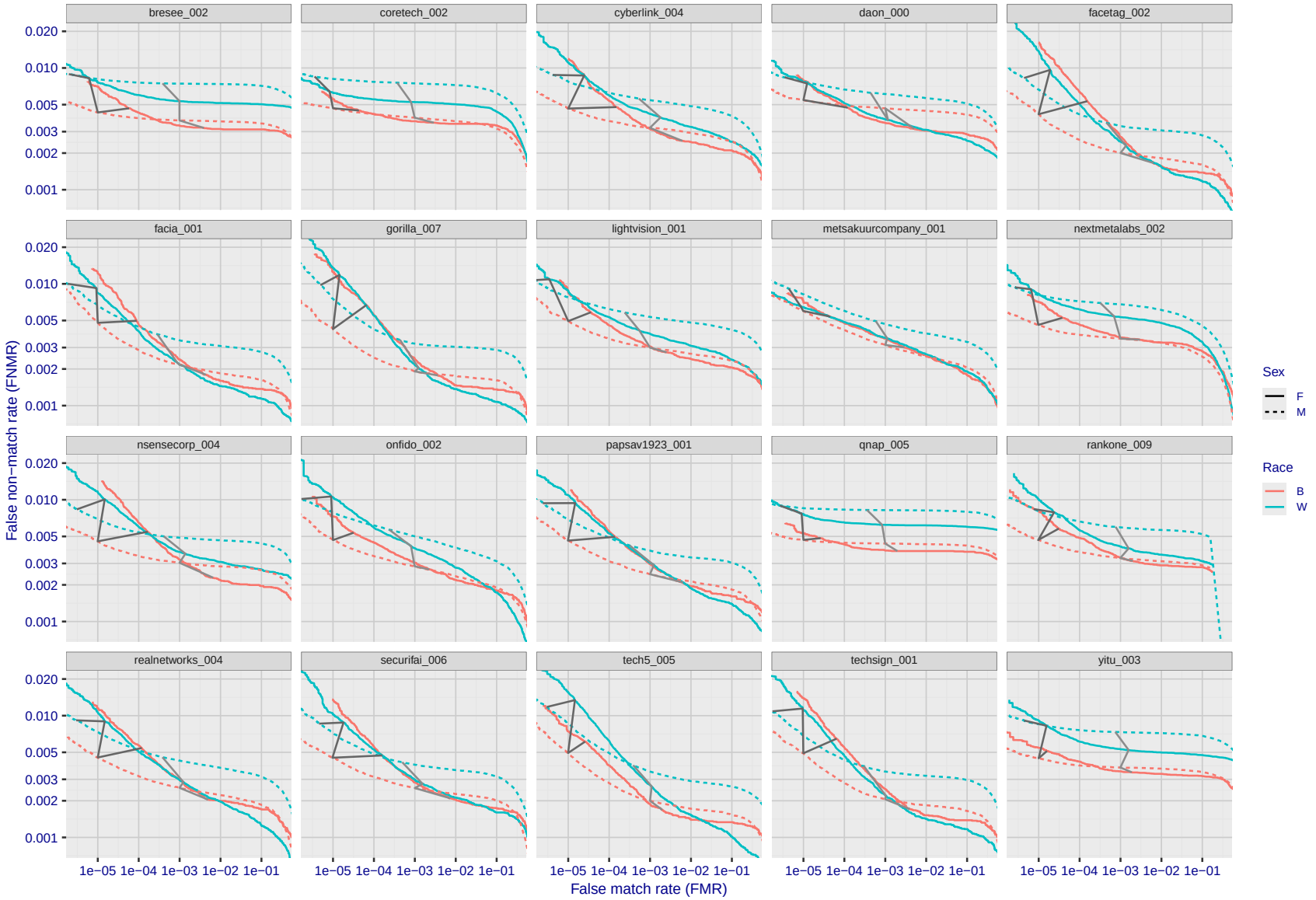


Figure 433: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

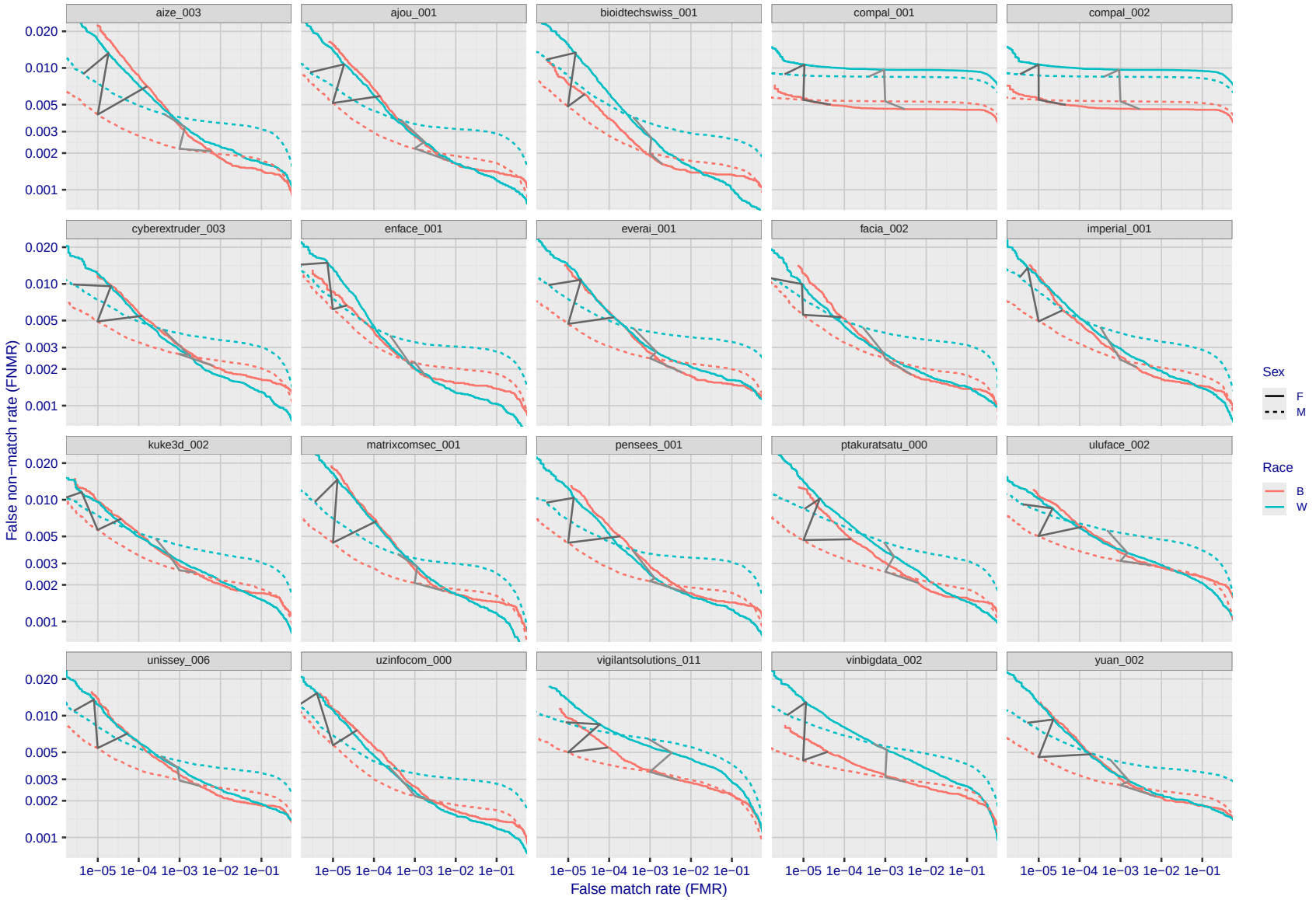


Figure 434: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

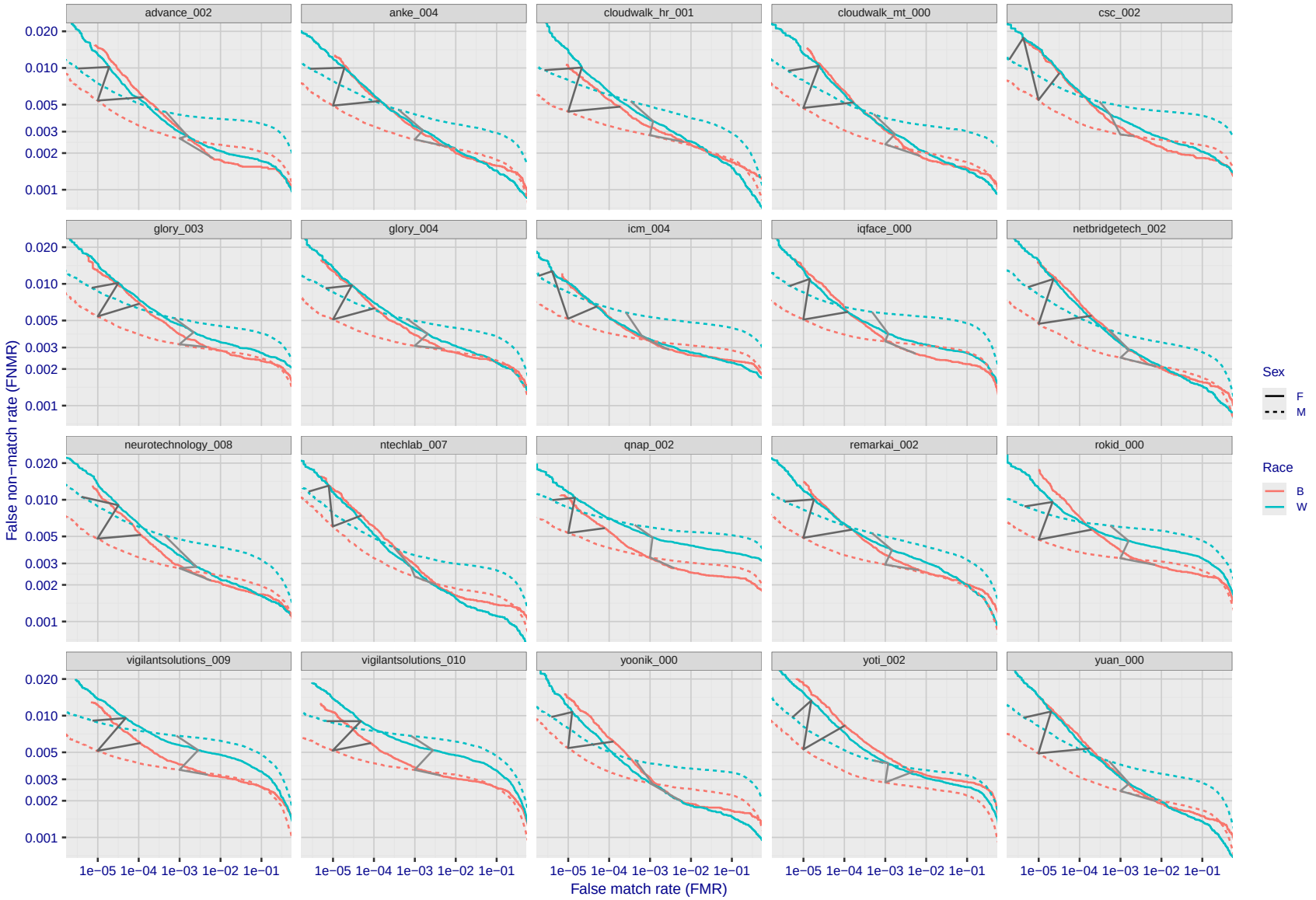


Figure 435: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

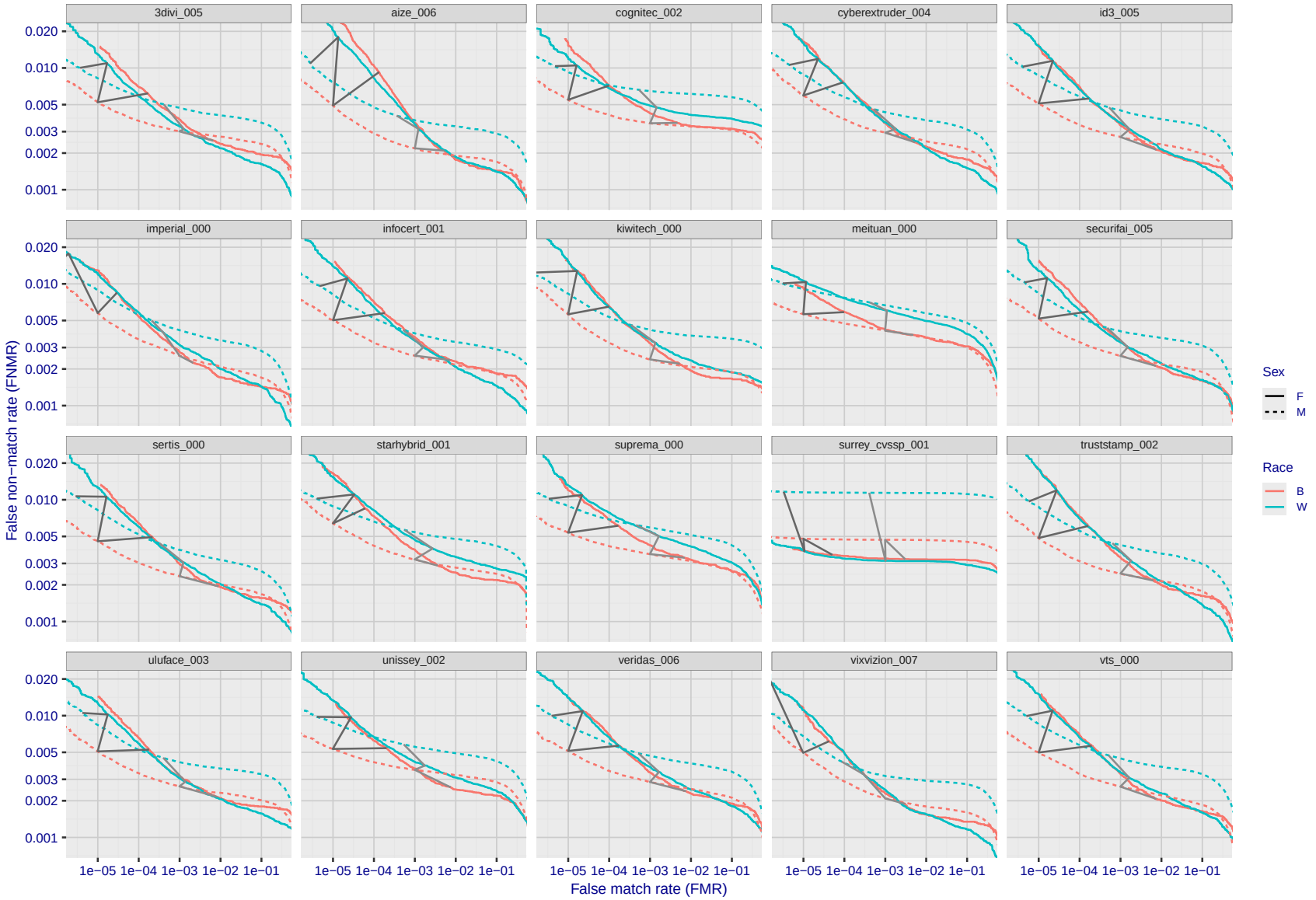


Figure 436: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

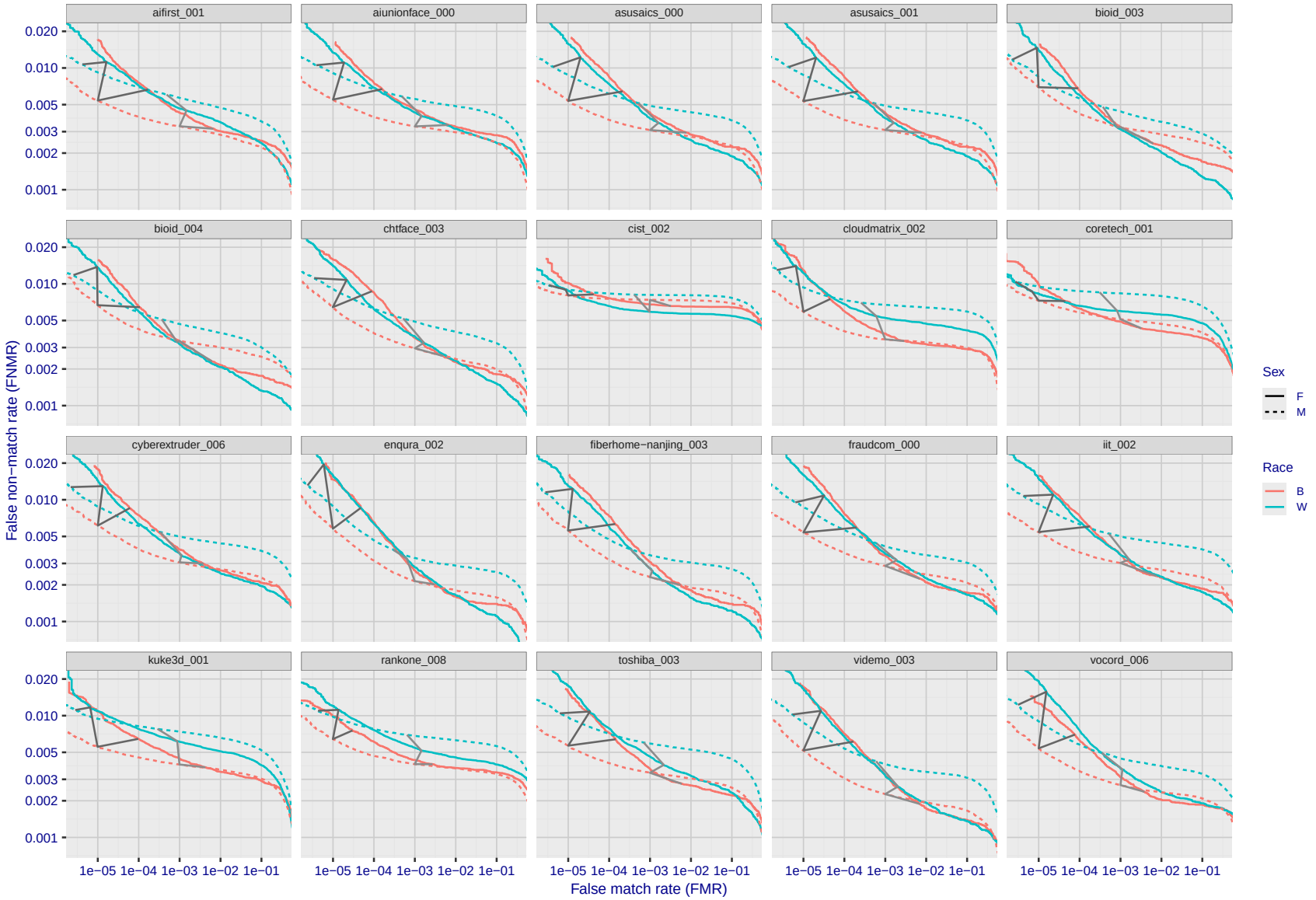


Figure 437: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

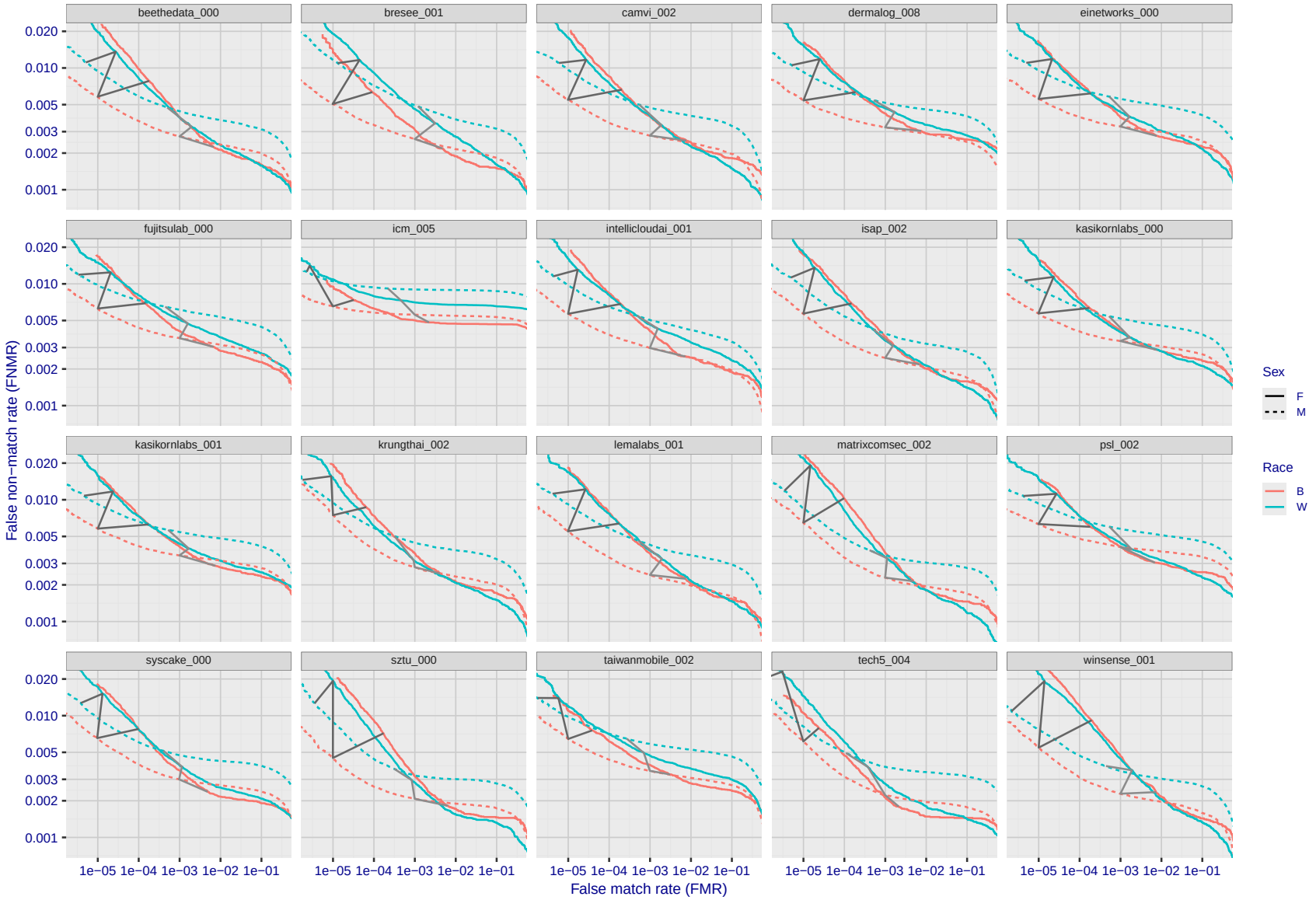


Figure 438: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

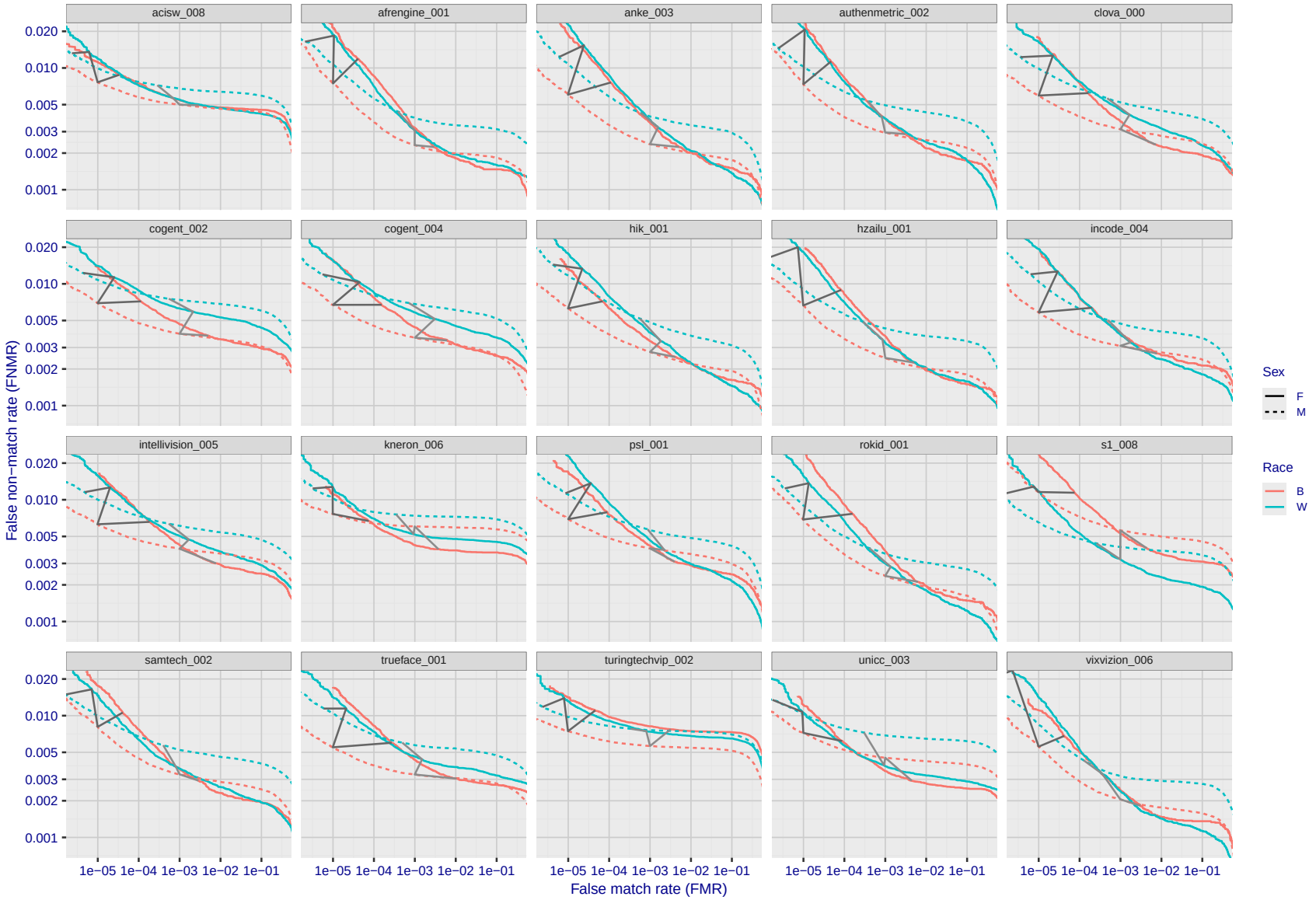


Figure 439: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

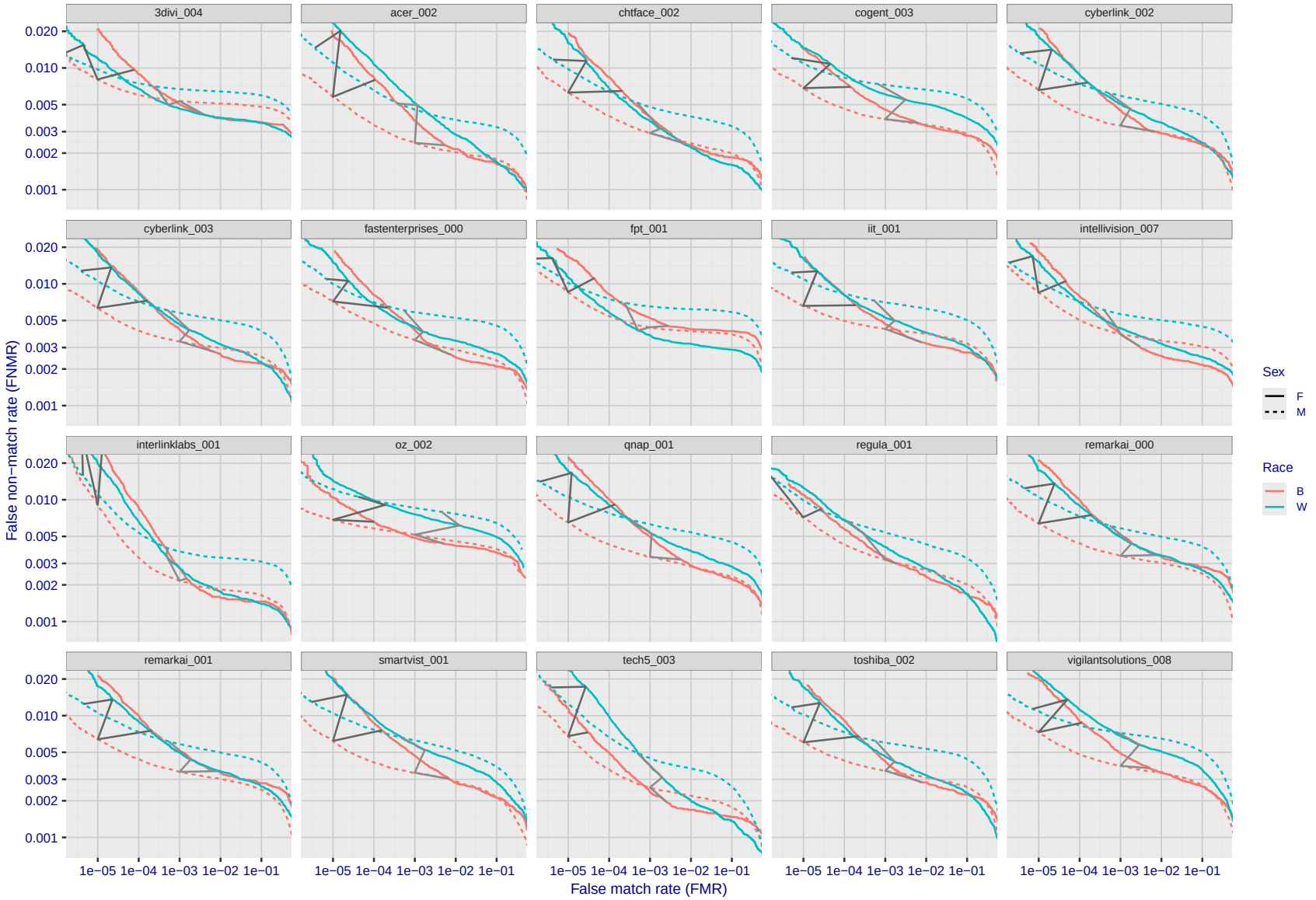
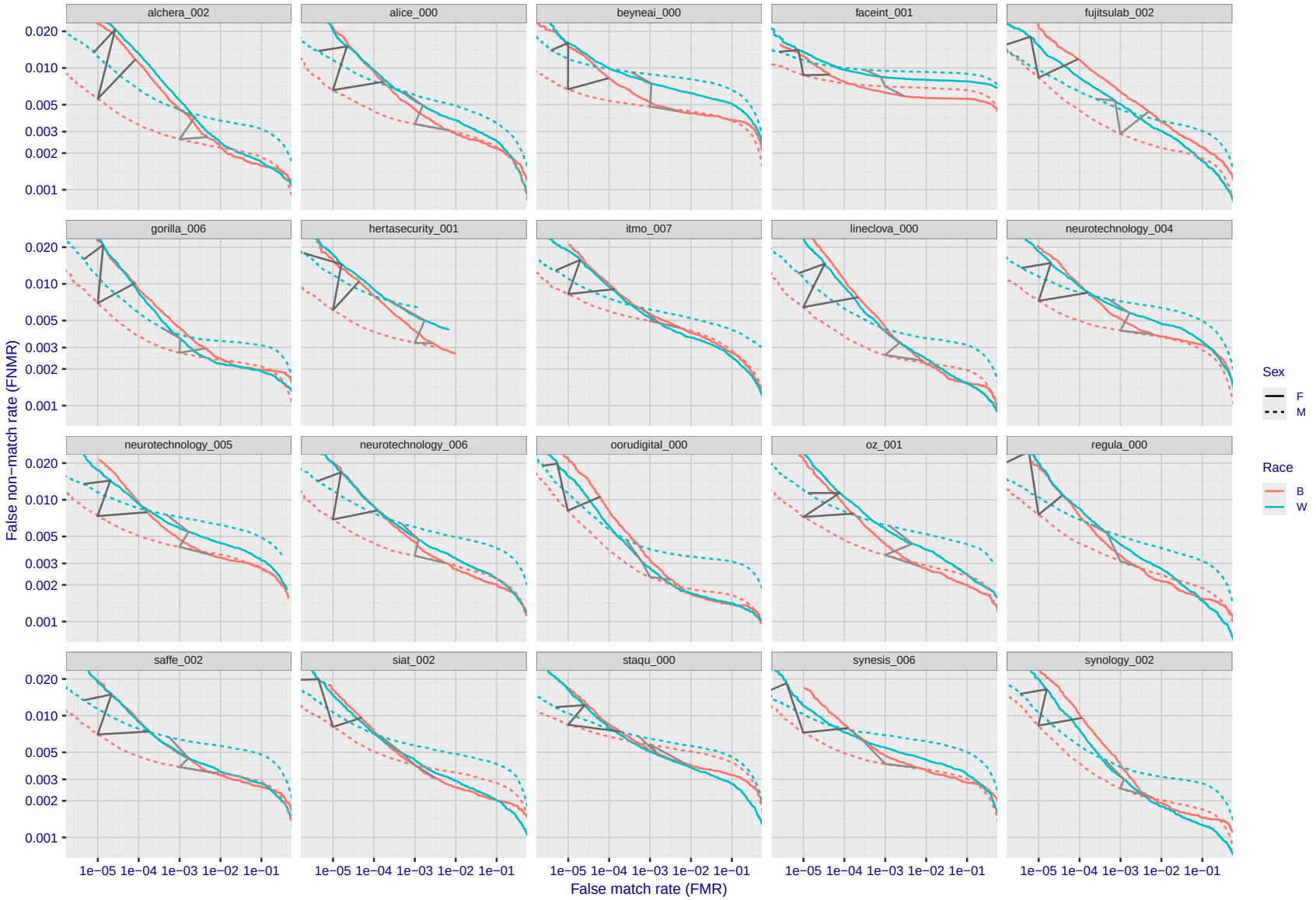


Figure 440: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*



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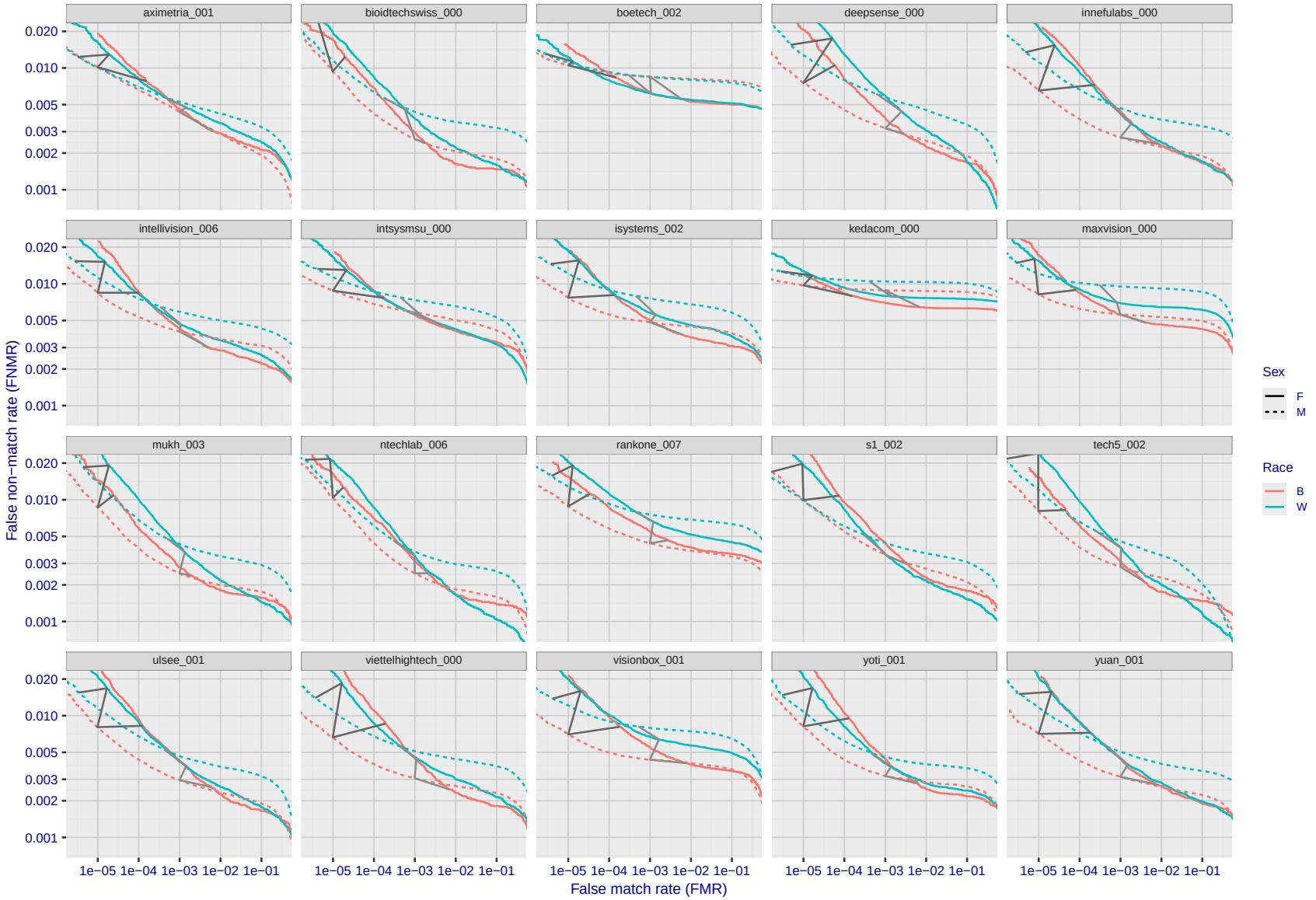


Figure 442: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

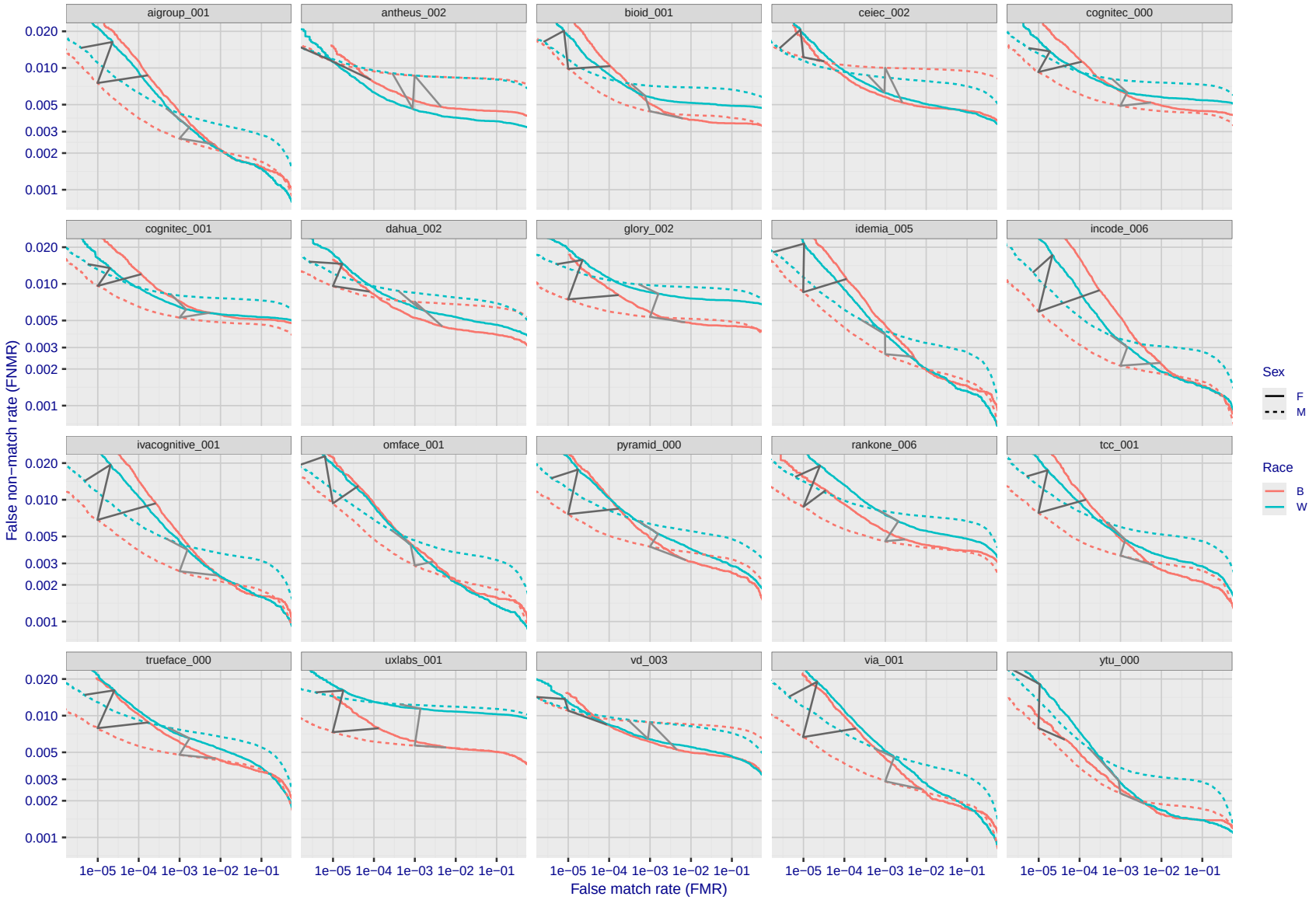
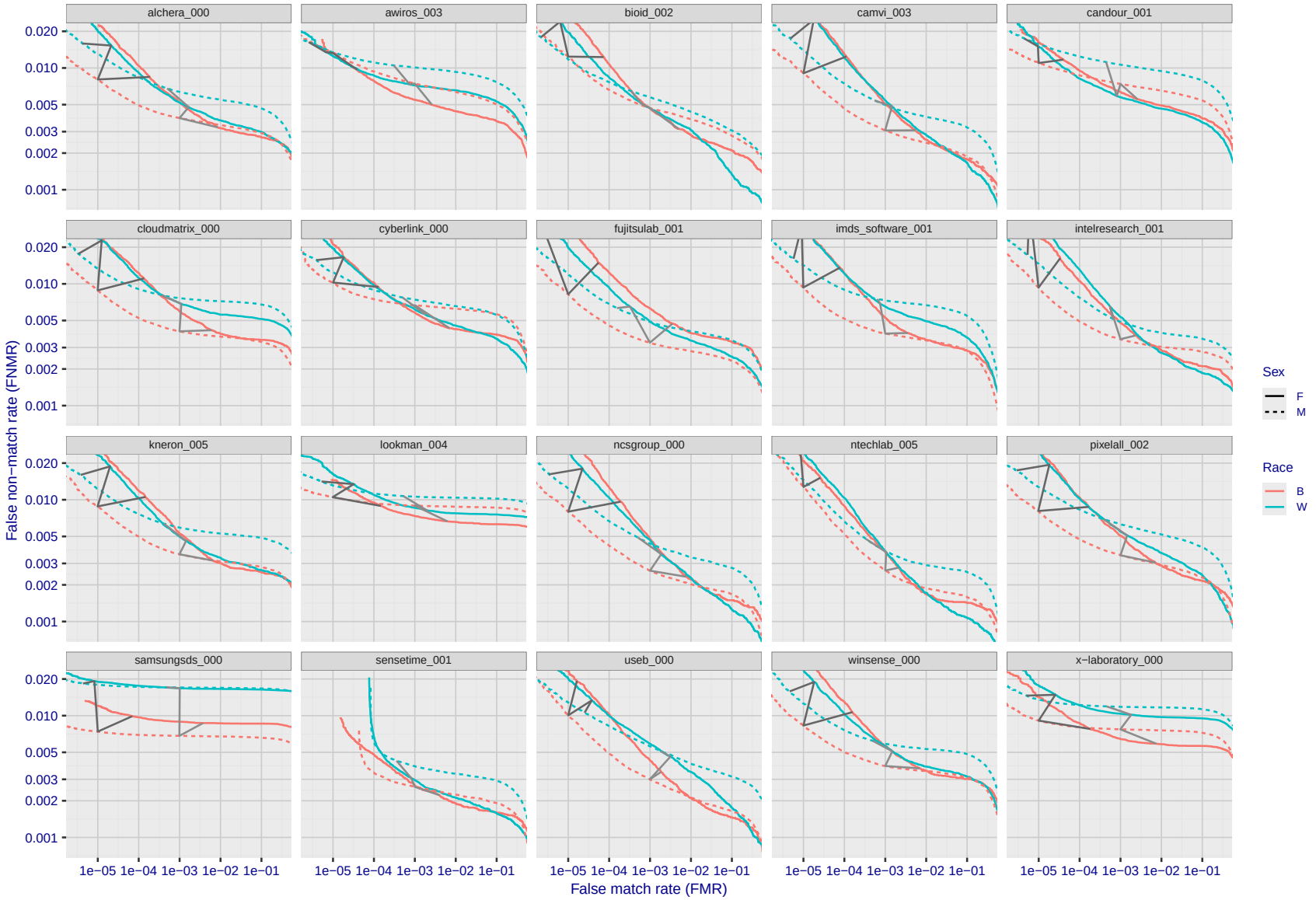


Figure 443: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*



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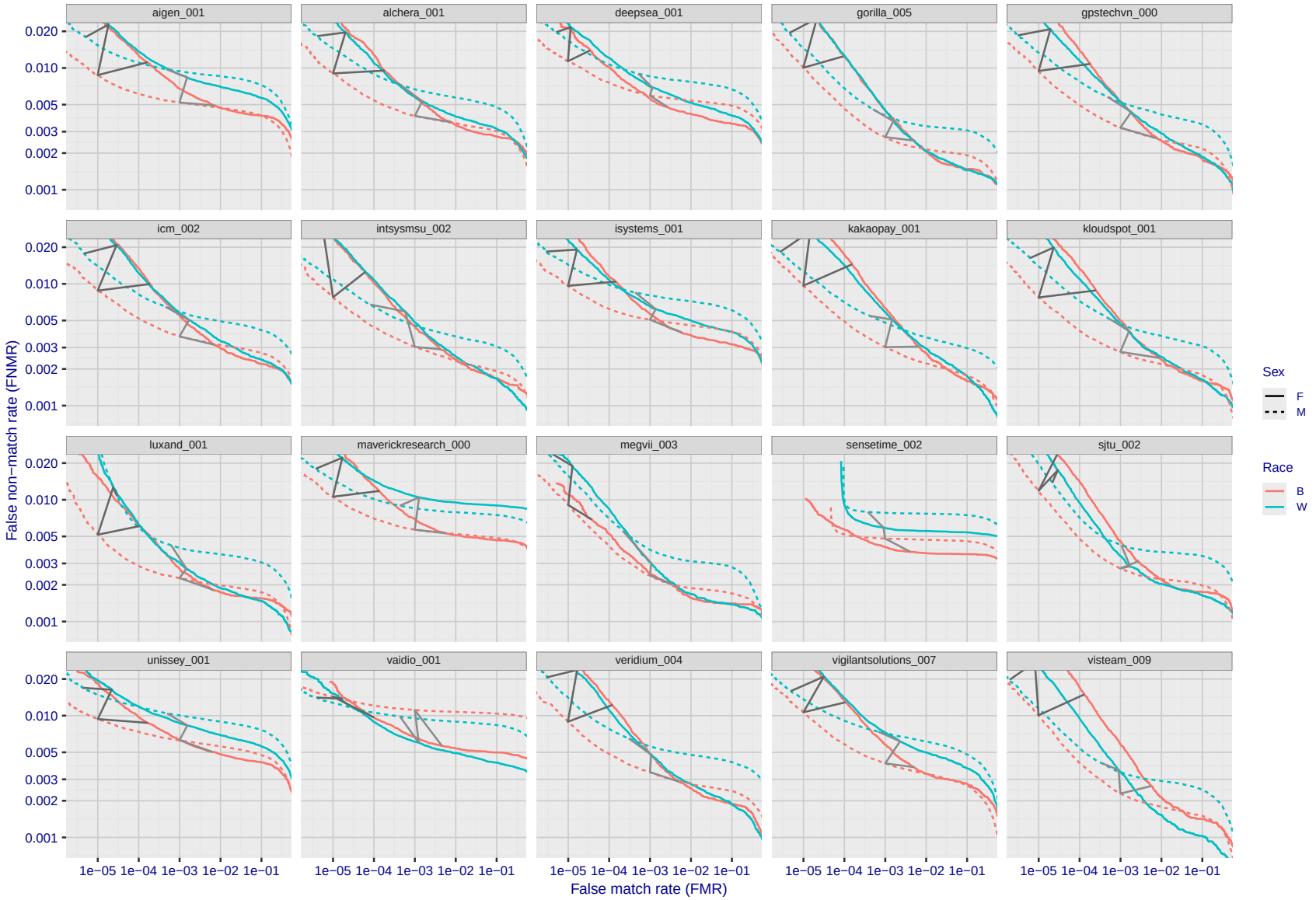

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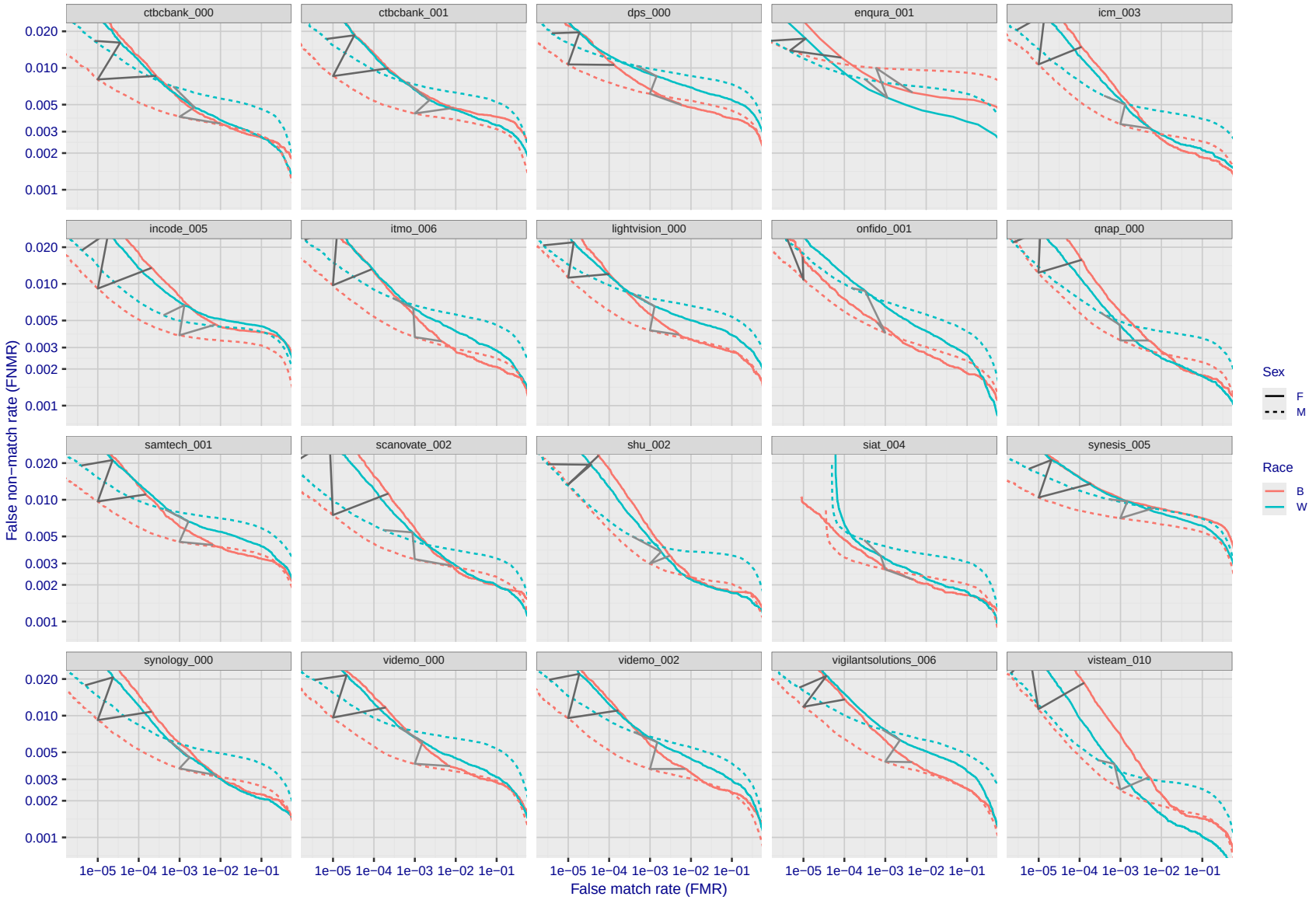

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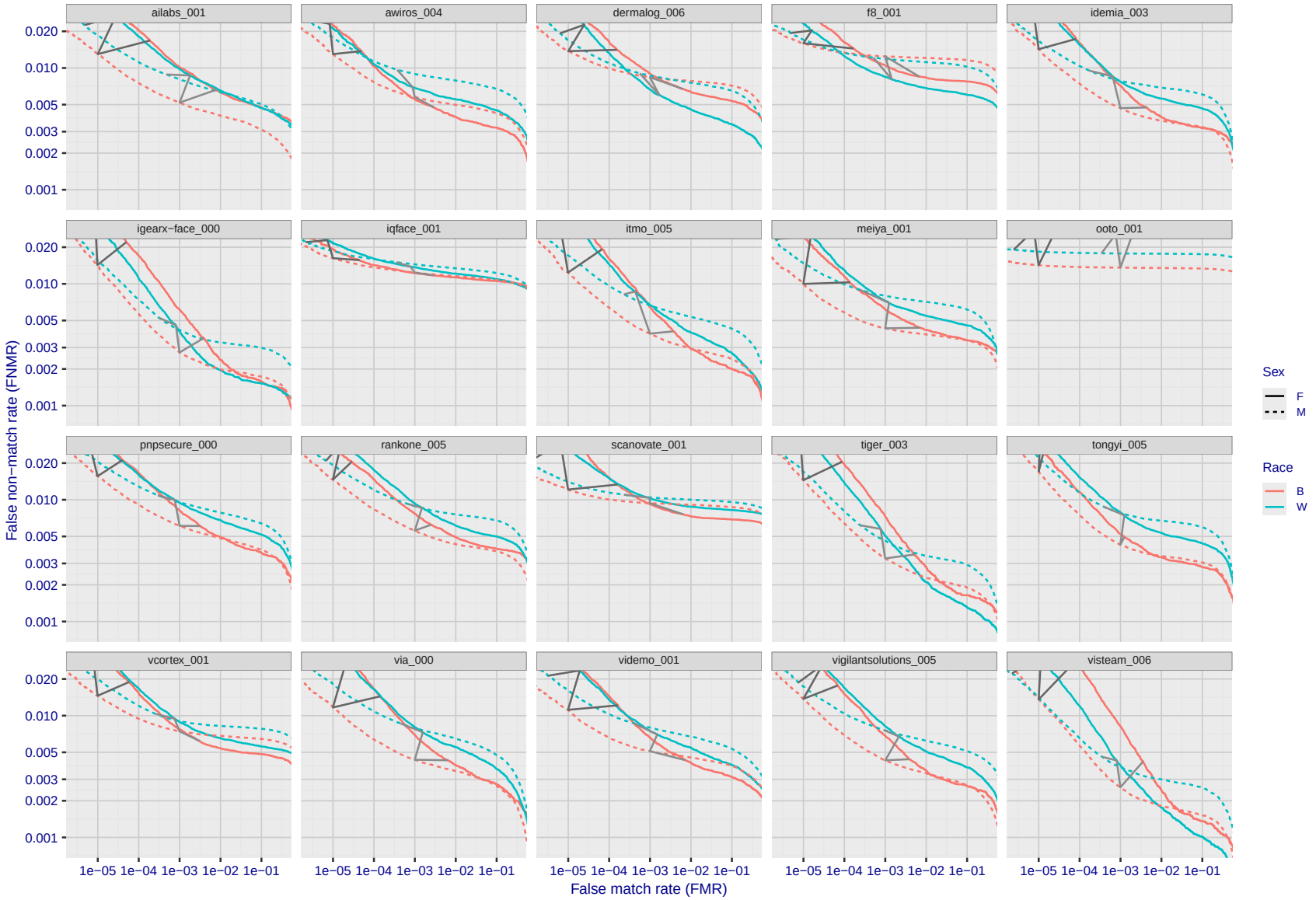
FNMR(T) "False non-match rate"
FMR(T) "False match rate"



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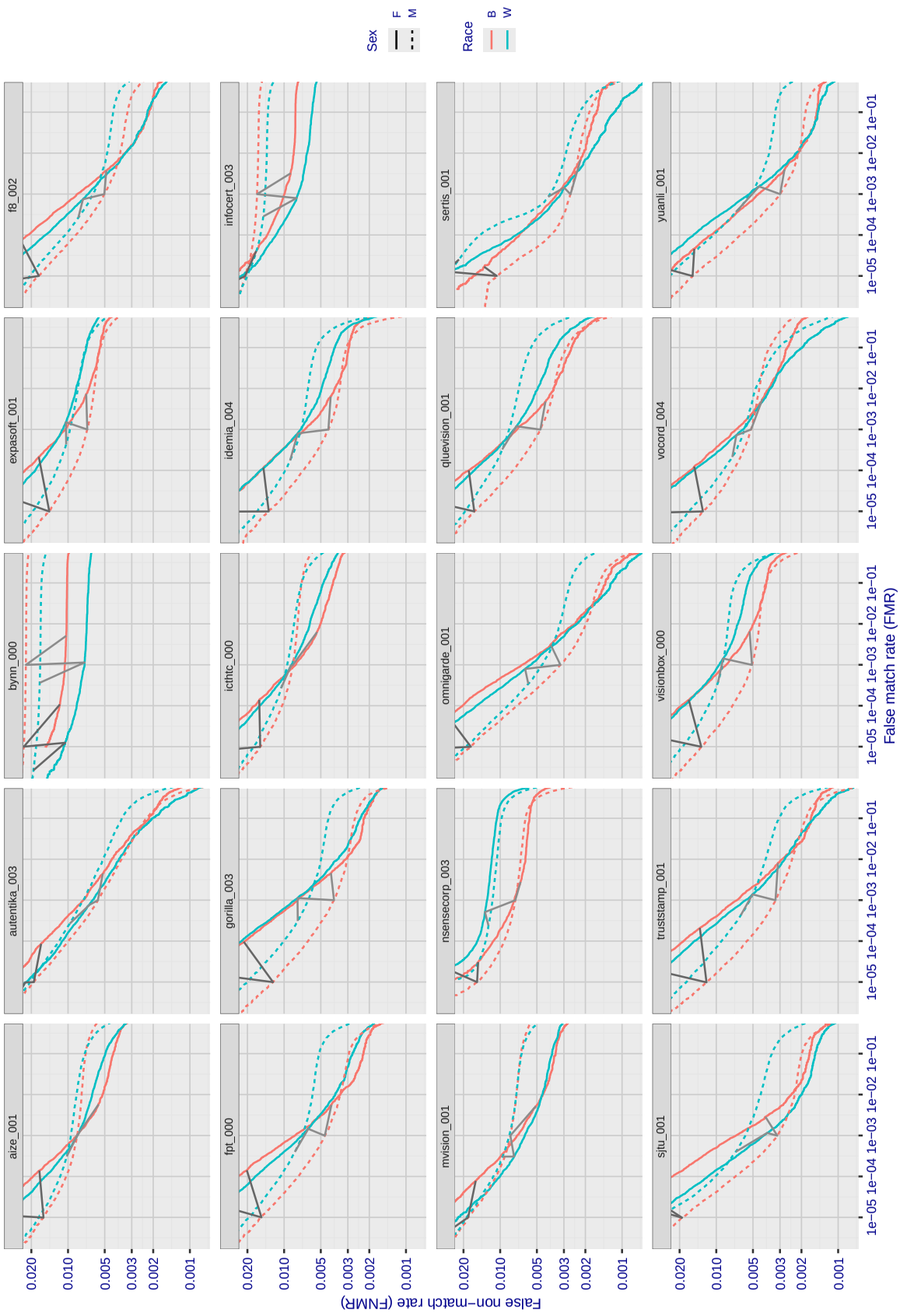

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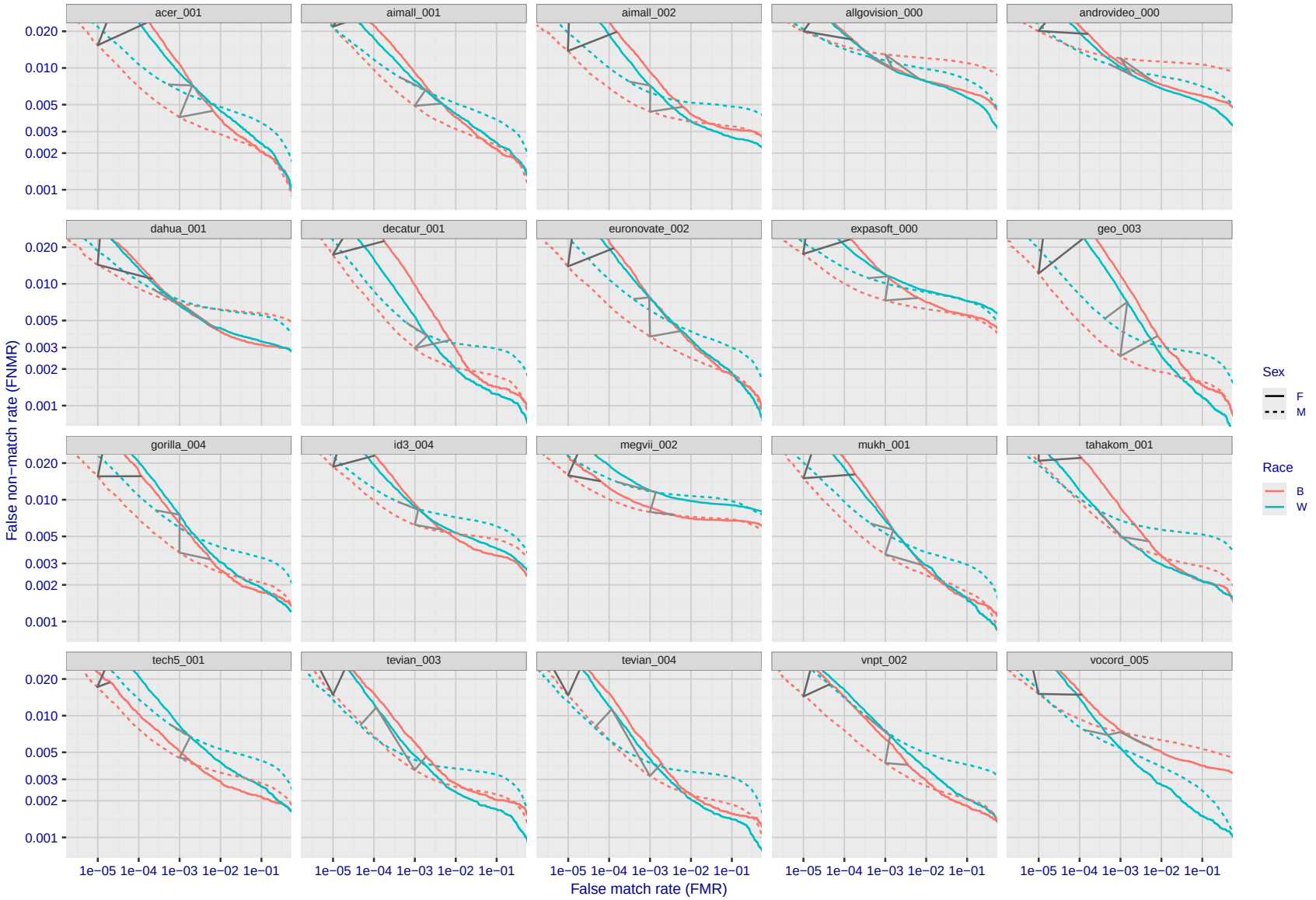

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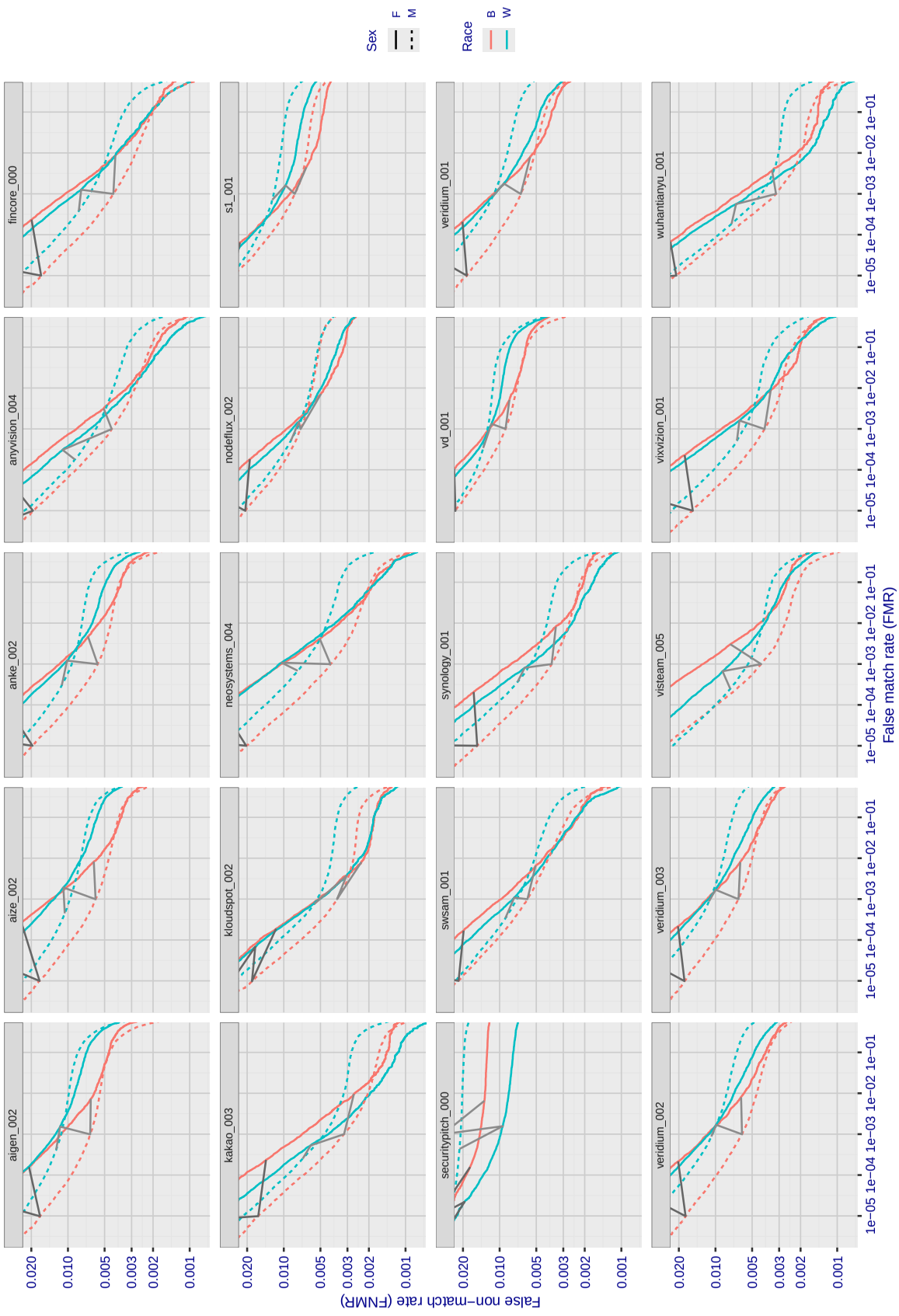

Figure 450: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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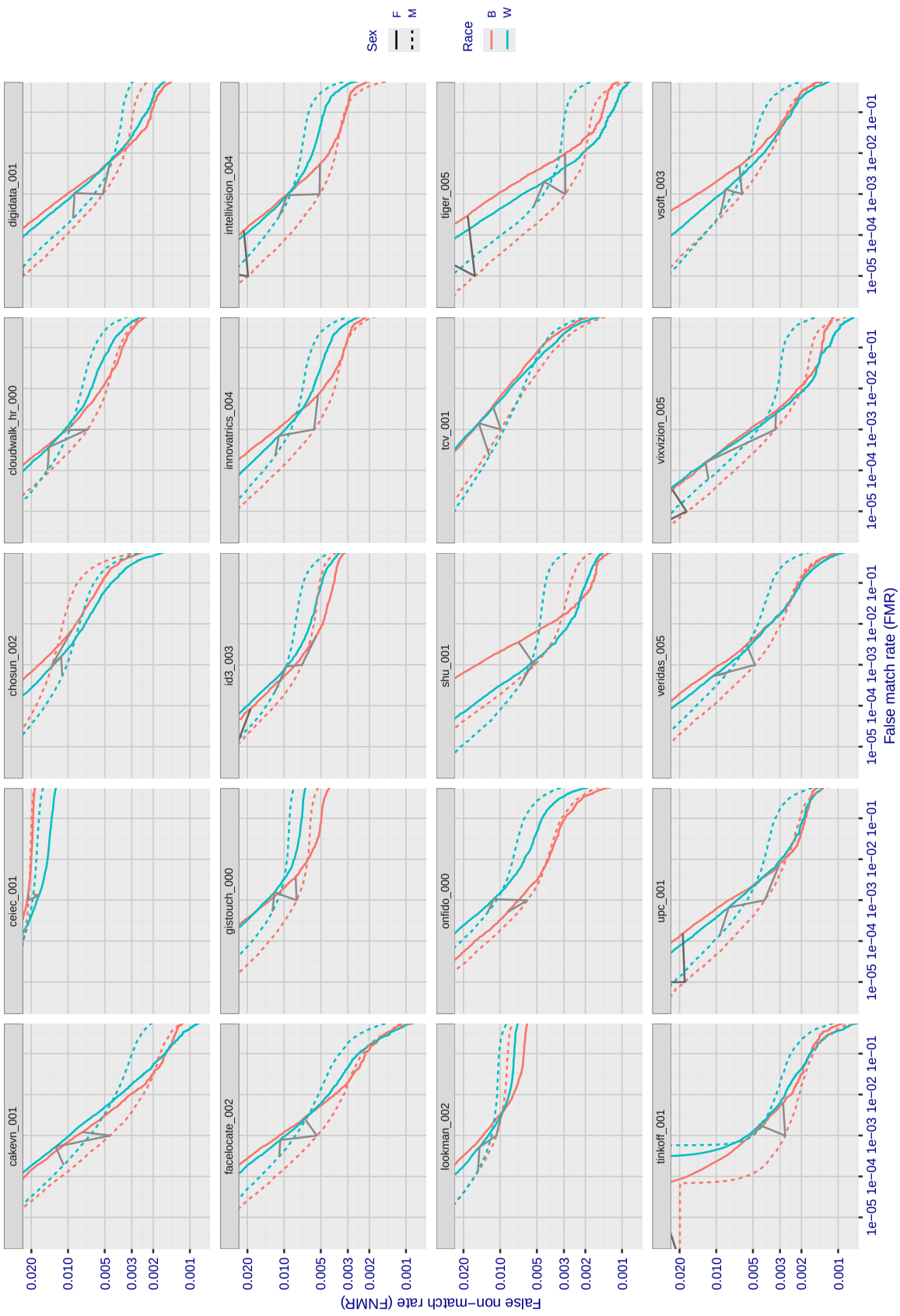

Figure 451: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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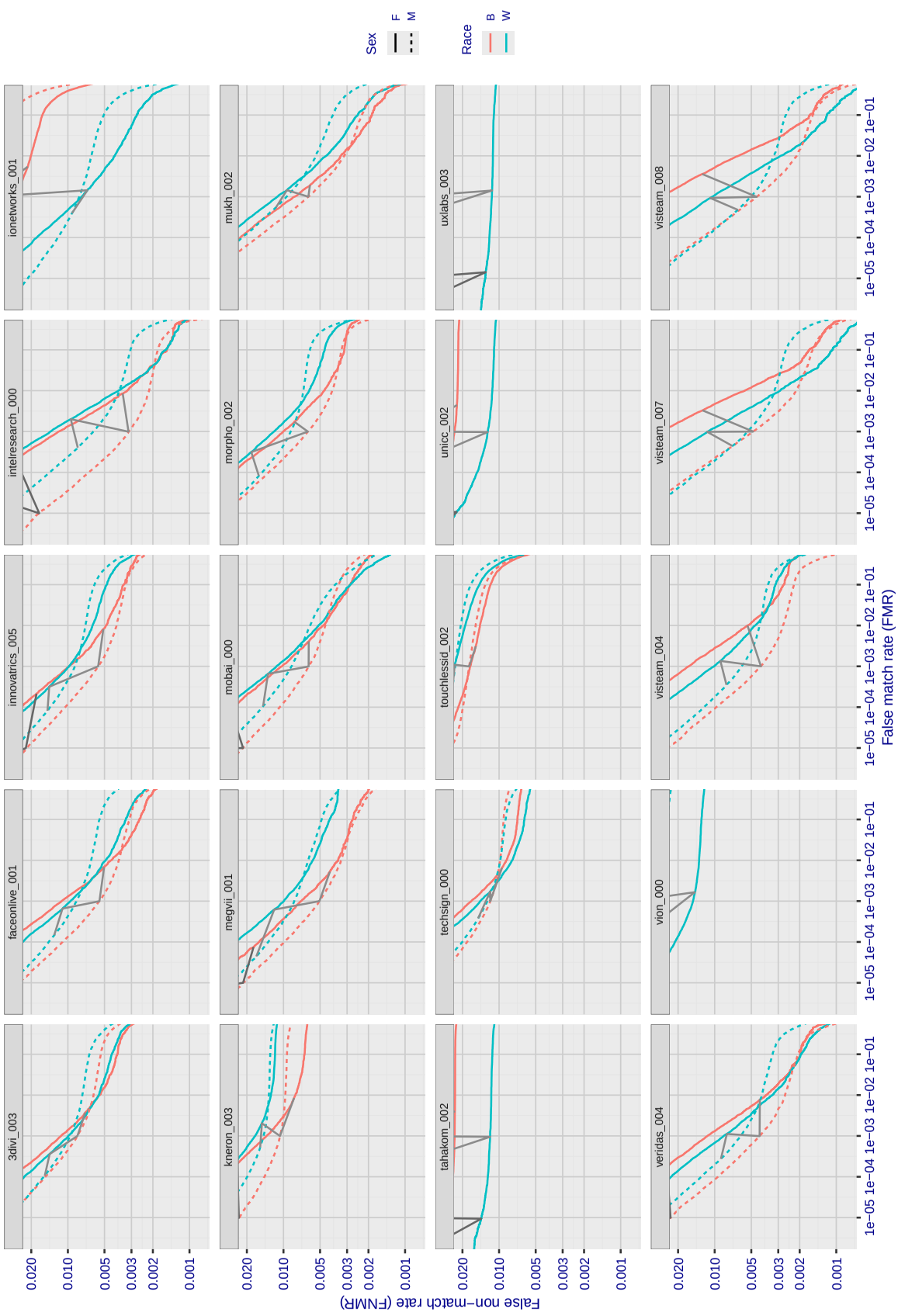

Figure 452: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

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FMR(T) "False match rate"

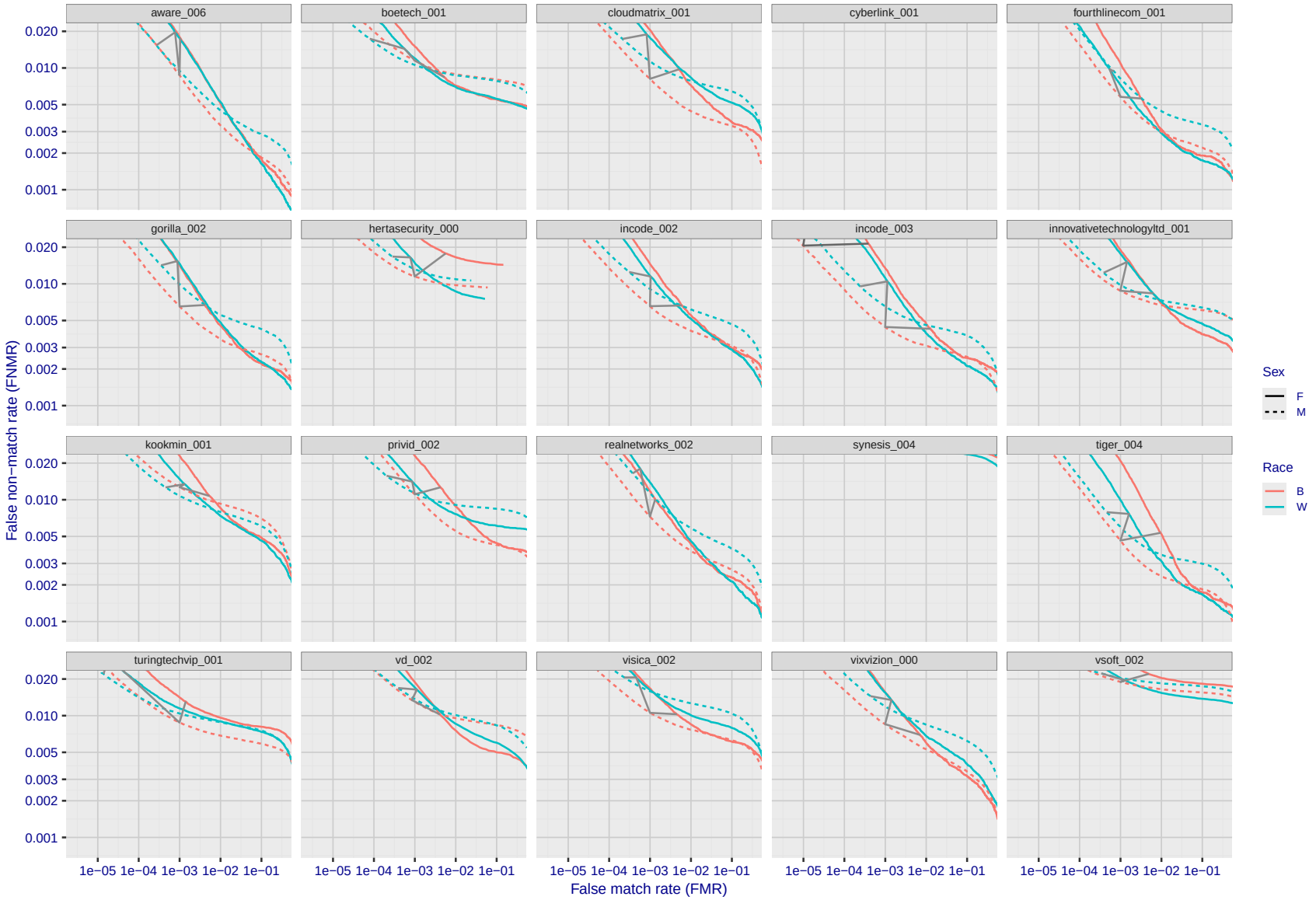

Figure 453: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

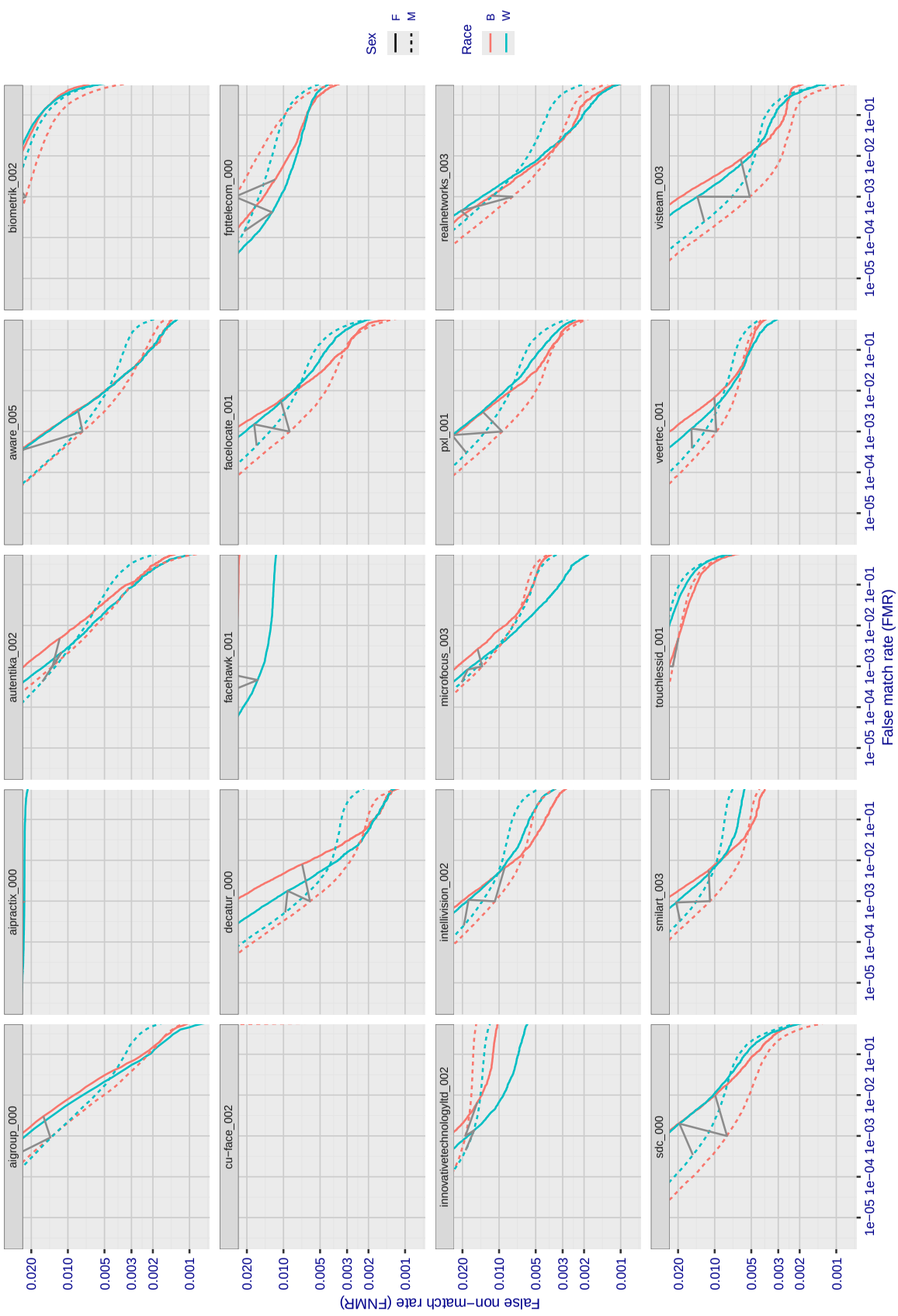

Figure 454: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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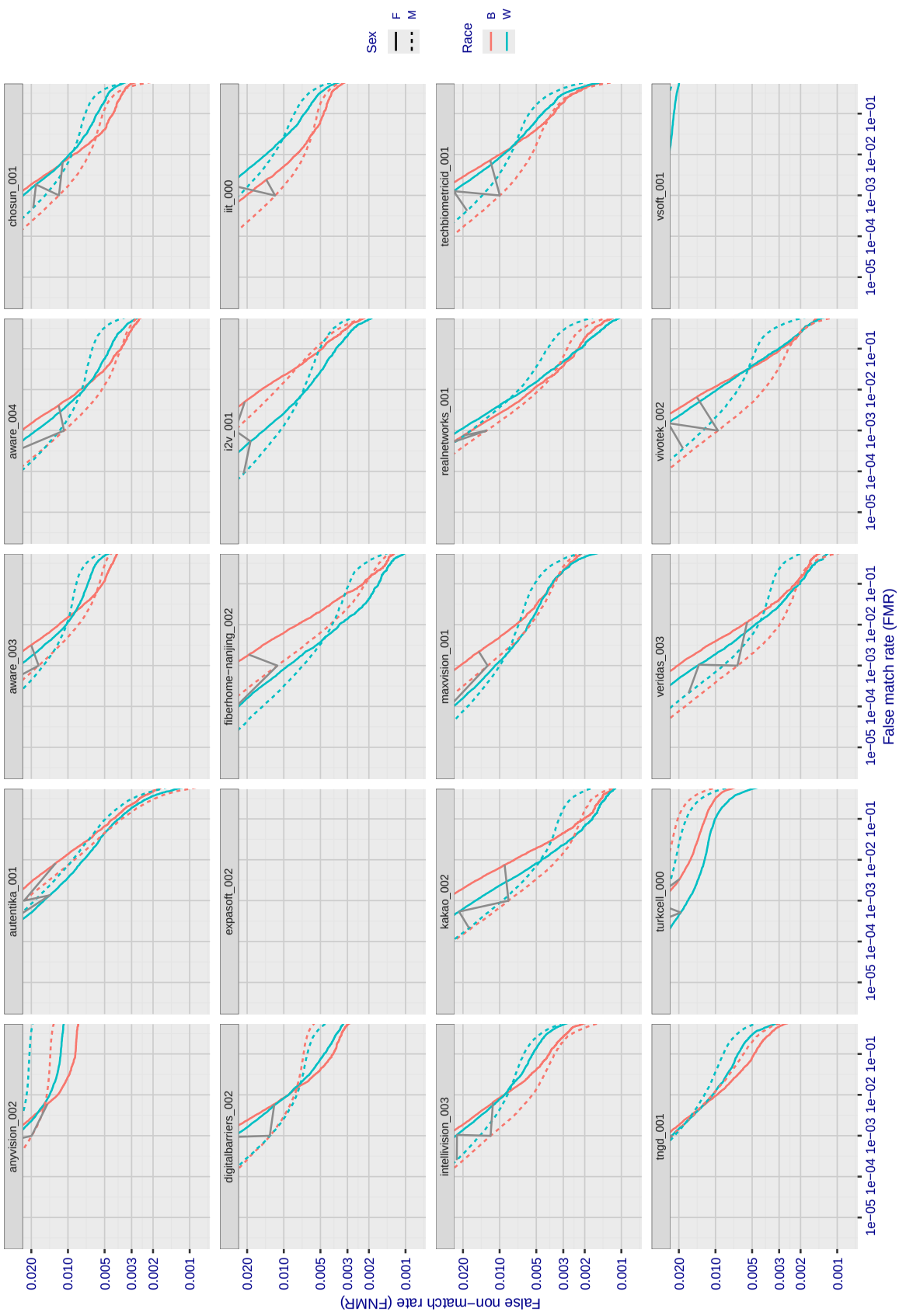

Figure 455: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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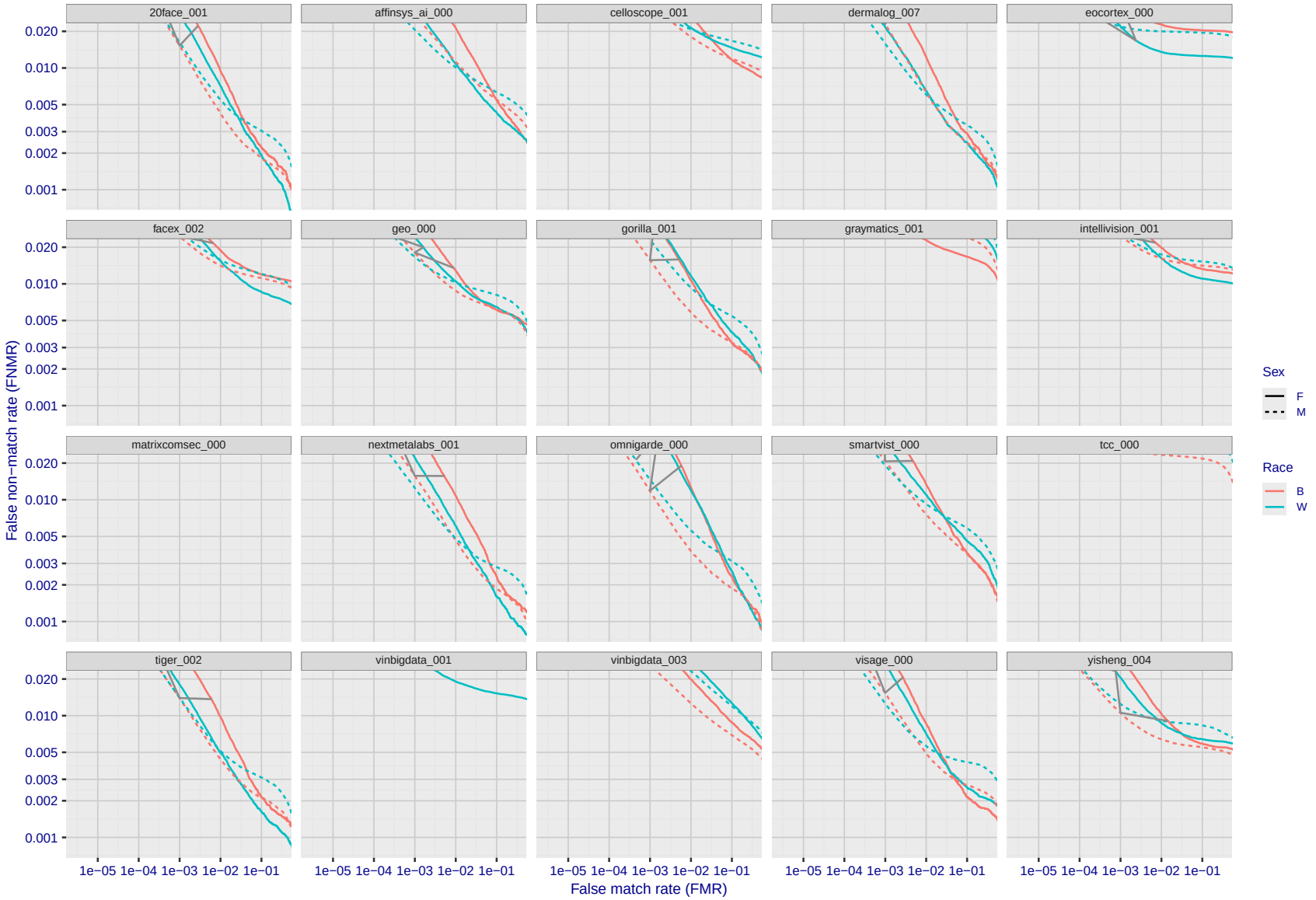

Figure 456: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

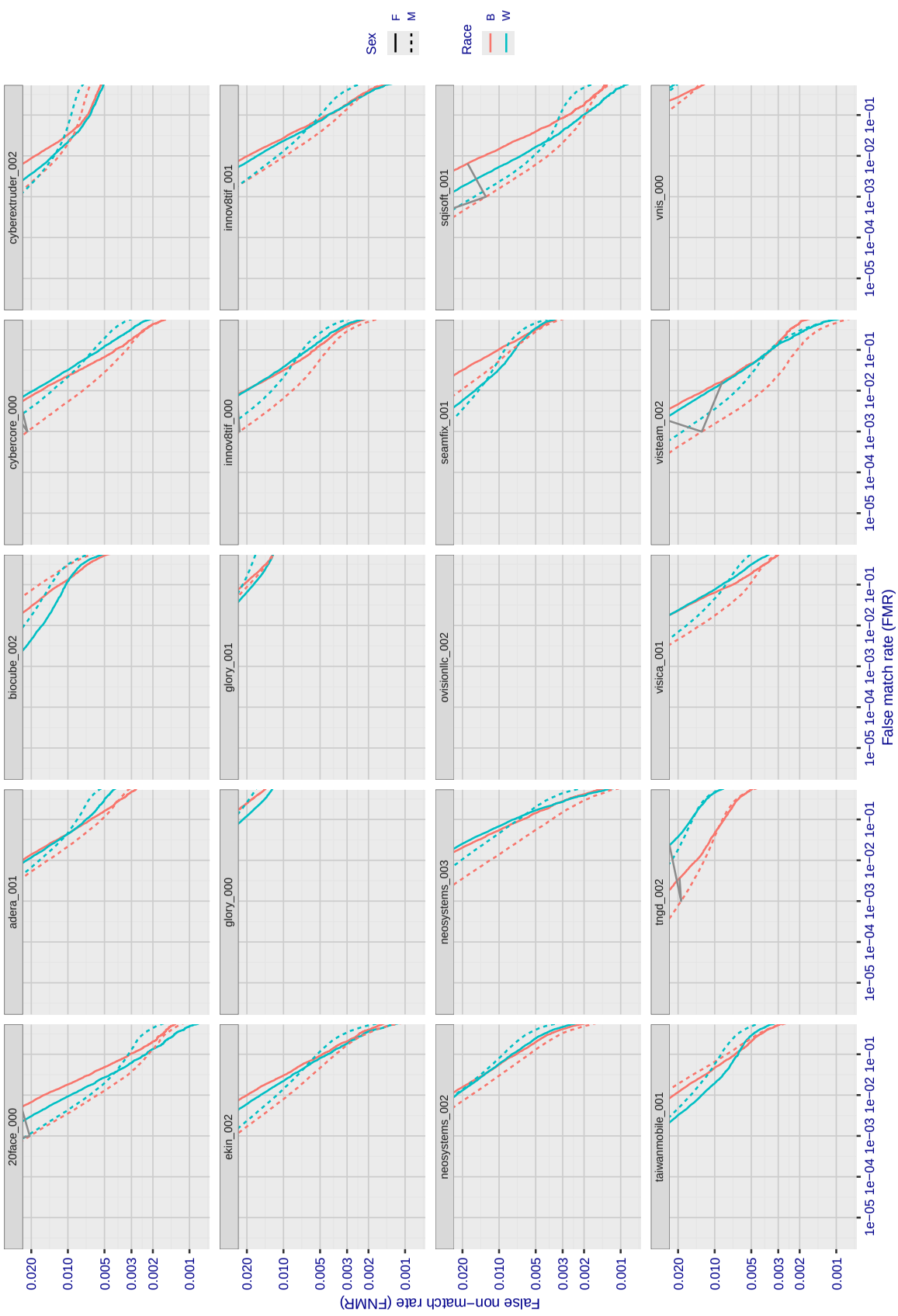

Figure 457: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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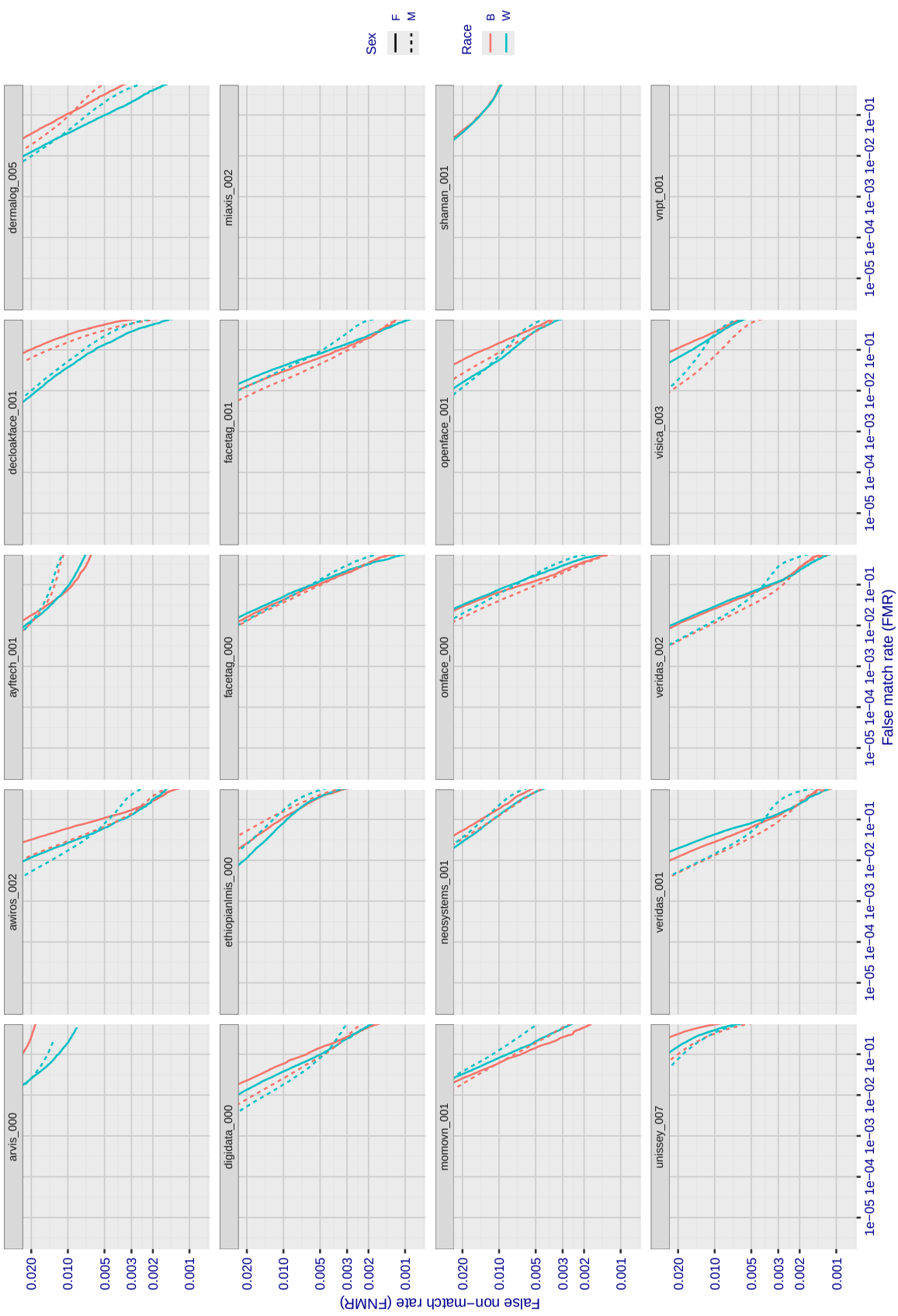

Figure 458: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

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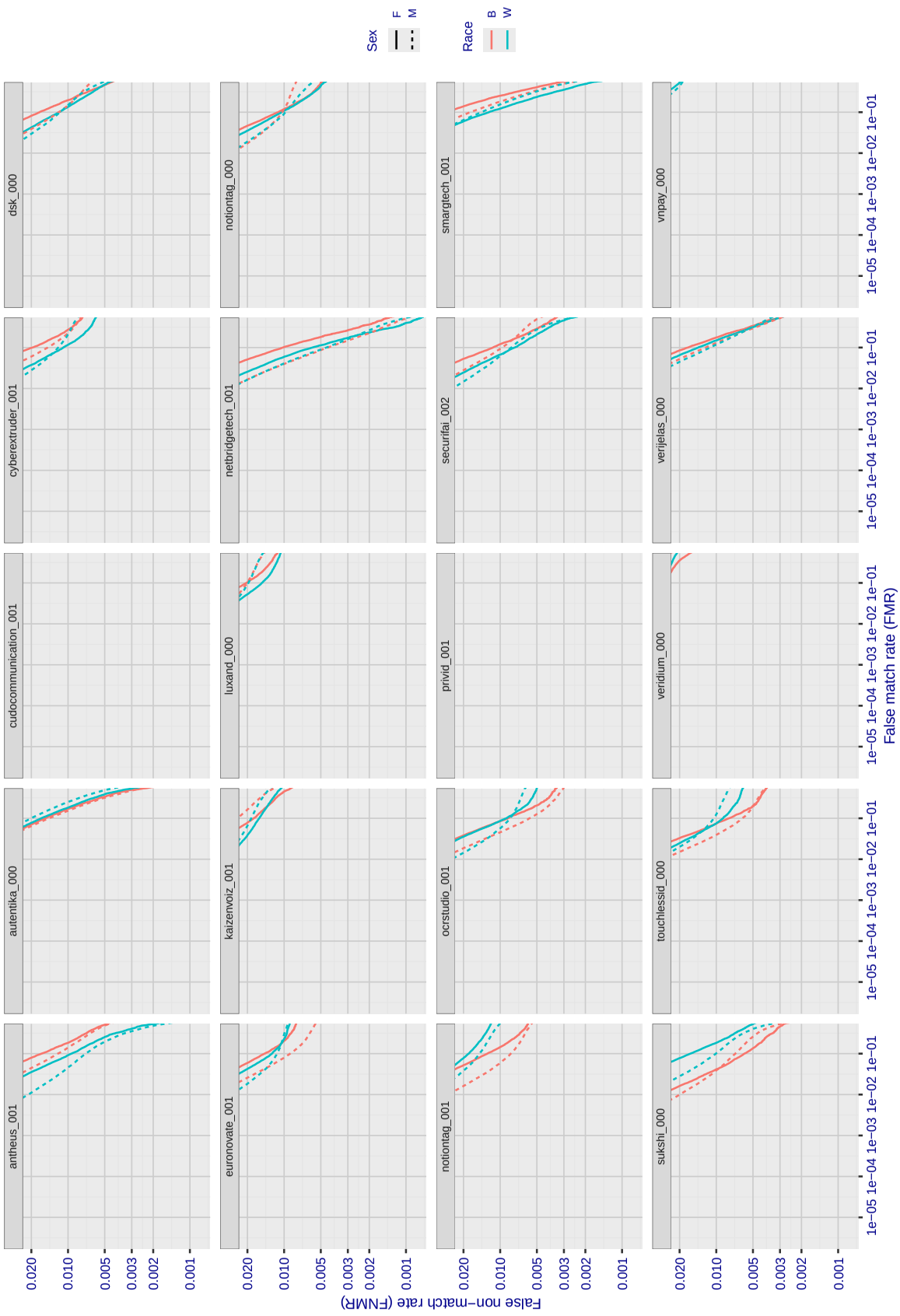

Figure 459: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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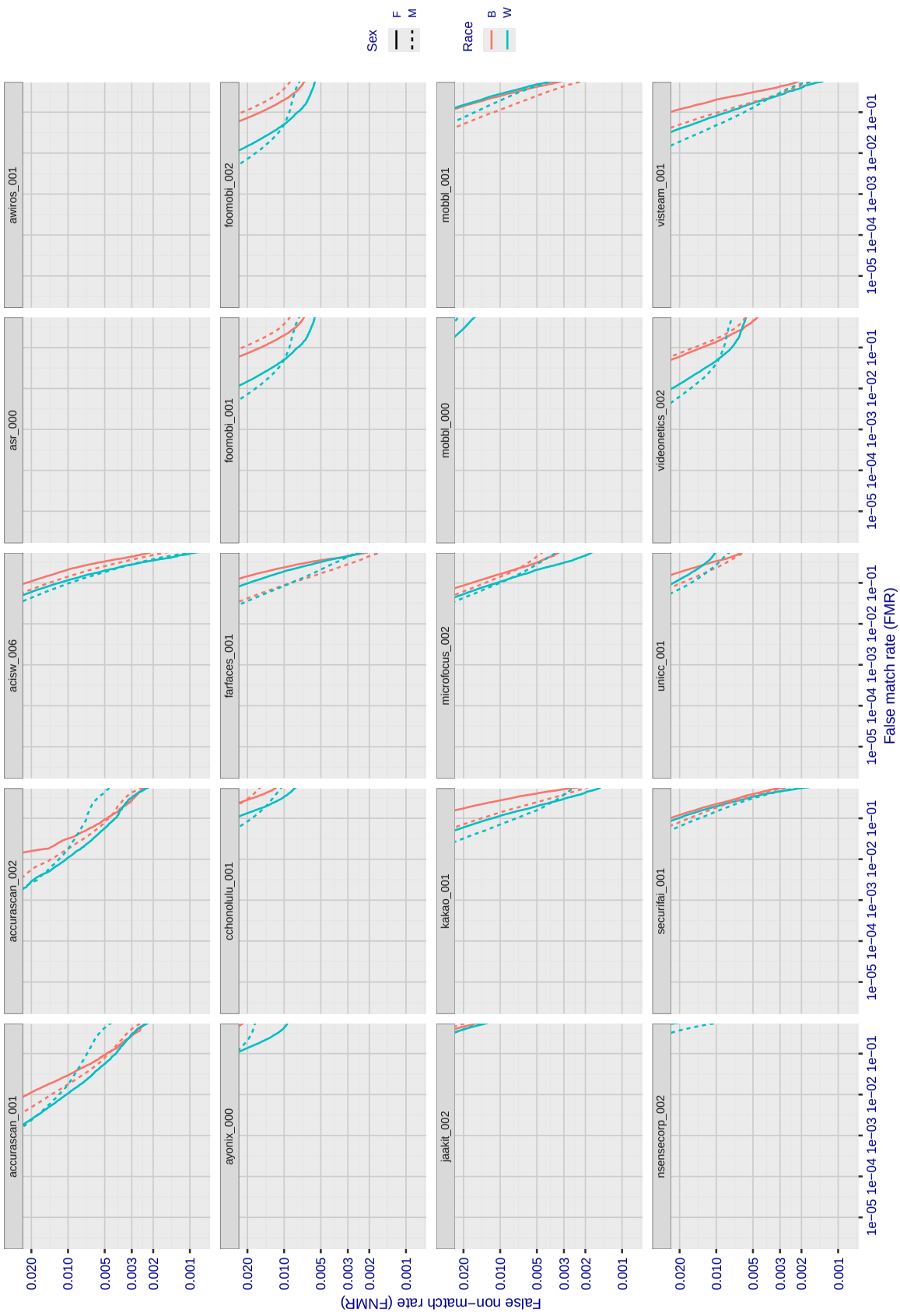

Figure 460: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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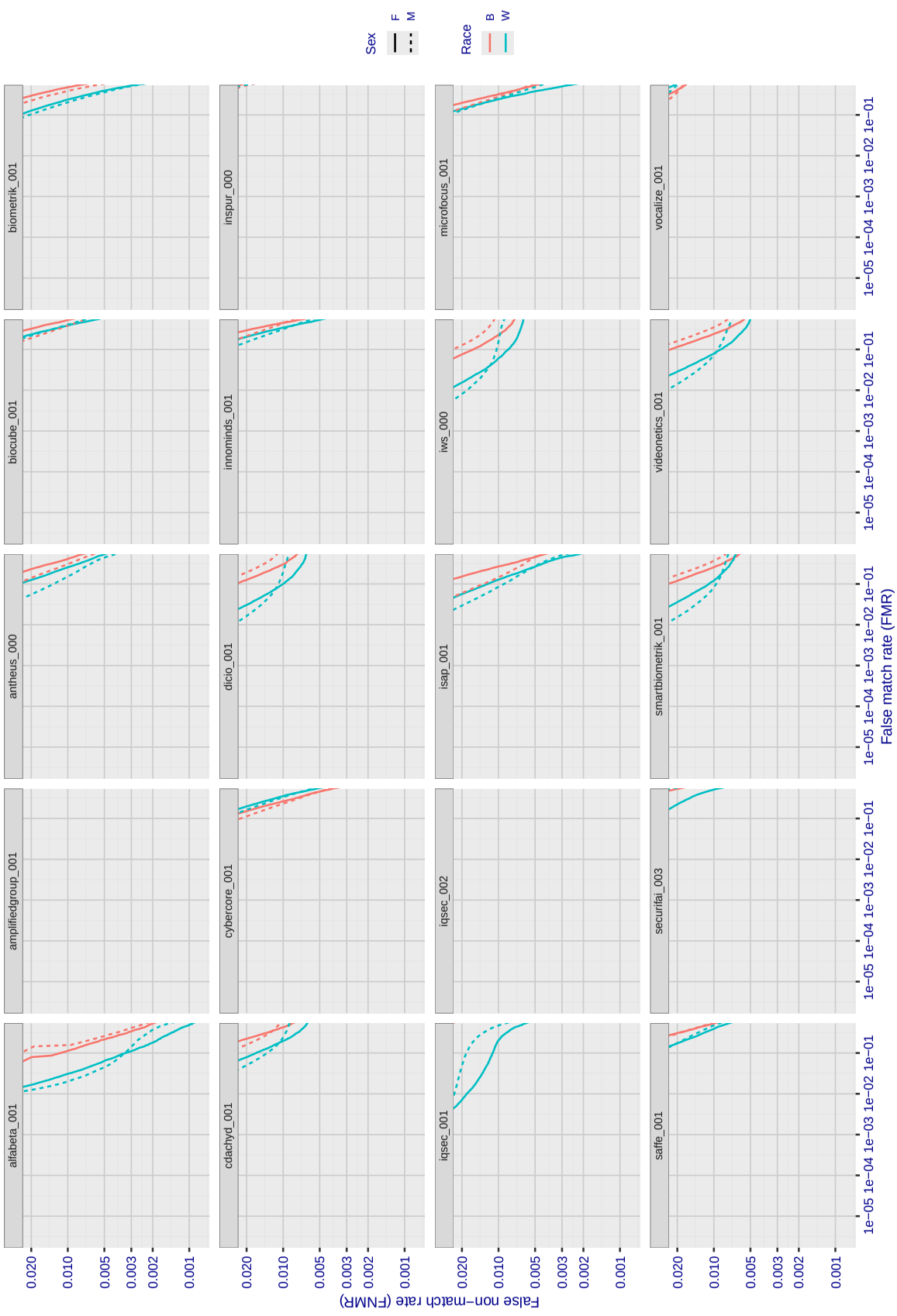

Figure 461: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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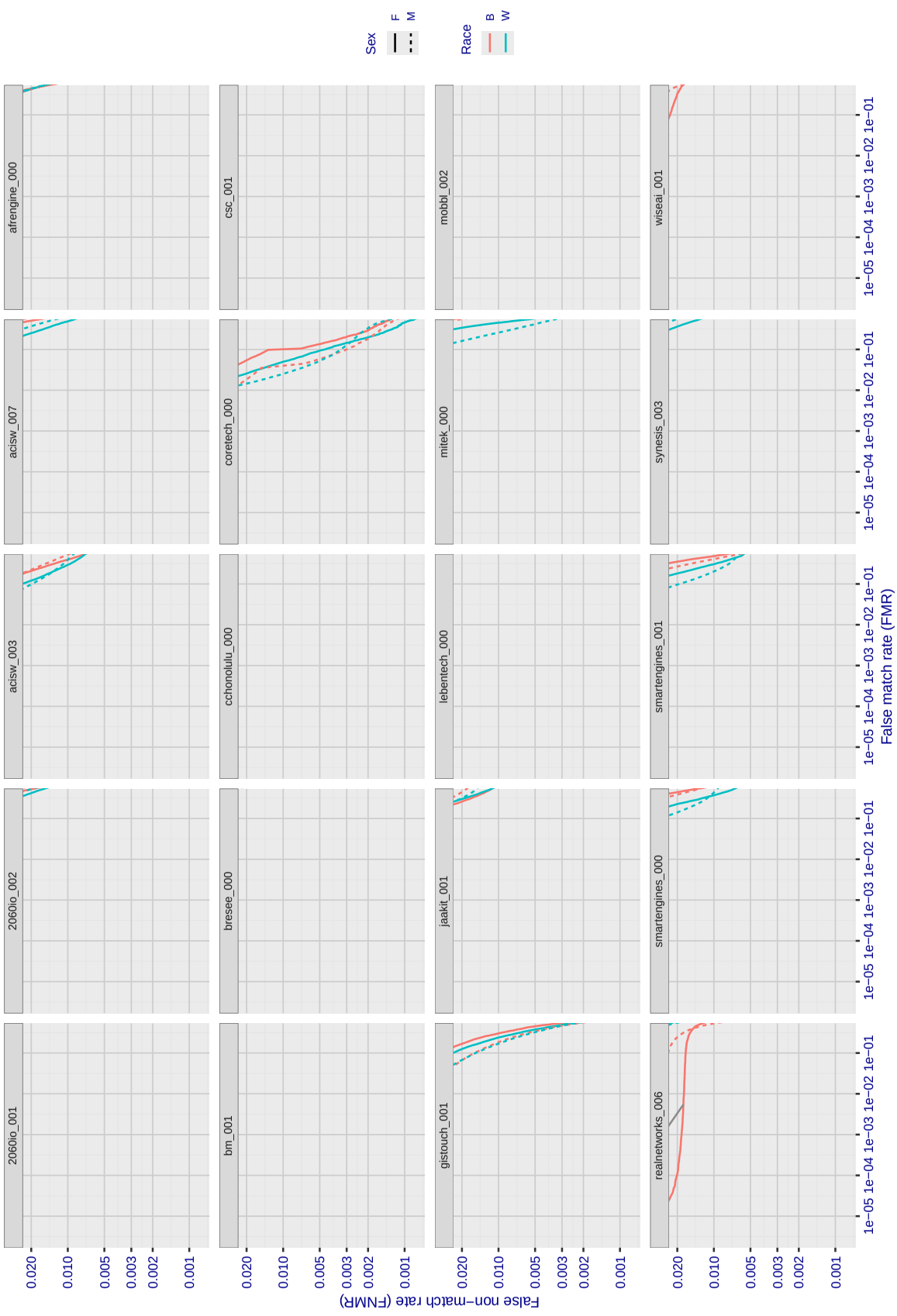

Figure 462: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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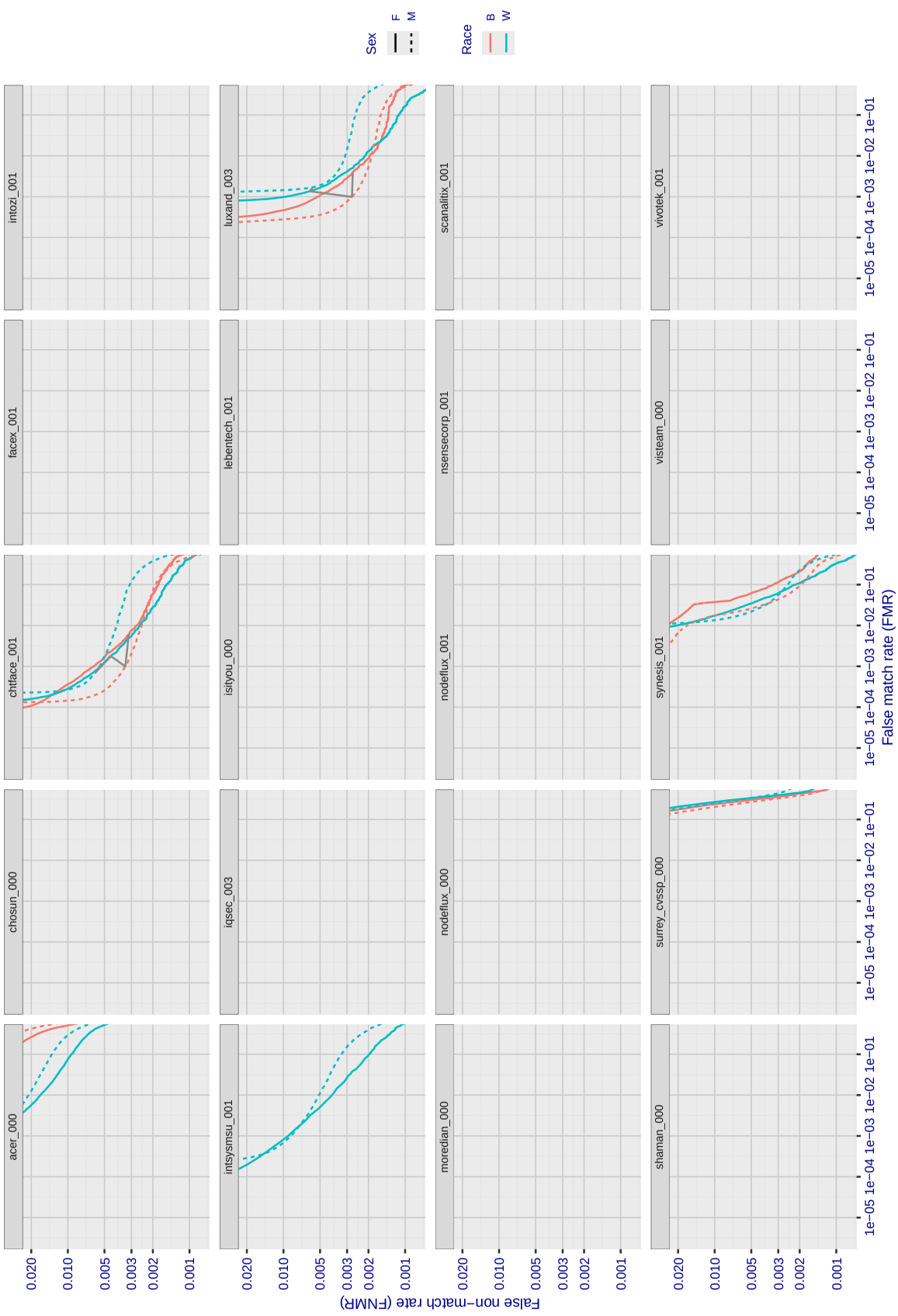

Figure 463: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 623. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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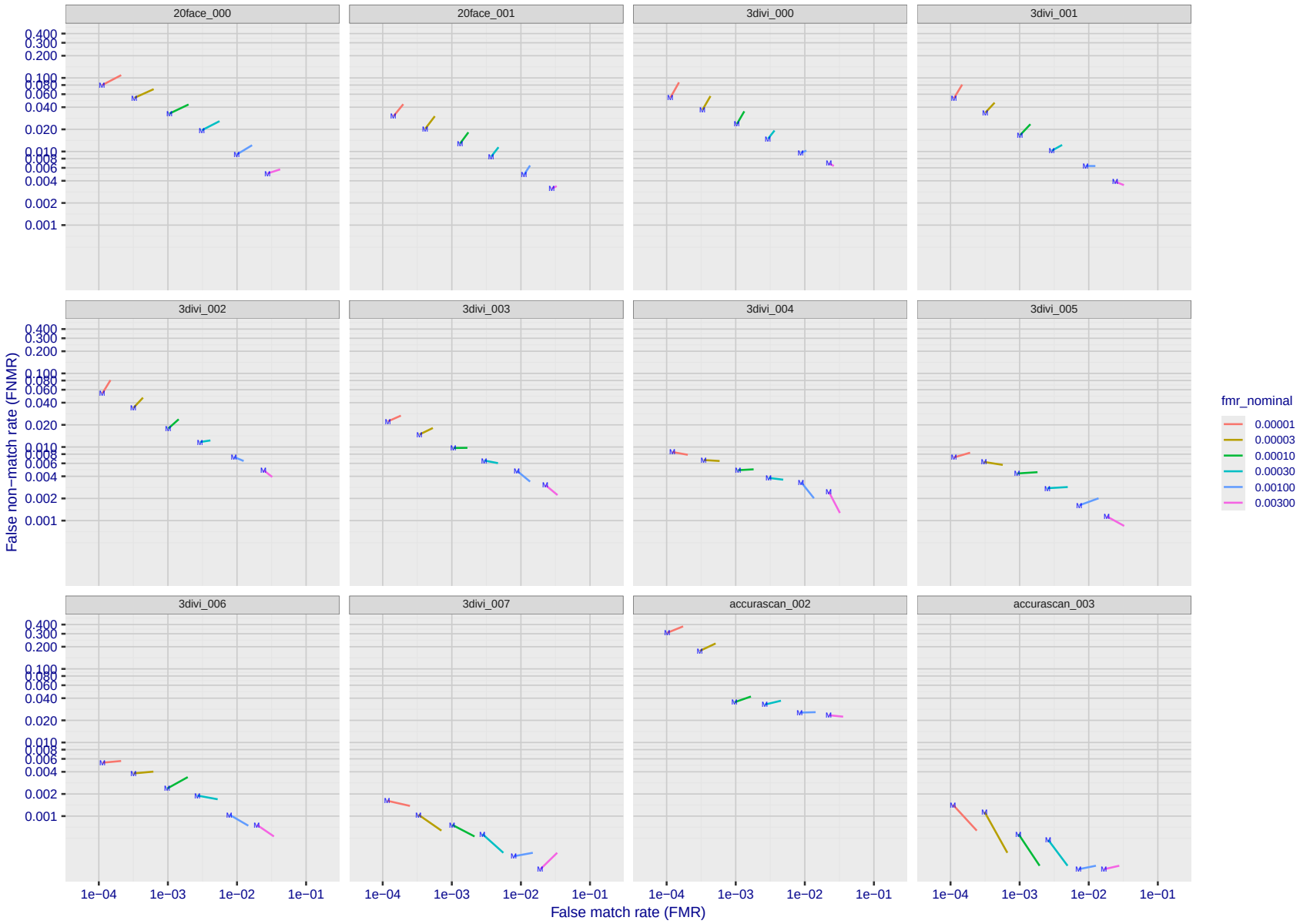

Figure 465: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

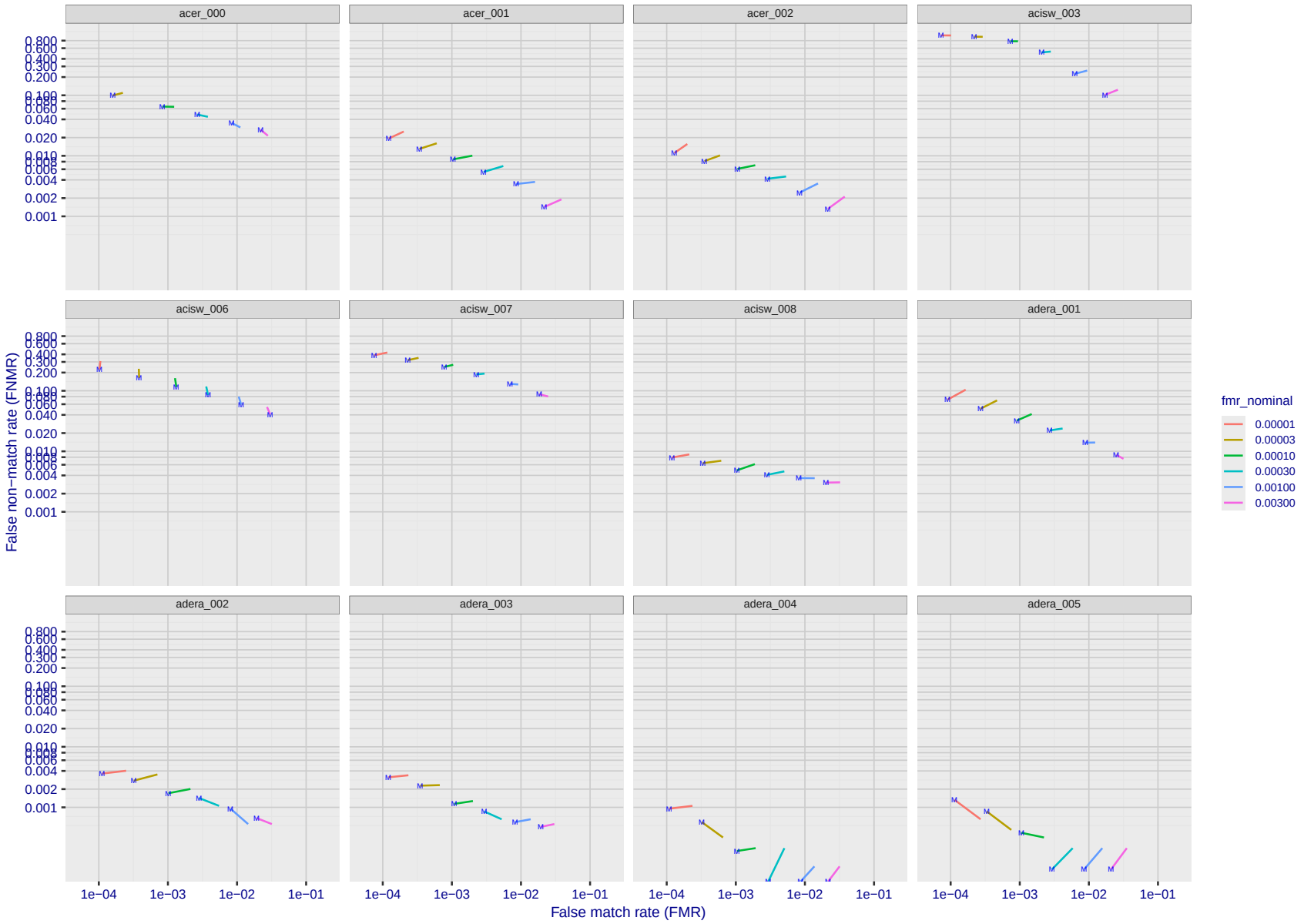

Figure 466: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

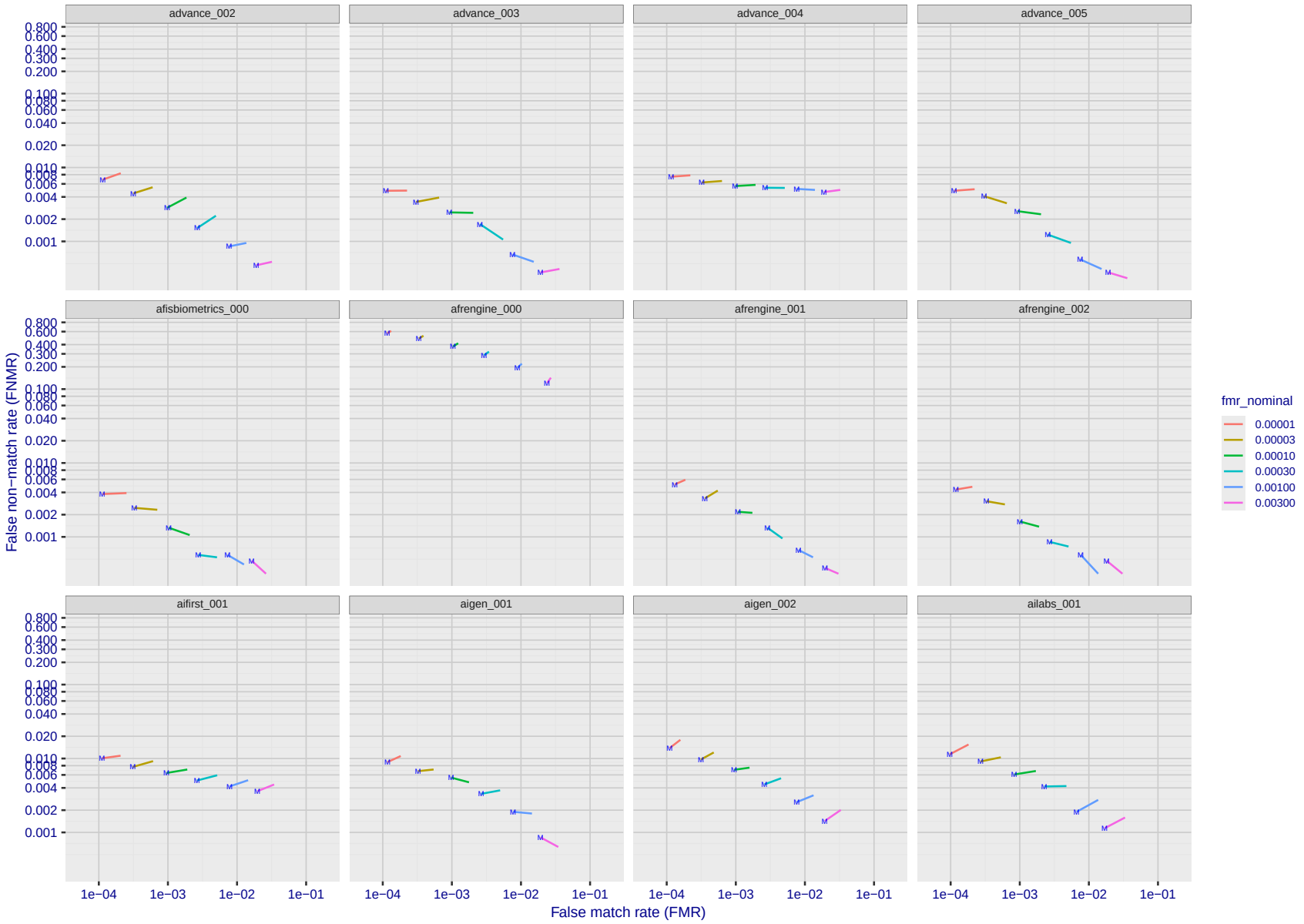

Figure 467: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

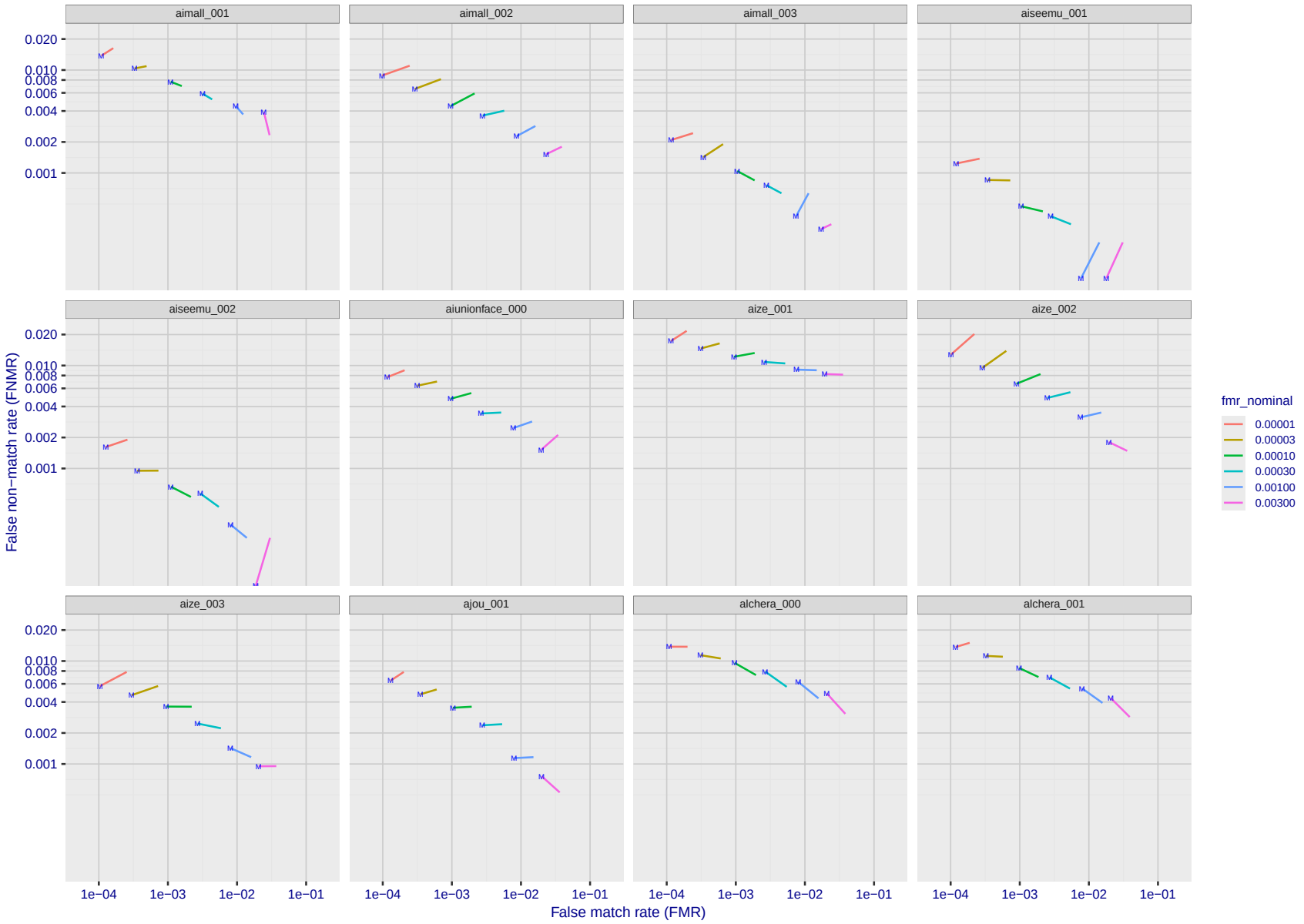

Figure 468: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

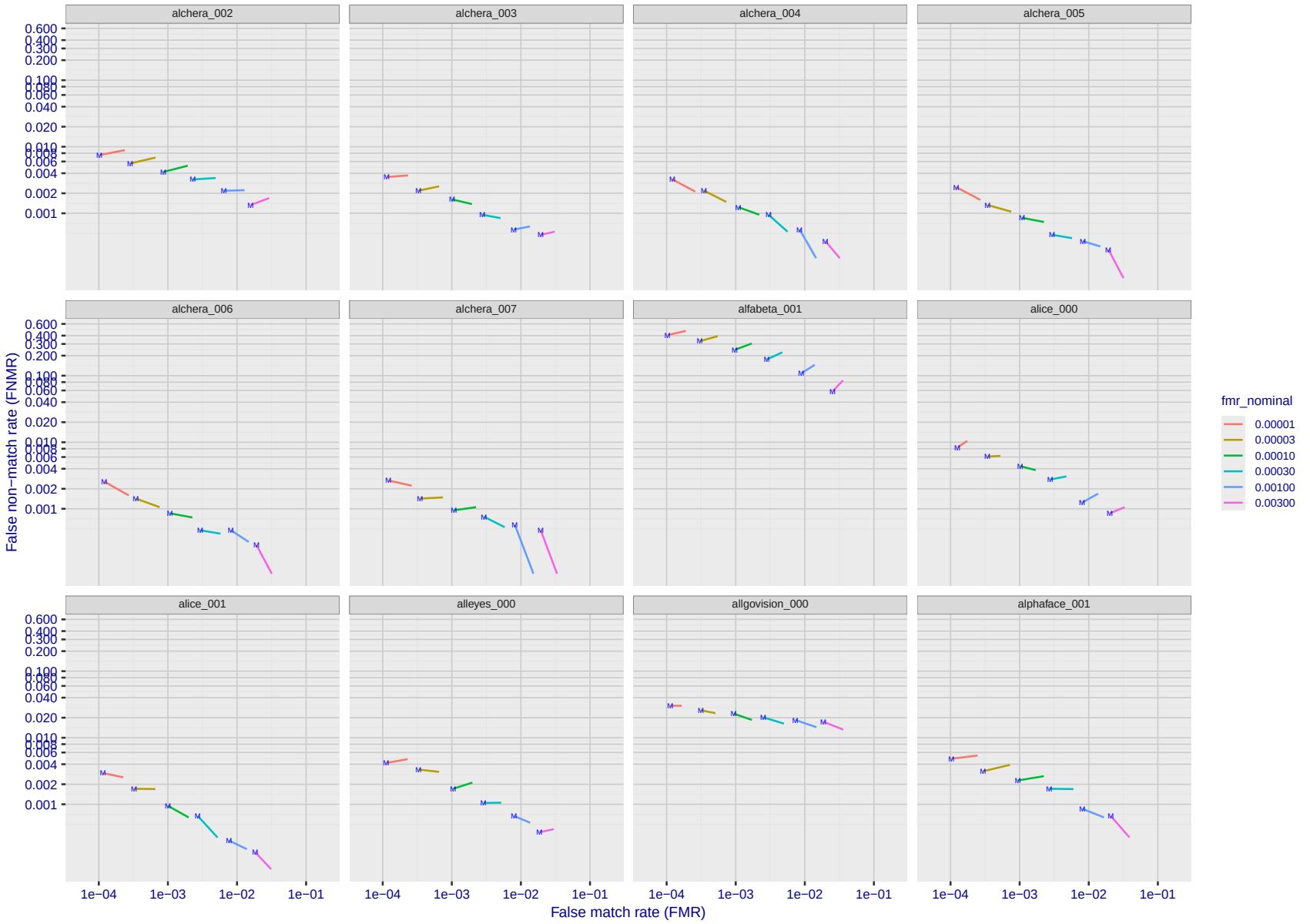

Figure 469: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

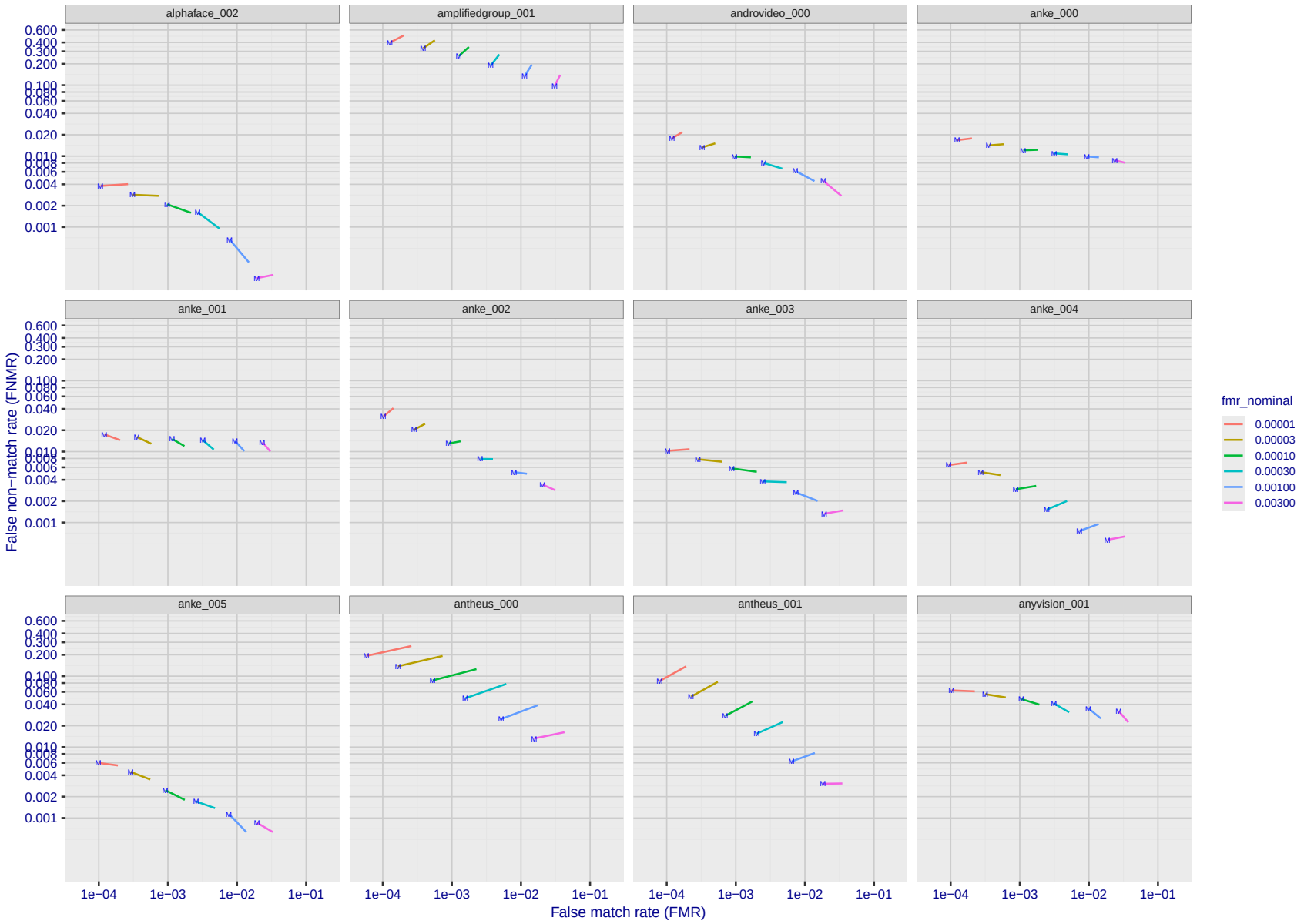

Figure 470: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

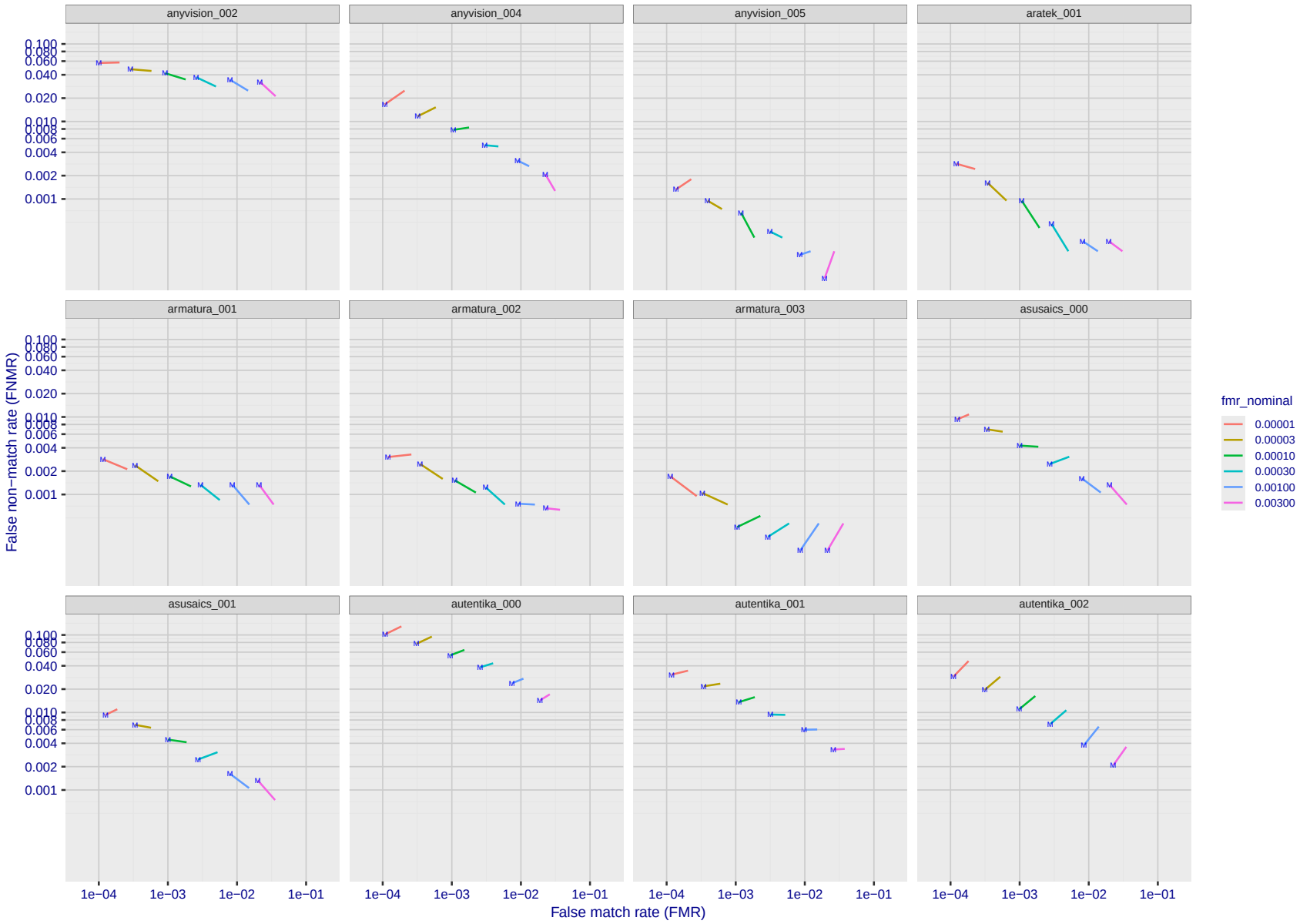

Figure 471: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

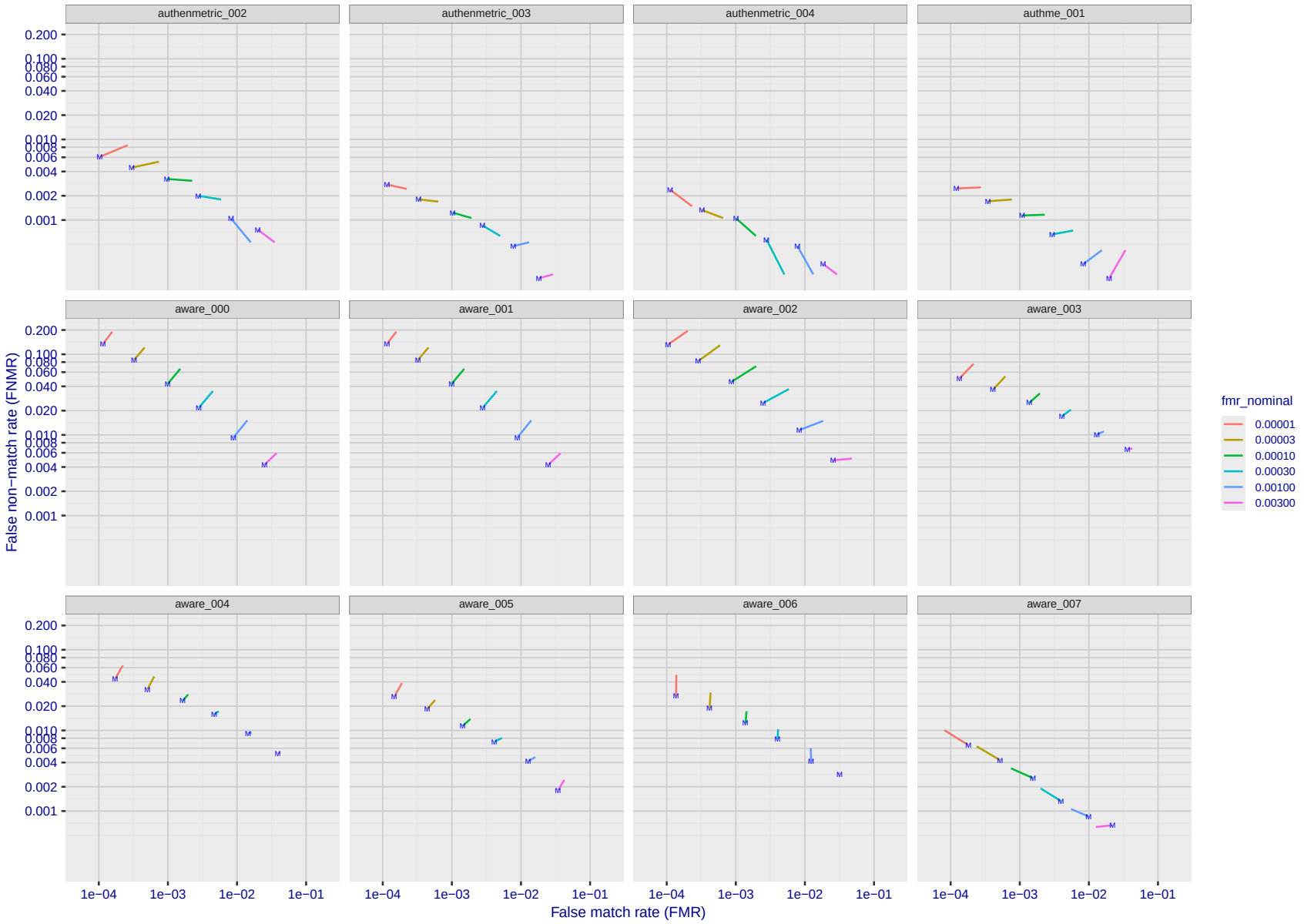

Figure 472: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

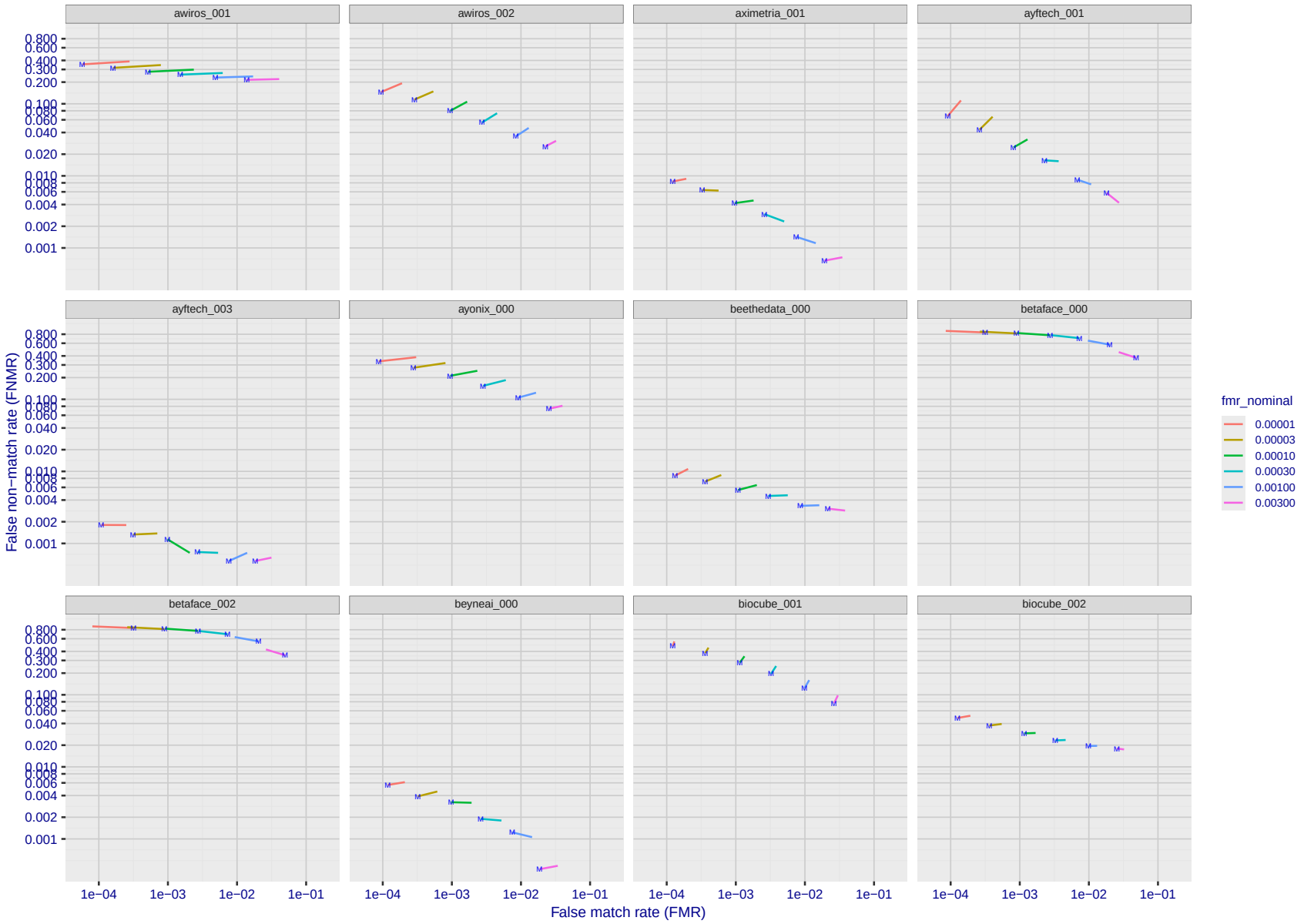

Figure 473: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

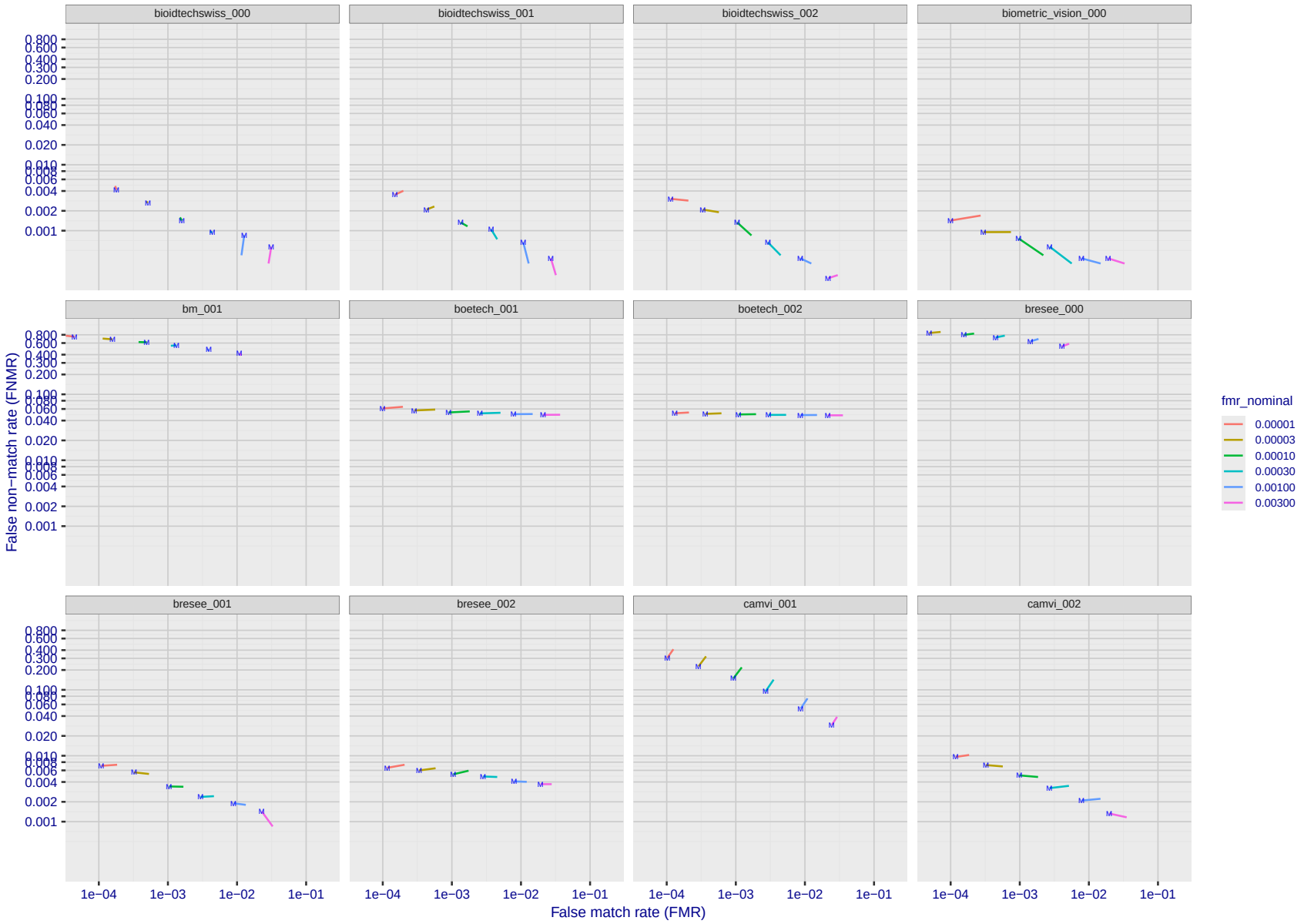

Figure 474: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

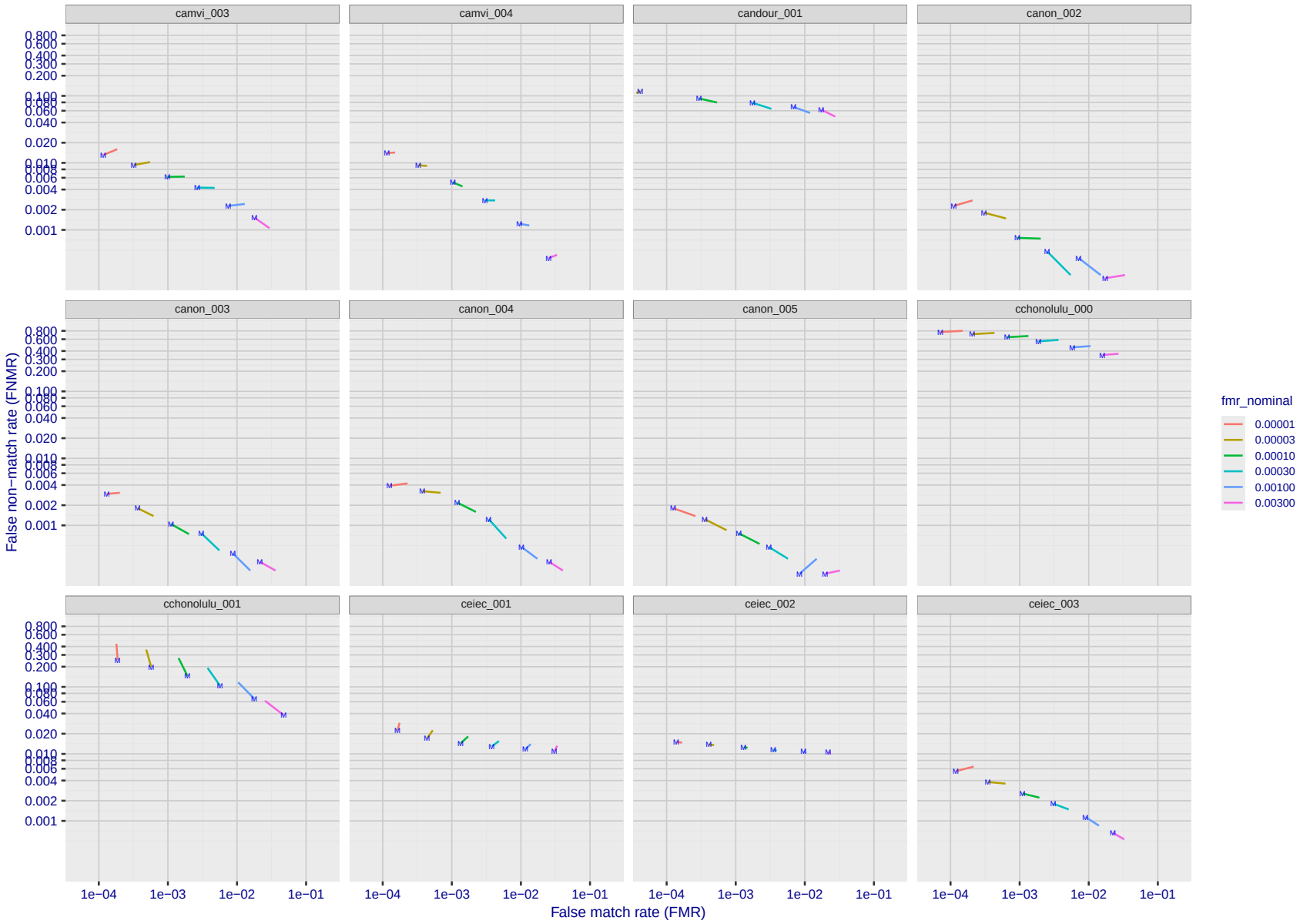

Figure 475: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

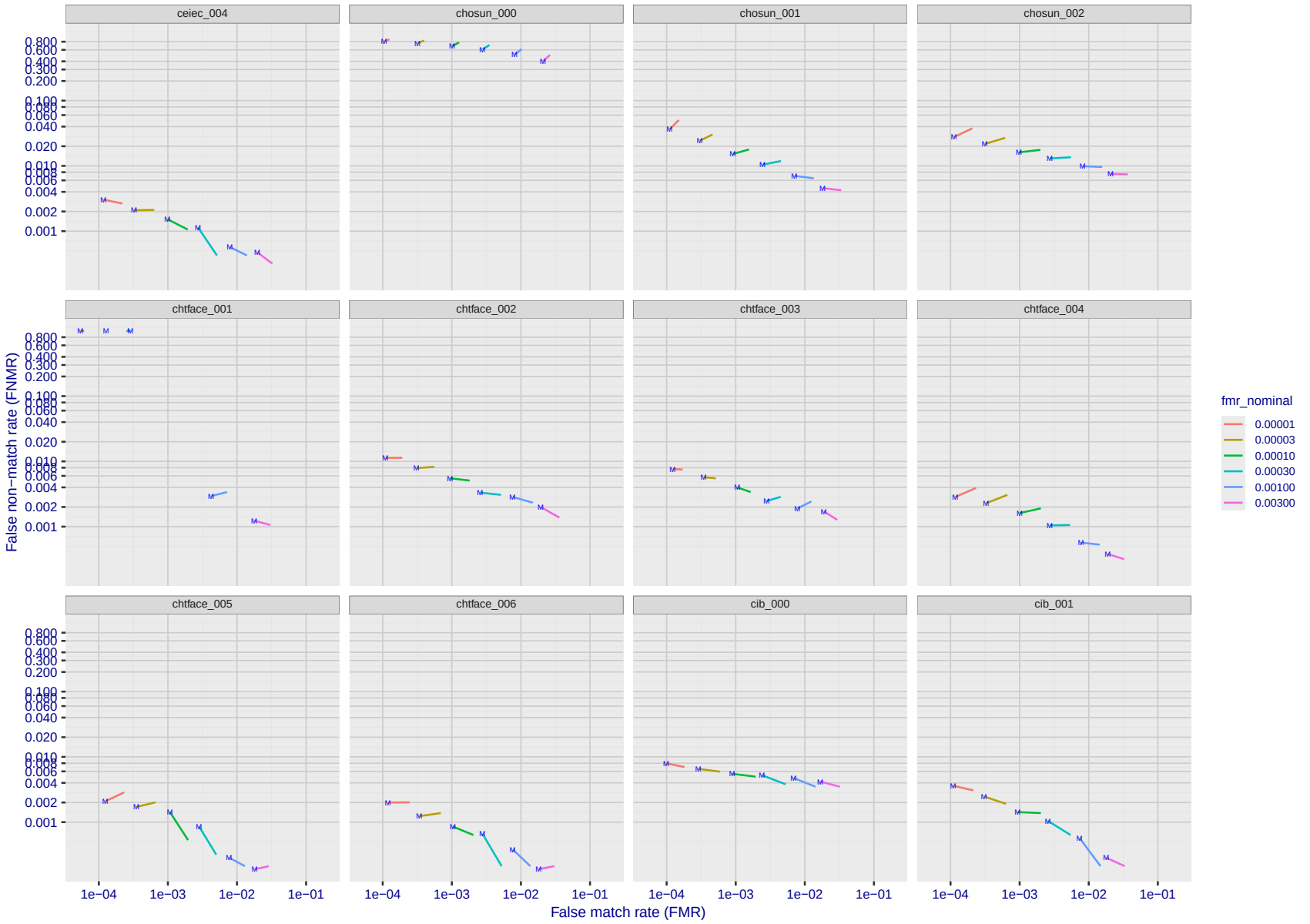

Figure 476: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

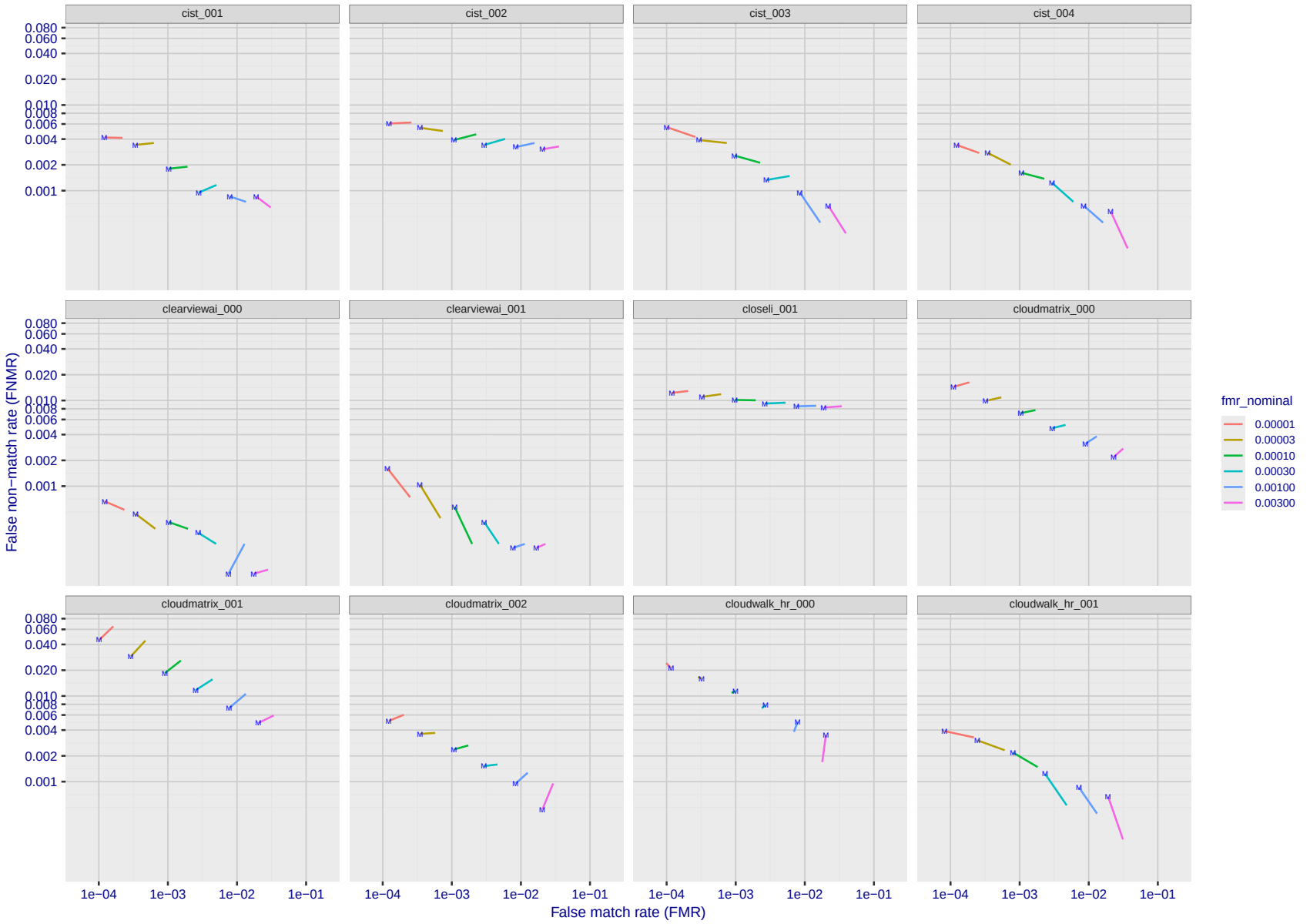

Figure 477: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

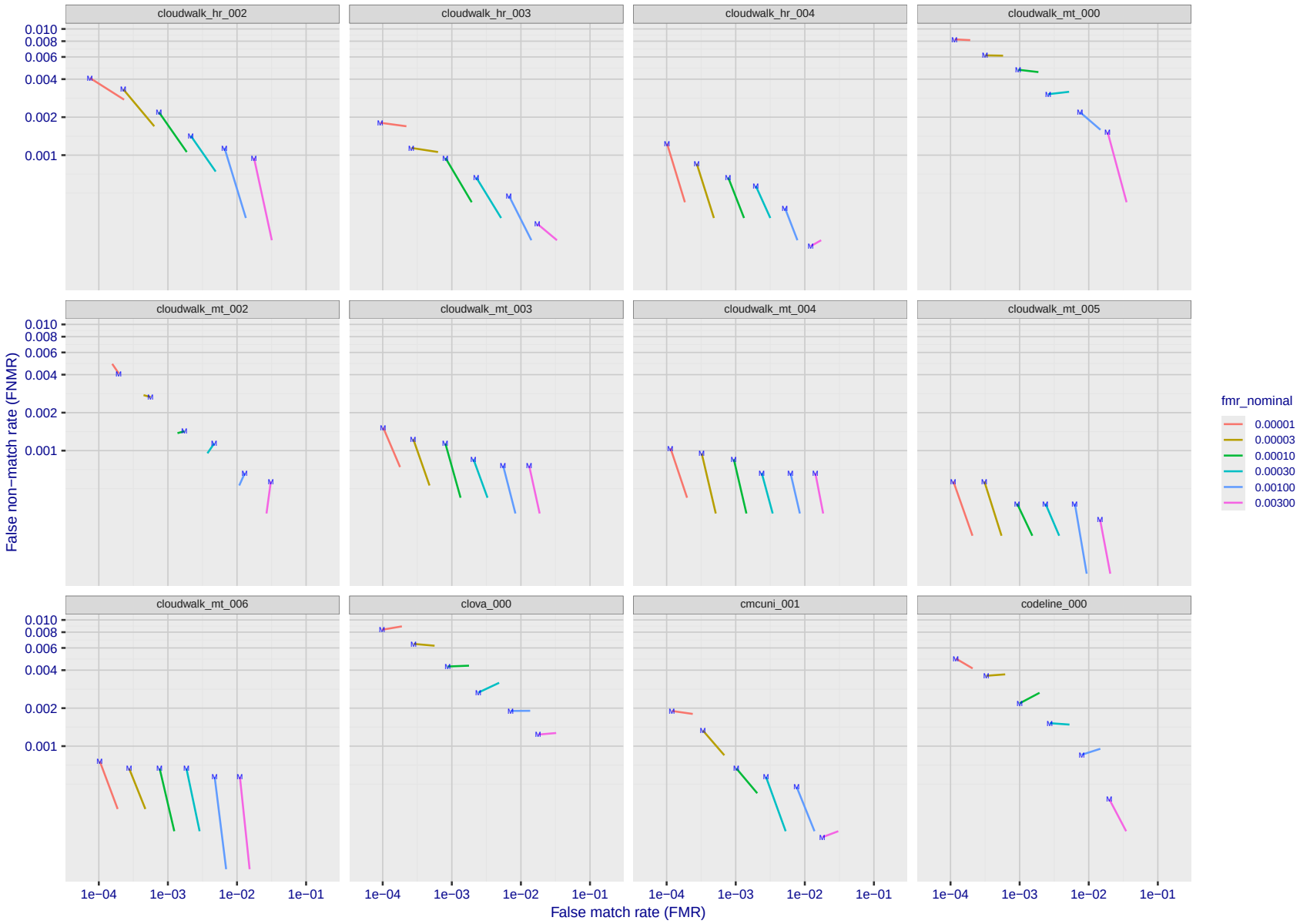

Figure 478: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

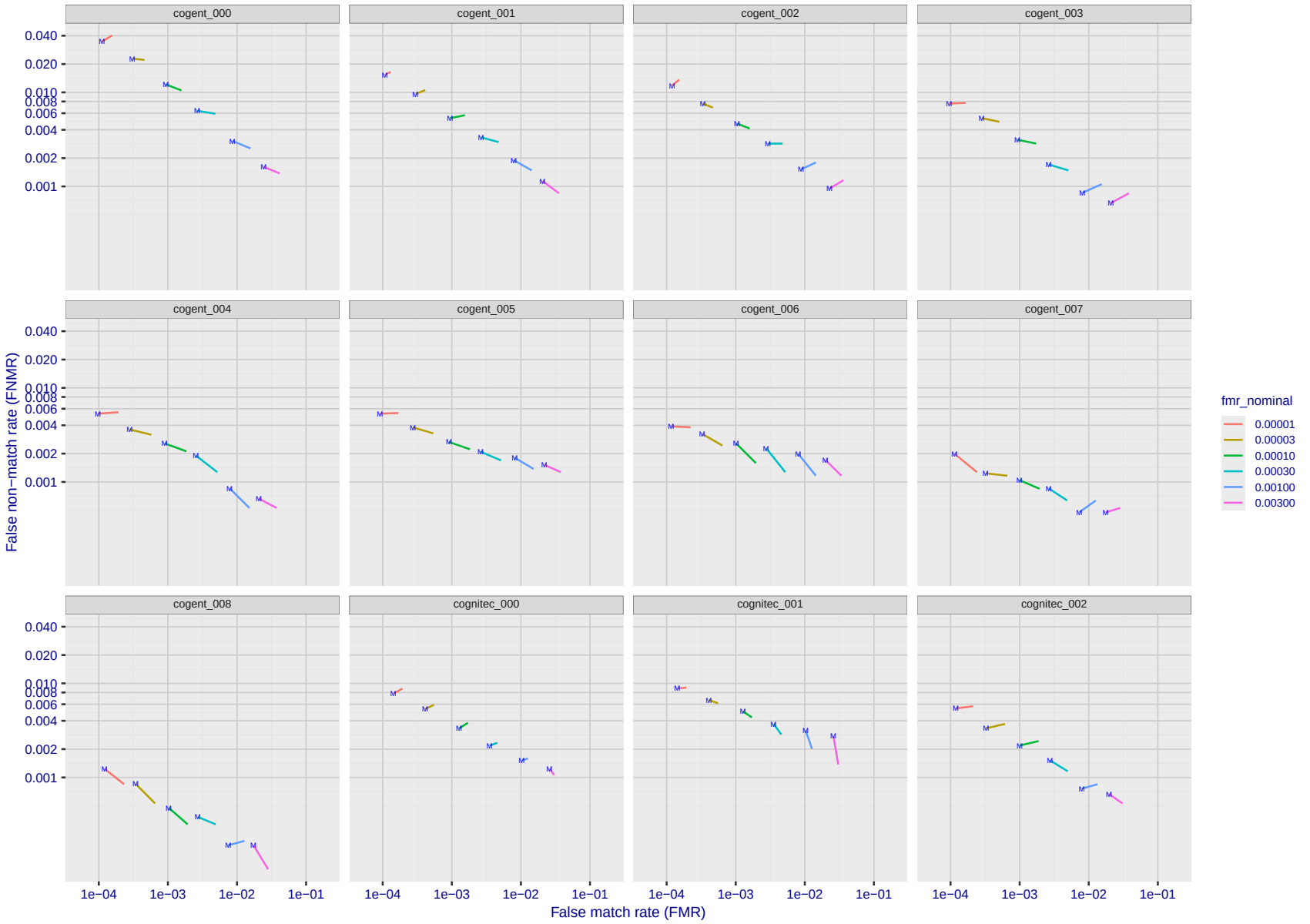

Figure 479: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

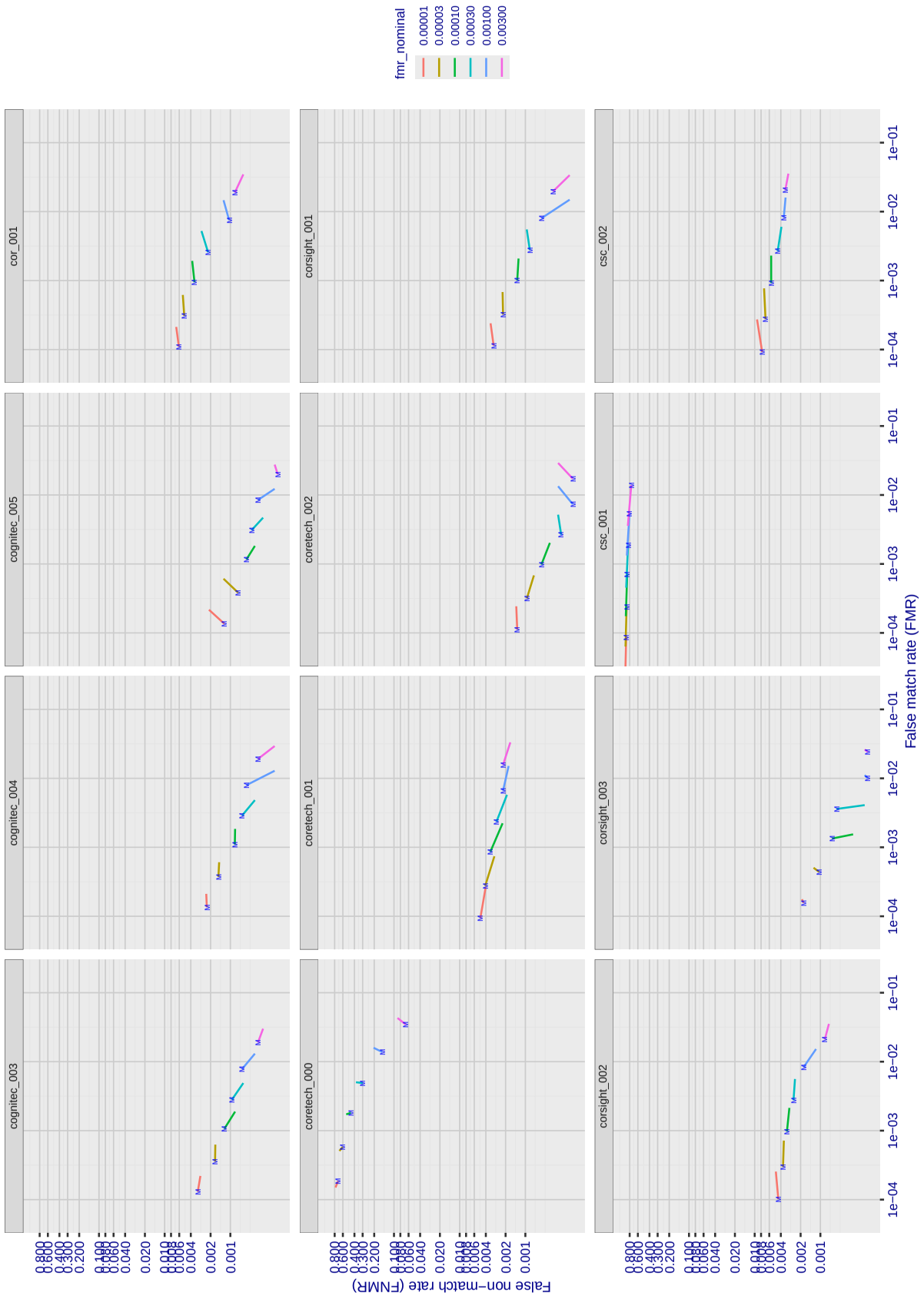

Figure 480: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

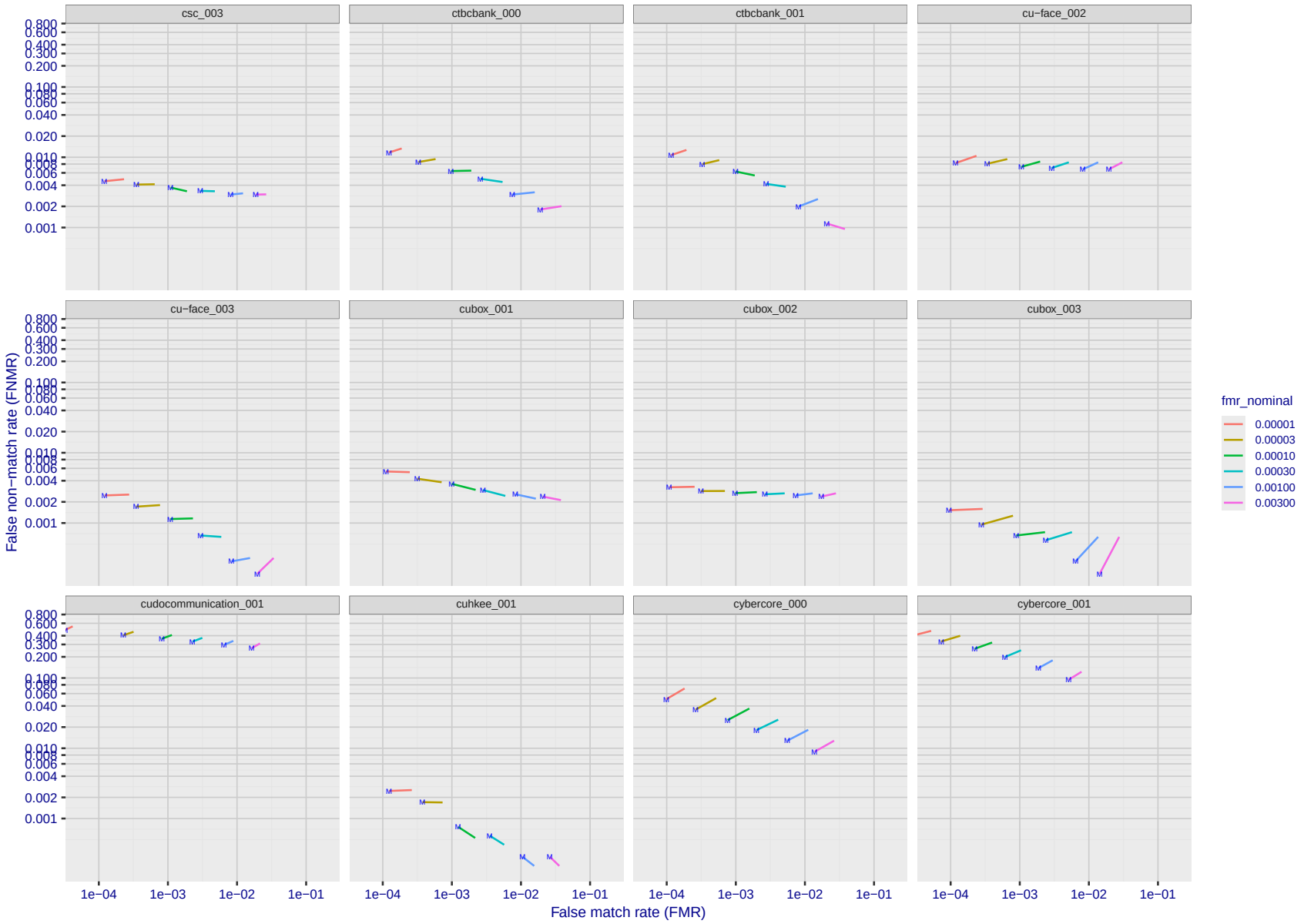

Figure 481: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

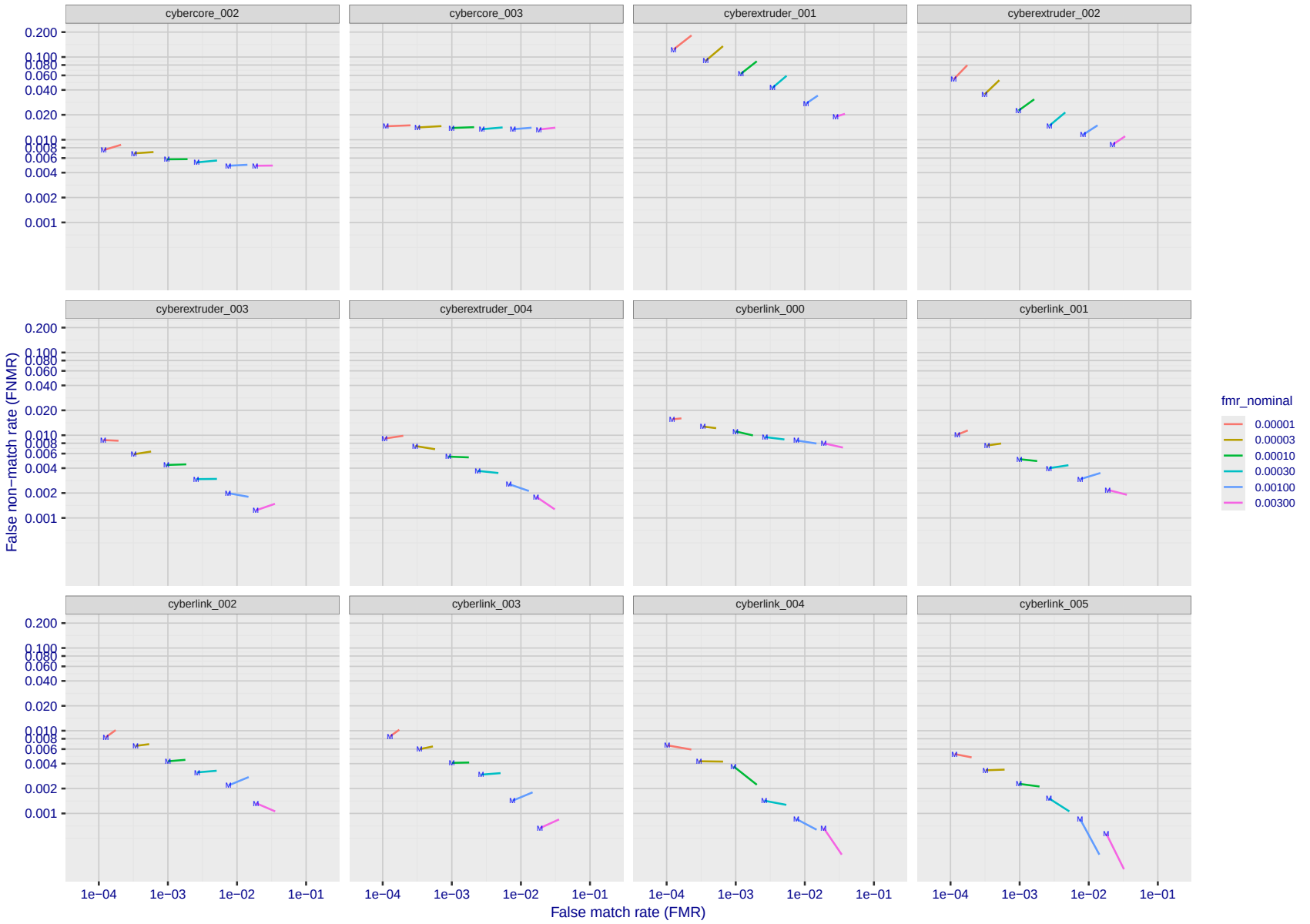

Figure 482: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

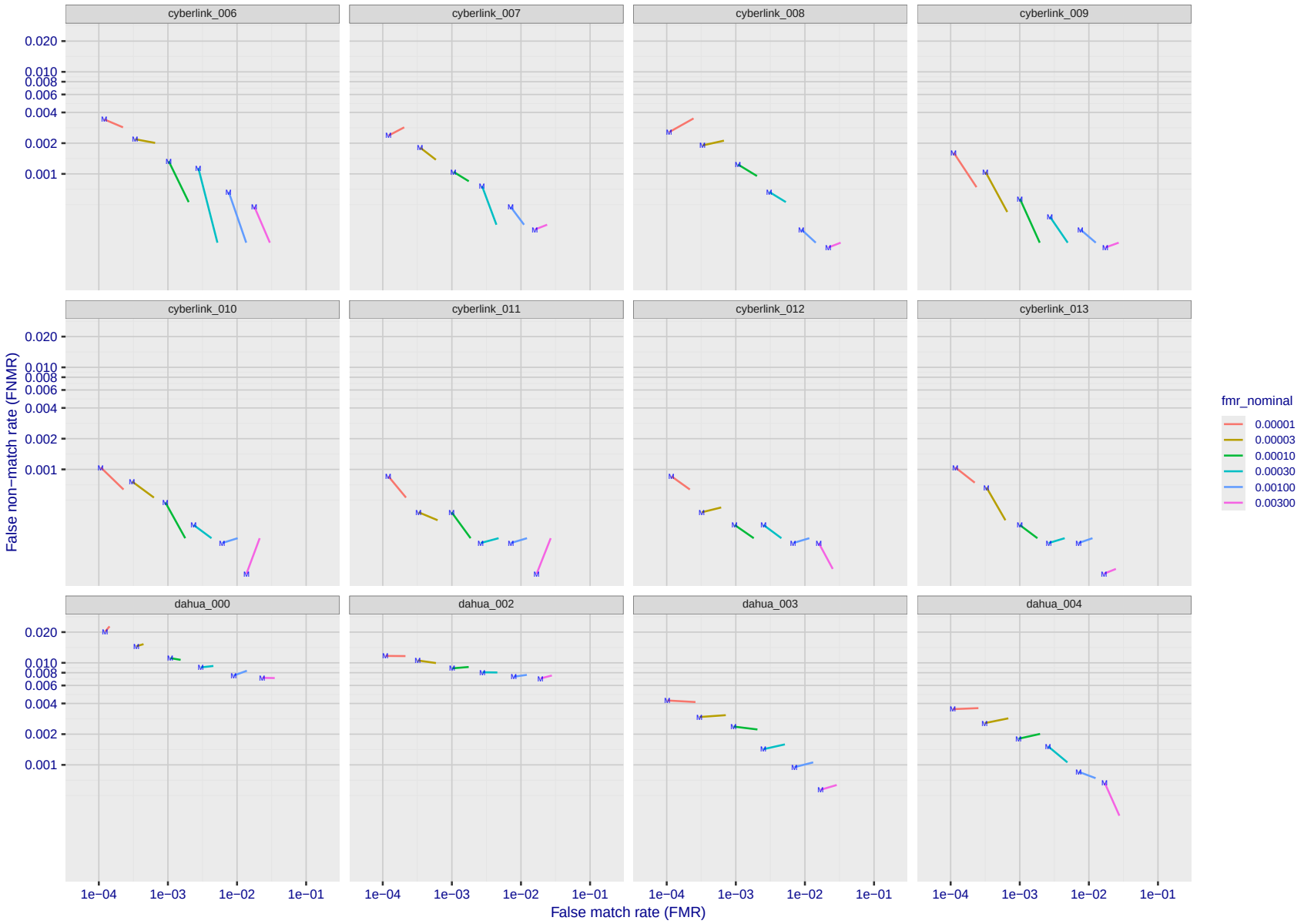

Figure 483: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

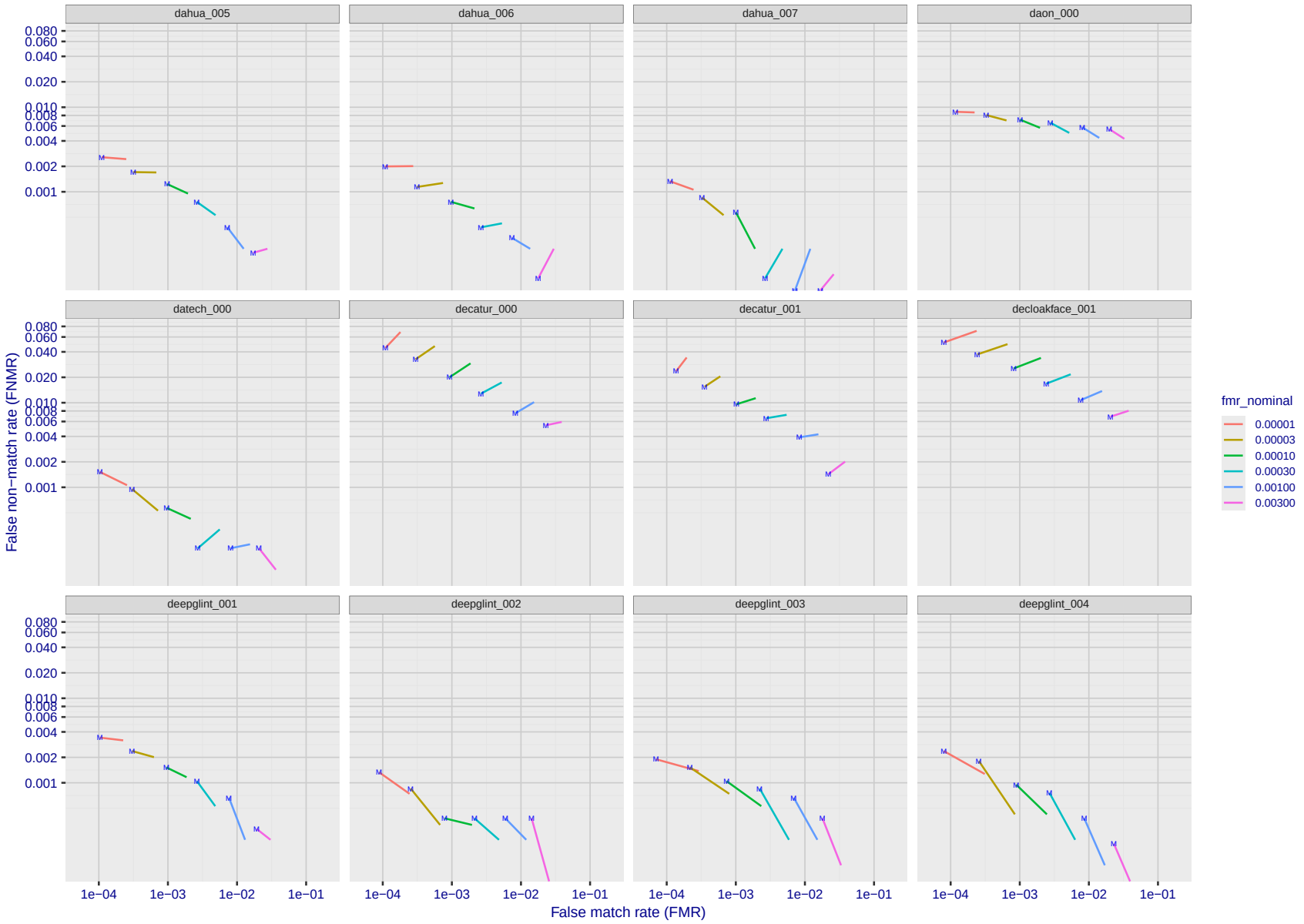

Figure 484: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

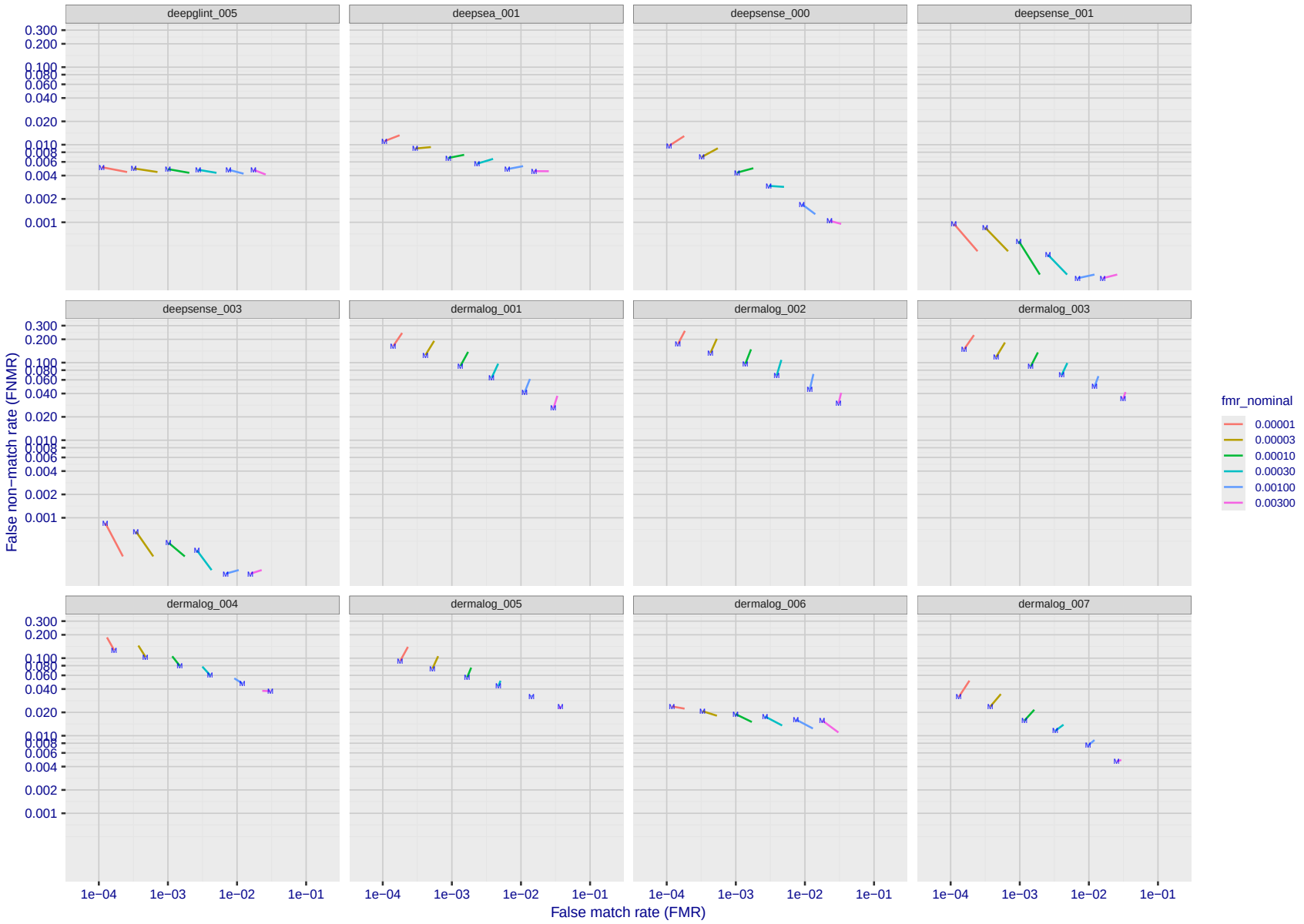

Figure 485: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

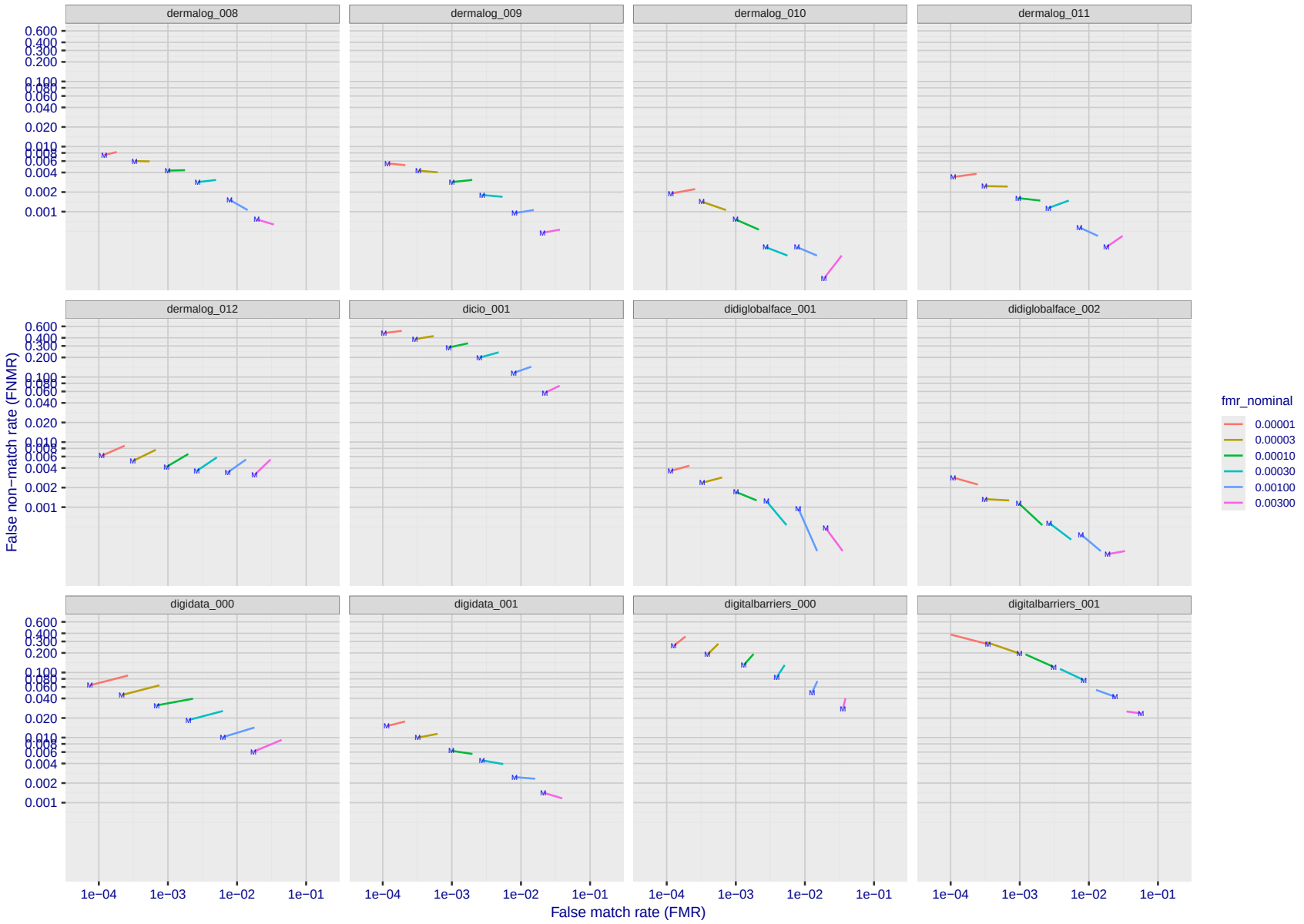

Figure 486: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

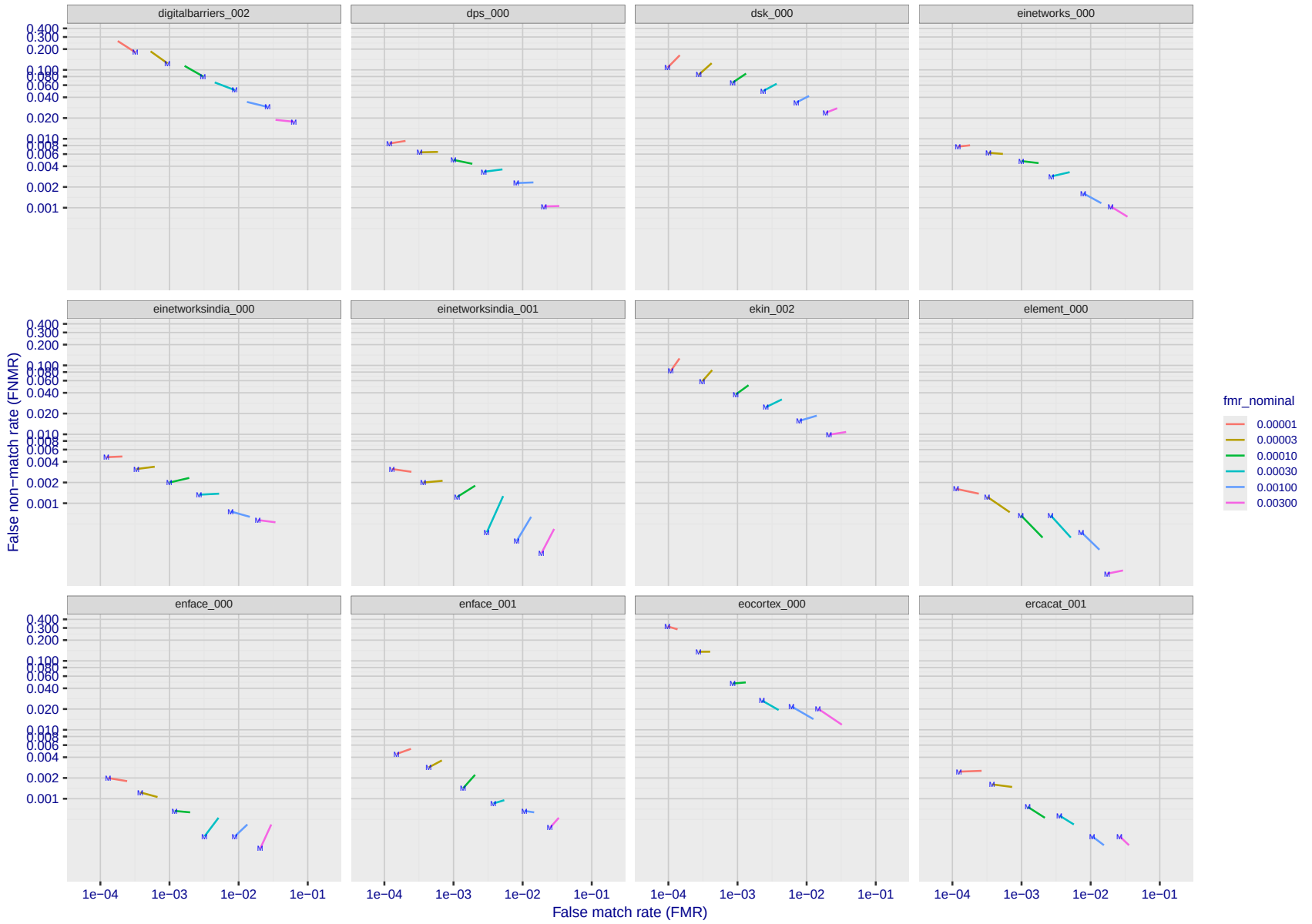

Figure 487: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

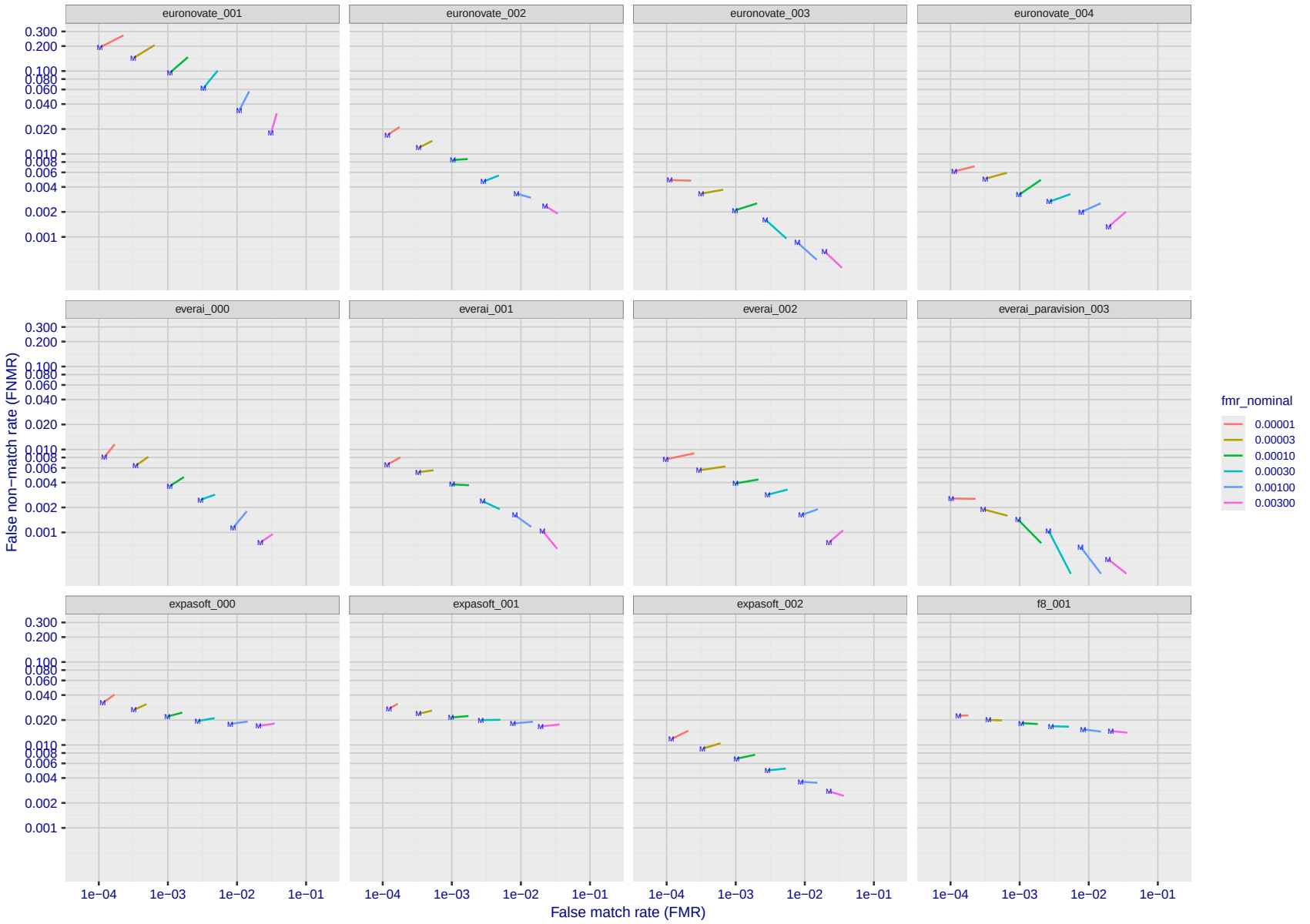

Figure 488: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

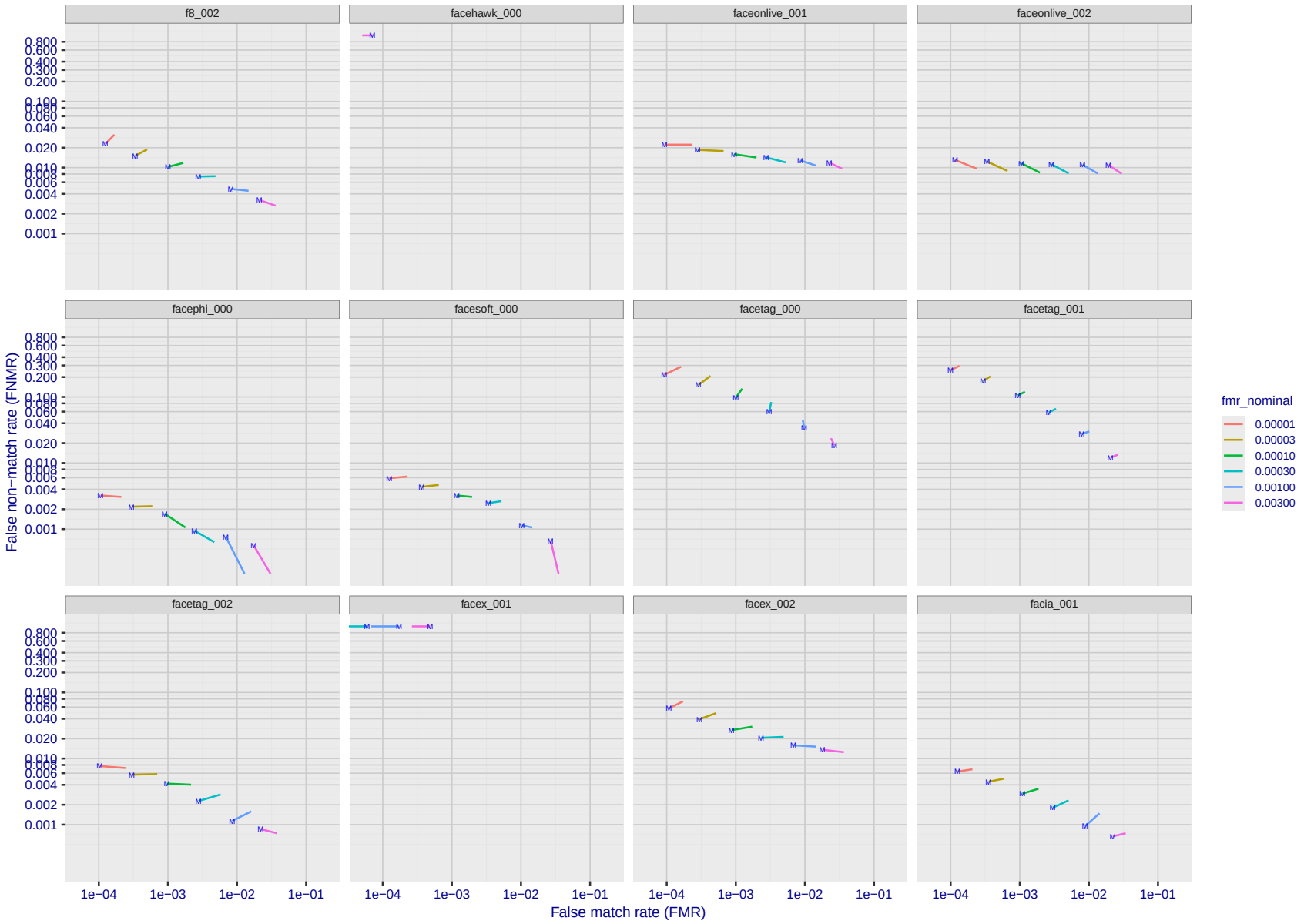

Figure 489: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

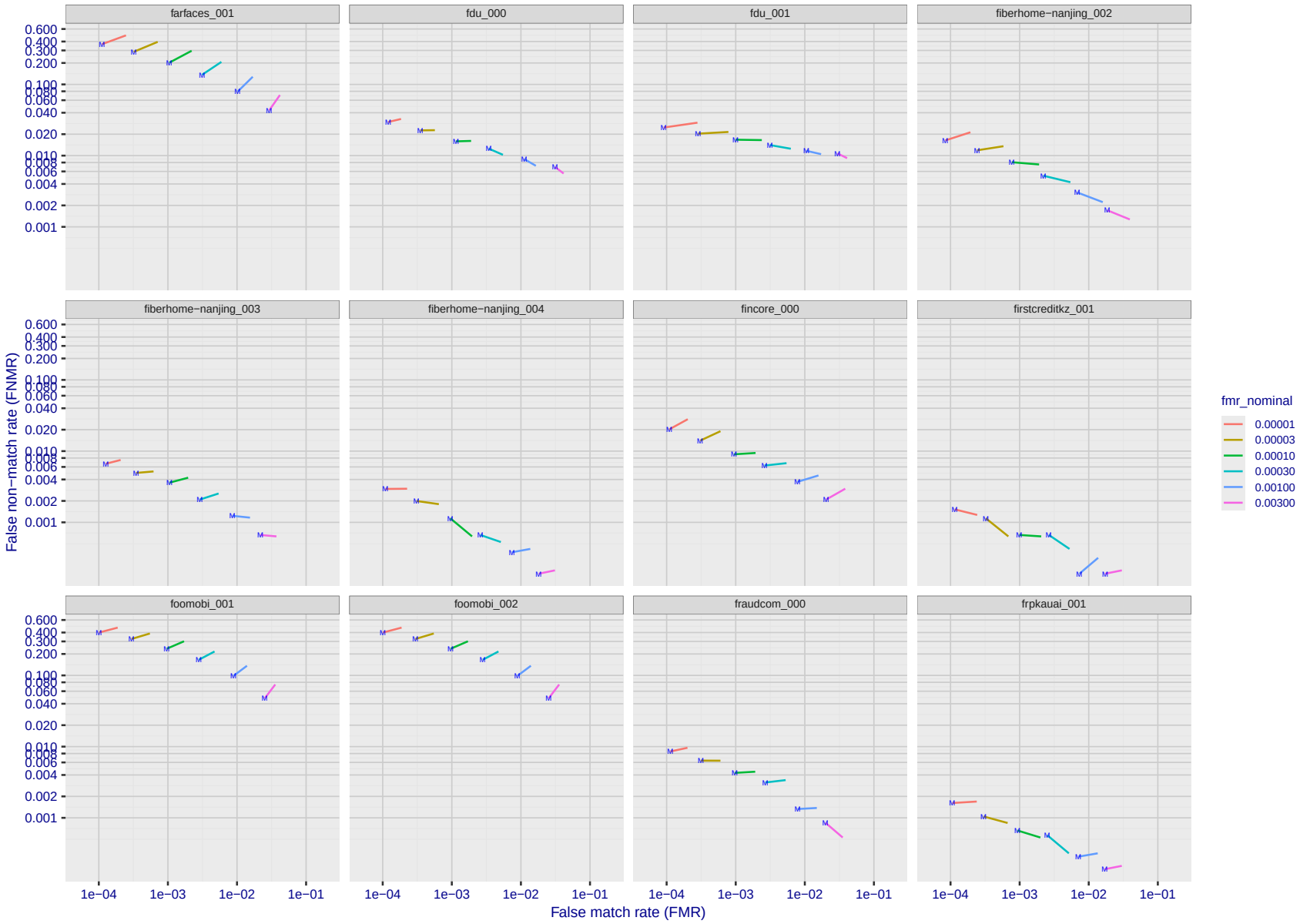

Figure 490: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

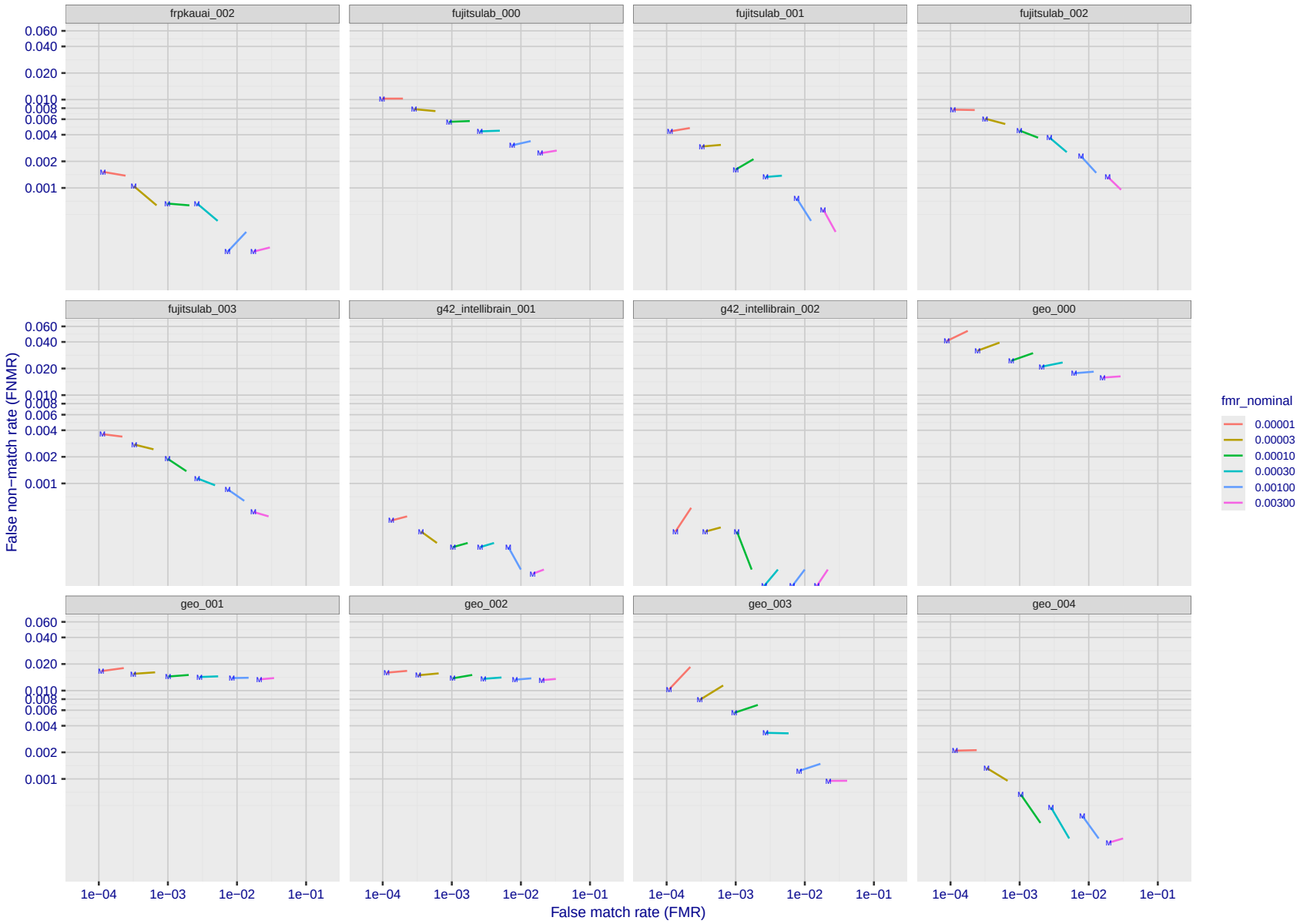

Figure 491: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

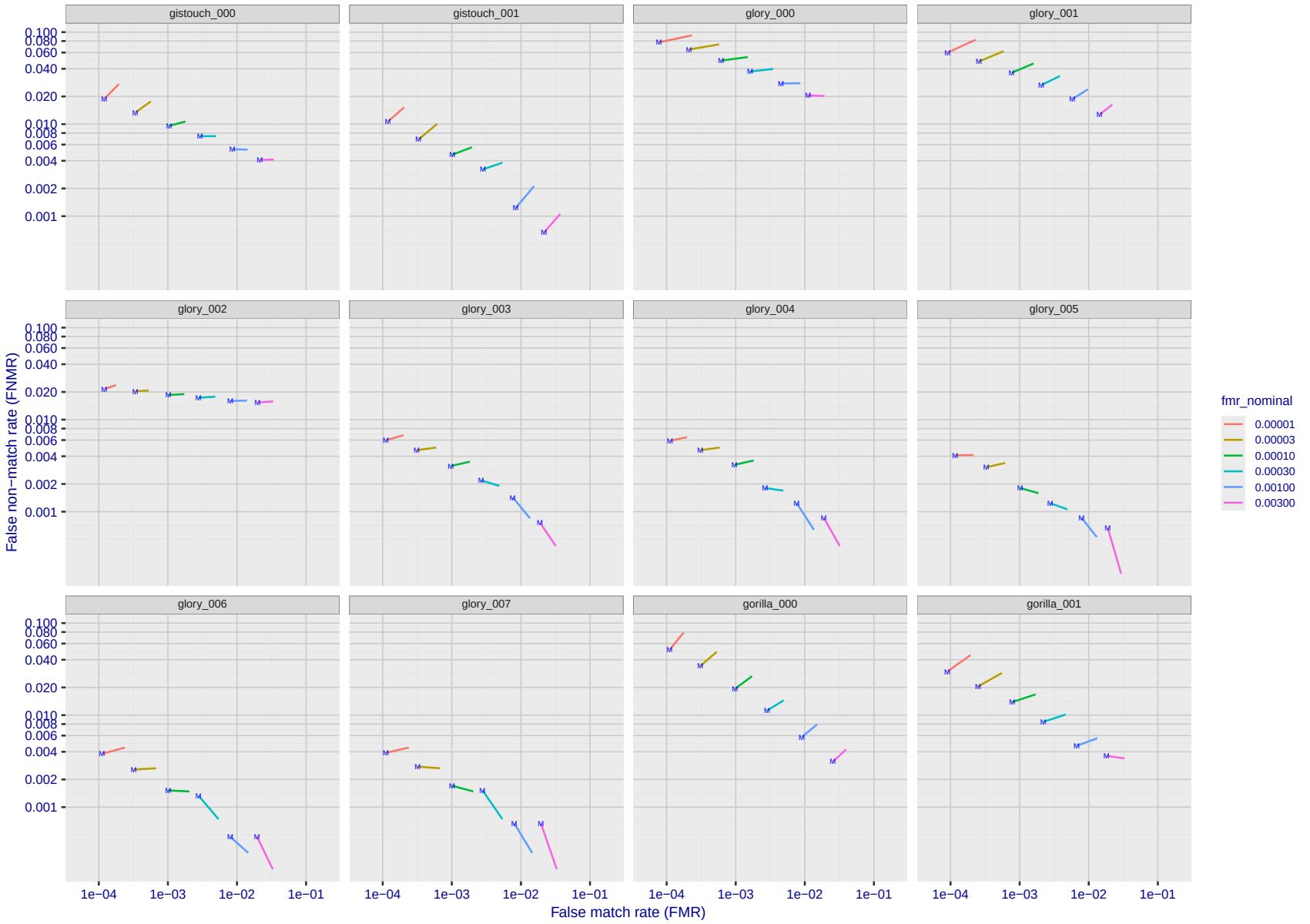

Figure 492: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

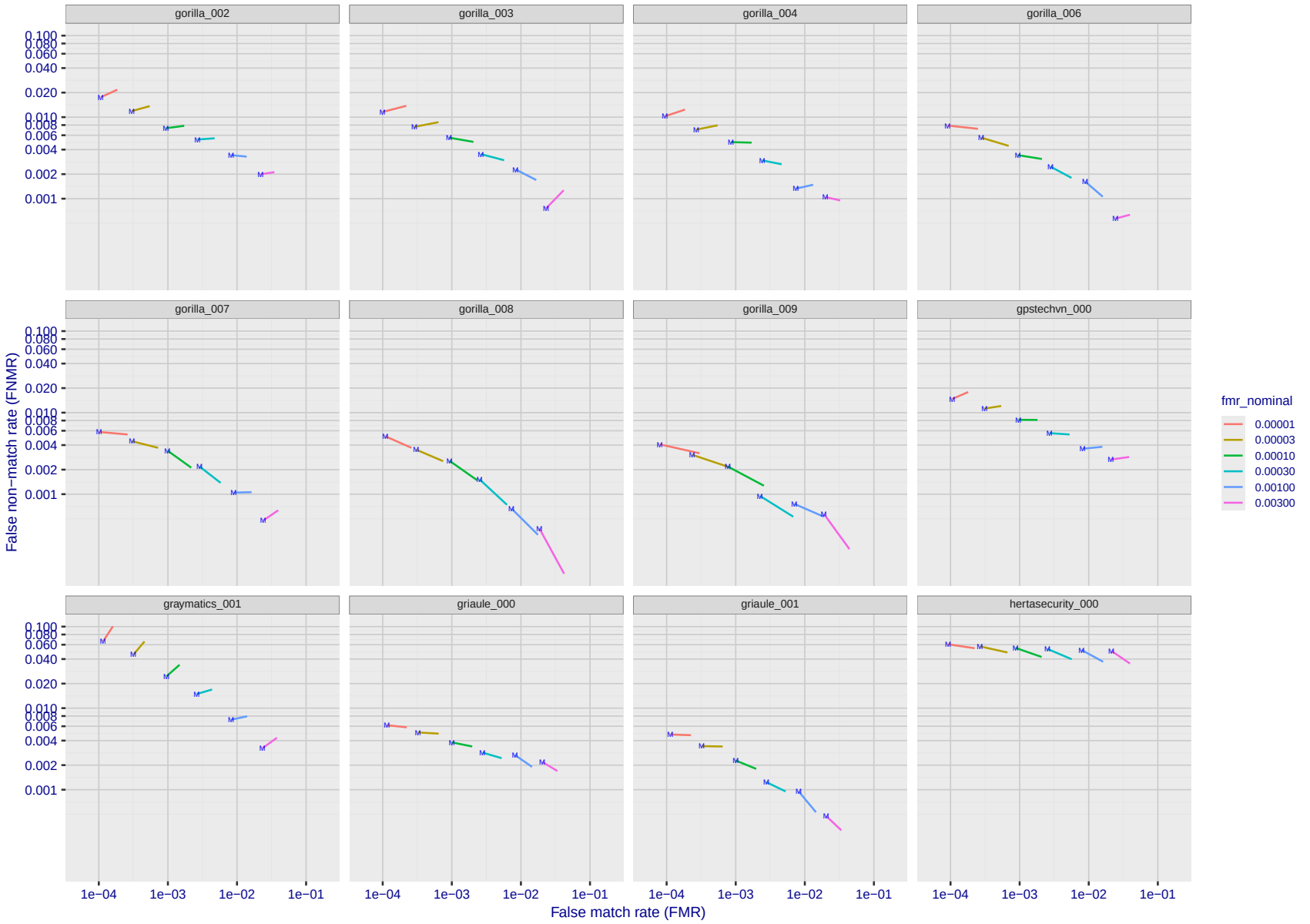

Figure 493: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

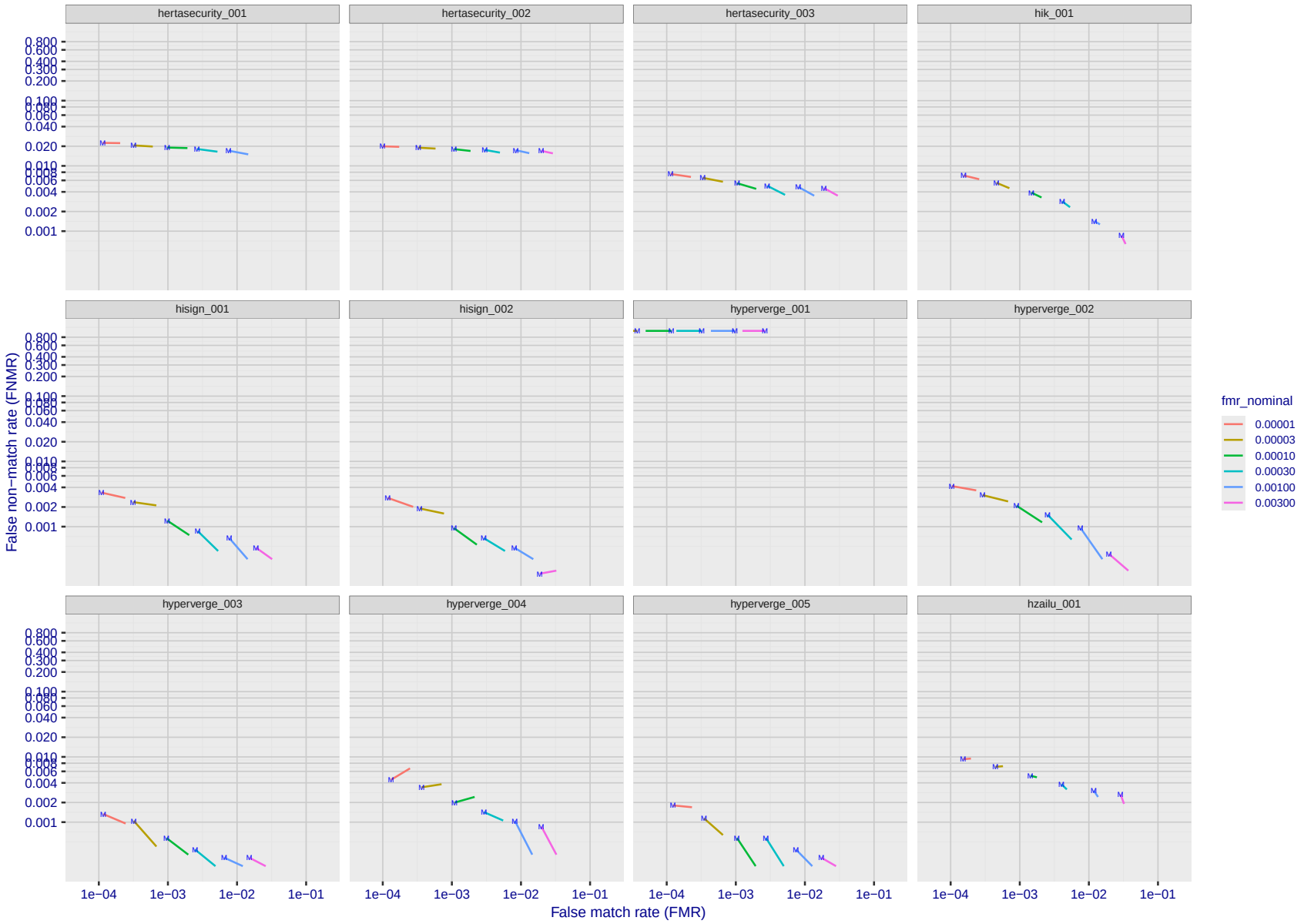

Figure 494: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

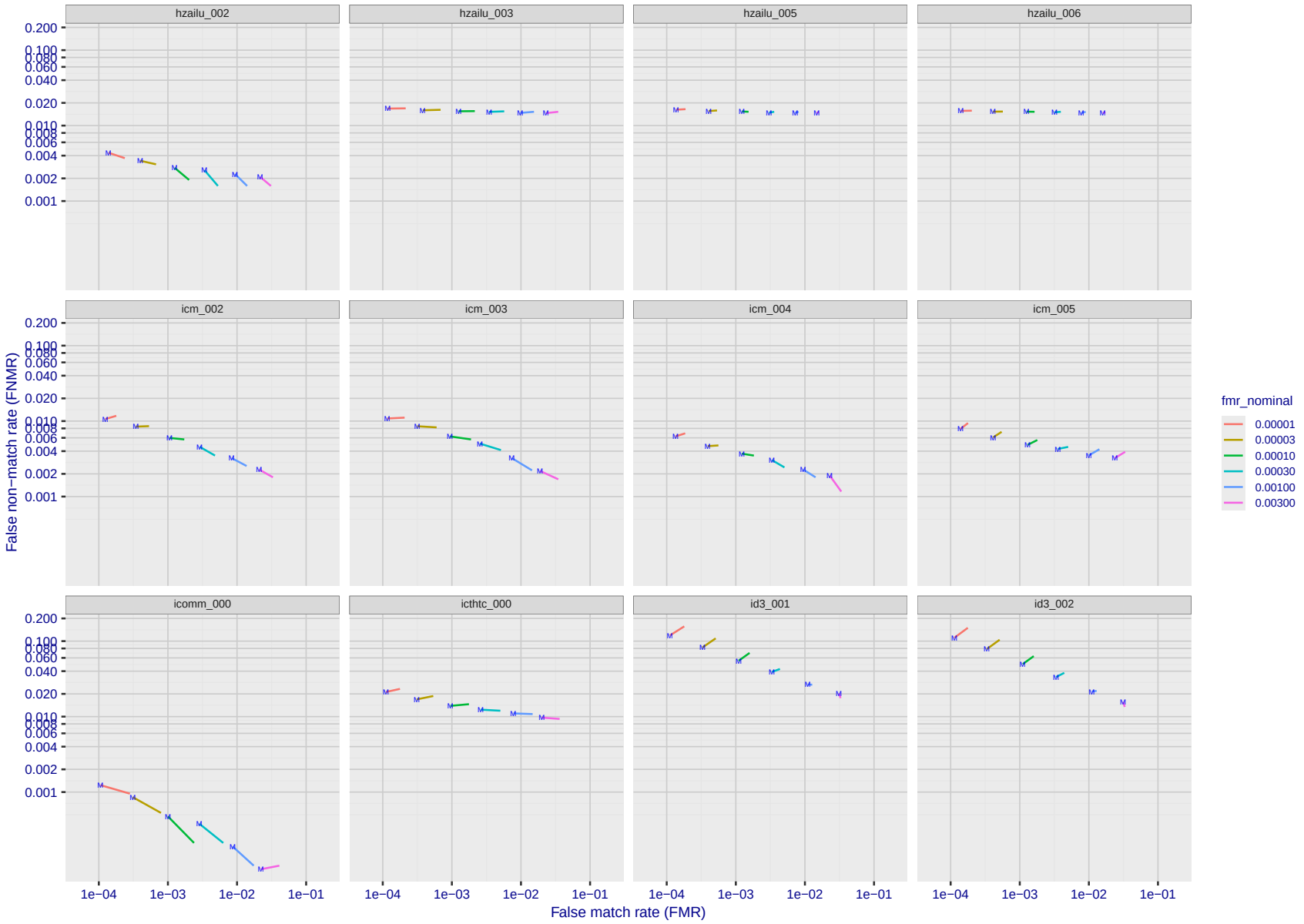

Figure 495: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

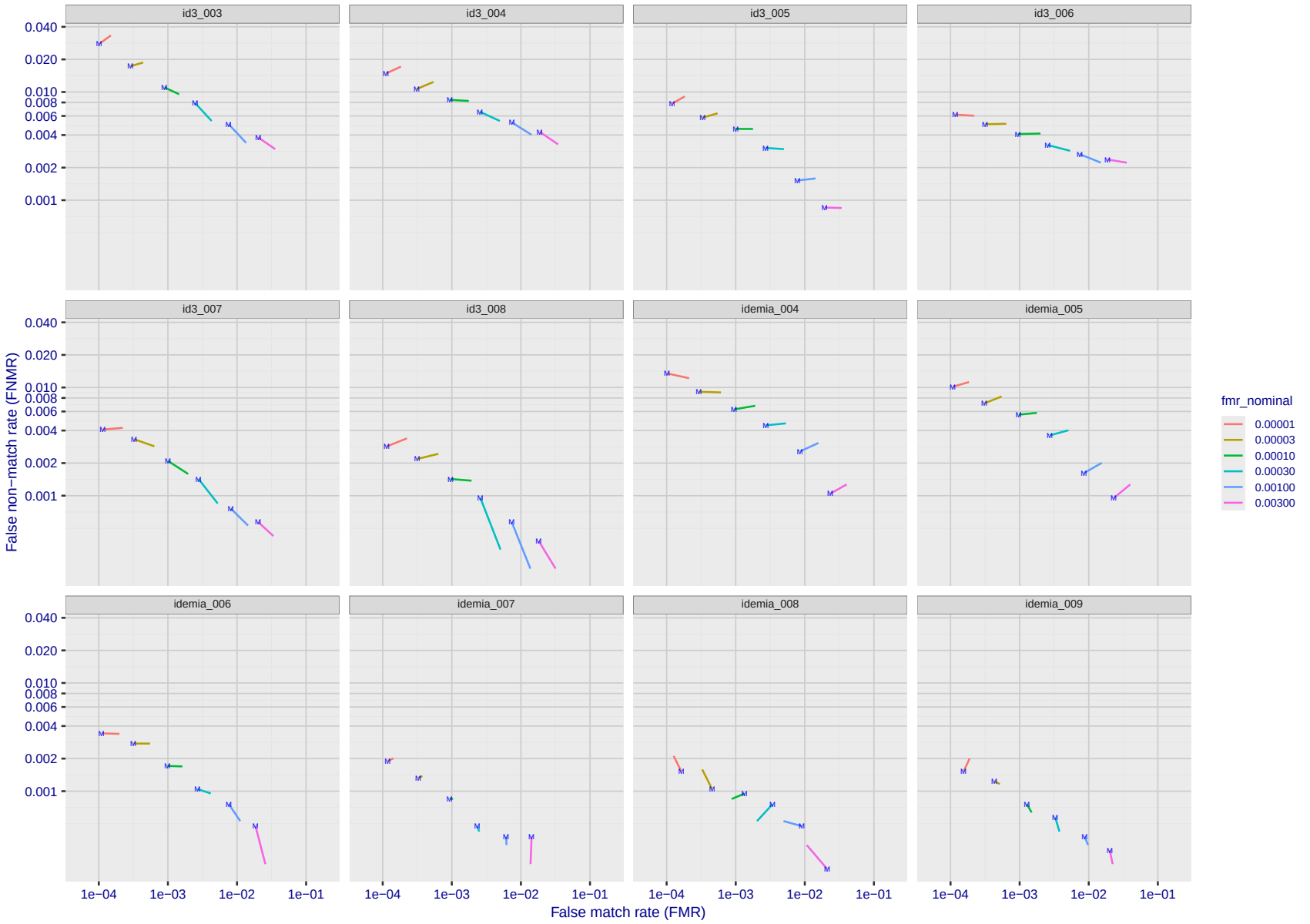

Figure 496: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

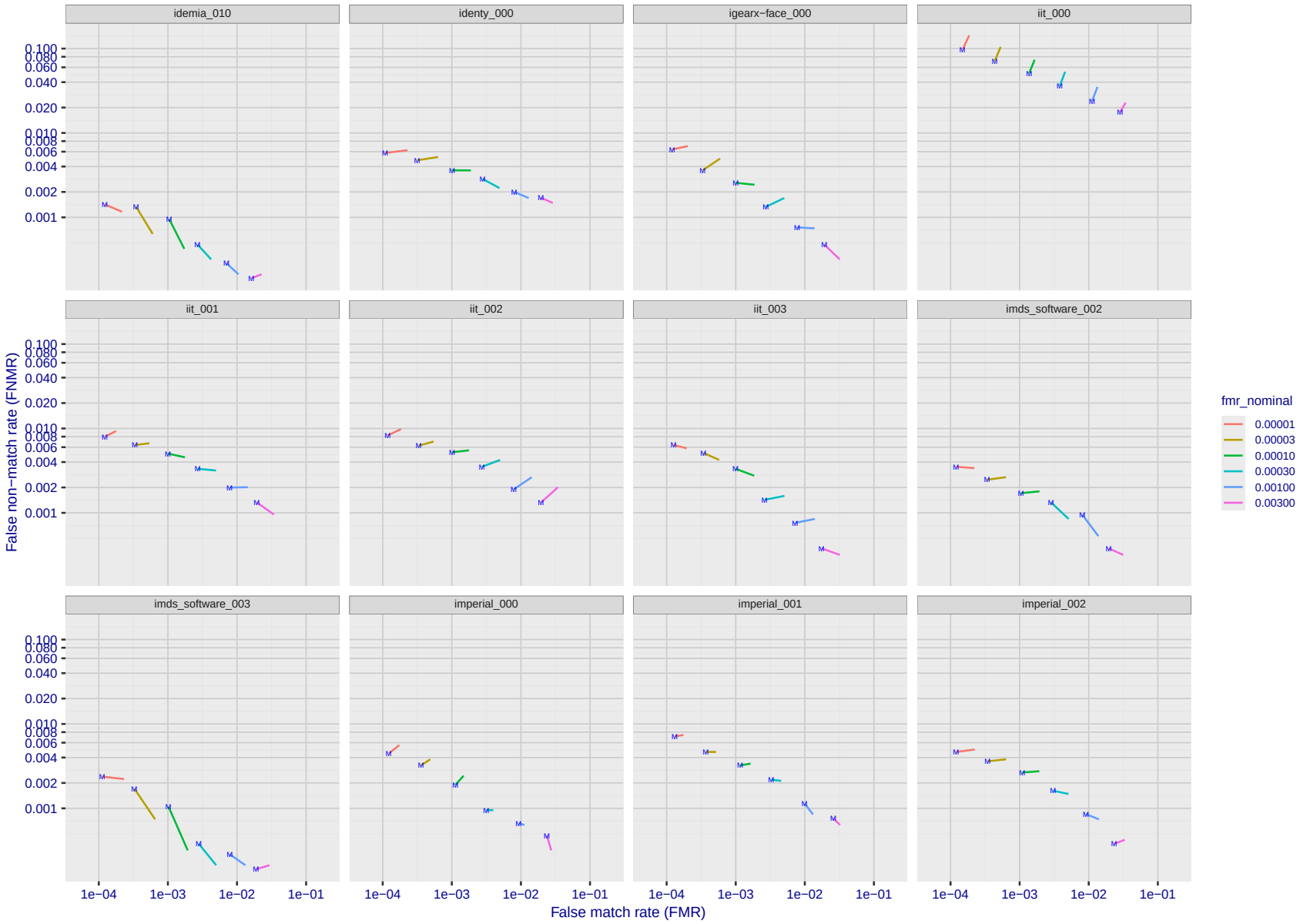

Figure 497: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

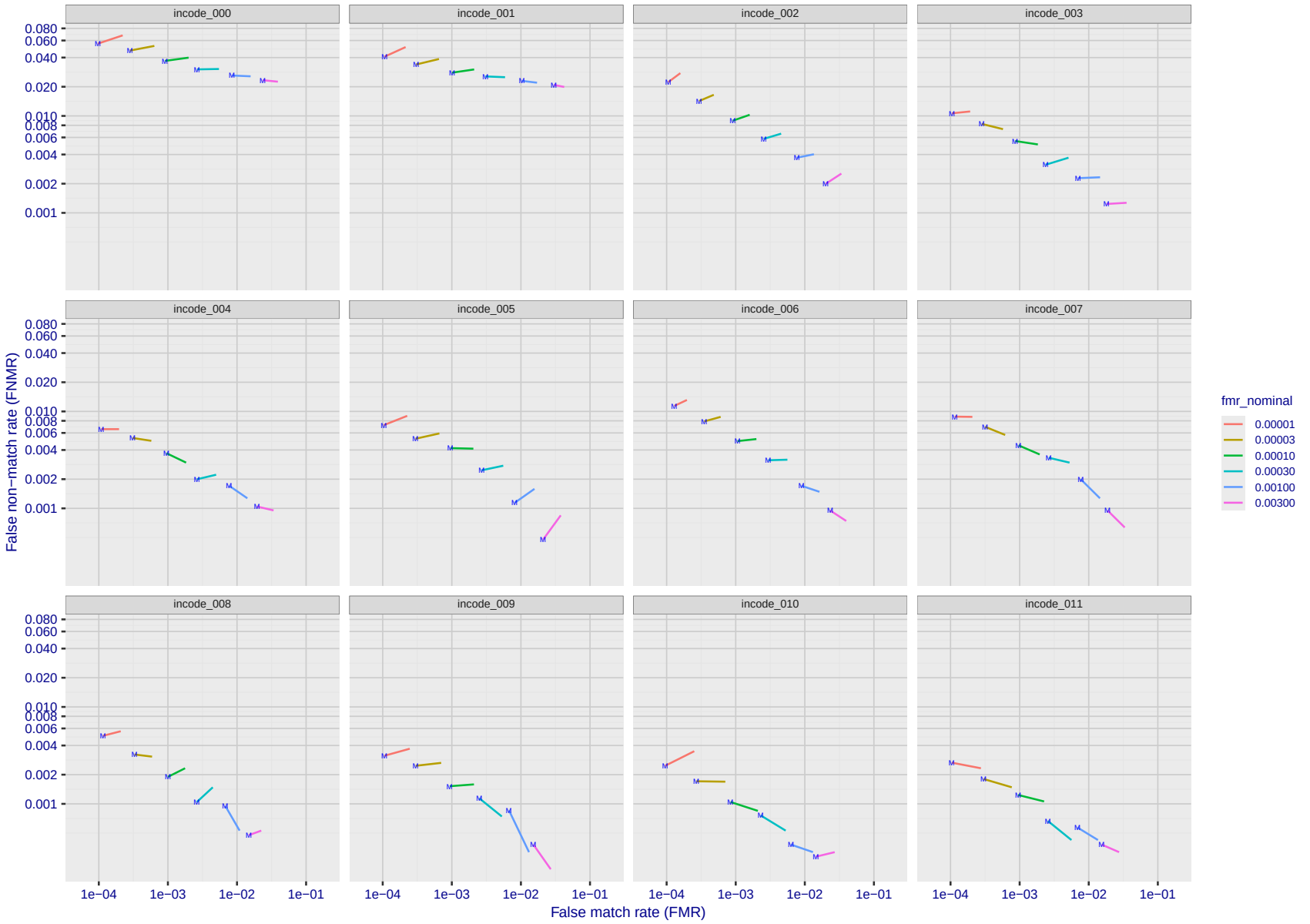

Figure 498: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

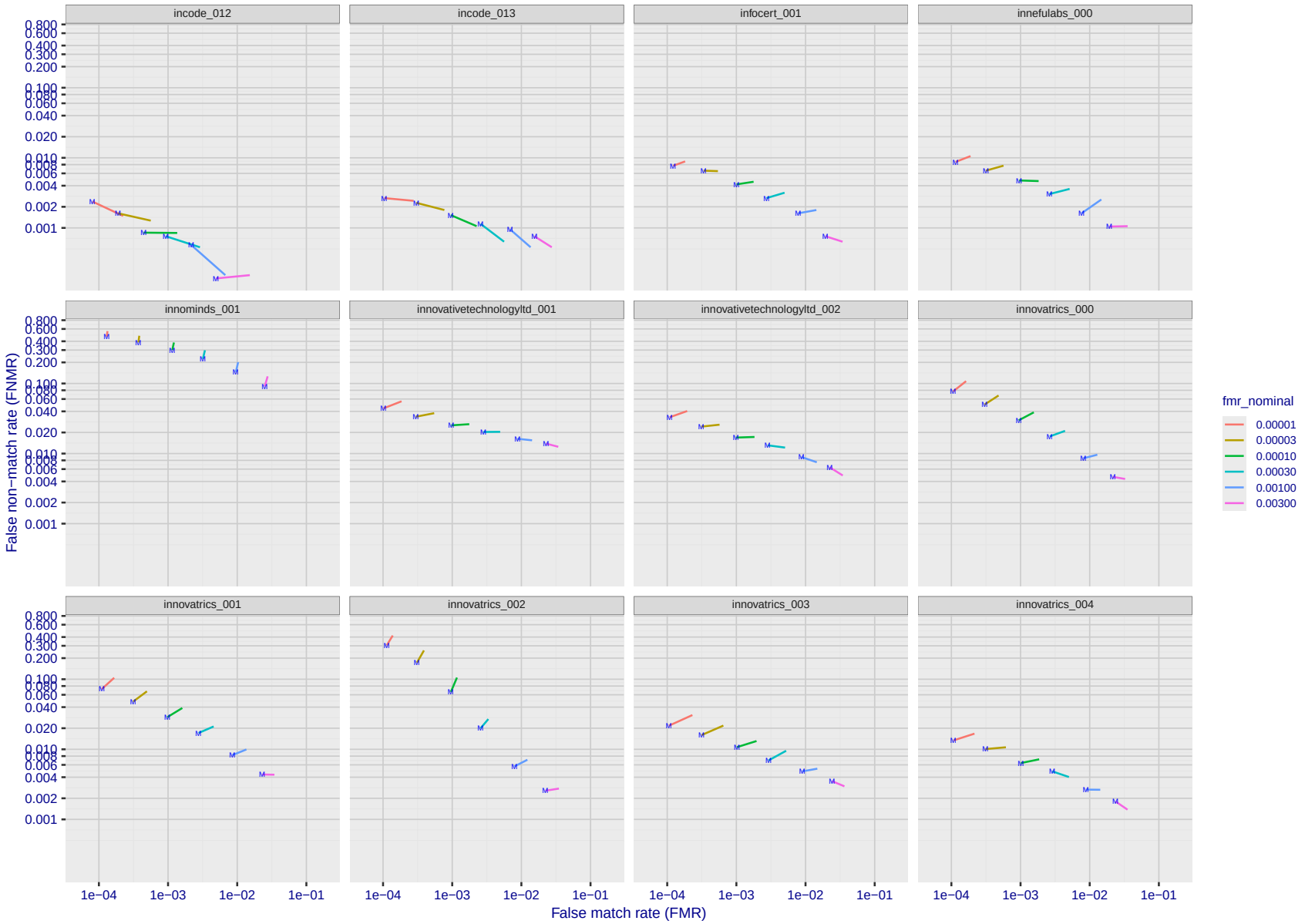

Figure 499: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

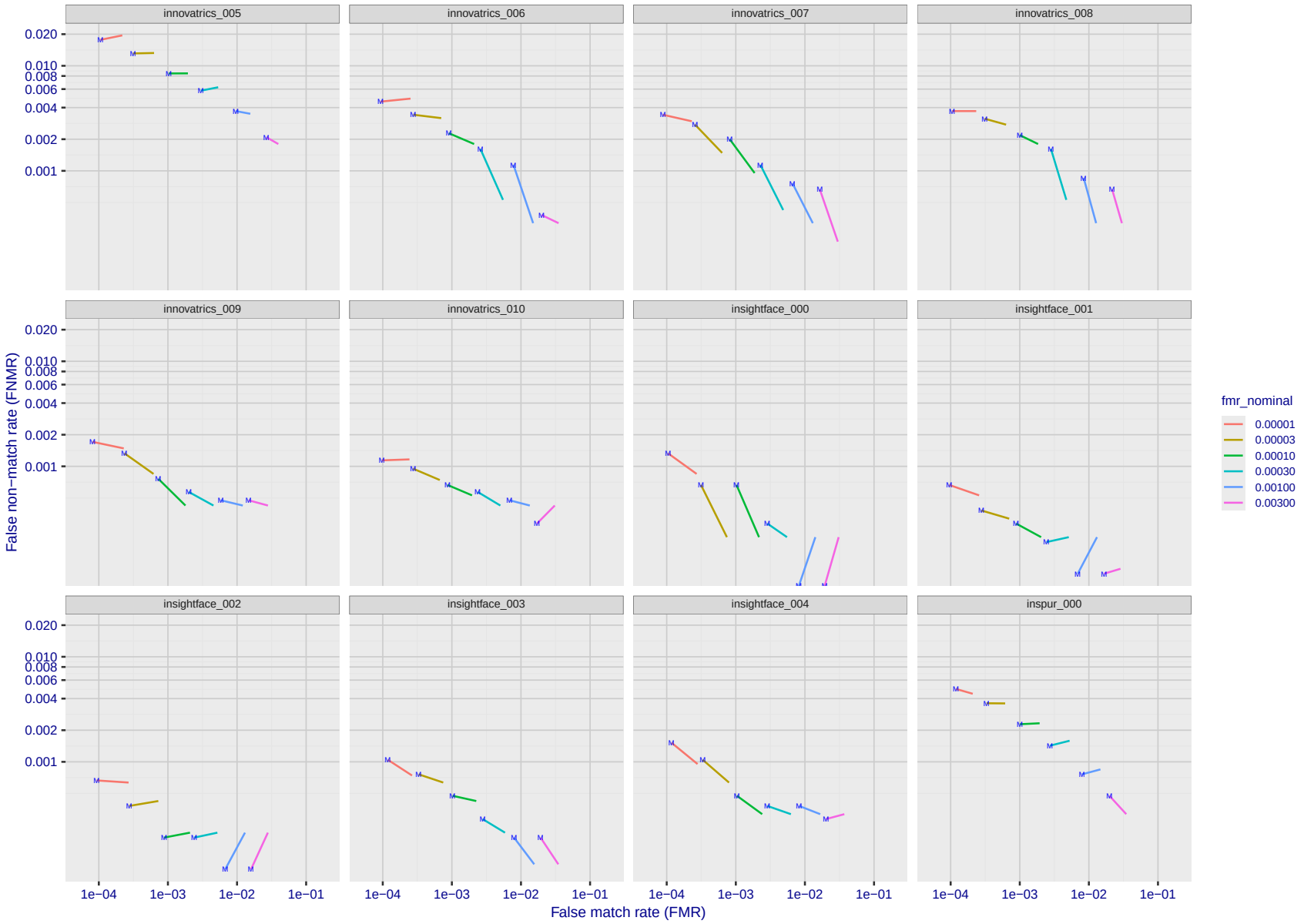

Figure 500: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

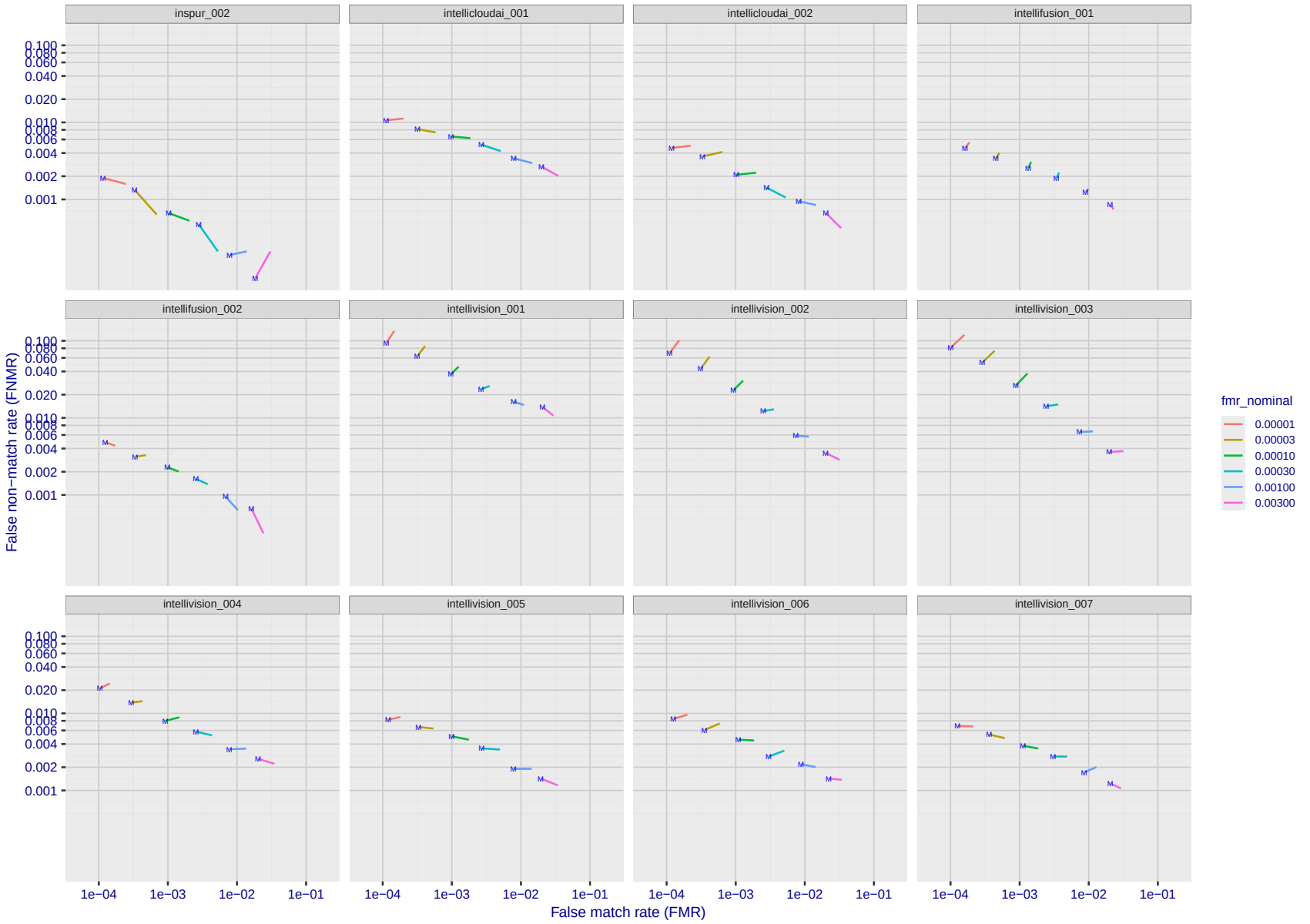

Figure 501: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

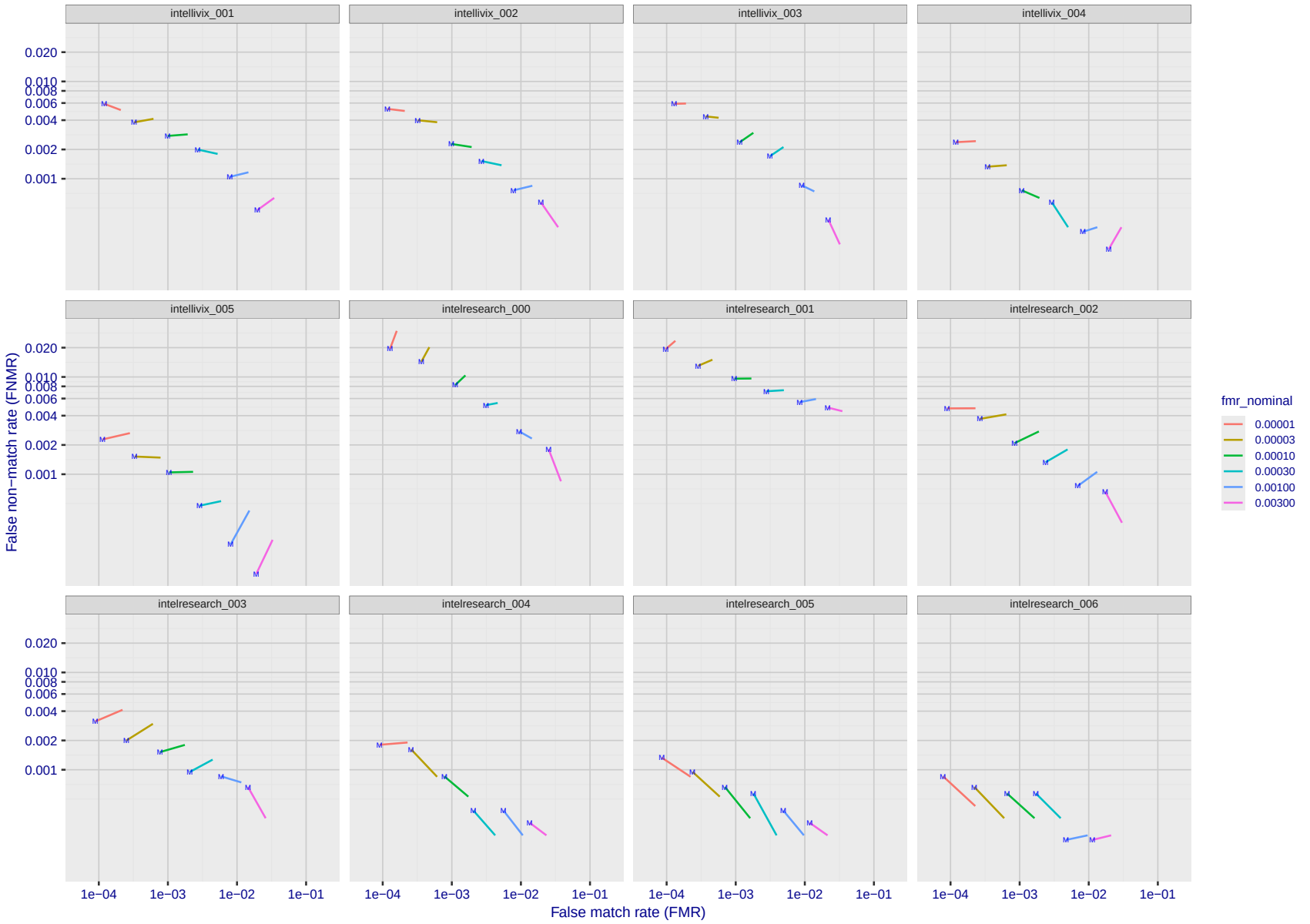

Figure 502: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

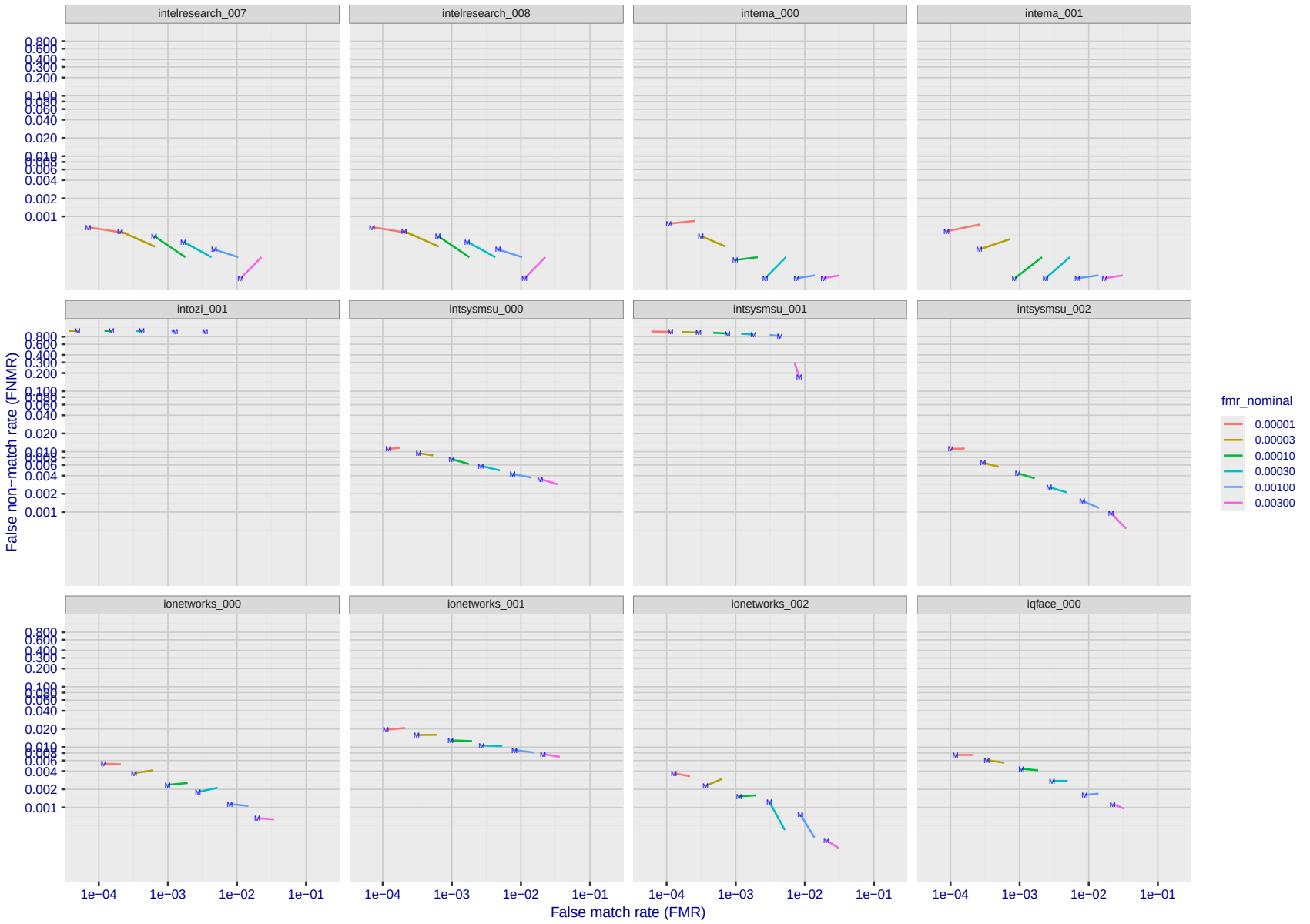

Figure 503: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

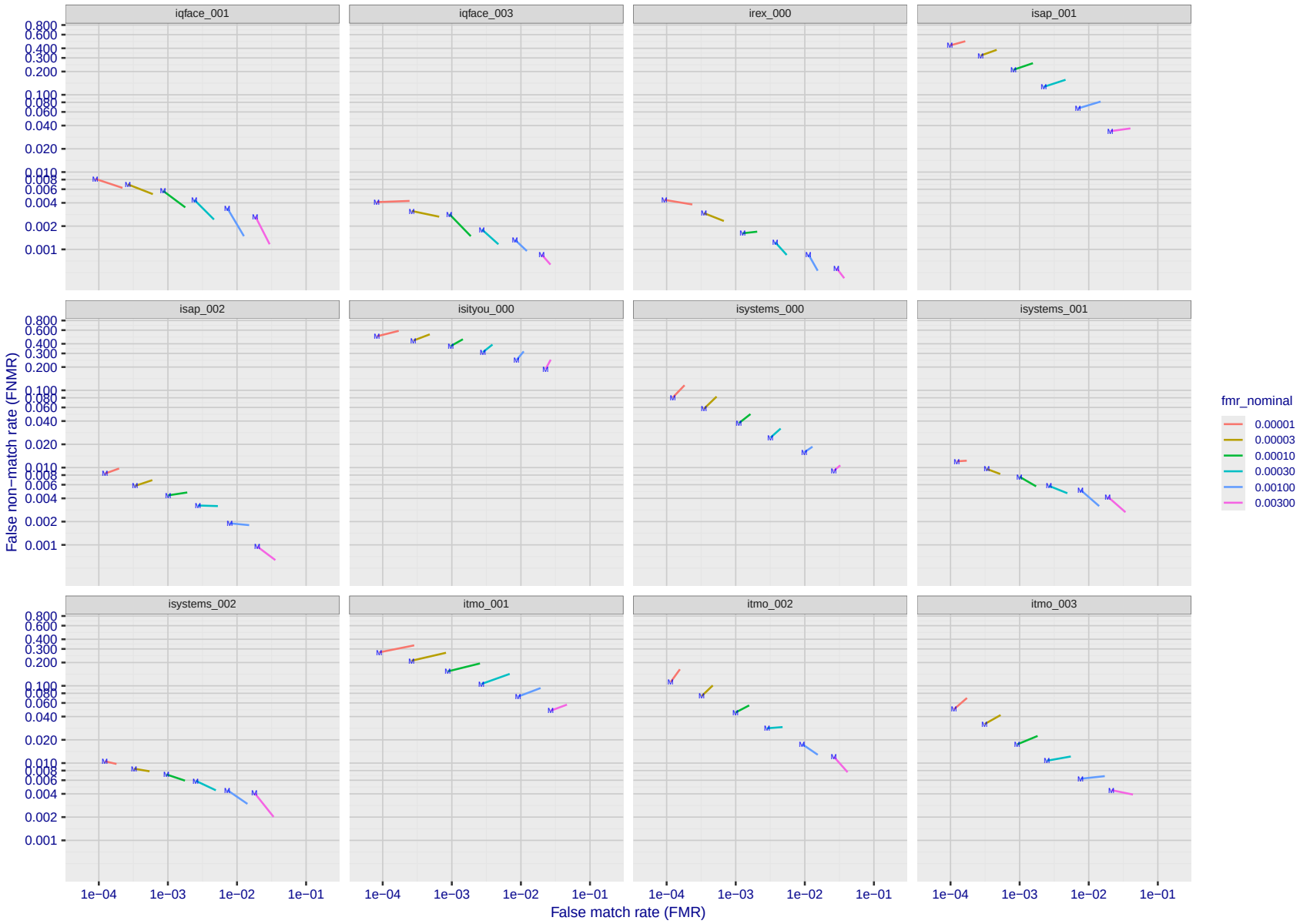

Figure 504: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

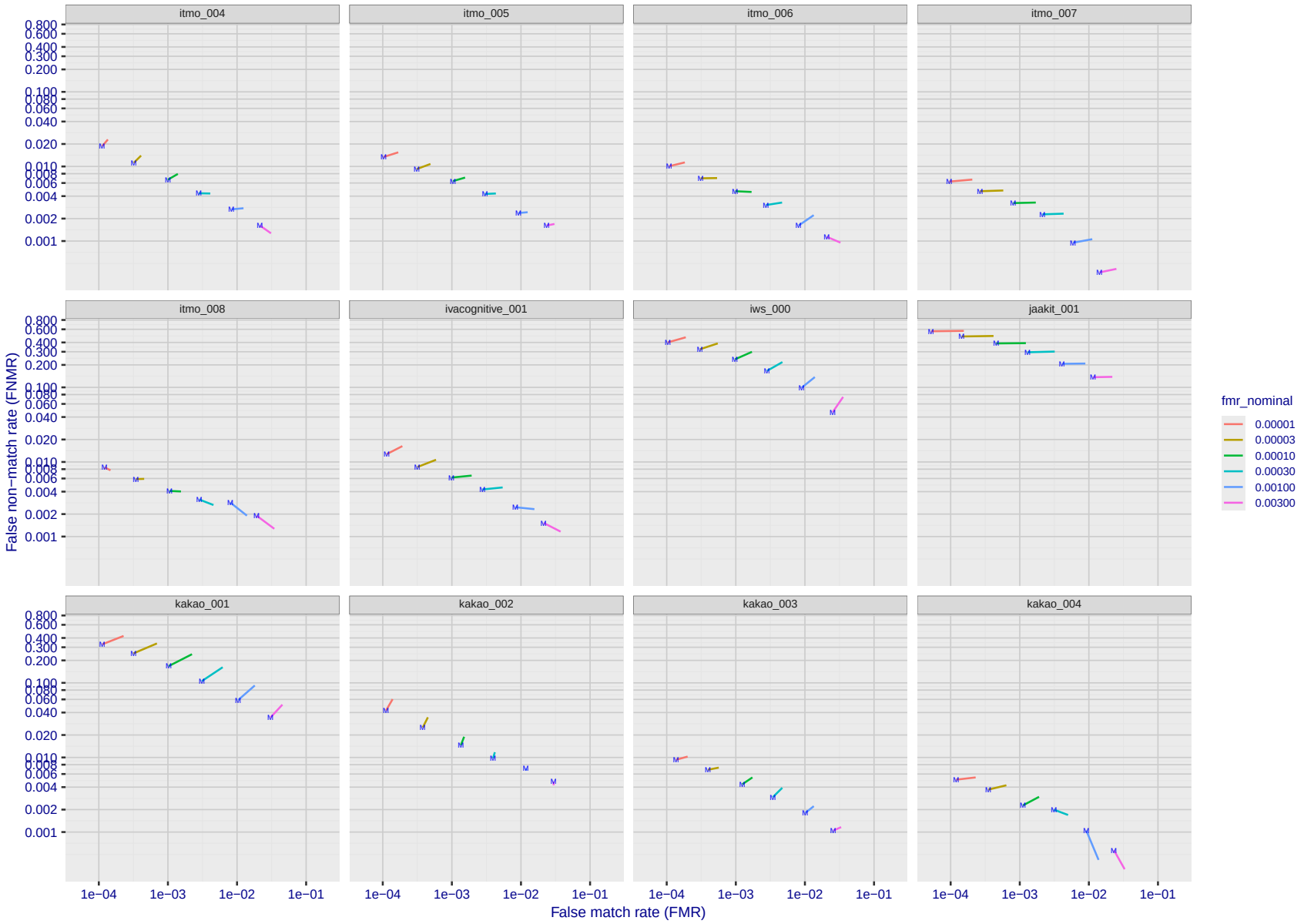

Figure 505: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

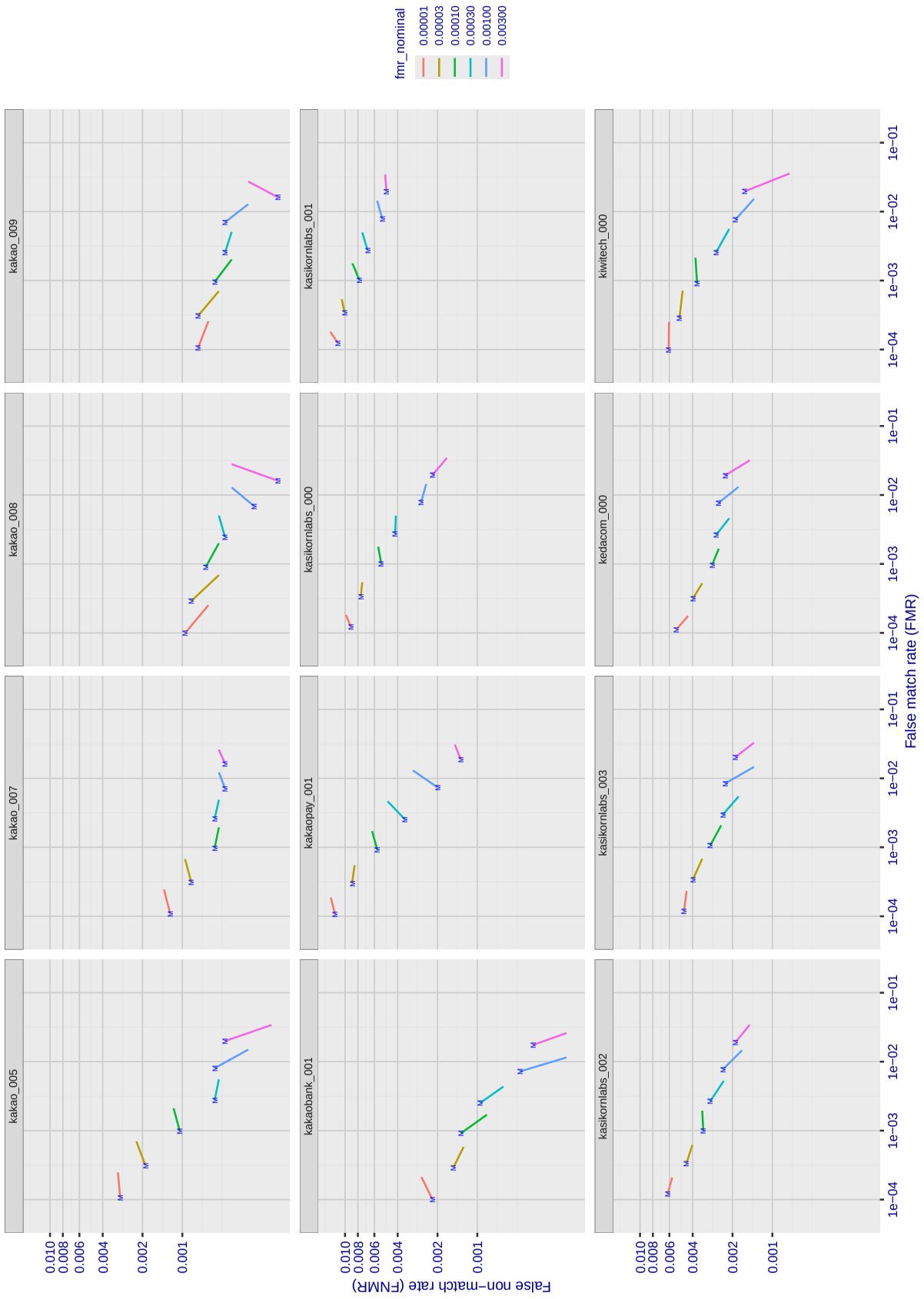

Figure 506: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

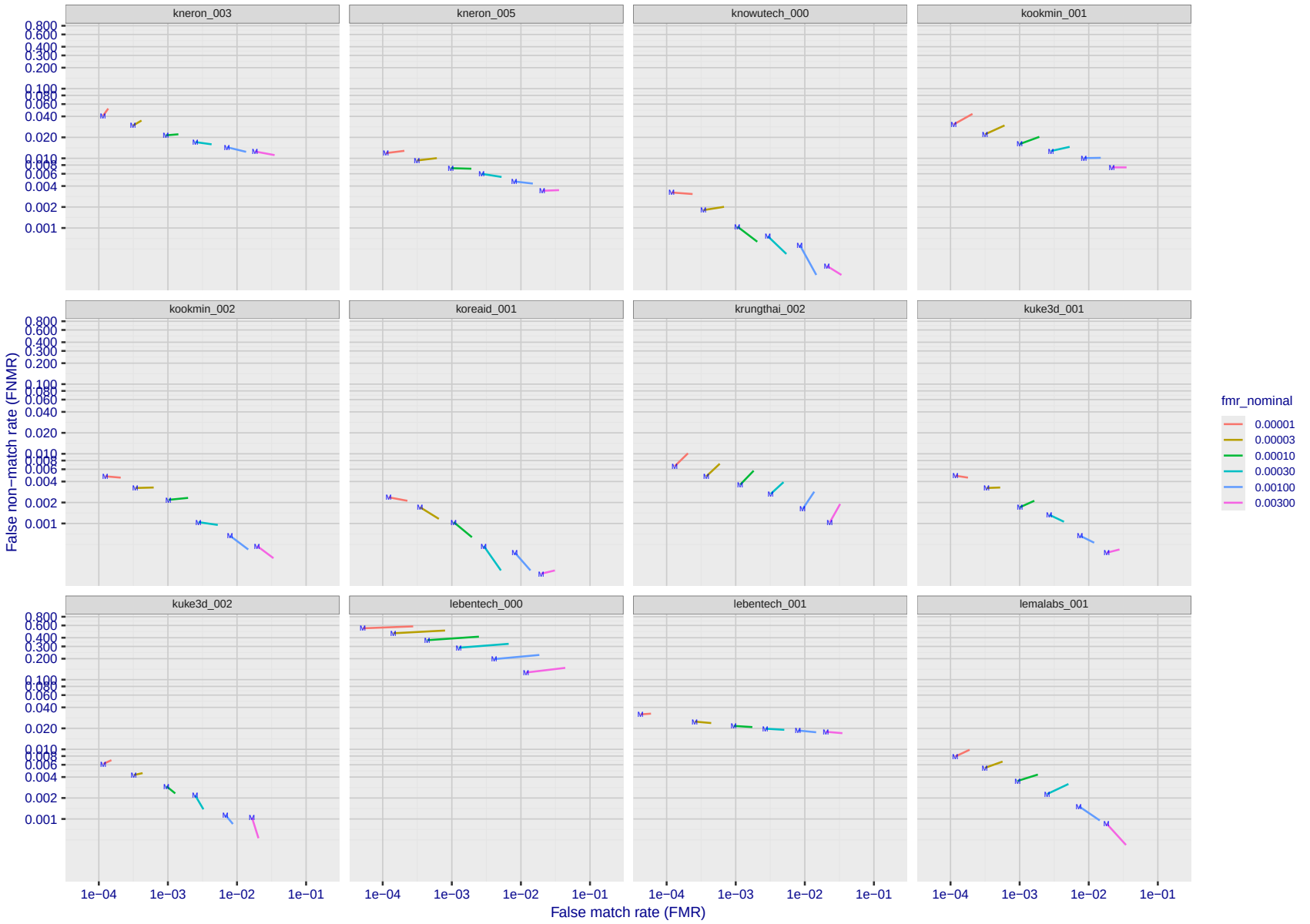

Figure 507: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

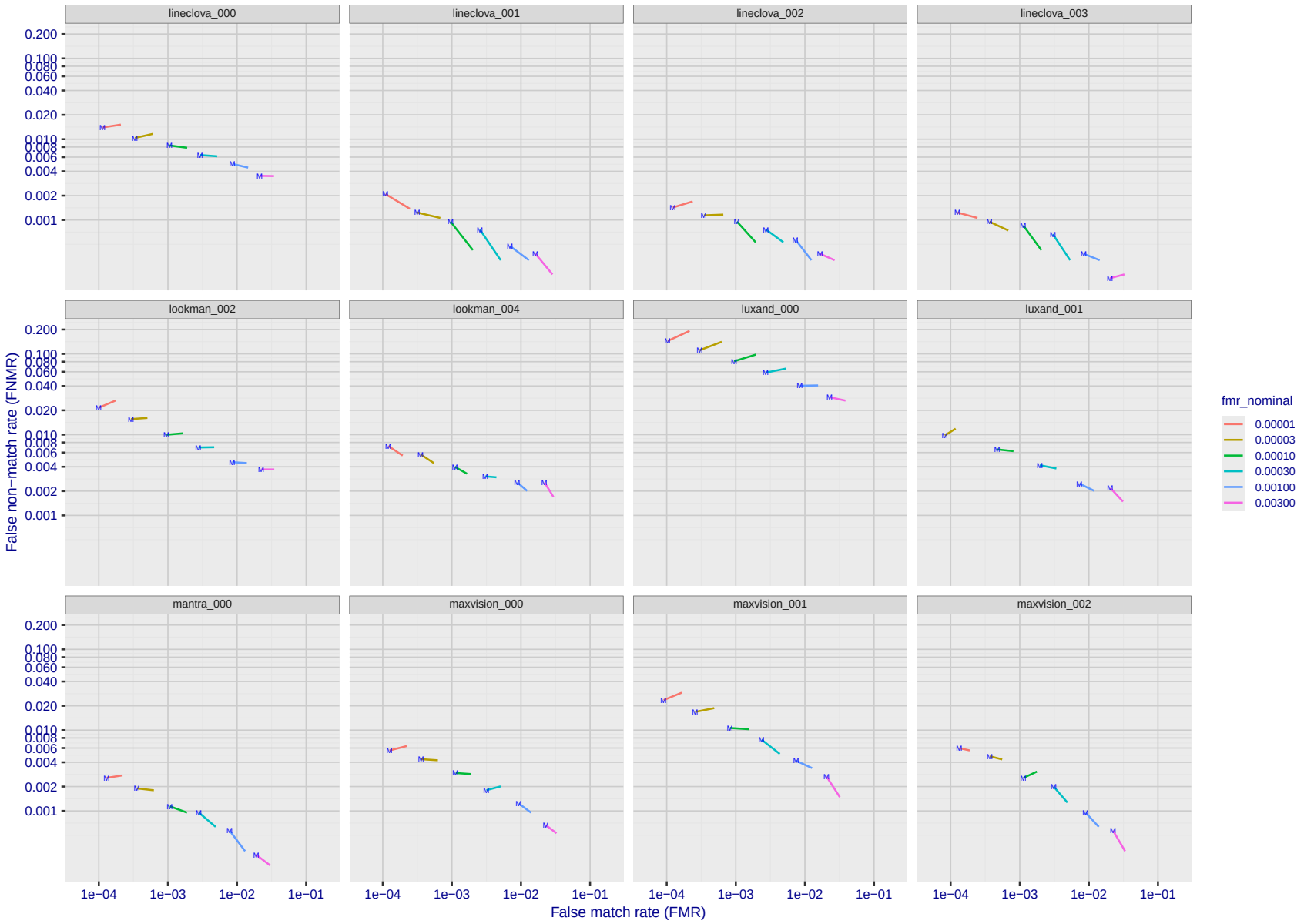

Figure 508: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

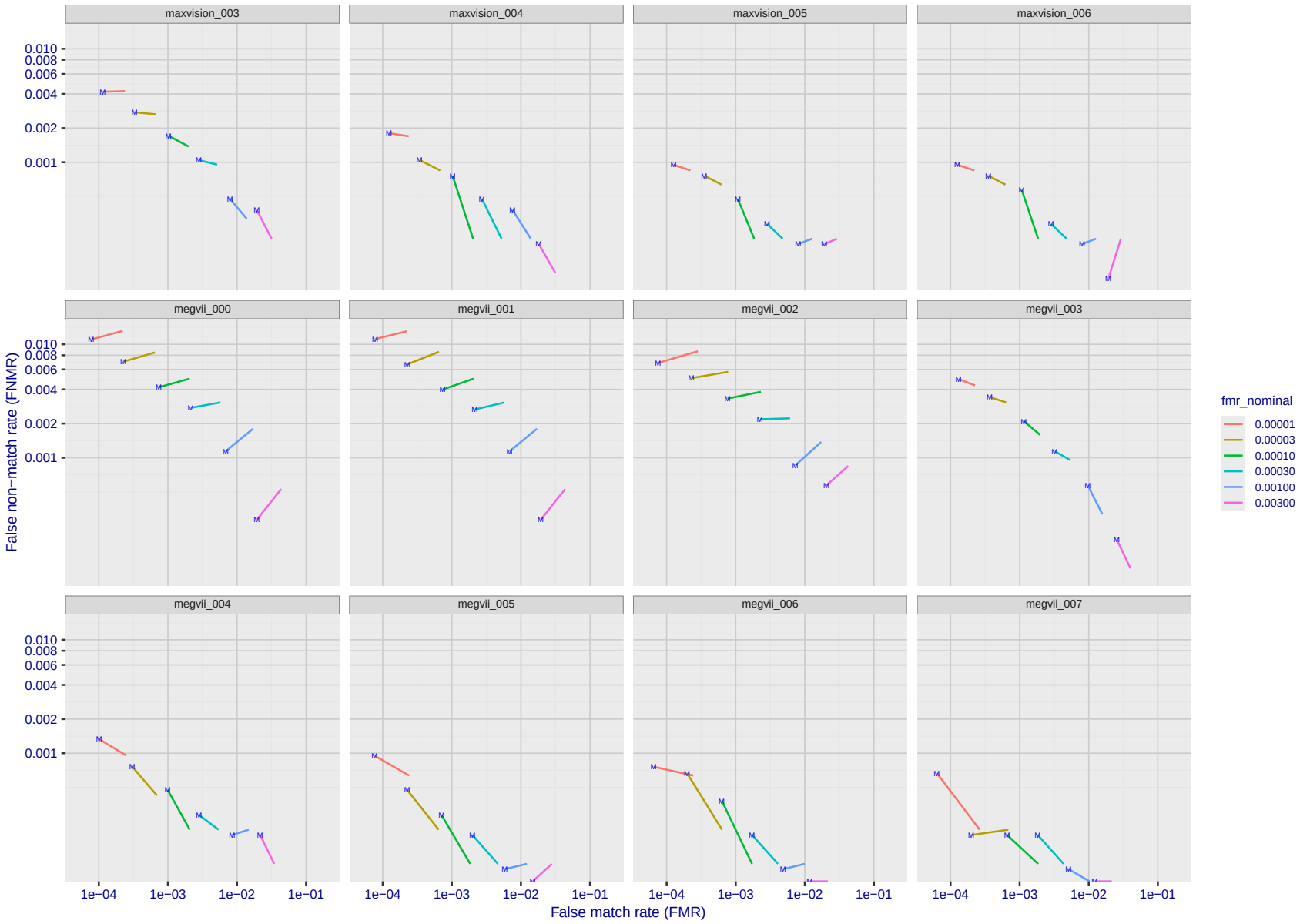

Figure 509: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

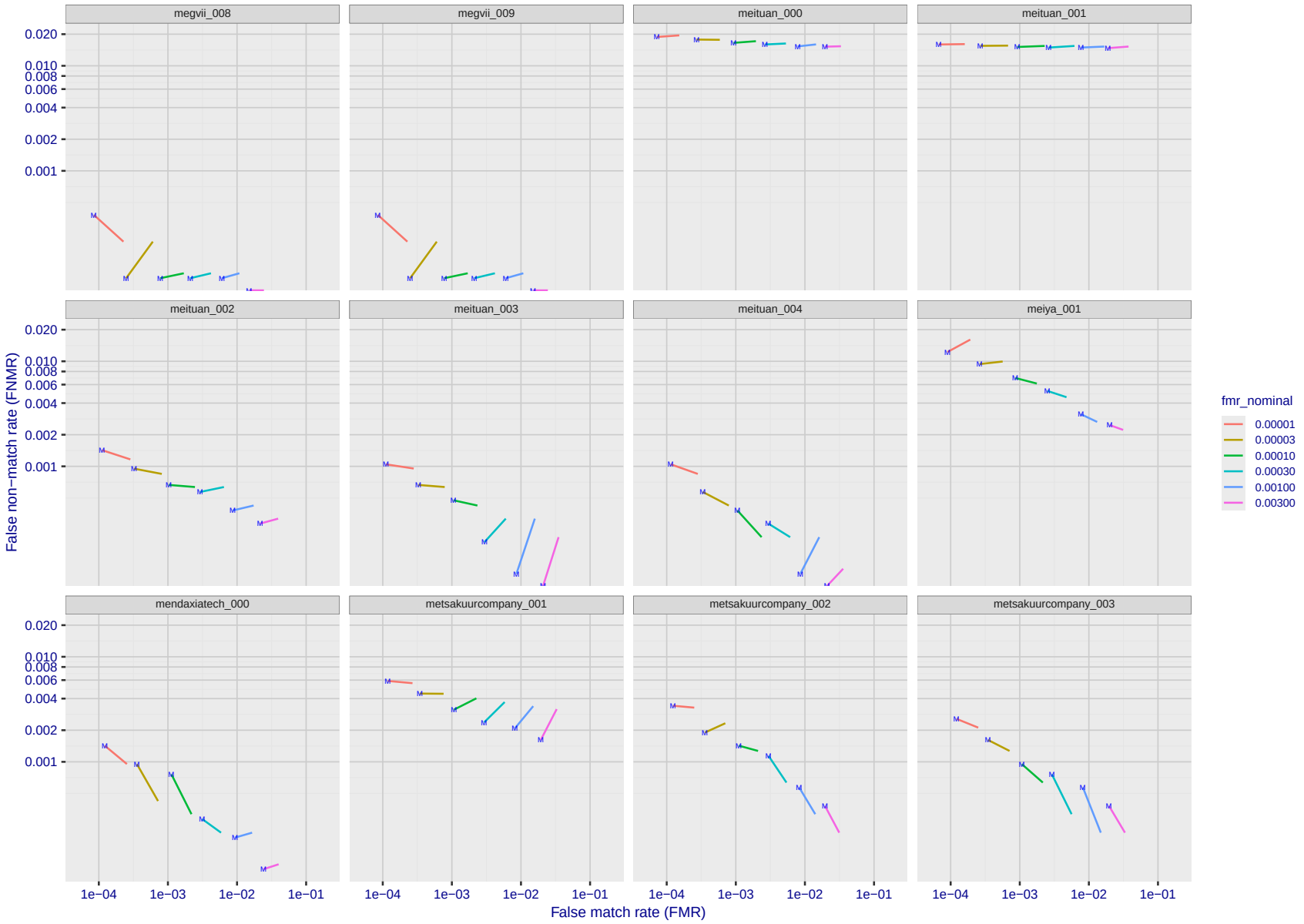

Figure 510: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

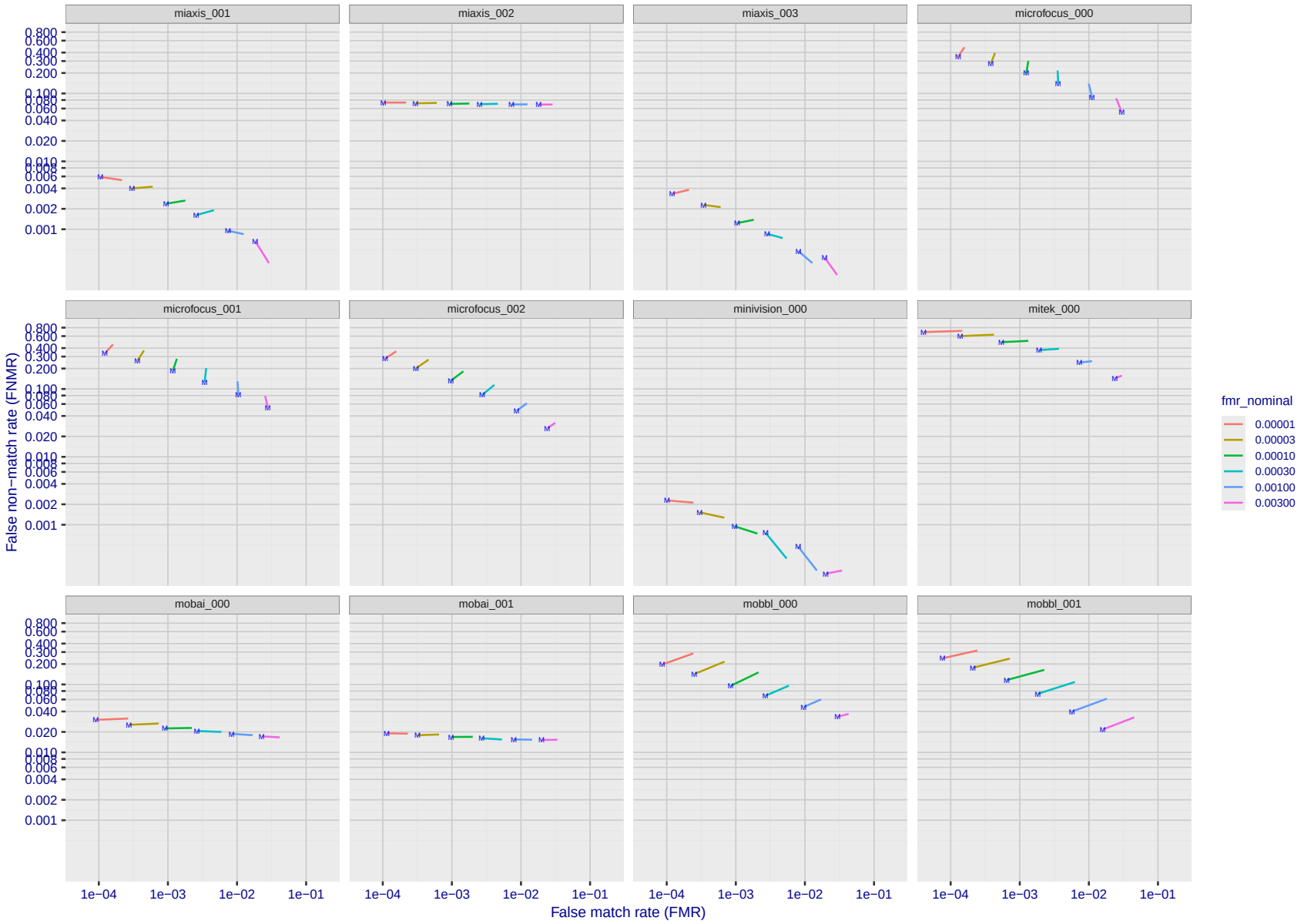

Figure 511: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

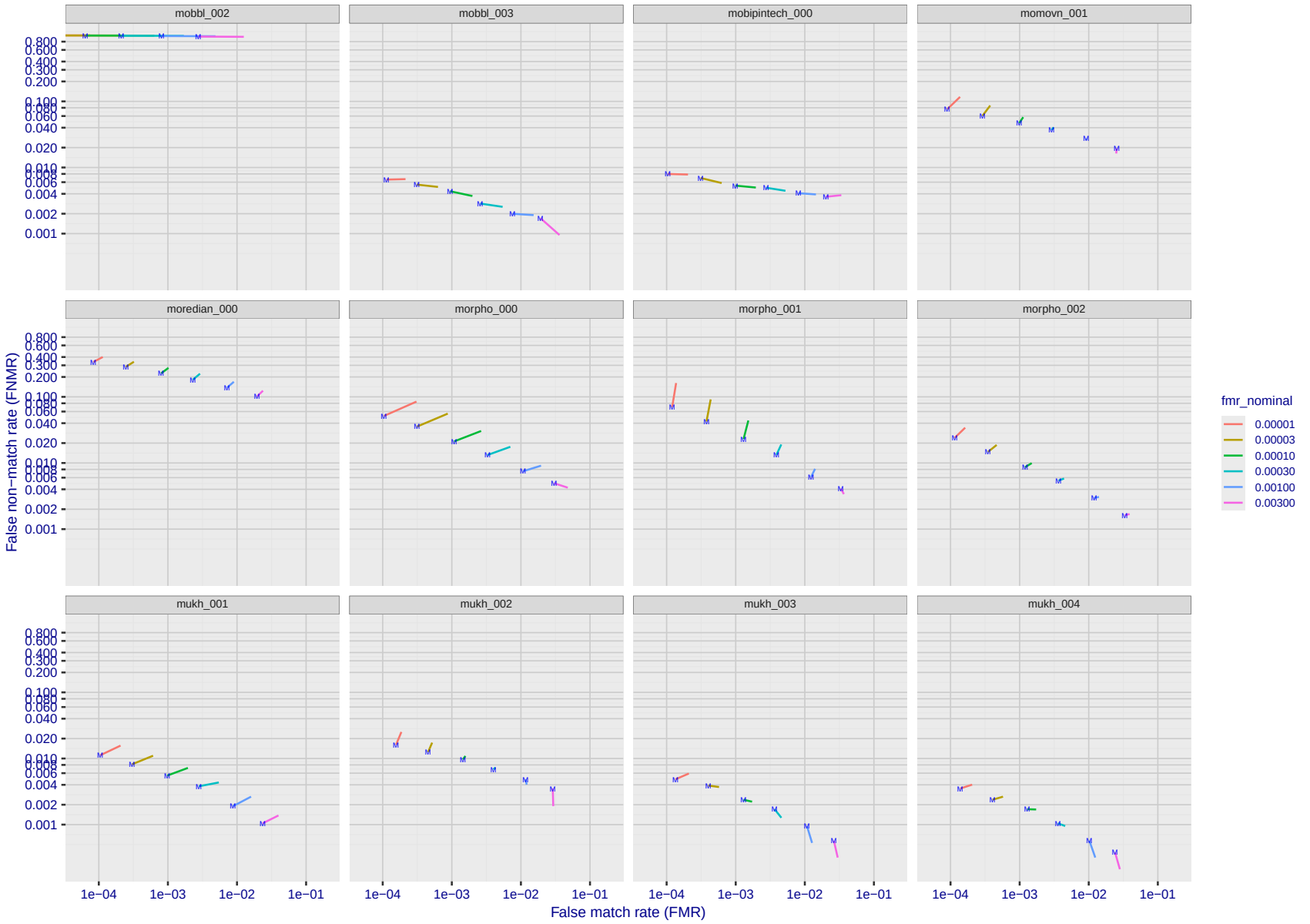

Figure 512: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

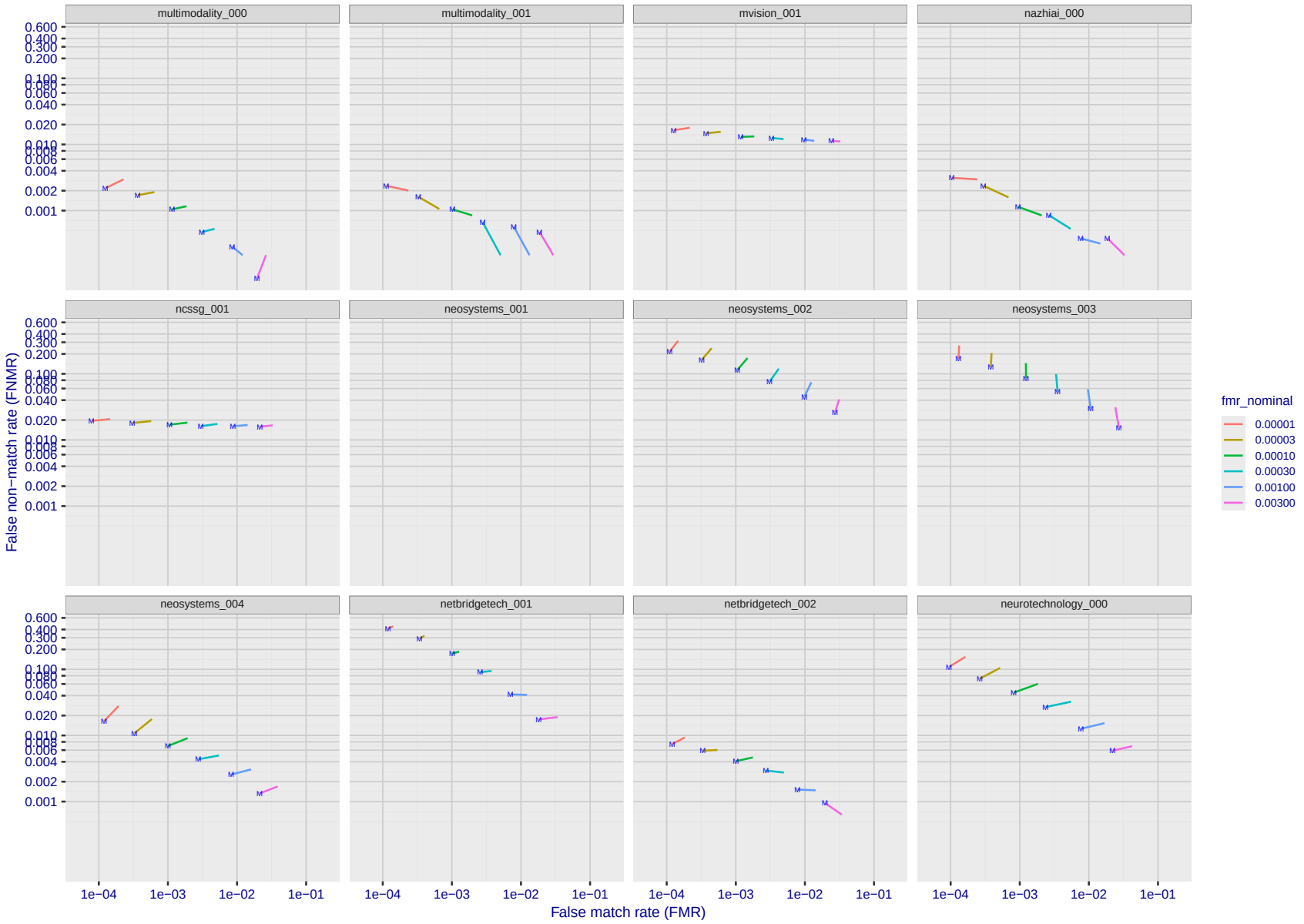

Figure 513: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

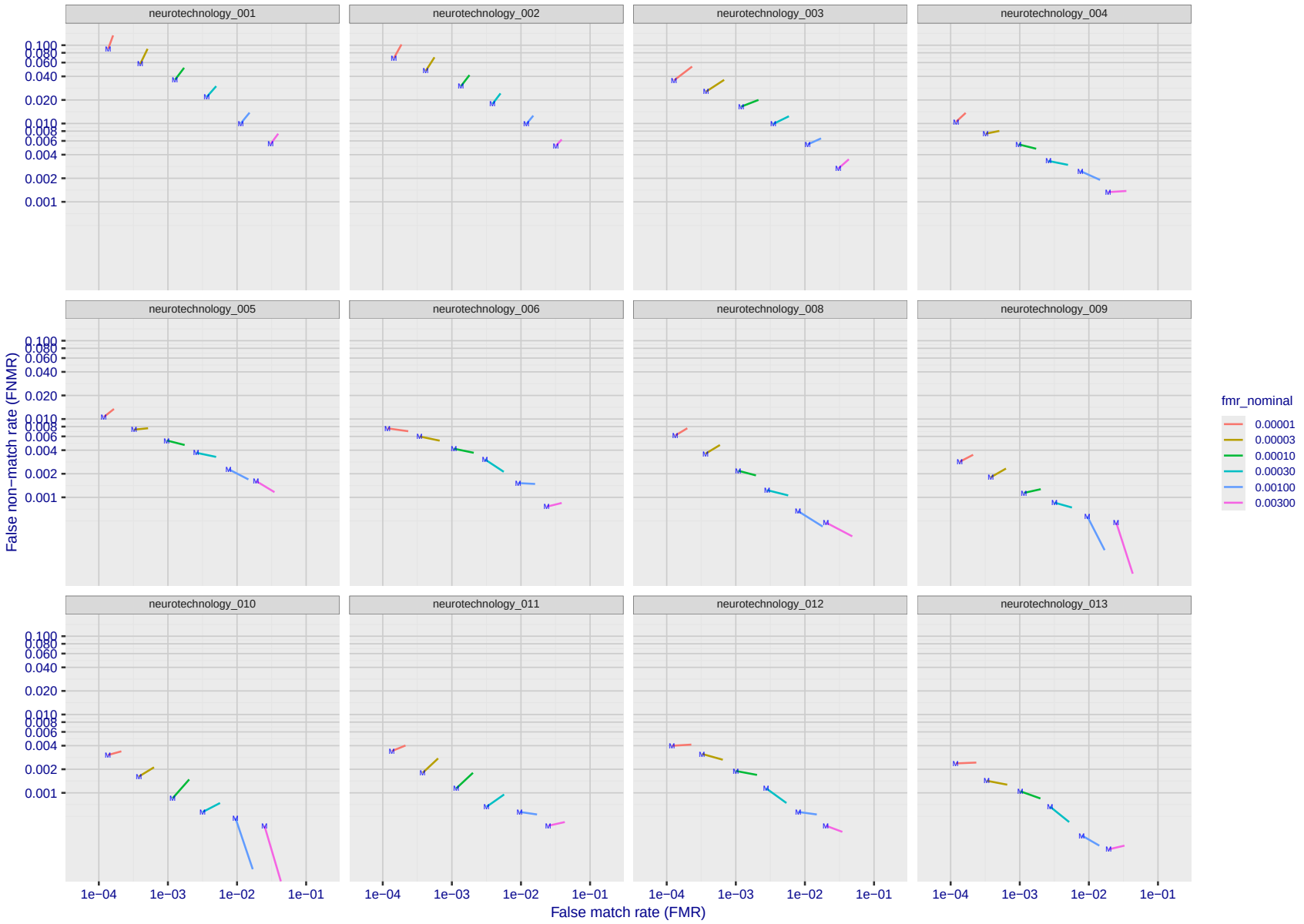

Figure 514: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

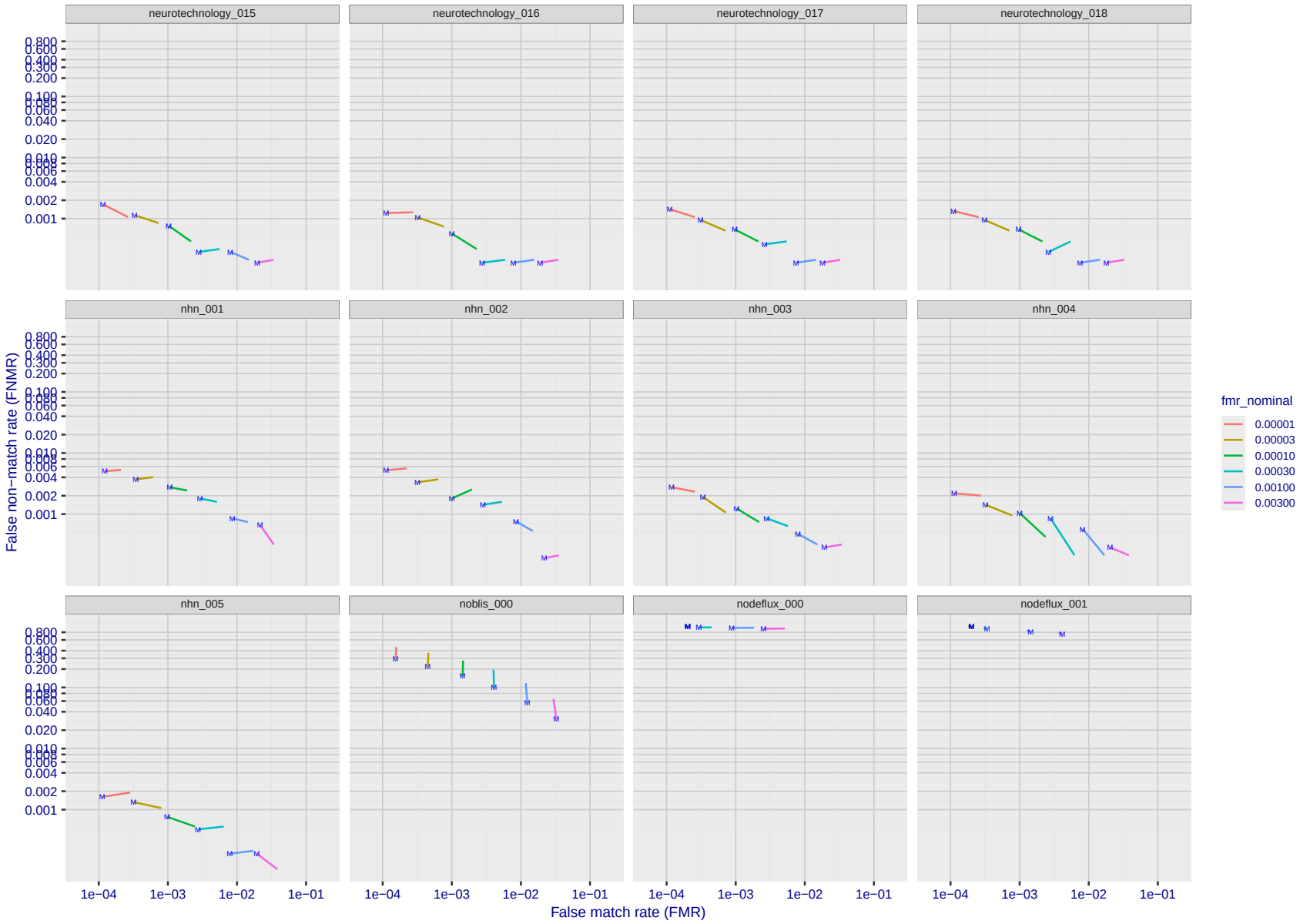

Figure 515: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

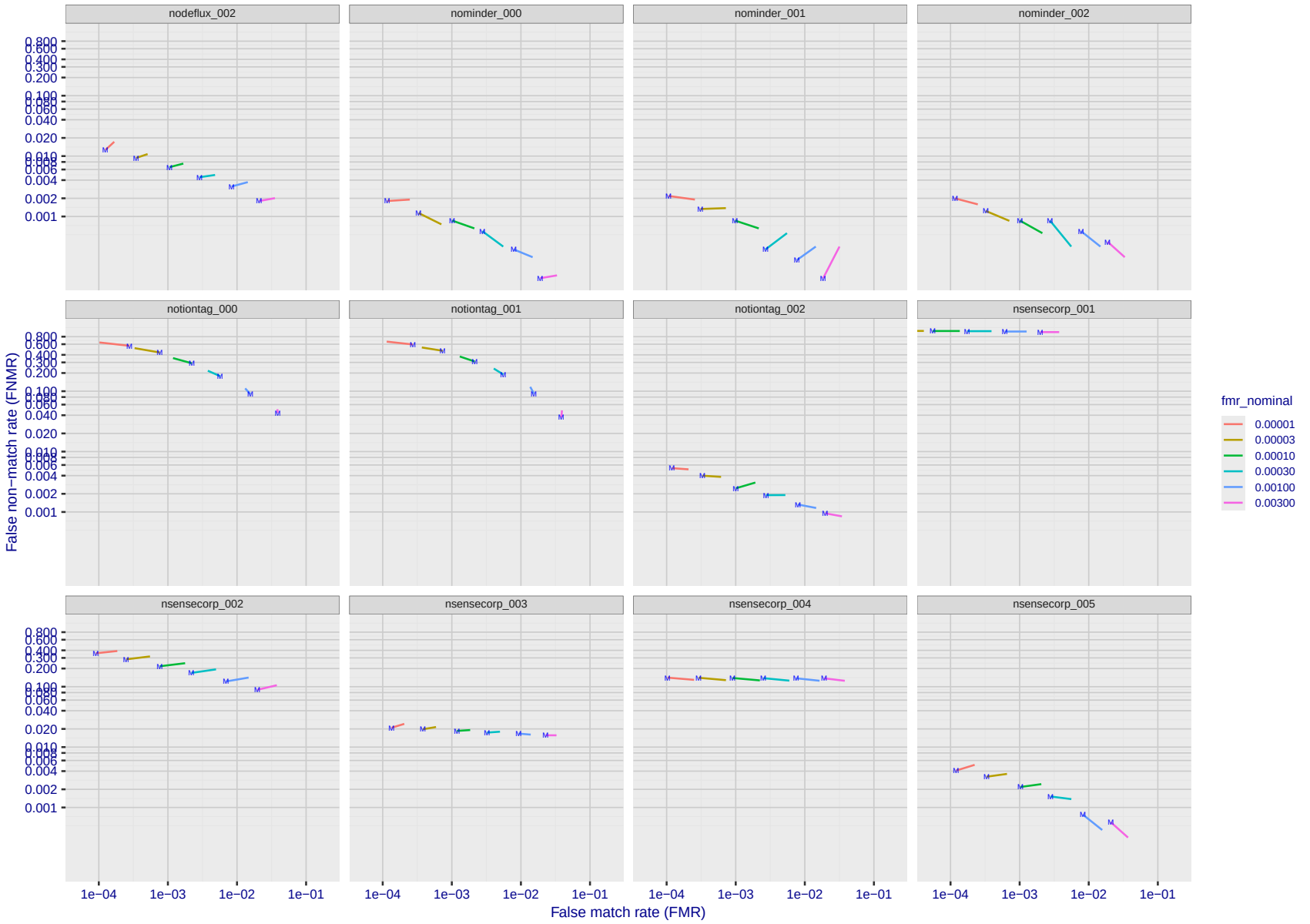

Figure 516: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

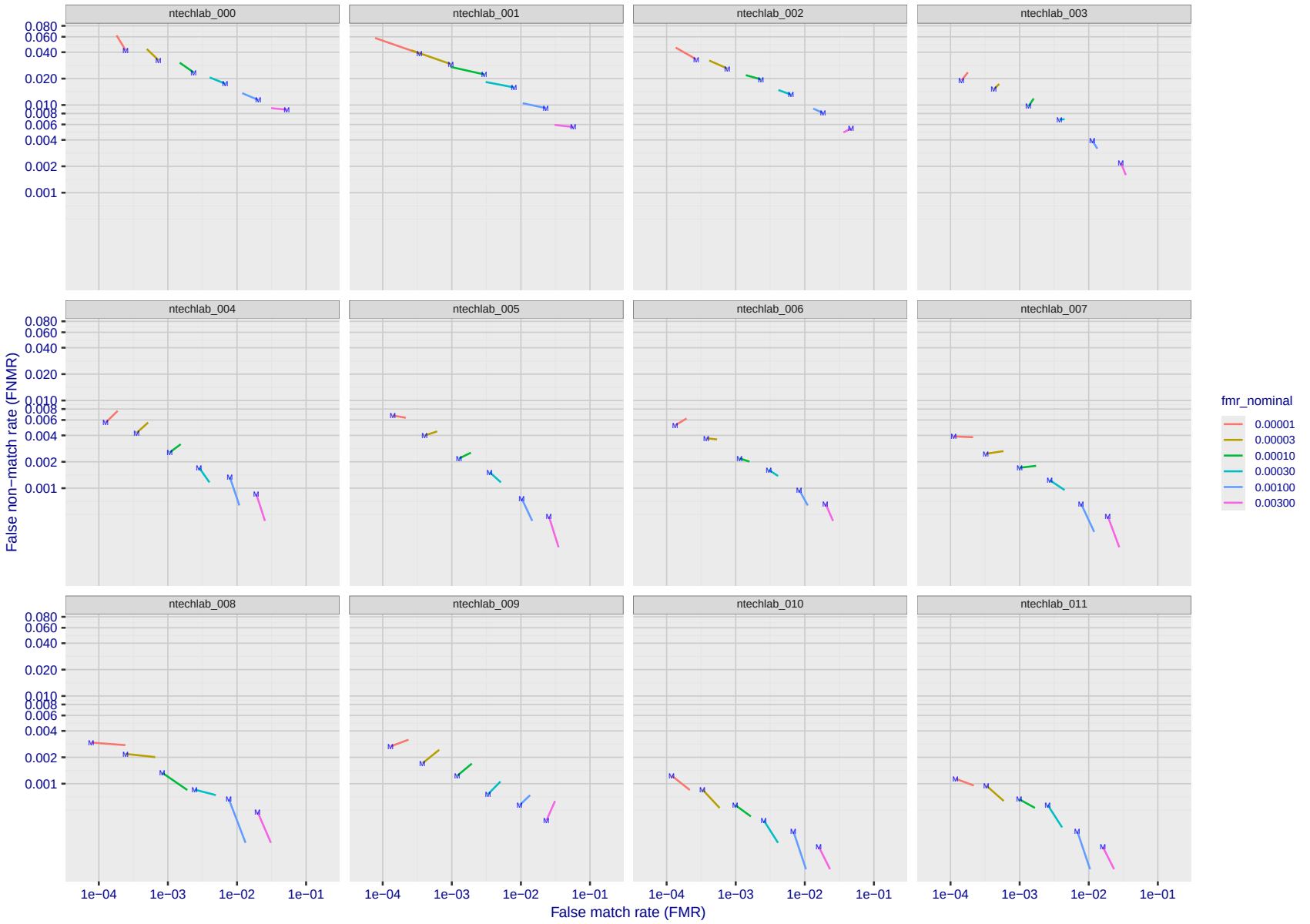

Figure 517: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

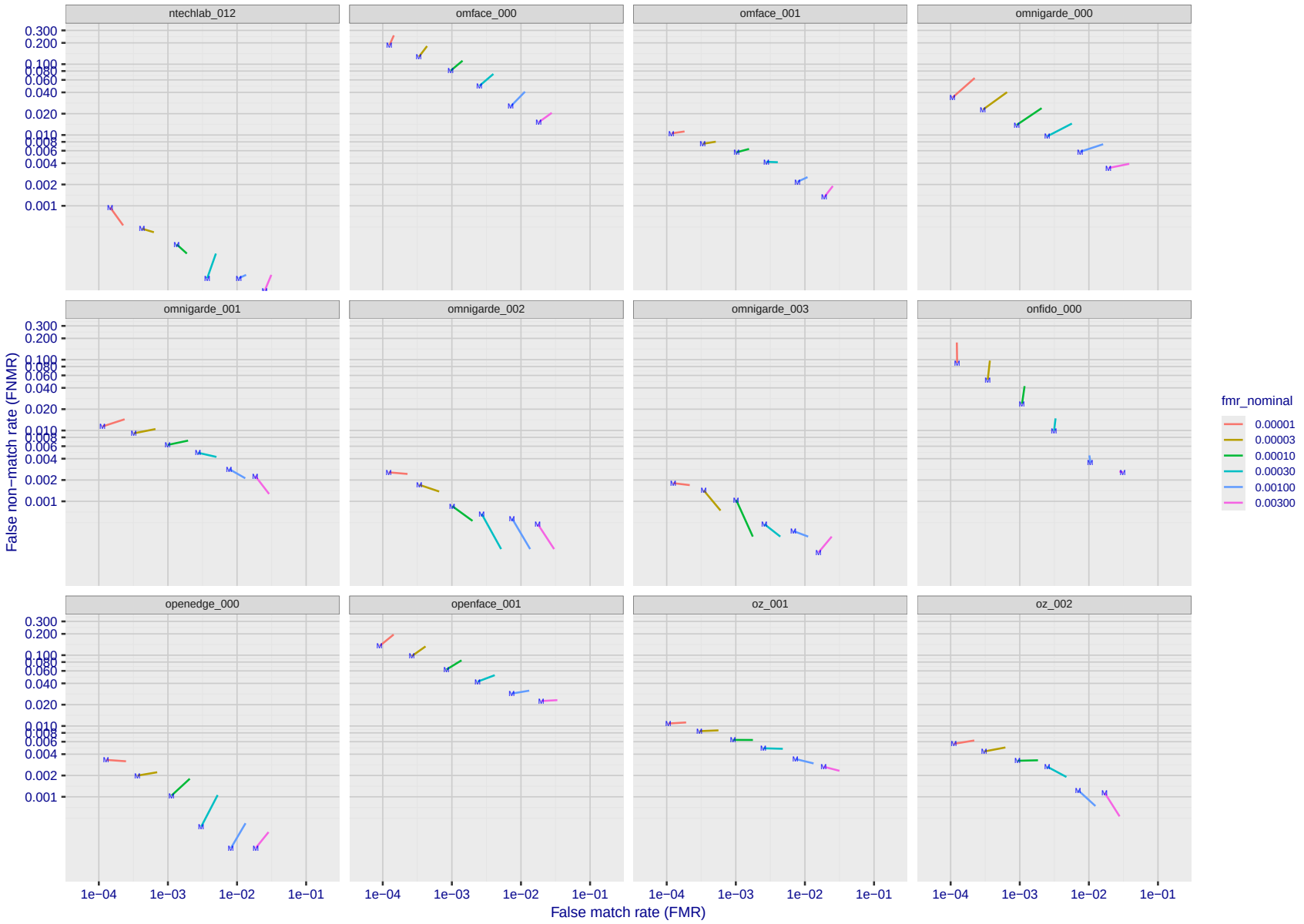

Figure 518: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

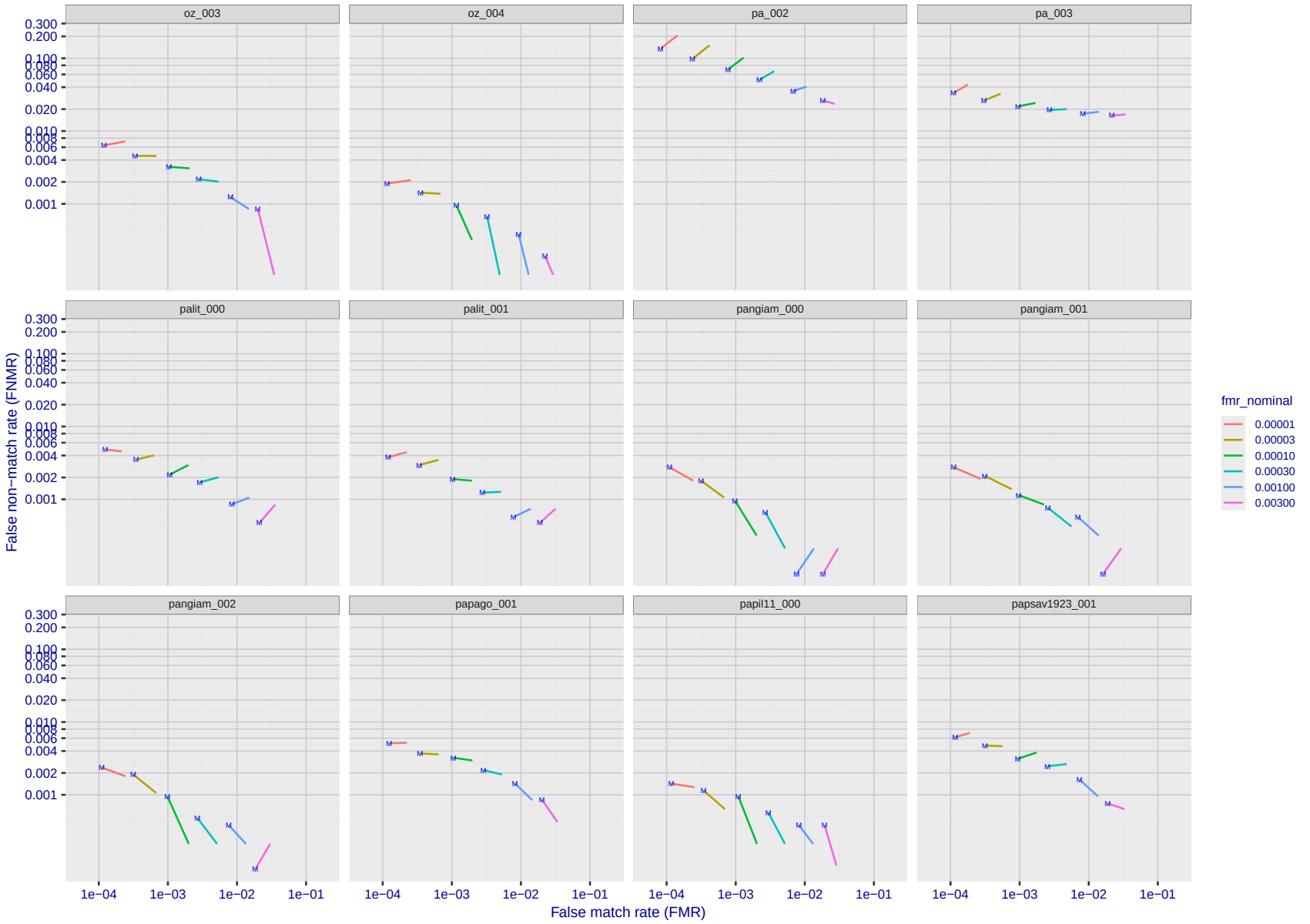

Figure 519: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

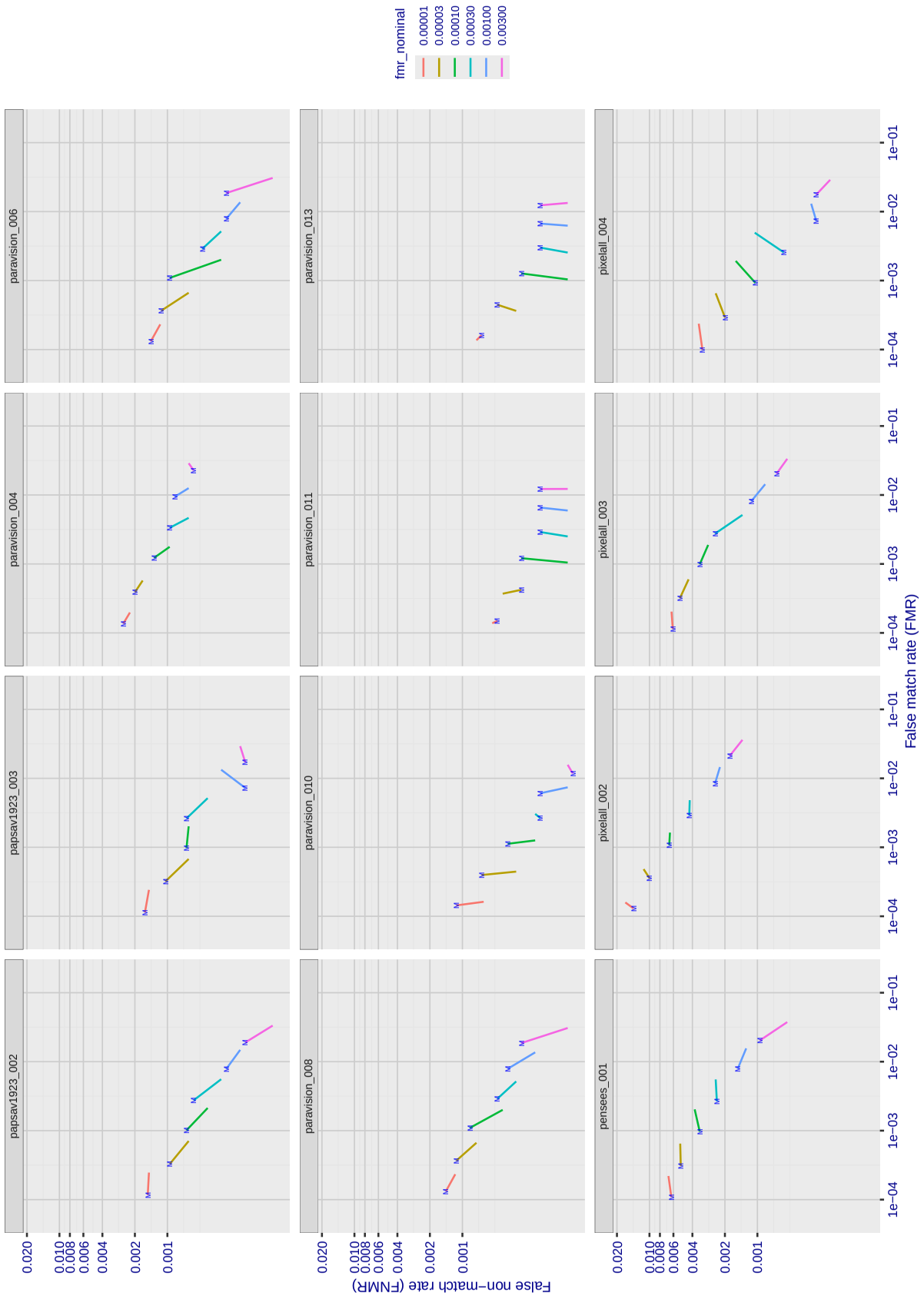

Figure 520: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
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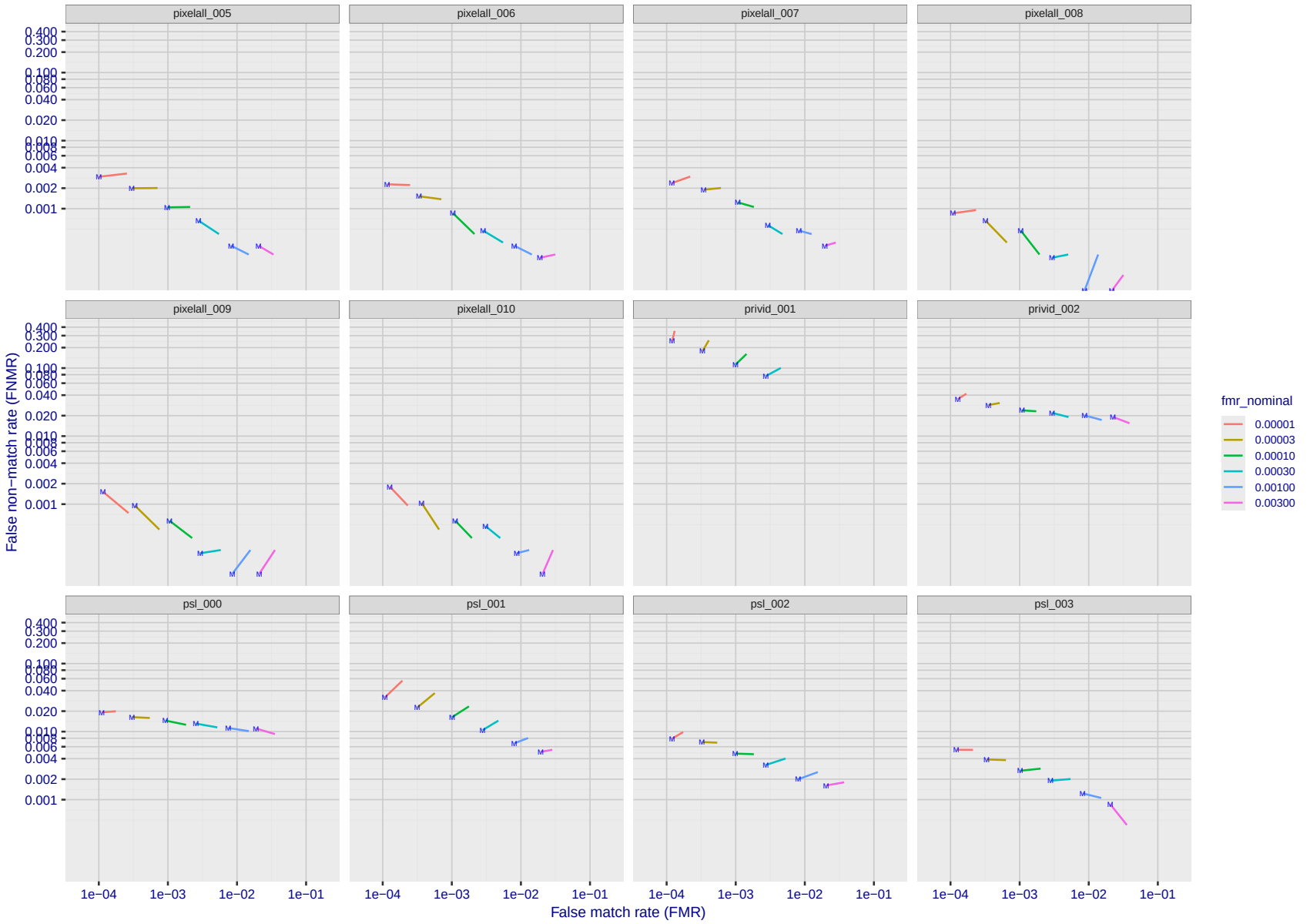

Figure 521: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

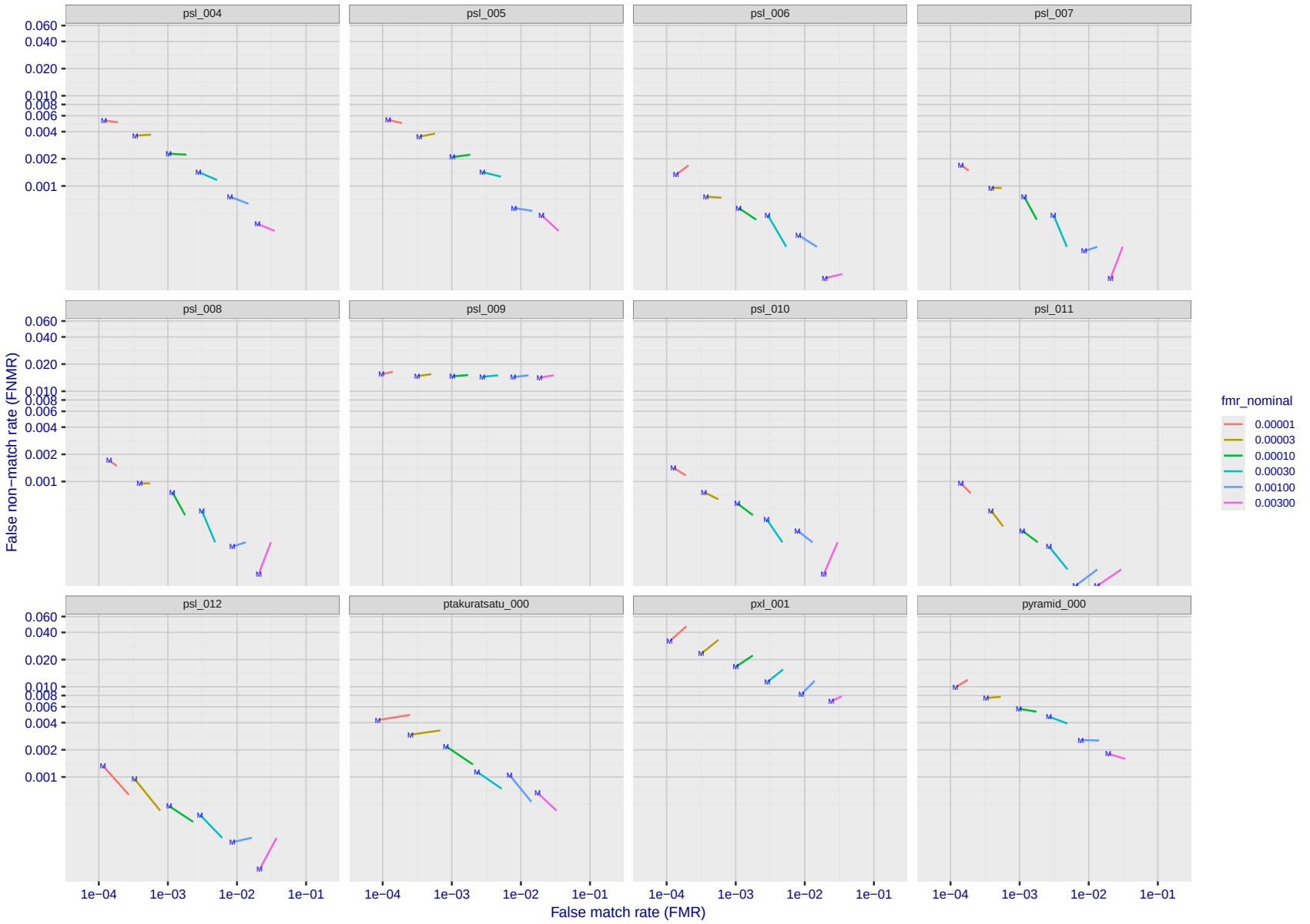

Figure 522: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

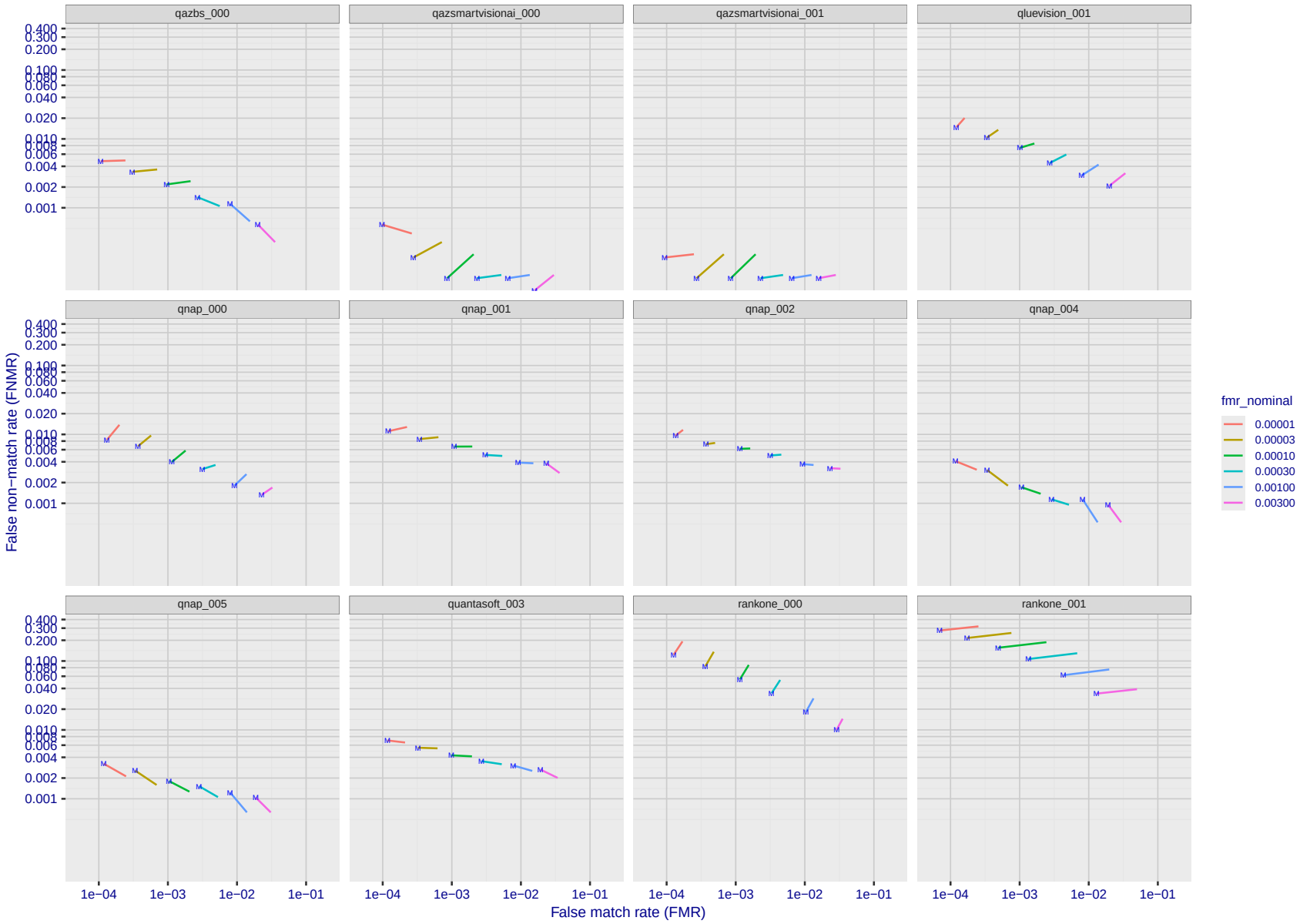

Figure 523: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

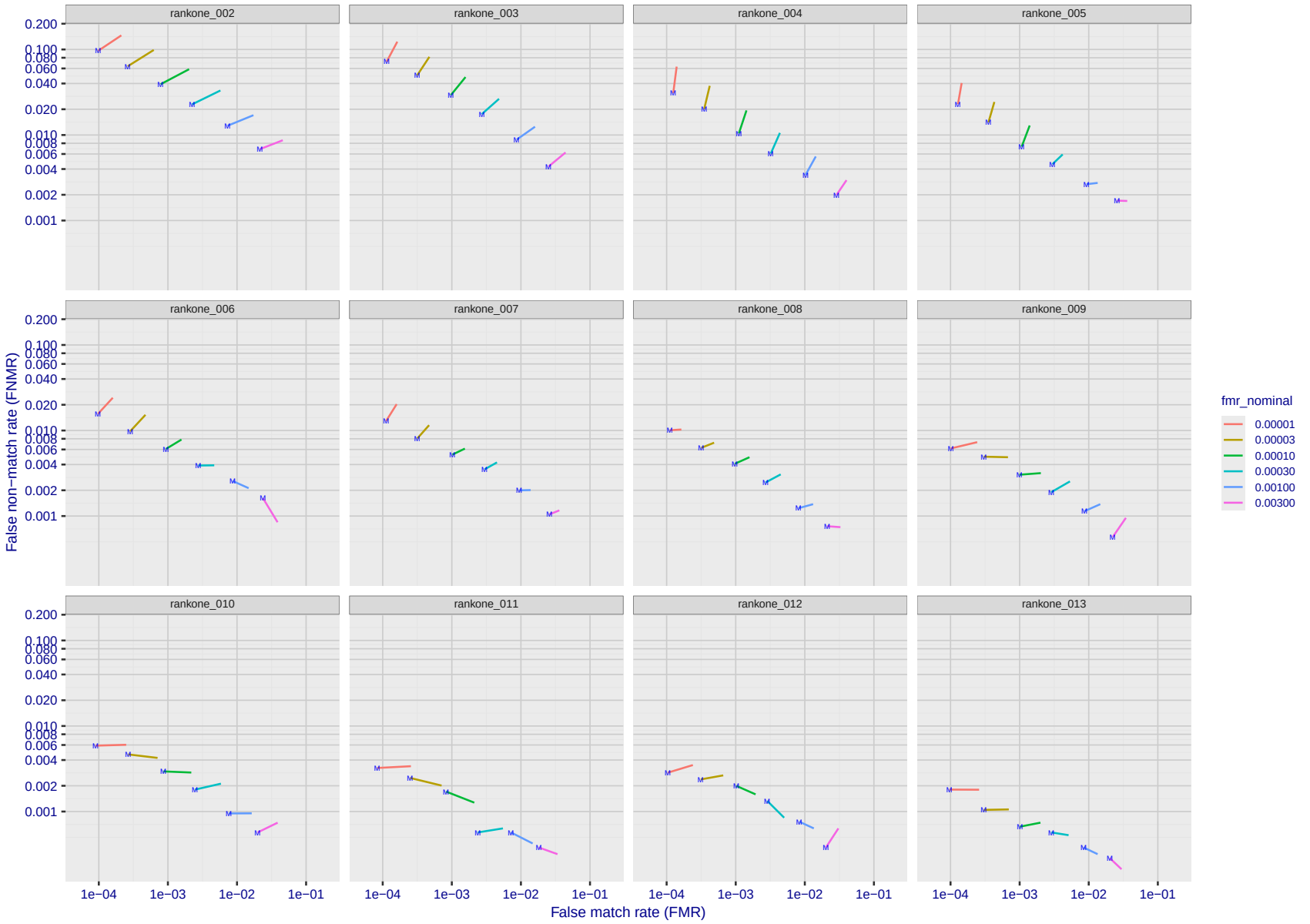

Figure 524: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

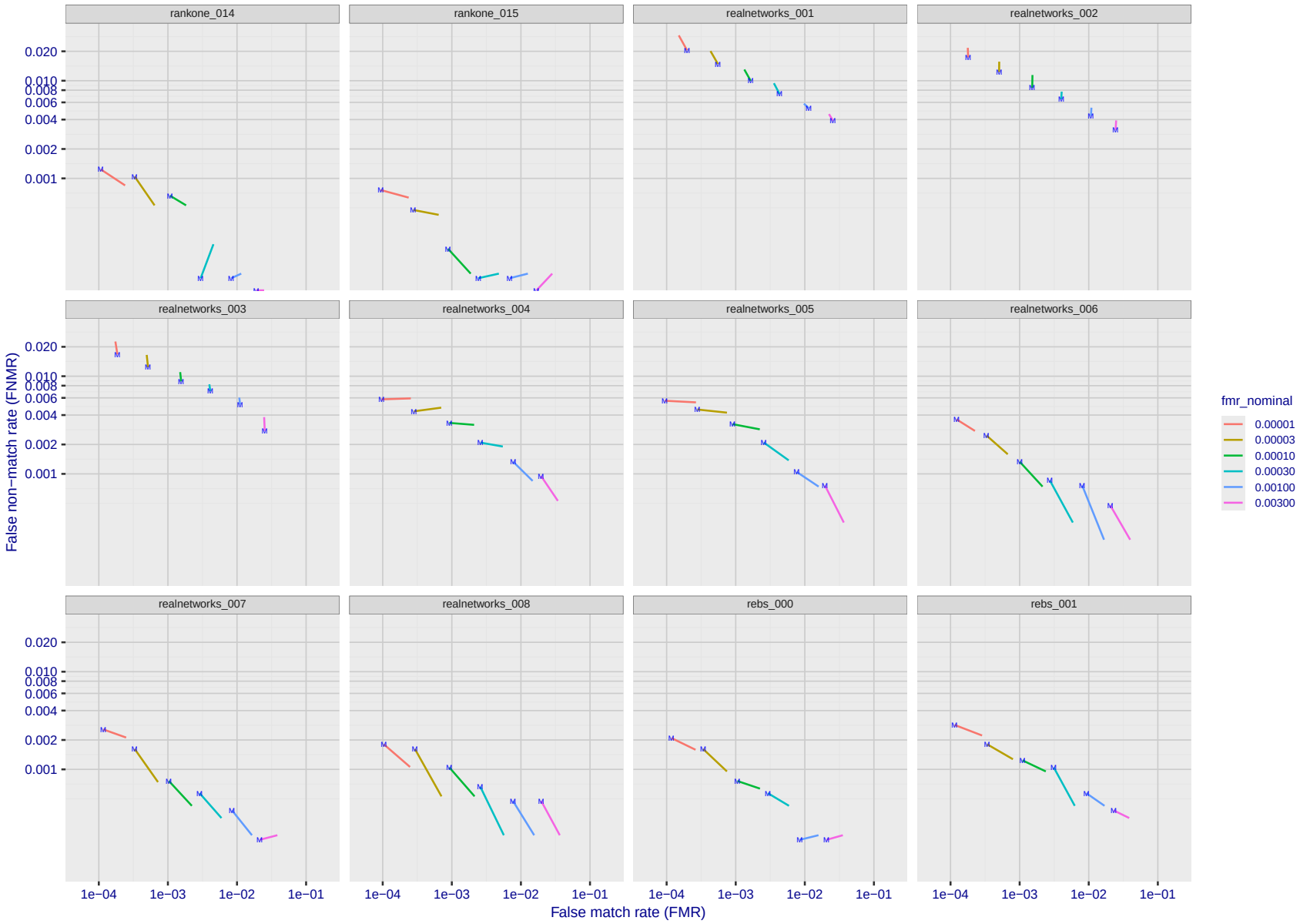

Figure 525: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

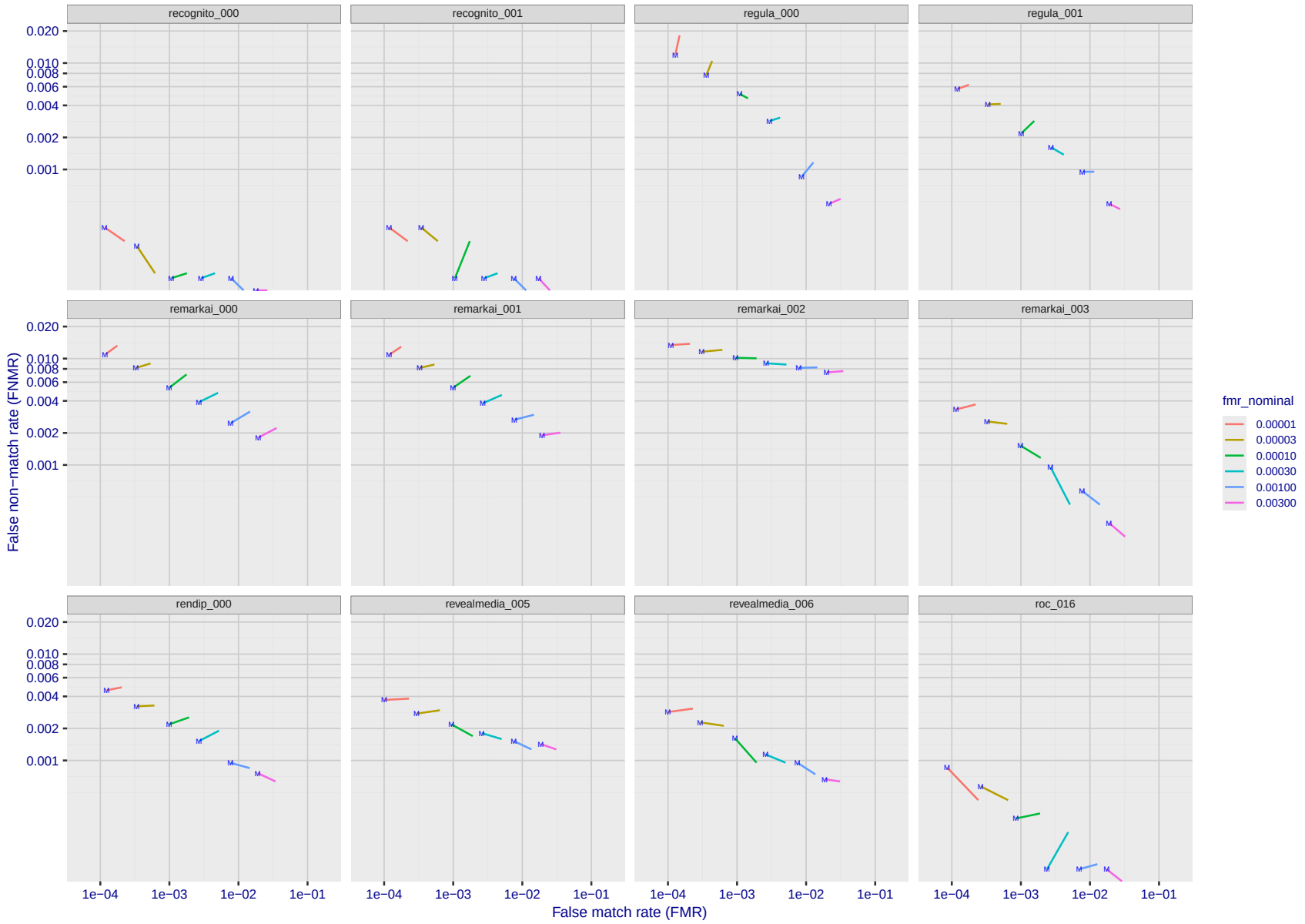

Figure 526: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

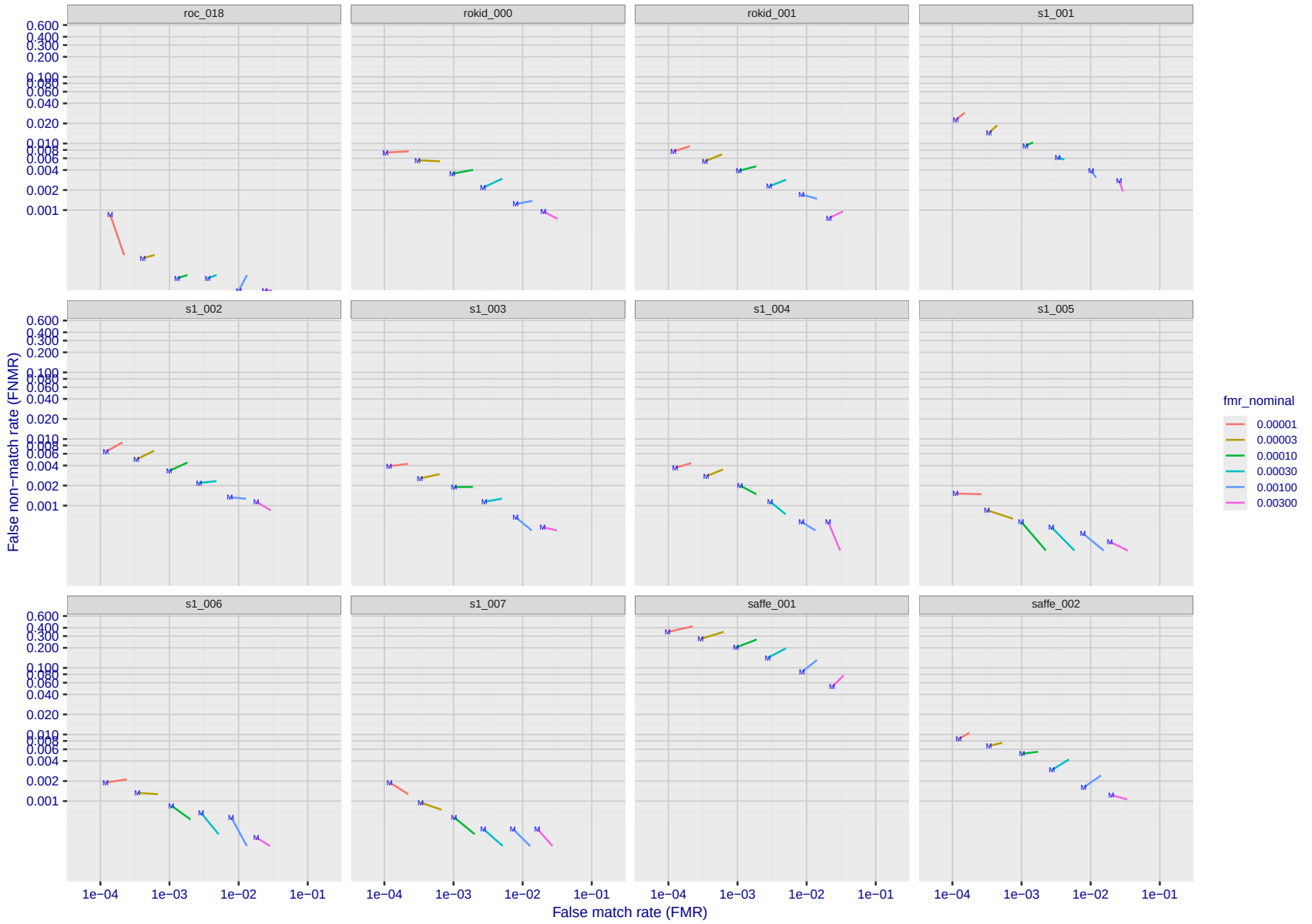

Figure 527: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

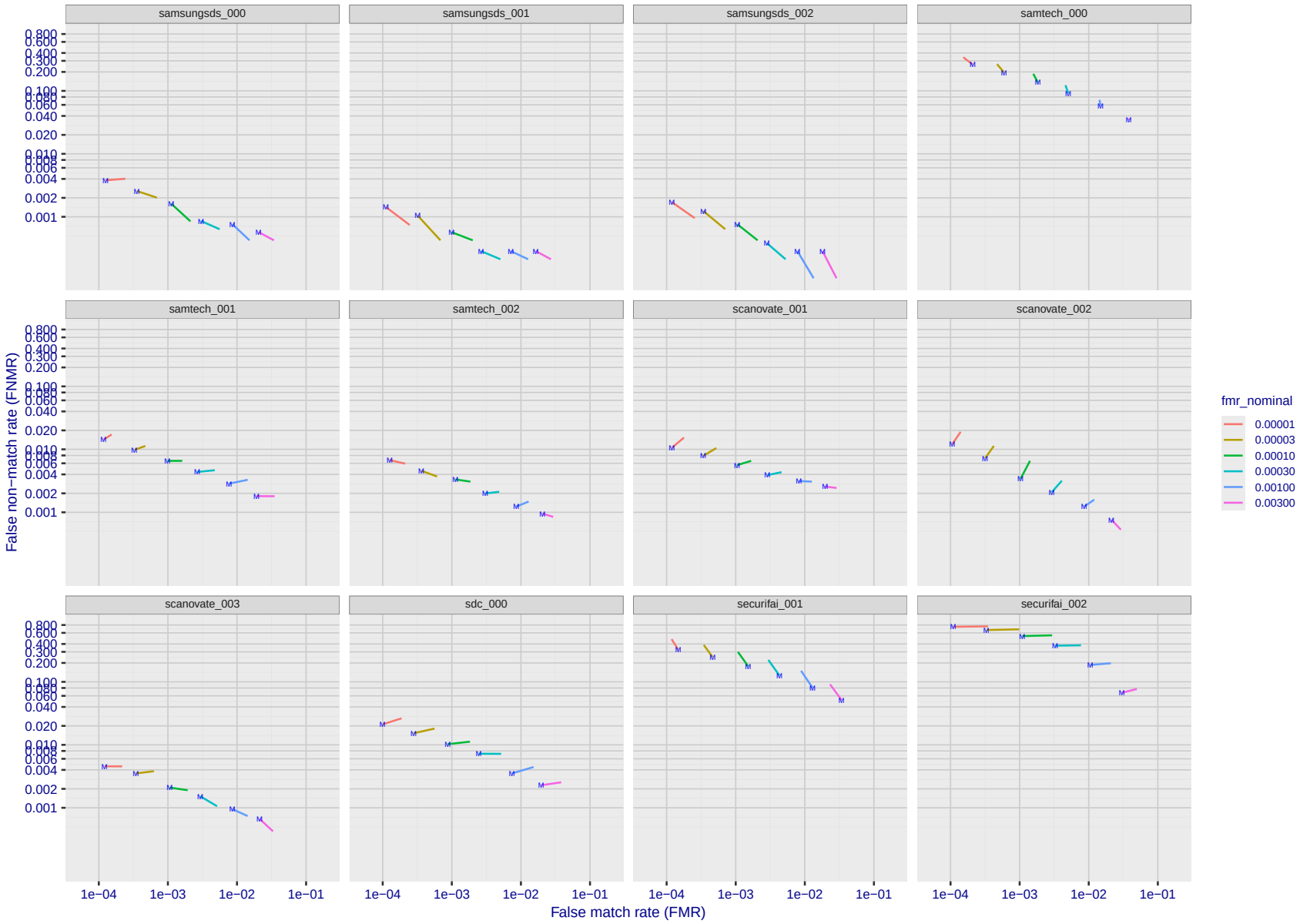

Figure 528: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

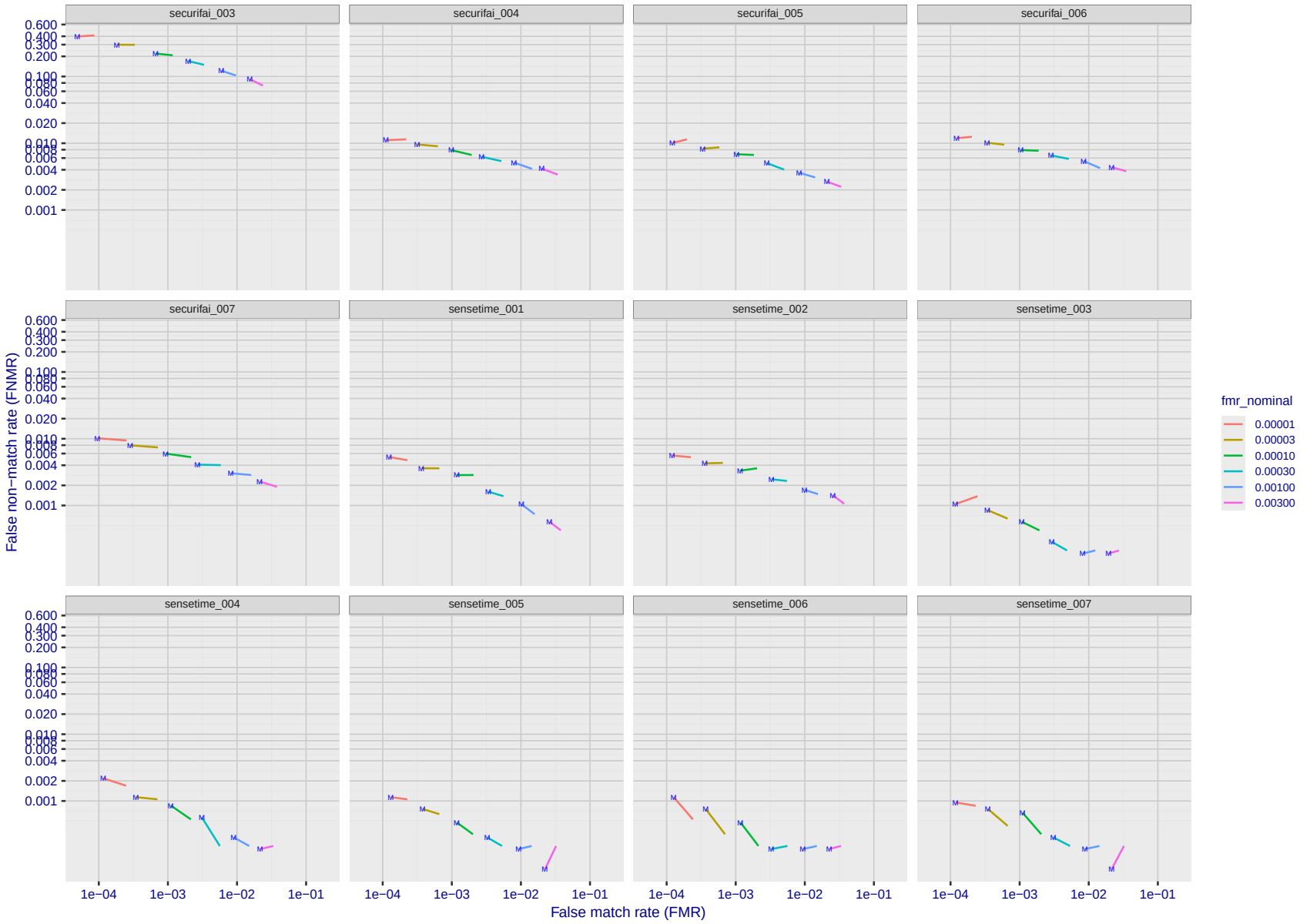

Figure 529: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

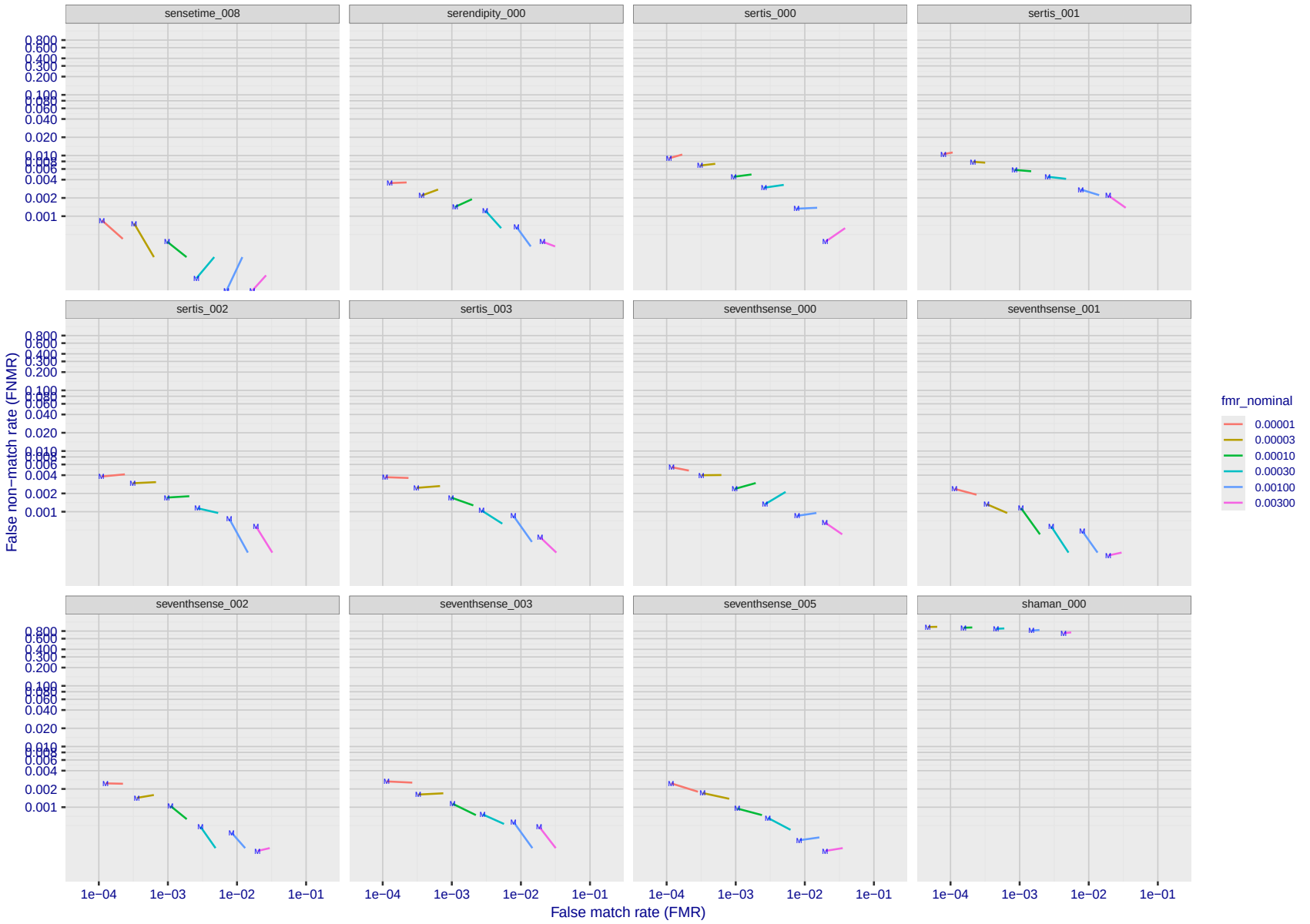

Figure 530: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

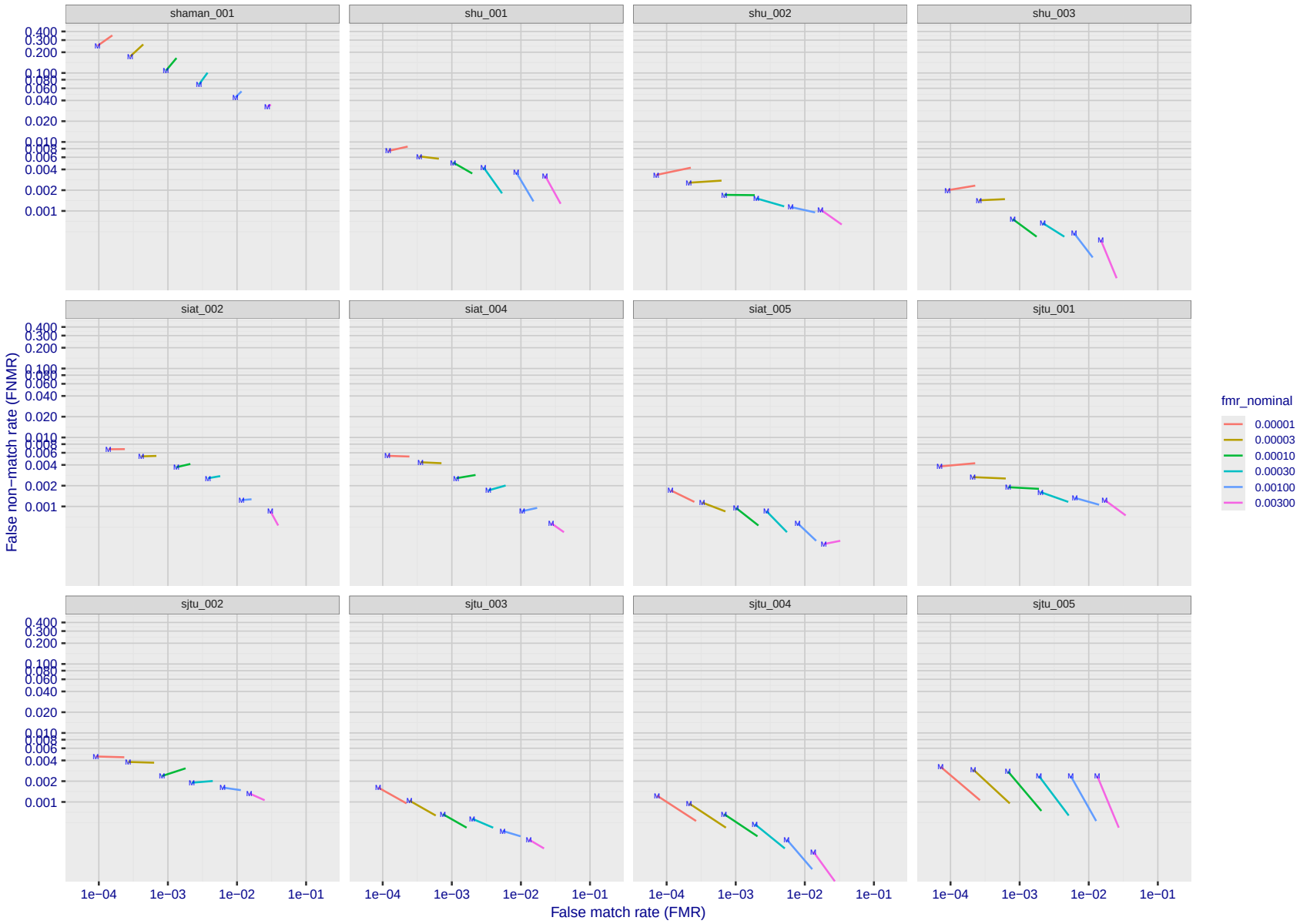

Figure 531: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

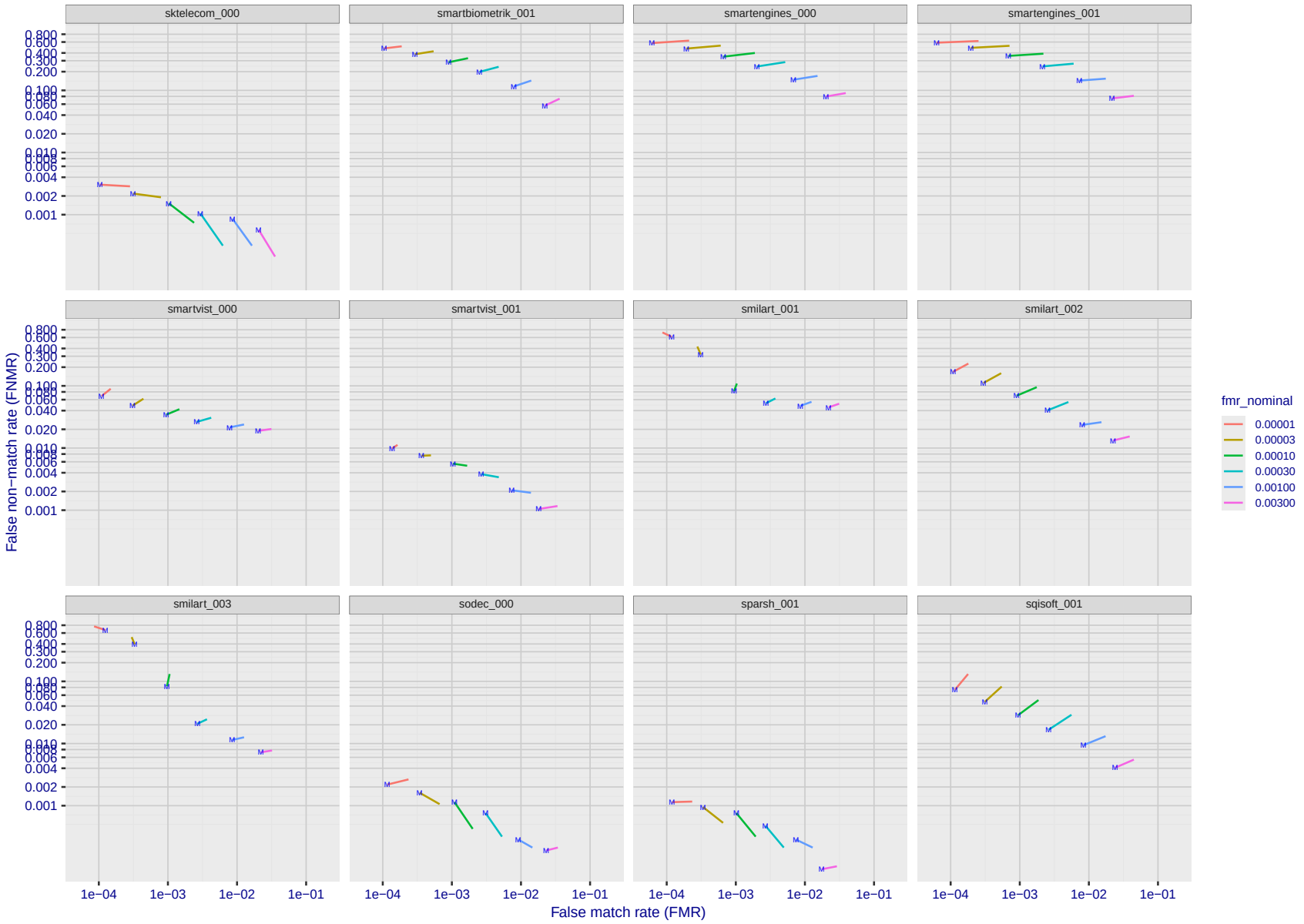

Figure 532: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

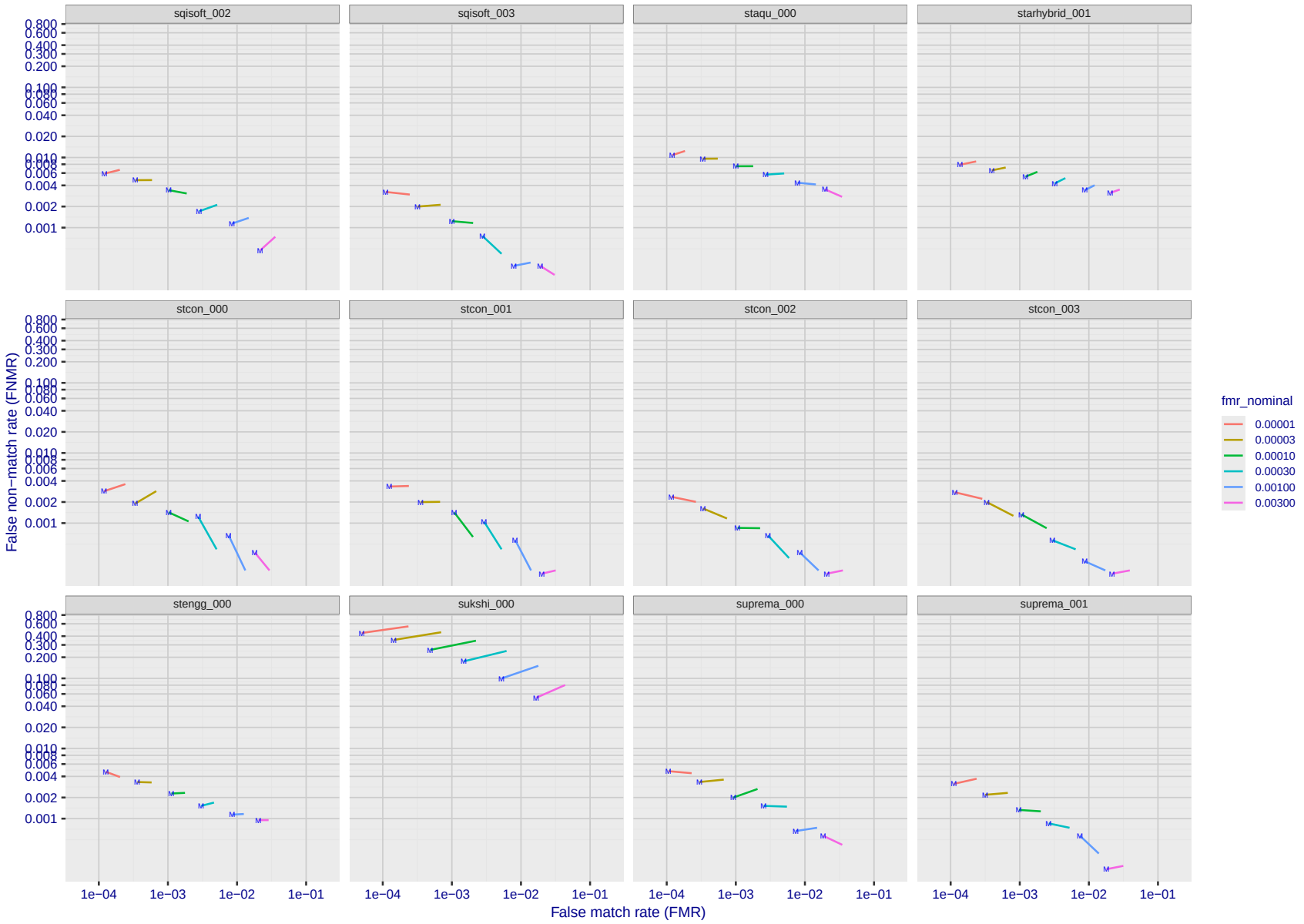

Figure 533: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

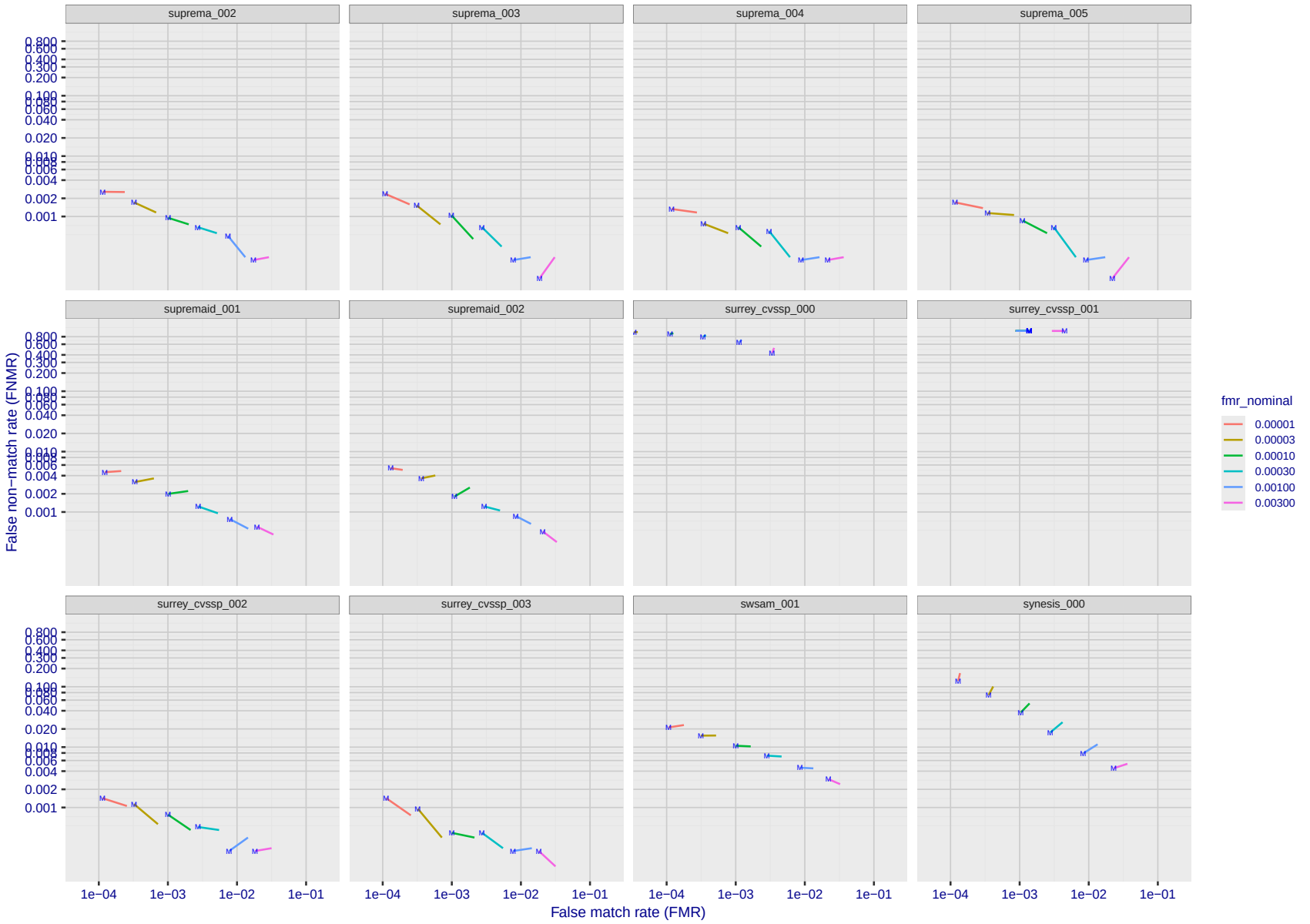

Figure 534: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

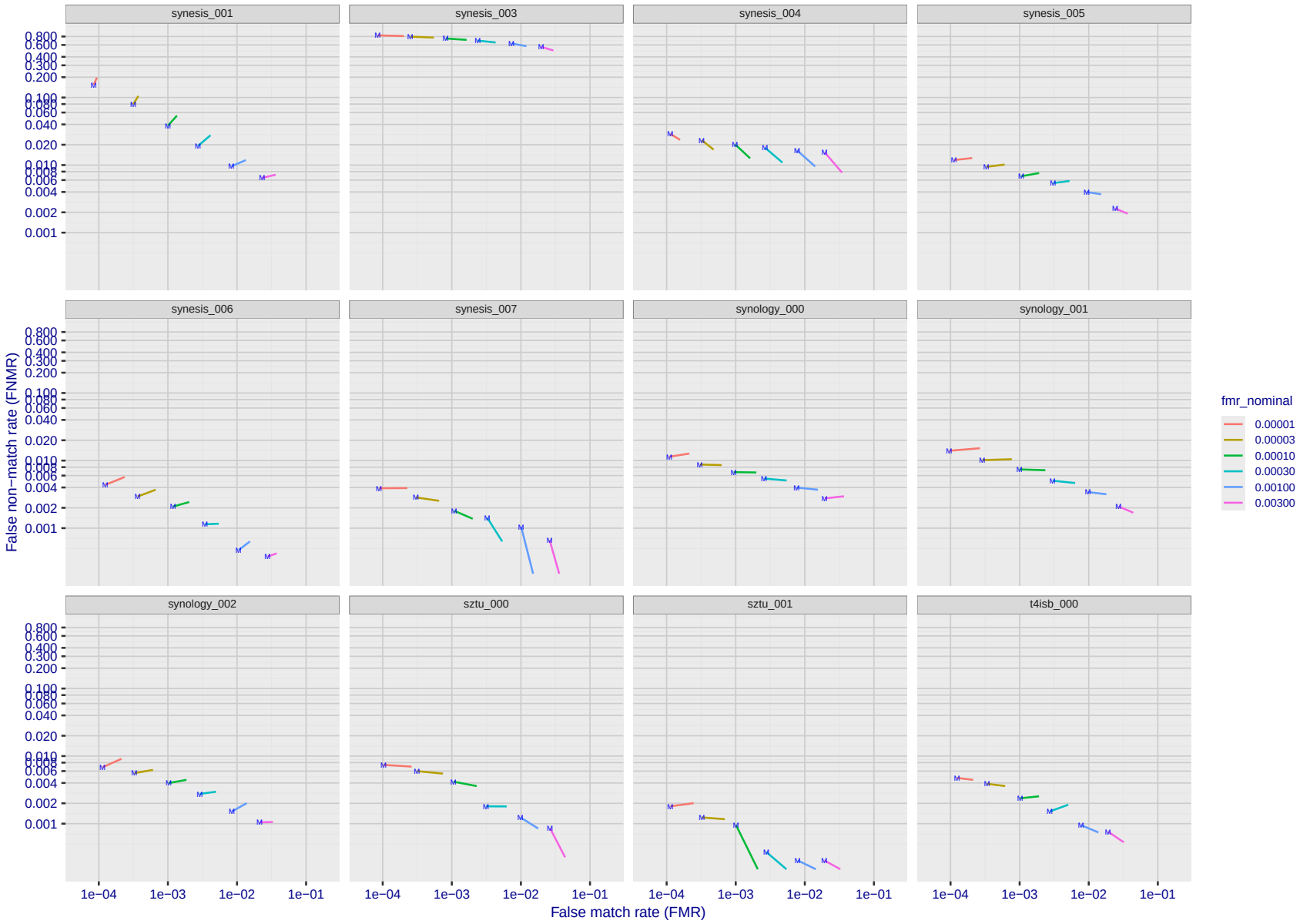

Figure 535: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

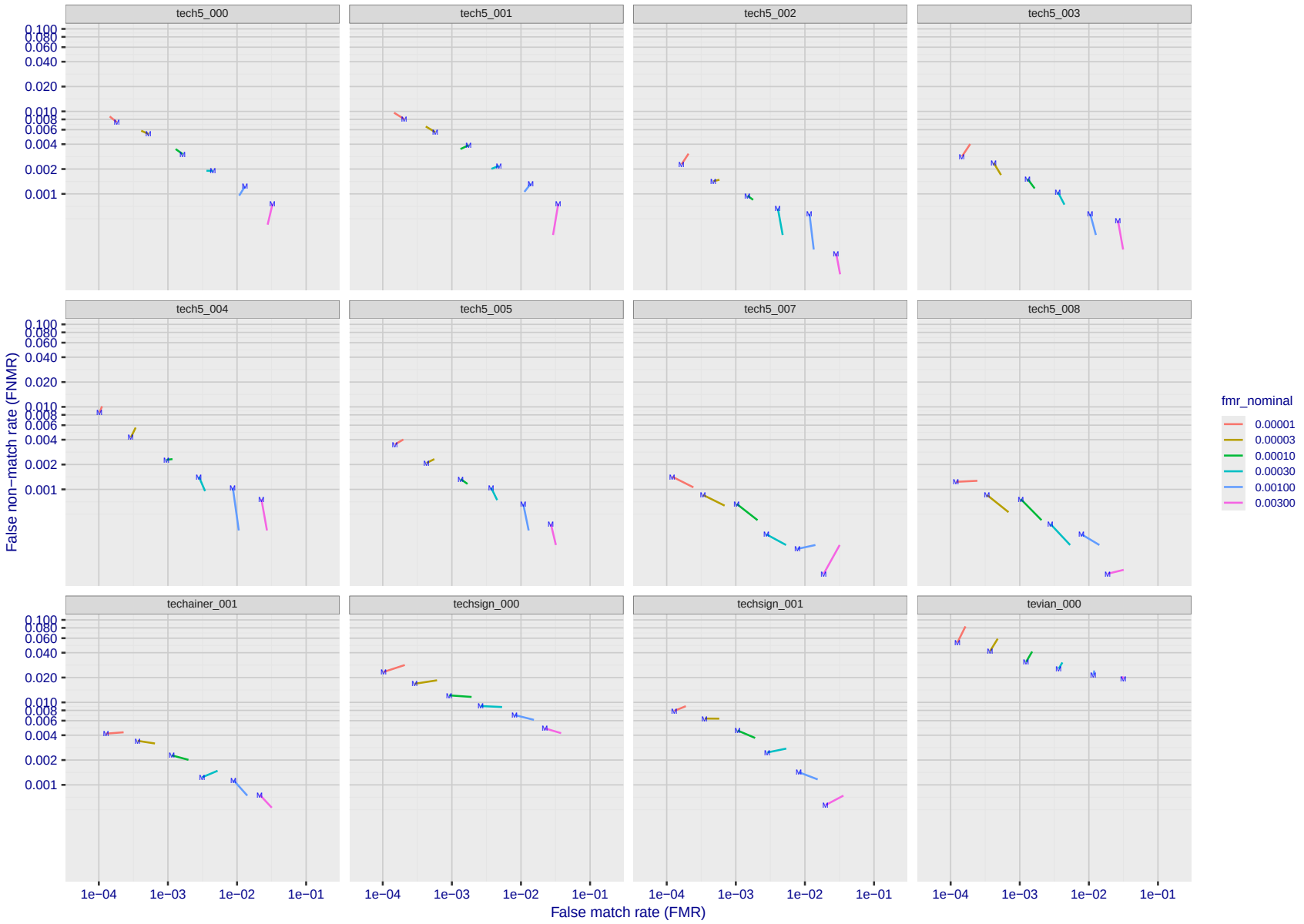

Figure 536: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

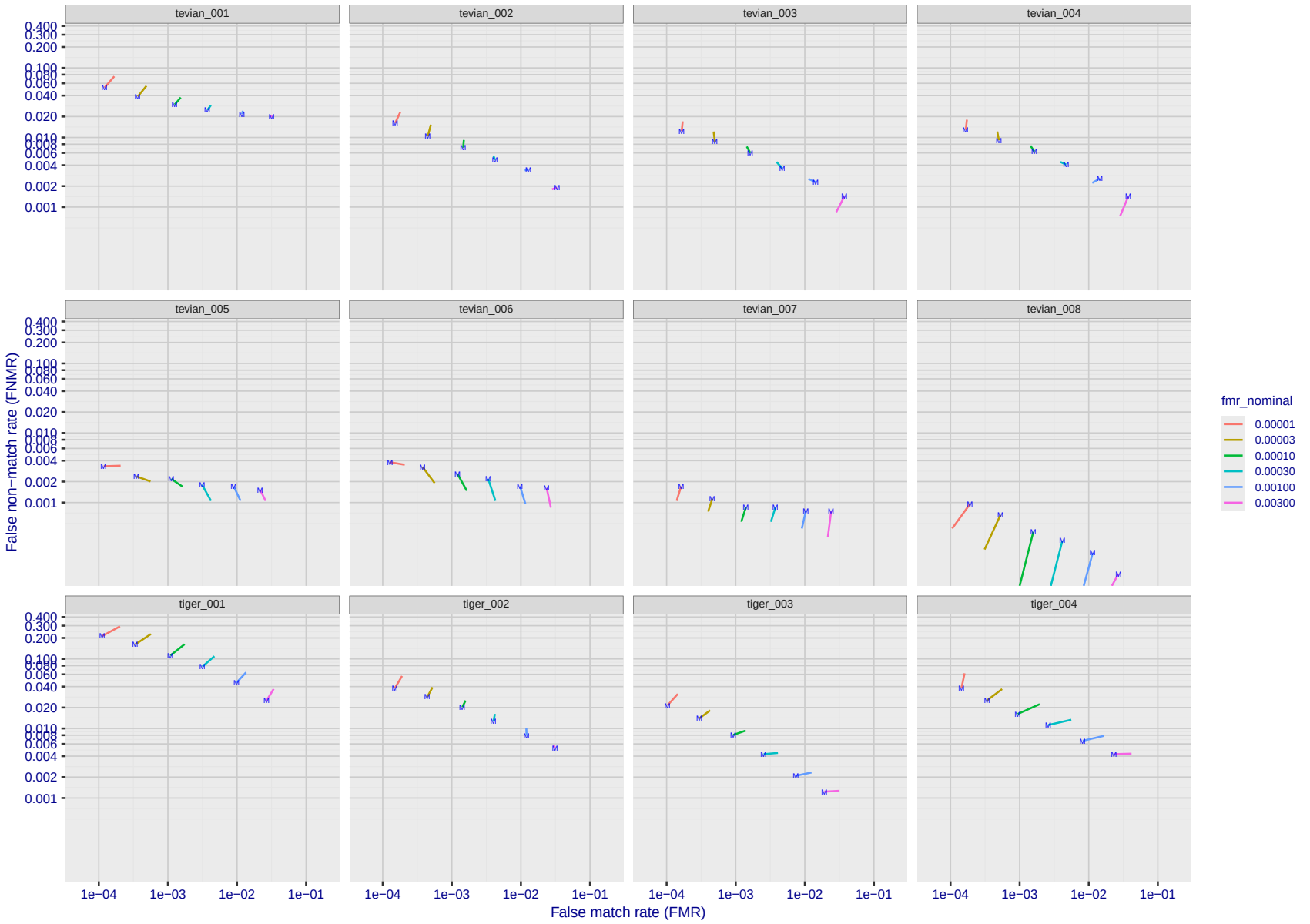

Figure 537: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

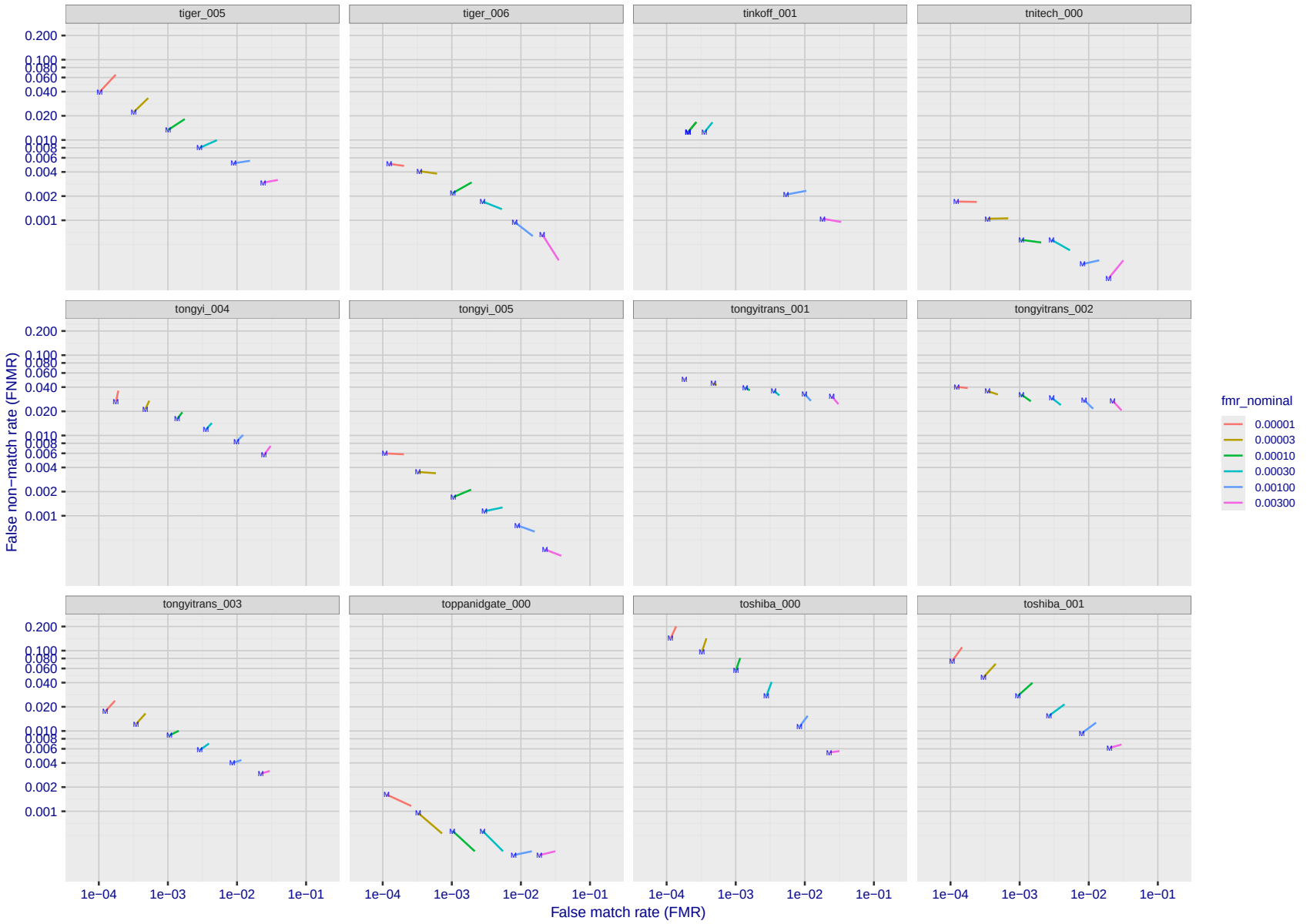

Figure 538: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

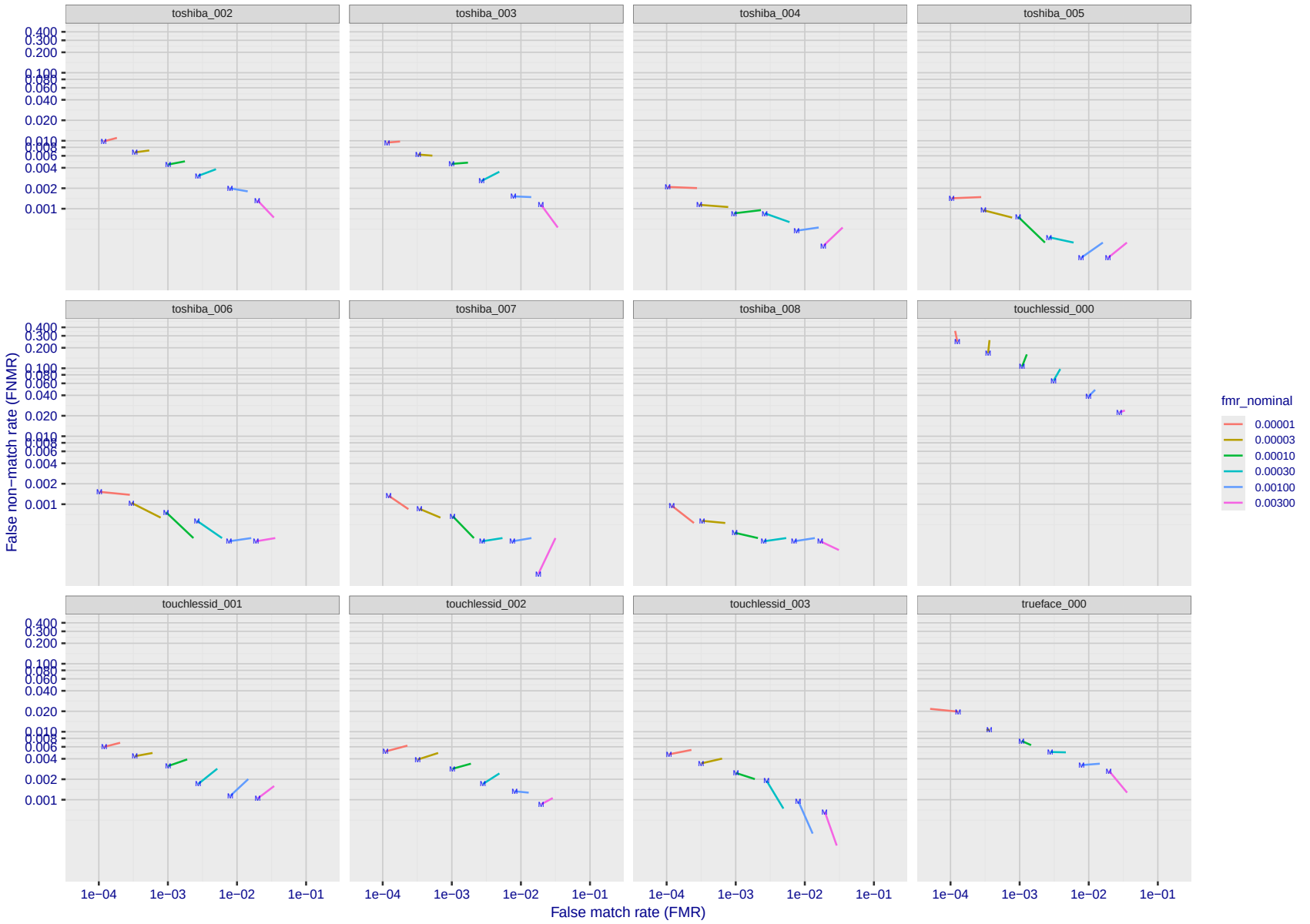

Figure 539: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

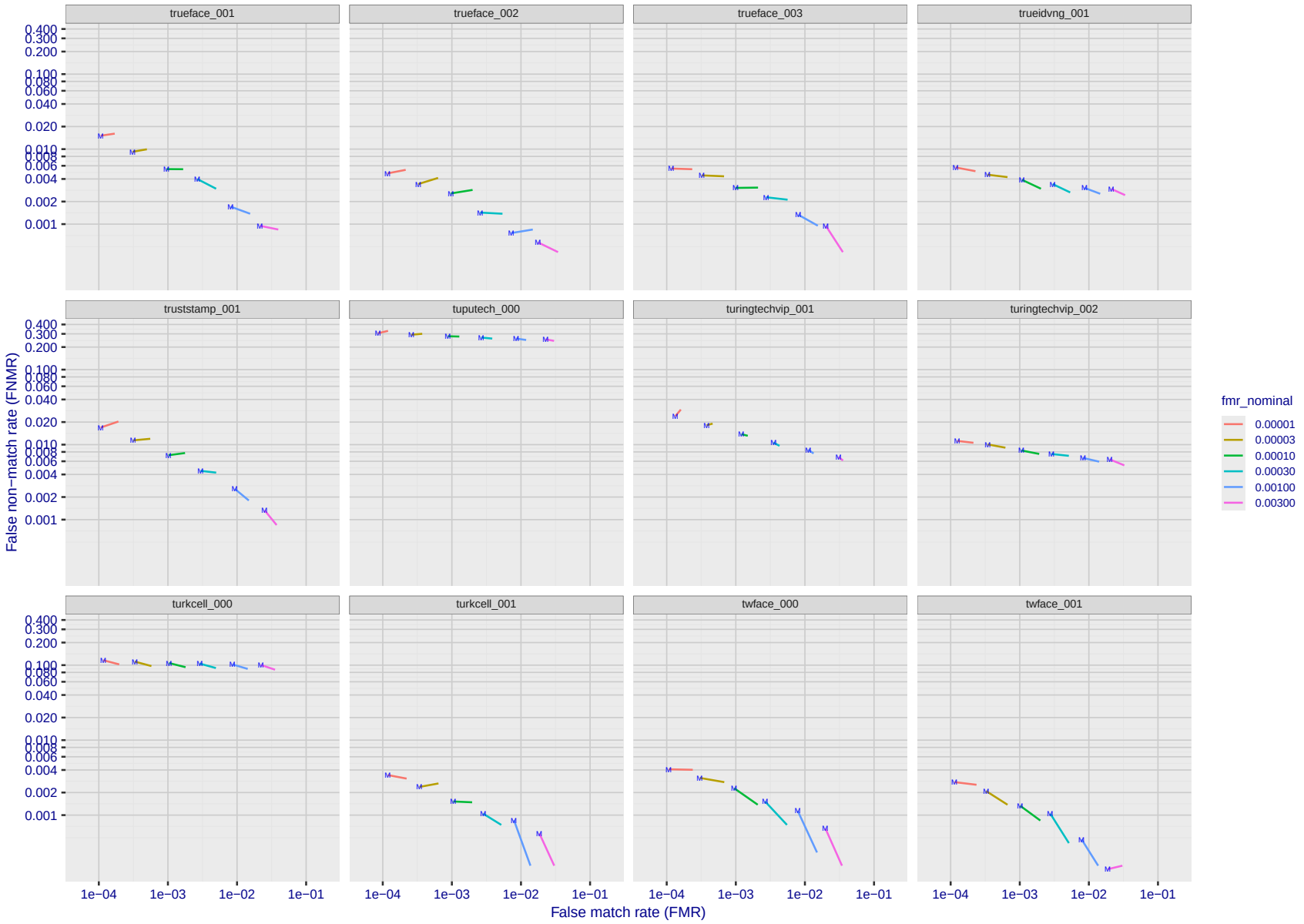

Figure 540: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

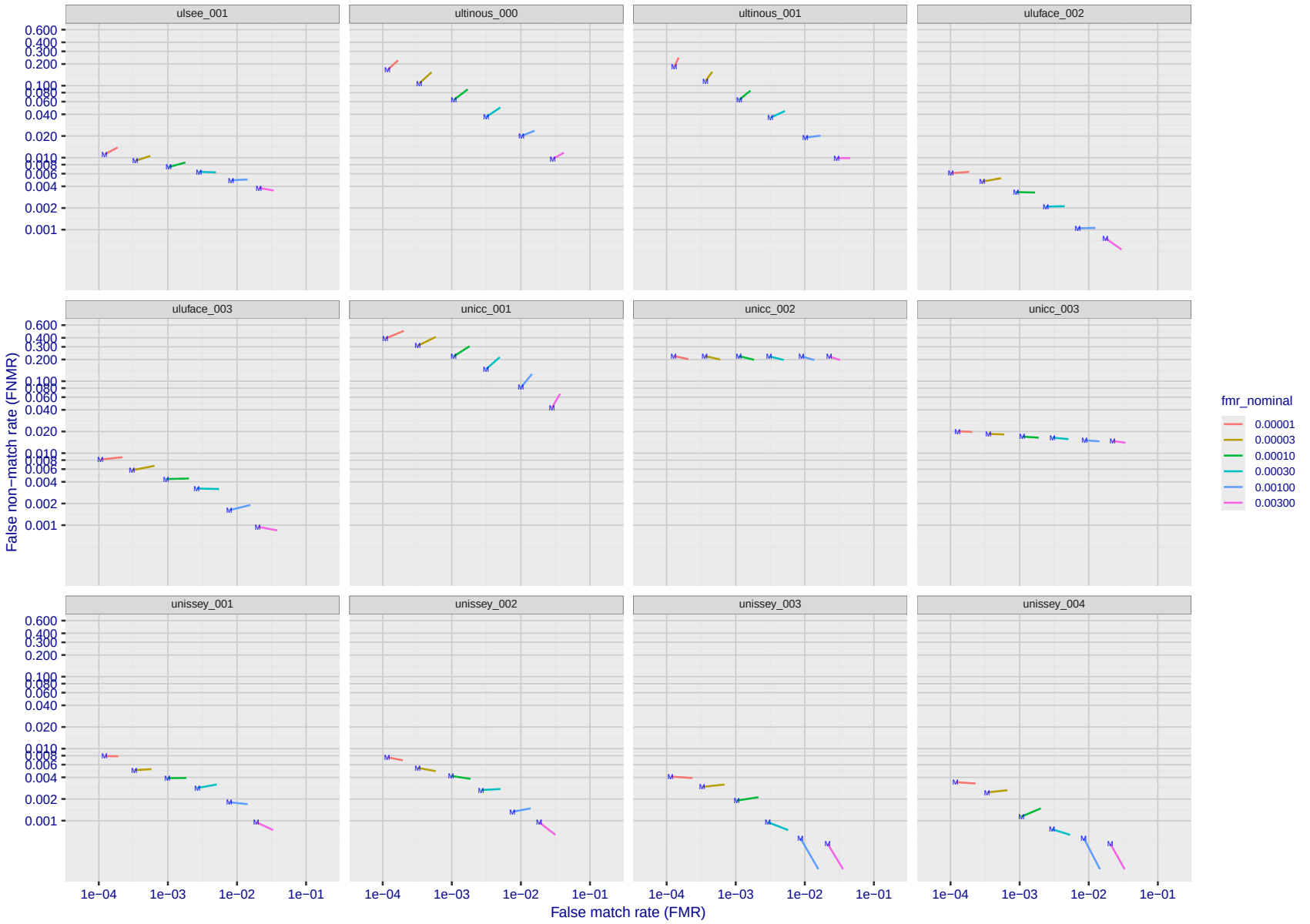

Figure 541: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

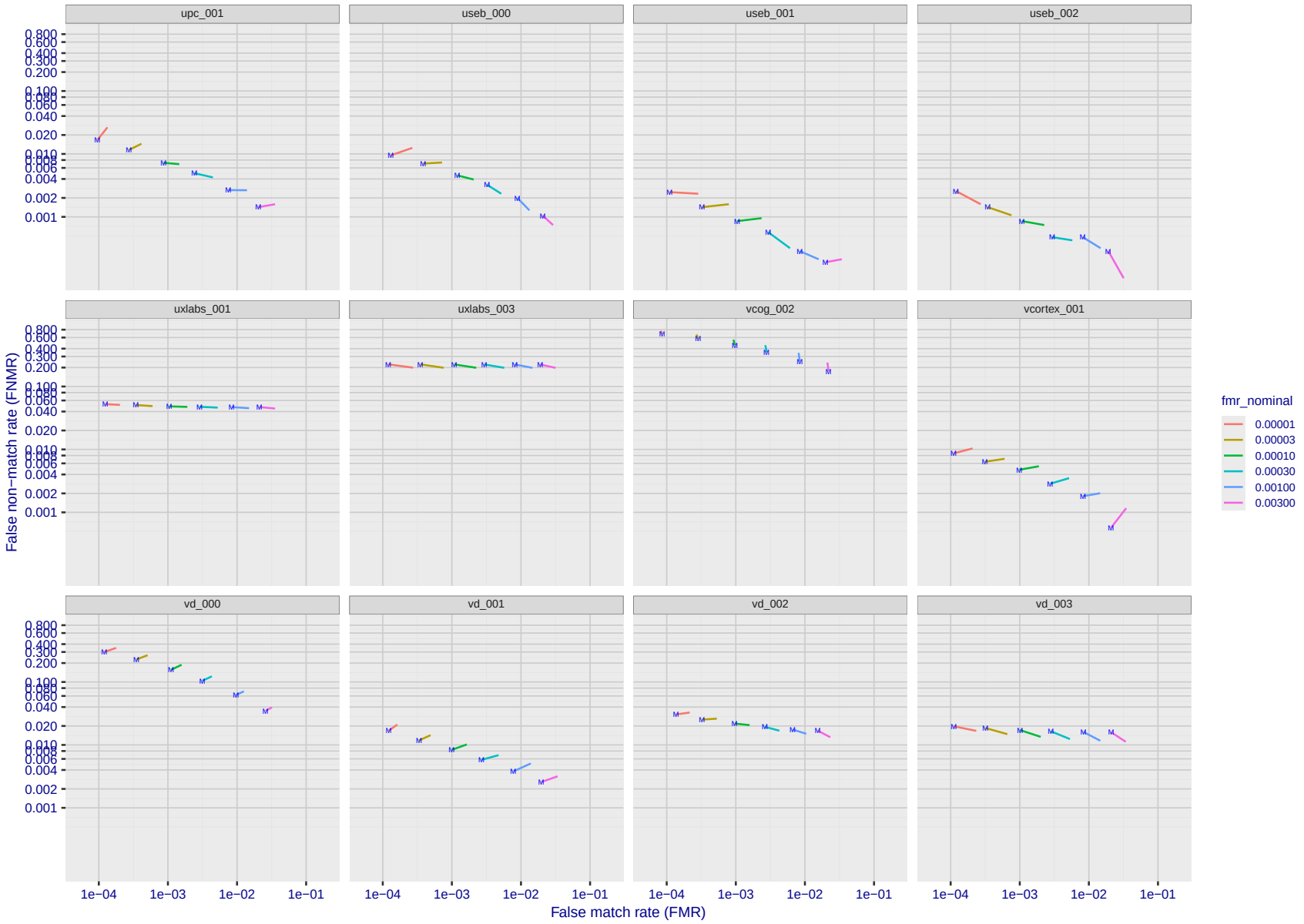

Figure 542: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

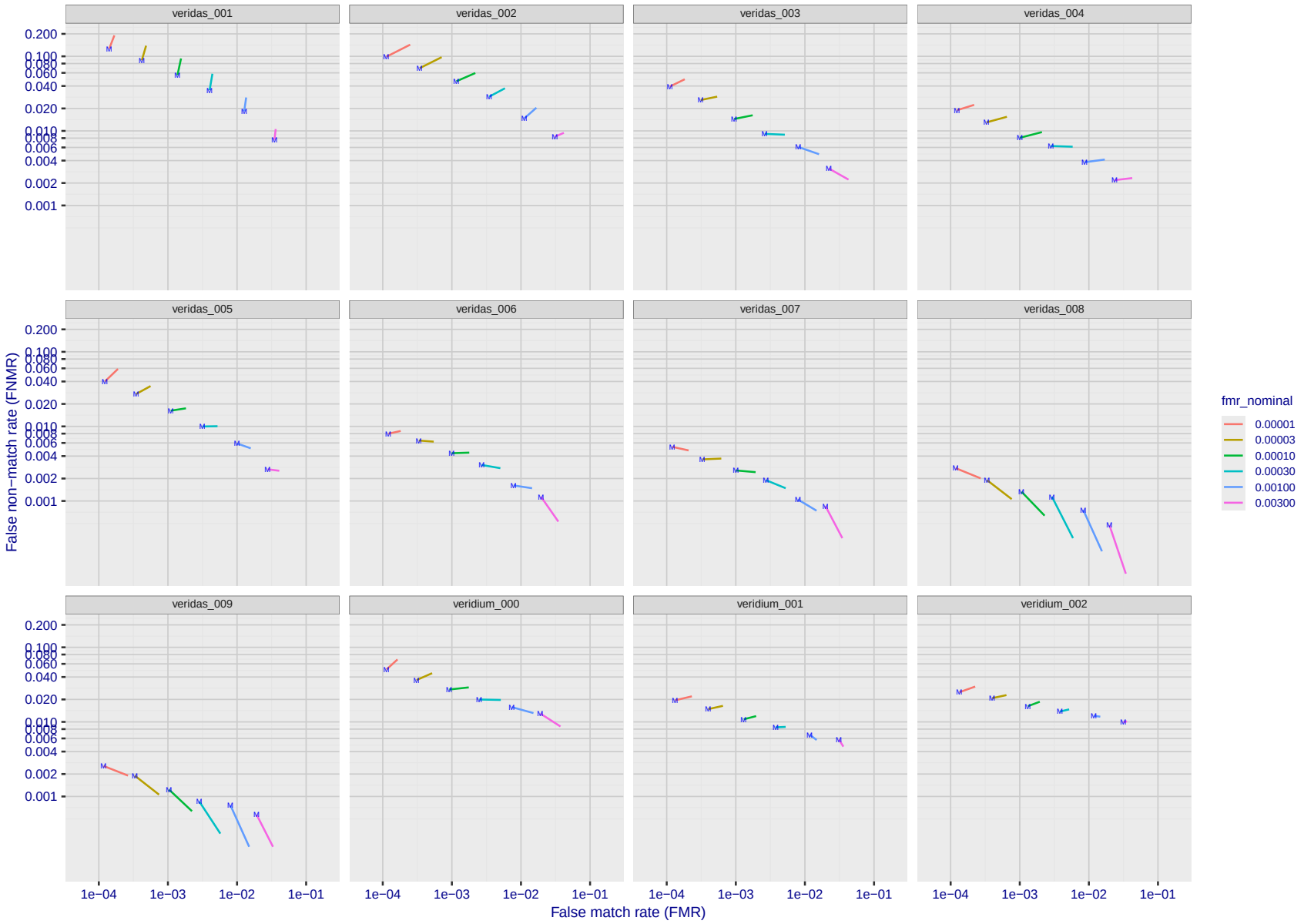

Figure 543: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

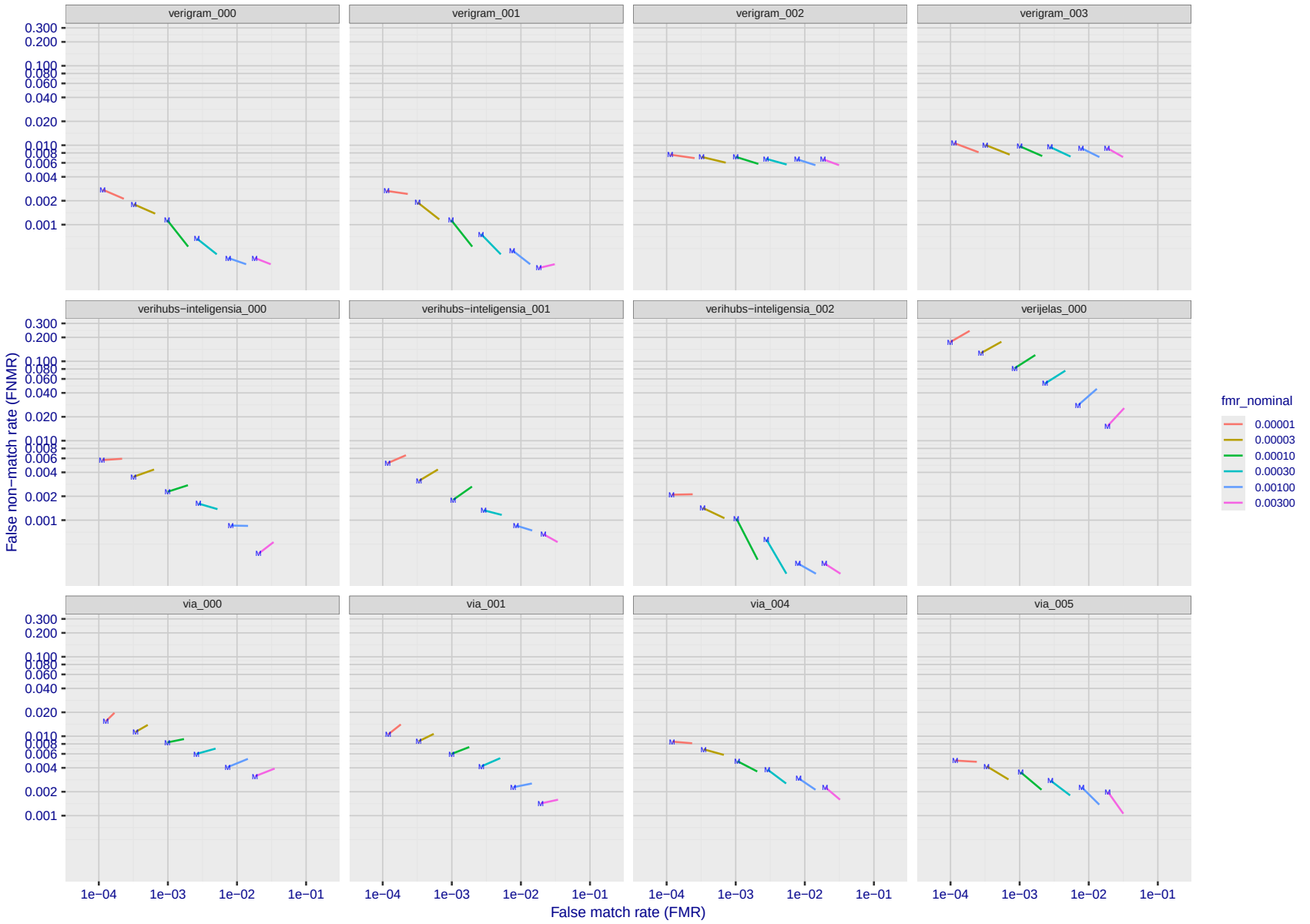

Figure 544: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

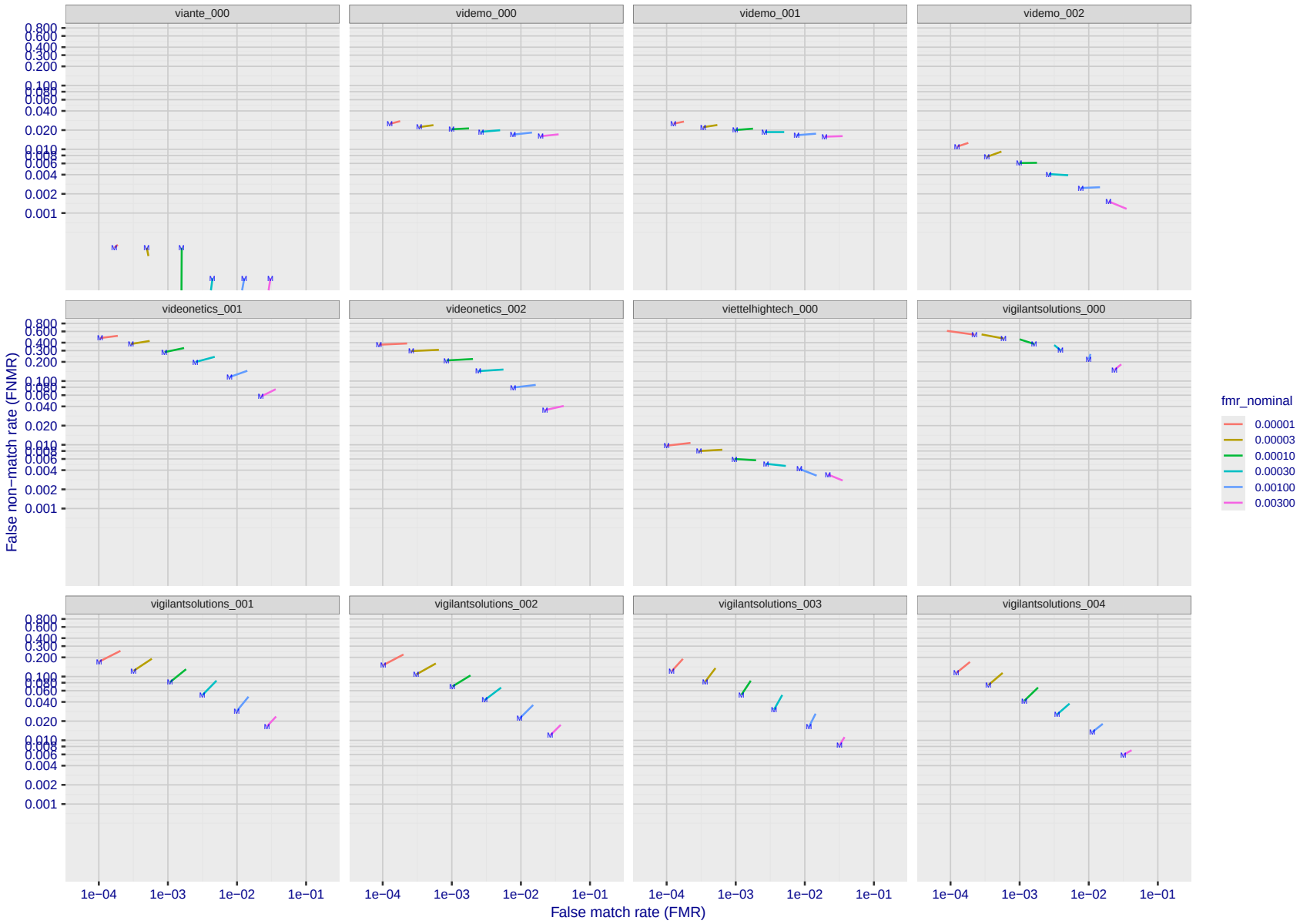

Figure 545: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

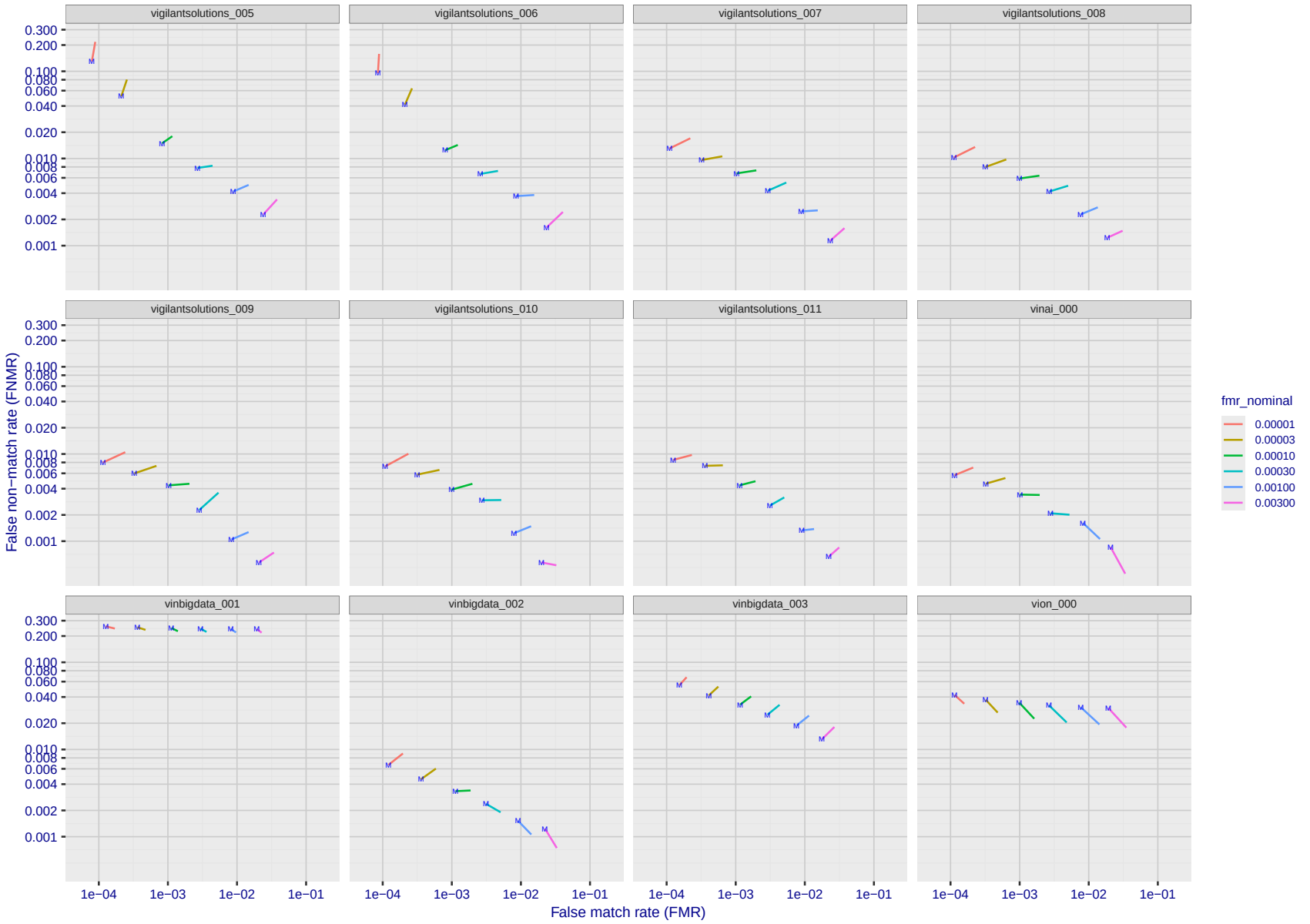

Figure 546: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

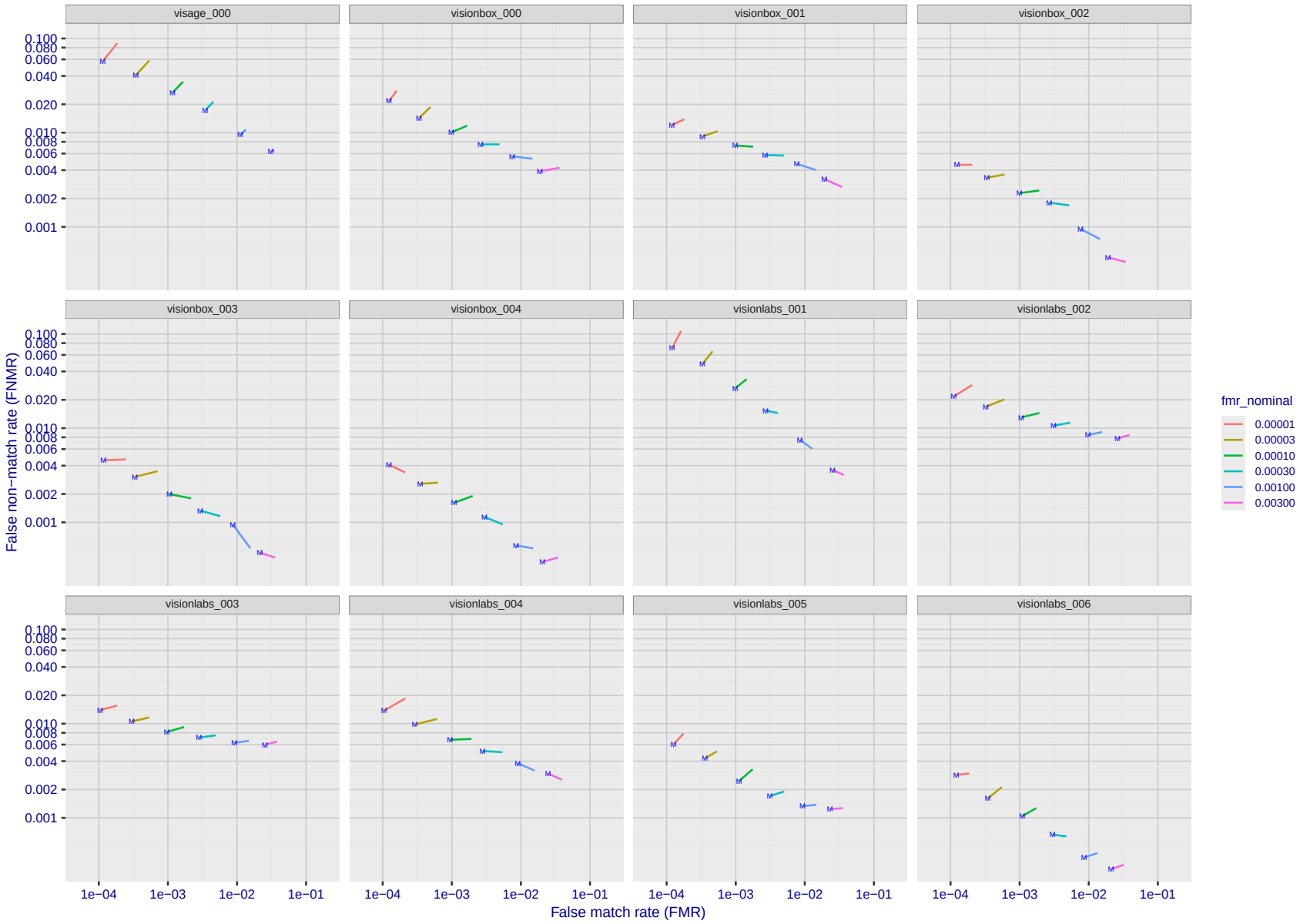

Figure 547: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

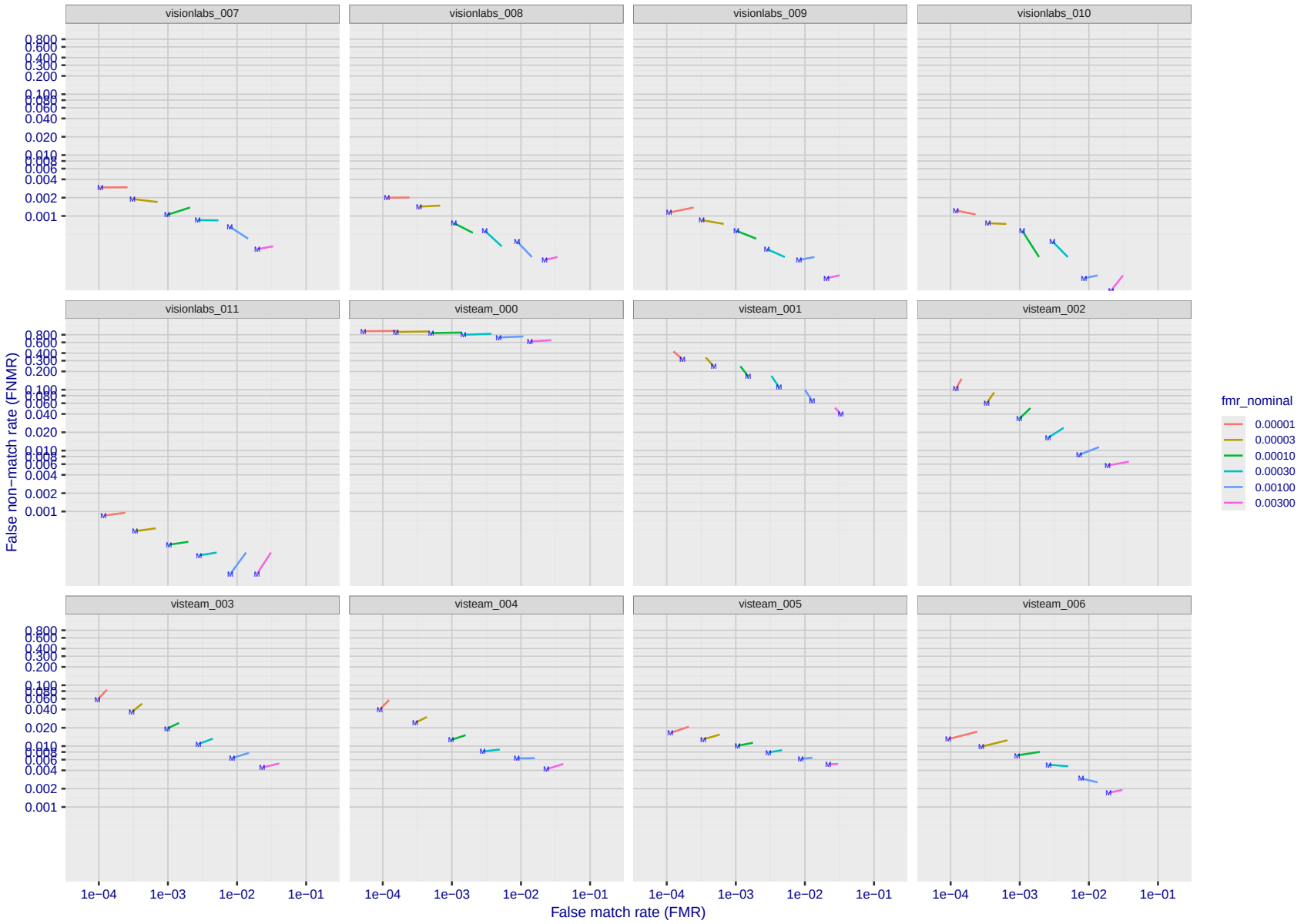

Figure 548: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

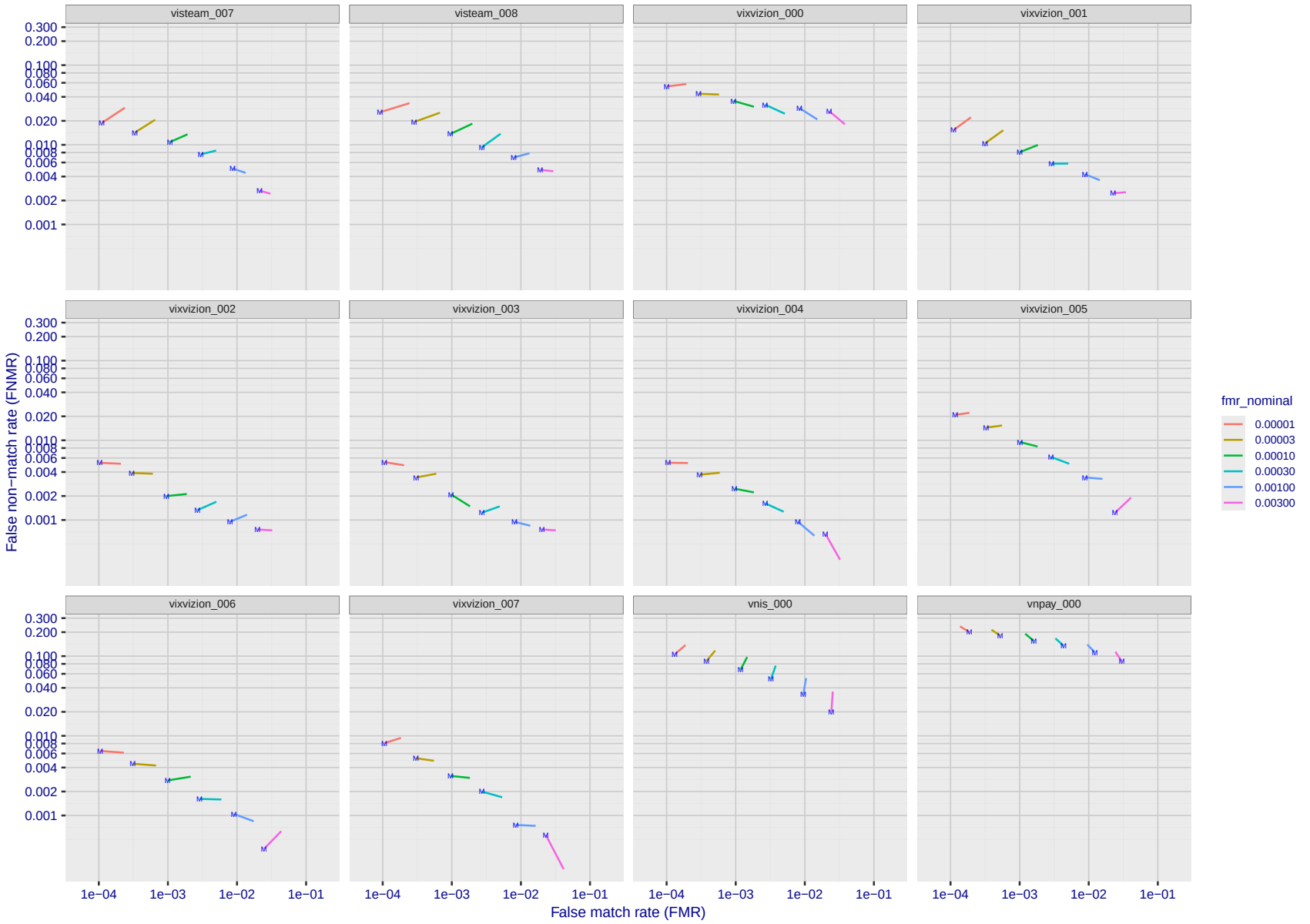

Figure 549: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

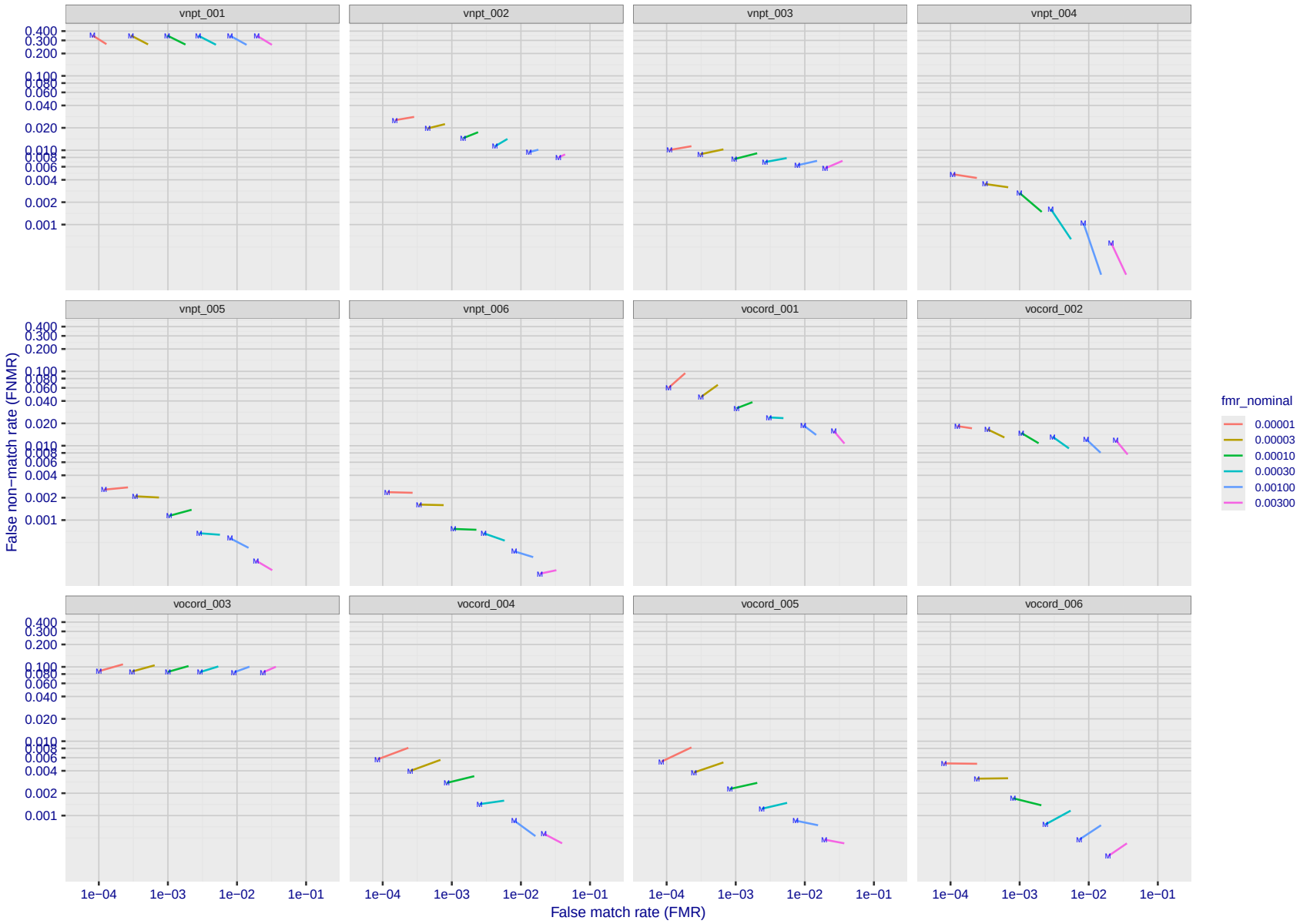

Figure 550: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

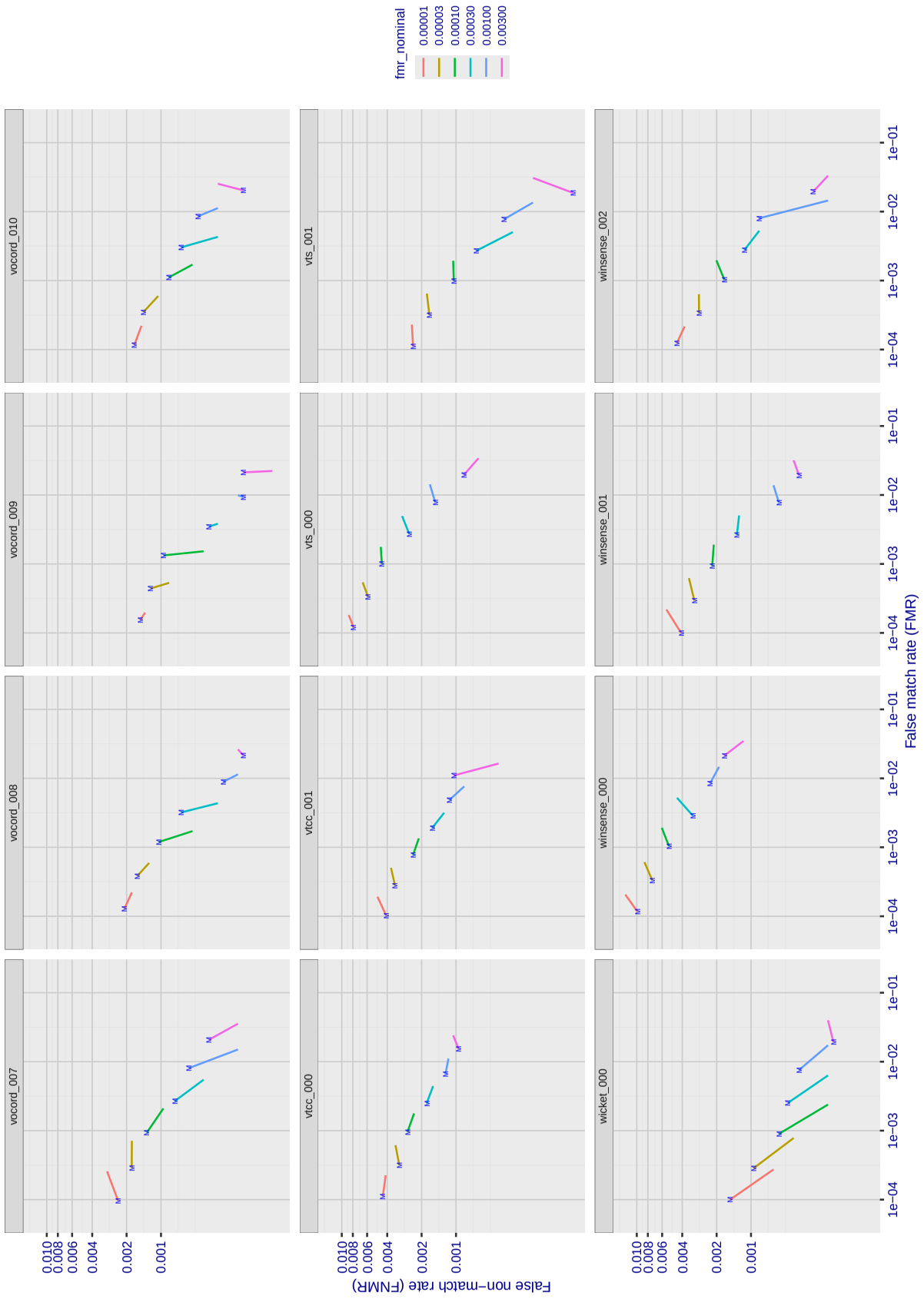

Figure 551: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

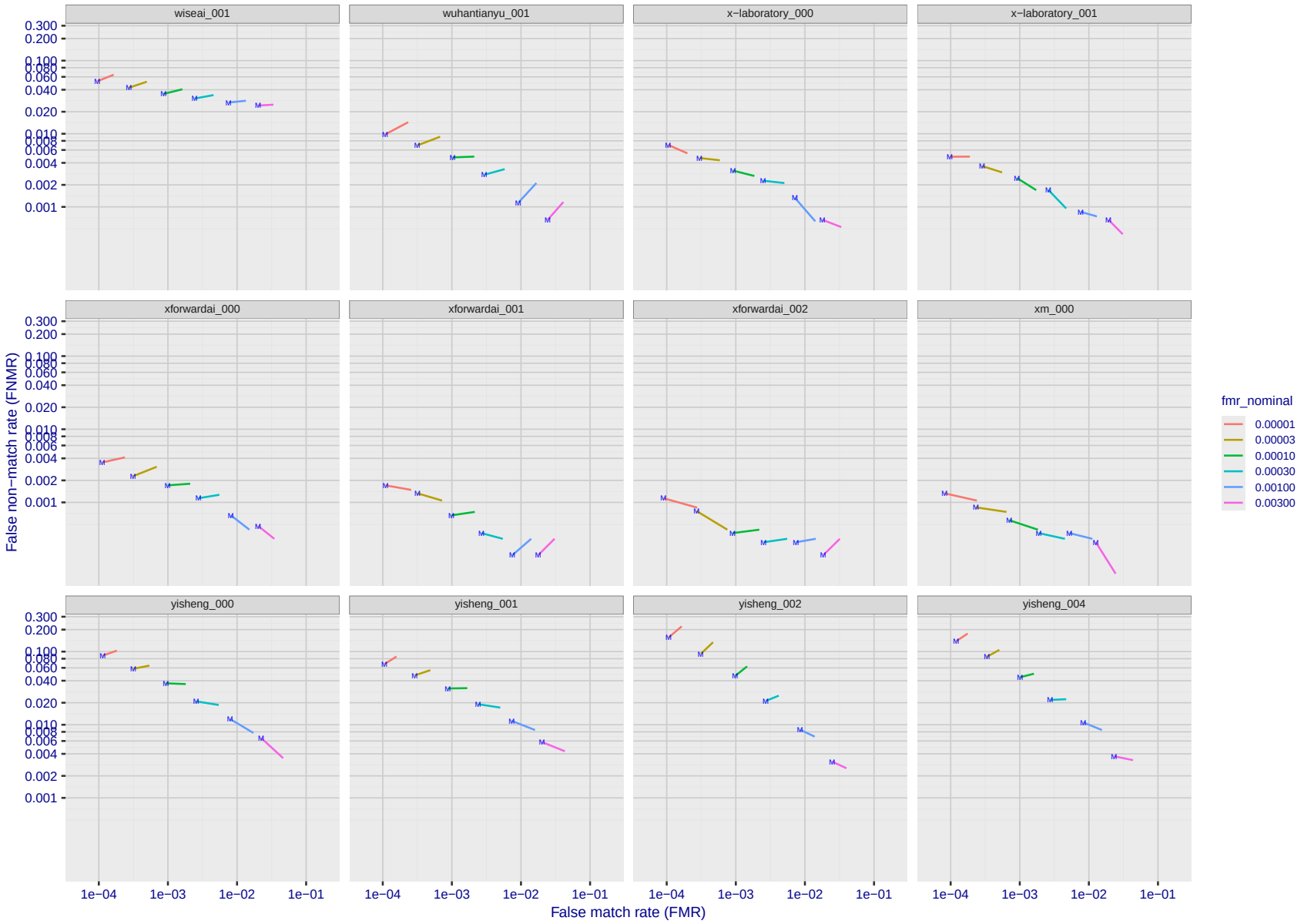

Figure 552: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

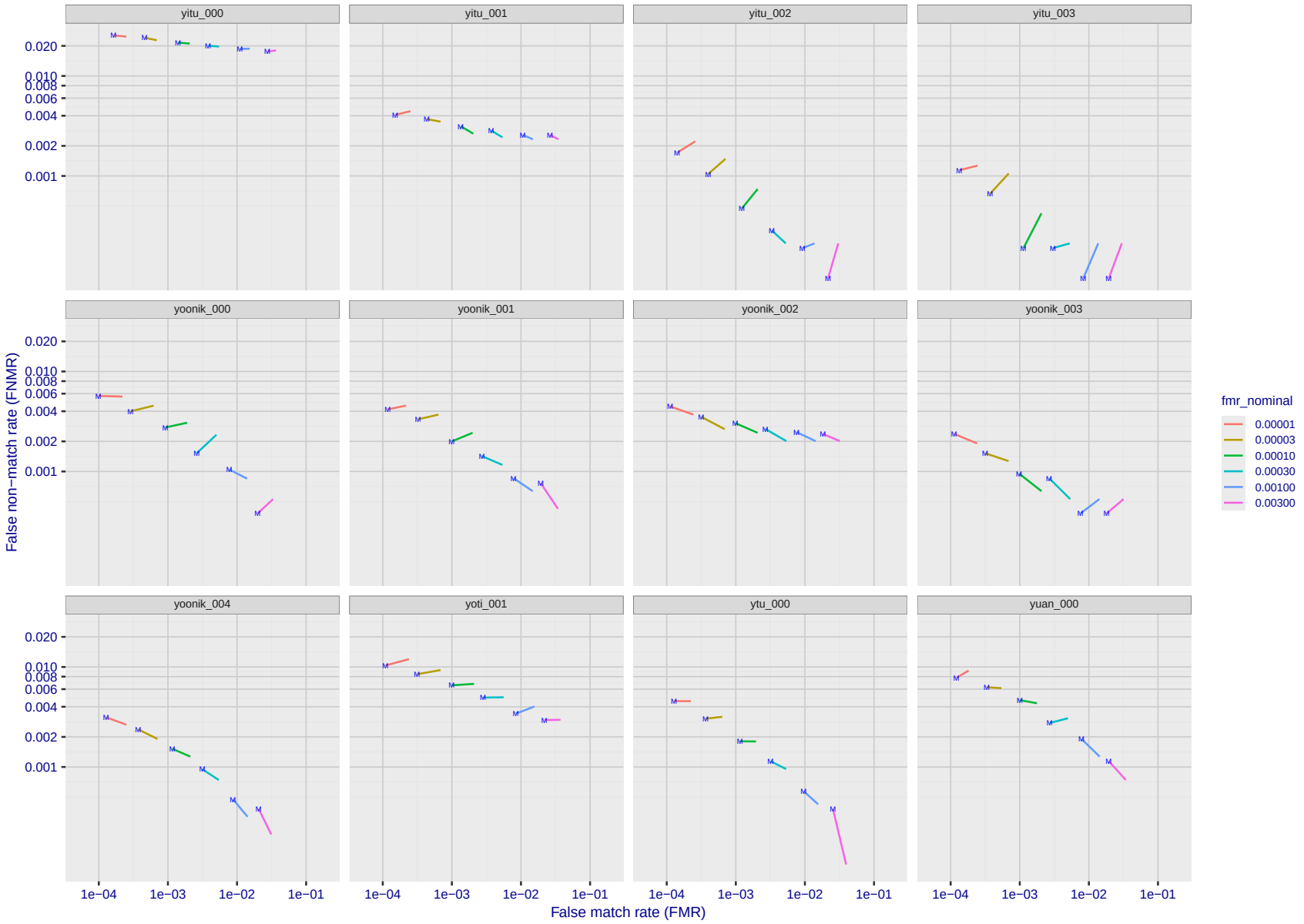

Figure 553: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

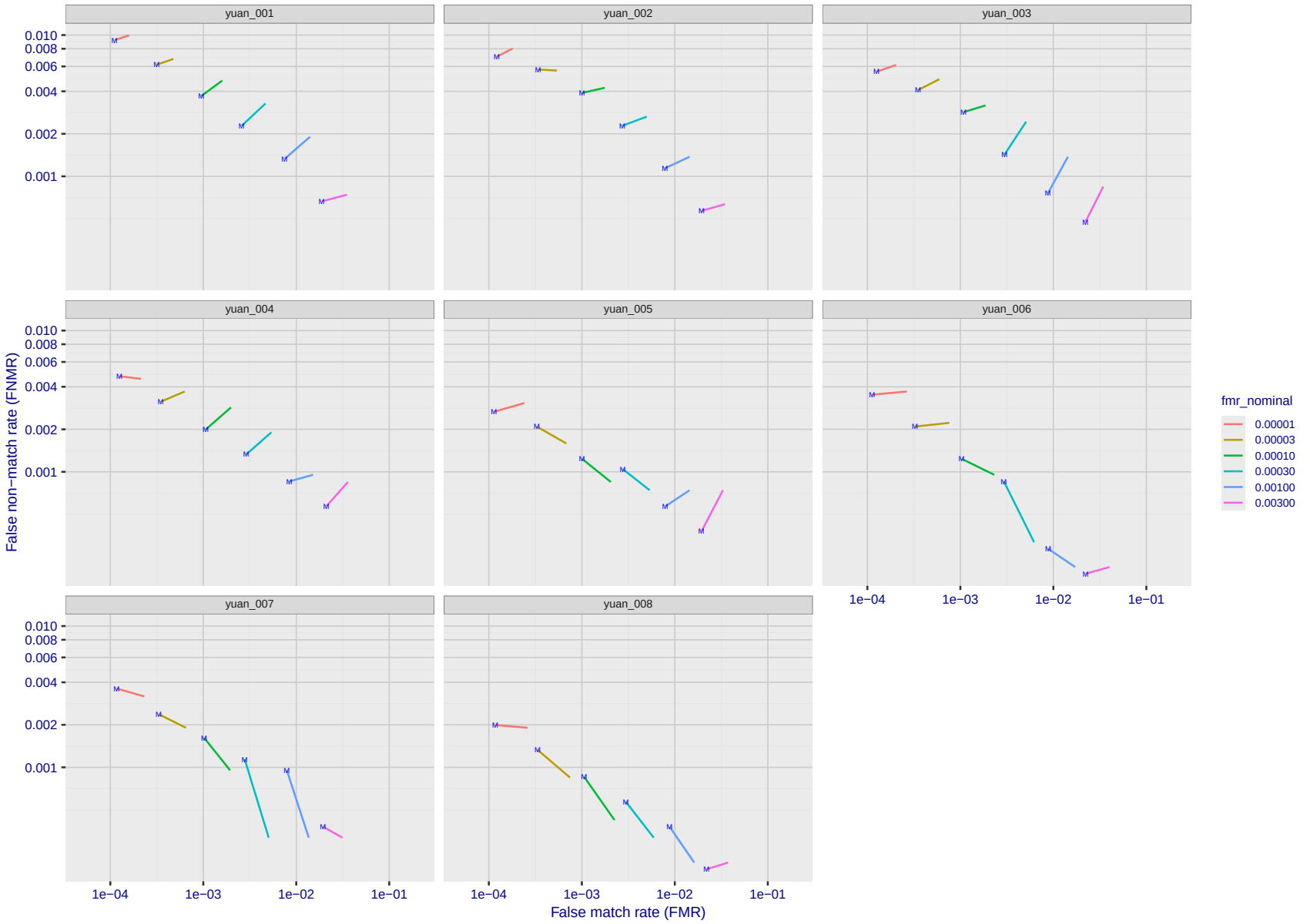

Figure 554: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

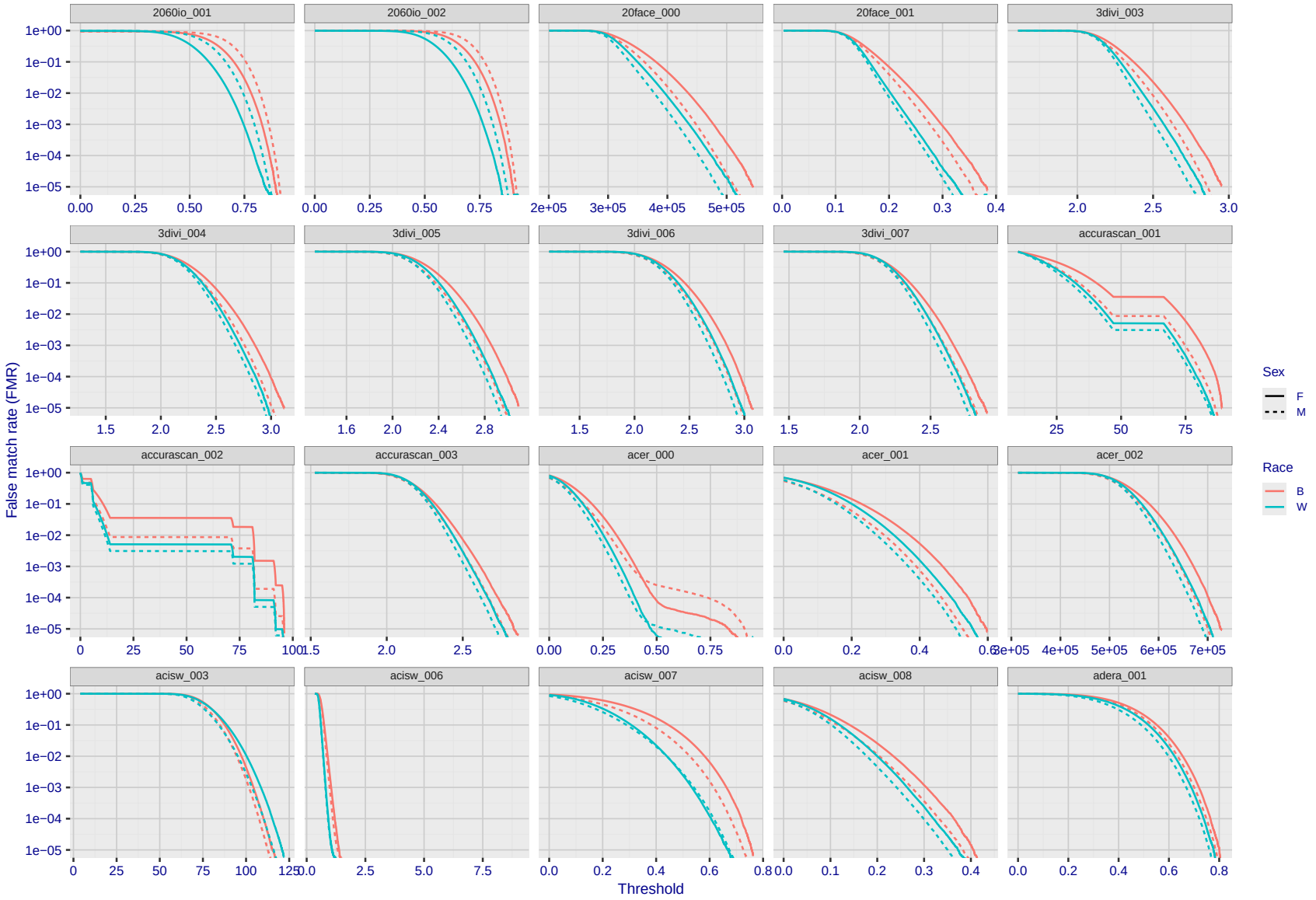

Figure 555: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

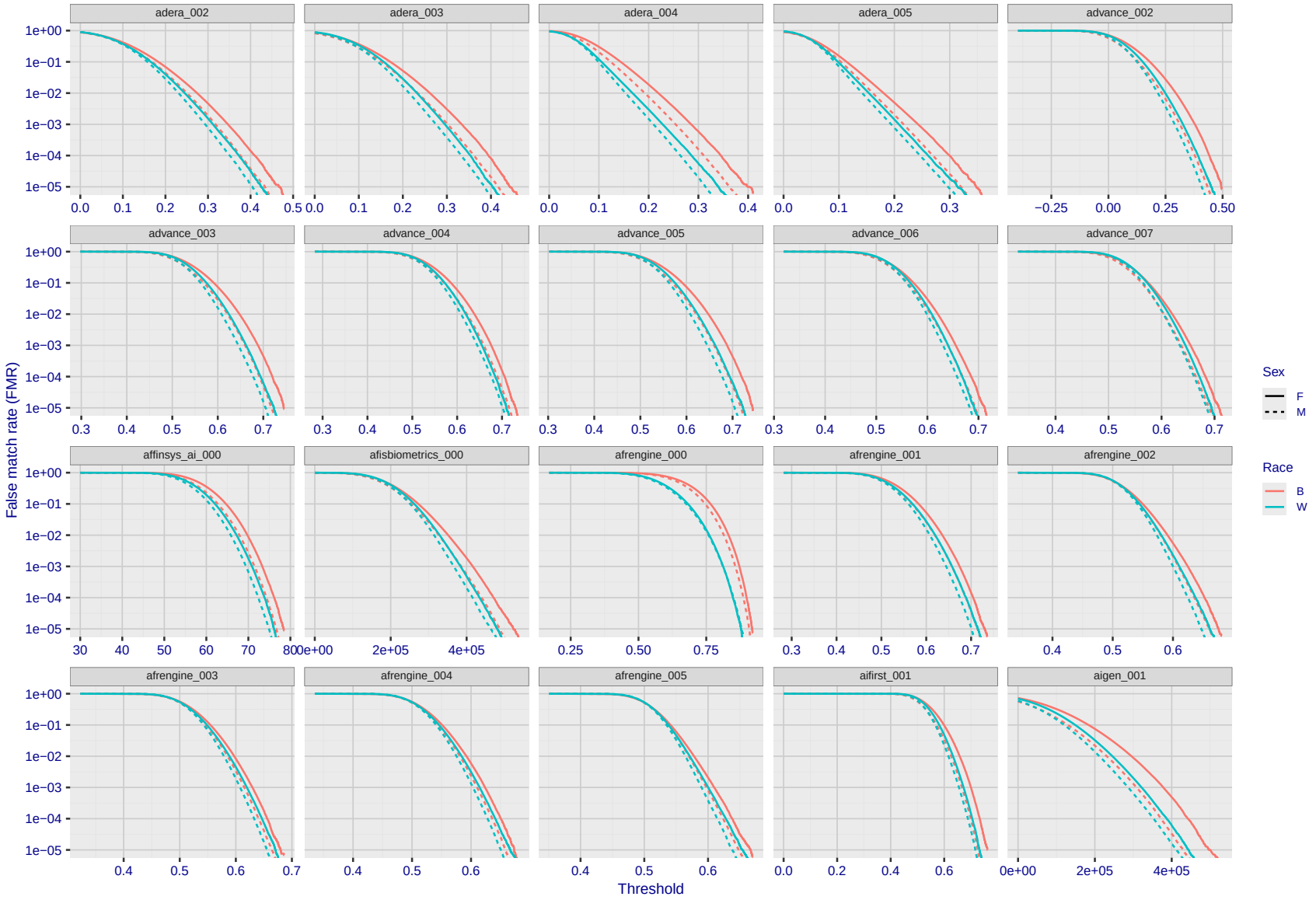

Figure 556: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

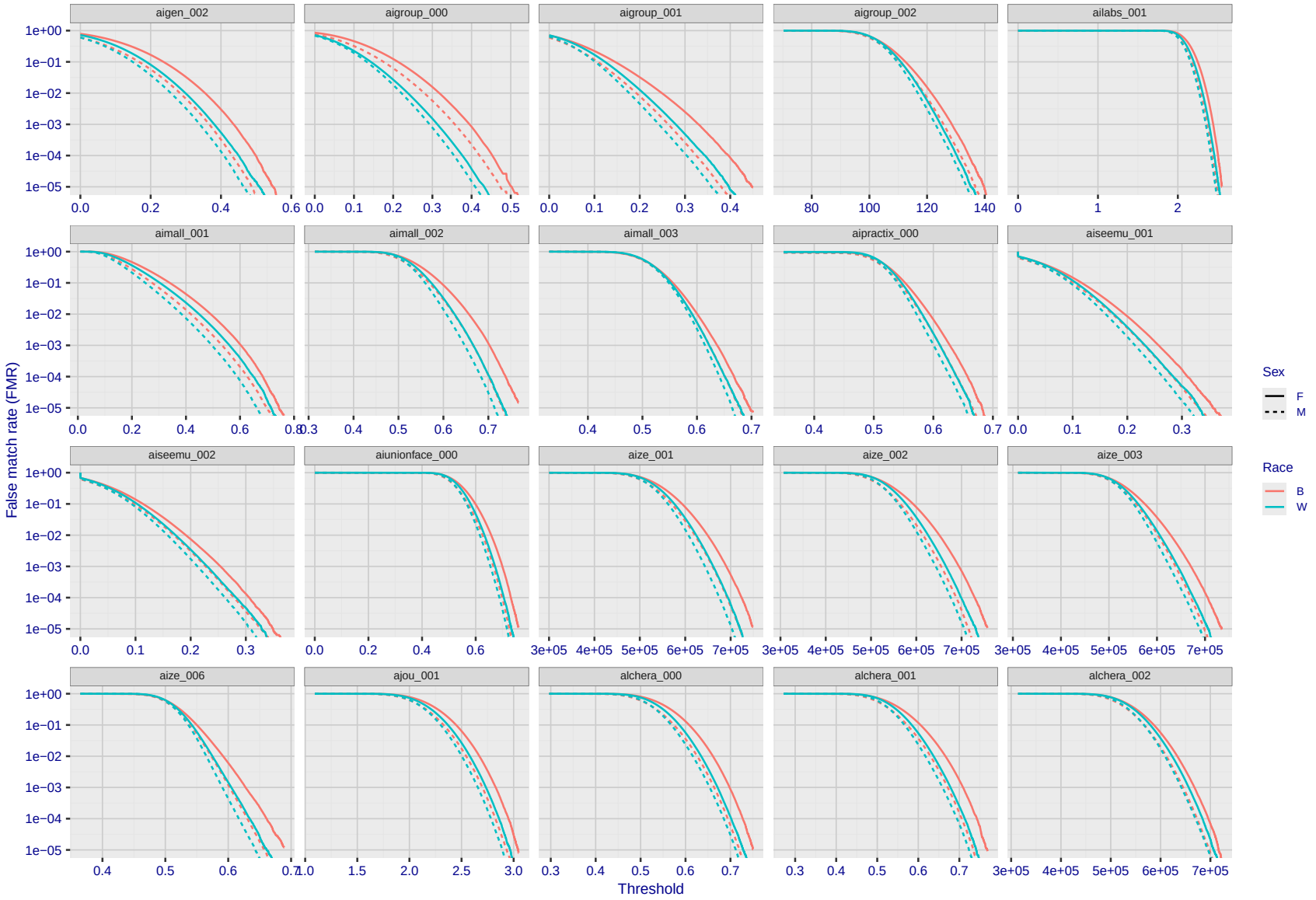

Figure 557: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

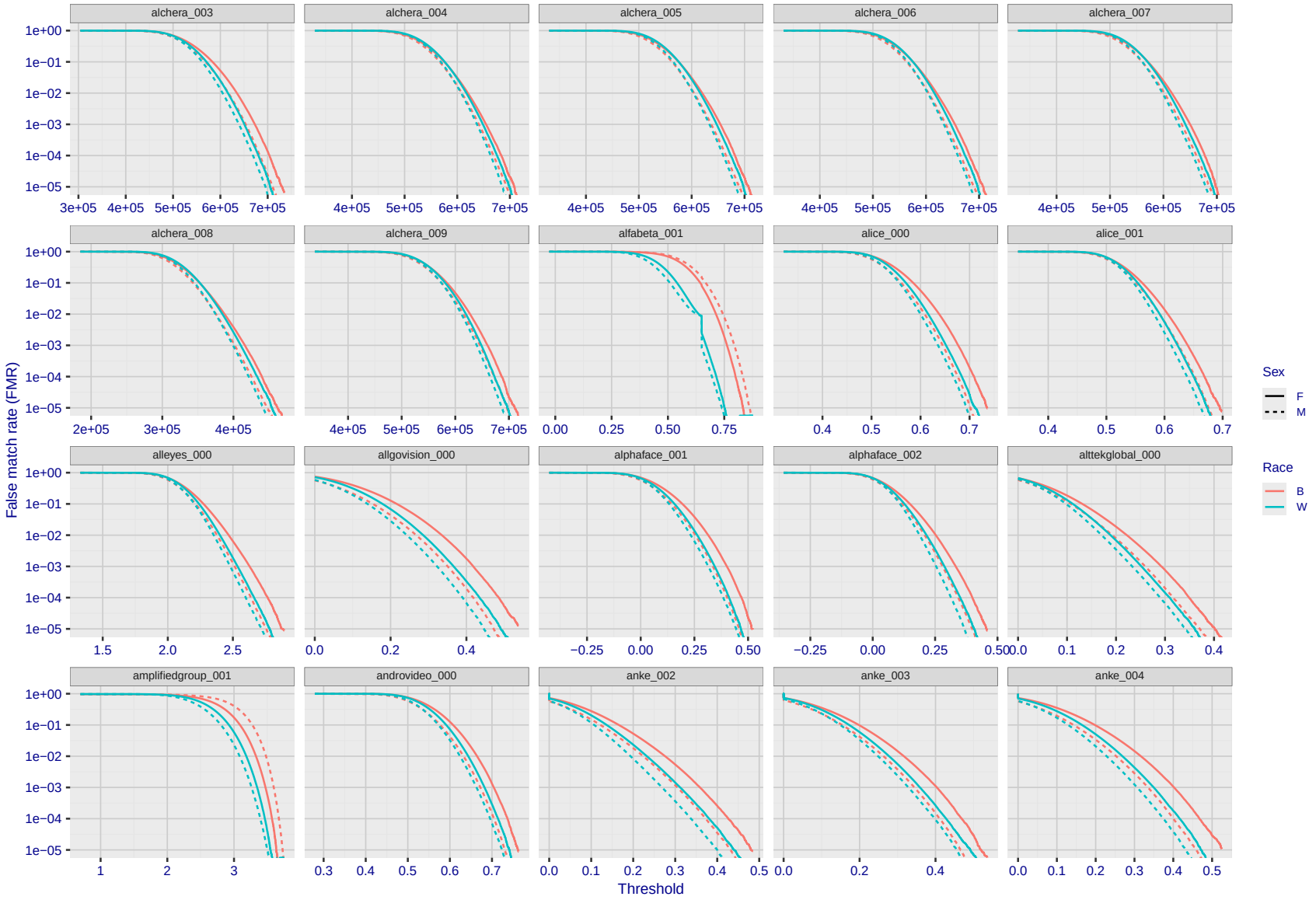

Figure 558: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

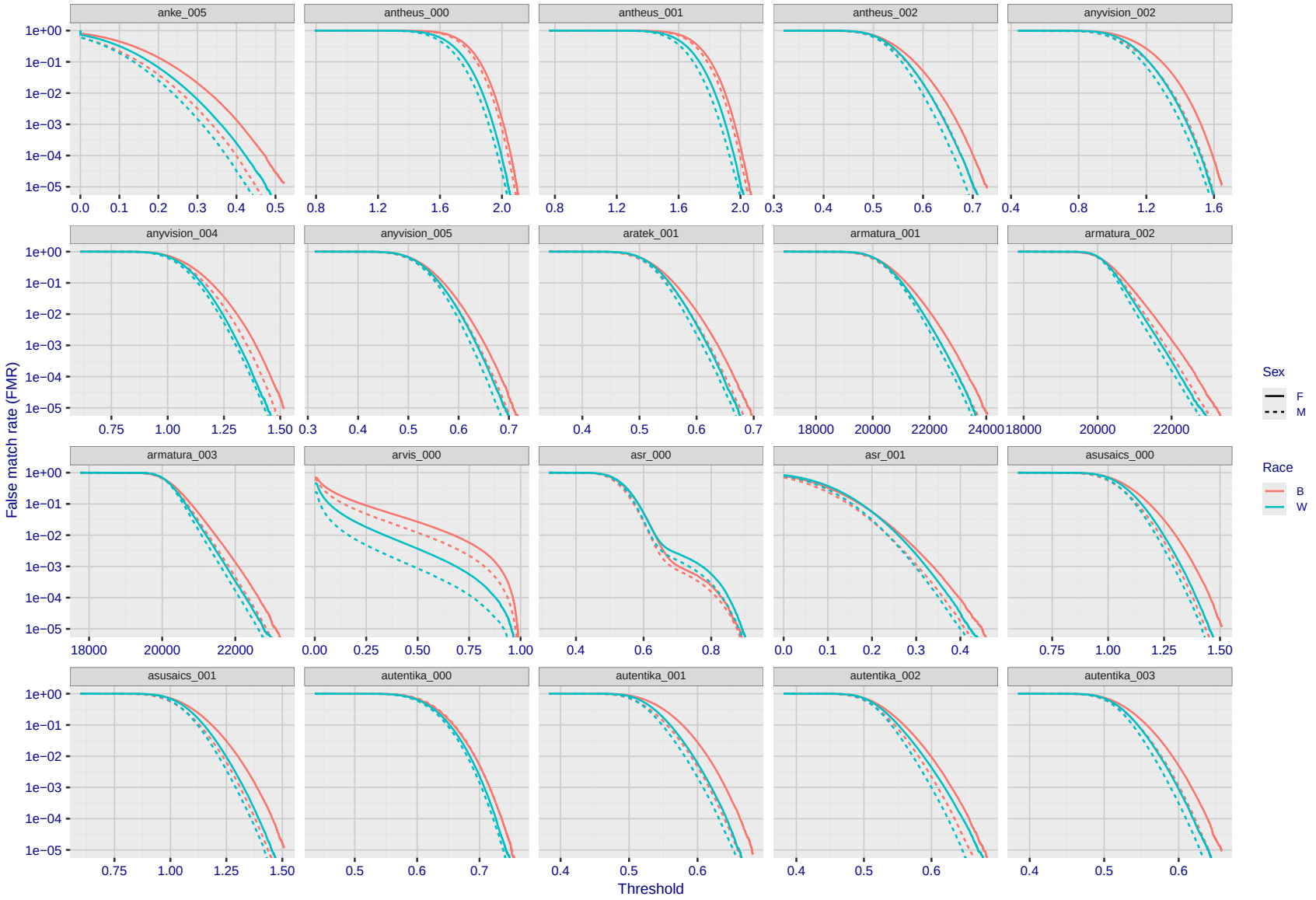

Figure 559: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

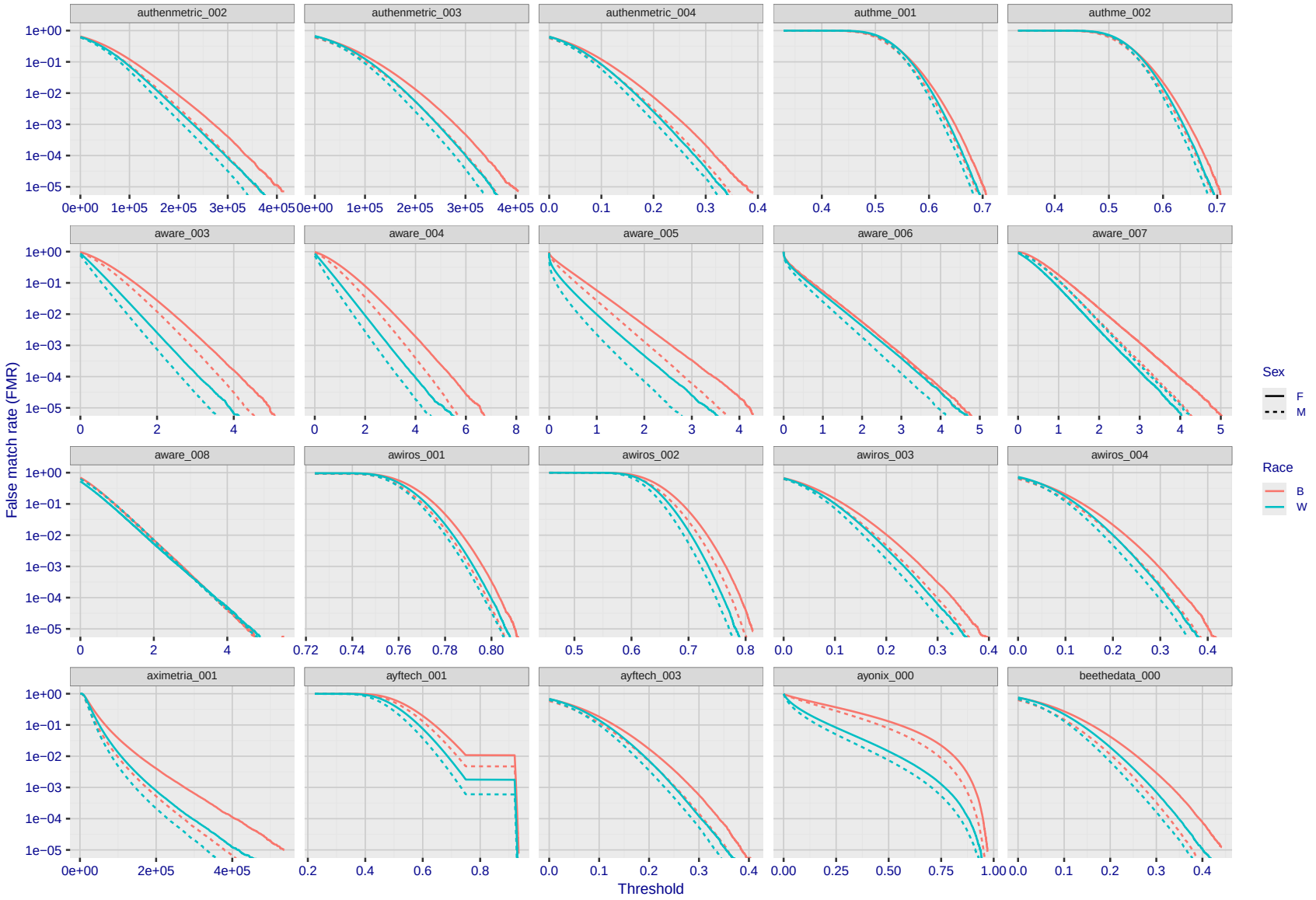

Figure 560: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

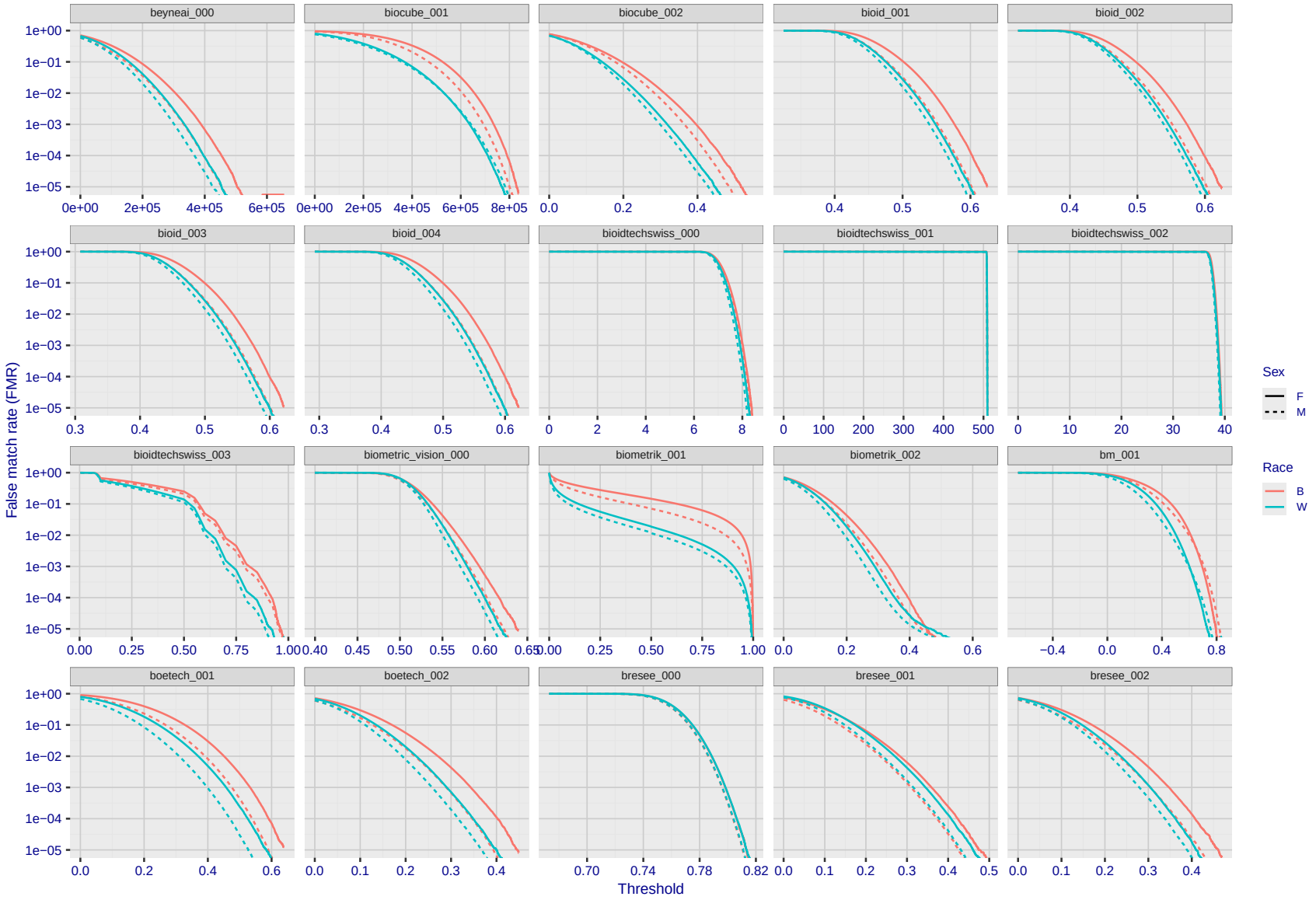

Figure 561: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

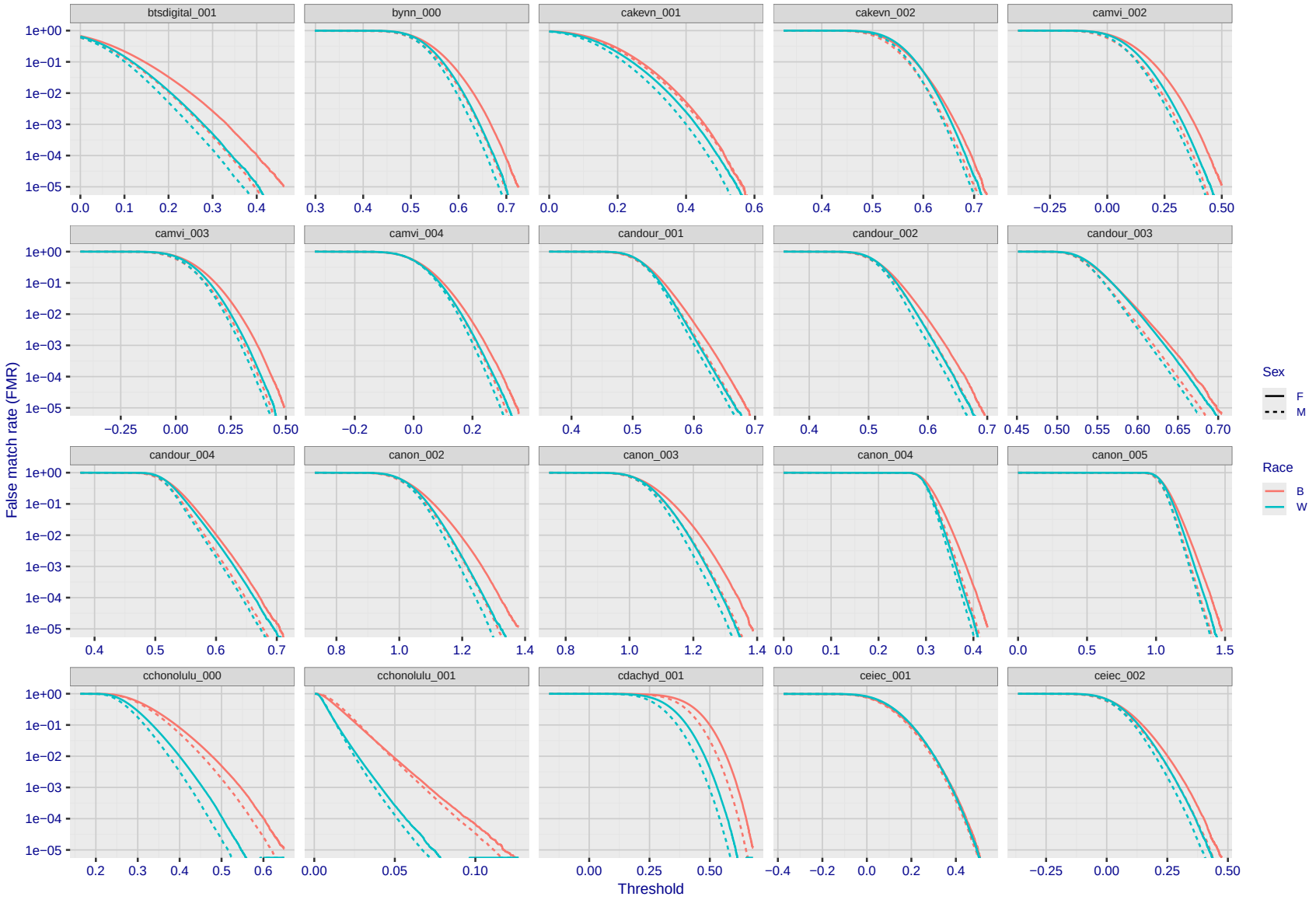

Figure 562: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

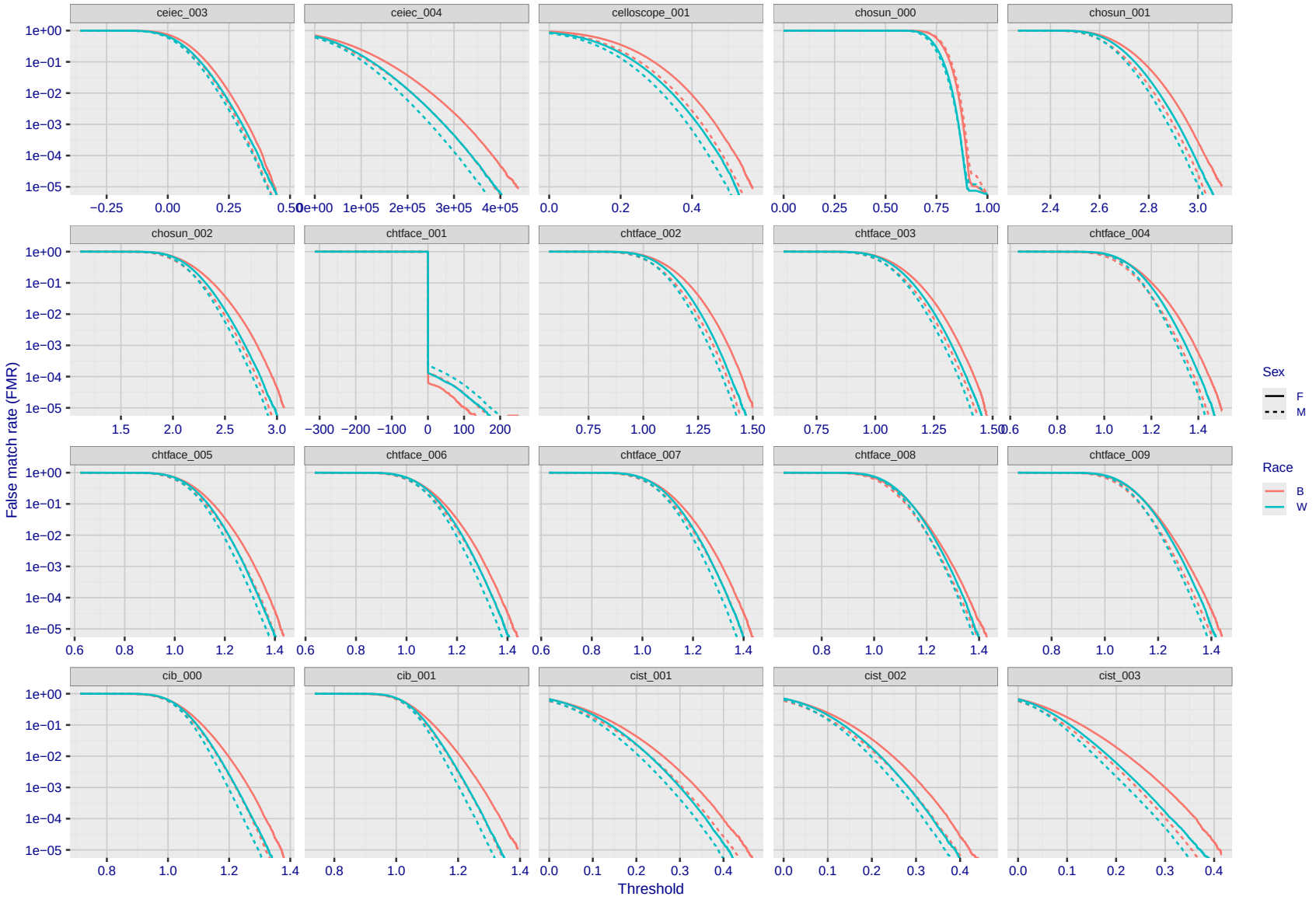

Figure 563: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

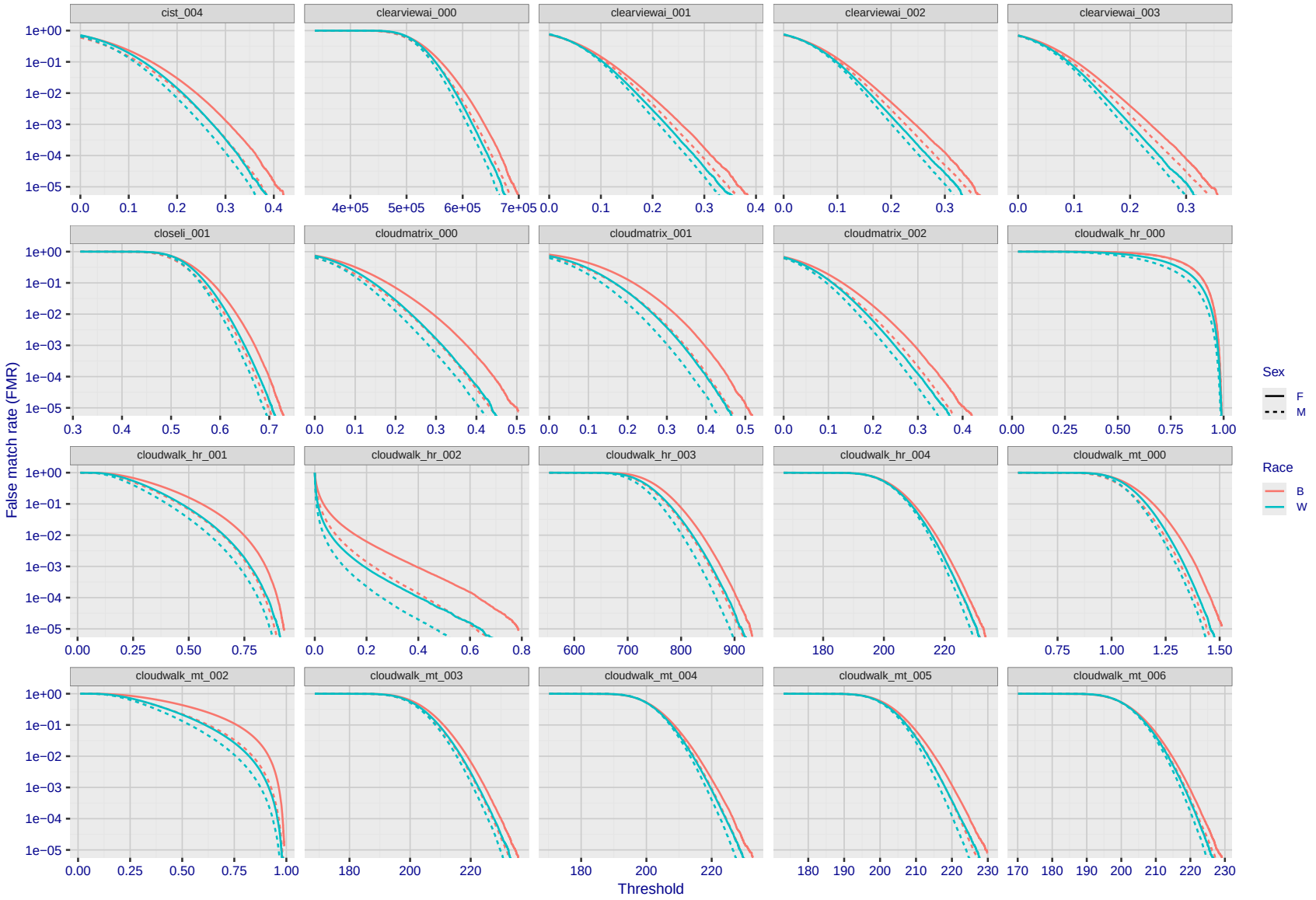

Figure 564: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

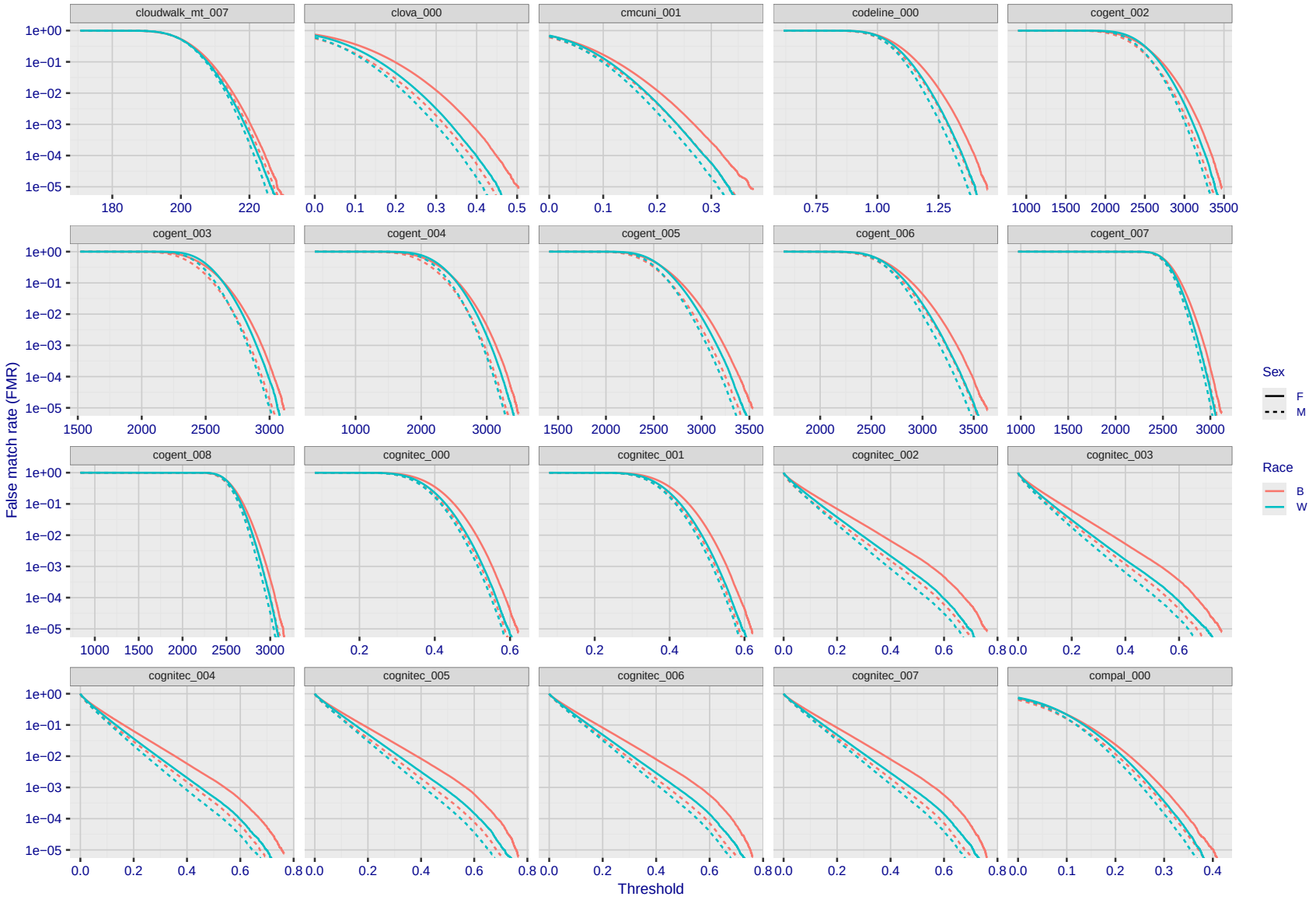

Figure 565: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

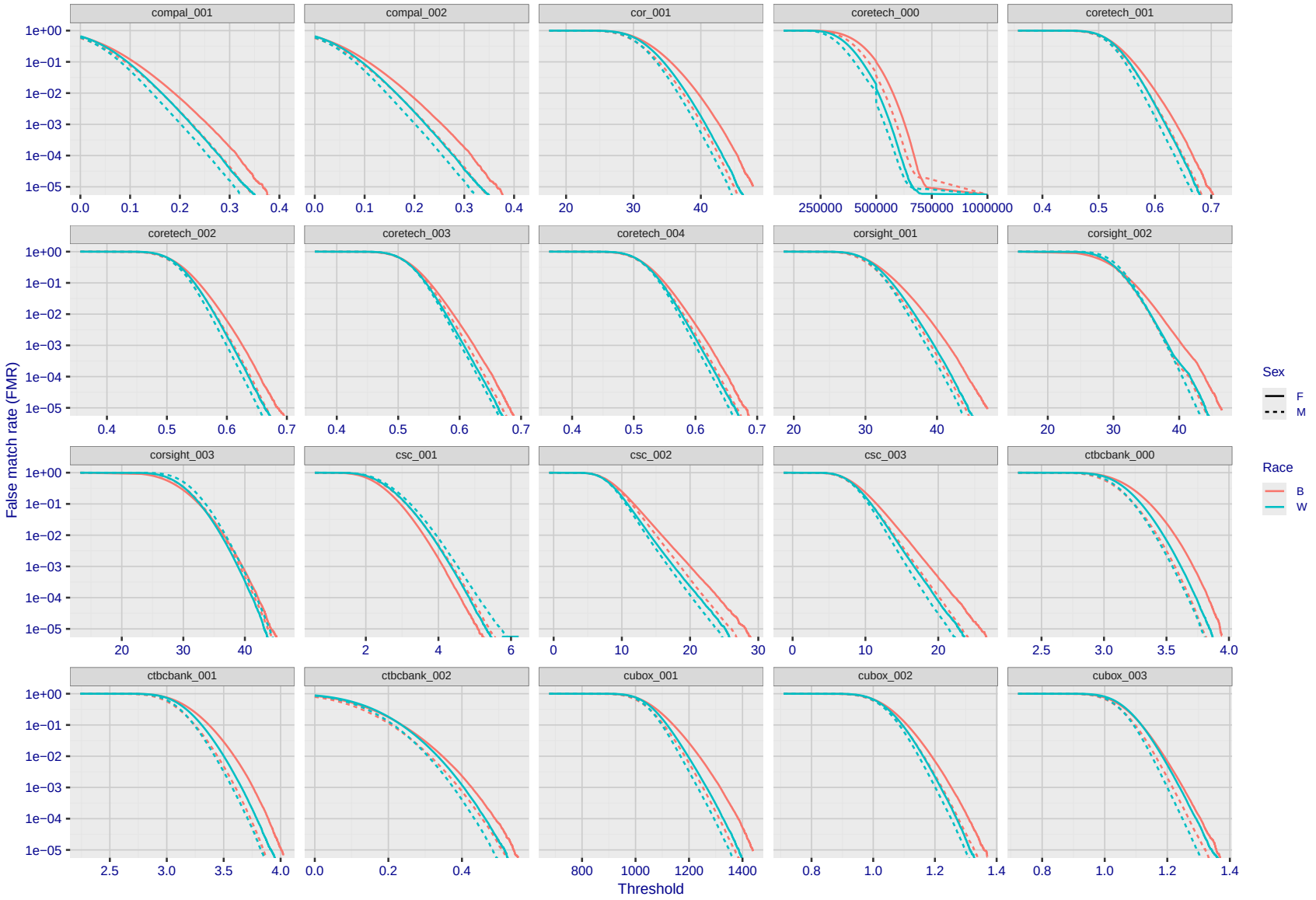

Figure 566: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

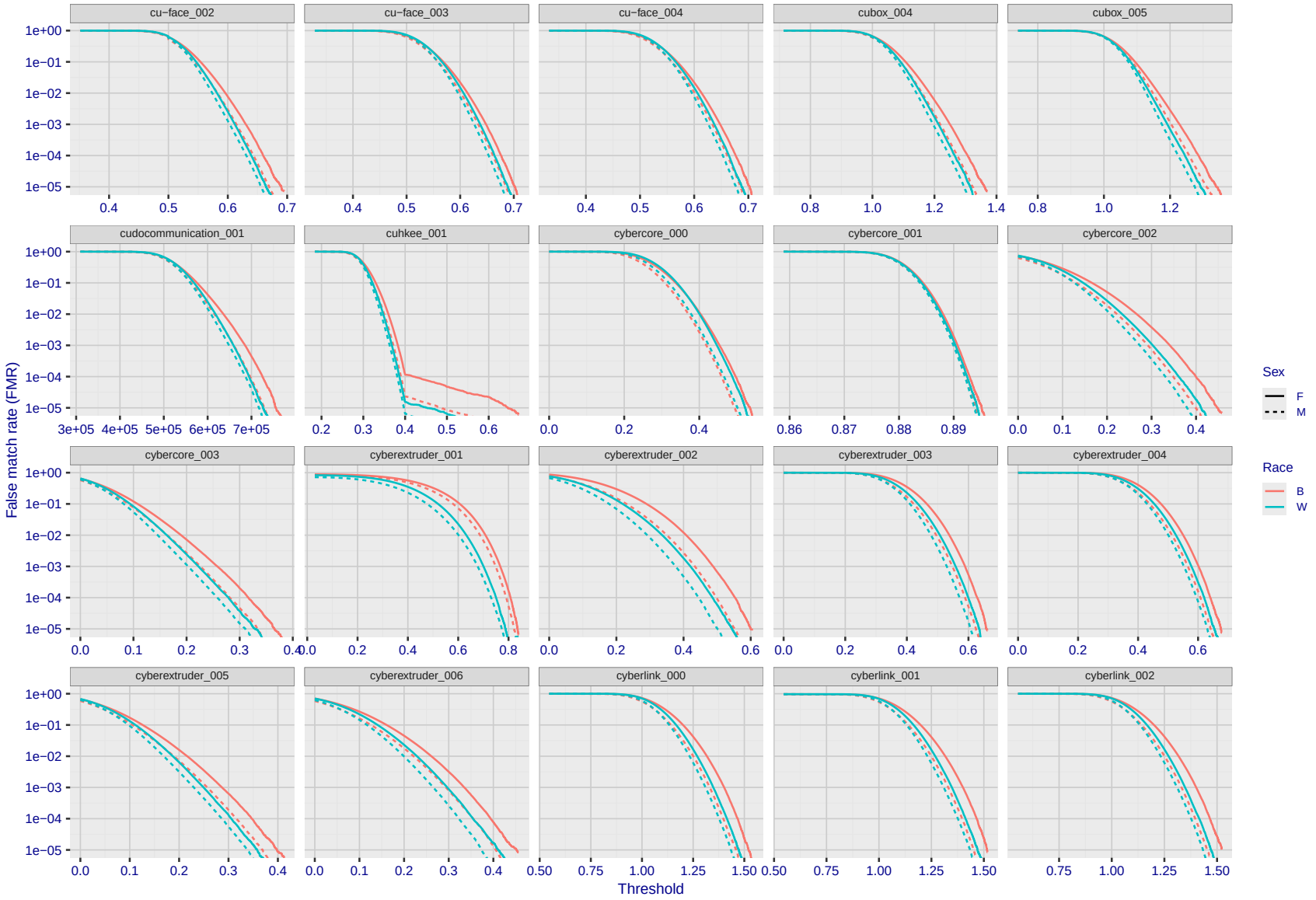

Figure 567: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

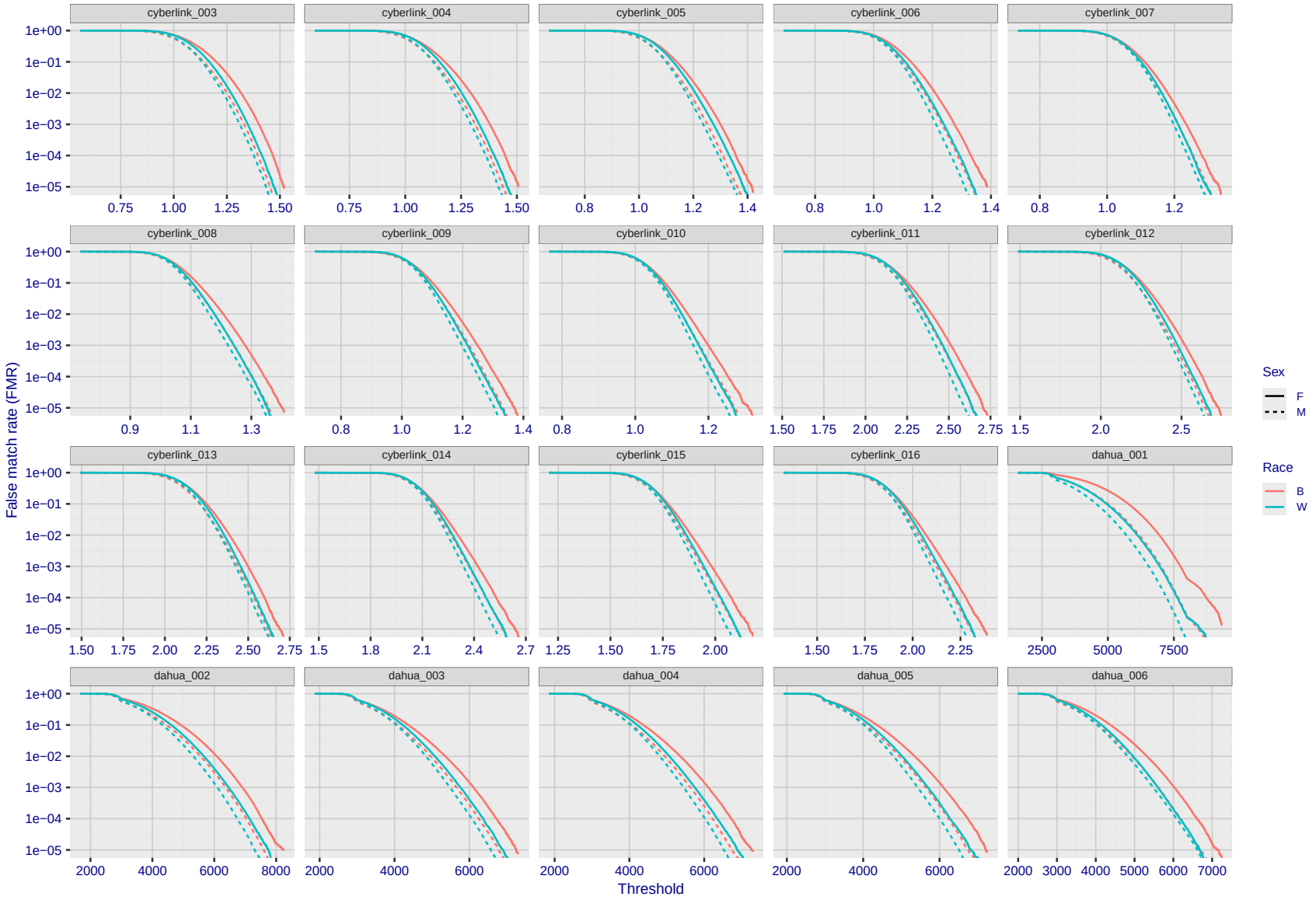

Figure 568: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

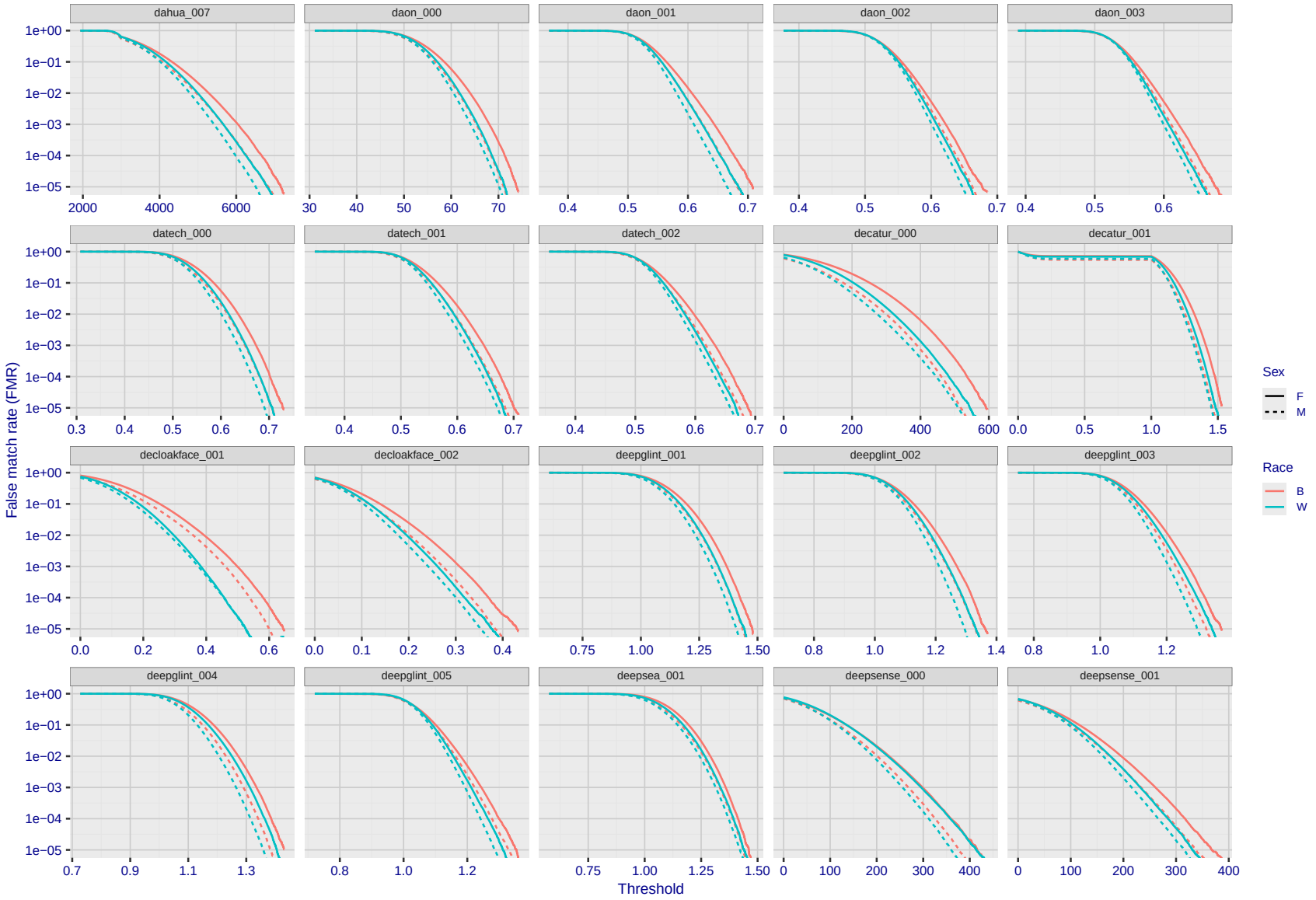

Figure 569: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

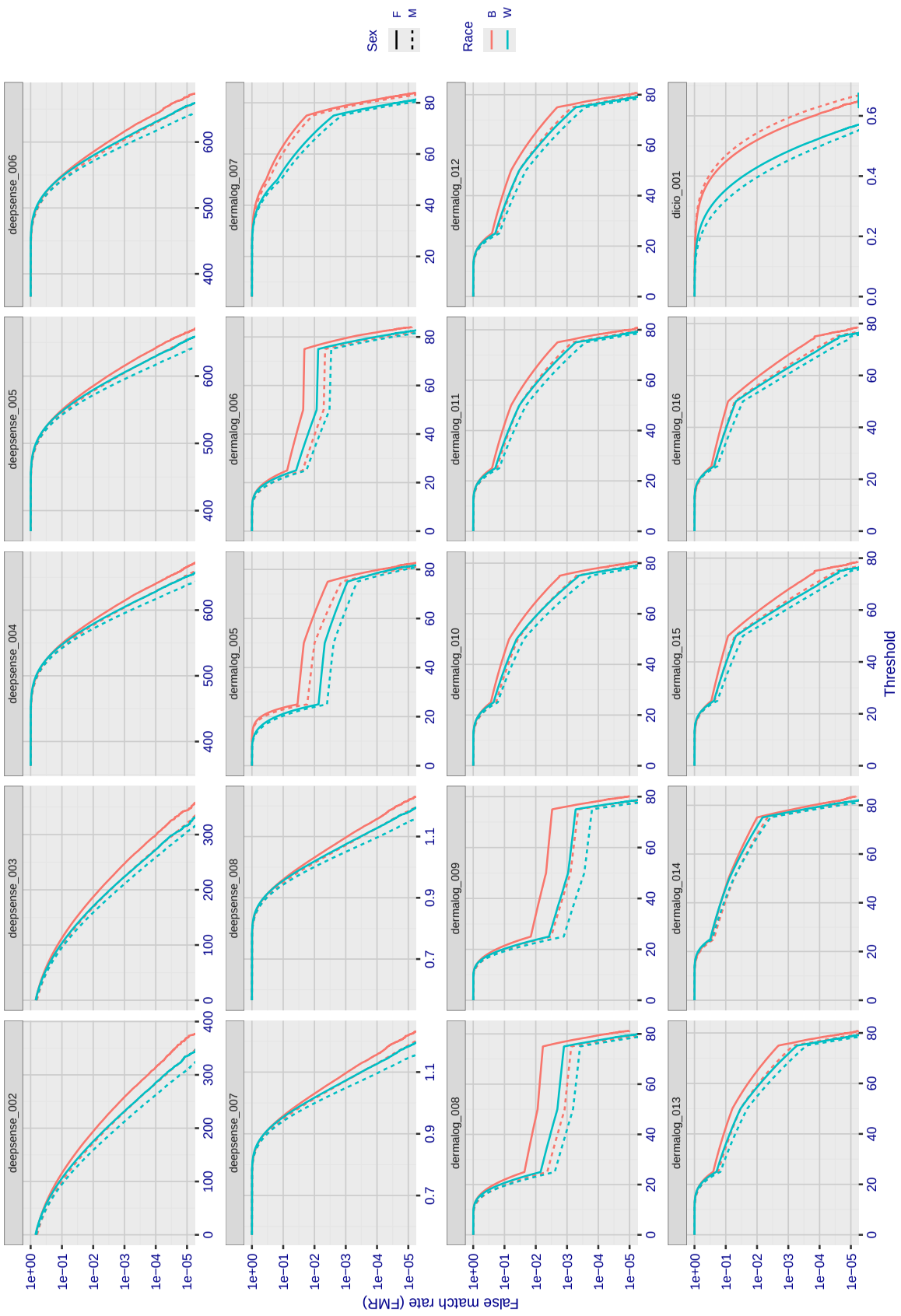

Figure 570: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

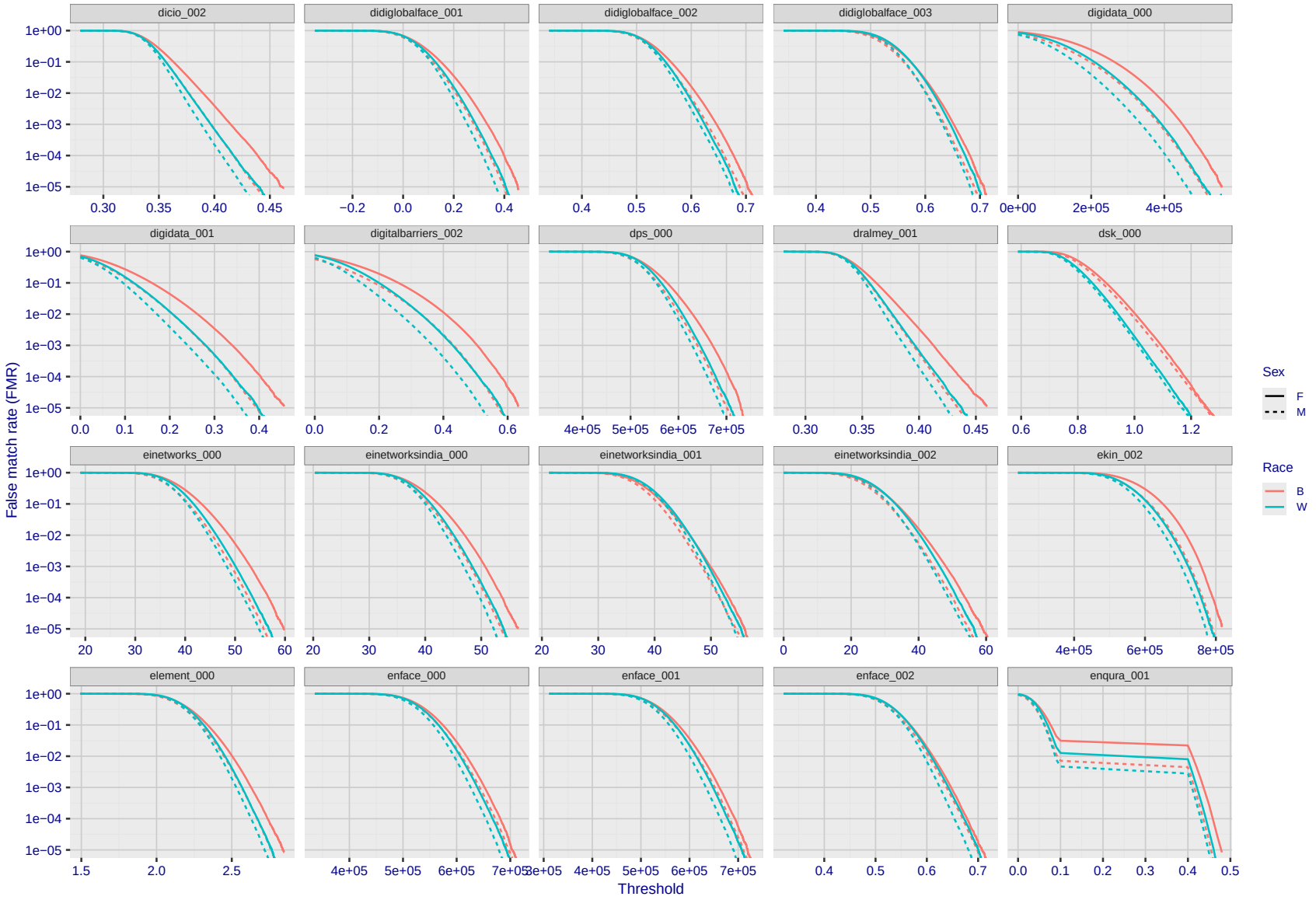

Figure 571: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

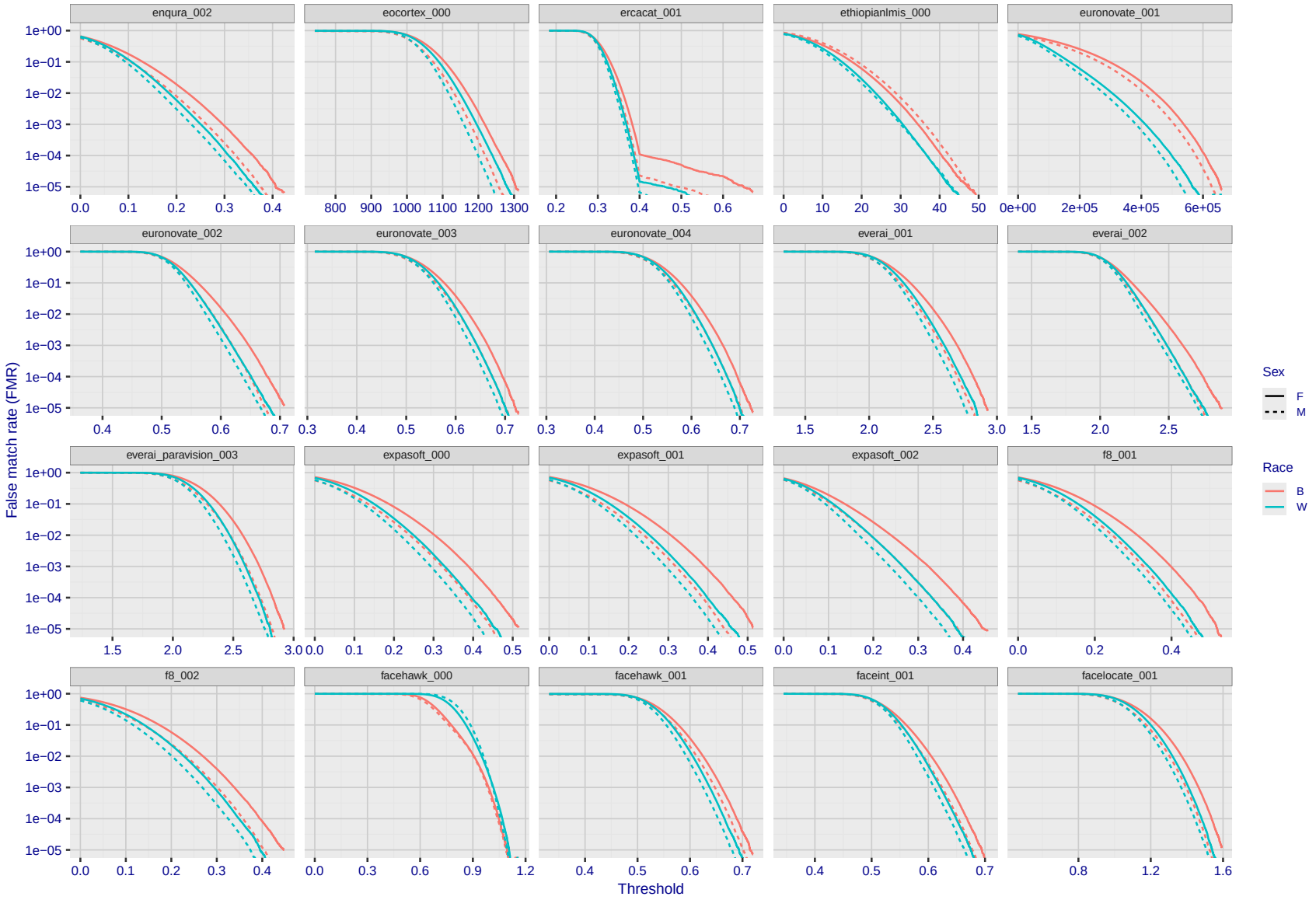

Figure 572: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

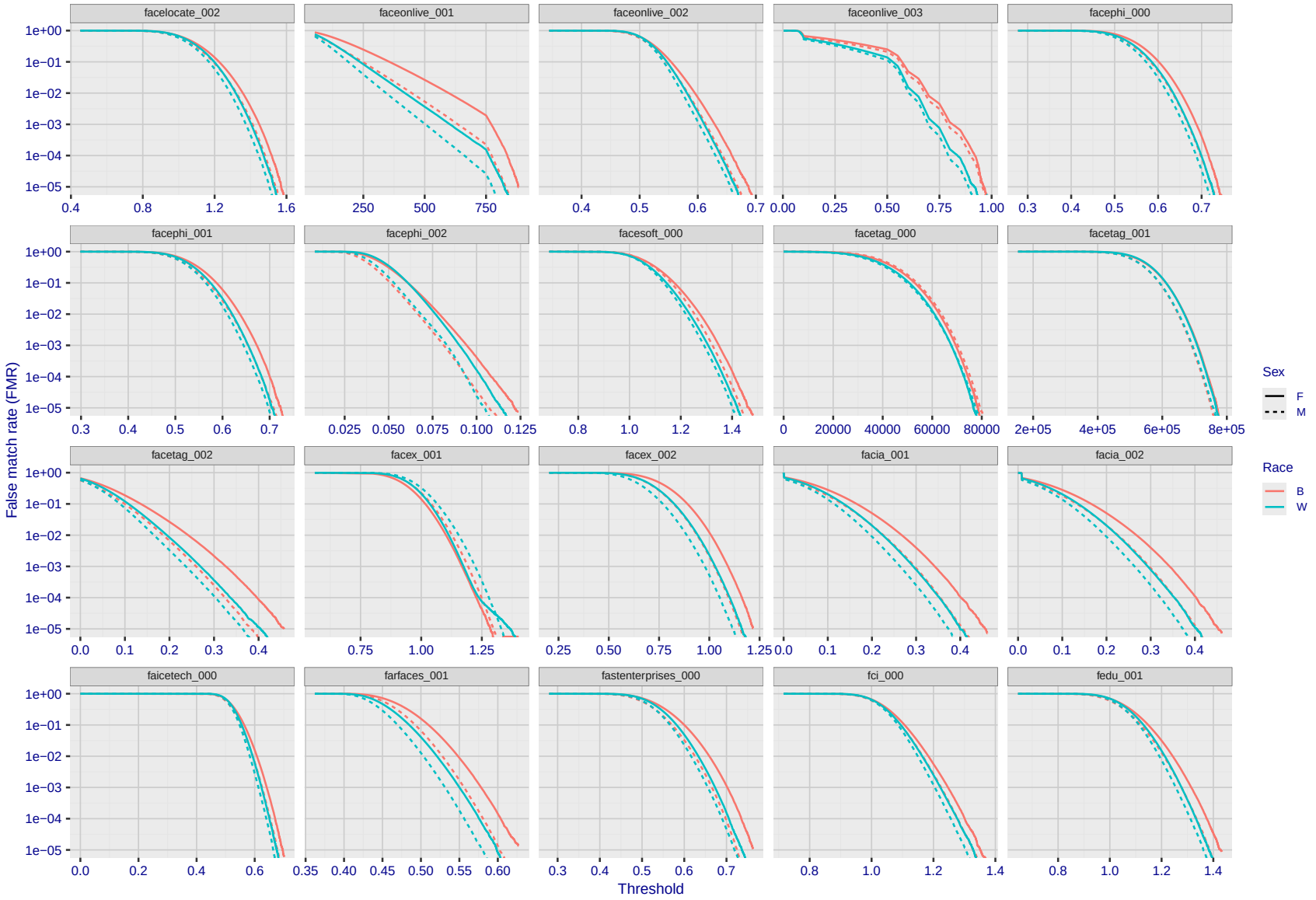

Figure 573: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

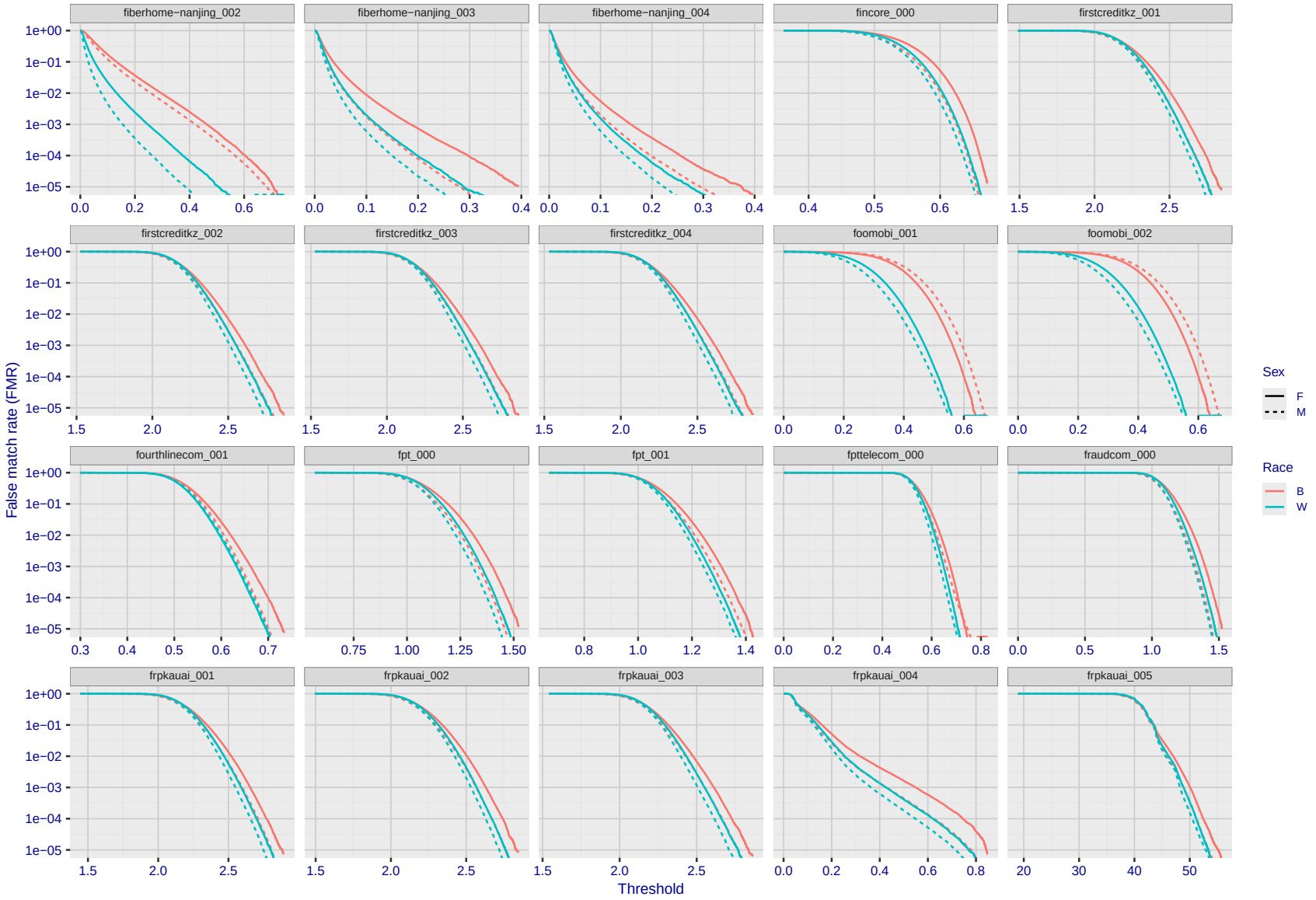

Figure 574: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

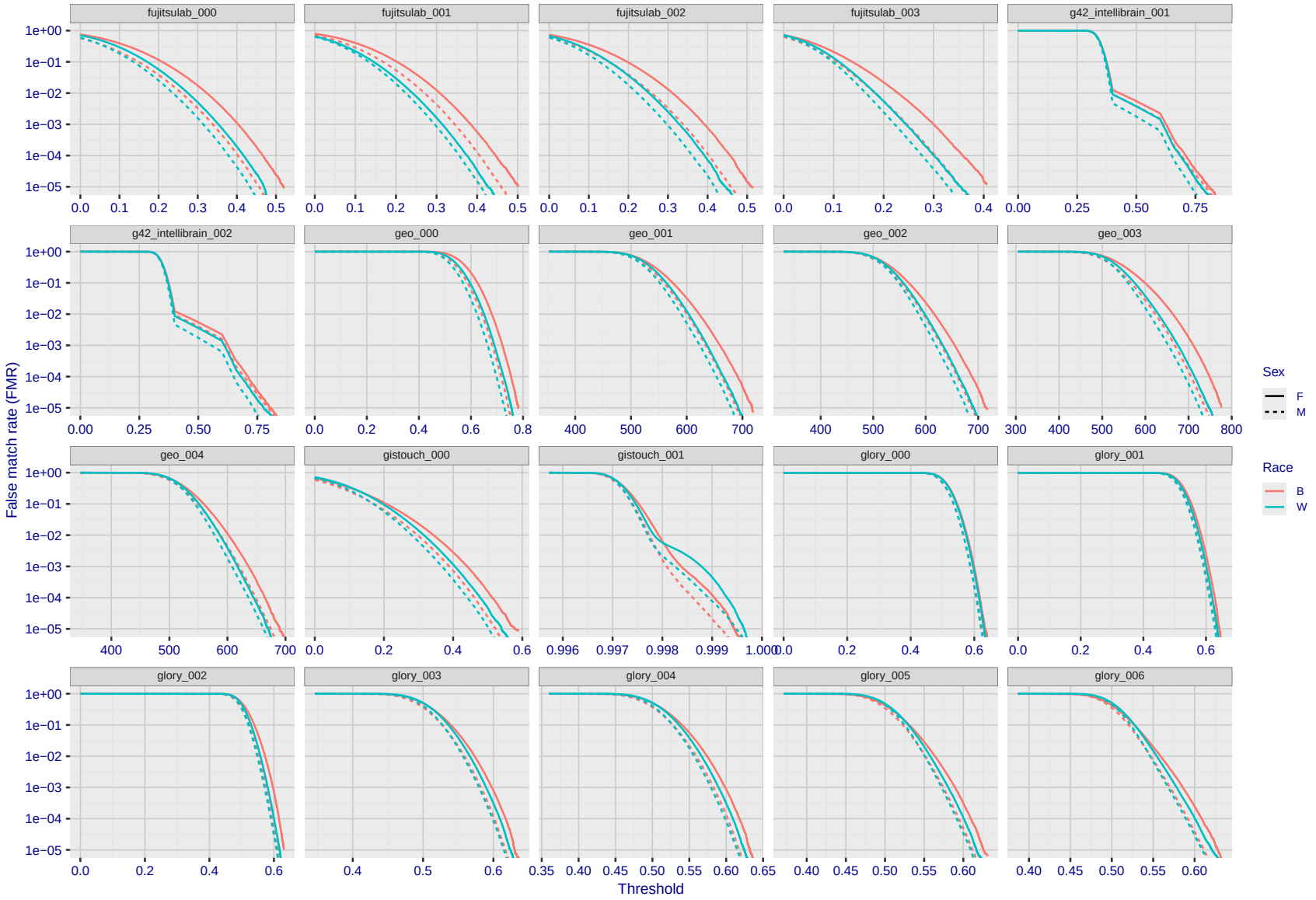

Figure 575: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

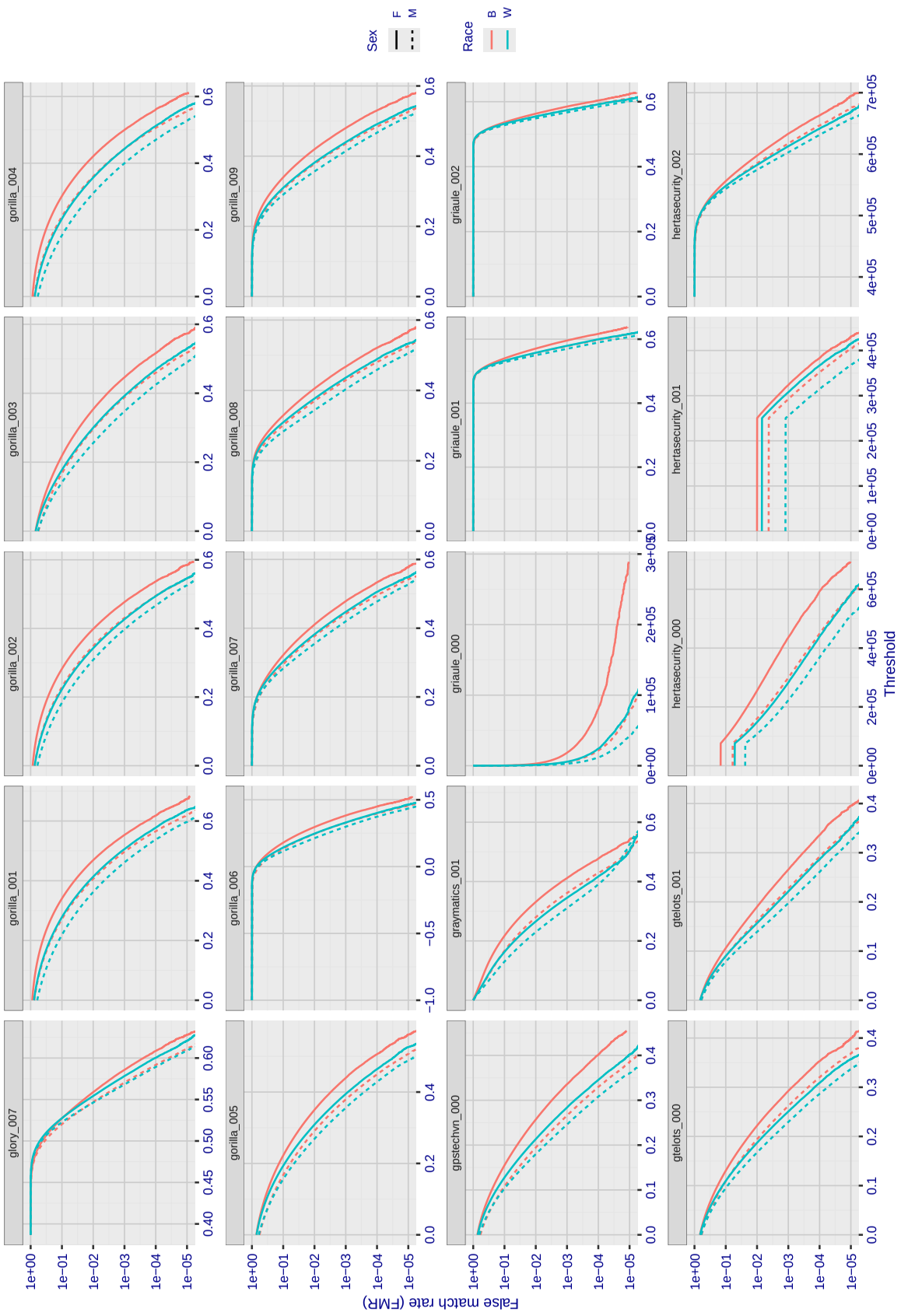

Figure 576: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

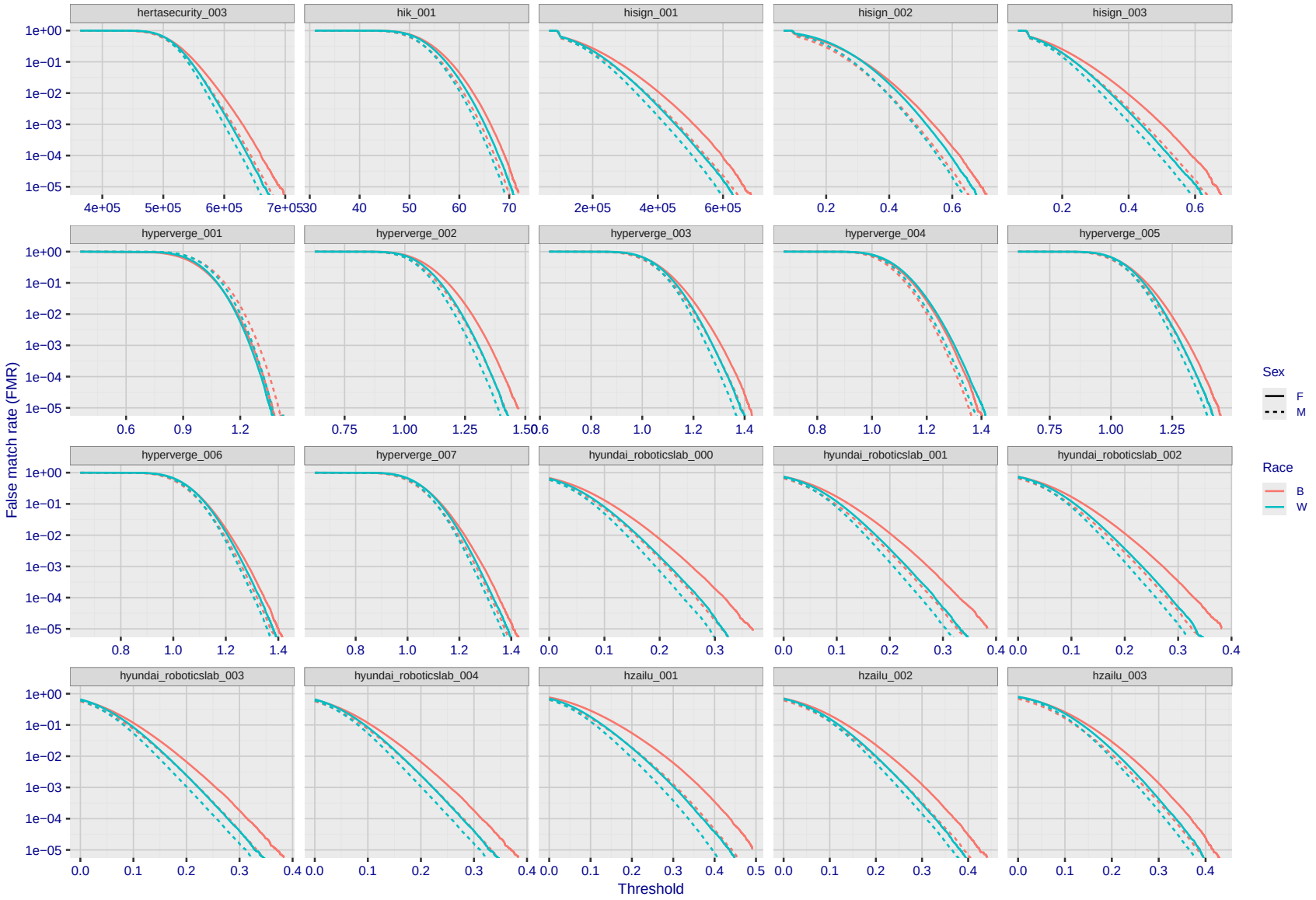

Figure 577: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

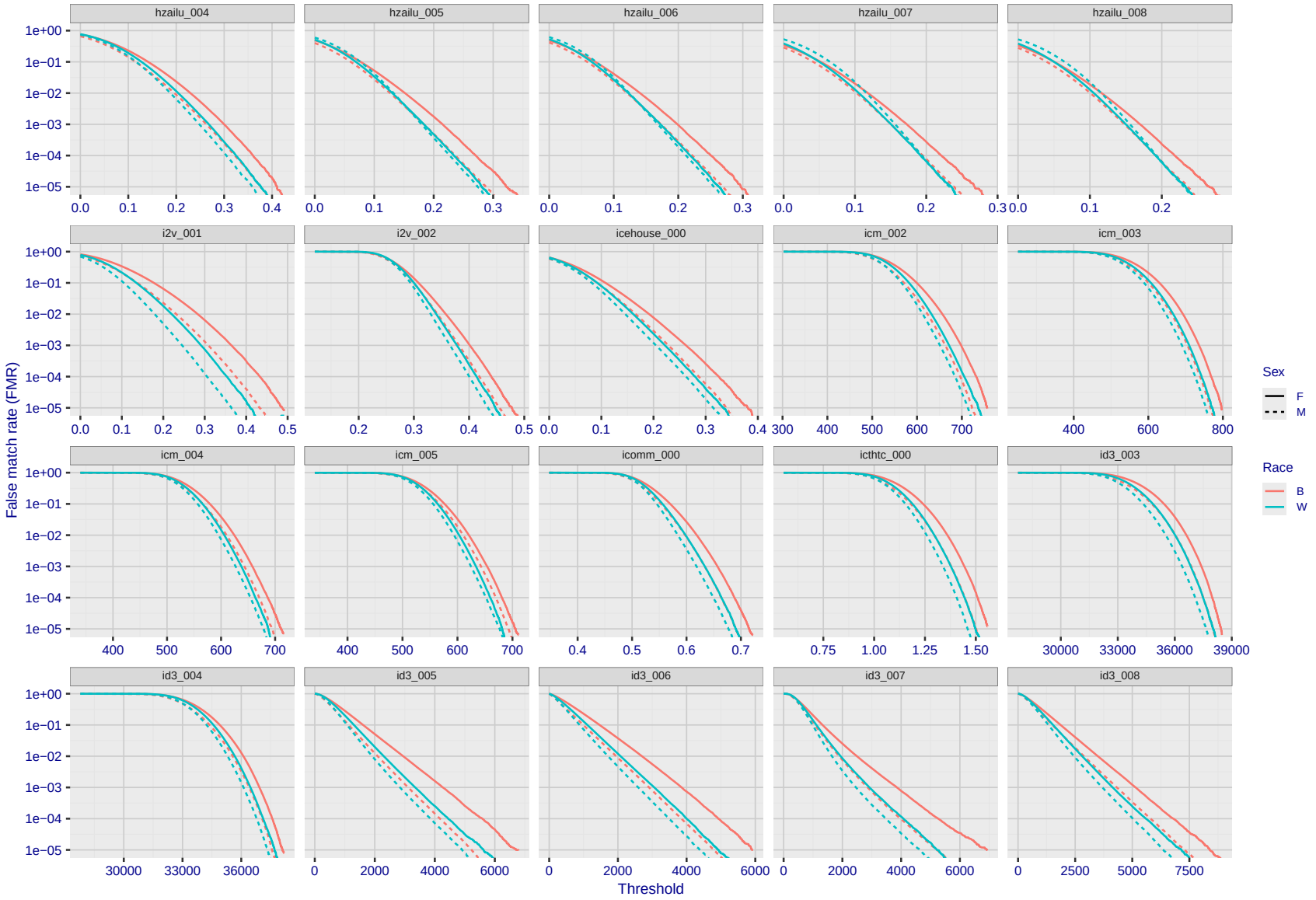

Figure 578: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

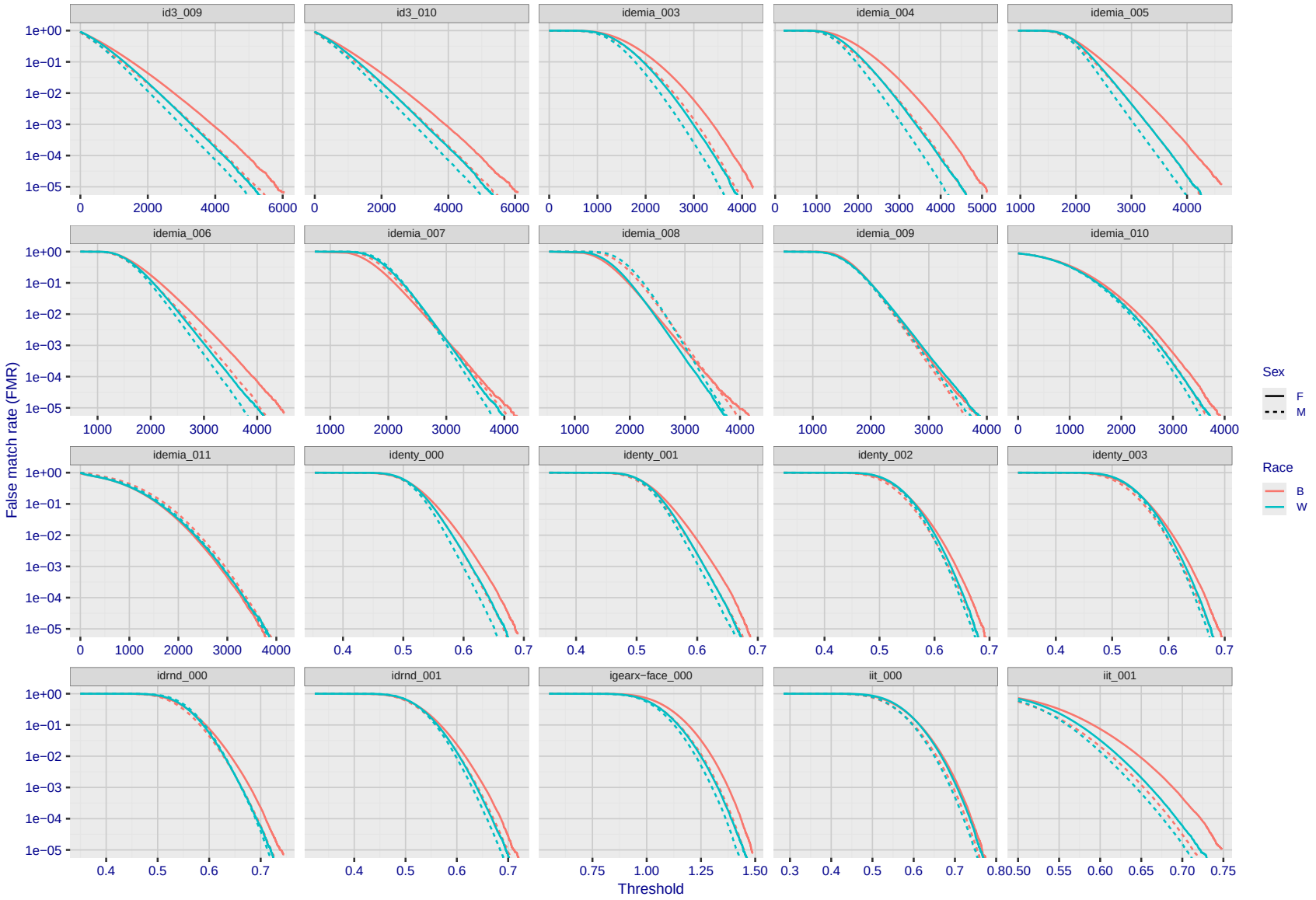

Figure 579: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

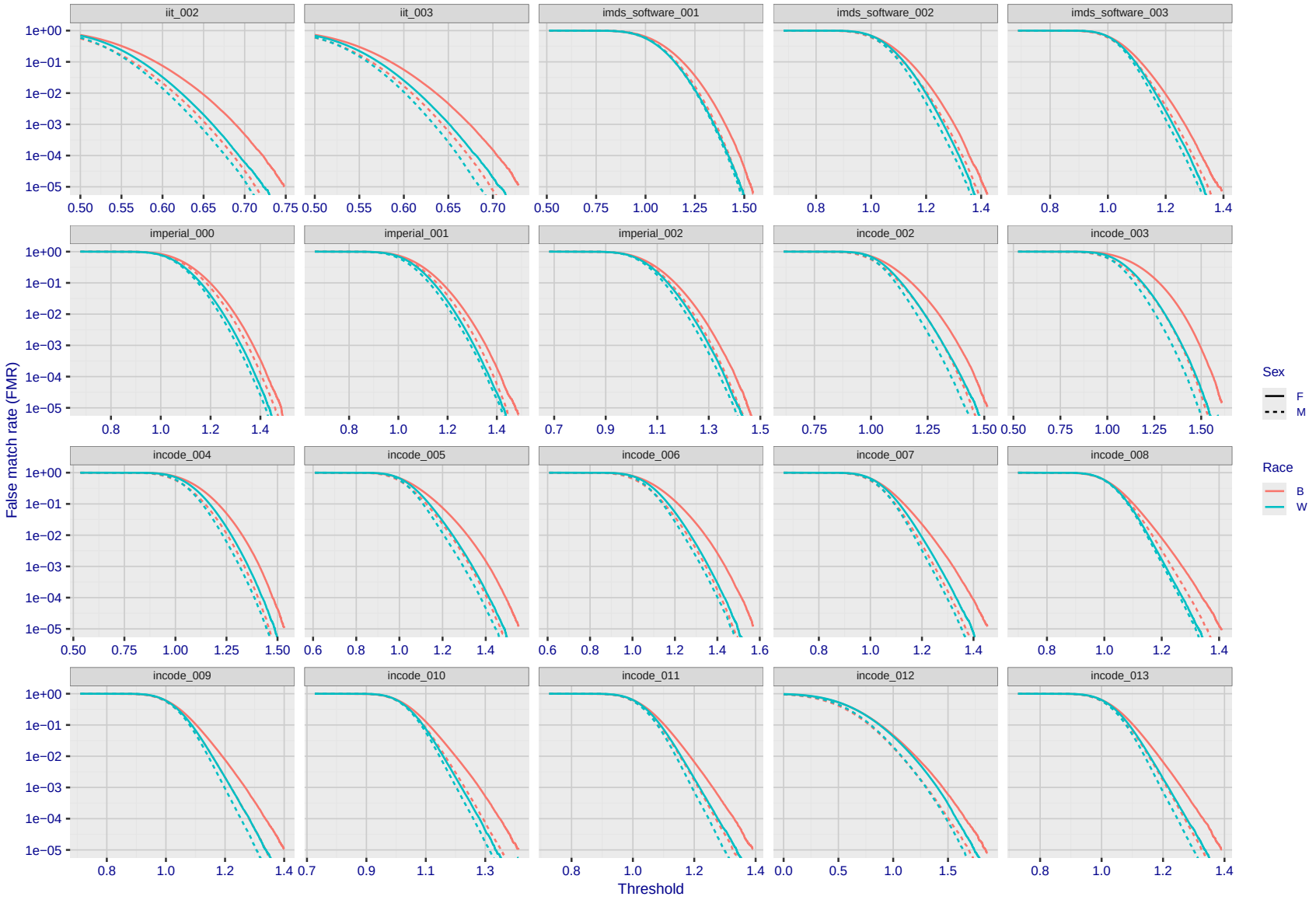

Figure 580: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

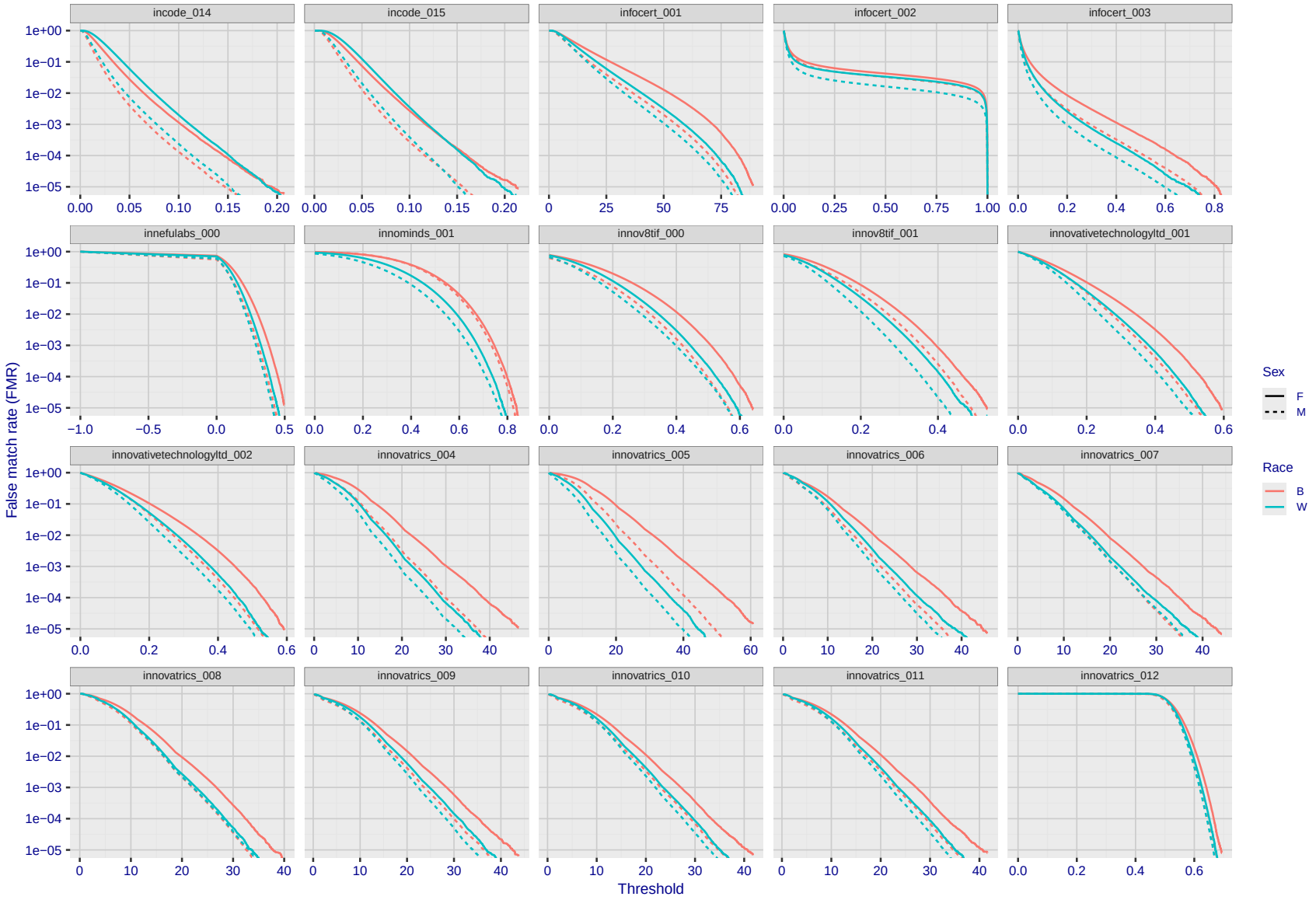

Figure 581: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

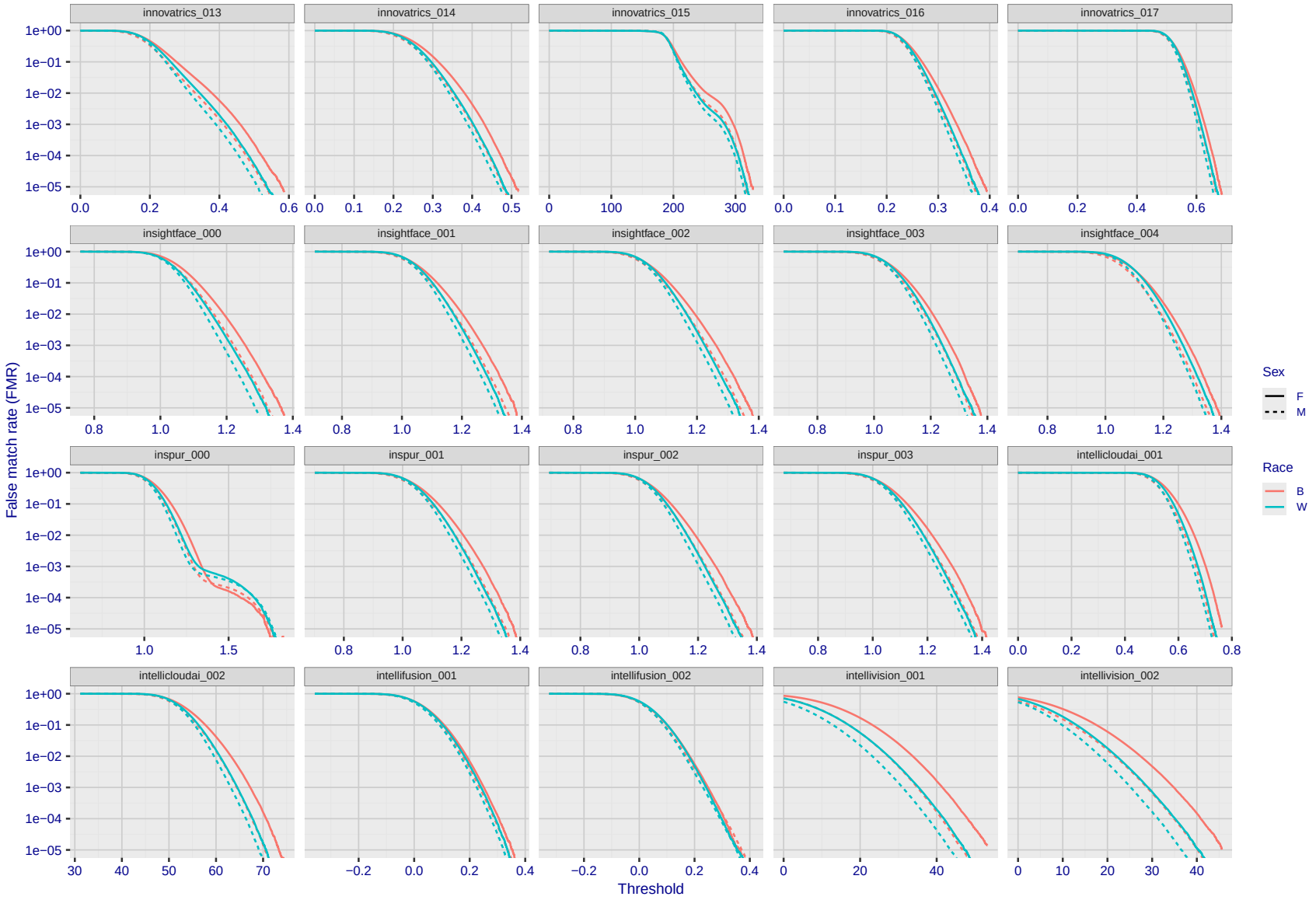

Figure 582: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

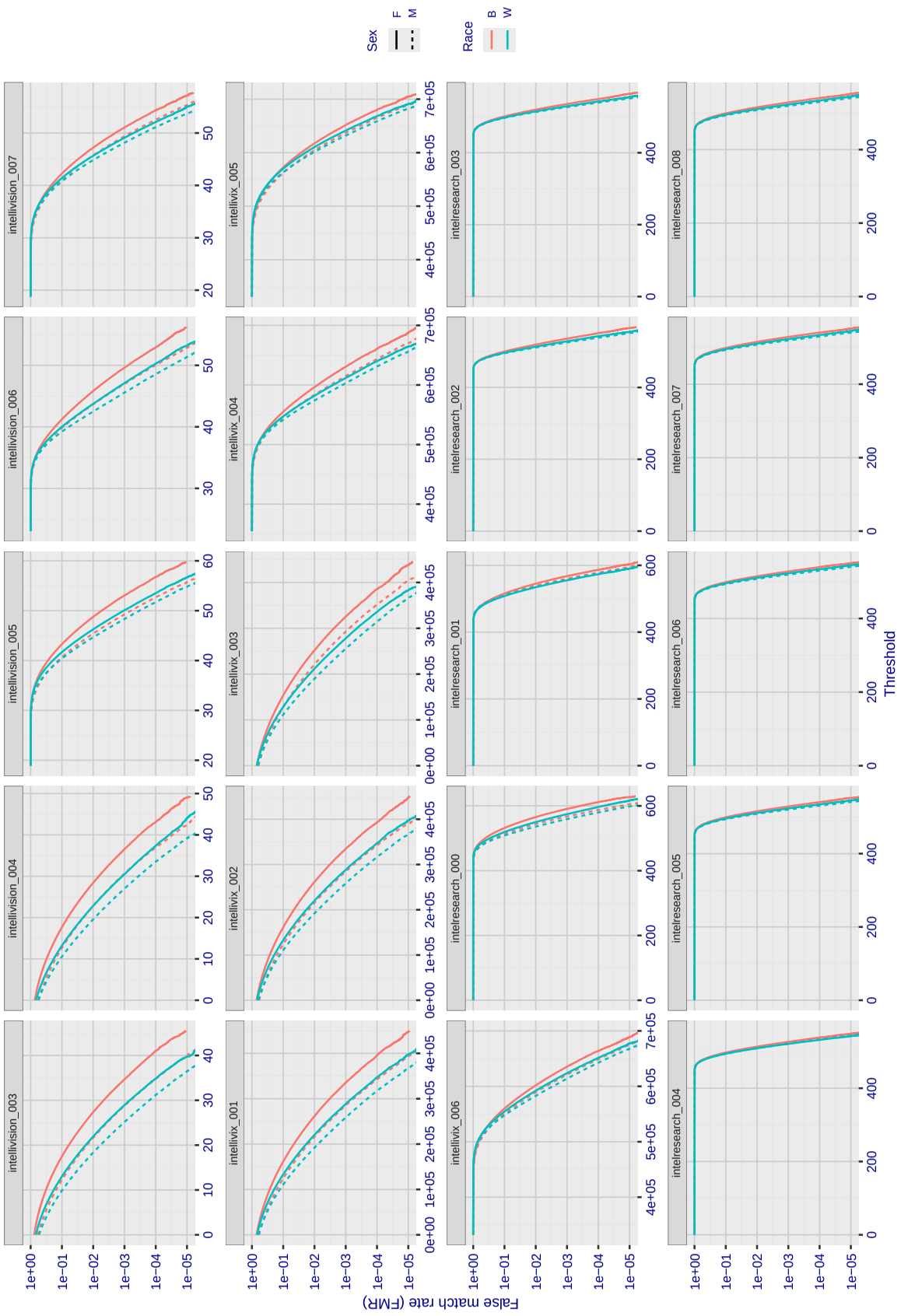

Figure 583: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

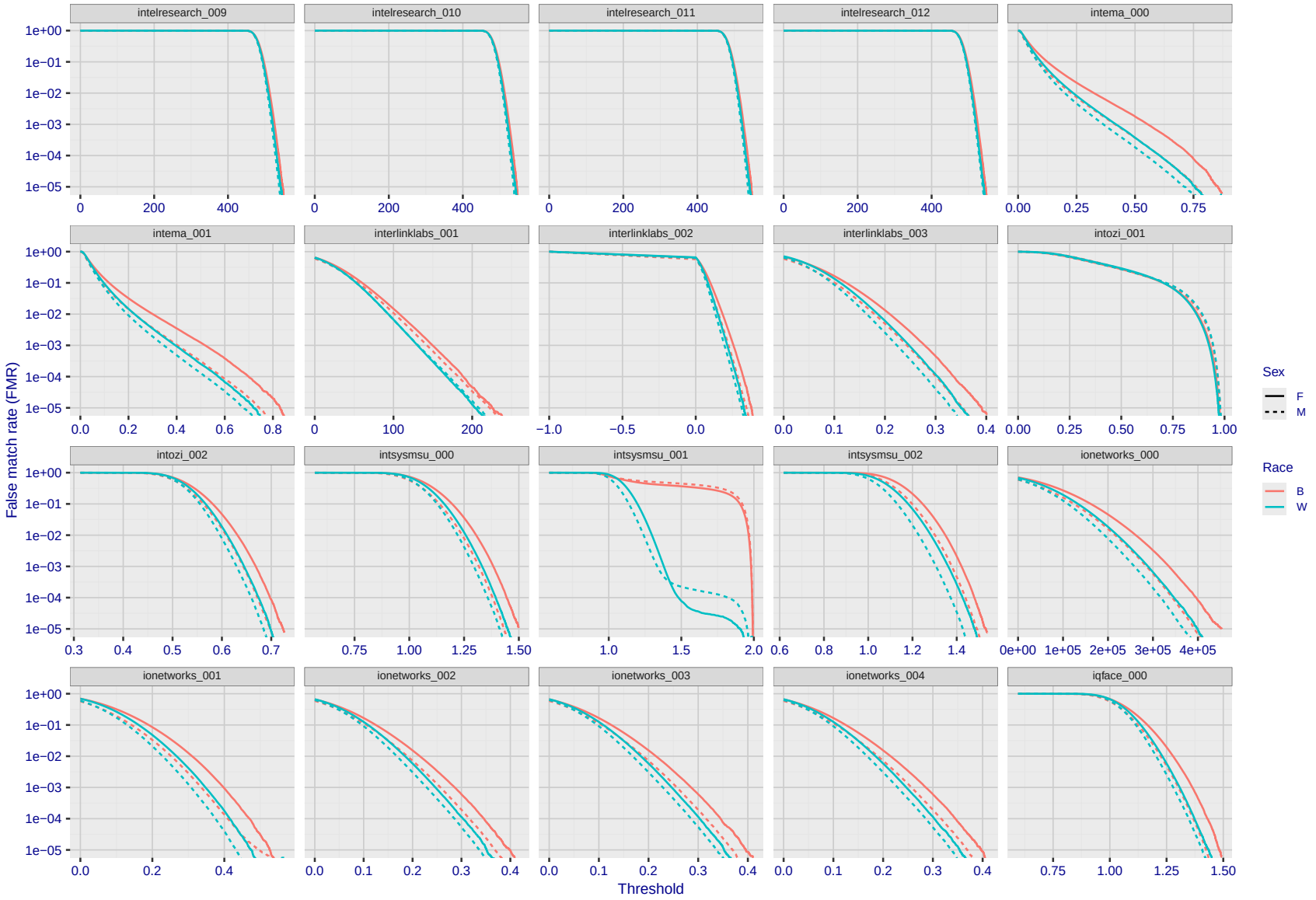

Figure 584: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

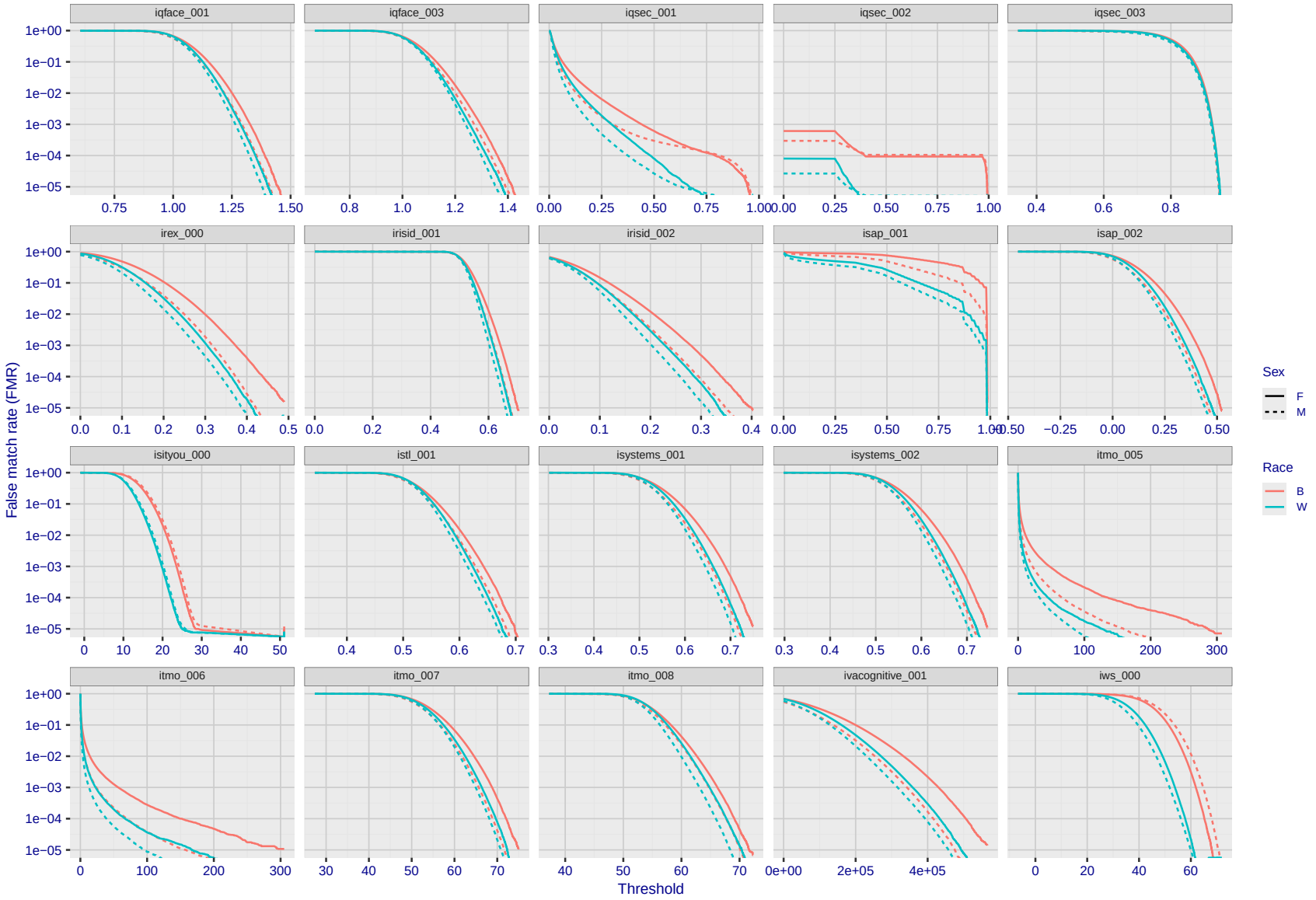

Figure 585: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

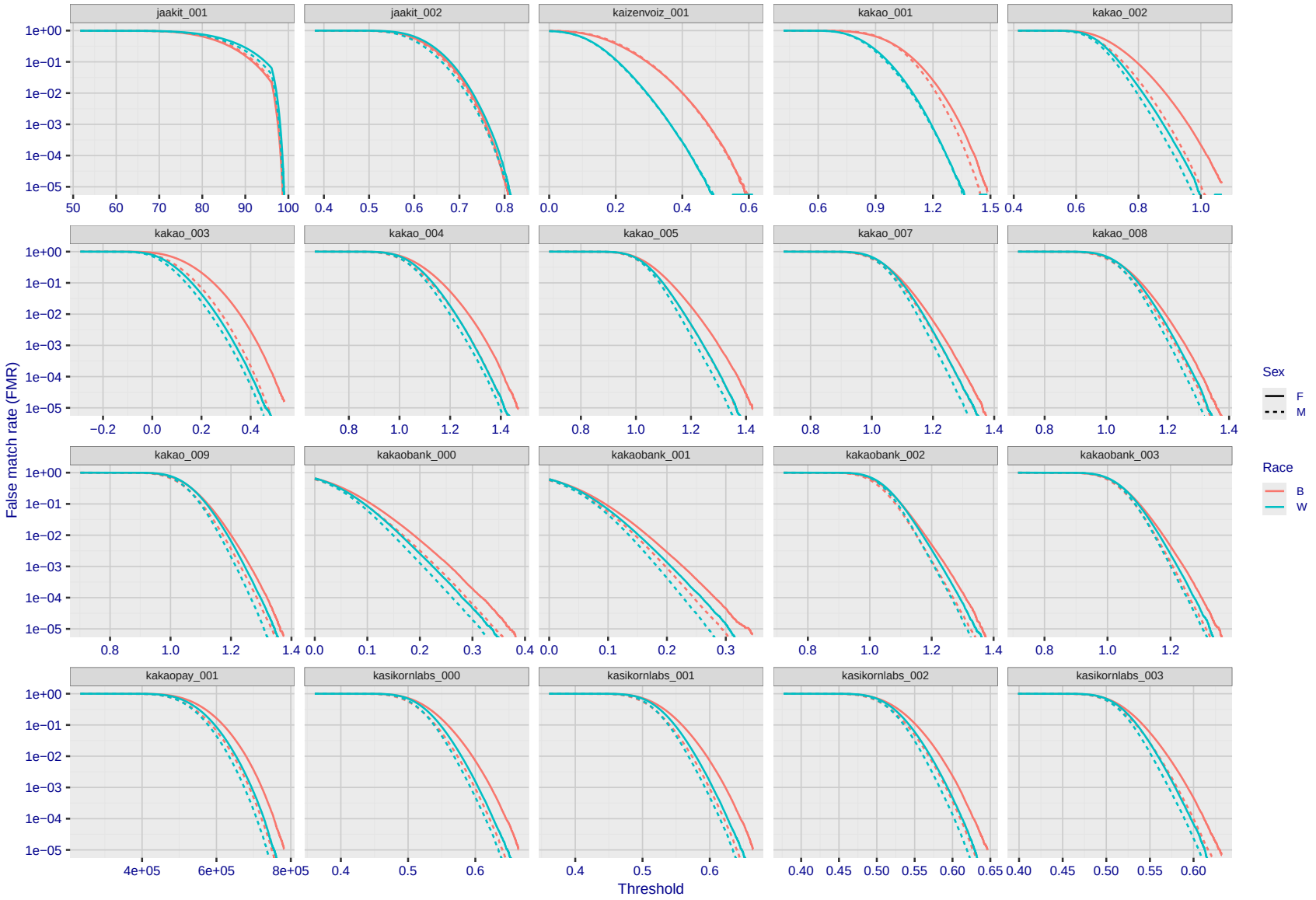

Figure 586: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

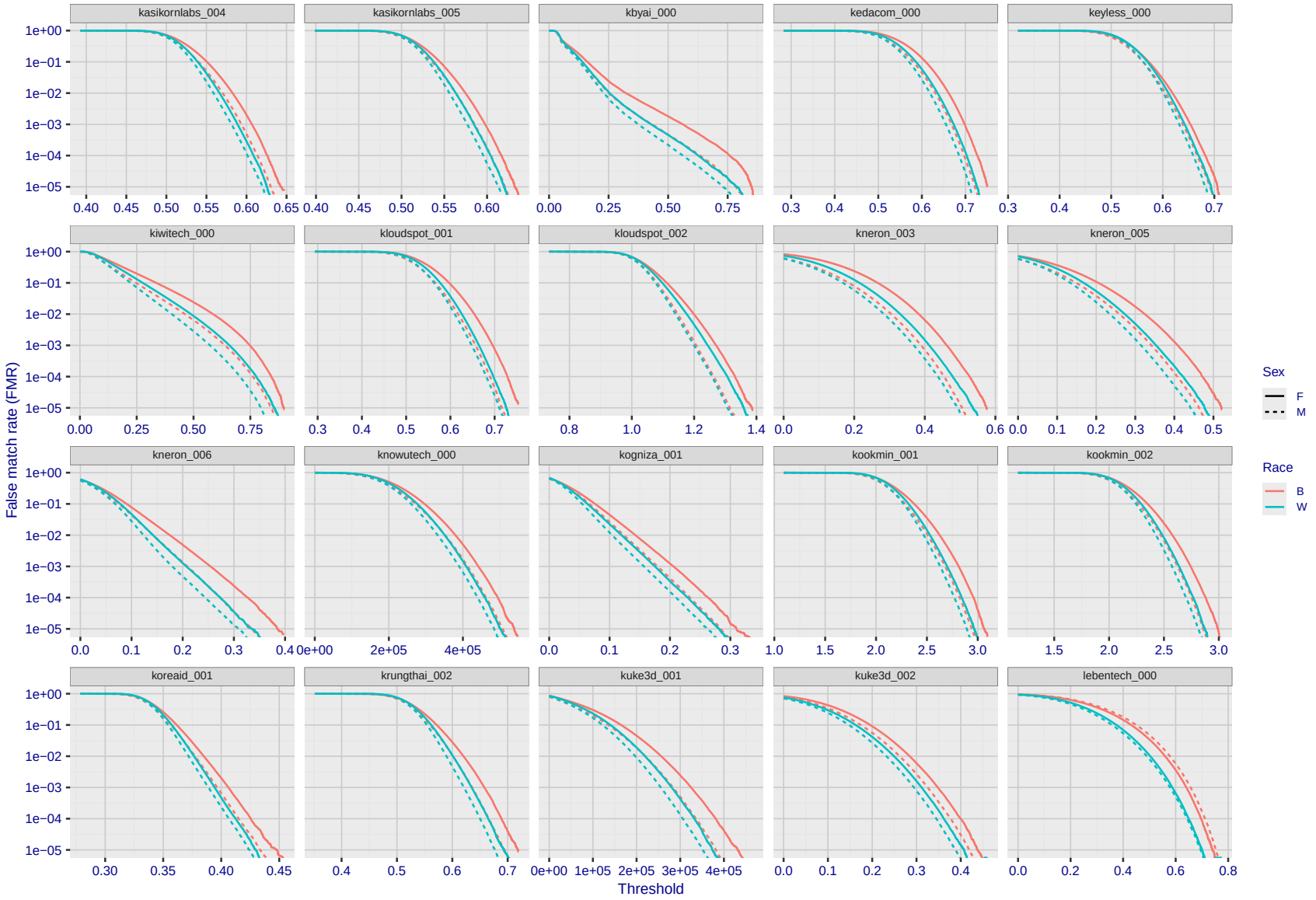

Figure 587: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

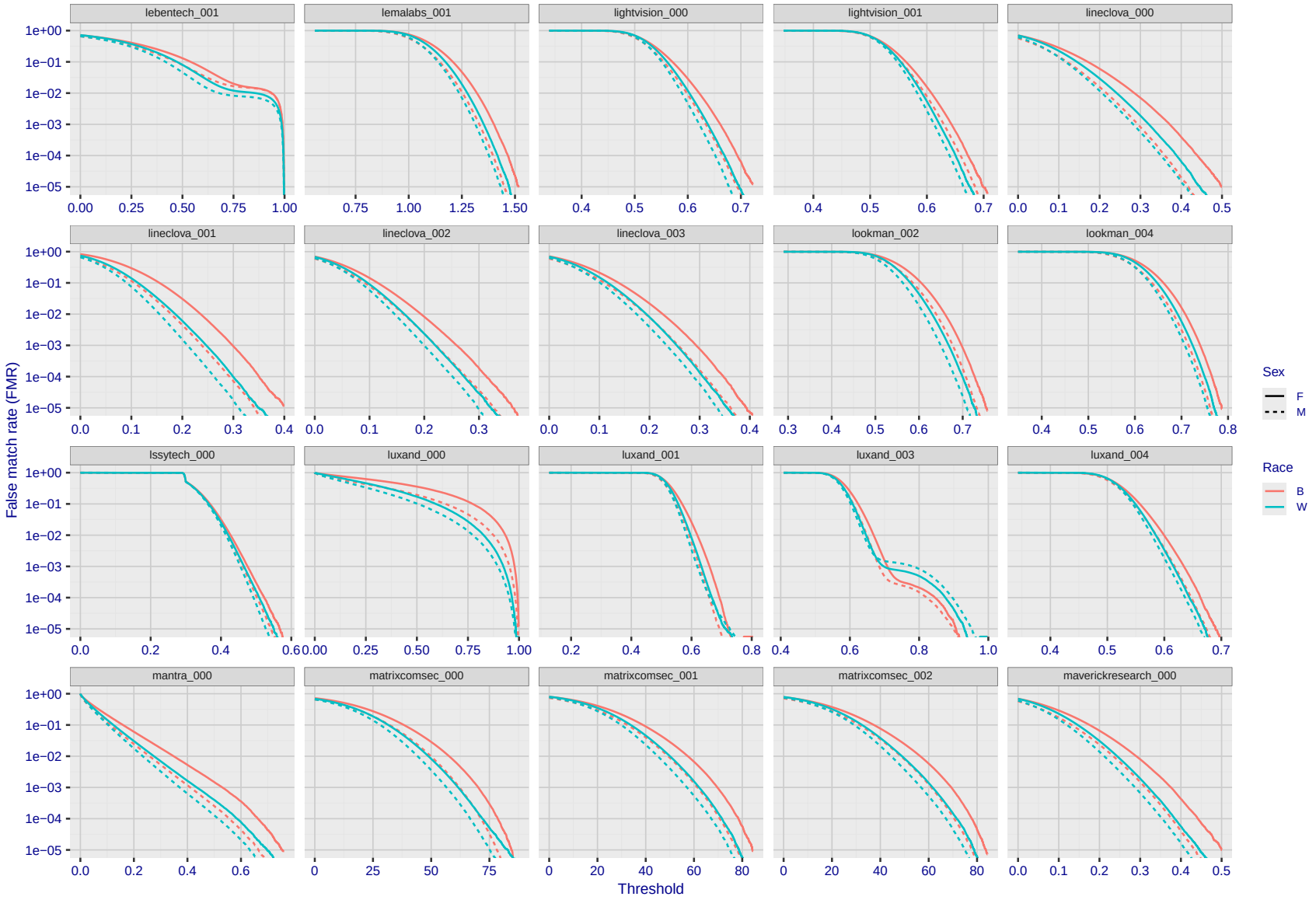

Figure 588: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

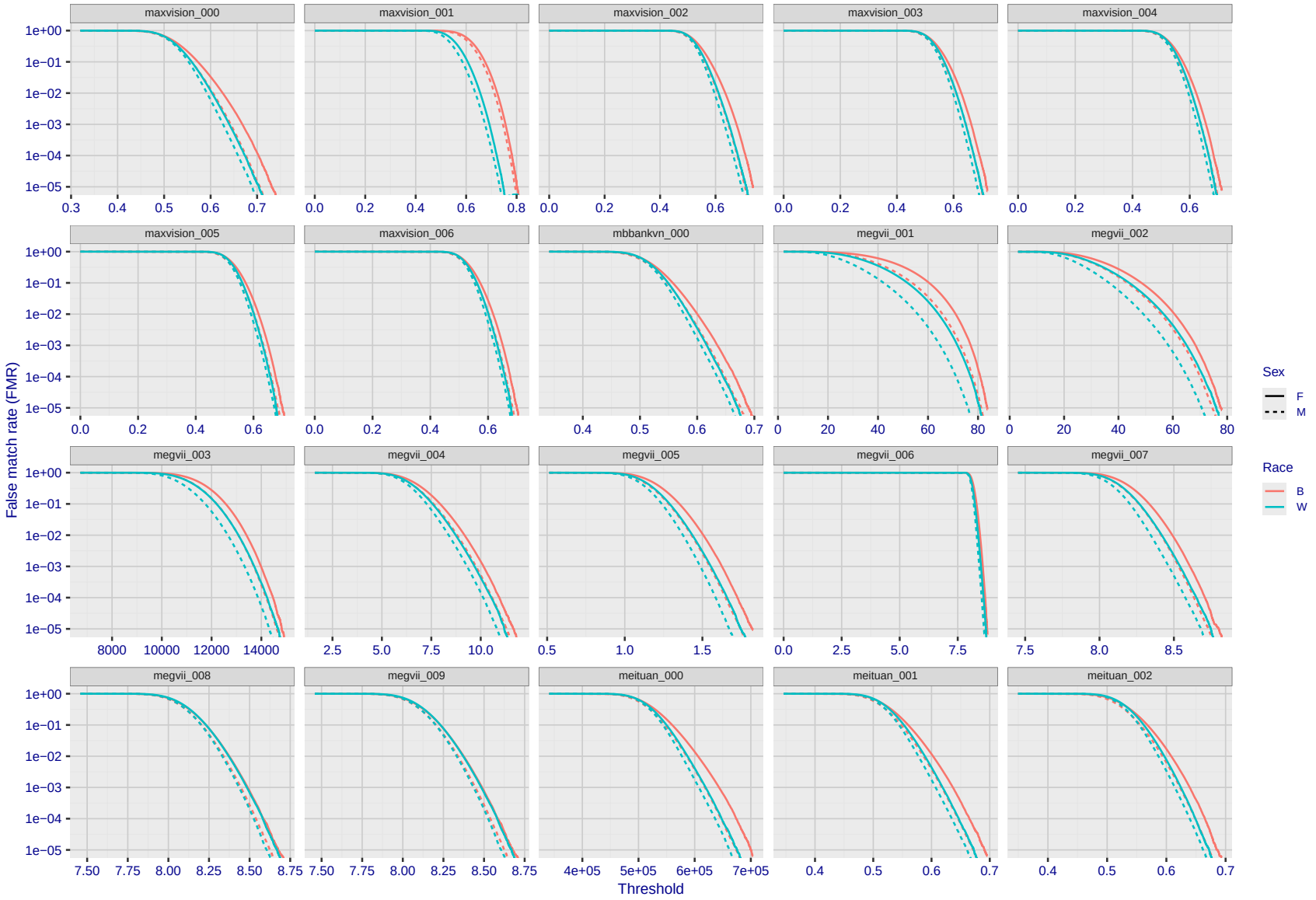

Figure 589: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

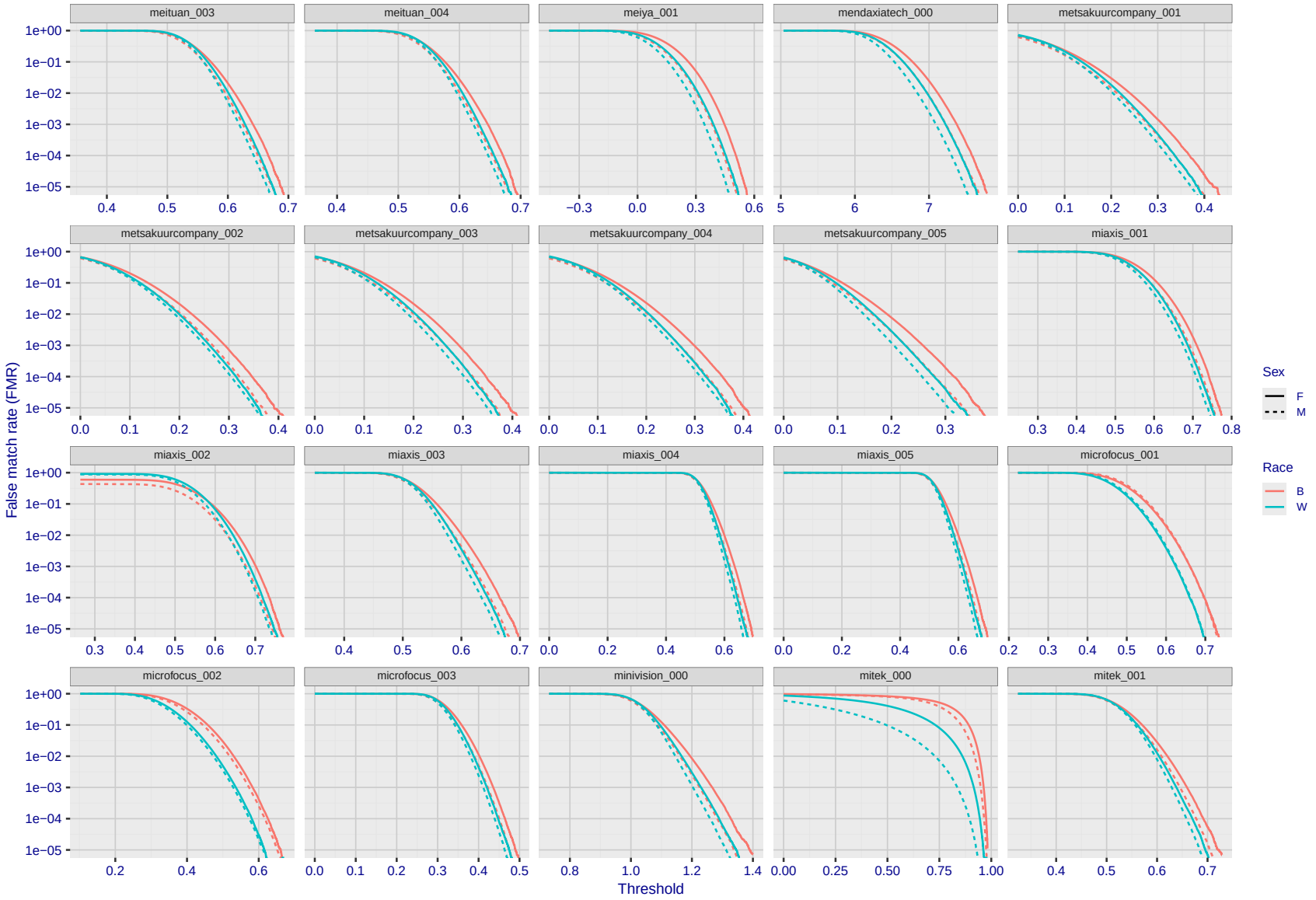

Figure 590: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

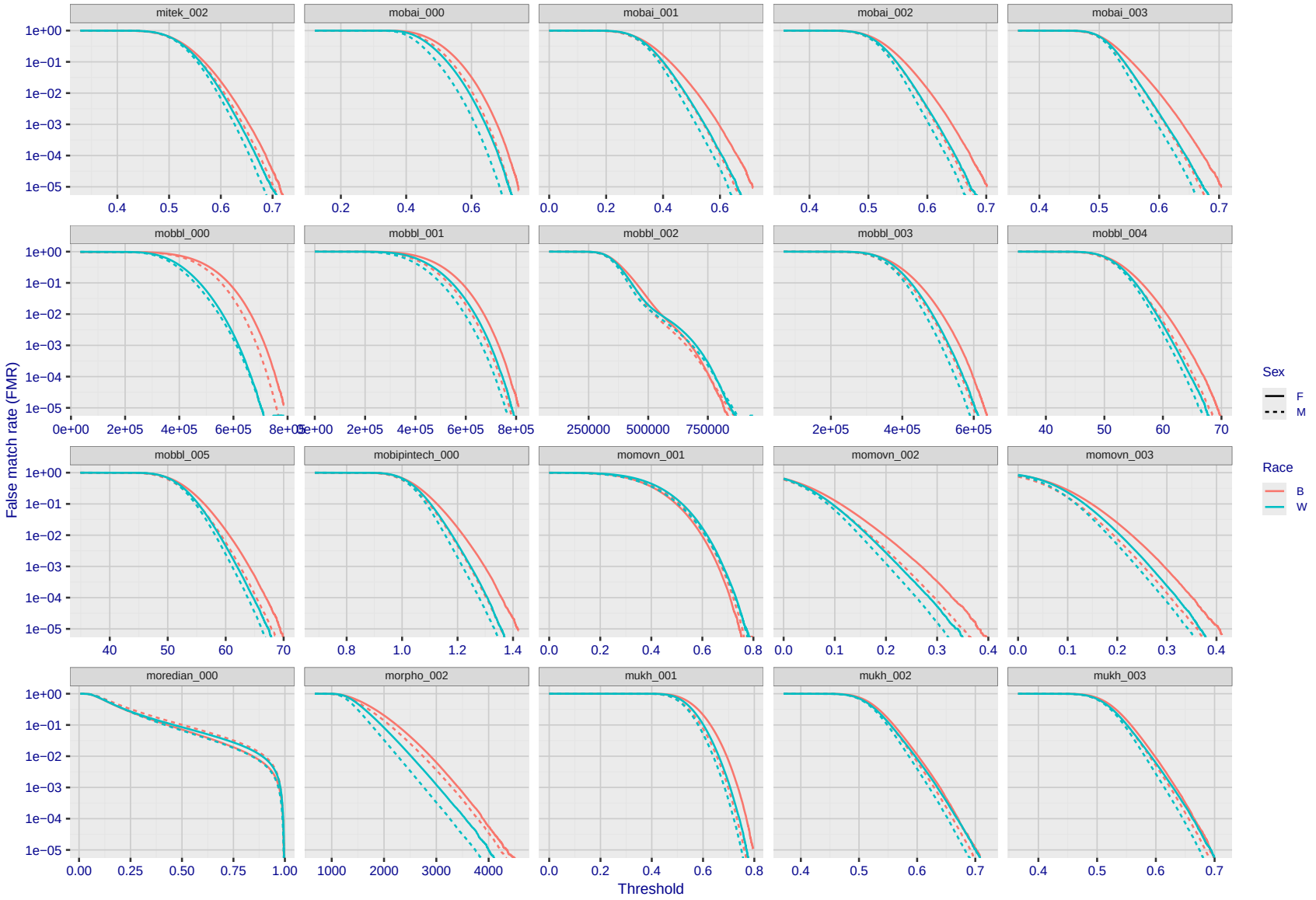

Figure 591: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

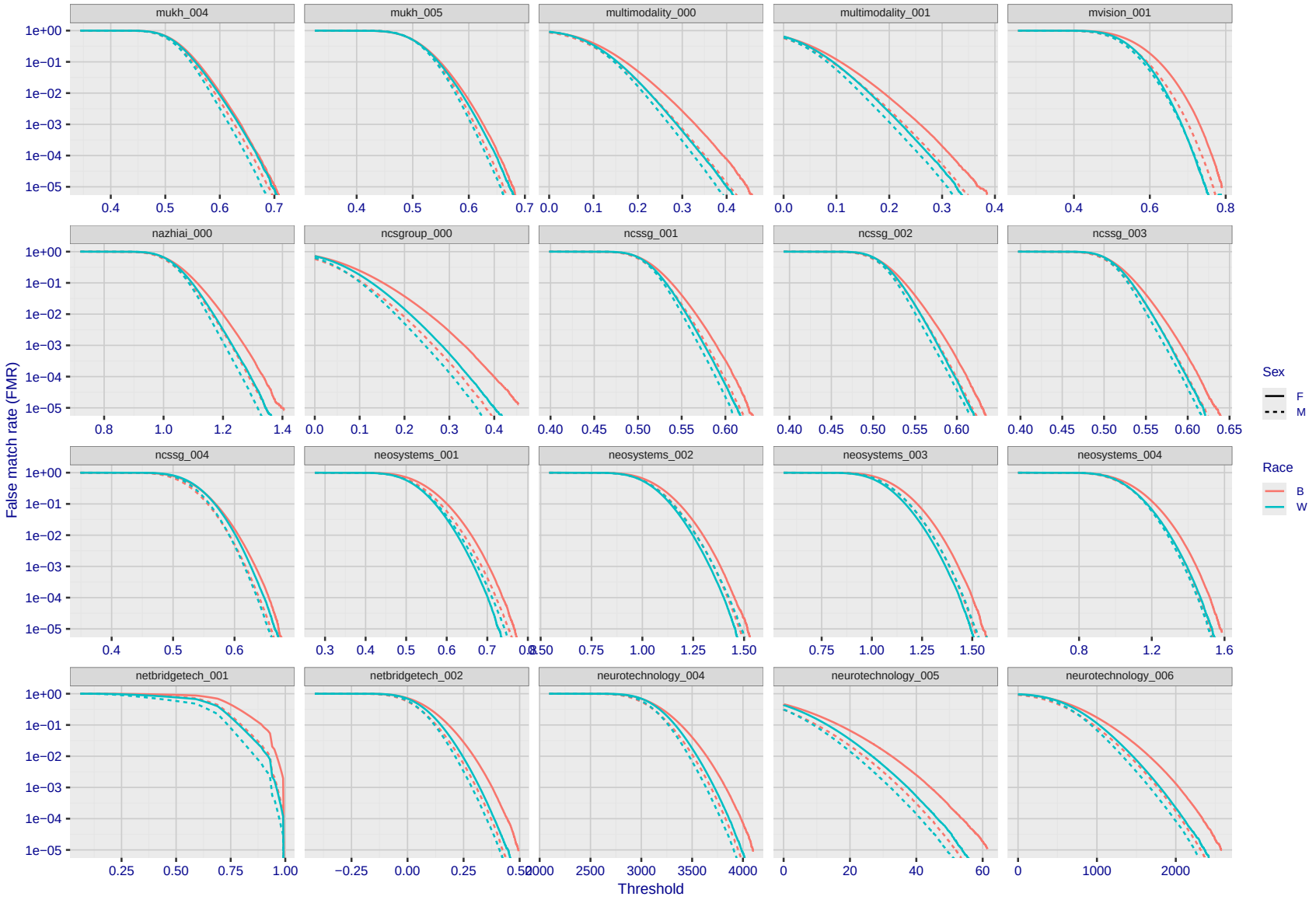

Figure 592: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

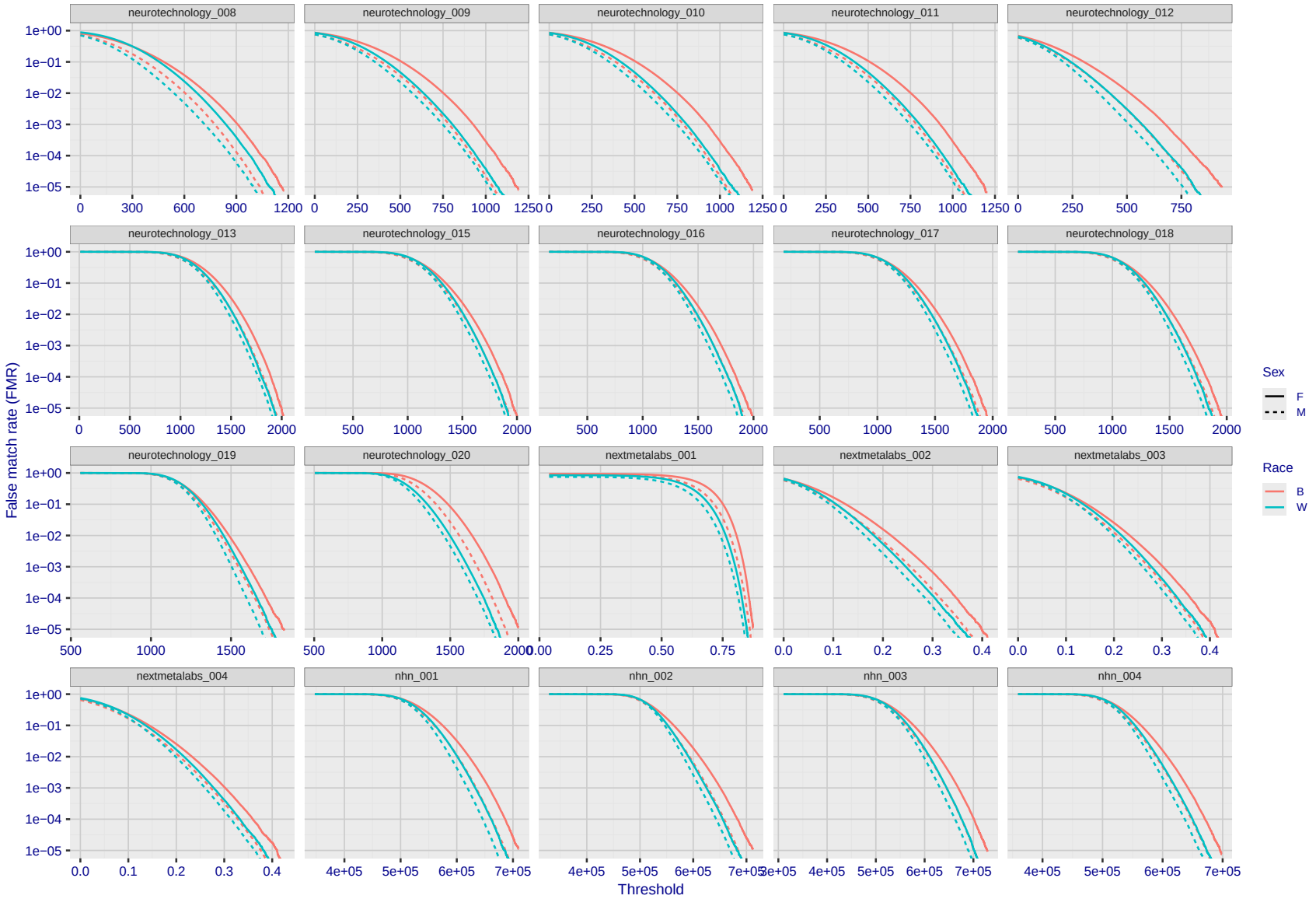

Figure 593: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

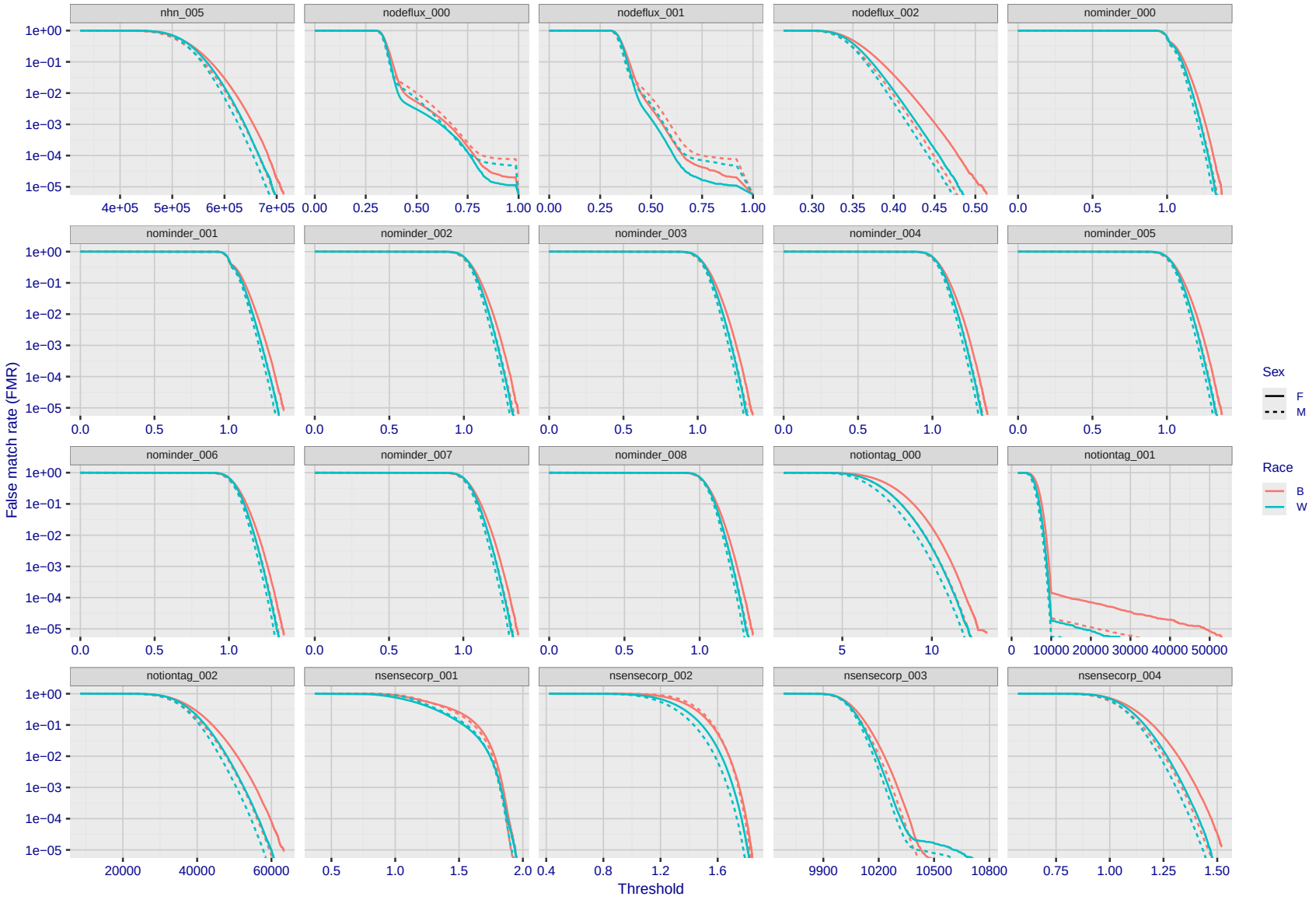

Figure 594: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

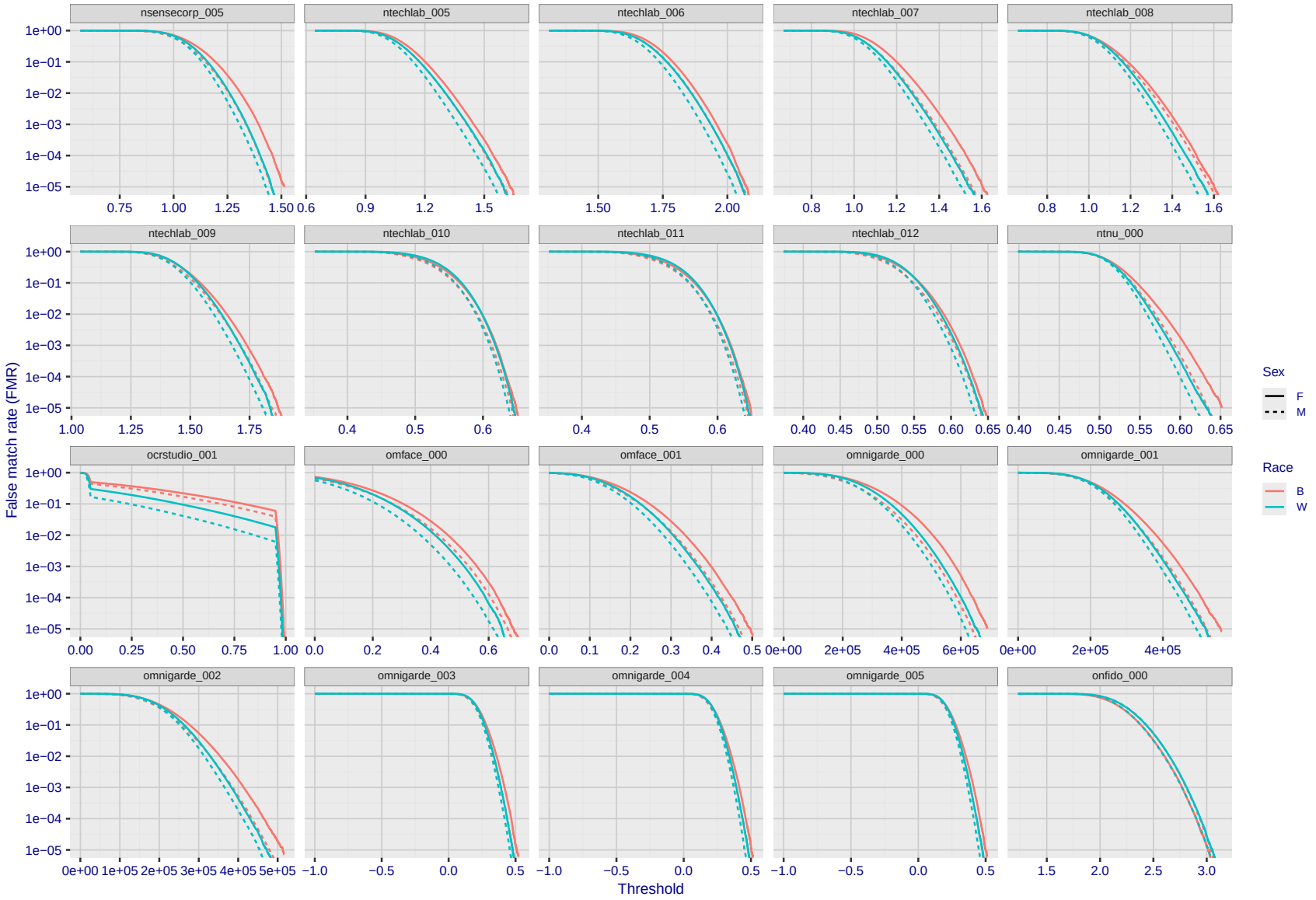

Figure 595: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

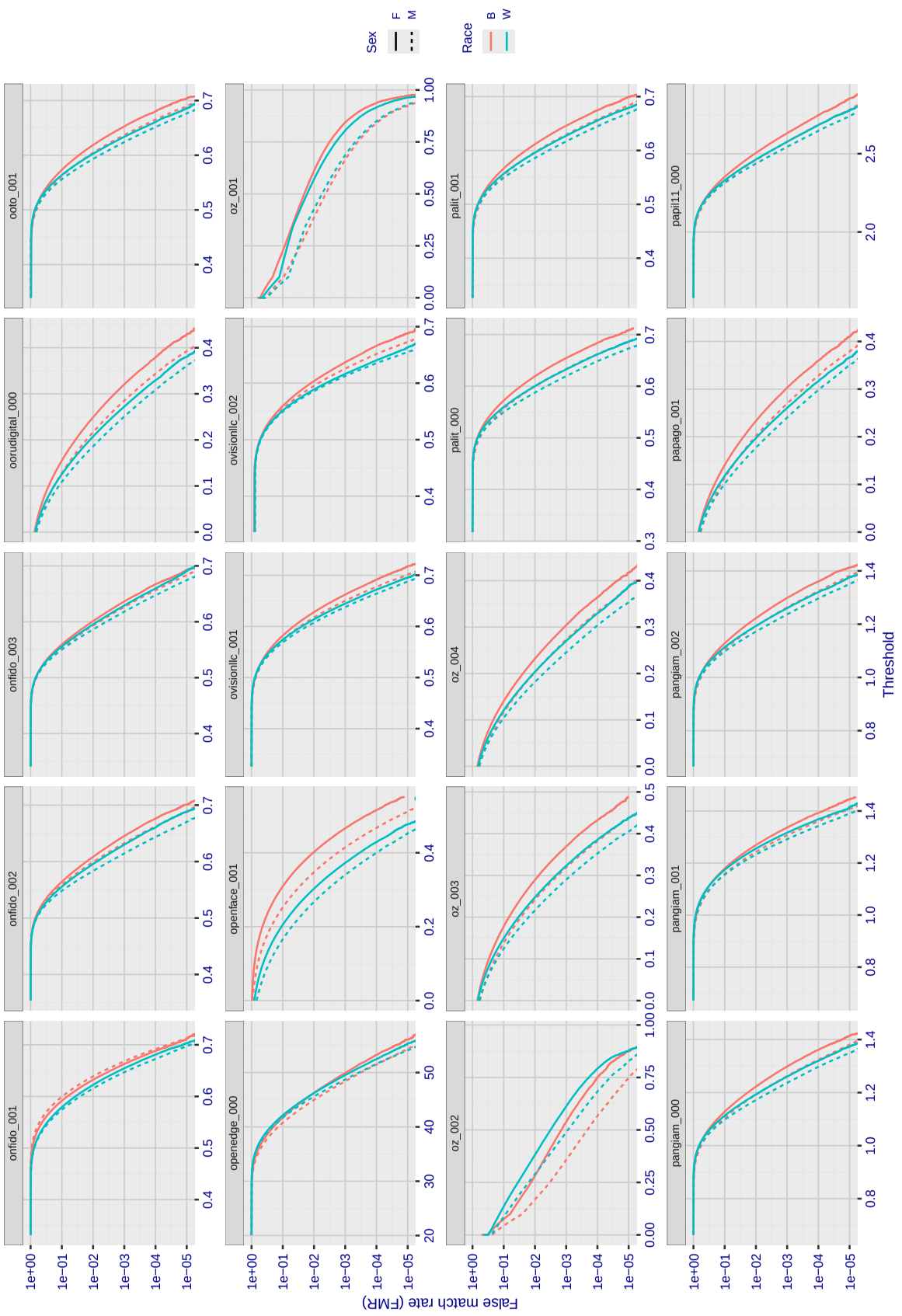

Figure 596: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

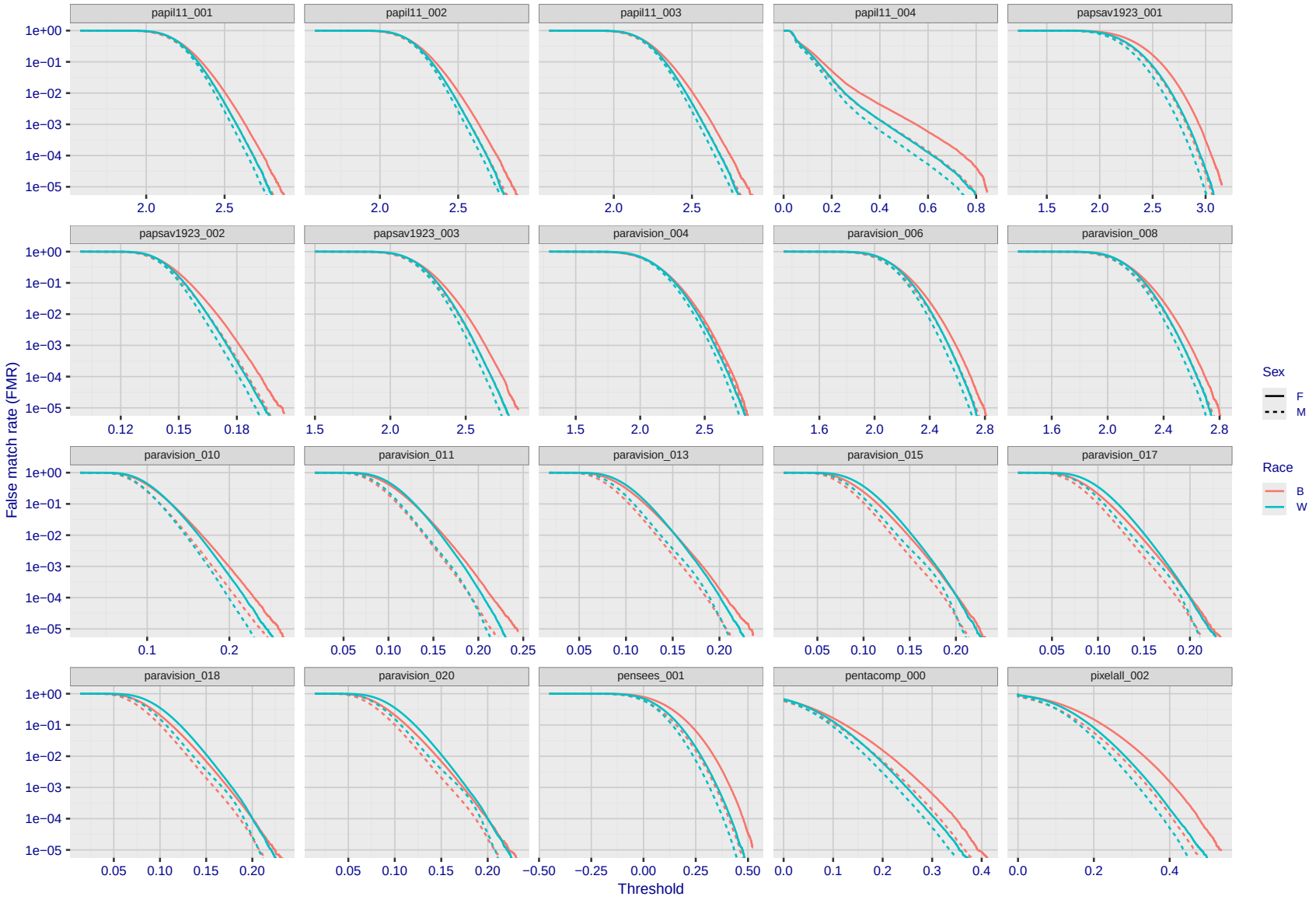

Figure 597: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

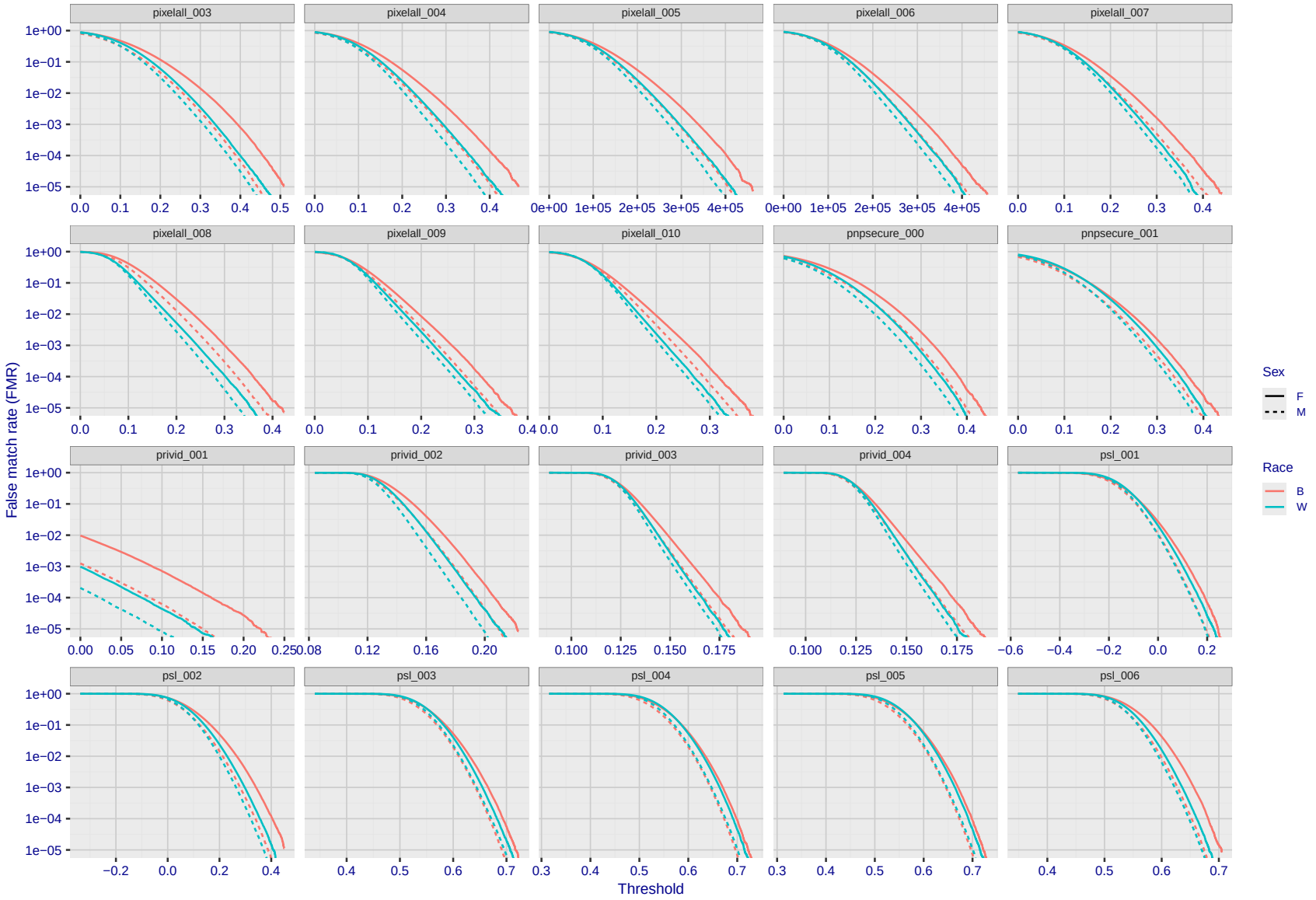

Figure 598: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

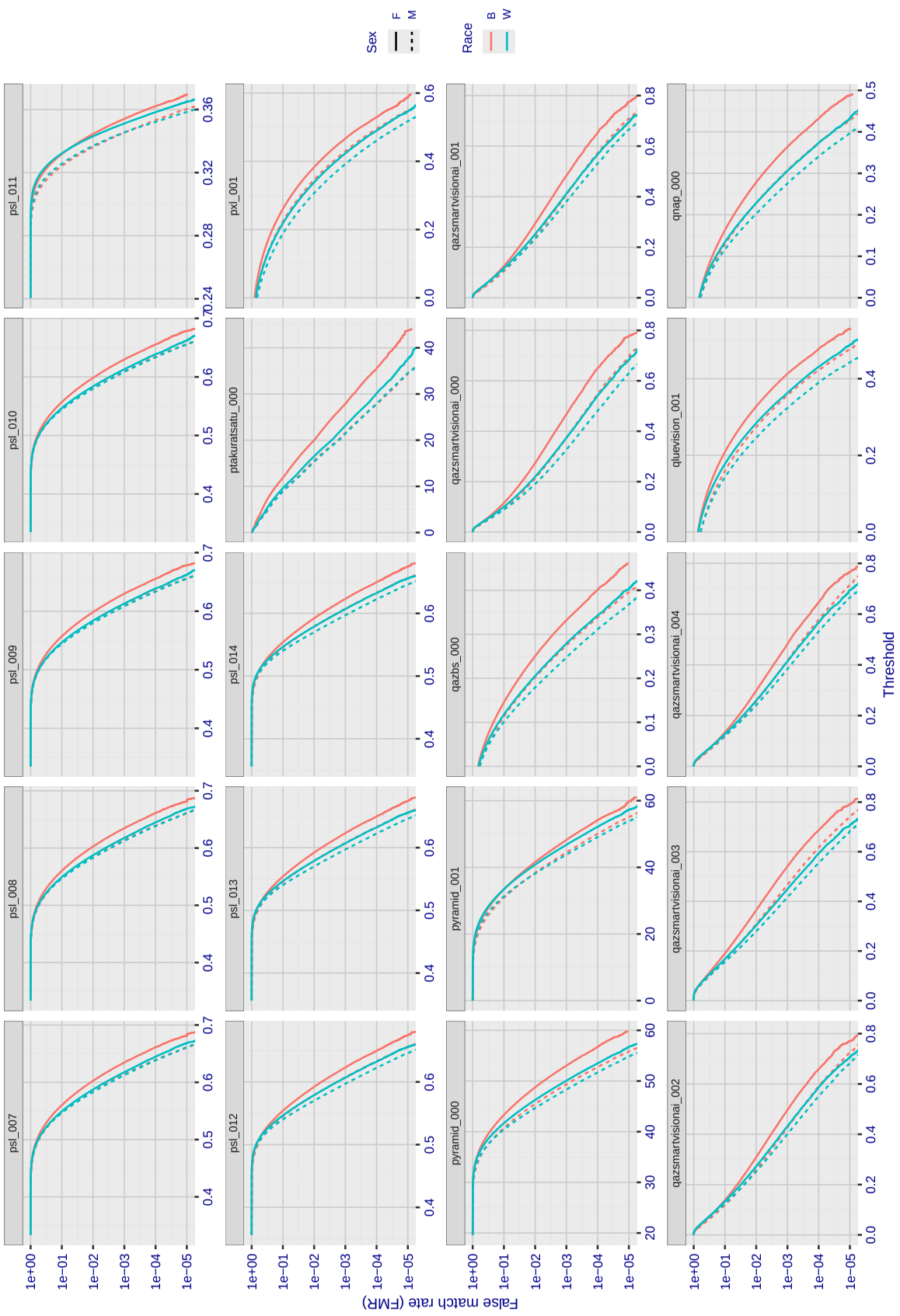

Figure 599: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

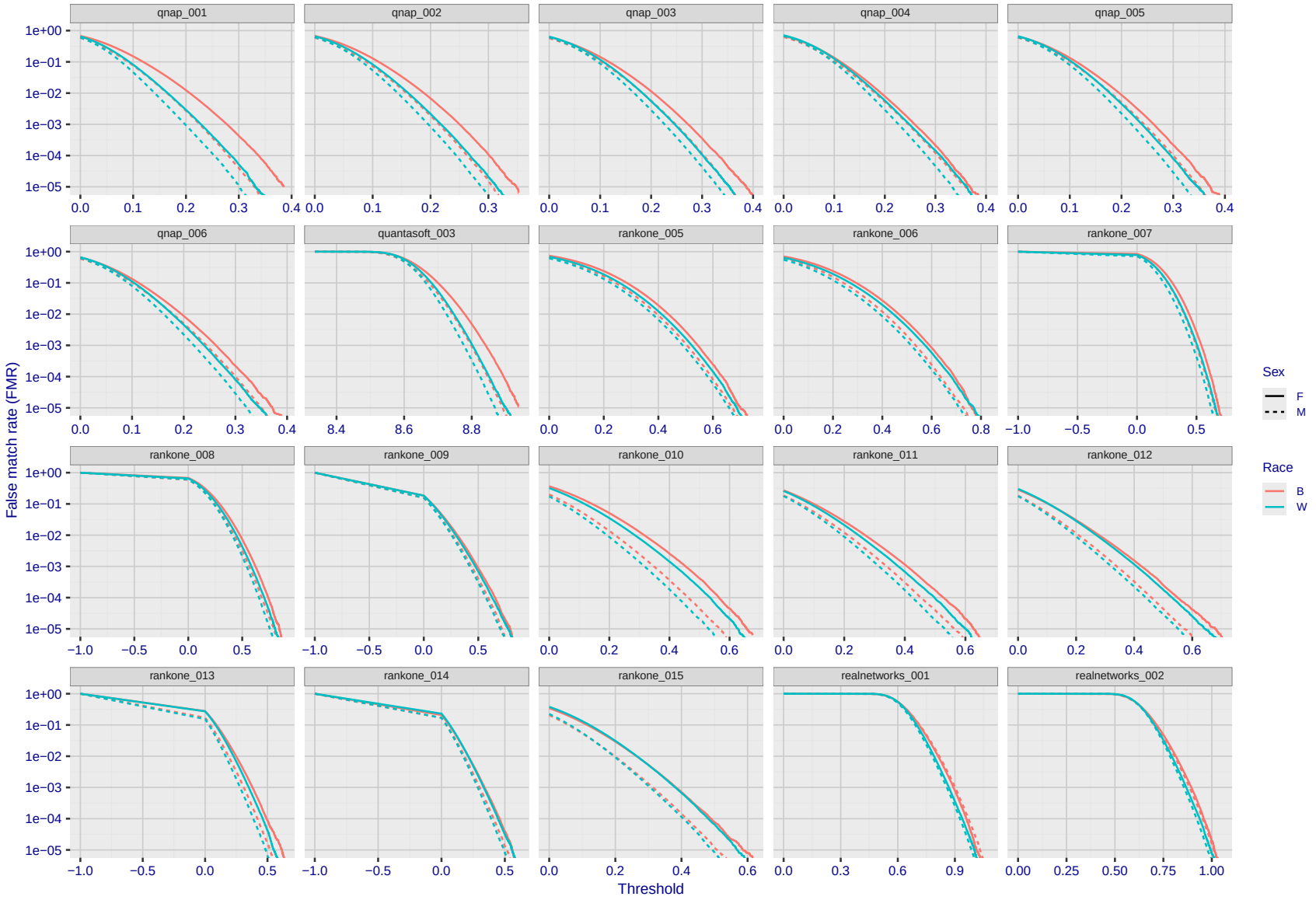

Figure 600: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

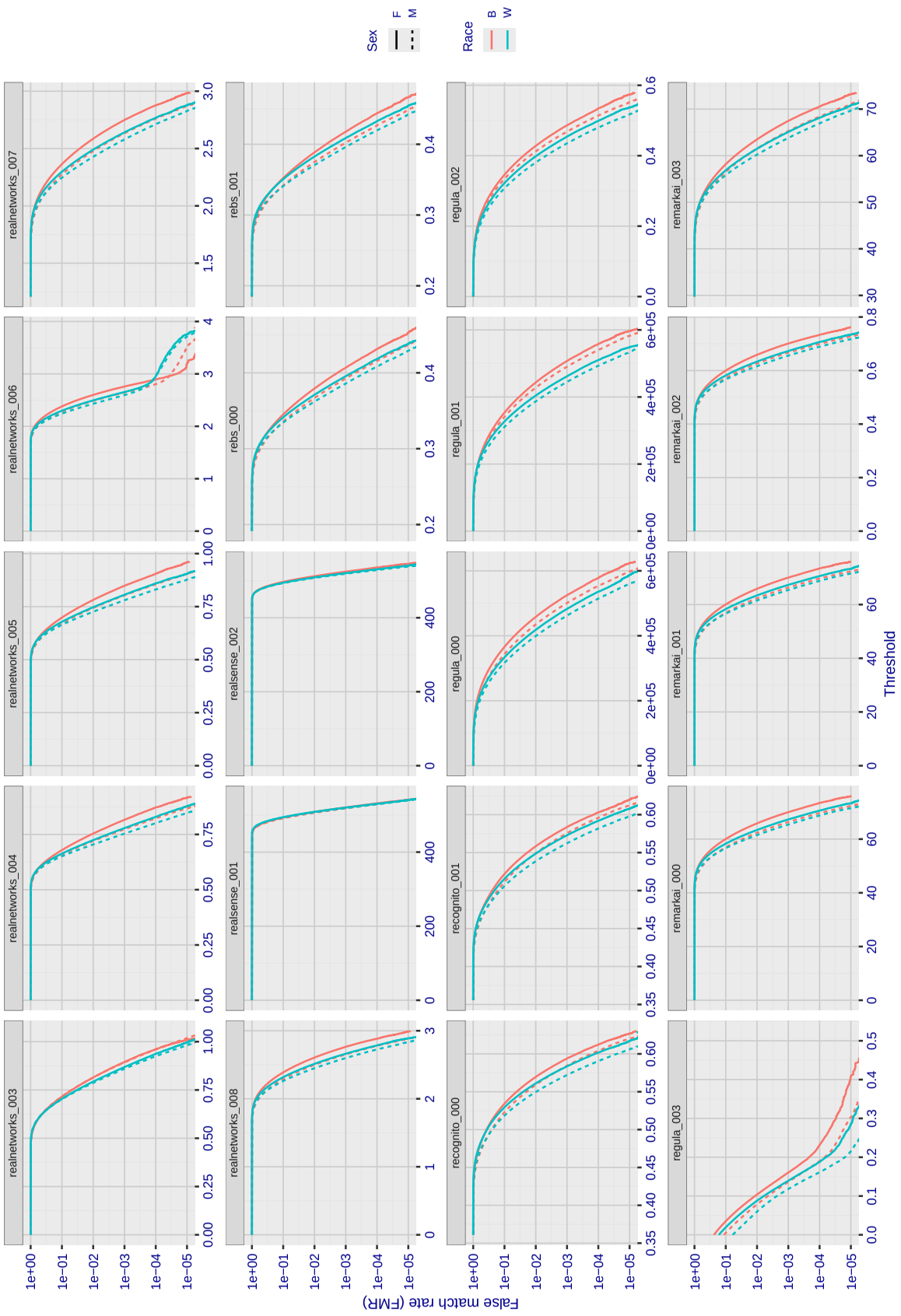

Figure 601: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

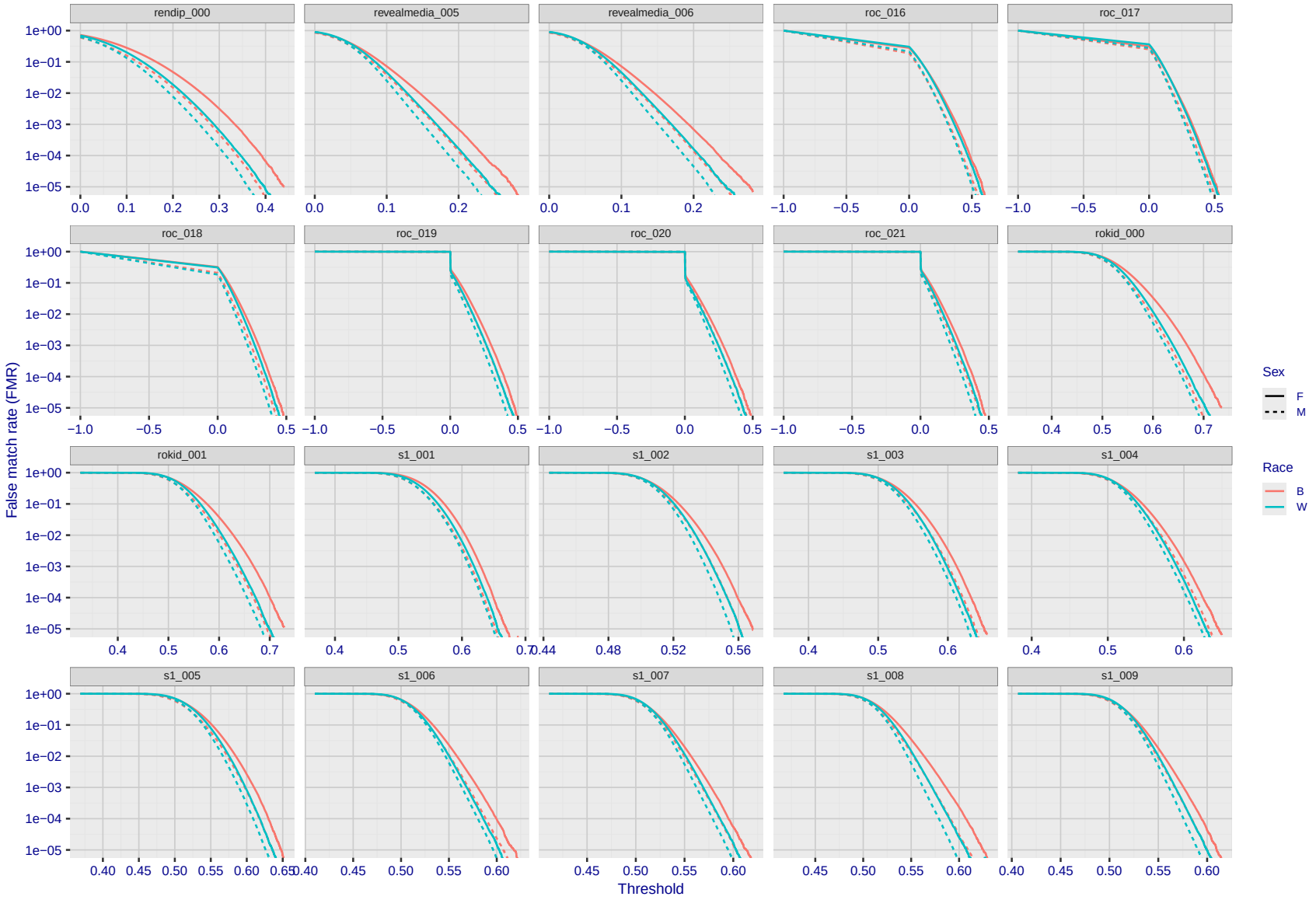

Figure 602: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

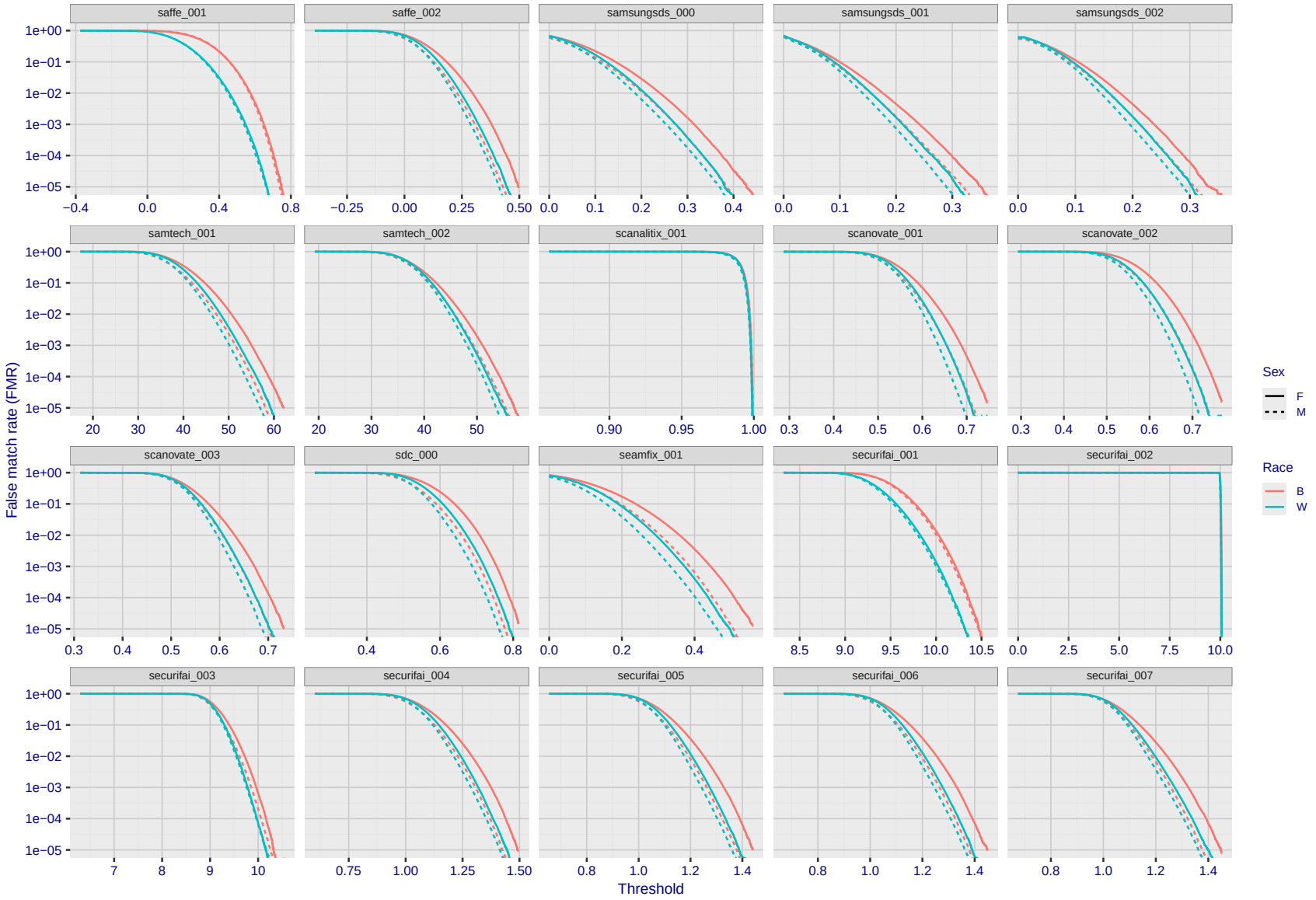

Figure 603: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

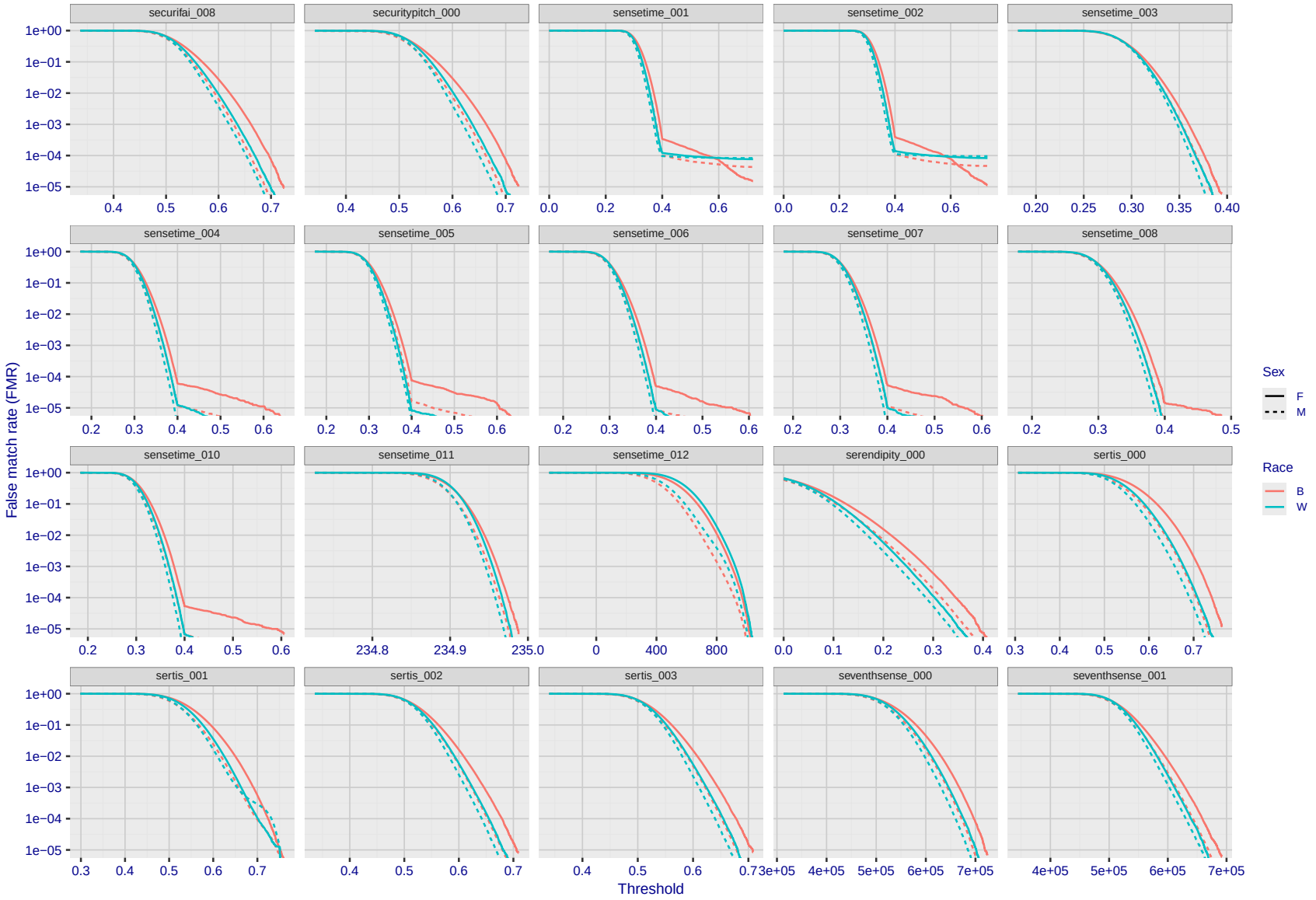

Figure 604: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

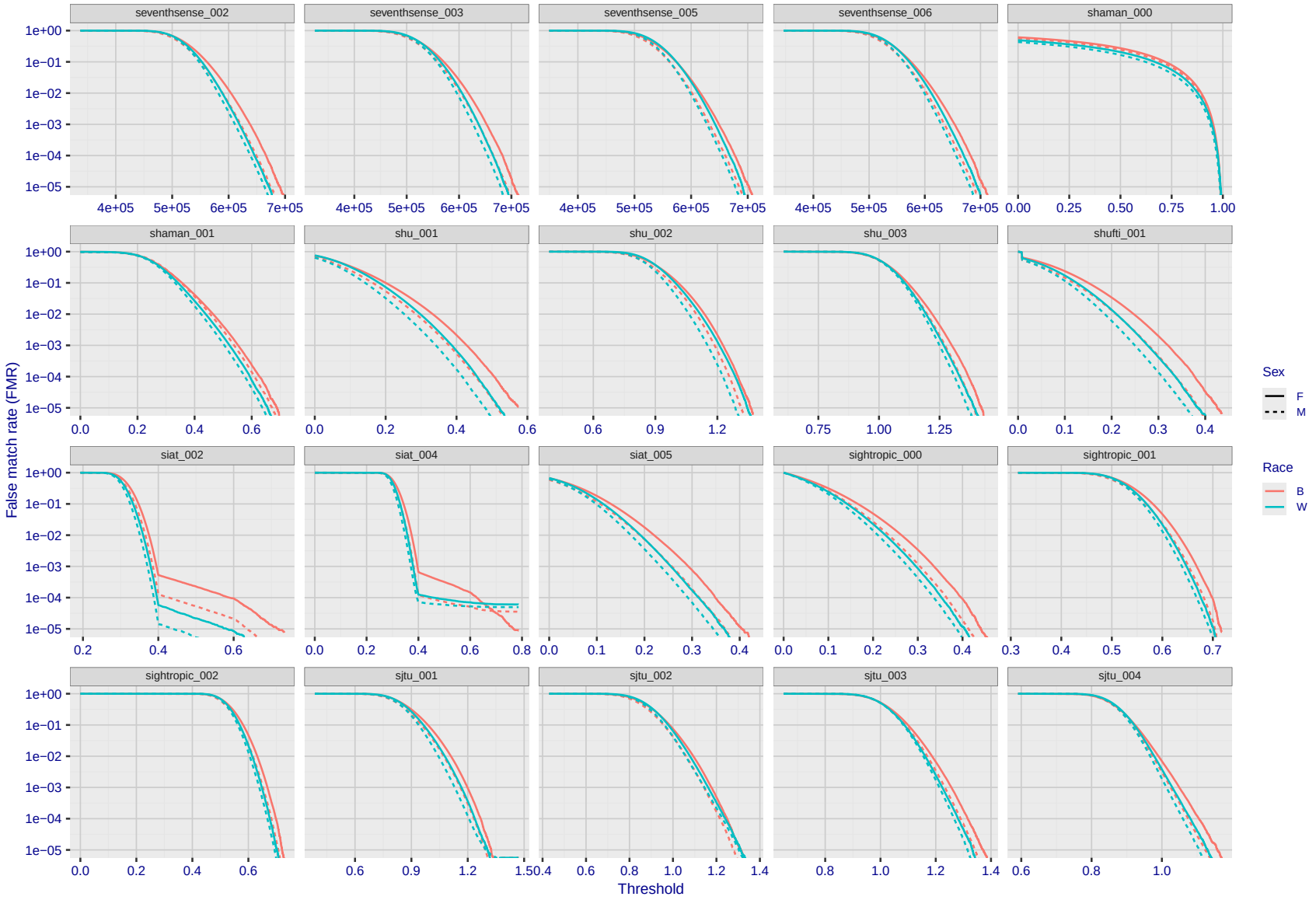

Figure 605: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

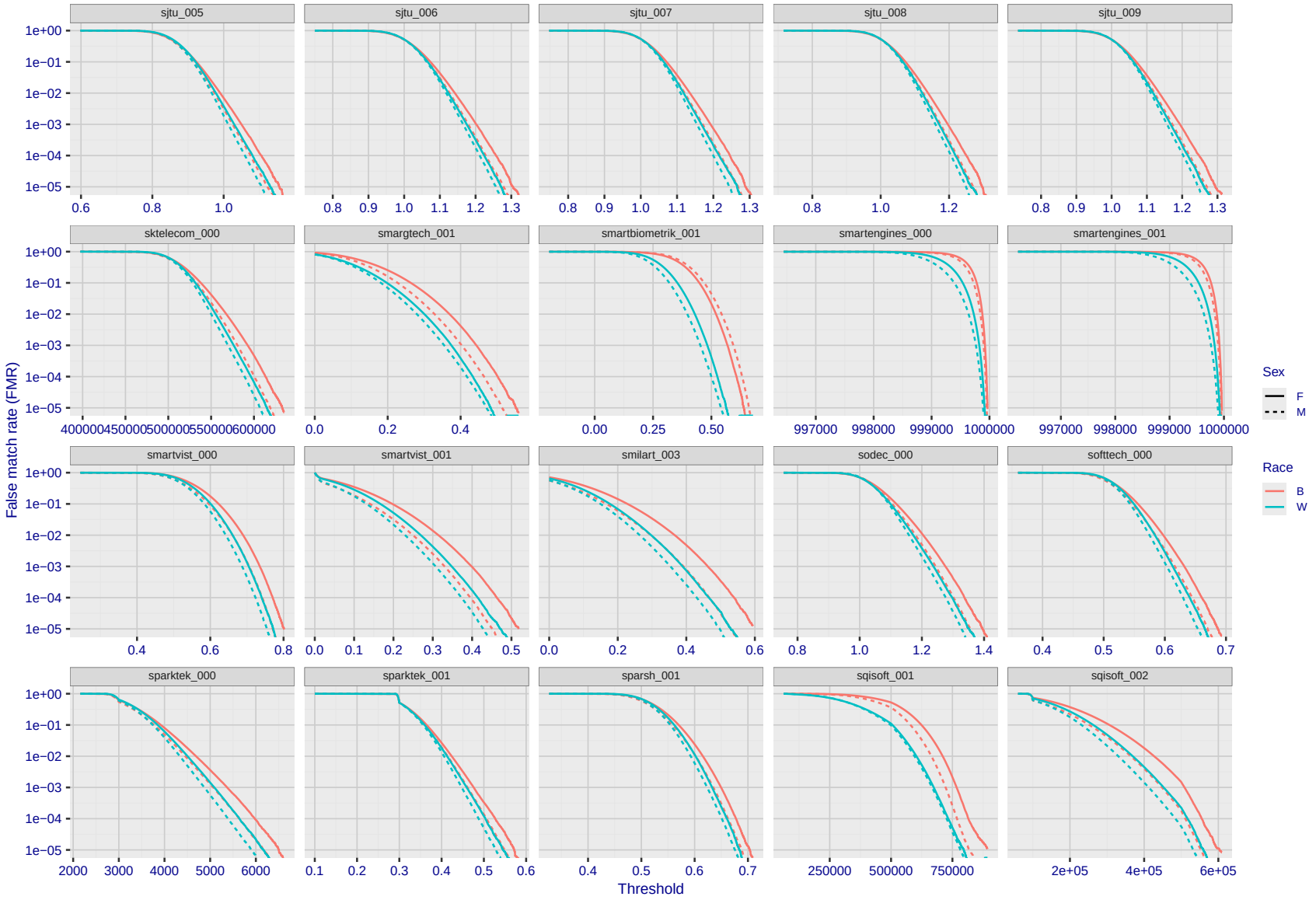

Figure 606: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

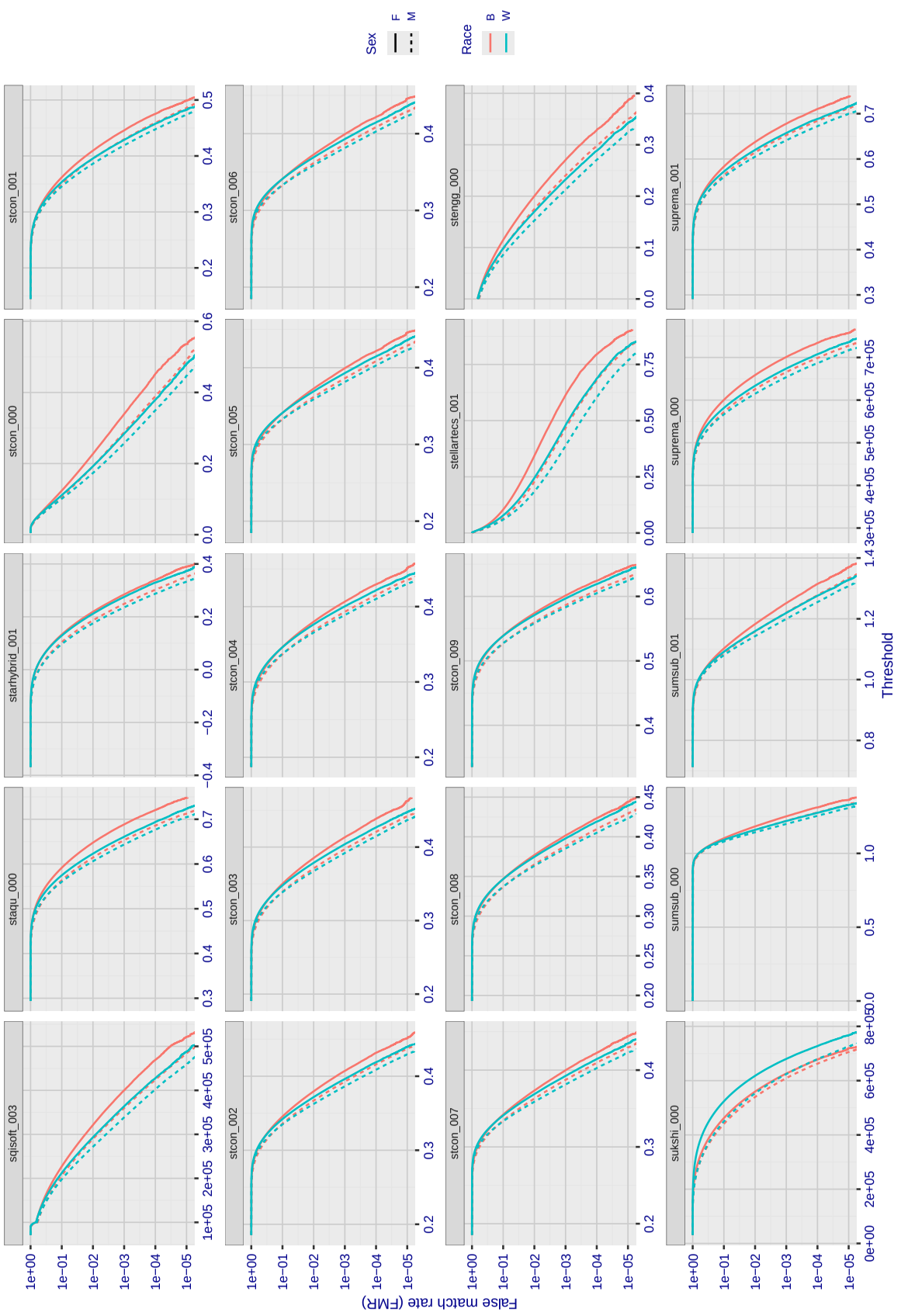

Figure 607: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

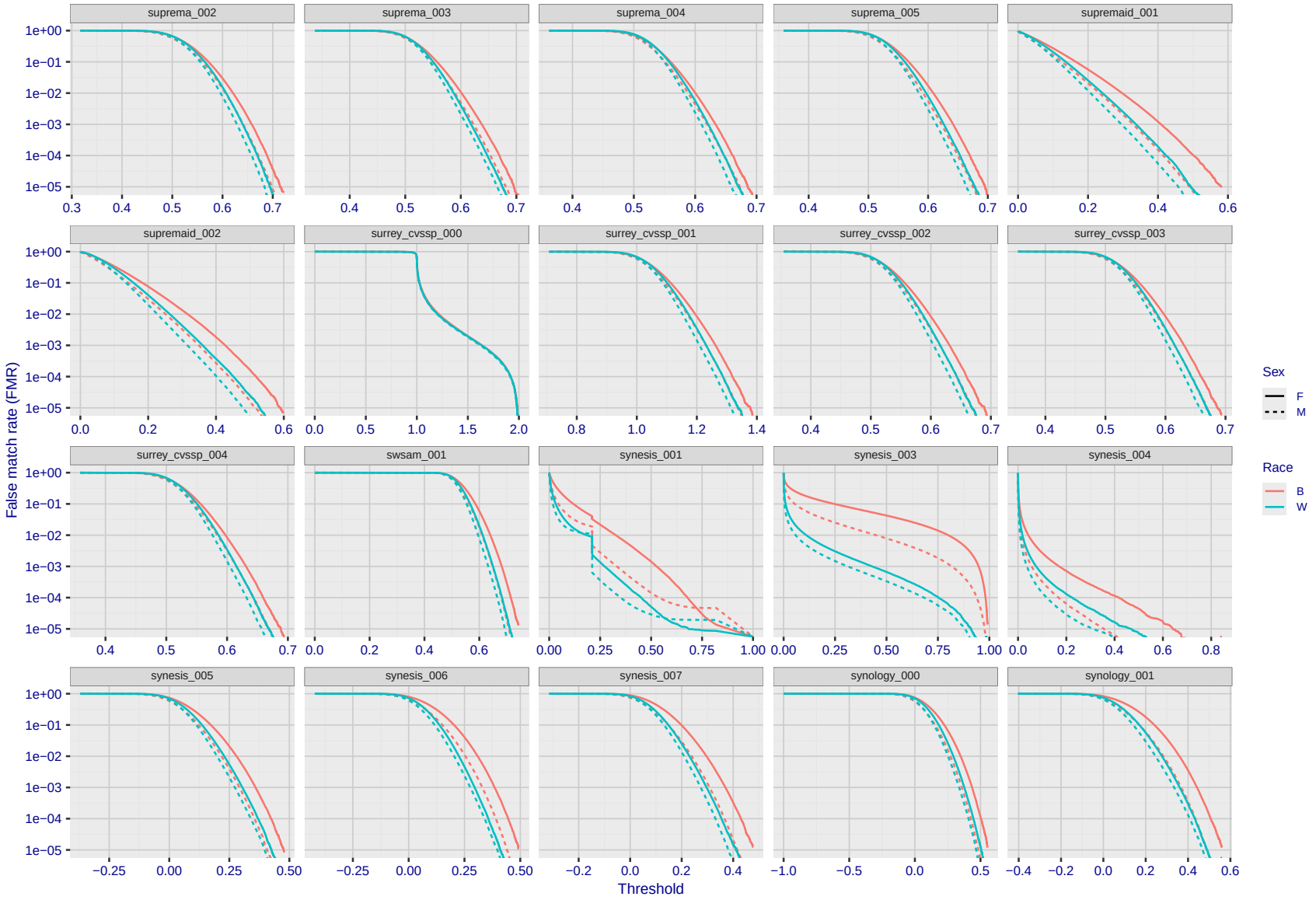

Figure 608: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

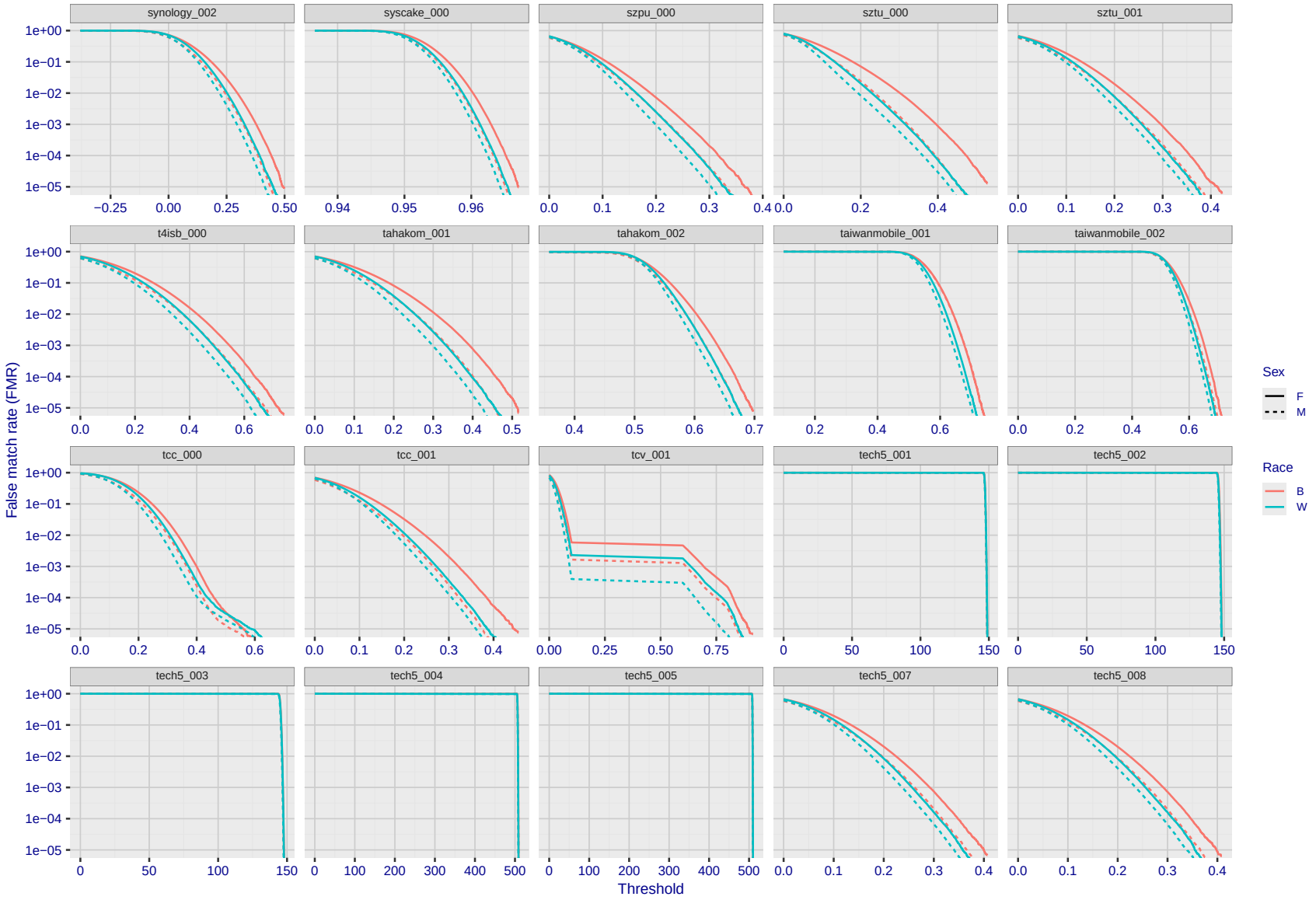

Figure 609: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

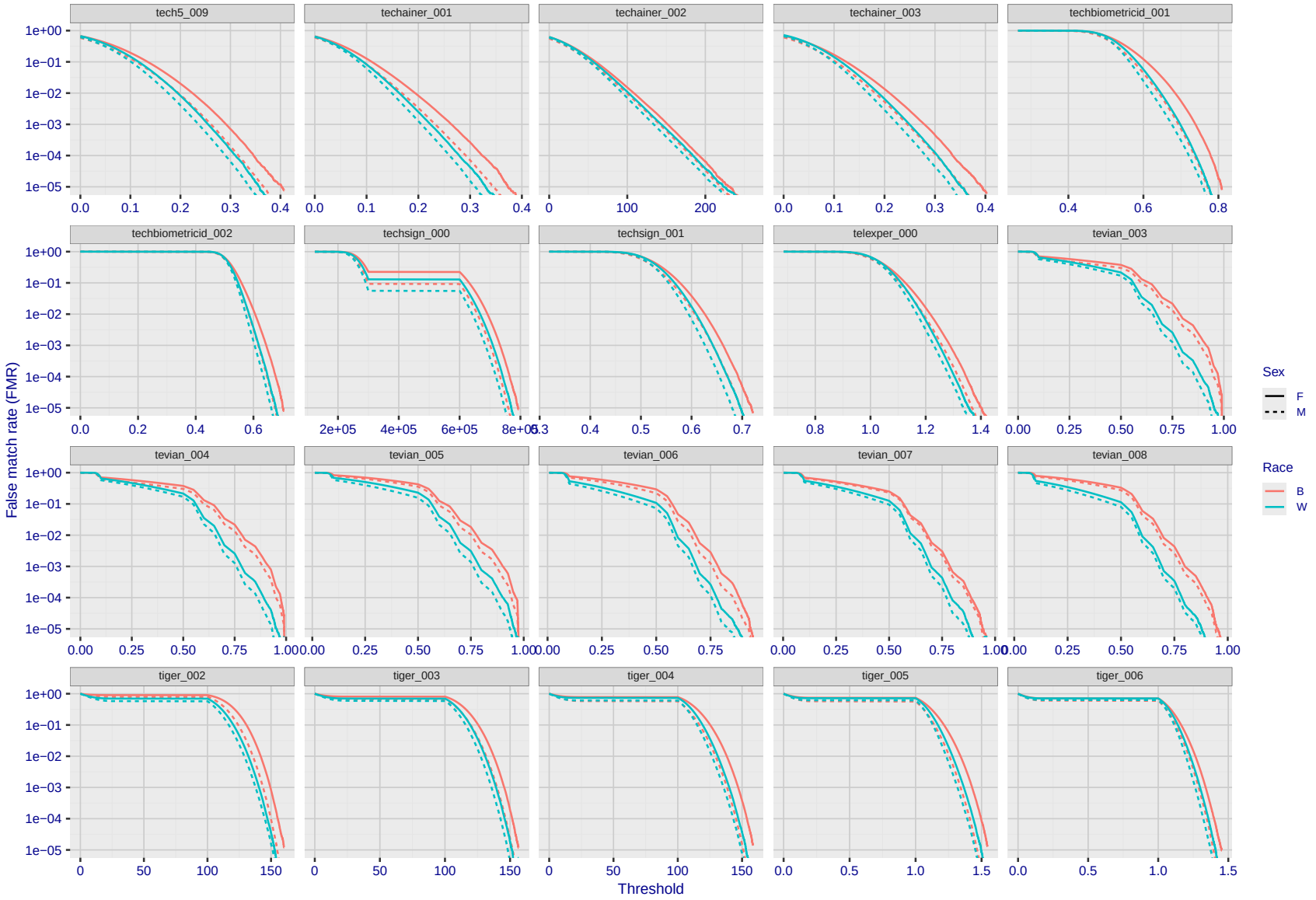

Figure 610: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

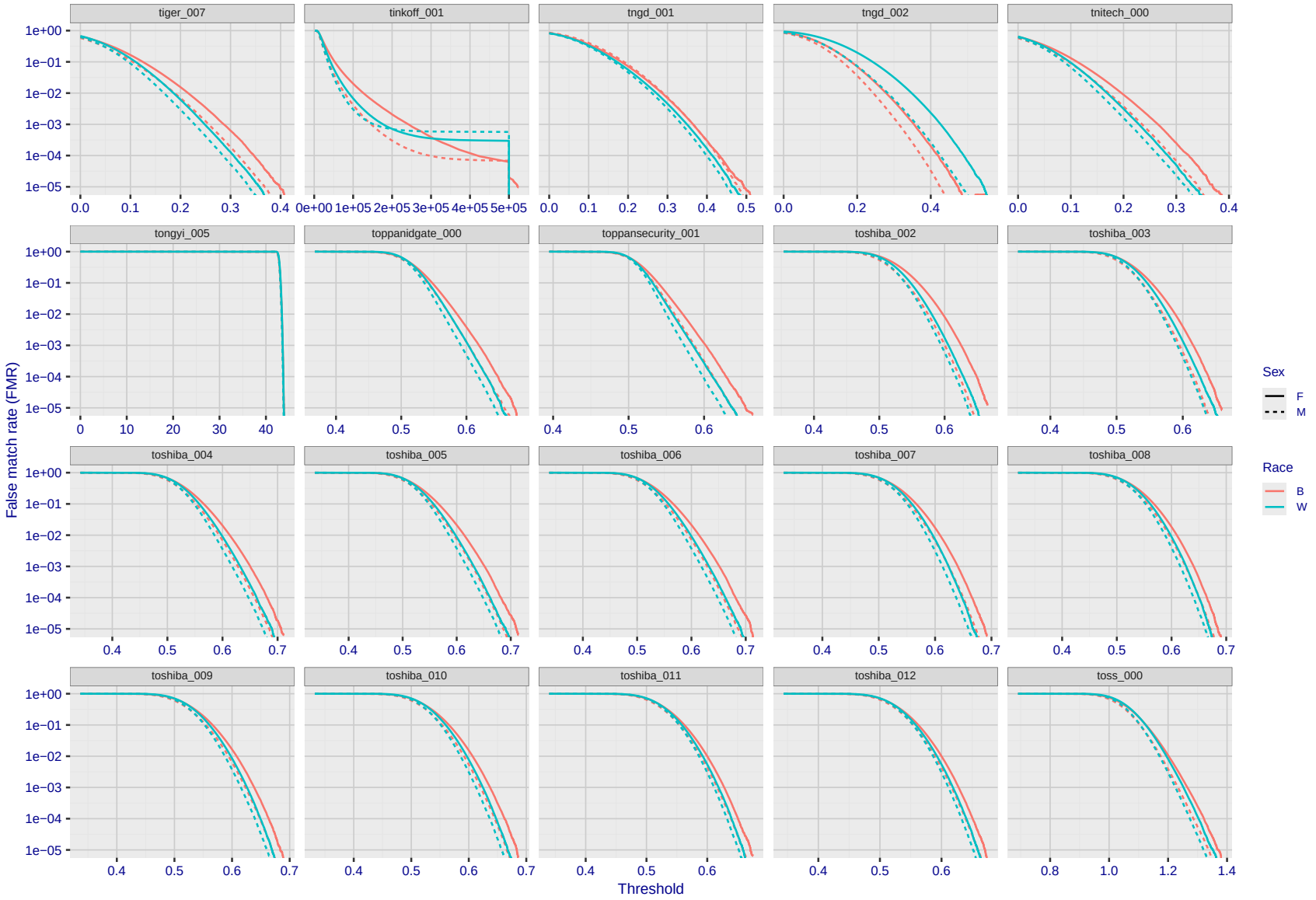

Figure 611: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

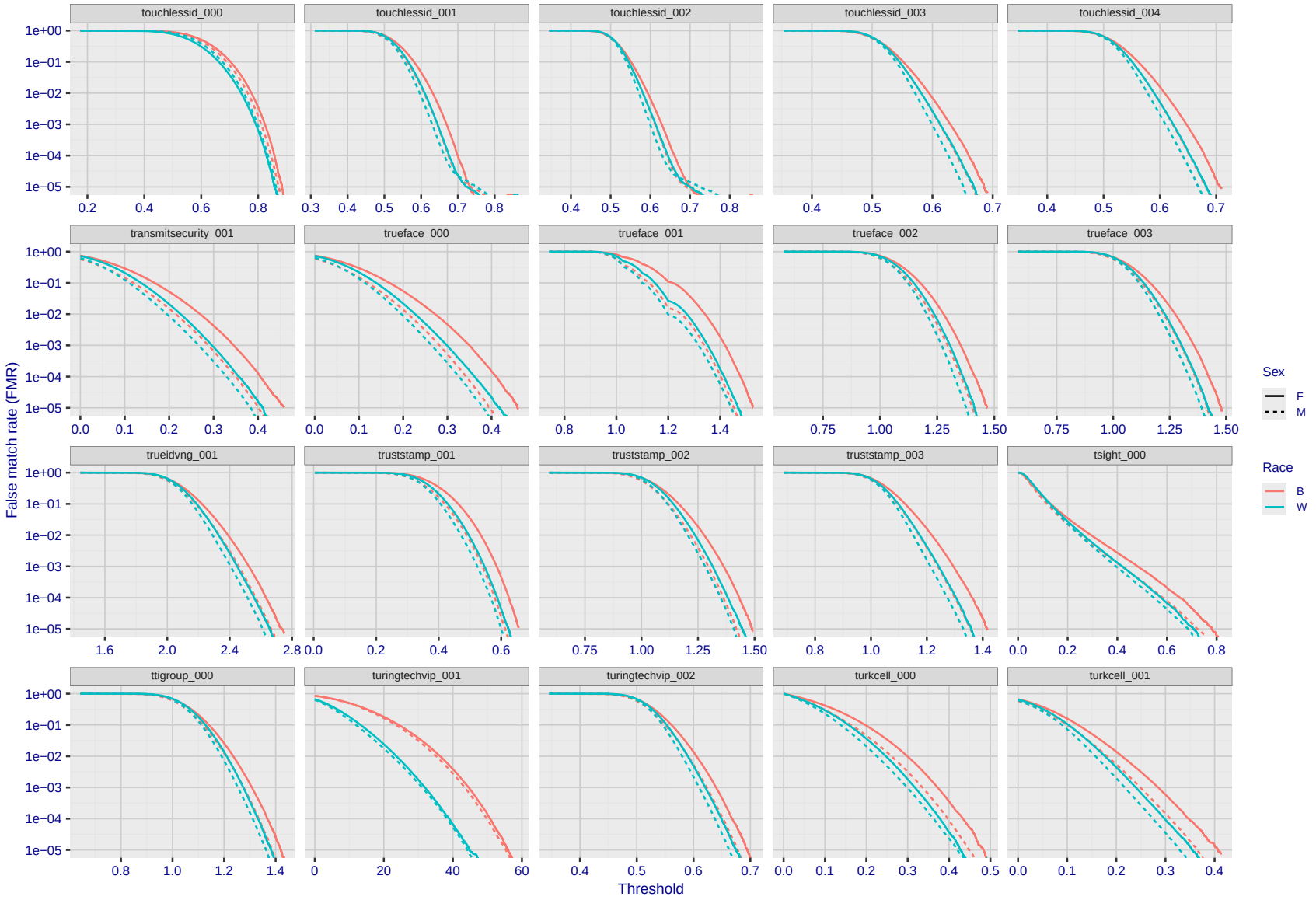

Figure 612: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

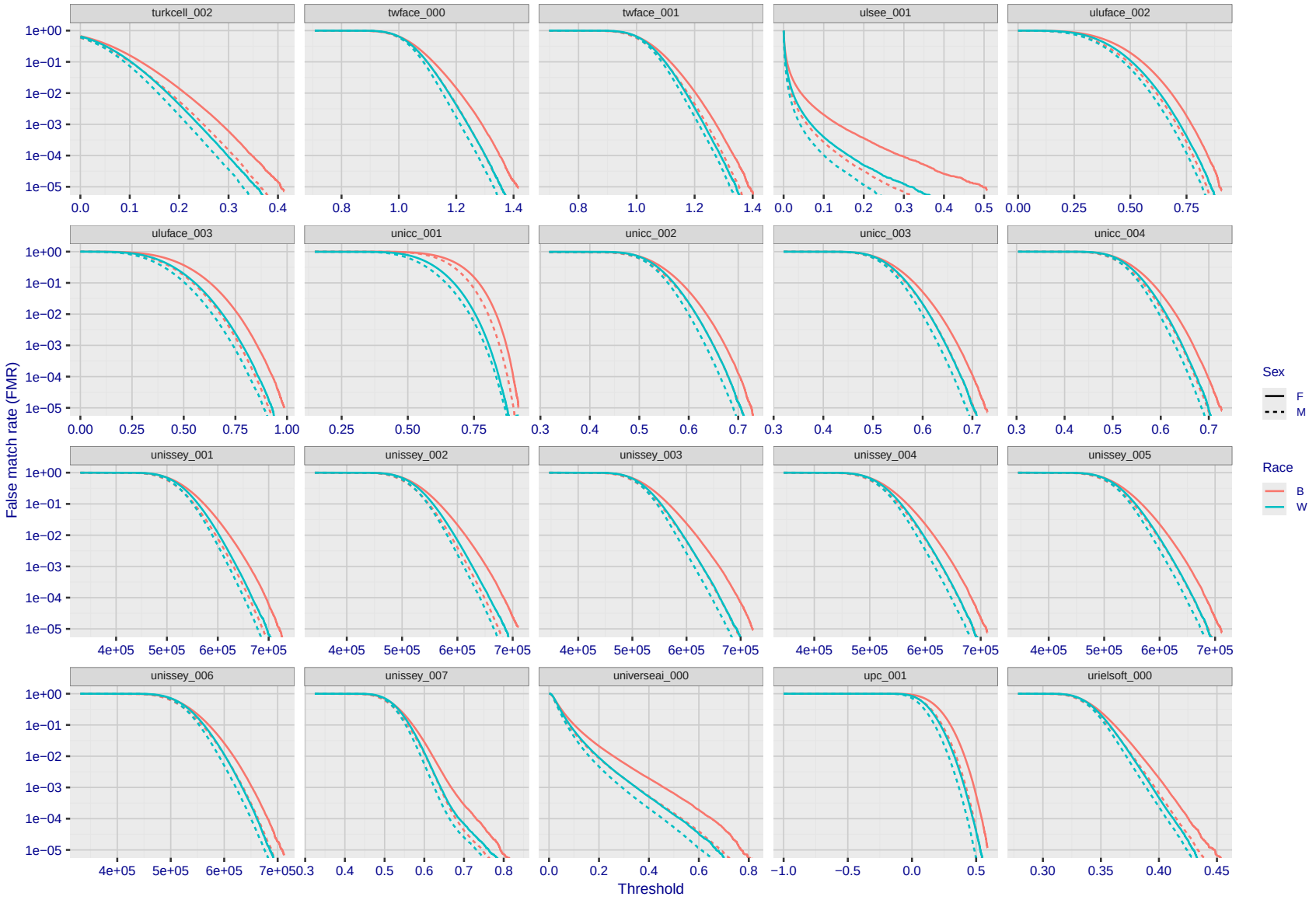

Figure 613: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

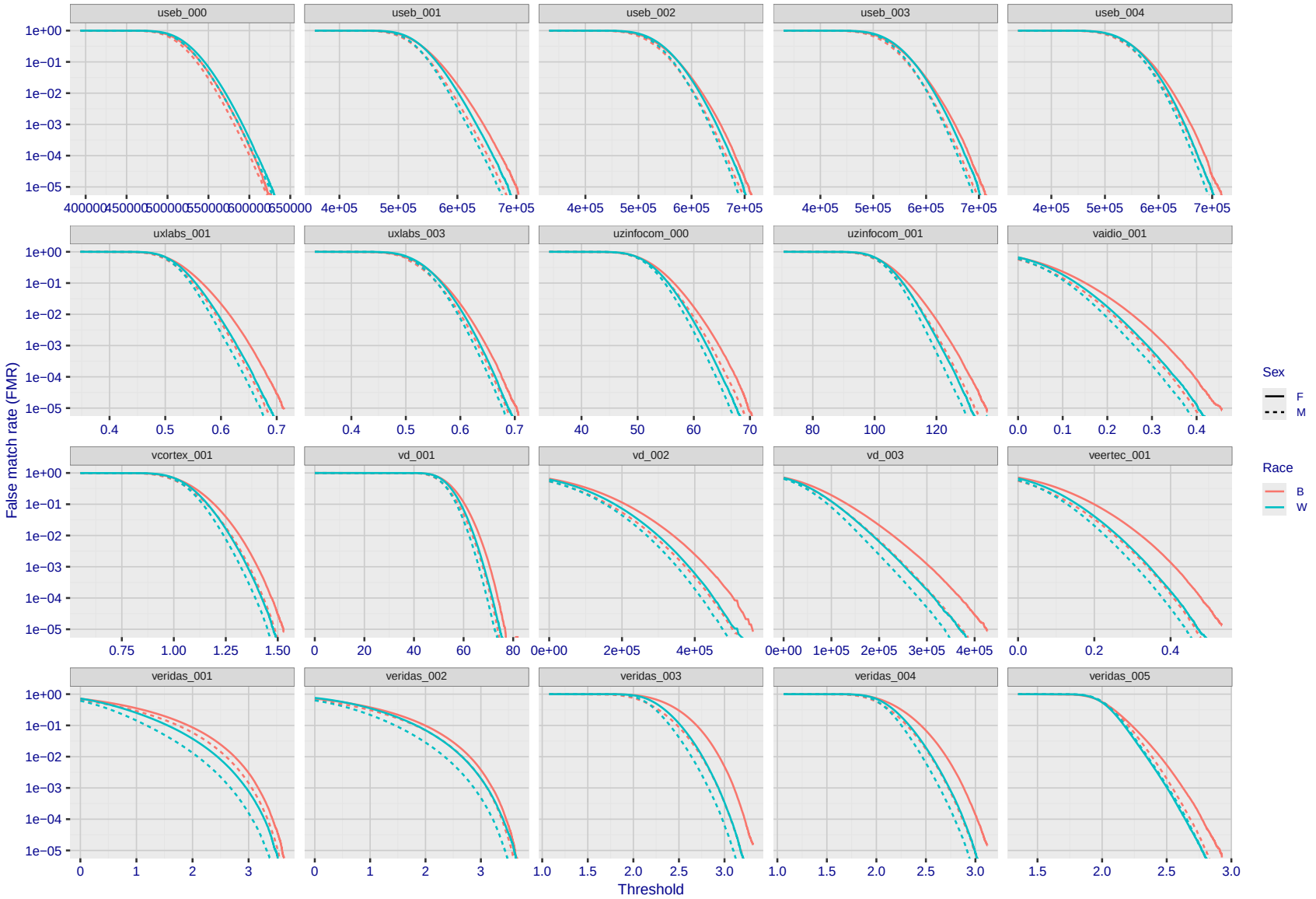

Figure 614: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

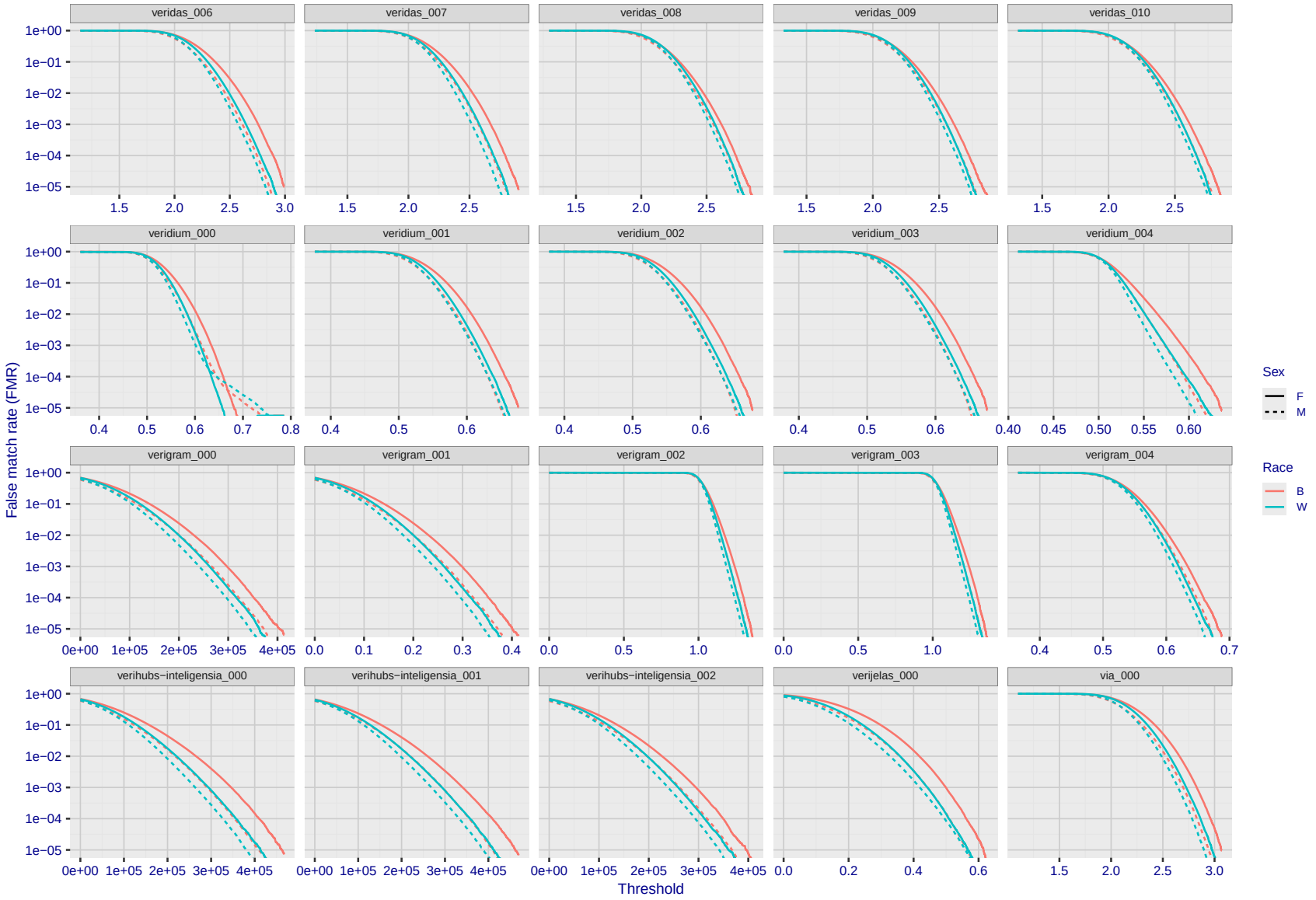

Figure 615: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

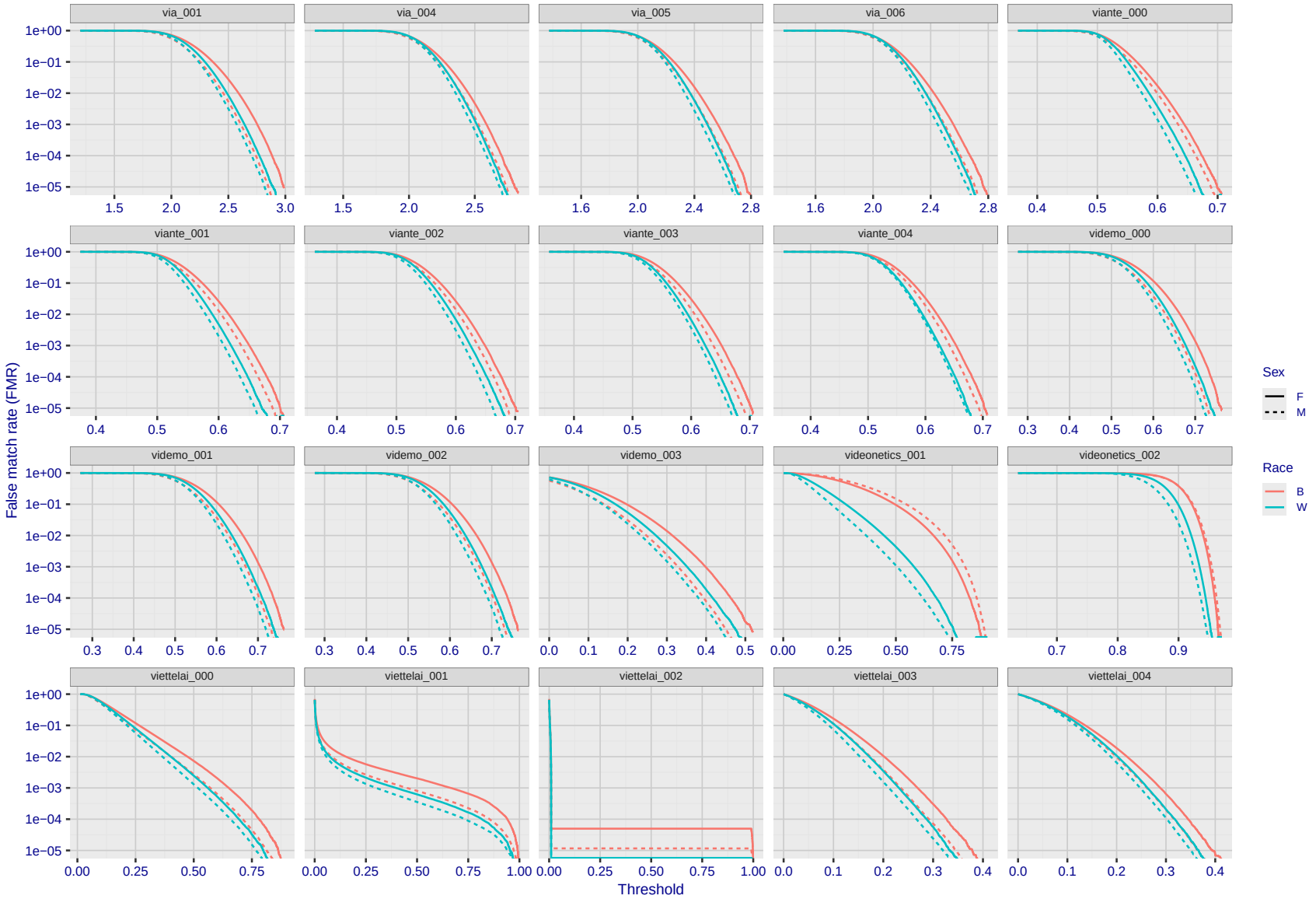

Figure 616: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

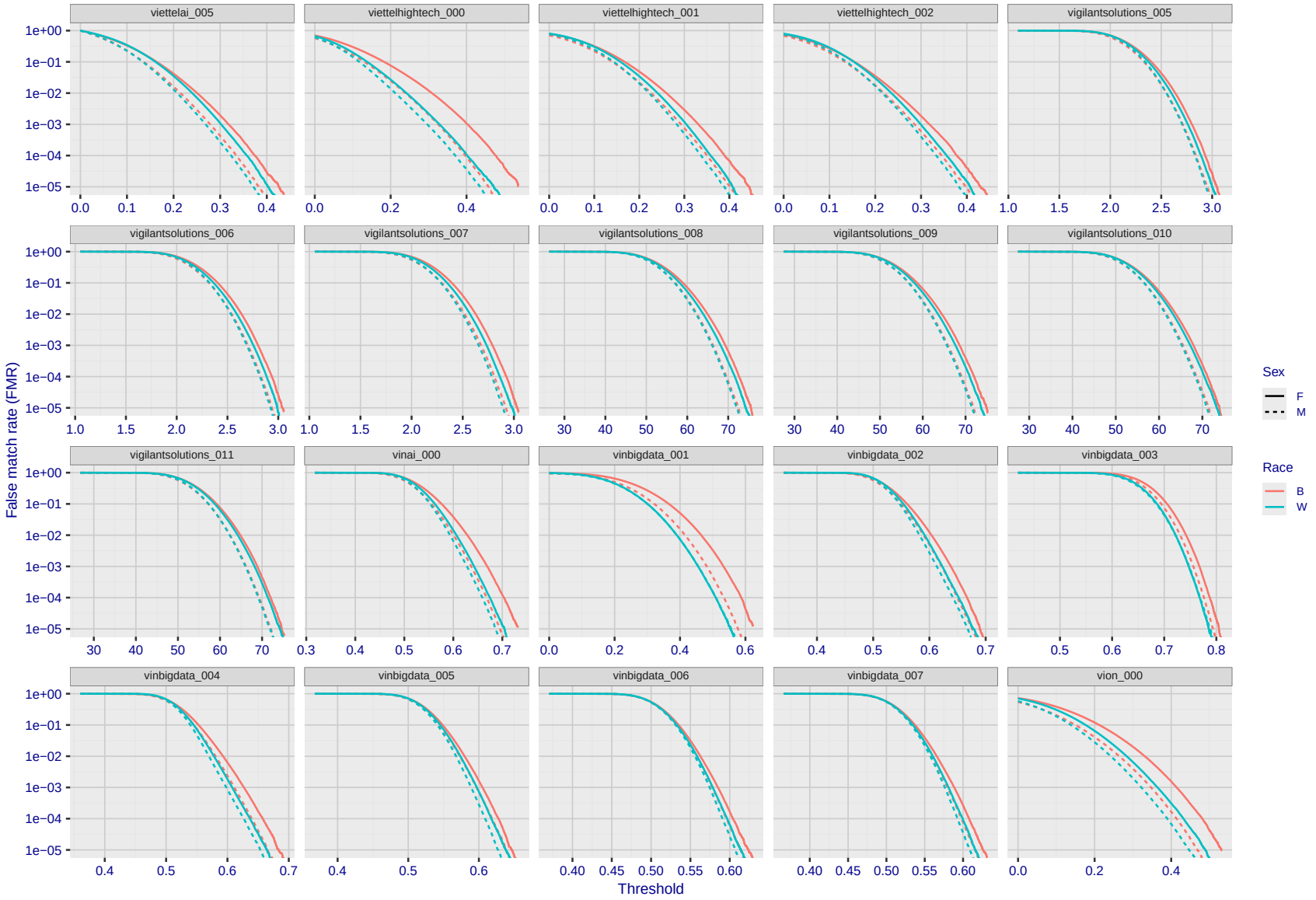

Figure 617: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

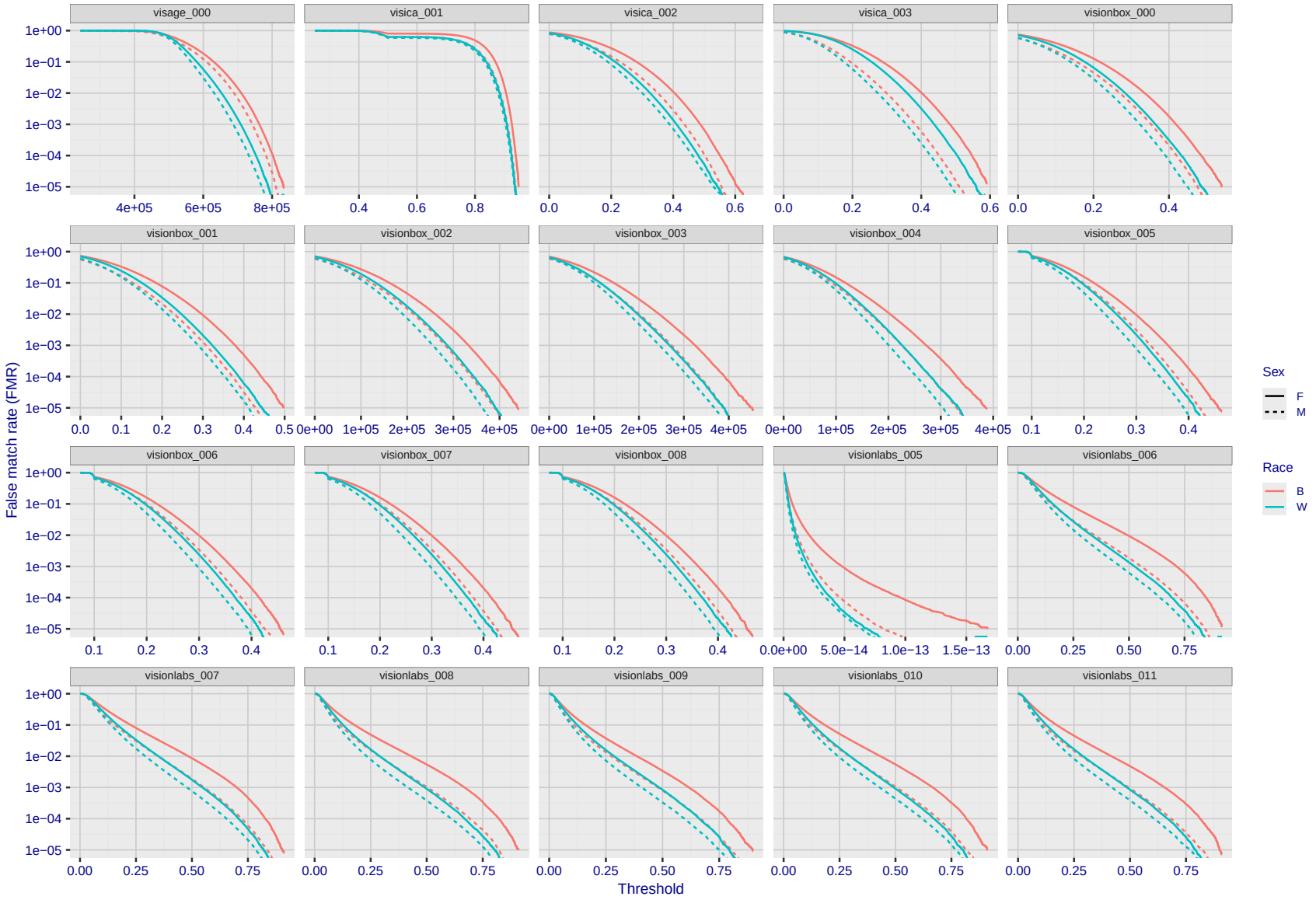

Figure 618: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

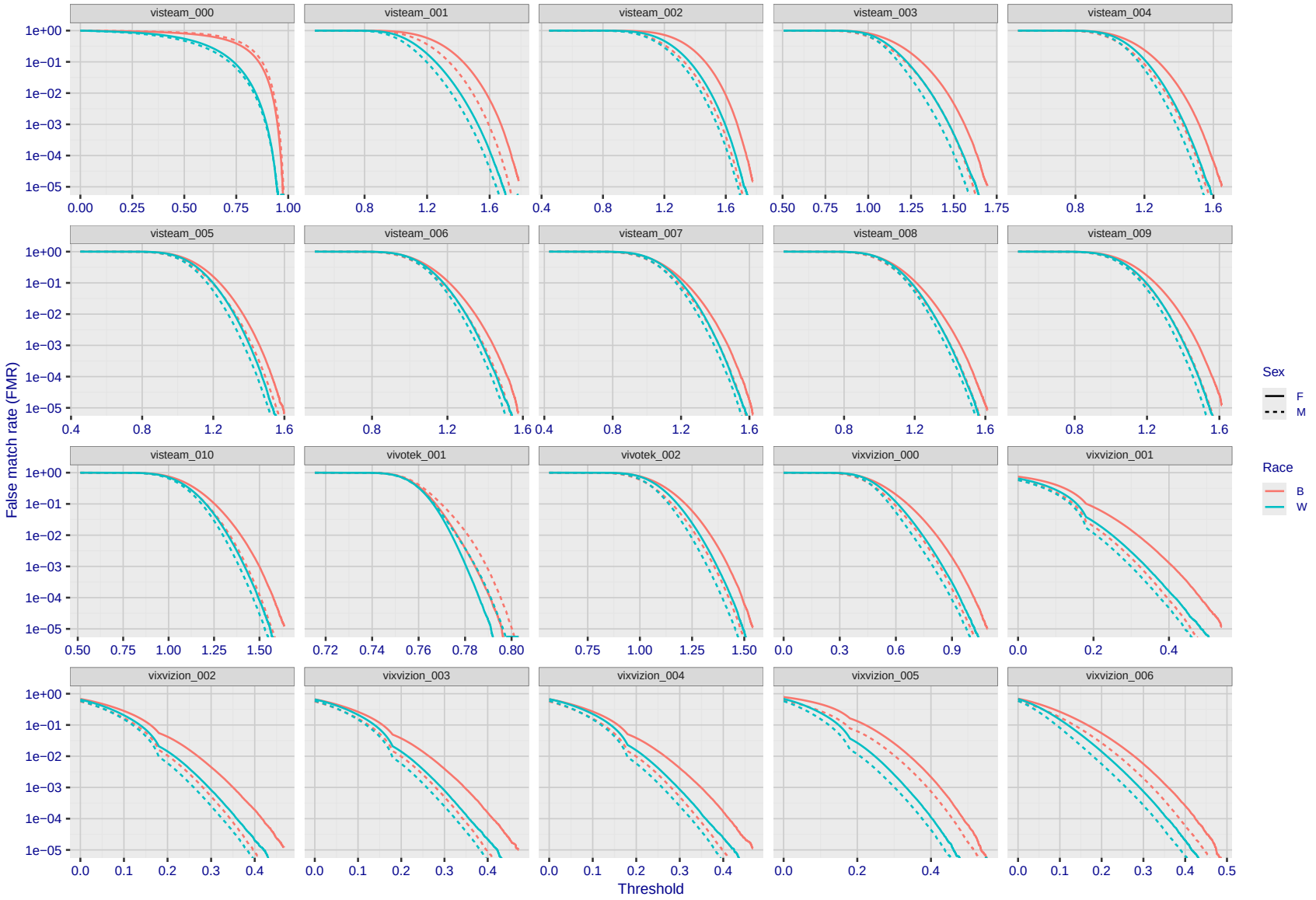

Figure 619: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

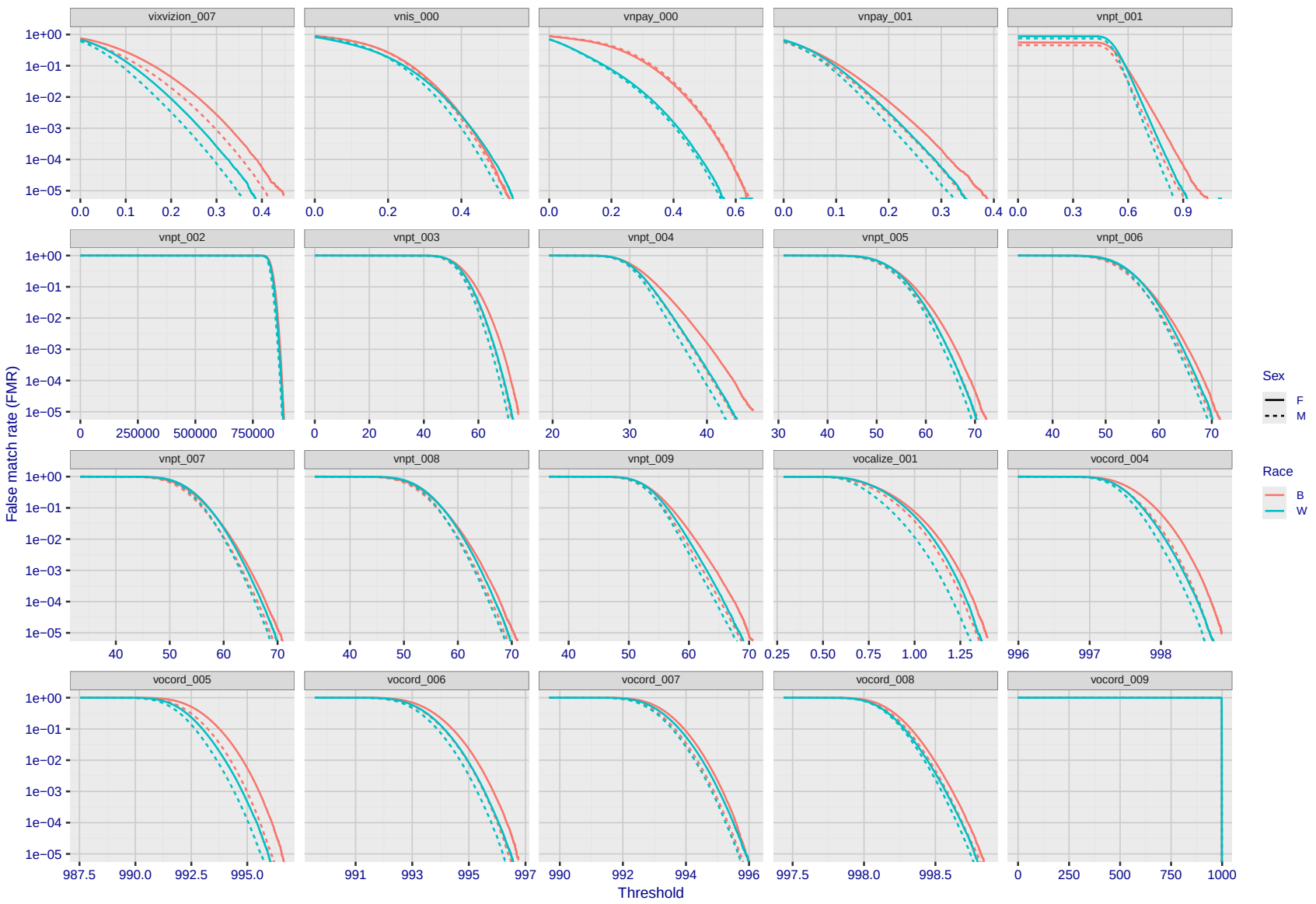

Figure 620: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

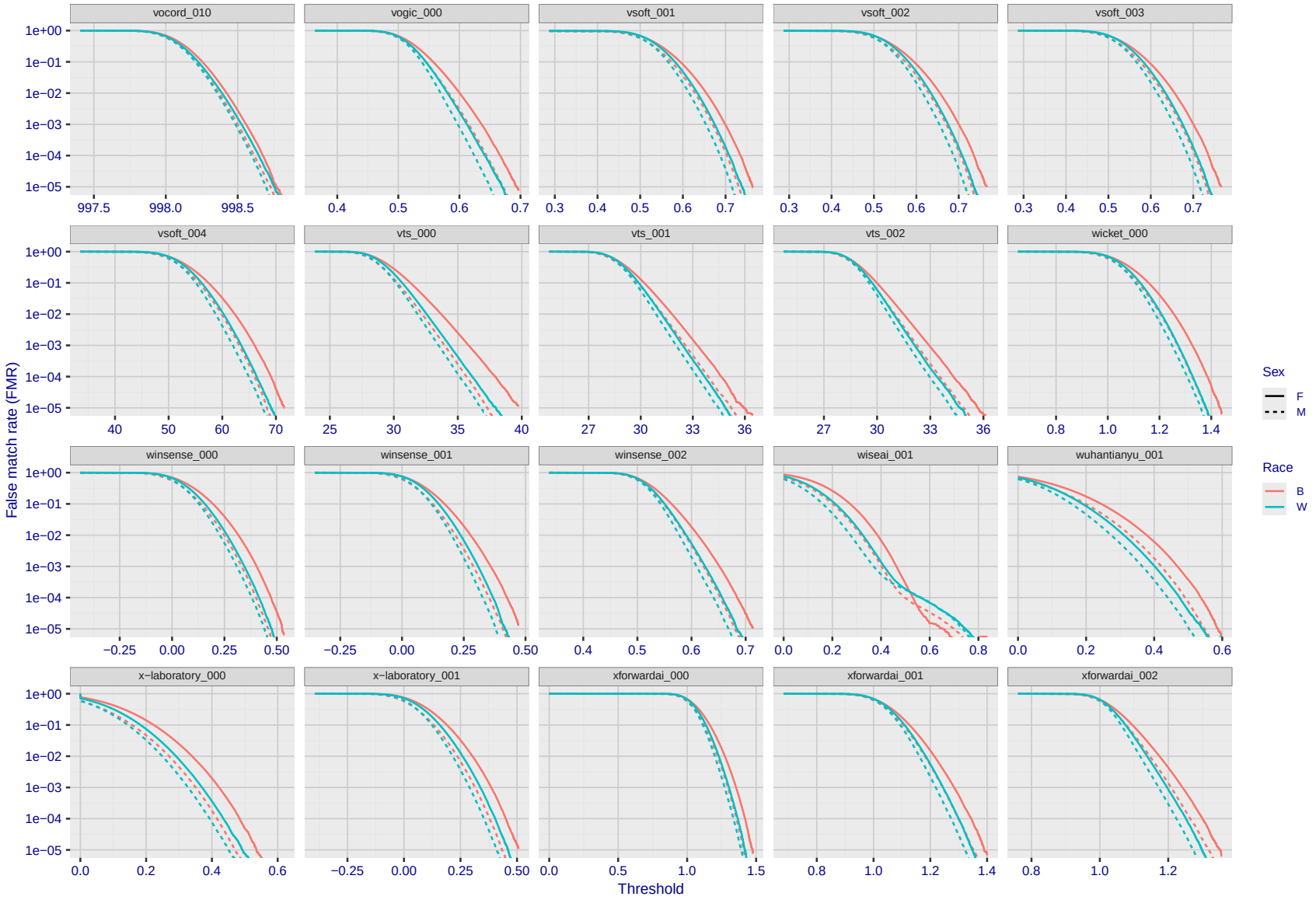

Figure 621: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

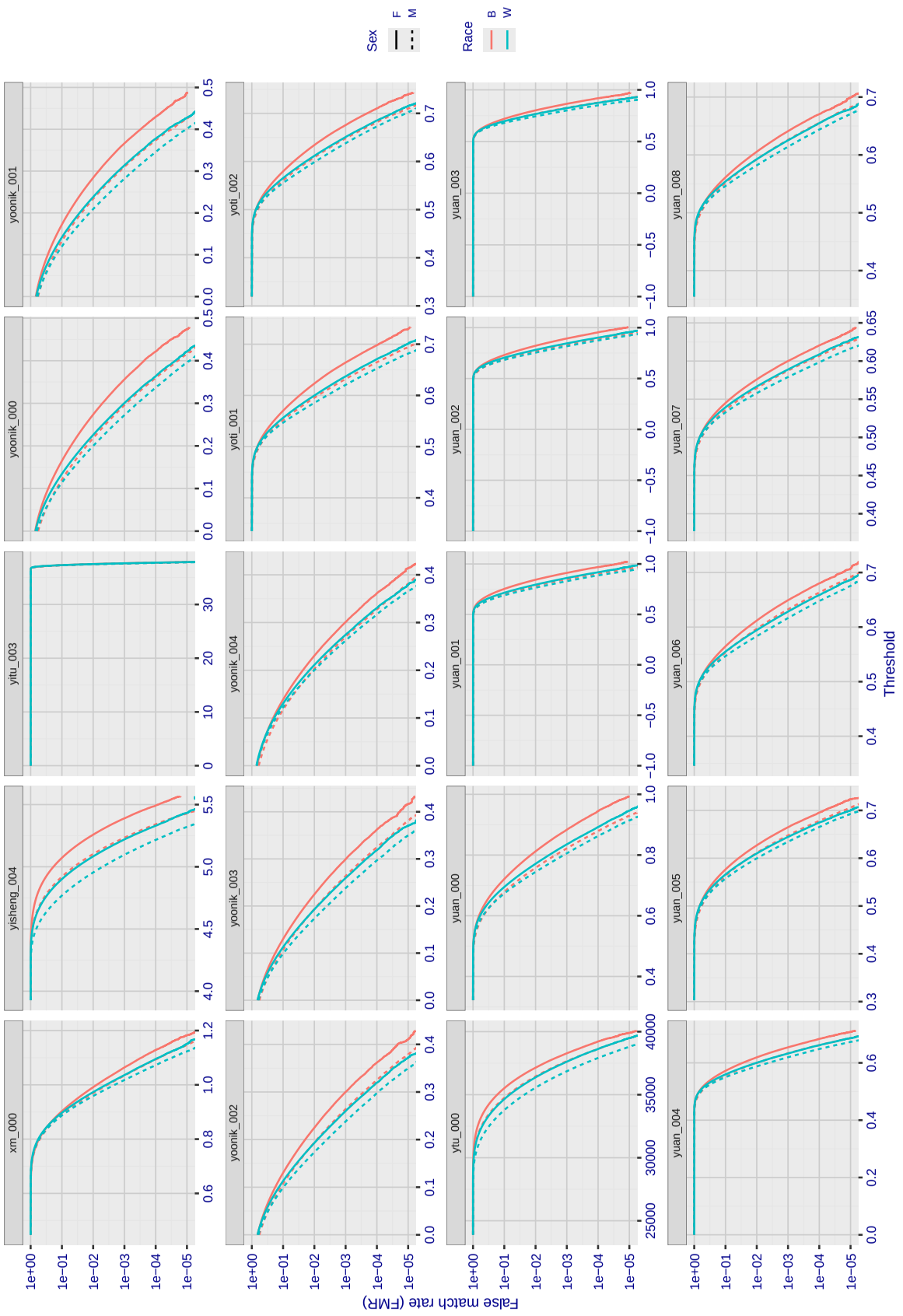

Figure 622: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

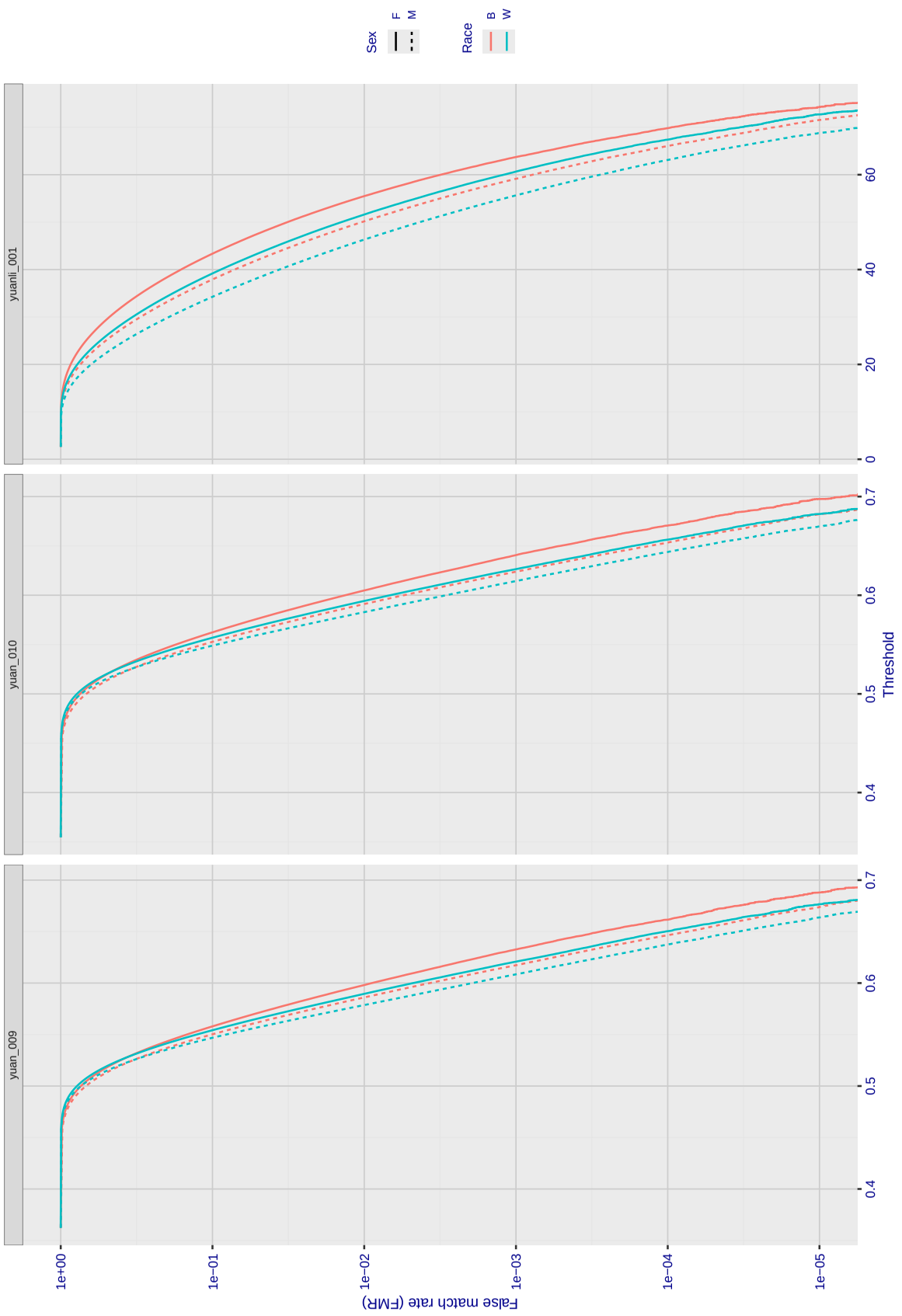

Figure 623: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 624: For the visa images, the false match calibration curves show FMR vs. threshold, T. The blue (lower) curves are for zero-effort impostors (i.e. comparing all images against all). The red (upper) curves are for persons of the same-sex, same-age, and same national-origin. This shows that FMR is underestimated (by a factor of 10 or more) by using a zero-effort impostor calculation to calibrate T. As shown later (sec. 3.6), FMR is higher for demographic-matched impostors.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.5 Genuine distribution stability

3.5.1 Effect of birth place on the genuine distribution

Background: Both skin tone and bone structure vary geographically. Prior studies have reported variations in FNMR and FMR.

Goal: To measure false non-match rate (FNMR) variation with country of birth.

Methods: Thresholds are determined that give FMR = $\{0.001, 0.0001\}$ over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores. Only those countries with at least 140 individuals are included in the analysis.

Results: Figure 714 shows FNMR by country of birth for the two thresholds.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

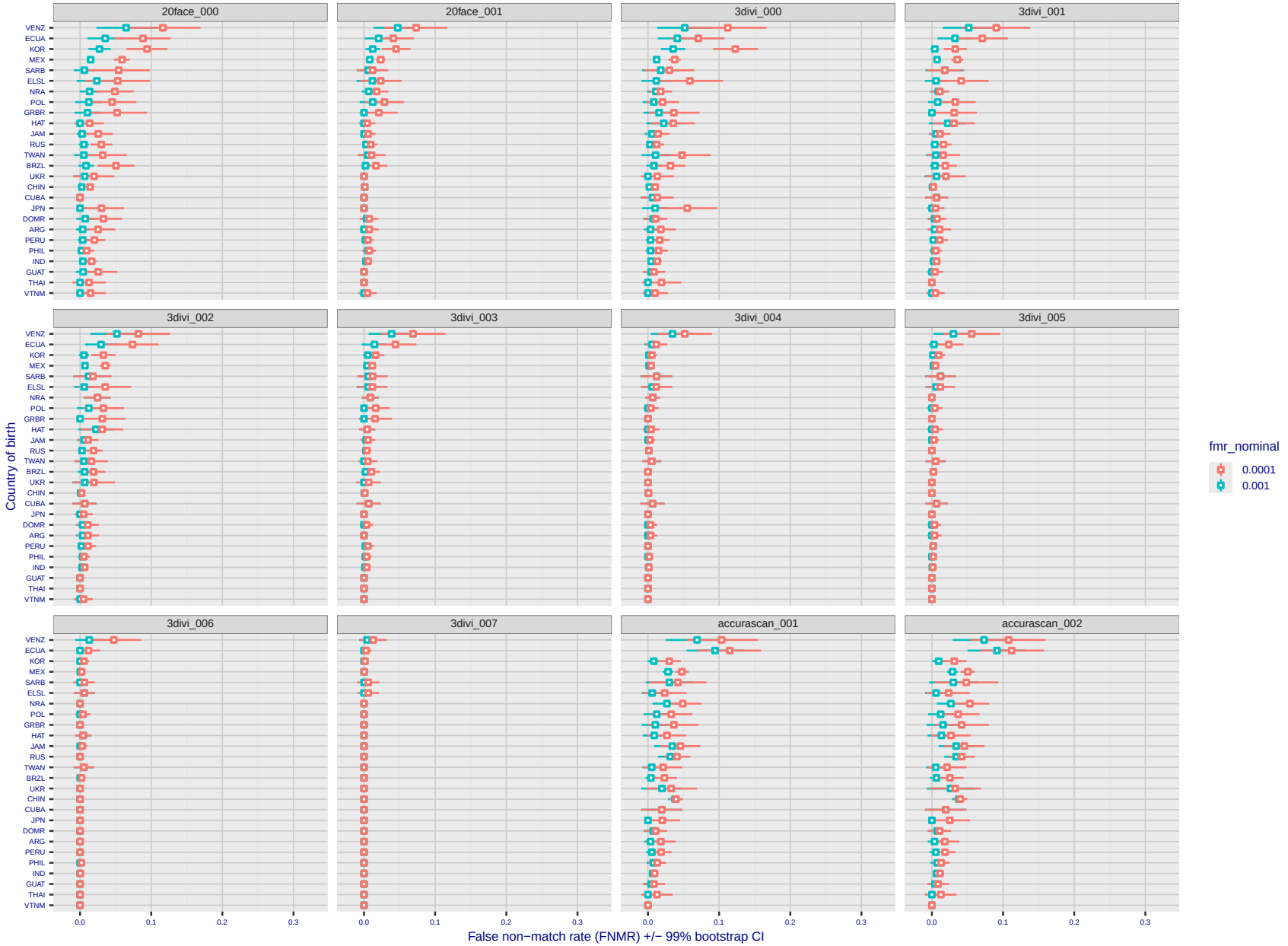

Figure 625: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 626: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

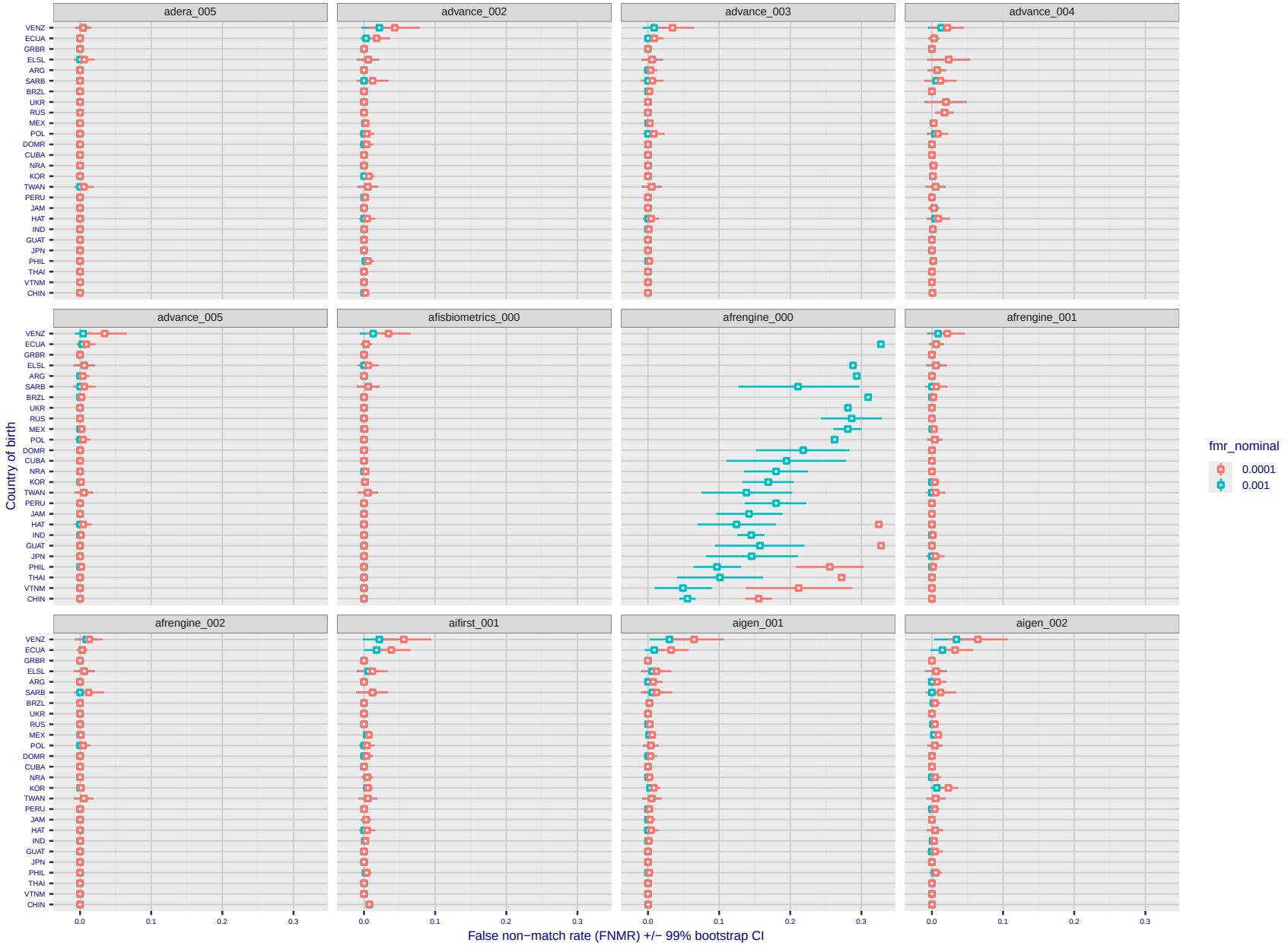

Figure 627: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

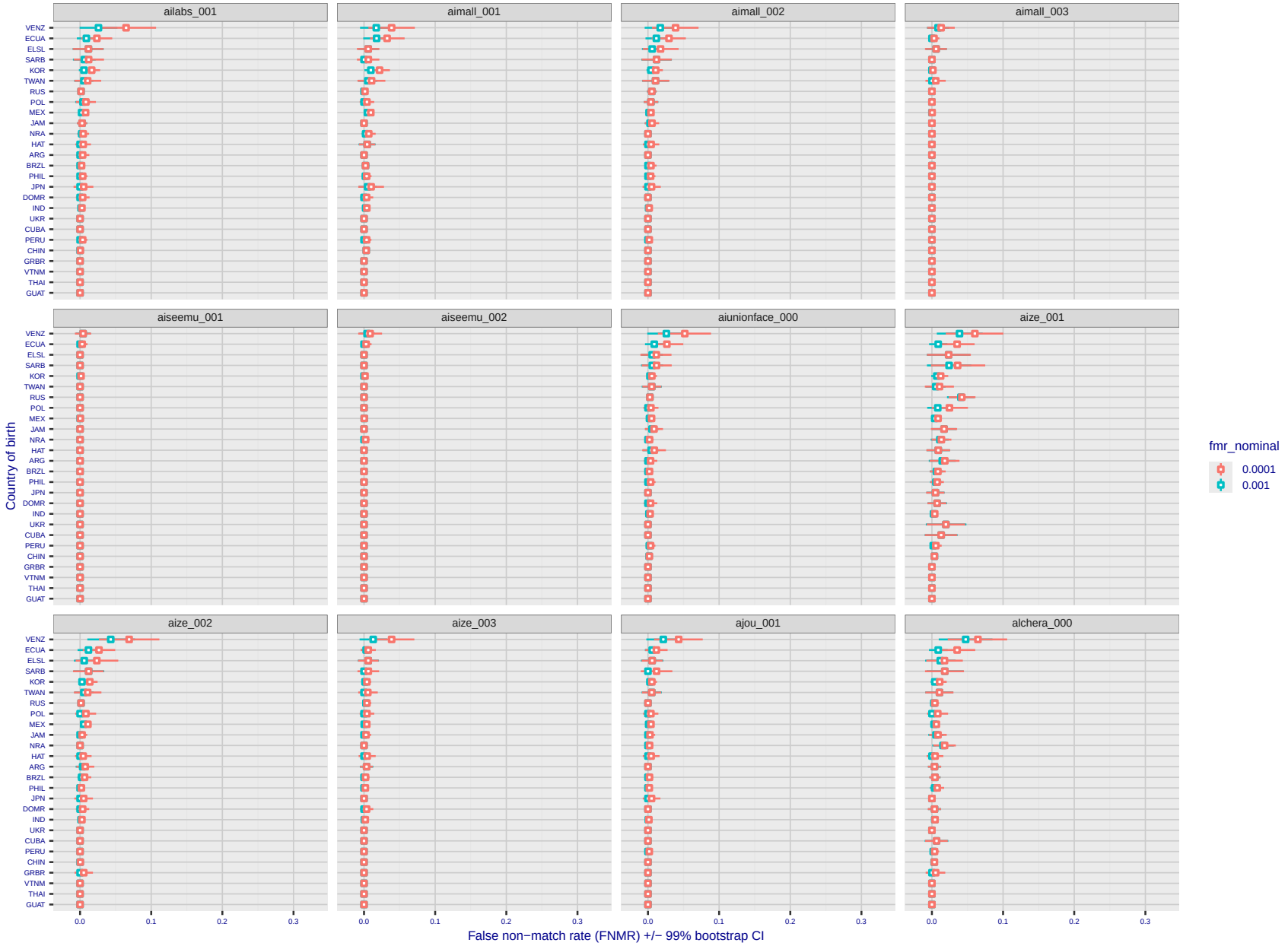

Figure 628: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

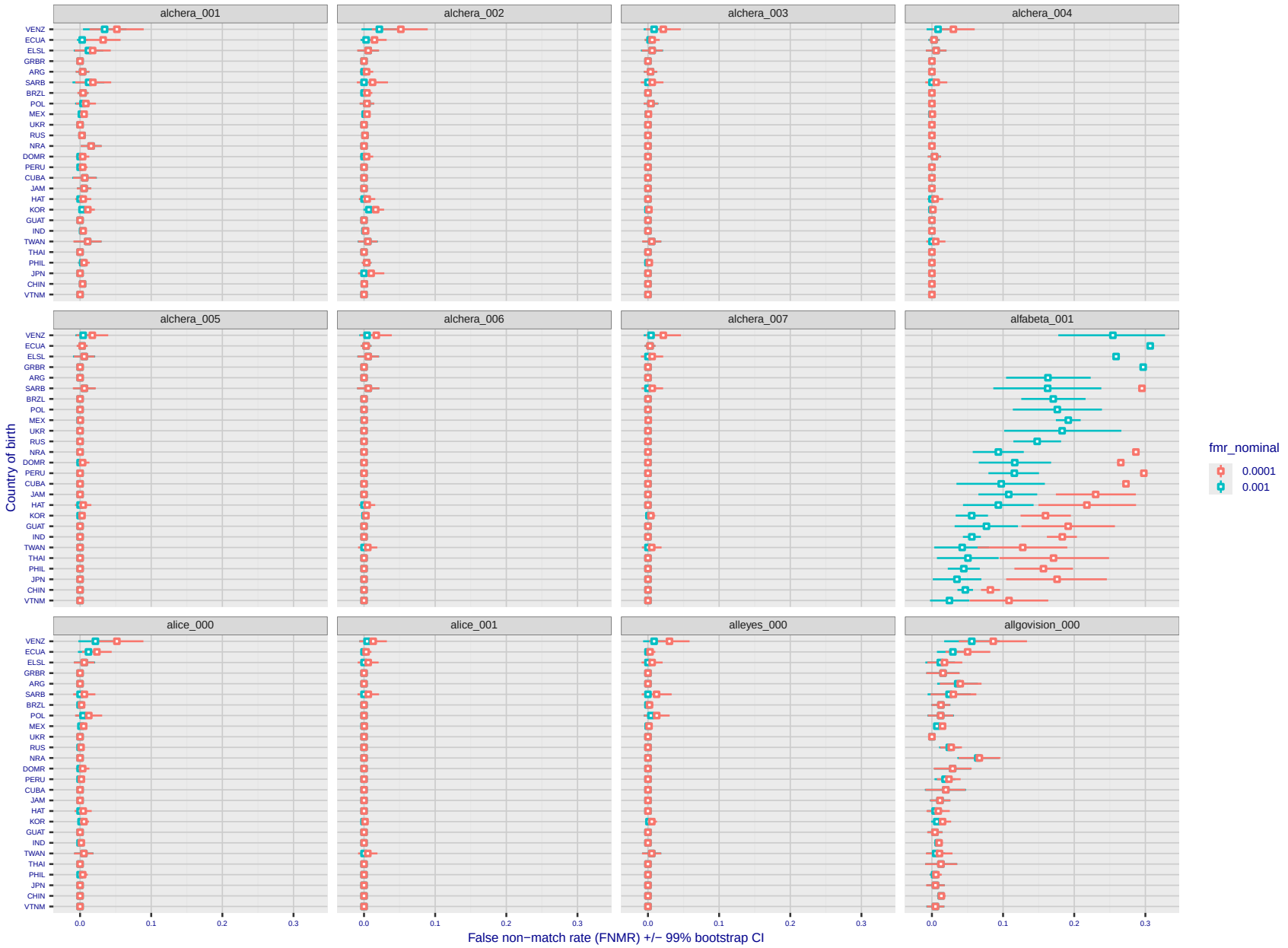

Figure 629: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

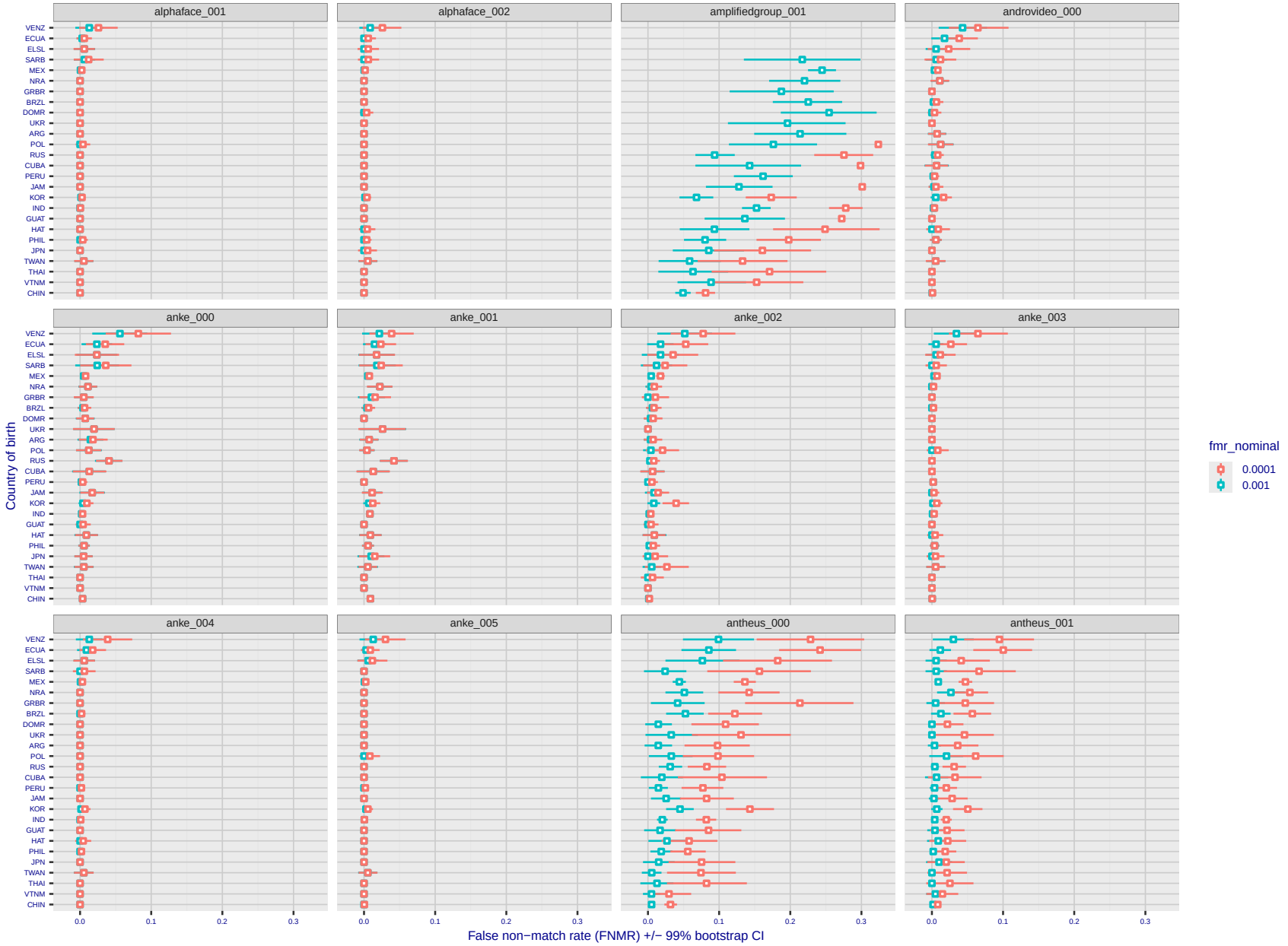

Figure 630: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 631: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 632: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

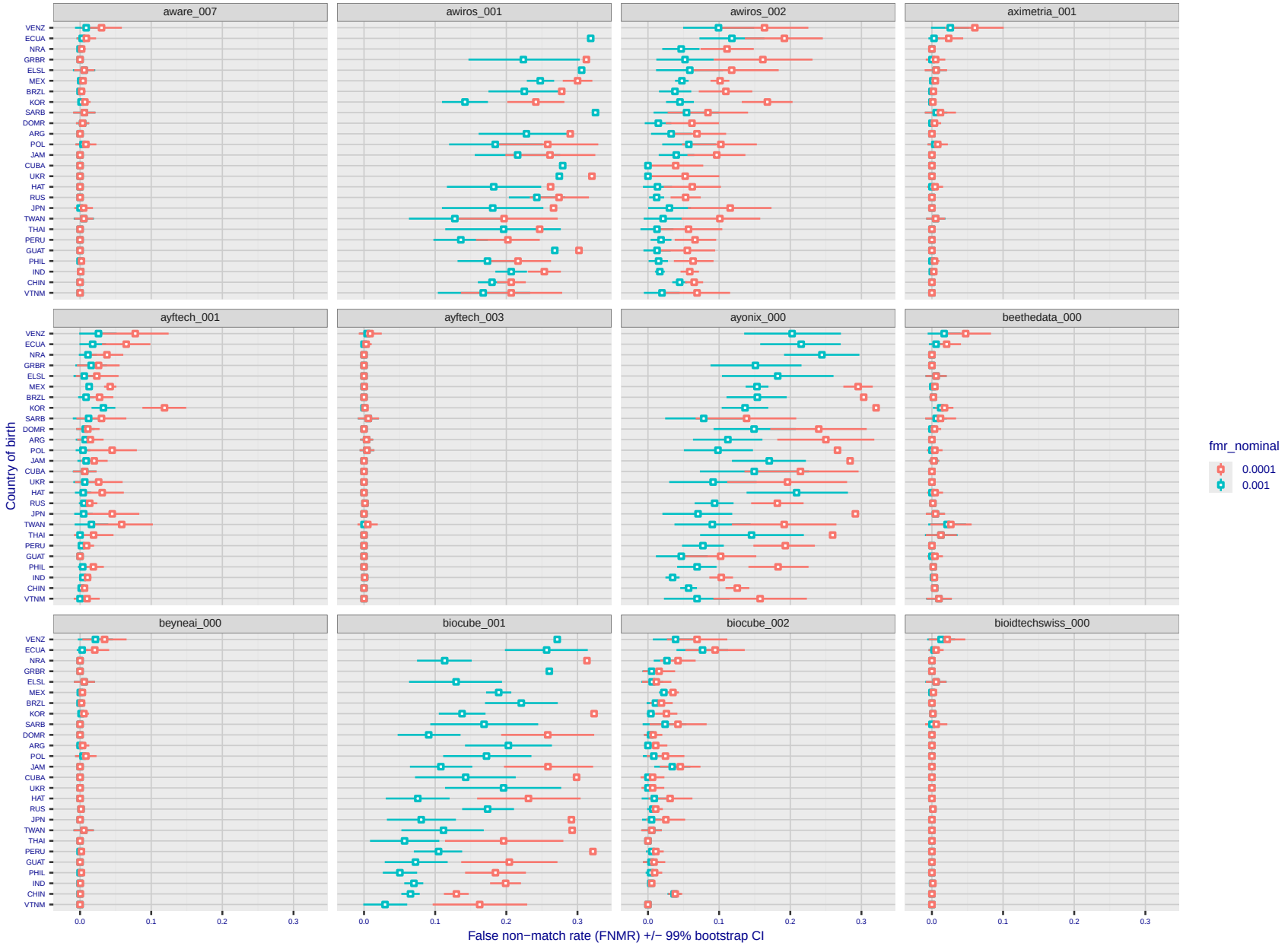

Figure 633: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 634: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 635: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

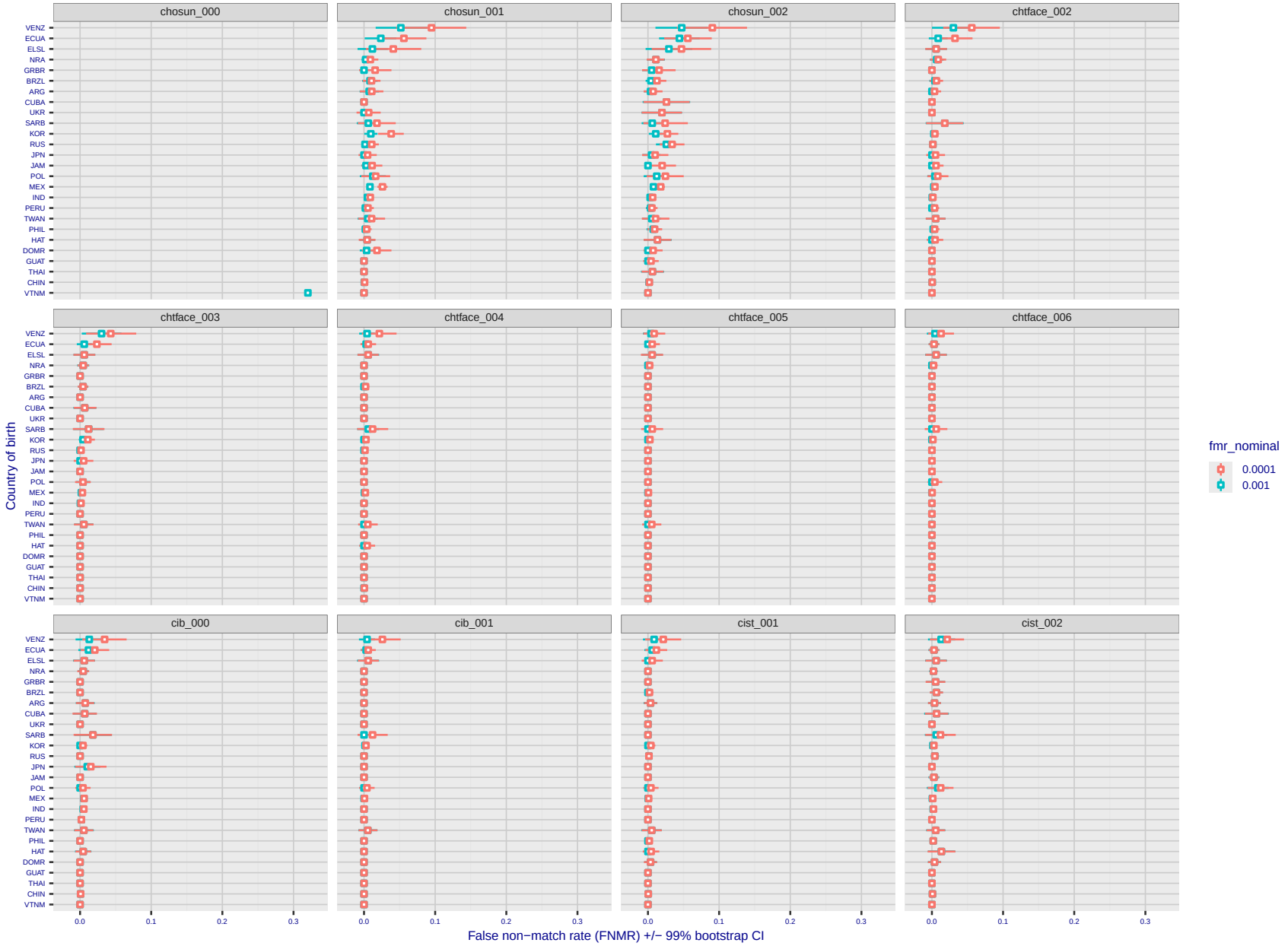

Figure 636: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

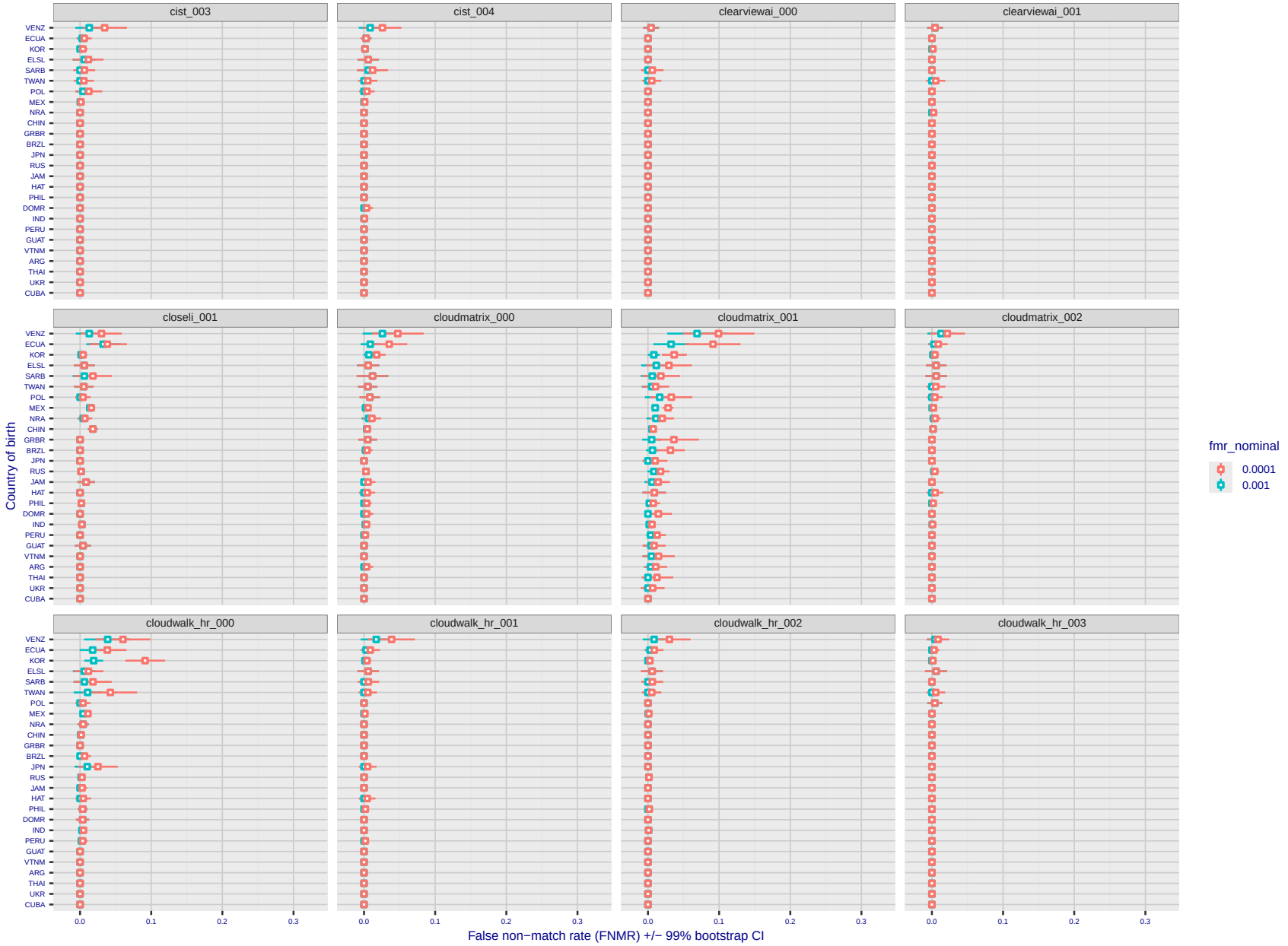

Figure 637: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

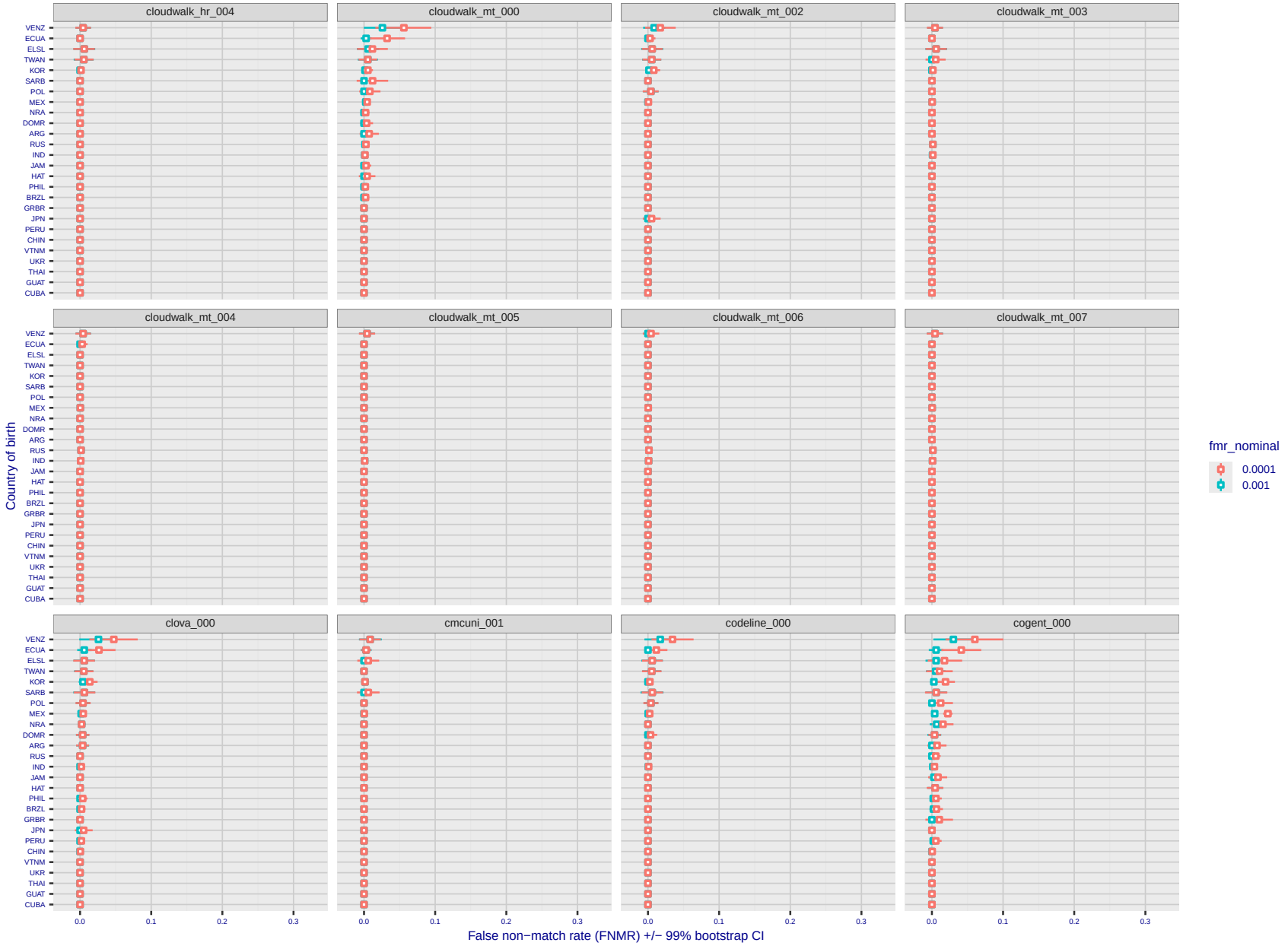

Figure 638: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

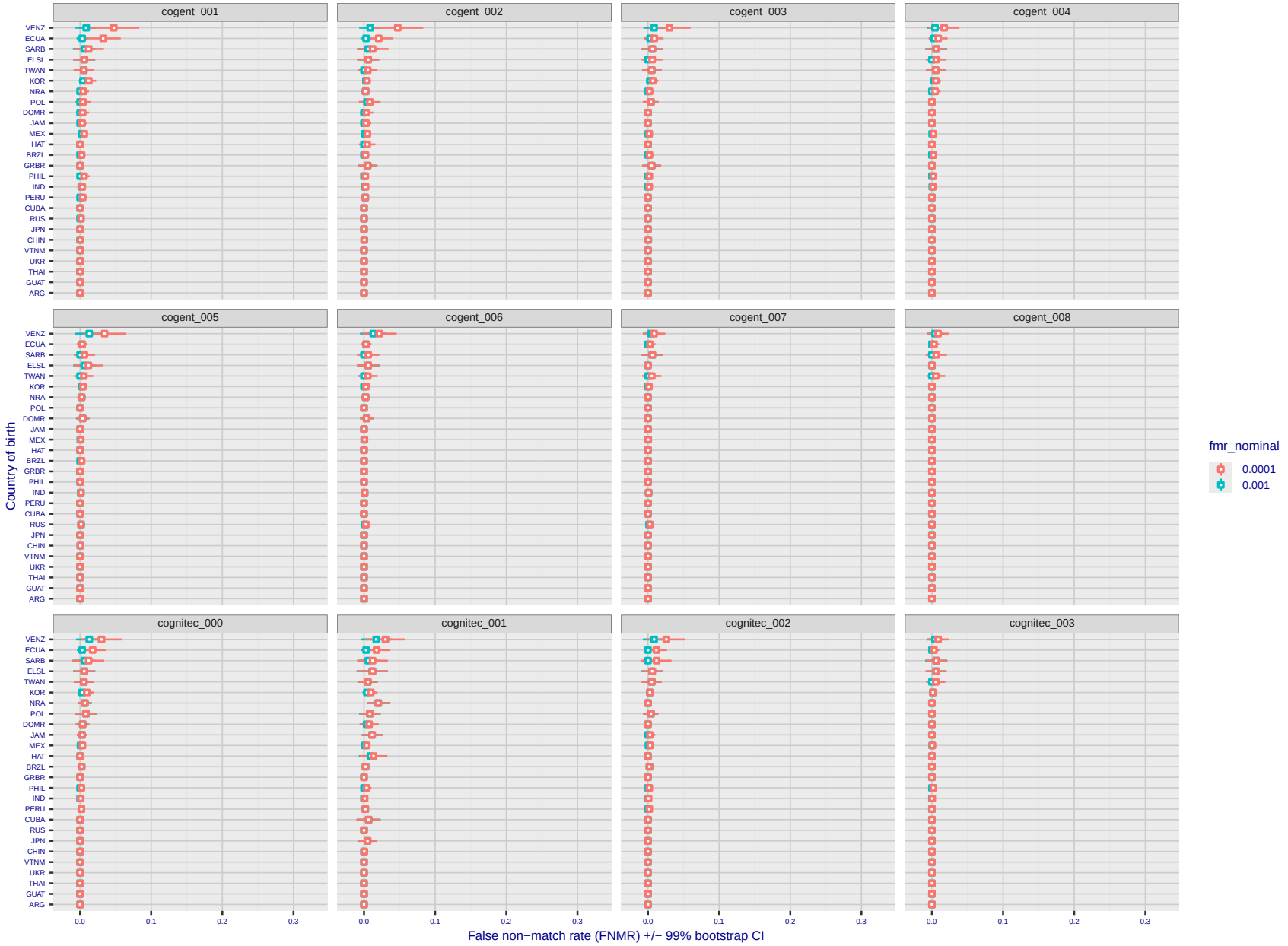

Figure 639: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

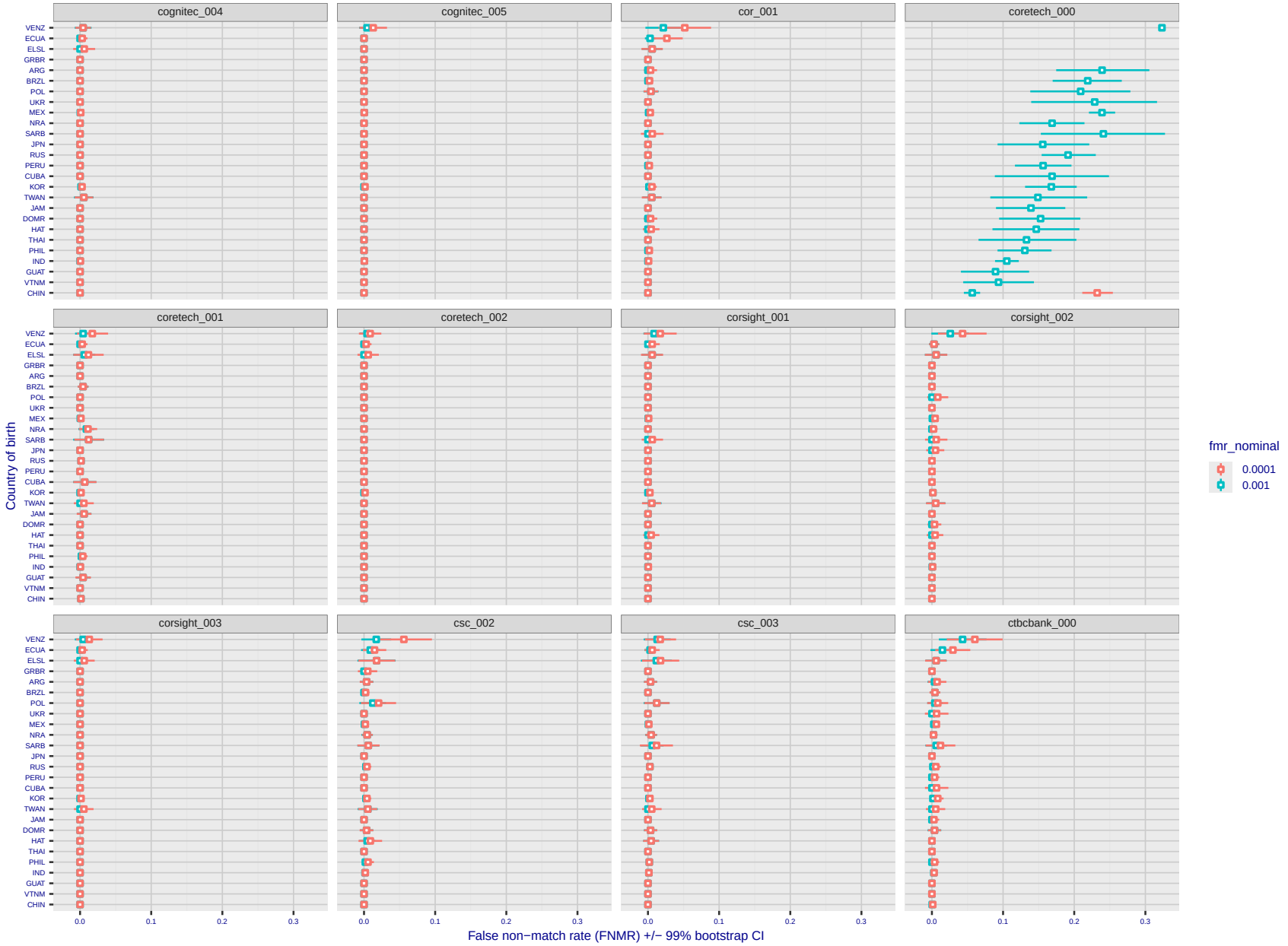

Figure 640: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

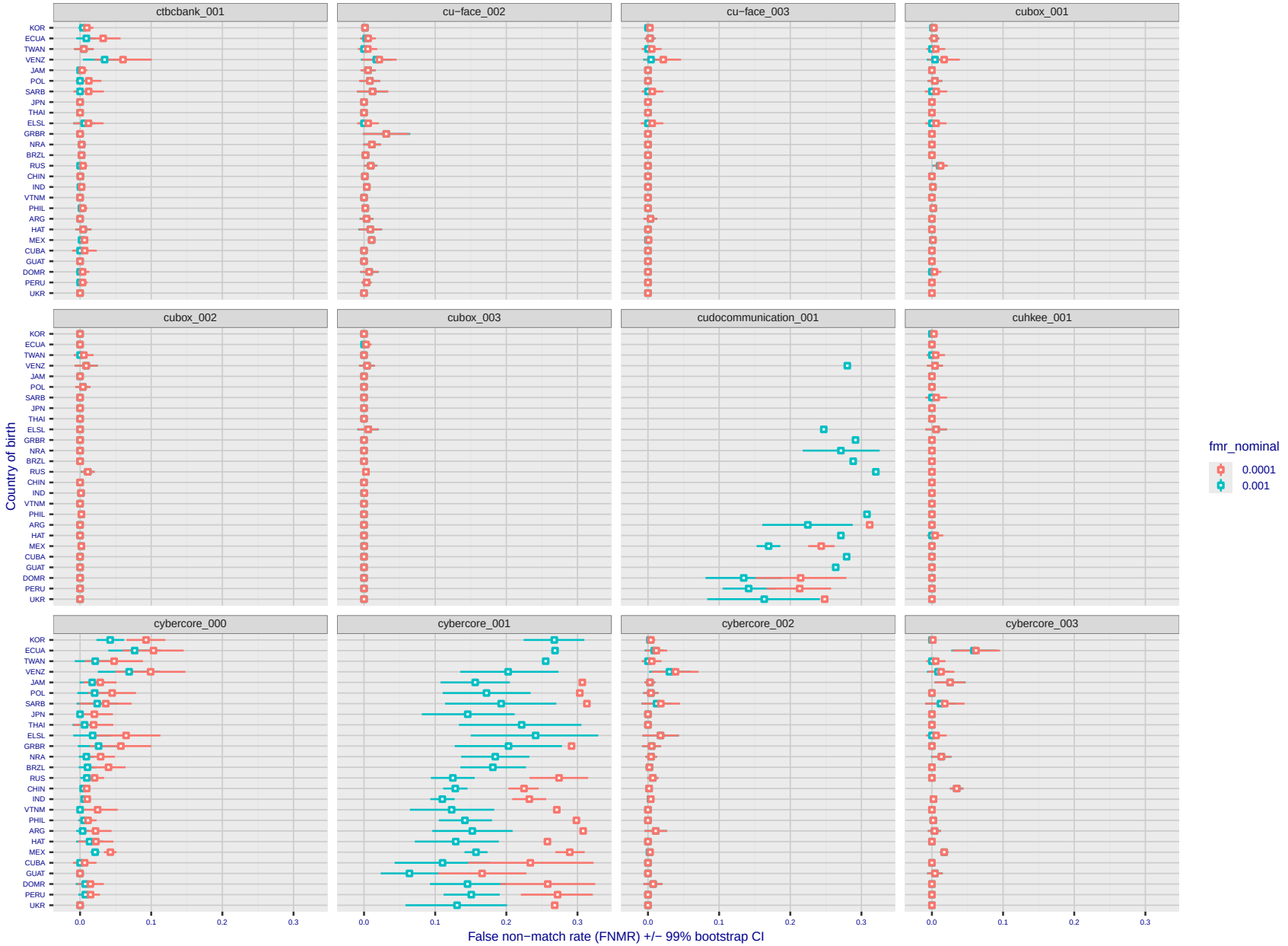

Figure 641: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 642: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

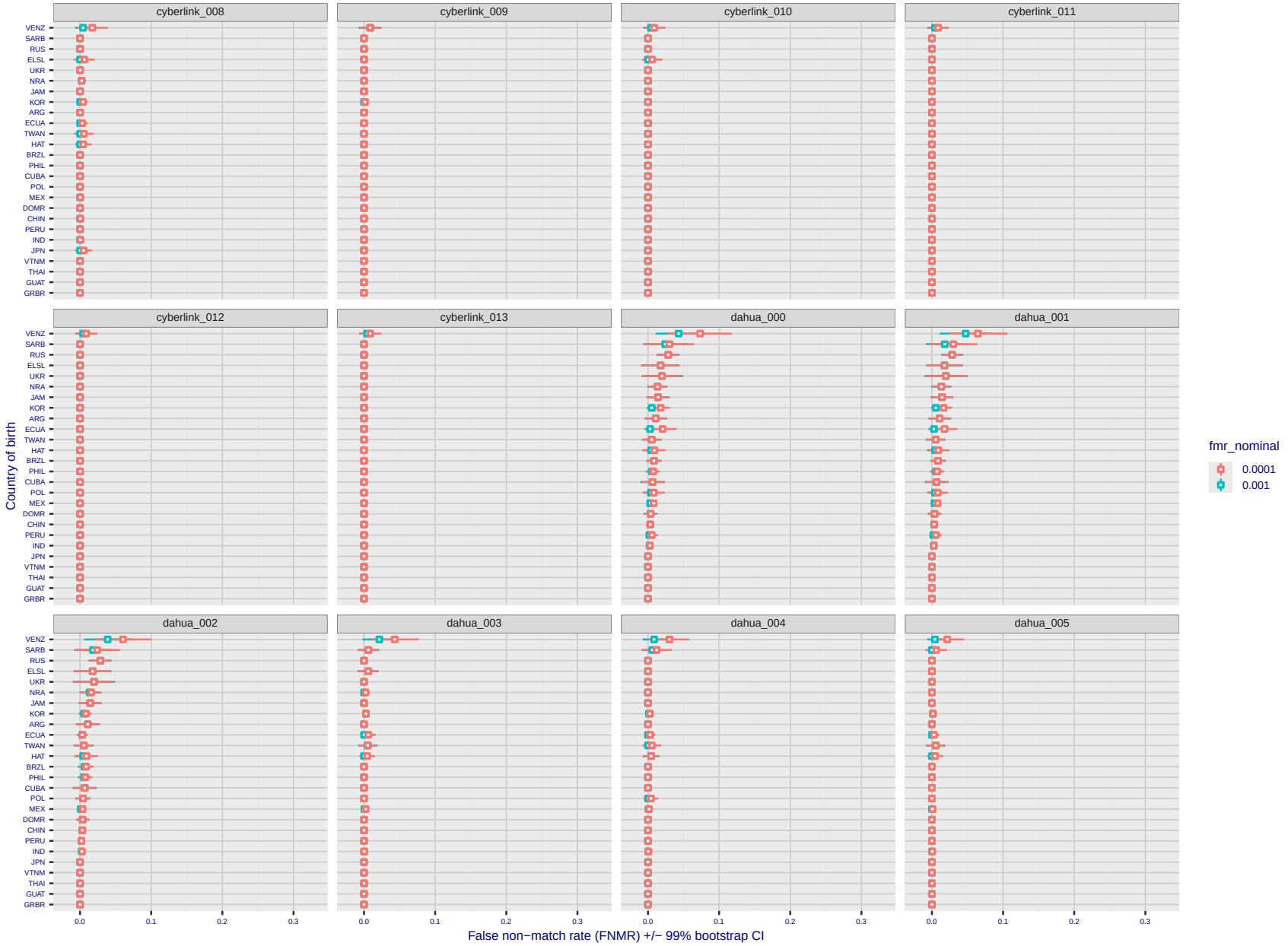

Figure 643: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 644: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

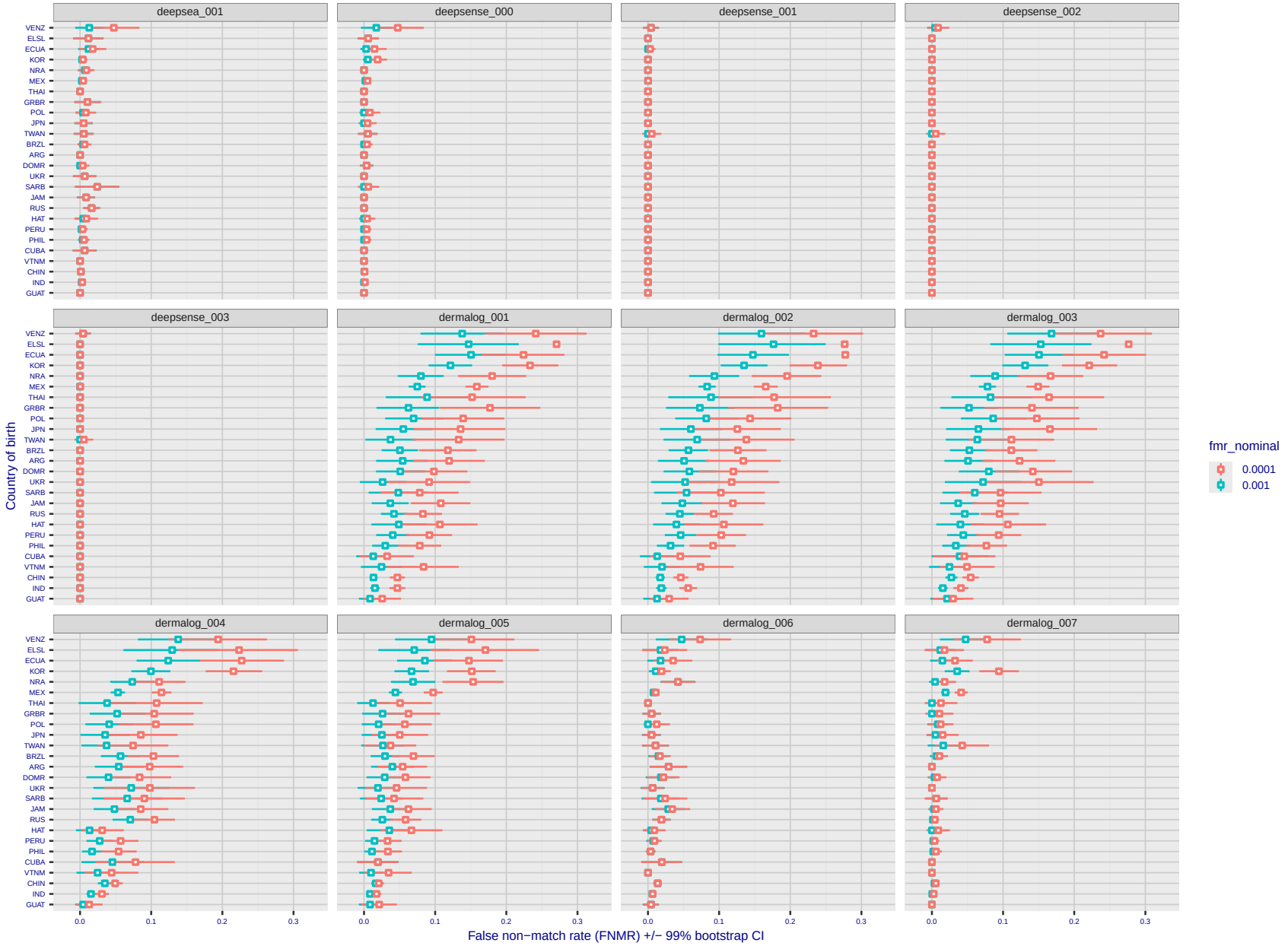

Figure 645: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

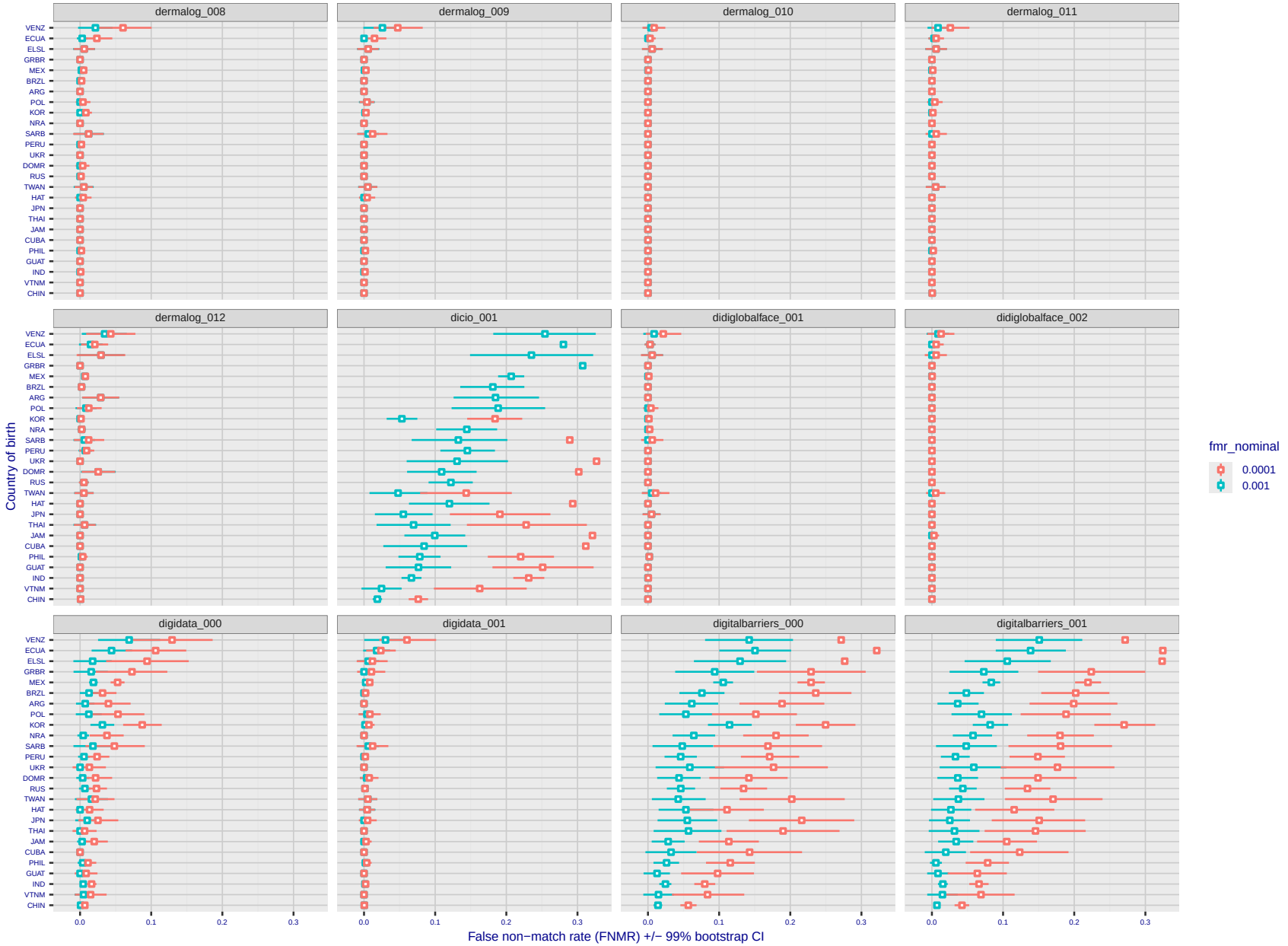

Figure 646: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 647: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 648: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 649: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

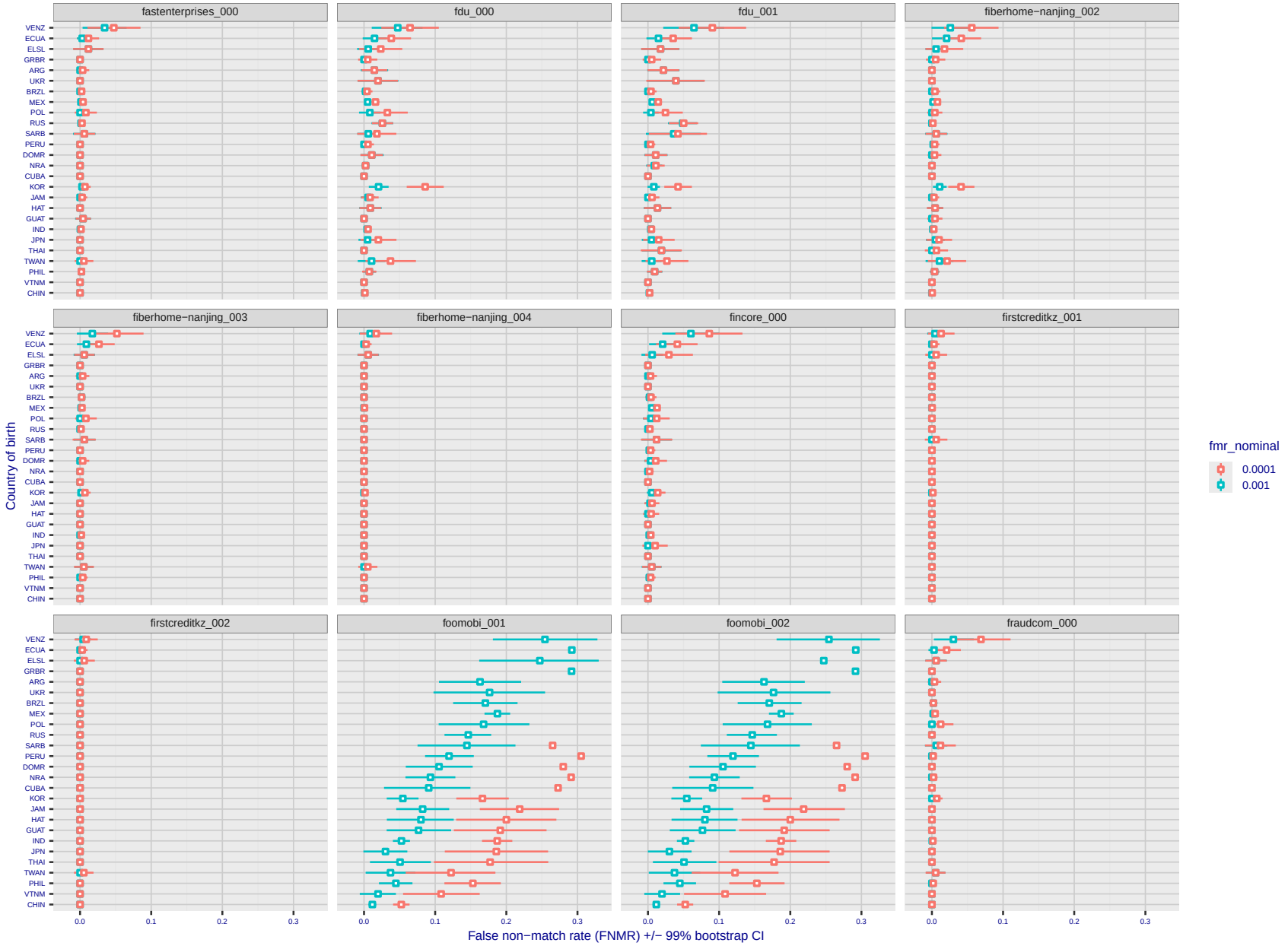

Figure 650: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

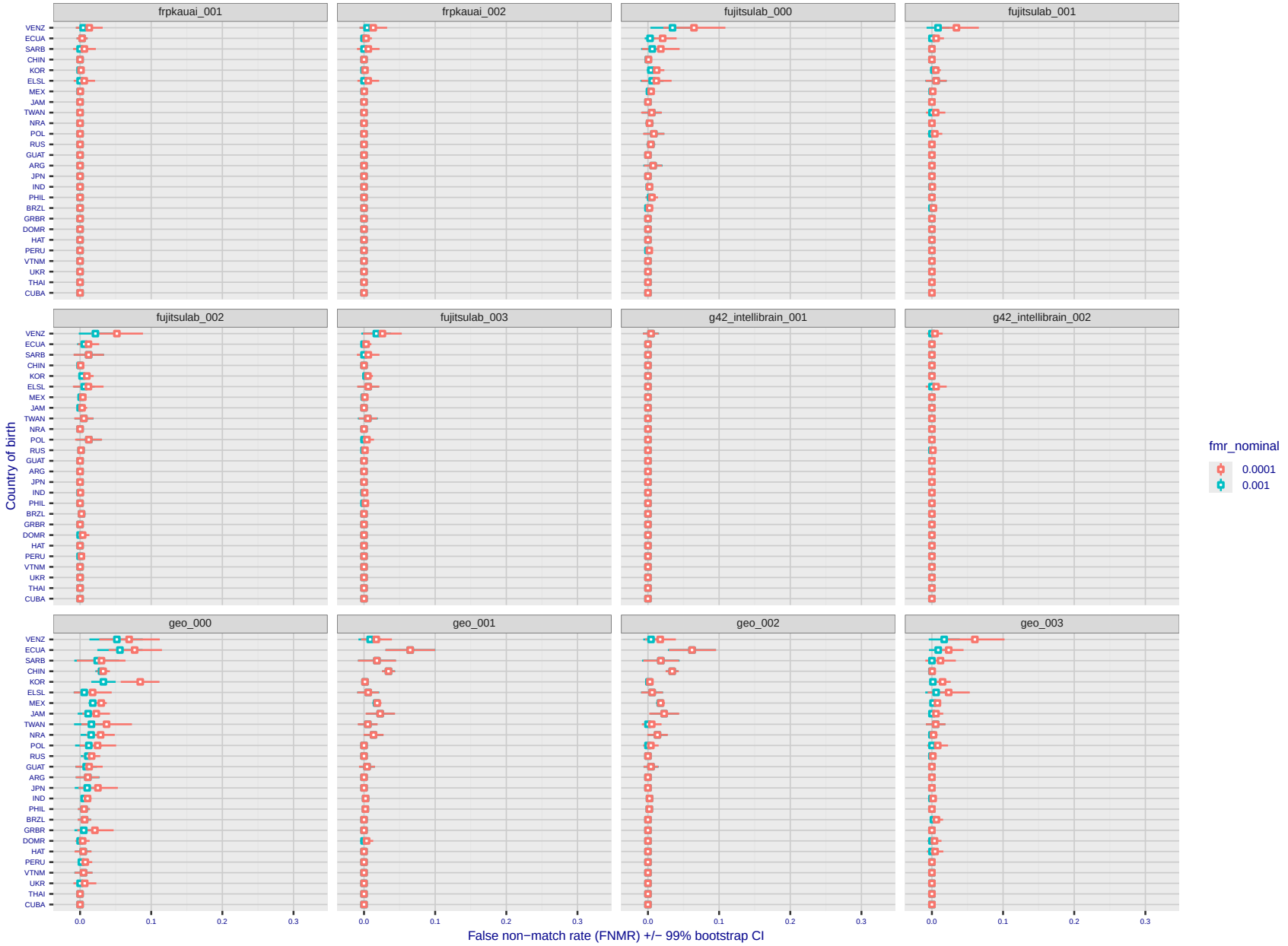

Figure 651: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 652: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

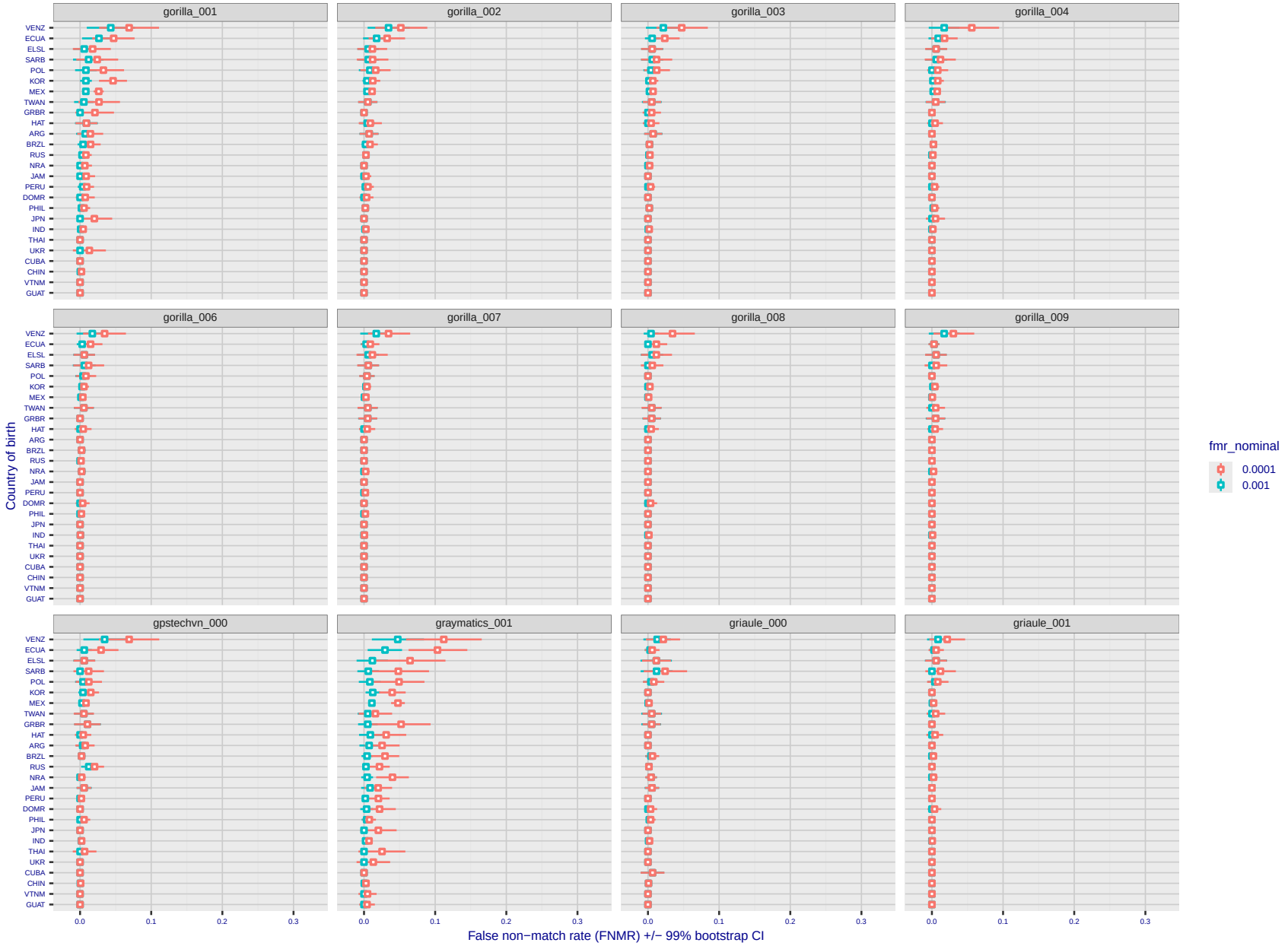

Figure 653: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 654: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 655: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

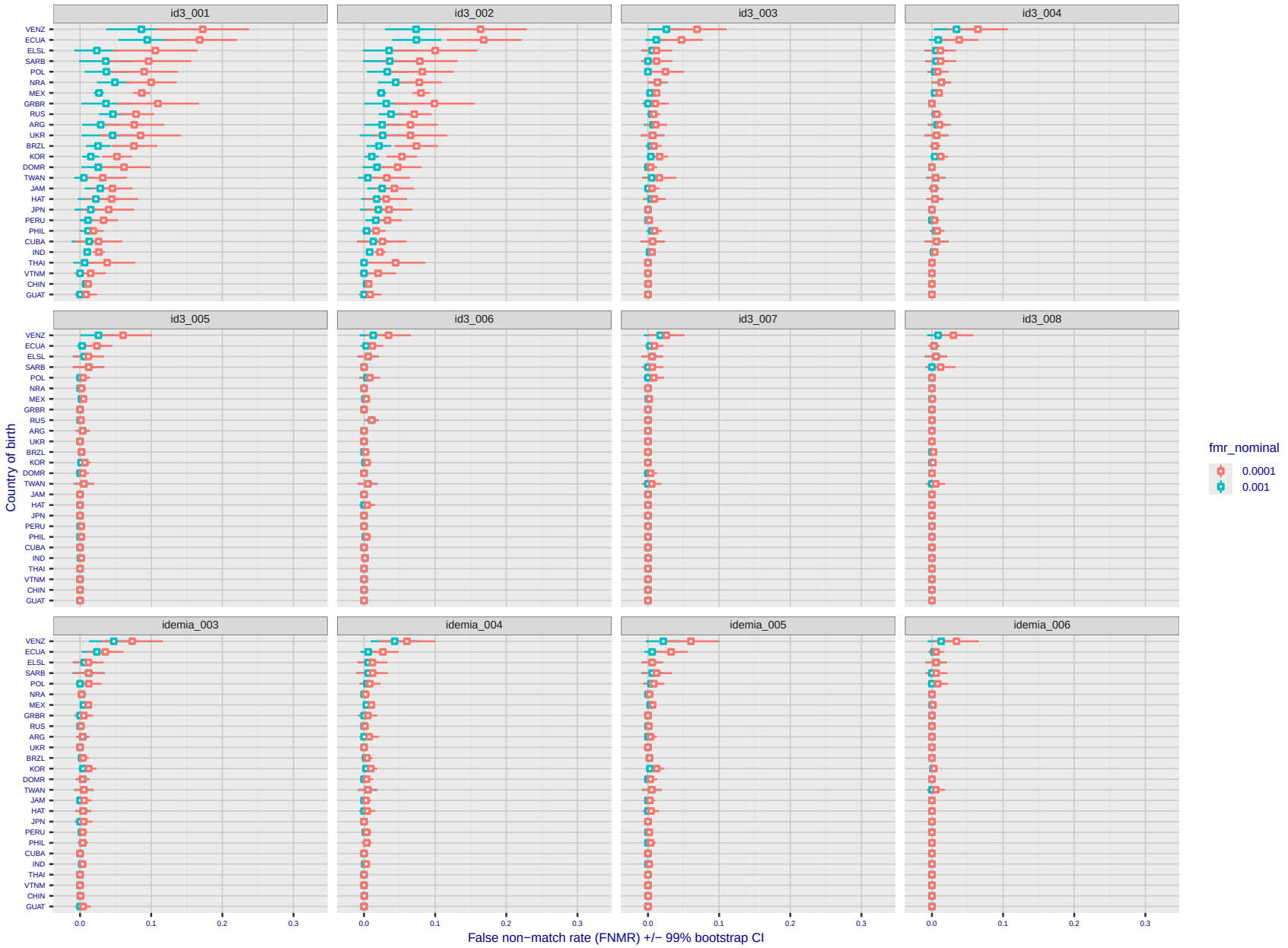

Figure 656: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

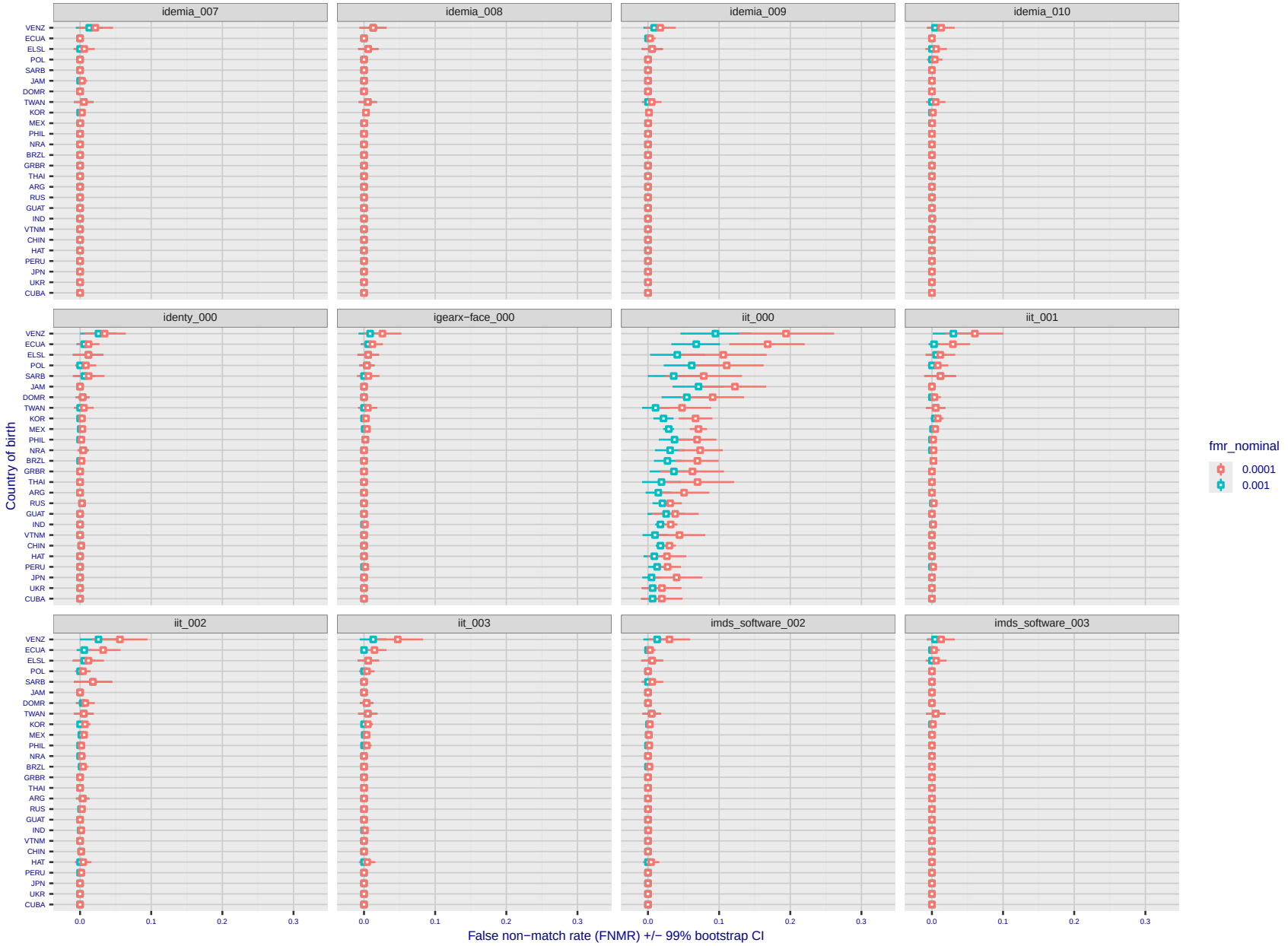

Figure 657: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

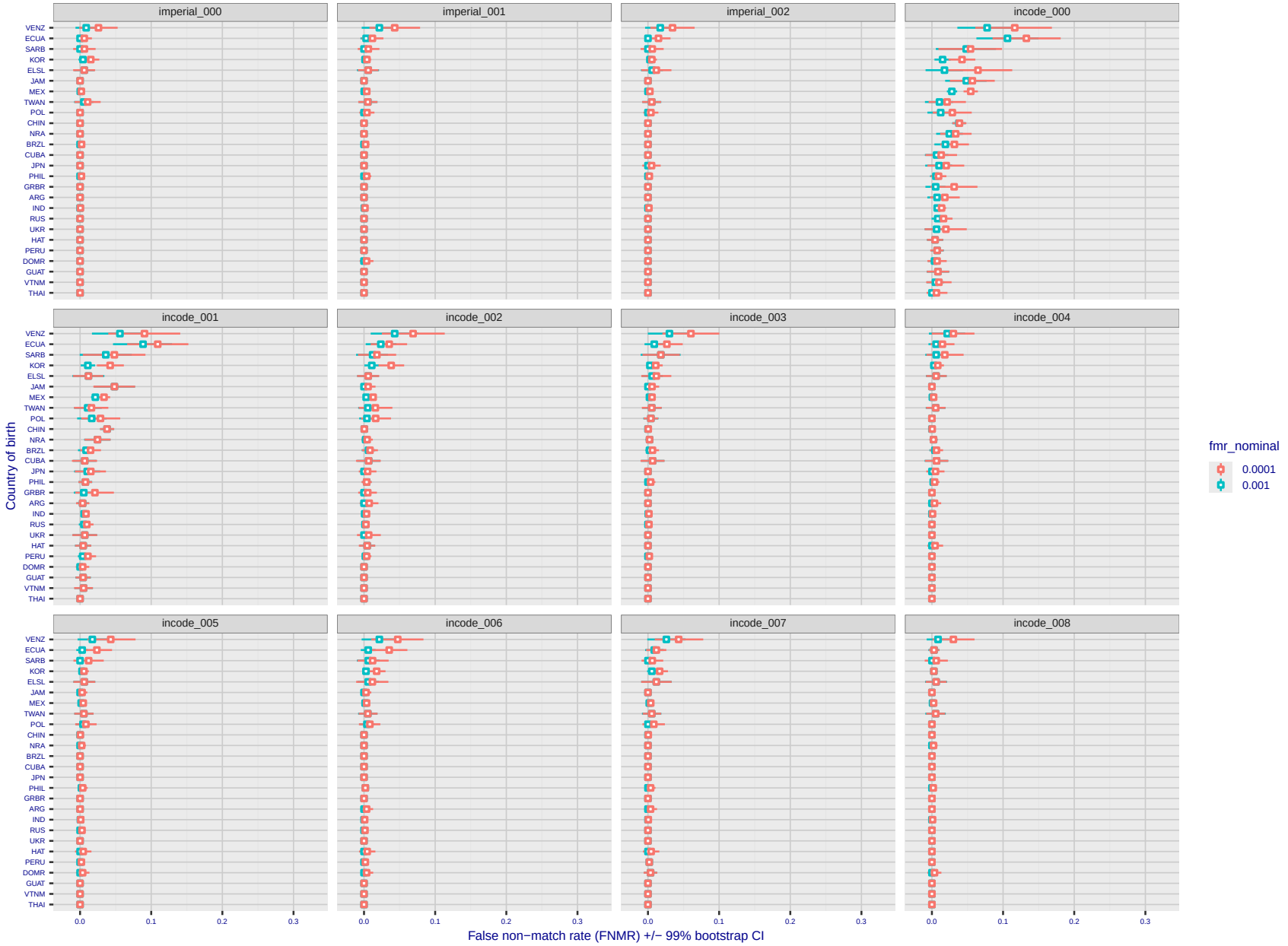

Figure 658: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

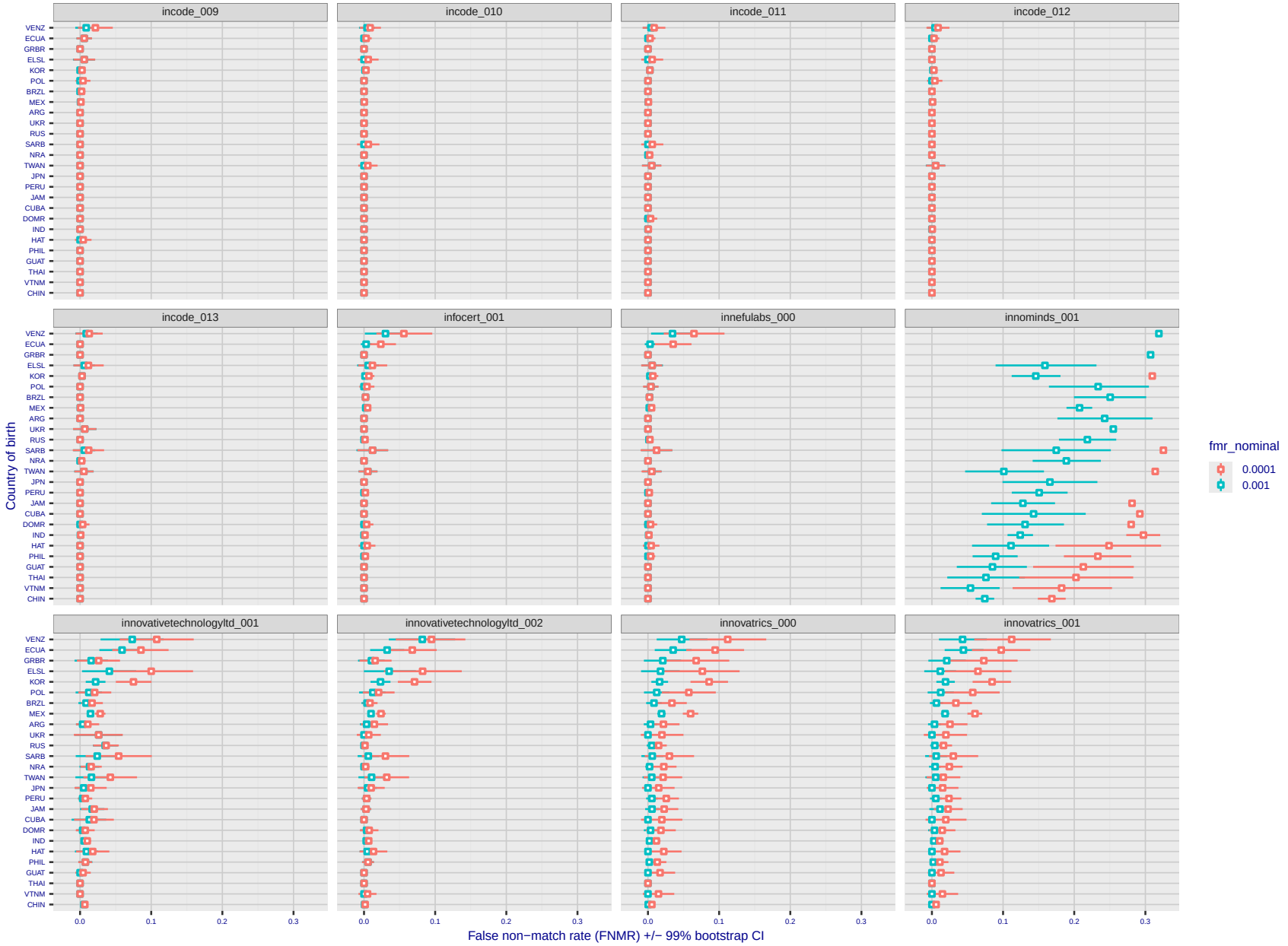

Figure 659: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

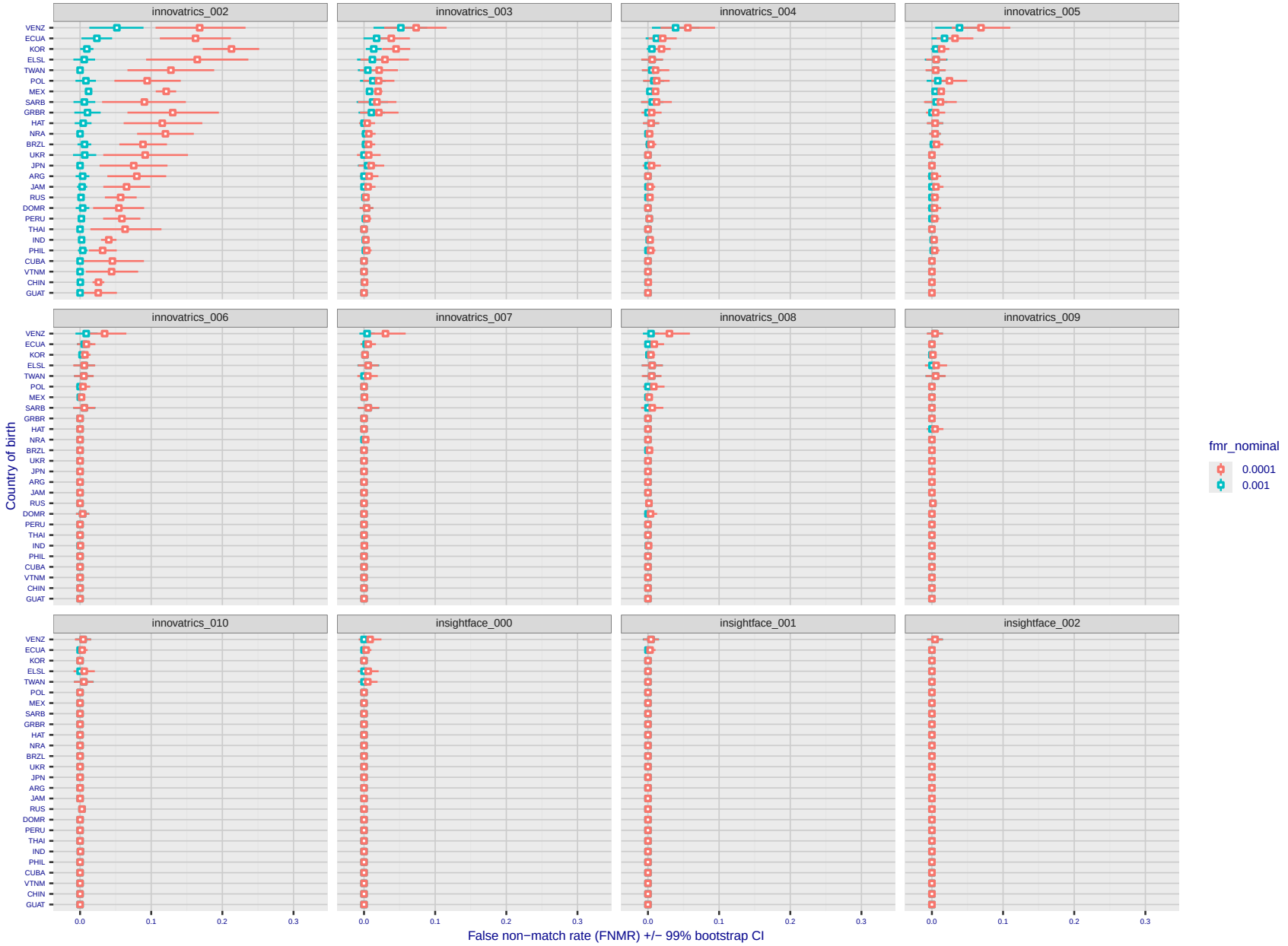

Figure 660: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

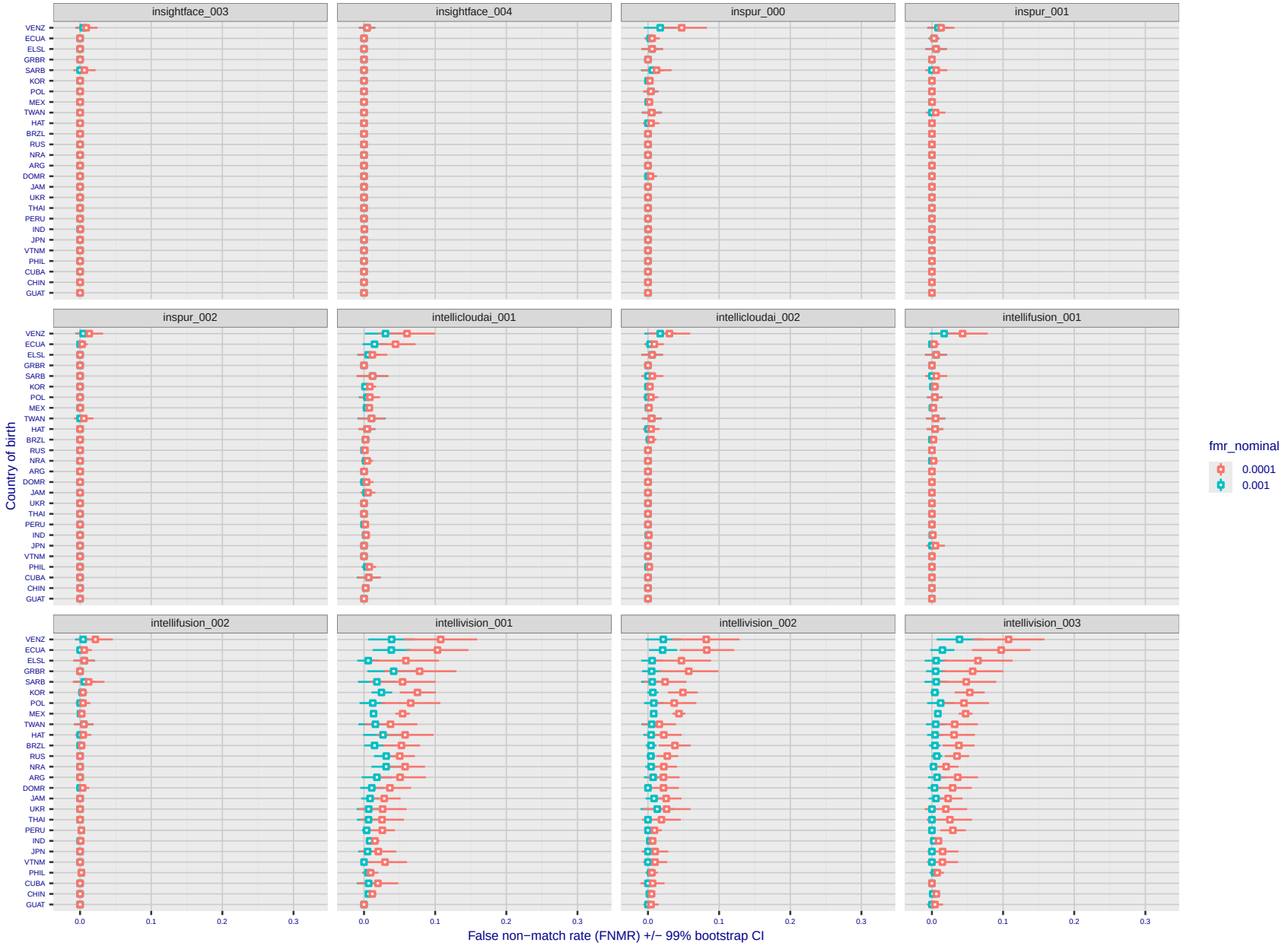

Figure 661: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

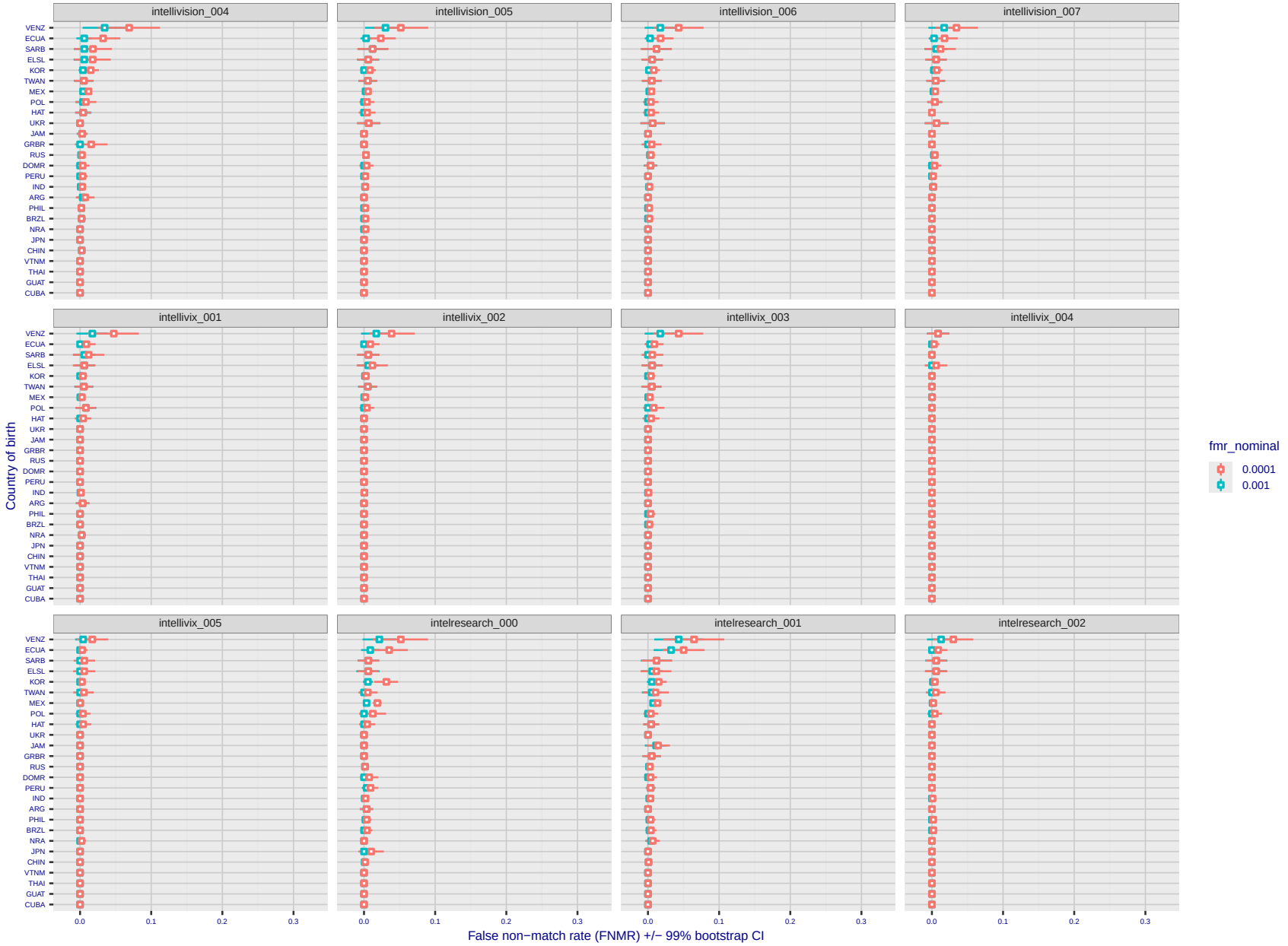

Figure 662: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 663: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

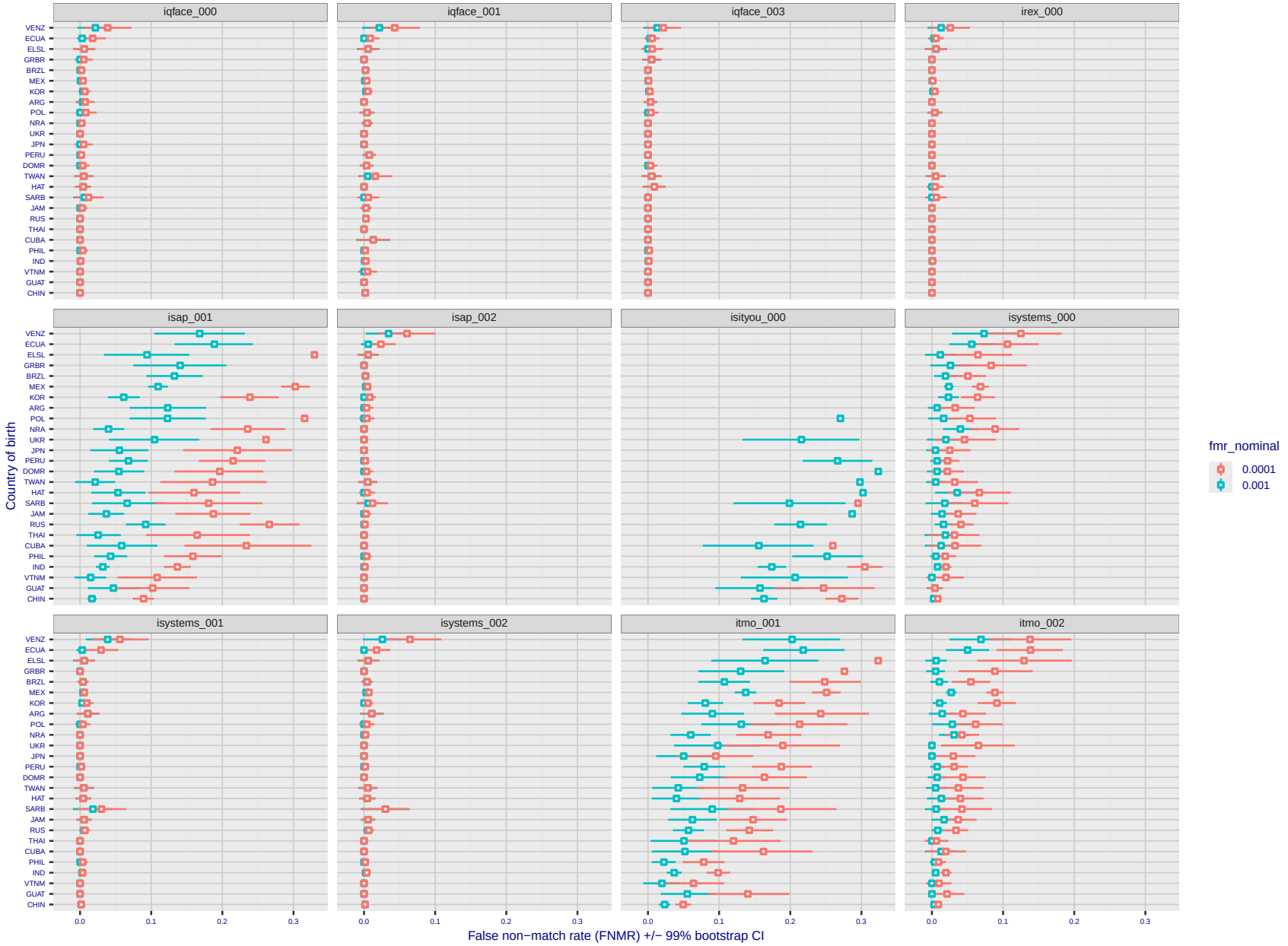

Figure 664: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

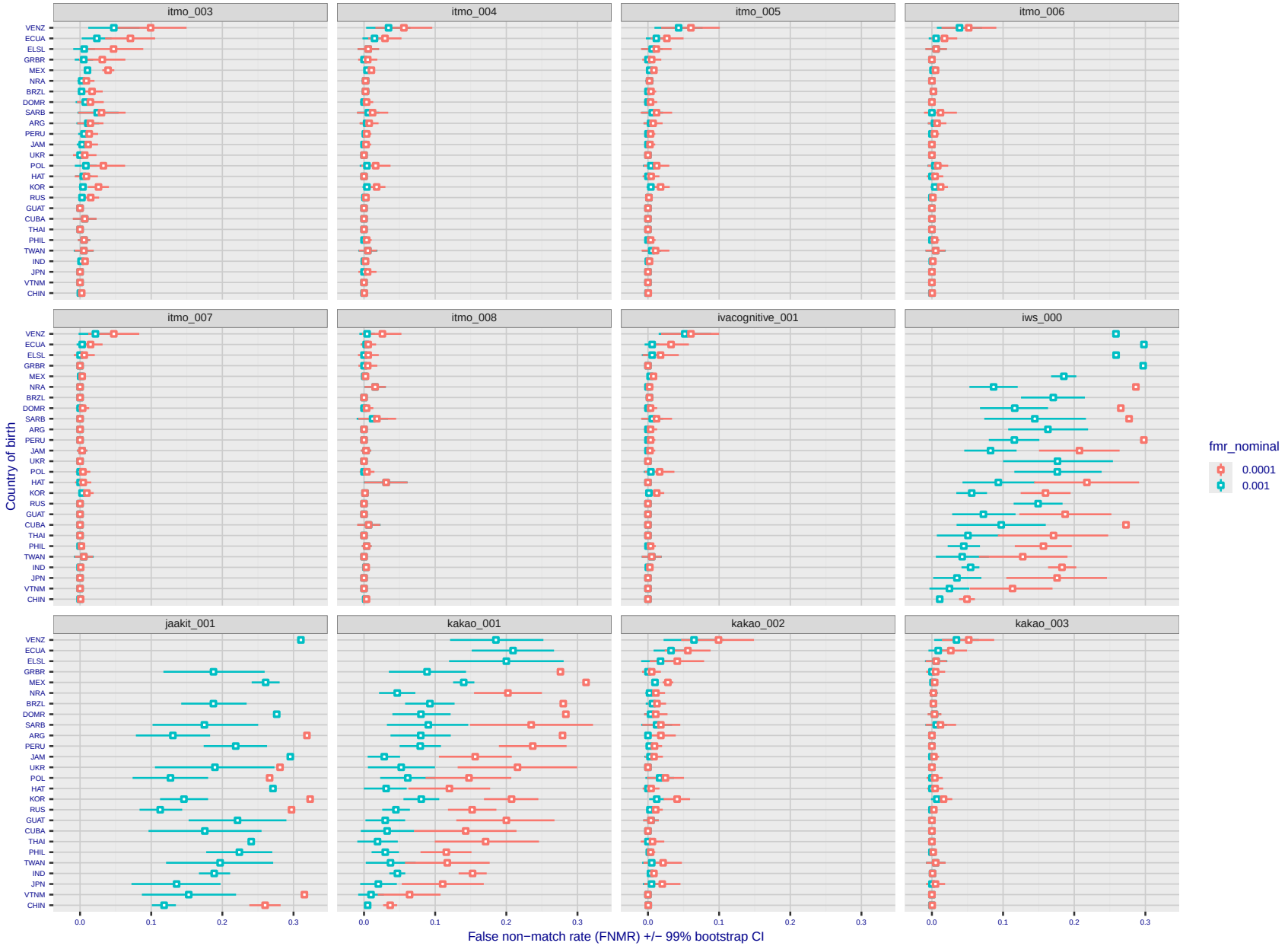

Figure 665: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

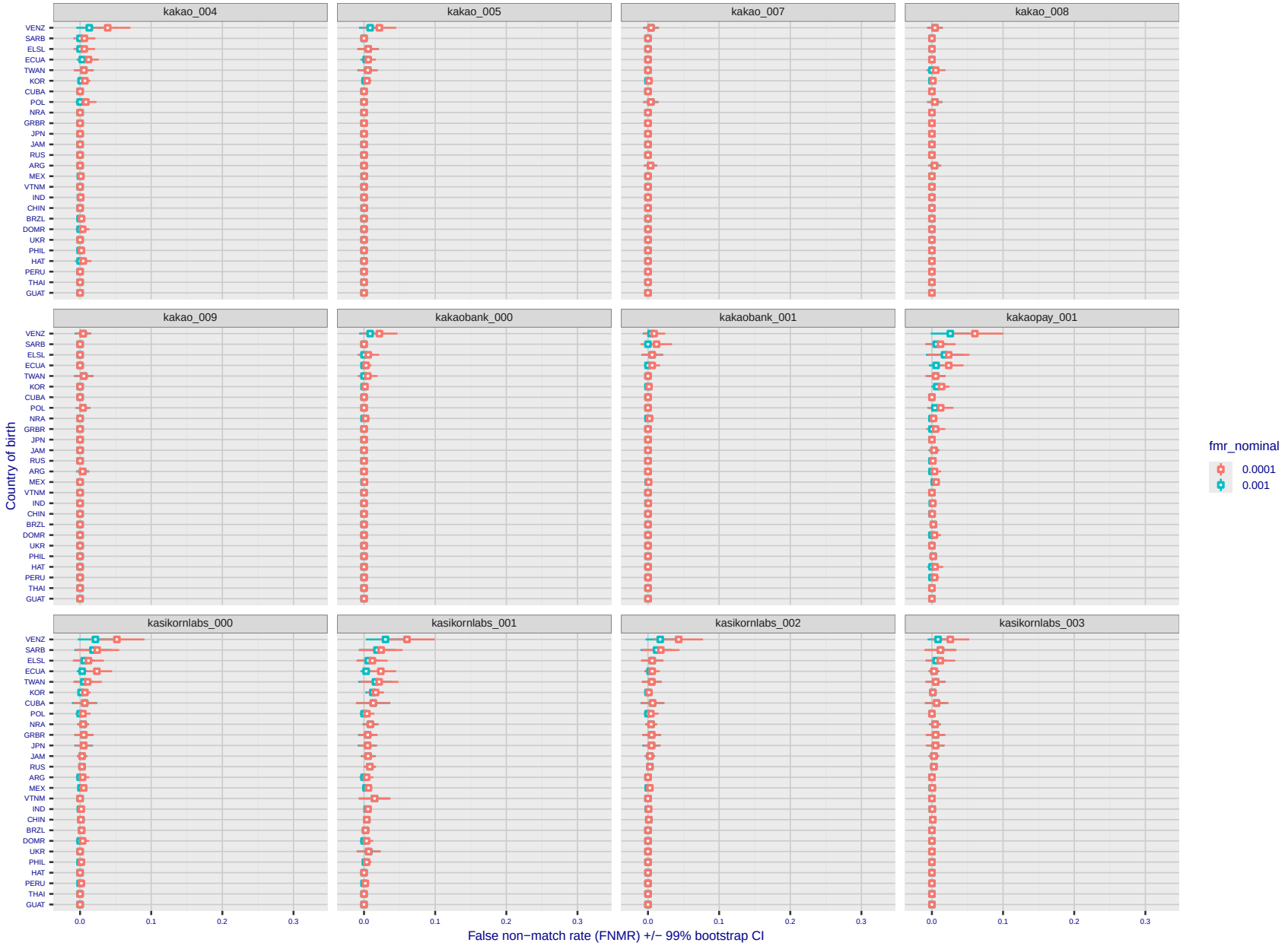

Figure 666: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

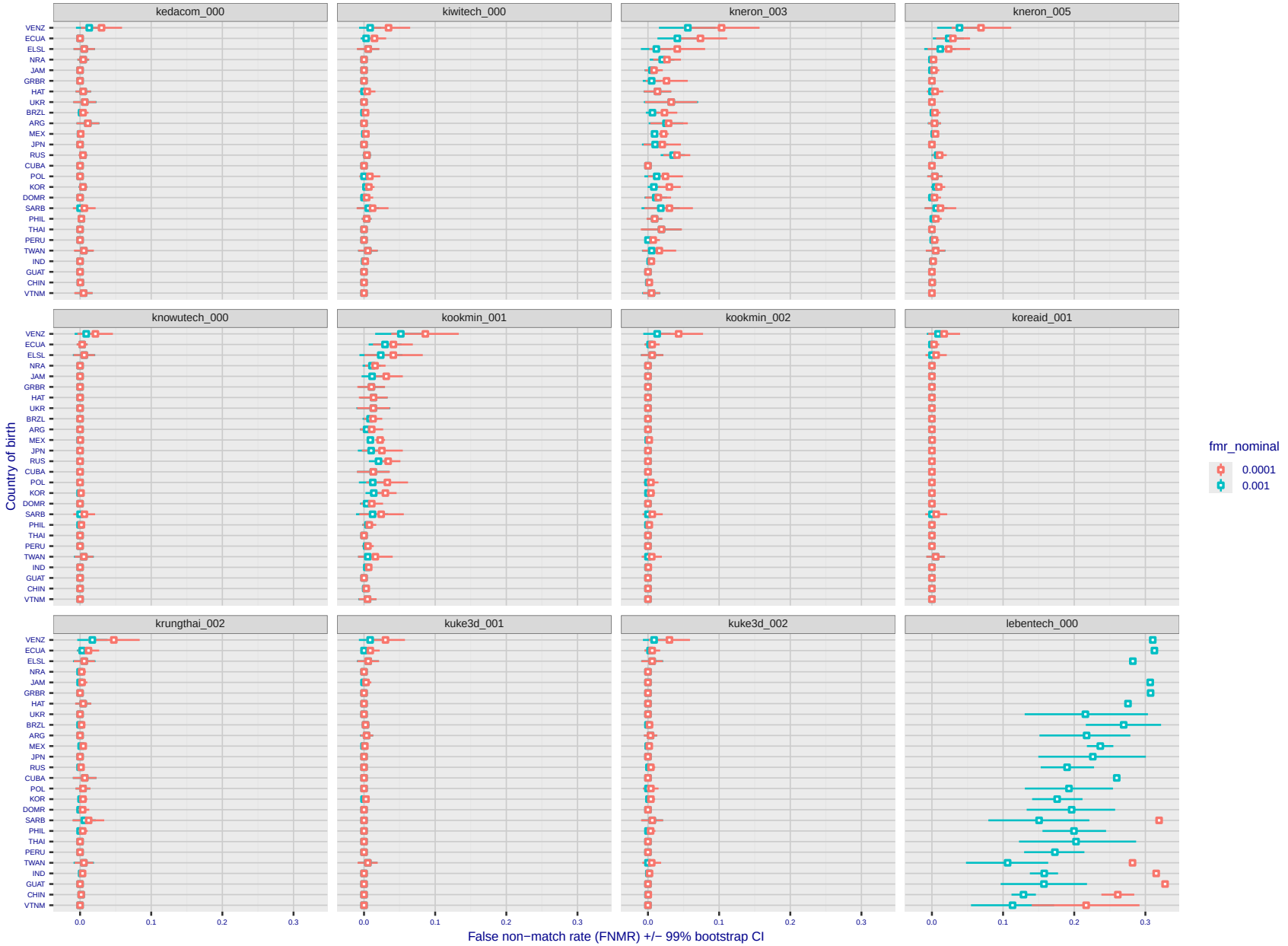

Figure 667: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

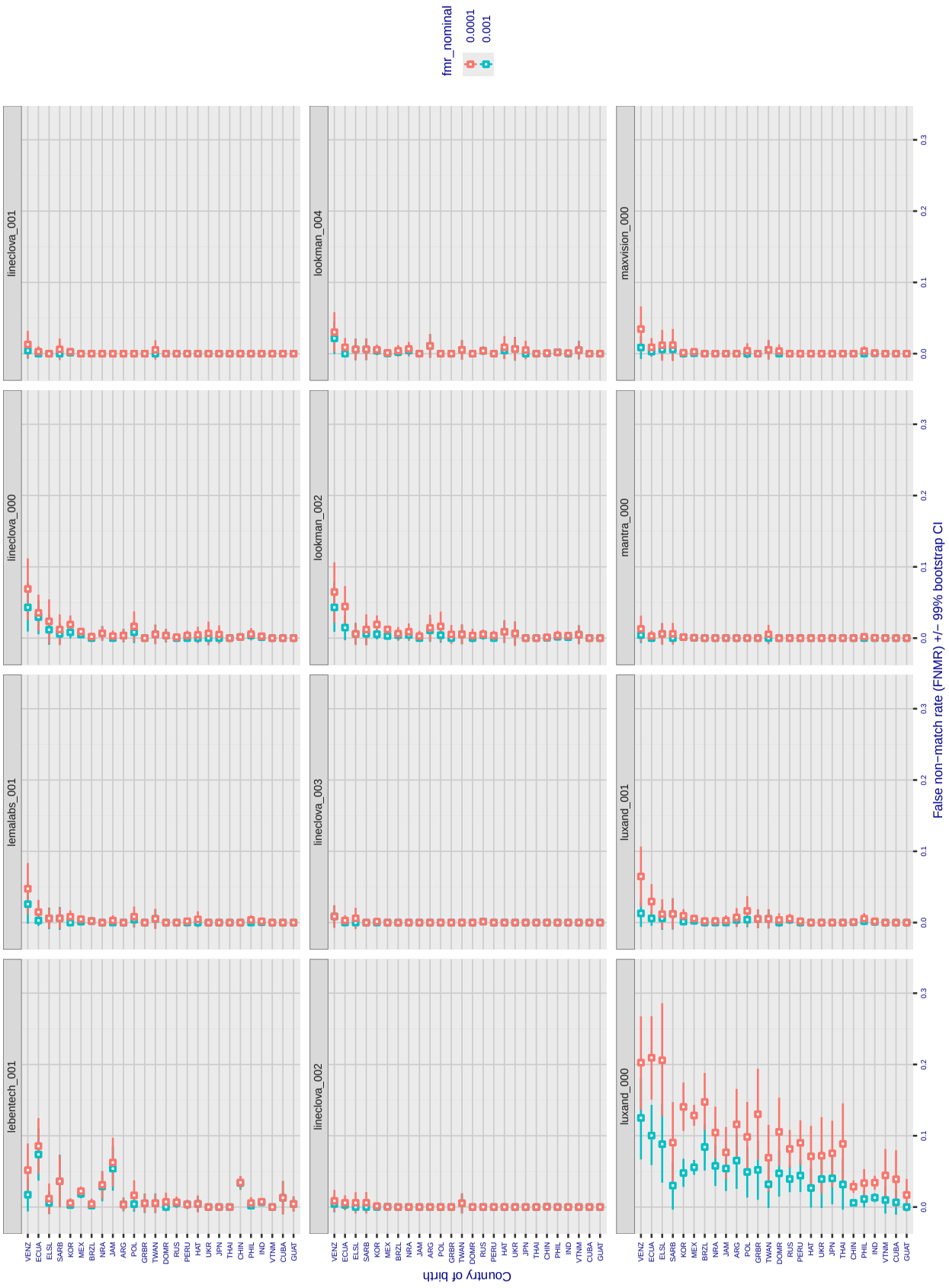

Figure 668: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = {0.001, 0.0001} computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

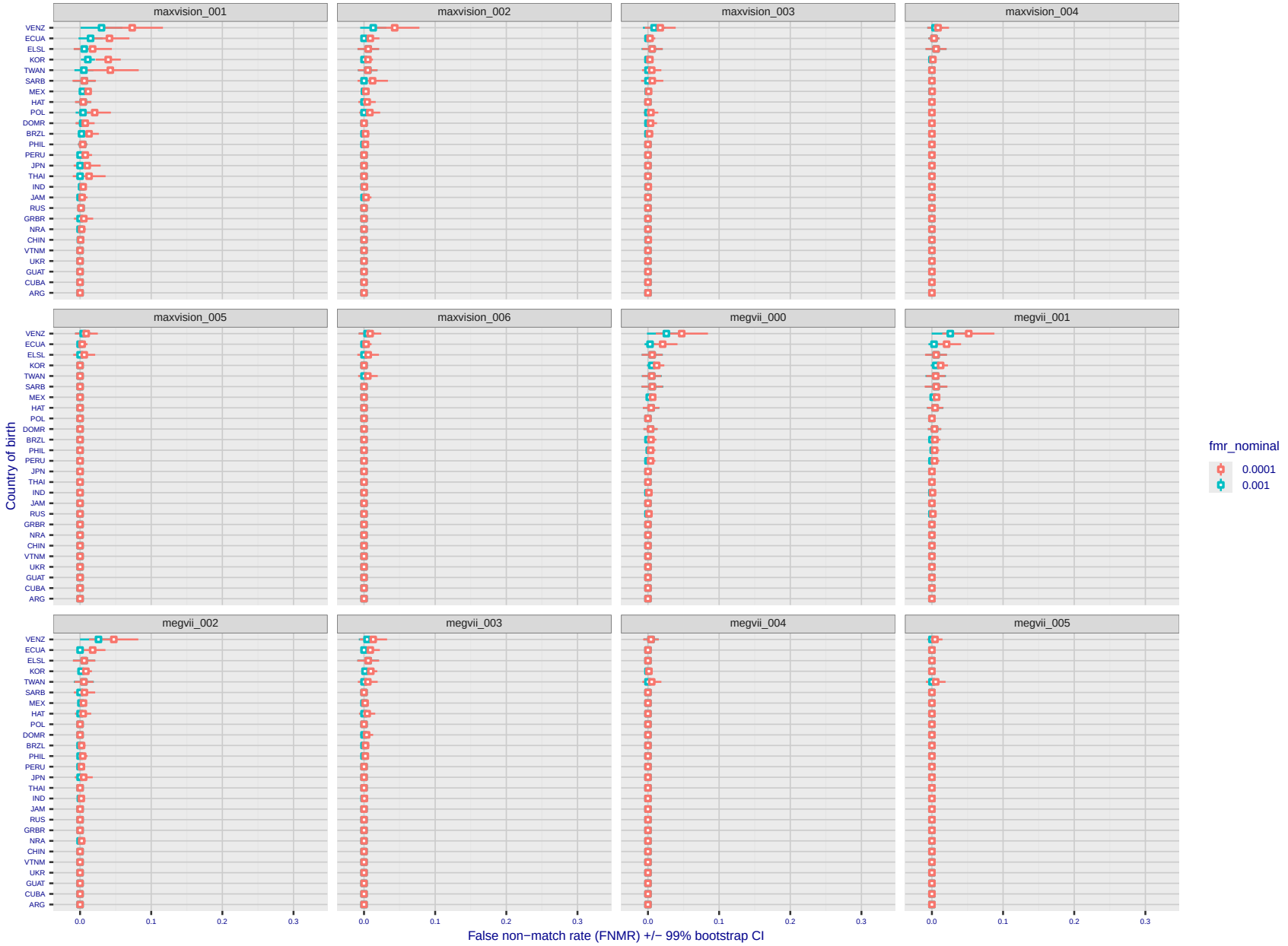

Figure 669: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

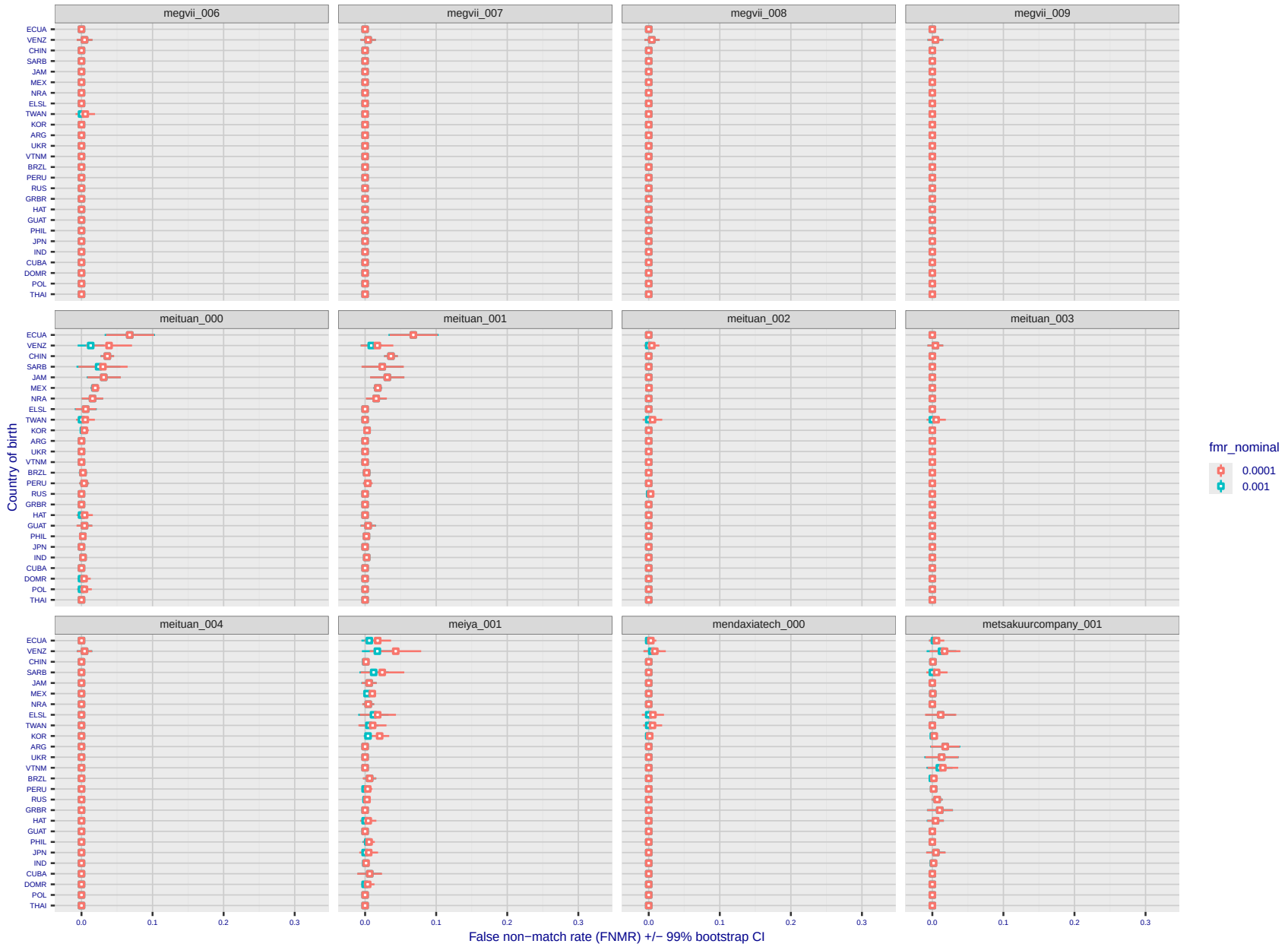

Figure 670: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 671: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 672: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

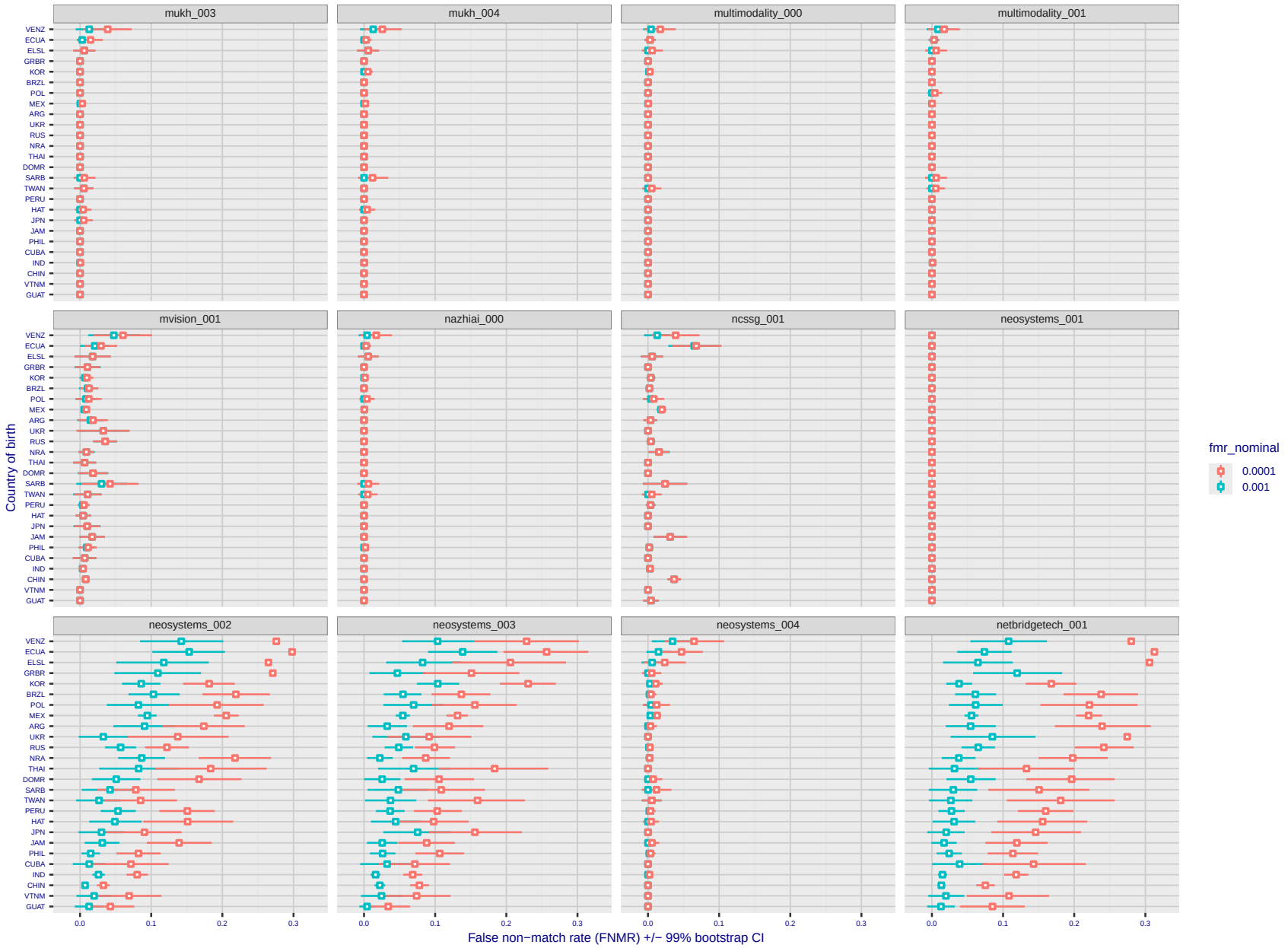

Figure 673: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

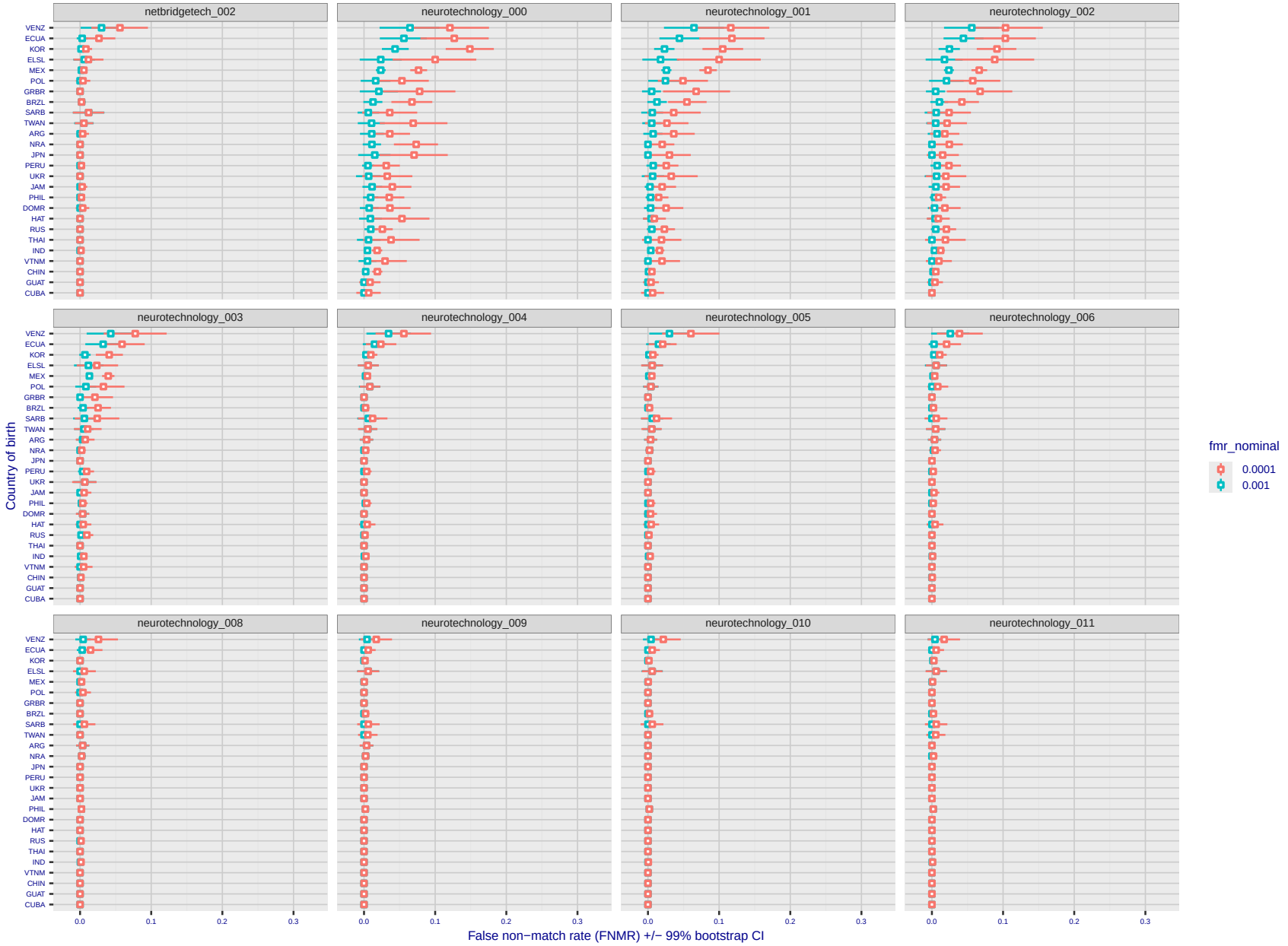

Figure 674: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

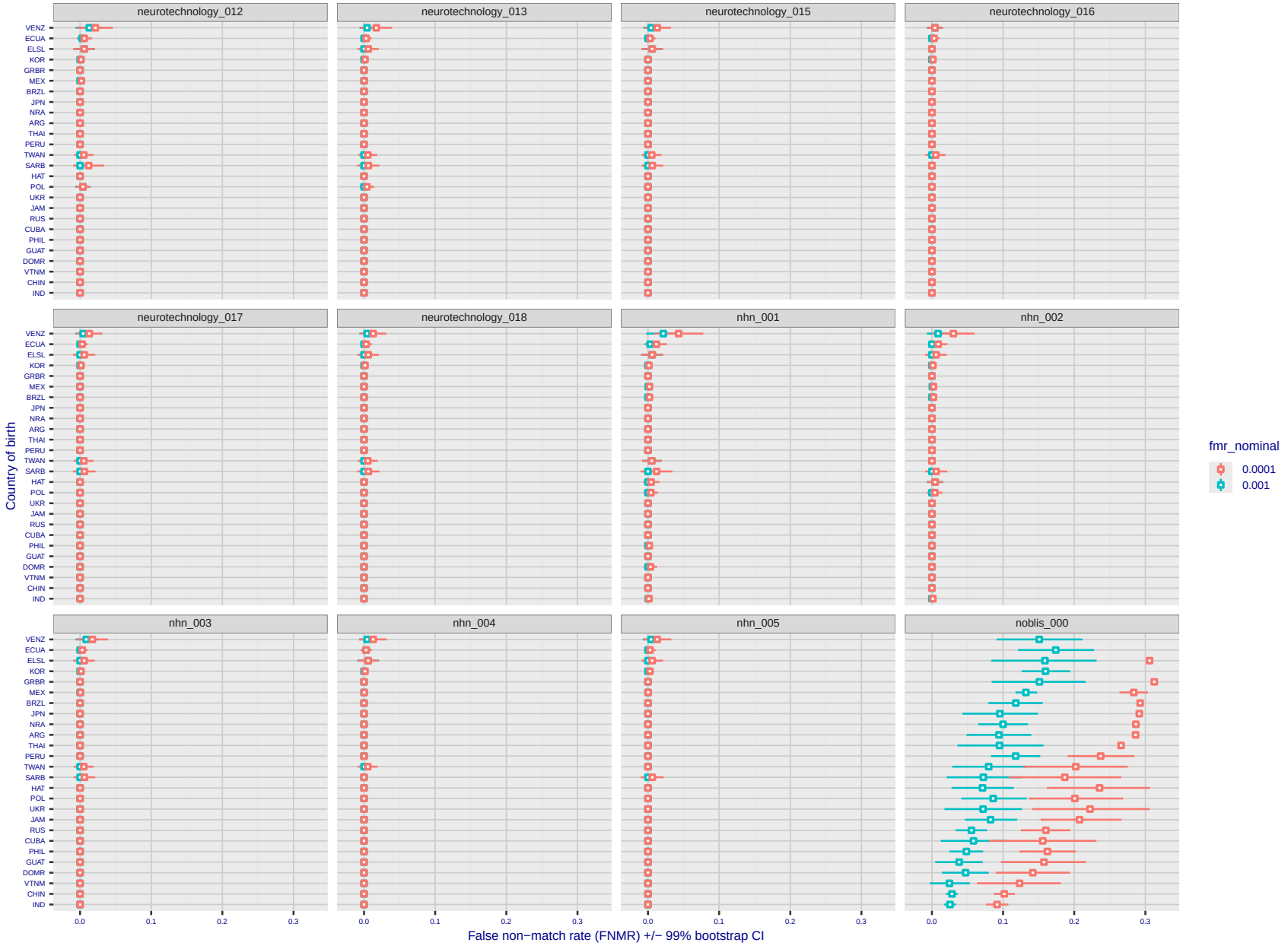

Figure 675: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 676: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

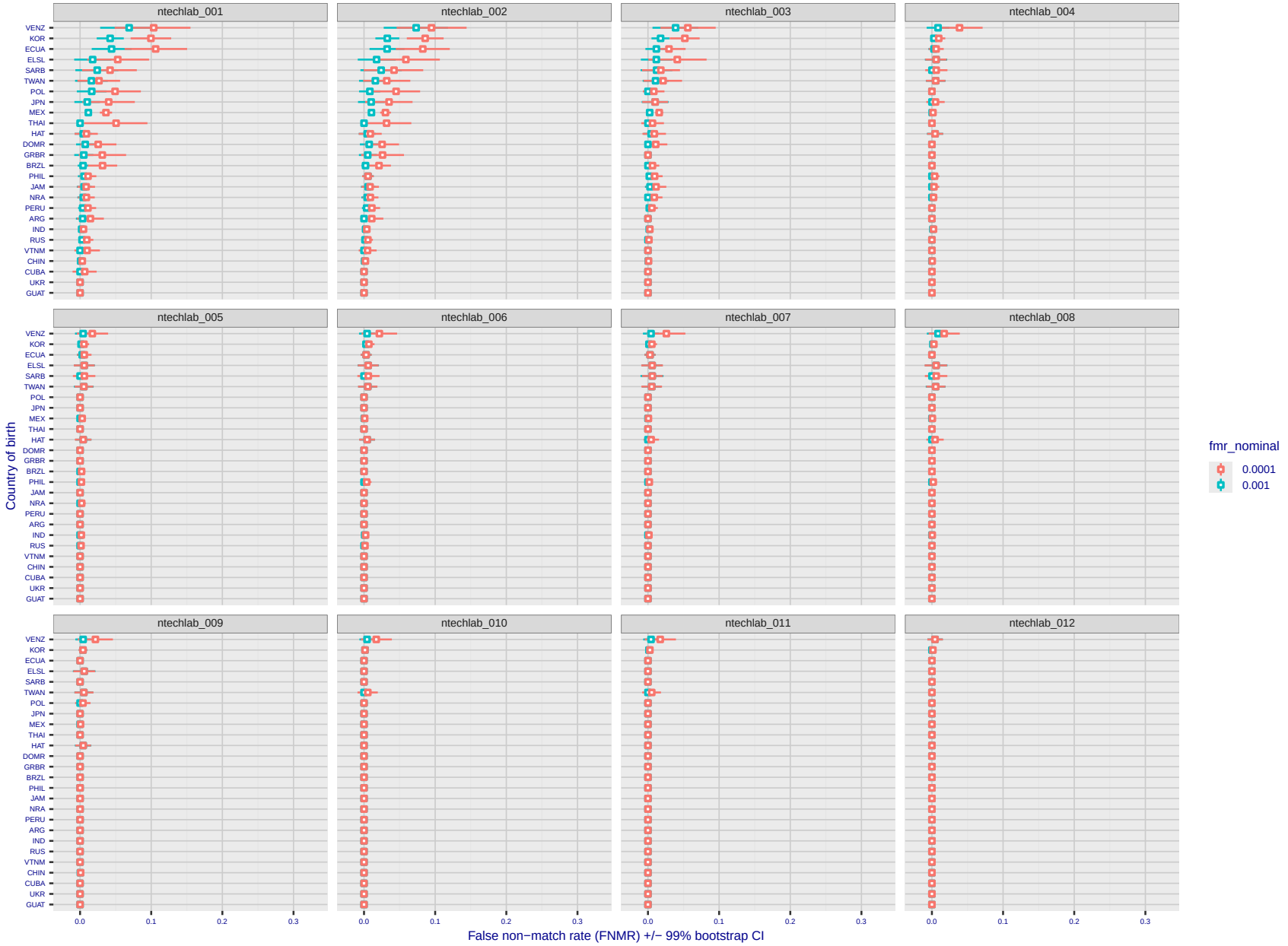

Figure 677: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 678: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 679: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

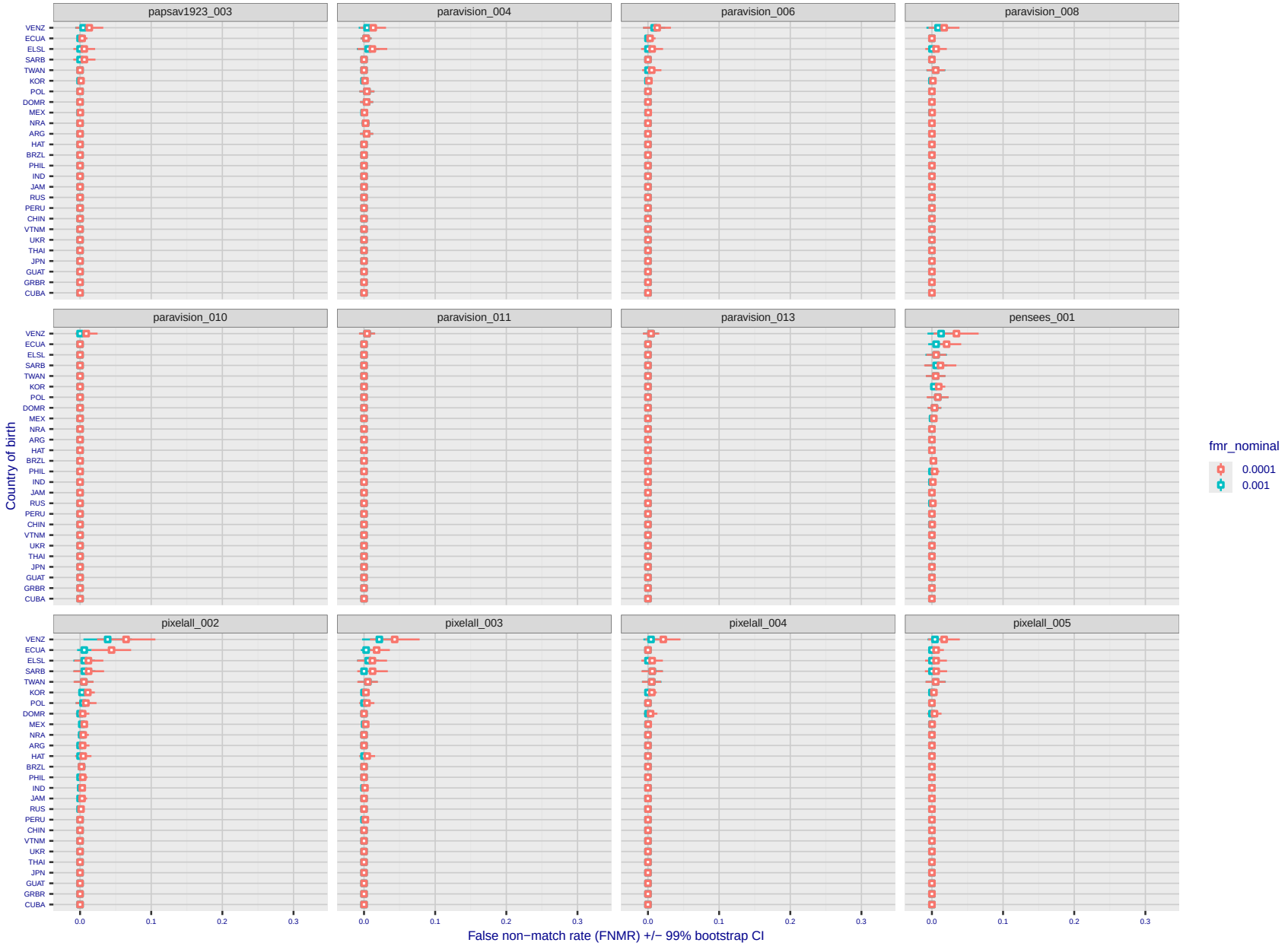

Figure 680: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

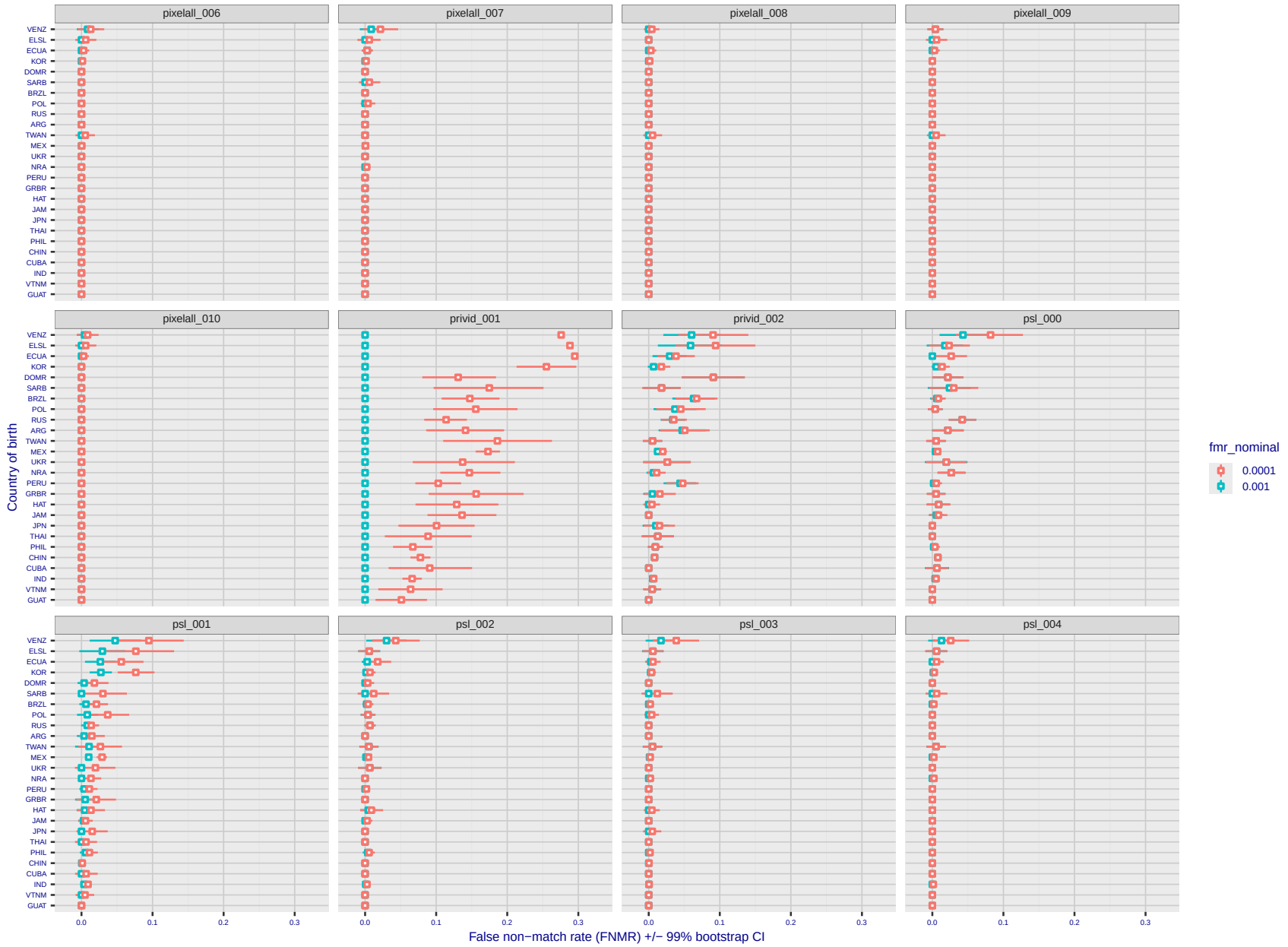

Figure 681: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

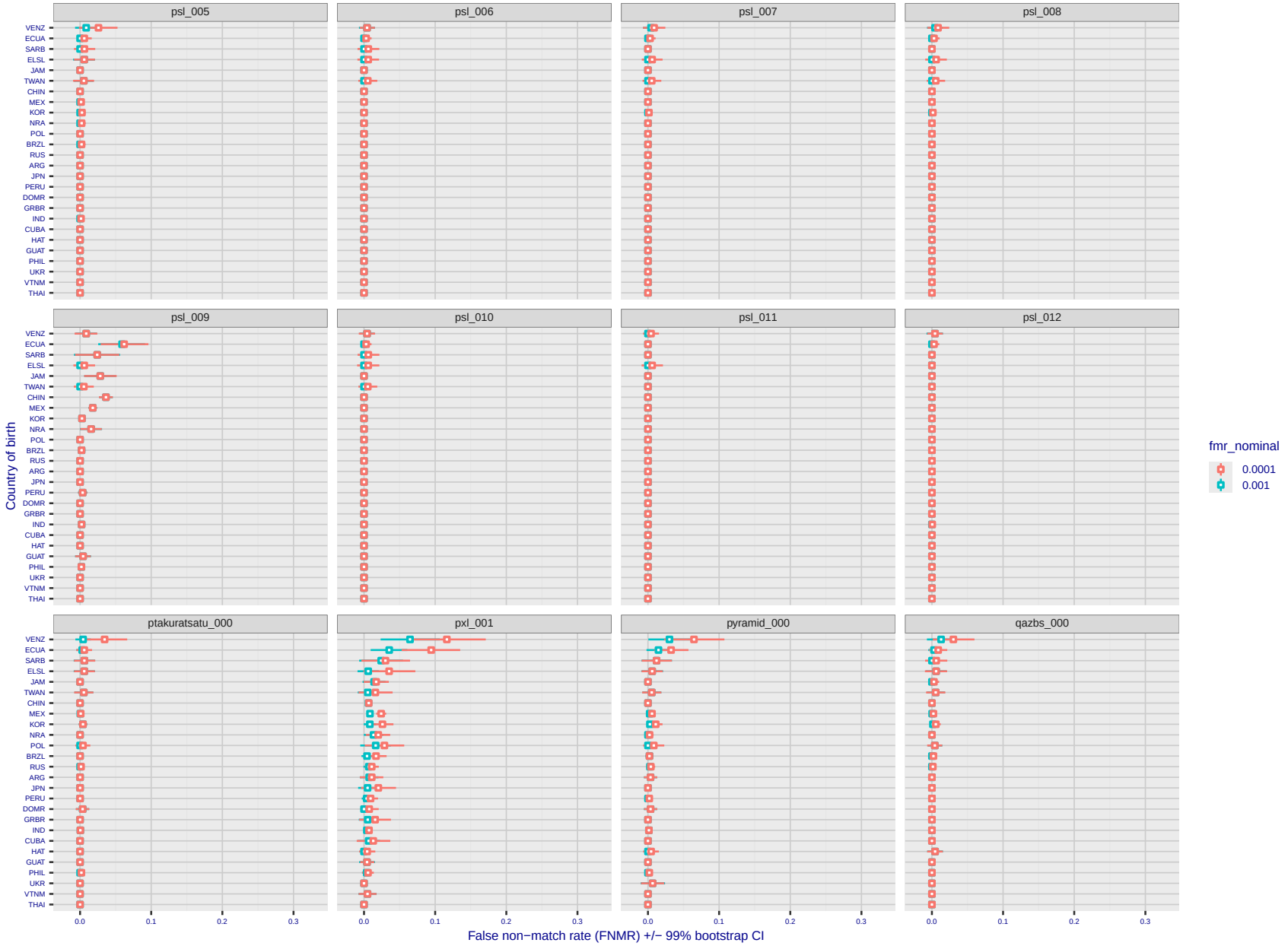

Figure 682: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 683: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

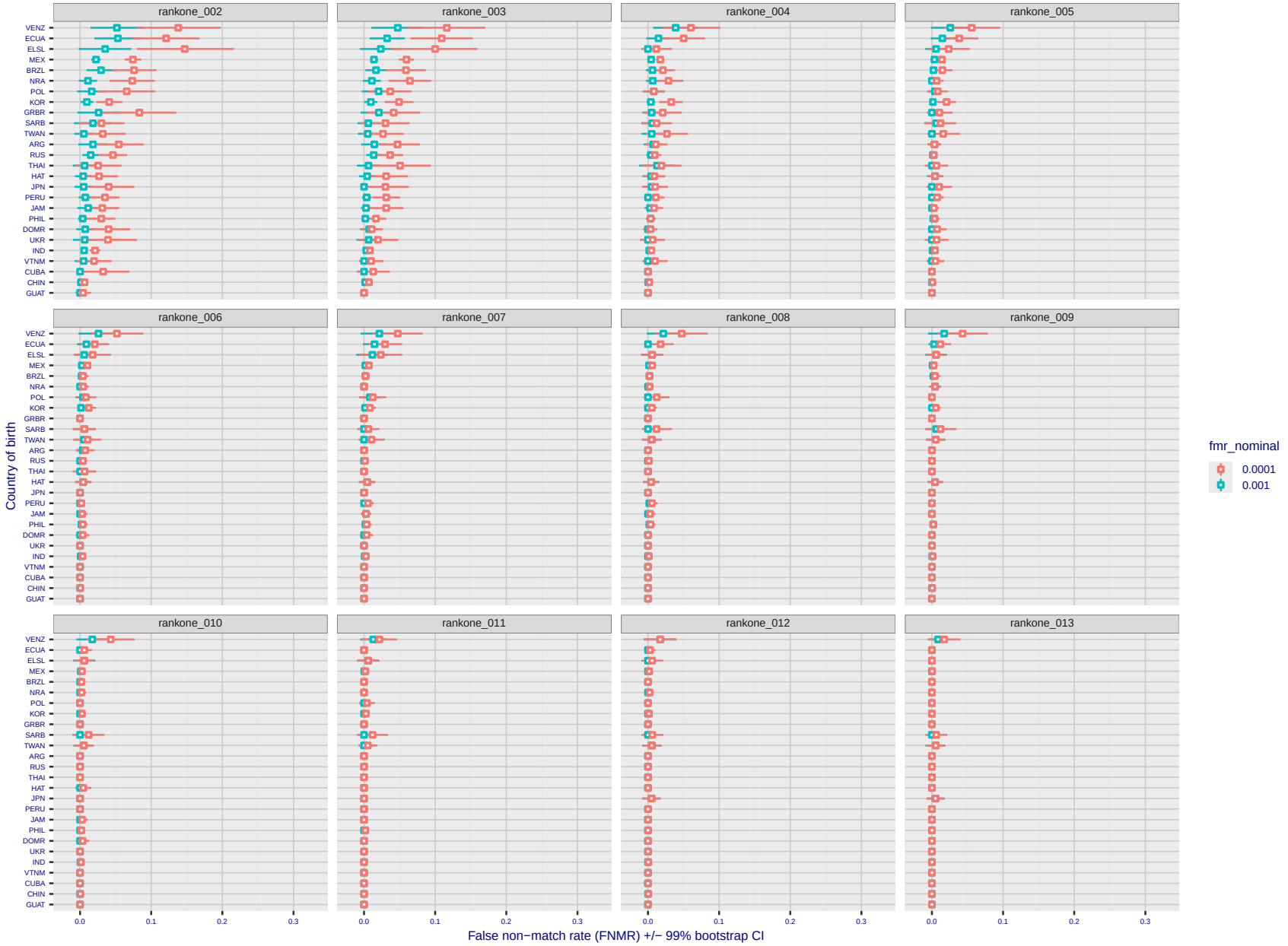

Figure 684: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

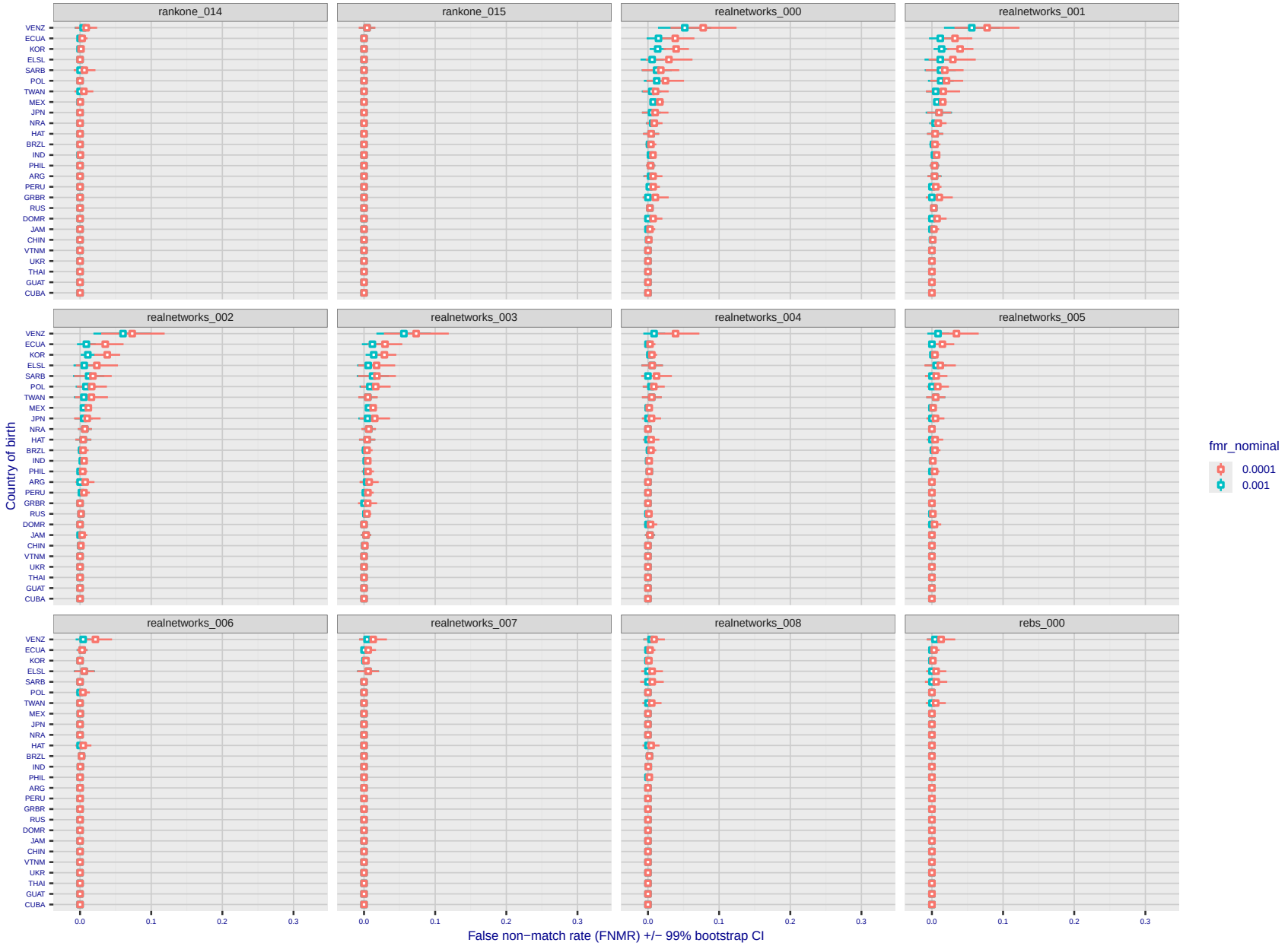

Figure 685: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

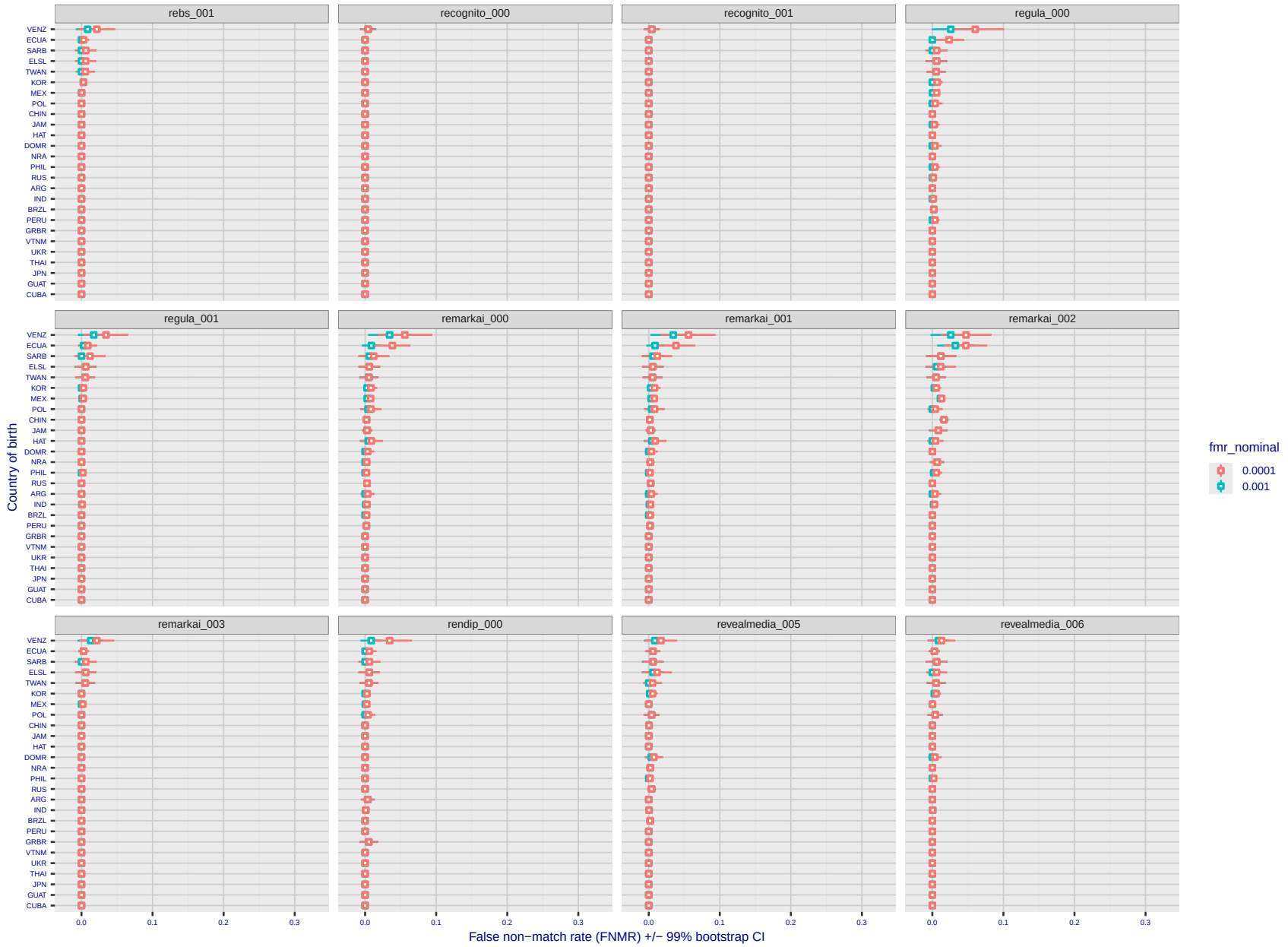

Figure 686: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

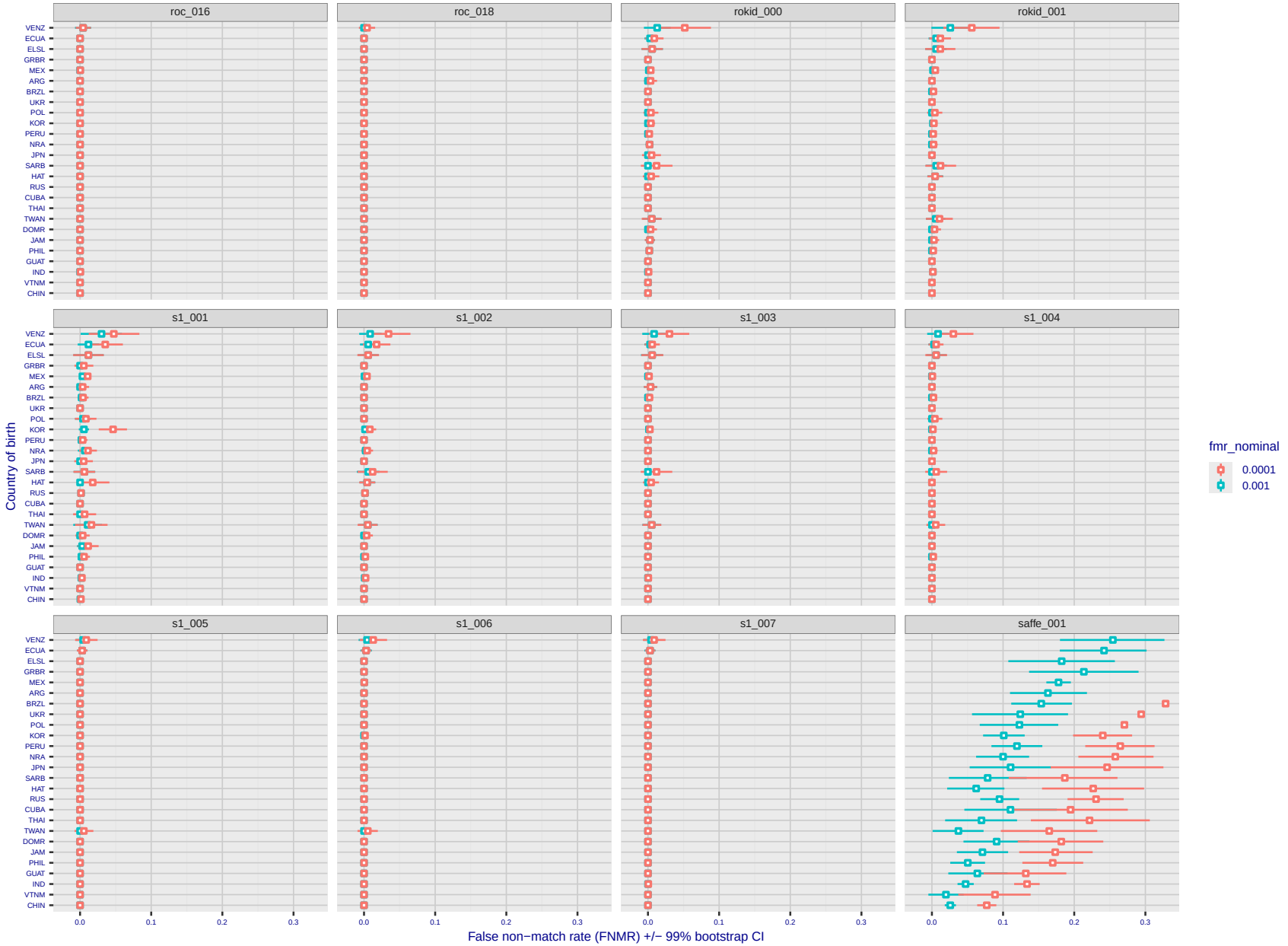

Figure 687: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

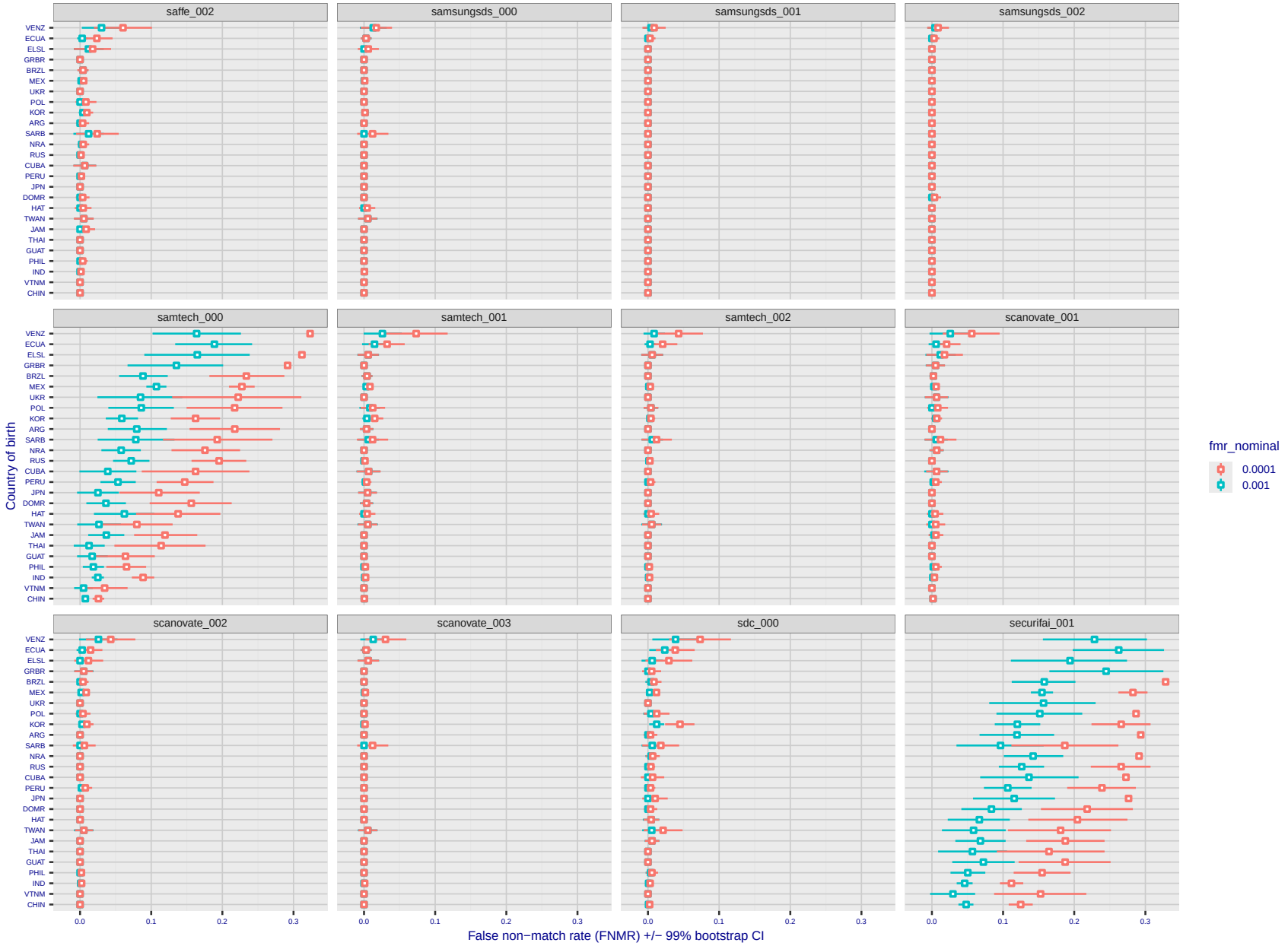

Figure 688: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 689: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

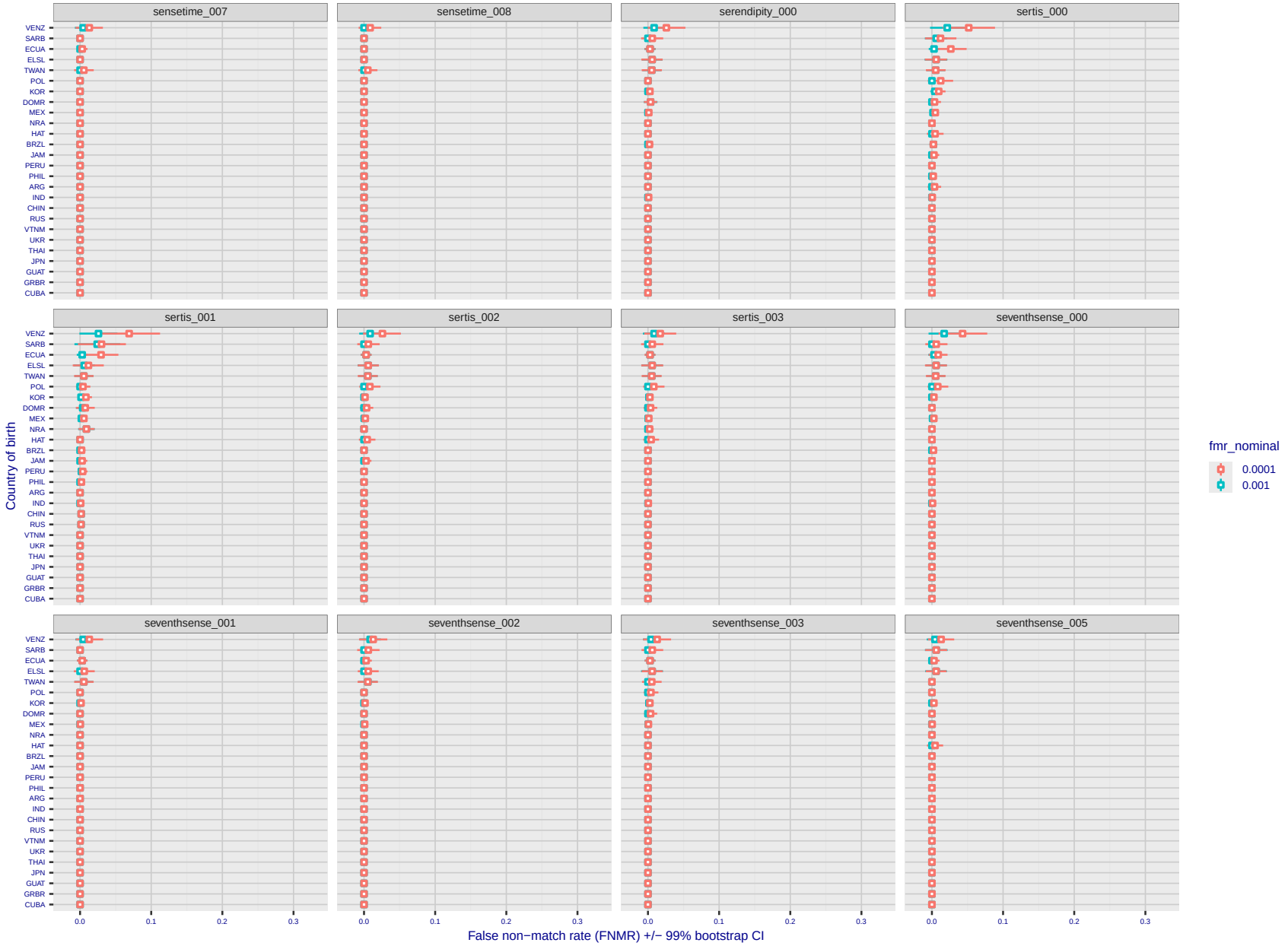

Figure 690: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 691: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 692: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

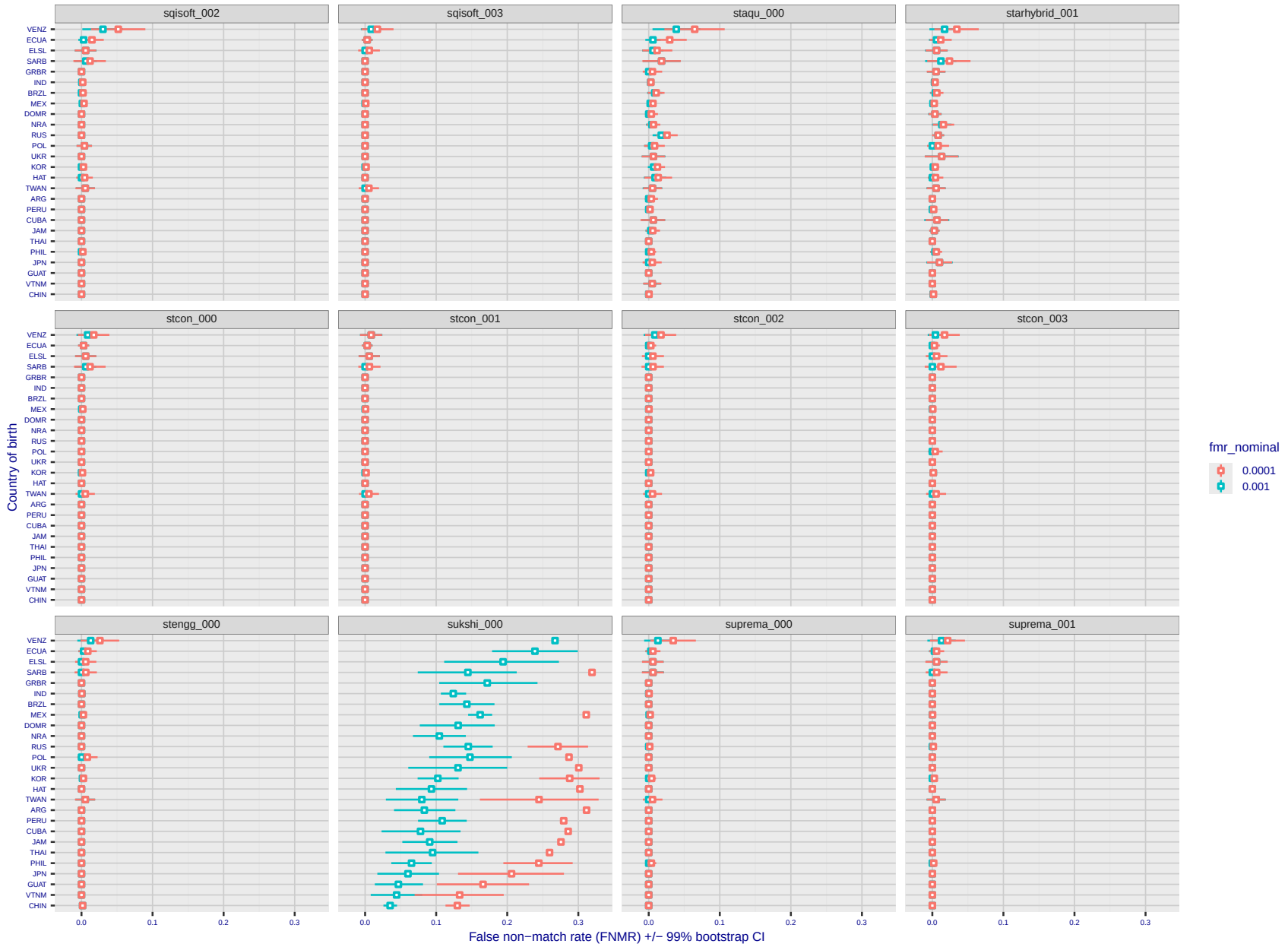

Figure 693: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

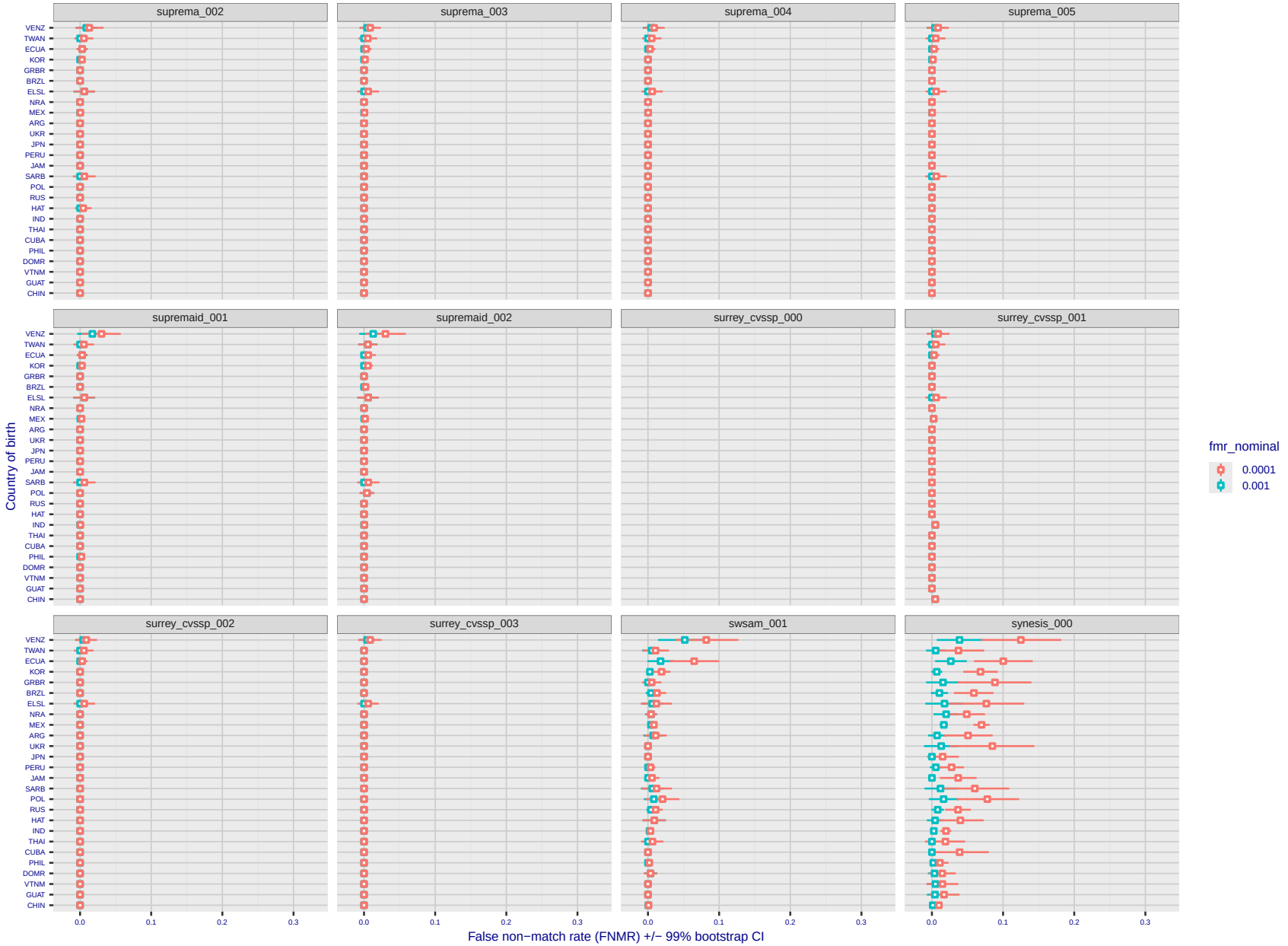

Figure 694: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 695: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

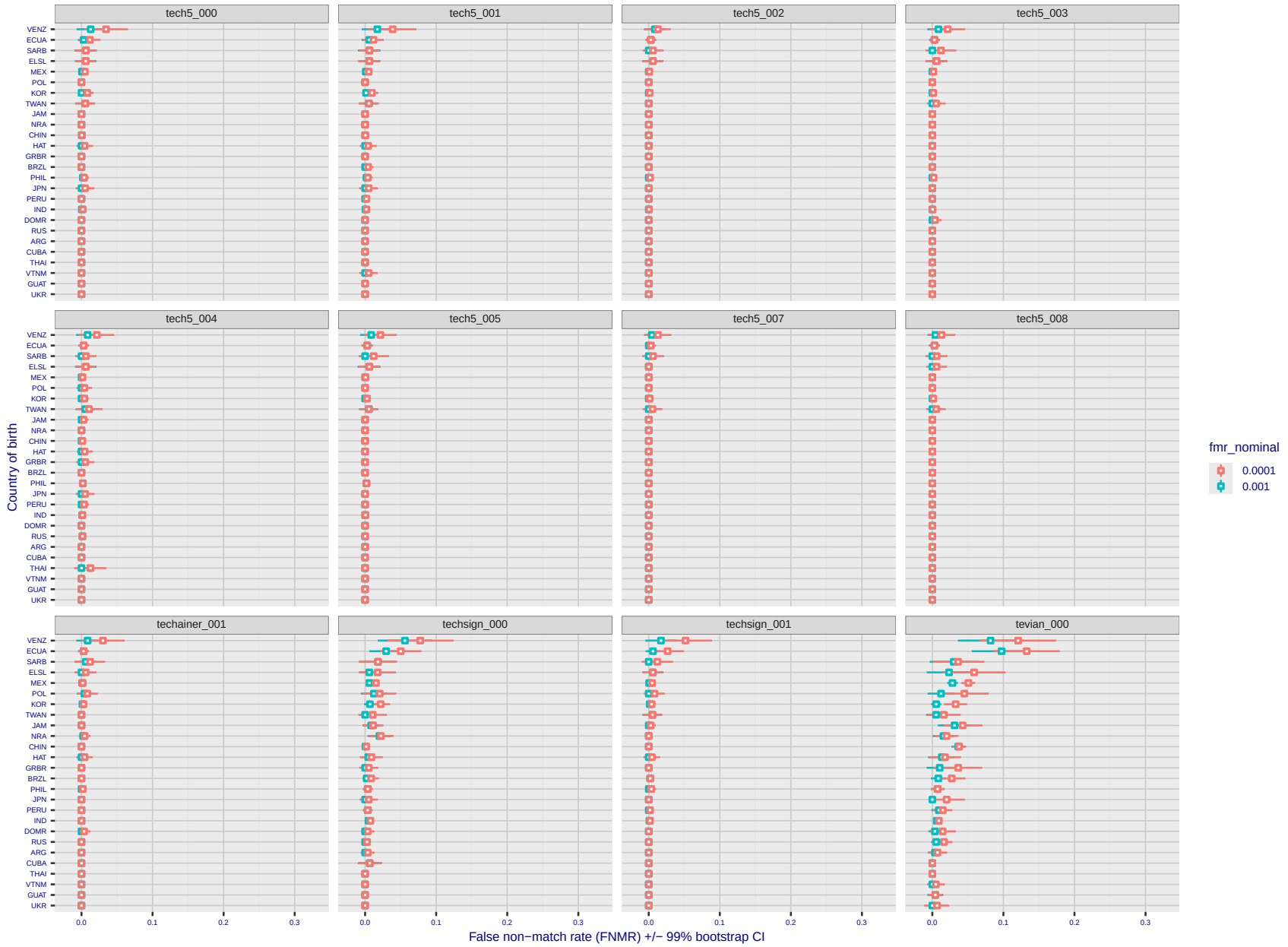

Figure 696: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

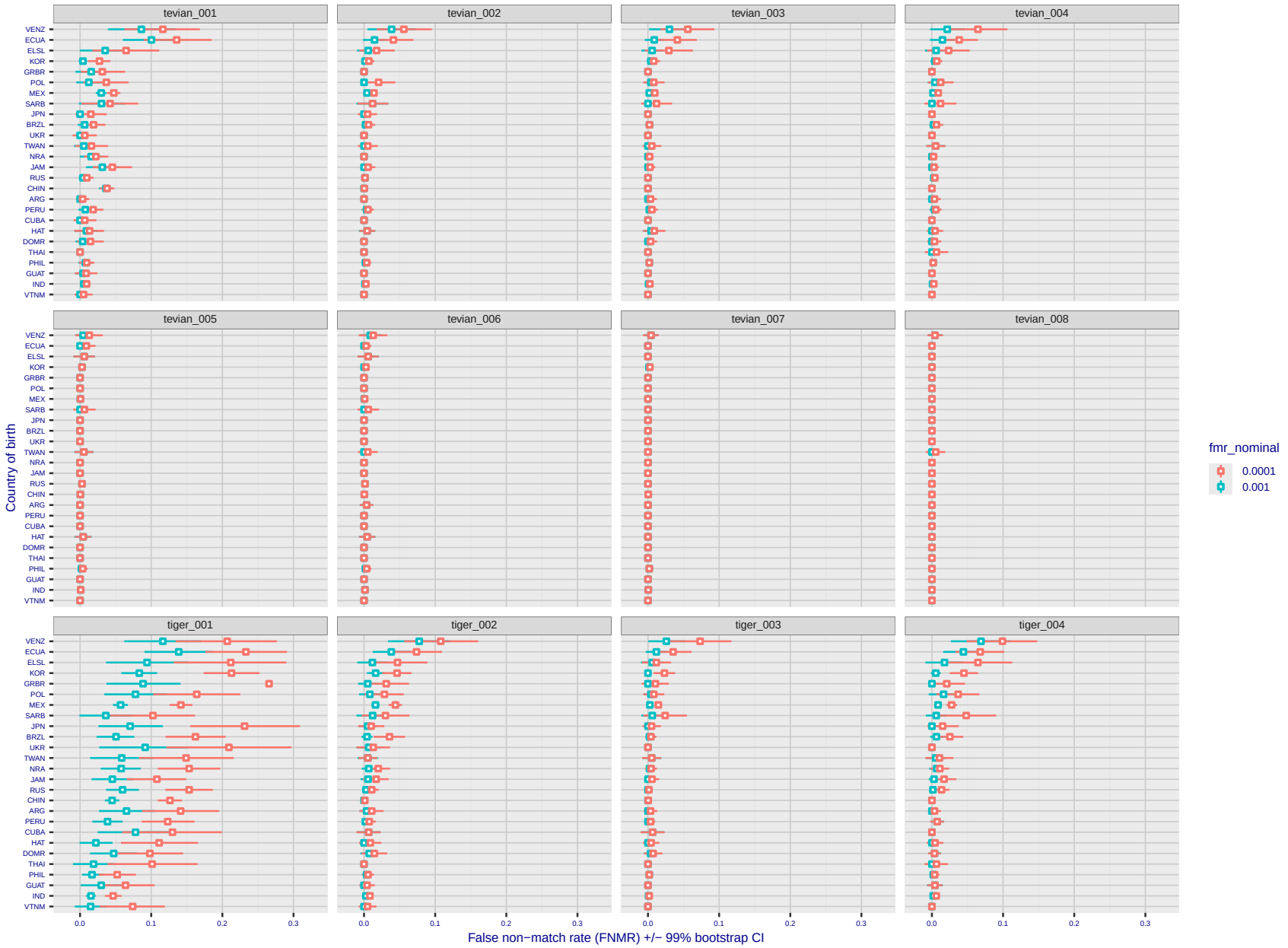

Figure 697: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

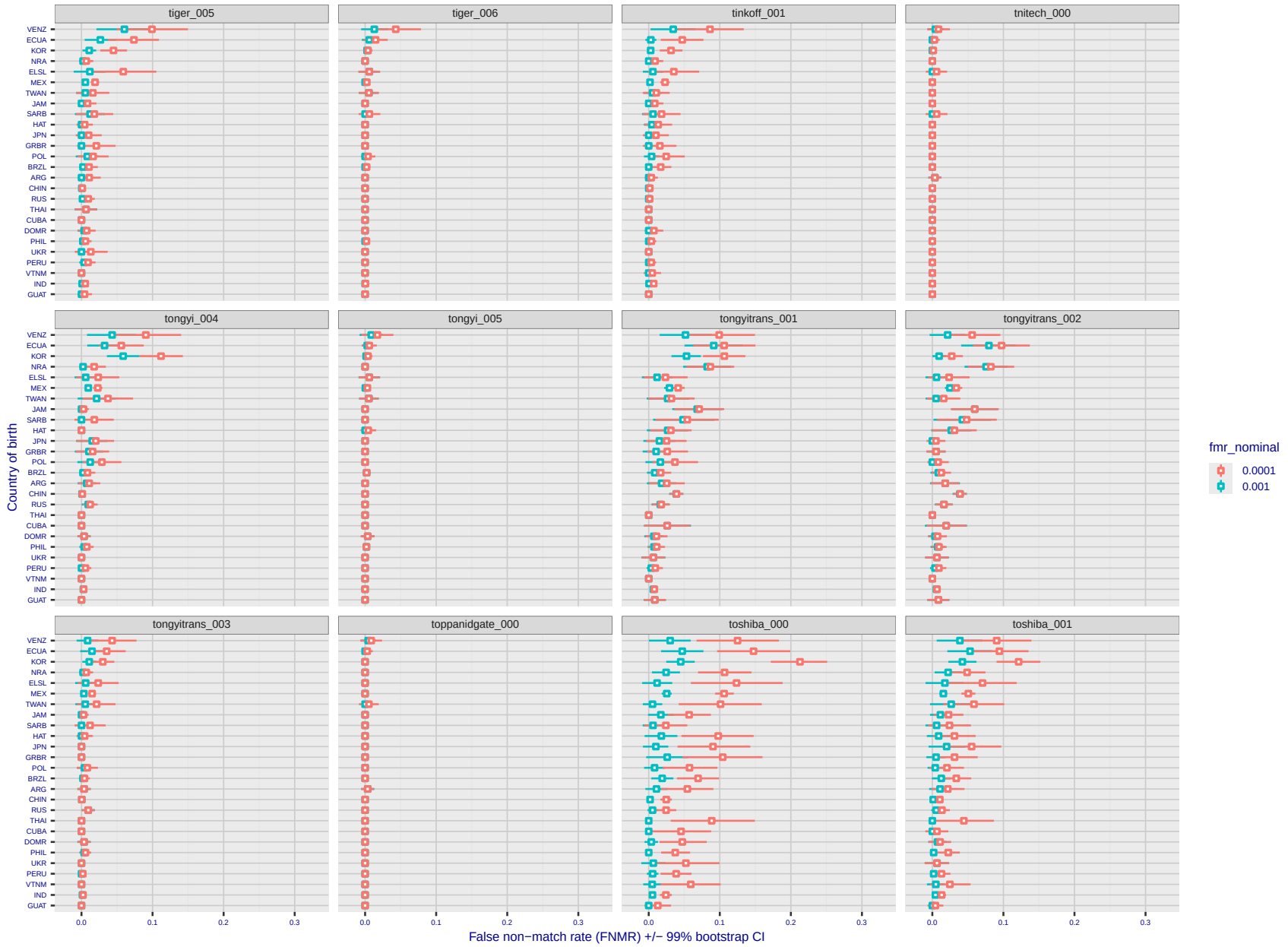

Figure 698: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

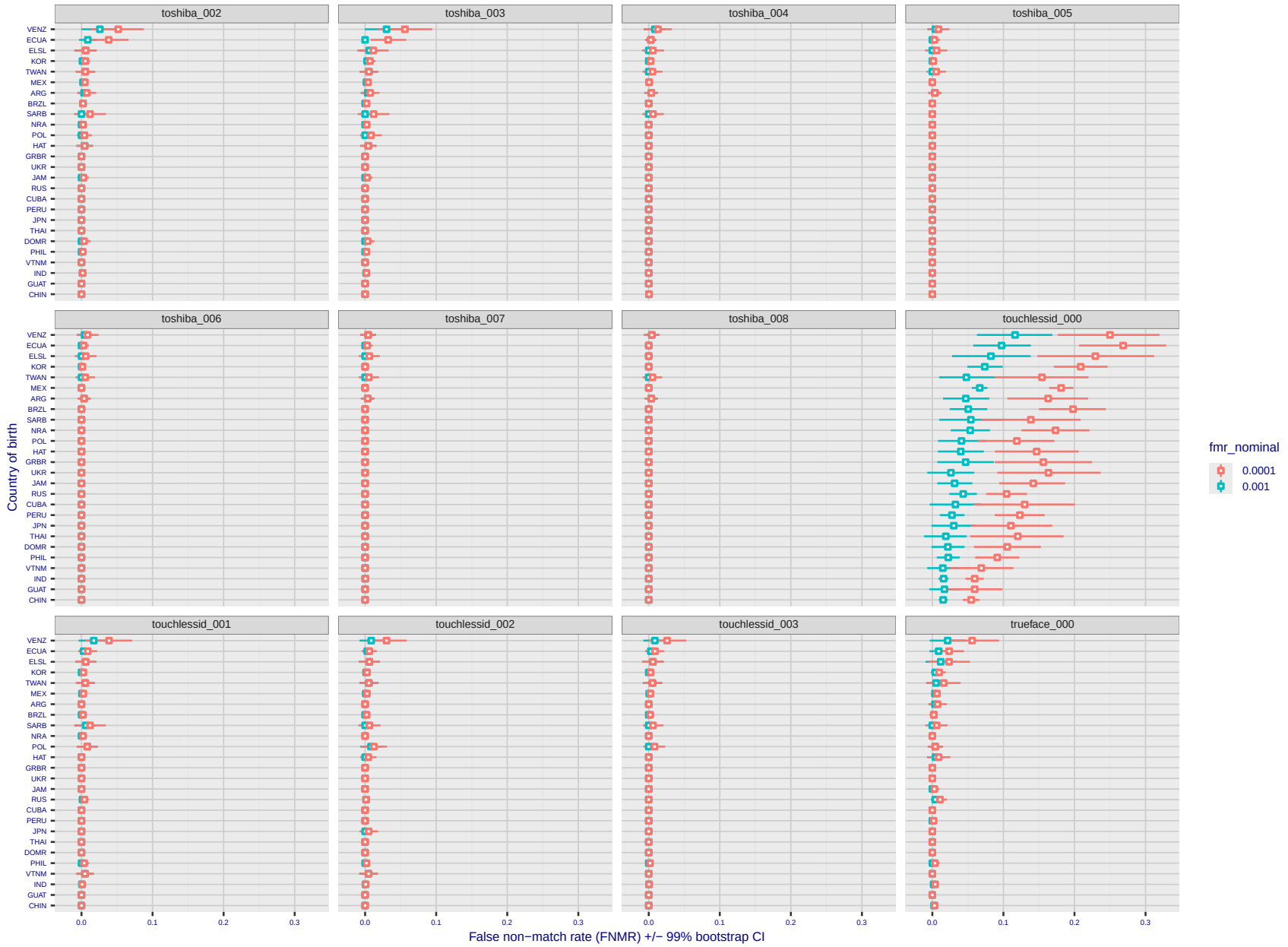

Figure 699: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 700: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

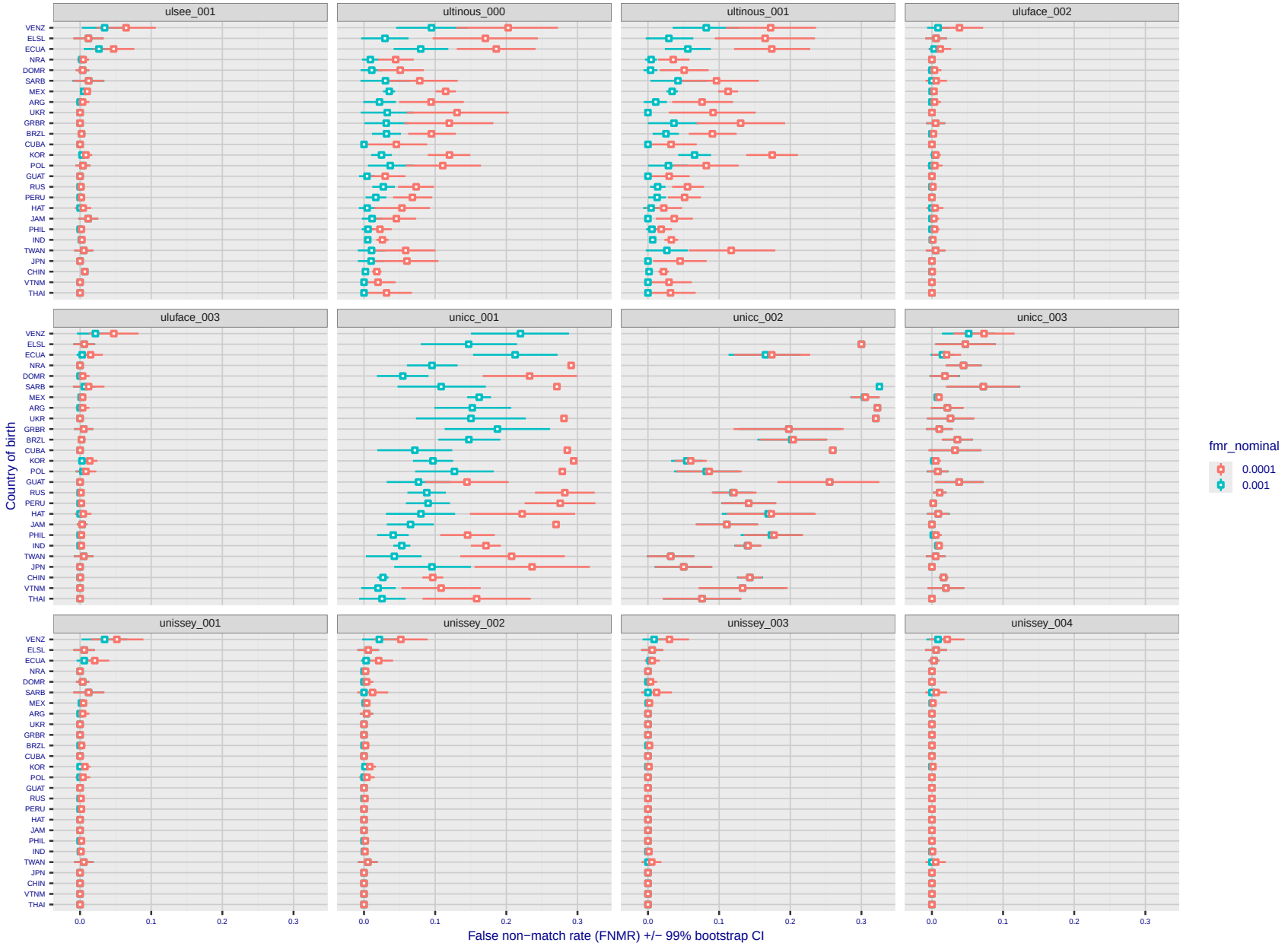

Figure 701: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 702: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

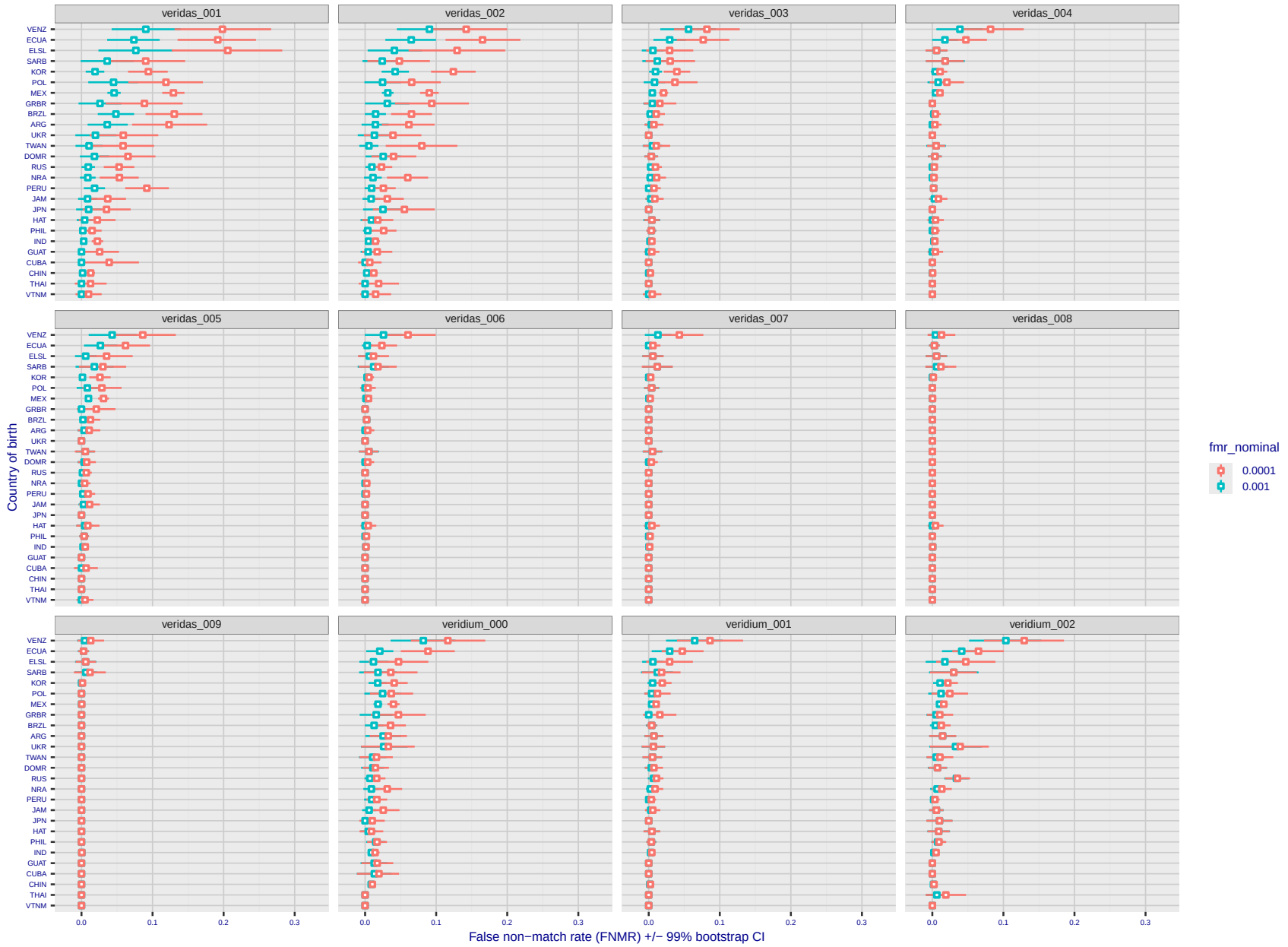

Figure 703: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

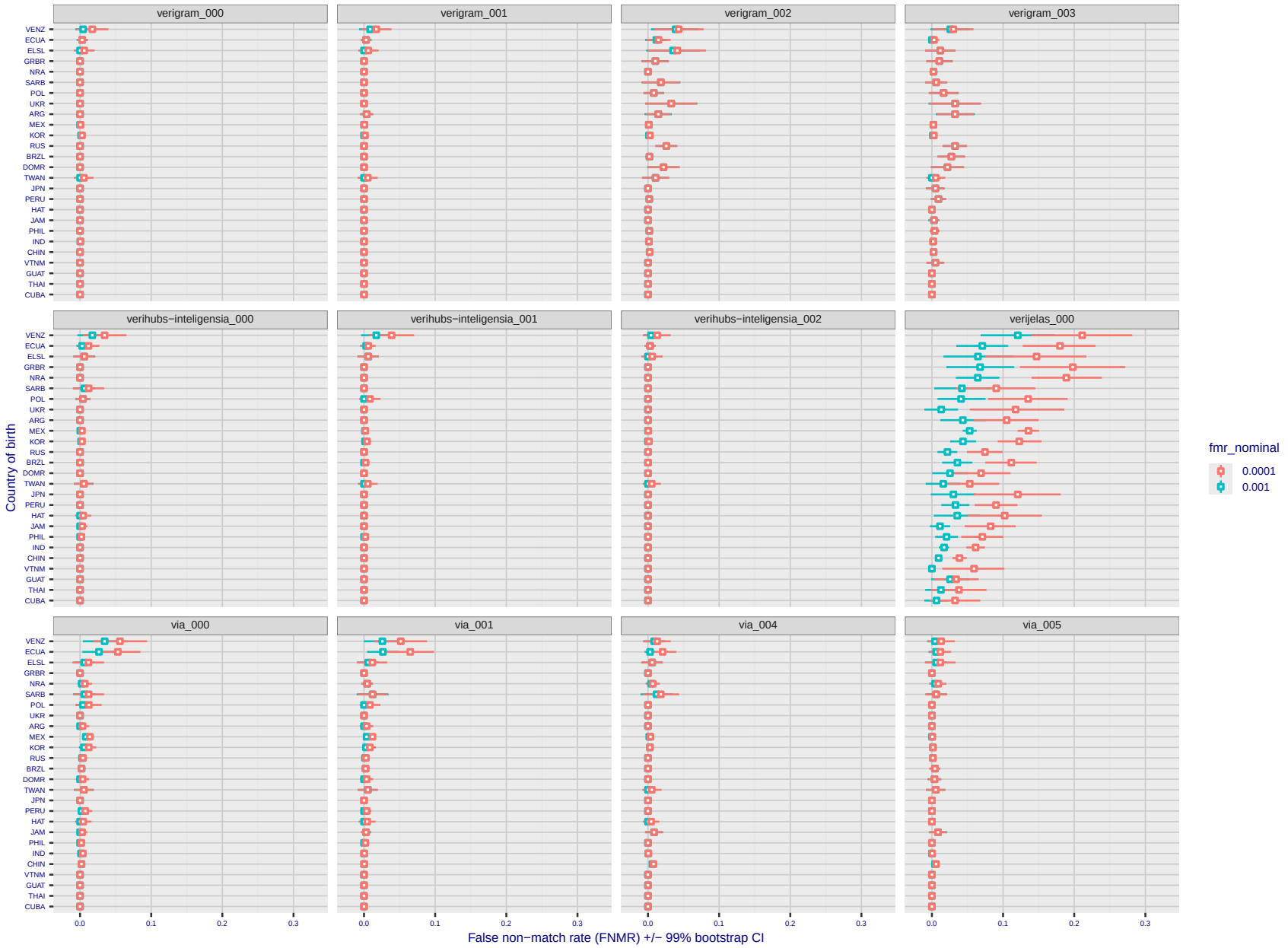

Figure 704: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

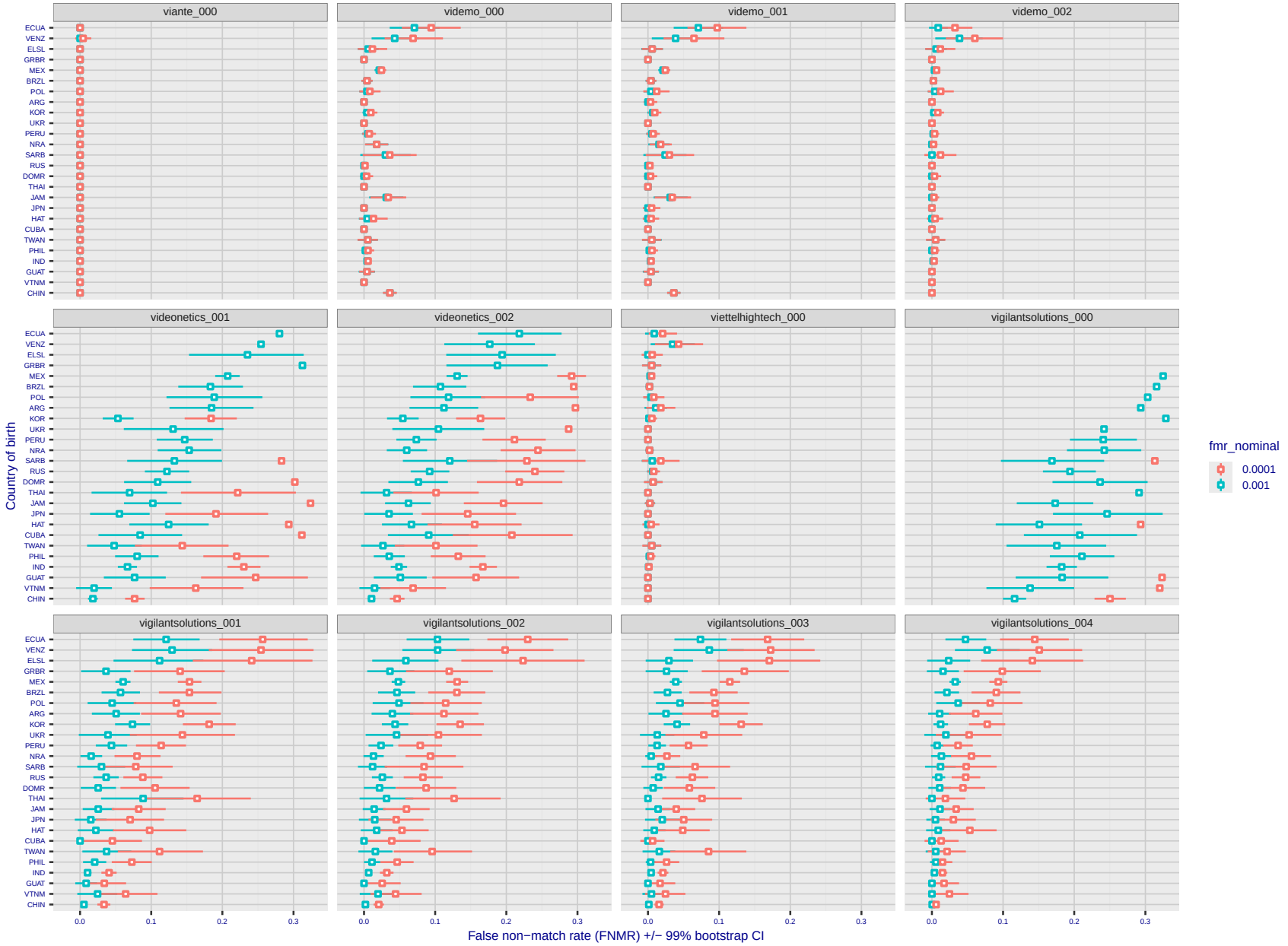

Figure 705: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 706: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

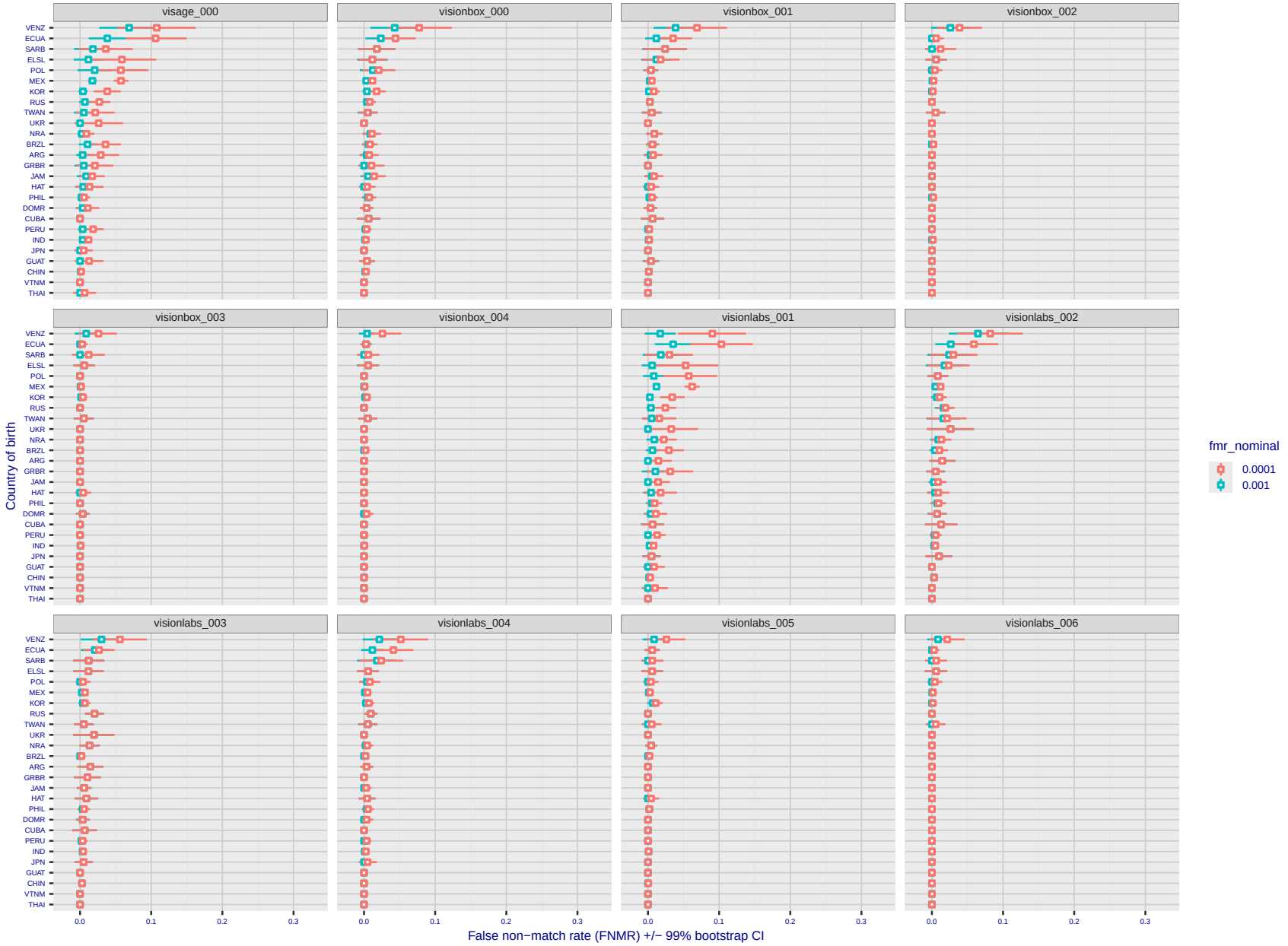

Figure 707: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

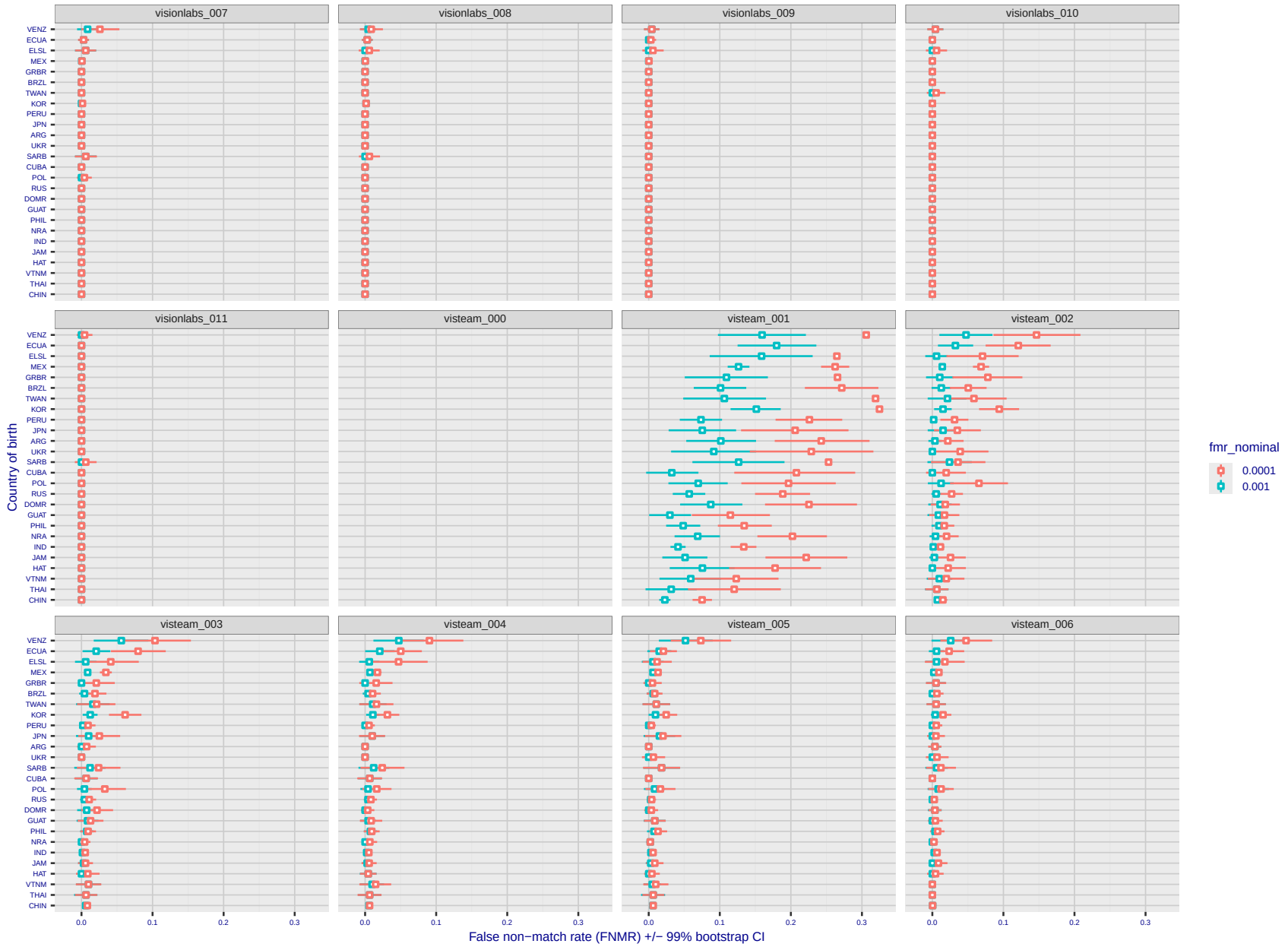

Figure 708: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 709: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

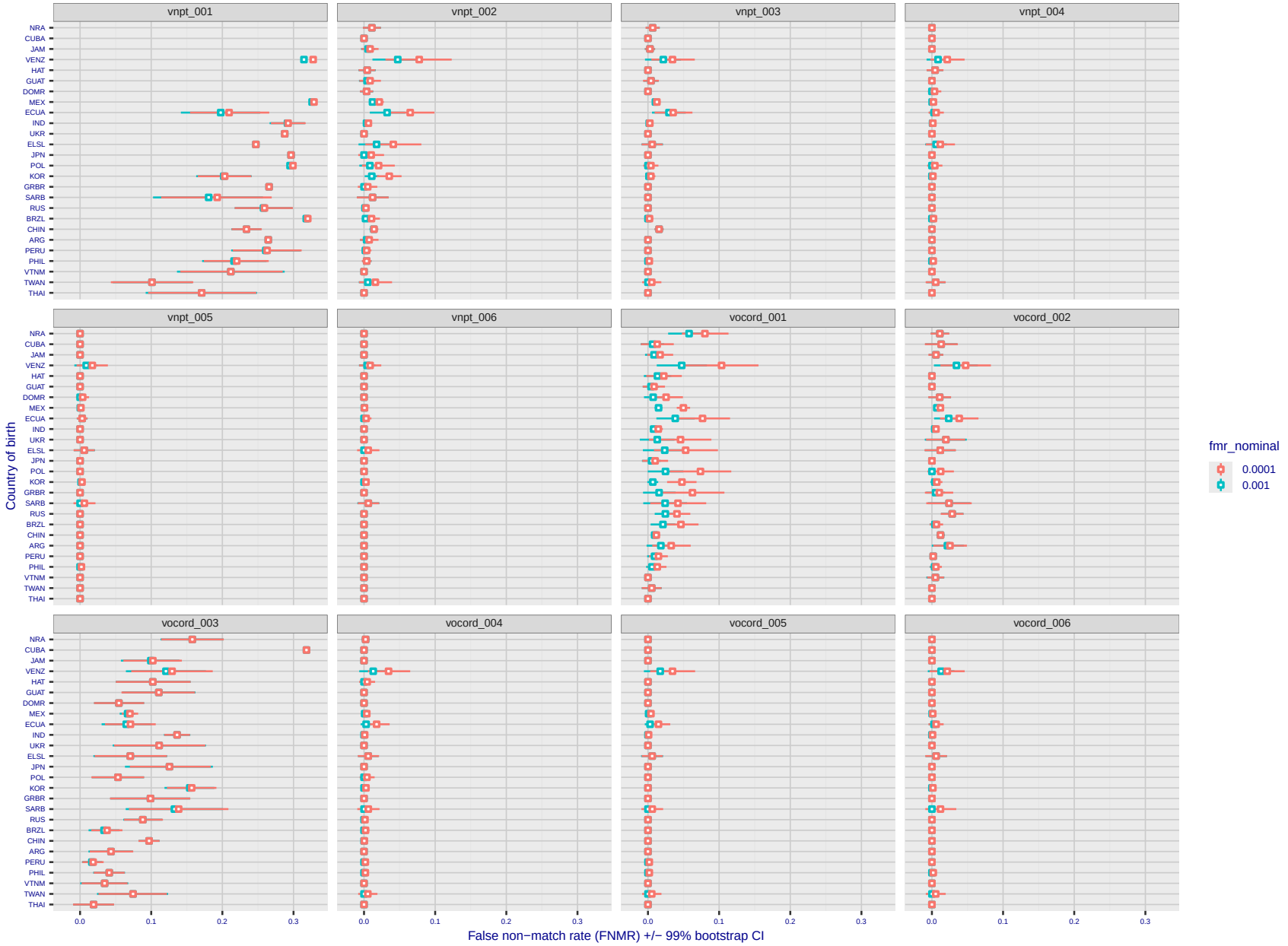

Figure 710: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

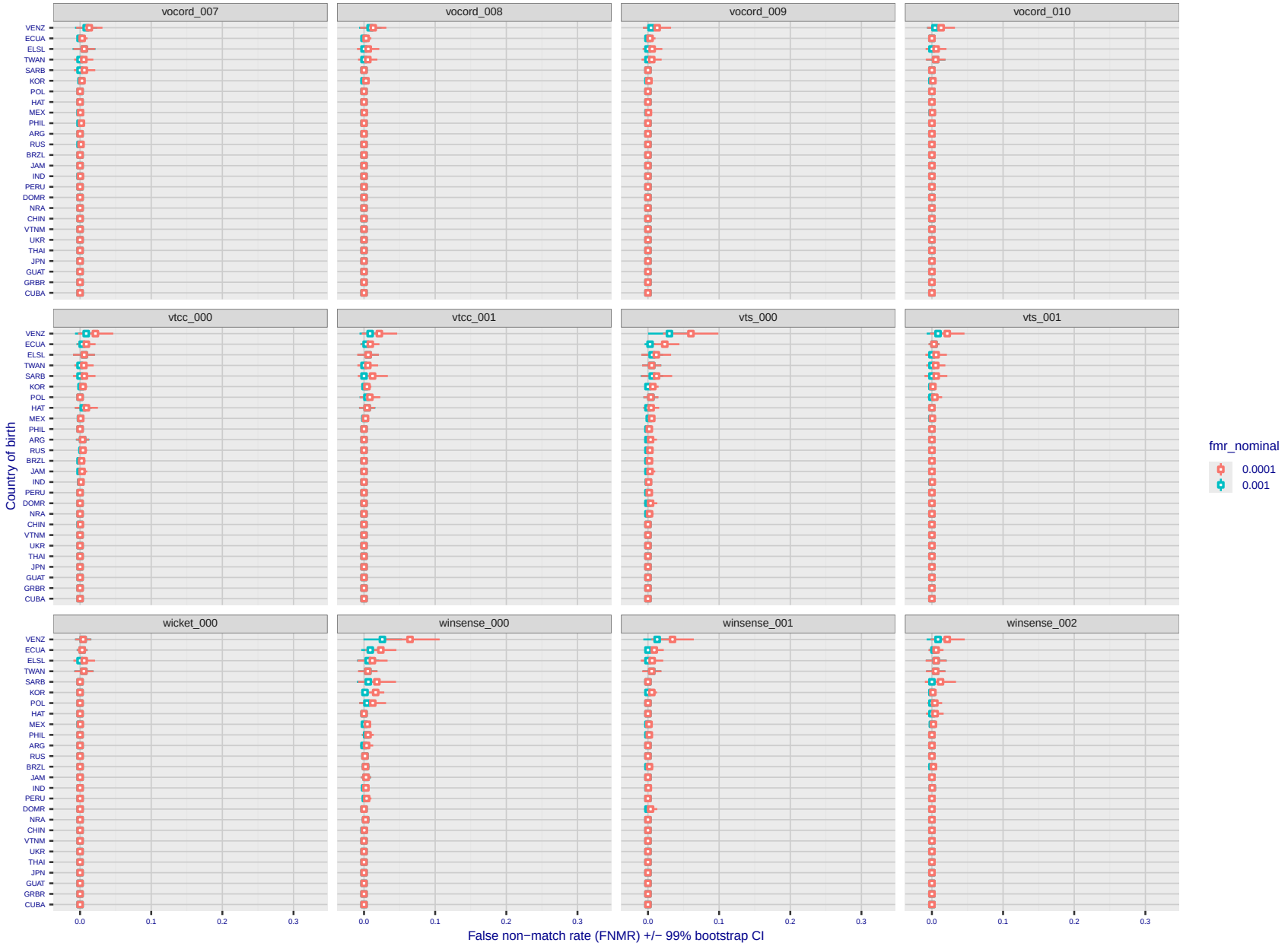

Figure 711: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

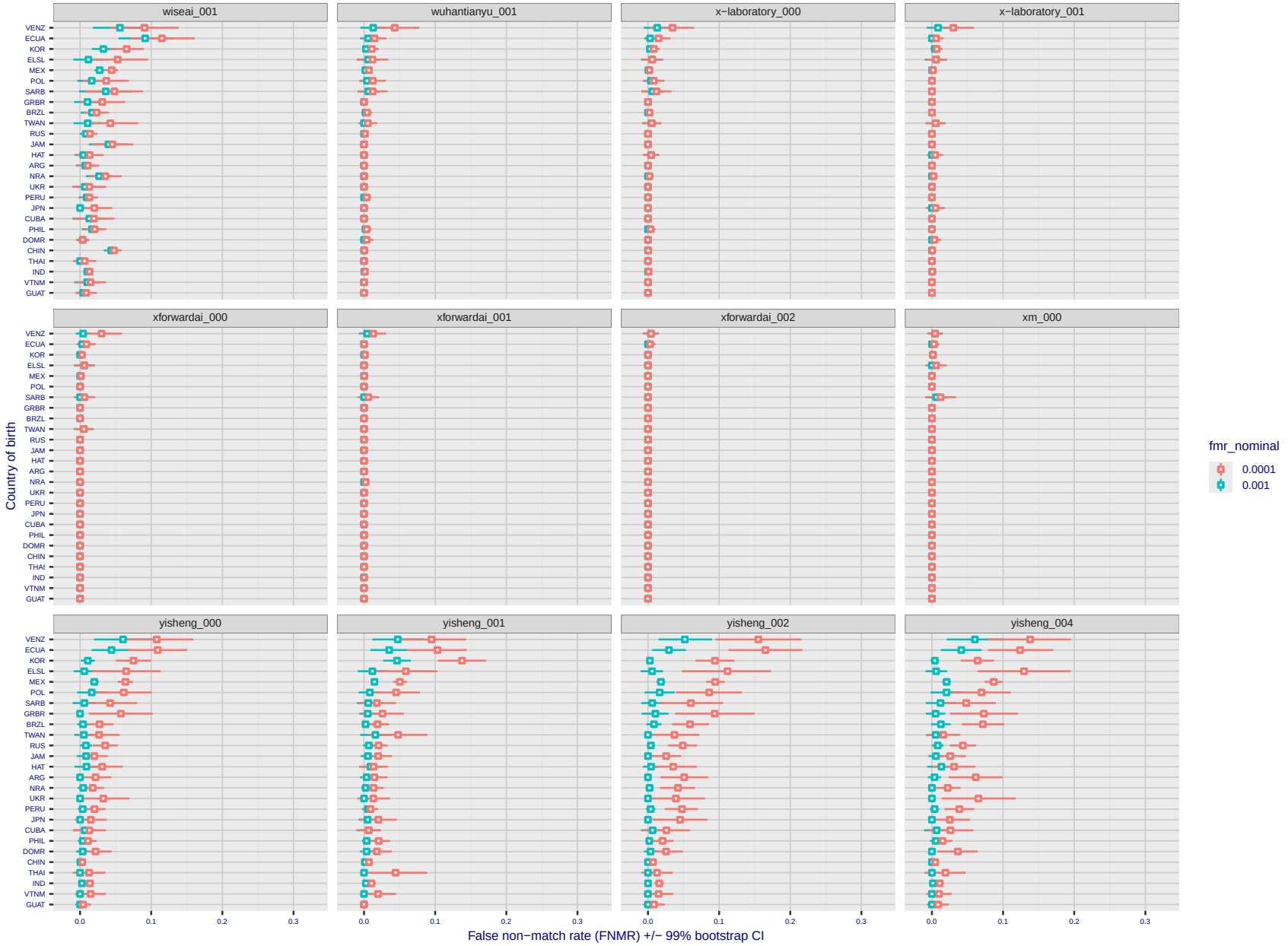

Figure 712: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 713: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

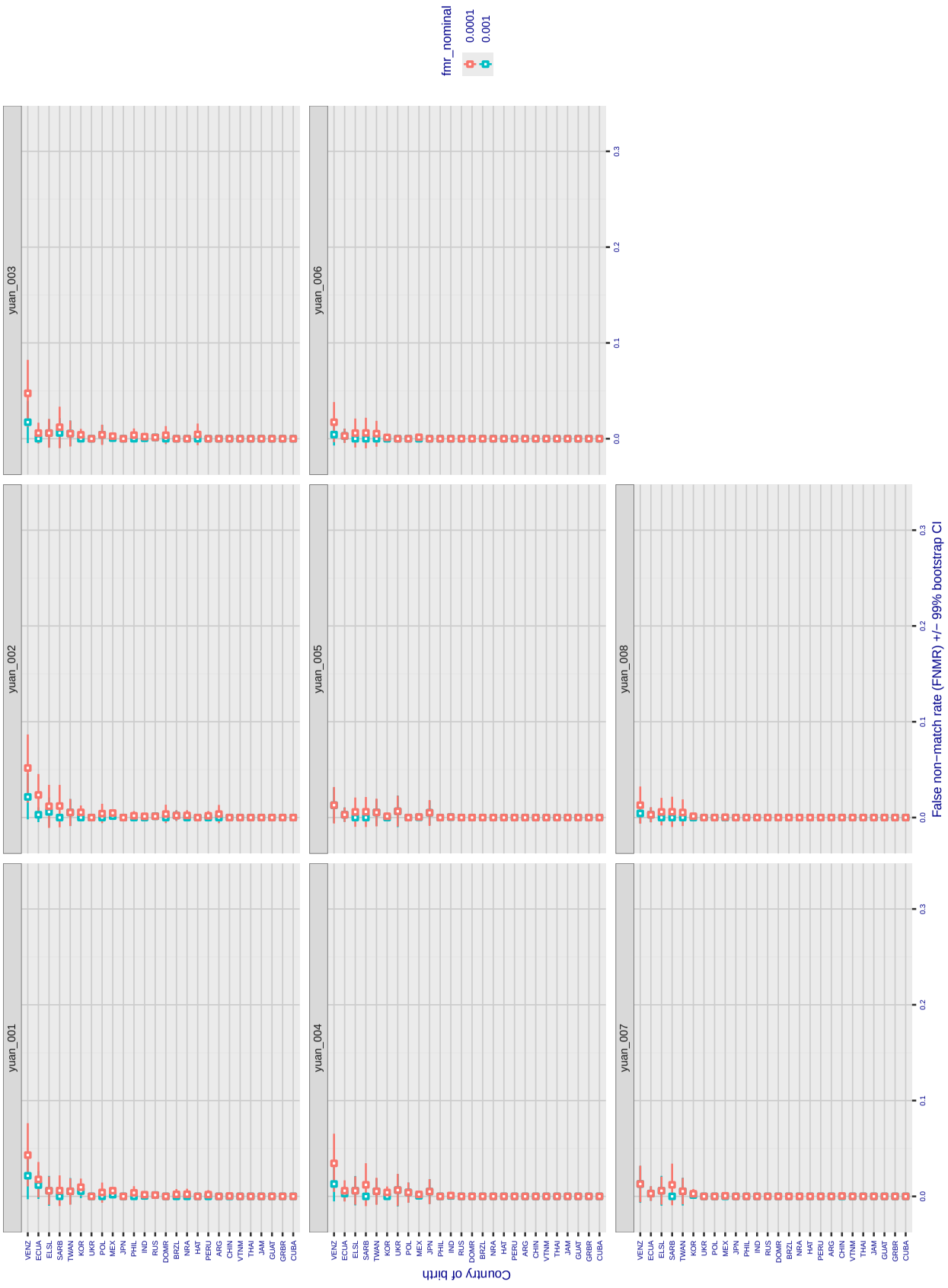

Figure 714: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Caveats: The results may not relate to subject-specific properties. Instead they could reflect image-specific quality differences, which could occur due to collection protocol or software processing variations.

3.5.2 Effect of ageing

Background: Faces change appearance throughout life. This change gradually reduces similarity of a new image to an earlier image. Face recognition algorithms give reduced similarity scores and more frequent false rejections.

Goal: To quantify false non-match rates (FNMR) as a function of elapsed time in an adult population.

Methods: Using the mugshot images, a threshold is set to give FMR = 0.00001 over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores.

Results: For the visa images, Figure 799 shows how false non-match rates for genuine users, as a function of age group.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

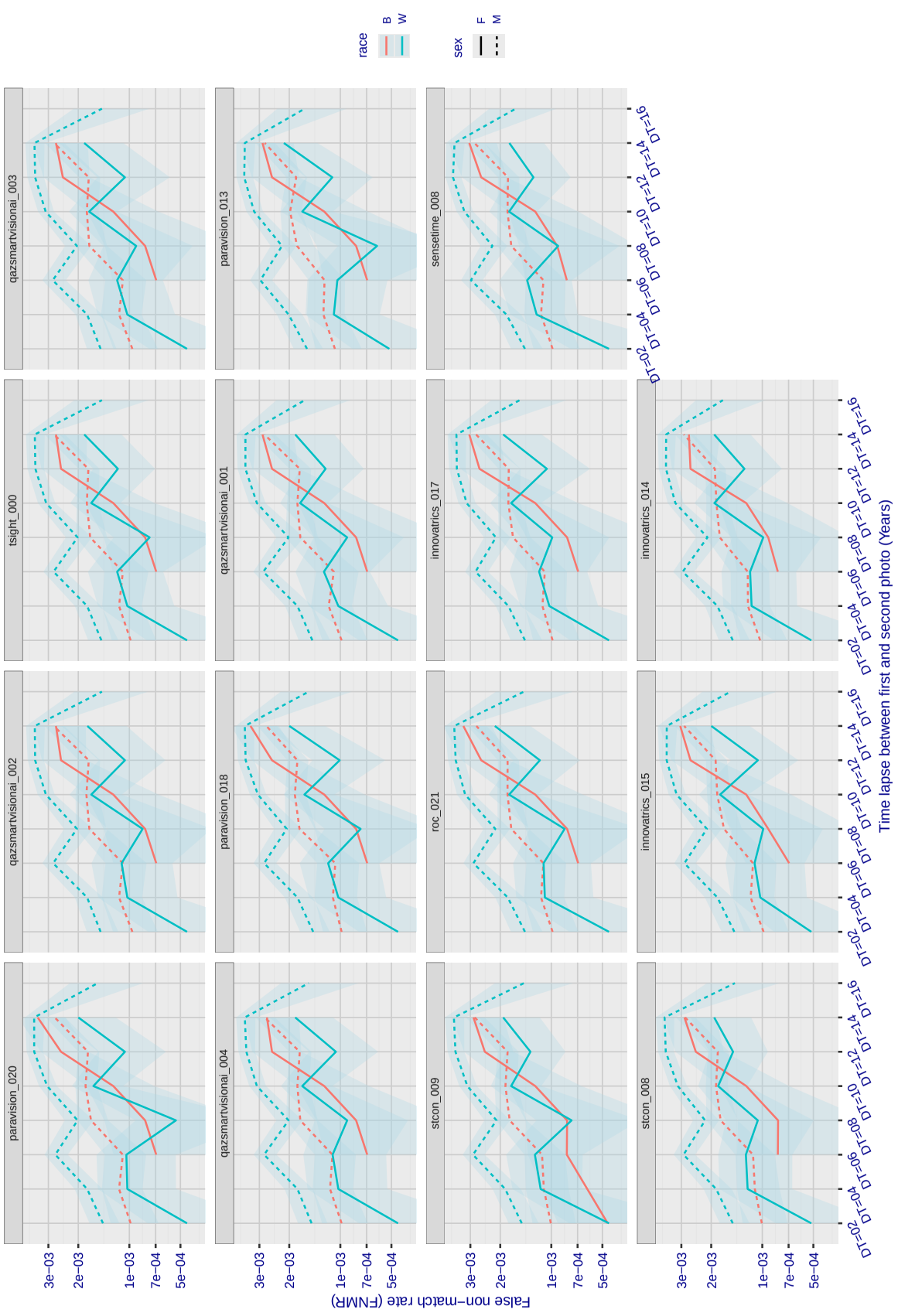

Figure 715: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

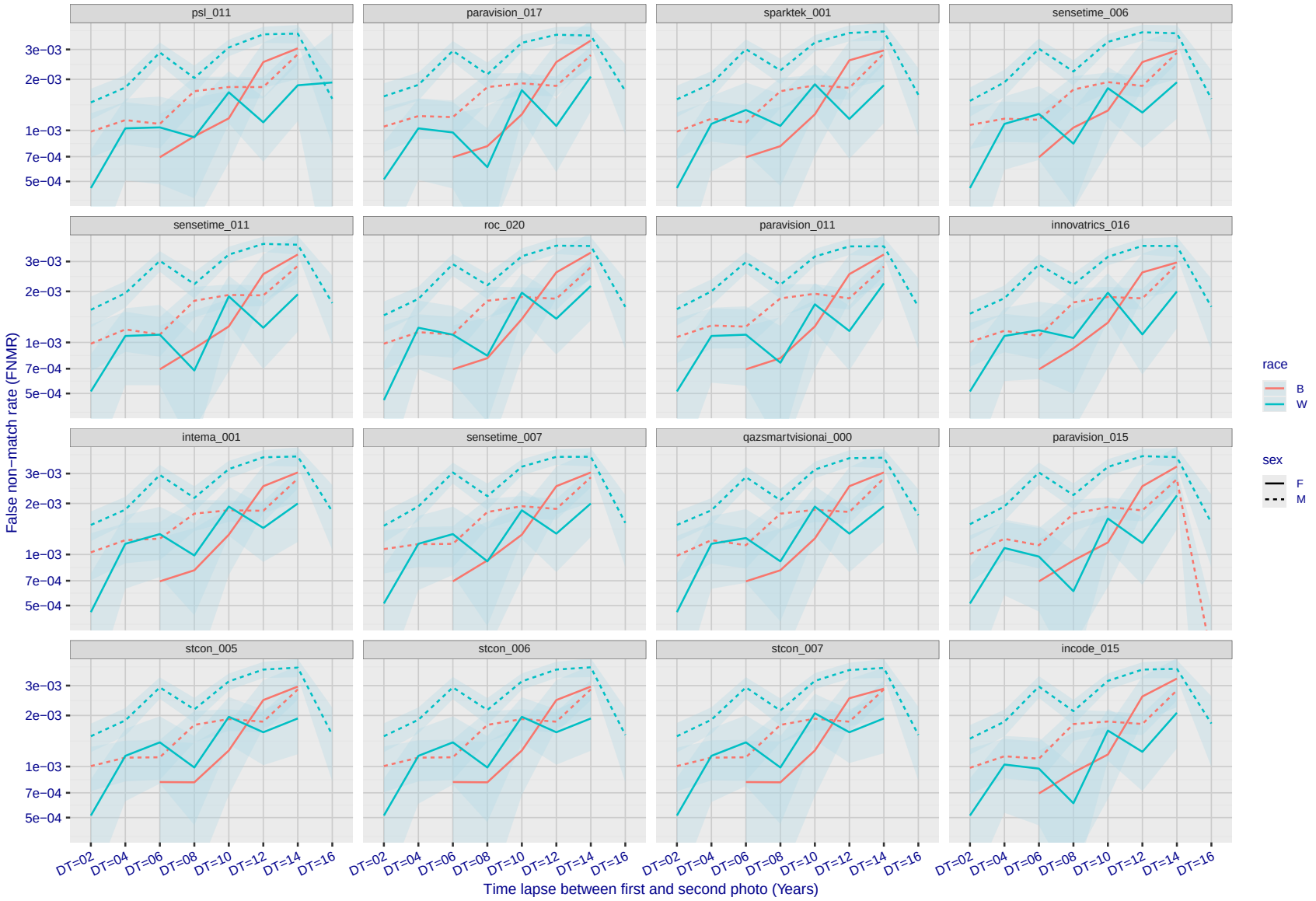

Figure 716: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

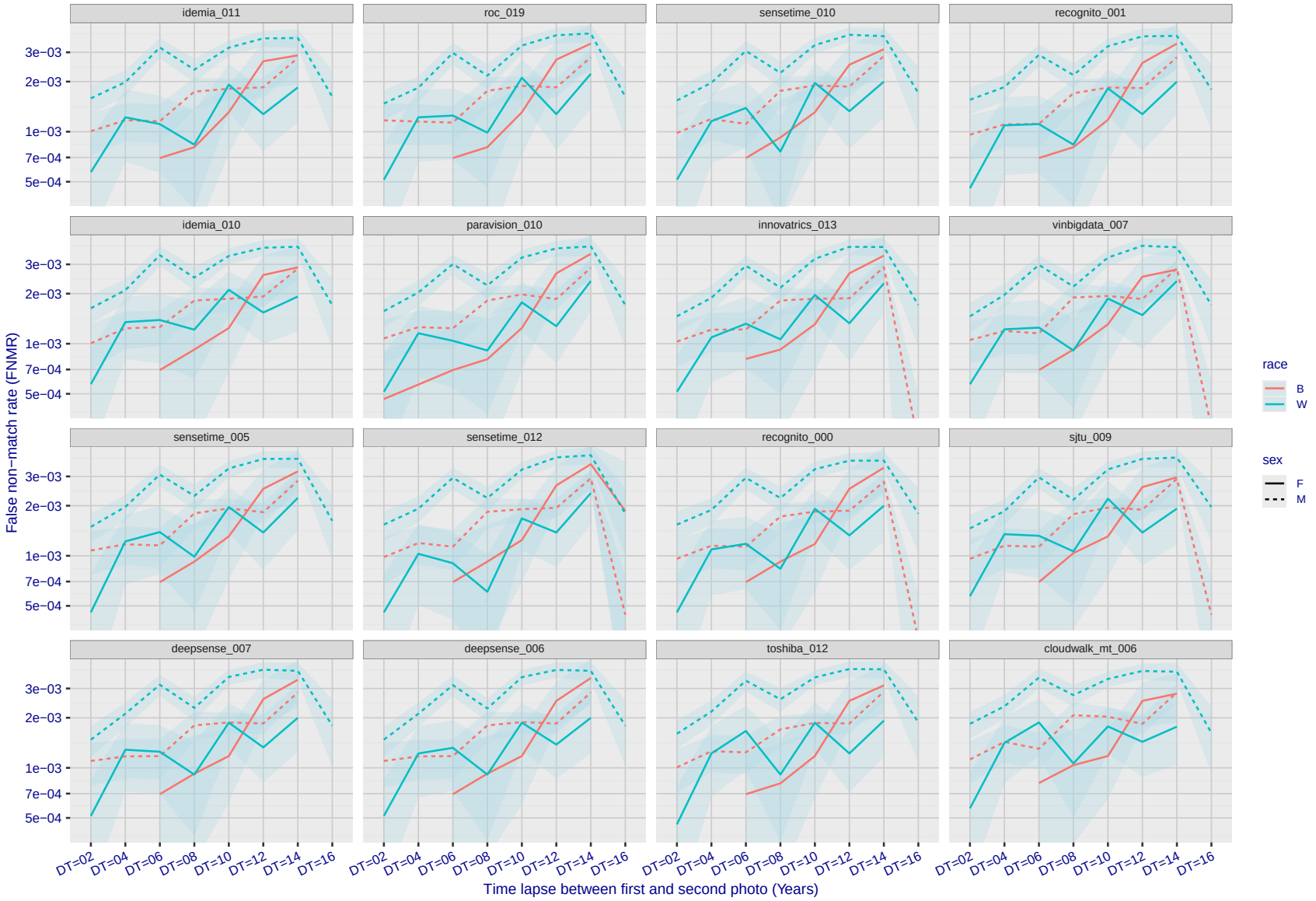

Figure 717: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

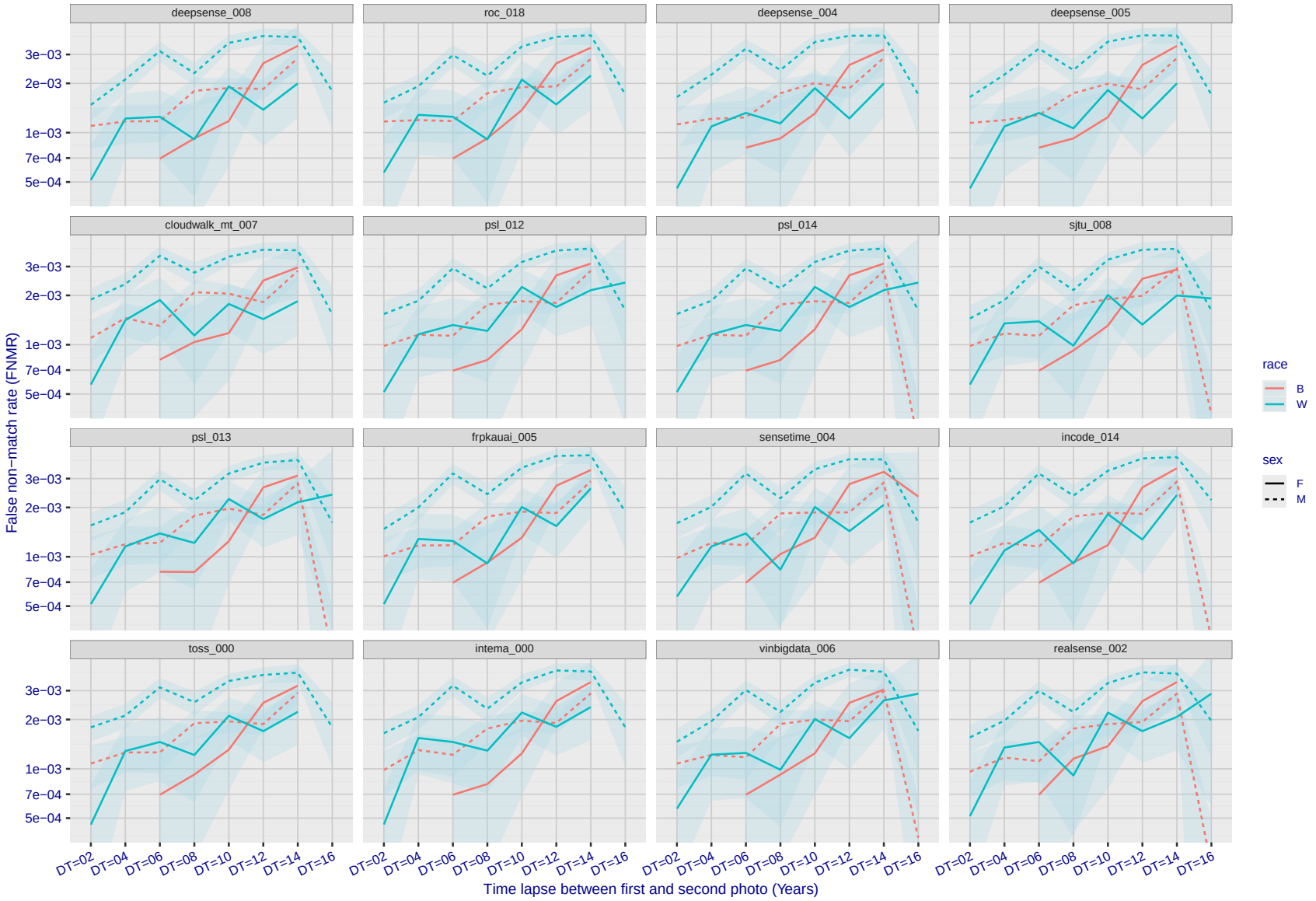

Figure 718: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

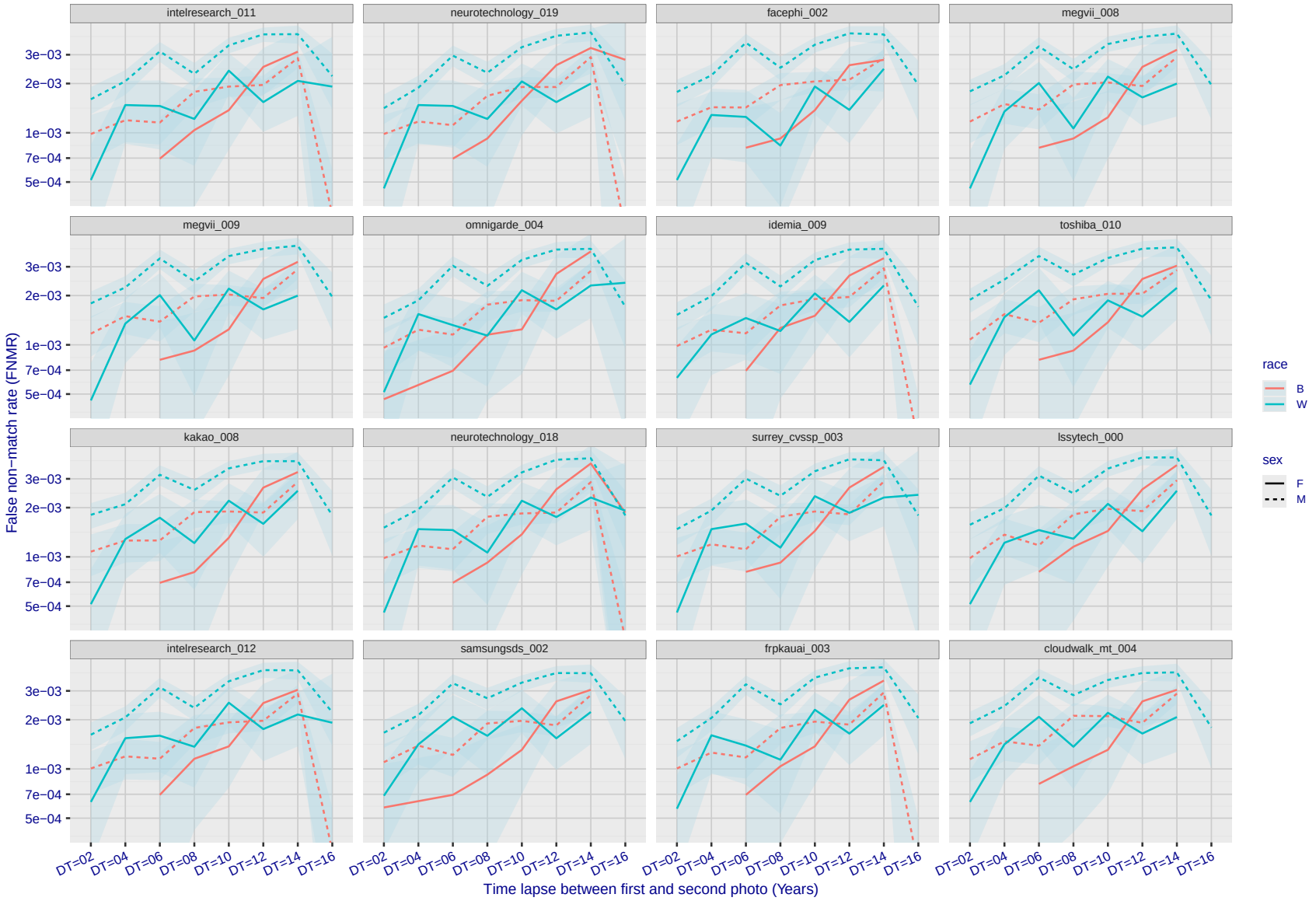

Figure 719: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

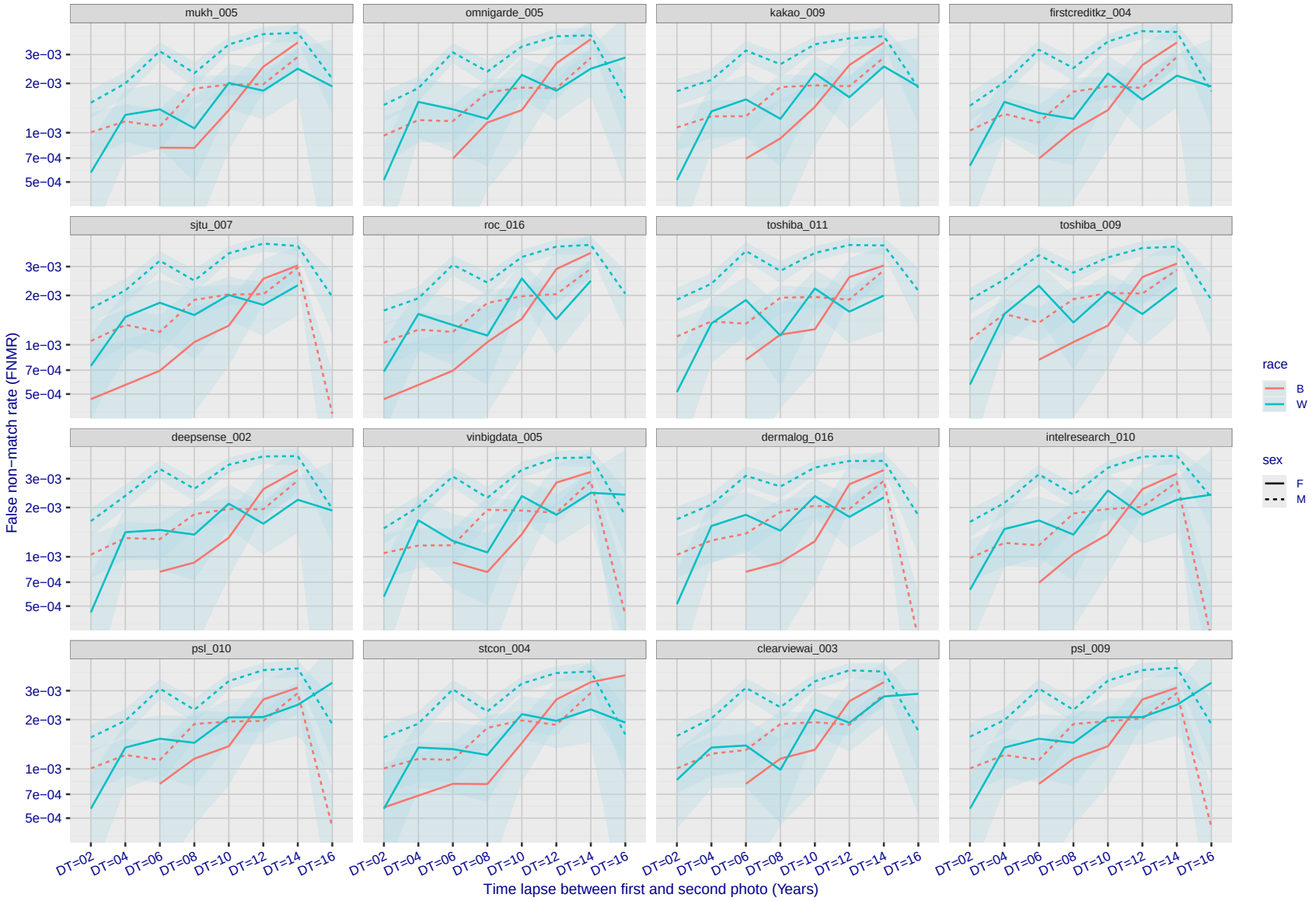

Figure 720: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

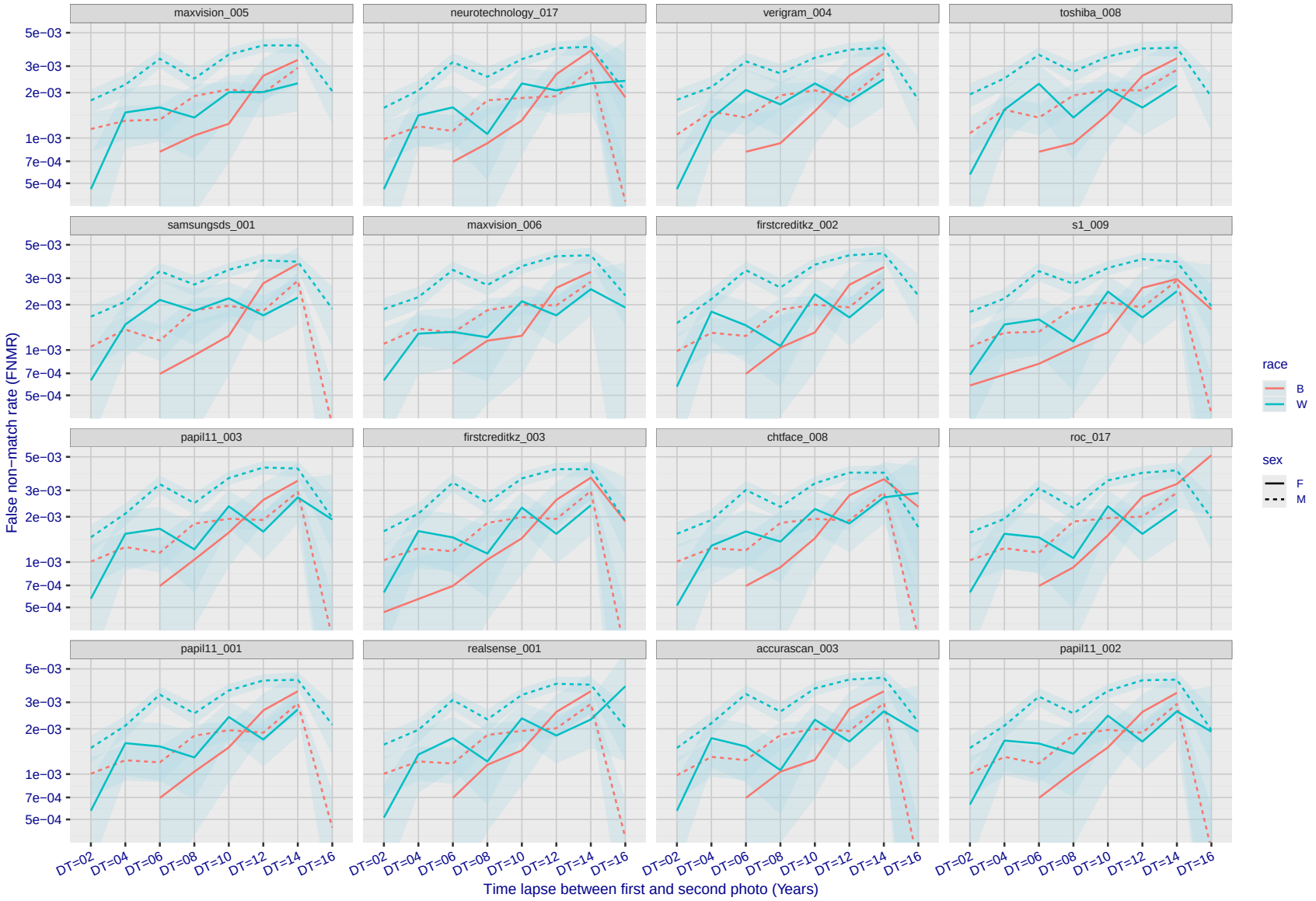

Figure 721: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\bar{1}0^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

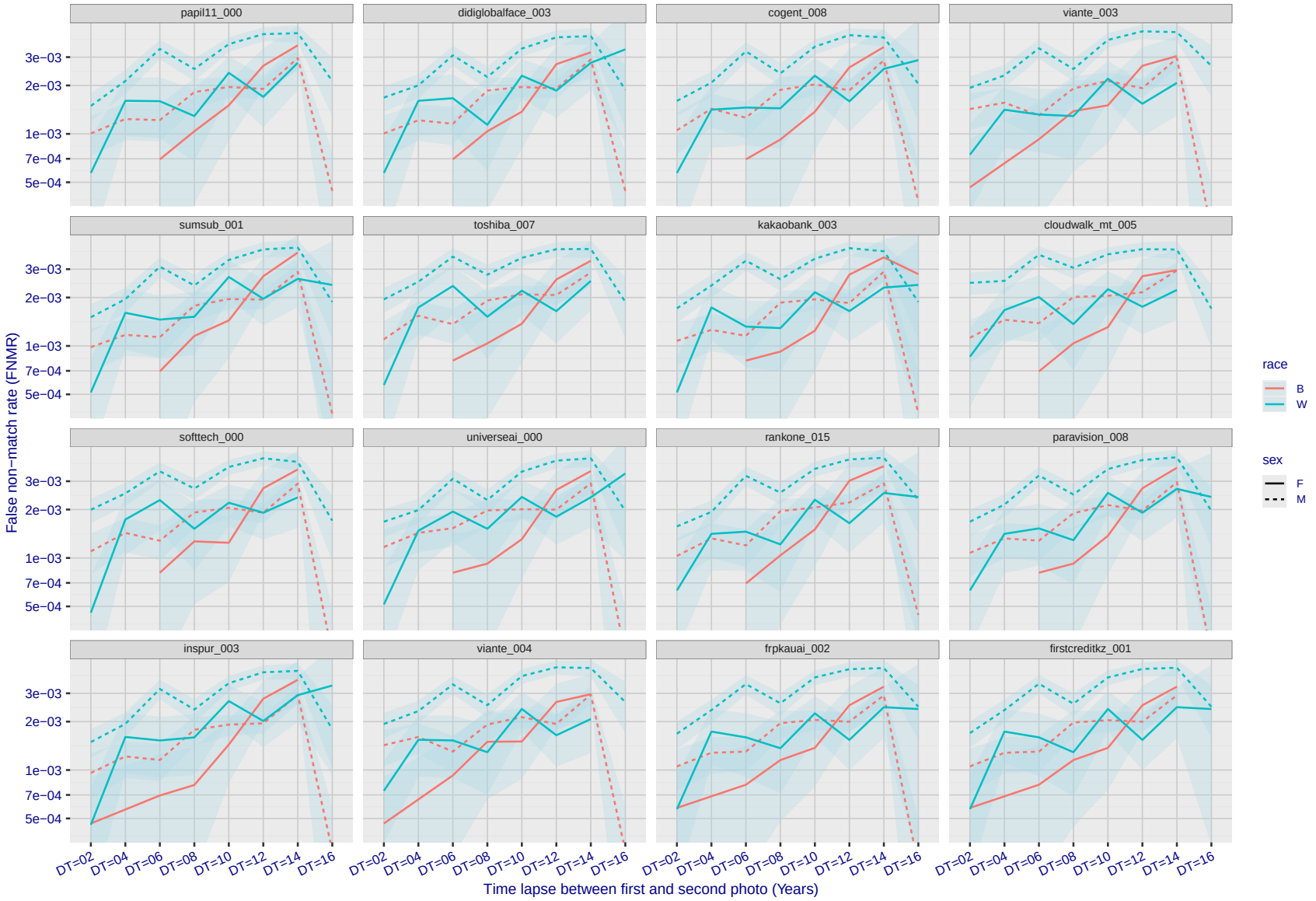

Figure 722: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

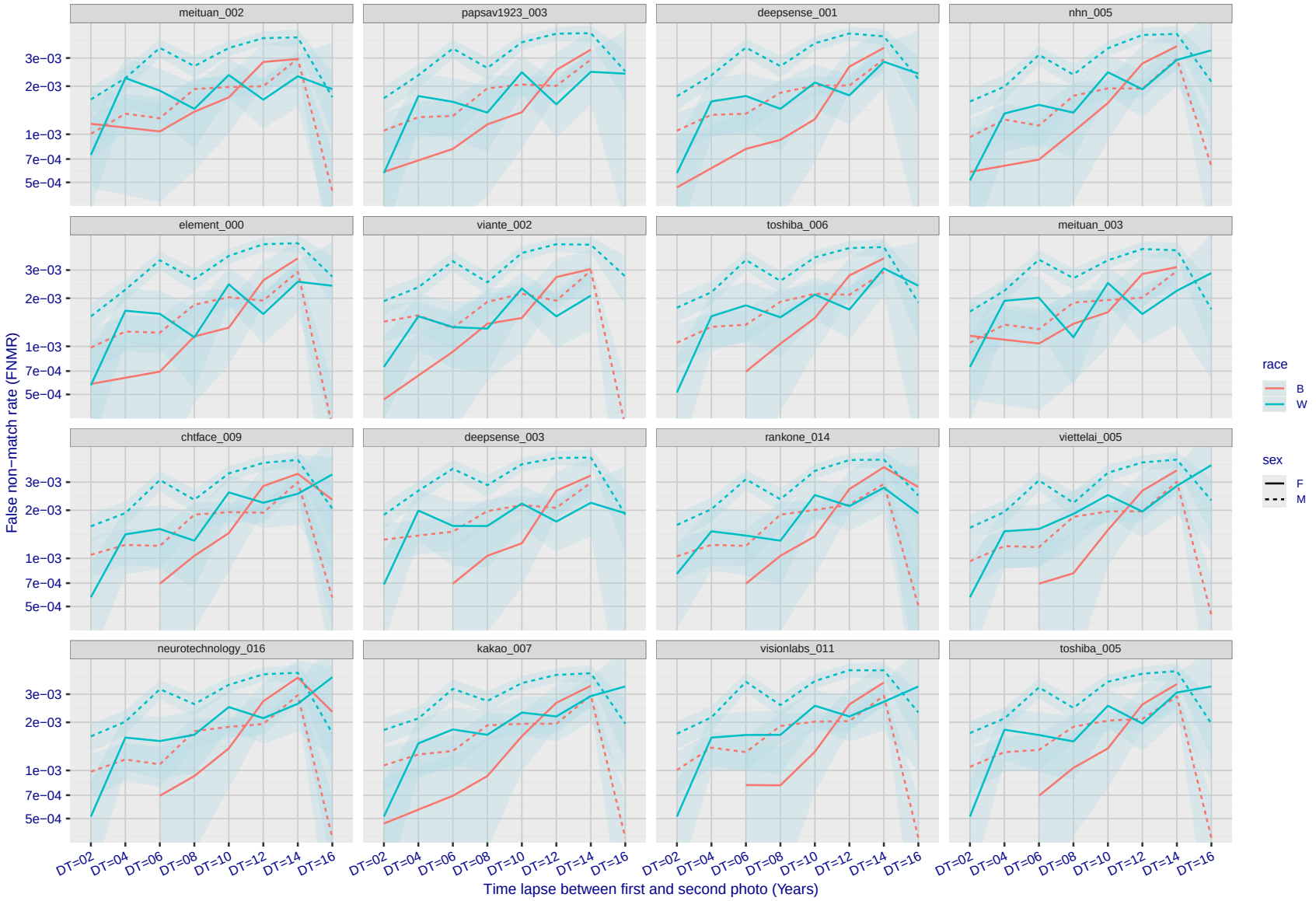

Figure 723: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

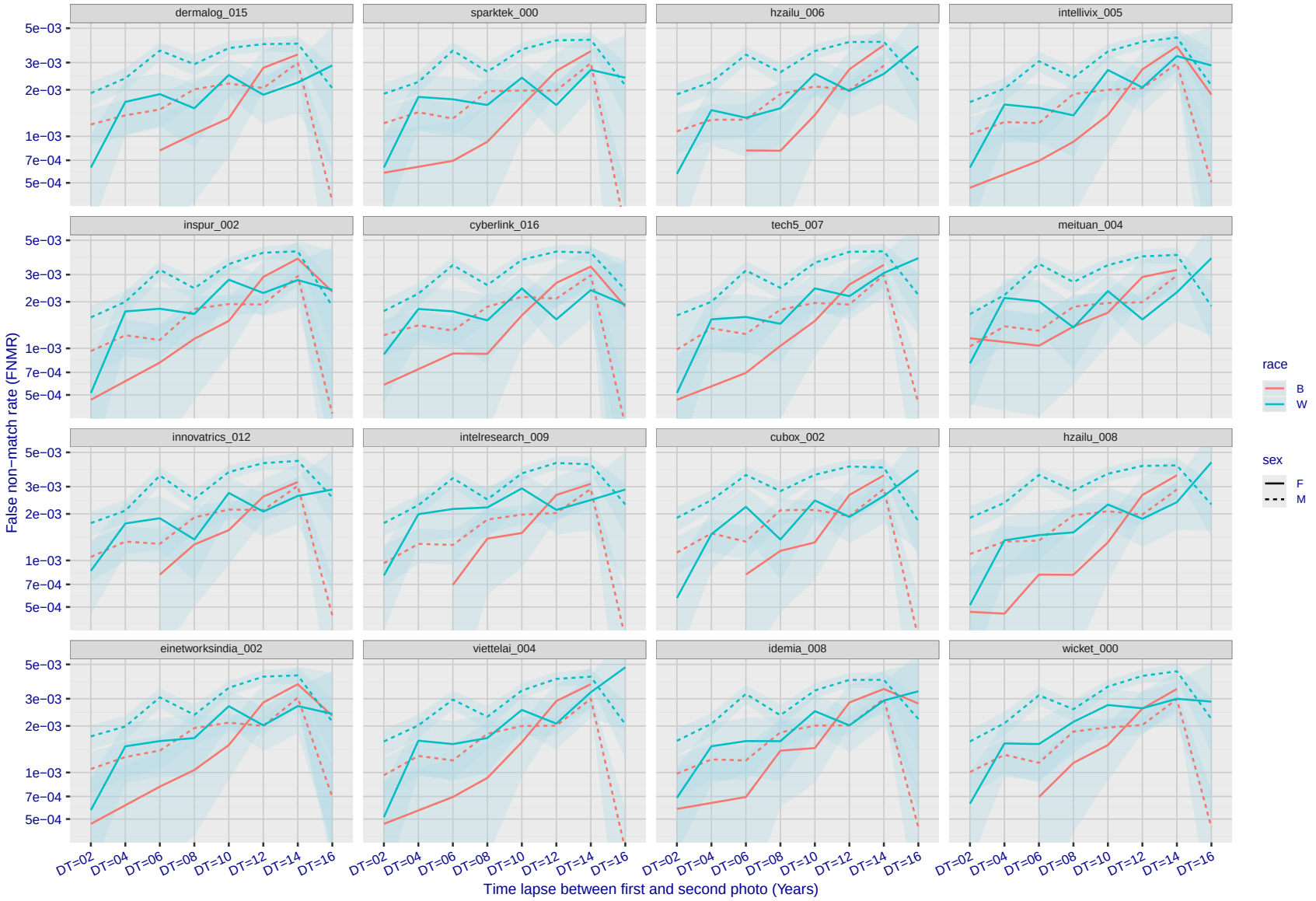

Figure 724: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

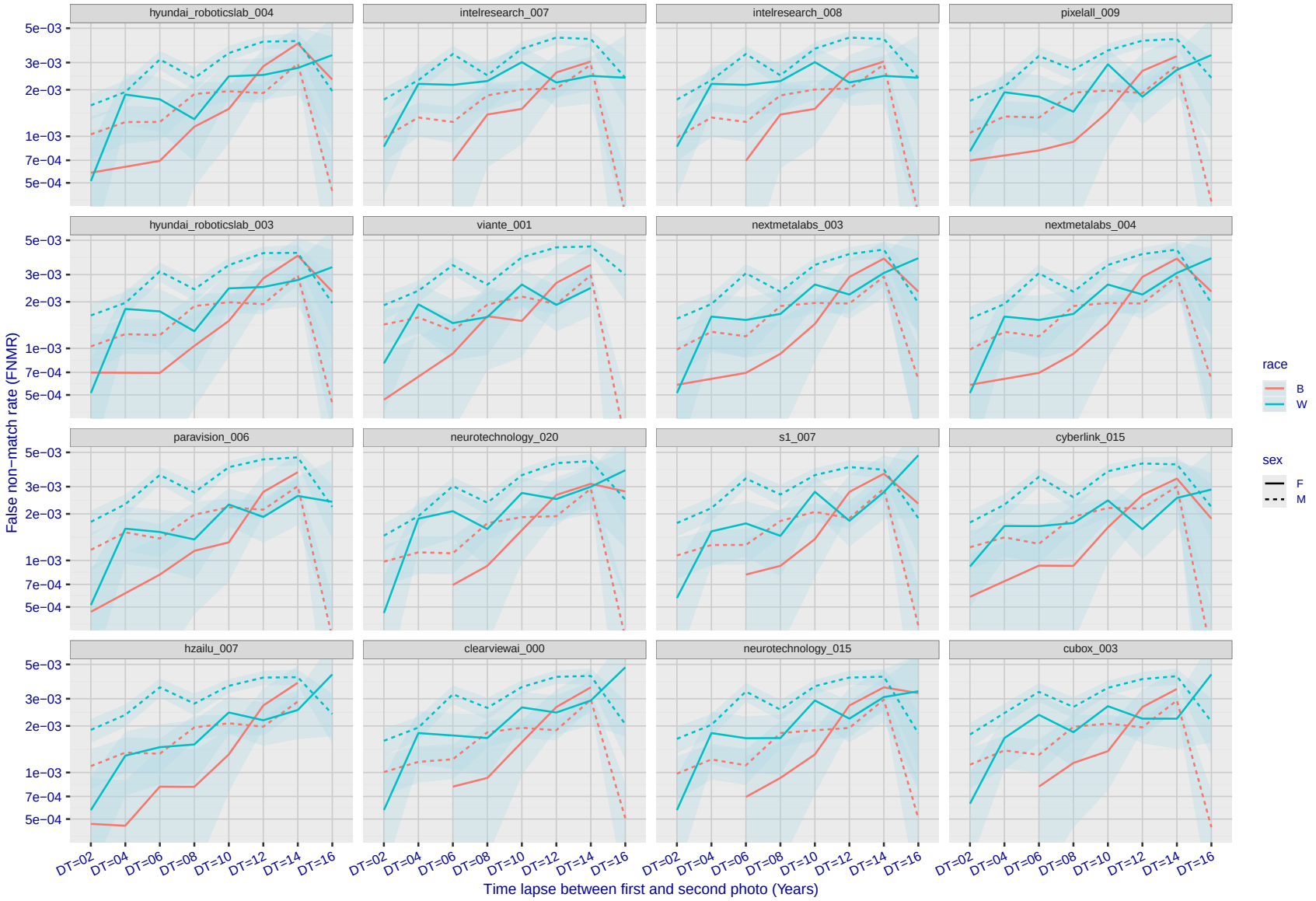

Figure 725: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

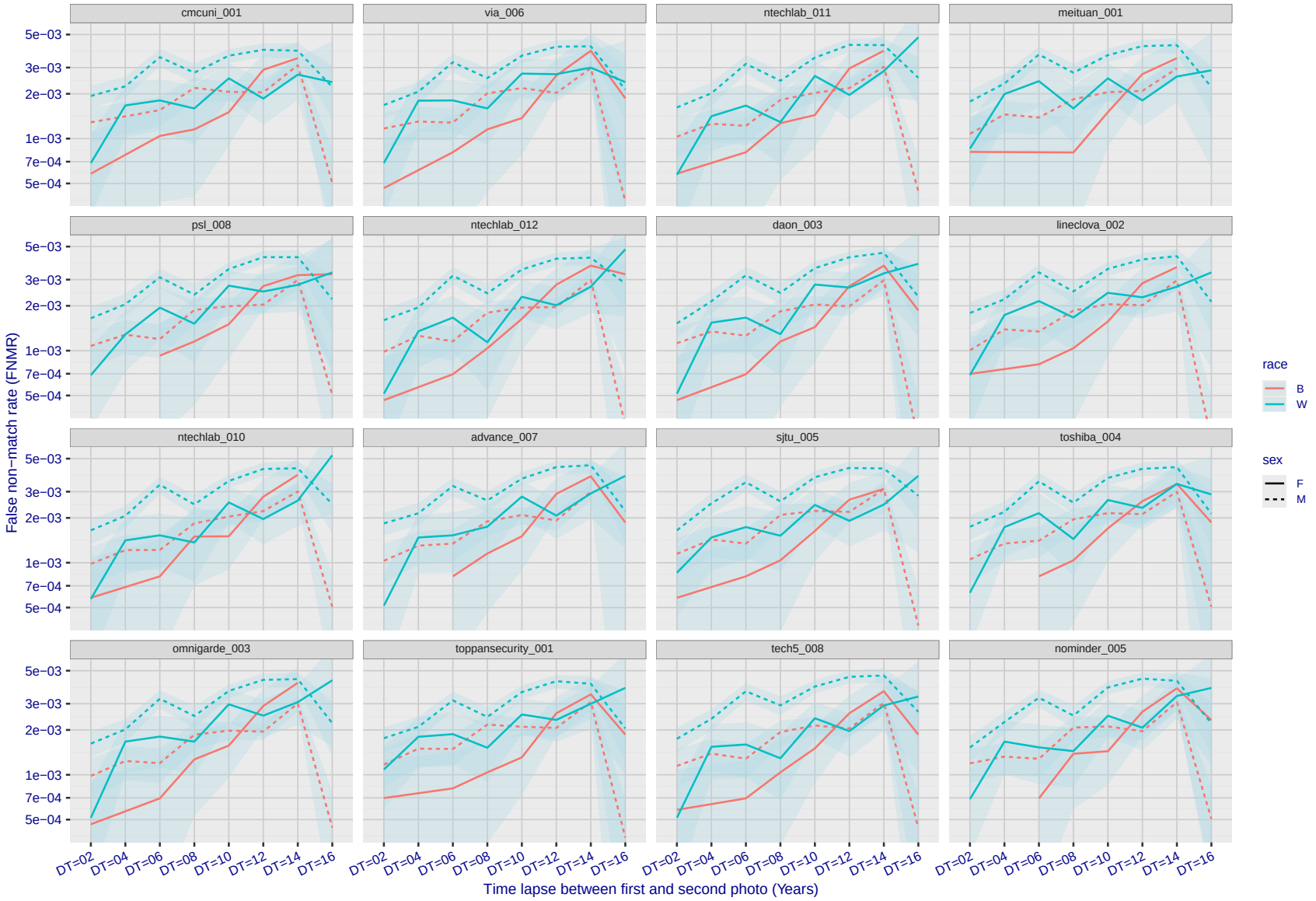

Figure 726: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

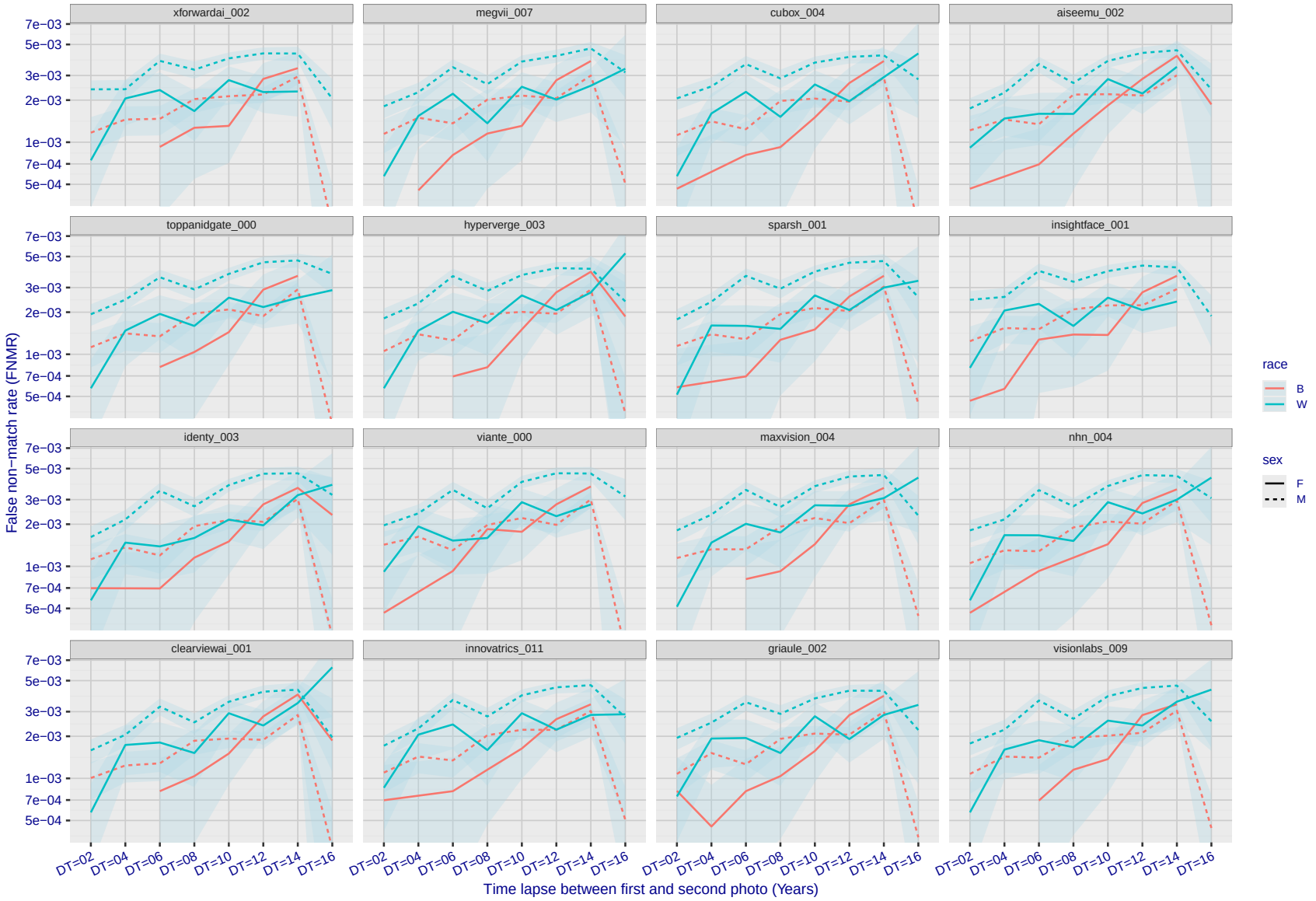

Figure 727: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

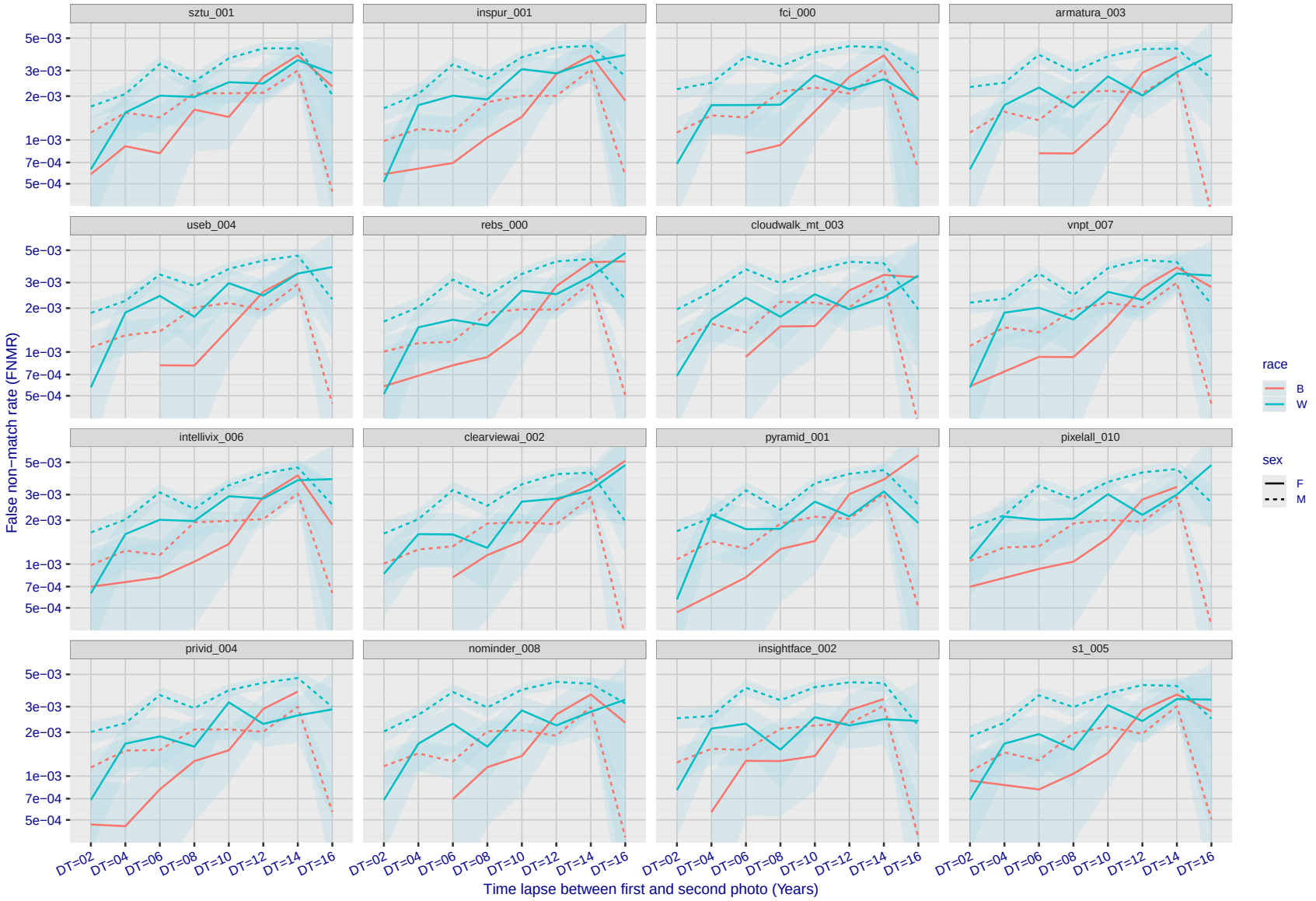

Figure 728: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

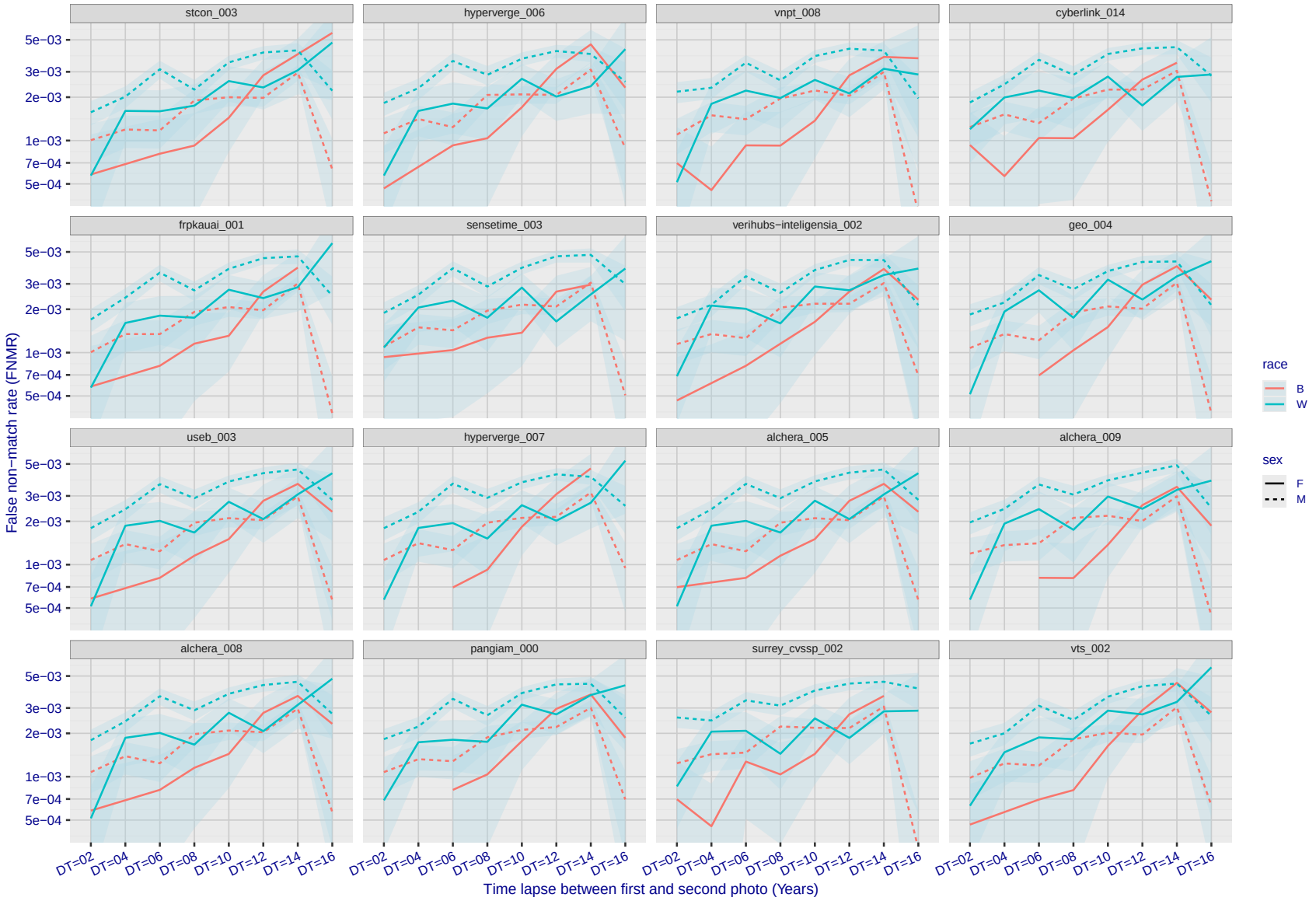

Figure 729: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widehat{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

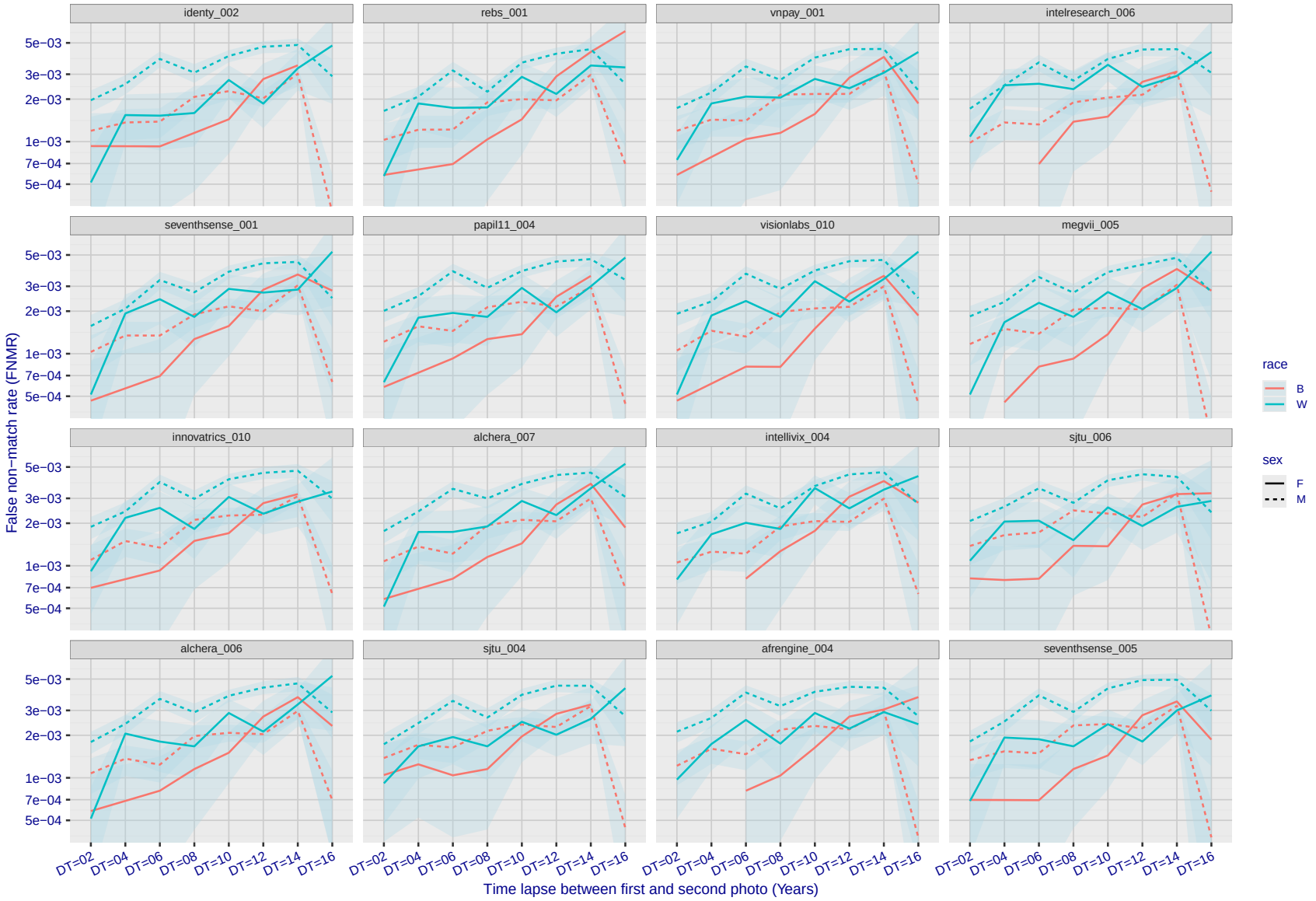

Figure 730: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

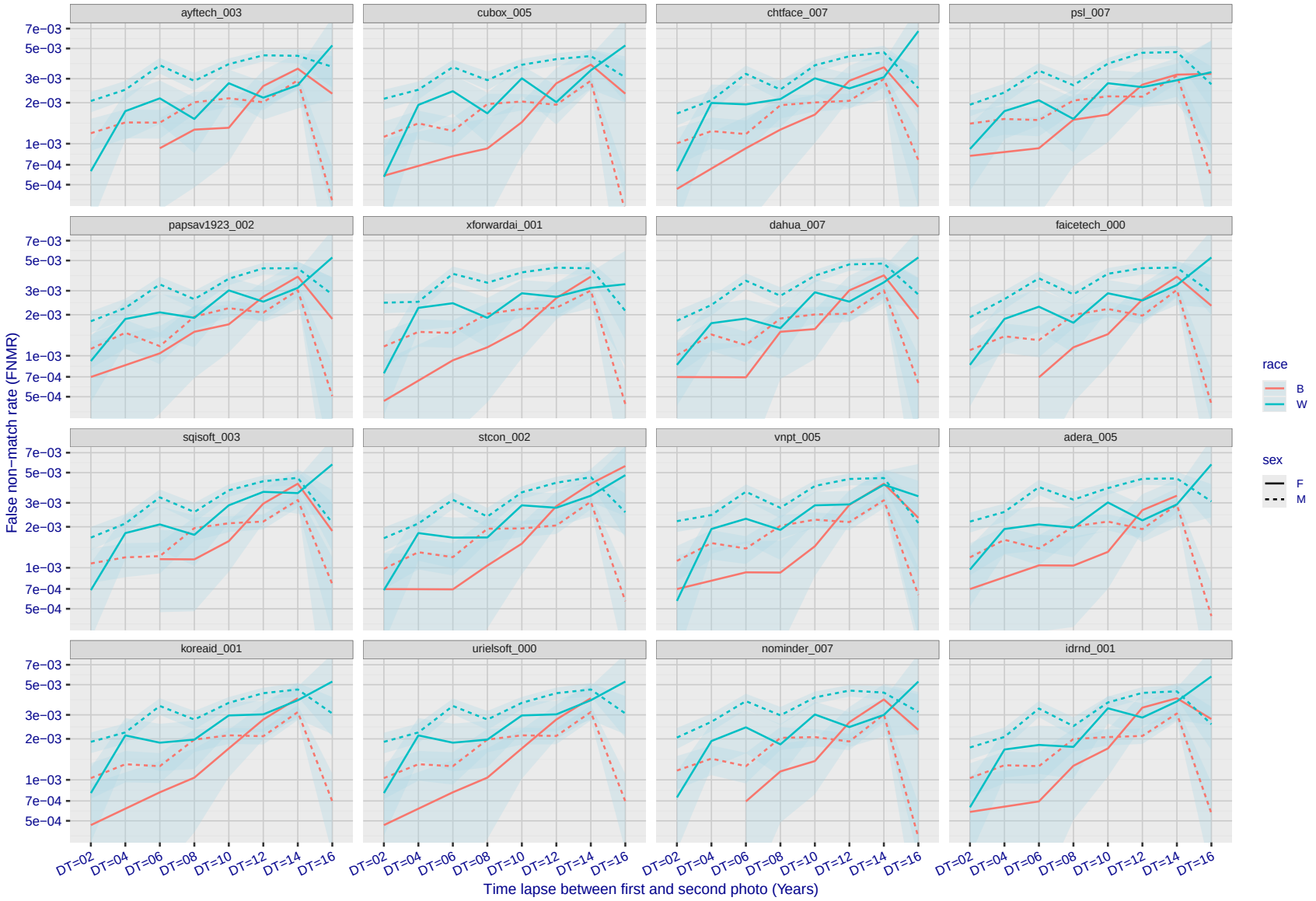

Figure 731: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

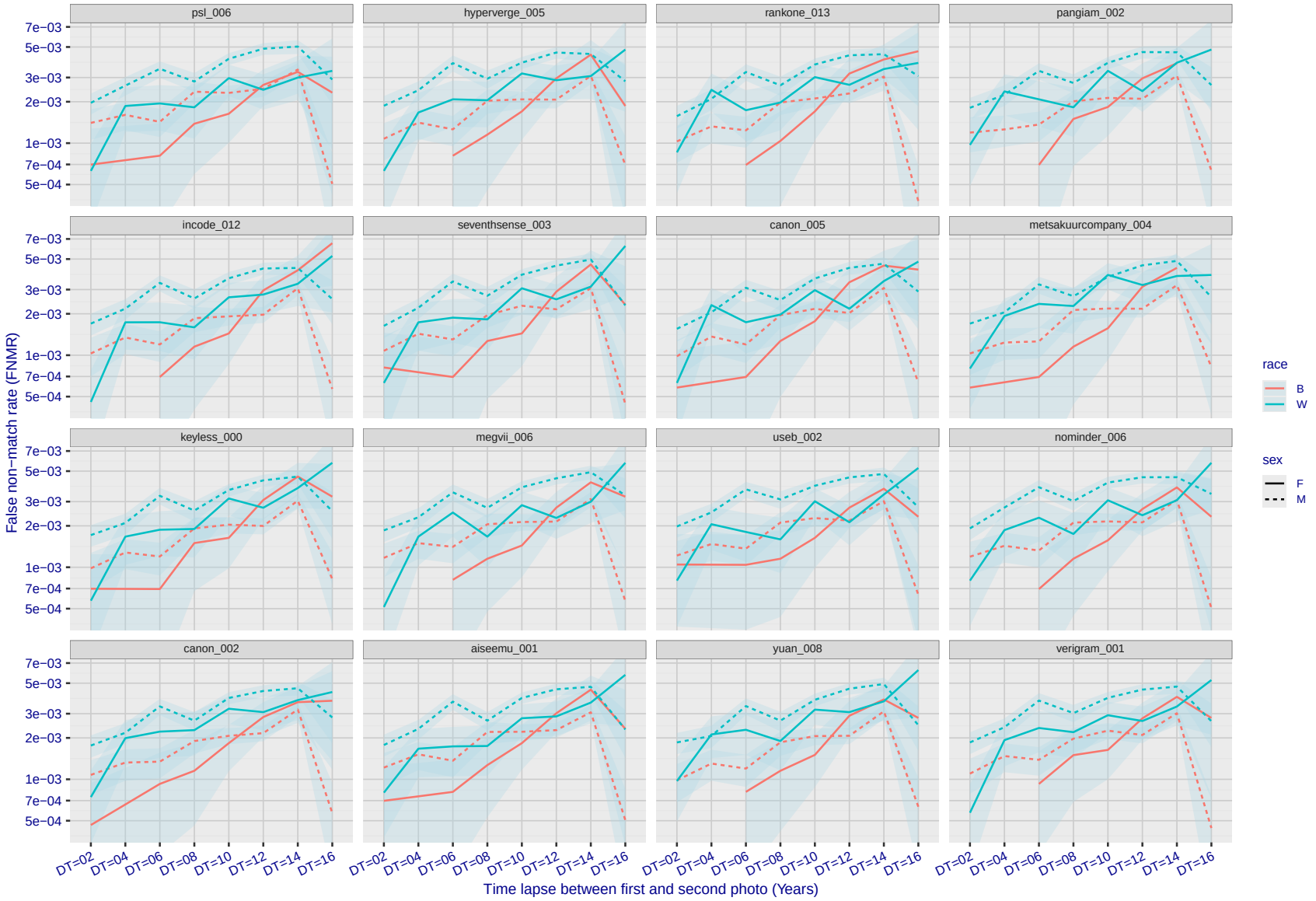

Figure 732: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

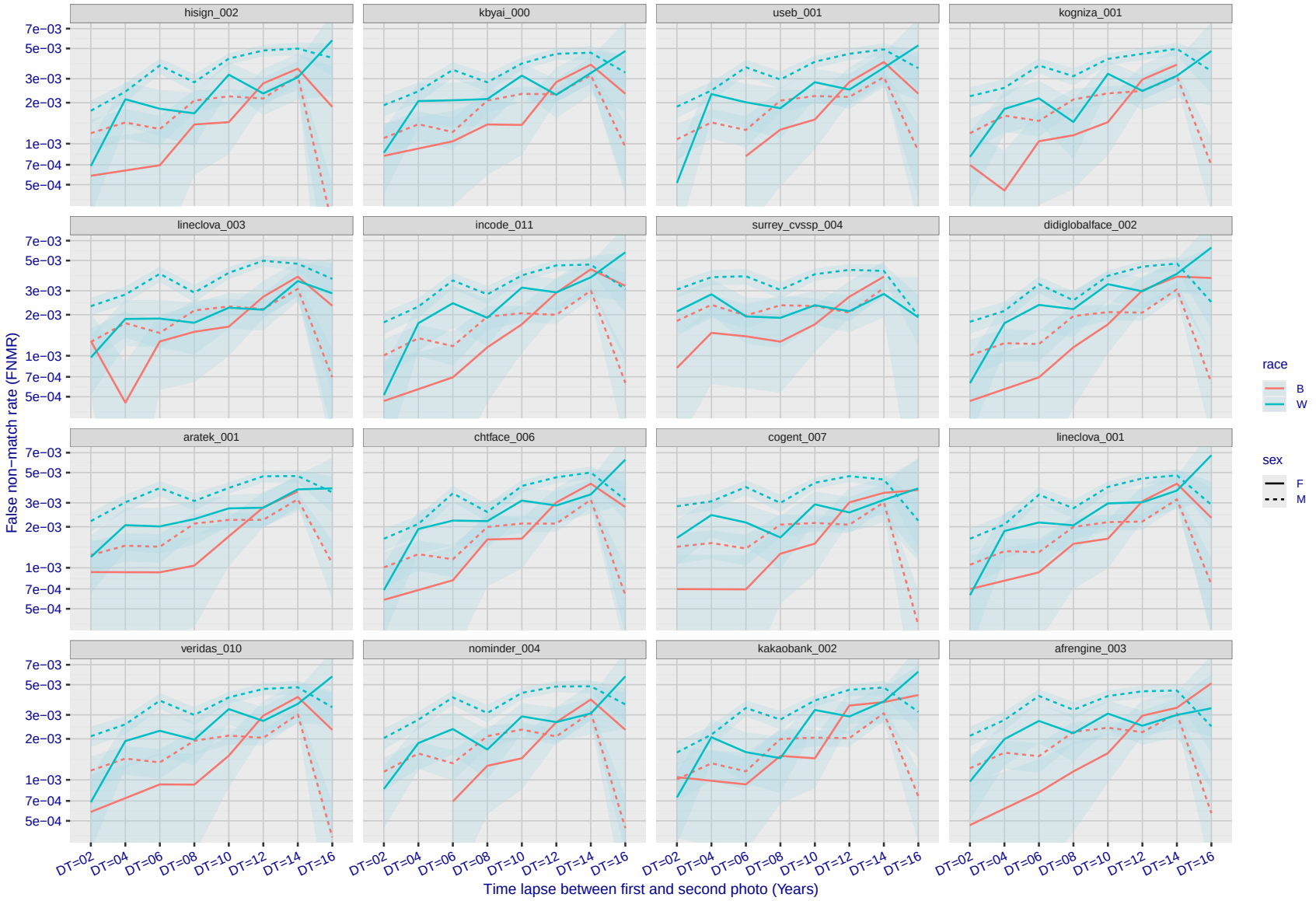

Figure 733: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

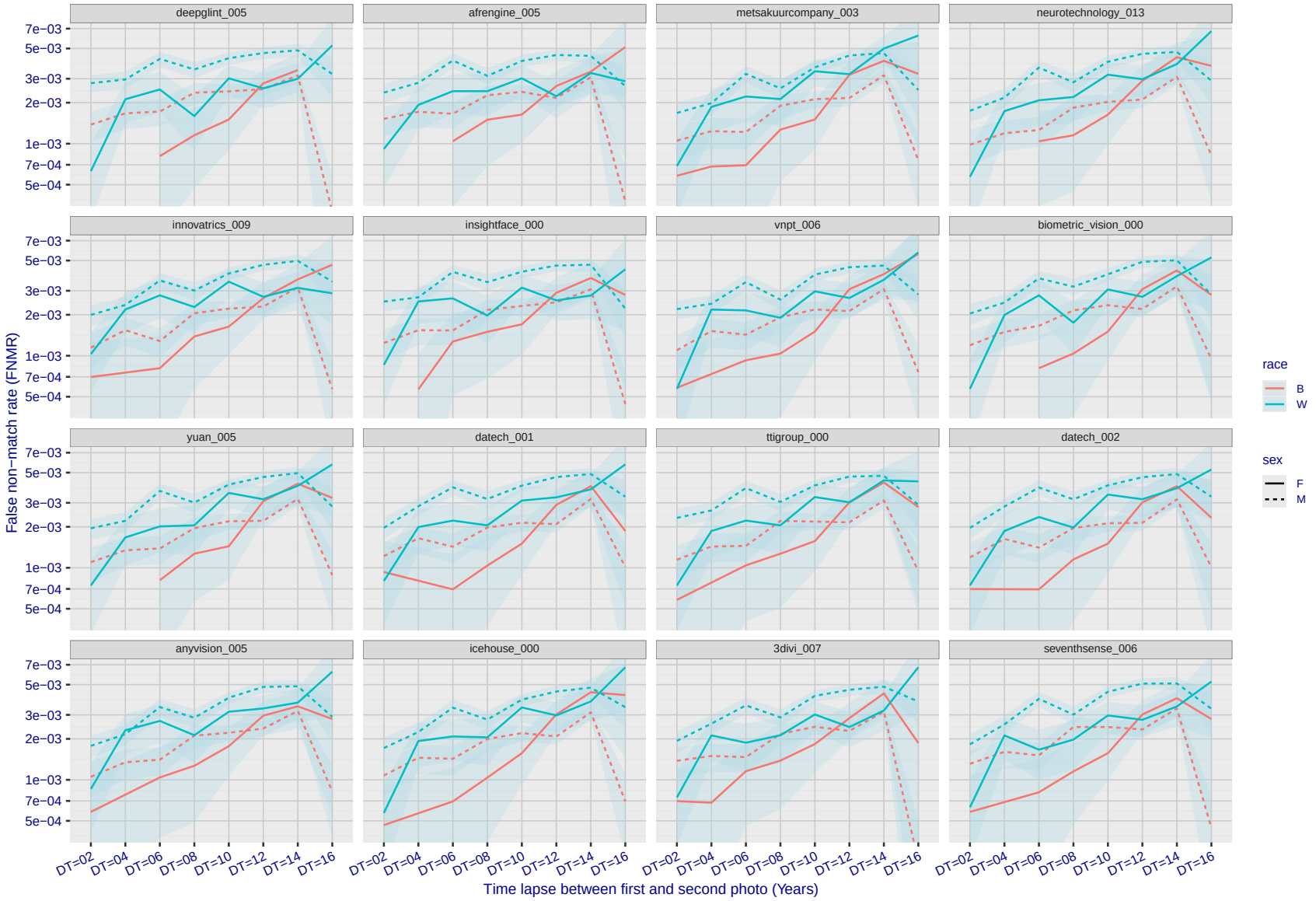

Figure 734: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

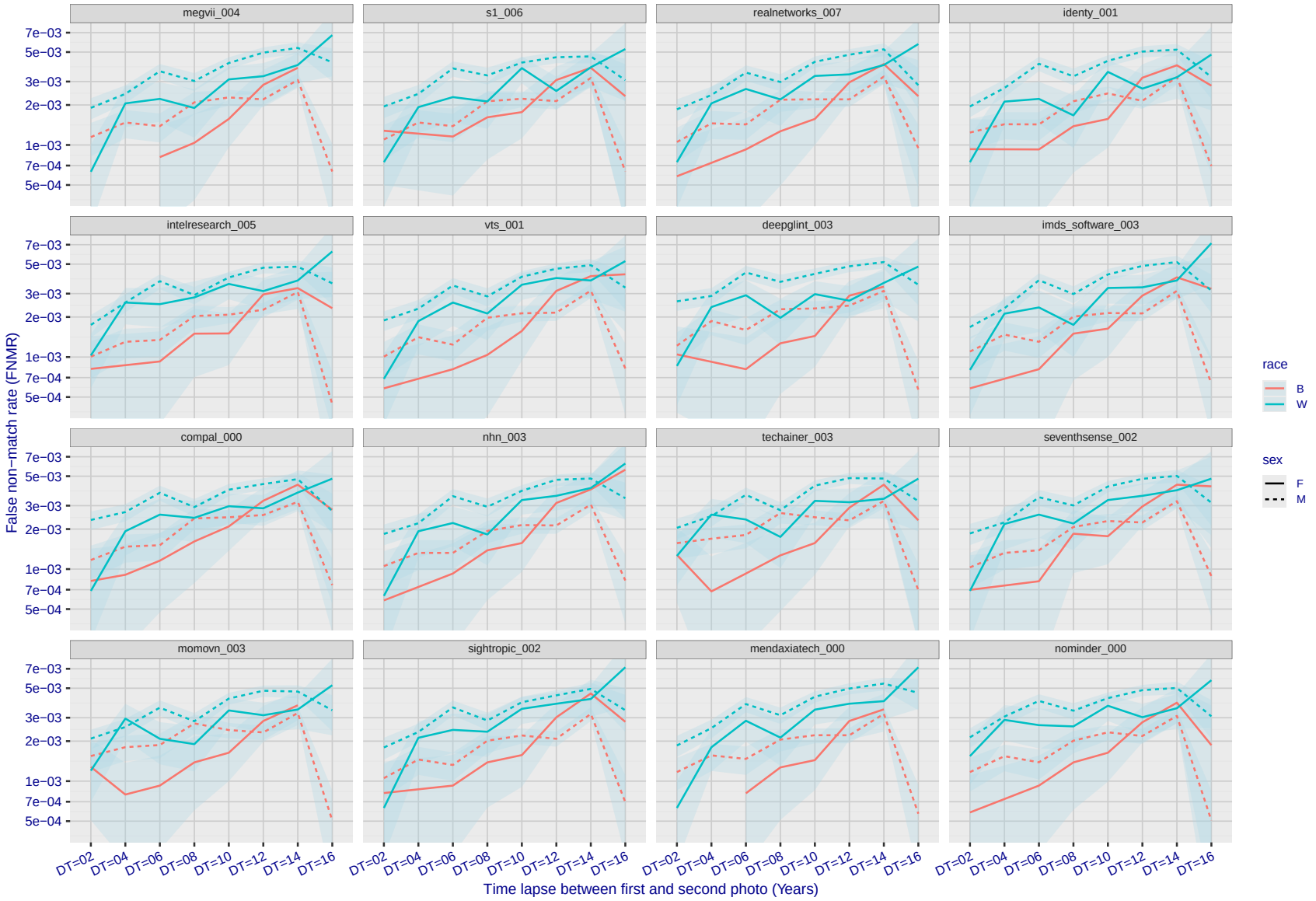

Figure 735: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

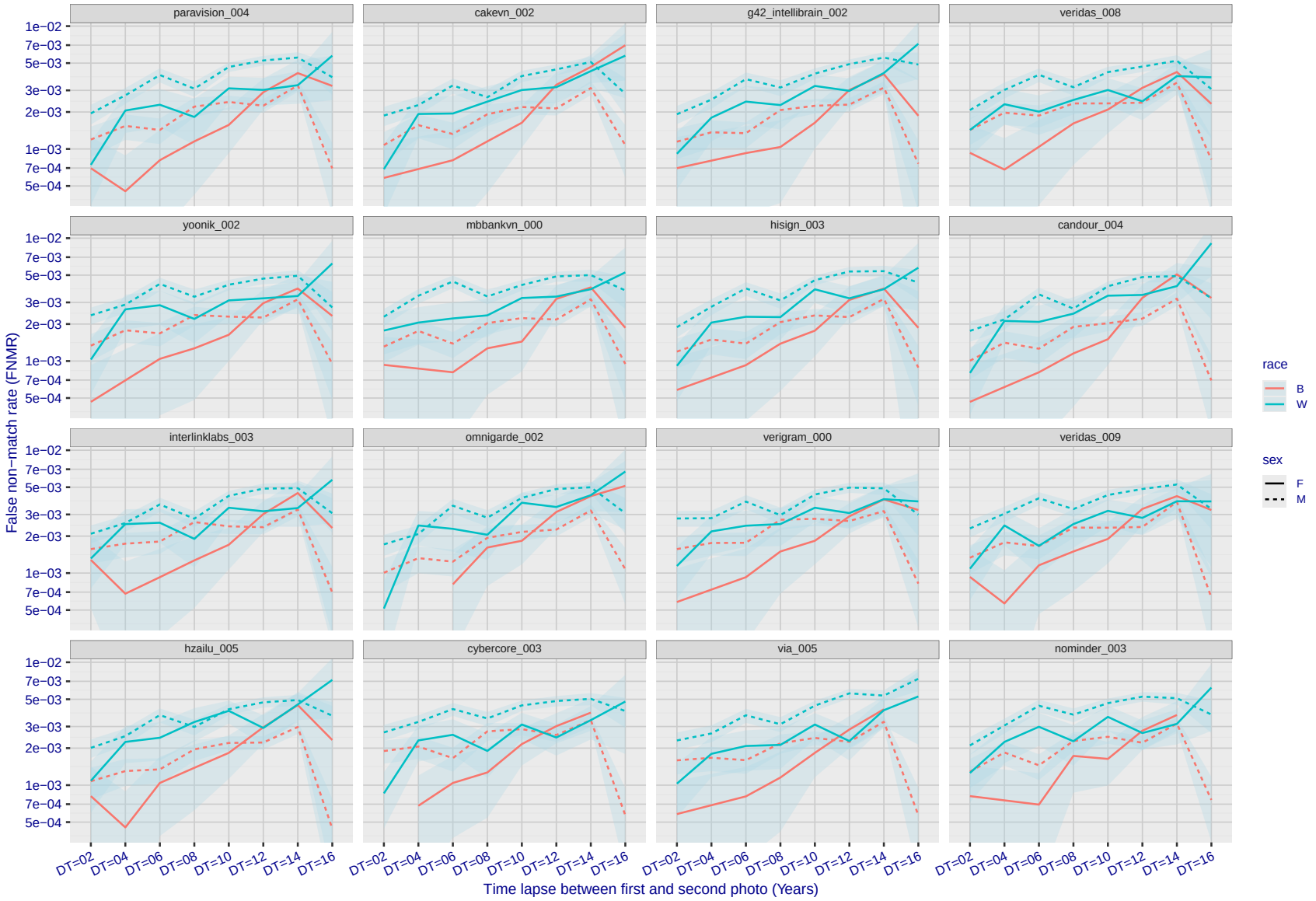

Figure 736: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

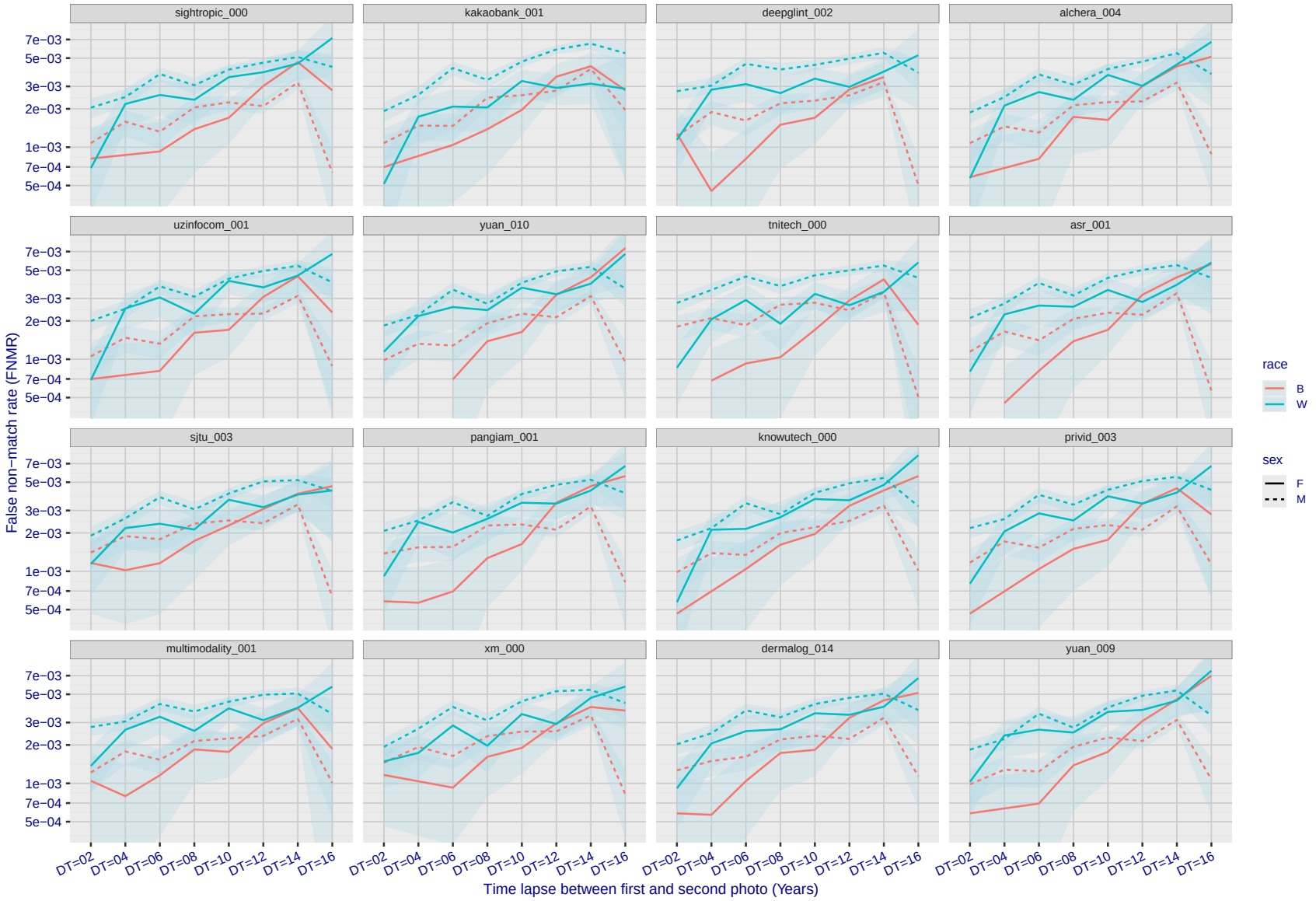

Figure 737: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

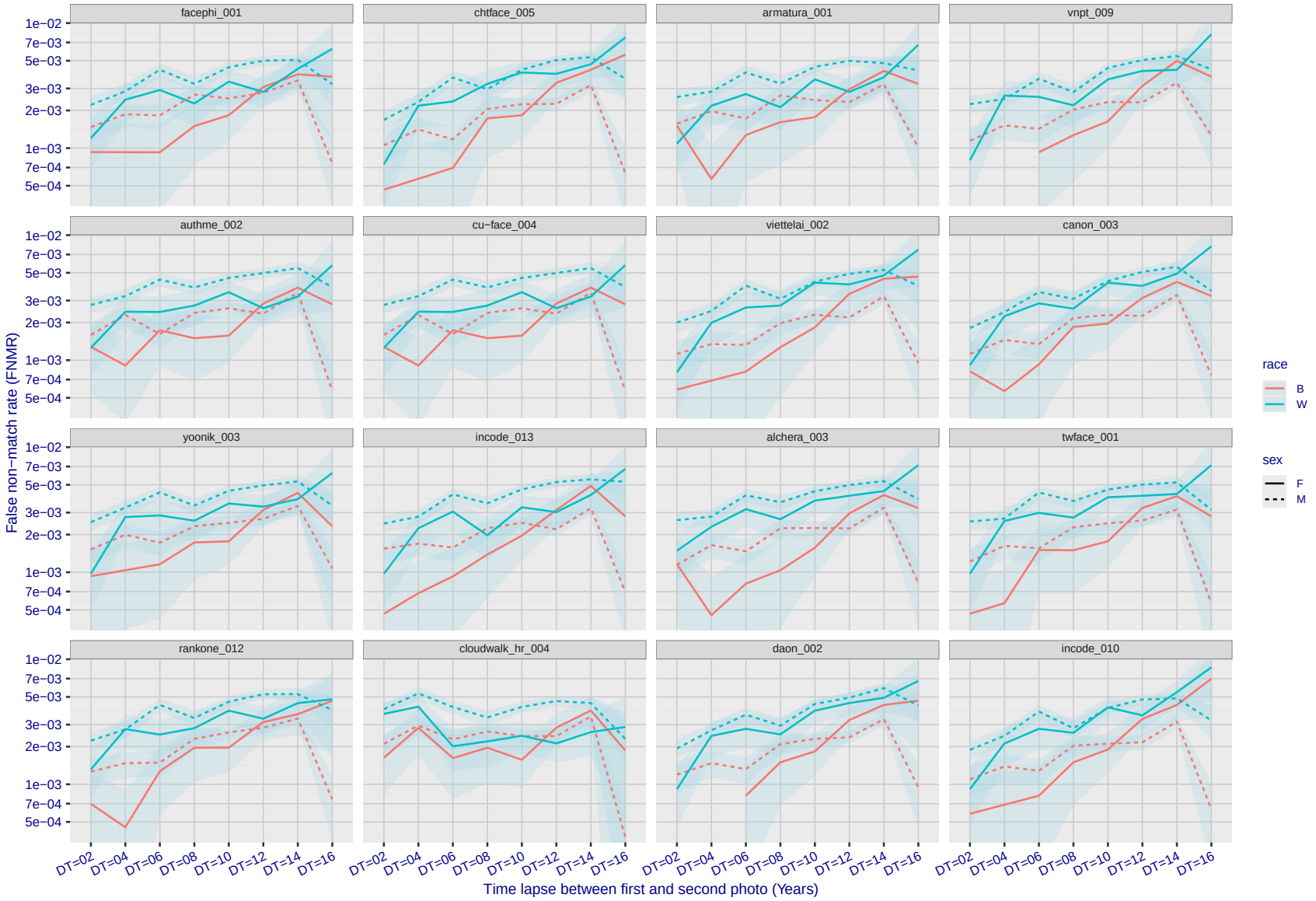

Figure 738: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

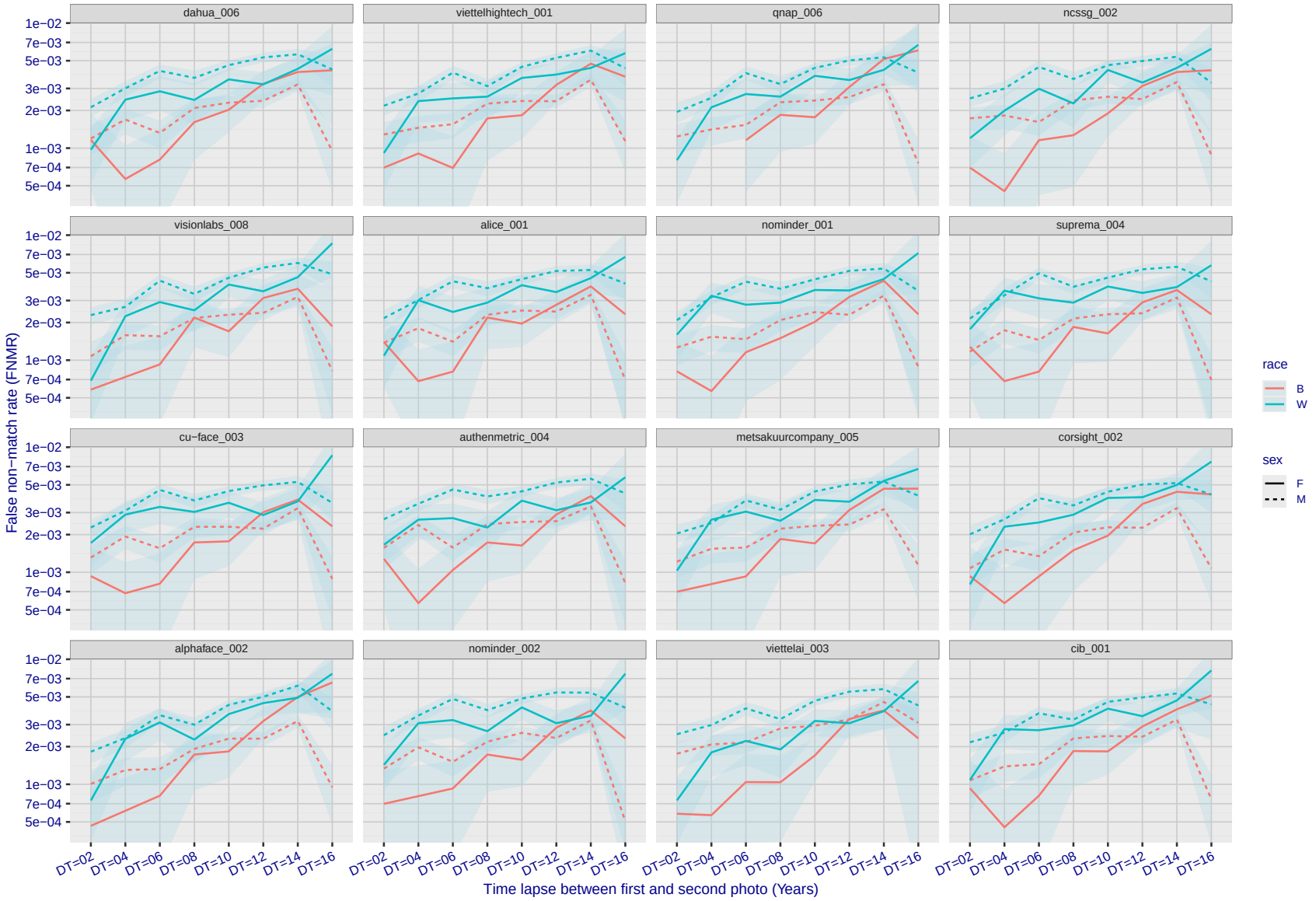

Figure 739: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

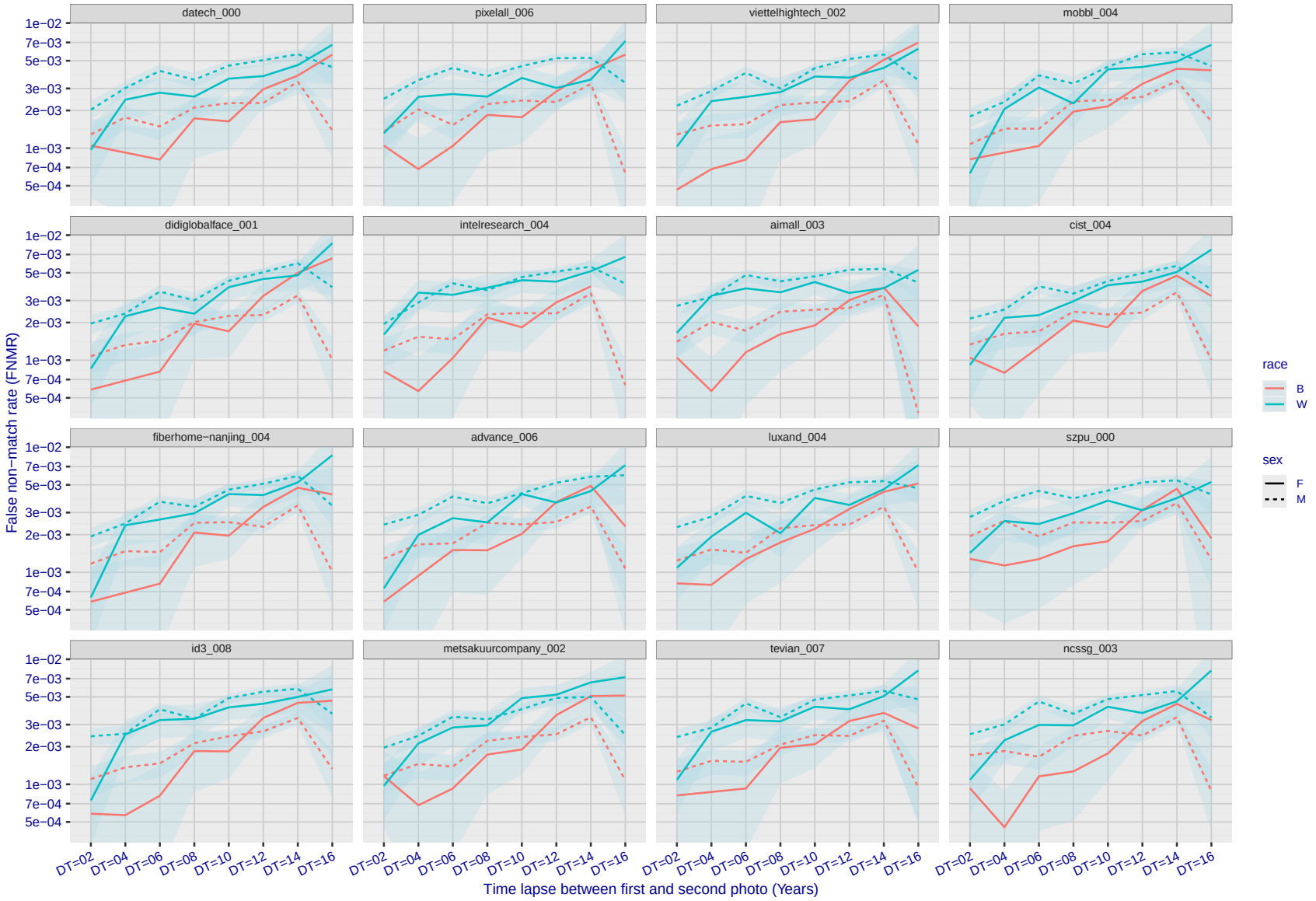

Figure 740: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widehat{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

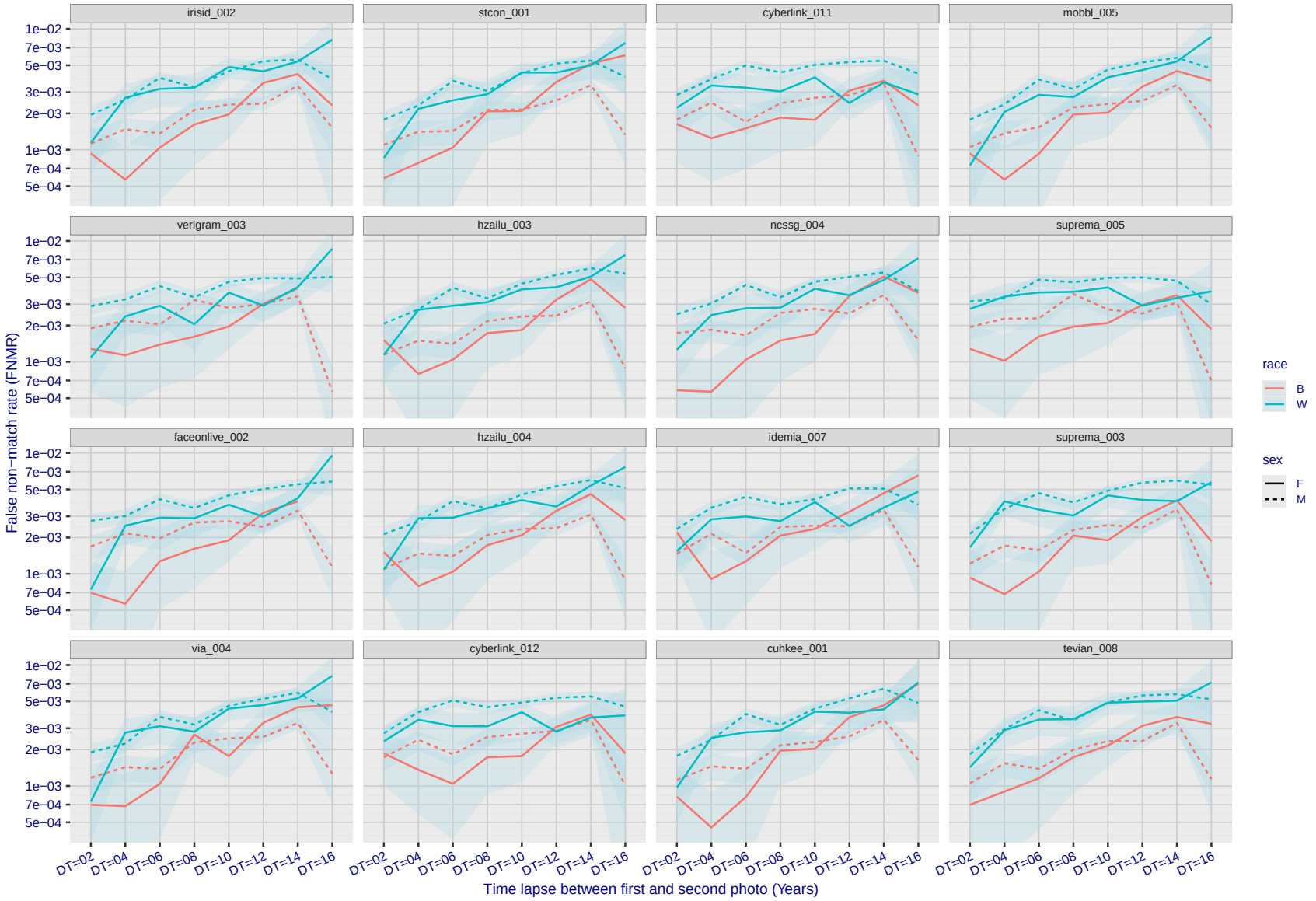

Figure 741: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

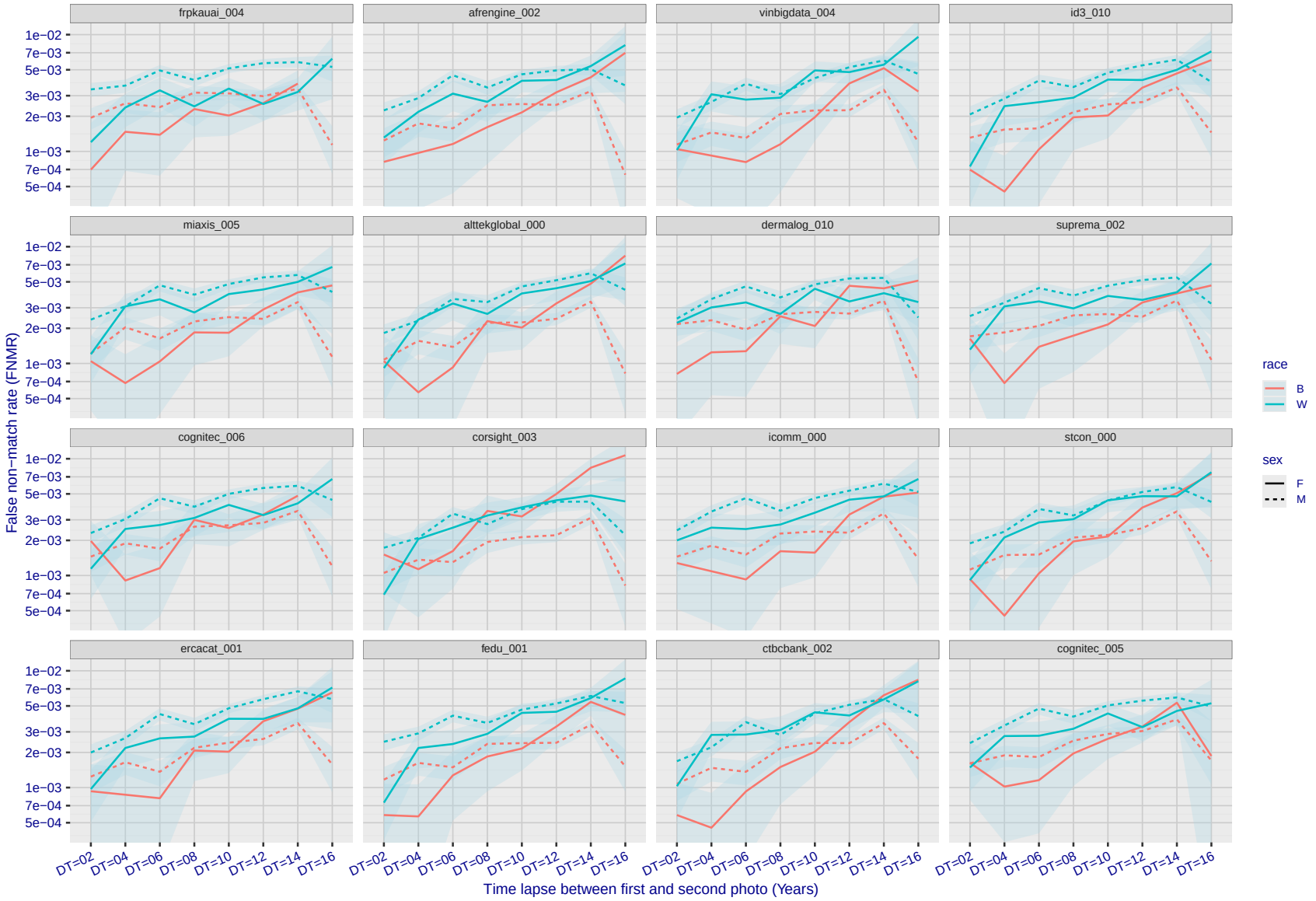

Figure 742: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

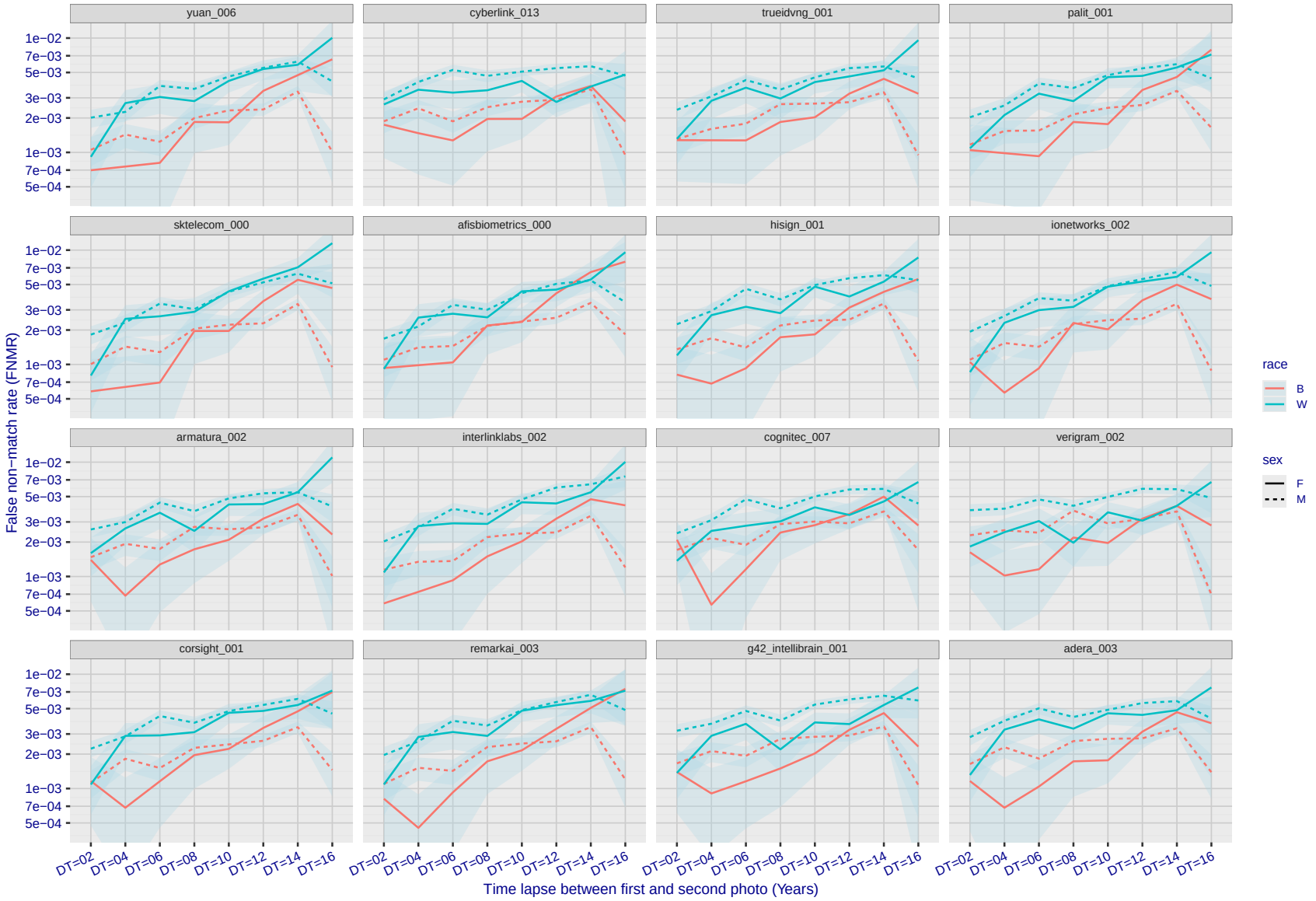

Figure 743: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

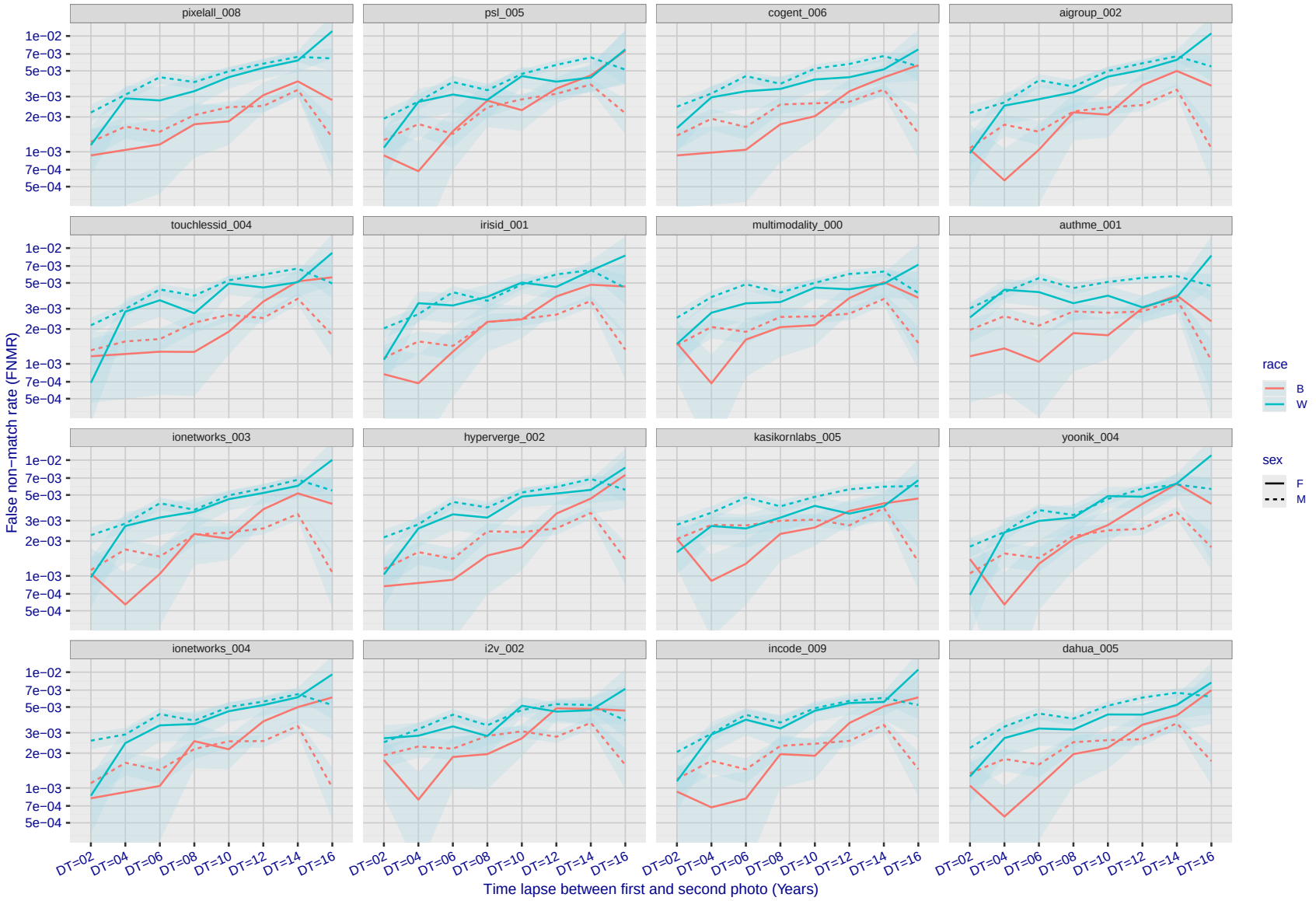

Figure 744: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

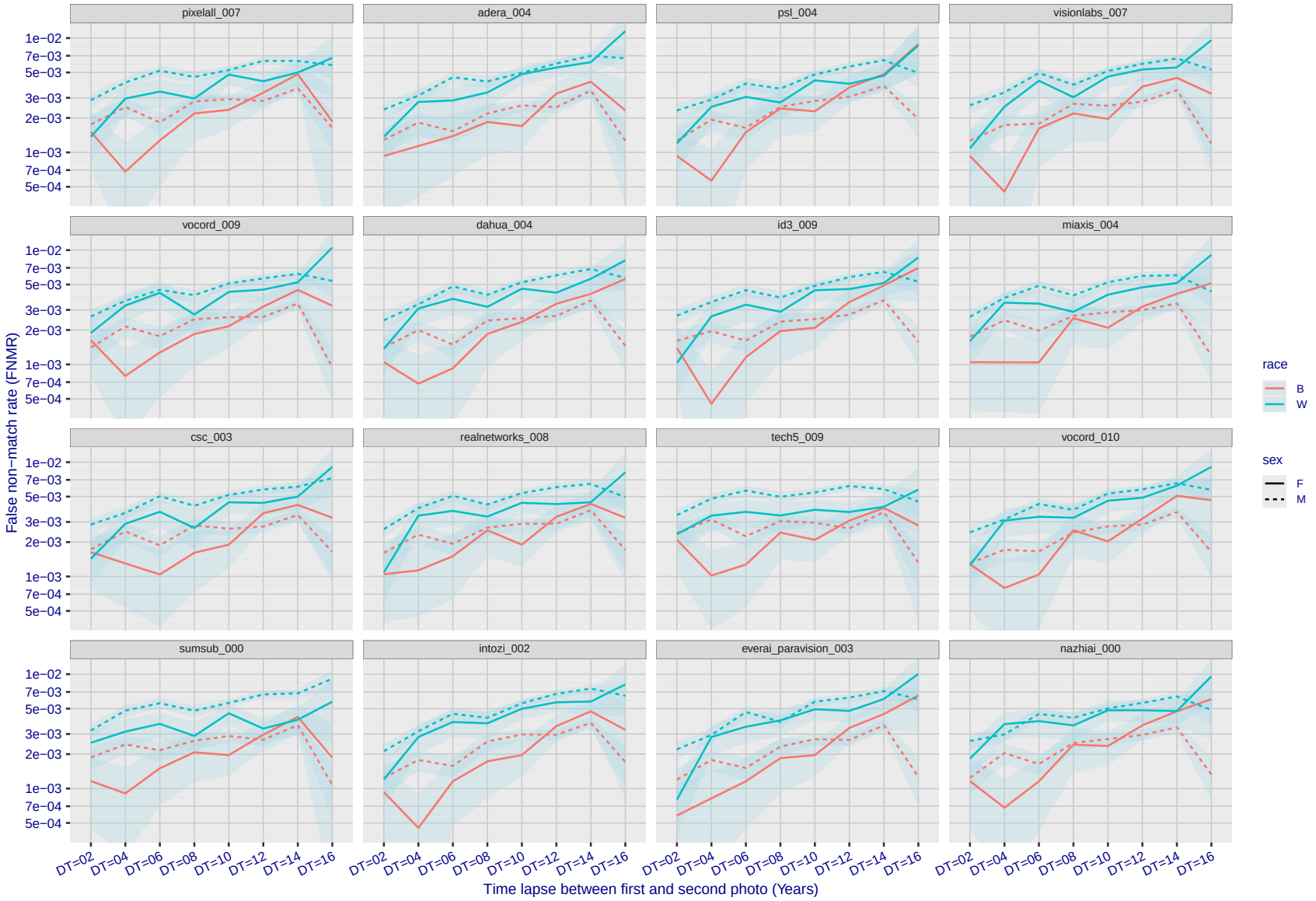

Figure 745: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

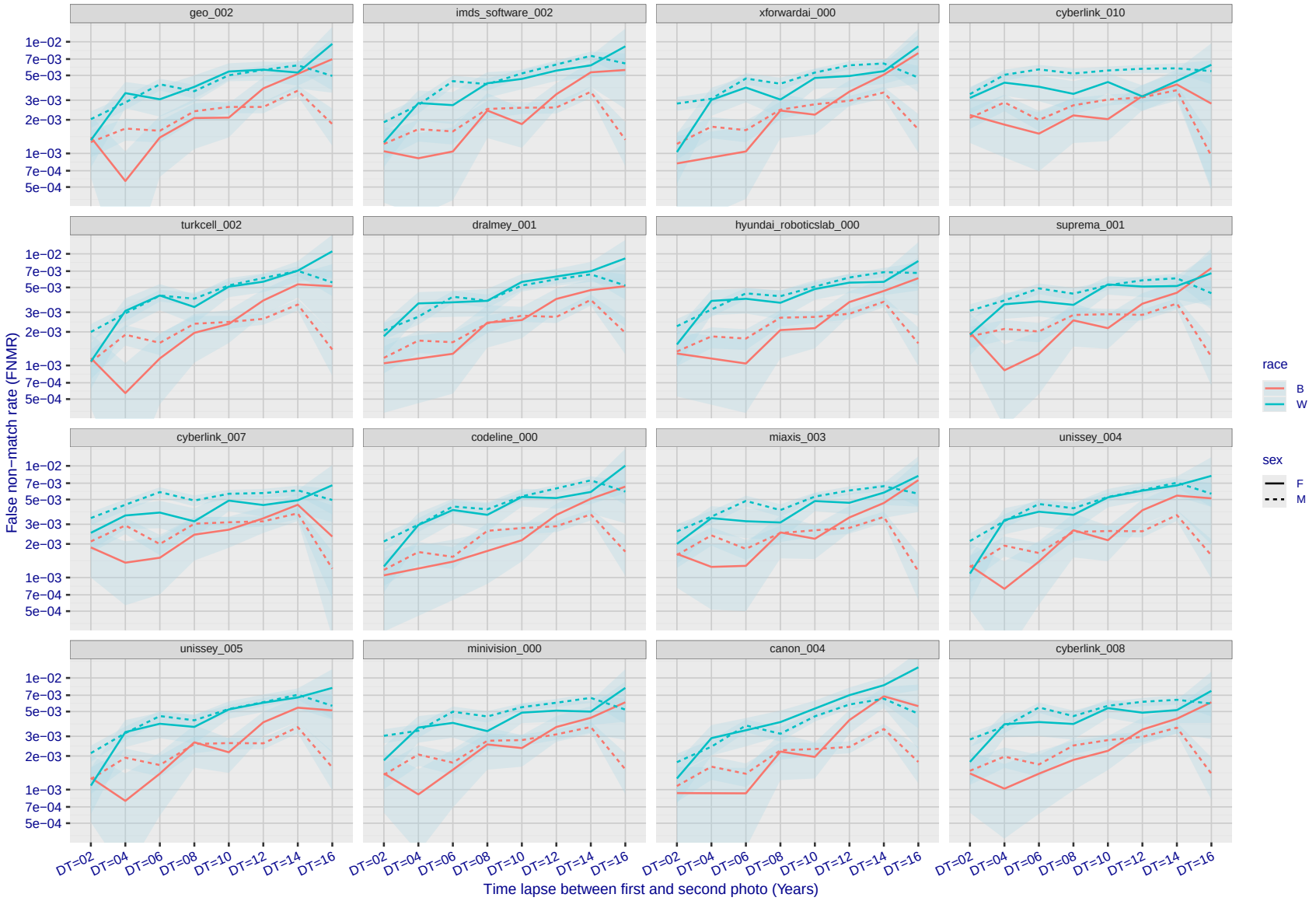

Figure 746: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

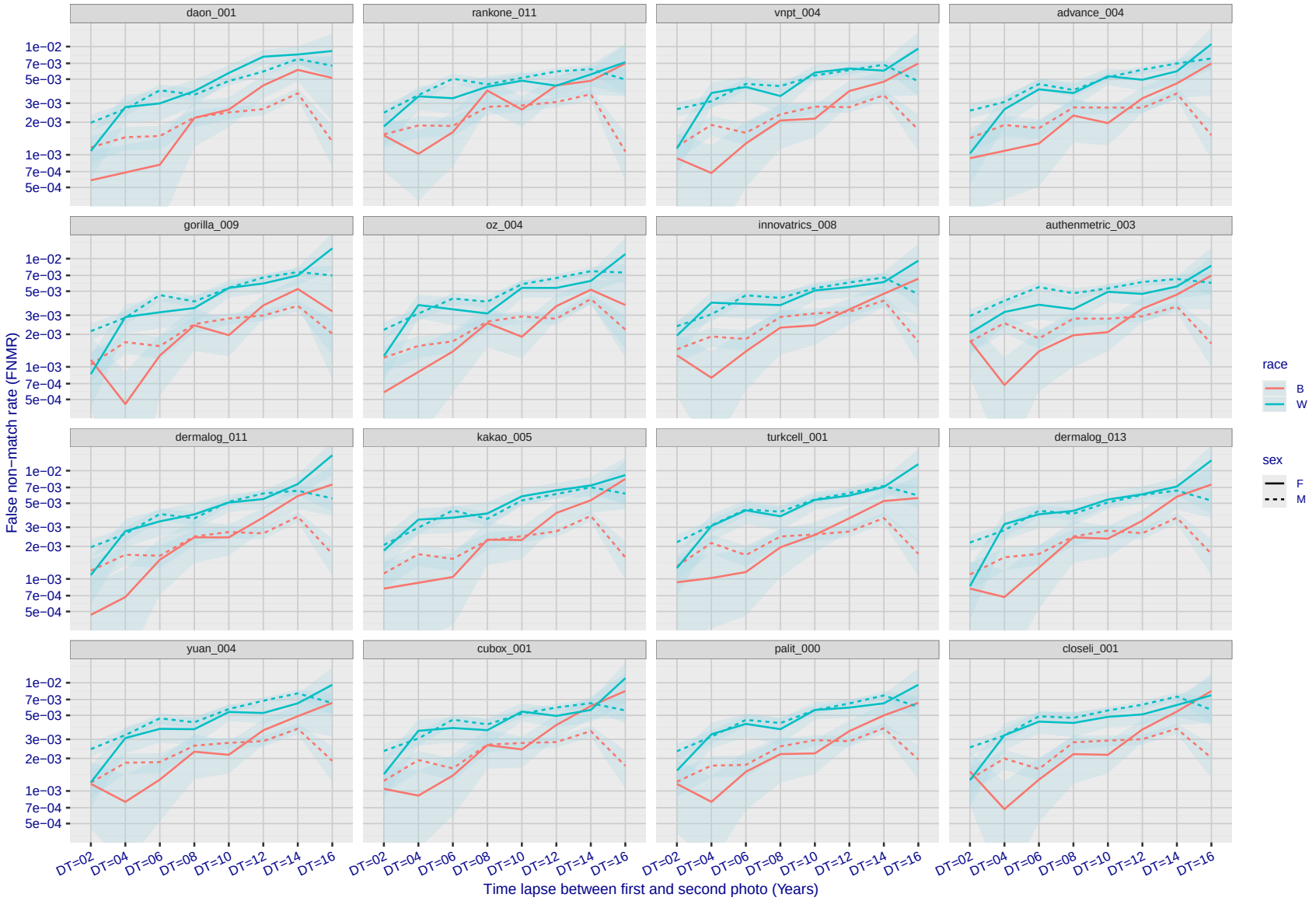

Figure 747: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

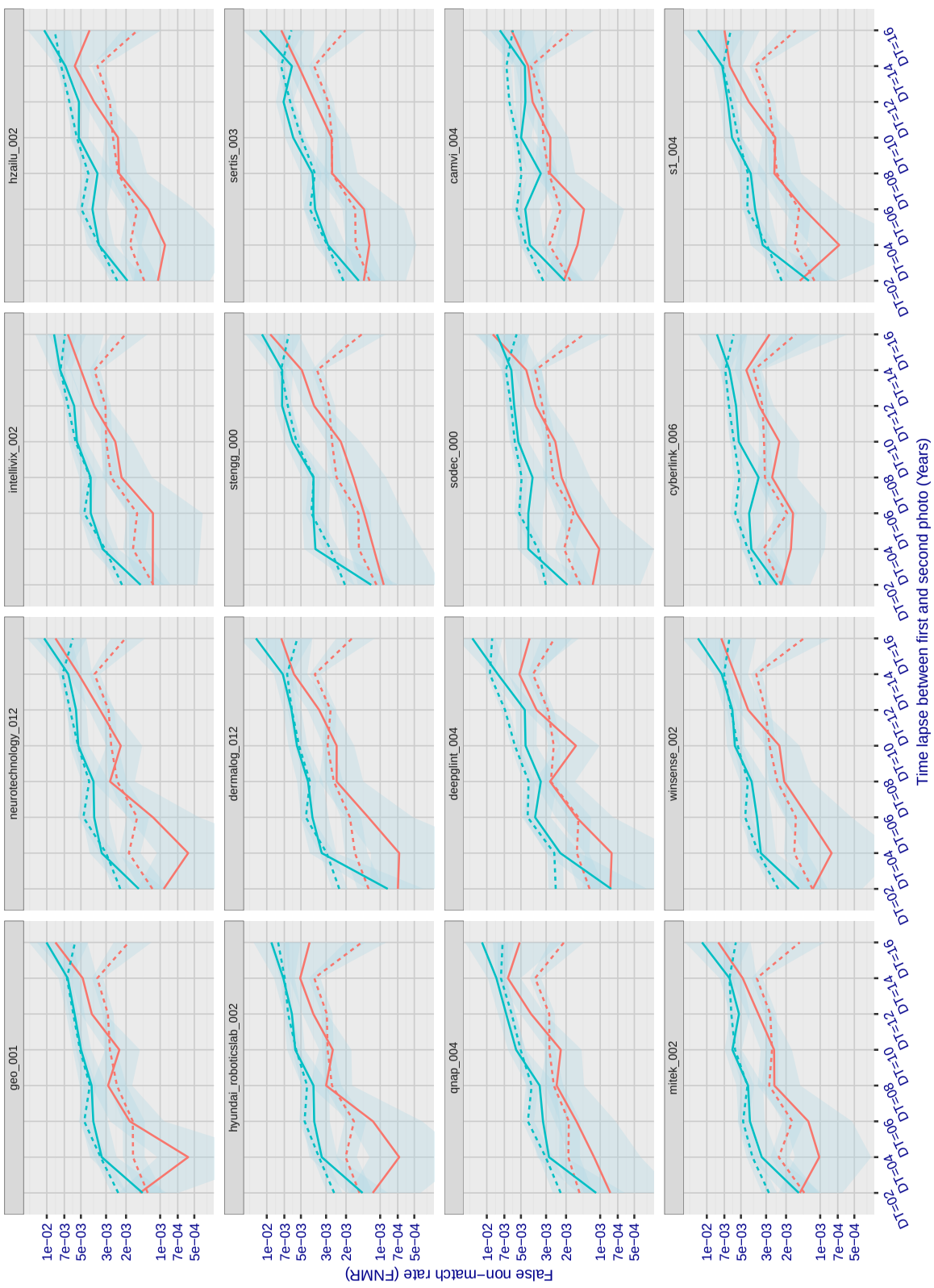

Figure 748: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

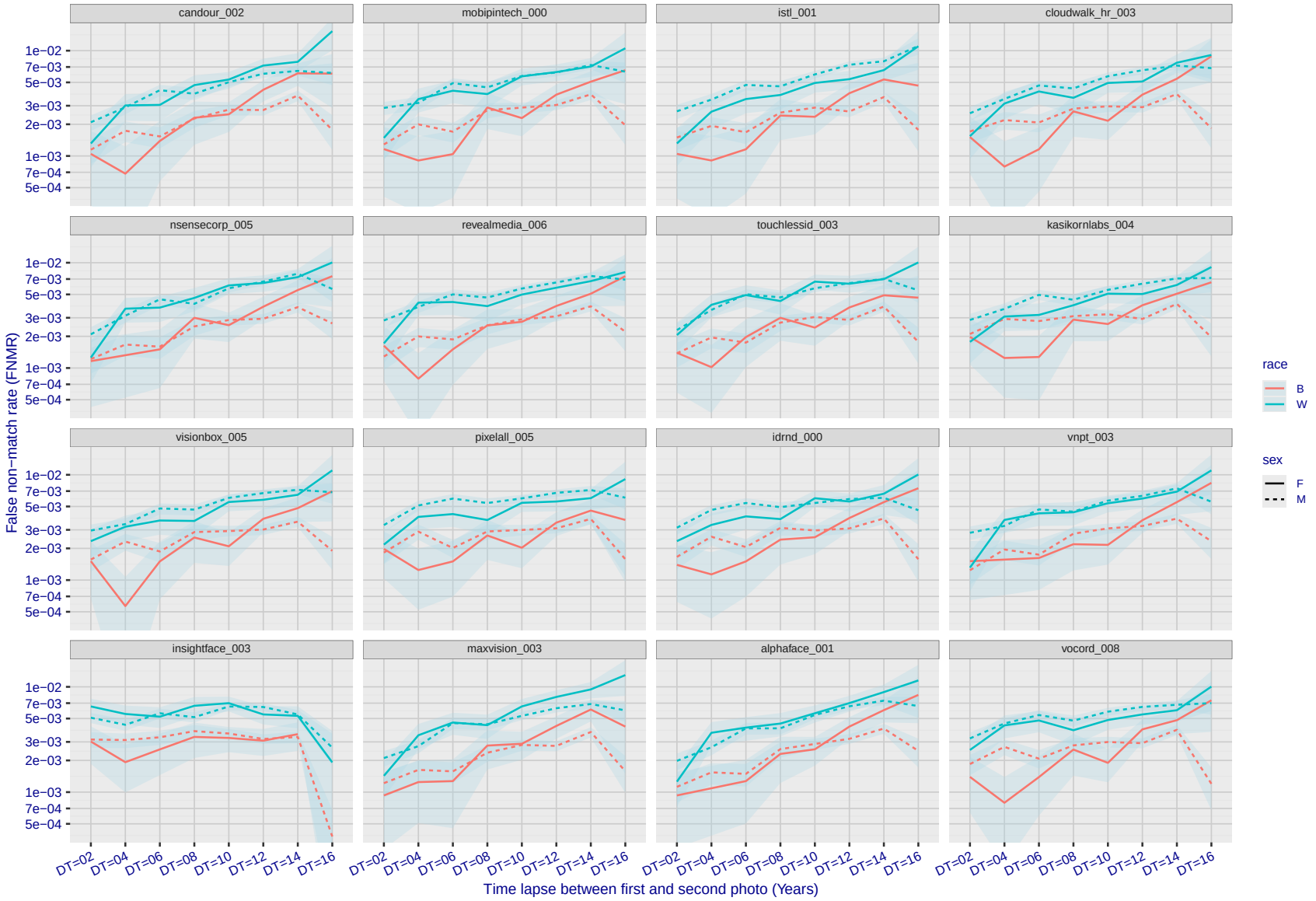

Figure 749: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

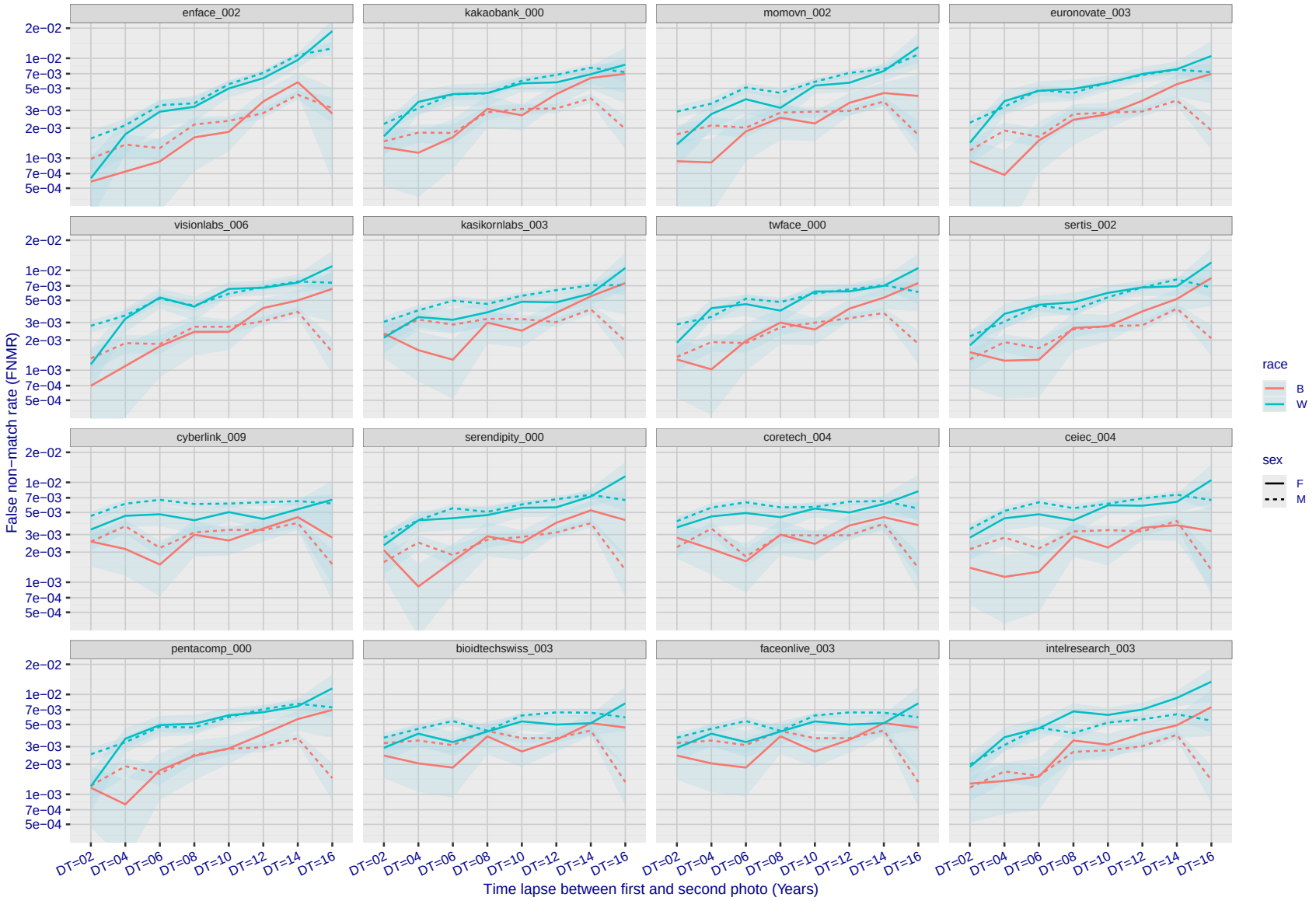

Figure 750: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

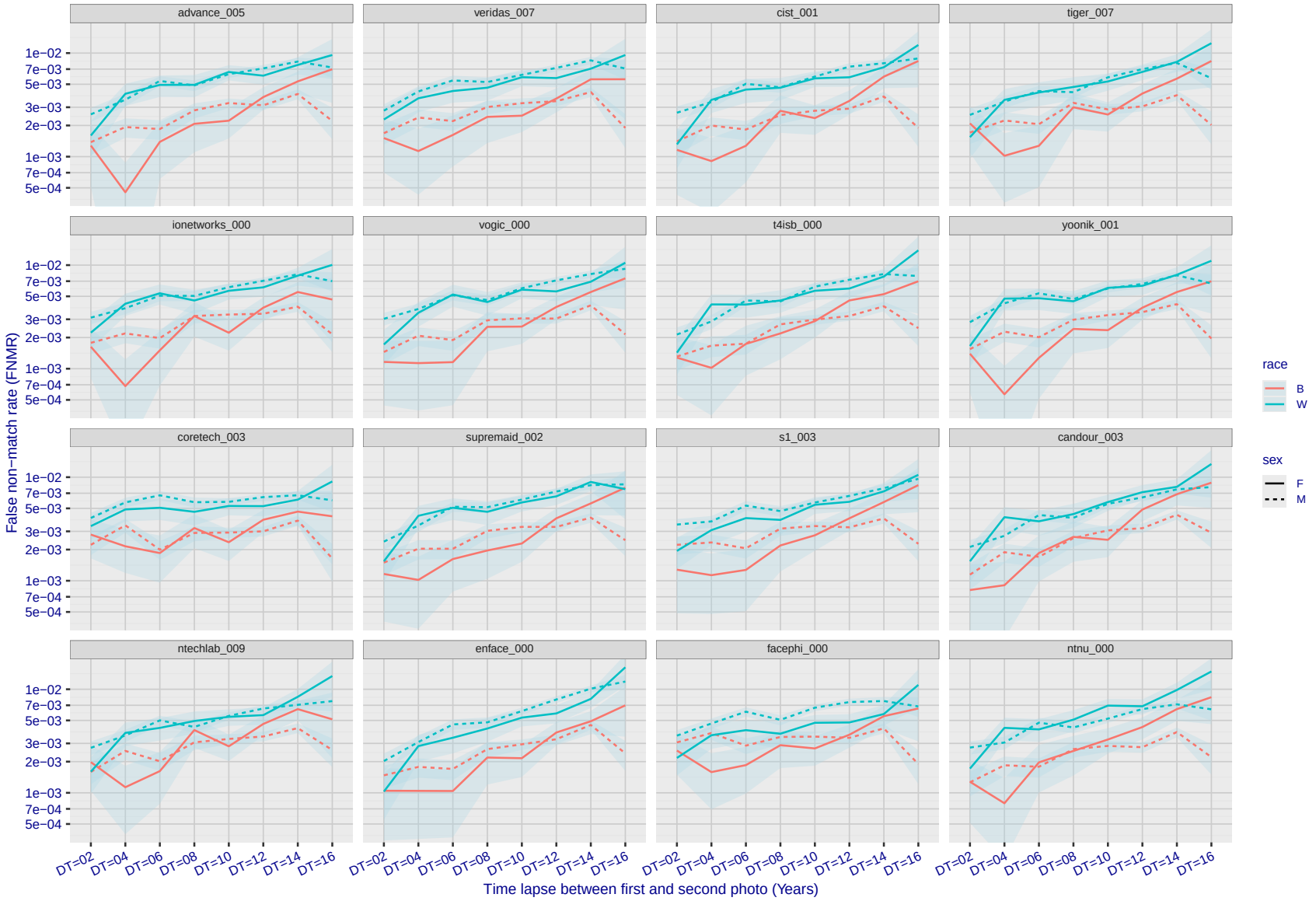

Figure 751: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

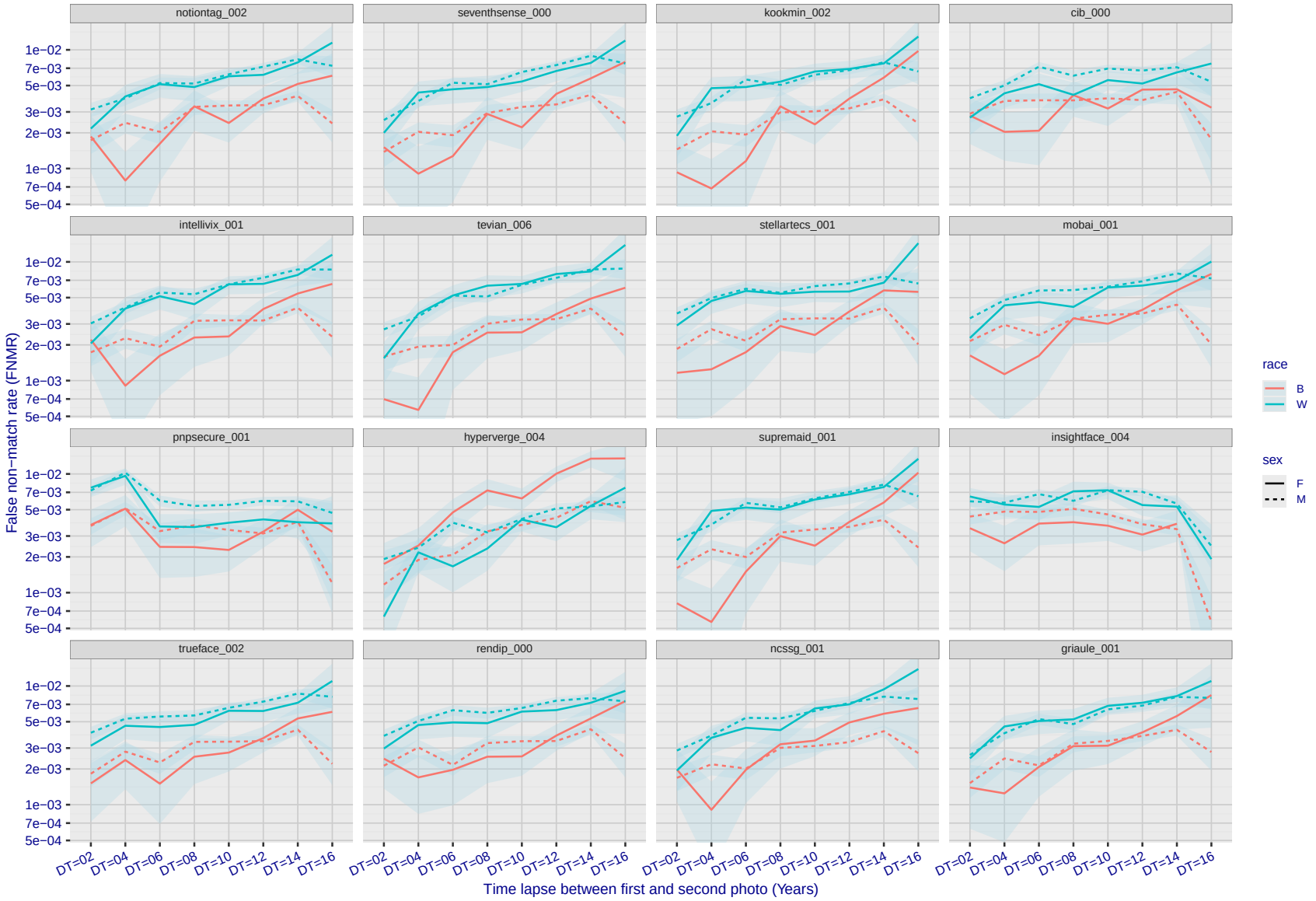

Figure 752: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

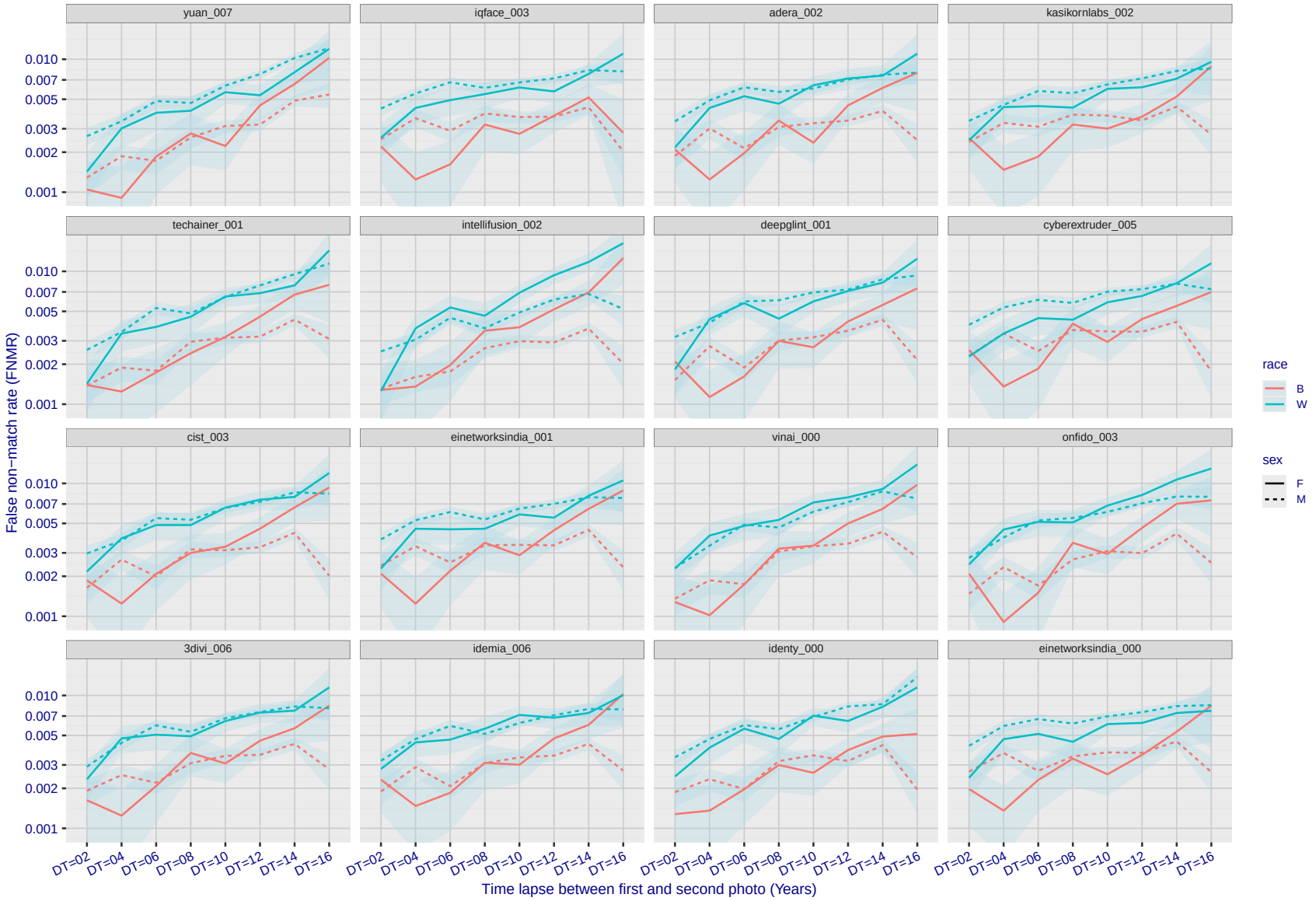

Figure 753: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

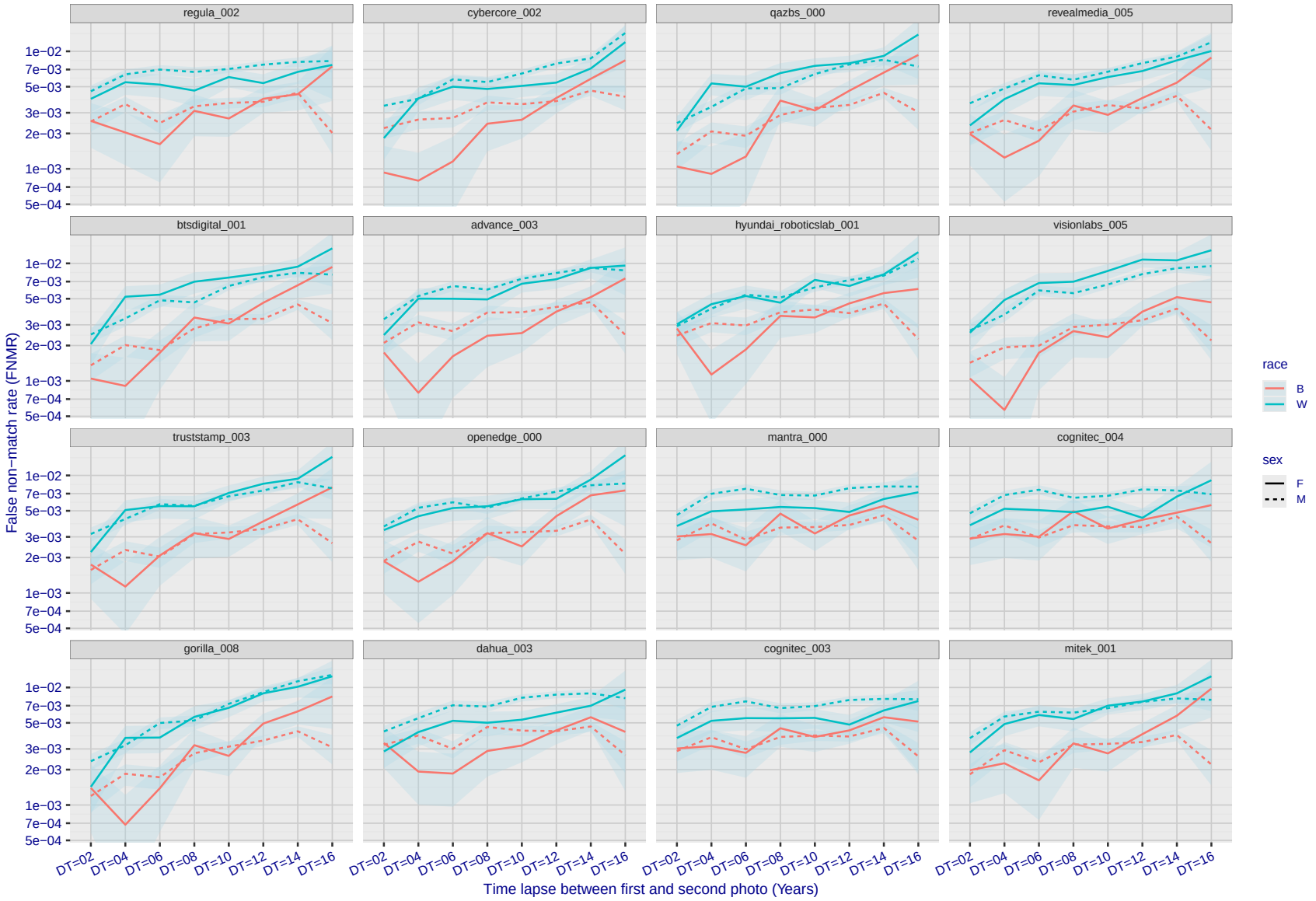

Figure 754: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

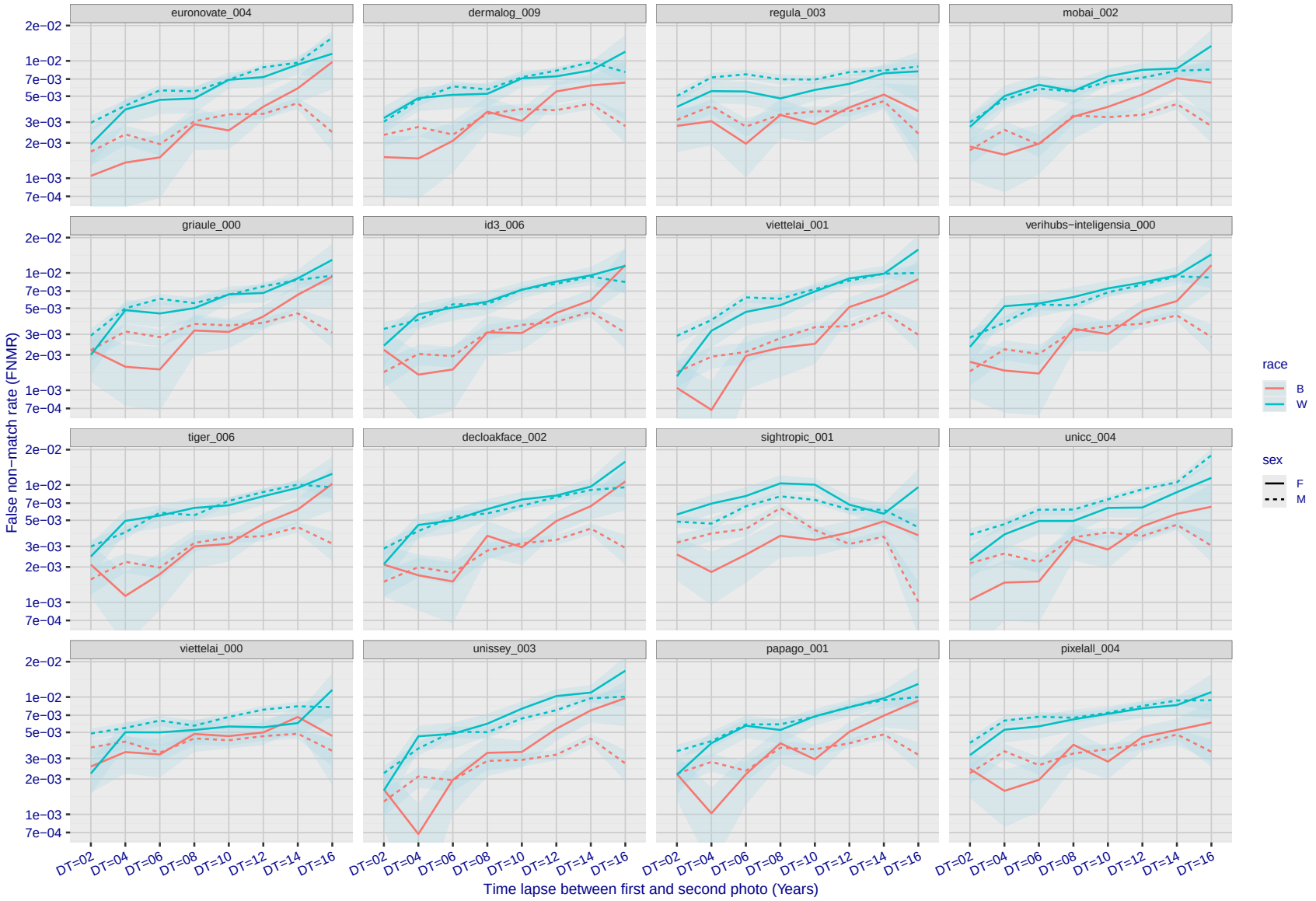

Figure 755: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

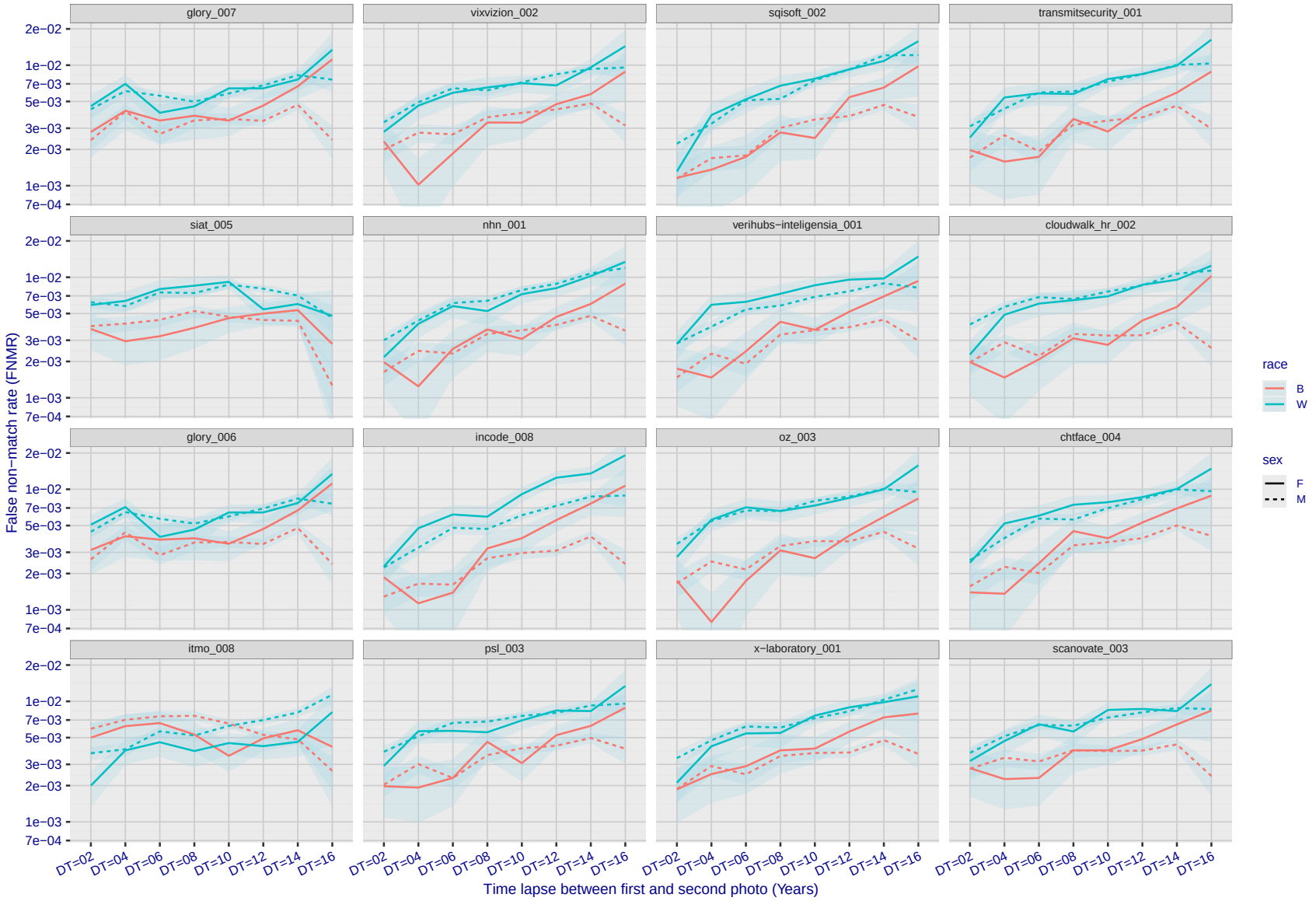

Figure 756: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

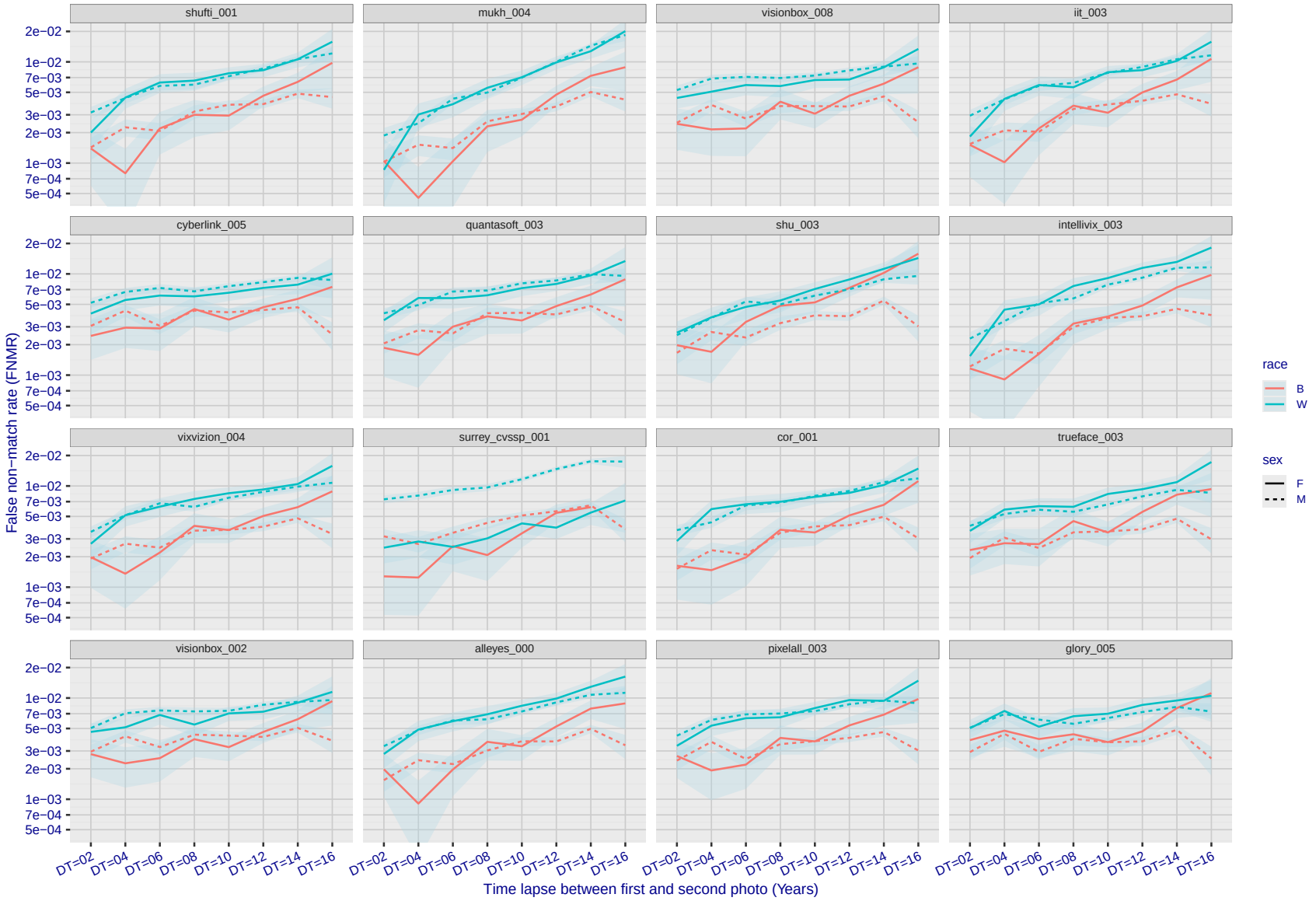

Figure 757: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

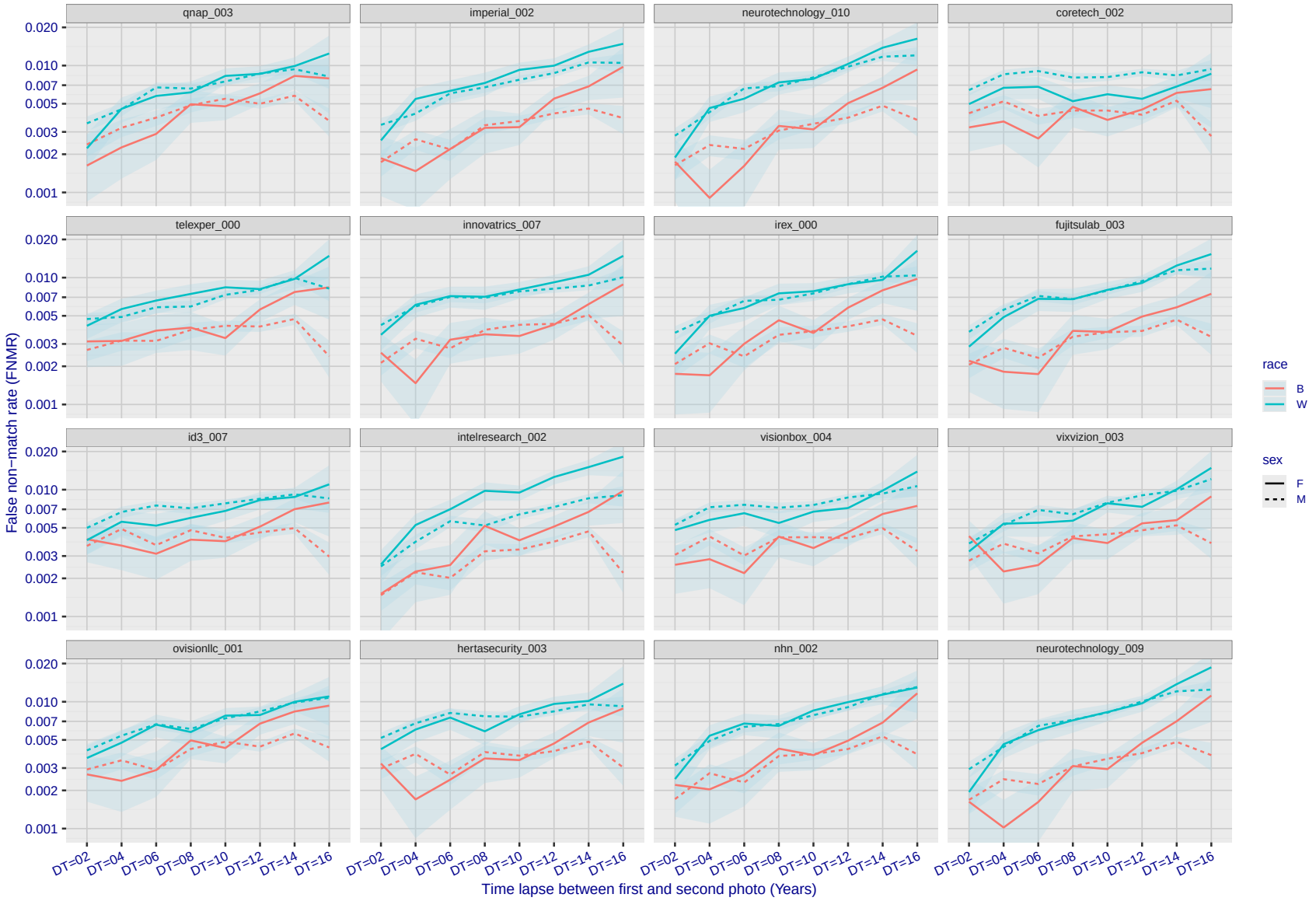

Figure 758: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

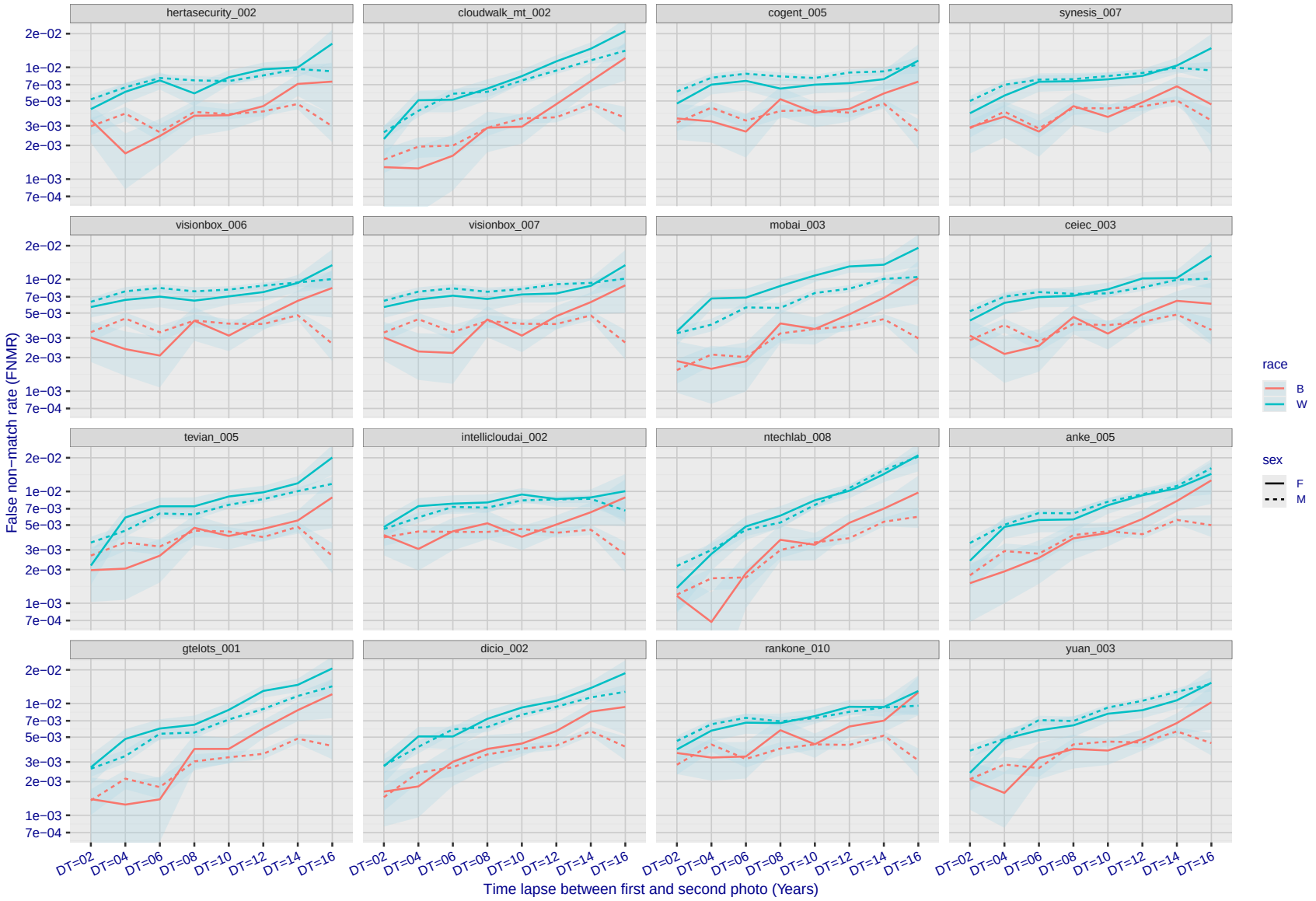

Figure 759: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

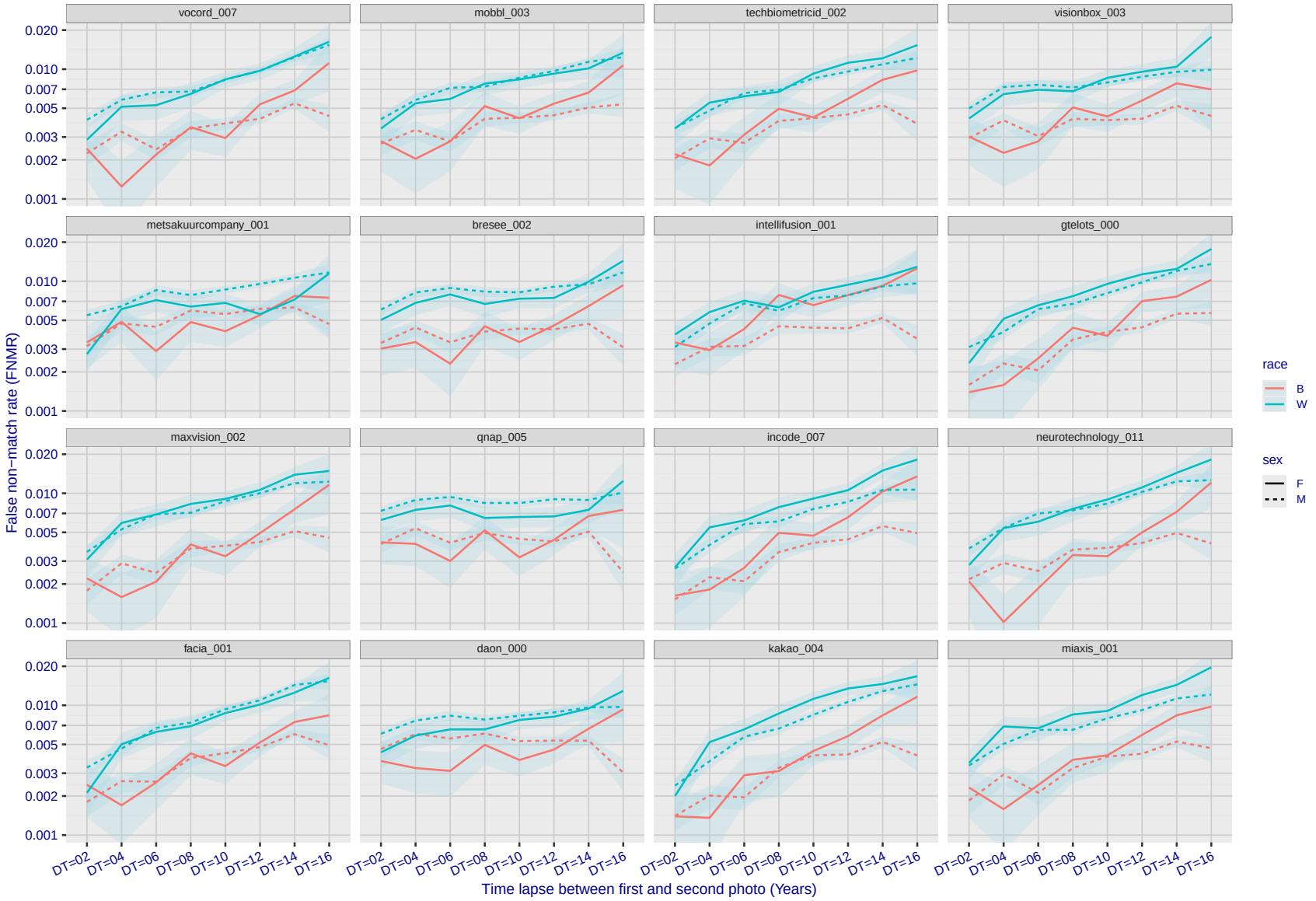

Figure 760: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

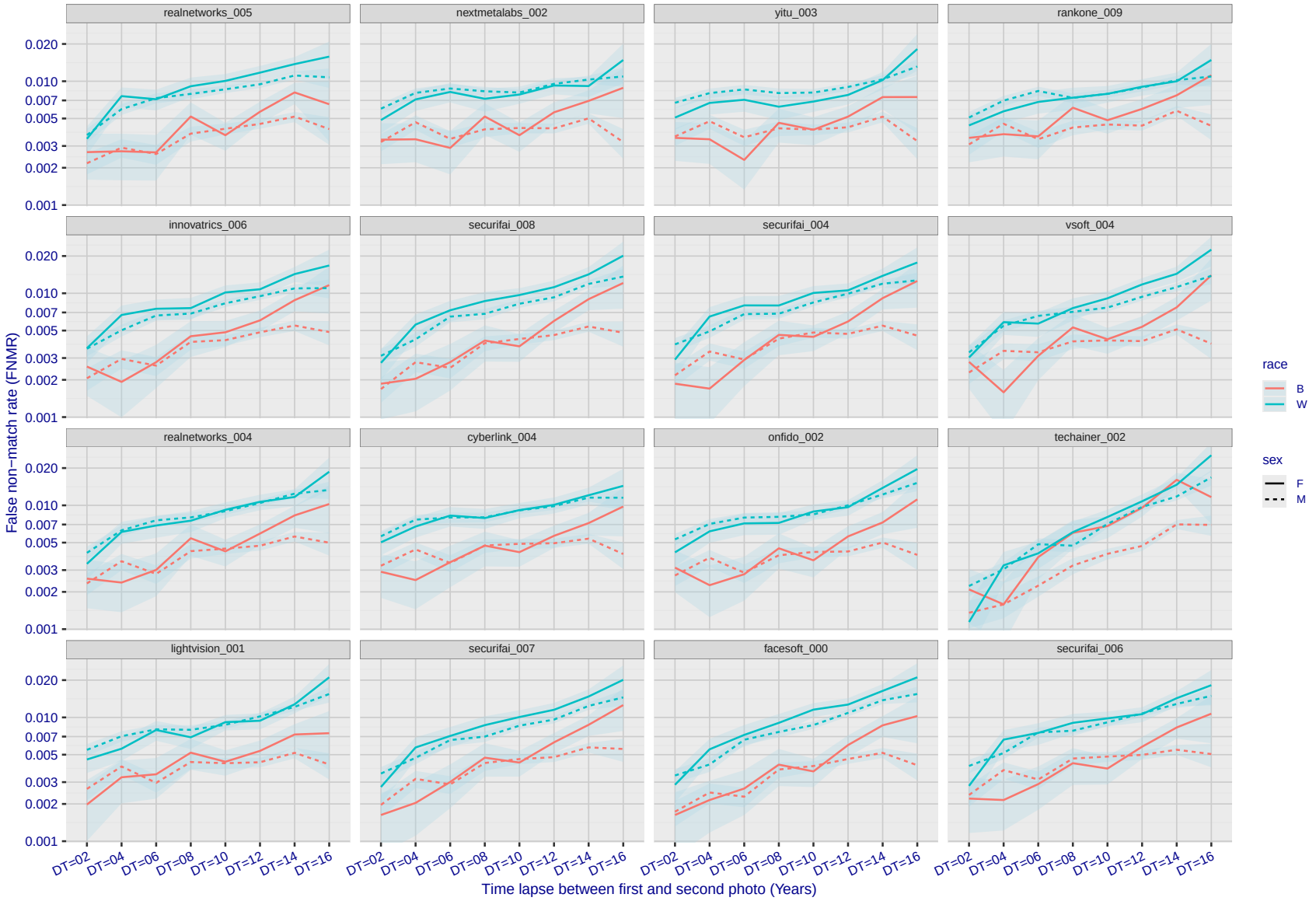

Figure 761: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

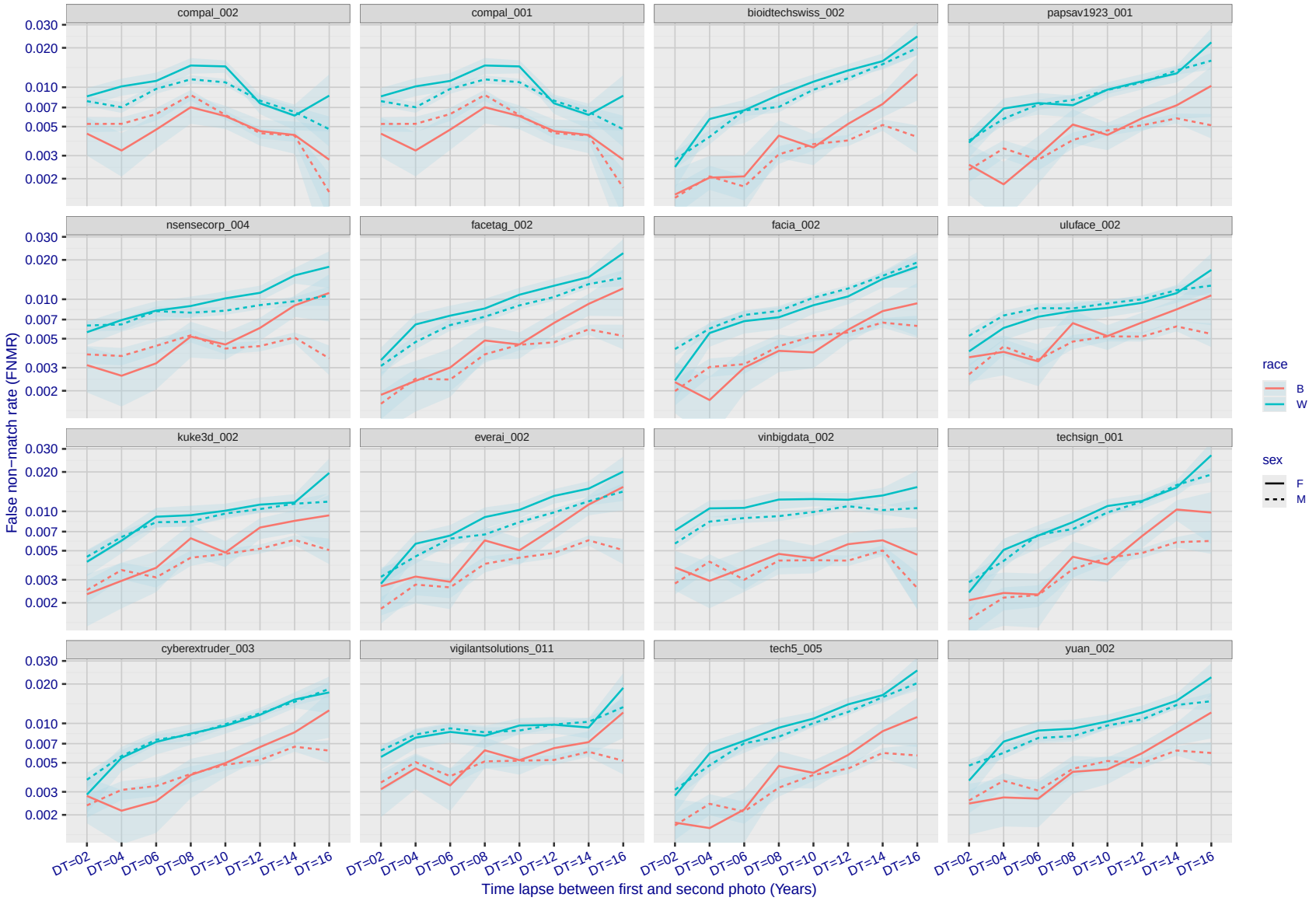

Figure 762: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

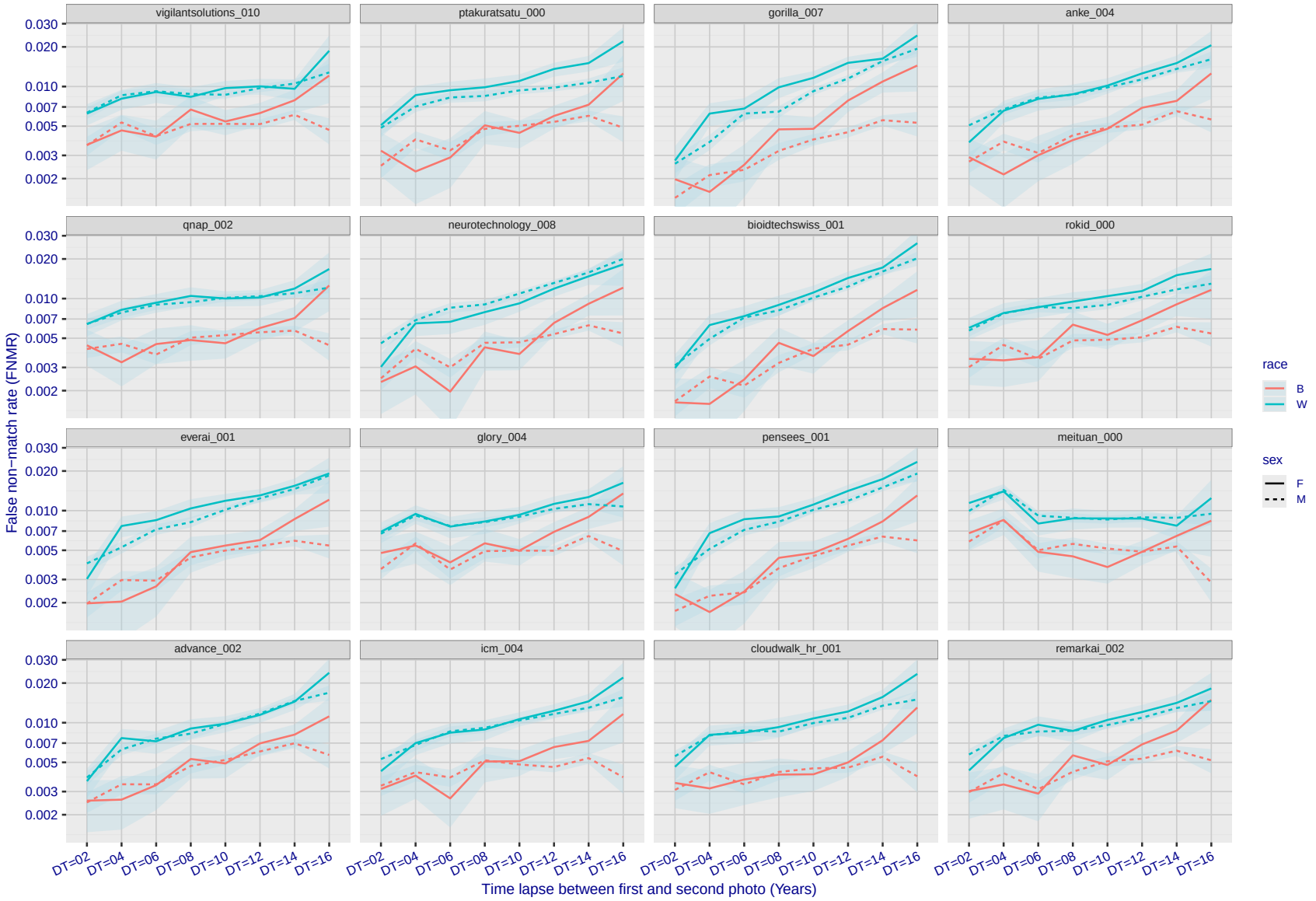

Figure 763: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

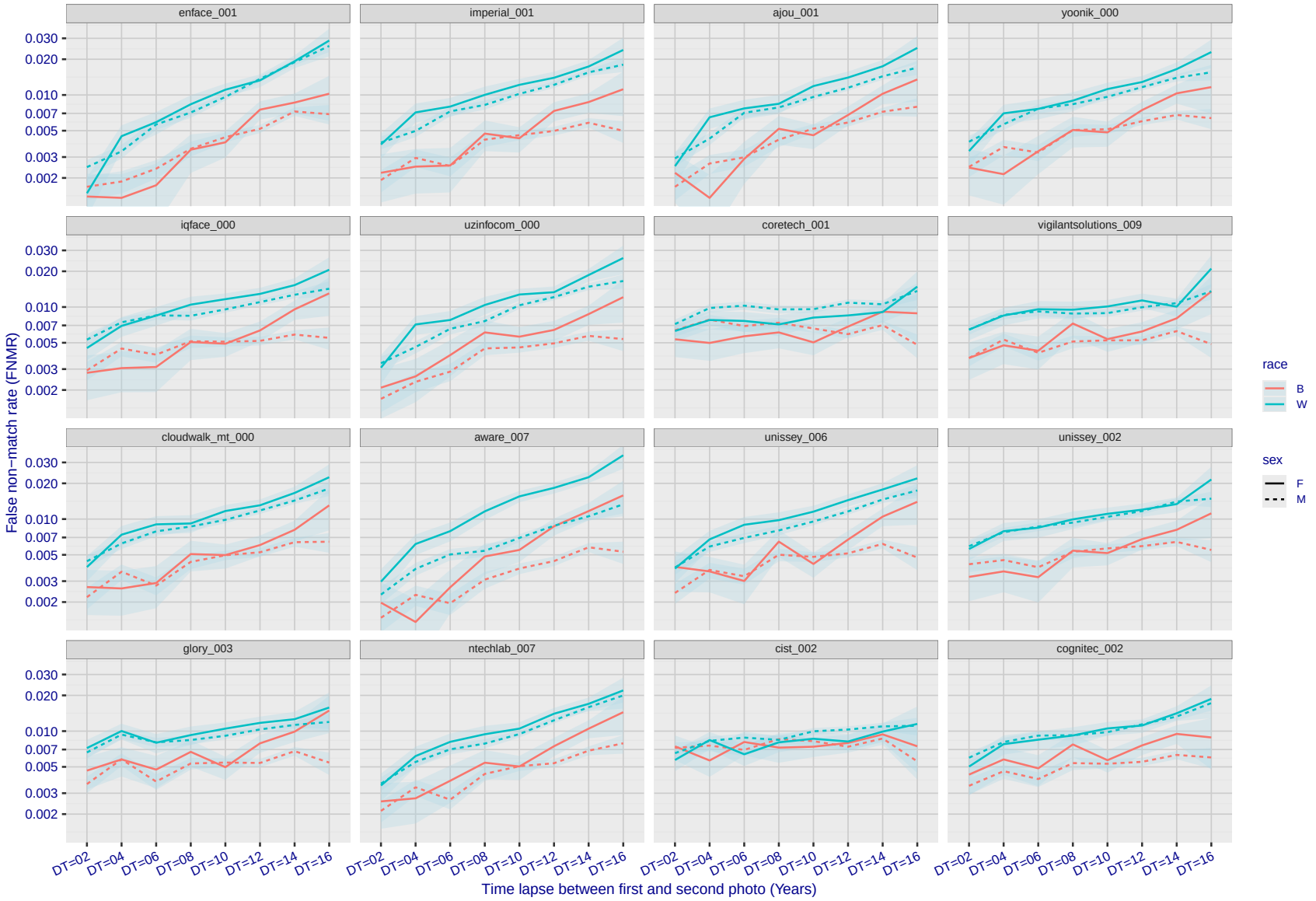

Figure 764: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

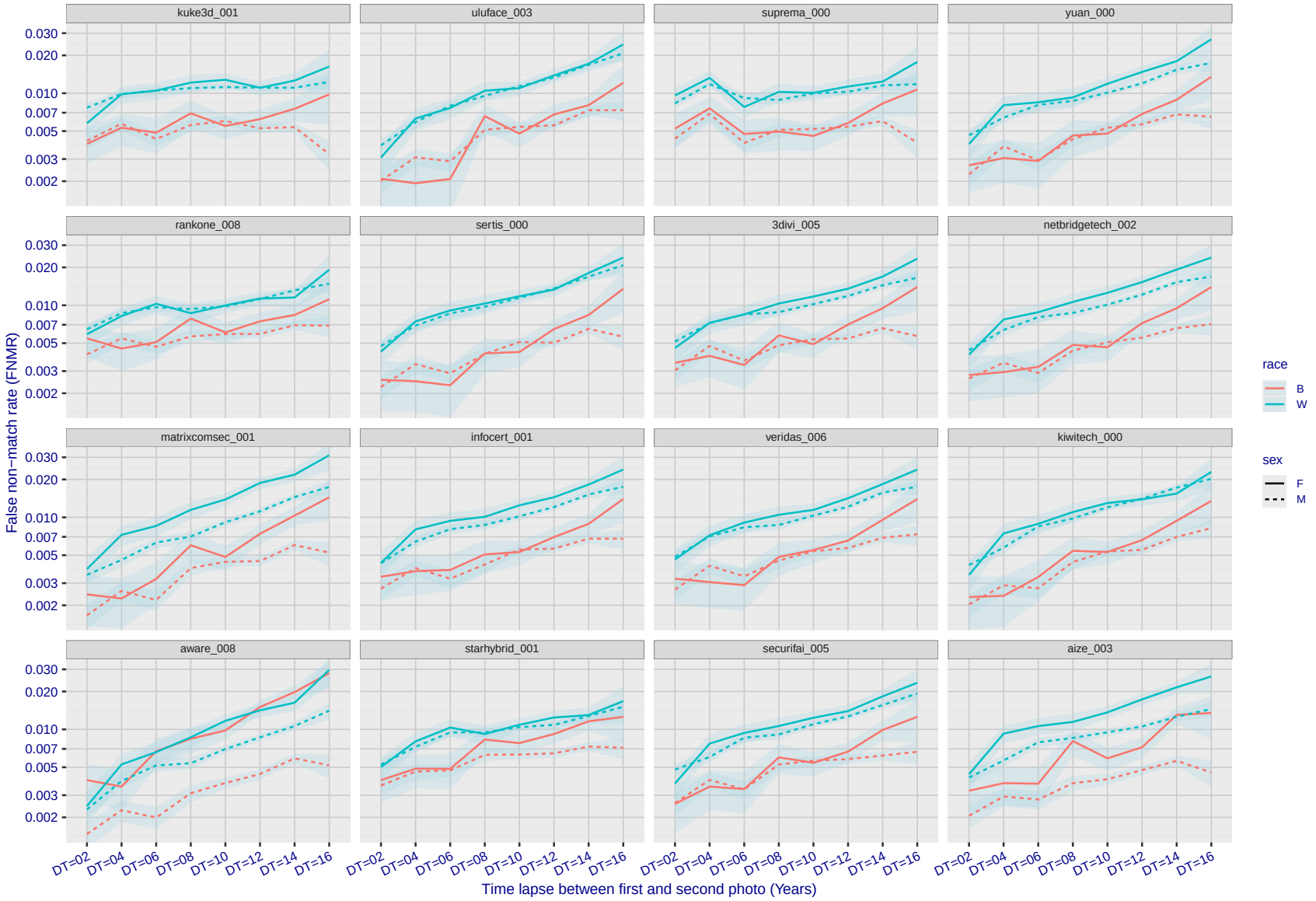

Figure 765: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

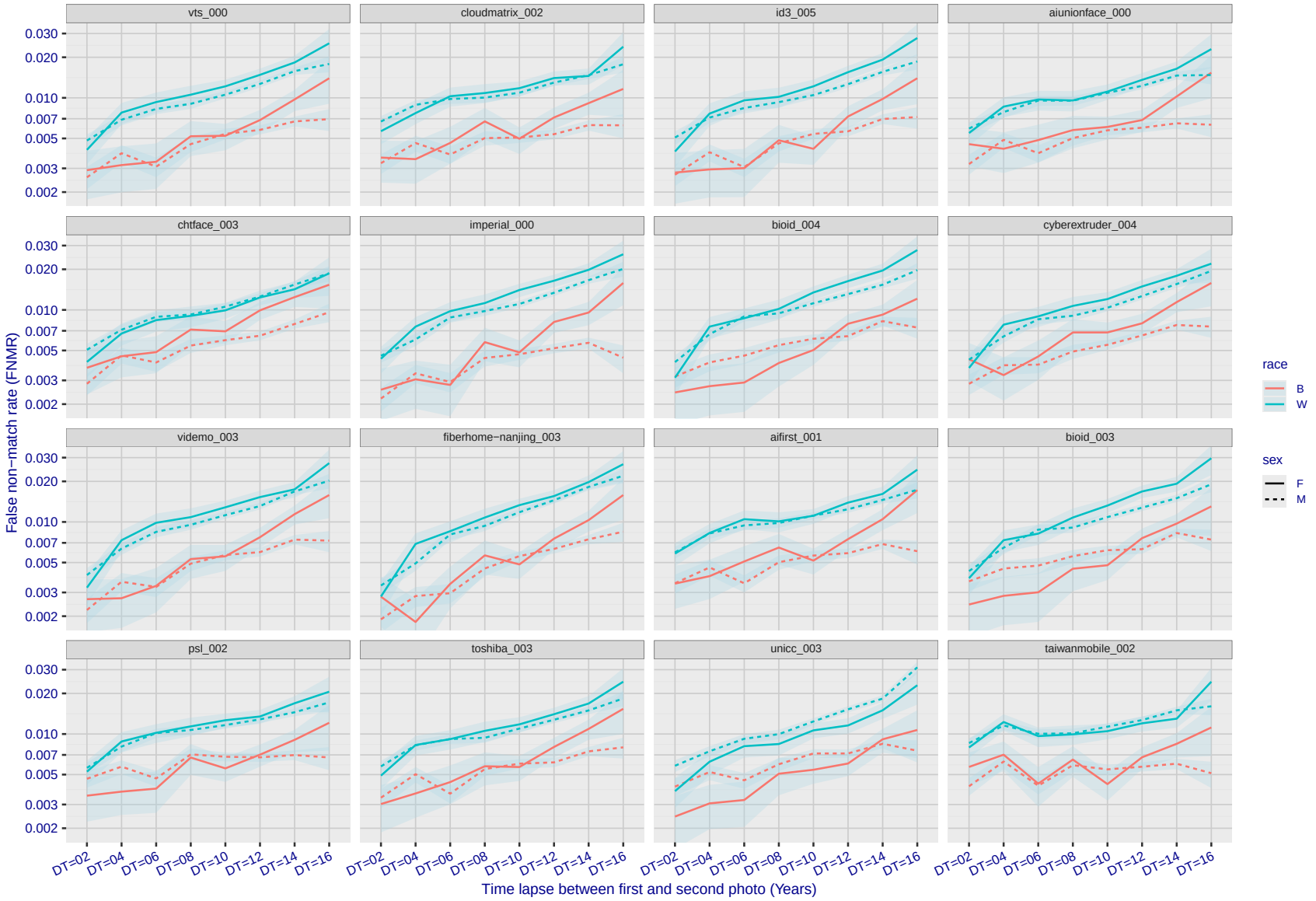

Figure 766: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

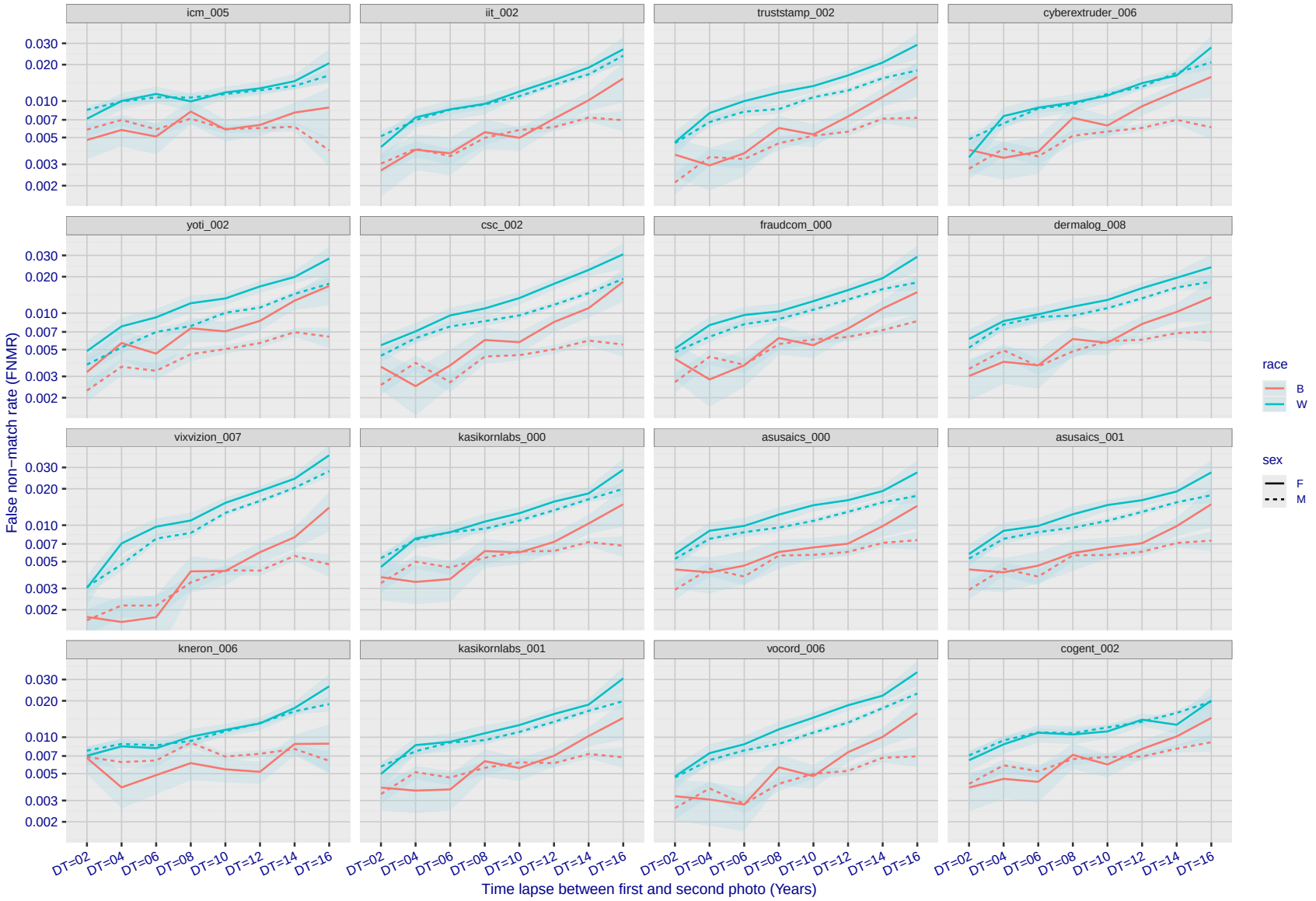

Figure 767: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

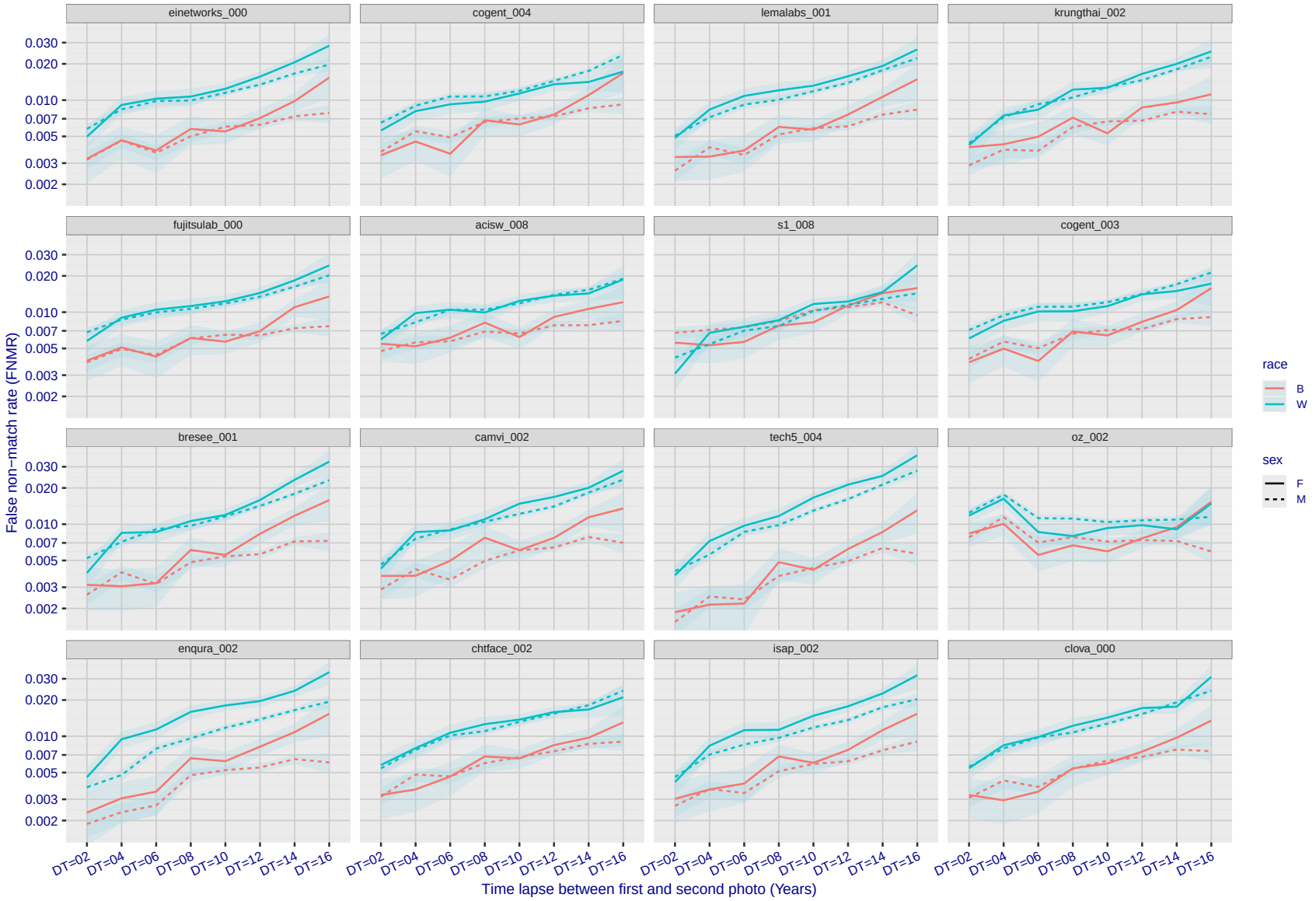

Figure 768: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

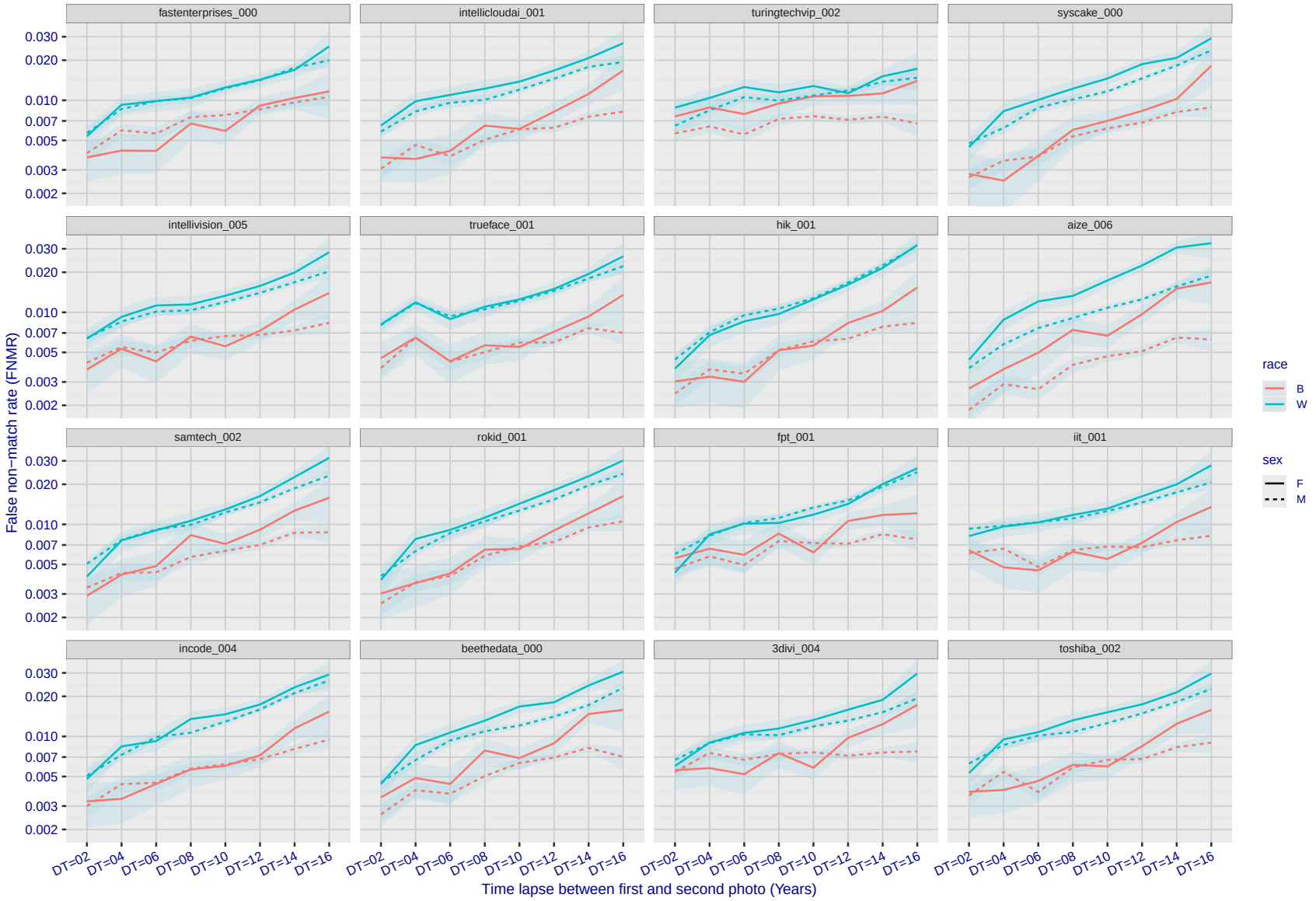

Figure 769: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

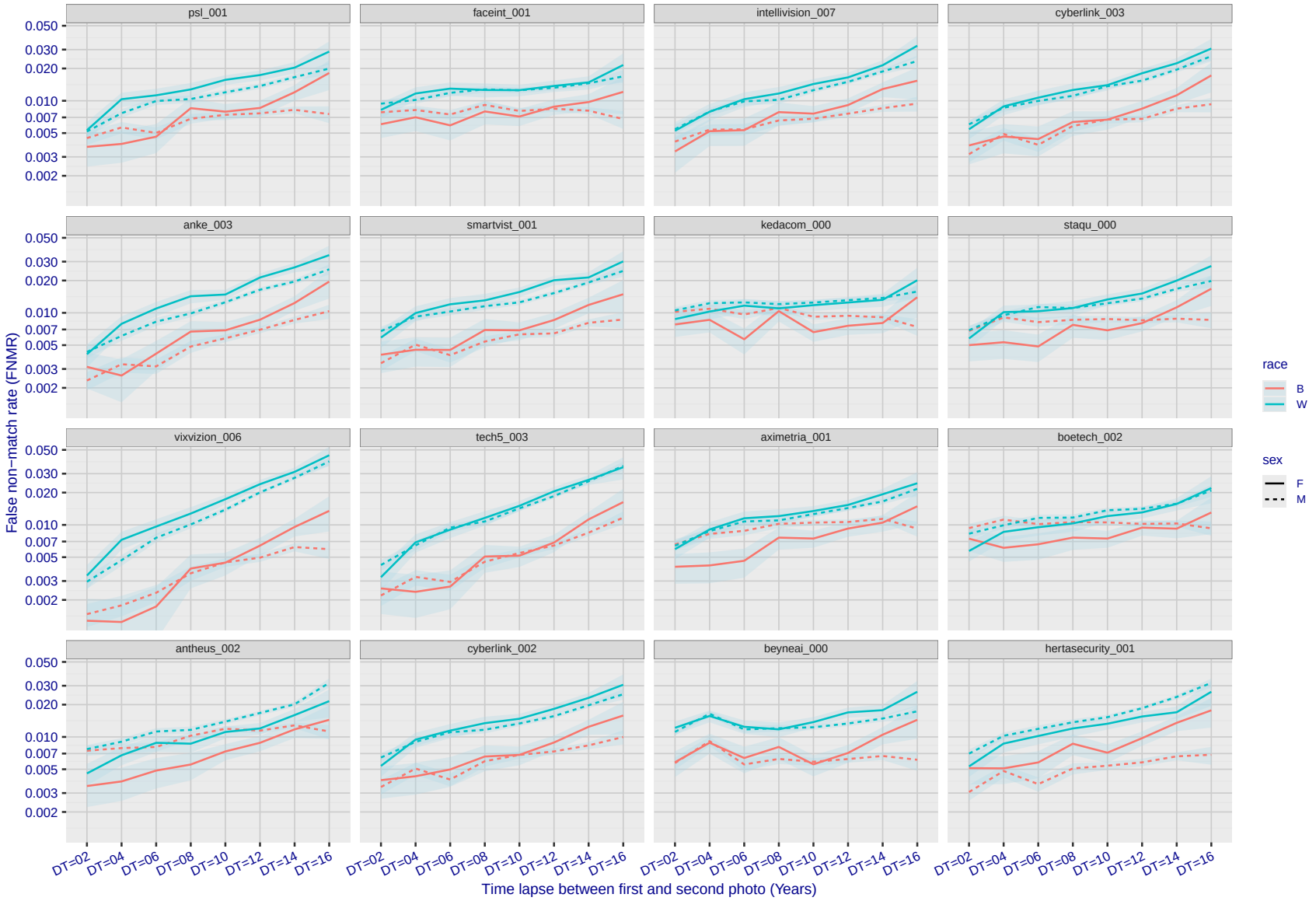

Figure 770: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

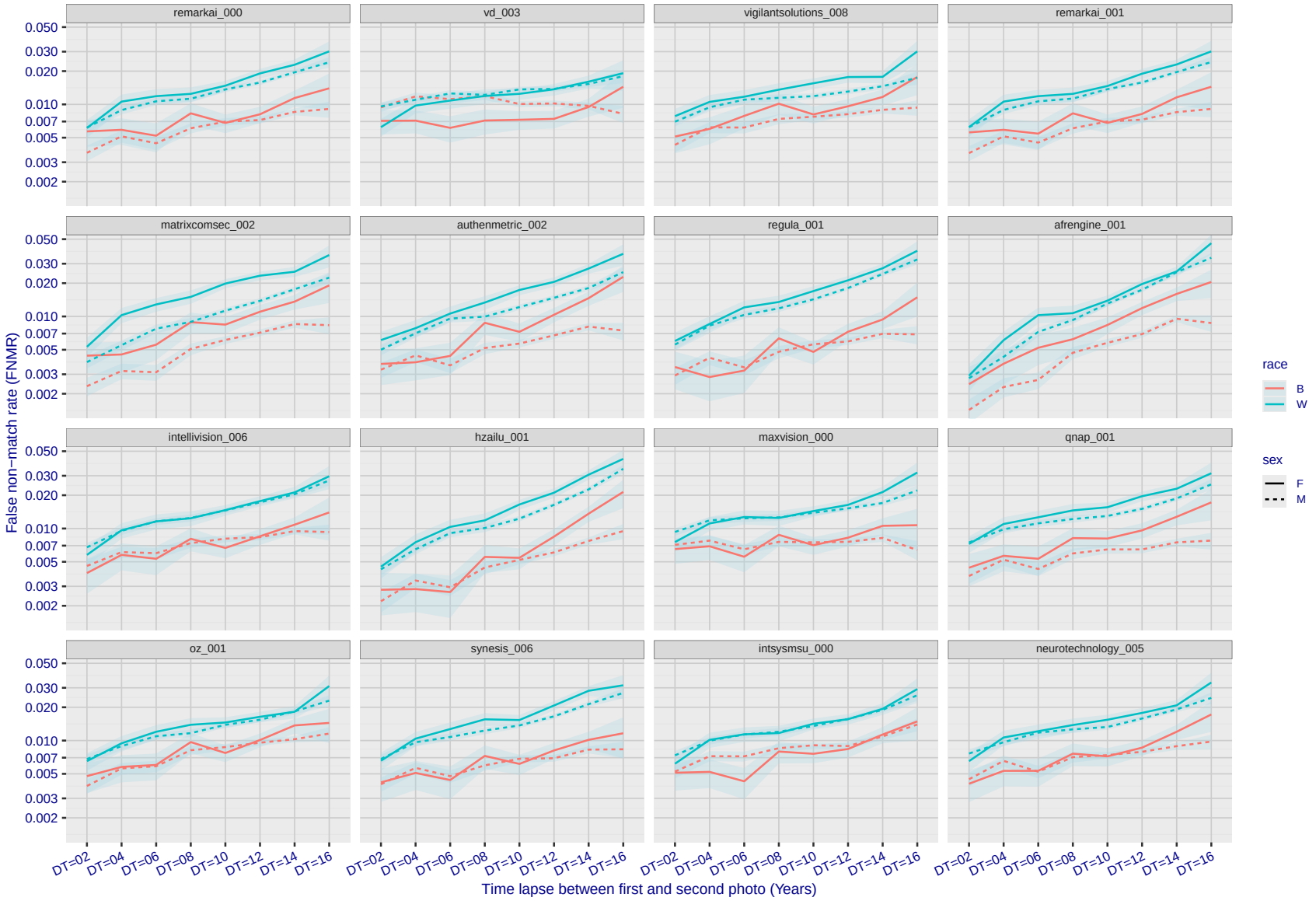

Figure 771: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

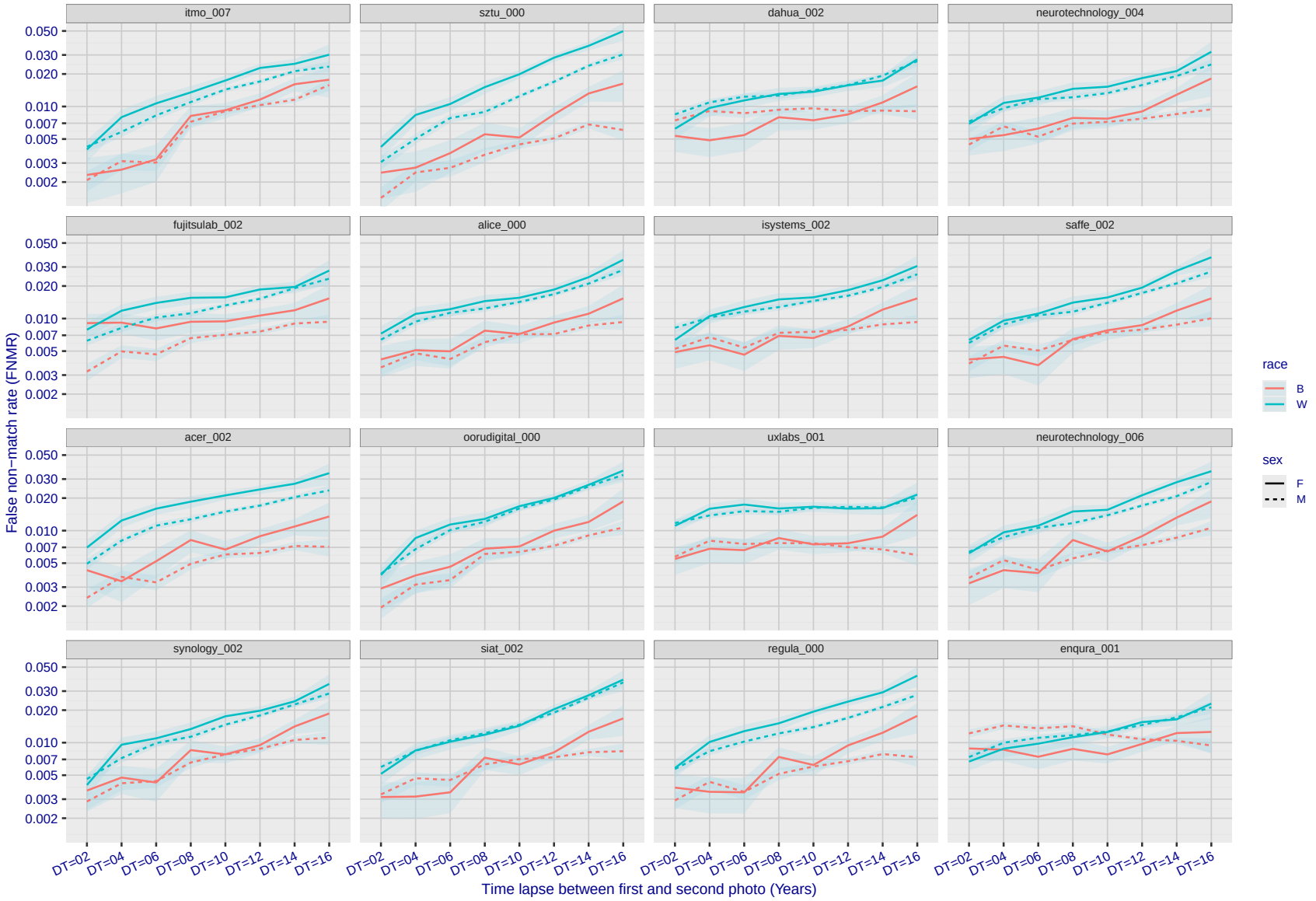

Figure 772: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

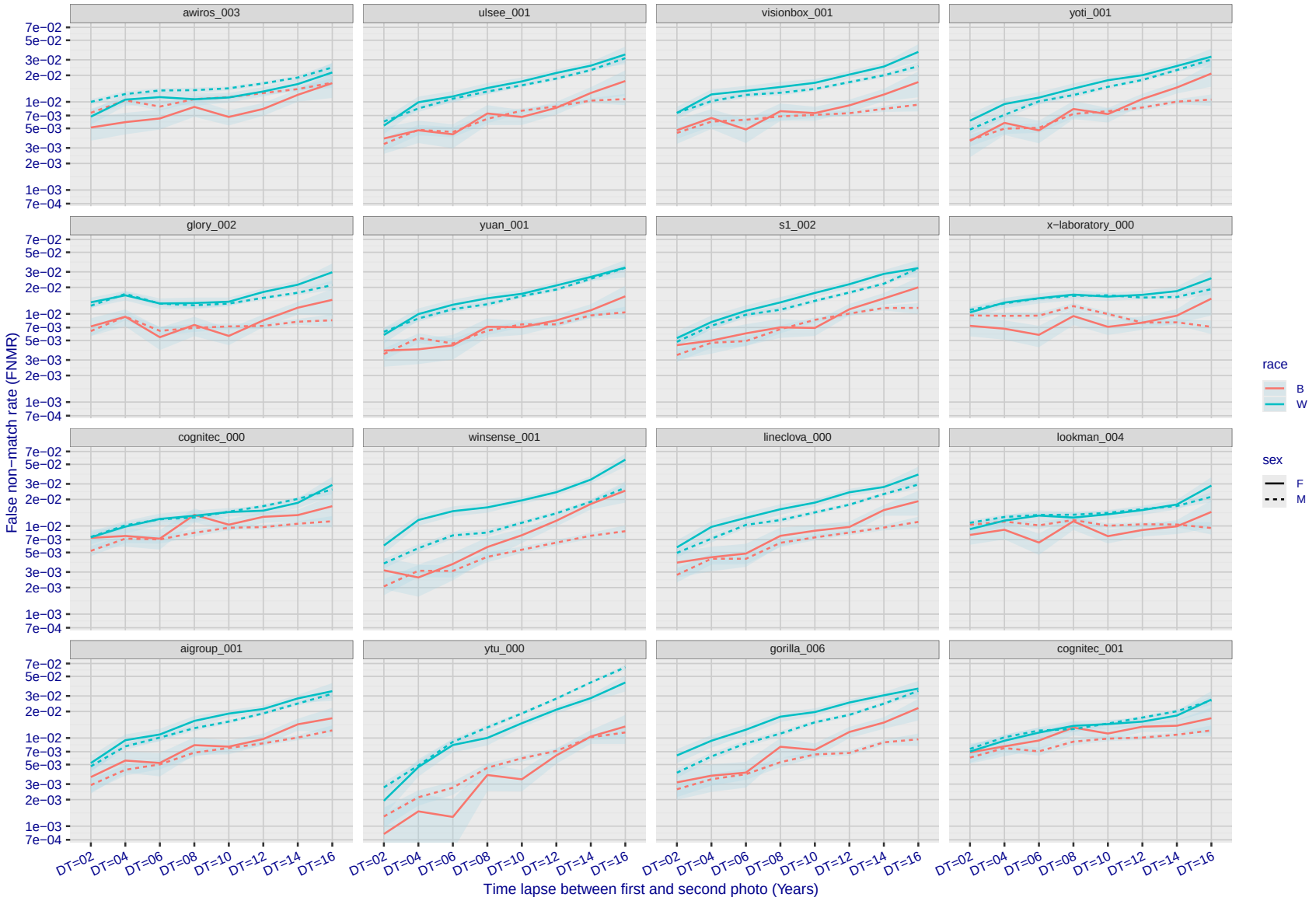

Figure 773: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

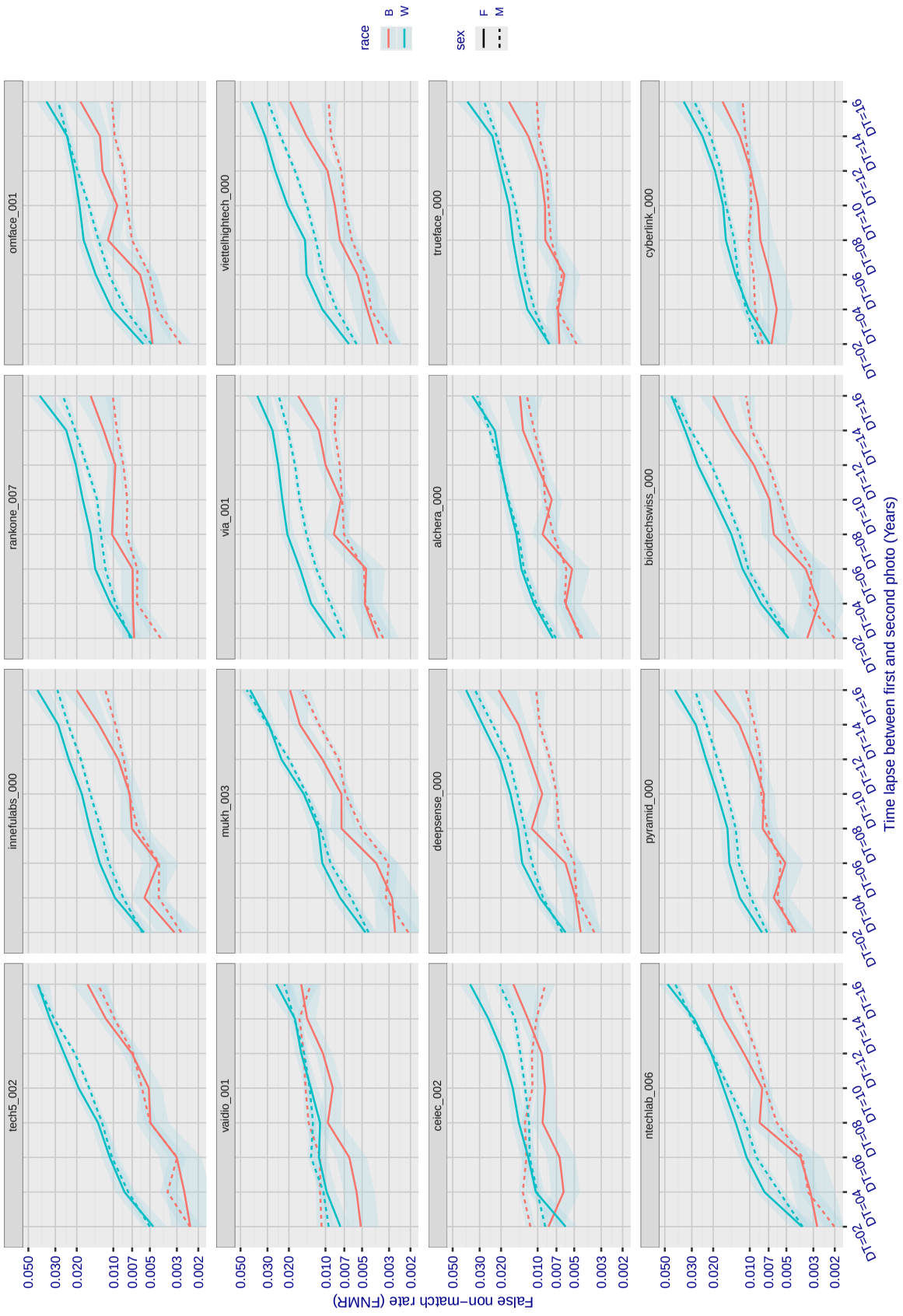

Figure 774: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (*$FNMR < 0.001$*) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

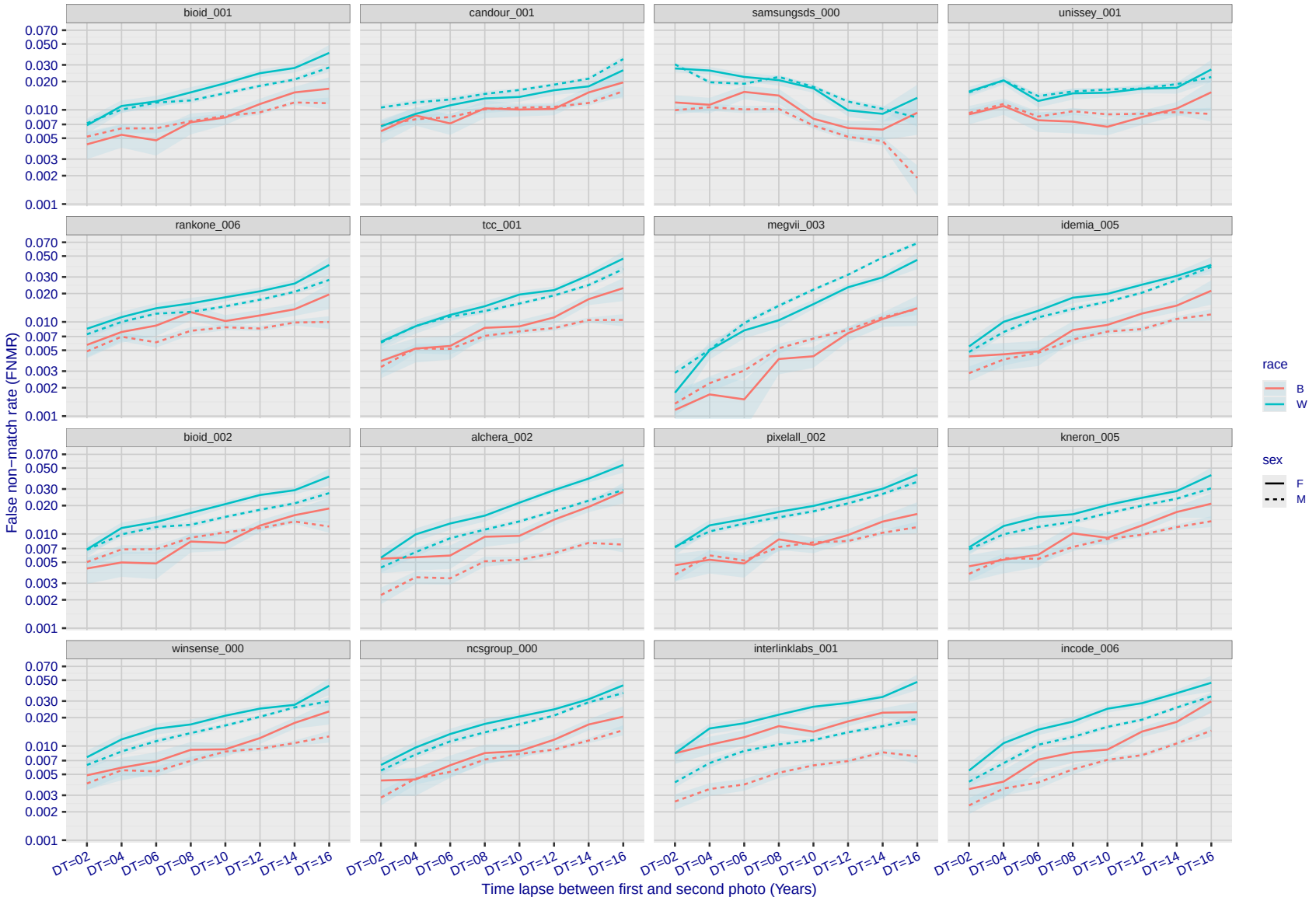

Figure 775: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

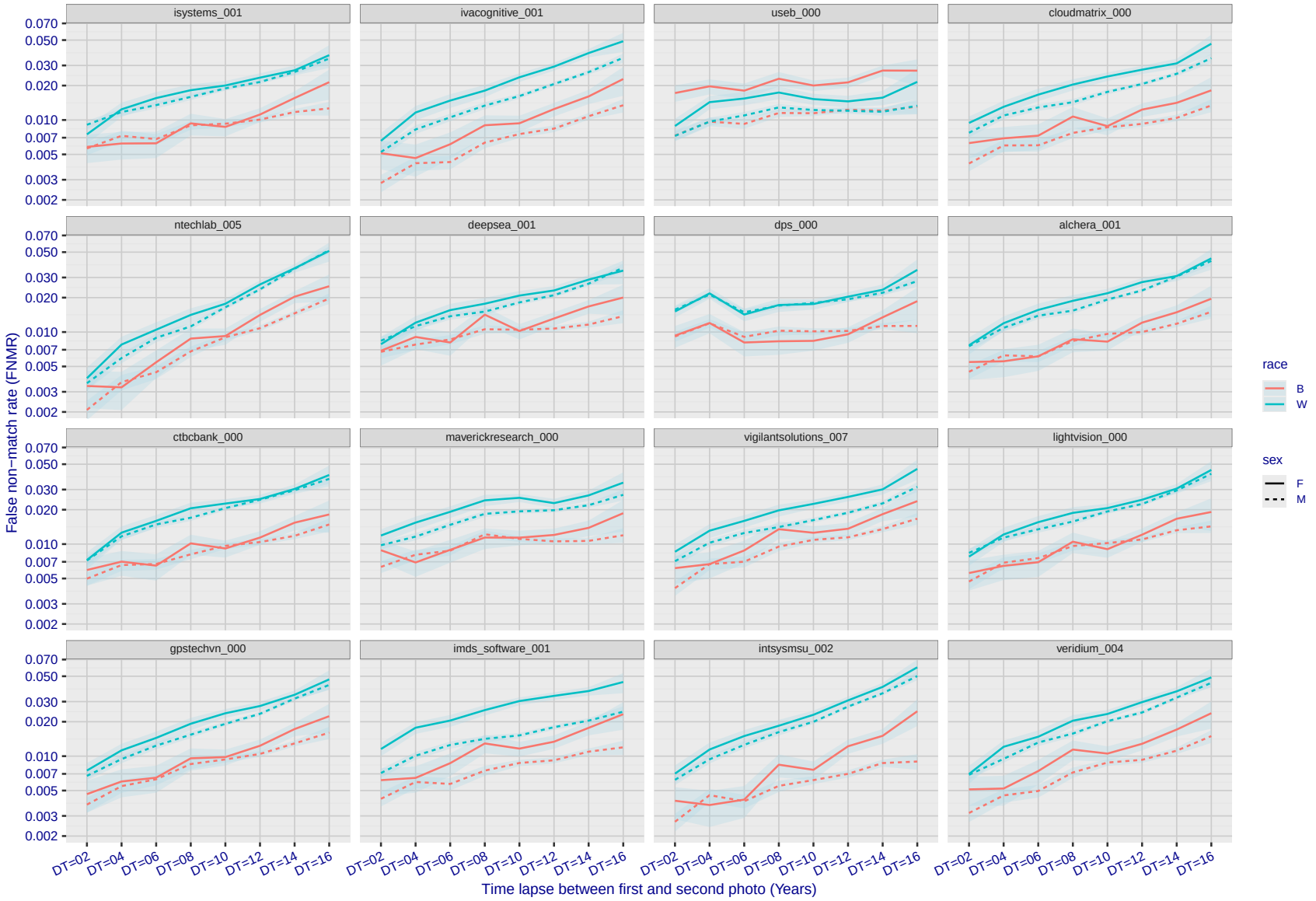

Figure 776: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

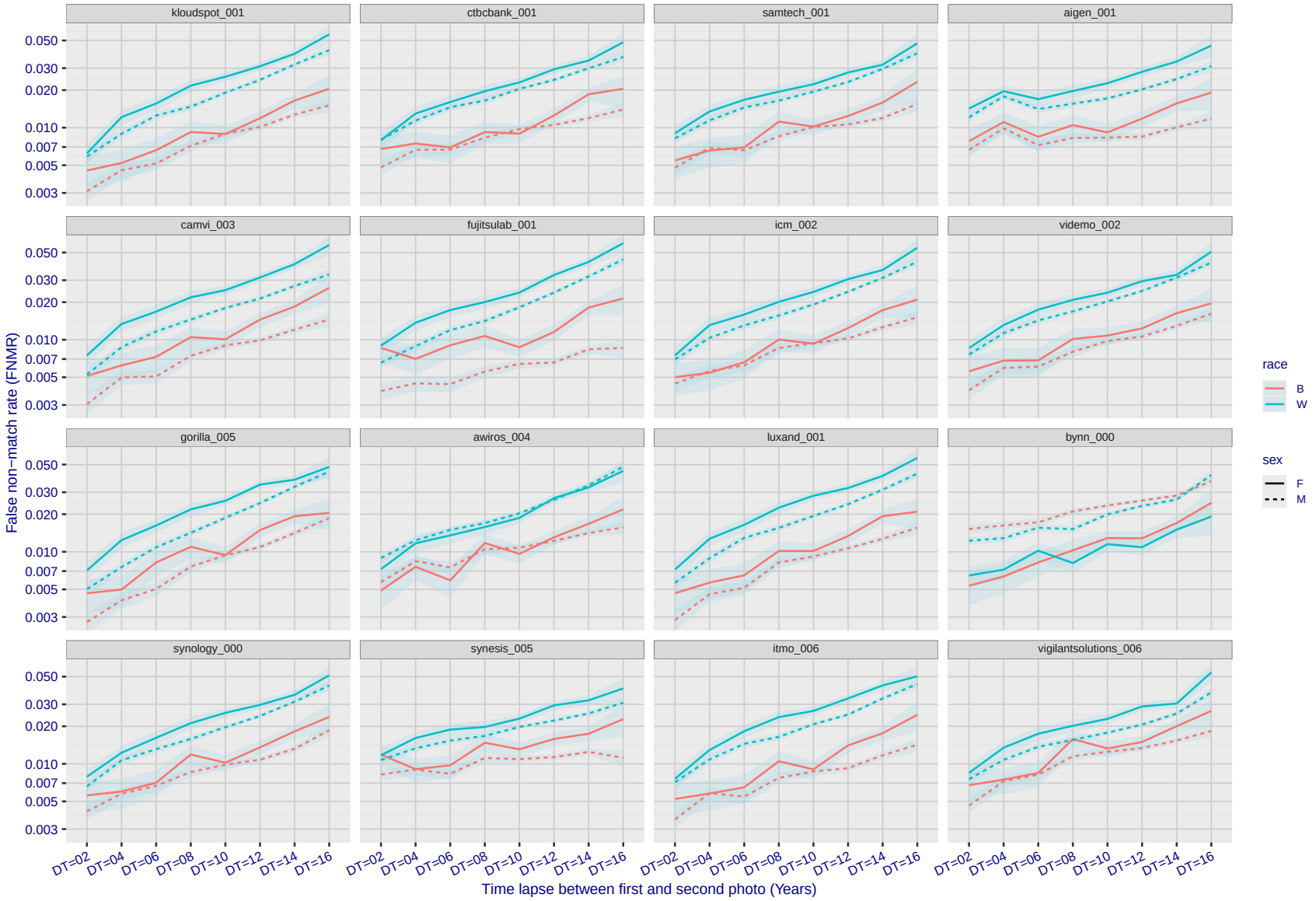

Figure 777: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

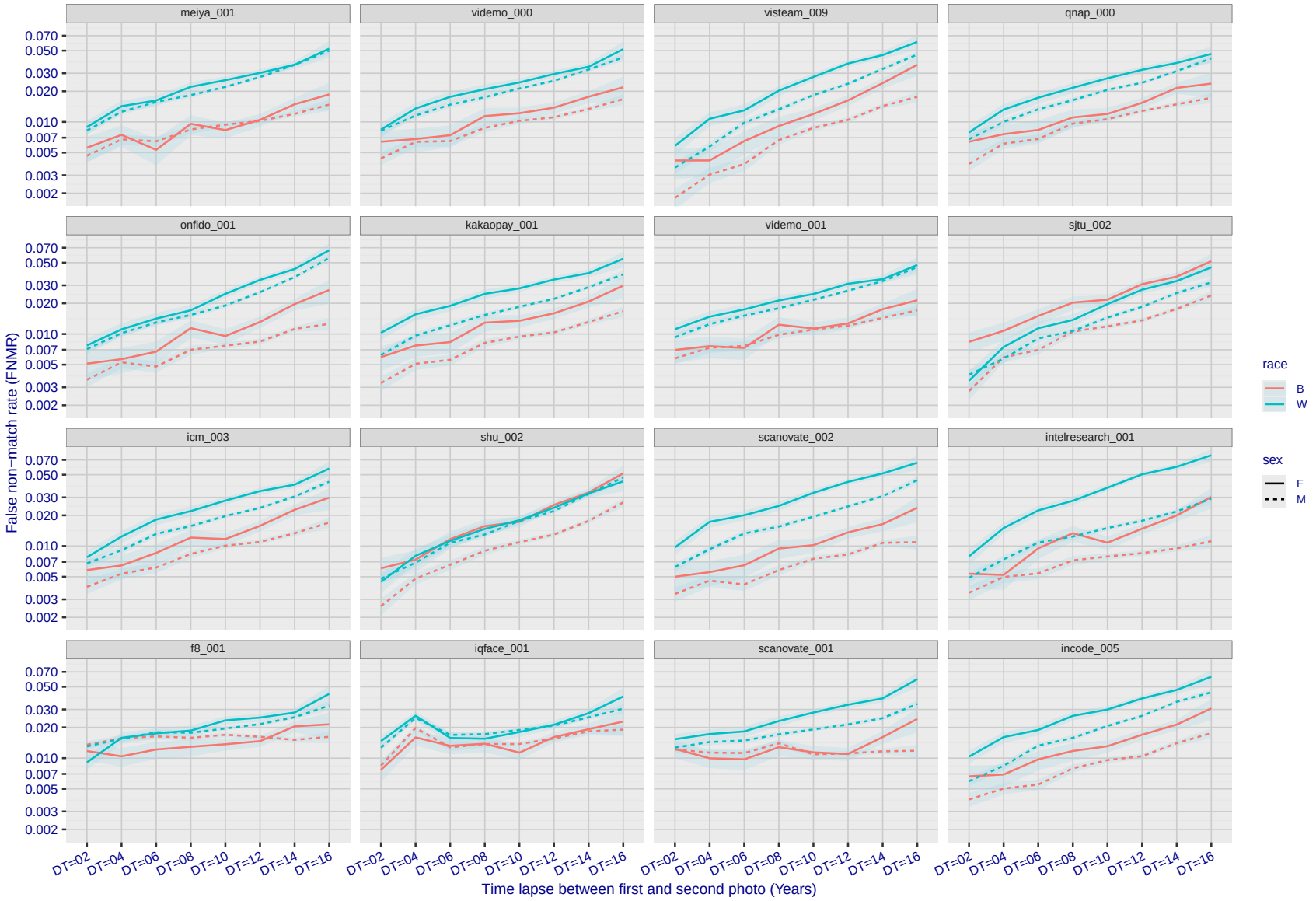

Figure 778: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

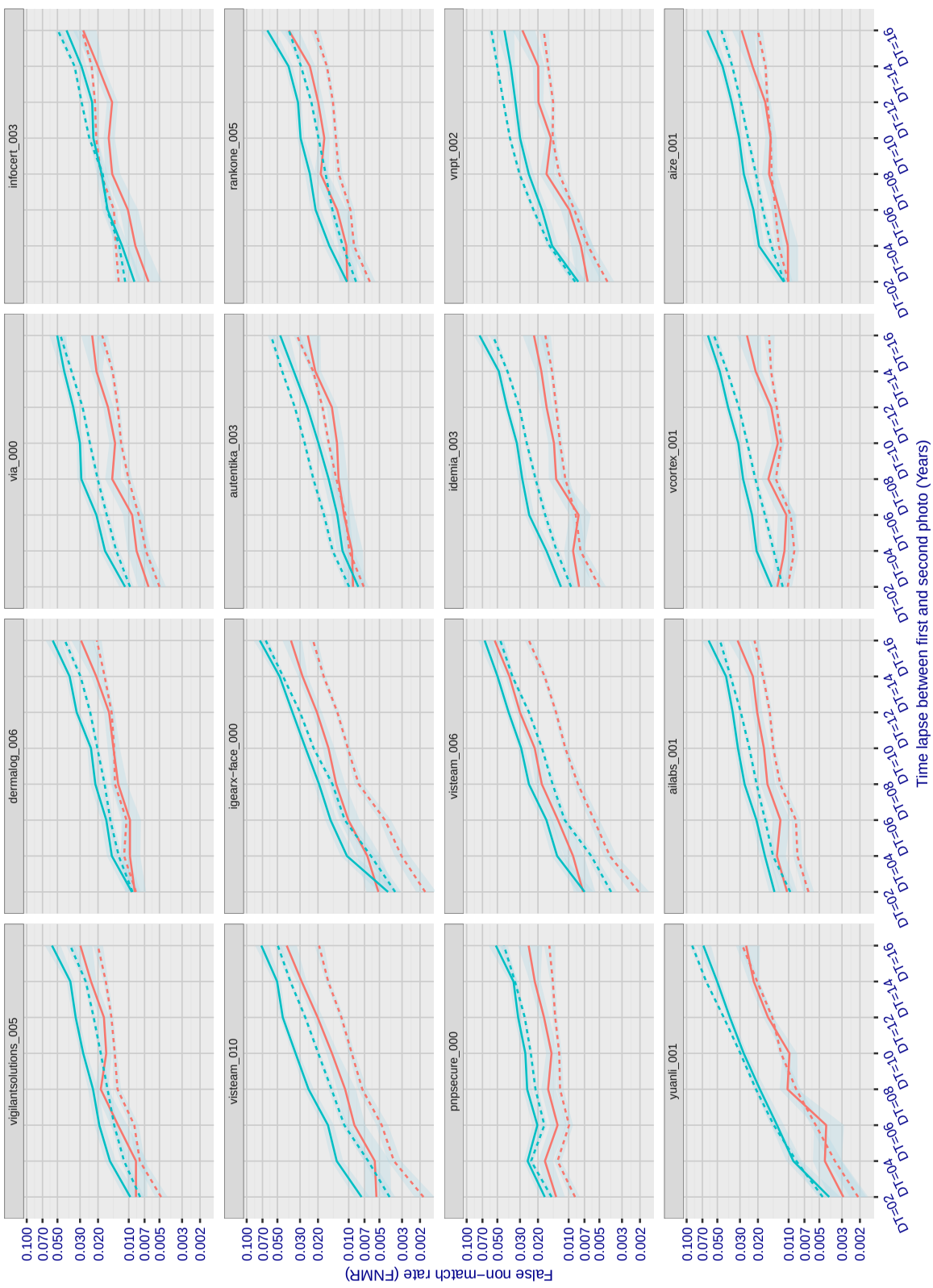

Figure 779: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

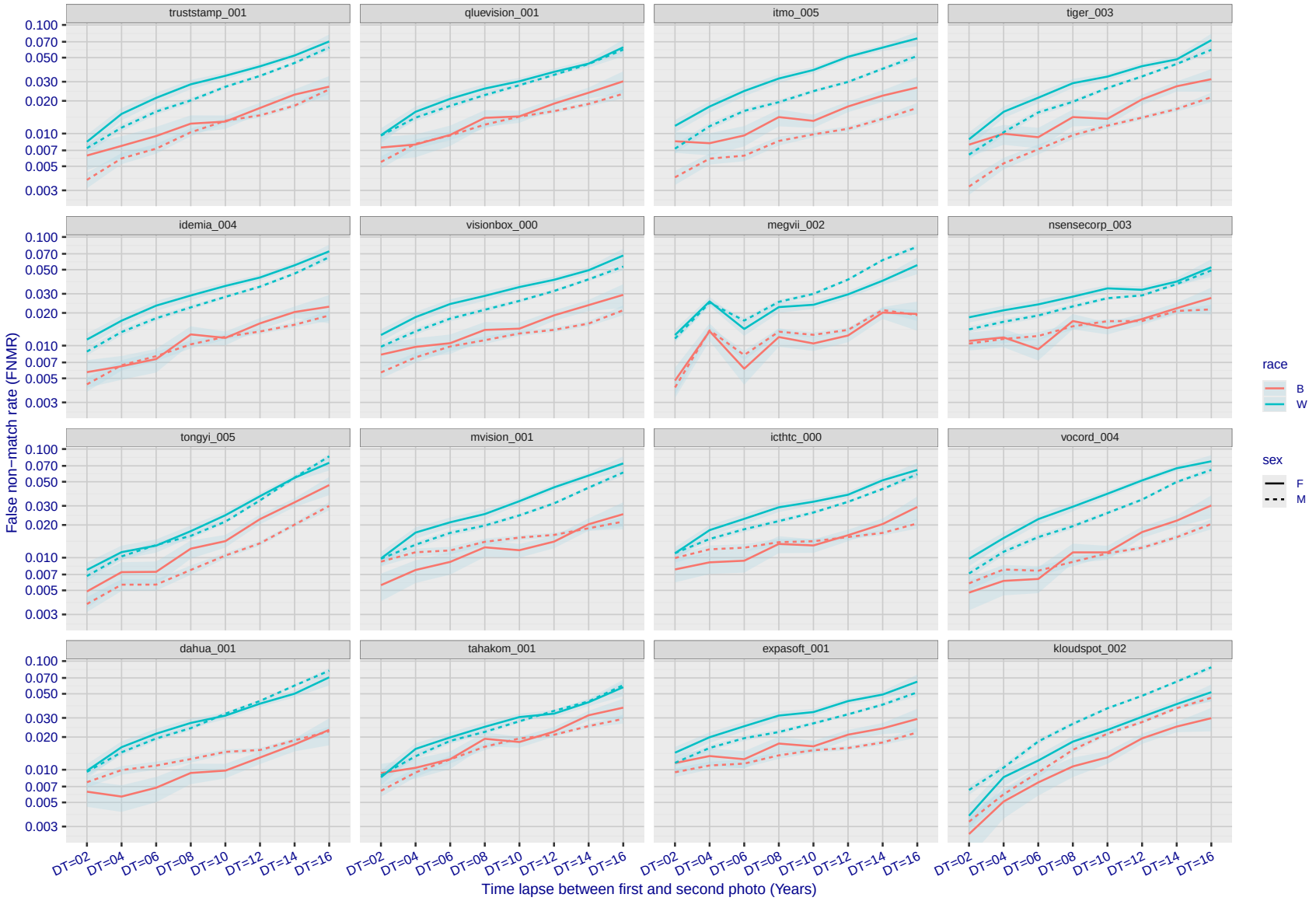

Figure 780: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

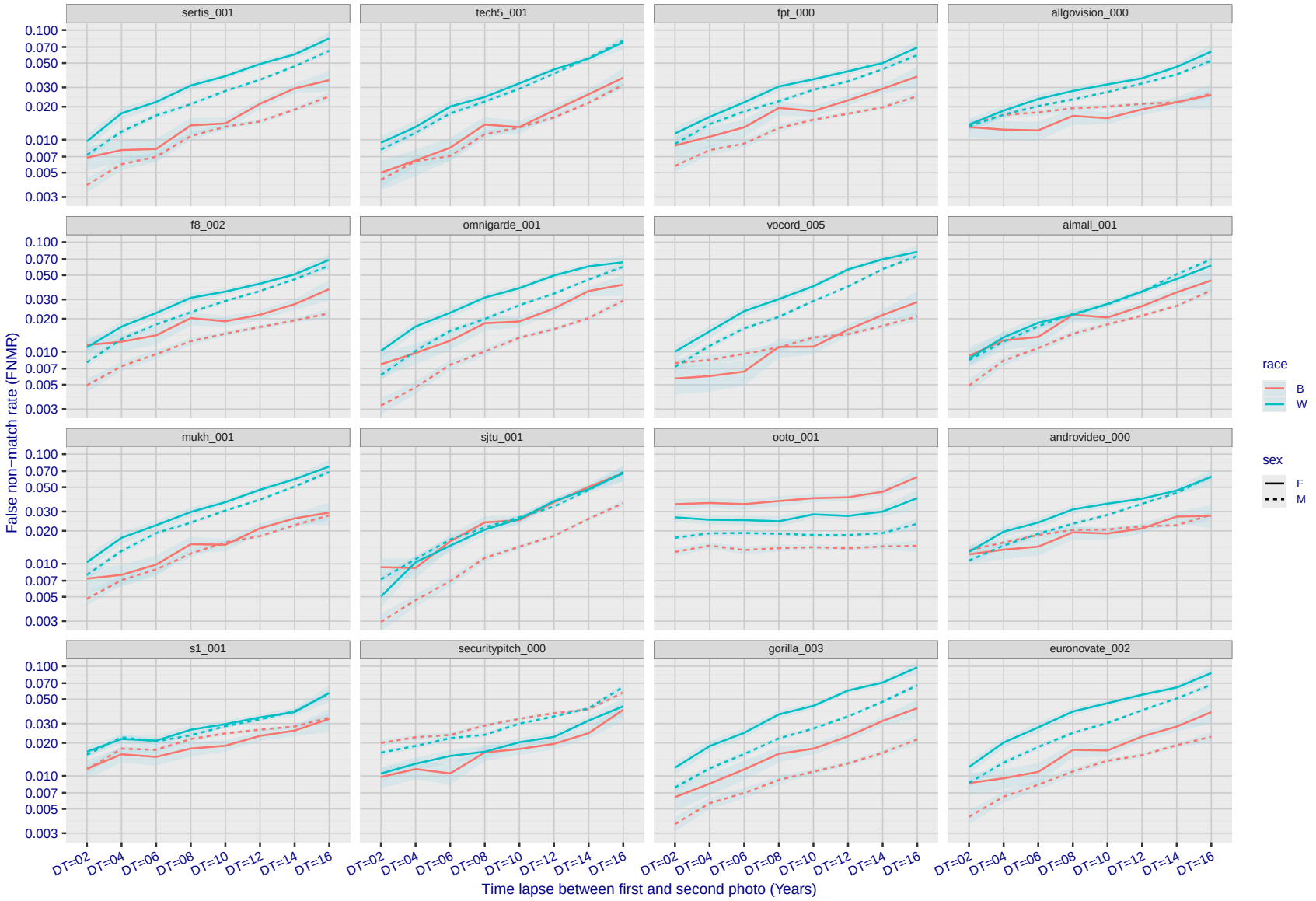

Figure 781: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

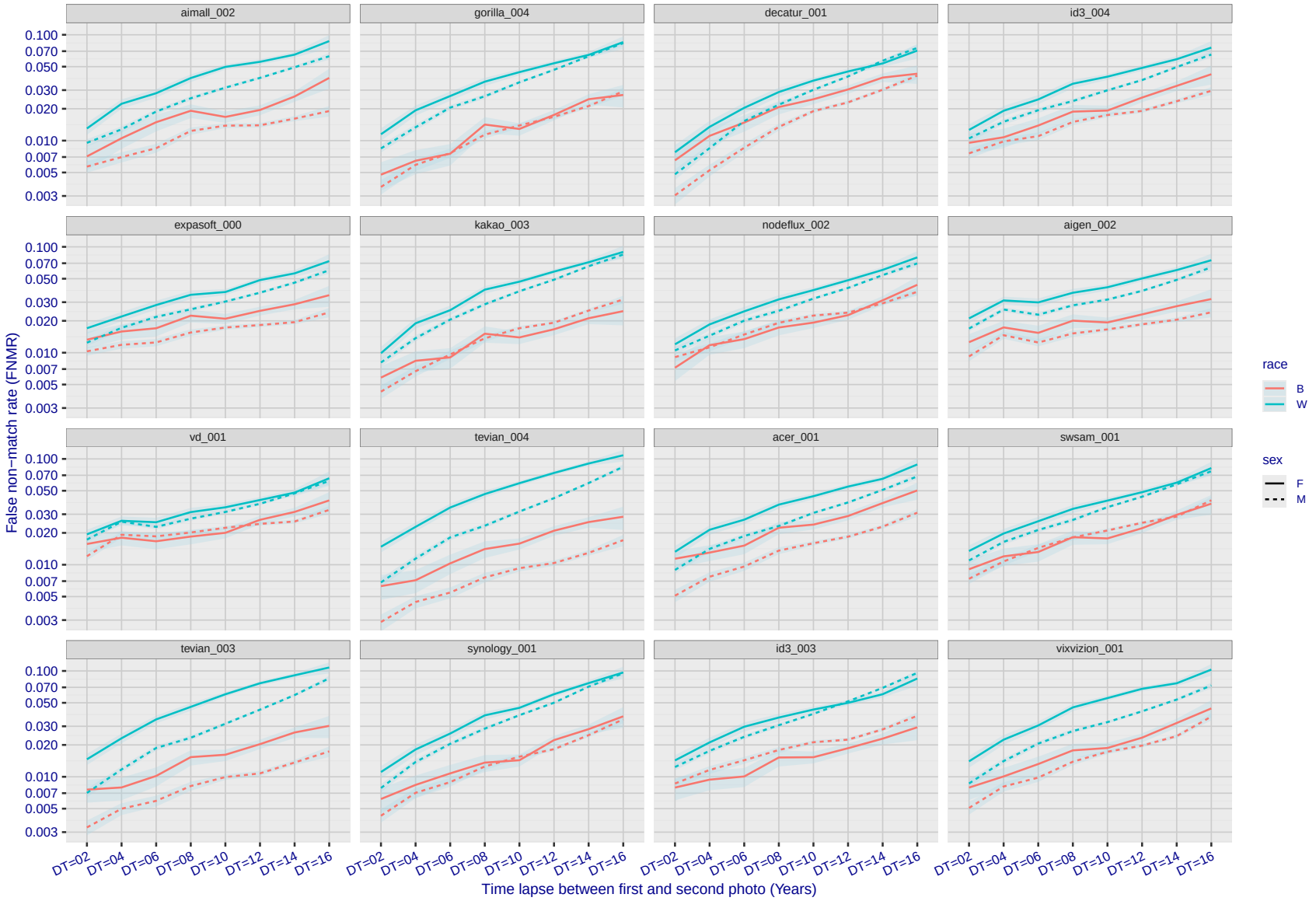

Figure 782: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

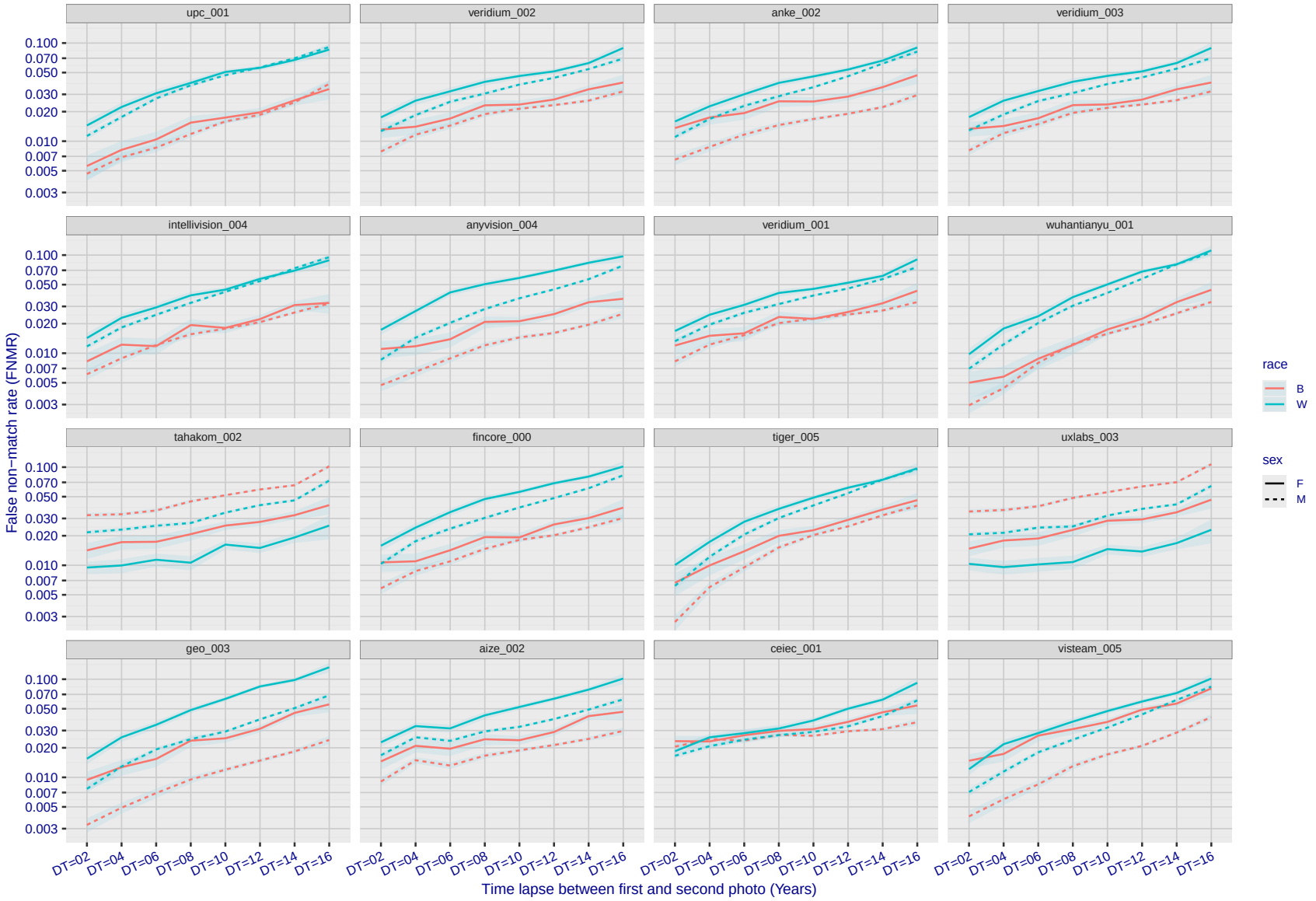

Figure 783: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

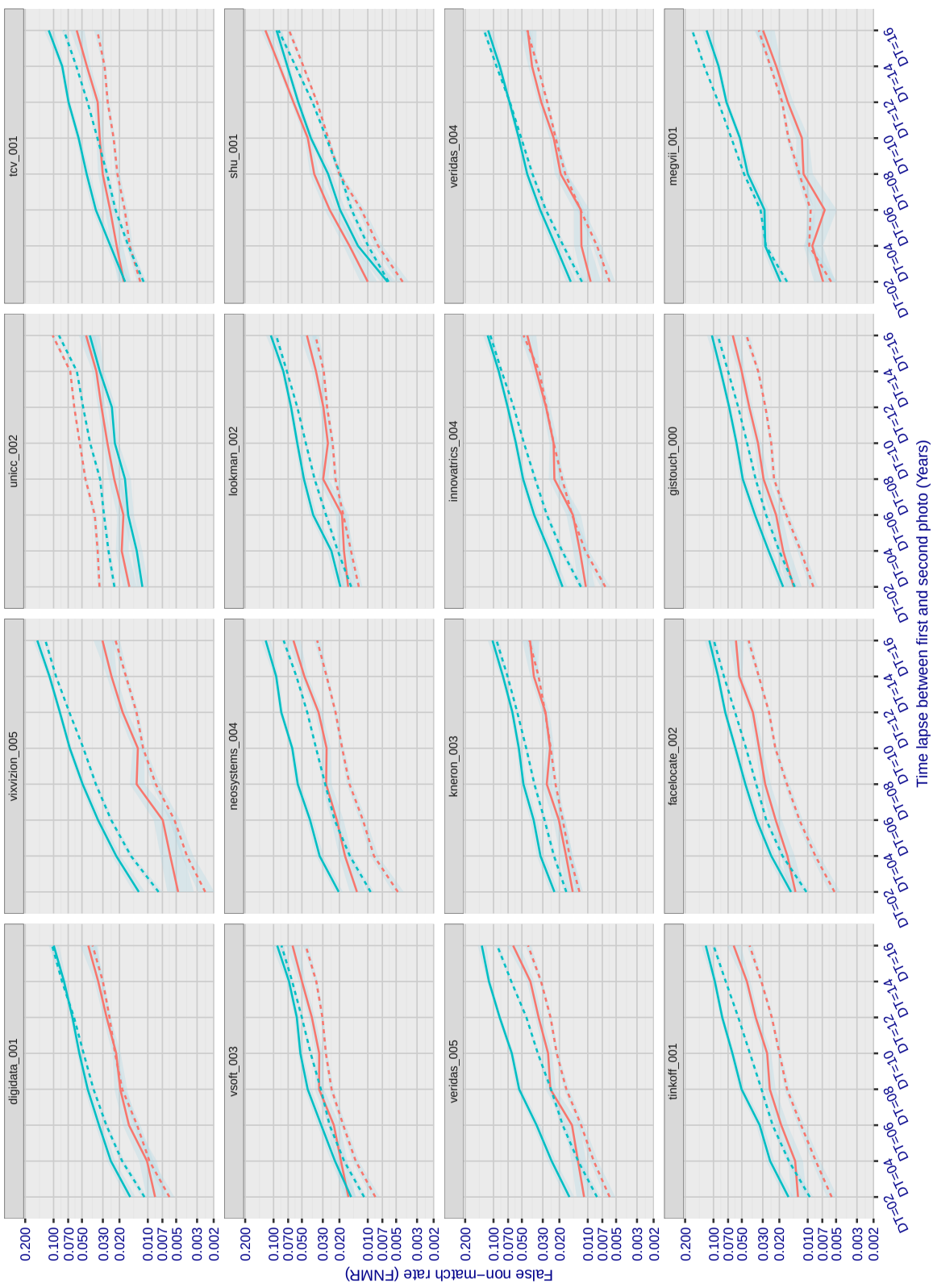

Figure 784: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

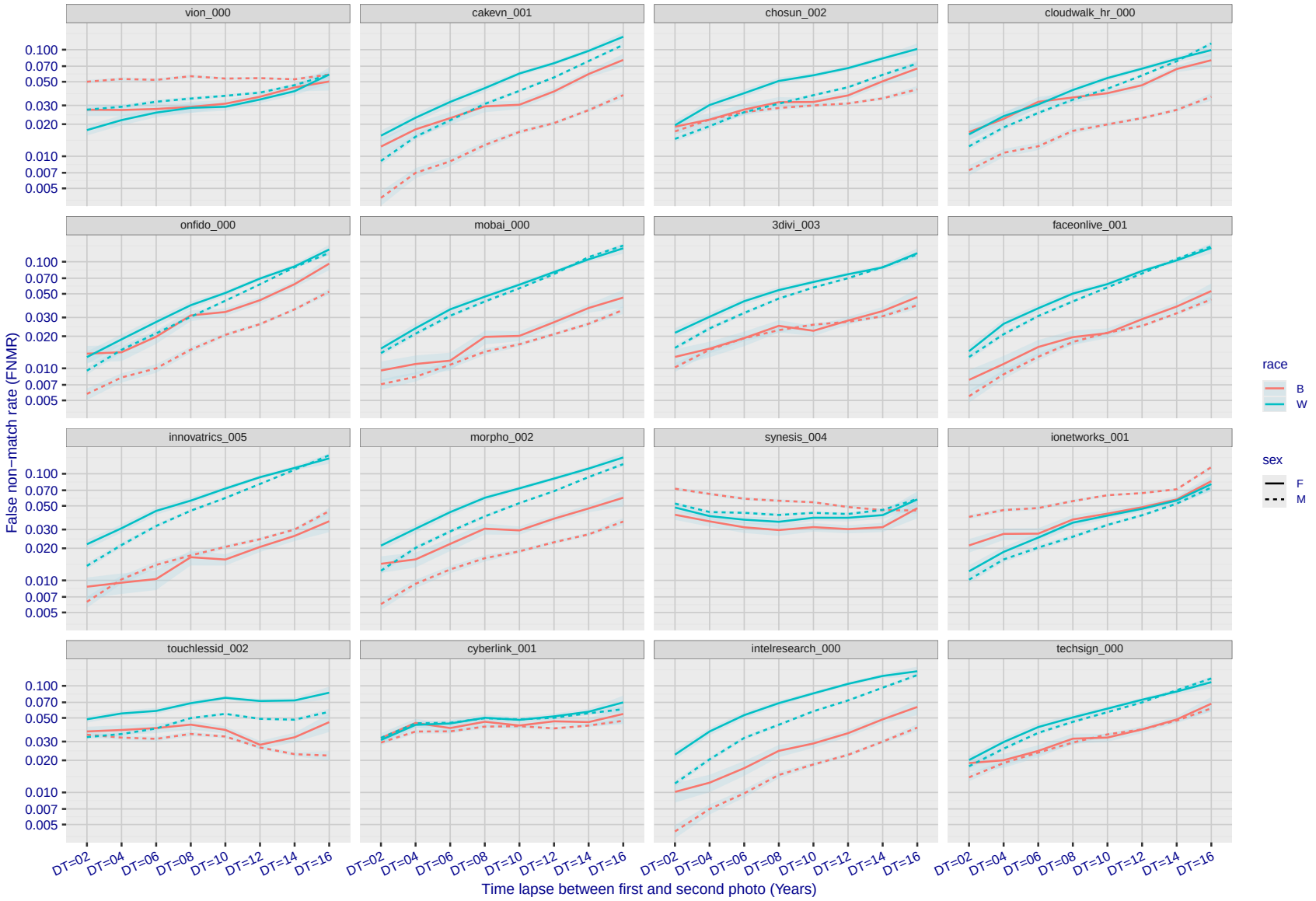

Figure 785: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

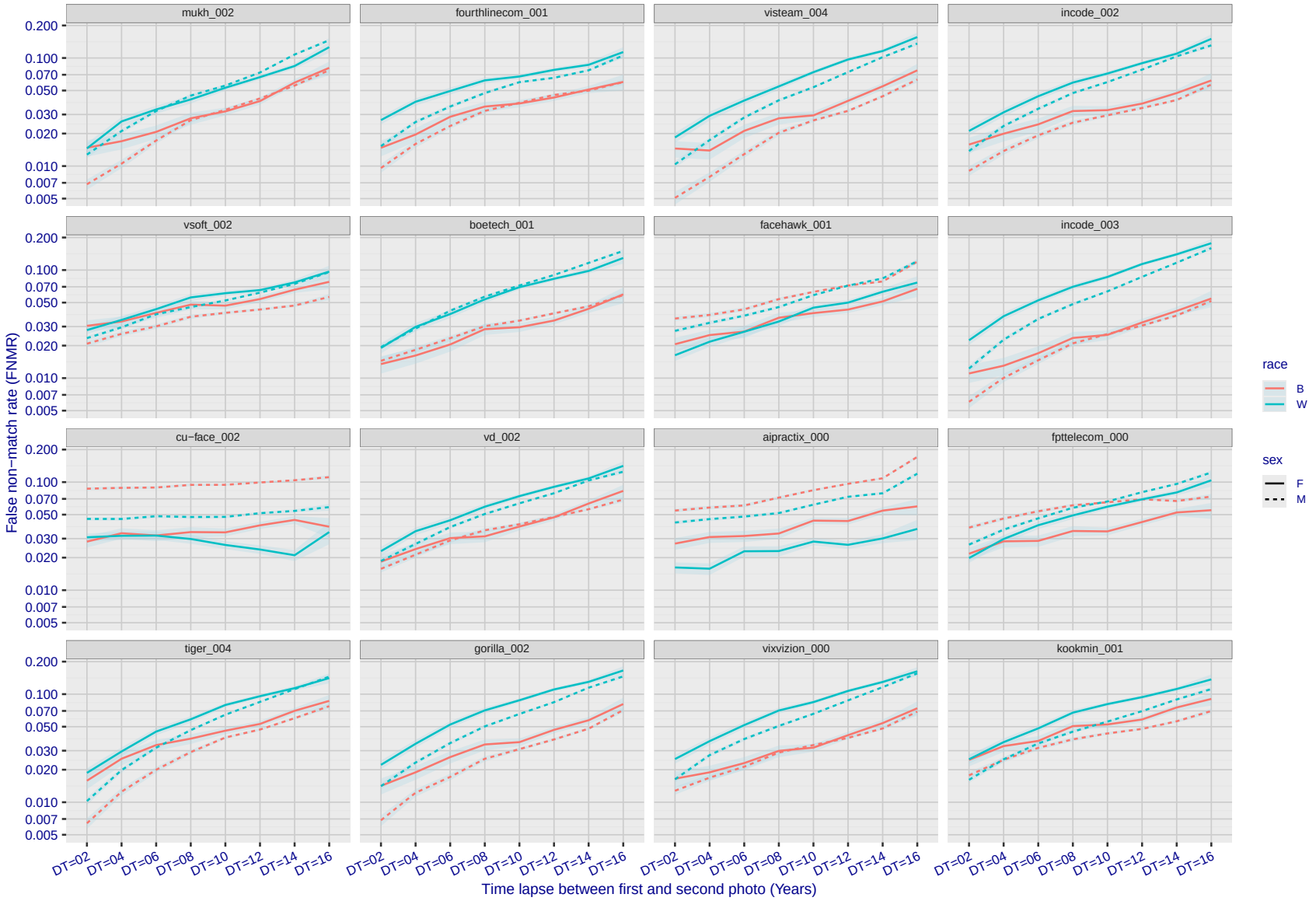

Figure 786: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

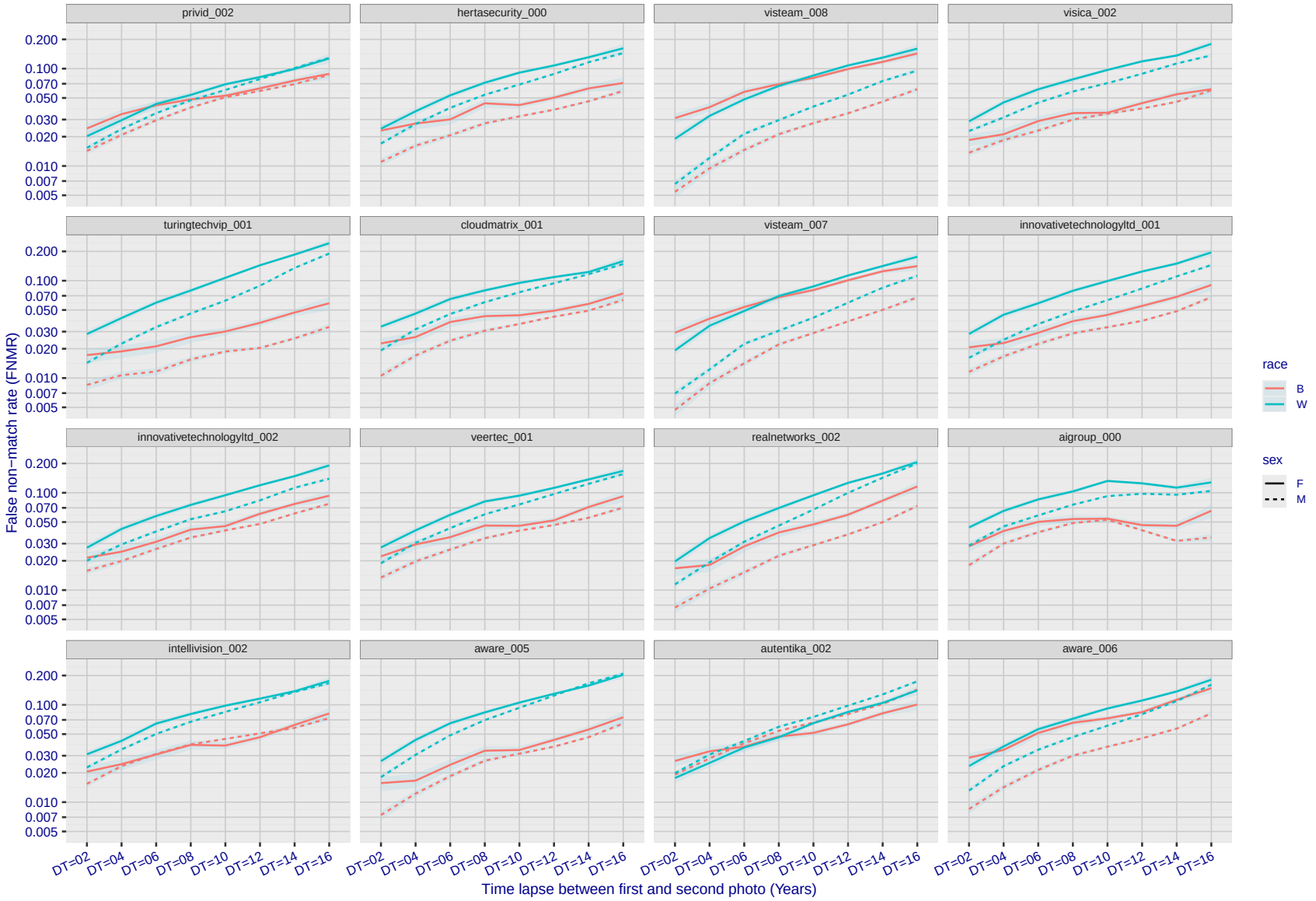

Figure 787: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

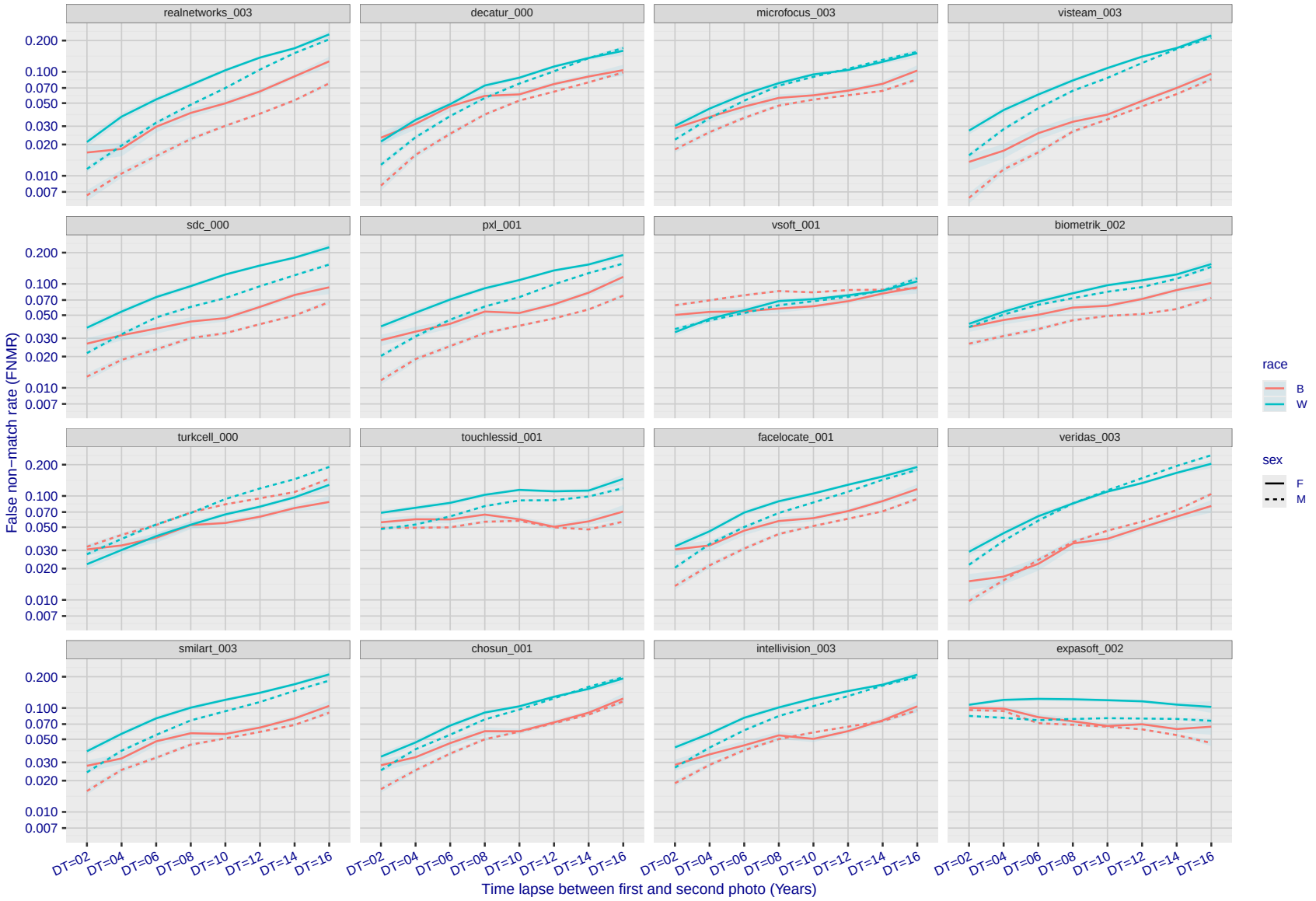

Figure 788: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

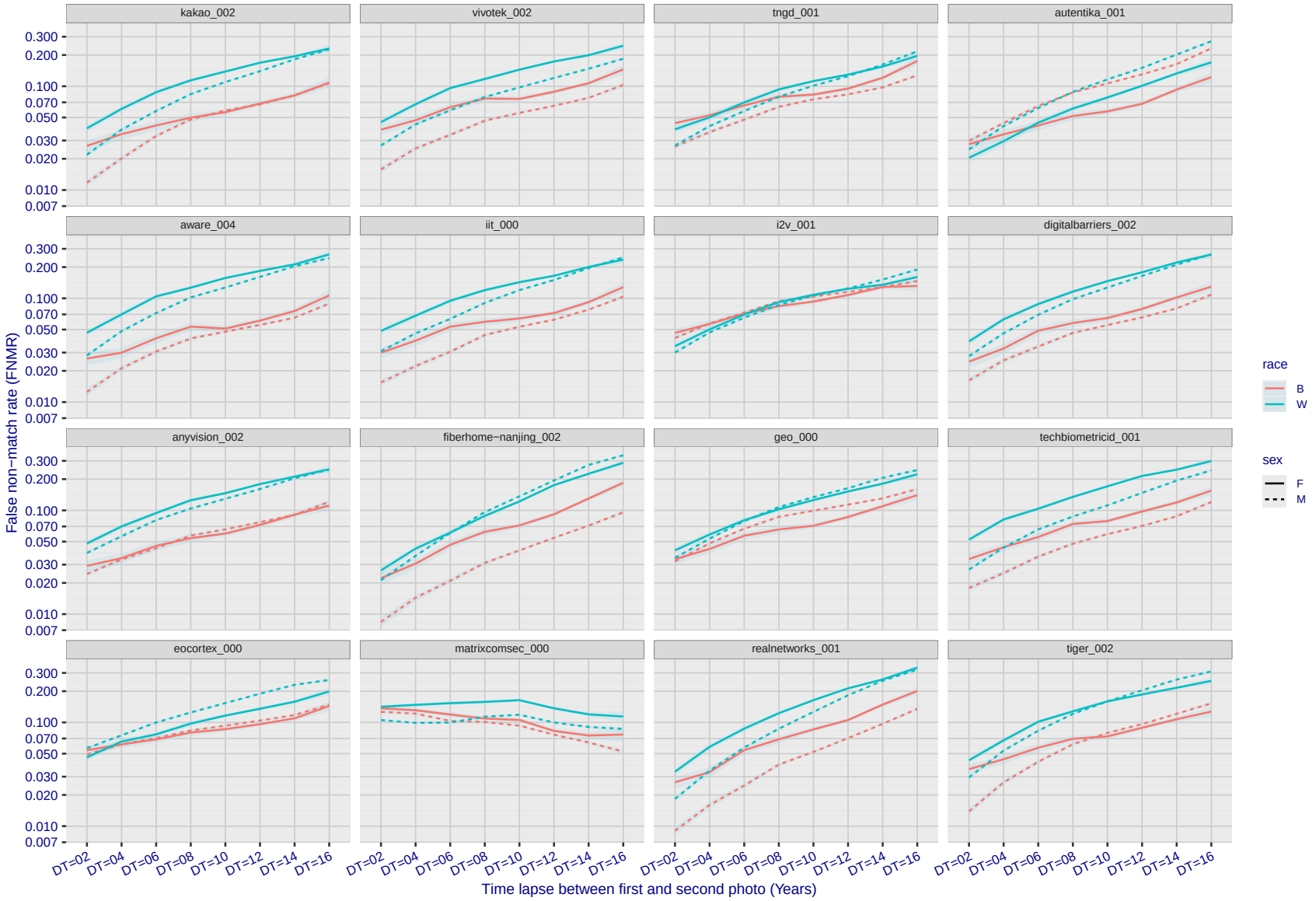

Figure 789: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

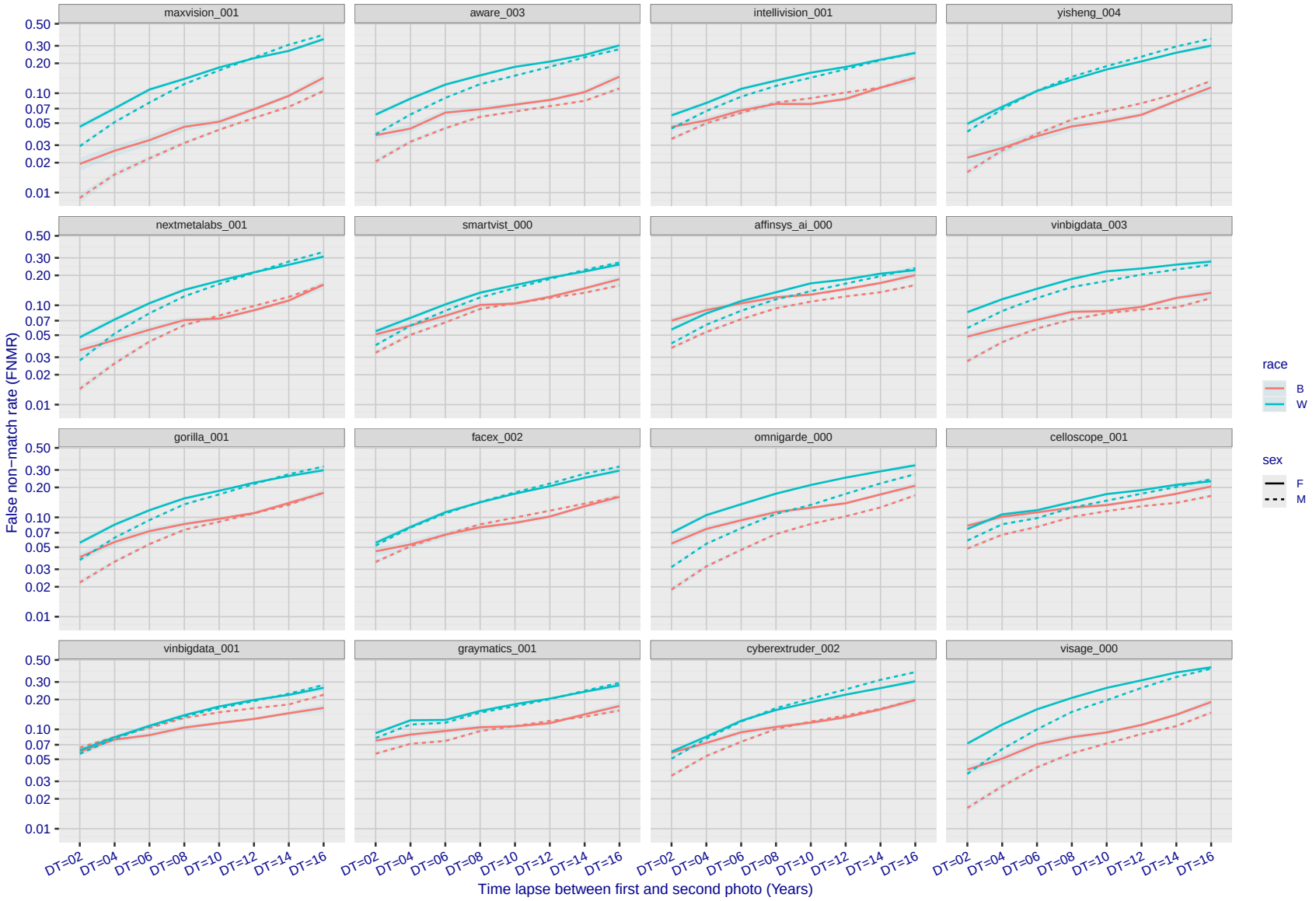

Figure 790: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

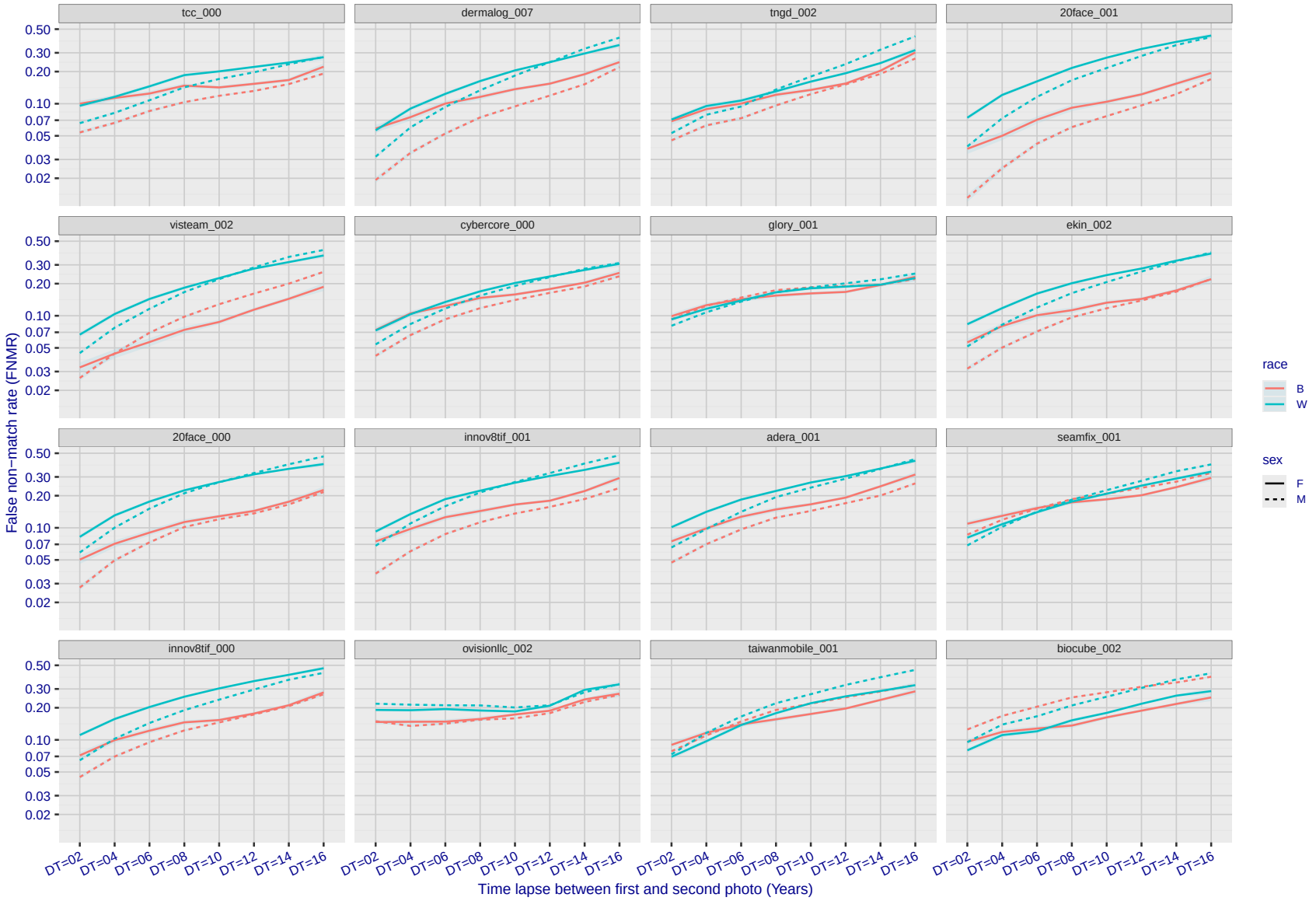

Figure 791: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

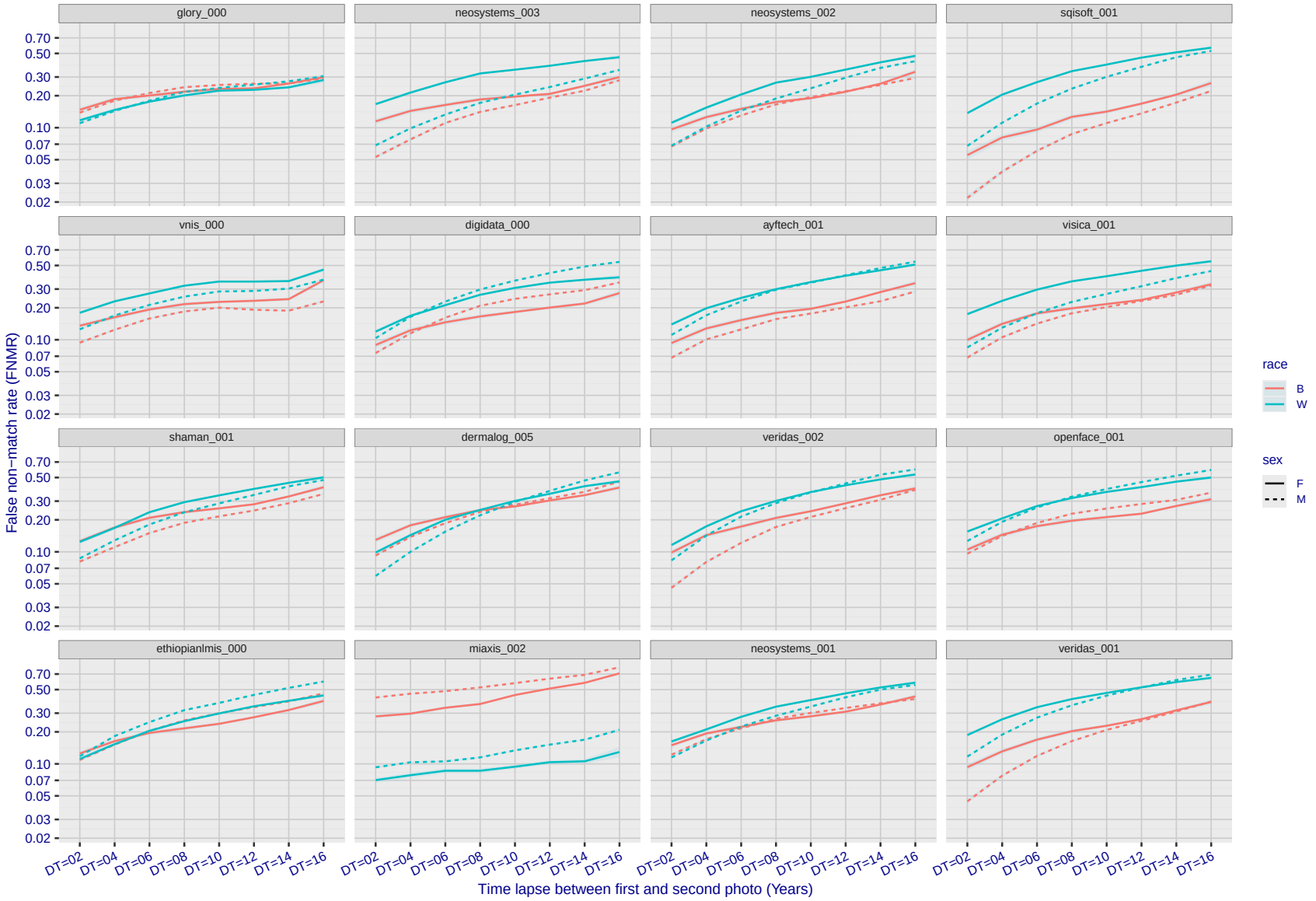

Figure 792: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

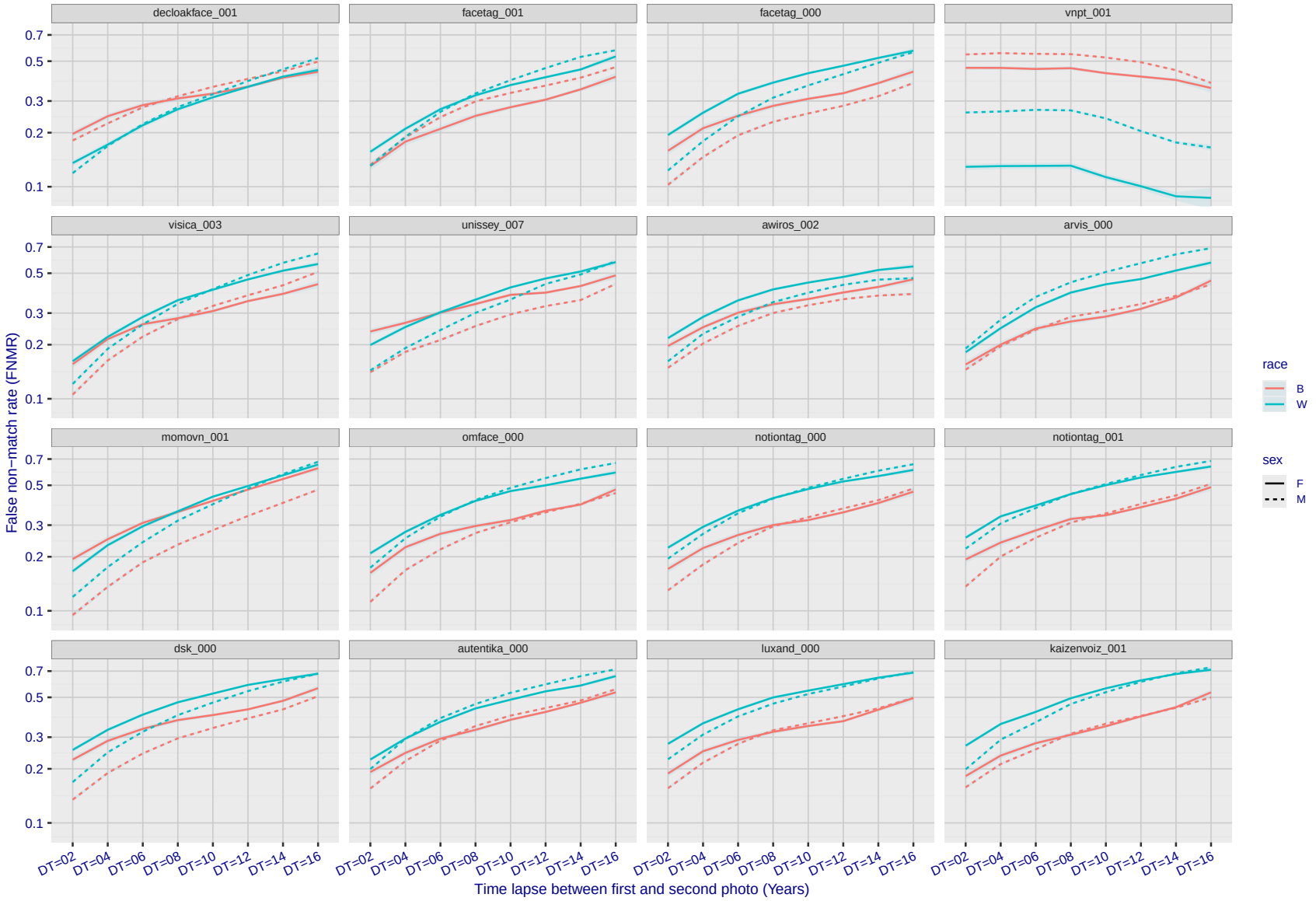

Figure 793: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

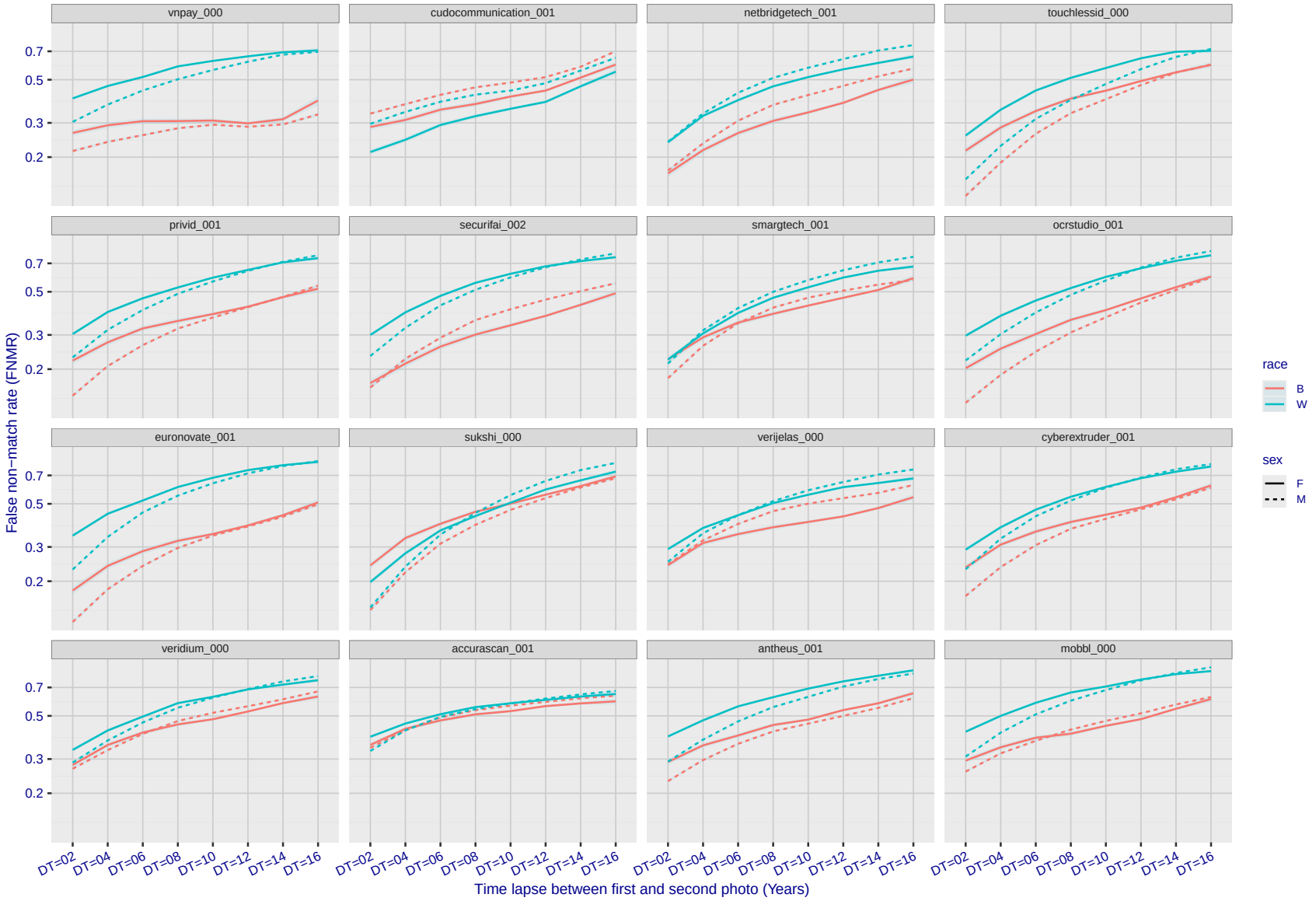

Figure 794: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

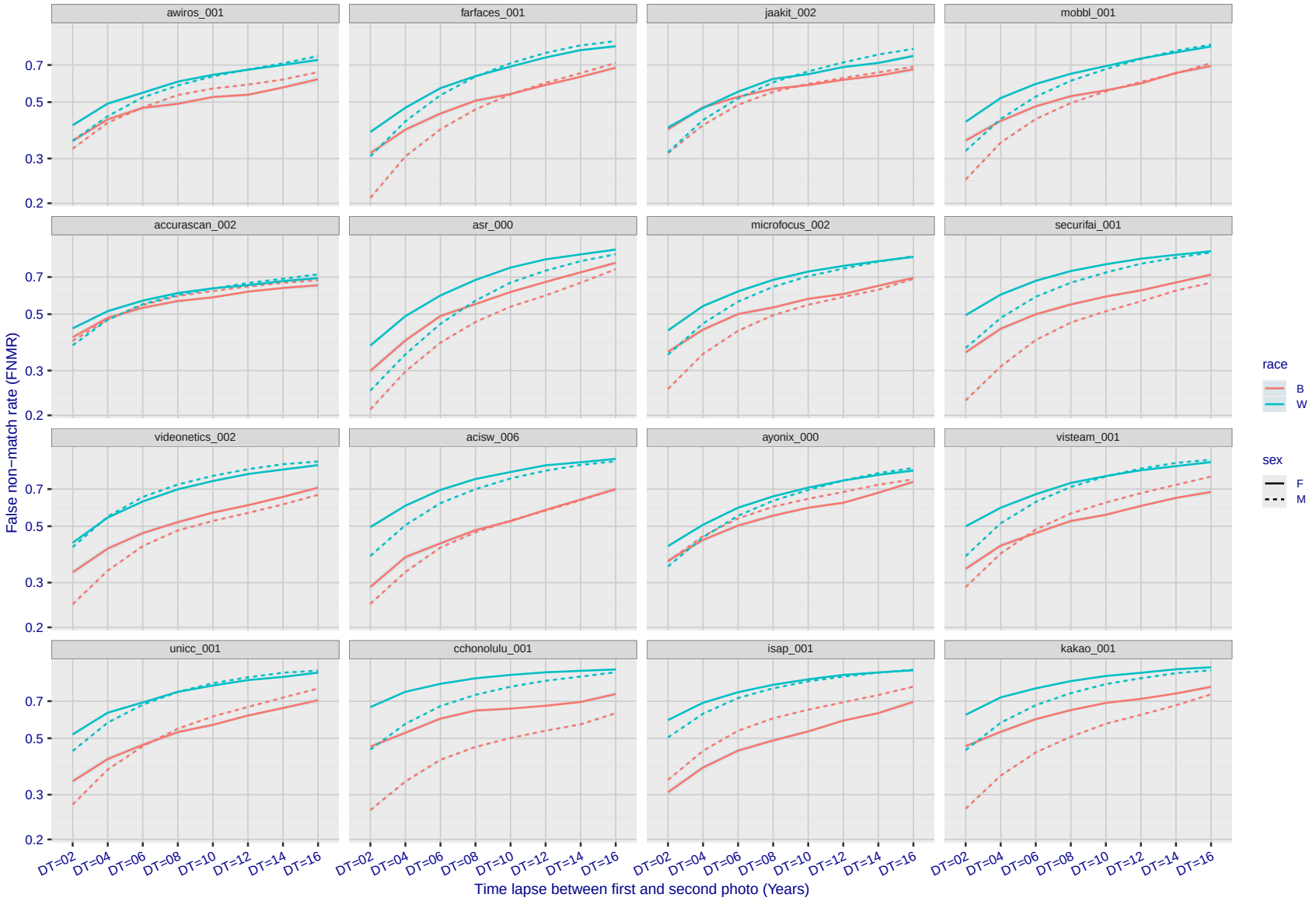

Figure 795: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

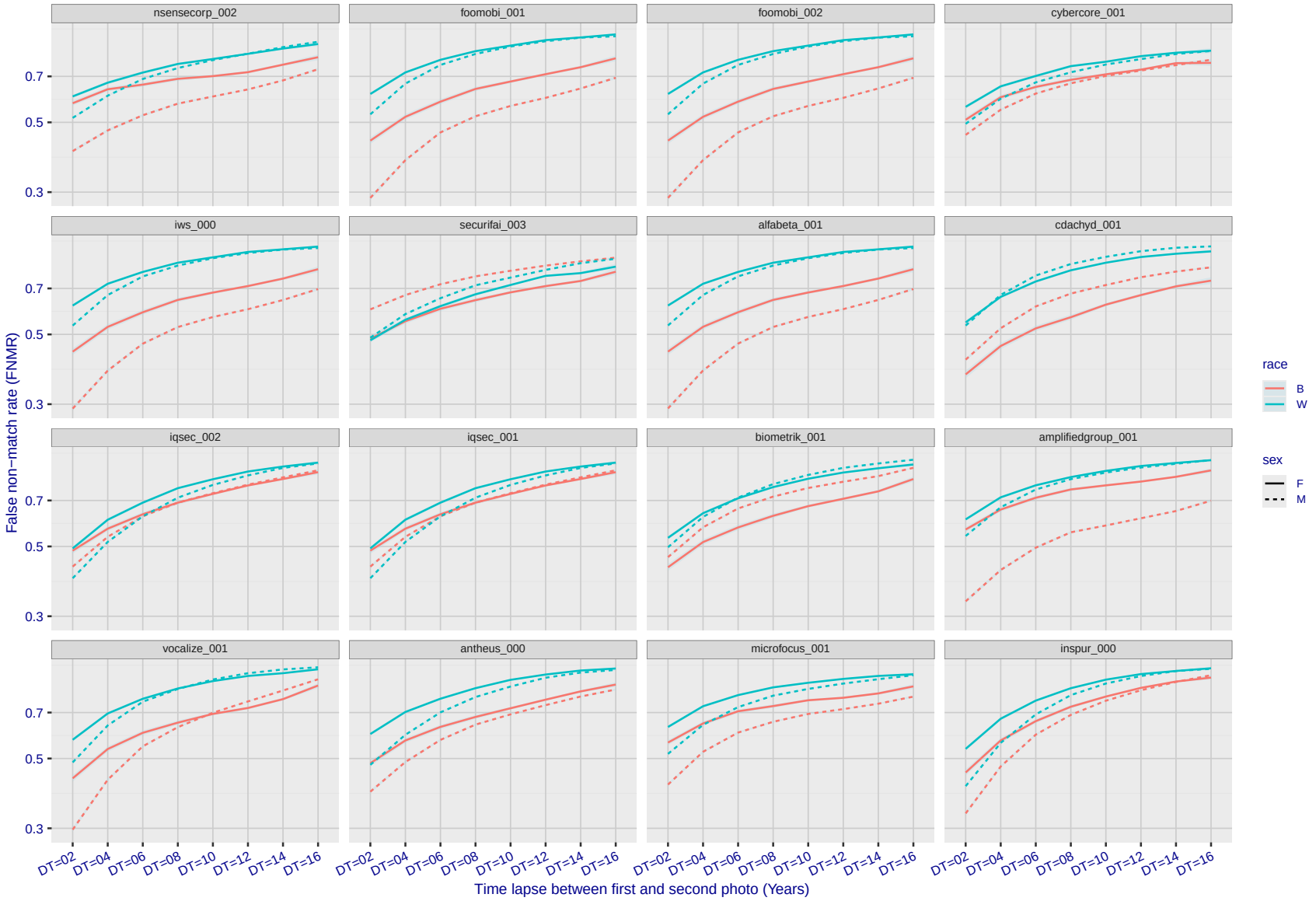

Figure 796: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

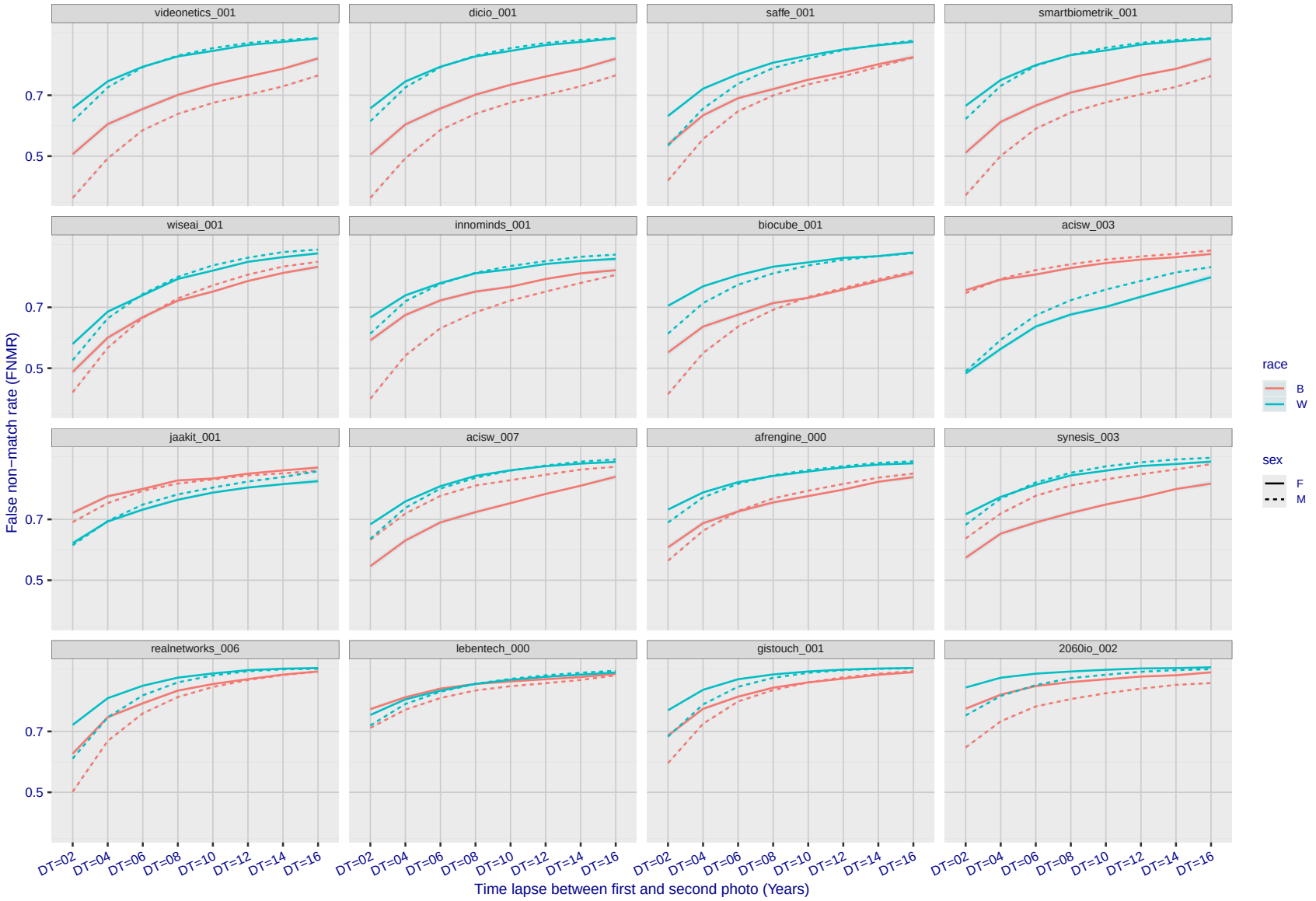

Figure 797: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

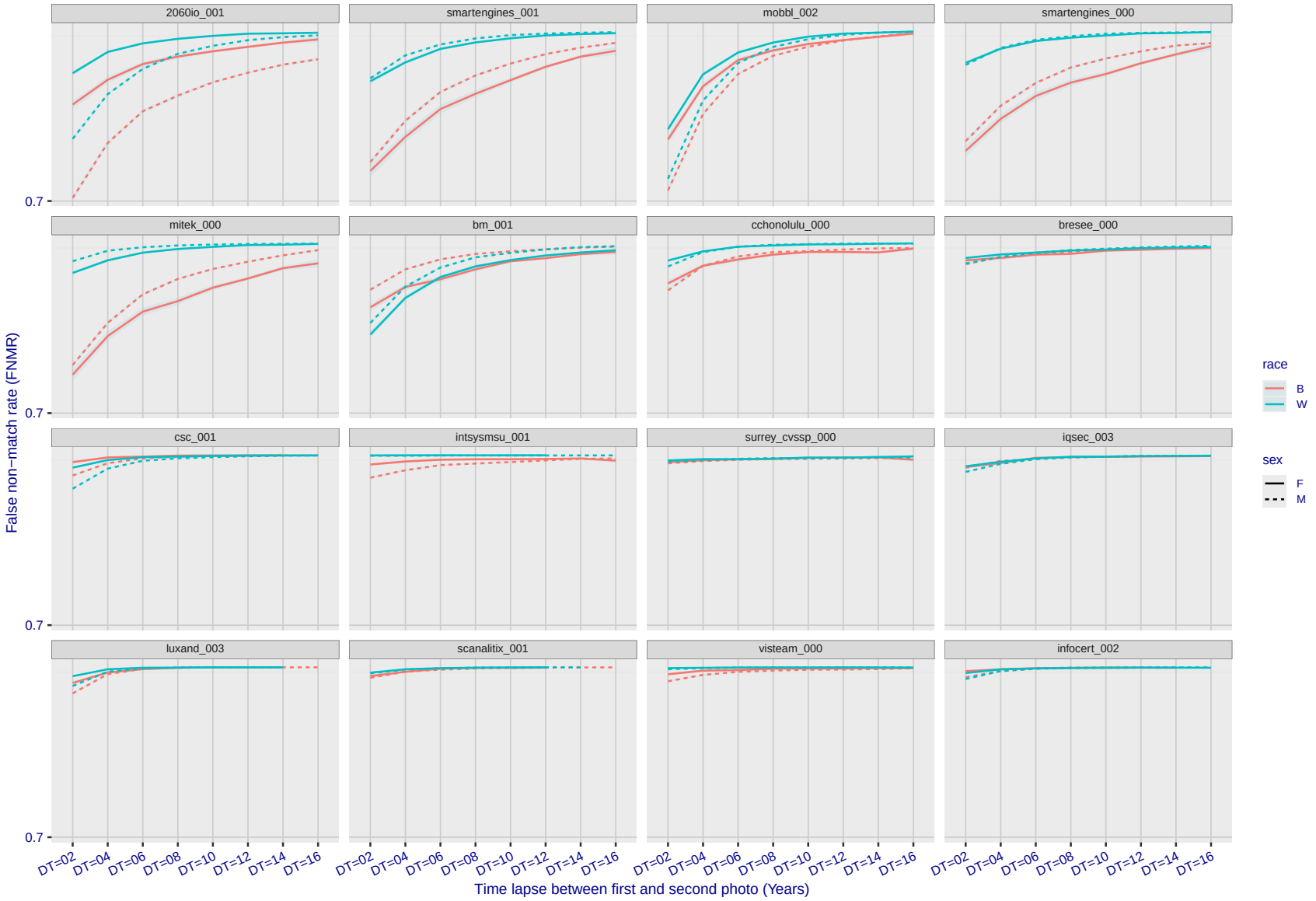

Figure 798: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

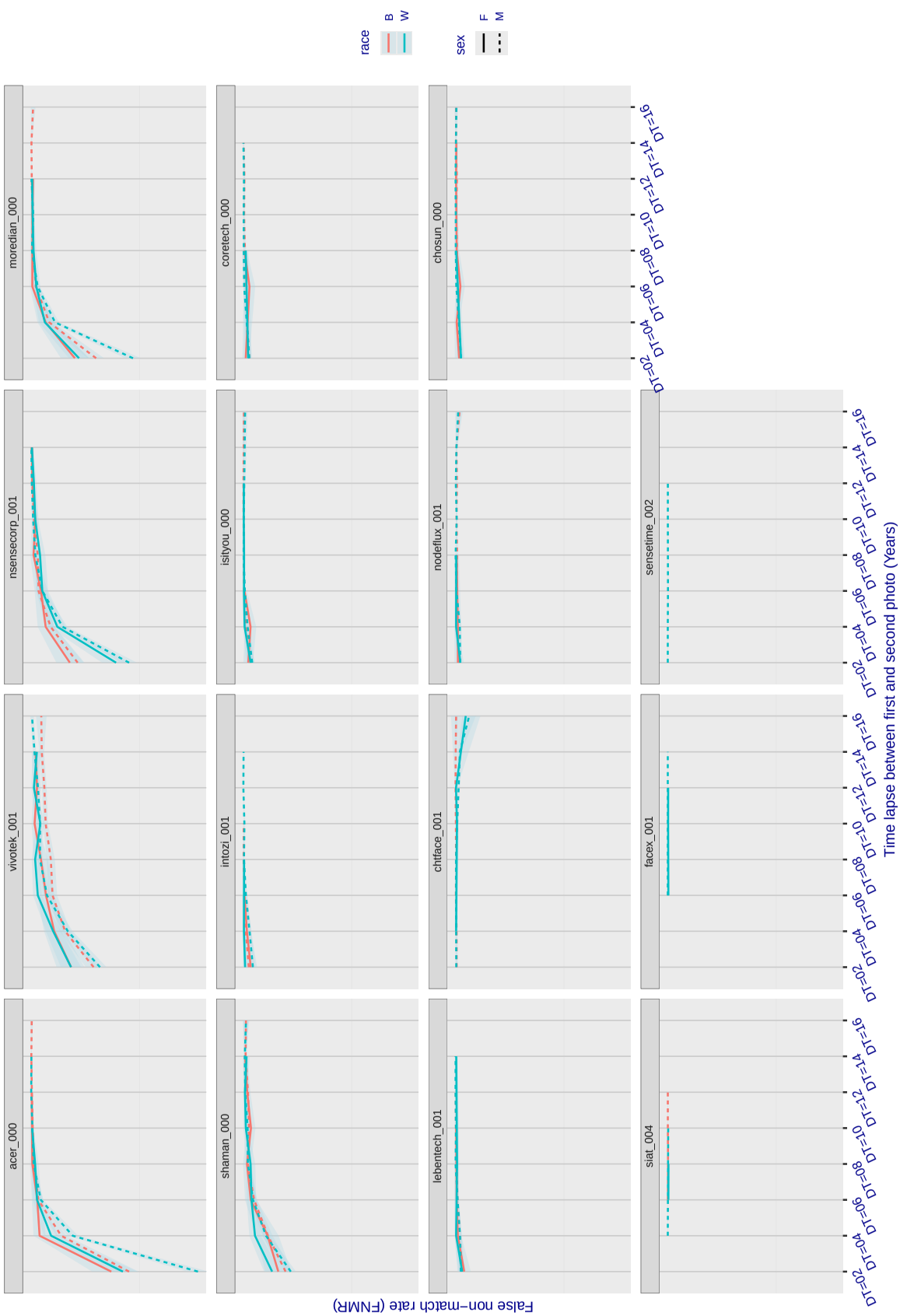

Figure 799: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.5.3 Effect of age on genuine subjects

Background: Faces change appearance throughout life. Face recognition algorithms have previously beeen reported to give better accuracy on older individuals (See NIST IR 8009).

Goal: To quantify false non-match rates (FNMR) as a function of age, without an ageing component.

Methods: Using the visa images, which span fewer than five years, thresholds are determined that give FMR = 0.001 and 0.0001 over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores.

Results: For the visa images, Figure 890 shows how false non-match rates for genuine users, as a function of age group.

The notable aspects are:

- Younger subjects give considerably higher FNMR. This is likely due to rapid growth and change in facial appearance.
- FNMR trends down throughout life. The last bin, AGE $>$ 72, contains fewer than 140 mated pairs, and may be affected by small sample size.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 800: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 801: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 802: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 803: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 804: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

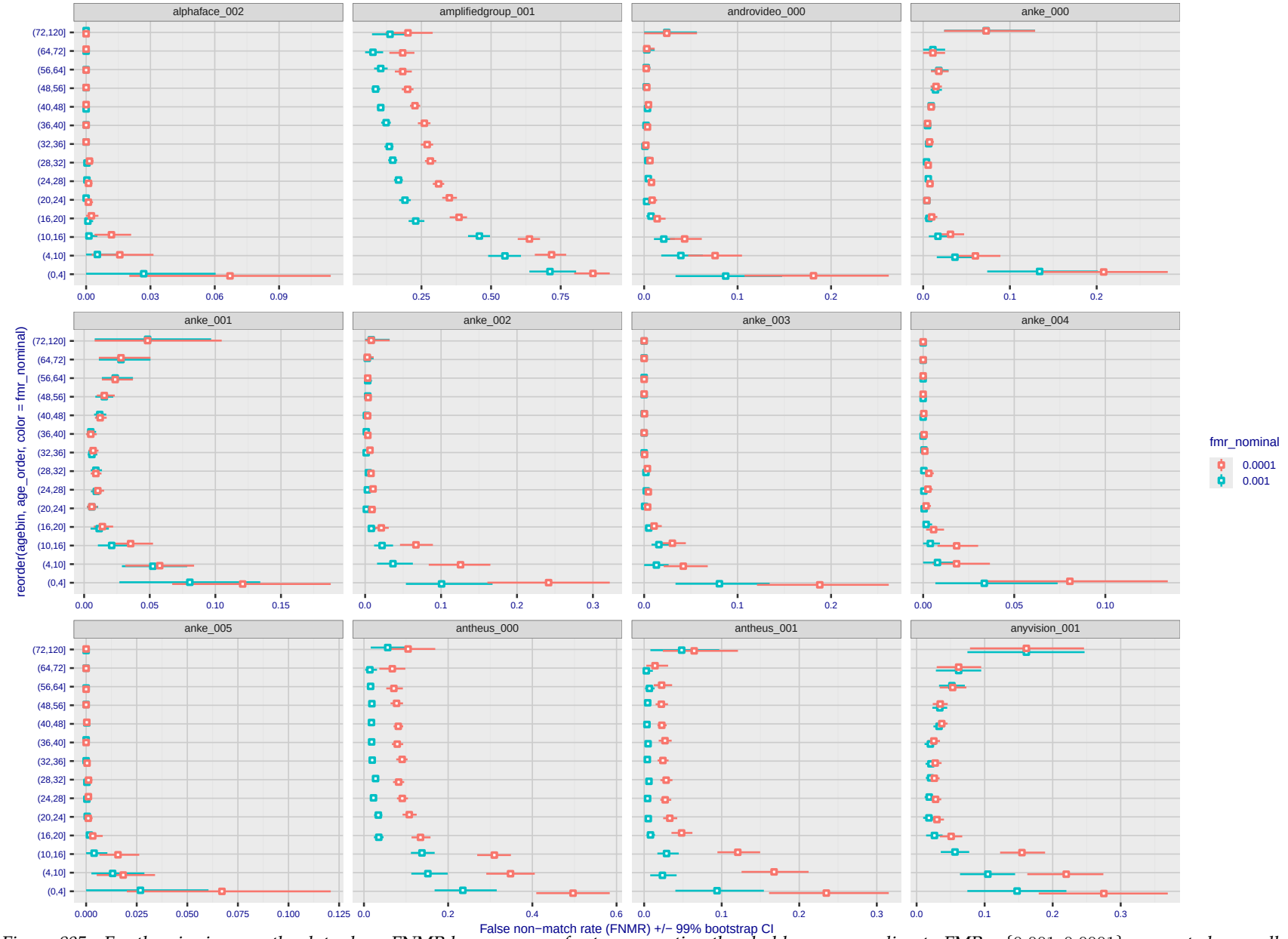

Figure 805: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 806: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 807: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 808: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 809: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 810: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

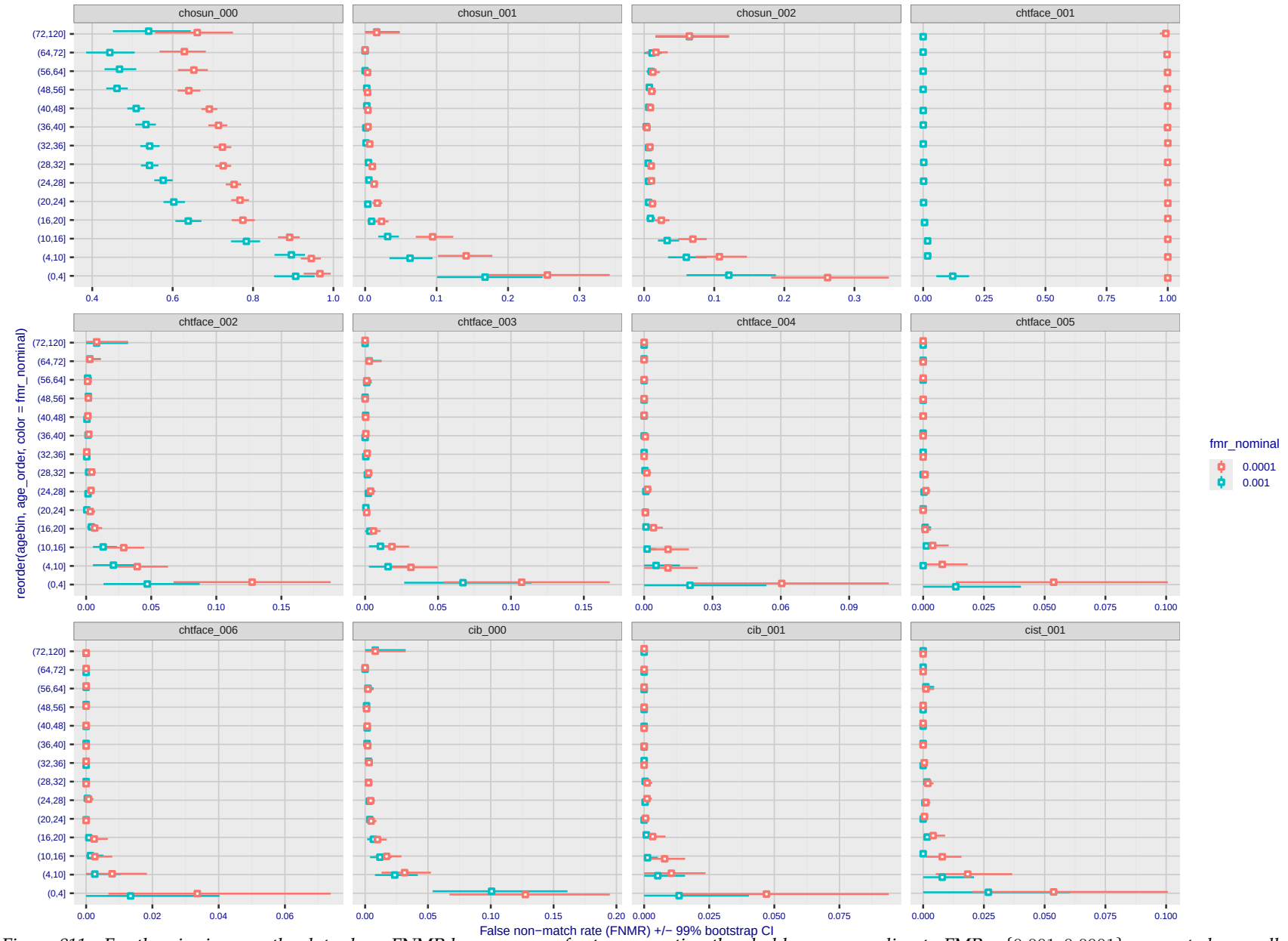

Figure 811: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 812: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 813: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

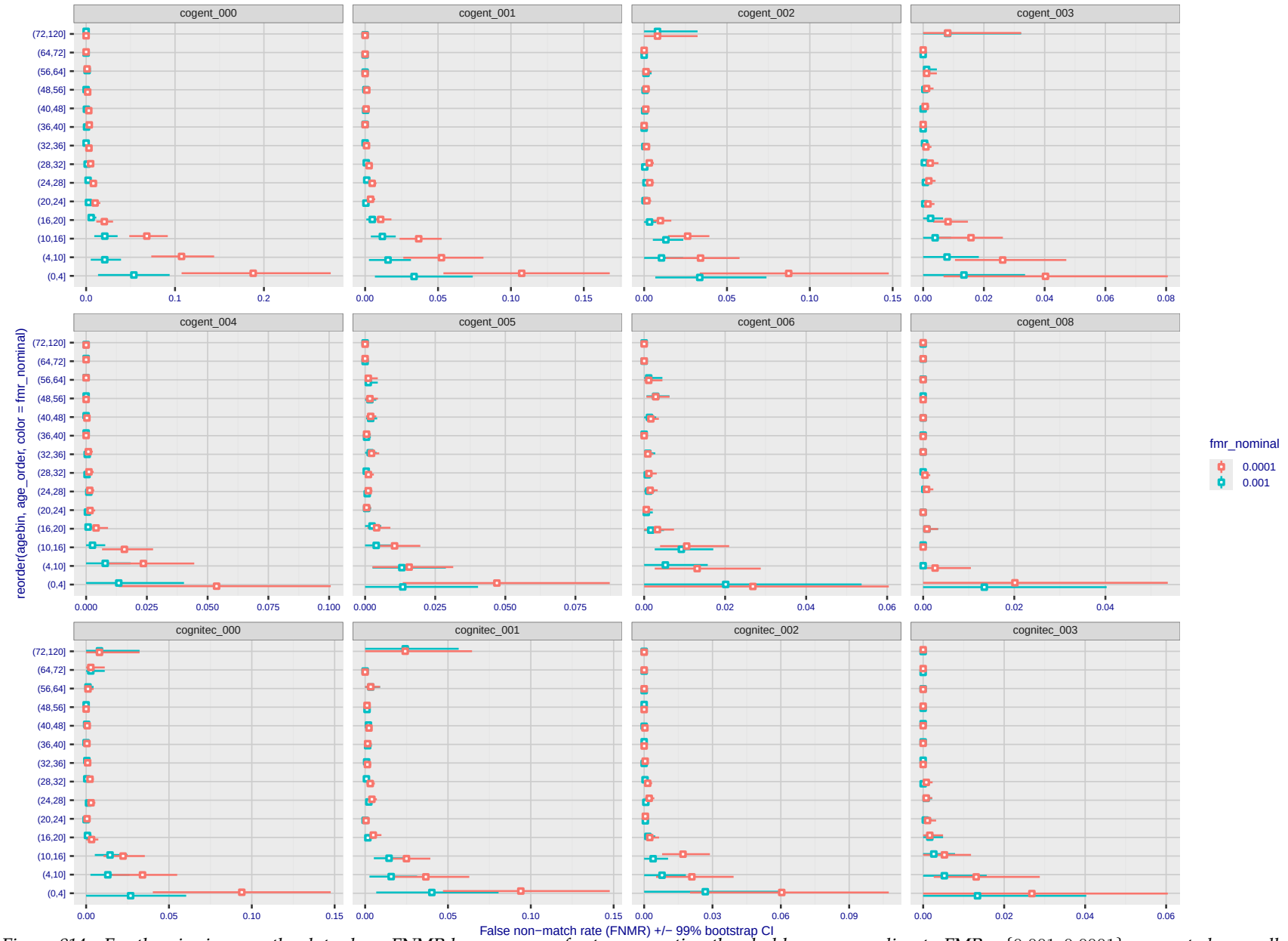

Figure 814: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 815: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 816: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 817: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

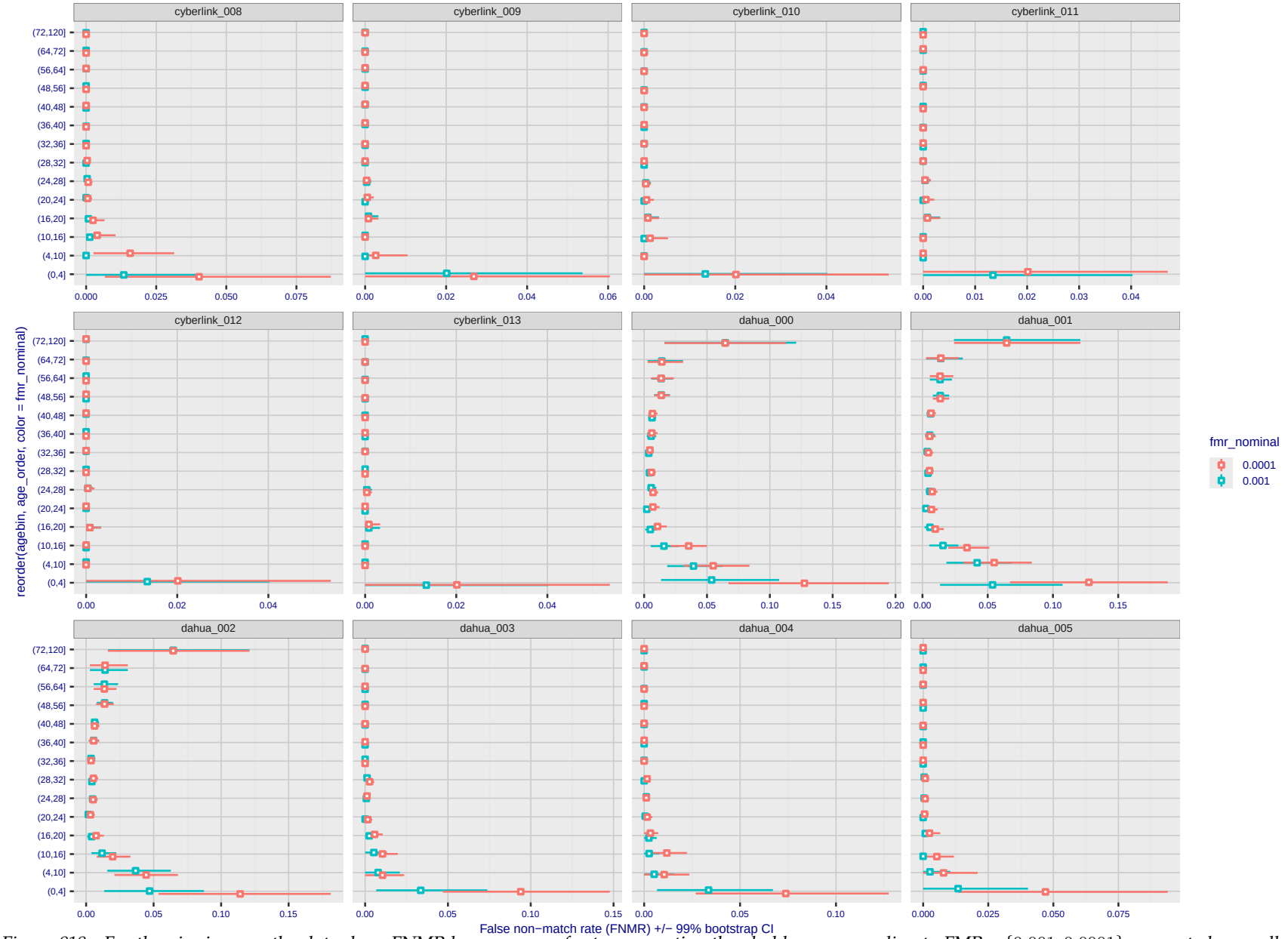

Figure 818: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

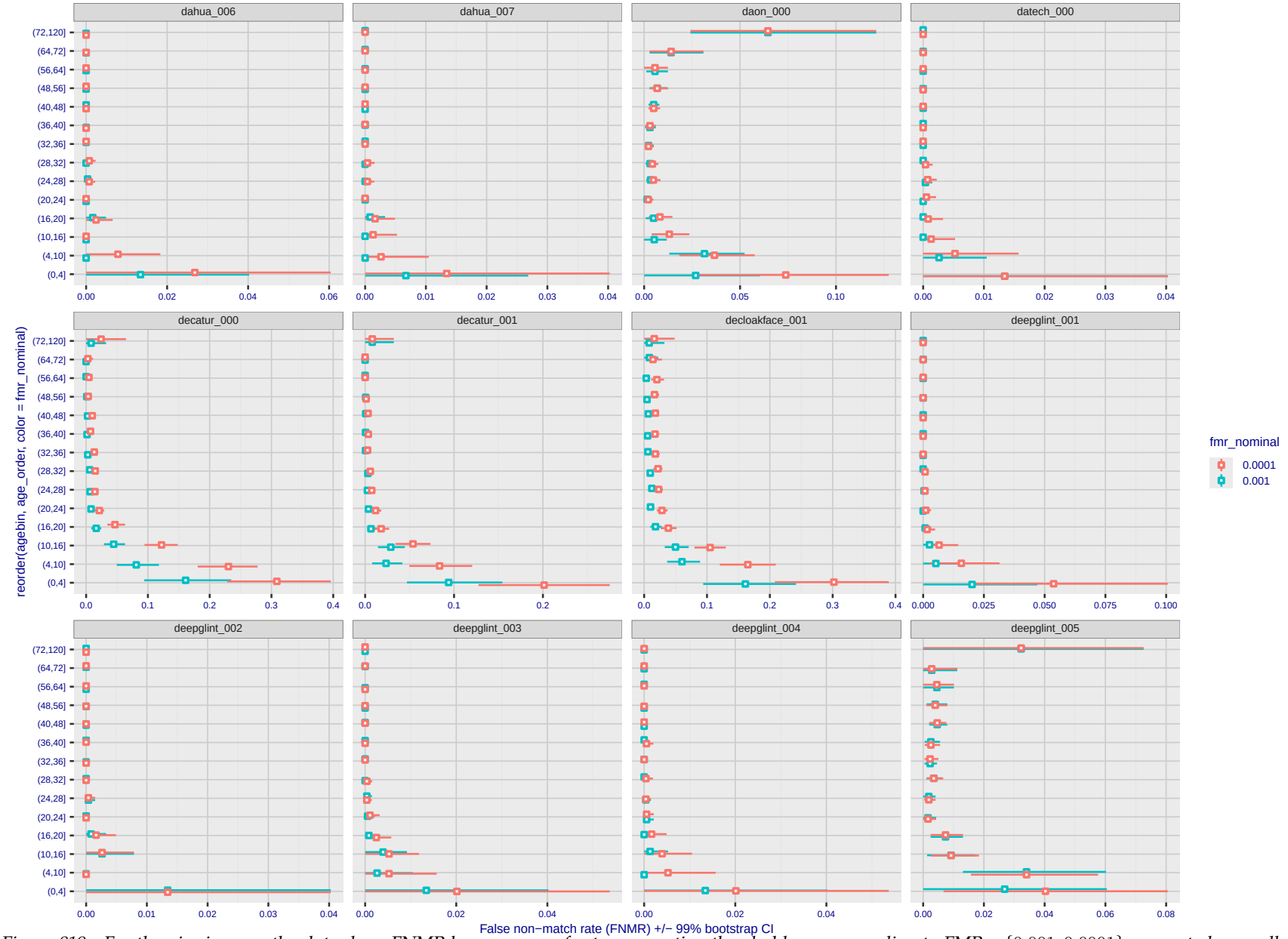

Figure 819: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 820: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

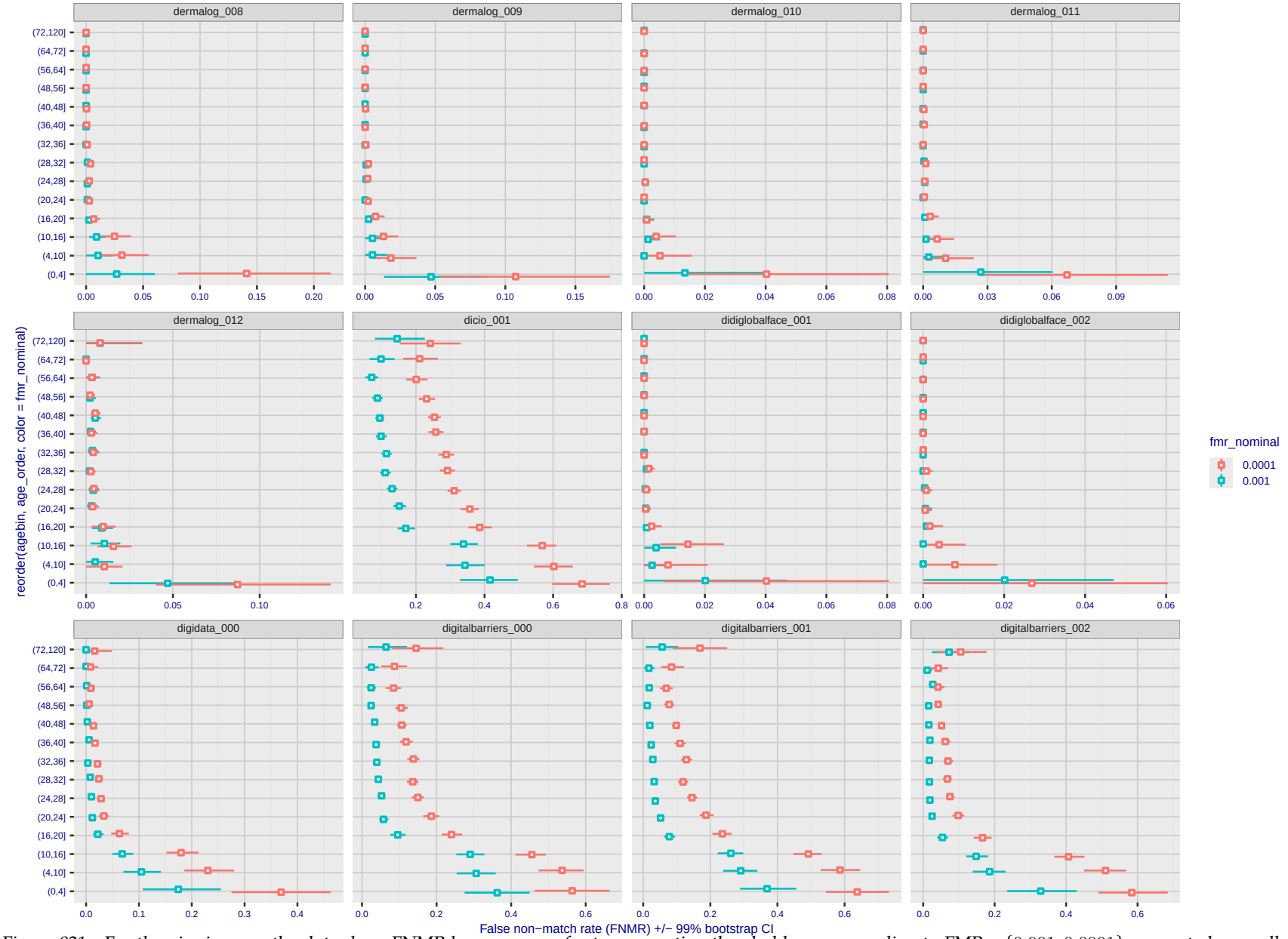

Figure 821: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 822: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

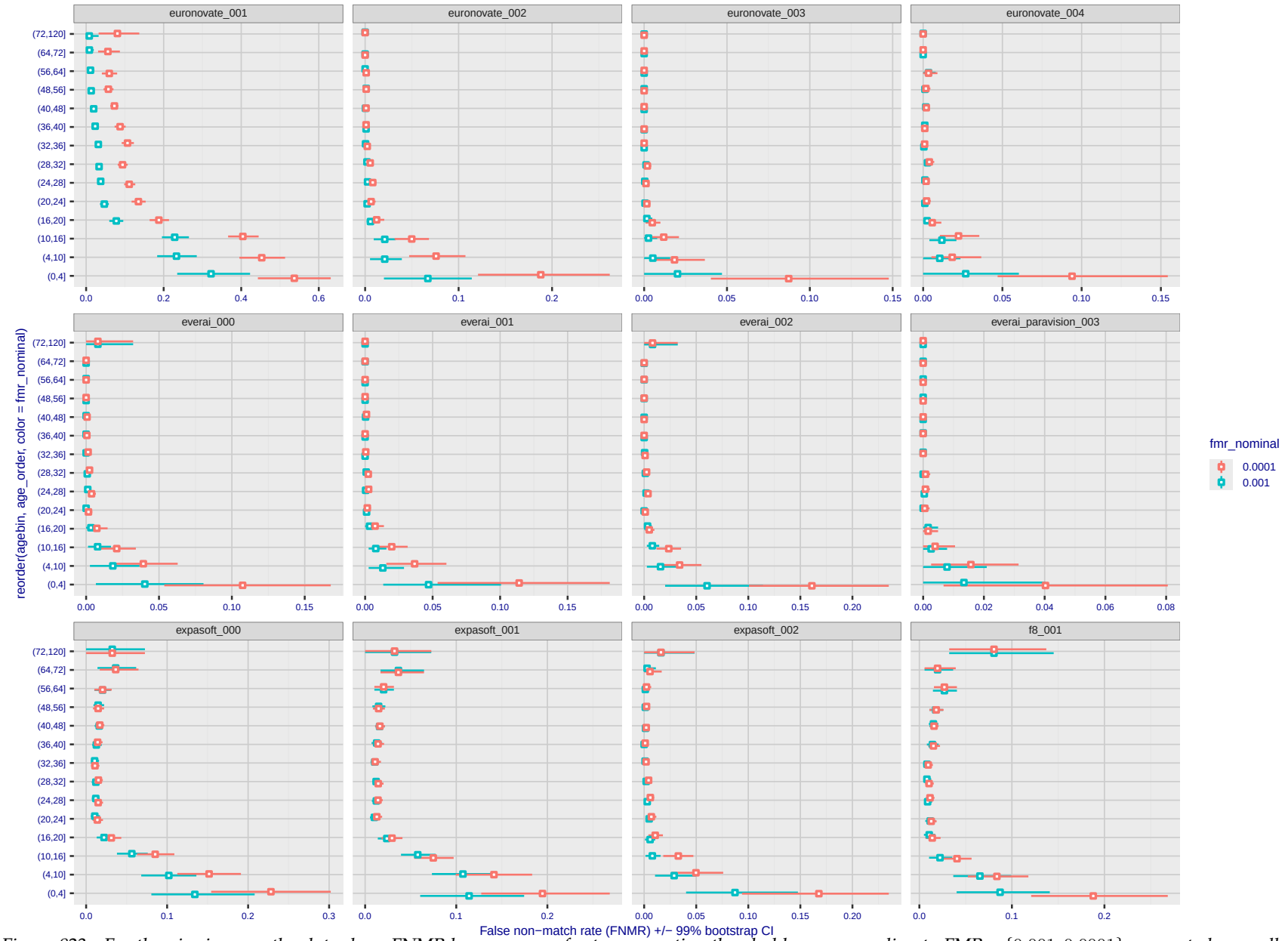

Figure 823: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 824: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

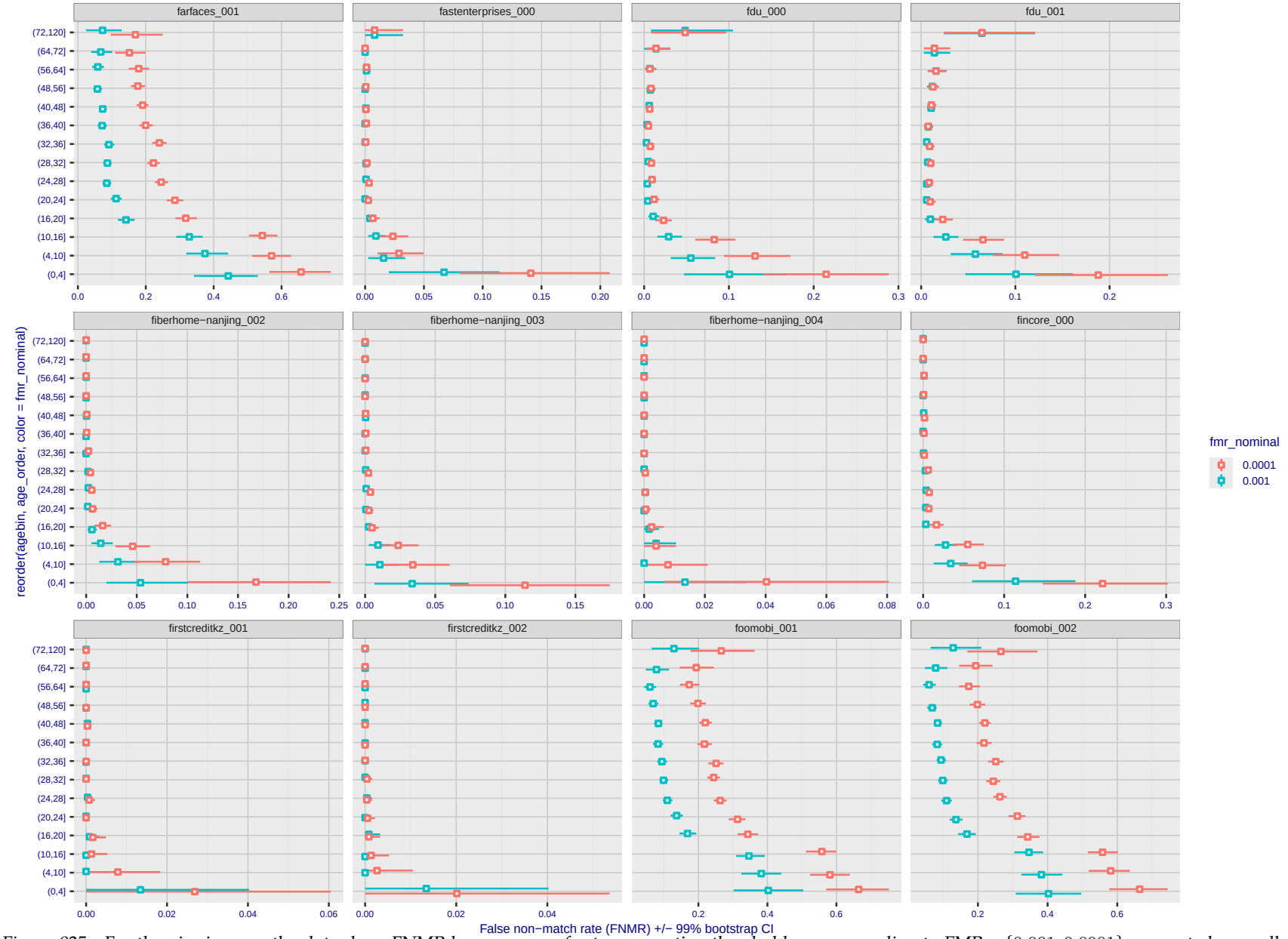

Figure 825: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 826: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 827: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 828: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 829: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 830: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

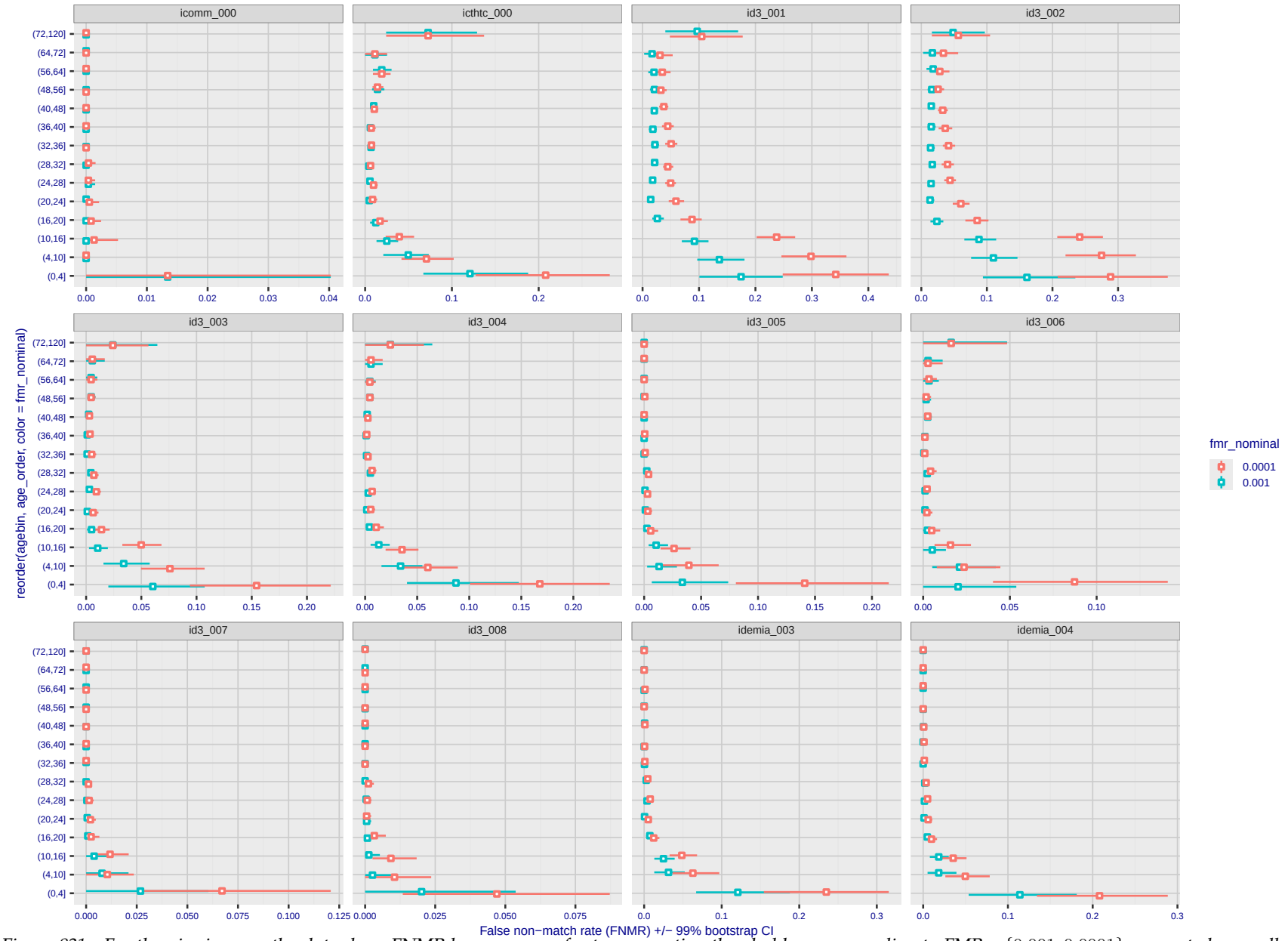

Figure 831: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

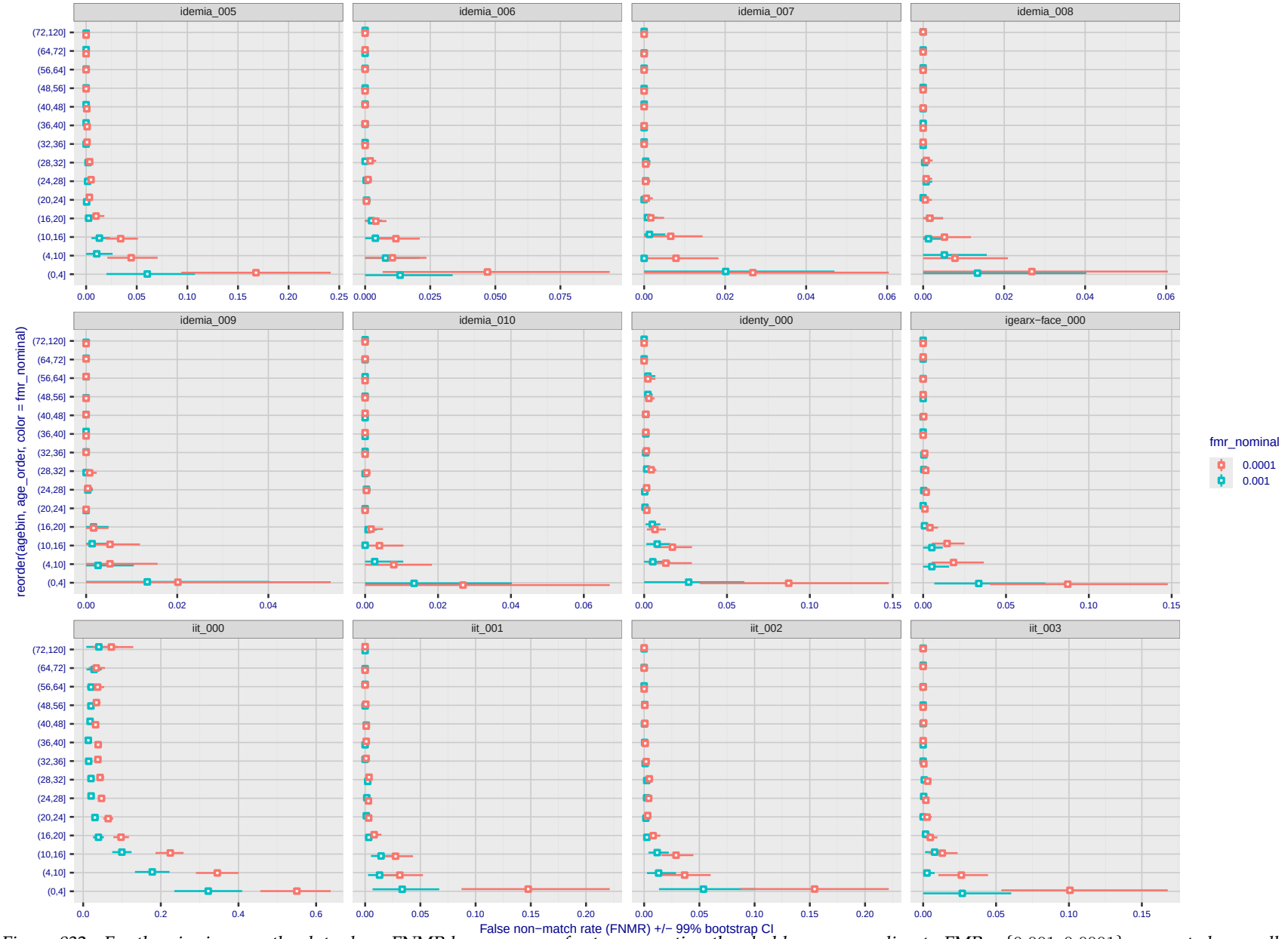

Figure 832: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 833: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

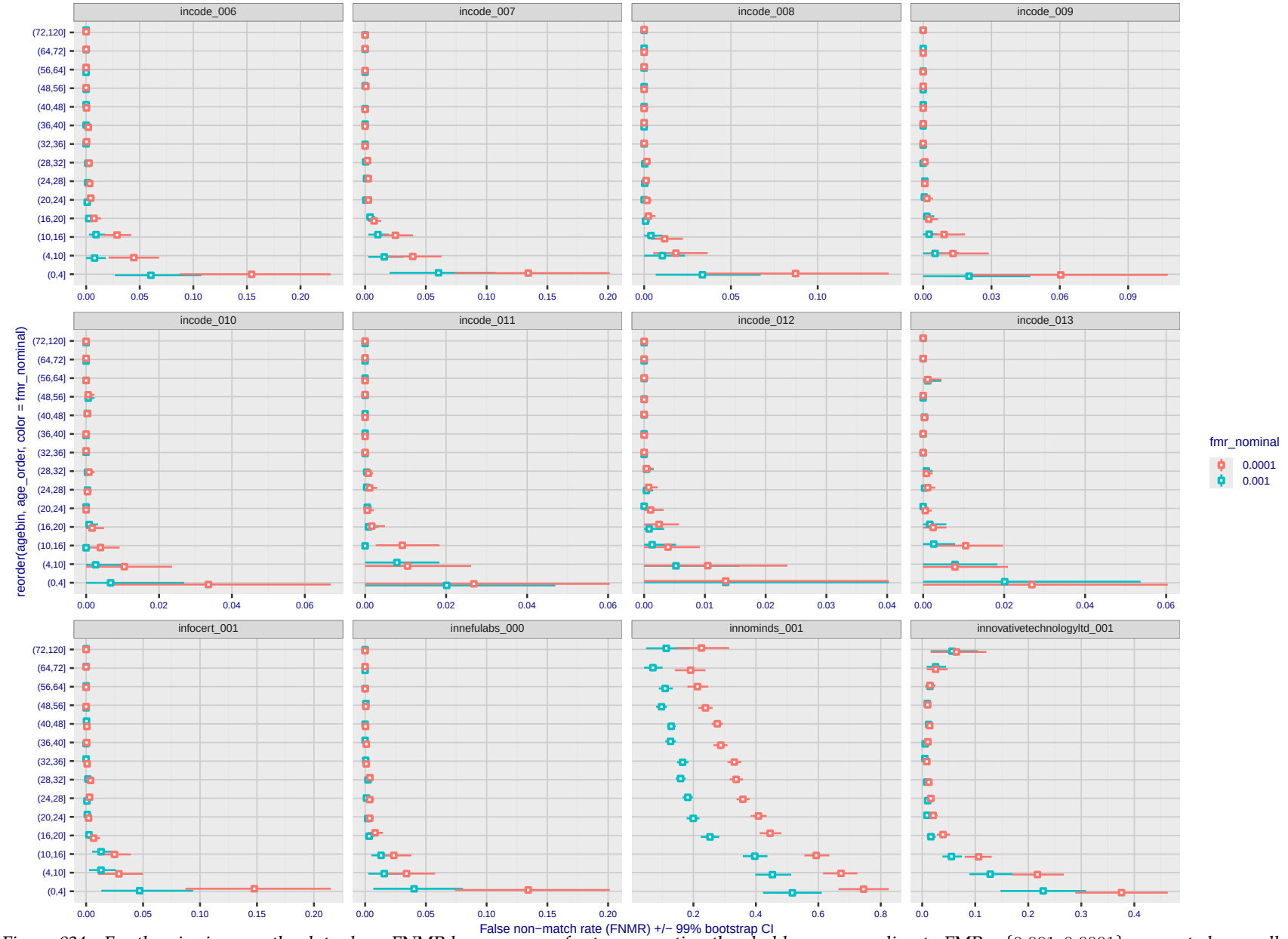

Figure 834: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 835: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

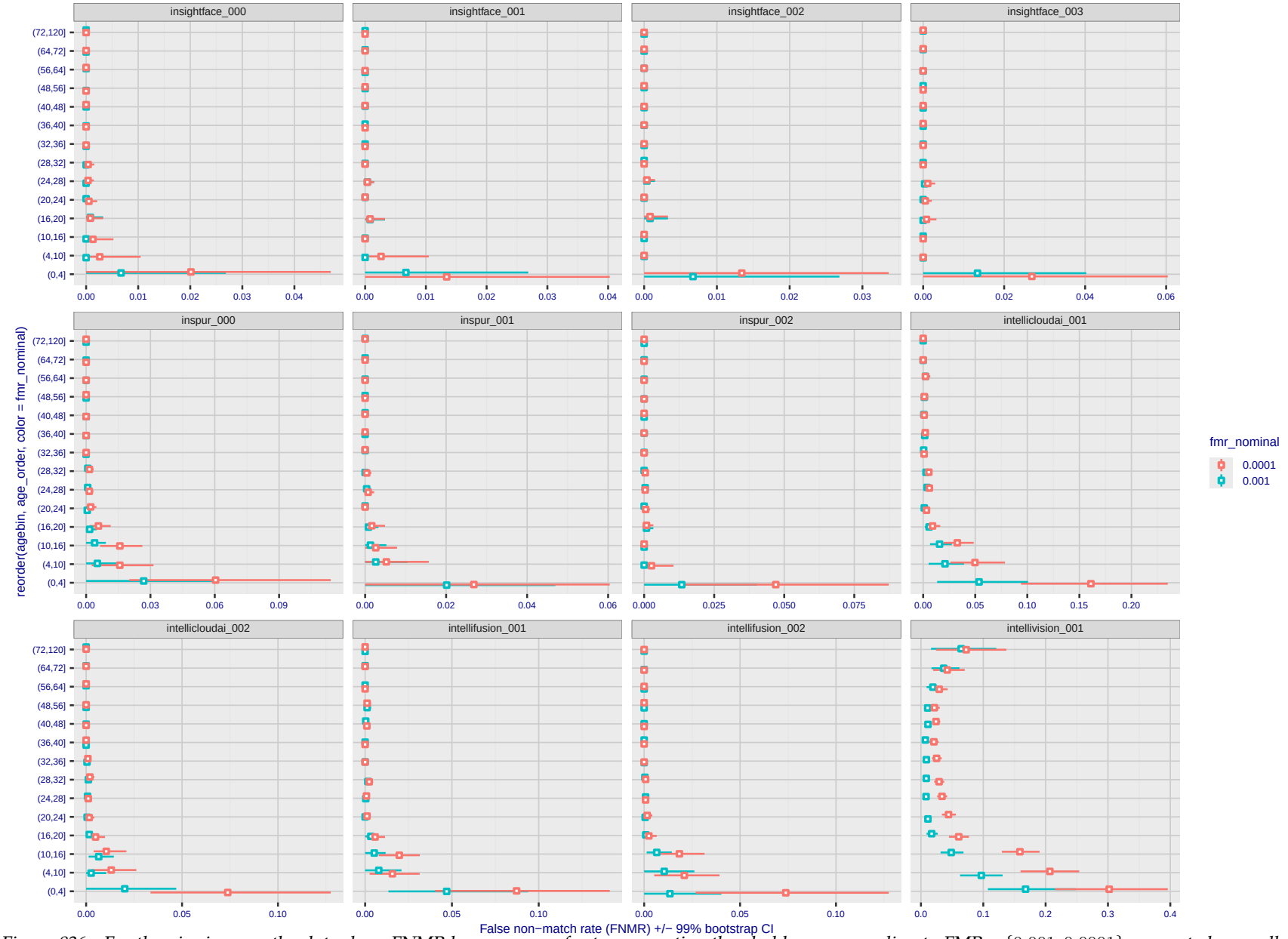

Figure 836: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

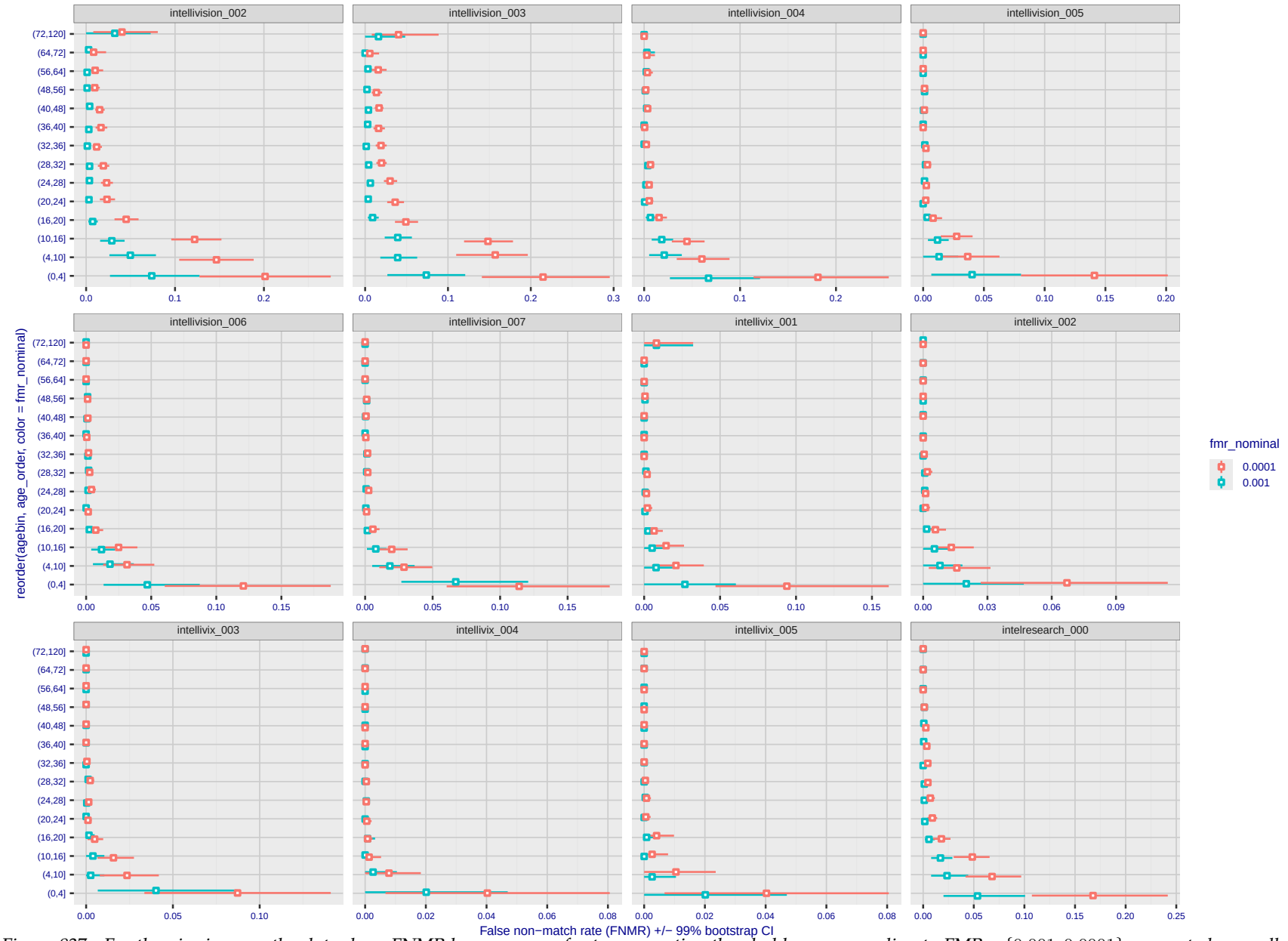

Figure 837: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

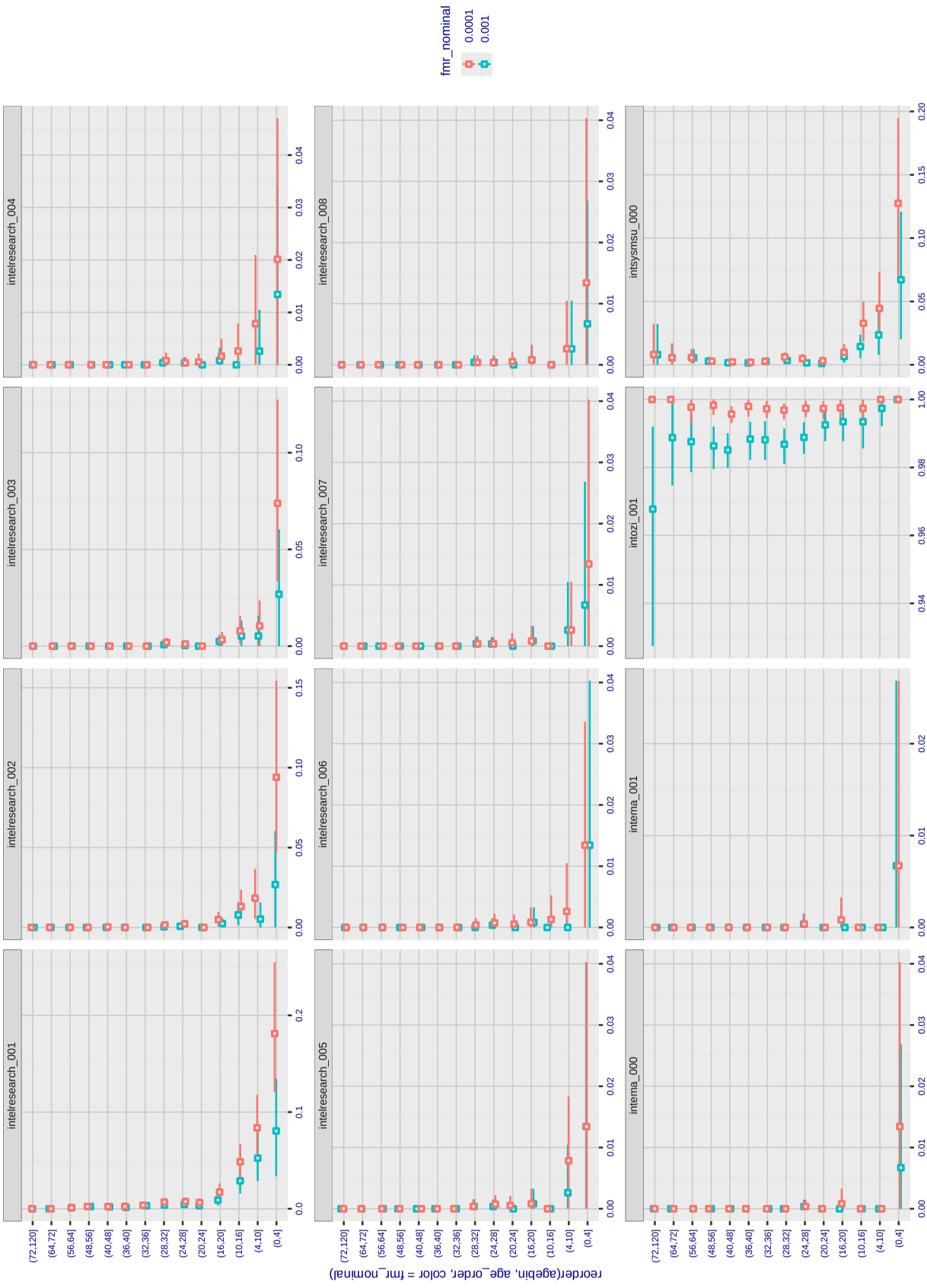

Figure 838: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 839: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 840: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 841: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 842: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 843: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

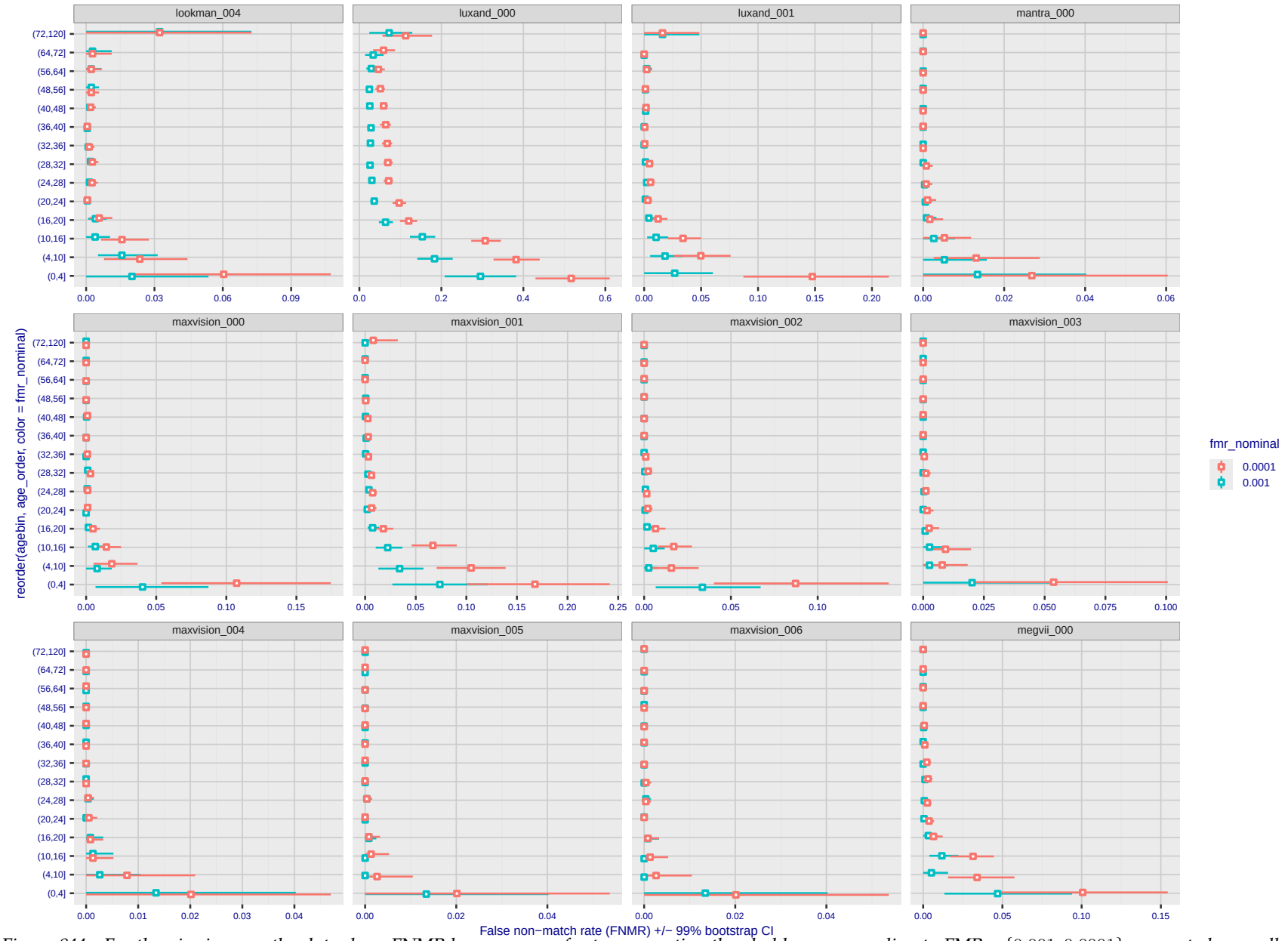

Figure 844: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

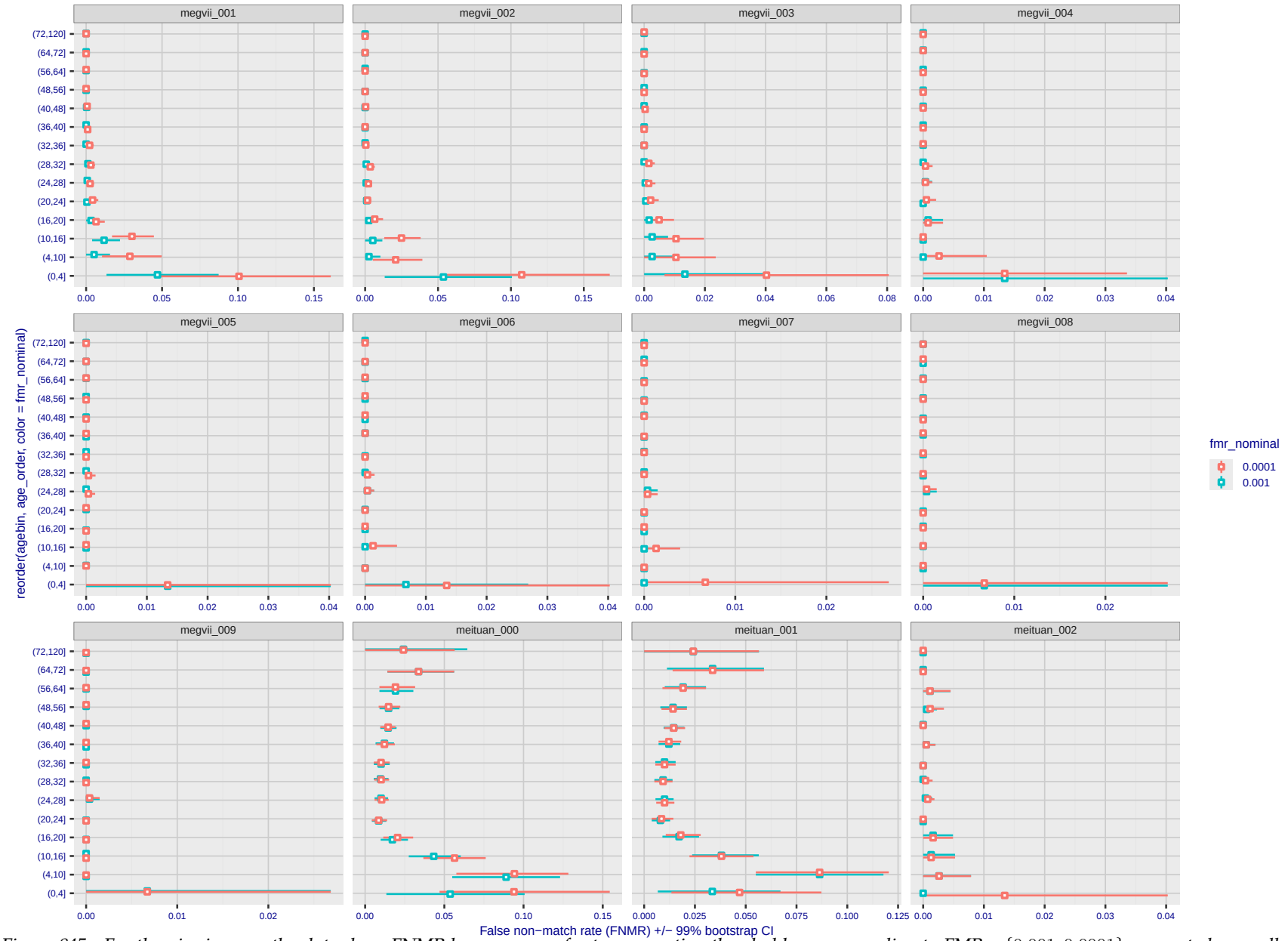

Figure 845: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 846: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

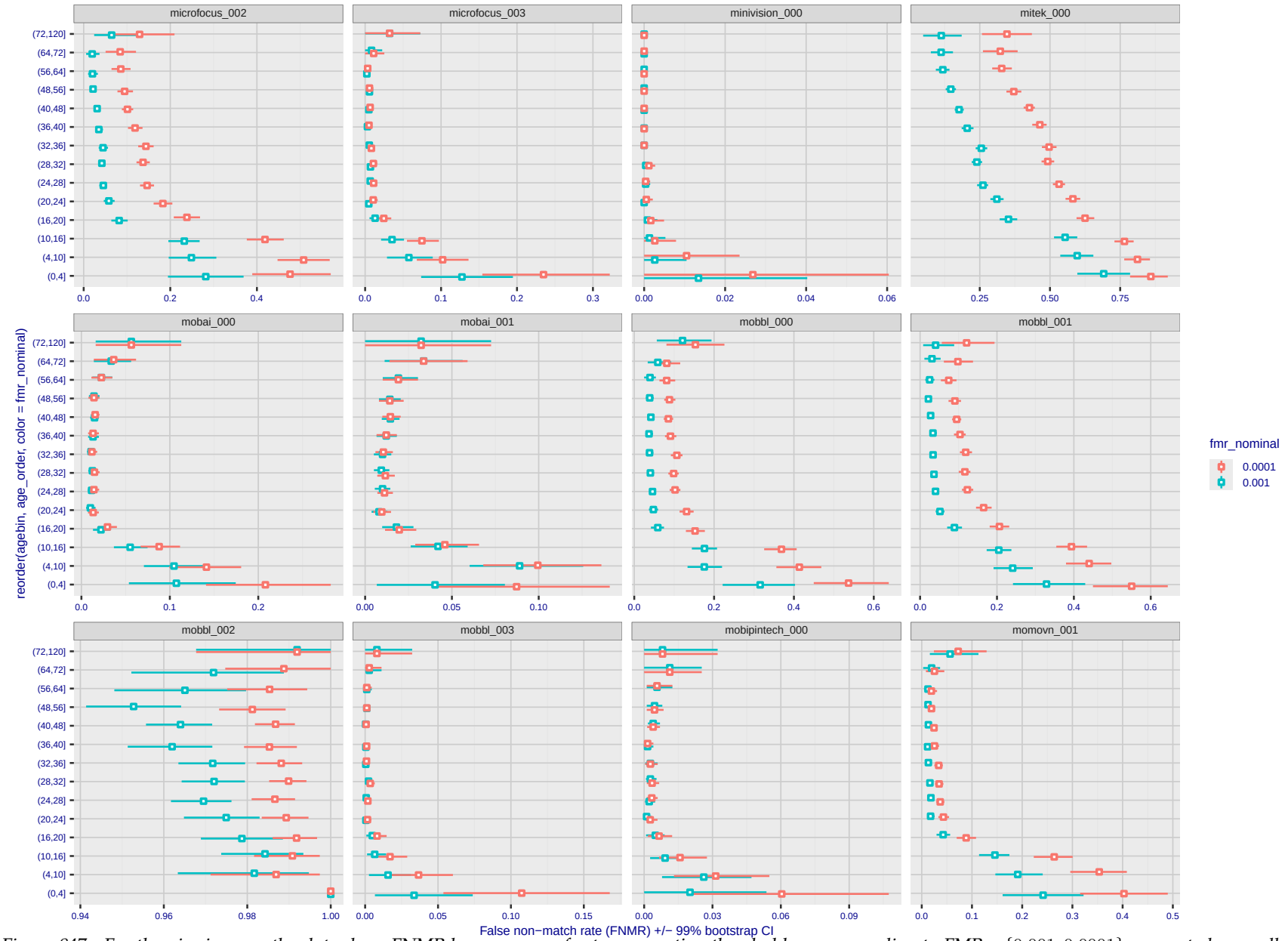

Figure 847: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 848: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

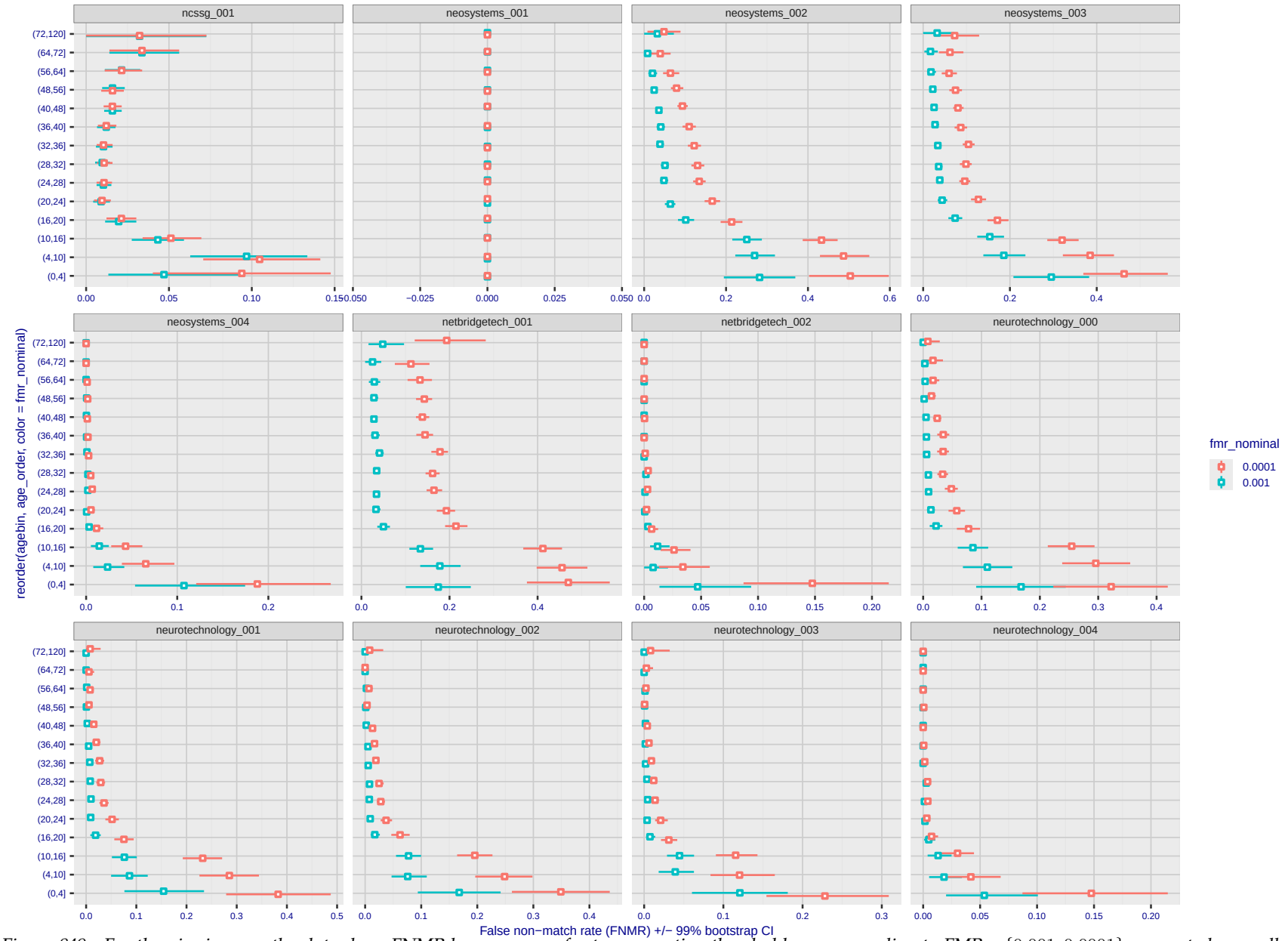

Figure 849: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

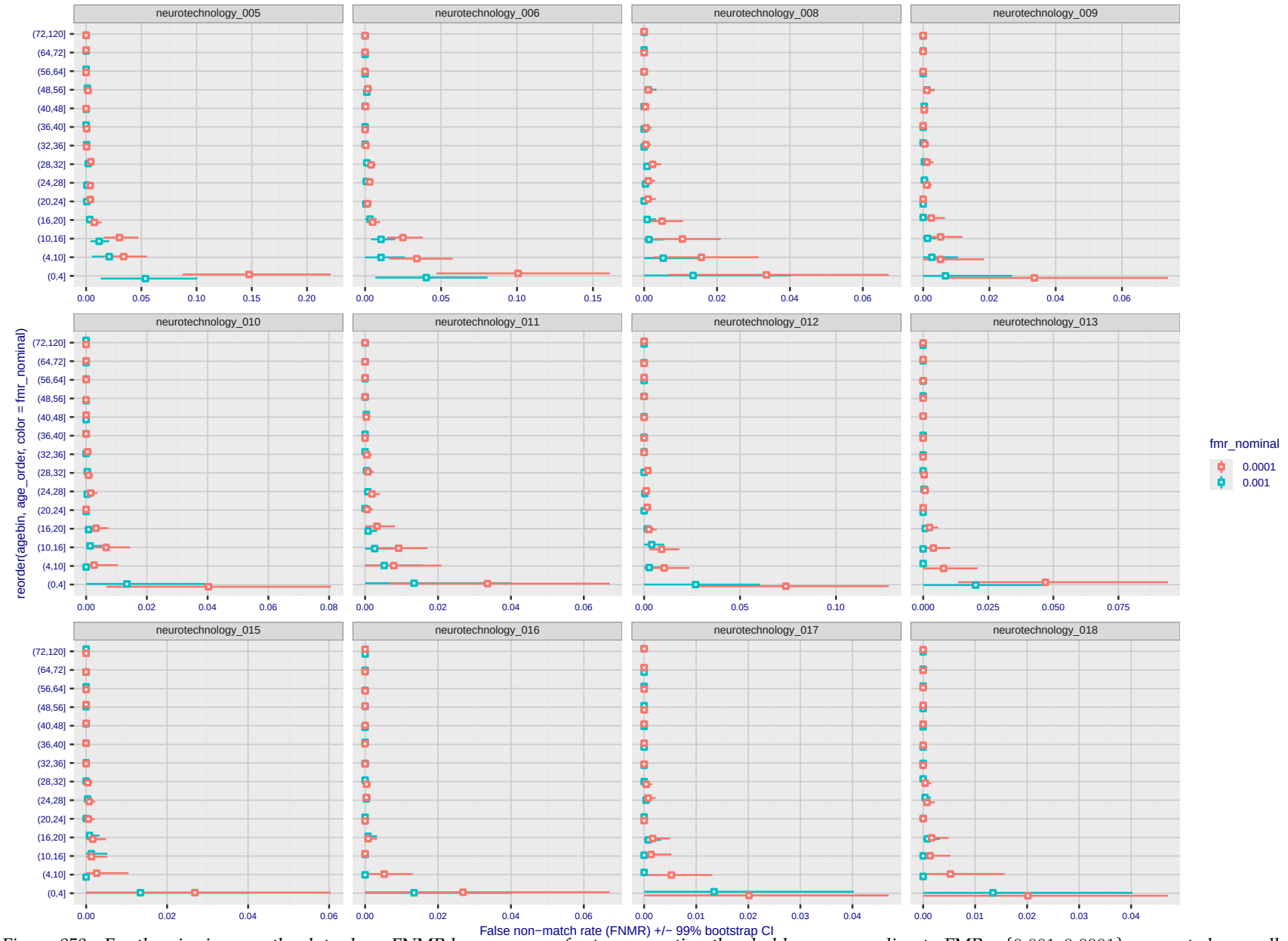

Figure 850: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 851: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 852: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

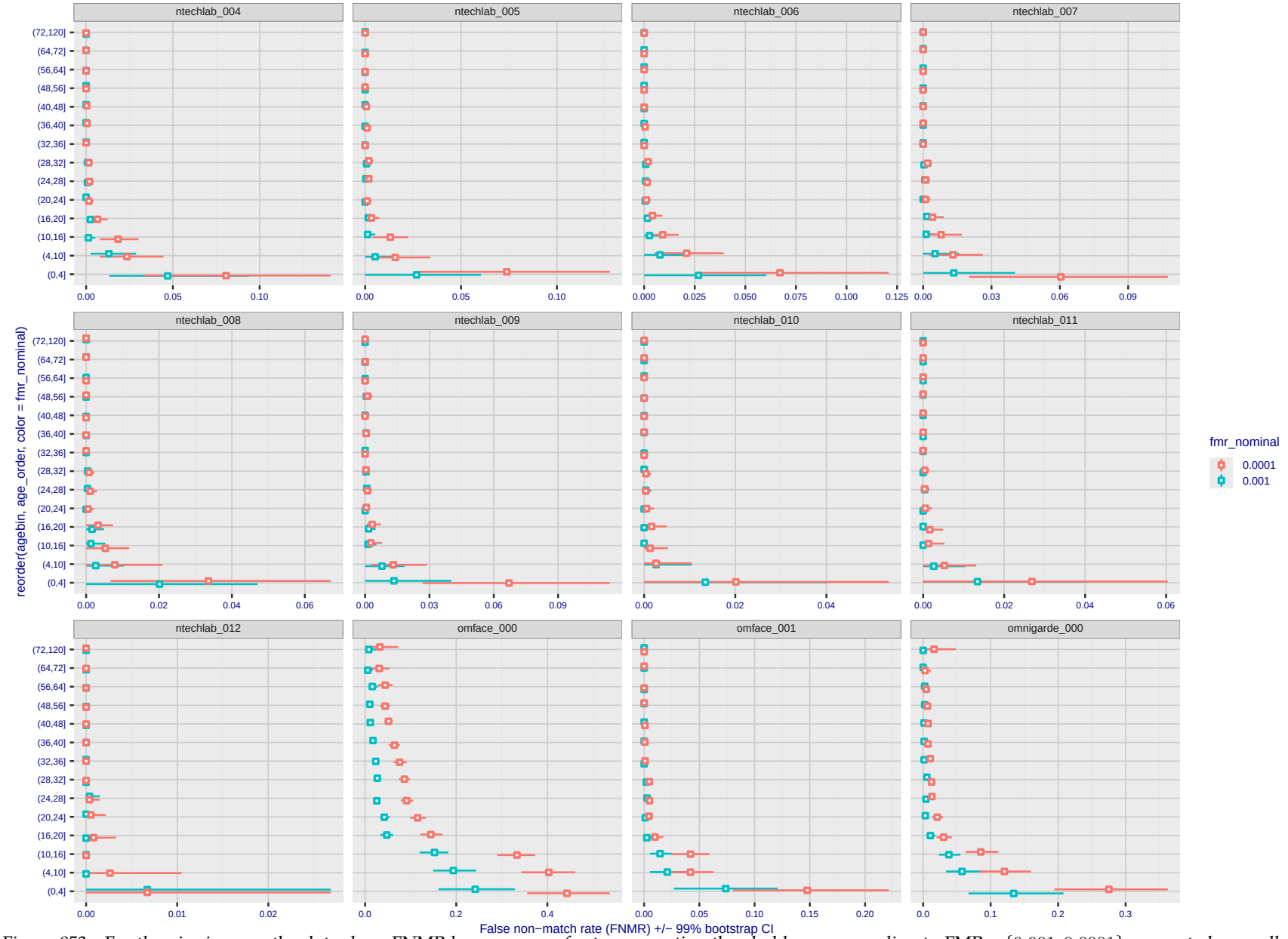

Figure 853: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 854: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

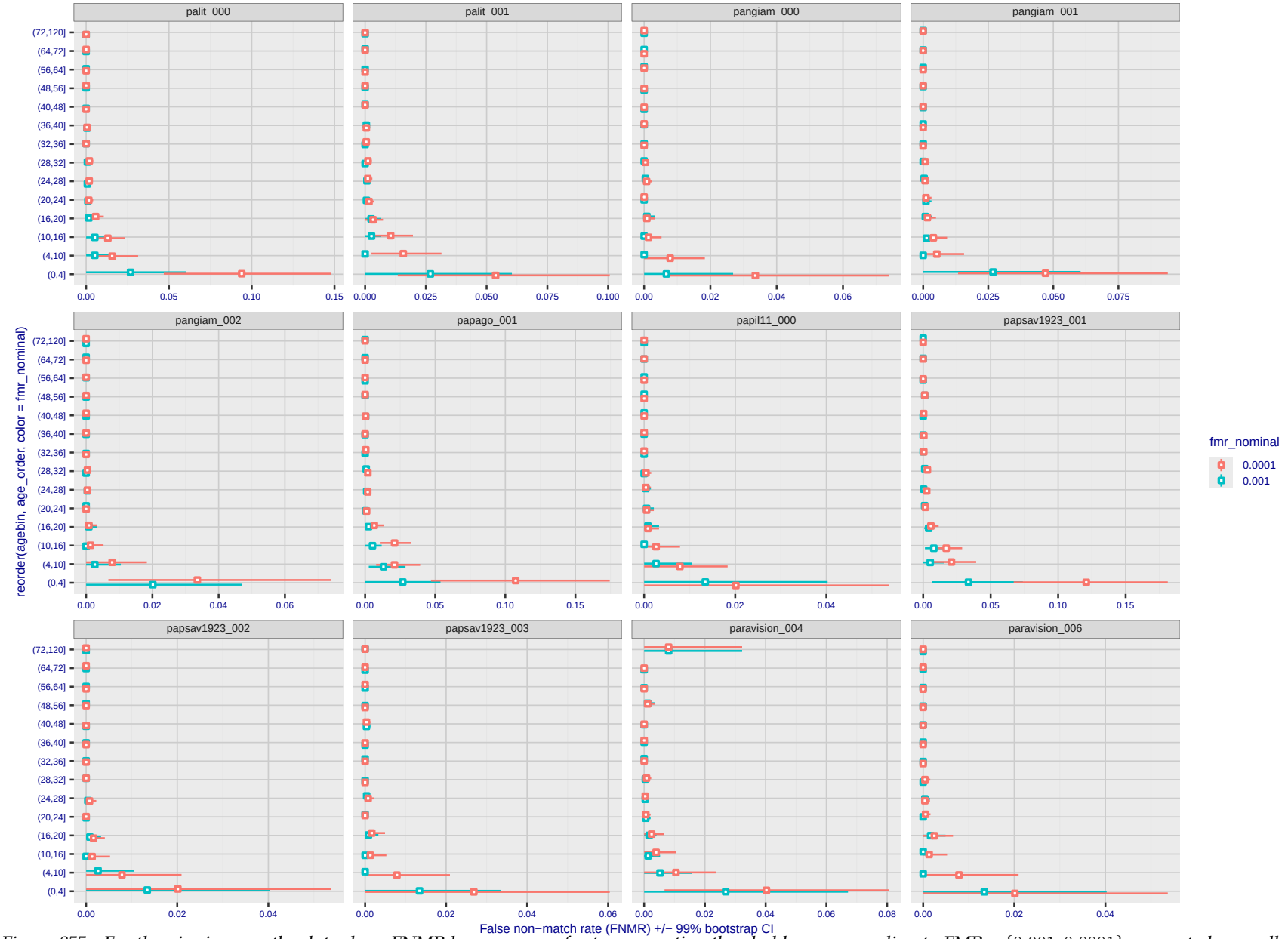

Figure 855: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

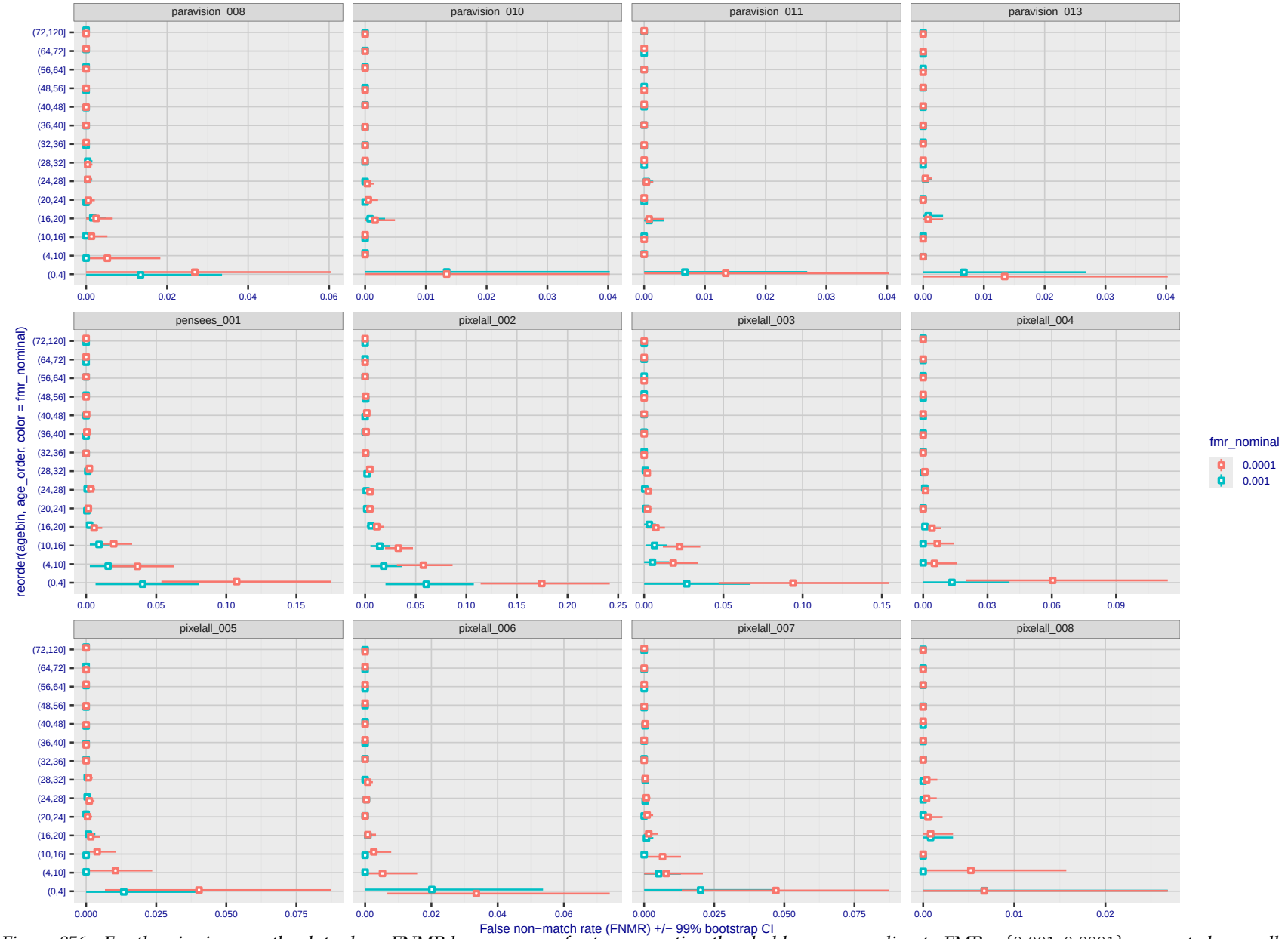

Figure 856: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 857: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

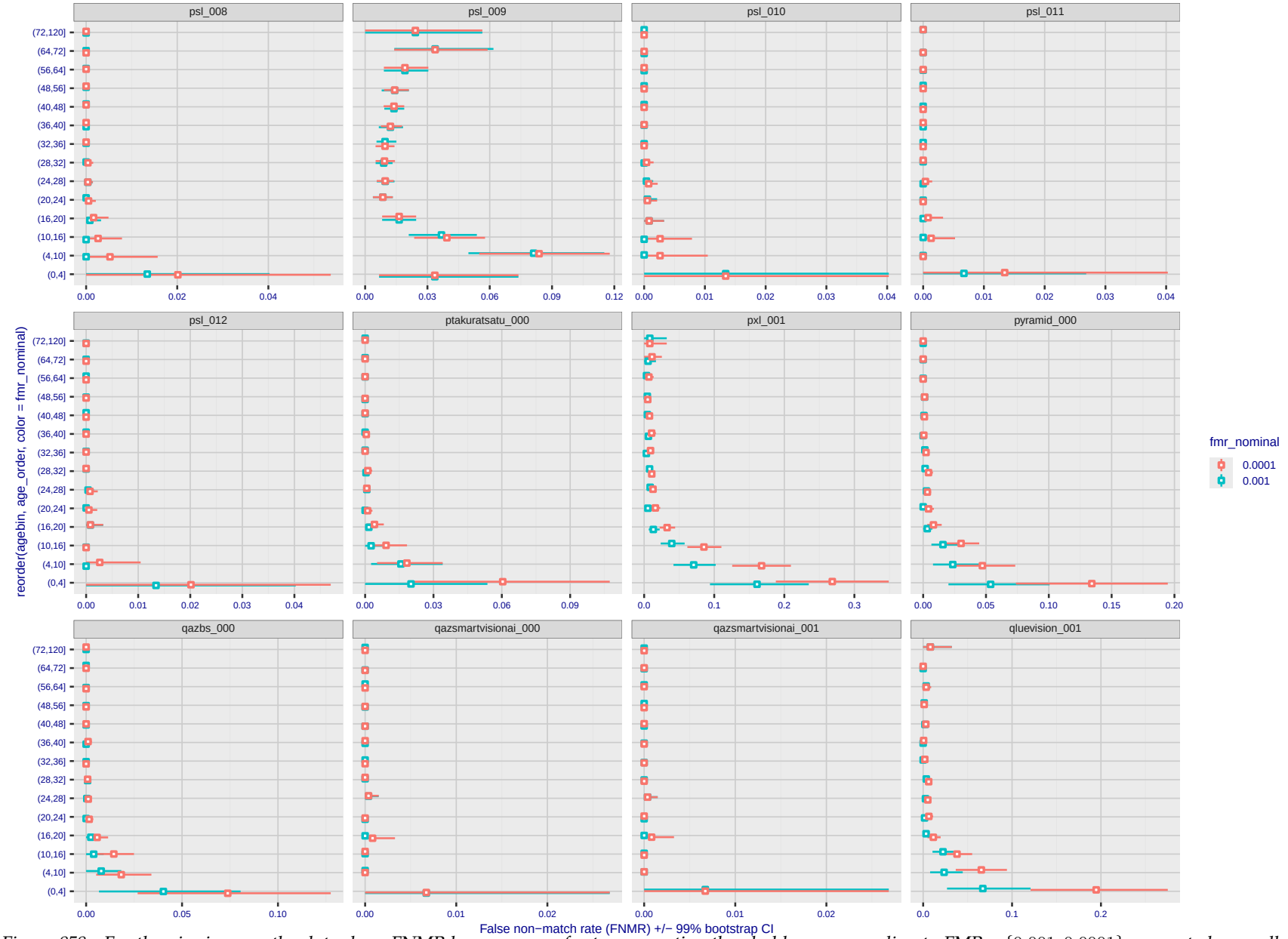

Figure 858: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 859: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

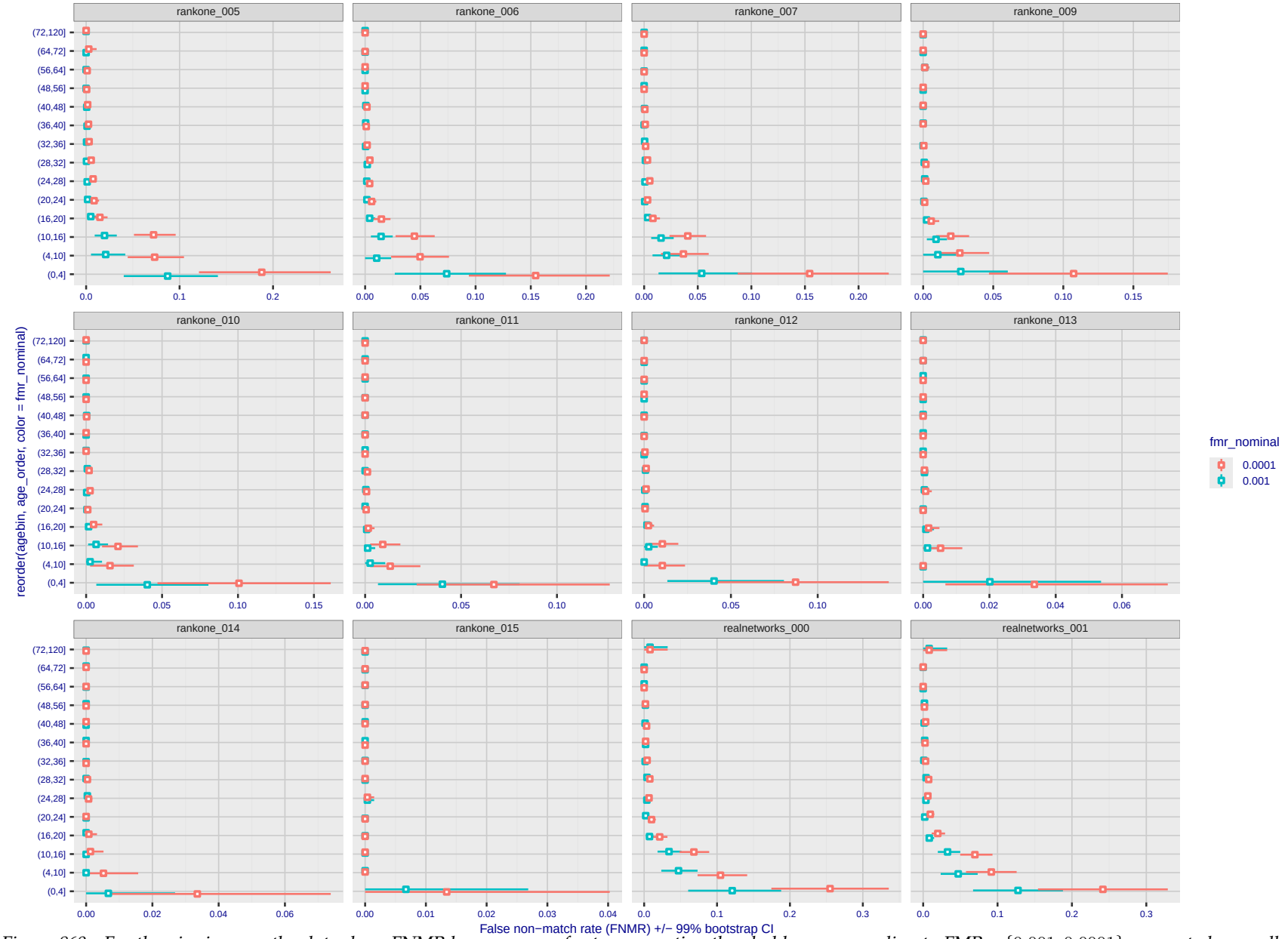

Figure 860: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

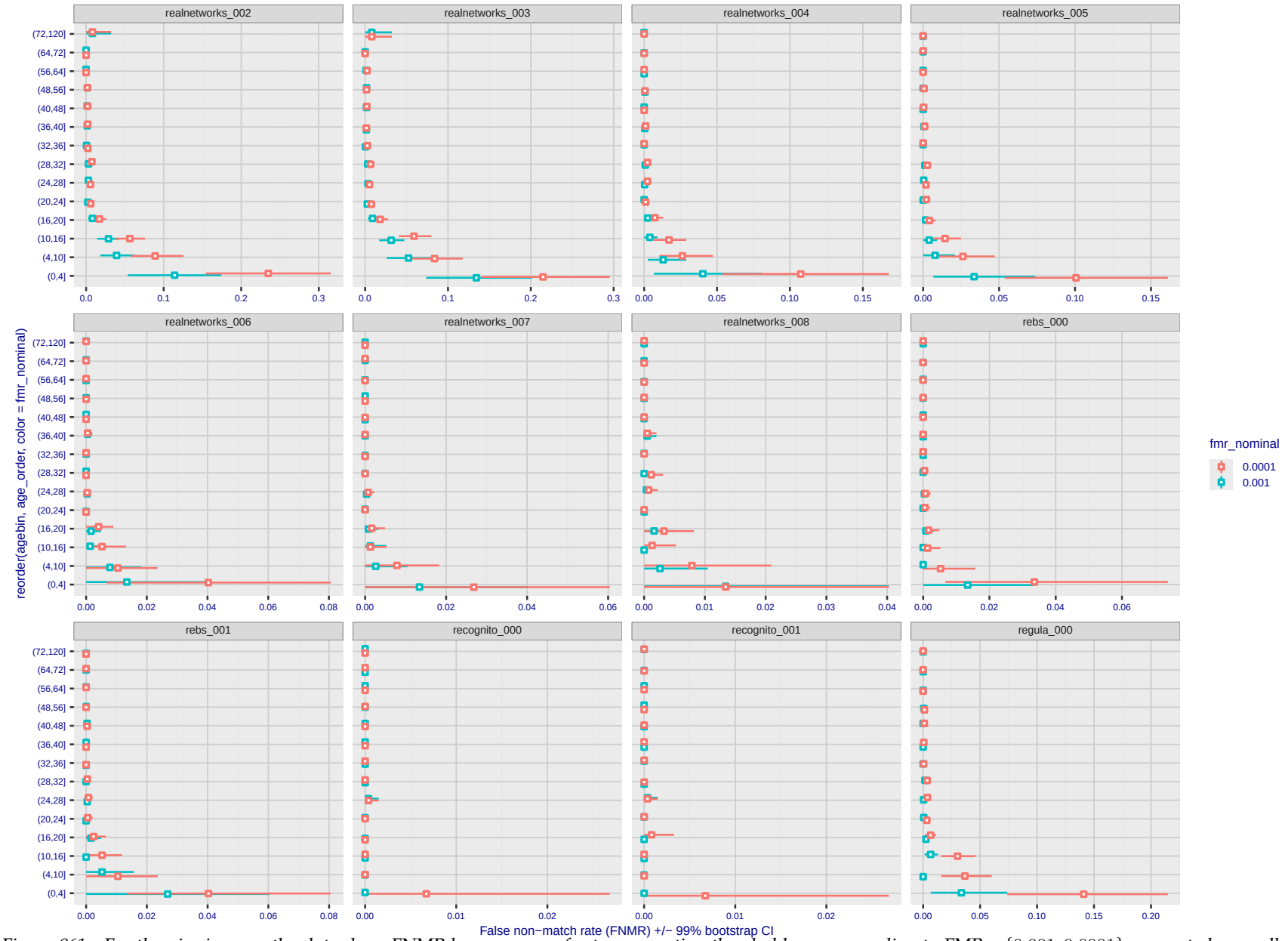

Figure 861: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 862: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

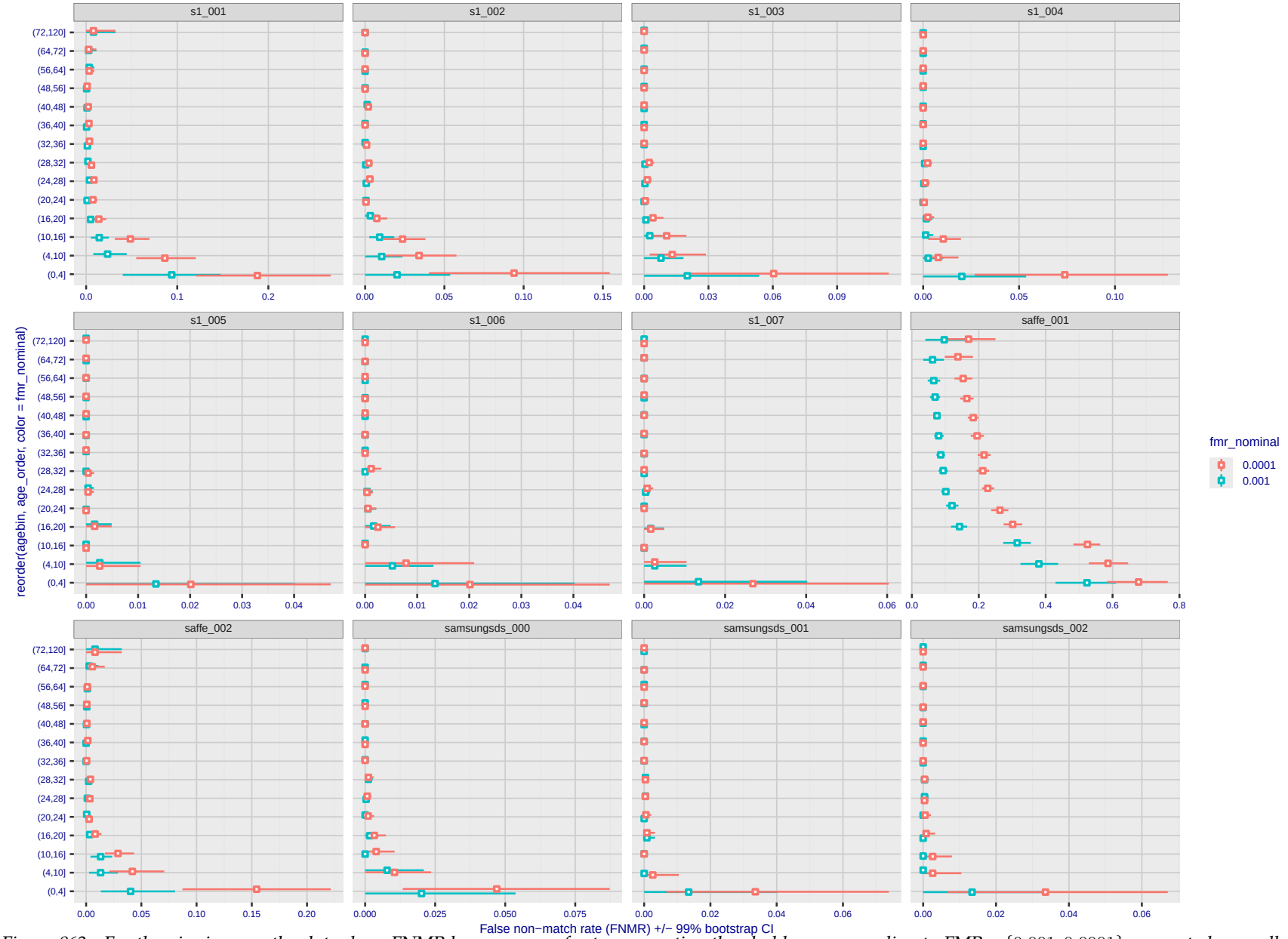

Figure 863: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

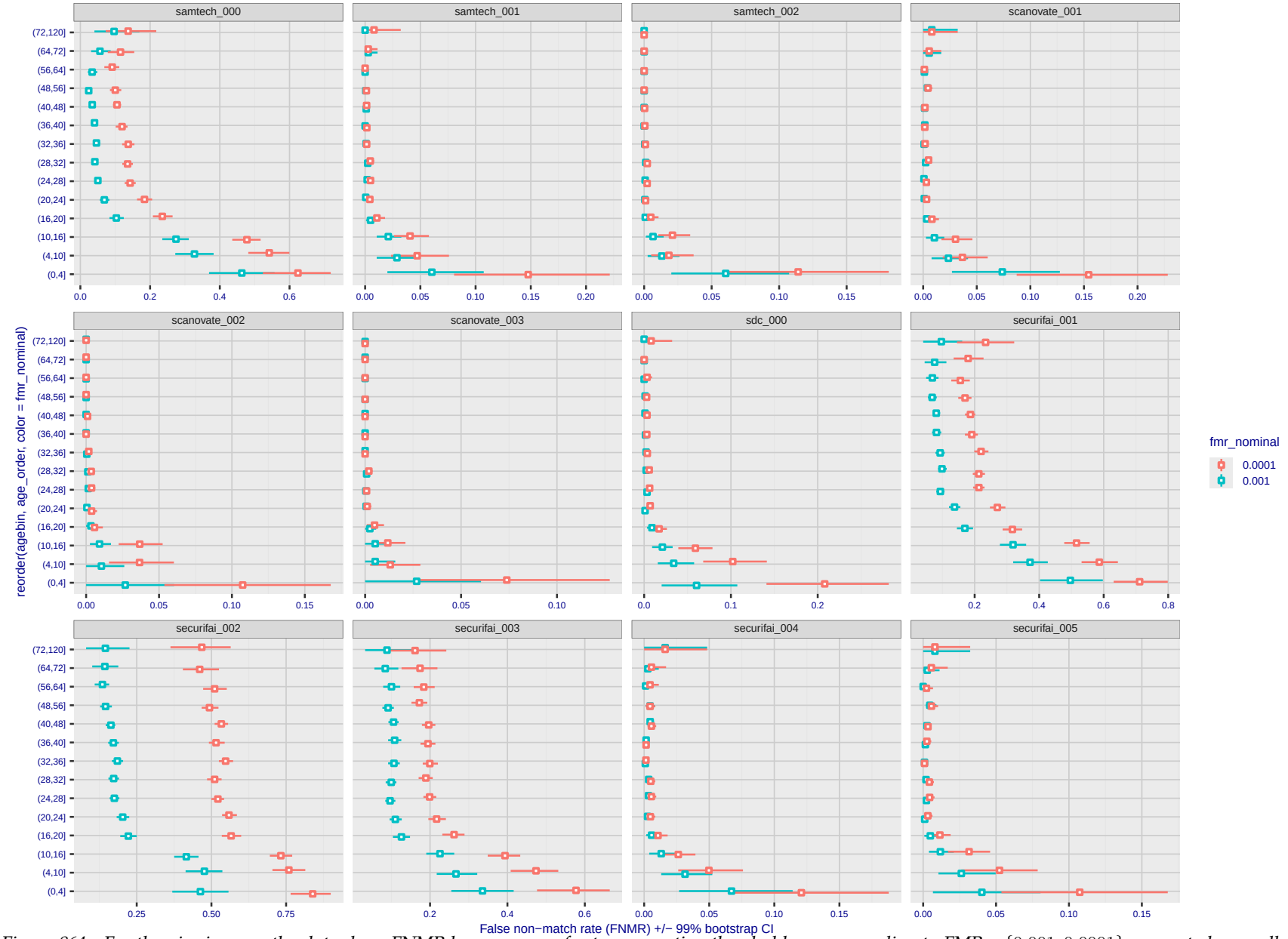

Figure 864: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

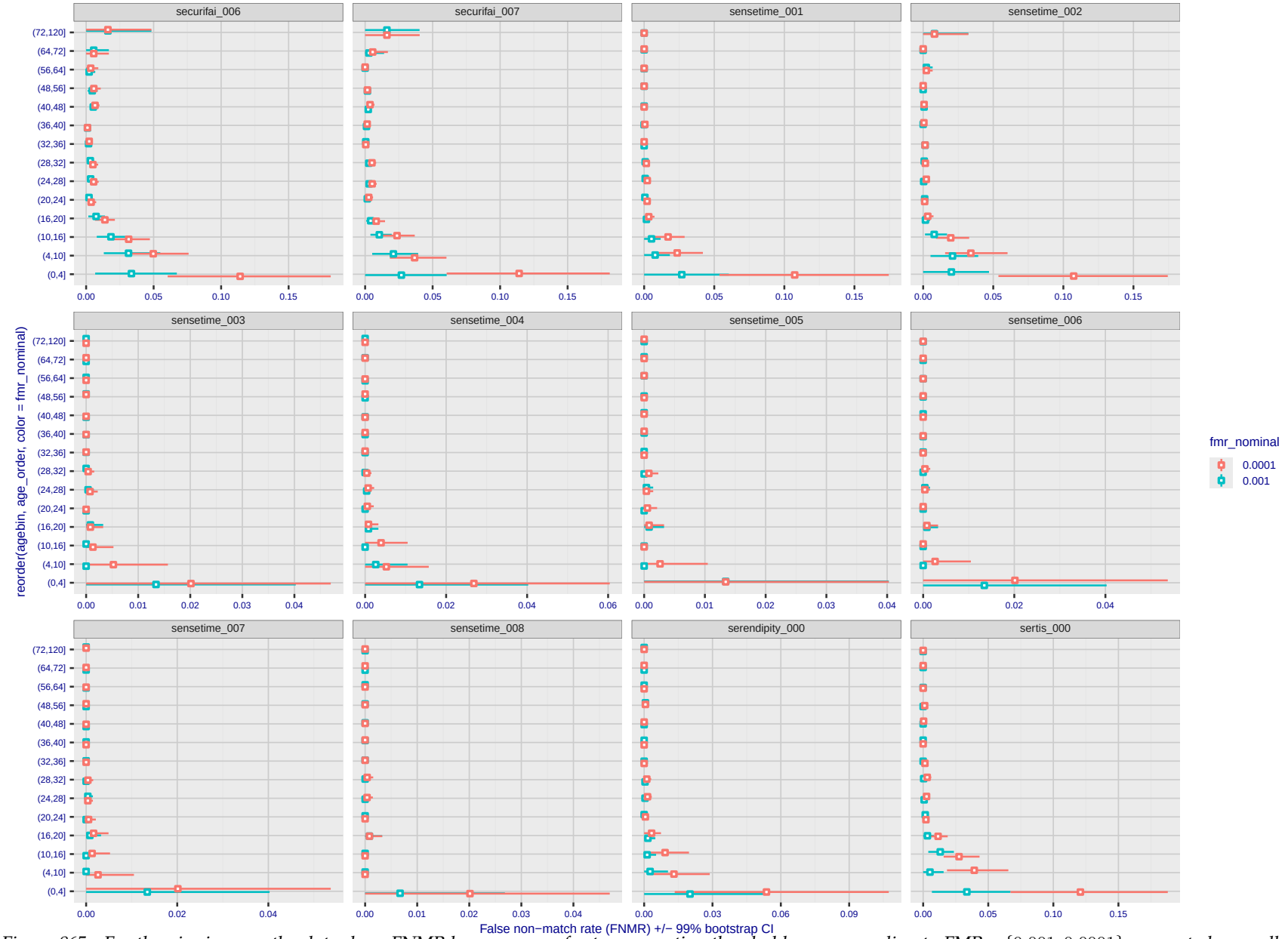

Figure 865: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

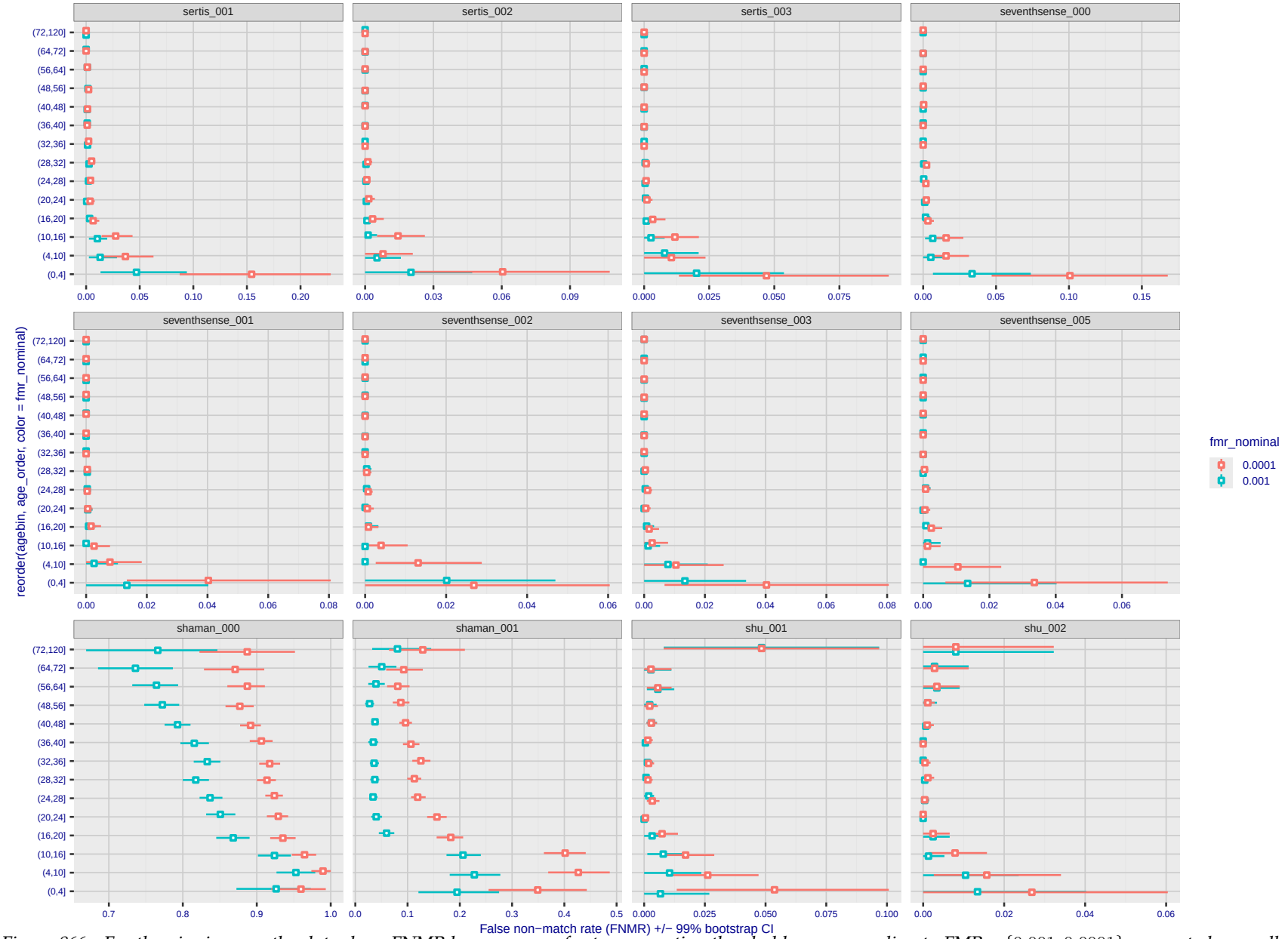

Figure 866: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 867: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 868: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

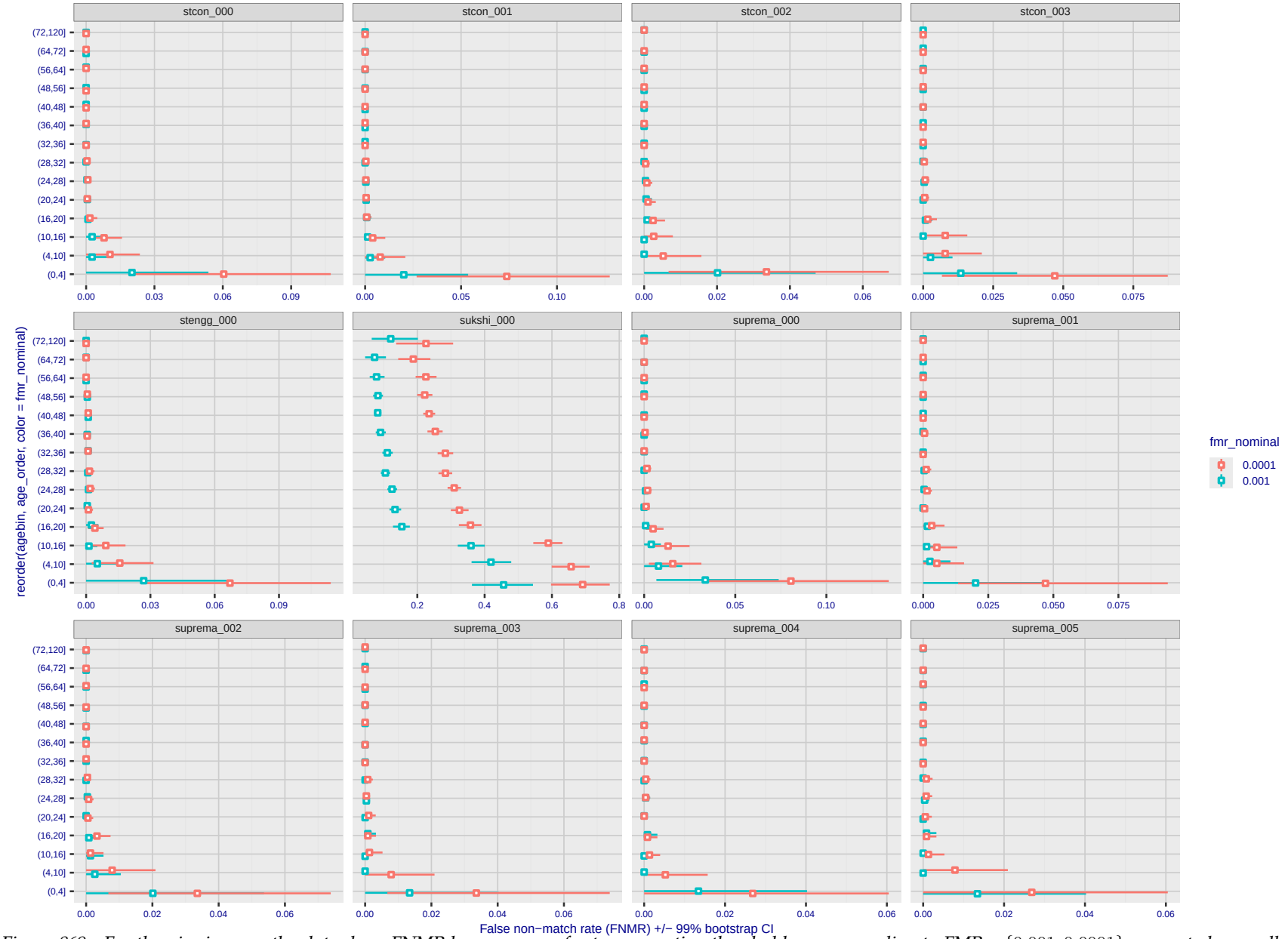

Figure 869: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

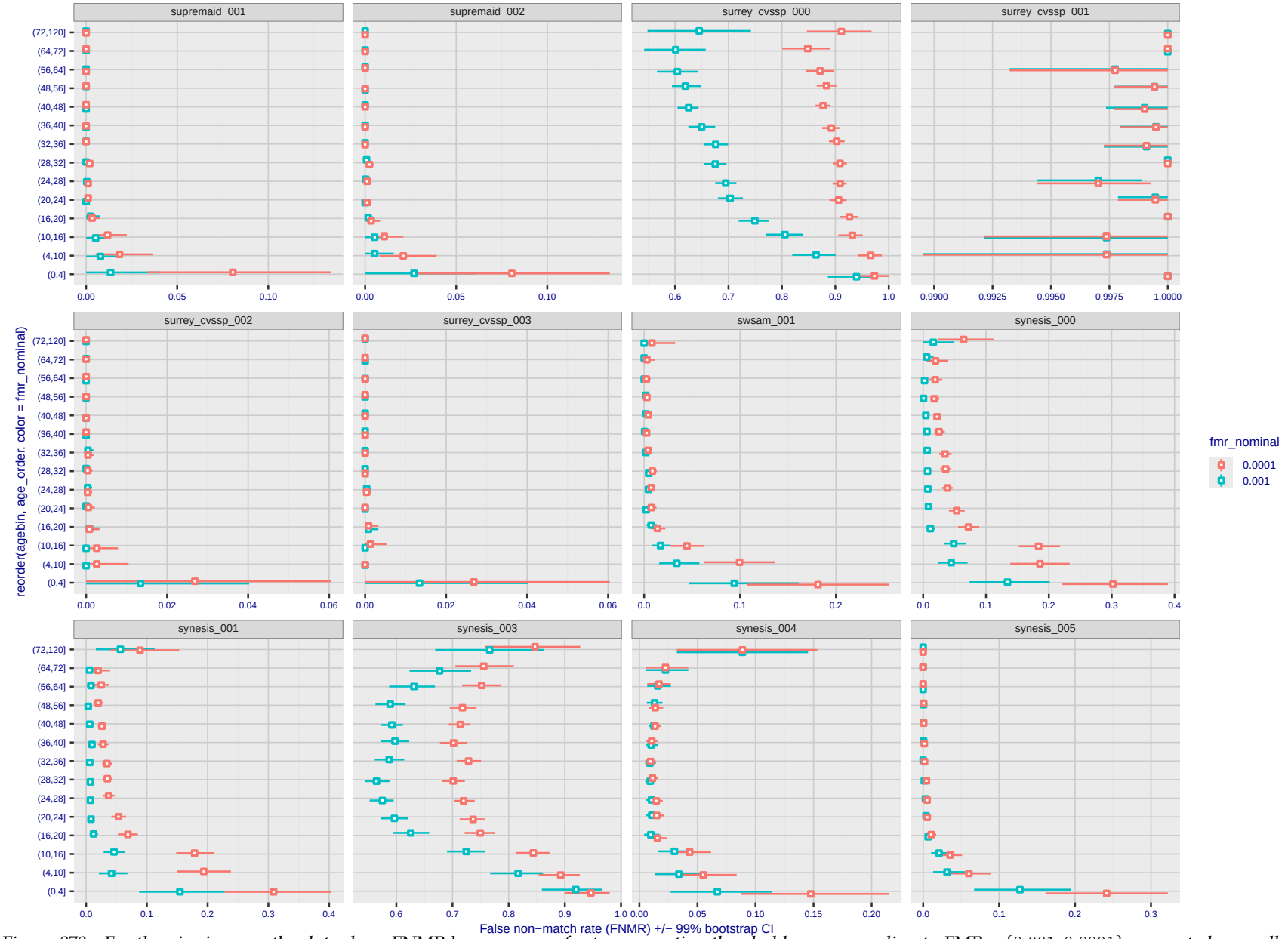

Figure 870: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 871: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 872: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 873: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

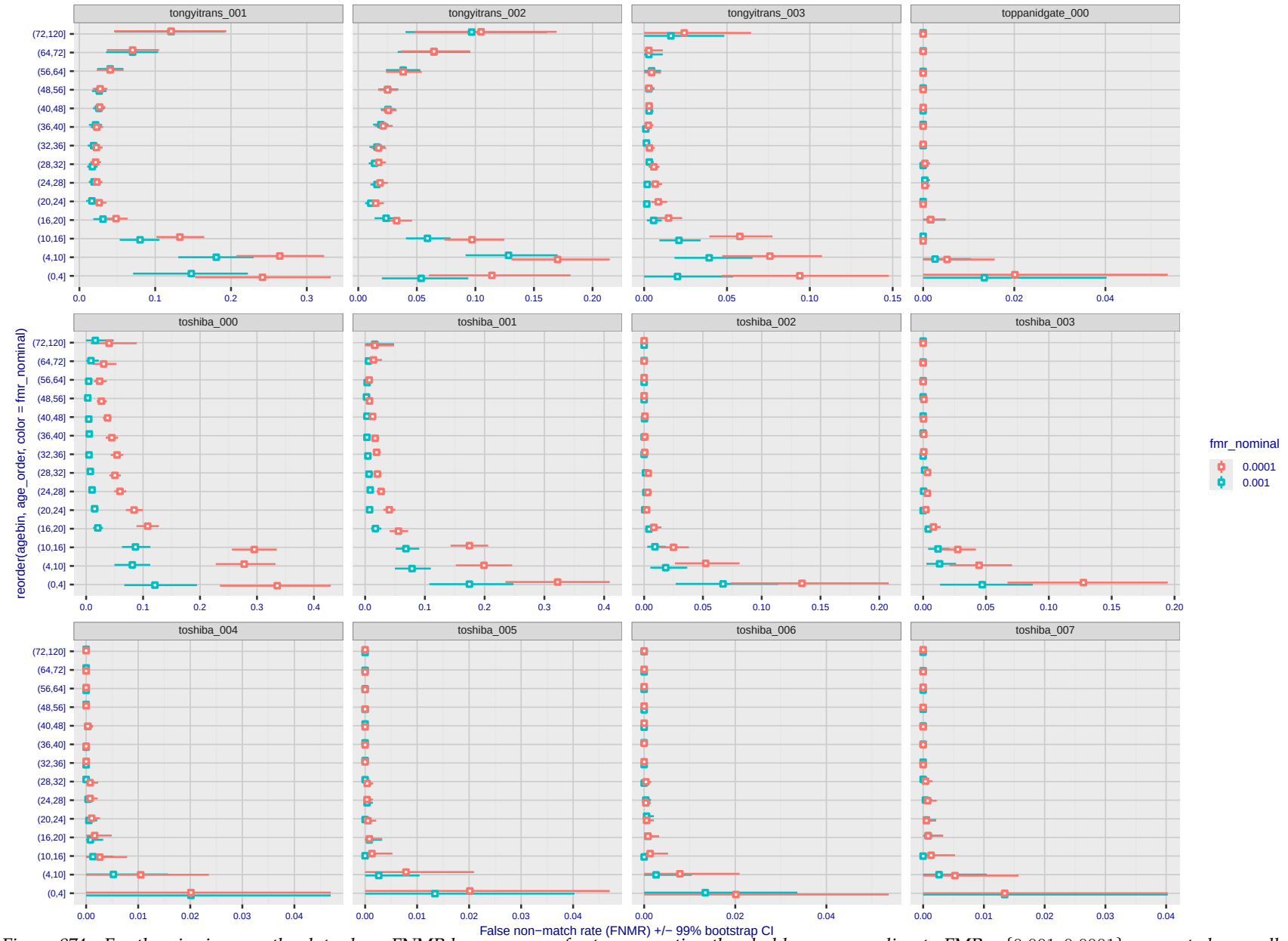

Figure 874: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

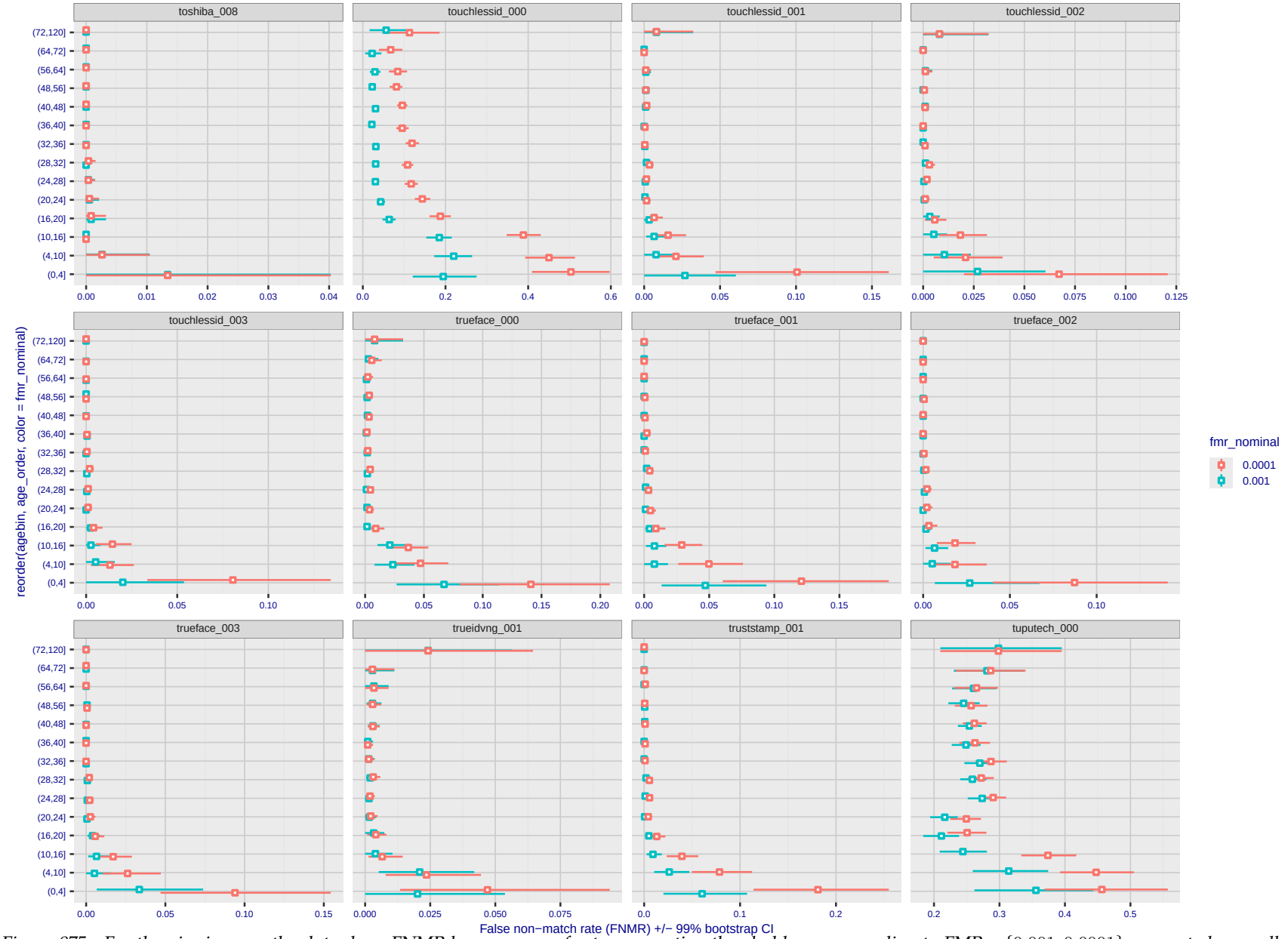

Figure 875: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 876: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

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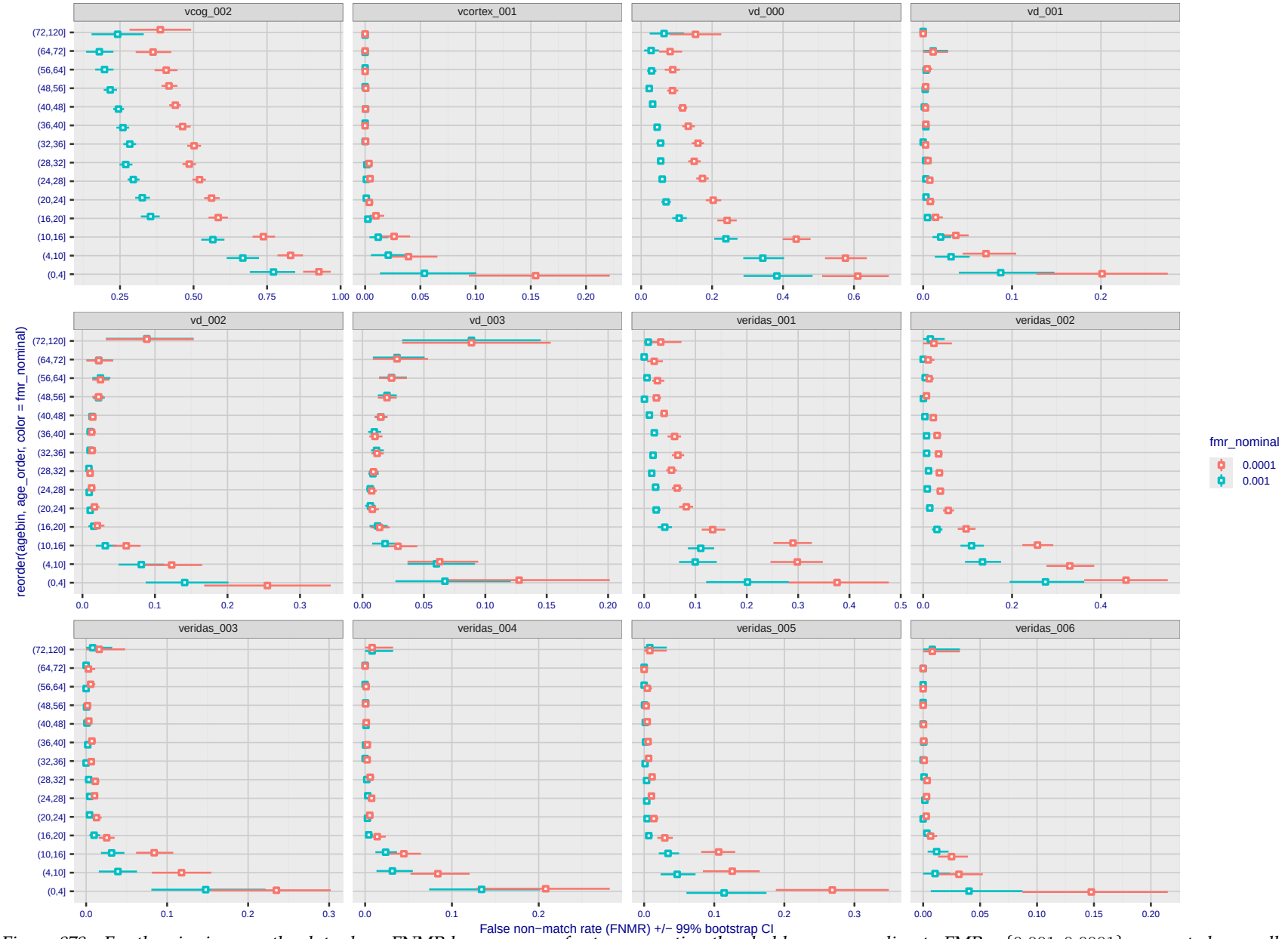


Figure 878: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 879: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 880: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

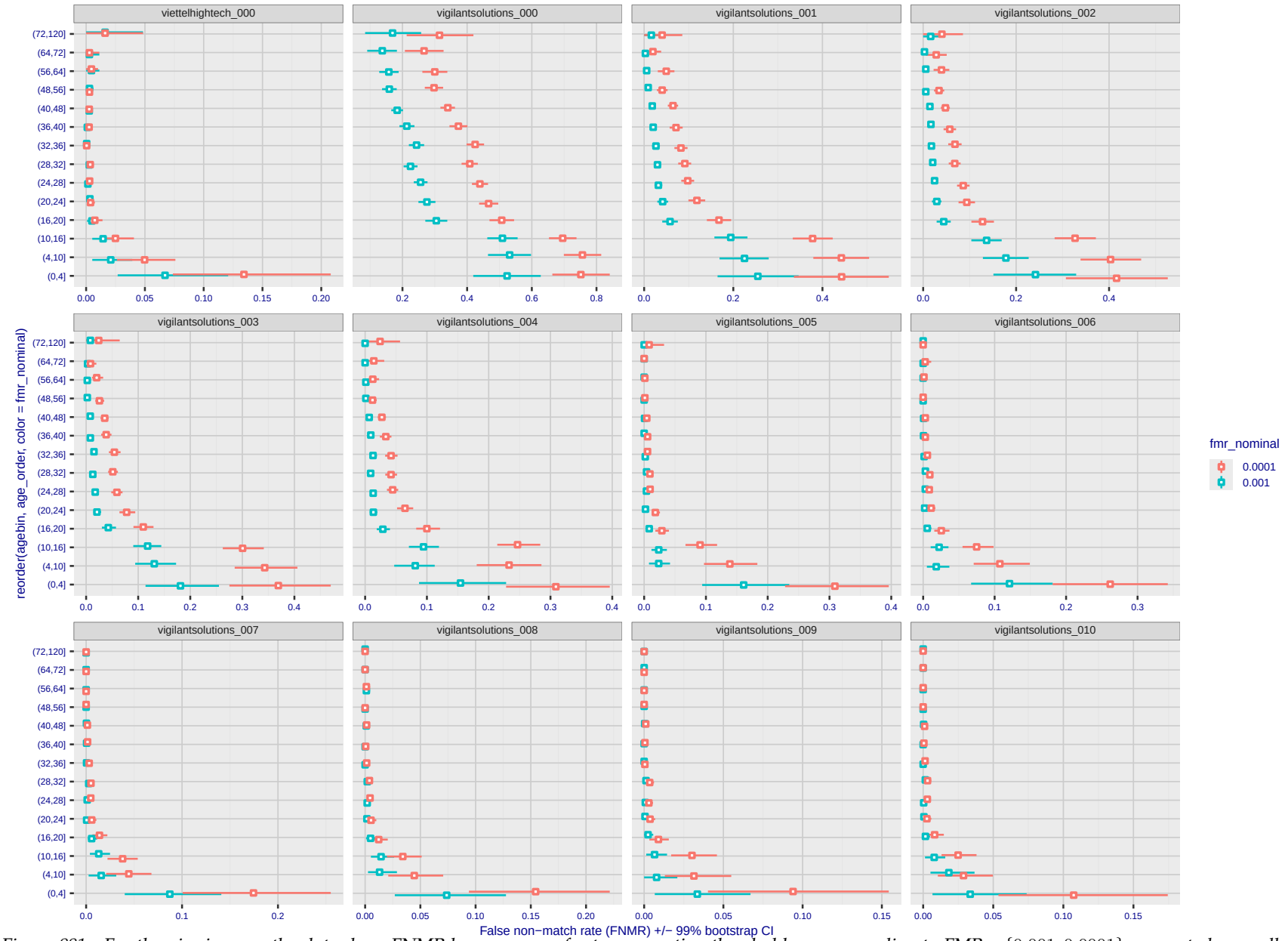


Figure 881: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 882: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

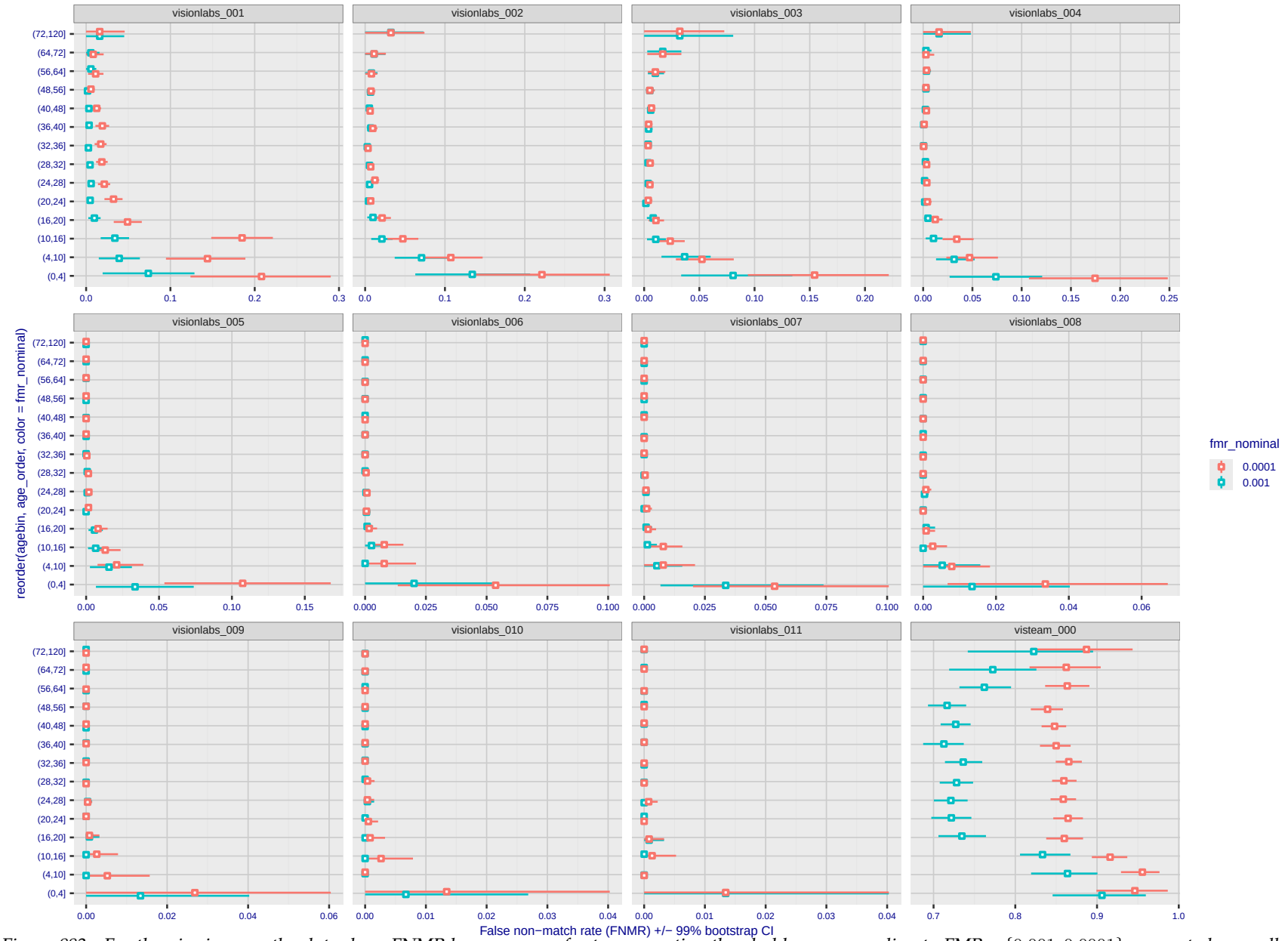


Figure 883: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 884: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 885: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 886: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

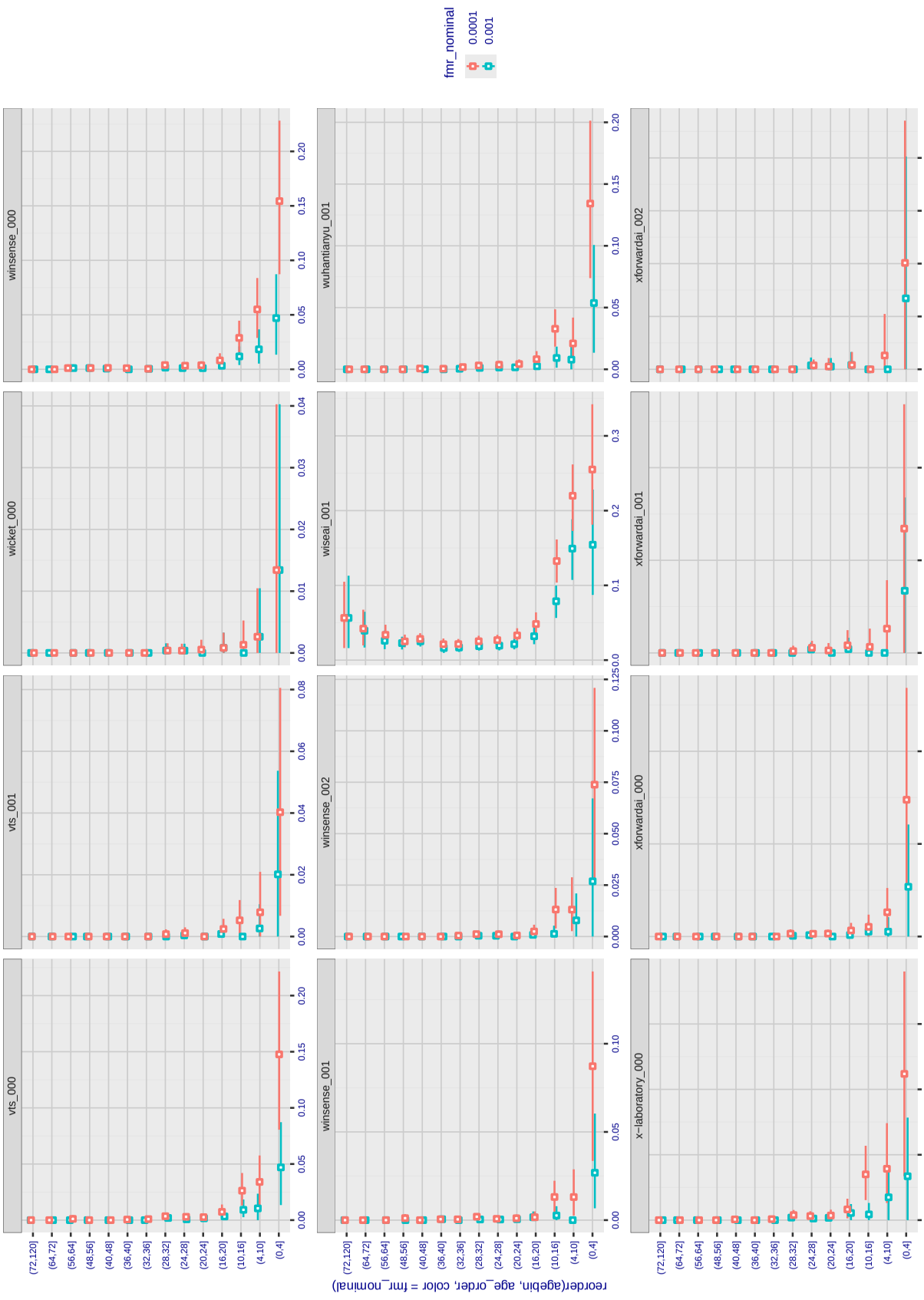


Figure 887: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 888: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

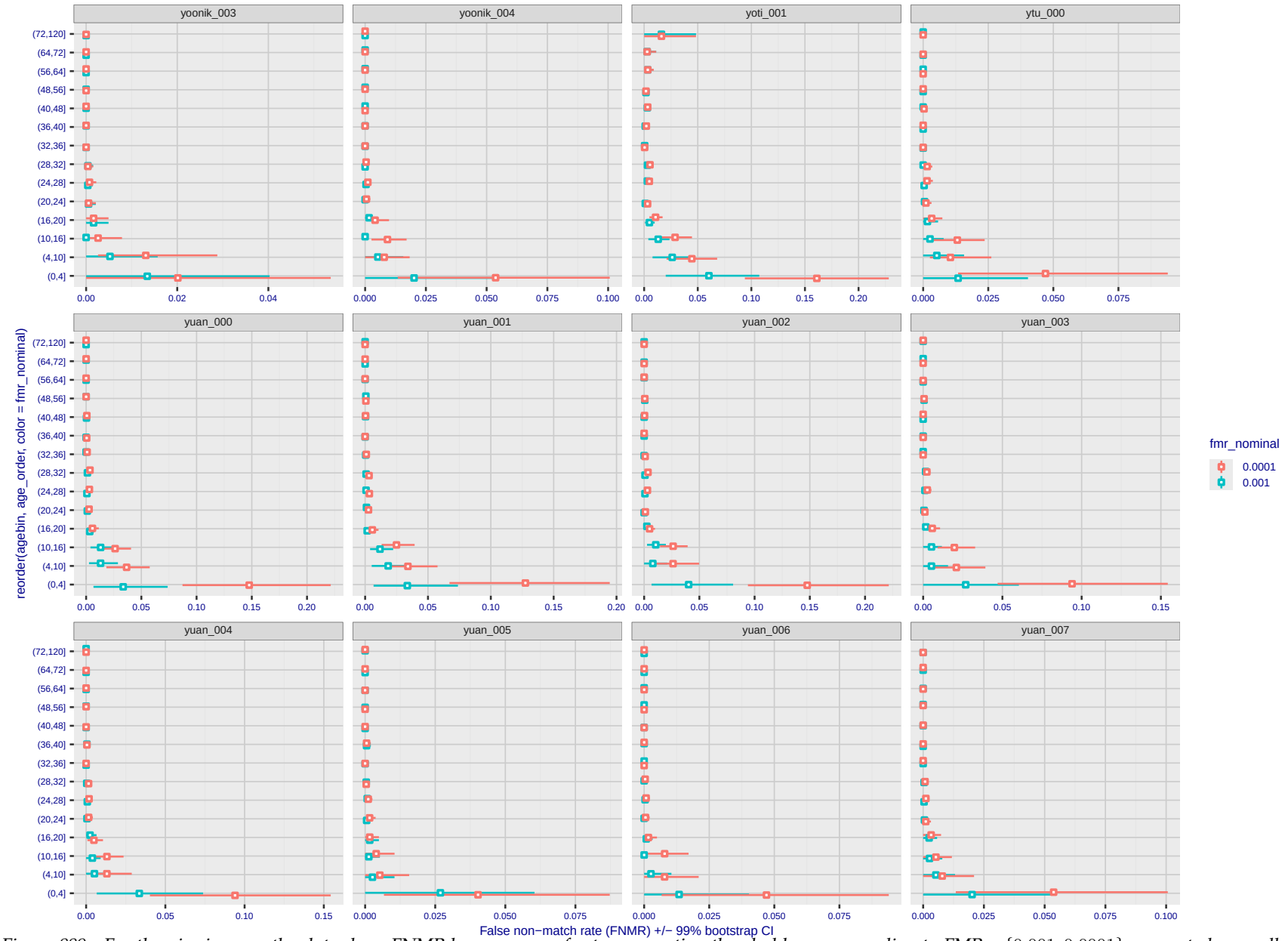


Figure 889: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

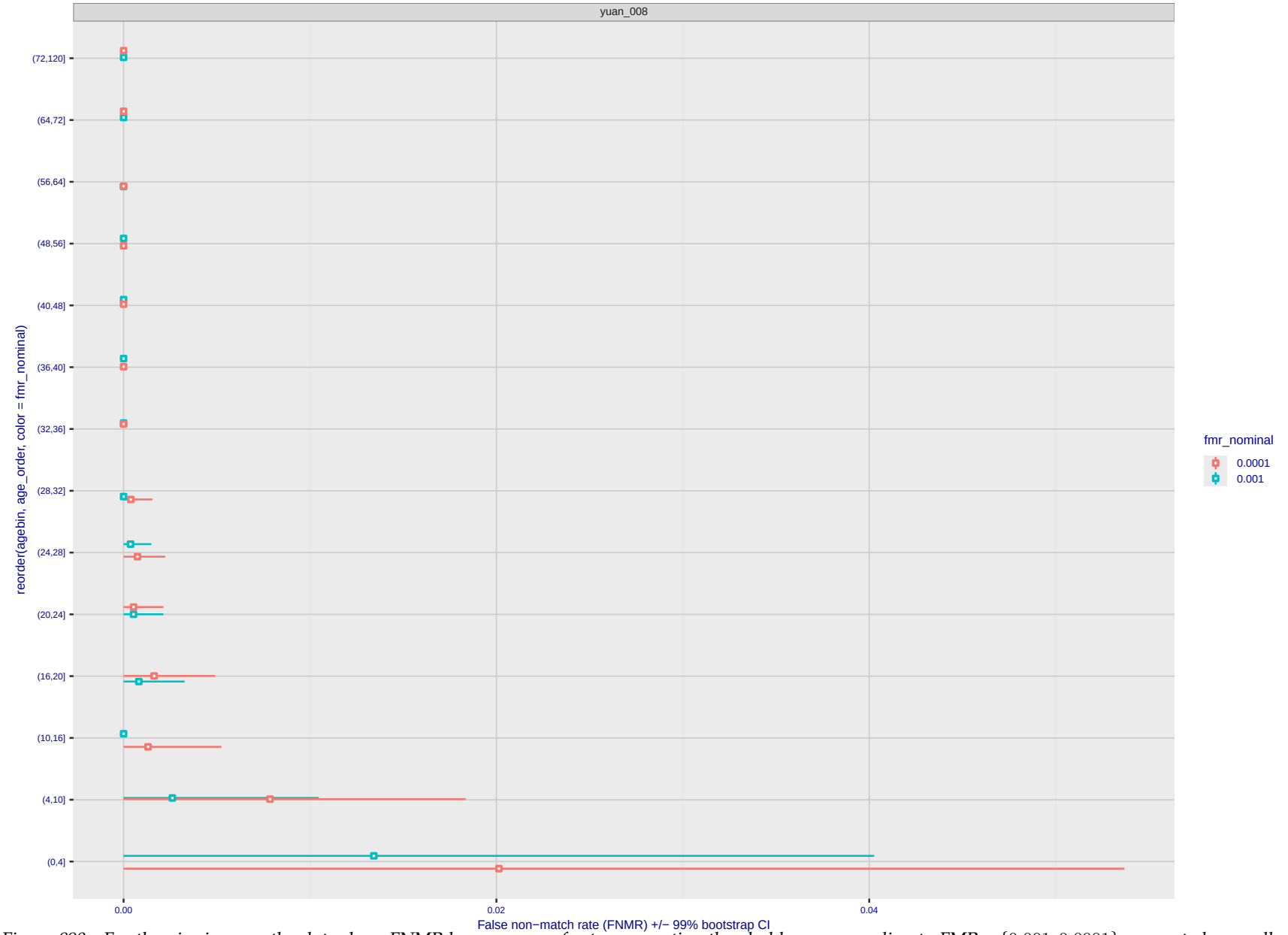


Figure 890: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Caveats: None.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.6 Impostor distribution stability

3.6.1 Effect of birth place on the impostor distribution

Background: Facial appearance varies geographically, both in terms of skin tone, cranio-facial structure and size. This section addresses whether false match rates vary intra- and inter-regionally.

Goals:

- ▷ To show the effect of birth region of the impostor and enrollee on false match rates.
- ▷ To determine whether some algorithms give better impostor distribution stability.

Methods:

- ▷ For the visa images, NIST defined 10 regions: Sub-Saharan Africa, South Asia, Polynesia, North Africa, Middle East, Europe, East Asia, Central and South America, Central Asia, and the Caribbean.
- ▷ For the visa images, NIST mapped each country of birth to a region. There is some arbitrariness to this. For example, Egypt could reasonably be assigned to the Middle East instead of North Africa. An alternative methodology could, for example, assign the Philippines to *both* Polynesia and East Asia.
- ▷ FMR is computed for cases where all face images of impostors born in region r_2 are compared with enrolled face images of persons born in region r_1.

$$\text{FMR}(r_1, r_2, T) = \frac{\sum_{i=1}^{N_{r_1,r_2}} H(s_i - T)}{N_{r_1,r_2}} \tag{5}$$

  where the same threshold, T, is used in all cells, and H is the unit step function. The threshold is set to give FMR(T) = 0.001 over the entire set of visa image impostor comparisons.
- ▷ This analysis is then repeated by country-pair, but only for those country pairs where both have at least 1000 images available. The countries[1] appear in the axes of graphs that follow.
- ▷ The mean number of impostor scores in any cross-region bin is 33 million. The smallest number of impostor scores in any bin is 135000, for Central Asia - North Africa. While these counts are large enough to support reasonable significance, the number of individual faces is much smaller, on the order of $N^{0.5}$.
- ▷ The numbers of impostor scores in any cross-country bin is shown in Figure 891.

Results: Subsequent figures show heatmaps that use color to represent the base-10 logarithm of the false match rate. Red colors indicate high (bad) false match rates. Dark colors indicate benign false match rates. There are two series of graphs corresponding to aggregated geographical regions, and to countries. The notable observations are:

- ▷ The on-diagonal elements correspond to within-region impostors. FMR is generally above the nominal value of FMR = 0.001. Particularly there is usually higher FMR in, Sub-Saharan Africa, South Asia, and the Caribbean. Europe and Central Asia, on the other hand, usually give FMR closer to the nominal value.
- ▷ The off-diagonal elements correspond to across-region impostors. The highest FMR is produced between the Caribbean and Sub-Saharan Africa.
- ▷ Algorithms vary.

[1] These are Argentina, Australia, Brazil, Chile, China, Costa Rica, Cuba, Czech Republic, Dominican Republic, Ecuador, Egypt, El Salvador, Germany, Ghana, Great Britain, Greece, Guatamala, Haiti, Hong Kong, Honduras, Indonesia, India, Israel, Jamaica, Japan, Kenya, Korea, Lebanon, Mexico, Malaysia, Nepal, Nigeria, Peru, Philippines, Pakistan, Poland, Romania, Russia, South Africa, Saudi Arabia, Thailand, Trinidad, Turkey, Taiwan, Ukraine, Venezuela, and Vietnam.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

▷ We computed the same quantities for a global FMR = 0.0001. The effects are similar.

Caveats:

▷ The effects of variable impostor rates on one-to-many identification systems may well differ from what's implied by these one-to-one verification results. Two reasons for this are a) the enrollment galleries are usually imbalanced across countries of birth, age and sex; b) one-to-many identification algorithms often implement techniques aimed at stabilizing the impostor distribution. Further research is necessary.

▷ In principle, the effects seen in this subsection could be due to differences in the image capture process. We consider this unlikely since the effects are maintained across geography - e.g. Caribbean vs. Africa, or Japan vs. China.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

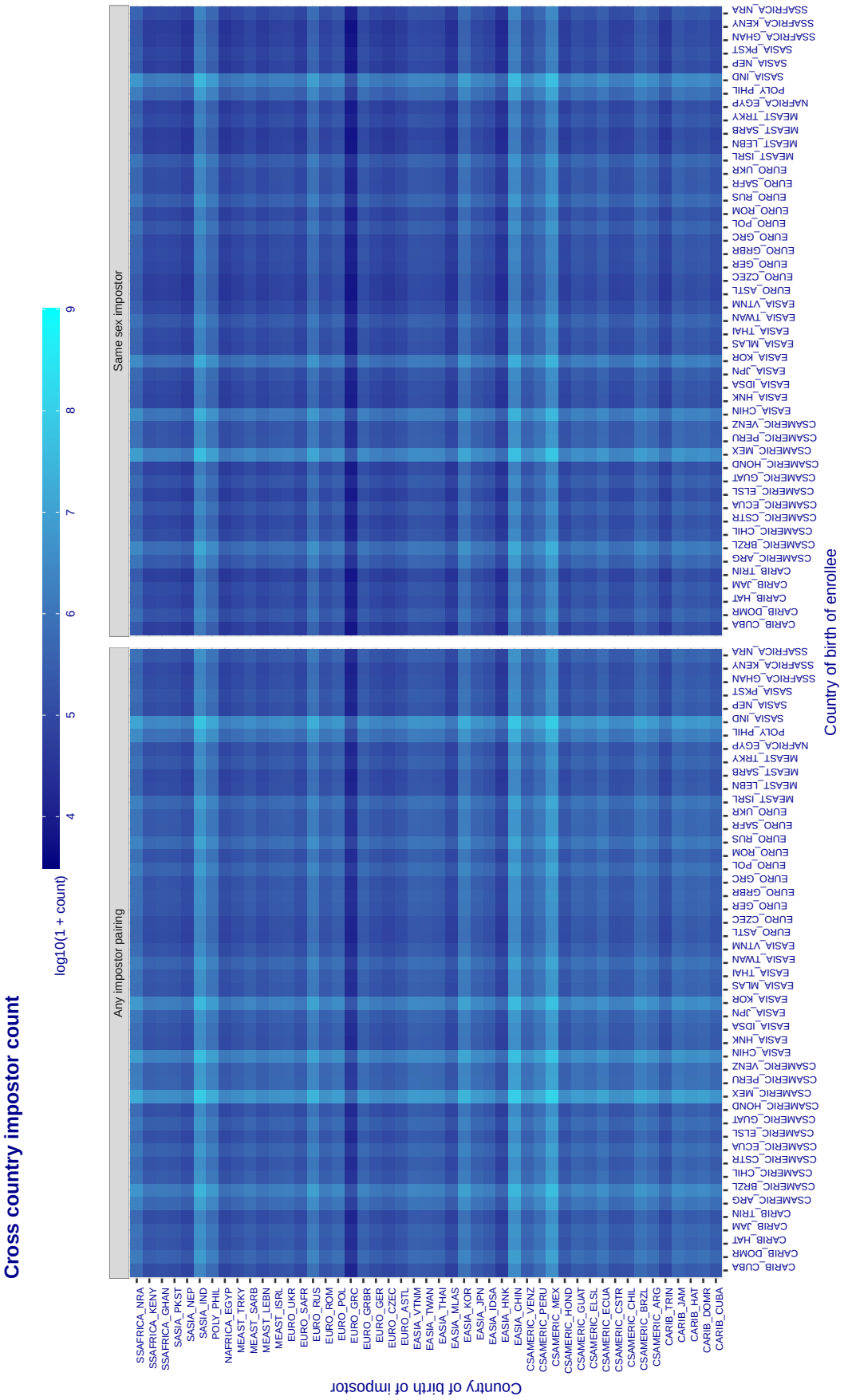


Figure 891: For visa images, the heatmap shows the count of impostor comparisons of faces from different individuals who were born in the given country pair. The FMR heatmaps themselves appear in the 1:1 report cards, for example, this one.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.6.2 Effect of age on impostors

Background: This section shows the effect of age on the impostor distribution. The ideal behaviour is that the age of the enrollee and the impostor would not affect impostor scores. This would support FMR stability over sub-populations.

Goals:

- ▷ To show the effect of relative ages of the impostor and enrollee on false match rates.
- ▷ To determine whether some algorithms have better impostor distribution stability.

Methods:

- ▷ Define 14 age group bins, spanning 0 to over 100 years old.
- ▷ Compute FMR over all impostor comparisons for which the subjects in the enrollee and impostor images have ages in two bins.
- ▷ Compute FMR over all impostor comparisons for which the subjects are additionally of the same sex, and born in the same geographic region.

Results:

The notable aspects are:

- ▷ Diagonal dominance: Impostors are more likely to be matched against their same age group.
- ▷ Same sex and same region impostors are more successful. On the diagonal, an impostor is more likely to succeed by posing as someone of the same sex. If $\Delta \log_{10}$ FMR $= 0.2$, then same-sex same-region FMR exceeds the all-pairs FMR by factor of $10^{0.2} = 1.6$.
- ▷ Young children impostors give elevated FMR against young children. Older adult impostor give elevated FMR against older adults. These effects are quite large, for example if $\Delta \log_{10}$ FMR $= 1.0$ larger than a 32 year old, then these groups have higher FMR by a factor of $10^1 = 10$. This would imply an FMR above 0.01 for a nominal (global) FMR = 0.001.
- ▷ Algorithms vary.
- ▷ We computed the same quantities for a global FMR = 0.0001. The effects are similar.

Note the calculations in this section include impostors paired across all countries of birth.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Accuracy Terms + Definitions

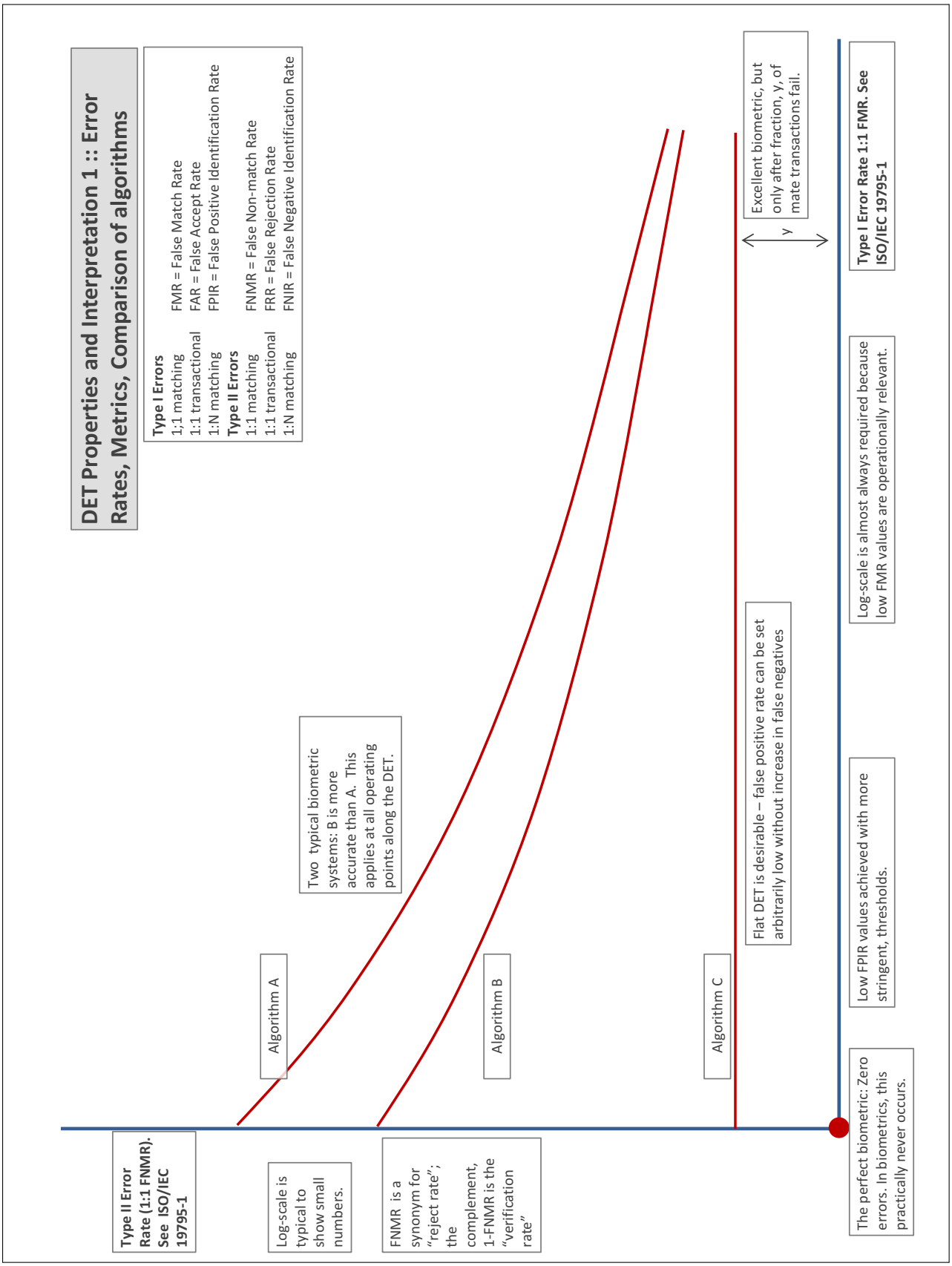
In biometrics, Type II errors occur when two samples of one person do not match – this is called a **false negative**. Correspondingly, Type I errors occur when samples from two persons do match – this is called a **false positive**. Matches are declared by a biometric system when the native comparison score from the recognition algorithm meets some **threshold**. Comparison scores can be either **similarity scores**, in which case higher values indicate that the samples are more likely to come from the same person, or **dissimilarity scores**, in which case higher values indicate different people. Similarity scores are traditionally computed by **fingerprint** and **face** recognition algorithms, while dissimilarities are used in **iris recognition**. In some cases, the dissimilarity score is a distance; this applies only when **metric** properties are obeyed. In any case, scores can be either **mate** scores, coming from a comparison of one person's samples, or **nonmate** scores, coming from comparison of different persons' samples. The words **genuine** or **authentic** are synonyms for mate, and the word **impostor** is used a synonym for nonmate. The words mate and nonmate are traditionally used in identification applications (such as law enforcement search, or background checks) while genuine and impostor are used in verification applications (such as access control).

A **error tradeoff** characteristic represents the tradeoff between Type II and Type I classification errors. For verification this plots false non-match rate (FNMR) vs. false match rate (FMR) parametrically with T.

The error tradeoff plots are often called **detection error tradeoff (DET)** characteristics or **receiver operating characteristic** (ROC). These serve the same function but differ, for example, in plotting the complement of an error rate (e.g. TMR = 1 – FNMR) and in transforming the axes most commonly using logarithms, to show multiple decades of FMR. More rarely, the function might be the inverse Gaussian function.

More detail and generality is provided in formal biometrics testing standards, see the various parts of ISO/IEC 19795 Biometrics Testing and Reporting. More terms, including and beyond those to do with accuracy, see ISO/IEC 2382-37 Information technology -- Vocabulary -- Part 37: Harmonized biometric vocabulary

FNMR(T) "False non-match rate"
FMR(T) "False match rate"



FNMR(T) "False non-match rate"
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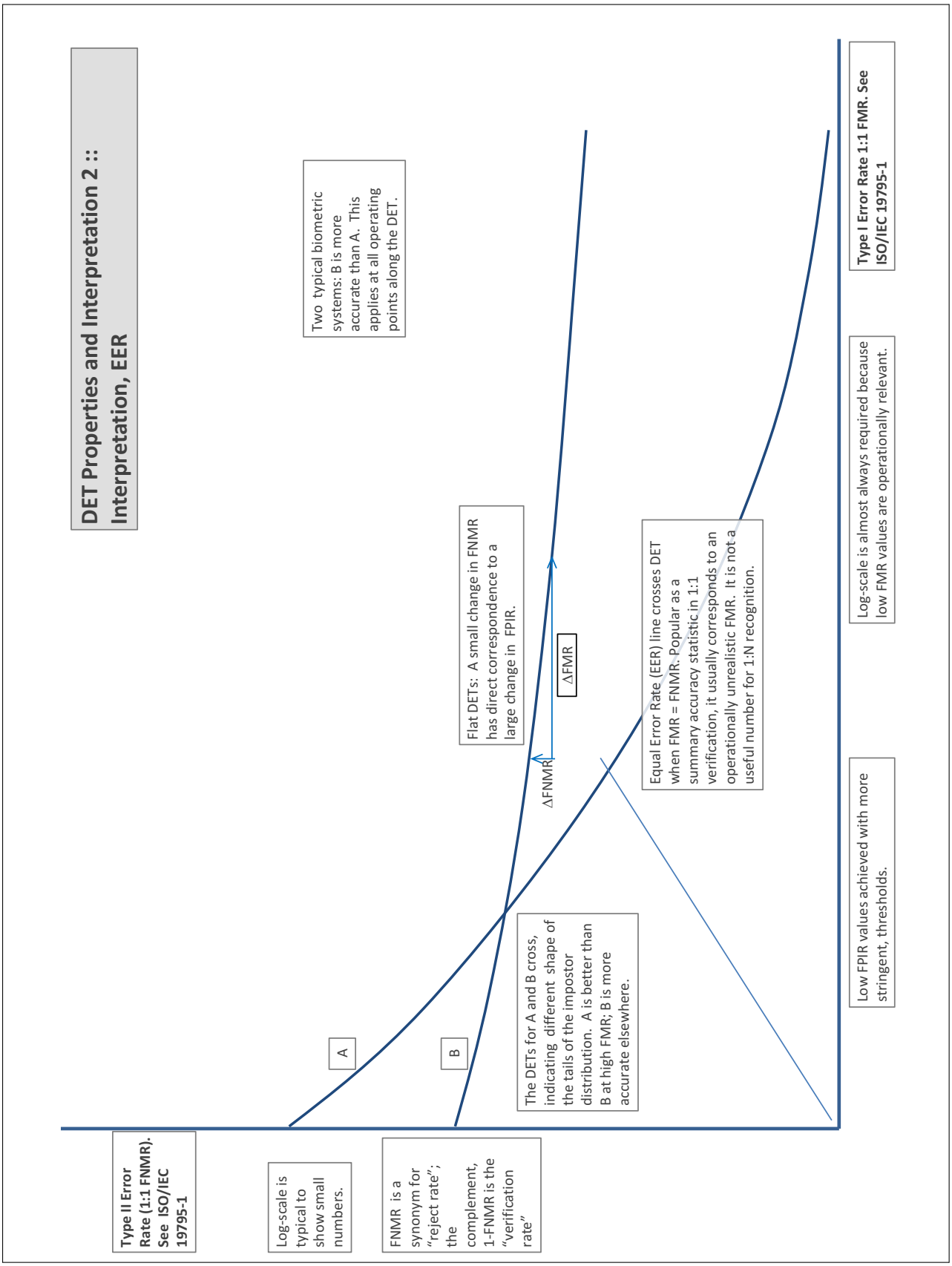

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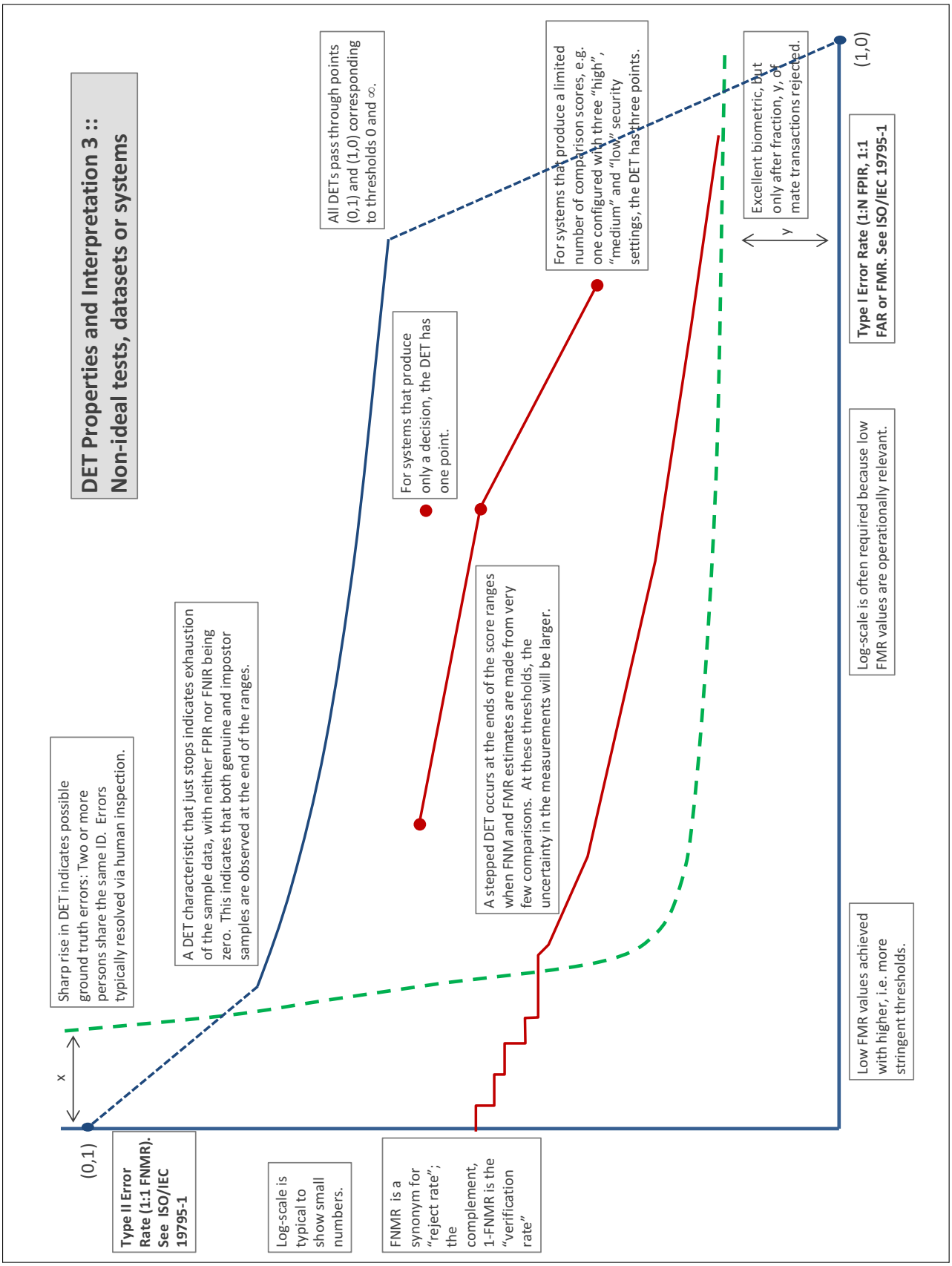
FMR(T) "False match rate"



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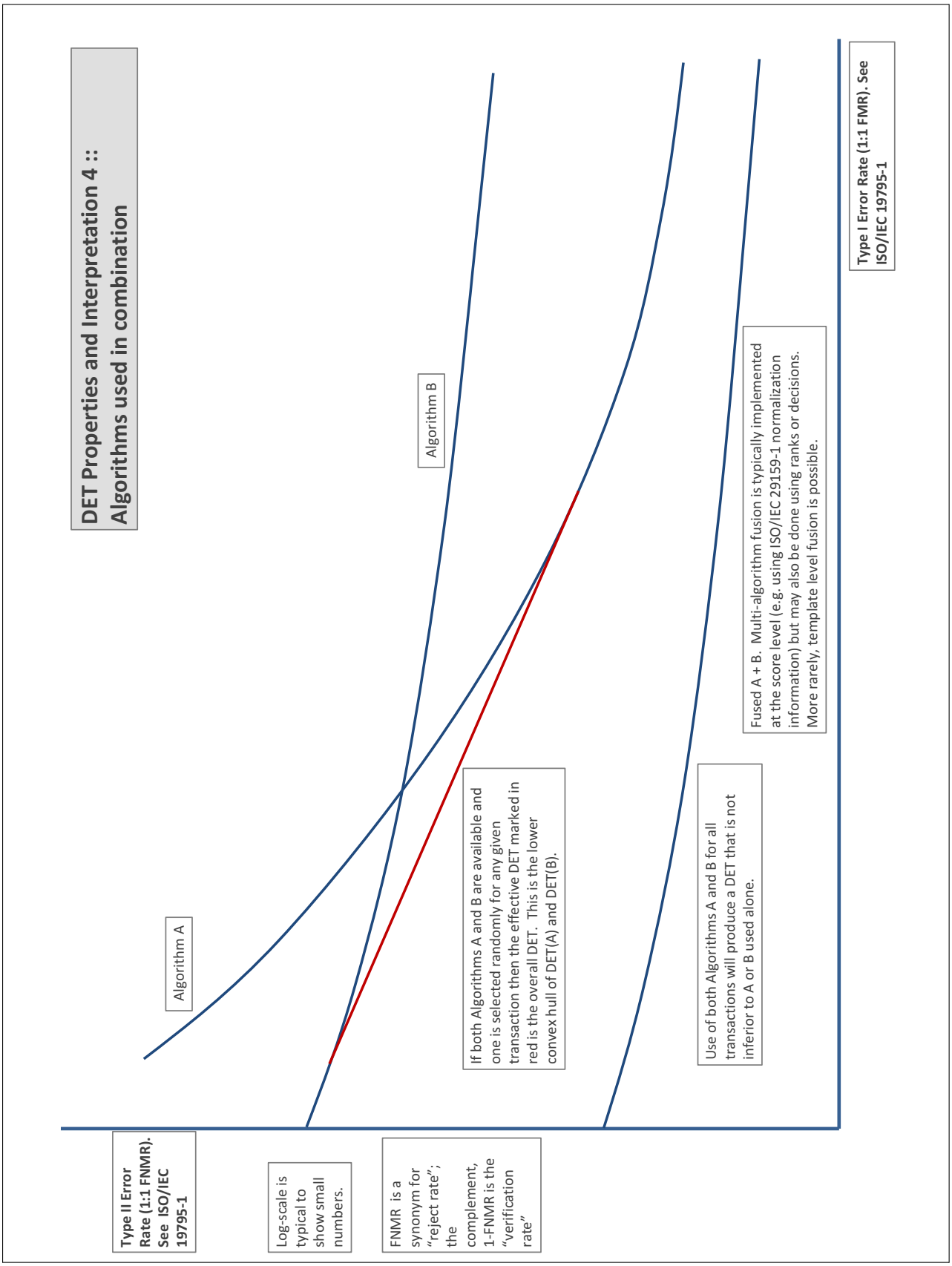

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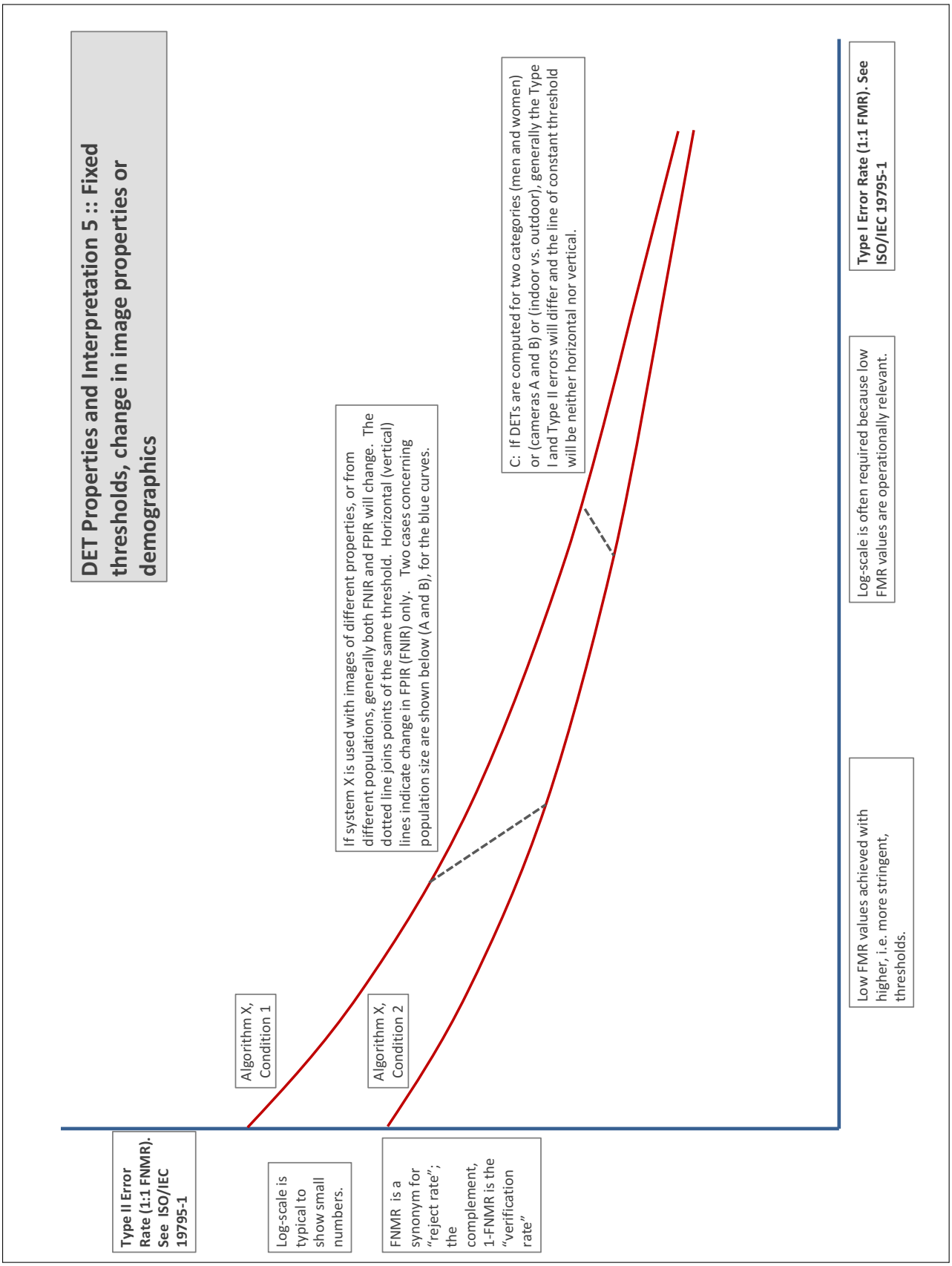

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FMR(T) "False match rate"

References

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"